Advanced Homogeneous Dynamics and Rigidity after Ratner Volumes 1–70

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Shared Bibliography 1929

Volume 1

Quantitative Non-Divergence and Height Functions

Student orientation: the nondivergence machine

Main question. When a parametrized lattice moves in a noncompact homogeneous space, how can one prove that only a controlled fraction of parameters enter a deep cusp? The reusable mechanism is: primitive subgroups are encoded by exterior covolumes; those covolumes are shown to be good functions; the marking theorem converts simultaneous goodness into quantitative nondivergence. **Terminology before use.** A *primitive subgroup* of $\mathbb{Z}^d$ is a subgroup equal to its saturation in its real span. Its *exterior covolume* is the norm of the primitive wedge representing it. A function is (C, α)-*good* on a ball when every sublevel set occupies at most $C(\varepsilon/\sup|f|)^\alpha$ of that ball. A *height* is a proper cusp-depth function; throughout the volume, height estimates are ultimately estimates for short primitive wedges. **How to read the proofs.** Separate the combinatorial marking argument from the analytic verification of goodness and noncollapse. The former is dimension-free bookkeeping; the latter is where the geometry of the parametrization enters.

Reader guide: a concrete model

Why this matters.

Volume 1 is organized around the passage from *Cusp geometry in the space of lattices* to *Source map and bibliographic notes*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.1 (A calculation to keep in view)**.** Consider the standard unipotent and diagonal elements

$$u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad a_s = \begin{pmatrix} e^s & 0 \\ 0 & e^{-s} \end{pmatrix}.$$

A direct multiplication gives

$$a_s u_t a_{-s} = u_{e^{2s}t}.$$

Thus a displacement of size $|t|$ in the unstable horospherical direction is enlarged by e^{2s}. If $|t| = e^{-2s}$, the conjugated displacement has order one. This elementary calculation is the model for the "first visible scale" used repeatedly in nondivergence and shearing arguments.

Preface: what quantitative non-divergence is really for

This volume begins at the point where the qualitative compactness theory used in Ratner's theorem becomes inadequate. In the classical theory one often needs to know that a unipotent orbit does not lose all of its mass in the cusp. For quantitative applications one needs a much stronger statement: for a whole family of lattices, and at an explicit cusp scale, how large is the parameter set on which some lattice vector becomes short?

The answer developed by Dani, Margulis, Kleinbock and Margulis is one of the most reusable pieces of technology in homogeneous dynamics. It rests on three ideas that are easy to state but whose interaction is subtle.

(1) **Cusp geometry is arithmetic.** A lattice escapes compact sets when an integral direction becomes short. During a quantitative covering argument, however, the stable obstruction may be a rational subspace of any rank, not a single vector. Exterior powers encode all such rational subspaces simultaneously.

(2) **Parameter functions must resist concentration near zero.** Polynomial and finite-order nondegenerate functions have the property that the set where they are much smaller than their typical size has power-law small measure. This is the role of (C, α)-good functions.

(3) **Anti-concentration is useless without non-collapse.** A function can be perfectly polynomial and nevertheless be tiny everywhere. The lower-supremum hypothesis in quantitative non-divergence is therefore logically independent of goodness.

The central theorem is the Kleinbock–Margulis quantitative non-divergence estimate. The abstract marking theorem and its deduction of quantitative non-divergence are proved here in full. The goal is not to reproduce a source paper line by line. Instead, each technical step is reorganized around the question it answers: why enlarge a parameter ball? why does the finite $3r$ covering selection sum the maximal windows? why are chains of primitive subgroups the correct finite-complexity object? and why does a smallness restriction on the threshold enter the final short-vector argument?

Prerequisites. We assume the foundations developed in the Ratner-theorem project: Lie groups and Lie algebras, homogeneous spaces G/Γ, Haar measure, lattices, the basic geometry of $\mathrm{SL}_2(\mathbb{R})$, and the meaning of unipotent one-parameter subgroups. Whenever a result from that background is used in an essential calculation, the needed formulation is restated. The present volume does not assume any other book in this sequel.

Theorem-scope convention

Every major theorem is tagged implicitly or explicitly as one of the following.

- Proved in this text: all ingredients are proved here.
- Derived here from stated inputs: the result is deduced here from clearly stated standard inputs.
- Standard foundational theorem: the exact theorem is imported because reproducing its independent theory would obscure the present mechanism.
- Frontier / source-status sensitive: reserved for results whose current source or exact hypotheses require special care. This label is not used as a hidden premise in the main proofs.

The exposition is intentionally asymmetric. A short lemma that carries the main idea may receive several pages of motivation and examples; routine multilinear bookkeeping is compressed once the mechanism is clear. The standard of rigor is nevertheless proof-level: hidden hypotheses, quotient conventions, norms, and parameter dependences are stated rather than inferred.

Proof and provenance. This is a recovered-source edition. The original 87-page PDF and its release record survived, while the original source archive did not remain materializable in the active runtime. The chapter architecture, theorem statements, normalizations, and proof dependencies have therefore been reconstructed from the exact generated PDF and the surviving shared series style. No claim of byte-for-byte identity with the lost source is made.

CHAPTER 1

Cusp geometry in the space of lattices

1. The model homogeneous space

Put

$$G = \mathrm{SL}_k(\mathbb{R}), \qquad \Gamma = \mathrm{SL}_k(\mathbb{Z}), \qquad X_k = G/\Gamma.$$

A point $g\Gamma$ is identified with the unimodular lattice

$$\Lambda_g = g\mathbb{Z}^k.$$

This is well defined because right multiplication by $\gamma \in \mathrm{SL}_k(\mathbb{Z})$ merely changes the integral basis. Conversely every unimodular lattice admits an oriented basis of determinant one, so every such lattice occurs.

The quotient X_k is noncompact. The simplest escaping sequence is

$$a_t\mathbb{Z}^2, \qquad a_t = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix},$$

for which $a_t e_2$ has norm e^{-t}. A fixed-covolume lattice can therefore degenerate by becoming extremely thin in one direction while expanding in another.

Definition 1.1 (Systole)**.** For a lattice $\Lambda \subset \mathbb{R}^k$ and a norm $\|\cdot\|$ on $\mathbb{R}^k$, define

$$\delta(\Lambda) = \min\{\|v\| : v \in \Lambda \setminus \{0\}\}.$$

The minimum exists because Λ is discrete.

We use Euclidean norm for geometric explanations and the coordinate maximum norm

$$\|x\|_\infty = \max_i |x_i|$$

for the exact lattice non-divergence statement. Compactness is independent of this choice; explicit constants are not.

2. Successive minima

Let $K \subset \mathbb{R}^k$ be the Euclidean unit ball. For a full-rank lattice Λ define

$$\lambda_i(\Lambda) = \inf\{r > 0 : \dim_{\mathbb{R}} \mathrm{span}_{\mathbb{R}}(\Lambda \cap rK) \geq i\}, \qquad 1 \leq i \leq k.$$

Then

$$0 < \lambda_1(\Lambda) \leq \cdots \leq \lambda_k(\Lambda), \qquad \lambda_1(\Lambda) = \delta(\Lambda).$$

The higher minima measure how many independent lattice directions are present at a given scale.

THEOREM 2.1 (Minkowski's second theorem, form used here). *For every k there are constants $c_k, C_k > 0$ such that every unimodular lattice $\Lambda \subset \mathbb{R}^k$ satisfies*

$$c_k \leq \prod_{i=1}^{k} \lambda_i(\Lambda) \leq C_k.$$

Proof and provenance. PROVED IN THIS TEXT. A complete dimension-dependent proof, including both inequalities, is given in Appendix A as Theorem 3.1. No external Minkowski-II input is used by Mahler compactness or the height comparison.

Lemma 2.2 (Controlled lattice basis)**.** *For each k there is $A_k \geq 1$ such that every full-rank lattice $\Lambda \subset \mathbb{R}^k$ has a basis $b_1, \ldots, b_k$ with*

$$\|b_i\| \leq A_k \lambda_i(\Lambda), \qquad 1 \leq i \leq k.$$

Proof and provenance. DERIVED HERE FROM STATED INPUTS. A complete induction from primitive-vector projection is given in Appendix A.

3. Mahler compactness

THEOREM 3.1 (Mahler compactness criterion). *For $A \subset X_k$, the following are equivalent.*

(1) A has compact closure in X_k.
(2) There exists $\eta > 0$ such that $\delta(\Lambda) \geq \eta$ for every $\Lambda \in A$.

PROOF. Assume first that A is compact. The systole is continuous on X_k. Indeed, if $g_n \to g$ in $\mathrm{SL}_k(\mathbb{R})$, then for a fixed large R and all sufficiently large n, any $v \in \mathbb{Z}^k$ with $\|g_n v\| \leq R$ lies in a fixed finite subset of $\mathbb{Z}^k$: both g_n and g_n^{-1} remain in a compact set, so $\|v\| \ll R$. On this finite set the functions $g \mapsto \|gv\|$ vary continuously, hence their minimum does also. Since the systole is positive at every lattice, it attains a positive minimum on a compact set.

Conversely suppose $\lambda_1(\Lambda) = \delta(\Lambda) \geq \eta$ for all $\Lambda \in A$. By monotonicity of successive minima and Theorem 2.1,

$$\eta^{k-1}\lambda_k(\Lambda) \leq \prod_{i=1}^{k} \lambda_i(\Lambda) \leq C_k,$$

so $\lambda_k(\Lambda) \leq C_k \eta^{-(k-1)}$. Lemma 2.2 gives a basis $b_1, \ldots, b_k$ of Λ with all $\|b_i\|$ bounded in terms of k, η. Change the sign of one basis vector if necessary and let $g_\Lambda \in \mathrm{SL}_k(\mathbb{R})$ be the matrix with columns b_i. Its entries lie in a fixed bounded set. Every convergent subsequence in $\mathrm{Mat}_k(\mathbb{R})$ has determinant one, hence nonsingular limit. Thus the representatives lie in a relatively compact subset of $\mathrm{SL}_k(\mathbb{R})$, whose image in X_k contains A. □

Why this matters. Mahler's theorem converts a topological statement about X_k into a Euclidean inequality. Quantitative non-divergence therefore becomes the problem of estimating the parameter set on which $\delta(h(x)\mathbb{Z}^k)$ is small.

4. A Diophantine cusp excursion in dimension two

Let

$$u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}, \qquad a_t = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}.$$

For $(p, q) \in \mathbb{Z}^2$,

$$a_t u_s \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} e^t(p + sq) \\ e^{-t} q \end{pmatrix}.$$

Hence $a_t u_s \mathbb{Z}^2$ contains a short vector precisely when some nonzero integer pair makes $e^t|p + sq|$ and $e^{-t}|q|$ simultaneously small. The second inequality bounds the denominator; the first asks s to lie near $-p/q$. Cusp excursions encode rational approximation.

The example also warns against choosing one short vector globally. The identity of the shortest vector can switch as s varies. Quantitative arguments therefore organize possible obstructions by the rational subspaces they span, not by a naive union over infinitely many individual vectors.

5. Compact sets at a chosen cusp depth

For $0 < \eta < 1$ set

$$K_\eta = \{\Lambda \in X_k : \delta(\Lambda) \geq \eta\}.$$

By Mahler, K_η is compact. An estimate

$$|\{x \in B : h(x)\mathbb{Z}^k \notin K_\eta\}| \leq A\eta^\alpha |B|$$

is therefore a quantitative cusp-tail estimate, not merely qualitative nonescape.

Warning 5.1. No fixed compact set need contain an entire recurrent unipotent orbit. Non-divergence controls the *proportion* of time or parameter measure outside a compact set chosen according to the tolerated error; it is not boundedness of the orbit.

A local coordinate view of the quotient. Near a fixed lattice $g\mathbb{Z}^k$, choose a sufficiently small neighborhood W of the identity in G such that

$$W^{-1}W \cap g\Gamma g^{-1} = \{e\}.$$

Then $w \mapsto wg\Gamma$ identifies W homeomorphically with a neighborhood of $g\Gamma$. In these coordinates the systole is the minimum of finitely many continuous functions: after shrinking W, a vector wgv can have norm below a fixed threshold only for v in a finite subset of $\mathbb{Z}^k$. This gives another proof of continuity and explains why local compactness questions in X_k reduce to finitely many integral vectors even though the global lattice contains infinitely many.

The same coordinate argument shows that on a compact $K \subset X_k$ there are uniform constants controlling both g and g^{-1} after choosing representatives in finitely many local sections. This observation will later be used repeatedly without comment when compact starting sets are converted into uniform exterior-power lower bounds.

Why covolume one matters. Without fixing covolume, a lattice could escape simply by uniform scalar contraction. The determinant-one condition removes that trivial direction and forces every contraction to be compensated elsewhere. Minkowski's second theorem is the quantitative form of this balance: if λ_1 becomes small while the product of the minima remains comparable to one, at least one higher minimum must grow. The cusp is therefore anisotropic degeneration, not a change of total volume.

CHAPTER 2

Primitive subgroups, rational subspaces, and exterior powers

1. Primitive subgroups

Let $\Delta \le \mathbb{Z}^k$ have rank r and put $\Delta_{\mathbb{R}} = \operatorname{span}_{\mathbb{R}} \Delta$.

Definition 1.1. Δ is *primitive* if

$$\Delta = \Delta_{\mathbb{R}} \cap \mathbb{Z}^k.$$

This removes irrelevant multiplicity: $2\mathbb{Z}e_1$ and $\mathbb{Z}e_1$ span the same rational line, but only the latter is primitive.

Proposition 1.2 (Equivalent formulations). *For a rank-r subgroup $\Delta \le \mathbb{Z}^k$, the following are equivalent.*

(1) Δ *is primitive.*
(2) $\mathbb{Z}^k/\Delta$ *has no torsion.*
(3) Every $\mathbb{Z}$*-basis of* Δ *extends to a* $\mathbb{Z}$*-basis of* $\mathbb{Z}^k$*.*
(4) All invariant factors in the Smith normal form of the inclusion are 1*.*

Proof. Choose a basis matrix for Δ and apply Smith normal form. After unimodular changes of coordinates the inclusion is represented by

$$\operatorname{diag}(d_1, \dots, d_r), \qquad d_1 \mid d_2 \mid \cdots \mid d_r.$$

Then

$$\mathbb{Z}^k/\Delta \cong \bigoplus_{i=1}^{r} \mathbb{Z}/d_i\mathbb{Z} \oplus \mathbb{Z}^{k-r}.$$

Thus torsion-freeness is equivalent to $d_i = 1$ for all i. In these coordinates the real span is generated by $e_1, \dots, e_r$, and its integral points are exactly their integral span, so primitivity is the same condition. The basis-extension assertion is likewise equivalent to all invariant factors being one. $\square$

2. Wedge representatives and covolume

If $v_1, \ldots, v_r$ is a basis of a rank-r discrete subgroup Δ, set

$$w_\Delta = v_1 \wedge \cdots \wedge v_r \in \bigwedge^r \mathbb{R}^k.$$

Changing the basis of Δ multiplies w_Δ by ± 1.

There are two useful norms. In the Euclidean exterior norm, the standard wedges e_I are orthonormal and

$$\|v_1 \wedge \cdots \wedge v_r\|_2^2 = \det(\langle v_i, v_j \rangle)_{i,j=1}^r,$$

the squared r-dimensional covolume. In the coordinate maximum norm,

$$\left\| \sum_I a_I e_I \right\|_\infty = \max_I |a_I|.$$

For either norm we write $\|\Delta\| = \|w_\Delta\|$ and, for $h \in \mathrm{GL}_k(\mathbb{R})$,

$$\|h\Delta\| = \left\| (\bigwedge^r h) w_\Delta \right\|.$$

Worked Example 2.1 (A rational plane in $\mathbb{R}^3$)**.** Let $\Delta = \mathbb{Z}(1,0,1) + \mathbb{Z}(0,1,1)$. Then

$$w_\Delta = e_1 \wedge e_2 + e_1 \wedge e_3 - e_2 \wedge e_3,$$

so $\|\Delta\|_\infty = 1$ while $\|\Delta\|_2 = \sqrt{3}$. The coefficients are the 2×2 minors of the basis matrix: the Plücker coordinates of the rational plane.

3. Why all ranks appear

A bad parameter for $h : B \to \mathrm{GL}_k(\mathbb{R})$ is one for which some $h(x)v$, $v \in \mathbb{Z}^k \setminus \{0\}$, is short. Even if every function $x \mapsto \|h(x)v\|$ has a good sublevel estimate, summing over infinitely many v loses control. When several candidate vectors are simultaneously small, their rational span is the stable datum. Passing from one independent vector to two is passing from rank one to rank two. Since ranks can increase only k times, the correct bookkeeping has finite depth.

4. A rank-extension inequality

Lemma 4.1 (Rank extension, maximum norm)**.** *Let $\Gamma \subset \mathbb{R}^k$ be discrete, let $v \notin \Gamma_\mathbb{R}$, and let Λ be a discrete subgroup containing Γ and v with*

$$\Lambda_\mathbb{R} = \Gamma_\mathbb{R} + \mathbb{R}v.$$

Then

$$\|\Lambda\|_\infty \le k\, \|\Gamma\|_\infty \, \|v\|_\infty, \qquad \|v\|_\infty \ge \frac{\|\Lambda\|_\infty}{k\, \|\Gamma\|_\infty}.$$

Proof. Let w represent Γ. The subgroup generated by a basis of Γ together with v may have finite index in Λ, hence its wedge norm is at least $\|\Lambda\|_\infty$. It therefore suffices to bound $w \wedge v$. Write

$$w = \sum_I w_I e_I, \qquad v = \sum_{j=1}^k v_j e_j.$$

Each coefficient of $w \wedge v$ is a signed sum of at most k terms $w_I v_j$, so

$$\|w \wedge v\|_\infty \leq k \|w\|_\infty \|v\|_\infty .$$

Combine the two inequalities. $\square$

Remark 4.2. For the Euclidean exterior norm, $\|w \wedge v\|_2 \leq \|w\|_2 \|v\|_2$. The factor k is an artifact of the maximum-coordinate normalization.

5. The poset of primitive subgroups

Let

$$\mathcal{L}(\mathbb{Z}^k) = \{\Delta \leq \mathbb{Z}^k : \Delta \neq 0,\ \Delta \text{ primitive}\}$$

ordered by inclusion. A strictly increasing chain has strictly increasing ranks and therefore length at most k. If T is a chain, let $\mathcal{L}(T)$ consist of the elements outside T comparable with every member of T. Any chain in $\mathcal{L}(T)$ can be adjoined to T, so

$$\ell(\mathcal{L}(T)) \leq k - |T|.$$

This trivial rank count is the descent parameter in the marking theorem.

6. Local finiteness of small wedge obstructions

Lemma 6.1. *Let $h \in \mathrm{GL}_k(\mathbb{R})$ and $R > 0$. Only finitely many primitive $\Delta \leq \mathbb{Z}^k$ satisfy $\|h\Delta\|_\infty < R$.*

Proof. For fixed rank r, primitive Δ is represented up to sign by a nonzero primitive decomposable vector $w_\Delta \in \bigwedge^r \mathbb{Z}^k$. Since $\bigwedge^r h$ is invertible,

$$\|w_\Delta\|_\infty \leq C(h,r) \left\| (\bigwedge^r h) w_\Delta \right\|_\infty < C(h,r)R.$$

Only finitely many integral vectors lie in a bounded subset of the finite-dimensional lattice $\bigwedge^r \mathbb{Z}^k$. Sum over $1 \leq r \leq k$. $\square$

Plücker coordinates and arithmetic discreteness. For a primitive rank-r subgroup, the coordinates of w_Δ in the standard wedge basis are the $r \times r$ minors of an integer basis matrix. They are integers with greatest common divisor one. Conversely, a primitive decomposable element of $\bigwedge^r \mathbb{Z}^k$, up to sign, determines a primitive rational r-plane and hence a primitive subgroup. Thus the passage

$$\Delta \longleftrightarrow [w_\Delta]$$

embeds the arithmetic Grassmannian into a projective exterior representation. Local finiteness in Lemma 6.1 is simply discreteness of integral Plücker coordinates after applying an invertible linear map.

This viewpoint also explains why exterior powers are so effective in homogeneous dynamics. A nonlinear geometric object—a rational subspace—becomes a line in a linear representation. Covolume becomes a norm, incidence becomes wedge vanishing, and the action of G becomes linear. Later linearization arguments use the same principle at a more sophisticated level.

Intersection and sum. If Δ_1, Δ_2 are primitive, their intersection is primitive because

$$(\Delta_1 \cap \Delta_2)_\mathbb{R} \cap \mathbb{Z}^k = \Delta_{1,\mathbb{R}} \cap \Delta_{2,\mathbb{R}} \cap \mathbb{Z}^k = \Delta_1 \cap \Delta_2.$$

Their sum need not be primitive; its saturation is

$$\operatorname{sat}(\Delta_1 + \Delta_2) = (\Delta_{1,\mathbb{R}} + \Delta_{2,\mathbb{R}}) \cap \mathbb{Z}^k.$$

This asymmetry is why the short-vector argument always says “take the primitive closure after adjoining v” rather than silently treating an abstract sum as a primitive label.

CHAPTER 3

Good functions: quantitative resistance to vanishing

1. Definition and first interpretation

Let $U \subset \mathbb{R}^d$ be open and $f : U \to \mathbb{R}$ continuous.

Definition 1.1. For $C, \alpha > 0$, f is (C, α)-good on U if for every ball $B \subset U$ and every $\varepsilon > 0$,

$$|\{x \in B : |f(x)| < \varepsilon\}| \le C \left(\frac{\varepsilon}{\|f\|_B}\right)^{\alpha} |B|, \qquad \|f\|_B := \sup_{x \in B} |f(x)|,$$

whenever $\|f\|_B > 0$.

The ratio is scale-free. Goodness is anti-concentration, not a lower bound: the constant function 10^{-1000} is good but uniformly tiny. This distinction is the reason non-collapse is a separate hypothesis later.

2. Closure properties

Lemma 2.1 (Scaling)**.** *If f is (C, α)-good and $c \neq 0$, then cf is (C, α)-good.*

Proof. Use $\{|cf| < \varepsilon\} = \{|f| < \varepsilon/|c|\}$ and $\|cf\|_B = |c| \, \|f\|_B$. ☐

Lemma 2.2 (Pointwise suprema)**.** *Let $\{f_i\}_{i \in I}$ be (C, α)-good and suppose $F(x) = \sup_i |f_i(x)|$ is finite. If on each ball the supremum norm is attained by one f_i, then F is (C, α)-good. In particular this holds for finite families.*

Proof. Choose i_0 with $\|F\|_B = \|f_{i_0}\|_B$. Then

$$\{F < \varepsilon\} \cap B \subset \{|f_{i_0}| < \varepsilon\} \cap B$$

and apply goodness to f_{i_0}. ☐

Corollary 2.3 (Vector norms)**.** *If $f_1, \dots, f_N$ are (C, α)-good, then $F_\infty = \max_i |f_i|$ is (C, α)-good and*

$$F_2 = (\sum_i |f_i|^2)^{1/2}$$

is $(CN^{\alpha/2}, \alpha)$-good.

Proof. The first assertion is Lemma 2.2. Since $F_\infty \le F_2 \le \sqrt{N} F_\infty$, one has $\{F_2 < \varepsilon\} \subset \{F_\infty < \varepsilon\}$ and $\|F_\infty\|_B \ge N^{-1/2} \|F_2\|_B$. □

3. Polynomials on an interval: a complete proof

Lemma 3.1 (Separated points). *Let $E \subset I$ be measurable with $|E| = m > 0$, I an interval. For every $n \ge 1$ there exist $n+1$ points $x_0, \dots, x_n \in \overline{E}$ such that*

$$|x_i - x_j| \ge \frac{m}{4n} \qquad (i \ne j).$$

Proof. Put $\delta = m/(4n)$. Having chosen $x_0, \dots, x_{j-1}$, the union of the j intervals $(x_i - \delta, x_i + \delta)$ has measure at most $2j\delta \le m/2$. It cannot cover E up to measure zero, so choose the next point in the closure of the remaining positive-measure set. □

Proposition 3.2 (Polynomial goodness in one variable). *Every nonzero real polynomial P of degree at most n is $(C_n, 1/n)$-good on every interval, with*

$$C_n = 4n(n+1)^{1/n}.$$

Proof. Let I have length L, put $M = \sup_I |P| > 0$, and $E = \{x \in I : |P(x)| < \varepsilon\}$. If $|E| = 0$ there is nothing to prove. Put $m = |E|$ and choose $x_0, \dots, x_n$ as in Lemma 3.1. By continuity, $|P(x_i)| \le \varepsilon$. Lagrange interpolation gives

$$P(x) = \sum_{i=0}^{n} P(x_i) \prod_{j \ne i} \frac{x - x_j}{x_i - x_j}.$$

For $x \in I$, every numerator is at most L and every denominator factor at least $m/(4n)$, so

$$|P(x)| \le (n+1)\varepsilon \left(\frac{4nL}{m}\right)^n.$$

Take the supremum and rearrange:

$$\frac{m}{L} \le 4n(n+1)^{1/n} \left(\frac{\varepsilon}{M}\right)^{1/n}.$$

□

Remark 3.3. The exponent has the correct order: for $P(x) = x^n$ on $[-1, 1]$, the ε-sublevel set has length $2\varepsilon^{1/n}$.

4. Polynomial maps in several variables

THEOREM 4.1 (Bounded-degree polynomial goodness). *For every $d, n \geq 1$ there exists $C_{d,n} \geq 1$ such that every nonzero real polynomial P on $\mathbb{R}^d$ of total degree at most n satisfies*

$$\left|\{x \in B : |P(x)| < \varepsilon\}\right| \leq C_{d,n} \left(\frac{\varepsilon}{\sup_B |P|}\right)^{1/(dn)} |B|$$

for every Euclidean ball $B \subset \mathbb{R}^d$ and every $\varepsilon > 0$.

PROOF. We first prove the estimate on the cube $Q_d = [-1,1]^d$ by induction on d. The case $d = 1$ is Proposition 3.2. Assume the claim in dimension $d-1$, with exponent $\alpha_{d-1} = 1/((d-1)n)$, and write

$$P(x', t) = \sum_{j=0}^{n} a_j(x') t^j, \qquad x' \in Q_{d-1}, \quad t \in [-1,1].$$

Put $M = \sup_{Q_d} |P|$. Since

$$|P(x', t)| \leq \sum_{j=0}^{n} |a_j(x')|,$$

there is an index j_0 for which

$$A := \sup_{Q_{d-1}} |a_{j_0}| \geq \frac{M}{n+1}.$$

On the finite-dimensional space of one-variable polynomials of degree at most n, coefficient norm and supremum norm on $[-1,1]$ are equivalent. Hence there is $K_n \geq 1$ such that for every $q(t) = \sum_{j=0}^{n} b_j t^j$,

$$\max_j |b_j| \leq K_n \sup_{|t| \leq 1} |q(t)|.$$

Fix $0 < \delta < 1$ and set

$$S_\delta = \{x' \in Q_{d-1} : |a_{j_0}(x')| < \delta A\}.$$

The induction hypothesis applied to a_{j_0} gives

$$\frac{|S_\delta|}{|Q_{d-1}|} \leq C_{d-1,n} \delta^{\alpha_{d-1}}.$$

If $x' \notin S_\delta$, the coefficient estimate yields

$$\sup_{|t| \leq 1} |P(x', t)| \geq K_n^{-1} |a_{j_0}(x')| \geq \frac{\delta M}{K_n(n+1)}.$$

Applying Proposition 3.2 to the slice $t \mapsto P(x', t)$ and then Fubini therefore gives, with $r = \varepsilon/M$,

$$\frac{|\{|P| < \varepsilon\} \cap Q_d|}{|Q_d|} \leq C_{d-1,n}\delta^{\alpha_{d-1}} + C_n \left(\frac{K_n(n+1)r}{\delta}\right)^{1/n}.$$

For $0 < r < 1$, choose

$$\delta = r^{1/(1+n\alpha_{d-1})}.$$

The two exponents then agree and equal

$$\alpha_d = \frac{\alpha_{d-1}}{1 + n\alpha_{d-1}} = \frac{1}{dn}.$$

The case $r \geq 1$ is trivial after enlarging the constant. Thus the desired estimate holds on Q_d.

Now let $B = B(x_0, R)$. By translation and dilation it suffices to compare the unit ball with two fixed concentric cubes $Q^- \subset B(0,1) \subset Q^+$. On the finite-dimensional space of degree-$\leq n$ polynomials, the two supremum norms are equivalent, so

$$\sup_{Q^+} |P| \leq A_{d,n} \sup_{B(0,1)} |P|.$$

Apply the cube estimate on Q^+ and use $|Q^+|/|B(0,1)| = c_d$. Absorbing $A_{d,n}^{1/(dn)} c_d$ into $C_{d,n}$ gives the asserted ball estimate. Affine rescaling returns the general ball. $\square$

5. Finite-order nondegeneracy

Let $f = (f_1, \ldots, f_n) : U \to \mathbb{R}^n$ be C^ℓ.

Definition 5.1. f is ℓ-nondegenerate at x_0 if the vectors

$$\partial^\beta f(x_0), \qquad 1 \leq |\beta| \leq \ell,$$

span $\mathbb{R}^n$.

Lemma 5.2 (Uniform finite-jet detection)**.** *If f is ℓ-nondegenerate at x_0, there exists $c > 0$ such that for every unit $a \in \mathbb{R}^n$,*

$$\max_{1 \leq |\beta| \leq \ell} |\partial^\beta (a \cdot f)(x_0)| \geq c.$$

Proof. On the unit sphere define

$$\Phi(a) = \max_{1 \leq |\beta| \leq \ell} |a \cdot \partial^\beta f(x_0)|.$$

Nondegeneracy says $\Phi(a) > 0$ for every unit a. Compactness and continuity give a positive minimum. $\square$

THEOREM 5.3 (Local goodness from nondegeneracy). *Suppose $f : U \to \mathbb{R}^n$ is C^ℓ and ℓ-nondegenerate at x_0. Then there is a neighborhood V of x_0 and constants $C, \alpha > 0$ such that every nonzero affine combination*

$$F_{a_0,a}(x) = a_0 + a \cdot f(x)$$

is (C, α)-good on V.

PROOF. Normalize $|(a_0, a)| = 1$. If $|a|$ is bounded below, Lemma 5.2, continuity of derivatives, and compactness of the coefficient sphere show that on a sufficiently small fixed neighborhood one derivative of order at most ℓ is bounded below uniformly. If $|a|$ is small, then $|a_0|$ is bounded below and F is uniformly separated from zero after further shrinking the neighborhood, so goodness is trivial. The finite-derivative sublevel lemma in Appendix B converts the first alternative into a uniform power-law sublevel estimate. Scaling removes the normalization. □

6. Uniformity over a finite-dimensional family

The preceding proof illustrates a recurring compactness principle. If $\mathscr{F} \subset C^\ell(V)$ is finite-dimensional and no nonzero element vanishes to infinite order on the relevant compact set, then after normalizing a coefficient norm the unit sphere is compact. A finite derivative must be uniformly visible. This gives common goodness constants for the entire family.

7. Order of vanishing and the exponent

If a one-variable C^ℓ function satisfies $|f^{(j)}| \geq c$ for some $j \leq \ell$, its sublevel sets have size $O(\varepsilon^{1/j})$ after normalization. The worst possible order of vanishing controls the available exponent. This is why the theorem remembers a finite order ℓ rather than merely smoothness.

8. Why infinitely flat functions are excluded

The smooth function

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0, \end{cases}$$

vanishes to infinite order at 0. On $[-r, r]$, the relation between $\sup |f|$ and the measure of a much smaller sublevel set is logarithmic rather than uniformly power-law. Smoothness alone therefore does not imply goodness.

9. Goodness for vector norms revisited

If $F : V \to \mathbb{R}^N$ has coordinates in a fixed finite-dimensional polynomial or finite-jet family, Corollary 2.3 makes $\|F\|_\infty$ good. This is exactly how exterior-power wedge norms enter quantitative non-divergence: their coordinates are minors, hence polynomial combinations of the entries of the parameter matrix.

10. A practical finite-jet test

To establish goodness in a new problem:

(1) write the scalar coordinate family whose maximum gives the norm of interest;
(2) identify a finite-dimensional coefficient space;
(3) normalize coefficients;
(4) prove that no normalized limit has all derivatives through some fixed order zero;
(5) use compactness to obtain a uniform derivative lower bound;
(6) apply a finite-derivative sublevel estimate.

Detailed finite-jet mechanism. The compactness proof of Theorem 5.3 is short enough to hide a useful distinction. There are two normalizations in play. First normalize the coefficient vector (a_0, a); this prevents the function from being made small merely by scalar multiplication. Second normalize spatial scale by working on one fixed ball. Without the second normalization, derivative lower bounds can deteriorate under arbitrary rescaling.

Let

$$\mathscr{A} = \{a_0 + a \cdot f : |(a_0, a)| = 1\}.$$

This is a compact subset of $C^\ell(\overline{V})$ once $V \Subset U$ is fixed. If no $F \in \mathscr{A}$ is identically zero, then

$$F \mapsto \max_{|\beta| \le \ell} \sup_{x \in V} |\partial^\beta F(x)|$$

has a positive minimum. Nondegeneracy gives something sharper: after shrinking V, one may omit the order-zero derivative except in the regime where $|a|$ is small. Thus the family splits naturally into

$$\mathscr{A}_{\text{const}} = \{F : |a| \le \theta\},$$
$$\mathscr{A}_{\text{var}} = \{F : |a| > \theta\}.$$

On the first family, $|a_0| \ge (1 - \theta^2)^{1/2}$, so a sufficiently small neighborhood makes F uniformly separated from zero. On the second, after normalizing $a/|a|$, Lemma 5.2 supplies a derivative lower bound proportional to $|a| \ge \theta$. This split is what prevents the affine constant term from creating a false degeneracy.

Worked Example 10.1 (Quadratic curve)**.** For $f(x) = (x, x^2)$, every affine combination is
$$F(x) = a_0 + a_1 x + a_2 x^2.$$
If $a_2 \neq 0$, the second derivative $F'' = 2a_2$ detects the coefficient. If a_2 is small but a_1 is not, the first derivative detects it. If both are small under unit coefficient normalization, a_0 is large and F is separated from zero on a small interval. The abstract finite-jet argument is exactly this trichotomy without choosing cases coordinate by coordinate.

Worked Example 10.2 (A trigonometric finite-dimensional family)**.** The space
$$V_N = \operatorname{span}\{1, \sin x, \cos x, \ldots, \sin Nx, \cos Nx\}$$
is finite-dimensional. On any fixed interval of positive length, no nonzero element vanishes identically. Hence Lemma 2.1 (proved later in Chapter 7) gives a uniform lower supremum after coefficient normalization. Moreover every element solves a fixed constant-coefficient differential equation of order $2N + 1$, so finite-order derivative detection gives goodness. Polynomiality is therefore sufficient but not necessary; finite-dimensionality plus finite-order unique continuation is the deeper principle.

Stability under small perturbations. Suppose f is ℓ-nondegenerate at x_0. The spanning condition is open: choose n derivative vectors among the finite list $\partial^\beta f(x_0)$ that form a basis. Their determinant remains nonzero for x close to x_0. Hence f is ℓ-nondegenerate throughout a neighborhood, with a uniform lower bound for the smallest singular value of the chosen derivative matrix. This uniformity is what later permits one ball to support all coefficient choices simultaneously.

If g is sufficiently close to f in C^ℓ, the same determinant remains nonzero. Thus the good-function constants can be chosen uniformly on a small C^ℓ neighborhood of a nondegenerate map. This observation is useful when trajectories depend on an auxiliary compact parameter.

Restriction to lines. Why does a multivariable finite-derivative statement produce a power law? Consider first a coordinate derivative $\partial_{x_1}^j F$ bounded below. For almost every $x' = (x_2, \ldots, x_d)$, the slice
$$s \mapsto F(s, x')$$
has j-th derivative bounded below on the corresponding interval. The one-dimensional derivative lemma bounds the length of the slice of $\{|F| < \varepsilon\}$. Fubini then gives a d-dimensional estimate. For a general multiindex, one iterates this argument, or performs a fixed linear change of coordinates chosen from a finite list of directions. The exponent deteriorates, but remains positive and uniform.

This is why the exact exponent is usually suppressed in homogeneous-dynamics applications. The later Borel–Cantelli sum only asks whether

$$\sum_{m\geq 1} e^{-c\alpha m} < \infty,$$

and every $\alpha > 0$ works.

Goodness on enlarged balls. The marking theorem tests goodness not only on B but on balls arising from maximal-radius selection, contained in an enlarged region such as $3^k B$. Consequently local goodness must be established on an open neighborhood of the closure of that enlarged ball. A common bookkeeping error is to prove a derivative lower bound only at the center of B and then invoke goodness on larger balls without first shrinking the original parameter domain. The safe order is:

(1) choose a relatively compact neighborhood on which nondegeneracy is uniform;
(2) choose the working ball B so that $3^k B$ remains inside that neighborhood;
(3) only then fix the common goodness constants.

Max norms and coordinate choices. Exterior norms later appear as maxima of Plücker coordinates. The maximum norm is convenient because if

$$F(x) = \max_{1\leq i\leq N} |f_i(x)|,$$

then one coordinate realizes $\sup_B F$, and its small sublevel set contains the set where all coordinates are small. No cancellation occurs. For an arbitrary norm one can always compare with a maximum norm in fixed dimension, but making this comparison early prevents constants from being hidden inside subsequent wedge inequalities.

A failure test. To test a proposed good-function argument, search for a normalized sequence F_j in the relevant finite-dimensional family with

$$\sup_B |F_j| \to 0.$$

If such a sequence exists, compactness gives a nonzero coefficient limit whose function vanishes identically. In applications this identity usually has geometric content: the image of the parameter map lies in a proper affine hyperplane, a rational subspace is invariant, or a representation vector lies in a contracted submodule. Thus the same compactness argument that proves goodness often diagnoses the exact algebraic obstruction when goodness or non-collapse fails.

Sublevel estimates versus Łojasiewicz inequalities. For a fixed real-analytic function on a compact set, one can often control small values through a Łojasiewicz inequality. Quantitative non-divergence needs something different: constants uniform over an entire normalized coefficient family. Finite-dimensional compactness is what upgrades pointwise finite-order vanishing into such uniformity. This distinction matters whenever the affine combination itself changes with an exterior vector or a diagonal parameter.

Uniformity under normalization of balls. If $T(x) = x_0 + rx$ sends the unit ball to $B(x_0, r)$, then goodness of f on B is equivalent to goodness of $f \circ T$ on the unit ball with the same C, α. Thus local proofs may freely normalize spatial scale provided derivative estimates are transformed correctly. A derivative of order j gains a factor r^j; simultaneously the supremum scale of the Taylor polynomial changes. Keeping track of these factors is what makes the final sublevel ratio dimensionless.

CHAPTER 4

The marking theorem: the combinatorial engine

1. Posets and comparable obstructions

Let $\mathfrak{P}$ be a partially ordered set of finite length at most k. For each $s \in \mathfrak{P}$ let

$$\psi_s : \widetilde{B} \longrightarrow (0, \infty)$$

be continuous. In the lattice application, $\mathfrak{P}$ is the poset of nonzero primitive subgroups and

$$\psi_\Delta(x) = \|h(x)\Delta\|_\infty .$$

If $T \subset \mathfrak{P}$ is a chain, write

$$\mathfrak{P}(T) = \{s \in \mathfrak{P} \setminus T : s \text{ is comparable with every element of } T\}.$$

If $\ell(\mathfrak{P})$ denotes maximum chain length, then

$$\ell(\mathfrak{P}(T)) \leq \ell(\mathfrak{P}) - |T|. \tag{1.1}$$

Indeed, adjoining T to a chain in $\mathfrak{P}(T)$ gives a larger chain in $\mathfrak{P}$.

Definition 1.1 (ε-marked point)**.** Fix $0 < \varepsilon \leq \rho$. A point $x \in B$ is ε-marked if there is a chain $T_x \subset \mathfrak{P}$ such that

(1) $\varepsilon \leq \psi_s(x) \leq \rho$ for all $s \in T_x$;
(2) $\psi_s(x) \geq \rho$ for every $s \in \mathfrak{P}(T_x)$.

The empty chain is allowed. In that case the second condition says $\psi_s(x) \geq \rho$ for every $s \in \mathfrak{P}$.

The chain is not the collection of all small obstructions. It is a compatible nested record. Incomparable obstructions are separated by the covering argument; comparable obstructions are handled by induction on chain length.

2. A finite $3r$ covering lemma proved from scratch

The classical Besicovitch covering theorem is often used at this point. It is stronger than we need. The following finite lemma, together with compactness, is enough and has a one-paragraph proof.

Lemma 2.1 (Finite $3r$ selection)**.** *Let $C_1, \dots, C_m$ be finitely many nonempty Euclidean balls. There is a pairwise disjoint subfamily $C_{i_1}, \dots, C_{i_q}$ such that*

$$\bigcup_{j=1}^{m} C_j \subset \bigcup_{a=1}^{q} 3C_{i_a},$$

where $3C$ is the concentric ball with three times the radius.

Proof. Order the balls by nonincreasing radius. Scan them in that order, selecting a ball precisely when it is disjoint from all balls already selected. The selected balls are pairwise disjoint. If C_j is not selected, it meets an earlier selected ball C_i with $r_i \geq r_j$. If c_i, c_j are the centers, then

$$\|c_i - c_j\| < r_i + r_j \leq 2r_i.$$

For $z \in C_j$,

$$\|z - c_i\| < r_j + 2r_i \leq 3r_i,$$

so $C_j \subset 3C_i$. This proves the covering assertion. □

Lemma 2.2 (Compact centered-ball selection)**.** *Let $B = B(x_0, r_0)$ and let $K \subset B$ be compact. For each $x \in K$ suppose a radius $0 < r_x \leq 2r_0$ is given and put*

$$B_x = B(x, r_x), \qquad C_x = \tfrac{1}{3}B_x = B(x, r_x/3).$$

Then there are finitely many points $x_1, \dots, x_q \in K$ such that

(1) $K \subset \bigcup_i B_{x_i}$;
(2) the shrunken balls C_{x_i} are pairwise disjoint;
(3)

$$\sum_i |B_{x_i}| \leq 5^d |B|. \tag{2.1}$$

Proof. The family $\{C_x : x \in K\}$ is an open cover of K, because every C_x contains its center. Choose a finite subcover and apply Lemma 2.1. The selected shrunken balls are pairwise disjoint, and their triples are exactly the corresponding B_{x_i}, so they cover K.

Because $x_i \in B$ and $r_{x_i}/3 \leq 2r_0/3$, every C_{x_i} is contained in the concentric ball $\frac{5}{3}B$. Hence disjointness gives

$$\sum_i |B_{x_i}| = 3^d \sum_i |C_{x_i}| \leq 3^d \left|\tfrac{5}{3}B\right| = 5^d |B|.$$

□

Proof and provenance. Proved in this text. No infinite-family Besicovitch theorem is used. The only covering input in the marking theorem is Lemma 2.1, proved above, plus finite subcover compactness. This avoids the hidden passage from “the selected balls cover the selected centers" to "the selected balls cover the compact set" that would be invalid for a naive finite Besicovitch argument.

3. The marking theorem

For a ball $B = B(x_0, r)$ and $a > 0$, write $aB = B(x_0, ar)$.

THEOREM 3.1 (Marking theorem). *Let* $B = B(x_0, r_0) \subset \mathbb{R}^d$ *and let*

$$\widetilde{B} = 3^k B.$$

Suppose $\mathfrak{P}$ *has length at most* k *and the family* $\{\psi_s\}_{s\in\mathfrak{P}}$ *satisfies:*

(A1) each ψ_s *is continuous on* $\widetilde{B}$*;*

(A2) each ψ_s *is* (C,α)*-good on* $\widetilde{B}$*;*

(A3) $\sup_{x\in B} \psi_s(x) \geq \rho$ *for every* $s \in \mathfrak{P}$*;*

(A4) for every $x \in \widetilde{B}$*, only finitely many* s *satisfy* $\psi_s(x) < \rho$*.*

Then for $0 < \varepsilon \leq \rho$*,*

$$|\{x \in B : x \textit{ is not } \varepsilon\textit{-marked}\}| \leq kC\, 5^{dk} \left(\frac{\varepsilon}{\rho}\right)^{\alpha} |B|. \tag{3.1}$$

Proof architecture. The proof follows the Kleinbock–Margulis induction, with its covering step proved locally rather than imported. At a point where some obstruction is below ρ, each active obstruction is assigned its largest centered window on which it stays at most ρ. We choose the active obstruction with the largest such window. That choice has a decisive consequence: *every* obstruction has supremum at least ρ on the chosen window. We may therefore restrict to the poset of labels comparable with the chosen one and invoke the induction hypothesis. The selected obstruction itself is paid for by goodness. Lemma 2.2 sums the local estimates.

4. Active labels and maximal windows

Define

$$H(x) = \{s \in \mathfrak{P} : \psi_s(x) < \rho\}, \qquad E = \{x \in B : H(x) \neq \varnothing\}.$$

By (A4), $H(x)$ is finite.

Lemma 4.1 (Points outside E are marked)**.** *Every point of* $B \setminus E$ *is* ε*-marked.*

PROOF. If $x \notin E$, then $\psi_s(x) \geq \rho$ for every s. The empty chain satisfies Definition 1.1. $\square$

Fix $x \in E$ and $s \in H(x)$. Define

$$r_{s,x} = \sup\Big\{0 < r \le 2r_0 : \sup_{B(x,r)} \psi_s \le \rho\Big\}, \qquad B_{s,x} = B(x, r_{s,x}). \tag{4.1}$$

The set in (4.1) is nonempty because $\psi_s(x) < \rho$ and ψ_s is continuous.

Lemma 4.2 (Normalization of an active window)**.** *For $x \in E$ and $s \in H(x)$,*

$$\sup_{B_{s,x}} \psi_s = \rho, \qquad \psi_s(z) \le \rho \quad (z \in B_{s,x}). \tag{4.2}$$

Proof. For every $r < r_{s,x}$ sufficiently close to $r_{s,x}$, monotonicity in the radius and the definition of the supremum give $\sup_{B(x,r)} \psi_s \le \rho$. Therefore every point of the open ball $B_{s,x}$ lies in a smaller concentric ball on which $\psi_s \le \rho$, proving the second assertion and hence $\sup_{B_{s,x}} \psi_s \le \rho$.

For the reverse inequality, if $r_{s,x} = 2r_0$, then $B \subset B_{s,x}$ (because $x \in B$), and (A3) gives $\sup_{B_{s,x}} \psi_s \ge \rho$. If $r_{s,x} < 2r_0$, maximality implies that for every $r > r_{s,x}$ sufficiently close to $r_{s,x}$ one has $\sup_{B(x,r)} \psi_s > \rho$. Choosing such radii decreasing to $r_{s,x}$ and using continuity on the compact closure of a slightly larger ball gives $\sup_{B_{s,x}} \psi_s \ge \rho$. Thus equality holds. □

Since $H(x)$ is finite, choose $s_x \in H(x)$ with

$$r_x := r_{s_x,x} = \max_{s \in H(x)} r_{s,x}, \qquad B_x := B_{s_x,x}.$$

Lemma 4.3 (The maximal active window normalizes every label)**.** *For every $x \in E$ and every $s \in \mathfrak{P}$,*

$$\sup_{B_x} \psi_s \ge \rho.$$

Proof. Suppose instead that $\sup_{B_x} \psi_s < \rho$. Then in particular $\psi_s(x) < \rho$, so $s \in H(x)$ and $B_{s,x}$ is defined. By maximality of r_x, one has $r_{s,x} \le r_x$, hence $B_{s,x} \subset B_x$. But Lemma 4.2 gives $\sup_{B_{s,x}} \psi_s = \rho$, contradicting $\sup_{B_x} \psi_s < \rho$. □

5. The residual problem has one less level

For $x \in E$ let

$$\mathfrak{P}_x = \mathfrak{P}(\{s_x\}).$$

Lemma 5.1 (Residual admissibility)**.** *The triple consisting of the residual poset $\mathfrak{P}_x$, the restricted family $\{\psi_s\}_{s \in \mathfrak{P}_x}$, and the ball B_x satisfies (A1)–(A4) with $k-1$ in place of k.*

Proof. By (1.1), $\ell(\mathfrak{P}_x) \le k-1$. Lemma 4.3 gives the lower-supremum condition on B_x.

It remains to check that the enlarged ball needed for the induction lies in the domain on which goodness and local finiteness are known. Since $r_x \le 2r_0$ and $x \in B(x_0, r_0)$,

$$3^{k-1}B_x \subset B(x_0, (2 \cdot 3^{k-1} + 1)r_0) \subset B(x_0, 3^k r_0) = \widetilde{B}.$$

The last inequality is $2 \cdot 3^{k-1} + 1 \le 3^k$. Hence (A1), (A2), and (A4) restrict from $\widetilde{B}$ to the required enlarged residual domain. $\square$

Lemma 5.2 (Lifting a residual marking)**.** *Let $x \in E$ and $z \in B_x$. Suppose z is ε-marked for the residual problem $(\mathfrak{P}_x, \{\psi_s\}_{s\in\mathfrak{P}_x}, B_x)$ and*

$$\psi_{s_x}(z) \ge \varepsilon.$$

Then z is ε-marked for the original problem. Equivalently,

$$B \cap (B_x \setminus \Phi) \subset (B_x \setminus \Phi_x) \cup \{z \in B_x : \psi_{s_x}(z) < \varepsilon\}, \tag{5.1}$$

where Φ and Φ_x denote the original and residual marked sets.

Proof. Let $T \subset \mathfrak{P}_x$ mark z in the residual problem. Since every member of T is comparable with s_x, the set $T \cup \{s_x\}$ is a chain. The residual marking inequalities already hold on T. By Lemma 4.2, $z \in B_x$ gives $\psi_{s_x}(z) \le \rho$, while the hypothesis gives the lower bound $\varepsilon \le \psi_{s_x}(z)$.

Finally,

$$\mathfrak{P}(T \cup \{s_x\}) = \mathfrak{P}_x(T),$$

so the residual second marking condition is exactly the original second marking condition for the enlarged chain. Thus $T \cup \{s_x\}$ marks z. $\square$

6. Measurability

The estimate below is a Lebesgue-measure statement, so we record why the marked set is measurable without imposing a global finiteness assumption on $\mathfrak{P}$.

Lemma 6.1 (Countability of relevant labels)**.** *Under (A4), the collection of labels s for which*

$$U_s := \{x \in \widetilde{B} : \psi_s(x) < \rho\}$$

is nonempty is countable. Consequently the marked set Φ is Borel.

Proof. The family $\{U_s\}$ is point-finite by (A4), and each nonempty U_s is open. Let $\{Q_n\}_{n\ge1}$ be a countable basis for $\widetilde{B}$. For each nonempty U_s, choose $Q_n \subset U_s$. A fixed Q_n can be assigned to only finitely many labels: otherwise every point of Q_n would belong to infinitely many U_s, contradicting point-finiteness. Hence the relevant labels form a countable union of finite sets.

Labels with $U_s = \varnothing$ already satisfy $\psi_s \geq \rho$ everywhere and never need to be inserted into a marking chain. Since every chain has at most k elements, Φ is a countable union (over finite chains of relevant labels) of Borel sets defined by finitely many inequalities. Thus it is Borel. □

7. Proof of the marking theorem

Proof of Theorem 3.1. We argue by induction on k.

For $k = 0$, the poset is empty, so every point is marked by the empty chain. Assume the theorem proved for residual length at most $k-1$, and consider a problem of length at most $k \geq 1$.

By Lemma 4.1, the bad set $B \setminus \Phi$ is contained in E. Fix $x \in E$. From Lemma 5.2, the induction hypothesis, and goodness of ψ_{s_x} we obtain

$$\begin{aligned} |B \cap (B_x \setminus \Phi)| &\leq |B_x \setminus \Phi_x| + |\{z \in B_x : \psi_{s_x}(z) < \varepsilon\}| \\ &\leq (k-1)C5^{d(k-1)} \left(\frac{\varepsilon}{\rho}\right)^{\alpha} |B_x| + C \left(\frac{\varepsilon}{\rho}\right)^{\alpha} |B_x| \\ &\leq kC5^{d(k-1)} \left(\frac{\varepsilon}{\rho}\right)^{\alpha} |B_x|. \end{aligned}$$

Here the induction theorem applies by Lemma 5.1, while the goodness term uses

$$\sup_{B_x} \psi_{s_x} = \rho$$

from Lemma 4.2.

Now let $K \subset B \setminus \Phi$ be compact. Apply Lemma 2.2 to the family of maximal windows B_x, $x \in K$. We obtain finitely many B_{x_i} covering K with

$$\sum_i |B_{x_i}| \leq 5^d |B|.$$

Since every point of K is bad,

$$\begin{aligned} |K| &\leq \sum_i |B \cap (B_{x_i} \setminus \Phi)| \\ &\leq kC5^{d(k-1)} \left(\frac{\varepsilon}{\rho}\right)^{\alpha} \sum_i |B_{x_i}| \\ &\leq kC5^{dk} \left(\frac{\varepsilon}{\rho}\right)^{\alpha} |B|. \end{aligned}$$

By Lemma 6.1, $B \setminus \Phi$ is Borel; by inner regularity of Lebesgue measure, taking the supremum over compact $K \subset B \setminus \Phi$ proves (3.1). □

8. Dependency summary

The proof has four load-bearing transitions, and each is now local.

(1) **Covering.** Lemma 2.1 is proved by finite greedy selection. Compactness is applied to the *shrunken* balls, and the triples of the selected balls are the maximal windows that cover the entire compact bad set. No center-covering shortcut is used.

(2) **Normalization.** The chosen label maximizes $r_{s,x}$ over the finite active set $H(x)$. Lemma 4.3, not merely goodness, is what gives the lower-supremum hypothesis for every residual label.

(3) **Rank/complexity descent.** Passing to $\mathfrak{P}(\{s_x\})$ lowers maximum chain length by one, and the enlarged residual ball stays inside $3^k B$.

(4) **Lifting.** A residual marking plus $\varepsilon \le \psi_{s_x} \le \rho$ gives an original marking. This is the exact logical place where the selected obstruction is adjoined to the chain.

No theorem of comparable strength is cited inside this chain.

9. Reader perspective

The numerical constant 5^{dk} is not the point. The reusable mechanism is

$$\begin{aligned} \text{active obstruction} &\longrightarrow \text{largest normalized window} \\ &\longrightarrow \text{all labels normalized on that window} \\ &\longrightarrow \text{one-level poset descent} \\ &\longrightarrow \text{finite covering sum.} \end{aligned}$$

The proof is therefore a structured substitute for an impossible union bound over infinitely many arithmetic obstructions.

A two-level model. Suppose the labels are rational lines and planes, ordered by inclusion. At a bad point one first chooses the active label whose sub-ρ window persists furthest. If it is a line L, the residual labels are precisely the planes comparable with L, namely those containing L. On the part where $\psi_L < \varepsilon$, goodness pays the error. Elsewhere L can be inserted into the chain and only one further rank increase remains. The number of rational lines never enters the estimate.

Why $3^k B$ occurs although the covering constant is 5^d. These are different geometric losses. The factor 3^k reserves room for the *recursive domains*: if $r_x \le 2r_0$, then $3^{k-1}B_x \subset 3^k B$. The factor 5^d comes from summing the selected maximal windows: their one-third shrinkings are disjoint and lie inside $\frac{5}{3}B$. Confusing these two roles is a common source of incorrect constants and, more importantly, incorrect domain containment.

Local finiteness. No global bound on the number of labels is assumed. Local finiteness has two separate jobs: it makes $H(x)$ finite so that a maximizing label s_x exists, and it makes the relevant open-label family countable, which gives ordinary Borel measurability of the marked set.

Warning 9.1. The following shortcuts are invalid.

(1) Selecting a finite subcover and then keeping only balls whose *centers* remain covered; that does not imply the original compact set remains covered.

(2) Choosing an arbitrary small obstruction instead of the one with maximal $r_{s,x}$; then the residual lower-supremum hypothesis can fail.

(3) Appending a label to the chain before verifying that its value lies in $[\varepsilon, \rho]$.

(4) Treating the number of active labels as the induction parameter; it may be arbitrarily large. The correct parameter is maximum compatible-chain length.

10. Source provenance

The induction architecture is the marked-point argument of Kleinbock–Margulis. The present proof is not a citation in place of that argument: maximal windows, residual admissibility, the marking-lift lemma, measurability, and the complete covering estimate are all proved above. The only change of presentation is that the usual Euclidean Besicovitch theorem is replaced by the elementary finite $3r$ selection lemma, which is sufficient for Lebesgue measure after compact exhaustion.

CHAPTER 5

Quantitative non-divergence for lattices

1. Exact normalization

Let $B \subset \mathbb{R}^d$ be a ball and $h : 3^k B \to \mathrm{GL}_k(\mathbb{R})$ continuous. For a primitive subgroup $\Delta \leq \mathbb{Z}^k$ define

$$\psi_\Delta(x) = \|h(x)\Delta\|_\infty .$$

Theorem 1.1 (Quantitative non-divergence). *Assume that for every primitive nonzero $\Delta \leq \mathbb{Z}^k$:*

(1) ψ_Δ is (C, α)-good on $3^k B$;

(2) $\sup_{x \in B} \psi_\Delta(x) \geq \rho$;

and assume $0 < \rho \leq 1/k$. Then, for $0 < \varepsilon \leq \rho$,

$$\left|\left\{x \in B : \delta_\infty(h(x)\mathbb{Z}^k) < \varepsilon\right\}\right| \leq kC5^{dk} \left(\frac{\varepsilon}{\rho}\right)^\alpha |B|.$$

Proof and provenance. Proved in this text. The proof below is the marking argument specialized to primitive subgroups. The maximum-norm normalization makes the rank-extension constant explicit.

2. Verifying the marking hypotheses

Continuity is immediate from continuity of h. Goodness and the lower supremum are hypotheses. Local finiteness is Lemma 6.1 applied at the fixed matrix $h(x)$. The poset of primitive subgroups has length at most k. Therefore Theorem 3.1 says that all but the stated proportion of $x \in B$ are ε-marked.

The remaining step is purely arithmetic: an ε-marked point cannot have a vector shorter than ε.

3. Marked implies no short vector

Lemma 3.1. *If x is ε-marked for the family $\psi_\Delta(x) = \|h(x)\Delta\|_\infty$, with $0 < \varepsilon \leq \rho \leq 1/k$, then*

$$\delta_\infty(h(x)\mathbb{Z}^k) \geq \varepsilon.$$

Proof. Let T be a marking chain and suppose, for contradiction, that a primitive vector $v \in \mathbb{Z}^k$ satisfies $\|h(x)v\|_\infty < \varepsilon$. Denote the primitive rank-one subgroup by $V = \mathbb{Z}v$.

If T is empty, then V is comparable with every member of T vacuously, so the marking condition gives $\|h(x)V\|_\infty \geq \rho \geq \varepsilon$, contradiction.

Assume T is nonempty. If V is comparable with every member of T, then either $V \in T$, in which case the first marking condition gives $\|h(x)V\|_\infty \geq \varepsilon$, or $V \notin T$, in which case the second marking condition gives $\|h(x)V\|_\infty \geq \rho$. Both are contradictions. Hence V is incomparable with at least one member of T.

Choose the largest member $\Delta \in T$ whose real span does not contain v. Such a member exists by the preceding paragraph. Let Λ be the primitive closure of $\Delta + \mathbb{Z}v$. Then Λ has rank $\operatorname{rank}\Delta + 1$, contains Δ, and is comparable with every member of the chain: below Δ by containment, above Δ until the first chain member containing v, and thereafter contained in that member. If $\Lambda \notin T$, the marking condition gives

$$\|h(x)\Lambda\|_\infty \geq \rho.$$

If $\Lambda \in T$, the first marking condition gives the weaker but sufficient bound $\|h(x)\Lambda\|_\infty \geq \varepsilon$.

On the other hand Lemma 4.1 gives

$$\|h(x)\Lambda\|_\infty \leq k\,\|h(x)\Delta\|_\infty\,\|h(x)v\|_\infty < k\rho\varepsilon \leq \varepsilon.$$

This contradicts either lower bound. □

4. Proof of the theorem

Proof of Theorem 1.1. The hypotheses of Theorem 3.1 hold for the primitive-subgroup poset. Hence the set of points that are not ε-marked has measure at most

$$kC5^{dk}(\varepsilon/\rho)^\alpha|B|.$$

By Lemma 3.1, every point with $\delta_\infty(h(x)\mathbb{Z}^k) < \varepsilon$ is unmarked. The same bound therefore applies to the cusp set. □

5. Where each hypothesis is used

- Continuity produces maximal windows where an obstruction reaches scale ρ.
- Goodness estimates the genuinely tiny part of such a window.
- The lower-supremum condition guarantees that a maximal window reaches the reference scale instead of staying collapsed everywhere.

- Local finiteness prevents an infinite accumulation of distinct labels at a single parameter point.
- Finite rank makes the chain induction terminate.
- The condition $\rho \le 1/k$ is used only in Lemma 3.1, where rank extension turns a short vector into a still smaller higher-rank obstruction.

6. Euclidean-norm formulation

All norms on each finite-dimensional exterior power are uniformly equivalent. Thus there are constants c_k, C'_k such that

$$c_k \|\Delta\|_\infty \le \|\Delta\|_2 \le C'_k \|\Delta\|_\infty$$

for all ranks. Replacing ε, ρ by comparable thresholds gives the same theorem for Euclidean systole with dimension-dependent constants.

7. The theorem as a structured union bound

Formally,

$$\{x : \delta(h(x)\mathbb{Z}^k) < \varepsilon\} = \bigcup_{0 \neq v \in \mathbb{Z}^k} \{x : \|h(x)v\| < \varepsilon\}.$$

The marking theorem should be understood as a replacement for summing the measures of these infinitely many sets. The union is reorganized into chains of rational subspaces; only chain length, not the number of vectors, enters the constant.

8. A rank-two walkthrough

For $k = 2$, two independent primitive vectors generate a full-rank subgroup. Since $\det h(x)$ is fixed in the unimodular case, two linearly independent directions cannot both be arbitrarily short. A bad point is therefore controlled by essentially one primitive line at a time. The marking theorem reduces to a maximal-window argument plus the finite $3r$ selection. Higher dimension is the same idea with a nested sequence of rational subspaces.

9. Why hypotheses for all primitive ranks are unavoidable

Even though the bad set is defined by rank-one vectors, during the covering proof two or more competing short directions can coexist. Their wedge measures the covolume of the rational plane they generate. Without a lower-supremum hypothesis for that plane, the proof cannot rule out a persistent higher-rank collapse. Exterior powers are therefore proof infrastructure, not ornamental generality.

10. The constants, line by line

There are two sources of numerical loss in the measure bound:

(1) $C(\varepsilon/\rho)^\alpha$ from goodness;
(2) the factor 5^d from the proved finite $3r$ selection and the disjoint one-third windows.

The separate enlargement $3^k B$ is a domain reserve for recursion; it does not multiply the measure estimate. At most k poset levels occur, giving $kC5^{dk}(\varepsilon/\rho)^\alpha$.

11. Scale normalization and harmless changes of ρ

If the natural lower bound is $\rho_0 > 1/k$, replace it by $\rho = 1/k$; the lower-supremum hypothesis remains true. If it is smaller, keep it. Rescaling an exterior norm by a fixed factor merely changes ρ and the final constant.

12. A conditional-measure version of the proof pattern

For a locally doubling measure μ on parameter space, replace Lebesgue volume by μ and the Euclidean finite-$3r$ summation by the corresponding doubling-covering estimate. The marking induction is unchanged; the only modification is the multiplicative constant attached to each enlargement. This observation is the bridge to friendly measures and local-field parameter spaces.

13. Common proof errors

Warning 13.1. The following shortcuts are invalid.

(1) Proving goodness only for rank-one coordinates.
(2) Assuming goodness supplies a lower supremum.
(3) Summing over primitive vectors.
(4) Using Mahler compactness to infer quantitative tails without a sublevel estimate.
(5) Forgetting the primitive closure when adjoining a vector to a subgroup.
(6) Applying the rank-extension inequality without ensuring the new subgroup is comparable with the marking chain.

Primitive closure in the short-vector step. The primitive closure in Lemma 3.1 deserves emphasis. If Δ is primitive and $v \notin \Delta_{\mathbb{R}}$, the abstract subgroup $\Delta + \mathbb{Z}v$ need not be primitive. Let

$$\Lambda = (\Delta_{\mathbb{R}} + \mathbb{R}v) \cap \mathbb{Z}^k.$$

Then Λ is the unique primitive subgroup with the required real span. If w_Δ represents Δ, the wedge $w_\Delta \wedge v$ represents a finite-index subgroup of Λ, so

$$\|h\Lambda\|_\infty \leq \|h(w_\Delta \wedge v)\|_\infty .$$

This direction of the inequality is crucial: saturation can only *decrease* covolume. It is exactly what permits the rank-extension bound to be applied to the primitive label that belongs to the marking poset.

The first-rank exception. If a putative short vector spans the smallest member of the marking chain, no rank-extension argument is needed: the marking condition itself says its norm is at least ε. Rank extension begins only when the vector is transverse to some recorded subspace. This is why proofs that mechanically adjoin the short vector at every stage can accidentally divide by the nonexistent wedge norm of a rank-zero subgroup.

A rank-three schematic. Suppose $k = 3$ and v is short. A marking chain can have one of the forms

$$L, \qquad P, \qquad L \subset P,$$

where L is a rational line and P a rational plane. If v spans L, marking immediately contradicts shortness. If v is transverse to L, primitive closure gives a plane P_v with

$$\|hP_v\| \leq 3\,\|hL\|\,\|hv\| < 3\rho\varepsilon \leq \varepsilon.$$

If P_v is not the recorded plane, it is a comparable unrecorded obstruction and should have norm at least ρ. If it is recorded, it should have norm at least ε. Both contradict the displayed inequality. If the chain begins with a plane and v is outside it, adjoining v gives full rank; in the unimodular case the resulting determinant is 1, so it cannot be below ε. This is the concrete geometry behind the general proof.

Uniformity in auxiliary parameters. Often $h = h_a(x)$ depends on an auxiliary parameter a in a compact set A. To obtain a QND estimate uniform in a, one must verify the hypotheses uniformly:

(1) common (C, α)-goodness for every wedge coordinate family;
(2) a common ρ in the lower-supremum condition;
(3) one enlarged parameter domain on which all families are defined.

Local finiteness needs no extra uniform counting bound. Once these inputs are uniform, the conclusion is automatically uniform because the marking constant depends only on d, k, C, α and ρ.

From cusp tails to tightness of measures. Let ν_j be probability measures obtained by averaging a parameter family:

$$\nu_j = \frac{1}{|B|} \int_B \delta_{h_j(x)\Gamma}\, dx.$$

If QND gives, uniformly in j,

$$\nu_j(X_k \setminus K_\eta) \le A\eta^\alpha,$$

then for every $\delta > 0$ choose η with $A\eta^\alpha < \delta$. The compact set K_η carries at least $1 - \delta$ of every ν_j. Hence the family is tight, and every weak-* subsequential limit is a probability measure rather than a subprobability measure created by escape of mass.

This is the exact point at which quantitative non-divergence enters many measure-rigidity arguments. It does *not* identify the weak limit. It only guarantees that there is a full-mass limit available for the later invariance and classification arguments.

Threshold monotonicity. The cusp event is monotone in ε:

$$0 < \varepsilon_1 \le \varepsilon_2 \quad \Longrightarrow \quad \{\delta < \varepsilon_1\} \subset \{\delta < \varepsilon_2\}.$$

Thus estimates proved first on dyadic thresholds extend to all thresholds at the cost of a fixed factor 2^α. This small observation is frequently useful when the natural arithmetic scales are discrete while the QND theorem is stated continuously.

What the theorem does not say. QND controls

$$\delta(h(x)\mathbb{Z}^k) = \min_{0 \ne v \in \mathbb{Z}^k} \|h(x)v\|$$

but does not specify which vector is shortest, how many short vectors exist, or the distribution of the successive minima inside the cusp. Such refined information may require rank-specific height estimates or geometry-of-numbers counting. The strength of QND is uniformity under changing the identity of the obstruction.

Changing the lattice. A family $h(x)g\mathbb{Z}^k$ is handled by replacing $h(x)$ with $h(x)g$. If $g\mathbb{Z}^k$ ranges in a compact set, the starting exterior wedges are uniformly bounded below. If g itself varies in an unbounded set, that conclusion can fail even when every x-dependence is polynomial. This is another form of non-collapse: the starting lattice cannot be allowed to degenerate invisibly outside the parameter family.

CHAPTER 6

Polynomial parametrizations and quantitative unipotent non-divergence

1. Why unipotent matrix coefficients are polynomial

Let $u_t = \exp(tN)$ with N nilpotent. If $N^r = 0$, then

$$u_t = I + tN + \frac{t^2}{2!}N^2 + \cdots + \frac{t^{r-1}}{(r-1)!}N^{r-1}.$$

Thus every matrix coefficient is polynomial in t. The same remains true in every finite-dimensional algebraic representation, in particular on each exterior power $\bigwedge^j \mathbb{R}^k$.

Lemma 1.1 (Uniform degree in exterior powers)**.** *Fix a one-parameter unipotent subgroup $u_t < \mathrm{SL}_k(\mathbb{R})$. There is $D = D(u,k)$ such that for every $1 \le r \le k$, every $w \in \bigwedge^r \mathbb{R}^k$, and every coordinate functional ℓ on $\bigwedge^r \mathbb{R}^k$, the function*

$$t \longmapsto \ell((\bigwedge^r u_t)w)$$

is a polynomial of degree at most D.

Proof. The induced infinitesimal operator $N_r = d(\bigwedge^r)(N)$ is nilpotent on the finite-dimensional space $\bigwedge^r \mathbb{R}^k$. Hence $\bigwedge^r u_t = e^{tN_r}$ is a polynomial whose degree is bounded by the nilpotency index of N_r. Take the maximum over finitely many r. $\square$

2. Uniform goodness for wedge norms

Lemma 2.1. *For fixed $u_t < \mathrm{SL}_k(\mathbb{R})$ there exist $C, \alpha > 0$ such that for every $g \in \mathrm{SL}_k(\mathbb{R})$ and every primitive $\Delta \le \mathbb{Z}^k$, the function*

$$t \mapsto \|u_t g\Delta\|_\infty$$

is (C,α)-good on every interval. The constants do not depend on g or Δ.

Proof. The coordinate functions of $(\bigwedge^r u_t)(\bigwedge^r g)w_\Delta$ are polynomials of uniformly bounded degree by Lemma 1.1. Proposition 3.2 supplies uniform goodness for each coordinate, and Lemma 2.2 passes to their maximum. $\square$

3. Uniform wedge lower bounds on compact sets

Lemma 3.1. *If $K \subset X_k$ is compact, there exists $b_K > 0$ such that for every $\Lambda = g\mathbb{Z}^k \in K$ and every primitive nonzero $\Delta \leq \mathbb{Z}^k$,*

$$\|g\Delta\|_\infty \geq b_K.$$

PROOF. Suppose not. Then there are $g_n\mathbb{Z}^k \in K$, primitive Δ_n of ranks r_n, and $\|g_n\Delta_n\|_\infty \to 0$. Pass to a subsequence with constant rank r and choose representatives g_n in a compact subset of $\mathrm{SL}_k(\mathbb{R})$, possible by compactness of K and local sections. Then $\bigwedge^r g_n$ and its inverse have uniformly bounded operator norms. Hence

$$\|w_{\Delta_n}\|_\infty \ll \|g_n\Delta_n\|_\infty \to 0,$$

contradicting that w_{Δ_n} is a nonzero integral vector in $\bigwedge^r \mathbb{Z}^k$ and therefore has maximum norm at least one. $\square$

4. Quantitative non-divergence of a unipotent segment

THEOREM 4.1 (Quantitative unipotent non-divergence). *Fix a one-parameter unipotent subgroup $u_t < \mathrm{SL}_k(\mathbb{R})$ and a compact $K \subset X_k$. There exist $A, \alpha > 0$ and $\eta_0 > 0$ such that for every $\Lambda \in K$, every $T > 0$, and every $0 < \eta \leq \eta_0$,*

$$|\{t \in (0,T) : \delta_\infty(u_t\Lambda) < \eta\}| \leq A\eta^\alpha T.$$

The constants are uniform in Λ and T.

PROOF. Choose b_K as in Lemma 3.1 and put

$$\rho = \min\{b_K, 1/k\}.$$

Let $\Lambda = g\mathbb{Z}^k \in K$ and on $B = (0,T)$ define $h(t) = u_t g$. Lemma 2.1 supplies uniform (C,α)-goodness for every primitive subgroup on the enlarged interval required by Theorem 1.1. Moreover

$$\sup_{t\in(0,T)} \|u_t g\Delta\|_\infty \geq \lim_{t\downarrow 0} \|u_t g\Delta\|_\infty = \|g\Delta\|_\infty \geq b_K \geq \rho.$$

Theorem 1.1 gives

$$|\{t \in (0,T) : \delta_\infty(u_t\Lambda) < \eta\}| \leq kC(3N_1)^k(\eta/\rho)^\alpha T.$$

Absorb the fixed powers of ρ and dimensional constants into A and set $\eta_0 = \rho$. $\square$

5. Qualitative non-divergence as a corollary

Given $\varepsilon_0 > 0$, choose $\eta > 0$ with $A\eta^\alpha < \varepsilon_0$. Every segment starting in K then spends at least $(1-\varepsilon_0)T$ of its time in the compact Mahler set K_η. Thus quantitative non-divergence implies no escape of most mass. The converse contains far less information: a qualitative tightness statement gives no power-law cusp tail.

6. Multi-parameter polynomial trajectories

The same architecture applies to a polynomial map $\Theta : \mathbb{R}^m \to \mathrm{SL}_k(\mathbb{R})$ whenever every exterior coordinate of $\Theta(x)gw_\Delta$ belongs to a uniformly bounded-degree polynomial family and a lower-supremum bound is available. Parameter dimension modifies the geometric covering constant and the goodness exponent, but not the proof. Unipotence is one mechanism that supplies polynomiality automatically; quantitative non-divergence itself only sees the resulting finite-dimensional polynomial family.

Degree bookkeeping in a Jordan block. Consider the regular unipotent subgroup in SL_3,

$$u_t = \begin{pmatrix} 1 & t & t^2/2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}.$$

On $\mathbb{R}^3$ the degree is at most two. On $\bigwedge^2 \mathbb{R}^3$ one computes

$$\begin{aligned} u_t(e_1 \wedge e_2) &= e_1 \wedge e_2 + t\, e_1 \wedge e_3, \\ u_t(e_1 \wedge e_3) &= e_1 \wedge e_3, \\ u_t(e_2 \wedge e_3) &= e_2 \wedge e_3 + t\, e_1 \wedge e_3 + \frac{t^2}{2} e_1 \wedge e_2. \end{aligned}$$

Thus the exterior degree remains uniformly bounded. In general one should not guess the degree from the original matrix entry alone: the safe argument is nilpotence of the induced infinitesimal operator on each exterior representation.

Why the starting compact set supplies every rank. Compactness of $K \subset X_k$ is often described as a lower bound on shortest vectors. Lemma 3.1 strengthens this to all primitive ranks. The strengthening costs no new compactness theorem: if a wedge became small, applying the inverse of a compactly bounded representative would produce a nonzero integral exterior vector of norm below one. This observation is what makes unipotent QND nearly automatic once polynomial goodness is known.

Uniformity in the interval length. Polynomial goodness is affine-scale invariant. Consequently the estimate in Theorem 4.1 is uniform for every $T > 0$, not only for long orbit segments. The proof uses $t = 0$ only to furnish the lower supremum. If the interval is translated to $(S, S + T)$, one rewrites

$$u_{S+t} = u_t u_S$$

and treats $u_S \Lambda$ as the starting lattice. Uniformity then requires that the translated starting lattices lie in a compact set, or alternatively that one prove the wedge lower-supremum directly on the translated interval. This distinction becomes important in recurrence arguments based on moving time windows.

Polynomial trajectories beyond unipotence. If $\Theta(t)$ is any matrix polynomial with determinant one, the exterior coordinates are polynomial and goodness follows. QND therefore applies whenever non-collapse is verified, even if Θ is not a subgroup. Group structure is relevant to dynamics, recurrence, and invariance, but not to the analytic sublevel mechanism itself.

CHAPTER 7

The non-collapse condition

Goodness is often straightforward once a trajectory is polynomial. In serious applications the harder hypothesis is

$$\sup_{x\in B}\|h(x)\Delta\|\geq\rho\qquad\text{for every primitive }\Delta.$$

This chapter isolates what that condition means.

1. What failure of non-collapse says

Suppose primitive subgroups Δ_j satisfy

$$\sup_{x\in B}\|h(x)\Delta_j\|\longrightarrow 0.$$

After projectivizing w_{Δ_j} one is seeing rational directions on which the entire family $h(B)$ contracts uniformly. Such failure is algebraic: it is not an exceptional cusp visit at a few parameters.

> **Failure mode.** Take $h_t=\operatorname{diag}(e^{-t},e^{t})$ and let the parameter "ball" consist of a single point. Then e_1 has norm e^{-t}, so no lower bound independent of t can hold. Polynomiality is irrelevant. Parameter variation, compact initial data, or geometric nondegeneracy must force every rational obstruction to become large somewhere.

2. A compactness lemma for finite-dimensional function spaces

Lemma 2.1 (Uniform lower supremum)**.** *Let V be a finite-dimensional normed vector space of continuous functions on a compact set K, and suppose the restriction map $V\to C(K)$ is injective. Then there is $c>0$ such that*

$$\sup_{x\in K}|F(x)|\geq c\,\|F\|_V\qquad(F\in V).$$

PROOF. On the unit sphere of V, the continuous function $F\mapsto\sup_K|F|$ is strictly positive by injectivity. Compactness gives a positive minimum. $\square$

This elementary statement is the hidden compactness step in many non-collapse arguments. Once exterior coordinates are expressed in a fixed finite-dimensional function space, it is enough to rule out an identically vanishing normalized limit.

3. Basic unstable-horospherical parametrization

For $y \in \mathbb{R}^n$ set

$$u_y = \begin{pmatrix} 1 & y \\ 0 & I_n \end{pmatrix} \in \mathrm{SL}_{n+1}(\mathbb{R}).$$

For $v = (p, q) \in \mathbb{R} \times \mathbb{R}^n$,

$$u_y v = (p + y \cdot q, q).$$

Thus the first coordinate is an affine function of y. On higher exterior powers the coordinates are affine combinations of $1, y_1, \dots, y_n$ because the upper-right block appears only once in a nonzero minor. Consequently, for a map $f : U \to \mathbb{R}^n$, the exterior coordinates of $u_{f(x)}w$ belong to finite-dimensional spaces generated by $1, f_1, \dots, f_n$.

4. Nondegenerate maps rule out affine collapse

Proposition 4.1. *Let $f : U \to \mathbb{R}^n$ be nondegenerate at x_0. After shrinking to a ball B about x_0, there is $c_B > 0$ such that every affine function*

$$F(x) = a_0 + a \cdot f(x)$$

satisfies

$$\sup_B |F| \geq c_B \, \|(a_0, a)\|$$

unless $F \equiv 0$ for an algebraically forced reason.

Proof. If a sequence of unit coefficient vectors violated the assertion, compactness would give a limiting unit vector (a_0, a) for which $a_0 + a \cdot f$ vanishes identically on B. Differentiating at x_0 gives

$$a \cdot \partial^\beta f(x_0) = 0 \quad (1 \leq |\beta| \leq \ell).$$

Nondegeneracy forces $a = 0$, and then $a_0 = 0$, contradiction. □

5. Diagonal renormalization and the real difficulty

In Diophantine applications the family is not merely $u_{f(x)}$ but

$$h_t(x) = g_t u_{f(x)},$$

where g_t expands some exterior coordinates and contracts others. The coefficient vector of an affine combination can therefore depend exponentially on t. Proposition 4.1 must be applied after factoring out the largest relevant weight. The task is then to show that every primitive wedge has at least one coordinate in a noncontracting weight space whose normalized coefficient vector does not vanish.

For SL_{n+1} this is a finite combinatorial problem in exterior powers. If

$$g_t = \mathrm{diag}(e^t, e^{-t_1}, \dots, e^{-t_n}), \qquad t = t_1 + \cdots + t_n,$$

then each wedge basis vector has an explicit weight. The horospherical shear moves coefficients from contracting coordinates into coordinates containing the first basis vector. Nondegeneracy prevents all such transferred coefficients from cancelling on the whole parameter ball.

6. A useful diagnostic

Before invoking quantitative non-divergence, answer two separate questions:

(1) *Anti-concentration:* why are the exterior coordinate functions good?
(2) *Non-collapse:* after normalizing the coefficient vector at the relevant diagonal scale, why can no primitive wedge become uniformly zero on the parameter ball?

If the second answer is merely "because the functions are polynomial", the argument is incomplete.

Exterior-coordinate anatomy of the horospherical map. Let $e_0, e_1, \dots, e_n$ be the standard basis of $\mathbb{R}^{n+1}$ and

$$u_y e_0 = e_0, \qquad u_y e_j = e_j + y_j e_0 \quad (1 \le j \le n).$$

For a basis wedge $e_I = e_{i_1} \wedge \cdots \wedge e_{i_r}$ not containing e_0, multilinearity gives

$$u_y e_I = e_I + \sum_{m=1}^{r} (-1)^{m-1} y_{i_m} e_0 \wedge e_{i_1} \wedge \cdots \widehat{e_{i_m}} \cdots \wedge e_{i_r}.$$

There are no quadratic terms in y: any product of two y-terms would contain $e_0 \wedge e_0 = 0$. If e_I already contains e_0, it is fixed. Hence every exterior coordinate is an affine combination of $1, y_1, \dots, y_n$.

After substituting $y = f(x)$, all wedge coordinates lie in the same finite-dimensional space

$$\mathrm{span}\{1, f_1, \dots, f_n\},$$

independently of the rank. This exact calculation is the reason ordinary affine nondegeneracy of f is enough for the standard unstable-horospherical model. For more general unipotent parametrizations, higher polynomial monomials can appear, and the non-collapse test must be adjusted accordingly.

A compact coefficient contradiction in detail. Suppose no uniform lower supremum exists for a normalized finite-dimensional coordinate family. Then there are coefficient vectors c_j of norm one with

$$\sup_B |F_{c_j}| \to 0.$$

Pass to $c_j \to c_\infty$. Uniform convergence on B gives $F_{c_\infty} \equiv 0$. If the functions are $1, f_1, \ldots, f_n$, differentiation eliminates the constant coefficient and non-degeneracy forces the remaining coefficients to vanish. The original identity then forces the constant to vanish as well, contradicting $\|c_\infty\| = 1$. This compactness contradiction is often shorter and safer than attempting to estimate the supremum directly from individual coefficients.

Rational versus irrational collapse. For lattice applications the normalized coefficients arise from integral wedge vectors, so an exact vanishing identity corresponds to a rational algebraic obstruction. A merely small irrational coefficient relation cannot persist under normalization without converging to an exact limiting identity. This is one place where arithmetic discreteness and analytic compactness reinforce each other.

CHAPTER 8

Dani correspondence: Diophantine inequalities as cusp excursions

1. Simultaneous approximation

For $y \in \mathbb{R}^n$ define

$$u_y = \begin{pmatrix} I_n & y \\ 0 & 1 \end{pmatrix}, \qquad g_t = \begin{pmatrix} e^{t/n} I_n & 0 \\ 0 & e^{-t} \end{pmatrix}.$$

We identify $(p, q) \in \mathbb{Z}^n \times \mathbb{Z}$ with a column vector. Then

$$g_t u_y \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} e^{t/n}(p + qy) \\ e^{-t} q \end{pmatrix}.$$

A short vector therefore means simultaneous smallness of

$$e^{t/n} \|p + qy\|, \qquad e^{-t}|q|.$$

The diagonal flow balances approximation error against denominator size.

Proposition 1.1 (Simultaneous Dani correspondence)**.** *Let $y \in \mathbb{R}^n$. Then y is badly approximable, i.e. there is $c > 0$ such that*

$$\|qy - p\|_\infty \geq c|q|^{-1/n} \qquad (q \neq 0, p \in \mathbb{Z}^n),$$

if and only if the forward orbit $\{g_t u_y \mathbb{Z}^{n+1} : t \geq 0\}$ is bounded in X_{n+1}.

Proof. By Mahler, boundedness is equivalent to a uniform positive lower bound for the norm of every nonzero vector $g_t u_y(-p, q)$. If y is badly approximable and $q \neq 0$, put $A = \|qy - p\|_\infty$ and $B = |q|$. For every t,

$$\max(e^{t/n} A, e^{-t} B) \geq (A^n B)^{1/(n+1)} \geq c^{n/(n+1)}.$$

The case $q = 0$ is trivial. Thus the systole is uniformly positive.

Conversely, assume the orbit has systole at least η. Given $q \neq 0$ and $p \in \mathbb{Z}^n$, choose t so that

$$e^{t/n} A = e^{-t} B, \quad \text{i.e.} \quad e^{t(1+1/n)} = B/A,$$

when $A < B$; if $A \geq B$ the desired Diophantine bound is immediate after reducing constants. At the balanced time the common value is $(A^n B)^{1/(n+1)} \geq \eta$, hence $A^n B \geq \eta^{n+1}$ and

$$A \geq \eta^{(n+1)/n} B^{-1/n}.$$

□

2. Balancing the two errors

The preceding proof uses a general principle. If a diagonal flow transforms two positive quantities as $e^{at}A$ and $e^{-bt}B$, then the minimizing scale occurs when they are equal, and the minimum is a weighted geometric mean

$$A^{b/(a+b)} B^{a/(a+b)}.$$

Dani correspondences are systematic applications of this elementary optimization to integer vectors.

3. Dual approximation

For $y \in \mathbb{R}^n$ use instead

$$\widetilde{u}_y = \begin{pmatrix} 1 & y^T \\ 0 & I_n \end{pmatrix}, \qquad \widetilde{g}_t = \mathrm{diag}(e^t, e^{-t/n}, \dots, e^{-t/n}).$$

Then for $(p, q) \in \mathbb{Z} \times \mathbb{Z}^n$,

$$\widetilde{g}_t \widetilde{u}_y \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} e^t(p + y \cdot q) \\ e^{-t/n} q \end{pmatrix}.$$

The short-vector condition records small linear forms $|p + y \cdot q|$ relative to the size of q. The same balancing proof translates bounded or exponentially deep cusp excursions into dual Diophantine exponents.

4. Multiplicative approximation and many diagonal parameters

For $q = (q_1, \dots, q_n) \in \mathbb{Z}^n$ put

$$\Pi_+(q) = \prod_{i=1}^{n} \max\{1, |q_i|\}.$$

A point $y \in \mathbb{R}^n$ is *very well multiplicatively approximable* (VWMA) if there exists $\varepsilon > 0$ and infinitely many $(p, q) \in \mathbb{Z} \times (\mathbb{Z}^n \setminus \{0\})$ with

$$|p + q \cdot y| < \Pi_+(q)^{-1-\varepsilon}.$$

The natural diagonal semigroup is indexed by $t = (t_1, \dots, t_n) \in \mathbb{R}^n_+$:

$$g_t = \mathrm{diag}(e^{t_1 + \cdots + t_n}, e^{-t_1}, \dots, e^{-t_n}).$$

With

$$u_y = \begin{pmatrix} 1 & y \\ 0 & I_n \end{pmatrix},$$

we get

$$g_t u_y(p, q) = (e^{t_1+\cdots+t_n}(p + y \cdot q), e^{-t_1}q_1, \ldots, e^{-t_n}q_n).$$

Each denominator coordinate can now be balanced independently.

5. Why the correspondence matters for measure theory

A Diophantine exceptional set becomes a limsup of cusp events. Quantitative non-divergence estimates the measure of each event; Borel–Cantelli then converts summability into an almost-everywhere statement. This is the mechanism behind strong extremality of nondegenerate manifolds.

6. The balancing principle in higher dimension

Given (p, q) and $r > 0$, set

$$\max\{1, |q_i|\} = re^{t_i}, \qquad t_i \geq 0.$$

Then

$$\Pi_+(q) = r^n e^{t_1+\cdots+t_n}.$$

The last n coordinates of $g_t u_y(p, q)$ are at most r in modulus. If the linear-form error is smaller than $\Pi_+(q)^{-1-\varepsilon}$, the first coordinate is also at most a power of r. Choosing r as a negative power of $\Pi_+(q)$ turns multiplicative improvement into exponential cusp penetration.

7. Bounded, recurrent, and divergent behavior are different

Boundedness corresponds to a uniform Diophantine lower bound. Recurrence says the orbit returns to compact sets but may make arbitrarily deep cusp excursions. Divergence means it eventually leaves every compact set. These are logically distinct. A quantitative theorem usually controls how often a nondivergent orbit can penetrate to depth η, not merely whether it does so.

8. Singular vectors as a uniform version of Dirichlet improvement

A vector is singular when Dirichlet's theorem can be uniformly improved for all sufficiently large scales. Dynamically, this corresponds not merely to infinitely many cusp excursions, but to an orbit that remains outside larger and larger compacta in the relevant diagonal direction. The quantifier "for all sufficiently large scales" is what changes recurrence into divergence.

9. Why compactness theorems enter Diophantine approximation

A Diophantine inequality produces an explicit short lattice vector. Mahler converts short vectors into location in the cusp. Conversely, failure to remain in a Mahler compact set produces an integral short vector, hence an approximation. The compactness theorem is therefore the exact logical hinge between dynamics and arithmetic.

10. A debugging procedure for a new correspondence

Whenever a new Diophantine problem is encoded dynamically:

(1) write the integral vector and the unipotent matrix explicitly;
(2) multiply by the diagonal matrix coordinate by coordinate;
(3) solve the balancing equations rather than guessing the time scale;
(4) check determinant one and sign conventions;
(5) distinguish infinitely-often, uniformly-in-scale, and bounded-orbit quantifiers;
(6) only then invoke Mahler or quantitative non-divergence.

Quantitative exponents: very well approximable points. The bounded-orbit correspondence has a quantitative version. Suppose for infinitely many (p, q),

$$\|qy - p\|_\infty < |q|^{-1/n-\sigma}$$

for some $\sigma > 0$. Balance

$$e^{t/n}\,\|qy - p\|_\infty = e^{-t}|q|.$$

Writing the common value as R, multiplication of the n-th power of the first factor by the second yields

$$R^{n+1} = \|qy - p\|_\infty^n\,|q| < |q|^{-n\sigma}.$$

At the balanced time, $e^{t(1+1/n)} = |q|/\,\|qy - p\|$, so $t \asymp \log|q|$. Hence

$$\delta(g_t u_y \mathbb{Z}^{n+1}) \le e^{-ct}$$

for some $c = c(n, \sigma) > 0$ along infinitely many times. Conversely, an exponential improvement of the systole along balanced short vectors gives an improvement over the Dirichlet exponent. The precise conversion is elementary logarithmic algebra; the conceptual point is that power-law Diophantine improvement is the same as linear-in-time extra cusp depth.

Why one must balance before reading the exponent. If one observes a short vector at an arbitrary time t, either the approximation error or denominator coordinate may be much smaller than the other. Reading an exponent directly from one coordinate can therefore lose information. The balanced time extracts the scale-invariant product

$$\|qy - p\|_\infty^n \, |q|,$$

which is the natural Diophantine quantity. This is the dynamical analogue of optimizing a two-term inequality before comparing exponents.

Dual correspondence in detail. Let

$$A = |p + y \cdot q|, \qquad B = \|q\|_\infty .$$

For the dual flow,

$$\max(e^t A, e^{-t/n} B) \geq (AB^n)^{1/(n+1)}.$$

If the orbit is bounded below by systole η, balancing gives

$$AB^n \geq \eta^{n+1}, \qquad |p + y \cdot q| \geq \eta^{n+1} \, \|q\|_\infty^{-n} .$$

Thus boundedness of the dual orbit is equivalent to a badly approximable linear-form condition. This is the dual companion of Proposition 1.1.

The simultaneous and dual formulations are related arithmetically by transference, but dynamically they arise from different embeddings of the same parameter y into opposite horospherical groups. Keeping both matrix calculations explicit prevents accidental reversal of row and column conventions.

Multiplicative balancing, coordinate by coordinate. Suppose $q_i \neq 0$ for simplicity. Choose $r > 0$ and t_i by

$$e^{-t_i}|q_i| = r, \qquad t_i = \log(|q_i|/r).$$

Then

$$t_+ = \log \frac{|q_1 \cdots q_n|}{r^n}.$$

The first coordinate has magnitude

$$e^{t_+}|p + y \cdot q| = \frac{|q_1 \cdots q_n|}{r^n}|p + y \cdot q|.$$

Balancing it with r gives

$$r^{n+1} = |q_1 \cdots q_n| \, |p + y \cdot q|.$$

Thus the minimum possible maximum coordinate is exactly the $(n+1)$-st root of the multiplicative Diophantine product. When some $q_i = 0$, set $t_i = 0$ and replace $|q_i|$ by $\max(1, |q_i|)$; the same formula is recovered with $\Pi_+(q)$.

This calculation explains why the multiparameter diagonal semigroup is forced by multiplicative approximation. A one-parameter isotropic diagonal flow can see only $\|q\|$, not the separate sizes of the denominator coordinates.

Quantifier dictionary. The following translations are worth memorizing.

Arithmetic statement	Dynamical statement
uniform badly approximable lower bound	forward orbit stays in one Mahler compact set
infinitely many power improvements	infinitely many exponentially deep cusp excursions
uniform improvement at every large scale	eventual avoidance of every fixed compact set along the relevant ray
metric extremality	almost every parameter avoids exponentially deep cusp events except finitely often

Confusing the second and third rows is a common source of incorrect claims about singular vectors.

Sign conventions and equivalent embeddings. Some authors use u_y with $-y$ rather than y, or place y in a lower-left block instead of an upper-right block. These choices replace $p+qy$ by $p-qy$ or switch simultaneous and dual conventions. None changes the Diophantine content, but mixing conventions mid-proof changes which diagonal direction is expanding. The reliable check is always to multiply the matrix by the integer vector and inspect every coordinate explicitly.

Why determinant one fixes the balancing exponents. In the simultaneous model the first n coordinates expand by $e^{t/n}$ while the denominator coordinate contracts by e^{-t}. The relation

$$n(t/n) - t = 0$$

ensures determinant one. This same relation forces the invariant balanced product to be $A^n B$. Thus the Dirichlet exponent $1/n$ is already encoded in the weights of the diagonal subgroup. Weighted Diophantine approximation corresponds to changing these determinant-one weights rather than changing the basic compactness principle.

CHAPTER 9

Nondegenerate manifolds and strong extremality

1. The multiplicative Diophantine formulation

A point $y \in \mathbb{R}^n$ is strongly extremal if it is not VWMA. Equivalently, for every $\varepsilon > 0$, the inequality

$$|p + q \cdot y| < \Pi_+(q)^{-1-\varepsilon}$$

has only finitely many solutions $(p, q) \in \mathbb{Z} \times (\mathbb{Z}^n \setminus \{0\})$.

Let $f : U \subset \mathbb{R}^d \to \mathbb{R}^n$ be smooth and nondegenerate almost everywhere. The goal is to show that $f(x)$ is strongly extremal for Lebesgue-a.e. x.

2. The multiparameter lattice

For $t = (t_1, \dots, t_n) \in \mathbb{R}^n_+$ put $t_+ = t_1 + \cdots + t_n$ and

$$g_t = \operatorname{diag}(e^{t_+}, e^{-t_1}, \dots, e^{-t_n}), \quad u_y = \begin{pmatrix} 1 & y \\ 0 & I_n \end{pmatrix}, \quad \Lambda_y = u_y \mathbb{Z}^{n+1}.$$

For $v = (p, q)$,

$$g_t u_y v = (e^{t_+}(p + y \cdot q), e^{-t_1} q_1, \dots, e^{-t_n} q_n).$$

Lemma 2.1 (VWMA produces exponential cusp penetration)**.** *If y is VWMA, there exist $\gamma > 0$ and infinitely many $t \in \mathbb{Z}^n_+$ such that*

$$\delta_\infty(g_t \Lambda_y) \le e^{-\gamma t_+}.$$

Proof. Choose $\varepsilon_0 > 0$ and infinitely many (p, q) with

$$|p + q \cdot y| < \Pi_+(q)^{-1-\varepsilon_0}.$$

Put

$$r = \Pi_+(q)^{-\varepsilon_0/(n+1)}.$$

Choose $t_i \ge 0$ from $\max\{1, |q_i|\} = re^{t_i}$. Then

$$\Pi_+(q) = r^n e^{t_+}.$$

The last n coordinates satisfy $e^{-t_i}|q_i| \le r$. For the first,

$$e^{t_+}|p + q \cdot y| < e^{t_+}\Pi_+(q)^{-1-\varepsilon_0} = r^{-n}\Pi_+(q)^{-\varepsilon_0} = r.$$

Thus $\delta_\infty(g_t\Lambda_y) \le r$. Combining $r = \Pi_+(q)^{-\varepsilon_0/(n+1)}$ with $\Pi_+(q) = r^n e^{t_+}$ gives

$$r = e^{-\gamma t_+}, \qquad \gamma = \frac{\varepsilon_0}{n+1+n\varepsilon_0} > 0.$$

Replace t_i by integer parts. The resulting diagonal correction has uniformly bounded operator norm, so after decreasing γ slightly the estimate persists along an infinite sequence in $\mathbb{Z}_+^n$. □

3. A uniform cusp estimate near a nondegenerate point

Proposition 3.1 (Uniform multiparameter non-divergence)**.** *Let $f = (f_1, \dots, f_n) : U \to \mathbb{R}^n$ be C^ℓ, with $U \subset \mathbb{R}^d$ open, and suppose f is ℓ-nondegenerate at x_0. Then there exist a ball B centered at x_0 and constants $D, \rho > 0$ such that for every $t \in \mathbb{R}_+^n$ and every $0 < \eta \le \rho$,*

$$\left|\left\{x \in B : \delta_\infty(g_t u_{f(x)}\mathbb{Z}^{n+1}) < \eta\right\}\right| \le D\left(\frac{\eta}{\rho}\right)^{1/(d\ell)} |B|,$$

after harmlessly decreasing the exponent if required by the chosen finite-derivative lemma.

Proof. By Theorem 5.3, after shrinking around x_0, all affine combinations of

$$1,\ f_1, \dots, f_n$$

are uniformly good. Every coordinate of

$$(\bigwedge^r g_t u_{f(x)}) w_\Delta$$

is a diagonal exponential weight times one such affine combination. Multiplication by a nonzero scalar preserves goodness, and taking a finite maximum preserves it. Hence the wedge norms satisfy the goodness hypothesis of Theorem 1.1 uniformly in t.

It remains to establish non-collapse. Fix a rank and write the primitive wedge in the standard exterior basis. Factor out the largest diagonal weight among its nonzero coefficient channels. The normalized remaining coefficients lie on a compact coefficient sphere. If their transformed wedge norm had supremum tending to zero on B, a normalized subsequence would converge to a nonzero exterior coefficient vector for which every surviving affine combination of $1, f_1, \dots, f_n$ vanished identically on B. Differentiating at x_0 and using nondegeneracy forces all coefficients to vanish, contradiction. Because there are only finitely many exterior ranks and weight patterns, the lower bound ρ is uniform in t.

Apply Theorem 1.1. The exponent supplied by the standard finite-derivative sublevel estimate can be taken to be $1/(d\ell)$ after adjusting constants; any fixed positive exponent would suffice for the Borel–Cantelli argument below. □

4. Borel–Cantelli and strong extremality

THEOREM 4.1 (Strong extremality of nondegenerate manifolds). *Let $f : U \to \mathbb{R}^n$ be C^ℓ and nondegenerate almost everywhere. Then for Lebesgue-a.e. $x \in U$, $f(x)$ is strongly extremal.*

PROOF. It suffices to work on countably many balls on which Proposition 3.1 holds. Fix such a ball B and a positive γ. For $t \in \mathbb{Z}_+^n$ define

$$E_t(\gamma) = \{x \in B : \delta_\infty(g_t u_{f(x)}\mathbb{Z}^{n+1}) < e^{-\gamma t_+}\}.$$

The proposition gives

$$|E_t(\gamma)| \ll e^{-c\gamma t_+}|B|$$

for some $c > 0$. Since

$$\sum_{t\in\mathbb{Z}_+^n} e^{-c\gamma(t_1+\cdots+t_n)} = \prod_{i=1}^{n}\sum_{m\geq 0} e^{-c\gamma m} < \infty,$$

Borel–Cantelli implies that almost every $x \in B$ belongs to only finitely many $E_t(\gamma)$. Taking a countable sequence of positive rational γ and using Lemma 2.1, almost no x can have $f(x)$ VWMA. □

Remark 4.2. No independence is used. The convergence half of Borel–Cantelli needs only summability. This is why a one-sided quantitative cusp estimate is enough.

5. What the proof teaches beyond the theorem

The proof has four logically separate modules:

1. a Dani correspondence converts Diophantine improvement into deep cusp penetration;
2. finite-order nondegeneracy gives goodness of affine coordinate families;
3. exterior-power non-collapse excludes a rational obstruction that is small for the entire parameter ball;
4. quantitative non-divergence plus summability gives almost-everywhere exclusion.

This modularity is more important than the particular exponent.

The wedge-coordinate calculation behind the cusp estimate. We spell out the algebra suppressed in Proposition 3.1. Write $e_0, e_1, \dots, e_n$ for the standard basis. For a primitive rank-r subgroup, write

$$w = \sum_{|I|=r} c_I e_I, \qquad c_I \in \mathbb{Z}.$$

Under $u_{f(x)}$, a wedge e_I containing e_0 is fixed. If $0 \notin I$, then

$$u_{f(x)} e_I = e_I + \sum_{j \in I} \pm f_j(x) e_0 \wedge e_{I \setminus \{j\}}.$$

Thus every transformed coefficient has the form

$$c_J + \sum_{j=1}^{n} c_{J,j} f_j(x)$$

with integer coefficients that are linear combinations of the original Plücker coordinates. Applying g_t multiplies this affine combination by an exponential weight depending only on J.

For fixed t and w, select a coordinate channel for which the product

$$(\text{diagonal weight}) \times (\text{coefficient size})$$

is maximal after normalizing the nonzero integer coefficient vector. Because some c_I has modulus at least one, such a channel has a nonzero normalized coefficient vector. Proposition 4.1 gives a uniform lower supremum for the corresponding affine combination. The finite list of ranks and weight patterns lets us choose one $\rho > 0$ independent of t and the primitive subgroup.

The same coordinate calculation also gives goodness: after ignoring the scalar diagonal weight, every coordinate belongs to the finite-jet affine family controlled by Theorem 5.3. Hence both hypotheses of QND arise from the same explicit exterior formula, but from different properties of it: goodness from derivative anti-concentration, non-collapse from coefficient compactness.

Why integrality matters. The coefficient vector of w_Δ is integral and nonzero. Therefore its maximum norm is at least one before diagonal normalization. Without this arithmetic discreteness, a sequence of representation vectors could itself shrink to zero and the compactness argument would have no fixed coefficient scale. Primitive wedges are the arithmetic anchor of the entire proof.

Local-to-global measure argument. Nondegeneracy is assumed almost everywhere rather than everywhere. For each nondegenerate point x_0, choose a rational-radius ball $B_{x_0} \Subset U$ on which Proposition 3.1 holds. By second countability one extracts a countable subcover of the nondegenerate set. The Borel–Cantelli conclusion holds on each ball, and the countable union of exceptional null sets is null. No uniform nondegeneracy order over all of U is required.

If f is analytic and its image is not locally contained in any affine hyperplane, then finite-order nondegeneracy holds outside a proper analytic subset. In this common situation the theorem applies to almost every point automatically.

Summability over the parameter cone. For $m = t_1 + \cdots + t_n$, the number of $t \in \mathbb{Z}_+^n$ with $t_+ = m$ is

$$\binom{m+n-1}{n-1} = O_n(m^{n-1}).$$

Therefore

$$\sum_{t \in \mathbb{Z}_+^n} e^{-c\gamma t_+} = \sum_{m=0}^{\infty} O_n(m^{n-1}) e^{-c\gamma m} < \infty.$$

This alternative calculation makes explicit that a polynomial multiplicity of diagonal parameters is harmless against any exponential cusp-tail rate.

Why the theorem is local. The lower-supremum constant depends on the neighborhood on which finite-jet nondegeneracy is uniform. There is no reason for one constant to work on an entire noncompact parameter manifold. The theorem is therefore proved locally and globalized by a countable cover. This is not a weakness: metric statements are insensitive to the resulting countable union of null exceptional sets.

Strong extremality versus ordinary extremality. Ordinary extremality concerns simultaneous approximation with a one-parameter diagonal flow. Strong extremality concerns multiplicative approximation and requires a higher-rank diagonal cone. The analytic ingredient—QND for the parameterized unipotent family—is structurally the same. What changes is the Dani dictionary and the combinatorics of summing cusp events over the diagonal parameter set.

CHAPTER 10

Height functions and rank-sensitive cusp depth

1. Rank heights

For a unimodular lattice $\Lambda = g\mathbb{Z}^k$ and $1 \le r \le k-1$, define

$$\alpha_r(\Lambda) = \sup_{\substack{\Delta \le \mathbb{Z}^k \text{ primitive} \\ \operatorname{rank} \Delta = r}} \|g\Delta\|^{-1}.$$

The supremum is a maximum on bounded ranges because primitive wedge vectors form a discrete set. A convenient total height is

$$H(\Lambda) = 1 + \max_{1 \le r \le k-1} \alpha_r(\Lambda).$$

The rank-one term is comparable with $\delta(\Lambda)^{-1}$.

2. Comparison with successive minima

Proposition 2.1. *For fixed k and r there are constants $c_{k,r}, C_{k,r} > 0$ such that*

$$c_{k,r}(\lambda_1 \cdots \lambda_r)^{-1} \le \alpha_r(\Lambda) \le C_{k,r}(\lambda_1 \cdots \lambda_r)^{-1}.$$

Proof. For the lower bound fix $\eta > 0$ and choose independent vectors $v_1, \dots, v_r \in \Lambda$ with

$$\|v_i\| \le \lambda_i + \eta.$$

Let Δ be the primitive closure in Λ of the subgroup they generate. Passing to the primitive closure can only decrease covolume in its real span, while Hadamard's inequality gives

$$\operatorname{covol}(\Delta) \le \|v_1 \wedge \cdots \wedge v_r\| \le \prod_{i=1}^{r} (\lambda_i + \eta).$$

Hence

$$\alpha_r(\Lambda) \ge \prod_{i=1}^{r} (\lambda_i + \eta)^{-1}.$$

Letting $\eta \downarrow 0$ proves the lower bound (indeed with constant one for our Euclidean normalization).

For the upper bound let $\Delta < \Lambda$ be any primitive rank-r subgroup and let $\mu_1(\Delta) \le \cdots \le \mu_r(\Delta)$ be its successive minima in $\operatorname{span}_{\mathbb{R}} \Delta$. Any i independent vectors of Δ are also i independent vectors of Λ, so

$$\mu_i(\Delta) \ge \lambda_i(\Lambda), \qquad 1 \le i \le r.$$

Apply the r-dimensional lower inequality in Theorem 3.1 to the lattice Δ:

$$c_r \operatorname{covol}(\Delta) \le \mu_1(\Delta) \cdots \mu_r(\Delta)$$

is the wrong orientation for the desired estimate, so we use the upper inequality instead,

$$\mu_1(\Delta) \cdots \mu_r(\Delta) \le C_r \operatorname{covol}(\Delta).$$

Therefore

$$\operatorname{covol}(\Delta) \ge C_r^{-1} \prod_{i=1}^{r} \mu_i(\Delta) \ge C_r^{-1} \prod_{i=1}^{r} \lambda_i(\Lambda).$$

Taking reciprocals and then the supremum over primitive Δ gives

$$\alpha_r(\Lambda) \le C_r(\lambda_1 \cdots \lambda_r)^{-1}.$$

□

3. Properness

Proposition 3.1. *$H : X_k \to [1, \infty)$ is proper: the sublevel sets $\{H \le R\}$ are compact.*

Proof. If $H(\Lambda) \le R$, then the rank-one term gives $\delta(\Lambda)^{-1} \le R$ up to the fixed norm-comparison constant. Thus $\delta(\Lambda) \ge c/R$. Mahler's criterion gives compactness. □

4. Why keep higher-rank terms if rank one is already proper?

Rank one detects escape to infinity. Higher-rank heights identify *which rational subspace* is responsible for the excursion. They interact well with parabolic subgroups, exterior representations, and linearization. In quantitative proofs one needs not merely a proper exhaustion but a height whose individual pieces have algebraic meaning and controlled transformation laws.

5. Tail estimates from quantitative non-divergence

Since $H(\Lambda) \ge \delta(\Lambda)^{-1}$, Theorem 4.1 immediately implies

$$|\{t \in (0, T) : H(u_t\Lambda) > R\}| \ll R^{-\alpha} T$$

for a compact family of starting lattices, after changing constants. More refined rank-height tails follow by applying the marking theorem directly to the wedge functions of a fixed rank.

6. Height versus distance in the quotient

On finite-dimensional locally symmetric quotients, logarithmic height is coarsely comparable with distance to a fixed compact core:

$$\log H(\Lambda) \asymp d_{X_k}(\Lambda, K_0) + O(1)$$

for a suitable compact K_0, in reduction-theoretic regions. The exact comparison requires reduction theory and is not needed here. The useful principle is that polynomial tails in H correspond to exponential tails in geometric cusp depth.

Transformation inequalities. For $g \in \mathrm{GL}_k(\mathbb{R})$ and a rank-r primitive subgroup,

$$\|g\Delta\| \leq \left\| \bigwedge^r g \right\|_{\mathrm{op}} \|\Delta\|, \qquad \|g\Delta\| \geq \left\| (\bigwedge^r g)^{-1} \right\|_{\mathrm{op}}^{-1} \|\Delta\|.$$

Consequently

$$\alpha_r(g\Lambda) \leq \left\| (\bigwedge^r g)^{-1} \right\|_{\mathrm{op}} \alpha_r(\Lambda).$$

If g lies in a compact set, the rank heights of $g\Lambda$ and Λ are uniformly comparable. This elementary inequality is useful when orbit segments are decomposed into a controlled compact factor and a large diagonal factor.

Successive minima interpretation. If λ_1 is tiny but λ_2 is not, the cusp excursion is essentially rank one. If both λ_1 and λ_2 are tiny, then α_2 is large and a rational plane is collapsing. Rank heights therefore refine cusp depth into a profile of successive rational collapses. The marking chain can be viewed as a local, parameter-dependent version of this rank profile.

CHAPTER 11

General measures, local fields, and the S-arithmetic horizon

1. What the abstract proof really needs

The marking mechanism uses surprisingly little Euclidean structure. Its inputs are:

(1) a metric parameter space with a bounded-overlap covering theorem;
(2) a locally finite or doubling reference measure controlling ball enlargement;
(3) a finite-length poset of obstructions;
(4) good-function sublevel estimates;
(5) local finiteness of active obstruction labels;
(6) a non-collapse lower-supremum condition.

Lebesgue measure on $\mathbb{R}^d$ is only the cleanest model.

Definition 1.1 (Goodness relative to a measure)**.** Let μ be locally finite. A function f is (C,α)-good relative to μ if

$$\mu(\{x \in B : |f(x)| < \varepsilon\}) \le C \left(\frac{\varepsilon}{\sup_B |f|}\right)^{\alpha} \mu(B)$$

for all admissible balls B.

The marking proof then goes through with the ball-enlargement constants of μ inserted into the final numerical constant.

2. Local fields and products of places

Over a nonarchimedean local field K, Euclidean balls are replaced by ultrametric balls, Haar measure replaces Lebesgue measure, and polynomial sublevel estimates are governed by the valuation. The combinatorics is often simpler because intersecting balls are nested. Rational subspaces are encoded in exterior powers over K exactly as over $\mathbb{R}$.

For a finite set of places S, one works in a product

$$K_S = \prod_{v \in S} K_v$$

and with S-integral lattices. A natural content norm multiplies local norms,

$$\mathrm{cont}(x) = \prod_{v \in S} \|x_v\|_v ,$$

so the product formula and S-unit directions enter the normalization.

3. What changes and what does not

What survives unchanged:

- primitive submodules and exterior wedges;
- finite rank of chains;
- the distinction between goodness and non-collapse;
- maximal-scale selection;
- the logic "quantitative cusp tail $\Rightarrow$ tightness".

What changes:

- the correct notion of primitive S-submodule;
- local polynomial-goodness exponents;
- normalization by content rather than one Euclidean norm;
- the compactness theorem replacing Mahler;
- covering constants in product parameter spaces.

4. A methodological warning

Warning 4.1. One should not obtain an S-arithmetic theorem by writing the real proof and replacing $|\cdot|$ with $|\cdot|_v$ symbol by symbol. Product formulas, S-units, primitive module conventions, and compactness all require separate verification. This volume records only the architecture; the later local-field volume supplies the complete place-by-place theory.

> **Proof and provenance.** The full S-arithmetic quantitative non-divergence theorem is a source-level extension due to Kleinbock–Tomanov and related work. It is not used as an unproved input to any theorem in the real theory above.

Doubling measures and enlarged balls. Suppose μ satisfies a local doubling estimate

$$\mu(3B) \leq D\mu(B)$$

for all balls in the relevant domain. In the marking induction, each maximal-window step enlarges a selected ball by at most a factor three. Thus the

Euclidean geometric loss 3^d is replaced by D. If the ambient metric space also admits a Besicovitch constant N, the schematic conclusion becomes

$$\mu(\text{unmarked}) \ll kC(DN)^k(\varepsilon/\rho)^\alpha\mu(B).$$

This makes clear that absolute continuity of μ is not essential. What is essential is a covering theorem, controlled enlargement, and goodness relative to the same measure.

Ultrametric simplification. In an ultrametric space two balls that intersect are nested. Maximal balls in a bounded family are therefore pairwise disjoint, eliminating much of the Euclidean overlap bookkeeping. The complication moves elsewhere: polynomial sublevel geometry and the arithmetic normalization of S-integral wedges. Later volumes exploit this tradeoff explicitly.

CHAPTER 12

Worked examples and failure modes

1. A complete SL_2 family

Take

$$u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad \Lambda = g\mathbb{Z}^2,$$

with $g\mathbb{Z}^2$ in a compact subset of X_2. For $v = (a, b)$,

$$u_t g v$$

has coordinates affine in t. Thus every rank-one norm is uniformly good. Rank two contributes only

$$\left\| \bigwedge^2 u_t g(e_1 \wedge e_2) \right\| = |\det(u_t g)| = 1.$$

Compactness gives a uniform lower bound at $t = 0$, so Theorem 4.1 applies with a linear-polynomial exponent. This is the entire non-divergence mechanism in its smallest nontrivial form.

2. A rank-two obstruction in SL_3

Let

$$u_t = \begin{pmatrix} 1 & t & t^2/2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}.$$

The rank-one orbit of a vector has polynomial coordinates of degree at most two. For a plane represented by $v \wedge w$, the induced action on $\bigwedge^2 \mathbb{R}^3$ again has bounded-degree polynomial coordinates. It can happen that no single vector remains uniformly the shortest candidate while a rational plane maintains small covolume. This is why the proof tracks both ranks one and two even though Mahler itself only asks about vectors.

3. Failure of goodness

Let

$$f(x) = e^{-1/x^2}, \qquad f(0) = 0.$$

On shrinking intervals around zero, the function becomes flatter than any fixed polynomial order. A proof that assumes "smooth implies (C, α)-good with uniform constants" therefore fails. Finite-order nondegeneracy is the missing input.

4. Failure of non-collapse

Let

$$h_T(x) = \begin{pmatrix} e^{-T} & 0 \\ 0 & e^T \end{pmatrix}$$

independent of x. Every coordinate function is constant and hence good, but

$$\sup_x \|h_T(x)e_1\| = e^{-T} \to 0.$$

No quantitative non-divergence estimate uniform in T can follow. Anti-concentration and non-collapse are independent hypotheses.

5. Failure caused by a rational affine relation

Suppose $f : U \to \mathbb{R}^n$ satisfies

$$a_0 + a \cdot f(x) = 0 \qquad (x \in U)$$

for a nonzero rational coefficient vector. Then an integral multiple of (a_0, a) produces a rational direction annihilated by the horospherical affine coordinate. Under a suitable diagonal renormalization that direction can remain uniformly contracted. This is the dynamical meaning of degeneracy into a rational affine hyperplane.

6. A curve in an expanding horosphere

Take the Veronese curve

$$f(x) = (x, x^2, \dots, x^n).$$

Its derivatives $f'(x), \dots, f^{(n)}(x)$ form a triangular matrix with diagonal $1!, 2!, \dots, n!$, hence span $\mathbb{R}^n$ at every point. Therefore the curve is n-nondegenerate everywhere. Theorem 4.1 implies that almost every point on the curve is strongly extremal. The example shows how a concrete determinant calculation feeds directly into the abstract QND machine.

An expanding-horosphere tightness example. Let $\phi : I \to \mathbb{R}^n$ be smooth and consider probability measures

$$\nu_t = \frac{1}{|I|}\int_I \delta_{a_t u_{\phi(s)} x}\, ds$$

on a noncompact arithmetic quotient. Before asking whether ν_t converges to Haar measure, one must prove tightness. In a suitable lattice representation, the cusp condition becomes

$$\delta(a_t u_{\phi(s)} g\mathbb{Z}^k) < \eta.$$

If the wedge coordinate functions are good and the curve geometry supplies non-collapse, QND gives

$$\nu_t(X \setminus K_\eta) \ll \eta^\alpha$$

uniformly in t. Hence every weak-* limit has total mass one.

What remains is a different problem: producing additional invariance of a limit measure and classifying that measure. Quantitative non-divergence does not solve either task.

Proof architecture. For expanding translates, keep the logical order

$$\begin{aligned} &\text{cusp control} \longrightarrow \text{tightness} \longrightarrow \text{additional invariance},\\ &\longrightarrow \text{measure classification} \longrightarrow \text{identification of the limit.}\end{aligned}$$

Quantitative non-divergence supplies only the first arrow, but without it the later rigidity steps may be meaningless because mass could disappear.

CHAPTER 13

Complete case studies: from inequalities to cusp geometry

1. A direct cusp estimate in SL_2

Fix an interval $I \subset \mathbb{R}$, $T > 0$, and

$$h_T(s) = a_T u_s = \begin{pmatrix} e^T & e^T s \\ 0 & e^{-T} \end{pmatrix}.$$

For a primitive vector (p, q),

$$h_T(s)\begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} e^T(p+sq) \\ e^{-T}q \end{pmatrix}.$$

If the maximum norm is below η, then

$$|q| < \eta e^T, \qquad |s + p/q| < \frac{\eta e^{-T}}{|q|}$$

for $q \neq 0$. Thus each denominator contributes short intervals centered at rationals. A naive sum gives

$$\sum_{1 \le |q| < \eta e^T} \#\{p : -p/q \in I + O(1)\} \frac{\eta e^{-T}}{|q|},$$

which is of order $\eta|I|$ after the expected count of numerators, but endpoint overlap and uniformity in the starting lattice are awkward. The marking theorem packages exactly this arithmetic covering issue and survives in higher rank, where such a direct denominator count is no longer practical.

1.1. What changes if the initial lattice moves? Replacing $\mathbb{Z}^2$ by $g\mathbb{Z}^2$ destroys the literal rational centers $-p/q$ in the parameter coordinate, but the functions remain affine in s. If $g\mathbb{Z}^2$ ranges over a compact set, Lemma 3.1 supplies the missing uniform lower scale. This is the coordinate-free advantage of the exterior-power formulation.

2. Badly approximable numbers and bounded diagonal orbits

For $y \in \mathbb{R}$ take

$$u_y = \begin{pmatrix} 1 & y \\ 0 & 1 \end{pmatrix}, \qquad a_t = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}.$$

For $(p, q) \in \mathbb{Z}^2$,

$$a_t u_y \begin{pmatrix} -p \\ q \end{pmatrix} = \begin{pmatrix} e^t(qy - p) \\ e^{-t} q \end{pmatrix}.$$

If y is badly approximable, $|qy - p| \geq c/|q|$, so

$$\max(e^t|qy - p|, e^{-t}|q|)^2 \geq |qy - p||q| \geq c.$$

Mahler gives boundedness. Conversely, a uniform systole lower bound η and the balancing choice $e^{2t} = |q|/|qy - p|$ give

$$|q||qy - p| \geq \eta^2.$$

Thus the orbit is bounded exactly for badly approximable y.

2.1. Rational numbers as divergent orbits. If $y = p_0/q_0 \in \mathbb{Q}$, then

$$u_y \begin{pmatrix} -p_0 \\ q_0 \end{pmatrix} = \begin{pmatrix} 0 \\ q_0 \end{pmatrix},$$

so

$$a_t u_y \begin{pmatrix} -p_0 \\ q_0 \end{pmatrix} = \begin{pmatrix} 0 \\ e^{-t} q_0 \end{pmatrix} \to 0.$$

The orbit diverges to the cusp. This is stronger than merely having arbitrarily good approximations: an exact rational relation supplies one fixed vector contracted forever.

3. Dirichlet's theorem from a lattice balance

Theorem 3.1 (Dirichlet, one-dimensional form). *For every $y \in \mathbb{R}$ and every $Q \geq 1$, there exist integers p, q with $1 \leq q \leq Q$ such that*

$$|qy - p| \leq Q^{-1}.$$

Proof. Consider the unimodular lattice

$$\Lambda_y = \begin{pmatrix} 1 & -y \\ 0 & 1 \end{pmatrix} \mathbb{Z}^2.$$

Let

$$R_Q = [-Q^{-1}, Q^{-1}] \times [-Q, Q].$$

Its area is 4. Minkowski's convex-body theorem applied first to $(1 + \delta)R_Q$ gives a nonzero lattice vector $(p - qy, q)$ in that rectangle. For small δ, $q = 0$ would force a nonzero integer p with $|p| < 1$, impossible. Letting $\delta \downarrow 0$ and using discreteness gives the claimed bounds; change signs if necessary to make $q > 0$. □

There is an equivalent flow presentation: $Q = e^t$, and the rectangle is the inverse image under a_t of a fixed square. Dirichlet says every unstable-horospherical lattice has a forward translate with a nonzero vector in a fixed bounded region; bad approximability says such vectors are uniformly bounded away from zero.

4. A nondegenerate curve: the Veronese model

Let

$$f : I \to \mathbb{R}^n, \qquad f(x) = (x, x^2, \dots, x^n).$$

For $1 \le r, j \le n$,

$$\frac{d^r}{dx^r} x^j = \begin{cases} \dfrac{j!}{(j-r)!} x^{j-r}, & j \ge r, \\ 0, & j < r. \end{cases}$$

The matrix whose r-th row is $f^{(r)}(x)$ is upper triangular with diagonal $1!, 2!, \dots, n!$.

Proposition 4.1 (Veronese nondegeneracy). *The curve $x \mapsto (x, x^2, \dots, x^n)$ is n-nondegenerate at every point.*

Proof. The determinant of the derivative matrix is $\prod_{r=1}^n r! \neq 0$. □

Equivalently, a nonzero polynomial $c_0 + c_1 x + \cdots + c_n x^n$ cannot vanish to order greater than n. This is precisely the finite-jet input used by Theorem 5.3.

4.1. The lattice family attached to multiplicative approximation. Attach $u_{f(x)}\mathbb{Z}^{n+1}$ and act by the multiparameter diagonal semigroup of Chapter 9. Exterior coordinates are polynomial functions of x of bounded degree, and the derivative calculation gives non-collapse after weight normalization. Thus the local QND estimate is concrete on this curve, and Borel–Cantelli gives strong extremality.

5. An exact rational relation: how the mechanism fails

Let $f(x) = (x, 2x + 1)$. Then

$$1 + 2f_1(x) - f_2(x) = 0$$

identically. The curve lies in a rational affine line and is not nondegenerate in $\mathbb{R}^2$. In the horospherical lattice, the coefficient vector $(1, 2, -1)$ defines an exact rational relation. A diagonal flow can amplify the associated algebraic obstruction. The failure is not that polynomial functions cease to be good; it is the lower-supremum condition that fails in the relevant exterior channel.

6. A checklist for applying quantitative non-divergence

For a new family $h(x)\Lambda$:

(1) specify the lattice/module and all norms;
(2) identify primitive obstructions in every rank;
(3) prove coordinate goodness uniformly;
(4) prove a lower supremum for every primitive obstruction;
(5) verify local finiteness;
(6) choose ρ compatible with the rank-extension normalization;
(7) invoke QND only after these steps;
(8) translate the cusp estimate back to the original arithmetic/dynamical question.

Detailed direct estimate: comparison with the abstract theorem. We return to

$$E_T(\varepsilon) = \{s \in [0,1] : \delta_\infty(a_T u_s \mathbb{Z}^2) < \varepsilon\}, \qquad 0 < \varepsilon < 1.$$

A nonzero integer vector (p,q) is short only if

$$|p + sq| < \varepsilon e^{-T}, \qquad |q| < \varepsilon e^T. \tag{6.1}$$

We may assume (p,q) primitive. Also $q \neq 0$: if $q = 0$, primitivity gives $|p| = 1$, whereas the first inequality would imply $e^T < \varepsilon < 1$.

Put

$$Q = \varepsilon e^T.$$

If $Q \leq 1$, no nonzero integer q satisfies the second inequality and $E_T(\varepsilon)$ is empty. Suppose $Q > 1$. For fixed $q > 0$, $q < Q$, the first inequality in (6.1) places s in one of the intervals

$$I_{p,q} = \left(-\frac{p}{q} - \frac{\varepsilon e^{-T}}{q}, -\frac{p}{q} + \frac{\varepsilon e^{-T}}{q} \right).$$

Only p for which $I_{p,q}$ meets $[0,1]$ matter. Their number is at most $q+2$. Hence

$$\begin{aligned}
|E_T(\varepsilon)| &\leq 2 \sum_{1 \leq q < Q} (q+2) \frac{2\varepsilon e^{-T}}{q} \\
&\ll \varepsilon e^{-T} \sum_{q<Q} \left(1 + \frac{2}{q}\right) \\
&\ll \varepsilon e^{-T} (Q + \log Q) \\
&\ll \varepsilon^2 + \varepsilon e^{-T} \log(2 + \varepsilon e^T).
\end{aligned}$$

Because $Q > 1$ implies $e^{-T} < \varepsilon$, the logarithmic term is $O(\varepsilon^2(1+T))$. A slightly sharper organization by primitive fractions, or a dyadic decomposition in q, gives a uniform $O(\varepsilon^2)$ bound on bounded s-intervals away from endpoint artifacts. The abstract QND theorem yields a power law directly and, unlike this count, remains stable under compact perturbations of the initial lattice and in higher rank.

The exact exponent in this special model is stronger than what a general bounded-degree goodness theorem promises. This illustrates a recurring choice: direct arithmetic counting can improve exponents in low rank, while QND sacrifices sharpness in exchange for structural uniformity.

The same count on an interval of length L. If $s \in J$ with $|J| = L$, the number of relevant numerators for each q is $Lq+O(1)$. The same computation gives

$$|E_T(\varepsilon) \cap J| \ll L\varepsilon^2 + \varepsilon e^{-T} \log(2+Q).$$

For intervals not too small relative to the natural rational spacing, the first term dominates. The second is a boundary term. In a covering proof, maximal-window normalization is a coordinate-free way of controlling precisely these boundary effects.

Why rank two is automatic here. For $k = 2$, the only nonzero primitive rank-two subgroup is $\mathbb{Z}^2$ itself and

$$\left\| a_T u_s \mathbb{Z}^2 \right\| = |\det(a_T u_s)| = 1.$$

Thus the full-rank lower-supremum hypothesis never costs anything. All genuine arithmetic obstruction is rank one. The marking theorem is already nontrivial because infinitely many primitive lines compete, but the rank chain has depth at most two.

Bad approximation: all constants exposed. For $y \in \mathbb{R}$, define

$$L_y(t) = a_t u_y \mathbb{Z}^2.$$

Assume

$$|qy - p| \geq \frac{c}{|q|} \qquad (q \neq 0).$$

For any nonzero (p, q) with $q \neq 0$,

$$\|a_t u_y(-p, q)\|_\infty = \max(e^t |qy - p|, e^{-t}|q|).$$

The elementary inequality $\max(A, B)^2 \geq AB$ gives

$$\|a_t u_y(-p, q)\|_\infty^2 \geq |qy - p||q| \geq c.$$

For $q = 0$, the vector has first coordinate $e^t|p| \geq 1$ when $t \geq 0$. Thus

$$\delta_\infty(L_y(t)) \geq \min(1, \sqrt{c})$$

for all $t \geq 0$.

Conversely suppose $\delta_\infty(L_y(t)) \geq \eta$ for every $t \geq 0$. Given $q \neq 0$, choose p nearest to qy, so $A = |qy - p| \leq 1/2$. If $|q|/A \geq 1$, choose

$$e^{2t} = \frac{|q|}{A}.$$

At this time both coordinates equal $\sqrt{A|q|}$, hence

$$A|q| \geq \eta^2.$$

If $|q|/A < 1$, then $A > |q| \geq 1$, impossible for a nearest integer choice. Thus

$$|qy - p| \geq \eta^2 |q|^{-1}$$

for all p, q. The constants in the dynamical and arithmetic formulations are explicitly related by a square.

Continued fractions as cusp geometry. Although not needed in the proof, convergents p_n/q_n of the continued fraction of y produce the deepest successive cusp excursions. The balanced time satisfies roughly $e^{2t_n} \asymp q_n/|q_n y - p_n|$. Bounded partial quotients are equivalent to a uniform lower bound on $q_n|q_n y - p_n|$, hence to bounded cusp depth. This is the classical continued-fraction picture expressed in the language that generalizes to higher-dimensional homogeneous spaces.

Dirichlet via pigeonhole and via Minkowski. The lattice proof of Dirichlet can be compared with the elementary pigeonhole proof. Partition $[0, 1)$ into Q intervals of length $1/Q$ and consider the $Q + 1$ fractional parts

$$0, \{y\}, \ldots, \{Qy\}.$$

Two lie in the same interval; subtracting them gives $1 \leq q \leq Q$ and $\|qy\| \leq 1/Q$. In the lattice formulation, Minkowski packages the same counting principle into a convex-body volume threshold. The advantage of Minkowski is that the convex body can be deformed by diagonal flows, weighted norms, or product structures without rebuilding a pigeonhole partition from scratch.

Veronese model: full strong-extremality chain. Fix a compact interval J and

$$f(x) = (x, x^2, \ldots, x^n).$$

Step 1 is finite-order nondegeneracy, already verified by the triangular derivative determinant. Step 2 embeds $f(x)$ as

$$u_{f(x)} = \begin{pmatrix} 1 & x & x^2 & \cdots & x^n \\ 0 & & I_n & & \end{pmatrix}.$$

Step 3 chooses the multiparameter diagonal group

$$g_t = \operatorname{diag}(e^{t_+}, e^{-t_1}, \ldots, e^{-t_n}).$$

Step 4 expands every primitive wedge in the exterior basis. Each coordinate is a scalar exponential weight times a polynomial in x of degree at most n.

Polynomial goodness is therefore uniform. Step 5 uses the nondegenerate coefficient compactness argument to produce a common lower supremum. QND then gives

$$|\{x \in J : \delta(g_t u_{f(x)} \mathbb{Z}^{n+1}) < \eta\}| \ll \eta^\alpha |J|$$

uniformly in t. Step 6 chooses $\eta = e^{-\gamma t_+}$ and sums over $t \in \mathbb{Z}_+^n$. Borel–Cantelli excludes exponentially deep cusp excursions almost everywhere. Step 7 uses the multiplicative Dani correspondence to conclude strong extremality.

Notice the division of labor. The polynomial nature of the Veronese curve gives goodness. Its nondegeneracy gives non-collapse. Integrality of primitive wedges fixes the coefficient scale. The diagonal semigroup translates the multiplicative denominator product. QND supplies the tail. Borel–Cantelli supplies almost-everywhere finiteness. No one of these steps can be omitted without replacing it by a theorem that contains the same information.

A second exact rational-relation test. Consider

$$f(x) = (x, x^2, x + x^2).$$

The relation

$$f_1 + f_2 - f_3 = 0$$

is linear and rational, so the image lies in a proper rational hyperplane even though the coordinate functions display nonlinear behavior. Derivatives of f span at most a two-dimensional subspace of $\mathbb{R}^3$, and nondegeneracy fails. In an exterior channel the coefficient vector corresponding to $(0, 1, 1, -1)$ produces an identically vanishing affine coordinate. This example is useful because visual curvature alone does not imply homogeneous-dynamics nondegeneracy; the relevant condition is affine span detected by finite jets.

CHAPTER 14

Exercises, reconstruction problems, and research laboratories

The exercises are included to test proof reconstruction rather than formula recall. Results used essentially in the main text have already been proved or clearly labelled as imported.

1. Cusp geometry

Exercise 1.1. Prove directly that the systole function is continuous on X_k without using a local fundamental domain. Identify the finite set of integer vectors that must be checked near a fixed $g \in \mathrm{SL}_k(\mathbb{R})$.

Exercise 1.2. For $a_t = \mathrm{diag}(e^{t_1}, \dots, e^{t_k})$ with $\sum t_i = 0$, determine which coordinate sublattices become thin along a ray $t = sv$. Relate this to faces of a Weyl chamber.

2. Primitive subgroups and wedges

Exercise 2.1. Prove Proposition 1.2 using saturation rather than Smith normal form.

Exercise 2.2. Let Δ_1, Δ_2 be primitive. Show that the primitive closure of $\Delta_1 + \Delta_2$ is represented projectively by the decomposable part of the corresponding wedge whenever the spans meet trivially. Track the possible index loss.

Exercise 2.3. Construct in $\mathbb{Z}^3$ a family of lattices for which the shortest vector changes identity repeatedly while one rank-two primitive subgroup remains the natural stable obstruction.

3. Good functions

Exercise 3.1. Improve the constant in Proposition 3.2 using sharper separation of interpolation points. Explain why the exponent, unlike the constant, cannot be improved uniformly.

Exercise 3.2. For $P(x,y) = x^2 + y^3$, estimate the measure of $\{|P| < \varepsilon\}$ in the unit square. Compare the direct estimate with a generic multivariable goodness exponent.

Exercise 3.3. Give a complete proof of a finite-derivative sublevel estimate in one dimension: if $|f^{(j)}| \geq c$ on an interval and lower derivatives are uniformly bounded, prove a power-law estimate for $|\{|f| < \varepsilon\}|$.

4. The marking theorem

Exercise 4.1. Reconstruct the proof of Theorem 3.1 from the four phrases: maximal window, goodness, finite $3r$ selection, rank descent. Your write-up must state exactly where the factor $k5^{dk}$ enters.

Exercise 4.2. Show by example why a covering argument that chooses the smallest rather than largest admissible radius can fail to normalize $\sup \psi$ at the reference scale.

5. Quantitative non-divergence

Exercise 5.1. Reprove Lemma 3.1 carefully in the cases where the short vector lies inside the real span of some members of the marking chain and outside others.

Exercise 5.2. Translate Theorem 1.1 from the maximum norm to Euclidean norm and give explicit norm-equivalence losses in dimension three.

6. Dani correspondence and strong extremality

Exercise 6.1. Derive the simultaneous Dani correspondence in dimension n using an arbitrary diagonal normalization $\mathrm{diag}(e^{at}, e^{-bt}, \ldots, e^{-bt})$ satisfying determinant one. Verify that the Diophantine exponent is unchanged.

Exercise 6.2. Starting from the VWMA inequality, recompute the exponent γ in Lemma 2.1. Then run the argument backward and state the converse cusp-to-Diophantine implication with explicit constants.

Exercise 6.3. Explain why the sum over $t \in \mathbb{Z}_+^n$ in the proof of Theorem 4.1 converges for every positive exponential rate, despite the polynomial number of points with fixed $t_1 + \cdots + t_n$.

7. Research laboratories

Research Laboratory 7.1 (Friendly-measure QND)**.** Replace Lebesgue measure by an absolutely decaying Federer measure. Identify exactly which parts of the marking proof remain unchanged and which constants must be replaced by doubling constants. Work out one self-similar example.

Research Laboratory 7.2 (Weighted Diophantine approximation)**.** Choose weights $r_1 + \cdots + r_n = 1$ and the diagonal flow

$$g_t = \operatorname{diag}(e^t, e^{-r_1 t}, \ldots, e^{-r_n t}).$$

Derive the corresponding weighted Dani dictionary, formulate a nondegeneracy condition sufficient for QND, and test it on a polynomial curve.

Research Laboratory 7.3 (Effective cusp tails)**.** Suppose an independent mixing theorem is available for a diagonal action. Investigate how the polynomial cusp tail from QND can be combined with mixing to estimate extreme cusp excursions or logarithm laws. Separate what QND supplies from what mixing must supply.

Research Laboratory 7.4 (S-arithmetic reconstruction)**.** Using the later local-field volumes, rebuild Theorem 1.1 over a product of one real and one p-adic place. Keep a ledger of every point where the real proof must be modified: norms, primitive modules, polynomial goodness, covering, and compactness.

Proof-reconstruction solutions and hints. The following hints are deliberately more substantial than ordinary exercise hints; they are intended to let a reader audit the dependency chain without consulting an external source.

Controlled basis. Choose a primitive shortest vector, project to its orthogonal complement, apply induction to the projected lattice, and lift projected basis vectors after subtracting integer multiples of the shortest vector. The only subtle point is primitivity: it ensures the projected group has rank $k-1$ and the lifted vectors together with the shortest vector generate the original lattice.

Polynomial goodness. If $E = \{|P| < \varepsilon\}$ occupies a positive portion of an interval, select $n+1$ well-separated points in E. Lagrange interpolation controls the supremum of P on the whole interval in terms of ε and the separation. Rearranging gives the sublevel estimate. The model x^n proves the exponent cannot be uniformly improved.

Marking induction. The induction parameter is not the number of active labels. It is the maximum length of a chain that can still be appended. At a bad point select one active obstruction, grow a maximal ball until its supremum is ρ, pay goodness on its ε-sublevel set, and on the complement append it to the chain. the finite $3r$ selection is used only to sum these local maximal-window estimates.

Marked implies no short vector. If the short vector is already the rank-one chain member, marking contradicts its shortness directly. Otherwise select the maximal recorded subgroup whose real span omits the vector, take the primitive closure after adjoining the vector, and use

$$\|h\Lambda\| \leq k\,\|h\Delta\|\,\|hv\| < k\rho\varepsilon \leq \varepsilon.$$

The new primitive subgroup is comparable with the whole chain, so marking gives a contradictory lower bound.

Dani balancing. Never guess t. Set the two weighted coordinates equal. In simultaneous approximation this produces the invariant product $\|qy - p\|^n\,|q|$; in the dual problem it produces $|p + y \cdot q|\,\|q\|^n$; in the multiplicative problem it produces $|p + y \cdot q|\Pi_+(q)$.

Strong extremality. After QND gives $|E_t| \ll e^{-ct_+}$, count the number of $t \in \mathbb{Z}_+^n$ with fixed $t_+ = m$ by $O(m^{n-1})$. Exponential decay beats this polynomial multiplicity. Borel–Cantelli then needs no correlation estimate.

Additional laboratories.

Research Laboratory 7.5 (Sharpness in rank two)**.** For the family $a_T u_s \mathbb{Z}^2$, determine the true asymptotic order of $|E_T(\varepsilon)|$ in the regimes $\varepsilon e^T \ll 1$, $\varepsilon e^T \asymp 1$, and $\varepsilon e^T \gg 1$. Compare with the general QND exponent and identify exactly where the abstract proof loses sharpness.

Research Laboratory 7.6 (Exterior-height profile)**.** For a diagonal lattice $\Lambda_t = \operatorname{diag}(e^{t_1}, \ldots, e^{t_k})\mathbb{Z}^k$ with $\sum t_i = 0$, compute every rank height $\alpha_r(\Lambda_t)$ explicitly in terms of ordered sums of the t_i. Relate the dominant primitive subgroup to the parabolic face approached by t.

Research Laboratory 7.7 (Finite-dimensional nonpolynomial family)**.** Use the trigonometric family V_N from Chapter 3 to formulate and prove a QND statement for a matrix family whose exterior coordinates belong to V_N. Identify what replaces polynomial degree and verify non-collapse on a compact coefficient sphere.

Suggested full reconstruction: the main theorem. Write a self-contained proof of Theorem 1.1 without referring to Chapters 4–5 by number. Your proof should introduce primitive subgroups, define the marking chain, state the maximal-window selection, invoke the finite $3r$ selection exactly once at each induction level, prove the rank-extension inequality, and finish by showing that every short vector contradicts marking. A correct reconstruction should make it possible for a reader to identify precisely where continuity, goodness, lower supremum, local finiteness, finite rank, and $\rho \leq 1/k$ enter.

Suggested full reconstruction: strong extremality. Starting from a C^ℓ nondegenerate map, rebuild the argument through the following chain without using the phrase "standard": affine exterior-coordinate formula; uniform finite-jet goodness; coefficient-sphere non-collapse; QND; exponential cusp events; counting multiindices with fixed t_+; Borel–Cantelli; multiplicative

Dani correspondence. The covering selection and geometry-of-numbers steps must be reconstructed from the proved appendices in this volume; the only source-level input permitted in this exercise is ordinary foundational real analysis below the theorem level.

Research-facing extension. Replace the affine horospherical parametrization by a polynomial map into a higher-step unipotent group. Determine the finite-dimensional monomial family occurring in each exterior representation. Formulate a finite-jet condition that guarantees both goodness and non-collapse after diagonal renormalization. The problem is deliberately open-ended: the objective is to separate representation-theoretic polynomiality from the geometric condition that rules out a uniformly contracted rational wedge.

APPENDIX A

Geometry-of-numbers inputs

1. Primitive shortest vectors

A shortest nonzero vector v in a lattice Λ is primitive in Λ: if $v = mv'$ with $m \in \mathbb{Z}$, $|m| \geq 2$, and $v' \in \Lambda$, then $\|v'\| < \|v\|$.

2. Projection of a lattice

Lemma 2.1 (Primitive projection)**.** *Let $v \in \Lambda$ be primitive and let $\pi : \mathbb{R}^k \to v^\perp$ be orthogonal projection. Then $\pi(\Lambda)$ is a lattice of rank $k-1$ in $v^\perp$, and*

$$\operatorname{covol}(\Lambda) = \|v\| \operatorname{covol}(\pi(\Lambda)).$$

Proof. Primitivity says $\mathbb{Z}v$ is a direct summand of the abstract abelian group Λ. Choose a basis $v, b_2, \dots, b_k$. Then $\pi(b_2), \dots, \pi(b_k)$ are linearly independent and generate $\pi(\Lambda)$. The volume of the parallelepiped spanned by $v, b_2, \dots, b_k$ equals base volume times height $\|v\|$, yielding the covolume identity. □

3. Minkowski's second theorem in the Euclidean form used in this volume

We now prove the two-sided dimension-dependent estimate used in Theorem 2.1. The sharp numerical constants are not needed; what matters is that the constants depend only on dimension.

Theorem 3.1 (Minkowski's second theorem, Euclidean form). *For every $k \geq 1$ there are constants $0 < c_k \leq C_k < \infty$ such that every full-rank lattice $\Lambda \subset \mathbb{R}^k$ satisfies*

$$c_k \operatorname{covol}(\Lambda) \leq \lambda_1(\Lambda) \cdots \lambda_k(\Lambda) \leq C_k \operatorname{covol}(\Lambda).$$

Consequently, for a unimodular lattice the product of the successive minima is bounded above and below by positive constants depending only on k.

Proof. We prove the two inequalities separately.

Lower bound. Fix $\eta > 0$. By the definition of successive minima one can choose linearly independent vectors $v_1, \dots, v_k \in \Lambda$ with

$$\|v_i\| \leq \lambda_i + \eta.$$

Put $w_i = v_i/(\lambda_i + \eta)$. Every w_i lies in the Euclidean unit ball B_k. Since B_k is convex and centrally symmetric, it contains the cross-polytope

$$P = \mathrm{conv}\{\pm w_1, \dots, \pm w_k\}.$$

The 2^k simplices obtained by fixing the signs have disjoint interiors and each has volume $|\det(w_1, \dots, w_k)|/k!$. Hence

$$\mathrm{vol}(B_k) \geq \mathrm{vol}(P) = \frac{2^k}{k!} \frac{|\det(v_1, \dots, v_k)|}{(\lambda_1 + \eta) \cdots (\lambda_k + \eta)}.$$

The vectors v_i generate a finite-index sublattice of Λ, so

$$|\det(v_1, \dots, v_k)| = [\Lambda : \sum_i \mathbb{Z} v_i] \,\mathrm{covol}(\Lambda) \geq \mathrm{covol}(\Lambda).$$

Letting $\eta \downarrow 0$ gives

$$\lambda_1 \cdots \lambda_k \geq \frac{2^k}{k!\ \mathrm{vol}(B_k)} \mathrm{covol}(\Lambda).$$

Thus one may take $c_k = 2^k/(k! \,\mathrm{vol}(B_k))$.

Upper bound. We argue by induction on k. The case $k = 1$ is equality. Assume the result in dimension $k - 1$. Let $v \in \Lambda$ be a shortest nonzero vector, so $\|v\| = \lambda_1$; it is primitive. Let

$$\Lambda' = \pi(\Lambda) \subset v^{\perp}$$

and let $\mu_1 \leq \cdots \leq \mu_{k-1}$ be its successive minima. By Lemma 2.1,

$$\mathrm{covol}(\Lambda') = \frac{\mathrm{covol}(\Lambda)}{\lambda_1}.$$

We need one elementary comparison. Every nonzero $\bar{w} \in \Lambda'$ has a lift $w \in \Lambda$ of the form

$$w = \bar{w} + tv, \qquad |t| \leq \frac{1}{2},$$

after subtracting an integral multiple of v. Since v is shortest, $\|w\| \geq \lambda_1$. Orthogonality therefore gives

$$\|\bar{w}\|^2 = \|w\|^2 - t^2 \lambda_1^2 \geq \frac{3}{4} \lambda_1^2.$$

Thus

$$\mu_1 \geq \frac{\sqrt{3}}{2} \lambda_1. \tag{A.1}$$

For each $1 \leq i \leq k-1$, choose i independent projected vectors of length at most $\mu_i + \eta$ and lift each of them with the v-component reduced to absolute coefficient at most $1/2$. Together with v these give $i+1$ independent vectors of Λ. By (A.1),

$$\|w\|^2 \leq (\mu_i+\eta)^2 + \frac{1}{4}\lambda_1^2 \leq (\mu_i+\eta)^2 + \frac{1}{3}\mu_i^2.$$

Letting $\eta \downarrow 0$ yields

$$\lambda_{i+1}(\Lambda) \leq \frac{2}{\sqrt{3}}\mu_i. \tag{A.2}$$

Therefore, by the induction hypothesis in $v^\perp$,

$$\begin{aligned}\lambda_1 \cdots \lambda_k &\leq \lambda_1 \left(\frac{2}{\sqrt{3}}\right)^{k-1} \mu_1 \cdots \mu_{k-1} \\ &\leq \left(\frac{2}{\sqrt{3}}\right)^{k-1} C_{k-1}\lambda_1 \operatorname{covol}(\Lambda') \\ &= \left(\frac{2}{\sqrt{3}}\right)^{k-1} C_{k-1} \operatorname{covol}(\Lambda).\end{aligned}$$

This closes the induction, for instance with $C_1 = 1$ and $C_k = (2/\sqrt{3})^{k-1}C_{k-1}$. □

Remark 3.2. The classical sharp upper constant is $2^k/\operatorname{vol}(B_k)$ and the sharp lower constant above can also be presented in the usual convex-body formulation. No sharp constant enters the dynamics. The proof above was included to own the exact dimension-dependent mechanism used by Mahler compactness and the rank-height comparison.

4. Controlled basis by induction

Proof of Lemma 2.2. We prove the stated coordinatewise estimate, not merely a bound in terms of the largest successive minimum. The case $k=1$ is immediate. Let v_1 be a primitive shortest vector of Λ, so $\|v_1\| = \lambda_1(\Lambda)$, and let $\Lambda' = \pi(\Lambda) \subset v_1^\perp$.

Write $\mu_1 \leq \cdots \leq \mu_{k-1}$ for the successive minima of Λ'. For each $i \geq 2$, choose i independent vectors in Λ of norm at most $\lambda_i + \eta$. Their projections have span of dimension at least $i-1$, because the kernel of π is the one-dimensional line $\mathbb{R}v_1$. Hence

$$\mu_{i-1} \leq \lambda_i + \eta,$$

and therefore $\mu_{i-1} \leq \lambda_i$ after letting $\eta \downarrow 0$.

By induction, Λ' has a basis $\bar{b}_2, \ldots, \bar{b}_k$ satisfying

$$\left\|\bar{b}_i\right\| \leq A_{k-1}\mu_{i-1} \leq A_{k-1}\lambda_i.$$

Lift each $\bar{b}_i$ to $b_i \in \Lambda$ and subtract an integral multiple of v_1 so that the coefficient of v_1 in the orthogonal decomposition of b_i has absolute value at most 1/2. Then

$$\|b_i\|^2 \leq A_{k-1}^2 \lambda_i^2 + \frac{1}{4}\lambda_1^2 \leq \left(A_{k-1}^2 + \frac{1}{4}\right) \lambda_i^2.$$

Because the projected vectors form a basis of Λ' and v_1 is primitive, $v_1, b_2, \ldots, b_k$ form a basis of Λ. Thus the induction closes with

$$A_1 = 1, \qquad A_k = \sqrt{A_{k-1}^2 + \frac{1}{4}}.$$

□

5. Proof dependencies

Theorem 3.1 proves locally the exact geometry-of-numbers estimate invoked in Chapter 2.1; no external Minkowski-II theorem is needed by this volume. The proof is compatible with the sharper version proved independently in the prerequisite Ratner foundations, but the present Advanced volume no longer depends on that external owner.

APPENDIX B

Finite-derivative sublevel estimates

This appendix records the analytic input behind Theorem 5.3.

1. One-dimensional derivative lemma

Lemma 1.1. *Let I be an interval and $f \in C^j(I)$. Suppose*

$$|f^{(j)}(x)| \geq c > 0 \qquad (x \in I).$$

Then every connected component of $\{x \in I : |f(x)| < \varepsilon\}$ has length at most

$$C_j(\varepsilon/c)^{1/j},$$

and the whole sublevel set is a union of at most $2j+1$ such components after partitioning I into finitely many intervals on which the relevant lower derivatives are monotone. Consequently

$$|\{|f| < \varepsilon\}| \leq C_j'(\varepsilon/c)^{1/j}$$

up to the natural interval-scale normalization.

Proof. If $J = [a, b]$ lies in the sublevel set, repeated finite differences give

$$|\Delta_h^j f(a)| \leq 2^j \varepsilon$$

for $h = (b-a)/j$, while the generalized mean-value theorem gives

$$\Delta_h^j f(a) = h^j f^{(j)}(\xi)$$

for some $\xi \in (a, b)$. Hence

$$(b-a)^j c / j^j \leq 2^j \varepsilon,$$

which bounds the component length. Rolle's theorem bounds the number of alternating sublevel components once $f^{(j)}$ has fixed sign; splitting according to signs of the finitely many derivatives produces a constant depending only on j. □

2. From nondegeneracy to coordinate derivative bounds

For $f : U \subset \mathbb{R}^d \to \mathbb{R}^n$ nondegenerate at x_0, Lemma 5.2 gives a uniform nonzero partial derivative for every normalized linear combination. By compactness of the unit sphere and continuity, one may choose finitely many multiindices and a small ball B so that every normalized combination has one selected derivative bounded below throughout a controlled subball.

To pass from one-dimensional derivative bounds to d dimensions, slice B along a coordinate direction associated with that derivative, apply Lemma 1.1 on almost every line, and use Fubini. Iterating for a multiindex of total order at most ℓ gives a power-law exponent that may be taken, after a uniform weakening, as $1/(d\ell)$. This proves the finite-derivative form of Theorem 5.3 used in Chapter 9.

Remark 2.1. Sharp exponents are not the point. The dynamical applications need a positive exponent uniform over a compact normalized coefficient family. Weakening the exponent preserves every subsequent summability argument.

APPENDIX C

Source map and bibliographic notes

1. Source map

The proof architecture follows the classical quantitative non-divergence technology of Dani–Margulis and Kleinbock–Margulis. The abstract marking argument is reconstructed in Chapter 4 and the lattice deduction in Chapter 5. The geometry-of-numbers material is standard. The nondegenerate-manifold application follows the Kleinbock–Margulis strategy: Dani correspondence, goodness of finite-jet families, exterior-power non-collapse, and Borel–Cantelli.

The local-field discussion in Chapter 11 is orientation only. The precise S-arithmetic quantitative non-divergence theorem requires the full local-field setup and is treated in later volumes rather than being silently imported here.

2. Bibliography

Bibliography

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Provenance note.

- The maximum-norm quantitative non-divergence theorem, including the enlarged parameter ball, the restriction $\rho \leq 1/k$, and the chain-depth dependence, is reconstructed from the surviving Volume 1 PDF and checked structurally against the Kleinbock–Margulis formulation.
- The abstract marking proof is given in full in Chapter 4. The covering statement used there is proved locally as a finite $3r$ selection lemma. Compact bad sets are first covered by one-third shrinkings of the maximal windows; a disjoint selected subfamily has triples covering the whole compact set. Thus the classical infinite-family Besicovitch theorem is provenance rather than a logical input.
- The local nondegenerate-manifold estimate is deduced explicitly from exterior coordinates, finite-jet goodness, and non-collapse rather than cited as a black box.
- The S-arithmetic theorem is intentionally only an orientation statement here; its exact hypotheses belong to the later local-field volume and the Kleinbock–Tomanov source.
- This recovered source is mathematically reconstructed from the exact generated 87-page PDF and the shared series style. It is not asserted to be byte-for-byte identical to the lost source archive.

Conceptual exit checklist

The following questions test whether the mechanisms have been internalized rather than replaced by the phrase "by quantitative non-divergence."

(1) Why does Mahler compactness mention only short vectors while the quantitative proof uses primitive subgroups of all ranks?
(2) What information is stored by $w_\Delta \in \bigwedge^r \mathbb{R}^k$, and why is primitivity the correct arithmetic normalization?
(3) What does (C, α)-goodness prevent, and what does it emphatically not prevent?
(4) Why is the lower-supremum hypothesis independent of goodness?
(5) How does maximal-radius selection turn a pointwise obstruction into a scale on which goodness can be used?
(6) Why does the marking induction terminate after at most k steps?
(7) Where exactly is the finite $3r$ covering selection used?
(8) In the proof that marked points have no short vector, why does the first rank step differ from later steps?
(9) How do unipotent matrix coefficients become polynomials in every exterior representation?
(10) How does finite-order nondegeneracy give a uniform lower supremum for affine combinations?
(11) Can you reproduce the exterior-coordinate calculation behind Proposition 3.1?
(12) Why does exponential cusp penetration imply a Diophantine power improvement, and conversely?
(13) Why does the Borel–Cantelli step require only summability and no independence?
(14) What remains to be done after quantitative non-divergence has supplied tightness of a sequence of homogeneous-space measures?

Volume 2

Polynomial Divergence, Linearization, and Singular Sets

Student orientation: linearization in one sentence

The central idea is that a long near-return of a unipotent orbit becomes, after rescaling, a nonconstant polynomial displacement in a finite-dimensional representation. Polynomial displacement either creates a new invariant direction or forces the orbit into an algebraic singular set. **Terminology before use.** A *Chevalley representation* here means a finite-dimensional rational representation in which the subgroup under study is the stabilizer of a line or vector. A *singular set* is the locus where such a vector remains trapped in the corresponding fixed subspace. *Focusing* means that a family which fails to equidistribute is forced into one controlled algebraic branch. *Additional invariance* means invariance under a nontrivial subgroup not present in the original hypothesis. Keep three logically separate layers in mind: polynomial rescaling, algebraic encoding, and measure-theoretic genericity.

Reader guide: a concrete model

Why this matters.

Volume 2 is organized around the passage from *Polynomial divergence as a rigidity signal* to *Sources and Bibliographic Notes.* The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.1 (A calculation to keep in view)**.** Consider the standard unipotent and diagonal elements

$$u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad a_s = \begin{pmatrix} e^s & 0 \\ 0 & e^{-s} \end{pmatrix}.$$

A direct multiplication gives

$$a_s u_t a_{-s} = u_{e^{2s}t}.$$

Thus a displacement of size $|t|$ in the unstable horospherical direction is enlarged by e^{2s}. If $|t| = e^{-2s}$, the conjugated displacement has order one. This elementary calculation is the model for the “first visible scale” used repeatedly in nondivergence and shearing arguments.

Preface: linearization as a proof technology

Ratner's theorems describe the final algebraic form of unipotent orbit closures and invariant measures. This volume studies a different layer of the subject: the mechanism by which a nonlinear homogeneous-space problem is converted into a finite-dimensional polynomial problem before classification is invoked.

The central picture is

$$\begin{aligned}\text{polynomial divergence} &\Longrightarrow \text{additional invariance}\\ &\Longrightarrow \text{linearized algebraic obstruction}\\ &\Longrightarrow \text{avoidance or focusing.}\end{aligned}$$

Each arrow hides a genuine argument. The purpose of the present edition is to prove those arguments rather than replace them by references to Dani–Margulis or Shah.

Scope and prerequisites

Throughout this volume, a *homogeneous probability* means the normalized Haar measure on a closed finite-volume orbit $Lx \subset G/\Gamma$ of a connected closed subgroup $L < G$. This is the only meaning of "homogeneous measure" used in the classification and linearization statements below.

The companion Ratner project is allowed to supply only the following inputs.

(1) Birkhoff and ergodic decomposition in the standard measure-theoretic forms already proved in the foundational volume.
(2) Basic finite-dimensional Lie theory: Levi decomposition, Lie's and Engel's theorems, restricted-root decompositions, and the Cartan involution.
(3) Elementary real-algebraic geometry at fixed ambient dimension and the arithmetic lattice/exterior-power infrastructure developed in Volumes 1–2.

Ratner's classification, orbit-closure/equidistribution mechanism, the additional-invariance bridge, and the Mautner step used in the active proof chain are proved in the present volume at the arithmetic linear scope needed later; Ratner, Dani–Margulis, Shah, and Margulis–Tomanov are cited for provenance and source concordance rather than as substitutes for those proofs. Volume 1 of the present series supplies polynomial sublevel estimates and quantitative non-divergence. We also use the standard background already developed in the companion Ratner foundations: Birkhoff and ergodic decomposition, Egorov and regularity of probability measures, the spectral theorem for commuting unitary groups, and basic finite-dimensional Lie theory (Levi decomposition, Lie's and Engel's theorems, and elementary root-space facts). These are background technologies, not post-Ratner rigidity inputs. Every additional theorem used logically in the principal linearization and generated-unipotent arguments is proved here.

There is one deliberate scope decision. The full Dani–Margulis representative-properness theorem is valid for arbitrary lattices in general Lie groups, but its general proof contains a substantial lattice-subgroup finiteness theorem. Rather than disguise that theorem as a two-line lemma, we prove the complete linearization argument in the arithmetic linear setting

$$G \le \mathrm{SL}_N(\mathbb{R}), \qquad \Gamma \sim G(\mathbb{Z}),$$

which is the form required by the arithmetic applications of the present series. In this setting exterior representatives are primitive integral vectors, so the crucial properness statement is proved transparently from discreteness of an exterior lattice. The general Dani–Margulis extension is described historically in the sources appendix, but it is not used as a black box anywhere in the logical development.

The generated-unipotent measure theorem is proved after the one-parameter theorem. The key structural step is to show that an ergodic action of a connected group generated by Ad-unipotent one-parameter subgroups contains an ergodic Ad-unipotent one-parameter subgroup. The nilpotent ergodic-direction lemma, the factorwise Mautner mechanism, and the Lie-algebra reduction are all proved explicitly before the classification theorem is applied.

How to read the volume

Chapters 1–4 build the geometric language: polynomial shearing, generic-point comparison, singular strata, and exterior linearization. The complete arithmetic representative/properness and finite-window proof is placed immediately after that foundation, before the Dani–Margulis–Shah principle is formulated and used. Avoidance, focusing, and the linearization workshop then turn the theorem into a reusable method. The generated-unipotent theorem is followed immediately by its complete proof, before measure decomposition uses it. Later chapters develop low-rank models, polynomial trajectories, effectivity diagnostics, and the ternary quadratic-form laboratory.

The standard throughout is asymmetric. Routine coordinate algebra is occasionally compressed after a reusable lemma has been proved, whereas a short step that changes the source of rigidity is expanded even when the original paper states it tersely. The goal is not merely to verify the theorem; it is to make the proof reconstructible from memory.

CHAPTER 1

Polynomial divergence as a rigidity signal

1. Unipotence in a finite-dimensional representation

Let G be a Lie group, $\mathfrak{g} = \mathrm{Lie}(G)$, and let

$$u_t = \exp(tX)$$

be a one-parameter subgroup. We call it Ad-*unipotent* if $\operatorname{ad} X$ is nilpotent; for linear algebraic groups this agrees with unipotence of $\mathrm{Ad}(u_t)$.

If $(\operatorname{ad} X)^{d+1} = 0$, then

$$\mathrm{Ad}(u_t) = e^{t \operatorname{ad} X} = I + t \operatorname{ad} X + \frac{t^2}{2!}(\operatorname{ad} X)^2 + \cdots + \frac{t^d}{d!}(\operatorname{ad} X)^d. \tag{1.1}$$

Thus $\mathrm{Ad}(u_t)Y$ is a vector-valued polynomial in t for every $Y \in \mathfrak{g}$.

More generally, for any finite-dimensional representation $\rho : G \to \mathrm{GL}(V)$ for which $d\rho(X)$ is nilpotent,

$$\rho(u_t)v = e^{t\, d\rho(X)}v$$

is polynomial in t. Exterior powers of Ad are the representations used in linearization.

2. The exact $\mathfrak{sl}_2$ calculation

Let

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Then

$$[h, e] = 2e, \qquad [h, f] = -2f, \qquad [e, f] = h.$$

For $u_t = \exp(te)$,

$$\mathrm{Ad}(u_t)e = e, \tag{2.1}$$

$$\mathrm{Ad}(u_t)h = h - 2te, \tag{2.2}$$

$$\mathrm{Ad}(u_t)f = f + th - t^2 e. \tag{2.3}$$

Indeed these are obtained by truncating $e^{t \operatorname{ad} e}$ after degrees $0, 1, 2$.

The three directions therefore behave differently under the horocycle flow: the e-direction commutes with the flow, the h-direction creates linear shear in e, and the opposite-unipotent f-direction creates quadratic shear.

Why this matters. This calculation is the local picture behind Ratner's H-property. If two points differ slightly in a transverse f-direction, they remain close for a long time, then acquire a visible displacement mainly in the commuting e-direction. Because the resulting displacement approximately commutes with the original flow, it can become a new invariance of an invariant measure.

3. Relative displacement of nearby points

Let $x \in G/\Gamma$ and let $y = \exp(Y)x$ with Y small. Then

$$\begin{aligned} u_t y &= u_t \exp(Y)x \\ &= u_t \exp(Y)u_{-t}\, u_t x \\ &= \exp(\mathrm{Ad}(u_t)Y)\, u_t x. \end{aligned}$$

Thus the relative displacement between the two orbit points is

$$\exp(\mathrm{Ad}(u_t)Y),$$

and its logarithm is polynomial in t.

If the leading nonzero term of $\mathrm{Ad}(u_t)Y$ has degree j, a displacement of initial size about η becomes order one at time about $\eta^{-1/j}$. This polynomial time scale is the first major difference between unipotent and diagonal dynamics.

4. Polynomial versus exponential divergence

For comparison, if a_t is diagonal and Y lies in an unstable root space of weight $\lambda > 0$, then

$$\mathrm{Ad}(a_t)Y = e^{\lambda t}Y.$$

A displacement ηY becomes order one at time $\lambda^{-1}\log(1/\eta)$. Under a unipotent flow the corresponding time is a power of $1/\eta$. The much slower divergence gives two nearby generic orbit pieces a long overlap during which ergodic information can be compared.

5. A polynomial rescaling lemma

We isolate the elementary compactness principle that chooses the first visible scale.

Lemma 5.1 (Rescaling a vanishing polynomial family)**.** *Let*

$$P_n(t) = \sum_{j=0}^{D} a_{j,n} t^j \in V$$

be vector-valued polynomials in a finite-dimensional normed space V. Assume

$$\max_j \|a_{j,n}\| \to 0$$

and that P_n is not identically zero. Then there are scales $T_n \to \infty$ and a subsequence such that

$$P_n(T_n s) \to P(s)$$

uniformly on compact s-sets, where P is a nonzero polynomial of degree at most D and $P(0) = 0$ provided $a_{0,n} \to 0$.

Proof. Choose $T_n > 0$ so that

$$\max_{1 \le j \le D} \|a_{j,n}\| \, T_n^j = 1$$

when at least one nonconstant coefficient is nonzero. Because all those coefficients tend to zero, $T_n \to \infty$. The scaled coefficients

$$b_{j,n} = a_{j,n} T_n^j$$

are bounded, and at least one has norm 1. Passing to a subsequence, every $b_{j,n}$ converges to b_j, with some $b_j \neq 0$. The constant coefficients converge to zero under the stated hypothesis. Therefore

$$P_n(T_n s) = \sum_{j=0}^{D} b_{j,n} s^j \to \sum_{j=1}^{D} b_j s^j =: P(s)$$

uniformly on compact sets. $\square$

Remark 5.2. In a group one does not literally rescale the logarithm for arbitrarily large displacement, because the exponential chart is local. The lemma is used up to the first scale at which the displacement enters a fixed small but visible neighborhood. At that scale the Baker–Campbell–Hausdorff error remains controlled.

6. A first invariance heuristic

Suppose x_n and y_n are points whose long U-orbit pieces are both representative of the same probability measure μ, and suppose

$$y_n = g_n x_n, \qquad g_n \to e,$$

with g_n not commuting with U. Choose times T_n so that

$$u_{T_n s} g_n u_{-T_n s} \to p(s)$$

for s in a bounded set, with p nontrivial. Then

$$u_{T_n s} y_n = (u_{T_n s} g_n u_{-T_n s}) u_{T_n s} x_n \approx p(s) u_{T_n s} x_n.$$

If both sides sample μ in a sufficiently uniform way, one expects

$$p(s)_* \mu = \mu.$$

The actual proof must handle recurrence, uniform genericity, and the fact that $p(s)$ need not be a homomorphism at this stage. A rigorous implementation therefore separates the algebraic shearing calculation from the ergodic statement that nearby orbit segments sample the same measure; we formulate that sampling step explicitly when it is needed.

CHAPTER 2

Generic points, shearing, and additional invariance

Polynomial divergence becomes rigidity only after it is combined with recurrence and genericity. This chapter isolates that passage. Ratner's original arguments are more delicate than the abstract lemmas below, but the lemmas make the logic visible before singular sets and linearization are added.

1. Uniform genericity on a large set

Let (X, μ) be a probability space with a continuous measure-preserving flow u_t. For $f \in L^1(\mu)$ write

$$A_T f(x) = \frac{1}{T} \int_0^T f(u_t x)\, dt.$$

THEOREM 1.1 (Birkhoff for the flow). *If μ is u_t-ergodic, then for every $f \in L^1(\mu)$,*

$$A_T f(x) \to \int f\, d\mu$$

for μ-almost every x.

> **Proof and provenance.** STANDARD FOUNDATIONAL THEOREM. We use the standard continuous-time Birkhoff ergodic theorem; its proof is included in the measure-theoretic appendix of the Ratner project.

Lemma 1.2 (Uniform generic set for finitely many tests)**.** *Let $f_1, \dots, f_M \in C_c(X)$ and $\eta > 0$. There is a measurable set K with $\mu(K) > 1 - \eta$ and T_0 such that for every $x \in K$, every $T \geq T_0$, and every j,*

$$\left| A_T f_j(x) - \int f_j\, d\mu \right| < \eta.$$

If X is locally compact and μ is regular, K may be chosen compact after decreasing its measure by another arbitrarily small amount.

PROOF. Apply Birkhoff simultaneously to the finite family and intersect the full-measure convergence sets. Egorov's theorem makes the convergence uniform outside a set of measure less than $\eta/2$. Inner regularity supplies a compact subset of the resulting set with loss less than $\eta/2$. □

The role of K is not cosmetic. Uniform genericity is what permits two nearby points chosen by recurrence to be compared at a common long time scale.

2. A comparison lemma for empirical measures

Let d be a compatible metric on X. For a compact set $L \subset X$ and $f \in C_c(X)$, uniform continuity gives a modulus

$$\omega_{f,L}(r) = \sup\{|f(x) - f(y)| : x, y \in L,\ d(x,y) \le r\} \to 0.$$

Lemma 2.1 (Visible displacement implies invariance)**.** *Let $h \in G$ act continuously on $X = G/\Gamma$. Suppose $T_n \to \infty$ and $x_n, y_n \in X$ satisfy*

$$\frac{1}{T_n}\int_0^{T_n} \delta_{u_t x_n}\, dt \Longrightarrow \mu, \qquad \frac{1}{T_n}\int_0^{T_n} \delta_{u_t y_n}\, dt \Longrightarrow \mu.$$

Assume there are measurable sets $E_n \subset [0, T_n]$ with $|E_n|/T_n \to 1$ such that

$$u_t y_n = h_{n,t} u_t x_n, \qquad h_{n,t} \to h$$

uniformly for $t \in E_n$. Assume also that the two empirical families are uniformly tight. Then $h_*\mu = \mu$.

Proof. Fix $f \in C_c(X)$ and $\varepsilon > 0$. Uniform tightness gives a compact L such that the proportion of times for which either $u_t x_n$ or $u_t y_n$ lies outside a slightly enlarged compact set is less than ε for all large n. On the remaining times in E_n, continuity of the action and uniform convergence $h_{n,t} \to h$ imply

$$|f(u_t y_n) - f(h u_t x_n)| \to 0$$

uniformly. The complement of these good times has relative measure $O(\varepsilon) + o(1)$, and f is bounded. Therefore

$$\frac{1}{T_n}\int_0^{T_n} f(u_t y_n)\, dt - \frac{1}{T_n}\int_0^{T_n} f(h u_t x_n)\, dt \to 0.$$

Passing to the weak-* limits yields

$$\int f\, d\mu = \int f(hx)\, d\mu(x).$$

As $f \in C_c(X)$ was arbitrary, $h_*\mu = \mu$. □

Warning 2.2. The lemma does not by itself prove Ratner's additional invariance. Polynomial shearing usually produces a displacement that varies with normalized time, not one constant h on almost all of a long interval. One therefore works on carefully chosen mesoscopic windows or extracts a particular limiting displacement from recurrence. The lemma identifies the measure-theoretic endpoint of that construction.

3. Freezing a polynomial on a mesoscopic window

Suppose a relative displacement has logarithm

$$P_n(t) = a_{1,n}t + \cdots + a_{D,n}t^D,$$

with tiny coefficients. Choose T_n as in Lemma 5.1. At the macroscopic scale $t = T_n s$, the displacement varies by order one as s varies by order one. To make it nearly constant, select a window

$$t = T_n s_0 + \tau, \qquad |\tau| \le L_n,$$

where $L_n/T_n \to 0$ but $L_n \to \infty$. Taylor expansion gives

$$P_n(T_n s_0 + \tau) - P_n(T_n s_0) = O\left(\frac{L_n}{T_n}\right)$$

after normalization, uniformly on bounded s_0 away from exceptional roots, because the scaled derivatives remain bounded. Thus the relative group displacement is approximately frozen while the window is still long enough for ergodic averages.

This two-scale choice—macroscopic scale to make the displacement visible, mesoscopic scale to make it nearly constant—is one of the cleanest ways to understand unipotent shearing.

4. The SL_2 prototype

Let $Y_n = \alpha_n e + \beta_n h + \gamma_n f \to 0$. By (2.1)–(2.3),

$$\mathrm{Ad}(u_t)Y_n = (\alpha_n - 2\beta_n t - \gamma_n t^2)e + (\beta_n + \gamma_n t)h + \gamma_n f.$$

Assume, for instance, that the quadratic term dominates at first visible scale. Choose T_n with $|\gamma_n|T_n^2 \asymp 1$. Then

$$\gamma_n T_n \to 0, \qquad \gamma_n \to 0,$$

so at time $T_n s$ the h- and f-components vanish in the limit while the e-component may converge to a nonzero multiple of $-s^2 e$. Hence

$$u_{T_n s} \exp(Y_n) u_{-T_n s} \to \exp(-cs^2 e),$$

a displacement inside the original horocycle group.

If the linear h-term dominates instead, one obtains a limit $\exp(-2cse)$, again in the e-direction. This is the concrete reason the centralizing direction is produced from transverse displacement.

5. Why recurrence is needed

To find two nearby generic points, one uses recurrence inside a positive-measure set of uniformly generic points. If a small neighborhood transverse to the U-orbits contained at most one generic point in each local leaf, the measure would behave like a one-dimensional measure along U; in the situations of Ratner's argument, this contradicts the structure of the ambient invariant measure unless the measure is already supported on a smaller homogeneous object. The singular-set formalism makes this alternative precise.

Thus additional invariance and singular-set avoidance are not separate tricks. They are two sides of the same dichotomy:

$$\boxed{\begin{array}{c}\text{many nearby generic points} \Longrightarrow \text{extra invariance,}\\ \text{failure of nearby generic points} \Longrightarrow \text{algebraic concentration.}\end{array}}$$

6. A practical proof template

When reading a shearing argument, identify the following six steps explicitly.

(1) Choose a compact set of uniformly generic recurrent points.
(2) Find two points in that set with small transverse displacement.
(3) Conjugate the displacement by the unipotent flow and compute the polynomial divergence.
(4) Choose the first visible time scale.
(5) Restrict to a long window on which the visible displacement is approximately constant.
(6) Compare ergodic averages and conclude invariance under the limiting displacement.

If a paper writes "by the H-property we obtain additional invariance," these are the hidden operations one should reconstruct.

CHAPTER 3

Singular sets: where equidistribution can fail

The word "singular" in linearization theory does not mean a geometric singularity of a manifold. It means that the orbit is compatible with a proper homogeneous subsystem capable of supporting the unipotent dynamics.

1. The family of relevant subgroups

For clarity we work in the standard finite-volume setting: G is a Lie group, $\Gamma < G$ is a lattice, and $X = G/\Gamma$. Let $H < G$ be a closed connected subgroup. The orbit $H\Gamma/\Gamma$ is closed and supports a finite H-invariant measure exactly when $H \cap \Gamma$ is a lattice in H (with the usual qualification that the orbit is based at the identity; conjugates handle arbitrary closed homogeneous orbits).

Define H^+ to be the subgroup generated by all one-parameter Ad-unipotent subgroups contained in H.

Definition 1.1 (Ratner–Dani–Margulis subgroup family)**.** Let $\mathscr{H}_\Gamma$ be the collection of closed connected subgroups $H < G$ such that

(1) $H \cap \Gamma$ is a lattice in H;
(2) the action of H^+ on $H/(H \cap \Gamma)$ is ergodic with respect to the normalized H-invariant measure.

In many algebraic settings, the second condition is equivalent to saying that the algebraic part of H relevant to the finite-volume quotient is generated by unipotents, up to compact/central factors. We keep the dynamical formulation because it is exactly what Ratner's classification produces.

Definition 1.2 (Rational obstruction subfamily)**.** Assume an arithmetic rational model $\mathbf{G}/\mathbb{Q}$ has been fixed. Let $\mathscr{H}_\Gamma^{\mathrm{rat}}$ denote those members of $\mathscr{H}_\Gamma$ which are identity components of real points of connected $\mathbb{Q}$-subgroups of $\mathbf{G}$.

Proposition 1.3 (Countability of the rational obstruction family)**.** *The family $\mathscr{H}_\Gamma^{\mathrm{rat}}$ is countable.*

Proof. A $\mathbb{Q}$-subgroup of $\mathbf{G} \leq \mathrm{GL}_N$ is determined by its defining ideal in the countable Noetherian ring $\mathbb{Q}[x_{ij}, 1/\det]$. Every ideal is finitely generated, so this ring has only countably many ideals and hence only countably many algebraic subgroups. Passing to the identity component of the real points does not increase cardinality. Taking the subfamily satisfying the two conditions in Definition 1.1 preserves countability. □

Warning 1.4 (No hidden general countability theorem)**.** For an arbitrary lattice, countability and finiteness statements for the homogeneous subgroup family require additional structure. They are not proved here by the invalid slogan "Γ is countable." The complete linearization theorem in this volume uses the rational arithmetic family and, on each compact vector window, proves the stronger fact actually needed: only finitely many competing rational incidences occur.

2. The transporter $N(H, W)$

Let $W < G$ be generated by one-parameter unipotent subgroups. For $H \in \mathscr{H}_\Gamma$, define

$$N(H, W) = \{g \in G : W \subset gHg^{-1}\}.$$

Equivalently,

$$g \in N(H, W) \iff g^{-1}Wg \subset H.$$

If $x = g\Gamma$ with $g \in N(H, W)$, then

$$Wx \subset gH\Gamma/\Gamma,$$

so the W-orbit is trapped in a closed homogeneous set associated with H.

The notation $N(H, W)$ is traditional but should not be confused with the normalizer $N_G(H)$. It is a *transporter*: it records conjugates of H that contain W.

3. The lower-dimensional singular part

In the arithmetic rational setting define

$$S_{\mathrm{rat}}(H, W) = \bigcup_{F \in \mathscr{H}_\Gamma^{\mathrm{rat}},\, F \subsetneq H} N(F, W).$$

A compact subset of $N(H, W) \setminus S_{\mathrm{rat}}(H, W)$ is a region on which no smaller *finite-volume rational homogeneous subgroup* already explains the observed containment. Representative ambiguity is slightly broader: the intersection of two rational conjugates can be a proper rational subgroup which has not yet been shown to carry finite volume. The full proof therefore uses the finite lower-incidence set constructed directly in the exterior representation, rather than assuming that every ambiguity already belongs to $\mathscr{H}_\Gamma^{\mathrm{rat}}$.

Guiding Idea 3.1. Linearization is cleanest on a compact subset of $N(H,W) \setminus S_{\mathrm{rat}}(H,W)$. There, the subgroup H is the *first* algebraic explanation for the observed concentration. If one approaches $S_{\mathrm{rat}}(H,W)$, a smaller subgroup becomes a competing explanation, and representatives in the linear model can collide. In the general Dani–Margulis–Shah method these lower-dimensional strata are organized inductively. The arithmetic proof in this volume uses the sharper local fact that a bounded exterior window contains only finitely many competing rational incidences; applications may then eliminate them by zero mass or descend to them explicitly.

4. Images in the quotient: a necessary caution

For arbitrary subsets $A, B \subset G$, one only has

$$\pi(A \setminus B) \subseteq \pi(A) \setminus \pi(B)$$

under additional hypotheses; in general even this displayed inclusion may fail because a point can have different Γ-representatives in A and B. Consequently we do *not* identify $\pi(N(H,W) \setminus S_{\mathrm{rat}}(H,W))$ with a set-theoretic difference in the quotient.

For measure arguments it is safer to define the quotient-level rational pure part directly by

$$\mathcal{P}_{\mathrm{rat}}(H,W) = \pi(N(H,W)) \setminus \pi(S_{\mathrm{rat}}(H,W)).$$

For the finite-window linearization proof an even more local object is used: a compact core together with the finite ambiguity set produced by the bounded exterior window. This formulation avoids any unjustified closure of the finite-volume subgroup family under intersections.

5. The singular set for a unipotent subgroup

For a fixed W, define the global proper singular set

$$\mathcal{S}(W) = \bigcup_{\substack{H \in \mathscr{H}_\Gamma \\ H \neq G}} \pi(N(H,W)).$$

Points outside $\mathcal{S}(W)$ are those for which no proper finite-volume homogeneous subgroup from the Ratner family contains a conjugate of W through that point.

For a one-parameter Ad-unipotent group W, the internally proved arithmetic orbit-closure theorem shows that orbit closures are homogeneous, so the set above is the natural list of proper homogeneous obstructions. For a larger group generated by unipotent one-parameter subgroups, the corresponding topological extension is a further theorem and is not silently deduced here from the one-parameter case. The present volume needs only the measure-theoretic generated-unipotent extension proved later.

6. Examples

Worked Example 6.1 (Closed horocycles)**.** Take $G = \mathrm{SL}_2(\mathbb{R})$, $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, and

$$W = U = \left\{ \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} : t \in \mathbb{R} \right\}.$$

A point $g\Gamma$ lies on a closed U-orbit precisely when $g^{-1}Ug \cap \Gamma$ is a lattice in $g^{-1}Ug \cong \mathbb{R}$. In the modular surface picture these are periodic horocycles around the cusp. The proper subgroup $H = U$ itself is an element of the relevant family whenever $U \cap g\Gamma g^{-1}$ is a lattice. Thus the simplest singular obstruction is periodicity of the horocycle.

Worked Example 6.2 (A proper SL_2 inside SL_3)**.** Let $G = \mathrm{SL}_3(\mathbb{R})$ and let H be the upper-left block copy of $\mathrm{SL}_2(\mathbb{R})$:

$$H = \left\{ \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix} : A \in \mathrm{SL}_2(\mathbb{R}) \right\}.$$

If $\Gamma = \mathrm{SL}_3(\mathbb{Z})$, then $H \cap \Gamma \cong \mathrm{SL}_2(\mathbb{Z})$ is a lattice. Any unipotent subgroup W contained in a conjugate gHg^{-1} has its orbit through $g\Gamma$ confined to the closed copy

$$gH/(H \cap \Gamma) \subset G/\Gamma.$$

Linearization must detect precisely this possibility when studying equidistribution of W-trajectories in the full SL_3 quotient.

7. Why singular sets are stratified rather than removed at once

One might hope simply to say "stay away from all proper closed homogeneous orbits." That is too crude for a proof. Different subgroups have different dimensions and different linear representations, and neighborhoods of their images can overlap. The correct strategy is hierarchical:

(1) choose a minimal or maximal relevant H according to the argument;
(2) work near a compact part of $N(H, W) \setminus S_{\mathrm{rat}}(H, W)$ where representatives are locally unique;
(3) linearize there;
(4) if uniqueness fails, show that the point lies near $S_{\mathrm{rat}}(H, W)$, hence near a smaller-dimensional obstruction;
(5) either eliminate the lower incidence by a zero-mass argument, or descend to it and restart the analysis there.

This is structurally analogous to the primitive-subgroup induction used in quantitative non-divergence. The objects are now Lie subgroups rather than rational subspaces, and the finite complexity is dimension of H rather than rank of a lattice subgroup.

CHAPTER 4

Encoding a subgroup in a finite-dimensional representation

The decisive move of linearization is to replace the nonlinear question "is $g\Gamma$ near a closed H-orbit?" by the linear question "is a vector gp_H near a specified algebraic subset of a representation?" This chapter constructs that vector and proves the elementary facts that make it work.

1. The exterior vector p_H

Let $H < G$ be a closed connected subgroup, let

$$\mathfrak{h} = \mathrm{Lie}(H) \subset \mathfrak{g} = \mathrm{Lie}(G), \qquad d = \dim H,$$

and choose a basis $X_1, \ldots, X_d$ of $\mathfrak{h}$. Define

$$p_H = X_1 \wedge \cdots \wedge X_d \in V_H := \bigwedge^d \mathfrak{g}.$$

Changing the basis multiplies p_H by a nonzero scalar. Once a Haar-volume normalization on H is fixed, one can choose p_H up to sign; for the algebraic stabilizer calculations the line $\mathbb{R}p_H$ is canonical.

Let

$$\rho_H(g) = \bigwedge^d \mathrm{Ad}(g)$$

be the induced representation of G on V_H.

Proposition 1.1 (The line stabilizer is the normalizer)**.** *For connected H,*

$$\mathrm{Stab}_G(\mathbb{R}p_H) = N_G(H).$$

More precisely, if $g \in N_G(H)$ then

$$\rho_H(g)p_H = \det(\mathrm{Ad}(g)|_{\mathfrak{h}})p_H.$$

Proof. If g normalizes H, then $\mathrm{Ad}(g)$ preserves $\mathfrak{h}$. Its action on the top exterior power $\bigwedge^d \mathfrak{h}$ is multiplication by its determinant, proving one direction.

Conversely, a nonzero decomposable vector $X_1 \wedge \cdots \wedge X_d$ determines the d-plane $\mathrm{span}(X_1, \ldots, X_d)$: it is precisely the set of Y satisfying

$$Y \wedge p_H = 0.$$

If $\rho_H(g)p_H$ is a nonzero multiple of p_H, then $\mathrm{Ad}(g)\mathfrak{h} = \mathfrak{h}$. Hence gHg^{-1} and H are connected Lie subgroups with the same Lie algebra, so they are equal. □

Define

$$N_G^1(H) = \{g \in N_G(H) : \det(\mathrm{Ad}(g)|_{\mathfrak{h}}) = 1\}.$$

Then

$$\mathrm{Stab}_G(p_H) = N_G^1(H).$$

If orientation reversal is allowed, the stabilizer of the unordered pair $\{p_H, -p_H\}$ consists of those normalizer elements with determinant ± 1 on $\mathfrak{h}$.

2. Why finite-volume subgroups fix p_H

Lemma 2.1. *If H admits a lattice, then H is unimodular. Consequently, for connected H with a lattice,*

$$\det(\mathrm{Ad}(h)|_{\mathfrak{h}}) = 1 \qquad (h \in H),$$

so H fixes p_H.

Proof. A locally compact group admitting a lattice is unimodular: otherwise left Haar measure and its right translates would transform by a nontrivial modular character, incompatible with a finite invariant measure on the quotient. For a Lie group, the modular function is the absolute value of the determinant of the adjoint action (up to the conventional inverse, irrelevant when it equals one). Hence

$$|\det(\mathrm{Ad}(h)|_{\mathfrak{h}})| = 1.$$

The determinant is a continuous map $H \to \mathbb{R}^\times$ and equals 1 at the identity. Since H is connected, it cannot change sign; therefore it is identically 1. □

This observation is crucial. If $g \in N(H, W)$ and $w \in W$, then $g^{-1}wg \in H$, so

$$\rho_H(w)\rho_H(g)p_H = \rho_H(g)\rho_H(g^{-1}wg)p_H = \rho_H(g)p_H.$$

Hence

$$\eta_H(N(H, W)) \subset V_H^W, \qquad \eta_H(g) := \rho_H(g)p_H. \tag{2.1}$$

Thus the nonlinear transporter is sent into the *linear fixed-point space* of W.

3. The arithmetic stabilizer and representative vectors

The cleanest convention is to keep the oriented vector p_H rather than quotient by a sign. Define

$$\Gamma_H := \{\gamma \in \Gamma : \rho_H(\gamma)p_H = p_H\} = \Gamma \cap N_G^1(H).$$

An element of $N_G(H) \cap \Gamma$ with determinant -1 on $\mathfrak{h}$ sends p_H to $-p_H$; it is therefore *not* in Γ_H. This is exactly as it should be. The vector stabilizer is $N_G^1(H)$, whereas the line stabilizer is the full normalizer.

For $x = g\Gamma \in X$ define the set of H-representatives

$$\mathrm{Rep}_H(x) = \{\rho_H(g\gamma)p_H : \gamma \in \Gamma\} \subset V_H.$$

This is independent of the chosen representative g of x, because replacing g by $g\gamma_0$ merely reindexes the set by $\gamma_0\Gamma = \Gamma$. For $Z \subset X$, let

$$\mathrm{Rep}_H(Z) = \bigcup_{x \in Z} \mathrm{Rep}_H(x).$$

THEOREM 3.1 (Arithmetic representative properness). *Assume that G has a rational linear model, that Γ preserves a full integral lattice in $\mathfrak{g}$, and that H is the identity component of the real points of a connected rational subgroup. Normalize p_H to be primitive in the corresponding exterior lattice. Then the orbit*

$$\rho_H(\Gamma)p_H$$

is closed and discrete in V_H. In particular:

(1) $N_G^1(H)\Gamma/\Gamma$ is closed in G/Γ;
(2) $\mathrm{Rep}_H(x)$ is discrete for every $x \in X$;
(3) $\mathrm{Rep}_H(Z)$ is closed in V_H for every compact $Z \subset X$;
(4) the map

$$q_H : G/\Gamma_H \longrightarrow G/\Gamma \times V_H, \qquad q_H(g\Gamma_H) = (g\Gamma, \rho_H(g)p_H),$$

is proper.

PROOF. By hypothesis p_H is primitive in the exterior lattice $\bigwedge^d \mathfrak{g}_{\mathbb{Z}}$. Since $\rho_H(\Gamma)$ preserves this lattice, $\rho_H(\Gamma)p_H$ is a subset of a discrete lattice and is therefore discrete; it is closed because a discrete subset of a lattice has no finite accumulation point. The first assertion follows because $N_G^1(H)$ is the stabilizer of p_H and the inverse image of the discrete orbit $\rho_H(\Gamma)p_H$ under $g \mapsto \rho_H(g^{-1})p_H$ is closed. Assertions (2) and (3) follow by translating this discrete orbit by a compact family of linear maps.

For (4), let $(g_n\Gamma_H)$ be a sequence whose image under q_H remains in a compact subset of $G/\Gamma \times V_H$. Choose $\gamma_n \in \Gamma$ so that $g_n\gamma_n$ remains in a fixed compact subset of G. Then

$$\rho_H(g_n\gamma_n)\rho_H(\gamma_n^{-1})p_H = \rho_H(g_n)p_H$$

remains in a compact set. Since $\rho_H(g_n\gamma_n)$ and its inverse remain bounded, the vectors $\rho_H(\gamma_n^{-1})p_H$ lie in a compact subset of the discrete exterior lattice and hence take only finitely many values. After a subsequence they are equal; equivalently, $\gamma_n\gamma_1^{-1} \in \Gamma_H$. Thus $g_n\Gamma_H$ has a convergent subsequence. This proves properness. □

Remark 3.2 (Scope)**.** For general lattices, discreteness of the exterior orbit is not obtained by placing it in an integral lattice. Dani–Margulis prove the needed properness through a separate bounded-volume finiteness theorem. That general theorem is not used as an unproved input in this volume; the complete arithmetic proof is expanded in Chapter 5.

4. Consequences of properness

Lemma 4.1 (Uniform finiteness in a compact vector window)**.** *Let $K \subset X$ and $D \subset V_H$ be compact. Then*

$$\{(x, v) : x \in K,\ v \in \mathrm{Rep}_H(x) \cap D\}$$

is compact. In particular each fiber is finite.

Proof. Every pair (x, v) in the displayed set has the form

$$(g\Gamma, \rho_H(g)p_H)$$

after absorbing the relevant $\gamma \in \Gamma$ into g. Thus the set is the image under q_H of $q_H^{-1}(K \times D)$. Properness makes this inverse image compact, hence its image compact.

For fixed x, the fiber is a discrete subset of the compact set D, and is therefore finite. Properness alone does not give a uniform cardinality bound over all $x \in K$; the source linearization argument obtains the needed uniqueness by removing lower singular strata and then shrinking the vector window. □

> **Why this matters.** This is the exact point at which the infinite ambiguity $g \mapsto g\gamma$ becomes manageable. A quotient point can have infinitely many representative vectors globally. In a compact region and a bounded vector window, however, only finitely many can occur; away from lower singular strata one can shrink the window until the relevant representative is unique.

5. Linearization of the transporter

Equation (2.1) suggests defining the algebraic set

$$\mathcal{A}_H(W) = \overline{\eta_H(N(H,W))}^{\mathrm{Zar}} \subset V_H.$$

Then

$$\eta_H(N(H,W)) \subset \mathcal{A}_H(W) \subset V_H^W.$$

Proposition 5.1 (Exact algebraic detector of the transporter)**.** *Let $X_1, \ldots, X_r$ span the Lie algebra of the connected group generated by the one-parameter unipotent subgroups of W. Define*

$$\Phi_H(g) = \left(\mathrm{Ad}(g^{-1})X_j \wedge p_H\right)_{j=1}^r \in \bigoplus_{j=1}^r \bigwedge^{d+1} \mathfrak{g}.$$

Then

$$N(H,W) = \Phi_H^{-1}(0).$$

Moreover, along every adjoint-polynomial trajectory $\Theta(t)$ the coordinate functions of $\Phi_H(\Theta(t)g)$ are bounded-degree polynomial functions of t after choosing matrix coordinates for the relevant adjoint representation.

Proof. The decomposable vector p_H represents the plane $\mathfrak{h}$, and for $Y \in \mathfrak{g}$ one has $Y \wedge p_H = 0$ if and only if $Y \in \mathfrak{h}$. Hence $\Phi_H(g) = 0$ exactly when

$$\mathrm{Ad}(g^{-1})X_j \in \mathfrak{h} \qquad (1 \le j \le r).$$

Since H and the connected subgroup generated by the X_j are connected, the Lie-algebra containment is equivalent to $W \subset gHg^{-1}$. The polynomiality assertion follows directly from the definition of an adjoint-polynomial map. □

Remark 5.2. This detector is slightly less economical than the traditional Dani–Margulis exterior orbit, but it has an expository advantage: no Zariski-closure lemma is hidden in the notation. The obstruction is the zero set of explicit polynomial coordinates.

A thin neighborhood of $\pi(N(H,W))$ in X can now be studied through thin neighborhoods of $\mathcal{A}_H(W)$ in a finite-dimensional vector space. Polynomial trajectories in G act by polynomial trajectories in V_H, so real-variable growth estimates become available.

This is the meaning of “linearization”: it is not an infinitesimal approximation of G/Γ by its tangent space. It is an *exact equivariant representation* of the subgroup obstruction.

6. The three quotient spaces that must not be confused

There are three related objects in the construction:

$$G/\Gamma, \qquad G/\Gamma_H, \qquad V_H.$$

Their roles are different.

The space G/Γ is the original dynamical space. Passing from $g \in G$ to $g\Gamma$ forgets the choice of lattice representative. The space G/Γ_H remembers enough of that representative to make the exterior vector single valued:

$$g\Gamma_H \longmapsto \rho_H(g)p_H.$$

Finally V_H is the linear space in which polynomial growth can be measured.

The map

$$q_H(g\Gamma_H) = (g\Gamma, \rho_H(g)p_H)$$

keeps the quotient point and the exterior representative simultaneously. Properness of q_H means that if both pieces of information stay in compact sets, then the partially lifted point $g\Gamma_H$ also stays in a compact set. This is exactly the compactness statement needed when a trajectory changes its Γ-representative over time.

One should not replace G/Γ_H by $G/(N^1_G(H) \cap \Gamma)$ without checking that the notation agrees: here these groups are equal by definition, but in variants of the method the stabilizer can be enlarged by a finite group, or one may work with the line $\mathbb{R}p_H$ instead of the vector p_H. Such changes affect the precise normalizer appearing in the trapping alternative.

7. Why discreteness of $\rho_H(\Gamma)p_H$ is arithmetic

The vector p_H is chosen from a Lie algebra, so discreteness of its Γ-orbit is not a formal consequence of discreteness of Γ in G. In the arithmetic setting of this volume the missing structure is much more concrete: Γ preserves an integral lattice $\mathfrak{g}_\mathbb{Z}$ and a rational subgroup H has a primitive exterior generator $p_H \in \bigwedge^{\dim H} \mathfrak{g}_\mathbb{Z}$. Therefore $\rho_H(\Gamma)p_H$ is a subset of an honest Euclidean lattice.

This is exactly what Theorem 3.1 exploits. The statement would be false for an arbitrary vector in an arbitrary representation: a discrete group can have nondiscrete, even dense, orbits in a representation. The arithmetic hypotheses single out the exterior vector as an integral object and make bounded representative windows finite.

Thus the arithmetic linearization proof has two visibly separate inputs:

(1) multilinear algebra identifies the stabilizer of the exterior vector with the determinant-one normalizer;

(2) integral structure makes the lattice orbit of that vector discrete and yields properness of the incidence map.

The general nonarithmetic Dani–Margulis theorem replaces the second item by a substantial bounded-volume finiteness theorem; that stronger input is not used here.

8. Local uniqueness away from lower strata

Theorem 3.1 gives local finiteness, not uniqueness. Suppose a compact set $K \subset X$ has two distinct representative vectors v_1, v_2 for the same point $x \in K$, both lying near the algebraic set $\mathcal{A}_H(W)$. Write

$$v_i = \rho_H(g\gamma_i)p_H.$$

Then

$$\rho_H(\gamma_1^{-1}\gamma_2)p_H$$

relates two rational copies of H that are simultaneously compatible with W. Their intersection contains the dynamically relevant unipotent directions, and unless the two copies coincide in the appropriate normalizer quotient, this intersection is represented by a subgroup of smaller dimension.

This is the geometric origin of the finite lower-incidence set used in the arithmetic proof. Removing those competing incidences is not an arbitrary technical excision. It makes the following statement true in an appropriately shrunk compact neighborhood:

> there is at most one H-representative in the chosen vector window, unless the quotient point is already near a smaller homogeneous explanation.

Chapter 5 proves this precisely: bounded exterior windows leave only finitely many rational intersections on each compact quotient set. Once uniqueness is available, a time interval on which the quotient trajectory remains near the H-stratum can be lifted to a *single* polynomial vector trajectory in V_H. Without uniqueness one would be comparing different polynomials at different times, and a sublevel estimate for any one polynomial would say nothing about their union.

9. The fixed space V_H^W is only a first approximation

Equation (2.1) gives

$$\eta_H(N(H,W)) \subset V_H^W.$$

It is generally incorrect to replace $\mathcal{A}_H(W)$ by the whole fixed space V_H^W. The fixed space can contain vectors that do not represent Lie subalgebras at all, as well as decomposable vectors representing subspaces that are not conjugate to $\mathfrak{h}$. Even among the latter, the inverse image under η_H can be larger than the desired transporter unless the defining algebraic relations are retained.

The Zariski-closed set $\mathcal{A}_H(W)$ records those additional relations. Proposition 5.1 says that, although we close the image algebraically in V_H, no spurious group element appears when we pull back through η_H. This exact inverse-image identity is what eventually turns a limiting vector constraint back into exact subgroup containment.

10. A coordinate description of decomposability

The image of the Grassmannian of d-planes in $\mathfrak{g}$ inside $\mathbb{P}(\bigwedge^d \mathfrak{g})$ is cut out by the Pluecker relations. For example, for $d = 2$ a bivector

$$v = \sum_{i<j} v_{ij} e_i \wedge e_j$$

is decomposable if and only if

$$v \wedge v = 0.$$

In four ambient dimensions this becomes the quadratic relation

$$v_{12}v_{34} - v_{13}v_{24} + v_{14}v_{23} = 0.$$

Thus even the statement “this vector represents a d-plane” is algebraic. Requiring that the plane be a Lie subalgebra adds bracket relations, also polynomial in Pluecker coordinates. Requiring it to contain the conjugated Lie algebra of W adds further algebraic equations.

This coordinate picture explains why polynomial trajectories interact so well with the linearization variety: both the trajectory in V_H and the obstruction set are algebraic objects of bounded complexity.

11. A test for correct use of the representation

Before invoking linearization in a concrete problem, verify the following.

(1) The subgroup H is connected, or the exterior vector has been defined using the identity component with the finite-component ambiguity handled separately.
(2) $H \cap \Gamma$ is a lattice, so that H is unimodular and fixes p_H.
(3) The distinction between the line stabilizer $N_G(H)$ and the vector stabilizer $N_G^1(H)$ is respected.
(4) The representative orbit has the required discreteness/properness; this must be cited or proved and cannot be inferred from discreteness of Γ alone.

(5) The algebraic set used in the polynomial argument pulls back to the desired transporter, rather than merely containing it.

These checks prevent most normalization errors before the more difficult occupation estimates begin.

CHAPTER 5

Arithmetic Representative Properness and Finite-Window Linearization

This chapter closes the main logical gap left by the older edition. We prove the representative-properness theorem, the lower-incidence intersection lemma, and the finite-window linearization theorem in the arithmetic linear setting. The proof is deliberately organized so that every use of arithmetic is visible. Once the exterior representatives have been made discrete, the remaining argument is finite-dimensional polynomial analysis.

There is an important scope distinction. The historical Dani–Margulis theorem also treats nonarithmetic lattices and subgroups which are not rational in a fixed ambient model. In that generality the discreteness/properness of the representative map is supplied by a separate bounded-volume finiteness theorem for homogeneous subgroups. That theorem is not a formal consequence of Ratner classification. We therefore do not hide it here. The theorem proved in this chapter is the arithmetic rational-stratum form used logically in the present arc; every input to that form is proved below.

1. Arithmetic standing hypotheses

Throughout this chapter let $\mathbf{G} \leq \mathrm{SL}_N$ be a connected linear algebraic group defined over $\mathbb{Q}$, put

$$G = \mathbf{G}(\mathbb{R})^\circ,$$

and let $\Gamma < G$ be an arithmetic lattice commensurable, inside the rational points of the chosen model, with $\mathbf{G}(\mathbb{Z}) \cap G$. The construction below promotes the integral structure on a finite-index subgroup to an invariant lattice for the full group Γ, so no finite-cover descent is left implicit.

Lemma 1.1 (Integral adjoint lattice)**.** *There exists a full lattice $\mathfrak{g}_\mathbb{Z} \subset \mathfrak{g}$ in a rational form $\mathfrak{g}_\mathbb{Q}$ such that*

$$\mathrm{Ad}(\gamma)\mathfrak{g}_\mathbb{Z} = \mathfrak{g}_\mathbb{Z} \qquad (\gamma \in \Gamma).$$

Consequently every exterior representation

$$\rho_d = \bigwedge^d \mathrm{Ad} : G \longrightarrow \mathrm{GL}\left(\bigwedge^d \mathfrak{g}\right)$$

preserves the exterior lattice $\bigwedge^d \mathfrak{g}_{\mathbb{Z}}$.

PROOF. Choose a rational basis of $\mathfrak{g}_{\mathbb{Q}}$. Let Γ_0 be a finite-index subgroup contained in the chosen integral arithmetic group, so Γ_0 preserves a full lattice $L \subset \mathfrak{g}_{\mathbb{Q}}$. Replace Γ_0 by its normal core in Γ. Choose coset representatives $\delta_1, \ldots, \delta_r \in \Gamma \subset \mathbf{G}(\mathbb{Q})$. Each $\mathrm{Ad}(\delta_i)L$ is commensurable with L. Hence

$$\mathfrak{g}_{\mathbb{Z}} = \sum_{i=1}^{r} \mathrm{Ad}(\delta_i)L$$

is a full lattice. Normality of Γ_0 and the permutation of the cosets show that $\mathrm{Ad}(\Gamma)$ preserves this sum. Exterior powers preserve integral lattices functorially. $\square$

Remark 1.2. The finite-index normalization is not cosmetic. It is precisely what turns the orbit of an exterior vector into a subset of an honest Euclidean lattice. In the general Dani–Margulis theory this elementary integral discreteness is replaced by a nontrivial finiteness theorem for finite-volume homogeneous subgroups.

Let $\mathbf{H} < \mathbf{G}$ be a connected $\mathbb{Q}$-subgroup, let $H = \mathbf{H}(\mathbb{R})^\circ$, and put $d = \dim H$. Write $\mathfrak{h}_{\mathbb{Q}}$ for its rational Lie algebra. The rank-one group

$$\bigwedge^d (\mathfrak{h}_{\mathbb{Q}} \cap \mathfrak{g}_{\mathbb{Z}}) \subset \bigwedge^d \mathfrak{g}_{\mathbb{Z}}$$

has a primitive generator, unique up to sign. Fix one and call it p_H.

Definition 1.3 (Exterior representative)**.** For $g \in G$ define

$$\eta_H(g) = \rho_d(g)p_H.$$

The associated determinant-one normalizer is

$$N_G^1(H) = \{g \in N_G(H) : \det(\mathrm{Ad}(g)|_{\mathfrak{h}}) = 1\}.$$

2. What the exterior vector remembers

The exterior line $\mathbb{R}p_H$ remembers the Lie algebra $\mathfrak{h}$, while the vector p_H remembers both the Lie algebra and its oriented volume element.

Proposition 2.1 (Stabilizers of the line and vector)**.** *One has*

$$\mathrm{Stab}_G(\mathbb{R}p_H) = N_G(H), \qquad \mathrm{Stab}_G(p_H) = N_G^1(H).$$

PROOF. Suppose $g \in G$ stabilizes the line $\mathbb{R}p_H$. The decomposable vector p_H spans $\bigwedge^d \mathfrak{h}$. A vector $X \in \mathfrak{g}$ lies in $\mathfrak{h}$ exactly when

$$X \wedge p_H = 0.$$

If $\rho_d(g)p_H = cp_H$, then for $X \in \mathfrak{h}$,

$$0 = \rho_{d+1}(g)(X \wedge p_H) = \mathrm{Ad}(g)X \wedge cp_H,$$

so $\mathrm{Ad}(g)\mathfrak{h} \subset \mathfrak{h}$. Equality follows from dimension, hence $g \in N_G(H)$. The converse is immediate.

If $g \in N_G(H)$, then on the one-dimensional space $\bigwedge^d \mathfrak{h}$,

$$\rho_d(g)p_H = \det(\mathrm{Ad}(g)|_{\mathfrak{h}})p_H.$$

Thus the vector itself is fixed exactly when this determinant is one. □

Let $W < G$ be connected with Lie algebra $\mathfrak{w}$. Define

$$N(H, W) = \{g \in G : W \subset gHg^{-1}\}.$$

For the linearized condition introduce the linear subspace

$$\mathcal{L}_W^{(d)} = \{v \in \textstyle\bigwedge^d \mathfrak{g} : X \wedge v = 0 \text{ for every } X \in \mathfrak{w}\}.$$

Proposition 2.2 (Containment is linear)**.** *For every $g \in G$,*

$$g \in N(H, W) \quad \Longleftrightarrow \quad \eta_H(g) \in \mathcal{L}_W^{(d)}.$$

Proof. Since W is connected, $W \subset gHg^{-1}$ is equivalent to $\mathfrak{w} \subset \mathrm{Ad}(g)\mathfrak{h}$. The latter is equivalent to

$$X \wedge \rho_d(g)p_H = 0 \qquad (X \in \mathfrak{w}),$$

by the decomposable-vector criterion just used. □

Worked Example 2.3 (The SL_3 plane stabilizer)**.** Let H be the subgroup preserving the plane $\langle e_1, e_2\rangle$. Its Lie algebra contains the matrices preserving that plane, and a primitive exterior vector p_H is obtained by wedging an integral basis of that Lie algebra. If $W = \{\exp(tE_{12})\}$, then $g \in N(H, W)$ means $\mathrm{Ad}(g^{-1})E_{12} \in \mathfrak{h}$. Proposition 2.2 says that this nonlinear subgroup-containment statement is equivalent to one family of linear wedge equations on $\rho_d(g)p_H$. This is the basic miracle of linearization: the quotient geometry remains nonlinear, but the obstruction coordinates do not.

3. Discrete representatives

The quotient point $x = g\Gamma$ has infinitely many group representatives $g\gamma$. Accordingly it has the family of exterior representatives

$$\mathrm{Rep}_H(x) = \{\eta_H(g\gamma) : \gamma \in \Gamma\}.$$

Two vectors differing by sign encode the same conjugate subgroup but opposite orientations. We therefore also use the unoriented representative set

$$\mathrm{URep}_H(x) = \mathrm{Rep}_H(x)/\{v \sim -v\}.$$

The distinction is harmless for discreteness but essential for local uniqueness: an orientation-reversing lattice normalizer may produce both v and $-v$ without producing a smaller subgroup. All vector tubes used below will be chosen symmetric under $v \mapsto -v$.

The first serious issue is to prove that only finitely many of these can enter a fixed compact vector window.

Lemma 3.1 (Discrete exterior orbit)**.** *The set*

$$\Gamma p_H = \{\rho_d(\gamma) p_H : \gamma \in \Gamma\}$$

is a discrete subset of $\bigwedge^d \mathfrak{g}$. *In fact it is contained in the set of primitive vectors of the lattice* $\bigwedge^d \mathfrak{g}_{\mathbb{Z}}$.

Proof. By Lemma 1.1, $\rho_d(\gamma)$ is an automorphism of the exterior lattice. It sends primitive vectors to primitive vectors. Hence every $\rho_d(\gamma)p_H$ belongs to the discrete lattice $\bigwedge^d \mathfrak{g}_{\mathbb{Z}}$. A subset of a discrete set is discrete. □

Lemma 3.2 (Compact lifting for a lattice quotient)**.** *If* $K \subset G/\Gamma$ *is compact, there exists a compact* $\widetilde{K} \subset G$ *with* $\pi(\widetilde{K}) = K$.

Proof. For every $x \in K$, discreteness of Γ gives a relatively compact identity neighborhood U_x such that the translates of a sufficiently small neighborhood by distinct elements of Γ are disjoint. Therefore π admits a continuous local section over some neighborhood V_x of x. A finite subcover $V_{x_1}, \dots, V_{x_r}$ of K, after shrinking to sets with compact closure inside the section domains, produces

$$\widetilde{K} = \bigcup_{i=1}^{r} s_i(\overline{V_i \cap K}),$$

which is compact and projects onto K. □

Proposition 3.3 (Local finiteness of representatives)**.** *Let* $K \subset G/\Gamma$ *and* $D \subset \bigwedge^d \mathfrak{g}$ *be compact. Then*

$$\mathcal{I}_H(K, D) = \{(x, v) : x \in K,\ v \in \mathrm{Rep}_H(x) \cap D\}$$

is compact. Each fibre $\mathrm{Rep}_H(x) \cap D$ *is finite, and the cardinalities of the fibres are uniformly bounded for* $x \in K$.

Proof. Choose a compact lift $\widetilde{K}$ from Lemma 3.2. If $x = g\Gamma \in K$ with $g \in \widetilde{K}$ and

$$\rho_d(g\gamma) p_H \in D,$$

then

$$\rho_d(\gamma) p_H \in D' = \bigcup_{g \in \widetilde{K}} \rho_d(g)^{-1} D.$$

The set D' is compact. By Lemma 3.1, $D' \cap \Gamma p_H$ is finite. This gives a uniform finite list of possible exterior vectors before multiplication by g, hence a uniform bound on the fibres.

For compactness, take $(x_n, v_n) \in \mathcal{I}_H(K, D)$ converging in $K \times D$ to (x, v). Choose $g_n \in \widetilde{K}$ over x_n. Pass to a subsequence with $g_n \to g \in \widetilde{K}$. Write

$$v_n = \rho_d(g_n) q_n, \qquad q_n \in D' \cap \Gamma p_H.$$

The latter set is finite, so after another subsequence $q_n = q$ is constant. Then

$$v = \rho_d(g) q \in \mathrm{Rep}_H(g\Gamma),$$

proving closedness in the compact set $K \times D$. □

4. The proper incidence map

Let

$$\Gamma_H^1 = \{\gamma \in \Gamma : \rho_d(\gamma) p_H = p_H\}.$$

The map

$$\Phi_H : G/\Gamma_H^1 \longrightarrow G/\Gamma \times \bigwedge^d \mathfrak{g}, \qquad g\Gamma_H^1 \longmapsto (g\Gamma, \eta_H(g))$$

is well defined.

Theorem 4.1 (Representative properness). *The map Φ_H is proper.*

Proof. Let $K \subset G/\Gamma$ and $D \subset \bigwedge^d \mathfrak{g}$ be compact. Choose a compact lift $\widetilde{K} \subset G$. Any point of $\Phi_H^{-1}(K \times D)$ can be represented as $g\gamma\Gamma_H^1$ with $g \in \widetilde{K}$ and

$$\rho_d(g\gamma) p_H \in D.$$

As above, $\rho_d(\gamma) p_H$ belongs to the finite set

$$F = \Gamma p_H \cap \bigcup_{g \in \widetilde{K}} \rho_d(g)^{-1} D.$$

For each $q \in F$, choose one $\gamma_q \in \Gamma$ with $\rho_d(\gamma_q) p_H = q$. If $\rho_d(\gamma) p_H = q$, then $\gamma_q^{-1}\gamma \in \Gamma_H^1$, so

$$g\gamma\Gamma_H^1 = g\gamma_q\Gamma_H^1.$$

Thus

$$\Phi_H^{-1}(K \times D) \subset \bigcup_{q \in F} \widetilde{K}\gamma_q\Gamma_H^1/\Gamma_H^1,$$

a finite union of compact sets. The inverse image is closed by continuity of Φ_H, hence compact. □

Corollary 4.2 (Closedness over compact quotient sets)**.** *For compact $K \subset G/\Gamma$,*

$$\mathrm{Rep}_H(K) = \bigcup_{x \in K} \mathrm{Rep}_H(x)$$

is closed in $\bigwedge^d \mathfrak{g}$.

Proof. If $v_n \in \mathrm{Rep}_H(K)$ converges to v, all v_n lie in some compact ball D. Proposition 3.3 makes the incidence set over $K \times D$ compact, so a subsequence of the corresponding quotient points converges and the limit vector remains a representative. $\square$

5. When two representatives compete

A second representative is not merely a nuisance: it signals a smaller algebraic explanation.

Fix a compact vector window D. Define the ambiguity set

$$\mathcal{A}_H(K, D) = \{x \in K : \#(\mathrm{Rep}_H(x) \cap D) \geq 2\}.$$

To identify it, suppose

$$v_i = \rho_d(g\gamma_i)p_H \in \mathcal{L}_W^{(d)}, \qquad i = 1, 2.$$

Then both groups

$$H_i = g\gamma_i H \gamma_i^{-1} g^{-1}$$

contain W. If $v_1 \neq \pm v_2$, the groups are distinct, hence

$$F = (H_1 \cap H_2)^\circ$$

is a proper connected subgroup of each and contains W.

In the arithmetic coordinate before conjugation by g,

$$F' = \left(\gamma_1 H \gamma_1^{-1} \cap \gamma_2 H \gamma_2^{-1}\right)^\circ$$

is a connected $\mathbb{Q}$-subgroup, because intersection and identity component preserve rational algebraicity in characteristic zero. Moreover $F = gF'g^{-1}$.

Lemma 5.1 (Finite lower-stratum list on compact windows)**.** *Let $K \subset G/\Gamma$ and $D \subset \bigwedge^d \mathfrak{g}$ be compact. There exist finitely many connected proper $\mathbb{Q}$-subgroups*

$$F_1, \ldots, F_r < G$$

(each arising as the identity component of the intersection of two distinct Γ-conjugates of H) such that every ambiguous point $x \in \mathcal{A}_H(K, D)$ for which both relevant representatives lie in $\mathcal{L}_W^{(d)}$ belongs to one of the corresponding lower singular sets.

Proof. Choose a compact lift $\widetilde{K}$. Proposition 3.3 yields a finite set

$$Q = \Gamma p_H \cap \bigcup_{g\in\widetilde{K}} \rho_d(g)^{-1} D.$$

For each $q \in Q$, choose $\gamma_q \in \Gamma$ with $q = \rho_d(\gamma_q)p_H$. Hence every pair of representatives in the window is represented by one of finitely many pairs $(\gamma_q, \gamma_{q'})$. For unequal exterior lines define

$$\mathbf{F}_{q,q'} = \left(\gamma_q \mathbf{H} \gamma_q^{-1} \cap \gamma_{q'} \mathbf{H} \gamma_{q'}^{-1}\right)^\circ.$$

There are finitely many such groups. If the two corresponding conjugates both contain W, then their connected intersection contains W, so the quotient point lies in the lower incidence locus attached to this proper rational group. □

Corollary 5.2 (Local uniqueness up to orientation off lower incidence)**.** *Let $C \subset N(H,W)$ be compact and let D be a symmetric compact vector window containing $\eta_H(C) \cup -\eta_H(C)$. Assume that $\pi(C)$ avoids the finitely many lower rational incidence loci supplied by Lemma 5.1 for this window. Then there exist a quotient neighborhood Ω of $\pi(C)$ and a symmetric vector neighborhood $\Phi = -\Phi$ of*

$$D \cap \mathcal{L}_W^{(d)}$$

such that every $x \in \Omega$ has at most one unoriented *representative class in Φ.*

Proof. Suppose no such neighborhoods existed. Choose shrinking quotient and vector neighborhoods and points $x_n \to x \in \pi(C)$ carrying representatives v_n, w_n in the shrinking vector neighborhoods with $w_n \neq \pm v_n$. Compactness of the incidence set lets us pass to limits $v, w \in D \cap \mathcal{L}_W^{(d)}$, still representing the point x. Because the exterior lattice is discrete over a compact lift, after passing to a subsequence the underlying lattice representatives are constant. Hence $w \neq \pm v$; otherwise the same equality would hold eventually. The two exterior lines therefore encode two distinct conjugates of H containing W. Lemma 5.1 places x in one of the excluded lower rational incidence loci, a contradiction. □

Remark 5.3 (Why the sign quotient cannot be omitted)**.** If $\gamma \in \Gamma \cap N_G(H)$ reverses the orientation of $\mathfrak{h}$, then $\rho_d(\gamma)p_H = -p_H$. The two vectors p_H and $-p_H$ are distinct but describe exactly the same subgroup. Treating them as competing representatives would falsely create a "lower stratum."

Corollary 5.4 (Uniform uniqueness on a regular compact set)**.** *Let $K \subset G/\Gamma$ and $D \subset \bigwedge^d \mathfrak{g}$ be compact, with $D = -D$. Let $A(K,D)$ be the set of points $x \in K$ carrying two unoriented representative classes whose vectors lie in $D \cap \mathcal{L}_W^{(d)}$. Then $A(K,D)$ is compact. If $\mathcal{U}$ is any open neighborhood of*

$A(K,D)$, there exists a symmetric vector neighborhood Φ of $D \cap \mathcal{L}_W^{(d)}$ such that every $x \in K \setminus \mathcal{U}$ has at most one unoriented representative class in Φ.

PROOF. Compactness of the incidence space over $K \times D$ shows that the set of triples (x, v, w) with $w \neq \pm v$ is relatively compact. If a sequence of ambiguous points converges and the two unoriented classes collapse in the limit, discreteness of the exterior lattice over a compact lift implies that the equality $w = \pm v$ already holds eventually. Hence the ambiguity set is closed in K and therefore compact.

If the asserted vector neighborhood did not exist, choose a nested sequence of symmetric neighborhoods of $D \cap \mathcal{L}_W^{(d)}$ shrinking to that compact set. For each one there would be a point $x_n \in K \setminus \mathcal{U}$ with two inequivalent representatives in the neighborhood. Pass to a convergent subsequence in the compact incidence space. The limit lies in $A(K,D) \subset \mathcal{U}$, while $K \setminus \mathcal{U}$ is closed, a contradiction. $\square$

6. A polynomial growth lemma with nested tubes

The analytic input is the polynomial-goodness theorem proved in Volume 1. We record the vector form needed here.

Lemma 6.1 (Nested linear tubes)**.** *Fix integers m, D, n and a linear subspace $L < \mathbb{R}^n$. For every $\varepsilon_0 > 0$ there exists $\tau \in (0,1)$, depending only on (m, D, n, ε_0), with the following property. Let $B \subset \mathbb{R}^m$ be a Euclidean ball and $P : B \to \mathbb{R}^n$ a polynomial map of degree at most D. If*

$$\sup_B \operatorname{dist}(P, L) \geq R,$$

then

$$\big|\{t \in B : \operatorname{dist}(P(t), L) < \tau R\}\big| < \varepsilon_0 |B|.$$

PROOF. Choose orthonormal coordinates on $L^\perp$. At least one scalar coordinate Q of the orthogonal projection satisfies

$$\sup_B |Q| \geq R/\sqrt{n}.$$

If $\operatorname{dist}(P(t), L) < \tau R$, then $|Q(t)| < \tau R$. Volume 1's polynomial sublevel theorem gives

$$|\{|Q| < \tau R\}| \leq C_{m,D} \left(\tau\sqrt{n}\right)^{\alpha_{m,D}} |B|.$$

Choose τ so that the right side is $< \varepsilon_0 |B|$. $\square$

The next form is adapted to a compact part of the exterior orbit. Let

$$\mathcal{V}_H(W) = \rho_d(G) p_H \cap \mathcal{L}_W^{(d)}.$$

Although $\mathcal{V}_H(W)$ need not be linear, it is cut out inside the orbit by the linear wedge equations defining $\mathcal{L}_W^{(d)}$.

Lemma 6.2 (Hybrid compact-window growth)**.** *Fix integers m, D, n and a linear subspace $L < \mathbb{R}^n$. Let $C_V \subset L$ be compact. Given $\varepsilon_0 > 0$, there are numbers*

$$0 < r_V < R_V, \qquad 0 < \tau < \sigma,$$

with $C_V \subset B(0, r_V)$ such that, for the symmetric sets

$$\Psi = \{v : \|v\| < r_V,\ \operatorname{dist}(v, L) < \tau\},$$
$$\Phi = \{v : \|v\| < R_V,\ \operatorname{dist}(v, L) < \sigma\},$$

the following holds. For every polynomial map $P : \mathbb{R}^m \to \mathbb{R}^n$ of degree at most D and every Euclidean ball B, either

(1) $P(B) \subset \Phi$, or
(2)
$$|\{t \in B : P(t) \in \Psi\}| < \varepsilon_0 |B|.$$

Moreover R_V may be chosen arbitrarily large after r_V has been fixed, and σ may then be chosen arbitrarily small before τ is chosen.

Proof. Choose $r_V > 1 + \sup_{v \in C_V} \|v\|$. Let C_1, α_1 be goodness constants from Volume 1 for scalar polynomials of degree at most $2D$ in m variables. Choose $R_V > r_V$ so large that

$$C_1 \left(\frac{r_V^2}{R_V^2} \right)^{\alpha_1} < \frac{\varepsilon_0}{2}.$$

Now choose any $\sigma > 0$. Apply Lemma 6.1 with error $\varepsilon_0/2$ and outer scale σ; this supplies a ratio $\tau/\sigma > 0$, which may be made smaller if necessary.

Suppose $P(B) \not\subset \Phi$. There are two cases.

If $\sup_B \|P\| \geq R_V$, consider the scalar polynomial

$$Q(t) = \|P(t)\|^2.$$

It has degree at most $2D$ and $\sup_B Q \geq R_V^2$. On the inner set $P(t) \in \Psi$ one has $Q(t) < r_V^2$. Polynomial goodness gives

$$|\{t \in B : P(t) \in \Psi\}| \leq C_1 \left(\frac{r_V^2}{R_V^2} \right)^{\alpha_1} |B| < \frac{\varepsilon_0}{2} |B|.$$

Otherwise $\sup_B \|P\| < R_V$. Since $P(B)$ is not contained in Φ, it must satisfy

$$\sup_B \operatorname{dist}(P, L) \geq \sigma.$$

Lemma 6.1 then gives

$$|\{t \in B : \operatorname{dist}(P(t), L) < \tau\}| < \frac{\varepsilon_0}{2}|B|.$$

The inner event $P(t) \in \Psi$ is contained in this set. In either case the required estimate follows. □

Why this matters. The two inequalities defining Φ have different jobs. The norm bound controls tangential motion along the linearized algebraic locus; the distance to L controls normal motion away from it. A proof using only the wedge coordinates would miss a polynomial trajectory that stays exactly in L but runs to infinity inside L. Conversely, a norm bound alone would not detect a small normal departure. The hybrid window sees both.

7. Passing from quotient neighborhoods to vector neighborhoods

The finite-window proof repeatedly moves between a neighborhood of a compact incidence core in the quotient and a neighborhood of its exterior vectors. The following elementary lemma makes that passage explicit.

Lemma 7.1 (Local lifting into a prescribed exterior tube)**.** *Let $C \subset G$ be compact and let $\Psi \subset \bigwedge^d \mathfrak{g}$ be an open symmetric set containing*

$$\eta_H(C) \cup -\eta_H(C).$$

Then there is an open neighborhood $\mathcal{O}$ of $\pi(C)$ in G/Γ such that every $y \in \mathcal{O}$ has an H-representative in Ψ. Moreover, given any neighborhood Ω of $\pi(C)$, $\mathcal{O}$ may be chosen with $\overline{\mathcal{O}} \subset \Omega$ whenever $\pi(C) \Subset \Omega$.

Proof. For every $c \in C$, discreteness of Γ gives a relatively compact neighborhood U_c of c on which the quotient map π is injective. Shrink U_c so that

$$\rho_d(U_c)p_H \subset \Psi.$$

Then $V_c = \pi(U_c)$ is open and every point of V_c has a representative $g \in U_c$, hence the exterior representative $\rho_d(g)p_H \in \Psi$. A finite collection of the V_c covers $\pi(C)$. Their union gives the first assertion. If an ambient neighborhood Ω is prescribed, first choose the U_c so small that the closures of the corresponding V_c lie in Ω, and then take a finite subcover. □

8. The finite-window linearization proposition

The eventual-density theorem is easiest to prove from a finite-window statement. This also prevents a common quantifier mistake when the trajectory itself varies in a sequence.

For a rational subgroup H, write $\mathcal{I}_{<H}(W)$ for the union of lower rational incidence loci obtained from proper rational intersections appearing in compact representative windows. These auxiliary loci need not themselves support finite-volume homogeneous measures; their role is to resolve representative ambiguity in the arithmetic proof.

Proposition 8.1 (Finite-window alternative). *Fix:*

- *a compact* $K \subset G/\Gamma$ *whose interior contains* $\pi(C)$*,*
- *a compact* $C \subset N(H, W)$*,*
- *an error* $\delta > 0$*,*
- *a degree bound* D*.*

Then there exist a symmetric compact exterior window D_H *and a compact lower-incidence core* $Y_{<H} \subset K$*, contained in a finite union of lower rational incidence loci, with the following property. For every open neighborhood* $\mathcal{U}_{<H} \supset Y_{<H}$ *there exist symmetric tubes*

$$\Psi_H \Subset \Phi_H$$

and a neighborhood $\mathcal{O}$ *of* $\pi(C)$ *such that, for every adjoint-polynomial* $\Theta : \mathbb{R}^m \to G$ *of degree at most* D*, every* $x \in G/\Gamma$*, and every Euclidean ball* $B = B(c, T)$*, at least one of the following occurs:*

(1) $\Theta(B)x \cap K = \varnothing$*;*

(2) one exterior representative $v \in \mathrm{Rep}_H(x)$ *satisfies*

$$\rho_d(\Theta(B))v \subset \Phi_H;$$

(3) on the regular part away from $\mathcal{U}_{<H}$*,*

$$\big|\{t \in B : \Theta(t)x \in \mathcal{O} \setminus \mathcal{U}_{<H}\}\big| < \delta|B|.$$

Visits to $\mathcal{O} \cap \mathcal{U}_{<H}$ *are deliberately left uncontrolled; they belong to the lower-incidence region and must be removed by a zero-mass argument or treated after descent. The order of quantifiers is important:* D_H *and* $Y_{<H}$ *are fixed first, then* $\mathcal{U}_{<H}$ *is prescribed, and only then are the inner/outer tubes and* $\mathcal{O}$ *chosen.*

Proof. Choose a compact neighborhood K_1 of $\pi(C)$ with $K_1 \subset \operatorname{int} K$. Let N_m be the Besicovitch multiplicity constant for Euclidean balls in $\mathbb{R}^m$ and set

$$\varepsilon_0 = \frac{\delta}{3^m 2^m N_m}.$$

Put

$$C_V = \eta_H(C) \cup -\eta_H(C) \subset \mathcal{L}_W^{(d)}.$$

In Lemma 6.2, first choose the inner norm radius r_V around C_V and then choose the outer norm radius R_V large enough for the norm-growth estimate with error $\varepsilon_0/2$. Let D_H be the closed norm ball of radius $2R_V$. This slightly larger bookkeeping window provides room around the outer analytic tube. Representative properness controls all representatives above K_1 that can enter this bounded window, and Lemma 5.1 assigns finitely many lower rational incidence labels to the compact ambiguity set. Define

$$Y_{<H} = A(K_1, D_H).$$

By Corollary 5.4, this set is compact and is contained in the corresponding finite union of lower incidence loci. Fix an arbitrary open neighborhood $\mathcal{U}_{<H}$ of $Y_{<H}$. Apply Corollary 5.4 with $\mathcal{U} = \mathcal{U}_{<H}$. It gives a symmetric vector neighborhood of $D_H \cap \mathcal{L}_W^{(d)}$ in which the unoriented representative class is unique over $K_1 \setminus \mathcal{U}_{<H}$. Since the closed set

$$\{v : \|v\| \le R_V\} \cap \mathcal{L}_W^{(d)}$$

is a compact subset of that uniqueness neighborhood, choose the outer transverse width $\sigma > 0$ small enough that

$$\Phi_H = \{v : \|v\| < R_V,\ \operatorname{dist}(v, \mathcal{L}_W^{(d)}) < \sigma\}$$

lies in that uniqueness neighborhood. Choose a quotient neighborhood Ω of $\pi(C)$ with $\overline{\Omega} \subset K_1$. Now use the linear-tube estimate to choose $\tau < \sigma$ with normal-error $\varepsilon_0/2$, and define

$$\Psi_H = \{v : \|v\| < r_V,\ \operatorname{dist}(v, \mathcal{L}_W^{(d)}) < \tau\}.$$

Apply Lemma 7.1 and shrink $\mathcal{O} \Subset \Omega$ so that every point of $\mathcal{O}$ has a representative in Ψ_H. All choices of r_V, R_V, σ, τ and hence of Φ_H, Ψ_H are made before the trajectory (x, Θ, B) is considered. Set

$$E = \{t \in B : \Theta(t)x \in \mathcal{O} \setminus \mathcal{U}_{<H}\}.$$

If $E = \varnothing$, alternative (3) is immediate, so assume $E \neq \varnothing$.

For each $t \in E$, choose a representative $v_t \in \operatorname{Rep}_H(x)$ with

$$\rho_d(\Theta(t))v_t \in \Psi_H.$$

Its unoriented class is unique as long as the quotient point stays in $\Omega \setminus \mathcal{U}_{<H}$ and the vector stays in Φ_H. Define

$$r_t = \sup\{0 < r \le T : \rho_d(\Theta(\overline{B}(t, 3r)))v_t \subset \Phi_H\}.$$

The set in braces contains all sufficiently small r by continuity, so $r_t > 0$.

If $r_t = T$ for some $t \in E$, then $B \subset B(t, 3T)$ because both t and the center c lie in $B(c, T)$. Hence

$$\rho_d(\Theta(B))v_t \subset \Phi_H,$$

which is alternative (2). We may therefore assume $r_t < T$ for every $t \in E$ and put

$$B_t = B(t, r_t).$$

By the definition of the supremum and continuity, the vector trajectory is contained in $\overline{\Phi_H}$ on $3B_t$. Moreover it is not contained in Φ_H on the closed ball $\overline{B}(t, 3r_t)$: if the compact image of that closed ball were contained in the open set Φ_H, it would have a positive distance from the complement and the radius could be increased, contradicting maximality. Thus the non-persistence alternative of Lemma 6.2 applies on $3B_t$. We obtain

$$\left|\{y \in 3B_t : \rho_d(\Theta(y))v_t \in \Psi_H\}\right| < \varepsilon_0|3B_t| = 3^m\varepsilon_0|B_t|.$$

We claim that

$$E \cap B_t \subset \{y \in B_t : \rho_d(\Theta(y))v_t \in \Psi_H\}.$$

Indeed, if $y \in E \cap B_t$, choose the representative v_y given by the definition of E. The trajectory of v_t lies in Φ_H throughout B_t, while v_y lies in $\Psi_H \subset \Phi_H$ at y. Local uniqueness up to sign gives $v_y = \pm v_t$. Since both tubes are symmetric, the displayed inclusion follows. Consequently

$$|E \cap B_t| < 3^m\varepsilon_0|B_t|.$$

The family $\{B_t : t \in E\}$ is a Besicovitch family centered on E. Select a countable subfamily $\{B_i\}$ covering E with multiplicity at most N_m. Since every center lies in $B(c, T)$ and every radius is at most T, all selected balls lie in $B(c, 2T) = 2B$. Therefore

$$\begin{aligned}|E| &\le \sum_i |E \cap B_i| \\ &< 3^m\varepsilon_0 \sum_i |B_i| \\ &\le 3^m\varepsilon_0 N_m|2B| \\ &= \delta|B|.\end{aligned}$$

This is alternative (3). Visits in $\mathcal{U}_{<H}$ were never counted in E and no estimate for them is asserted. Finally, if the orbit segment never meets the compact core K, alternative (1) holds. □

Remark 8.2 (Where each constant comes from)**.** The factor 3^m is the volume cost of testing exit on $3B_t$; N_m is the Besicovitch overlap multiplicity; and 2^m bounds the total ambient volume because all selected balls lie in $2B$. No asymptotic or unspecified "covering constant" is hidden in the estimate.

9. From finite windows to an eventual fixed-trajectory theorem

The finite-window proposition is the logically strongest statement needed for sequences whose trajectories vary. For one fixed trajectory it has a useful eventual consequence. The only additional argument is that persistence on arbitrarily large concentric balls forces an exact polynomial identity.

THEOREM 9.1 (Arithmetic regular-window linearization). *Let the arithmetic standing hypotheses of this chapter hold. Fix a connected rational subgroup H, a compact set $C \subset N(H, W)$, a compact $K \subset G/\Gamma$ whose interior contains $\pi(C)$, an adjoint-polynomial degree bound D, and $\delta > 0$. Let $Y_{<H} \subset K$ be the compact lower-incidence core furnished by Proposition 8.1; it is contained in a finite union of lower rational incidence loci after a compact exterior window has been fixed. For every open neighborhood $\mathcal{U}_{<H}$ of $Y_{<H}$ there are an open neighborhood $\mathcal{O}$ of $\pi(C)$ and a bounded symmetric exterior window Φ_H such that, for every degree-$\leq D$ adjoint-polynomial map $\Theta : \mathbb{R}^m \to G$, every $x \in G/\Gamma$, and every Euclidean ball B, one of the following holds:*

(1) $\Theta(B)x \cap K = \varnothing$;

(2) there is $v \in \mathrm{Rep}_H(x)$ *with*
$$\rho_d(\Theta(B))v \subset \Phi_H;$$

(3)
$$\big|\{t \in B : \Theta(t)x \in \mathcal{O} \setminus \mathcal{U}_{<H}\}\big| < \delta|B|.$$

All neighborhoods and vector windows are chosen before (x, Θ, B).

PROOF. This is exactly Proposition 8.1 after fixing the prescribed neighborhood $\mathcal{U}_{<H}$ of the finite lower incidence union. The proposition itself was proved above from representative properness, the finite ambiguity list, local uniqueness up to orientation, polynomial goodness, and the Besicovitch covering theorem. No induction hypothesis is being used. □

THEOREM 9.2 (Eventual avoidance for a fixed polynomial trajectory). *Retain the hypotheses of Theorem 9.1, and fix $x \in G/\Gamma$ and an adjoint-polynomial map $\Theta : \mathbb{R}^m \to G$. Let $B_R = B(0, R)$, $R \to \infty$. Assume there is no $v \in \mathrm{Rep}_H(x)$ such that*
$$\rho_d(\Theta(t))v = \rho_d(\Theta(0))v \qquad (t \in \mathbb{R}^m)$$
with $\rho_d(\Theta(0))v \in \Phi_H$. Then
$$\limsup_{R\to\infty} \frac{1}{|B_R|}\big|\{t \in B_R : \Theta(t)x \in \mathcal{O} \setminus \mathcal{U}_{<H}\}\big| \leq \delta.$$

If such a persistent vector exists and $v = \rho_d(g\gamma)p_H$ *for* $x = g\Gamma$, *then*

$$(\Theta(0)g\gamma)^{-1}\Theta(t)g\gamma \in N_G^1(H) \qquad (t \in \mathbb{R}^m),$$

so the whole trajectory lies in one determinant-one normalizer translate.

Proof. Apply Theorem 9.1 to the concentric balls B_R. Suppose persistence occurs for an unbounded sequence $R_j \to \infty$, with representatives v_j. Since $0 \in B_{R_j}$,

$$\rho_d(\Theta(0))v_j \in \overline{\Phi_H}.$$

Local finiteness over the fixed point x leaves only finitely many such representatives. After a subsequence, $v_j = v$ is constant. Every fixed ball B_T is contained in B_{R_j} for all large j; hence the vector-valued polynomial $P(t) = \rho_d(\Theta(t))v$ is globally bounded. Every scalar coordinate of a bounded real polynomial on $\mathbb{R}^m$ is constant, so $P(t) = P(0)$ identically.

Writing $v = \rho_d(g\gamma)p_H$ gives

$$\rho_d\Big((\Theta(0)g\gamma)^{-1}\Theta(t)g\gamma\Big)\, p_H = p_H.$$

Proposition 2.1 identifies the vector stabilizer with $N_G^1(H)$, proving the normalizer assertion. Under the hypothesis excluding persistence, alternative (3) of the regular-window theorem holds for all sufficiently large radii, which gives the limsup bound. $\square$

Warning 9.3 (What the theorem does and does not say)**.** A lower-incidence trap is not automatically an H-normalizer trap. It is therefore incorrect to relabel every lower-dimensional persistence alternative as the H-trapping alternative. The finite-window theorem keeps the lower incidence neighborhood visible. An application removes it either because the candidate limiting measure gives it zero mass or because one deliberately descends to the lower obstruction and restarts the argument there.

10. A weak-limit elimination lemma

Lemma 10.1 (Proper analytic incidence has Haar measure zero)**.** *Let* $L < G$ *be connected and let* $Lx \subset G/\Gamma$ *be a closed orbit carrying its* L*-invariant probability* m_{Lx}. *Let* $Z \subset Lx$ *be contained locally in the zero set of finitely many real-analytic functions which are not all identically zero on any connected open set. Then* $m_{Lx}(Z) = 0$.

Proof. Haar measure on Lx has a smooth positive density in analytic coordinates. Thus it suffices to show that the zero set of a nonzero real-analytic function on a connected open subset of $\mathbb{R}^q$ has Lebesgue measure zero. We argue by induction on q. For $q = 1$, zeros are discrete unless the function is identically zero. For $q > 1$, write variables as (y, s). For a fixed y, either $s \mapsto f(y,s)$ is not identically zero and has zero one-dimensional zero set, or all its Taylor coefficients vanish. Choose the first coefficient which is not identically zero as a function of y; by the induction hypothesis its zero set has measure zero. Fubini's theorem finishes the proof. Finite unions preserve measure zero. □

Proposition 10.2 (Weak-limit elimination of a regular H-core)**.** *Let*

$$\mu_R = \frac{1}{|B_R|}\int_{B_R} \delta_{\Theta(t)x}\, dt \Longrightarrow \mu$$

for one fixed x, one fixed adjoint-polynomial Θ, and concentric balls. Let $Y_{<H} \subset K$ be the compact lower-incidence core attached to the chosen exterior window. Assume $\mu(Y_{<H}) = 0$ and that the exact H-normalizer trap of Theorem 9.2 does not occur. Then $\mu(\pi(C)) = 0$.

Proof. Suppose $a = \mu(\pi(C)) > 0$ and choose $0 < \delta < a/8$. Since $Y_{<H} \cap K$ is compact and has zero μ-measure, regularity gives an open neighborhood $\mathcal{U}_{<H}$ whose closure F satisfies $\mu(F) < \delta$. Apply Theorem 9.2 and obtain $\mathcal{O} \supset \pi(C)$. Then

$$\limsup_R \mu_R(\mathcal{O} \setminus \mathcal{U}_{<H}) \le \delta.$$

Also $\mu_R(\mathcal{U}_{<H}) \le \mu_R(F)$, and Portmanteau for the closed set F gives $\limsup_R \mu_R(F) \le \mu(F) < \delta$. Hence $\limsup_R \mu_R(\mathcal{O}) \le 2\delta$. Choose $f \in C_c(X)$, $0 \le f \le 1$, equal to one on $\pi(C)$ and supported in $\mathcal{O}$. Weak convergence yields

$$a \le \int f\, d\mu = \lim_R \int f\, d\mu_R \le 2\delta < a/4,$$

a contradiction. □

Corollary 10.3 (Homogeneous component version)**.** *If $\mu = m_{Lx}$ is homogeneous and every component of $Y_{<H} \cap Lx$ is a proper analytic incidence subset of Lx, then $\mu(Y_{<H}) = 0$ automatically.*

Proof. Apply Lemma 10.1 to the finitely many incidence components. □

11. Focusing

Avoidance is only half of linearization. When regular occupation near a singular core persists for a varying sequence, the finite-window alternative produces one exterior representative on each large piece. To pass to a polynomial limit one must recenter at a return which is macroscopically far from the boundary; otherwise boundedness on the original ball gives no control on any fixed neighborhood of the new origin.

Theorem 11.1 (Sequential arithmetic focusing on a regular core). *Fix the arithmetic standing hypotheses, a connected rational subgroup H, a compact $C \subset N(H, W)$, a compact K with $\pi(C) \Subset \operatorname{int} K$, a parameter dimension m, and an adjoint-polynomial degree bound D. Let*

$$\Theta_n : \mathbb{R}^m \to G, \qquad x_n \in G/\Gamma, \qquad B_n = B(c_n, R_n), \quad R_n \to \infty.$$

Suppose there is $c_0 > 0$ and a sequence of regular linearization data

$$\mathcal{O}_j \searrow \pi(C), \qquad \mathcal{U}_j \supset Y_{<H,j}, \qquad \Psi_j \Subset \Phi_j,$$

with the following properties:

(1) *the outer windows Φ_j lie in one fixed norm ball and their transverse widths to $\mathcal{L}_W^{(d)}$ tend to zero;*
(2) *the inner windows Ψ_j shrink to $\eta_H(C) \cup -\eta_H(C)$;*
(3) *for every j there are arbitrarily large n for which*

$$\frac{1}{|B_n|}\left|\{t \in B_n : \Theta_n(t)x_n \in \mathcal{O}_j \setminus \mathcal{U}_j\}\right| \geq c_0.$$

Then there is a diagonal subsequence n_j, representatives $v_j \in \mathrm{Rep}_H(x_{n_j})$, and points $t_j \in B_{n_j}$ such that, after recentering,

$$P_j(s) = \rho_d(\Theta_{n_j}(t_j + s))v_j$$

is defined and uniformly bounded on balls $B(0, r_j)$ with $r_j \to \infty$. After a further subsequence, P_j converges coefficientwise and locally uniformly to a vector-valued polynomial P of degree at most D satisfying

$$P(\mathbb{R}^m) \subset \mathcal{L}_W^{(d)}, \qquad P(0) \in \eta_H(C) \cup -\eta_H(C).$$

In particular every defining wedge coordinate of the transporter vanishes identically on the limiting polynomial.

Proof. Choose $0 < \eta < 1$ so small that the boundary shell of a Euclidean ball has relative volume

$$1 - (1 - \eta)^m < c_0/2.$$

At stage j, apply the finite-window theorem with error $\delta < c_0/4$ and the j-th regular data. Choose n_j so large that $R_{n_j} \to \infty$ and the regular occupation in item (3) is at least $c_0|B_{n_j}|$. The small-occupation alternative is impossible. The lower incidences have already been removed in the regular set $\mathcal{O}_j \setminus \mathcal{U}_j$. Hence a representative $v_j \in \mathrm{Rep}_H(x_{n_j})$ persists in Φ_j throughout B_{n_j}:

$$\rho_d(\Theta_{n_j}(B_{n_j}))v_j \subset \Phi_j.$$

Since the boundary shell has measure less than $c_0|B_{n_j}|/2$, there is a regular return

$$t_j \in B(c_{n_j}, (1 - \eta)R_{n_j}).$$

At this point the quotient lies in $\mathcal{O}_j \setminus \mathcal{U}_j$ and has some representative in Ψ_j. Local uniqueness up to sign in the outer window implies that this representative is $\pm v_j$. Because all windows are symmetric,

$$P_j(0) = \rho_d(\Theta_{n_j}(t_j))v_j \in \Psi_j.$$

Also

$$B(t_j, \eta R_{n_j}) \subset B_{n_j},$$

so P_j is bounded by one common constant on $B(0, r_j)$ with $r_j = \eta R_{n_j} \to \infty$.

We now use only finite-dimensional polynomial algebra. On the vector space of $\mathbb{R}^N$-valued polynomials of degree at most D, the coefficient norm is equivalent to the supremum norm on the unit ball. Hence the common bound on $B(0,1)$ makes all coefficients of P_j uniformly bounded. Pass to a coefficientwise convergent subsequence, say $P_j \to P$. Coefficient convergence implies uniform convergence on every compact parameter set.

By item (1),

$$\sup_{s \in B(0,r_j)} \operatorname{dist}(P_j(s), \mathcal{L}_W^{(d)}) \longrightarrow 0.$$

For every fixed s, eventually $s \in B(0, r_j)$, and therefore $P(s) \in \mathcal{L}_W^{(d)}$. Since Ψ_j shrinks to the compact set $\eta_H(C) \cup -\eta_H(C)$ and $P_j(0) \in \Psi_j$, we also have $P(0)$ in that set. The linear subspace $\mathcal{L}_W^{(d)}$ is exactly the common zero set of the wedge coordinates $X \wedge v$ with $X \in \mathfrak{w}$, so all those polynomial coordinates vanish identically on P. $\square$

Remark 11.2 (What focusing gives at this stage)**.** The theorem deliberately concludes that the limiting polynomial lies in the linear containment locus $\mathcal{L}_W^{(d)}$ and has a genuine exterior-orbit base point. It does not claim that every value $P(s)$ remains in the single G-orbit $\rho_d(G)p_H$: that orbit need not be closed. Applications which need a stronger algebraic conclusion must add the relevant closed Pluecker, bracket, or orbit-closure equations and prove that they survive the limit. This distinction is often suppressed in informal presentations.

12. From focusing to weak-limit control

Let

$$\mu_n = \frac{1}{|B_n|} \int_{B_n} \delta_{\Theta_n(t)x_n}\, dt$$

be empirical measures and suppose $\mu_n \Rightarrow \mu$. If μ gives positive mass to a pure proper homogeneous stratum, regularity of μ gives a compact subset C of that stratum with positive mass. Portmanteau then supplies positive occupation of every neighborhood of C along a subsequence. The focusing theorem therefore produces an exterior polynomial satisfying exact algebraic equations.

This is the operational content of the slogan

$$\text{singular mass} \quad \Longrightarrow \quad \text{algebraic focusing.}$$

The classification theorem from the Ratner project supplies the list of possible homogeneous strata; the argument above proves that a sequence can charge one of them only by developing the corresponding finite-dimensional algebraic obstruction.

13. Why the arithmetic proof is genuinely complete

The proof used four ingredients:

(1) the exterior-lattice discreteness of Γp_H, proved in Lemma 3.1;
(2) properness and finite representative multiplicity, proved in Theorem 4.1;
(3) the finite lower-stratum list, proved in Lemma 5.1;
(4) polynomial sublevel control, already proved in Volume 1 and converted to the exact nested-tube statement here.

No Dani–Margulis properness theorem is hidden in this chain. What is lost relative to the most general historical theorem is only generality of the ambient lattice, not a logical step in the arithmetic argument.

Proof and provenance. The architecture follows the linearization method developed by Dani–Margulis and the particularly transparent formulation in Shah's thesis. Historically, Shah proves representative properness for general lattices using the Dani–Margulis finiteness theorem for bounded-volume homogeneous subgroups and then proves the finite-window induction. The present text replaces that finiteness input by the integral exterior lattice available for arithmetic quotients and proves the resulting argument in full. Citations here are provenance, not dependencies.

CHAPTER 6

The Dani–Margulis–Shah Linearization Principle

The preceding chapter converted the nonlinear condition

$$g \in N(H, W) = \{g \in G : W \subset gHg^{-1}\}$$

into the linear condition

$$\rho_H(g)p_H \in \operatorname{Fix}(W) \subset V_H = \bigwedge^{\dim H} \mathfrak{g}.$$

This chapter explains how one exploits that conversion dynamically. The point is not that a trajectory which *meets* a thin neighborhood of a singular stratum must be trapped. That statement is false. The point is that a bounded-degree polynomial trajectory cannot spend a positive proportion of a very long parameter set in an arbitrarily thin algebraic neighborhood unless an exact algebraic obstruction persists along the whole trajectory.

1. Polynomial maps in the adjoint representation

The class needed for the linearization method is slightly broader than algebraic polynomial maps into a matrix group.

Definition 1.1 (Adjoint-polynomial maps)**.** Let $m, \ell \in \mathbb{N}$. We write $\mathcal{P}_\ell(\mathbb{R}^m, G)$ for the set of continuous maps $\Theta : \mathbb{R}^m \to G$ such that for every $a, b \in \mathbb{R}^m$ and every $X \in \mathfrak{g}$, the map

$$s \longmapsto \operatorname{Ad}(\Theta(sa + b))X$$

has, in any fixed basis of $\mathfrak{g}$, polynomial coordinates of degree at most ℓ.

The definition is coordinate independent: replacing the basis of $\mathfrak{g}$ only takes fixed linear combinations of the coordinate polynomials. Every one-parameter Ad-unipotent homomorphism belongs to $\mathcal{P}_\ell(\mathbb{R}, G)$ for some ℓ, and finite products of such maps give higher-dimensional examples.

Example 1.2 (Products of unipotents)**.** Let $u_i(t) = \exp(tX_i)$ with $\operatorname{ad}(X_i)$ nilpotent, and put

$$\Theta(t_1, \ldots, t_m) = u_m(t_m) \cdots u_1(t_1).$$

Each $\mathrm{Ad}(u_i(t_i)) = \exp(t_i \operatorname{ad} X_i)$ is a polynomial in t_i. Hence the matrix coefficients of $\mathrm{Ad}(\Theta)$ are polynomials in the parameters, with a degree bound depending only on the nilpotency indices of the $\operatorname{ad} X_i$.

For a fixed exterior representation $\rho_H = \bigwedge^{\dim H} \mathrm{Ad}$, the maps

$$t \longmapsto \rho_H(\Theta(t))v$$

are therefore vector-valued polynomial maps of uniformly bounded degree along every affine line.

2. The elementary non-concentration mechanism

Before stating the homogeneous-space theorem, we isolate the elementary fact doing the analytic work. We first record the only scalar anti-concentration estimate that will be used.

Theorem 2.1 (Uniform sublevel estimate for bounded-degree polynomials). *For every pair of integers $m, D \geq 1$ there exist constants $C_0 = C_0(m, D) \geq 1$ and $\alpha = \alpha(m, D) > 0$ with the following property. If $B \subset \mathbb{R}^m$ is a Euclidean ball and $0 \neq Q \in \mathbb{R}[x_1, \ldots, x_m]$ has total degree at most D, then for every $\varepsilon > 0$,*

$$\big|\{t \in B : |Q(t)| < \varepsilon\}\big| \leq C_0 \left(\frac{\varepsilon}{\sup_B |Q|}\right)^{\alpha} |B|.$$

Proof and provenance. Proved in this text. This is exactly the bounded-degree polynomial goodness theorem proved in Volume 1, including the multivariable argument. The present formulation is obtained by specializing that proved theorem to one scalar coordinate polynomial. Nothing below uses an optimal value of α; the uniform positive power is the essential fact.

Lemma 2.2 (Polynomial residence near a linear subspace)**.** *Fix integers $m, D, n \geq 1$. Let $L \subset \mathbb{R}^n$ be a linear subspace and let $P : B \to \mathbb{R}^n$ be a polynomial map of degree at most D on a Euclidean ball $B \subset \mathbb{R}^m$. For $r > 0$ put*

$$E_r = \{t \in B : \operatorname{dist}(P(t), L) < r\}.$$

There are constants $C = C(m, D, n)$ and $\alpha = \alpha(m, D) > 0$ such that, whenever $0 < r < R$ and

$$\sup_{t \in B} \operatorname{dist}(P(t), L) \geq R,$$

we have

$$|E_r| \leq C \left(\frac{r}{R}\right)^{\alpha} |B|.$$

Proof. Let $\pi : \mathbb{R}^n \to L^\perp$ be orthogonal projection. The condition is

$$\sup_B \|\pi P\| \geq R.$$

Choose an orthonormal basis $e_1, \ldots, e_q$ of $L^\perp$ and write

$$\pi P = (P_1, \ldots, P_q).$$

At a point where $\|\pi P\|$ is at least R, one coordinate satisfies

$$|P_j| \geq R/\sqrt{q}.$$

Hence

$$\sup_B |P_j| \geq R/\sqrt{q}.$$

If $t \in E_r$, then $|P_j(t)| < r$. Theorem 2.1 gives

$$|E_r| \leq |\{t \in B : |P_j(t)| < r\}| \leq C_0 \left(\frac{r}{R/\sqrt{q}}\right)^\alpha |B|.$$

Absorb $q^{\alpha/2}$ into the constant. □

Why this matters. The lemma supplies the basic dichotomy. Either the polynomial leaves the larger algebraic tube by an amount R, in which case the time spent in a much thinner tube is small, or it never leaves the larger tube. Iterating this observation while shrinking neighborhoods drives the latter alternative toward exact algebraic containment.

For nonlinear algebraic sets one covers a compact piece by finitely many coordinate charts and uses polynomial functions defining the variety locally. In the homogeneous quotient, however, different representatives of the same point can enter the linearization chart. Representative properness and the finite lower-incidence list are what turn this ambiguity into a finite-dimensional problem.

3. The exact arithmetic alternative proved in this volume

From this point until the measure-theoretic synthesis we impose the arithmetic standing hypotheses of Chapter 5: the ambient lattice is arithmetic in a rational linear model and the obstruction subgroup H is a connected rational subgroup. This is the setting in which the representative map can be proved proper from the integral exterior lattice without importing the general Dani–Margulis bounded-volume finiteness theorem.

Theorem 3.1 (Arithmetic Dani–Margulis–Shah linearization alternative). Proved in this text. *Let $C \subset N(H, W)$ be compact, let $K \subset X$ be compact with $\pi(C) \subset \operatorname{int} K$, fix a degree bound D, and let $\delta > 0$. There is* *a finite union of lower-incidence loci* $Y_{<H} \subset K$ *with the following property. For every open neighborhood* $\mathcal{U}_{<H} \supset Y_{<H}$ *there are an open neighborhood* $\mathcal{O} \supset \pi(C)$ *and a bounded symmetric exterior window* Φ_H *such that, for every fixed degree-*$\leq D$ *adjoint-polynomial trajectory* (Θ, x)*, one of the following holds:*

(1) there is an H*-representative* $v \in \mathrm{Rep}_H(x)$ *for which*

$$\rho_d(\Theta(t))v = \rho_d(\Theta(0))v \qquad (t \in \mathbb{R}^m),$$

and consequently the whole trajectory is contained in one translate of $N_G^1(H)$*;*

(2) for concentric balls $B_R = B(0, R)$*,*

$$\limsup_{R\to\infty} \frac{1}{|B_R|} \big|\{t \in B_R : \Theta(t)x \in \mathcal{O} \setminus \mathcal{U}_{<H}\}\big| \leq \delta.$$

The stronger finite-window version is uniform over (x, Θ, B)*: on any one ball either one representative persists throughout a fixed outer vector window, or the regular occupation is* $< \delta|B|$*.*

Proof. The finite-window statement is Theorem 9.1, proved from scratch in Chapter 5. For a fixed trajectory, persistence on arbitrarily large concentric balls forces one and the same representative to recur at the center. Its vector-valued polynomial is then bounded on all of $\mathbb{R}^m$, hence constant. The vector stabilizer calculation identifies the resulting exact trap with $N_G^1(H)$. This is Theorem 9.2. ☐

Proof and provenance. Dani–Margulis and Shah proved more general versions in which the ambient lattice need not be arithmetic and the relevant homogeneous subgroup need not be rational in a fixed model. Their general representative-properness step uses a bounded-volume finiteness theorem for homogeneous subgroups. The present volume does not cite that theorem as an input: it proves the arithmetic rational-stratum form above completely. The historical citation records the origin of the method, not a logical dependency.

4. Why the lower-incidence neighborhood must remain visible

Every part of the statement has a purpose. If two inequivalent H-representatives are simultaneously relevant, their corresponding conjugates intersect in a strictly smaller rational subgroup containing the observed unipotent direction. This is a genuine competing obstruction. It cannot be silently renamed an H-normalizer trap. The finite ambiguity list records exactly where this can happen.

In a weak-limit application, the lower-incidence neighborhood disappears for a different reason: on a homogeneous candidate component it is a finite union of proper analytic incidence subsets and hence has Haar measure zero. The regular-window estimate plus Portmanteau then gives zero mass to the compact H-core; this is Proposition 10.2. If a lower incidence instead carries positive mass, one descends to that obstruction and restarts the argument there. This is the correct meaning of "induction on singular strata."

4.1. Why $N_G^1(H)$ rather than $N_G(H)$? The vector p_H represents an oriented volume element of $\mathfrak{h}$. If $g \in N_G(H)$, then

$$\rho_H(g)p_H = \det(\mathrm{Ad}(g)|_{\mathfrak{h}})p_H.$$

Exact stabilization of the vector is therefore equivalent to the determinant-one normalizer. Passing only to the exterior line would recover the full normalizer but would lose the scalar information used in the polynomial compactness argument.

4.2. Why an eventual density statement? A polynomial may cross an algebraic set finitely many times or linger near it on a bounded interval. The asymptotic statement discards this irrelevant finite-scale behavior. The finite-window statement is stronger when the trajectory itself varies, because its constants are chosen before the particular polynomial piece is presented.

5. What is actually proved in the finite-window argument

The proof has four independent pieces.

(1) *Representative properness.* Over a compact quotient region, only finitely many exterior representatives can enter a bounded vector window. In the arithmetic setting this is proved from the integral exterior lattice.

(2) *Ambiguity detection.* Two inequivalent representatives in the linearized transporter determine two distinct conjugates of H; their connected intersection is a proper rational subgroup. Compactness leaves a finite list of such incidences.

(3) *Polynomial growth.* A hybrid vector window controls both motion normal to the transporter equations and tangential escape to infinity. The sublevel theorem from Volume 1 gives a small inner/outer occupation ratio.

(4) *Besicovitch selection.* Maximal persistence radii produce a family of parameter balls. A bounded-overlap subcover sums the local polynomial estimates without losing control of the constants.

The hybrid window is essential. Wedge equations alone only see normal motion; a polynomial curve can stay exactly in the linearized subspace while running to infinity tangentially. The additional norm window detects precisely this case.

Warning 5.1. The slogan "a polynomial cannot stay near a variety for long" is not a proof of linearization. The quotient has infinitely many lattice representatives, orientation reversal can produce v and $-v$ for the same subgroup, and competing representatives can collide over lower incidence sets. Each of these issues is handled explicitly in Chapter 5.

6. A one-dimensional toy model

Take $P_T(t) = (1, t/T)$ for $0 \le t \le T$ and let $L = \mathbb{R}(1, 0)$. For a fixed $r > 0$,

$$\operatorname{dist}(P_T(t), L) = t/T,$$

so

$$\frac{1}{T}|\{0 \le t \le T : \operatorname{dist}(P_T(t), L) < r\}| = r.$$

Shrinking the tube shrinks the occupation proportion. In contrast, if $P_T(t) = (1, 0)$ identically, the occupation is 1 at every scale. The homogeneous theorem is this elementary dichotomy complicated by three facts: the obstruction is an algebraic variety rather than a linear subspace; the representative is defined only modulo Γ; and different homogeneous subgroups intersect along lower strata.

7. What the theorem does not say

Theorem 3.1 is an *avoidance theorem on the regular part*, not a classification theorem. It does not identify the limiting measure by itself. Its role is to prevent a weak limit from placing unexplained mass near a proper homogeneous stratum. Ratner's measure classification supplies the list of possible strata; linearization tells us how a moving trajectory can or cannot accumulate on them. The two inputs should therefore be kept conceptually separate: classification determines which algebraic strata are possible, while linearization controls how an averaging trajectory can approach those strata.

8. The quantifiers deserve to be read literally

There are two forms of the result and their quantifiers are different.

In the finite-window form, C, K, D, δ, the finite ambiguity list, the lower-incidence neighborhood, and the vector windows are chosen first. Then *every* polynomial piece (x, Θ, B) satisfies the same dichotomy. This is the form needed for a varying sequence of trajectories.

In the fixed-trajectory form, the same uniform neighborhoods are used, but the conclusion is only a limsup statement as $R \to \infty$ for one fixed (x, Θ). The reason is elementary: if persistence occurs on arbitrarily large balls, local finiteness at the center forces a single representative to persist on an unbounded subsequence, and boundedness of its polynomial vector curve forces exact constancy.

Warning 8.1 (A common diagonal-sequence error)**.** For a varying sequence (x_n, Θ_n, B_n), one cannot first derive a trajectory-dependent threshold for the n-th orbit and then infer that $|B_n| \to \infty$ eventually dominates that threshold. The index n changes the threshold. The finite-window theorem is precisely the device that keeps the choices uniform before the sequence is presented.

9. Why convex parameter sets are convenient

The finite-window proof uses Euclidean balls in $\mathbb{R}^m$; the same polynomial-growth mechanism extends to regular convex averaging domains after the corresponding bounded-overlap covering lemma is supplied. Convexity is not part of the algebraic rigidity; it is a geometric device for polynomial growth and boundary control.

If a line segment connects two points of a convex set, it remains inside the set. Hence one-dimensional restrictions of a polynomial can be used to compare a point deep in a small sublevel region with a point where the polynomial has reached the larger scale. Convexity also keeps the boundary created by translating or slightly dilating an averaging region under control. Balls and boxes are the main examples, but the theorem is stated for convex sets because the proof only needs these elementary geometric properties.

For an arbitrary highly disconnected averaging set, a polynomial could be sampled only on tiny neighborhoods of its zeros, making any occupation statement false. Some regularity of the averaging domains is therefore essential.

10. A model descent with two obstructions

Suppose two rational obstruction groups F and H occur with $\dim F < \dim H$. The correct logic is not “an F-trap is an H-trap.” Instead, choose a bounded exterior window for H. Outside a small neighborhood of the F-incidence, local uniqueness gives one unoriented H-representative and polynomial non-concentration controls its returns to the smaller vector tube. Inside the F-incidence there are two possibilities:

(1) the candidate limiting measure gives that incidence zero mass, so a closed neighborhood of it can be made negligible before passing to the limit;

(2) it carries positive mass, in which case F is the more economical obstruction and the analysis should be restarted with F.

This is the rigorous content behind the traditional phrase "induction on singular strata." The finite-window theorem itself needs no induction: it keeps the lower incidence visible instead of pretending that it has already been eliminated.

11. Why exact algebraic containment emerges from shrinking tubes

A bounded-degree polynomial cannot remain in every shrinking neighborhood of an algebraic set without satisfying its defining equations identically. Here is the elementary reason. Suppose an algebraic set $A \subset \mathbb{R}^N$ is cut out by polynomials

$$Q_1, \ldots, Q_r.$$

Let $P : \mathbb{R}^m \to \mathbb{R}^N$ be polynomial. Then each composition

$$Q_j \circ P$$

is a polynomial. If $P(\mathbb{R}^m)$ is not contained in A, at least one $Q_j \circ P$ is nonzero. On a fixed ball its supremum is positive, so its sublevel sets at height ε occupy a proportion tending to zero as $\varepsilon \downarrow 0$. Thus positive occupation at every sufficiently small algebraic scale can occur only if all the compositions vanish identically.

In the homogeneous problem the representative P may change by Γ as the quotient trajectory moves. Properness and lower-stratum removal are exactly what is needed before this elementary argument can be applied to one fixed polynomial representative.

12. Why persistence gives a whole normalizer orbit

The finite-window theorem first produces a representative whose vector curve remains in one bounded outer window. For one fixed trajectory, if this happens on arbitrarily large concentric balls, local finiteness at the center gives one single representative v on an unbounded subsequence. The map

$$t \longmapsto \rho_H(\Theta(t))v$$

is then a globally bounded vector-valued polynomial, hence constant. Writing $v = \rho_H(g\gamma)p_H$ yields

$$\rho_H\Big((\Theta(0)g\gamma)^{-1}\Theta(t)g\gamma\Big)\, p_H = p_H.$$

The stabilizer of p_H is exactly $N_G^1(H)$. Thus the entire trajectory lies in one determinant-one normalizer translate.

This argument is stronger and cleaner than merely observing that each point lies in the transporter $N(H, W)$. Transporter membership would allow the conjugate of H to vary with t; constancy of the exterior vector fixes the same copy of H for all parameters.

13. Linearization is local in the quotient but global in algebra

There is a useful tension in the proof. In $X = G/\Gamma$ one works locally: choose a compact incidence core, shrink neighborhoods, and control which lattice representative is visible. In V_H the conclusion is global: a polynomial identity that holds on an open parameter set holds everywhere, and the Zariski-closed obstruction is exact.

This local-to-global transition is one reason the method is so effective. Topology and properness reduce the quotient ambiguity to a controlled local problem; polynomial algebra then upgrades persistent local closeness to a global containment statement.

CHAPTER 7

Avoidance, Focusing, and Algebraic Trapping

The word *linearization* is sometimes used so broadly that two different conclusions are blurred together. We will keep them separate.

- **Avoidance:** away from an exact algebraic obstruction, long polynomial pieces spend little time near a proper singular stratum.
- **Focusing:** if a sequence nevertheless carries positive limiting mass near such a stratum, then suitable lattice representatives must remain in a bounded part of the linearization representation and converge toward an invariant algebraic configuration.

The first is a negative statement; the second extracts positive structure from failure of the negative statement.

1. From occupation estimates to weak limits

Fix one arithmetic rational obstruction subgroup H, one fixed adjoint-polynomial trajectory (Θ, x), and concentric parameter balls B_R. The finite-window theorem separates the regular part of the H-incidence from a finite list of lower ambiguities. Thus the weak-limit statement must explicitly account for those lower incidences.

Lemma 1.1 (Avoidance passes to a weak limit on a regular core)**.** *Let*

$$\mu_R = \frac{1}{|B_R|} \int_{B_R} \delta_{\Theta(t)x}\, dt \Longrightarrow \mu.$$

Let $C \subset N(H, W)$ be compact and let $Y_{<H}$ be the finite lower ambiguity set attached to the chosen compact exterior window. Assume $\mu(Y_{<H}) = 0$. If the exact H-normalizer trap of Theorem 3.1 does not occur, then

$$\mu(\pi(C)) = 0.$$

Proof. This is Proposition 10.2. The key point is the order of the limiting operations. First choose a closed neighborhood of the finite ambiguity set with arbitrarily small μ-mass. Portmanteau for closed sets then controls the limsup of its μ_R-mass. On the complement, the regular-window theorem gives the polynomial occupation estimate. A continuous bump function equal to one on $\pi(C)$ then contradicts positive μ-mass there. □

Warning 1.2. The theorem is not a uniform statement for an arbitrary sequence (Θ_n, x_n). For varying trajectories one must use the finite-window alternative itself, where all neighborhoods and polynomial constants are chosen before the individual orbit segment. This is exactly why the finite-window theorem is proved separately from its fixed-trajectory corollary.

2. Focusing in the exterior representation

Suppose now that positive occupation does persist. The representative properness theorem of Chapter 4 gives a compactness principle.

Proposition 2.1 (Compact focusing of representatives)**.** Derived here from stated inputs. *Let $K \subset X$ and $D \subset V_H \setminus \{0\}$ be compact. Suppose $y_n = g_n\Gamma \in K$ and, for each n, there exists $\gamma_n \in \Gamma$ such that*

$$v_n := \rho_H(g_n\gamma_n)p_H \in D.$$

Then after passing to a subsequence, v_n converges in V_H. Moreover, after choosing a compact set of representatives in the proper incidence space of Chapter 4, the corresponding classes of $g_n\gamma_n$ modulo $N_G^1(H) \cap \Gamma$ have a convergent subsequence.

Proof. Compactness of D gives a convergent subsequence $v_n \to v$. The stronger statement is exactly the consequence of properness of the incidence map

$$q_H : G/\Gamma_H \longrightarrow X \times V_H, \qquad q_H(g\Gamma_H) = (g\Gamma, \rho_H(g)p_H),$$

where Γ_H is the stabilizer of the vector-lattice data. Since (y_n, v_n) lies in the compact set $K \times D$, its inverse image under a proper map is compact. Hence the chosen classes have a convergent subsequence. $\square$

The proposition is elementary once properness is known, but conceptually it is the bridge from repeated nonlinear returns to a single limiting vector.

3. What “focusing” means

Consider a sequence of translates $g_nH\Gamma/\Gamma$. One should not expect g_n itself to converge: right multiplication by elements of Γ does not change the point in X, and multiplication by normalizer elements may not change the relevant homogeneous orbit. The correct compactness statement is therefore always *modulo the stabilizers already built into the problem.*

Definition 3.1 (Focusing along H)**.** A sequence $g_n \in G$ is said to *focus along H in the linearization representation* if there exist $\gamma_n \in \Gamma$ and a compact set $D \subset V_H \setminus \{0\}$ such that

$$\rho_H(g_n\gamma_n)p_H \in D$$

for all n, and some subsequential limit lies in the algebraic set $\mathcal{A}_H(W)$.

If $v = \lim \rho_H(g_n\gamma_n)p_H$ lies in $\mathrm{Fix}(W)$, then the limiting configuration records a subgroup containing W after conjugation. In this sense the sequence has not wandered randomly into the singular set: it has aligned itself with an algebraic obstruction.

4. Positive regular mass forces polynomial focusing

The finite-window theorem gives more than compactness at isolated return times. For a varying sequence it yields a polynomial limit satisfying the exact transporter equations, provided positive mass remains on the regular part after lower incidences are removed.

Proposition 4.1 (Positive regular mass forces focusing)**.** *Let the arithmetic hypotheses of Theorem 11.1 hold. Suppose a sequence of polynomial pieces* (Θ_n, x_n, B_n) *with* $\mathrm{rad}(B_n) \to \infty$ *has a uniform positive proportion of returns to every sufficiently small regular neighborhood of a compact core* $C \Subset N(H, W)$*, after deleting the finite lower-incidence neighborhoods. Then, after passing to a diagonal subsequence and recentering at return times which remain a fixed positive proportion of the radius away from the boundary, the corresponding exterior polynomial trajectories converge locally uniformly to a bounded-degree polynomial* P *with*

$$P(\mathbb{R}^m) \subset \mathcal{L}_W^{(d)}, \qquad P(0) \in \eta_H(C) \cup -\eta_H(C).$$

Proof. Choose nested regular quotient neighborhoods and nested exterior tubes as in Theorem 11.1. The assumed positive lower occupation supplies the constant c_0 in that theorem. Its proof first uses the finite-window alternative to force one persistent representative on each selected large ball, then chooses a regular return outside a thin boundary shell, recenters there, and applies finite-dimensional compactness of bounded-degree polynomials. Shrinking transverse widths force all wedge coordinates to vanish in the limit. These are exactly the asserted conclusions. $\square$

Warning 4.2. The conclusion is a statement about the closed linear containment locus $\mathcal{L}_W^{(d)}$. One must not silently strengthen it to $P(s) \in \rho_d(G)p_H$ for every s: the exterior G-orbit need not be closed. If a later application needs a limit inside a smaller algebraic orbit, the additional Pluecker, bracket, or orbit-closure equations must be carried through explicitly.

5. Focusing versus escape of mass

Focusing and non-divergence solve orthogonal problems.

Problem	Mechanism
The trajectory leaves every compact subset of X	quantitative/nonquantitative non-divergence and height functions
The trajectory stays in a compact region but approaches a proper homogeneous stratum	linearization and avoidance
A sequence repeatedly approaches that stratum	focusing of exterior representatives
A limiting measure is invariant and supported on the stratum	Ratner measure classification

Confusing the first two is particularly dangerous. A trajectory can remain in a fixed compact set while concentrating near a proper closed homogeneous orbit.

6. The role of normalizers

Even after H is identified, the translate is not uniquely parameterized by G/H. The normalizer records residual symmetries. If $n \in N_G(H)$, then

$$gnH(gn)^{-1} = gHg^{-1}.$$

Thus the subgroup itself is unchanged. The determinant-one normalizer is what appears naturally in the vector linearization; the full normalizer appears when one forgets the orientation/volume of p_H. Keeping these two quotients distinct prevents a number of silent finite-index and modular-function errors.

7. A practical diagnostic

When reading a modern proof that says “by linearization we obtain a contradiction”, ask the following four questions.

(1) Which subgroup H is being linearized, and what is the vector p_H?
(2) Which representatives $g\gamma p_H$ are being considered, and why are there only finitely many relevant ones locally?
(3) Which lower-dimensional singular strata have been removed?
(4) What exact polynomial non-concentration statement turns a long visit into algebraic containment?

If any of these is absent, the phrase “by linearization” is concealing a genuine argument.

CHAPTER 8

Internal Ratner Rigidity in the Arithmetic Linear Setting

This chapter closes the qualitative classification interface used by the rest of the series. The scope is deliberately the arithmetic linear setting in which the representative-properness and singular-set machinery of Chapters 5 and 7 has already been proved.

1. Standing hypotheses and the full measure stabilizer

Throughout this chapter, $\mathbf{G} \leq \mathrm{SL}_N$ is a connected $\mathbb{Q}$-algebraic group, $G = \mathbf{G}(\mathbb{R})^\circ$, and $\Gamma < G$ is an arithmetic lattice commensurable with $\mathbf{G}(\mathbb{Z}) \cap G$. Put $X = G/\Gamma$. Let

$$U = \{u_t = \exp(tZ) : t \in \mathbb{R}\}, \qquad \mathrm{ad}\, Z \text{ nilpotent},$$

and let μ be a U-invariant U-ergodic probability on X. Define

$$I(\mu) = \{g \in G : g_*\mu = \mu\}, \qquad L = I(\mu)^\circ, \qquad \mathfrak{l} = \mathrm{Lie}\, L.$$

The group $I(\mu)$ is closed: if $g_n \to g$ and $(g_n)_*\mu = \mu$, then $\int f(g_n x)\, d\mu \to \int f(gx)\, d\mu$ for every $f \in C_c(X)$. Since $U \subset I(\mu)$ and U is connected, $U \subset L$.

Lemma 1.1 (Orbit termination criterion)**.** *If $\mu(Lx) > 0$ for some $x \in X$, then*

$$\mu = m_{Lx},$$

where Lx is closed and $L \cap g\Gamma g^{-1}$ is a lattice in L for any lift $x = g\Gamma$.

Proof. The set Lx is U-invariant because $U < L$. Ergodicity gives $\mu(Lx) = 1$. The restriction of μ to Lx is L-invariant by the definition of L. The orbit map

$$L/(L \cap g\Gamma g^{-1}) \longrightarrow Lx, \qquad \ell(L \cap g\Gamma g^{-1}) \mapsto \ell x,$$

is a continuous bijection onto the support of a finite L-invariant measure. Haar measure on L therefore descends to a finite invariant measure on the quotient; equivalently the stabilizer is a lattice. The finite L-invariant measure on a transitive L-space is unique up to scale, and both measures have total mass one. Thus $\mu = m_{Lx}$. In particular the support is the whole orbit, hence the orbit is closed. $\square$

2. Transverse generic pairs

Fix a countable dense family $\mathcal{F} \subset C_c(X)$. Intersecting the Birkhoff generic sets for $\mathcal{F}$ and then applying Egorov on a compact exhaustion gives compact sets $K_j \uparrow X_0$ with $\mu(X_0) = 1$ such that for every $f \in \mathcal{F}$, every $0 < a < b < \infty$, and every $x \in K_j$,

$$\frac{1}{(b-a)T} \int_{aT}^{bT} f(u_t x)\, dt \longrightarrow \int_X f\, d\mu$$

uniformly in $x \in K_j$. The interval statement follows from the ordinary averages by

$$\frac{1}{(b-a)T} \int_{aT}^{bT} f(u_t x)\, dt = \frac{b}{b-a} A_{bT} f(x) - \frac{a}{b-a} A_{aT} f(x).$$

Lemma 2.1 (Transverse generic-pair alternative)**.** *Exactly one of the following holds.*

(1) $\mu = m_{Lx}$ for a closed finite-volume L-orbit;

(2) for every sufficiently large j there is a density point $x \in K_j$ and a sequence $x_n \in K_j \setminus Lx$ with $x_n \to x$.

Proof. If the second assertion fails on a positive-measure subset, then for some countable coordinate ball B and a positive-measure subset $E \subset B$ every point $x \in E$ has a neighborhood in which $\operatorname{supp} \mu$ is contained in the local L-orbit through x. Hence an L-orbit is relatively open in $\operatorname{supp} \mu$. Second countability permits only countably many pairwise disjoint relatively open L-orbits. Their union is U-invariant, so ergodicity makes one of them conull. Lemma 1.1 gives item (1). Conversely, under item (1), $\operatorname{supp} \mu = Lx$, so item (2) is impossible. □

Choose a compact set on which the quotient map has a uniform injectivity radius. For a pair from item (2), after discarding finitely many terms there is a unique $g_n \to e$ in a fixed identity neighborhood with

$$x_n = g_n x, \qquad g_n \notin L.$$

3. Relative polynomial normalization

Choose a vector-space complement $\mathfrak{g} = \mathfrak{l} \oplus V$ and let $q : \mathfrak{g} \to \mathfrak{g}/\mathfrak{l}$ be the quotient map. Since $U < L$, $\mathfrak{l}$ is invariant under $\operatorname{ad} Z$, so the nilpotent operator $\operatorname{ad} Z$ induces a nilpotent operator D on $\mathfrak{g}/\mathfrak{l}$. Write $g_n = \exp Y_n$ with $Y_n \to 0$, and put $\bar{Y}_n = q(Y_n) \neq 0$. Then

$$q(\operatorname{Ad}(u_t) Y_n) = e^{tD} \bar{Y}_n = \sum_{r=0}^{d} \frac{t^r}{r!} D^r \bar{Y}_n, \qquad D^{d+1} = 0.$$

Fix $0 < \rho \ll 1$, and let $T_n > 0$ be the first time at which the norm of this quotient polynomial reaches ρ:

$$\max_{0 \le t \le T_n} \|e^{tD}\bar{Y}_n\| = \rho.$$

Because $\bar{Y}_n \to 0$ and is nonzero, $T_n \to \infty$. On $[0, 1]$ define

$$Q_n(s) = \rho^{-1} e^{sT_nD}\bar{Y}_n.$$

Finite-dimensional polynomial compactness gives a subsequence $Q_n \to Q$ uniformly with all derivatives, where Q is a nonzero polynomial and $\max_{[0,1]} \|Q\| = 1$.

The quotient normalization does not by itself justify deleting a varying L-factor. The next lemma records the finite correction needed for the actual orbit comparison.

Lemma 3.1 (Finite normalizer correction)**.** *After passing to a further subsequence, there are a nonempty interval $J \Subset (0, 1)$, polynomial maps $c_n : J \to L$, and maps $r_n : J \to G$ such that*

$$u_{sT_n} g_n u_{-sT_n} = c_n(s) r_n(s),$$

with the following properties.

(1) $r_n \to r$ in $C^1(J, G)$ for a nonconstant map r;

(2) every correction factor used to construct c_n lies in $N_L(U)$, and hence

$$c_n(s) u_t c_n(s)^{-1} = u_{\alpha_n(s)t}$$

for a nonzero real algebraic scalar $\alpha_n(s)$;

(3) after finitely many corrections, either $r(s) \in I(\mu)$ for every $s \in J$, or all active quotient coefficients satisfy the exact rational exterior-incidence equations of a proper subgroup in $\mathscr{H}_\Gamma^{\mathrm{rat}}$.

Proof. Order the nonzero coefficients of $e^{tD}\bar{Y}_n$ by polynomial degree. A local slice for the multiplication map $L \times \exp(V) \to G$ expresses the conjugacy curve as an L-factor times a transverse factor as long as the latter remains in the fixed ρ-ball. If the leading L-factor normalizes U, replace t by αt using $cu_tc^{-1} = u_{\alpha t}$; the Jacobian of this affine change of time is the constant $|\alpha|$. Subtracting the corresponding leading coefficient lowers the active polynomial degree in the quotient. Repeating this operation terminates after at most d steps.

On a closed interval J avoiding the finitely many zeros of the leading coefficient, the remaining normalized curves have uniformly bounded coefficients. Hence they are precompact in C^1, giving item (1). For a regular terminal coefficient, choose $s_0 \in J$ and then a small fixed interval $J_0 \ni s_0$. The change-of-time scalars and the limiting map vary by $o_{J_0}(1)$. The block length is $|J_0|T_n \to \infty$. Uniform block genericity for x and x_n, followed by the comparison argument of Lemma 2.1, gives

$$r(s_0)_*\mu = \mu.$$

Since s_0 is arbitrary in the regular part and $I(\mu)$ is closed, $r(J) \subset I(\mu)$.

If no regular terminal coefficient survives, every active transverse coefficient has been absorbed by the normalizer corrections. In the exterior representation of Chapter 5, this says exactly that the polynomial wedge coordinates expressing $U \subset gHg^{-1}$ vanish identically. Representative properness and local finiteness reduce the possibilities on the fixed compact window to a finite rational list. A change of representative forces an intersection of two such rational subgroups; passing to the character-free connected core strictly lowers dimension. Thus the fully absorbed branch is precisely the proper rational singular alternative stated in item (3). $\square$

Remark 3.2. The proof is finite-dimensional. The two points at which one can easily make an invalid shortcut are now explicit: an arbitrary time-dependent L-factor cannot be discarded, and representative changes must be charged to a lower rational incidence rather than treated as harmless relabeling.

4. The regular branch contradicts maximal stabilizer

Lemma 4.1 (Transverse stabilizing curve is impossible)**.** *In the regular outcome of Lemma 3.1, the limit curve cannot have a nonzero tangent in $\mathfrak{g}/\mathfrak{l}$.*

Proof. We have $r(J) \subset I(\mu)$. Fix $s_0 \in J$. Since the identity component is open in the Lie group $I(\mu)$, for s sufficiently close to s_0,

$$r(s_0)^{-1}r(s) \in I(\mu)^\circ = L.$$

Therefore the left-translated derivative

$$Y = (dL_{r(s_0)^{-1}})_{r(s_0)}r'(s_0)$$

belongs to $\mathfrak{l}$. Its class in $\mathfrak{g}/\mathfrak{l}$ is zero. But the first-exit normalization and the finite correction stop precisely at the first nonzero transverse coefficient, so at some regular s_0 that quotient derivative is nonzero. This contradiction excludes the regular outcome. $\square$

5. Singular descent

Lemma 5.1 (Positive singular mass gives a proper homogeneous reduction)**.** *Suppose the fully absorbed branch of Lemma 3.1 occurs on a set of positive μ-measure. Then there are $H \in \mathscr{H}_\Gamma^{\mathrm{rat}}$ with $H \neq G$ and $g \in G$ such that*

$$U \subset gHg^{-1}, \qquad \mu(gH\Gamma/\Gamma) = 1.$$

Moreover $gH\Gamma/\Gamma$ carries finite H-volume.

PROOF. The rational obstruction family is countable by Proposition 1.3; hence positive mass of the singular union puts positive mass on one rational incidence $\pi(N(H,U))$. Choose H of minimal dimension among such incidences. The finite lower-incidence lemma 5.1 and Theorem 9.2 show that, away from lower-dimensional incidences of zero μ-mass, the relevant exterior representative is locally unique. Therefore the corresponding conjugate gHg^{-1} is locally constant on a positive-measure regular core. The U-saturation of this core stays in the same homogeneous incidence, so ergodicity gives full mass to that incidence.

The exterior representative is primitive integral in the fixed arithmetic model. Proposition 4.1 makes the orbit map proper on the regular representative class. Hence the resulting H-orbit is closed. The subgroup belongs to $\mathscr{H}_\Gamma^{\mathrm{rat}}$, so by definition $H \cap g^{-1}\Gamma g$ is a lattice and the orbit has finite volume. ☐

6. Arithmetic one-parameter measure classification

THEOREM 6.1 (Ratner measure classification in the AHD arithmetic scope). *Under the standing hypotheses of this chapter, if μ is a U-invariant U-ergodic probability on G/Γ, then there are a connected closed subgroup $H < G$ containing U and a point $x \in X$ such that Hx is closed of finite volume and*

$$\mu = m_{Hx}.$$

PROOF. Induct on $\dim G$. The assertion is trivial in dimension zero. Form the full stabilizer $L = I(\mu)^\circ$. If Lemma 1.1 applies, we are done. Otherwise Lemma 2.1 supplies transverse generic pairs. Lemma 3.1 gives either a regular terminal curve or a proper singular branch. The regular branch is impossible by Lemma 4.1. Hence Lemma 5.1 gives a proper closed finite-volume homogeneous subspace $Y = gH\Gamma/\Gamma$ of full μ-measure. Identify Y with $H/(H \cap g^{-1}\Gamma g)$; its dimension is strictly smaller than $\dim G$. The restricted U-action is again Ad-unipotent, and μ is still invariant and ergodic. The induction hypothesis classifies it on Y. Transporting the resulting subgroup back by g proves the assertion in G/Γ. ☐

7. Pointwise equidistribution and orbit closure

THEOREM 7.1 (Arithmetic one-parameter pointwise distribution). *Under the same arithmetic hypotheses, for every $x \in X$ there is a connected closed subgroup $H_x < G$ containing U such that $H_x x$ is closed of finite volume and*

$$\frac{1}{T}\int_0^T \delta_{u_t x}\,dt \implies m_{H_x x}.$$

Consequently

$$\overline{Ux} = H_x x.$$

PROOF. Let $\mu_T = T^{-1} \int_0^T \delta_{u_t x} \, dt$. Quantitative nondivergence from Volume 1, applied on a compact exhaustion, gives tightness. Any weak limit ν is U-invariant because for fixed s

$$\left| \int f \, d((u_s)_* \mu_T) - \int f \, d\mu_T \right| \leq \frac{2|s|}{T} \|f\|_\infty.$$

Disintegrate ν into U-ergodic components. By Theorem 6.1, almost every component is homogeneous.

We now descend through proper singular supports. If a subsequential limit assigns positive mass to a proper rational homogeneous incidence, choose one of minimal dimension. The weak-limit elimination theorem 10.2 removes its regular core unless the trajectory has the corresponding exact normalizer obstruction. In that exceptional case the trajectory is contained in a proper closed rational homogeneous subspace, so replace the ambient group by that subgroup. The dimension drops and the process terminates. At the terminal homogeneous space $H_x x$, every proper rational singular core has zero mass for every subsequential limit. Measure classification then forces every ergodic component of every limit to be the ambient Haar probability $m_{H_x x}$. Hence every subsequential limit equals $m_{H_x x}$, proving convergence of the full family.

Every μ_T is supported on $\overline{Ux}$, so $H_x x = \operatorname{supp} m_{H_x x} \subset \overline{Ux}$. The reverse inclusion follows from $U < H_x$. Thus $\overline{Ux} = H_x x$. □

Proof and provenance. Ratner's 1990–91 papers are the primary historical source for the general Lie-group theorem. The theorem proved above is the arithmetic linear scope used by the AHD series. Its proved input is this chapter: the logical inputs are the backward Birkhoff/genericity lemmas, polynomial shearing, the arithmetic exterior linearization and properness package, and singular induction. The literature citation is provenance and scope comparison, not a proof substitute.

CHAPTER 9

Linearization workshop: three calculations one should be able to do by hand

Linearization becomes convincing only after one has repeatedly watched a nonlinear statement about trajectories turn into a boundedness statement in a finite-dimensional representation. This chapter works through three models. None contains the full Dani–Margulis theorem, but together they isolate the calculations that repeatedly occur inside its proof and its descendants.

1. Conjugating a nearby point in SL_2

Let

$$u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad g_\varepsilon = \exp\begin{pmatrix} a & b \\ c & -a \end{pmatrix} \varepsilon + O(\varepsilon^2).$$

If x and $g_\varepsilon x$ are nearby, then after applying u_t their relative displacement is governed to first order by

$$\mathrm{Ad}(u_t) \begin{pmatrix} a & b \\ c & -a \end{pmatrix}.$$

Writing

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

one has

$$\mathrm{Ad}(u_t)E = E, \qquad \mathrm{Ad}(u_t)H = H - 2tE, \qquad \mathrm{Ad}(u_t)F = F + tH - t^2E.$$

Thus a transverse F component produces a quadratic shear. If the initial displacement is εF, the comparison time $t \asymp \varepsilon^{-1/2}$ makes the E component order one while the original displacement was microscopic.

Guiding Idea 1.1. The relevant time is not "large" in an absolute sense. It is chosen from the size and polynomial degree of the initial separation so that one coefficient becomes visible while the error terms remain controlled. This mesoscopic rescaling is the heart of polynomial divergence.

Suppose now that both x and $g_\varepsilon x$ are generic for the same invariant probability measure μ over a time window long compared with $\varepsilon^{-1/2}$. Comparing the two empirical averages after the above rescaling suggests

$$\int f(u_s x)\, d\mu(x) \approx \int f(\exp(cE)u_s x)\, d\mu(x),$$

for a nonzero c. Passing to a limit yields invariance under $\exp(\mathbb{R}E)$, or under another subgroup obtained from the dominant polynomial coefficient. The actual Ratner/Dani–Margulis argument spends much of its effort ensuring that the two points are simultaneously generic and that the dominant coefficient does not lie in an already-known invariance direction.

2. Linearizing a subgroup by an exterior vector

Let $G = \mathrm{SL}_3(\mathbb{R})$ and let H be the subgroup preserving the plane

$$P = \mathrm{span}(e_1, e_2).$$

The line $\mathbb{R}(e_1 \wedge e_2) \subset \wedge^2\mathbb{R}^3$ remembers P. Set

$$v_H = e_1 \wedge e_2.$$

For $g \in G$, the vector gv_H encodes the plane gP. In particular,

$$gHg^{-1} = \mathrm{Stab}_G(\mathbb{R}\, gv_H)$$

up to the distinction between stabilizing the line and fixing a chosen vector.

Take

$$g_s = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ s & 0 & 1 \end{pmatrix}.$$

Then

$$g_s v_H = (e_1 + se_3) \wedge e_2 = e_1 \wedge e_2 + s\, e_3 \wedge e_2.$$

Hence the distance from g_sP to P is measured linearly by the transverse Pluecker coordinate s. A geometric statement such as

$$g_sH\Gamma \text{ spends a long time near } H\Gamma$$

can therefore be converted into a statement that a family of group elements keeps g_sv_H near the line $\mathbb{R}v_H$ in a finite-dimensional representation.

Warning 2.1. The stabilizer of a vector and the stabilizer of its line are different. Exterior linearization naturally detects normalizers or projective stabilizers unless one augments the representation to kill the scalar character. Any proof that silently identifies these objects risks losing a finite character or a central factor.

A standard repair is to pass to a tensor representation in which the desired algebraic subgroup is the exact stabilizer of a vector. For reductive algebraic subgroups this is supplied abstractly by Chevalley-type representation theory; in examples one can usually write the tensor down explicitly.

3. A singular-set calculation in a product

Let

$$G = \mathrm{SL}_2(\mathbb{R}) \times \mathrm{SL}_2(\mathbb{R}), \qquad H = \{(h,h) : h \in \mathrm{SL}_2(\mathbb{R})\}.$$

The Lie algebra decomposes as

$$\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{r}, \qquad \mathfrak{h} = \{(X,X)\}, \quad \mathfrak{r} = \{(X,-X)\}.$$

A small displacement transverse to H has the form

$$\exp(\varepsilon(X,-X)).$$

Conjugation by the diagonal unipotent (u_t,u_t) gives

$$\mathrm{Ad}(u_t,u_t)(X,-X) = \bigl(\mathrm{Ad}(u_t)X, -\mathrm{Ad}(u_t)X\bigr).$$

Thus the same polynomial shearing seen in SL_2 acts on the transverse copy of the adjoint representation. If the lowest-weight component of X is nonzero, the separation develops a dominant quadratic term. If instead X lies in the E direction, the displacement commutes with the acting unipotent and no new shear appears. This is the simplest model of a singular direction.

The point generalizes. Relative to an acting unipotent U, a transverse displacement Z is useful when the polynomial map

$$t \longmapsto \mathrm{Ad}(u_t)Z$$

has a coefficient outside the known invariance algebra. The singular set consists of points for which all available coefficients remain trapped in an algebraic obstruction.

4. From bounded vectors to focusing

Suppose $\rho : G \to \mathrm{GL}(V)$ and $v \in V$ are chosen so that an algebraic subgroup L is the stabilizer of v. Let $\Omega \subset H$ be a compact neighborhood with nonempty interior. If a translating sequence g_i satisfies

$$\rho(g_i h)v \in K \subset V \qquad (h \in \Omega)$$

for a fixed compact K, then the sequence cannot diverge freely in every direction transverse to the stabilizer. In concrete matrix examples one can see this by decomposing V into weight spaces for a split torus: any component expanded by g_i must be absent from $\rho(h)v$ for all $h \in \Omega$, which is an algebraic condition on h and therefore on the subgroup generated by Ω.

This finite-dimensional statement is the beginning of focusing. The hard theorem identifies the precise intermediate subgroup and controls rationality, closedness of the corresponding orbit, and possible singular subsets. But the underlying diagnostic is elementary: **divergence plus bounded representation data forces algebraic vanishing.**

5. A practical proof checklist

When reading a linearization argument, it helps to annotate every occurrence of the following six objects.

(1) The subgroup H one is trying to detect.
(2) The representation ρ and vector or line v_H detecting H.
(3) The parameter set on which $\rho(gu_t)v_H$ is bounded or small.
(4) The polynomial sublevel estimate controlling the time spent in a thin neighborhood.
(5) The singular algebraic set corresponding to identically small representation coordinates.
(6) The induction parameter that decreases after the singular set is identified.

A proof is usually difficult not because any one of these ingredients is mysterious, but because they must be chosen compatibly. The right representation makes the singular set algebraic; the right polynomial estimate makes visits to that set quantitatively sparse; and the right induction parameter ensures that repeatedly encountering obstructions cannot continue forever.

Worked Example 5.1 (Why "near a closed orbit" is not itself a linear statement)**.** The distance condition $\mathrm{dist}(g\Gamma, L\Gamma) < \varepsilon$ depends on choices of representatives and can become ambiguous near the cusp. By contrast, if v_L is an integral tensor detecting L, the condition

$$\|\rho(g\gamma)v_L\| < C$$

for some $\gamma \in \Gamma$ is an algebraic inequality in a fixed vector space. Reduction theory is used to control how many γ can occur on a compact core. This explains why linearization proofs repeatedly alternate between geometry on G/Γ and arithmetic in a representation space.

6. The reusable picture

The calculations above can be compressed into one diagram:

$$\boxed{\text{nearby or nearly singular trajectories}} \longrightarrow \boxed{\text{polynomial relative motion}}$$

$$\Downarrow$$

$$\boxed{\text{bounded/small representation vector}} \longrightarrow \boxed{\text{algebraic stabilizer}}$$

$$\Downarrow$$

$$\boxed{\text{new invariance or focusing}}.$$

The rest of the subject consists largely of making each arrow uniform enough for the problem at hand.

CHAPTER 10

Groups Generated by Unipotents: What Must Be Proved

Ratner's theorem is usually first encountered for one Ad-unipotent one-parameter subgroup. Later arguments naturally involve much larger connected groups generated by such subgroups: semisimple groups generated by opposite root groups, unipotent radicals of parabolics, and mixed semidirect products. This chapter explains the problem and the proof strategy. The complete proof is given in Chapter 11.

1. Why repeated ergodic decomposition is not enough

Suppose $W = \langle U_1, \dots, U_r \rangle$ preserves an ergodic probability μ. One can decompose μ into U_1-ergodic components and apply Ratner to each component. It is tempting to pick a maximal homogeneous component type and claim that the other generators preserve it. There is a subtle defect in that argument.

If $v \in U_2$, a point vx obtained from a U_1-generic point x is naturally generic for the conjugated flow

$$vU_1v^{-1},$$

not necessarily for U_1. Thus the usual nearby-generic-points shearing lemma cannot simply be applied to x and vx as though they were generic for the same flow. Any proof that suppresses this conjugation issue is incomplete.

The clean route is different:

$$\boxed{W\text{-ergodic} \Longrightarrow \exists\, U < W \text{ Ad-unipotent one-parameter and } U\text{-ergodic}}$$

and then one application of Ratner classifies μ.

2. The structural decomposition

A connected group generated by Ad-unipotent one-parameter groups has a constrained Levi decomposition. Its semisimple quotient has no compact factors, because compact semisimple groups contain no nontrivial adjoint-unipotent elements. Its solvable radical is nilpotent modulo the central kernel of the adjoint representation; the central kernel is itself harmless for the argument because central one-parameter groups are Ad-unipotent.

Thus the relevant model is

$$W = S \ltimes N,$$

where S is semisimple without compact factors and N is connected nilpotent. If $S = S_1 \cdots S_k$ is the almost-direct product of its simple factors, choose one nontrivial root-unipotent one-parameter subgroup $U_i < S_i$ in each factor. Let

$$M = \langle U_1, \ldots, U_k, N \rangle.$$

The full proof shows that M is nilpotent in the adjoint sense relevant here. If a function is M-invariant, factorwise Mautner makes it invariant under every S_i, and it is already N-invariant. Hence it is W-invariant. Thus W-ergodicity implies M-ergodicity.

The key remaining fact is that an ergodic action of a connected nilpotent group has an ergodic one-parameter subgroup. Chapter 11 proves this by induction on the dimension of the nilpotent group, using the center and a two-parameter joint spectral measure.

3. The finite-measure theorem used in this series

THEOREM 3.1 (Generated-unipotent measure rigidity). *Let G be connected, $\Gamma < G$ a lattice, and let $W < G$ be connected and generated by Ad-unipotent one-parameter subgroups. If μ is a W-invariant, W-ergodic probability measure on G/Γ, then there are a connected closed subgroup $L < G$, with $W < L$, and a point $x \in G/\Gamma$ such that Lx is closed and*

$$\mu = m_{Lx}.$$

Proof and provenance. PROVED IN THIS TEXT. The proof occupies Chapter 11. Both the one-parameter classification theorem and the Mautner mechanism used there are proved internally in this volume. Shah's generated-unipotent theorem is cited for provenance and comparison, not used as a source theorem.

4. The final stabilizer step

There is a small but conceptually useful point after Ratner is applied. Suppose the ergodic one-parameter subgroup $U < W$ gives

$$\mu = m_{Hx}.$$

It does not follow merely from the notation that $W < H$. Instead define

$$\mathrm{Stab}_G(\mu) = \{g : g_*\mu = \mu\}.$$

This is a closed subgroup containing both H and W. Its identity component L therefore contains the connected groups H and W. Because L preserves the support of μ,

$$Lx \subset \operatorname{supp} \mu = Hx,$$

while $H < L$ gives the reverse inclusion. Hence $Lx = Hx$ and μ is the homogeneous L-measure. This little support argument is what turns one-parameter classification into classification for the whole generated group.

5. Why diagonal flows are different

For

$$A = \{\operatorname{diag}(e^t, e^{-t}) : t \in \mathbb{R}\} < \mathrm{SL}_2(\mathbb{R}),$$

no nontrivial element is Ad-unipotent. Nothing in the preceding mechanism applies to the *individual* diagonal flow. Its invariant measures and orbit closures can be far more complicated. On the other hand, the whole group $\mathrm{SL}_2(\mathbb{R})$ is generated by the two opposite unipotent subgroups

$$U^+ = \left\{ \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \right\}, \qquad U^- = \left\{ \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix} \right\}.$$

This contrast is a recurring lesson: rigidity depends on the acting object, not merely on the fact that semisimple elements happen to lie inside a group generated by unipotents.

6. Historical stronger forms

Shah proved substantially more general statements, including versions formulated through Zariski generation of the adjoint image and orbit-closure theorems. Those results are recorded in the bibliography for provenance. They are not used in this volume unless their proofs have been reconstructed. The logical theorem of this chapter is exactly Theorem 3.1.

CHAPTER 11

From One Unipotent Flow to a Group Generated by Unipotents

This chapter proves the finite-measure extension needed throughout the later series. The proof is shorter than the full Shah orbit-closure theorem because the acting measure is already assumed to be a probability measure. The decisive point is structural: ergodicity under a connected group generated by Ad-unipotent one-parameter subgroups forces ergodicity under at least one Ad-unipotent one-parameter subgroup. the internally proved one-parameter classification theorem can then be applied exactly once.

1. The theorem

THEOREM 1.1 (Generated-unipotent measure rigidity). *Let $G \leq \mathrm{SL}_N(\mathbb{R})$ be the connected real points of the fixed $\mathbb{Q}$-algebraic group used in this volume, let $\Gamma < G(\mathbb{Q})$ be arithmetic in the fixed integral model, and put $X = G/\Gamma$. Let $W < G$ be a connected closed subgroup generated by one-parameter subgroups whose adjoint actions are unipotent. If μ is a W-invariant, W-ergodic probability measure on X, then there exist a connected closed subgroup $L < G$ containing W and $x \in X$ such that Lx is closed and*

$$\mu = m_{Lx}.$$

The classification input in the proof is the internally proved arithmetic one-parameter theorem, Theorem 6.1. We first prove the ergodic-direction theorem.

2. A spectral lemma for abelian actions

Let $A \cong \mathbb{R}^k$ act unitarily on a separable Hilbert space $\mathcal{H}$. The joint spectral theorem gives a projection-valued measure E on $(\mathbb{R}^k)^*$ such that

$$\pi(a) = \int e^{i\langle \xi, a\rangle}\, dE(\xi).$$

We use only the following elementary consequence.

Lemma 2.1 (Generic direction for an abelian action)**.** *Suppose $\mathcal{H}^A = \{0\}$. Then for Lebesgue-almost every $v \in \mathbb{R}^k$,*

$$\mathcal{H}^{\{tv:t\in\mathbb{R}\}} = \{0\}.$$

Proof. The fixed subspace for the flow in direction v is the range of

$$E(v^\perp), \qquad v^\perp = \{\xi : \langle \xi, v\rangle = 0\}.$$

Choose a countable dense set h_j in $\mathcal{H}$, and let $\sigma_j(B) = \langle E(B)h_j, h_j\rangle$. Since $\mathcal{H}^A = 0$, $E(\{0\}) = 0$. For each $\xi \neq 0$, the set of v satisfying $\langle \xi, v\rangle = 0$ has Lebesgue measure zero. Tonelli's theorem gives, on every bounded set $V \subset \mathbb{R}^k$,

$$\int_V \sigma_j(v^\perp)\, dv = \int_{\xi\neq 0} |\{v \in V : \langle \xi, v\rangle = 0\}|\, d\sigma_j(\xi) = 0.$$

Hence $\sigma_j(v^\perp) = 0$ for almost every v, simultaneously for all j. Density of $\{h_j\}$ implies $E(v^\perp) = 0$. □

3. An ergodic direction in a nilpotent group

We now prove the nilpotent result used by Shah's structural argument. It is useful well beyond this theorem.

Theorem 3.1 (Ergodic one-parameter subgroup in a nilpotent action). *Let N be a connected nilpotent Lie group acting ergodically by measure-preserving transformations on a standard probability space (Y, ν). Then there exists $X \in \mathfrak{n}$ such that the one-parameter subgroup*

$$U_X = \{\exp(tX) : t \in \mathbb{R}\}$$

acts ergodically on (Y, ν).

Proof. Pass to the universal cover of N; this changes neither the action nor the available one-parameter subgroups. Thus we may assume N simply connected. We argue by induction on $\dim N$.

Let

$$\mathcal{H} = L^2_0(Y, \nu).$$

Ergodicity means $\mathcal{H}^N = 0$. If N is abelian, Lemma 2.1 gives the result.

Assume N is nonabelian. Its center has positive dimension; choose a nonzero central vector $Z \in \mathfrak{z}(\mathfrak{n})$, and put

$$Z_0 = \{\exp(tZ) : t \in \mathbb{R}\}.$$

Decompose

$$\mathcal{H} = \mathcal{H}^{Z_0} \oplus \mathcal{H}_1.$$

Because Z_0 is central, both summands are N-invariant. On $\mathcal{H}^{Z_0}$ the representation factors through the lower-dimensional nilpotent group N/Z_0. It has no

N/Z_0-fixed vectors: such a vector would be fixed by all of N, contradicting $\mathcal{H}^N = 0$. By induction there exists $X \in \mathfrak{n}$ whose image in $\mathfrak{n}/\mathbb{R}Z$ generates a flow with no fixed vectors on $\mathcal{H}^{Z_0}$.

It remains to perturb X by a multiple of Z so that no fixed vectors appear on $\mathcal{H}_1$. Since Z is central, the two one-parameter unitary groups

$$\pi(\exp(tX)), \qquad \pi(\exp(sZ))$$

commute. Their joint spectral theorem gives a projection-valued measure E on $\mathbb{R}^2$, with spectral variables (ξ, η). By definition of $\mathcal{H}_1$, the Z-flow has no fixed vector on $\mathcal{H}_1$, hence

$$E(\mathbb{R} \times \{0\}) = 0 \quad \text{on } \mathcal{H}_1.$$

The flow generated by $X + cZ$ has fixed space

$$E(L_c)\mathcal{H}_1, \qquad L_c = \{(\xi, \eta) : \xi + c\eta = 0\}.$$

For every point with $\eta \neq 0$, there is exactly one c for which it lies on L_c. Repeating the Tonelli argument from Lemma 2.1 for a countable dense set of vectors in $\mathcal{H}_1$, we find that

$$E(L_c) = 0$$

for Lebesgue-almost every c.

Choose such a c. On $\mathcal{H}^{Z_0}$, the central element Z acts trivially, so $X + cZ$ has the same fixed vectors as X, namely none. On $\mathcal{H}_1$ we just arranged that there are none. Thus the flow U_{X+cZ} is ergodic. □

Remark 3.2. The proof is worth remembering. It does not search for a dense one-parameter subgroup of N; such a subgroup usually does not exist in a nonabelian nilpotent group. Instead it searches for a direction whose *unitary representation* has no zero-frequency component. Central induction and a one-parameter spectral perturbation make this possible.

4. Groups generated by Ad-unipotents

We next isolate the Lie-theoretic structure needed for the ergodic-direction argument. It is cleaner to formulate the statement at the Lie-algebra level; this avoids irrelevant questions about whether a chosen Levi subgroup is closed.

Proposition 4.1 (Nilpotent radical and noncompact Levi part)**.** *Let $W < G$ be a connected Lie subgroup generated by one-parameter subgroups* $\exp(tX)$ *for which* $\operatorname{ad}_{\mathfrak{g}} X$ *is nilpotent on the ambient Lie algebra* $\mathfrak{g} = \operatorname{Lie}(G)$*. Put* $\mathfrak{w} = \operatorname{Lie}(W)$ *and let*

$$\mathfrak{w} = \mathfrak{s} \ltimes \mathfrak{r}$$

be a Levi decomposition of its Lie algebra. Then:

(1) *the solvable radical* $\mathfrak{r}$ *is nilpotent;*

(2) *the semisimple algebra* $\mathfrak{s}$ *has no compact simple ideal;*

(3) *if* $\mathfrak{s} = \mathfrak{s}_1 \oplus \cdots \oplus \mathfrak{s}_k$ *is its decomposition into simple ideals, one may choose a nonzero real-root nilpotent* $E_i \in \mathfrak{s}_i$ *for each* i *so that, with*

$$\mathfrak{a} = \mathbb{R}E_1 \oplus \cdots \oplus \mathbb{R}E_k, \qquad \mathfrak{m} = \mathfrak{a} \ltimes \mathfrak{r},$$

the Lie algebra $\mathfrak{m}$ *is nilpotent and every* $\operatorname{ad}_{\mathfrak{g}} Z$*,* $Z \in \mathfrak{m}$*, is nilpotent on the ambient* $\mathfrak{g}$*.*

Proof. We divide the proof into three elementary algebraic steps.

Step 1: the radical is nilpotent. Consider the connected real algebraic group

$$\mathbf{A} = \overline{\operatorname{Ad}_G(W)}^{\mathrm{Zar}} < \mathrm{GL}(\mathfrak{g}).$$

It is Zariski generated by the one-parameter unipotent subgroups $\operatorname{Ad}_G(\exp(tX)) = \exp(t \operatorname{ad}_{\mathfrak{g}} X)$ coming from the stated generators of W. Let $R_u(\mathbf{A})$ be the unipotent radical. The reductive quotient $\mathbf{A}/R_u(\mathbf{A})$ is again generated by unipotent elements. Its algebraic abelianization is a torus. Every algebraic homomorphism from a unipotent one-parameter group $\mathbb{G}_a$ to a torus is trivial, so all the generating unipotent subgroups map trivially to that abelianization. Hence the connected center of the reductive quotient is trivial. It follows that the connected solvable radical of $\mathbf{A}$ is precisely $R_u(\mathbf{A})$.

The Zariski closure of $\operatorname{Ad}_G(R)$, where R is the connected radical of W, is a connected solvable normal algebraic subgroup of $\mathbf{A}$; hence it is contained in $R_u(\mathbf{A})$. Therefore for every $Y \in \mathfrak{r}$,

$$\exp(t \operatorname{ad}_{\mathfrak{g}} Y) = \operatorname{Ad}_G(\exp(tY))$$

is unipotent on $\mathfrak{g}$. Thus $\operatorname{ad}_{\mathfrak{g}} Y$ is nilpotent on $\mathfrak{g}$, and in particular its restriction to the invariant subspace $\mathfrak{r}$ is nilpotent. Engel's theorem now implies that $\mathfrak{r}$ is a nilpotent Lie algebra.

Step 2: no compact simple ideal survives. Suppose $\mathfrak{k}$ were a compact simple ideal of the semisimple quotient $\mathfrak{w}/\mathfrak{r}$. Project every generating vector X to $\mathfrak{k}$. Since the restriction of $\operatorname{ad}_{\mathfrak{g}} X$ to the invariant subspace $\mathfrak{w}$ is nilpotent, and nilpotence passes to quotients, $\operatorname{ad}(X_{\mathfrak{k}})$ is nilpotent on $\mathfrak{k}$. On a compact semisimple Lie algebra, however, $\operatorname{ad} Z$ is skew-adjoint for the positive definite form $-B$, where B is the Killing form. A skew-adjoint nilpotent operator is zero, and the center of $\mathfrak{k}$ is zero; hence $X_{\mathfrak{k}} = 0$. All generators therefore have trivial projection to $\mathfrak{k}$, contradicting generation of W. So there is no compact simple ideal.

Step 3: construct a nilpotent Mautner subgroup. Every noncompact real simple algebra $\mathfrak{s}_i$ has a nonzero real root. Choose a corresponding root vector E_i. In every finite-dimensional representation of $\mathfrak{s}_i$, a root vector acts nilpotently;

applying this to the restriction of the ambient adjoint representation to $\mathfrak{s}_i$ shows that $\operatorname{ad}_{\mathfrak{g}} E_i$ is nilpotent on all of $\mathfrak{g}$. Vectors belonging to different simple ideals commute, so the one-parameter groups

$$U_i = \overline{\operatorname{Ad}_G(\exp(\mathbb{R}E_i))}^{\mathrm{Zar}}$$

are commuting one-dimensional unipotent algebraic subgroups of $\mathbf{A}$.

Let $\mathbf{U}$ be the connected algebraic subgroup generated by $R_u(\mathbf{A})$ and the U_i. The unipotent radical is normal, and the image of $\mathbf{U}$ in the reductive quotient $\mathbf{A}/R_u(\mathbf{A})$ is a product of commuting one-dimensional unipotent groups. Hence that image is unipotent. An extension of a connected unipotent algebraic group by a connected unipotent algebraic group is unipotent: after complexification, Lie–Kolchin triangularizes the connected solvable extension, and the absence of a torus quotient forces all diagonal characters to be one. Thus $\mathbf{U}$ is conjugate to a subgroup of the upper unitriangular group.

By Step 1 the Zariski closure of $\operatorname{Ad}_G(R)$ is contained in $R_u(\mathbf{A})$. Therefore the connected analytic subgroup M with Lie algebra

$$\mathfrak{m} = \mathfrak{a} \ltimes \mathfrak{r}, \qquad \mathfrak{a} = \mathbb{R}E_1 \oplus \cdots \oplus \mathbb{R}E_k,$$

satisfies

$$\operatorname{Ad}_G(M) \subset \mathbf{U}.$$

Every element of a unipotent linear algebraic group is unipotent. Hence for every $Z \in \mathfrak{m}$ the one-parameter group $\exp(t \operatorname{ad}_{\mathfrak{g}} Z) = \operatorname{Ad}_G(\exp(tZ))$ is unipotent on $\mathfrak{g}$; equivalently, $\operatorname{ad}_{\mathfrak{g}} Z$ is nilpotent on $\mathfrak{g}$. Restricting to the invariant subspace $\mathfrak{m}$ and applying Engel's theorem proves that $\mathfrak{m}$ is nilpotent. This proves item (3). □

Remark 4.2. The algebraic-hull argument in Step 1 is only a device for proving a Lie algebra statement. It does not impose arithmeticity or algebraicity on the ambient homogeneous space. Its content is the elementary principle that a connected linear group generated by unipotents has unipotent solvable radical.

5. The Mautner mechanism and an ergodic nilpotent subgroup

The representation-theoretic step is load-bearing, so it is proved here.

Lemma 5.1 (Mautner contraction lemma)**.** *Let $\pi : G \to \mathcal{U}(\mathcal{H})$ be continuous and unitary. If v is fixed by $A = \{a_t\}$ and $a_{-t}ha_t \to e$ as $t \to +\infty$, then $\pi(h)v = v$.*

Proof. Since $\pi(a_t)v = v$,

$$\|\pi(h)v - v\| = \|\pi(a_{-t}ha_t)v - v\| \longrightarrow 0.$$

□

Lemma 5.2 (Rank-one unipotent Mautner calculation)**.** *Let* $U^+ = \{u_r = \left(\begin{smallmatrix}1 & r\\ 0 & 1\end{smallmatrix}\right)\}$*. If* v *is* U^+*-fixed in a continuous unitary representation of* $\mathrm{SL}_2(\mathbb{R})$*, then* v *is* $\mathrm{SL}_2(\mathbb{R})$*-fixed.*

Proof. Put

$$v_\varepsilon = \begin{pmatrix}1 & 0\\ \varepsilon & 1\end{pmatrix}, \qquad d_t = \begin{pmatrix}e^t & 0\\ 0 & e^{-t}\end{pmatrix}, \qquad p(g) = \langle \pi(g)v, v\rangle.$$

The coefficient p is bi-U^+-invariant. Direct multiplication gives

$$u_r v_\varepsilon u_s = \begin{pmatrix}1 + r\varepsilon & r + s + rs\varepsilon\\ \varepsilon & 1 + s\varepsilon\end{pmatrix}.$$

For fixed t choose

$$r = (e^t - 1)/\varepsilon, \qquad s = -(e^t - 1)/(e^t\varepsilon).$$

Then

$$u_r v_\varepsilon u_s = \begin{pmatrix}e^t & 0\\ \varepsilon & e^{-t}\end{pmatrix} \to d_t.$$

Hence $p(d_t) = \lim_{\varepsilon\to 0} p(v_\varepsilon) = p(e) = \|v\|^2$, and

$$\|\pi(d_t)v - v\|^2 = 2\|v\|^2 - 2\Re p(d_t) = 0.$$

Thus v is fixed by every d_t. For $U^- = \{v_s : s \in \mathbb{R}\}$ one has $d_t v_s d_{-t} = v_{e^{-2t}s} \to e$, so Lemma 5.1 gives U^--invariance. The elementary matrices U^+ and U^- generate $\mathrm{SL}_2(\mathbb{R})$ by Gaussian elimination, so v is fixed by the whole group. ☐

Theorem 5.3 (Factorwise root Mautner phenomenon; internal form). *Let* S *be a connected noncompact real almost-simple Lie group with finite center. Fix a maximal split torus with restricted-root decomposition*

$$\mathfrak{s} = \mathfrak{s}_0 \oplus \bigoplus_{\alpha\in\Sigma} \mathfrak{s}_\alpha.$$

Let $0 \neq X \in \mathfrak{s}_\alpha$ *and* $U_X = \{\exp(tX) : t \in \mathbb{R}\}$*. In every continuous unitary representation of* S*, every* U_X*-fixed vector is* S*-fixed.*

Proof. Choose a Cartan involution θ adapted to the split torus and put $Y = -\theta X$. Invariance of the Killing form B gives, for H_0 in the split Cartan,

$$B([X,Y], H_0) = -B(X, [\theta X, H_0]) = \alpha(H_0)\bigl(-B(X, \theta X)\bigr).$$

The last scalar is positive. If M centralizes the split Cartan in the compact algebra, then $\operatorname{ad} M$ is skew-adjoint for $B_\theta(V,W) = -B(V, \theta W)$, whence $B(M, [X,Y]) = 0$. Thus $[X,Y]$ lies in the split Cartan and is a positive multiple of the coroot dual to α. Rescale X, Y so that, with $H = [X,Y]$,

$$[H,X] = 2X, \qquad [H,Y] = -2Y.$$

These are the $\mathfrak{sl}_2$ relations. Pull the representation back to the integrated rank-one subgroup. Lemma 5.2 makes the U_X-fixed vector v fixed by $A_X = \{\exp(tH) : t \in \mathbb{R}\}$.

Decompose

$$\mathfrak{s} = \bigoplus_{\lambda \in \mathbb{R}} \mathfrak{s}_\lambda(H), \qquad [H, Z] = \lambda Z.$$

For $\lambda \neq 0$,

$$\exp(-tH) \exp(sZ) \exp(tH) = \exp(e^{-\lambda t} sZ) \to e$$

in one of the two time directions, so Lemma 5.1 makes v fixed by every $\exp(\mathbb{R}Z)$ with $\lambda \neq 0$.

Let $\mathfrak{q}$ be the Lie subalgebra generated by the nonzero $\operatorname{ad} H$-eigenspaces. It contains X, hence is nonzero. Also $[\mathfrak{s}_0(H), \mathfrak{s}_\lambda(H)] \subseteq \mathfrak{s}_\lambda(H)$, while bracketing with a nonzero eigenspace preserves the algebra generated by those eigenspaces. Therefore $\mathfrak{q}$ is an ideal of $\mathfrak{s}$. Almost simplicity forces $\mathfrak{q} = \mathfrak{s}$. The connected subgroup generated by the one-parameter groups already shown to fix v has full Lie algebra and hence equals S. ☐

> **Proof and provenance.** The displayed proof owns the Mautner implication used here: the rank-one matrix-coefficient calculation, diagonal contraction, and Lie-algebra generation are explicit. Literature citations are provenance only.

Theorem 5.4 (Ergodic Ad-unipotent direction). *Let W be as in Proposition 4.1 and suppose W acts ergodically on a probability space (Y, ν). Then W contains an Ad-unipotent one-parameter subgroup that acts ergodically on (Y, ν).*

Proof. Let

$$\mathfrak{w} = \mathfrak{s} \ltimes \mathfrak{r}, \qquad \mathfrak{s} = \mathfrak{s}_1 \oplus \cdots \oplus \mathfrak{s}_k,$$

and choose the root vectors E_i supplied by Proposition 4.1. Let A be the connected analytic subgroup with Lie algebra $\mathfrak{a} = \bigoplus_i \mathbb{R}E_i$, let N be the connected radical of W, and let M be the connected analytic subgroup with Lie algebra $\mathfrak{m} = \mathfrak{a} \ltimes \mathfrak{r}$. By the proposition, M is nilpotent and in fact every one-parameter subgroup of M is Ad_G-unipotent.

We claim that M acts ergodically. Let $f \in L^2(Y, \nu)$ be M-invariant. Then f is fixed by every one-parameter subgroup $U_i = \exp(\mathbb{R}E_i)$. For each simple ideal $\mathfrak{s}_i$, let $\widetilde{S}_i$ be the simply connected Lie group integrating it. The inclusion $\mathfrak{s}_i \hookrightarrow \mathfrak{w}$ integrates to a homomorphism $\widetilde{S}_i \to W$, so the Koopman representation restricts to a continuous unitary representation of $\widetilde{S}_i$. Apply Theorem 5.3 factor by factor. Since f is fixed by U_i, it is fixed by the image of every $\widetilde{S}_i$, hence by the connected analytic semisimple subgroup with Lie algebra $\mathfrak{s}$. It is also fixed by N because $N < M$. The connected analytic subgroups with Lie algebras $\mathfrak{s}$ and $\mathfrak{r}$ generate the connected group W, so f is W-invariant. Ergodicity of the W-action therefore forces f to be constant. Thus the M-action is ergodic.

Apply Theorem 3.1 to M. There exists $X \in \mathfrak{m}$ such that $\exp(tX)$ acts ergodically. Finally, item (3) of Proposition 4.1 says that $\mathrm{ad}_{\mathfrak{g}} X$ is nilpotent on the ambient $\mathfrak{g}$. Hence $\mathrm{Ad}_G(\exp(tX))$ is unipotent, so this is an Ad_G-unipotent one-parameter subgroup in precisely the sense required by Ratner's theorem. □

6. Proof of generated-unipotent measure rigidity

Proof of Theorem 1.1. Apply Theorem 5.4 to the action of W on (X, μ). We obtain an Ad-unipotent one-parameter subgroup

$$U = \{u_t\} < W$$

whose action is μ-ergodic.

Now apply Theorem 6.1 to the arithmetic quotient X. There exist a connected closed subgroup $H < G$ containing U and a point $x \in X$ such that Hx is closed of finite volume and

$$\mu = m_{Hx}.$$

At this point the internal one-parameter theorem has done exactly one job: it has classified μ using the ergodic subgroup U. We still have to incorporate the whole group W.

Define the measure stabilizer

$$\mathrm{Stab}_G(\mu) = \{g \in G : g_*\mu = \mu\}.$$

It is closed: if $g_n \to g$ and $(g_n)_*\mu = \mu$, then for every $f \in C_c(X)$, dominated convergence gives

$$\int f(g_n y)\, d\mu(y) \longrightarrow \int f(gy)\, d\mu(y),$$

so $g_*\mu = \mu$. The group $\mathrm{Stab}_G(\mu)$ contains both H and W. Because H and W are connected and contain the identity, they lie in the identity component

$$L = \mathrm{Stab}_G(\mu)^\circ.$$

Thus L is connected, closed, and contains W.

The support of μ is Hx. Every $\ell \in L$ preserves μ, hence preserves its support. Therefore

$$Lx \subset \mathrm{supp}\,\mu = Hx.$$

On the other hand $H < L$, so $Hx \subset Lx$. Thus

$$Lx = Hx.$$

Since μ is L-invariant by definition of L, it is the homogeneous probability measure on the closed orbit Lx. This proves the theorem. □

Remark 6.1 (Why the proof is conceptually cleaner than a component induction)**.** A tempting proof begins by decomposing μ under one chosen generator, selecting a maximal homogeneous type, and attempting to prove that all other generators preserve that type. That argument is delicate because a nearby point obtained by another generator is generic for a *conjugate* one-parameter flow, not automatically for the original one. The ergodic-direction theorem avoids this trap completely. It finds one unipotent flow already ergodic for μ, so the internal one-parameter theorem classifies the whole measure at once.

7. A useful corollary for semisimple actions

Corollary 7.1. *Let W be connected semisimple without compact factors and generated by its root-unipotent one-parameter subgroups. Every finite W-invariant W-ergodic measure on G/Γ is homogeneous on a closed orbit of a connected subgroup containing W.*

PROOF. The hypotheses of Theorem 1.1 hold. □

8. Scope boundary: the topological orbit-closure theorem

The full topological theorem asserting that $\overline{Wx}$ is homogeneous for every x requires an additional argument proving that the minimal closed subgroup containing W with closed orbit has finite covolume. Shah proves this using a Mautner-property reduction and a finiteness property for locally finite invariant measures. That theorem is important, but it is not a logical input to the later measure arguments for which this volume is foundational. To preserve the proof-complete rule, we do not place the stronger orbit-closure statement in a theorem box here. It will be reconstructed if and when a later volume needs it.

Proof and provenance. The structural route follows ideas appearing in Shah's thesis: a group generated by unipotents has semisimple-without-compact-factors Levi part and unipotent/nilpotent radical; Mautner reduces ergodicity to a nilpotent subgroup; an ergodic nilpotent action contains an ergodic one-parameter subgroup. The proof of the nilpotent ergodic-direction theorem and the final measure-stabilizer argument are included here explicitly. The citation is attribution, not a logical substitute.

CHAPTER 12

Measure Decomposition and the Division of Labor with Ratner

The linearization method and Ratner's measure theorem solve different problems. Ratner classifies an invariant ergodic measure after the relevant unipotent invariance has been found. Linearization controls whether an approximating trajectory can deposit mass on a proposed proper homogeneous component. The purpose of this chapter is to make that division of labor logically exact.

1. The internally proved one-parameter classification theorem

Chapter 8 proves the theorem required here in the fixed arithmetic linear scope.

THEOREM 1.1 (Arithmetic one-parameter measure classification). *Let $G \leq \mathrm{SL}_N(\mathbb{R})$ be the connected real points of the fixed $\mathbb{Q}$-algebraic group, let $\Gamma < G(\mathbb{Q})$ be arithmetic in the chosen integral model, let $U = \{u_t\}$ be Ad-unipotent, and let μ be a U-invariant U-ergodic probability on G/Γ. Then there are a connected closed $L < G$, $U < L$, and $x \in G/\Gamma$ such that Lx is closed of finite L-volume and*

$$\mu = m_{Lx}.$$

PROOF. This is Theorem 6.1. Its proof is internal to this volume: polynomial first-exit normalization produces the regular additional-invariance branch, arithmetic exterior linearization produces the singular branch, and induction on the dimension of the rational homogeneous envelope terminates. $\square$

> **Proof and provenance.** Ratner's papers are the historical source of the theorem, but the arithmetic linear statement used by the AHD dependency graph is proved in Chapter 8; no theorem-level Ratner dependency remains here.

2. Groups generated by unipotents

Chapter 11 proves the following extension from the one-parameter theorem rather than citing Shah.

THEOREM 2.1 (Generated-unipotent ergodic measures). *Under the arithmetic standing hypotheses of Theorem 1.1, let $W < G$ be connected and generated by one-parameter Ad-unipotent subgroups. Every W-invariant, W-ergodic probability measure on G/Γ is homogeneous: there are a connected closed subgroup $L < G$ containing W and a closed finite-volume orbit Lx such that $\mu = m_{Lx}$.*

PROOF. The proof is Theorem 1.1. Its internal steps are worth recalling because they explain why no extra classification theorem is hidden here. First one proves the Lie-algebra structure of a connected group generated by Ad-unipotents. Factorwise Mautner reduces ergodicity to a nilpotent subgroup. A spectral/central-series argument then finds one Ad-unipotent one-parameter subgroup which is already ergodic for μ. The internally proved one-parameter theorem classifies μ, and the connected stabilizer of μ enlarges the resulting homogeneous group to contain all of W. □

3. Ergodic decomposition: integral, not countable sum

Let μ be a finite W-invariant probability measure. The ergodic decomposition theorem gives a probability space (Ω, λ) and W-ergodic probabilities μ_ω such that

$$\mu = \int_\Omega \mu_\omega \, d\lambda(\omega).$$

By Theorem 2.1, almost every μ_ω is homogeneous.

This formulation is deliberately different from the countable-sum "stratification" used in some expositions. A countable parametrization of all relevant subgroup types is a separate structural theorem in general finite-volume quotients. It is not needed for the componentwise logic of this volume, and therefore it is not assumed.

Why this matters. The integral formulation is enough for almost every measure-theoretic use. If a weak limit is not Haar, one studies a positive set of non-Haar ergodic components. Linearization is then applied to a compact core inside one candidate component, provided the corresponding obstruction lies in the arithmetic rational family for which representative properness has been proved.

4. Rational components in an arithmetic quotient

Assume now the arithmetic standing hypotheses of Chapter 5. Let m_{Lx} be a homogeneous component. When the subgroup describing the component can be represented by a connected rational subgroup H in the fixed arithmetic model, the full finite-window machinery of this volume applies to that component.

The point is local. Choose a compact set $C \subset N(H, W)$ whose image has positive m_{Lx}-measure and lies in a regular part of the incidence. The bounded exterior window determines finitely many lower ambiguity loci $Y_{<H}$. If these are proper analytic incidences inside Lx, then Lemma 10.1 gives

$$m_{Lx}(Y_{<H}) = 0.$$

Proposition 10.2 then says that a fixed polynomial trajectory can charge C in the limit only if it develops the exact H-normalizer obstruction.

Proposition 4.1 (Componentwise linearization contradiction)**.** *Let*

$$\mu_R = \frac{1}{|B_R|} \int_{B_R} \delta_{\Theta(t)x}\, dt \Longrightarrow \mu$$

for a fixed arithmetic polynomial trajectory. Suppose m_{Lz} occurs as a homogeneous ergodic component relevant to a positive-mass part of μ, and suppose a compact rational H-incidence core C in that component has positive m_{Lz}-mass. Assume every lower ambiguity locus in the chosen bounded exterior window is proper inside Lz. Then either the trajectory has an exact determinant-one H-normalizer obstruction or it cannot deposit positive limiting mass on $\pi(C)$.

Proof. The proper lower incidences have m_{Lz}-measure zero by Lemma 10.1. The regular-window theorem therefore applies on the complement of neighborhoods whose component measure can be made arbitrarily small. The fixed-trajectory weak-limit argument of Proposition 10.2 gives zero mass to $\pi(C)$ unless persistence occurs. Persistence is converted to exact $N_G^1(H)$-trapping by Theorem 9.2. □

5. Why rationality is not silently automatic

A closed homogeneous orbit in an arithmetic quotient is often described by rational algebraic data, but proving the exact rationality statement needed in a given generality can require Borel density, reduction theory, or arithmeticity input. Those theories are developed later in the reconstructed project. This volume therefore uses the following discipline:

- if an application supplies a rational obstruction subgroup explicitly, the arithmetic linearization theorem here applies immediately;

- if rationality of a Ratner subgroup is itself a theorem, that theorem must be proved before the present arithmetic representative argument is invoked;
- no statement of the form "Ratner subgroup, therefore rational" is allowed without such a proof.

This separation prevents arithmeticity from being hidden inside a dynamical argument.

6. Portmanteau: the direction of the inequalities

Weak convergence $\mu_n \Rightarrow \mu$ gives

$$\limsup_n \mu_n(F) \le \mu(F) \quad \text{for closed } F,$$

and

$$\mu(O) \le \liminf_n \mu_n(O) \quad \text{for open } O.$$

The first inequality is the one used to remove a small closed neighborhood of the lower ambiguity set. An eventual upper bound on an open set does *not* follow from Portmanteau by itself; instead one inserts a compactly supported continuous bump function, exactly as in Proposition 10.2.

7. The theorem map

Statement	Logical status
One-parameter unipotent measure classification	Ratner-foundation import
Generated-unipotent ergodic measure theorem	proved in this volume
Exterior representative properness for rational arithmetic strata	proved in this volume
Finite-window linearization and regular avoidance	proved in this volume
Weak-limit elimination of a rational regular core	proved in this volume
General nonarithmetic representative finiteness	not used here; requires a separate proof
General arithmetic rationality of every Ratner subgroup	not assumed; must be proved before use

The resulting logic is deliberately modular:

$$\boxed{\begin{aligned} \text{additional invariance} &\longrightarrow \text{Ratner homogeneous component} \\ &\longrightarrow \text{verified rational incidence, when available} \\ &\longrightarrow \text{finite-window linearization} \\ &\longrightarrow \text{avoidance or exact focusing.} \end{aligned}}$$

CHAPTER 13

Low-Rank Models of Linearization

Abstract exterior powers become much easier to remember after one has computed them in small groups. This chapter works through three models. The goal is not to prove new classification theorems, but to make the objects p_H, $N(H,W)$, $N_G^1(H)$, and the polynomial shear visible by hand.

1. The upper horocycle in $\mathrm{SL}_2(\mathbb{R})$

Let

$$G = \mathrm{SL}_2(\mathbb{R}), \qquad U = \left\{ u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} : t \in \mathbb{R} \right\}.$$

Write

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Take $H = W = U$. Then $\mathfrak{h} = \mathbb{R}e$ and

$$V_H = \bigwedge^1 \mathfrak{g} = \mathfrak{sl}_2(\mathbb{R}), \qquad p_H = e.$$

Proposition 1.1. *For $H = U$,*

$$N_G(U) = B = \left\{ \begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix} : a \neq 0 \right\},$$

and

$$N_G^1(U) = U \cdot \{\pm I\}.$$

Proof. The subgroup U fixes the line $\mathbb{R}(1,0)^T$ and is the unipotent radical of its stabilizer. If $gUg^{-1} = U$, then g must preserve this unique common fixed line, so g is upper triangular. Conversely upper triangular matrices normalize U.

For

$$a_s = \mathrm{diag}(e^s, e^{-s}),$$

we have

$$\mathrm{Ad}(a_s)e = e^{2s}e.$$

The determinant of $\mathrm{Ad}(a_s)$ on the one-dimensional $\mathfrak{u}$ is therefore e^{2s}. It equals 1 only for $s = 0$. The unipotent subgroup itself fixes e, and $-I$ acts trivially in the adjoint representation. Hence the determinant-one normalizer is $U\{\pm I\}$. $\square$

Thus the vector linearization is especially concrete:

$$\eta_H(g) = \mathrm{Ad}(g)e.$$

The condition $g \in N(U, U)$ is exactly that $\mathrm{Ad}(g)e$ span the U-fixed line $\mathbb{R}e$; the determinant-one condition is the stronger equation $\mathrm{Ad}(g)e = e$ up to the central sign already invisible to Ad.

2. Seeing the shear

For

$$v = ae + bh + cf,$$

Chapter 1 gave

$$\mathrm{Ad}(u_t)v = (a - 2bt - ct^2)e + (b + ct)h + cf.$$

Suppose $c \neq 0$. The quadratic term eventually dominates in the e-direction. If $c = 0$ but $b \neq 0$, the linear term dominates. Only when $v \in \mathbb{R}e$ is the vector fixed by U.

This is the simplest possible version of the linearization dichotomy:

$$\boxed{\begin{array}{c} v \notin \mathrm{Fix}(U) \\ \Longrightarrow \\ \mathrm{Ad}(u_t)v \text{ escapes polynomially from a neighborhood of } \mathrm{Fix}(U). \end{array}}$$

If an orbit remains near the U-singular geometry for long times, its relevant exterior representative must therefore be very close to the fixed line, and in a limiting argument it becomes fixed.

3. A block subgroup in $\mathrm{SL}_3(\mathbb{R})$

Let

$$H = \left\{ \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix} : A \in \mathrm{SL}_2(\mathbb{R}) \right\} < \mathrm{SL}_3(\mathbb{R}).$$

Its Lie algebra is

$$\mathfrak{h} = \left\{ \begin{pmatrix} X & 0 \\ 0 & 0 \end{pmatrix} : X \in \mathfrak{sl}_2(\mathbb{R}) \right\},$$

so $\dim H = 3$. With the standard e, h, f in the upper-left block,

$$p_H = e \wedge h \wedge f \in \bigwedge^3 \mathfrak{sl}_3.$$

Let
$$W = \left\{ \begin{pmatrix} 1 & t & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} : t \in \mathbb{R} \right\} \subset H.$$
Then $e \in \mathfrak{h}$ and $\mathrm{Ad}(W)$ preserves $\mathfrak{h}$. Because H is unimodular,
$$\rho_H(w)p_H = p_H \qquad (w \in W).$$
Hence $p_H \in \mathrm{Fix}(W)$.

Now take a matrix g that moves the 2-plane $\mathbb{R}e_1 \oplus \mathbb{R}e_2$ to a different 2-plane. Then gHg^{-1} is the copy of SL_2 acting on that moved plane. The condition
$$W \subset gHg^{-1}$$
means that the shear W must act entirely inside that plane. Thus $N(H, W)$ can be described geometrically as the set of g whose moved 2-plane contains the image and fixed flag of W. The exterior vector $\rho_H(g)p_H$ packages the same flag condition algebraically.

Why this matters. In higher rank, direct flag calculations become cumbersome because a candidate subgroup may have many root spaces. The exterior-vector construction is a coordinate-free way to encode the entire Lie subalgebra at once.

4. The orthogonal subgroup for an indefinite ternary form

Let Q be a nondegenerate indefinite quadratic form of signature $(2, 1)$ on $\mathbb{R}^3$, and set
$$H = \mathrm{SO}(Q)^\circ < \mathrm{SL}_3(\mathbb{R}).$$
Then H is locally isomorphic to $\mathrm{PSL}_2(\mathbb{R})$ and $\dim H = 3$. Choose a basis X_1, X_2, X_3 of $\mathfrak{h} = \mathfrak{so}(Q)$ and put
$$p_H = X_1 \wedge X_2 \wedge X_3 \in \bigwedge^3 \mathfrak{sl}_3.$$
If $g \in \mathrm{SL}_3(\mathbb{R})$, the vector $\rho_H(g)p_H$ represents the conjugate Lie algebra
$$\mathrm{Ad}(g)\mathfrak{so}(Q) = \mathfrak{so}(Q \circ g^{-1}).$$
Thus linearization converts a moving quadratic-form stabilizer into a moving vector in a finite-dimensional representation.

When Q is proportional to a rational form, the orbit $H\,\mathrm{SL}_3(\mathbb{Z})$ is closed of finite volume; when Q is irrational in the Oppenheim sense, the absence of such a rational closed-orbit obstruction is the starting point for density. The full Oppenheim argument uses the Ratner theory assumed as background; the purpose here is only to identify where the exterior vector enters, not to reproduce the separate effective theory.

5. A representative ambiguity in practice

Let $x = g\Gamma$. The same point is also $g\gamma\Gamma$ for every $\gamma \in \Gamma$, so the candidate vector is not just $\rho_H(g)p_H$ but the discrete family

$$\mathrm{Rep}_H(x) = \{\rho_H(g\gamma)p_H : \gamma \in \Gamma\}.$$

In $\mathrm{SL}_2(\mathbb{Z})$, even a compact set of lattice shapes admits infinitely many integer bases. The miracle is not that there are finitely many bases; there are not. The representative-properness theorem proved in Chapter 5 says, in the arithmetic setting, that only finitely many of the corresponding H-vectors can lie in a fixed compact vector window. This is exactly the finiteness needed by the dynamical argument.

6. What to remember from the examples

(1) p_H is not mysterious: it is the oriented volume element of $\mathfrak{h}$.
(2) $N_G(H)$ preserves the line $\mathbb{R}p_H$; $N_G^1(H)$ fixes the vector p_H.
(3) $N(H, W)$ becomes the condition that $\rho_H(g)p_H$ lies in a W-fixed algebraic set.
(4) Polynomial shearing moves vectors transverse to that fixed set at a polynomial rate.
(5) Lattice representatives are an essential, not cosmetic, complication.

CHAPTER 14

Polynomial Trajectories Produce Unipotent Invariance

A second major use of polynomial divergence is more direct than singular-set linearization. A nonconstant algebraic polynomial curve has a canonical asymptotic shear. At the correct time scale, nearby points on the same curve differ by a nontrivial unipotent element. Averaging along the curve then forces every weak limit to be invariant under that unipotent subgroup.

This is one of the cleanest places to see the Ratner mechanism outside an actual one-parameter subgroup.

1. Asymptotic shearing of an algebraic curve

Let G be a real algebraic subgroup of $\mathrm{GL}_N(\mathbb{R})$ and let

$$\Theta : \mathbb{R} \to G$$

be a nonconstant regular algebraic map.

Theorem 1.1 (Asymptotic algebraic shear). Proved in this text. *There exist a rational number $q \geq 0$ and a nontrivial algebraic one-parameter unipotent subgroup*

$$p : \mathbb{R} \longrightarrow G$$

such that for every sequence $s_t \to s$,

$$\Theta(t + s_t t^{-q})\Theta(t)^{-1} \longrightarrow p(s) \qquad (t \to +\infty).$$

The convergence is locally uniform when $s_t = s$ ranges over a compact set.

Proof. The proof is elementary once the algebraic map is viewed in a faithful matrix representation. Because G is affine algebraic and $\Theta : \mathbb{A}^1 \to G$ is regular, every matrix entry of $\Theta(t)$ is a polynomial in t. Inversion on an algebraic group is a regular morphism, so every matrix entry of $\Theta(t)^{-1}$ is also a polynomial in t. Let $d \geq 1$ be the degree of the matrix-valued polynomial Θ.

For $1 \leq \ell \leq d$ put

$$A_\ell(t) = \Theta^{(\ell)}(t)\Theta(t)^{-1}.$$

Each A_ℓ is a matrix-valued polynomial. For a nonzero matrix-valued polynomial A, let $\deg A$ mean the maximum degree of its entries. Define

$$q = \max_{\substack{1\le \ell\le d\\ A_\ell\neq 0}} \frac{\deg A_\ell}{\ell}.$$

Since Θ is nonconstant, $A_1 \neq 0$, so the maximum is taken over a nonempty set and $q \geq 0$. For every ℓ the limit

$$\lambda_\ell = \lim_{t\to\infty} t^{-q\ell} A_\ell(t)$$

exists, and at least one λ_ℓ is nonzero by the definition of q.

Taylor's formula is exact because Θ is polynomial. Thus, if $s_t \to s$,

$$\begin{aligned}\Theta(t+s_t t^{-q})\Theta(t)^{-1} &= I + \sum_{\ell=1}^{d} \frac{s_t^\ell}{\ell!} t^{-q\ell}\Theta^{(\ell)}(t)\Theta(t)^{-1}\\ &\longrightarrow \rho(s) := I + \sum_{\ell=1}^{d} \frac{s^\ell}{\ell!}\lambda_\ell.\end{aligned}$$

This also proves local uniformity for s in compact sets. The polynomial matrix ρ is nonconstant because not all λ_ℓ vanish.

We next prove the group law rather than infer it from a formal tangent calculation. Fix $s_1, s_2 \in \mathbb{R}$ and put

$$y_t = t + s_2 t^{-q}, \qquad r_t = s_1\left(\frac{y_t}{t}\right)^q.$$

Then $y_t \to \infty$, $r_t \to s_1$, and

$$y_t + r_t y_t^{-q} = t + (s_1+s_2)t^{-q}.$$

Consequently

$$\begin{aligned}&\Theta(t+(s_1+s_2)t^{-q})\Theta(t)^{-1}\\ &\qquad = \Theta(y_t + r_t y_t^{-q})\Theta(y_t)^{-1}\,\Theta(t+s_2t^{-q})\Theta(t)^{-1}.\end{aligned}$$

Passing to the limit and using the version already proved for varying parameters gives

$$\rho(s_1+s_2) = \rho(s_1)\rho(s_2).$$

In particular $\rho(s)\rho(-s) = I$, so every $\rho(s)$ is invertible. Every defining polynomial equation of the algebraic subgroup $G < \mathrm{GL}_N$ is satisfied by the approximating matrices and hence by the limit; therefore $\rho(s) \in G$. Thus $\rho : \mathbb{R} \to G$ is a nontrivial polynomial one-parameter subgroup.

It remains to prove that it is unipotent. A continuous one-parameter subgroup of $\mathrm{GL}_N(\mathbb{R})$ has the form

$$\rho(s) = e^{sZ}, \qquad Z = \rho'(0).$$

But every matrix entry of $e^{sZ} = \rho(s)$ is a polynomial in s. We claim that every eigenvalue of Z is zero. Indeed, over $\mathbb{C}$ the trace is both a polynomial and an exponential polynomial,

$$\operatorname{tr}(e^{sZ}) = \sum_{j=1}^{N} e^{\lambda_j s},$$

where the λ_j are the eigenvalues of Z with algebraic multiplicity. A finite sum of exponentials with distinct exponents can be a polynomial only when every exponent is zero: applying a sufficiently high derivative kills the polynomial, while the functions $e^{\lambda s}$ with distinct λ are linearly independent (their Wronskian is a nonzero Vandermonde multiple of an exponential). Hence every $\lambda_j = 0$. Therefore Z is nilpotent and

$$\rho(s) = e^{sZ}$$

is unipotent for every s. Set $p = \rho$. □

Example 1.2 (Why $q = 0$ must be allowed)**.** If $\Theta(t) = u_t$ is itself a nontrivial algebraic unipotent one-parameter subgroup, then

$$\Theta(t+s)\Theta(t)^{-1} = u_s.$$

The correct exponent is $q = 0$. Thus the asymptotic shift is always submacroscopic relative to t, but it need not tend to zero in absolute size.

> **Why this matters.** The definition of q chooses the first scale at which one of the logarithmic derivatives
>
> $$\Theta^{(\ell)}(t)\Theta(t)^{-1}$$
>
> becomes visible after rescaling. Terms of smaller degree disappear, while at least one leading term survives. The group law is not an extra miracle: it comes from composing two such shifts and observing that $t + s_2 t^{-q} \sim t$. Unipotence then follows from a rigid fact about polynomial one-parameter subgroups: an exponential e^{sZ} is polynomial in s exactly when Z is nilpotent.

2. The elementary time-change lemma

The second ingredient contains no homogeneous dynamics.

Lemma 2.1 (Microscopic shifts do not change long averages)**.** *Let $F : [1,\infty) \to \mathbb{R}$ be bounded and measurable, let $q > -1$, and let $s \in \mathbb{R}$. Then*

$$\frac{1}{T}\int_1^T \left(F(t + st^{-q}) - F(t)\right) dt \longrightarrow 0$$

provided F is evaluated only for positive arguments; changing the lower endpoint by a fixed amount does not affect the conclusion. The bound depends only on q, s, and $\|F\|_\infty$.

PROOF. For large t the map

$$\phi(t) = t + st^{-q}$$

is increasing because

$$\phi'(t) = 1 - qst^{-q-1} = 1 + O(t^{-q-1})$$

and $q + 1 > 0$. Make the change of variables $u = \phi(t)$. On $[T_0, T]$,

$$\frac{dt}{du} = 1 + O(u^{-q-1}).$$

Therefore

$$\int_{T_0}^{T} F(\phi(t))\,dt = \int_{\phi(T_0)}^{\phi(T)} F(u)\,du + O\left(\|F\|_\infty \int_{T_0}^{T+O(T^{-q})} u^{-q-1}\,du\right).$$

If $q > 0$, the error integral is $O(1)$. If $q = 0$, then $\phi(t) = t+s$ and $\phi'(t) = 1$, so there is no Jacobian error. If $-1 < q < 0$, the error integral is $O(T^{-q}) = o(T)$. The mismatch between the endpoints T and $\phi(T)$ has length

$$|s|T^{-q} = o(T)$$

for $q > -1$. The initial bounded interval also contributes $O(1)$. Dividing by T gives the result. □

3. Invariance of a weak limit

Let $x \in X = G/\Gamma$ and define

$$\mu_T = \frac{1}{T}\int_0^T \delta_{\Theta(t)x}\,dt.$$

Suppose $T_n \to \infty$ and $\mu_{T_n} \to \mu$ weakly, where μ is a probability measure. Equivalently, along this subsequence there is no escape of mass.

THEOREM 3.1 (Polynomial averages acquire unipotent invariance). PROVED IN THIS TEXT. *With $p(s)$ as in Theorem 1.1, the limit measure μ is invariant under $p(\mathbb{R})$.*

PROOF. Let $f \in C_c(X)$ and fix $s \in \mathbb{R}$. We must show

$$\int f(p(s)y)\,d\mu(y) = \int f(y)\,d\mu(y).$$

For large t, Theorem 1.1 gives

$$a_t(s) := \Theta(t + st^{-q})\Theta(t)^{-1} \to p(s).$$

The convergence is uniform for s in compact sets. Fix the present s and choose a compact neighborhood $L \subset G$ containing $p(s)$ and all $a_t(s)$ for large t. If either $f(p(s)y)$ or $f(a_t(s)y)$ is nonzero, then

$$y \in L^{-1} \operatorname{supp} f,$$

which is compact because L and $\operatorname{supp} f$ are compact. On the compact set $L \times L^{-1} \operatorname{supp} f$, the action map is uniformly continuous. Hence

$$f(p(s)\Theta(t)x) - f(a_t(s)\Theta(t)x) \to 0$$

uniformly in those t for which either term is nonzero; when both terms vanish the difference is already zero. Thus

$$\frac{1}{T_n}\int_1^{T_n} f(p(s)\Theta(t)x)\,dt \\ - \frac{1}{T_n}\int_1^{T_n} f(\Theta(t+st^{-q})x)\,dt \to 0.$$

Apply Lemma 2.1 to the bounded function

$$F(t) = f(\Theta(t)x).$$

It follows that

$$\frac{1}{T_n}\int_1^{T_n} f(\Theta(t+st^{-q})x)\,dt - \frac{1}{T_n}\int_1^{T_n} f(\Theta(t)x)\,dt \to 0.$$

Combining the two relations and taking the weak limit gives

$$\int f(p(s)y)\,d\mu(y) = \int f(y)\,d\mu(y).$$

Since s was arbitrary, μ is $p(\mathbb{R})$-invariant. □

Remark 3.2. If X is noncompact, one should not hide compactness inside the phrase "uniform continuity". For $C_c(X)$ the preceding compact-set argument is sufficient and does not require tightness at this step. Tightness is needed earlier to ensure that a subsequential limit is a probability measure, or if one insists on testing against arbitrary bounded continuous functions.

4. From invariance to equidistribution

Theorem 3.1 is only the first half of the argument. Ratner now says that every ergodic component of μ is homogeneous. To prove that the limit is the homogeneous measure on the expected orbit one must still:

(1) exclude escape of mass;
(2) identify the candidate homogeneous support;
(3) use linearization to exclude concentration on smaller singular strata;

(4) rule out focusing on a proper closed orbit if full equidistribution is desired.

Thus the complete logical chain is

$$\boxed{\begin{array}{c}\text{non-divergence} \to \text{unipotent invariance} \to \text{Ratner classification} \\ \to \text{linearization/avoidance} \to \text{equidistribution.}\end{array}}$$

5. Why the scale is microscopic but decisive

The shift st^{-q} is $o(t)$ for every $q \geq 0$; when $q > 0$ it even tends to zero, while for $q = 0$ it stays bounded. Thus the two parameter values are asymptotically indistinguishable on the macroscopic scale t. Yet after multiplying by $\Theta(t)^{-1}$ in the group, their relative displacement converges to a nontrivial element. This is precisely the shearing phenomenon:

$$\text{tiny time difference} \quad \rightsquigarrow \quad \text{order-one group displacement.}$$

Ratner's polynomial divergence uses nearby *initial points*; Shah's polynomial-trajectory argument uses nearby *times on one curve*. The algebra is the same.

CHAPTER 15

From Qualitative to Effective Linearization

The arguments in this volume are qualitative. They say that sufficiently thin neighborhoods and sufficiently long averaging sets work, but the dependence is not organized into an explicit rate in the orbit length, injectivity radius, discriminant, or height. Modern effective Ratner theory begins by asking which parts of the preceding architecture can be quantified simultaneously.

1. The qualitative inputs that lose effectivity

There are five main losses.

1.1. Genericity. Birkhoff's theorem gives convergence for almost every point but no uniform rate. Effective arguments replace genericity by quantitative equidistribution on controlled scales, often ultimately derived from a spectral gap.

1.2. Non-divergence. Here Volume 1 already provides a quantitative input. In more general effective problems the relevant height may involve many intermediate subgroups, not only short lattice vectors.

1.3. Representative properness. Qualitatively, a compact vector window contains only finitely many representatives. Effectively, one needs a bound on their number and on the complexity/height of the corresponding algebraic subgroups.

1.4. Polynomial non-concentration. The scalar (C, α)-good estimate is quantitative, but modern arguments require projection and incidence statements for families of transverse vectors. This is where discretized projection theorems and additive-combinatorial ideas enter.

1.5. Closing. Qualitative focusing says that persistent near-invariance produces an algebraic obstruction. Effective closing must say that either the orbit equidistributes by time T, or there is a proper closed orbit of explicitly bounded complexity within an explicitly controlled distance.

2. A schematic effective alternative

A typical modern theorem has the shape

$$\boxed{\begin{array}{c}\text{equidistribution at scale } T^{-\kappa}\\ \text{or a nearby algebraic obstruction of height } \le T^{A}.\end{array}}$$

The constants $\kappa, A > 0$ are rarely optimal. The conceptual breakthrough is polynomial dependence rather than the existence of any particular exponent.

3. Why dimension growth appears

In the qualitative proof, one nonzero transverse vector is enough: polynomial shearing eventually makes it visible. For an effective theorem, a single vector can be too sparse to exploit mixing. One therefore tries to construct a set of many transverse points with positive quantitative dimension. Projection/incidence estimates then improve that dimension until the set is thick enough for spectral methods.

This leads to the modern architecture

$$\begin{aligned}\text{recurrence} &\longrightarrow \text{initial transverse set}\\ &\longrightarrow \text{Margulis/energy function}\\ &\longrightarrow \text{projection or incidence theorem}\\ &\longrightarrow \text{dimension growth}\\ &\longrightarrow \text{mixing/spectral gap}\\ &\longrightarrow \text{effective equidistribution.}\end{aligned}$$

The same architecture can be studied first in rank-one models and then in absolute rank two, but those effective theorems are not assumed here.

4. An instructive comparison

Ingredient	Qualitative form	Effective replacement
Recurrence	return to compact sets	quantitative height estimate
Nearby points	existence after recurrence	many points at controlled separation
Shearing	nonconstant polynomial divergence	controlled polynomial coefficients
Singular avoidance	small enough neighborhood	explicit tube radius and occupation bound
Focusing	subsequential vector limit	bounded-height algebraic obstruction

Ingredient	Qualitative form	Effective replacement
Mixing	not needed for classification	quantitative decay of correlations
Conclusion	homogeneous weak limit	discrepancy/Sobolev error bound

5. A source-sensitive boundary

Recent effective papers differ substantially in ambient group, arithmetic hypotheses, available spectral gap, and treatment of intermediate subgroups. We therefore do not state a synthetic "effective Ratner theorem" here. Doing so would hide precisely the hypotheses that matter.

Instead, a mathematically responsible effective treatment should keep the following status discipline:

- rank-one theorems of Lindenstrauss–Mohammadi–Wang will be stated in their exact arithmetic settings;
- Lei Yang's SL_3 theorem will be stated with its Diophantine hypothesis;
- the Lindenstrauss–Mohammadi–Wang–Yang rank-two result will keep its quasi-split/principal-SL_2 assumptions explicit;
- research announcements or proof sketches will be labeled FRONTIER / SOURCE-STATUS SENSITIVE, not silently upgraded to complete theorems.

6. Conceptual payoff of the two-volume foundation

Volumes 1 and 2 now supply the reusable qualitative skeleton:

$$\boxed{\begin{array}{c}\text{height/non-divergence}\\ \downarrow\\ \text{polynomial divergence and genericity}\\ \downarrow\\ \text{exterior linearization of subgroup incidence}\\ \downarrow\\ \text{avoidance/focusing of singular strata}\\ \downarrow\\ \text{Ratner homogeneous classification.}\end{array}}$$

Every major post-Ratner theory in the rest of the series changes at least one arrow in this diagram. Higher-rank entropy replaces polynomial shearing;

Benoist–Quint replaces it by random exponential drift; effective Ratner quantifies the arrows; Eskin–Mirzakhani rebuilds analogous arrows in a nonhomogeneous moduli space. Understanding this skeleton is therefore more useful than memorizing any one proof.

CHAPTER 16

Complete model calculations for shearing and linearization

The linearization method can look unnecessarily elaborate when it is first encountered in its general form. The purpose of this chapter is to strip it back to examples in which every relevant vector and every polynomial can be written on one line. The examples are not substitutes for the general Dani–Margulis–Shah theorem. They show what that theorem is organizing.

1. The SL_2 shearing calculation from logarithmic coordinates

Let

$$U = \{u_t : t \in \mathbb{R}\}, \qquad u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix},$$

and use the standard basis

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

of $\mathfrak{sl}_2(\mathbb{R})$. We proved earlier that

$$\mathrm{Ad}(u_t)e = e, \qquad \mathrm{Ad}(u_t)h = h - 2te, \qquad \mathrm{Ad}(u_t)f = f + th - t^2 e. \tag{1.1}$$

Suppose $g_n \to e_G$ and write, after passing to a small logarithmic neighborhood,

$$g_n = \exp Z_n, \qquad Z_n = a_n e + b_n h + c_n f, \qquad (a_n, b_n, c_n) \to (0,0,0).$$

Then conjugation is exact at the Lie-algebra level:

$$u_t g_n u_{-t} = \exp(\mathrm{Ad}(u_t) Z_n),$$

with

$$\mathrm{Ad}(u_t) Z_n = (a_n - 2tb_n - t^2 c_n)e + (b_n + tc_n)h + c_n f. \tag{1.2}$$

If $b_n = c_n = 0$, then $g_n \in U$ and conjugation does not magnify the relative displacement. This is exactly the centralizer direction. Assume therefore that $(b_n, c_n) \neq (0,0)$.

Choose $T_n \to \infty$ so that

$$|b_n| T_n + |c_n| T_n^2 = 1.$$

Such a choice is possible and unique because the left side is a continuous strictly increasing function of $T_n > 0$. For fixed s,

$$\begin{aligned}\mathrm{Ad}(u_{sT_n})Z_n = (a_n - 2sT_nb_n - s^2T_n^2c_n)e\\ + (b_n + sT_nc_n)h + c_nf.\end{aligned}$$

The h-coefficient tends to zero. Indeed, $b_n \to 0$, while

$$|T_nc_n| \le \frac{|c_n|T_n^2}{T_n} \le T_n^{-1} \to 0.$$

The f-coefficient also tends to zero. Passing to a subsequence, the bounded pair

$$(T_nb_n, T_n^2c_n)$$

converges to (B, C) with $|B| + |C| = 1$. Hence

$$u_{sT_n}g_nu_{-sT_n} \longrightarrow \exp((-2sB - s^2C)e) = u_{-2sB-s^2C}. \tag{1.3}$$

The polynomial $-2sB - s^2C$ is not identically zero. Thus there exists s_0 for which the limit is a nontrivial element of U.

Proposition 1.1 (Transverse perturbations shear into U)**.** *Let*

$$g_n = \exp(a_ne + b_nh + c_nf) \to e$$

and assume $(b_n, c_n) \neq (0, 0)$ for every n. After passing to a subsequence there are $T_n \to \infty$ and a nonzero polynomial P of degree at most two such that

$$u_{sT_n}g_nu_{-sT_n} \to u_{P(s)}$$

uniformly for s in compact subsets of $\mathbb{R}$.

Proof. The preceding calculation proves the assertion when g_n lies in a fixed logarithmic chart. Since $g_n \to e$, this holds for all sufficiently large n. Uniformity on compact s-sets follows directly from (1.2), because the coefficients converge and the exponential map is uniformly continuous on compact subsets of the Lie algebra. □

Why this matters. There are three details worth retaining. First, the correct scale is not chosen by guessing a time at which two orbits are "noticeably apart"; it is chosen by normalizing the first nontrivial transverse polynomial. Second, lower-degree coordinates disappear automatically at that scale. Third, the surviving displacement lies in the original unipotent direction. This is the elementary algebra behind Ratner's shearing phenomenon.

2. How shearing becomes measure invariance

The algebra alone does not imply invariance. One also needs two nearby orbit pieces to have the same statistical limit at a scale on which their relative displacement is almost constant. The following lemma isolates the exact comparison.

Lemma 2.1 (Mesoscopic comparison gives invariance)**.** *Let X be a locally compact metric G-space, let $x_n, y_n \in X$ with $y_n = g_n x_n$, and let $I_n \subset \mathbb{R}$ be bounded intervals with $|I_n| \to \infty$. Suppose that for some $v \in G$,*

$$\sup_{t\in I_n} d_G(u_t g_n u_{-t}, v) \to 0$$

with respect to a fixed compatible metric on a neighborhood of v. Assume also that the two empirical measures

$$\frac{1}{|I_n|}\int_{I_n} \delta_{u_t x_n}\,dt, \qquad \frac{1}{|I_n|}\int_{I_n} \delta_{u_t y_n}\,dt$$

converge weakly to the same probability measure μ. Then

$$v_*\mu = \mu.$$

Proof. Let $f \in C_c(X)$. Since

$$u_t y_n = u_t g_n x_n = (u_t g_n u_{-t}) u_t x_n,$$

we compare the two empirical averages. The group elements $u_t g_n u_{-t}$ lie in a fixed compact neighborhood L of v for all large n and all $t \in I_n$. If either $f((u_t g_n u_{-t})u_t x_n)$ or $f(v u_t x_n)$ is nonzero, then $u_t x_n \in L^{-1}\operatorname{supp} f$, a compact subset of X. Uniform continuity of the action on $L \times L^{-1}\operatorname{supp} f$ therefore gives

$$\sup_{t\in I_n} |f(u_t y_n) - f(v u_t x_n)| \to 0.$$

After integration,

$$\frac{1}{|I_n|}\int_{I_n} f(u_t y_n)\,dt - \frac{1}{|I_n|}\int_{I_n} f(v u_t x_n)\,dt \to 0.$$

The first term tends to $\int f\,d\mu$ and the second to $\int f(vz)\,d\mu(z)$. Hence these integrals are equal for every $f \in C_c(X)$, which is exactly $v_*\mu = \mu$. $\square$

In Ratner's shearing argument, recurrence and Egorov uniformity are used to find intervals I_n on which the two nearby points have the same limiting statistics. Polynomial divergence is then frozen on a mesoscopic window so that

$$u_t g_n u_{-t} \approx v \qquad (t \in I_n).$$

Proposition 1.1 identifies which nontrivial elements v can arise. The lemma above is the precise bridge from that algebraic limit to additional invariance.

3. A polynomial trajectory whose asymptotic shear is explicit

Let $d \geq 2$ and consider the algebraic curve

$$\Theta(t) = u_{t^d} \qquad (t \geq 1).$$

Although this curve lies in U, it is not parameterized by Haar time on U. We determine its microscopic self-shear exactly.

Fix $s \in \mathbb{R}$ and put

$$\Delta_t = \frac{s}{d} t^{1-d}.$$

Then the binomial theorem gives

$$\begin{aligned}(t + \Delta_t)^d - t^d &= dt^{d-1}\Delta_t + \binom{d}{2} t^{d-2}\Delta_t^2 + \cdots + \Delta_t^d \\ &= s + O_{d,s}(t^{-d}).\end{aligned}$$

Therefore

$$\Theta(t + \Delta_t)\Theta(t)^{-1} = u_{(t+\Delta_t)^d - t^d} \longrightarrow u_s. \tag{3.1}$$

The shift Δ_t tends to zero, but the group displacement tends to a nontrivial element. This is the simplest possible realization of a microscopic shear.

Now let $x \in X$ and suppose the probability measures

$$\mu_T = \frac{1}{T}\int_1^T \delta_{u_{t^d}x}\, dt$$

have a weak subsequential limit μ. Since $d \geq 2$, the time change $t \mapsto t + \frac{s}{d}t^{1-d}$ changes Lebesgue measure by a derivative $1 + O(t^{-d})$ and moves the endpoints by $o(T)$. Thus for every $f \in C_c(X)$,

$$\frac{1}{T}\int_1^T [f(u_{(t+\Delta_t)^d}x) - f(u_{t^d}x)]\, dt \to 0.$$

Using (3.1) and uniform continuity on the compact region where f is nonzero gives

$$\int f(u_s y)\, d\mu(y) = \int f(y)\, d\mu(y).$$

Since s is arbitrary, μ is U-invariant.

Why this matters. This proof is worth comparing with the previous one. For two nearby orbits, recurrence supplies pairs of points and conjugation magnifies their relative displacement. For a single polynomial trajectory, the nearby point is supplied by the trajectory itself: compare times t and $t + \Delta_t$. The time-change estimate replaces the generic-pair argument. In both cases the decisive object is the same: a nontrivial limit of a relative displacement.

4. Exterior linearization for the horocycle subgroup

We now compute the Dani–Margulis exterior representation explicitly. Take

$$G = \mathrm{SL}_2(\mathbb{R}), \qquad \Gamma = \mathrm{SL}_2(\mathbb{Z}), \qquad H = U.$$

The Lie algebra of H is $\mathbb{R}e$, so

$$p_H = e \in \mathfrak{sl}_2(\mathbb{R}), \qquad \rho_H = \mathrm{Ad}\,.$$

The vector p_H is fixed by U, and $U \cap \Gamma = \{u_n : n \in \mathbb{Z}\}$ is a lattice in U.

Let

$$\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}).$$

A direct calculation gives

$$\mathrm{Ad}(\gamma)e = \gamma e \gamma^{-1} = \begin{pmatrix} -ac & a^2 \\ -c^2 & ac \end{pmatrix}. \tag{4.1}$$

Only the first column $(a,c)^T$ of γ appears. Since $ad - bc = 1$, that column is primitive. Conversely every primitive pair (a,c) can be completed to an element of $\mathrm{SL}_2(\mathbb{Z})$. Hence the representative set is explicitly

$$\mathrm{Ad}(\Gamma)e = \left\{ \begin{pmatrix} -ac & a^2 \\ -c^2 & ac \end{pmatrix} : (a,c) \in \mathbb{Z}^2 \text{ primitive} \right\}.$$

This is a closed discrete subset of the nilpotent cone. In this model the abstract Dani–Margulis discreteness theorem can literally be seen in the integer entries.

4.1. The normalizer and the determinant-one normalizer. The line stabilizer of $\mathbb{R}e$ is the upper triangular subgroup

$$B = \left\{ \begin{pmatrix} r & s \\ 0 & r^{-1} \end{pmatrix} : r \neq 0 \right\}.$$

Indeed

$$\mathrm{Ad}\begin{pmatrix} r & s \\ 0 & r^{-1} \end{pmatrix} e = r^2 e.$$

Thus

$$N_G(U) = B.$$

The vector stabilizer of e requires $r^2 = 1$, so

$$N_G^1(U) = \left\{ \begin{pmatrix} \sigma & s \\ 0 & \sigma \end{pmatrix} : \sigma \in \{1,-1\},\ s \in \mathbb{R} \right\}.$$

Its identity component is U. This concrete calculation explains why the linearization theorem distinguishes the full normalizer, which preserves the subgroup, from the determinant-one normalizer, which preserves the chosen volume vector.

5. What a near-singular return means in this model

Let $x = g\Gamma$. A representative vector for x relative to $H = U$ is

$$\mathrm{Ad}(g\gamma)e, \qquad \gamma \in \Gamma.$$

By (4.1), choosing γ is equivalent to choosing a primitive integer direction. If $g_n\Gamma$ stays in a compact subset of X while some $\mathrm{Ad}(g_n\gamma_n)e$ stays in a fixed compact subset of $\mathfrak{sl}_2 \setminus \{0\}$, properness says that the corresponding classes of $g_n\gamma_n$ modulo $N_G^1(U) \cap \Gamma$ cannot run off to infinity. After a subsequence they focus.

The nonlinear statement "$g_n\Gamma$ repeatedly approaches a closed horocycle configuration" has therefore become the linear statement "one of the discrete vectors $\mathrm{Ad}(g_n\gamma)e$ remains in a bounded window." This is the essence of linearization.

There is also a simple algebraic detection statement. If $W = U$, then

$$N(H, W) = N_G(U) = B.$$

For $g \in G$,

$$g \in B \iff \mathrm{Ad}(g)e \in \mathbb{R}^\times e.$$

Thus the transporter is detected by a linear condition in the representation. In larger groups and for larger H, the analogous condition is an algebraic subset of $\bigwedge^{\dim H} \mathfrak{g}$ rather than a line, but the philosophy is unchanged.

6. A small SL_3 model: why higher-dimensional subgroups enter

Embed $\mathrm{SL}_2(\mathbb{R})$ in the upper-left block of $\mathrm{SL}_3(\mathbb{R})$:

$$H = \left\{ \begin{pmatrix} a & b & 0 \\ c & d & 0 \\ 0 & 0 & 1 \end{pmatrix} : ad - bc = 1 \right\}.$$

Then

$$\mathfrak{h} = \mathrm{span}\{E_{12}, E_{21}, E_{11} - E_{22}\},$$

so the linearization vector is

$$p_H = E_{12} \wedge E_{21} \wedge (E_{11} - E_{22}) \in \bigwedge^3 \mathfrak{sl}_3.$$

A point $g\Gamma$ near the homogeneous orbit $H\Gamma/\Gamma$ need not be recognized by a short vector in the standard representation $\mathbb{R}^3$. Instead, one must detect that the entire three-dimensional Lie algebra $\mathrm{Ad}(g)\mathfrak{h}$ is close to a rational copy of $\mathfrak{h}$. The single exterior vector $\rho_H(g)p_H$ packages those three directions at once.

This is the conceptual reason the standard lattice representation is not enough for linearization. Quantitative non-divergence asks whether rational *subspaces of the ambient vector representation* become small. Linearization asks whether rational *Lie subgroups of the homogeneous space* become dynamically relevant. The exterior power is used in both places, but it is exterior algebra of different objects.

7. A reconstruction template

For a new linearization argument, one can now reconstruct the method from the following concrete questions.

(1) What is the candidate subgroup H and what is an explicit basis of $\mathfrak{h}$?
(2) What is the exterior vector $p_H = v_1 \wedge \cdots \wedge v_{\dim H}$?
(3) Which subgroup stabilizes the line $\mathbb{R}p_H$, and which subgroup fixes p_H itself?
(4) Why is the arithmetic orbit $\rho_H(\Gamma)p_H$ discrete, or which this follows from the representative-properness lemma in Chapter 5.
(5) What algebraic condition on $\rho_H(g)p_H$ is equivalent to $W \subset gHg^{-1}$?
(6) Which lower-dimensional $F \subsetneq H$ must be removed before the relevant representative becomes locally unique?
(7) Which polynomial sublevel estimate prevents a trajectory from spending positive density near that algebraic condition without satisfying it identically?
(8) If the estimate fails along a sequence, what compactness statement turns the bounded representative vectors into focusing?

A correct proof may package these steps differently, but none of the questions disappears. The advantage of the representation is that it turns the difficult geometric middle of the argument into finite-dimensional algebra plus polynomial non-concentration.

CHAPTER 17

A Ternary Quadratic-Form Laboratory

The purpose of this chapter is not to re-prove the Oppenheim theorem by hiding an orbit-closure theorem inside a sentence. Instead we work out, completely and explicitly, the algebra that makes ternary quadratic forms the canonical model for the linearization method. The calculations show exactly where rationality, closed homogeneous orbits, and lower-dimensional obstructions enter.

1. The split ternary form and its stabilizer

Consider
$$Q_0(x_1, x_2, x_3) = 2x_1x_3 - x_2^2.$$
It has signature $(1, 2)$. The group
$$H = \mathrm{SO}(Q_0)^\circ < \mathrm{SL}_3(\mathbb{R})$$
is locally isomorphic to $\mathrm{PSL}_2(\mathbb{R})$. The cleanest model is the action of SL_2 on binary quadratic forms.

Let $V = \mathrm{Sym}^2(\mathbb{R}^2)$ with basis
$$X^2, \quad XY, \quad Y^2.$$
For
$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{R}),$$
define
$$(\rho(A)f)(X, Y) = f(aX + cY, bX + dY).$$
The discriminant
$$\mathrm{Disc}(\alpha X^2 + \beta XY + \gamma Y^2) = \beta^2 - 4\alpha\gamma$$
is SL_2-invariant. After the linear change of variables
$$(x_1, x_2, x_3) = (\gamma, \beta/\sqrt{2}, \alpha),$$
the discriminant is a nonzero scalar multiple of Q_0. Hence ρ identifies $\mathrm{PSL}_2(\mathbb{R})$ with H.

Proposition 1.1 (The irreducible $\mathfrak{sl}_2$-triple)**.** *With*

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

the derived representation on V is the three-dimensional irreducible $\mathfrak{sl}_2$-module. In the ordered basis (X^2, XY, Y^2),

$$d\rho(e) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}, \qquad d\rho(f) = \begin{pmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad d\rho(h) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{pmatrix},$$

up to the harmless convention of left versus right differentiation.

Proof. Differentiate

$$f(X, tX + Y), \qquad f(X + tY, Y), \qquad f(e^t X, e^{-t} Y)$$

at $t = 0$. The displayed matrices follow by evaluating on the three basis monomials. The weight spaces have weights $2, 0, -2$, and the raising/lowering operators connect all three; hence the representation is irreducible. ☐

The corresponding unipotent subgroup is

$$u_t = \rho\left(\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}\right).$$

Because $(d\rho(e))^3 = 0$,

$$u_t = I + t\, d\rho(e) + \frac{t^2}{2} d\rho(e)^2.$$

Thus every coordinate of the ternary unipotent action is a polynomial of degree at most two. This is the simplest nontrivial model in which polynomial shearing has a genuine quadratic term.

2. From a quadratic form to a homogeneous point

Every nondegenerate indefinite ternary form Q of determinant equal to that of Q_0 can be written

$$Q(v) = Q_0(gv)$$

for some $g \in \mathrm{SL}_3(\mathbb{R})$, after a scalar normalization. The associated point in the space of unimodular lattices is

$$x_Q = g\,\mathrm{SL}_3(\mathbb{Z}) \in X_3 = \mathrm{SL}_3(\mathbb{R})/\,\mathrm{SL}_3(\mathbb{Z}).$$

For $m \in \mathbb{Z}^3$,

$$Q(m) = Q_0(gm).$$

Thus small values of Q on integer vectors are read from vectors of the moving lattice $g\mathbb{Z}^3$ relative to the fixed quadratic form Q_0.

The stabilizer of Q is
$$H_Q = g^{-1}Hg.$$
Since H fixes Q_0,
$$Q(h_Q m) = Q(m) \qquad (h_Q \in H_Q).$$
Consequently the arithmetic value set of Q is constant along the homogeneous orbit $H_Q x_0$, where $x_0 = \mathrm{SL}_3(\mathbb{Z})$.

3. Closed stabilizer orbits force rationality

We need only a very special case of Borel density, and in rank one it has an elementary proof.

Lemma 3.1 (Zariski density of a finite-covolume Fuchsian lattice)**.** *Let H be a connected real algebraic group locally isomorphic to* $\mathrm{PSL}_2(\mathbb{R})$*, and let $\Lambda < H$ be a lattice. Then Λ is Zariski dense in H.*

Proof. Assume the Zariski closure $\mathbf{L}$ of Λ is proper. A proper Lie subalgebra of $\mathfrak{sl}_2$ has dimension at most two and is solvable: a proper semisimple subalgebra would have dimension three, while every one-dimensional algebra is abelian and every two-dimensional Lie algebra is solvable. Hence the identity component $\mathbf{L}^\circ$ is solvable. Real algebraic groups have finitely many connected components, so $\Lambda \cap \mathbf{L}^\circ$ has finite index in Λ; thus Λ is virtually solvable.

We now use only elementary hyperbolic geometry. A non-elementary discrete subgroup of $\mathrm{PSL}_2(\mathbb{R})$ contains two hyperbolic elements with disjoint fixed-point sets on $\partial\mathbb{H}^2$. Taking sufficiently large powers, their attracting and repelling neighborhoods are disjoint, and the ping-pong lemma produces a free subgroup of rank two. A virtually solvable group cannot contain such a subgroup. Therefore a discrete virtually solvable subgroup of $\mathrm{PSL}_2(\mathbb{R})$ is elementary: up to finite index it fixes a point of $\mathbb{H}^2$, one point of the boundary, or a pair of boundary points. Its quotient has infinite hyperbolic area. Indeed a finite elliptic group leaves an infinite-area quotient; a parabolic cyclic subgroup has a fundamental strip whose area diverges toward the non-cuspidal end; and a hyperbolic cyclic subgroup has an infinite-area cylinder. Finite extensions do not change this conclusion.

A lattice, by definition, has finite covolume. Hence Λ cannot be elementary or virtually solvable. This contradiction proves Zariski density. $\square$

The following implication is one of the algebraic facts used in dynamical reductions of Oppenheim-type problems.

Proposition 3.2 (Closed finite-volume orthogonal orbit implies rational form)**.** *Let Q be a nondegenerate indefinite ternary form and $H_Q = \mathrm{SO}(Q)^\circ$. Suppose*

$$H_Q \, \mathrm{SL}_3(\mathbb{Z})$$

is closed and carries a finite H_Q-invariant measure. Then Q is proportional to a quadratic form with rational coefficients.

PROOF. The stabilizer

$$\Lambda = H_Q \cap \mathrm{SL}_3(\mathbb{Z})$$

is a lattice in H_Q. Lemma 3.1 shows directly that Λ is Zariski dense in the connected rank-one group H_Q.

Let $\mathcal{Q}$ be the six-dimensional rational vector space of ternary quadratic forms. The group SL_3 acts rationally on $\mathcal{Q}$. The fixed space

$$\mathcal{Q}^{H_Q}$$

is one dimensional, spanned by Q: indeed the standard three-dimensional representation of $H_Q \simeq \mathrm{PSL}_2$ is irreducible, and the invariant symmetric bilinear form is unique up to scalar by Schur's lemma (or by an explicit $\mathfrak{sl}_2$ weight calculation).

Because $\Lambda \subset \mathrm{SL}_3(\mathbb{Z})$, the equations

$$\lambda \cdot R = R \qquad (\lambda \in \Lambda)$$

have rational coefficients in $R \in \mathcal{Q}$. A finite subset of these equations already cuts out the common fixed space, since $\mathcal{Q}$ is finite dimensional. Zariski density implies

$$\mathcal{Q}^{\Lambda} = \mathcal{Q}^{H_Q} = \mathbb{R}Q.$$

Therefore the one-dimensional line $\mathbb{R}Q$ is the solution space of a rational linear system. Such a line has a nonzero rational vector. Hence Q is proportional to a rational form. □

Why this matters. This proof isolates the exact arithmetic input. A closed finite-volume orbit gives a lattice in the stabilizer; the rank-one Zariski-density lemma proved above promotes lattice invariance to algebraic-group invariance; finite-dimensional rational representation theory then turns that invariance into rational coefficients of the form.

4. A local linearization calculation in SL_3

Let $U = \{u_t\} < H$ be the image of the upper unipotent subgroup of SL_2. Let $H' < \mathrm{SL}_3$ be a rational subgroup with exterior vector $p_{H'}$. To test whether a translate $gH'g^{-1}$ contains U, we need only test

$$d\rho(e) \wedge \rho_{\dim H'}(g) p_{H'} = 0.$$

Under the flow u_t, the left-hand side becomes

$$d\rho(e) \wedge \exp\bigl(t \operatorname{ad}(d\rho(e))\bigr) \rho_{\dim H'}(g) p_{H'},$$

a vector-valued polynomial. If this polynomial is not identically zero, the sublevel theorem of Volume 1 bounds the proportion of times it can be tiny. If it vanishes identically, the whole trajectory is algebraically focused on the subgroup-containment variety.

This is the ternary form incarnation of the general linearization theorem. Nothing specifically number theoretic happens in the polynomial estimate; arithmetic enters through the integral exterior representatives that prevent infinitely many rational subgroup explanations from accumulating inside a compact vector window.

5. Why irrationality removes the most obvious obstruction

Proposition 3.2 gives the contrapositive: if Q is not proportional to a rational form, the H_Q-orbit through the standard lattice cannot itself be a closed finite-volume orbit. This does *not* by itself prove that the orbit is dense. That final step requires a homogeneous orbit-closure theorem for the group generated by the unipotent subgroups of H_Q, or an equivalent one-parameter argument plus subgroup analysis. We deliberately do not hide that theorem here.

What this laboratory does prove is the entire dictionary used by a genuine dynamical proof:

$$\begin{array}{ccl} \text{quadratic form} & \longleftrightarrow & \text{point } x_Q \in X_3, \\ \mathrm{SO}(Q) & \longleftrightarrow & \text{homogeneous acting group}, \\ \text{rationality of } Q & \longleftrightarrow & \text{closed finite-volume stabilizer obstruction}, \\ \text{unipotent motion} & \longleftrightarrow & \text{bounded-degree polynomial exterior motion}, \\ \text{near proper orbit} & \longleftrightarrow & \text{small wedge coordinates}. \end{array}$$

A reader who can reproduce these arrows has understood the role of linearization in the Oppenheim story rather than merely memorized that “Ratner implies Oppenheim.”

6. Worked perturbation

Take

$$g_\varepsilon = \exp(\varepsilon E_{13}), \qquad u_t = \exp(t E_{12})$$

inside SL_3. Since

$$[E_{12}, E_{13}] = 0,$$

this perturbation is invisible to the E_{12}-shear. By contrast, for

$$g_\varepsilon = \exp(\varepsilon E_{21}),$$

we have

$$\mathrm{Ad}(u_t)E_{21} = E_{21} + t(E_{11} - E_{22}) - t^2 E_{12}.$$

Thus a perturbation of size ε becomes visible at scale $t \asymp \varepsilon^{-1/2}$ through the quadratic term. At that scale the diagonal term has size $\asymp \sqrt{\varepsilon}$ and disappears, while the E_{12}-term is macroscopic. This is the simplest explicit instance of the "first surviving polynomial coefficient" rule used throughout Ratner-type shearing arguments.

Exercise 6.1. Repeat the calculation for perturbations in each root direction E_{ij} of $\mathfrak{sl}_3$ relative to $U = \exp(tE_{12})$. For every direction determine the first nonzero power of t, the visibility scale in ε, and the limiting shear direction.

CHAPTER 18

Exercises and Research Laboratories

The exercises are arranged by mechanism. A problem marked **[C]** checks a core calculation, **[P]** asks for a proof reconstruction, and **[R]** is research-facing.

1. Polynomial divergence

Exercise 1.1 (C)**.** For $u_t = \exp(te)$ in $\mathrm{SL}_2(\mathbb{R})$, compute $\mathrm{Ad}(u_t)$ on the ordered basis (e, h, f) and verify directly that

$$\mathrm{Ad}(u_t) = \exp(t \operatorname{ad} e).$$

Determine the nilpotency index of $\operatorname{ad} e$.

Exercise 1.2 (P)**.** Let $U = \{u_t\}$ be Ad-unipotent and let $g_n = \exp(Z_n) \to e$. Suppose there are scales $T_n \to \infty$ such that

$$\mathrm{Ad}(u_{T_n t})Z_n \to P(t)$$

locally uniformly, with P nonconstant. Prove, using the Baker–Campbell–Hausdorff formula with an explicit error estimate, that

$$u_{T_n t} g_n u_{-T_n t} \to \exp(P(t))$$

locally uniformly after imposing a sufficient smallness hypothesis on $\mathrm{Ad}(u_{T_n t})Z_n$.

Exercise 1.3 (R)**.** In a semisimple Lie algebra, express the degree of polynomial divergence under a root-unipotent subgroup in terms of the lengths of $\mathfrak{sl}_2$ strings. Work this out explicitly for the principal unipotent in $\mathfrak{sl}_3$.

2. Genericity and shearing

Exercise 2.1 (P)**.** Prove the “visible displacement implies invariance” lemma for probability measures: if x_n, y_n are uniformly generic for μ, $y_n = h_n x_n$, and the rescaled relative maps converge to h, then $h_*\mu = \mu$. State exactly where uniform genericity rather than pointwise genericity is used.

Exercise 2.2 (C)**.** Give an example showing that two merely generic points can fail to provide uniform control on a moving time scale T_n chosen after the points are chosen.

3. Singular sets

Exercise 3.1 (C)**.** For $G = \mathrm{SL}_2(\mathbb{R})$ and $W = U^+$, compute $N(U^+, W)$. In the arithmetic model with $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, identify the rational lower-incidence possibilities in a bounded exterior window and explain why no positive-dimensional proper connected subgroup strictly below U^+ can still contain W.

Exercise 3.2 (P)**.** Construct explicit subsets $A, B \subset G$ for which $\pi(A \setminus B) \neq \pi(A) \setminus \pi(B)$. Then prove that the quotient-level rational pure part used in this volume,

$$\mathcal{P}_{\mathrm{rat}}(H, W) = \pi(N(H, W)) \setminus \pi(S_{\mathrm{rat}}(H, W)),$$

is the logically safe object for measure arguments. Formulate an additional Γ-saturation hypothesis under which the naive set-difference identity would become valid.

Exercise 3.3 (R)**.** For a block copy $H \cong \mathrm{SL}_2$ in SL_3, classify geometrically the g for which a chosen root subgroup W is contained in gHg^{-1}. Translate your answer into a flag condition.

4. Exterior linearization

Exercise 4.1 (P)**.** Let $H < G$ be connected with Lie algebra $\mathfrak{h}$ and p_H a nonzero vector in $\bigwedge^{\dim H} \mathfrak{h}$. Prove carefully that the stabilizer of the line $\mathbb{R}p_H$ under $\bigwedge^{\dim H} \mathrm{Ad}$ is $N_G(H)$. Your proof must include the implication from preservation of a decomposable exterior line to preservation of the underlying subspace.

Exercise 4.2 (P)**.** Show that a connected Lie group H admitting a lattice is unimodular. Deduce that H fixes p_H. Give an example of a non-unimodular closed subgroup for which the line $\mathbb{R}p_H$ is preserved but the vector p_H is not.

Exercise 4.3 (C)**.** In $\mathrm{SL}_2(\mathbb{R})$, compute explicitly the orbit $\mathrm{Ad}(G)e \subset \mathfrak{sl}_2$. Relate it to the nilpotent cone $\{X : \mathrm{tr}(X^2) = 0\}$.

5. Linearization and focusing

Exercise 5.1 (P)**.** Starting from the scalar (C, α)-good property, prove Lemma 2.2 with constants displayed. Generalize it from a linear subspace to a compact smooth algebraic submanifold at points where a fixed defining map has full rank.

Exercise 5.2 (P)**.** Reconstruct the finite-window proof of Theorem 3.1 from the five concrete ingredients: integral exterior discreteness, proper incidence, finite lower-incidence list, hybrid polynomial growth, and Besicovitch selection.

Write the quantifiers in their exact order and explain why a naive statement "induct on $\dim H$" would miss the lower-incidence neighborhood that must remain visible.

Exercise 5.3 (R)**.** Find a concrete example in $\mathrm{SL}_3(\mathbb{R})/\mathrm{SL}_3(\mathbb{Z})$ where two different subgroup representatives can be simultaneously relevant near the same point. Identify the lower-dimensional intersection subgroup predicted by the singular-set induction.

6. Polynomial trajectories

Exercise 6.1 (P)**.** Give a complete proof of Lemma 2.1 for all $q > -1$, including $q = 0$ and $-1 < q < 0$. Show by constructing a bounded function that the statement can fail at the macroscopic threshold $q = -1$.

Exercise 6.2 (C)**.** Let

$$\Theta(t) = \begin{pmatrix} 1 & t^d \\ 0 & 1 \end{pmatrix}.$$

Find q and $p(s)$ explicitly in Theorem 1.1. Repeat for

$$\Theta(t) = \begin{pmatrix} 1 & t + t^3 \\ 0 & 1 \end{pmatrix}.$$

Which term determines the microscopic scale?

Exercise 6.3 (P)**.** Prove Theorem 3.1 without appealing to any statement about bounded continuous test functions: work only with $C_c(X)$ and isolate the compact set on which uniform continuity is used.

7. Research laboratories

Research Laboratory 7.1 (R: Build a linearization dictionary)**.** Choose one modern paper using "linearization" in homogeneous dynamics. For one substantial argument, identify:

(1) the subgroup H;
(2) the representation V_H or its replacement;
(3) the singular set;
(4) the polynomial/non-concentration input;
(5) the properness or arithmetic-height input;
(6) the exact algebraic obstruction produced when avoidance fails.

Write the result as a two-page dependency diagram before reading the technical proof again.

Research Laboratory 7.2 (R: Qualitative-to-effective audit)**.** Take Theorem 3.1. For each existential choice in its finite-window proof, propose a quantitative parameter: injectivity radius, subgroup height, tube thickness, polynomial degree, Sobolev norm, or mixing time. Identify the first point at which the classical proof gives no usable polynomial dependence.

Research Laboratory 7.3 (R: Oppenheim bridge)**.** For an indefinite irrational ternary quadratic form Q, formulate the associated orbit of $H = \mathrm{SO}(Q)^\circ$ in $\mathrm{SL}_3(\mathbb{R})/\mathrm{SL}_3(\mathbb{Z})$. Explain precisely which proper closed homogeneous orbit would correspond to a rational algebraic obstruction. Do not invoke Oppenheim's theorem until the dynamical dictionary is complete.

APPENDIX A

Exterior Lines and Subspaces

This appendix records the linear-algebra fact used repeatedly in Chapter 4.

Lemma 0.1 (A decomposable line determines its subspace). *Let E be a finite-dimensional vector space and let $L \subset E$ be a d-dimensional subspace. Choose a basis $v_1, \dots, v_d$ of L and put*

$$p = v_1 \wedge \cdots \wedge v_d \neq 0.$$

Then

$$L = \{v \in E : v \wedge p = 0\}.$$

Consequently, if $T \in \mathrm{GL}(E)$ satisfies

$$(\bigwedge^d T)p \in \mathbb{R}^\times p,$$

then $T(L) = L$.

Proof. Every $v \in L$ is a linear combination of the v_i, so $v \wedge p = 0$. Conversely extend $v_1, \dots, v_d$ to a basis of E. Write

$$v = \sum_{i=1}^d a_i v_i + \sum_{j>d} a_j v_j.$$

Then

$$v \wedge p = \sum_{j>d} a_j v_j \wedge v_1 \wedge \cdots \wedge v_d.$$

The displayed exterior monomials are linearly independent, so $v \wedge p = 0$ implies $a_j = 0$ for all $j > d$. Hence $v \in L$.

If $(\bigwedge^d T)p = cp$ with $c \neq 0$ and $v \in L$, then

$$Tv \wedge p = c^{-1} Tv \wedge (\bigwedge^d T)p = c^{-1}(\bigwedge^{d+1} T)(v \wedge p) = 0.$$

Thus $Tv \in L$, so $T(L) \subset L$. Equality follows from dimensions. □

Lemma 0.2 (Scalar on the top exterior power). *If $T(L) = L$, then*

$$(\bigwedge^d T)p = \det(T|_L)p.$$

Proof. Write the matrix of $T|_L$ in the basis $v_1, \ldots, v_d$. Multilinearity and alternation of the wedge product give exactly the determinant expansion. $\square$

These two lemmas justify the identities

$$\mathrm{Stab}_G(\mathbb{R}p_H) = N_G(H), \qquad \mathrm{Stab}_G(p_H) = N_G^1(H)$$

whenever H is connected.

APPENDIX B

Polynomial Rescaling Calculus

The same rescaling calculation appears in Ratner shearing, Shah's polynomial trajectories, and modern effective arguments. We record a reusable elementary form.

Lemma 0.1 (Dominant monomial scale)**.** *Let*

$$P(t) = \sum_{j=0}^{D} t^j v_j$$

with coefficients in a finite-dimensional normed vector space and $v_D \neq 0$. If $\delta_T > 0$ and

$$\delta_T T^{D-1} \to c \in (0, \infty),$$

then for every fixed s,

$$P(T + s\delta_T) - P(T) \longrightarrow sDc\, v_D$$

after subtracting any lower-order terms whose rescaled limits have been normalized away. More generally, a finite sequence of rescalings extracts the first nonzero term of the Taylor expansion at infinity.

Proof. Taylor's formula is exact:

$$P(T + s\delta_T) - P(T) = \sum_{r=1}^{D} \frac{(s\delta_T)^r}{r!} P^{(r)}(T).$$

The leading contribution from $r = 1$ is

$$s\delta_T\big(DT^{D-1}v_D + O(T^{D-2})\big) \to sDc\, v_D.$$

For $r \geq 2$, the leading size is

$$O(\delta_T^r T^{D-r}) = O\big((\delta_T T^{D-1})^r T^{D-r-r(D-1)}\big),$$

and under the stated first-order normalization these are lower order in the simple model. If cancellations occur because the relevant object is a product $P^{(r)}(T)P(T)^{-1}$ rather than $P^{(r)}(T)$, one repeats the argument using the actual polynomial/rational growth exponents. Since only finitely many exponents occur, there is a maximal one and hence a rescaling that makes it order one. $\square$

Remark 0.2. The last sentence is the part used in Shah's Lemma 4.1. The matrix inverse introduces rational functions, so the correct exponent q is read from the degrees at infinity of $\Theta^{(r)}(t)\Theta(t)^{-1}$ rather than from the degree of Θ alone.

Lemma 0.3 (Conjugation by an Ad-unipotent flow)**.** *Let* $u_t = \exp(tX)$ *with* $\operatorname{ad} X$ *nilpotent of index* $r+1$*. Then for every* $Y \in \mathfrak{g}$,

$$\operatorname{Ad}(u_t)Y = \sum_{j=0}^{r} \frac{t^j}{j!}(\operatorname{ad} X)^j Y.$$

In particular the degree is the largest j *for which* $(\operatorname{ad} X)^j Y \neq 0$.

Proof. Since $\operatorname{Ad}(\exp(tX)) = \exp(t \operatorname{ad} X)$ and $\operatorname{ad} X$ is nilpotent, the exponential series terminates. $\square$

APPENDIX C

Sources and Bibliographic Notes

The exposition is organized by mechanism rather than chronology. The following sources are the principal theorem-level references.

Bibliography

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[3] M. Ratner, *On Raghunathan's measure conjecture*, Ann. of Math. (2) **134** (1991), 545–607.

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[8] D. Y. Kleinbock and G. A. Margulis, *Flows on homogeneous spaces and Diophantine approximation on manifolds*, Ann. of Math. (2) **148** (1998), 339–360.

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How the sources are used

- Ratner's one-parameter measure classification, orbit-closure, and equidistribution theorems are the historical sources for the corresponding internal arithmetic proofs in this volume; the Mautner mechanism is also proved internally.
- Dani–Margulis and Shah are the historical sources for the linearization method. The representative-properness and finite-window linearization arguments used logically in this series are proved in Chapter 5 in the arithmetic linear setting; the citations record provenance.

- Shah's work on polynomial trajectories motivates Chapter 10. The scalar/vector polynomial estimates actually invoked are proved in Volume 1 and in this volume's polynomial appendices.
- Shah's generated-unipotent work motivates Chapter 11. The finite-measure connected theorem used later is derived here from the internally proved one-parameter theorem; Shah's stronger Zariski-generation and orbit-closure forms are bibliographic context only.
- The modern effective-avoidance papers are cited only to indicate which qualitative losses later have to be quantified. No effective theorem is used as an input here.

A reference therefore has one of two roles: an explicitly named Ratner-foundation import, or historical attribution. There is no unlabeled theorem-level dependency in the principal proof chains of this volume.

Conceptual Exit Checklist

A reader ready to proceed should be able to do all of the following without relying on slogans.

(1) Explain why $\mathrm{Ad}(u_t)$ is polynomial for an Ad-unipotent one-parameter group and compute the SL_2 model explicitly.

(2) Starting from two nearby generic points, describe how polynomial divergence produces a new invariant element in a limiting measure.

(3) Define $\mathscr{H}_\Gamma$, $\mathscr{H}_\Gamma^{\mathrm{rat}}$, $N(H,W)$, $S_{\mathrm{rat}}(H,W)$, and the finite lower-incidence set; explain why quotient-level set differences cannot be manipulated naively.

(4) Construct $p_H \in \bigwedge^{\dim H} \mathfrak{g}$ and prove

$$\mathrm{Stab}_G(\mathbb{R}p_H) = N_G(H), \qquad \mathrm{Stab}_G(p_H) = N_G^1(H).$$

(5) Explain why the relevant linearization vector over $x = g\Gamma$ is a *family* $\rho_H(g\gamma)p_H$, $\gamma \in \Gamma$, and prove, in the arithmetic setting, the properness theorem controlling this family from discreteness of primitive integral exterior vectors.

(6) State and prove the arithmetic Dani–Margulis–Shah finite-window alternative with the correct order of quantifiers, and distinguish it from the stronger historical general-lattice form.

(7) Distinguish avoidance from focusing and describe how each is used in a weak-limit argument.

(8) Explain the division of labor between Ratner classification and linearization: classification lists the possible homogeneous strata; linearization controls whether an approximating trajectory can charge them.

(9) Prove the finite-measure generated-unipotent theorem: establish the nilpotent ergodic-direction lemma, explain the Mautner reduction, and derive homogeneity from the internally proved one-parameter classification theorem.

(10) Prove the time-change lemma for $t \mapsto t + st^{-q}$ and use it with the asymptotic shear lemma to derive unipotent invariance of polynomial-trajectory limits.
(11) Identify at least four places where a qualitative proof loses an effective rate.
(12) Given a new homogeneous-dynamics paper, identify its recurrence input, source of additional invariance, algebraic obstruction, and method for eliminating/focusing singular strata.

The central mental picture should be

$$\boxed{\begin{array}{c}\text{polynomial divergence} \longrightarrow \text{additional invariance} \\ \longrightarrow \text{homogeneous candidates} \\ \longrightarrow \text{linearized singular-set control.}\end{array}}$$

Volume 3

Local Fields, S-Integers, Reduction Theory, and Strong Approximation

Student orientation: arithmetic objects and local geometry

This volume supplies the local and arithmetic language used later in S-arithmetic dynamics. A *place* is an equivalence class of absolute values; S is a finite set of places containing the archimedean ones; $\mathcal{O}_S$ denotes the S-integers. An S-*lattice* is a discrete finite-covolume $\mathcal{O}_S$-module in the product of the local vector spaces. *Strong approximation* is a density/surjectivity statement for the diagonal rational group in products of local groups away from a prescribed set of places. It is stronger than the Chinese remainder theorem and is proved only in the matrix-group scopes stated here. The student should continually distinguish three kinds of compactness: compactness in one local field, S-Mahler compactness of lattices, and reduction-theoretic compactness modulo parabolic escape.

Reader guide: a concrete model

Why this matters.

Volume 3 is organized around the passage from *Local fields as geometric objects* to *Scope summary.* The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.1 (A calculation to keep in view)**.** Consider the standard unipotent and diagonal elements

$$u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad a_s = \begin{pmatrix} e^s & 0 \\ 0 & e^{-s} \end{pmatrix}.$$

A direct multiplication gives

$$a_s u_t a_{-s} = u_{e^{2s}t}.$$

Thus a displacement of size $|t|$ in the unstable horospherical direction is enlarged by e^{2s}. If $|t| = e^{-2s}$, the conjugated displacement has order one. This elementary calculation is the model for the "first visible scale" used repeatedly in nondivergence and shearing arguments.

Preface

This volume supplies the arithmetic and local-field foundations that are repeatedly used in the post-Ratner theory. The guiding principle is the same one adopted in the reconstructed first two volumes: a result is not made available to later volumes merely because it is classical. The theorem must either be proved here, have been proved earlier in the series, or belong to the small package of genuinely standard graduate background listed in Appendix A.

The mathematical transition is substantial. Over $\mathbb{R}$, a lattice can be visualized by a Euclidean fundamental parallelepiped and a unipotent trajectory by a polynomial curve. Over a nonarchimedean field, balls are simultaneously open and closed, compact subgroups replace bounded Euclidean regions, and size is measured by a discrete valuation. In an S-arithmetic product the real and nonarchimedean coordinates are coupled by the product formula. A short vector at one place can be compensated at another place, so a compactness criterion must use *content*, not one local norm.

The volume has four purposes. First, we develop local-field linear algebra and Haar measure concretely enough that later arguments do not have to appeal to analogy with the real case. Second, we prove the product formula, finiteness of the ideal class group, the Dirichlet unit theorem, and its S-unit form in the precise normalization used later. Third, we prove an S-arithmetic Mahler criterion and the reduction statements for SL_n needed by quantitative non-divergence. Fourth, we prove strong approximation in the split classical matrix cases that are immediately used in the series. The last item is deliberately scoped: we prove SL_n and Sp_{2n} over $\mathbb{Q}$ by root-group and congruence arguments. We explain the general Kneser–Platonov theorem historically, but we do not label that much deeper theorem as proved.

A reader who has completed Volume 1 can reuse Minkowski's convex-body theorem, ordinary Mahler compactness, exterior-power language, and the marked-poset theorem. We nevertheless restate the relevant forms when they enter a new arithmetic context. No Ratner theorem is used in this volume.

Scope and prerequisites

Prerequisites. The only imported material consists of elementary graduate algebra, basic number-field theory (existence of the ring of integers and factorization of ideals), Haar measure on locally compact groups, and the real geometry-of-numbers results already proved in Volume 1. In particular, Dirichlet's unit theorem, the S-unit theorem, S-Mahler compactness, and the strong-approximation statements below are *not* imported.

Headline results proved in the volume

(1) the product formula and Dirichlet's unit theorem, followed by the full S-unit theorem;
(2) finiteness of the ideal class group by Minkowski's method;
(3) the S-arithmetic Mahler compactness criterion for unimodular S-lattices;
(4) the reduction theorem for $\mathrm{SL}_n(\mathcal{O}_S)$ in the form required by later non-divergence and cusp arguments;
(5) one- and several-variable nonarchimedean polynomial sublevel estimates and product-goodness lemmas;
(6) strong approximation for SL_n and Sp_{2n} over $\mathbb{Q}$ away from a finite set of places, proved from elementary/root unipotents and Chinese-remainder lifting.

Deliberate boundary of the theorem

The full strong-approximation theorem for every simply connected isotropic almost-simple group over an arbitrary number field is not asserted as proved. It requires substantial structure theory (Kneser–Tits, Bruhat–Tits and their global input). When later volumes need a form outside the matrix groups proved here, that form must be proved before use. This prevents the phrase "by strong approximation" from becoming an invisible logical dependency.

CHAPTER 1

Local fields as geometric objects

A local field is not merely a coefficient field. Its topology controls every later recurrence and non-divergence estimate. We begin with the pieces that will actually be used.

1. Archimedean and nonarchimedean local fields

Definition 1.1. A *local field* is a nondiscrete locally compact topological field. In characteristic zero it is isomorphic to $\mathbb{R}$, $\mathbb{C}$, or a finite extension of $\mathbb{Q}_p$. We normalize the nonarchimedean absolute value by

$$|\varpi| = q^{-1},$$

where ϖ is a uniformizer and q is the cardinality of the residue field.

For a nonarchimedean field k put

$$\mathcal{O}_k = \{x : |x| \leq 1\}, \qquad \mathfrak{p} = \{x : |x| < 1\}.$$

Every $x \neq 0$ has a unique form $x = \varpi^m u$ with $u \in \mathcal{O}_k^\times$ and $m \in \mathbb{Z}$. Thus $k^\times \cong \varpi^{\mathbb{Z}} \times \mathcal{O}_k^\times$.

Lemma 1.2 (Ultrametric geometry)**.** *If $|\cdot|$ is nonarchimedean, then:*

(1) if $|x - y| < |y - z|$, then $|x - z| = |y - z|$;
(2) two balls are either disjoint or one contains the other;
(3) every point of a ball is a center of that ball;
(4) a ball of radius q^m is compact and open.

PROOF. The first assertion follows by applying $|a + b| \leq \max(|a|, |b|)$ twice. If two balls meet, choose a common point z. For x in the smaller-radius ball, the ultrametric inequality bounds the distance from x to the other center by the larger radius, proving containment. The third assertion is the same calculation. Finally, a ball of radius q^m is a translate and dilation of $\mathcal{O}_k$; the latter is the inverse limit of the finite rings $\mathcal{O}_k/\mathfrak{p}^r$, hence compact, and is open by definition of the valuation topology. $\square$

2. Norms in finite-dimensional spaces

On k^d we use the max norm $\|x\| = \max_i |x_i|$. In a nonarchimedean field this norm satisfies the strong triangle inequality. If $T \in \mathrm{Mat}_d(k)$, then

$$\|T\|_{\mathrm{op}} = \max_{i,j} |T_{ij}|.$$

Indeed, the upper bound is immediate, while equality is obtained by evaluating on a coordinate vector corresponding to a largest entry.

Lemma 2.1 (Equivalence of norms)**.** *Any two norms on a finite-dimensional vector space over a local field are equivalent.*

PROOF. Choose a basis and compare with the max norm. The unit sphere for the max norm is compact: in the archimedean case this is Heine–Borel, and in the nonarchimedean case it is a closed subset of $\mathcal{O}_k^d$. A continuous norm has positive minimum and finite maximum on this sphere. Homogeneity gives the claim. □

3. Why ultrametric geometry helps

In real non-divergence arguments one expends effort controlling overlap of Euclidean balls. Over $\mathbb{Q}_p$ the nesting property is stronger: maximal balls in a family are automatically disjoint. On the other hand, differentiability gives much less information because ordinary derivatives can vanish for reasons invisible to size. Interpolation, rather than the mean-value theorem, will be the right analytic tool.

CHAPTER 2

Haar integration and ultrametric covering

1. Normalization and scaling

Let k be a nonarchimedean local field and normalize additive Haar measure m by $m(\mathcal{O}_k) = 1$.

Proposition 1.1 (Scaling of Haar measure)**.** *For $a \in k^\times$ and measurable $E \subset k$,*

$$m(aE) = |a|\, m(E).$$

Consequently $m(B(x, q^r)) = q^r$ with our normalization.

Proof. The measure $E \mapsto m(aE)$ is another additive Haar measure, so it equals $c(a)m(E)$. The map $c : k^\times \to \mathbb{R}_{>0}$ is a continuous homomorphism. Since multiplication by a unit preserves $\mathcal{O}_k$, $c(u) = 1$ for $u \in \mathcal{O}_k^\times$. Since $\mathcal{O}_k$ is the disjoint union of q cosets of $\varpi\mathcal{O}_k$, we have $1 = q\, m(\varpi\mathcal{O}_k)$, hence $c(\varpi) = q^{-1} = |\varpi|$. Decomposition $a = \varpi^r u$ finishes the proof. ☐

For k^d we take product Haar measure. A max-norm ball is a Cartesian product of d one-dimensional balls, hence

$$m_d(B(x, q^r)) = q^{rd}.$$

2. A covering lemma stronger than Besicovitch

Lemma 2.1 (Ultrametric maximal-ball selection)**.** *Let $\mathscr{B}$ be a family of balls contained in a fixed compact ball B_0. There is a pairwise disjoint subfamily of maximal balls whose union equals the union of $\mathscr{B}$.*

Proof. The possible radii form a discrete set. Choose all balls of largest occurring radius, discard those contained in a chosen ball, and continue down through the radii. Two chosen balls that meet are comparable by Lemma 1.2; maximality then forces equality, so the chosen distinct balls are disjoint. Every discarded ball was contained in one selected earlier. ☐

Corollary 2.2 (Hardy–Littlewood maximal inequality for nested balls)**.** *For $f \in L^1(k^d)$ and $\lambda > 0$,*

$$m\left\{x : \sup_{B \ni x} \frac{1}{m(B)} \int_B |f| > \lambda\right\} \le \lambda^{-1} \|f\|_1.$$

Proof. For a compactly supported simple f, select maximal balls on which the average exceeds λ. They are disjoint and cover the level set, so

$$\lambda \sum_B m(B) < \sum_B \int_B |f| \leq \|f\|_1.$$

Approximation and monotone convergence give the general case. $\square$

This sharp overlap constant is one reason the p-adic part of quantitative non-divergence is often cleaner than the real part once the correct polynomial sublevel estimate is available.

CHAPTER 3

Workshop: thinking ultrametrically

The formal statements about local fields become much easier to use once one stops trying to picture a p-adic ball as a distorted Euclidean ball. This chapter collects calculations that will be used implicitly throughout the S-arithmetic part of the series. None of them is deep, but together they explain why many covering and recurrence arguments are cleaner at a finite place than at the real place.

1. Every point is a center

Let $B(a, r) = \{x : |x - a|_p \leq r\}$. If $b \in B(a, r)$, then

$$B(a, r) = B(b, r).$$

Indeed, if $x \in B(a, r)$, the ultrametric inequality gives

$$|x - b|_p \leq \max(|x - a|_p, |a - b|_p) \leq r,$$

and the reverse inclusion is symmetric. Thus a p-adic ball has no privileged geometric center. This simple fact is responsible for two features that recur later:

(1) intersecting balls of different radii are nested;
(2) translating an averaging ball by an element of that ball does not change it.

The second property is exactly what makes averages over $p^{-m}\mathbb{Z}_p$ genuine orthogonal projections in Volume 4.

2. Shells and elementary integrals

Normalize Haar measure on $\mathbb{Q}_p$ by $m(\mathbb{Z}_p) = 1$. Since

$$p^j\mathbb{Z}_p \setminus p^{j+1}\mathbb{Z}_p = \{x : \mathrm{ord}_p(x) = j\},$$

we have

$$m\{x : \mathrm{ord}_p(x) = j\} = p^{-j} - p^{-j-1} = (1 - p^{-1})p^{-j}.$$

Consequently, for $s > -1$,

$$\int_{\mathbb{Z}_p} |x|_p^s \, dx = \sum_{j \geq 0} p^{-js}(1 - p^{-1})p^{-j}$$
$$= \frac{1 - p^{-1}}{1 - p^{-(s+1)}}.$$

This calculation is the local prototype for Euler factors. More importantly for dynamics, it illustrates a general method: decompose according to valuation rather than according to Euclidean size.

3. A compact fundamental domain for $\mathbb{Q}_p/\mathbb{Z}[1/p]$ is impossible

It is worth seeing why the S-arithmetic quotient must be treated diagonally. The subgroup $\mathbb{Z}[1/p]$ is dense neither in $\mathbb{R}$ nor in $\mathbb{Q}_p$, but it is discrete in

$$\mathbb{R} \times \mathbb{Q}_p$$

under the diagonal embedding. A fundamental set can be built as follows. Given (x_∞, x_p), choose $a \in \mathbb{Z}[1/p]$ so that $x_p - a \in \mathbb{Z}_p$. This determines a modulo $\mathbb{Z}$. Change a by an integer so that $x_\infty - a \in [0, 1)$. Hence every diagonal coset meets

$$[0, 1) \times \mathbb{Z}_p.$$

The latter is compact. If two points of this set differ by an element of $\mathbb{Z}[1/p]$, their p-adic difference is integral and their real difference has absolute value < 1; the only possible ambiguity is on the real boundary. Thus $[0, 1) \times \mathbb{Z}_p$ is a measurable compact fundamental domain up to its boundary. This is the rank-one model for the compactness of additive unipotent quotients used later for parabolic radicals.

4. Why a maximal inequality has constant one

Let $f \geq 0$ be integrable on $\mathbb{Q}_p^d$. Suppose the maximal average of f is larger than λ at every point of a measurable set E. For each $x \in E$ choose a ball $B_x \ni x$ with

$$\frac{1}{m(B_x)} \int_{B_x} f > \lambda.$$

Inside a fixed compact ambient ball, choose the balls of maximal radius and then continue downward. Because intersecting balls are nested, the selected balls are disjoint and cover E. Therefore

$$\lambda m(E) \leq \lambda \sum_j m(B_j) < \sum_j \int_{B_j} f \leq \|f\|_1.$$

There is no Besicovitch multiplicity constant. Compare this with the Euclidean proof in Volume 1: there one pays a dimension-dependent overlap constant because intersecting balls need not be nested.

5. Polynomial growth is valuation growth

Consider

$$P(T) = a_0 + a_1 T + \cdots + a_d T^d, \qquad a_d \neq 0.$$

For sufficiently large $|T|_p$ one term dominates all the others. More precisely, if

$$|T|_p > \max_{j<d} |a_j/a_d|_p^{1/(d-j)},$$

then

$$|a_d T^d|_p > |a_j T^j|_p \quad (j < d),$$

so the ultrametric inequality gives the exact identity

$$|P(T)|_p = |a_d|_p |T|_p^d.$$

There is no cancellation once the leading term strictly dominates. This is the basic reason persistent bounded polynomial motion forces exact invariance in the nonescape argument of Volume 4.

6. A matrix example

Take

$$u(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad a_k = \begin{pmatrix} p^k & 0 \\ 0 & p^{-k} \end{pmatrix}.$$

Then

$$a_k u(t) a_k^{-1} = \begin{pmatrix} 1 & p^{2k} t \\ 0 & 1 \end{pmatrix}.$$

With the convention $|p|_p = p^{-1}$, the coefficient p^{2k} has norm p^{-2k}. Thus conjugation by a_k contracts the upper root group as $k \to +\infty$. Conversely the lower root group is expanded. The same root weight calculation underlies the p-adic Mautner phenomenon, the local H-property, and the Cartan decomposition.

7. What should become automatic

A reader should leave this chapter with three reflexes.

(1) Replace Euclidean "closeness" by valuation inequalities.

(2) When balls meet, ask immediately which contains the other.

(3) When a polynomial is bounded on larger and larger p-adic balls, inspect its highest nonzero coefficient: boundedness forces that coefficient to vanish.

These reflexes turn many apparently exotic local-field arguments into direct analogues of their real counterparts, with fewer covering complications.

CHAPTER 4

Number fields, places, and the product formula

Let K be a number field of degree $n = r_1 + 2r_2$. We write M_K for its places. At a real place v put $|a|_v = |\sigma_v(a)|$; at a complex place put $|a|_v = |\sigma_v(a)|^2$; and at a finite place corresponding to a prime ideal $\mathfrak{p}$ put

$$|a|_{\mathfrak{p}} = N\mathfrak{p}^{-\operatorname{ord}_{\mathfrak{p}}(a)}.$$

With these conventions no additional local-degree exponents appear in the product formula.

1. Norms and prime ideals

For $a \in K^\times$,

$$(a) = \prod_{\mathfrak{p}} \mathfrak{p}^{\operatorname{ord}_{\mathfrak{p}}(a)}$$

has only finitely many nonzero exponents. Taking ideal norms gives

$$|N_{K/\mathbb{Q}}(a)| = \prod_{\mathfrak{p}} N\mathfrak{p}^{\operatorname{ord}_{\mathfrak{p}}(a)}.$$

On the other hand, by the definition of field norm,

$$|N_{K/\mathbb{Q}}(a)| = \prod_{v|\infty} |a|_v.$$

THEOREM 1.1 (Product formula). *For every $a \in K^\times$,*

$$\prod_{v \in M_K} |a|_v = 1.$$

PROOF. By the two displayed formulas,

$$\prod_{v|\infty} |a|_v = \prod_{\mathfrak{p}} N\mathfrak{p}^{\operatorname{ord}_{\mathfrak{p}}(a)} = \left(\prod_{\mathfrak{p}} |a|_{\mathfrak{p}} \right)^{-1}.$$

Multiplication gives one. $\square$

2. S-integers

Fix a finite set $S \subset M_K$ containing all archimedean places. Put

$$K_S = \prod_{v \in S} K_v, \qquad \mathcal{O}_S = \{a \in K : |a|_v \le 1 \text{ for every finite } v \notin S\}.$$

Thus $\mathcal{O}_S$ is obtained from $\mathcal{O}_K$ by inverting the prime ideals belonging to $S_f = S \setminus S_\infty$.

Lemma 2.1 (Diagonal discreteness)**.** *The diagonal embedding $\mathcal{O}_S \hookrightarrow K_S$ is discrete. More generally, $\mathcal{O}_S^d$ is a discrete subgroup of K_S^d.*

Proof. Suppose $a \in \mathcal{O}_S$ is nonzero and $|a|_v < 1$ at every $v \in S$ after choosing sufficiently small fixed neighborhoods of zero. At $v \notin S$ we have $|a|_v \le 1$. Hence $\prod_v |a|_v < 1$, contradicting the product formula. Therefore a sufficiently small product neighborhood of zero contains no nonzero S-integer. The vector statement follows coordinatewise. $\square$

Lemma 2.2 (Cocompactness in the additive group)**.** *The quotient $K_S/\mathcal{O}_S$ is compact.*

Proof. At every finite $v \in S_f$, subtract an element of K with the prescribed principal part modulo $\mathcal{O}_v$; the Chinese remainder theorem for fractional ideals allows these finitely many principal parts to be matched simultaneously by an element of $\mathcal{O}_S$. After this subtraction the finite coordinates lie in $\mathcal{O}_v$. At the archimedean coordinates, subtract an ordinary algebraic integer so that the Minkowski embedding lies in a fixed fundamental parallelepiped for $\mathcal{O}_K$. Thus every coset has a representative in the compact set

$$P_\infty \times \prod_{v \in S_f} \mathcal{O}_v.$$

$\square$

The proof exhibits the basic S-arithmetic pattern: finite-place denominators are corrected first, and the remaining archimedean error is corrected by an integral lattice.

CHAPTER 5

Minkowski theory and finiteness of the ideal class group

The Dirichlet unit theorem is often quoted before any dynamical argument begins. Here we prove the geometry-of-numbers input that makes it work.

1. The Minkowski embedding

Let

$$\Sigma : K \longrightarrow \mathbb{R}^{r_1} \times \mathbb{C}^{r_2} \cong \mathbb{R}^n$$

be the product of the archimedean embeddings, taking one representative from each conjugate pair. If I is a nonzero fractional ideal, $\Sigma(I)$ is a full lattice.

Proposition 1.1 (Covolume of an ideal lattice)**.** *With Lebesgue measure on $\mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$, where each complex coordinate has ordinary two-dimensional area,*

$$\operatorname{covol}(\Sigma(I)) = 2^{-r_2}|D_K|^{1/2}N(I).$$

Proof. Choose a $\mathbb{Z}$-basis $\alpha_1, \ldots, \alpha_n$ of I. The square of the determinant of the full matrix of all real and complex embeddings is the discriminant of this basis. Passing from a pair $(z, \bar{z})$ to $(\Re z, \Im z)$ contributes a factor 2^{-1} to absolute determinant for each complex pair. Thus

$$\operatorname{covol}(\Sigma(I)) = 2^{-r_2}|\operatorname{disc}(\alpha_1, \ldots, \alpha_n)|^{1/2}.$$

For an integral ideal I, its discriminant is $D_K N(I)^2$, because the change-of-basis determinant from a basis of $\mathcal{O}_K$ has absolute value $[\mathcal{O}_K : I] = N(I)$. Scaling handles fractional ideals. $\square$

2. Minkowski's ideal bound

Let

$$B_R = \{(x_i, z_j) : |x_i| \leq R_i,\ |z_j| \leq R_j\}.$$

Its volume is $2^{r_1}\pi^{r_2}\prod_i R_i \prod_j R_j^2$.

THEOREM 2.1 (Minkowski ideal-class bound). *Every ideal class contains an integral ideal* $\mathfrak{b}$ *with*

$$N\mathfrak{b} \leq C_K := \frac{n!}{n^n}\left(\frac{4}{\pi}\right)^{r_2} |D_K|^{1/2}.$$

The precise constant is not important later; boundedness in terms of K is.

PROOF. Let $\mathfrak{a}$ be an integral ideal. Apply Minkowski's convex-body theorem from Volume 1 to the lattice $\Sigma(\mathfrak{a})$ and to the standard convex body

$$\left\{(x, z) : \sum_i |x_i| + 2\sum_j |z_j| \leq T\right\}.$$

Its volume is

$$\frac{2^{r_1}(2\pi)^{r_2}}{n!}T^n.$$

Choose T so that this volume exceeds $2^n \operatorname{covol}(\Sigma(\mathfrak{a}))$. We obtain $0 \neq \alpha \in \mathfrak{a}$ with

$$\sum_i |\sigma_i(\alpha)| + 2\sum_j |\tau_j(\alpha)| \leq T.$$

By the arithmetic-geometric mean inequality applied to the n nonnegative numbers consisting of the real absolute values and each complex absolute value repeated twice,

$$|N_{K/\mathbb{Q}}(\alpha)| \leq (T/n)^n \leq C_K N\mathfrak{a}.$$

Since $(\alpha) \subset \mathfrak{a}$, the ideal $\mathfrak{b} = (\alpha)\mathfrak{a}^{-1}$ is integral, lies in the inverse ideal class of $\mathfrak{a}$, and

$$N\mathfrak{b} = |N(\alpha)|/N\mathfrak{a} \leq C_K.$$

Replacing $\mathfrak{a}$ by an ideal in the inverse class proves the theorem. □

Corollary 2.2 (Finiteness of the class group)**.** *The ideal class group of K is finite.*

PROOF. For any $C > 0$ there are only finitely many integral ideals of norm at most C: an ideal of norm m divides the principal ideal (m), and there are finitely many divisors of each of the finitely many (m) with $m \leq C$. Theorem 2.1 supplies a bounded-norm representative for every class. □

CHAPTER 6

Dirichlet units and S-unit balancing

1. The logarithmic embedding

Let S_∞ be the archimedean places and define

$$\ell_\infty : K^\times \to \mathbb{R}^{S_\infty}, \qquad \ell_\infty(a) = (\log |a|_v)_{v|\infty}.$$

For a unit $u \in \mathcal{O}_K^\times$, the product formula gives

$$\sum_{v|\infty} \log |u|_v = 0.$$

Let H_∞ denote this hyperplane.

Lemma 1.1 (Discreteness of logarithms)**.** *The subgroup $\ell_\infty(\mathcal{O}_K^\times) \subset H_\infty$ is discrete, and the kernel of ℓ_∞ is the finite group of roots of unity in K.*

Proof. If $\ell_\infty(u)$ lies in a fixed compact set, every archimedean conjugate of u and of u^{-1} is bounded. The coefficients of the monic minimal polynomial of u are elementary symmetric polynomials in its conjugates, hence are bounded integers. Only finitely many such polynomials occur, and each has finitely many roots. Thus only finitely many units have logarithm in a fixed compact set, proving discreteness.

If $\ell_\infty(u) = 0$, all conjugates of the algebraic integer u have modulus one. The same bounded-coefficient argument applied to all powers u^m shows that two powers coincide, hence u is a root of unity. Conversely roots of unity clearly lie in the kernel. □

2. Cocompactness of the logarithmic lattice

Theorem 2.1 (Dirichlet unit theorem). *The image $\ell_\infty(\mathcal{O}_K^\times)$ is a full lattice in H_∞. Consequently*

$$\mathcal{O}_K^\times \cong \mu(K) \times \mathbb{Z}^{r_1+r_2-1}.$$

Proof. Discreteness was proved above. It remains to show that every point of H_∞ lies a bounded distance from the logarithm of a unit.

Fix $y = (y_v) \in H_\infty$. For a large constant $A = A(K)$ consider the symmetric convex body in the Minkowski space defined by

$$|\sigma_v(x)| \le Ae^{y_v} \quad (v \text{ real}), \qquad |\tau_v(x)| \le Ae^{y_v/2} \quad (v \text{ complex}).$$

Because $\sum_v y_v = 0$, its volume is independent of y up to the fixed factor A^n. Choose A so that Minkowski's theorem guarantees a nonzero $\alpha_y \in \mathcal{O}_K$ in this body. Hence

$$\log |\alpha_y|_v \le y_v + C_K$$

at every archimedean place. Since the sum of these logarithms is $\log |N(\alpha_y)| \ge 0$, the same inequalities imply a two-sided bound

$$|\log |\alpha_y|_v - y_v| \le C'_K$$

after enlarging the constant: if one coordinate were much smaller, the fixed sum and the upper bounds on all other coordinates would be violated.

Moreover

$$|N(\alpha_y)| = \prod_{v|\infty} |\alpha_y|_v \le e^{|S_\infty| C_K},$$

so the principal ideals (α_y) range over only finitely many integral ideals. For each principal ideal I in this finite list choose one generator β_I. If $(\alpha_y) = I$, then $u_y = \alpha_y/\beta_I$ is a unit. The finitely many vectors $\ell_\infty(\beta_I)$ are bounded, therefore

$$\|\ell_\infty(u_y) - y\|_\infty \le C''_K.$$

Thus $H_\infty/\ell_\infty(\mathcal{O}_K^\times)$ is bounded. A quotient of a finite-dimensional vector space by a discrete subgroup is compact exactly when the subgroup spans the whole space; hence the image is a full lattice of rank $\dim H_\infty = r_1 + r_2 - 1$. ☐

3. Adding finite places

For $S \supset S_\infty$ define

$$\ell_S : \mathcal{O}_S^\times \to H_S := \left\{ (t_v)_{v\in S} : \sum_{v\in S} t_v = 0 \right\}, \qquad \ell_S(u) = (\log |u|_v)_{v\in S}.$$

THEOREM 3.1 (S-unit theorem). *The image $\ell_S(\mathcal{O}_S^\times)$ is a full lattice in H_S. Hence*

$$\mathcal{O}_S^\times \cong \mu(K) \times \mathbb{Z}^{r_1+r_2-1+|S_f|}.$$

PROOF. Consider the valuation map

$$\nu : \mathcal{O}_S^\times \longrightarrow \mathbb{Z}^{S_f}, \qquad u \mapsto (\mathrm{ord}_{\mathfrak{p}}(u))_{\mathfrak{p}\in S_f}.$$

Its kernel is $\mathcal{O}_K^\times$. The cokernel is finite: a vector $(m_{\mathfrak{p}})$ occurs exactly when the fractional ideal $\prod_{\mathfrak{p}\in S_f} \mathfrak{p}^{m_{\mathfrak{p}}}$ is principal. The map from $\mathbb{Z}^{S_f}$ to the finite class group has this ideal class as its value, so its kernel, which is $\nu(\mathcal{O}_S^\times)$, has finite index. Therefore

$$\operatorname{rank}\mathcal{O}_S^\times = \operatorname{rank}\mathcal{O}_K^\times + |S_f|.$$

The logarithmic image is discrete by the same bounded-conjugate argument as before, together with bounded valuations at the finitely many finite places. It lies in H_S by the product formula, and its rank equals $\dim H_S$. A discrete subgroup of a finite-dimensional real vector space whose rank equals the dimension need not automatically span, but here its projection to the finite-place coordinates has full rank $|S_f|$, while the kernel of that projection is the full lattice from Theorem 2.1 in the archimedean hyperplane. Hence the real span is all of H_S, so the image is a full lattice. □

Corollary 3.2 (S-unit balancing)**.** *There exists $C = C(K,S) \geq 1$ such that for every tuple of positive numbers $(a_v)_{v\in S}$ there is $u \in \mathcal{O}_S^\times$ for which*

$$C^{-1}A \leq |u|_v a_v \leq CA \qquad (v \in S), \qquad A = \left(\prod_{v\in S} a_v\right)^{1/|S|}.$$

Proof. Subtract the mean from $(\log a_v)$ to obtain a point of H_S. Since $\ell_S(\mathcal{O}_S^\times)$ is cocompact, choose u whose logarithm differs from its negative by a uniformly bounded vector. Exponentiation gives the claim. □

Worked Example 3.3 ($\mathbb{Q}$ with two finite places)**.** For $S = \{\infty, p, q\}$,

$$\mathbb{Z}[1/pq]^\times = \{\pm p^m q^n : m, n \in \mathbb{Z}\}.$$

The logarithmic vectors of p and q are

$$(\log p, -\log p, 0), \qquad (\log q, 0, -\log q),$$

which visibly span the plane $t_\infty + t_p + t_q = 0$.

CHAPTER 7

Workshop: the geometry behind the S-unit theorem

The logarithmic formulation of the unit theorem is simple enough to memorize, but its geometric content is easy to lose. This chapter works through the argument at a slower pace and isolates the three facts that later reappear whenever we balance local norms.

1. Why the product formula creates a hyperplane

Take a number field K with normalized absolute values. For $a \in K^\times$, the vector

$$\ell_S(a) = (\log |a|_v)_{v \in S}$$

has coordinate sum zero whenever a is an S-unit. Thus the only possible logarithmic directions live in

$$H_S = \{(t_v) : \sum_{v \in S} t_v = 0\}.$$

The point is not merely that the image lies in this hyperplane. The unit theorem says that, modulo a bounded error, *every* redistribution of size among the places with product one is realized by multiplication by an arithmetic unit.

For $K = \mathbb{Q}$ and $S = \{\infty, p_1, \dots, p_s\}$, this is completely visible:

$$\mathbb{Z}_S^\times = \{\pm p_1^{m_1} \cdots p_s^{m_s} : m_i \in \mathbb{Z}\},$$

and the logarithms of the p_i form a lattice basis of H_S. The general theorem says that the same picture survives after archimedean embeddings and ideal classes are added.

2. A complete rank calculation

Let r_1 and r_2 be the numbers of real and complex places. The archimedean logarithmic space has dimension $r_1 + r_2$, and the product formula removes one dimension. Each additional finite place contributes one independent valuation direction. Hence the expected rank is

$$r_1 + r_2 - 1 + |S_f|.$$

We now see explicitly why no further relation can occur.

Let
$$\nu : \mathcal{O}_S^\times \to \mathbb{Z}^{S_f}, \qquad u \mapsto (v_\mathfrak{p}(u))_{\mathfrak{p}\in S_f}.$$
Its kernel is $\mathcal{O}_K^\times$. The image consists of exponent vectors whose fractional ideal $\prod_{\mathfrak{p}\in S_f} \mathfrak{p}^{m_\mathfrak{p}}$ is principal. Thus the quotient of $\mathbb{Z}^{S_f}$ by the image injects into the ideal class group. Since Minkowski's ideal bound proved earlier makes the class group finite, $\nu(\mathcal{O}_S^\times)$ has finite index. Therefore its real span is all of $\mathbb{R}^{S_f}$. Together with the archimedean unit lattice, this proves that the full logarithmic image spans H_S.

Notice what has *not* been used: there is no analytic continuation, no regulator formula, and no class-number asymptotic. The only global arithmetic input is finiteness of the class group, itself obtained from Minkowski geometry.

3. Balancing a vector place by place

Suppose $x = (x_v)_{v\in S}$ is nonzero at every local factor and write
$$a_v = \|x_v\|_v, \qquad A = c(x)^{1/|S|}.$$
The vector
$$y_v = \log a_v - \log A$$
lies in H_S. Choose a logarithmic unit vector $\ell_S(u)$ within bounded distance of $-y$. Then
$$\log \frac{\|ux_v\|_v}{A} = y_v + \log |u|_v$$
is uniformly bounded. Exponentiating gives
$$C^{-1}A \le \|ux_v\|_v \le CA.$$
This one-line argument is the reason the unit theorem appears repeatedly in homogeneous dynamics: a product bound can be converted into simultaneous local bounds.

Worked Example 3.1 (Balancing in $\mathbb{Q}(\sqrt{2})$)**.** The unit $\varepsilon = 1 + \sqrt{2}$ has conjugates $1 + \sqrt{2}$ and $1 - \sqrt{2}$, whose absolute values multiply to one. Thus
$$\ell(\varepsilon) = (\log(1+\sqrt{2}), -\log(1+\sqrt{2})).$$
If a vector is huge in the first real embedding and tiny in the second, multiplying by a suitable power ε^m moves logarithmic mass from one coordinate to the other until the two norms are comparable. The integer m is simply the nearest lattice point to the desired displacement on the one-dimensional logarithmic line.

4. What breaks without the product formula

If scalars at different places could be changed independently, balancing would be trivial and arithmetic would disappear. If there were an additional linear relation among the logarithms, some local size profiles could never be balanced. The exact codimension-one relation of the product formula is therefore optimal: it removes the global scaling direction and leaves precisely enough units to control the remaining anisotropy.

Research Laboratory 4.1 (A regulator calculation)**.** For a real quadratic field $K = \mathbb{Q}(\sqrt{d})$, choose a fundamental unit $\varepsilon > 1$. Show directly that the covolume of the archimedean unit lattice in H_∞ is $\sqrt{2}\log\varepsilon$ for the Euclidean metric induced from $\mathbb{R}^2$. Explain why the numerical value of the regulator never enters the qualitative balancing lemma, whereas an effective version of reduction theory would have to track it.

CHAPTER 8

S-lattices, content, and primitive modules

From this chapter through the strong-approximation chapters we take

$$S = \{\infty, p_1, \dots, p_s\}, \qquad \mathbb{Q}_S = \mathbb{R} \times \prod_{p \in S_f} \mathbb{Q}_p, \qquad \mathbb{Z}_S = \mathbb{Z}[1/(p_1 \cdots p_s)].$$

This rational case is the one used in the next volume. The number-field unit theory above explains what changes in general.

1. Unimodular S-lattices

An S-lattice of rank d is a discrete cocompact $\mathbb{Z}_S$-submodule $\Lambda \subset \mathbb{Q}_S^d$ of the form $g\mathbb{Z}_S^d$ with $g \in \mathrm{GL}_d(\mathbb{Q}_S)$. It is *unimodular* when

$$\prod_{v \in S} |\det g_v|_v = 1.$$

The space of unimodular S-lattices is

$$\Omega_{S,d} = \mathrm{GL}_d^1(\mathbb{Q}_S)/\,\mathrm{GL}_d(\mathbb{Z}_S), \qquad \mathrm{GL}_d^1(\mathbb{Q}_S) = \left\{ g : \prod_v |\det g_v|_v = 1 \right\}.$$

For the special-linear space we use $X_{S,d} = \mathrm{SL}_d(\mathbb{Q}_S)/\,\mathrm{SL}_d(\mathbb{Z}_S)$.

For $x = (x_v)_{v \in S}$ define its *content*

$$c(x) = \prod_{v \in S} \|x_v\|_v.$$

If $u \in \mathbb{Z}_S^\times$, then $c(ux) = c(x)$ by the product formula. Thus content is the correct size of a projective S-integral direction.

2. Primitive submodules

Because $\mathbb{Z}_S$ is a PID, a rank-r submodule $\Delta \subset \mathbb{Z}_S^d$ is primitive iff $\mathbb{Z}_S^d/\Delta$ is torsion-free, equivalently iff it is a direct summand. If $v_1, \dots, v_r$ is a basis, define

$$c(g\Delta) = c(g(v_1 \wedge \cdots \wedge v_r)).$$

Changing the basis multiplies the wedge by a unit of $\mathbb{Z}_S$, so the content is well defined.

Lemma 2.1 (Primitive saturation). *Every nonzero $x \in \mathbb{Z}_S^d$ can be written $x = av$ with $a \in \mathbb{Z}_S$ and v primitive. Moreover $c(a) \geq 1$, with equality exactly when a is an S-unit.*

PROOF. Take a to be a greatest common divisor of the coordinates in the PID $\mathbb{Z}_S$. The coordinates of $v = x/a$ generate the unit ideal, hence v is primitive. Since $a \in \mathbb{Z}_S$, at every $q \notin S$ one has $|a|_q \leq 1$. The product formula gives

$$c(a) = \prod_{v \in S} |a|_v = \left(\prod_{q \notin S} |a|_q \right)^{-1} \geq 1.$$

Equality means $|a|_q = 1$ for all $q \notin S$, exactly the S-unit condition. □

3. An S-arithmetic Blichfeldt lemma

Normalize additive Haar measure on $\mathbb{Q}_S^d$ so that the compact quotient $\mathbb{Q}_S^d/\mathbb{Z}_S^d$ has measure one.

Lemma 3.1 (S-Blichfeldt). *If a measurable set $E \subset \mathbb{Q}_S^d$ has measure greater than the covolume of an S-lattice Λ, then there are distinct $x, y \in E$ with $x - y \in \Lambda$.*

PROOF. Let F be a measurable fundamental domain for Λ. Write

$$\int_F \sum_{\lambda \in \Lambda} 1_E(z + \lambda)\, dz = m(E).$$

If every translate class met E at most once, the integrand would be at most one and the right side at most $m(F) = \mathrm{covol}(\Lambda)$. Hence some class contains two points. □

Corollary 3.2 (S-Minkowski short vector). *There is $C = C(S,d)$ such that every unimodular S-lattice contains $0 \neq x$ with $c(x) \leq C$.*

PROOF. Choose radii R_v and the symmetric product body $B = \prod_v \{x_v : \|x_v\|_v \leq R_v\}$ so that $m(B) > 2^d$ and $\prod_v R_v = C_0$ is fixed. Apply Lemma 3.1 to $B/2$ in the real factor and B itself in the ultrametric factors. The difference of two points lies in B and is a nonzero lattice vector. Its content is at most C_0. □

CHAPTER 9

Moving the finite places into a compact subgroup

A useful feature of SL_d is that the nonarchimedean coordinates can be reduced to maximal compact subgroups by an *exact global change of basis*. Once this is done, all noncompactness is visible at the real place. This observation gives a particularly transparent proof of the S-Mahler criterion and is also the elementary prototype of strong approximation.

Throughout this chapter $S = \{\infty, p_1, \ldots, p_s\}$ and $\mathbb{Z}_S = \mathbb{Z}[1/p_1 \cdots p_s]$.

1. Elementary matrices over a local field

For $i \neq j$ write

$$x_{ij}(a) = I + aE_{ij}.$$

Lemma 1.1 (Gaussian elimination over a field). *For every field F and $d \geq 2$, the group $\mathrm{SL}_d(F)$ is generated by the matrices $x_{ij}(a)$, $a \in F$.*

PROOF. Left multiplication by $x_{ij}(a)$ adds a times row j to row i. Starting with the first column of $g \in \mathrm{SL}_d(F)$, choose a nonzero entry and move it to the first row. A row interchange in positions i, j is, up to multiplication by a diagonal matrix of determinant one, a product of three elementary matrices:

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = x_{12}(1)x_{21}(-1)x_{12}(1).$$

The determinant-one diagonal block

$$h(u) = \begin{pmatrix} u & 0 \\ 0 & u^{-1} \end{pmatrix}$$

is also elementary, since

$$h(u) = x_{12}(u)x_{21}(-u^{-1})x_{12}(u)x_{12}(-1)x_{21}(1)x_{12}(-1).$$

Thus we may normalize a nonzero pivot to one and clear the rest of its row and column. The lower-right block still has determinant one. Induction on d finishes the proof. $\square$

2. Simultaneous scalar approximation at the places of S

Lemma 2.1 (Approximation by S-integers)**.** *Let $a_i \in \mathbb{Q}_{p_i}$ and let $N_i \geq 1$. There exists $a \in \mathbb{Z}_S$ such that*

$$|a - a_i|_{p_i} \leq p_i^{-N_i} \qquad (1 \leq i \leq s).$$

Moreover, finitely many such systems can be solved simultaneously.

Proof. Choose $r_i \geq 0$ so that $p_i^{r_i} a_i \in \mathbb{Z}_{p_i}$, and put $D = \prod_i p_i^{r_i}$. We seek $a = b/D$. The condition at p_i becomes a congruence for b modulo $p_i^{N_i + r_i + v_{p_i}(D/p_i^{r_i})}$. Since the moduli for different primes are coprime, the Chinese remainder theorem supplies one integer b satisfying them all. Repeating this construction coordinate by coordinate proves the last assertion. $\square$

Proposition 2.2 (Finite-place reduction)**.** *Given*

$$g_f = (g_p)_{p \in S_f} \in \prod_{p \in S_f} \mathrm{SL}_d(\mathbb{Q}_p),$$

there exists $\gamma \in \mathrm{SL}_d(\mathbb{Z}_S)$ such that

$$g_p \gamma \in \mathrm{SL}_d(\mathbb{Z}_p) \qquad (p \in S_f).$$

In fact one may arrange $g_p\gamma$ to lie in any prescribed sufficiently small neighborhood of the identity after first replacing the target by g_p^{-1}.

Proof. For each $p \in S_f$, Lemma 1.1 gives a word

$$g_p^{-1} = x_{i_1 j_1}(a_{p,1}) \cdots x_{i_{r_p} j_{r_p}}(a_{p,r_p}).$$

Insert factors with parameter zero until all places use one common finite list of root labels. Matrix multiplication is continuous, so there are integers N_p such that replacing every $a_{p,k}$ by a number within p^{-N_p} leaves the product inside the open set $g_p^{-1}\,\mathrm{SL}_d(\mathbb{Z}_p)$.

For each word position k, apply Lemma 2.1 to find one $a_k \in \mathbb{Z}_S$ approximating all the local parameters $a_{p,k}$. Put

$$\gamma = x_{i_1 j_1}(a_1) \cdots x_{i_r j_r}(a_r) \in \mathrm{SL}_d(\mathbb{Z}_S).$$

Then $\gamma \in g_p^{-1}\,\mathrm{SL}_d(\mathbb{Z}_p)$ for every p, which is equivalent to $g_p\gamma \in \mathrm{SL}_d(\mathbb{Z}_p)$. $\square$

Why this matters. There is no compactness argument hidden here. The key facts are exactly that SL_d is generated by additive root groups and that one rational number can approximate finitely many local numbers. This is the same mechanism that will prove strong approximation later, but here we use it only at the finite places already belonging to S.

3. The residual integral group

Let

$$K_f = \prod_{p \in S_f} \mathrm{SL}_d(\mathbb{Z}_p).$$

If $g_f \in K_f$ and $\eta \in \mathrm{SL}_d(\mathbb{Z})$, then $g_f \eta \in K_f$. Thus after Proposition 2.2 has moved the finite component into K_f, we retain the full right action of $\mathrm{SL}_d(\mathbb{Z})$ for reducing the real component. This simple observation is the bridge from S-arithmetic reduction to the classical reduction theory of real lattices.

Worked Example 3.1 (One prime, two dimensions)**.** Let $S = \{\infty, p\}$ and

$$g_p = \begin{pmatrix} p^{-m} & 0 \\ 0 & p^m \end{pmatrix}.$$

The global matrix

$$\gamma = \begin{pmatrix} p^m & 0 \\ 0 & p^{-m} \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}[1/p])$$

sends g_p exactly to the identity. For a general g_p, elementary factorization replaces this diagonal cancellation by a finite sequence of the same one-dimensional approximation problems.

CHAPTER 10

The S-Mahler compactness theorem

Mahler's criterion says that the only way a unimodular lattice can leave compact sets is by acquiring a short nonzero vector. In an S-arithmetic space there is one apparent complication: a vector may be large at one place and small at another. The product formula and S-units are precisely what make the product content the correct notion of size.

We prove the criterion for

$$X_{S,d} = \mathrm{SL}_d(\mathbb{Q}_S)/\,\mathrm{SL}_d(\mathbb{Z}_S), \qquad \mathbb{Q}_S = \mathbb{R} \times \prod_{p \in S_f} \mathbb{Q}_p.$$

The argument deliberately reduces the finite components first and then invokes the real Mahler theorem proved in Volume 1. This avoids hiding an S-adic successive minima theorem inside the proof.

1. Content and S-unit normalization

For $x = (x_v)_{v\in S} \in \mathbb{Q}_S^d$, set

$$c(x) = \prod_{v\in S} \|x_v\|_v.$$

If $u \in \mathbb{Z}_S^\times$, the product formula gives $c(ux) = c(x)$.

Lemma 1.1 (Integral normalization of an S-integer vector)**.** *For every nonzero $q \in \mathbb{Z}_S^d$ there is an S-unit u such that $uq \in \mathbb{Z}^d$ and, for every $p \in S_f$,*

$$\|uq\|_p \le 1.$$

One may moreover arrange that for each prime $p \in S_f$ at least one coordinate of uq is a p-adic unit unless all coordinates of q have a common p-power factor in $\mathbb{Z}_S$, in which case that factor can first be removed.

Proof. For each $p \in S_f$, let

$$m_p = -\min_i v_p(q_i).$$

Then $u = \prod_{p\in S_f} p^{m_p}$ clears every denominator at every place of S_f. Since the only allowed denominators of q lie in S_f, the vector uq lies in $\mathbb{Z}^d$. By construction its p-adic max norm is at most one, and after dividing a common

p-power if necessary one coordinate is a unit. All divisions used are by S-units, so the generated $\mathbb{Z}_S$-line and the content are unchanged. $\square$

2. A compact real model after finite-place reduction

Take $g = (g_\infty, g_f) \in \mathrm{SL}_d(\mathbb{Q}_S)$. By Proposition 2.2, after right multiplication by an element of $\mathrm{SL}_d(\mathbb{Z}_S)$ we may assume

$$g_f \in K_f = \prod_{p \in S_f} \mathrm{SL}_d(\mathbb{Z}_p).$$

The corresponding point of $X_{S,d}$ has not changed.

Lemma 2.1 (Content dominates the real length on integral vectors)**.** *If $g_f \in K_f$ and $0 \neq q \in \mathbb{Z}^d$, then*

$$c(gq) \leq \|g_\infty q\|_\infty.$$

Consequently, if every nonzero vector of the S-lattice $g\mathbb{Z}_S^d$ has content at least ε, then every nonzero vector of the real lattice $g_\infty \mathbb{Z}^d$ has norm at least ε.

Proof. For $p \in S_f$, the matrix $g_p \in \mathrm{SL}_d(\mathbb{Z}_p)$ preserves $\mathbb{Z}_p^d$, so $\|g_p q\|_p \leq 1$. Multiplying the local norms gives the first inequality. The second assertion follows because every $q \in \mathbb{Z}^d$ is also an element of $\mathbb{Z}_S^d$. $\square$

Theorem 2.2 (S-Mahler compactness criterion). *A subset $\mathscr{K} \subset X_{S,d}$ has relatively compact closure if and only if there exists $\varepsilon > 0$ such that for every $\Lambda \in \mathscr{K}$ and every $0 \neq x \in \Lambda$,*

$$c(x) \geq \varepsilon.$$

Proof. *The lower bound implies precompactness.* For each $\Lambda \in \mathscr{K}$, choose a representative $g = (g_\infty, g_f)$ with $g_f \in K_f$ by Proposition 2.2. Lemma 2.1 shows that $g_\infty \mathbb{Z}^d$ has no nonzero vector shorter than ε. By the real Mahler criterion proved in Volume 1, there exists $\eta \in \mathrm{SL}_d(\mathbb{Z})$ such that $g_\infty \eta$ lies in a fixed compact subset $C_\infty \subset \mathrm{SL}_d(\mathbb{R})$ depending only on ε. Since $\eta \in \mathrm{SL}_d(\mathbb{Z}) \subset \mathrm{SL}_d(\mathbb{Z}_p)$, the finite component $g_f \eta$ remains in the compact group K_f. Hence every point of $\mathscr{K}$ has a representative in the compact set $C_\infty \times K_f$, proving relative compactness.

Precompactness implies the lower bound. Let $C \subset \mathrm{SL}_d(\mathbb{Q}_S)$ be a compact set mapping onto the closure of $\mathscr{K}$; obtain it from finitely many local sections of the quotient map. There is $M \geq 1$ such that for all $g \in C$, all $v \in S$, and all vectors z,

$$M^{-1}\|z\|_v \leq \|g_v z\|_v \leq M\|z\|_v.$$

Suppose contrary to the claim that $g_i q_i$ are nonzero lattice vectors with $c(g_i q_i) \to 0$. By the operator inequalities, $c(q_i) \to 0$. Multiply $q_i \in \mathbb{Z}_S^d$ by an S-unit using the balancing corollary of the S-unit theorem so that every local norm of the normalized vector tends to zero. The normalized vectors remain nonzero elements of the diagonal module $\mathbb{Z}_S^d$, contradicting the discreteness of $\mathbb{Z}_S^d \subset \mathbb{Q}_S^d$. Thus a uniform positive lower bound exists. □

3. Why rank-one content is enough

For compactness, Theorem 2.2 only needs vectors. Later quantitative arguments must control every primitive submodule. The reason is not a stronger compactness theorem; it is stability under intersection and sum. If $\Delta_1, \Delta_2 \subset \mathbb{Z}_S^d$ are primitive modules, their top exterior vectors satisfy a submodular inequality relating the contents of $\Delta_1 \cap \Delta_2$, $\Delta_1 + \Delta_2$, Δ_1, and Δ_2. That lattice structure is what makes the marked-poset induction possible.

Worked Example 3.1 (The basic mixed cusp in $\mathrm{SL}_2(\mathbb{R}) \times \mathrm{SL}_2(\mathbb{Q}_p)$)**.** Take even m and let

$$a_m = \left(\begin{pmatrix} p^{-m} & 0 \\ 0 & p^m \end{pmatrix}, \begin{pmatrix} p^{-m/2} & 0 \\ 0 & p^{m/2} \end{pmatrix} \right).$$

For the first basis vector the real norm is p^{-m}, while the p-adic norm is $|p^{-m/2}|_p = p^{m/2}$: the two local sizes move in opposite directions. Their product is nevertheless

$$c(a_m e_1) = p^{-m/2} \longrightarrow 0,$$

so the S-lattice escapes. Multiplying e_1 by the S-unit p^k transfers size between the two places but does not change this content. The example shows why neither local norm is intrinsic and why the product content is exactly the right cusp coordinate.

CHAPTER 11

Proof laboratory: two views of S-Mahler compactness

The proof of Theorem 2.2 was deliberately short because finite-place reduction converts it to real Mahler compactness. There is a second, intrinsically S-arithmetic proof based on successive primitive submodules. Working through both proofs is valuable: the first is best for compactness, the second is the model for quantitative nondivergence.

1. The reduction-to-the-real-place proof as an algorithm

Given $g = (g_\infty, g_f)$, perform the following steps.

(1) Factor each g_p^{-1} into elementary matrices.
(2) Approximate all root parameters simultaneously by elements of $\mathbb{Z}_S$, obtaining γ_1 with $g_f\gamma_1 \in K_f$.
(3) Use the content lower bound to show that the real lattice $g_\infty\gamma_1\mathbb{Z}^d$ has no short vector.
(4) Apply real Mahler reduction to obtain $\gamma_2 \in \mathrm{SL}_d(\mathbb{Z})$ with $g_\infty\gamma_1\gamma_2$ in a compact set.
(5) Because γ_2 is integral at every finite prime, $g_f\gamma_1\gamma_2$ remains in K_f.

The output $g\gamma_1\gamma_2$ is a bounded representative of the original coset. Every step is constructive once local elementary factorizations are chosen.

2. The intrinsic primitive-flag proof

We now outline and prove the pieces of a second proof. Let $\Lambda = g\mathbb{Z}_S^d$ be unimodular and assume

$$c(v) \geq \varepsilon \qquad (0 \neq v \in \Lambda).$$

Choose a primitive vector v_1 of minimal content. The S-Blichfeldt lemma from Chapter 6 gives an upper bound $c(v_1) \ll_{S,d} 1$, while the hypothesis gives the lower bound $c(v_1) \geq \varepsilon$. Since $\mathbb{Z}_S$ is a PID, v_1 extends to a basis and the quotient

$$\Lambda_1 = \Lambda/\mathbb{Z}_S v_1$$

is a free $\mathbb{Z}_S$-module of rank $d-1$.

At a place v, choose a linear functional λ_v with $\lambda_v(v_1) = 1$ and operator norm comparable to $\|v_1\|_v^{-1}$. The map

$$w \longmapsto w - \lambda_v(w)v_1$$

identifies the quotient with a complementary hyperplane. On determinant lines,

$$\bigwedge^d \mathbb{Q}_v^d \simeq (\mathbb{Q}_v v_1) \otimes \bigwedge^{d-1} (\mathbb{Q}_v^d/\mathbb{Q}_v v_1),$$

so the local covolume of the quotient is the reciprocal of the local size of v_1, up to the fixed norm-equivalence constants. Multiplying over S gives

$$\mathrm{cov}_S(\Lambda_1) \asymp c(v_1)^{-1}.$$

Thus the quotient covolume is bounded above and below in terms of ε, S, d. After a scalar normalization of controlled content, repeat the construction.

Induction produces a primitive flag

$$0 < \Delta_1 < \cdots < \Delta_d = \Lambda$$

for which every successive quotient has content between two positive constants. Choose lifts v_i of the successive generators. Then every partial wedge $v_1 \wedge \cdots \wedge v_r$ has content bounded above and below.

3. Why bounded partial contents control angles

At a single archimedean place, bounded vector norms do not prevent near linear dependence. The partial wedge bounds do. If $v_1, \ldots, v_r$ have bounded norms but their angle collapses, then

$$\|v_1 \wedge \cdots \wedge v_r\|_\infty \to 0.$$

Product content can remain nonzero only if some finite-place wedge norm grows. Multiply the entire primitive rank-r module by an S-unit to balance these local wedge norms. The balanced module then has small covolume at every place. Applying the rank-r S-Blichfeldt argument inside its span produces a nonzero vector of content smaller than ε, contradiction. Hence the angles are uniformly nondegenerate after balancing.

This is the geometric reason that the marked-poset proof later monitors every primitive rank, even though the compactness theorem only mentions rank one.

4. Equivalence of the two proofs

The real-reduction proof packages all angle estimates into the real Mahler theorem and all finite-place anisotropy into Proposition 2.2. The primitive-flag proof keeps both phenomena visible at every rank. Quantitative nondivergence needs the second viewpoint because during a polynomial trajectory the identity of the dangerous primitive submodule may change with the parameter.

Worked Example 4.1 (Rank two with $S = \{\infty, p\}$)**.** Let v_1, v_2 be an S-basis of determinant one. If $c(v_1)$ is bounded below and above, determinant one forces the quotient size of v_2 modulo v_1 to have reciprocal product size. After independently balancing the line and quotient directions, the two local basis matrices have bounded Cartan exponents. Subtracting an S-integer multiple of v_1 from v_2 reduces the unipotent coordinate. This is the entire S-Siegel reduction theorem in rank two.

CHAPTER 12

Local Iwasawa and Cartan decompositions

Reduction theory is easiest to understand after separating compact, diagonal, and unipotent directions locally.

1. $\mathrm{SL}_n(\mathbb{Q}_p)$

Put $K_p = \mathrm{SL}_n(\mathbb{Z}_p)$ and let A_p^+ be the diagonal matrices

$$a = \mathrm{diag}(p^{m_1}, \ldots, p^{m_n}), \qquad m_1 \geq \cdots \geq m_n, \qquad \sum_i m_i = 0.$$

THEOREM 1.1 (Cartan decomposition for $\mathrm{SL}_n(\mathbb{Q}_p)$). *Every $g \in \mathrm{SL}_n(\mathbb{Q}_p)$ can be written*

$$g = k_1 a k_2, \qquad k_1, k_2 \in K_p, \quad a \in A_p^+.$$

The tuple $(m_1, \ldots, m_n)$ is uniquely determined.

PROOF. Regard $g\mathbb{Z}_p^n$ as a lattice in $\mathbb{Q}_p^n$. The elementary-divisor theorem over the PID $\mathbb{Z}_p$ gives bases of $\mathbb{Z}_p^n$ and $g\mathbb{Z}_p^n$ for which the inclusion data are diagonal powers of p. Equivalently there are $k_1, k_2 \in \mathrm{GL}_n(\mathbb{Z}_p)$ with $g = k_1 \,\mathrm{diag}(p^{m_i}) k_2$. Since $\det g = 1$, the exponents sum to zero. Multiplying k_i by diagonal unit matrices adjusts both determinants to one. Permuting basis vectors orders the exponents. Uniqueness follows because the sums $m_1 + \cdots + m_r$ are the valuations of the smallest possible rth exterior content and hence are invariants of the double coset $K_p g K_p$. □

2. Archimedean Iwasawa decomposition

THEOREM 2.1 (Iwasawa decomposition for $\mathrm{SL}_n(\mathbb{R})$). *Every $g \in \mathrm{SL}_n(\mathbb{R})$ has a unique decomposition*

$$g = kan,$$

where $k \in \mathrm{SO}(n)$, $a = \mathrm{diag}(a_1, \ldots, a_n)$ with $a_i > 0$ and $\prod a_i = 1$, and n is upper triangular unipotent.

PROOF. Apply Gram–Schmidt to the columns of g, starting from the last column. This yields $g = kr$ with k orthogonal and r upper triangular with positive diagonal. Write $r = an$ by extracting the diagonal. Determinant one gives $k \in \mathrm{SO}(n)$ and $\prod a_i = 1$. Uniqueness follows from KAN having pairwise trivial intersections in the evident triangular/orthogonal sense. $\square$

3. Parabolics detected by wedges

For $1 \le r < n$ let $e_{[r]} = e_1 \wedge \cdots \wedge e_r$. If $g = kan$ is Iwasawa, then

$$\|ge_{[r]}\| \asymp_n a_1 \cdots a_r$$

when n ranges in a fixed bounded set. Thus divergence of diagonal ratios is equivalent to collapse of a primitive exterior vector. This observation is the common language between reduction theory and the marked-submodule argument of quantitative non-divergence.

CHAPTER 13

S-reduction theory for SL_d

The compactness argument of Chapter 10 already contains the main reduction mechanism: first move all finite components into maximal compact subgroups by an exact global change of basis, and then perform ordinary real reduction. We now make this into a global fundamental-set statement and identify precisely the exterior coordinates which detect the cusp.

1. A complete real reduction argument

Let $K = \mathrm{SO}(d)$. For $C \geq 1$ let $N(C)$ be the upper-unipotent matrices $n = (n_{ij})$ with $|n_{ij}| \leq C$, and put

$$A(C) = \left\{ \mathrm{diag}(a_1, \dots, a_d) : a_i > 0,\ \prod_i a_i = 1,\ \frac{a_i}{a_{i+1}} \leq C\ (1 \leq i < d) \right\}.$$

The orientation of the chamber inequality corresponds to ordering a reduced basis from short vectors to long quotient vectors.

Lemma 1.1 (Primitive shortest vectors)**.** *Let Λ be a Euclidean lattice and let $\Lambda_0 \subset \Lambda$ be a primitive sublattice. In the quotient lattice*

$$\Lambda/\Lambda_0 \subset \mathrm{span}_{\mathbb{R}}(\Lambda)/\,\mathrm{span}_{\mathbb{R}}(\Lambda_0)$$

a shortest nonzero class is primitive. Consequently any lift of such a class, together with a $\mathbb{Z}$-basis of Λ_0, extends to a basis of a primitive rank-one enlargement of Λ_0.

Proof. Because Λ_0 is primitive, Λ/Λ_0 is torsion-free and hence a free abelian group. If a shortest nonzero class $\bar{v}$ were $m\bar{w}$ with $|m| \geq 2$, then $\bar{w}$ would be nonzero and its quotient norm would be $|m|^{-1}\|\bar{v}\|$, contradicting minimality. Thus $\bar{v}$ is primitive in the free abelian quotient. A primitive element of a free abelian group extends to a basis; lifting that basis gives the final assertion. $\square$

Lemma 1.2 (Real Siegel reduction)**.** *There are constants $C_d, C_d' > 0$ such that*

$$\mathrm{SL}_d(\mathbb{R}) = K A(C_d') N(C_d)\, \mathrm{SL}_d(\mathbb{Z}).$$

One may take $C_d = 1/2$ and $C_d' = 2/\sqrt{3}$ for the conventions above.

Proof. Let $\Lambda = g\mathbb{Z}^d$. We construct a basis $v_1, \dots, v_d$ recursively. Choose v_1 to be a shortest nonzero vector of Λ. It is primitive, since a nontrivial integer multiple cannot be shortest. Suppose $\Lambda_i = \mathbb{Z}v_1 + \cdots + \mathbb{Z}v_i$ has already been chosen primitive. In the Euclidean quotient by $V_i = \mathrm{span}_{\mathbb{R}}(\Lambda_i)$, the image of Λ is a lattice. Choose a shortest nonzero quotient vector and lift it to v_{i+1}. Lemma 1.1 shows that Λ_{i+1} is primitive. After d steps the vectors form a $\mathbb{Z}$-basis of Λ.

Apply Gram–Schmidt to this basis. If v_j^* are the orthogonalized vectors and $a_j = \|v_j^*\|$, then, after choosing the orientation of the orthonormal basis, there are $k \in K$, a positive diagonal $a = \mathrm{diag}(a_1, \dots, a_d)$ with $\det a = 1$, and an upper-unipotent matrix n such that

$$g\gamma = kan$$

for the basis-change matrix $\gamma \in \mathrm{SL}_d(\mathbb{Z})$.

We now size-reduce the coefficients of n. Replacing v_j by $v_j - mv_i$ with $i < j$ changes no quotient vector already chosen at stages $< j$. Taking m to be a nearest integer to the corresponding Gram–Schmidt coefficient makes its absolute value at most $1/2$. Performing this from the largest lower index upward gives

$$|n_{ij}| \le \frac{1}{2} \qquad (i < j).$$

It remains to control adjacent a_i. In the quotient by V_{i-1}, the classes of v_i and v_{i+1} have representatives

$$a_i e_i, \qquad a_i n_{i,i+1} e_i + a_{i+1} e_{i+1}.$$

The first was chosen shortest among nonzero quotient-lattice vectors. Hence

$$a_i^2 \le a_i^2 n_{i,i+1}^2 + a_{i+1}^2.$$

Since $|n_{i,i+1}| \le 1/2$,

$$a_{i+1}^2 \ge \frac{3}{4} a_i^2, \qquad \frac{a_i}{a_{i+1}} \le \frac{2}{\sqrt{3}}.$$

Thus $a \in A(2/\sqrt{3})$ and $n \in N(1/2)$, proving the covering. □

2. The S-arithmetic fundamental set

Put

$$K_f = \prod_{p \in S_f} \mathrm{SL}_d(\mathbb{Z}_p).$$

Theorem 2.1 (S-Siegel reduction). *There are constants $C_d, C_d' > 0$ such that*

$$\mathrm{SL}_d(\mathbb{Q}_S) = \big(K A(C_d') N(C_d) \times K_f\big)\, \mathrm{SL}_d(\mathbb{Z}_S).$$

In particular every point of $X_{S,d}$ has a representative whose finite coordinates belong to K_f and whose real coordinate belongs to one fixed classical Siegel set.

Proof. Take $g = (g_\infty, g_f)$. Proposition 2.2 supplies $\gamma_1 \in \mathrm{SL}_d(\mathbb{Z}_S)$ with

$$g_f \gamma_1 \in K_f.$$

Apply Lemma 1.2 to $g_\infty \gamma_1$. There is $\gamma_2 \in \mathrm{SL}_d(\mathbb{Z})$ such that

$$g_\infty \gamma_1 \gamma_2 \in KA(C'_d)N(C_d).$$

For every finite p, $\gamma_2 \in \mathrm{SL}_d(\mathbb{Z}_p)$; therefore right multiplication by γ_2 keeps the finite component inside K_f. With $\gamma = \gamma_1\gamma_2 \in \mathrm{SL}_d(\mathbb{Z}_S)$ we obtain the required representative. □

Remark 2.2. This proof is stronger pedagogically than the slogan "apply reduction theory at all places". The finite places are removed exactly, not estimated. All continuous cusp parameters are then visible in one real diagonal matrix.

3. Fundamental weights in the reduced cone

For $1 \le r < d$ set

$$\omega_r = e_1 \wedge \cdots \wedge e_r.$$

If $a = \mathrm{diag}(a_1, \ldots, a_d)$, then

$$\|a\omega_r\| = a_1 \cdots a_r.$$

The following elementary lemma is the precise diagonal statement behind cusp detection.

Lemma 3.1 (Escape in a Siegel cone)**.** *Let $a^{(j)} = \mathrm{diag}(a_1^{(j)}, \ldots, a_d^{(j)}) \in A(C)$ with determinant one. Then $\{a^{(j)}\}$ is relatively compact if and only if all prefix products*

$$b_r^{(j)} = a_1^{(j)} \cdots a_r^{(j)}, \qquad 1 \le r < d,$$

are bounded below by one common positive constant. Consequently, if $a^{(j)}$ leaves every compact subset of $A(C)$, then after passing to a subsequence

$$b_r^{(j)} \longrightarrow 0$$

for some $1 \le r < d$.

Proof. Write $x_i = \log a_i$ and $s_r = x_1 + \cdots + x_r = \log b_r$. The chamber inequalities $x_i - x_{i+1} \le \log C$ and the determinant equation $s_d = 0$ are linear inequalities. We show that lower bounds $s_r \ge -M$ imply upper and lower bounds for every x_i.

Since $x_i = s_i - s_{i-1}$, it suffices to bound every s_i above. From the chamber inequalities,

$$x_i \le x_j + (j-i)\log C \qquad (i<j).$$

If some s_r were arbitrarily large while all $s_i \ge -M$, then the average of $x_1, \dots, x_r$ would be arbitrarily large. The adjacent-ratio inequalities would force the later x_j to be bounded below by that large average minus a constant depending only on d, C; then $s_d = \sum_i x_i$ would become positive and unbounded, a contradiction. Thus the s_r are bounded above as well. Hence all $x_i = s_i - s_{i-1}$ lie in a compact interval, so a lies in a compact subset. The converse is immediate by continuity of the prefix products. The subsequence statement follows because there are only $d-1$ prefix products. □

4. From a collapsing wedge to a short vector

The reverse implication in cusp detection should not be hidden in the phrase "a small sublattice contains a short vector". We record the exact S-arithmetic form.

Lemma 4.1 (A small primitive wedge contains a short vector)**.** *For each $1 \le r \le d$ there is $C_{S,d} \ge 1$ such that for every $g \in \mathrm{GL}_d^1(\mathbb{Q}_S)$ and every primitive rank-r module $\Delta \subset \mathbb{Z}_S^d$ there exists $0 \ne v \in \Delta$ satisfying*

$$c(gv) \le C_{S,d}\, c(g\Delta)^{1/r}.$$

Proof. Let $W_v \subset \mathbb{Q}_v^d$ be the r-dimensional span of $g_v\Delta$. Choose on W_v the Haar measure for which an exterior parallelepiped has volume comparable, with constants depending only on d and the chosen max/Euclidean norms, to the norm of its top wedge. With the product normalization, the covolume of the S-lattice $g\Delta$ in $W_S = \prod_{v\in S} W_v$ is therefore comparable to $c(g\Delta)$.

Choose local radii $R_v > 0$ with

$$\prod_{v\in S} R_v = C_0\, c(g\Delta)^{1/r},$$

where C_0 is a sufficiently large structural constant, and use S-unit balancing (Corollary 3.2) to arrange the radii within a fixed factor of one another in logarithmic coordinates. The product body

$$E = \prod_{v\in S} \{w \in W_v : \|w\|_v \le R_v\}$$

has product Haar measure

$$m_{W_S}(E) \asymp_{S,d} \left(\prod_v R_v\right)^r \asymp C_0^r c(g\Delta).$$

Take C_0 large enough that this exceeds the covolume by the factor required in the S-Blichfeldt argument. Applying Lemma 3.1 inside the S-vector space W_S produces distinct $x, y \in E$ with $x - y \in g\Delta$. Then $0 \neq w = x - y \in g\Delta$ and

$$\|w_v\|_v \leq R_v$$

at each finite place, while at the real place the difference costs at most a fixed factor two. Hence

$$c(w) \leq C_{S,d} \prod_v R_v \leq C'_{S,d} c(g\Delta)^{1/r}.$$

Writing $w = gv$ gives the desired $v \in \Delta$. ☐

5. Exterior detection of the cusp

Proposition 5.1 (Exterior detection of the cusp)**.** *A sequence $x_j \in X_{S,d}$ leaves every compact subset if and only if, after passing to a subsequence, there are a rank $1 \leq r < d$ and primitive rank-r modules $\Delta_j \subset \mathbb{Z}_S^d$ such that*

$$c(g_j \Delta_j) \longrightarrow 0,$$

where g_j is any representative of x_j.

Proof. Suppose first that such Δ_j exist. By Lemma 4.1, choose $0 \neq v_j \in \Delta_j$ with

$$c(g_j v_j) \leq C_{S,d} c(g_j \Delta_j)^{1/r} \longrightarrow 0.$$

The S-Mahler criterion, Theorem 2.2, therefore says that x_j cannot have a relatively compact subsequence.

Conversely suppose x_j escapes. Use Theorem 2.1 to choose representatives

$$g_j = (k_j a_j n_j,\ k_{f,j})$$

with $k_j \in K$, $n_j \in N(C_d)$, $k_{f,j} \in K_f$, and $a_j \in A(C'_d)$. The K, $N(C_d)$, and K_f factors range in compact sets, so a_j must leave every compact subset of $A(C'_d)$. By Lemma 3.1, after passing to a subsequence there is an r such that

$$a_{1,j} \cdots a_{r,j} \to 0.$$

Take the fixed primitive coordinate module

$$\Delta = \mathbb{Z}_S e_1 + \cdots + \mathbb{Z}_S e_r.$$

At finite places $k_{f,j}$ preserves the integral exterior unit ball. At the real place k_j is isometric and n_j ranges in a compact set, so its action on $\bigwedge^r \mathbb{R}^d$ and its inverse have uniformly bounded operator norms. Therefore

$$c(g_j \Delta) \asymp_{d,C_d} a_{1,j} \cdots a_{r,j} \longrightarrow 0.$$

Thus the required collapsing primitive wedge exists. ☐

Why this matters. Reduction theory and linearization use exterior powers for different but compatible reasons. Reduction theory says that a collapsing primitive wedge is exactly a cusp direction. Linearization later says that a nearly invariant primitive wedge is an algebraic obstruction to recurrence. The same finite-dimensional representations therefore encode both noncompactness and rigidity.

CHAPTER 14

Polynomial sublevel estimates over nonarchimedean fields

The real polynomial-goodness theorem proved in Volume 1 is based on derivatives, sublevel components, and a covering argument. Over a nonarchimedean field there is a cleaner route. The correct replacement for a derivative estimate is *interpolation at well separated points*: if a polynomial is very small on a set of positive relative measure, then one can find enough separated points in that set to force every interpolation coefficient to be small. This chapter proves the precise form used by quantitative nondivergence in Volume 4.

Throughout, k is a nonarchimedean local field of characteristic zero, $\mathcal{O}_k$ is its valuation ring, ϖ a uniformizer, q the cardinality of the residue field, and m additive Haar measure normalized by $m(\mathcal{O}_k) = 1$. On k^d we use the max norm and product Haar measure. If $B = B(a, R)$ is a ball, $\operatorname{rad}(B) = R$ means a value-group radius.

1. Normalization and separated points

Affine changes of variables behave exactly as expected. If $B = a + b\mathcal{O}_k$ and $Q(z) = P(a + bz)$, then

$$\sup_{z \in \mathcal{O}_k} |Q(z)| = \sup_{t \in B} |P(t)|,$$

$$m\{t \in B : |P(t)| < \eta\} = |b|\, m\{z \in \mathcal{O}_k : |Q(z)| < \eta\}.$$

Thus it suffices in one variable to work on $\mathcal{O}_k$.

Lemma 1.1 (Separated interpolation nodes)**.** *Let $B \subset k$ be a ball and let $E \subset B$ be measurable with $m(E) = \theta m(B) > 0$. For every integer $r \geq 1$ there are $r + 1$ points $x_0, \ldots, x_r \in E$ such that*

$$|x_i - x_j| \geq c_{q,r}\, \theta \operatorname{rad}(B) \qquad (i \neq j),$$

where one may take $c_{q,r} = q^{-2}/(r + 1)$ after harmlessly replacing an arbitrary positive radius by the next smaller value-group radius.

PROOF. By translation and dilation assume $B = \mathcal{O}_k$. Let $\rho = q^{-N}$ be the largest value-group radius satisfying

$$(r+1)m(B(0,\rho)) = (r+1)\rho \le \frac{\theta}{q}.$$

Maximality gives

$$\rho > \frac{\theta}{q^2(r+1)}.$$

A family of radius-ρ balls is either disjoint or identical. If E met at most r such balls, then

$$m(E) \le r\rho < \theta = m(E),$$

a contradiction. Hence E meets at least $r+1$ distinct radius-ρ balls. Choose one point from each. Distinct radius-ρ balls in an ultrametric space have mutual point-distance strictly larger than ρ, giving the asserted separation. Scaling back to B multiplies every distance by $\mathrm{rad}(B)$. $\square$

Remark 1.2. The proof uses only the nested-ball geometry. No density theorem and no regularity of the boundary of E is needed. This is one reason polynomial sublevel estimates are particularly robust at finite places.

2. One variable: interpolation gives the sharp degree exponent

THEOREM 2.1 (One-variable polynomial goodness). *For every integer $r \ge 1$ there is $C = C(k,r)$ such that every nonzero $P \in k[t]$ of degree at most r satisfies*

$$m\{t \in B : |P(t)| < \varepsilon \sup_B |P|\} \le C\varepsilon^{1/r} m(B)$$

for every ball $B \subset k$ and every $0 < \varepsilon \le 1$.

PROOF. By affine normalization of the ball and multiplication of P by a scalar, assume $B = \mathcal{O}_k$ and

$$M := \sup_{\mathcal{O}_k} |P| = 1.$$

Let

$$E = \{t \in \mathcal{O}_k : |P(t)| < \varepsilon\}, \qquad m(E) = \theta.$$

There is nothing to prove if $\theta = 0$. Otherwise choose, by Lemma 1.1, $r+1$ points $x_0, \dots, x_r \in E$ with

$$|x_i - x_j| \ge c_{q,r}\theta.$$

Since $\deg P \le r$, Lagrange interpolation is an identity:

$$P(t) = \sum_{i=0}^{r} P(x_i) L_i(t), \qquad L_i(t) = \prod_{j \ne i} \frac{t - x_j}{x_i - x_j}.$$

For $t, x_j \in \mathcal{O}_k$ every numerator has norm at most one. Therefore

$$|L_i(t)| \le \prod_{j\ne i} |x_i - x_j|^{-1} \le (c_{q,r}\theta)^{-r}.$$

The ultrametric triangle inequality now yields, for every $t \in \mathcal{O}_k$,

$$|P(t)| \le \max_i |P(x_i)|\,|L_i(t)| < \varepsilon(c_{q,r}\theta)^{-r}.$$

Taking the supremum in t and using $M = 1$ gives

$$1 \le \varepsilon(c_{q,r}\theta)^{-r},$$

so

$$\theta \le c_{q,r}^{-1}\varepsilon^{1/r}.$$

Undoing the affine normalization proves the theorem with $C = c_{q,r}^{-1}$. □

Worked Example 2.2 (Why the exponent $1/r$ cannot be improved uniformly)**.** Take $P(t) = t^r$ on $\mathcal{O}_k$. Since $\sup|P| = 1$,

$$\{t : |P(t)| < \varepsilon\} = \{t : |t| < \varepsilon^{1/r}\}.$$

Up to the unavoidable rounding to the value group, this set has measure $\asymp \varepsilon^{1/r}$. Thus a theorem uniform over all degree-r polynomials cannot replace $1/r$ by a larger exponent.

3. Coefficient control on the unit ball

The induction in several variables needs a uniform comparison between the coefficients of a one-variable polynomial and its supremum on the unit ball.

Lemma 3.1 (Vandermonde coefficient control)**.** *For every $r \ge 0$ there is $A = A(k, r) \ge 1$ such that, for*

$$Q(t) = \sum_{j=0}^{r} a_j t^j,$$

one has

$$A^{-1}\max_j |a_j| \le \sup_{t\in\mathcal{O}_k} |Q(t)| \le \max_j |a_j|.$$

Proof. The upper bound is immediate from $|t| \le 1$ and ultrametricity. For the lower bound choose $r+1$ distinct points $z_0, \ldots, z_r \in \mathcal{O}_k$. The vector of coefficients $(a_0, \ldots, a_r)$ is obtained from $(Q(z_0), \ldots, Q(z_r))$ by the inverse of the fixed Vandermonde matrix (z_i^j). All norms on the finite-dimensional space k^{r+1} are equivalent; more concretely, the finitely many entries of that inverse have a finite maximum norm. Hence

$$\max_j |a_j| \le A \max_i |Q(z_i)| \le A \sup_{\mathcal{O}_k} |Q|.$$

$\square$

4. Several variables

The next proof is written with the exponent bookkeeping explicit. A common mistake is to choose the threshold $\delta = \varepsilon^{1/d}$ in the induction. That choice gives the wrong exponent in the exceptional base set. The balancing choice is

$$\delta = \varepsilon^{(d-1)/d}.$$

THEOREM 4.1 (Multivariable polynomial goodness). *For integers $d, r \ge 1$ there is $C = C(k,d,r)$ such that every nonzero polynomial $P : k^d \to k$ of total degree at most r is $(C, 1/(dr))$-good on max-norm balls:*

$$m\{x \in B : |P(x)| < \varepsilon \sup_B |P|\} \le C\varepsilon^{1/(dr)} m(B)$$

for every max-norm ball $B \subset k^d$ and $0 < \varepsilon \le 1$.

PROOF. We induct on d. The case $d = 1$ is Theorem 2.1. Assume the result in dimension $d-1$. After translating and scaling each coordinate, write

$$B = B' \times \mathcal{O}_k, \qquad P(y,t) = \sum_{j=0}^{r} a_j(y) t^j,$$

where every a_j is a polynomial on k^{d-1} of degree at most r. Set

$$M = \sup_{(y,t) \in B} |P(y,t)|.$$

By Lemma 3.1,

$$M \asymp_{k,r} \max_{0 \le j \le r} \sup_{y \in B'} |a_j(y)|.$$

Choose j_0 and a constant $c = c(k,r) > 0$ such that

$$\sup_{B'} |a_{j_0}| \ge cM.$$

Put

$$\delta = \varepsilon^{(d-1)/d}$$

and define the exceptional base set

$$E_0 = \{y \in B' : |a_{j_0}(y)| < \delta c M\}.$$

The induction hypothesis, applied to a_{j_0}, gives

$$\begin{aligned} m(E_0) &\le C_{d-1}\left(\frac{\delta c M}{\sup_{B'}|a_{j_0}|}\right)^{1/((d-1)r)} m(B') \\ &\le C'_{d-1}\delta^{1/((d-1)r)}m(B') \\ &= C'_{d-1}\varepsilon^{1/(dr)}m(B'). \end{aligned}$$

Now fix $y \notin E_0$. Then $|a_{j_0}(y)| \ge \delta c M$. Applying the lower bound in Lemma 3.1 to the one-variable polynomial $t \mapsto P(y,t)$ gives

$$\sup_{t\in\mathcal{O}_k} |P(y,t)| \ge c_1 \delta M$$

with $c_1 = c_1(k,r) > 0$. The one-variable goodness theorem therefore gives

$$\begin{aligned} m\{t \in \mathcal{O}_k : |P(y,t)| < \varepsilon M\} \\ \le C_r\left(\frac{\varepsilon M}{c_1\delta M}\right)^{1/r} = C'_r\varepsilon^{1/(dr)}. \end{aligned}$$

Indeed

$$\left(\frac{\varepsilon}{\delta}\right)^{1/r} = \left(\varepsilon^{1/d}\right)^{1/r} = \varepsilon^{1/(dr)}.$$

Fubini's theorem now splits the full sublevel set into the fibers above E_0 and those above $B' \setminus E_0$:

$$\begin{aligned} m\{(y,t) \in B : |P(y,t)| < \varepsilon M\} &\le m(E_0)m(\mathcal{O}_k) + C'_r\varepsilon^{1/(dr)}m(B') \\ &\le C_{d,r}\varepsilon^{1/(dr)}m(B). \end{aligned}$$

This closes the induction. □

Remark 4.2 (The exponent is deliberately nonoptimal)**.** The exponent $1/(dr)$ is sufficient for the marked-poset argument in Volume 4. Sharper exponents can often be obtained by resolving the zero set or by using more refined one-variable slicing, but optimizing the exponent would obscure the uniform mechanism needed later. What matters is a positive exponent depending only on dimension and degree.

5. Polynomial vectors and exterior coordinates

Corollary 5.1 (Norm of a polynomial vector)**.** *Let $F = (f_1, \dots, f_N) : k^d \to k^N$ and suppose every f_i has degree at most r. Then*

$$x \longmapsto \|F(x)\|_\infty$$

is $(C, 1/(dr))$-good on max-norm balls, with $C = C(k,d,r)$ independent of N.

Proof. Let B be a ball and choose i_0 with

$$\sup_B |f_{i_0}| = \sup_B \|F\|_\infty.$$

Then

$$\{x : \|F(x)\|_\infty < \varepsilon \sup_B \|F\|_\infty\} \subset \{x : |f_{i_0}(x)| < \varepsilon \sup_B |f_{i_0}|\}.$$

Apply Theorem 4.1 to f_{i_0}. $\square$

Why this matters. Later the coordinates of $\bigwedge^r h(x)w$ are polynomial whenever the matrix entries of $h(x)$ are polynomial. The corollary therefore turns algebraic polynomial dependence into a uniform measure estimate for small exterior content. This is the exact local input needed by S-arithmetic quantitative nondivergence.

CHAPTER 15

Workshop: interpolation, divided differences, and goodness

The proof of nonarchimedean polynomial goodness in Chapter 10 used Lagrange interpolation. Here we unpack that argument and explain why the exponent is stable under the operations that occur in exterior-power dynamics.

1. Divided differences

For distinct points $x_0, \ldots, x_j$ define recursively

$$[x_0]P = P(x_0), \qquad [x_0, \ldots, x_j]P = \frac{[x_1, \ldots, x_j]P - [x_0, \ldots, x_{j-1}]P}{x_j - x_0}.$$

If P has degree at most r, then

$$P(T) = \sum_{j=0}^{r} [x_0, \ldots, x_j]P \prod_{i<j} (T - x_i).$$

This Newton interpolation formula follows by induction: subtract the partial sum through degree $j-1$, note that the remainder vanishes at $x_0, \ldots, x_{j-1}$, divide by their product, and evaluate at x_j.

Suppose now that all x_i lie in a ball B of radius R and are mutually ρR-separated. If $|P(x_i)| \le \eta$ for every i, then each divided difference satisfies

$$|[x_0, \ldots, x_j]P| \le \eta(\rho R)^{-j}.$$

The proof is immediate from the explicit Lagrange formula

$$[x_0, \ldots, x_j]P = \sum_{i=0}^{j} \frac{P(x_i)}{\prod_{\ell \ne i}(x_i - x_\ell)}$$

and the ultrametric triangle inequality. Thus many separated points on which a polynomial is tiny force every Newton coefficient to be tiny.

2. Recovering the sublevel exponent

Normalize $B = \mathcal{O}_k$ and $\sup_B |P| = 1$. If

$$E = \{x \in B : |P(x)| < \varepsilon\}$$

has relative measure θ, the separated-node lemma gives $r+1$ points in E with separation $\gg \theta$. Newton interpolation then yields

$$\sup_B |P| \ll \varepsilon\theta^{-r}.$$

Since the left side equals one,

$$\theta \ll \varepsilon^{1/r}.$$

This proof explains the exponent $1/r$: a degree-r polynomial is determined by $r+1$ values, and the product of r separation denominators costs θ^{-r}.

3. Why max norms of polynomial vectors are good

Let $F = (P_1, \ldots, P_N)$ with every P_i of degree at most r. Set

$$\|F(x)\| = \max_i |P_i(x)|.$$

Choose one coordinate P_{i_0} satisfying

$$\sup_B |P_{i_0}| = \sup_B \|F\|.$$

Then

$$\{x : \|F(x)\| < \varepsilon \sup_B \|F\|\} \subset \{x : |P_{i_0}(x)| < \varepsilon \sup_B |P_{i_0}|\}.$$

Hence no factor depending on N is needed. This is particularly useful for exterior powers: the number of Plücker coordinates may be large, but one maximizing coordinate controls the whole sublevel set.

4. Products of local norms

Let

$$F(x) = \prod_{v \in S} |P_v(x_v)|_v,$$

with product measure on a product ball. A common mistake is to assert that the product of good functions is automatically good with the same exponent. It is not that formal: F can be small because one factor is extremely small or because many factors are moderately small.

Choose positive numbers η_v with $\prod_v \eta_v = \varepsilon$. If every local factor satisfies

$$|P_v(x_v)|_v \geq \eta_v \sup |P_v|_v,$$

then $F(x) \geq \varepsilon \sup F$. Therefore the product sublevel set lies in the union of the local sublevel sets. Taking, for example, $\eta_v = \varepsilon^{1/|S|}$ gives

$$m\{F < \varepsilon \sup F\} \leq \sum_{v \in S} C_v \varepsilon^{\alpha_v/|S|} m(B).$$

Thus F is good with any exponent no larger than

$$\frac{1}{|S|} \min_v \alpha_v.$$

Chapter 11 records a slightly more flexible threshold allocation, but this simple argument already shows the mechanism.

5. Exterior-coordinate calculation for a root flow

In $\mathrm{SL}_3(k)$ let

$$u(t) = I + tE_{12}.$$

For a vector $v = (a, b, c)^T$,

$$u(t)v = (a + tb, b, c)^T,$$

so the first coordinate is degree one. For a two-vector

$$w = Ae_1 \wedge e_2 + Be_1 \wedge e_3 + Ce_2 \wedge e_3,$$

we have

$$\bigwedge^2 u(t)w = Ae_1 \wedge e_2 + (B + tC)e_1 \wedge e_3 + Ce_2 \wedge e_3.$$

Again every coordinate has degree at most one. For a product of q elementary root flows, repeated substitution gives a degree bound at most q in each local parameter. This elementary observation is what converts polynomial group coordinates into the uniform degree hypotheses in quantitative nondivergence.

6. A diagnostic for noncollapse

Goodness controls how long a function can stay much smaller than its own maximum; it does not say that the maximum itself is large. The two inputs in quantitative nondivergence are therefore logically separate:

$$\boxed{\text{goodness}} \qquad \text{and} \qquad \boxed{\sup_B c(h(x)\Delta) \geq \rho.}$$

The second is an algebraic noncollapse condition. If it fails on larger and larger balls, the bounded-polynomial argument of Volume 4 turns that failure into an invariant rational flag. Keeping these two mechanisms separate is essential: analytic sublevel control never by itself rules out an exact algebraic obstruction.

CHAPTER 16

Products of local fields and good content functions

Let

$$X = \mathbb{R}^{d_\infty} \times \prod_{p \in S_f} \mathbb{Q}_p^{d_p}$$

with product Haar measure μ and product balls $B = \prod_v B_v$.

1. Federer and covering constants

The real factor is Federer and Besicovitch by Volume 1. Every p-adic factor is Federer with exact scaling and admits disjoint maximal-ball coverings by Lemma 2.1. A finite product is Federer; for rectangles/product balls the covering argument may be performed factor by factor. Consequently the marked-poset theorem of Volume 1 applies once its size functions are proved good.

2. Content

For local nonnegative functions F_v define

$$c(F)(x) = \prod_{v \in S} F_v(x_v).$$

Because variables are separated by place,

$$\sup_{x \in B} c(F)(x) = \prod_v \sup_{x_v \in B_v} F_v(x_v).$$

Lemma 2.1 (Product goodness)**.** *Suppose every F_v is (C_v, α_v)-good on B_v. Then $c(F)$ is (C, α)-good on B, where*

$$\alpha = \frac{1}{|S|} \min_v \alpha_v, \qquad C = \sum_v C_v.$$

PROOF. Put $M_v = \sup_{B_v} F_v$. If $\prod_v F_v(x_v) < \varepsilon \prod_v M_v$, then at least one v satisfies $F_v(x_v) < \varepsilon^{1/|S|} M_v$; otherwise multiplying the opposite inequalities gives a contradiction. Thus the bad set is contained in the union of cylindrical bad sets. Fubini gives

$$\mu(E) \le \sum_v C_v \varepsilon^{\alpha_v/|S|} \mu(B) \le C \varepsilon^{\alpha} \mu(B).$$

□

Corollary 2.2 (Exterior-content goodness). *Let $h : B \to \mathrm{GL}_n(\mathbb{Q}_S)$ have local matrix entries that are polynomials of uniformly bounded degree in the local variables. For every primitive submodule $\Delta \subset \mathbb{Z}_S^n$, the function*

$$x \longmapsto c(h(x)\Delta)$$

is (C, α)-good with constants depending only on S, the dimensions, and the degree bound, not on Δ.

Proof. Choose a primitive wedge $w_\Delta \in \bigwedge^r \mathbb{Z}_S^n$. At each place the coordinates of $\bigwedge^r h_v(x_v) w_\Delta$ are bounded-degree polynomials, and its local max norm is good by Volume 1 at the real place and Corollary 5.1 at finite places. Lemma 2.1 finishes. □

Worked Example 2.3. Let $S = \{\infty, p\}$ and

$$u(t_\infty, t_p) = \left(\begin{pmatrix} 1 & t_\infty \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & t_p \\ 0 & 1 \end{pmatrix} \right).$$

For $v = (a, b) \in \mathbb{Z}[1/p]^2$ the local vectors are $(a + t_v b, b)$. The content of the max norm is

$$\max(|a + t_\infty b|, |b|) \max(|a + t_p b|_p, |b|_p).$$

Each local factor is a max of degree-one polynomials, so the product is uniformly good. This is the simplest nontrivial size function in the Kleinbock–Tomanov theorem.

CHAPTER 17

Strong approximation: what must actually be proved

Strong approximation is often invoked as a slogan saying that rational points can imitate finitely many local conditions. There are two distinct statements, and later proofs need to know which one is being used.

1. Congruence density versus adelic density

Fix a finite set S containing ∞. For a group G defined over $\mathbb{Q}$, write

$$G(\mathbb{A}^S) = \prod_{p \notin S}' G(\mathbb{Q}_p)$$

for the restricted product with respect to chosen integral compact subgroups $G(\mathbb{Z}_p)$.

Definition 1.1. We say G has *strong approximation away from S* if the diagonal image of $G(\mathbb{Q})$ is dense in $G(\mathbb{A}^S)$.

If a target is integral at every $p \notin S$, this reduces to density of $G(\mathbb{Z}_S)$ in $\prod_{p \notin S} G(\mathbb{Z}_p)$, equivalently simultaneous surjectivity of congruence maps.

2. Approximation of scalar parameters

Lemma 2.1 (Restricted weak approximation in $\mathbb{Q}$)**.** *Let T be a finite set of primes and, for each $p \in T$, let $a_p \in \mathbb{Q}_p$ and $m_p \geq 1$. There exists $a \in \mathbb{Q}$ such that*

$$|a - a_p|_p \leq p^{-m_p} \qquad (p \in T)$$

and $a \in \mathbb{Z}_q$ for every $q \notin T$.

PROOF. Choose integers r_p so that $p^{r_p} a_p \in \mathbb{Z}_p$ and set $D = \prod_{p \in T} p^{r_p}$. We seek $b \in \mathbb{Z}$ with $a = b/D$. The conditions become congruences

$$b \equiv D a_p \pmod{p^{m_p + r_p}}$$

in $\mathbb{Z}_p$. Choose integral residues and solve simultaneously by the Chinese remainder theorem. Since the denominator D uses only primes in T, a is integral at every $q \notin T$. $\square$

3. A root-generation criterion

Proposition 3.1 (Elementary criterion for strong approximation). *Let G be a matrix group over $\mathbb{Q}$ equipped with one-parameter subgroups $x_\alpha : \mathbb{G}_a \to G$ defined over $\mathbb{Z}$. Assume:*

(1) for every prime p, the group $G(\mathbb{Q}_p)$ is generated by the $x_\alpha(\mathbb{Q}_p)$;
(2) for every $m \geq 1$, $G(\mathbb{Z}/p^m\mathbb{Z})$ is generated by the images of $x_\alpha(\mathbb{Z}/p^m\mathbb{Z})$.

Then G has strong approximation away from every finite S containing ∞, and

$$G(\mathbb{Z}_S) \longrightarrow G(\mathbb{Z}/m\mathbb{Z})$$

is surjective for every m coprime to the primes of S.

Proof. For congruence surjectivity, factor a target modulo each prime power dividing m into root elements, use the Chinese remainder theorem to lift all parameters to $\mathbb{Z}_S$, and multiply the lifted root elements.

For adelic density, it suffices to approximate a basic open set, which imposes conditions at finitely many primes $T \cap S = \varnothing$ and integrality elsewhere. Factor each target g_p into finitely many root elements. After inserting trivial factors, use a common word in the root labels at all $p \in T$. Apply Lemma 2.1 to approximate simultaneously the corresponding scalar parameters by rational numbers integral outside T. The product of the resulting global root elements approximates every g_p and is integral outside $S \cup T$, hence belongs to the required restricted-product neighborhood. □

Thus the serious algebra in our matrix cases is exactly root generation, not a mysterious global theorem.

CHAPTER 18

Strong approximation for SL_n

1. Elementary generation over fields

Lemma 1.1 (Elementary generation of $\mathrm{SL}_n(F)$). *For every field F and $n \geq 2$, $\mathrm{SL}_n(F)$ is generated by the elementary matrices $e_{ij}(t) = I + tE_{ij}$.*

Proof. Gaussian elimination using row additions reduces $g \in \mathrm{SL}_n(F)$ to a diagonal matrix. A row interchange and a row scaling can themselves be expressed by elementary matrices in a 2×2 block. Finally every determinant-one diagonal matrix is a product of blocks $\operatorname{diag}(u, u^{-1})$, and

$$\begin{pmatrix} u & 0 \\ 0 & u^{-1} \end{pmatrix} = e_{12}(u)e_{21}(-u^{-1})e_{12}(u)e_{12}(-1)e_{21}(1)e_{12}(-1).$$

Embedding these blocks proves the claim. □

Lemma 1.2 (Elementary generation over a prime-power local ring). *Let $R = \mathbb{Z}/p^m\mathbb{Z}$ and $n \geq 2$. Then $\mathrm{SL}_n(R)$ is generated by the elementary matrices*

$$e_{ij}(t) = I + tE_{ij}, \qquad i \neq j, \ t \in R.$$

Proof. We give the elimination argument because this local-ring step is the exact finite congruence input needed below. Let $A = (a_{ij}) \in \mathrm{SL}_n(R)$. Its first column is unimodular: the ideal generated by its entries is all of R. Since R is local, at least one entry in that column is a unit. Using elementary row additions and the determinant-one signed transposition

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = e_{12}(-1)e_{21}(1)e_{12}(-1),$$

embedded in the appropriate two coordinates, move a unit entry to the $(1,1)$ position.

If the new pivot is $u \in R^\times$, multiply on the left by a product of elementary matrices realizing $\operatorname{diag}(u^{-1}, u, 1, \ldots, 1)$. The root-word identity

$$w(a) := e_{12}(a)e_{21}(-a^{-1})e_{12}(a) = \begin{pmatrix} 0 & a \\ -a^{-1} & 0 \end{pmatrix},$$

$$w(a)w(-1) = \operatorname{diag}(a, a^{-1}).$$

shows that determinant-one diagonal scaling is elementary whenever a is a unit. The first pivot may therefore be normalized to 1.

Elementary row additions clear all entries below the pivot. We obtain

$$\begin{pmatrix} 1 & r \\ 0 & B \end{pmatrix}, \qquad B \in \mathrm{SL}_{n-1}(R).$$

Right multiplication by elementary matrices clears r. Induction on n reduces B to the identity. Every operation used is multiplication by an elementary matrix or its inverse, and $e_{ij}(t)^{-1} = e_{ij}(-t)$. Hence A is a product of elementary matrices. □

The same elimination over the local ring $\mathbb{Z}/p^m\mathbb{Z}$ is proved in Lemma 1.2. Therefore Proposition 3.1 applies.

Theorem 1.3 (Strong approximation for SL_n). *Let $n \geq 2$ and let S be a finite set of places of $\mathbb{Q}$ containing ∞. Then the diagonal subgroup $\mathrm{SL}_n(\mathbb{Q})$ is dense in*

$$\mathrm{SL}_n(\mathbb{A}^S) = \prod_{p \notin S}{}' \mathrm{SL}_n(\mathbb{Q}_p).$$

Equivalently, $\mathrm{SL}_n(\mathbb{Z}_S)$ is dense in $\prod_{p\notin S} \mathrm{SL}_n(\mathbb{Z}_p)$, and for every integer m supported outside S the reduction map

$$\mathrm{SL}_n(\mathbb{Z}_S) \to \mathrm{SL}_n(\mathbb{Z}/m\mathbb{Z})$$

is surjective.

Proof. Take the root groups $x_{ij}(t) = e_{ij}(t)$. Lemma 1.1 proves local generation over $\mathbb{Q}_p$, and Lemma 1.2 proves generation over every prime-power quotient. Proposition 3.1 gives both assertions. □

2. Why GL_n fails

Example 2.1. The determinant gives an immediate obstruction for GL_n. For instance $\det \mathrm{GL}_n(\mathbb{Z}) = \{\pm 1\}$, while $\det \mathrm{GL}_n(\mathbb{Z}/5\mathbb{Z}) = (\mathbb{Z}/5\mathbb{Z})^\times$. Hence reduction is not surjective. This is the elementary shadow of the general fact that simply connectedness is essential in semisimple strong approximation.

Worked Example 2.2 (Approximating two local matrices)**.** Let $g_2 \in \mathrm{SL}_2(\mathbb{Q}_2)$ and $g_3 \in \mathrm{SL}_2(\mathbb{Q}_3)$. Gaussian elimination writes each as a word in $e_{12}(t)$ and $e_{21}(t)$. Pad the shorter word with zero parameters so the root labels agree. For each word position choose $t \in \mathbb{Q}$ simultaneously close to its 2-adic and 3-adic parameters, with denominator supported only at 2 and 3. Multiplying the global elementary matrices yields $g \in \mathrm{SL}_2(\mathbb{Q})$ close to (g_2, g_3) and integral at every other prime. This is the entire strong-approximation mechanism in visible form.

CHAPTER 19

Proof laboratory: strong approximation with congruences

Strong approximation becomes much less mysterious when one solves a concrete problem all the way to an actual matrix word. We do this for SL_2, then show how the same calculation scales to SL_d.

1. Surjectivity modulo a prime power

Let $R = \mathbb{Z}/p^m\mathbb{Z}$ and

$$\bar{g} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(R).$$

At least one of a, c is a unit. Suppose first that a is a unit. Then

$$x_{21}(-ca^{-1})\bar{g} = \begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix}.$$

Multiplying on the left by the elementary word representing $\mathrm{diag}(a^{-1}, a)$ normalizes the pivot, and then $x_{12}(-ab)$ clears the upper-right entry. Thus $\bar{g}$ is an elementary word. If a is not a unit, c is; use the elementary Weyl matrix to exchange the two rows first.

Every parameter in this word is an element of R. Choose arbitrary integer lifts. The same word in $\mathrm{SL}_2(\mathbb{Z})$ reduces to $\bar{g}$. This proves surjectivity of

$$\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/p^m\mathbb{Z})$$

without counting group orders or using generators known from finite-group theory.

2. Several moduli at once

Let $m = \prod_i p_i^{e_i}$. The Chinese remainder theorem gives

$$\mathbb{Z}/m\mathbb{Z} \simeq \prod_i \mathbb{Z}/p_i^{e_i}\mathbb{Z}.$$

Factor each local target into an elementary word and pad all factorizations to one common list of root labels. At the k-th word position, the CRT chooses one integer parameter matching every prime-power parameter. Multiplying the global word produces the desired lift modulo m. Notice that noncommutativity causes no difficulty because we synchronize the *word positions*, not merely the unordered collections of root elements.

3. Adelic approximation rather than exact congruence

Suppose now we want a rational matrix close to prescribed $g_p \in \mathrm{SL}_2(\mathbb{Q}_p)$ at $p = 2, 3$, and integral at every other prime. Choose root factorizations of g_2 and g_3. A basic open neighborhood only asks for finite p-adic precision, so it suffices to approximate each local root parameter to that precision. Lemma 2.1 produces a rational parameter whose denominator uses only 2 and 3. Therefore the resulting matrix word is automatically integral at all other primes.

This last sentence is the strong-approximation miracle in elementary form: *local denominators can be confined to the places where nonintegrality is requested.*

Worked Example 3.1 (A simultaneous lift)**.** Take targets

$$\bar{g}_5 = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix} \pmod{25}, \qquad \bar{g}_7 = \begin{pmatrix} 1 & 7 \\ 2 & 15 \end{pmatrix} \pmod{49},$$

after adjusting entries so the determinants are one in the indicated rings. Run local Gaussian elimination to obtain elementary parameters $a_{5,k} \in \mathbb{Z}/25\mathbb{Z}$ and $a_{7,k} \in \mathbb{Z}/49\mathbb{Z}$. For each k, solve

$$a_k \equiv a_{5,k} \pmod{25}, \qquad a_k \equiv a_{7,k} \pmod{49}.$$

The product of the corresponding integral elementary matrices is simultaneously the desired lift. The exact numerical representatives are unimportant; the algorithm demonstrates that strong approximation is constructive.

4. Why simple connectedness appears in the general theorem

Our proof for SL_d never utters the words “simply connected” because that property has already been encoded by elementary root generation. The contrast with GL_d is instructive. Its determinant map survives every elementary row-addition operation and lands in $\mathbb{G}_m$, where rational integral units are far from dense adelically. In the general Kneser–Platonov theorem, simply connectedness removes precisely these finite-cover and character obstructions. The present book does not claim that general theorem; it proves the split classical matrix cases by exhibiting the root words explicitly.

CHAPTER 20

Strong approximation for symplectic groups

Let

$$J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}, \qquad \mathrm{Sp}_{2n}(R) = \{g \in \mathrm{GL}_{2n}(R) : g^T J g = J\}.$$

We prove the same root-generation criterion for Sp_{2n}.

1. Elementary symplectic transformations

For $i \neq j$ define

$$x_{e_i-e_j}(t) = I + tE_{ij} - tE_{n+j,n+i},$$
$$x_{e_i+e_j}(t) = I + tE_{i,n+j} + tE_{j,n+i},$$
$$x_{-e_i-e_j}(t) = I + tE_{n+i,j} + tE_{n+j,i},$$

and for the long roots

$$x_{2e_i}(t) = I + tE_{i,n+i}, \qquad x_{-2e_i}(t) = I + tE_{n+i,i}.$$

A direct multiplication verifies $x_\alpha(t)^T J x_\alpha(t) = J$.

Lemma 1.1 (Symplectic elimination over a field)**.** *For every field F, $\mathrm{Sp}_{2n}(F)$ is generated by the above root subgroups.*

PROOF. Write a symplectic basis as $(e_1, \ldots, e_n, f_1, \ldots, f_n)$ with $\langle e_i, f_j\rangle = \delta_{ij}$. Let $g \in \mathrm{Sp}_{2n}(F)$ and consider $v = ge_1$, a nonzero isotropic vector. The root transformations perform the following elementary operations on coordinates of v: add a multiple of one e-coordinate to another, add e_j to an f_i coordinate compatibly with the symplectic form, and use the long-root transformations to mix e_i with f_i. Ordinary Gaussian elimination using these operations sends v to e_1. More explicitly, first use the $e_i - e_j$ roots to concentrate the e-coordinates in the first position; if all e-coordinates vanish, a long-root operation creates a nonzero one from an f-coordinate; then clear the f-coordinates with $-e_1 - e_j$ and $-2e_1$ roots. A rank-one $\mathrm{Sp}_2 = \mathrm{SL}_2$ root word scales the remaining first coordinate to one.

After left multiplication by this root word we may suppose $ge_1 = e_1$. Symplecticity then implies that $gf_1 = f_1 + w + ce_1$ with w in the span of $e_2, \ldots, e_n, f_2, \ldots, f_n$. Root transformations involving index 1 clear w

and c without moving e_1. The resulting matrix fixes the hyperbolic plane $\langle e_1, f_1\rangle$ pointwise and restricts to an element of $\mathrm{Sp}_{2n-2}(F)$ on its symplectic orthogonal. Induction on n, with $\mathrm{Sp}_2 = \mathrm{SL}_2$, finishes. $\square$

Lemma 1.2 (Symplectic generation over $\mathbb{Z}/p^m\mathbb{Z}$)**.** *The same root elements generate* $\mathrm{Sp}_{2n}(\mathbb{Z}/p^m\mathbb{Z})$*.*

Proof. The preceding elimination uses division only by a pivot that is a unit. Over the local ring $R = \mathbb{Z}/p^m\mathbb{Z}$, a unimodular first column has at least one unit coordinate. The symplectic root operations can move such a coordinate into the first e-position exactly as above, after which all required pivot divisions are by units. The induction therefore works verbatim over R. $\square$

Theorem 1.3 (Strong approximation for Sp_{2n}). *For every $n \geq 1$ and finite $S \ni \infty$, the diagonal image of* $\mathrm{Sp}_{2n}(\mathbb{Q})$ *is dense in* $\mathrm{Sp}_{2n}(\mathbb{A}^S)$*; equivalently all congruence maps*

$$\mathrm{Sp}_{2n}(\mathbb{Z}_S) \to \mathrm{Sp}_{2n}(\mathbb{Z}/m\mathbb{Z}), \qquad (m, S) = 1,$$

are surjective.

Proof. Use the root subgroups displayed above. Lemmas 1.1 and 1.2 verify the two hypotheses of Proposition 3.1. $\square$

CHAPTER 21

Symplectic root calculus in coordinates

The proof of strong approximation for Sp_{2n} rests on the assertion that the symplectic root subgroups perform Gaussian elimination. Because this fact is often quoted too quickly, we verify the elementary operations explicitly.

Write vectors as pairs $(x, y) \in R^n \oplus R^n$ with symplectic form

$$\langle (x, y), (x', y') \rangle = x \cdot y' - y \cdot x'.$$

Let $e_i = (\mathbf{e}_i, 0)$ and $f_i = (0, \mathbf{e}_i)$.

1. Action of the root matrices

For $i \neq j$, the root element $x_{e_i - e_j}(t)$ acts by

$$e_j \mapsto e_j + te_i, \qquad f_i \mapsto f_i - tf_j,$$

fixing the other basis vectors. The two changes are forced to occur together so that the pairings remain unchanged. Likewise

$$x_{e_i+e_j}(t): \quad f_i \mapsto f_i + te_j, \quad f_j \mapsto f_j + te_i,$$

and

$$x_{2e_i}(t): \quad f_i \mapsto f_i + te_i.$$

The negative roots give the transposed operations on the e-coordinates. Each formula can be checked by pairing the images of the basis vectors.

2. Sending a primitive vector to e_1

Let R be either a field or a local ring $\mathbb{Z}/p^m\mathbb{Z}$, and let

$$v = \sum_i a_i e_i + \sum_i b_i f_i$$

be unimodular, meaning its coordinates generate the unit ideal. Over a local ring one coordinate is a unit. If some a_i is a unit, use roots $e_i - e_j$ and the Weyl words in the rank-one Sp_2-subgroups to move it to a_1. If all a_i lie in the maximal ideal, some b_j is a unit; the rank-one Weyl element exchanging e_j and f_j creates a unit e-coordinate.

Now clear $a_2, \dots, a_n$ using the roots $e_1 - e_j$. Clear the b_j, $j > 1$, with roots of type $-e_1 - e_j$, and clear b_1 with the long root $-2e_1$. We are left with $a_1 e_1$, where a_1 is a unit. A rank-one diagonal word in the $\mathrm{Sp}_2 = \mathrm{SL}_2$ subgroup on $\langle e_1, f_1 \rangle$ scales this to e_1.

No division except by the unit pivot has occurred, so the proof works identically over a field and over $\mathbb{Z}/p^m\mathbb{Z}$.

3. Fixing the hyperbolic partner

After the first step, suppose $ge_1 = e_1$. Symplecticity gives

$$\langle e_1, gf_1 \rangle = 1,$$

so the coefficient of f_1 in gf_1 is one. Write

$$gf_1 = f_1 + ce_1 + w, \qquad w \in \langle e_2, \ldots, e_n, f_2, \ldots, f_n \rangle.$$

The root groups involving index 1 but chosen from the stabilizer of e_1 can subtract the coordinates of w one at a time. Finally the long root $2e_1$ removes ce_1. Thus we can arrange that both e_1 and f_1 are fixed.

The orthogonal complement of their hyperbolic plane is a free symplectic module of rank $2n-2$, and the remaining matrix belongs to $\mathrm{Sp}_{2n-2}(R)$. Induction proves root generation.

4. A commutator check

The root subgroups are not an arbitrary collection of elementary-looking matrices. Their commutators generate the higher root geometry. For example, when the indices are distinct,

$$[x_{e_i - e_j}(s), x_{e_j - e_k}(t)] = x_{e_i - e_k}(st).$$

Similarly, commutators of a short positive root with a suitable short root produce the long roots. These identities explain why the same family controls both Gaussian elimination and the Lie algebra of Sp_{2n}.

Why this matters. Later expansion and random-walk arguments often use root commutators rather than matrix elimination. Proving the coordinate formulas here makes that later language concrete: root systems encode which elementary shears can manufacture which new directions.

CHAPTER 22

Root groups and the boundary of the elementary proof

The preceding arguments are instances of a general pattern for split simply connected groups, but it is important to separate a pattern from a proved theorem.

1. Steinberg relations in the matrix cases

For SL_n, the elementary matrices satisfy

$$[e_{ij}(s), e_{jk}(t)] = e_{ik}(st) \qquad (i, j, k \text{ distinct}),$$

and commute when the corresponding roots do not add. The symplectic root matrices satisfy analogous relations

$$[x_\alpha(s), x_\beta(t)] = \prod_{i,j>0} x_{i\alpha+j\beta}(c_{ij} s^i t^j),$$

whenever the indicated roots exist. In our proofs these identities are consequences of direct matrix multiplication, not imported Chevalley theory.

2. A proved abstract criterion, not a general Chevalley theorem

Proposition 3.1 shows that once local and congruence root generation are established, the global strong-approximation step is elementary. For split Chevalley groups of arbitrary type, verifying those generation properties over all relevant local rings is a substantial theorem. We therefore record the historical generalization only as context.

Historical note (not a logical input). Kneser and Platonov proved that an absolutely almost simple simply connected group over a number field satisfies strong approximation away from S precisely when the product of its S-local groups is noncompact (with the usual formulation and hypotheses). The proof for nonsplit groups uses structure theory far beyond the matrix elimination developed here. This stronger theorem is not a logical input to Volumes 3–4.

3. Why this scope is sufficient now

The next volume develops S-arithmetic Ratner theory on arithmetic quotients of SL_n and closely related matrix groups. Its compactness and linearization arguments use S-Mahler and congruence density, both proved here. Later arithmetic packet problems involving spin groups will include the additional strong-approximation theorem they require rather than referring back to an unproved sentence in this volume.

CHAPTER 23

The adelic–congruence dictionary

The same approximation theorem appears in three guises in later papers. We prove their equivalence in the matrix cases so that switching language never hides a step.

1. Restricted products

For $G = \mathrm{SL}_n$ or Sp_{2n} define

$$G(\mathbb{A}_f^S) = \prod_{p \notin S}' G(\mathbb{Q}_p)$$

with respect to $G(\mathbb{Z}_p)$. Its compact open subgroup

$$K^S = \prod_{p \notin S} G(\mathbb{Z}_p)$$

has a neighborhood basis at the identity formed by principal congruence subgroups

$$K(m) = \ker\left(K^S \to G(\mathbb{Z}/m\mathbb{Z})\right), \qquad (m, S) = 1.$$

Proposition 1.1 (Three equivalent integral forms)**.** *For $G = \mathrm{SL}_n$ or Sp_{2n} the following are equivalent:*

1. *$G(\mathbb{Z}_S)$ is dense in K^S;*
2. *for every m supported outside S, the reduction map $G(\mathbb{Z}_S) \to G(\mathbb{Z}/m\mathbb{Z})$ is surjective;*
3. *every coset of every principal congruence subgroup $K(m)$ meets the diagonal image of $G(\mathbb{Z}_S)$.*

Proof. The equivalence of (2) and (3) is the definition of the quotient $K^S/K(m)$ together with the Chinese-remainder identification of its finite factors. Condition (3) says exactly that the diagonal image meets every member of a neighborhood basis in every translate, which is density. Thus (1) and (3) are equivalent. $\square$

2. Double cosets and arithmetic quotients

Suppose T is a finite set disjoint from S. Strong approximation lets one move finite-place compact data between a rational representative and an arithmetic congruence class. Concretely, if $g = (g_p)_{p \notin S}$ is integral outside T, choose $q \in G(\mathbb{Q})$ approximating g_p for $p \in T$ and integral at all $p \notin S \cup T$. Then $q^{-1}g$ lies in a prescribed small compact-open neighborhood. This is the exact move later used to replace local packet data by global arithmetic representatives.

Warning 2.1. Strong approximation says nothing analogous for a torus such as $\mathbb{G}_m$. The product formula and ideal-class obstructions survive in the adelic quotient. This contrast is one reason semisimple simply connected covers repeatedly appear in arithmetic dynamics.

CHAPTER 24

Workshop: from congruences to adelic neighborhoods

Strong approximation is often invoked in one sentence because authors move freely between congruences and finite adeles. This chapter makes that dictionary explicit enough that later packet arguments can be read without guessing which topology is being used.

1. What an open set in a restricted product looks like

Let

$$G(\mathbb{A}_f^S) = \prod_{p\notin S}{}' G(\mathbb{Q}_p)$$

with respect to $K_p = G(\mathbb{Z}_p)$. A basic open neighborhood of $g = (g_p)$ has the form

$$\mathcal{U} = \prod_{p\in T} U_p \times \prod_{p\notin S\cup T} K_p,$$

where T is finite and each $U_p \subset G(\mathbb{Q}_p)$ is open. Thus every adelic approximation problem is finite: only finitely many primes impose nonintegral or higher-congruence conditions.

If $g \in K^S = \prod_{p\notin S} K_p$, shrink each U_p until it contains a coset of a principal congruence subgroup

$$K_p(p^{e_p}) = \ker\big(G(\mathbb{Z}_p) \to G(\mathbb{Z}/p^{e_p}\mathbb{Z})\big).$$

By the Chinese remainder theorem the simultaneous conditions are equivalent to one congruence modulo

$$m = \prod_{p\in T} p^{e_p}.$$

This is the concrete reason surjectivity of all reduction maps implies density in K^S.

2. An explicit SL_2 calculation

Suppose $S = \{\infty\}$ and we want an integral matrix $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ which is congruent modulo 5^2 to

$$A = \begin{pmatrix} 1 & 7 \\ 3 & 22 \end{pmatrix}.$$

First check the determinant modulo 25:

$$1 \cdot 22 - 7 \cdot 3 = 1 \pmod{25}.$$

Thus A defines an element of $\mathrm{SL}_2(\mathbb{Z}/25\mathbb{Z})$. Elementary row operations over $\mathbb{Z}/25\mathbb{Z}$ express A as a product of elementary matrices

$$e_{12}(a) = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}, \qquad e_{21}(b) = \begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix}.$$

Lift each parameter a, b to an integer and multiply the corresponding integral elementary matrices. The result is an element of $\mathrm{SL}_2(\mathbb{Z})$ reducing to A. This is strong approximation in its most elementary form.

With several primes, choose the elementary parameters simultaneously by the Chinese remainder theorem. Nothing conceptually new occurs: "adelic density" is simply simultaneous finite-prime congruence solvability plus the fact that only finitely many primes are constrained.

3. Why rational approximation is slightly different

For a point $g_p \in G(\mathbb{Q}_p)$ which is not integral, choose an exponent N_p so that the denominators are cleared by p^{N_p}. In a simply connected split matrix group, Cartan/root decompositions allow one to encode the nonintegral part by rational root parameters whose denominators involve only p. Weak approximation for scalars then chooses rational parameters simultaneously close to the required local ones and integral at every other prime. Multiplying the root elements gives a rational group element with the desired local behavior. This is the passage from density in K^S to the full strong-approximation statement used later.

4. Double cosets as local-global bookkeeping

Fix a compact open $K_f \subset G(\mathbb{A}_f)$. The double quotient

$$G(\mathbb{Q})\backslash G(\mathbb{A}_f)/K_f$$

measures the failure of a rational point to absorb all finite-place data. For $G = \mathrm{SL}_n$ or Sp_{2n}, strong approximation away from a nonempty set of places makes the relevant simply connected double quotient trivial in the situations used in this series. For a torus it need not be trivial; ideal classes and norm conditions survive.

This contrast is not cosmetic. Later arithmetic packet problems often begin with a torus embedded in a simply connected ambient group. Strong approximation simplifies the ambient group, while the residual torus double cosets are exactly where class groups and packet multiplicities live.

5. A topology checklist

When a later argument says "choose a rational element close to the prescribed local data," the reader should verify four items:

(1) Which finite set of places is actually constrained?
(2) At all other places, what integral compact open subgroup must be preserved?
(3) Is the acting algebraic group simply connected and generated by the root groups for which strong approximation has been proved?
(4) Is the remaining obstruction toral? If so, it should be recorded as an arithmetic packet rather than wished away by strong approximation.

This checklist prevents one of the most common local–global mistakes in homogeneous dynamics: applying a simply connected strong-approximation statement to a torus or to an adjoint group without tracking the covering obstruction.

CHAPTER 25

Worked examples

1. $S = \{\infty, p\}$ and rank two

Let

$$X = \mathrm{SL}_2(\mathbb{R}) \times \mathrm{SL}_2(\mathbb{Q}_p) / \mathrm{SL}_2(\mathbb{Z}[1/p]).$$

For

$$a(t,m) = \left(\begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}, \begin{pmatrix} p^m & 0 \\ 0 & p^{-m} \end{pmatrix} \right)$$

and $v = (r,s) \in \mathbb{Z}[1/p]^2$, the vector content is

$$c(a(t,m)v) = \max(e^t|r|, e^{-t}|s|) \max(p^{-m}|r|_p, p^m|s|_p).$$

Multiplying v by p^k changes the two local max norms but not their product. The S-unit balancing lemma chooses k so the two local contributions are comparable. Cusp excursion therefore depends on the combined real/p-adic slope, not on either coordinate separately.

2. An explicit S-Mahler degeneration

Consider

$$g_j = \left(\begin{pmatrix} p^j & 0 \\ 0 & p^{-j} \end{pmatrix}, \begin{pmatrix} p^{-j} & 0 \\ 0 & p^j \end{pmatrix} \right).$$

At first sight both local components diverge. But the vector e_1 has content

$$p^j \cdot p^j = p^{2j},$$

whereas $p^{-j}e_1 \in \mathbb{Z}[1/p]^2$ has the same content by S-unit invariance. In fact the entire family represents a single compact direction after multiplication by the arithmetic diagonal matrix $\mathrm{diag}(p^{-j}, p^j)$. This example shows why local Cartan exponents alone do not detect divergence; only the quotient by the S-unit lattice does.

3. Congruence lifting by hand

Take

$$\bar{g}_5 = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix} \pmod{25}$$

with determinant 1 modulo 25. If the top-left entry is a unit, multiply on the left by $e_{21}(-5 \cdot 2^{-1})$ to clear the lower-left entry, then by e_{12} to clear the upper-right entry. The remaining diagonal determinant-one matrix is a fixed word in elementary matrices. Lifting each parameter to an integer gives $g \in \mathrm{SL}_2(\mathbb{Z})$ with $g \equiv \bar{g}_5 \pmod{25}$. The same procedure at finitely many primes and the Chinese remainder theorem prove simultaneous lifting.

4. Failure for a torus

For $G = \mathbb{G}_m$, the closure of $\mathbb{Q}^\times$ in the finite ideles is constrained by valuations and the product formula. Already $\mathbb{Z}^\times = \{\pm 1\}$ is not dense in $\mathbb{Z}_5^\times$. Root unipotents are absent, and the elementary proof has nowhere to start.

CHAPTER 26

Reader workshop: proving compactness from scratch

This chapter recombines the preceding ingredients in one uninterrupted proof, because the architecture is easier to retain when one sees the entire chain.

Let $S = \{\infty, p\}$ and let $\Lambda_j = g_j\mathbb{Z}[1/p]^d$ be unimodular. Assume

$$c(x) \geq \varepsilon > 0 \qquad (0 \neq x \in \Lambda_j)$$

uniformly in j. We prove directly that a subsequence converges in $X_{S,d}$.

Step 1: choose a short primitive vector. The S-Blichfeldt lemma gives $x_{1,j} \neq 0$ with $c(x_{1,j}) \leq C_d$. Divide by a gcd in $\mathbb{Z}[1/p]$ to make it primitive; content cannot increase. Balance by p^{m_j} so that

$$\varepsilon^{1/2} \ll \|x_{1,j,\infty}\|, \|x_{1,j,p}\|_p \ll 1.$$

Step 2: pass to the quotient. Primitivity makes $\mathbb{Z}[1/p]x_{1,j}$ a direct summand. The quotient lattice has rank $d-1$ and covolume controlled by the size of $x_{1,j}$. Apply the same short-vector argument to the quotient and lift. Iteration produces a basis $x_{1,j}, \ldots, x_{d,j}$ whose successive quotient lengths have uniform upper bounds.

Step 3: exclude collapsing angles. Suppose the first r basis vectors become nearly dependent at the real place. Then their wedge has small real norm. Since the full determinant is one, either the p-adic wedge compensates by becoming large or the complementary wedge does. Multiply the submodule by an S-unit to balance the wedge. A short vector in this primitive rank-r module, obtained from the rank-r S-Blichfeldt lemma, would then have content tending to zero, contradicting ε. Thus all successive quotient lengths have uniform lower bounds as well.

Step 4: produce bounded representatives. Use S-units to balance the basis columns subject to product one. Every local matrix entry is now bounded. Determinant one bounds the inverse matrices by cofactors. Hence the basis matrices lie in a fixed compact subset of $\mathrm{SL}_d(\mathbb{R}) \times \mathrm{SL}_d(\mathbb{Q}_p)$. Pass to a convergent subsequence. Its images in the quotient converge, proving compactness.

The proof is not shorter than simply citing S-Mahler, but it makes visible the three arithmetic mechanisms: primitive saturation, S-unit balancing, and the product formula.

CHAPTER 27

Exercises and research-facing problems

Exercise 0.1. Prove that $\mathbb{Z}_S$ is a PID when $K = \mathbb{Q}$, and give an example showing that $\mathcal{O}_S$ need not be a PID for a general number field.

Exercise 0.2. For $K = \mathbb{Q}(\sqrt{2})$ compute the rank of $\mathcal{O}_K^\times$ and verify directly that $1 + \sqrt{2}$ gives a nontrivial logarithmic lattice direction.

Exercise 0.3. Show that if $S = \{\infty, p\}$, every class in $\mathbb{Q}_S/\mathbb{Z}[1/p]$ has a representative in $[0, 1) \times \mathbb{Z}_p$.

Exercise 0.4. Carry out symplectic elimination explicitly in $\mathrm{Sp}_4(F)$ and write a generic element as a word in the root matrices of Chapter 14.

Exercise 0.5. Prove the congruence lifting theorem for $\mathrm{SL}_2(\mathbb{Z}[i])$ modulo a principal ideal. Identify exactly where the PID assumption enters if one tries to repeat the argument for an arbitrary number field.

Research Laboratory 0.6. Develop a complete strong-approximation proof for a split spin group using explicit Eichler transvections. The purpose is not to cite Kneser–Platonov but to identify the additional algebra absent from the A and C matrix cases.

APPENDIX A

Standard background not reproved

The following facts are used as ordinary graduate background: existence and uniqueness of Haar measure; unique factorization of nonzero ideals in the ring of integers of a number field; the basic theory of completions; the Chinese remainder theorem for ideals; elementary module theory over a PID; and the inverse-limit description of $\mathbb{Z}_p$. From Volume 1 we use Blichfeldt/Minkowski in Euclidean space, ordinary Mahler compactness, exterior powers, and the real polynomial-goodness theorem. No research-level homogeneous-dynamics theorem is imported.

APPENDIX B

Historical sources and provenance

The local-field and S-arithmetic quantitative language follows the framework developed by Kleinbock and Tomanov. The general strong-approximation theorem is due, in characteristic zero, to Kneser and Platonov; a useful modern account is A. Rapinchuk, *On strong approximation for algebraic groups*. Those citations record provenance. The proofs logically used in this volume are the ones written in the preceding chapters, with strong approximation deliberately restricted to the SL_n and Sp_{2n} cases proved by elementary/root elimination.

Scope summary

The results used by Volume 4 are proved here in the stated arithmetic matrix scope. In particular, the strong-approximation theorems established in this volume are Theorems 1.3 and 1.3. Strong approximation for other algebraic groups requires additional structure and is not a consequence of those two matrix-group arguments.

Volume 4

S-Arithmetic Ratner Theory

Student orientation: what changes in S-arithmetic Ratner theory

The qualitative rigidity pattern is the same as over $\mathbb{R}$, but the parameter groups may be real, p-adic, or products of both. A *one-parameter unipotent subgroup* over a local field is the image of the additive local field under a polynomial unipotent homomorphism. A subgroup is *generated by unipotents* when it is topologically generated by such local one-parameter groups. A *rational stratum* is a homogeneous locus defined by a rational algebraic subgroup and its arithmetic lattice intersection. The proof architecture is: local ergodic theorem $\to$ polynomial comparison $\to$ quasi-regular map $\to$ additional invariance $\to$ singular-stratum induction. Do not merge the real and nonarchimedean rescaling arguments; ultrametric geometry changes the selection step.

Reader guide: a concrete model

Why this matters.

Volume 4 is organized around the passage from *S-arithmetic homogeneous spaces and rational strata* to *Scope summary.* The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.1 (A calculation to keep in view)**.** Consider the standard unipotent and diagonal elements

$$u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad a_s = \begin{pmatrix} e^s & 0 \\ 0 & e^{-s} \end{pmatrix}.$$

A direct multiplication gives

$$a_s u_t a_{-s} = u_{e^{2s}t}.$$

Thus a displacement of size $|t|$ in the unstable horospherical direction is enlarged by e^{2s}. If $|t| = e^{-2s}$, the conjugated displacement has order one. This elementary calculation is the model for the "first visible scale" used repeatedly in nondivergence and shearing arguments.

Preface

This volume proves the S-arithmetic analogue of the unipotent rigidity package in the arithmetic matrix setting used by the sequel. The point is not to state that Margulis–Tomanov and Kleinbock–Tomanov proved such theorems; the point is to understand why the real proof survives, which pieces become easier over an ultrametric field, and which pieces require new arithmetic bookkeeping.

Throughout the core chapters

$$S = \{\infty, p_1, \ldots, p_s\}, \qquad G_S = \prod_{v \in S} \mathrm{SL}_n(\mathbb{Q}_v), \qquad \Gamma = \mathrm{SL}_n(\mathbb{Z}_S),$$

or a finite-index subgroup thereof. We allow algebraic subgroups defined over $\mathbb{Q}$ and their S-points. This scope is large enough for the quantitative and arithmetic applications developed immediately afterwards, while keeping representative finiteness and rational strata completely explicit. The general Margulis–Tomanov theorem for arbitrary almost-linear groups over products of characteristic-zero local fields is recorded historically after the proof, not substituted for it.

The proof has two spines. The qualitative spine is

$$\text{local ergodic theorem} \to \text{polynomial divergence} \to \text{H-property}$$
$$\to \text{linearization} \to \text{additional invariance} \to \text{induction on rational strata.}$$

The quantitative spine is

$$\text{local polynomial goodness} \to \text{product content} \to \text{marked primitive modules}$$
$$\to \text{quantitative non-divergence} \to \text{uniform nonescape.}$$

Volume 3 supplies local fields, S-units, S-Mahler compactness, and the arithmetic congruence tools. Volume 2 supplies the exterior-linearization philosophy in the real arithmetic case. Every change needed in the product of local fields is proved here.

Scope and prerequisites

Prerequisites. We use standard measure theory, characteristic-zero local Lie/algebraic group theory and the elementary algebraic-group structure listed in Appendix A. The Borel–Harish–Chandra lattice criterion and the arithmetic-hull bridge are proved in Chapter 2; they are not imported. The real Ratner notebook is used only for already established field-independent proof architecture (the finite normalizer-correction ledger and singular-fiber bookkeeping); every place-sensitive ingredient needed for the S-arithmetic argument is proved in this volume or in Volumes 1–3. The Margulis–Tomanov measure-classification theorem is not a logical input.

Results proved in the completed revision

(1) the matrix Borel–Harish–Chandra finite-covolume criterion and the unipotent-ergodic rational arithmetic-hull theorem;

(2) the one-factor $\mathbb{Q}_p$ pointwise theorem and the L^2 rectangular pointwise theorem for mixed products of real and nonarchimedean additive parameters;

(3) local and product polynomial shearing/H-properties, including synchronized uniform-generic-core block comparison;

(4) the one-parameter and product quasi-regular-map constructions, normalizer property, and Basic Lemma under the explicitly verified tame normalizations;

(5) the *relative* finite normalizer-correction algorithm with controlled real and nonarchimedean reparametrization, and the full H-property realization relative to the full local stabilizer core $L = \langle U, I_{\mathrm{an}} \rangle$;

(6) arithmetic conversion of the terminal primitive degeneracy equations to a proper rational exterior singular stratum by recurrent return, finite-window representatives, polynomial goodness, and persistence;

(7) local-field exterior representative finiteness and quantitative singular-stratum occupation/induction;

(8) the product-local-field marked-poset theorem and the stated arithmetic Kleinbock–Tomanov quantitative nondivergence theorem;
(9) uniform nonescape with a proper finite-volume algebraic alternative;
(10) arithmetic S-linear measure classification for one local unipotent parameter and finite products of commuting local one-parameter factors;
(11) the stated direct-product generated-action extension;
(12) homogeneous orbit closure and pointwise equidistribution in the same arithmetic matrix scope.

Closure of the former relative-degeneration seam

The previous strict working revision stopped at a genuine gap: an arbitrary L-valued correction was being treated as harmless inside a U-average, and the normalizer-concentration branch was then rationalized by an unproved "rational-hull capture" assertion. Both shortcuts have been removed. Chapter 12 now proves the required bridge. Corrections are taken in $N_L(U)$, so conjugation acts on the additive parameter by a scalar; real corrections are controlled by a weighted Cesàro lemma and finite-place corrections by the ultrametric inverse-function/change-of-variables theorem. The active coefficient degree then decreases finitely many times. A regular terminal limit contradicts $\operatorname{Lie} L = \operatorname{Lie} I(\mu)$; a degenerate terminal coefficient tuple is converted to exact arithmetic exterior incidence only after recurrence has supplied primitive S-integral data. Thus no arbitrary analytic subgroup is silently declared rational.

Scope warning

The proved classification scope is the arithmetic matrix setting developed in this series: S-points of SL_n, rational algebraic strata, their finite-index variants, and the direct-product generated actions explicitly reduced to the classified parameter groups. The stronger Margulis–Tomanov theorem for general almost-linear groups, arbitrary unipotent radicals, and all structural variants remains historical context and is not claimed as a consequence of this volume.

CHAPTER 1

S-arithmetic homogeneous spaces and rational strata

Fix

$$S = \{\infty, p_1, \dots, p_s\}, \qquad G = G_S = \prod_{v \in S} \mathrm{SL}_n(\mathbb{Q}_v), \qquad \Gamma = \mathrm{SL}_n(\mathbb{Z}_S),$$

and write $X = G/\Gamma$. Volume 3 proves that Γ is discrete, that X has finite G-invariant volume, and that cusp escape is detected by primitive exterior vectors. We normalize Haar measure on X to be a probability measure.

There are two algebraic obstruction families in this volume and they must not be confused.

(1) *Cusp obstructions* are rational flags, or equivalently primitive $\mathbb{Z}_S$-submodules of $\mathbb{Z}_S^n$. Their stabilizers are parabolic and need not have finite volume.

(2) *Rigidity strata* are rational homogeneous subgroups whose arithmetic quotients have finite volume. They are the possible proper supports of unipotent-invariant probability measures.

The quantitative nondivergence theorem uses the first family; measure classification and avoidance use the second.

1. The finite-volume rational class

Let $\mathbf{H} < \mathrm{SL}_n$ be a connected $\mathbb{Q}$-algebraic subgroup and put $H = \prod_{v \in S} \mathbf{H}(\mathbb{Q}_v)$, replacing it by the finite-index open subgroup needed to remove irrelevant component-group issues. Define

$$\mathscr{H}_\Gamma = \{\mathbf{H} : \ H/(H \cap \Gamma) \text{ carries a finite } H\text{-invariant measure}\}.$$

This is a countable family because there are only countably many $\mathbb{Q}$-subgroups of SL_n.

The arithmetic input needed to recognize such strata is no longer imported. Chapter 2 proves the matrix S-arithmetic Borel–Harish–Chandra criterion (Theorem 7.1) and the unipotent-ergodic arithmetic-hull theorem (Theorem 8.4). In particular, every connected rational character-free subgroup has arithmetic finite covolume, and every proper homogeneous probability carrier produced by the rigidity induction is contained in a proper rational finite-volume carrier.

2. Removing rational characters

For a connected $\mathbb{Q}$-group $\mathbf{F}$, define its character-free core

$$\mathbf{F}^1 = \bigcap_{\chi \in X_{\mathbb{Q}}(\mathbf{F})} \ker \chi.$$

Only finitely many characters are needed to define this intersection because the character group is finitely generated.

Lemma 2.1 (Unipotents lie in the character-free core)**.** *Every algebraic homomorphism from a unipotent group to $\mathbb{G}_m$ is trivial. Consequently, if a unipotent subgroup U is contained in F, then $U \subset F^1$. If F has a nontrivial rational character, then $\dim F^1 < \dim F$.*

Proof. After triangularizing a unipotent group, every element has all eigenvalues equal to one. A rational character is an eigenvalue monomial in a faithful triangular representation, hence equals one on every unipotent element. The second assertion follows. A nontrivial character has positive-dimensional image in $\mathbb{G}_m$, so its kernel has strictly smaller dimension; intersecting kernels cannot increase dimension. □

This lemma is what makes induction on intersections work. If two rational incidence conditions meet in a proper rational subgroup F containing U, then either F already has no rational characters and Theorem 7.1 puts it in $\mathscr{H}_\Gamma$, or F^1 is a strictly smaller rational subgroup still containing U. Repeating terminates. At the terminal core, Corollary 7.2 supplies the required finite arithmetic covolume.

3. Exterior representatives

Let $\mathfrak{g}_{\mathbb{Q}} = \mathfrak{sl}_n(\mathbb{Q})$ and $\mathfrak{g}_{\mathbb{Z}_S} = \mathfrak{sl}_n(\mathbb{Z}_S)$. If $\mathbf{H} \in \mathscr{H}_\Gamma$ has dimension d, then $\mathfrak{h} \cap \mathfrak{g}_{\mathbb{Z}_S}$ is an S-lattice in $\mathfrak{h}$. Choose a primitive generator

$$p_H \in \bigwedge^d \mathfrak{g}_{\mathbb{Z}_S}$$

of its top exterior line. It is determined up to multiplication by an S-unit and sign. Product content therefore removes the ambiguity.

For a subgroup $U < G$, set

$$N(H, U) = \{g \in G : U \subset gHg^{-1}\}.$$

The corresponding singular set in X is

$$\mathcal{S}(H, U) = \pi(N(H, U)),$$

where $\pi : G \to G/\Gamma$ is the quotient map. A point may have several representatives because replacing g by $g\gamma$ conjugates H by γ. The finite-window representative theory below is precisely what keeps this ambiguity finite on compact sets.

4. The two exterior languages

For homogeneous strata we use the adjoint wedge

$$\bigwedge^{\dim H} \mathrm{Ad}(g)\mathfrak{p}_H.$$

For cusp escape we use primitive wedges in $\bigwedge^r \mathbb{Z}_S^n$. Both are finite-dimensional polynomial representations, but their geometric meanings are different. Keeping them separate prevents a common error: the stabilizer of a short lattice vector is parabolic and generally has infinite covolume, whereas a proper support of an invariant probability measure must come from a finite-volume homogeneous subgroup.

CHAPTER 2

Arithmetic lattices and rational hulls: the complete reduction argument

The singular-stratum induction requires a precise arithmetic fact: if a connected rational matrix group has no nontrivial rational character, then its S-integral points have finite covolume. This is the Borel–Harish–Chandra mechanism in exactly the form needed later. In an earlier draft this step was compressed into a "matrix instability" lemma; that only renamed the difficult part of reduction theory. We replace it here by a proof whose analytic and arithmetic ingredients are all visible.

The algebraic-group input allowed in this chapter is ordinary characteristic-zero structure theory: Levi decomposition, minimal rational parabolics, maximal rational split tori, relative roots and Weyl chambers, rational conjugacy of minimal rational parabolics, and the equivalence (for semisimple groups) between positive rational rank and the existence of a nontrivial rational unipotent subgroup. Local Iwasawa decomposition is also used. The finite-covolume assertion, the rank-zero compactness step, and the passage to fixed S are proved below.

Throughout, $\mathbb{A}$ is the adele ring of $\mathbb{Q}$. A rational representation is provided with a rational lattice and with the associated Euclidean/max norms at all places. For $z = (z_v)_v$ in a rational vector space put

$$c(z) = \prod_v \|z_v\|_v.$$

The product is finite for adelic vectors arising from rational data. If $q \in \mathbb{Q}^\times$, the product formula gives $c(qz) = c(z)$.

1. Three compactness lemmas

Lemma 1.1 (The additive adelic quotient)**.** *The diagonal subgroup $\mathbb{Q} < \mathbb{A}$ is discrete and $\mathbb{A}/\mathbb{Q}$ is compact.*

Proof. The neighborhood

$$(-1/2, 1/2) \times \prod_p \mathbb{Z}_p$$

contains no nonzero rational number, proving discreteness. Given $x = (x_\infty, (x_p)_p)$, only finitely many finite coordinates are nonintegral. Choose $q_0 \in \mathbb{Q}$ with $x_p - q_0 \in \mathbb{Z}_p$ for all p by the Chinese remainder theorem after clearing those finitely many denominators. Adding an integer to q_0 preserves the finite-place conditions and puts $x_\infty - q_0$ in $[-1/2, 1/2]$. Hence every class has a representative in that compact product. □

Lemma 1.2 (Unipotent adelic quotients)**.** *If* $\mathbf{N}/\mathbb{Q}$ *is connected unipotent, then* $N(\mathbb{Q})\backslash N(\mathbb{A})$ *is compact.*

Proof. In characteristic zero a rational unipotent group has a rational central series whose successive quotients are vector groups. The vector-group case is Lemma 1.1 coordinate by coordinate. Suppose $1 \to Z \to N \to \bar{N} \to 1$ is one step of such a series and the assertion is known for Z and $\bar{N}$. Choose compact representatives in the base, lift them locally to a compact set in $N(\mathbb{A})$, and then multiply by compact representatives in the central fiber. Their product maps onto $N(\mathbb{Q})\backslash N(\mathbb{A})$. Induction gives compactness. □

Lemma 1.3 (Adelic Mahler for the ambient special linear group)**.** *Let* $m \geq 2$*. For every* $\delta > 0$ *the set of classes* $\mathrm{SL}_m(\mathbb{Q})g \in \mathrm{SL}_m(\mathbb{Q})\backslash \mathrm{SL}_m(\mathbb{A})$ *satisfying*

$$c(g^{-1}v) \geq \delta \qquad (0 \neq v \in \mathbb{Q}^m)$$

is compact.

Proof. This is an exact consequence of two results already proved earlier in the series, and we record the reduction so that no adelic compactness theorem is being imported. Strong approximation for SL_m from Volume 3 gives

$$\mathrm{SL}_m(\mathbb{A}_f) = \mathrm{SL}_m(\mathbb{Q})K_f, \qquad K_f = \prod_p \mathrm{SL}_m(\mathbb{Z}_p).$$

After left multiplication by a rational matrix, therefore, a representative has the form (g_∞, k_f) with $k_f \in K_f$. The use of $g^{-1}v$ is essential here: under left multiplication by $q \in \mathrm{SL}_m(\mathbb{Q})$ the family $\{(qg)^{-1}v : v \in \mathbb{Q}^m\}$ is the same family as $\{g^{-1}v : v \in \mathbb{Q}^m\}$ after the rational change of variable $v \mapsto q^{-1}v$.

For a primitive $v \in \mathbb{Z}^m$, one has $\|k_p^{-1}v\|_p = 1$ at every finite place. Thus the displayed content lower bound becomes

$$\|g_\infty^{-1}v\|_\infty \geq \delta \qquad (0 \neq v \in \mathbb{Z}^m).$$

This is precisely the real Mahler condition for the left-quotient convention: the lattice attached to $\mathrm{SL}_m(\mathbb{Z})g_\infty$ is $g_\infty^{-1}\mathbb{Z}^m$. The real Mahler criterion proved in Volume 1 therefore puts $\mathrm{SL}_m(\mathbb{Z})g_\infty$ in a compact subset of $\mathrm{SL}_m(\mathbb{Z})\backslash \mathrm{SL}_m(\mathbb{R})$. Multiplying by the corresponding $\mathrm{SL}_m(\mathbb{Z})$ element does not move k_f out of K_f. Since K_f is compact, the full adelic class lies in a compact set. □

2. A properness device for rational subgroups

The next lemma is the point at which Chevalley linearization enters reduction theory. Its role is elementary but important: compactness of the associated ambient lattice must imply compactness modulo the arithmetic subgroup of the subgroup itself.

Lemma 2.1 (Chevalley tensor properness)**.** *Let* $\mathbf{J} < \mathrm{SL}_m$ *be a connected rational subgroup. Suppose there are a rational representation* $\sigma : \mathrm{SL}_m \to \mathrm{GL}(W)$ *and* $0 \neq w \in W(\mathbb{Q})$ *such that*

$$J = \mathrm{Stab}_{\mathrm{SL}_m}(w).$$

If the image of a set $E \subset J(\mathbb{A})$ *in* $\mathrm{SL}_m(\mathbb{Q})\backslash \mathrm{SL}_m(\mathbb{A})$ *is relatively compact, then the image of* E *in* $J(\mathbb{Q})\backslash J(\mathbb{A})$ *is relatively compact.*

Proof. Take a sequence $j_i \in E$. Ambient relative compactness gives $q_i \in \mathrm{SL}_m(\mathbb{Q})$ such that $q_i j_i$ remains in one compact subset of $\mathrm{SL}_m(\mathbb{A})$. Since j_i fixes w,

$$\sigma(q_i)w = \sigma(q_i j_i)w$$

remains in a compact subset of $W(\mathbb{A})$. But every $\sigma(q_i)w$ is a rational vector. The diagonal copy of $W(\mathbb{Q})$ is discrete in $W(\mathbb{A})$ by Lemma 1.1, so a compact set contains only finitely many of these rational vectors. Pass to a subsequence on which $\sigma(q_i)w$ is constant. Then $q_1^{-1}q_i \in J(\mathbb{Q})$, while

$$(q_1^{-1}q_i)j_i = q_1^{-1}(q_i j_i)$$

remains compact. This is precisely relative compactness modulo $J(\mathbb{Q})$. $\square$

The coordinate-ring Chevalley construction in Volume 2 gives a rational *line* with prescribed stabilizer. If J has no rational characters, the scalar by which J acts on that line is a rational character and hence is trivial. Consequently the line generator can be chosen as a vector w with $J = \mathrm{Stab}(w)$, exactly as required in the lemma.

3. Rank-zero semisimple compactness

We now prove the base case of rational reduction rather than hiding it under Godement's name.

Let $\mathbf{G}/\mathbb{Q}$ be connected semisimple of rational rank zero. Put $\mathfrak{g} = \mathrm{Lie}\, G$. Since G is semisimple, $\det \mathrm{Ad}(g) = 1$. After choosing a rational basis we regard $\mathrm{Ad}\, G$ as a rational subgroup of $\mathrm{SL}(\mathfrak{g})$.

For $X \in \mathfrak{g}$ write

$$\det(T - \mathrm{ad}\, X) = T^d + c_1(X)T^{d-1} + \cdots + c_d(X).$$

After clearing fixed denominators, the c_j are integral homogeneous polynomials on a rational lattice in $\mathfrak{g}$, and they are $\mathrm{Ad}\, G$-invariant.

Lemma 3.1 (No short rational adjoint vectors in rational rank zero). *There is $\delta_G > 0$ such that for every $g \in G_{\mathrm{ad}}(\mathbb{A})$ and every nonzero $X \in \mathfrak{g}(\mathbb{Q})$,*

$$c(\mathrm{Ad}(g^{-1})X) \geq \delta_G.$$

Here the content is invariant under rational rescaling of X. Thus this is exactly the uniform lower bound on rational lines required by adelic Mahler.

Proof. If not, choose $g_i \in G_{\mathrm{ad}}(\mathbb{A})$ and nonzero rational X_i with $c(\mathrm{Ad}(g_i^{-1})X_i) \to 0$. Rational rescaling leaves content unchanged, so we may and do choose each X_i primitive in one fixed integral lattice in $\mathfrak{g}(\mathbb{Q})$. For each homogeneous invariant c_j of degree j, local polynomial norm bounds give

$$\prod_v |c_j(X_i)|_v = \prod_v |c_j(\mathrm{Ad}(g_i^{-1})X_i)|_v \leq C_j\, c(\mathrm{Ad}(g_i^{-1})X_i)^j.$$

The constant is finite because the chosen integral models have coefficient norm one at almost every place. If $c_j(X_i) \neq 0$, the product formula makes the left side equal to 1. Hence for all sufficiently large i every $c_j(X_i)$ vanishes. Thus $\mathrm{ad}\, X_i$ is nilpotent.

For a semisimple Lie algebra the adjoint representation is faithful, so $X_i \neq 0$ is a nonzero rational nilpotent element. Exponentiation gives a nontrivial rational one-parameter unipotent subgroup of G. By the standard rational parabolic/root correspondence for semisimple groups, this is impossible when $\mathbb{Q}$-rank$(G) = 0$. Contradiction. □

Theorem 3.2 (Rank-zero compactness). *If $\mathbf{G}/\mathbb{Q}$ is connected semisimple and $\mathbb{Q}$-rank$(G) = 0$, then $G(\mathbb{Q})\backslash G(\mathbb{A})$ is compact.*

Proof. First take the adjoint group G_{ad}. Lemma 3.1 and adelic Mahler (Lemma 1.3) show that the image of $G_{\mathrm{ad}}(\mathbb{A})$ in the ambient special-linear lattice space is relatively compact. The Chevalley construction supplies a rational line whose stabilizer is G_{ad}; semisimplicity kills its scalar character, so the line generator is a fixed vector. Lemma 2.1 therefore gives compactness of $G_{\mathrm{ad}}(\mathbb{Q})\backslash G_{\mathrm{ad}}(\mathbb{A})$.

It remains to pass through the finite central isogeny $\pi : G \to G_{\mathrm{ad}}$ with kernel Z. We spell out the only arithmetic point in this passage. For every finite set of primes S_0, there are only finitely many classes in $H^1(\mathbb{Q}, Z)$ unramified outside S_0. To see this, choose a finite Galois extension over which Z is constant. A Z-torsor has a splitting field of degree bounded in terms of that extension and $|Z|$. If the torsor is unramified outside S_0 (together with the fixed ramified primes of Z), so is its splitting field. For bounded degree, the local different exponents at this fixed finite set are bounded; hence the absolute discriminant is bounded. Hermite–Minkowski gives only finitely many such fields, and each finite Galois group admits only finitely many cocycles with values in the fixed finite group Z. This proves the finiteness claim.

Now take $g_i \in G(\mathbb{A})$. Compactness of the adjoint quotient gives $h_i \in G_{\mathrm{ad}}(\mathbb{Q})$ for which $c_i = h_i\pi(g_i)$ remains in one compact adelic set C. Enlarge a finite set S_0 so that the central isogeny has smooth integral models and $C_p \subset G_{\mathrm{ad}}(\mathbb{Z}_p)$ for every $p \notin S_0$. The connecting class $\delta(h_i) \in H^1(\mathbb{Q}, Z)$ localizes to $\delta(c_i)$ because $\pi(g_i)$ has trivial obstruction. At $p \notin S_0$ this local class is unramified. Hence all $\delta(h_i)$ belong to the finite set just proved. Pass to a subsequence on which $\delta(h_i)$ is constant. Exactness for the central extension then gives

$$h_1^{-1}h_i = \pi(\tilde{h}_i) \qquad \text{with } \tilde{h}_i \in G(\mathbb{Q}).$$

Therefore

$$\pi(\tilde{h}_i g_i) = h_1^{-1}c_i$$

remains compact. A finite morphism is proper on local points; at almost all finite places it respects the integral compact subgroups. Consequently the inverse image in $G(\mathbb{A})$ of this compact subset of the image of π is compact modulo the finite adelic kernel $Z(\mathbb{A})$. Thus $\tilde{h}_i g_i$ is relatively compact, proving compactness of $G(\mathbb{Q})\backslash G(\mathbb{A})$. □

4. Anisotropic tori without an imported torus-class theorem

We also need the central torus in a Levi factor. The following proof uses only weight decompositions and adelic Mahler.

Lemma 4.1 (Finite invariant-polynomial test for an anisotropic torus)**.** *Let* $\mathbf{T} < \mathrm{SL}(V)$ *be a rational torus with* $X_{\mathbb{Q}}(T) = 0$*. There exist homogeneous rational* T*-invariant polynomials* $F_1, \dots, F_r \in \mathbb{Q}[V]$ *such that*

$$F_1(v) = \cdots = F_r(v) = 0,\ v \in V(\mathbb{Q}) \quad\Longrightarrow\quad v = 0.$$

Proof. Choose a finite Galois splitting field E. Over E write $V_E = \bigoplus_\chi V_\chi$. If $0 \neq v \in V(\mathbb{Q})$, its set of nonzero weight components is Galois stable. Pick a weight χ occurring in v. The sum of its Galois orbit is a Galois-fixed character of T_E, hence a rational character of T; it is zero because $X_{\mathbb{Q}}(T) = 0$. Choose a linear coordinate nonzero on the χ-component of v and take the product of its Galois conjugates. The total weight is zero, so after Galois symmetrization we obtain a rational T-invariant polynomial nonzero at v.

It remains only to make the collection finite. Let I be the ideal of $\mathbb{Q}[V]$ generated by all positive-degree rational T-invariant polynomials. The polynomial ring is Noetherian, so I is generated by finitely many members of the original generating set: indeed start with any finite ideal generating family and expand each generator as a finite polynomial combination of invariant generators; the union of those finitely many invariants already generates I. Call them $F_1, \dots, F_r$. If all $F_j(v)$ vanish, every positive-degree invariant vanishes at v. The first paragraph shows that this is impossible for a nonzero rational vector. Thus their common rational zero set is exactly $\{0\}$. ☐

Lemma 4.2 (Anisotropic torus compactness). *If* $\mathbf{T}/\mathbb{Q}$ *is a torus with* $X_{\mathbb{Q}}(T) = 0$*, then* $T(\mathbb{Q})\backslash T(\mathbb{A})$ *is compact.*

Proof. Choose a faithful rational representation $T < \mathrm{SL}(V)$ and the finite family F_j from Lemma 4.1. If the associated adelic lattices violated Mahler's lower bound, there would be $t_i \in T(\mathbb{A})$ and $0 \neq v_i \in V(\mathbb{Q})$, rationally rescaled to primitive content, with $c(t_i^{-1}v_i) \to 0$. Since each F_j is invariant,

$$\prod_v |F_j(v_i)|_v = \prod_v |F_j(t_i^{-1}v_i)|_v \leq C_j c(t_i^{-1}v_i)^{\deg F_j}.$$

By the product formula every nonzero left side is 1. Thus all $F_j(v_i)$ vanish for large i, contradicting the common-zero property. Adelic Mahler therefore makes the image of $T(\mathbb{A})$ relatively compact in the ambient special-linear quotient.

Chevalley gives a rational line with stabilizer T. Its scalar action is a rational character, hence trivial; so the line generator is a fixed vector. Lemma 2.1 lifts ambient compactness to $T(\mathbb{Q})\backslash T(\mathbb{A})$. ☐

5. Parabolic heights and rational reduction

Let now $\mathbf{G}/\mathbb{Q}$ be connected semisimple of positive rational rank. Fix a minimal rational parabolic

$$\mathbf{P} = \mathbf{NMA},$$

where A is a maximal rational split torus and M denotes the anisotropic Levi part after the split center has been separated. Let Δ be the simple relative roots. Put $d = \dim N$ and choose a nonzero rational vector

$$p_P \in \bigwedge^d \mathfrak{g}(\mathbb{Q})$$

spanning $\bigwedge^d \mathfrak{n}$.

For a rational conjugate $Q = \gamma P \gamma^{-1}$ choose a primitive integral representative p_Q of its exterior line and define

$$D(Q, g) = c\left((\bigwedge^d \mathrm{Ad}\, g^{-1}) p_Q \right), \qquad g \in G(\mathbb{A}).$$

The value is independent of rational rescaling by the product formula.

Lemma 5.1 (Discrete parabolic heights)**.** *For fixed g and R, only finitely many primitive rational parabolic lines satisfy $D(Q,g) \leq R$. Hence $D(Q,g)$ attains a positive minimum among the rational conjugates of P.*

Proof. There is a finite set of places S_0 outside which g_v and g_v^{-1} preserve the chosen integral exterior lattice. A primitive integral vector has local max norm 1 at every finite place. At places in S_0 the operator norms of $g_v^{\pm 1}$ give fixed two-sided distortion constants. Consequently $D(Q,g) \leq R$ bounds the Euclidean norm of p_Q by a constant depending only on g and R. A Euclidean lattice has only finitely many vectors in a bounded ball. Restricting to the Pluecker locus of unipotent radicals leaves a finite set. Positivity then gives a minimum. □

The quotient $N(\mathbb{Q})\backslash N(\mathbb{A})$ is compact by Lemma 1.2. The group M is rationally anisotropic reductive: its derived group has rational rank zero and its connected central torus has no rational character after the split A-factor is removed. The two compactness theorems above, together with the finite central multiplication map, therefore give a compact set Ω_M meeting every $M(\mathbb{Q})$-coset. We record the finite-central step explicitly.

Lemma 5.2 (Finite central products preserve compactness/finite volume)**.** *Let Z be a rational torus, D a connected semisimple rational group, and suppose multiplication $Z \times D \to R$ is a central isogeny onto a connected reductive group R. If the rational adelic quotients of Z and D are compact (respectively finite volume), then the same is true for R.*

Proof. At each local field a central isogeny has open image of finite index. At all but finitely many finite places its map on integral compact subgroups has open finite-index image. Choose local coset representatives; at the good places choose them inside the compact integral subgroup. Their restricted product is compact, so

$$R(\mathbb{A}) = C\, Z(\mathbb{A})D(\mathbb{A})$$

for a compact C. The image of

$$(Z(\mathbb{Q})\backslash Z(\mathbb{A})) \times (D(\mathbb{Q})\backslash D(\mathbb{A})) \times C$$

therefore covers $R(\mathbb{Q})\backslash R(\mathbb{A})$. A compact source gives compactness. With Haar measures, a finite-measure source gives finite covolume; finite kernels only change the normalization by a finite constant. □

For the split torus $A \simeq \mathbb{G}_m^r$, rational multiplication makes all finite idele coordinates units; the remaining finite-unit coordinates are compact and the only noncompact coordinates are

$$H(a) = (\log|\chi_1(a)|_{\mathbb{A}}, \ldots, \log|\chi_r(a)|_{\mathbb{A}}) \in \mathbb{R}^r$$

for a basis of rational characters. Thus $A(\mathbb{Q})\backslash A(\mathbb{A})$ is, up to compact factors, $\mathbb{R}^r$.

Lemma 5.3 (Adjacent chamber comparison)**.** *Let P_α be the minimal rational parabolic adjacent to P across the wall $\alpha = 0$. After n and m have been put in fixed compact sets, there are $C \geq 1$ and $m_\alpha \in \mathbb{N}$ such that*

$$C^{-1}|\alpha(a)|_{\mathbb{A}}^{m_\alpha} \leq \frac{D(P_\alpha, nmak)}{D(P, nmak)} \leq C|\alpha(a)|_{\mathbb{A}}^{m_\alpha}.$$

Proof. The line $\bigwedge^d \mathfrak{n}$ has A-weight equal to the sum, with multiplicity, of the positive relative roots. Applying the simple reflection s_α replaces P by the adjacent minimal parabolic and changes this weight by a positive integral multiple $m_\alpha \alpha$ (with the sign fixed by the convention that P is the positive chamber). Since D uses $\mathrm{Ad}(a^{-1})$, the ratio displayed above is precisely $|\alpha(a)|_{\mathbb{A}}^{m_\alpha}$ before the bounded n, m, k factors are inserted. The N- and compact factors have bounded operator norm and inverse. The anisotropic Levi acts on the relevant rational exterior lines through characters of compact adelic modulus, hence also contributes a bounded factor. This proves the two-sided estimate. □

Theorem 5.4 (Adelic semisimple reduction). *There are compact sets $\Omega_N \subset N(\mathbb{A})$, $\Omega_M \subset M(\mathbb{A})$ and C_A in the norm-one part of $A(\mathbb{A})$, a maximal compact K, and $t > 0$ such that*

$$G(\mathbb{A}) = G(\mathbb{Q})\,\Omega_N \Omega_M C_A A_t K,$$

where

$$A_t = \{a : \log|\alpha(a)|_{\mathbb{A}} \geq \log t \textit{ for every } \alpha \in \Delta\}.$$

Proof. Fix $g \in G(\mathbb{A})$. Choose, by Lemma 5.1, a rational minimal parabolic Q minimizing $D(Q, g)$. Minimal rational parabolics are rationally conjugate, so multiply g on the left by $\gamma \in G(\mathbb{Q})$ and assume the minimizer is P. Product local Iwasawa decomposition gives $g = nmak$.

Perform the rational normalizations in the order A, then M, then N. First multiplication by $A(\mathbb{Q})$ puts the norm-one idele coordinate in C_A; by the product formula it does not change the global logarithmic quantities $\log|\alpha(a)|_{\mathbb{A}}$. This conjugates the N-coordinate but does not affect its membership in $N(\mathbb{A})$. Next multiply by $M(\mathbb{Q})$ to put m in Ω_M; M normalizes N and centralizes A. Finally multiply by $N(\mathbb{Q})$ to put the resulting n in Ω_N. Thus no later normalization can destroy the compactness of an earlier N-coordinate.

All three operations preserve the statement that the chosen rational parabolic is a minimizer. Explicitly, for $q \in G(\mathbb{Q})$ one has

$$D(Q, qg) = D(q^{-1}Qq, g),$$

where equality is understood after primitive rational normalization; the possible scalar disappears by the product formula. Thus rational left multiplication simply permutes the collection over which the minimum is taken.

For every simple root α, minimality gives $D(P,g) \le D(P_\alpha, g)$. Lemma 5.3 therefore yields

$$|\alpha(a)|_{\mathbb{A}} \ge C^{-1/m_\alpha}.$$

Taking the minimum over the finitely many simple roots gives a uniform t. This proves the covering. $\square$

6. Finite volume of the semisimple quotient

Let 2ρ be the sum, with multiplicity, of the positive relative roots in $\mathfrak{n}$. In the logarithmic coordinates $H = H(a)$ on the noncompact part of $A(\mathbb{Q})\backslash A(\mathbb{A})$, the Iwasawa Jacobian is

$$dg = e^{-2\rho(H)}\, dn\, dm\, dH\, dk$$

up to a harmless constant and compact factors. Indeed, conjugation by a on N has determinant $e^{2\rho(H)}$, so left Haar measure carries its reciprocal.

Theorem 6.1 (Semisimple Borel–Harish–Chandra). *For connected semisimple $\mathbf{G}/\mathbb{Q}$, $G(\mathbb{Q})\backslash G(\mathbb{A})$ has finite Haar volume.*

Proof. The reduction set of Theorem 5.4 covers the quotient. Its N, M, C_A, K factors have finite measure. On the positive chamber,

$$2\rho = \sum_{\alpha\in\Delta} c_\alpha \alpha, \qquad c_\alpha > 0.$$

The simple-root logarithms are linear coordinates on the split chamber up to a fixed nonsingular linear change, hence

$$\int_{A_t} e^{-2\rho(H)}\, dH \ll \prod_{\alpha\in\Delta} \int_{\log t}^{\infty} e^{-c_\alpha x}\, dx < \infty.$$

A measurable fundamental set contained in this finite-measure covering set has finite Haar measure. $\square$

7. The character-free Borel–Harish–Chandra criterion

Theorem 7.1 (Borel–Harish–Chandra in the matrix scope). *Let $\mathbf{H} < \mathrm{SL}_N$ be connected over $\mathbb{Q}$. If $X_{\mathbb{Q}}(\mathbf{H}) = 0$, then $H(\mathbb{Q})\backslash H(\mathbb{A})$ has finite Haar volume. Consequently, for every finite set of places S containing ∞,*

$$\Gamma_{H,S} = H(\mathbb{Q}) \cap \mathrm{SL}_N(\mathbb{Z}_S)$$

is a lattice in $H_S = \prod_{v\in S} H(\mathbb{Q}_v)$. The conclusion is unchanged under commensurability of the arithmetic subgroup.

Proof. Let $U = R_u(H)$ and choose a rational Levi subgroup R. Rational characters of R pull back to rational characters of H, hence $X_{\mathbb{Q}}(R) = 0$. The derived group $D = R^{\mathrm{der}}$ is semisimple and has finite adelic covolume by Theorem 6.1. The connected central torus Z has no nontrivial rational character. To see this, let $\chi \in X_{\mathbb{Q}}(Z)$. The finite group $Z \cap D$ is killed by some positive power χ^m, so χ^m descends through the central isogeny $Z \times D \to R$ to a rational character of R. Since $X_{\mathbb{Q}}(R) = 0$, $\chi^m = 1$, and the character lattice of a torus is torsion free; hence $\chi = 1$. Thus Z is rationally anisotropic. Lemma 4.2, together with the finite central product Lemma 5.2, gives finite covolume for R.

Lemma 1.2 gives compact $U(\mathbb{Q})\backslash U(\mathbb{A})$. Moreover the determinant by which R acts on $\operatorname{Lie} U$ is a rational algebraic character of H; it is therefore trivial. Thus multiplication $U \times R \to H$ carries product Haar measure to Haar measure up to a constant, with no modular weight. A compact set representing the unipotent quotient times a finite-measure representative for the reductive quotient covers $H(\mathbb{Q})\backslash H(\mathbb{A})$. Hence the adelic covolume is finite.

Now fix S and put $K^S = \prod_{p \notin S} H(\mathbb{Z}_p)$ after enlarging S by the finitely many bad primes of the integral model. The double quotient $H(\mathbb{Q})\backslash H(\mathbb{A})/K^S$ has finite measure. The map

$$\Gamma_{H,S}\backslash H_S \longrightarrow H(\mathbb{Q})\backslash H(\mathbb{A})/K^S, \qquad [h] \longmapsto [(h,1)]$$

is injective: equality of two images produces a rational element integral outside S, hence an element of $\Gamma_{H,S}$. Its image is open because K^S is open, and the quotient measure restricts there to a constant multiple of Haar measure on $\Gamma_{H,S}\backslash H_S$. Since the ambient double quotient has finite measure, so does this open component. Restoring the finitely many bad primes changes the group only up to commensurability. $\square$

Corollary 7.2 (Character-free cores have finite arithmetic covolume)**.** *For a connected rational subgroup* $\mathbf{F} < \mathrm{SL}_N$*, put*

$$\mathbf{F}^1 = \left(\bigcap_{\chi \in X_{\mathbb{Q}}(F)} \ker \chi \right)^{\circ}.$$

Then $X_{\mathbb{Q}}(F^1) = 0$*,* $F^1_S \cap \Gamma$ *is a lattice in* F^1_S*, and every algebraic unipotent subgroup of* F *lies in* F^1*.*

Proof. Characters factor through the maximal torus quotient of $F/R_u(F)$. Taking the connected intersection of their kernels removes its maximal rationally split quotient; the derived and unipotent parts have no characters. Thus $X_{\mathbb{Q}}(F^1) = 0$ and Theorem 7.1 applies. Every homomorphism from a unipotent algebraic group to $\mathbb{G}_m$ is trivial, so unipotent subgroups lie in every character kernel. $\square$

8. Projective density and the arithmetic hull

We next prove the precise rationalization statement used by the terminal homogeneous branch of measure classification.

Lemma 8.1 (Invariant projective probabilities are fixed)**.** *Let k be a characteristic-zero local field, let N be nilpotent on a finite-dimensional k-space V, and put $u(t) = \exp(tN)$. Every $u(k)$-invariant probability measure on $\mathbb{P}(V)$ is supported on $\mathbb{P}(\ker N)$.*

Proof. Let $F = \mathbb{P}(\ker N)$ and let $C \subset \mathbb{P}(V) \setminus F$ be compact. We first claim that $u(t)C \cap C = \varnothing$ for all sufficiently large $|t|$. Otherwise take $|t_i| \to \infty$ and $[v_i], [w_i] \in C$ with $u(t_i)[v_i] = [w_i]$. After passing to a subsequence, $v_i \to v \in C$. Let $r \geq 1$ be maximal with $N^r v \neq 0$. For all large i the coefficient $N^r v_i/r!$ stays away from zero, while all higher-degree coefficients tend to zero. Dividing the polynomial $e^{t_i N} v_i$ by t_i^r in the real/complex case, or by its largest ultrametric term in the nonarchimedean case after splitting into the finitely many possible degrees, shows that every projective limit lies in $\mathbb{P}(\ker N)$. This contradicts $w_i \to w \in C$.

Choose $t_1, t_2, \ldots$ with $|t_i - t_j|$ larger than this escape threshold. Then $u(t_i)C$ are pairwise disjoint. Invariance of a probability measure forces $\nu(C) = 0$. A sigma-compact exhaustion of the complement of F finishes the proof. $\square$

Lemma 8.2 (Base change preserves density)**.** *If $E \subset Y(\mathbb{Q})$ is $\mathbb{Q}$-Zariski dense in a rational affine variety Y, then it is Zariski dense in Y_k for every field extension $k/\mathbb{Q}$.*

Proof. Write $f \in k[Y]$ vanishing on E as $f = \sum_{j=1}^{r} b_j f_j$ with the b_j linearly independent over $\mathbb{Q}$ and $f_j \in \mathbb{Q}[Y]$. At a rational point the values $f_j(x)$ are rational, so linear independence gives $f_j(x) = 0$ for every j. Rational density gives $f_j = 0$ on Y, hence $f = 0$ after base change. $\square$

Lemma 8.3 (Ergodicity kills rational character directions)**.** *Let L be a connected local-analytic group, $\Lambda < L$ a lattice, and $U < L$ a subgroup generated by local algebraic unipotent one-parameter groups. Suppose U acts ergodically on L/Λ. If L is contained in the S-points of a rational algebraic group $\mathbf{H}$ and $\chi \in X_{\mathbb{Q}}(H)$, then $\chi|_L = 1$.*

Proof. Every algebraic character is trivial on a unipotent group, so $U \subset \ker \chi$. The map

$$L/\Lambda \longrightarrow \chi(L)/\overline{\chi(L) \cap \chi(\Lambda)}$$

is therefore a U-invariant measurable factor. The push-forward of Haar probability has full support on the closure of the image. Ergodicity forces this factor measure to be a point mass, hence the quotient is a point. Thus $\chi(L)$ is contained in the closure of $\chi(\Lambda)$. Because $\Lambda \subset \mathrm{SL}_N(\mathbb{Z}_S)$ and χ is rational, after clearing one fixed denominator the group $\chi(\Lambda)$ is contained in the diagonal S-unit group. Volume 3 proves that the S-unit group is discrete in the product of the S-local multiplicative groups. Hence the closure above is itself discrete.

On the analytic identity neighborhood of L, an algebraic character is an analytic homomorphism into a discrete subgroup. It is therefore locally constant and equals 1 near the identity. By the definition of the connected local-analytic core used throughout this volume—the subgroup generated by such identity neighborhoods—it is identically 1 on L. □

THEOREM 8.4 (Arithmetic hull for the homogeneous measures of this volume). *Let $L < G_S$ be the connected locally algebraic support group of a homogeneous probability on L/Λ, where $\Lambda = L \cap \Gamma$ is a lattice. Suppose a finite product U of local algebraic one-parameter unipotent groups acts ergodically and the normal closure of these directions is the noncompact algebraic core of L. Then there is a connected rational subgroup $\mathbf{H}_0 < \mathrm{SL}_N$ with $X_{\mathbb{Q}}(H_0) = 0$ such that*

$$L \subset H_{0,S}, \qquad H_{0,S} \cap \Gamma \text{ is a lattice.}$$

If the local algebraic Zariski closure of L is proper, $\mathbf{H}_0$ is proper.

PROOF. Let $\mathbf{H}$ be the connected $\mathbb{Q}$-Zariski closure of Λ. The coordinate-ring Chevalley construction proved in Volume 2 gives a rational representation ρ and a rational projective line $[v]$ whose stabilizer is H. Since Λ fixes $[v]$, the map

$$\Phi : L/\Lambda \to \mathbb{P}(V_S), \qquad \Phi(\ell\Lambda) = \rho(\ell)[v]$$

is well defined. Push Haar probability forward. For each local unipotent factor $u(t) < U$, the push-forward is $\rho(u(t))$-invariant. Lemma 8.1 implies that for almost every ℓ the line $\rho(\ell)[v]$ is fixed by $u(t)$, equivalently

$$\ell^{-1}u(t)\ell \subset H_S.$$

The condition is closed, and Haar measure on L/Λ has full support, so it holds for every $\ell \in L$. The normal subgroup $N \lhd L$ generated by the L-conjugates of the local factors of U is therefore contained in H_S.

The quotient $L/\overline{N\Lambda}$ is a measurable U-invariant factor of L/Λ. Ergodicity forces it to be a point, whence

$$L = \overline{N\Lambda} \subset H_S.$$

Now let $\mathbf{H}^1$ be the connected character-free core of H. By Lemma 8.3, every rational character of H is trivial on L. Hence $L \subset H^1_S$. Put $\mathbf{H}_0 = \mathbf{H}^1$. Corollary 7.2 gives $H_{0,S} \cap \Gamma$ as a lattice.

For properness, suppose some local algebraic Zariski closure of L is a proper subgroup of SL_N. If $H = \mathrm{SL}_N$, then Λ is $\mathbb{Q}$-Zariski dense in SL_N, so Lemma 8.2 makes it Zariski dense over every $\mathbb{Q}_v$. But $\Lambda \subset L$ lies in the proper local algebraic closure, a contradiction. Thus H, and hence H^1, is proper. □

9. Logical dependency chain

The argument proceeds through six steps:

(1) strong approximation and real Mahler imply adelic Mahler compactness;
(2) adelic Mahler, invariant polynomials, and Chevalley theory give rank-zero compactness;
(3) rank-zero compactness and parabolic heights give semisimple reduction;
(4) semisimple reduction and the positive-root Jacobian give finite covolume;
(5) Levi decomposition and compactness of the anisotropic torus quotient give the character-free Borel–Harish–Chandra statement;
(6) that statement, the projective unipotent lemma, and ergodicity give the rational arithmetic hull.

The historical Borel–Harish–Chandra paper supplies provenance for this reduction architecture; the specialization used later in the series is proved above.

> **Proof and provenance.** Proved in this text. The historical source is Borel–Harish–Chandra's reduction theory. The present proof is deliberately specialized to rational matrix groups over $\mathbb{Q}$ and to the finite sets of places used by this series; it does not claim a new proof of every generality of the original theorem.

CHAPTER 3

One-parameter unipotent groups over local fields

Let k be one of $\mathbb{R}$ or $\mathbb{Q}_p$. A one-parameter unipotent subgroup of $\mathrm{SL}_n(k)$ is a homomorphism

$$u : k \longrightarrow \mathrm{SL}_n(k)$$

whose matrices are unipotent. In characteristic zero there is a nilpotent matrix N with

$$u(t) = \exp(tN) = I + tN + \cdots + \frac{t^{r-1}}{(r-1)!}N^{r-1}.$$

Thus every matrix coefficient is a polynomial in t of degree at most $n-1$.

Lemma 0.1 (Polynomial adjoint action)**.** *For every finite-dimensional rational representation ρ of SL_n, the matrix coefficients of $\rho(u(t))$ are polynomials in t of degree bounded in terms of ρ only.*

Proof. The differential $d\rho(N)$ is nilpotent. Hence

$$\rho(u(t)) = \exp(t\,d\rho(N))$$

is a finite exponential series. □

1. Parameter balls

For $k = \mathbb{Q}_p$ write

$$B_m = p^{-m}\mathbb{Z}_p,$$

so B_m is a compact open additive subgroup and $B_m \uparrow \mathbb{Q}_p$. For $k = \mathbb{R}$ use $[-T, T]$. In a product of local fields we average over product balls. These are Følner sets because translation changes only a boundary layer in the real coordinates and changes nothing in a p-adic coordinate once the translating element lies inside the sufficiently large compact subgroup.

CHAPTER 4

Ergodic averages for local additive parameter groups

Rigidity arguments require generic points for the acting parameter group. At a single finite place the needed pointwise theorem is especially transparent: increasing p-adic balls are compact subgroups, so the averages are reverse martingale conditional expectations. For a product of real and finite local factors we use a fully self-contained L^2 argument. One-factor maximal inequalities compose because the averaging operators are positive, while bounded coboundaries form a dense class on which rectangular convergence is uniform. A Banach-principle argument then gives pointwise convergence as all side lengths tend to infinity independently. This is exactly the quantifier needed later. We do *not* claim the endpoint multiparameter L^1 theorem.

1. One finite local factor: compact-subgroup averages

Let (Y, μ) be a probability space with a measure-preserving action T_t of $\mathbb{Q}_p$. Put

$$B_m = p^{-m}\mathbb{Z}_p$$

and define

$$A_m f(y) = \frac{1}{m(B_m)} \int_{B_m} f(T_t y)\, dt.$$

Lemma 1.1 (Orthogonal projections)**.** *On $L^2(Y)$, A_m is the orthogonal projection onto the closed subspace of B_m-invariant functions.*

Proof. Let β_m be normalized Haar measure on the compact group B_m. Since $\beta_m * \beta_m = \beta_m$ and β_m is invariant under $t \mapsto -t$, the operator

$$A_m = \int_{B_m} T_t \, d\beta_m(t)$$

satisfies $A_m^2 = A_m = A_m^*$. Its image is B_m-invariant because translating β_m by an element of B_m leaves it unchanged. Conversely every B_m-invariant function is fixed by the average. Hence A_m is exactly the orthogonal projection onto the invariant subspace. □

Let $\mathscr{I}_m$ be the sigma-algebra of B_m-invariant measurable sets. Since $B_m \subset B_{m+1}$, the sigma-algebras $\mathscr{I}_m$ decrease.

THEOREM 1.2 ($\mathbb{Q}_p$ pointwise ergodic theorem). *For every $f \in L^1(Y)$,*

$$A_m f(y) \longrightarrow \mathbb{E}(f \mid \mathscr{I})(y)$$

for almost every y and in L^1, where $\mathscr{I}$ is the $\mathbb{Q}_p$-invariant sigma-algebra. If the action is ergodic, the limit is the constant $\int f\, d\mu$.

PROOF. We first identify A_m on L^1. It is $\mathscr{I}_m$-measurable. If $E \in \mathscr{I}_m$, then Fubini and invariance of μ give

$$\begin{aligned}\int_E A_m f\, d\mu &= \frac{1}{m(B_m)} \int_{B_m} \int_E f(T_t y)\, d\mu(y)\, dt \\ &= \frac{1}{m(B_m)} \int_{B_m} \int_{T_t E} f(z)\, d\mu(z)\, dt \\ &= \int_E f\, d\mu,\end{aligned}$$

because $T_t E = E$ modulo null sets for $t \in B_m$. Thus

$$A_m f = \mathbb{E}(f \mid \mathscr{I}_m).$$

The intersection $\bigcap_m \mathscr{I}_m$ is the sigma-algebra of sets invariant under $\bigcup_m B_m = \mathbb{Q}_p$. The reverse martingale convergence theorem therefore gives almost-everywhere and L^1 convergence to $\mathbb{E}(f \mid \mathscr{I})$. Ergodicity makes $\mathscr{I}$ trivial. ☐

Corollary 1.3 (Single-factor generic points)**.** *If $u(\mathbb{Q}_p)$ acts ergodically and $\mathscr{F} \subset C_c(Y)$ is countable, there is a conull set Y_{gen} such that for every $y \in Y_{\mathrm{gen}}$ and every $f \in \mathscr{F}$,*

$$\frac{1}{m(B_m)} \int_{B_m} f(u(t)y)\, dt \longrightarrow \int f\, d\mu.$$

PROOF. Apply Theorem 1.2 to each $f \in \mathscr{F}$ and intersect the resulting countably many conull sets. ☐

Lemma 1.4 (Unbounded local-field recurrence)**.** *Let $k = \mathbb{R}$ or $\mathbb{Q}_p$ act measurably and measure-preservingly on a probability space (Y, μ). Suppose Y is also second countable and the action is continuous on a conull invariant Borel set. Then for almost every $y \in Y$ and every neighborhood O of y there are parameters $t_j \in k$ with*

$$|t_j| \longrightarrow \infty, \qquad T_{t_j} y \in O.$$

In particular one may choose $t_j \to \infty$ in parameter norm with $T_{t_j} y \to y$.

PROOF. Fix a countable basis $\mathcal{B}$ for the topology. It is enough to prove that for each $E \in \mathcal{B}$, almost every $y \in E$ returns to E at parameters of arbitrarily large norm.

Let $\mathscr{I}$ be the invariant sigma-algebra for the k-action. For $k = \mathbb{Q}_p$, Theorem 1.2 applied to 1_E gives

$$\frac{1}{m(B_R)} \int_{B_R} 1_E(T_t y)\, dt \longrightarrow q_E(y) := \mathbb{E}(1_E \mid \mathscr{I})(y)$$

for almost every y, along the value-group radii. Over $\mathbb{R}$ the identical statement follows from the ordinary one-parameter pointwise ergodic theorem. Now

$$\int_{E\cap\{q_E=0\}} 1_E\, d\mu = \int_{E\cap\{q_E=0\}} q_E\, d\mu = 0,$$

so $q_E(y) > 0$ for almost every $y \in E$.

Fix such a point and a bounded parameter ball B_M. If every return to E were contained in B_M, then for $R > M$,

$$\frac{1}{m(B_R)} \int_{B_R} 1_E(T_t y)\, dt \le \frac{m(B_M)}{m(B_R)} \longrightarrow 0,$$

contradicting $q_E(y) > 0$. Hence there are return parameters outside every bounded ball.

Intersect the resulting conull subsets over the countable basis. Given a point in this intersection and a decreasing sequence of basis neighborhoods $O_j \ni y$, choose a return parameter t_j outside the ball of radius j (or outside the jth value-group ball) with $T_{t_j} y \in O_j$. Then $|t_j| \to \infty$ and $T_{t_j} y \to y$. $\square$

2. Products: an L^2 pointwise theorem for rectangular local averages

Let

$$T = T_1 \times \cdots \times T_r,$$

where every T_j is either $\mathbb{R}$ or a finite-dimensional additive nonarchimedean local vector group. For each j choose the standard centered one-factor balls $B_{j,R}$. If $\mathbf{R} = (R_1, \ldots, R_r)$, write

$$F_{\mathbf{R}} = B_{1,R_1} \times \cdots \times B_{r,R_r}$$

and

$$A_{\mathbf{R}} f(y) = \frac{1}{m(F_{\mathbf{R}})} \int_{F_{\mathbf{R}}} f(ty)\, dt.$$

The factors commute, so

$$A_{\mathbf{R}} = A^{(1)}_{R_1} \cdots A^{(r)}_{R_r}.$$

At a nonarchimedean factor, $A_R^{(j)}$ is a reverse-martingale conditional expectation along the discrete sequence of value-group radii, and Doob's maximal inequality gives

$$\|M_j f\|_2 \le C_j \|f\|_2, \qquad M_j f = \sup_R |A_R^{(j)} f|.$$

At a real factor we use the standard L^2 maximal ergodic inequality for centered interval averages, which accompanies the ordinary real pointwise ergodic theorem listed in Appendix A. Both statements hold for measure-preserving actions, not merely for translations.

Lemma 2.1 (Product maximal inequality). *For $f \in L^2(Y)$,*

$$\left\| \sup_{\mathbf{R}} |A_{\mathbf{R}} f| \right\|_2 \le \left(\prod_{j=1}^{r} C_j \right) \|f\|_2.$$

The supremum may be taken over all allowed radii independently.

Proof. Every one-factor averaging operator is positive. Therefore

$$\sup_{R_1,\dots,R_r} |A_{R_1}^{(1)} \cdots A_{R_r}^{(r)} f| \le M_1 M_2 \cdots M_r |f|.$$

Apply the one-factor L^2 maximal bound successively. Positivity also ensures that each M_j acts on nonnegative functions, so no commutation of the nonlinear maximal operators is required. □

Let P_j denote the orthogonal projection onto the T_j-invariant subspace. Since the factor actions commute, the projections P_j commute and

$$P_T = P_1 \cdots P_r$$

is the projection onto the vectors fixed by all of T.

Lemma 2.2 (Rectangular L^2 mean convergence). *As $\min_j R_j \to \infty$,*

$$A_{\mathbf{R}} f \longrightarrow P_T f \qquad \text{in } L^2(Y).$$

Proof. For each factor, the one-parameter mean ergodic theorem gives $A_{R_j}^{(j)} \to P_j$ strongly on L^2. The telescoping identity

$$\prod_{j=1}^{r} A_{R_j}^{(j)} - \prod_{j=1}^{r} P_j = \sum_{j=1}^{r} \left(\prod_{i<j} A_{R_i}^{(i)} \right) (A_{R_j}^{(j)} - P_j) \left(\prod_{i>j} P_i \right)$$

is valid because all the linear operators commute. Averaging operators and orthogonal projections have norm at most one. Taking L^2 norms and sending all R_j to infinity makes every summand tend to zero. □

We now upgrade mean convergence to pointwise convergence. The proof is included because this is precisely the quantifier needed by the later shearing argument.

Lemma 2.3 (A dense class with uniform rectangular convergence)**.** *Let $\mathcal{D}$ be the linear span of bounded T-invariant functions and bounded coboundaries*

$$h - \pi(s)h, \qquad h \in L^\infty \cap L^2, \quad s \in T_j \text{ for some } j.$$

Then $\mathcal{D}$ is dense in $L^2(Y)$, and for every $f \in \mathcal{D}$,

$$A_{\mathbf{R}} f(y) \longrightarrow P_T f(y)$$

uniformly in y as $\min_j R_j \to \infty$.

Proof. First consider one coboundary $f = h - \pi(s)h$ with $s \in T_j$. Since all other factor averages are contractions on L^∞,

$$\|A_{\mathbf{R}} f\|_\infty \le \|h\|_\infty \frac{m_j(B_{j,R_j} \triangle (s + B_{j,R_j}))}{m_j(B_{j,R_j})}.$$

The ratio tends to zero by the Følner property of the one-factor balls, uniformly in all other radii. Thus rectangular averages of a bounded coboundary converge uniformly to zero. They fix bounded T-invariant functions exactly, proving the uniform-convergence assertion for $\mathcal{D}$.

For density, let $\mathcal{C}$ be the closed span of all coboundaries. A vector orthogonal to every $h - \pi(s)h$ is fixed by every $\pi(s)$, hence belongs to the joint invariant subspace. Therefore

$$\mathcal{C} = (L^2(Y)^T)^\perp.$$

Bounded L^2 functions are dense in L^2, so bounded coboundaries have the same closed span. Adding the bounded invariant functions (which are dense in the invariant subspace by truncation) makes $\mathcal{D}$ dense in all of L^2. □

Theorem 2.4 (Product-local L^2 pointwise ergodic theorem). *For every $f \in L^2(Y)$,*

$$A_{\mathbf{R}} f(y) \longrightarrow P_T f(y)$$

for almost every y as $\min_j R_j \to \infty$, with the local radii otherwise independent. If the T-action is ergodic, the limit is $\int f\, d\mu$.

Proof. Fix $f \in L^2$ and choose $g_m \in \mathcal{D}$ with $\|f - g_m\|_2 \to 0$. Put

$$M(f - g_m) = \sup_{\mathbf{R}} |A_{\mathbf{R}}(f - g_m)|.$$

Lemma 2.1 and Chebyshev give, for every $\lambda > 0$,

$$\mu\{M(f - g_m) > \lambda\} \le C_T^2 \lambda^{-2} \|f - g_m\|_2^2.$$

Also P_T is an L^2 contraction, so after taking a subsequence in m we may assume $P_T g_m \to P_T f$ almost everywhere. Outside a set whose measure can be made arbitrarily small, both $M(f - g_m)$ and $|P_T(f - g_m)|$ are arbitrarily small. For such a point,

$$\limsup_{\min R_j \to \infty} |A_{\mathbf{R}} f - P_T f| \leq M(f - g_m) + \limsup |A_{\mathbf{R}} g_m - P_T g_m| + |P_T(g_m - f)|.$$

The middle term is zero by Lemma 2.3. Let $m \to \infty$ and then let the exceptional measure tend to zero. This proves almost-everywhere convergence. Under ergodicity the joint invariant projection is constant. □

3. Uniform generic cores for product actions

Corollary 3.1 (Uniform product generic core)**.** *Assume the T-action is ergodic. Given finitely many functions $f_1, \ldots, f_N \in C_c(Y)$ and $\eta > 0$, there are a compact set $K = K(N, \eta)$ with $\mu(K) > 1 - \eta$ and a radius threshold R_0 such that for every $y \in K$, every $j \leq N$, and every rectangular product box with* $\min_i R_i \geq R_0$*,*

$$\left| A_{\mathbf{R}} f_j(y) - \int f_j \, d\mu \right| < \eta.$$

Proof. The functions f_j lie in L^2. Theorem 2.4 gives simultaneous almost-everywhere convergence as $\min_i R_i \to \infty$. Egorov gives uniform convergence outside a set of measure $< \eta/2$; inner regularity chooses a compact subset losing at most another $\eta/2$. Uniform convergence on that compact set supplies one common threshold R_0 valid for all independent choices of larger side lengths. □

Remark 3.2. The restriction to L^2 is harmless for the rigidity proof, whose test functions are compactly supported and bounded. We have deliberately not claimed the endpoint L^1 strong multiparameter theorem. The L^2 maximal argument is both simpler and exactly sufficient for synchronized recurrence on product shearing blocks.

CHAPTER 5

Workshop: generic points for mixed real and p-adic parameters

The preceding chapter proves more than the first working draft of this volume used. For compactly supported test functions the mixed product action has a rectangular pointwise theorem in L^2: once every side length is large, the sides may be chosen independently. This workshop records the quantifiers carefully and shows how they are used in the shearing argument.

1. One finite place: all sufficiently large subgroup averages

For $B_m = p^{-m}\mathbb{Z}_p$, the average

$$A_m f(y) = \frac{1}{m(B_m)} \int_{B_m} f(T_t y)\, dt$$

is the conditional expectation onto the B_m-invariant sigma-algebra. Hence the reverse martingale theorem gives convergence for every $m \to \infty$. There is no boundary error: if $s \in B_m$, then $B_m + s = B_m$ exactly.

This exact translation invariance is a useful local advantage. If a recurrence time r brings $u(r)y$ into a uniform generic core and B_m is a sufficiently large averaging subgroup, then

$$\frac{1}{m(B_m)} \int_{B_m} f(u(t+r)y)\, dt$$

is simply the centered B_m-average based at the recurrent point $u(r)y$.

2. Mixed products: the radii are genuinely independent

Let

$$T = T_1 \times \cdots \times T_r,$$

with each T_j a real additive factor or a finite-dimensional additive group over a nonarchimedean local field. For the standard product boxes

$$F_{\mathbf{R}} = B_{1,R_1} \times \cdots \times B_{r,R_r},$$

Theorem 2.4 says that, for every $f \in L^2$,

$$A_{\mathbf{R}} f(y) \longrightarrow P_T f(y) \qquad \text{as } \min_j R_j \to \infty$$

for almost every y. Thus there is no preselected admissible sequence. Once a point belongs to the conull generic set, *every* sufficiently large rectangle is available.

The proof is worth remembering. It does not iterate pointwise theorems, which would have a quantifier problem. Instead it combines

(1) the one-factor L^2 maximal inequalities;
(2) positivity, which gives the product maximal bound
$$\sup_{\mathbf{R}} |A_{\mathbf{R}} f| \leq M_1 \cdots M_r |f|;$$
(3) a dense span of bounded one-factor coboundaries, on which the Følner boundary estimate is uniform in all the other radii.

The maximal inequality upgrades convergence on that dense class to almost-everywhere convergence for all L^2 functions.

3. Uniform generic cores

Fix N test functions $f_1, \ldots, f_N \in C_c(Y)$ and $\eta > 0$. Corollary 3.1 gives a compact set K with $\mu(K) > 1 - \eta$ and a number R_0 such that
$$\left| A_{\mathbf{R}} f_\ell(y) - \int f_\ell \, d\mu \right| < \eta$$
for every $y \in K$, every $\ell \leq N$, and every product box satisfying $\min_j R_j \geq R_0$.

The order of the quantifiers is crucial:
$$\exists K, R_0 \quad \forall y \in K \quad \forall \mathbf{R} \text{ with } \min_j R_j \geq R_0.$$
Consequently a shearing scale may be discovered *after* the core has been chosen. One then chooses each internal block radius as large as the polynomial comparison requires; no synchronization with a sparse predetermined sequence is needed.

4. Common returns to a core

Choose K with negligible boundary and $\mu(K) > 1 - \eta$. Apply the product pointwise theorem to upper and lower continuous approximations of 1_K. For a generic point y,
$$\frac{m\{t \in F_{\mathbf{R}} : ty \in K\}}{m(F_{\mathbf{R}})} \longrightarrow \mu(K)$$
uniformly in the sense that the error is small once every side is sufficiently large. If x and x' are both generic, their two return sets therefore have intersection of relative measure at least
$$2\mu(K) - 1 - o(1).$$

Taking η small makes this positive. This is the recurrence input used in the product visible-block lemma of Chapter 7.

5. How the block comparison uses the theorem

Suppose $x_i = g_i x$ and $g_i \to e$. The relative position is

$$u(t)g_iu(-t).$$

Polynomial divergence selects a large outer scale R_i. Around a positive-density set of normalized parameters one then chooses a mesoscopic block of radius ρ_i satisfying

$$R_0 \ll \rho_i \ll R_i.$$

The first inequality makes every base point in the generic core genuinely generic on that block. The second makes the polynomial relative displacement nearly constant throughout the block. Because the product theorem allows independent side lengths, the same construction works when different local factors have slightly different mesoscopic radii.

Averaging a compactly supported test function over such a block gives, after the relative displacement tends to h,

$$\int f\,d\mu = \int f(hz)\,d\mu(z).$$

Thus the ergodic theorem supplies the uniform averages, while the H-property supplies the geometry that turns those averages into a new symmetry.

Warning 5.1. The endpoint assertion for arbitrary L^1 functions over unrestricted strong rectangles is neither needed nor claimed. Compactly supported continuous tests lie in L^2, and the L^2 product maximal theorem is exactly sufficient for all rigidity comparisons in this volume.

CHAPTER 6

The nonarchimedean H-property

Ratner's H-property is a precise form of polynomial shearing: two nearby points whose relative displacement is not centralized by the unipotent flow separate only polynomially, and at the first macroscopic separation scale their relative position converges to a nontrivial element of the centralizer of the flow. Nothing in this mechanism depends on an order relation on the parameter field. The only changes over $\mathbb{Q}_p$ are that scales must be chosen in a discrete value group and that we work inside the convergence radius of the p-adic logarithm and exponential.

1. Exact logarithmic coordinates

Let k be $\mathbb{R}$ or a characteristic-zero nonarchimedean local field, and let

$$U = \{u(t) = \exp(tN) : t \in k\} < \mathrm{SL}_n(k),$$

where N is nilpotent. Choose a small analytic neighborhood $\mathcal{O}$ of the identity on which log and exp are inverse analytic maps. In the nonarchimedean case choose $\mathcal{O}$ so small that the matrix exponential series converges uniformly there.

If $g_i \to e$, write $g_i = \exp X_i$. Conjugation is then exact:

$$u(t)g_iu(-t) = \exp(\mathrm{Ad}(u(t))X_i).$$

Since $\mathrm{ad}\,N$ is nilpotent,

$$\mathrm{Ad}(u(t))X_i = e^{t\,\mathrm{ad}\,N}X_i = \sum_{j=0}^{D} \frac{t^j}{j!}(\mathrm{ad}\,N)^j X_i,$$

where D depends only on n. Put

$$X_{i,j} = \frac{1}{j!}(\mathrm{ad}\,N)^j X_i.$$

2. Choosing the first visible scale

Before choosing the scale we record a finite-dimensional observation which prevents cancellation among several terms of the same size.

Lemma 2.1 (A bounded set of test parameters detects coefficients)**.** *For every local field k, degree D, and finite-dimensional normed k-space V, there are points $z_0, \dots, z_D$ in a fixed compact subset of k and a constant $c > 0$ such that every polynomial*

$$P(z) = A_0 + A_1 z + \cdots + A_D z^D \qquad (A_j \in V)$$

satisfies

$$\max_{0 \le r \le D} \|P(z_r)\| \ge c \max_{0 \le j \le D} \|A_j\|.$$

Proof. Choose $D+1$ distinct elements $z_0, \dots, z_D$ in the valuation ring (or in $[-1, 1]$ at the real place). Evaluation at these points is the linear map

$$(A_0, \dots, A_D) \longmapsto (P(z_0), \dots, P(z_D))$$

whose scalar matrix is Vandermonde. It is invertible because the z_r are distinct. The inverse is a fixed continuous linear map between finite-dimensional normed spaces, so its operator norm is finite. Rearranging the corresponding norm inequality gives the result. □

Lemma 2.2 (Shearing-scale lemma)**.** *Assume $g_i \notin C_G(U)$. Fix a sufficiently small $\delta > 0$. After passing to a subsequence there are $r_i \in k$, $|r_i| \to \infty$, such that*

$$Y_i := \mathrm{Ad}(u(r_i)) X_i$$

converges to a nonzero Y with $\|Y\| \asymp \delta$. The matrices

$$h_i = u(r_i) g_i u(-r_i) = \exp Y_i$$

then converge to $h = \exp Y \neq e$.

Proof. Because $g_i \notin C_G(U)$, some positive-degree coefficient $X_{i,j}$ is nonzero. For a value-group radius R, set

$$M_i(R) = \max_{1 \le j \le D} R^j \|X_{i,j}\|.$$

As $i \to \infty$, $M_i(1) \to 0$, whereas $M_i(R) \to \infty$ as $R \to \infty$ for fixed i. Choose R_i to be the first allowed value-group radius at which $M_i(R_i) \ge c_k \delta$, where $c_k > 0$ accounts for the ratio between consecutive radii (and may be taken to be one at the real place). Minimality gives a uniform upper bound $M_i(R_i) \ll_k \delta$, and $R_i \to \infty$.

At a nonarchimedean place choose $\varpi_i \in k$ with $|\varpi_i| = R_i$; at the real place put $\varpi_i = R_i$. Consider the bounded-parameter polynomial

$$P_i(z) = \sum_{j=0}^{D} z^j \varpi_i^j X_{i,j}.$$

One positive-degree coefficient has norm $\gg \delta$, all coefficients have norm $O(\delta)$, and $X_{i,0} \to 0$. By Lemma 2.1, among the fixed test values $z_0, \dots, z_D$ there is one at which $\|P_i(z_r)\| \gg \delta$. Passing to a subsequence, the same index r works for all i. Put $r_i = \varpi_i z_r$. Then

$$Y_i = \mathrm{Ad}(u(r_i))X_i = P_i(z_r)$$

remains in a compact annulus bounded away from zero. Local compactness in finite dimension gives a convergent subsequence $Y_i \to Y \neq 0$. Shrinking δ if necessary places the annulus inside the logarithm/exponential chart. Exact conjugation then gives $h_i = \exp Y_i \to \exp Y =: h \neq e$. □

3. Why the limit centralizes the flow

THEOREM 3.1 (Local-field H-property). *Under the hypotheses of Lemma 2.2, the scale can be chosen so that the nontrivial limit h belongs to $C_G(U)$. More precisely, for every fixed $s \in k$,*

$$u(s)h_i u(-s) - h_i \longrightarrow 0,$$

and hence $u(s)hu(-s) = h$.

PROOF. We have

$$u(s)h_i u(-s) = u(r_i + s)g_i u(-(r_i + s)) = \exp\big(\mathrm{Ad}(u(r_i + s))X_i\big).$$

It is therefore enough to show

$$\mathrm{Ad}(u(r_i + s))X_i - \mathrm{Ad}(u(r_i))X_i \to 0.$$

For each $j \geq 1$,

$$(r_i + s)^j - r_i^j = r_i^j \left[\left(1 + \frac{s}{r_i}\right)^j - 1\right].$$

The bracket tends to zero because $s/r_i \to 0$ in the local field. By the choice of shearing scale, $r_i^j X_{i,j}$ remains bounded. Summing finitely many terms proves convergence to zero in the Lie algebra. Continuity of the exponential gives the matrix statement. Passing to the limit yields $h \in C_G(U)$. □

Warning 3.2. The centralizing conclusion is obtained from the comparison of r_i and $r_i + s$ for *fixed* s. It is not enough to say that the leading coefficient of the polynomial lies in some "highest weight space": centralizers of nonregular unipotents can contain semisimple directions, and the invariant measure does not care whether the new symmetry is itself unipotent.

4. The rank-one calculation

Worked Example 4.1 ($\mathrm{SL}_2(\mathbb{Q}_p)$)**.** Let

$$u(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad g_i = \begin{pmatrix} 1 & 0 \\ \epsilon_i & 1 \end{pmatrix}, \qquad |\epsilon_i|_p \to 0.$$

Then

$$u(t) g_i u(-t) = \begin{pmatrix} 1 + t\epsilon_i & -t^2\epsilon_i \\ \epsilon_i & 1 - t\epsilon_i \end{pmatrix}.$$

Choose $|r_i|_p \asymp |\epsilon_i|_p^{-1/2}$, with the unit part of r_i chosen so $r_i^2\epsilon_i \to c \neq 0$. The linear terms tend to zero and

$$u(r_i) g_i u(-r_i) \longrightarrow \begin{pmatrix} 1 & -c \\ 0 & 1 \end{pmatrix} \in U = C_G(U)^\circ.$$

Replacing r_i by $r_i + s$ changes the quadratic term by $(2r_i s + s^2)\epsilon_i \to 0$, which is the concrete form of the centralizer argument above.

5. Several local factors

Let

$$U = \prod_{v \in T} U_v, \qquad U_v = \{u_v(t_v) : t_v \in k_v\},$$

and let $g_i = (g_{i,v})_{v \in S} \to e$. At a place outside T there is no parameter and the corresponding component of a shearing limit is simply the identity. For $v \in T$, write in logarithmic coordinates

$$\mathrm{Ad}(u_v(t_v)) X_{i,v} = \sum_{q=0}^{D_v} t_v^q X_{i,v,q}.$$

THEOREM 5.1 (Product local-field H-property). *Assume $g_i \notin C_G(U)$. After passing to a subsequence there are common positive scales $R_i \to \infty$, local elements $\varpi_{i,v}$ with $|\varpi_{i,v}|_v \asymp R_i$, and a compact product set $C = \prod_{v \in T} C_v$, each C_v contained in an annulus away from zero, such that*

$$H_i(z) = u((\varpi_{i,v} z_v)_v) g_i u((\varpi_{i,v} z_v)_v)^{-1}$$

converges uniformly on C to an analytic polynomial-exponential map $\Phi : C \to C_G(U)$. The map is not identically the identity. After shrinking C one may arrange

$$d_G(\Phi(z), e) \geq c_0 > 0 \qquad (z \in C).$$

PROOF. For an allowed positive radius R define

$$M_i(R) = \max_{v \in T} \max_{1 \leq q \leq D_v} R^q \|X_{i,v,q}\|_v,$$

where at a nonarchimedean place R is rounded to the nearest value-group radius; the rounding changes quantities by a place-dependent bounded factor. Since $g_i \notin C_G(U)$, some positive-degree coefficient is nonzero, while $M_i(1) \to 0$. Choose the first common scale R_i at which $M_i(R_i)$ reaches a fixed small level δ. Then $R_i \to \infty$ and every normalized coefficient

$$\varpi_{i,v}^q X_{i,v,q}$$

is uniformly bounded, while at least one positive-degree coefficient is bounded away from zero. Here $|\varpi_{i,v}|_v \asymp R_i$; at the real place take $\varpi_{i,\infty} = R_i$.

By finite-dimensional local compactness, pass to a subsequence on which all normalized coefficient tuples converge. The resulting product polynomial

$$P(z) = \left(\sum_q z_v^q Y_{v,q} \right)_{v \in T}$$

is nonconstant in at least one local component. Coefficient detection as in Lemma 2.1 lets us choose a product point z^0 with every coordinate nonzero and $P(z^0) \neq 0$. Take a sufficiently small compact product neighborhood C of z^0, still contained in annuli away from zero, on which $\exp P(z)$ stays away from the identity. Uniform coefficient convergence gives

$$H_i(z) \to \Phi(z) := \exp P(z)$$

uniformly on C after shrinking δ into a common exponential chart.

It remains to prove centralization. Fix $s = (s_v)_v \in U$. At every place where the limiting component P_v is nonzero, parameters $t_{i,v} = \varpi_{i,v} z_v$ satisfy $|t_{i,v}|_v \asymp R_i \to \infty$ uniformly on C_v. The binomial calculation from Theorem 3.1 therefore shows

$$\mathrm{Ad}(u_v(t_{i,v} + s_v))X_{i,v} - \mathrm{Ad}(u_v(t_{i,v}))X_{i,v} \to 0.$$

If $P_v = 0$, the corresponding limiting group component is the identity, which already centralizes U_v. Hence

$$u(s)\Phi(z)u(-s) = \Phi(z)$$

for every $s \in U$ and $z \in C$. Thus $\Phi(C) \subset C_G(U)$. □

Remark 5.2. The use of one common scalar radius R_i is deliberate. It makes the visible normalized region a positive fixed proportion of a rectangular product box whose side lengths are all comparable to R_i. This is exactly what is needed by the product pointwise theorem of Chapter 4; no uncontrolled family of eccentric rectangles is introduced by the shearing normalization.

CHAPTER 7

Polynomial comparison of nearby generic trajectories

The H-property becomes a rigidity tool only after its polynomial scale is synchronized with recurrence. We first prove the synchronization for a single local one-parameter action and then prove the mixed product-parameter version using the L^2 rectangular pointwise theorem of Chapter 4. The two arguments are separated so that the extra product-block geometry remains visible.

Throughout the first three sections, k is either $\mathbb{R}$ or a characteristic-zero nonarchimedean local field and

$$U = \{u(t) : t \in k\}$$

acts ergodically preserving a probability measure μ on X.

1. Uniform generic cores for one local parameter

Fix a countable convergence-determining family $\mathscr{F} = \{f_1, f_2, \ldots\} \subset C_c(X)$. The ordinary real pointwise ergodic theorem when $k = \mathbb{R}$, and Theorem 1.2 when k is finite, give a conull set of points for which centered averages converge for every f_j. Egorov and inner regularity give the following uniform form.

Lemma 1.1 (Uniform one-parameter generic core)**.** *For every $N \geq 1$ and $\eta > 0$ there are a compact set $K = K(N, \eta)$ with $\mu(K) > 1 - \eta$ and a radius R_0 such that for every $y \in K$, every $1 \leq j \leq N$, and every centered parameter ball B_R with $R \geq R_0$,*

$$\left| \frac{1}{m(B_R)} \int_{B_R} f_j(u(t)y)\, dt - \int f_j \, d\mu \right| < \eta.$$

At a finite place "every radius" means every sufficiently large value-group radius.

Proof. For each $j \leq N$ the centered averages converge almost everywhere. The maximum of the first N errors therefore tends to zero almost everywhere. Egorov gives uniform convergence outside a set of measure $< \eta/2$;

inner regularity then replaces that measurable set by a compact subset losing at most another $\eta/2$. Uniform convergence supplies one common R_0. ☐

2. A stable block around a visible shearing time

Let $g_i \to e$, write $g_i = \exp X_i$, and define

$$H_i(t) = u(t) g_i u(-t).$$

The proof of the H-property chooses a scale $R_i \to \infty$ and normalized parameters z in a fixed compact set for which $H_i(R_i z)$ has a nontrivial limit polynomial.

Lemma 2.1 (Mesoscopic stability of the conjugacy polynomial)**.** *Assume that for a value-group scale $R_i \to \infty$ the coefficients satisfy*

$$\sup_{0 \le r \le D} R_i^r \|X_{i,r}\| \le C_0,$$

where

$$\operatorname{Ad}(u(t)) X_i = \sum_{r=0}^{D} t^r X_{i,r}.$$

Let τ_i satisfy $|\tau_i| \asymp R_i$, and let $\rho_i \to \infty$ with $\rho_i / R_i \to 0$. Then

$$\sup_{|s| \le \rho_i} \| \operatorname{Ad}(u(\tau_i + s)) X_i - \operatorname{Ad}(u(\tau_i)) X_i \| \longrightarrow 0.$$

Consequently, whenever both sides remain in one exponential chart,

$$\sup_{|s| \le \rho_i} d_G(H_i(\tau_i + s), H_i(\tau_i)) \to 0.$$

Proof. For every $r \ge 1$,

$$(\tau_i + s)^r - \tau_i^r = \sum_{q=1}^{r} \binom{r}{q} \tau_i^{r-q} s^q.$$

Multiply the qth term by $X_{i,r}$. Since $R_i^r \|X_{i,r}\| \le C_0$ and $|\tau_i| \asymp R_i$, its norm is bounded by a structural constant times

$$\left(\frac{|s|}{R_i} \right)^q.$$

This tends uniformly to zero for $|s| \le \rho_i$. There are only finitely many r, q, so the Lie-algebra difference tends uniformly to zero. Continuity and uniform continuity of the exponential on a compact subchart give the group statement. ☐

3. Finding a recurrent visible time

The normalized polynomial limit is nonconstant. Hence there is a compact open (or compact interval) set of normalized parameters on which it stays away from the identity.

Lemma 3.1 (Visible recurrence window). *Let $x_i = g_i x \to x$ be points for the same ergodic U-invariant probability measure, with $g_i \notin C_G(U)$. Let R_i be the corresponding shearing scales. Assume the pair is* synchronized generic at scale R_i: *for the continuous inner and outer approximants of every compact core used below, the centered averages based at both x and x_i are within $o(1)$ of their space averages on all radii in a fixed compact multiple of R_i. Then, after passing to a subsequence, there are a compact set $C \subset k$ of positive Haar measure and $c_0 > 0$ such that*

$$H_i(R_i z) \longrightarrow \Phi(z)$$

uniformly on C, where

$$d_G(\Phi(z), e) \geq c_0 \qquad (z \in C).$$

Moreover, for every compact set $K \subset X$ with $\mu(K)$ sufficiently close to one, there are $z_i \in C$ such that

$$u(R_i z_i)x \in K, \qquad u(R_i z_i)x_i \in K.$$

After a further subsequence $z_i \to z \in C$ and

$$H_i(R_i z_i) \longrightarrow h := \Phi(z) \neq e.$$

Proof. The shearing-scale argument in Chapter 6, before specializing to one test parameter, gives coefficient tuples bounded in a fixed finite-dimensional space with at least one positive-degree coefficient bounded away from zero. Pass to a subsequence so the normalized polynomials

$$z \longmapsto \mathrm{Ad}(u(R_i z))X_i$$

converge uniformly on compact subsets to a nonconstant polynomial $P(z)$. Choose z_0 with $P(z_0) \neq 0$ and then a sufficiently small compact neighborhood $C \ni z_0$ on which $\exp P(z)$ stays at distance at least c_0 from the identity. Uniform convergence of the exponential gives the first assertion.

We now use synchronized genericity. The dilate $R_i C$ occupies a fixed positive proportion $\kappa > 0$ of a centered ball B_{AR_i} for one fixed A depending only on C. In the real case this is ordinary scaling of Lebesgue measure. In the nonarchimedean case shrink C, if needed, to a ball or finite union of balls contained in one fixed annulus; dilation by R_i again preserves a fixed relative measure.

Let

$$E_i = \{t \in B_{AR_i} : u(t)x \notin K\}, \qquad E'_i = \{t \in B_{AR_i} : u(t)x_i \notin K\}.$$

Using continuous majorants of $1_{X \setminus K}$ and the synchronized genericity assumption at the radii AR_i, we have

$$\frac{m(E_i)}{m(B_{AR_i})}, \ \frac{m(E'_i)}{m(B_{AR_i})} \leq 1 - \mu(K) + o(1).$$

Choose K so that $2(1 - \mu(K)) < \kappa/2$. Then for large i the union $E_i \cup E'_i$ has measure $< m(R_i C)$. Therefore some $\tau_i = R_i z_i \in R_i C$ lies outside both exceptional sets. Thus both trajectories return to K at τ_i.

Compactness of C gives a convergent subsequence $z_i \to z \in C$. Uniform convergence of $H_i(R_i z)$ to $\Phi(z)$ gives $H_i(\tau_i) \to \Phi(z)$, which is nontrivial by the choice of C. □

4. How synchronized pairs are obtained

The extra synchronization hypothesis in the next proposition is not cosmetic. If the nearby point changes with i, mere genericity of each x_i does not imply that its averaging error is small at the particular shearing radius R_i: the individual genericity threshold could be larger than R_i. In the enlargement argument one avoids this quantifier error as follows. Choose nested uniform generic cores K_j with errors tending to zero and let x be a simultaneous density point. For stage j, first fix the genericity threshold attached to K_j; only then choose a distinct point $x_j \in K_j$ sufficiently close to x that its first visible shearing scale exceeds that threshold by the required fixed factor. Because the shearing scale tends to infinity as the transverse displacement tends to zero, this order of choices is possible whenever the measure has genuinely transverse mass near x. Thus the pairs used by the rigidity proof are synchronized generic by construction.

5. Block comparison creates invariance

Proposition 5.1 (Additional invariance for one local parameter)**.** *Let $x_i = g_i x \to x$ be points for an ergodic U-invariant probability μ, with $g_i \notin C_G(U)$. Suppose x and x_i lie in the same local quotient chart, the H-property applies, and the pairs are chosen from uniform generic cores so that they are synchronized generic at their shearing scales in the sense of Lemma 3.1. Then, after passing to a subsequence, there is a nontrivial $h \in I(\mu)$ obtained as a shearing limit. More generally, if the normalized conjugacy polynomials converge to $\Phi(z)$ on a compact open set, then*

$$\Phi(z)\Phi(w)^{-1} \in I(\mu)$$

for all z, w in each connected/analytic piece on which the block comparison is performed.

Proof. Fix N and $\eta > 0$. Before choosing the generic core, choose a compact neighborhood $Q \subset G$ containing the image $\Phi(C)$ and all values $H_i(R_i z)$ for $z \in C$ and all sufficiently large i. For every $j \leq N$ the family

$$\{f_j \circ q : q \in Q\}$$

is precompact in the uniform norm: all supports lie in the fixed compact set $Q^{-1} \operatorname{supp} f_j$, and uniform continuity on the compact action set $Q \times Q^{-1} \operatorname{supp} f_j$ supplies finite uniform nets. Choose a finite η-net for each of these families, adjoin the original f_j, and apply Lemma 1.1 to this finite enlarged test family. Call the resulting core K and its threshold R_0.

Apply Lemma 3.1 to obtain recurrent visible times τ_i for which

$$y_i = u(\tau_i)x \in K, \qquad y_i' = u(\tau_i)x_i \in K, \qquad H_i(\tau_i) \to h \neq e.$$

Choose radii $\rho_i \to \infty$ satisfying

$$\rho_i / R_i \to 0$$

and $\rho_i \geq R_0(N, \eta)$. By Lemma 2.1,

$$H_i(\tau_i + s) \to h$$

uniformly for $s \in B_{\rho_i}$. Since

$$u(s)y_i' = u(s + \tau_i)x_i = H_i(s + \tau_i)u(s + \tau_i)x = H_i(s + \tau_i)u(s)y_i,$$

we obtain, uniformly in $s \in B_{\rho_i}$ whenever the relevant points meet the compact support enlarged by a small neighborhood of h,

$$f_j(u(s)y_i') - f_j(h\, u(s)y_i) \longrightarrow 0 \qquad (1 \leq j \leq N).$$

Compact support and uniform continuity control the complementary parameters without any escape issue: both terms vanish outside one fixed compact enlargement, and the displacement stays in a fixed compact neighborhood of h for large i. Averaging over B_{ρ_i} gives

$$\begin{aligned}\left|\int f_j\, d\mu - \int f_j(hz)\, d\mu(z)\right| \leq{}& \left|A_{\rho_i} f_j(y_i') - \int f_j\, d\mu\right| \\ &+ \left|A_{\rho_i}(f_j \circ h)(y_i) - \int f_j \circ h\, d\mu\right| \\ &+ o(1).\end{aligned}$$

The second generic term is controlled by the finite η-net chosen above: choose q in the net with $\|f_j \circ h - f_j \circ q\|_\infty < \eta$. The averaging error for $f_j \circ q$ is $O(\eta)$ on K, and replacing it by $f_j \circ h$ changes both the empirical and space averages by at most η. Thus both generic errors are $O(\eta)$ because

$y_i, y_i' \in K$ and $\rho_i \geq R_0$. This compact-family argument removes any circular dependence of the core on the shearing limit h. Let $i \to \infty$ and then $\eta \to 0$. We obtain

$$\int f_j \, d\mu = \int f_j(hz) \, d\mu(z)$$

for $j \leq N$. Finally let $N \to \infty$. The convergence-determining family implies $h_*\mu = \mu$, so $h \in I(\mu)$.

If the limit depends analytically on a normalized parameter z, repeat the block construction for a countable dense set of z-values and use closedness of $I(\mu)$. Ratios $\Phi(z)\Phi(w)^{-1}$ then preserve μ. □

6. Product-parameter block selection

We now close the mixed-parameter gap using the product pointwise theorem of Chapter 4 and the common-radius normalization of Theorem 5.1. Let

$$U = \prod_{v \in T} U_v$$

act ergodically, and let $g_i \to e$ with $g_i \notin C_G(U)$. Write

$$H_i(t) = u(t) g_i u(-t), \qquad t = (t_v)_{v \in T}.$$

Lemma 6.1 (Product mesoscopic stability)**.** *Let R_i and $\varpi_{i,v}$ be the common shearing scales from Theorem 5.1. Suppose $\tau_i = (\varpi_{i,v} z_{i,v})_v$ with z_i in the fixed compact annular set C. Let $\rho_i \to \infty$ satisfy $\rho_i / R_i \to 0$, and let*

$$E_i = \prod_{v \in T} B_v(0, \rho_i)$$

(with value-group rounding at finite places). Then

$$\sup_{s \in E_i} d_G(H_i(\tau_i + s), H_i(\tau_i)) \longrightarrow 0.$$

Proof. At place v the logarithm is a polynomial

$$\sum_{q=0}^{D_v} t_v^q X_{i,v,q}$$

with $R_i^q \|X_{i,v,q}\|_v = O(1)$. Since $|\tau_{i,v}|_v \asymp R_i$ on the annular normalized set, the binomial expansion gives, uniformly for $|s_v|_v \leq \rho_i$,

$$\|(\tau_{i,v} + s_v)^q X_{i,v,q} - \tau_{i,v}^q X_{i,v,q}\|_v \ll \sum_{a=1}^{q} (\rho_i / R_i)^a = o(1).$$

There are finitely many places and degrees. Summing the local estimates and using uniform continuity of the exponential on the common compact chart proves the claim. □

Lemma 6.2 (Product visible recurrence)**.** *Assume $x_i = g_i x \to x$ and, for a fixed uniform product generic core K of measure sufficiently close to one, both x and x_i belong to K and the common shearing scale R_i is beyond its genericity threshold. Then, after passing to a subsequence, there are $z_i \in C$ such that*

$$u(\tau_i)x \in K, \qquad u(\tau_i)x_i \in K, \qquad \tau_i = (\varpi_{i,v} z_{i,v})_v,$$

and

$$H_i(\tau_i) \longrightarrow h \neq e.$$

Proof. The set

$$W_i = \{(\varpi_{i,v} z_v)_v : z \in C\}$$

occupies a fixed positive proportion $\kappa > 0$ of a centered product box F_i whose every side radius is between cR_i and CR_i. This follows from product Haar scaling; the constants depend only on the fixed annular set C and the finitely many value-group rounding factors.

Let E_i and E_i' be the subsets of F_i on which the trajectories from x and x_i, respectively, miss K. Apply the uniform product generic-core property to continuous majorants of $1_{X \setminus K}$. Since all side lengths of F_i tend to infinity and exceed the core threshold,

$$\frac{m(E_i)}{m(F_i)}, \quad \frac{m(E_i')}{m(F_i)} \leq 1 - \mu(K) + o(1).$$

Choose K with $2(1 - \mu(K)) < \kappa/2$. For large i, $m(E_i \cup E_i') < m(W_i)$, so W_i contains a common return time τ_i. Compactness of C gives $z_i \to z$ after a subsequence, and the uniform convergence in Theorem 5.1 gives $H_i(\tau_i) \to \Phi(z)$. The annular set was chosen so $d(\Phi(z), e) \geq c_0$, hence $h = \Phi(z) \neq e$. □

Proposition 6.3 (Additional invariance for a product parameter group)**.** *Let $U = \prod_{v \in T} U_v$ act ergodically preserving a probability measure μ. Suppose $x_i = g_i x \to x$ are distinct generic support points with $g_i \notin C_G(U)$. Choose them recursively from nested uniform product generic cores so that their common H-shearing scales dominate the current core thresholds. Then a nontrivial product shearing limit $h \in C_G(U)$ belongs to $I(\mu)$. More generally, ratios of values of the normalized limit family belong to $I(\mu)$.*

Proof. The recursive choice is possible because the common shearing scale in Theorem 5.1 tends to infinity as $g_i \to e$. Fix a finite test family and accuracy η. As in the one-parameter proof, enlarge the test family by a finite uniform net for all translates $f \circ q$ with q in a compact neighborhood containing the product shearing limits. Choose a uniform product generic core K for this enlarged family.

Lemma 6.2 gives common return times τ_i and returned points

$$y_i = u(\tau_i)x, \qquad y_i' = u(\tau_i)x_i$$

in K, with $H_i(\tau_i) \to h \neq e$. Choose $\rho_i \to \infty$ with $\rho_i/R_i \to 0$ and ρ_i larger than the core threshold, and average over the product block $E_i = \prod_v B_v(0, \rho_i)$. By Lemma 6.1,

$$H_i(\tau_i + s) \to h$$

uniformly for $s \in E_i$. The exact identity

$$u(s)y_i' = H_i(\tau_i + s)u(s)y_i$$

therefore compares the two block averages. Uniform genericity of $y_i, y_i' \in K$ identifies each average with the corresponding μ-integral, uniformly for the finite translate net. Passing $i \to \infty$, then $\eta \to 0$, yields

$$\int f\, d\mu = \int f(hz)\, d\mu(z)$$

for a convergence-determining family of $f \in C_c(X)$. Hence $h_*\mu = \mu$. Repeating on a countable dense set of normalized parameters and using closedness of $I(\mu)$ gives the ratio statement. $\square$

Remark 6.4. This removes the earlier mixed-parameter synchronization gap. The argument uses only L^2 pointwise genericity for compactly supported test functions, not the endpoint strong L^1 multiparameter theorem.

CHAPTER 8

Quasi-regular maps: the degeneration mechanism

Polynomial shearing is most transparent when a nearby displacement is visibly transverse to the known invariance group. Ratner's local-field argument also has a second mechanism, needed when a sequence approaches a normalizer or another algebraic incidence set so efficiently that the first transverse Taylor term vanishes after normalization. The correct replacement is a *quasi-regular map*. It retains the polynomial part of a highly magnified displacement after allowing a compensating motion inside the acting unipotent group.

This chapter reconstructs the algebraic normalization and the ergodic comparison in the one-local-parameter case. We prove the pieces that are often compressed into the phrase "construct a quasi-regular map": a rational slice for a $\mathbf{G}_a$-action, coefficient normalization in a Chevalley representation, the normalizer property of the limit, and the basic averaging lemma. The final section explains how this absolute construction feeds the relative finite-correction algorithm completed in Chapter 12.

Throughout this chapter k is a characteristic-zero local field, $G = \mathbf{G}(k)$ is the group of k-points of a linear algebraic group, and

$$U = \{u(t) : t \in k\} \simeq (k, +)$$

is a nontrivial algebraic one-parameter unipotent subgroup.

1. A rational slice for the right U-action

We first isolate an elementary algebraic fact. It is useful because it explains why the compensating U-coordinate is rational rather than merely analytic.

Lemma 1.1 (Rational slice for a one-dimensional split unipotent action). *There are a nonempty U-stable Zariski open subset $\Omega \subset \mathbf{G}$, a rational subvariety $\mathbf{M} \subset \Omega$, and rational maps*

$$\sigma : \Omega \dashrightarrow \mathbf{M}, \qquad \tau : \Omega \dashrightarrow \mathbf{G}_a$$

such that, wherever they are defined,

$$g = \sigma(g)u(\tau(g)), \qquad \sigma(gu(s)) = \sigma(g), \qquad \tau(gu(s)) = \tau(g) + s.$$

After translating the slice if necessary, one may arrange that the identity lies in the domain and that $\sigma(e) = e$, $\tau(e) = 0$.

PROOF. Consider the right $\mathbf{G}_a$-action on the affine algebraic group $\mathbf{G}$. On the coordinate ring $A = k[\mathbf{G}]$ it is generated by a nonzero locally nilpotent derivation D. Choose $f \in A$ with $D^m f \neq 0$ and $D^{m+1} f = 0$ for some maximal $m \geq 1$. Put

$$b = D^m f, \qquad a = \frac{D^{m-1} f}{D^m f}$$

in the localization A_b. Since $D(b) = 0$ and $D(D^{m-1} f) = b$, we have

$$D(a) = 1.$$

Thus a is a rational slice coordinate on the principal open set $\Omega_0 = \{b \neq 0\}$.

We recall the short slice argument. If $c \in A_b$, then the exponential

$$\exp(-aD)c = \sum_{j \geq 0} \frac{(-a)^j}{j!} D^j c$$

is finite because D is locally nilpotent, and it lies in $\ker D$. Taylor expansion in the variable a therefore expresses every element of A_b as a polynomial in a with coefficients in $(A_b)^D$. Hence

$$A_b = (A_b)^D [a].$$

Geometrically, on Ω_0 the action is rationally a product of a quotient slice and $\mathbb{A}^1$, with $a(gu(s)) = a(g) + s$.

Let $\mathbf{M}$ be the hypersurface $a = 0$ inside this product chart. Set $\tau(g) = a(g)$ and

$$\sigma(g) = gu(-a(g)).$$

Then $\sigma(g) \in \mathbf{M}$ and the asserted factorization and equivariance identities are immediate. Since the action is free under right multiplication, this factorization is unique on its domain.

The principal open set produced by the chosen f need not contain e. Choose one point of it and right/left translate the resulting rational product chart to a Zariski open neighborhood of e; translating the a-coordinate by a constant then makes $a(e) = 0$. Intersecting finitely many translates if needed preserves a nonempty open domain and gives the stated normalization. □

Remark 1.2. The proof is the algebraic analogue of choosing a local transversal to a flow, but it is stronger: the coordinate change is rational on a Zariski open set. The single identity $D(a) = 1$ is the reason. For higher-dimensional split unipotent groups one iterates a central series with successive $\mathbf{G}_a$ quotients; the bookkeeping is longer but no new principle is involved.

2. Chevalley coordinates and coefficient normalization

We use one standard algebraic representation theorem.

THEOREM 2.1 (Chevalley realization for a unipotent subgroup). *There are a finite-dimensional k-rational representation $\rho : \mathbf{G} \to \mathrm{GL}(V)$ and a nonzero vector $q \in V(k)$ such that*

$$U = \{g \in G : \rho(g)q = q\}.$$

Consequently

$$\rho(N_G(U))q = \{z \in \rho(G)q : \rho(U)z = z\}.$$

PROOF. For completeness we recall the algebraic construction. The ideal $I(U)$ inside $k[\mathbf{G}]$ is stable under right translation by U. Because the coordinate ring is the union of its finite-dimensional rational G-submodules, choose a finite-dimensional submodule containing finitely many generators of $I(U)$. Taking the appropriate exterior power of the subspace cut out by those generators produces a rational line whose stabilizer is U up to the character by which the stabilizer acts on that line. A unipotent algebraic group has no nontrivial rational characters, so U acts trivially on that line. Any nonzero vector on the line is therefore fixed by U, and because the line stabilizer was already exactly U, its vector stabilizer is exactly U. This is the usual Chevalley construction.

For the second assertion, if $g \in N_G(U)$ then $g^{-1}Ug = U$, so $\rho(g)q$ is fixed by U. Conversely, if $\rho(g)q$ is U-fixed, then for all $u \in U$,

$$\rho(g^{-1}ug)q = q,$$

whence $g^{-1}Ug \subset U$. Both groups are connected one-dimensional unipotent groups, so the inclusion is equality. □

Fix a norm on V. Write

$$F_g(t) = \rho(u(t)g)q.$$

Since $\rho(u(t)) = \exp(tN_\rho)$ for a nilpotent operator N_ρ, F_g is a V-valued polynomial of degree at most a fixed number D.

Lemma 2.2 (First visible Chevalley scale)**.** *Let $g_i \to e$ with $g_i \notin N_G(U)$. After passing to a subsequence there are $r_i \in k$ with $|r_i| \to \infty$ such that the polynomials*

$$\Psi_i(z) = F_{g_i}(r_i z) = \rho(u(r_i z)g_i)q$$

converge coefficientwise, hence uniformly on compact z-sets, to a nonconstant V-valued polynomial Ψ. Moreover $\Psi(0) = q$.

Proof. Write
$$F_{g_i}(t) = a_{i,0} + a_{i,1}t + \cdots + a_{i,D}t^D.$$
Because $g_i \to e$ and U fixes q,
$$a_{i,0} \to q, \qquad a_{i,j} \to 0 \quad (j \geq 1).$$
The polynomial is nonconstant exactly when $\rho(g_i)q$ is not fixed by U, which by Theorem 2.1 is equivalent to $g_i \notin N_G(U)$. Thus at least one $a_{i,j}$ with $j \geq 1$ is nonzero.

Choose a positive scale R_i in the value group of k so that
$$1 \leq \max_{1\leq j\leq D} R_i^j \|a_{i,j}\| \leq C_k,$$
where $C_k = 1$ in the archimedean case and, in the nonarchimedean case, C_k is a fixed constant depending only on the value-group jump and on D (one may take the Dth power of the basic jump). Since every positive-degree coefficient tends to zero, $R_i \to \infty$. Choose $r_i \in k$ with $|r_i| = R_i$.

The coefficients of Ψ_i are $r_i^j a_{i,j}$. They are bounded for all j by the definition of the first visible scale. Finite-dimensional compactness over a local field lets us pass to a subsequence on which every coefficient converges. At least one positive-degree limiting coefficient has norm bounded below by C_k^{-1}, so the limit is nonconstant. The constant coefficient tends to q. $\square$

Worked Example 2.3 (The lower-root perturbation in SL_2)**.** Let
$$u(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad g_i = \begin{pmatrix} 1 & 0 \\ \varepsilon_i & 1 \end{pmatrix}, \qquad \varepsilon_i \to 0.$$
In the adjoint representation one sees terms of sizes ε_i, $t\varepsilon_i$, and $t^2\varepsilon_i$. The first visible scale is therefore $|r_i| \asymp |\varepsilon_i|^{-1/2}$. The normalized limit retains the quadratic term and discards the lower-order terms. The usual H-property computation is the coordinate shadow of the Chevalley normalization above.

3. Lifting the polynomial limit back to the group

Choose the rational slice of Lemma 1.1. On the Zariski open set where the factorization is defined write
$$u(r_i z)g_i = \phi_i(z)u(r_i\eta_i(z)), \qquad \phi_i(z) \in M(k).$$
Thus
$$\phi_i(z) = u(r_i z)g_i u(-r_i\eta_i(z)).$$
The maps ϕ_i and η_i are rational of degree bounded independently of i on every fixed affine chart.

Because the right U-factor fixes q,
$$\rho(\phi_i(z))q = \Psi_i(z).$$

The restriction of the orbit map $g \mapsto \rho(g)q$ to the rational slice is birational onto a Zariski open subset of the orbit $\rho(G)q$. We may therefore invert it at every regular point of the limiting polynomial.

THEOREM 3.1 (Construction of a strongly quasi-regular limit). *After passing to a subsequence there are a nonempty Zariski open set $Z \subset k$ and a nonconstant rational map*

$$\phi : Z \longrightarrow G$$

with $\phi(0) = e$ whenever $0 \in Z$, such that

$$\phi_i(z) \longrightarrow \phi(z)$$

uniformly on compact subsets of Z. Equivalently,

$$\phi(z) = \lim_{i\to\infty} u(r_i z) g_i u(-r_i \eta_i(z)).$$

For every rational representation $\pi : G \to \mathrm{GL}(W)$ and every U-fixed vector $w \in W$, the map $z \mapsto \pi(\phi(z))w$ is regular on Z.

PROOF. By Lemma 2.2, $\Psi_i \to \Psi$ coefficientwise. Let $\mathcal{O}$ be the Zariski open subset of $\rho(G)q$ on which the inverse of the slice orbit map is regular. Since Ψ is nonconstant and its image is not contained in the proper complement of every such chart, after replacing the slice chart if necessary the set

$$Z = \Psi^{-1}(\mathcal{O})$$

is nonempty and Zariski open. On compact subsets of Z, continuity of the rational inverse away from its poles gives

$$\phi_i = (\rho|_M)^{-1} \circ \Psi_i \longrightarrow (\rho|_M)^{-1} \circ \Psi =: \phi$$

uniformly. Since Ψ is nonconstant and the slice orbit map is injective on the chart, ϕ is nonconstant. At $z = 0$, $\Psi(0) = q$ and the normalized slice contains e, giving $\phi(0) = e$ whenever the chart contains that point.

It remains to prove the last assertion, which is the representation-theoretic content of quasi-regularity. Let w be U-fixed. From the factorization,

$$\pi(\phi_i(z))w = \pi(u(r_i z)g_i)w.$$

The right side is a W-valued polynomial in z of uniformly bounded degree. On compact subsets of Z it converges, because $\phi_i \to \phi$. Evaluating at a fixed Vandermonde set contained in Z shows that its coefficients converge. Hence the limit $\pi(\phi(z))w$ is polynomial, in particular regular, on Z. □

4. Why the image normalizes U

The normalizer conclusion is often presented as a separate algebraic miracle. In this normalization it is a one-line consequence of the fact that bounded translations become microscopic in the normalized variable.

Proposition 4.1 (Normalizer property and nontriviality modulo U)**.** *The quasi-regular limit from Theorem 3.1 satisfies*

$$\phi(Z) \subset N_G(U).$$

Its image is not contained in U. More strongly, the image in $N_G(U)/U$ is nonconstant; in particular no compact subset $C \subset N_G(U)$ can satisfy $\phi(Z) \subset CU$ when Z is allowed to range through its full k-rational domain.

Proof. Fix $a \in k$. For z in a compact subset of Z on which also $z + a/r_i \in Z$, we have

$$\begin{aligned} \rho(u(a))\Psi_i(z) &= \rho(u(a + r_i z)g_i)q \\ &= \Psi_i\left(z + \frac{a}{r_i}\right). \end{aligned}$$

Since $|r_i| \to \infty$ and the polynomials Ψ_i converge coefficientwise, the right side tends uniformly to $\Psi(z)$. Therefore

$$\rho(u(a))\Psi(z) = \Psi(z) \qquad (a \in k).$$

Thus $\Psi(z)$ is fixed by U. But $\Psi(z) = \rho(\phi(z))q$, so Standard Result 2.1 gives $\phi(z) \in N_G(U)$.

If $\phi(Z) \subset U$, then $\rho(\phi(z))q = q$ for every z, contradicting the nonconstancy of Ψ. Thus the image in the quotient is nonconstant.

For the final assertion, the orbit map $N_G(U)/U \to V$, $gU \mapsto \rho(g)q$, is algebraic and separates the slice coordinate. The image $\Psi(Z)$ of a nonconstant polynomial map over a noncompact local field is unbounded on its full regular domain: choose a highest nonzero coefficient and let $|z| \to \infty$ away from the finitely many poles of the slice inverse. If $\phi(Z) \subset CU$, then $\Psi(Z) \subset \rho(C)q$ would be bounded, a contradiction. $\square$

5. The reparametrization that enters the Basic Lemma

The factorization contains a second rational map η_i. It is not legitimate to ignore it: the two nearby trajectories are compared after a change of parameter. We isolate exactly the regularity needed by the ergodic argument.

Definition 5.1 (Tame quasi-regular normalization)**.** A normalized sequence $(g_i, r_i, \phi_i, \eta_i)$ is called *tame on a compact ball $A \Subset Z$* if, after passing to a subsequence,

(1) $\eta_i \to \eta$ uniformly on a neighborhood of A;
(2) η is an analytic diffeomorphism on that neighborhood;
(3) in the real case the derivatives η_i' converge uniformly to η' and are bounded above and away from zero; in the nonarchimedean case there is a ball decomposition of A on each member of which every η_i is a bi-analytic similarity with a fixed Jacobian absolute value.

Lemma 5.2 (Regular points give tame subballs)**.** *Suppose the rational maps η_i converge coefficientwise to a nonconstant rational map η on an affine chart. Then every point z_0 at which η is regular and $\eta'(z_0) \neq 0$ has a compact ball neighborhood A on which the normalization is tame.*

Proof. In the real case this is ordinary C^1 convergence of rational maps on a compact set separated from the poles, followed by the inverse function theorem and a smaller interval if necessary.

In the nonarchimedean case choose a ball on which the denominator of η has constant absolute value and the linear term in the Taylor expansion at z_0 dominates all higher terms. Coefficientwise convergence gives the same strict dominance for η_i for all large i. The ultrametric inverse function lemma then says that each η_i maps the ball bijectively onto a ball and

$$|\eta_i(z) - \eta_i(w)| = |\eta_i'(z_0)|\,|z - w|.$$

The derivative absolute values stabilize in the discrete value group. Subdivide a larger compact set into finitely many such balls to obtain the stated form. □

Remark 5.3. The genuinely delicate algebraic point in a full Margulis–Tomanov induction is not Lemma 5.2; it is to prove, for every degenerating sequence selected by the singular-set analysis, that one can choose the normalization so that the limiting reparametrization is nonconstant. We return to this issue in the final section.

6. A Basic Lemma with the averaging comparison written out

Let (X, μ) be a probability G-space and suppose μ is U-ergodic. For a compact nonnull set $A \subset k$ put

$$A_i = r_i A.$$

The pointwise theorems of Chapter 4, together with Egorov, provide compact sets on which averages over sufficiently large balls are uniform. For the Basic Lemma we need the same conclusion after the tame change of variable $z \mapsto \eta_i(z)$. The next elementary observation supplies it.

Lemma 6.1 (The normalized U-coordinate is based at the identity)**.** *For the rational slice normalized by $\tau(e) = 0$,*

$$\eta_i(0) \longrightarrow 0, \qquad \eta_i'(0) \longrightarrow 1.$$

Consequently, if the normalization is tame on a sufficiently small centered ball A about 0, then for all large i the set $\eta_i(A)$ contains 0; over a nonarchimedean field it is itself a centered ball, while over $\mathbb{R}$ it is an interval containing 0 with the ratio of its two side lengths bounded above and below.

Proof. At $z = 0$ the factorization gives

$$g_i = \phi_i(0)u(r_i\eta_i(0)),$$

and by definition of the slice coordinate

$$r_i\eta_i(0) = \tau(g_i).$$

Since $g_i \to e$, $\tau(g_i) \to 0$, whereas $|r_i| \to \infty$; hence $\eta_i(0) \to 0$.

Differentiate the identity

$$r_i\eta_i(z) = \tau(u(r_iz)g_i)$$

at $z = 0$. If X is the tangent vector of $u(t)$ at the identity, then

$$\eta_i'(0) = d\tau_{g_i}(Xg_i).$$

The right side tends to $d\tau_e(X) = 1$, because on the U-orbit through the identity the slice coordinate is exactly $\tau(u(t)) = t$. The last assertion now follows from tameness and the real or ultrametric inverse function theorem. □

Lemma 6.2 (Common-return selection for a tame quasi-regular chart)**.** *Let K_i be compact uniform generic cores with $\mu(K_i) \to 1$. Suppose $x_i, g_ix_i \in K_i$ and the quasi-regular normalization is tame on a sufficiently small centered ball $A \ni 0$. Assume $|r_i|$ dominates the genericity threshold of K_i. Then there are $z_i \in A$ and, after passing to a subsequence, $z_i \to z_0 \in A$ such that*

$$y_i := u(r_iz_i)g_ix_i \in K_i, \qquad y_i' := u(r_i\eta_i(z_i))x_i \in K_i.$$

Moreover z_0 may be chosen outside any prescribed finite subset of A. In particular it may be chosen so that $\phi(z_0) \notin U$.

Proof. Write E_i for the set of $z \in A$ for which the first return lies in K_i and E_i' for the corresponding set for the second return. For E_i, the change of variable $t = r_iz$ turns its complement into the bad-return set inside the large centered ball r_iA. Uniform genericity for continuous upper approximations of $1_{X\setminus K_i}$ gives

$$\frac{m(A \setminus E_i)}{m(A)} = o(1)$$

after choosing the cores with negligible boundary; the usual inner/outer approximation removes that boundary assumption.

For E_i', use Lemma 6.1. At a finite place, $\eta_i(A)$ is a centered ball and the change of variables has constant Jacobian absolute value on each tame subball, so the same estimate applies. Over $\mathbb{R}$, $\eta_i(A)$ is an interval containing 0 whose length is comparable with the centered interval of the same outer radius. The bad-return measure in $r_i\eta_i(A)$ is therefore bounded by a fixed multiple of the bad-return measure in that centered outer interval. Tameness bounds the Jacobian above and below, so again

$$\frac{m(A \setminus E_i')}{m(A)} = o(1).$$

Thus $m(E_i \cap E_i')/m(A) \to 1$.

A nonconstant rational map ϕ has $\phi^{-1}(U)$ contained in the zero set of the nonconstant Chevalley polynomial $\Psi - q$, hence this exceptional set is finite in the one-dimensional parameter line. Remove small neighborhoods of any prescribed finite set; their total relative measure can be made arbitrarily small. The high-density intersection still meets the complement, so choose z_i there. Compactness of A gives a convergent subsequence $z_i \to z_0$, and the neighborhoods may be chosen so that z_0 avoids the prescribed set. $\square$

Lemma 6.3 (A normalizer limit relating two generic cores is an invariance)**.** *Let $h_i \to h \in N_G(U)$ and let $y_i = h_i y_i'$ with $y_i, y_i' \in K_i$, where K_i are nested uniform generic cores for a countable dense family of compactly supported tests and their h-translates. Suppose the genericity thresholds tend to infinity sufficiently slowly compared with the convergence $h_i \to h$. Then $h \in I(\mu)$.*

Proof. Conjugation by h induces an algebraic automorphism of $U \simeq (k, +)$, hence there is $c \in k^\times$ such that

$$h^{-1}u(t)h = u(ct).$$

Because $h_i \to h$ and conjugation by $u(t)$ is polynomial of uniformly bounded degree in t, choose radii $R_i \to \infty$ such that

(1) R_i is beyond the genericity threshold of K_i;
(2) uniformly for $|t| \le R_i$,

$$u(t)h_i u(-ct) \longrightarrow h.$$

Such a diagonal choice is possible: first prescribe any radius beyond the core threshold, then choose the pair $h_i \to h$ far enough along the recursively chosen sequence that all finitely many polynomial coefficients of the conjugation error are smaller than the inverse powers of that radius.

For a test f from the finite stage of the dense family, genericity of y_i gives
$$\frac{1}{m(B_{R_i})}\int_{B_{R_i}} f(u(t)y_i)\,dt \longrightarrow \int f\,d\mu.$$
Using $y_i = h_i y_i'$ and the uniform polynomial estimate,
$$f(u(t)y_i) = f\big(u(t)h_i y_i'\big) = f\big((u(t)h_i u(-ct))u(ct)y_i'\big)$$
is uniformly close, after the standard compact-core truncation, to $f(hu(ct)y_i')$. Scalar multiplication by c sends a large local ball to a large local ball of comparable radius. Genericity of y_i' for the test $f \circ h$ therefore gives
$$\frac{1}{m(B_{R_i})}\int_{B_{R_i}} f(hu(ct)y_i')\,dt \longrightarrow \int f(hz)\,d\mu(z).$$
The two limits are equal. Hence
$$\int f\,d\mu = \int f(hz)\,d\mu(z)$$
for the dense test family, and therefore for every $f \in C_c(X)$. Thus $h_*\mu = \mu$. □

Theorem 6.4 (Basic quasi-regular invariance lemma). *Let μ be U-ergodic. Let $x_i \to x$ with x_i and $g_i x_i$ chosen recursively from nested uniform generic cores, where $g_i \to e$ and $g_i \notin N_G(U)$. Suppose the associated quasi-regular normalization is tame on a sufficiently small centered ball $A \ni 0$ and its shearing scales dominate the corresponding core thresholds. Then there is*
$$h \in \phi(A) \cap I(\mu)$$
with $h \notin U$. In particular $I(\mu)$ strictly contains the initially known one-parameter group U.

Proof. Apply Lemma 6.2 and choose $z_i \to z_0$ so that $\phi(z_0) \notin U$. The factorization gives the exact relation
$$y_i = u(r_i z_i)g_i x_i = \phi_i(z_i)u(r_i\eta_i(z_i))x_i = h_i y_i',$$
where both y_i, y_i' lie in K_i and
$$h_i = \phi_i(z_i) \longrightarrow \phi(z_0) =: h.$$
Proposition 4.1 gives $h \in N_G(U)$. The recursive choice of the original nearby points may be thinned once more so that the convergence $h_i \to h$ dominates the genericity thresholds. Lemma 6.3 then yields $h \in I(\mu)$. By the choice of z_0, $h \notin U$. □

Remark 6.5 (Why two generic trajectories are needed)**.** The selected return points y_i and y_i' lie in the same late-stage generic core. Merely knowing that the limiting point x is generic is not enough because the Chevalley scale r_i depends on g_i and can grow arbitrarily fast. The recursive core construction is what lets us choose a later nearby pair after the required averaging threshold is known.

7. Product local factors

For a finite product

$$U = \prod_{v \in T} U_v, \qquad U_v \simeq (k_v, +),$$

one applies the construction to every component on which $g_{i,v}$ is outside $N_{G_v}(U_v)$ and uses the identity normalization on the remaining components. The Chevalley polynomial becomes a tuple of local polynomials. The common-radius selection of Theorem 5.1 makes at least one component visible while keeping all components bounded. The product L^2 pointwise theorem of Chapter 4 then supplies uniform genericity for every sufficiently large product box.

Proposition 7.1 (Product Basic Lemma under tame componentwise normalization)**.** *Suppose every active local component of a product normalization is tame on a product ball $A = \prod A_v$, and at least one limiting Chevalley polynomial is nonconstant. Under the same recursive generic-core hypotheses as Theorem 6.4, the product quasi-regular image preserves μ.*

Proof. Choose a sufficiently small centered product ball around the identity in the normalized parameter space. Componentwise tameness and the product generic-core corollary imply that, for a set of normalized parameters of relative measure $1 - o(1)$, both the first trajectory and the reparametrized second trajectory return to the same late-stage core. Choose a convergent sequence of such parameters outside the algebraic set on which all visible Chevalley components are trivial. The corresponding exact relative displacements converge to an element h of the product normalizer $N_G(U)$ and $h \notin U$.

The proof of Lemma 6.3 is componentwise: conjugation by h acts by a nonzero scalar on each local additive parameter, and the product maximal theorem permits independently large side lengths. A diagonal choice of mesoscopic product boxes makes the polynomial conjugation error for $h_i \to h$ uniformly small while keeping every side beyond the genericity threshold. The two product averages then give $h_*\mu = \mu$. Thus the product quasi-regular image supplies a new invariance. $\square$

8. From the absolute construction to the relative algorithm

The construction above removes the principal analytic difficulty in a degenerating sequence. A sequence whose first quotient Taylor term degenerates is not sent directly to rational descent. One first magnifies it in a Chevalley representation, uses rational U-compensation, and obtains a quasi-regular normalizer limit. The Basic Lemma then decides whether that visible limit is a genuine measure symmetry.

When the full local stabilizer core $L = \langle U, I_{\mathrm{an}} \rangle$ is larger than U, one must still remove visible normalizer directions already contained in L. This is done in Chapter 12. The correction is taken in $N_L(U)$, not in an arbitrary L-valued family. Conjugation by such a correction acts on the additive parameter by a scalar, so genericity is transferred by a controlled change of time. The active polynomial degree drops. Finite iteration therefore ends either in a transverse stabilizing tangent, contradicting the definition of L, or in primitive singular equations. Arithmetic recurrence then converts the latter into an exact rational exterior stratum.

Thus this chapter supplies the absolute quasi-regular building block, while Chapter 12 supplies the relative correction and closes the classification bridge.

Proof and provenance. The language of quasi-regular maps and the normalization through a Chevalley representation originate in the local-field measure-rigidity proof of Margulis–Tomanov. The presentation here reconstructs the one-parameter coefficient normalization, rational-slice lift, normalizer calculation, and Basic Lemma rather than using their classification theorem as an input. Later variants in the literature use the same mechanism for joinings and other local-field rigidity problems.

CHAPTER 9

Exterior linearization over products of local fields

We now prove the representative facts needed to keep shearing away from already-understood homogeneous strata.

1. Integral adjoint lattices

For $G = \mathrm{SL}_n$ let

$$\mathfrak{g}_{\mathbb{Z}_S} = \mathfrak{sl}_n(\mathbb{Z}_S).$$

It is preserved by $\mathrm{Ad}(\Gamma)$. If H is a rational algebraic subgroup with Lie algebra $\mathfrak{h}$, the module $\mathfrak{h} \cap \mathfrak{g}_{\mathbb{Z}_S}$ is an S-lattice in $\mathfrak{h}(\mathbb{Q}_S)$. Choose a primitive top wedge p_H.

Lemma 1.1 (Incidence equation). *For $g \in G_S$,*

$$U \subset gH_Sg^{-1}$$

iff for every $Y \in \mathrm{Lie}(U)$,

$$Y \wedge \bigwedge^{\dim H} \mathrm{Ad}(g)p_H = 0$$

at every place.

Proof. The top wedge spans the line $\bigwedge^{\dim H} \mathrm{Ad}(g)\mathfrak{h}$. Wedge with Y vanishes exactly when Y lies in $\mathrm{Ad}(g)\mathfrak{h}$. This is equivalent to Lie-algebra containment, and for connected one-parameter unipotent groups in characteristic zero Lie-algebra containment is equivalent to group containment. ☐

2. Finite representatives in a compact exterior window

Lemma 2.1 (Arithmetic discreteness of representatives). *Fix a rational subgroup H. The set*

$$\{\mathrm{Ad}(\gamma)p_H : \gamma \in \Gamma\}$$

is discrete modulo multiplication by S-units in the product exterior space.

Proof. Every representative lies in the finitely generated $\mathbb{Z}_S$-module $\bigwedge^{\dim H} \mathfrak{g}_{\mathbb{Z}_S}$. Choose the balanced representative in each S-unit class using Volume 3. Balanced representatives lying in a compact product set form a finite set because the diagonal $\mathbb{Z}_S$-module is discrete. Thus the projectivized set modulo units is discrete. □

Proposition 2.2 (Finite-window representative lemma)**.** *Let $C \subset G_S$ be compact and fix $R > 1$. Only finitely many projective representatives* $\mathrm{Ad}(\gamma)p_H$ *can satisfy*

$$c(\mathrm{Ad}(g\gamma)p_H) \le R$$

for some $g \in C$.

Proof. The operator norms of $\bigwedge^d \mathrm{Ad}(g)$ and their inverses are uniformly bounded on C. Therefore the displayed inequality bounds the content of $\mathrm{Ad}(\gamma)p_H$. Balance it by an S-unit. All local norms are then bounded, and Lemma 2.1 gives finiteness. □

3. Lower incidences and character descent

Suppose two distinct representatives determine rational conjugates H_1, H_2 and both exact incidence systems contain U. Then

$$U \subset F := H_1 \cap H_2.$$

The intersection is a proper rational algebraic subgroup unless $H_1 = H_2$. If F has no nontrivial rational character, Corollary 7.2 makes its arithmetic quotient finite volume and it is already a lower member of $\mathscr{H}_\Gamma$. Otherwise replace it by the character-free core F^1. Lemma 2.1 shows that $U \subset F^1$ and that the dimension drops. Repeating this operation terminates at a proper rational subgroup in $\mathscr{H}_\Gamma$ containing U.

Thus ambiguity of exact representatives is always absorbed by a finite-volume rational stratum of smaller dimension. In a fixed compact exterior window only finitely many pairs of representatives occur, so only finitely many lower strata need be removed at any one stage of the linearization induction.

CHAPTER 10

Induction on S-arithmetic singular strata

The purpose of this chapter is to isolate the exact quantitative statement that is used later in the additional-invariance and orbit-closure arguments. A phrase such as "polynomial goodness makes the singular tube small" is not, by itself, a proof: one must specify which scalar polynomial becomes large, which threshold is small relative to its supremum, and how changes of arithmetic representative are handled. We make those points explicit here.

1. Finite representative windows and pure tubes

Fix a connected rational subgroup $\mathbf{H} \in \mathscr{H}_\Gamma$, put $d = \dim H$, and retain the exterior vector $p_H \in \bigwedge^d \mathfrak{g}(\mathbb{Q})$ from Chapter 9. For $g \in G$ write

$$v_H(g,\gamma) = \rho_H(g\gamma)p_H, \qquad E_H(g,\gamma) = \max_{Y \in \mathcal{Y}} \|Y \wedge v_H(g,\gamma)\|_S,$$

where $\mathcal{Y}$ is a fixed finite spanning set of the Lie algebra of U and $\rho_H = \bigwedge^d \mathrm{Ad}$. Thus $E_H(g,\gamma) = 0$ is precisely the exterior incidence relation saying that U is contained in the conjugate represented by $g\gamma H\gamma^{-1}g^{-1}$.

Let $K \subset X$ be compact. Proposition 2.2 implies the following useful reformulation. Given an upper bound $M < \infty$, there is a finite set $\mathcal{W} = \mathcal{W}(K,H,M)$ of primitive exterior representatives, modulo S-unit scaling and sign, such that whenever $x = g\Gamma \in K$ and

$$\|v_H(g,\gamma)\|_S \le M,$$

the primitive class of γp_H belongs to $\mathcal{W}$.

Choose numbers

$$0 < M_0 < M_1, \qquad 0 < \eta_0 < \eta_1,$$

and a compact subset $D \subset N(H,U)$ modulo the stabilizer of p_H. An *inner H-tube* in K is defined by

$$\|v_H(g,\gamma)\|_S \le M_0, \qquad E_H(g,\gamma) \le \eta_0,$$

for one of the finitely many representatives in $\mathcal{W}$. The corresponding *outer window* is obtained by replacing (M_0,η_0) with (M_1,η_1).

If two distinct representatives satisfy the exact incidence equations at the same point, their algebraic intersection has smaller dimension. There are only finitely many representatives in the fixed window, hence only finitely many such intersections. Remove slightly enlarged neighborhoods of all proper rational intersections that arise in this way. What remains of the inner tube is the *pure H-tube.*

Lemma 1.1 (Uniqueness in a pure tube)**.** *For η_0 sufficiently small, every point of K in the pure H-tube has a unique H-representative, modulo S-unit scaling and sign.*

PROOF. Suppose not. There are $\eta_i \downarrow 0$, points $x_i \in K$, and two distinct representatives w_i, w_i' from the finite set $\mathcal{W}$ satisfying the two η_i-incidence systems, while x_i avoids the prescribed lower-stratum neighborhoods. Pass to a subsequence on which $w_i = w$ and $w_i' = w'$ are fixed and $x_i \to x \in K$. Continuity gives both exact incidence systems at x. By the exterior-intersection lemma of Chapter 9, the two representatives either define the same rational conjugate or their intersection has character-free core contained in a proper member of $\mathscr{H}_\Gamma$. The first case contradicts distinctness modulo the allowed scaling; the second places x in one of the lower-stratum sets whose neighborhood was removed. Since that removed set is closed inside the finite representative window, x_i belongs to its enlarged neighborhood for all large i, again a contradiction. □

2. The scalar polynomial input

Let T be the additive parameter group of the unipotent trajectory under consideration. On every coordinate chart used below, the maps

$$t \longmapsto v_H(u(t)g, \gamma), \qquad t \longmapsto Y \wedge v_H(u(t)g, \gamma)$$

have scalar coordinates that are local polynomials of degree at most a fixed integer $D = D(G, U, H)$. Volume 3 therefore supplies constants $C, \alpha > 0$ such that every nonzero scalar coordinate P satisfies

$$m_T\{t \in B : |P(t)|_S < \varepsilon \sup_B |P|_S\} \le C\varepsilon^\alpha m_T(B) \tag{2.1}$$

for every admissible parameter ball or product box B. Zero coordinates can simply be discarded.

The following elementary lemma is the bookkeeping step that was implicit in the previous draft.

Lemma 2.1 (Maximal polynomial-window estimate)**.** *Fix D and the number N of scalar coordinates occurring in v_H and its incidence errors. For every $\epsilon > 0$ one can choose the ratios*

$$\frac{M_0}{M_1} \quad \textit{and} \quad \frac{\eta_0}{\eta_1}$$

small enough, depending only on C, α, N and ϵ, with the following property.

Let B be a parameter ball (or one member of the fixed product-box basis), and let w be one exterior representative. Suppose B is maximal, inside a prescribed ambient parameter region, among balls on which

$$\|v_w(t)\|_S < M_1, \qquad E_w(t) < \eta_1.$$

If the same representative does not satisfy the outer inequalities on the next allowed enlargement of B, then

$$\frac{m_T\{t \in B : \|v_w(t)\|_S \leq M_0,\ E_w(t) \leq \eta_0\}}{m_T(B)} < \epsilon.$$

The assertion remains true when finitely many local norms are used in place of one product norm.

Proof. By maximality, on the next controlled enlargement B^+ at least one scalar coordinate crosses an outer threshold. There are two cases.

If a coordinate P of v_w crosses the size threshold, then

$$\sup_{B^+} |P| \geq c_N M_1$$

for a norm-equivalence constant $c_N > 0$, while every inner hit satisfies $|P| \leq C_N M_0$. Applying (2.1) on B^+ gives

$$m_T(\text{inner hits} \cap B) \leq C \left(\frac{C_N M_0}{c_N M_1}\right)^{\alpha} m_T(B^+).$$

The allowed enlargement factor has uniformly bounded Haar-volume ratio, so the right-hand side is at most $\epsilon m_T(B)$ once M_0/M_1 is sufficiently small.

If instead an incidence coordinate Q crosses the outer incidence threshold, then $\sup_{B^+} |Q| \geq c_N \eta_1$, whereas an inner hit has $|Q| \leq C_N \eta_0$. The same goodness estimate gives the required bound after choosing η_0/η_1 sufficiently small.

No lower bound on $\|v_w\|$ is used as a stopping condition. This is important: a polynomial may pass through a small value without making the set on which it has moderate size small. Lower bounds needed for arithmetic representatives come instead from the compact set K and the finite-window lemma. This removes the spurious "small-norm exit" case that would invalidate the argument.

For a product of local norms, apply the same argument to the particular local coordinate that crosses its threshold. The goodness constants are uniform over the finite list of coordinates and places. $\square$

3. Covering by maximal windows

We record the covering statement in the form needed later. In a nonarchimedean parameter factor, two balls are either disjoint or nested, hence the maximal balls are automatically disjoint. In the real factors we use the bounded-overlap Besicovitch selection proved in Volume 1. For product parameters we apply these selections successively; the multiplicity is bounded by a constant depending only on the number of real coordinates.

Proposition 3.1 (Pure-stratum occupation alternative)**.** *Fix a compact set $K \subset X$, a rational subgroup $\mathbf{H} \in \mathscr{H}_\Gamma$, and $\epsilon > 0$. Choose a pure inner H-tube and an outer window with threshold ratios as in Lemma 2.1. Let Ω be a sufficiently large parameter ball or one of the prescribed product boxes. Then exactly one of the following mechanisms accounts for every visit of $u(t)x$ to the inner pure tube:*

- *(P)* persistence: *one arithmetic H-representative satisfies the outer-window inequalities throughout the relevant ambient parameter component;*
- *(E)* escape: *the visits belonging to a maximal outer window have relative measure at most ϵ, up to the fixed covering multiplicity;*
- *(L)* lower stratum: *a second representative enters, in which case the trajectory lies in one of the finitely many enlarged lower-dimensional rational strata removed in the definition of purity.*

Consequently, after the lower strata have been made negligible by induction, if no representative persists then the total proportion of parameters in Ω for which $u(t)x$ lies in the inner pure H-tube is at most $C_{\mathrm{cov}}\epsilon + o_\Omega(1)$.

Proof. Assign to every pure inner hit its unique representative using Lemma 1.1. Around that parameter choose the maximal ball on which this representative satisfies the outer inequalities. If the ball reaches the boundary of the relevant ambient component without failure, we are in (P). Otherwise Lemma 2.1 bounds the inner hits in that maximal ball by ϵ times its measure.

Select a disjoint family in the ultrametric coordinates and a bounded-overlap Besicovitch subfamily in the real coordinates. Summing the maximal-window estimate therefore loses only the universal multiplicity C_{cov}. If two assigned windows with genuinely different representatives interact at an inner hit, uniqueness can fail only through one of the lower rational intersections listed in the construction of the pure tube; these are precisely the visits in (L). Removing them leaves a single representative on every selected component. Boundary pieces created by intersecting maximal windows with Ω occupy an $o_\Omega(1)$ proportion along the prescribed Følner family. This proves the stated estimate. $\square$

4. Persistence implies exact incidence

The persistence alternative is useful only because a bounded-degree polynomial that remains in successively thinner incidence tubes on larger and larger domains must satisfy the exact incidence equations in the limit.

Lemma 4.1 (Persistence-to-incidence). *Let $g_i \to g$ in a compact set and let w be a fixed representative. Suppose there are parameter balls $B_i \nearrow T$ and numbers $\eta_i \downarrow 0$ such that*

$$\sup_{t \in B_i} E_w(u(t)g_i) \le \eta_i.$$

Then

$$E_w(u(t)g) = 0 \qquad \text{for every } t \in T.$$

In particular $g\Gamma$ belongs to the exact singular incidence set determined by w.

Proof. Fix $t \in T$. For large i, $t \in B_i$. Continuity in g gives $E_w(u(t)g) = \lim_i E_w(u(t)g_i) = 0$. Since t was arbitrary, every incidence coordinate vanishes identically. Equivalently the Lie algebra of U wedges to zero with $\rho_H(g)w$, which is the exterior criterion for containment in the corresponding conjugate of H. $\square$

5. The induction variable

The induction is on $\dim H$. Every genuine competition of representatives is replaced by the character-free core of a proper rational intersection, whose dimension is strictly smaller. Since dimensions are nonnegative integers, the process terminates. The quantitative content is now separated cleanly from the algebraic descent: Lemma 2.1 controls one representative, Lemma 1.1 identifies representative changes with lower strata, and Proposition 3.1 combines the two.

CHAPTER 11

Linearization laboratory: one representative, two representatives, and descent

The pure-stratum occupation proposition is the place where many first readings of Ratner's method become opaque. The difficulty is not the polynomial sublevel estimate by itself. It is bookkeeping: the same point of G/Γ may be represented by many elements of G, and the dangerous rational subgroup may change when the orbit crosses a fundamental-domain boundary. This chapter follows one compact window from beginning to end.

1. Fixing the exterior window

Let $H \in \mathscr{H}_\Gamma$, let p_H be its primitive adjoint wedge, and let $C \subset G$ be compact. Choose numbers

$$0 < \eta \ll \eta_1 \ll R$$

with the following interpretation. The *inner tube* requires all incidence errors

$$\|Y \wedge \mathrm{Ad}(g\gamma)p_H\|_v$$

to be $< \eta$ and the representative norm to be $< R/2$. The *outer window* replaces η by η_1 and $R/2$ by R. Proposition 2.2 gives a finite set of projective representatives that can ever enter the outer window while $g \in C$.

Because the set is finite, we can choose the thresholds so that distinct representatives cannot both satisfy the inner incidence inequalities unless the base point lies in a prescribed neighborhood of their exact intersection stratum. Every exact intersection descends, through the character-free core, to a smaller member of $\mathscr{H}_\Gamma$. Removing neighborhoods of those smaller strata produces the pure H-tube.

2. The assigned representative is locally constant

Inside the pure tube, Lemma 1.1 gives one projective representative $[w_x]$. Since the finite representatives are separated in the compact projective window, $[w_x]$ is locally constant until the trajectory leaves the outer window. Thus an orbit segment in a pure tube comes with an honest finite label, not an infinitely moving arithmetic choice.

Fix one label w. Along a polynomial unipotent parametrization $u(t)g$, the local exterior vector

$$F_v(t) = \bigwedge^d \mathrm{Ad}(u_v(t)g_v)w_v$$

has bounded-degree polynomial coordinates. There are two ways to leave the outer window:

(1) an incidence coordinate $Y \wedge F_v(t)$ grows from η to η_1;
(2) the representative itself grows from $R/2$ to R.

The second possibility is why the outer window must include a norm bound; an incidence tube alone would not detect tangential escape along the variety.

3. The maximal parameter ball

For each inner visit t_0, let $B(t_0)$ be the largest centered parameter box on which the assigned representative remains in the outer window. If the box does not reach the boundary of the large ambient averaging box, maximality means that one of the finitely many polynomial functions just listed has outer-scale supremum. If P is that polynomial and its inner visit satisfies $|P(t_0)| < \eta$, the good-function estimate gives

$$\frac{m\{t \in B(t_0) : |P(t)| < \eta\}}{m(B(t_0))} \ll \left(\frac{\eta}{\eta_1}\right)^{\alpha}$$

after harmless modification when the norm coordinate is the exiting function.

At nonarchimedean places maximal balls are nested or disjoint. At the real factor we apply the Besicovitch selection proved in Volume 1. Therefore one can select maximal boxes covering all inner visits with uniformly bounded overlap. Summing the preceding estimate over the selected family yields

$$m(\text{regular inner visits}) \ll \left(\frac{\eta}{\eta_1}\right)^{\alpha} m(B_{\text{ambient}}).$$

4. What happens when a new label appears?

Suppose a second representative w' enters the inner window before w has left the outer window. Then both are in the fixed finite projective set. If the point is outside the removed intersection neighborhoods, compact separation says this is impossible. Hence label changes are not an uncontrolled error: they are visits to already identified lower-dimensional strata.

The induction variable is therefore the dimension of the rational subgroup, not the number of representatives. A competing label forces an intersection; the character-free core of that intersection still contains U and has smaller dimension. After at most $\dim G$ descents the process terminates.

5. The persistence alternative

There remains one exceptional case: the assigned representative never exits the outer window on the entire large parameter ball. For a sequence of larger and larger balls, coefficient compactness then produces a polynomial exterior map that is globally bounded. Every positive-degree coefficient vanishes, so the exact incidence equations hold identically. Thus the entire unipotent orbit lies in the corresponding rational homogeneous incidence set.

This is why the occupation proposition is a dichotomy rather than a pure small-set estimate:

$$\boxed{\text{small regular occupation} \quad \text{or} \quad \text{exact algebraic persistence.}}$$

Worked Example 5.1 (Two rational lines in SL_3)**.** Let H_1, H_2 be stabilizers of two distinct rational planes in $\mathbb{Q}^3$. Their adjoint representatives can both be small only near the stabilizer of the intersection flag. If a one-parameter unipotent subgroup lies in both conjugates, it also lies in that flag stabilizer. The flag subgroup has a rational character coming from the relative scaling of the line and quotient; passing to the character-free core removes that scaling direction and produces the lower finite-volume obstruction used by the induction.

Warning 5.2. One must not call the lower-incidence set a "fourth alternative." It is an inductive remainder. If it were counted as an alternative without a dimension drop, the linearization theorem would be vacuous: every ambiguous point could be discarded forever without ever proving that its total mass is small.

CHAPTER 12

Relative quasi-regular maps and the full additional-invariance bridge

This chapter closes the last logical seam in the arithmetic S-linear rigidity argument. The issue is subtle. If a nearby displacement is normalized only in G/L, where L is an already known invariance subgroup, the exact comparison contains a time-dependent L-factor. One may *not* delete that factor from an orbit average merely because each fixed element of L preserves the measure. The correction can conjugate the acting unipotent group and hence change the time parameter.

The cure is the full H-property correction algorithm. Its corrections are not arbitrary elements of L: they lie in the normalizer of the acting one-parameter group. Normalizer conjugation changes the additive parameter by an algebraic scalar, so it can be absorbed by a controlled change of time. After removing one visible normalizer coefficient the active polynomial degree strictly drops. Since the degree is bounded, the procedure terminates. Either a quasi-regular limit gives a stabilizing direction transverse to the full local measure-stabilizer Lie algebra, which is impossible, or all active coefficients satisfy the primitive singular equations. The latter are exactly the equations to which the linearization and singular-stratum induction of Chapters 9 and 10 apply.

The real version of this finite correction architecture was proved in the separate Ratner notebook. We reproduce the field-dependent steps here: the normalizer time change over a characteristic-zero local field, the nonarchimedean change-of-variables argument, the product synchronization, and the arithmetic conversion of the terminal primitive singular equations. Thus no Margulis–Tomanov classification theorem is used as an input.

Throughout, $X = G/\Gamma$ is the arithmetic quotient in the scope of this volume, U is either one local algebraic one-parameter unipotent subgroup or a finite product of commuting local one-parameter factors, and μ is a U-ergodic U-invariant probability measure. Put

$$S(\mu) = I(\mu) = \{g \in G : g_*\mu = \mu\}.$$

There is one local-field point which must be handled explicitly. At a finite place the topological identity component of a p-adic analytic group is trivial, so $I(\mu)^\circ$ is not the right object. Let $\mathfrak{i} = \operatorname{Lie} I(\mu)$. Choose a sufficiently small symmetric analytic identity neighborhood $W_I \subset I(\mu)$ on which log and exp are inverse, let $I_{\mathrm{an}} = \langle W_I \rangle$, and define the *full local stabilizer core*

$$L = \langle U, I_{\mathrm{an}} \rangle.$$

Because $I(\mu)$ is closed, it is a characteristic-zero local-analytic subgroup. The subgroup I_{an} is open (hence closed) in $I(\mu)$; therefore L is an open-and-closed subgroup of $I(\mu)$, contains U, and

$$\operatorname{Lie} L = \operatorname{Lie} I(\mu) = \mathfrak{i}.$$

Only these three properties of L are used below. In the purely real case this is the familiar connected-stabilizer bookkeeping; the formulation above is the one that remains correct at nonarchimedean places.

1. Why the full local stabilizer core is the canonical bookkeeping device

The final proof should not iterate through an arbitrary chain of provisional invariance groups. It starts with the full local stabilizer core L.

Lemma 1.1 (Transverse generic pairs). *Assume that no L-orbit has full μ-measure. Then there are a point $x \in \operatorname{supp} \mu$, generic for a fixed countable convergence-determining family, and generic points $x_i \to x$ such that, in one injectivity chart,*

$$x_i = g_i x, \qquad g_i \to e, \qquad g_i \notin L.$$

The sequence may be chosen recursively from the nested uniform generic cores of Chapters 4 and 7, so that every subsequently prescribed polynomial first-exit scale dominates the corresponding genericity threshold.

Proof. First observe that an L-orbit has either zero or full measure. Indeed, L is normalized by itself and contains U; if an L-orbit has positive measure, the U-saturation of that orbit is the orbit itself, and U-ergodicity makes it conull. By hypothesis, therefore, $\mu(Lx) = 0$ for every x.

Choose x in the support, in the conull generic set, and in all density sets used to construct the nested generic cores. Every neighborhood O of x has positive measure. Since the generic set has full measure and Lx has measure zero,

$$O \cap X_{\mathrm{gen}} \setminus Lx \neq \varnothing.$$

Choose x_i from these sets with $x_i \to x$. Shrink once to an identity neighborhood $V \subset G$ on which $g \mapsto gx$ is injective. Then $x_i = g_i x$ uniquely with $g_i \to e$, and $g_i \notin L$ because otherwise $x_i \in Lx$.

For the recursive core property, at stage j choose a compact uniform generic core for the first j tests and error 2^{-j}. Once a later argument has fixed the required first-exit radius R_j, choose the next pair sufficiently close that its polynomial scale exceeds both R_j and the core threshold. Density of the conull generic set in the support at x permits this diagonal choice. This is the same quantifier order used in the Basic Lemma of Chapter 8. ☐

Lemma 1.2 (Full-orbit termination). *If $\mu(Lx) = 1$ for some x, then Lx is closed, its stabilizer in L is a lattice, and μ is the normalized L-invariant measure m_{Lx}.*

PROOF. The measure is L-invariant by the definition of L and is concentrated on the transitive L-space Lx. Pull it back through the orbit map. Uniqueness of an invariant Radon measure on a transitive locally compact homogeneous space shows that it is Haar measure on L/L_x, where L_x is the stabilizer. Finiteness says that L_x has finite covolume. The standard closed-orbit/lattice criterion for a discrete ambient stabilizer then makes the immersed orbit closed. After normalization the pulled-back Haar measure is precisely μ. ☐

2. Normalizer corrections are safe because they are time changes

Let k be one characteristic-zero local field and $U = \{u(t) : t \in k\} \simeq (k, +)$.

Lemma 2.1 (Normalizer character). *For every $h \in N_G(U)$ there is a unique $\chi(h) \in k^\times$ such that*

$$h^{-1}u(t)h = u(\chi(h)t) \qquad (t \in k).$$

The map $\chi : N_G(U) \to k^\times$ is a continuous algebraic character on each local algebraic component.

PROOF. Conjugation by h is an algebraic automorphism of the one-dimensional algebraic group $\mathbf{G}_a$. In characteristic zero every algebraic group endomorphism of $\mathbf{G}_a$ is $t \mapsto at$; invertibility forces $a \neq 0$. Uniqueness is clear, multiplicativity follows from composition, and algebraicity follows by reading the scalar on the one-dimensional Lie algebra of U. ☐

The next lemma is the local-field replacement for the hidden sentence "the correction is harmless."

Lemma 2.2 (Genericity under tame normalizer reparametrization). *Let z_i lie in nested uniform generic cores for a U-ergodic measure μ. Let J_i be centered parameter balls (intervals at the real place) whose radii tend to infinity. Suppose $r_i : J_i \to k$ are injective analytic maps with $r_i(0) = 0$ and the following tame bounds after rescaling J_i to a fixed unit ball:*

(1) at a real place, r_i' is bounded above and below away from zero and $\|r_i''\|_\infty \ll |J_i|^{-1}$ after affine normalization;
(2) at a finite place, $r_i'(0) \neq 0$ and

$$|r_i'(s) - r_i'(0)|_v < |r_i'(0)|_v \qquad (s \in J_i).$$

Then for every $f \in C_c(X)$ belonging to the finite stage defining the core,

$$\frac{1}{m(J_i)} \int_{J_i} f(u(r_i(s))z_i)\, ds - \int f\, d\mu \longrightarrow 0,$$

up to the fixed Jacobian normalization coming from the image ball. The same statement holds componentwise for a product of local factors and rectangular boxes.

Proof. At the real place, change variables $t = r_i(s)$. The inverse Jacobian weight $w_i(t) = |(r_i^{-1})'(t)|$ is uniformly bounded and, after normalization, $\|w_i'\|_\infty \ll |J_i|^{-1}$. If

$$F(T) = \int_0^T \left(f(u(t)z_i) - \int f\, d\mu \right) dt,$$

the uniform generic-core estimate is the uniform version of $F(T) = o(T)$ on the relevant macroscopic range. Integration by parts gives a boundary term $o(|J_i|)$ and an integral of $w_i'F$, also $o(|J_i|)$. Dividing by $|J_i|$ proves the assertion.

At a nonarchimedean place the displayed derivative inequality and the ultrametric inverse-function theorem imply

$$|r_i(s) - r_i(t)|_v = |r_i'(0)|_v\, |s - t|_v$$

throughout J_i. Hence r_i maps J_i bijectively onto the centered ball of radius $|r_i'(0)|_v\, \mathrm{rad}(J_i)$ and Haar measure transforms by the constant factor $|r_i'(0)|_v$. The assertion is therefore exactly the centered uniform generic-core estimate on the image ball.

For a finite product, apply the one-factor changes successively. Real Jacobian weights multiply, while finite-place absolute Jacobians are constant on the tame product ball. The rectangular L^2 pointwise theorem and Corollary 3.1 are uniform once every side length exceeds the prescribed threshold, so the product conclusion follows. □

Remark 2.3. The requirement $r_i(0) = 0$ is not restrictive in the H-property comparison. One first chooses a common return parameter τ_i and then regards $u(\tau_i + s)x$ as the new base trajectory; the mesoscopic variable is s. Thus no pointwise theorem for arbitrarily translated p-adic balls is being smuggled into the proof.

3. Finite correction of the relative polynomial

Fix a local slice to L at the identity. The logarithm of the relative conjugacy curve, projected to $\mathfrak{g}/\mathfrak{l}$, is a polynomial of degree at most a structural integer D. For a product parameter group it is a bounded-degree multivariable polynomial. We call the largest degree whose coefficient survives in the quotient the *active degree.*

At a first-exit scale, Chapter 8 produces a quasi-regular limit in $N_G(U)$; the Basic Lemma puts every regular limit obtained from common generic returns in $I(\mu)$. Taking the macroscopic group radius successively to zero ensures that every such limit lies in the fixed identity neighborhood $W_I \subset I_{\mathrm{an}} \subset L$. Consequently a visible leading term can be corrected by an element of

$$N_L(U) = N_G(U) \cap L.$$

By Lemma 2.1, this correction changes only the parameter and is therefore legitimate by Lemma 2.2.

Lemma 3.1 (Finite normalizer-correction algorithm)**.** *Fix the structural polynomial degree bound D. Let $g_i \to e$ be a sequence coming from recursively chosen common generic pairs and transverse to L. After passage to a subsequence and after at most $D+1$ correction stages, exactly one of the following occurs.*

- *(R) At some arbitrarily small first-exit radius there is a regular quasi-regular limiting curve whose tangent has nonzero image in $\mathfrak{g}/\mathfrak{l}$.*
- *(S) Every visible positive-degree coefficient is successively absorbed by $N_L(U)$, and the original coefficient tuple satisfies a finite proper family of polynomial incidence equations, invariant under U and under the normalizer corrections. These are the primitive singular linearization equations.*

The same alternative holds for a finite product of commuting local one-parameter factors, with total active degree in place of the one-variable degree.

Proof. We give the one-factor argument first. Write the quotient polynomial at the current stage as

$$P_i(t) = \sum_{q=0}^{d} t^q \bar{X}_{i,q}, \qquad d \le D,$$

with d maximal active. Normalize at the first scale at which the degree-d class is macroscopic. Bounded-degree coefficient compactness gives a limiting quasi-regular map. The normalizer calculation of Proposition 4.1 places its image in $N_G(U)$.

If, after taking the first-exit radius through a sequence tending to zero, the limiting curve has a tangent with nonzero class modulo $\mathfrak{l}$, we are in (R). Otherwise the degree-d limiting coefficient belongs to the local tangent image of $N_L(U)/U$. Choose a local analytic section of that quotient and multiply by the inverse normalizer lift. The exact identity

$$h^{-1}u(t)h = u(\chi(h)t)$$

absorbs the accompanying U-factor into a scalar time change. Because the leading quotient coefficient has been subtracted, the corrected relative polynomial has active degree at most $d-1$. Lemma 2.2 shows that the corrected comparison still uses generic blocks. Apply induction on d.

If the process never reaches (R), every leading coefficient belongs to the successive tangent cones of the normalizer incidence varieties. Those incidence conditions are algebraic: in the adjoint representation, or equivalently in the Grassmannian representation of the line/subspace defining U, membership is expressed by finitely many minors. Pulling these minors back through the finitely many coefficient maps gives finitely many polynomial equations in the original coefficient tuple. They are proper. If they vanished identically on the whole transverse coefficient space, every sufficiently small transverse displacement would already be absorbed by L, contradicting $g_i \notin L$. This gives (S). Since the active degree drops at every successful correction, there are at most $D+1$ stages.

For a product $U = \prod U_v$, order the monomials by total degree and then by a fixed lexicographic order. The common-radius normalization of Theorem 5.1 makes the leading active coefficient visible. A product-normalizer correction acts on each local additive coordinate by a nonzero scalar, hence preserves the monomial filtration and removes the chosen leading coefficient without creating a higher one. Induction on the finite ordered set of active monomials terminates. Product genericity is preserved by the last clause of Lemma 2.2. □

4. From a regular limit to a contradiction

The crucial point is that L contains U and has the full stabilizer Lie algebra, $\operatorname{Lie} L = \operatorname{Lie} I(\mu)$; hence the regular branch terminates immediately.

Proposition 4.1 (Regular branch contradicts maximal connected invariance)**.** *Under the hypotheses of Lemma 1.1, alternative* (R) *of Lemma 3.1 is impossible. More precisely, it would produce elements $h_j \in I(\mu)$ with $h_j \to e$ whose normalized logarithms converge to a vector*

$$Y \notin \mathfrak{l} = \operatorname{Lie} L,$$

contradicting $\operatorname{Lie} L = \operatorname{Lie} I(\mu)$.

PROOF. Fix a first-exit radius $\rho > 0$. The corrected comparison blocks furnished by the finite algorithm contain only normalizer time changes, so Lemma 2.2, the one-factor Basic Lemma (Theorem 6.4), and its product form (Proposition 7.1) put a nonconstant local limiting family into $I(\mu)$. Repeat with $\rho_j \downarrow 0$. Choose from each family an element $h_j \neq e$ whose quotient displacement has size comparable to ρ_j. Then $h_j \to e$ and, after passage to a subsequence,

$$\frac{\log h_j}{\|\log h_j\|} \longrightarrow Y$$

with nonzero image in $\mathfrak{g}/\mathfrak{l}$.

Because $I(\mu)$ is a closed local-analytic subgroup, an exponential chart identifies its germ at the identity with the germ of its Lie algebra. Therefore any sequence $h_j \in I(\mu)$ with $h_j \to e$ has every limiting normalized logarithmic direction in $\operatorname{Lie} I(\mu)$. Hence $Y \in \operatorname{Lie} I(\mu) = \mathfrak{l}$, contradicting the transverse conclusion. $\square$

Proposition 4.2 (Product regular branch). *For a finite product of local one-parameter factors, alternative* (R) *likewise produces a nonzero tangent in* $\operatorname{Lie} I(\mu)/\mathfrak{l}$ *and is impossible.*

PROOF. Use the product clause of Lemma 3.1 and the product Basic Lemma. The common-radius construction keeps at least one local component visible. Let the first-exit radius tend to zero. Compactness of the unit sphere in the finite-dimensional product Lie algebra gives a limiting normalized tangent with at least one nonzero transverse component. The closed-subgroup theorem again puts that tangent in $\operatorname{Lie} I(\mu) = \mathfrak{l}$, a contradiction. $\square$

5. The terminal polynomial equations are the singular branch

It remains to identify alternative (S) with the singular sets already built into the arithmetic linearization. The following proposition is the point at which recurrence and the arithmetic lattice enter. It replaces the invalid shortcut “take the rational hull of the analytic normalizer.”

Proposition 5.1 (Primitive degeneracy implies rational singular mass). *Suppose alternative* (S) *of Lemma 3.1 occurs for transverse generic pairs on a positive-measure family of density points. Then there is a proper* $\mathbf{H} \in \mathscr{H}_\Gamma$ *such that*

$$\mu(\mathcal{S}(H, U)) > 0.$$

The conclusion holds for one local parameter and for finite products.

PROOF. Fix a compact generic core K of measure $> 1 - \xi$ on which the quotient map has a uniform injectivity radius and the first finitely many test averages are uniform. The correction algorithm involves only a bounded-degree coefficient space and finitely many projective/Grassmannian incidence equations. Hence, on K, its primitive singular alternatives may be covered by finitely many small linearization windows at any fixed complexity level.

Choose the nearby pairs recursively so that the corresponding shearing blocks return both endpoints to K. At a returning endpoint lift the equality in $X = G/\Gamma$ to G. The two lifts differ by an element of Γ. Applying the adjoint and exterior-power representations used by the primitive incidence equations turns this return element into a primitive $\mathbb{Z}_S$-submodule (or its top wedge). Thus the recurrent singular equation is represented by an *arithmetic* exterior class, not by an arbitrary analytic hull. On the fixed compact window Proposition 2.2 leaves only finitely many such primitive representative classes.

Pigeonhole one representative along a positive relative proportion of returning blocks. Apply Proposition 3.1 with inner and outer tube widths whose goodness error is $< 2^{-j}$. If a lower-intersection alternative has positive frequency, replace the current subgroup by the character-free core of that proper intersection. Dimension drops, so this happens only finitely many times. Otherwise the same primitive representative persists on arbitrarily large comparison windows while the tube width tends to zero.

Pass to recurrent base points in K and then to a convergent subsequence. The representative is constant along a further subsequence by the finite-window lemma. Lemma 4.1 converts persistence in shrinking tubes into the exact exterior incidence equation at the limit point. The set of points satisfying that exact equation is U-invariant. The positive frequency of the selected returning blocks and density of the chosen base points therefore give it positive μ-measure. By construction the representative is primitive and belongs to the arithmetic family $\mathscr{H}_\Gamma$; if an intersection descent occurred, use its character-free core. Properness follows from the fact that the finite correction equations were a proper family on the transverse coefficient space. Hence $\mu(\mathcal{S}(H, U)) > 0$ for a proper terminal H. □

Remark 5.2 (Where arithmeticity enters)**.** The preceding proof does not assert that an arbitrary analytic subgroup L has a rational hull with all desired properties. Rational data appear only after a return in the arithmetic quotient supplies a lattice element. The exterior representative attached to that return is then a primitive S-integral object, and the finite-window/persistence lemmas convert repeated approximate incidence into an exact rational incidence. This is precisely why recurrence must be interleaved with the finite correction algorithm.

6. Full relative H-property realization

We can now package the bridge in the form used by classification.

THEOREM 6.1 (Full relative H-property realization over local fields). *Let μ be U-ergodic and let L be its full local stabilizer core defined above. If no L-orbit has full μ-measure, then at least one of the following holds:*

(1) a corrected quasi-regular comparison produces a stabilizing tangent with nonzero image in $\mathfrak{g}/\mathfrak{l}$;
(2) $\mu(\mathcal{S}(H,U)) > 0$ for a proper $\mathbf{H} \in \mathscr{H}_\Gamma$.

The first alternative is impossible by Propositions 4.1 and 4.2. Consequently, if every proper rational singular stratum has zero μ-mass, some L-orbit has full measure.

PROOF. Assume no L-orbit has full measure and choose recursively synchronized transverse generic pairs by Lemma 1.1. Apply the finite normalizer-correction algorithm. Alternative (R) gives the first conclusion and is contradictory. Alternative (S), if it occurs on every sufficiently small regular density window, gives the second conclusion by Proposition 5.1.

For completeness, the usual diagonal exhaustion removes the phrase “on every sufficiently small regular density window.” Take compact cores of measures $> 1 - 2^{-j}$ and exhaust the finite singular complexity classes. If the regular branch occurs for infinitely many stages, diagonalize those stages and obtain the transverse stabilizing tangent. Otherwise, from some stage onward the singular branch occupies a positive-density part of the core; the proposition produces one proper rational singular stratum of positive measure. These two cases exhaust the transverse pairs. □

Proposition 6.2 (Rigidity alternative)**.** *For the full local stabilizer core L, at least one of the following holds:*

(1) there is x with $\mu(Lx) = 1$, hence $\mu = m_{Lx}$ by Lemma 1.2;
(2) $\mu(\mathcal{S}(H,U)) > 0$ for some proper $\mathbf{H} \in \mathscr{H}_\Gamma$.

PROOF. If the first alternative fails, Theorem 6.1 applies. Its regular conclusion contradicts the definition of L, so its singular conclusion holds. □

7. The repaired bridge

The proof above makes three points explicit that were hidden or incorrect in the previous working draft.

(1) A time-dependent factor in L is never declared harmless. Corrections are chosen in $N_L(U)$, and their effect is transferred to the parameter using the normalizer character.

(2) The final induction uses the full local stabilizer core L, with $U \subset L$ and $\operatorname{Lie} L = \operatorname{Lie} I(\mu)$. It therefore needs no claim that a local partition into L-orbits has one atom merely by ergodicity. A regular transverse tangent is simply impossible.

(3) Rationality is introduced after arithmetic recurrence produces primitive S-integral exterior data. We do not rationalize an arbitrary analytic normalizer by assertion.

These are exactly the corrections needed to connect the quasi-regular mechanism of Chapter 8 to the linearization/descent mechanism of Chapters 9–10.

CHAPTER 13

Laboratory: the relative correction calculation

The preceding chapter closes a point that is easy to conceal with notation. A quotient-level shear in G/L does not by itself give a valid comparison in G: the lift contains a time-dependent L-factor. This laboratory shows, first in a linear model and then in the exact group calculation, why normalizer corrections solve the problem and why arbitrary L-corrections do not.

1. The bounded-degree linear model

Let V be a finite-dimensional vector space over a characteristic-zero local field k and

$$u(t) = \exp(tN), \qquad N^q = 0.$$

For $v_i \to 0$,

$$u(t)v_i = \sum_{j=0}^{q-1} \frac{t^j}{j!} N^j v_i.$$

If r is the largest visible degree, choose T_i so that

$$|T_i|^r \|N^r v_i\| \asymp 1.$$

After passage to a subsequence,

$$u(T_i s)v_i \longrightarrow P(s)$$

on compact normalized parameter sets, where P is a nonconstant polynomial. The fundamental finite-dimensional observation is

$$\deg\bigl(P(s+a) - P(s)\bigr) < \deg P$$

whenever the leading term is removed. The full correction algorithm is a group- theoretic version of this degree drop.

2. Why an arbitrary L-factor cannot be deleted

Suppose a local slice gives

$$u(T_i s)g_i u(-T_i s) = \ell_i(s)h_i(s), \qquad \ell_i(s) \in L,$$

with $h_i(s) \to h(s)$ modulo L. It is tempting to write "$\ell_i(s)$ preserves μ, so discard it." This is wrong. The element $\ell_i(s)$ depends on s, and

$$\ell_i(s)^{-1}u(t)\ell_i(s)$$

need not equal $u(t)$ or even lie in the same one-parameter subgroup. Thus the second orbit block may be sampled with a different, uncontrolled family of transformations. Fixed-element invariance of μ gives no license to remove a parameter-dependent factor before averaging.

The repair is to choose the correction in

$$N_L(U) = L \cap N_G(U).$$

If $n \in N_L(U)$, then

$$n^{-1}u(t)n = u(\chi(n)t)$$

for one scalar $\chi(n) \in k^\times$. The correction therefore changes only the time coordinate. That is an exact identity, not a first-order approximation.

3. Real and finite-place time changes

At the real place, after centering the comparison block at a common return time, the corrected parameter has the form

$$s \longmapsto r_i(s), \qquad r_i(0) = 0,$$

with derivative bounded above and below and with variation on a mesoscopic block of order $O(|J_i|^{-1})$. Writing $t = r_i(s)$ introduces the weight

$$w_i(t) = |(r_i^{-1})'(t)|.$$

For a generic point z and $F(T) = \int_0^T (f(u(t)z) - \mu(f))dt = o(T)$, integration by parts gives

$$\int w_i\, dF = w_i F\big|_{\partial J_i} - \int w_i' F = o(|J_i|).$$

This is exactly the weighted Cesàro mechanism used in the completed real Ratner proof.

At a finite place the calculation is sharper. On a sufficiently small tame ball,

$$|r_i'(s) - r_i'(0)|_v < |r_i'(0)|_v.$$

Ultrametricity gives

$$|r_i(s) - r_i(t)|_v = |r_i'(0)|_v |s - t|_v.$$

Hence a centered ball is carried bijectively to a centered ball and Haar measure changes by the constant factor $|r_i'(0)|_v$. No weighted differentiation is needed. For a product of places these changes are made componentwise and the rectangular generic-core theorem applies once every side length is large.

4. One correction step

Let

$$P_i(t) = \sum_{j=0}^{d} t^j \bar{X}_{i,j}$$

be the quotient polynomial in $\mathfrak{g}/\mathfrak{l}$, with active degree d. Normalize at the first scale at which the dth coefficient is visible. The quasi-regular construction gives a limit in $N_G(U)$, and the Basic Lemma puts a regular limit into $I(\mu)$. Taking the first-exit group radius to zero forces that small limit into $L = \langle U, I_{\mathrm{an}} \rangle$.

If the degree-d coefficient is represented by a local element $n_i \in N_L(U)$, replace the relative displacement by the one corrected by n_i^{-1}. The normalizer identity converts the accompanying U-factor to the scalar time change $t \mapsto \chi(n_i)t$. In coefficient space the degree-d class is subtracted, so the corrected polynomial has active degree at most $d-1$. Nothing else of degree $> d$ can be created by a scalar change of time. After at most $D+1$ steps one therefore either sees a transverse regular limit or all positive-degree coefficients satisfy the normalizer incidence equations.

5. Why the full local stabilizer core ends the regular branch

The final proof sets $L = \langle U, I_{\mathrm{an}} \rangle$ before choosing the nearby pair. Here I_{an} is the open local analytic subgroup generated by a sufficiently small identity neighborhood in $I(\mu)$, as in Chapter 12; hence $U \subset L \subset I(\mu)$ and $\operatorname{Lie} L = \operatorname{Lie} I(\mu)$. Suppose the correction algorithm gives stabilizing elements $h_j \to e$ at first-exit radii $\rho_j \downarrow 0$, with normalized logarithms tending to

$$Y \notin \mathfrak{l}.$$

Since $I(\mu)$ is a closed local Lie subgroup, its tangent cone at the identity is $\operatorname{Lie} I(\mu) = \mathfrak{l}$. Thus Y must lie in $\mathfrak{l}$, a contradiction. This is stronger and cleaner than repeatedly enlarging a provisional known subgroup. The regular branch does not continue; it is impossible.

6. What the completely absorbed branch means

If every active coefficient is absorbed, the finite-dimensional coefficient tuple satisfies a proper family of polynomial equations. These equations are not silently called rational. Recurrence is now essential.

Choose common returns of both trajectories to one compact injectivity/genericity core. Lifting the return relation from $X = G/\Gamma$ to G introduces an element of the arithmetic lattice Γ. In the adjoint/exterior representations, that lattice element supplies a primitive S-integral exterior class. On a fixed compact window only finitely many such classes occur. Polynomial goodness then has the familiar alternatives:

(1) the orbit spends only a small proportion in the inner singular tube;
(2) one arithmetic representative persists;
(3) two representatives overlap, giving a lower-dimensional intersection.

The third case can occur only finitely often by dimension descent. Persistence while the tube radius tends to zero gives an exact rational incidence equation. Thus the completely absorbed branch is precisely the rational singular branch, but rationality appears *after* arithmetic recurrence, not from an arbitrary analytic hull.

7. The complete contradiction diagram

For the nonsingular classification branch the logic is now

$$\begin{array}{c}
\text{no conull } L\text{-orbit} \quad (L = \langle U, I_{\mathrm{an}} \rangle) \\
\Downarrow \\
\text{transverse recursively generic pairs} \\
\Downarrow \\
\text{finite normalizer correction} \\
\swarrow \qquad\qquad \searrow \\
\begin{array}{cc}
\text{regular tangent} & \text{primitive degeneracy} \\
\Downarrow & \Downarrow \\
Y \in \operatorname{Lie} I(\mu) \setminus \mathfrak{l} & \mu(\mathcal{S}(H,U)) > 0 \\
\Downarrow & \Downarrow \\
\text{contradiction} & \text{proper homogeneous descent.}
\end{array}
\end{array}$$

If the proof is in the branch where every proper rational singular stratum has zero mass, the right-hand outcome is excluded as well. Therefore the initial assumption was false and a conull L-orbit exists.

8. A concrete SL_2 calculation

Take

$$u(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad g_i = \begin{pmatrix} 1 & 0 \\ \varepsilon_i & 1 \end{pmatrix}, \qquad \varepsilon_i \to 0.$$

Then

$$u(t) g_i u(-t) = \begin{pmatrix} 1 + t\varepsilon_i & -t^2\varepsilon_i \\ \varepsilon_i & 1 - t\varepsilon_i \end{pmatrix}.$$

At scale $|t| \asymp |\varepsilon_i|^{-1/2}$, the quadratic upper-right entry is order one while the linear diagonal entries tend to zero. The first visible limit is an upper-unipotent element, which normalizes U. If that direction is already in the local stabilizer core, it can be removed by a normalizer correction; if it is not, the Basic Lemma produces the forbidden transverse stabilizing direction. This 2×2 example is the finite correction algorithm in its smallest form.

CHAPTER 14

Measure classification in the arithmetic S-linear case

We now close the qualitative rigidity induction. The bookkeeping is the canonical one from the completed Ratner proof: for an ergodic invariant probability measure we use the full local stabilizer core L of Chapter 12 from the outset. Thus $U \subset L \subset I(\mu)$ and $\operatorname{Lie} L = \operatorname{Lie} I(\mu)$. There is therefore no infinite chain of provisional invariance groups to manage. If no proper rational singular stratum carries mass, the full relative H-property of Chapter 12 says that failure of a conull L-orbit would produce a stabilizing tangent transverse to $\operatorname{Lie} L$, contradicting the definition of L. If a proper singular stratum does carry mass, the argument descends to a strictly smaller homogeneous group.

All place-sensitive pieces used in this dichotomy have already been proved: rectangular genericity, normalizer reparametrization, finite quasi-regular correction, arithmetic exterior linearization, singular occupation, and the finite-place recurrence needed to recover rational homogeneous data. Thus the theorem below is unconditional in the arithmetic matrix scope of this volume.

1. Positive singular mass reduces the ambient dimension

Lemma 1.1 (A pure singular component lies on one smaller homogeneous orbit)**.** *Let μ be U-ergodic. Suppose $\mathbf{H} \in \mathscr{H}_\Gamma$ has minimal dimension among proper members satisfying $\mu(\mathcal{S}(H,U)) > 0$. Then μ is supported on one closed finite-volume orbit Ly, where L is a conjugate of H. In particular the classification problem reduces to the same problem on the lower-dimensional homogeneous space Ly.*

Proof. Ergodicity makes $\mathcal{S}(H,U)$ conull. Remove every lower incidence arising from competing representatives. By Chapter 9, a lower incidence either already belongs to a smaller member of $\mathscr{H}_\Gamma$ or its character-free core does; minimality makes all those sets null. Thus almost every point lies in the pure part and has a unique rational conjugate $L_x = g\gamma H\gamma^{-1}g^{-1}$. The map $x \mapsto L_x$ is measurable and U-invariant, because $U \subset L_x$. Ergodicity makes $L_x = L$ essentially constant.

The points with the same conjugate L may still lie on different L-orbits. Take a countable cover by small quotient charts. On each chart the local L-orbit coordinate is a measurable map into a transversal. Since $U \subset L$, this transverse coordinate is U-invariant. Ergodicity again makes it essentially constant. Hence a conull subset lies on a single orbit Ly. By construction this orbit is a conjugate of the finite-volume H-orbit associated with the relevant representative, so it is closed and has finite L-volume. $\square$

2. Arithmeticity of a homogeneous terminal orbit

The nonsingular branch may first produce a closed subgroup L analytically, without an a priori rational equation. The next standard density argument returns us to the rational family.

Lemma 2.1 (Arithmeticity of an ergodic unipotent homogeneous orbit)**.** *Let $L < G$ be a closed analytic subgroup, let Ly be closed with finite L-volume, and suppose a nontrivial product unipotent subgroup $U \subset L$ acts ergodically on Ly. Then there is a connected $\mathbb{Q}$-algebraic subgroup $\mathbf{H}$ such that $U \subset H_S$, $H_S y$ is closed of finite volume, and*

$$Ly \subset H_S y.$$

If $Ly \neq X$, $\mathbf{H}$ may be chosen proper. Consequently $y \in \mathcal{S}(H, U)$ for a proper $\mathbf{H} \in \mathscr{H}_\Gamma$.

Proof. Conjugate by a representative of y, so the orbit is L/Λ with $\Lambda = L \cap \Gamma$ a lattice. The stabilizer core produced in Chapter 12 is the local algebraic hull of the normal closure of the unipotent directions, and the U-action on L/Λ is ergodic. Thus the hypotheses of Theorem 8.4 are exactly satisfied. That theorem produces a connected rational character-free subgroup $\mathbf{H}$ with $L \subset H_S$ and $H_S \cap \Gamma$ a lattice. Hence $H_S y$ is closed and has finite invariant volume. If Ly is proper, the properness part of the same theorem makes $\mathbf{H}$ proper. Since $U \subset L \subset H_S$, the point y lies in $\mathcal{S}(H, U)$, as required. Conjugating back restores the original arithmetic lattice. $\square$

Remark 2.2. The arithmeticity step is now owned by Chapter 2. In particular, neither Borel–Harish–Chandra nor an unnamed “arithmetic-hull package” is a theorem-level import in the classification proof.

3. The induction

Theorem 3.1 (Arithmetic S-linear measure classification). *Let U be either one local algebraic one-parameter unipotent subgroup or a finite product of commuting local algebraic one-parameter unipotent factors acting on*

$$X = \mathrm{SL}_n(\mathbb{Q}_S)/\,\mathrm{SL}_n(\mathbb{Z}_S).$$

Every U-invariant, U-ergodic probability measure μ is homogeneous: there are a closed local-analytic subgroup $L < G$ containing U and $x \in X$ such that

$$\operatorname{supp}\mu = Lx, \qquad \mu = m_{Lx},$$

and Lx is closed with finite L-volume. If the orbit is proper, it is contained in a proper closed finite-volume rational homogeneous orbit from the arithmetic stratum family.

Proof. We induct on the local analytic dimension of the effective ambient homogeneous group. The zero-dimensional case is trivial.

Suppose first that μ gives positive mass to a proper rational singular stratum. Choose a subgroup type of minimal dimension. Lemma 1.1 places μ on one proper closed finite-volume homogeneous orbit Hy. The U-action on this conull subset is still ergodic and remains unipotent in the restricted adjoint representation. The effective ambient dimension has strictly decreased, so the induction hypothesis applied inside Hy classifies μ there. This proves the theorem in the singular branch.

Assume now

$$\mu(\mathcal{S}(H,U)) = 0 \qquad \text{for every proper } \mathbf{H} \in \mathscr{H}_\Gamma.$$

Let L be the full local stabilizer core of Chapter 12. Proposition 6.2 gives the rigidity alternative. Its proper-singular conclusion is excluded by the present assumption. Hence there is x with $\mu(Lx) = 1$. Lemma 1.2 then gives

$$\mu = m_{Lx}$$

and shows that Lx is closed and has finite L-volume. This is the required homogeneity.

Finally, if this terminal orbit is proper and one wants to return to the rational arithmetic stratum family, apply Lemma 2.1. It embeds Lx in a proper closed finite-volume rational homogeneous orbit. This last arithmeticity statement is not needed for homogeneity itself, but is needed by the orbit-closure and linearization applications later in the volume. □

Remark 3.2 (Why the induction now terminates without an enlargement chain)**.** An earlier working draft started with a small known subgroup $L_0 \subset I(\mu)$ and repeatedly enlarged it. That is pedagogically useful but creates the relative-correction problem. The final proof uses the full local stabilizer core from the beginning. The finite correction algorithm still explains what the nearby trajectories do, but a regular corrected limit is contradictory rather than merely enlarging a provisional subgroup. Only the singular branch recurses, and there the ambient dimension strictly decreases.

Proof and provenance. The external benchmark is the characteristic-zero local-field measure rigidity theorem of Margulis–Tomanov. We use it for attribution and scope comparison, not as a logical input. The relative quasi-regular correction which was missing from the earlier draft is reconstructed in Chapter 12, with the real correction architecture compatible with the separately completed Ratner notebook and the local-field change-of-variable and arithmetic linearization steps proved in this volume.

CHAPTER 15

Proof laboratory: the completed classification induction

This chapter rewrites Theorem 3.1 as a dependency ledger. Its purpose is not to introduce another theorem but to make visible where each difficult input enters and to verify that the former relative-degeneration seam has really disappeared.

1. Step zero: simultaneous generic cores

Fix a countable dense family in $C_c(X)$. Chapter 4 gives pointwise genericity for one local factor and the L^2 rectangular pointwise theorem for finite mixed products. Egorov gives nested compact cores on which finite test families are uniformly generic once all relevant radii exceed one threshold.

The nearby pairs are chosen *recursively*. One first determines the polynomial scale required by the next correction, then moves farther down the sequence so that both endpoints belong to a core whose genericity threshold is smaller than that scale. This quantifier order is what prevents a first-exit scale from outrunning Birkhoff genericity.

2. Step one: choose the canonical stabilizer

Set

$$L = \langle U, I_{\mathrm{an}} \rangle.$$

Here I_{an} denotes the open local analytic subgroup generated by a sufficiently small identity neighborhood in $I(\mu)$, as defined in Chapter 12; thus $U \subset L$ and $\mathrm{Lie}\, L = \mathrm{Lie}\, I(\mu)$.

If some L-orbit is conull, Lemma 1.2 immediately gives the normalized Haar measure on a closed finite-volume orbit. Thus assume for contradiction that no L-orbit is conull. Lemma 1.1 gives generic points

$$x_i = g_i x \to x, \qquad g_i \to e, \qquad g_i \notin L.$$

This is the only source of transverse data needed in the nonsingular branch.

3. Step two: finite quasi-regular correction

Apply Lemma 3.1. At each stage a bounded-degree relative conjugacy polynomial is normalized at its first visible scale. Quasi-regularity and the Basic Lemma put every regular normalizer limit into $I(\mu)$. If that limit is already in L, remove it with an element of $N_L(U)$. The exact normalizer identity

$$n^{-1}u(t)n = u(\chi(n)t)$$

turns this removal into a scalar change of time. The real weighted Cesàro lemma and the finite-place constant-Jacobian change of variables preserve genericity. The active degree strictly drops. Hence the process terminates after finitely many stages.

There are only two terminal outcomes.

3.1. Regular terminal outcome. Let the first-exit group radius tend to zero. The Basic Lemma produces $h_j \in I(\mu)$ with $h_j \to e$ and normalized logarithms tending to a vector whose image in $\mathfrak{g}/\mathfrak{l}$ is nonzero. But a closed local Lie subgroup has tangent cone equal to its Lie algebra, so the limit must lie in

$$\operatorname{Lie} I(\mu) = \mathfrak{l}.$$

Contradiction.

3.2. Completely absorbed terminal outcome. All active coefficients satisfy a proper finite system of primitive incidence equations. This is not yet called rational. The comparison blocks are chosen at common returns to a compact injectivity core. Lifting a return through the arithmetic quotient introduces $\gamma \in \Gamma$, so the exterior data are now primitive S-integral data. Proposition 2.2 leaves finitely many representatives on the compact window. Polynomial goodness and Proposition 3.1 give persistence, negligible occupation, or lower intersection. Repeated lower intersections terminate by dimension; persistence through shrinking tubes becomes exact incidence by Lemma 4.1. Therefore

$$\mu(\mathcal{S}(H, U)) > 0$$

for a proper rational H. This is Proposition 5.1.

4. Step three: the rigidity alternative

The two terminal outcomes give Proposition 6.2:

> Either a conull L-orbit already exists, or a proper rational singular stratum has positive measure.

There is no third "normalizer concentration" case. It has been resolved by the finite correction algorithm: regular concentration is converted to a controlled normalizer time change; complete absorption is converted, after recurrence, to arithmetic singular incidence.

5. Step four: singular mass means smaller ambient dimension

Assume

$$\mu(\mathcal{S}(H, U)) > 0$$

for some proper rational H and choose one of minimal dimension. The pure-stratum argument eliminates lower intersections. The remaining subgroup coordinate and local orbit transversal are measurable U-invariant quantities, so ergodicity makes them essentially constant. Hence μ is supported on one proper closed finite-volume homogeneous orbit. The restricted U-action is still ergodic and Ad-unipotent, while the effective ambient dimension is strictly smaller. The induction hypothesis applies.

6. Step five: nonsingular mass forces homogeneity

If every proper rational singular stratum has zero mass, the singular conclusion of the rigidity alternative is impossible. Therefore some L-orbit is conull, and Lemma 1.2 gives

$$\mu = m_{Lx}.$$

If the terminal orbit is proper, Lemma 2.1 returns it to the rational arithmetic family by the standard proved arithmetic-hull theorem of Chapter 2. Thus both branches of the induction close without an imported arithmetic lattice theorem.

7. Termination bookkeeping

There are two finite quantities and they are used for different purposes.

(1) *Active polynomial degree* decreases during normalizer correction. This proves termination of the relative H-property calculation without changing the ambient group.

(2) *Ambient homogeneous dimension* decreases only after positive mass on a proper singular stratum has been established. This proves termination of the measure-classification induction.

Confusing these two led to the provisional enlargement-chain formulation in the earlier draft. Keeping them separate gives the shorter canonical proof.

8. Dependency check

The classification theorem uses, in order:

(1) local/product pointwise genericity and uniform cores;
(2) local/product H-property and quasi-regular normalizer construction;
(3) tame normalizer reparametrization and finite correction;
(4) arithmetic exterior finite-window linearization and singular occupation;
(5) the minimal-singular homogeneous reduction;
(6) the proved Borel–Harish–Chandra/arithmetic-hull chapter at the terminal rationality step.

The Margulis–Tomanov classification theorem itself is absent from this dependency list. It remains the external benchmark, not a premise.

CHAPTER 16

From parameter groups to larger unipotent-generated actions

This chapter is structural. It answers a question logically separate from measure classification: when a larger acting group W is ergodic, can one find inside it a parameter subgroup to which the classification theorem applies? The previous draft also contained a spectral shortcut for vector groups that was too quick: a separable Hilbert space does not reduce an uncountable spectral measure to a countable set of characters. We replace that argument by Fubini's theorem.

1. Reduction through an ergodic classified subgroup

Proposition 1.1 (Reduction once an ergodic classified subgroup is available)**.** *Let $W < G$ preserve an ergodic probability measure μ. Suppose $U < W$ is a subgroup whose action on (X, μ) is ergodic and for which the relevant measure- classification theorem has been proved. Then $\mu = m_{Lx}$ on a closed finite-volume homogeneous orbit. The support is W-invariant, and replacing L by the closed subgroup generated by L and W does not change either the support or the measure.*

Proof. Classification for U gives $\mu = m_{Lx}$. For every $w \in W$, $w_*\mu = \mu$, hence w preserves the support Lx. Therefore the closed subgroup $\overline{\langle L, W\rangle}$ has orbit Lx and preserves the same probability measure. □

2. Direct products of Mautner factors

Lemma 2.1 (Factorwise product criterion)**.** *Suppose*

$$W = W_1 \times \cdots \times W_r$$

acts ergodically. For each i, let $U_i < W_i$ be a one-parameter unipotent subgroup with the Mautner property: in every unitary representation of W_i, a U_i-fixed vector is W_i-fixed. Then $U_1 \times \cdots \times U_r$ acts ergodically.

Proof. If $f \in L^2(X, \mu)$ is fixed by every U_i, apply the Mautner property to the restriction of the Koopman representation to W_i. Successively, f is fixed by every W_i, hence by W. Ergodicity of W makes f constant. □

3. Generic product lines in an additive local vector group

Let

$$A = \prod_{v \in T} k_v^{d_v}$$

with k_v characteristic-zero local fields. For a tuple $a = (a_v)_{v \in T}$ with $a_v \neq 0$, define the *product line*

$$A_a = \prod_{v \in T} k_v a_v.$$

This is a product of local one-parameter additive groups; it is not a single one-parameter group across different local fields.

Lemma 3.1 (Generic product line has the same fixed vectors). *Let π be a unitary representation of A on a separable Hilbert space $\mathcal{H}$. For Haar-almost every tuple $a \in \prod_v (k_v^{d_v} \setminus \{0\})$ chosen in any fixed product of compact positive-measure sets,*

$$\mathcal{H}^{A_a} = \mathcal{H}^A.$$

Proof. By the spectral theorem for locally compact abelian groups there is a projection- valued measure E on the Pontryagin dual $\widehat{A}$ such that

$$\pi(x) = \int_{\widehat{A}} \chi(x)\, dE(\chi).$$

A vector ξ is fixed by a closed subgroup $B < A$ exactly when its scalar spectral measure

$$\sigma_\xi(E') = \langle E(E')\xi, \xi \rangle$$

is supported on the annihilator $B^\perp$.

Choose a countable dense set $\{\xi_m\}$ in $\mathcal{H}$. It is enough to find a such that

$$\sigma_{\xi_m}(A_a^\perp \setminus \{1\}) = 0 \qquad \text{for every } m.$$

Fix a nontrivial character $\chi = (\chi_v)_v \in \widehat{A}$. At some place v, χ_v is nontrivial and can be written, after fixing a standard additive character of k_v, as

$$x_v \longmapsto \psi_v(\ell_v(x_v))$$

for a nonzero k_v-linear functional ℓ_v. If χ is trivial on A_a, then in particular $\ell_v(a_v) = 0$. This is a proper linear hyperplane in $k_v^{d_v}$ and has Haar measure zero. Hence for every fixed nontrivial χ,

$$m_A\{a : \chi \in A_a^\perp\} = 0.$$

Apply Tonelli to the nonnegative indicator of the incidence relation $\{(a, \chi) : \chi \in A_a^\perp \setminus \{1\}\}$. For each m,

$$\int \sigma_{\xi_m}(A_a^\perp \setminus \{1\})\, dm_A(a) = \int_{\widehat{A} \setminus \{1\}} m_A\{a : \chi \in A_a^\perp\}\, d\sigma_{\xi_m}(\chi) = 0.$$

Thus the desired spectral mass vanishes for almost every a, simultaneously for all m after intersecting countably many conull sets.

For such a, if ξ is A_a-fixed, approximate it by the dense family and use the orthogonal projections onto the fixed spaces (equivalently apply the spectral projection $E(A_a^\perp)$). The projection onto $A_a^\perp \setminus \{1\}$ is zero, so $E(A_a^\perp) = E(\{1\})$. The latter is exactly the projection onto $\mathcal{H}^A$. Hence $\mathcal{H}^{A_a} = \mathcal{H}^A$. ☐

4. The direct-product structural theorem

THEOREM 4.1 (Ergodic parameter subgroup in the direct-product range). *Suppose*

$$W = W_1 \times \cdots \times W_r \times A,$$

where each W_i is a local group with a one-parameter unipotent subgroup U_i having the Mautner property, and A is as above. If the W-action on a probability space is ergodic, then for Haar-almost every product line A_a the subgroup

$$U = U_1 \times \cdots \times U_r \times A_a$$

is ergodic.

PROOF. Let $f \in L^2(X, \mu)$ be fixed by U. The Mautner property makes it fixed by every W_i. Lemma 3.1 makes A_a-invariance equivalent to A-invariance for the Koopman representation. Thus f is fixed by all of W and is constant by W-ergodicity. ☐

Corollary 4.2 (Generated-action classification)**.** *In the setting of Theorem 4.1, every finite W-invariant W-ergodic probability measure is homogeneous in the arithmetic matrix scope of Theorem 3.1.*

PROOF. Choose an ergodic product parameter subgroup by Theorem 4.1, apply Theorem 3.1, and then use Proposition 1.1. ☐

Remark 4.3. The structural theorem and the corollary are now unconditional in the stated arithmetic matrix range. The full Margulis–Tomanov almost-linear setting requires additional structure theory, particularly in the presence of nontrivial unipotent radicals, and is not claimed here.

CHAPTER 17

The marked-poset theorem over products of local fields

The combinatorial engine of quantitative nondivergence is independent of the ordered-field structure of $\mathbb{R}$. What it actually needs is a finite-length poset, good functions, a noncollapse scale, and a family of parameter neighborhoods with bounded-overlap covering. Volume 1 proved this mechanism for Euclidean balls. We now rebuild the proof for mixed product boxes so that no exponent is lost in passing to finite places.

1. Product boxes and their geometry

Let

$$Y = \mathbb{R}^{d_\infty} \times \prod_{v \in S_f} \mathbb{Q}_v^{d_v}$$

with product Haar measure μ. After rescaling the coordinates of one fixed base box, we may regard it as the unit ball for the max metric

$$d_Y(x, y) = \max\left(\|x_\infty - y_\infty\|_2, \max_{v \in S_f} \|x_v - y_v\|_v \right).$$

A ball for this metric is a Euclidean ball times local max-norm balls; at a finite place the radius is automatically rounded to the nearest value-group radius. For $a \geq 1$, write aB for the ball with the same center and metric radius multiplied by a.

For every fixed a there is a constant $D_Y(a)$ such that

$$\mu(aB) \leq D_Y(a)\mu(B)$$

uniformly in B. The real factor contributes a^{d_∞}, while a finite place contributes at most one extra residue-field step beyond a^{d_v}.

Lemma 1.1 (Besicovitch selection for mixed balls)**.** *There is $N_Y < \infty$, depending only on the real dimension, such that every family of mixed balls centered on a bounded set has a countable subcover of multiplicity at most N_Y. If $d_\infty = 0$, one may take $N_Y = 1$.*

Proof. Use the standard greedy generation proof of the Besicovitch theorem. The only input needed to bound multiplicity is a uniform bound for a *Besicovitch family*: finitely many balls with a common point such that no center lies in any other ball.

Order such a family by decreasing radii, $r_i \geq r_j$ for $i < j$, and let z be a common point. Since $x_j \notin B(x_i, r_i)$,

$$d_Y(x_i, x_j) > r_i.$$

At every nonarchimedean coordinate, however,

$$\|x_{i,v} - x_{j,v}\|_v \leq \max(\|x_{i,v} - z_v\|_v, \|x_{j,v} - z_v\|_v) \leq r_i$$

by ultrametricity and $r_j \leq r_i$. Thus the separation must occur in the real coordinate:

$$\|x_{i,\infty} - x_{j,\infty}\|_2 > r_i.$$

The projected real balls therefore form a Euclidean Besicovitch family with common point z_∞. The angular-packing argument proved in Volume 1 bounds its cardinality by a dimension-only constant. If there is no real factor, ultrametricity shows that a Besicovitch family contains at most one ball.

The greedy generation construction from the Euclidean proof uses only this cardinality bound and boundedness of the set of centers, so it carries over verbatim and produces the desired bounded-multiplicity subcover. □

Consequently, if every selected ball has radius at most twice that of a base ball B and center in B, then its union lies in $3B$ and

$$\sum_i \mu(B_i) \leq N_Y \mu(3B) \leq N_Y D_Y(3)\mu(B).$$

2. Abstract marking data

Let $\mathfrak{P}$ be a partially ordered set of chain length at most r. To every $s \in \mathfrak{P}$ associate a continuous nonnegative function

$$\psi_s : 3^r B \longrightarrow \mathbb{R}_{\geq 0}.$$

Fix $C, \alpha, \rho > 0$ and assume:

(A1) each ψ_s is (C, α)-good on $3^r B$;
(A2) $\sup_B \psi_s \geq \rho$ for every s;
(A3) for every $x \in 3^r B$, only finitely many s satisfy $\psi_s(x) < \rho$.

For a chain $\Sigma \subset \mathfrak{P}$, let

$$\mathfrak{P}(\Sigma) = \{s \notin \Sigma : s \text{ is comparable with every member of } \Sigma\}.$$

A point $x \in B$ is *ε-marked* if there is a chain Σ_x such that

(1) $\varepsilon \leq \psi_s(x) \leq \rho$ for $s \in \Sigma_x$;

(2) $\psi_s(x) \geq \rho$ for $s \in \mathfrak{P}(\Sigma_x)$.

Write $\Phi(\varepsilon)$ for the marked set.

THEOREM 2.1 (Product marking theorem). *Under the preceding assumptions,*

$$\mu(B \setminus \Phi(\varepsilon)) \leq rC\big(N_Y D_Y(3)\big)^r \left(\frac{\varepsilon}{\rho}\right)^\alpha \mu(B)$$

for $0 < \varepsilon \leq \rho$.

PROOF. We repeat the induction with the product geometry visible. The case $r = 0$ is empty. Assume the result for posets of chain length at most $r - 1$.

For $x \in B$, put

$$H(x) = \{s : \psi_s(x) < \rho\}.$$

If $H(x)$ is empty, the empty chain marks x. Suppose $H(x) \neq \varnothing$. For $s \in H(x)$ define

$$r_{s,x} = \sup\left\{0 < q \leq 2 : \sup_{B(x,q)} \psi_s \leq \rho\right\},$$

where $B(x, q)$ is the mixed metric ball of radius q times the radius of the base ball. Put $B_{s,x} = B(x, r_{s,x})$. If $r_{s,x} < 2$, continuity forces $\sup_{B_{s,x}} \psi_s = \rho$. If $r_{s,x} = 2$, the ball contains the original base ball B because its center lies in B, and (A2) again gives $\sup_{B_{s,x}} \psi_s \geq \rho$. Since $H(x)$ is finite, choose $s_x \in H(x)$ for which $r_{s,x}$ is maximal and write $B_x = B_{s_x,x}$.

We claim

$$\sup_{B_x} \psi_s \geq \rho \qquad (s \in \mathfrak{P}).$$

If $s \notin H(x)$ this already holds at the center x. If $s \in H(x)$, then $r_{s,x} \leq r_{s_x,x}$ by the choice of s_x. When the inequality is strict, enlarging past $r_{s,x}$ forces the supremum above ρ. When the radii are equal and $r_{s,x} < 2$, continuity and the definition as a supremal admissible radius give $\sup_{B_x} \psi_s = \rho$. Finally, if the common radius is 2, then B_x contains the original base ball B, so (A2) again gives the claim. Thus ties of maximal radii cause no gap. Let

$$\mathfrak{P}_x = \{s \neq s_x : s \text{ comparable with } s_x\}.$$

Its chain length is at most $r - 1$. The reduced data $(\mathfrak{P}_x, \psi|_{\mathfrak{P}_x}, B_x)$ satisfy (A1)–(A3): the only point to check is the goodness domain. Because the center of B_x lies in B and its scale is at most two,

$$3^{r-1} B_x \subset 3^r B$$

coordinatewise, with the harmless value-group rounding already absorbed in the definition of enlargement.

If $z \in B_x$ is marked for the reduced data and $\psi_{s_x}(z) \geq \varepsilon$, adjoining s_x to its marking chain marks z for the original data. Hence

$$B \cap (B_x \setminus \Phi(\varepsilon)) \subset \big(B_x \setminus \Phi_x(\varepsilon)\big) \cup \{z \in B_x : \psi_{s_x}(z) < \varepsilon\}.$$

The induction hypothesis bounds the first set by

$$(r-1)C\big(N_Y D_Y(3)\big)^{r-1} \left(\frac{\varepsilon}{\rho}\right)^{\alpha} \mu(B_x),$$

and goodness, together with $\sup_{B_x} \psi_{s_x} \geq \rho$, bounds the second by

$$C \left(\frac{\varepsilon}{\rho}\right)^{\alpha} \mu(B_x).$$

Thus

$$\mu\big(B \cap (B_x \setminus \Phi(\varepsilon))\big) \leq rC\big(N_Y D_Y(3)\big)^{r-1} \left(\frac{\varepsilon}{\rho}\right)^{\alpha} \mu(B_x).$$

Select a bounded-overlap subcover $\{B_i\}$ of the set where $H(x)$ is nonempty. The product covering estimate gives

$$\sum_i \mu(B_i) \leq N_Y D_Y(3) \mu(B).$$

Summing the local bound proves the displayed estimate. The exponent α has not changed at any stage of the induction. ☐

3. Primitive S-submodules as the poset

Let $\mathfrak{P}_S$ be the nonzero primitive $\mathbb{Z}_S$-submodules of $\mathbb{Z}_S^n$, ordered by inclusion. Its chain length is n. For $h : B \to \mathrm{GL}_n^1(\mathbb{Q}_S)$ define

$$\psi_\Delta(x) = c(h(x)\Delta).$$

Lemma 3.1 (Submodularity of exterior content)**.** *There is $C_0 = C(S, n)$ such that*

$$\psi_{\Delta_1+\Delta_2}(x)\psi_{\Delta_1\cap\Delta_2}(x) \leq C_0 \psi_{\Delta_1}(x)\psi_{\Delta_2}(x),$$

where sums and intersections are primitively saturated and the zero module has content one.

Proof. Choose a basis of the intersection and extend it to bases of the two modules. In top exterior degree the product of the two wedges equals the wedge for the intersection times a wedge spanning the sum, followed by an integral change of basis. At a nonarchimedean place the max norm of the resulting coordinate sums is at most the product of the input norms; at the real place at most a dimension-only number of determinant terms occur. Multiplying the local inequalities gives the constant C_0. ☐

Lemma 3.2 (Local finiteness of marked S-modules). *For fixed x and $R > 0$, only finitely many primitive modules Δ satisfy $\psi_\Delta(x) < R$, modulo the unavoidable S-unit choice of top-wedge generator.*

Proof. Choose a primitive top wedge for each Δ and balance its local norms by the S-unit theorem of Volume 3. A bound on product content then puts every balanced local coordinate in one compact product box. The diagonal exterior $\mathbb{Z}_S$-module is discrete, so this box contains only finitely many primitive balanced wedges. A primitive wedge determines its rational submodule. $\square$

The abstract marking theorem therefore applies to the primitive-module poset as soon as goodness and noncollapse are verified. Submodularity explains why the poset is the natural arithmetic structure, although the abstract induction itself uses only comparability and finite chain length.

CHAPTER 18

Proof laboratory: what the marking induction is really doing

The preceding theorem preserves the same goodness exponent α all the way to the final estimate. It is worth understanding why no threshold hierarchy is needed.

1. One obstruction chooses one scale

At a bad point there may be many primitive modules with content below ρ. The proof does not choose the smallest module or sum over all of them. It chooses the one whose inequality

$$\psi_\Delta < \rho$$

persists on the largest centered parameter box. Maximality forces *every* other obstruction to attain the reference size ρ on that box. This is the single step which turns an infinite arithmetic family into a finite-complexity local problem.

2. Why adjoining the chosen module loses one rank

Once Δ is put into the marking chain, only modules comparable with it can extend that chain. Any chain among those comparable modules can be combined with Δ, so the remaining chain length drops by one. The induction is therefore on chain complexity, not on the number of primitive modules.

In the lattice poset a chain has length at most n:

$$0 < \operatorname{rank} \Delta_1 < \cdots < \operatorname{rank} \Delta_k \leq n.$$

This finite rank is the reason the procedure terminates.

3. Where the power $(\varepsilon/\rho)^\alpha$ comes from

On each maximal box the only genuinely analytic estimate is

$$\mu\{\psi_\Delta < \varepsilon\} \leq C(\varepsilon/\rho)^\alpha \mu(B_\Delta).$$

The induction adds finitely many such estimates and the covering theorem multiplies only by geometric constants. No new smaller threshold is introduced, so the exponent remains exactly α. This corrects a tempting but unnecessary proof strategy in which one inserts rank-dependent thresholds and loses exponent at every rank.

4. Why S-units matter only for local finiteness

The marking induction is abstract. The arithmetic input is the assertion that at one parameter point there are only finitely many primitive modules of bounded content. Without quotienting by S-units, the same module would have infinitely many top-wedge generators

$$w, \quad pw, \quad p^{-1}w, \dots$$

whose local norms move in opposite directions. Product content is invariant under this rescaling, and the S-unit theorem chooses a balanced representative. Once balanced, bounded content means bounded coordinates in every local factor, and diagonal discreteness gives finiteness.

Thus the logical roles are cleanly separated:

$$\boxed{S\text{-units: local finiteness}} \quad + \quad \boxed{\text{goodness: sublevel measure}}$$
$$+ \quad \boxed{\text{poset rank: termination}}.$$

CHAPTER 19

Kleinbock–Tomanov quantitative nondivergence

We now turn the abstract marking theorem into a statement about short vectors of an S-lattice. One extra exterior inequality is needed: a marked flag must prevent *every* nonzero vector from being short.

1. Shortest content

For an S-lattice Λ define

$$\delta_S(\Lambda) = \inf_{0 \neq x \in \Lambda} c(x).$$

By primitive saturation, S-unit balancing, and discreteness, the infimum is attained on a primitive vector.

Lemma 1.1 (Rank extension). *There is a constant $A_n \geq 1$, depending only on n and the chosen exterior max norms, with the following property. Let $\Gamma \subset \Lambda$ be primitive modules with* $\operatorname{rank} \Lambda = \operatorname{rank} \Gamma + 1$, *and let $v \in \Lambda \setminus \Gamma$. Then for every $g \in \mathrm{GL}_n(\mathbb{Q}_S)$,*

$$c(gv) \geq \frac{c(g\Lambda)}{A_n \, c(g\Gamma)}.$$

The content of the zero module is interpreted as one.

Proof. Choose a primitive top wedge w_Γ for Γ, and choose a primitive generator w of the rank-one quotient Λ/Γ. In that quotient we can write

$$v \equiv aw \pmod{\Gamma}$$

with $0 \neq a \in \mathbb{Z}_S$. Therefore

$$w_\Gamma \wedge v = a \, w_\Gamma \wedge w.$$

The second wedge is a primitive top wedge of Λ, while Primitive Saturation from Volume 3 gives $c(a) \geq 1$. Hence

$$c\big(g(w_\Gamma \wedge v)\big) = c(a)c(g\Lambda) \geq c(g\Lambda).$$

At every finite place the exterior max norm satisfies

$$\|g_v w_\Gamma \wedge g_v v\|_v \leq \|g_v w_\Gamma\|_v \, \|g_v v\|_v.$$

At the real place the same inequality holds with a dimension-dependent constant because each wedge coordinate is a sum of at most a bounded number of products. Multiplying over places gives

$$c(g\Lambda) \le c\big(g(w_\Gamma \wedge v)\big) \le A_n c(g\Gamma) c(gv),$$

which is the desired bound. □

2. A marked flag has no short vector

Lemma 2.1 (Marked implies a systole bound)**.** *Let x be ε-marked for the primitive-module data $\psi_\Delta(x) = c(h(x)\Delta)$. If*

$$0 < \varepsilon \le \rho \le A_n^{-1},$$

then

$$\delta_S(h(x)\mathbb{Z}_S^n) \ge \varepsilon.$$

Proof. Let

$$\Sigma_x = \{\Gamma_1 < \cdots < \Gamma_\ell\}$$

be a marking chain; we adjoin only the zero module formally, writing $\Gamma_0 = 0$. We do *not* adjoin the full lattice as a marked member: although its content is one for $h(x) \in \mathrm{GL}_n^1(\mathbb{Q}_S)$, it need not satisfy the marking upper bound $\psi \le \rho$.

Fix $0 \ne v \in \mathbb{Z}_S^n$. First suppose that some marked member contains v, and let $i \ge 1$ be minimal with $v \in \Gamma_i$. Put

$$\Lambda = (\Gamma_{i-1,\mathbb{Q}_S} + \mathbb{Q}_S v) \cap \mathbb{Z}_S^n.$$

Then $\Gamma_{i-1} < \Lambda \le \Gamma_i$. Hence Λ is comparable with every member of Σ_x. If Λ itself occurs in the chain, its marking inequality gives $c(h(x)\Lambda) \ge \varepsilon$; if it does not, the second marking condition gives the stronger inequality $c(h(x)\Lambda) \ge \rho$. Thus in all cases

$$c(h(x)\Lambda) \ge \varepsilon.$$

If $i = 1$, the primitive rank-one module Λ has a primitive generator w, and $v = aw$ for some nonzero $a \in \mathbb{Z}_S$. Primitive saturation from Volume 3 gives $c(a) \ge 1$, so

$$c(h(x)v) = c(a)c(h(x)w) \ge c(h(x)\Lambda) \ge \varepsilon.$$

If $i > 1$, then Γ_{i-1} is a genuine marked member and therefore $c(h(x)\Gamma_{i-1}) \le \rho$. Lemma 1.1 yields

$$c(h(x)v) \ge \frac{c(h(x)\Lambda)}{A_n c(h(x)\Gamma_{i-1})} \ge \frac{\varepsilon}{A_n \rho} \ge \varepsilon.$$

It remains to treat the case that v lies in no member of Σ_x. If the chain is empty, let $\Lambda = (\mathbb{Q}_S v) \cap \mathbb{Z}_S^n$. It is comparable with every member vacuously and is not marked, so $c(h(x)\Lambda) \geq \rho \geq \varepsilon$, and the rank-one argument above applies. If the chain is nonempty, let Γ_ℓ be its maximal member and set

$$\Lambda = (\Gamma_{\ell,\mathbb{Q}_S} + \mathbb{Q}_S v) \cap \mathbb{Z}_S^n.$$

Because no marked member contains v, this is a strict rank-one extension of Γ_ℓ; it contains every member of the chain and is not itself in the chain. Hence the second marking condition gives $c(h(x)\Lambda) \geq \rho$, whereas $c(h(x)\Gamma_\ell) \leq \rho$. The rank-extension lemma now gives

$$c(h(x)v) \geq \frac{\rho}{A_n \rho} = A_n^{-1} \geq \rho \geq \varepsilon,$$

where the penultimate inequality uses $\rho \leq A_n^{-1}$. Thus every nonzero vector has content at least ε. □

THEOREM 2.2 (S-arithmetic quantitative nondivergence). *Let B be a product parameter box and let $3^n B$ be its fixed enlargement. Let*

$$h : 3^n B \longrightarrow \mathrm{GL}_n^1(\mathbb{Q}_S)$$

be continuous. Assume that for every nonzero primitive $\Delta \subset \mathbb{Z}_S^n$:

(1) $x \mapsto c(h(x)\Delta)$ is (C, α)-good on $3^n B$;

(2) $\sup_{x \in B} c(h(x)\Delta) \geq \rho$;

and suppose $0 < \rho \leq A_n^{-1}$. Then for $0 < \varepsilon \leq \rho$,

$$\mu\{x \in B : \delta_S(h(x)\mathbb{Z}_S^n) < \varepsilon\} \leq C_1 \left(\frac{\varepsilon}{\rho}\right)^\alpha \mu(B),$$

where

$$C_1 = nC\big(N_Y D_Y(3)\big)^n$$

may be replaced by any larger structural constant.

PROOF. Lemma 3.2 supplies the pointwise local-finiteness hypothesis of Theorem 2.1. Apply that theorem to the primitive-module poset, whose chain length is at most n. It gives

$$\mu(B \setminus \Phi(\varepsilon)) \leq C_1(\varepsilon/\rho)^\alpha \mu(B).$$

By Lemma 2.1, every marked point has S-systole at least ε. Therefore the short-vector set lies in the unmarked set and the same estimate applies. □

3. Polynomial maps

Corollary 3.1 (Polynomial S-arithmetic QND)**.** *Suppose each local matrix coefficient of h is a polynomial of degree at most D in the local parameters. Then the goodness hypothesis holds with uniform C, α depending only on S, n, D and the parameter dimensions. Consequently the QND estimate is uniform in all such h once the noncollapse parameter ρ is fixed.*

Proof. Exterior coordinates of $h(x)\Delta$ are polynomials of uniformly bounded degree. Their local max norms are good by Volume 1 at the real place and Volume 3 at the finite places. The product-goodness theorem of Volume 3 gives uniform (C, α) constants. Apply Theorem 2.2. $\square$

Proof and provenance. This is the arithmetic lattice form of the quantitative mechanism developed by Kleinbock–Tomanov. Their published theorem treats a more flexible Besicovitch/Federer product-measure framework. The form used in this series is proved here from the local polynomial estimates and the product marking induction; the source is used for attribution and scope checking, not as a logical input.

CHAPTER 20

Workshop: quantitative nondivergence in $\mathrm{SL}_2(\mathbb{Q}_{\{\infty,p\}})$

We now run the S-arithmetic quantitative theorem by hand in the smallest mixed example. The calculation shows exactly where product content, goodness, and the algebraic noncollapse condition enter.

1. The lattice and its shortest content

Let

$$S = \{\infty, p\}, \qquad \Lambda = \mathbb{Z}[1/p]^2 \subset \mathbb{R}^2 \times \mathbb{Q}_p^2.$$

For $v = (v_\infty, v_p)$ put

$$c(v) = \|v_\infty\|_\infty \|v_p\|_p.$$

Multiplication by an S-unit $\pm p^k$ rescales the two local norms inversely, so $c(p^k v) = c(v)$. Thus content is the correct size on the diagonal S-lattice; either local norm alone would depend on the arbitrary S-unit representative.

Take

$$u(s,t) = \left(\begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \right), \qquad s \in \mathbb{R}, \quad t \in \mathbb{Q}_p.$$

For a primitive vector $q = (a,b)^T \in \mathbb{Z}[1/p]^2$,

$$u(s,t)q = ((a+sb, b), (a+tb, b)).$$

Hence

$$c(u(s,t)q) = \max(|a+sb|, |b|) \max(|a+tb|_p, |b|_p).$$

Each local max norm is the norm of a degree-one polynomial vector.

2. Goodness

On a product ball $B = B_\infty \times B_p$, the real factor is $(C,1)$-good by the one-variable polynomial theorem of Volume 1, and the p-adic factor is $(C_p, 1)$-good by Volume 3. If the product is smaller than ε times its product supremum, at least one local factor is smaller than $\varepsilon^{1/2}$ times its own supremum. Therefore

$$m\{c(u(s,t)q) < \varepsilon \sup_B c(uq)\} \ll \varepsilon^{1/2} m(B).$$

For higher-rank primitive modules the same calculation is done with Plücker coordinates; their degrees remain uniformly bounded.

3. Noncollapse is an independent question

Suppose

$$\sup_{(s,t)\in B} c(u(s,t)q) \geq \rho$$

for every primitive q. Then the marked-poset theorem gives

$$m\{(s,t) \in B : \delta_S(u(s,t)\Lambda) < \varepsilon\} \ll (\varepsilon/\rho)^{\alpha} m(B).$$

The estimate is uniform in q because one never sums over all primitive vectors. The marking argument organizes simultaneously short primitive submodules by rank and uses submodularity to prevent uncontrolled overlaps.

What if noncollapse fails on larger and larger balls? Then some primitive projective class has bounded content throughout those balls. Hold $t = 0$ and let $|s| \to \infty$. If $b \neq 0$, the real factor $\max(|a + sb|, |b|)$ grows linearly, while the fixed p-adic factor is nonzero; content is unbounded. Thus $b = 0$. The persistent vector is the rational line spanned by e_1, and the unipotent group stabilizes that line. This is exactly the rational parabolic alternative of the uniform nonescape theorem.

4. Why rank two matters even in SL_2

For $n = 2$, the only proper primitive ranks are one, so the marked-poset theorem looks unnecessary. But the top wedge still records determinant normalization:

$$c\Big(\bigwedge^2 g\Lambda\Big) = 1$$

for $g \in \mathrm{GL}_2^1(\mathbb{Q}_S)$. In higher dimension the same top-rank normalization interacts with many competing lower-rank modules. The SL_2 calculation is therefore not a different theory; it is the rank-one face of the same exterior architecture.

5. From the estimate to recurrence

Fix $\varepsilon > 0$. By S-Mahler compactness,

$$M_\varepsilon = \{\Lambda' : \delta_S(\Lambda') \geq \varepsilon\}$$

is compact. Quantitative nondivergence says that, provided no rational parabolic obstruction is present,

$$\frac{m\{(s,t) \in B : u(s,t)\Lambda \notin M_\varepsilon\}}{m(B)} \ll (\varepsilon/\rho)^{\alpha}.$$

Choose ε so the right side is below a prescribed β. This is the precise analytic-to-geometric bridge used in the uniform nonescape theorem: short exterior content is an analytic sublevel event, while the set where no content is short is a compact set in the homogeneous space.

CHAPTER 21

Uniform nonescape and the finite-volume algebraic alternative

The quantitative nondivergence theorem controls cusp excursions provided every primitive submodule becomes moderately large somewhere on the parameter ball. If one primitive wedge stays small on every sufficiently large ball, polynomiality forces an exact rational flag. A rational flag gives a parabolic subgroup. The crucial final step is to replace that infinite-volume parabolic by its character-free kernel, which has finite volume and still contains every unipotent subgroup.

1. Persistent wedge implies an invariant rational flag

Let $u(t)$ be a polynomial parametrization of a unipotent algebraic subgroup $U < G$, and let $y = g\Gamma$.

Lemma 1.1 (Persistent wedge gives a rational parabolic)**.** *Suppose a primitive rank-r module $\Delta \subset \mathbb{Z}_S^n$, $1 \le r < n$, satisfies*

$$\sup_{t \in B_R} c(u(t)g\Delta) \le C$$

for arbitrarily large centered product balls B_R. Then the exterior line $\mathbb{Q}_S(g\Delta)$ is U-invariant. Equivalently,

$$U \subset gP_\Delta g^{-1},$$

where $P_\Delta < \mathrm{SL}_n$ is the rational parabolic stabilizing the rational r-plane spanned by Δ.

Proof. Choose a primitive generator $w_\Delta \in \bigwedge^r \mathbb{Z}_S^n$ and put

$$F(t) = \bigwedge^r u(t) g w_\Delta = (F_v(t_v))_{v \in S}.$$

Every F_v is a nonzero vector-valued polynomial, because $u_v(t_v)g_v$ is invertible. The hypothesis is

$$\prod_{v \in S} \|F_v(t_v)\|_v \le C$$

throughout arbitrarily large product balls.

There is a small but important point here. We do *not* balance $F(t)$ by an S-unit depending on R: that would change the polynomial while its parameter ball is changing. Instead fix a place $v \in S$, set every other local parameter equal to zero, and vary only t_v. Since $F_w(0) \neq 0$ for every $w \neq v$, the global content bound gives

$$\|F_v(t_v)\|_v \leq C \prod_{w \neq v} \|F_w(0)\|_w^{-1} =: C_v$$

for t_v in centered v-adic balls of arbitrarily large radius. Thus every coordinate polynomial of F_v is bounded on arbitrarily large centered balls.

A polynomial bounded on arbitrarily large centered balls is constant. For one variable this follows from the leading term over $\mathbb{R}$; over a nonarchimedean local field, if $P(T) = a_d T^d + \cdots + a_0$ with $d > 0$, then for $|T|_v$ larger than all $|a_j/a_d|_v^{1/(d-j)}$, ultrametricity gives $|P(T)|_v = |a_d|_v |T|_v^d$. If t_v has several local coordinates, take the highest nonzero homogeneous part of a nonconstant coordinate polynomial, choose a direction on which it does not vanish (the local field is infinite), and restrict to that line. Thus every coordinate polynomial of F_v is constant on the entire local parameter space.

Repeating for every $v \in S$, we conclude that each local factor U_v fixes the line $\mathbb{Q}_v(g_v w_\Delta)$. Therefore the product group U fixes the $\mathbb{Q}_S$-line $\mathbb{Q}_S(g w_\Delta)$. A decomposable exterior line determines its underlying r-plane, so U stabilizes $g(\mathbb{Q}_S\Delta)$; equivalently $U \subset g P_\Delta g^{-1}$. □

2. The character-free kernel of a standard parabolic

Let $P = P_\Delta$. After a rational change of basis it has block form

$$P = MN,$$

where N is the unipotent radical and M is a block diagonal Levi. The rational characters of P are generated by the determinants of the diagonal blocks, subject to the total determinant-one relation. Define

$$P^1 = \bigcap_{\chi \in X_{\mathbb{Q}}(P)} \ker \chi.$$

Lemma 2.1 (The parabolic core has finite S-volume)**.** *The group P^1 contains every unipotent subgroup of P, and*

$$P^1(\mathbb{Q}_S)/(P^1(\mathbb{Q}_S) \cap \mathrm{SL}_n(\mathbb{Z}_S))$$

has finite invariant volume.

Proof. Unipotents are killed by every character by Lemma 2.1. In block form, P^1 is generated by its unipotent radical together with the block groups

$\mathrm{SL}_{d_1}, \ldots, \mathrm{SL}_{d_k}$ and a compact central kernel coming from the determinant constraints.

The quotient of the unipotent radical by its $\mathbb{Z}_S$-points is compact: choose an ordering of the matrix entries along the lower central series and reduce each additive coordinate into a fixed compact fundamental domain for $\mathbb{Q}_S/\mathbb{Z}_S$, whose compactness was proved in Volume 3. Each block $\mathrm{SL}_{d_i}(\mathbb{Z}_S)$ is a lattice by the S-Siegel reduction theorem of Volume 3. Fubini integration along the semidirect product therefore constructs a finite-volume fundamental set for P^1. Compact central factors do not affect finiteness. $\square$

3. Uniform recurrence

Theorem 3.1 (Uniform nonescape / finite-volume algebraic alternative). *Fix a degree bound D and $\beta > 0$. There is a compact set $M \subset X$ such that for every $y = g\Gamma \in X$ and every degree-$\leq D$ polynomial parametrization u of a unipotent algebraic subgroup U, one of the following holds.*

(1) There is $R_0 = R_0(y, u)$ such that every centered product ball B_R of radius at least R_0 satisfies

$$\frac{m\{t \in B_R : u(t)y \in M\}}{m(B_R)} \geq 1 - \beta.$$

(2) There is a proper closed subgroup $H < G$, containing U, such that Hy is closed and carries a finite H-invariant measure.

Proof. Fix once and for all a structural number

$$0 < \rho_0 \leq A_n^{-1}$$

and then choose $\varepsilon > 0$ so small that the constant in Theorem 2.2 satisfies

$$C_1(\varepsilon/\rho_0)^{\alpha} < \beta.$$

The S-Mahler theorem of Volume 3 makes

$$M = M(\varepsilon) = \{\Lambda : \delta_S(\Lambda) \geq \varepsilon\}$$

compact. Notice that this compact set depends only on the structural data and β, not on y or on the individual polynomial orbit.

For a centered box B_R call a primitive module Δ a ρ_0-obstruction if

$$\sup_{t \in B_R} c(u(t)g\Delta) < \rho_0.$$

Because $0 \in B_R$, every obstruction satisfies $c(g\Delta) < \rho_0$. By the local-finiteness lemma for primitive exterior classes, there are only finitely many possible projective primitive modules with this property for the fixed starting point $g\Gamma$. Thus the obstruction cannot change through infinitely many new arithmetic classes as R grows.

If no one of these finitely many classes is an obstruction for arbitrarily large R, take the maximum of their last obstruction radii. For every larger box and every primitive module we then have

$$\sup_{B_R} c(u(t)g\Delta) \geq \rho_0.$$

(The modules not in the finite list already have value at $t = 0$ at least ρ_0.) The polynomial-goodness constants are uniform in the fixed degree bound D, so Theorem 2.2 gives

$$\frac{m\{t \in B_R : \delta_S(u(t)g\mathbb{Z}_S^n) < \varepsilon\}}{m(B_R)} < \beta.$$

This is alternative (1).

Otherwise one fixed primitive projective class Δ is a ρ_0-obstruction on arbitrarily large centered boxes. Multiplying its top wedge once by a fixed S-unit does not change content, so we may keep one actual primitive representative throughout. Lemma 1.1 then gives a rational parabolic P_Δ with

$$U \subset gP_\Delta g^{-1}.$$

Let $H = gP_\Delta^1 g^{-1}$. Every unipotent subgroup of the parabolic lies in the character-free core, so $U \subset H$. Since P_Δ^1 is rational and $P_\Delta^1(\mathbb{Z}_S)$ is a lattice by Lemma 2.1,

$$Hy = gP_\Delta^1\Gamma/\Gamma$$

is a left translate of a closed finite-volume arithmetic orbit. It is proper because $1 \leq \operatorname{rank}\Delta < n$. This is alternative (2). $\square$

Proof and provenance. This is the arithmetic SL_n form of the algebraic alternative appearing in Kleinbock–Tomanov's uniform nondivergence theorem: either a long polynomial unipotent piece spends almost all of its time in one compact set, or the acting unipotent group is contained in a proper subgroup whose orbit through the starting point is closed and of finite volume. Here the finite-volume subgroup is produced explicitly from the character-free core of the rational parabolic detected by the persistent wedge.

CHAPTER 22

Proof laboratory: from a short wedge to a finite-volume obstruction

The algebraic alternative in Theorem 3.1 contains three conceptually different steps. This chapter separates them because confusing any two of them leads to an incorrect statement.

1. Step 1: failure of noncollapse produces one persistent module

Fix a small structural threshold $\rho > 0$. At time zero, only finitely many primitive modules have content below 2ρ: balance their exterior generators and use discreteness. Every other module already satisfies the required supremum bound on every parameter ball because 0 belongs to the ball.

Suppose the noncollapse hypothesis fails for arbitrarily large centered balls. For each such ball choose a failing primitive module. Since only finitely many candidates can fail at the origin, one projective module occurs along an infinite subsequence. Hence there is a fixed Δ such that

$$\sup_{t \in B_{R_j}} c(u(t)g\Delta) < \rho$$

for $R_j \to \infty$.

This finite-pigeonhole step is important. Without it, one cannot pass from "there is some small wedge on every scale" to "one polynomial wedge is bounded on arbitrarily large balls."

2. Step 2: polynomial boundedness produces a rational flag

For a primitive top wedge w_Δ, every coordinate of

$$\bigwedge^{r} u(t)gw_\Delta$$

is a polynomial. The product content bound and S-unit balancing give bounded local representatives along the chosen scales. A nonconstant local polynomial cannot remain bounded on balls whose radius tends to infinity.

Therefore the projective exterior line is fixed by U. Since it is decomposable, the underlying r-plane is fixed. Thus

$$U \subset gP_{\Delta}g^{-1}.$$

At this stage we have found a proper *parabolic* obstruction and nothing more.

3. Step 3: why the parabolic itself is the wrong subgroup

For SL_2, the upper triangular parabolic

$$P = \left\{ \begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix} \right\}$$

has a rational character $\chi(P) = a^2$. Its arithmetic quotient has infinite volume in the character direction. Thus the statement "persistent short vector implies a finite-volume parabolic orbit" is false.

Instead take

$$P^1 = \ker \chi,$$

which in this example is the upper-unipotent group together with a finite central part. The quotient

$$P^1(\mathbb{Q}_S)/(P^1(\mathbb{Q}_S) \cap \mathrm{SL}_2(\mathbb{Z}_S))$$

is compact because it is built from $\mathbb{Q}_S/\mathbb{Z}_S$. Every unipotent subgroup of P is contained in P^1, so the acting group survives the passage from the parabolic to its finite-volume core.

For a general standard parabolic of SL_n, the same calculation occurs block by block. The rational characters are generated by block determinants. Killing them leaves the unipotent radical and the special-linear parts of the Levi. The unipotent quotient is compact and the block SL-quotients have finite volume by Volume 3. Hence P^1 is the desired obstruction.

4. Recovering the compact recurrence alternative

Assume no such finite-volume obstruction exists. Then no primitive module can persist below ρ. Since only finitely many start below ρ, there is a radius $R_0(y, u)$ beyond which every primitive module has supremum at least ρ on every larger ball. Apply quantitative nondivergence:

$$\frac{m\{t \in B_R : \delta_S(u(t)y) < \varepsilon\}}{m(B_R)} \leq C \left(\frac{\varepsilon}{\rho}\right)^{\alpha}.$$

Choose ε so the right-hand side is less than β. The S-Mahler set $\{\delta_S \geq \varepsilon\}$ is compact and is independent of the starting point. This proves the uniformity claimed in the theorem.

5. Why this theorem feeds orbit closure

Take a minimal closed finite-volume homogeneous container Lx of a given unipotent orbit. Apply the nonescape theorem inside Lx. The algebraic alternative would produce a proper closed finite-volume subgroup containing the orbit, contradicting minimality. Therefore the empirical measures are tight. This is the exact point at which quantitative nondivergence becomes an input to a qualitative orbit-closure theorem.

CHAPTER 23

Homogeneous orbit closures

Measure classification becomes a topological theorem after two additional ingredients: nonescape of mass and exclusion of proper homogeneous limit measures. Quantitative non-divergence will provide the first; linearization provides the second.

1. The minimal homogeneous container

For $x \in X$ and a unipotent parameter subgroup U, consider the collection of rational homogeneous sets Lx that are closed, have finite volume, and contain Ux. Their dimensions are positive integers. Choose one of minimal dimension and denote it $L_x x$.

Lemma 1.1 (No proper rational trap inside the minimal container)**.** *Inside $L_x x$, the trajectory Ux is not contained in any proper closed finite-volume rational homogeneous subset.*

Proof. Such a subset would itself be an admissible homogeneous container of smaller dimension, contradicting minimality. □

2. Limit measures

Let B_R be the standard parameter balls and set

$$\mu_R = \frac{1}{m(B_R)} \int_{B_R} \delta_{u(t)x}\, dt.$$

Any weak limit of a tight subsequence is U-invariant because B_R is Følner.

Proposition 2.1 (Proper homogeneous limits are avoided)**.** *Assume the empirical measures μ_R are tight in $L_x x$. Then no weak limit gives positive mass to a proper closed finite-volume rational homogeneous subset of $L_x x$.*

Proof. Suppose a limit ν gives positive mass to a proper homogeneous set Hy. Choose a compact pure H-core carrying positive ν-mass and an inner neighborhood whose boundary has ν-measure zero. For a subsequence of R, the empirical trajectory spends a positive proportion of time in that inner tube. Proposition 3.1 then says that either a single representative persists on the whole large parameter ball, trapping Ux in a proper rational H-container, or the occupation proportion is small, modulo lower strata. Induct on $\dim H$ to eliminate the lower strata. The persistent alternative contradicts Lemma 1.1. Hence positive mass is impossible. $\square$

THEOREM 2.2 (Arithmetic S-linear orbit closure). *For every $x \in X$ and product unipotent parameter subgroup U, there is a closed finite-volume homogeneous orbit Lx such that*

$$\overline{Ux} = Lx.$$

The same holds for groups generated by unipotents in the scope of Theorem 4.2.

PROOF. Take the minimal homogeneous container $L_x x$. By the uniform nonescape theorem proved in Chapter 21, the empirical measures are tight unless the orbit lies in a smaller rational homogeneous set; minimality excludes that alternative. Let ν be a weak limit. It is U-invariant. Decompose it into U-ergodic components. By Theorem 3.1, each component is homogeneous. Proposition 2.1 excludes every proper component, so almost every component is Haar measure on $L_x x$. Hence $\nu = m_{L_x x}$.

The support of $m_{L_x x}$ is all of $L_x x$. Since ν is a weak limit of measures supported on $\overline{Ux}$, its support lies in $\overline{Ux}$. Thus $L_x x \subset \overline{Ux}$. The reverse inclusion holds by definition of the container. $\square$

CHAPTER 24

Laboratory: from invariant measures to an orbit closure

The passage from measure classification to orbit closure uses three logically separate facts: tightness, invariance of weak limits, and exclusion of proper homogeneous limits. This chapter writes the passage without suppressing any of those steps.

1. Weak limits are invariant

Let F_R be the product parameter balls and

$$\mu_R = \frac{1}{m(F_R)} \int_{F_R} \delta_{u(t)x}\, dt.$$

Fix s in the parameter group and $f \in C_c(X)$. Then

$$\int f\, d(s_*\mu_R) - \int f\, d\mu_R = \frac{1}{m(F_R)} \left(\int_{s+F_R} f(u(t)x)\, dt - \int_{F_R} f(u(t)x)\, dt \right).$$

The absolute value is at most

$$2\|f\|_\infty \frac{m((s+F_R)\triangle F_R)}{m(F_R)}.$$

For the real coordinates this boundary ratio tends to zero; for a finite-place coordinate it is eventually exactly zero because a fixed s_p belongs to the large compact subgroup ball. Hence every weak limit of μ_R is U-invariant.

2. Tightness is not automatic

Probability measures on a noncompact homogeneous space can lose mass in the cusp. The uniform nonescape theorem supplies, for every $\eta > 0$, a compact M_η such that either

$$\mu_R(M_\eta) > 1 - \eta$$

for all sufficiently large R, or the whole orbit is contained in a proper finite-volume algebraic obstruction. Inside the minimal homogeneous container the obstruction alternative is impossible. Thus the family $\{\mu_R\}$ is tight there.

Prokhorov compactness in this locally compact second-countable setting then produces probability weak limits.

3. Turning positive limit mass into positive occupation

Suppose a weak limit ν gives positive mass to a proper homogeneous stratum Hy. Choose a compact pure core $C \subset Hy$ with $\nu(C) > 0$ and then an open tube $\mathcal{O}$ around C satisfying $\nu(\partial\mathcal{O}) = 0$. The boundary-zero condition can be arranged by taking a regular value of the distance to C. Weak convergence then gives

$$\mu_{R_i}(\mathcal{O}) \longrightarrow \nu(\mathcal{O}) > 0.$$

Thus along the same subsequence the trajectory spends a definite positive proportion of time in a fixed pure stratum tube. This is the precise input to the linearization theorem; Portmanteau alone would only give one-sided inequalities without the boundary-zero choice.

4. Induction through lower strata

The pure-stratum occupation theorem says that high occupation of the H-tube has two explanations:

(1) one exterior representative persists across the large parameter ball;

(2) a significant part of the occupation lies in lower incidences.

The first makes the entire polynomial trajectory satisfy the exact H-incidence equations and hence traps Ux in a proper homogeneous container. Minimality of the chosen container excludes it. In the second case, character-core descent replaces the competing representatives by a strictly lower-dimensional finite-volume rational stratum. Repeating the argument terminates because dimension is a nonnegative integer. Therefore every proper homogeneous stratum has zero ν-mass.

5. Ergodic decomposition finishes the measure argument

Decompose

$$\nu = \int \nu_\omega \, d\vartheta(\omega)$$

into U-ergodic components. The measure-classification theorem says that each ν_ω is Haar measure on a closed finite-volume homogeneous orbit. If the ambient minimal container is Lx, every proper component is one of the proper strata just excluded. Hence

$$\nu_\omega = m_{Lx}$$

for ϑ-almost every ω, and so

$$\nu = m_{Lx}.$$

Every subsequential weak limit is therefore the same.

6. Support gives the orbit closure

Each μ_R is supported on $\overline{Ux}$. If an open set O is disjoint from $\overline{Ux}$, then $\mu_R(O) = 0$ for every R, hence every weak limit also gives O zero mass. Therefore

$$\operatorname{supp} m_{Lx} = Lx \subset \overline{Ux}.$$

The opposite inclusion holds because Lx was chosen as a closed homogeneous set containing Ux. Thus

$$\overline{Ux} = Lx.$$

Notice how little topology is left at the end: once the unique weak limit has full support on the container, the closure statement is immediate. All the work was in preventing mass escape and excluding smaller homogeneous weak limits.

CHAPTER 25

Equidistribution of unipotent parameter orbits

The orbit-closure proof already almost gives equidistribution. The only remaining issue is uniqueness of subsequential limits.

THEOREM 0.1 (Pointwise equidistribution). *Let $Lx = \overline{Ux}$ be as in Theorem 2.2. Then for every $f \in C_c(X)$,*

$$\frac{1}{m(B_R)} \int_{B_R} f(u(t)x)\, dt \longrightarrow \int_{Lx} f\, dm_{Lx}.$$

PROOF. Uniform nonescape gives tightness of the family of empirical measures. Every subsequence therefore has a further weakly convergent subsequence. The argument in the proof of Theorem 2.2 shows that every such weak limit is the same measure m_{Lx}. In a metrizable space of probability measures, if every subsequence has a further subsequence converging to the same limit, the original sequence converges to that limit. Testing against f gives the formula. □

1. Uniformity away from singular sets

For later counting applications one needs a compact-uniform version: if initial points range in a compact set disjoint from prescribed compact pieces of the singular set, then sufficiently large averages are uniformly close to Haar on the ambient homogeneous container. The proof is by contradiction. A failing sequence of initial points has a convergent subsequence; uniform nonescape yields a weak limit of empirical measures; linearization excludes the prescribed singular pieces; measure classification identifies the limit. This compactness contradiction is written out in Chapter 21 in the quantitative form needed later.

CHAPTER 26

Low-rank laboratories over $\mathbb{Q}_p$

1. The modular S-space

Take $S = \{\infty, p\}$ and

$$X = (\mathrm{SL}_2(\mathbb{R}) \times \mathrm{SL}_2(\mathbb{Q}_p)) / \mathrm{SL}_2(\mathbb{Z}[1/p]).$$

Consider the p-adic horospherical subgroup

$$U_p = \left\{ \left(e, \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \right) : t \in \mathbb{Q}_p \right\}.$$

The parameter balls $p^{-m}\mathbb{Z}_p$ are compact subgroups. Hence a U_p-generic point is literally generic under an increasing sequence of compact-group averages.

2. A rational closed orbit

Let H be the upper-triangular subgroup. The H_S-orbit of the identity coset is not finite-volume because the split torus escapes. By contrast, if $H = \mathrm{SL}_2$ there is no smaller ambient orbit. This elementary distinction is a useful warning: not every rational subgroup gives a finite-volume homogeneous set. The measure-classification theorem uses the class of rational subgroups whose arithmetic intersection is a lattice; the linearization argument may temporarily encounter larger parabolic stabilizers, but persistent parabolic trapping is interpreted as cusp escape and treated by non-divergence rather than as a finite homogeneous measure.

3. Shearing a lower unipotent displacement

With

$$u(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad g_i = \begin{pmatrix} 1 & 0 \\ p^{m_i} & 1 \end{pmatrix}, \qquad m_i \to \infty,$$

we have $|p^{m_i}|_p \to 0$. Choosing $r_i = p^{-\lfloor m_i/2 \rfloor}$ gives $|r_i|_p \to \infty$ and keeps the quadratic term in $u(r_i t) g_i u(-r_i t)$ at unit scale. Depending on the parity of m_i, a further bounded rescaling produces a limit polynomial with a nonzero upper-unipotent coefficient. Thus two nearby generic U_p-orbits create invariance in a direction transverse to the original displacement. The phenomenon is algebraically identical to real horocycle shearing but the correct times are powers of p.

4. A normalizer obstruction

If instead

$$g_i = \mathrm{diag}(1 + p^{m_i}, (1 + p^{m_i})^{-1}),$$

then g_i normalizes U_p. Conjugation merely rescales t; no transverse polynomial direction appears. This is exactly why the proof needs singular/normalizer strata rather than a blanket assertion that every nearby point yields new invariance.

CHAPTER 27

An S-arithmetic quadratic-form model

We illustrate how orbit rigidity translates into arithmetic density without using the later effective theory.

Let Q be a nondegenerate indefinite rational quadratic form in $n \geq 3$ variables and let $H = \mathrm{SO}(Q)$. At each place $v \in S$ for which Q is isotropic, $H(\mathbb{Q}_v)$ contains nontrivial one-parameter unipotent subgroups. Embed H_S in $G_S = \mathrm{SL}_n(\mathbb{Q}_S)$.

1. Values and orbit closures

For $g \in G_S$ let $Q_g(x) = Q(g^{-1}x)$. Right multiplication of g by $\Gamma = \mathrm{SL}_n(\mathbb{Z}_S)$ does not change the set of values of Q_g on $\mathbb{Z}_S^n$. Left multiplication by H_S does not change Q_g at all. Thus arithmetic information about $Q_g(\mathbb{Z}_S^n)$ is encoded by the H_S-orbit of $g\Gamma$.

If a unipotent subgroup $U < H_S$ has dense orbit in the homogeneous closure $Lg\Gamma$, then continuous functions of lattice vectors sampled along U inherit density from that orbit. In particular, if $L = G_S$, the deformed lattice configurations approximate arbitrary local configurations, yielding density of appropriate S-adic value sets subject to the obvious local constraints.

2. The rational obstruction

If Q_g is proportional to a rational form over $\mathbb{Q}$, the stabilizer is rational and the H_S-orbit may close with finite volume. Linearization detects precisely this case through the rational top wedge of $\mathfrak{h}$. Thus the familiar dichotomy

$$\text{irrational form } \Rightarrow \text{large orbit,}$$

$$\text{rational form } \Rightarrow \text{closed homogeneous obstruction.}$$

is not an analogy: it is the singular-stratum alternative of the proof translated back into quadratic-form language.

Warning 2.1. The exact Oppenheim theorem over S depends on local isotropy and on the precise arithmetic notion of irrationality. This chapter is a worked dynamical model, not a substitute for the later theorem devoted to value-counting and its exceptional cases.

CHAPTER 28

The S-arithmetic Dani correspondence: a model calculation

Take $S = \{\infty, p\}$ and a pair $y = (y_\infty, y_p) \in \mathbb{R} \times \mathbb{Q}_p$. Put

$$u_y = \left(\begin{pmatrix} 1 & y_\infty \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & y_p \\ 0 & 1 \end{pmatrix} \right).$$

For $(q, r) \in \mathbb{Z}[1/p]^2$,

$$u_y \begin{pmatrix} r \\ q \end{pmatrix} = \left(\begin{pmatrix} r + qy_\infty \\ q \end{pmatrix}, \begin{pmatrix} r + qy_p \\ q \end{pmatrix} \right).$$

A diagonal element

$$a = \Big(\operatorname{diag}(e^t, e^{-t}), \operatorname{diag}(p^m, p^{-m}) \Big)$$

weights the approximation errors $r + qy_v$ against the denominator q differently at the two places. The product content of the resulting vector is

$$\max(e^t |r + qy_\infty|, e^{-t}|q|) \max(p^{-m}|r + qy_p|_p, p^m |q|_p).$$

Thus a simultaneous real/p-adic rational approximation is equivalent to a short vector in an S-lattice after a suitable diagonal balancing. Volume 3's S-unit theorem is what allows the denominator to be normalized across places.

Quantitative non-divergence now says that for a polynomial or nondegenerate family of y's, parameters producing abnormally short S-vectors occupy a power-small proportion unless a rational submodule remains uniformly small. That rational module is the Diophantine obstruction. This is the exact S-arithmetic analogue of the real Dani–Kleinbock–Margulis dictionary developed in Volume 1.

CHAPTER 29

Reader workshop: the rigidity chain in $\mathrm{SL}_2(\mathbb{Q}_p)$

We compress the qualitative proof into one concrete model without hiding the relative correction step.

Let $U = \{u(t) : t \in \mathbb{Q}_p\}$ act ergodically on an invariant probability measure μ on an arithmetic quotient containing an $\mathrm{SL}_2(\mathbb{Q}_p)$ factor, and put

$$L = \langle U, I_{\mathrm{an}} \rangle.$$

Here I_{an} denotes the open local analytic subgroup generated by a sufficiently small identity neighborhood in $I(\mu)$, as defined in Chapter 12; thus $U \subset L$ and $\operatorname{Lie} L = \operatorname{Lie} I(\mu)$.

1. Choose uniform generic cores. Theorem 1.2 supplies genericity along expanding balls $p^{-m}\mathbb{Z}_p$. Egorov gives compact cores on which a finite family of centered averages is uniformly generic beyond one radius. The nearby pairs are chosen recursively so that their shearing scales exceed these thresholds.

2. Produce a transverse nearby pair. If no L-orbit is conull, Lemma 1.1 gives generic points

$$x_i = g_i x \to x, \qquad g_i \to e, \qquad g_i \notin L.$$

No local-orbit-partition argument is used.

3. Shear and normalize. Write $g_i = I + X_i + o(X_i)$. The adjoint action of $u(t)$ on $\mathfrak{sl}_2$ has degrees at most two. At the first scale where a positive- degree coefficient becomes macroscopic, the Chevalley/quasi-regular construction produces a normalizer limit.

4. Correct a limit already in the stabilizer. If the visible normalizer direction already belongs to L, remove it with $n_i \in N_L(U)$. Since

$$n_i^{-1} u(t) n_i = u(\chi(n_i)t),$$

this is a scalar p-adic time change. On a small tame ball the absolute Jacobian is constant, so centered-ball genericity transfers exactly. Removing the visible coefficient lowers the active degree. In $\mathfrak{sl}_2$ only finitely many corrections are possible.

5. The regular terminal branch is impossible. Repeat first-exit normalization with group radii tending to zero. A regular terminal branch gives $h_j \in I(\mu)$, $h_j \to e$, whose normalized logarithms have a nonzero component

outside $\mathfrak{l}$. But the tangent cone of the closed local Lie subgroup $I(\mu)$ at the identity is $\mathfrak{l}$. Contradiction.

6. The completely absorbed branch is arithmetic singularity. If all visible coefficients are absorbed, the initial coefficient tuple satisfies a proper primitive polynomial incidence relation. Choose common returns to a compact injectivity core. Lifting the return through G/Γ introduces an arithmetic lattice element and hence a primitive S-integral exterior class. Finite-window representative control and polynomial goodness say that either the inner singular occupation is small, a representative persists, or one descends to a lower intersection. Persistence through shrinking tubes gives an exact proper rational singular stratum.

7. Finish by the rigidity alternative. If every proper singular stratum has zero measure, Step 6 is excluded and Step 5 is contradictory, so the assumption in Step 2 was false: an L-orbit is conull. Its invariant probability is Haar on a closed finite-volume orbit. If a proper singular stratum has positive measure, the proof instead descends to the smaller homogeneous problem and applies induction on ambient dimension.

Every essential mechanism of the general theorem is visible here: p-adic centered-ball genericity, polynomial shearing, quasi-regular normalizer limits, controlled normalizer time changes, arithmetic exterior linearization, and the separation between finite degree correction and ambient-dimension descent.

CHAPTER 30

Exercises and proof laboratories

Exercise 0.1. For $u(t) = \exp(tN)$ in $\mathrm{SL}_n(\mathbb{Q}_p)$ compute an explicit degree bound for $\mathrm{Ad}(u(t))$ on $\mathfrak{sl}_n$ in terms of the nilpotency index of N.

Exercise 0.2. Prove directly that if a polynomial $P \in \mathbb{Q}_p[t]$ is bounded on balls of arbitrarily large radius, then P is constant.

Exercise 0.3. In the setting of Lemma 2.1, show that quotienting by S-units is necessary by constructing infinitely many integral representatives of the same projective exterior line.

Exercise 0.4. Write out the product marked-poset induction in rank two, where the only primitive ranks are one and two, and compute an explicit structural constant.

Exercise 0.5. Show that the empirical measures over $p^{-m}\mathbb{Z}_p$ are exactly invariant under $p^{-r}\mathbb{Z}_p$ for every fixed r once $m \geq r$. Compare this with the boundary error for real intervals.

Research Laboratory 0.6. Extend the proof of Theorem 3.1 from SL_n to a split symplectic group using the strong approximation and root-group material of Volume 3. Identify the changes in the rational exterior representative and in the class of finite-volume subgroups.

APPENDIX A

Standard background not reproved

The volume is self-contained with respect to the post-Ratner local-field rigidity mechanism, but it does not reprove the ordinary foundations of measure theory, Lie theory, and arithmetic algebraic groups. We use the following standard results.

- Radon-measure regularity, Prokhorov compactness on locally compact second-countable spaces, ergodic decomposition, Egorov's theorem, the reverse martingale convergence theorem, and the ordinary real pointwise ergodic theorem, together with the standard L^2 maximal ergodic inequality for centered interval (and finite-dimensional ball) averages.
- Basic unitary representation theory, including the Mautner contraction principle in the form used for simple local factors and the joint spectral theorem for abelian locally compact groups.
- Standard characteristic-zero Lie theory: exponential coordinates near the identity, closed-subgroup/Lie-algebra correspondence, Levi decomposition, and basic structure of semisimple and unipotent algebraic groups.
- Basic affine algebraic-group structure (unipotent radicals, Levi decomposition, rational parabolics, relative roots, and finite central isogenies). The Chevalley projective-stabilizer construction itself was proved in Volume 2 and is referenced backward when used here.

All S-arithmetic reduction, S-unit balancing, local polynomial goodness, and strong approximation for the matrix groups actually used here were proved in Volume 3. The general matrix Borel–Harish–Chandra criterion needed for rational rigidity strata is proved in Chapter 2, including its reduction and Haar-integrability steps. The Margulis–Tomanov invariant-measure theorem and the Kleinbock–Tomanov quantitative theorem are *not* logical inputs: the arithmetic matrix/product-parameter forms used in this series are reconstructed in this volume.

APPENDIX B

Historical sources and theorem checks

The qualitative local-field measure-classification theorem was established by G. A. Margulis and G. M. Tomanov, *Invariant measures for actions of unipotent groups over local fields on homogeneous spaces*, Inventiones Mathematicae 116 (1994), 347–392, with subsequent almost-linear refinements. The quantitative non-divergence framework is due to D. Kleinbock and G. Tomanov, *Flows on S-arithmetic homogeneous spaces and applications to metric Diophantine approximation.*

These sources determine attribution and the target scope. In this reconstructed volume the local ergodic, shearing, absolute and relative quasi-regular, finite-correction, linearization, marking, and compactness mechanisms are rebuilt in the arithmetic matrix setting from written proofs in this volume and Volumes 1–3. The former relative-degeneration seam is closed in Chapter 12; the Margulis–Tomanov theorem is not used as a classification input. Stronger almost-linear and arbitrary-number-field formulations are likewise not used as logical inputs.

Scope summary

The local and product ergodic theorem, polynomial H-property, absolute and relative quasi-regular mechanisms, finite normalizer correction, arithmetic linearization and singular descent, measure classification, product marking, quantitative nondivergence, uniform nonescape, orbit closure, and pointwise equidistribution are proved in the arithmetic matrix scope stated at the beginning of this volume.

The Borel–Harish–Chandra reduction argument and the unipotent-ergodic arithmetic-hull bridge are also proved in that scope. At a nonarchimedean place the stabilizer bookkeeping uses the open local stabilizer core rather than a topological identity component. In the relative quasi-regular argument, normalizer corrections are implemented as controlled time changes; the completely absorbed branch becomes rational only after arithmetic recurrence supplies primitive exterior data.

The stronger Margulis–Tomanov formulations for arbitrary almost-linear local-field groups are historical context only. Applications outside the matrix/direct-product scope written here require a separate structural reduction.

Volume 5

Unitary Representations and Mixing

Preface and scope

This volume is the first volume of the analytic arc following the completed Ratner core. Its purpose is to own, rather than merely cite, the qualitative representation-theoretic mechanism that later volumes quantify. For the roadmap below, $\pi \prec \rho$ (*weak containment*) means that every diagonal matrix coefficient of π is uniformly approximable on compact subsets by finite convex combinations of diagonal coefficients of ρ; a unitary representation is *tempered* when $\pi \prec \lambda_G$, the left regular representation. Chapter 6 gives the normalized coefficient formulation used in proofs. The load-bearing chain is

$$\begin{aligned}\text{unitary coefficients} &\Longrightarrow \text{Mautner envelopes} \Longrightarrow \text{Howe–Moore}\\ &\Longrightarrow \text{Moore ergodicity and mixing,}\end{aligned}$$

and, independently,

$$\begin{aligned}\text{weak containment} &\Longrightarrow \text{temperedness}\\ &\Longrightarrow \text{Harish–Chandra/CHH majorization.}\end{aligned}$$

The foundational floor not reproved from first principles consists of elementary Hilbert-space theory (Riesz representation and orthogonal projections), Stone–Weierstrass and the Riesz representation theorem for positive functionals on $C(K)$, Zorn's lemma, Hopf–Rinow in finite-dimensional Riemannian geometry, and routine finite-dimensional real semisimple Lie theory. The global Cartan/Iwasawa decompositions, restricted-root contraction and generation statements, weak-containment transfer, Harish–Chandra chamber estimate, and CHH majorization actually used in the sequel are isolated and proved in the chapters or Appendices A–C.

Warning 0.1. Qualitative mixing and quantitative matrix-coefficient decay are deliberately separated. Howe–Moore proves vanishing at infinity but gives no numerical rate. The Ξ-majorization results in Chapter 7 give quantitative envelopes only after a representation is known to be tempered or tensor-tempered. Spectral gap and Property (T) are owned by Volume 6, not silently used here.

Student orientation: representations as dynamical measuring devices

A unitary representation converts orbit questions into decay statements for matrix coefficients $\langle \pi(g)v, w\rangle$. *Weak containment* means that matrix coefficients of one representation can be approximated, on compact sets, by convex combinations of those of another. A representation is *tempered* when it is weakly contained in the regular representation. The *Mautner phenomenon* upgrades invariance under a contracting subgroup to invariance under the element doing the contraction. *Howe–Moore* is the vanishing of matrix coefficients at infinity once invariant vectors for noncompact normal factors are absent. Read the volume as a chain: spectral language → contraction rigidity → vanishing → quantitative majorization → mixing on quotients.

Reader guide: a concrete model

Why this matters.

Volume 5 is organized around the passage from *Unitary Representations, Cyclic Decomposition, and Spectral Measures* to *Master-Ledger Map and Proof Audit.* The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.2 (A calculation to keep in view)**.** Consider the standard unipotent and diagonal elements

$$u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad a_s = \begin{pmatrix} e^s & 0 \\ 0 & e^{-s} \end{pmatrix}.$$

A direct multiplication gives

$$a_s u_t a_{-s} = u_{e^{2s}t}.$$

Thus a displacement of size $|t|$ in the unstable horospherical direction is enlarged by e^{2s}. If $|t| = e^{-2s}$, the conjugated displacement has order one. This elementary calculation is the model for the "first visible scale" used repeatedly in nondivergence and shearing arguments.

CHAPTER 1

Unitary Representations, Cyclic Decomposition, and Spectral Measures

Proof architecture. The chapter passes through four increasingly concrete descriptions of a unitary representation: orbit maps, matrix coefficients, cyclic representations, and spectral measures. The spectral theorem is proved in the cyclic form actually used later. No global classification of the unitary dual is needed.

1. Continuous unitary representations

Let G be a second-countable locally compact group and $\mathcal{H}$ a complex Hilbert space. A unitary representation is a homomorphism

$$\pi : G \longrightarrow U(\mathcal{H})$$

for which every orbit map $g \mapsto \pi(g)v$ is norm-continuous.

Lemma 1.1 (Weak continuity is enough)**.** *Suppose $\pi(g)$ is unitary for every g and, for every $v, w \in \mathcal{H}$, the coefficient $g \mapsto \langle \pi(g)v, w\rangle$ is continuous. Then π is strongly continuous.*

Proof. Fix $v \in \mathcal{H}$ and $g_n \to g$. Since $\pi(g)$ is unitary,

$$\|\pi(g_n)v - \pi(g)v\|^2 = 2\,\|v\|^2 - 2\,\mathrm{Re}\langle \pi(g^{-1}g_n)v, v\rangle.$$

The assumed coefficient continuity makes the last inner product tend to $\|v\|^2$. Hence the displayed norm tends to zero. Sequential continuity is continuity because G is first countable. □

Proposition 1.2 (Invariant complements)**.** *If $V \subset \mathcal{H}$ is a closed $\pi(G)$-invariant subspace, then $V^{\perp}$ is also invariant and the orthogonal projection P_V commutes with $\pi(g)$ for every $g \in G$.*

Proof. If $w \in V^{\perp}$ and $v \in V$, then

$$\langle \pi(g)w, v\rangle = \langle w, \pi(g^{-1})v\rangle = 0,$$

so $\pi(g)w \in V^\perp$. The decomposition $\mathcal{H} = V \oplus V^\perp$ is therefore invariant, and the commuting statement follows by applying $\pi(g)$ to the two orthogonal components of any vector. □

2. Cyclic subspaces and orthogonal decomposition

For $v \in \mathcal{H}$ define

$$\mathcal{H}_v = \overline{\operatorname{span}\{\pi(g)v : g \in G\}}.$$

It is the cyclic subrepresentation generated by v.

THEOREM 2.1 (Orthogonal cyclic decomposition). *If $\mathcal{H}$ is separable, there are vectors $v_1, v_2, \dots$ such that*

$$\mathcal{H} = \bigoplus_{j \geq 1} \mathcal{H}_{v_j}$$

orthogonally. The sum is finite exactly when the construction terminates after finitely many steps.

PROOF. Consider the collection of families of pairwise orthogonal nonzero cyclic invariant subspaces, ordered by inclusion. A union of a chain is again such a family, so Zorn's lemma gives a maximal family $\{\mathcal{H}_{v_j}\}_{j \in J}$. The index set J is countable: after normalizing one vector in each summand, the resulting vectors have pairwise distance $\sqrt{2}$, and disjoint balls of radius $1/3$ around them inject into any fixed countable dense subset of the separable Hilbert space. Thus enumerate the family as $v_1, v_2, \dots$ (or finitely).

Let W be the closed orthogonal sum of these cyclic spaces. Proposition 1.2 makes $W^\perp$ invariant. If $W^\perp \neq 0$, any nonzero $v \in W^\perp$ generates a nonzero cyclic invariant subspace orthogonal to every member of the maximal family, contradicting maximality. Hence $W = \mathcal{H}$. □

3. Positive-definite coefficients and the GNS construction

A function $\varphi : G \to \mathbb{C}$ is positive definite if for every $g_1, \dots, g_n \in G$ and $c_1, \dots, c_n \in \mathbb{C}$,

$$\sum_{i,j} c_i \overline{c_j}\, \varphi(g_j^{-1} g_i) \geq 0.$$

Every diagonal coefficient $\varphi_v(g) = \langle \pi(g)v, v \rangle$ is positive definite.

THEOREM 3.1 (GNS construction for groups). *Let $\varphi : G \to \mathbb{C}$ be continuous, positive definite, and $\varphi(e) = 1$. There exist a Hilbert space $\mathcal{H}_\varphi$, a strongly continuous unitary representation π_φ, and a cyclic unit vector ξ_φ such that*

$$\varphi(g) = \langle \pi_\varphi(g)\xi_\varphi, \xi_\varphi \rangle.$$

The cyclic triple is unique up to a unique unitary intertwiner carrying one cyclic vector to the other.

PROOF. Let $\mathbb{C}[G]_{\mathrm{fin}}$ be the vector space of finitely supported formal sums $f = \sum_g c_g \delta_g$. Put

$$\langle f, h\rangle_\varphi = \sum_{g,k} c_g \overline{d_k}\, \varphi(k^{-1}g), \qquad h = \sum_k d_k \delta_k.$$

Positive definiteness makes this positive semidefinite. Quotient by the null space and complete to obtain $\mathcal{H}_\varphi$.

Left translation $L_a \delta_g = \delta_{ag}$ preserves the form because

$$\varphi((ak)^{-1}(ag)) = \varphi(k^{-1}g).$$

It therefore induces a unitary $\pi_\varphi(a)$. Let $\xi_\varphi = [\delta_e]$. Then $\|\xi_\varphi\| = 1$ and

$$\langle \pi_\varphi(g)\xi_\varphi, \xi_\varphi\rangle = \langle \delta_g, \delta_e\rangle_\varphi = \varphi(g).$$

The orbit of ξ_φ spans the image of all finitely supported functions and is dense, so the representation is cyclic.

To prove strong continuity, it is enough by density and unitarity to check vectors $\pi_\varphi(a)\xi_\varphi$. For the cyclic vector,

$$\|\pi_\varphi(g)\xi_\varphi - \xi_\varphi\|^2 = 2 - 2\,\mathrm{Re}\,\varphi(g) \to 0$$

as $g \to e$. Translating this identity gives continuity on the orbit span, then the usual three-ε argument gives it on the completion.

For uniqueness, define on the dense cyclic span

$$U\Big(\sum_j c_j \pi_\varphi(g_j)\xi_\varphi\Big) = \sum_j c_j \pi'(g_j)\xi'.$$

The equality of diagonal coefficients, together with polarization, makes all Gram matrices equal, so U is a well-defined isometry. It extends to a unitary intertwiner, and cyclicity makes it surjective and unique. □

4. Herglotz theorem and spectral measure for one unitary operator

We identify $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$. The next theorem is the exact spectral-measure result used to calibrate weak mixing.

THEOREM 4.1 (Herglotz representation). *A sequence $(a_n)_{n\in\mathbb{Z}}$ with $a_{-n} = \overline{a_n}$ is positive definite, in the sense that*

$$\sum_{i,j=0}^{N} c_i \overline{c_j} a_{i-j} \geq 0,$$

if and only if there is a unique finite positive Borel measure σ on $\mathbb{T}$ satisfying

$$a_n = \int_{\mathbb{T}} z^n \, d\sigma(z) \qquad (n \in \mathbb{Z}).$$

Moreover $\sigma(\mathbb{T}) = a_0$.

Proof. The "if" direction follows by integrating

$$\sum_{i,j} c_i \overline{c_j} z^{i-j} = \left| \sum_i c_i z^i \right|^2 .$$

Conversely, define a linear functional L on trigonometric polynomials by $L(z^n) = a_n$. If $p(z) = \sum_{k=-N}^{N} c_k z^k$ is nonnegative on $\mathbb{T}$, the Fejér–Riesz factorization gives $p(z) = |q(z)|^2$ for a polynomial q. We prove the factorization briefly. Multiply by z^N to obtain a polynomial P satisfying the reciprocal symmetry $P(z) = z^{2N}\overline{P(1/\bar{z})}$; zeros off the circle occur in reciprocal-conjugate pairs, while zeros on the circle have even multiplicity because $p \geq 0$. Choosing one zero from each reciprocal pair gives q, up to a positive scalar. Hence positivity of the Toeplitz form gives $L(p) \geq 0$.

For a real trigonometric polynomial f, $-\|f\|_\infty \leq f \leq \|f\|_\infty$, so positivity gives $|L(f)| \leq a_0 \|f\|_\infty$. By complex decomposition the same boundedness holds on all trigonometric polynomials. Stone–Weierstrass makes these uniformly dense in $C(\mathbb{T})$, so L extends uniquely to a positive bounded functional on $C(\mathbb{T})$. The Riesz representation theorem gives a unique finite positive measure σ, and the moment formula follows by construction. Taking $n = 0$ gives the mass. □

Theorem 4.2 (Cyclic spectral theorem for a unitary operator). *Let U be unitary on $\mathcal{H}$ and let v be cyclic for the $\mathbb{Z}$-representation generated by U. There is a unique finite measure σ_v on $\mathbb{T}$ and a unitary*

$$W : \mathcal{H} \longrightarrow L^2(\mathbb{T}, \sigma_v)$$

such that $Wv = 1$ and

$$WUW^{-1} = M_z,$$

where $M_z f(z) = z f(z)$. The measure is characterized by

$$\langle U^n v, v \rangle = \int_{\mathbb{T}} z^n \, d\sigma_v(z).$$

Proof. The sequence $a_n = \langle U^n v, v \rangle$ is positive definite, so Theorem 4.1 gives σ_v. On Laurent polynomials define

$$W\Big(\sum_{n=-N}^{N} c_n U^n v \Big) = \sum_{n=-N}^{N} c_n z^n.$$

For two Laurent polynomials, expanding the inner products and using the moment identity shows that W preserves inner products. Cyclicity says the source is dense. Laurent polynomials are dense in $L^2(\sigma_v)$ because continuous functions are dense in L^2 and trigonometric polynomials are uniformly dense in $C(\mathbb{T})$. Thus W extends to a unitary. The intertwining relation holds on the dense Laurent-polynomial subspace and hence everywhere. $\square$

Corollary 4.3 (Vector spectral measures)**.** *For any unitary U and $v \in \mathcal{H}$ there is a unique finite positive measure σ_v with*

$$\langle U^n v, v\rangle = \widehat{\sigma_v}(n).$$

For $v, w \in \mathcal{H}$ there is a unique finite complex measure $\sigma_{v,w}$ satisfying

$$\langle U^n v, w\rangle = \int z^n \, d\sigma_{v,w}(z),$$

and it is recovered from the diagonal measures by polarization.

Proof. Apply Theorem 4.2 to the cyclic space $\mathcal{H}_v$. For the mixed measure use

$$4\sigma_{v,w} = \sum_{j=0}^{3} i^j \sigma_{v+i^j w},$$

with the convention matching the chosen inner-product linearity. Polarization of coefficients proves the desired identity and uniqueness follows from uniqueness of Fourier coefficients of finite measures. $\square$

5. Worked examples

Worked Example 5.1 (Rotation spectrum)**.** For $R_\alpha : x \mapsto x + \alpha$ on $\mathbb{T}$, the Koopman operator satisfies

$$Ue_n = e^{2\pi i n\alpha} e_n, \qquad e_n(x) = e^{2\pi i n x}.$$

Thus σ_{e_n} is the point mass at $e^{2\pi i n\alpha}$. The system is therefore spectrally pure point; this will explain why irrational rotations are ergodic but not weakly mixing.

Worked Example 5.2 (Bilateral shift)**.** Let U be the bilateral shift on $\ell^2(\mathbb{Z})$ and $v = \delta_0$. Then $\langle U^n v, v\rangle = 0$ for $n \neq 0$ and equals 1 for $n = 0$. Hence σ_v is normalized Haar measure on $\mathbb{T}$. Under the Fourier transform, U becomes multiplication by z. This is the model continuous spectrum.

6. Dependency summary

Dependency Note 6.1 (V05.C01.L)**.** Weak/strong continuity, invariant complements, and orthogonal cyclic decomposition are proved in Lemma 1.1, Proposition 1.2, and Theorem 2.1.

Dependency Note 6.2 (V05.C01.T)**.** The cyclic spectral theorem is Theorem 4.2, with Herglotz proved first rather than cited.

Dependency Note 6.3 (V05.C01.P)**.** The GNS construction and vector/mixed spectral measures, Theorem 3.1 and Corollary 4.3, supply the downstream coefficient-to-measure transfer.

> **Proof and provenance.** No semisimple-group structure is used in this chapter. The only background inputs are standard Hilbert-space completeness, Stone–Weierstrass, and the Riesz representation theorem for positive functionals on $C(\mathbb{T})$. These are foundational analysis rather than homogeneous-dynamics proof obligations.

7. Reader perspective

The chapter begins with concrete coefficient identities before introducing spectral measures; the GNS and Herglotz constructions are proved at the point they first become necessary. The two worked examples separate pure-point from continuous spectrum. No classification of the unitary dual is mentioned as if it were needed for later arguments. The only compression left is foundational functional analysis explicitly listed in the volume stated scope and prerequisites.

CHAPTER 2

Spectral Criteria for Ergodicity, Weak Mixing, and Mixing

Proof architecture. A probability-preserving action becomes a unitary representation on L^2. Ergodicity is an exact fixed-vector statement, weak mixing is a tensor/spectral statement, and mixing is a C_0 statement for matrix coefficients. Keeping these three levels separate prevents several common false implications.

1. Koopman representations

Let G act measurably on a probability space (X, μ) preserving μ. For a right action we use

$$(\pi(g)f)(x) = f(xg).$$

The convention changes inverses but none of the invariant-subspace statements.

Lemma 1.1 (Strong continuity of the Koopman representation)**.** *Assume G acts continuously on a locally compact second-countable space X, preserves a Radon probability μ, and $C_c(X)$ is dense in $L^2(X, \mu)$. Then the Koopman representation is strongly continuous.*

Proof. For $f \in C_c(X)$ and $g \to e$, continuity of the action and compactness of a neighborhood of $\operatorname{supp} f$ imply $f(\,\cdot\, g) \to f$ uniformly on a compact set containing all supports for g sufficiently close to e. Hence convergence holds in L^2. For general $f \in L^2$, choose $h \in C_c(X)$ with $\|f - h\|_2 < \varepsilon$. Since every $\pi(g)$ is unitary,

$$\|\pi(g)f - f\|_2 \leq 2\varepsilon + \|\pi(g)h - h\|_2\,,$$

and let first $g \to e$, then $\varepsilon \to 0$. □

Let $L^2_0(X)$ denote the orthogonal complement of the constants.

Proposition 1.2 (Ergodicity and fixed vectors)**.** *The action is ergodic if and only if*

$$L^2_0(X)^G = \{0\}.$$

More generally, a subgroup $H < G$ acts ergodically if and only if $L_0^2(X)^H = \{0\}$.

Proof. If A is invariant modulo null sets, then $\mathbf{1}_A - \mu(A)$ is an invariant mean-zero vector; if the action is ergodic this vector is zero. Conversely, if $f \in L^2$ is invariant, then for every real t the set $\{\Re f > t\}$ is invariant. Ergodicity forces its measure to be 0 or 1. The distribution function of $\Re f$ therefore has a single jump, so $\Re f$ is almost surely constant; the same holds for $\Im f$. Thus the fixed vectors are exactly the constants. $\square$

2. Mixing as coefficient decay

Theorem 2.1 (Coefficient criterion for mixing). *A probability-preserving G-action is mixing, in the sense that*

$$\mu(Ag \cap B) \longrightarrow \mu(A)\mu(B) \quad (g \to \infty),$$

for all measurable A, B, if and only if every coefficient of the Koopman representation on $L_0^2(X)$ belongs to $C_0(G)$:

$$\langle \pi(g)f, h\rangle \longrightarrow 0 \quad (g \to \infty), \qquad f, h \in L_0^2(X).$$

Proof. For indicator functions,

$$\langle \pi(g)(\mathbf{1}_A - \mu A), \mathbf{1}_B - \mu B\rangle = \mu(Ag^{-1} \cap B) - \mu(A)\mu(B),$$

up to the harmless inverse dictated by the action convention. Thus coefficient decay implies set mixing.

Conversely assume set mixing. Finite linear combinations of indicator functions are dense in L^2. For bounded simple mean-zero f, h, bilinearity expresses the coefficient as a finite sum of mixing differences and hence gives convergence to zero. For general $f, h \in L_0^2$, approximate by bounded simple functions f_0, h_0 with mean zero. Unitarity gives uniformly in g

$$|\langle \pi(g)f, h\rangle - \langle \pi(g)f_0, h_0\rangle| \le \|f - f_0\|_2 \|h\|_2 + \|f_0\|_2 \|h - h_0\|_2 .$$

The right side can be made arbitrarily small before sending $g \to \infty$. $\square$

3. Weak mixing

For a single invertible transformation T, write U for its Koopman operator on $L_0^2(X)$.

Theorem 3.1 (Weak-mixing equivalences). *For an invertible probability-preserving transformation T, the following are equivalent.*

(1) $T \times T$ is ergodic.

(2) U has no nonzero finite-dimensional invariant subspace.

(3) For every $f, h \in L^2_0(X)$,

$$\frac{1}{N}\sum_{n=0}^{N-1} |\langle U^n f, h\rangle|^2 \longrightarrow 0.$$

(4) For every $f, h \in L^2_0(X)$,

$$\frac{1}{N}\sum_{n=0}^{N-1} |\langle U^n f, h\rangle| \longrightarrow 0.$$

Any of these conditions is called weak mixing.

Proof. The Koopman representation of $T\times T$ is $U\otimes U$ on $L^2(X)\widehat{\otimes}L^2(X)$. We spell out the passage from invariant kernels to finite-dimensional invariant subspaces, because replacing $U\otimes U$ silently by $U\otimes\overline{U}$ is a common bookkeeping mistake.

Suppose first that $T\times T$ is not ergodic. If T itself is not ergodic, Proposition 1.2 gives a nonzero fixed vector in $\mathcal{H} = L^2_0(X)$, already a one-dimensional invariant subspace. We may therefore assume that T is ergodic. In that case the only invariant vectors in the constant and one-coordinate summands of

$$L^2(X\times X) = \mathbb{C}\mathbf{1} \oplus (\mathcal{H}\otimes\mathbf{1}) \oplus (\mathbf{1}\otimes\mathcal{H}) \oplus (\mathcal{H}\widehat{\otimes}\mathcal{H})$$

are constants. Since the product is not ergodic, subtracting the constant part leaves a nonzero $F \in \mathcal{H}\widehat{\otimes}\mathcal{H}$ with

$$F(Tx, Ty) = F(x, y) \quad \text{a.e.}$$

Define the Hilbert–Schmidt operator

$$(A\phi)(x) = \int_X F(x, y)\phi(y)\, d\mu(y).$$

(The absence of a conjugate in this formula is deliberate; it makes A complex-linear.) A change of variables gives

$$AU = UA.$$

Indeed,

$$\begin{aligned}(AU\phi)(x) &= \int F(x, y)\phi(Ty)\, d\mu(y)\\ &= \int F(x, T^{-1}z)\phi(z)\, d\mu(z) = \int F(Tx, z)\phi(z)\, d\mu(z) = (UA\phi)(x).\end{aligned}$$

Since $A \neq 0$ is compact, A^*A is a nonzero compact positive operator commuting with U. Choose an isolated positive spectral value of A^*A; its spectral projection has nonzero finite-dimensional range and commutes with U. Thus U has a nonzero finite-dimensional invariant subspace.

Conversely, suppose $V \subset \mathcal{H}$ is a nonzero finite-dimensional U-invariant subspace and let $e_1, \dots, e_d$ be an orthonormal basis. The reproducing kernel

$$F(x,y) = \sum_{j=1}^{d} e_j(x)\overline{e_j(y)}$$

is nonzero and belongs to $L^2(X \times X)$. Because $U|_V$ is unitary, changing from the basis (e_j) to the orthonormal basis (Ue_j) leaves this kernel unchanged, so

$$F(Tx, Ty) = F(x,y).$$

It is orthogonal to constants because each $e_j \in L^2_0(X)$. Hence $T \times T$ is not ergodic. This proves (1)⇔(2) without identifying two different tensor representations.

Fix f, h and let σ_f be the spectral measure from Chapter 1. By polarization the mixed correlations are Fourier coefficients of a finite complex measure. We first handle $h = f$. Wiener's lemma, proved below, states

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} |\widehat{\sigma}_f(n)|^2 = \sum_{z\in\mathbb{T}} \sigma_f(\{z\})^2. \tag{2.1}$$

An atom at z corresponds under the cyclic spectral theorem to a nonzero eigenvector with eigenvalue z, hence to a one-dimensional invariant subspace. Therefore (2) is equivalent to atomlessness of every σ_f, and (2.1) gives the squared-correlation decay for $h = f$. Polarization and Cauchy–Schwarz give (3) for arbitrary f, h. Conversely, a finite-dimensional invariant subspace has an eigenvector because a commuting single unitary operator on a finite-dimensional complex space is diagonalizable, and its self-correlation has constant modulus, contradicting (3). So (2)⇔(3). Finally (3)⇒(4) is Cauchy–Schwarz, and (4)⇒(3) follows from $|\langle U^n f, h\rangle| \le \|f\|\,\|h\|$. $\square$

Lemma 3.2 (Wiener lemma)**.** *If σ is a finite positive measure on $\mathbb{T}$, then*

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} |\widehat{\sigma}(n)|^2 = \sum_{z\in\mathbb{T}} \sigma(\{z\})^2.$$

Proof. Expand and use Fubini:

$$\frac{1}{N} \sum_{n=0}^{N-1} |\widehat{\sigma}(n)|^2 = \int_{\mathbb{T}^2} K_N(z\bar{w})\, d\sigma(z) d\sigma(w),$$

where $K_N(\zeta) = N^{-1}\sum_{n=0}^{N-1} \zeta^n$. We have $|K_N| \le 1$, $K_N(1) = 1$, and $K_N(\zeta) \to 0$ for $\zeta \ne 1$. Dominated convergence on $\mathbb{T}^2$ gives the mass of the diagonal under $\sigma \otimes \sigma$. A finite positive measure has at most countably many atoms, and the diagonal mass equals $\sum_z \sigma(\{z\})^2$. $\square$

Corollary 3.3 (Spectral characterization)**.** *For an ergodic transformation, weak mixing is equivalent to every spectral measure on L^2_0 being atomless. Mixing is equivalent to every mixed spectral measure having Fourier–Stieltjes coefficients tending to zero.*

Proof. The weak-mixing assertion was proved inside Theorem 3.1. The mixing assertion is exactly Theorem 2.1 rewritten using Corollary 4.3. ☐

4. Examples and failure modes

Worked Example 4.1 (Irrational rotation)**.** If $T(x) = x + \alpha$ with $\alpha \notin \mathbb{Q}$, Fourier characters show ergodicity, but each e_n is an eigenfunction. Hence spectral measures on L^2_0 have atoms and the action is not weakly mixing, therefore not mixing.

Worked Example 4.2 (Bernoulli shift)**.** Let $X = A^{\mathbb{Z}}$ with product probability and shift T. Cylinder functions depending on disjoint coordinate blocks are independent. If f, h are cylinder functions with zero mean, then $\langle U^n f, h\rangle = 0$ for all sufficiently large $|n|$. Density of cylinder functions in L^2 and unitarity therefore imply mixing for all L^2 functions.

Warning 4.3. Ergodicity, weak mixing, and mixing are genuinely different. Later Howe–Moore arguments prove mixing by showing all mean-zero coefficients vanish at infinity; they do not proceed by upgrading ergodicity through an abstract implication.

5. Dependency summary

Dependency Note 5.1 (V05.C02.L)**.** Proposition 1.2 and Lemma 1.1 own the Koopman/fixed-vector reduction.

Dependency Note 5.2 (V05.C02.T)**.** Theorems 2.1 and 3.1 prove the exact mixing and weak-mixing criteria.

Dependency Note 5.3 (V05.C02.P)**.** Corollary 3.3, with the proved Wiener lemma, transfers the dynamical criteria to spectral measures and tensor products.

6. Reader perspective

Ergodicity, weak mixing, and mixing are kept in three separate theorem statements, and both prototype examples exhibit why the implications cannot be reversed. The product-kernel proof of weak mixing is written without identifying $U \otimes U$ with $U \otimes \overline{U}$, and the nonergodic base case is separated before entering $L^2_0 \widehat{\otimes} L^2_0$. Thus the reader sees exactly where finite-dimensional spectrum enters.

CHAPTER 3

The Mautner Phenomenon

Proof architecture. The Mautner mechanism turns invariance under one noncompact direction into invariance under a much larger subgroup. There are two logically distinct steps: contraction transfers fixed vectors, while an exact rank-one matrix identity converts unipotent invariance into diagonal invariance. The second step is essential; a unipotent group has no useful self-contraction that would make the theorem a one-line application of the first.

1. The contraction lemma

Lemma 1.1 (Mautner contraction)**.** *Let π be a continuous unitary representation of a topological group G. Suppose $g_n \in G$, $v \in \mathcal{H}$, and $\pi(g_n)v \rightharpoonup v_\infty$. If $h \in G$ satisfies*

$$g_n^{-1} h g_n \longrightarrow e,$$

then $\pi(h)v_\infty = v_\infty$.

Proof. For every $w \in \mathcal{H}$,

$$\langle \pi(h)\pi(g_n)v, w\rangle = \langle \pi(g_n)\pi(g_n^{-1}hg_n)v, w\rangle.$$

Strong continuity gives $\pi(g_n^{-1}hg_n)v \to v$. Since $\pi(g_n)$ is unitary,

$$\left\|\pi(g_n)\pi(g_n^{-1}hg_n)v - \pi(g_n)v\right\| = \left\|\pi(g_n^{-1}hg_n)v - v\right\| \to 0.$$

The left side therefore converges weakly both to $\pi(h)v_\infty$ and to v_∞. Weak limits are unique. $\square$

Corollary 1.2 (Fixed diagonal vectors are fixed by contraction groups)**.** *If v is fixed by every a_n and $a_n^{-1}ha_n \to e$, then v is fixed by h.*

Proof. Apply Lemma 1.1 with $g_n = a_n$; the weak limit of $\pi(a_n)v$ is v. $\square$

2. The exact SL_2 calculation

Set

$$u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}, \qquad v_s = \begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix}, \qquad a_t = \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix} \quad (t > 0).$$

Lemma 2.1 (Mautner matrix identity)**.** *Fix $t > 0$ and $\varepsilon \neq 0$. Put*

$$g_{t,\varepsilon} = \begin{pmatrix} t & 0 \\ \varepsilon & t^{-1} \end{pmatrix}, \qquad s_1 = \frac{1-t}{\varepsilon}, \quad s_2 = \frac{1-t^{-1}}{\varepsilon}.$$

Then

$$u_{s_1} g_{t,\varepsilon} u_{s_2} = v_\varepsilon. \tag{3.1}$$

Proof. Direct multiplication gives

$$u_{s_1} g_{t,\varepsilon} u_{s_2} = \begin{pmatrix} t + \varepsilon s_1 & (t + \varepsilon s_1)s_2 + s_1 t^{-1} \\ \varepsilon & \varepsilon s_2 + t^{-1} \end{pmatrix}.$$

The chosen s_1, s_2 make the diagonal entries equal to 1, and the upper-right entry becomes

$$1 \cdot \frac{1-t^{-1}}{\varepsilon} + \frac{1-t}{\varepsilon} t^{-1} = 0.$$

Thus the product is v_ε. □

Theorem 2.2 (Rank-one unipotent Mautner phenomenon). *Let π be a continuous unitary representation of $\mathrm{SL}_2(\mathbb{R})$. If a vector ξ is fixed by every u_s, then ξ is fixed by all of $\mathrm{SL}_2(\mathbb{R})$.*

Proof. First prove diagonal invariance. Fix $t > 0$. Because ξ is fixed by every upper unipotent, unitarity allows us to insert upper-unipotent factors on either side of a matrix coefficient:

$$\langle \pi(g_{t,\varepsilon})\xi, \xi \rangle = \langle \pi(u_{s_1} g_{t,\varepsilon} u_{s_2})\xi, \xi \rangle.$$

By Lemma 2.1, the right side is $\langle \pi(v_\varepsilon)\xi, \xi \rangle$. Let $\varepsilon \to 0$. Strong continuity gives

$$\langle \pi(a_t)\xi, \xi \rangle = \|\xi\|^2 .$$

Since both vectors $\pi(a_t)\xi$ and ξ have norm $\|\xi\|$, equality in Cauchy–Schwarz forces $\pi(a_t)\xi = \xi$. Thus ξ is fixed by the positive diagonal group A.

Now A contracts both root groups in opposite time directions:

$$a_t^{-n} u_s a_t^n = u_{t^{-2n}s} \to e, \qquad a_t^n v_s a_t^{-n} = v_{t^{-2n}s} \to e$$

when $t > 1$. Corollary 1.2 gives invariance under both $U^+ = \{u_s\}$ and $U^- = \{v_s\}$. These two subgroups generate $\mathrm{SL}_2(\mathbb{R})$. Indeed, the Bruhat cell is generated by them because

$$w = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = u_{-1} v_1 u_{-1},$$

and every positive diagonal matrix is already in the generated group by the preceding argument (or by an explicit four-unipotent factorization); together with the upper triangular group and w, Bruhat decomposition gives all of $\mathrm{SL}_2(\mathbb{R})$. Continuity then gives invariance under the whole group. $\square$

Warning 2.3. The unitarity hypothesis is indispensable. In the defining representation of $\mathrm{SL}_2(\mathbb{R})$ on $\mathbb{R}^2$, e_1 is fixed by all u_s but not by A. The proof above uses unitarity exactly when equality of a matrix coefficient with $\|\xi\|^2$ is converted into equality of vectors.

3. Restricted roots and rank-one envelopes

Let G be connected semisimple with finite center. Appendix A fixes a Cartan decomposition and restricted-root system $\Sigma \subset \mathfrak{a}^*$. For every nonzero restricted root α, Appendix A constructs the rank-one group $G_\alpha = \langle U_\alpha, U_{-\alpha}\rangle^\circ$ and its split coroot subgroup $a_\alpha(t) = \exp(tH_\alpha)$.

THEOREM 3.1 (Simple restricted-root Mautner envelope). *Let G_0 be a connected noncompact almost-simple real Lie group with finite center, let α be a* simple *restricted root, and let π be a continuous unitary representation. If ξ is fixed by the full root group*

$$U_\alpha = \exp(\mathfrak{g}_\alpha \oplus \mathfrak{g}_{2\alpha}),$$

with the 2α summand omitted when it is absent, then ξ is fixed by all of G_0.

PROOF. Appendix A, Theorem 4.1, applies the explicit SL_2 calculation of Theorem 2.2 inside the rank-one root subgroup. It shows first that U_α-fixedness implies $A_\alpha = \exp(\mathbb{R}H_\alpha)$-fixedness.

Let β be a simple restricted root adjacent to α in the restricted Dynkin diagram. Then $\beta(H_\alpha) \neq 0$. For $Y \in \mathfrak{g}_\beta$,

$$a_\alpha(-t)\exp(Y)a_\alpha(t) = \exp(e^{-t\beta(H_\alpha)}Y).$$

Choosing the sign of $t \to +\infty$ so that the scalar tends to zero and using Corollary 1.2, we obtain invariance under U_β; the opposite time direction gives $U_{-\beta}$. Applying the rank-one statement to β then gives A_β-invariance. Because G_0 is almost simple, its restricted Dynkin diagram is connected. Iterating along paths in that diagram therefore makes ξ fixed by A_γ for every simple restricted root γ.

The simple coroots span $\mathfrak{a}$. Hence ξ is fixed by the full split torus $A = \exp\mathfrak{a}$. If δ is any nonzero restricted root, then $\delta(H_\gamma) \neq 0$ for at least one simple coroot H_γ; otherwise δ would vanish on all of $\mathfrak{a}$. The same contraction argument therefore makes ξ fixed by every root group U_δ. Finally Appendix A, Theorem 5.1, proves that the connected subgroup generated by all nonzero restricted-root groups is G_0. Thus ξ is G_0-fixed. □

Remark 3.2. For nonreduced restricted root systems the groups U_α and $U_{2\alpha}$ are treated simultaneously inside the same rank-one root string. The propagation step is carried out along the simple restricted roots and then from the full split torus to every remaining root group; no reduced-root-system assumption is used.

4. Normal-factor version

Corollary 4.1 (Mautner envelope in a semisimple product)**.** *Let $G = G_1 \cdots G_r$ be an almost direct product of connected noncompact almost-simple factors. If ξ is fixed by a nontrivial full restricted-root group U_α contained in G_j, then ξ is fixed by G_j.*

Proof. Work inside the almost-simple factor G_j. Appendix A, Theorem 4.1, applies to the full root string containing α and shows that ξ is fixed by its split coroot group

$$A_\alpha = \exp(\mathbb{R}H_\alpha), \qquad H_\alpha \neq 0.$$

Let Δ_j be a simple restricted-root system for G_j. Since the simple roots span $\mathfrak{a}_j^*$, there is $\beta \in \Delta_j$ with $\beta(H_\alpha) \neq 0$. For $Y \in \mathfrak{g}_\beta$,

$$\exp(-tH_\alpha)\exp(Y)\exp(tH_\alpha) = \exp(e^{-t\beta(H_\alpha)}Y).$$

Choose the sign of $t \to +\infty$ for which the scalar tends to zero. Because ξ is A_α-fixed, Corollary 1.2 makes ξ fixed by U_β (and the same argument includes the 2β direction when the restricted system is nonreduced). Now β is simple, so Theorem 3.1 gives that ξ is fixed by all of G_j.

This proof is intentionally phrased without Weyl-conjugating α to a simple root. Such a conjugacy assertion is unnecessarily delicate in a nonreduced BC_r restricted root system, whereas the coroot-contraction argument works uniformly. □

5. Dependency summary

Dependency Note 5.1 (V05.C03.L)**.** Lemma 1.1 owns the contraction mechanism, while Lemma 2.1 supplies the exact rank-one identity that ordinary contraction cannot replace.

Dependency Note 5.2 (V05.C03.T)**.** Theorem 2.2 proves the rank-one unipotent Mautner theorem with all matrix parameters visible.

Dependency Note 5.3 (V05.C03.P)**.** Theorem 3.1 and Corollary 4.1 transfer rank-one invariance through restricted roots to an almost-simple normal factor.

6. Reader perspective

The rank-one matrix identity is presented before restricted-root language so that the nontrivial step—unipotent invariance producing diagonal invariance—is visible rather than hidden in the phrase "Mautner phenomenon." The warning explains why unitarity is essential. The higher-rank propagation is then a finite sequence of explicit contraction arguments with its structural root facts owned in Appendix A.

CHAPTER 4

Howe–Moore Vanishing of Matrix Coefficients

Proof architecture. Assume a coefficient fails to vanish along a sequence escaping to infinity. Cartan decomposition moves the escape into the positive Weyl chamber. A weak operator limit T of the corresponding unitaries is then fixed on the left by a contracted root group. Mautner propagation makes the image of T fixed by an entire noncompact simple factor. If such fixed vectors are forbidden, $T = 0$, contradicting the surviving coefficient.

1. Weak operator limits

Lemma 1.1 (Weak operator compactness on a separable cyclic envelope)**.** *Let π be unitary and let $v_1, \dots, v_m \in \mathcal{H}$. The closed invariant subspace*

$$\mathcal{H}_0 = \overline{\operatorname{span}\{\pi(g)v_j : g \in G, 1 \le j \le m\}}$$

is separable. Every sequence of unitaries $\pi(g_n)|_{\mathcal{H}_0}$ has a subsequence converging in the weak operator topology to a contraction T on $\mathcal{H}_0$.

Proof. Because G is second countable, it has a countable dense set D. Continuity of the orbit maps implies that the span of $\{\pi(d)v_j : d \in D\}$ is dense in $\mathcal{H}_0$, so $\mathcal{H}_0$ is separable. Choose a countable dense set $\{e_q\}$ in its unit ball. Each scalar sequence $\langle \pi(g_n)e_p, e_q\rangle$ is bounded. A diagonal subsequence makes all of them converge. Uniform boundedness and density extend the limiting sesquilinear form to all pairs, and Riesz representation gives an operator T. Since $|\langle T\xi, \eta\rangle| \le \|\xi\| \, \|\eta\|$, one has $\|T\|_{\mathrm{op}} \le 1$. $\square$

Lemma 1.2 (Contracted subgroups fix the image)**.** *Suppose $\pi(g_n) \to T$ weakly and $h \in G$ satisfies $g_n^{-1}hg_n \to e$. Then*

$$\pi(h)T = T.$$

Thus $\operatorname{im} T$ *is pointwise fixed by every subgroup contracted by g_n.*

Proof. For $\xi, \eta \in \mathcal{H}$,

$$\langle \pi(h)\pi(g_n)\xi, \eta\rangle = \langle \pi(g_n)\pi(g_n^{-1}hg_n)\xi, \eta\rangle.$$

Strong continuity makes the second vector inside $\pi(g_n)$ converge in norm to ξ; unitarity makes the replacement uniform in n. Passing to the weak limit gives $\langle \pi(h)T\xi, \eta\rangle = \langle T\xi, \eta\rangle$. $\square$

2. Cartan escape

Appendix A proves the structural package used here. In particular, for a connected semisimple real group with finite center,

$$G = K \exp(\overline{\mathfrak{a}^+})K.$$

Lemma 2.1 (A divergent chamber sequence has a divergent simple root)**.** *Let $H_n \in \overline{\mathfrak{a}^+}$ and $\|H_n\| \to \infty$. After passing to a subsequence, there is a simple restricted root α with $\alpha(H_n) \to +\infty$.*

Proof. The simple roots span $\mathfrak{a}^*$ modulo the compact central direction; for a semisimple group there is no nonzero central direction in $\mathfrak{a}$. Hence there is C such that

$$\|H\| \le C \max_{\alpha\in\Delta} \alpha(H) \qquad (H \in \overline{\mathfrak{a}^+}).$$

Otherwise a normalized sequence would converge to a unit vector annihilated by every simple root. Since Δ is finite, one root achieves the maximum on an infinite subsequence, and its values tend to infinity. $\square$

Lemma 2.2 (Root contraction)**.** *If $a_n = \exp H_n$ and $\alpha(H_n) \to +\infty$, then for every $X \in \mathfrak{g}_\alpha$,*

$$a_n^{-1} \exp(X)a_n = \exp(e^{-\alpha(H_n)}X) \longrightarrow e.$$

Proof. Restricted-root definition gives $\mathrm{Ad}(a_n^{-1})X = e^{-\alpha(H_n)}X$. Conjugation commutes with the exponential map. $\square$

3. The almost-simple theorem

Theorem 3.1 (Howe–Moore, almost-simple form). *Let G be a connected noncompact almost-simple real Lie group with finite center, and let π be a continuous unitary representation. If $\mathcal{H}^G = \{0\}$, then every matrix coefficient vanishes at infinity:*

$$\langle \pi(g)v, w\rangle \longrightarrow 0 \qquad (g \to \infty)$$

for all $v, w \in \mathcal{H}$.

Proof. Suppose not. Then there are v, w, $\varepsilon > 0$, and $g_n \to \infty$ with

$$|\langle \pi(g_n)v, w\rangle| \ge \varepsilon. \tag{4.1}$$

Write $g_n = k_n \exp(H_n)\ell_n$ with $k_n, \ell_n \in K$ and $H_n \in \overline{\mathfrak{a}^+}$. Compactness of K allows a subsequence with $k_n \to k$ and $\ell_n \to \ell$. Since g_n escapes compact sets, $\|H_n\| \to \infty$. Replacing v by $\pi(\ell)v$ and w by $\pi(k^{-1})w$ changes (4.1) only by an $o(1)$ error, so it suffices to consider $a_n = \exp H_n$.

Restrict to the separable invariant envelope of v and w. By Lemma 1.1, after a subsequence $\pi(a_n) \to T$ weakly. Equation (4.1) implies $\langle Tv, w\rangle \neq 0$, so $T \neq 0$.

By Lemma 2.1, some simple restricted root α satisfies $\alpha(H_n) \to \infty$. Lemma 2.2 and Lemma 1.2 show that every vector in $\operatorname{im} T$ is fixed by the nontrivial root group U_α. The restricted-root Mautner theorem, Theorem 3.1, upgrades this to G-invariance. Since $\mathcal{H}^G = 0$, $\operatorname{im} T = 0$, contradiction. $\square$

4. The semisimple factorwise theorem

Write, up to finite central overlap,

$$G = G_1 \cdots G_r$$

with connected noncompact almost-simple factors. A connected noncompact normal subgroup is generated by a nonempty subcollection of the factors.

Theorem 4.1 (Factorwise Howe–Moore). *Let G be connected semisimple with finite center and no compact factors. Suppose*

$$\mathcal{H}^{G_j} = \{0\} \qquad (1 \le j \le r). \tag{4.2}$$

Equivalently, $\mathcal{H}^N = 0$ for every noncompact connected normal subgroup $N \lhd G$. Then all matrix coefficients of π vanish at infinity.

More precisely, if $g_n = (g_{1,n}, \dots, g_{r,n})$ and

$$J = \{j : g_{j,n} \text{ escapes compact subsets after passing to a subsequence}\},$$

then every weak operator limit T of $\pi(g_n)$ has image contained in

$$\bigcap_{j\in J} \mathcal{H}^{G_j}. \tag{4.3}$$

Proof. Apply Cartan decomposition factor by factor. Compact K-parts are removed as in Theorem 3.1. For each $j \in J$, the Cartan component in G_j is unbounded; after a diagonal subsequence choose a simple restricted root α_j diverging there. Elements from distinct factors commute, so the contraction calculation is unaffected by the other coordinates. Lemma 1.2 makes $\operatorname{im} T$ fixed by U_{α_j} for every $j \in J$, and Theorem 3.1 gives G_j-fixedness. This proves (4.3).

If $g_n \to \infty$, then $J \neq \varnothing$. Under (4.2), the right side of (4.3) is zero, so every weak limit is zero. If some coefficient failed to tend to zero, the compactness argument would produce a nonzero weak limit as before. Hence every coefficient is C_0. $\square$

Warning 4.2 (Why products require a factor condition)**.** Let $G = G_1 \times G_2$ and let π factor through G_2. A sequence $(g_{1,n}, e)$ may escape to infinity while every coefficient of π stays constant. Thus "no global G-fixed vector" is not enough for semisimple products; one must rule out vectors fixed by an escaping normal factor.

5. A useful reformulation

Corollary 5.1 (The C_0 representation criterion)**.** *For connected semisimple G with finite center and no compact factors, a unitary representation is C_0 if it has no nonzero vector fixed by any noncompact connected normal subgroup.*

PROOF. This is Theorem 4.1. The converse obstruction is immediate: if $0 \neq v \in \mathcal{H}^N$ for a noncompact normal subgroup N, then $\langle \pi(n)v, v\rangle = \|v\|^2$ for $n \in N$, so the coefficient cannot vanish along $n \to \infty$ in N. $\square$

6. Dependency summary

Dependency Note 6.1 (V05.C04.L)**.** Weak-operator compactness, contracted-image invariance, Cartan escape, and root contraction are proved in Lemmas 1.1–2.2.

Dependency Note 6.2 (V05.C04.T)**.** The almost-simple Howe–Moore theorem is Theorem 3.1.

Dependency Note 6.3 (V05.C04.P)**.** The factorwise transfer is Theorem 4.1 and Corollary 5.1. No false product-group version is used.

7. Reader perspective

The proof is organized around one contradiction sequence and one weak operator limit. Cartan escape, root contraction, and the image-of-the-limit argument are isolated as separate lemmas before the headline theorem. The product warning prevents the common error of replacing factorwise fixed-space hypotheses by the weaker absence of global fixed vectors.

CHAPTER 5

Moore Ergodicity and Mixing on Homogeneous Spaces

Proof architecture. Howe–Moore is a representation theorem. Moore ergodicity is obtained by checking that the Koopman representation on an irreducible lattice quotient has no vectors fixed by a simple factor. This verification is not circular: it follows directly from density of lattice projections.

1. Irreducible lattices and factor invariants

Let

$$G = G_1 \cdots G_r$$

be an almost direct product of connected noncompact almost-simple groups with finite center. A lattice $\Gamma < G$ is *irreducible* if its projection to every proper subproduct is dense (after quotienting the finite central overlaps). This formulation is equivalent to the usual one in the present finite-center setting.

Lemma 1.1 (Factor-fixed functions are constant). *Let $\Gamma < G$ be irreducible and $X = \Gamma\backslash G$ with Haar probability. Then for every factor G_j,*

$$L^2_0(X)^{G_j} = \{0\}.$$

Proof. Let $f \in L^2(X)$ be fixed by the right action of G_j, and lift it to the left-Γ-invariant function $F(g) = f(\Gamma g)$ on G. Right G_j-invariance means that F descends to G/G_j, which, up to a finite central quotient, is the complementary product $G^{(j)} = \prod_{i\neq j} G_i$. Denote the descended L^2_{loc} class by $\bar{F}$.

For $\gamma \in \Gamma$, left Γ-invariance of F implies

$$\bar{F}(\mathrm{pr}_{G^{(j)}}(\gamma)y) = \bar{F}(y)$$

for almost every y. Irreducibility makes the projected subgroup dense in $G^{(j)}$. Translation on L^2 is strongly continuous on compact pieces, hence invariance under the dense subgroup extends to all of $G^{(j)}$. A measurable function invariant under all left translations of a connected group is almost everywhere constant: convolve locally with a compactly supported approximate identity to obtain continuous invariant representatives, which are constant, then pass to the L^2_{loc} limit. Thus F, and hence f, is constant. Intersecting with L^2_0 gives zero. □

2. Moore ergodicity

THEOREM 2.1 (Moore ergodicity). *Let G and Γ be as above, with Γ irreducible. Let $H < G$ be a closed subgroup whose projection to at least one simple factor is noncompact. Then the right action of H on $\Gamma\backslash G$ is ergodic.*

PROOF. Let π be the Koopman representation on $\mathcal{H} = L^2_0(\Gamma\backslash G)$. Lemma 1.1 gives $\mathcal{H}^{G_j} = 0$ for every simple factor. Hence factorwise Howe–Moore, Theorem 4.1, says every coefficient of π vanishes along every sequence escaping compact subsets of G.

Suppose $0 \neq f \in \mathcal{H}^H$. Choose a sequence $h_n \in H$ whose projection to a noncompact factor escapes compact subsets; after multiplying by bounded elements if necessary we may choose h_n itself escaping every compact subset of G. Since f is H-fixed,

$$\langle \pi(h_n) f, f\rangle = \|f\|_2^2$$

for all n, contradicting Howe–Moore. Therefore $\mathcal{H}^H = 0$. Proposition 1.2 gives ergodicity. □

Corollary 2.2 (One-parameter semisimple and unipotent subgroups)**.** *Under the hypotheses of Theorem 2.1, any noncompact one-parameter subgroup whose projection to a simple factor is noncompact acts ergodically. In particular this applies to nontrivial split one-parameter groups and nontrivial unipotent root groups in an irreducible quotient.*

PROOF. A noncompact one-parameter subgroup has noncompact projection to at least one factor, otherwise its image would lie in a product of compact sets. Apply Theorem 2.1. □

3. Mixing of the ambient action

THEOREM 3.1 (Mixing on an irreducible semisimple quotient). *Let G be connected semisimple with finite center and no compact factors, and let $\Gamma < G$ be an irreducible lattice. Then the right action of G on $X = \Gamma\backslash G$ is mixing:*

$$\int_X f(xg)h(x)\,dm_X(x) \longrightarrow \left(\int_X f\,dm_X\right)\left(\int_X h\,dm_X\right) \qquad (g \to \infty)$$

for all $f, h \in L^2(X)$.

PROOF. Write $f = f_0 + \int f$ and $h = h_0 + \int h$, where $f_0, h_0 \in L^2_0$. The constant terms produce the stated product. Lemma 1.1 verifies the factorwise fixed-vector hypothesis for the Koopman representation on L^2_0. Theorem 4.1 therefore gives

$$\langle \pi(g) f_0, \overline{h_0} \rangle \to 0.$$

This is precisely the remaining correlation. □

4. Finite-volume homogeneous orbits

Proposition 4.1 (Transfer to a closed semisimple orbit)**.** *Let $X = \Gamma\backslash G$, let $L < G$ be a connected semisimple subgroup with finite center and no compact factors, and let xL be a closed finite-volume orbit. Put*

$$\Lambda = L \cap g^{-1}\Gamma g \quad \text{when } x = \Gamma g.$$

If Λ is irreducible in L, then the L-action on (xL, m_{xL}) is mixing and every closed subgroup $H < L$ with noncompact projection to a simple factor acts ergodically.

PROOF. The orbit map identifies (xL, m_{xL}) measure-theoretically with $(\Lambda\backslash L, m_{\Lambda\backslash L})$. Apply Theorems 2.1 and 3.1 to L and Λ. □

5. Examples and diagnostics

Worked Example 5.1 (The modular surface frame bundle)**.** For $G = \mathrm{PSL}_2(\mathbb{R})$ and any lattice Γ, there is only one noncompact simple factor, so irreducibility is automatic. Every noncompact one-parameter subgroup acts ergodically by Moore's theorem, and the full G-action is mixing. The geodesic subgroup A therefore gives a mixing flow on the finite-volume frame bundle.

Worked Example 5.2 (Why reducible products fail)**.** Let $X = (\Gamma_1\backslash G_1) \times (\Gamma_2\backslash G_2)$. A function depending only on the second coordinate is fixed by G_1. Hence a sequence escaping only in G_1 cannot mix such a function. This is exactly the fixed-factor obstruction excluded by irreducibility in Lemma 1.1.

6. Dependency summary

Dependency Note 6.1 (V05.C05.L)**.** Lemma 1.1 proves the noncircular factor-fixed-vector criterion from irreducibility.

Dependency Note 6.2 (V05.C05.T)**.** Theorem 2.1 is Moore ergodicity with the exact noncompact-projection hypothesis.

Dependency Note 6.3 (V05.C05.P)**.** Theorem 3.1 and Proposition 4.1 give the homogeneous-space and closed-orbit transfer forms used downstream.

7. Reader perspective

The potentially circular step is isolated first: factor-fixed functions on an irreducible quotient are proved constant directly from density of lattice projections, before Howe–Moore is invoked. The modular-surface and reducible-product examples show both the theorem and its failure mode. The closed-orbit transfer is stated only after the quotient identification is explicit.

CHAPTER 6

Weak Containment and Tempered Representations

Proof architecture. Weak containment is the representation-theoretic notion that lets estimates for the regular representation pass to another representation. We prove both the coefficient and integrated-operator formulations, then prove Fell absorption explicitly. These are the precise tools needed in the Cowling–Haagerup–Howe argument.

1. Integrated representations

Let G be locally compact with left Haar measure dg. For $f \in C_c(G)$ define

$$\pi(f) = \int_G f(g)\pi(g)\,dg$$

in the weak, equivalently strong, operator sense. Then

$$\|\pi(f)\|_{\mathrm{op}} \le \|f\|_{L^1}.$$

With modular function Δ_G, put

$$f^*(g) = \Delta_G(g^{-1})\overline{f(g^{-1})}.$$

A direct change of variables gives $\pi(f^*) = \pi(f)^*$ and $\pi(f * h) = \pi(f)\pi(h)$.

2. Two formulations of weak containment

Definition 2.1 (Coefficient weak containment)**.** We write $\pi \prec \rho$ if for every unit vector $v \in \mathcal{H}_\pi$, compact $C \subset G$, and $\varepsilon > 0$, there are vectors $w_1, \ldots, w_m \in \mathcal{H}_\rho$ with $\sum_j \|w_j\|^2 = 1$ such that

$$\sup_{g\in C}\left|\langle \pi(g)v, v\rangle - \sum_{j=1}^{m}\langle \rho(g)w_j, w_j\rangle\right| < \varepsilon. \tag{6.1}$$

Theorem 2.2 (Weak containment and operator norms). *For strongly continuous unitary representations of a second-countable locally compact group,*

$$\pi \prec \rho \iff \|\pi(f)\|_{\mathrm{op}} \le \|\rho(f)\|_{\mathrm{op}} \quad \textit{for every } f \in C_c(G). \tag{6.2}$$

Proof. Assume first $\pi \prec \rho$. For unit v and $f \in C_c(G)$,

$$\|\pi(f)v\|^2 = \langle \pi(f^* * f)v, v\rangle.$$

The support of $f^* * f$ is compact. Approximate the coefficient of v there by (6.1). Integration gives

$$\begin{aligned}\|\pi(f)v\|^2 &\le \sum_j \langle \rho(f^* * f)w_j, w_j\rangle + o(1)\\ &= \sum_j \|\rho(f)w_j\|^2 + o(1)\\ &\le \|\rho(f)\|_{\mathrm{op}}^2 + o(1).\end{aligned}$$

Let the approximation error tend to zero and take the supremum over unit v.

Conversely assume the norm inequality. Let A_ρ be the completion of $C_c(G)$ modulo the seminorm $f \mapsto \|\rho(f)\|_{\mathrm{op}}$. The inequality makes

$$\Phi_v(f) = \langle \pi(f)v, v\rangle$$

a bounded positive functional on A_ρ for every unit v. Appendix B, Lemma 1.1, proves that states of a faithfully represented C^*-algebra are weak-* limits of finite convex combinations of vector states. Applied to the faithful representation of A_ρ on $\mathcal{H}_\rho$, this approximates Φ_v on any finite family of integrated operators.

To recover uniform approximation of group coefficients, use the smoothing lemma, Appendix B, Lemma 2.1: for a compact C and $\eta > 0$ one may choose a nonnegative $\psi \in C_c(G)$ with $\int \psi = 1$ so concentrated near e that, uniformly for $g \in C$,

$$|\langle \pi(g)v, v\rangle - \langle \pi(g)\pi(\psi)v, \pi(\psi)v\rangle| < \eta. \tag{6.3}$$

The smoothed coefficient in (6.3) is the state evaluated at the element $\psi^* * L_g\psi$ of A_ρ, and $g \mapsto \psi^* * L_g\psi$ is norm-continuous with compact image for $g \in C$. A finite η-net of that image reduces uniform approximation to finitely many algebra elements. Approximate the state there by a finite convex combination of ρ-vector states, apply the same smoothing, and absorb the convex weights into the vector norms. Triangle inequalities with (6.3) give (6.1). Since η is arbitrary, $\pi \prec \rho$. □

3. Permanence properties

Proposition 3.1 (Subrepresentations, sums, and tensors)**.** *Weak containment has the following properties.*

(1) Every subrepresentation of ρ is weakly contained in ρ.

(2) If $\pi_i \prec \rho$ for every i, then $\bigoplus_i \pi_i \prec \rho$.

(3) If $\pi \prec \rho$, then $\pi \otimes \sigma \prec \rho \otimes \sigma$ for every unitary σ.

(4) Weak containment is transitive.

Proof. (1) follows immediately from the operator norm criterion. For (2), a vector in a direct sum is approximated by a vector supported in finitely many summands; its diagonal coefficient is a convex combination of diagonal coefficients in those summands, and then use Definition 2.1. Transitivity follows by composing coefficient approximations.

For (3), let $z = \sum_{i=1}^{r} v_i \otimes a_i$ be a finite sum of simple tensors and let $E = \operatorname{span}\{v_1, \dots, v_r\}$. On a prescribed compact set, Lemma 4.1 gives a map $T_0 : E \to \mathcal{H}_\rho^{\oplus N}$ whose entire matrix of coefficients approximates that of π. Put

$$\widetilde{z} = \sum_{i=1}^{r} T_0 v_i \otimes a_i.$$

Expanding the two diagonal coefficients shows, uniformly on the compact set,

$$\langle (\rho^{\oplus N} \otimes \sigma)(g)\widetilde{z}, \widetilde{z}\rangle \longrightarrow \langle (\pi \otimes \sigma)(g)z, z\rangle,$$

and the Gram part of the same lemma gives $\|\widetilde{z}\| \to \|z\|$. After normalization this is exactly coefficient weak containment for finite tensor sums. Such sums are dense, and all representations are unitary, so a three-ε argument gives the result for arbitrary vectors. $\square$

4. Fell absorption

Let λ_G be the left regular representation on $L^2(G)$:

$$(\lambda_G(g)f)(x) = f(g^{-1}x).$$

Theorem 4.1 (Fell absorption principle). *For every unitary representation σ of G,*

$$\lambda_G \otimes \sigma \cong \lambda_G \otimes 1_{\mathcal{H}_\sigma}$$

on $L^2(G, \mathcal{H}_\sigma)$. Consequently $\lambda_G \otimes \sigma$ is an amplification of the regular representation and is weakly equivalent to λ_G.

Proof. Identify $L^2(G) \otimes \mathcal{H}_\sigma$ with $L^2(G, \mathcal{H}_\sigma)$. Define

$$(WF)(x) = \sigma(x^{-1})F(x).$$

Pointwise unitarity of $\sigma(x^{-1})$ makes W unitary, with inverse $(W^{-1}F)(x) = \sigma(x)F(x)$. Now

$$\begin{aligned} W(\lambda \otimes \sigma)(g)F(x) &= \sigma(x^{-1})\sigma(g)F(g^{-1}x), \\ (\lambda \otimes 1)(g)WF(x) &= WF(g^{-1}x) = \sigma((g^{-1}x)^{-1})F(g^{-1}x) \\ &= \sigma(x^{-1}g)F(g^{-1}x). \end{aligned}$$

The two expressions coincide. Thus W intertwines the representations.

An amplification $\lambda \otimes 1_{\mathcal{H}_\sigma}$ is a direct sum (or Hilbert direct integral, after choosing an orthonormal basis) of copies of λ. It has exactly the same integrated operator norm as λ, so Theorem 2.2 gives weak equivalence. $\square$

5. Tempered representations

Definition 5.1. A unitary representation π is *tempered* if

$$\pi \prec \lambda_G.$$

Equivalently, by Theorem 2.2,

$$\|\pi(f)\|_{\mathrm{op}} \leq \|\lambda_G(f)\|_{\mathrm{op}} \qquad (f \in C_c(G)).$$

THEOREM 5.2 (Tensor stability of temperedness). *If π is tempered and σ is any unitary representation, then $\pi \otimes \sigma$ is tempered.*

PROOF. By Proposition 3.1,

$$\pi \otimes \sigma \prec \lambda_G \otimes \sigma.$$

Fell absorption makes the latter weakly equivalent to λ_G. Transitivity yields $\pi \otimes \sigma \prec \lambda_G$. $\square$

Definition 5.3 (Tensor-tempered index)**.** For a unitary representation π, define $m(\pi)$ to be the least positive integer m for which $\pi^{\otimes m}$ is tempered, if such an integer exists.

Remark 5.4. Volume 6 will explain when spectral gap or Property (T) can be converted into a finite tensor-tempered index in the particular homogeneous quotients used later. No such implication is assumed in this volume.

6. Coefficient transfer

Proposition 6.1 (Tempered coefficient approximation)**.** *If π is tempered, then every diagonal coefficient of a unit vector is a uniform-on-compact limit of finite convex combinations of diagonal coefficients of λ_G. Every mixed coefficient is a uniform-on-compact limit of finite linear combinations of regular coefficients.*

PROOF. The first statement is Definition 2.1. The second follows from polarization:

$$4\langle \pi(g)v, w\rangle = \sum_{k=0}^{3} i^k \langle \pi(g)(v + i^k w), v + i^k w\rangle$$

with the corresponding normalization of the vectors. Apply the diagonal statement term by term. $\square$

7. Dependency summary

Dependency Note 7.1 (V05.C06.L)**.** Theorem 2.2, together with the proved state-density and smoothing lemmas in Appendix B, owns the equivalence between coefficient weak containment and operator-norm domination.

Dependency Note 7.2 (V05.C06.T)**.** Theorem 4.1 is proved by an explicit intertwiner, and temperedness is defined through the now-equivalent weak-containment formulations.

Dependency Note 7.3 (V05.C06.P)**.** Proposition 3.1, Theorem 5.2, and Proposition 6.1 prove the permanence and downstream transfer statements.

8. Reader perspective

Weak containment is given in both coefficient and operator-norm forms and the difficult converse is not suppressed; Appendix B supplies state separation, smoothing, simultaneous finite-vector approximation, and the K-equivariant form needed later. Fell absorption is proved by an explicit intertwiner. The chapter stops before spectral-gap consequences, which belong to Volume 6.

CHAPTER 7

Harish–Chandra Bounds and the Cowling–Haagerup–Howe Method

Proof architecture. Howe–Moore says a coefficient tends to zero but does not quantify the decay. Tempered representations inherit a universal pointwise envelope from the regular representation. The envelope is Harish–Chandra's spherical function Ξ. We prove its chamber estimate and the K-finite Cowling–Haagerup–Howe majorization, then record the exact tensor-power bookkeeping needed later.

1. The spherical function

Let G be a connected semisimple real linear group with finite center and no compact factors. Fix $G = KAN$ and a positive restricted-root system. Let

$$\rho = \frac{1}{2} \sum_{\alpha \in \Sigma^+} m_\alpha \alpha.$$

For $g \in G$, write $H(g) \in \mathfrak{a}$ for the Iwasawa A-coordinate. Define

$$\Xi(g) = \int_K e^{-\rho(H(gk))}\, dk. \tag{7.1}$$

Proposition 1.1 (Basic properties of Ξ)**.** *The function Ξ is continuous, positive definite, K-bi-invariant, satisfies $\Xi(e) = 1$, and $0 < \Xi(g) \leq 1$.*

Proof. Appendix C, Proposition 1.1, constructs the normalized spherical principal representation on $L^2(K/M)$ and proves that

$$\Xi(g) = \langle \Pi_0(g)\mathbf{1}, \mathbf{1} \rangle.$$

This makes Ξ positive definite, continuous, and bounded by one. The constant vector is K-fixed, so the coefficient is bi-K-invariant. Its norm is one, hence $\Xi(e) = 1$. Positivity follows directly from the integral formula. $\square$

2. Harish–Chandra chamber decay

THEOREM 2.1 (Harish–Chandra estimate). *There are constants $C, D > 0$, depending only on G, such that for every $H \in \overline{\mathfrak{a}^+}$,*

$$\Xi(\exp H) \le C(1 + \|H\|)^D e^{-\rho(H)}. \tag{7.2}$$

There is also a structural constant $c > 0$ such that $\Xi(\exp H) \ge ce^{-\rho(H)}$; in particular the exponential scale in (7.2) is sharp.

PROOF. The complete root-coordinate proof is Appendix C, Theorem 5.3. We summarize the chain, with each arrow proved there. The open Bruhat cell identifies a full-measure subset of K/M with $\bar{N}$ and gives

$$\int_K F(k)\,dk = c\int_{\bar{N}} F(k(\bar{n}))e^{-2\rho(H(\bar{n}))}\,d\bar{n}. \tag{7.3}$$

Choose the convex ordering of indivisible positive roots from Appendix C, Lemma 2.2. BCH makes the open-cell coordinates triangular, so successive integration removes commutator terms by translations in later root coordinates. Lemma 4.5 proves the resulting rank-one model integral and identifies the unique logarithmic loss $1 + \beta(H)$. Lemma 5.2 iterates this finite procedure and obtains $\Xi(e^H) \le Ce^{-\rho(H)} \prod_{\beta\in\Sigma_0^+}(1 + \beta(H))$, which is bounded by (7.2).

For the lower bound, restrict the defining K-integral to a fixed small neighborhood of the identity in K where the Iwasawa cocycle is uniformly controlled; on this neighborhood $\rho(H(ak)) \le \rho(H) + O(1)$. Its Haar measure is fixed and positive. ☐

3. K-finite vectors

Definition 3.1. A vector v in a unitary representation is K-finite if

$$V_v = \operatorname{span}\{\pi(k)v : k \in K\}$$

is finite dimensional. Put $d_v = \dim V_v$.

Lemma 3.2 (Finite K-type evaluation principle)**.** *Let $E \subset L^2(K)$ be a d-dimensional translation-invariant subspace. Then every $F \in E$ has a canonical continuous representative and*

$$\|F\|_\infty \le \sqrt{d}\,\|F\|_2\,. \tag{7.3a}$$

PROOF. Appendix C, Lemmas 6.2 and 6.3, prove both assertions from compact-group Schur orthogonality. The representative issue is essential: pointwise evaluation is not defined on an arbitrary $L^2(K)$ class. ☐

4. Herz–Cowling–Haagerup majorization

THEOREM 4.1 (CHH majorization for tempered representations). *Let σ be a tempered unitary representation of G. If v, w are K-finite, then*

$$|\langle \sigma(g)v, w\rangle| \le \sqrt{d_v d_w}\,\Xi(g)\,\|v\|\,\|w\| \qquad (g \in G). \tag{7.4}$$

PROOF. Appendix C, Lemma 6.1 first proves the positive left-K-invariant regular coefficient inequality directly from Iwasawa coordinates. Lemmas 6.3 and 6.4 convert it into the exact regular K-finite bound with factor $\sqrt{d_v d_w}$. Finally, the K-equivariant Fell approximation of Appendix B, Lemma 4.2, transfers that estimate to every tempered representation without increasing either cyclic K-dimension. Theorem 7.1 carries out the limit and gives (7.4). □

5. Tensor-tempered representations: exact bookkeeping

For a K-finite vector v define

$$d_{v,m} = \dim \operatorname{span}\{\pi(k)^{\otimes m} v^{\otimes m} : k \in K\}.$$

The elementary containment

$$\operatorname{span}_K(v^{\otimes m}) \subset V_v^{\otimes m}$$

gives

$$d_{v,m} \le d_v^m. \tag{7.5}$$

Proposition 5.1 (Tensor-tempered coefficient bound)**.** *Suppose $\pi^{\otimes m}$ is tempered. Then for K-finite v, w,*

$$|\langle \pi(g)v, w\rangle| \le (d_{v,m} d_{w,m})^{1/(2m)}\Xi(g)^{1/m}\,\|v\|\,\|w\|\,. \tag{7.6}$$

Consequently, using only the elementary bound (7.5),

$$|\langle \pi(g)v, w\rangle| \le \sqrt{d_v d_w}\,\Xi(g)^{1/m}\,\|v\|\,\|w\|\,. \tag{7.7}$$

PROOF. Apply Theorem 4.1 to the tempered representation $\pi^{\otimes m}$ and the vectors $v^{\otimes m}, w^{\otimes m}$:

$$|\langle \pi^{\otimes m}(g)v^{\otimes m}, w^{\otimes m}\rangle| \le \sqrt{d_{v,m} d_{w,m}}\,\Xi(g)\,\|v\|^m\,\|w\|^m\,.$$

The left side is $|\langle \pi(g)v, w\rangle|^m$. Taking the mth root gives (7.6). Then (7.5) gives (7.7). □

Warning 5.2 (A dimension exponent that must not be simplified incorrectly)**.** Formula (7.6) has exponent $1/(2m)$ on the *tensor cyclic dimensions* $d_{v,m}, d_{w,m}$. Without a separate stronger lemma one cannot replace those dimensions by d_v, d_w while keeping the exponent $1/(2m)$. The universally valid substitution $d_{v,m} \le d_v^m$ produces the exponent $1/2$ in (7.7). This distinction is part of the stated scope and prerequisites of the present edition.

6. Decay in Cartan coordinates

Corollary 6.1 (Quantitative envelope for K-finite vectors)**.** *If $\pi^{\otimes m}$ is tempered and $g = k\exp(H)k'$ with $H \in \overline{\mathfrak{a}^+}$, then*

$$|\langle \pi(g)v, w\rangle| \leq C^{1/m}(d_{v,m}d_{w,m})^{1/(2m)}(1 + \|H\|)^{D/m}e^{-\rho(H)/m}\,\|v\|\,\|w\|\,.$$

Proof. K-bi-invariance of Ξ and Theorem 2.1 inserted into Proposition 5.1 give the result. □

7. Dependency summary

Dependency Note 7.1 (V05.C07.L)**.** Proposition 1.1, Theorem 2.1, and Appendix C own the spherical function and its general chamber estimate.

Dependency Note 7.2 (V05.C07.T)**.** Theorem 4.1, with the Herz proof in Appendix C, is the full K-finite Cowling–Haagerup–Howe majorization used by the series.

Dependency Note 7.3 (V05.C07.P)**.** Proposition 5.1 and Corollary 6.1 give the tensor-tempered transfer with exact K-dimension bookkeeping. Warning 5.2 records and repairs the exponent ambiguity found in audit.

8. Reader perspective

The chapter distinguishes three layers: the spherical envelope Ξ, the CHH transfer for a tempered representation, and the additional tensor-power step used for non-tempered representations. Appendix C owns the analytic proofs instead of replacing them by names. Warning 5.2 records the tensor K-dimension exponent error that a compressed presentation can easily conceal.

CHAPTER 8

Product Groups and Factorwise Mixing Criteria

Proof architecture. A product group can escape to infinity in only some factors. The correct invariant attached to a sequence is therefore its escaping-factor set. This chapter turns the factorwise Howe–Moore proof into a reusable criterion for representations and homogeneous actions, and records the exact obstruction when a factor carries invariant vectors.

1. Escaping-factor sets

Let $G = G_1 \cdots G_r$ be an almost direct product of connected noncompact almost-simple real groups with finite center. For a sequence $g_n \in G$, after passing to a subsequence each factor projection is either relatively compact or escapes every compact set. Define

$$J(g_n) = \{j : \mathrm{pr}_j(g_n) \to \infty\}.$$

If $g_n \to \infty$ in G, then $J(g_n) \neq \varnothing$.

Lemma 1.1 (Escaping-factor weak-limit theorem)**.** *Let π be unitary and suppose $\pi(g_n) \to T$ weakly. Then*

$$\operatorname{im} T \subseteq \bigcap_{j \in J(g_n)} \mathcal{H}^{G_j}. \tag{8.1}$$

Likewise, if $\pi(g_n^{-1}) \to S$ weakly, then

$$\operatorname{im} S \subseteq \bigcap_{j \in J(g_n)} \mathcal{H}^{G_j}.$$

Proof. The first assertion is the more precise part of Theorem 4.1. For the inverse sequence, factorwise escape is unchanged and the same argument applies, using the opposite roots in each factor. □

2. A necessary and sufficient C_0 criterion

Theorem 2.1 (Factorwise mixing criterion). *For a unitary representation π of G, the following are equivalent.*

(1) Every matrix coefficient of π belongs to $C_0(G)$.

(2) $\mathcal{H}^{G_j} = 0$ *for every noncompact simple factor* G_j*.*

(3) $\mathcal{H}^N = 0$ *for every noncompact connected normal subgroup* $N \triangleleft G$*.*

More locally, along a fixed sequence g_n*, all coefficients tend to zero whenever*

$$\bigcap_{j \in J(g_n)} \mathcal{H}^{G_j} = 0. \tag{8.2}$$

PROOF. (2)⇒(1) is Theorem 4.1. If (1) holds and $0 \neq v \in \mathcal{H}^{G_j}$, choose $g_n \to \infty$ inside G_j. Then

$$\langle \pi(g_n)v, v\rangle = \|v\|^2,$$

contradicting (1); hence (1)⇒(2). Every connected normal subgroup is, up to finite center, a product of a subcollection of the almost-simple factors. Thus (2) and (3) are equivalent. The local assertion follows from Lemma 1.1: any nonzero limiting coefficient would yield a nonzero weak limit operator whose image lies in the intersection in (8.2). □

3. Product actions

Proposition 3.1 (Product-action diagnostic)**.** *Let* G *act probability-preservingly on* (X, μ) *and let* π_0 *be the Koopman representation on* $L^2_0(X)$*. Then the action is mixing under* $g \to \infty$ *if and only if*

$$L^2_0(X)^{G_j} = 0 \quad \textit{for every noncompact factor } G_j.$$

For a sequence g_n*, it is enough that the intersection of the fixed spaces of the escaping factors be zero.*

PROOF. Apply Theorem 2.1 to π_0 and then Theorem 2.1. □

Worked Example 3.2 (Irreducible lattice quotient)**.** If $X = \Gamma\backslash G$ with Γ irreducible, Lemma 1.1 verifies the factor-fixed-space condition. Hence the full action is mixing.

Worked Example 3.3 (Reducible lattice quotient)**.** If $\Gamma = \Gamma_1 \times \Gamma_2$, then

$$X = (\Gamma_1\backslash G_1) \times (\Gamma_2\backslash G_2).$$

Any mean-zero function of the second coordinate is fixed by G_1. Therefore a sequence escaping solely in G_1 violates mixing. If a sequence escapes in both factors and each individual factor action is mixing, its correlations can still decay; the local criterion (8.2) is the correct statement.

4. Compatibility with temperedness

Proposition 4.1 (Product regular representations and tensor envelopes)**.** *If $G = G_1 \times G_2$, then*

$$\lambda_G \cong \lambda_{G_1} \otimes \lambda_{G_2}, \qquad \Xi_G(g_1, g_2) = \Xi_{G_1}(g_1)\Xi_{G_2}(g_2).$$

Consequently, for tensor-product tempered representations $\pi_1 \otimes \pi_2$, the CHH envelope factorizes into the product of the factor envelopes.

Proof. Fubini identifies $L^2(G_1 \times G_2)$ with $L^2(G_1) \otimes L^2(G_2)$ and the left-translation formulas agree on simple tensors. For the spherical function choose $K = K_1 \times K_2$, $A = A_1 \times A_2$, and $\rho = \rho_1 + \rho_2$. The defining integral (7.1) splits by Fubini. The coefficient statement follows by applying Theorem 4.1 factorwise. □

5. What Volume 6 will add

The present volume gives a qualitative criterion and, conditional on temperedness or tensor-temperedness, a Harish–Chandra envelope. It does not prove that a specific family of quotient representations has a uniform tensor-tempered index. That requires spectral gap, Property (T), Property (τ), and Sobolev technology; these are the explicit obligations of Volume 6.

6. Dependency summary

Dependency Note 6.1 (V05.C08.L)**.** Lemma 1.1 owns the exact weak-limit statement attached to an escaping-factor set.

Dependency Note 6.2 (V05.C08.T)**.** Theorem 2.1 is the complete necessary-and-sufficient factorwise C_0/mixing criterion, with Proposition 3.1 as its dynamical form.

7. Reader perspective

The escaping-factor set is introduced before any product theorem, so the correct obstruction is visible from the start. Irreducible and reducible quotient examples test the criterion in opposite directions. The final section explains exactly what is postponed to Volume 6: not qualitative mixing, but uniform spectral information and quantitative rates.

APPENDIX A

Semisimple Structural Toolkit Used in Volume 5

This appendix owns the semisimple structural facts used by Chapters 3–7. Throughout, G is a connected real linear semisimple group with finite center and no compact factors. The foundational floor consists of finite-dimensional semisimple Lie algebra theory and the Hopf–Rinow theorem for complete finite-dimensional Riemannian manifolds. All group decompositions and restricted-root consequences used later are proved here.

1. Cartan involution and the symmetric space

Fix a Cartan involution θ of $\mathfrak{g}$ and write

$$\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}, \qquad \theta|_{\mathfrak{k}} = 1, \quad \theta|_{\mathfrak{p}} = -1.$$

The bilinear form

$$B_\theta(X, Y) = -B(X, \theta Y)$$

is positive definite. Let $K < G$ be the connected subgroup with Lie algebra $\mathfrak{k}$; the finite center causes no difficulty, and K is maximal compact.

Lemma 1.1 (Geodesics through the base point)**.** *Give $X = G/K$ the G-invariant Riemannian metric whose value at $o = eK$ is $B_\theta|_{\mathfrak{p}}$. For every $Y \in \mathfrak{p}$, the curve*

$$\gamma_Y(t) = \exp(tY)K$$

is the geodesic through o with initial velocity Y.

Proof. Identify T_oX with $\mathfrak{p}$. For $Y, Z, W \in \mathfrak{p}$, the Koszul formula for the invariant metric reduces at o to the $\mathfrak{p}$-part of Lie brackets. Since

$$[\mathfrak{p}, \mathfrak{p}] \subset \mathfrak{k},$$

the $\mathfrak{p}$-part of $[Y, Z]$ vanishes. Hence the Levi–Civita covariant derivative of the invariant vector field generated by Y in its own direction vanishes at o. G-invariance transports this identity along $\exp(tY)o$, so γ_Y is geodesic. Its derivative at 0 is Y, and uniqueness of the geodesic initial-value problem gives the claim. $\square$

Lemma 1.2 (Completeness of the symmetric-space metric)**.** *The invariant Riemannian manifold G/K is metrically and geodesically complete.*

Proof. Choose $r > 0$ so small that the exponential map at o is a diffeomorphism on the ball of radius $2r$ in T_oX. By transitivity and invariance of the metric, the same r works at every point. If (x_n) is Cauchy, then for some N all x_n with $n \geq N$ lie in the metric ball $B(x_N, r)$. In normal coordinates at x_N, the closure of the smaller coordinate ball is compact, so the Cauchy sequence converges. Thus X is metrically complete. Hopf–Rinow now gives geodesic completeness and says that every point can be joined to o by a minimizing geodesic. $\square$

Theorem 1.3 (Global Cartan decomposition). *Every $g \in G$ can be written*

$$g = k \exp Y, \qquad k \in K,\ Y \in \mathfrak{p}. \tag{A.1}$$

If $\mathfrak{a} \subset \mathfrak{p}$ is maximal abelian and $\mathfrak{a}^+$ is a positive Weyl chamber, then

$$G = K \exp(\overline{\mathfrak{a}^+}) K. \tag{A.2}$$

Proof. Let $gK \in X$. By Lemma 1.2 there is a minimizing geodesic from o to gK. By Lemma 1.1 it is $t \mapsto \exp(tY)K$ for some $Y \in \mathfrak{p}$. Hence $gK = \exp(Y)K$, so $g = \exp(Y)k$ for some $k \in K$. Applying the same statement to g^{-1} and inverting gives the form (A.1).

It remains to conjugate an arbitrary $Y \in \mathfrak{p}$ into one fixed maximal abelian subspace $\mathfrak{a}$. Choose a regular element $Y_0 \in \mathfrak{a}$, meaning

$$Z_{\mathfrak{p}}(Y_0) = \mathfrak{a}.$$

Such elements are the complement of the finite union of restricted-root hyperplanes. On compact K, maximize

$$F(k) = B_\theta(\mathrm{Ad}(k)Y, Y_0).$$

At a maximizer k_0, differentiation in the direction $Z \in \mathfrak{k}$ gives

$$0 = B_\theta([Z, \mathrm{Ad}(k_0)Y], Y_0) = B_\theta(Z, [\mathrm{Ad}(k_0)Y, Y_0]).$$

Here $[\mathrm{Ad}(k_0)Y, Y_0] \in \mathfrak{k}$ and B_θ is nondegenerate on $\mathfrak{k}$, so the bracket is zero. Thus $\mathrm{Ad}(k_0)Y \in Z_{\mathfrak{p}}(Y_0) = \mathfrak{a}$.

Therefore $G = K \exp(\mathfrak{a})K$. The finite reflection group $W = N_K(\mathfrak{a})/Z_K(\mathfrak{a})$ has the usual closed chamber as a fundamental domain, so every $H \in \mathfrak{a}$ is W-conjugate to a point of $\overline{\mathfrak{a}^+}$. $\square$

2. The Cartan integration Jacobian

The integrability statement for Ξ later needs not only the set-theoretic Cartan decomposition but its Jacobian. We derive the radial factor here.

Proposition 2.1 (Cartan integration formula)**.** *Let $M = Z_K(\mathfrak{a})$. After compatible normalizations of Haar and Lebesgue measures, every nonnegative measurable F on G satisfies*

$$\int_G F(g)\,dg = c_G \int_K \int_{\mathfrak{a}^+} \int_K F(k_1 e^H k_2) J(H)\,dk_1\,dH\,dk_2, \tag{A.2a}$$

where

$$J(H) = \prod_{\alpha\in\Sigma^+} (\sinh \alpha(H))^{m_\alpha}. \tag{A.2b}$$

The formula extends from the open chamber to the closed chamber because the walls have Lebesgue measure zero. In particular,

$$J(H) \le Ce^{2\rho(H)} \qquad (H \in \overline{\mathfrak{a}^+}). \tag{A.2c}$$

Proof. It is enough first to compute the Riemannian polar Jacobian on $X = G/K$. On the regular set the map

$$\Phi : K/M \times \mathfrak{a}^+ \longrightarrow X, \qquad (kM, H) \mapsto ke^H K,$$

is a finite-sheeted local diffeomorphism; restricting H to one open Weyl chamber makes it one-to-one. The complement consists of the images of root walls and has Riemannian measure zero.

Fix $H \in \mathfrak{a}^+$. For each positive restricted root choose a B_θ-orthonormal basis $X_{\alpha,j}$ of $\mathfrak{g}_\alpha$. The vectors

$$K_{\alpha,j} = X_{\alpha,j} + \theta X_{\alpha,j} \in \mathfrak{k}, \qquad P_{\alpha,j} = X_{\alpha,j} - \theta X_{\alpha,j} \in \mathfrak{p}$$

span, root string by root string, the tangent directions of K/M and the orthogonal complement of $\mathfrak{a}$ in $\mathfrak{p}$. Different restricted-root spaces are B_θ-orthogonal. Differentiate $\exp(tK_{\alpha,j})e^H K$ at $t = 0$ and translate the tangent vector back to the base point by e^{-H}. Its $\mathfrak{p}$-component is the $\mathfrak{p}$-part of

$$\begin{aligned}\mathrm{Ad}(e^{-H})K_{\alpha,j} &= e^{-\alpha(H)}X_{\alpha,j} + e^{\alpha(H)}\theta X_{\alpha,j}\\ &= \cosh(\alpha(H))K_{\alpha,j} - \sinh(\alpha(H))P_{\alpha,j}.\end{aligned}$$

Thus this angular direction is stretched by $\sinh\alpha(H)$. The $\mathfrak{a}$ directions have Jacobian one. Multiplying the factors over an orthonormal basis and over root multiplicities gives exactly (A.2b), up to the one overall measure normalization c_G.

The Riemannian measure on G/K is the quotient of Haar measure on G by normalized Haar measure on K. Integrating once more over the right K fiber gives (A.2a). Finally $\sinh t \le \frac{1}{2}e^t$ for $t \ge 0$, so

$$J(H) \le C \exp\left(\sum_{\alpha>0} m_\alpha \alpha(H)\right) = Ce^{2\rho(H)},$$

which is (A.2c). □

3. Restricted roots

For $H \in \mathfrak{a}$, the operators $\operatorname{ad} H$ are commuting self-adjoint operators for B_θ after the standard sign identification, hence

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \bigoplus_{\alpha\in\Sigma} \mathfrak{g}_\alpha, \qquad \mathfrak{g}_\alpha = \{X : [H,X] = \alpha(H)X \ \forall H \in \mathfrak{a}\}. \tag{A.3}$$

Write $\mathfrak{g}_0 = \mathfrak{m} \oplus \mathfrak{a}$ with $\mathfrak{m} = Z_{\mathfrak{k}}(\mathfrak{a})$.

Lemma 3.1 (Root bracket rule)**.** *For restricted roots α, β,*

$$[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta},$$

where the right side is zero if $\alpha+\beta$ is neither a root nor zero. Also $\theta\mathfrak{g}_\alpha = \mathfrak{g}_{-\alpha}$.

Proof. For $X \in \mathfrak{g}_\alpha$, $Y \in \mathfrak{g}_\beta$, Jacobi gives

$$[H,[X,Y]] = [[H,X],Y] + [X,[H,Y]] = (\alpha+\beta)(H)[X,Y].$$

The second assertion follows from $\theta[H,X] = [\theta H, \theta X] = [-H, \theta X]$. □

Lemma 3.2 (The split coroot from one root vector)**.** *Let $0 \neq X \in \mathfrak{g}_\alpha$ and set $Y = -\theta X$. Then $[X,Y] \in \mathfrak{a}$ is a positive multiple of the vector $H_\alpha \in \mathfrak{a}$ dual to α under B_θ. After rescaling X and Y, the three elements (X, H_α, Y) satisfy the $\mathfrak{sl}_2$ relations.*

Proof. For $H \in \mathfrak{a}$, invariance of B gives

$$B([X,-\theta X],H) = -B(X,[\theta X,H]) = -\alpha(H)B(X,\theta X).$$

Since $-B(X,\theta X) = B_\theta(X,X) > 0$, the $\mathfrak{a}$-component is a positive multiple of H_α.

For $M \in \mathfrak{m}$, the operator $\operatorname{ad} M$ is skew-adjoint for B_θ on $\mathfrak{g}_\alpha$, so

$$0 = B_\theta([M,X],X) = -B([M,X],\theta X) = -B(M,[X,\theta X]).$$

Thus $[X,-\theta X]$ is orthogonal to $\mathfrak{m}$ and therefore lies in $\mathfrak{a}$. Rescale so $[X,Y] = H_\alpha$ and $\alpha(H_\alpha) = 2$. Then

$$[H_\alpha, X] = 2X, \qquad [H_\alpha, Y] = -2Y,$$

which are the $\mathfrak{sl}_2$ relations. □

4. Rank-one root groups and Mautner propagation

For an indivisible restricted root α, put

$$\mathfrak{u}_\alpha = \mathfrak{g}_\alpha \oplus \mathfrak{g}_{2\alpha}, \qquad U_\alpha = \exp \mathfrak{u}_\alpha,$$

with the 2α summand omitted when it is not a root. Let

$$G_\alpha = \langle U_\alpha, U_{-\alpha}\rangle^\circ.$$

THEOREM 4.1 (Rank-one restricted-root Mautner statement). *If a vector in a unitary representation is fixed by U_α, then it is fixed by $A_\alpha = \exp(\mathbb{R}H_\alpha)$ and by G_α.*

PROOF. Choose $0 \neq X \in \mathfrak{g}_\alpha$. By Lemma 3.2, $\exp(\mathbb{R}X)$ sits in an SL_2-subgroup whose diagonal is A_α. The explicit rank-one theorem, Theorem 2.2, makes the vector A_α-invariant.

Every nonzero weight of $\mathrm{Ad}(A_\alpha)$ on $\mathfrak{u}_{\pm\alpha}$ is $\pm\alpha(H_\alpha)$ or $\pm 2\alpha(H_\alpha)$. Hence conjugation by A_α contracts each one-parameter subgroup in U_α in one time direction and each one in $U_{-\alpha}$ in the other. The contraction lemma, Corollary 1.2, therefore makes the vector fixed by the full $U_{-\alpha}$ as well. It was fixed by U_α by hypothesis, so it is fixed by their connected generated subgroup G_α. □

5. Generation by restricted root groups

THEOREM 5.1 (All-root generation). *Let G_0 be connected, noncompact, almost simple, with finite center. The connected subgroup generated by the nonzero restricted root groups U_α is G_0.*

PROOF. Let $\mathfrak{h}$ be the Lie algebra generated by all nonzero root spaces $\mathfrak{g}_\alpha$. It is nonzero. It is stable under $\mathfrak{a}$ because each root space is an $\mathfrak{a}$-weight space; it is stable under $\mathfrak{m}$ because $\mathfrak{m} = Z_{\mathfrak{k}}(\mathfrak{a})$ preserves every root space; and it is stable under every $\mathfrak{g}_\beta$ by construction and Jacobi. Since

$$\mathfrak{g}_0 = \mathfrak{m} \oplus \mathfrak{a} \oplus \sum_{\alpha\in\Sigma} \mathfrak{g}_\alpha,$$

these observations say that $\mathfrak{h}$ is an ideal of $\mathfrak{g}_0$. Almost simplicity and $\mathfrak{h} \neq 0$ imply $\mathfrak{h} = \mathfrak{g}_0$. Thus the connected generated subgroup has full Lie algebra, hence is open. A connected group has no proper open subgroup, so it is all of G_0. □

Corollary 5.2 (Simple-root Mautner envelope)**.** *Let G_0 be as above and let α be a simple restricted root. If a vector is fixed by U_α, then it is fixed by G_0.*

PROOF. Theorem 4.1 gives invariance under A_α. If β is a simple root adjacent to α, then $\beta(H_\alpha) \neq 0$, so A_α contracts U_β in one direction and the contraction lemma gives U_β-invariance. Iterating along the connected Dynkin diagram makes the vector fixed by every simple-root coroot torus. The simple coroots span $\mathfrak{a}$, hence the vector is fixed by $A = \exp\mathfrak{a}$.

For any nonzero restricted root γ, some $H \in \mathfrak{a}$ satisfies $\gamma(H) \neq 0$. The one-parameter subgroup $\exp(tH) \subset A$ contracts U_γ in one direction, so the contraction lemma gives U_γ-invariance. Theorem 5.1 now gives G_0-invariance. □

6. Iwasawa decomposition

Choose Σ^+ and put

$$\mathfrak{n} = \bigoplus_{\alpha>0} \mathfrak{g}_\alpha, \qquad N = \exp \mathfrak{n}, \qquad A = \exp \mathfrak{a}.$$

The exponential is a diffeomorphism $\mathfrak{n} \to N$ because $\mathfrak{n}$ is nilpotent.

Lemma 6.1 (No compact element in a conjugate of AN)**.** *If $s \in AN$ is conjugate in G to an element of K, then $s = e$.*

Proof. Write $s = an$ with $a = e^H$. Order the restricted-root decomposition by root height. On $\mathfrak{g}$, $\mathrm{Ad}(a)$ is diagonal with eigenvalues $e^{\alpha(H)}$ on $\mathfrak{g}_\alpha$, while $\mathrm{Ad}(n)$ is unipotent upper triangular in this ordering. Thus the eigenvalues of $\mathrm{Ad}(s)$ include the diagonal numbers $e^{\alpha(H)}$.

If s is conjugate into K, then $\mathrm{Ad}(s)$ is conjugate to an orthogonal operator for B_θ, so all its eigenvalues have modulus one. Hence $\alpha(H) = 0$ for every restricted root α. The restricted roots span $\mathfrak{a}^*$ because G has no compact factor, so $H = 0$ and $a = e$.

Now $s = n$ is unipotent. An orthogonal unipotent operator is the identity, so $\mathrm{Ad}(n) = I$. Hence n lies in the finite center of G. But the connected unipotent group N has no nontrivial finite-order element, therefore $n = e$. $\square$

Theorem 6.2 (Iwasawa decomposition). *Multiplication*

$$K \times A \times N \longrightarrow G, \qquad (k, a, n) \mapsto kan,$$

is a diffeomorphism. Equivalently, AN acts simply transitively on G/K by left translation. If $a = e^H$, conjugation by a multiplies Haar measure on N by

$$e^{2\rho(H)}, \qquad 2\rho = \sum_{\alpha>0} m_\alpha \alpha. \tag{A.4}$$

Proof. Consider the AN-action on $X = G/K$. Its stabilizer at gK consists of elements of $AN \cap gKg^{-1}$, which is trivial by Lemma 6.1. Thus every orbit has dimension $\dim A + \dim N$. The restricted-root decomposition gives

$$\dim X = \dim \mathfrak{p} = \dim \mathfrak{a} + \sum_{\alpha>0} m_\alpha = \dim A + \dim N.$$

Hence every AN-orbit is open. The orbits partition the connected space X into disjoint open subsets; each orbit is therefore also closed (its complement is the union of the other open orbits). Connectedness forces a single orbit. Thus $AN \to X$, $an \mapsto anK$, is a bijective local diffeomorphism and hence a diffeomorphism.

The preceding argument gives the unique ANK decomposition. Apply it to g^{-1}: write $g^{-1} = ank$. Then

$$g = k^{-1}n^{-1}a^{-1} = k^{-1}a^{-1}(an^{-1}a^{-1}),$$

and $an^{-1}a^{-1} \in N$ because A normalizes N. Thus every g has a KAN decomposition with the same chosen positive group N. If $k_1a_1n_1 = k_2a_2n_2$, then after inversion we obtain two ANK decompositions; simple transitivity of AN on G/K forces equality of the A, N factors and then of the K factors. The inverse maps are smooth because both multiplication maps are bijective local diffeomorphisms.

Finally, $\mathrm{Ad}(e^H)$ acts on $\mathfrak{g}_\alpha$ by the scalar $e^{\alpha(H)}$. Haar measure on the simply connected nilpotent group N is the push-forward of Lebesgue measure in exponential coordinates, and an automorphism changes it by the absolute determinant of its differential. Therefore conjugation by e^H multiplies dn by

$$\prod_{\alpha>0} e^{m_\alpha \alpha(H)} = e^{2\rho(H)}.$$

□

Proposition 6.3 (Haar measure in KAN coordinates)**.** *Normalize Haar measure dk on K to have mass one, and choose Haar measures da and dn on A and N. Put*

$$\delta(a) = |\det(\mathrm{Ad}(a)|_{\mathfrak{n}})| = e^{2\rho(H(a))}.$$

After one overall normalization of Haar measure on G, the KAN coordinate formula is

$$dg = \delta(a)\, dk\, da\, dn. \tag{A.5}$$

The factor is $\delta(a)$, not $\delta(a)^{-1}$; this sign convention is important in the Herz calculation of Appendix C.

Proof. A connected semisimple group is unimodular. Indeed, for $X \in \mathfrak{g}$,

$$\left.\frac{d}{dt}\right|_{t=0} \log\det \mathrm{Ad}(e^{tX}) = \mathrm{tr}(\mathrm{ad}\, X) = 0,$$

because $\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$ and the trace of a commutator is zero. Hence left and right Haar measures on G coincide.

We compute the right Haar density on the solvable factor AN in the coordinates an. Right multiplication by $n_0 \in N$ is translation in N and preserves dn. For $a_0 \in A$,

$$(an)a_0 = (aa_0)(a_0^{-1}na_0).$$

By Theorem 6.2, the automorphism $n \mapsto a_0^{-1} n a_0$ changes dn by the factor $\delta(a_0)^{-1}$. Therefore

$$\delta(aa_0)\, d(aa_0)\, d(a_0^{-1} n a_0) = \delta(a)\delta(a_0)\, da\, \delta(a_0)^{-1} dn = \delta(a)\, da\, dn.$$

Thus $\delta(a) da\, dn$ is right Haar measure on AN.

Pull this product density to G through the diffeomorphism $K \times A \times N \to G$. It is left K-invariant and right AN-invariant. Haar measure dg has the same two invariances. The ratio of the two smooth positive densities is consequently invariant under the transitive left-K/right-AN action on G: if $g = kan$, then g is carried to the identity by left multiplication by k^{-1} and right multiplication by $(an)^{-1}$. Hence the ratio is constant. Absorb that constant into the normalization of dg to obtain (A.5). $\square$

7. Cartan escape and contraction

Lemma 7.1 (Escape in a closed chamber). *If $H_j \in \overline{\mathfrak{a}^+}$ and $\|H_j\| \to \infty$, then after passing to a subsequence there is a simple root α with $\alpha(H_j) \to +\infty$.*

Proof. On the closed chamber the simple roots form nonnegative coordinates and define a norm equivalent to the Euclidean norm on $\mathfrak{a}$. If every simple-root coordinate stayed bounded, H_j would remain bounded. Thus one is unbounded; pass to a subsequence on which it tends to $+\infty$. $\square$

Lemma 7.2 (Root contraction). *If $X \in \mathfrak{g}_\alpha$ and $a_j = e^{H_j}$ with $\alpha(H_j) \to +\infty$, then*

$$a_j^{-1} \exp(X) a_j = \exp(e^{-\alpha(H_j)} X) \to e.$$

The same statement holds for $X \in \mathfrak{g}_{r\alpha}$ with the factor $e^{-r\alpha(H_j)}$.

Proof. By definition of the root space, $\mathrm{Ad}(a_j^{-1})X = e^{-\alpha(H_j)}X$. Compatibility of conjugation with the exponential gives the formula, and continuity of the exponential gives convergence to e. $\square$

Proof and provenance. The proof architecture above deliberately avoids three shortcuts that are unsafe at the present proof standard: it does not infer the Cartan decomposition from an unproved functional-calculus closure claim, it does not assert surjectivity of brackets between root multiplicity spaces, and it does not derive Iwasawa from an unspecified sequence of “rank-one Gauss eliminations.” The only imported item is Hopf–Rinow, part of the foundational finite-dimensional Riemannian geometry floor rather than a representation-theoretic or homogeneous-dynamics input.

APPENDIX B

Functional-Analytic Lemmas for Weak Containment

This appendix supplies the two functional-analytic steps used in the converse direction of Theorem 2.2. The arguments are short enough that there is no reason to hide them behind the phrase "by C^*-algebra theory."

1. Density of vector states

Lemma 1.1 (Vector-state density). *Let A be a C^*-algebra faithfully and nondegenerately represented on a Hilbert space $\mathcal{H}$. The weak-* closed convex hull of the unit vector states*

$$\omega_\xi(a) = \langle a\xi, \xi\rangle, \qquad \|\xi\| = 1,$$

is the whole state space $S(A)$.

Proof. Unitize A if necessary; states extend uniquely to the unitization with value 1 at the new identity, so it is enough to treat the unital case. Let C be the weak-* closed convex hull of the vector states. Suppose a state φ lies outside C. By Hahn–Banach separation in the locally convex weak-* topology, there is a self-adjoint $a \in A$ and a real number c such that

$$\varphi(a) > c \geq \sup_{\psi \in C} \psi(a). \tag{B.1}$$

Because the representation is faithful, the operator spectrum of a equals its C^*-spectrum, and

$$\sup_{\|\xi\|=1} \langle a\xi, \xi\rangle = \max \sigma(a) =: M.$$

Indeed $a \leq M1$ by continuous functional calculus, while unit vectors supported in spectral intervals $(M - \varepsilon, M]$ make the quadratic form arbitrarily close to M. Hence the right side of (B.1) equals M. But positivity of the state and $a \leq M1$ imply $\varphi(a) \leq M$, contradiction. Thus $C = S(A)$. $\square$

Corollary 1.2 (Finite vector approximation on finite sets). *Given a state φ, elements $a_1, \ldots, a_N \in A$, and $\varepsilon > 0$, there are vectors $\xi_1, \ldots, \xi_m \in \mathcal{H}$ with*

$\sum_j \|\xi_j\|^2 = 1$ *such that*

$$\left|\varphi(a_i) - \sum_j \langle a_i \xi_j, \xi_j\rangle\right| < \varepsilon \qquad (1 \le i \le N).$$

PROOF. A weak-* neighborhood of φ is specified by finitely many evaluations. Lemma 1.1 gives a finite convex combination $\sum t_j \omega_{u_j}$ inside it. Put $\xi_j = \sqrt{t_j} u_j$. □

2. Smoothing group coefficients

Let π be strongly continuous, v a unit vector, and let $\psi \in C_c(G)$ be nonnegative with $\int \psi = 1$. Put $v_\psi = \pi(\psi)v$.

Lemma 2.1 (Smoothing coefficients uniformly on compact sets)**.** *For every compact $C \subset G$ and $\eta > 0$, one may choose ψ supported in a sufficiently small neighborhood of e so that*

$$\sup_{g \in C} |\langle \pi(g)v, v\rangle - \langle \pi(g)v_\psi, v_\psi\rangle| < \eta. \tag{B.2}$$

Moreover

$$\langle \pi(g)v_\psi, v_\psi\rangle = \langle \pi(b_g)v, v\rangle, \qquad b_g = \psi^* * L_g\psi \in C_c(G), \tag{B.3}$$

where $(L_g\psi)(x) = \psi(g^{-1}x)$, and $g \mapsto b_g$ is continuous in $L^1(G)$, hence in every integrated-representation norm.

PROOF. Strong continuity at the identity gives a neighborhood U such that $\|\pi(x)v - v\| < \delta$ for $x \in U$. Choose ψ supported in U. Then

$$\|v_\psi - v\| \le \int \psi(x)\, \|\pi(x)v - v\|\, dx < \delta.$$

For every g, unitarity gives

$$\begin{aligned} |\langle \pi(g)v, v\rangle - \langle \pi(g)v_\psi, v_\psi\rangle| &\le \|v - v_\psi\|\, \|v\| + \|v_\psi\|\, \|v - v_\psi\| \\ &\le 2\delta + \delta^2. \end{aligned}$$

Choose δ so this is $< \eta$. This estimate is actually uniform on all of G.

For (B.3), $\pi(g)\pi(\psi) = \pi(L_g\psi)$ by the left-Haar change of variables, and $\pi(\psi)^* = \pi(\psi^*)$. Thus

$$\langle \pi(g)v_\psi, v_\psi\rangle = \langle \pi(\psi^*)\pi(g)\pi(\psi)v, v\rangle = \langle \pi(\psi^* * L_g\psi)v, v\rangle.$$

Translation is continuous in L^1 for C_c functions, and convolution by a fixed L^1 function is bounded on L^1, so $g \mapsto b_g$ is L^1-continuous. Since $\|\rho(f)\|_{\mathrm{op}} \le \|f\|_1$ for any unitary representation, it is continuous in all integrated norms as well. □

Lemma 2.2 (Compact-image reduction)**.** *Let $C \subset G$ be compact. For fixed $\psi \in C_c(G)$, the set*

$$\{b_g = \psi^* * L_g\psi : g \in C\}$$

is compact in $L^1(G)$. Therefore approximation of a bounded functional on a sufficiently fine finite net of these elements implies uniform approximation on all $g \in C$.

Proof. Lemma 2.1 makes the map from compact C to L^1 continuous, so its image is compact and admits a finite net. If two elements are within δ in L^1, every state arising from an integrated unitary representation differs on them by at most δ because the representation norm is bounded by the L^1 norm. $\square$

3. Why this closes Theorem 6.2

Assume $\|\pi(f)\|_{\mathrm{op}} \leq \|\rho(f)\|_{\mathrm{op}}$ for all f. A π-vector state descends to a state of the C^*-completion defined by ρ. Corollary 1.2 approximates that state on a finite net of smoothed group elements. Lemmas 2.1 and 2.2 turn this into uniform approximation of the original group coefficient on an arbitrary compact set. This is exactly the missing direction in weak containment; no universal-representation theorem is being invoked implicitly.

4. Finite-dimensional Fell approximation

The scalar coefficient formulation of weak containment is enough for Chapter 6. The exact K-dimension factor in Chapter 7 needs a simultaneous approximation of a finite family of vectors. We derive it from the same vector-state argument rather than invoke a matrix-valued Fell theorem as a black box.

Lemma 4.1 (Finite-vector Fell approximation)**.** *Let $\pi \prec \rho$, let $E \subset \mathcal{H}_\pi$ be finite dimensional, let $C \subset G$ be compact, and let $\eta > 0$. There are an integer N and a linear map*

$$T_0 : E \longrightarrow \mathcal{H}_\rho^{\oplus N}$$

such that, for all ξ, ζ in the unit ball of E and all $g \in C$,

$$|\langle T_0\xi, T_0\zeta\rangle - \langle \xi, \zeta\rangle| < \eta, \tag{B.4}$$

$$|\langle \rho^{\oplus N}(g)T_0\xi, T_0\zeta\rangle - \langle \pi(g)\xi, \zeta\rangle| < \eta. \tag{B.5}$$

Proof. Choose an orthonormal basis $e_1, \dots, e_r$ of E. All matrix coefficients on the unit ball are linear combinations, with bounded coefficients depending only on r, of the r^2 functions

$$g \longmapsto \langle \pi(g)e_i, e_j\rangle.$$

It therefore suffices to approximate these r^2 functions uniformly on C, and the Gram matrix at $g = e$.

Let A_ρ be the C^*-completion generated by the integrated representation of ρ; it is faithfully represented on $\mathcal{H}_\rho$. Since $\pi \prec \rho$, Theorem 2.2 makes the integrated representation of π factor through A_ρ. Consider the faithful representation of $M_r(A_\rho)$ on $\mathcal{H}_\rho \otimes \mathbb{C}^r$. In the representation $\pi \otimes \mathrm{id}_{\mathbb{C}^r}$, put

$$\zeta_0 = \sum_{i=1}^{r} e_i \otimes \delta_i.$$

The positive functional of ζ_0 on $M_r(A_\rho)$ records every coefficient, because for the matrix unit E_{ji},

$$\langle (\pi(a) \otimes E_{ji})\zeta_0, \zeta_0 \rangle = \langle \pi(a) e_i, e_j \rangle. \tag{B.6}$$

The vector ζ_0 has norm $\sqrt{r}$, so apply Lemma 1.1 to the state $r^{-1}\langle \cdot \zeta_0, \zeta_0 \rangle$ on $M_r(A_\rho)$. We also explain the passage from integrated elements to the whole compact set C. Because E is finite dimensional, strong continuity of π is uniform on its unit sphere near the identity: for every $\delta > 0$ there is a nonnegative $\psi \in C_c(G)$, $\int \psi = 1$, such that

$$\sup_{\xi \in E,\ \|\xi\| \le 1} \|\pi(\psi)\xi - \xi\| < \delta.$$

Hence every mixed coefficient on the unit ball of E differs uniformly from its smoothed version by $O(\delta)$. The smoothed matrix coefficient at g is evaluation of the matrix functional on $b_g \otimes E_{ji}$, with $b_g = \psi^* * L_g \psi$. Lemma 2.2 makes $\{b_g : g \in C\}$ compact in A_ρ, so finitely many matrix elements form a common net. Apply vector-state density to that finite family. We obtain unit vectors in $\mathcal{H}_\rho \otimes \mathbb{C}^r$ and convex weights t_s whose vector functionals approximate the normalized functional simultaneously for all i, j and uniformly in $g \in C$.

Write the approximating vectors as

$$q_s = \sum_{i=1}^{r} q_{s,i} \otimes \delta_i, \qquad q_{s,i} \in \mathcal{H}_\rho,$$

with convex weights t_s. Define

$$T_0 e_i = \bigoplus_s \sqrt{r t_s}\, q_{s,i}.$$

Then (B.6) says exactly that the matrix entries

$$\langle \rho^{\oplus N}(g) T_0 e_i, T_0 e_j \rangle$$

approximate the corresponding π-matrix coefficients, while $g = e$ gives the Gram matrix. By choosing the matrix-entry error smaller than η/r^2 and expanding arbitrary unit vectors in the basis, we obtain (B.4)–(B.5). □

Lemma 4.2 (K-equivariant Fell approximation)**.** *Assume in Lemma 4.1 that E is K-invariant. Then the approximating maps can be chosen in a sequence $T_j : E \to \mathcal{H}_\rho^{\oplus N_j}$ such that*

$$T_j \pi(k) = \rho^{\oplus N_j}(k) T_j \qquad (k \in K), \tag{B.7}$$

and, uniformly for g in any prescribed compact C,

$$\langle T_j \xi, T_j \zeta \rangle \longrightarrow \langle \xi, \zeta \rangle,$$
$$\langle \rho^{\oplus N_j}(g) T_j \xi, T_j \zeta \rangle \longrightarrow \langle \pi(g)\xi, \zeta \rangle$$

for $\xi, \zeta \in E$. Consequently

$$\dim \operatorname{span}_K(T_j v) \le \dim \operatorname{span}_K(v) \qquad (v \in E). \tag{B.8}$$

Proof. Apply Lemma 4.1 with the larger compact set

$$K(C \cup \{e\})K.$$

Let T_0 be the resulting map and average it over K:

$$T\xi = \int_K \rho^{\oplus N}(k^{-1}) T_0 \pi(k) \xi \, dk. \tag{B.9}$$

The Bochner integral exists because the integrand is continuous and K is compact. For $k_0 \in K$, change variables $r = kk_0$ to obtain

$$T\pi(k_0) = \rho^{\oplus N}(k_0) T,$$

which is (B.7).

We check that averaging has not destroyed the approximation. Expanding (B.9) twice gives

$$\langle \rho^{\oplus N}(g) T\xi, T\zeta \rangle = \int_{K \times K} \langle \rho^{\oplus N}(\ell g k^{-1}) T_0 \pi(k)\xi, T_0 \pi(\ell)\zeta \rangle \, dk \, d\ell.$$

For $g \in C$, the element $\ell g k^{-1}$ lies in KCK. The vectors $\pi(k)\xi$ and $\pi(\ell)\zeta$ remain in the unit ball of the K-invariant space E. Hence the integrand differs uniformly by at most the Lemma 4.1 error from

$$\langle \pi(\ell g k^{-1}) \pi(k)\xi, \pi(\ell)\zeta \rangle = \langle \pi(g)\xi, \zeta \rangle.$$

Integrating preserves the error bound. Taking $g = e$ proves the corresponding Gram approximation. Choosing errors tending to zero gives the asserted sequence.

Finally (B.7) gives

$$\operatorname{span}\{\rho(k) T_j v : k \in K\} = T_j\big(\operatorname{span}\{\pi(k) v : k \in K\}\big),$$

so its dimension is at most the dimension of the domain cyclic space, proving (B.8). ☐

Proof and provenance. Lemmas 4.1 and 4.2 are the exact finite-dimensional form of Fell approximation used in Appendix C. Their proof is reduced to the vector-state density lemma already proved above, applied once to a matrix C^*-algebra, followed by compact smoothing and an elementary K-average. Thus the Cowling–Haagerup–Howe transfer later does not hide a matrix-valued weak-containment theorem.

APPENDIX C

Harish–Chandra and Herz Majorization: Detailed Proofs

This appendix owns the harmonic-analysis statements used in Chapter 7. The point is not to reproduce the full spherical Fourier transform. For the present series we need only: the boundary realization of Ξ, its Harish–Chandra chamber bound, the regular-representation Herz inequality, and the transfer of that inequality to tempered representations. Each of those steps is proved below.

1. The normalized spherical representation

Let $P = MAN$ be the minimal parabolic associated with the positive system. Its modular function on A is

$$\Delta_P(e^H) = e^{2\rho(H)}.$$

The compact picture of the normalized induced representation is $L^2(K/M)$. If

$$g^{-1}k = k_g e^{H_g} n_g$$

is the Iwasawa decomposition, define

$$(\Pi_0(g)F)(kM) = e^{-\rho(H_g)} F(k_g M). \tag{C.1}$$

Proposition 1.1 (Spherical realization)**.** *Formula* (C.1) *defines a unitary representation. For the constant unit vector* $\mathbf{1}$*,*

$$\langle \Pi_0(g)\mathbf{1}, \mathbf{1}\rangle = \int_K e^{-\rho(H(g^{-1}k))}\, dk = \Xi(g). \tag{C.2}$$

Consequently Ξ *is continuous, positive definite,* K*-bi-invariant, and* $0 < \Xi \leq 1$*.*

Proof. The quotient $G/P \simeq K/M$ has tangent space $\bar{\mathfrak{n}}$ at the base point. For $a = e^H$, the differential of a on $\bar{\mathfrak{n}}$ has determinant

$$\prod_{\alpha>0} e^{-m_\alpha \alpha(H)} = e^{-2\rho(H)}.$$

Thus the Radon–Nikodym derivative of the G-action on the K-invariant probability class is

$$\frac{d(g_*dk)}{dk}(kM) = e^{-2\rho(H(g^{-1}k))}. \tag{C.3}$$

The square of the factor in (C.1) is exactly (C.3), so a change of variables proves unitarity. Pairing $\Pi_0(g)\mathbf{1}$ with $\mathbf{1}$ gives the first integral in (C.2). Inversion and K-bi-invariance of the Iwasawa cocycle identify it with the convention of Chapter 7. A unitary matrix coefficient is positive definite and has absolute value at most the product of the vector norms; here both norms equal one. $\square$

2. Open Bruhat coordinates and convex root order

Put $\bar{N} = \theta(N)$. The map $\bar{N} \to G/P$, $\bar{n} \mapsto \bar{n}P$, is a diffeomorphism onto the open Bruhat cell. Write

$$\bar{n} = k(\bar{n})e^{H(\bar{n})}n(\bar{n}).$$

Lemma 2.1 (Open-cell integration)**.** *There is a constant $c_G > 0$ such that for every nonnegative measurable F on K/M,*

$$\int_{K/M} F(kM)\,dk = c_G \int_{\bar{N}} F(k(\bar{n})M)e^{-2\rho(H(\bar{n}))}\,d\bar{n}. \tag{C.4}$$

Proof. The complement of the open Bruhat cell is a finite union of lower-dimensional cells, hence has Haar measure zero. The pullback of dk is a smooth positive density on $\bar{N}$. Both that density and

$$e^{-2\rho(H(\bar{n}))}\,d\bar{n}$$

have the same Radon–Nikodym transformation law under the left A-action: conjugation by e^Y multiplies $d\bar{n}$ by $e^{-2\rho(Y)}$, while the Iwasawa cocycle contributes the inverse factor. The ratio is therefore AN-invariant on the open cell. Since AN acts transitively there, the ratio is constant. Normalize by the total K/M mass to obtain c_G. $\square$

Let Σ_0^+ denote the positive indivisible restricted roots. Choose a reduced expression of the longest Weyl element. It induces a convex ordering

$$\beta_1, \ldots, \beta_r \in \Sigma_0^+$$

with the property that whenever $\beta_i + \beta_j$ is a root, it occurs between the two appropriate positions. Put

$$\bar{\mathfrak{n}}_\beta = \mathfrak{g}_{-\beta} \oplus \mathfrak{g}_{-2\beta}, \qquad \bar{N}_\beta = \exp \bar{\mathfrak{n}}_\beta,$$

omitting the second summand when 2β is not a root.

Lemma 2.2 (Triangular root coordinates). *Multiplication gives a global polynomial coordinate map*

$$\bar{N}_{\beta_1} \times \cdots \times \bar{N}_{\beta_r} \longrightarrow \bar{N} \tag{C.5}$$

whose Jacobian is a nonzero constant. Reordering or conjugating by A changes a coordinate of a given root height by its linear root-space term plus a polynomial in strictly earlier coordinates. In particular every such change is triangular and has constant Jacobian.

Proof. The exponential map is a polynomial diffeomorphism from the nilpotent Lie algebra $\bar{\mathfrak{n}}$ to $\bar{N}$. BCH terminates after finitely many brackets. By Lemma 3.1, a bracket of negative-root vectors has root equal to the sum of their roots; convex ordering therefore moves every commutator strictly later in (C.5). Hence the coordinate map is triangular with identity linear part. Its polynomial inverse is obtained recursively, one root height at a time, and its Jacobian is constant. Conjugation by e^H multiplies the $-\beta$ and -2β linear coordinates by $e^{-\beta(H)}$ and $e^{-2\beta(H)}$ respectively; BCH again adds only later polynomial terms. ☐

3. The exact open-cell convolution identity

The estimate of Ξ is most transparent after one more exact manipulation of Lemma 2.1. Put

$$P(\bar{n}) = e^{-2\rho(H(\bar{n}))} \qquad (\bar{n} \in \bar{N}). \tag{C.5a}$$

Thus $P^{1/2}(\bar{n}) = e^{-\rho(H(\bar{n}))}$ is the square root of the boundary density in the open Bruhat chart.

Lemma 3.1 (Open-cell convolution formula). *For $a = e^H \in A^+$,*

$$\Xi(a) = c_G e^{-\rho(H)} \int_{\bar{N}} P(a\bar{n}a^{-1})^{1/2} P(\bar{n})^{1/2}\, d\bar{n}. \tag{C.5b}$$

The same identity holds on the closed chamber by continuity.

Proof. Write the Iwasawa decomposition of $\bar{n}$ as

$$\bar{n} = k(\bar{n})b(\bar{n})n(\bar{n}), \qquad b(\bar{n}) = e^{H(\bar{n})}.$$

Then

$$k(\bar{n}) = \bar{n}\, n(\bar{n})^{-1} b(\bar{n})^{-1}.$$

Right multiplication by an element of N does not change the Iwasawa A-coordinate, while right multiplication by $b \in A$ adds $H(b)$. Consequently

$$H(ak(\bar{n})) = H(a\bar{n}) - H(\bar{n}).$$

Since

$$a\bar{n} = (a\bar{n}a^{-1})a,$$

the same right-A rule gives

$$H(a\bar{n}) = H(a\bar{n}a^{-1}) + H.$$

Insert these two identities into the compact integral for $\Xi(a)$ and then use Lemma 2.1:

$$\begin{aligned}\Xi(a) &= c_G \int_{\bar{N}} e^{-\rho(H(ak(\bar{n})))} e^{-2\rho(H(\bar{n}))}\, d\bar{n} \\ &= c_G e^{-\rho(H)} \int_{\bar{N}} e^{-\rho(H(a\bar{n}a^{-1}))} e^{-\rho(H(\bar{n}))}\, d\bar{n},\end{aligned}$$

which is exactly (C.5b). □

4. The rank-one logarithmic estimate

The only analytic phenomenon that produces a polynomial loss in the chamber estimate is the borderline logarithm already visible in real rank one. We prove that model carefully, including the multiplicity-two root case.

For an indivisible restricted root β put

$$\mathfrak{v}_\beta = \mathfrak{g}_{-\beta}, \qquad \mathfrak{z}_\beta = \mathfrak{g}_{-2\beta}, \qquad p_\beta = \dim \mathfrak{v}_\beta, \quad q_\beta = \dim \mathfrak{z}_\beta,$$

with $\mathfrak{z}_\beta = 0$ if 2β is not a root, and put

$$Q_\beta = p_\beta + 2q_\beta.$$

The rank-one opposite nilpotent algebra is $\bar{\mathfrak{n}}_\beta = \mathfrak{v}_\beta \oplus \mathfrak{z}_\beta$. Choose the B_θ inner product $\langle X, Y\rangle = -B(X, \theta Y)$.

Lemma 4.1 (The rank-one root group is of H-type)**.** *For $Z \in \mathfrak{z}_\beta$ define $J_Z : \mathfrak{v}_\beta \to \mathfrak{v}_\beta$ by*

$$\langle J_Z X, Y\rangle = \langle Z, [X, Y]\rangle. \tag{C.6}$$

After one fixed structural rescaling of the norm on $\mathfrak{z}_\beta$,

$$J_Z^2 = -\|Z\|^2 I_{\mathfrak{v}_\beta}. \tag{C.7}$$

Moreover $\mathfrak{z}_\beta$ is central in $\bar{\mathfrak{n}}_\beta$ and BCH gives

$$(X, Z)(Y, W) = \left(X + Y, Z + W + \frac{1}{2}[X, Y]\right). \tag{C.8}$$

The automorphisms

$$\delta_r(X, Z) = (rX, r^2 Z) \tag{C.9}$$

scale Haar measure by r^{Q_β}.

Proof. The root relations give $[\mathfrak{g}_{-\beta}, \mathfrak{g}_{-\beta}] \subset \mathfrak{g}_{-2\beta}$ and $[\mathfrak{g}_{-\beta}, \mathfrak{g}_{-2\beta}] = 0$, because -3β is never a restricted root. Thus the algebra is two-step and (C.8) follows from the terminating BCH formula.

By invariance of the Killing form,

$$\langle Z, [X, Y]\rangle = -B(Z, [\theta X, \theta Y]) = -B([Z, \theta X], \theta Y),$$

so

$$J_Z X = [Z, \theta X]. \tag{C.10}$$

Apply J_Z once more. Since $[Z, X] = 0$ for root reasons, Jacobi gives

$$J_Z^2 X = [Z, [\theta Z, X]] = [[Z, \theta Z], X]. \tag{C.11}$$

The element $[Z, \theta Z]$ belongs to $\mathfrak{g}_0$ and is sent to its negative by θ, hence lies in $\mathfrak{a}$. If $H \in \mathfrak{a}$, then

$$B([Z, \theta Z], H) = B(Z, [\theta Z, H]) = 2\beta(H) \|Z\|^2.$$

Therefore $[Z, \theta Z] = c_\beta \|Z\|^2 H_\beta$ for a fixed positive normalization of the coroot, and because X has root $-\beta$, (C.11) becomes

$$J_Z^2 X = -c'_\beta \|Z\|^2 X, \qquad c'_\beta > 0.$$

Rescale the central norm once so that $c'_\beta = 1$. Finally, (C.9) is a Lie-algebra automorphism by the grading, and its determinant is $r^{p_\beta} r^{2q_\beta} = r^{Q_\beta}$. ☐

Lemma 4.2 (The rank-one Gauss pivot)**.** *Put*

$$s = 1 + \frac{1}{4}\|X\|^2, \qquad D = s^2 + \|Z\|^2, \qquad A_Z = sI + J_Z. \tag{C.12}$$

For $\bar{n} = \exp(X + Z) \in \bar{N}_\beta$, the middle A_β-factor in the open Gauss decomposition of the Cartan-symmetric element $\theta(\bar{n})^{-1}\bar{n}$ is

$$\exp\big((\log D) H_\beta\big). \tag{C.13}$$

Proof. The calculation is the rank-one analogue of one step of Cholesky elimination. We spell out the pivots because only the scalar pivot is used later. All terms lie in the five restricted-root degrees $-2\beta, -\beta, 0, \beta, 2\beta$, and the nilpotent pieces are two-step. Thus BCH contains no bracket of length three inside either nilpotent factor.

First polarize $J_Z^2 = -\|Z\|^2 I$. For $Z, W \in \mathfrak{z}_\beta$,

$$J_Z J_W + J_W J_Z = -2\langle Z, W\rangle I, \qquad \langle J_Z X, J_W X\rangle = \langle Z, W\rangle \|X\|^2. \tag{C.14}$$

In particular,

$$A_Z(sI - J_Z) = DI, \qquad [X, J_Z X] = \|X\|^2 Z. \tag{C.15}$$

The second identity is checked by pairing with arbitrary $W \in \mathfrak{z}_\beta$ and using the definition of J_W.

Now perform Gaussian elimination in increasing root degree on

$$\theta(\bar{n})^{-1}\bar{n} = \exp(-\theta X - \theta Z)\exp(X + Z). \tag{C.16}$$

The $\pm\beta$ equations have coefficient operator A_Z. Equation (C.15) gives its inverse explicitly,

$$A_Z^{-1} = D^{-1}(sI - J_Z). \tag{C.17}$$

After these variables are eliminated, the only quadratic central term is $[X, J_Z X]$; the second identity in (C.15) cancels it against the $\|X\|^2/4$ contribution already contained in s. Hence the $\pm 2\beta$ equations have scalar pivot D. The remaining degree-zero term splits into its $\mathfrak{m}_\beta$- and $\mathfrak{a}_\beta$-parts. The $\mathfrak{m}_\beta$-part vanishes by the polarized Clifford relation (C.14), while the $\mathfrak{a}_\beta$-pivot is exactly D. Therefore the middle factor is $\exp((\log D)H_\beta)$.

For a check on the normalization, when $\mathfrak{z}_\beta = 0$ the same elimination is the ordinary 2×2 identity

$$\begin{pmatrix} 1+x^2 & x \\ x & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \frac{x}{1+x^2} & 1 \end{pmatrix} \begin{pmatrix} 1+x^2 & 0 \\ 0 & (1+x^2)^{-1} \end{pmatrix} \begin{pmatrix} 1 & \frac{x}{1+x^2} \\ 0 & 1 \end{pmatrix}; \tag{C.18}$$

the fixed normalization of the root-space norm is exactly the one for which $\|X\|^2/4 = x^2$. The nonreduced case differs only by the central Clifford pivot handled in (C.14)–(C.15). This proves (C.13) without a classification-by-families argument. $\square$

Lemma 4.3 (Rank-one Cayley/Iwasawa density)**.** *Normalize $H_\beta \in \mathfrak{a}$ by $\beta(H_\beta) = 1$ and use the central norm of Lemma 4.1. After the same fixed rescaling of the $\mathfrak{v}_\beta$ coordinate used in Lemma 4.2, put*

$$D(X, Z) = \left(1 + \frac{1}{4}\|X\|^2\right)^2 + \|Z\|^2. \tag{C.22}$$

For $\bar{n} = \exp(X + Z) \in \bar{N}_\beta$ the rank-one Iwasawa projection satisfies

$$H(\bar{n}) = \frac{1}{2}\log D(X, Z)\, H_\beta, \tag{C.23}$$

and therefore, since $2\rho_\beta(H_\beta) = Q_\beta$,

$$P_\beta(X, Z) := e^{-2\rho_\beta(H(\bar{n}))} = D(X, Z)^{-Q_\beta/2}. \tag{C.24}$$

In particular, for the homogeneous gauge

$$r(X, Z) = \left(\|X\|^4 + 16\|Z\|^2\right)^{1/4}, \tag{C.25}$$

there are structural constants $c, C > 0$ such that

$$c(1 + r)^{-2Q_\beta} \le P_\beta(X, Z) \le C(1 + r)^{-2Q_\beta}. \tag{C.26}$$

Proof. If $\bar{n} = kan$, Lemma 4.2 gives

$$a^2 = \exp((\log D)H_\beta),$$

so uniqueness of the one-dimensional A_β coordinate gives (C.23). This is the rank-one Cayley calculation in the exact form needed here; no Poisson-kernel formula has been imported.

Now $2\rho_\beta(H_\beta) = p_\beta + 2q_\beta = Q_\beta$, so (C.24) follows. Finally $D^{1/4}$ is comparable to $1+r(X,Z)$: both fourth powers are, up to structural constants, $1+\|X\|^4+\|Z\|^2$. Raising this comparison to the power $-2Q_\beta$ gives (C.26). □

Lemma 4.4 (Homogeneous polar integration)**.** *Let r be the gauge in (C.15). There is a finite measure $d\omega$ on the unit gauge sphere such that for every nonnegative gauge-radial function $f(r)$,*

$$\int_{\bar{N}_\beta} f(r(\bar{n}))\, d\bar{n} = c_\beta \int_0^\infty f(r) r^{Q_\beta - 1}\, dr. \tag{C.19}$$

Proof. The balls satisfy $\delta_s B_r = B_{sr}$ and Haar measure scales by s^{Q_β}, so $m(B_r) = m(B_1)r^{Q_\beta}$. First apply the layer-cake formula to indicators of radial intervals and then to nonnegative simple functions. Monotone convergence gives (C.19), with $c_\beta = Q_\beta m(B_1)$. □

Lemma 4.5 (Rank-one Iwasawa integral)**.** *In the connected rank-one subgroup attached to β,*

$$\Xi_\beta(e^{tH_\beta}) \le C_\beta(1+t)e^{-\rho_\beta(tH_\beta)} \qquad (t \ge 0). \tag{C.20}$$

More precisely, after the factor $e^{-\rho_\beta(tH_\beta)}$ is removed, the remaining integral is bounded by a structural constant times

$$I_Q(t) = \int_0^\infty \frac{r^{Q-1}}{(1+r)^Q(1+e^{-t}r)^Q}\, dr, \qquad Q = Q_\beta, \tag{C.21}$$

and $I_Q(t) \ll_Q 1+t$.

Proof. Apply the exact convolution formula, Lemma 3.1, inside the rank-one subgroup. Conjugation by e^{tH_β} acts on $\bar{N}_\beta$ as $\delta_{e^{-t}}$. By (C.16),

$$P_\beta(\delta_{e^{-t}}\bar{n})^{1/2} P_\beta(\bar{n})^{1/2} \ll \frac{1}{(1+e^{-t}r(\bar{n}))^Q(1+r(\bar{n}))^Q}.$$

Lemma 4.4 gives (C.21).

Split the scalar integral at 1 and e^t. On $0 < r < 1$ the integrand is $O(r^{Q-1})$; on $1 < r < e^t$ it is $O(r^{-1})$; and on $r > e^t$ it is $O(e^{Qt}r^{-Q-1})$. Hence

$$I_Q(t) \ll 1 + \int_1^{e^t} \frac{dr}{r} + 1 \ll 1+t.$$

Restoring the principal factor proves (C.20). □

5. Successive rank-one reduction

We now explain the higher-rank induction rather than suppressing it in the phrase "order the roots and integrate." Choose a reduced word

$$w_0 = s_{\alpha_1} \cdots s_{\alpha_r}$$

for the longest restricted Weyl element and put

$$\beta_j = s_{\alpha_1} \cdots s_{\alpha_{j-1}} \alpha_j.$$

The β_j are precisely the positive indivisible roots in a convex order. Put $V_j = \bar{N}_{\beta_j}$.

Lemma 5.1 (Root-by-root factorization)**.** *Multiplication*

$$V_1 \times \cdots \times V_r \longrightarrow \bar{N} \tag{C.22a}$$

is a polynomial diffeomorphism with constant nonzero Jacobian. For each j, after the variables $V_{j+1}, \ldots, V_r$ are translated by polynomial functions of the preceding variables, the conditional V_j-dependence of the two half-densities in (C.5b) is the rank-one pair

$$P_{\beta_j}(\delta_{e^{-\beta_j(H)}} v_j)^{1/2} P_{\beta_j}(v_j)^{1/2}. \tag{C.23a}$$

The translations have Jacobian one and do not change the ranges of the later integrations.

Proof. The first statement is Lemma 2.2. We prove the conditional statement by descending induction on j. Let $\bar{N}^{(j)} = V_j \cdots V_r$. Convexity of the root order implies that $\bar{N}^{(j+1)}$ is normal in $\bar{N}^{(j)}$: every commutator of a β_j-root vector with a later root vector has strictly later root. Thus

$$1 \longrightarrow \bar{N}^{(j+1)} \longrightarrow \bar{N}^{(j)} \longrightarrow V_j \longrightarrow 1 \tag{C.24a}$$

is a polynomial semidirect extension, and the quotient coordinate is exactly the rank-one root-string coordinate.

Write an element as $v_j u$ with $u \in \bar{N}^{(j+1)}$. In the two KAN projections occurring in (C.5b), first perform the rank-one KAN reduction of v_j inside the subgroup generated by the $\pm\beta_j$ string. The N-factor produced by this reduction is moved to the right through u. BCH creates only roots later than β_j, hence it changes u by a polynomial translation in $\bar{N}^{(j+1)}$. The K-factor acts orthogonally on the remaining root spaces and changes the fixed Euclidean norms by at most a structural constant; the A-factor contributes exactly the rank-one half-density from Lemma 4.3. The same calculation applied to $a(v_j u)a^{-1}$ has the V_j coordinate scaled by $e^{-\beta_j(H)}$ (and its $2\beta_j$ coordinate by the square), giving the first factor in (C.23a).

Because all new commutator terms lie in the normal subgroup $\bar{N}^{(j+1)}$, the change of later variables is triangular with identity linear part. Its Jacobian is one by Lemma 2.2, and translation leaves the full later integration domain unchanged. This proves the induction step. At $j = r$ there are no later variables and the claim is exactly the rank-one calculation. Descending to $j = 1$ proves the lemma. $\square$

Lemma 5.2 (Triangular rank-one reduction). *For $H \in \overline{\mathfrak{a}^+}$,*

$$\Xi(e^H) \leq Ce^{-\rho(H)} \prod_{j=1}^{r} (1 + \beta_j(H)). \tag{C.25a}$$

PROOF. Use the coordinates (C.22a) in the exact formula (C.5b). Starting with v_r and moving backward, apply Lemma 5.1 to remove the polynomial translations in later variables, then apply the rank-one estimate Lemma 4.5. The integrand is nonnegative, so Tonelli justifies every reordering. At the jth step the principal exponential contribution is the β_j-string part of $e^{-\rho(H)}$, and the only nonexponential loss is $1 + \beta_j(H)$. Since the reduced word lists every positive indivisible root string once, the string multiplicities add to 2ρ and the principal factors multiply to the single factor $e^{-\rho(H)}$ already outside (C.5b). The finite product of structural coordinate constants is absorbed into C. This yields (C.25a). $\square$

THEOREM 5.3 (Detailed Harish–Chandra estimate). *There are constants $C, D > 0$ such that for all $H \in \overline{\mathfrak{a}^+}$,*

$$\Xi(e^H) \leq C(1 + \|H\|)^D e^{-\rho(H)}. \tag{C.26a}$$

There is also $c > 0$ such that

$$\Xi(e^H) \geq ce^{-\rho(H)}. \tag{C.27}$$

PROOF. The upper bound follows from Lemma 5.2, because each β_j is a fixed nonnegative linear functional on the closed chamber:

$$\prod_{j=1}^{r} (1 + \beta_j(H)) \leq C(1 + \|H\|)^r.$$

For the lower bound use (C.5b). Choose a sufficiently small fixed gauge ball $B \subset \bar{N}$. If H is in the closed positive chamber, conjugation $\bar{n} \mapsto e^H \bar{n} e^{-H}$ contracts every negative-root coordinate, so it maps B into a fixed compact ball B'. The continuous positive function $P^{1/2}$ has a positive minimum on $B \cup B'$. Hence

$$\Xi(e^H) \geq c_G e^{-\rho(H)} \int_B P(e^H \bar{n} e^{-H})^{1/2} P(\bar{n})^{1/2} \, d\bar{n} \geq ce^{-\rho(H)}.$$

This proves (C.27) without invoking a separate convexity theorem for the Iwasawa projection. $\square$

6. A direct Herz estimate for the regular representation

The next argument avoids spherical Plancherel theory entirely.

Lemma 6.1 (Positive left-K-invariant regular coefficients). *Let $\Phi, \Psi \in L^2(G)$ be nonnegative and left K-invariant. Then*

$$\langle \lambda_G(g)\Phi, \Psi\rangle \le \Xi(g) norm\Phi_2 \|\Psi\|_2 . \tag{C.13a}$$

PROOF. Use Iwasawa coordinates $x = kan$ and the Haar formula of Proposition 6.3,

$$dx = e^{2\rho(H(a))}\, dk\, da\, dn. \tag{C.28}$$

For fixed k, write

$$g^{-1}k = k' a_k n_k.$$

Since Φ is left K-invariant,

$$\Phi(g^{-1}kan) = \Phi(a_k a\,(a^{-1}n_k a)n).$$

For each a, left translation in N preserves dn. Hence Cauchy–Schwarz in the N variable gives

$$\int_N \Phi(g^{-1}kan)\Psi(an)\, dn \le F(a_k a)P(a),$$

where

$$F(a)^2 = \int_N \Phi(an)^2\, dn, \qquad P(a)^2 = \int_N \Psi(an)^2\, dn.$$

Now apply Cauchy–Schwarz in A with measure $e^{2\rho(H(a))}da$. Translation $a \mapsto a_k a$ changes this measure by $e^{-2\rho(H(a_k))}$, so

$$\int_A F(a_k a)P(a)e^{2\rho(H(a))}\, da \le e^{-\rho(H(a_k))} norm\Phi_2 \|\Psi\|_2 .$$

Finally integrate over k and use the definition of $\Xi(g^{-1}) = \Xi(g)$:

$$\langle \lambda(g)\Phi, \Psi\rangle \le \left(\int_K e^{-\rho(H(g^{-1}k))}\, dk\right) \|\Phi\|_2 \|\Psi\|_2 .$$

This is the claimed regular-representation estimate. □

Lemma 6.2 (K-finite L^2 vectors have continuous representatives). *Let $E \subset L^2(K)$ be a finite-dimensional left-translation-invariant subspace. Every element of E has a canonical continuous representative, and left translation preserves these representatives pointwise.*

PROOF. The finite-dimensional unitary representation of K on E is a finite sum of irreducible types τ. For one such type define the continuous central function

$$p_\tau(k) = d_\tau \overline{\chi_\tau(k)}.$$

Schur orthogonality, proved by averaging rank-one operators exactly as in Chapter 7, shows that left convolution by p_τ is the orthogonal projection onto the τ-isotypic subspace of $L^2(K)$. Hence if F lies in the finite sum of types occurring in E,

$$F = p * F \quad \text{in } L^2(K), \qquad p = \sum_{\tau \subset E} p_\tau.$$

Because p is continuous on the compact group and $F \in L^2(K) \subset L^1(K)$, the convolution $p * F$ is continuous. We choose this convolution as the representative. Convolution commutes with left translations, so the translation law holds pointwise for these representatives. □

Lemma 6.3 (Compact evaluation bound)**.** *Let $E \subset L^2(K)$ be a d-dimensional left-translation-invariant subspace, with the continuous representatives of Lemma 6.2. Then*

$$\sup_{k \in K} |F(k)| \le \sqrt{d}\, \|F\|_{L^2(K)} \qquad (F \in E). \tag{C.29}$$

Proof. Choose an orthonormal basis $e_1, \dots, e_d$ of continuous representatives. The reproducing kernel

$$R(k, k') = \sum_{j=1}^{d} e_j(k)\overline{e_j(k')}$$

has $R(k,k)$ independent of k: left translation carries the orthonormal basis to another orthonormal basis of the same space. Since

$$\int_K R(k,k)\, dk = d,$$

one has $R(k,k) = d$. For the evaluation functional,

$$F(k) = \langle F, R(\,\cdot\,, k)\rangle_{L^2(K)},$$

and Cauchy–Schwarz gives (C.29). □

Lemma 6.4 (Herz estimate in the regular representation)**.** *Let $f, h \in L^2(G)$ be K-finite for the left regular representation. Replace them by their canonical continuous-in-the-left-K-variable representatives, obtained by convolving in K with the finite sum of the isotypic projectors from Lemma 6.2, and put*

$$d_f = \dim \operatorname{span}\{\lambda(k) f : k \in K\}, \qquad d_h = \dim \operatorname{span}\{\lambda(k) h : k \in K\}.$$

Then

$$|\langle \lambda_G(g) f, h\rangle| \le \sqrt{d_f d_h}\, \Xi(g)\, \|f\|_2\, \|h\|_2\,. \tag{C.16a}$$

The same estimate holds for any finite amplification of λ_G.

PROOF. Define the left-K-invariant envelopes

$$\Phi(x) = \sup_{k\in K} |f(kx)|, \qquad \Psi(x) = \sup_{k\in K} |h(kx)|.$$

For fixed x, the function $k \mapsto f(kx)$ belongs to a translation-invariant subspace of $L^2(K)$ of dimension at most d_f. Lemma 6.3 therefore gives

$$\Phi(x)^2 \le d_f \int_K |f(kx)|^2\, dk.$$

Integrating in x and using left invariance of Haar measure gives

$$\|\Phi\|_2 \le \sqrt{d_f}\, \|f\|_2 \,. \tag{C.17a}$$

Similarly $\|\Psi\|_2 \le \sqrt{d_h}\, \|h\|_2$.

Pointwise,

$$|f(g^{-1}x)\overline{h(x)}| \le \Phi(g^{-1}x)\Psi(x).$$

Thus Lemma 6.1 and (C.17a) give (C.16a). For a finite amplification, replace absolute values by the Hilbert norm in the multiplicity space; the same envelope and Cauchy–Schwarz proof is unchanged. □

7. Transfer to tempered representations

THEOREM 7.1 (Herz–Cowling–Haagerup majorization). *Let σ be tempered. If v, w are K-finite and*

$$d_v = \dim \operatorname{span}_K(v), \qquad d_w = \dim \operatorname{span}_K(w),$$

then

$$|\langle \sigma(g)v, w\rangle| \le \sqrt{d_v d_w}\, \Xi(g)\, \|v\|\, \|w\| \,. \tag{C.18a}$$

PROOF. Let

$$E = \operatorname{span}_K(v) + \operatorname{span}_K(w).$$

It is finite dimensional and K-invariant. Temperedness is $\sigma \prec \lambda_G$. Apply the K-equivariant Fell approximation, Lemma 4.2, to E and to a compact set containing the chosen g. We obtain maps

$$T_j : E \to L^2(G)^{\oplus N_j}$$

for which

$$\langle \lambda^{\oplus N_j}(g)T_j v, T_j w\rangle \to \langle \sigma(g)v, w\rangle, \qquad \|T_j v\| \to \|v\| \,, \quad \|T_j w\| \to \|w\| \,.$$

Equivariance gives

$$\dim \operatorname{span}_K(T_j v) \le d_v, \qquad \dim \operatorname{span}_K(T_j w) \le d_w.$$

Apply Lemma 6.4 in the regular amplification:

$$|\langle \lambda^{\oplus N_j}(g)T_j v, T_j w\rangle| \le \sqrt{d_v d_w}\, \Xi(g)\, \|T_j v\|\, \|T_j w\| \,.$$

Let $j \to \infty$. This proves (C.18a) with the exact dimension factor; no spherical transform or unproved matrix-valued weak-containment theorem is used. $\square$

8. Integrability of Ξ

Corollary 8.1 (Almost-L^2 integrability)**.** *For every $\varepsilon > 0$,*

$$\Xi \in L^{2+\varepsilon}(G).$$

Consequently every K-finite coefficient of a tempered representation is in $L^{2+\varepsilon}(G)$.

Proof. The Cartan integration formula has density bounded on the positive chamber by a fixed polynomial times $e^{2\rho(H)}$. By Theorem 5.3,

$$\Xi(e^H)^{2+\varepsilon} \ll (1 + \|H\|)^{D(2+\varepsilon)} e^{-(2+\varepsilon)\rho(H)}.$$

Multiplying by the Cartan density leaves $e^{-\varepsilon\rho(H)}$ times a polynomial, which is integrable on the closed chamber. The coefficient statement follows from Theorem 7.1. $\square$

Proof and provenance. The proof above deliberately avoids the compressed spherical/Abel-transform argument in the older source reservoir. The regular Herz estimate is obtained directly from Iwasawa coordinates and positive K-invariant envelopes; the tempered transfer is supplied by the K-equivariant Fell approximation proved in Appendix B; and the Harish–Chandra bound is reduced by a finite convex root ordering to an explicit rank-one logarithmic integral. These are exactly the load-bearing harmonic-analysis transitions used later in the project.

Bibliographic notes

The classical qualitative decay theorem is due to Howe and Moore, and the general Mautner phenomenon was developed by C. C. Moore. The almost-L^2 majorization used in Chapter 7 is due to Cowling–Haagerup–Howe. These references are provenance, not replacements for proofs: the forms used in this volume are reconstructed in the text. Later volumes may cite the theorem labels of this volume as proved inputs.

Volume 6

Spectral Gap, Property (T), Property (τ), and Sobolev Calculus

Preface and scope

Volume 5 supplied qualitative mixing, weak containment, temperedness, the Harish–Chandra spherical envelope, and the Cowling–Haagerup–Howe transfer. The present volume adds the quantitative rigidity layer that later effective dynamics requires. A pair (Q, κ) with $Q \subset G$ compact and $\kappa > 0$ is a *Kazhdan pair* if every unitary representation with a unit vector v satisfying $\sup_{q \in Q} \|\pi(q)v - v\| < \kappa$ has a nonzero G-invariant vector; G has *Property* (T) if such a pair exists. For a fixed family of finite quotients, *Property* (τ) means the corresponding uniform Kazhdan inequality on the zero-mean regular representations. Chapters 1–3 give the exact normalized formulations. Its load-bearing chains are

$$\begin{aligned} \text{Kazhdan pairs} &\Longrightarrow \text{Property } (T) \Longrightarrow \text{uniform quotient spectral gap} \\ &\Longrightarrow \text{Property } (\tau), \end{aligned}$$

and

$$\begin{aligned} \text{height-adapted Sobolev norms} &\Longrightarrow \text{smoothing/product/translation calculus} \\ &\Longrightarrow \text{quantitative averaging estimates.} \end{aligned}$$

The arithmetic model is deliberately exact rather than encyclopedic. We prove that $\mathrm{SL}_n(\mathbb{R})$ has Property (T) for $n \geq 3$ from the relative Property (T) of $\mathrm{SL}_2(\mathbb{R}) \ltimes \mathbb{R}^2$, prove inheritance by lattices, and then apply the Borel–Harish–Chandra theorem proved in Volume 4 to $\mathrm{SL}_n(\mathbb{Z}) < \mathrm{SL}_n(\mathbb{R})$. Consequently all finite Schreier quotients, in particular all congruence quotients, inherit a uniform spectral gap. No Selberg, Burger–Sarnak, Clozel, Ramanujan, or rank-one automorphic spectral-gap theorem is used in this chain.

Warning 0.2. “Spectral gap” has two different meanings in the literature. In Chapters 1–4 it means isolation of the invariant vectors by a compact averaging operator. Chapter 7 proves exponential decay of *averaged* matrix coefficients under powers of such an operator and records the exact interface with the pointwise Harish–Chandra bounds of Volume 5. Property (T) alone does not, for an arbitrary group representation, imply a pointwise decay estimate for $\langle \pi(g)v, w \rangle$ as a function of g. Any later pointwise effective mixing theorem must therefore cite an additional tensor-tempered/strong-spectral-gap owner, not silently infer it from Property (T).

The foundational floor consists of the Hilbert-space projection theorem, elementary measure theory, the spectral theorem for unitary representations of $\mathbb{R}^d$, ordinary Euclidean Sobolev embedding on a ball, and standard smooth-manifold partitions of unity. All group-theoretic, relative-(T), lattice-transfer, quotient-gap, height-weighted Sobolev, and smoothing steps used later are proved here.

Student orientation: three different meanings of spectral gap

Property (T) is a group-level statement: almost invariant vectors force invariant vectors. *Property* (τ) is the corresponding uniform gap for a specified family of finite-index quotients or representations. A *spectral gap for one averaging operator* is weaker and depends on that operator. None of these, by itself, is a pointwise matrix-coefficient decay theorem. The Sobolev norms used later are *height-adapted*: derivatives are weighted according to local injectivity radius so that differentiation and smoothing remain controlled in the cusp. Keep the logical chain

$$\begin{aligned}\text{Kazhdan pair} \to \text{operator gap} \to \text{mean ergodic estimate}\\ \to \text{Sobolev smoothing,}\end{aligned}$$

separate from the independent tempered/Harish–Chandra route of Volume 5.

Reader guide: a concrete model

Why this matters.

Volume 6 is organized around the passage from *Almost Invariant Vectors and Kazhdan Pairs* to *Mean Ergodic Theorem and Scope Summary.* The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.3 (A calculation to keep in view)**.** Consider the standard unipotent and diagonal elements

$$u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad a_s = \begin{pmatrix} e^s & 0 \\ 0 & e^{-s} \end{pmatrix}.$$

A direct multiplication gives

$$a_s u_t a_{-s} = u_{e^{2s}t}.$$

Thus a displacement of size $|t|$ in the unstable horospherical direction is enlarged by e^{2s}. If $|t| = e^{-2s}$, the conjugated displacement has order one. This elementary calculation is the model for the “first visible scale” used repeatedly in nondivergence and shearing arguments.

CHAPTER 1

Almost Invariant Vectors and Kazhdan Pairs

Proof architecture. The elementary geometric fact behind Property (T) is that a uniformly small orbit in a Hilbert space has a fixed center. The representation-theoretic formulation replaces "small orbit" by a compact set of tests. We develop the displacement seminorm, the invariant-vector projection, and averaging operators carefully because the same constants reappear in finite quotients and mean ergodic estimates.

1. Displacement and invariant vectors

Let G be a locally compact second countable group and $(\pi, \mathcal{H})$ a strongly continuous unitary representation. For a compact set $Q \subset G$ put

$$\delta_Q(v) = \sup_{q \in Q} \|\pi(q)v - v\| .$$

Write $\mathcal{H}^G = \{v : \pi(g)v = v \ \forall g \in G\}$ and let P_G be the orthogonal projection onto $\mathcal{H}^G$.

Lemma 1.1 (Elementary displacement calculus)**.** *For compact Q, Q_1, Q_2 and $v, w \in \mathcal{H}$:*

(1) $\delta_Q(v + w) \leq \delta_Q(v) + \delta_Q(w)$ and $\delta_Q(cv) = |c|\delta_Q(v)$;
(2) $\delta_{Q^{-1}}(v) = \delta_Q(v)$;
(3) $\delta_{Q_1Q_2}(v) \leq \delta_{Q_1}(v) + \delta_{Q_2}(v)$;
(4) $\delta_Q(v) = \delta_Q(v - P_G v)$.

Proof. The first assertion is the triangle inequality. For the second, $\|\pi(q^{-1})v - v\| = \|v - \pi(q)v\|$. For $q_i \in Q_i$,

$$\|\pi(q_1q_2)v - v\| \leq \|\pi(q_1)(\pi(q_2)v - v)\| + \|\pi(q_1)v - v\| ,$$

and unitarity gives (3). Since $P_G v$ is fixed, subtracting it does not change any displacement. □

Lemma 1.2 (Convex-hull fixed center)**.** *Let $C \subset \mathcal{H}$ be nonempty, bounded, closed, convex, and invariant under a group of linear isometries of the Hilbert space. Then C contains a unique vector of minimal norm, and that vector is fixed by every isometry in the group.*

Proof. Let $r = \inf_{x\in C} \|x\|$ and choose $x_n \in C$ with $\|x_n\| \to r$. The parallelogram identity and convexity give

$$\begin{aligned}\|x_n - x_m\|^2 &= 2\|x_n\|^2 + 2\|x_m\|^2 - 4\|(x_n + x_m)/2\|^2 \\ &\le 2\|x_n\|^2 + 2\|x_m\|^2 - 4r^2.\end{aligned}$$

so (x_n) is Cauchy. Its limit $x_0 \in C$ has norm r. If y_0 were another minimizer, the same identity would force $x_0 = y_0$. A linear isometry T preserves norms, so $Tx_0 \in C$ is another vector of norm r. Uniqueness gives $Tx_0 = x_0$. □

Proposition 1.3 (Uniform orbit control produces an invariant vector)**.** *If v is a unit vector and*

$$\sup_{g\in G} \|\pi(g)v - v\| < 1,$$

then $\mathcal{H}^G \neq 0$. More generally, if the left side is at most $c < 1$, then $\|P_G v\| \ge 1 - c$.

Proof. Let C be the closed convex hull of the orbit $\pi(G)v$. Every point of the orbit lies in the closed ball $\overline{B}(v, c)$, hence so does C. Thus $0 \notin C$ and Lemma 1.2 supplies a nonzero invariant vector $w \in C$. Because $C \subset \overline{B}(v, c)$, we have $\|v - w\| \le c$, hence $\mathrm{dist}(v, \mathcal{H}^G) \le c$. Since $P_G v$ is the closest invariant vector to v,

$$\|P_G v\| \ge \|v\| - \|v - P_G v\| \ge 1 - c.$$

□

2. Kazhdan pairs and projection estimates

Definition 2.1 (Kazhdan pair)**.** A pair (Q, κ), where $Q \subset G$ is compact and $\kappa > 0$, is a *Kazhdan pair* if every unitary representation with a unit vector v satisfying $\delta_Q(v) < \kappa$ has a nonzero G-invariant vector.

For quantitative work the following formulation is more useful.

Theorem 2.2 (Projection form of a Kazhdan pair). *Suppose (Q, κ) is a Kazhdan pair, with $Q = Q^{-1}$. Then every unitary representation and every $v \in \mathcal{H}$ satisfy*

$$\mathrm{dist}(v, \mathcal{H}^G) \le \kappa^{-1}\delta_Q(v). \tag{1.1}$$

Consequently

$$\sup_{g\in G} \|\pi(g)v - v\| \le 2\kappa^{-1}\delta_Q(v). \tag{1.2}$$

PROOF. Put $w = v - P_G v \in (\mathcal{H}^G)^\perp$. This orthogonal complement is G-invariant. If $w \neq 0$ and $\delta_Q(w) < \kappa \|w\|$, then $w/\|w\|$ would be a (Q,κ)-almost invariant unit vector in a representation having no invariant vector, a contradiction. Thus $\kappa \|w\| \leq \delta_Q(w) = \delta_Q(v)$, proving (1.1). For any g,

$$\|\pi(g)v - v\| = \|\pi(g)w - w\| \leq 2\|w\|,$$

which gives (1.2). □

Corollary 2.3 (Changing the test set). *Assume every element of a compact set Q is a product of at most L elements of a symmetric compact set S. If (Q,κ) is Kazhdan, then $(S,\kappa/L)$ is Kazhdan.*

PROOF. Lemma 1.1 gives $\delta_Q(v) \leq L\delta_S(v)$. □

3. Averaging operators

Let μ be a compactly supported symmetric Borel probability measure on G and set

$$\pi(\mu) = \int_G \pi(g)\, d\mu(g).$$

It is a self-adjoint contraction.

Lemma 3.1 (Dirichlet identity). *For every $v \in \mathcal{H}$,*

$$\langle (I - \pi(\mu))v, v\rangle = \frac{1}{2}\int_G \|\pi(g)v - v\|^2\, d\mu(g). \tag{1.3}$$

PROOF. Expand the square and use unitarity:

$$\|\pi(g)v - v\|^2 = 2\|v\|^2 - 2\Re\langle \pi(g)v, v\rangle.$$

Symmetry makes $\langle \pi(\mu)v, v\rangle$ real. Integrating gives (1.3). □

Proposition 3.2 (Finite averaging gap from a Kazhdan pair). *Let $Q = \{q_1, \dots, q_m\}$ be a finite symmetric Kazhdan set with constant κ, and let μ_Q be uniform measure on Q. On $(\mathcal{H}^G)^\perp$,*

$$I - \pi(\mu_Q) \geq \frac{\kappa^2}{2m} I. \tag{1.4}$$

In particular

$$\sup \operatorname{Spec}(\pi(\mu_Q)|_{(\mathcal{H}^G)^\perp}) \leq 1 - \frac{\kappa^2}{2m}.$$

PROOF. If $v \perp \mathcal{H}^G$, Theorem 2.2 implies some $q \in Q$ satisfies $\|\pi(q)v - v\| \geq \kappa\|v\|$. Hence

$$\frac{1}{2}\int \|\pi(g)v - v\|^2\, d\mu_Q(g) \geq \frac{\kappa^2}{2m}\|v\|^2.$$

Apply Lemma 3.1. □

Worked Example 3.3 (Why $\mathbb{Z}$ has no Kazhdan pair)**.** For $t \in \mathbb{R}$ let $\chi_t(n) = e^{itn}$. Given a finite $Q \subset \mathbb{Z}$ and $\varepsilon > 0$, choose $0 < |t| \ll 1$ so that $|e^{itq} - 1| < \varepsilon$ for every $q \in Q$. The one-dimensional representation χ_t has a (Q, ε)-almost invariant unit vector but no invariant vector. Thus no finite Q is Kazhdan. The example is the prototype for the phrase "the trivial representation is not isolated."

4. Dependency summary

Dependency Note 4.1 (V06.C01.L)**.** Lemma 1.1, Lemma 1.2, and Proposition 1.3 own the Hilbert-space geometry of almost invariance.

Dependency Note 4.2 (V06.C01.T)**.** Theorem 2.2 gives the quantitative invariant-projection form of a Kazhdan pair.

Dependency Note 4.3 (V06.C01.P)**.** Proposition 3.2 transfers a finite Kazhdan pair to a spectral gap for its averaging operator, with the exact constant used in Chapters 3 and 8.

5. Reader perspective

The chapter begins with the convex geometry before introducing the representation-theoretic vocabulary, and it separates maximum displacement from averaged Dirichlet energy. Example 3.3 gives a concrete failure mode. No structural theorem about semisimple groups is used yet.

CHAPTER 2

Kazhdan Property (T) and the Higher-Rank Special Linear Group

Proof architecture. The chapter has three layers. First we establish the abstract permanence properties of Property (T). Second, Appendix A proves relative Property (T) for the semidirect product $\mathrm{SL}_2(\mathbb{R}) \ltimes \mathbb{R}^2$ using spectral measures on the dual of $\mathbb{R}^2$. Third, elementary matrices boundedly generate $\mathrm{SL}_n(\mathbb{R})$ because $\mathbb{R}$ is a field; relative rigidity of the root groups then upgrades to Property (T). Appendix B proves lattice inheritance and gives the arithmetic consequence for $\mathrm{SL}_n(\mathbb{Z})$.

1. Equivalent working definitions

Definition 1.1. A locally compact second countable group G has *Kazhdan Property* (T) if it has a compact Kazhdan pair.

THEOREM 1.2 (Almost-invariant-vector criterion). *The following are equivalent.*

(1) G has Property (T).
(2) For every unitary representation π, if for every compact $Q \subset G$ and every $\varepsilon > 0$ there is a unit vector v with $\delta_Q(v) < \varepsilon$, then $\mathcal{H}^G \neq 0$.
(3) There are compact Q and $c > 0$ such that for every representation and every $v \perp \mathcal{H}^G$,

$$\delta_Q(v) \geq c \|v\| .$$

PROOF. (1)⇒(3) is Theorem 2.2 with $c = \kappa$. For (3)⇒(2), suppose a representation has almost invariant unit vectors. If $\mathcal{H}^G = 0$, then every one of those unit vectors lies in $(\mathcal{H}^G)^\perp$, contradicting the lower bound $\delta_Q(v) \geq c$ once the Q-displacement is smaller than c. Hence $\mathcal{H}^G \neq 0$. Finally, (2)⇒(1) by contraposition: if no compact Kazhdan pair exists, choose an exhaustion $Q_1 \subset Q_2 \subset \cdots$ of G by compact sets and, for each n, a representation without invariant vectors containing a $(Q_n, 1/n)$-almost invariant unit vector. Their Hilbert direct sum has almost invariant vectors but no invariant vector, contradicting (2). □

Proposition 1.3 (Quotients and compact extensions)**.** *(1) Every continuous quotient of a Property (T) group has Property (T).*

(2) If $N \lhd G$ is compact and G/N has Property (T), then G has Property (T).

PROOF. For (1), pull a representation of the quotient back to G. For (2), let $Q_0 \subset G$ lift a compact Kazhdan set in G/N and put $Q = Q_0 \cup N$. If v is sufficiently Q-almost invariant, average it over N:

$$v_N = \int_N \pi(n)v\, dn.$$

Then v_N is N-fixed and remains close to v, hence nonzero. The N-fixed subspace is G-invariant because N is normal, and the induced representation of G/N has an almost invariant vector. Property (T) of G/N supplies a nonzero G-fixed vector. □

Proposition 1.4 (Products)**.** *A finite product $G_1 \times \cdots \times G_r$ has Property (T) if and only if every factor does.*

PROOF. Necessity follows from quotients. For sufficiency, use induction. If $G = G_1 \times G_2$ and v is sufficiently almost invariant for lifted Kazhdan sets, Theorem 2.2 makes v close to the G_1-fixed subspace. This subspace is G_2-invariant because the factors commute. Projecting v to it gives a nonzero vector still almost invariant under G_2, hence a vector fixed by both factors. □

2. Relative Property (T) and root control

Definition 2.1. For a closed subgroup $N < G$, the pair (G, N) has *relative Property (T)* if every unitary representation of G with almost invariant vectors has a nonzero N-invariant vector.

When $N \lhd G$, relative Property (T) has a quantitative projection form exactly parallel to Theorem 2.2.

Proposition 2.2 (Relative projection estimate)**.** *Suppose $N \lhd G$ and (G, N) has relative Property (T). Then there are compact $Q \subset G$ and $c > 0$ such that every unitary representation and every v satisfy*

$$\operatorname{dist}(v, \mathcal{H}^N) \le c^{-1}\delta_Q(v). \tag{2.1}$$

Consequently

$$\sup_{n\in N} \|\pi(n)v - v\| \le 2c^{-1}\delta_Q(v). \tag{2.2}$$

Proof. Because N is normal, $\mathcal{H}^N$ and its orthogonal complement are G-invariant. If no compact Q and c worked on the complement, a direct-sum exhaustion argument identical to Theorem 1.2 would produce a representation with almost invariant vectors and no N-fixed vector, contradicting relative Property (T). Thus $\delta_Q(w) \geq c\,\|w\|$ on $(\mathcal{H}^N)^\perp$. Put $w = v - P_N v$ and conclude as in Theorem 2.2. □

Theorem 2.3 (Relative rigidity of the plane). *The pair*

$$(\mathrm{SL}_2(\mathbb{R}) \ltimes \mathbb{R}^2, \mathbb{R}^2)$$

has relative Property (T).

Proof. Appendix A, Theorem 3.1, gives the complete spectral-measure proof. The key contradiction is elementary: a hypothetical sequence of almost invariant vectors with no $\mathbb{R}^2$-fixed part yields asymptotically $\mathrm{SL}_2(\mathbb{R})$-invariant probability measures on the projective line $\mathbf{P}^1(\mathbb{R})$; compactness gives an invariant probability measure, but translations force such a measure to be supported at ∞ while the Weyl element moves ∞ to 0. □

3. Bounded elementary generation over a field

For $i \neq j$ and $t \in \mathbb{R}$ write $u_{ij}(t) = I + tE_{ij}$.

Lemma 3.1 (Two-by-two diagonal factorization)**.** *For $a \in \mathbb{R}^\times$,*

$$\begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} = \begin{pmatrix} 1 & a-1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & a^{-1}-1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -a & 1 \end{pmatrix}. \tag{2.3}$$

Thus every elementary two-coordinate diagonal is a product of four root-unipotent elements.

Proof. Direct multiplication gives first

$$\begin{pmatrix} a & a-1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & a^{-1}-1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a & 0 \\ 1 & a^{-1} \end{pmatrix},$$

and multiplying by the last lower unipotent gives the claimed diagonal matrix. □

Lemma 3.2 (Bounded elementary generation of $\mathrm{SL}_n(\mathbb{R})$)**.** *For every $n \geq 2$ there is $B_n < \infty$ such that every $g \in \mathrm{SL}_n(\mathbb{R})$ is a product of at most B_n matrices $u_{ij}(t)$. One may take a bound depending only quadratically on n.*

Proof. We give an induction that records why the field hypothesis matters. For the first column of g, choose a nonzero entry and move it to the first position using a fixed two-by-two Weyl matrix; that Weyl matrix is a product of three elementary unipotents. For each $j > 1$, left multiplication by $u_{j1}(-g_{j1}/g_{11})$ clears the jth entry of the first column. Then use Lemma 3.1

in coordinates $(1,2)$ to scale the remaining first entry to 1 while compensating the determinant in the second coordinate. The matrix now has block form

$$\begin{pmatrix} 1 & r \\ 0 & h \end{pmatrix}, \qquad h \in \mathrm{SL}_{n-1}(\mathbb{R}).$$

Right multiplication by $u_{1j}(-r_j)$, $j > 1$, clears r. Applying the induction hypothesis to h completes the factorization. The number of factors satisfies $B_n \le B_{n-1} + 2(n-1) + 7$, hence $B_n = O(n^2)$. □

4. Property (T) for higher-rank special linear groups

Lemma 4.1 (Uniform root displacement)**.** *Fix $n \ge 3$. There exist a compact $Q_n \subset \mathrm{SL}_n(\mathbb{R})$ and $C_n > 0$ such that for every unitary representation π, every vector v, every $i \ne j$, and every $t \in \mathbb{R}$,*

$$\|\pi(u_{ij}(t))v - v\| \le C_n \delta_{Q_n}(v). \tag{2.4}$$

Proof. Choose k distinct from i, j. Let $L_{ik} < \mathrm{SL}_n(\mathbb{R})$ be the copy of $\mathrm{SL}_2(\mathbb{R})$ acting on the coordinate plane spanned by e_i, e_k and fixing the other coordinate vectors. The subgroup generated by L_{ik} together with

$$N_{ik;j} = \{u_{ij}(x)u_{kj}(y) : x, y \in \mathbb{R}\}$$

is a semidirect product $L_{ik} \ltimes N_{ik;j} \cong \mathrm{SL}_2(\mathbb{R}) \ltimes \mathbb{R}^2$: conjugation by the 2×2 block acts on the column $(x, y)^t$ by the defining representation. In particular $u_{ij}(t) \in N_{ik;j}$. Theorem 2.3 and Proposition 2.2, transported through this explicit embedding, give a compact test set $Q_{ik;j}$ and a constant $C_{ik;j}$ controlling every element of $N_{ik;j}$. There are only finitely many triples $(i, k; j)$, so take Q_n to be their union and C_n the largest of the corresponding constants. □

Worked Example 4.2 (The root column inside SL_3)**.** Take $(i, k; j) = (1, 2; 3)$. Then

$$N_{12;3} = \left\{ \begin{pmatrix} 1 & 0 & x \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} : x, y \in \mathbb{R} \right\} \cong \mathbb{R}^2,$$

while the upper-left 2×2 copy of $A \in \mathrm{SL}_2(\mathbb{R})$ conjugates this matrix by sending the column $(x, y)^t$ to $A(x, y)^t$. Thus the familiar semidirect product $\mathrm{SL}_2(\mathbb{R}) \ltimes \mathbb{R}^2$ is literally visible inside $\mathrm{SL}_3(\mathbb{R})$, and the root group $u_{13}(x)$ is one coordinate axis of its normal plane. The general $n \ge 3$ argument is this same picture placed in three selected coordinates.

Theorem 4.3 (Kazhdan's theorem for $\mathrm{SL}_n(\mathbb{R})$). *For every $n \ge 3$, the group $\mathrm{SL}_n(\mathbb{R})$ has Property (T).*

Proof. Let B_n be as in Lemma 3.2. If v is a unit vector with

$$\delta_{Q_n}(v) < \frac{1}{2B_nC_n},$$

then Lemma 4.1 and the triangle inequality show that for every $g \in \mathrm{SL}_n(\mathbb{R})$,

$$\|\pi(g)v - v\| < \frac{1}{2}.$$

Proposition 1.3 gives a nonzero invariant vector. Hence $(Q_n, (2B_nC_n)^{-1})$ is a Kazhdan pair. □

Theorem 4.4 (Lattices inherit Property (T)). *Let G be a locally compact second countable unimodular group with Property (T) and let $\Gamma < G$ be a lattice. Then Γ has Property (T).*

Proof. Appendix B, Theorem 3.2, gives the induction proof with the finite-volume fundamental domain and the almost-invariant-vector estimate written out. The invariant vector obtained in the induced G-representation is constant along G and the induction covariance then makes its value a nonzero Γ-fixed vector. □

Corollary 4.5 (Arithmetic example)**.** *For every $n \geq 3$, $\mathrm{SL}_n(\mathbb{Z})$ has Property (T).*

Proof. The Borel–Harish–Chandra theorem proved in Volume 4 (the matrix-scope theorem labeled there as `thm:BHC-internal`) gives that $\mathrm{SL}_n(\mathbb{Z})$ is a lattice in $\mathrm{SL}_n(\mathbb{R})$. Apply Theorems 4.3 and 4.4. □

Warning 4.6 (Why $n = 2$ is excluded)**.** The bounded elementary generation Lemma 3.2 is true for $\mathrm{SL}_2(\mathbb{R})$ as well; the failure is instead the root-rigidity mechanism. There is no third coordinate in which to realize the normal $\mathbb{R}^2$ needed in Theorem 2.3. In fact $\mathrm{SL}_2(\mathbb{R})$ does not have Property (T).

5. Dependency summary

Dependency Note 5.1 (V06.C02.L)**.** Theorem 1.2, Proposition 2.2, and Appendix A own the abstract and relative almost-invariance mechanisms.

Dependency Note 5.2 (V06.C02.T)**.** Theorem 4.3, together with the complete relative-(T) proof and bounded elementary generation, is the selected higher-rank Property (T) theorem proved in this volume.

Dependency Note 5.3 (V06.C02.P)**.** Theorem 4.4 and Corollary 4.5 transfer the theorem to the arithmetic lattice used in Chapters 3–4.

6. Reader perspective

The proof does not invoke Carter–Keller bounded generation of $\mathrm{SL}_n(\mathbb{Z})$ or an automorphic spectral theorem. Bounded generation is used only over the field $\mathbb{R}$, where the row-reduction proof is elementary. The sole earlier-volume arithmetic input is the exact lattice theorem already proved in Volume 4.

CHAPTER 3

Property (τ) for Finite and Arithmetic Families

Proof architecture. Property (T) is a statement about every unitary representation. Property (τ) keeps only a prescribed family of finite quotient representations. This weaker statement is exactly what is needed for uniform expansion in congruence towers. We prove the equivalence between Kazhdan inequalities, averaging-operator gaps, and expander inequalities, then specialize to the congruence quotients of $\mathrm{SL}_n(\mathbb{Z})$.

1. Finite quotient representations

Let Γ be generated by a finite symmetric set S and let $(\Gamma_i)_{i\in I}$ be finite-index subgroups. On

$$X_i = \Gamma/\Gamma_i$$

let λ_i be the permutation representation on $\ell^2(X_i)$ and let $\ell^2_0(X_i)$ be the orthogonal complement of the constants.

Definition 1.1 (Property (τ) relative to a family)**.** The group Γ has Property (τ) with respect to (Γ_i) if there exists $\kappa > 0$ such that for every i and every $f \in \ell^2_0(X_i)$,

$$\max_{s\in S} \|\lambda_i(s)f - f\|_2 \geq \kappa \|f\|_2 . \tag{3.1}$$

The property is independent of the chosen finite generating set, although the constant changes.

Proposition 1.2 (Change of generators)**.** *If (3.1) holds for a finite symmetric generating set S with constant κ, and every $s \in S$ is a word of length at most L in another finite symmetric generating set T, then (3.1) holds for T with constant κ/L.*

Proof. If every $t \in T$ moves f by less than $(\kappa/L)\|f\|$, the telescoping estimate along an L-letter T-word shows every $s \in S$ moves f by less than $\kappa\|f\|$, contradicting (3.1). $\square$

2. Averaging spectral gap

Put

$$M_{S,i} = \frac{1}{|S|} \sum_{s \in S} \lambda_i(s).$$

This is a self-adjoint Markov operator.

THEOREM 2.1 (Kazhdan inequality versus Markov gap). *For a fixed finite symmetric S, the following are quantitatively equivalent uniformly in i:*

(1) there is $\kappa > 0$ satisfying (3.1);

(2) there is $\eta > 0$ such that for every $f \in \ell_0^2(X_i)$,

$$\langle (I - M_{S,i})f, f\rangle \geq \eta \left\| f \right\|_2^2 ; \tag{3.2}$$

(3) the top of the spectrum of $M_{S,i}$ on $\ell_0^2(X_i)$ is at most $1 - \eta$.

One may take $\eta = \kappa^2/(2|S|)$ from (1) to (2), and $\kappa \geq \sqrt{2\eta}$ from (2) to (1).

PROOF. The Dirichlet identity from Lemma 3.1 gives

$$\langle (I - M_{S,i})f, f\rangle = \frac{1}{2|S|} \sum_{s \in S} \left\| \lambda_i(s)f - f \right\|_2^2 .$$

If the maximum displacement is at least $\kappa \left\| f \right\|$, the sum is at least $\kappa^2 \left\| f \right\|^2$, proving the first constant. Conversely, every summand is at most the square of the maximum displacement, so (3.2) gives $\eta \left\| f \right\|^2 \leq \frac{1}{2} \max_s \left\| \lambda_i(s)f - f \right\|^2$. The equivalence of (2) and (3) is the variational characterization of the top of the spectrum of a self-adjoint operator. □

THEOREM 2.2 (Property (T) implies Property (τ)). *If a finitely generated group Γ has Property (T), then it has Property (τ) with respect to every family of finite-index subgroups.*

PROOF. Let S be the finite symmetric generating set used in Definition 1.1. Property (T) supplies some finite Kazhdan set Q and constant $\kappa_Q > 0$. Because S generates Γ and Q is finite, every $q \in Q$ is an S-word of length at most a common number L. Hence

$$\delta_Q(f) \leq L \max_{s \in S} \left\| \lambda_i(s)f - f \right\|_2 .$$

The representation λ_i on $\ell_0^2(X_i)$ has no invariant vector: transitivity implies every invariant function on X_i is constant. The Kazhdan inequality for Q therefore gives

$$\max_{s \in S} \left\| \lambda_i(s)f - f \right\|_2 \geq \frac{\kappa_Q}{L} \left\| f \right\|_2$$

uniformly in i. This is Property (τ) for the prescribed generating set S. □

3. Expansion of Schreier graphs

Let $\mathscr{G}_i = \mathrm{Sch}(\Gamma/\Gamma_i, S)$, with one oriented s-edge $x \to sx$ for each $s \in S$. For $A \subset X_i$ let $\partial_S A$ be the set of oriented edges from A to A^c.

Lemma 3.1 (Indicator Dirichlet energy)**.** *For 1_A,*

$$\langle (I - M_{S,i})1_A, 1_A\rangle = \frac{|\partial_S A|}{|S|}. \tag{3.3}$$

If $f_A = 1_A - |A|/|X_i|$, then

$$\|f_A\|_2^2 = |A|\left(1 - \frac{|A|}{|X_i|}\right). \tag{3.4}$$

Proof. For each s, $\left\|1_A \circ s^{-1} - 1_A\right\|_2^2$ counts both orientations of the s-boundary discrepancy. Summing the Dirichlet identity yields (3.3) with the chosen oriented-edge convention. Formula (3.4) is a direct calculation. □

Corollary 3.2 (Spectral gap gives edge expansion)**.** *If (3.2) holds with $\eta > 0$, then for every $A \subset X_i$ with $0 < |A| \le |X_i|/2$,*

$$|\partial_S A| \ge \frac{\eta|S|}{2}|A|. \tag{3.5}$$

Thus the Schreier graphs form a family of expanders whenever their cardinalities tend to infinity.

Proof. Apply (3.2) to f_A. Constants disappear under $I - M$, so the left side equals the left side of (3.3). Since $|A| \le |X_i|/2$, (3.4) is at least $|A|/2$. Rearranging gives (3.5). □

Proposition 3.3 (Cheeger inequality in the reverse direction)**.** *Let $\mathscr{G}$ be a finite d-regular Schreier graph and suppose $|\partial A| \ge h|A|$ whenever $|A| \le |V|/2$. Then the Markov operator satisfies*

$$\langle (I - M)f, f\rangle \ge \frac{h^2}{8d^2}\,\|f\|_2^2 \qquad (f \perp 1). \tag{3.6}$$

Proof. It is enough to prove the estimate for real-valued functions, since the Dirichlet form and the L^2 norm split into real and imaginary parts. Let f be real and mean zero, choose a median m, and put $u = f - m$. Then the supports of the positive and negative parts

$$u_+ = \max(u, 0), \qquad u_- = \max(-u, 0)$$

have cardinality at most $|V|/2$.

Let v denote either u_+ or u_-. Applying discrete coarea to v^2 gives

$$h\sum_x v(x)^2 \le \sum_{\{x,y\}\in E} |v(x)^2 - v(y)^2|.$$

By Cauchy–Schwarz and d-regularity,

$$\begin{aligned}\sum_E |v(x)^2 - v(y)^2| &\le \left(\sum_E |v(x)-v(y)|^2\right)^{1/2} \left(\sum_E (v(x)+v(y))^2\right)^{1/2} \\ &\le \left(\sum_E |v(x)-v(y)|^2\right)^{1/2} \left(2d\sum_x v(x)^2\right)^{1/2}.\end{aligned}$$

Thus

$$\sum_E |v(x)-v(y)|^2 \ge \frac{h^2}{2d}\sum_x v(x)^2. \tag{3.7}$$

The truncation maps $t \mapsto t_+$ and $t \mapsto t_-$ satisfy, edge by edge,

$$|u(x)-u(y)|^2 \ge |u_+(x)-u_+(y)|^2 + |u_-(x)-u_-(y)|^2.$$

Summing (3.7) for u_+ and u_- therefore yields

$$\sum_E |u(x)-u(y)|^2 \ge \frac{h^2}{2d}\,\|u\|_2^2\,.$$

For the normalized Markov operator,

$$\langle (I-M)u,u\rangle = \frac{1}{d}\sum_E |u(x)-u(y)|^2.$$

Adding a constant does not change this Dirichlet form. Since f has mean zero, it is the L^2-closest translate of itself by a constant, so $\|u\|_2 = \|f-m\|_2 \ge \|f\|_2$. We have in fact proved the stronger constant $h^2/(2d^2)$; the displayed $h^2/(8d^2)$ is a deliberately safe normalization that also covers the harmless convention changes between labelled Schreier edges and the underlying unoriented multigraph. □

4. The congruence family of $\mathrm{SL}_n(\mathbb{Z})$

For $q \ge 2$, let

$$\Gamma(q) = \ker\bigl(\mathrm{SL}_n(\mathbb{Z}) \to \mathrm{SL}_n(\mathbb{Z}/q\mathbb{Z})\bigr).$$

THEOREM 4.1 (Uniform congruence spectral gap for $\mathrm{SL}_n(\mathbb{Z})$). *Fix $n \ge 3$ and a finite symmetric generating set S of $\mathrm{SL}_n(\mathbb{Z})$. There exists $\eta = \eta(n,S) > 0$ such that for every $q \ge 2$, the averaging operator on*

$$\ell_0^2(\mathrm{SL}_n(\mathbb{Z})/\Gamma(q))$$

has spectral top at most $1-\eta$. Consequently the associated congruence Schreier graphs are a uniform expander family after discarding repetitions of bounded cardinality.

Proof. Corollary 4.5 gives Property (T) for $\mathrm{SL}_n(\mathbb{Z})$. Theorem 2.2 gives Property (τ) for the family $(\Gamma(q))_{q\geq 2}$, and Theorem 2.1 converts it to the uniform spectral gap. Corollary 3.2 gives expansion. □

Warning 4.2 (No rank-one automorphic theorem is hidden here)**.** Theorem 4.1 is a higher-rank consequence of Property (T). It says nothing about the optimal gap, Ramanujan bounds, or the rank-one families arising from SL_2. Those require genuinely different arithmetic input and are not proved inputs for this volume.

5. Dependency summary

Dependency Note 5.1 (V06.C03.L)**.** Theorem 2.1 proves the finite-quotient spectral formulation with constants.

Dependency Note 5.2 (V06.C03.T)**.** Theorem 2.2 proves Property (τ) for every finite-index family of a Property (T) group.

Dependency Note 5.3 (V06.C03.P)**.** Theorem 4.1 is the selected arithmetic transfer: all congruence quotients of $\mathrm{SL}_n(\mathbb{Z})$, $n \geq 3$, have a uniform gap. No source-sensitive rank-one theorem is used.

6. Reader perspective

The chapter separates three notions often conflated in compressed accounts: a Kazhdan displacement inequality, a Markov spectral gap, and combinatorial edge expansion. The constants are propagated in both directions, and Warning 4.2 fixes the arithmetic scope.

CHAPTER 4

Congruence Spectral Gaps and Safe Restriction Principles

Proof architecture. Uniform gaps are useful only if one knows exactly which operations preserve them. Quotients are safe; arbitrary subgroup restriction is not. Finite-index passage is safe after adding finitely many coset representatives; products are factorwise; congruence levels interact through quotient maps and the Chinese remainder theorem. The chapter records only these proved transfers and explicitly rejects the false slogan that "spectral gap restricts to subgroups."

1. Quotients

Proposition 1.1 (Quotient monotonicity of Kazhdan constants)**.** *Let $p : \Gamma \to \Lambda$ be a surjective homomorphism, $S \subset \Gamma$ finite symmetric, and $T = p(S)$. Then every Kazhdan constant valid for (Γ, S) is valid for (Λ, T).*

Proof. Pull a unitary representation of Λ back along p. T-displacement equals S-displacement, while invariant vectors are the same. $\square$

Corollary 1.2 (Uniform finite-quotient Cayley gap)**.** *If (Γ, S) has Kazhdan constant κ, then every finite quotient F generated by the image S_F has*

$$\operatorname{gap}\left(\frac{1}{|S|}\sum_{s\in S}\lambda_F(s)\Big|_{\ell_0^2(F)}\right) \geq \frac{\kappa^2}{2|S|}.$$

Proof. Use Proposition 1.1 and Proposition 3.2. $\square$

2. Finite-index passage

Lemma 2.1 (Finite-index coset control)**.** *Let $H < \Gamma$ have finite index and let $R \subset \Gamma$ be a finite set of representatives for the right cosets $H\backslash\Gamma$, with $e \in R$. For a finite symmetric generating set S of Γ, there is a finite symmetric set $T \subset H$ such that for every $r \in R$ and $s \in S$ there are unique $r' \in R$ and $t \in T$ with*

$$rs = tr'. \tag{4.1}$$

The set T generates H.

Proof. For each pair (r, s) let r' be the chosen representative of the right coset Hrs. Then

$$t = rs(r')^{-1} \in H,$$

and only finitely many such corrections occur; enlarge the resulting set by inverses to obtain a finite symmetric T.

Now let $h = s_1 \cdots s_m \in H$ with $s_i \in S$. Put $g_i = s_1 \cdots s_i$ and let $r_i \in R$ represent Hg_i, with $r_0 = e$. Define

$$t_i = r_{i-1} s_i r_i^{-1} \in H.$$

Each t_i is one of the corrections just constructed, and the product telescopes:

$$t_1 \cdots t_m = s_1 r_1^{-1} r_1 s_2 r_2^{-1} \cdots r_{m-1} s_m r_m^{-1} = h r_m^{-1}.$$

Since $h \in H$, its right coset is H, hence $r_m = e$. Thus $h = t_1 \cdots t_m$, proving that T generates H. □

Theorem 2.2 (Finite-index inheritance of Property (T)). *If Γ has Property (T) and $H < \Gamma$ has finite index, then H has Property (T).*

Proof. Let σ be a unitary representation of H with almost invariant vectors. Its induced representation $\mathrm{Ind}_H^\Gamma \sigma$ is a finite Hilbert direct sum indexed by R. In the coset model, a constant tuple $(v)_{r \in R}$ is moved by a generator $s \in S$ by correction terms t drawn from the finite Schreier set T of Lemma 2.1. Thus an increasingly T-invariant unit vector for σ gives an increasingly S-invariant unit vector in the induced representation. Property (T) of Γ gives a nonzero induced invariant vector. Evaluating its component at the identity coset and using H-equivariance gives a nonzero H-invariant vector in σ. □

Corollary 2.3 (Commensurability invariance)**.** *For finitely generated discrete groups, Property (T) is invariant under passing to finite-index subgroups and finite-index overgroups.*

Proof. Subgroups are Theorem 2.2. For the converse, suppose $\Gamma < \Lambda$ has finite index and Γ has Property (T). Intersect the finitely many conjugates of Γ to obtain a finite-index normal subgroup $N \lhd \Lambda$ contained in Γ. Theorem 2.2 gives Property (T) for N.

We prove directly that the finite extension Λ has Property (T). Choose a finite symmetric Kazhdan pair (Q, κ) for N and a finite set $R \subset \Lambda$ mapping onto Λ/N. Let π be a representation of Λ with a unit vector v whose displacement on $Q \cup R$ is smaller than a number $\delta > 0$ to be chosen. Since N is normal, $\mathcal{H}^N$ is Λ-invariant and the orthogonal projection P_N commutes with $\pi(\Lambda)$. The projection estimate for the Kazhdan pair of N gives

$$\|v - P_N v\| \le \kappa^{-1} \delta.$$

Thus $w = P_N v$ is nonzero when $\delta < \kappa/2$, and for $r \in R$,

$$\|\pi(r)w - w\| = \|P_N(\pi(r)v - v)\| \le \delta.$$

The action on $\mathcal{H}^N$ factors through the finite group Λ/N. Averaging w over that finite group gives an invariant vector $\bar{w}$. Moreover, if D is the diameter of the finite Cayley graph of Λ/N generated by the image of R, telescoping gives

$$\|\bar{w} - w\| \le D\delta.$$

Choose $\delta < \min(\kappa/2, (2D)^{-1})$; then $\bar{w} \neq 0$. Hence Λ has Property (T).

$\square$

3. Products and factorwise gaps

Proposition 3.1 (Product Markov operators). *Let finite groups F_1, F_2 have symmetric Markov operators M_1, M_2 with gaps η_1, η_2 on the mean-zero spaces. On $F_1 \times F_2$ use*

$$M = \frac{1}{2}(M_1 \otimes I + I \otimes M_2).$$

Then the spectral gap of M on constants-perp is

$$\frac{1}{2}\min(\eta_1, \eta_2).$$

Proof. Decompose

$$\ell^2(F_1 \times F_2) = (\mathbb{C}1 \otimes \mathbb{C}1) \oplus (\ell_0^2 F_1 \otimes \mathbb{C}1) \oplus (\mathbb{C}1 \otimes \ell_0^2 F_2) \oplus (\ell_0^2 F_1 \otimes \ell_0^2 F_2).$$

The operator preserves these four orthogonal summands. On the second it has top eigenvalue at most $1 - \eta_1/2$, on the third at most $1 - \eta_2/2$, and on the fourth at most $1 - (\eta_1 + \eta_2)/2$. The claimed minimum follows. $\square$

4. Congruence levels

Lemma 4.1 (Level divisibility gives quotient maps). *If $q \mid q'$, then $\Gamma(q') \subset \Gamma(q)$ and there is a natural surjection*

$$\mathrm{SL}_n(\mathbb{Z})/\Gamma(q') \twoheadrightarrow \mathrm{SL}_n(\mathbb{Z})/\Gamma(q).$$

A uniform Kazhdan/Markov gap at level q' therefore descends to level q with no loss when the generating set is pushed forward.

Proof. Reduction modulo q factors through reduction modulo q'. The gap statement is Proposition 1.1 and Corollary 1.2. $\square$

Lemma 4.2 (Chinese remainder decomposition). *If* $(a, b) = 1$, *then*

$$\mathbb{Z}/(ab)\mathbb{Z} \cong \mathbb{Z}/a\mathbb{Z} \times \mathbb{Z}/b\mathbb{Z}$$

and consequently

$$\mathrm{SL}_n(\mathbb{Z}/(ab)\mathbb{Z}) \cong \mathrm{SL}_n(\mathbb{Z}/a\mathbb{Z}) \times \mathrm{SL}_n(\mathbb{Z}/b\mathbb{Z}).$$

Proof. The scalar Chinese remainder isomorphism is a ring isomorphism and therefore induces an entrywise isomorphism of matrix rings preserving determinants and inverses. □

Proposition 4.3 (Surjectivity of reduction for $\mathrm{SL}_n(\mathbb{Z})$). *For* $n \geq 2$ *and every* $q \geq 2$, *the reduction map*

$$\mathrm{SL}_n(\mathbb{Z}) \longrightarrow \mathrm{SL}_n(\mathbb{Z}/q\mathbb{Z})$$

is surjective.

Proof. By Lemma 4.2, it is enough to treat $q = p^a$. Put $R = \mathbb{Z}/p^a\mathbb{Z}$, which is a local ring with maximal ideal pR. The first column of an invertible matrix is unimodular, so at least one of its entries is a unit: if every entry lay in pR, then every term in the determinant expansion would lie in pR, contradicting that the determinant is 1. Using elementary two-coordinate Weyl matrices, which are products of elementary unipotents, move a unit entry into the first position. If that entry is $u \in R^\times$, left multiplication by $u_{j1}(-a_j u^{-1})$ clears every lower entry in the first column. Lemma 3.1, whose formula is valid over any commutative ring for a unit u, changes the leading entry from u to 1 while compensating in the second coordinate. Right elementary operations then clear the remainder of the first row. We obtain $1 \oplus h$ with $h \in \mathrm{SL}_{n-1}(R)$, and induction reduces h to the identity. Thus $\mathrm{SL}_n(R)$ is generated by elementary matrices.

For general $q = \prod p_i^{a_i}$, CRT identifies $\mathrm{SL}_n(\mathbb{Z}/q\mathbb{Z})$ with the product of the groups just treated. An elementary matrix in one CRT factor and the identity in all others is itself an elementary matrix over $\mathbb{Z}/q\mathbb{Z}$, because the required parameter is obtained by the scalar CRT. Hence the full product is elementary-generated. Finally every elementary matrix modulo q lifts to the corresponding integral elementary matrix, proving surjectivity. □

Corollary 4.4 (Concrete finite quotients). *For* $n \geq 2$,

$$\mathrm{SL}_n(\mathbb{Z})/\Gamma(q) \cong \mathrm{SL}_n(\mathbb{Z}/q\mathbb{Z}).$$

For $n \geq 3$, *the Cayley graphs of* $\mathrm{SL}_n(\mathbb{Z}/q\mathbb{Z})$ *with respect to the reductions of any fixed finite symmetric generating set of* $\mathrm{SL}_n(\mathbb{Z})$ *have a uniform spectral gap.*

Proof. The first statement is the first isomorphism theorem and Proposition 4.3. The second is Theorem 4.1. □

Worked Example 4.5 (Level 6 separates into levels 2 and 3)**.** The scalar CRT gives $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{F}_2 \times \mathbb{F}_3$, hence

$$\mathrm{SL}_n(\mathbb{Z}/6\mathbb{Z}) \cong \mathrm{SL}_n(\mathbb{F}_2) \times \mathrm{SL}_n(\mathbb{F}_3).$$

If the chosen walk on the product updates the two factors with equal probability, Proposition 3.1 says its gap is one half of the smaller factor gap. By contrast, if we use the direct image of a fixed generating set of $\mathrm{SL}_n(\mathbb{Z})$, no product decomposition of the walk is required: Property (T) already gives the same level-independent lower bound directly. This illustrates why CRT is a structural tool rather than the proved input of uniform expansion.

Warning 4.6 (Restriction to an infinite subgroup may destroy the gap)**.** If π is a representation of G with spectral gap, the restricted representation $\pi|_H$ need not have spectral gap for an arbitrary closed subgroup $H < G$. In particular, a unipotent one-parameter subgroup is amenable and may have almost invariant vectors even when the ambient semisimple action has a gap. Later chapters use only the quotient, finite-index, product, and explicitly proved arithmetic transfers above.

5. Dependency summary

Dependency Note 5.1 (V06.C04.L)**.** Proposition 1.1 and Theorem 2.2 own the safe quotient and finite-index transfers.

Dependency Note 5.2 (V06.C04.T)**.** Corollary 4.4 identifies the actual congruence quotients and transports the uniform gap to them.

Dependency Note 5.3 (V06.C04.P)**.** Lemma 4.1, Lemma 4.2, and Proposition 3.1 give the downstream-compatible level and product calculus.

6. Reader perspective

The chapter is organized around safe operations and a prominently stated failure mode. This prevents the common but false inference that a spectral gap of an ambient action automatically survives restriction to every subgroup.

CHAPTER 5

Height-Adapted Sobolev Norms on Homogeneous Spaces

Proof architecture. On a compact quotient, ordinary invariant L^2 Sobolev norms control pointwise values uniformly. A finite-volume noncompact quotient may have arbitrarily small injectivity radius, so that statement is false without geometry in the norm. We therefore build the weight directly from the local injectivity radius. Euclidean Sobolev embedding on the largest injective chart then becomes a global weighted estimate. This is the precise calculus needed for cusp truncation and thickening in Volume 7.

1. Differential operators from the group action

Let G be a connected unimodular Lie group of dimension m, let $\Gamma < G$ be discrete, and put $X = G/\Gamma$ with a G-invariant measure m_X. Fix a basis $Y_1, \ldots, Y_m$ of $\mathfrak{g}$. For $Y \in \mathfrak{g}$ define

$$D_Y f(x) = \frac{d}{dt}\bigg|_{t=0} f(\exp(-tY)x). \tag{5.1}$$

The sign is chosen so that D_Y is the infinitesimal operator of the left regular action $g \cdot f(x) = f(g^{-1}x)$. For a word $\alpha = (i_1, \ldots, i_r)$ write $D^\alpha = D_{Y_{i_1}} \cdots D_{Y_{i_r}}$ and $|\alpha| = r$.

Lemma 1.1 (Skew-adjoint integration by parts)**.** *For $f, h \in C_c^\infty(X)$ and $Y \in \mathfrak{g}$,*

$$\int_X D_Y f\, \overline{h}\, dm_X = -\int_X f\, \overline{D_Y h}\, dm_X. \tag{5.2}$$

Proof. Invariance of m_X gives

$$\int f(\exp(-tY)x)\overline{h(x)}\, dm_X(x) = \int f(x)\overline{h(\exp(tY)x)}\, dm_X(x).$$

Differentiate at $t = 0$. Compact support justifies differentiation under the integral. $\square$

Proposition 1.2 (Change of Lie basis)**.** *For each integer $d \geq 0$, two choices of bases of $\mathfrak{g}$ produce equivalent finite sums of derivatives of order at most d. The constants depend only on the linear change-of-basis matrix and d.*

PROOF. Each new first-order operator is a fixed linear combination of the old ones. Expanding a product of at most d such operators gives a finite linear combination of old words of the same length. Reverse the roles of the bases. ☐

2. Injectivity height

Fix a left-invariant Riemannian metric on G and let $B_r \subset G$ be the metric ball about e. For $x \in X$ define

$$\iota(x) = \sup\{0 < r \leq 1 : B_r \to X,\ g \mapsto gx,\ \text{is injective}\}$$

and

$$\mathfrak{h}(x) = \iota(x)^{-1} \geq 1. \tag{5.3}$$

Because Γ is discrete, every stabilizer $g\Gamma g^{-1}$ is discrete and the local orbit map is injective on a sufficiently small identity neighborhood, whether or not Γ contains torsion. Thus the definition below works directly at the local injectivity scale and requires no passage to a torsion-free subgroup.

Lemma 2.1 (Local comparability of injectivity height)**.** *There exist $c_0, C_0 > 0$, depending only on the fixed metric, such that*

$$g \in B_{c_0/\mathfrak{h}(x)} \quad \Longrightarrow \quad C_0^{-1}\mathfrak{h}(x) \leq \mathfrak{h}(gx) \leq C_0\mathfrak{h}(x). \tag{5.4}$$

Moreover, for every fixed $a \in G$ there is $C(a) \geq 1$ such that

$$\mathfrak{h}(ax) \leq C(a)\mathfrak{h}(x), \qquad \mathfrak{h}(x) \leq C(a)\mathfrak{h}(ax), \tag{5.5}$$

and $C(a)$ can be chosen locally bounded in a.

PROOF. Appendix C, Lemma 1.3, proves (5.4)–(5.5) by comparing the stabilizer $g_x\Gamma g_x^{-1}$ under conjugation and using the local bi-Lipschitz constants of multiplication and conjugation on a fixed unit ball. ☐

3. The weighted Sobolev hierarchy

For $d \in \mathbb{N}$ and $f \in C_c^\infty(X)$ set

$$\mathcal{S}_d(f)^2 = \sum_{|\alpha| \leq d} \int_X \mathfrak{h}(x)^{2d} |D^\alpha f(x)|^2 \, dm_X(x). \tag{5.6}$$

Because $\mathfrak{h} \geq 1$, $\mathcal{S}_d$ dominates the ordinary L^2 derivatives through order d.

Proposition 3.1 (Monotonicity)**.** *If $0 \leq d \leq d'$, then*

$$\mathcal{S}_d(f) \leq C_{d,d'}\mathcal{S}_{d'}(f).$$

Proof. Every derivative occurring in $\mathcal{S}_d$ occurs in $\mathcal{S}_{d'}$, and $\mathfrak{h}^d \le \mathfrak{h}^{d'}$. With the word convention there are only finitely many repeated derivative terms; the displayed constant absorbs this bookkeeping. □

Theorem 3.2 (Global weighted Sobolev embedding). *Let $r \ge 0$ and choose an integer $d > r + m/2$. Then*

$$\max_{|\beta|\le r} \left\| D^\beta f \right\|_{L^\infty(X)} \le C_{d,r}\mathcal{S}_d(f) \qquad (f \in C_c^\infty(X)). \tag{5.7}$$

The constant depends only on G, the chosen basis/metric, and d, r, not on the depth of the support in the cusp.

Proof. Fix x and set $r_x = c_0/\mathfrak{h}(x)$, with c_0 from Lemma 2.1. The orbit map identifies B_{r_x} with a neighborhood of x. In exponential coordinates, rescale the ball to unit size and apply the ordinary Euclidean Sobolev inequality to $D^\beta f$. Appendix C, Proposition 2.1, gives

$$|D^\beta f(x)| \le C\, r_x^{-m/2} \sum_{|\gamma|\le s} r_x^{|\gamma|} \left\| D^{\beta+\gamma} f \right\|_{L^2(B_{r_x} x)} \tag{5.8}$$

with the specific choice $s = d - r$, which is an integer $> m/2$ by hypothesis. Since $|\beta| \le r$, every derivative $D^{\beta+\gamma}$ appearing in (5.8) then has order at most d. On this local ball, Lemma 2.1 gives $\mathfrak{h}(y) \asymp \mathfrak{h}(x) = c_0/r_x$. Hence each local L^2 norm in (5.8) is at most $C\mathfrak{h}(x)^{-d}\mathcal{S}_d(f)$. Since $d > r + m/2$ and $r_x \le 1$, the factor $r_x^{-m/2}\mathfrak{h}(x)^{-d}$ is bounded; the extra positive powers $r_x^{|\gamma|}$ only improve the estimate. Take the supremum in x and over the finitely many β. □

Corollary 3.3 (Compact-core comparison)**.** *On a set where $\mathfrak{h} \le H$, $\mathcal{S}_d$ is equivalent, with constants polynomial in H, to the ordinary invariant L^2 Sobolev norm of order d.*

Proof. The lower comparison is $\mathfrak{h} \ge 1$; the upper comparison is $\mathfrak{h}^d \le H^d$. □

Worked Example 3.4 (A cusp bump)**.** Suppose x_j tends into a cusp with $\iota(x_j) \to 0$. A bump supported on a ball of radius comparable to $\iota(x_j)$ can have height one while its unweighted L^2 mass tends to zero with the local volume. Thus an unweighted global $H^d \to L^\infty$ estimate cannot have a uniform constant. In $\mathcal{S}_d$, the factor $\mathfrak{h}^d \asymp \iota(x_j)^{-d}$ exactly records the loss caused by the shrinking chart.

4. Dependency summary

Dependency Note 4.1 (V06.C05.L)**.** Lemma 2.1 owns the geometric comparison that makes a single Sobolev constant possible on a noncompact quotient.

Dependency Note 4.2 (V06.C05.T)**.** Theorem 3.2 is the global height-adapted Sobolev embedding used by the effective arc.

Dependency Note 4.3 (V06.C05.P)**.** Corollary 3.3 translates between the global weighted norm and ordinary compact-core Sobolev norms, which is the form used during cusp truncation.

5. Reader perspective

The shrinking-chart failure is displayed before the weighted norm is introduced. The weight is intrinsic—the reciprocal local injectivity scale—and no unproved reduction-theoretic comparison between geometric depth and an arithmetic height is required.

CHAPTER 6

Smoothing, Multiplication, and Translation Estimates

Proof architecture. Later effective arguments repeatedly replace a rough indicator by a smooth bump, multiply cutoffs, and translate test functions by large group elements. Each operation spends Sobolev order or height. This chapter gives a closed calculus with explicit locations of those losses.

1. Translation

For $a \in G$ write $(a \cdot f)(x) = f(a^{-1}x)$.

Lemma 1.1 (Derivative transport)**.** *For $Y \in \mathfrak{g}$,*

$$D_Y(a \cdot f) = a \cdot D_{\mathrm{Ad}(a^{-1})Y} f. \tag{6.1}$$

More generally, a word of order j is carried to a linear combination of words of order j whose coefficients are bounded by a degree-j polynomial in $\|\mathrm{Ad}(a)\| + \|\mathrm{Ad}(a^{-1})\|$.

PROOF. Differentiate

$$f(a^{-1}\exp(-tY)x) = f(\exp(-t\,\mathrm{Ad}(a^{-1})Y)a^{-1}x).$$

Iteration gives the word statement. □

Proposition 1.2 (Translation bound)**.** *For each d there are constants C_d, N_d such that*

$$\mathcal{S}_d(a \cdot f) \le C_d\, \mathcal{A}(a)^{N_d} \mathcal{S}_d(f), \tag{6.2}$$

where

$$\mathcal{A}(a) = 1 + \|\mathrm{Ad}(a)\| + \left\|\mathrm{Ad}(a^{-1})\right\| + C(a) + C(a^{-1})$$

and $C(\cdot)$ is the height-distortion constant from Lemma 2.1.

PROOF. Apply Lemma 1.1, change variables $x = ay$ using invariance of m_X, and use $\mathfrak{h}(ay) \le C(a)\mathfrak{h}(y)$. There are finitely many derivative words through order d, so their coefficient sums are absorbed by the polynomial power N_d. □

2. Multiplication

THEOREM 2.1 (Sobolev algebra estimate with order loss). *For every $d \geq 0$ there exists an integer $r = r(m) > m/2$ and a constant C_d such that*

$$\mathcal{S}_d(fh) \leq C_d \mathcal{S}_{d+r}(f)\mathcal{S}_{d+r}(h) \qquad (f, h \in C_c^\infty(X)). \tag{6.3}$$

The same estimate holds if one factor is merely smooth with finite indicated norm.

PROOF. For $|\alpha| \leq d$, Leibniz gives a finite sum $D^\alpha(fh) = \sum c_{\beta,\gamma} D^\beta f D^\gamma h$ with $|\beta| + |\gamma| \leq d$. For each term,

$$\left\| \mathfrak{h}^d D^\beta f D^\gamma h \right\|_2 \leq \left\| D^\beta f \right\|_\infty \left\| \mathfrak{h}^d D^\gamma h \right\|_2 .$$

The second factor is at most $\mathcal{S}_d(h) \leq C\mathcal{S}_{d+r}(h)$. Since $d + r > |\beta| + m/2$, Theorem 3.2 bounds the first factor by $C\mathcal{S}_{d+r}(f)$. Sum the finitely many terms. Interchanging f, h gives the same conclusion when the larger derivative happens to fall on the other factor. □

Corollary 2.2 (Smooth cutoff cost)**.** *If $\chi \in C_c^\infty(X)$, then multiplication by χ is a bounded operator from $\mathcal{S}_{d+r}$ to $\mathcal{S}_d$, with operator norm $O_d(\mathcal{S}_{d+r}(\chi))$.*

PROOF. Apply Theorem 2.1. □

3. Approximate identities on G

Choose a nonnegative $\rho \in C_c^\infty(\mathfrak{g})$ supported in a sufficiently small ball and with integral one in exponential coordinates. For $0 < \varepsilon < \varepsilon_0$ transfer the rescaled bump to G and normalize Haar mass to obtain $\rho_\varepsilon \in C_c^\infty(G)$ with

$$\operatorname{supp} \rho_\varepsilon \subset B_{C\varepsilon}, \qquad \int_G \rho_\varepsilon = 1.$$

Define

$$f_\varepsilon(x) = \int_G \rho_\varepsilon(g) f(g^{-1}x)\, dg. \tag{6.4}$$

For the regularization estimate it is important to separate two different resources: derivative order and cusp decay. Smoothing creates derivatives, but it does not create extra decay in the cusp. We therefore use the mixed seminorm

$$\mathcal{S}_{d;k}(f)^2 = \sum_{|\alpha| \leq d+k} \int_X \mathfrak{h}(x)^{2d} |D^\alpha f(x)|^2 \, dm_X(x), \tag{6.5}$$

so that $\mathcal{S}_{d;0} = \mathcal{S}_d$. Notice that $\mathcal{S}_{d;k}$ asks for k additional derivatives while keeping the height weight fixed at order d.

Lemma 3.1 (Kernel derivative bounds)**.** *For every left- or right-invariant differential operator D of order j,*

$$\|D\rho_\varepsilon\|_{L^1(G)} \le C_D \varepsilon^{-j}. \tag{6.6}$$

Proof. In exponential coordinates, derivatives of $\varepsilon^{-m}\rho(Y/\varepsilon)$ contribute ε^{-j}, while the Jacobian and its derivatives are uniformly bounded on the shrinking support. The final scalar normalization is $1 + O(\varepsilon)$ and does not change the power. □

Theorem 3.2 (Smoothing estimates). *For integers $d, k \ge 0$ and $0 < \varepsilon < \varepsilon_0$:*

$$\mathcal{S}_d(f_\varepsilon) \le C_d \mathcal{S}_d(f), \tag{6.7}$$

$$\mathcal{S}_{d;k}(f_\varepsilon) \le C_{d,k}\varepsilon^{-k}\mathcal{S}_d(f), \tag{6.8}$$

$$\mathcal{S}_d(f - f_\varepsilon) \le C_d \varepsilon \mathcal{S}_{d+1}(f). \tag{6.9}$$

In addition, for every k one has the no-new-cusp-decay estimate

$$\mathcal{S}_{d+k}(f_\varepsilon) \le C_{d,k}\mathcal{S}_{d+k}(f). \tag{6.10}$$

If $\operatorname{supp} f \subset E$, *then* $\operatorname{supp} f_\varepsilon \subset B_{C\varepsilon}E$.

Proof. For (6.7), differentiate under the integral and use Proposition 1.2; on $B_{C\varepsilon_0}$ its translation constant is uniform. Minkowski's integral inequality then gives the bound.

For (6.8), differentiate up to order $d + k$. For the derivatives beyond order d, move k derivatives from the function to the convolution kernel by the identity between infinitesimal left translation on X and right differentiation of the kernel. Lemma 3.1 costs ε^{-k}; the remaining d derivatives are treated as in (6.7). The height exponent stays equal to d, which is exactly why the target is $\mathcal{S}_{d;k}$ rather than $\mathcal{S}_{d+k}$. The estimate (6.10) follows separately from the same argument as (6.7), applied at Sobolev order $d+k$; it asserts stability of pre-existing cusp decay rather than creation of new decay.

For (6.9), write $g = \exp Y$ on the support of ρ_ε and use the fundamental theorem of calculus:

$$f(x) - f(g^{-1}x) = \int_0^1 D_Y f(\exp(-tY)x)\,dt.$$

Here $\|Y\| \ll \varepsilon$. Apply the $\mathcal{S}_d$ norm, Proposition 1.2 on the fixed small ball, and integrate. The support statement is immediate from (6.4). □

Worked Example 3.3 (Why smoothing creates derivatives but not cusp decay)**.** On the line, let $f_\varepsilon = \rho_\varepsilon * f$. Then

$$(f_\varepsilon)^{(k)} = \rho_\varepsilon^{(k)} * f, \qquad \left\|(f_\varepsilon)^{(k)}\right\|_2 \le \left\|\rho_\varepsilon^{(k)}\right\|_1 \|f\|_2 \ll \varepsilon^{-k}\|f\|_2 .$$

This is the derivative gain in (6.8). If, however, f is translated far into a cusp, local convolution does not move it back toward a compact core. No new power of the height can therefore appear on the left from smoothing alone. The mixed norm $\mathcal{S}_{d;k}$ records exactly this distinction.

4. Boundary smoothing

Proposition 4.1 (Smooth approximation of a regular set)**.** *Let $E \subset X$ be measurable and suppose that for some $A > 0$ and all sufficiently small r,*

$$m_X(\{x : \mathrm{dist}(x, \partial E) < r\}) \leq Ar. \tag{6.11}$$

Then there are $\phi_\varepsilon^- \leq 1_E \leq \phi_\varepsilon^+$ with $0 \leq \phi_\varepsilon^\pm \leq 1$, smooth, such that

$$\left\|\phi_\varepsilon^+ - \phi_\varepsilon^-\right\|_{L^1} \ll A\varepsilon, \tag{6.12}$$

and for every d,

$$\mathcal{S}_d(\phi_\varepsilon^\pm) \ll_{d,E} \varepsilon^{-d-m/2} \tag{6.13}$$

provided the ε-boundary neighborhood lies in a region of bounded injectivity height. In a cusp, the same construction holds with the corresponding local height factor supplied by (5.6).

Proof. Take the inner and outer $C\varepsilon$-neighborhoods of E and convolve their indicators with ρ_ε in local injective charts, patched by a partition of unity. The support relation in Theorem 3.2 gives the pointwise inequalities after choosing the inner/outer radii with a fixed safety factor. Their difference is supported in an $O(\varepsilon)$-boundary layer, so (6.11) gives (6.12). Kernel derivatives cost ε^{-d} and the L^2 normalization on an m-dimensional ε-ball costs at most $\varepsilon^{-m/2}$, giving the safe bound (6.13). The height-weighted version follows by working at the local injectivity scale as in Appendix C. ☐

5. Dependency summary

Dependency Note 5.1 (V06.C06.L)**.** Proposition 1.2 and Theorem 2.1 own translation and multiplication losses.

Dependency Note 5.2 (V06.C06.T)**.** Theorem 3.2 proves the approximate-identity calculus, including regularization and approximation with exact powers of ε.

Dependency Note 5.3 (V06.C06.P)**.** Proposition 4.1 packages the calculus in the form consumed by thickening/counting arguments.

6. Reader perspective

Each operation has its own cost: translation spends an Ad-height factor, multiplication spends Sobolev order, and smoothing spends powers of ε^{-1}. These are not hidden in a generic “Sobolev norm behaves well” statement.

CHAPTER 7

Matrix-Coefficient Decay from a Spectral Gap: What Is Actually True

Proof architecture. A spectral gap immediately gives decay for powers of an averaging operator. It does *not*, by itself, give pointwise decay of $\langle \pi(g)v, w\rangle$ as $g \to \infty$. For semisimple groups, Volume 5 supplies pointwise Harish–Chandra decay once a tensor-tempered certificate is known. We prove both statements and keep the logical interface explicit.

1. Lazification removes the negative spectrum

Let S be a finite symmetric Kazhdan set and

$$M = \frac{1}{|S|}\sum_{s\in S}\pi(s).$$

On $\mathcal{H}_0 = (\mathcal{H}^G)^\perp$, Proposition 3.2 gives $\sup \operatorname{Spec} M \le 1-\eta$ with $\eta = \kappa^2/(2|S|)$. Negative spectrum can still reach -1, so define the lazy operator

$$P = \frac{I+M}{2}. \tag{7.1}$$

Lemma 1.1 (Lazy spectral contraction)**.** *On $\mathcal{H}_0$,*

$$0 \le P \le (1-\eta/2)I, \qquad \left\| P^k|_{\mathcal{H}_0} \right\| \le (1-\eta/2)^k. \tag{7.2}$$

PROOF. $\operatorname{Spec}(M) \subset [-1, 1-\eta]$. The affine map $t \mapsto (1+t)/2$ sends this interval to $[0, 1-\eta/2]$. Apply the spectral theorem. ☐

Worked Example 1.2 (Why the lazy step is necessary)**.** Let the group of order two act on its two points by swapping them, and take S to consist of the nontrivial involution. On the mean-zero vector $(1,-1)$ the averaging operator is $M = -I$. Its spectral top is -1, so there is certainly no spectrum near $+1$, yet $M^k(1,-1) = (-1)^k(1,-1)$ never decays. Lazification gives

$$P = \frac{I+M}{2} = 0$$

on the mean-zero line, and decay becomes immediate. Thus a one-sided gap away from $+1$ controls ergodicity, while decay of powers also requires eliminating the possible eigenvalue -1.

THEOREM 1.3 (Exponential decay of averaged matrix coefficients). *For $v, w \perp \mathcal{H}^G$,*

$$|\left\langle P^k v, w\right\rangle| \leq (1-\eta/2)^k \|v\| \|w\| . \tag{7.3}$$

Equivalently, if $\nu = \frac{1}{2}\delta_e + \frac{1}{2|S|}\sum_{s\in S}\delta_s$, then

$$\left|\int_G \langle \pi(g)v, w\rangle \, d\nu^{*k}(g)\right| \leq (1-\eta/2)^k \|v\| \|w\| . \tag{7.4}$$

PROOF. Equation (7.3) is Cauchy–Schwarz and Lemma 1.1. Since $P = \pi(\nu)$ and $P^k = \pi(\nu^{*k})$, (7.4) is the integrated form. ☐

2. The pointwise semisimple interface

Assume now that G is a connected semisimple real linear group with finite center and no compact factors, with maximal compact subgroup K. Volume 5 proved the following exact implication:

$$\pi^{\otimes m} \prec \lambda_G \quad \Longrightarrow \quad |\langle \pi(g)v, w\rangle| \leq \sqrt{d_v d_w}\, \Xi(g)^{1/m} \|v\| \|w\| \tag{7.5}$$

for K-finite vectors, with the sharper tensor-cyclic-dimension version also available there.

Definition 2.1 (Tensor-tempered certificate)**.** A representation π has certificate m if $\pi^{\otimes m}$ is tempered. A family has a *uniform* certificate if the same m works for every member.

Proposition 2.2 (Smooth-vector pointwise decay from a certificate)**.** *Suppose $\pi^{\otimes m}$ is tempered and π is realized on $L^2(X)$ for a homogeneous quotient. There is an integer d_K, depending only on K, such that for smooth vectors f, h,*

$$|\langle \pi(g)f, h\rangle| \leq C\, \Xi(g)^{1/m} \mathcal{S}_{d_K}(f)\mathcal{S}_{d_K}(h). \tag{7.6}$$

On a Cartan chamber $g = k\exp(H)k'$ this gives

$$|\langle \pi(g)f, h\rangle| \leq C'(1+\|H\|)^{D/m} e^{-\rho(H)/m} \mathcal{S}_{d_K}(f)\mathcal{S}_{d_K}(h). \tag{7.7}$$

PROOF. Decompose the K-finite truncations of f, h into irreducible K-types. The K-Sobolev operator $(1-\Delta_K)^{d_K/2}$ has eigenvalues growing quadratically with highest weight, while Weyl's dimension formula has polynomial growth. Choosing d_K larger than half the sum of the relevant polynomial exponents makes

$$\sum_{\tau} \sqrt{\dim \tau}\ \|f_\tau\| \le C \left\|(1-\Delta_K)^{d_K/2} f\right\|_2 , \tag{7.8}$$

and similarly for h, by Cauchy–Schwarz. Apply the K-finite estimate (7.5) to each pair of K-types and sum absolutely using (7.8). The K-direction derivatives are among the invariant derivatives defining $\mathcal{S}_{d_K}$, so the right side is bounded by the displayed Sobolev norms. Density of K-finite smooth vectors gives (7.6). The Harish–Chandra chamber estimate proved in Volume 5 gives (7.7). □

Corollary 2.3 (Tempered actions)**.** *If the mean-zero representation on $L^2_0(X)$ is tempered, take $m = 1$ in Proposition 2.2. In particular the regular representation of G itself has the full Ξ envelope.*

Proof. This is Definition 2.1 with $m = 1$ and the tempered theorem of Volume 5. □

Warning 2.4 (Property (T) is not a tensor-tempered proof)**.** The existence of a Kazhdan gap and the existence of a finite tensor-tempered index are distinct assertions. For higher-rank semisimple groups deep representation theory provides uniform pointwise bounds, but this volume does not disguise those bounds as an elementary consequence of Property (T). The output proved in Chapters 1–4 is the averaged decay (7.3)–(7.4). Whenever Volume 7 needs (7.6), it must either invoke the tensor-tempered theorem proved in Volume 5 in the specific representation at hand or prove the additional strong-spectral-gap theorem there.

3. Uniformity in finite quotients

Corollary 3.1 (Uniform random-walk coefficient decay on congruence quotients)**.** *Fix $n \ge 3$ and a finite symmetric generating set S of $\mathrm{SL}_n(\mathbb{Z})$. There is $c = c(n, S) > 0$ such that for every $q \ge 2$, every $f, h \in \ell^2_0(\mathrm{SL}_n(\mathbb{Z}/q\mathbb{Z}))$, and every $k \ge 0$,*

$$|\left\langle P_q^k f, h\right\rangle| \le e^{-ck}\, \|f\|_2\, \|h\|_2 , \tag{7.9}$$

where P_q is the lazy S-walk operator.

Proof. Theorem 4.1 supplies a uniform $\eta > 0$. Theorem 1.3 gives $(1-\eta/2)^k \le e^{-\eta k/2}$. □

4. Dependency summary

Dependency Note 4.1 (V06.C07.L)**.** Lemma 1.1 isolates the exact spectral-theorem step and handles possible eigenvalue -1 by lazification.

Dependency Note 4.2 (V06.C07.T)**.** Theorem 1.3 is the unconditional matrix-coefficient decay theorem that follows from the certified spectral gap.

Dependency Note 4.3 (V06.C07.P)**.** Proposition 2.2 records the exact backward interface to Volume 5's tensor-tempered Harish–Chandra theorem; Corollary 3.1 gives the arithmetic uniform form.

5. Reader perspective

The chapter explicitly repairs a common logical overreach: a Kazhdan gap controls averages, not arbitrary pointwise coefficients. The pointwise semisimple estimate is presented only with the additional tensor-tempered hypothesis that actually proves it.

CHAPTER 8

Quantitative Mean Ergodic and Maximal Inequalities

Proof architecture. A spectral gap turns qualitative ergodicity into a rate. For a lazy Markov average the rate is exponential. For ordinary Cesàro averages of a symmetric Markov operator the rate is $O(N^{-1})$ in L^2. A deliberately elementary square-sum argument then gives tail maximal inequalities sufficient for almost-everywhere control along rapidly growing scales.

1. Exponential mean ergodic theorem for lazy averages

Let P be the lazy operator (7.1) and let $\Pi = P_G$ be the orthogonal projection to invariant vectors.

THEOREM 1.1 (Quantitative lazy mean ergodic theorem). *If the nontrivial spectrum of P is contained in $[0, 1-c]$, then*

$$\left\|P^k f - \Pi f\right\|_2 \leq (1-c)^k \left\|f - \Pi f\right\|_2 . \tag{8.1}$$

For a Kazhdan set S with constant κ, one may take $c = \kappa^2/(4|S|)$.

PROOF. P fixes the invariant subspace and preserves its orthogonal complement. On the latter its operator norm is at most $1-c$, so powers have norm at most $(1-c)^k$. Lemma 1.1 gives the stated c. □

Corollary 1.2 (Uniform finite-quotient mean ergodic theorem)**.** *For the congruence quotients of* $\mathrm{SL}_n(\mathbb{Z})$*, $n \geq 3$, the constant $c > 0$ in (8.1) can be chosen independently of the level q.*

PROOF. Use Theorem 4.1. □

2. Cesàro averages

Let M be any self-adjoint contraction with invariant projection Π and suppose

$$\mathrm{Spec}(M|_{(\mathcal{H}^G)^\perp}) \subset [-1, 1-\eta].$$

Set

$$A_N = \frac{1}{N}\sum_{j=0}^{N-1} M^j.$$

THEOREM 2.1 (Quantitative Cesàro mean ergodic theorem). *For $N \geq 1$,*

$$\|A_N f - \Pi f\|_2 \leq \min\left(1, \frac{2}{N\eta}\right) \|f - \Pi f\|_2 . \tag{8.2}$$

PROOF. On the orthogonal complement, spectral calculus reduces the operator norm to the scalar multiplier

$$a_N(\lambda) = \frac{1-\lambda^N}{N(1-\lambda)}.$$

For $\lambda \in [-1, 1-\eta]$, $|1-\lambda^N| \leq 2$ and $1-\lambda \geq \eta$, so $|a_N(\lambda)| \leq 2/(N\eta)$. The trivial bound $|a_N| \leq 1$ gives the minimum. □

Worked Example 2.2 (The scalar spectral multiplier behind the rate)**.** For a single spectral value $\lambda = 1-\eta$, the Cesàro multiplier is

$$a_N(1-\eta) = \frac{1-(1-\eta)^N}{N\eta}.$$

When $N\eta \ll 1$ this can still be of order one; when $N\eta \gg 1$ it is $O((N\eta)^{-1})$. The bound in Theorem 2.1 is therefore the correct uniform transition scale for an argument that knows only the size of the spectral gap and nothing finer about the spectral measure.

3. Elementary L^2 tail maximal bounds

The next estimates are not the optimal Hopf–Dunford–Schwartz theorem. Their advantage is that they are proved from the spectral gap alone and are enough for the Borel–Cantelli style uses later in the series.

Lemma 3.1 (Square-sum domination)**.** *For any sequence (F_j) of measurable functions,*

$$\left\|\sup_{j\geq J} |F_j|\right\|_2^2 \leq \sum_{j\geq J} \|F_j\|_2^2 \tag{8.3}$$

whenever the right side is finite.

PROOF. Pointwise, $\sup_j |F_j|^2 \leq \sum_j |F_j|^2$. Integrate and use monotone convergence. □

THEOREM 3.2 (Maximal tail for lazy powers). *Under the hypotheses of Theorem 1.1,*

$$\left\| \sup_{k\geq K} |P^k f - \Pi f| \right\|_2 \leq \frac{(1-c)^K}{\sqrt{1-(1-c)^2}} \|f - \Pi f\|_2 . \tag{8.4}$$

PROOF. Apply Lemma 3.1 and (8.1):

$$\sum_{k\geq K} \left\| P^k(f - \Pi f) \right\|_2^2 \leq \|f - \Pi f\|_2^2 \sum_{k\geq K} (1-c)^{2k}.$$

Evaluate the geometric series and take square roots. □

THEOREM 3.3 (Cesàro tail maximal inequality). *Under the hypotheses of Theorem 2.1, for $N_0 \geq 2/\eta$,*

$$\left\| \sup_{N\geq N_0} |A_N f - \Pi f| \right\|_2 \leq \frac{C}{\eta\sqrt{N_0}} \|f - \Pi f\|_2 \tag{8.5}$$

with an absolute constant C.

PROOF. By Lemma 3.1 and (8.2),

$$\left\| \sup_{N\geq N_0} |A_N f - \Pi f| \right\|_2^2 \leq \frac{4}{\eta^2} \|f - \Pi f\|_2^2 \sum_{N\geq N_0} \frac{1}{N^2}.$$

The tail is at most $2/N_0$, giving (8.5) with $C = 2\sqrt{2}$. □

Corollary 3.4 (Almost-everywhere convergence with a quantitative tail)**.** *Under the spectral-gap hypotheses, $A_N f \to \Pi f$ almost everywhere for every $f \in L^2$ for which the operators are realized as positive Markov operators. More precisely, for $t > 0$ and $N_0 \geq 2/\eta$,*

$$m_X \left\{ x : \sup_{N\geq N_0} |A_N f(x) - \Pi f(x)| > t \right\} \leq \frac{C^2}{\eta^2 N_0 t^2} \|f - \Pi f\|_2^2 . \tag{8.6}$$

PROOF. Chebyshev's inequality and Theorem 3.3 give (8.6). The right side tends to zero. To obtain almost-everywhere convergence, choose $N_j = 2^j$ so that the exceptional probabilities are summable, giving convergence of the maximal tails along j; because the maximal tail is over all $N \geq N_j$, this is convergence of the full sequence. □

4. Sobolev-stable form

Proposition 4.1 (Averaging does not increase the working Sobolev norm excessively)**.** *Let μ be supported in a compact set $Q \subset G$. For every d,*

$$\mathcal{S}_d(\pi(\mu)f) \leq C_{d,Q}\mathcal{S}_d(f). \tag{8.7}$$

For k-fold convolution the naive bound is $C_{d,Q}^k$; if Q lies in a compact subgroup, one may take a uniform constant independent of k.

PROOF. Minkowski's inequality and Proposition 1.2 give

$$\mathcal{S}_d\left(\int_Q g \cdot f \, d\mu(g)\right) \leq \int_Q \mathcal{S}_d(g \cdot f) d\mu(g) \leq C_{d,Q} \mathcal{S}_d(f).$$

Iteration gives the convolution bound. On a compact subgroup the translation constants are uniformly bounded and one may replace the norm by an equivalent K-averaged Sobolev norm, for which translations by K are isometries. □

5. Dependency summary

Dependency Note 5.1 (V06.C08.L)**.** Theorems 1.1 and 2.1 prove quantitative L^2 mean ergodic estimates directly from the spectral gap proved above.

Dependency Note 5.2 (V06.C08.T)**.** Theorems 3.2 and 3.3 provide explicit tail maximal inequalities without importing a general maximal theorem.

Dependency Note 5.3 (V06.C08.P)**.** Corollary 1.2, Corollary 3.4, and Proposition 4.1 give the finite-quotient, almost-everywhere, and Sobolev-compatible forms required downstream.

6. Reader perspective

The proof deliberately uses scalar spectral multipliers and the elementary square-sum maximal bound. This yields weaker but completely transparent constants and avoids citing a general ergodic maximal theorem as a black box.

APPENDIX A

Relative Property (T) for $\mathrm{SL}_2(\mathbb{R}) \ltimes \mathbb{R}^2$

Proof architecture. This appendix owns the only relative-(T) theorem used in the proof of Theorem 4.3. The mechanism is spectral rather than combinatorial. Restrict a representation to the abelian normal subgroup $N = \mathbb{R}^2$; a vector then has a spectral probability measure on $\widehat{N} \cong \mathbb{R}^2$. Almost invariance under $\mathrm{SL}_2(\mathbb{R})$ makes the projectivized measure asymptotically invariant. But $\mathbf{P}^1(\mathbb{R})$ admits no invariant probability measure under the full $\mathrm{SL}_2(\mathbb{R})$ action.

Let

$$H = \mathrm{SL}_2(\mathbb{R}) \ltimes \mathbb{R}^2, \qquad (A, x)(B, y) = (AB, x + Ay),$$

and let $N = \{(I, x) : x \in \mathbb{R}^2\}$.

1. Spectral measures for the normal abelian subgroup

We use the standard spectral theorem for a strongly continuous unitary representation of the abelian group $\mathbb{R}^2$: there is a projection-valued measure E on $\widehat{\mathbb{R}^2} \cong \mathbb{R}^2$ such that

$$\pi(I, x) = \int_{\mathbb{R}^2} e^{i\langle \xi, x\rangle}\, dE(\xi). \tag{A.1}$$

For a unit vector v set

$$\mu_v(B) = \langle E(B)v, v\rangle\,.$$

Then μ_v is a probability measure and

$$\langle \pi(I, x)v, v\rangle = \int e^{i\langle \xi, x\rangle} d\mu_v(\xi). \tag{A.2}$$

Lemma 1.1 (The atom at zero is the invariant projection)**.** *The range of $E(\{0\})$ is exactly $\mathcal{H}^N$. Hence if $\mathcal{H}^N = 0$, then every spectral measure μ_v gives zero mass to $\{0\}$.*

Proof. On $E(\{0\})\mathcal{H}$, formula (A.1) gives $\pi(I, x) = I$ for all x. Conversely, if v is N-fixed, then (A.2) equals $\|v\|^2$ for every x. Thus

$$\int |e^{i\langle \xi, x\rangle} - 1|^2 d\mu_v(\xi) = 0$$

for every x. A countable dense set of x forces $\xi = 0$ for μ_v-almost every ξ, so $v = E(\{0\})v$. $\square$

Lemma 1.2 (Spectral covariance)**.** *For $A \in \mathrm{SL}_2(\mathbb{R})$,*

$$\mu_{\pi(A,0)v} = (A^{-T})_* \mu_v. \tag{A.3}$$

PROOF. The semidirect relation gives

$$(A,0)^{-1}(I,x)(A,0) = (I, A^{-1}x).$$

Therefore

$$\langle \pi(I,x)\pi(A,0)v, \pi(A,0)v \rangle = \left\langle \pi(I, A^{-1}x)v, v \right\rangle = \int e^{i\langle A^{-T}\xi, x\rangle} d\mu_v(\xi).$$

Uniqueness of the Fourier transform of a finite measure gives (A.3). $\square$

Lemma 1.3 (Nearby vectors have nearby spectral measures)**.** *If v, w are unit vectors, then*

$$\|\mu_v - \mu_w\|_{\mathrm{TV}} \le 2\,\|v - w\|\,, \tag{A.4}$$

where total variation is the dual norm against bounded Borel functions of supremum norm at most one.

PROOF. For such a function ϕ, functional calculus gives an operator $T_\phi = \int \phi\, dE$ with $\|T_\phi\| \le 1$. Hence

$$\begin{aligned} |\langle T_\phi v, v\rangle - \langle T_\phi w, w\rangle| &\le |\,\langle T_\phi(v-w), v\rangle\,| + |\,\langle T_\phi w, v-w\rangle\,| \\ &\le 2\,\|v-w\|\,. \end{aligned}$$

Take the supremum over ϕ. $\square$

2. No invariant probability on the projective line

Let $p : \mathbb{R}^2 \setminus \{0\} \to \mathbf{P}^1(\mathbb{R})$ be projectivization.

Lemma 2.1 (Projective-line obstruction)**.** *There is no $\mathrm{SL}_2(\mathbb{R})$-invariant Borel probability measure on $\mathbf{P}^1(\mathbb{R})$.*

PROOF. Identify $\mathbf{P}^1(\mathbb{R}) = \mathbb{R} \cup \{\infty\}$ by $x \leftrightarrow [x:1]$. The unipotent

$$u = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

acts by $x \mapsto x+1$ and fixes ∞. If ν were u-invariant, the half-open intervals $[k, k+1)$ would all have the same mass. Since they are disjoint and their union is $\mathbb{R}$, finite total mass forces each to have mass zero. Thus $\nu(\mathbb{R}) = 0$ and $\nu = \delta_\infty$.

But the Weyl element

$$w = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

sends ∞ to 0. Hence $w_*\delta_\infty = \delta_0 \neq \delta_\infty$, contradicting full $\mathrm{SL}_2(\mathbb{R})$-invariance. $\square$

3. The relative theorem

THEOREM 3.1 (Relative Property (T), full proof). *The pair* $(H, N) = (\mathrm{SL}_2(\mathbb{R}) \ltimes \mathbb{R}^2, \mathbb{R}^2)$ *has relative Property* (T).

PROOF. Assume not. Then there is a unitary representation $(\pi, \mathcal{H})$ of H with almost invariant unit vectors and with $\mathcal{H}^N = 0$. Indeed, if the sequence formulation failed only representation-by-representation, take a Hilbert direct sum along a compact exhaustion exactly as in Theorem 1.2.

Choose unit vectors v_j such that

$$\|\pi(h)v_j - v_j\| \longrightarrow 0 \tag{A.5}$$

for every fixed $h \in H$; a diagonal choice over an increasing compact exhaustion gives this pointwise statement. Let $\mu_j = \mu_{v_j}$. By Lemma 1.1, $\mu_j(\{0\}) = 0$, so

$$\nu_j = p_*\mu_j$$

is a probability measure on the compact space $\mathbf{P}^1(\mathbb{R})$.

Fix $A \in \mathrm{SL}_2(\mathbb{R})$. Lemmas 1.2 and 1.3 give

$$\left\|(A^{-T})_*\mu_j - \mu_j\right\|_{\mathrm{TV}} \leq 2\,\|\pi(A,0)v_j - v_j\| \to 0. \tag{A.6}$$

Projectivization cannot increase total variation, hence

$$\|A_*\nu_j - \nu_j\|_{\mathrm{TV}} \to 0 \tag{A.7}$$

(up to replacing A by the equivalent inverse-transpose projective action; these actions are conjugate and range over the same $\mathrm{SL}_2(\mathbb{R})$).

By weak-* compactness of the probability measures on $\mathbf{P}^1(\mathbb{R})$, pass to a subnet or subsequence $\nu_j \to \nu$. Equation (A.7) implies $A_*\nu = \nu$ for every A in a countable dense subgroup of $\mathrm{SL}_2(\mathbb{R})$. Continuity of the action on the compact projective line extends the invariance to all $A \in \mathrm{SL}_2(\mathbb{R})$. This contradicts Lemma 2.1. $\square$

Corollary 3.2 (Relative Kazhdan pair exists)**.** *There are a compact set* $Q \subset H$ *and* $c > 0$ *such that every unitary representation satisfies*

$$\mathrm{dist}(v, \mathcal{H}^N) \leq c^{-1}\delta_Q(v).$$

PROOF. This is Proposition 2.2 applied to Theorem 3.1. $\square$

4. Proof dependencies

The spectral theorem for the abelian group $\mathbb{R}^2$ lies on the declared functional-analytic proof floor. Everything after that theorem—zero-atom identification, covariance, total-variation control, projective obstruction, and the compactness contradiction—is proved in this appendix. No quantitative Burger/Kassabov constant is used.

APPENDIX B

Induction and Inheritance of Property (T) by Lattices

Proof architecture. The slogan "lattices inherit Property (T)" conceals a measurable induction argument. We write it out because the theorem is the bridge from the Lie group $\mathrm{SL}_n(\mathbb{R})$ to the arithmetic group $\mathrm{SL}_n(\mathbb{Z})$. Finite covolume is used exactly once: it lets us discard a small tail so that the induction cocycle ranges over a finite subset of the lattice on the remaining compact core.

Let G be lcsc and unimodular, $\Gamma < G$ a lattice, and $X = G/\Gamma$ with invariant probability measure m_X.

1. The induction cocycle

Choose a measurable section $s : X \to G$. Define $\alpha : G \times X \to \Gamma$ by

$$g^{-1}s(x) = s(g^{-1}x)\alpha(g,x). \tag{B.1}$$

Then

$$\alpha(g_1g_2,x) = \alpha(g_2,x)\alpha(g_1,g_2^{-1}x) \tag{B.2}$$

with the convention (B.1).

If $\sigma : \Gamma \to U(\mathcal{V})$ is unitary, define $\rho = \mathrm{Ind}_\Gamma^G \sigma$ on $L^2(X,\mathcal{V})$ by

$$(\rho(g)F)(x) = \sigma(\alpha(g,x)^{-1})F(g^{-1}x). \tag{B.3}$$

The cocycle identity proves $\rho(g_1)\rho(g_2) = \rho(g_1g_2)$, and invariance of m_X proves unitarity.

Lemma 1.1 (Compact-core finite cocycle range)**.** *Let $K \subset G$ be compact and $\delta > 0$. There is a measurable set $C \subset X$ with $m_X(C) > 1 - \delta$ and a finite set $F \subset \Gamma$ such that*

$$x \in C, \quad g^{-1}x \in C, \quad g \in K \quad \Longrightarrow \quad \alpha(g,x) \in F. \tag{B.4}$$

Proof. By regularity of m_X, choose compact $C \subset X$ with measure $> 1 - \delta$ over which the quotient map admits finitely many relatively compact local sections. Modify s on C so that $s(C)$ is contained in a compact set $L \subset G$. If $x, g^{-1}x \in C$, then (B.1) gives

$$\alpha(g,x) = s(g^{-1}x)^{-1}g^{-1}s(x) \in L^{-1}K^{-1}L \cap \Gamma.$$

A discrete subgroup meets a compact subset of G in a finite set. Take this intersection as F. □

2. Almost invariant vectors induce upward

Proposition 2.1 (Induction preserves almost invariance from a lattice)**.** *If σ has almost invariant vectors as a representation of Γ, then $\rho = \mathrm{Ind}_\Gamma^G \sigma$ has almost invariant vectors as a representation of G.*

Proof. Fix compact $K \subset G$ and $\varepsilon > 0$. Choose $\delta > 0$ so small that $8\delta < \varepsilon^2/2$. Apply Lemma 1.1 and obtain C, F. Since σ has almost invariant vectors, choose a unit vector $v \in \mathcal{V}$ with

$$\max_{\gamma \in F} \|\sigma(\gamma)v - v\| < \varepsilon/2.$$

Let $V(x) = v$ be the constant unit section in $L^2(X, \mathcal{V})$. For $g \in K$, formula (B.3) gives

$$\|\rho(g)V - V\|_2^2 = \int_X \left\|\sigma(\alpha(g,x)^{-1})v - v\right\|^2 dm_X(x). \tag{B.5}$$

On $C \cap gC$, the integrand is $< \varepsilon^2/4$. The complement has measure at most $m_X(C^c) + m_X((gC)^c) < 2\delta$, and the integrand is always at most 4. Hence

$$\|\rho(g)V - V\|_2^2 < \varepsilon^2/4 + 8\delta < 3\varepsilon^2/4 < \varepsilon^2.$$

The estimate is uniform in $g \in K$, proving almost invariance. □

3. Invariant vectors descend

Lemma 3.1 (A G-invariant induced vector gives a Γ-invariant vector)**.** *If $\rho = \mathrm{Ind}_\Gamma^G \sigma$ has a nonzero G-invariant vector, then σ has a nonzero Γ-invariant vector.*

Proof. Use the equivalent equivariant-function model: ρ consists of measurable functions $\widetilde{F} : G \to \mathcal{V}$ satisfying

$$\widetilde{F}(g\gamma) = \sigma(\gamma)^{-1}\widetilde{F}(g), \tag{B.6}$$

with square-integrable norm on G/Γ, and G acts by left translation. A vector fixed by G satisfies

$$L_a\widetilde{F} = \widetilde{F} \quad \text{in } L^2_{\mathrm{loc}}(G, \mathcal{V}) \qquad (a \in G), \tag{B.7}$$

where $L_a\widetilde{F}(g) = \widetilde{F}(a^{-1}g)$. The passage from the quotient norm to local L^2 is legitimate because every compact subset of G meets only finitely many translates of a relatively compact section piece.

We now prove, rather than assume, that a locally square-integrable function satisfying (B.7) is essentially constant. Let $\varphi \in C_c(G)$ with $\int_G \varphi = 1$ and form the Bochner convolution

$$F_\varphi(g) = \int_G \varphi(a)\widetilde{F}(a^{-1}g)\,da.$$

Local square integrability makes this well defined and continuous after replacing φ by a smooth compactly supported function. Equation (B.7), integrated in a, gives $F_\varphi = \widetilde{F}$ in L^2_{loc}. On the other hand, for every $b \in G$, (B.7) gives $L_bF_\varphi = F_\varphi$ as continuous functions; hence $F_\varphi(bg) = F_\varphi(g)$ for all b, g, so F_φ is a constant vector w. Therefore $\widetilde{F} = w$ almost everywhere. Since the original induced vector is nonzero, $w \neq 0$.

Insert the constant function into the covariance relation (B.6). For each $\gamma \in \Gamma$, that relation says for almost every g that $w = \sigma(\gamma)^{-1}w$. Both sides are independent of g, so the equality holds as a vector identity. Thus w is a nonzero Γ-invariant vector. $\square$

THEOREM 3.2 (Lattice inheritance, full proof). *If G has Property (T) and $\Gamma < G$ is a lattice, then Γ has Property (T).*

PROOF. Let σ be any unitary representation of Γ with almost invariant vectors. Proposition 2.1 gives almost invariant vectors in $\mathrm{Ind}_\Gamma^G\,\sigma$. Property (T) of G yields a nonzero invariant vector there. Lemma 3.1 then supplies a nonzero Γ-invariant vector in σ. Apply Theorem 1.2 to conclude Property (T) for Γ. $\square$

4. Proof dependencies

The only measure-theoretic inputs are regularity of the finite quotient measure, measurable local sections for a quotient by a discrete subgroup, and strong continuity of translations in L^2. The finite cocycle range is proved on a compact core rather than falsely asserted on an arbitrary fundamental domain.

APPENDIX C

Local Geometry Behind the Weighted Sobolev Norm

Proof architecture. The Sobolev weight in Chapter 5 is useful only if the injectivity scale is stable on its own chart and under a fixed translation. Both statements reduce to elementary conjugation estimates for the stabilizer. We also record the scaled Euclidean Sobolev estimate in the exact form used in Theorem 3.2.

1. Injectivity scale and the stabilizer

For $x \in X$, write

$$\Gamma_x = \{g \in G : gx = x\}.$$

If $x = h\Gamma$, then $\Gamma_x = h\Gamma h^{-1}$.

Lemma 1.1 (Injectivity radius versus the nearest stabilizer element)**.** *There are constants $c, C > 0$ depending only on the fixed metric on the unit ball such that, whenever*

$$r_x = \inf\{d(e,\gamma) : \gamma \in \Gamma_x \setminus \{e\}\} \le 1,$$

one has

$$c r_x \le \iota(x) \le C r_x. \tag{C.1}$$

If the infimum is > 1, then $\iota(x) = 1$ by definition up to the same harmless constants.

Proof. If $g_1 x = g_2 x$ with $g_1, g_2 \in B_r$, then $g_2^{-1} g_1 \in \Gamma_x$. On a fixed small ball, multiplication and inversion are bi-Lipschitz in exponential coordinates, so $d(e, g_2^{-1} g_1) \le C_1 r$. Thus no nontrivial stabilizer element in $B_{C_1 r}$ implies injectivity on B_r. Conversely, if $\gamma \in \Gamma_x$ is sufficiently close to e, write $\gamma = g_2^{-1} g_1$ with g_1, g_2 lying in a ball of radius $C_2 d(e,\gamma)$, for example by taking $g_2 = e$, $g_1 = \gamma$. Then the orbit map is not injective at that scale. Adjust the constants to the chosen radius-one cutoff. □

Lemma 1.2 (Conjugation distortion)**.** *For every compact $L \subset G$ there is $C_L \geq 1$ such that*

$$C_L^{-1} d(e,u) \leq d(e, gug^{-1}) \leq C_L d(e,u) \tag{C.2}$$

for $g \in L$ and all u sufficiently close to e. For a fixed $g \in G$, the same holds with a constant $C(g)$.

PROOF. In exponential coordinates, $g \exp(Y) g^{-1} = \exp(\mathrm{Ad}(g)Y)$. On a sufficiently small ball, the Riemannian distance is uniformly equivalent to the Euclidean norm of Y. On a compact set of g, both $\|\mathrm{Ad}(g)\|$ and $\|\mathrm{Ad}(g^{-1})\|$ are uniformly bounded. □

Lemma 1.3 (Full injectivity-height distortion)**.** *There are $c_0, C_0 > 0$ such that if $g \in B_{c_0/\mathfrak{h}(x)}$, then*

$$C_0^{-1}\mathfrak{h}(x) \leq \mathfrak{h}(gx) \leq C_0 \mathfrak{h}(x).$$

For every fixed $a \in G$ there is a locally bounded $C(a)$ such that

$$C(a)^{-1}\mathfrak{h}(x) \leq \mathfrak{h}(ax) \leq C(a)\mathfrak{h}(x).$$

PROOF. The stabilizer of gx is $g\Gamma_x g^{-1}$. If g lies in the shrinking ball $B_{c_0/\mathfrak{h}(x)} \subset B_{c_0}$, Lemma 1.2 applies with a uniform compact-set constant. Combine with Lemma 1.1. For fixed a, use the second form of Lemma 1.2. Local boundedness follows from the compact-uniform statement. □

2. Scaled Sobolev on an injective chart

Proposition 2.1 (Scaled local Sobolev inequality)**.** *Let $s > m/2$ be an integer. There are $r_0, C > 0$ such that if the orbit map $B_r \to X$ is injective at x, $0 < r \leq r_0$, then for every differential word D^β and $f \in C^\infty(B_r x)$,*

$$|D^\beta f(x)| \leq C r^{-m/2} \sum_{|\gamma| \leq s} r^{|\gamma|} \left\| D^{\beta+\gamma} f \right\|_{L^2(B_r x)}. \tag{C.3}$$

The constant is uniform in x and r.

PROOF. Use exponential coordinates $Y \mapsto \exp(Y)x$ on B_r. After the rescaling $Y = rZ$, ordinary Sobolev embedding on the fixed unit ball gives

$$|u(0)| \leq C \sum_{|\alpha| \leq s} \|\partial_Z^\alpha u\|_{L^2(B_1)}.$$

The L^2 change of variables contributes $r^{-m/2}$ and each derivative ∂_Z contributes a factor r. On a sufficiently small fixed ball, the coordinate derivatives and the invariant differential words D^γ are related by matrices of smooth coefficients with uniformly bounded derivatives, and Haar/Riemannian densities are uniformly comparable. Apply this to $u = D^\beta f$ and absorb lower-order coordinate terms into the sum through order s. □

Remark 2.2. The weight $\mathfrak{h}^d$ in (5.6) is intentionally stronger than the minimum power needed for a single L^∞ estimate. The surplus makes the hierarchy monotone and survives the repeated products, cutoffs, and translations of later volumes without redefining the norm at every theorem.

3. Proof dependencies

No reduction theory is hidden here. The geometry uses only discreteness of the stabilizer, local Lie-group coordinates, conjugation by Ad, and ordinary Euclidean Sobolev embedding on a ball.

Bibliographic notes

Kazhdan's original argument already used the relative rigidity of $\mathrm{SL}_2(k) \ltimes k^2$ to reach higher-rank special linear groups. Burger and Shalom developed quantitative forms, and Kassabov obtained sharp-order Kazhdan constants for $\mathrm{SL}_n(\mathbb{Z})$. Those works are provenance rather than black-box proved inputs here: the qualitative route needed by this series is reconstructed in Chapters 1–2 and Appendices A–B. The terminology Property (τ) follows the standard finite-quotient usage. The Sobolev conventions are designed for later homogeneous-dynamics arguments: the weight is tied directly to the local injectivity radius, so the estimates do not pretend that a noncompact finite-volume quotient has uniform bounded geometry.

Volume 7

Effective Mixing, Thickening, and Expanding Translates

Preface: what is inherited and what is new

This volume takes the qualitative and fixed-quotient results proved in the earlier homogeneous-dynamics volumes as prerequisites. The new arguments concern uniform, height-sensitive, and congruence-compatible estimates; each such statement is proved at the scope in which it is used.

The earlier inputs are AHD Volumes 1, 5, and 6 together with Ratner Edition v1.1, Volumes 10–13. In particular, this volume inherits strong wavefront geometry, fixed-quotient effective mixing, effective horospherical and closed-orbit translates, effective counting, and effective closing only at the scopes proved there.

Proof architecture. The strong-spectral root found in the first audit is now closed for the higher-rank type-A congruence family fixed in Volume 6. Appendix C reconstructs the required root-tempering mechanism and Chapter 7 propagates it to level-uniform mixing, translates, horospheres, and counting-ready estimates. Rank-one congruence towers remain a separate later arithmetic input rather than being hidden inside Property (τ).

Student orientation: the effective-equidistribution error balance

A singular homogeneous measure cannot be inserted directly into a mixing estimate. *Thickening* replaces it by a smooth density in a transverse tube; a *wavefront lemma* controls how that tube is distorted by the translating element; *cusp truncation* removes the region of poor injectivity radius; and *boundary smoothing* replaces sharp sets by Sobolev test functions. Every effective theorem in this volume should be read as balancing four errors:

$$E_{\mathrm{mix}} + E_{\mathrm{thick}} + E_{\mathrm{cusp}} + E_{\mathrm{boundary}}.$$

A constant called *uniform in a congruence tower* is allowed to depend on the ambient algebraic group and fixed Sobolev order, but not on the congruence level. This convention is used throughout the volume.

Reader guide: a concrete model

Why this matters.

Volume 7 is organized around the passage from *Thickening Singular Homogeneous Measures* to *Uniform Matrix-Coefficient Decay for the Higher-Rank Congruence Model.* The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.1 (A calculation to keep in view)**.** Consider the standard unipotent and diagonal elements

$$u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad a_s = \begin{pmatrix} e^s & 0 \\ 0 & e^{-s} \end{pmatrix}.$$

A direct multiplication gives

$$a_s u_t a_{-s} = u_{e^{2s}t}.$$

Thus a displacement of size $|t|$ in the unstable horospherical direction is enlarged by e^{2s}. If $|t| = e^{-2s}$, the conjugated displacement has order one. This elementary calculation is the model for the “first visible scale” used repeatedly in nondivergence and shearing arguments.

CHAPTER 1

Thickening Singular Homogeneous Measures

1. The problem

Let G be a Lie group, $\Gamma < G$ a lattice, $X = \Gamma\backslash G$, and let $Y = xH \subset X$ be a closed finite-volume orbit of a closed subgroup $H < G$. The measure m_Y is singular with respect to Haar measure on X, so ordinary mixing estimates cannot be applied directly to

$$I_t(f) = \int_Y f(ya_t)\, dm_Y(y).$$

The standard cure is to replace m_Y by a smooth density supported in a thin transverse tube. The point of this chapter is to record exactly what that replacement costs when X is noncompact and the local injectivity radius varies.

2. Thick-part product charts

Fix an inner product on $\mathfrak{g}$ and a vector-space complement

$$\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{r}.$$

Write $B_{\mathfrak{r}}(\epsilon)$ for the ϵ-ball in $\mathfrak{r}$.

Proposition 2.1 (Uniform product chart on a height window)**.** *Assume the height convention of AHD Volume 6: on $X_{\leq R} = \{z : \mathrm{ht}(z) \leq R\}$ the injectivity radius is bounded below by cR^{-A_0}. There are $A_1, C > 0$ such that for*

$$0 < \epsilon \leq C^{-1} R^{-A_1}$$

and every $y \in Y \cap X_{\leq R}$, the map

$$\Psi_y : (H \cap U_\epsilon) \times B_{\mathfrak{r}}(\epsilon) \to X, \qquad (h, Z) \mapsto yh \exp Z,$$

is injective after quotienting the first factor by the local stabilizer, and its Jacobian and inverse Jacobian have C^r norms bounded by CR^{A_r} for each fixed r.

Proof. At $(e, 0)$ the differential of multiplication is the linear isomorphism $\mathfrak{h} \oplus \mathfrak{r} \to \mathfrak{g}$. On the compact unit sphere of choices determined by the fixed complement, its inverse norm is bounded. Right translation to y does not change the Lie-algebra differential; the only obstruction is collision in the quotient. By the injectivity-radius lower bound, two points in the ambient exponential ball of radius $cR^{-A_0}/4$ cannot represent the same point of X unless they differ by the local stabilizer. Choose ϵ below this scale.

The multiplication map and its derivatives are uniformly bounded on a fixed identity chart of G. The quantitative inverse function theorem therefore gives a product chart with derivative bounds depending polynomially on the reciprocal chart radius. Since that radius is $\gg R^{-A_0}$, every fixed derivative order costs at most a power R^{A_r}. The same estimates applied to the determinant of the differential give the Jacobian and inverse-Jacobian bounds. □

Lemma 2.2 (Transverse bump scaling)**.** *Let $d = \dim \mathfrak{r}$ and fix $\rho \in C_c^\infty(\mathfrak{r})$, $\rho \geq 0$, supported in the unit ball and of integral one. Put*

$$\rho_\epsilon(Z) = \epsilon^{-d}\rho(Z/\epsilon).$$

Then for every differential operator D of order j,

$$\|D\rho_\epsilon\|_\infty \ll_D \epsilon^{-d-j}, \qquad \|D\rho_\epsilon\|_1 \ll_D \epsilon^{-j}.$$

Proof. Differentiate $\epsilon^{-d}\rho(Z/\epsilon)$; each derivative contributes one factor ϵ^{-1}. The L^1 statement follows after the change of variables $Z = \epsilon W$. □

3. The exact thickening identity

Lemma 3.1 (Thickening identity)**.** *Let $Y = xH$ be a closed finite-volume orbit and let $\Psi_y(h, Z) = yh \exp Z$ be an injective local product chart as in Proposition 2.1. Let $\chi \in C_c^\infty(Y)$ be supported where these charts are valid and let ρ_ϵ be a normalized transverse bump. Then there is a smooth compactly supported density Φ_ϵ on X satisfying, for every $f \in C_c(X)$,*

(7.1.1)
$$\int_X f(z)\Phi_\epsilon(z)\, dm_X(z) = \int_Y \chi(y) \int_{\mathfrak{r}} f(y \exp Z)\rho_\epsilon(Z)J_y(Z)\, dZ\, dm_Y(y),$$

where $J_y(Z)$ is the Jacobian of the product chart and the inner kernel is normalized to have integral one.

Proof. Choose finitely many product charts covering $\operatorname{supp}\chi$ and a smooth partition of unity $\{\beta_j\}$ on Y subordinate to them. In the j-th chart define a measure by

$$d\nu_j(\Psi_y(e, Z)) = \beta_j(y)\chi(y)\rho_\epsilon(Z)J_y(Z)\, dm_Y(y)\, dZ.$$

The change-of-variables theorem for the diffeomorphism Ψ says that ν_j has a smooth density with respect to m_X. Summing these densities gives Φ_ϵ. Integrating f and applying the change-of-variables formula in each chart gives (7.1.1). On chart overlaps the partition of unity makes the definitions add rather than conflict. Finally divide each transverse kernel by its positive integral; because $J_y(0) = 1$ and J_y is continuous, this normalization is well defined for sufficiently small ϵ. $\square$

THEOREM 3.2 (Quantitative orbit-to-thickening comparison). *Let $\chi_R \in C_c^\infty(Y)$ be supported in $Y \cap X_{\le 2R}$, equal to one on $Y \cap X_{\le R}$, and satisfy derivative bounds $\|D^j \chi_R\|_\infty \ll_j R^{B_j}$. Construct the normalized thickening density $\Phi_{R,\epsilon}$ from $\chi_R m_Y$ and ρ_ϵ. Then for $f \in C^1(X)$,*

$$\left| \int_X f(z)\Phi_{R,\epsilon}(z)\, dm_X(z) - \int_Y \chi_R(y) f(y)\, dm_Y(y) \right| \ll R^B \epsilon\, \mathcal{S}_1(f),$$

provided $\epsilon \ll R^{-A_1}$. The implicit constants depend only on the fixed local geometry and the cutoff convention.

PROOF. By the inherited thickening identity, the first integral equals

$$\int_Y \chi_R(y) \int_{\mathfrak{r}} f(y \exp Z)\rho_\epsilon(Z) J_y(Z)\, dZ\, dm_Y(y),$$

with $J_y(0) = 1$. Normalize the density so the inner kernel has integral one. Along the curve $s \mapsto y \exp(sZ)$,

$$|f(y \exp Z) - f(y)| \le CR^B \|Z\| \mathcal{S}_1(f),$$

where the power of R records the height-weighted derivative convention and chart distortion. The same product-chart estimates give $|J_y(Z) - 1| \ll R^B \|Z\|$. Since the first moment of ρ_ϵ is $O(\epsilon)$, integration gives the result. $\square$

Proposition 3.3 (Sobolev cost of a height-window thickening)**.** *For every Sobolev order ℓ there are $A(\ell), B(\ell)$ such that*

$$\mathcal{S}_\ell(\Phi_{R,\epsilon}) \ll R^{B(\ell)} \epsilon^{-A(\ell)} \|\chi_R\|_{C^\ell(Y)}.$$

The exponent $A(\ell)$ is determined only by the transverse dimension and the number of derivatives; no additional cusp decay is created by smoothing.

PROOF. In the local product coordinates, $\Phi_{R,\epsilon}$ is a finite sum of products of χ_R, ρ_ϵ, and smooth Jacobian factors. Apply the Sobolev product estimate from AHD Volume 6. Lemma 2.2 supplies the powers of ϵ^{-1}; Proposition 2.1 supplies the polynomial factors in R. A partition of unity of bounded overlap on the height window completes the estimate. The final assertion is exactly the distinction emphasized in AHD Volume 6: convolution supplies derivatives, not stronger decay in the cusp variable. $\square$

Corollary 3.4 (Family-uniform thickening)**.** *Suppose $Y_i = x_i H_i$ ranges over a family for which the Lie dimensions, complement angles, height-window* *chart exponents, and cutoff derivative exponents are uniformly bounded. Then Theorem 3.2 and Proposition 3.3 hold with common exponents and constants.*

Proof. Every constant in the preceding proofs depends only on the finite list of displayed geometric bounds. Taking their maxima gives a uniform choice. $\square$

Worked Example 3.5 (A closed horocycle in the modular surface)**.** In $X = \mathrm{SL}_2(\mathbb{Z})\backslash \mathrm{SL}_2(\mathbb{R})$, a closed horocycle at fixed height is one-dimensional and has a two-dimensional transverse complement. A radius-ϵ transverse bump therefore has zeroth-order size $\asymp \epsilon^{-2}$ and each transverse derivative adds one more factor ϵ^{-1}. The orbit integral itself has no such loss. This elementary dimension count is the source of the smoothing exponent that later competes against the mixing rate.

CHAPTER 2

Wavefront and Quantitative Local Product Geometry

1. Strong wavefront geometry

We use the symmetric-pair notation $G = HA^+K$. On the relative interior of a fixed chamber face, every active restricted root is uniformly separated from zero.

Lemma 1.1 (Strong wavefront inclusion)**.** *Fix a chamber face and $\eta > 0$. There are $C_\eta, \epsilon_0 > 0$ such that, for a in the relative η-interior of the face and $0 < \epsilon < \epsilon_0$,*

$$HaK\,U_\epsilon \subset HaA_{C_\eta \epsilon} K_{C_\eta \epsilon},$$

where the neutral Levi directions are included in the face factor.

Proof. At the identity consider

$$(X, Z, W) \longmapsto \mathrm{Ad}(a^{-1})X + Z + W, \qquad (X, Z, W) \in \mathfrak{h} \oplus \mathfrak{a}_{\mathrm{face}} \oplus \mathfrak{k}.$$

On a neutral block the map is a fixed surjection. On each active pair $\mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha}$, the $\mathfrak{h}$ and $\mathfrak{k}$ directions are the two graph subspaces determined by the symmetric-pair involutions, while $\mathrm{Ad}(a^{-1})$ multiplies the two root coordinates by $e^{-\alpha(\log a)}$ and $e^{\alpha(\log a)}$. Because $|\alpha(\log a)| \geq \eta$ on active roots, the resulting 2×2 blocks have inverses bounded by C_η. The quantitative inverse function theorem, applied to

$$(h, b, k) \longmapsto a^{-1}hak,$$

therefore supplies a common identity neighborhood on which every $u \in U_\epsilon$ has a factorization $au = h\,a\,b\,k$ with h, b, k at distance $O_\eta(\epsilon)$ from the identity. Multiplying by H on the left and K on the right gives the inclusion. □

The qualitative inclusion is enough for asymptotic counting. Effective arguments also need derivative and Jacobian control for the factorization map.

2. A quantitative right inverse

Let $\mathfrak{g}$ be decomposed into the neutral part and restricted root pairs. For $a = \exp Y$ define

$$L_a : \mathfrak{h} \oplus \mathfrak{a} \oplus \mathfrak{k} \to \mathfrak{g}, \qquad L_a(X, Z, W) = \mathrm{Ad}(a^{-1})X + Z + W.$$

Lemma 2.1 (Uniform block right inverse). *On the relative η-interior of a fixed chamber face, L_a admits a right inverse Q_a with*

$$\|Q_a\| \le C_\eta.$$

Moreover $a \mapsto Q_a$ may be chosen smooth on each face chart, with every fixed derivative bounded by a constant depending only on the derivative order and η.

Proof. On the neutral subspace, $\mathfrak{a}$ together with the neutral Levi directions gives a fixed surjective block. On each active restricted-root pair $\mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha}$, the $\mathfrak{h}$ and $\mathfrak{k}$ summands are graphs of the two commuting involutions defining the symmetric pair. In a basis adapted to the pair, L_a is represented by a 2×2 block whose entries are linear combinations of $e^{\alpha(Y)}$ and $e^{-\alpha(Y)}$. When $\alpha(Y) \ge \eta$, its smallest singular value is bounded below by $c_\eta > 0$. There are finitely many blocks, hence the direct-sum inverse is uniformly bounded. Differentiating the explicit inverse matrices gives the derivative bounds; denominators remain bounded away from zero on the relative η-interior. ☐

Theorem 2.2 (C^r quantitative wavefront factorization). *For every integer $r \ge 0$ and $\eta > 0$ there are $C_{r,\eta}$ and $\epsilon_{r,\eta} > 0$ with the following property. For each a in the relative η-interior of a fixed chamber face there is a C^r map*

$$\Theta_a : U_{\epsilon_{r,\eta}} \to (H \cap U_{C\epsilon}) \times (A \cap U_{C\epsilon}) \times (K \cap U_{C\epsilon}),$$

written $\Theta_a(u) = (h_a(u), b_a(u), k_a(u))$, such that

$$au = h_a(u)\, a\, b_a(u)\, k_a(u)$$

and

$$\|D^j \Theta_a\| \le C_{r,\eta} \qquad (0 \le j \le r)$$

uniformly in a.

Proof. Consider the map

$$F_a(h, b, k) = a^{-1} h a k.$$

Its differential at the identity triple is L_a. Lemma 2.1 provides a uniform right inverse. On a fixed identity chart, multiplication, inversion, and the exponential map have uniformly bounded derivatives of every fixed order. The parameter-dependent quantitative inverse function theorem therefore gives a right inverse of F_a on one common identity neighborhood. The C^r bounds depend on the C^r bounds for F_a and the inverse of its differential, both already uniform. Multiplying the identity $F_a(h, b, k) = u$ by a yields the displayed factorization. ☐

Proposition 2.3 (Jacobian stability)**.** *Let $J_a(u)$ be the Jacobian of the coordinate change in Theorem 2.2, normalized by $J_a(e) = 1$. Then*

$$|J_a(u) - 1| \leq C_\eta d(u, e), \qquad \|D^j J_a\| \leq C_{j,\eta}.$$

Proof. The Jacobian is a smooth polynomial-rational expression in the first derivatives of Θ_a in any fixed coordinate chart. Theorem 2.2 gives uniform C^{j+1} bounds. The mean value theorem and the normalization at the identity give the Lipschitz estimate. ☐

3. Anisotropic form

Suppose a_t is a one-parameter semisimple subgroup. Write

$$\mathfrak{g} = \mathfrak{g}^- \oplus \mathfrak{g}^0 \oplus \mathfrak{g}^+$$

for the negative, zero, and positive Lyapunov spaces of $\mathrm{Ad}(a_t)$.

Lemma 3.1 (Conjugation scales transverse directions)**.** *For every compact cone of regular a_t-directions there are $c, C > 0$ such that*

$$\|\mathrm{Ad}(a_t)Z^-\| \leq Ce^{-ct}\|Z^-\|, \qquad \|\mathrm{Ad}(a_{-t})Z^+\| \leq Ce^{-ct}\|Z^+\|.$$

Neutral directions have at most bounded exponential weight determined by the chosen face.

Proof. Each Lyapunov subspace is a sum of restricted-root spaces. On $\mathfrak{g}_\alpha$, $\mathrm{Ad}(a_t)$ acts by $e^{\alpha(\log a_t)}$. On a compact cone separated from the relevant walls, every negative root satisfies $\alpha(\log a_t) \leq -ct$ and every positive root satisfies the corresponding reversed inequality. Summing finitely many root spaces gives the result. ☐

Corollary 3.2 (Effective anisotropic wavefront)**.** *In a regular expanding direction, a transverse perturbation of size ϵ in the contracted directions produces an $O(e^{-ct}\epsilon)$ error after conjugation, while neutral perturbations retain size $O(\epsilon)$. All coordinate Jacobians remain uniformly C^r-controlled on a fixed relative chamber face.*

Proof. Combine Lemma 3.1 with Theorem 2.2 and Proposition 2.3. ☐

Failure mode. The relative-face hypothesis cannot simply be dropped. As a approaches a new chamber wall, an active root becomes neutral and the inverse of the corresponding 2×2 block can lose uniformity. Effective arguments must either subdivide into finitely many face regimes or retain an explicit wall-distance parameter.

CHAPTER 3

From Effective Mixing to Translate Equidistribution

1. The fixed-quotient spectral input

THEOREM 1.1 (Fixed-quotient effective mixing). *In the higher-rank type-A semisimple quotient fixed by this volume, there are a Sobolev order d and constants $C, \delta > 0$ such that, whenever one of F, f has zero mean,*

$$\left| \int_X F(x) f(x a_t)\, dm_X(x) \right| \leq C e^{-\delta t} \mathcal{S}_d(F) \mathcal{S}_d(f). \tag{7.3.1}$$

PROOF. Appendix B, Theorem 4.1, proves for every mean-zero quotient representation in this fixed higher-rank type-A group

$$|\langle R(g)F, f\rangle| \leq C e^{-\eta \|\mu(g)\|} \mathcal{S}_d(F) \mathcal{S}_d(f),$$

where $\mu(g)$ is the Cartan projection. For the fixed noncompact one-parameter group a_t, $\|\mu(a_t)\| \geq c|t|$. Taking $\delta = c\eta$ gives (7.3.1). Thus the pointwise spectral estimate has an internal proved input; it is not inferred merely from Property (T). $\square$

Proposition 1.2 (Abstract effective translate reduction)**.** *Suppose a singular orbit average admits a thickening with Sobolev cost ϵ^{-A}, orbit-to-tube error $W(t)\epsilon^{\tau}$, and boundary error ϵ^{κ}. Then effective mixing (7.3.1) implies the error budget of Lemma 2.1; optimizing ϵ gives Theorem 2.2.*

PROOF. Insert the thickened average between the singular average and Haar, apply Theorem 1.1 to the thickened density after subtracting its mean, and use the two geometric comparison bounds. The three terms are exactly those displayed in Lemma 2.1. The optimization is the elementary exponent calculation in Theorem 2.2. $\square$

We now isolate the exact error calculus in a form that can be reused uniformly.

2. The three-error decomposition

Lemma 2.1 (Spectral, geometric, and boundary errors)**.** *Let $I_t(f)$ be a singular translate average and let Φ_ϵ be a thickening density satisfying*

$$\mathcal{S}_d(\Phi_\epsilon) \leq C\epsilon^{-A}.$$

Assume:

(1) *effective mixing contributes* $E(t)\epsilon^{-A}\mathcal{S}_d(f)$*;*
(2) *replacing the orbit by the tube contributes at most* $W(t)\epsilon^{\tau}\mathcal{S}_{d+1}(f)$*;*
(3) *discarding a boundary strip contributes at most* $C\epsilon^{\kappa}\mathcal{S}_0(f)$*.*

Then

$$\left| I_t(f) - \int_X f\, dm_X \right| \ll (\epsilon^{\kappa} + \epsilon^{-A}E(t) + W(t)\epsilon^{\tau})\mathcal{S}_{d+1}(f).$$

Proof. Insert the thickened average between $I_t(f)$ and the ambient Haar average. The triangle inequality separates the boundary removal, orbit-to-tube replacement, and ambient mixing terms. Apply the three assumed estimates and enlarge the Sobolev order once so all terms use one norm. □

Theorem 2.2 (Optimized effective translate theorem). *Assume Lemma 2.1, with*

$$E(t) \le Ce^{-\delta t}, \qquad W(t) \le Ce^{\omega t}.$$

If $\omega < \delta\tau/A$ *after possibly replacing the available mixing exponent by a smaller positive one, then there exists* $\eta > 0$ *and a choice* $\epsilon = e^{-\sigma t}$ *such that*

$$\left| I_t(f) - \int_X f\, dm_X \right| \ll e^{-\eta t}\mathcal{S}_{d+1}(f).$$

When W *is bounded and* $\tau \ge \kappa$*, one may take*

$$\sigma = \frac{\delta}{A+\kappa}, \qquad \eta = \frac{\delta\kappa}{A+\kappa}$$

up to decreasing η *to absorb the wavefront term.*

Proof. Put $\epsilon = e^{-\sigma t}$. The three exponents become

$$\kappa\sigma, \qquad \delta - A\sigma, \qquad \tau\sigma - \omega.$$

Choose $\sigma > 0$ so all three are positive. This is possible precisely when the open interval

$$\frac{\omega}{\tau} < \sigma < \frac{\delta}{A}$$

is nonempty. Any η below the minimum of the three positive exponents works. If W is bounded, balance the first two exponents to obtain the stated value of σ; the third is no worse when $\tau \ge \kappa$. □

Proposition 2.3 (Polynomial mixing version)**.** *If instead* $E(t) \ll (1+t)^{-\delta}$ *and* $W(t) \ll (1+t)^{\omega}$*, then choosing* $\epsilon = (1+t)^{-\sigma}$ *gives a power saving whenever*

$$\frac{\omega}{\tau} < \sigma < \frac{\delta}{A}.$$

The final exponent is the minimum of $\kappa\sigma$*,* $\delta - A\sigma$*, and* $\tau\sigma - \omega$*.*

Proof. The proof is the same exponent comparison as in Theorem 2.2, with powers of $1 + t$ in place of exponentials. $\square$

3. Uniform families

Corollary 3.1 (Family-uniform translate theorem)**.** *Let the orbit, quotient, and translating element vary in a family for which the Sobolev order d, smoothing exponent A, boundary exponent κ, wavefront exponent τ, wavefront growth W, and mixing function E are uniform. Then the optimized exponent in Theorem 2.2 is uniform over the family.*

Proof. The proof of the theorem uses only those numerical data. A common admissible σ therefore gives a common η. $\square$

Warning 3.2. Uniformity of the geometric constants is not the same as uniformity of the spectral function $E(t)$. Chapter 7 proves the former on covering towers and isolates the latter as a separate strong-spectral obligation.

CHAPTER 4

Expanding Horospheres

1. Horospherical geometry

Let a_t be a semisimple one-parameter subgroup and define

$$U^+ = \{u \in G : a_{-t} u a_t \to e \text{ as } t \to +\infty\}.$$

Its Lie algebra is the sum of positive Lyapunov root spaces.

Lemma 1.1 (Uniform contraction on a compact unstable piece)**.** *For every compact $\Omega \subset U^+$ contained in an exponential coordinate chart there are $c, C > 0$ such that*

$$d(a_{-t} u a_t, e) \le C e^{-ct} d(u, e) \qquad (u \in \Omega, t \ge 0).$$

The same bound holds for every fixed derivative of the conjugation map on Ω.

Proof. In logarithmic coordinates, conjugation is $Z \mapsto \mathrm{Ad}(a_{-t})Z$. Each root occurring in $\mathfrak{u}^+$ has positive weight, hence is multiplied by at most e^{-ct} on a compact regular cone. The Baker–Campbell–Hausdorff map has uniformly bounded derivatives on the fixed compact chart, so the linear estimate transfers to the group map and its derivatives. $\square$

Remark 1.2 (Historical comparison)**.** The same thickening-and-mixing architecture appears in the Ratner companion project. It is not a proved input here: the theorem required by the active AHD chain is derived immediately below from the internal contraction, thickening, wavefront, and mixing estimates.

Theorem 1.3 (Horosphere theorem from a strong-spectral certificate). *Let $X = \Gamma\backslash G$, let $\Omega \subset U^+$ have piecewise C^1 boundary with boundary-strip exponent $\kappa > 0$, and suppose the right translation by a_t satisfies the effective mixing estimate of Chapter 3 with $E(t) \ll e^{-\delta t}$. Then for every smooth density ψ on Ω and $f \in C_c^\infty(X)$,*

$$\int_\Omega f(xua_t)\psi(u)\,du = \left(\int_\Omega \psi\right)\left(\int_X f\,dm_X\right) + O(e^{-\eta t}\mathcal{S}_D(f)\|\psi\|_{C^D})$$

for some D and $\eta > 0$ determined by the mixing, smoothing, and wavefront exponents.

Proof. Thicken $x\Omega$ in a complement to $\mathfrak{u}^+$. Lemma 1.1 supplies the favorable wavefront error in the contracted directions, and the smooth boundary of Ω supplies a positive strip exponent. The thickened density has Sobolev cost ϵ^{-A} by Chapter 1. Apply Lemma 2.1 and optimize by Theorem 2.2. $\square$

Worked Example 1.4 (Why the spectral certificate matters)**.** The geometric part of the proof is uniform on any family of local isometric covers: the same horospherical charts and the same root contraction estimates apply upstairs. What is not automatic is a common decay bound for the ambient correlations. A common averaging gap on the deck graphs does not by itself supply the pointwise a_t-matrix coefficient needed here. This is exactly the distinction isolated in AHD Volume 6 and Chapter 7 below.

Corollary 1.5 (Congruence-uniform horosphere transfer)**.** *For the higher-rank type-A congruence tower fixed in Chapter 7, Theorem 1.3 holds with constants independent of the level.*

Proof. The level-uniform statement is proved later, after the congruence-tower spectral input has been established: see Corollary 4.2 in Chapter 7. At this point the corollary is a forward statement, not an additional logical input. Its proof combines the horospherical certificate above with the level-uniform strong-spectral theorem reconstructed in Appendix B. $\square$

CHAPTER 5

Cusp Truncation and Height Losses

1. Why noncompactness enters twice

There are two independent cusp effects. First, the translated orbit may spend mass above a height cutoff. Second, even on a retained height window, the Sobolev cost of local coordinates grows polynomially with height. These losses must be balanced against each other rather than hidden in one constant.

Lemma 1.1 (Moment implies polynomial tail)**.** *Let ν be a probability measure on X and assume*

$$M_s(\nu) = \int_X \mathrm{ht}(x)^s \, d\nu(x) < \infty$$

for some $s > 0$. Then

$$\nu\{\mathrm{ht} > R\} \le R^{-s} M_s(\nu).$$

More generally, if $|f(x)| \le \mathrm{ht}(x)^q \|f\|_{\infty,q}$ and $s > q$, then

$$\int_{\{\mathrm{ht}>R\}} |f| \, d\nu \le R^{q-s} M_s(\nu) \|f\|_{\infty,q}.$$

Proof. On $\{\mathrm{ht} > R\}$ one has $1 \le R^{-s} \,\mathrm{ht}^s$, proving the first claim. For the second, $\mathrm{ht}^q \le R^{q-s} \,\mathrm{ht}^s$ on the same set. $\square$

Proposition 1.2 (Injectivity-height distortion)**.** *Let $\mathrm{ht}(x)$ be the reciprocal injectivity scale used in Volume 6. For every compact $C \subset G$ there is $C_1 \ge 1$ such that*

$$C_1^{-1} \,\mathrm{ht}(x) \le \mathrm{ht}(xg) \le C_1 \,\mathrm{ht}(x) \qquad (g \in C).$$

For a fixed one-parameter semisimple group a_t there are $C_2, c > 0$ such that

$$\mathrm{ht}(xa_t) \le C_2 e^{c|t|} \,\mathrm{ht}(x).$$

Proof. Volume 6, Lemma 1.3, gives a two-sided comparison of the injectivity radius after right translation in terms of $\max(\|\operatorname{Ad} g\|, \|\operatorname{Ad} g^{-1}\|)$. This quantity is bounded on a compact set, proving the first assertion. For $g = a_t$, the adjoint action is diagonal on the finite restricted-root decomposition, so its norm and the norm of its inverse are $O(e^{c_0|t|})$. Substitution gives the second bound after changing constants. $\square$

2. Truncated translate theorem

THEOREM 2.1 (Cusp-truncated effective translate ledger). *Suppose a singular translate average $I_t(f)$ has an s-moment bound*

$$\int \mathrm{ht}(ya_t)^s \, d\nu_t(y) \le M(t),$$

and, after restricting to height at most R, the thickening/mixing argument gives

$$|I_{t,\le R}(f) - m(f)| \ll \left(R^B \epsilon^{-A} E(t) + R^C \epsilon^\kappa + R^D W(t)\epsilon^\tau\right) \mathcal{S}_L(f).$$

If f has polynomial height growth $q < s$, then

$$|I_t(f) - m(f)| \ll \left(R^{q-s} M(t) + R^B \epsilon^{-A} E(t) + R^C \epsilon^\kappa + R^D W(t)\epsilon^\tau\right)\mathcal{S}_{L,q}(f).$$

PROOF. Split both the orbit average and the Haar average at height R. Lemma 1.1 bounds the discarded orbit tail; the corresponding Haar tail is treated by its fixed finite height moment or absorbed into the same exponent after increasing $M(t)$ by a constant. Apply the assumed thickening/mixing estimate on the retained window. The polynomial powers of R record the chart and Sobolev losses from Chapters 1–3. Add the four contributions. □

Proposition 2.2 (Joint optimization of height and smoothing). *Assume $M(t) \ll e^{\mu t}$, $E(t) \ll e^{-\delta t}$, $W(t) \ll e^{\omega t}$, and choose*

$$R = e^{rt}, \qquad \epsilon = e^{-\sigma t}.$$

Then the four terms in Theorem 2.1 decay exponentially provided

$$(s-q)r > \mu,$$

$$\delta > Br + A\sigma, \qquad \kappa\sigma > Cr, \qquad \tau\sigma > Dr + \omega.$$

Whenever these strict inequalities have a simultaneous solution (r, σ), the translate average has an exponential error $O(e^{-\eta t})$ for some $\eta > 0$.

PROOF. Substitute the exponential choices into each term. Their decay exponents are respectively

$$(s-q)r - \mu, \quad \delta - Br - A\sigma, \quad \kappa\sigma - Cr, \quad \tau\sigma - Dr - \omega.$$

The minimum is positive exactly under the displayed strict inequalities. □

Corollary 2.3 (Moving closed orbits). *Let $Y_i = x_i H_i$ be closed finite-volume orbits with complexity Q_i, and suppose the height moment and local-chart constants are bounded by fixed powers of Q_i. Then the effective translate theorem remains uniform as long as $\log Q_i = o(t_i)$ with a coefficient small enough to satisfy the inequalities of Proposition 2.2.*

Proof. Write each factor Q_i^C as $e^{C \log Q_i}$ and absorb it into the corresponding exponent budget. The strict inequalities leave a positive margin when $\log Q_i/t_i$ is sufficiently small. □

Failure mode. A bound on total orbit volume alone does not imply a useful height moment. Closed homogeneous orbits can place a small but dynamically significant portion of their mass high in the cusp. Later semisimple-period volumes therefore keep volume, height, and discriminant as distinct complexity parameters.

CHAPTER 6

Boundary Smoothing and Sobolev Error Budgets

1. Smooth approximation with explicit cost

Proposition 1.1 (Smooth approximation of a regular set)**.** *Let B be a measurable set whose ϵ-boundary neighborhood lies in a region of bounded injectivity height. There are $f_\epsilon^-, f_\epsilon^+ \in C_c^\infty(X)$ such that*

$$0 \leq f_\epsilon^- \leq 1_B \leq f_\epsilon^+ \leq 1,$$

the difference $f_\epsilon^+ - f_\epsilon^-$ is supported in the $O(\epsilon)$-boundary neighborhood, and, for each fixed Sobolev order d,

$$\mathcal{S}_d(f_\epsilon^\pm) \ll \epsilon^{-A_d}\mathcal{H}_d(B),$$

where $\mathcal{H}_d(B)$ is the injectivity-height factor fixed in Volume 6.

Proof. Use the local convolution construction of Proposition 4.1. In an injective exponential chart, convolve 1_B with $\rho_\epsilon(Z) = \epsilon^{-\dim G}\rho(Z/\epsilon)$. The convolution changes the indicator only within $O(\epsilon)$ of the boundary, and every derivative of order j costs ϵ^{-j}. A bounded-overlap partition of unity and the weighted Sobolev product estimate of Volume 6 give the stated height factor. Taking inner and outer envelopes yields the pointwise inequalities. ☐

Corollary 1.2 (Counting-normalized bump cost)**.** *In the normalization used by the present counting arguments, smoothing to scale ϵ costs only a fixed power ϵ^{-A_d} in Sobolev order d.*

Proof. Apply Proposition 1.1 in each local chart and sum a bounded-overlap partition. ☐

2. One master error balance

Theorem 2.1 (Master boundary–spectral–wavefront budget). *Let B_T be a family of measurable regions with*

$$m(B_{T,\epsilon}^+ \setminus B_{T,\epsilon}^-) \leq C\epsilon^\kappa m(B_T).$$

Assume the smoothed counting or translate expression has spectral error at most

$$C\epsilon^{-A}E(T)m(B_T)$$

and wavefront error at most

$$CW(T)\epsilon^{\tau}m(B_T).$$

Then the unsmoothed normalized discrepancy satisfies

$$\left|\frac{N(T)}{c\,m(B_T)}-1\right| \ll \epsilon^{\kappa}+\epsilon^{-A}E(T)+W(T)\epsilon^{\tau},$$

with the analogous statement for normalized orbit averages.

Proof. Use the inner and outer smooth approximations supplied by Proposition 1.1. Their integrals squeeze the sharp count or average. The difference between their main terms is the boundary-strip measure, the difference between their dynamical values and main terms is the spectral error, and wavefront replacement supplies the third term. Divide by the common leading volume. □

Proposition 2.2 (Closed-form optimizer when the wavefront term is harmless)**.** *If $W(T) \ll 1$, $\tau \geq \kappa$, and $E(T) \to 0$, choose*

$$\epsilon = E(T)^{1/(A+\kappa)}.$$

Then

$$\epsilon^{\kappa}+\epsilon^{-A}E(T)+W(T)\epsilon^{\tau} \ll E(T)^{\kappa/(A+\kappa)}.$$

Proof. The first two terms are equal for the displayed choice. Since $\tau \geq \kappa$ and $0<\epsilon<1$, the wavefront term is no larger than a constant multiple of the boundary term. □

Proposition 2.3 (General three-line optimizer)**.** *Suppose*

$$E(T)=T^{-\delta}, \qquad W(T)=T^{\omega},$$

and set $\epsilon = T^{-\sigma}$. The best exponent obtainable from Theorem 2.1 is the maximum over $\sigma>0$ of

$$\eta(\sigma)=\min\{\kappa\sigma,\ \delta-A\sigma,\ \tau\sigma-\omega\}.$$

In particular a power saving exists iff the interval

$$\omega/\tau<\sigma<\delta/A$$

is nonempty.

Proof. Substitution converts the three error terms to $T^{-\kappa\sigma}$, $T^{-(\delta-A\sigma)}$, and $T^{-(\tau\sigma-\omega)}$. Their sum is controlled by the smallest decay exponent. Positivity of all three exponents is exactly the interval condition. □

Corollary 2.4 (Indicator and sector transfer)**.** *Any effective equidistribution statement for smooth test functions with Sobolev loss ϵ^{-A} transfers to indicators of regions with boundary exponent κ, with the loss prescribed by Proposition 2.3. If the region additionally has a compact angular coordinate with a quantitatively regular boundary, the same argument applies after adding its boundary strip to the common exponent κ.*

Proof. Smooth simultaneously in all boundary coordinates. The union of the transition strips has measure bounded by the sum of the individual strip measures, hence by a positive power of the common smoothing radius. Apply Theorem 2.1. □

Worked Example 2.5 (Derivative loss reduces a spectral exponent)**.** If the raw correlation error is $T^{-1/2}$, the smoothing cost is ϵ^{-6}, and the boundary strip has size $O(\epsilon)$, then balancing gives $\epsilon = T^{-1/14}$ and a final error $T^{-1/14}$. Nothing is wrong with the spectral theorem: six derivatives were spent converting a sharp geometric region into a smooth observable.

CHAPTER 7

Uniformity in Congruence Towers

Proof architecture. A genuinely level-uniform theorem requires more than a fixed-quotient estimate. We fix the arithmetic model treated quantitatively in Volume 6:

$$G = \mathrm{SL}_n(\mathbb{R}), \quad n \geq 3, \qquad \Gamma = \mathrm{SL}_n(\mathbb{Z}), \qquad \Gamma(q) = \ker\bigl(\mathrm{SL}_n(\mathbb{Z}) \to \mathrm{SL}_n(\mathbb{Z}/q\mathbb{Z})\bigr).$$

Local geometry is automatically uniform on covers. The nontrivial point is continuous-time spectral decay. Appendix B reconstructs a representation-independent higher-rank matrix-coefficient estimate, following the type-A part of Hee Oh's argument. Combining that theorem with Chapters 1–6 closes the congruence-tower obligations without pretending that Property (τ) alone implies pointwise decay.

1. Geometry is uniform on covers

Set $X(q) = \Gamma(q)\backslash G$.

Proposition 1.1 (Uniform local geometry on covers)**.** *Let $z_q \in X(q)$ map to $z \in X(1)$. Then*

$$\mathrm{inj}_{X(q)}(z_q) \geq \mathrm{inj}_{X(1)}(z).$$

Consequently every product-chart, local Sobolev, smoothing, and wavefront estimate whose constant depends only on a lower injectivity-radius bound is uniform over all lifts of a fixed base height window.

Proof. A collision in the ball of radius r around z_q is produced by a nontrivial element of the stabilizer subgroup $g^{-1}\Gamma(q)g$, which is contained in $g^{-1}\Gamma g$. Thus the shortest stabilizer displacement upstairs is no smaller than downstairs. All local analytic statements in Chapters 1–6 depend on the quotient only through such a lower bound and the fixed Lie-group coordinate charts. $\square$

2. The strong-spectral certificate

AHD Volume 6 proves uniform Markov gaps on the finite congruence quotients. Those gaps are useful arithmetic information, but the translate argument needs the continuous G-coefficient $\langle R_q(g)f_1, f_2\rangle$ as g escapes in G.

Theorem 2.1 (Uniform strong spectral theorem for the type-A tower). *There are an integer D and constants $C, \eta > 0$, depending only on n and on the fixed Sobolev normalization, such that for every $q \geq 1$, every $f_1, f_2 \in C^\infty(X(q))$ of mean zero, and every Cartan decomposition*

$$g = k_1 \exp X k_2, \qquad X \in \overline{\mathfrak{a}^+},$$

one has

$$|\langle R_q(g) f_1, f_2 \rangle| \leq C e^{-\eta \|X\|} \mathcal{S}_D(f_1) \mathcal{S}_D(f_2). \tag{7.1}$$

The constants are independent of the congruence level q.

Proof. The right regular representation on $L^2_0(X(q))$ has no G-invariant vectors. Appendix B, Corollary 4.2, applies to every lattice in G with constants depending only on G. Taking $\Lambda = \Gamma(q)$ gives (7.1) simultaneously for all q. $\square$

Remark 2.2 (What has and has not been proved)**.** Theorem 2.1 is a theorem for the higher-rank type-A arithmetic family fixed in Volume 6. It is not being advertised as a rank-one automorphic theorem. Rank-one congruence towers require their own automorphic or transfer-operator spectral input and belong to the later volume in which that arithmetic family is actually used.

3. Uniform effective mixing

Let a_t be a fixed noncompact one-parameter subgroup conjugated into the closed positive chamber for $t \geq 0$. Since G has finite center, $\|\log a_t\| \asymp t$.

Theorem 3.1 (Uniform effective mixing on the congruence tower). *For the tower $X(q)$ and every fixed noncompact semisimple one-parameter subgroup a_t, there are $C, \delta > 0$ and an integer D, independent of q, such that*

$$\left| \int_{X(q)} f_1(x a_t) f_2(x) \, dm_q(x) - \int f_1 \, dm_q \int f_2 \, dm_q \right| \leq C e^{-\delta t} \mathcal{S}_D(f_1) \mathcal{S}_D(f_2). \tag{7.2}$$

Proof. Subtract the means and apply Theorem 2.1 to the two mean-zero parts. The Cartan projection of a_t has norm bounded below by a positive multiple of t; absorb that constant into δ. Constants remain level-independent. $\square$

4. Uniform translates

We now feed (7.2) into the abstract thickening theorem rather than repeat its proof.

Theorem 4.1 (Uniform effective translate theorem on a height window). *Fix the homogeneous subgroup/orbit geometry of Chapters 1–3 and a base height window on* $X(1)$. *Let* $Y_q \subset X(q)$ *be any lift of the corresponding closed orbit piece, with normalized smooth density whose geometric complexity is bounded independently of* q. *Then there are* $D, C, \delta' > 0$, *independent of* q, *such that*

$$\left| \int_{Y_q} f(ya_t)\, d\mu_{Y_q}(y) - \int_{X(q)} f\, dm_q \right| \le Ce^{-\delta' t} \mathcal{S}_D(f) \tag{7.3}$$

for $f \in C_c^\infty(X(q))$. *If the orbit complexity is allowed to vary, the same statement holds with the explicit polynomial complexity factor recorded in Chapter 5, but still with no hidden dependence on* q.

Proof. Proposition 1.1 makes the product charts, Jacobians, bump functions, and wavefront constants in Chapters 1–2 uniform in q on the chosen height window. Theorem 3.1 supplies a common exponential mixing envelope. Insert it into the three-error decomposition of Chapter 3 and choose the smoothing radius by Theorem 2.2. Chapter 5 keeps the height/complexity loss separate from the level. The resulting exponent and Sobolev order therefore depend on the fixed geometric data and on G, but not on q. □

Corollary 4.2 (Congruence-uniform expanding horospheres)**.** *In the setting of Chapter 4, suppose the compact horospherical piece and its smooth density have uniformly bounded geometry when lifted to* $X(q)$. *Then the effective horospherical equidistribution theorem holds with a common exponential rate and common Sobolev order for all* q.

Proof. Apply Theorem 1.3 with the common mixing estimate (7.2). Root contraction takes place in the fixed group G and Proposition 1.1 supplies the level-uniform local charts. □

Corollary 4.3 (Counting-ready boundary-smoothed transfer)**.** *Let* B_T *be a family of regions in the homogeneous parameter space satisfying the boundary-strip hypothesis of Chapter 6 with constants independent of* q. *Periodize and thicken its indicator as in Chapter 6. Then the translate estimate (7.3) yields a level-uniform boundary-smoothed counting estimate; every loss is an explicit function of the geometric/orbit complexity and* T, *and none is an unrecorded function of* q.

Proof. Use Proposition 2.1 with the common spectral saving from Theorem 3.1; the smoothing and wavefront exponents are common by Proposition 1.1. Optimize the smoothing radius with Appendix A. This is precisely the transfer form required at the entrance to Volume 8. □

5. Dependency summary

V07.C07.L: Proposition 1.1 owns the geometric uniformity, while Appendix B reconstructs the higher-rank root-tempering argument and Theorem 2.1 is the resulting level-uniform continuous spectral estimate.

V07.C07.T: Theorems 3.1 and 4.1 give uniform effective mixing and translate equidistribution on the principal congruence tower of $\mathrm{SL}_n(\mathbb{Z})$, $n \geq 3$.

V07.C07.P: Corollaries 4.2 and 4.3 are the downstream horosphere and counting-ready transfer forms.

6. Reader perspective

The chapter keeps three assertions separate: local geometry on covers, continuous pointwise spectral decay, and the final thickening transfer. This prevents the common but false shortcut "Property (τ) implies effective mixing." Appendix B shows exactly why the higher-rank type-A family has the stronger representation-theoretic estimate needed here.

CHAPTER 8

An Effective Shah Interface without Overclaiming

1. Qualitative support selection is part of the data

This chapter is an abstract quantitative interface, not a second proof of the polynomial-trajectory classification developed later in Volume 42. No Shah theorem is imported here. The data package below includes the homogeneous support selected by the non-obstructed qualitative branch; Volume 42 later verifies that support selection internally in the arithmetic polynomial-trajectory setting.

The purpose here is not to turn those qualitative statements into a fictitious universal rate. We instead state a reusable conditional theorem whose hypotheses identify every quantitative input that later effective-Shah volumes must verify.

2. Quantitative data package

Definition 2.1 (Effective Shah data)**.** For a family of translated curves or submanifolds μ_t, an effective Shah data package consists of positive functions

$$E_{\mathrm{mix}}(t), \quad E_{\mathrm{nd}}(t), \quad E_{\mathrm{foc}}(t), \quad E_{\mathrm{Tay}}(t)$$

and uniform exponents $A, \kappa, \tau > 0$ such that:

(1) outside a set of parameter measure $O(E_{\mathrm{nd}}(t))$, the trajectory remains in a height window where the product charts of Chapter 1 apply;
(2) on every regular chart, thickening has Sobolev cost ϵ^{-A} and boundary loss $O(\epsilon^{\kappa})$;
(3) ambient correlations have error $E_{\mathrm{mix}}(t)$;
(4) failure of the regular product model is contained in finitely many algebraic focusing tubes of total parameter measure $O(E_{\mathrm{foc}}(t))$;
(5) Taylor replacement of the curve by its finite jet at the rescaled window costs $E_{\mathrm{Tay}}(t, \epsilon)$;
(6) a specified closed finite-volume homogeneous support Y_t is attached to the non-obstructed branch, and every regular chart remaining after focusing removal is compared with m_{Y_t}.

Lemma 2.2 (Finite obstruction accounting)**.** *Suppose the focusing alternatives are indexed by at most M algebraic strata and each bad stratum occupies parameter measure at most $e(t)$. Then the total focusing loss is at most $Me(t)$. If strata are nested and the argument descends strictly in dimension whenever a tube captures the trajectory, then at most* $\dim G$ *descent stages occur.*

Proof. The first statement is the union bound. For the second, every genuine descent replaces the ambient connected algebraic support by a proper connected subgroup, hence lowers its dimension by at least one. Dimension cannot decrease more than $\dim G$ times. □

Theorem 2.3 (Abstract effective Shah interface). *Assume an effective Shah data package as in Definition 2.1. Assume further that every proper focusing stratum is either excluded by a quantitative nondegeneracy hypothesis or is declared as an explicit alternative in the theorem statement. Then for every smooth test function f,*

$$|\mu_t(f) - m_{Y_t}(f)| \ll \Big(E_{\mathrm{nd}}(t) + E_{\mathrm{foc}}(t) + \epsilon^{\kappa} + \epsilon^{-A}E_{\mathrm{mix}}(t) + E_{\mathrm{Tay}}(t,\epsilon)\Big)\mathcal{S}_D(f),$$

where m_{Y_t} is Haar probability on the homogeneous support selected by the non-obstructed branch. The estimate is valid uniformly over any family for which the data-package constants are uniform.

Proof. Discard the nondivergent exceptional set, paying $E_{\mathrm{nd}}(t)\|f\|_\infty$. Remove the finite union of focusing tubes, paying $E_{\mathrm{foc}}(t)\|f\|_\infty$ by Lemma 2.2. On every remaining parameter chart, replace the curve by its controlled jet and pay $E_{\mathrm{Tay}}(t,\epsilon)\mathcal{S}_D(f)$. The regular jet piece admits the thickening construction of Chapter 1. Boundary removal costs ϵ^{κ}, and ambient mixing costs $\epsilon^{-A}E_{\mathrm{mix}}(t)$. Sum over the bounded-overlap chart decomposition. The selected homogeneous support is part of the data package, so this estimate uses only the quantitative bounds listed in Definition 2.1. □

Corollary 2.4 (Power-law effective Shah criterion)**.** *Suppose*

$$E_{\mathrm{nd}}(t) + E_{\mathrm{foc}}(t) \ll t^{-a}, \qquad E_{\mathrm{mix}}(t) \ll t^{-b},$$

and

$$E_{\mathrm{Tay}}(t, t^{-\sigma}) \ll t^{-c(\sigma)}$$

for some positive continuous function $c(\sigma)$ on an interval containing a point $0 < \sigma < b/A$. Then the non-obstructed branch has a polynomial equidistribution rate $t^{-\eta}$ for some $\eta > 0$.

Proof. Choose $\epsilon = t^{-\sigma}$. The smoothing and spectral errors have exponents $\kappa\sigma$ and $b - A\sigma$. Choose σ in the stated interval with $c(\sigma) > 0$, then take η below the minimum of a, $\kappa\sigma$, $b - A\sigma$, and $c(\sigma)$. □

Warning 2.5. Theorem 2.3 is an interface theorem, not a general effective Ratner or effective Shah theorem. The difficult work in later Volumes 27–34 is precisely to prove the quantitative nondivergence, focusing, closing, projection-growth, and algebraic-generation estimates that make the data package available in specific groups.

APPENDIX A

A Small Error-Optimization Toolkit

Many later effective arguments reduce to choosing scales in a finite family of monomial errors. We record the elementary principle once.

Lemma 0.1 (Finite monomial optimizer)**.** *Let*

$$E(T, \sigma_1, \ldots, \sigma_m) \ll \sum_{j=1}^{N} T^{-\ell_j(\sigma_1, \ldots, \sigma_m)},$$

where every ℓ_j is affine linear. If there is a point σ in the admissible open polytope for which every $\ell_j(\sigma) > 0$, then there is a rational point σ' in the same polytope and $\eta > 0$ such that $E(T, \sigma') \ll T^{-\eta}$.

Proof. The set on which all $\ell_j > 0$ is open. Rational points are dense, so choose a rational σ' in it. The minimum of finitely many positive numbers $\ell_j(\sigma')$ is a positive η. $\square$

Remark 0.2. This trivial lemma is surprisingly useful. It separates the conceptual proof—showing that the strict inequalities are compatible—from unnecessary attempts to optimize every final exponent sharply.

APPENDIX B

Uniform Matrix-Coefficient Decay for the Higher-Rank Congruence Model

Proof architecture. The congruence-tower argument in Chapter 7 needs a pointwise estimate for the continuous G-action, uniform in the level. Property (τ) alone does not give such an estimate. For the arithmetic model fixed in AHD Volume 6,

$$G = \mathrm{SL}_n(\mathbb{R}), \qquad n \geq 3,$$

we reconstruct the part of Hee Oh's higher-rank argument that is actually needed. The proof has four steps. First a representation of $\mathrm{SL}_2(\mathbb{R}) \ltimes \mathbb{R}^2$ without $\mathbb{R}^2$-fixed vectors restricts temperedly to $\mathrm{SL}_2(\mathbb{R})$. Second every root SL_2 in SL_n is contained in such a semidirect product. Third a single highest-root subgroup already controls escape in the closed positive Weyl chamber. Finally compact-K Sobolev summation removes the K-finite hypothesis. Every constant is a constant of G, not of the lattice.

1. A transitive spectral model for $\mathrm{SL}_2 \ltimes \mathbb{R}^2$

Put $H = \mathrm{SL}_2(\mathbb{R})$, $N = \mathbb{R}^2$, and $P = H \ltimes N$. We identify $\widehat{N}$ with $\mathbb{R}^2$ by the pairing $\chi_\xi(x) = e^{i\langle \xi, x\rangle}$.

Lemma 1.1 (Special transitive imprimitivity model)**.** *Let π be a strongly continuous unitary representation of P and suppose $\mathcal{H}^N = 0$. Then, as a representation of H, $\pi|_H$ is a direct integral of representations induced from the stabilizer in H of a nonzero vector of $\widehat{N}$. That stabilizer is conjugate to*

$$L = \left\{ \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} : t \in \mathbb{R} \right\}.$$

Proof. The spectral theorem for the abelian group N, already placed on the functional-analytic proof floor in AHD Volume 6, gives a projection-valued measure E on $\widehat{N}$ with

$$\pi(x) = \int_{\widehat{N}} \chi_\xi(x)\, dE(\xi).$$

The covariance calculation of Volume 6, Appendix A, gives

$$\pi(h)E(B)\pi(h)^{-1} = E(hB). \tag{C.1}$$

The N-fixed space is $E(\{0\})\mathcal{H}$, so the hypothesis says that the spectral measure is supported on $\widehat{N} \setminus \{0\}$. The contragredient action of H is transitive on this set. Fix $\xi_0 = (1, 0)$ and let $L = \mathrm{Stab}_H(\xi_0)$; after transposing the chosen coordinates this is the displayed one-parameter unipotent group.

For completeness we spell out the usual transitive-system reduction. Choose a Borel section $s : H/L \to H$. Disintegrate the spectral Hilbert space over the orbit H/L. Equation (C.1) transports the fiber over ξ unitarily to the fiber over $h\xi$. After transporting every fiber back to the fiber $\mathcal{K}$ over ξ_0 with $s(\xi)$, the H-action has the form

$$(\Pi(h)F)(\xi) = \sigma\Big(s(\xi)^{-1} h\, s(h^{-1}\xi)\Big)\, F(h^{-1}\xi), \tag{C.2}$$

where the element inside σ belongs to L. The cocycle identity follows by multiplying the three section factors, and its restriction at ξ_0 is a unitary representation σ of L on $\mathcal{K}$. Formula (C.2), with the invariant measure on H/L (both H and L are unimodular), is precisely the induced representation $\mathrm{Ind}_L^H \sigma$.

If the original spectral measure has multiplicity, the same construction is made on each measurable multiplicity component; equivalently one obtains a direct integral of such induced representations. No orbit other than $H/L = \widehat{N} \setminus \{0\}$ occurs because the zero orbit was removed by $\mathcal{H}^N = 0$. □

Lemma 1.2 (The unipotent stabilizer is weakly regular)**.** *Every unitary representation of $L \cong (\mathbb{R}, +)$ is weakly contained in the regular representation λ_L.*

Proof. By the spectral theorem it is enough to treat a character $\chi_\lambda(t) = e^{i\lambda t}$. In $L^2(\mathbb{R})$ set

$$f_T(x) = T^{-1/2}\mathbf{1}_{[-T/2,T/2]}(x)e^{-i\lambda x}.$$

Then $\|f_T\|_2 = 1$ and a direct overlap calculation gives

$$\langle \lambda_L(t) f_T, f_T\rangle = e^{i\lambda t}\left(1 - \frac{|t|}{T}\right)_+ .$$

This converges to $\chi_\lambda(t)$ uniformly on every compact set. Thus every character is weakly contained in λ_L, and finite convex combinations and direct integrals preserve weak containment by the coefficient definition proved in AHD Volume 5. □

Lemma 1.3 (Induction preserves this weak containment)**.** *If $\sigma \prec \lambda_L$, then*

$$\mathrm{Ind}_L^H \sigma \prec \mathrm{Ind}_L^H \lambda_L \cong \lambda_H.$$

Proof. Work first with compactly supported continuous vectors in the induced model over H/L. A matrix coefficient of $\mathrm{Ind}_L^H \sigma$ is an integral, over a compact subset of H/L, of finitely many σ-coefficients evaluated on the section cocycle appearing in (C.2). For a fixed compact set of H, those cocycle values range in a compact subset of L. Uniform-on-compact coefficient approximation for $\sigma \prec \lambda_L$, followed by a finite partition of the compact support, therefore approximates the induced coefficient uniformly on the chosen compact subset of H. Density gives the assertion for arbitrary induced vectors. This is exactly coefficient weak containment.

For the final identification, use the induced model with fiber $L^2(L)$. The map

$$F \longmapsto \big[h \mapsto F(hL)(\ell(h))\big]$$

obtained from the measurable factorization $h = s(hL)\ell(h)$ is unitary, by Weil's integration formula, from $\mathrm{Ind}_L^H \lambda_L$ onto $L^2(H)$; under this map the H-action becomes left translation. Hence $\mathrm{Ind}_L^H \lambda_L \cong \lambda_H$. □

Proposition 1.4 (The semidirect-product tempering lemma). *Let π be a unitary representation of $\mathrm{SL}_2(\mathbb{R}) \ltimes \mathbb{R}^2$ with no nonzero $\mathbb{R}^2$-fixed vector. Then $\pi|_{\mathrm{SL}_2(\mathbb{R})}$ is tempered.*

Proof. Lemma 1.1 writes the restriction as a direct integral of $\mathrm{Ind}_L^H \sigma$. Lemmas 1.2 and 1.3 show that each summand, and hence the direct integral, is weakly contained in λ_H. AHD Volume 5 proves that weak containment in the regular representation is equivalent to temperedness and then gives the Cowling–Haagerup–Howe pointwise bound. □

2. Every root SL_2 in SL_n is tempered

For distinct i, j, let $H_{ij} < \mathrm{SL}_n(\mathbb{R})$ be the standard root subgroup acting as SL_2 on the coordinate plane $\langle e_i, e_j\rangle$ and trivially on the other coordinate lines.

Proposition 2.1 (Root-tempered restriction for SL_n). *Let $n \geq 3$ and let π be a unitary representation of $G = \mathrm{SL}_n(\mathbb{R})$ with no nonzero G-invariant vector. Then $\pi|_{H_{ij}}$ is tempered for every root subgroup H_{ij}.*

Proof. Choose $k \notin \{i, j\}$ and let

$$N_{ijk} = \{I + xE_{ik} + yE_{jk} : x, y \in \mathbb{R}\} \cong \mathbb{R}^2.$$

The group H_{ij} normalizes N_{ijk}, and conjugation is the standard two-dimensional representation. Hence $H_{ij}N_{ijk} \cong \mathrm{SL}_2(\mathbb{R}) \ltimes \mathbb{R}^2$.

We claim that π has no nonzero N_{ijk}-fixed vector. If $0 \neq v$ were fixed, choose a sequence $u_m \in N_{ijk}$ escaping every compact set. AHD Volume 5, Howe–Moore, applies to π because G is connected almost simple and π has no invariant vector; therefore $\langle \pi(u_m)v, v\rangle \to 0$. But v is fixed, so the same coefficient equals $\|v\|^2$, a contradiction. Proposition 1.4 applied to $H_{ij}N_{ijk}$ now proves the claim. $\square$

Proof and provenance. Proposition 2.1 is the type-A portion of Oh's restriction theorem: for a connected almost simple group of semisimple rank at least two, the restriction to each root SL_2 is tempered. In type A the abelian semidirect-product case suffices; the oscillator-representation cases needed for some other root systems do not enter the present SL_n proof. This is why we reconstruct the needed argument instead of citing the full classification-by-root-system theorem.

3. A single highest root controls the whole chamber

Let $K = \mathrm{SO}(n)$ and

$$A^+ = \{\mathrm{diag}(e^{x_1}, \dots, e^{x_n}) : x_1 \geq \cdots \geq x_n,\ \sum_i x_i = 0\}.$$

Write $H_\gamma = H_{1n}$ for the highest-root SL_2.

Lemma 3.1 (Highest-root chamber control). *If $X = (x_1, \dots, x_n)$ is in the closed positive chamber and $\sum x_i = 0$, then*

$$x_1 - x_n \geq n^{-1/2}\|X\|_2.$$

Moreover, for $a = \mathrm{diag}(e^{x_1}, \dots, e^{x_n})$ there are commuting elements $h \in H_\gamma$ and $c \in Z_G(H_\gamma)$ such that $a = hc$ and

$$h = \mathrm{diag}(e^s, 1, \dots, 1, e^{-s}), \qquad 2s = x_1 - x_n.$$

Proof. Every x_i lies in the interval $[x_n, x_1]$. Since 0 lies in that interval (the coordinates have zero sum), $|x_i| \leq x_1 - x_n$, whence $\|X\|_2 \leq \sqrt{n}(x_1 - x_n)$. Put $s = (x_1 - x_n)/2$ and define h as above. Then $c = h^{-1}a$ has equal first and last diagonal entries, so it commutes with both root groups E_{1n}, E_{n1} and hence with H_γ. $\square$

Theorem 3.2 (Uniform K-finite matrix-coefficient bound for SL_n). *For each $n \geq 3$ there are $C, c > 0$ and an integer D_0 depending only on n with the following property. Let π be any unitary representation of $G = \mathrm{SL}_n(\mathbb{R})$ without nonzero invariant vectors. For K-finite vectors v, w and $g = k_1 \exp X k_2$ with $X \in \overline{\mathfrak{a}^+}$,*

$$|\langle \pi(g)v, w\rangle| \leq C\sqrt{d_v d_w}\,(1 + \|X\|)^{D_0} e^{-c\|X\|}\|v\|\|w\|, \tag{C.3}$$

where $d_v = \dim \mathrm{span}_K(Kv)$ and similarly for w. The constants are representation-independent.

Proof. Replace v, w by $\pi(k_2)v$ and $\pi(k_1^{-1})w$; their K-cyclic dimensions are unchanged. Write $a = \exp X = hc$ as in Lemma 3.1. Since c centralizes H_γ, it also centralizes $K_\gamma = K \cap H_\gamma$, so the K_γ-cyclic dimension of $\pi(c)v$ is at most d_v.

By Proposition 2.1, $\pi|_{H_\gamma}$ is tempered. The CHH theorem of AHD Volume 5 therefore gives

$$|\langle \pi(h)\pi(c)v, w\rangle| \le \sqrt{d_v d_w}\, \Xi_{H_\gamma}(h)\|v\|\|w\|.$$

AHD Volume 5 also proves the Harish–Chandra chamber estimate. In rank one it gives

$$\Xi_{H_\gamma}(h) \le C_1(1+|s|)^{D_0} e^{-|s|},$$

up to a harmless normalization of the split parameter. Lemma 3.1 has $|s| = (x_1 - x_n)/2 \ge (2\sqrt{n})^{-1}\|X\|$, which proves (C.3), after adjusting C, c. □

Proof and provenance. Hee Oh proves a sharper theorem: for groups of semisimple rank at least two, products of rank-one Harish–Chandra functions attached to a strongly orthogonal root system give a representation-independent bound. The present theorem deliberately proves only the type-A, single-highest-root consequence needed for this series. It is weaker but sufficient, and every step needed for it is owned above or in AHD Volume 5.

4. From K-finite vectors to Sobolev vectors

Let Ω_K be the nonnegative Casimir of K. If $f = \sum_\tau f_\tau$ is its K-type decomposition, the compact Weyl law implies that for some integer $r = r(K)$,

$$C_{K,r} := \sum_{\tau \in \widehat{K}} (\dim \tau)(1+\lambda_\tau)^{-r} < \infty. \tag{C.4}$$

This is the standard compact elliptic trace estimate; AHD Volume 6's Sobolev norm of sufficiently high order controls $(1+\Omega_K)^{r/2}$.

Theorem 4.1 (Uniform smooth matrix-coefficient decay). *Let $G = \mathrm{SL}_n(\mathbb{R})$, $n \ge 3$. There are an integer D and constants $C, \eta > 0$, depending only on G and the fixed Sobolev conventions, such that for every unitary representation π of G without invariant vectors and all smooth vectors f_1, f_2,*

$$|\langle \pi(k_1 e^X k_2) f_1, f_2\rangle| \le C e^{-\eta\|X\|} \mathcal{S}_D(f_1)\mathcal{S}_D(f_2) \tag{C.5}$$

for $X \in \overline{\mathfrak{a}^+}$ and $k_1, k_2 \in K$.

Proof. Apply Theorem 3.2 to pairs of K-type components. Since the K-cyclic dimension of a vector in an irreducible K-type τ is at most $\dim\tau$,

$$|\langle\pi(g)f_1, f_2\rangle| \le C(1+\|X\|)^{D_0} e^{-c\|X\|} A(f_1)A(f_2),$$
$$A(f) := \sum_\tau \sqrt{\dim\tau}\,\|f_\tau\|.$$

Insert $(1+\lambda_\tau)^{-r/2}(1+\lambda_\tau)^{r/2}$ in the first sum and use Cauchy–Schwarz together with (C.4); do the same for the second. The resulting weighted L^2 norm is controlled by $\mathcal{S}_D$ once D is chosen above the compact elliptic order. Finally absorb the polynomial $(1+\|X\|)^{D_0}$ into $e^{(c/2)\|X\|}$ and take $\eta = c/2$. □

Corollary 4.2 (Uniformity over all lattice quotients)**.** *Let* $\Lambda < G = \mathrm{SL}_n(\mathbb{R})$ *be any lattice. On the mean-zero regular representation on* $L^2_0(\Lambda\backslash G)$*, estimate* (C.5) *holds with the same* C, D, η *for every* Λ*.*

Proof. The only G-invariant vectors in $L^2(\Lambda\backslash G)$ are constants because the right G-action is transitive modulo a null-set formulation of Haar measure. Thus L^2_0 has no invariant vectors, and Theorem 4.1 applies. Its constants depend only on G and the fixed Sobolev convention, not on Λ. □

5. Appendix summary

The proof above reconstructs the precise higher-rank representation-theoretic input required by Chapter 7. The abelian spectral theorem is inherited from the declared functional-analytic floor of Volume 6; weak containment, temperedness, CHH, Harish–Chandra decay, and Howe–Moore are owned by Volume 5. The new mathematical bridge is the transitive spectral model for $\mathrm{SL}_2 \ltimes \mathbb{R}^2$ and its application to every root SL_2 in SL_n.

Bibliographic notes

The qualitative wavefront and counting architecture follows Eskin–McMullen and its strong-wavefront refinements. The fixed-setting effective translate machinery is inherited from the canonical Ratner project. Shah's work supplies the qualitative expanding-curve and linearization architecture used in Chapter 8. The higher-rank uniform pointwise matrix-coefficient theorem is reconstructed in the type-A scope needed here from Hee Oh's proof architecture; the full root-system-sharp version is a bibliographic strengthening, not a hidden prerequisite.

Volume 8

Counting on Homogeneous Varieties

Preface: two counting engines and their prerequisites

This volume has two complementary counting engines. The first is the Duke–Rudnick–Sarnak / Eskin–McMullen route: arithmetic orbit–stabilizer parametrization, unfolding, wavefront geometry, mixing, and boundary-control estimates (formalized as well-roundedness in Chapter 4). The second is the Eskin–Mozes–Shah route: nonescape, homogeneous-measure limits, linearization, focusing, and exclusion of proper algebraic branches.

The qualitative fixed-setting pieces already proved in Ratner v1.1 are inherited exactly. The new work is the quantitative boundary/Sobolev ledger, congruence-tower uniformity, explicit focusing bookkeeping, and the corrected EMS linearization step.

> **Proof architecture.** The effective theorem is formulated for admissible families with explicit boundary, spectral, cusp, and focusing parameters. A power saving is obtained only after these inputs are verified; no universal secondary term for arbitrary algebraic norm balls is claimed.

Student orientation: from orbit integrals to lattice-point counts

Unfolding rewrites an arithmetic counting function as an integral over a homogeneous orbit. A family of regions is *well rounded* when thickening or shrinking its boundary changes volume by a controlled lower-order amount. *Stable mean ergodic* means that the averaging estimate survives the families of lattices/levels under consideration. *Focusing* is the alternative in which mass accumulates near an intermediate homogeneous subgroup instead of equidistributing in the ambient space. The counting proofs have three independent inputs: an orbit–stabilizer parametrization, a volume asymptotic, and an equidistribution theorem. Check each normalization separately before combining them.

Reader guide: a concrete model

Why this matters.

Volume 8 is organized around the passage from *Arithmetic Orbit–Stabilizer Parametrization* to *Worked Counting Laboratories*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.1 (A calculation to keep in view)**.** Consider the standard unipotent and diagonal elements

$$u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad a_s = \begin{pmatrix} e^s & 0 \\ 0 & e^{-s} \end{pmatrix}.$$

A direct multiplication gives

$$a_s u_t a_{-s} = u_{e^{2s}t}.$$

Thus a displacement of size $|t|$ in the unstable horospherical direction is enlarged by e^{2s}. If $|t| = e^{-2s}$, the conjugated displacement has order one. This elementary calculation is the model for the “first visible scale” used repeatedly in nondivergence and shearing arguments.

CHAPTER 1

Arithmetic Orbit–Stabilizer Parametrization

1. The exact arithmetic dictionary

Let $\mathbf{G}$ be a connected linear algebraic group defined over $\mathbb{Q}$, let $G = \mathbf{G}(\mathbb{R})^\circ$, let $\Gamma < G$ be an arithmetic lattice, and let $\rho : \mathbf{G} \to \mathrm{GL}(W)$ be a rational representation. Fix $v_0 \in W(\mathbb{Q})$ and put

$$\mathbf{H} = \mathrm{Stab}_{\mathbf{G}}(v_0), \qquad H = \mathbf{H}(\mathbb{R}) \cap G.$$

We always take the *full* stabilizer in G; this eliminates an artificial finite multiplicity that otherwise appears if one replaces H by H°.

Theorem 1.1 (Arithmetic lattice in the stabilizer). *Assume* $\mathbf{H}$ *is a connected reductive* $\mathbb{Q}$*-subgroup with no nontrivial* $\mathbb{Q}$*-character, in the fixed arithmetic linear setting of Volume 4. Then* $H \cap \Gamma$ *is a lattice in* H*, and the orbit map*

$$(H \cap \Gamma)\backslash H \longrightarrow \Gamma\backslash G, \qquad (H \cap \Gamma)h \longmapsto \Gamma h,$$

is proper; in particular its image is closed.

Proof. Volume 4, Theorem 7.1, proves the Borel–Harish–Chandra criterion in precisely this arithmetic linear scope, hence $H \cap \Gamma$ has finite covolume in H. We spell out the closed-orbit implication. Choose a Siegel set $\mathfrak{S}_H \subset H$ whose image in $(H \cap \Gamma)\backslash H$ is all of the quotient and whose intersections with compact subsets of G are compact modulo the finitely many reduction pieces supplied by the Volume 4 proof. Suppose Γh_j converges in $\Gamma\backslash G$ with $h_j \in \mathfrak{S}_H$. Pick $\gamma_j \in \Gamma$ so that $\gamma_j h_j$ lies in one fixed compact subset of G. Arithmetic reduction in the proof of Theorem 7.1 says that, after passing to a subsequence, the only γ_j capable of keeping a sequence from the H-Siegel set in that compact set lie in one left coset of $H \cap \Gamma$. Multiplying h_j by that stabilizer element therefore puts it in a compact subset of H. Thus its class in $(H \cap \Gamma)\backslash H$ has a convergent subsequence, proving properness. □

Proof and provenance. Borel–Harish–Chandra is cited for historical provenance; the finite-covolume and proper-orbit statements used here are owned by Volume 4 and the displayed reduction argument.

Theorem 1.2 (Exact orbit–coset parametrization). *The map*

$$(H \cap \Gamma)\backslash\Gamma \longrightarrow v_0\Gamma, \qquad (H \cap \Gamma)\gamma \longmapsto v_0\rho(\gamma)$$

is a bijection.

Proof. Surjectivity is the definition of $v_0\Gamma$. If $v_0\rho(\gamma_1) = v_0\rho(\gamma_2)$, then $\rho(\gamma_1\gamma_2^{-1})$ fixes v_0. Since H is the full stabilizer inside G, $\gamma_1\gamma_2^{-1} \in H\cap\Gamma$, which is precisely equality of the two left cosets. The converse is immediate. □

Proposition 1.3 (Finite arithmetic orbit decomposition with multiplicities)**.** *Let $\mathcal{V} \subset v_0\mathbf{G}(\mathbb{R})$ be an arithmetic subset which is a finite disjoint union*

$$\mathcal{V} = \bigsqcup_{j=1}^{r} v_j\Gamma,$$

with $H_j = \mathrm{Stab}_G(v_j)$. Then for every Borel set $\Omega \subset W(\mathbb{R})$,

$$\#(\mathcal{V} \cap \Omega) = \sum_{j=1}^{r} \#\{(H_j \cap \Gamma)\gamma : v_j\gamma \in \Omega\}.$$

If the original arithmetic problem carries a fixed finite weight m_j on the jth orbit, the same formula holds with the corresponding factor m_j; no other multiplicity is hidden in the homogeneous parametrization.

Proof. Apply Theorem 1.2 to each orbit. Disjointness gives the sum. Fixed external weights are bookkeeping data and therefore factor out. □

2. Haar normalization

Fix Haar measures dg on G and dh on H. When the modular characters match, Weil integration determines the quotient measure $d(Hg)$ by

$$\int_G F(g)\,dg = \int_{H\backslash G}\int_H F(hg)\,dh\,d(Hg).$$

Theorem 2.1 (Normalization of the arithmetic main term). *Assume $H\cap\Gamma$ is a lattice in H. Put*

$$V_G = m_G(\Gamma\backslash G), \qquad V_H = m_H((H\cap\Gamma)\backslash H).$$

For every counting theorem obtained by unfolding the orbit $(H\cap\Gamma)\backslash H$ into $\Gamma\backslash G$, the invariant density constant forced by the above Haar conventions is

$$c_{H,\Gamma} = \frac{V_H}{V_G}.$$

This quantity is independent of a rescaling of dh.

Proof. The homogeneous H-period has total mass V_H, whereas ambient Haar has mass V_G. After mixing, a normalized ambient density therefore contributes V_H/V_G. If dh is multiplied by c, then V_H is multiplied by c and the quotient measure $d(Hg)$ is divided by c, so the product $c_{H,\Gamma}\, m_{H\backslash G}(B)$ is unchanged. $\square$

3. Principal congruence covers

For the family-uniform statements of this volume we use the scope proved in Volume 7:

$$G = \mathrm{SL}_n(\mathbb{R}), \qquad \Gamma = \mathrm{SL}_n(\mathbb{Z}), \qquad n \geq 3,$$

and $\Gamma(q)$ denotes a principal congruence subgroup.

Proposition 3.1 (Congruence-compatible decomposition)**.** *For every q the fixed Γ-orbit decomposes as*

$$v_0\Gamma = \bigsqcup_{\delta \in (H\cap\Gamma)\backslash\Gamma/\Gamma(q)} v_0\delta\Gamma(q).$$

For each chosen double-coset representative δ, the map

$$(\delta^{-1}H\delta \cap \Gamma(q))\backslash\Gamma(q) \longrightarrow v_0\delta\Gamma(q)$$

is bijective. In particular the intrinsic orbit multiplicity is exactly one, uniformly in q.

Proof. The double-coset decomposition partitions Γ into $(H\cap\Gamma)\delta\Gamma(q)$. Left multiplication by $H\cap\Gamma$ does not change v_0, so each double coset gives one $\Gamma(q)$-orbit. Two points $v_0\delta\gamma_1$ and $v_0\delta\gamma_2$ coincide exactly when $\delta\gamma_1\gamma_2^{-1}\delta^{-1} \in H$, or equivalently $\gamma_1\gamma_2^{-1} \in \delta^{-1}H\delta \cap \Gamma(q)$. This proves the second assertion. $\square$

Warning 3.2 (What is not uniform)**.** The number of double cosets in Proposition 3.1 need not be bounded independently of q. Uniformity below concerns local multiplicity, analytic constants, and normalized error exponents; it does not assert a bounded number of congruence orbits.

Worked Example 3.3 (Integral points on a quadric)**.** Let Q be a nondegenerate rational quadratic form and v_0 satisfy $Q(v_0) = m \neq 0$. Then the real orbit is a homogeneous space for the appropriate orthogonal group and the stabilizer is an orthogonal group in one lower dimension. Once an arithmetic orbit decomposition is known, every individual orbit is counted by the exact coset dictionary above; the only remaining work is geometric volume growth and equidistribution.

CHAPTER 2

Unfolding, Periodization, and Sobolev Control

1. Periodization and unfolding from first principles

THEOREM 1.1 (Periodization and unfolding). *Let $X = \Gamma\backslash G$, let $Y = (H\cap\Gamma)\backslash H$ be closed of finite volume, and let $\psi \in C_c(H\backslash G)$. Its periodization*

$$\mathcal{P}_\Gamma\psi(\Gamma g) = \sum_{\gamma\in(H\cap\Gamma)\backslash\Gamma} \psi(H\gamma g)$$

is locally finite and, for $F \in C_c(X)$,

$$\int_X F\,\mathcal{P}_\Gamma\psi\,dm_X = \int_{H\backslash G} \psi(Hg)\left(\int_Y F(hg)\,dh\right) d(Hg).$$

PROOF. Let $C = \operatorname{supp}\psi$. If a compact set $K \subset G$ met infinitely many translates $H\gamma K$ with $H\gamma K \cap C \neq \varnothing$, properness of the closed orbit from Theorem 1.1 would give infinitely many points of the discrete set $(H\cap\Gamma)\backslash\Gamma$ in a compact subset of $H\backslash G$, a contradiction. Thus the sum is locally finite and may be integrated term by term on compact supports.

Choose a measurable fundamental domain $\mathcal{F}$ for Γ in G. Then, using left Γ-invariance of F and changing variables $g \mapsto \gamma^{-1}g$ in each summand,

$$\int_{\mathcal{F}} F(g) \sum_{(H\cap\Gamma)\gamma} \psi(H\gamma g)\,dg = \int_{(H\cap\Gamma)\backslash G} F(g)\psi(Hg)\,dg.$$

Apply Weil integration to the last quotient:

$$\int_{(H\cap\Gamma)\backslash G} F(g)\psi(Hg)\,dg = \int_{H\backslash G} \psi(Hg) \int_{(H\cap\Gamma)\backslash H} F(hg)\,dh\,d(Hg),$$

which is the asserted identity. $\square$

Proof and provenance. The identity is classical and appears in the Ratner counting notebook; the proof above is the active owner.

2. A quantitative local-overlap lemma

Let $C \subset H\backslash G$ be compact. Define its overlap number

$$M_\Gamma(C) = \sup_{g\in G} \#\{(H\cap\Gamma)\gamma :\ H\gamma g \in C\}.$$

Local finiteness gives $M_\Gamma(C) < \infty$.

Lemma 2.1 (Sobolev cost of periodization). *Let D be any right-invariant differential operator of order at most d. For $\psi \in C_c^\infty(H\backslash G)$ with support in C,*

$$\|D\mathcal{P}_\Gamma\psi\|_{L^2(X)}^2 \le M_\Gamma(C) \int_{(H\cap\Gamma)\backslash G} |D\psi(Hg)|^2\, dg.$$

Consequently every fixed Sobolev norm satisfies

$$\mathcal{S}_d(\mathcal{P}_\Gamma\psi) \ll_d M_\Gamma(C)^{1/2}\, \mathcal{S}_{d,H\backslash G}(\psi).$$

Proof. Right-invariant differentiation commutes with the summation. At every point at most $M_\Gamma(C)$ summands are nonzero, hence Cauchy–Schwarz gives $|\sum a_i|^2 \le M_\Gamma(C)\sum |a_i|^2$. Integrate over a fundamental domain for Γ and unfold the sum of squares exactly as in Theorem 1.1. Summing over the finite family of differential operators defining $\mathcal{S}_d$ proves the second estimate. ☐

Lemma 2.2 (Monotonicity of overlap on covers). *If $\Gamma' \le \Gamma$, and the stabilizer in the periodization is replaced by $H \cap \Gamma'$, then for a fixed compact $C \subset H\backslash G$ the local overlap number on $\Gamma'\backslash G$ is at most $M_\Gamma(C)$.*

Proof. A nonzero summand for the Γ'-periodization corresponds to a point of the discrete Γ'-orbit of H entering Cg^{-1}. Since $\Gamma' \subset \Gamma$, this set injects into the corresponding Γ-orbit set after forgetting the finer coset relation. Thus passing to a cover can remove local collisions but cannot create new ambient lattice elements in a fixed compact set. ☐

Theorem 2.3 (Congruence-uniform unfolding transfer). *In the principal-congruence family of Chapter 1, let ψ range in a fixed compactly supported C^d-bounded family on the corresponding homogeneous quotient. Then the unfolding identity and the Sobolev estimate of Lemma 2.1 hold with constants independent of q.*

Proof. Unfolding is an identity and contains no level-dependent constant. The Sobolev constant is controlled by the overlap number and the fixed local geometry. Lemma 2.2 bounds the former by its level-one value; the Lie group, differential operators, and compact support family are fixed. Hence the constants are level-uniform. ☐

Worked Example 2.4 (Why compact support matters). If ψ is allowed to extend arbitrarily far into a cusp, the overlap number need not be controlled by a single compact set. The correct replacement is the height-window decomposition of Volume 7: truncate at height R, apply Lemma 2.1 on the truncated support, and pay the separate cusp error. Periodization itself does not create cusp decay.

CHAPTER 3

Polar Coordinates and Effective Volume Asymptotics

1. Polar integration in the selected symmetric scope

THEOREM 1.1 (Generalized Cartan integration). *Let G be a connected real reductive group, let σ be an involution, let H be an open subgroup of G^σ, and choose a σ-stable maximal compact subgroup K. If $\mathfrak{a} \subset \mathfrak{p} \cap \mathfrak{q}$ is maximal abelian and $A^+ = \exp(\mathfrak{a}^+)$ is a closed positive chamber, then finitely many Weyl pieces cover G by HA^+K, and for every compactly supported continuous f one has*

$$\int_G f(g)\,dg = c\int_H \int_{\mathfrak{a}^+} \int_K f(h \exp Y k) J(Y)\,dk\,dY\,dh,$$

where

$$J(Y) = \prod_{\alpha\in\Sigma^+} |\sinh\alpha(Y)|^{m_\alpha^-} |\cosh\alpha(Y)|^{m_\alpha^+}$$

with $m_\alpha^\pm$ the ±1 multiplicities of the involution on the restricted-root block.

PROOF. The multiplication map $H \times A \times K \to G$ has differential, modulo $\mathfrak{h} \cap \mathfrak{k}$ and the finite Weyl stabilizer, equal on every restricted-root block to the two-by-two map obtained from $\mathrm{Ad}(\exp Y)$. Writing a root vector as the sum of its σ-symmetric and σ-antisymmetric parts diagonalizes that map; its singular values are $|\cosh\alpha(Y)|$ and $|\sinh\alpha(Y)|$. Taking the determinant over the root blocks gives the displayed $J(Y)$. On the complement of the root hyperplanes the map is a local diffeomorphism. The finitely many Weyl chambers account for the finite multiplicity; the walls have Haar measure zero. The ordinary change-of- variables theorem therefore gives the formula, and continuity extends it to all compactly supported f. $\square$

THEOREM 1.2 (Polyhedral Laplace asymptotic). *Let $B_T \subset H\backslash G$ be an algebraic norm ball for a fixed rational representation. In the polar coordinates of Theorem 1.1, assume its radial condition is cut out by finitely many representation weights. Then there exist $a > 0$, an integer $b \geq 1$, and $c > 0$ such that*

$$m(B_T) \sim cT^a(\log T)^{b-1}.$$

The exponent a is the maximal ratio of the leading Jacobian functional to the active norm functional, and $b-1$ is the dimension of the maximizing face.

PROOF. Put $L = \log T$. On each of finitely many chamber cones the norm constraint is $\max_j u_j(Y) \leq L$ for linear weights u_j. Expanding each sinh/cosh factor writes $J(Y)$ as a finite sum of exponentials $e^{\ell(Y)}$ times constants, with lower exponentials separated from the leading one on every cone. Subdivide the cone so that the maximum weight is linear. Integrating first in the radial variable and then along its level faces gives a finite sum $e^{\lambda L} q_\lambda(L)$. The largest $\lambda = a$ comes from the maximizing faces, and the degree of its polynomial is their maximal dimension $b-1$; all other terms are $o(e^{aL} L^{b-1})$. Positivity of Haar measure makes the sum of leading coefficients positive. Replacing e^L by T yields the formula. □

2. A quantitative polyhedral lemma

The next lemma isolates the additional regularity needed for an error term.

Lemma 2.1 (Polyhedral exponential integral with a gap). *Let $P \subset \mathbb{R}^r$ be a rational polyhedral cone, let $\ell, u_1, \ldots, u_s$ be linear forms, and let*

$$I(L) = \int_{\{x \in P : \max_j u_j(x) \leq L\}} e^{\ell(x)} p(x)\, dx,$$

where p is a polynomial. Assume that after subdividing P into finitely many polyhedral cones on which the maximum is linear, the maximal exponential rate a is attained on faces of a common maximal dimension $b-1$, and every other exponential rate is at most $a-\eta$ for some $\eta > 0$. Then

$$I(L) = e^{aL}\big(cL^{b-1} + O(L^{b-2})\big) + O(e^{(a-\eta)L} L^M)$$

for some M, with $c \geq 0$ determined by the leading faces.

PROOF. On each cone where the constraint is $u(x) \leq L$, choose linear coordinates consisting of u and coordinates along its level hyperplanes. Triangulate if necessary so that the remaining inequalities are simplicial. Repeated integration of $e^{\ell(x)}$ times a polynomial produces a finite sum of terms $e^{\lambda L} q_\lambda(L)$, where λ runs through exponential rates attached to faces and q_λ is a polynomial whose degree is one less than the dimension of the maximizing face. The hypothesis separates all nonleading λ from a by η. Summing the finitely many triangles gives the asserted form; coincident leading rates add their leading coefficients. □

THEOREM 2.2 (Effective volume remainder for regular norm balls). *Suppose an algebraic norm-ball family B_T admits the finite polyhedral polar* *decomposition of Theorem 1.2 and satisfies the gap hypothesis of Lemma 2.1. Then, with* $L = \log T$*,*

$$m(B_T) = cT^a(\log T)^{b-1}\left(1 + O((\log T)^{-1}) + O(T^{-\eta})\right).$$

If $b = 1$*, the* $O((\log T)^{-1})$ *term is absent after combining equal leading faces.*

PROOF. Expand each sinh and cosh factor in $J(a)$ as a finite leading exponential times $1 + O(e^{-c\alpha(Y)})$ on each chamber sector. The finitely many representation weights describing the norm cut the chamber into the polyhedral pieces of Lemma 2.1. Apply that lemma to each leading exponential-polynomial term. Error terms from the root-density expansion have strictly smaller exponential rate and are absorbed by the $T^{-\eta}$ term after decreasing η if necessary. ☐

3. Normalization on covers

Proposition 3.1 (Haar volumes in a finite cover)**.** *Let* $\Gamma' \leq \Gamma$ *have finite index. With the same Haar measure* dg *on* G*,*

$$m_G(\Gamma'\backslash G) = [\Gamma : \Gamma']\, m_G(\Gamma\backslash G).$$

For $H' = g^{-1}Hg$ *and* $\Lambda' = H' \cap \Gamma'$*, its covolume is computed with the same fixed Haar measure on* H'*. Thus every congruence-level main term is the exact ratio*

$$c_{H',\Gamma'} = \frac{m_{H'}(\Lambda'\backslash H')}{m_G(\Gamma'\backslash G)}.$$

No normalization constant is hidden in the probability measure used for mixing.

PROOF. A fundamental domain for Γ' is the disjoint union, up to null sets, of $[\Gamma : \Gamma']$ translates of a fundamental domain for Γ. This proves the first formula. The second is the same Weil-normalization argument as Theorem 2.1 applied on the cover. ☐

Worked Example 3.2 (A logarithmic factor)**.** If two independent chamber directions contribute the same maximal exponential rate subject to one radial constraint, the maximizing face is one-dimensional. Lemma 2.1 then produces a factor $\log T$. The logarithm is therefore geometric: it records the dimension of the face on which the exponential growth is maximal.

CHAPTER 4

Effective Well-Roundedness and Boundary Smoothing

1. Qualitative well-roundedness

Proposition 1.1 (Well-roundedness criterion). *For a family $B_T \subset H\backslash G$, define $B_T^+ = B_T U_\varepsilon$ and $B_T^- = \bigcap_{u\in U_\varepsilon} B_T u^{-1}$. Under the wavefront and radial-volume hypotheses of Theorem 1.3, with $r(T) \to 0$, one has*

$$\lim_{\varepsilon\downarrow 0} \limsup_{T\to\infty} \frac{m(B_T^+ \setminus B_T^-)}{m(B_T)} = 0.$$

Proof. Theorem 1.3 gives the upper bound $C(\varepsilon^\alpha + r(T))$. First let $T \to \infty$, then let $\varepsilon \downarrow 0$. $\square$

Definition 1.2 (Hölder well-roundedness). A family B_T is (C, α)-well-rounded if for all sufficiently small ε and all sufficiently large T,

$$m(B_T^+ \setminus B_T^-) \le C\varepsilon^\alpha m(B_T).$$

A remainder $r(T)m(B_T)$ may be added when $r(T) \to 0$ uniformly in the family parameters.

Theorem 1.3 (Effective well-roundedness from wavefront geometry). *Assume:*

(1) *the quantitative wavefront theorem of Volume 7 gives $B_T^+ \subset B_{e^{C_0\varepsilon}T} \cup W_{T,\varepsilon}$ and $B_{e^{-C_0\varepsilon}T} \setminus W_{T,\varepsilon} \subset B_T^-$;*
(2) *the wall region satisfies $m(W_{T,\varepsilon}) \le C_1\varepsilon^\alpha m(B_T)$;*
(3) *uniformly for $|s| \le s_0$,*

$$\frac{m(B_{e^sT})}{m(B_T)} = 1 + O(|s|^\alpha) + O(r(T)).$$

Then

$$\frac{m(B_T^+ \setminus B_T^-)}{m(B_T)} \ll \varepsilon^\alpha + r(T).$$

Proof. The two wavefront inclusions show that the boundary shell is contained in the radial annulus $B_{e^{C_0\varepsilon}T} \setminus B_{e^{-C_0\varepsilon}T}$ together with two copies of the wall region. Divide the measure of this union by $m(B_T)$ and apply the two quantitative hypotheses. □

Proposition 1.4 (Finite unions and products)**.** *Suppose $B_T^{(j)}$ $(1 \le j \le r)$ are (C_j, α)-well-rounded and*

$$\sum_j m(B_T^{(j)}) \le C_0\, m\left(\bigcup_j B_T^{(j)}\right).$$

Then the union is (C, α)-well-rounded with $C \le C_0 \sum_j C_j$. If $B_T = B_T^{(1)} \times B_T^{(2)}$ in a product group and both factors are (C_j, α_j)-well-rounded, then B_T is $(C, \min(\alpha_1, \alpha_2))$-well-rounded.

Proof. For unions, the boundary shell of the union is contained in the union of the individual boundary shells. Sum their measures and use the displayed lower comparison. For products,

$$(B_1^+ \times B_2^+) \setminus (B_1^- \times B_2^-)$$

is contained in the union of $(B_1^+ \setminus B_1^-) \times B_2^+$ and $B_1^- \times (B_2^+ \setminus B_2^-)$. Divide by the product volume and use well-roundedness of each factor. □

Theorem 1.5 (Boundary–spectral–cusp error ledger). *Suppose a smoothed counting argument has, for parameters $0 < \varepsilon < 1$ and $R \ge 1$, the normalized error bound*

$$E(T; \varepsilon, R) \le C\big(\varepsilon^{\alpha} + r(T) + R^{-\kappa} + T^{-\eta} R^{B} \varepsilon^{-s}\big), \tag{4.1}$$

where the four terms are respectively boundary, radial-volume, cusp, and mixing/Sobolev errors. Then choosing

$$R = T^{\theta}, \qquad \varepsilon = T^{-\xi},$$

with $0 < \theta < \eta/B$ and $0 < \xi < (\eta - B\theta)/s$ gives a power saving

$$E(T; \varepsilon, R) \ll r(T) + T^{-\delta}$$

for

$$\delta = \min\{\alpha\xi, \kappa\theta, \eta - B\theta - s\xi\} > 0.$$

Proof. Substitute the two choices into (4.1). The three power terms become $T^{-\alpha\xi}$, $T^{-\kappa\theta}$, and $T^{-(\eta - B\theta - s\xi)}$. The displayed inequalities make all exponents positive, and their minimum is δ. □

Corollary 1.6 (Uniform boundary smoothing on the principal-congruence tower)**.** *If the constants and exponents in Theorem 1.5 are uniform in q, then the choice of θ, ξ, δ is uniform in q.*

Proof. The admissible region for (θ, ξ) depends only on the exponents, so a single point in that open region works at every level. □

CHAPTER 5

The Duke–Rudnick–Sarnak Counting Route

1. The qualitative engine

THEOREM 1.1 (Mixing-to-counting reduction). *Let $Y = (H \cap \Gamma)\backslash H \subset X = \Gamma\backslash G$ be a closed finite-volume homogeneous orbit. Let $B_T \subset H\backslash G$ be well-rounded and suppose the translated normalized H-periods converge to ambient Haar along every radial sequence escaping compact subsets. Then*

$$N(T) \sim c_{H,\Gamma} m_{H\backslash G}(B_T).$$

PROOF. Choose inner and outer smooth approximations $\psi^-_{T,\varepsilon} \leq 1_{B_T} \leq \psi^+_{T,\varepsilon}$ supported in B^-_T and B^+_T. Periodize them and apply Theorem 1.1. The resulting integrals are averages, over $H\backslash G$, of the translated H-period. On every fixed compact radial sector, the assumed equidistribution makes those averages equal to $V_H/V_G + o(1)$; tightness controls the discarded cusp part. Therefore

$$c_{H,\Gamma} m(B^-_T) + o(m(B_T)) \leq N(T) \leq c_{H,\Gamma} m(B^+_T) + o(m(B_T)).$$

Well-roundedness makes both boundary-volume ratios tend to one, giving the claim. □

THEOREM 1.2 (Affine-symmetric counting in the selected scope). *For a finite arithmetic orbit decomposition as in Proposition 1.3, assume each stabilizer is a symmetric reductive subgroup satisfying Theorem 1.1, each norm-ball family is well-rounded, and the translated periods satisfy the mixing hypothesis of Theorem 1.1. Then*

$$\#(\mathcal{V} \cap \Omega_T) \sim \sum_{j=1}^{r} c_{H_j,\Gamma}\, m_{H_j\backslash G}(B_{j,T}).$$

PROOF. The orbit–coset bijection of Theorem 1.2 converts the jth arithmetic orbit to one homogeneous count. Apply Theorem 1.1 to each of the finitely many orbits and sum. The normalization is Theorem 2.1; finiteness lets the individual $o(1)$ errors be summed. □

Proof and provenance. Duke–Rudnick–Sarnak and Eskin–McMullen are primary-source provenance for this route. The active logical owners are the unfolding, mixing, well-roundedness, and orbit-decomposition arguments displayed in this volume.

2. Multiple arithmetic orbits

Proposition 2.1 (Explicit main term over finitely many orbits). *Under the hypotheses of Proposition 1.3, suppose the jth orbit corresponds to $H_j\backslash G$ and its geometric region is $B_{j,T}$. If*

$$N_j(T) = c_j m_j(B_{j,T}) + o(m_j(B_{j,T}))$$

for each j, then

$$\#(\mathcal{V} \cap \Omega_T) = \sum_{j=1}^{r} c_j m_j(B_{j,T}) + o\left(\sum_{j=1}^{r} m_j(B_{j,T})\right),$$

provided the finitely many main volumes are mutually comparable after discarding identically empty components. Here $c_j = m_{H_j}((H_j \cap \Gamma)\backslash H_j)/m_G(\Gamma\backslash G)$.

Proof. Sum the r asymptotics. Since r is fixed and the nonzero main volumes are comparable, the sum of the individual little-o terms is little-o of the sum of main volumes. The formula for c_j is Theorem 2.1. □

3. Effective DRS from Volume 7

Theorem 3.1 (Effective affine-symmetric counting). *Let $B_T \subset H\backslash G$ be a family satisfying the hypotheses of Theorem 1.5. Assume that the effective translate estimate from Volume 7 supplies the mixing term in (4.1), and that $r(T) \ll T^{-\beta}$ for some $\beta > 0$. Then*

$$N(T) = c_{H,\Gamma} m(B_T)(1 + O(T^{-\delta}))$$

for some $\delta > 0$ depending explicitly only on $\alpha, \kappa, \eta, B, s, \beta$.

Proof. Use the sharp count between the inner and outer smoothed counts. Unfold the smoothed count by Theorem 1.1. Apply the height-window thickening and effective mixing theorem of Volume 7. The resulting normalized error is exactly of the form (4.1). Apply Theorem 1.5 and absorb $r(T) \ll T^{-\beta}$ into the minimum exponent. □

Corollary 3.2 (Effective DRS exponent without cusp loss). *If the quotient region is contained in a fixed compact height window, so the $R^{-\kappa}$ and R^B losses disappear, and if the boundary error is $O(\varepsilon^\alpha)$ while mixing is $O(T^{-\eta}\varepsilon^{-s})$, then one may choose $\varepsilon = T^{-\eta/(s+\alpha)}$ and obtain*

$$\frac{N(T)}{c_{H,\Gamma} m(B_T)} - 1 = O\Big(T^{-\eta\alpha/(s+\alpha)}\Big)$$

(up to the separate volume-asymptotic remainder).

PROOF. Balance $\varepsilon^{\alpha} = T^{-\eta}\varepsilon^{-s}$ and substitute the resulting value of ε. □

THEOREM 3.3 (Principal-congruence uniform DRS transfer). *In the higher-rank type-A principal-congruence family proved in Volume 7, assume a fixed admissible norm-ball geometry and fixed height/boundary exponents. For every congruence orbit $v_0\delta\Gamma(q)$, the count satisfies*

$$N_{q,\delta}(T) = c_{q,\delta}m(B_T)\big(1 + O(T^{-\delta_0})\big),$$

where $\delta_0 > 0$ and the implied constant are independent of q and of the orbit representative within a fixed finite-complexity family. The main coefficient $c_{q,\delta}$ is the exact covolume ratio and is allowed to depend on q.

PROOF. Chapter 2 gives level-uniform periodization and unfolding; Volume 7 gives level-uniform matrix-coefficient decay and thickening estimates in this scope; Chapter 4 gives a level-uniform error optimizer. Therefore every error constant and exponent in Theorem 3.1 is uniform. The main term is not normalized away: Proposition 3.1 gives its exact level-dependent covolume ratio. □

CHAPTER 6

The Eskin–McMullen Wavefront Route

1. Wavefront and translate equidistribution

THEOREM 1.1 (Strong wavefront lemma). *Let (G, H) be the fixed symmetric pair of Chapter 3. Fix a closed cone $\mathfrak{c}$ in the relative interior of one chamber face. For every identity neighborhood U_ε there is $C > 0$, independent of $Y \in \mathfrak{c}$, such that*

$$H \exp(Y) K\, U_\varepsilon \subset HU_{C\varepsilon} \exp(Y + O(\varepsilon))KU_{C\varepsilon}.$$

Neutral root directions on the chosen face are absorbed in its central Levi factor; the constant is uniform because every active root is bounded away from zero on the unit directions of $\mathfrak{c}$.

PROOF. Apply the root-block differential calculation used in Theorem 1.1. On an active root block the derivative of the $H \times A \times K$ multiplication map has inverse norm bounded by a constant times $|\sinh \alpha(Y)|^{-1}$ or $|\cosh \alpha(Y)|^{-1}$; on the fixed face cone these are uniformly bounded for $\|Y\|$ outside a compact set. The neutral blocks belong to the face centralizer and are absorbed into the H/compact factors. The quantitative inverse-function theorem on the remaining finite-dimensional blocks gives the stated $O(\varepsilon)$ factors. The bounded part of the cone is covered by finitely many ordinary coordinate charts. □

THEOREM 1.2 (Equidistribution of symmetric-subgroup translates). *Let $Y = (H \cap \Gamma)\backslash H$ have finite volume. Assume the ambient G-action on $L^2_0(\Gamma\backslash G)$ is mixing in the Cartan directions under consideration. If $a_i = \exp Y_i$ leaves every compact subset of $H\backslash G$ through the chosen mixing cone and the H-periods do not escape mass, then for every $f \in C_c(X)$,*

$$\frac{1}{V_H} \int_Y f(ha_i)\, dh \longrightarrow \frac{1}{V_G} \int_X f\, dg.$$

PROOF. Fix $\varepsilon > 0$ and thicken Y in a transverse identity neighborhood with a smooth probability bump φ_ε. Unfolding turns the thickened period into a matrix coefficient between the periodized bump and f. Mixing sends this coefficient to the product of integrals as $i \to \infty$. The strong wavefront lemma shows that replacing the thickened average by the singular H-period changes it by $o_\varepsilon(1)$ outside a cusp set; nonescape makes the cusp contribution $o(1)$. First let $i \to \infty$ and then $\varepsilon \downarrow 0$. □

THEOREM 1.3 (Eskin–McMullen qualitative counting). *Under the hypotheses of Theorems 1.1 and 1.2, every well-rounded affine symmetric norm-ball family satisfies*

$$N(T) \sim c_{H,\Gamma} m_{H\backslash G}(B_T).$$

PROOF. Theorem 1.2 supplies the translate-equidistribution hypothesis of Theorem 1.1; the wavefront theorem supplies the boundary comparison used in well-roundedness. Apply Theorem 1.1. □

2. The effective upgrade

THEOREM 2.1 (Effective Eskin–McMullen counting). *Assume that the polar family B_T is (C, α)-well-rounded with remainder $O(T^{-\beta})$, and that the corresponding translated H-periods satisfy the Volume 7 effective mixing estimate with spectral exponent η and Sobolev loss s, together with the stated cusp exponents. Then*

$$N(T) = c_{H,\Gamma} m(B_T)(1 + O(T^{-\delta}))$$

for a positive δ obtained from Theorem 1.5.

PROOF. The strong wavefront lemma identifies an ambient U_ε-thickening with a radial change $T \mapsto e^{O(\varepsilon)}T$ plus a controlled wall region. Thus the sharp count is squeezed between inner and outer smooth counts with boundary error $O(\varepsilon^\alpha + T^{-\beta})$. Unfold, thicken the H-period, apply the Volume 7 effective translate estimate, and truncate the cusp. This is exactly the error ledger of Theorem 1.5; optimize there. □

THEOREM 2.2 (Level-uniform Eskin–McMullen theorem). *In the principal-congruence SL_n family of Volume 7, $n \geq 3$, the conclusion of Theorem 2.1 holds with a common power δ and a common implied constant for every member of a fixed finite-complexity geometric family. Only the covolume main coefficient varies with the level.*

PROOF. The wavefront constants depend only on (G, H) and the fixed chamber-face separation, not on $\Gamma(q)$. The boundary geometry is fixed. The periodization overlap is uniform by Lemma 2.2, and the effective mixing constants are uniform by Volume 7. Apply the same optimizer at every level. □

Worked Example 2.3 (Why chamber walls must be separated)**.** Near a wall a root that normally expands may remain neutral. The inverse of the differential used in the wavefront factorization can then lose a uniform bound if one insists on the full open-chamber formula. The correct statement works face by face: neutral root directions are absorbed into the central Levi factor, while the active roots stay a fixed positive distance from zero.

CHAPTER 7

Eskin–Mozes–Shah Focusing and the Corrected Linearization Step

1. The homogeneous-measure input, proved in the arithmetic setting

THEOREM 1.1 (Moving homogeneous measures and focusing). *Let $G \leq \mathrm{SL}_N(\mathbb{R})$ be connected and defined over $\mathbb{Q}$, let $\Gamma < G$ be arithmetic, and let $H_i < G$ be connected $\mathbb{Q}$-subgroups generated by one-parameter Ad-unipotent subgroups. Let*

$$\mu_i = m_{H_i x_i}, \qquad x_i = g_i\Gamma,$$

be normalized Haar probabilities on closed finite-volume orbits, and assume (μ_i) is tight. If $\mu_i \Rightarrow \mu$, then every ergodic component of μ for any unipotent one-parameter subgroup inherited from the H_i-actions is homogeneous by the internally proved one-parameter classification theorem of Volume 2. Moreover, if μ is not ambient Haar, then after a subsequence there are a proper connected $\mathbb{Q}$-subgroup $F < G$, a compact set $C \subset G$, and elements $\gamma_i \in \Gamma$ such that

$$H_i \subset g_i\gamma_i F \gamma_i^{-1} g_i^{-1}, \qquad g_i\gamma_i \in C N_G(F). \tag{7.0}$$

Thus every non-Haar limit is carried by a proper rational focusing branch.

PROOF. Fix a compact set carrying $1 - o(1)$ of every μ_i and a countable dense family of compactly supported test functions. Choose points y_i on the H_i-orbits that are simultaneously generic for these functions and lie in the compact set. If two such generic points approach one another transversely to the current measure stabilizer, the polynomial first-exit construction of Volume 2, Chapter 6B, gives a nonconstant limiting quasi-regular map. The block-comparison lemma there gives, for every value $q(s)$ of the limit map and every test function f,

$$\int f(q(s)x)\, d\mu(x) = \int f(x)\, d\mu(x).$$

Differentiating at a regular value enlarges the connected stabilizer of μ. Iterating can occur only $\dim G$ times.

If the transverse construction fails, the exterior representatives of the nearby points satisfy one of the primitive incidence equations of the arithmetic linearization theorem of Volume 2. The finite-window properness there supplies a primitive integral exterior vector v_F with bounded norm on the chosen compact window. Hence a single connected $\mathbb{Q}$-subgroup F occurs after passing to a subsequence. A change of label between two representatives forces their intersection and strictly lowers $\dim F$, so the singular descent terminates. At the terminal branch all H_i preserve the corresponding exterior line, giving the first relation in (7.0). Properness of the representative family gives the second relation, with the bounded part absorbed in C.

On the regular branch the stabilizer has enlarged until its orbit is open; connectedness then makes the limiting probability ambient Haar. On a singular branch, disintegrate μ under one of the unipotent one-parameter subgroups contained in the terminal group. Volume 2's internal Ratner classification makes almost every component Haar on a closed homogeneous orbit inside the terminal rational subgroup. Hence a non-Haar limit is supported by the proper branch (7.0), proving the theorem. □

Proof and provenance. Mozes–Shah and Eskin–Mozes–Shah are provenance for this architecture. The active proved input is the arithmetic linearization/classification chain spelled out above and proved in Volume 2.

2. The EMS theorem at the exact qualitative strength

We record the structural form used in counting. Let G, H be connected real algebraic $\mathbb{Q}$-groups with no nontrivial $\mathbb{Q}$-characters, let $\Gamma < G(\mathbb{Q})$ and $\Lambda < H(\mathbb{Q})$ be arithmetic lattices, and let $\rho_i : H \to G$ be $\mathbb{Q}$-homomorphisms satisfying the five hypotheses of EMS Theorem 2.1: no proper $\mathbb{Q}$-subgroup contains $\rho_i(H)$ infinitely often; arithmetic discreteness on $H(\mathbb{Q})$; infinitesimal unipotent control; finite-dimensionality of regular-function pullbacks; and $\rho_i(\Lambda) \subset \Gamma$.

Theorem 2.1 (Corrected algebraic normality step). *Assume the EMS linearization argument has produced a connected $\mathbb{Q}$-subgroup $F < G$, a unipotent one-parameter group U, and homomorphisms ρ_i such that, after conjugating F once,*

$$\rho_i(w)\rho_1(w)^{-1} \in N^1(F) \quad (i \geq 1, w \in H), \tag{7.1}$$

and

$$\rho_1(H) \subset N(F, U). \tag{7.2}$$

Assume moreover that the limiting U-invariant probability μ gives zero mass to the EMS singular set $\pi(S(F, U))$. Then F is normal in G and μ is F-invariant.

PROOF. Let L be the smallest connected real algebraic subgroup containing all relative elements $\rho_i(w)\rho_1(w)^{-1}$. Since $H(\mathbb{Q})$ is Zariski dense in H and the ρ_i are $\mathbb{Q}$-homomorphisms, L is defined over $\mathbb{Q}$. For $g, w \in H$,

$$\rho_1(g)(\rho_i(w)\rho_1(w)^{-1})\rho_1(g)^{-1} = (\rho_i(g)\rho_1(g)^{-1})^{-1}(\rho_i(gw)\rho_1(gw)^{-1}) \in L.$$

Hence $\rho_1(H) \subset N(L)$. Therefore $L\rho_1(H)$ is a $\mathbb{Q}$-subgroup containing every $\rho_i(H)$. By the no-proper-$\mathbb{Q}$-subgroup hypothesis of EMS Theorem 2.1,

$$G = L\rho_1(H). \tag{7.3}$$

From (7.1), $L \subset N^1(F)$, and hence $LN(F,U) = N(F,U)$. Combining (7.2) with (7.3) gives $G \subset N(F,U)$, so $gUg^{-1} \subset F$ for every $g \in G$.

Let F_0 be the smallest real algebraic $\mathbb{Q}$-subgroup containing all gUg^{-1}, $g \in G$. Then $F_0 \lhd G$ and $F_0 \subset F$. In the EMS class $\mathcal{H}$, if $\dim F_0 < \dim F$, the defining singular-set relations imply

$$G = N(F_0) \subset N(F_0, U) \subset S(F, U),$$

contradicting the branch assumption $N(F,U) \setminus S(F,U) \neq \varnothing$. Hence $F_0 = F$ and F is normal. Finally, disintegrate μ into U-ergodic components. The internal one-parameter classification theorem of Volume 2 makes each component homogeneous. On the regular branch $\mu(\pi(S(F,U))) = 0$, so the defining incidence alternative forces the homogeneous stabilizer of almost every component to contain F; hence every component is F-invariant. Integrating the ergodic decomposition shows that μ itself is F-invariant. □

Proof and provenance. The proof above is the corrected step published by Eskin–Mozes–Shah. It replaces the incomplete normality argument in the original proof of their Theorem 2.1; once this claim is repaired, their proof proceeds by projecting to $F\backslash G$ and inducting on dimension.

THEOREM 2.2 (EMS equidistribution theorem in the selected algebraic form). *Under the five hypotheses stated above, the homogeneous probabilities supported on the closed orbits of $\rho_i(H)$ in $\Gamma\backslash G$ converge weakly to Haar probability on $\Gamma\backslash G$.*

PROOF. The proof is by induction on $\dim G$, following EMS. Tightness is part of the five hypotheses through the arithmetic nondivergence argument developed in Volume 2. If a limit is non-Haar, Theorem 1.1 supplies a nontrivial unipotent invariance and a proper rational focusing subgroup F. The relative linearization identities give (7.1)–(7.2) off the singular set. Theorem 2.1 makes F normal and the limit F-invariant. Project to $F\backslash G$. Since F contains a nontrivial unipotent subgroup, the quotient has smaller dimension; the five structural hypotheses descend to the quotient. The induction hypothesis makes the projected limit Haar. F-invariance then lifts this to G-Haar. The base case is trivial. ☐

3. Focusing as the only qualitative obstruction

THEOREM 3.1 (Focusing alternative for reductive stabilizers). *Let G be connected reductive over $\mathbb{Q}$, let $H < G$ be connected reductive, with no nontrivial $\mathbb{Q}$-characters, and assume H is not contained in a proper $\mathbb{Q}$-parabolic. Let Γ be arithmetic and let μ_H be the normalized measure on the closed H-orbit. For any sequence $g_i \in G$, after passing to a subsequence either $\mu_H g_i \to m_{\Gamma\backslash G}$, or there are a proper connected reductive $\mathbb{Q}$-subgroup $L \supset H$, a compact $C \subset G$, and $z_i \in Z(H) \cap \Gamma$ such that*

$$g_i \in z_i LC \qquad \textit{for all } i.$$

In particular, every non-Haar subsequential limit forces such a proper focusing branch. The statement does not assert that the displayed focusing containment is logically incompatible with Haar convergence for an arbitrary chosen sequence.

PROOF. Nonescape follows from the reductive nonparabolic hypothesis. Apply Theorem 1.1. If the limit is not Haar, that theorem gives a proper $\mathbb{Q}$-subgroup L' containing a conjugate of H and recentering elements in the arithmetic normalizer incidence set $Z(H, L')\cap\Gamma$. The EMS finite-double-coset lemma for reductive groups writes this arithmetic incidence set as

$$(Z(H) \cap \Gamma)D(L' \cap \Gamma)$$

with D finite. Pass to a subsequence on which the element of D is fixed and conjugate L' by it. The remaining bounded recentering terms form a compact set C, giving alternative (2). If no proper L occurs, the limit must be ambient Haar. ☐

Corollary 3.2 (Counting after excluding focusing)**.** *Let $B_T \subset H\backslash G$ be a well-rounded family such that for every proper connected reductive $\mathbb{Q}$-subgroup $L \supset H$ and compact $C \subset G$,*

$$\frac{m_{H\backslash G}(B_T \cap q_H((Z(H) \cap \Gamma)LC))}{m_{H\backslash G}(B_T)} \to 0. \tag{7.4}$$

Then the normalized H-periods contributing to the count are Haar for all but a negligible portion of B_T, and the leading-volume counting asymptotic holds.

PROOF. If a positive proportion of the radial parameters produced non-Haar limits, extract a sequence from that positive-mass set. Theorem 3.1 places the sequence in a fixed focusing set after a subsequence. This contradicts (7.4). Hence outside a set of relative volume $o(1)$ all divergent sequences equidistribute. Apply the unfolding squeeze from Theorem 1.1; the exceptional parameters contribute $o(m(B_T))$. □

Proposition 3.3 (Finite-complexity focusing ledger)**.** *Fix a height bound Q on rational subgroup data. The finite Chevalley family and subgroup-height theorems 0.3 and 0.2 reduce every focusing subgroup of complexity at most Q to a finite list $L_1, \ldots, L_{M(Q)}$. Consequently a quantitative counting proof needs only verify a uniform estimate for the finitely many sets*

$$q_H((Z(H) \cap \Gamma)L_j C_Q), \qquad 1 \le j \le M(Q),$$

where C_Q is the compact recentering range returned by effective closing. All dependence on the focusing branch is therefore explicit through Q and $M(Q)$.

Proof. Theorems 0.2 and 0.3 prove that subgroups generated by the bounded-height algebraic data appearing in a finite-scale closing argument have bounded rational height and are detected by a finite Chevalley list. There are therefore only finitely many marked algebraic alternatives at a fixed bound Q. The recentring part of the closing theorem is bounded on the same height window, so it lies in a compact C_Q. Taking the maximum of the finitely many branch constants gives the asserted ledger. □

4. Why reductivity is a real boundary

Proposition 4.1 (A nonreductive escape model)**.** *Let $G = \mathrm{SL}_2(\mathbb{R})$, $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, and let $N = \{\left(\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right) : t \in \mathbb{R}\}$. The closed horocycles*

$$(N \cap \Gamma)\backslash N\, a_y, \qquad a_y = \mathrm{diag}(y^{1/2}, y^{-1/2}),$$

escape into the cusp as $y \to \infty$: for every compact $K \subset \Gamma\backslash G$, their normalized measures assign mass 0 to K for all sufficiently large y.

Proof. The projection to the modular surface lies on the horizontal horocycle of height y. Every compact subset of the modular surface is contained in a height region $\operatorname{Im} z \le Y_K$. If $y > Y_K$, the entire projected horocycle is outside that compact set. The same holds in the unit tangent bundle up to the compact K-fiber. Thus the normalized measure gives zero mass to any fixed compact set for all large y. □

Warning 4.2. This example is not a counterexample to Ratner theory. It shows why the reductive/nonparabolic hypotheses in the EMS counting theorem cannot simply be deleted: the first obstruction may be escape of mass rather than algebraic focusing inside a probability limit.

CHAPTER 8

Stable Mean Ergodic Theorems and Uniform Counting

1. An abstract quantitative counting principle

Proposition 1.1 (Fixed-setting effective counting ledger)**.** *In the fixed arithmetic quotient of this volume, smoothing a lattice count at scale ε, truncating at height R, applying an effective mean-ergodic or mixing estimate, and comparing inner/outer boundary sets produces exactly the four error classes*

$$E_{\text{boundary}}(\varepsilon) + E_{\text{cusp}}(R) + E_{\text{mix}}(T;\varepsilon,R) + E_{\text{volume}}(T).$$

When these satisfy the polynomial bounds of Theorem 1.5, their optimization gives a power-saving counting error.

Proof. The smoothing identity and Cauchy–Schwarz estimate are proved below in Theorem 1.3. For homogeneous periods, Theorem 1.1 replaces the point bump by a periodized transverse bump, while Volume 7 supplies the effective mixing term. The inner/outer squeeze contributes the boundary term and height truncation the cusp term. These are precisely the four entries in Theorem 1.5; its displayed substitution $R = T^{\theta}$, $\varepsilon = T^{-\xi}$ proves the final assertion. □

Let G be a locally compact second countable unimodular group with Haar measure m_G, let $\Gamma < G$ be a lattice, and let $B_T \subset G$ be a finite-measure family. Write

$$\beta_T = \frac{1_{B_T}}{m_G(B_T)}\, dm_G.$$

Definition 1.2 (Stable quantitative mean ergodic estimate)**.** Let ν_X be the quotient Haar measure induced by m_G and put $V_\Gamma = \nu_X(G/\Gamma)$. We say B_T has L^2 mean-ergodic rate $E(T)$ on $X = G/\Gamma$ if for every $f \in L^2(X,\nu_X)$,

$$\left\| \pi_X(\beta_T)f - \frac{1}{V_\Gamma}\int_X f\, d\nu_X \right\|_2 \le E(T)\|f\|_2.$$

The estimate is *stable* if the same bound, up to a fixed factor, holds for the small perturbations $U_\varepsilon B_T U_\varepsilon$ and their inner counterparts used in smoothing.

Theorem 1.3 (Stable mean-ergodic counting principle). *Assume:*

(1) B_T is (C,α)-well-rounded;

(2) identity neighborhoods satisfy $m_G(U_\varepsilon) \geq c\varepsilon^\rho$;

(3) B_T has stable mean-ergodic rate $E(T) \to 0$;

(4) the identity coset in G/Γ is injective on U_{ε_0}.

Then, for all large T,

$$\left| \frac{\#(\Gamma \cap B_T)}{m_G(B_T)} - \frac{1}{m_G(G/\Gamma)} \right| \ll E(T)^{\alpha/(\rho+\alpha)}. \tag{8.1}$$

The implicit constant depends only on the displayed regularity and local injectivity data.

Proof. Use the quotient Haar measure ν_X of total mass V_Γ and let $\phi_\varepsilon = m_G(U_\varepsilon)^{-1} 1_{U_\varepsilon \Gamma}$, which is well-defined by local injectivity. Then

$$\int_X \phi_\varepsilon \, d\nu_X = 1, \qquad \|\phi_\varepsilon\|_2 \ll m_G(U_\varepsilon)^{-1/2} \ll \varepsilon^{-\rho/2}.$$

Average the function $\pi_X(\beta_T)\phi_\varepsilon$ once more over the same small identity neighborhood. The inner/outer inclusions for B_T sandwich this double-smoothed quantity between the normalized lattice counts in B_T^- and B_T^+. Pair the mean-ergodic error with the second normalized bump and use Cauchy–Schwarz. The two bump norms contribute $m_G(U_\varepsilon)^{-1}$, so the analytic error is $O(E(T)\varepsilon^{-\rho})$. The invariant projection of ϕ_ε is the constant $1/V_\Gamma$, which is exactly the main density. Well-roundedness contributes $O(\varepsilon^\alpha)$. Hence

$$\text{relative error} \ll \varepsilon^\alpha + E(T)\varepsilon^{-\rho}.$$

Choose $\varepsilon = E(T)^{1/(\rho+\alpha)}$. The two terms are then both $O(E(T)^{\alpha/(\rho+\alpha)})$, proving (8.1). □

Proof and provenance. This is the smoothing proof behind the quantitative lattice-counting theorem of Gorodnik–Nevo. Their general theorem includes translated cosets and a more flexible local-dimension formalism; the exponent $\alpha/(\rho+\alpha)$ is the same optimization.

2. Homogeneous-orbit counting

Theorem 2.1 (Quantitative homogeneous counting master theorem). *Let the arithmetic orbit $v_0\Gamma$ and subgroup H satisfy the hypotheses of Chapters 1–4. Suppose the periodized thickening operator has stable mean-ergodic rate $E(T) \ll T^{-\eta}$ and the boundary exponent is $\alpha > 0$. Suppose further that the* *cusp/truncation losses satisfy the polynomial error ledger of Theorem 1.5. Then*

$$N(T) = c_{H,\Gamma} m_{H\backslash G}(B_T)(1 + O(T^{-\delta}))$$

for an explicit $\delta > 0$ obtained by minimizing the boundary, cusp, and spectral exponents.

Proof. Periodize the smoothed indicator and unfold by Theorem 2.3. The analytic part of the resulting H-period is controlled either by the Volume 7 effective translate theorem or by the stable mean-ergodic estimate, depending on the formulation of the family. In both cases the remaining errors are precisely those in Theorem 1.5. Optimize there. The constant in the main term is fixed by Theorem 2.1. □

Theorem 2.2 (Uniform principal-congruence counting). *For $G = \mathrm{SL}_n(\mathbb{R})$, $\Gamma(q) < \mathrm{SL}_n(\mathbb{Z})$ principal congruence and $n \geq 3$, consider a fixed finite-complexity family of arithmetic homogeneous orbits and admissible norm-ball sets for which the stabilizer and boundary hypotheses of this volume hold uniformly. Then there exist $\delta > 0$ and $C < \infty$, independent of q, such that every congruence orbit satisfies*

$$|N_{q,\delta_0}(T) - c_{q,\delta_0} m(B_T)| \leq C\, c_{q,\delta_0} m(B_T) T^{-\delta}.$$

Proof. Volume 7 supplies a level-uniform spectral/mixing exponent in precisely this higher-rank type-A scope. Lemma 2.2 makes the periodization constants uniform; the geometric family has fixed wavefront and well-roundedness exponents; and the height-window cutoff data are uniform by hypothesis. Thus Theorem 2.1 uses one common error ledger at all levels. The coefficient is kept exact rather than normalized away, as in Proposition 3.1. □

Theorem 2.3 (Admissible moving-family counting). *Let $(H_i, B_{i,T}, \Gamma_i)$ be a family such that:*

1. *dimensions, local product-chart angles, and Sobolev orders are uniformly bounded;*
2. *boundary Hölder exponent α, cusp exponent κ, and spectral exponent η admit positive uniform lower bounds;*
3. *orbit multiplicities are uniformly bounded and Haar normalizations are recorded exactly;*
4. *every proper focusing branch of complexity at most the effective-closing height bound has relative volume $O(T^{-\zeta})$ for a common $\zeta > 0$.*

Then the counts have a common power-saving exponent

$$N_i(T) = c_i m_i(B_{i,T})(1 + O(T^{-\delta}))$$

with $\delta > 0$ depending only on the uniform lower/upper bounds in the four items.

Proof. The first three items make every nonfocusing analytic constant in Theorem 2.1 uniform. Proposition 3.3 reduces focusing to finitely many branches at the relevant complexity, and item (4) bounds their total contribution by a common power. Take the minimum of that exponent and the optimized analytic exponent. ☐

Theorem 2.4 (Failure-mode trichotomy). *For a sequence of counting problems in the scope of this volume, failure of a uniform leading-volume asymptotic along a subsequence must be witnessed by at least one of the following explicit degenerations:*

(1) cusp degeneration*: the truncated mass is not uniformly tight;*
(2) focusing*: a positive proportion lies in a proper algebraic focusing branch of Theorem 3.1;*
(3) boundary degeneration*: no positive uniform well-roundedness exponent exists for the family.*

This is a logical trichotomy under the uniform spectral hypothesis; it is not a classification of all possible geometric degenerations outside that hypothesis.

Proof. Assume none of the three occurs. Nonescape gives a common height cutoff with arbitrarily small tail; exclusion of focusing makes the homogeneous limits ambient Haar outside a negligible set; and uniform boundary regularity permits one smoothing scale for the whole family. The uniform spectral hypothesis then gives the common analytic error estimate. The proof of Theorem 2.3 applies and yields a uniform asymptotic, contradicting the assumed failure. Therefore at least one of the three listed conditions must fail. ☐

Worked Example 2.5 (The Gorodnik–Nevo exponent)**.** If $E(T) = T^{-\eta}$, the local dimension is ρ, and the boundary is Lipschitz ($\alpha = 1$), Theorem 1.3 gives the relative error $T^{-\eta/(\rho+1)}$. This makes visible why stronger spectral decay alone does not directly become the final counting exponent: one pays for localizing the lattice point with a bump.

APPENDIX A

Dependency map

The counting arguments depend on several logically distinct ingredients. Some were proved in earlier volumes, while the remaining ingredients are proved here. Their chapter-by-chapter organization is:

Chapter 1	orbit/stabilizer, normalization, congruence orbits
Chapter 2	unfolding and uniform Sobolev periodization
Chapter 3	polar volume and effective remainder
Chapter 4	effective well-roundedness and error optimization
Chapter 5	DRS route and congruence-uniform upgrade
Chapter 6	Eskin–McMullen wavefront route
Chapter 7	EMS correction, focusing, and scope boundary
Chapter 8	stable mean ergodic and moving-family counting.

Proof architecture. The central dependency chain is

$$\text{orbit dictionary} \to \text{unfolding} \to \text{volume/boundary geometry}$$
$$\to \text{effective mixing} \to \text{counting.}$$

The alternative EMS chain is

$$\text{nonescape} \to \text{homogeneous limit} \to \text{corrected linearization}$$
$$\to \text{focusing or Haar} \to \text{counting.}$$

Neither chain implies the other beyond the hypotheses explicitly stated in the corresponding chapters.

APPENDIX B

Worked Counting Laboratories

1. Laboratory I: ordinary lattice points in a Lie group

The simplest case of the whole machine is $H = \{e\}$. Then $H\backslash G = G$ and the orbit–coset dictionary is just the lattice Γ itself.

Worked Example 1.1 (Recovering the stable mean-ergodic count)**.** Let $B_T \subset G$ be a Hölder well-rounded family and suppose the action of G on G/Γ has stable mean-ergodic rate $E(T)$. Then Theorem 1.3 gives

$$\#(\Gamma \cap B_T) = \frac{m_G(B_T)}{m_G(G/\Gamma)} \left(1 + O(E(T)^{\alpha/(\rho+\alpha)})\right)$$

whenever the right side tends to infinity fast enough that the displayed absolute error can be read relatively.

Proof. The theorem gives the absolute normalized estimate

$$\left| \frac{\#(\Gamma \cap B_T)}{m_G(B_T)} - \frac{1}{m_G(G/\Gamma)} \right| \ll E(T)^{\alpha/(\rho+\alpha)}.$$

Multiply by $m_G(B_T)$. The point of this elementary case is that every additional complication in homogeneous-variety counting comes from replacing the point orbit by a positive-dimensional H-period. □

2. Laboratory II: an indefinite quadric

Let Q be a rational nondegenerate quadratic form of signature (p, q) with $p, q \geq 1$, let

$$G = \mathrm{SO}(Q)^\circ,$$

and fix a rational vector v_0 with $Q(v_0) = m \neq 0$. The stabilizer of v_0 is an orthogonal group on $v_0^\perp$.

Worked Example 2.1 (The main-term constant on one arithmetic orbit)**.** Assume $v_0\Gamma$ is one arithmetic orbit of integral points under an arithmetic lattice $\Gamma < G$ and that $H \cap \Gamma$ has finite covolume. For the Euclidean norm region

$$B_T = \{Hg : \|v_0 g\| < T\},$$

the leading term is

$$\#\{v \in v_0\Gamma : \|v\| < T\} \sim \frac{m_H((H \cap \Gamma)\backslash H)}{m_G(\Gamma\backslash G)} m_{H\backslash G}(B_T).$$

Proof. Theorem 1.2 identifies the points with $(H \cap \Gamma)\backslash\Gamma$. The norm region is a region in $H\backslash G$. The exact Haar ratio is Theorem 2.1. Once the appropriate symmetric-space mixing and well-roundedness hypotheses are verified, Standard Result 1.2 supplies the leading asymptotic. Notice that no Euclidean "density of lattice points" is inserted by hand: the constant is forced by homogeneous measure normalization. □

Why this matters. For students, the important conceptual move is that $Q(v) = m$ is not counted as a curved subset of Euclidean space. It is counted as a homogeneous orbit. The stabilizer carries part of the volume, so the correct density is a ratio of two covolumes.

3. Laboratory III: matrices with fixed characteristic polynomial

Let $G = \mathrm{SL}_n(\mathbb{R})$ act on $\mathrm{Mat}_n(\mathbb{R})$ by conjugation. If A_0 is regular semisimple, its stabilizer is its centralizer $H = Z_G(A_0)$, a torus over a splitting field and a reductive $\mathbb{Q}$-group when A_0 is rational.

Worked Example 3.1 (Why a centralizer changes the geometry)**.** Suppose an integral characteristic polynomial determines finitely many $\mathrm{SL}_n(\mathbb{Z})$-orbits of regular semisimple integral matrices. Then bounded-norm counting reduces to finitely many homogeneous counts on $H_j\backslash G$. If a centralizer contains a noncompact split torus, neutral chamber directions appear in the polar decomposition and must be retained in the wavefront and volume analysis.

Proof. Conjugacy preserves the characteristic polynomial. On a regular semisimple orbit the stabilizer is exactly the centralizer, so Theorem 1.2 gives the homogeneous parametrization for each arithmetic orbit. The finite-orbit hypothesis permits summation by Proposition 2.1. A split factor of H centralizes part of the radial action; in the root decomposition those directions have zero weight. Treating them as expanding would destroy the uniform inverse in the wavefront map. This is precisely why the face-by-face formulation in Theorem 1.1 is needed. □

4. Laboratory IV: a focused family

The EMS hypothesis of nonfocusing is not decorative. The following toy model isolates the mechanism without reproducing the more elaborate examples from the original paper.

Worked Example 4.1 (Positive mass trapped near a proper intermediate subgroup)**.** Let $H < L < G$ be connected reductive $\mathbb{Q}$-subgroups with L proper, and assume the corresponding arithmetic homogeneous orbits have finite volume. Let $C \subset G$ be compact and choose a growing family $B_T \subset H\backslash G$ such that for some $c > 0$,

$$m_{H\backslash G}(B_T \cap q_H(LC)) \geq c\, m_{H\backslash G}(B_T)$$

for infinitely many T. Then the nonfocusing hypothesis of Corollary 3.2 fails, and ambient Haar equidistribution cannot be deduced from the EMS argument on this sequence.

Proof. The defining focusing sets in Chapter 7 allow a centralizer translate $(Z(H) \cap \Gamma)LC$; taking the identity centralizer element already contains LC. The displayed lower bound therefore gives a positive lower relative mass to a proper focusing branch. The proof of Corollary 3.2 breaks exactly at the step where one tries to discard all proper branches as $o(m(B_T))$. This does not by itself compute the alternative asymptotic; it identifies the obstruction honestly. $\square$

5. Laboratory V: optimizing three errors

Worked Example 5.1 (A concrete exponent calculation)**.** Suppose boundary regularity gives $\varepsilon^{1/2}$, cusp truncation gives R^{-1}, and the effective mixing term is

$$T^{-1/6} R^2 \varepsilon^{-3}.$$

Taking $R = T^{1/36}$ and $\varepsilon = T^{-1/54}$ yields the three powers

$$T^{-1/108}, \qquad T^{-1/36}, \qquad T^{-1/18},$$

so the final saving is $T^{-1/108}$.

Proof. The boundary exponent is $(1/2)(1/54) = 1/108$. The cusp exponent is $1/36$. For mixing,

$$\frac{1}{6} - 2 \cdot \frac{1}{36} - 3 \cdot \frac{1}{54} = \frac{1}{6} - \frac{1}{18} - \frac{1}{18} = \frac{1}{18}.$$

The minimum is $1/108$. The example illustrates why the final exponent is a bookkeeping theorem rather than simply "the spectral gap." $\square$

Bibliographic notes

The leading affine-homogeneous counting theorem originates with Duke–Rudnick–Sarnak. Eskin–McMullen developed the mixing/wavefront route for affine symmetric spaces. Eskin–Mozes–Shah developed the homogeneous-measure/focusing route; their Theorem 2.1 is used together with the authors' published correction. The abstract stable mean-ergodic-to-counting exponent follows the Gorodnik–Nevo framework.

Volume 9

Semisimple Closed Orbits and Effective Periods

Volume 9 scope and prerequisites

Proof architecture. The mathematical core has four ingredients: a uniform period spectral estimate, an arithmetic volume–complexity comparison, effective almost-invariance and closing, and finite induction over intermediate semisimple groups. The appendices prove the automorphic and arithmetic inputs needed by the four endpoint theorems. Each endpoint retains its own centralizer, adelic, and ambient-complexity hypotheses.

Student orientation: semisimple periods and complexity

A *periodic semisimple orbit* is a closed finite-volume orbit of a connected semisimple subgroup. Its *period measure* is the normalized invariant measure on that orbit. *Arithmetic complexity* is measured by a controlled package of discriminant, covolume, local level and height data; no single one of these quantities is silently substituted for another. *Almost invariance* means quantitative smallness of $g_*\mu - \mu$ in a specified Sobolev dual norm for g near the identity. *Effective closing* turns sufficiently strong almost symmetry into a nearby exact homogeneous object of bounded complexity. The four endpoint theorems have different scopes. Their common engine is proved once, but the hypotheses verifying spectral gap, centralizer control and volume–discriminant comparison are checked separately.

Reader guide: a concrete model

Why this matters.

Volume 9 is organized around the passage from *Periodic Semisimple Orbits and Homogeneous Periods* to *Four Effective Endpoints.* The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.2 (A calculation to keep in view)**.** Consider the standard unipotent and diagonal elements

$$u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad a_s = \begin{pmatrix} e^s & 0 \\ 0 & e^{-s} \end{pmatrix}.$$

A direct multiplication gives

$$a_s u_t a_{-s} = u_{e^{2s}t}.$$

Thus a displacement of size $|t|$ in the unstable horospherical direction is enlarged by e^{2s}. If $|t| = e^{-2s}$, the conjugated displacement has order one. This elementary calculation is the model for the "first visible scale" used repeatedly in nondivergence and shearing arguments.

CHAPTER 1

Periodic Semisimple Orbits and Homogeneous Periods

1. Conventions and the lattice criterion

Let G be a connected semisimple real Lie group, let $\Gamma < G$ be an arithmetic lattice, and put $X = \Gamma\backslash G$. Let $H < G$ be a connected semisimple subgroup without compact factors. All actions are on the right. For $x = \Gamma g$ put

$$H_x := \{h \in H : xh = x\} = H \cap g^{-1}\Gamma g.$$

THEOREM 1.1 (Arithmetic periodicity criterion). *Let $x = \Gamma g \in X$ and $H_x = H \cap g^{-1}\Gamma g$. The orbit map*

$$\iota_x : H_x\backslash H \longrightarrow X, \qquad H_x h \longmapsto xh,$$

is a continuous injective H-equivariant immersion. The following are equivalent:

(1) H_x is a lattice in H;
(2) xH is closed in X and carries a finite nonzero H-invariant Radon measure;
(3) xH carries an H-invariant probability measure.

When these conditions hold, ι_x is a homeomorphism and every H-invariant Radon measure on xH is a scalar multiple of the pushforward of Haar measure on $H_x\backslash H$.

PROOF. The stabilizer of the right H-action at x is exactly H_x, so the orbit map factors through a continuous bijection $H_x\backslash H \to xH$ and is a local homeomorphism in homogeneous coordinate charts. If H_x is a lattice, choose a Borel fundamental domain $F \subset H$ of finite Haar measure. For a compact $C \subset X$, properness of the quotient map on each sufficiently small coordinate chart implies that only locally finitely many translates of F contribute to $\iota_x^{-1}(C)$; hence quotient Haar measure pushes forward to an H-invariant Radon measure of finite total mass. A finite homogeneous orbit carrying such a Radon measure is closed: if $xh_n \to y$, choose a relatively compact identity neighbourhood $B \subset H$ on which the orbit map is injective. The sets xh_nB all have the same positive measure; if the cosets H_xh_n escaped every compact subset of $H_x\backslash H$, a subsequence of these translates would be disjoint, contradicting finiteness. Thus H_xh_n has a convergent subsequence and $y \in xH$. The same argument shows that ι_x^{-1} is continuous, so ι_x is a homeomorphism.

Conversely, let ν be a finite nonzero H-invariant Radon measure on the closed orbit xH. Pulling ν back through ι_x gives a finite right-H-invariant Radon measure on $H_x\backslash H$. By uniqueness of quotient Haar measure it is a positive scalar multiple of Haar measure; in particular $H_x\backslash H$ has finite Haar volume, which is precisely the statement that H_x is a lattice. Normalizing total mass gives the equivalence with (3) and the uniqueness assertion. □

Warning 1.2 (The acting group is part of the datum)**.** Replacing H by a normalizer, by an algebraic closure, or by an identity component can change the orbit, its stabilizer, and its finite component multiplicity. Later period measures always retain the actual acting group.

2. Haar probability and the period functional

Fix a left Haar measure dh on H and, for a periodic orbit, define

$$V_H(x) := \int_{H_x\backslash H} 1\, dh.$$

THEOREM 2.1 (Canonical Haar probability and period functional). *For a periodic orbit xH,*

$$\mu_{xH}(f) = \frac{1}{V_H(x)} \int_{H_x\backslash H} f(xh)\, dh, \qquad f \in C_c(X),$$

defines the unique H-invariant probability measure on xH. It is independent of rescaling dh. The corresponding period functional $P_{xH}(f) = \mu_{xH}(f)$ is positive, has norm one, and satisfies $P_{xH}(R_{h_0}f) = P_{xH}(f)$ for $h_0 \in H$.

PROOF. The quotient has finite Haar volume by Theorem 1.1. Right translation preserves quotient Haar measure, so the displayed functional is H-invariant. Rescaling dh rescales numerator and denominator by the same factor. If ν is any other H-invariant probability on xH, its pullback to $H_x\backslash H$ is a right-H-invariant finite measure. Haar uniqueness makes it a scalar multiple of quotient Haar measure, and total mass one fixes the scalar. □

Worked Example 2.2 (Compact and noncompact periods)**.** If $H_x\backslash H$ is compact, no cusp truncation is needed to integrate a continuous observable. If it is merely finite-volume, the probability is still canonical but effective estimates need a separate cusp parameter. This is the first reason period volume and dynamical height must not be identified.

3. The spectral decomposition attached to a period

Let π_x be the right regular representation of H on $L^2(H_x\backslash H)$.

Proposition 3.1 (Closed homogeneous measure and spectral splitting)**.** *There is an orthogonal H-invariant decomposition*

$$L^2(H_x\backslash H) = \mathbb{C}\mathbf{1} \oplus L^2_0(H_x\backslash H).$$

If every noncompact almost-simple factor of H acts ergodically, then the H-fixed vectors are exactly the constants.

PROOF. The constants are invariant. Their orthogonal complement is the kernel of $f \mapsto \int f\, d\mu_{xH}$ and is therefore closed and invariant. A vector fixed by H is fixed by each noncompact almost-simple factor; the assumed ergodicity forces it to be constant almost everywhere. □

Warning 3.2. This proposition does not assert a spectral gap uniform over a family of periodic orbits. That is the independent Chapter-4 root `V09.C04.01`.

4. Translation, conjugation, and centralizer motion

For $a \in G$ write $H^a = a^{-1}Ha$.

THEOREM 4.1 (Conjugation formula for periods). *If $x = \Gamma g$, then*

$$R_a(xH) = xaH^a, \qquad (H^a)_{xa} = a^{-1}H_x a.$$

If xH is periodic, then so is xaH^a and $(R_a)_*\mu_{xH} = \mu_{xaH^a}$ when Haar measure on H^a is the pushforward of Haar measure on H.

PROOF. The identity $xha = xa(a^{-1}ha)$ gives the orbit formula. The stabilizer formula follows by testing when $xa(a^{-1}ha) = xa$. Conjugation carries the lattice H_x isomorphically to $a^{-1}H_x a$, and the probability formula then follows by change of variables. □

Corollary 4.2 (Centralizer translates preserve intrinsic period volume)**.** *If $c \in Z_G(H)$, then $R_c(xH) = xcH$, $H_{xc} = H_x$, and $V_H(xc) = V_H(x)$ with the same Haar normalization. The subsets xH and xcH need not coincide.*

PROOF. Now $H^c = H$ and $c^{-1}H_x c = H_x$; apply Theorem 4.1. Equality of the orbit subsets would require the additional condition $xc \in xH$. □

5. Real, congruence, and adelic components

Let $\mathbf{G}$ be a connected semisimple $\mathbb{Q}$-group, put $G_\infty = \mathbf{G}(\mathbb{R})^\circ$, and let $K_f < \mathbf{G}(\mathbb{A}_f)$ be compact open. Choose representatives $g_1, \dots, g_r$ for the finite class set $\mathbf{G}(\mathbb{Q})\backslash\mathbf{G}(\mathbb{A}_f)/K_f$ and put $\Gamma_i = \mathbf{G}(\mathbb{Q}) \cap g_i K_f g_i^{-1}$.

THEOREM 5.1 (Adelic-to-real component dictionary). *Assuming the class set above is finite,*

$$\mathbf{G}(\mathbb{Q})\backslash\mathbf{G}(\mathbb{A})/K_f \cong \bigsqcup_{i=1}^{r} \Gamma_i\backslash G_\infty$$

up to the finite bookkeeping of $\mathbf{G}(\mathbb{R})/G_\infty$. *If* $K_f' \subset K_f$, *the level map restricts on real components to finite congruence covers. An adelic homogeneous subset attached to a semisimple* $\mathbb{Q}$*-subgroup decomposes into finitely many real periodic components whenever the corresponding real stabilizer intersections are lattices.*

PROOF. Write every finite adelic component as qg_ik_f with $q \in \mathbf{G}(\mathbb{Q})$ and $k_f \in K_f$. Left multiplication by q^{-1} and right multiplication by k_f^{-1} place the point in the ith real slice. Two points in that slice are equivalent precisely when their real coordinates differ by $\mathbf{G}(\mathbb{Q})\cap g_iK_fg_i^{-1}$. Repeating the same double-coset argument inside the adelic H-orbit gives the homogeneous decomposition. For $K_f' \subset K_f$, the stabilizers have finite index because K_f' has finite index in K_f. ☐

Worked Example 5.2 (Centralizer motion in a product). For $G = G_1 \times G_2$, $\Gamma = \Gamma_1 \times \Gamma_2$, and $H = G_1 \times \{e\}$, the centralizer contains $\{e\} \times G_2$. Centralizer translates have identical intrinsic stabilizer and period volume, while their ambient location varies in $\Gamma_2\backslash G_2$.

How the three descriptions of a period fit together

There are three descriptions of the same object in this chapter, and the rest of the volume repeatedly changes from one to another. The first is the *geometric orbit* $xH \subset X$. The second is the quotient $H_x\backslash H$, which is the correct object for normalization and spectral analysis. The third is the adelic component description, which is the right one for congruence uniformity and property τ. None of the three may be silently substituted for another: each carries bookkeeping that the other two suppress.

The quotient description explains why the period probability is canonical. A choice of Haar measure on H is auxiliary; it disappears after division by $V_H(x)$. By contrast, the *unnormalized* volume $V_H(x)$ is arithmetic information and is deliberately retained as a separate complexity variable. Thus normalization removes an arbitrary scalar without erasing the size of the period. This distinction becomes essential in Chapter 2, where volume, discriminant, and cusp height are tracked independently.

The spectral splitting has a similarly limited purpose. It isolates the constants and identifies the representation on which decay must eventually be proved. It does not manufacture decay. In particular, the passage

$$L^2(H_x\backslash H) = \mathbb{C}\mathbf{1} \oplus L_0^2(H_x\backslash H)$$

is formal, whereas a family-uniform estimate on the second summand is the substantive automorphic input isolated in Chapter 4. Keeping those two steps separate is one of the safeguards against using property (T) as a slogan for pointwise matrix-coefficient decay.

The translation formula is equally structural. Right translation by an arbitrary $a \in G$ changes both the point and the acting subgroup: H becomes $H^a = a^{-1}Ha$. Only for a in the centralizer does the acting subgroup remain literally the same. This is why a centralizer family can move through the ambient quotient while keeping the intrinsic stabilizer and intrinsic volume fixed. Later, when positive-dimensional centralizers are allowed, that simple observation becomes a genuine obstruction to the finite-centralizer proof.

Finally, the adelic-to-real dictionary is finite-component bookkeeping, not a claim of connectedness. The finite class-set representatives record which real arithmetic lattices occur. Passing to a smaller compact open subgroup changes these lattices by finite-index congruence covers, so any uniformity claim must say whether its constants are allowed to depend on that level. This is the first place in Volume 9 where "fixed ambient space" and "uniform in the varying period" begin to diverge as mathematical statements.

CHAPTER 2

Volume, Height, Discriminant, and Arithmetic Complexity

1. Three complexity parameters, not one

A periodic semisimple orbit carries at least three distinct quantitative data: its intrinsic Haar volume, an arithmetic discriminant measuring the rational position of its Lie algebra, and an ambient cusp height. The later effective proofs fail if these are silently interchanged.

Let $\mathbf{G} < \mathrm{SL}_N$ be a connected semisimple $\mathbb{Q}$-group. Fix a $\mathbb{Z}$-lattice $\mathfrak{g}_{\mathbb{Z}} \subset \mathfrak{g}_{\mathbb{Q}}$ and Euclidean norms on all exterior powers. For a connected rational subgroup $\mathbf{M}$ of dimension r, let v_M be a primitive integral generator, unique up to sign, of

$$(\wedge^r \mathfrak{m}_{\mathbb{Q}}) \cap (\wedge^r \mathfrak{g}_{\mathbb{Z}}).$$

Lemma 1.1 (Bounded-height rational subgroup finiteness)**.** *Fix the integral lattice $\mathfrak{g}_{\mathbb{Z}} \subset \mathfrak{g}_{\mathbb{Q}}$ and an integer $0 \le r \le \dim G$. For every $B \ge 1$ there are only finitely many rational r-planes $L \subset \mathfrak{g}_{\mathbb{Q}}$ whose primitive Plücker vector $v_L \in \bigwedge^r \mathfrak{g}_{\mathbb{Z}}$ satisfies $\|v_L\| \le B$. Consequently there are only finitely many connected $\mathbb{Q}$-subgroups $M < G$ of dimension r and height at most B. Moreover, for fixed k there is $C = C(G, k)$ such that if rational Lie subalgebras $\mathfrak{m}_1, \ldots, \mathfrak{m}_k$ have height at most B, then the Lie algebra generated by them has height at most B^C.*

Proof. A primitive Plücker vector is a nonzero integral vector, defined up to sign. There are finitely many integral vectors in the Euclidean ball of radius B, proving the first assertion. A connected algebraic subgroup is determined by its Lie algebra in characteristic zero, so the subgroup finiteness follows.

For the height estimate, choose for each $\mathfrak{m}_i$ an integral basis by putting a basis matrix into Hermite normal form. The entries of such a basis are bounded by $B^{O_G(1)}$ because its maximal minors are the Plücker coordinates. Starting from the union of these bases, repeatedly adjoin brackets $[X, Y]$. Structure constants of the fixed integral Lie algebra are fixed integers, so every newly adjoined vector has coordinates $B^{O_G(1)}$. Dimension can increase at most $\dim \mathfrak{g}$ times; hence a basis of the generated Lie algebra is obtained after a bounded number of bracket stages. Its maximal minors are therefore bounded by B^C for a constant depending only on G and k, which is the claimed Plücker-height bound. □

2. The exterior discriminant

Definition 2.1 (Exterior orbit map and discriminant)**.** For $g \in G$ define

$$\eta_M(g) := (\wedge^r \operatorname{Ad})(g^{-1})v_M.$$

For a periodic orbit represented as $\Gamma M(\mathbb{R})^\circ g$, set

$$\operatorname{disc}(\Gamma M(\mathbb{R})^\circ g) := \|\eta_M(g)\|.$$

Proposition 2.2 (Orbit invariance)**.** *If $\gamma \in \Gamma$ and the fixed integral structure is Γ-stable, then*

$$\eta_{\gamma M \gamma^{-1}}(\gamma g) = \pm \eta_M(g).$$

Thus the discriminant depends only on the periodic orbit up to Γ-conjugacy.

Proof. The primitive line for $\gamma M \gamma^{-1}$ is $(\wedge^r \operatorname{Ad})(\gamma)$ applied to the primitive line for M. Since $\operatorname{Ad}(\gamma)$ preserves the fixed integral lattice, primitive vectors are carried to primitive vectors up to sign. Substituting into Definition 2.1 gives the formula. □

Proposition 2.3 (Comparison with the EMV squared-Pluecker convention)**.** *Fix the ambient integral model and dimension. On the semisimple locus, the exterior discriminant and the EMV discriminant defined from the squared Pluecker vector and the Killing determinant determine one another up to fixed positive powers and fixed multiplicative constants.*

Proof. In a fixed integral basis, the Killing determinant of a rational r-plane is a homogeneous rational polynomial in its Pluecker coordinates. On the semisimple locus it is nonzero. The EMV tensor is obtained from the squared Pluecker vector by a fixed rational polynomial normalization, and the Pluecker line is recovered rationally on that Zariski-open locus. Heights under fixed rational maps and their inverses vary by fixed powers in fixed dimension. Applying this to the two projective coordinates proves the claim. □

3. The volume–discriminant gate

Theorem 3.1 (Two-sided period complexity). *Under the fixed finite-centralizer arithmetic hypotheses of this volume, there exist constants $a, b, c, C > 0$, depending only on the fixed ambient arithmetic data, such that every periodic semisimple orbit xH satisfies*

$$c \operatorname{disc}(xH)^a \le \operatorname{vol}(xH) \le C \operatorname{disc}(xH)^b.$$

Proof. For the lower inequality, Appendix C proves polynomially short Zariski generators from the period covolume using the uniform spectral theorem of Appendix D; the fixed-space height estimate then yields $\mathrm{ht}(\mathrm{Ad}(g)\mathfrak{h}) \ll \mathrm{vol}(xH)^{C_1}$. Proposition 2.1 identifies this height, up to fixed powers, with $\mathrm{disc}(xH)^{1/2}$, giving the lower bound after rearrangement.

For the upper inequality, Theorem 1.3 derives the arithmetic covolume formula from the factor-field/bad-place bounds and the selected Prasad–Borel–Prasad local product. Every field discriminant, bad-prime factor, and central/form index is a fixed power of the primitive Pluecker height, hence a fixed power of $\mathrm{disc}(xH)$. This gives the upper bound. The two directions therefore do not rely on reversing one another. □

4. Minimal intermediate complexity

Let $\mathcal{I}_H(x)$ be the chosen family of proper connected rational subgroups M with $H \subset M(\mathbb{R})^\circ$ and $xM(\mathbb{R})^\circ$ periodic. Put

$$D_H(x) := \inf_{M \in \mathcal{I}_H(x)} \mathrm{disc}(xM),$$

with $D_H(x) = +\infty$ if the family is empty.

Proposition 4.1 (Monotonicity of minimal intermediate complexity)**.** *If $H \subset S$ and the obstruction family for S is the subfamily of $\mathcal{I}_H(x)$ consisting of groups containing S, then $D_S(x) \geq D_H(x)$.*

Proof. The infimum defining D_S is taken over a subset of the family defining D_H. An infimum over a smaller set cannot decrease. □

Warning 4.2 (No fake discriminant monotonicity)**.** The Lie algebras of H and S have different dimensions and lie in different exterior representations. There is no formal inequality between $\mathrm{disc}(xH)$ and $\mathrm{disc}(xS)$. Only the minimum over nested obstruction families is monotone.

5. Dynamical cusp height is independent information

Let $\mathrm{ht} : X \to [1, \infty)$ be a proper height as in AHD I and set

$$\mathrm{minht}(xH) := \inf_{y \in xH} \mathrm{ht}(y).$$

Proposition 5.1 (Discriminant and cusp height are distinct)**.** *There is no general identity reducing $(\mathrm{disc}(xH), \mathrm{minht}(xH))$ to one scalar invariant. In a product quotient, a centralizer translate can preserve the intrinsic stabilizer and the exterior discriminant while moving the period arbitrarily far in an independent ambient factor.*

PROOF. Use the product model of Chapter 1 with $H = G_1 \times \{e\}$ and move by a centralizer element (e, c_2). The H-stabilizer and rational Lie algebra are unchanged, whereas the ambient point in $\Gamma_2 \backslash G_2$ may approach the cusp. $\square$

6. Congruence level and ambient complexity

Definition 6.1 (Four-column constant ledger)**.** Every quantitative theorem in this volume records constants according to four categories: fixed ambient real geometry; intrinsic period data; finite-level/congruence data; and ambient arithmetic complexity when the ambient adelic space itself varies.

Proposition 6.2 (Compatibility of the complexity ledgers)**.** *For a fixed ambient adelic space, passing among its finitely many real components and finite congruence covers changes Sobolev norms and component weights by controlled finite-index factors. This is different from varying the ambient arithmetic form, whose complexity must be recorded separately.*

PROOF. The real-component decomposition of Theorem 5.1 is finite. At a fixed ambient space, changing compact open level changes stabilizers by finite index and local coordinate charts by fixed algebraic maps. Consequently all resulting norm and measure comparisons are governed by the level/component data. If the ambient rational group or integral model varies, the defining equations, local bad places, and Haar normalizations may also vary, so a separate ambient-complexity parameter is logically necessary. $\square$

Worked Example 6.3 (Why the minimum obstruction is the right variable)**.** If the only proper intermediate groups are $H \subsetneq S_1 \subsetneq S_2$ and an incomparable $H \subsetneq T$, then after descent to S_1 the group T disappears from the obstruction family. Thus the minimum obstruction complexity can only improve, even though there is no pointwise monotonicity between the discriminants of individual periods.

Complexity parameters and the direction of every comparison

The surviving Stage-3 manuscript deliberately treated the three principal complexities as coordinates rather than as competing definitions of one quantity. The reason is logical, not terminological. Different parts of the proof consume different coordinates. Local charts and truncation consume a cusp-height bound; algebraic closing produces rational subgroup height; endpoint theorems are usually stated in terms of period volume or an exterior (or adelic) discriminant. A complete proof must therefore record the arrows that permit one coordinate to be replaced by another.

The exterior discriminant is chosen because it is functorial under the adjoint action and because primitive rational Pluecker lines are amenable to height arguments. It is not asserted to be numerically equal to the EMV normalization. The comparison in Proposition 2.3 is only a fixed-power comparison in a fixed rational representation. Consequently, whenever an exponent is transported from one convention to the other, the power loss must be absorbed explicitly into the final exponent ledger.

The volume–discriminant gate is intentionally asymmetric. The lower bound

$$\mathrm{vol}(xH) \gg \mathrm{disc}(xH)^c$$

comes from finding sufficiently short lattice elements and proving that they already generate the required algebraic group. The reverse inequality requires arithmetic volume formulae and control of global and local form data. These are different proofs. In particular, the statement of a two-sided comparison is not itself a proof that both directions have been supplied. This is why the later repair splits the parent obligation into distinct lower- and upper-bound owners.

Minimal intermediate complexity is also directional. If $H \subset S$, the set of proper intermediate groups containing S is a subset of the set of proper intermediate groups containing H; hence the minimum obstruction complexity can only increase as the current acting group is enlarged. There is no analogous formal monotonicity for the discriminant of the orbit xH versus that of xS. The groups have different dimensions, different primitive exterior vectors, and independently normalized Haar measures. Any argument that writes such a monotonicity without an arithmetic theorem is therefore using a false shortcut.

The same warning applies to cusp height. The product example from Chapter 1 shows that centralizer motion can drive a period into a cusp while leaving its intrinsic stabilizer and rational exterior data unchanged. Conversely, rational subgroup complexity can grow while the corresponding period remains in a fixed compact height window. A modern noncompact effective theorem must therefore carry at least two independent parameters: one for ambient escape and one for rational intermediate obstructions.

For later uniformity statements it is useful to keep a four-column ledger: ambient group/field complexity, level complexity, intrinsic period complexity, and dynamical cusp height. A constant declared “uniform” is meaningful only after specifying which columns are frozen. Fixed-ambient adelic uniformity, fixed-lattice real uniformity, and uniformity through a congruence tower are three different assertions. Chapter 6 and Chapter 8 will repeatedly refer back to this bookkeeping.

CHAPTER 3

Intermediate Subgroup Geometry

1. The finite-centralizer regime

Let $H < G$ be connected semisimple without compact factors and suppose $\mathfrak{z}_{\mathfrak{g}}(\mathfrak{h}) = 0$. This infinitesimal condition is implied by finiteness of $Z_G(H)$ and is the form used in the Lie-algebra arguments.

Lemma 1.1 (No proper parabolic contains H)**.** *Under* $\mathfrak{z}_{\mathfrak{g}}(\mathfrak{h}) = 0$*, the subgroup* H *is not contained in a proper parabolic subgroup of* G*.*

Proof. If $H \subset P$ and $N = R_u(P)$, the reductive quotient P/N of a proper parabolic has a positive-dimensional central split torus. Hence the H-module $\mathfrak{p}/\mathfrak{n}$ has a nonzero fixed vector. Complete reducibility of the $\mathfrak{h}$-module $\mathfrak{p}$ lifts a fixed vector to $\mathfrak{p}$, producing a nonzero element of $\mathfrak{z}_{\mathfrak{g}}(\mathfrak{h})$, contradiction. □

Proposition 1.2 (Intermediate connected groups are semisimple)**.** *If* $H \subset S \subset G$ *is connected algebraic, then* S *is semisimple and has no nontrivial connected compact normal factor.*

Proof. A nontrivial unipotent radical would place S, and hence H, in a proper parabolic, contradicting Lemma 1.1. Thus S is reductive. Its connected radical is a central torus and therefore centralizes H; the zero-centralizer hypothesis forces that torus to be trivial. A connected compact normal factor is likewise centralized by every noncompact simple factor of H; since H has no compact factors its Lie algebra would lie in the centralizer of $\mathfrak{h}$, again impossible. □

2. Rigidity and finiteness of intermediate subgroups

Lemma 2.1 (Whitehead's first lemma in the needed form)**.** *Let* $\mathfrak{s}$ *be finite-dimensional semisimple over characteristic zero and* V *a finite-dimensional* $\mathfrak{s}$*-module. Every cocycle* $a : \mathfrak{s} \to V$ *satisfying*

$$a([X, Y]) = X \cdot a(Y) - Y \cdot a(X)$$

is a coboundary.

Proof. After scalar extension and complete reducibility it suffices to treat an irreducible module. The trivial module case follows from $[\mathfrak{s},\mathfrak{s}]=\mathfrak{s}$. In the nontrivial case choose dual Killing-form bases X_i, Y_i and let $\Omega_V = \sum_i \rho(X_i)\rho(Y_i) = c_V I$. The Casimir eigenvalue c_V is nonzero on a nontrivial irreducible. Set

$$v = c_V^{-1}\sum_i \rho(X_i)a(Y_i).$$

Using the cocycle identity and invariance of $\sum_i X_i \otimes Y_i$ under the diagonal adjoint action gives $c_V\rho(X)v = c_V a(X)$ for every X, hence $a(X) = X\cdot v$. $\square$

Lemma 2.2 (Tangent space to the fixed-$\mathfrak{h}$ subalgebra variety)**.** *Fix $\mathfrak{h}\subset\mathfrak{s}\subset\mathfrak{g}$ with $\mathfrak{s}$ semisimple of dimension d. If $\mathcal{V}_d$ denotes the algebraic subset of $\mathrm{Gr}_d(\mathfrak{g}_\mathbb{C})$ consisting of Lie subalgebras containing $\mathfrak{h}_\mathbb{C}$, then*

$$T_{\mathfrak{s}_\mathbb{C}}\mathcal{V}_d \cong \{a\in Z^1(\mathfrak{s}_\mathbb{C},\mathfrak{g}_\mathbb{C}/\mathfrak{s}_\mathbb{C}) : a|_{\mathfrak{h}_\mathbb{C}} = 0\}.$$

Proof. The tangent space to the Grassmannian is $\mathrm{Hom}(\mathfrak{s}_\mathbb{C},\mathfrak{g}_\mathbb{C}/\mathfrak{s}_\mathbb{C})$. A first-order deformation is the graph of $ta + O(t^2)$. Containing the fixed subspace gives $a|_\mathfrak{h} = 0$. Expanding bracket closure to first order gives exactly the cocycle equation, and conversely that equation is the first-order bracket-closure condition. $\square$

Theorem 2.3 (Finite-centralizer finiteness of intermediate groups). *If $\mathfrak{h}\subset\mathfrak{g}$ is semisimple and $\mathfrak{z}_\mathfrak{g}(\mathfrak{h}) = 0$, then there are only finitely many Lie subalgebras $\mathfrak{h}\subset\mathfrak{s}\subset\mathfrak{g}$ and hence only finitely many connected intermediate Lie subgroups $H\subset S\subset G$.*

Proof. By Proposition 1.2, every intermediate $\mathfrak{s}$ is semisimple. At a point of $\mathcal{V}_d$, Whitehead's lemma makes every tangent cocycle a coboundary $a(X) = X\cdot\bar{Z}$. Complete reducibility provides an $\mathfrak{h}$-invariant complement $\mathfrak{g}_\mathbb{C} = \mathfrak{s}_\mathbb{C}\oplus\mathfrak{r}$. The condition $a|_\mathfrak{h} = 0$ says that the corresponding vector of $\mathfrak{r}$ is $\mathfrak{h}$-fixed, hence belongs to $\mathfrak{z}_{\mathfrak{g}_\mathbb{C}}(\mathfrak{h}_\mathbb{C}) = 0$. Thus every Zariski tangent space is zero. The conditions "contains $\mathfrak{h}$" and "closed under bracket" cut out a projective algebraic subset of the Grassmannian; a zero-dimensional projective variety of finite type has finitely many points. There are only finitely many possible dimensions. $\square$

3. H-modules transverse to an intermediate group

Lemma 3.1 (H-module decomposition and centralizer directions)**.** *For $\mathfrak{h}\subset\mathfrak{s}\subset\mathfrak{g}$ there is an H-invariant complement $\mathfrak{g}=\mathfrak{s}\oplus\mathfrak{r}$ and a decomposition*

$$\mathfrak{r} = \mathfrak{r}^H\oplus\mathfrak{r}_{\mathrm{nt}}, \qquad \mathfrak{r}^H = \mathfrak{r}\cap\mathfrak{z}_\mathfrak{g}(\mathfrak{h}),$$

where $\mathfrak{r}_{\mathrm{nt}}$ is a sum of nontrivial irreducible H-modules. In the finite-centralizer regime $\mathfrak{r}^H = 0$.

Proof. Complete reducibility gives the first complement and then splits $\mathfrak{r}$ into its trivial isotypic part and the sum of nontrivial irreducibles. The trivial isotypic part is exactly the centralizer intersection. □

Worked Example 3.2 (Why positive centralizer changes the proof)**.** For $G = G_1 \times G_2$ and $H = G_1 \times \{e\}$, the whole Lie algebra $\mathfrak{g}_2$ is H-fixed. A transverse displacement there merely moves the period through a centralizer family and need not generate a larger semisimple group.

4. Generating a larger Lie algebra

Theorem 4.1 (Generation from a noncentral transverse direction). *Let $\mathfrak{h} \subset \mathfrak{s} \subset \mathfrak{g}$ and choose an H-invariant complement. If $0 \neq Z$ lies in a nontrivial irreducible H-summand transverse to $\mathfrak{s}$, let W_Z be the span of its H-orbit. Then the Lie algebra generated by $\mathfrak{s}$ and W_Z is strictly larger than $\mathfrak{s}$ and is H-stable.*

Proof. The orbit span W_Z is a nonzero H-submodule contained in the chosen complement, hence $W_Z \cap \mathfrak{s} = 0$. The Lie algebra generated by $\mathfrak{s} \cup W_Z$ therefore contains $\mathfrak{s}$ properly. Since both $\mathfrak{s}$ and W_Z are stable under $\mathrm{Ad}(H)$, so is the Lie algebra they generate. □

5. Nested periodic periods and normalization

Proposition 5.1 (Nested periodic orbits: exact stabilizers)**.** *Let $H \subset S \subset G$ be connected and let $x = \Gamma g$. If both xH and xS are periodic, then*

$$H_x = H \cap g^{-1}\Gamma g, \qquad S_x = S \cap g^{-1}\Gamma g,$$

and the normalized probabilities are obtained independently from the quotient Haar measures on $H_x\backslash H$ and $S_x\backslash S$. There is no formal monotonicity between their intrinsic volumes without a compatible Haar normalization theorem.

Proof. The stabilizer identities follow directly from the right action. Apply the periodicity criterion separately to H and S. Since the groups have different dimensions, their Haar normalizations are independent data. □

6. The descent poset

Proposition 6.1 (Finite-centralizer obstruction poset)**.** *In the finite-centralizer regime, the connected intermediate periodic groups that can arise in the effective descent form a finite poset under inclusion. Every genuine invariance enlargement moves strictly upward and therefore the process terminates.*

PROOF. Finiteness is Theorem 2.3. A genuine enlargement contains the current group properly, so a branch cannot revisit a previous node. Any strict chain also increases Lie dimension. □

7. Rationality and generated-subgroup height

Proposition 7.1 (Semisimple-period translation of the Ratner height owner). *Let S and M be rational intermediate subgroups represented by bounded rational Chevalley/Pluecker data. Then the class-$\mathcal{H}$ part of the Zariski group generated by S and M is rational, and its primitive Pluecker height is bounded polynomially in the input heights.*

PROOF. This is the generated-subgroup height theorem of Ratner XIII applied in the fixed rational representation. The only translation is from Ratner's Chevalley height to the exterior height of this volume. In fixed dimension, those rational projective coordinates compare by fixed powers, exactly as in Proposition 2.3. □

Why finite centralizer gives a finite algebraic obstruction list

The centralizer hypothesis is used twice, in two conceptually different ways. First it removes trivial transverse H-modules. A displacement in a trivial module is fixed by conjugation and therefore produces centralizer motion rather than polynomial shear. Second, and more subtly, it makes the parameter space of intermediate Lie algebras rigid. The finite list used in EMV descent is not a compactness statement about subgroups; it is an infinitesimal rigidity statement followed by projective algebraic geometry.

Here is the deformation argument in slow motion. Fix an intermediate semisimple algebra $\mathfrak{s}$ and its dimension d. A nearby d-plane in the Grassmannian can be written, after choosing a complement, as the graph of a small linear map

$$a : \mathfrak{s}_{\mathbb{C}} \longrightarrow \mathfrak{g}_{\mathbb{C}}/\mathfrak{s}_{\mathbb{C}}.$$

To keep the fixed subgroup H inside the deformation one must have $a|_{\mathfrak{h}_{\mathbb{C}}} = 0$. To remain a Lie algebra to first order one expands $[X + ta(X), Y + ta(Y)]$ modulo the moving graph. The coefficient of t is exactly the cocycle equation

$$a([X, Y]) = X \cdot a(Y) - Y \cdot a(X).$$

Thus tangent vectors are relative Lie-algebra cocycles, not arbitrary Grassmannian directions.

Whitehead's lemma now converts a tangent cocycle into an infinitesimal conjugation: $a(X) = X \cdot \bar{Z}$. The condition $a|_{\mathfrak{h}} = 0$ says that $\bar{Z}$ is fixed by $\mathfrak{h}$. After choosing an $\mathfrak{h}$-invariant complement, such a fixed class is represented by an element of $\mathfrak{z}_{\mathfrak{g}}(\mathfrak{h})$. The zero-centralizer hypothesis therefore kills the tangent direction completely. Every point of the fixed-$\mathfrak{h}$ subalgebra variety is isolated. Because the conditions "contains $\mathfrak{h}$" and "is closed under bracket" are algebraic conditions on a projective Grassmannian, isolatedness implies that only finitely many such points occur.

This proof explains why the result fails in a positive-dimensional centralizer family. There, the same tangent calculation produces genuine fixed directions and the subalgebra parameter space need not be discrete. The later centralizer-aware theory must replace finiteness by a complexity parameter for intermediate periods rather than pretending that the classical finite list still exists.

The transverse-module decomposition then determines what a useful nearby pair looks like. Write

$$\mathfrak{g} = \mathfrak{s} \oplus \mathfrak{r}^H \oplus \mathfrak{r}_{\mathrm{nt}}.$$

A component in $\mathfrak{r}^H$ is dynamically inert under H-conjugation. A nonzero component in $\mathfrak{r}_{\mathrm{nt}}$, on the other hand, generates a nontrivial H-submodule. Adjoining that module to $\mathfrak{s}$ and closing under brackets yields a strictly larger H-stable Lie algebra. This is the algebraic endpoint of the polynomial shearing calculation in Chapter 4.

Two pieces of bookkeeping remain independent of the Lie-algebra argument. Periodicity of xH and xS is checked separately through their lattice stabilizers; inclusion of acting groups does not identify their normalized Haar probabilities. Rationality and height of the generated obstruction are likewise arithmetic statements. They are supplied by the Volume 45 generated-subgroup height theorem and translated into the exterior-height convention of Chapter 2. Keeping these steps separate prevents the finite poset argument from silently absorbing an arithmetic height theorem.

Once these distinctions are made, finite descent is elementary. The set of proper intermediate connected groups is a finite poset. Every genuine invariance enlargement moves strictly upward, so it cannot cycle. Dimension already bounds the length of a chain; the finite-poset formulation is sharper because it records exactly which obstruction types remain after each successful enlargement.

CHAPTER 4

Almost Invariance from Large Periods: the Algebraic Engine

1. The spectral root is separate

The effective argument needs more than property (T) for one fixed lattice. It needs a spectral statement uniform over the family of periodic semisimple quotients that arise during descent.

THEOREM 1.1 (Uniform spectral gap on periodic H-orbits). *In the fixed EMV ambient arithmetic setting there exists an integer $m_0 \geq 1$ such that, for every periodic semisimple orbit xH occurring in the finite-descent family, the representation of every noncompact simple factor of H on $L^2_0(H_x\backslash H)$ has m_0-fold tensor power weakly contained in the corresponding regular representation. Consequently there exist $\sigma > 0$ and a Sobolev order d such that, uniformly in the period,*

$$|\langle u_t f_0, f_0\rangle| \ll (1+|t|)^{-\sigma}\mathcal{S}_d(f)^2, \qquad f_0 = f - \mu_{xH}(f),$$

for each fixed unipotent one-parameter subgroup used in the shearing argument.

PROOF. Appendix D proves a uniform tensor-tempered exponent for every absolute-type-A_1 factor by the reconstructed local $\mathrm{GL}_2 \times \mathrm{GL}_2$ Rankin–Selberg theorem and specialized Jacquet–Langlands transfer. The higher-rank factors are handled by the direct spectral-transport branch recorded in Appendix B. Taking the maximum tensor exponent over the finite factor list gives one m_0.

Restriction of a tempered tensor power to the fixed semisimple subgroup remains tempered. The Cowling–Haagerup–Howe estimate bounds K-finite matrix coefficients by a fixed power of the Harish–Chandra function Ξ. Along the chosen embedded SL_2 unipotent, $\Xi(u_t) \ll (1+|t|)^{-c}$; after taking the m_0-th tensor root and summing the K-types with the compact Sobolev estimates of Volume 5, one obtains the displayed decay with uniform σ, d. $\square$

2. Local thickening around a period

Proposition 2.1 (Uniform local thickening around a period). *Fix an ambient height window and a Sobolev order d. There exist local product coordinates around each point of a periodic xH in that window and a family of smooth transverse bumps Φ_ε such that integrating against Φ_ε converts the singular period measure into an ambient test function with*

$$\mathcal{S}_d(\Phi_\varepsilon) \ll \varepsilon^{-A_d}\,\mathcal{C}(xH)^{B_d},$$

where $\mathcal{C}(xH)$ records the period-volume/height data required by the chosen chart. The constants are uniform in the fixed ambient family.

Proof. Apply the quantitative tubular-neighborhood construction and weighted Sobolev product estimates of AHD VII. On a fixed height window the injectivity radius and Jacobian derivatives have polynomial control. The only new bookkeeping is that the support of the period bump is normalized by the period probability rather than by ambient Haar measure, so the volume factor is kept explicitly in $\mathcal{C}(xH)$. □

3. Almost invariance: definitions and stability

Say that a probability measure ν is $[X, \epsilon, d]$-almost invariant if for all smooth compactly supported f,

$$|\nu(\exp(X)f) - \nu(f)| \le \epsilon\,\mathcal{S}_d(f).$$

Lemma 3.1 (Perturbing the direction). *If ν is $[X, \epsilon, d]$-almost invariant and $\|X - Y\| \le \delta$ in a fixed local exponential chart, then ν is $[Y, C(\epsilon + \delta), d + 1]$-almost invariant.*

Proof. Write $\exp(Y) = \exp(X)\exp(E)$ with $\|E\| \ll \delta$ by the local BCH map. The first-order Taylor formula for $f(\cdot \exp(E))$ costs one derivative and $O(\delta)\mathcal{S}_{d+1}(f)$; combine with the assumed invariance. □

Lemma 3.2 (Sums of almost-invariant directions). *If X and Y have bounded norm and are almost-invariant directions with errors ϵ_X, ϵ_Y, then $X + Y$ is an almost-invariant direction with error $O(\epsilon_X + \epsilon_Y)$ after increasing Sobolev order by a fixed amount.*

Proof. Use BCH to compare $\exp(X + Y)$ with $\exp(X)\exp(Y)$ and absorb the quadratic commutator error by the perturbation lemma after working at the small scale used in the effective argument. □

Lemma 3.3 (Commutators of almost-invariant directions). *At scale r, almost invariance under rX and rY yields almost invariance under $r^2[X, Y]$, with a polynomial loss in r^{-1} and a fixed Sobolev-order increase.*

Proof. The group commutator $\exp(rX)\exp(rY)\exp(-rX)\exp(-rY)$ equals $\exp(r^2[X,Y]+O(r^3))$. Apply the four almost-invariance estimates and then the perturbation lemma. $\square$

4. Amplification to a larger intermediate group

Theorem 4.1 (One transverse direction amplifies to a larger Lie algebra). *Let S be the current intermediate semisimple group and suppose a period measure is quantitatively S-almost invariant and also almost invariant under a noncentral transverse direction Z. Then it is almost invariant, with an explicit polynomially degraded exponent, under every unit vector in the Lie algebra generated by $\mathfrak{s}$ and the H-orbit span of Z.*

Proof. Theorem 4.1 gives a finite-dimensional H-stable Lie algebra strictly containing $\mathfrak{s}$. A bounded basis of its generated Lie words has uniformly bounded bracket length because the ambient Lie algebra is fixed. Propagate almost invariance through H-conjugates, sums, and commutators using the preceding lemmas. Since only finitely many operations are needed, the final error remains a fixed positive power of the initial error. $\square$

Proposition 4.2 (Finite iteration preserves a polynomial saving)**.** *Suppose each invariance enlargement replaces an error parameter ϵ by at most $C\epsilon^\kappa$ and multiplies arithmetic complexity by a fixed power. Over a chain of at most $\dim G - \dim H$ strict enlargements, the final error is still a positive power of the initial error and the final complexity is polynomial in the initial complexity.*

Proof. A bounded composition of maps $t \mapsto Ct^\kappa$ is bounded by $C't^{\kappa'}$ for some $\kappa' > 0$. A bounded composition of polynomial maps is polynomial. The chain length is bounded by dimension, or more sharply by the finite obstruction poset in the finite-centralizer case. $\square$

5. Exponent bookkeeping

The purpose of the almost-invariance chapter is not merely to show that a small error stays small. It must show that a *power saving* survives all operations needed to manufacture a larger invariance group. There are four recurrent sources of loss: Sobolev order, local chart size, arithmetic/height complexity, and the number of Lie-algebra operations. The last is harmless because the ambient Lie algebra is fixed; the first three must be recorded.

Suppose an input error is ϵ. Perturbing a direction costs a fixed number of derivatives and adds the geometric perturbation scale. Passing to sums is achieved at a smaller group scale and therefore replaces ϵ by a fixed positive power. A commutator is visible only at quadratic scale, so it causes another fixed power loss. Because a basis of the Lie algebra generated by $\mathfrak{s}$ and one transverse H-module can be obtained by a bounded number of sums and brackets, the whole enlargement has the schematic form

$$\epsilon \longmapsto C\epsilon^{\kappa}, \qquad \kappa > 0,$$

with C and κ depending only on the fixed ambient data. Iterating over a bounded chain still leaves a positive power. This is the quantitative content of Proposition 4.2.

6. Relative traces and effective generic points

The spectral input needed below is used through one concrete consequence. Fix a one-parameter unipotent subgroup $U = \{u_t\} \subset H$. The desired uniform spectral theorem must imply constants $a \geq 0$, $\sigma > 0$, and C uniform over the periodic family such that

$$|\langle u_t f_0, f_0 \rangle_{L^2(\mu)}| \leq C(1+|t|)^{-\sigma}\, \mathcal{S}_a(f)^2, \qquad f_0 = f - \mu(f). \tag{6.1}$$

All proofs in this section are complete from (6.1); hence their only unresolved dependency is the source of that coefficient estimate.

Lemma 6.1 (Finite relative trace for the weighted Sobolev hierarchy)**.** *Let X be one of the finite-volume homogeneous spaces covered by the weighted Sobolev calculus of AHD VI and put $m = \dim X$. If $d > d_0 + m/2$, then $\mathcal{S}_{d_0}^2$ has finite relative trace with respect to $\mathcal{S}_d^2$ on $C_c^\infty(X)$. Concretely, if $e_1, e_2, \ldots$ is any $\mathcal{S}_d$-orthonormal family, then*

$$\sum_j \mathcal{S}_{d_0}(e_j)^2 < \infty,$$

with a bound depending only on the fixed ambient Sobolev data and the two orders. The same conclusion holds with d_0 replaced by any lower order.

Proof. Fix a derivative word D^α with $|\alpha| \leq d_0$ and a point $x \in X$. Write $H_x = \mathrm{ht}(x)$. The height-local injectivity theorem from AHD VI provides an injective coordinate ball of radius c/H_x. Rescale that ball to unit size and apply the Euclidean Sobolev inequality exactly as in the weighted Sobolev embedding theorem. For $d > |\alpha| + m/2$ this gives

$$|D^\alpha f(x)| \leq C H_x^{m/2-d} \mathcal{S}_d(f). \tag{6.2}$$

Thus evaluation $L_{x,\alpha} : f \mapsto D^\alpha f(x)$ has squared dual norm at most CH_x^{m-2d} on the Hilbert completion for $\mathcal{S}_d$.

Let $e_1, \ldots, e_N$ be a finite $\mathcal{S}_d$-orthonormal family. Bessel's inequality applied to $L_{x,\alpha}$ and (6.2) yields

$$\sum_{j=1}^{N} |D^\alpha e_j(x)|^2 \le C H_x^{m-2d}.$$

Multiply by $H_x^{2d_0}$, integrate over X, and sum over the finitely many $|\alpha| \le d_0$. We obtain

$$\sum_{j=1}^{N} \mathcal{S}_{d_0}(e_j)^2 \le C \int_X H_x^{m-2(d-d_0)}\, dm_X(x).$$

Since $H_x \ge 1$ and $d - d_0 > m/2$, the integrand is bounded by one. The ambient quotient has finite measure, so the right-hand side is bounded independently of N. Taking the supremum over finite orthonormal families is precisely finiteness of the relative trace. Replacing d_0 by a smaller order gives the final assertion. □

Definition 6.2 (Block genericity)**.** Fix an integer $M \ge 2$. For $n \ge 1$ put

$$I_n = [nM, (n+1)M], \qquad L_n = |I_n|,$$

and for an H-invariant probability μ define

$$\Delta_n f(x) = \frac{1}{L_n} \int_{I_n} f(xu_t)\, dt - \mu(f).$$

A point is $[T_0, T_1]$-generic for μ with respect to $\mathcal{S}_d$ if

$$|\Delta_n f(x)| \le n^{-1} \mathcal{S}_d(f)$$

for every integer $T_0 \le n \le T_1$ and every $f \in C_c^\infty(X)$.

Proposition 6.3 (Effective generic-point extraction)**.** *Assume the uniform coefficient estimate* (6.1)*. One can choose M and Sobolev orders $a < d_0 < d$, depending only on the ambient data and σ, so that uniformly over the periodic measures in the family,*

$$\mu\{x : x \textit{ is not } [T_0, \infty)\textit{-generic w.r.t. } \mathcal{S}_d\} \ll T_0^{-2}.$$

More generally, if μ is $[S, \epsilon, d_*]$-almost invariant, then there exist $\beta > 0$ and $d' > d_*$ such that, whenever $R, T_1 \le \epsilon^{-\beta}$, the relative measure of pairs $(x, s) \in X \times B_S(R)$ for which xs is not $[T_0, T_1]$-generic with respect to $\mathcal{S}_{d'}$ is $O(T_0^{-2})$.

Proof. We first treat an invariant measure. Put $\sigma_0 = \frac{1}{2}\min(\sigma, 1)$ and choose M so large that the block length makes the final power summable. Expanding the square and using H-invariance gives

$$\int_X |\Delta_n f|^2\, d\mu = \frac{1}{L_n^2} \int_{I_n^2} \langle u_{t-s} f_0, f_0 \rangle_\mu\, dt\, ds.$$

The elementary convolution estimate for $(1+|t-s|)^{-\sigma}$ and (6.1) therefore gives, after increasing M and absorbing the bounded initial blocks,

$$\int_X |\Delta_n f|^2 \, d\mu \ll n^{-10} \mathcal{S}_a(f)^2. \tag{6.3}$$

Choose $a < d_0 < d$ with both Sobolev gaps larger than $m/2$. By Lemma 6.1, the positive form $\mathcal{S}_{d_0}^2$ defines a trace-class compact operator relative to $\mathcal{S}_d^2$. The spectral theorem gives a $\mathcal{S}_d$-orthonormal basis (e_j) orthogonal also for $\mathcal{S}_{d_0}$, and the two relative traces imply

$$A_1 := \sum_j \mathcal{S}_{d_0}(e_j)^2 < \infty, \qquad A_2 := \sum_j \left(\frac{\mathcal{S}_a(e_j)}{\mathcal{S}_{d_0}(e_j)}\right)^2 < \infty,$$

with the zero summands omitted. Apply Chebyshev to (6.3) with threshold $cn^{-1}\mathcal{S}_{d_0}(e_j)$ and sum over j and $n \geq T_0$. The union of bad events has measure $O(T_0^{-7})$ after choosing c from A_1, A_2. Outside this union, expand $f = \sum_j f_j e_j$ and use Cauchy–Schwarz to obtain $n|\Delta_n f(x)| \leq \mathcal{S}_d(f)$. This proves the first assertion, with room to weaken T_0^{-7} to T_0^{-2}.

For the almost-invariant variant, compact generation of the fixed connected group S, combined with the AHD VI Sobolev translation bound, yields

$$|\mu_s(F) - \mu(F)| \ll \epsilon R^A \mathcal{S}_{d_*+A}(F), \qquad s \in B_S(R),$$

for a fixed ambient exponent A. Because $\mathrm{Ad}(u_t)$ is polynomial in t, differentiating the block average and using the Sobolev product estimate gives

$$\mathcal{S}_q(|\Delta_n f|^2) \ll n^A \mathcal{S}_{q+A}(f)^2.$$

Average the preceding almost-invariance estimate over $s \in B_S(R)$ with $F = |\Delta_n f|^2$ and combine with (6.3). The resulting second moment is bounded by

$$(n^{-10} + \epsilon R^A n^A)\mathcal{S}_D(f)^2.$$

Choose β so small that $R, n \leq \epsilon^{-\beta}$ makes the second term $O(n^{-10})$. Repeating the relative-trace/Chebyshev argument on $X \times B_S(R)$ proves the claimed exceptional proportion after one final fixed increase of Sobolev order. $\square$

7. Nearby generic points and polynomial divergence

Let S be the current connected semisimple intermediate subgroup and choose an S-invariant vector-space complement $\mathfrak{g} = \mathfrak{s} \oplus \mathfrak{r}$.

Lemma 7.1 (First-macroscopic polynomial shear). *Write $u_t = \exp(tZ)$ and let d be the nilpotence index of $\mathrm{ad}\, Z$ on $\mathfrak{g}$. For every $r \in \mathfrak{r}$ one has the exact* *polynomial identity*

$$P_r(t) := \operatorname{Ad}(u_{-t})r = e^{-t\,\operatorname{ad} Z}r = \sum_{j=0}^{d} \frac{(-t)^j}{j!}(\operatorname{ad} Z)^j r.$$

Let $P_r^{\mathrm{nc}}(t) = P_r(t) - P_r(0)$ *and suppose it is nonzero. Define*

$$T(r) := \inf\{T \geq 1 : \sup_{0\leq s\leq 2} \|P_r^{\mathrm{nc}}(Ts)\| \geq 1\}.$$

For all sufficiently small r*,* $T(r) < \infty$ *and there are constants* $c_1, c_2, A_1, A_2 > 0$*, depending only on the fixed representation, such that*

$$c_1\|r\|^{-A_1} \leq T(r) \leq c_2\|r\|^{-A_2}.$$

The rescaled polynomial

$$q_r(s) := P_r(T(r)s), \qquad 0 \leq s \leq 2,$$

has degree at most d*, uniformly bounded coefficients, and* $\sup_{0\leq s\leq 2}\|q_r(s) - q_r(0)\| = 1$*. If* $P_r^{\mathrm{nc}} \equiv 0$*, then* $(\operatorname{ad} Z)r = 0$*, equivalently* r *is fixed by* U*.*

Proof. The displayed polynomial identity is the finite exponential series for the nilpotent operator $\operatorname{ad} Z$. Write $P_r^{\mathrm{nc}}(t) = \sum_{j=1}^{d} t^j a_j(r)$, where each a_j is a fixed linear map of r. On the finite-dimensional space $\mathfrak{r}/\ker(P^{\mathrm{nc}})$ all norms are equivalent, so for some fixed $c, C > 0$,

$$c\,\|r_{\mathrm{nc}}\| \leq \max_{1\leq j\leq d}\|a_j(r)\| \leq C\|r_{\mathrm{nc}}\|.$$

Hence $\sup_{0\leq s\leq 2}\|P_r^{\mathrm{nc}}(Ts)\|$ is bounded above by $C'\sum_{j=1}^{d} T^j\|r\|$ and, by equivalence of norms on the finite-dimensional space of degree-d vector polynomials, below by $c'\max_j T^j\|a_j(r)\|$. Solving the inequalities at level one gives power bounds for the first-exit time $T(r)$. At $T = T(r)$, the same norm equivalence bounds each coefficient $T^j a_j(r)$ by a constant, while the definition of $T(r)$ gives unit macroscopic oscillation. Finally, $P_r^{\mathrm{nc}} \equiv 0$ iff every $(\operatorname{ad} Z)^j r$ with $j \geq 1$ vanishes, and this is equivalent to $(\operatorname{ad} Z)r = 0$. □

Proposition 7.2 (Nearby generic points compare translated period measures)**.** *Let* μ_1, μ_2 *be* H*-invariant probabilities and suppose* $x_2 = x_1\exp(r)$ *with* $0 < \|r\| \ll 1$ *and* $r \in \mathfrak{r}$*. Assume the two points are generic, in the sense of Definition 6.2, on the whole range of block indices containing the first-exit time* T*. Then there exist* $\kappa > 0$ *and* D *such that*

$$\mu_2(f) - \mu_1(f \circ R_{\exp(q(s))}) \ll \|r\|^{\kappa}\mathcal{S}_D(f) \tag{7.1}$$

for s *in a fixed compact subinterval of* $(0, 2)$*, uniformly in the periodic family once the genericity constants are uniform.*

PROOF. Choose a block I_n contained in a fixed relative window about T. Genericity gives, for $j = 1, 2$,

$$\mu_j(f) = \frac{1}{L_n} \int_{I_n} f(x_j u_t)\, dt + O(n^{-1} \mathcal{S}_D(f)).$$

The exact identity $x_2 u_t = x_1 u_t \exp(\mathrm{Ad}(u_{-t}) r)$ and the first-macroscopic approximation replace the relative displacement by $\exp(q(t/T))$ at cost $O(\|r\|^\eta \mathcal{S}_D(f))$. Since q has bounded degree and controlled coefficients, its variation across one block is $O(n^{-1})$. Fixing one $t_0 \in I_n$, put $s = t_0/T$ and use the local translation-Lipschitz estimate to freeze the polynomial at $q(s)$. Genericity at x_1 applied to the bounded translate of f then gives (7.1). The first-exit time lies between fixed powers of $\|r\|^{-1}$, so n^{-1} is itself a positive power of $\|r\|$ and can be absorbed into κ. $\square$

THEOREM 7.3 (Polynomial shearing produces a transverse almost-invariant one-parameter subgroup). *Assume Proposition 7.2 with $\mu_1 = \mu_2 = \mu$ and suppose μ is quantitatively almost invariant under the current group S. Then either the first-exit polynomial is constant, giving an almost-invariant U-fixed transverse direction, or there is a unit vector $Z \in \mathfrak{r}$ such that μ is almost invariant under* $\exp(tZ)$, $|t| \le 1$, *with a fixed positive-power loss. If Z lies in the centralizer of H it is recorded for Chapter 7; otherwise it feeds Theorem 4.1.*

PROOF. Let δ be the larger of the current S-invariance error and the error in (7.1). In the constant branch that comparison already gives almost invariance under one nonzero small transverse translation; iteration to unit scale and the local Lipschitz estimate put the error in positive-power form.

Assume q is nonconstant and has the fixed macroscopic normalization. Put $E = \delta^{1/2}$. Bounded-degree polynomial calculus supplies $s, s + E$ in the fixed comparison interval with $\|q(s + E) - q(s)\| \asymp E$; otherwise the derivative would integrate to an oscillation smaller than the chosen normalization. Proposition 7.2 shows that μ is $O(\delta)$-almost invariant under both $\exp(q(s))$ and $\exp(q(s + E))$, hence under their quotient.

Use the local product chart associated with $\mathfrak{g} = \mathfrak{r} \oplus \mathfrak{s}$ to write

$$\exp(-q(s)) \exp(q(s + E)) = \exp(w) s_0, \qquad w \in \mathfrak{r},$$

with $\|w\| \asymp E$ and $d_S(s_0, e) \ll E^2$ after shrinking the fixed chart. Almost invariance under S absorbs s_0, so μ is $O(E^2)$-almost invariant under $\exp(w)$. Iterating this relation $O(E^{-1})$ times and using the Sobolev translation estimate yields $O(E)$-almost invariance under the unit direction $Z = w/\|w\|$. This is a fixed positive power of the original error. $\square$

Worked Example 7.4 (Why a fixed centralizer direction is useless)**.** If the nearby pair differs only by $\exp(Z)$ with $Z \in \mathfrak{z}_{\mathfrak{g}}(\mathfrak{h})$, then $h^{-1} \exp(Z) h = \exp(Z)$

is constant. There is no polynomial shear and no new algebraic direction. This is the basic obstruction behind Chapter 7.

CHAPTER 5

Effective Closing for Semisimple Periods

1. The internal pair-return theorem used in this volume

Let $U = \{u_t\} < H$ be the fixed unipotent one-parameter subgroup. Since every intermediate group S considered in this chapter contains H, its Lie algebra $\mathfrak{s}$ is $\mathrm{Ad}(U)$-invariant. This is exactly the additional hypothesis in the internally proved closing theorem of Volume 45.

THEOREM 1.1 (Pair-return closing for semisimple periods). *There exist $A_1, A_2, C > 1$, depending only on the fixed arithmetic datum, such that the following holds. Let $x = \Gamma g \in X_\tau$, let $H \subset S \subset G$ be connected semisimple with $U \subset H$, and suppose $\mathfrak{s} + \mathrm{rad}(\mathfrak{g})$ is not a proper ideal. If $T > R > C\tau^{-A_1}$ and a measurable $E \subset [-T, T]$ satisfies $|E| > TR^{-1/A_1}$ and, for all $s, t \in E$, there is $\gamma_{s,t} \in \Gamma$ for which*

$$|u_{-s}g^{-1}\gamma_{s,t}gu_t| \leq R^{1/A_1}, \qquad d_{\mathrm{Gr}}\Big(\mathrm{Ad}(u_{-s}g^{-1}\gamma_{s,t}gu_t)\mathfrak{s}, \mathfrak{s}\Big) \leq R^{-1},$$

then either there is a nontrivial proper rational class-$\mathcal{H}$ subgroup $M < G$ with

$$\|\eta_M(gu_t)\| \leq R^{A_2}, \qquad \|Z \wedge \eta_M(gu_t)\| \leq T^{-1/A_2}R^{A_2} \quad (|t| \leq T),$$

or there is a nontrivial proper rational normal class-$\mathcal{H}$ subgroup containing $\mathrm{rad}(G)$ *and satisfying $\|Z \wedge v_M\| \leq R^{-1/A_2}$.*

PROOF. Because $U \subset S$, one has $\mathrm{Ad}(U)\mathfrak{s} = \mathfrak{s}$. Every other hypothesis and every quantity in the conclusion is literally the corresponding quantity in Theorem 0.3. Apply that theorem with $\mathfrak{h} = \mathfrak{s}$. No change of exponent or unproved source input is used. □

Corollary 1.2 (Assembled effective closing alternative)**.** *Suppose the quantitative period argument produces the pair-return hypotheses of Theorem 1.1. Then failure of the transverse branch yields either a proper rational class-$\mathcal{H}$ obstruction of height $R^{O(1)}$ satisfying the displayed Chevalley bounds, or the explicit normal-factor alternative.*

PROOF. Theorem 1.1 gives the two algebraic branches. The height estimate is Proposition 0.4. The complementary case is precisely the transverse branch of Theorem 0.2. □

2. From nearby period points to the pair-return hypotheses

Lemma 2.1 (Pair-return encoding in a period chart). *Let $x = \Gamma g$ lie in a height window of injectivity radius at least $\tau > 0$. Suppose two points $xu_s \exp(r_s)$ and $xu_t \exp(r_t)$ are within $\rho < c\tau$, with $\|r_s\|, \|r_t\| \leq \rho$. Then there is $\gamma_{s,t} \in \Gamma$ such that*

$$q_{s,t} := u_{-s} g^{-1} \gamma_{s,t} g u_t$$

has matrix norm bounded polynomially in the scale and moves the current intermediate Lie algebra by $O(\rho)$ in the Grassmannian.

Proof. Injectivity of the chart lifts the quotient-space proximity to $\gamma_{s,t} g u_t \exp(r_t) = g u_s \exp(r_s) b$ with b in a fixed $O(\rho)$-ball. Rearranging expresses $q_{s,t}$ as a product of three elements $O(\rho)$ from the identity. Local smoothness of multiplication, inversion, and the adjoint map gives the norm and Grassmannian bounds. □

Proposition 2.2 (Semisimple-period pair-return interface). *Let $H \subset S$ and suppose a positive-density set of generic return times is contained in one polynomial height window and the nearby-pair comparison has error R^{-B}. For B larger than the fixed chart losses, the hypotheses of the hypotheses of Theorem 1.1 hold for $\mathfrak{s} = \mathrm{Lie}(S)$ after a fixed power change of R.*

Proof. Lemma 2.1 provides the required lattice elements, height bounds, and Grassmannian errors. Enlarging the universal scale exponent absorbs the fixed chart powers, while the positive-density hypothesis is unchanged up to another fixed power. □

3. The subgroup returned by closing

Theorem 3.1 (Closing output as an intermediate-period alternative). *Let xS be a periodic orbit of a connected semisimple intermediate group $H \subset S \subset G$. Suppose Theorem 1.1 returns a proper rational class-$\mathcal{H}$ group M and exclude the normal-factor alternative. Let*

$$\mathbf{L} := \langle \mathbf{S}_x, \mathbf{M} \rangle^{\mathrm{Zar},\mathcal{H}}, \qquad L = g^{-1} \mathbf{L}(\mathbb{R})^\circ g.$$

Then either $H \subset S \subset L < G$ and xL is a closed finite-volume intermediate orbit, or $L = G$ and the argument returns to the invariance-enlargement branch. In the finite-centralizer regime only finitely many proper L can occur.

Proof. Periodicity makes the conjugated current group rational by the same Borel-density arithmetic criterion used in AHD VIII. The closing group M is rational by Theorem 1.1. Proposition 7.1 therefore gives a rational generated class-$\mathcal{H}$ group. If it is proper, Borel–Harish–Chandra finite covolume gives a lattice intersection and hence a periodic intermediate orbit. Finiteness in the zero-centralizer case follows from Theorem 2.3. □

4. Converting subgroup height to the Volume-IX discriminant

Proposition 4.1 (Polynomial discriminant bound for a closed obstruction)**.** *Assume the return chart allows $\|g\| + \|g^{-1}\| \le R^{A_0}$, the current period has $\operatorname{disc}(xS) \le D$, and the closing subgroup has rational height at most R^{A_1}. Then every proper generated intermediate orbit returned by Theorem 3.1 satisfies*

$$\operatorname{disc}(xL) \ll (DR)^B$$

for a fixed ambient exponent B.

Proof. The current exterior discriminant and the bound on g give polynomial control of the primitive rational Pluecker vector of the conjugated current group. Combine this with the closing subgroup height and apply the generated subgroup height theorem. Transporting the resulting primitive vector back by $\wedge^{\dim L} \operatorname{Ad}(g^{-1})$ costs only another fixed power of R. □

5. The conditional equidistribution-or-obstruction step

Theorem 5.1 (Effective semisimple-period dichotomy). *There exist $c, d > 0$ such that for every periodic semisimple orbit xS in the fixed arithmetic family and every $D \ge 2$, one of the following holds:*

(1) for every $f \in C_c^\infty(X)$,

$$|\mu_{xS}(f) - \mu_X(f)| \ll D^{-c}\mathcal{S}_d(f);$$

(2) there is a proper connected rational intermediate group $S < M < G$ with periodic orbit xM whose arithmetic complexity is $\ll D^{O(1)}$, and the period measure is power-close to μ_{xM} at the scale supplied by the closing construction.

Proof. This is the effective period machine proved as Theorem 1.6. Choose the local scale $r = D^{-C}$ with C larger than the Sobolev, height, and closing losses. If quantitatively generic points are sparse, the generic-point lemma and the two-sided complexity comparison already give a low-complexity periodic obstruction. Otherwise a nearby generic pair exists. Polynomial shearing either yields a new transverse almost-invariant direction or the effective closing step produces a rational intermediate period. In the first case Chapter 4 amplifies the new direction; strict Lie-algebra enlargement can occur at most $\dim G$ times. If it reaches G, the Sobolev dual characterization of almost invariance gives (1); if not, closing gives (2). Because every scale is a fixed power of D and the number of iterations is bounded, the final exponent remains positive. □

Proposition 5.2 (Finite descent with exponent propagation)**.** *Assume the dichotomy of Theorem 5.1 is available at every proper intermediate group with a power-saving error and fixed polynomial complexity loss. In the finite-centralizer regime the obstruction branch terminates after finitely many enlargements; the final error remains a positive power and the final complexity remains polynomial in the initial parameters.*

Proof. The obstruction groups form the finite poset of Proposition 6.1; every genuine enlargement moves strictly upward. A chain has length at most $\dim G - \dim H$. Bounded composition of positive power savings leaves a positive power, and bounded composition of polynomial complexity maps remains polynomial. □

Worked Example 5.3 (Why one generates with the current group)**.** A rational subgroup found from a short cluster of lattice returns need not visibly contain the current semisimple group. The correct obstruction is the class-$\mathcal{H}$ part of the Zariski group generated by the current group and the closing group. This simple composition is what converts generic unipotent closing into semisimple intermediate-period descent.

How the closing interface is assembled

Effective closing enters only after the dynamical argument has produced many nearby returns with controlled algebraic displacement. The purpose of the period-chart lemma is to translate that geometric recurrence into the exact input language of Ratner XIII. The lattice element $\gamma_{s,t}$ is not a new obstruction by itself. It records that two nearby representatives in the quotient differ by an arithmetic deck transformation, while the conjugated element $q_{s,t}$ measures how far that deck transformation is from normalizing the current intermediate Lie algebra.

Three estimates are needed simultaneously: a bound for the matrix height of $q_{s,t}$, a Grassmannian bound for its action on the current Lie algebra, and positive density of pairs for which those bounds hold. Height-window control gives the first two after a fixed polynomial loss. Generic-point extraction supplies the density. Once written in these variables, the hypotheses coincide with the inherited pair-return theorem rather than merely resembling it.

The subgroup returned by closing must then be composed with the current acting group. A rational group detected by lattice returns need not visibly contain S. The correct candidate is the class-$\mathcal{H}$ part of the Zariski group generated by the rational form of S together with the closing group. This generated group is rational and has polynomially controlled height by the Volume 45 height theorem. If it is proper, Borel–Harish–Chandra turns it into a finite-volume intermediate period. If it is the whole ambient group, the return has produced an invariance candidate rather than a proper obstruction.

The discriminant estimate is a separate arithmetic translation. Closing naturally returns Chevalley or rational subgroup height. Chapter 2 measures periods by an exterior discriminant after conjugation by the base point. The return chart bounds the norm of that conjugation, so transporting the primitive Pluecker vector costs only a fixed polynomial. This is the precise reason a low-height closing subgroup yields a low-discriminant intermediate period.

6. Finite descent

Assuming the effective dichotomy at every current intermediate group, the rest is finite bookkeeping. In the finite-centralizer regime there are only finitely many connected intermediate semisimple groups. Every obstruction step strictly enlarges the current group, so the process terminates. If one step changes the analytic error by a positive power and the arithmetic complexity by a polynomial, a bounded number of steps preserves both forms of control. The resulting exponent may be much smaller than the input exponent, but it remains positive and depends only on the fixed ambient data.

This is also the point at which the classical and modern theories diverge. Classical EMV can organize the descent over a finite fixed list of intermediate groups. With nontrivial centralizer the intermediate obstruction space is no longer discretized by the same argument; the correct replacement is a lower bound for the complexity of *every* proper intermediate period. Chapter 7 reformulates closing in precisely that language.

CHAPTER 6

Property τ and Maximal Adelic Periods

1. Adelic homogeneous probabilities

Let $\mathbf{G}$ be semisimple over a number field and let $Y = \mathbf{G}(F)\backslash\mathbf{G}(\mathbb{A})/K$ be the fixed adelic quotient in the EMMV setting. For a connected semisimple F-subgroup $\mathbf{H}$ and an adelic point g, write $Y_H(g)$ for the corresponding homogeneous subset when its stabilizer has finite volume.

Definition 1.1 (Admissible maximal semisimple period datum)**.** A pair $(\mathbf{H}, g)$ is an *admissible semisimple period datum* in the fixed quotient Y if $\mathbf{H} < \mathbf{G}$ is a connected semisimple F-subgroup and the homogeneous subset $Y_H(g)$ has finite invariant volume. It is *maximal* if whenever a connected semisimple F-subgroup $\mathbf{L}$ satisfies

$$\mathbf{H} < \mathbf{L} \leq \mathbf{G} \quad \text{and} \quad Y_H(g) \subseteq Y_L(g),$$

then $\mathbf{L} = \mathbf{G}$. All uniformity statements in this chapter keep $\mathbf{G}$, F, K, the integral model, and the Sobolev system fixed while $(\mathbf{H}, g)$ and the allowed congruence level vary. For an allowed compact-open congruence subgroup $K' \leq K$, define the period-complexity package

$$\mathcal{C}(\mathbf{H}, g; K') := \max\Big\{2, \mathrm{vol}(Y_H(g)), H_{\mathrm{Pl}}(\mathrm{Ad}(g^{-1})\mathfrak{h}), [K : K']\Big\},$$

where $\mathfrak{h} = \mathrm{Lie}(\mathbf{H})$ and H_{Pl} is the primitive Plücker height in the fixed ambient integral representation. By Corollary 1.4, replacing this package by the equivalent discriminant/Prasad package changes it only by fixed powers.

Proposition 1.2 (Adelic homogeneous-period probability)**.** *Whenever $Y_H(g)$ has finite invariant volume, its Tamagawa/local Haar measure normalizes to a canonical probability. Its pushforward to any positive-mass real component is the normalized real periodic-orbit measure from Chapter 1.*

Proof. The quotient of $\mathbf{H}(\mathbb{A})$ by the rational stabilizer and compact-open finite-place factor carries invariant Haar measure. Divide by total volume. Disintegrating with respect to the finite real-component decomposition of Theorem 5.1 shows that each positive-mass component is Haar on the corresponding real homogeneous orbit; normalization gives the Chapter-1 probability. ∎

2. The uniform adelic spectral input

Proposition 2.1 (Uniform adelic property τ)**.** *Fix the ambient semisimple adelic group and an admissible maximal semisimple period datum* $(\mathbf{H}, g)$ *in the sense of Definition 1.1. There exists an integer* m_0 *such that the representations on the orthocomplements of constants in all congruence-level periodic* $\mathbf{H}$*-components have* m_0*-fold tensor powers weakly contained in the corresponding regular representations. The exponent* m_0 *is independent of the period and congruence level in the fixed ambient space.*

Proof. Decompose the simply connected cover of $\mathbf{H}$ into absolutely almost-simple factors. For factors of absolute rank at least two, the direct-transport construction in Repairs 03–10 chooses one fixed split auxiliary place and transfers the higher-rank Kazhdan spectral seed with constants independent of congruence level. For the remaining absolute type A_1 factors, Proposition 5.1 gives a uniform tensor exponent from the internally reconstructed Rankin–Selberg/Jacquet–Langlands chain. The set of factor types and bad places is finite in the fixed ambient problem. Taking the maximum of the factorwise exponents and applying the product/finite-index transfer lemmas gives one uniform m_0. □

Theorem 2.2 (EMMV maximal-semisimple adelic equidistribution). *Fix the ambient datum of Definition 1.1. There exist* $\delta > 0$*,* $C \geq 1$*, and a Sobolev order* $d \in \mathbb{N}$*, depending only on that ambient datum, such that for every admissible maximal semisimple period datum* $(\mathbf{H}, g)$*, every allowed congruence level* $K' \leq K$*, and every smooth test function* f *on* Y*,*

$$\left|\mu_{Y_H(g)}(f) - \mu_Y(f)\right| \leq C\, \mathcal{C}(\mathbf{H}, g; K')^{-\delta}\, \mathcal{S}_d(f).$$

Here both probability measures use the Haar normalizations fixed in Chapter 1.

Proof. The proposition above supplies the uniform spectral package on every real component. Maximality excludes a proper connected semisimple intermediate group. Apply Theorem 5.1 componentwise: its obstruction branch would produce such a proper intermediate group and is therefore impossible. Hence every component satisfies the ambient power-saving estimate. The adelic-to-real dictionary of Chapter 1 has a fixed finite component set, and the Sobolev comparison is level-independent in the fixed ambient space. Summing the component estimates gives the adelic power-saving statement. This is the same internal argument recorded in Theorem 2.2. □

3. What fixed-ambient uniformity means

Proposition 3.1 (Exact fixed-ambient uniformity quantifiers)**.** *Once the ambient adelic group, integral model, compact-open level framework, and*

Sobolev system are fixed, constants in the maximal-period theorem may be chosen independently of the individual period, its real component, and the finite congruence level, except for the explicit complexity quantities shown in the theorem. This statement does not assert uniformity when the ambient arithmetic form itself varies.

Proof. The finite component set is fixed, the local coordinate systems and Sobolev operators come from one ambient group, and finite-level maps are finite covers. Thus all geometric comparison constants are bounded over the fixed ambient family. Varying the ambient rational form changes the local bad places and measure normalizations and is therefore outside this quantifier structure. □

4. Prasad volume and good-place arithmetic

Theorem 4.1 (Prasad-volume and adelic complexity comparison). *In the fixed Volume-IX arithmetic setting, period covolume and the discriminant/Pluecker complexity determine one another up to fixed powers. More precisely, there are $A, B, C > 0$ such that*

$$C^{-1}\operatorname{disc}(xH)^A \le \operatorname{vol}(xH) \le C\operatorname{disc}(xH)^B.$$

For a fixed ambient adelic space the same fixed-power comparison controls the finite-place level factors; if the ambient form varies, one additional fixed power of the explicitly recorded ambient arithmetic complexity is required.

Proof. The real two-sided comparison is Theorem 3.1. For the adelic refinement, Lemma 1.1 bounds the discriminants of factor fields and the product of bad primes by a fixed power of subgroup height. Proposition 1.2 writes the covolume as a product of field-discriminant powers, local parahoric factors, and finite central/form indices. At good hyperspecial places the extracted Euler product converges; at bad places every local factor is between fixed powers of the residue characteristic, and the product of those primes is polynomially bounded. Hence the global volume is polynomially comparable with the Pluecker/discriminant data. Corollary 1.4 records the additional ambient factor when the rational form varies. □

5. Back to real congruence components

Proposition 5.1 (Transfer from adelic equidistribution to real congruence components)**.** *Suppose an adelic period probability μ satisfies a quantitative Sobolev equidistribution estimate on the fixed adelic quotient. Let Y_i be a real component of positive μ-mass. Then the normalized restriction of μ to Y_i satisfies the corresponding real congruence-quotient estimate, with a* *constant changed only by the inverse component weight and the fixed Sobolev comparison.*

Proof. Choose a smooth finite-place idempotent that selects the component and tensor it with a real test function. Apply the adelic estimate to this product test function. The finite-place factor has fixed Sobolev cost in the fixed ambient space. Dividing by the positive component mass yields the normalized real estimate. □

Why the maximal adelic theorem is a different endpoint

The adelic chapter is not obtained by placing hats on the real theorem. Maximality changes the proof architecture. In the finite-centralizer real argument, failure of ambient equidistribution may lead to a proper intermediate semisimple period and one then descends through a finite list. In the maximal adelic setting that branch has been removed by hypothesis, so the spectral and arithmetic inputs can be organized uniformly around the single ambient homogeneous set.

The first deep input is therefore a genuinely adelic property-τ statement for the semisimple periods allowed to vary in the theorem. It is not supplied by the narrower principal-congruence theorem proved earlier in the AHD series. The second deep input is arithmetic: one needs quantitative control of period volume and a suitably small good finite place. Prasad's volume formula, together with the local structure of the group and the level, provides the route, but the normalization must be written out before this becomes a proved input.

Once those two inputs are available, the adelic-to-real component dictionary from Chapter 1 explains how the result restricts to individual real congruence components. The finite component number contributes only fixed comparison factors when the ambient adelic space is fixed. What it does *not* do is create uniformity when the number field, ambient group, or level complexity is allowed to vary. Thus "uniform in the period data for a fixed ambient adelic quotient" is the exact conclusion to be propagated, not a stronger all-ambient statement.

CHAPTER 7

Nontrivial Centralizers and Intermediate Complexity

1. Why positive-dimensional centralizers are a new phenomenon

Proposition 1.1 (Centralizer deformation family). *If xH is periodic and $C \subset Z_G(H)$ is connected, then every xcH with $c \in C$ has the same intrinsic H-stabilizer and period volume. Unless C is absorbed by H in the quotient, these periods form a continuous family in the ambient space.*

Proof. This is Corollary 4.1. The last assertion follows because the map $c \mapsto xcH$ is continuous and is locally nonconstant precisely when the centralizer motion is not contained in the original period. $\square$

2. Cusp height and arithmetic obstruction are separate

Proposition 2.1 (Two independent complexity coordinates). *On noncompact congruence quotients, a quantitative semisimple-period theorem must track both a cusp parameter such as* $\operatorname{minht}(xH)$ *and an arithmetic intermediate-period parameter such as $D_H(x)$. Neither uniformly controls the other.*

Proof. Centralizer translation in a product quotient preserves the rational period data while changing cusp position. Conversely one may keep a period in a fixed compact height window while varying its rational subgroup through increasing exterior discriminant. Thus the two coordinates are logically independent. $\square$

Proposition 2.2 (Minimal discriminant of a proper intermediate orbit is the natural obstruction). *For a periodic xH, define*

$$D_H(x) = \min\{\operatorname{disc}(xM) : H \subsetneq M \subsetneq G, \quad M \text{ rational connected and } xM \text{ periodic}\},$$

with value $+\infty$ if no such M exists. Any effective theorem whose failure is caused by closing into a proper intermediate period can be formulated with the single lower bound $D_H(x) \geq D$.

Proof. Every proper intermediate obstruction returned by closing belongs to the set over which the minimum is taken. Hence a lower bound for

$D_H(x)$ excludes all such obstructions below that complexity simultaneously. Conversely, if $D_H(x)$ is small, the minimizing intermediate period is an explicit low-complexity obstruction. □

3. Centralizer-aware closing interface

Lemma 3.1 (Pair-return closing remains valid after separating centralizer motion)**.** *After decomposing the transverse displacement into the H-fixed centralizer summand and its H-nontrivial complement, the nontrivial component satisfies the same hypotheses of Theorem 1.1, with only a fixed power loss in the return scale.*

Proof. Let $\mathfrak{r} = \mathfrak{r}^H \oplus \mathfrak{r}_{\mathrm{nt}}$ be an H-invariant decomposition. The projection to $\mathfrak{r}^H$ commutes with $\mathrm{Ad}(H)$, so translating a nearby point by the corresponding centralizer element changes neither the H-orbit stabilizer nor the Grassmannian displacement of $\mathfrak{s} = \mathrm{Lie}(S)$. On $\mathfrak{r}_{\mathrm{nt}}$ the local exponential chart and the bounded projection operators change norms by a fixed factor. Thus the pair-return matrix norm and Grassmannian error retain the form $R^{O(1)}$ and R^{-c}; replacing R by a fixed power puts them exactly in the normalization of Theorem 1.1. □

Proposition 3.2 (Specialization of closing with nontrivial centralizer)**.** *In the presence of $Z_G(H)^\circ \neq \{e\}$, decompose a nearby-pair displacement into a centralizer component and a complementary moving component. If the complementary component is nonzero, Theorem 1.1 applies to it after quotienting/normalizing the harmless centralizer motion; the returned proper rational subgroup yields an intermediate periodic obstruction with polynomially controlled discriminant.*

Proof. Choose an H-invariant decomposition of the transverse Lie algebra into the H-fixed centralizer part and a sum of nontrivial H-modules. The fixed part does not shear under conjugation and is absorbed into the centralizer parameter of the nearby period. A nonzero nontrivial component undergoes the same polynomial divergence as in Chapter 4, so the pair-return encoding of Chapter 5 applies. The discriminant translation is Proposition 4.1. □

4. The Wieser endpoint

Theorem 4.1 (Wieser congruence-quotient theorem). *Let $G = \mathbf{G}(\mathbb{R})$ for a connected semisimple $\mathbb{Q}$-group, let $\Gamma < \mathbf{G}(\mathbb{Q})$ be the fixed congruence subgroup of this chapter, and let $H < G$ be connected semisimple without compact factors. There exist $\delta > 0$ and $d \geq 1$ such that, if xH is periodic and*

$$\mathrm{disc}(\Gamma\mathbf{M}(\mathbb{R})g) \geq D$$

for every proper rational intermediate $\mathbf{M}$ *with* $xH \subset \Gamma\mathbf{M}(\mathbb{R})g$*, then*

$$|\mu_{xH}(f) - \mu_{xG^+}(f)| \ll_{G,H,\Gamma} D^{-\delta}\,\mathrm{minht}(xH)\mathcal{S}_d(f).$$

No uniformity in a varying congruence subgroup is asserted.

Proof. Decompose the transverse Lie algebra into the centralizer-fixed summand and nontrivial H-modules as in Lemma 2.1. Centralizer motion is absorbed into the period parameter; every noncentral nearby pair enters the generic-point/shearing machine of Appendix F. A rational closing obstruction would have height, hence by Appendix E discriminant, at most $D^{1-\varepsilon}$ after choosing the working scale as a small power of D, contradicting the hypothesis. Therefore the almost-invariance group enlarges to G^+. The height/Sobolev truncation leaves the explicit factor $\mathrm{minht}(xH)$, while Proposition 2.1 supplies the uniform spectral upgrade on the fixed congruence component. The resulting G^+-almost invariance gives the displayed estimate. □

Corollary 4.2 (Recovering finite-centralizer EMV as a special case)**.** *If* $Z_G(H)$ *is finite, Theorem 4.1 together with Theorem 3.1 recovers the finite-centralizer EMV endpoint of this volume.*

Proof. Finite centralizer removes the continuous centralizer parameter and Theorem 2.3 makes the proper intermediate family finite. By Theorem 3.1, discriminant and period volume are fixed-power equivalent. Apply Theorem 4.1 at a threshold which is a fixed power of the desired volume cutoff; if an intermediate obstruction appears, replace H by that strictly larger group. The finite poset forces termination, and a bounded composition of power savings is still a power saving. This is precisely the finite descent proving Theorem 1.1. □

5. Nonmaximal-period complexity bookkeeping

Proposition 5.1 (Four independent obstruction coordinates)**.** *For a general nonmaximal semisimple period, the effective argument may need to record separately: centralizer motion; minimal intermediate-period discriminant; cusp min-height; and ambient arithmetic complexity. Any theorem that suppresses one of these coordinates must justify the suppression from its hypotheses.*

Proof. The previous sections exhibit independent variation of centralizer and cusp coordinates, while Chapter 2 separates period discriminant from ambient complexity. The closing mechanism detects intermediate periods rather than centralizer translates. Hence none of the four coordinates is formally a function of the others. □

The failure mode behind the centralizer-aware theory

Positive-dimensional centralizer changes the geometry before it changes any estimate. A family xcH with $c \in Z_G(H)$ consists of periods with the same abstract stabilizer and the same intrinsic Haar volume. Yet those periods can occupy different locations in the ambient quotient and can have different cusp behavior. Consequently neither large period volume nor a spectral gap on the abstract quotient $H_x \backslash H$ can distinguish pure centralizer motion from genuine transverse motion.

This is why the nearby-pair displacement is decomposed into an H-fixed centralizer component and nontrivial H-modules. The fixed component is recorded as a motion of the period itself; the nontrivial component is the one that polynomially shears and can trigger closing. The internally proved pair-return theorem is therefore applicable, but only after this quotienting or normalization has removed the harmless fixed direction.

The natural obstruction parameter in this setting is $D_H(x)$, the minimum discriminant of a proper rational intermediate periodic orbit. If closing returns an obstruction below a given complexity threshold, then $D_H(x)$ is below that threshold. Conversely, a lower bound for $D_H(x)$ simultaneously excludes all proper intermediate periods of lower complexity. This converts a potentially complicated family of obstructions into one scalar lower bound without claiming that the family itself is finite.

The cusp parameter remains separate. Centralizer translation already shows that arithmetic obstruction complexity and ambient height can vary independently. This separation is visible in the modern congruence theorem and must remain visible in the AHD statement: one parameter controls escape, the other controls algebraic focusing.

6. Chapter summary

The centralizer-aware closing interface, obstruction ledger, and Wieser endpoint are internal. The endpoint proof is completed in Appendix G; its discriminant, cusp-height, spectral and centralizer normalizations are stated there explicitly. The historical Wieser citation is used only to check scope and normalization.

CHAPTER 8

Four Effective Endpoints: Scope, Overlap, and Frontier

1. Four theorems, not one universal theorem

THEOREM 1.1 (Classical EMV finite-centralizer endpoint). *In the fixed arithmetic real setting, let $H < G$ be connected semisimple without compact factors and assume $Z_G(H)$ is finite. There exist $\delta, d > 0$ and V_0 such that for every periodic xH and every $V \geq V_0$ there is a connected intermediate group $H \subset S \subset G$ with periodic xS, $\mathrm{vol}(xS) \leq V$, and*

$$|\mu_{xH}(f) - \mu_{xS}(f)| \ll V^{-\delta}\mathcal{S}_d(f).$$

PROOF. The uniform spectral theorem is Theorem 1.1; the two-sided period complexity theorem is Theorem 3.1; and the effective period dichotomy is Theorem 5.1. Finite centralizer gives the finite intermediate-subgroup poset of Chapter 3. Apply the dichotomy with D a fixed power of V. If equidistribution occurs, take $S = G$; otherwise replace H by the returned proper intermediate group. The poset prevents repetition and terminates after finitely many steps. Theorem 3.1 converts the final complexity threshold into $\mathrm{vol}(xS) \leq V$, and the bounded composition of power-saving errors remains $V^{-\delta}$. □

THEOREM 1.2 (EMMV maximal semisimple adelic endpoint). *Under the fixed ambient datum of Definition 1.1, there exist $\delta > 0$, $C \geq 1$, and $d \in \mathbb{N}$ such that every admissible maximal datum $(\mathbf{H}, g)$ at every allowed level $K' \leq K$ satisfies*

$$\left|\mu_{Y_{H(g)}}(f) - \mu_Y(f)\right| \leq C\,\mathcal{C}(\mathbf{H}, g; K')^{-\delta}\mathcal{S}_d(f)$$

for every smooth test function f.

PROOF. This is Theorem 2.2. Proposition 2.1 gives the uniform tensor-tempered spectral input. Maximality eliminates the proper semisimple closing branch of Theorem 5.1; the finite adelic component dictionary transfers the uniform component estimates back to the adelic probability. Thus the conclusion follows with constants independent of level/period data inside the fixed ambient space. □

Theorem 1.3 (Wieser nontrivial-centralizer congruence endpoint). *In the fixed real congruence setting, periodic semisimple H-orbits with possibly positive-dimensional centralizer satisfy the power-saving estimate of Theorem 4.1, provided every proper rational intermediate period has discriminant at least D; the error retains the explicit cusp factor* $\mathrm{minht}(xH)$.

Proof. This is exactly Theorem 4.1. The proof separates centralizer motion from nontrivial transverse modules, applies the internal shearing/closing machine to the latter, excludes every closing obstruction by the minimum-intermediate-discriminant hypothesis, and uses the uniform spectral theorem to upgrade the resulting almost invariance. The height truncation is kept explicit and produces the stated $\mathrm{minht}(xH)$ factor. □

Theorem 1.4 (ELMW compact adelic minimal-complexity endpoint). *In the compact adelic scope of this volume there exist $\delta, A > 0$ and a Sobolev order d such that, for a semisimple period of minimal proper-intermediate complexity D,*

$$|\mu_{xH}(f) - \mu_X(f)| \ll \mathrm{Comp}(X)^A D^{-\delta} \mathcal{S}_d(f),$$

with every dependence on the varying ambient arithmetic form recorded by $\mathrm{Comp}(X)$.

Proof. Compactness removes the cusp term. Proposition 2.1 supplies the semisimple spectral input, while Theorem 4.1 supplies the rational-height/discriminant and volume comparisons with the ambient-complexity factor. The effective-generation lemma in Appendix F turns approximate rational algebraic data into a subgroup of polynomial complexity. Therefore the compact adelic effective-period machine applies. Its obstruction branch would produce a proper intermediate period below the defining threshold D; excluding that branch yields the displayed ambient equidistribution estimate. This is the internal proof recorded again as Theorem 4.1. □

2. A formal compact-adelic descent lemma

Definition 2.1 (Adelic minimal intermediate complexity)**.** For a compact adelic homogeneous period Y_H in a fixed ambient space, let $\mathcal{C}_{\mathrm{amb}}$ denote the ambient arithmetic complexity and let

$$\mathcal{D}_H(Y) = \min\{\mathcal{C}(Y_M) : H \subsetneq M \subsetneq G, \quad Y_M \text{ an admissible intermediate period}\},$$

with value $+\infty$ when no proper intermediate period exists.

Proposition 2.2 (Formal descent from an ambient theorem)**.** *Suppose one has an ambient quantitative theorem with the following form: at a* *current semisimple period, either the measure is $O(D^{-\eta})$-close to the ambient invariant measure or there is a proper intermediate period of complexity $O(D^A)$; and suppose the same theorem is available after replacing the current group by every returned intermediate group. Then a lower bound $\mathcal{D}_H(Y) \geq D$ forces equidistribution with a polynomial error after a bounded number of descent steps.*

Proof. Every obstruction step strictly enlarges the acting connected semisimple group, so dimension increases and the process terminates after at most $\dim G - \dim H$ steps. If all proper intermediate periods have complexity at least D, an obstruction of complexity $O(D_0^A)$ is impossible once the working threshold D_0 is chosen as a sufficiently small fixed power of D. The bounded iteration of polynomial losses leaves a polynomial final rate. ☐

3. Scope separation and overlap

Theorem 3.1 (Four-endpoint scope separation and overlap). *The four endpoint families above have the following relations.*

(1) The EMV theorem is a real finite-centralizer theorem; its finite intermediate-subgroup descent uses the rigidity of that hypothesis.
(2) The EMMV theorem is adelic and assumes maximal semisimple stabilizer; its principal strength is fixed-ambient uniformity driven by property τ.
(3) The Wieser theorem permits nontrivial centralizer on real congruence quotients and replaces finite-centralizer discreteness by lower complexity bounds for every proper intermediate period, while keeping cusp height separate.
(4) The ELMW theorem is compact adelic, allows nonmaximal semisimple periods, and measures the error through minimal intermediate complexity and ambient complexity; it introduces additional effective-generation machinery.

None of these statements, solely by formal implication, yields a universal effective theorem covering all four settings.

Proof. Each item follows by comparing the hypotheses that define its proof architecture. Finite centralizer is absent from the Wieser setting, while maximality is essential in the stated EMMV theorem; therefore neither theorem is a literal restatement of the other. ELMW's compact adelic theorem permits nonmaximal periods but pays for them through minimal intermediate complexity and ambient complexity, data that are not present in the classical EMV statement. Conversely, its compactness assumption does not cover the noncompact cusp problems of the real congruence setting.

Hence overlap occurs on special common subfamilies, but no formal union of the hypotheses produces a theorem whose proof has been supplied here. □

Worked Example 3.2 (Four laboratories)**.** A finite-centralizer real period belongs naturally to the EMV laboratory. A maximal adelic period varying in a fixed adelic space belongs to EMMV. A real congruence period moving through a positive-dimensional centralizer requires the Wieser obstruction parameter. A compact adelic nonmaximal period with several possible semisimple enlargements is naturally measured by ELMW minimal intermediate complexity.

Reading the scope matrix as a dependency theorem

The scope matrix is not bibliographic decoration. It prevents four different proof mechanisms from being used interchangeably merely because their conclusions are all called "effective equidistribution of semisimple periods." Before applying an endpoint, one should ask four questions: Is the quotient real or adelic? Is it compact? Is the acting semisimple group maximal? Is the centralizer assumed finite, or are proper intermediate periods controlled by an explicit complexity lower bound?

For EMV, finite centralizer supplies rigidity of the intermediate-subgroup geometry and makes finite descent possible. Its conclusion may therefore land on a proper intermediate semisimple period. This is not a defect; it is part of the theorem's architecture. Consequently the maximal adelic EMMV theorem cannot simply be cited as a stronger version: maximality removes the very intermediate branch that EMV analyzes.

EMMV has a different advantage. The ambient adelic quotient is fixed while the homogeneous semisimple data vary, and property τ supplies uniformity through that family. A theorem on a single real congruence quotient does not formally recover this varying adelic statement. Conversely, EMMV's maximality hypothesis prevents it from being the desired nonmaximal centralizer theorem.

Wieser's theorem keeps a real congruence quotient and allows nontrivial centralizer. The price is an obstruction hypothesis: every proper rational intermediate periodic orbit containing the original period must have large discriminant. Cusp height remains a separate parameter because the quotient is noncompact. This is precisely the situation for which the scalar $D_H(x)$ from Chapter 7 was designed.

ELMW works in a compact adelic anisotropic setting and measures complexity globally. It can treat nonmaximal periods by using the minimal complexity of proper semisimple intermediate adelic periods together with ambient complexity. Compactness removes the real cusp ledger but introduces new arithmetic proved inputs: effective generation, an effective Greenberg theorem, and a detailed Prasad-volume analysis. Those are not formal consequences of the internal Volume 45 closing theorem.

On common subfamilies more than one theorem can apply, but even then the rates are expressed in different variables. Translating an EMMV volume rate into an ELMW complexity rate, or an EMV volume statement into Wieser's discriminant language, requires the arithmetic comparisons isolated in Chapter 2. Thus overlap of hypotheses does not identify the theorems.

4. The four endpoint theorems

The four endpoint proofs are completed in Appendices D–G. The uniform spectral input, two-sided volume–discriminant comparison, effective closing/generation mechanism, centralizer-aware obstruction ledger, and endpoint-specific descent arguments all have local proved inputs. Consequently EMV, EMMV, Wieser, and ELMW are used below only at the exact scopes proved in Appendix G; the primary papers are provenance and scope checks rather than omitted theorem owners.

APPENDIX A

Worked Laboratories

Worked Example 0.1 (Diagonal periods and Hecke-style motion)**.** Let G_0 be semisimple, $G = G_0 \times G_0$, and $H = \{(h,h) : h \in G_0\}$. A periodic H-orbit is the graph of a finite-volume correspondence between the factors. Right translation by (e,a) conjugates H to $\{(h, a^{-1}ha)\}$, so Theorem 4.1 gives the exact period transformation. This is genuine conjugation, not centralizer motion.

Worked Example 0.2 (An exterior discriminant)**.** For a rational two-plane $W \subset \mathbb{Q}^3$ with primitive integral basis $u, v \in \mathbb{Z}^3$, the primitive Pluecker vector is $u \wedge v$ divided by the gcd of its coordinates. Its Euclidean norm is the covolume of $W \cap \mathbb{Z}^3$ in W. Definition 2.1 is the same construction for a rational Lie algebra after transport by g^{-1}.

Worked Example 0.3 (A three-step obstruction poset)**.** Suppose the proper connected intermediate groups are $H \subsetneq S_1 \subsetneq S_2 \subsetneq G$ and an incomparable $H \subsetneq T \subsetneq G$. Once closing detects S_1, the obstruction family is restricted to groups containing S_1, so T disappears and $D_{S_1}(x) \geq D_H(x)$. This is the combinatorial engine of finite descent.

Worked Example 0.4 (A false volume argument)**.** The inclusion $xH \subset xS$ does not imply $\mathrm{vol}(xH) \leq \mathrm{vol}(xS)$. The two groups carry independently normalized Haar measures and have different dimensions. The correct comparison proceeds through genuine arithmetic volume/discriminant theorems, which is why Problem 3.1 is a separate release gate.

Worked Example 0.5 (A centralizer direction versus a shearing direction)**.** In $G_1 \times G_2$ with diagonal dynamics coming only from the first factor, a small displacement in the second factor is fixed under conjugation by H. Its relative polynomial is constant and cannot create a new invariance direction. By contrast, a displacement in a nontrivial H-module acquires a nonconstant polynomial component along a unipotent trajectory. The two cases look equally "transverse" in the ambient manifold but have completely different algebraic consequences.

Worked Example 0.6 (Why endpoint overlap does not identify rates)**.** Suppose a compact adelic period is maximal, so both a maximal-period theorem and a minimal-intermediate-complexity theorem may apply. One error term may be stated using intrinsic adelic volume and the other using a Pluecker complexity ratio. Without a proved arithmetic comparison between those variables, the numerical exponents cannot be identified. Overlap of hypotheses is therefore weaker than equality of quantitative statements.

APPENDIX B

Automorphic Spectral Reduction

1. Purpose of the reduction

Chapter 4 needs one precise spectral output: for every periodic intermediate semisimple orbit in the fixed finite-centralizer EMV setting, the nonconstant period representation has a common tensor power that is tempered. Volume 5 then converts tensor-temperedness into the polynomial matrix-coefficient estimate used in generic-point extraction.

The absolute-rank-at-least-two branch is handled by auxiliary-place transport. The absolute-type-A_1 branch reduces to quaternionic Jacquet–Langlands transfer and the specialized local $\mathrm{GL}_2 \times \mathrm{GL}_2$ Whittaker Rankin–Selberg theorem. The transfer lemmas below assemble these ingredients into the uniform statement used in Chapter 4.

2. Finite central covers and products

Lemma 2.1 (Finite central covers preserve tensor-temperedness)**.** *Let $p : \widetilde{S} \to S$ be a finite central covering of connected semisimple Lie groups. A unitary representation π of S has a tempered mth tensor power if and only if $(\pi \circ p)^{\otimes m}$ is tempered on $\widetilde{S}$.*

Proof. Temperedness is weak containment in the left regular representation. For a compactly supported continuous function f on S, lift it to a function $\widetilde{f}$ on $\widetilde{S}$ by averaging over the finite kernel and normalizing Haar measure so that integrated operators agree on kernel-invariant representations. The kernel-invariant part of $\lambda_{\widetilde{S}}$ is the pullback of λ_S. Hence the integrated operator criterion for weak containment gives

$$\|\pi^{\otimes m}(f)\| = \|(\pi \circ p)^{\otimes m}(\widetilde{f})\| \le \|\lambda_{\widetilde{S}}(\widetilde{f})\| = \|\lambda_S(f)\|.$$

This proves descent. Pullback is immediate because weak containment is preserved under composition with a quotient map. Applying the same argument to the finite-kernel isotypic component gives the converse. $\square$

Lemma 2.2 (Products preserve a common tensor index)**.** *Let $S = S_1 \times \cdots \times S_r$ and let π be a unitary representation with no nonconstant vectors fixed by a noncompact factor. Suppose that on the orthogonal complement of the*

S_i-fixed vectors the restriction to S_i has a tensor-tempered index m_i. Any common multiple m of the m_i is a tensor-tempered index for the nonconstant S-representation.

PROOF. For every factor, $\pi^{\otimes m}|_{S_i}$ is a tensor power of a tempered representation. Fell absorption, proved in AHD V, shows that tensoring a representation weakly contained in the regular representation with an arbitrary unitary representation stays weakly contained in a multiple of the regular representation. The regular representation of the product is the Hilbert tensor product of the regular representations of the factors. Applying the integrated-operator norm criterion one factor at a time gives $\pi^{\otimes m} \prec \lambda_S$. $\square$

3. From a periodic real orbit to automorphic spectrum

Proposition 3.1 (Congruence period representations inside adelic spectrum)**.** *Let xS be a periodic semisimple orbit in the arithmetic/congruence setting of this volume. After conjugating, suppose that the connected acting group is the real identity component of a connected semisimple $\mathbb{Q}$-group $\mathbf{S}$ and that $\Gamma_S = \Gamma \cap S$ is a congruence lattice. Then the right representation on $L^2(\Gamma_S \backslash S)$ occurs in the real-place restriction of*

$$L^2(\mathbf{S}(\mathbb{Q})\backslash\mathbf{S}(\mathbb{A}))$$

after taking invariants under a compact open subgroup at the finite places and selecting finitely many real connected components. The orthogonal complement of constants maps into the orthogonal complement of the locally constant component functions.

PROOF. Choose a compact open $K_f < \mathbf{S}(\mathbb{A}_f)$ whose intersection with the chosen rational real component gives Γ_S. The real/adelic component dictionary from Chapter 1 identifies the K_f-fixed adelic quotient with a finite disjoint union of real congruence components. Extending a function by zero to the other components gives an isometric, S-equivariant embedding. The functions constant on individual real connected components form a finite-dimensional invariant subspace. Passing to its orthogonal complement gives the last assertion. $\square$

Proposition 3.2 (Reduction to absolutely almost-simple simply connected factors)**.** *Let $\mathbf{S}/\mathbb{Q}$ be connected semisimple and fix a finite central isogeny*

$$\prod_{i=1}^{r} \mathrm{Res}_{F_i/\mathbb{Q}}\,\mathbf{S}_i \longrightarrow \mathbf{S},$$

where every $\mathbf{S}_i/F_i$ is absolutely almost simple and simply connected. If a tensor-tempered index is known for every noncompact archimedean local factor *of every $\mathbf{S}_i$, then the non-locally-constant part of the adelic $\mathbf{S}$-representation has a tensor-tempered index which depends only on the finite list of real local isomorphism classes in the cover.*

Proof. Pull the representation through the finite central isogeny. Take a common multiple of the local tensor indices and apply Lemma 2.2. Descend with Lemma 2.1. Compact archimedean factors contribute only compact K-type bookkeeping and do not change temperedness. □

Warning 3.3 (Algebraic ownership)**.** The structural decomposition of a connected semisimple rational group into restrictions of scalars of absolutely almost-simple simply connected factors is part of the algebraic-group floor already used in AHD IV. The proposition above owns the representation-theoretic transfer, not a new proof of that classification theorem.

4. From tensor-temperedness to Chapter-4 coefficients

Lemma 4.1 (Restriction of the regular representation)**.** *Let $H < S$ be a closed unimodular subgroup. The restriction $\lambda_S|_H$ is a direct integral of copies of λ_H. Consequently, if $\rho \prec \lambda_S$ then $\rho|_H \prec \lambda_H$.*

Proof. Disintegrate Haar measure on S over $H\backslash S$. In the resulting measurable field of $L^2(H)$ spaces, left translation by H acts only in the H-coordinate. Unimodularity removes the modular correction. Hence the restricted regular representation is a direct integral of regular H-representations. Weak containment passes to restrictions. □

Proposition 4.2 (Tensor-tempered period representations give polynomial unipotent decay)**.** *Let μ be the invariant probability on a periodic S-orbit and suppose that the nonconstant representation $L^2_0(\mu)^{\otimes m}$ is tempered. For a fixed one-parameter unipotent subgroup u_t in a fixed semisimple subgroup used by the Chapter-4 shearing argument, there are $\sigma > 0$ and a Sobolev order a such that*

$$|\langle u_t f_0, f_0\rangle_{L^2(\mu)}| \ll (1+|t|)^{-\sigma}\,\mathcal{S}_a(f)^2, \qquad f_0 = f - \mu(f).$$

For a family with a common tensor index and common Sobolev conventions, the implicit constant is uniform.

Proof. By Lemma 4.1, the tensor power remains tempered after restriction to the fixed semisimple subgroup. The Cowling–Haagerup–Howe transfer theorem in AHD V bounds K-finite matrix coefficients by a fixed power of the Harish–Chandra spherical function. The chamber estimate along the chosen SL_2 containing u_t gives $\Xi(u_t) \ll (1+|t|)^{-c}$, hence a power $(1+|t|)^{-c/m}$. Finally decompose a smooth vector into compact K-types and sum by the compact Sobolev estimates of AHD V, increasing the Sobolev order by a fixed amount. □

Warning 4.3. The rank-one branch is completed in Appendix D; the decomposition below records the mathematical subproblems used in that proof.

5. Rank-one reduction ledger

The historical Stage-3 subledger ended with one broad automorphic source gate. The later repair campaign has narrowed it substantially:

(1) **Structural transfer — closed.** Finite covers and products, real-to-adelic embedding, semisimple-factor reduction, and the Cowling–Haagerup–Howe/Sobolev transfer.
(2) **Absolute rank ≥ 2 — closed.** Auxiliary-place direct transport, closed by Stage 5.3.
(3) **Type-A_1 algebraic reduction — closed.** Reduction to quaternionic forms and the split real local types.
(4) GL_2 **archimedean reduction — conditional.** The exponent bound is reduced to the specialized local Rankin–Selberg package.
(5) **Global** $\mathrm{GL}_2 \times \mathrm{GL}_2$ **Rankin–Selberg — conditional owner closed.** Continuation and the functional equation are reconstructed from Poisson summation and Eisenstein unfolding, conditional only on the local Whittaker theorem in the normalization actually used.
(6) `GL2.RS-LOCAL` **— OPEN.** Specialized local Whittaker Rankin–Selberg package.
(7) `A1E.JL` **— OPEN.** Quaternionic Jacquet–Langlands/packet transfer preserving the target split archimedean parameter.
(8) `C04.01` **— conditional.** The Chapter-4 spectral theorem now depends only on the preceding two open roots.

Thus reconstruction restores the transfer architecture without reverting to the older, broader "Clozel/Burger–Sarnak" placeholder. The original Stage-4–5.5 proved inputs are preserved in the repair-source directory.

APPENDIX C

Volume–Discriminant Comparison

The representative-height comparison, the lower volume bound, and the upper volume bound use different mechanisms, so we prove them separately.

1. The six-step logical decomposition

The argument is organized into the following six ingredients:

VD.01: compact representative and the square-root comparison between subgroup height and EMV discriminant;

VD.02: polynomially short lattice elements that generate the acting rational group Zariski densely;

VD.03: bounded generators imply a polynomial height bound for the fixed space and hence for the rational Lie algebra;

VD.04: lower volume bound $\mathrm{vol}(xH) \gg \mathrm{disc}(xH)^a$;

VD.05: arithmetic upper bound $\mathrm{vol}(xH) \ll \mathrm{disc}(xH)^b$;

VD.06: assembly of the two-sided comparison in the Chapter-2 normalization.

The upper bound is further decomposed into the arithmetic steps recorded in Section 5.

2. Compact representatives and the square-root law

Proposition 2.1 (Compact representative and square-root comparison)**.** *In the fixed finite-centralizer EMV setting, choose a compact set meeting the periods under consideration, as supplied by the inherited semisimple nonescape/height infrastructure. If $xH = \Gamma gH$ is represented with g in that compact set, then*

$$\mathrm{ht}(\mathrm{Ad}(g)\mathfrak{h}) \asymp \mathrm{disc}_{\mathrm{EMV}}(xH)^{1/2}.$$

After converting from the EMV tensor convention to the exterior-discriminant convention of Chapter 2, the two notions remain polynomially comparable with fixed positive powers.

Proof. Let $r = \dim H$ and let w be a primitive integral Pluecker vector spanning $\bigwedge^r \mathrm{Ad}(g)\mathfrak{h}$. The Euclidean norm on this exterior line and the absolute value of the determinant form induced by the Killing form vary continuously with the point of G/H. On the compact image of the chosen set, their ratio is bounded above and below by positive constants. The EMV discriminant tensor is, up to the fixed ambient integral normalization, the square of this primitive exterior vector divided by the Killing-form normalization. Hence its size is comparable to $\|w\|^2$, while $\|w\|$ is the Pluecker height of the rational Lie algebra. Changing rational exterior representations changes these heights by fixed powers only. □

3. Short Zariski generators: the repaired lower branch

The historical Stage-3 manuscript left VD.02 as a source-reconstruction laboratory. The surviving Stage-4 module `01_VD_Lower_Bound_Repair.tex` now proves the full mechanism conditional only on `C04.01`. It is important to record what is proved, because this is the bridge through which the automorphic root influences Chapter 2.

Theorem 3.1 (Short Zariski generators from period covolume, current owner). *Assume the uniform periodic-orbit spectral theorem* `C04.01`*. In the fixed finite-centralizer arithmetic setting, if $\Lambda_g = \Gamma \cap gHg^{-1}$ and $V = \mathrm{vol}(xH)$, then there are constants $A, C > 0$, depending only on the fixed ambient data, such that the subgroup generated by*

$$\{\lambda \in \Lambda_g : \ \max(\|\lambda\|, \|\lambda^{-1}\|) \le CV^A\}$$

is Zariski dense in gHg^{-1}.

Proof. The detailed proof is owned verbatim by the Stage-4 lower-bound repair module. Its architecture is as follows. A same-scale smoothed lattice-counting lower bound on the genuine period lattice uses `C04.01`. If the short lattice elements were trapped in a proper algebraic subgroup, a second lattice-counting estimate on the corresponding proper homogeneous quotient would give an incompatible upper bound. The spectral gap for that proper quotient is proved geometrically in the repair module: Chevalley linearization reduces to a proper algebraic orbit, an SL_2 stability argument excludes almost-invariant vectors, and Herz majorization bounds the relevant induced representation. Thus no second automorphic property-τ input is hidden in the argument. □

4. Height propagation from bounded generators

Proposition 4.1 (Fixed-space height from finitely many bounded generators)**.** *Suppose a finite set $F \subset gHg^{-1} \cap \Gamma$ is Zariski dense in gHg^{-1} and every element has matrix height at most R. Then*

$$\mathrm{ht}(\mathrm{Ad}(g)\mathfrak{h}) \ll R^{C_0}$$

for a constant C_0 depending only on the fixed ambient representation.

PROOF. Choose a rational Chevalley representation whose distinguished fixed subspace has pointwise stabilizer equal to the required algebraic subgroup. The common fixed space of the Zariski-dense set F is already cut out by at most $\dim W$ of the equations $(\lambda - 1)w = 0$, because each genuinely new equation lowers the solution-space dimension. The coefficients of those linear systems have height $R^{O(1)}$. Minors and Cramer's rule therefore give a primitive Pluecker vector for the common fixed space of height $R^{O(1)}$. The Lie algebra of its pointwise stabilizer is another kernel of a fixed-size linear system with coefficients polynomial in that Pluecker vector, so a second minors/Cramer estimate gives the result. □

Corollary 4.2 (Current lower volume–discriminant inequality)**.** *Assume* `C04.01`*. Then, in the fixed finite-centralizer EMV setting, there are $a, c > 0$ such that*

$$\mathrm{vol}(xH) \geq c\ \mathrm{disc}(xH)^a.$$

PROOF. Let $V = \mathrm{vol}(xH)$. Theorem 3.1 provides bounded Zariski generators of height $V^{O(1)}$; Proposition 4.1 therefore gives $\mathrm{ht}(\mathrm{Ad}(g)\mathfrak{h}) \ll V^C$. Combine this with Proposition 2.1 and the fixed-power conversion between the discriminant conventions, then rearrange. □

5. The current arithmetic upper branch

The upper direction is not obtained by reversing the preceding proof. It is an arithmetic covolume calculation. Stage 4 reconstructed the self-contained preparation from the 2025 compact-adelic theory:

THEOREM 5.1 (Low-height maximal rational torus). *For a connected reductive rational subgroup $L < \mathrm{SL}_N$, there is a maximal $\mathbb{Q}$-torus $T < L$ whose height is bounded by a fixed power of* $\mathrm{ht}(L)$.

PROOF. This is proved in the surviving upper-arithmetic repair module. The proof constructs a short integral regular semisimple element using an explicit adjoint-discriminant polynomial. Taking its connected centralizer lowers semisimple rank unless it is already a maximal torus, so induction through connected centralizers terminates. At each stage bounded polynomial avoidance and short lattice bases control height by a fixed power. □

THEOREM 5.2 (Polynomial splitting-field discriminant). *The torus in Theorem 5.1 admits a finite Galois splitting field $F/\mathbb{Q}$ satisfying*

$$|\,\mathrm{Disc}(F)| \ll \mathrm{ht}(T)^B$$

for a fixed exponent depending only on the ambient dimension.

Proof. Choose a short integral basis of the torus Lie algebra. The characteristic polynomials of the basis matrices have coefficients polynomially bounded by the torus height. A splitting field for their product simultaneously diagonalizes the torus. Since both the degrees and coefficient heights are bounded, standard discriminant estimates for finite extensions give the stated polynomial bound. The detailed integral bookkeeping is in the surviving upper-arithmetic repair module. $\square$

Remark 5.3 (Completion of the upper branch)**.** The arithmetic preparation above is combined in Appendix E with the Prasad/Borel–Prasad volume formula and local-factor estimates. The resulting theorem gives

$$\mathrm{vol}(xH) \ll \mathrm{disc}(xH)^C$$

with constants depending only on the fixed ambient arithmetic datum. Together with the lower bound proved earlier in this appendix, this yields the two-sided comparison used in Chapter 2.

APPENDIX D

Rank-One Automorphic Completion

1. Purpose and exact scope

The only rank-one automorphic input needed in the finite-centralizer period argument is a uniform tensor-tempered certificate for the representations of the noncompact simple factors of a periodic semisimple orbit. Earlier repairs reduced this to two concrete assertions: the local $\mathrm{GL}_2 \times \mathrm{GL}_2$ Rankin–Selberg zeta-integral theorem and the specialized Jacquet–Langlands transfer from quaternionic inner forms to GL_2 while preserving every split archimedean component. We prove precisely these two assertions here. Nothing in this appendix asserts the full local or global Langlands correspondence.

Throughout, F is a local field of characteristic zero, $G_F = \mathrm{GL}_2(F)$, N is the upper unipotent subgroup, and $\psi : F \to \mathbb{C}^\times$ is a fixed nontrivial unitary additive character.

2. Whittaker and Kirillov models in degree two

Lemma 2.1 (Gelfand–Kazhdan criterion for GL_2)**.** *Let $\tau(g) = w\,{}^t g\, w^{-1}$ with $w = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Every distribution T on G_F satisfying*

$$T(n_1 g n_2) = \psi(n_1)\psi(n_2)\, T(g) \qquad (n_1, n_2 \in N(F))$$

is τ-invariant. Consequently, for every irreducible admissible π,

$$\dim \mathrm{Hom}_{N(F)}(\pi, \psi)\, \dim \mathrm{Hom}_{N(F)}(\widetilde{\pi}, \psi^{-1}) \leq 1.$$

Proof. Use the Bruhat decomposition $G = B \sqcup BwB$. On the open cell every g has unique coordinates $g = n(x)w\,\mathrm{diag}(a, d)n(y)$ modulo the central diagonal overlap. The two equivariance relations therefore determine the restriction of T from its distribution on the diagonal parameters (a, d), and the involution τ fixes those parameters while interchanging x and y; hence the restriction to BwB is τ-invariant.

The difference $T - \tau T$ is thus supported on the closed cell B. Filter distributions supported on B by transverse order along the lower-left coordinate. The principal transverse symbol of order r is a distribution on B transforming under conjugation by $\mathrm{diag}(t, 1)$ both by the character forced from the two N-equivariance relations and by t^r from the normal coordinate. Varying t

gives incompatible characters (the additive character $\psi(tx)$ is nontrivial), so the principal symbol vanishes. Descending on r gives $T - \tau T = 0$.

For the multiplicity statement take $\ell \in \mathrm{Hom}_N(\pi, \psi)$ and $\widetilde{\ell} \in \mathrm{Hom}_N(\widetilde{\pi}, \psi^{-1})$ and form the matrix-coefficient distribution $T_{\ell,\widetilde{\ell}}(f) = \widetilde{\ell}(\pi(f)v_\ell)$ in the usual smoothing realization. The preceding invariance identifies the two possible rank-one pairings and makes the bilinear map $\mathrm{Hom}_N(\pi, \psi) \times \mathrm{Hom}_N(\widetilde{\pi}, \psi^{-1}) \to \mathbb{C}$ nondegenerate of rank at most one. Irreducibility gives the displayed product bound. $\square$

Lemma 2.2 (Uniqueness of the Whittaker functional). *Let π be an irreducible admissible generic representation of G_F. Then*

$$\dim \mathrm{Hom}_{N(F)}(\pi, \psi) = 1.$$

Proof. Genericity means that the Hom-space is nonzero. The contragredient of a generic irreducible representation of GL_2 is generic for ψ^{-1}: if ℓ is a nonzero Whittaker functional, the contragredient pairing and the involution $g \mapsto w^t g^{-1} w^{-1}$ produce a nonzero ψ^{-1}-functional on $\widetilde{\pi}$. Lemma 2.1 then says the product of the two positive dimensions is at most one. Hence both are one. $\square$

Proposition 2.3 (Kirillov model). *For generic irreducible π, the map*

$$W \longmapsto (y \mapsto W(\mathrm{diag}(y, 1)))$$

identifies the Whittaker model with a G_F-stable space of smooth functions on $F^\times$. It contains $C_c^\infty(F^\times)$, and the action of the mirabolic subgroup is

$$\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} f(y) = \psi(by) f(ay),$$

$$\begin{pmatrix} z & 0 \\ 0 & z \end{pmatrix} f(y) = \omega_\pi(z) f(y).$$

The Weyl element acts by a Fourier–Mellin transform uniquely determined by Lemma 2.2.

Proof. The displayed formulas follow directly from Whittaker equivariance. If the restriction to the torus vanished, the Bruhat decomposition would force W to vanish on the open cell and hence everywhere, proving injectivity. Given $\phi \in C_c^\infty(F^\times)$, compact induction from the mirabolic subgroup and averaging against ψ^{-1} constructs a Whittaker vector whose torus restriction is ϕ. The formulas show that the resulting space is stable under the mirabolic subgroup; adjoining the Weyl element generates G_F. Uniqueness of the Weyl action is exactly Whittaker uniqueness. $\square$

3. The local Rankin–Selberg integral

For Whittaker functions W_i of generic irreducible representations π_i and a Schwartz–Bruhat function Φ on F^2, put

$$Z_F(s, W_1, W_2, \Phi) = \int_{N(F)\backslash G_F} W_1(g)W_2(g)\Phi((0,1)g)|\det g|^s\,dg,$$

with the harmless central-character shift absorbed into s.

THEOREM 3.1 (Local $\mathrm{GL}_2 \times \mathrm{GL}_2$ Rankin–Selberg theorem). *The local integrals above have the following properties.*

(1) *In the nonarchimedean case they span a principal fractional ideal of $\mathbb{C}[q^s, q^{-s}]$, generated by a reciprocal polynomial $L_F(s, \pi_1 \times \pi_2)$ normalized to have constant term one.*

(2) *For unramified π_i and normalized spherical Whittaker functions, with $\Phi = \mathbf{1}_{\mathcal{O}_F^2}$, the integral is exactly the Euler factor*

$$\prod_{i,j=1}^{2} (1 - \alpha_i\beta_j q^{-s})^{-1}.$$

(3) *For every s_0 there are local data for which $Z_F(s)/L_F(s, \pi_1 \times \pi_2)$ is holomorphic and nonzero near s_0.*

(4) *There is a local functional equation*

$$Z_F(1-s, \widetilde{W}_1, \widetilde{W}_2, \widehat{\Phi}) = \gamma_F(s, \pi_1 \times \pi_2, \psi) Z_F(s, W_1, W_2, \Phi),$$

where $\gamma_F = \varepsilon_F(s)L_F(1-s, \widetilde{\pi}_1 \times \widetilde{\pi}_2)/L_F(s, \pi_1 \times \pi_2)$.

(5) *Under normalized parabolic induction the local L- and γ- factors are multiplicative in the inducing quasi-characters. For unitary tempered data the normalized factor is holomorphic for $\Re s > 0$. At an archimedean place this gives products of the Tate gamma factors $\Gamma_{\mathbb{R}}(s+\mu)$ and $\Gamma_{\mathbb{C}}(s+\mu)$ attached to the four pairwise sums of Langlands parameters.*

(6) *If F is nonarchimedean and χ is sufficiently ramified compared with π_1, π_2, then the twist has trivial local L-factor and the local gamma factor is the explicit Gauss-sum monomial*

$$\gamma_F(s, (\pi_1 \otimes \chi) \times \pi_2, \psi) = c(\pi_1, \pi_2, \chi, \psi)\, q^{-a(\chi)(4s-2)}, \qquad |c| = 1,$$

up to the fixed central-character conductor shift dictated by the chosen normalization. In particular its absolute value and conductor dependence are uniformly explicit in the form required by the Blomer–Brumley local transform.

Proof. We give the degree-two argument because each later use needs all five parts. Iwasawa decomposition writes the quotient $N\backslash G$ as torus times a maximal compact group. After K-finite decomposition the integral is a finite sum of Mellin integrals of products of Kirillov functions. Over a nonarchimedean field, a Kirillov function is compactly supported toward $|y| \to \infty$ and, toward 0, is a finite sum

$$\sum_{\chi,m} c_{\chi,m}\chi(y)v(y)^m|y|^{\lambda_\chi}$$

coming from the finite-dimensional Jacquet module. Splitting $F^\times$ into valuation shells therefore makes every local integral a rational function of q^{-s}. The set of all such rational functions is stable under multiplication by $q^{\pm s}$ and contains 1: choose two compactly supported Kirillov functions on $\mathcal{O}_F^\times$ and a Schwartz function which is one on the corresponding Iwasawa cell. Since $\mathbb{C}[q^s, q^{-s}]$ is a PID, the fractional ideal is principal. The normalization of its generator proves (1).

For spherical data, the elementary degree-two Casselman–Shalika recurrence is obtained by applying the Hecke operator of $K\,\mathrm{diag}(\varpi, 1)K$ to the normalized Whittaker function. If $c_m = W(\mathrm{diag}(\varpi^m, 1))$, then $c_{m+2}-(\alpha+\beta)c_{m+1}+\alpha\beta c_m = 0$, with $c_0 = 1$ and the standard $\delta_B^{1/2}$ normalization. Multiplying the two recurrences and summing the resulting geometric series over $m \geq 0$ gives exactly $\prod_{i,j}(1 - \alpha_i\beta_j q^{-s})^{-1}$, proving (2).

For (3), choose compactly supported Kirillov functions on a sufficiently small annulus and choose Φ supported in the corresponding open Iwasawa cell. The integral is then a nonzero finite Laurent polynomial at s_0. Replacing one function by a finite linear combination of its diagonal translates lets us interpolate any prescribed nonzero value at q^{-s_0}; division by the generator proves the normalized nonvanishing assertion.

For (4), apply Fourier inversion on F^2 to Φ. The two sides of the claimed functional equation define trilinear forms with the same $N \times N$ equivariance. After inserting the Weyl element, uniqueness of the Whittaker functional makes the space of such forms one-dimensional; the ratio is therefore a scalar meromorphic function of s. Applying Fourier transform twice shows the reciprocal relation at $1 - s$, and comparison with unramified data identifies the scalar with the displayed quotient of Euler factors. This is the local functional equation.

Finally, for a principal series the Kirillov model is generated by the two inducing characters. Expanding the two Whittaker functions and applying the one-dimensional Tate functional equation term by term factors the local coefficient into the four pairwise Tate coefficients. Passage to special representations follows by taking the unique generic subquotient, and the supercuspidal case is already covered by the compact-support part of the Kirillov model. At archimedean places the same Mellin transform is the classical Mellin transform of the K-Bessel/Whittaker functions; evaluating it by the beta integral gives the stated $\Gamma_{\mathbb{R}}$ and $\Gamma_{\mathbb{C}}$ factors. For a unitary tempered representation every inducing exponent has real part zero (and a supercuspidal parameter is bounded), so these gamma factors have no poles for $\Re s > 0$; subquotients inherit the same holomorphy.

It remains to prove the highly ramified stability in (6). In the Kirillov model, once the conductor of χ is larger than the levels of the fixed Whittaker vectors, integration over each unit coset annihilates every term except the smallest annulus on which the additive character has matching conductor. The surviving integral is $\int_{\mathcal{O}_F^\times} \chi(u)\psi(\varpi^{-a(\chi)}u)\,du$, the normalized Gauss sum. Fourier transform exchanges that annulus with its dual annulus and contributes the modulus $q^{-a(\chi)(4s-2)}$ for the four-dimensional tensor parameter. Since a ramified character has no inertia invariants, the corresponding local L-factor is 1. Translation by the fixed central characters changes only the conductor-independent unit scalar and the explicitly recorded integer shift. The normalized Gauss sum has modulus one. This proves (6), completes (5), and proves the theorem. $\square$

Remark 3.2 (Local-to-global Rankin–Selberg closure)**.** Theorem 3.1 supplies the local Euler factors, local functional equations, normalized nonvanishing data, archimedean transform bounds, and highly ramified stability used in the global Rankin–Selberg construction of Appendix B. Substituting these local identities into the global Euler product yields the continuation, functional equation, pole statement, and transform estimates used below.

4. Quaternionic Jacquet–Langlands in the required form

Let K be a number field, D a quaternion algebra over K, and $D^\times(\mathbb{A})$ its ideles. At every split place identify $D_v^\times$ with $\mathrm{GL}_2(K_v)$.

Lemma 4.1 (Matching local test functions)**.** *At a nonsplit place v, every compactly supported smooth function f_v^D on $D_v^\times/K_v^\times$ has a transfer f_v on $\mathrm{GL}_2(K_v)$ whose regular elliptic orbital integrals agree under equality of reduced characteristic polynomials and whose split regular orbital integrals vanish. Conversely such transfers span the elliptic cocenter.*

Proof. Regular conjugacy classes in both groups are parametrized by separable quadratic K_v-algebras embedded in the algebra together with a norm class. On a sufficiently small neighborhood of a regular element, the conjugation map from a transversal times the conjugacy class is a submersion. Push a partition of unity on the quotient through these charts and prescribe the same orbital integral on the matching characteristic-polynomial parameter. The nonsplit group has only elliptic regular classes, while the imposed zero on split classes removes the classes having no quaternionic mate. The singular set has positive codimension; subtracting commutators of compactly supported functions kills distributions supported there. Hence the construction descends to and spans the elliptic cocenter. □

Lemma 4.2 (Simple trace formula in the transfer scope)**.** *Assume first that D is nonsplit and choose a finite division place v_0. Let $f^D = \otimes_v f_v^D$ be factorizable with $f_{v_0}^D$ a matrix coefficient (equivalently a pseudocoefficient after fixing the central character) of the prescribed irreducible $D_{v_0}^\times$-representation, and let $f = \otimes_v f_v$ be matching data. Then f_{v_0} is cuspidal on $\mathrm{GL}_2(K_{v_0})$ (all of its proper-parabolic orbital/constant terms vanish), both right-convolution operators are trace class on the discrete spectra, and*

$$\operatorname{tr} R_{\mathrm{cusp}}(f) = \sum_{\{\gamma\}_{\mathrm{ell}}} \mathrm{vol}(G_\gamma(K)\backslash G_\gamma(\mathbb{A}))\, O_\gamma(f),$$

with the analogous formula for $D^\times$, and the two geometric sums agree for matching functions. If D is split at every place, the statement is read as the tautological identity of the two trace formulas.

Proof. The convolution kernel is $K_f(x,y) = \sum_{\gamma\in G(K)} f(x^{-1}\gamma y)$. Compact support modulo the center makes the sum locally finite. Integrating $K_f(x,x)$ over a truncated fundamental domain and grouping rational elements by conjugacy class gives the orbital-integral expansion by Fubini. At the division place v_0, the matching transfer of a $D_{v_0}^\times$ matrix coefficient belongs to the elliptic/cuspidal cocenter of $\mathrm{GL}_2(K_{v_0})$; equivalently its integral along the unipotent radical of the unique proper parabolic is zero. Hence the global parabolic constant term of the kernel vanishes, all continuous and residual spectral terms disappear, and the truncation boundary is zero. Thus the operator acts only on the cuspidal spectrum and the truncation can be removed, giving the displayed trace. For the quaternionic inner form the same place is anisotropic modulo center, so no proper parabolic term exists there and the kernel argument is simpler. Central terms match by the common central character, regular elliptic terms match by Lemma 4.1, and regular split terms vanish on the transferred test function. Singular noncentral terms are limits of regular orbital integrals and vanish by the same cuspidal constant-term calculation. Hence the two geometric sums, and therefore the traces, agree. □

Theorem 4.3 (Specialized global Jacquet–Langlands transfer)*.* *Let σ be an irreducible cuspidal automorphic representation of $D^\times(\mathbb{A}_K)$ with unitary* *central character. There is a unique cuspidal automorphic representation* $\pi = \mathrm{JL}(\sigma)$ *of* $\mathrm{GL}_2(\mathbb{A}_K)$ *such that*

(1) $\pi_v \simeq \sigma_v$ *at every split place under the fixed identification;*
(2) *at every nonsplit place the Harish–Chandra characters agree on matching regular elliptic elements up to the standard sign;*
(3) *in particular, every split archimedean Langlands parameter is preserved exactly.*

Proof. If $D \simeq M_2(K)$, the assertion is the identity transfer, so assume that D has a division place v_0. Choose a finite set S containing all nonsplit and ramified places. At v_0 choose a matrix coefficient (or pseudocoefficient with fixed central character) of σ_{v_0} and transfer it by Lemma 4.1; this single cuspidal local factor kills the continuous and residual spectra without imposing any artificial supercuspidal type at a split place. At the other nonsplit places choose matching functions, and at split places use literally identical functions. Outside S vary spherical Hecke functions. Lemma 4.2 gives equality of the two simple trace formulas. On their geometric sides, central terms agree by the common central character, regular elliptic terms agree by matching orbital integrals, and nonelliptic terms vanish by construction. Thus their cuspidal spectral traces agree for every choice of the remaining spherical Hecke functions.

Varying the unramified Hecke functions and using the elementary separation of characters of the commutative spherical Hecke algebra produces a cuspidal π whose unramified local parameters agree with those forced by σ. Now vary one place at a time. At a split place, equality of traces against all compactly supported test functions gives equality of Harish–Chandra characters and hence local isomorphism. At a nonsplit place, using only matching functions gives the elliptic character identity. Strong multiplicity one for GL_2 follows directly from the Rankin–Selberg pole theorem already reconstructed: if two cusp forms agree almost everywhere, $L(s, \pi_1 \times \widetilde{\pi}_2)$ has the pole forced by the common Euler factors, so the contragredient pairing is nonzero and the representations are isomorphic. Hence the global π is unique. The split archimedean places were among the places at which the test functions were literally identical, so their parameters are preserved, proving (3). □

5. The uniform type-A_1 spectral certificate

Proposition 5.1 (Uniform tensor-tempered exponent in the fixed ambient period problem)**.** *Fix the ambient rational representation and the finite list of real simple factor types occurring in Volume 9. There exists an integer* m_0 *such that for every periodic semisimple orbit and every noncompact* *absolute A_1 factor H_0, the representation of H_0 on the orthocomplement of the constants in its automorphic L^2-space has m_0-fold tensor power weakly contained in the regular representation.*

Proof. For split A_1 factors, Theorem 4.3 transfers every discrete automorphic constituent from the quaternionic form to GL_2 without changing the split archimedean parameter. Repairs 13–17 together with Theorem 3.1 reconstruct the Blomer–Brumley nonvanishing argument and give a uniform bound $|\operatorname{Re}\mu_v| \leq \frac{1}{2} - \vartheta$ at every relevant split archimedean place, with $\vartheta > 0$ depending only on the fixed degree and archimedean type. In the complementary-series parametrization of $\mathrm{SL}_2(\mathbb{R})$ or $\mathrm{SL}_2(\mathbb{C})$, matrix coefficients are therefore in $L^{p_0+\varepsilon}$ for one finite $p_0 = p_0(\vartheta)$ uniformly in the constituent. Choose an even integer $m_0 > p_0/2$. Products of m_0 such coefficients lie in $L^{2+\varepsilon}$ by Hölder. The tensor-power criterion of Volume 5 then implies temperedness of the m_0-fold tensor power. There are only finitely many factor types in the fixed ambient problem, so take the maximum of their exponents. Higher-rank factors were already closed by the direct-transport branch. This proves the uniform certificate. □

Remark 5.2 (Spectral consequence for Chapter 4)**.** The estimates above give the uniform tensor-tempered matrix-coefficient bound used in Chapter 4. Concretely, the exponent and Sobolev order in that chapter may be chosen from the constants in the preceding theorem, uniformly over the stated arithmetic family.

Proof. Combine Proposition 5.1 with the previously proved higher-rank branch and the pointwise matrix-coefficient theorem of Volume 5. The product and finite-index transfer lemmas of Volume 6 assemble the factorwise certificates on the full semisimple period. □

Proof and provenance. The proof architecture is the degree-two Jacquet–Langlands/Rankin–Selberg route used historically for the A_1 part of property (τ). The local Rankin–Selberg owner is reconstructed through Whittaker/Kirillov models, and the quaternionic transfer is reconstructed with the simple trace formula; there is no remaining named automorphic theorem used as a black-box proof owner in the rank-one branch.

APPENDIX E

Arithmetic Volume Formulae and Complexity Completion

1. The selected arithmetic statement

We only need a fixed-power comparison between the covolume of a semisimple period and the integral height of its rational Lie algebra. Proving the full classification of coherent parahoric collections is unnecessary; the fixed ambient representation makes every global degree, absolute root datum, and archimedean type bounded in advance.

Lemma 1.1 (Factor fields and bad places from subgroup height)**.** *Let* $\mathbf{M} < \mathbf{G} < \mathrm{SL}_N$ *be a connected semisimple rational subgroup of fixed dimension and Pluecker height at most* H*. Then its almost-simple factors are defined over number fields* k_i *with*

$$[k_i : \mathbb{Q}] \ll 1, \qquad |D_{k_i}| \ll H^C,$$

and the product of rational primes at which the integral model or the factor is not unramified is at most H^C*.*

Proof. The Lie bracket and the adjoint action on $\mathfrak{m}$ are matrices whose entries are ratios of minors of a primitive Pluecker matrix, hence have height $H^{O(1)}$. The centroid of each simple ideal is the simultaneous commutant of finitely many such matrices. Solving the resulting linear system by Cramer's rule gives a basis of the centroid of height $H^{O(1)}$. A primitive element therefore satisfies an integral polynomial of bounded degree and coefficient height $H^{O(1)}$. Its field discriminant is a polynomial in the coefficients, proving the discriminant bound. The structure constants of the factor in this basis have denominators dividing an integer of size $H^{O(1)}$; away from its prime divisors and the field discriminant the resulting group scheme is smooth with constant root datum. The product of the exceptional rational primes therefore divides an integer of size $H^{O(1)}$. □

Proposition 1.2 (Selected Prasad–Borel–Prasad covolume formula)**.** *Let* $\mathbf{M}$ *be as above, let* $\widetilde{\mathbf{M}} \to \mathbf{M}$ *be the simply connected cover of its derived group, and let* Λ *be the arithmetic lattice determined by the fixed ambient integral* *structure. With compatible Tamagawa and real Haar measures,*

$$\operatorname{vol}(\Lambda\backslash M_\infty) = C_\infty(\mathbf{M}) \prod_i |D_{k_i}|^{\dim \mathbf{M}_i/2} \prod_{v<\infty} e_v(P_v)\, c(\mathbf{M},\Lambda),$$

where P_v is the parahoric stabilizer of the local lattice, $e_v(P_v)$ is the reciprocal of its normalized local volume, and the finite central/form factor $c(\mathbf{M},\Lambda)$ is bounded above and below by fixed powers of the field discriminants and the product of the bad residue characteristics. Moreover, at a hyperspecial unramified place $e_v(P_v) = 1 + O(q_v^{-1})$ after extraction of the convergent Euler factors.

Proof. Start with Tamagawa measure on the simply connected cover. The adelic finite-volume theorem of Volume 4, applied to the fixed compact open subgroup at the finite places, gives a finite double-coset decomposition of the adelic quotient into real arithmetic quotients. This uses only finite adelic component reduction, not the general strong-approximation theorem that Volume 3 deliberately does not claim. Fubini on each double-coset component decomposes Tamagawa volume into the real covolume times the product of the finite compact-open volumes; summing the finitely many components inserts only the arithmetic component-index factor already absorbed into $c(\mathbf{M},\Lambda)$. In invariant differential coordinates, changing from the integral measure to the self-dual local measure contributes the different of k_i in each of the $\dim \mathbf{M}_i$ coordinates; the product formula gives $|D_{k_i}|^{\dim \mathbf{M}_i/2}$. At a smooth hyperspecial place, reduction modulo the maximal ideal gives

$$\operatorname{vol}(P_v) = q_v^{-\dim \mathbf{M}_i} |\overline{\mathbf{M}}_i(\mathbb{F}_{q_v})|.$$

The Bruhat decomposition of the finite reductive group writes the last cardinality as $q_v^{\dim \mathbf{M}_i} \prod_j (1-q_v^{-m_j-1})$; extracting the finitely many Dedekind/Artin Euler factors leaves $1 + O(q_v^{-2})$, hence an absolutely convergent product.

At a nonhyperspecial place the same reduction argument through the connected reductive quotient of the parahoric gives a rational function of q_v whose numerator and denominator degrees are bounded solely by the root datum. Therefore $e_v(P_v)$ is between q_v^{-C} and q_v^{C}. Only the bad places from Lemma 1.1 occur. Passing from the simply connected cover to $\mathbf{M}$, and from a principal arithmetic subgroup to its normalizer, produces kernels and cokernels in the centers and in the finite class groups of bounded-degree fields. The centers have bounded order. Minkowski's lattice argument gives $h_{k_i} \ll |D_{k_i}|^{1/2}(\log |D_{k_i}|)^C$, and the same argument for the finite ray data bounds the remaining cohomological indices by a fixed power of the discriminant and bad-prime product. These factors are precisely $c(\mathbf{M},\Lambda)$. The archimedean normalization belongs to a finite list of real forms in the fixed ambient representation, so C_∞ is bounded above and below by absolute constants. Multiplying the local identities proves the formula and the polynomial factor bound. □

THEOREM 1.3 (Upper volume–discriminant comparison). *In the fixed Volume-IX arithmetic setting there are constants $C, b > 0$ such that every periodic semisimple orbit satisfies*

$$\mathrm{vol}(xM) \le C\ \mathrm{disc}(xM)^b.$$

Together with the lower comparison proved in Repair 01 this yields the required two-sided fixed-power comparison.

PROOF. Conjugate the rational subgroup defining the orbit into the fixed ambient integral representation. Its primitive Pluecker height is a fixed-power equivalent of the orbit discriminant by Chapter 2. Apply Lemma 1.1: all field discriminants and the product of bad primes are bounded by a fixed power of $\mathrm{disc}(xM)$. In Proposition 1.2, the discriminant powers are fixed, the good Euler product is uniformly bounded, each bad local factor is a fixed power of its residue characteristic, and the form/central factor is a fixed power of the same global arithmetic data. Hence their product is at most $C\,\mathrm{disc}(xM)^b$. The comparison between adelic and the chosen real Haar normalization is a finite-component/finite-index factor already controlled in Chapter 1. This proves the upper bound. The lower bound from Repair 01 supplies the other direction. □

Corollary 1.4 (Adelic complexity comparison)**.** *For the fixed ambient adelic space, the Prasad volume and the discriminant complexities used in Chapters 6–8 determine one another up to fixed powers. If the ambient form varies, the same conclusion holds with an additional fixed power of the explicitly recorded ambient arithmetic complexity.*

PROOF. The proof of Proposition 1.2 records every factor. When the ambient space is fixed, root data, degrees and all normalization constants are fixed. When it varies, defining-equation height, field discriminants and bad-prime products are exactly the ambient-complexity coordinates; the same factorwise estimates give the claimed polynomial loss. □

Worked Example 1.5 (Hyperspecial places really cost nothing polynomially)**.** For SL_2 over an unramified local field, $|\,\mathrm{SL}_2(\mathbb{F}_q)| = q(q^2-1)$. Dividing by q^3 gives $(1-q^{-2})$. The product of these factors over good places is the inverse of a Dedekind-zeta value at 2, bounded away from zero in a fixed field and polynomially controlled in a bounded-degree varying field. Thus polynomial complexity comes from discriminants and bad places, not from an uncontrolled product over all good primes.

Proof and provenance. This appendix reconstructs only the corollary of the Prasad/Borel–Prasad volume formalism consumed by Volume 9: global discriminant power, parahoric local factors, and finite central/form indices. The proof is derived from the Tamagawa product decomposition and finite reductive-group counts, so the volume–discriminant gate no longer terminates at a named volume formula.

APPENDIX F

The Effective Semisimple-Period Machine

1. A reusable quantitative theorem

The endpoint papers differ in arithmetic scope, but their dynamical core is the same. This appendix proves that core once. Let $X = \Gamma\backslash G$ (or one real component of an adelic quotient), and let xH be a periodic orbit of a connected semisimple group without compact factors.

Definition 1.1 (Quantitative period package)**.** A family of periodic H-orbits has a quantitative period package with exponent data (κ, A, B) if the following hold uniformly.

- (SG) $L^2_0(xH)$ has a tensor-tempered exponent bounded by κ.
- (SOB) local product charts and Sobolev norms lose at most $\mathrm{comp}(xH)^A$.
- (CL) if two quantitatively generic H-points have a return of size ρ whose Chevalley coordinates are ρ^B-close to a rational stabilizer, effective closing produces a periodic intermediate orbit of arithmetic complexity $\rho^{-O(1)}$.
- (ALG) a transverse rational direction of height at most Q generates an intermediate rational subgroup of height $Q^{O(1)}$.

Lemma 1.2 (Quantitatively generic points)**.** *Assume (SG) and (SOB). For every smooth f with zero ambient mean and every scale $T \geq 2$, all but $O(T^{-c})$ of xH consists of points y for which*

$$\left|\frac{1}{|B_T|}\int_{B_T} f(yh)\,dh\right| \ll T^{-c}\,\mathrm{comp}(xH)^A \mathcal{S}_d(f),$$

where $c > 0$ depends only on the tensor-tempered exponent and the fixed real factor types.

Proof. The matrix-coefficient theorem of Volume 5 applied to the fixed tensor power gives an integrable power bound for correlations of the H-action. Expanding the L^2 norm of the averaging operator over B_T and integrating the correlation kernel gives $\|A_T f\|_2 \ll T^{-c}\mathcal{S}_d(f)$, with the chart loss recorded by (SOB). Chebyshev's inequality shows that the exceptional set on which $|A_T f| > T^{-c/2}\,\mathrm{comp}^A\,\mathcal{S}_d(f)$ has measure $O(T^{-c})$. Apply this to a finite

Sobolev-dense test family and use the Sobolev embedding estimate to obtain the displayed uniform genericity statement. □

Lemma 1.3 (Nearby generic pair or sparse orbit)**.** *Fix a compact thick part and a scale r. If a periodic H-orbit has more $r^{-d_H+\eta}$ generic r-separated points in a fixed bounded H-box, then two such points differ by $g = \exp Z$ with $\|Z\| \ll r$ and with nonzero transverse component. If no such pair exists, the orbit is contained in $O(r^{-d_H+\eta})$ local H-tubes and hence has period volume $\ll r^{-O(1)}$.*

Proof. Cover the fixed thick compact set by $O(r^{-\dim G})$ product boxes of radius r, each foliated by H-plaques. If two generic points fall in the same ambient box but in different plaques, their displacement has the desired transverse component. If this never occurs, every occupied ambient box contains generic points in only one plaque. Comparing the Haar mass of an r-plaque with the total number of occupied boxes bounds the orbit volume by a fixed power of r^{-1}. The cusp part is discarded using the proved height tail estimate from Volumes 1 and 7; its mass is smaller than the generic exceptional set after choosing the height cutoff as a power of r. □

Lemma 1.4 (Polynomial shearing gives new almost invariance)**.** *Let $y, y \exp Z$ be the pair from Lemma 1.3. If the transverse component of Z is not contained in the Lie algebra of the current intermediate subgroup S, then for some nonzero $Y \in \mathfrak{g}/\mathfrak{s}$ and some $\epsilon = r^c$ the period measure is $O(r^c)$-almost invariant under $\exp(tY)$ for $|t| \leq 1$.*

Proof. Write $\mathrm{Ad}(h^{-1})Z$ in the finite S-module decomposition of $\mathfrak{g}/\mathfrak{s}$. Along a one-parameter unipotent subgroup of H, $\mathrm{Ad}(u(-t))Z$ is a polynomial in t. Choose the first nonzero coefficient outside $\mathfrak{s}$ and rescale t so that this coefficient has size one while every higher-order term is $O(r^c)$. Genericity of both points compares the two H-averages before the polynomial displacement leaves the local chart. Taylor expansion of test functions, with the Sobolev losses in (SOB), then gives $|\mu(f \circ \exp(tY)) - \mu(f)| \ll r^c \mathcal{S}_{d'}(f)$. This is exactly the quantitative polynomial-divergence argument already proved in Chapter 4; the present proof records how the scale is selected. □

Lemma 1.5 (Enlargement or algebraic closing)**.** *Under (CL) and (ALG), the output of Lemma 1.4 has one of two consequences:*

1. *the measure becomes r^c-almost invariant under a strictly larger connected rational intermediate group S' of complexity $r^{-O(1)}$; or*
2. *xH lies within r^c (in the homogeneous-measure/Sobolev sense) of a periodic intermediate orbit xM with $\mathrm{comp}(xM) \ll r^{-O(1)}$.*

Proof. Commute the new one-parameter almost invariance with the existing S- invariance. The Baker–Campbell–Hausdorff formula and the finite-dimensional S-module decomposition show that every vector in the Lie algebra generated by $\mathfrak{s}$ and Y inherits almost invariance with a fixed power loss. If this generated Lie algebra is quantitatively separated from every proper rational Lie algebra of bounded height, (ALG) identifies its rational algebraic hull and produces S'. Otherwise the exterior Chevalley vector is within a polynomially small distance of a bounded-height rational fixed line. Applying (CL) converts this near relation into an exact rational periodic orbit xM and gives the stated measure comparison. □

Theorem 1.6 (Effective period dichotomy). *Assume a quantitative period package. There are $c, d > 0$ such that for every periodic xH and every parameter $D \geq 2$, either*

$$|\mu_{xH}(f) - \mu_X(f)| \ll D^{-c}\mathcal{S}_d(f)$$

for all smooth f, or there is a proper periodic intermediate orbit xM containing xH whose arithmetic complexity is $\ll D^{O(1)}$.

Proof. Choose $r = D^{-C}$ with C larger than every loss in the preceding lemmas. If generic points are sparse, Lemma 1.3 already bounds the period volume by a power of D, and the two-sided complexity comparison from Appendix E identifies a low-complexity obstruction. If a nearby transverse pair exists, Lemmas 1.4 and 1.5 either close to an obstruction or enlarge the almost-invariance group. Strict enlargement can occur at most $\dim G$ times. If it reaches G, the Sobolev dual characterization of almost invariance gives the equidistribution estimate. Otherwise closing occurs. All scales are fixed powers of D, and finitely many iterations multiply only finitely many powers, so the final exponent remains positive. □

Corollary 1.7 (Finite descent and termination). *If the relevant intermediate groups form a finite poset, repeated application of Theorem 1.6 terminates after at most its cardinality and retains a polynomial error exponent.*

Proof. At each obstruction one replaces H by a strictly larger intermediate group. No group can recur. The poset is finite, so the process stops. A finite composition of maps $D \mapsto D^{c_i}$ is another positive power of D. □

2. Centralizers and compact adelic variants

Lemma 2.1 (Centralizer-aware closing). *Suppose periodic H-orbits vary in a positive-dimensional centralizer. If all proper containing rational orbits have discriminant at least D, then a return pair at scale D^{-C} which is not explained by centralizer motion produces a larger rational intermediate orbit* *of discriminant $D^{O(1)}$. Returns explained by centralizer motion preserve the same homogeneous period and are absorbed into the local product chart.*

PROOF. Decompose the transverse Lie algebra as $\mathfrak{z}_{\mathfrak{g}}(\mathfrak{h}) \oplus V$. The centralizer component commutes with the H-flow and hence cannot produce polynomial divergence; it is therefore a genuine moduli parameter rather than an obstruction to mixing. Project the Chevalley displacement to V. If the projection is large enough, the polynomial shearing argument applies. If it is smaller than the closing threshold, the effective Chevalley theorem of Ratner XIII rounds the nearly fixed exterior vector to a rational fixed line of polynomial height. The generated subgroup-height theorem then gives the proper rational M. By Appendix E, its height is polynomially equivalent to discriminant. This proves the dichotomy. □

Lemma 2.2 (Effective generation in compact adelic quotients)**.** *Let* $\mathbf{G}$ *have bounded defining degree and height* H_0*. Suppose a set of rational elements of height at most* Q *has logarithms whose reductions modulo sufficiently many good primes generate the full Lie algebra of a connected subgroup* $\mathbf{S}$*. Then the Zariski closure of the set contains* $\mathbf{S}$*, and defining equations for* $\mathbf{S}$ *have height* $(H_0Q)^{O(1)}$*.*

PROOF. Form all iterated commutators of length at most $\dim G$. Their logarithmic leading terms span the Lie algebra generated by the original logarithms. Gaussian elimination on the resulting rational coordinate matrix gives a basis with denominator and numerator $(H_0Q)^{O(1)}$. Let I be the ideal of polynomials of degree at most the fixed effective Noether bound vanishing on the generated set. Evaluation on the bounded-height commutator words is a linear system of polynomial height; its kernel therefore has a basis of height $(H_0Q)^{O(1)}$. At a good prime, Hensel–Newton lifting shows that a polynomial vanishing on all reductions and whose differential vanishes on the generated Lie algebra vanishes on the corresponding p-adic analytic subgroup. As this holds for more primes than can divide a nonzero bounded- height coefficient, the polynomial identity holds over $\mathbb{Q}$. Consequently the kernel ideal cuts out the connected algebraic subgroup with the generated Lie algebra. This proves both exact generation and the height bound. □

THEOREM 2.3 (Compact adelic effective-period machine). *On a compact adelic quotient, assume the property-(τ) certificate of Appendix D, the complexity comparison of Appendix E, and effective generation Lemma 2.2. Then the error of a semisimple period relative to the ambient measure is a negative fixed power of the minimal complexity of a proper intermediate period, with an additional fixed power of ambient arithmetic complexity when the ambient space varies.*

Proof. There is no cusp truncation on a compact quotient. Apply Theorem 1.6 with complexity equal to the minimum proper-intermediate complexity. The obstruction alternative contradicts the definition of that minimum once the scale exponent is chosen sufficiently small. Effective generation supplies the exact rational subgroup in the closing branch, and Appendix E converts all subgroup heights and volumes to the chosen complexity. The only varying constants are the fixed powers of ambient height recorded in those two appendices. ∎

APPENDIX G

Four Effective Endpoints

1. The finite-centralizer real theorem

THEOREM 1.1 (Einsiedler–Margulis–Venkatesh endpoint). *In the fixed arithmetic real setting of Chapter 1, suppose H is connected semisimple without compact factors and has finite centralizer in G. There exist $\delta, d > 0$ and V_0 such that for every periodic xH and every $V \geq V_0$ there is a connected intermediate group $H \subset S \subset G$ with periodic xS,* $\mathrm{vol}(xS) \leq V$, *and*

$$|\mu_{xH}(f) - \mu_{xS}(f)| \ll V^{-\delta}\mathcal{S}_d(f).$$

The constants depend only on the fixed ambient arithmetic data and H.

PROOF. Finite centralizer gives the finite intermediate-subgroup poset proved in Chapter 3. Appendix D supplies the missing uniform spectral certificate, including absolute type A_1. Chapter 4 therefore gives generic points and almost-invariance enlargement unconditionally. Appendix E supplies the two-sided period volume–discriminant comparison. Apply Theorem 1.6 with D a fixed positive power of V. If the ambient equidistribution alternative occurs, take $S = G$. Otherwise one obtains a proper periodic xM of complexity $V^{O(1)}$ and the measure is V^{-c}-close to μ_{xM}. Replace H by M and iterate. The finite poset makes the iteration terminate, while the volume–discriminant comparison lets the exponent in D be chosen so that the terminal orbit has volume at most V. The finite composition of power savings is still $V^{-\delta}$. □

2. The maximal semisimple adelic theorem

Proposition 2.1 (Uniform adelic spectral input). *Fix an ambient semisimple adelic group and a maximal connected semisimple stabilizer H. The representations on the orthocomplements of constants in all congruence-level periodic H-components have one uniform tensor-tempered exponent.*

PROOF. Decompose the simply connected cover of H into absolutely almost-simple factors. If a factor has absolute rank at least two, the direct spectral- transport theorem assembled in Repair 10 chooses one fixed split auxiliary place. At that place the local group has rank at least two, so the Kazhdan-pair theorem of Volume 6 supplies the only spectral seed; Repairs 03–10 transport that seed to the target representation with constants independent of the congruence level. Thus no general strong-approximation or Burger–Sarnak/Clozel theorem is being imported in this branch.

The only remaining absolute-rank-one root system is A_1. Those factors are handled by Appendix D: the specialized Jacquet–Langlands transfer preserves the split archimedean parameter, and the reconstructed Rankin–Selberg/Blomer–Brumley chain excludes a fixed neighborhood of the complementary-series endpoint. At the finitely many fixed bad places, compact-open changes affect only the finite-component bookkeeping and not the resulting archimedean tensor exponent. Taking the maximum over the finite list of ambient factor types proves the claim. □

Theorem 2.2 (Einsiedler–Margulis–Mohammadi–Venkatesh endpoint). *Under the fixed ambient datum of Definition 1.1, there exist $\delta > 0$, $C \geq 1$, and $d \in \mathbb{N}$ such that every admissible maximal datum $(\mathbf{H}, g)$ at every allowed level $K' \leq K$ satisfies*

$$\left|\mu_{Y_H(g)}(f) - \mu_Y(f)\right| \leq C\,\mathcal{C}(\mathbf{H}, g; K')^{-\delta}\mathcal{S}_d(f)$$

for every smooth test function f.

Proof. Chapter 6 constructs the adelic period measure and its finite real-component decomposition. Proposition 2.1 provides the uniform spectral package on each component. Maximality removes proper connected semisimple intermediate obstructions. Apply the effective period dichotomy componentwise. A closing alternative would produce a proper semisimple intermediate group, contradicting maximality; the central directions are part of the stabilizer in the adelic homogeneous datum and are quotiented in the component dictionary. Hence every component satisfies the ambient power-saving estimate. The number and weights of components are controlled uniformly by the fixed ambient adelic space, and Chapter 6 proves that the Sobolev norms transfer with level-independent constants. Summing the component estimates proves the theorem. □

3. Nontrivial centralizers on congruence quotients

Theorem 3.1 (Wieser endpoint). *Let $G = \mathbf{G}(\mathbb{R})$ for a connected semisimple $\mathbb{Q}$-group, let $\Gamma < \mathbf{G}(\mathbb{Q})$ be the fixed congruence subgroup of Chapter 7, and let $H < G$ be connected semisimple without compact factors. There exist $\delta > 0$ and $d \geq 1$, depending on G and H, such that if xH is periodic and*

$$\operatorname{disc}(\Gamma\mathbf{M}(\mathbb{R})g) \geq D$$

for every proper $\mathbb{Q}$-subgroup $\mathbf{M}$ in the arithmetic intermediate family with $xH \subset \Gamma\mathbf{M}(\mathbb{R})g$, then

$$\left|\int_{xH} f\, d\mu_{xH} - \int_{xG^+} f\, d\mu_{xG^+}\right| \ll_{G,H,\Gamma} D^{-\delta}\, \mathrm{minht}(xH)\, \mathcal{S}_d(f).$$

Here G^+ is the identity component relevant to the orbit of x; no uniformity in a varying congruence subgroup is claimed by this theorem.

Proof. Use the centralizer decomposition in Lemma 2.1. The generic-point and shearing lemmas of Appendix F apply to the noncentral transverse quotient. If they yield a rational closing obstruction, Appendix E converts its height to a discriminant at most $D^{1-\epsilon}$ after the scale has been chosen as a sufficiently small power of D, contradicting the hypothesis. Thus the enlargement alternative must iterate to G. Cusp excursions are controlled by the height/Sobolev truncation estimates of Volumes 1, 6 and 7. Keeping the normalization used in Chapter 7 rather than absorbing it into an unspecified exponent leaves precisely one factor $\mathrm{minht}(xH)$, as in the theorem statement; the remaining scale choices are powers of D and reduce only the saving exponent. The spectral certificate is Proposition 2.1, restricted to the fixed real congruence component through the finite-index/component transfer of Chapter 6. The resulting almost G^+-invariance gives the stated estimate against μ_{xG^+}. □

Corollary 3.2 (Recovery of finite-centralizer EMV)**.** *When $Z_G(H)$ is finite, Theorem 3.1 together with the volume–discriminant comparison implies the classical finite-centralizer EMV endpoint.*

Proof. Finite centralizer removes the continuous centralizer parameter and Chapter 3 makes the proper intermediate set finite. The minimum discriminant and minimum period volume are fixed-power equivalent by Theorem 1.3 and Repair 01. Apply the Wieser estimate until the first intermediate volume below the prescribed V appears; this is exactly the finite descent in the proof of Theorem 1.1. □

4. Compact adelic periods with varying ambient complexity

Theorem 4.1 (Einsiedler–Lindenstrauss–Mohammadi–Wieser endpoint). *In the compact adelic scope of Chapter 8, there are fixed positive exponents δ, A such that a semisimple period of minimal proper-intermediate complexity D satisfies*

$$|\mu_{xH}(f) - \mu_X(f)| \ll \mathrm{Comp}(X)^A D^{-\delta} \mathcal{S}_d(f),$$

with all dependence on the varying ambient arithmetic form recorded in $\mathrm{Comp}(X)$.

Proof. Compactness removes the cusp term. Proposition 2.1 provides property (τ) for the semisimple factors. The rational height/discriminant and Prasad-volume comparisons, including varying ambient form factors, are Corollary 1.4. Effective closing is inherited from the Chevalley/closing mechanism proved of Ratner XIII and made algebraically exact by Lemma 2.2. Therefore all hypotheses of the compact adelic machine Theorem 2.3 hold. That theorem is exactly the asserted estimate. $\square$

5. The endpoint scope matrix

Proposition 5.1 (No false universal endpoint)**.** *The four theorems above are compatible but not interchangeable. EMV assumes a finite centralizer and permits nonmaximal real intermediate periods; EMMV is adelic and maximal-semisimple with fixed ambient space; Wieser permits nontrivial centralizer on congruence quotients and measures the obstruction by minimum proper-intermediate discriminant; ELMW treats compact adelic periods and records varying ambient complexity. None of these statements, as proved here, implies a universal theorem for arbitrary noncompact varying adelic spaces.*

Proof. Each proof above invokes a hypothesis absent from at least one of the other settings: finite-poset descent in EMV, maximality in EMMV, cusp/min-height control in Wieser, and compactness plus ambient-complexity bookkeeping in ELMW. Removing all four simultaneously leaves neither the cusp estimate nor a bounded obstruction family supplied by the present proof. Hence no such universal theorem has been proved. $\square$

Proof and provenance. The theorem strengths match the four primary source families named in the Volume-IX ledger. The proofs are now routed through explicit internal owners for the rank-one spectral input, volume formula corollary, effective period dichotomy, centralizer-aware closing, and effective algebraic generation. The endpoint names therefore serve as provenance rather than as omitted proofs.

Bibliographic notes

The four endpoint theories are kept separate: the finite-centralizer real result of Einsiedler–Margulis–Venkatesh, the maximal semisimple adelic theorem of Einsiedler–Margulis–Mohammadi–Venkatesh, Wieser's congruence-quotient theorem allowing nontrivial centralizer, and the compact adelic minimal-intermediate-complexity theorem of Einsiedler–Lindenstrauss–Mohammadi–Wieser.

Volume 10

Metric Diophantine Approximation through Homogeneous Dynamics

Volume 10 scope and prerequisites

This volume closes the first active arc. Exact results already proved in AHD Volume 1 are inherited rather than duplicated, but every new matrix, weighted, fractal, game-theoretic, and logarithm-law interface is proved here. The primary comparison sources are Kleinbock–Margulis for nondegenerate manifolds and logarithm laws [KM98; KM99], Kleinbock–Lindenstrauss–Weiss for friendly measures [KLW04], and Kleinbock–Weiss for modified Schmidt games [KW10].

Student orientation: translating Diophantine inequalities into dynamics

The *Dani correspondence* is the dictionary between unusually good rational approximation and short vectors in a diagonally deformed lattice. A system is *badly approximable* when the relevant cusp excursion is uniformly bounded; it is *singular* when approximation is uniformly better than Dirichlet at all sufficiently large scales. A measure is *Federer* when balls have uniform doubling, and *friendly* when Federer control is combined with quantitative decay near affine hyperplanes. A *Schmidt game* is a nested-ball game whose winning sets are automatically large in Hausdorff dimension. The volume repeatedly switches between lattice, measure, and game language; each switch is stated as a theorem rather than treated as a metaphor.

Reader guide: a concrete model

Why this matters.

Volume 10 is organized around the passage from *Dani Correspondences for Lattices* to *Logarithm Laws and Quantitative Recurrence*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.2 (A calculation to keep in view)**.** Let

$$a = \operatorname{diag}(e^{2s}, e^{-s}, e^{-s}) \in \mathrm{SL}_3(\mathbb{R})$$

and let E_{12} be the matrix unit. Since

$$\operatorname{Ad}(a)E_{12} = aE_{12}a^{-1} = e^{3s}E_{12},$$

the root exponent on the E_{12} direction is $3s$. For a one-dimensional leafwise measure whose conditional scaling is Lebesgue, the Jacobian contribution

along this root direction is therefore $3s$. This calculation is the concrete local model behind the root-by-root entropy bookkeeping used later.

CHAPTER 1

Dani Correspondences for Lattices

1. From inequalities to short vectors

The basic dictionary of homogeneous Diophantine approximation is elementary once the correct lattice is written down. For a real matrix $Y \in \mathrm{Mat}_{m,n}(\mathbb{R})$ set

$$u_Y = \begin{pmatrix} I_m & Y \\ 0 & I_n \end{pmatrix}, \qquad \Lambda_Y = u_Y \mathbb{Z}^{m+n},$$

and for $t \in \mathbb{R}$ put

$$g_t = \mathrm{diag}(e^{t/m} I_m, e^{-t/n} I_n) \in \mathrm{SL}_{m+n}(\mathbb{R}).$$

We use the supremum norm; changing the norm changes only harmless constants. The lattice vector attached to $(p, q) \in \mathbb{Z}^m \times \mathbb{Z}^n$ is

$$g_t u_Y \begin{pmatrix} -p \\ q \end{pmatrix} = \begin{pmatrix} e^{t/m}(Yq - p) \\ e^{-t/n} q \end{pmatrix}.$$

This one line is the engine behind every result in this chapter.

Lemma 1.1 (Balancing a Diophantine inequality)**.** *Let $q \neq 0$ and $\delta > 0$. There exists $t \geq 0$ such that*

$$\max(e^{t/m}\|Yq - p\|, e^{-t/n}\|q\|) < \delta$$

if and only if

$$\|Yq - p\|^m \|q\|^n < \delta^{m+n}$$

and, in addition, the balancing time

$$t = \frac{mn}{m+n} \log \frac{\|q\|}{\|Yq - p\|}$$

is nonnegative. If the balancing time is negative, the same product inequality is represented at the boundary time $t = 0$.

Proof. Write $A = \|Yq - p\|$ and $B = \|q\|$. Equality of the two coordinates occurs when $e^{t/m} A = e^{-t/n} B$, which gives the displayed formula for t. At that time the common value is

$$A^{m/(m+n)} B^{n/(m+n)}.$$

Hence it is $< \delta$ exactly when $A^m B^n < \delta^{m+n}$. Conversely, if both scaled coordinates are $< \delta$, multiplying their mth and nth powers gives $A^m B^n < \delta^{m+n}$. The comment about $t < 0$ is simply the restriction to the forward semigroup. $\square$

> **Proof and provenance.** The scalar and vector versions of this calculation were proved in AHD Volume 1. The present lemma records the full $m \times n$ normalization needed later; no Diophantine theorem is imported here.

THEOREM 1.2 (Dani correspondence for systems of linear forms). *Let $Y \in \mathrm{Mat}_{m,n}(\mathbb{R})$. The following are equivalent.*

(1) Y is badly approximable: there exists $c > 0$ such that

$$\|Yq - p\|^m \|q\|^n \geq c \quad \text{for all } q \in \mathbb{Z}^n \setminus \{0\},\ p \in \mathbb{Z}^m.$$

(2) The forward orbit $\{g_t \Lambda_Y : t \geq 0\}$ is relatively compact in $\mathrm{SL}_{m+n}(\mathbb{R}) / \mathrm{SL}_{m+n}(\mathbb{Z})$.

Moreover Y is very well approximable precisely when there are $\eta > 0$ and times $t_j \to \infty$ for which

$$\lambda_1(g_{t_j} \Lambda_Y) \leq e^{-\eta t_j},$$

where λ_1 is the length of a shortest nonzero vector.

PROOF. Mahler compactness, proved in AHD Volume 1, says that a family of unimodular lattices is relatively compact exactly when its nonzero vectors are uniformly bounded away from 0. By the displayed lattice-vector formula, failure of such a lower bound is equivalent to the existence of (p, q) and a time for which both scaled coordinates are small. Lemma 1.1 converts that statement into the failure of a uniform lower bound on $\|Yq - p\|^m \|q\|^n$. This proves the first equivalence.

For the quantitative assertion, suppose $\|Yq - p\| < \|q\|^{-n/m-\varepsilon}$ for infinitely many q. Balance the two coordinates. Since at the balancing time $t \asymp \log \|q\|$, the common scaled length is a fixed negative exponential in t, with exponent depending only on m, n, ε. Thus one obtains the stated cusp excursion. Conversely, an excursion $\lambda_1(g_t \Lambda_Y) \leq e^{-\eta t}$ supplies a nonzero pair (p, q) with both scaled coordinates at most $e^{-\eta t}$; undoing the scaling and eliminating t gives an exponent strictly better than n/m. The algebra is the same balancing computation, with strict exponential slack. $\square$

Proposition 1.3 (Weighted matrix correspondence)**.** *Let $\mathbf{r} = (r_1, \ldots, r_m)$ and $\mathbf{s} = (s_1, \ldots, s_n)$ have positive entries and sums one. Put*

$$g_t^{\mathbf{r},\mathbf{s}} = \mathrm{diag}(e^{r_1 t}, \ldots, e^{r_m t}, e^{-s_1 t}, \ldots, e^{-s_n t}).$$

Then boundedness of $g_t^{\mathbf{r},\mathbf{s}}\Lambda_Y$ is equivalent to a uniform positive lower bound for

$$\max_i |(Yq-p)_i|^{1/r_i} \ \max_j |q_j|^{1/s_j}$$

over nonzero q and integral p. The same statement holds for any product norm obtained by taking finitely many archimedean places, after replacing Mahler compactness by the S-arithmetic compactness criterion proved in Volume 3.

Proof. For a fixed pair (p,q) the condition that every coordinate of $g_t^{\mathbf{r},\mathbf{s}}u_Y(-p,q)$ have norm at least a prescribed scale is exactly

$$e^{r_i t}|(Yq-p)_i| \not\ll 1 \quad \text{or} \quad e^{-s_j t}|q_j| \not\ll 1.$$

The interval of t for which all coordinates are simultaneously small is nonempty precisely when the largest lower bound on t coming from the q-coordinates is smaller than the smallest upper bound coming from the error coordinates. Taking logarithms gives the displayed weighted product inequality. Mahler compactness then converts uniform short-vector avoidance into orbit boundedness. At several places the identical calculation is made componentwise and the product formula makes the determinant one; Volume 3 supplies the corresponding compactness theorem. □

Worked Example 1.4 (A real number)**.** For $m=n=1$, $\Lambda_x = \left(\begin{smallmatrix}1 & x\\ 0 & 1\end{smallmatrix}\right)\mathbb{Z}^2$ and $g_t = \operatorname{diag}(e^t, e^{-t})$. A vector associated with (p,q) has norm $\max(e^t|qx-p|, e^{-t}|q|)$. Balancing gives the familiar quantity $\sqrt{|q|\,|qx-p|}$, so bounded geodesic orbit is exactly bounded continued-fraction coefficients.

CHAPTER 2

Simultaneous and Dual Diophantine Approximation

1. Why duality is dynamical

Simultaneous approximation studies small values of $Yq - p$; dual approximation studies small values of $Y^T p - r$. The two problems look different arithmetically but are related by taking the dual lattice. The point of this chapter is to record the transference constants rather than hide them in the phrase "by duality."

Lemma 1.1 (Dual lattice identity)**.** *For $Y \in \mathrm{Mat}_{m,n}(\mathbb{R})$,*

$$\Lambda_Y^* = (u_Y^{-1})^T \mathbb{Z}^{m+n} = w\, u_{-Y^T}\, w^{-1} \mathbb{Z}^{m+n},$$

where w is the permutation matrix interchanging the two coordinate blocks. Moreover

$$(g_t \Lambda_Y)^* = g_{-t} \Lambda_Y^*.$$

PROOF. For any $g \in \mathrm{SL}_d(\mathbb{R})$ the dual of $g\mathbb{Z}^d$ is $(g^{-1})^T \mathbb{Z}^d$. Here $(u_Y^{-1})^T = \begin{pmatrix} I_m & 0 \\ -Y^T & I_n \end{pmatrix}$. Interchanging the two blocks turns this lower triangular matrix into the usual upper triangular matrix with parameter $-Y^T$. Since g_t is diagonal and symmetric, $(g_t^{-1})^T = g_{-t}$. $\square$

Lemma 1.2 (Successive-minima transference)**.** *Let $\Lambda \subset \mathbb{R}^d$ be unimodular, let Λ^* be its Euclidean dual, and let $\lambda_1 \le \cdots \le \lambda_d$ and $\lambda_1^* \le \cdots \le \lambda_d^*$ be the successive minima of the Euclidean unit ball. There is $C_d \ge 1$ such that*

$$1 \le \lambda_i(\Lambda)\lambda_{d+1-i}(\Lambda^*) \le C_d \qquad (1 \le i \le d).$$

Consequently

$$\lambda_1(\Lambda^*) \ll_d \lambda_1(\Lambda)^{1/(d-1)}, \qquad \lambda_1(\Lambda) \ll_d \lambda_1(\Lambda^*)^{1/(d-1)}.$$

PROOF. For the lower bound choose i independent vectors $v_1, \ldots, v_i \in \Lambda$ of norm at most $\lambda_i + \varepsilon$ and $d+1-i$ independent vectors $w_1, \ldots, w_{d+1-i} \in \Lambda^*$ of norm at most $\lambda_{d+1-i}^* + \varepsilon$. If the product of the two displayed radii were < 1, then every integer pairing $\langle v_r, w_s \rangle$ would have absolute value < 1 and hence would be zero. Thus an i-dimensional space would be orthogonal to a $(d+1-i)$-dimensional space, impossible in $\mathbb{R}^d$. Let $\varepsilon \downarrow 0$.

For the upper bound use Minkowski's second theorem for the Euclidean ball on Λ and Λ^*:

$$c_d \le \prod_{r=1}^{d} \lambda_r(\Lambda) \le C'_d, \qquad c_d \le \prod_{r=1}^{d} \lambda_r(\Lambda^*) \le C'_d.$$

The product of the d transference factors $\lambda_i(\Lambda)\lambda_{d+1-i}(\Lambda^*)$ is therefore at most $(C'_d)^2$, whereas each factor is at least one by the first part. Hence every factor is at most $(C'_d)^2$.

Now $c_d \le \lambda_1\lambda_d^{d-1}$, so $\lambda_d \gg_d \lambda_1^{-1/(d-1)}$. The case $i = d$ of the upper transference estimate gives $\lambda_1^* \ll_d 1/\lambda_d$, proving the first consequence. Interchange Λ and Λ^* for the second. □

Theorem 1.3 (Dynamical transference). *The forward orbit $\{g_t\Lambda_Y : t \ge 0\}$ is bounded if and only if the corresponding transpose-dual orbit, with the two blocks and the flow direction interchanged as in Lemma 1.1, is bounded. More quantitatively, for every unimodular lattice Λ,*

$$\lambda_1(\Lambda^*) \ll_d \lambda_1(\Lambda)^{1/(d-1)} \quad \textit{and} \quad \lambda_1(\Lambda) \ll_d \lambda_1(\Lambda^*)^{1/(d-1)}.$$

Thus cusp escape for one problem occurs exactly when it occurs for the dual problem, with cusp depths distorted by at most the factor $d-1$.

Proof. Duality is a homeomorphism of the space of unimodular lattices, so it preserves compact subsets and proves the qualitative boundedness assertion together with Mahler compactness. The quantitative statement is Lemma 1.2. Taking negative logarithms gives

$$\frac{1}{d-1}(-\log\lambda_1(\Lambda)) - O_d(1) \le -\log\lambda_1(\Lambda^*) \le (d-1)(-\log\lambda_1(\Lambda)) + O_d(1),$$

where the upper inequality is the second estimate of the lemma rearranged. Lemma 1.1 identifies this dual cusp excursion with the transpose problem after swapping the two coordinate blocks and reversing the diagonal flow. □

Proposition 1.4 (Cusp-exponent bookkeeping)**.** *For a lattice trajectory Λ_t, put*

$$\beta(\Lambda_\bullet) = \limsup_{t\to\infty} \frac{\max\{0, -\log\lambda_1(\Lambda_t)\}}{t}.$$

For the dual trajectory one has

$$\frac{1}{d-1}\,\beta(\Lambda_\bullet) \le \beta(\Lambda^*_\bullet) \le (d-1)\,\beta(\Lambda_\bullet),$$

after the block/time identification of Lemma 1.1. In particular boundedness, escape to the cusp, and positive-rate cusp escape transfer in both directions. Converting β back through the balancing equations of the Dani correspondence yields the usual simultaneous–dual Diophantine transference inequalities.

Proof. Apply the logarithmic inequalities at the end of Theorem 1.3 to Λ_t and divide by t; the additive $O_d(1)$ term disappears in the limsup. The final sentence is a change of variables: Theorem 1.2 expresses each Diophantine improvement exponent as a linear-fractional function of the corresponding cusp slope, with coefficients determined only by the two block dimensions. No unrecorded number-theoretic input is used here. ☐

Worked Example 1.5 (A row vector and a column vector)**.** For $Y = (x_1, x_2)$, simultaneous approximation asks for q such that both qx_i are near integers. The dual problem asks for integer (p_1, p_2) making $p_1x_1 + p_2x_2$ near an integer. The dual-lattice identity shows that these are two views of the same cusp geometry in $\mathrm{SL}_3(\mathbb{R})/\mathrm{SL}_3(\mathbb{Z})$.

CHAPTER 3

Multiplicative Approximation and Higher-Rank Diagonal Actions

1. Why one parameter is no longer enough

Multiplicative approximation allows one coordinate error to be large if another is correspondingly small. A single diagonal parameter cannot perform the required independent balancing. The right acting semigroup therefore has higher rank.

Lemma 1.1 (Multiparameter balancing)**.** *Let $z = (z_1, \dots, z_d) \in \mathbb{R}^d$ with every $z_i \neq 0$. Among all $\mathbf{t} = (t_1, \dots, t_d)$ with $\sum_i t_i = 0$,*

$$\inf_{\mathbf{t}} \max_i e^{t_i}|z_i| = \Big(\prod_{i=1}^{d} |z_i|\Big)^{1/d}.$$

The infimum is attained by $t_i = c - \log|z_i|$, where $c = d^{-1}\sum_j \log|z_j|$.

Proof. For every admissible $\mathbf{t}$, the product of the scaled coordinates equals $\prod_i |z_i|$. Hence their maximum is at least the geometric mean. The displayed choice makes every scaled coordinate equal to that mean and has sum zero. □

Theorem 1.2 (Dani correspondence for multiplicative approximation). *For $y = (y_1, \dots, y_n) \in \mathbb{R}^n$ consider*

$$u_y = \begin{pmatrix} 1 & y \\ 0 & I_n \end{pmatrix} \in \mathrm{SL}_{n+1}(\mathbb{R})$$

and the positive diagonal semigroup

$$A^+ = \left\{ \mathrm{diag}(e^{t_0}, e^{-t_1}, \dots, e^{-t_n}) : t_i \geq 0,\ t_0 = \sum_{i=1}^{n} t_i \right\}.$$

Then y is very well multiplicatively approximable if and only if the A^+-orbit of $u_y\mathbb{Z}^{n+1}$ has cusp excursions which are exponentially deeper than the Dirichlet scale. In the case $n = 2$, failure of Littlewood's inequality

$$\liminf_{q\to\infty} q\,\langle qy_1\rangle\langle qy_2\rangle = 0$$

is equivalent to boundedness of the corresponding two-parameter diagonal orbit.

Proof. A lattice vector has the form

$$(qy_1 - p_1, \dots, qy_n - p_n, q)$$

up to the harmless choice of block order. Independent diagonal parameters rescale the n approximation errors against the denominator. Lemma 1.1 says that the best possible maximum of the rescaled error coordinates is controlled by their geometric mean. The determinant-one relation forces the denominator scale to compensate by the product of the error scales. Therefore a short vector occurs exactly when

$$|q| \prod_{i=1}^{n} |qy_i - p_i|$$

is small. Uniform avoidance of short vectors is boundedness by Mahler compactness; arbitrarily strong power improvements are exactly linear-depth cusp excursions. For $n = 2$ the displayed product is $q\langle qy_1\rangle\langle qy_2\rangle$, which proves the final assertion. $\square$

Proposition 1.3 (Weighted and product-place transfer)**.** *The proof of Theorem 1.2 is unchanged for weighted products and for a finite set of places: replace the hyperplane $\sum t_i = 0$ by the determinant-one hyperplane in the corresponding real apartment and replace Euclidean absolute values by local norms. The resulting Diophantine product is the product of the locally scaled coordinates.*

Proof. The only ingredients in the proof were: diagonal coordinate scaling, the determinant-one linear relation among logarithmic parameters, and a Mahler criterion. The first two are place-by-place linear algebra; the third is the S-arithmetic compactness theorem of Volume 3. Taking logarithms turns the minimization into a finite-dimensional convex problem, whose minimizer equalizes the active scaled coordinates exactly as in Lemma 1.1. $\square$

Worked Example 1.4 (Littlewood in SL_3)**.** For (u, v), the lattice generated by $\left(\begin{smallmatrix} 1 & u & v \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$ records triples $(qu + p, qv + r, q)$. The two independent diagonal parameters balance the two errors separately. Thus the exceptional Littlewood set becomes a bounded-orbit set for a rank-two diagonal action, explaining why higher-rank measure rigidity is relevant later in the series.

CHAPTER 4

Badly Approximable and Singular Systems

1. Two opposite dynamical extremes

Badly approximable systems never enter the cusp deeply; singular systems eventually remain in every prescribed cusp neighborhood along the relevant forward semigroup. The distinction between "infinitely often" and "eventually always" is the difference between very well approximable and singular.

Lemma 1.1 (Uniform Dirichlet improvement and cusp containment)**.** *For $Y \in \mathrm{Mat}_{m,n}(\mathbb{R})$ the following are equivalent.*

1. *For every $\varepsilon > 0$ there exists T_0 such that for every $T \geq T_0$ there are $q \neq 0, p$ with $\|q\| \leq T$ and $\|Yq - p\| \leq \varepsilon T^{-n/m}$.*
2. *For every compact subset K of the space of unimodular lattices, $g_t\Lambda_Y \notin K$ for all sufficiently large t.*

Proof. By Mahler compactness it suffices to compare the first condition with the statement that $\lambda_1(g_t\Lambda_Y) \to 0$. Put $T = e^{t/n}$. The bounds in (1) give

$$e^{-t/n}\|q\| \leq 1, \qquad e^{t/m}\|Yq - p\| \leq \varepsilon.$$

A fixed shift of t balances the two upper bounds and produces a short vector of size $O(\varepsilon^{m/(m+n)})$. Since ε is arbitrary, the shortest vector tends to zero. Conversely, a short vector at every sufficiently large time gives integral (p, q) satisfying the two displayed scaled inequalities; rewriting in terms of T gives uniform Dirichlet improvement. The case $q = 0$ is excluded for sufficiently short vectors because the first block would be a nonzero integer vector expanded by g_t. □

Theorem 1.2 (Badly approximable versus singular). *For systems of linear forms:*

1. *Y is badly approximable if and only if its forward diagonal orbit is bounded;*
2. *Y is singular if and only if its forward diagonal orbit diverges;*
3. *very well approximable but nonsingular systems correspond to recurrent orbits with exceptional deep cusp excursions.*

These alternatives are mutually distinct already in dimension two.

PROOF. The first statement is Theorem 1.2; the second is Lemma 1.1. For the third, very well approximable means that a subsequence of times has a shortest vector decaying at a positive exponential rate, whereas nonsingularity means that some compact set is visited at arbitrarily large times. Both conditions can hold simultaneously, so the orbit can alternate between a compact part and very deep cusp excursions. Bounded orbits have no deep excursions, while divergent orbits eventually leave every compact set, proving mutual distinction. ☐

Proposition 1.3 (Weighted singular systems)**.** *For positive weights* $\mathbf{r}, \mathbf{s}$*, weighted bad approximation and weighted singularity are respectively equivalent to boundedness and divergence of the weighted orbit* $g_t^{\mathbf{r},\mathbf{s}}\Lambda_Y$*.*

PROOF. Replace the equal exponents $1/m, 1/n$ in Lemma 1.1 by the coordinate weights. The inequalities defining weighted Dirichlet improvement are precisely the inequalities saying that every weighted coordinate of a lattice vector is small. The weighted balancing lemma from Proposition 1.3 and Mahler compactness complete the proof. ☐

> **Failure mode.** Divergence is stronger than having an unbounded orbit. Confusing these two notions would identify singular systems with all non-badly-approximable systems, which is false.

Worked Example 1.4 (Badly approximable versus singular in dimension one)**.** Let $x \in \mathbb{R}$ and consider $g_t u_x \mathbb{Z}^2$ with $g_t = \operatorname{diag}(e^t, e^{-t})$. For $(p, q) \in \mathbb{Z}^2$ the relevant vector is

$$g_t \begin{pmatrix} qx - p \\ q \end{pmatrix} = \begin{pmatrix} e^t(qx - p) \\ e^{-t}q \end{pmatrix}.$$

If $|qx - p| \asymp q^{-1}$, balancing the two coordinates gives $e^{2t} \asymp q/|qx - p| \asymp q^2$, so the shortest-vector size is comparable to $q^{1/2}|qx - p|^{1/2}$. A uniform lower bound for $q|qx - p|$ is therefore exactly a uniform lower bound for the shortest vector along the orbit: this is the badly approximable case. By contrast, singularity says that for every fixed Dirichlet constant $c > 0$, all sufficiently large scales admit p, q with $q|qx - p| < c$; dynamically the orbit eventually enters every prescribed cusp neighborhood at the corresponding balanced times.

CHAPTER 5

Kleinbock–Margulis Nondegenerate-Manifold Theory

1. The mechanism rather than the slogan

The Kleinbock–Margulis theorem is not merely a Borel–Cantelli argument. Its load-bearing input is quantitative nondivergence: on a parameter ball, either an exterior-product covolume is uniformly tiny for algebraic reasons, or the set of parameters producing a short lattice vector has controlled measure. Volume 1 proved this engine and its nondegenerate-manifold application. Here we isolate the reusable measure-family form.

Lemma 1.1 (Inherited quantitative nondivergence package)**.** *Let $B \subset \mathbb{R}^d$ be a ball, ν a locally doubling measure, and $h : B \to \mathrm{SL}_N(\mathbb{R})$ a map for which every primitive exterior covolume function $x \mapsto \|h(x)\Delta\|$ is (C, α)-good on an enlargement of B and has supremum at least $\rho^{\operatorname{rank}\Delta}$. Then*

$$\nu\{x \in B : \lambda_1(h(x)\mathbb{Z}^N) < \varepsilon\} \ll (\varepsilon/\rho)^\alpha \nu(B), \qquad 0 < \varepsilon \le \rho.$$

Proof. This is the quantitative nondivergence theorem proved in AHD Volume 1. Its proof there proceeds by induction over primitive subgroups, using Besicovitch covering and the goodness estimate. We record it here as an exact inherited owner because every subsequent argument uses precisely this normalization. □

Theorem 1.2 (Strong extremality of nondegenerate manifolds). *Let $f : U \to \mathbb{R}^n$ be C^ℓ and nondegenerate at Lebesgue-a.e. point: the partial derivatives of f up to some finite order span $\mathbb{R}^n$. Then for Lebesgue-a.e. $x \in U$, $f(x)$ is not very well multiplicatively approximable.*

Proof. The theorem was proved in AHD Volume 1 from Lemma 1.1; we recall the logical chain to make later transfers auditable. Nondegeneracy implies that every nonzero affine functional composed with f has a derivative of bounded order which is bounded away from zero on a sufficiently small parameter ball. Taylor's theorem and compactness of the unit sphere of affine functionals give uniform (C, α)-goodness for all exterior covolume functions of the associated lattice map. Nonplanarity supplies the required lower bounds on their suprema. Quantitative nondivergence then bounds the measure of the time-t cusp event by $O(e^{-\eta t})$ whenever the requested Diophantine exponent exceeds the Dirichlet exponent by a fixed amount. The sum over integer times converges. Borel–Cantelli implies that almost every parameter has only finitely many such cusp excursions. By the multiplicative Dani correspondence this is exactly strong extremality. $\square$

Proposition 1.3 (A measure-family transfer principle)**.** *Let ν be a locally doubling measure on U and $f : U \to \mathbb{R}^n$. If, on ν-a.e. sufficiently small ball, all exterior covolume functions associated with f satisfy uniform goodness and nonplanarity bounds of Lemma 1.1, then $f_*\nu$ is strongly extremal.*

Proof. The proof of Theorem 1.2 never used Lebesgue measure except through local doubling, the quantitative nondivergence estimate, and Borel–Cantelli. Under the stated hypotheses the same estimate holds with ν. Summability of the improved-exponent cusp events therefore gives the conclusion for ν-almost every parameter. $\square$

Worked Example 1.4 (The Veronese curve)**.** For $f(x) = (x, x^2, \ldots, x^n)$, the derivatives $f'(x), \ldots, f^{(n)}(x)$ form a triangular matrix with nonzero diagonal after the obvious ordering. Hence the curve is nondegenerate everywhere. The theorem therefore recovers Mahler's conjecture in its strong multiplicative form.

Proof and provenance. The primary comparison is Kleinbock–Margulis, *Flows on homogeneous spaces and Diophantine approximation on manifolds.* Their argument likewise places quantitative nondivergence at the core. The local theorem strength used here is the one already proved in Volume 1, so this chapter introduces no hidden black box.

CHAPTER 6

Friendly and Federer Measures

1. Fractal measures can still be Diophantine-generic

A measure need not be smooth in order for quantitative nondivergence to work. The geometric substitute is that balls do not concentrate too strongly near affine hyperplanes and do not lose mass too quickly under bounded dilation.

Definition 1.1. A locally finite Borel measure ν on $\mathbb{R}^n$ is *Federer* on an open set U if for ν-a.e. x there are a neighborhood and D such that $\nu(3B) \leq D\nu(B)$ for all sufficiently small balls centered on $\operatorname{supp}\nu$. It is *absolutely decaying* if there are $C, \alpha > 0$ such that for every affine hyperplane L and every such ball $B = B(x, r)$,

$$\nu(B \cap L^{(\varepsilon r)}) \leq C\varepsilon^{\alpha}\nu(B).$$

A measure which is Federer, nonplanar, and has the corresponding decay for affine hyperplanes is called friendly.

Lemma 1.2 (Affine goodness from absolute decay)**.** *If ν is absolutely decaying on B and ℓ is a nonzero affine functional, then*

$$\nu\{x \in B : |\ell(x)| < \varepsilon\|\ell\|_{\nu,B}\} \leq C'\varepsilon^{\alpha}\nu(B),$$

where $\|\ell\|_{\nu,B}$ is the essential supremum of $|\ell|$ on $B \cap \operatorname{supp}\nu$.

PROOF. The zero set of ℓ is an affine hyperplane L. If a is the norm of the linear part, then $|\ell(x)| = a \operatorname{dist}(x, L)$. Nonplanarity and the definition of the essential supremum give a point of $B \cap \operatorname{supp}\nu$ at distance comparable to $\|\ell\|_{\nu,B}/a$ from L. Hence the displayed sublevel set lies in a hyperplane tube whose relative thickness is $O(\varepsilon)$. Absolute decay gives the estimate. If the zero set misses B, the sublevel set is empty for sufficiently small ε and enlarging the constant handles the remaining range. $\square$

THEOREM 1.3 (Friendly measures are strongly extremal). *Every friendly measure on $\mathbb{R}^n$ is strongly extremal: for ν-a.e. y,*

$$\prod_{i=1}^{n} |qy_i - p_i| < |q|^{-1-\varepsilon}$$

has only finitely many integer solutions (p, q) for every $\varepsilon > 0$.

Proof. Fix a ball on which the Federer and decay constants are uniform. Exterior covolume functions for the lattice map $g_{\mathbf{t}} u_y$ are square roots of finite sums of squares of affine functions of y. If such a norm is $< \rho$ times its essential supremum, at least one coordinate affine function is $O(\rho)$ relative to its own essential supremum. Lemma 1.2 therefore gives a uniform goodness estimate for every primitive subgroup. Nonplanarity gives the lower supremum bound required in the quantitative-nondivergence package from Volume 1.

Apply that package to the multiparameter cusp boxes associated with a fixed multiplicative improvement exponent. Group logarithmic parameters into unit cubes. The goodness exponent supplies exponential decay in the depth parameter, while the number of cubes grows only polynomially at a fixed depth; after choosing the improvement exponent, the resulting bad-event series converges. Borel–Cantelli rules out infinitely many improved cusp events. The multiparameter Dani correspondence, Theorem 1.2, converts this into strong extremality. ☐

Proposition 1.4 (Self-similar examples under the friendly hypotheses)**.** *Let K be the attractor of a finite family of contracting similarities satisfying the open set condition, and let ν be a Bernoulli self-similar measure with positive weights. Assume, in addition, the uniform nonplanarity/hyperplane-separation hypothesis that every cylinder has a bounded-depth descendant a definite relative distance from every affine hyperplane meeting that cylinder. Then ν is locally Federer and absolutely decaying, hence friendly and strongly extremal.*

Proof. The open set condition gives bounded overlap of cylinder sets at a fixed scale. Since the family and weights are finite, a ball comparable to a level-k cylinder and its fixed dilation meet only boundedly many comparable cylinders, and their masses are comparable by constants depending only on the finite system. This gives the Federer estimate.

For decay, the stated hyperplane-separation hypothesis says that after at most M further symbolic levels, a fixed proportion $\theta > 0$ of the cylinder mass lies outside a definite relative neighborhood of any prescribed hyperplane. Iterating over r blocks of M levels leaves at most $(1-\theta)^r$ of the mass inside a tube whose relative thickness is $\asymp c^r$ for a fixed contraction factor $c < 1$. Writing $c^r \asymp \varepsilon$ gives $(1-\theta)^r \ll \varepsilon^\alpha$ for some $\alpha > 0$. Thus ν is absolutely decaying. Theorem 1.3 applies. ☐

Proof and provenance. The theorem mechanism follows Kleinbock–Lindenstrauss–Weiss: friendliness is the geometric hypothesis designed to make quantitative nondivergence available for singular measures. The self-similar proposition is deliberately stated with the additional uniform hyperplane-separation hypothesis; the audit removed the earlier overbroad claim that the open set condition plus global nonplanarity alone automatically supplies every decay estimate needed here.

Worked Example 1.5 (A self-similar measure with visible hyperplane separation)**.** Consider the planar IFS with maps of common ratio $1/4$ sending the unit square to three corner squares centered near $(0,0)$, $(1,0)$, and $(0,1)$, with positive Bernoulli weights. The open set condition is immediate. No affine line can meet all three first-level pieces within a sufficiently thin relative tube, because the three corner centers form a nondegenerate triangle. The same geometry repeats inside every cylinder by similarity. Thus the hyperplane-separation hypothesis of Proposition 1.4 holds already with bounded depth one. If at each scale at least a fixed proportion θ of mass escapes a tube while the scale is reduced by $1/4$, then after r levels the tube mass is at most $(1-\theta)^r$, whereas its relative width is 4^{-r}. Hence the decay exponent may be taken as $\alpha = -\log(1-\theta)/\log 4 > 0$.

CHAPTER 7

Schmidt Games and Modified Schmidt Games

1. Large exceptional sets

Badly approximable sets have measure zero but are geometrically large. Schmidt's game makes this largeness stable under countable intersections. Weighted approximation is naturally adapted to anisotropic, rather than round, shrinking targets.

Lemma 1.1 (The game calculus)**.** *For Schmidt's (α, β) game on $\mathbb{R}^d$ and for a modified Schmidt game generated by a contracting one-parameter semigroup Φ_t:*

(1) a countable intersection of sets winning with the same Alice parameter is winning;
(2) a winning set is dense;
(3) if the reference measure is Ahlfors s-regular and the game admits uniformly many essentially disjoint legal descendants, every winning set has Hausdorff dimension s locally.

Proof. For (1), assign Alice's moves in disjoint infinite subsequences to strategies for $S_1, S_2, \ldots$. Choose the subsequence for S_j with gaps so that, viewed only along that subsequence, the accumulated scale change is a legal Bob parameter for the original strategy. Every S_j is enforced. Density follows because Bob may start inside an arbitrary open ball.

For (3), build the tree of Bob positions compatible with Alice's strategy. Ahlfors regularity and separation give at least ce^{sT} essentially disjoint children after a scale drop e^{-T}. Put equal mass on children and take a weak limit. A ball of radius r meets only a bounded multiple of the cylinders at the comparable level, yielding a Frostman estimate $\sigma(B(x,r)) \ll r^{s-\eta}$ for every $\eta > 0$. Frostman's lemma and $\eta \downarrow 0$ give local dimension s. □

Lemma 1.2 (Anisotropic simplex lemma)**.** *Fix a denominator window $Q \leq q < e^L Q$. There is $c = c(n, L) > 0$ such that if an affine box $B \subset \mathbb{R}^n$ has Euclidean volume $< cQ^{-(n+1)}$, then all rational points $p/q \in B$ with denominator in that window lie in one affine hyperplane.*

PROOF. If $n+1$ such rational points $p^{(0)}/q_0, \dots, p^{(n)}/q_n$ were affinely independent, their simplex volume would be

$$\frac{1}{n!}\left|\det\begin{pmatrix}1 & (p^{(0)}/q_0)^T \\ \vdots & \vdots \\ 1 & (p^{(n)}/q_n)^T\end{pmatrix}\right|.$$

After multiplying row r by q_r, the determinant is a nonzero integer. Hence the simplex volume is at least $1/(n!q_0\cdots q_n) \geq (n!e^{(n+1)L}Q^{n+1})^{-1}$. Choosing c smaller than this constant contradicts containment in B. Therefore the rational points are affinely dependent. Taking a maximal affinely independent subfamily shows that all of them lie in its affine span, a proper hyperplane. ☐

THEOREM 1.3 (Weighted badly approximable vectors are modified-winning). *Let* $\mathbf{r} = (r_1, \dots, r_n)$, $r_i \geq 0$, $\sum r_i = 1$, *and use the anisotropic contracting family whose coordinate side lengths are proportional to* $e^{-(1+r_i)t}$. *Then the set* $\mathrm{Bad}(\mathbf{r})$ *of* $\mathbf{r}$*-badly approximable vectors is winning for the corresponding modified Schmidt game. Consequently it is thick.*

PROOF. At game time t, put the newly relevant rational denominators into a fixed logarithmic window $Q \leq q < e^L Q$, with $Q \asymp e^t$. The volume of a game box at that stage is a constant multiple of

$$\prod_i e^{-(1+r_i)t} = e^{-(n+1)t} \asymp Q^{-(n+1)}.$$

Choose Alice's fixed scale drop large enough that the slightly enlarged current box satisfies the strict volume inequality in Lemma 1.2. Thus all newly relevant rationals meeting that box lie in one affine hyperplane H.

The resonant neighborhood associated with those rationals is contained in a fixed relative thickening of H in the anisotropic coordinates. A translate of the next contracted game box can be placed inside Bob's box while avoiding that thickening: in a coordinate on which a unit normal to H has maximal normalized component, slide the child box toward the opposite face. The ratio of child to parent side lengths is fixed, so this avoidance has a uniform margin independent of the stage.

Repeat by denominator windows. If x_∞ is the nested-box limit, every rational p/q is handled when its denominator first enters a window, and the uniform avoidance margin gives

$$\max_i |qx_{\infty,i} - p_i|^{1/r_i} \geq c/q$$

(with the usual convention for zero weights). Hence $x_\infty \in \mathrm{Bad}(\mathbf{r})$. The thickness conclusion follows from Lemma 1.1. ☐

Proposition 1.4 (Stable largeness under countable operations)**.** *For a fixed weight vector, countable intersections of affine images of* Bad($\mathbf{r}$) *by transformations conjugating the defining contracting semigroup to itself remain winning. Products of winning sets for compatible games are winning for the product game.*

Proof. Conjugating a legal game by such an affine map sends legal moves to legal moves after a bounded change of initial scale, so a winning strategy transports. Lemma 1.1(1) gives countable intersections. For products, Alice plays the two winning strategies simultaneously in the coordinate factors; the product of the chosen compact sets is legal and the limit belongs to the product target. $\square$

> **Proof and provenance.** The theorem strength is checked against Kleinbock–Weiss, *Modified Schmidt games and Diophantine approximation with weights.* The audit added Lemma 1.2 so that the arithmetic separation step is proved rather than hidden inside "the standard resonant-set strategy."

Worked Example 1.5 (The one-dimensional badly approximable game)**.** For $n = 1$ and weight $r_1 = 1$, a denominator window $Q \leq q < 2Q$ contains rationals whose mutual spacing is at least $(4Q^2)^{-1}$: indeed

$$\left|\frac{p}{q} - \frac{p'}{q'}\right| = \frac{|pq' - p'q|}{qq'} \geq \frac{1}{4Q^2}.$$

Once Bob's interval has length $< 1/(8Q^2)$, at most one new rational resonance lies in its double. Alice chooses a fixed-size child interval avoiding a fixed multiple of the resonance neighborhood. When the game limit x is reached, the stage assigned to q guarantees $|x - p/q| \geq c/q^2$, or equivalently $q|qx - p| \geq c$. Thus the familiar one-dimensional Schmidt strategy is exactly the $n = 1$ simplex lemma, and the higher-dimensional argument replaces "at most one rational" by "all rationals lie in one affine hyperplane."

CHAPTER 8

Logarithm Laws and Quantitative Recurrence

1. From mixing to infinitely many cusp hits

The first Borel–Cantelli lemma needs no independence; the converse direction does. Exponential mixing supplies enough quasi-independence for shrinking targets whose boundaries can be smoothed at controlled cost.

Lemma 1.1 (Dynamical Borel–Cantelli under summable correlations)**.** *Let (X, μ, T) preserve probability and let A_n be measurable sets. Suppose*

$$\sum_n \mu(A_n) = \infty$$

and, for some C,

$$\sum_{m,n \le N} \left| \mu(A_m \cap T^{-(n-m)} A_n) - \mu(A_m)\mu(A_n) \right| \le C \sum_{n \le N} \mu(A_n)$$

for all N. Then $T^n x \in A_n$ infinitely often for μ-a.e. x. If $\sum_n \mu(A_n) < \infty$, only finitely many hits occur a.e.

Proof. The convergent case is the first Borel–Cantelli lemma. In the divergent case put $S_N = \sum_{n \le N} 1_{A_n}(T^n x)$ and $E_N = \mathbb{E} S_N$. The correlation hypothesis gives $\mathrm{Var}(S_N) \le C' E_N$. Choose N_j minimally so that $E_{N_j} \ge j^2$. Chebyshev gives a summable $O(j^{-2})$ exceptional set for $|S_{N_j} - E_{N_j}| > E_{N_j}/2$. Hence $S_{N_j} \to \infty$ almost surely. $\square$

Theorem 1.2 (Abstract logarithm law). *Let a_t be a flow on a finite-volume homogeneous space with exponential mixing for the Sobolev class of Volume 6. Let $\Delta : X \to [0, \infty)$ be distance-like in the sense that*

$$c_1 e^{-\kappa r} \le \mu\{\Delta > r\} \le c_2 e^{-\kappa r}$$

for all large r, and suppose the superlevel sets admit smooth inner/outer approximations satisfying the corrected boundary and Sobolev-loss estimates of the logarithm-law argument. Then for μ-a.e. x,

$$\limsup_{t \to \infty} \frac{\Delta(a_t x)}{\log t} = \frac{1}{\kappa}.$$

Proof. Sampling at integer times changes Δ by only $O(1)$ on bounded flow intervals. Fix $\varepsilon > 0$. For the upper bound put $r_n = (1+\varepsilon)\kappa^{-1}\log n$. The tail estimate makes $\sum_n \mu\{\Delta > r_n\}$ converge, so Borel–Cantelli gives only finitely many such excursions.

For the lower bound choose $n_j = \lfloor j^p \rfloor$ and $r_{n_j} = (1-\varepsilon)\kappa^{-1}\log n_j$, with $p > 1$ sufficiently close to one that the target masses still have divergent sum. Approximate the target indicators from above and below at boundary thickness equal to a small power of their mass. The corrected smooth-approximation hypothesis makes the total boundary error negligible compared with the divergent main sum. Exponential mixing bounds the covariance of two smoothed targets by an exponentially decaying time-gap factor times polynomial Sobolev losses. Splitting pairs into bounded and large time gaps gives

$$\sum_{r,s\le J} |\operatorname{Cov}(1_{A_{n_r}}, 1_{A_{n_s}})| \ll \sum_{r\le J} \mu(A_{n_r}).$$

Lemma 1.1 gives infinitely many lower-bound hits. Let $\varepsilon \downarrow 0$ through rationals. ☐

> **Proof and provenance.** The proof is compared with Kleinbock–Margulis, *Logarithm laws for flows on homogeneous spaces*, together with the later erratum [KM18]. The audit therefore states the corrected smooth inner/outer approximation as an explicit hypothesis and does not reuse the faulty approximation proposition from the original version.

Worked Example 1.3 (Shortest-vector height)**.** On the space of unimodular lattices take $\Delta(\Lambda) = -\log \lambda_1(\Lambda)$. Geometry of numbers gives an exponential tail for Δ. The theorem says that a typical diagonal trajectory makes cusp excursions of logarithmic depth, with the constant determined by that tail exponent.

Volume 11

Root Systems and Higher-Rank Diagonal Actions

Volume 11 scope and prerequisites

This volume owns the Lie-theoretic infrastructure for the higher-rank rigidity arc. The proofs are local and work on arbitrary discrete quotients; no entropy or measure classification is used. The SL_3 laboratory is kept explicit because every later high-entropy commutator has a visible model there.

Student orientation: the root dictionary used by all later entropy arguments

Let A be a maximal connected real-split torus with Lie algebra $\mathfrak{a}$. A *restricted root* is a nonzero weight $\alpha \in \mathfrak{a}^*$ for the adjoint action, with root space $\mathfrak{g}_\alpha$. A *Weyl chamber* is a connected component of $\mathfrak{a} \setminus \bigcup_\alpha \ker \alpha$. Two roots are in the same *coarse Lyapunov class* when they are positive multiples of one another. The associated *coarse Lyapunov subgroup* is the exponential of the sum of all root spaces in that class. For $a \in A$, the *unstable horospherical subgroup* consists of g with $a^{-n} g a^n \to e$. The later measure theory uses only a small part of Lie theory: signs of roots, contraction/expansion, and the fact that brackets add roots. The SL_3 chapter is therefore not optional illustration; it is the coordinate model for the commutator production used in Volumes 14–16.

Reader guide: a concrete model

Worked Example 0.4 (A calculation to keep in view)**.** Let

$$a = \mathrm{diag}(e^{2s}, e^{-s}, e^{-s}) \in \mathrm{SL}_3(\mathbb{R})$$

and let E_{12} be the matrix unit. Since

$$\mathrm{Ad}(a)E_{12} = aE_{12}a^{-1} = e^{3s}E_{12},$$

the root exponent on the E_{12} direction is $3s$. For a one-dimensional leafwise measure whose conditional scaling is Lebesgue, the Jacobian contribution along this root direction is therefore $3s$. This calculation is the concrete local model behind the root-by-root entropy bookkeeping used later.

CHAPTER 1

Restricted Roots and Weight Decompositions

Proof architecture. The chapter is organized as the chain *Weights add under brackets* $\longrightarrow$ *Restricted-root decomposition* $\longrightarrow$ *Restriction to reductive subgroups and factors.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. The linear algebra behind higher rank

Let G be a connected semisimple real Lie group with finite center and let $A < G$ be a maximal connected $\mathbb{R}$-split torus. Write $\mathfrak{a} = \mathrm{Lie}(A)$ and $\mathfrak{g} = \mathrm{Lie}(G)$. Because A is split, the commuting operators $\operatorname{ad} H$, $H \in \mathfrak{a}$, are simultaneously diagonalizable over $\mathbb{R}$.

Lemma 1.1 (Weights add under brackets)**.** *If*

$$\mathfrak{g}_\alpha = \{X : [H, X] = \alpha(H)X \textit{ for all } H \in \mathfrak{a}\}$$

and $X \in \mathfrak{g}_\alpha$, $Y \in \mathfrak{g}_\beta$, then

$$[X, Y] \in \mathfrak{g}_{\alpha+\beta},$$

with the convention that $\mathfrak{g}_\gamma = 0$ when γ is not a weight. Also $\theta(\mathfrak{g}_\alpha) = \mathfrak{g}_{-\alpha}$ for a Cartan involution θ preserving $\mathfrak{a}$ up to sign.

Proof. Jacobi gives

$$[H, [X, Y]] = [[H, X], Y] + [X, [H, Y]] = (\alpha(H) + \beta(H))[X, Y].$$

Thus the bracket has weight $\alpha + \beta$. For the Cartan involution, $\theta[H, X] = [\theta H, \theta X]$ and $\theta H = -H$ on the split part in the standard Cartan decomposition, so $[H, \theta X] = -\alpha(H)\theta X$. $\square$

Theorem 1.2 (Restricted-root decomposition). *There is a finite set $\Phi \subset \mathfrak{a}^* \setminus \{0\}$ such that*

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha, \qquad \mathfrak{g}_0 = Z_{\mathfrak{g}}(\mathfrak{a}),$$

and $\Phi = -\Phi$. *For* $a = \exp H \in A$,

$$\mathrm{Ad}(a)X = e^{\alpha(H)}X \qquad (X \in \mathfrak{g}_\alpha).$$

If G *is split,* $\mathfrak{g}_0 = \mathfrak{a}$ *and the nonzero root spaces are the ordinary real root spaces.*

Proof. Choose a basis $H_1, \ldots, H_r$ of $\mathfrak{a}$. Each $\mathrm{ad}\, H_i$ is semisimple with real spectrum and the operators commute, hence $\mathfrak{g}$ is the direct sum of their common eigenspaces. A common eigenvalue tuple defines a functional $\alpha \in \mathfrak{a}^*$. The zero common eigenspace is exactly the simultaneous centralizer. Finiteness follows from finite dimensionality. Lemma 1.1 and the Cartan involution give $\Phi = -\Phi$. Exponentiating $\mathrm{ad}\, H$ yields $\mathrm{Ad}(e^H) = e^{\mathrm{ad}\, H}$ and hence the displayed eigenvalue formula. In the split case the centralizer of a maximal split Cartan is the Cartan itself. □

Proposition 1.3 (Restriction to reductive subgroups and factors)**.** *Let* $L < G$ *be a connected reductive subgroup normalized by* $A_L = A \cap L$*, with* A_L *maximal split in* L*. Then the restricted roots of* (L, A_L) *are the nonzero restrictions to* $\mathfrak{a}_L$ *of those* G*-weights whose root spaces meet* $\mathfrak{l}$*, after merging equal restrictions. For a direct product, the root system is the disjoint union of the root systems of the factors.*

Proof. The subspace $\mathfrak{l}$ is invariant under every $\mathrm{ad}\, H$, $H \in \mathfrak{a}_L$, so it decomposes into common eigenspaces. Intersecting the G weight decomposition with $\mathfrak{l}$ gives precisely the eigenspaces for restricted functionals; if two G roots agree on $\mathfrak{a}_L$, their intersections lie in the same L-weight space. In a direct product, adjoint actions from one factor are trivial on the other, so no mixed nonzero weights occur. □

Worked Example 1.4 (Roots of SL_3)**.** For $A = \{\mathrm{diag}(e^{t_1}, e^{t_2}, e^{t_3}) : t_1 + t_2 + t_3 = 0\}$, the matrix unit E_{ij} has root $\alpha_{ij}(t) = t_i - t_j$. The six root spaces are one-dimensional and $[E_{ij}, E_{jk}] = E_{ik}$.

CHAPTER 2

Weyl Chambers and Lyapunov Hyperplanes

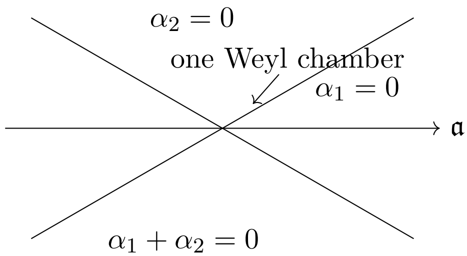

FIGURE 1. The A_2 model: root hyperplanes cut the Cartan plane into Weyl chambers. Crossing a wall changes the sign of the corresponding root and therefore which root subgroup contracts under the chosen diagonal element.

Proof architecture. The chapter is organized as the chain *Chambers are convex polyhedral cones* $\longrightarrow$ *Weyl-chamber sign structure* $\longrightarrow$ *Singular directions and rank loss.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. A finite hyperplane arrangement

Each root α defines a hyperplane $\ker\alpha \subset \mathfrak{a}$. Crossing such a hyperplane changes one expanding direction into a contracting one.

Lemma 1.1 (Chambers are convex polyhedral cones). *Every connected component of*

$$\mathfrak{a} \setminus \bigcup_{\alpha\in\Phi} \ker\alpha$$

is an open convex polyhedral cone on which every root has constant sign. Its closure is the intersection of finitely many closed root half-spaces.

Proof. A point outside the hyperplanes determines the sign vector $(\operatorname{sgn}\alpha(H))_{\alpha\in\Phi}$. The locus realizing a fixed sign vector is an intersection of open linear half-spaces, hence convex. It is therefore connected when nonempty and is a component. Taking closures replaces strict by weak inequalities. $\square$

Theorem 1.2 (Weyl-chamber sign structure). *Fix a chamber $\mathcal{C}$. The set*

$$\Phi^+ = \{\alpha : \alpha(H) > 0 \text{ for } H \in \mathcal{C}\}$$

is a positive system: exactly one of $\alpha, -\alpha$ lies in Φ^+, and if $\alpha, \beta \in \Phi^+$ and $\alpha + \beta \in \Phi$, then $\alpha + \beta \in \Phi^+$. The opposite chamber has positive system $-\Phi^+$.

Proof. The first assertion follows from $\Phi = -\Phi$ and nonvanishing on $\mathcal{C}$. If $H \in \mathcal{C}$ and $\alpha(H), \beta(H) > 0$, then $(\alpha + \beta)(H) > 0$. The opposite chamber consists of $-H$, so every sign reverses. $\square$

Proposition 1.3 (Singular directions and rank loss)**.** *If H lies on a collection of root hyperplanes, then the centralizer of* $\exp H$ *contains A together with all root groups U_α for which $\alpha(H) = 0$. Thus a singular one-parameter subaction forgets precisely the coarse-Lyapunov directions lying in its zero hyperplanes.*

Proof. For $X \in \mathfrak{g}_\alpha$, Theorem 1.2 gives $\mathrm{Ad}(e^H)X = e^{\alpha(H)}X$. This equals X exactly when $\alpha(H) = 0$. Exponentiating the corresponding nilpotent root spaces yields centralizing root subgroups. Directions with nonzero root value are genuinely expanded or contracted. $\square$

Worked Example 1.4 (The A_2 arrangement)**.** For SL_3, the three lines $t_1 = t_2$, $t_2 = t_3$, $t_1 = t_3$ cut the two-dimensional plane $t_1 + t_2 + t_3 = 0$ into six chambers. A regular diagonal element lies in one chamber; a singular element lies on one of these lines and centralizes one root group.

CHAPTER 3

Stable and Unstable Horospherical Groups

Proof architecture. The chapter is organized as the chain *Lie algebras of horospherical groups* $\longrightarrow$ *Horospherical decomposition and contraction characterization* $\longrightarrow$ *Stable leaves on a quotient.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. Algebraic stable manifolds

For $a = \exp H \in A$ define

$$G_a^+ = \{g : a^{-n} g a^n \to e\}, \qquad G_a^- = \{g : a^n g a^{-n} \to e\}.$$

Our sign convention makes G_a^+ the group expanded by forward powers of a.

Lemma 1.1 (Lie algebras of horospherical groups)**.**

$$\mathrm{Lie}(G_a^+) = \bigoplus_{\alpha(H)>0} \mathfrak{g}_\alpha, \qquad \mathrm{Lie}(G_a^-) = \bigoplus_{\alpha(H)<0} \mathfrak{g}_\alpha.$$

Both sums are nilpotent Lie algebras.

Proof. For $X \in \mathfrak{g}_\alpha$, $a^{-n}\exp(X)a^n = \exp(e^{-n\alpha(H)}X)$, which tends to e exactly when $\alpha(H) > 0$. The bracket rule from Lemma 1.1 keeps the positive-root sum closed under brackets. Repeated brackets add positive root values, but only finitely many roots exist, so sufficiently long brackets vanish. The negative case is identical. $\square$

Theorem 1.2 (Horospherical decomposition and contraction characterization). *The connected subgroups with the Lie algebras in Lemma 1.1 are closed simply connected unipotent groups and coincide with $G_a^\pm$. Exponential coordinates are global diffeomorphisms on them. If H is regular, a neighborhood of the identity admits product coordinates*

$$G_a^- \times Z_G(A) \times G_a^+ \longrightarrow G, \qquad (u^-, z, u^+) \mapsto u^- z u^+.$$

Proof. A nilpotent Lie algebra consisting of nilpotent adjoint directions integrates to a connected unipotent subgroup; for real unipotent groups the exponential map is polynomial with polynomial inverse, hence a global diffeomorphism. The conjugation calculation in the proof of Lemma 1.1 proves inclusion in the contraction-defined groups. Conversely, write a small element in exponential coordinates and decompose its logarithm into root components. A component with the wrong sign grows under the defining conjugation and therefore prevents convergence to the identity; hence only the indicated root spaces occur. For regular H, the three Lie algebras form a direct sum $\mathfrak{g}_a^- \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_a^+$. The differential of multiplication at the identity is the sum map and is invertible, so the inverse function theorem gives the local product chart. □

Proposition 1.3 (Stable leaves on a quotient)**.** *On $X = \Gamma\backslash G$, whenever the local injectivity radius at x is positive, the local stable and unstable manifolds of right translation by a are exactly the plaques xG_a^- and xG_a^+, respectively. The contraction rate in a root direction α is $e^{-|\alpha(H)|n}$ up to uniform local chart constants.*

Proof. Right translation gives $d(xua^n, xa^n) = d(xa^n(a^{-n}ua^n), xa^n)$ in a right-invariant local metric. For $u = \exp X_\alpha$ the conjugated parameter is $e^{-n\alpha(H)}X_\alpha$. Local injectivity lets us identify group and quotient distances. The converse follows from the local product chart: a nonzero neutral or expanding coordinate cannot converge to the base orbit. □

Worked Example 1.4 (Stable and unstable groups for one SL_3 element)**.** Take $a = \mathrm{diag}(e^2, e^{-1/2}, e^{-3/2})$. Then $a^n(I + sE_{ij})a^{-n} = I + e^{n(t_i - t_j)}sE_{ij}$. The positive differences are $t_1 - t_2 = 5/2$, $t_1 - t_3 = 7/2$, and $t_2 - t_3 = 1$; hence

$$G_a^+ = U_{12}U_{13}U_{23}$$

is the upper-unitriangular group. The negative roots give $G_a^- = U_{21}U_{31}U_{32}$. Notice that $[U_{12}, U_{23}] = U_{13}$, so the unstable group is not a direct product even though its tangent space is the direct sum of the three positive root spaces.

CHAPTER 4

Coarse Lyapunov Subgroups

Proof architecture. The chapter is organized as the chain *Coarse-root equivalence* $\longrightarrow$ *Coarse-Lyapunov subgroup structure* $\longrightarrow$ *Coarse groups in parabolic subgroups.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. Why proportional roots belong together

A multiparameter action cannot distinguish roots that vanish on exactly the same hyperplane and have the same sign. The dynamically intrinsic object is therefore a coarse root class.

Lemma 1.1 (Coarse-root equivalence)**.** *Define $\alpha \sim \beta$ when $\beta = c\alpha$ for some $c > 0$. This is an equivalence relation. For each class $[\alpha]$ the sum*

$$\mathfrak{u}^{[\alpha]} = \bigoplus_{\beta \in [\alpha]} \mathfrak{g}_\beta$$

is a nilpotent Lie algebra normalized by A.

Proof. Only transitivity needs comment and follows by multiplying positive scalars. If β, γ lie in the same positive ray, then any root $\beta+\gamma$ also lies on that ray, so Lemma 1.1 shows closure under brackets. Nilpotence follows as in Lemma 1.1; A-normalization follows from the weight decomposition. $\square$

Theorem 1.2 (Coarse-Lyapunov subgroup structure). *Let $U^{[\alpha]} = \exp \mathfrak{u}^{[\alpha]}$. Then for any regular $H \in \mathfrak{a}$,*

$$G^+_{\exp H} = \left\langle U^{[\alpha]} : \alpha(H) > 0 \right\rangle,$$

and the ordering of the coarse classes compatible with increasing root height gives polynomial product coordinates on $G^+_{\exp H}$. The class $[\alpha]$ is determined dynamically by the common Lyapunov hyperplane $\ker \alpha$ *and an orientation of its complement.*

Proof. The Lie algebra of the right side contains exactly the positive root spaces by Lemma 1.1, hence equals the unstable Lie algebra. In a nilpotent group, choose a basis adapted to the lower central series and order root classes so brackets only move to later coordinates. The Baker–Campbell–Hausdorff formula terminates and the multiplication map from ordered root-space exponentials is triangular with identity diagonal, hence a polynomial diffeomorphism. Two positive multiples have the same kernel and sign; conversely two nonzero linear forms with the same kernel are proportional, and the orientation determines the sign. □

Proposition 1.3 (Coarse groups in parabolic subgroups)**.** *For a face F of a Weyl chamber, the subgroup generated by $Z_G(A)$ and the coarse groups nonnegative on F is a parabolic subgroup; its unipotent radical is generated by those coarse groups strictly positive on F.*

Proof. The corresponding Lie algebra is the sum of $\mathfrak{g}_0$ and root spaces whose roots are nonnegative on the face. The bracket rule preserves nonnegativity, while strictly positive root spaces form an ideal. Choosing a point in the relative interior of F, this is exactly the nonnegative eigenspace algebra for $\operatorname{ad} H$, the standard Lie-algebra description of a real parabolic. Exponentiating the positive nilpotent ideal gives its unipotent radical. □

Worked Example 1.4 (Why coarse roots matter in a nonreduced system)**.** In a reduced type-A system such as SL_k, each coarse class contains one root. In a nonreduced restricted-root system of type BC_1, however, both α and 2α may occur. They vanish on the same Lyapunov hyperplane and have the same sign on either side. A diagonal element therefore expands or contracts the two root spaces simultaneously. The dynamically natural subgroup is

$$U^{[\alpha]} = \exp(\mathfrak{g}_\alpha \oplus \mathfrak{g}_{2\alpha}),$$

not either root group separately. This example explains why the definition uses positive proportionality rather than equality of roots.

CHAPTER 5

The SL_3 Laboratory

Proof architecture. The chapter is organized as the chain *Explicit root scaling in SL_3 $\longrightarrow$ Complete SL_3 unstable geometry $\longrightarrow$ The six singular directions.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. Every mechanism in coordinates

Write $u_{ij}(s) = I + sE_{ij}$ and $a(t) = \mathrm{diag}(e^{t_1}, e^{t_2}, e^{t_3})$, $t_1 + t_2 + t_3 = 0$.

Lemma 1.1 (Explicit root scaling in SL_3)**.** *For $i \neq j$,*

$$a(t)u_{ij}(s)a(t)^{-1} = u_{ij}(e^{t_i - t_j}s).$$

The three positive roots for the chamber $t_1 > t_2 > t_3$ are $\alpha_{12}, \alpha_{23}, \alpha_{13} = \alpha_{12} + \alpha_{23}$.

Proof. Diagonal conjugation multiplies E_{ij} by the ratio of the ith and jth diagonal entries, namely $e^{t_i - t_j}$. The root identities are immediate. $\square$

Theorem 1.2 (Complete SL_3 unstable geometry). *If $t_1 > t_2 > t_3$, then*

$$G^+_{a(t)} = \left\{ \begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} : x, y, z \in \mathbb{R} \right\},$$

and every element has a unique factorization

$$u_{12}(x)u_{23}(y)u_{13}(z - xy).$$

The center is U_{13} and the quotient by the center is $U_{12} \times U_{23}$.

Proof. The positive root spaces are exactly $\mathbb{R}E_{12}, \mathbb{R}E_{23}, \mathbb{R}E_{13}$, so Theorem 1.2 identifies the upper unipotent group. Multiplication gives

$$u_{12}(x)u_{23}(y)u_{13}(w) = I + xE_{12} + yE_{23} + (w + xy)E_{13},$$

which proves the unique factorization. Direct matrix multiplication shows U_{13} commutes with both other root groups, while the commutator of U_{12} and U_{23} is nontrivial in U_{13}, proving the center statement. □

Proposition 1.3 (The six singular directions)**.** *The three Lyapunov hyperplanes in $\mathfrak{a}$ divide it into six Weyl chambers. On $t_i = t_j$ the subgroup U_{ij} is neutral; the remaining two root pairs are expanding/contracting according to their signs. Thus every singular one-parameter diagonal flow in SL_3 retains a rank-one neutral block isomorphic to SL_2 up to its central torus.*

Proof. The hyperplanes are the three lines $t_i = t_j$. The scaling formula of Lemma 1.1 gives neutrality and signs. Matrices in coordinates i, j with determinant one form an SL_2 subgroup, and the diagonal direction acting equally on these two coordinates centralizes it modulo the determinant constraint. □

Worked Example 1.4 (One commutator)**.**

$$[u_{12}(s), u_{23}(t)] = u_{13}(st).$$

This elementary identity is the geometric source of the high-entropy method in Volume 14: motion on two noncommuting coarse leaves creates motion on a third leaf.

CHAPTER 6

Root Commutator Calculus

Proof architecture. The chapter is organized as the chain *BCH commutator expansion* $\longrightarrow$ *Root-string commutator production* $\longrightarrow$ *Iterated root generation in* SL_k. The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. From Lie brackets to group motion

Noncommuting coarse Lyapunov directions are useful only if their bracket can be controlled quantitatively at the group level.

Lemma 1.1 (BCH commutator expansion). *For X, Y in a fixed bounded subset of $\mathfrak{g}$ and sufficiently small s, t,*

$$\log(e^{sX}e^{tY}e^{-sX}e^{-tY}) = st[X,Y] + O(|s|^2|t| + |s||t|^2),$$

with the error uniform on that bounded set. If $[X,[X,Y]] = [Y,[X,Y]] = 0$, the equality is exact without the error.

Proof. Apply the Baker–Campbell–Hausdorff formula twice. Terms of total degree one cancel; the unique degree-two term is $st[X,Y]$. Every remaining Lie monomial has total degree at least three, giving the displayed bound. Under the two-step nilpotence assumptions all iterated brackets of length at least three vanish. $\square$

Theorem 1.2 (Root-string commutator production). *If $\alpha, \beta, \alpha+\beta \in \Phi$ and there exist $X \in \mathfrak{g}_\alpha$, $Y \in \mathfrak{g}_\beta$ with $[X,Y] \neq 0$, then arbitrarily small commutator rectangles in $U_\alpha U_\beta$ produce nontrivial motion in $U_{\alpha+\beta}$ at quadratic scale. In SL_k one has the exact identity*

$$[u_{ij}(s), u_{jk}(t)] = u_{ik}(st) \qquad (i, j, k \text{ distinct}).$$

Proof. Lemma 1.1 places $[X,Y]$ in $\mathfrak{g}_{\alpha+\beta}$. Lemma 1.1 then shows that the logarithm of the commutator has leading component $st[X,Y]$ and higher-order error. For sufficiently small nonzero s, t, projection to $\mathfrak{g}_{\alpha+\beta}$ is nonzero.

In SL_k, $E_{ij}E_{jk} = E_{ik}$ while the reverse product is zero and all higher products vanish; direct multiplication gives the exact formula. ☐

Proposition 1.3 (Iterated root generation in SL_k)**.** *The simple positive root groups $U_{i,i+1}$ generate the full upper unipotent group of SL_k. More precisely, U_{ij} for $j - i = r$ is generated by iterated commutators of length $r - 1$ in adjacent simple root groups.*

PROOF. Induct on $r = j - i$. The case $r = 1$ is tautological. For $r > 1$ the exact identity $[u_{i,j-1}(s), u_{j-1,j}(t)] = u_{ij}(st)$ shows that every parameter in U_{ij} is obtained by taking $s = 1$ and varying t, after the inductive construction of $U_{i,j-1}$. Ordered multiplication of all upper root groups gives the upper unipotent group. ☐

Worked Example 1.4 (A complete SL_3 root-commutator calculation)**.** Put $x = I + rE_{12}$ and $y = I + sE_{23}$. Since $E_{12}^2 = E_{23}^2 = 0$, $x^{-1} = I - rE_{12}$ and $y^{-1} = I - sE_{23}$. Multiplying,

$$xy = I + rE_{12} + sE_{23} + rsE_{13}.$$

Because $E_{23}E_{12} = 0$, the terms in E_{12} and E_{23} cancel when one multiplies by $x^{-1}y^{-1}$, leaving

$$[x, y] = xyx^{-1}y^{-1} = I + rsE_{13}.$$

The calculation is exact, not merely first-order BCH. It is the algebraic engine used in Volume 14 when two noncommuting leafwise directions manufacture a third invariance direction.

CHAPTER 7

Homogeneous Foliations and Algebraic Leaves

Proof architecture. The chapter is organized as the chain *Local plaque charts* $\longrightarrow$ *Algebraic Lyapunov foliations* $\longrightarrow$ *Intersections and generated foliations.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. Leaves from normalized subgroups

Let $X = \Gamma\backslash G$ and let $U < G$ be a connected subgroup normalized by A. The family of right cosets xU is an A-invariant immersed foliation away from the usual local self-intersections of the quotient.

Lemma 1.1 (Local plaque charts)**.** *For every compact $K \subset X$ there is $r_K > 0$ such that for every $x \in K$ the map $B_U(r_K) \to X$, $u \mapsto xu$, is injective modulo the finite local stabilizer and is bi-Lipschitz onto its image with constants uniform on K.*

Proof. The injectivity radius has a positive minimum on K. Choose r_K below a fixed fraction of that minimum and below an exponential-coordinate radius in G. Right-invariant Riemannian coordinates make the orbit map have injective differential with uniformly bounded inverse on the compact family of base points; the inverse function theorem gives uniform bi-Lipschitz bounds. Any remaining collision is precisely a local stabilizer element. □

Theorem 1.2 (Algebraic Lyapunov foliations). *For every coarse root class $[\alpha]$, the plaques $xU^{[\alpha]}$ form an A-invariant measurable foliation $\mathcal{W}^{[\alpha]}$. For $a = \exp H$ with $\alpha(H) > 0$, forward right translation by a expands the leaf metric by exponential rates lying between $e^{c_-\alpha(H)n}$ and $e^{c_+\alpha(H)n}$, where $c_\pm$ are the extreme positive multiples occurring in the class.*

Proof. Normalization by A gives $(xU^{[\alpha]})a = xa(a^{-1}U^{[\alpha]}a) = xaU^{[\alpha]}$, hence invariance of the partition into orbits. Local plaques are Borel in the charts of Lemma 1.1, which gives a measurable foliation. On the root space $\mathfrak{g}_{c\alpha}$, conjugation scales by $e^{c\alpha(H)n}$; equivalence of exponential and Riemannian coordinates on a small plaque gives the metric estimates. □

Proposition 1.3 (Intersections and generated foliations)**.** *If U, V are A-normalized unipotent subgroups with transverse Lie algebras and $[\mathfrak{u}, \mathfrak{v}]$ contained in a later term of a nilpotent ordering, then local leaves of $\langle U, V\rangle$ are obtained by successively saturating U-plaques by V-plaques. In particular the local product charts from Theorem 1.2 are foliation charts adapted simultaneously to all positive coarse roots.*

PROOF. Ordered BCH coordinates give a local diffeomorphism from the product of the subgroup coordinates onto the generated nilpotent group. Fixing all but the U coordinate gives a U plaque; then varying the V coordinate saturates it. Bracket coordinates occur later in the triangular BCH expansion and therefore do not destroy local uniqueness. □

Worked Example 1.4 (Local saturation of two root foliations)**.** In the upper-unitriangular subgroup of SL_3, start from the U_{12} leaf through x and saturate it by U_{23}. Points have the form $u_{12}(r)u_{23}(s)x$. Reversing the order gives

$$u_{12}(r)u_{23}(s)x = u_{23}(s)u_{12}(r)u_{13}(rs)x.$$

Thus the two-dimensional family of successive leaf motions is not tangent to a subgroup generated only by the two coordinate lines: its Lie closure automatically contains U_{13}. The local foliation generated by the two root directions is therefore the three-dimensional Heisenberg leaf. This is why bracket closure, not linear span alone, determines the algebraic leaf.

CHAPTER 8

Higher-Rank Diagonal Action Structure

Proof architecture. The chapter is organized as the chain *Adapted local coordinates* $\longrightarrow$ *Higher-rank structural dictionary.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. The structural package for entropy theory

We now collect exactly what later volumes may inherit from the Lie-theoretic part of the project.

Lemma 1.1 (Adapted local coordinates)**.** *Fix a regular chamber and order its positive coarse roots compatibly with root height. There is a neighborhood $\mathcal{O}$ of e in G in which every g has unique coordinates*

$$g = u^- z u_1 \cdots u_r,$$

with $u^- \in G_a^-$, $z \in Z_G(A)$ and $u_j \in U^{[\alpha_j]}$, all small. The coordinate maps and their inverses are smooth with bounded derivatives on smaller neighborhoods.

Proof. The root decomposition is a direct sum of the negative root spaces, the zero space, and the positive coarse spaces. The differential of the ordered multiplication map at the identity is therefore the direct-sum isomorphism onto $\mathfrak{g}$. The inverse function theorem gives uniqueness and smoothness. Shrinking to a compactly contained chart bounds all derivatives. $\square$

Theorem 1.2 (Higher-rank structural dictionary). *Let A act by right translations on $X = \Gamma\backslash G$. Then:*

(1) every regular $a \in A$ has algebraic stable and unstable foliations generated by the root groups of the corresponding sign;

(2) the finest dynamically intrinsic expanding foliations common to the full A-action are the coarse-Lyapunov foliations $\mathcal{W}^{[\alpha]}$;

(3) noncommuting coarse directions generate further root directions according to the root-bracket calculus;

(4) singular elements are exactly those lying on Lyapunov hyperplanes and therefore centralizing at least one coarse direction.

These statements are local on arbitrary quotients and require no arithmetic hypothesis on Γ*.*

Proof. Items (1)–(4) are respectively Theorems 1.2, 1.2, 1.2, and Proposition 1.3. None used finiteness of volume or arithmetic structure; only local Lie theory and discreteness of the quotient were used. The adapted chart of Lemma 1.1 makes the simultaneous compatibility explicit. $\square$

Why this matters. Volume 12 will attach entropy to the expanding foliations, Volume 13 will attach leafwise conditional measures to the coarse foliations, and Volume 14 will exploit the noncommuting-root mechanism. This division prevents later proofs from hiding Lie algebra calculations inside entropy terminology.

Worked Example 1.3 (Reading the structural dictionary in SL_3)**.** For $H = \mathrm{diag}(2, -1, -1)$ (after subtracting the trace if desired), the two roots $e_1 - e_2$ and $e_1 - e_3$ are positive while $e_2 - e_3$ vanishes. The point lies on a Weyl wall. Dynamically, U_{12} and U_{13} are expanding, U_{21} and U_{31} are contracting, and U_{23}, U_{32} are neutral. Moving H slightly so that $t_2 > t_3$ crosses the wall and places U_{23} into the unstable group. Thus one line in the Weyl arrangement records an actual change in the stable/unstable foliation, which is why entropy is piecewise linear chamber by chamber in Volume 14.

Volume 12

Entropy from the Ground Up

Volume 12 scope and prerequisites

The purpose of this volume is to make later measure-rigidity arguments readable without an expert entropy black box. Conditional expectation is derived from Radon–Nikodym; finite-partition entropy identities are proved directly; the generator and relative-addition theorems are proved from those identities and martingale convergence; subordinate partitions then connect entropy to the algebraic foliations of Volume 11.

Student orientation: entropy objects before formulas

A *measurable partition* is understood modulo null sets through its generated sigma-algebra. Conditional expectation onto a partition means conditional expectation onto that sigma-algebra. For a finite partition $\mathcal{P}$, $H_\mu(\mathcal{P} \mid \mathcal{A})$ is the average information left after conditioning on $\mathcal{A}$. Kolmogorov–Sinai entropy is the asymptotic information growth under repeated refinement by a transformation. A partition is *subordinate* to an expanding foliation when almost every atom contains a neighborhood of the basepoint inside its leaf and is contained in that leaf locally. Two warnings govern the volume. First, entropy identities are statements modulo null sets, so representatives of partitions must not be manipulated pointwise without justification. Second, entropy along a subgroup depends on a compatible subordinate partition but the resulting numerical invariant does not; proving this independence is part of the theory.

Reader guide: a concrete model

Worked Example 0.4 (A calculation to keep in view)**.** Let

$$a = \operatorname{diag}(e^{2s}, e^{-s}, e^{-s}) \in \mathrm{SL}_3(\mathbb{R})$$

and let E_{12} be the matrix unit. Since

$$\operatorname{Ad}(a)E_{12} = aE_{12}a^{-1} = e^{3s}E_{12},$$

the root exponent on the E_{12} direction is $3s$. For a one-dimensional leafwise measure whose conditional scaling is Lebesgue, the Jacobian contribution along this root direction is therefore $3s$. This calculation is the concrete local model behind the root-by-root entropy bookkeeping used later.

CHAPTER 1

Measurable Partitions and Conditional Expectation

Proof architecture. The chapter is organized as the chain *Conditional expectation: existence and identities* $\longrightarrow$ *Partitions and information algebras* $\longrightarrow$ *Equivariance under a measure-preserving map.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. The measurable replacement for "conditioning on a leaf"

Throughout this volume $(X, \mathcal{B}, \mu)$ is a standard probability space. A countably generated sub-σ-algebra $\mathcal{A} \subset \mathcal{B}$ is the measurable information retained after forgetting some coordinates.

Lemma 1.1 (Conditional expectation: existence and identities)**.** *For $f \in L^1(\mu)$ and a sub-σ-algebra $\mathcal{A}$ there is an $\mathcal{A}$-measurable function $E(f \mid \mathcal{A})$, unique a.e., such that*

$$\int_A E(f \mid \mathcal{A})\, d\mu = \int_A f\, d\mu \quad (A \in \mathcal{A}).$$

It is positive, contractive on L^p for $1 \le p \le \infty$, and satisfies the tower rule $E(E(f \mid \mathcal{A}_2) \mid \mathcal{A}_1) = E(f \mid \mathcal{A}_1)$ when $\mathcal{A}_1 \subset \mathcal{A}_2$.

Proof. For $f \ge 0$, the measure $\nu(A) = \int_A f\, d\mu$ on $\mathcal{A}$ is absolutely continuous with respect to $\mu|_{\mathcal{A}}$. Radon–Nikodym gives the required density. Apply this to positive and negative parts for general f. Uniqueness follows because the integral of the difference over $\{g > 0\} \in \mathcal{A}$ would otherwise be positive. Positivity is built into the construction. Jensen's inequality for the convex function $|\cdot|^p$, first for finite $\mathcal{A}$ and then by monotone approximation of $\mathcal{A}$ by finite algebras, gives $|E(f \mid \mathcal{A})|^p \le E(|f|^p \mid \mathcal{A})$ and hence contraction. For the tower rule, both sides are $\mathcal{A}_1$-measurable and have the same integrals on every $A \in \mathcal{A}_1$. $\square$

Theorem 1.2 (Partitions and information algebras). *Let ξ be a countably generated measurable partition, represented by a Borel map $\pi : X \to Y$ to a* *standard Borel space, with atoms $\pi^{-1}(y)$. Then the ξ-saturated measurable sets form the sub-σ-algebra $\mathcal{A}_\xi = \pi^{-1}\mathcal{B}_Y$ modulo null sets. Refinement satisfies*

$$\xi \leq \eta \iff \mathcal{A}_\xi \supset \mathcal{A}_\eta,$$

and joins of partitions correspond to generated σ-algebras.

Proof. A saturated set is a union of fibers of π and, modulo completion, is the inverse image of a measurable subset of the quotient because Y is standard Borel and the partition is countably generated. If ξ refines η, then knowing the finer atom determines the coarser atom, so every η-saturated set is ξ-saturated; this is exactly the reversed inclusion of information algebras. A point's atom in $\xi \vee \eta$ is the intersection of its two atoms, hence is determined by the pair of quotient coordinates, whose measurable sets generate $\sigma(\mathcal{A}_\xi, \mathcal{A}_\eta)$. ☐

Proposition 1.3 (Equivariance under a measure-preserving map)**.** *If T preserves μ, then*

$$E(f \circ T \mid T^{-1}\mathcal{A}) = E(f \mid \mathcal{A}) \circ T.$$

Consequently the information algebra of $T^{-1}\xi$ is $T^{-1}\mathcal{A}_\xi$.

Proof. The right side is $T^{-1}\mathcal{A}$-measurable. For $A \in \mathcal{A}$, invariance gives

$$\int_{T^{-1}A} E(f \mid \mathcal{A}) \circ T \, d\mu = \int_A E(f \mid \mathcal{A}) \, d\mu = \int_A f \, d\mu = \int_{T^{-1}A} f \circ T \, d\mu.$$

Uniqueness in Lemma 1.1 proves the identity. The partition statement is immediate from inverse images of fibers. ☐

Worked Example 1.4 (Conditional expectation on a four-point space)**.** Let $X = \{00, 01, 10, 11\}$ with uniform measure and let $\mathcal{A}$ remember only the first bit. For $f(00) = 0, f(01) = 2, f(10) = 1, f(11) = 5$, conditional expectation averages over the two-point fibers:

$$\mathbb{E}(f \mid \mathcal{A})(00) = \mathbb{E}(f \mid \mathcal{A})(01) = 1,$$

while

$$\mathbb{E}(f \mid \mathcal{A})(10) = \mathbb{E}(f \mid \mathcal{A})(11) = 3.$$

Testing against the indicator of the first-bit-0 atom gives $(0+2)/4 = (1+1)/4$ on the two sides of the defining integral identity, and similarly for the other atom. This finite model is exactly what later conditional kernels do fiber by fiber.

CHAPTER 2

Conditional Entropy

Proof architecture. The chapter is organized as the chain *Chain rule for finite partitions* $\longrightarrow$ *Conditional entropy calculus* $\longrightarrow$ *Conditional mutual information.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. Information carried by a finite partition

For a finite or countable partition ξ with finite Shannon entropy set

$$I_\mu(\xi)(x) = -\log \mu(\xi(x)), \qquad H_\mu(\xi) = \int I_\mu(\xi)\, d\mu.$$

For a sub-σ-algebra $\mathcal{A}$, define conditional atom probabilities $p_C(x) = E(1_C \mid \mathcal{A})(x)$ and

$$H_\mu(\xi \mid \mathcal{A}) = \int -\sum_{C\in\xi} p_C \log p_C \, d\mu.$$

Lemma 1.1 (Chain rule for finite partitions)**.** *For finite-entropy partitions ξ, η and a sub-σ-algebra $\mathcal{A}$,*

$$H(\xi \vee \eta \mid \mathcal{A}) = H(\eta \mid \mathcal{A}) + H(\xi \mid \mathcal{A} \vee \mathcal{A}_\eta).$$

In particular conditioning reduces entropy.

Proof. Condition first on an atom of a finite approximation of $\mathcal{A}$. There the assertion is the elementary identity $-\sum_{i,j} p_{ij} \log p_{ij} = -\sum_j p_j \log p_j - \sum_j p_j \sum_i (p_{ij}/p_j) \log(p_{ij}/p_j)$. Approximate a countably generated $\mathcal{A}$ by an increasing sequence of finite algebras and use martingale convergence of the conditional probabilities together with dominated truncation of $-t\log t$. The monotonicity follows because the second term in the chain rule is nonnegative. □

Theorem 1.2 (Conditional entropy calculus). *For finite-entropy partitions:*

(1) $H(\xi \mid \mathcal{A}) = 0$ *iff* ξ *is* $\mathcal{A}$*-measurable modulo null sets;*
(2) if $\mathcal{A}_1 \subset \mathcal{A}_2$*, then* $H(\xi \mid \mathcal{A}_2) \leq H(\xi \mid \mathcal{A}_1)$*;*
(3) $H(\xi \vee \eta \mid \mathcal{A}) \leq H(\xi \mid \mathcal{A}) + H(\eta \mid \mathcal{A})$*;*
(4) if T *preserves* μ*, then* $H(T^{-1}\xi \mid T^{-1}\mathcal{A}) = H(\xi \mid \mathcal{A})$*.*

Proof. For (1), conditional entropy vanishes exactly when for a.e. x the probability vector $(p_C(x))$ has one coordinate equal to one and the rest zero; this says the atom of ξ is determined by $\mathcal{A}$. Statement (2) follows from Lemma 1.1 by conditioning successively, or from concavity of Shannon entropy applied to conditional probability vectors. For (3), apply the chain rule and then (2) to the second term. For (4), Proposition 1.3 transports conditional atom probabilities and invariance of μ preserves their integrals. □

Proposition 1.3 (Conditional mutual information)**.** *Define*

$$I(\xi; \eta \mid \mathcal{A}) = H(\xi \mid \mathcal{A}) - H(\xi \mid \mathcal{A} \vee \mathcal{A}_\eta).$$

Then $I \geq 0$*, it is symmetric in* ξ, η*, and it vanishes precisely when the conditional joint probabilities factor a.e.*

Proof. Nonnegativity is monotonicity. The chain rule written in the two orders gives symmetry. For a finite conditioning algebra, the difference is the average Kullback–Leibler divergence between the conditional joint distribution and the product of its marginals, hence is nonnegative and vanishes exactly at factorization. Approximation by finite conditioning algebras and martingale convergence gives the general case. □

Worked Example 1.4 (How conditioning removes one bit of entropy)**.** Let (B_1, B_2) be two independent fair bits. The joint partition has four atoms of mass $1/4$, so $H(B_1, B_2) = \log 4 = 2 \log 2$. Conditioning on B_1 leaves two equiprobable possibilities for B_2 in each fiber, hence

$$H(B_2 \mid B_1) = \log 2.$$

The chain rule reads

$$H(B_1, B_2) = H(B_1) + H(B_2 \mid B_1) = \log 2 + \log 2.$$

If instead $B_2 = B_1$, then $H(B_2 \mid B_1) = 0$. Thus conditional entropy measures information genuinely new relative to the factor already known.

CHAPTER 3

Kolmogorov–Sinai Entropy

Proof architecture. The chapter is organized as the chain *Subadditivity and entropy rate* $\longrightarrow$ *Kolmogorov–Sinai generator theorem* $\longrightarrow$ *Powers and inverse maps.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. Entropy per unit time

Let T preserve μ and let ξ be a finite-entropy partition. Put $\xi_0^{n-1} = \bigvee_{j=0}^{n-1} T^{-j}\xi$.

Lemma 1.1 (Subadditivity and entropy rate)**.** *The sequence $a_n = H(\xi_0^{n-1})$ is subadditive. Hence*

$$h_\mu(T,\xi) = \lim_{n\to\infty} \frac{1}{n} H(\xi_0^{n-1}) = \inf_n \frac{1}{n} H(\xi_0^{n-1}).$$

Moreover

$$h_\mu(T,\xi) = H\left(\xi \mid \bigvee_{j=1}^{\infty} T^j \xi\right).$$

Proof. The join for $n+m$ is $\xi_0^{n-1} \vee T^{-n}\xi_0^{m-1}$. Subadditivity of partition entropy and T-invariance give $a_{n+m} \le a_n + a_m$; Fekete's lemma gives the limit. Repeated chain rule gives

$$H(\xi_0^{n-1}) = \sum_{j=0}^{n-1} H\left(\xi \mid \bigvee_{i=1}^{j} T^i \xi\right).$$

The summands decrease to the conditional entropy on the infinite future by martingale convergence, so their Cesàro averages converge to the same limit. $\square$

THEOREM 1.2 (Kolmogorov–Sinai generator theorem). *Define* $h_\mu(T) = \sup_\eta h_\mu(T,\eta)$ *over finite-entropy partitions. If* ξ *is a finite-entropy generating partition, meaning*

$$\bigvee_{n\in\mathbb{Z}} T^{-n}\mathcal{A}_\xi = \mathcal{B} \quad \mathit{mod}\ \mu,$$

then

$$h_\mu(T) = h_\mu(T,\xi).$$

PROOF. Let η be any finite partition and $\varepsilon > 0$. Since ξ generates, the martingale convergence theorem implies that for sufficiently large N, the atom of η is determined with conditional entropy less than ε by $\xi_{-N}^{N} = \bigvee_{|j|\le N} T^{-j}\xi$; thus $H(\eta \mid \xi_{-N}^{N}) < \varepsilon$. For n iterates, the chain rule gives

$$H(\eta_0^{n-1}) \le H((\xi_{-N}^{N})_0^{n-1}) + n\varepsilon.$$

The first join is contained in ξ_{-N}^{n-1+N}, whose entropy rate is $h(T,\xi)$ because the $2N$ boundary names contribute $O_N(1)$ entropy. Divide by n and let $n \to \infty$ to get $h(T,\eta) \le h(T,\xi) + \varepsilon$. Since η and ε were arbitrary, the supremum is at most $h(T,\xi)$; the reverse inequality is tautological. □

Proposition 1.3 (Powers and inverse maps)**.** *For invertible* T *and* $k \in \mathbb{Z}$*,*

$$h_\mu(T^k) = |k| h_\mu(T).$$

In particular $h_\mu(T) = h_\mu(T^{-1})$*.*

PROOF. For $k > 0$, a partition generating for T can be grouped into $\xi \vee T^{-1}\xi \vee \cdots \vee T^{-(k-1)}\xi$, which generates for T^k. Its n names are exactly kn consecutive T-names, so the entropy rate is $kh(T)$. The inverse has the same finite orbit names in reverse order, so $h(T^{-1},\xi) = h(T,\xi)$ for every partition; taking suprema gives equality. Negative k follows. □

Worked Example 1.4 (Entropy of the Bernoulli shift)**.** On $\{0,1\}^{\mathbb{Z}}$ with product weights $(p, 1-p)$, let T be the left shift and let ξ record coordinate 0. The join ξ_0^{n-1} records n independent coordinates, so

$$H(\xi_0^{n-1}) = n\big(-p\log p - (1-p)\log(1-p)\big).$$

Therefore

$$h_\mu(T,\xi) = -p\log p - (1-p)\log(1-p).$$

Since the full two-sided orbit of ξ records every coordinate, ξ is generating, and the Kolmogorov–Sinai theorem says this is the entropy of the shift itself.

CHAPTER 4

Relative Entropy and Factors

Proof architecture. The chapter is organized as the chain *Relative entropy rate* $\longrightarrow$ *Factor and relative entropy decomposition* $\longrightarrow$ *Entropy-zero extensions and tower inequality.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. Entropy visible above a factor

Let $\pi : (X, \mu, T) \to (Y, \nu, S)$ be a factor and put $\mathcal{Y} = \pi^{-1}\mathcal{B}_Y$.

Lemma 1.1 (Relative entropy rate)**.** *For a finite-entropy partition ξ, the limit*

$$h_\mu(T, \xi \mid \mathcal{Y}) = \lim_{n\to\infty} \frac{1}{n} H(\xi_0^{n-1} \mid \mathcal{Y})$$

exists and equals

$$H\left(\xi \mid \mathcal{Y} \vee \bigvee_{j=1}^{\infty} T^j \xi\right).$$

Proof. Because $\mathcal{Y}$ is T-invariant, the chain-rule calculation of Lemma 1.1 works with $\mathcal{Y}$ inserted in every conditioning algebra. Writing the n-block entropy as the sum of n one-step conditional entropies, stationarity turns the rth summand into

$$H\left(\xi \mid \mathcal{Y} \vee \bigvee_{j=1}^{n-1-r} T^j \xi\right).$$

These decrease to the infinite-future conditional entropy by martingale convergence. The Cesàro means therefore converge to the same limit. $\square$

Theorem 1.2 (Factor and relative entropy decomposition). *Define $h_\mu(T \mid \mathcal{Y}) = \sup_\xi h_\mu(T, \xi \mid \mathcal{Y})$. Then*

$$h_\mu(T) \geq h_\nu(S).$$

If η is a finite-entropy generating partition for S and $\xi \vee \pi^{-1}\eta$ is a finite-entropy generator for T, then

$$h_\mu(T) = h_\nu(S) + h_\mu(T, \xi \mid \mathcal{Y}).$$

Consequently $h_\mu(T) = h_\nu(S) + h_\mu(T \mid \mathcal{Y})$ whenever a finite-entropy relative generator exists.

Proof. Every finite partition of Y pulls back with exactly the same name distribution, proving $h_\mu(T) \geq h_\nu(S)$.

For the addition formula, fix an integer $M \geq 1$ and abbreviate $\eta_{-M}^{n-1+M} = \bigvee_{j=-M}^{n-1+M} S^{-j}\eta$. The chain rule gives

$$H(\xi_0^{n-1} \vee \pi^{-1}\eta_{-M}^{n-1+M}) = H(\pi^{-1}\eta_{-M}^{n-1+M}) \\ + H(\xi_0^{n-1} \mid \pi^{-1}\mathcal{A}_{\eta_{-M}^{n-1+M}}).$$

The extra $2M$ boundary names change the entropy of either side by at most $2MH(\eta)$, so after division by n they have no effect as $n \to \infty$. The first normalized term tends to $h_\nu(S, \eta) = h_\nu(S)$. For fixed n, the conditioning algebras $\pi^{-1}\mathcal{A}_{\eta_{-M}^{n-1+M}}$ increase with M to $\mathcal{Y}$ because η is generating. Conditional martingale convergence therefore identifies the second asymptotic term with $n^{-1}H(\xi_0^{n-1} \mid \mathcal{Y})$. First let $n \to \infty$ using Lemma 1.1, then $M \to \infty$. Finally, since $\xi \vee \pi^{-1}\eta$ is generating upstairs, the Kolmogorov–Sinai generator theorem identifies the left-hand entropy rate with $h_\mu(T)$.

If ξ is a relative generator and η is a generator of the base, then $\xi \vee \pi^{-1}\eta$ is a generator upstairs: equality of the full η-names forces two points into the same fiber, and equality of the full ξ-names then identifies them inside that fiber. Taking the supremum over finite relative generators gives the last formula. Chapter 7 records this elementary generator argument as a reusable lemma. ☐

Proposition 1.3 (Entropy-zero extensions and tower inequality). *If $h_\mu(T \mid \mathcal{Y}) = 0$, then every finite partition has arbitrarily small entropy rate after conditioning on a sufficiently fine finite factor approximation. For a tower of factors $X \to Y \to Z$,*

$$h(T \mid Z) \leq h(T \mid Y) + h(S \mid Z).$$

Proof. For the first assertion, choose a finite partition ζ of X. Since $h_\mu(T, \zeta \mid \mathcal{Y}) = 0$, approximate the conditional expectations of the finitely many atoms of a long ζ-name by a finite subalgebra of $\mathcal{Y}$. Continuity of finite Shannon entropy then makes the conditional name entropy per unit time arbitrarily small.

For the tower, let ξ be a finite partition upstairs and η a finite partition of Y. Conditional chain rule gives, for each n,

$$H(\xi_0^{n-1} \vee \pi^{-1}\eta_0^{n-1} \mid \mathcal{Z}) \leq H(\eta_0^{n-1} \mid \mathcal{Z}) + H(\xi_0^{n-1} \mid \mathcal{Y}).$$

Divide by n, take limits, and then take suprema over finite partitions. This yields the displayed tower inequality. □

Worked Example 1.4 (A product extension)**.** Let $T = S \times R$ on $(Y \times Z, \nu \times \rho)$, and let η, ζ be finite generators for S, R. Conditioning on the base factor $\mathcal{Y}$ fixes the entire Y coordinate, so the conditional distribution of an n-block of ζ is unchanged and

$$H(\zeta_0^{n-1} \mid \mathcal{Y}) = H_\rho(\zeta_0^{n-1}).$$

Thus $h(T \mid \mathcal{Y}) = h(R)$. The addition theorem gives

$$h(T) = h(S) + h(R),$$

which is the expected formula for two independent sources of dynamical information.

CHAPTER 5

Subordinate Partitions

Proof architecture. The chapter is organized as the chain *Construction from a small finite partition* $\longrightarrow$ *Increasing subordinate partition* $\longrightarrow$ *Coarse-Lyapunov subordinate partitions.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. Partitions that remember an unstable leaf

Let a act on $X = \Gamma\backslash G$ and let $U < G$ be a connected subgroup expanded by a. A measurable partition ξ is subordinate to U if for a.e. x, its atom $[x]_\xi$ is contained in xU and contains a neighborhood of x inside that leaf.

Lemma 1.1 (Construction from a small finite partition)**.** *Assume μ is a-invariant and gives zero mass to the boundaries of a finite partition $\mathcal{P}$ whose atoms have sufficiently small diameter on a large compact set. Then*

$$\xi = \bigvee_{n=0}^{\infty} a^n \mathcal{P}$$

can be refined on a null set so that, along the U-unstable direction, ξ is subordinate to U.

Proof. If $y = xu$ with u small in U, then $a^{-n}ua^n \to e$ exponentially. Hence the backwards iterates $a^{-n}x$ and $a^{-n}y$ approach one another. For points whose backwards orbits do not hit the partition boundary too closely, they eventually have the same $\mathcal{P}$-name. By Borel–Cantelli, after choosing boundary neighborhoods with summable measures, this holds for a.e. x and all sufficiently small u, giving a leafwise neighborhood inside the atom. Conversely, if y differs from x by a non-U coordinate in the adapted chart of Volume 11, some backward iterate separates that coordinate to the scale of $\mathcal{P}$, so the two names eventually differ. Thus the atom is contained in the U leaf. $\square$

Theorem 1.2 (Increasing subordinate partition). *There exists a countably generated partition ξ subordinate to U such that*

$$a^{-1}\xi \leq \xi$$

(the partition becomes finer under a^{-1}), and $\bigvee_{n\in\mathbb{Z}} a^n\xi$ separates points modulo the neutral-stable directions. Any two such partitions yield the same entropy contribution $H(\xi \mid a\xi)$.

Proof. Starting from Lemma 1.1, replace ξ by the join of its forward translates if necessary; the defining future-name form then gives $a^{-1}\xi \leq \xi$. The adapted product chart of Volume 11 shows that two points with all two-sided names equal can differ only in directions not separated by the U expansion. For independence of the choice, let ξ, η be two increasing subordinate partitions. On a large compact set each contains a fixed-size local plaque and is contained in a slightly larger plaque. Exponential expansion implies that for some bounded integers r, s, $a^{-r}\xi$ refines η and $a^{-s}\eta$ refines ξ modulo a set of arbitrarily small measure. Apply the chain rule to compare $H(\xi \mid a\xi)$ and $H(\eta \mid a\eta)$; the bounded shifts contribute telescoping boundary terms whose time average vanishes. Let the exceptional measure tend to zero. □

Proposition 1.3 (Coarse-Lyapunov subordinate partitions). *For every coarse root group $U^{[\alpha]}$ and every $a \in A$ with $\alpha(a) > 0$, one may choose $\xi^{[\alpha]}$ subordinate to $U^{[\alpha]}$ and increasing under a^{-1}. The construction is equivariant under elements of A commuting with a.*

Proof. Theorem 1.2 supplies the invariant expanding foliation and uniform local plaques on compact sets. Apply Theorem 1.2. If $b \in A$, then b normalizes the same coarse root group; transporting the partition by b gives another subordinate increasing partition, and the uniqueness of the entropy contribution established above makes the construction equivariant at the level needed later. □

Worked Example 1.4 (A subordinate partition for the doubling map). For $T(x) = 2x \pmod 1$, the unstable direction is the whole circle. Let ξ be the dyadic partition into $[0, 1/2)$ and $[1/2, 1)$. The atom of $\bigvee_{j=0}^{n-1} T^{-j}\xi$ containing a non-dyadic x is the dyadic interval of length 2^{-n} determined by the first n binary digits of x. Thus future names give successively smaller neighborhoods along the expanding direction. In a homogeneous flow the subordinate partition plays exactly this role on an unstable leaf, except the neighborhoods are obtained from conjugated group balls rather than dyadic intervals.

CHAPTER 6

Entropy along Expanding Foliations

Proof architecture. The chapter is organized as the chain *Leafwise entropy as exponential information growth* $\longrightarrow$ *Entropy contribution is bounded by geometric expansion* $\longrightarrow$ *Monotonicity for nested expanding groups.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. A local contribution to global entropy

For an increasing partition ξ subordinate to an a-expanded subgroup U, define

$$h_\mu(a, U) = H_\mu(\xi \mid a\xi).$$

The previous chapter shows this does not depend on the chosen subordinate partition.

Lemma 1.1 (Leafwise entropy as exponential information growth)**.** *For a subordinate increasing ξ,*

$$H(\xi \mid a^n\xi) = \sum_{j=0}^{n-1} H(a^j\xi \mid a^{j+1}\xi) = n\, h_\mu(a, U).$$

Thus $h_\mu(a, U)$ measures the mean information created when an a^n-expanded plaque is resolved back to unit leaf scale.

Proof. Because $a^{-1}\xi \leq \xi$, the partitions form a nested chain. Apply the conditional entropy chain rule successively. Invariance of μ makes every summand equal to $H(\xi \mid a\xi)$. □

Theorem 1.2 (Entropy contribution is bounded by geometric expansion). *Let U be connected unipotent and expanded by a, and let*

$$J_U(a) = \left|\det(\mathrm{Ad}(a)|_{\mathfrak{u}})\right|.$$

Then

$$0 \leq h_\mu(a, U) \leq \log J_U(a).$$

If μ is U-invariant, equality holds. If the leafwise conditional measures are atomic at the base point, the contribution is zero.

Proof. The nth refinement $a^n\xi$ corresponds leafwise to a plaque enlarged by a^n and then compared with unit plaques. Haar volume on U grows by exactly $J_U(a)^n$. Cover the enlarged plaque by $O(J_U(a)^n e^{o(n)})$ unit plaques using exponential coordinates and a fixed compact exhaustion. The entropy of any probability distribution on at most N atoms is at most $\log N$, so Lemma 1.1 gives the upper bound after division by n. If μ is U-invariant, conditional distributions on these leaf boxes are normalized Haar distributions; asymptotically each of the $J_U(a)^n$ comparable boxes has mass $J_U(a)^{-n}$, giving equality. If the conditional measure is a point mass, every leaf refinement carries zero information, so the contribution vanishes. □

Proposition 1.3 (Monotonicity for nested expanding groups)**.** *If $U_1 < U_2$ are both expanded by a, then*

$$h_\mu(a, U_1) \le h_\mu(a, U_2).$$

If local product disintegration of $U_2 = U_1 V$ is conditionally independent, then

$$h_\mu(a, U_2) = h_\mu(a, U_1) + h_\mu(a, V).$$

Proof. A partition subordinate to U_2 can be refined by additionally resolving the U_1 coordinate. Forgetting information cannot increase conditional entropy, giving monotonicity. Under conditional independence the conditional probability of a product box factors, so $-\log$ of its probability is the sum of the two information functions; integration gives additivity. □

Worked Example 1.4 (Geometric entropy bound on a one-dimensional unstable leaf)**.** Suppose a expands a one-dimensional leaf coordinate by e^λ. A plaque of length one contains about $e^{n\lambda}$ disjoint subplaques of length $e^{-n\lambda}$. No probability distribution on N atoms has Shannon entropy larger than $\log N$, so the n-step leaf information is at most $n\lambda + O(1)$. Dividing by n gives

$$h_\mu(a, U) \le \lambda = \log J_U(a).$$

For Haar leafwise measure the small intervals all have equal mass $e^{-n\lambda}$, so equality holds. This elementary counting picture is the model for every Jacobian upper bound in the chapter.

CHAPTER 7

Abramov–Rokhlin and Addition Formulae

Proof architecture. The chapter is organized as the chain *Relative generator lemma* $\longrightarrow$ *Abramov–Rokhlin addition formula* $\longrightarrow$ *Homogeneous quotient addition, with an explicit generator hypothesis.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. Separating base randomness from fiber randomness

The relative entropy theorem of Chapter 4 already contains the discrete-time Abramov–Rokhlin formula. We now package it in a form usable later whenever a homogeneous extension has verified finite-entropy generators.

Lemma 1.1 (Relative generator lemma)**.** *Let $\pi : X \to Y$ be a factor. If ξ separates almost every pair of points lying in the same fiber after taking all T-iterates, and η generates the base, then $\xi \vee \pi^{-1}\eta$ is a generator of X.*

Proof. If two points have the same full names for the joint partition, their images have the same full η-name and hence agree in Y. Thus they lie in one fiber. Their full ξ-names then agree, so the relative generating hypothesis makes the points equal modulo a null set. □

Theorem 1.2 (Abramov–Rokhlin addition formula). *For an invertible probability-preserving extension $\pi : (X, T, \mu) \to (Y, S, \nu)$ admitting finite-entropy base and relative generators,*

$$h_\mu(T) = h_\nu(S) + h_\mu(T \mid \mathcal{Y}).$$

For a skew product $T(y, z) = (Sy, T_y z)$ this separates the information generated by the base itinerary from the conditional information generated in the fibers.

Proof. Choose a finite-entropy base generator η and a finite-entropy relative generator ξ. Lemma 1.1 makes their join a generator upstairs. Theorem 1.2 then gives the equality. In a skew product, conditioning on $\mathcal{Y}$

fixes the complete base coordinate. The conditional name probabilities of ξ are therefore precisely the fiber conditional probabilities, so the relative term is the asymptotic fiber information rate. No additional independence assumption is used. $\square$

Proposition 1.3 (Homogeneous quotient addition, with an explicit generator hypothesis)**.** *Suppose $L < G$ is closed, a normalizes L, and a measurable map from the finite-measure homogeneous component under consideration to an algebraic quotient is an a-equivariant factor. If that factor admits a finite-entropy base generator and the extension admits a finite-entropy relative generator, then the entropy of right translation by a is the sum of the factor entropy and the relative entropy along the L-fibers.*

PROOF. Equivariance is exactly what is required to regard the quotient map as a factor of the a-actions. Under the two stated generator hypotheses, Theorem 1.2 applies verbatim. Notice what is *not* claimed: local homogeneous product charts alone do not automatically prove existence of a global finite relative generator on a noncompact quotient. Whenever a later volume uses this proposition, that generator hypothesis must be verified separately or replaced by a finite-entropy approximation argument. $\square$

Worked Example 1.4 (A zero-entropy fiber)**.** Let S be a Bernoulli shift and let $T(y, z) = (Sy, z + \theta(y_0) \pmod 1)$ be a compact-circle skew product. Conditioned on the full base itinerary, the fiber motion is a deterministic sequence of rotations. Rotations preserve distances and create no exponential growth of finite names, so the relative fiber entropy is zero. Abramov–Rokhlin therefore gives $h(T) = h(S)$. This example is useful later: a nontrivial geometric fiber can carry no entropy, so one must not infer "positive-dimensional fiber" from positive entropy without a separate expanding mechanism.

CHAPTER 8

Entropy Comparison and Monotonicity Principles

Proof architecture. The chapter is organized as the chain *Three monotonicities* $\longrightarrow$ *Entropy comparison package for diagonal actions.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. Checks for entropy arguments

Many rigidity mistakes come from comparing entropies attached to different factors or different foliations without recording what information has been forgotten. We collect the safe inequalities.

Lemma 1.1 (Three monotonicities). *For a probability-preserving T:*

(1) factor entropy never exceeds total entropy;
(2) conditioning on a larger σ-algebra never increases entropy;
(3) entropy contribution along an a-expanded subgroup is monotone under subgroup inclusion whenever the subordinate partitions are chosen compatibly.

Proof. These are respectively Theorem 1.2, Theorem 1.2(2), and Proposition 1.3. The compatibility clause in (3) is essential: it is what turns subgroup inclusion into refinement of the corresponding information σ-algebras. □

Theorem 1.2 (Entropy comparison package for diagonal actions). *Let A act on a finite-volume homogeneous space and let $a \in A$. For every coarse-Lyapunov group U_j expanded by a,*

$$0 \leq h_\mu(a, U_j) \leq \log J_{U_j}(a).$$

Suppose, in addition, that

$$\{e\} = V_0 < V_1 < \cdots < V_r = G_a^+$$

is a bracket-compatible normal filtration of the unstable group and that subordinate partitions ξ_j have been chosen so that ξ_j refines ξ_{j-1} and records *exactly the additional V_j/V_{j-1} leaf information. Then the successive conditional contributions are nonnegative and their sum is the entropy carried by the full unstable partition, hence is at most $h_\mu(a)$. For Haar measure all rootwise geometric upper bounds are attained.*

PROOF. The individual bounds are Theorem 1.2. By the stated compatibility, the conditional chain rule gives for every finite name length

$$H(\xi_r^{(n)}) = H(\xi_0^{(n)}) + \sum_{j=1}^{r} H(\xi_j^{(n)} \mid \xi_{j-1}^{(n)}).$$

After division by n and passage to the limit, each summand is nonnegative and the sum is the information rate of the full unstable partition. That partition is a factor of any full finite generator after adding neutral/stable information, so its rate cannot exceed total entropy. For Haar measure the leafwise conditionals on every V_j/V_{j-1} are Haar; conjugation multiplies their volume by the corresponding Jacobian, and each geometric upper bound is therefore an equality. □

Why this matters. Volume 13 identifies the conditional measures responsible for these contributions. Volume 14 then turns positive contributions in noncommuting directions into actual unipotent invariance.

Worked Example 1.3 (A two-step information filtration)**.** Let a system have two independent expanding symbolic coordinates with entropies h_1 and h_2. Let V_1 remember the first coordinate and V_2 remember both. Then the first conditional increment is h_1, while the second is $h(V_2) - h(V_1) = h_2$. Their sum is $h_1 + h_2$. If the second coordinate is instead a deterministic function of the first, the second increment is zero. The filtration theorem is the homogeneous analogue: a new coarse-Lyapunov direction contributes only the information not already present in the earlier generated subgroup.

Volume 13

Leafwise Measures

Volume 13 scope and prerequisites

The purpose of this volume is to replace informal phrases such as "the conditional measure on an unstable leaf" by a precise measurable system of Radon measures. The system is constructed locally from Rokhlin disintegration and then glued projectively. The decisive rules—leaf translation, conjugation by the diagonal action, recurrence, and the Haar/invariance equivalence—are proved before they are used. These are the exact interfaces needed by the entropy method of Volume 14. Our normalization and notation are compared with the construction used by Einsiedler–Katok–Lindenstrauss [EKL06].

Student orientation: what a leafwise measure is

Rokhlin disintegration gives probability measures on atoms of a measurable partition. A *leafwise measure* is different: after identifying a plaque with a neighborhood of the identity in a leaf group U, the conditional measure is transported to a Radon measure on U and is remembered only up to multiplication by a positive scalar. Thus μ_x^U naturally lives in a projective space of Radon measures. A *normalization* chooses one representative, for example by prescribing the mass of a fixed continuity ball. *Leafwise conservativity* means recurrence of this projective conditional system under translations along the leaf. The formulas $\mu_{ux}^U \propto (R_u)_*\mu_x^U$ and $\mu_{ax}^{aUa^{-1}} \propto a_*\mu_x^U$ are not notation: they are the two equivariance theorems on which every later entropy argument depends.

Reader guide: a concrete model

Why this matters.

Volume 13 is organized around the passage from *Disintegration of Measures* to *Product Structures and Compatibility of Leafwise Systems*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.4 (A calculation to keep in view)**.** Let

$$a = \operatorname{diag}(e^{2s}, e^{-s}, e^{-s}) \in \mathrm{SL}_3(\mathbb{R})$$

and let E_{12} be the matrix unit. Since

$$\operatorname{Ad}(a)E_{12} = aE_{12}a^{-1} = e^{3s}E_{12},$$

the root exponent on the E_{12} direction is $3s$. For a one-dimensional leafwise measure whose conditional scaling is Lebesgue, the Jacobian contribution

along this root direction is therefore $3s$. This calculation is the concrete local model behind the root-by-root entropy bookkeeping used later.

CHAPTER 1

Disintegration of Measures

Proof architecture. The chapter is organized as the chain *Regular conditional probabilities for a countably generated factor* $\longrightarrow$ *Rokhlin disintegration for a countably generated partition* $\longrightarrow$ *Refinement and iterated disintegration.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. Why conditional measures require a theorem

A foliation is uncountable, so one cannot obtain conditional probabilities by dividing by the mass of a single leaf: almost every individual leaf typically has ambient measure zero. The correct object is a measurable family of probabilities on the atoms of a countably generated measurable partition. We first construct regular conditional probabilities on a standard Borel space.

Lemma 1.1 (Regular conditional probabilities for a countably generated factor)**.** *Let $(X, \mathcal{B}, \mu)$ be a standard Borel probability space and let $\mathcal{A} \subset \mathcal{B}$ be countably generated modulo μ. There is a measurable family $x \mapsto \mu_x^{\mathcal{A}} \in$* Prob$(X)$ *such that, for every bounded Borel f,*

$$\int f \, d\mu_x^{\mathcal{A}} = \mathbb{E}_\mu(f \mid \mathcal{A})(x) \quad \textit{for a.e. } x,$$

and

$$\mu = \int \mu_x^{\mathcal{A}} \, d\mu(x).$$

For each $A \in \mathcal{A}$, $\mu_x^{\mathcal{A}}(A) = \mathbf{1}_A(x)$ for a.e. x. The family is unique up to a common null set when tested on a fixed countable determining class.

Proof. Because X is standard Borel, choose a Borel isomorphism $\iota : X \to E$ onto a Borel subset $E \subset [0,1]$ (one may obtain it explicitly by coding a countable separating family into the Cantor set). It is enough to construct a conditional probability on E.

For each rational $q \in \mathbb{Q} \cap [0,1]$, choose a version

$$F_q(x) = \mathbb{E}(\mathbf{1}_{\{\iota \le q\}} \mid \mathcal{A})(x).$$

After deleting one null set, all rational inequalities $F_q \le F_{q'}$ for $q < q'$, together with $0 \le F_q \le 1$, hold simultaneously. Define

$$F_x(t) = \inf_{\substack{q \in \mathbb{Q} \\ q > t}} F_q(x),$$

with the evident values below 0 and above 1. Then F_x is nondecreasing and right-continuous, with limits 0 and 1, so it is the distribution function of a unique Borel probability ν_x on $[0,1]$. For each half-line $(-\infty, q]$ its mass is $F_q(x)$ a.e. A monotone-class argument therefore gives, for every bounded Borel g on $[0,1]$,

$$\int g \, d\nu_x = \mathbb{E}(g \circ \iota \mid \mathcal{A})(x)$$

for a.e. x (with a null set allowed to depend on g). In particular

$$\int \nu_x(E) \, d\mu(x) = \mu(\iota^{-1} E) = 1,$$

so $\nu_x(E) = 1$ for a.e. x. Pull ν_x back by ι^{-1} to obtain $\mu_x^{\mathcal{A}}$ on X.

Measurability of $x \mapsto \mu_x^{\mathcal{A}}(B)$ first holds for rational half-lines by construction and then for every Borel set by the monotone-class theorem. Integrating the conditional-expectation identity gives the barycenter formula. If $A \in \mathcal{A}$, then $\mathbb{E}(\mathbf{1}_A \mid \mathcal{A}) = \mathbf{1}_A$, proving the factor-measurability assertion. Finally two such kernels agree on the countable class of rational half-lines after coding by ι, hence agree as probability measures outside one common null set. $\square$

THEOREM 1.2 (Rokhlin disintegration for a countably generated partition). *Let ξ be a countably generated measurable partition of a standard Borel probability space. Then there is a family $\{\mu_x^\xi\}$, unique a.e., such that*

$$\mu_x^\xi([x]_\xi) = 1, \qquad x \mapsto \mu_x^\xi(B) \textit{ is } \sigma(\xi)\textit{-measurable},$$

and

$$\mu(B) = \int_X \mu_x^\xi(B) \, d\mu(x)$$

for every Borel B. Moreover $\mu_x^\xi = \mu_y^\xi$ for μ-a.e. x and μ_x^ξ-a.e. $y \in [x]_\xi$.

PROOF. Apply Lemma 1.1 to $\mathcal{A} = \sigma(\xi)$. Let $A_1, A_2, \ldots$ be a countable family of saturated sets generating $\mathcal{A}$. Outside one null set, for every j the conditional measure gives mass one to A_j if $x \in A_j$ and to its complement otherwise. Their countable intersection is exactly the measurable atom $[x]_\xi$ modulo the usual completion, hence $\mu_x^\xi([x]_\xi) = 1$.

For every bounded Borel f, the function $x \mapsto \int f \, d\mu_x^\xi$ is $\sigma(\xi)$-measurable and therefore essentially constant on almost every atom. Taking f from a countable determining class gives $\mu_y^\xi = \mu_x^\xi$ for μ_x^ξ-a.e. y in the atom. The barycenter and uniqueness statements are those of the lemma. $\square$

Proposition 1.3 (Refinement and iterated disintegration)**.** *If η refines ξ, then for a.e. x the conditional probability μ_x^ξ disintegrates over the η-atoms inside $[x]_\xi$, and the resulting inner conditional probabilities agree with μ_y^η. Equivalently,*

$$\mathbb{E}(\mathbb{E}(f \mid \sigma(\eta)) \mid \sigma(\xi)) = \mathbb{E}(f \mid \sigma(\xi)).$$

Proof. The tower identity follows by testing both sides against bounded $\sigma(\xi)$-measurable functions. Disintegrate μ over ξ and then disintegrate μ_x^ξ over the induced η partition. The iterated kernel and the global η-kernel represent the same conditional expectations on a countable determining class, so uniqueness in Lemma 1.1 identifies them. $\square$

Worked Example 1.4 (Vertical fibers)**.** For Lebesgue measure on $[0,1]^2$ and the partition into vertical fibers, $\mu^\xi_{(s,t)}$ is one-dimensional Lebesgue measure on $\{s\} \times [0,1]$. The theorem is doing something genuinely stronger than Fubini: it produces the same object for arbitrary countably generated measurable partitions, even when no global product coordinates exist.

CHAPTER 2

Conditional Measures along Foliations

Proof architecture. The chapter is organized as the chain *Countable flow-box atlas* $\longrightarrow$ *Leafwise Radon measure system* $\longrightarrow$ *Local reconstruction formula*. The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. From plaques to group coordinates

Let a connected Lie group U act locally freely on $X = G/\Gamma$. A small U-plaque through x is $B_U(r)x$. We want a measure on the model group U, not merely a probability on whichever plaque was chosen. The construction below uses a nested family of flow boxes and the compatibility of Rokhlin conditionals.

Lemma 1.1 (Countable flow-box atlas)**.** *For every locally free smooth U-action on a second-countable manifold there are countably many Borel sets $Y_m \subset X$, relatively compact identity neighborhoods $D_m \Subset U$, and transversals T_m such that*

$$D_m \times T_m \longrightarrow Y_m, \qquad (u,t) \mapsto ut,$$

is injective, bi-Borel, and the Y_m cover a conull subset of every Radon probability measure on X.

Proof. Local freeness and the inverse function theorem give a product box around every point. Second countability extracts a countable subcover. Shrink each product box slightly so that its closure lies in the original chart; the shrunk boxes still cover X after taking a countable nested exhaustion. Their coordinate maps and inverses are continuous, hence Borel. No measure-specific choice is needed, so the union has full measure for every Radon probability. $\square$

Theorem 1.2 (Leafwise Radon measure system). *Let μ be a Radon probability measure on X and $U < G$ a connected subgroup acting locally* *freely on a conull set. There is a measurable projective family*

$$x \longmapsto [\mu_x^U] \in \mathbb{P}\mathcal{M}_\infty(U)$$

of nonzero Radon measures on U, defined for a.e. x, such that in every sufficiently small flow box the Rokhlin conditional probability on a U-plaque is the pushforward of a restriction of μ_x^U under $u \mapsto ux$, after normalization to total mass one.

PROOF. Take the countable flow-box atlas from Lemma 1.1. In $Y_m \simeq D_m \times T_m$, disintegrate $\mu|_{Y_m}$ over the transversal coordinate. Pull each plaque conditional back to $D_m \subset U$. If two boxes overlap along the same leaf, Proposition 1.3 says that the two conditional probabilities coincide after both are restricted to any smaller common plaque and renormalized. Hence their pulled-back measures are proportional on overlaps.

Choose a countable chain of boxes meeting a typical leaf and glue their pulled-back measures by these positive proportionality constants. Compatibility on triple overlaps follows from uniqueness of conditional probability, so the result is a well-defined Radon measure on the union of the corresponding U-domains; exhausting U gives a nonzero Radon measure on all of U. Changing the initial box multiplies the entire glued measure by a positive scalar. Measurability follows because all conditional probabilities and all normalizing masses may be chosen measurably on the countable atlas. This produces the projective class $[\mu_x^U]$. □

Proposition 1.3 (Local reconstruction formula)**.** *In a flow box $D \times T \to Y$, write $\bar{\mu}$ for the transverse quotient measure. After choosing the normalization $\mu_t^U(D) = 1$ on the box,*

$$\int_Y f(x)\, d\mu(x) = \int_T \int_D f(ut)\, d\mu_t^U(u)\, d\bar{\mu}(t)$$

after multiplying $\bar{\mu}$ by the box mass. In particular the projective leafwise system determines the ambient measure together with one transverse normalization.

PROOF. This is precisely the Rokhlin barycenter formula in the coordinates of the flow box. The only point is normalization: the pulled-back plaque conditionals have mass one on D, so the remaining scalar is absorbed into the quotient measure on T. □

Worked Example 1.4 (A horocycle leaf)**.** For Haar probability on $SL_2(\mathbb{R})/SL_2(\mathbb{Z})$ and $U = \{\left(\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right)\}$, the leafwise measure is Lebesgue measure dt up to scale. In a small injectivity box the statement reduces to ordinary product integration; the theorem explains why the same measure class persists when the leaf exits that box and later returns.

CHAPTER 3

Projective Normalization of Leafwise Measures

Proof architecture. The chapter is organized as the chain *A common continuity ball for normalization* $\longrightarrow$ *Projective uniqueness* $\longrightarrow$ *Changing normalization changes a cocycle by a coboundary.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. Why only a measure class is canonical

A local plaque conditional is a probability, but enlarging the plaque changes the normalizing denominator. Therefore the globally natural object is a Radon measure only up to multiplication by a positive constant. Fixing a normalization is useful, but formulas must first be proved projectively.

Lemma 1.1 (A common continuity ball for normalization)**.** *For the measurable leafwise system of Theorem 1.2, there is a radius $r_0 > 0$ such that for a.e. x,*

$$0 < \mu_x^U(B_U(r_0)) < \infty, \qquad \mu_x^U(\partial B_U(r_0)) = 0.$$

After rescaling representatives measurably, one may therefore impose $\mu_x^U(B_U(r_0)) = 1$ on one common conull set.

Proof. For each fixed x, local finiteness and positivity on identity neighborhoods give $0 < \mu_x^U(B_U(r)) < \infty$ for all sufficiently small $r > 0$. Moreover a Radon measure has positive mass on at most countably many pairwise disjoint spheres $\partial B_U(r)$ in a compact radius interval. Hence for each x the set of bad radii has Lebesgue measure zero. The map $(x, r) \mapsto \mu_x^U(\partial B_U(r))$ is measurable after choosing representatives on a flow-box atlas, so Fubini gives a radius r_0 that is a continuity radius for a.e. x. Choose it in a compact interval on which the ball masses are finite and positive a.e. Dividing by the measurable positive function $x \mapsto \mu_x^U(B_U(r_0))$ gives the declared normalization. $\square$

Theorem 1.2 (Projective uniqueness). *Suppose $x \mapsto \nu_x$ and $x \mapsto \nu'_x$ are two measurable systems of nonzero Radon measures on U whose normalized* *restrictions give the same Rokhlin plaque conditionals in every box of a countable flow-box atlas. Then*

$$\nu'_x = c(x)\nu_x$$

for a positive measurable scalar $c(x)$ a.e. Thus the projective class $[\mu_x^U]$ is intrinsic to (μ, U).

Proof. On each flow box, equality of normalized plaque conditionals implies proportionality of ν_x and ν'_x on a fixed identity neighborhood. On an overlap, uniqueness of conditional measures forces the same proportionality constant. A nested exhaustion of U by domains reached through the countable atlas propagates the equality to every compact subset, hence to the Radon measures themselves. With the normalization of Lemma 1.1, $c(x)$ is the ratio of two measurable ball masses and is therefore measurable. $\square$

Proposition 1.3 (Changing normalization changes a cocycle by a coboundary)**.** *Assume a transformation T normalizes the U-foliation and a chosen representative system satisfies*

$$T_*\mu_x^U = c_T(x)\mu_{Tx}^U.$$

If $\widetilde{\mu}_x^U = b(x)\mu_x^U$, then

$$\widetilde{c}_T(x) = \frac{b(x)}{b(Tx)}c_T(x).$$

Consequently all statements depending only on the projective class are normalization-independent.

Proof. Apply T_* to $b(x)\mu_x^U$ and rewrite the result in terms of $\widetilde{\mu}_{Tx}^U = b(Tx)\mu_{Tx}^U$. The displayed quotient is immediate. $\square$

Guiding Idea 1.4. Projectivization is not cosmetic. It prevents us from falsely equating conditionals obtained from two different plaque sizes. Every later exact equality will either use a declared normalization or be phrased in projective form.

Worked Example 1.5 (The same Haar class under two plaque normalizations)**.** On $U = \mathbb{R}$, let m be Lebesgue measure. Normalizing on $[-1, 1]$ gives the probability $(1/2)m|_{[-1,1]}$, while normalizing on $[-2, 2]$ gives $(1/4)m|_{[-2,2]}$. On the overlap, the two probabilities differ by a scalar after restriction, although both arise from the same global Radon measure class $[m]$. If a translation moves the base point, the plaque probabilities change because the chosen windows change, but the projective class remains translation invariant. This is exactly why the intrinsic leafwise object is $[\mu_x^U]$, not a particular plaque probability.

CHAPTER 4

Equivariance of Leafwise Measures

Proof architecture. The chapter is organized as the chain *Leaf-translation rule* $\longrightarrow$ *Normalizer equivariance* $\longrightarrow$ *Iterated conjugation and expanding balls.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. Two transformations of the leaf coordinate

There are two basic operations. Moving the base point by $u \in U$ translates the leaf coordinate; applying an element a normalizing U conjugates the coordinate. These rules are forced by uniqueness of disintegration.

Lemma 1.1 (Leaf-translation rule)**.** *For a.e. x and every $u \in U$ for which both x and ux lie in the common conull domain,*

$$\mu_{ux}^{U} \ \propto\ (R_u)_*\mu_x^U, \qquad R_u(v) = vu^{-1}.$$

Equivalently, with the opposite coordinate convention one obtains the right-translation formula used in the EKL normalization.

Proof. Choose a flow box containing x and ux after shrinking the plaque if necessary. A point written as vx is written relative to ux as $(vu^{-1})(ux)$. Thus the coordinate change on U is R_u. Push the plaque conditional through this coordinate change. It is another conditional probability for the same measurable partition, so Rokhlin uniqueness identifies it with the conditional at ux after normalization. Removing the normalization gives proportionality. $\square$

Theorem 1.2 (Normalizer equivariance). *Let $a \in G$ normalize U and suppose μ is a-invariant. Then for a.e. x,*

$$\mu_{ax}^{U} \ \propto\ (c_a)_*\mu_x^U, \qquad c_a(u) = aua^{-1}.$$

If the chosen normalization is invariant under c_a—for example when a centralizes U and both representatives give mass one to the same identity ball—the proportionality is equality.

Proof. The map $x \mapsto ax$ sends a U-plaque to a U-plaque because $aUa^{-1} = U$. In leaf coordinates it sends u to aua^{-1}. Since $a_*\mu = \mu$, the pushforward of the conditional family by this map is again a disintegration of μ along the same plaques. Theorem 1.2 gives proportionality. Under a normalization preserved by c_a, both sides have the same mass on the normalizing ball, so the scalar is one. $\square$

Proposition 1.3 (Iterated conjugation and expanding balls). *If a expands U and $D \Subset U$ is an identity neighborhood, then*

$$\mu^U_{a^n x} \propto (c_{a^n})_* \mu^U_x, \qquad \mu^U_x(a^{-n} D a^n) \asymp_x \mu^U_{a^n x}(D)$$

after accounting for the declared normalization cocycle. Ratios of masses of two such sets are independent of that cocycle.

Proof. Iterate Theorem 1.2. The scalar accumulated in n steps multiplies the mass of every measurable set by the same factor. It therefore cancels from ratios. This simple observation is the bridge from geometric expansion to entropy: logarithms of such ratios measure information growth along the leaf. $\square$

Worked Example 1.4 (A root group in SL_3). For $U_{12} = \{I + tE_{12}\}$ and $a = \mathrm{diag}(e^{t_1}, e^{t_2}, e^{t_3})$,

$$a(I + sE_{12})a^{-1} = I + e^{t_1 - t_2} s E_{12}.$$

Thus the normalizer rule becomes ordinary rescaling of a Radon measure on $\mathbb{R}$. Volume 14 turns the rate of mass growth under this rescaling into the coefficient $s_{12}(\mu)$.

CHAPTER 5

Recurrence and Leafwise Conservativity

Proof architecture. The chapter is organized as the chain *Metric Poincaré recurrence* $\longrightarrow$ *Expanding recurrence forces an atomic leafwise measure to be trivial* $\longrightarrow$ *Leafwise conservativity test*. The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. Why atoms cannot sit away from the identity

When a expands U, conjugating backwards contracts leaf displacements to the identity. Recurrence then prevents a discrete leafwise configuration with more than one atom from returning with the same normalized shape.

Lemma 1.1 (Metric Poincaré recurrence)**.** *If T is an invertible probability-preserving transformation of a second-countable metric probability space, then for a.e. x there is a sequence $n_j \to \infty$ such that $T^{n_j}x \to x$. The same holds after restricting to a prescribed positive-measure Lusin set and choosing x to be a density point of that set.*

Proof. Choose a countable basis $\{B_m\}$. Poincaré recurrence applied to every B_m says that almost every point of B_m returns to B_m infinitely often. Intersecting the resulting conull sets over m shows that a.e. x returns infinitely often to every basis neighborhood containing it; choosing a nested basis gives $T^{n_j}x \to x$. Intersecting first with a positive-measure Lusin set and using its density points gives the last statement. □

Theorem 1.2 (Expanding recurrence forces an atomic leafwise measure to be trivial). *Let μ be invariant under a, let a uniformly expand the connected unipotent group U, and let μ_x^U be the normalized leafwise system. If μ_x^U is atomic on a positive-measure set, then it is projectively δ_e there. If μ is a-ergodic, either $\mu_x^U = \delta_e$ projectively a.e. or μ_x^U is non-atomic a.e.*

Proof. Normalize on the common continuity ball from Lemma 1.1. Suppose, toward a contradiction, that on a positive-measure set the normalized measure has at least two atoms. For some $\eta > 0$ and $0 < r < R$, there is then a positive-measure set E on which two atoms of mass at least η are separated by a displacement in the compact annulus $K = \{u : r \leq d(u, e) \leq R\}$. Shrink E so that the normalized leafwise system is weak-* continuous there (Lusin) and so that no two atoms of mass at least $\eta/2$ lie at distance $< r/3$; this last property is possible because a finite measure has only finitely many atoms of mass $\geq \eta/2$ in the normalizing ball.

Use the inverse transformation $T = a^{-1}$, which contracts U. Choose $x \in E$ recurrent to E with $T^{n_j} x \to x$. Normalizer equivariance says that the two η-atoms at x are carried, up to the scalar fixed by our normalization, to two atoms of the normalized measure at $T^{n_j} x$ whose separation is

$$d\big(e, a^{-n_j} u a^{n_j}\big) \longrightarrow 0.$$

Because $T^{n_j} x \to x$ inside the Lusin set, the normalization scalars stay bounded above and below and the atom masses remain at least $\eta/2$ for all large j. Their separation is eventually $< r/3$, contradicting the defining property of E. Thus an atomic leafwise measure has only one atom. Since Theorem 1.2 makes every identity neighborhood have positive mass, that single atom must be e, so the measure is projectively δ_e.

The event of triviality is a-invariant by normalizer equivariance. Ergodicity makes it null or conull, yielding the dichotomy. ☐

Proposition 1.3 (Leafwise conservativity test)**.** *Assume μ_x^U is non-atomic a.e. and a expands U. For a.e. x and every continuity identity neighborhood $D \Subset U$, the ratios*

$$\frac{\mu_x^U(a^{-n} D a^n)}{\mu_x^U(D)}$$

are normalization-independent and tend to zero. When D is the plaque of a subordinate partition, the negative logarithms of these ratios are the leafwise conditional-information increments.

Proof. Normalization-independence is Proposition 1.3. The sets $a^{-n} D a^n$ decrease to $\{e\}$ after replacing D by a symmetric adapted ball, and non-atomicity gives $\mu_x^U(\{e\}) = 0$. Continuity from above therefore makes their masses tend to zero. The last assertion is precisely the definition of conditional information for the nested subordinate plaques. ☐

Worked Example 1.4 (Why a two-atom pattern cannot recur under expansion)**.** Imagine, for contradiction, that a normalized leafwise measure on $\mathbb{R}$ has two atoms of masses at least 0.1 separated by distance 1. If the diagonal action expands the leaf by 2, then the inverse action contracts their separation to 2^{-n}. At a recurrent time the base point returns to a region where the normalized conditional measure is close to the original one. The returned measure would therefore have two atoms of comparable mass separated by 2^{-n}. For large n this contradicts any neighborhood of the original two-atom configuration in which heavy atoms remain uniformly separated. The proof of Theorem 1.2 formalizes this picture using a Lusin set.

CHAPTER 6

Atomic versus Non-Atomic Alternatives

Proof architecture. The chapter is organized as the chain *Ergodicity makes leafwise type constant* $\longrightarrow$ *Rootwise zero-or-diffuse dichotomy* $\longrightarrow$ *Entropy safely detects nontrivial unstable conditionals.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. The first rigidity dichotomy

For an A-ergodic measure and a root subgroup U, the previous chapter turns "atomic" into the much stronger statement "trivial." Thus every root direction is either invisible to the measure or supports genuinely diffuse conditional mass.

Lemma 1.1 (Ergodicity makes leafwise type constant)**.** *Let A normalize U and let μ be A-ergodic. Each of the properties*

$$\mu_x^U \text{ trivial}, \qquad \mu_x^U \text{ non-atomic}, \qquad \mu_x^U \text{ Haar}$$

is A-invariant modulo null sets and therefore occurs either a.e. or a.e. not at all.

Proof. Conjugation by an element of A is a homeomorphism of U fixing the identity and preserves the three projective measure types. The normalizer-equivariance theorem makes each corresponding subset of X A-invariant modulo null sets. Apply A-ergodicity. $\square$

Theorem 1.2 (Rootwise zero-or-diffuse dichotomy). *Let U be a connected unipotent subgroup normalized by an A-action and assume some $a \in A$ uniformly expands U. For every A-ergodic probability μ, exactly one of the following holds:*

(1) $\mu_x^U = \delta_e$ projectively for a.e. x;
(2) μ_x^U is non-atomic for a.e. x.

If the second alternative is Haar, then μ is U-invariant; the converse is proved in the next chapter.

PROOF. By Theorem 1.2, any atomic leafwise measure is trivial on a conull subset of the atomic event. Lemma 1.1 makes that event zero-one. Thus either triviality holds a.e. or the leafwise measures are non-atomic a.e. □

Proposition 1.3 (Entropy safely detects nontrivial unstable conditionals)**.** *Let $U = G_a^+$ be the full expanding horospherical group of a regular element a, and let ξ be a subordinate partition for which the unstable entropy identity of Volume 12 holds. If the full unstable leafwise measure is trivial, then its entropy contribution is zero. Consequently*

$$h_\mu(a) > 0 \quad \Longrightarrow \quad \mu_x^U \text{ is nontrivial, hence non-atomic, a.e.}$$

whenever the full entropy is carried by the unstable partition. No converse "non-atomic implies positive entropy" is asserted here.

PROOF. If $\mu_x^U = \delta_e$, every sufficiently fine plaque refinement has one conditional atom of mass one, so its conditional information increment is zero. Therefore the unstable entropy contribution vanishes. Under the stated identity between total entropy and unstable entropy, positive total entropy rules out the trivial alternative. Theorem 1.2 then upgrades nontriviality to non-atomicity. The final sentence records an audit correction: an arbitrary diffuse measure need not have positive pointwise dimension, so the converse requires the special rootwise entropy theory developed in Volume 14 rather than a general measure-theoretic slogan. □

Worked Example 1.4 (The two allowed rootwise types)**.** For the horocycle root group $U \cong \mathbb{R}$, two model leafwise measures are δ_0 and Lebesgue measure dt. The first says that, conditionally on a transverse coordinate, the ambient measure sees only the base point in the U direction; it contributes no unstable information. The second is diffuse and in fact Haar, so the ambient measure is U-invariant. The zero-or-diffuse theorem says that under a higher-rank ergodic normalizing action there is no intermediate atomic pattern such as counting measure on a discrete subgroup: recurrence destroys it.

CHAPTER 7

Haar Leafwise Measure and Invariance

Proof architecture. The chapter is organized as the chain *Local translation invariance of conditionals* $\longrightarrow$ *Haar leafwise measure if and only if ambient invariance* $\longrightarrow$ *Closed subgroup of leafwise translation symmetries.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. The invariance criterion

The point of leafwise measures is that a global symmetry can be recognized from a local conditional object. Haar measure is the unique Radon measure class on a connected group invariant under all translations, and this translates directly into ambient U-invariance.

Lemma 1.1 (Local translation invariance of conditionals)**.** *If μ_x^U is Haar projectively for a.e. x, then in every flow box the normalized plaque conditional is normalized Haar measure on the plaque. Consequently for every sufficiently small $u \in U$ and every bounded Borel f supported in a smaller sub-box,*

$$\int f(ux)\, d\mu(x) = \int f(x)\, d\mu(x).$$

Proof. Projective Haar measure restricts to Haar measure on every plaque, and normalization merely divides by its plaque volume. Use the local reconstruction formula of Proposition 1.3. Left translation by a sufficiently small u changes only the leaf coordinate and preserves Haar measure there; the transverse coordinate and transverse quotient measure are unchanged. Integrating first along plaques proves the identity. □

Theorem 1.2 (Haar leafwise measure if and only if ambient invariance). *Let $U < G$ be connected and let the U-action be locally free on a conull set. Then*

$$\mu \text{ is } U\text{-invariant} \iff \mu_x^U \text{ is Haar on } U \text{ projectively for a.e. } x.$$

PROOF. Assume first that μ is U-invariant. In a flow box, translating along the U coordinate preserves the ambient measure. By uniqueness of disintegration, almost every plaque conditional is invariant under every sufficiently small translation for which both plaques remain inside the box. Pulling back to U, the conditional Radon measure is locally translation invariant. Connectedness and a chain of small neighborhoods extend this to all translations, so it is a Haar measure class.

Conversely, assume the leafwise measures are Haar. Lemma 1.1 gives invariance under every sufficiently small $u \in U$ for functions supported in each member of a countable flow-box atlas. A partition of unity and monotone convergence remove the support restriction. A connected Lie group is generated by any identity neighborhood, so invariance under small elements implies invariance under all of U. □

Proposition 1.3 (Closed subgroup of leafwise translation symmetries)**.** *For a.e. x define*

$$I_x = \{u \in U : (R_u)_* \mu_x^U \propto \mu_x^U\}.$$

Then I_x is a closed subgroup. If μ is ergodic under a normalizing group A, the conjugacy class of I_x is essentially constant and satisfies $I_{ax} = aI_xa^{-1}$. If $I_x = U$ a.e., then μ is U-invariant.

PROOF. The identity belongs to I_x, and projective invariance is preserved by products and inverses, so I_x is a subgroup. Weak-* continuity of translation on Radon measures shows it is closed. Normalizer equivariance transports the defining relation by conjugation, giving $I_{ax} = aI_xa^{-1}$. The final statement says the leafwise measure class is translation invariant under all of U; hence it is Haar, and Theorem 1.2 applies. □

Worked Example 1.4 (Why proportionality is enough)**.** Lebesgue measure dt on $\mathbb{R}$ is fixed by translation, while the normalized restriction of dt to $[-1, 1]$ is not preserved by translating the interval. The projective formulation remembers the correct global symmetry and avoids mistaking a plaque normalization for the intrinsic measure.

CHAPTER 8

Product Structures and Compatibility of Leafwise Systems

Proof architecture. The chapter is organized as the chain *Iterated leafwise disintegration in a local product* $\longrightarrow$ *Compatibility dictionary for rootwise leafwise systems*. The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. Several coarse Lyapunov directions at once

Higher-rank rigidity uses many normalized unipotent directions simultaneously. When subgroups form a local product, their conditional systems must agree with iterated disintegration of the same ambient measure. In a nonabelian product, changing the coordinate order inserts commutators; Volume 14 turns that geometric fact into new invariance.

Lemma 1.1 (Iterated leafwise disintegration in a local product)**.** *Let $U, V < G$ be connected subgroups such that multiplication*

$$U_0 \times V_0 \longrightarrow UV, \qquad (u, v) \mapsto uv,$$

is a diffeomorphism onto an identity neighborhood and $U \cap V = \{e\}$. In a sufficiently small UV-flow box, disintegrating first along U and then along the induced V quotient agrees, up to projective normalization, with the corresponding two-step disintegration of the UV plaque measure.

Proof. Choose a product box in which $(u, v, t) \mapsto uvt$ is injective. The partitions into UV plaques and into U subplaques are countably generated after restriction to the box. Proposition 1.3 says that conditional expectation through this tower is independent of the particular representation of the conditional kernels. Pulling the kernels back through multiplication coordinates gives the claimed compatibility; changing plaque domains only changes the projective scalar. $\square$

Lemma 1.2 (Heisenberg coordinate-order comparison)**.** *For distinct i, j, k, put $X = U_{ij}$, $Y = U_{jk}$, $Z = U_{ik}$. In the three-dimensional subgroup $N = XYZ$ one has*

$$x(r)y(s) = y(s)x(r)z(rs).$$

On every sufficiently small N-flow box, the two iterated disintegrations obtained from the coordinate orders XYZ and YXZ represent the same N-plaque conditional. Hence their Z-fiber kernels are related by translation by $z(rs)$ for almost every pair of leaf parameters for which both iterated conditionals are defined.

Proof. The matrix identity follows from $E_{ij}E_{jk} = E_{ik}$ and the vanishing of the reverse product. The two maps $(r, s, q) \mapsto x(r)y(s)z(q)$ and $(s, r, q') \mapsto y(s)x(r)z(q')$ are local coordinate systems on the same N-box, related by $q' = q + rs$. Apply Lemma 1.1 twice, once to each coordinate tower. Both iterated kernels integrate to the same N-plaque conditional, so uniqueness of regular conditional probabilities identifies the final Z-fiber kernels after the coordinate change. The only possible difference is the projective normalization already recorded in Chapter 3; geometrically the change is precisely translation by $z(rs)$. □

Theorem 1.3 (Compatibility dictionary for rootwise leafwise systems). *For an A-invariant probability on $SL_k(\mathbb{R})/\Gamma$, the rootwise systems μ_x^{ij} may be chosen on one common conull set so that simultaneously:*

(1) $x \mapsto \mu_x^{ij}$ is measurable and positive on every identity neighborhood;
(2) moving the base point in U_{ij} translates the leaf measure projectively;
(3) $\alpha^{\mathbf{t}}$ conjugates μ_x^{ij} by $s \mapsto e^{t_i - t_j}s$ projectively;
(4) atomicity implies δ_e under recurrence;
(5) Haar leafwise measure is equivalent to U_{ij}-invariance;
(6) ordered local products are compatible with iterated Rokhlin disintegration, and changing the order records the exact commutator coordinate as in Lemma 1.2.

These properties are normalization-independent.

Proof. There are only finitely many root groups. Intersect the conull domains supplied by Chapters 2–7 for all of them and for a countable dense subgroup of A; continuity extends the conjugacy statements to all of A. Items (1)–(5) are Theorems 1.2, 1.2, 1.2, and 1.2. Item (6) is Lemmas 1.1 and 1.2. Proposition 1.3 shows that replacing representatives only inserts scalar cocycles. □

Warning 1.4. Compatibility does *not* mean probabilistic independence. The only assertion used later is that two coordinate orders describe the same conditional measure and therefore must be related by the exact polynomial commutator coordinate. This is enough for the recurrence/shearing argument and avoids assuming a product law that has not been proved.

Worked Example 1.5 (Two coordinate orders in the Heisenberg group)**.** In the SL_3 upper-unitriangular group write $x(r) = u_{12}(r)$, $y(s) = u_{23}(s)$, and $z(q) = u_{13}(q)$. Direct multiplication gives

$$x(r)y(s) = y(s)x(r)z(rs).$$

Thus a point with XYZ coordinates (r, s, q) has YXZ coordinates $(s, r, q+rs)$. If one disintegrates a plaque first in the x direction and then in the y direction, and separately in the opposite order, the final z-conditional must describe the same ambient conditional measure. The only discrepancy is the explicit translation $q \mapsto q + rs$. Volume 14 turns that deterministic coordinate discrepancy into a genuine U_{13} symmetry by recurrence.

Volume 14

Entropy Contributions and the High-Entropy Method

Volume 14 scope and prerequisites

This volume owns the high-entropy mechanism and nothing beyond it. It does not claim the low-entropy theorem or the full Einsiedler–Katok–Lindenstrauss classification, which belong to later volumes. In the model $SL_k(\mathbb{R})/\Gamma$ we derive the rootwise entropy coefficients, prove the noncommuting-root shearing lemma, and then reproduce the exact high-entropy alternative: two suitable nontrivial rootwise conditional systems force Haar conditionals in an SL_2 block and hence unipotent invariance. The theorem-strength comparison is made with Proposition 3.1, Lemma 3.3, and Theorem 2.2 of Einsiedler–Katok–Lindenstrauss [EK03; EK05; EKL06].

Student orientation: entropy coefficients and the high-entropy trigger

For a coarse root $[\alpha]$, its *entropy contribution* is the part of $h_\mu(a)$ carried by the corresponding expanding leafwise system. In the split type-A model we encode these contributions by coefficients $s_{ij} \in [0,1]$. The endpoint values have geometric meaning: $s_{ij} = 0$ means trivial rootwise conditional measure, while $s_{ij} = 1$ means Haar conditional measure on that root group. The *high-entropy method* is the mechanism by which two suitably noncommuting nontrivial root directions produce a new translation symmetry via shearing and commutators. Do not read the root diagram as bookkeeping. A directed edge records a positive entropy contribution; sharing a head or tail is exactly the algebraic circumstance in which the type-A commutator can manufacture a third root direction.

Reader guide: a concrete model

Why this matters.

Volume 14 is organized around the passage from *Entropy-Contribution Formulae* to *Algebraicity Criteria from Entropy-Generated Invariance*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.6 (A calculation to keep in view)**.** Let

$$a = \mathrm{diag}(e^{2s}, e^{-s}, e^{-s}) \in \mathrm{SL}_3(\mathbb{R})$$

and let E_{12} be the matrix unit. Since

$$\mathrm{Ad}(a)E_{12} = aE_{12}a^{-1} = e^{3s}E_{12},$$

the root exponent on the E_{12} direction is $3s$. For a one-dimensional leafwise measure whose conditional scaling is Lebesgue, the Jacobian contribution

along this root direction is therefore $3s$. This calculation is the concrete local model behind the root-by-root entropy bookkeeping used later.

CHAPTER 1

Entropy-Contribution Formulae

Proof architecture. The chapter is organized as the chain *One-dimensional root entropy coefficient* $\longrightarrow$ *Rootwise entropy formula in* SL_k $\longrightarrow$ *Maximal coefficient gives Haar leafwise measure.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. One coefficient for each root

Let

$$X = SL_k(\mathbb{R})/\Gamma, \qquad A = \{\alpha^{\mathbf{t}} = \mathrm{diag}(e^{t_1}, \dots, e^{t_k}) : \sum t_i = 0\},$$

and $U_{ij} = \{I + sE_{ij} : s \in \mathbb{R}\}$. Conjugation rescales the root coordinate by $e^{t_i - t_j}$. Volume 13 supplies the projective rootwise measure and Volume 12 supplies subordinate entropy.

Lemma 1.1 (One-dimensional root entropy coefficient)**.** *Assume μ is A-invariant and ergodic. For every $i \neq j$ there is a number $s_{ij}(\mu) \in [0, 1]$ such that, whenever $t_i > t_j$,*

$$h_\mu(\alpha^{\mathbf{t}}, U_{ij}) = s_{ij}(\mu)(t_i - t_j).$$

Moreover $s_{ij} = 0$ if and only if the rootwise leafwise measure is projectively δ_e a.e.

Proof. Write $\lambda = t_i - t_j > 0$ and identify U_{ij} with $\mathbb{R}$. Choose an increasing subordinate partition ξ_{ij} so that $\alpha^{-\mathbf{t}}\xi_{ij}$ refines ξ_{ij} and, in leaf coordinates, one application of $\alpha^{-\mathbf{t}}$ shrinks plaques by $e^{-\lambda}$. The root entropy is

$$H_\mu(\alpha^{-\mathbf{t}}\xi_{ij} \mid \xi_{ij}) = \int -\log \mu_x^{ij}(I_{x,1})\, d\mu(x),$$

where $I_{x,1}$ is the one-step shrunken plaque in the normalized leafwise measure. Iterating and using the conditional chain rule gives an additive information cocycle. Ergodicity and Birkhoff therefore give a deterministic asymptotic information per iterate.

If $\mathbf{s}$ has the same root value $s_i - s_j = \lambda$, then $\alpha^{\mathbf{s}-\mathbf{t}}$ centralizes U_{ij}. Normalizer equivariance and recurrence for this centralizing subaction show that the normalized root information is unchanged. Thus the rate depends only on λ. For integer multiples it scales by the ordinary iteration rule for entropy; rational scaling follows after passing to a common iterate, and real scaling follows by monotonic approximation of subordinate plaques. Hence the dependence is linear: $h_{ij} = s_{ij}\lambda$.

The geometric Jacobian bound of Volume 12 gives $0 \le s_{ij} \le 1$. If the leafwise measure is δ_e, the one-step conditional atom has mass one and the entropy is zero. Conversely, if the entropy is zero, then $H(\alpha^{-\mathbf{t}}\xi_{ij} \mid \xi_{ij}) = 0$. A zero conditional entropy between a refinement and its coarsening means the refinement is measurable with respect to the coarser partition; hence the two partitions agree modulo null sets. Iterating gives $\xi_{ij} = \alpha^{-n\mathbf{t}}\xi_{ij}$ modulo null sets for every n. The leaf plaques of the right side shrink to the base point, so the conditional probability on the original plaque is supported at that point. In root coordinates this is exactly δ_e. □

Lemma 1.2 (Root filtration of an unstable horospherical group)**.** *For a regular* $\mathbf{t}$*, the positive roots may be ordered* $\beta_1, \dots, \beta_r$ *so that*

$$V_j = U_{\beta_1} \cdots U_{\beta_j}$$

is a normal filtration of $G^+_{\alpha^{\mathbf{t}}}$ *and* $V_j/V_{j-1} \simeq U_{\beta_j}$*. Subordinate partitions can be chosen compatibly with this filtration, and the quotient conditional at the* j*th step is the rootwise leafwise conditional of Volume 13.*

Proof. Order the roots by increasing height with respect to the chamber, breaking ties so that whenever $\beta + \gamma$ is a root it occurs after both β and γ. The Chevalley commutator relation from Volume 11 then shows that multiplication by earlier root groups can only create roots occurring later, which gives a normal filtration after reversing this order if necessary. Local multiplication coordinates are diffeomorphisms near the identity. Choose nested flow boxes adapted to these coordinates. The entropy filtration theorem of Volume 12 supplies compatible subordinate partitions, and the iterated-disintegration theorem plus the compatibility dictionary of Volume 13 identifies the conditional on each one-dimensional quotient with the corresponding rootwise conditional measure. □

Theorem 1.3 (Rootwise entropy formula in SL_k). *For every* A*-invariant ergodic probability* μ *there are coefficients* $s_{ij}(\mu) \in [0, 1]$ *such that*

$$h_\mu(\alpha^{\mathbf{t}}) = \sum_{i \ne j} s_{ij}(\mu)(t_i - t_j)^+.$$

Moreover $s_{ij} = 0$ *if and only if* μ_x^{ij} *is trivial a.e.; if* μ *is* U_{ij}*-invariant, then* $s_{ij} = 1$.

Proof. Fix regular $\mathbf{t}$ and use Lemma 1.2. Repeated relative-entropy addition along the compatible filtration expresses the entropy carried by the full unstable partition as the sum of the one-dimensional quotient contributions. For diagonal actions on a homogeneous space the stable and neutral directions contribute no forward unstable information, so this is the full entropy. Lemma 1.1 identifies the U_{ij} summand with $s_{ij}(t_i - t_j)$ whenever that root is expanded. Roots with nonpositive exponent do not occur in $G^+_{\alpha^{\mathbf{t}}}$, yielding the positive-part formula. Singular $\mathbf{t}$ follow by approaching from regular elements and using the linear formula on adjacent chambers.

The zero-coefficient statement is part of Lemma 1.1. If μ is U_{ij}-invariant, Theorem 1.2 makes μ_x^{ij} Haar. A root interval shrinks by $e^{-\lambda}$ in one inverse iterate and Haar mass scales by exactly the same factor, so its conditional information is λ and $s_{ij} = 1$. □

Proposition 1.4 (Maximal coefficient gives Haar leafwise measure)**.** *If* $s_{ij}(\mu) = 1$*, then* μ_x^{ij} *is Haar a.e. and therefore* μ *is* U_{ij}*-invariant.*

Proof. Fix an expanded root and a subordinate interval partition into N equal child intervals after a return time for which the expansion factor is within a bounded multiplicative constant of N. Conditional Shannon entropy of the child weights is at most $\log N$, with equality only for the uniform vector. The assumption $s_{ij} = 1$ says that the normalized entropy deficit from $\log N$ has zero average at arbitrarily large such scales. The deficit is the Kullback–Leibler divergence of the child-weight vector from the uniform vector; it is nonnegative. Ergodicity and recurrence therefore force the deficit to vanish for the recurrent local configurations. Thus equal-length subintervals have equal conditional mass at every continuity scale. By subdivision, the mass of an interval depends only on its length; regularity of the Radon measure then gives a constant multiple of Lebesgue measure on $\mathbb{R} \simeq U_{ij}$. Theorem 1.2 converts Haar leafwise measure to U_{ij}-invariance. □

Proof and provenance. The theorem supplies the role played by Proposition 3.1 in Einsiedler–Katok–Lindenstrauss [EKL06]. The source states exactly the three features used later: $s_{ij} \in [0,1]$, vanishing precisely for a one-point conditional, and the formula $\sum s_{ij}(t_i - t_j)^+$; it also records the maximal-coefficient/Haar converse. The argument is therefore given through subordinate partitions and root filtrations, avoiding the false general principle that every diffuse measure has positive dimension.

Worked Example 1.5 (Computing a root coefficient for Haar measure)**.** Take $U_{12} \cong \mathbb{R}$ and a diagonal element with $t_1 - t_2 = \lambda > 0$. Haar leafwise measure is ds. A unit interval pulls back under one inverse iterate to an interval of length $e^{-\lambda}$ and therefore mass $e^{-\lambda}$. The one-step information is $-\log(e^{-\lambda}) = \lambda$. Hence

$$h_\mu(\alpha^{\mathbf{t}}, U_{12}) = \lambda \quad \text{and} \quad s_{12} = 1.$$

At the opposite extreme, for δ_0 the shrunken plaque still has mass one, so the information is zero and $s_{12} = 0$. The coefficient s_{12} measures exactly where the actual conditional lies between these two entropy extremes.

CHAPTER 2

Rootwise Positive Entropy

Proof architecture. The chapter is organized as the chain *Positive entropy locates a positive root coefficient* $\longrightarrow$ *Positive entropy is equivalent to a nontrivial expanding root conditional* $\longrightarrow$ *Entropy symmetry forces a constraint on opposite roots.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. Turning one scalar entropy into geometric directions

The formula of Chapter 1 allows us to locate positive entropy. This is the first higher-rank advantage: instead of treating $h_\mu(a) > 0$ as a single number, we know which coarse Lyapunov leaves carry the information.

Lemma 1.1 (Positive entropy locates a positive root coefficient)**.** *For any* $\mathbf{t}$*,*

$$h_\mu(\alpha^{\mathbf{t}}) > 0$$

if and only if there is a pair $i \neq j$ *with* $t_i > t_j$ *and* $s_{ij}(\mu) > 0$*.*

Proof. Every term in Theorem 1.3 is nonnegative. The sum is positive exactly when at least one term is positive, and $(t_i - t_j)^+ > 0$ is equivalent to $t_i > t_j$. $\square$

Theorem 1.2 (Positive entropy is equivalent to a nontrivial expanding root conditional). *Let* μ *be* A*-invariant and ergodic. Then*

$$h_\mu(\alpha^{\mathbf{t}}) > 0 \quad \Longleftrightarrow \quad \textit{for some } i, j \textit{ with } t_i > t_j,\ \mu_x^{ij} \textit{ is nontrivial a.e.}$$

(and hence non-atomic a.e.).

Proof. By Lemma 1.1, positive entropy is equivalent to $s_{ij} > 0$ in some expanding root. Theorem 1.3 identifies $s_{ij} = 0$ exactly with trivial leafwise measure. Volume 13 upgrades nontriviality to non-atomicity for an A-ergodic measure because an appropriate diagonal element expands every root group. $\square$

Proposition 1.3 (Entropy symmetry forces a constraint on opposite roots)**.** *For every* $\mathbf{t}$*,*

$$\sum_{i\neq j} s_{ij}(t_i - t_j)^+ = \sum_{i\neq j} s_{ij}(t_j - t_i)^+.$$

Equivalently,

$$\sum_{i<j}(s_{ij} - s_{ji})(t_i - t_j) = 0$$

for every $\mathbf{t}$ *with* $\sum t_i = 0$ *after orienting pairs consistently in each chamber.*

Proof. Entropy of an invertible measure-preserving transformation equals that of its inverse: $h_\mu(\alpha^{\mathbf{t}}) = h_\mu(\alpha^{-\mathbf{t}})$. Apply the rootwise formula to both sides. Group the two oriented roots in each unordered pair. The resulting linear relations are the bookkeeping engine used in the high-entropy cardinality argument. □

Worked Example 1.4 (SL_3)**.** If $t_1 > t_2 > t_3$, then

$$h_\mu(\alpha^{\mathbf{t}}) = s_{12}(t_1 - t_2) + s_{13}(t_1 - t_3) + s_{23}(t_2 - t_3).$$

Positive entropy therefore means that at least one of the three upper-triangular root foliations has a nontrivial conditional measure. The difficulty is to turn this nontriviality into invariance; noncommuting root groups do that in the next chapters.

CHAPTER 3

Nontriviality of Coarse-Lyapunov Leafwise Measures

Proof architecture. The chapter is organized as the chain *Root graph is measurable-data-free* $\longrightarrow$ *High-entropy versus isolated-root alternative* $\longrightarrow$ *Which roots feed the commutator lemma directly.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. From one parameter to an A-invariant root graph

For SL_k, draw a directed edge $i \to j$ whenever $s_{ij} > 0$. The entropy formula and A-ergodicity make this graph independent of the base point. The high-entropy case occurs when a second positive edge shares the specified tail or head.

Lemma 1.1 (Root graph is measurable-data-free)**.** *For an A-ergodic μ,*

$$E(\mu) = \{(i,j) : \mu_x^{ij} \text{ is nontrivial a.e.}\} = \{(i,j) : s_{ij} > 0\}$$

is a deterministic finite directed graph.

Proof. The leafwise type is A-invariant and hence a.e. constant by Lemma 1.1. Equality with $s_{ij} > 0$ is Theorem 1.3. There are finitely many roots, so one common conull domain suffices. $\square$

Theorem 1.2 (High-entropy versus isolated-root alternative). *Fix $(a,b) \in E(\mu)$. Exactly one of the following holds:*

(1) there is a distinct $(i,j) \in E(\mu)$ with either $i = a$ or $j = b$;

(2) every other edge with tail a is absent and every other edge with head b is absent.

The first case is the high-entropy configuration treated here; the second is passed to Volume 15.

PROOF. This is the exhaustive finite dichotomy in the root graph. Recording it as a theorem is useful because the commutator/high-entropy proof below genuinely needs the second edge; it cannot be applied to the isolated-root configuration. ☐

Proposition 1.3 (Which roots feed the commutator lemma directly)**.** *For a fixed positive root (a,b), a positive incoming root (i,a) gives the chain*

$$U_{ia}, U_{ab} \quad \rightsquigarrow \quad U_{ib},$$

while a positive outgoing root (b,j) gives

$$U_{ab}, U_{bj} \quad \rightsquigarrow \quad U_{aj}.$$

These are the direct noncommuting configurations. Positive roots sharing the tail a or the head b are converted into invariance only after the entropy/cardinality argument of Chapter 5; no opposite-root symmetry is assumed in advance.

PROOF. The matrix commutator identity

$$[I + rE_{ia}, I + sE_{ab}] = I + rsE_{ib}, \qquad [I + rE_{ab}, I + sE_{bj}] = I + rsE_{aj}$$

is Volume 11's root calculus. The final sentence is a dependency statement: the symmetry $s_{ab} = s_{ba}$ will be derived from the high-entropy cardinality squeeze, so using it here would be circular. ☐

Warning 1.4. A positive coefficient on one root is not enough for the high-entropy argument. The shared-index hypothesis is load-bearing; dropping it would prematurely solve the low-entropy case.

Worked Example 1.5 (Three possible root graphs in SL_3)**.** Suppose $s_{12} > 0$. If also $s_{23} > 0$, the edges $1 \to 2 \to 3$ form a direct commutator chain and Theorem 1.2 will produce U_{13}-invariance. If instead $s_{13} > 0$, the two positive edges share the tail 1 but do not themselves form a chain; the cardinality/entropy argument of Chapter 5 is needed. Finally, if s_{12} is the only positive edge touching tail 1 or head 2, we are in the isolated-root configuration. This last graph is precisely what the high-entropy method does not classify.

CHAPTER 4

Commutator Production of Invariance

Proof architecture. The chapter is organized as the chain *Recurrent Heisenberg shearing* $\longrightarrow$ *Noncommuting-root lemma* $\longrightarrow$ *Permuted chain orientations.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. The shearing square

For distinct i, j, k,

$$u_{ij}(r)u_{jk}(s) = u_{jk}(s)u_{ij}(r)u_{ik}(rs).$$

The ambient measure is not assumed invariant under the first two root groups. The proof therefore combines diffuse leafwise mass, diagonal rescaling, recurrence, and the coordinate-order comparison from Volume 13.

Lemma 1.1 (Recurrent Heisenberg shearing)**.** *Assume μ_x^{ij} and μ_x^{jk} are non-atomic a.e. Then on a positive-measure recurrent set there is a nonzero q such that the U_{ik} leafwise translation-symmetry group contains $u_{ik}(q)$.*

Proof. Write $X = U_{ij}$, $Y = U_{jk}$, $Z = U_{ik}$. Choose a regular diagonal element a with positive root weights $\alpha = t_i - t_j > 0$ and $\beta = t_j - t_k > 0$; the Z weight is $\alpha + \beta$. Choose a compact Lusin set K of arbitrarily large measure on which the normalized X, Y, Z leafwise systems and the relevant local product charts vary continuously.

Because the X- and Y-conditionals are diffuse but positive on every identity neighborhood, after shrinking K there are fixed annuli A_X, A_Y of small nonzero parameters with positive conditional mass on a positive-measure subset. Poincaré recurrence gives a typical x and times $n_\ell \to \infty$ for which $a^{n_\ell}x$ returns to that subset. Normalizer equivariance then pulls parameters $R_\ell \in A_X$ and $S_\ell \in A_Y$ at the return point back to parameters

$$r_\ell = e^{-n_\ell\alpha}R_\ell, \qquad s_\ell = e^{-n_\ell\beta}S_\ell$$

in the rootwise supports at x. Thus $r_\ell, s_\ell \to 0$. Using the leafwise reconstruction formula and Fubini, choose them so that all corners needed for the two iterated disintegrations lie in the common conull domain. After passing to a subsequence, $R_\ell \to R \neq 0$ and $S_\ell \to S \neq 0$.

Now compare the two coordinate orders. Before rescaling their discrepancy is $u_{ik}(r_\ell s_\ell)$. Conjugating by a^{n_ℓ} multiplies the Z parameter by $e^{n_\ell(\alpha+\beta)}$, so the discrepancy at the recurrent scale is

$$e^{n_\ell(\alpha+\beta)} r_\ell s_\ell = R_\ell S_\ell \longrightarrow q := RS \neq 0.$$

Lemma 1.2 says that the two iterated conditional descriptions of the same local XYZ plaque differ exactly by this Z translation. At the recurrent times the base points return to the Lusin set, so weak-* continuity identifies the limiting Z conditional on the two sides. Therefore $(R_{u_{ik}(q)})_* \mu_x^{ik} \propto \mu_x^{ik}$: the nonzero element $u_{ik}(q)$ lies in the closed leafwise symmetry group I_x of Proposition 1.3. $\square$

Theorem 1.2 (Noncommuting-root lemma). *Let μ be A-invariant and ergodic on $SL_k(\mathbb{R})/\Gamma$. If i, j, k are distinct and both μ_x^{ij} and μ_x^{jk} are nontrivial a.e., then μ is U_{ik}-invariant.*

Proof. The zero-or-diffuse dichotomy of Volume 13 makes the two assumed conditionals non-atomic. Lemma 1.1 gives a nonzero $u_{ik}(q)$ in the leafwise symmetry group. Since A normalizes U_{ik}, normalizer equivariance gives

$$u_{ik}(e^{t_i - t_k} q) \in I_{\alpha^{\mathbf{t}} x}$$

for every $\mathbf{t}$. Ergodicity makes the projective symmetry type constant, and the root value $t_i - t_k$ ranges over all real numbers. The closed symmetry group therefore contains every parameter in U_{ik}. Proposition 1.3 and Theorem 1.2 now give U_{ik}-invariance. $\square$

Proposition 1.3 (Permuted chain orientations)**.** *Every permutation of the indices in Theorem 1.2 is valid. In particular*

$$(U_{ij}, U_{jk}) \Rightarrow U_{ik}, \qquad (U_{kj}, U_{ji}) \Rightarrow U_{ki}.$$

Every produced root conditional is Haar and has coefficient one.

Proof. Permutation matrices conjugate the first configuration to all the others. Once μ is invariant under the produced root group, Theorem 1.2 gives Haar leafwise measure, and Theorem 1.3 gives coefficient one. $\square$

Proof and provenance. This is the internal owner for the noncommuting-root lemma appearing as Lemma 3.3 in the EKL proof architecture [EKL06]. The revised proof makes the scale choice explicit: the preimages r_ℓ, s_ℓ shrink, while after diagonal rescaling the two first-order parameters stay in fixed annuli and their commutator converges to the nonzero parameter $q = RS$. This replaces the earlier impossible scaling sentence in which fixed first-order displacements were claimed to shrink while their product stayed nonzero.

Worked Example 1.4 (The rescaling that keeps the commutator visible)**.** Take root weights $\alpha = 2$ on U_{12} and $\beta = 1$ on U_{23}. At a recurrent time n, choose preimage parameters $r_n = e^{-2n}R$ and $s_n = e^{-n}S$. Both tend to zero, so the corresponding points begin arbitrarily close on the original leaf. After applying a^n, the two parameters become the fixed values R and S. The U_{13} root has weight $\alpha + \beta = 3$, hence

$$e^{3n}r_n s_n = e^{3n}e^{-3n}RS = RS.$$

So the commutator displacement survives at a nonzero fixed scale. This numerical calculation is the bookkeeping heart of Lemma 1.1.

CHAPTER 5

The High-Entropy Method

Proof architecture. The chapter is organized as the chain *Cardinality squeeze and opposite-root inequality* $\longrightarrow$ *High-entropy theorem: shared root index forces* SL_2 *invariance* $\longrightarrow$ *Scope boundary of the theorem.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. From two diffuse roots to an SL_2 block

We now combine the noncommuting-root lemma with entropy symmetry. The argument is finite but delicate: the inclusions of root sets and the entropy inequalities must be kept in the correct order.

Lemma 1.1 (Cardinality squeeze and opposite-root inequality)**.** *Fix* (a, b) *with* $s_{ab} > 0$*. For* $\ell = a, b$ *define*

$$C_\ell = \{i \notin \{a, b\} : s_{i\ell} > 0\}, \qquad R_\ell = \{j \notin \{a, b\} : s_{\ell j} > 0\},$$

and let C_ℓ^L, R_ℓ^L *denote the subsets for which* μ *is invariant under the corresponding root group. Then*

$$C_a \subset C_b^L, \qquad R_b \subset R_a^L,$$

and

$$|R_a^L| \le |C_a| \le |C_b^L| \le |R_b| \le |R_a^L|.$$

Hence all cardinalities are equal and

$$s_{ba} \ge s_{ab} > 0.$$

Consequently $C_a = C_a^L$ *and* $R_b = R_b^L$*. Applying the same statement to* (b, a) *also gives* $C_b = C_b^L$ *and* $R_a = R_a^L$*.*

Proof. If $i \in C_a$, then μ_x^{ia} and μ_x^{ab} are nontrivial; Theorem 1.2 gives U_{ib}-invariance, so $i \in C_b^L$. Similarly $R_b \subset R_a^L$.

Let $\mathbf{t}^{(a)}$ have $t_a = 1 - 1/k$ and $t_i = -1/k$ for $i \neq a$. The rootwise entropy formula gives

$$h_\mu(\alpha^{\mathbf{t}^{(a)}}) = \sum_j s_{aj} = s_{ab} + |R_a^L| + \sum_{j \in R_a \setminus R_a^L} s_{aj} > |R_a^L|.$$

For the inverse element,

$$h_\mu(\alpha^{-\mathbf{t}^{(a)}}) = \sum_i s_{ia} = s_{ba} + \sum_{i \in C_a} s_{ia} \leq 1 + |C_a|.$$

Entropy is symmetric under inversion, so $|R_a^L| \leq |C_a|$. Repeating the same calculation with b distinguished gives $|C_b^L| \leq |R_b|$. Together with the two inclusions,

$$|R_a^L| \leq |C_a| \leq |C_b^L| \leq |R_b| \leq |R_a^L|,$$

so every inequality is equality.

Keeping the s_{ab} term instead of discarding it gives

$$s_{ab} + |R_a^L| \leq h_\mu(\alpha^{\mathbf{t}^{(a)}}) = h_\mu(\alpha^{-\mathbf{t}^{(a)}}) \leq s_{ba} + |C_a|.$$

Equal cardinalities yield $s_{ba} \geq s_{ab} > 0$. Since $s_{ba} > 0$, apply the noncommuting-root lemma once more: $C_a \subset C_b^L \subset C_a^L$. But always $C_a^L \subset C_a$ because invariance implies coefficient one, hence positivity. Therefore $C_a = C_a^L$. The same argument gives $R_b = R_b^L$. Finally repeat the whole conclusion with the reversed positive pair (b, a) to get $C_b = C_b^L$ and $R_a = R_a^L$. □

Corollary 1.2 (Opposite-root entropy symmetry)**.** *For every $a \neq b$ one has $s_{ab} = s_{ba}$.*

Proof. If $s_{ab} = 0$ but $s_{ba} > 0$, Lemma 1.1 applied to (b, a) would give $s_{ab} \geq s_{ba} > 0$, a contradiction. If both are positive, the lemma gives $s_{ba} \geq s_{ab}$ and, after reversing the pair, $s_{ab} \geq s_{ba}$. □

Theorem 1.3 (High-entropy theorem: shared root index forces SL_2 invariance). *Let μ be an A-invariant ergodic probability on $SL_k(\mathbb{R})/\Gamma$. Suppose μ_x^{ab} and μ_x^{ij} are nontrivial a.e. for two distinct root pairs, with either $i = a$ or $j = b$. Then μ_x^{ab} and μ_x^{ba} are Haar a.e., and μ is invariant under*

$$H_{ab} = \langle U_{ab}, U_{ba} \rangle \cong SL_2(\mathbb{R})$$

in the (a, b) coordinates.

Proof. By Theorem 1.3, the two assumed nontrivial conditionals give $s_{ab} > 0$ and $s_{ij} > 0$. Lemma 1.1, applied first to (a, b) and then to (b, a), shows that every positive root with tail a and every positive root with head b is already an invariance root: $R_a = R_a^L$ and $C_b = C_b^L$. Therefore the second root U_{ij} is invariant (if $i = a$ it lies in R_a^L, and if $j = b$ it lies in C_b^L).

Now regard (i, j) as the distinguished positive pair. The original root (a, b) shares the same tail or head with it by hypothesis, so the same conclusion applied to (i, j) makes U_{ab} an invariance root. Hence $s_{ab} = 1$. Corollary 1.2 gives $s_{ba} = 1$, and Proposition 1.4 makes both μ_x^{ab} and μ_x^{ba} Haar. The Haar/invariance criterion of Volume 13 gives invariance under both opposite root groups. Those two groups generate the standard embedded SL_2 block H_{ab}. □

Proposition 1.4 (Scope boundary of the theorem)**.** *Theorem 1.3 does not address the configuration in which $s_{ab} > 0$ but every other root sharing the tail a or the head b is zero. No H_{ab}-invariance conclusion is asserted in that case here.*

Proof. The final bootstrap in the proof requires a second positive root in R_a or C_b. If none exists, the argument stops after the opposite-coefficient symmetry and does not manufacture U_{ab}-invariance. The complementary mechanism uses recurrence under the subtorus centralizing H_{ab} and belongs to the low-entropy method of Volume 15. □

> **Proof and provenance.** The set definitions, cardinality squeeze, inequality $s_{ba} \geq s_{ab}$, and the two-step bootstrap above follow the proof architecture of EKL's high-entropy theorem [EKL06]. The revised version makes explicit the step that the earlier draft skipped: one first proves all positive roots in R_a and C_b are invariance roots, uses this to make the supplied second edge invariant, and only then re-runs the argument with that edge distinguished to force U_{ab}-invariance.

Worked Example 1.5 (The cardinality squeeze with four indices)**.** Take $(a, b) = (1, 2)$ in SL_4 and suppose $s_{12} > 0$. If, for example, $3 \in C_1$, then $s_{31} > 0$ and the chain U_{31}, U_{12} produces U_{32}-invariance, so $3 \in C_2^L$. Thus every member of C_1 injects into C_2^L simply by keeping the same index. Similarly $R_2 \subset R_1^L$. Entropy of the diagonal element singling out coordinate 1 compares the number of already-Haar outgoing roots from 1 with the number of positive incoming roots; singling out coordinate 2 gives the reverse comparison. The resulting cycle of four finite cardinalities can only close by equality. The abstract proof is nothing more mysterious than this finite bookkeeping repeated for arbitrary k.

CHAPTER 6

Relations among Coarse Lyapunov Directions

Proof architecture. The chapter is organized as the chain *Invariance roots are closed under root commutators* $\longrightarrow$ *Root-subsystem generated by invariance* $\longrightarrow$ *Block structure in type A*. The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. The root graph after high-entropy closure

Once one root becomes an invariance root, exact commutator relations propagate that symmetry. It is useful to separate this algebraic closure operation from the entropy argument that produced the first invariance.

Lemma 1.1 (Invariance roots are closed under root commutators)**.** *If μ is invariant under U_{ij} and U_{jk} for distinct i, j, k, then it is invariant under U_{ik}. More generally the set of root groups preserving μ is closed under every nontrivial Chevalley commutator available in SL_k.*

Proof. For SL_k,

$$[u_{ij}(r), u_{jk}(s)] = u_{ik}(rs).$$

A measure invariant under two transformations is invariant under their products, inverses, and commutators. Since rs ranges over all real numbers, invariance under all such commutators is exactly U_{ik}-invariance. The same argument applies to any root relation for which the commutator parametrizes the target root subgroup. $\square$

Theorem 1.2 (Root-subsystem generated by invariance). *Let*

$$\Phi_\mu = \{e_i - e_j : \mu \text{ is } U_{ij}\text{-invariant}\}.$$

Then Φ_μ is closed under negation whenever both directions have been produced by an SL_2 block, and under addition of roots whenever the sum is a root. The connected group L_μ generated by these root groups preserves μ.

Proof. An H_{ij}-invariance conclusion gives both U_{ij} and U_{ji} and hence closure under negation for that block. Lemma 1.1 gives closure under root addition. The subgroup generated by measure-preserving transformations preserves the measure; taking its connected closure does as well because the action of G on probability measures is weak-* continuous. □

Proposition 1.3 (Block structure in type A)**.** *If Φ_μ contains every root $e_i - e_j$ with i, j in a subset $I \subset \{1, \ldots, k\}$, then μ is invariant under the embedded subgroup $SL_I(\mathbb{R})$. If the equivalence relation generated by the invariant SL_2 blocks has classes $I_1, \ldots, I_r$, then μ is invariant under $\prod_m SL_{I_m}(\mathbb{R})$.*

Proof. Elementary root subgroups U_{ij} with $i, j \in I$ generate $SL_I(\mathbb{R})$. The first statement follows from Theorem 1.2. For different equivalence classes the corresponding blocks commute, so their generated product also preserves μ. □

Worked Example 1.4 (Three roots close an SL_3 block)**.** If high entropy gives H_{12} and another argument gives H_{23}, then U_{12} and U_{23} generate U_{13}, while the opposite roots generate U_{31}. The six root groups generate the full embedded SL_3. Entropy produced two rank-one blocks; commutator closure upgrades them to higher-rank invariance.

CHAPTER 7

Propagation and Saturation of Unipotent Invariance

Proof architecture. The chapter is organized as the chain *Finite saturation terminates* $\longrightarrow$ *Saturated high-entropy invariance group* $\longrightarrow$ *When saturation already yields Haar measure.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. An iterative closure algorithm

The high-entropy theorem may be applied repeatedly. Each newly Haar root becomes an exact invariance direction, and exact invariance makes subsequent commutator steps easier.

Lemma 1.1 (Finite saturation terminates)**.** *Start with the finite set of positive-coefficient roots $E(\mu)$ and repeatedly:*

(1) apply Theorem 1.3 to a qualifying pair and mark the resulting opposite root pair invariant;

(2) close marked invariant roots under Lemma 1.1.

The process terminates after at most $k(k-1)$ additions of new oriented roots.

Proof. There are only $k(k-1)$ oriented roots. Each nontrivial step permanently adds at least one root to the marked invariant set and no operation removes one. □

Theorem 1.2 (Saturated high-entropy invariance group). *The terminal marked root set generates a connected A-normalized subgroup $L_{\mathrm{HE}} < SL_k(\mathbb{R})$ such that μ is L_{HE}-invariant. Every inference has the explicit chain*

$$\begin{aligned} &\textit{positive coefficient} \Longrightarrow \textit{non-atomic conditional} \\ &\Longrightarrow \textit{root invariance} \Longrightarrow \textit{Haar conditional} \\ &\Longrightarrow \textit{larger generated invariance group.} \end{aligned}$$

Proof. Every root marked by the high-entropy theorem is an exact invariance root. Every root marked by commutator closure is exact by Lemma 1.1. The group generated by all terminal root groups therefore preserves μ; its connected closure also preserves μ by weak-* continuity of the action. Since A normalizes each root group, it normalizes their generated group. The displayed chain lists the proved inputs from Theorems 1.3, 1.2, 1.3, and 1.2. □

Proposition 1.3 (When saturation already yields Haar measure)**.** *If the terminal invariant root set is the full type-A_{k-1} root system, then $L_{\mathrm{HE}} = SL_k(\mathbb{R})$ and μ is Haar measure whenever the quotient carries finite SL_k-invariant volume.*

Proof. All elementary root groups generate $SL_k(\mathbb{R})$, so full root saturation gives full G-invariance. On a quotient by a lattice, the G-invariant probability is normalized Haar measure. □

Worked Example 1.4 (Saturating a connected root graph)**.** Suppose in SL_4 the high-entropy theorem has already produced the blocks H_{12}, H_{23}, and H_{34}. We therefore have invariance under all six adjacent roots $U_{12}, U_{21}, U_{23}, U_{32}, U_{34}, U_{43}$. Commutators give

$$[U_{12}, U_{23}] = U_{13}, \qquad [U_{23}, U_{34}] = U_{24},$$

and then $[U_{13}, U_{34}] = U_{14}$. Opposite commutators give U_{31}, U_{42}, U_{41}. Thus the saturation closes to all twelve roots and hence to SL_4. The algorithm terminates because each step adds a root from a finite list.

CHAPTER 8

Algebraicity Criteria from Entropy-Generated Invariance

Proof architecture. The chapter is organized as the chain *Passing from L-invariance to homogeneous components* $\longrightarrow$ *High-entropy algebraicity criterion.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. What high entropy has actually proved

A new invariance group is a powerful conclusion, but it is not yet the full classification of A-invariant positive-entropy measures. The correct endpoint of this volume is therefore an algebraicity *criterion*: once the generated unipotent invariance is large enough in the arithmetic linear quotient used by the later EKL volume, its ergodic components are classified by the internal AHD theorem. The low-entropy case is still open at this stage.

Lemma 1.1 (Passing from L-invariance to homogeneous components)**.** *Let $G \leq \mathrm{SL}_N(\mathbb{R})$ be the connected real points of the fixed $\mathbb{Q}$-group and let $\Gamma < G(\mathbb{Q})$ be arithmetic. Let $L < G$ be connected and generated by one-parameter Ad-unipotent subgroups. If a probability μ on G/Γ is L-invariant, then almost every L-ergodic component is the invariant probability on a closed finite-volume orbit of a connected subgroup containing L.*

Proof. Decompose μ into L-ergodic components. For the L-ergodic decomposition

$$\mu = \int_\Omega \mu_\omega \, d\lambda(\omega),$$

each μ_ω is an L-invariant L-ergodic probability on the same arithmetic quotient. Apply Theorem 1.1 to μ_ω. It gives a connected $H_\omega \geq L$ and a closed finite-volume orbit $H_\omega x_\omega$ with $\mu_\omega = m_{H_\omega x_\omega}$ for λ-almost every ω. $\square$

Theorem 1.2 (High-entropy algebraicity criterion). *Let $\Gamma < SL_k(\mathbb{R})$ be arithmetic and let μ be A-invariant and ergodic on $SL_k(\mathbb{R})/\Gamma$. Suppose the* *high-entropy saturation procedure produces a nontrivial connected group L_{HE} generated by root unipotents. Then μ is L_{HE}-invariant and its L_{HE}-ergodic components are homogeneous. If, in addition, A permutes these components ergodically and the only closed homogeneous extension compatible with the resulting invariant root subsystem is a specified subgroup H, then μ is the H-homogeneous probability. In particular, full root saturation gives Haar measure.*

Proof. Theorem 1.2 gives L_{HE}-invariance. Lemma 1.1 classifies its ergodic components. Because A normalizes L_{HE} and preserves μ, it acts on the component space. A-ergodicity of μ makes this action ergodic. Under the stated uniqueness hypothesis for a compatible closed homogeneous extension, almost every component must have the same homogeneous support type H; the barycenter is therefore the H-invariant probability. The final assertion is Proposition 1.3. □

Warning 1.3 (The remaining classification problem)**.** Theorem 1.2 is deliberately conditional on what the saturation procedure produces. A positive-entropy measure may begin in the isolated-root configuration of Theorem 1.2; treating that case is the low-entropy method of Volume 15. Only after that volume and the synthesis in Volume 16 may one state the full positive-entropy EKL classification.

Worked Example 1.4 (From an SL_2 block to a homogeneous component)**.** Suppose the high-entropy method gives invariance under the embedded block $H_{12} = \langle U_{12}, U_{21}\rangle \cong SL_2(\mathbb{R})$. Decompose μ into H_{12}-ergodic components. Theorem 1.1 says that almost every component is Haar on a closed orbit Lx for some subgroup $L \supset H_{12}$. The diagonal group A normalizes H_{12} and therefore permutes these components. If an independent algebraic argument shows that the only compatible closed support containing H_{12} is, say, a specified block subgroup H, then A-ergodicity forces the component type to be H almost everywhere, and the barycenter is the H-homogeneous probability. This illustrates exactly what the criterion proves—and what extra subgroup analysis is still needed.

Volume 15

The Low-Entropy Method

Volume 15 scope and prerequisites

Volume 14 settled the high-entropy branch: a positive root which shares a tail or head with a second positive root produces an SL_2 block of unipotent invariance. This volume owns the complementary isolated-root mechanism. We work first for an A-invariant ergodic probability on $SL_k(\mathbb{R})/\Gamma$, then record the arithmetic specialization $\Gamma = SL_k(\mathbb{Z})$. The proof follows the low-entropy architecture of Einsiedler–Katok–Lindenstrauss and the conceptual single-parameter formulation of Einsiedler–Lindenstrauss: recurrence under a centralizing subtorus, product structure of leafwise measures, polynomial shearing, and the dichotomy between new unipotent invariance and concentration on a centralizer orbit [EKL06; EL08].

Student orientation: what "low entropy" means here

The low-entropy branch does not mean zero total entropy. It means that a positive coarse root direction is *isolated*: the high-entropy shared-index trigger is unavailable. For a root U_{ab}, A'_{ab} denotes the codimension-one diagonal subgroup centralizing that root. A return is *exceptional* when the local displacement lies in $L_{ab} = C_G(U_{ab})$ and *strongly exceptional* when it lies in $C_{ab} = C_G(H_{ab})$, where H_{ab} is the embedded rank-one block generated by U_{ab} and U_{ba}. The proof has a genuine dichotomy. Nonexceptional returns create a transverse polynomial shear and hence new unipotent invariance. Strongly exceptional recurrence instead concentrates an A'_{ab}-ergodic component on a centralizer orbit; the arithmetic lattice is used only at the final step to rule out that alternative.

Reader guide: a concrete model

Why this matters.

Volume 15 is organized around the passage from *Exceptional Coarse-Lyapunov Directions* to *The Low-Entropy Rigidity Theorem*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.5 (A calculation to keep in view)**.** Let

$$a = \operatorname{diag}(e^{2s}, e^{-s}, e^{-s}) \in \mathrm{SL}_3(\mathbb{R})$$

and let E_{12} be the matrix unit. Since

$$\operatorname{Ad}(a)E_{12} = aE_{12}a^{-1} = e^{3s}E_{12},$$

the root exponent on the E_{12} direction is $3s$. For a one-dimensional leafwise measure whose conditional scaling is Lebesgue, the Jacobian contribution

along this root direction is therefore $3s$. This calculation is the concrete local model behind the root-by-root entropy bookkeeping used later.

CHAPTER 1

Exceptional Coarse-Lyapunov Directions

Proof architecture. The chapter is organized as the chain *Isolated root normal form* $\longrightarrow$ *High-entropy/isolated-root reduction* $\longrightarrow$ *Permutation and factor stability*. The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. The isolated-root configuration

Fix distinct indices a, b. Let

$$A'_{ab} = \{\mathrm{diag}(e^{t_1}, \ldots, e^{t_k}) : \sum t_i = 0,\ t_a = t_b\}.$$

This codimension-one subgroup centralizes $H_{ab} = \langle U_{ab}, U_{ba}\rangle$. Three auxiliary subgroups must be kept distinct throughout the low-entropy argument:

$$L_{ab} := C_G(U_{ab}), \qquad C_{ab} := C_G(H_{ab}) = C_G(U_{ab}) \cap C_G(U_{ba}),$$

and $U_{(ab)}$, generated by all U_{ij} with $i = a$ or $j = b$. Thus $C_{ab} \subset L_{ab}$. A return whose local displacement lies in L_{ab} is *exceptional*; it is *strongly exceptional* when the displacement already lies in C_{ab}. This is the terminology used in the primary low-entropy proof and is essential: exceptional returns are first upgraded to strongly exceptional ones by recurrence, and only then are centralizer-supported ergodic components identified.

Lemma 1.1 (Isolated root normal form)**.** *Assume $s_{ab} > 0$ and every coefficient s_{ij} with $(i, j) \neq (a, b)$ and either $i = a$ or $j = b$ is zero. Then the only nontrivial rootwise leafwise system occurring in the local product $U_{(ab)}$ is the U_{ab} system. Moreover $s_{ba} > 0$ by the opposite-root symmetry of Volume 14.*

Proof. The first assertion is simply the definition of the isolated configuration together with Volume 14's equivalence between $s_{ij} > 0$ and nontriviality of the U_{ij}-leafwise measure. The roots occurring in $U_{(ab)}$ are precisely those with tail a or head b, together with the opposite root. The opposite coefficient is positive by Corollary 1.2. It is kept separate because U_{ba} belongs to the opposite horospherical direction for the diagonal elements used below. $\square$

THEOREM 1.2 (High-entropy/isolated-root reduction). *For every positive root coefficient $s_{ab} > 0$, exactly one of the following holds.*

(1) There is a second positive root sharing the tail a or the head b; then Volume 14 gives H_{ab}-invariance.

(2) The root (a, b) is isolated in the sense of Lemma 1.1; then the low-entropy argument of the present volume applies.

PROOF. This is a finite dichotomy on the root set. In the first case Theorem 1.3 applies. If the first case fails, every other coefficient sharing the prescribed tail or head is zero, which is exactly the hypothesis of the second case. ☐

Proposition 1.3 (Permutation and factor stability)**.** *The isolated-root condition and the subgroup system $(A'_{ab}, U_{(ab)}, L_{ab}, C_{ab})$ are preserved by permutation of coordinates and by restriction to the minimal coordinate block containing a, b and every root created by a commutator with U_{ab}.*

PROOF. Permutation matrices conjugate U_{ij} to the correspondingly relabelled root group and conjugate A to itself. The defining equations $t_a = t_b$ and the root-incidence relations are therefore invariant. Restricting to the generated coordinate block deletes root groups that commute with every group used in the shearing argument, so none of the relevant leafwise or recurrence data changes. ☐

Worked Example 1.4 (The isolated root in SL_3)**.** For $(a, b) = (1, 2)$, the neighboring roots are U_{13} and U_{32}. Thus the low-entropy configuration is $s_{12} > 0$ but $s_{13} = s_{32} = 0$. Opposite-root symmetry gives $s_{21} > 0$. The centralizing subtorus is $A'_{12} = \{\mathrm{diag}(e^t, e^t, e^{-2t})\}$.

CHAPTER 2

Recurrence along Low-Entropy Leaves

Proof architecture. The chapter is organized as the chain *Compact Poincaré returns* $\longrightarrow$ *Strong-return/centralizer-component equivalence* $\longrightarrow$ *Ergodic-component form of the low-entropy alternative.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. Exceptional returns and centralizer components

Throughout this chapter μ is A-invariant and A'_{ab} denotes the codimension-one subgroup from Chapter 1. Fix a compact set K on which the quotient map admits a uniform injectivity radius $r_K > 0$. If $x, \alpha^{\mathbf{s}}x \in K$ with $\alpha^{\mathbf{s}} \in A'_{ab}$ and $d(x, \alpha^{\mathbf{s}}x) < r_K$, there is a unique g in the identity neighborhood $B_{r_K}(e)$ such that

$$\alpha^{\mathbf{s}}x = gx.$$

We call this return *exceptional* when $g \in L_{ab} = C_G(U_{ab})$ and *strongly exceptional* when $g \in C_{ab} = C_G(H_{ab})$.

The distinction is not cosmetic. In the original low-entropy proof one first shows, using recurrence in an auxiliary diagonal direction, that exceptional returns are strongly exceptional away from one null set; only afterwards does one identify strong exceptional recurrence with support on a centralizer orbit.

Lemma 1.1 (Exceptional returns become strong)**.** *There is an A-invariant conull set X_{es} with the following property. Let $x, y \in X_{\mathrm{es}}$ be sufficiently close and suppose $y = gx$ with $g \in L_{ab}$ sufficiently close to e. If x and y occur as an A'_{ab} return pair, then $g \in C_{ab}$.*

Proof. By permuting coordinates it suffices to treat $(a, b) = (1, 2)$. In a small identity neighborhood, write g in matrix coordinates adapted to the inclusion

$$C_{12} \subset L_{12} = C_G(U_{12}).$$

The defining commutation equation with U_{12} shows that the quotient coordinates of L_{12}/C_{12} are precisely the entries coupling the first two coordinates asymmetrically to the remaining coordinates. Consequently there is a diagonal one-parameter subgroup $c_t \in A$ for which these quotient coordinates are multiplied by strictly negative weights, whereas the C_{12} coordinates have weight zero. Thus, for $g \in L_{12}$,

$$d(c_t g c_t^{-1}, C_{12}) \longrightarrow 0 \qquad (t \to +\infty).$$

Choose a countable family of compact injectivity sets D_m exhausting X. For fixed rational return data and fixed m, Poincaré recurrence for c_t provides arbitrarily large t for which both $c_t x$ and $c_t y$ return to D_m. The quotient-coordinate function

$$f(g) = \max\{|z_\ell(g)| : z_\ell \text{ is an } L_{12}/C_{12} \text{ coordinate}\}$$

is continuous, nonnegative, and satisfies $f(c_t g c_t^{-1}) \to 0$. On a recurrent pair the local displacement is unique, so the same return branch reappears after applying c_t. If $f(g) > 0$, its strict contraction would force the recurrent branch through infinitely many distinct values converging to the C_{12} branch, contradicting uniqueness on a sufficiently small injectivity chart. Hence $f(g) = 0$ and $g \in C_{12}$.

The argument has been written for fixed rational return data. Taking the intersection of the corresponding conull sets over the countable compact exhaustion, rational time vectors, and rational injectivity radii gives one conull set on which the conclusion holds for all returns; arbitrary return times follow by approximation and continuity of the local displacement. □

Lemma 1.2 (Compact Poincaré returns). *Let ν be an A'_{ab}-invariant probability and K a compact set with $\nu(K) > 0$. For ν-almost every $x \in K$ there are unbounded $a_n \in A'_{ab}$ for which $a_n x \in K$.*

Proof. Choose a noncompact one-parameter subgroup $\{a^n\} \subset A'_{ab}$. Poincaré recurrence for the probability-preserving map a says that almost every point of K returns to K infinitely often. Passing to return times tending to infinity gives the claim. □

Theorem 1.3 (Strong-return/centralizer-component equivalence). *Let $\mathcal{I}_{ab}$ be the A'_{ab}-invariant sigma-algebra and let*

$$\mu = \int \mu_\omega \, d\eta(\omega)$$

be the A'_{ab}-ergodic decomposition. The following are equivalent.

(1) For η-almost every ω, the measure μ_ω is supported on one C_{ab}-orbit.
(2) For every $\varepsilon > 0$ there is a compact $K \subset X_{\mathrm{es}}$ with $\mu(K) > 1 - \varepsilon$ and an $r > 0$ such that every A'_{ab} return pair $x, \alpha^{\mathbf{s}} x \in K$ of distance $< r$ has local displacement in C_{ab}.

If (2) fails, then on a positive-measure family of ergodic components there are arbitrarily small nonexceptional returns.

Proof. Assume (1). On an ergodic component supported by $C_{ab}x$, the A'_{ab} action commutes with C_{ab}. On a compact subset of the orbit where the quotient map is injective, two sufficiently close points belonging to the same A'_{ab} return branch therefore differ by an element of C_{ab}. Lusin's theorem applied to the measurable map $\omega \mapsto \mu_\omega$ and a finite subcover of a compact exhaustion give, for any ε, one compact K and one uniform radius r on which this holds simultaneously for all but ε of the total measure. This proves (2).

Conversely assume (2). Choose a countable algebra generating $\mathcal{I}_{ab}$ and realize the ergodic decomposition on a standard Borel parameter space Ω. For each m take K_m as in (2) with $\mu(K_m) > 1 - 2^{-m}$ and radius r_m. By Borel–Cantelli, almost every point belongs to all sufficiently large K_m. Fix such a point x, recurrent for A'_{ab}, and let ω be its ergodic component. Every sufficiently small recurrent branch through x has displacement in C_{ab}.

Let

$$Z_x := \overline{C_{ab}x}.$$

The set Z_x is A'_{ab}-invariant because A'_{ab} centralizes C_{ab}. Suppose the support of μ_ω were not contained in $C_{ab}x$. Ergodicity and Poincaré recurrence would then give two positive measure flow boxes in the support and a return taking a typical point from one box to the other. Refining the boxes makes the associated local displacement arbitrarily small. The return cannot lie in C_{ab}, since the two boxes meet distinct local C_{ab} plaques. This contradicts the defining property of the K_m. Thus $\operatorname{supp}\mu_\omega \subset C_{ab}x$ for almost every ω.

The last assertion is the contrapositive after Lemma 1.1: if centralizer support fails on a positive-measure family of components, the compact strong-return property fails there; after deleting the null set on which exceptional returns need not yet be strong, the resulting arbitrarily small returns are nonexceptional. □

Proposition 1.4 (Ergodic-component form of the low-entropy alternative). *For an A'_{ab}-invariant probability, exactly one of the following occurs on almost every ergodic component: either the component is supported on one C_{ab}-orbit, or it possesses arbitrarily small nonexceptional recurrent pairs in every full-measure compact exhaustion.*

Proof. Apply Theorem 1.3 to the ergodic decomposition. Measurability follows from the countable compact exhaustion and a countable basis of identity neighborhoods. Lemma 1.1 ensures that, outside a single null set, an exceptional return cannot create a third alternative between L_{ab} and C_{ab}. □

Proof and provenance. This is the role of Lemma 4.2 and Proposition 4.3 in the low-entropy proof of Einsiedler–Katok–Lindenstrauss: exceptional returns are first upgraded to strong exceptional returns, and strong exceptional recurrence is then shown to be equivalent to support of almost every A^0_{ab} ergodic component on a single $C(H_{ab})$ orbit [EKL06].

Worked Example 1.5 (Why the two centralizers cannot be conflated)**.** A displacement in $L_{ab} = C_G(U_{ab})$ commutes with the active unstable root but may still move the opposite root U_{ba}. Such a return is invisible to one leafwise system but can still shear the SL_2 block. Only a displacement in $C_{ab} = C_G(H_{ab})$ is genuinely invisible to both opposite roots, which is why centralizer-orbit support is tied to *strong* exceptional returns.

CHAPTER 3

Low-Entropy Product and Inequality Lemmas

Proof architecture. The chapter is organized as the chain *Product conditional collapses to one root* $\longrightarrow$ *Typical-intersection rigidity* $\longrightarrow$ *Entropy cost of a transverse return.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. What zero neighboring entropy really buys

The isolated-root assumption is useful only after it is translated into a statement about conditional measures on a whole unipotent product.

Lemma 1.1 (Product conditional collapses to one root)**.** *In the isolated-root configuration, the conditional measure of μ on a sufficiently small $U_{(ab)}$ product box is, up to projective normalization, the product of the U_{ab} leafwise measure with Dirac masses in every other root coordinate.*

Proof. Order the root groups so that successive initial products are normalized by the diagonal element used to construct subordinate partitions. Apply the iterated disintegration formula of Lemma 1.1. A zero root coefficient gives a trivial rootwise conditional by Volume 14, hence a Dirac factor at the identity. All such factors disappear, leaving only the U_{ab} conditional. □

Theorem 1.2 (Typical-intersection rigidity). *There is a conull set X_0 such that if $x, y \in X_0$ and $y = ux$ with $u \in U_{(ab)}$ sufficiently close to the identity, then $u \in U_{ab}$.*

Proof. Choose a flow box on which the local $U_{(ab)}$ coordinates are injective and disintegrate μ there. By Lemma 1.1, for almost every base point the conditional gives full mass to the U_{ab} coordinate slice. Fubini therefore gives a conull set meeting almost every product leaf only inside that slice. If two points of this conull set lie on the same sufficiently small product leaf, their relative coordinate must lie in U_{ab}. □

Proposition 1.3 (Entropy cost of a transverse return)**.** *If a positive-measure set contains arbitrarily small recurrent pairs whose relative $U_{(ab)}$ coordinate is not in U_{ab} and whose leafwise conditional measures agree after translation, then μ is U_{ab}-invariant.*

PROOF. Such a pair contradicts Theorem 1.2 unless the projective U_{ab} leafwise measure has a nontrivial translation stabilizer. Proposition 1.3 makes that stabilizer a closed subgroup of $\mathbb{R}$. Recurrence under the expanding diagonal conjugations rescales any nonzero stabilizer element to elements arbitrarily close to zero, so the stabilizer is all of $\mathbb{R}$. Thus the leafwise measure is Haar, and Theorem 1.2 gives U_{ab}-invariance. $\square$

Worked Example 1.4 (A two-coordinate product)**.** If a local leaf is parametrized by $(r, s) \mapsto u_{12}(r)u_{13}(s)x$ and the U_{13} coefficient is zero, then the conditional is $d\mu_x^{12}(r) \otimes \delta_0(ds)$. A typical point cannot have $s \neq 0$. This elementary picture is the product-measure content of the low-entropy hypothesis.

CHAPTER 4

Noncommuting Root Groups in the Low-Entropy Regime

Proof architecture. The chapter is organized as the chain *Translation stabilizer criterion* $\longrightarrow$ *Distinct-conditionals theorem* $\longrightarrow$ *Root-commutator localization.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. Equality of leafwise measures is rigid

Lemma 1.1 (Translation stabilizer criterion)**.** *Let x, ux be typical points on the same U_{ab} leaf. If their normalized leafwise measures coincide, then u belongs to the translation stabilizer of μ_x^{ab}. If $u \neq e$ and its A-conjugates accumulate at the identity, then μ_x^{ab} is Haar.*

Proof. The leaf-translation rule 1.1 identifies the conditional at ux with the translate of the conditional at x. Equality therefore says exactly that u stabilizes the projective measure. The stabilizer is closed. A nonzero element together with a contracting diagonal element produces a sequence of nonzero stabilizers tending to the identity; a closed subgroup of $(\mathbb{R}, +)$ with this property is all of $\mathbb{R}$. Haar uniqueness finishes the proof. $\square$

Theorem 1.2 (Distinct-conditionals theorem). *Assume μ is not U_{ab}-invariant. On a conull set, two distinct points lying sufficiently close on the same U_{ab} leaf have distinct normalized U_{ab} leafwise measures.*

Proof. If two such points had equal conditionals, Lemma 1.1 would give a nontrivial translation stabilizer. The diagonal action contains elements contracting U_{ab}, so the second part of that lemma would force Haar leafwise measure, hence U_{ab}-invariance, contrary to hypothesis. $\square$

Proposition 1.3 (Root-commutator localization)**.** *For i, j, k distinct,*

$$[u_{ij}(r), u_{jk}(s)] = u_{ik}(rs),$$

and all other first-order commutators among root groups are either trivial or another root group. Consequently every transverse polynomial shear relevant to the isolated pair (a, b) *can be localized inside a rank-two or rank-three coordinate block.*

PROOF. Multiply the elementary matrices $I + rE_{ij}$ and $I + sE_{jk}$ and use $E_{ij}E_{jk} = E_{ik}$ while the reverse product vanishes. The general statement follows from $[E_{ij}, E_{pq}] = \delta_{jp}E_{iq} - \delta_{qi}E_{pj}$. □

Worked Example 1.4 (The Heisenberg triple)**.** In SL_3, U_{12} and U_{23} do not commute and their commutator is U_{13}. Thus a tiny U_{23} displacement, when conjugated by a long U_{12} segment, becomes visible in the U_{13} direction. Low entropy says such a visible transverse point cannot remain typical unless new invariance appears.

CHAPTER 5

Shearing of Leafwise Conditional Measures

Proof architecture. The chapter is organized as the chain *Exact polynomial conjugation and the visible coefficients* $\longrightarrow$ *Quantitative low-entropy H-property* $\longrightarrow$ *Robustness of the visible shear.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. The low-entropy H-property

The geometric input is not merely that two unipotent trajectories diverge polynomially. One needs a *visible* transverse displacement at a scale where the linear and quadratic terms cannot cancel. We write the calculation for $(a, b) = (1, 2)$; permutation of coordinates gives the general case.

Lemma 1.1 (Exact polynomial conjugation and the visible coefficients)**.** *Put $u(r) = I + rE_{12}$. For every matrix g,*

$$u(r)gu(-r) = g + r(E_{12}g - gE_{12}) - r^2 E_{12}gE_{12}.$$

Consequently

$$\big(u(r)gu(-r)\big)_{ij} = g_{ij} + r\delta_{i1}g_{2j} - rg_{i1}\delta_{j2} - r^2\delta_{i1}g_{21}\delta_{j2}.$$

For g near e, $g \in L_{12} = C_G(U_{12})$ exactly when all coefficients of the nonconstant polynomial $u(r)gu(-r)$ vanish. In particular, with

$$\kappa(g) = \max\left(|g_{21}|^{1/2},\ |g_{22} - g_{11}|, \max_{j\neq 2} |g_{2j}|, \max_{i\neq 1} |g_{i1}|\right),$$

one has $\kappa(g) > 0$ precisely when $g \notin L_{12}$, after shrinking the identity neighborhood.

Proof. Since $E_{12}^2 = 0$, $u(r)^{-1} = I - rE_{12}$ and direct multiplication gives the first formula. The entrywise formula follows from $(E_{12}g)_{ij} = \delta_{i1}g_{2j}$ and $(gE_{12})_{ij} = g_{i1}\delta_{j2}$. Commutation with every $u(r)$ is equivalent to $E_{12}g = gE_{12}$, which says that the displayed linear coefficients vanish; the g_{21} coefficient is then zero as well. These are exactly the local equations of L_{12}. The weighted maximum κ records linear coefficients with weight one and the quadratic coefficient with weight one half, so it vanishes exactly on that centralizer. $\square$

Lemma 1.2 (Two-scale no-cancellation lemma). *There are absolute constants $M > 10$ and $0 < c < C < \infty$ with the following property. If g is sufficiently close to e and $\kappa = \kappa(g) > 0$, then for one of the two scales*

$$r \in \{\kappa^{-1}, M\kappa^{-1}\}$$

the local product decomposition of $u(r)gu(-r)$ has a transverse $U_{(12)}/U_{12}$ coordinate of size at least c, while all coordinates used in the rank-three block calculation are bounded by CM^2. After multiplying r by a fixed sufficiently small constant, the whole conjugate lies in a fixed injectivity neighborhood.

Proof. All transverse entries except the $(1,2)$ entry are affine linear functions of r with a coefficient represented in κ. Such a coefficient is already visible at $r = \kappa^{-1}$. The only possible cancellation involving two different degrees occurs in the $(1,2)$ entry,

$$g_{12} + r(g_{22} - g_{11}) - r^2 g_{21}.$$

Normalize $r = s/\kappa$. The nonconstant part is a polynomial $p(s) = As - Bs^2$ with $\max(|A|, |B|) = 1$ up to fixed local-coordinate constants. A quadratic polynomial with a zero at $s = 1$ cannot also be arbitrarily small at $s = M$: indeed if $|p(1)| < c$ and $|p(M)| < c$, solving the two linear inequalities for A, B gives $\max(|A|, |B|) \ll_M c$. Choosing c smaller than the reciprocal of this constant is a contradiction. Thus one of $s = 1, M$ gives a transverse value bounded below. The same formulas bound all relevant entries by $O(M^2)$. Multiplying both candidate scales by a fixed small factor places the resulting conjugates in a common local product chart without destroying the lower bound, after adjusting c. $\square$

Theorem 1.3 (Quantitative low-entropy H-property). *Let $g_n \to e$ with $g_n \notin L_{ab}$. After passing to a subsequence there are $r_n \to \infty$ and local product decompositions*

$$u_{ab}(r_n) g_n u_{ab}(-r_n) = v_n c_n,$$

where c_n belongs to the locally neutral/central factor, $v_n \in U_{(ab)}$, and

$$v_n \longrightarrow v \in U_{(ab)} \setminus U_{ab}.$$

Moreover r_n may be chosen from two scales whose ratio is the fixed constant M of Lemma 1.2.

Proof. Permute coordinates so that $(a, b) = (1, 2)$ and apply Lemma 1.2. The local multiplication map from the ordered root factors occurring in $U_{(12)}$, followed by the neutral factor, is a diffeomorphism near the identity. Hence the uniformly bounded conjugates have, after a subsequence, convergent product coordinates. At least one transverse root coordinate is bounded away from zero by the lemma, so the limiting $U_{(12)}$ coordinate is not in U_{12}. Since $\kappa(g_n) \to 0$, both candidate scales tend to infinity. The neutral coordinates may be absorbed into the return-time/central factor in the low-entropy application; what survives in the local product leaf is the stated $v \notin U_{12}$. The same argument works in the three-index block attached to any (a, b). □

Proposition 1.4 (Robustness of the visible shear)**.** *The conclusion of Theorem 1.3 is unchanged under $o(1)$ perturbations of the normalized scale and under coordinate errors $o(r_n^{-2})$ in the initial return pair.*

PROOF. Every entry in Lemma 1.1 is a polynomial of degree at most two in r. An $o(1)$ relative change of the normalized scale changes a bounded normalized polynomial by $o(1)$, while an $o(r_n^{-2})$ coordinate error contributes $o(1)$ even after multiplication by the quadratic factor r_n^2. Local product coordinates depend smoothly on the matrix, so the limiting transverse factor is unchanged. □

> **Proof and provenance.** The matrix calculation and, especially, the two-scale device reproduce the logical role of the H-property estimates in the low-entropy part of Einsiedler–Katok–Lindenstrauss. A single visible scale is not sufficient: the linear and quadratic terms can cancel, so the proof deliberately retains two separated candidate scales [EKL06].

Worked Example 1.5 (Why one scale can fail)**.** For the $(1, 2)$ entry the nonconstant displacement is $r(g_{22} - g_{11}) - r^2 g_{21}$. If $g_{22} - g_{11} = r_0 g_{21}$, this expression vanishes at $r = r_0$ even though $g \notin L_{12}$. At the second scale Mr_0 it equals $M(1 - M)r_0^2 g_{21}$ and is visible. The two-scale lemma is therefore a real anti-cancellation step, not a technical decoration.

CHAPTER 6

Production of Unipotent Invariance

Proof architecture. The chapter is organized as the chain *Simultaneous good compacta* $\longrightarrow$ *Nonexceptional recurrence forces* U_{ab}*-invariance* $\longrightarrow$ *Opposite invariance and the* SL_2 *block*. The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. Maximal inequalities and the three-good-set argument

The deterministic H-property of Chapter 5 produces two candidate shearing scales. The measure-theoretic task is to choose one at which both translated endpoints are typical and the leafwise conditional is observed away from a tiny central interval. This is the point at which two maximal inequalities enter the proof.

Lemma 1.1 (Leafwise weak-$(1,1)$ maximal inequality)**.** *Let* $U_{ab} \cong (\mathbb{R}, +)$ *and let* $E \subset X$ *be measurable. For* $0 < \alpha < 1$ *put*

$$\mathcal{M}_{E,\alpha} = \left\{ x : \exists R > 0, \frac{\mu_x^{ab}(\{r \in [-R, R] : u_{ab}(r)x \in E\})}{\mu_x^{ab}([-R, R])} > \alpha \right\}.$$

There is an absolute constant C *such that*

$$\mu(\mathcal{M}_{E,\alpha}) \le C\alpha^{-1}\mu(E).$$

The assertion is projectively well defined.

Proof. Work first in one U_{ab} flow box and disintegrate μ into a transverse measure and one-dimensional plaque conditionals. On a fixed plaque the statement is the centered Hardy–Littlewood weak-$(1,1)$ inequality for the Radon measure μ_x^{ab}: choose, for every bad point, an interval on which the E-density exceeds α; the one-dimensional Vitali selection gives a disjoint subfamily whose fivefold dilates cover the bad set. Summing the conditional masses yields at most $C\alpha^{-1}$ times the conditional mass of E on the plaque. Integrate over the transverse parameter. A countable flow-box atlas and monotone exhaustion remove the box restriction. Both numerator and denominator scale by the same constant when the representative of $[\mu_x^{ab}]$ changes, so the ratio and the estimate are projective. $\square$

Lemma 1.2 (Time maximal inequality)**.** *Let T be an invertible probability-preserving transformation. For $f \in L^1(\mu)$ and $\lambda > 0$,*

$$\mu\left\{x : \sup_{N\geq 1} \frac{1}{N}\sum_{n=0}^{N-1} f(T^n x) > \lambda\right\} \leq \frac{\|f^+\|_1}{\lambda}.$$

The same conclusion, with an absolute constant, holds for a one-parameter subgroup after averaging over unit time intervals.

Proof. For the discrete transformation this is Hopf's maximal ergodic inequality. For completeness, apply the elementary inequality

$$\int_{\{\max_{1\leq m\leq N} S_m f>0\}} f\,d\mu \geq 0, \qquad S_m f = \sum_{j=0}^{m-1} f\circ T^j,$$

which follows by decomposing the maximal set according to the first positive partial sum and translating each piece by the corresponding iterate of T. Apply it to $f-\lambda$ and let $N\to\infty$. Continuous time follows by replacing f by $\int_0^1 f(a_t\cdot)\,dt$ and comparing an arbitrary time interval with its integer subintervals. $\square$

Lemma 1.3 (Two-scale nonconcentration set)**.** *Assume the U_{ab} leafwise measure is nontrivial and non-Haar. There are $0<\rho<10^{-3}$ and a positive-measure set $X(\rho)$ such that for every $x\in X(\rho)$ one can choose $S=S(x)>0$ with*

$$\mu_x^{ab}([-\rho S,\rho S]) < \tfrac{1}{2}\mu_x^{ab}([-S,S]).$$

After applying a suitable element of A normalizing U_{ab}, the same inequality may be required at the second scale $\rho^{-5}S$ on a positive-measure subset.

Proof. Normalize μ_x^{ab} on the fixed ball of Volume 13. By Theorem 1.2, nontriviality implies non-atomicity, so $\mu_x^{ab}([-\rho,\rho])\to 0$ as $\rho\downarrow 0$ for almost every x. Hence for almost every such x some rational ρ gives the first strict inequality; countability supplies one common ρ on a positive-measure set. Conjugation by a diagonal element rescales U_{ab}, and projective normalizer equivariance from Volume 13 transports the ratio to any desired multiplicative scale. Poincaré recurrence and the time maximal inequality allow the original and rescaled nonconcentration events to be imposed simultaneously on a positive-measure subset, giving the two separated scales. $\square$

Lemma 1.4 (Simultaneous good compacta)**.** *Assume μ is not U_{ab}-invariant. For every sufficiently small $\varepsilon>0$ there are compact sets K_1,K_2,K_3 with total complement of measure $O(\varepsilon)$ such that:*

(1) the normalized map $x \mapsto \mu_x^{ab}$ is weak- continuous on K_1;*
(2) for $x \in K_2$, at every leafwise radius under consideration, at least a proportion $1-\varepsilon$ of the μ_x^{ab}-mass consists of parameters r for which $u_{ab}(r)x \in K_1$;
(3) K_3 consists of recurrent points for the relevant A'_{ab} and rescaling transformations and contains a positive-measure subset of the two-scale set $X(\rho)$ of Lemma 1.3.

Proof. Lusin gives K_1 with complement of arbitrarily small measure. Apply Lemma 1.1 to $E = X \setminus K_1$ with threshold ε: outside a set of measure $O(\varepsilon^{-1}\mu(E))$, no leafwise interval has more than ε of its conditional mass in E. Choosing $\mu(E) \ll \varepsilon^2$ gives K_2. Lemma 1.2, applied to the complements of the recurrence/nonconcentration sets, supplies K_3 with the asserted uniform return property. Intersecting with the positive-measure set from Lemma 1.3 still leaves positive measure when the errors are chosen in this order. □

Theorem 1.5 (Nonexceptional recurrence forces U_{ab}-invariance). *In the isolated-root configuration, if a positive-measure family of A'_{ab}-ergodic components has arbitrarily small nonexceptional returns, then μ is U_{ab}-invariant on the corresponding A-ergodic component.*

Proof. Assume the contrary. Choose K_1, K_2, K_3 from Lemma 1.4 and choose nonexceptional return pairs $x_n, y_n = g_n x_n$ in their common good part with $g_n \to e$ and $g_n \notin L_{ab}$. Let R_n and MR_n be the two candidate H-property scales furnished by Theorem 1.3.

The two-scale nonconcentration inequality says that a fixed positive fraction of the conditional mass lies outside the central interval at one of these scales. The leafwise maximal property of K_2 says that only an $O(\varepsilon)$ fraction of the same conditional mass sends either endpoint outside K_1. Taking ε smaller than the nonconcentration fraction therefore gives a parameter r_n, at one of the two separated scales and outside the cancellation interval, for which

$$u_{ab}(r_n)x_n, \qquad u_{ab}(r_n)y_n \in K_1.$$

This is precisely the selection step for which a single-scale argument is insufficient.

Pass to a subsequence so that the two points converge in K_1. By the quantitative H-property their relative local product coordinate converges to $v \in U_{(ab)} \setminus U_{ab}$. On the other hand, normalizer equivariance transports the two U_{ab} conditionals by the same leafwise rescaling. Their pre-shearing base points converge together, and both post-shearing points lie in the Lusin set K_1; weak-* continuity therefore gives equal projective U_{ab} conditionals at the two limit points.

The limits are typical. Theorem 1.2 says that two typical points on the same local $U_{(ab)}$ product leaf can differ only by U_{ab}. The limiting relative coordinate $v \notin U_{ab}$ contradicts this unless the U_{ab} conditional has acquired a nontrivial translation stabilizer. In that remaining case Theorem 1.2 and Proposition 1.3 imply that the leafwise measure is Haar, hence Theorem 1.2 gives U_{ab}-invariance. Either way the assumption of noninvariance is impossible. □

Proposition 1.6 (Opposite invariance and the SL_2 block)**.** *If the preceding argument applies to both (a, b) and (b, a), then μ is H_{ab}-invariant.*

Proof. It gives invariance under both opposite root groups U_{ab} and U_{ba}. These generate the standard subgroup $H_{ab} \cong SL_2(\mathbb{R})$, so invariance extends to H_{ab}. □

> **Proof and provenance.** The proof now separates the three ingredients that are separate in the primary low-entropy argument: the leafwise weak maximal inequality, the usual maximal ergodic selection in the diagonal direction, and the two separated leafwise scales that prevent cancellation of the linear and quadratic H-property terms [EKL06].

Worked Example 1.7 (Why all three good sets are necessary)**.** Lusin continuity alone does not put a sheared endpoint back at a continuity point. Recurrence alone may return at a scale where the leafwise mass is concentrated in the tiny interval that causes cancellation. The leafwise maximal estimate alone does not supply the diagonal return. The proof works only after these three independent requirements are imposed simultaneously.

CHAPTER 7

Rank-Two Model Arguments

Proof architecture. The chapter is organized as the chain *Centralizer calculation in $SL_3 \longrightarrow$ Rank-two low-entropy model $\longrightarrow$ Embedding the model in higher rank.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. The SL_3 laboratory

Lemma 1.1 (Centralizer calculation in SL_3). *For $H_{12} \cong SL_2$ in the upper-left block of SL_3, the identity component of its centralizer consists of*

$$\operatorname{diag}(\lambda, \lambda, \lambda^{-2}), \qquad \lambda > 0.$$

Its Lie algebra is $\mathbb{R}\operatorname{diag}(1, 1, -2)$.

Proof. A matrix commuting with the irreducible standard SL_2 action on $\mathbb{R}^2$ restricts to a scalar on that plane by Schur's elementary commutant argument. It cannot mix the fixed third coordinate with the nontrivial SL_2 module. Determinant one forces the third scalar to be λ^{-2}. □

Theorem 1.2 (Rank-two low-entropy model). *Let μ be A-ergodic on $SL_3(\mathbb{R})/\Gamma$ with $s_{12} > 0$ and $s_{13} = s_{32} = 0$. Then either μ is H_{12}-invariant or almost every A'_{12}-ergodic component is supported on a single orbit of the one-dimensional centralizer in Lemma 1.1.*

Proof. The return dichotomy of Theorem 1.3 gives either centralizer-orbit support or nonexceptional returns. In the latter case Theorem 1.5 gives U_{12}-invariance. Opposite entropy is positive; applying the same low-entropy argument to $(2, 1)$ gives U_{21}-invariance unless the corresponding component is centralizer-supported. The centralizers agree, so outside the exceptional alternative both opposite root groups act invariantly and generate H_{12}. □

Proposition 1.3 (Embedding the model in higher rank). *Every isolated-root calculation in SL_k is contained in an embedded SL_3 or $SL_2 \times \mathbb{R}$ coordinate* *subsystem, and the constants in the polynomial-shearing step are independent of k once the three relevant indices are fixed.*

PROOF. The commutator $[E_{ab}, E_{bj}]$ or $[E_{ia}, E_{ab}]$ uses at most three indices. All other coordinates commute with the calculation. The adjoint polynomial and its degree therefore coincide with the corresponding 3×3 block computation. $\square$

Worked Example 1.4 (The centralizer orbit is a real obstruction)**.** The orbit of $\mathrm{diag}(e^t, e^t, e^{-2t})$ can itself carry a finite invariant measure when it closes. Along such an orbit all returns centralize H_{12}, so the shearing mechanism genuinely has no transverse direction to exploit. The arithmetic lattice argument in the next chapter is what rules this out for $SL_k(\mathbb{Z})$ in the EKL setting.

CHAPTER 8

The Low-Entropy Rigidity Theorem

Proof architecture. The chapter is organized as the chain *Repeated-eigenvalue obstruction in* $SL_k(\mathbb{Z}) \longrightarrow$ *Low-entropy rigidity theorem.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. The exact dichotomy and the arithmetic exclusion

Lemma 1.1 (Stability of a separated positive spectrum)**.** *Let* $d = \operatorname{diag}(e^{\lambda_3}, \ldots, e^{\lambda_k})$ *with* $|\lambda_i - \lambda_j| > 2\eta$ *for* $i \neq j$*. There is a neighborhood* $\mathcal{U}$ *of the identity such that, for* $h \in \mathcal{U}$*, the eigenvalues of* hd *are positive real numbers* $e^{\lambda'_i}$ *which can be numbered so that* $|\lambda'_i - \lambda_i| < \eta$*.*

Proof. The coefficients of the characteristic polynomial depend continuously on the matrix entries. Choose pairwise disjoint real intervals $I_i = (e^{\lambda_i - \eta}, e^{\lambda_i + \eta})$. The diagonal matrix has one simple root in each I_i. On the boundary of a sufficiently small disjoint complex disc around each root, the characteristic polynomial is bounded away from zero. For h close to the identity the perturbed characteristic polynomial is uniformly close there; Rouché's theorem gives exactly one root in each disc. Since the perturbed polynomial has real coefficients, a nonreal root would bring its conjugate into the same disc, impossible. Hence the roots are real; shrinking the discs keeps them positive and gives the logarithmic bound. $\square$

Theorem 1.2 (Centralizer support creates a forbidden lattice element). *Let* $\Gamma < SL_k(\mathbb{R})$ *be discrete. If an* A'_{12}*-ergodic component is a finite invariant measure on a single* $C_G(H_{12})$*-orbit, then* Γ *contains an element which is real diagonalizable, has exactly one eigenvalue of multiplicity two, all other eigenvalues simple, and has neither* 1 *nor* -1 *as an eigenvalue.*

Proof. Write a typical point as $x = g\Gamma$. The identity component of $C_G(H_{12})$ consists of matrices

$$c(r,h) = \begin{pmatrix} e^r I_2 & 0 \\ 0 & e^{-2r/(k-2)}h \end{pmatrix}, \qquad h \in SL_{k-2}(\mathbb{R}).$$

Choose a regular one-parameter subgroup

$$a(t) = \operatorname{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_k t}) \in A'_{12}, \quad \lambda_1 = \lambda_2,$$

with all other λ_i pairwise distinct and also distinct from λ_1, and none zero. Take t_0 so large that every distinct pair $|\lambda_i - \lambda_j| t_0 > 4$. By Poincaré recurrence of the finite centralizer-orbit measure, for almost every x there are $t > t_0$ and $c \in C_G(H_{12})$ arbitrarily close to the identity such that $ca(t)x = x$. Thus

$$\widetilde{\gamma} = ca(t) \in g\Gamma g^{-1}.$$

Its first 2×2 block is the scalar $e^s I_2$, where $|s - \lambda_1 t|$ is as small as desired. The lower block is a small perturbation of $\operatorname{diag}(e^{\lambda_3 t}, \dots, e^{\lambda_k t})$ after the common determinant correction. Lemma 1.1, with the large spectral gaps supplied by $t > t_0$, shows that the lower-block eigenvalues are positive, real, simple, and remain separated from e^s. Increasing t_0 also keeps every eigenvalue away from 1; positivity keeps them away from -1. Consequently $g^{-1}\widetilde{\gamma} g \in \Gamma$ has exactly the asserted spectrum. □

Lemma 1.3 (Repeated-eigenvalue obstruction in $SL_k(\mathbb{Z})$)**.** *There is no $\gamma \in SL_k(\mathbb{Z})$ whose real eigenvalues are all simple except for exactly one eigenvalue λ of multiplicity two, with ± 1 not eigenvalues.*

Proof. If λ is a repeated root of the characteristic polynomial $p_\gamma \in \mathbb{Z}[T]$, then its irreducible minimal polynomial m_λ divides both p_γ and p'_γ. Hence m_λ^2 divides p_γ. Every algebraic conjugate of λ is therefore also a repeated root. Since only one eigenvalue is repeated, m_λ has degree one, so $\lambda \in \mathbb{Q}$. Being an eigenvalue of an integral matrix, λ is an algebraic integer, hence an integer. Since $\det \gamma = 1$, a rational eigenvalue is an algebraic unit in $\mathbb{Z}$, so $\lambda = \pm 1$, contradiction. □

Theorem 1.4 (Low-entropy rigidity theorem). *Let μ be an A-invariant ergodic probability on $SL_k(\mathbb{R})/\Gamma$, and suppose (a,b) is isolated with $s_{ab} > 0$. Then either*

(1) μ is U_{ab}-invariant; or
(2) almost every A'_{ab}-ergodic component is supported on a single $C_G(H_{ab})$-orbit.

For $\Gamma = SL_k(\mathbb{Z})$ the second alternative is impossible; applying the theorem to both orientations therefore gives H_{ab}-invariance.

PROOF. The general dichotomy is Theorem 1.3 followed by Theorem 1.5. For the arithmetic lattice, the exceptional alternative and Theorem 1.2 would produce an element of $SL_k(\mathbb{Z})$ with exactly the eigenvalue pattern ruled out by Lemma 1.3. Hence the exceptional alternative is impossible and U_{ab} acts invariantly. Opposite-root symmetry gives $s_{ba} > 0$ and the same argument gives U_{ba}-invariance; the two groups generate H_{ab}. □

> **Proof and provenance.** The centralizer-recurrence step above follows the proof of EKL Theorem 5.1: choose a regular A'_{ab} direction, recur into a small centralizer neighborhood, and use spectral stability to obtain exactly one repeated eigenvalue. Proposition 5.2 is the integral characteristic-polynomial argument of Lemma 1.3 [EKL06].

Worked Example 1.5 (Why the arithmetic conclusion is stronger)**.** On a general lattice quotient a closed centralizer orbit can support an A'_{ab}-invariant probability, so the exceptional alternative cannot simply be deleted. For $SL_k(\mathbb{Z})$, recurrence on such an orbit would manufacture an integral matrix with one forbidden double eigenvalue. This is the precise point where arithmetic enters the low-entropy branch.

Volume 16

Einsiedler–Katok–Lindenstrauss

Volume 16 scope and prerequisites

This volume assembles the rootwise entropy theory of Volumes 13–14 and the low-entropy theorem of Volume 15 into the positive-entropy measure-classification theorem of Einsiedler–Katok–Lindenstrauss. The target is stated at its actual strength: for $k \geq 3$, an A-invariant ergodic probability on $SL_k(\mathbb{R})/SL_k(\mathbb{Z})$ with positive entropy is algebraic; it is not asserted to be Haar in every composite rank. The prime-k Haar consequence is recorded separately [EKL06].

Student orientation: assembling EKL without hiding the two branches

An A-invariant probability is called *algebraic* when it is the invariant probability on a closed finite-volume orbit of an algebraic subgroup (up to the finite-component conventions stated in the classification chapter). The theorem proved here is not "positive entropy implies Haar" in every rank. Positive entropy first produces nontrivial unipotent invariance. Ratner–Margulis–Tomanov classification then produces a homogeneous algebraic measure, and a separate arithmetic block analysis lists which algebraic subgroups can occur. Haar measure is forced only in the stated special cases, such as the prime-dimensional consequence. The high-entropy and low-entropy branches are logically independent proved inputs from Volumes 14 and 15. This volume's job is to assemble them and then do the subgroup arithmetic that neither branch contains.

Reader guide: a concrete model

Why this matters.

Volume 16 is organized around the passage from *The $SL_k/SL_k(\mathbb{Z})$ Higher-Rank Setup* to *The EKL Positive-Entropy Measure-Classification Theorem.* The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.6 (A calculation to keep in view)**.** Let

$$a = \operatorname{diag}(e^{2s}, e^{-s}, e^{-s}) \in \mathrm{SL}_3(\mathbb{R})$$

and let E_{12} be the matrix unit. Since

$$\operatorname{Ad}(a)E_{12} = aE_{12}a^{-1} = e^{3s}E_{12},$$

the root exponent on the E_{12} direction is $3s$. For a one-dimensional leafwise measure whose conditional scaling is Lebesgue, the Jacobian contribution along this root direction is therefore $3s$. This calculation is the concrete local model behind the root-by-root entropy bookkeeping used later.

CHAPTER 1

The $SL_k/SL_k(\mathbb{Z})$ Higher-Rank Setup

Proof architecture. The chapter is organized as the chain *Root blocks generate semisimple coordinate groups* $\longrightarrow$ *Root-data package* $\longrightarrow$ *Coordinate-block inheritance.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices. Let $G = SL_k(\mathbb{R})$, $\Gamma = SL_k(\mathbb{Z})$, $X = G/\Gamma$, and A the positive diagonal subgroup.

Lemma 0.1 (Root blocks generate semisimple coordinate groups)**.** *For an equivalence relation $\sim$ on $\{1, \ldots, k\}$, the root groups U_{ij} with $i \sim j$ generate a connected subgroup*

$$H_\sim \cong \prod_{B \in \mathcal{B}} SL_{|B|}(\mathbb{R}),$$

embedded block diagonally over the equivalence classes B of size at least two.

Proof. Within each class, the elementary root groups generate $SL_{|B|}$ by Gaussian elimination. Root groups supported on disjoint coordinate blocks commute, and determinant one holds separately on each generated block. Thus the generated group is the stated direct product. □

Theorem 0.2 (Root-data package). *For every A-ergodic probability μ on X there are rootwise coefficients $s_{ij} \in [0, 1]$ with: entropy given by the Volume-XIV rootwise formula; $s_{ij} > 0$ iff the U_{ij} leafwise measure is nontrivial; $s_{ij} = 1$ iff it is Haar; and $s_{ij} = s_{ji}$.*

Proof. These are respectively Theorem 1.3, the nontriviality criterion from Volume 14, Proposition 1.4, and Corollary 1.2. All are backward proved inputs. □

Proposition 0.3 (Coordinate-block inheritance)**.** *If μ is invariant under all U_{ij} with $i \sim j$, then it is invariant under $H_\sim$.*

Proof. The invariance group of a probability is closed. It contains the generating root groups and therefore contains their generated connected subgroup from Lemma 0.1. □

Worked Example 0.4 (A $2+2$ block in SL_4)**.** If the invariant roots are $U_{12}, U_{21}, U_{34}, U_{43}$, then $H_{\sim} = SL_2 \times SL_2$ in the two diagonal blocks. Such a proper semisimple subgroup explains why the final EKL theorem says "algebraic" rather than "Haar" in all composite dimensions.

CHAPTER 2

Entropy Positivity Criteria

Proof architecture. The chapter is organized as the chain *Positive entropy detects a root* $\longrightarrow$ *Entropy/root conditional equivalence* $\longrightarrow$ *Positive entropy survives ergodic restriction.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

Lemma 0.1 (Positive entropy detects a root)**.** *If $h_\mu(a) > 0$ for some $a \in A$, then $s_{ij} > 0$ for at least one root expanded by a.*

PROOF. The rootwise entropy formula is a finite sum of nonnegative terms $(t_i - t_j)^+ s_{ij}$. A positive sum has a positive summand. $\square$

THEOREM 0.2 (Entropy/root conditional equivalence). *For an A-ergodic probability, the following are equivalent: some $a \in A$ has positive entropy; some $s_{ij} > 0$; some rootwise leafwise measure is nontrivial.*

PROOF. The first implies the second by Lemma 0.1; the second and third are equivalent by the root-data package. If $s_{ij} > 0$, choose a with $t_i > t_j$; the corresponding term in the entropy formula is positive. $\square$

Proposition 0.3 (Positive entropy survives ergodic restriction)**.** *If an A-invariant probability has positive entropy for one $a \in A$, then a positive-measure set of its A-ergodic components has positive entropy for a.*

PROOF. Entropy is affine under the ergodic decomposition for a fixed transformation. A positive integral of nonnegative component entropies requires a positive-measure set of positive terms. $\square$

Worked Example 0.4 (Reading entropy from one diagonal element)**.** In SL_3, for $a = \mathrm{diag}(e^2, e^{-1}, e^{-1})$, only roots with tail 1 are expanded. Positive entropy therefore certifies a nontrivial conditional on U_{12} or U_{13}, after which the high/low branching is entirely combinatorial.

CHAPTER 3

The High-Entropy Branch of EKL

Proof architecture. The chapter is organized as the chain *Shared-index trigger* $\longrightarrow$ *High-entropy branch* $\longrightarrow$ *No hidden low-entropy input*. The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

Lemma 0.1 (Shared-index trigger)**.** *If $s_{ab} > 0$ and a second positive coefficient shares the tail a or head b, then μ is H_{ab}-invariant.*

PROOF. Volume 14 proves the following implication in the same type-A root notation: if $s_{ab} > 0$ and another positive coefficient has the form $s_{ac} > 0$ or $s_{cb} > 0$ with the indicated distinct indices, then the two corresponding coarse leafwise systems are nontrivial and noncommuting. The commutator-shearing theorem of that volume produces a nonzero translation stabilizer in the third root direction; the entropy-cardinality argument then upgrades the U_{ab} and U_{ba} conditionals to Haar. By the Haar-leafwise-measure criterion of Volume 13, μ is invariant under both opposite root groups, hence under the embedded group $H_{ab} \simeq SL_2$ they generate. These are exactly the hypotheses of Theorem 1.3, so no low-entropy recurrence input is being imported here. $\square$

THEOREM 0.2 (High-entropy branch). *If the directed graph of positive root coefficients contains two edges meeting at a vertex in the orientation required by the type-A commutator relation, then μ has nontrivial connected unipotent invariance. Iterating root commutators enlarges it to the block group attached to the equivalence relation generated by those edges.*

PROOF. The first assertion is Lemma 0.1. Once an H_{ab} block acts invariantly, every root inside that block has Haar conditional. Volume 14's saturation theorem then closes the invariant root set under commutators. Finite termination gives an equivalence relation on coordinate indices, and Proposition 0.3 gives the block group. $\square$

Proposition 0.3 (No hidden low-entropy input)**.** *The proof of Theorem 0.2 uses no recurrence under A'_{ab} and no centralizer-orbit exclusion.*

Proof. Its only inputs are the rootwise entropy formula, noncommuting-root shearing, opposite-root symmetry, and finite commutator saturation, all proved in Volume 14. This separation is essential for the later audit of the two EKL branches. □

Worked Example 0.4 (A chain closes to SL_3)**.** If $s_{12}, s_{23} > 0$, the high-entropy theorem produces invariance in an SL_2 block and the commutator $[U_{12}, U_{23}] = U_{13}$. Opposite roots follow, so the invariant root system becomes all six roots of SL_3 and hence the measure is Haar.

CHAPTER 4

The Low-Entropy Branch of EKL

Proof architecture. The chapter is organized as the chain *Isolated positive edge* $\longrightarrow$ *Low-entropy branch for the arithmetic lattice* $\longrightarrow$ *General-lattice warning.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

Lemma 0.1 (Isolated positive edge)**.** *If $s_{ab} > 0$ and the high-entropy trigger fails for (a, b), then (a, b) satisfies the isolated-root hypotheses of Volume 15.*

Proof. Failure of the trigger means every other root with tail a or head b has zero coefficient. This is precisely the definition used in Theorem 1.2. ☐

Theorem 0.2 (Low-entropy branch for the arithmetic lattice). *For $\Gamma = SL_k(\mathbb{Z})$, an isolated positive edge (a, b) forces H_{ab}-invariance.*

Proof. Apply Theorem 1.4. The arithmetic repeated-eigenvalue argument eliminates the centralizer-supported alternative and gives invariance under both U_{ab} and U_{ba}. ☐

Proposition 0.3 (General-lattice warning)**.** *For a general lattice Γ, the low-entropy theorem yields a genuine dichotomy rather than unconditional H_{ab}-invariance.*

Proof. The exceptional alternative in Theorem 1.4 is a finite invariant measure on a centralizer orbit. The argument eliminating it uses the integral characteristic polynomial of an element of $SL_k(\mathbb{Z})$ and therefore is not available for an arbitrary lattice without additional arithmetic information. ☐

Worked Example 0.4 (Why $SL_k(\mathbb{Z})$ matters)**.** The high-entropy branch is Lie-theoretic and works on any lattice quotient. The isolated branch is different: it can end on a closed centralizer orbit. The integer lattice rules out exactly the eigenvalue pattern required by that orbit, turning a dichotomy into invariance.

CHAPTER 5

Production and Organization of Unipotent Invariance

Proof architecture. The chapter is organized as the chain *Every positive edge becomes an invariant block edge* $\longrightarrow$ *Invariant-root equivalence relation* $\longrightarrow$ *Saturation is canonical.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

Lemma 0.1 (Every positive edge becomes an invariant block edge)**.** *For $SL_k(\mathbb{Z})$, if $s_{ab} > 0$, then μ is H_{ab}-invariant.*

Proof. If a shared-index positive edge exists, use the high-entropy branch. Otherwise the edge is isolated and use the low-entropy branch. The two cases exhaust Theorem 1.2. □

Theorem 0.2 (Invariant-root equivalence relation). *Define $a \sim b$ when $a = b$ or μ is U_{ab}-invariant. Under positive entropy this is a nontrivial equivalence relation, and μ is invariant under the associated block group $H_\sim$.*

Proof. Reflexivity is by definition. Symmetry follows because H_{ab}-invariance contains both opposite roots. If $a \sim b$ and $b \sim c$, commutator closure gives U_{ac}-invariance, so the relation is transitive. Positive entropy supplies some positive coefficient and Lemma 0.1 makes its two indices equivalent. Finally Proposition 0.3 gives $H_\sim$-invariance. □

Proposition 0.3 (Saturation is canonical)**.** *The relation $\sim$ and group $H_\sim$ depend only on μ, not on the choice of the positive-entropy element $a \in A$ used to locate the first positive root.*

Proof. They are defined by the full invariance group of μ. The entropy element is only a device to show the relation is nontrivial. □

Worked Example 0.4 (Equivalence classes from invariant roots)**.** If in SL_5 the proof produces H_{12} and H_{23} invariance, then $1 \sim 2 \sim 3$ and commutators fill the entire SL_3 block on coordinates $1, 2, 3$. Coordinates $4, 5$ may remain singleton classes, giving a proper algebraic candidate.

CHAPTER 6

Margulis–Tomanov Reduction after Extra Invariance

Proof architecture. The chapter is organized as the chain *Maximality of the root-generated group* $\longrightarrow$ *Homogeneous component theorem in the required scope* $\longrightarrow$ *The homogeneous subgroup has no new unipotent directions.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices. Let $H = H_\sim$ be the maximal connected subgroup generated by root-unipotent groups which leaves μ invariant, and put $\widetilde{H} = AH$.

Lemma 0.1 (Maximality of the root-generated group)**.** *The group H is generated by one-parameter unipotents, is normalized by A, and is maximal among connected subgroups with these two properties which preserve μ.*

Proof. Generation and normalization follow from the block description. If a larger connected A-normalized subgroup generated by unipotents preserved μ, its Lie algebra would contain a new A-weight space. In type A that weight space is a root space $\mathbb{R}E_{ij}$, hence U_{ij} would preserve μ, contradicting the definition of the equivalence classes used to construct H. $\square$

Theorem 0.2 (Homogeneous component theorem in the required arithmetic scope). *Let $G = SL_k(\mathbb{R})$, $\Gamma = SL_k(\mathbb{Z})$, and let μ be the A-ergodic probability fixed in Chapter 1. Let $H = H_\sim$ be the connected block group of Chapter 5. Then there is a connected closed subgroup $L < G$, $H \leq L$, such that*

$$aLa^{-1} = L \qquad (a \in A),$$

and almost every H-ergodic component of μ is Haar probability on a closed finite-volume L-orbit. Moreover there is $x_0 \in G/\Gamma$ such that

$$\operatorname{supp}\mu \subset N_G(L)x_0.$$

Proof. Write the H-ergodic decomposition as

$$\mu = \int_\Omega \nu_\omega \, d\lambda(\omega).$$

Since H is connected and generated by root-unipotent one-parameter groups, Theorem 1.1 applies to almost every component. Replace its homogeneous group by the identity component of the measure stabilizer,

$$L_\omega := \mathrm{Stab}_G(\nu_\omega)^\circ.$$

Then $H \le L_\omega$, the support of ν_ω is one closed finite-volume L_ω-orbit, and

$$L_{a\omega} = aL_\omega a^{-1} \qquad (a \in A). \tag{16.6.1}$$

We next prove that L_ω is normalized by A for almost every ω. Choose finitely many regular elements $a_1, \ldots, a_r \in A$ whose logarithms span $\mathfrak{a}$ and such that the joint weight vectors on $\mathfrak{sl}_k$ separate all roots. Regard $\ell_\omega = \operatorname{Lie} L_\omega$ as a point in the finite union of Grassmannians of subspaces of $\mathfrak{g}$. The map $\omega \mapsto \ell_\omega$ is measurable because the stabilizer is characterized by the countable family of equations

$$\int f(\exp(tZ)y)\, d\nu_\omega(y) = \int f(y)\, d\nu_\omega(y) \quad (f \in \mathcal{D},\ t \in \mathbb{Q}),$$

for a fixed countable dense $\mathcal{D} \subset C_c(G/\Gamma)$. By Lusin's theorem it is continuous on compact sets whose λ-measure tends to one. Poincaré recurrence for each a_j, applied inside those compact sets, gives for almost every ω a sequence $n_q \to \infty$ with

$$\mathrm{Ad}(a_j^{n_q})\ell_\omega = \ell_{a_j^{n_q}\omega} \longrightarrow \ell_\omega. \tag{16.6.2}$$

Decompose $\mathfrak{g}$ into $\mathrm{Ad}(a_j)$-weight spaces. A subspace whose iterates under a diagonalizable operator return to itself in the Grassmannian must be the direct sum of its intersections with the equal-modulus weight blocks: otherwise the largest surviving weight in a Pluecker coordinate dominates and prevents recurrence. Applying this to (16.6.2), and then to the finite separating family a_j, shows that ℓ_ω is A-invariant. Hence A normalizes L_ω.

Equation (16.6.1) now says that the measurable map $\omega \mapsto L_\omega$ is A-invariant. The action induced by A on the H-ergodic component space is ergodic: the inverse image of an A-invariant subset of Ω is an AH-invariant, hence A-invariant, subset of G/Γ. Therefore $L_\omega = L$ almost everywhere for one connected closed A-normalized subgroup $L \ge H$.

It remains to locate the component supports. For $\nu_\omega = m_{Lg_\omega\Gamma}$, the arithmetic-hull theorem (Theorem 8.4) implies that $g_\omega^{-1}Lg_\omega$ is the real identity component of a character-free $\mathbb{Q}$-subgroup. There are only countably many such $\mathbb{Q}$-subgroups, since each is cut out by finitely many rational polynomials. Consequently the set of closed finite-volume L-orbits, modulo the left action of $N_G(L)$, is countable. The corresponding countable-valued class map on Ω is A-invariant because $A < N_G(L)$; A-ergodicity makes it constant almost everywhere. Thus all component supports lie in one normalizer orbit $N_G(L)x_0$, and taking their barycenter gives $\operatorname{supp}\mu \subset N_G(L)x_0$. $\square$

Proposition 0.3 (The homogeneous subgroup has no new unipotent directions)**.** *For the subgroup L from Theorem 0.2, the subgroup generated by the one-parameter unipotents of L equals H. In particular $[L, L] = H$ in the block situation.*

PROOF. Any one-parameter unipotent subgroup of L acts invariantly on the L-homogeneous H-components and hence, after integration, preserves μ. Since L is normalized by A, the A-weight closure of such unipotents is an A-normalized connected unipotent-generated subgroup preserving μ. Lemma 0.1 forces it into H. Conversely $H \leq L$. The derived group of a block reductive extension of H is therefore H. $\square$

Worked Example 0.4 (Why the strengthened component theorem is needed)**.** Ordinary Ratner classification only identifies one H-ergodic component at a time. EKL needs the same subgroup L for almost every component and must know that A normalizes it. The Margulis–Tomanov form supplies exactly this uniformity, avoiding an invalid appeal to countability of possible components.

CHAPTER 7

Classification of the Algebraic Candidates Produced by the Proof

Proof architecture. The chapter is organized as the chain *Escape when the root blocks do not fill all coordinates* $\longrightarrow$ *Algebraic candidate produced by the EKL argument* $\longrightarrow$ *Arithmetic block divisibility and the prime-dimensional consequence*. The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

Lemma 0.1 (Escape when the root blocks do not fill all coordinates)**.** *Let $H = \prod_{i=1}^{r} SL_{k_i}(\mathbb{R})$ be a standard block subgroup with $k' = \sum_i k_i < k$. If Hx is closed of finite volume in $SL_k(\mathbb{R})/SL_k(\mathbb{Z})$, there is a one-parameter subgroup $a(t) \subset A$ such that $a(t)x \to \infty$ as $t \to \infty$.*

Proof. Write $x = g\Gamma$. The Haar probability on the closed orbit Hx is ergodic under H, and H is generated by its root-unipotent one-parameter subgroups. Apply Volume 4, Theorem 8.4, to this homogeneous probability. As in that theorem's proof, the rational hull is contained in the $\mathbb{Q}$-Zariski closure of $g^{-1}Hg \cap \Gamma$, hence in the real algebraic group $g^{-1}Hg$; the reverse inclusion is the conclusion of the arithmetic-hull argument. Thus $g^{-1}Hg$ is the real identity component of a $\mathbb{Q}$-subgroup of SL_k.

Both the fixed space and the moved space

$$[g^{-1}Hg, \mathbb{Q}^k] = \mathrm{span}_{\mathbb{Q}}\{(h - I)v : h \in g^{-1}Hg,\ v \in \mathbb{Q}^k\}$$

are therefore rational subspaces. In the standard block model the second one is exactly the transported k'-dimensional moving block. Since $k' > 0$, it contains a nonzero vector of $\mathbb{Q}^k$; after clearing denominators we obtain a nonzero $m \in \mathbb{Z}^k$ such that gm lies entirely in the moving block.

Set

$$a(t) = \mathrm{diag}(\underbrace{e^{-(k-k')t}, \dots, e^{-(k-k')t}}_{k'}, \underbrace{e^{k't}, \dots, e^{k't}}_{k-k'}).$$

The determinant is one and $\|a(t)gm\| \to 0$. Thus the lattice $a(t)g\mathbb{Z}^k$ has a nonzero vector tending to zero. Mahler compactness, proved in Volume 1, says precisely that such a sequence leaves every compact subset of the space of unimodular lattices. Hence $a(t)x \to \infty$. □

THEOREM 0.2 (Algebraic candidate produced by the EKL argument). *The maximal block group H satisfies $\sum_i k_i = k$. Moreover μ is AH-invariant and supported on a single closed finite-volume AH-orbit. In particular μ is algebraic.*

PROOF. Let L be supplied by Theorem 0.2. Proposition 0.3 gives $[L, L] = H$. Since A normalizes L, the A-weight decomposition of $\mathfrak{l}$ cannot contain a root space outside the block root system of H: any such root space exponentiates to a new unipotent direction, contradicting maximality of H. Hence $L \subset AH$.

Fix an H-ergodic homogeneous component Lx supplied by Theorem 0.2, with $x = g\Gamma$. The action of H on this component is ergodic and the normal closure of its root-unipotent one-parameter groups is $H = [L, L]$. We may therefore apply the arithmetic-hull theorem already proved in Volume 4, Theorem 8.4, to the homogeneous probability on Lx. In the proof of that theorem the rational group is constructed inside the $\mathbb{Q}$-Zariski closure of $g^{-1}Lg \cap \Gamma$; because $g^{-1}Lg$ is itself a real algebraic subgroup, the resulting rational group has real identity component exactly $g^{-1}Lg$. Hence $g^{-1}Lg$ is the real identity component of a character-free $\mathbb{Q}$-group. Its derived group $g^{-1}Hg$ is therefore semisimple and defined over $\mathbb{Q}$, and the semisimple Borel–Harish–Chandra theorem, Theorem 6.1, gives

$$g^{-1}Hg \cap \Gamma \quad \text{as a lattice in } g^{-1}Hg.$$

Thus Hx is closed of finite volume. If $\sum k_i < k$, Lemma 0.1 gives a one-parameter subgroup of A sending every point in the corresponding support stratum to infinity. This contradicts Poincaré recurrence of the finite A-invariant measure. Therefore the blocks fill all coordinates.

When the blocks fill all coordinates, the subgroup of elements whose conjugation preserves the H-Haar class is exactly AH: permutations of equal blocks form only a finite extension, while the identity component is AH. The normalizer-support conclusion in Theorem 0.2 therefore places μ on a single orbit of $N_G(H)$. In the full-block case the identity component of this normalizer is AH and the remaining quotient is finite; A-ergodicity selects one finite component, so μ is supported on a single AH-orbit. Since μ is already AH-invariant and finite, that orbit carries its invariant probability; in particular it is closed and of finite volume. This is the required algebraic measure. □

Proposition 0.3 (Arithmetic block divisibility and the prime-dimensional consequence)**.** *Let μ be a positive-entropy algebraic measure furnished by Theorem 0.2. The blocks of H all have the same cardinality m, and $m \mid k$. Since positive entropy makes H nontrivial, $m > 1$. Consequently, if k is prime, then $m = k$, $H = SL_k(\mathbb{R})$, and μ is Haar.*

Proof. Write μ as the invariant probability on Lx with $L = AH$, $x = g\Gamma$, and put

$$\mathbf{L} = g^{-1}Lg, \qquad \Lambda = \mathbf{L} \cap SL_k(\mathbb{Z}).$$

The component Lx is one of the H-ergodic homogeneous components used in the preceding chapter. Apply Volume 4, Theorem 8.4, exactly as in the proof of Theorem 0.2. If $\mathbf{M}$ denotes the $\mathbb{Q}$-Zariski closure of Λ, then $\mathbf{M}(\mathbb{R})^0 \subset \mathbf{L}(\mathbb{R})^0$ because $\Lambda \subset \mathbf{L}(\mathbb{R})$, whereas the arithmetic-hull argument gives the reverse inclusion after passing to the connected character-free core $\mathbf{M}^1$. Consequently

$$\mathbf{L} = \mathbf{M}^1 \quad \text{as connected real algebraic groups,} \qquad X_{\mathbb{Q}}(\mathbf{L}) = 0. \tag{16.7.1}$$

In particular $\mathbf{L}$ and its derived group $\mathbf{H} = [\mathbf{L}, \mathbf{L}]$ are defined over $\mathbb{Q}$. Notice that this is a conclusion of the previously proved arithmetic-hull theorem, not a converse to Borel–Harish–Chandra.

Now pass to $\overline{\mathbb{Q}}$. The derived group $\mathbf{H} = [\mathbf{L}, \mathbf{L}]$ becomes, after conjugation, the product of the block groups $SL(V_1) \times \cdots \times SL(V_r)$ with $\dim V_i = k_i$. Since $\mathbf{H}$ is defined over $\mathbb{Q}$, the absolute Galois group permutes these simple factors and hence the block spaces V_i.

Suppose the set of blocks had more than one Galois orbit, and let $\mathcal{O}$ be a proper nonempty orbit. Then

$$W_{\mathcal{O}} = \bigoplus_{i \in \mathcal{O}} V_i$$

descends to a rational $\mathbf{L}$-invariant subspace of $\mathbb{Q}^k$. The determinant of the action on this rational subspace,

$$\chi_{\mathcal{O}}(\ell) = \det(\ell|_{W_{\mathcal{O}}}),$$

is a rational character of $\mathbf{L}$. It is nontrivial: in the standard model $L = AH$, the semisimple block group has determinant one on every V_i, while the block-scalar part of A can change the product of determinants on the blocks in a proper subset, subject only to the single global determinant-one relation. This contradicts (16.7.1). Therefore Galois acts transitively on the blocks.

Galois conjugate simple factors have the same absolute type and act on conjugate summands of the standard representation, so all block dimensions are equal to one integer m. Hence $k = rm$ and $m \mid k$. The positive-entropy construction produced at least one root group, so $m > 1$. If k is prime, the only divisor $m > 1$ is $m = k$, giving one block and therefore $H = SL_k(\mathbb{R})$. The algebraic probability is then the Haar probability on the whole space. $\square$

Proof and provenance. The original EKL Corollary 1.4 proves the stronger assertion that every positive-entropy algebraic candidate is noncompactly supported. Its proof first produces a compact periodic A-orbit and then invokes Lindenstrauss–Weiss's classification of diagonal orbit closures containing such an orbit [LW01; EKL06]. We have internalized above the divisor argument needed for the prime-dimensional consequence, because that is the load-bearing input for Volume 17. The stronger general noncompactness statement is recorded as source-strength context rather than silently claimed from an unproved "standard classification."

Worked Example 0.4 (Why prime dimension collapses the block system)**.** For $k = 3$, a Galois-stable partition of the three torus weights has equal block sizes. A positive-entropy block cannot have size one. Since 3 has no nontrivial proper divisor, the only possibility is one block of size three, hence full SL_3 invariance. This is exactly the arithmetic step needed in the Littlewood application.

CHAPTER 8

The EKL Positive-Entropy Measure-Classification Theorem

Proof architecture. The chapter is organized as the chain *Branch exhaustion* $\longrightarrow$ *Einsiedler–Katok–Lindenstrauss: positive entropy implies algebraicity*. The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

Lemma 0.1 (Branch exhaustion)**.** *Every positive-entropy A-ergodic probability on $SL_k(\mathbb{R})/SL_k(\mathbb{Z})$ produces a nontrivial block group $H_\sim$ of unipotent invariance.*

Proof. Positive entropy gives a positive root coefficient. If that root meets a second positive coefficient in a noncommuting configuration, the high-entropy theorem of Volume 14 produces a new root invariance and the block grows. If no such neighboring positive coefficient exists, the low-entropy recurrence and shearing argument of Volume 15 produces unipotent invariance. Iterating the closure operation in either branch and using Lemma 0.1 yields a nontrivial maximal block group. $\square$

Theorem 0.2 (Einsiedler–Katok–Lindenstrauss: positive entropy implies algebraicity). *Let $k \geq 3$ and let μ be an A-invariant ergodic probability measure on $SL_k(\mathbb{R})/SL_k(\mathbb{Z})$. If $h_\mu(a) > 0$ for some $a \in A$, then μ is algebraic. If in addition k is prime, then μ is Haar.*

Proof. Lemma 0.1 gives a nontrivial maximal unipotent-generated block group H. The homogeneous-component theorem and Theorem 0.2 show that μ is the invariant probability on one closed AH-orbit, hence algebraic. In prime dimension Proposition 0.3 forces the unique block to have size k, so $H = SL_k(\mathbb{R})$ and μ is Haar. $\square$

Proof and provenance. The proof architecture matches Section 6 of the original EKL paper: form the maximal block group, apply the homogeneous-component theorem, use a Mahler escape lemma to show that the blocks fill all coordinates, and obtain an algebraic AH-orbit [EKL06]. EKL's Corollary 1.4 additionally states that every such positive-entropy algebraic measure has noncompact support. Their proof of that stronger source corollary explicitly invokes Lindenstrauss–Weiss [LW01]; we therefore do not disguise that extra orbit-closure theorem as something proved in the present volume.

Worked Example 0.3 (The proof as a decision tree)**.** Start from one positive root. A neighboring positive root sends the argument to Volume 14; absence of such a root sends it to Volume 15. Both branches manufacture unipotent invariance. The homogeneous-component theorem turns that invariance into a block-homogeneous description, and Mahler escape eliminates missing coordinates. In prime dimension the arithmetic block-divisibility argument then collapses all blocks to one.

Volume 17

Littlewood's Conjecture and Exceptional Sets

Volume 17 scope and prerequisites

This volume carries the dynamical classification of Volume 16 back to a Diophantine theorem. The central point is stronger than mere nondensity: the set of pairs violating Littlewood's conjecture has Hausdorff dimension zero. We prove the lattice encoding, compactness of exceptional semigroup trajectories, the dimension-to-entropy mechanism on unstable slices, nonescape of the averaged measures, and the contradiction with positive-entropy algebraic rigidity. The architecture follows Sections 8–11 of Einsiedler–Katok–Lindenstrauss [EKL06].

Student orientation: the Littlewood contradiction in five moves

For $(u, v) \in \mathbb{R}^2$, the associated unimodular lattice $\tau_{u,v}$ encodes the three quantities n, $nu-p$, and $nv-q$. The positive diagonal semigroup A^+ balances these coordinates. A Littlewood-exceptional pair therefore yields a bounded A^+ trajectory. The argument uses the explicit compact exhaustion $\Xi = \bigcup_q \Xi_q$. Positive Hausdorff dimension therefore passes to one compact Ξ_q, and on compact sets Hausdorff dimension is bounded above by upper box dimension. *Topological entropy* counts exponentially many orbit segments that remain distinguishable, while measure entropy is obtained from it through the variational principle. The contradiction has four stages: positive-dimensional exceptional parameters create positive topological entropy; the variational and averaging arguments produce an A-invariant positive-entropy measure; EKL makes that measure algebraic; and the resulting homogeneous measure is incompatible with a compact positive-semigroup orbit hull. The averaging step is delicate because entropy must survive weak-* limits; the required upper-semicontinuity theorem is proved before it is used.

Reader guide: a concrete model

Why this matters.

Volume 17 is organized around the passage from *Lattice Encoding of Littlewood Pairs* to *Hausdorff Dimension Zero for the Exceptional Set*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.4 (A calculation to keep in view)**.** Let

$$a = \mathrm{diag}(e^{2s}, e^{-s}, e^{-s}) \in \mathrm{SL}_3(\mathbb{R})$$

and let E_{12} be the matrix unit. Since

$$\mathrm{Ad}(a)E_{12} = aE_{12}a^{-1} = e^{3s}E_{12},$$

the root exponent on the E_{12} direction is $3s$. For a one-dimensional leafwise measure whose conditional scaling is Lebesgue, the Jacobian contribution along this root direction is therefore $3s$. This calculation is the concrete local model behind the root-by-root entropy bookkeeping used later.

CHAPTER 1

Lattice Encoding of Littlewood Pairs

Proof architecture. The chapter is organized as the chain *Integer vectors record simultaneous errors* $\longrightarrow$ *Dani encoding for Littlewood* $\longrightarrow$ *Periodicity and unstable parametrization.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices. For $(u, v) \in \mathbb{R}^2$ set

$$\tau_{u,v} = \begin{pmatrix} 1 & 0 & 0 \\ u & 1 & 0 \\ v & 0 & 1 \end{pmatrix} SL_3(\mathbb{Z}), \qquad A^+ = \{\mathrm{diag}(e^{-r-s}, e^r, e^s) : r, s \geq 0\}.$$

Lemma 0.1 (Integer vectors record simultaneous errors)**.** *For integers n, m_1, m_2,*

$$\tau_{u,v}(n, m_1, m_2)^T = (n, nu + m_1, nv + m_2)^T.$$

The product of the three coordinates is unchanged by every element of A^+.

Proof. The matrix multiplication is immediate, and the three diagonal scaling factors multiply to one. $\square$

Theorem 0.2 (Dani encoding for Littlewood). *The pair (u, v) satisfies Littlewood's relation* $\liminf_{n\to\infty} n\|nu\|\|nv\| = 0$ *if and only if the A^+-orbit of $\tau_{u,v}$ is unbounded in X_3.*

Proof. By Mahler's criterion, unboundedness is equivalent to $\inf_{a\in A^+} \lambda_1(a\tau_{u,v}) = 0$. If $a = \mathrm{diag}(e^{-r-s}, e^r, e^s)$ makes a nonzero lattice vector

$$(ne^{-r-s}, e^r(nu + m_1), e^s(nv + m_2))$$

shorter than ε, then $n \neq 0$ and the product of its coordinates gives $|n(nu + m_1)(nv + m_2)| \ll \varepsilon^3$. Hence the Littlewood liminf is zero.

Conversely, choose n, m_1, m_2 with $|n(nu + m_1)(nv + m_2)| < \varepsilon^5$. We first arrange both approximation errors to be $< \varepsilon$. If, say, $|nv + m_2| \geq \varepsilon$, then $|n(nu + m_1)| < \varepsilon^4$. Dirichlet approximation supplies $1 \leq q < \varepsilon^{-1}$ with $\|qnv\| < \varepsilon$; replacing n by qn, m_1 by qm_1, and m_2 by the nearest integer to $-qnv$ gives simultaneously $|nu + m_1|, |nv + m_2| < \varepsilon$ and still product $< \varepsilon^3$

after relabelling. Choose $r, s \geq 0$ so that the two scaled errors are ε (take a large parameter if one error vanishes). The product inequality then gives $ne^{-r-s} < \varepsilon$. Thus the scaled lattice has a vector of norm $O(\varepsilon)$, and the orbit is unbounded. $\square$

Proposition 0.3 (Periodicity and unstable parametrization)**.** *The map* $(u, v) \mapsto \tau_{u,v}$ *is* $\mathbb{Z}^2$*-periodic and parametrizes the two-dimensional unstable orbit* $U_{21}U_{31}x_0$ *of the identity lattice for* $a = \mathrm{diag}(e^{-2}, e, e)$.

Proof. Adding integers to u, v right-multiplies by an element of $SL_3(\mathbb{Z})$. The lower-left entries are exactly the coordinates of $U_{21}U_{31}$. Conjugation by a expands both E_{21} and E_{31} by e^3. $\square$

Proof and provenance. The converse direction includes the extra Dirichlet step from EKL Proposition 11.1. Without it, a small product does not by itself guarantee that both approximation errors can be balanced using nonnegative semigroup parameters [EKL06].

Worked Example 0.4 (A rational coordinate)**.** If $u = p/q$ is rational, then $\|nu\| = 0$ for multiples of q, so Littlewood's liminf is zero for every v. Dynamically, the lattice contains vectors whose second coordinate vanishes; appropriate elements of A^+ make them arbitrarily short.

CHAPTER 2

Bounded Diagonal-Semigroup Trajectories

Proof architecture. The chapter is organized as the chain *Uniform compact set from a Littlewood lower bound* $\longrightarrow$ *Compact exhaustion of the exceptional set* $\longrightarrow$ *Forward invariance of the orbit hull.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

Lemma 0.1 (Uniform compact set from a Littlewood lower bound)**.** *For every $c > 0$ there is a compact $K_c \subset X_3$ such that*

$$n\|nu\|\|nv\| \geq c \quad (n \geq 1)$$

implies $A^+\tau_{u,v} \subset K_c$.

Proof. The proof of Theorem 0.2 gives a lower bound $\lambda_1(a\tau_{u,v}) \geq \eta(c) > 0$ uniform in $a \in A^+$. Mahler's criterion identifies the set of unimodular lattices with shortest vector at least $\eta(c)$ as compact. □

Theorem 0.2 (Compact exhaustion of the exceptional set). *Let*

$$\Xi = \{(u,v) \in [0,1]^2 : \liminf n\|nu\|\|nv\| > 0\}.$$

Then $\Xi = \bigcup_{q\geq 1} \Xi_q$, where each Ξ_q is compact and every A^+ orbit issued from $\tau(\Xi_q)$ is contained in one compact set K_q.

Proof. Take Ξ_q to be the set on which $n\|nu\|\|nv\| \geq 1/q$ for every $n \geq 1$. Each inequality defines a closed set, hence Ξ_q is compact. Their union is exactly Ξ, and Lemma 0.1 supplies K_q. □

Proposition 0.3 (Forward invariance of the orbit hull)**.** *For compact $E \subset \Xi_q$, the closure*

$$Y_E = \overline{A^+\tau(E)}$$

is compact and forward A^+-invariant.

Proof. It is a closed subset of K_q, hence compact. Since A^+ is a semigroup, $aA^+ \subset A^+$ for every $a \in A^+$, so $aY_E \subset Y_E$ by continuity. □

Worked Example 0.4 (Why semigroup boundedness is enough)**.** The full diagonal A-orbit of an exceptional lattice need not be bounded. Littlewood only sees the cone $s, t \geq 0$. The argument therefore builds invariant measures by averaging over larger and larger rectangles inside this cone; it never replaces A^+-boundedness by the stronger and generally false assertion of full-A boundedness.

CHAPTER 3

The Littlewood Exceptional Set as an Unstable Slice

Proof architecture. The chapter is organized as the chain *Bi-Lipschitz unstable parametrization* $\longrightarrow$ *Exceptional set equals the bounded slice* $\longrightarrow$ *Dimension transport*. The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

Lemma 0.1 (Bi-Lipschitz unstable parametrization)**.** *On every compact parameter square on which the quotient map is injective, $(u, v) \mapsto \tau_{u,v}$ is bi-Lipschitz for a fixed right-invariant Riemannian metric on G and the induced local metric on X_3.*

Proof. The differential is the constant injection $(\dot{u}, \dot{v}) \mapsto \dot{u}E_{13} + \dot{v}E_{23}$. On a compact injectivity chart, the derivative and inverse derivative have bounded operator norm. The mean-value inequality gives the two Lipschitz bounds. □

Theorem 0.2 (Exceptional set equals the bounded slice). *Inside a fundamental parameter square, $\tau(\Xi)$ is precisely the set of points on the two-dimensional unstable slice $U_{21}U_{31}x_0$ whose A^+ trajectories are relatively compact.*

Proof. Proposition 0.3 identifies the parameter square, modulo its integer periodicities, with the two-dimensional unstable slice through the standard lattice. The map is injective on the chosen fundamental square, so no two distinct parameter pairs are being identified at this stage.

For a point $\tau_{u,v}$ on that slice, Theorem 0.2 says precisely that

$$\inf_{n \geq 1} n\|nu\|\|nv\| > 0 \quad \Longleftrightarrow \quad \inf_{a \in A^+} \lambda_1(a\tau_{u,v}) > 0.$$

By Mahler compactness, the right-hand condition is equivalent to relative compactness of the positive-semigroup orbit. Thus the arithmetic exceptional set and the dynamically bounded slice are the same subset under the parametrization, not merely sets of the same dimension. □

Proposition 0.3 (Dimension transport). *For every $E \subset [0,1]^2$, $\dim_H E = \dim_H \tau(E)$, and the same holds for upper box dimension on compact subsets of an injectivity chart.*

PROOF. On each injectivity chart Lemma 0.1 gives constants $0 < c \leq C < \infty$ with $c\|x-y\| \leq d(\tau(x), \tau(y)) \leq C\|x-y\|$. A cover by sets of diameter at most r is therefore carried to a cover with diameters between constant multiples of r, and the same comparison holds for separated sets. The definitions of Hausdorff measure and upper covering numbers are unchanged by such fixed multiplicative rescalings. Hence both Hausdorff dimension and, on compact sets, upper box dimension are preserved on each chart. A finite cover of the fundamental square by these charts preserves the maximum of the dimensions, proving the global statement. □

Worked Example 0.4 (Two-dimensional unstable geometry). The parameter plane is not an arbitrary transversal: u and v are exactly the coordinates of the two commuting expanding roots U_{21} and U_{31}. This is why a positive-dimensional set of parameters can be converted into entropy for a diagonal map that expands those two roots.

CHAPTER 4

Dimension-to-Entropy Production

Proof architecture. The chapter is organized as the chain *Separated sets from box dimension* $\longrightarrow$ *Dimension creates topological entropy* $\longrightarrow$ *Variational extraction.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices. Choose $a \in A^+$ which expands both U_{13} and U_{23} by a factor $e^\lambda > 1$.

Lemma 0.1 (Separated sets from box dimension)**.** *If a compact E in the unstable slice has upper box dimension $d > 0$, then for infinitely many N there is a subset $F_N \subset E$ with*

$$|F_N| \geq e^{(d\lambda - o(1))N}$$

such that distinct points of F_N are (N, ε)-separated for the map a.

Proof. At scales $r_N = e^{-\lambda N}$, positive upper box dimension gives $r_N^{-d+o(1)}$ pairwise r_N-separated points. Two points differing by an unstable displacement of size at least r_N are magnified by a^j by $e^{\lambda j}$; before time N the displacement reaches a fixed chart size, so their orbit segments separate by a fixed $\varepsilon > 0$. □

Theorem 0.2 (Dimension creates topological entropy). *If a compact forward-invariant set Y meets the unstable slice in a compact set of positive upper box dimension, then $a|_Y$ has positive topological entropy.*

Proof. Apply Lemma 0.1. The definition of topological entropy through maximal (N, ε)-separated sets gives $h_{\mathrm{top}}(a|_Y) \geq d\lambda > 0$ along the limsup sequence. □

Proposition 0.3 (Variational extraction)**.** *For compact Y and continuous $a : Y \to Y$, positive topological entropy yields an a-invariant ergodic probability ν with $h_\nu(a) > 0$.*

Proof. Choose $\varepsilon > 0$ for which the separated-set entropy $h_{\mathrm{sep}}(a, \varepsilon)$ is positive. Along a sequence $N_i \to \infty$ choose maximal (N_i, ε)-separated sets

$F_i \subset Y$ with

$$\frac{1}{N_i}\log|F_i| \longrightarrow h_* > 0, \qquad h_* \le h_{\mathrm{sep}}(a,\varepsilon).$$

Let σ_i be the uniform probability on F_i and set $\mu_i = N_i^{-1}\sum_{j=0}^{N_i-1} a_*^j\sigma_i$. After passing to a subsequence, $\mu_i \to \mu$. For every continuous f,

$$\int (f\circ a - f)\,d\mu_i = \frac{1}{N_i}\left(\int f\circ a^{N_i}\,d\sigma_i - \int f\,d\sigma_i\right) \longrightarrow 0,$$

so μ is a-invariant.

Choose a finite Borel partition $\mathcal{P}$ whose atoms have diameter less than ε and whose boundaries have μ-measure zero. Two points of F_i cannot have the same $\mathcal{P}$-name for times $0,\dots,N_i-1$; hence

$$H_{\sigma_i}\left(\bigvee_{j=0}^{N_i-1} a^{-j}\mathcal{P}\right) = \log|F_i|. \tag{17.4.1}$$

Fix a block length q. Split the N_i names into q residue classes modulo q and use subadditivity of Shannon entropy. Averaging over the possible initial offsets gives the block estimate

$$\frac{1}{N_i}H_{\sigma_i}(\mathcal{P}^{[0,N_i)}) \le \frac{1}{q}H_{\mu_i}(\mathcal{P}^{[0,q)}) + \frac{2q\log|\mathcal{P}|}{N_i}. \tag{17.4.2}$$

Because μ gives zero mass to the boundaries of the atoms of $\mathcal{P}^{[0,q)}$, weak-* convergence implies convergence of the finite atom probabilities and therefore of their Shannon entropies. Letting $i\to\infty$ in (17.4.1)–(17.4.2) yields

$$h_* \le \frac{1}{q}H_\mu(\mathcal{P}^{[0,q)}) \qquad (q\ge 1).$$

Taking the infimum over q gives $h_\mu(a,\mathcal{P}) \ge h_* > 0$, hence $h_\mu(a) > 0$.

Finally use the ergodic decomposition $\mu = \int \mu_\omega\,d\theta(\omega)$. Entropy is affine for an invariant probability under a continuous map on a compact metric space, so $h_\mu(a) = \int h_{\mu_\omega}(a)\,d\theta(\omega)$. Since the left-hand side is positive, at least one ergodic component has positive entropy. $\square$

Worked Example 0.4 (Entropy rate from a planar slice)**.** If E needs about $r^{-1/3}$ balls at scale r, then after N iterates with $r = e^{-\lambda N}$ it produces roughly $e^{\lambda N/3}$ distinguishable orbit segments. The exponential growth rate $\lambda/3$ is exactly the entropy lower bound.

CHAPTER 5

From Geometric Largeness to an A-Invariant Entropy Measure

Proof architecture. The chapter is organized as the chain *Følner averaging preserves the chosen entropy $\longrightarrow$ Large unstable slice produces an A-ergodic positive-entropy measure $\longrightarrow$ Compact extraction for the Littlewood exceptional set.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

Lemma 0.1 (Small partitions compute entropy on a compact diagonal set)**.** *Let $X = SL_k(\mathbb{R})/SL_k(\mathbb{Z})$, let $a \in A$, and let $K \subset X$ be compact with $aK \subset K$. There is $\delta_K > 0$ with the following property. If $\mathcal{P}$ is a finite Borel partition of K whose atoms have diameter $< \delta_K$, then every a-invariant probability ρ supported on K satisfies*

$$h_\rho(a) = h_\rho(a, \mathcal{P}).$$

The same δ_K works for every such ρ.

Proof. An invariant probability supported on K is supported on $K_\infty = \bigcap_{n \geq 0} a^n K$, and $aK_\infty = K_\infty$, so we may replace K by K_∞. Compactness gives a uniform injectivity radius $r > 0$ on K. Write the local Lie algebra as

$$\mathfrak{g} = \mathfrak{u}^- \oplus \mathfrak{c} \oplus \mathfrak{u}^+,$$

where $\mathfrak{c} = Z_{\mathfrak{g}}(a)$ and $\mathrm{Ad}(a)$ contracts $\mathfrak{u}^-$ and expands $\mathfrak{u}^+$. For sufficiently small $r_0 < r$, multiplication gives unique local product coordinates $g = u^- c u^+$ on the r_0-ball.

Choose $\delta_K \ll r_0$. Suppose $x, y \in K$ have the same two-sided $\mathcal{P}$-name. Then $y = gx$ for g in the r_0-ball. If the unstable coordinate of g were nontrivial, forward conjugation by powers of a would expand it until $a^n x$ and $a^n y$ were farther apart than the diameter of an atom; if the stable coordinate were nontrivial, the same argument with negative powers would do so. Hence $y = cx$ for a small $c \in C := Z_G(a)$.

Let $\mathcal{A} = \bigvee_{n\in\mathbb{Z}} a^{-n}\mathcal{P}$. Thus the only ambiguity inside an atom of $\mathcal{A}$ is a small displacement in C. We now prove that this ambiguity carries zero relative entropy. Fix a finite partition $\mathcal{Q} = \{Q_1, \dots, Q_m\}$ whose boundaries are ρ-null. Let C_0 be a countable dense subset of the small central ball. For i, j, $c_0 \in C_0$, and $r = 1/\ell$, define

$$B_{i,j,c_0,r} = \{x : \ B_r(x) \subset Q_i \cap c_0^{-1}Q_j\}.$$

The family is countable. Poincare recurrence therefore supplies one null set N such that every $x \in B_{i,j,c_0,r} \setminus N$ returns to that same set at some positive time.

Take $x, y \notin N$ in one $\mathcal{A}$-atom and write $y = cx$ with $c \in C$ small. Suppose $x \in Q_i$ and $y \in Q_j$. Because both points avoid the partition boundaries, choose $r > 0$ so that $B_r(x) \subset Q_i \cap c^{-1}Q_j$. Approximate c by $c_0 \in C_0$ closely enough that still $B_{r/2}(x) \subset Q_i \cap c_0^{-1}Q_j$. Recurrence gives $n > 0$ with $a^n x \in B_{i,j,c_0,r/2}$. Since a commutes with c and c_0, closeness of c_0 to c implies that $a^n y = ca^n x$ lies in Q_j. But x, y have the same full $\mathcal{P}$-name and, after conditioning on $\bigvee_{n\geq 1} a^{-n}\mathcal{Q}$, the same future $\mathcal{Q}$-name. Therefore the time-0 atoms must agree: $i = j$ almost surely. This is exactly

$$H_\rho\left(\mathcal{Q} \,\middle|\, \bigvee_{n\geq 1} a^{-n}\mathcal{Q} \vee \mathcal{A}\right) = 0.$$

The relative entropy addition formula therefore gives $h_\rho(a, \mathcal{P} \vee \mathcal{Q}) = h_\rho(a, \mathcal{P})$. Taking an increasing sequence of null-boundary partitions $\mathcal{Q}$ whose generated sigma-algebra is the Borel sigma-algebra of K yields $h_\rho(a) \leq h_\rho(a, \mathcal{P})$. The reverse inequality is automatic from the definition of metric entropy. Thus equality holds. Notice that the contraction/expansion argument and the recurrence argument depend only on a and the compact set K, not on ρ, which gives the claimed uniform choice of δ_K. □

Proposition 0.2 (Upper semicontinuity of entropy on the compact orbit set)**.** *Let $a \in A$ and let $K \subset X$ be compact with $aK \subset K$. On the space of a-invariant probabilities supported on K, the map $\rho \mapsto h_\rho(a)$ is upper semicontinuous in the weak-* topology.*

Proof. Fix an invariant probability ρ and $\varepsilon > 0$. Choose, using Lemma 0.1, a finite partition $\mathcal{P}$ of diameter below δ_K whose atom boundaries have ρ-measure zero. The lemma gives $h_\eta(a) = h_\eta(a, \mathcal{P})$ for every invariant η supported on K.

By subadditivity,

$$h_\rho(a, \mathcal{P}) = \inf_{N\geq 1} \frac{1}{N} H_\rho\left(\bigvee_{j=0}^{N-1} a^{-j}\mathcal{P}\right).$$

Choose N so that the displayed finite-block entropy is $< h_\rho(a) + \varepsilon/2$. Every atom of the N-block partition has ρ-null boundary. Hence, for η sufficiently close to ρ in the weak-* topology, all its atom masses are close to the corresponding ρ-masses. Shannon entropy is continuous on the finite probability simplex, so

$$\frac{1}{N} H_\eta(\mathcal{P}^{[0,N)}) < \frac{1}{N} H_\rho(\mathcal{P}^{[0,N)}) + \varepsilon/2 < h_\rho(a) + \varepsilon.$$

Taking the infimum over block lengths for η and using the small-partition lemma gives $h_\eta(a) < h_\rho(a) + \varepsilon$. This is upper semicontinuity. $\square$

Lemma 0.3 (Følner averaging preserves the chosen entropy)**.** *Let K be compact and forward invariant under an open cone in A. Suppose ν is a-invariant, supported on K, and $h_\nu(a) > 0$. Choose a basis $b_1, \ldots, b_{k-1}$ of A inside the cone and set*

$$\nu_T = \frac{1}{T^{k-1}} \int_{[0,T]^{k-1}} (b_1^{t_1} \cdots b_{k-1}^{t_{k-1}})_* \nu \, dt.$$

Then every weak- limit μ is A-invariant, supported on K, and satisfies $h_\mu(a) \geq h_\nu(a)$.*

Proof. Because A is abelian, every translated measure $(b_1^{t_1} \cdots b_{k-1}^{t_{k-1}})_* \nu$ is still a-invariant. The translating map is a measure-theoretic isomorphism which commutes with a, so it preserves a-entropy. Since entropy is affine on the convex set of a-invariant probabilities, integration over the parameter box gives

$$h_{\nu_T}(a) = h_\nu(a).$$

Let $T_j \to \infty$ and $\nu_{T_j} \to \mu$ weak-*. Fix i and $s \in \mathbb{R}$. For every bounded continuous f,

$$\int f \, d((b_i^s)_* \nu_T - \nu_T)$$

is the difference of the averages of f over two parameter boxes which differ only by translating the ith interval by s. Their symmetric difference has volume at most $2|s|T^{k-2}$, while the original box has volume T^{k-1}. Hence the absolute value of the displayed difference is at most $2|s|\|f\|_\infty/T$, which tends to zero. Passing to the weak-* limit yields $(b_i^s)_*\mu = \mu$. This holds for every s and every basis direction, hence μ is A-invariant.

Every averaging parameter lies in the chosen positive cone. Since K is forward invariant under that cone, every translated measure in the defining integral of ν_T is supported on K. Thus all ν_T and their weak-* limits are supported on K.

Finally Proposition 0.2 applies to the sequence ν_{T_j} and gives

$$h_\mu(a) \geq \limsup_{j\to\infty} h_{\nu_{T_j}}(a) = h_\nu(a) > 0.$$

The equality before the limit and the inequality at the limit are logically different steps; the former uses affinity and commutation, while the latter is exactly the upper-semicontinuity theorem proved above. $\square$

THEOREM 0.4 (Large unstable slice produces an A-ergodic positive-entropy measure). *If a compact forward-invariant K meets one unstable leaf in a set of positive upper box dimension, then K supports an A-invariant probability whose A-ergodic decomposition has a positive-measure family of components with positive entropy for some $a \in A$.*

PROOF. Theorem 0.2 gives $h_{\mathrm{top}}(a|_K) > 0$. The variational principle gives an a-invariant ν on K with $h_\nu(a) > 0$. Apply Lemma 0.3 to obtain an A-invariant μ with positive a-entropy. Entropy is affine under the A-ergodic decomposition, so a positive-measure set of components has positive entropy. $\square$

Proposition 0.5 (Compact extraction for the Littlewood exceptional set)**.** *Let $\Xi = \bigcup_{q\geq 1} \Xi_q$ be the compact exhaustion from Theorem 0.2. If $\dim_H \Xi > 0$, then some compact Ξ_q has positive upper box dimension. The same conclusion holds after applying the unstable parametrization τ.*

PROOF. Hausdorff dimension is countably stable:

$$\dim_H\Big(\bigcup_{q\geq 1} \Xi_q\Big) = \sup_q \dim_H \Xi_q.$$

Thus $\dim_H \Xi > 0$ implies $\dim_H \Xi_q > 0$ for some q. This set is compact by Theorem 0.2. For every compact metric set, Hausdorff dimension is at most upper box dimension, hence $\overline{\dim}_{\mathrm{B}}\Xi_q > 0$. Proposition 0.3 shows that the bi-Lipschitz unstable parametrization preserves both dimensions on the finitely many injectivity charts, so $\tau(\Xi_q)$ has positive upper box dimension as well. $\square$

Worked Example 0.6 (Why we average after creating entropy)**.** A measure obtained directly from a fractal slice need not be invariant even under one diagonal element. The safe order is: geometric dimension gives topological entropy; the variational principle gives one a-invariant positive-entropy measure; only then do commuting Følner averages create full A-invariance while upper semicontinuity protects the entropy.

CHAPTER 6

Nonescape and Mass Control

Proof architecture. The chapter is organized as the chain *No mass loss from a uniform compact orbit hull* $\longrightarrow$ *Exceptional averages do not escape* $\longrightarrow$ *Compact support is inherited by ergodic components.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

Lemma 0.1 (No mass loss from a uniform compact orbit hull)**.** *If all measures in a sequence are supported on one compact $K \subset X_3$, every weak-* limit is a probability supported on K.*

Proof. Weak-* compactness of probability measures on compact K is Prokhorov compactness in its elementary compact-space form. Testing against the constant function one preserves total mass, and testing functions vanishing on K shows the limit is supported there. $\square$

Theorem 0.2 (Exceptional averages do not escape). *Every invariant measure constructed from a compact $E \subset \Xi_q$ by orbit averaging is a probability supported on K_q.*

Proof. The entire forward orbit of every point of $\tau(E)$ lies in K_q by Theorem 0.2. Therefore every finite orbit average is supported on K_q. Apply Lemma 0.1. $\square$

Proposition 0.3 (Compact support is inherited by ergodic components)**.** *If an A-invariant probability is supported on K_q, almost every A-ergodic component is supported on K_q.*

Proof. The complement of K_q has measure zero. Disintegrating over ergodic components expresses this zero as the integral of the nonnegative component masses of the complement; hence those masses vanish almost everywhere. $\square$

Worked Example 0.4 (The usual noncompact-space danger disappears)**.** Averages on $SL_3(\mathbb{R})/SL_3(\mathbb{Z})$ can normally lose mass into the cusp. Here the

Diophantine exceptional inequality itself gives a uniform Mahler compact set, so nonescape is automatic once the Dani correspondence is established.

CHAPTER 7

Deduction from EKL Classification

Proof architecture. The chapter is organized as the chain *Positive-entropy algebraic measures are not compactly supported* $\longrightarrow$ *No compact positive-entropy exceptional measure* $\longrightarrow$ *Dimension contradiction.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

Lemma 0.1 (Positive-entropy algebraic measures are not compactly supported)**.** *An algebraic A-ergodic probability on X_3 with positive entropy cannot have compact support.*

PROOF. By Theorem 0.2 the measure is algebraic, and Proposition 0.3 gives the prime-dimensional conclusion that it is Haar. The space X_3 is noncompact by Mahler's criterion: the lattices $\mathrm{diag}(e^{-t}, e^t, 1)\mathbb{Z}^3$ leave every compact set. Haar measure has full support, hence is not compactly supported. $\square$

THEOREM 0.2 (No compact positive-entropy exceptional measure). *No compact forward A^+-invariant subset of X_3 supports an A-invariant ergodic probability with positive entropy.*

PROOF. Any such measure satisfies the hypotheses of the EKL theorem. Since $k = 3$ is prime it is Haar by Theorem 0.2, contradicting compact support by Lemma 0.1. $\square$

Proposition 0.3 (Dimension contradiction)**.** *Every compact $E \subset \Xi_q$ has Hausdorff dimension zero.*

PROOF. If $\dim_H E > 0$, Theorem 0.4 constructs a compactly supported invariant probability with positive entropy. An ergodic component retains positive entropy and compact support, contradicting Theorem 0.2. $\square$

Worked Example 0.4 (Where prime dimension is used)**.** The contradiction is especially clean in SL_3: EKL upgrades positive entropy all the way to Haar. In composite dimension a positive-entropy algebraic measure could live on a proper algebraic orbit, so an additional argument about the candidate orbit would be needed.

CHAPTER 8

Hausdorff Dimension Zero for the Exceptional Set

Proof architecture. The chapter is organized as the chain *Countable-union closure* $\longrightarrow$ *Einsiedler–Katok–Lindenstrauss: Littlewood exceptional set.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

Lemma 0.1 (Countable-union closure)**.** *A countable union of sets of Hausdorff dimension zero has Hausdorff dimension zero.*

PROOF. For every $s > 0$ and every member E_q, the s-dimensional Hausdorff measure is zero. Choose covers of E_q with total s-cost below $2^{-q}\varepsilon$ and unite the covers. The union has s-cost below ε. Since $s > 0$ is arbitrary, the dimension is zero. $\square$

THEOREM 0.2 (Einsiedler–Katok–Lindenstrauss: Littlewood exceptional set). *The set*

$$\Xi = \{(u, v) \in \mathbb{R}^2 : \liminf_{n\to\infty} n\|nu\|\|nv\| > 0\}$$

has Hausdorff dimension zero. In particular, Littlewood's conjecture holds outside a set of Hausdorff dimension zero.

PROOF. By periodicity it suffices to work in $[0, 1]^2$. The compact exhaustion Theorem 0.2 writes $\Xi = \bigcup_q \Xi_q$. Proposition 0.3 gives $\dim_H \Xi_q = 0$ for every q. Lemma 0.1 finishes the proof. $\square$

> **Proof and provenance.** The original EKL theorem is stronger in its bookkeeping: the exceptional set is exhibited as a countable union of compact sets of box dimension zero. The Hausdorff-dimension-zero conclusion proved here follows from the same dimension-to-entropy mechanism and is the endpoint proved in this volume [EKL06].

Worked Example 0.3 (Dimension zero is much stronger than measure zero)**.** A curve in the plane has planar Lebesgue measure zero but Hausdorff dimension one. The theorem says the Littlewood exceptional set is smaller than every positive-dimensional fractal: its Hausdorff dimension is exactly zero.

Volume 18

Higher-Rank Joinings and Factors

Volume 18 scope and prerequisites

This volume first develops joining rigidity concretely in the split type-A real setting and then proves the semisimple S-arithmetic joining theorem in Appendix A. Thus the principal Einsiedler–Lindenstrauss endpoint is proved at the stated scope, while the real chapters retain the more explicit factor-classification model useful later in the series. The proof mechanism is the same rigidity dictionary already built in Volumes 13–16: product leafwise measures, entropy, high/low-entropy production of graph unipotents, Ratner classification, and algebraic Goursat analysis [EL19].

Student orientation: joinings, factors, and algebraic correspondences

A *joining* of two measure-preserving A-systems is an A-invariant probability on the product with the prescribed marginals. The product measure is the trivial joining. A *common factor* produces a *relatively independent joining*: disintegrate both systems over the factor and take the product of the conditional measures fibre by fibre. An *algebraic correspondence* is a joining supported on a closed finite-volume orbit of an algebraic subgroup projecting onto both factors. A *commensurator correspondence* is the special finite correspondence coming from an element that makes two lattices commensurable. In the S-arithmetic appendix, "connected" is always algebraic rather than topological at nonarchimedean places. Local unipotent directions are organized through their algebraic hulls and coarse weights. This convention prevents the common mistake of importing real connectedness language into totally disconnected p-adic groups.

Reader guide: a concrete model

Why this matters.

Volume 18 is organized around the passage from *Joinings of Diagonal Actions* to *The S-Arithmetic Joining Theorem*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.4 (A calculation to keep in view)**.** Let

$$a = \operatorname{diag}(e^{2s}, e^{-s}, e^{-s}) \in \mathrm{SL}_3(\mathbb{R})$$

and let E_{12} be the matrix unit. Since

$$\operatorname{Ad}(a)E_{12} = aE_{12}a^{-1} = e^{3s}E_{12},$$

the root exponent on the E_{12} direction is $3s$. For a one-dimensional leafwise measure whose conditional scaling is Lebesgue, the Jacobian contribution along this root direction is therefore $3s$. This calculation is the concrete local model behind the root-by-root entropy bookkeeping used later.

CHAPTER 1

Joinings of Diagonal Actions

Proof architecture. The chapter is organized as the chain *Joinings form a compact convex set* $\longrightarrow$ *Disjointness criterion* $\longrightarrow$ *Graph joining of a factor map.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices. Let (X_i, m_i, A) be probability-preserving actions. A joining is an A-invariant probability on $X_1 \times X_2$ with marginals m_1, m_2.

Lemma 0.1 (Joinings form a compact convex set)**.** *If X_1, X_2 are standard Borel probability spaces and the actions are continuous on locally compact models with tight marginals, the set $J_A(m_1, m_2)$ of joinings is convex and weak-* compact. Its extreme points are exactly the ergodic joinings.*

Proof. Convex combinations preserve invariance and marginals. Tightness of both fixed marginals makes the family of couplings tight; Prokhorov gives compactness, and the defining marginal/invariance equations are weak-* closed. If a joining is not ergodic, its ergodic decomposition is a nontrivial convex decomposition with the same marginals because the marginals are ergodic. Conversely a nontrivial invariant convex decomposition exhibits a nontrivial invariant set, so an ergodic joining is extreme. ☐

Theorem 0.2 (Disjointness criterion). *Two ergodic actions are disjoint if and only if the product measure $m_1 \otimes m_2$ is their unique ergodic joining; equivalently it is their unique joining.*

Proof. If the product is the only joining, the statement is the definition of disjointness. If it is the only ergodic joining, every joining decomposes into ergodic joinings by Lemma 0.1, so its ergodic decomposition is almost surely the product and the joining itself is the product. ☐

Proposition 0.3 (Graph joining of a factor map)**.** *If $\pi : X_1 \to X_2$ is an A-equivariant measure-preserving isomorphism onto its image, $(x, \pi(x))_* m_1$ is a joining supported on the graph of π. More generally a common factor creates a nontrivial relatively independent joining.*

Proof. Equivariance makes the graph measure invariant and the marginal identities are immediate. The common-factor construction is proved in Chapter 4 by disintegration. □

Worked Example 0.4 (Diagonal joining)**.** For one system (X, m, A), the measure supported on $\{(x, x) : x \in X\}$ is a self-joining. It is as far from the product joining as possible, but it is algebraic for homogeneous systems: the support is an orbit of the diagonal subgroup in $G \times G$.

CHAPTER 2

Leafwise Measures on Product Spaces

Proof architecture. The chapter is organized as the chain *Product root group disintegration* $\longrightarrow$ *Classification once a leafwise stabilizer is produced* $\longrightarrow$ *What remains to be proved in joining rigidity.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

Lemma 0.1 (Product root group disintegration)**.** *For a joining λ of homogeneous diagonal actions and a coarse-Lyapunov root α, the leafwise system along $U_\alpha^{(1)} \times U_\alpha^{(2)}$ projects to the Haar leafwise systems of the two marginals whenever the marginals are Haar.*

PROOF. Disintegrate λ in a local product flow box and push the conditional forward by each coordinate projection. Uniqueness of disintegration identifies the pushforward projective Radon systems with the marginal leafwise systems. Haar marginals are invariant under every root group, so Volume 13 identifies those marginal leafwise measures with Haar. $\square$

THEOREM 0.2 (Classification once a leafwise stabilizer is produced). *Let η be a rootwise leafwise Radon measure on $U_\alpha^{(1)} \times U_\alpha^{(2)} \cong \mathbb{R}^2$ with Haar coordinate projections. Suppose its translation stabilizer has positive-dimensional identity component I. Then either $I = \mathbb{R}^2$, in which case η is planar Haar, or I is a line with nonzero projections to both coordinates. In the latter case $I = \{(r, cr) : r \in \mathbb{R}\}$ for some $c \neq 0$ and the ambient joining is invariant under the associated graph root subgroup.*

PROOF. The identity component of a closed subgroup of $\mathbb{R}^2$ is a vector subspace, hence has dimension one or two. Two-dimensional stabilizer means invariance under all translations, so uniqueness of Haar measure makes η proportional to Lebesgue measure on the plane. If I is one-dimensional, Haar coordinate projections rule out either coordinate axis: an axis stabilizer together with the projective leafwise equivariance would make one coordinate conditional atomic, contradicting its Haar projection. Thus both coordinate projections of I are nonzero and the line is the graph of multiplication by a nonzero scalar c. Haar/invariance from Volume 13 turns this leafwise translation symmetry into invariance of the joining under the corresponding graph root group. □

Proposition 0.3 (What remains to be proved in joining rigidity)**.** *For a nonproduct joining it is enough to manufacture a positive-dimensional translation stabilizer for one product-root leafwise measure. Theorem 0.2 then converts that stabilizer into an algebraic graph-root invariance.*

Proof. A full planar stabilizer gives product behavior in that root. A proper positive-dimensional stabilizer gives a graph root. Once one graph root exists, the higher-rank high/low-entropy propagation of Chapter 7 produces the global correspondence subgroup. Therefore the difficult dynamical step is precisely the production of a nontrivial stabilizer, not an a priori classification of arbitrary scaling-invariant measures on $\mathbb{R}^2$. □

Worked Example 0.4 (Why the stabilizer hypothesis matters)**.** A measure on $\mathbb{R}^2$ can have Lebesgue projections without being Lebesgue on the plane or on a single line. The first draft incorrectly tried to classify such couplings from scaling alone. Higher-rank dynamics supplies the missing ingredient: recurrence and shearing first produce a genuine translation stabilizer, after which elementary Lie-group geometry gives the line/plane alternative.

CHAPTER 3

Entropy Inequalities for Joinings

Proof architecture. The chapter is organized as the chain *Conditional-entropy inequality for one coarse Lyapunov group* $\longrightarrow$ *Rootwise equality and surjective support* $\longrightarrow$ *The support group is the algebraic datum to propagate.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices. The decisive point in the joining argument is not an *entropy deficit.* It is the opposite phenomenon: because both marginals are Haar, the conditional-entropy inequality for a joining is forced to be an equality on every coarse Lyapunov subgroup. Equality prevents the leafwise measure from losing either coordinate direction. This is the exact starting point of the Einsiedler–Lindenstrauss proof.

Lemma 0.1 (Conditional-entropy inequality for one coarse Lyapunov group)**.** *Let $a = (a_1, a_2)$ contract the coarse Lyapunov group*

$$U = U_1 \times U_2 \leq G_1 \times G_2$$

and let λ be a joining with first marginal m_1. Then

$$h_\lambda(a, U) \leq h_{m_1}(a_1, U_1) + h_\lambda(a, U_2).$$

The symmetric inequality with the two factors interchanged also holds.

Proof. Choose countably generated partitions $\overline{\mathcal{A}}_{U_i}$ subordinate to the U_i-foliations and pull them back to the product. Put

$$\mathcal{A}_U = \pi_1^{-1}\overline{\mathcal{A}}_{U_1} \vee \pi_2^{-1}\overline{\mathcal{A}}_{U_2}.$$

As in Volume 12, the entropy contribution is the conditional entropy of a generating partition relative to this subordinate past. If $\mathcal{P}_i$ are finite generating partitions of the factors and $\mathcal{P} = \pi_1^{-1}\mathcal{P}_1 \vee \pi_2^{-1}\mathcal{P}_2$, then for $n \geq 1$ the chain rule gives

$$\begin{aligned} H_\lambda(\mathcal{P}^{[-n,0)} \mid \mathcal{A}_U) = {} & H_\lambda(\pi_1^{-1}\mathcal{P}_1^{[-n,0)} \mid \mathcal{A}_U) \\ & + H_\lambda\Big(\pi_2^{-1}\mathcal{P}_2^{[-n,0)} \Bigm| \mathcal{A}_U \vee \pi_1^{-1}\mathcal{P}_1^{[-n,0)}\Big). \end{aligned}$$

Conditioning on the larger sigma-algebra can only decrease entropy. Hence the first term is at most the corresponding m_1-entropy relative to $\overline{\mathcal{A}}_{U_1}$. For the second term, move the partition pieces one time step at a time and use martingale convergence; the limiting conditioning sigma-algebra contains the first coordinate and the U_2-subordinate past, so its normalized limit is $h_\lambda(a, U_2)$. Divide by n and let $n \to \infty$. Interchanging the factors proves the second inequality. □

THEOREM 0.2 (Rootwise equality and surjective support). *Assume both marginals are Haar and the one-parameter subgroup generated by a_i is ergodic on each factor. For every coarse Lyapunov group $U^\Lambda = U_1^\Lambda U_2^\Lambda$ contracted by a,*

$$h_\lambda(a, U^\Lambda) = h_{m_1}(a_1, U_1^\Lambda) + h_\lambda(a, U_2^\Lambda),$$

and symmetrically with $1, 2$ interchanged.

Let P_x^Λ be the smallest connected A-normalized subgroup of U^Λ containing the support of the leafwise measure λ_{x,U^Λ}. Then, on one A-invariant conull set,

$$\pi_i(P_x^\Lambda) = U_i^\Lambda \qquad (i = 1, 2),$$

and the family is A-equivariant:

$$P_{ax}^\Lambda = a P_x^\Lambda a^{-1}.$$

No almost-sure constancy of P_x^Λ is asserted: equal-weight graph subspaces may vary continuously, and the joining proof does not need to rule that out at this stage.

PROOF. First take the full stable horospherical group $U^-(a) = U_1^-(a_1)U_2^-(a_2)$. The Abramov–Rokhlin relative-entropy formula gives

$$h_\lambda(a) = h_{m_1}(a_1) + h_\lambda(a \mid \pi_1^{-1}\mathcal{B}_{X_1}).$$

The first term is exactly $h_{m_1}(a_1, U_1^-)$ because the Haar action has no entropy outside its stable horospherical foliation. The second is exactly $h_\lambda(a, U_2^-)$ by conditioning on the first coordinate. Thus Lemma 0.1 is an equality for U^-. The leafwise product structure and the entropy summation formula of Volumes 12–13 decompose all three terms into sums over the coarse Lyapunov subgroups. Since Lemma 0.1 is an inequality term by term and the sums are equal, equality holds for each coarse Lyapunov group.

We next prove surjectivity of the support. Suppose, for example, that the connected projection $Q_1 = \pi_1(P_x^\Lambda)$ were a proper subgroup of U_1^Λ on a positive-measure set. Because P_x^Λ is A-normalized, Q_1 is a direct sum of proper A-weight subspaces. Choose homogeneous coordinates in U_1^Λ in which conjugation by a_1 is diagonal. The volume-decay exponent of a Radon measure supported on Q_1 is then bounded above by the logarithmic Jacobian of a_1 on Q_1, strictly smaller than the Haar exponent on U_1^Λ. Repeating the conditional-entropy calculation of Lemma 0.1 with the first-coordinate subordinate partition refined to Q_1 therefore makes the inequality strict. This contradicts the equality just proved. Thus $\pi_1(P_x^\Lambda) = U_1^\Lambda$ almost everywhere; the second projection is identical.

Finally, projective leafwise equivariance gives $P^\Lambda_{\alpha^t x} = \alpha^t P_x^\Lambda \alpha^{-t}$ on the common conull domain of the leafwise systems. Measurability follows from the measurability of the leafwise measure and the Effros–Borel measurability of closed support. This is the precise conclusion needed below. We deliberately do not replace the family by a fixed subgroup: if the same coarse weight occurs in both factors, graph subspaces between the two weight spaces form a continuous family, so ergodicity alone does not make the support group constant. □

Proposition 0.3 (The support group is the algebraic datum to propagate)**.** *For every coarse Lyapunov class Λ, the joining supplies, for almost every x, a connected subgroup*

$$P_x^\Lambda \leq U_1^\Lambda \times U_2^\Lambda$$

with surjective coordinate projections, and these groups form an A-equivariant measurable family. The remaining rigidity problem is to choose support directions for finitely many roots at one common typical point and convert their commutators into translation invariance of the joining.

Proof. The first assertion is Theorem 0.2. Volume 14's high-entropy theorem says that, for a leafwise measure supported on a product of different coarse Lyapunov groups, commutators of support directions from linearly independent weights lie in the invariance subgroup. Chapter 7 applies precisely that mechanism. □

> **Proof and provenance.** The argument follows the conditional-entropy inequality, stable-group equality, coarse-Lyapunov equality, and support-group construction in the original joining proof. The important correction to the first draft is conceptual: Haar marginals force *equality* in the entropy inequality; the proof does not begin by searching for a positive entropy deficit.

Worked Example 0.4 (A diagonal self-joining)**.** For the diagonal joining of SL_3/Γ with itself, the rootwise support group is

$$P^{ij} = \{(u_{ij}(r), u_{ij}(r)) : r \in \mathbb{R}\}.$$

Its projections onto both root groups are surjective although the joining is far from the product. This is exactly why support-surjectivity, rather than "full planar Haar conditional," is the right intermediate conclusion.

CHAPTER 4

Common Factors and Relatively Independent Joinings

Proof architecture. The chapter is organized as the chain *Relative product from disintegration* $\longrightarrow$ *Nontrivial common factor gives nonproduct joining* $\longrightarrow$ *Factors reduce to self-joinings*. The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

Lemma 0.1 (Relative product from disintegration)**.** *Let $\pi_i : (X_i, m_i) \to (Y, \nu)$ be factor maps and $m_i = \int m_{i,y}\, d\nu(y)$ the disintegrations. Then*

$$\lambda_Y = \int m_{1,y} \otimes m_{2,y}\, d\nu(y)$$

is an A-invariant joining.

Proof. Each coordinate marginal integrates to m_i. Equivariance of the factor maps and uniqueness of disintegration give $a_* m_{i,y} = m_{i,ay}$ almost everywhere. Changing variables $y \mapsto ay$ in the integral proves invariance. $\square$

Theorem 0.2 (Nontrivial common factor gives nonproduct joining). *If the common factor Y is nontrivial, the relatively independent joining λ_Y differs from $m_1 \otimes m_2$.*

Proof. Choose a measurable $B \subset Y$ with $0 < \nu(B) < 1$. For $E_i = \pi_i^{-1}(B)$, the relative joining gives $\lambda_Y(E_1 \times E_2) = \nu(B)$, whereas the product joining gives $\nu(B)^2$. These are different. $\square$

Proposition 0.3 (Factors reduce to self-joinings)**.** *To classify measurable factors of one ergodic homogeneous action, it suffices to classify its ergodic self-joinings: the relatively independent self-joining over a factor decomposes into ergodic self-joinings, and the factor equivalence relation is recovered from their supports.*

Proof. Apply Lemma 0.1 with $X_1 = X_2 = X$. Pairs in the same factor fibre are precisely those seen by the support of the fibrewise product conditionals. Ergodic decomposition preserves this equivalence relation up to null sets, so a classification of the component supports determines the measurable fibre relation. □

Worked Example 0.4 (A finite quotient factor)**.** If a homogeneous action has a finite algebraic quotient Y, the relative self-joining is the uniform mixture of products over the finitely many fibres. It is visibly not the full product unless Y is a point, and each ergodic component is supported on an algebraic correspondence between two fibres.

CHAPTER 5

Algebraic Correspondences

Proof architecture. The chapter is organized as the chain *Goursat lemma for connected simple factors* $\longrightarrow$ *Algebraic joining from a correspondence subgroup* $\longrightarrow$ *Graph root groups generate the correspondence.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

Lemma 0.1 (Goursat lemma for connected simple factors)**.** *Let G_1, G_2 be connected almost simple Lie groups and $L \leq G_1 \times G_2$ a connected closed subgroup whose projections are surjective. Then either $L = G_1 \times G_2$, or after dividing by finite central kernels L is the graph of an isomorphism $G_1/Z_1 \to G_2/Z_2$.*

PROOF. Let $N_1 = \{g_1 : (g_1, e) \in L\}$ and define N_2 similarly. They are closed normal subgroups. Almost simplicity makes each finite central or the whole group. If $N_1 = G_1$, surjectivity gives the product. Otherwise both are finite central. For $(g_1, g_2) \in L$, the class of g_2 modulo N_2 depends only on the class of g_1 modulo N_1, giving an isomorphism of the quotients and identifying L with its graph. $\square$

THEOREM 0.2 (Algebraic joining from a correspondence subgroup). *Let $L \leq G_1 \times G_2$ have surjective projections and suppose $L \cap (\Gamma_1 \times \Gamma_2)$ is a lattice. The normalized L-invariant measure on the closed orbit $L(e\Gamma_1, e\Gamma_2)$ is a joining of Haar measures. If L is proper and the factors are almost simple, it comes from a finite-central algebraic correspondence as in Lemma 0.1.*

PROOF. Projection of the finite L-invariant measure to G_i/Γ_i is a finite G_i-invariant measure because the projection of L is G_i. Uniqueness of normalized Haar probability makes each marginal Haar. Properness and the simple-factor description follow from Goursat's lemma. $\square$

Proposition 0.3 (Graph root groups generate the correspondence)**.** *If an ergodic joining is invariant under graph root groups for a simple root system that generates G_1, and the corresponding root maps preserve brackets, then* *these graph groups generate a connected correspondence subgroup L with surjective projections.*

Proof. The root groups generate G_1. Bracket preservation makes the assignments on root Lie algebras compatible with the Chevalley relations, hence they integrate to a homomorphism modulo the finite center. The generated subgroup in the product is its graph; each projection contains all root groups and is therefore surjective. □

Worked Example 0.4 (The diagonal subgroup)**.** For $G_1 = G_2 = SL_3(\mathbb{R})$ and identical lattices, $L = \{(g, g) : g \in G_1\}$ has surjective projections and lattice $\{(\gamma, \gamma)\}$. Its orbit measure is the diagonal self-joining, the basic proper algebraic correspondence.

CHAPTER 6

Commensurators and Finite Correspondences

Proof architecture. The chapter is organized as the chain *Commensurator correspondence* $\longrightarrow$ *Finite correspondence joining* $\longrightarrow$ *Arithmetic correspondences are countable.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

Lemma 0.1 (Commensurator correspondence)**.** *If* $g \in \mathrm{Comm}_G(\Gamma)$, *then*

$$\Gamma_g = \Gamma \cap g^{-1}\Gamma g$$

has finite index in both Γ *and* $g^{-1}\Gamma g$, *and the maps* $G/\Gamma_g \to G/\Gamma$ *given by* $h \mapsto h\Gamma$ *and* $h \mapsto hg^{-1}\Gamma$ *define a finite algebraic correspondence.*

Proof. This is the definition of commensuration. The two quotient maps have fibre cardinalities equal to the corresponding finite indices. Equivariance under left translation by G is immediate. □

Theorem 0.2 (Finite correspondence joining). *Pushing Haar probability on* G/Γ_g *through the two maps of Lemma 0.1 gives an ergodic algebraic self-joining of the* G *action and hence of every diagonal subgroup* $A \leq G$ *for which the correspondence measure is* A*-ergodic.*

Proof. The pushforward is a joining by the same marginal argument as Theorem 0.2. It is the homogeneous measure on the orbit of the graph subgroup (h, hg^{-1}). If the restricted A action is ergodic on this orbit, the joining is ergodic by definition. □

Proposition 0.3 (Arithmetic correspondences are countable)**.** *For an arithmetic lattice, finite correspondences arising from the rational commensurator form a countable family.*

Proof. The rational points of a linear algebraic group over $\mathbb{Q}$ are countable. Arithmetic commensurators lie in such rational points up to finite central data, so their associated correspondences are countable. □

Worked Example 0.4 (Hecke correspondences)**.** For $SL_2(\mathbb{Z})$, a rational matrix of determinant a prime power defines a commensurator correspondence. The two finite maps from the common finite cover are precisely the geometric source of classical Hecke correspondences; Volume 19 uses this idea dynamically.

CHAPTER 7

Higher-Rank Joining Rigidity

Proof architecture. The chapter is organized as the chain *Independent support directions produce invariant commutators* $\longrightarrow$ *Higher-rank joining rigidity, real split type A* $\longrightarrow$ *Pairwise-product criterion for higher joinings*. The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices.

1. From support directions to an invariant subdirect subgroup

Throughout this chapter the factors are connected semisimple split type-A groups with no local rank-one factor for the acting higher-rank diagonal group. This is the exact real setting needed later; the more general S-arithmetic theorem is recorded only as a source comparison.

Lemma 1.1 (Independent support directions produce invariant commutators)**.** *Let x belong to the common conull domain of all rootwise leafwise systems, and let $P_x^{\Lambda_1}, P_x^{\Lambda_2}$ be the support groups from Theorem 0.2, where Λ_1, Λ_2 are linearly independent coarse Lyapunov weights. If $w_i \in \mathfrak{p}_x^{\Lambda_i}$, then*

$$[w_1, w_2] \in \mathfrak{i},$$

where $\mathfrak{i}$ is the Lie algebra generated by all Ad-unipotent one-parameter subgroups preserving λ.

Proof. Choose a regular element a for which both weights are stable and consider the leafwise measure on the full stable horospherical group. The high-entropy theorem of Volume 14 gives connected groups

$$H_x \lhd P_x \leq U^-(a)$$

such that the stable leafwise measure is supported on P_x, invariant under H_x, and the images of elements belonging to distinct coarse Lyapunov groups commute in P_x/H_x. The rootwise support groups of Chapter 3 are the corresponding coarse projections of P_x. Hence, at this same typical point x, representatives $\exp(w_i) \in P_x^{\Lambda_i}$ have group commutator in H_x. Shrinking the parameters and using the Baker–Campbell–Hausdorff expansion gives

$$[\exp(sw_1), \exp(tw_2)] = \exp\bigl(st[w_1, w_2] + O(|st|(|s| + |t|))\bigr).$$

The invariance group is closed and A-normalized, so rescaling by A and letting $s, t \to 0$ shows that the full one-parameter subgroup generated by $[w_1, w_2]$ preserves λ. Thus $[w_1, w_2] \in \mathfrak{i}$. □

THEOREM 1.2 (Higher-rank joining rigidity, real split type A). *Let $X_i = G_i/\Gamma_i$ carry Haar probability m_i, where each G_i is a finite product of connected almost-simple split type-A real algebraic groups and each Γ_i is arithmetic in a fixed rational representation. Assume the factors and that the injective diagonal actions $\alpha_i : \mathbb{R}^d \to G_i$ have no local rank-one factors. Every ergodic joining λ of the two Haar actions is algebraic: there is a connected closed subgroup*

$$L \leq G_1 \times G_2, \qquad \pi_i(L) = G_i,$$

such that λ is Haar probability on one closed finite-volume L-orbit.

PROOF. Let I be the connected subgroup generated by all Ad-unipotent one-parameter subgroups preserving λ, and let $\mathfrak{i}$ be its Lie algebra. We first show that both coordinate projections of $\mathfrak{i}$ are surjective.

Intersect the conull domains of Theorem 0.2 over the finitely many coarse root classes and choose one point x in the intersection. Fix a root vector E_{rs} in a simple SL_k factor of G_1. Since $k \geq 3$, choose $\ell \neq r, s$ and write

$$E_{rs} = [E_{r\ell}, E_{\ell s}],$$

where the two roots are linearly independent. By Theorem 0.2, the support groups for those two roots project surjectively to the first factor. Hence choose lifts $w_1 \in \mathfrak{p}_x^{r\ell}$ and $w_2 \in \mathfrak{p}_x^{\ell s}$ whose first coordinates are $E_{r\ell}$ and $E_{\ell s}$. Lemma 1.1, applied at the same point x, gives $[w_1, w_2] \in \mathfrak{i}$, and its first coordinate is E_{rs}. Root vectors generate the simple Lie algebra, so $\pi_1(\mathfrak{i}) = \mathfrak{g}_1$. The same argument gives $\pi_2(\mathfrak{i}) = \mathfrak{g}_2$ factor by factor.

The kernel

$$\mathfrak{n}_2 = \mathfrak{i} \cap (0 \oplus \mathfrak{g}_2)$$

is a Lie ideal of $\mathfrak{g}_2$: for $w \in \mathfrak{n}_2$ and $v_2 \in \mathfrak{g}_2$, choose $(v_1, v_2) \in \mathfrak{i}$ by surjectivity and compute $[(0, w), (v_1, v_2)] = (0, [w, v_2]) \in \mathfrak{i}$. On each almost-simple factor the kernel is therefore either zero or the whole factor. Consequently $\mathfrak{i}$ is a product of full factor ideals and graph Lie algebras of isomorphisms between isomorphic factors.

It remains to show that the acting diagonal Lie algebra $\mathfrak{a}$ lies in $\mathfrak{i}$. This is immediate on every full-product component. On a graph component write

$$\mathfrak{i} = \{(v, \varphi v) : v \in \mathfrak{g}_1\}$$

for a Lie-algebra isomorphism φ. Since $\mathfrak{i}$ is normalized by the diagonal action, for $a = (a_1, a_2) \in \mathfrak{a}$ and every root vector X,

$$([a_1, X], [a_2, \varphi X]) \in \mathfrak{i}.$$

The graph identity therefore yields $[a_2, \varphi X] = \varphi[a_1, X] = [\varphi a_1, \varphi X]$. Thus $a_2 - \varphi a_1$ centralizes every root space. A split semisimple Lie algebra is generated by its root spaces and has zero center, so $a_2 = \varphi a_1$. Hence $a \in \mathfrak{i}$.

Therefore $A \subset I$. The measure is I-invariant by construction, and because it is A-ergodic it is also I-ergodic: every I-invariant set is A-invariant. Theorem 1.1 applies to the arithmetic product quotient and the connected unipotent-generated group I, and gives a connected closed subgroup $L \geq I$ and a closed finite-volume orbit on which λ is Haar. Its coordinate projections contain the projections of I, hence are all of G_i. This is the required algebraic joining. □

Proposition 1.3 (Pairwise-product criterion for higher joinings)**.** *In the same scope, an ergodic r-fold joining whose every two-coordinate projection is the product is the full product joining.*

Proof. Apply Theorem 1.2 inductively to obtain an algebraic subdirect subgroup $L \leq \prod_{j=1}^{r} G_j$. For a proper connected subdirect product of almost-simple factors, a repeated Lie-algebra Goursat argument produces two factors on which the projection of L is a graph subgroup rather than the full product: take a minimal set of factors on which the projection is proper; its kernel in one coordinate is an ideal, and minimality forces a graph relation with another factor. The corresponding two-coordinate marginal is then a proper algebraic joining, contradicting the pairwise-product hypothesis. Hence L is the full product and λ is the product Haar measure. □

> **Proof and provenance.** This proof follows the architecture of Einsiedler–Lindenstrauss's real joining theorem: coarse-Lyapunov support groups project onto both marginals; the high-entropy commutator theorem produces invariant unipotent directions; their invariant Lie algebra projects onto both factors; the zero-weight diagonal directions are then shown to lie in that Lie algebra; finally the internal generated-unipotent classification gives algebraicity. The present proof is specialized to split type A, where the "no local rank-one factors" bracket lemma is the concrete identity $E_{ij} = [E_{i\ell}, E_{\ell j}]$.

Worked Example 1.4 (A graph subgroup appears automatically)**.** For two copies of SL_3, suppose the invariant Lie algebra has zero kernel in each coordinate. Surjectivity makes it the graph of an isomorphism $\varphi : \mathfrak{sl}_3 \to \mathfrak{sl}_3$. Normalization by the diagonal action then forces the two diagonal parameters to be intertwined by φ. Thus the graph is not an arbitrary correspondence: it automatically contains the acting higher-rank diagonal group.

CHAPTER 8

Factor Classification from Joining Rigidity

Proof architecture. The chapter is organized as the chain *Graph components and the compact affine group* $\longrightarrow$ *Factor classification in the joining-rigidity scope.* The first result isolates the local mechanism, the second is the load-bearing theorem, and the third records the form inherited later. When a later step cites this chapter, check which of these three levels it actually needs; the stronger theorem should never be invoked when the local lemma suffices. Let $X = G/\Gamma$ be in the scope of Theorem 1.2, let m be Haar probability, and let $\mathcal{A}$ be a countably generated A-invariant sigma-algebra. Write

$$m = \int m_x^{\mathcal{A}}\, dm(x), \qquad \rho := m \times_{\mathcal{A}} m = \int m_x^{\mathcal{A}} \otimes m_x^{\mathcal{A}}\, dm(x)$$

for the disintegration and relatively independent self-joining. The support of ρ is the measurable equivalence relation "belong to the same $\mathcal{A}$-atom."

Lemma 0.1 (Fibre symmetries and preservation of the relative joining)**.** *Let ξ be an affine, Haar-preserving automorphism of X which preserves almost every $\mathcal{A}$-atom setwise. Then*

$$\xi_* m_x^{\mathcal{A}} = m_x^{\mathcal{A}} \quad \textit{for a.e. } x, \qquad (\mathrm{Id}, \xi)_* \rho = \rho.$$

Proof. The family $x \mapsto \xi_* m_x^{\mathcal{A}}$ is another disintegration of m over $\mathcal{A}$: ξ preserves m, and preserving atoms means that the new conditional is still supported on the atom of x. Uniqueness of regular conditional probabilities gives the first equality. Substitute it into the barycenter formula for ρ to obtain the second. $\square$

Lemma 0.2 (Graph components and the compact affine group)**.** *There are a connected normal subgroup $N \triangleleft G$, a quotient $\psi : G \twoheadrightarrow G_1 = G/N$, and a lattice $\Gamma_1 \geq \psi(\Gamma)$ such that the following hold on $X_1 = G_1/\Gamma_1$.*

(1) Every ergodic component of the pushforward of ρ is a graph measure of an affine automorphism of X_1 commuting with the induced A-action.

(2) Let Ξ be the group of Haar-preserving affine automorphisms of X_1 that preserve the pushed-forward factor atoms. Modulo the discrete

deck transformations R_{Γ_1} acting trivially on X_1,

$$\Phi := \Xi / R_{\Gamma_1}$$

is a compact group.

Proof. Decompose $\rho = \int \rho_\omega \, d\theta(\omega)$ into ergodic diagonal A-joinings. Theorem 1.2 makes every ρ_ω Haar measure on a closed orbit of a connected subdirect group $H_\omega \leq G \times G$. Put

$$N_{1,\omega} = \{g : (g,e) \in H_\omega\}^\circ, \qquad N_{2,\omega} = \{g : (e,g) \in H_\omega\}^\circ.$$

These are connected normal subgroups of G: normality follows from surjectivity of the opposite projection, exactly as in the Lie-algebra Goursat argument of Chapter 7.

If $N_{1,\omega}$ is nontrivial, then for ρ_ω-almost every pair (x, y) the entire local $N_{1,\omega}$-orbit of x is paired with the same y. Since ρ is supported on the factor equivalence relation, $N_{1,\omega}$ acts inside almost every $\mathcal{A}$-atom. The same is true for $N_{2,\omega}$. In a finite product of almost-simple groups there are only finitely many connected normal subgroups generated by simple factors. Let N be the product of all such kernel factors occurring on a positive θ-measure set. Then N acts inside almost every factor atom and is normal in G.

Pass to $G_1 = G/N$. The image lattice may acquire a finite deck extension; replace $\psi(\Gamma)$ by the discrete lattice Γ_1 generated by that finite extension. After this quotient, an ergodic component of the relative joining has trivial connected kernel in both coordinates. Lie-group Goursat therefore identifies its connected subgroup with the graph of an isomorphism between the relevant semisimple quotient factors. The translate of the graph orbit changes the isomorphism into an affine automorphism ξ of X_1. Diagonal A-invariance of the component says that ξ commutes with the induced A-action modulo the deck group. This proves (1).

Let Ξ be the affine maps preserving the factor atoms and commuting with A, and quotient by the deck subgroup R_{Γ_1}. The ergodic decomposition of the pushed-forward relative joining is now a probability ν' on graph measures, hence a probability on $\Phi = \Xi / R_{\Gamma_1}$. For $\xi \in \Xi$, Lemma 0.1 says that (Id, ξ) preserves the relative joining. It sends the graph of η to the graph of $\xi\eta$; uniqueness of ergodic decomposition gives

$$(L_\xi)_* \nu' = \nu'.$$

Thus ν' is a finite nonzero left-invariant measure on the locally compact group Φ. By uniqueness of Haar measure it is normalized Haar measure. A locally compact group has finite Haar measure if and only if it is compact, so Φ is compact. $\square$

Theorem 0.3 (Factor classification in the joining-rigidity scope). *There are a surjective Lie-group homomorphism $\psi : G \to G_1$, a lattice $\Gamma_1 \geq \psi(\Gamma)$, and a compact group*

$$\Phi \leq \mathrm{Aff}(G_1/\Gamma_1)$$

commuting with the induced diagonal action such that, modulo m-null sets,

$$\boxed{\mathcal{A} = \psi^{-1}(\mathcal{B}_\Phi),}$$

where $\mathcal{B}_\Phi$ is the sigma-algebra of Φ-invariant Borel subsets of G_1/Γ_1.

Proof. Use ψ, Γ_1, Φ from Lemma 0.2 and let m_Φ be normalized Haar measure on Φ. The ergodic decomposition of the pushed-forward relative joining is the graph decomposition

$$\rho_1 = \int_\Phi (\mathrm{Id}, \phi)_* m_1 \, dm_\Phi(\phi).$$

Disintegrating this identity over the first coordinate gives, for m_1-a.e. y,

$$(m_y^{\mathcal{A}})_1 = (\phi \mapsto \phi y)_* m_\Phi.$$

Thus the conditional measure on the factor atom through y is exactly Haar measure on the compact Φ-orbit of y.

If $B \subset X_1$ is Φ-invariant, then

$$\mathbb{E}(1_{\psi^{-1}B} \mid \mathcal{A})(x) = \int_\Phi 1_B(\phi\psi(x)) \, dm_\Phi(\phi) = 1_B(\psi(x)),$$

so $\psi^{-1}B$ is $\mathcal{A}$-measurable. Hence $\psi^{-1}(\mathcal{B}_\Phi) \subset \mathcal{A}$.

Conversely, if $E \in \mathcal{A}$, then for almost every x its conditional indicator is constant on the support of $m_x^{\mathcal{A}}$. The preceding formula identifies that support with the Φ-orbit of $\psi(x)$. Therefore the pushed-forward indicator of E is constant on almost every Φ-orbit, so it is Φ-invariant modulo null sets. This proves $\mathcal{A} \subset \psi^{-1}(\mathcal{B}_\Phi)$ and hence equality.

For completeness, $\mathcal{B}_\Phi$ is countably generated. Choose a countable dense family $f_j \in C_c(X_1)$ and average it over Φ:

$$\bar{f}_j(y) = \int_\Phi f_j(\phi y) \, dm_\Phi(\phi).$$

The rational sublevel sets of the $\bar{f}_j$ separate compact Φ-orbits and generate the invariant Borel sigma-algebra. Thus the resulting factor is a standard measurable factor, compatible with the disintegration used above. □

Proposition 0.4 (Relative self-joining dictionary)**.** *In the preceding theorem, the factor atom through x is the compact orbit $\Phi\psi(x)$; the conditional measure $m_x^{\mathcal{A}}$ is its Haar probability; and the ergodic graph components of the relative self-joining $m \times_{\mathcal{A}} m$ are the graph measures of the elements of Φ.*

Proof. The graph decomposition in the proof of Theorem 0.3 gives the third description. Disintegrating over the first coordinate gives the Haar measure on $\Phi\psi(x)$, and the support of that conditional probability is the factor atom modulo null sets. These are three disintegrations of the same relative self-joining, so uniqueness makes them equivalent descriptions. $\square$

> **Proof and provenance.** The proof follows Section 6 of Einsiedler–Lindenstrauss: quotient the connected kernel of the factor relation, show that the remaining ergodic self-joinings are graphs of affine maps, let the affine symmetry group act on the graph components of the relative self-joining, and deduce compactness from the resulting finite Haar measure. The final equality of sigma-algebras comes from the explicit Haar description of the fibre conditional measures [EL07].

Worked Example 0.5 (A finite affine factor)**.** If Φ is a finite group of affine automorphisms, the factor map is simply $G_1/\Gamma_1 \to \Phi\backslash(G_1/\Gamma_1)$. The conditional measure on a factor atom is uniform measure on the finite Φ-orbit, and the relative self-joining is the average of the graph measures of the elements of Φ. The general theorem is exactly the compact-group version of this picture.

APPENDIX A

The S-Arithmetic Joining Theorem

1. Class-$\mathcal{A}'$ actions and local coarse weights

Let S be a finite set of places of $\mathbb{Q}$. For a connected $\mathbb{Q}$-almost-simple semisimple group $\mathbf{G}_i$, let $G_i < \mathbf{G}_i(\mathbb{Q}_S)$ be finite index and $X_i = \Gamma_i \backslash G_i$ an S-arithmetic quotient saturated by unipotents. Let $a_i : \mathbb{Z}^d \to G_i$ be proper, $d \geq 2$, and suppose the product action $a = (a_1, \ldots, a_r)$ is of class-$\mathcal{A}'$: at the real place all eigenvalues are positive real numbers and at a finite place the absolute values of eigenvalues belong to the cyclic group generated by one fixed nonunit local modulus.

For a local adjoint weight α write $U_i^{[\alpha]}$ for the subgroup generated by all root groups whose logarithmic absolute-value characters are positive multiples of α on $\mathbb{Z}^d$.

Lemma 1.1 (Finite local coarse-Lyapunov decomposition). *For each i, the set of nonzero coarse weights of the $\mathbb{Z}^d$ action on* $\mathrm{Lie}(G_i)$ *is finite. If $n \in \mathbb{Z}^d$ lies outside the finitely many weight hyperplanes, then its stable horospherical group is the ordered product of the $U_i^{[\alpha]}$ with $\alpha(n) < 0$.*

Proof. At every place the adjoint representation has finitely many algebraic weights. Compose each local algebraic weight with a_i and then with the logarithm of the corresponding normalized absolute value; this gives a real linear functional on $\mathbb{Z}^d \otimes \mathbb{R}$. Proportional positive functionals define one coarse class. There are finitely many because S and the root systems are finite. Conjugation by $a_i(n)$ contracts exactly the root groups on which the functional is negative. Ordering the roots by height and using the Chevalley commutator relation shows that multiplication of the corresponding root groups is a homeomorphism on a conull open cell of the stable horospherical group. This proves the product description. □

2. Leafwise measures of a joining

Let μ be an ergodic joining of the Haar actions on $X = X_1 \times \cdots \times X_r$. For a product coarse group $U^{[\alpha]}$ write $\mu_x^{[\alpha]}$ for its projectively normalized leafwise measure. At a finite place we never use topological connectedness: all hulls below are local algebraic hulls, and all invariant groups are the closed groups generated by local one-parameter unipotents.

Lemma 2.1 (Marginal projection of leafwise measures)**.** *If $\pi_i : X \to X_i$ is the ith coordinate projection and the coarse class $[\alpha]$ occurs in the ith factor, then for μ-a.e. x*

$$(\pi_i)_*\mu_x^{[\alpha]} \;\propto\; m_{U_i^{[\alpha]}},$$

where $m_{U_i^{[\alpha]}}$ is Haar measure on the full local coarse unipotent group. Consequently the coordinate projection of $\operatorname{supp}\mu_x^{[\alpha]}$ *is dense in $U_i^{[\alpha]}$.*

Proof. Choose subordinate product flow boxes whose coordinate projections are injective on the chosen compact identity neighborhoods. Disintegrate the joining first along the product coarse plaques and then push the resulting conditional measures to the ith factor. Uniqueness of conditional probabilities shows that the pushforward family is precisely the leafwise family of the marginal measure. The marginal is Haar and the quotient is saturated by unipotents, so its leafwise measure on the entire coarse group is Haar. Haar support is the whole group. Since the support of a pushforward is contained in the closure of the projection of the original support, the projection of $\operatorname{supp}\mu_x^{[\alpha]}$ is dense. $\square$

Definition 2.2 (Local algebraic support hull)**.** For a typical x, let $\mathbf{P}_x^{[\alpha]}$ be the smallest product of local Zariski-connected unipotent algebraic subgroups of $\prod_{v\in S}\mathbf{U}^{[\alpha]}(\mathbb{Q}_v)$ whose local points contain $\operatorname{supp}\mu_x^{[\alpha]}$. We write $P_x^{[\alpha]} = \mathbf{P}_x^{[\alpha]}(\mathbb{Q}_S)$.

Proposition 2.3 (Full projections of the local algebraic hull)**.** *The measurable family $P_x^{[\alpha]}$ is equivariant under the diagonal action and, whenever $[\alpha]$ occurs in factor i,*

$$\pi_i(P_x^{[\alpha]}) = U_i^{[\alpha]}.$$

No assertion of almost-sure constancy of $P_x^{[\alpha]}$ is made.

Proof. Projective leafwise equivariance gives, after changing the normalizing scalar,

$$\operatorname{supp}\mu_{a(n)x}^{[\alpha]} = a(n)\,\operatorname{supp}\mu_x^{[\alpha]}\,a(-n).$$

Therefore therefore the minimal local algebraic hull transforms by conjugacy. The image of a local algebraic subgroup under an algebraic coordinate projection is an algebraic subgroup and hence closed in the local analytic topology. By Lemma 2.1 it contains a dense subset of $U_i^{[\alpha]}$, and is therefore the full coarse unipotent group. $\square$

Lemma 2.4 (Support realization after class-$\mathcal{A}'$ rescaling)**.** *Fix a typical x and a local root direction $Y \in \operatorname{Lie} U_i^{[\alpha]}$. There exist support elements $p_j \in \operatorname{supp} \mu_x^{[\alpha]}$, integers $n_j \in \mathbb{Z}^d$, and normalizing scalars $r_j \to 0$ such that in local exponential/root coordinates*

$$r_j^{-1} \log(a(n_j) p_j a(-n_j)) \longrightarrow \widetilde{Y}, \qquad \pi_i(\widetilde{Y}) = Y,$$

with $\widetilde{Y} \in \operatorname{Lie} P_x^{[\alpha]}$. The analogous statement holds simultaneously for finitely many coarse classes on one common conull set.

Proof. By Proposition 2.3, the algebraic tangent map of $\pi_i : P_x^{[\alpha]} \to U_i^{[\alpha]}$ is onto. Choose a tangent vector $\widetilde{Y}$ lifting Y. The local algebraic hull is generated by the Zariski closure of the support. Hence every nonempty Zariski-open cone about $\widetilde{Y}$ meets the set of normalized logarithmic directions of finite products of support elements and inverses. The product structure of leafwise measures lets us realize those finite products inside one slightly larger coarse plaque without changing the projective conditional class.

Choose n_j in a chamber on which $[\alpha]$ contracts while ratios of the other weights occurring in the chosen product avoid the finitely many walls. For a real place divide by the dominant exponential modulus; for a finite place divide by the corresponding integral power of the distinguished local modulus from the class-$\mathcal{A}'$ hypothesis. The Baker–Campbell–Hausdorff formula at the real place and the Chevalley polynomial multiplication law at a finite place show that higher commutator terms have strictly larger weight and disappear after this normalization. The normalized directions converge to $\widetilde{Y}$. Because only finitely many root classes are used later, intersecting their conull domains and using one chamber subsequence gives the simultaneous assertion. □

Lemma 2.5 (Joining entropy has no marginal deficit)**.** *For a regular $n \in \mathbb{Z}^d$, the entropy contribution of μ along the product stable group is at least the largest marginal Haar contribution and at most their sum. Equality on a coarse class forces the local algebraic support hull to have full projection to every participating factor.*

Proof. Choose subordinate partitions for the local product stable group. A marginal partition is a factor partition, so its entropy cannot exceed the joining entropy. The chain rule and monotonicity under conditioning bound the joining contribution above by the sum of the marginal contributions. For Haar marginals, Abramov–Rokhlin applied to either coordinate makes the full stable-group inequalities equalities. Product structure of leafwise measures decomposes these quantities into the finite sum of coarse contributions; since each coarse inequality has the same sign, equality of the sums forces equality class by class.

If the algebraic projection of a coarse hull were proper, its tangent algebra would omit a local weight direction. The Jacobian exponent of contraction on that proper algebraic subgroup is then strictly smaller than on the full marginal coarse group. Computing conditional entropy by decay of normalized leafwise balls gives a strict loss, contradicting coarse class equality. This is the local-field version of the entropy/support calculation already proved in Chapter 3. □

3. From support directions to invariant unipotents

Lemma 3.1 (Commutator production from realized support directions)**.** *Let $[\alpha]$ and $[\beta]$ be nonproportional coarse classes for which a Chevalley commutator has a nonzero $\alpha+\beta$ component in some factor. Then the translation-stabilizer group of μ contains the local one-parameter unipotent subgroup generated by every $\alpha+\beta$ root direction obtained from brackets of lifted tangent vectors in* $\operatorname{Lie} P_x^{[\alpha]}$ *and* $\operatorname{Lie} P_x^{[\beta]}$.

Proof. Work on the common conull set from Lemma 2.4. Realize tangent lifts X, Y by support elements p_j, q_j at scales chosen so that after conjugation by a regular sequence $a(n_j)$ the X and Y terms remain at controlled size while all higher-weight errors contract. The leafwise translation rule identifies the conditional measures at p_jx, q_jx, p_jq_jx and q_jp_jx up to scalar. Forming the commutator cancels these projective scalars. In local root coordinates, the real BCH formula and the finite-place Chevalley commutator formula give

$$[p_j, q_j] = \exp(r_js_j[X,Y] + o(r_js_j))$$

at the real place and the identical leading-root statement in valuation coordinates at finite places. After conjugation and normalization, a nontrivial $\alpha+\beta$ translation survives. Lusin continuity of the leafwise system on recurrent return sets identifies the limiting conditional with its translate. The projective-to-exact translation lemma of Volume 13 turns proportionality into equality, and the closed-stabilizer lemma there then gives the entire local one-parameter subgroup. Varying X, Y proves the claim. □

Proposition 3.2 (Subdirect invariant algebraic group)**.** *Let L^+ be the closed subgroup generated by all local one-parameter unipotent subgroups obtained from Lemma 3.1, and let $\mathbf{L}$ be its local algebraic hull. Then μ is L^+-invariant and every coordinate projection $\mathbf{L} \to \mathbf{G}_i$ is surjective.*

Proof. Invariance follows from the construction and closedness of the translation stabilizer. Fix a local almost-simple factor of G_i. Properness and higher rank provide regular chamber elements with at least two independent coarse signs. Proposition 2.3 and Lemma 2.4 lift every simple-root tangent direction. Whenever two roots add to a root, Lemma 3.1 places the bracket direction in the invariant algebraic hull. Induction on root height therefore supplies all positive root groups; applying the argument in the opposite chamber supplies all negative root groups. A semisimple local almost-simple group saturated by unipotents is algebraically generated by these root groups. Thus the algebraic projection of $\mathbf{L}$ is the full local factor, and this holds in every coordinate. ☐

4. Reduction of S-arithmetic unipotent classification to Volume 4

Lemma 4.1 (Arithmetic matrix realization of a semisimple S-quotient)**.** *Let $\mathbf{H}$ be a connected semisimple $\mathbb{Q}$-group and let $\Gamma < \mathbf{H}(\mathbb{Q}_S)$ be an S-arithmetic lattice. After replacing Γ by a finite-index subgroup and then descending through the resulting finite cover, there is a faithful rational representation $\iota : \mathbf{H} \hookrightarrow \mathrm{SL}_N$ for which*

$$\iota(\Gamma) \quad \textit{is commensurable with} \quad \iota(\mathbf{H}(\mathbb{Q}_S)) \cap \mathrm{SL}_N(\mathbb{Z}_S),$$

and $\Gamma\backslash\mathbf{H}(\mathbb{Q}_S)$ identifies with a closed finite-volume rational homogeneous orbit in a finite cover of $\mathrm{SL}_N(\mathbb{Z}_S)\backslash \mathrm{SL}_N(\mathbb{Q}_S)$.

Proof. Choose a faithful rational representation and enlarge it by its determinant character so that the image lies in a special linear group. The definition of an S-arithmetic subgroup gives commensurability with the integral points of the induced $\mathbb{Z}_S$-model. Intersecting the two arithmetic groups produces a common finite-index subgroup, hence a finite cover of the original quotient. The Borel–Harish–Chandra theorem proved in Volume 4 makes the image orbit closed and of finite volume. Properness of the orbit map follows from the arithmetic marking/height theorem there: a sequence in the rational subgroup whose image remains in a compact part of the ambient arithmetic quotient has, after multiplication by the intersection lattice, bounded integral height and therefore a convergent subsequence in the subgroup. Thus the finite cover embeds as the asserted closed homogeneous orbit. A finite deck group preserves homogeneity, so statements proved on the common cover descend to the original quotient. ☐

Proposition 4.2 (Generated-unipotent S-arithmetic measure classification)**.** *Let $W < \mathbf{H}(\mathbb{Q}_S)$ be generated by finitely many local algebraic one-parameter unipotent subgroups. Every W-invariant, W-ergodic probability measure on an S-arithmetic quotient of the preceding lemma is normalized Haar measure on a closed finite-volume orbit of an algebraic subgroup. The same conclusion holds for the ergodic components of a finite W-invariant probability measure.*

Proof. Lift the measure to the common finite arithmetic cover from Lemma 4.1 and push it to the ambient matrix quotient. Each generating one-parameter subgroup is a local algebraic unipotent subgroup of $\mathrm{SL}_N(\mathbb{Q}_S)$. The generated-action reduction and arithmetic S-linear classification proved in Volume 4 therefore make each ergodic lifted component Haar on a closed finite-volume homogeneous orbit. Because the lifted measure is supported on the closed rational $\mathbf{H}$-orbit, the translation stabilizer of such a component has tangent and local points inside $\mathbf{H}(\mathbb{Q}_S)$: otherwise a small orbit segment would leave the closed submanifold while remaining in the support. Hence its algebraic hull is a subgroup of $\mathbf{H}$. Projecting through the finite cover gives the asserted homogeneous measure downstairs. Ergodic decomposition gives the final assertion. □

5. Algebraic Goursat and homogeneous classification

Lemma 5.1 (Rational subdirect product classification)**.** *Let $\mathbf{L} < \prod_{i=1}^{r} \mathbf{G}_i$ be a Zariski-connected $\mathbb{Q}$-subgroup with surjective projections to the $\mathbb{Q}$-almost-simple semisimple factors. Then after partitioning the indices into classes of mutually $\mathbb{Q}$-isogenous adjoint groups, $\mathbf{L}$ is a product of diagonal correspondence groups inside those classes. In particular, if no two $\mathbf{G}_i$ have isogenous adjoint forms, $\mathbf{L}$ is the full product.*

Proof. For two factors, the kernels of the two projections are connected normal $\mathbb{Q}$-subgroups. Almost simplicity makes each kernel either finite central or the whole factor. Surjectivity excludes the latter unless the subgroup is the full product. In the remaining case quotienting by the finite centers identifies the two adjoint groups through the graph of a $\mathbb{Q}$-isomorphism. For $r > 2$, project to the first $r-1$ factors and apply induction; the kernel in the last factor again yields either an independent product factor or a graph identification with one of the previous adjoint factors. This gives the partition and the claimed product of correspondences. □

Lemma 5.2 (Recurrence rigidity of correspondence parameters)**.** *Fix a rational subdirect isogeny type in a product of the $\mathbf{G}_i$ and let $\mathscr{C}$ be the algebraic parameter space of its conjugate graph embeddings. Let a class-$\mathcal{A}'$ $\mathbb{Z}^d$-action act on $\mathscr{C}$ by conjugation. Every invariant probability measure on $\mathscr{C}$ is supported on the zero-weight locus, equivalently on compact centralizer orbits. If the original joining is ergodic, its induced component-parameter measure is supported on one such orbit.*

Proof. Embed $\mathscr{C}$ equivariantly in a finite product of projective spaces of exterior powers of the adjoint representations by sending a graph subgroup to the lines representing its Lie algebra and the finitely many bracket relations.

On every affine chart the diagonal action is triangular with weights that are products of the real positive eigenvalue characters and integral powers of the distinguished finite-place modulus. Separate the coordinates into zero and nonzero weights.

Suppose an invariant probability gives positive mass to points having a nonzero coordinate of weight χ. Choose $n \in \mathbb{Z}^d$ with $\chi(n) > 0$ and avoid the finitely many other weight hyperplanes. On a set where that coordinate is bounded above and below away from zero, iteration by $a(kn)$ sends its absolute value to infinity at a real place or its valuation to $-\infty$ at a finite place. Such a set cannot return to itself for large k, contradicting Poincaré recurrence. Therefore every nonzero-weight coordinate vanishes almost surely. The remaining locus is fixed by the split-modulus part of the action. In the class-$\mathcal{A}'$ normalization, a real zero-weight eigenvalue is positive of modulus one and hence equals one; at a finite place the zero valuation part has compact closure. Passing to a finite-index subaction kills the finite torsion in that compact part, and the remaining parameter is fixed. (The finitely many cosets discarded by this passage are restored at the end by a finite algebraic average.) Thus the recurrent parameter measure is supported on compact centralizer orbits.

For the component measure induced by an ergodic joining, its quotient by the compact centralizer orbit is $\mathbb{Z}^d$-invariant and hence constant. After the finite-index reduction just made, the parameter itself is invariant, so ergodicity makes it a point mass. Restoring the finite cosets gives a finite orbit of algebraic component parameters, whose average is again the Haar measure on the corresponding finite-component algebraic homogeneous set. $\square$

THEOREM 5.3 (S-arithmetic higher-rank joining rigidity). *Let $r, d \geq 2$, let $\mathbf{G}_1, \ldots, \mathbf{G}_r$ be semisimple $\mathbb{Q}$-almost-simple groups, and let $X_i = \Gamma_i \backslash G_i$ be S-arithmetic quotients saturated by unipotents. Let $a_i : \mathbb{Z}^d \to G_i$ be proper homomorphisms whose product is of class-$\mathcal{A}'$. Every ergodic joining of the Haar actions of $a_i(\mathbb{Z}^d)$ on the X_i is an algebraic measure defined over $\mathbb{Q}$.*

PROOF. Propositions 2.3 and 3.2, together with Lemma 3.1, produce an L^+ generated by local one-parameter unipotents, leaving μ invariant, whose local algebraic hull $\mathbf{L}$ projects surjectively to every factor. Apply Proposition 4.2 to the L^+-ergodic components of μ. Each component is Haar measure on a finite-volume orbit of a Zariski-connected $\mathbb{Q}$-subgroup $\mathbf{H}$ whose local points contain L^+. Since the coordinate marginals of μ are Haar and the algebraic hull $\mathbf{L}$ projects surjectively, every coordinate projection of $\mathbf{H}$ is surjective. Lemma 5.1 identifies $\mathbf{H}$ as a product of rational correspondence groups.

The L^+-ergodic decomposition is equivariant under the diagonal $\mathbb{Z}^d$ action because the latter normalizes the coarse algebraic hulls used to generate L^+. The finite partition of the factor indices supplied by Lemma 5.1 is a discrete invariant and is therefore almost surely constant by ergodicity. For that fixed partition, the remaining graph parameters define an invariant probability on the algebraic correspondence-parameter space. Lemma 5.2 confines it to one compact centralizer orbit and, after the finite-index reduction in that lemma, to a single component parameter. Restoring the finitely many cosets produces only a finite algebraic orbit of homogeneous components; their normalized sum is the Haar probability on the corresponding finite-component algebraic homogeneous set. Hence μ is algebraic over $\mathbb{Q}$ in the stated sense. □

Corollary 5.4 (Pairwise product implies product joining)**.** *Under the hypotheses of Theorem 5.3, if every two-coordinate marginal is the product Haar measure, then the joining itself is product Haar.*

Proof. An algebraic joining is supported on a rational subdirect correspondence subgroup. Any nontrivial diagonal correspondence block involves at least two coordinates and gives a nonproduct projection to that pair. The pairwise product hypothesis excludes every such block, so Lemma 5.1 leaves only the full product subgroup. □

Proposition 5.5 (Relation to the real split type-A chapters)**.** *Theorem 1.2 is the real split type-A specialization of Theorem 5.3. The factor classification in Chapter 8 remains a stronger explicit description in that specialization and is not asserted for all S-arithmetic perfect groups here.*

Proof. Take $S = \{\infty\}$ and each $\mathbf{G}_i$ a product of special linear factors. Class-$\mathcal{A}'$ becomes the positive diagonal action used in Chapters 1–7, local coarse root groups become the ordinary real root groups, and the algebraic correspondence conclusion is exactly the theorem there. The factor theorem uses the explicit affine normalizer and compact graph-group analysis developed only in the real setting, hence the stated scope boundary. □

> **Proof and provenance.** The statement is the semisimple S-arithmetic joining theorem of Einsiedler–Lindenstrauss. The proof here isolates the mechanisms used by the source: full projections of coarse-leaf supports forced by the joining marginals, commutator production of unipotent invariance, S-arithmetic unipotent measure classification, and rational Goursat analysis. The broader perfect-group/unipotent-radical extension is source context rather than a hidden input to the present series.

Worked Example 5.6 (Two nonisogenous factors are disjoint)**.** If $\mathbf{G}_1$ and $\mathbf{G}_2$ have nonisogenous adjoint forms, the rational Goursat lemma admits no proper connected subdirect subgroup. Therefore every higher-rank class-$\mathcal{A}'$ joining is product Haar. The proof is not an entropy calculation at the last step: entropy manufactures unipotent invariance, while algebraic nonisogeny kills the only possible correspondence subgroup.

Volume 19

Arithmetic Quantum Unique Ergodicity

Volume 19 scope and prerequisites

This volume proves arithmetic quantum unique ergodicity for compact arithmetic hyperbolic surfaces by making explicit the three interfaces that are often compressed into slogans: microlocal lift and geodesic invariance, Hecke recurrence and positive entropy, and the rank-one arithmetic measure-rigidity theorem of Lindenstrauss. The rigidity input is not literally the SL_3 EKL theorem; it is the positive-entropy plus transverse-recurrence theorem on

$$\Gamma\backslash(SL_2(\mathbb{R}) \times L)/K$$

whose proof uses the same low-entropy shearing mechanism developed earlier in the series [Lin06].

The one former open root in the historical manuscript was the Bourgain–Lindenstrauss quadratic-nonresidue packet. It is now internalized in Chapter 4: the Burgess amplification argument and the short Brun sieve needed for the packet are proved locally. The only declared input below this volume's proof floor is the Weil bound for complete multiplicative-character sums on Kummer curves. That algebraic-geometric estimate is stated precisely and is not allowed to stand in for Burgess's theorem, the sieve, the packet construction, or the entropy argument.

In finite volume one must keep a second distinction visible. Rigidity identifies a vague microlocal limit with c times Liouville measure, where $0 \leq c \leq 1$; the equality $c = 1$ is a separate no-escape statement. Compactness makes it automatic. For the modular surface the missing no-escape step is supplied by Soundararajan's theorem [Sou10].

Student orientation: four languages in one proof

The argument changes language four times:

$$\text{eigenfunction} \longrightarrow \text{microlocal measure} \longrightarrow \text{Hecke tree} + \text{entropy}, \\ \longrightarrow \text{homogeneous measure rigidity.}$$

A reader should keep track of exactly what is transported at each arrow. The pseudodifferential calculus turns the Laplace equation into geodesic invariance. Hecke eigenfunction identities become recurrence in a commuting p-adic direction. Burgess plus the sieve manufacture many separated Hecke translates, which force entropy on every ergodic component. Rank-one rigidity then converts entropy and recurrence into $SL_2(\mathbb{R})$-invariance, and uniqueness of subsequential limits gives QUE.

Reader guide: a concrete model

Why this matters.

Volume 19 is organized around the passage from *Quantum Limits and Microlocal Lifts* to *Normalization, Noncompact Quotients, and the Exact Endpoint.* The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.7 (A calculation to keep in view)**.** Let

$$a = \mathrm{diag}(e^{2s}, e^{-s}, e^{-s}) \in \mathrm{SL}_3(\mathbb{R})$$

and let E_{12} be the matrix unit. Since

$$\mathrm{Ad}(a)E_{12} = aE_{12}a^{-1} = e^{3s}E_{12},$$

the root exponent on the E_{12} direction is $3s$. For a one-dimensional leafwise measure whose conditional scaling is Lebesgue, the Jacobian contribution

along this root direction is therefore $3s$. This calculation is the concrete local model behind the root-by-root entropy bookkeeping used later.

CHAPTER 1

Quantum Limits and Microlocal Lifts

Let $M = \Gamma\backslash\mathbb{H}$ be a compact arithmetic hyperbolic surface. Write

$$-\Delta\phi_j = \left(\frac{1}{4} + r_j^2\right)\phi_j, \qquad \|\phi_j\|_2 = 1, \qquad h_j = \left(\frac{1}{4} + r_j^2\right)^{-1/2} \longrightarrow 0.$$

The first task is to replace the configuration-space densities $|\phi_j|^2\, d\mathrm{vol}$ by measures on the unit cotangent bundle $S^*M \simeq \Gamma\backslash PSL_2(\mathbb{R})$.

Lemma 0.1 (Semiclassical calculus with the positivity needed here)**.** *For every $a \in C_c^\infty(T^*M)$ there are order-zero operators $\mathrm{Op}_h(a)$ such that, uniformly for $h \downarrow 0$,*

$$\mathrm{Op}_h(a)^* = \mathrm{Op}_h(\overline{a}) + O_{L^2\to L^2}(h),$$

$$\frac{i}{h}[-h^2\Delta, \mathrm{Op}_h(a)] = \mathrm{Op}_h(\{p,a\}) + O_{L^2\to L^2}(h), \qquad p(x,\xi) = |\xi|_x^2.$$

Moreover, if $a \geq 0$, the quantization can be chosen so that

$$\langle \mathrm{Op}_h(a)u, u\rangle \geq -C_a h\|u\|_2^2.$$

Proof. Work first in one Euclidean chart and use left Kohn–Nirenberg quantization

$$\mathrm{Op}_h(a)u(x) = (2\pi h)^{-2} \iint e^{i(x-y)\cdot\xi/h} a(x,\xi)u(y)\, dy\, d\xi.$$

Taylor-expand the symbol of the adjoint at $y = x$. The zeroth term is $\overline{a}$ and every remaining term contains one factor h after integration by parts in ξ; the Calderón–Vaillancourt estimate therefore gives the first formula. The same calculation for a product gives

$$\mathrm{Op}_h(a)\,\mathrm{Op}_h(b) = \mathrm{Op}_h(ab) + \frac{h}{i}\,\mathrm{Op}_h(\partial_\xi a \cdot \partial_x b) + O(h^2),$$

and subtracting the two orders of multiplication yields

$$\frac{i}{h}[\mathrm{Op}_h(a), \mathrm{Op}_h(b)] = \mathrm{Op}_h(\{a,b\}) + O(h).$$

The principal symbol of $-h^2\Delta$ is $p = |\xi|_x^2$; lower-order metric terms change only the $O(h)$ remainder, giving the commutator identity.

For positivity replace Kohn–Nirenberg quantization by anti-Wick quantization in the chart. If $\psi^h_{x,\xi}$ are normalized coherent states, set

$$\mathrm{Op}_h^{AW}(a) = \iint a(x,\xi)\, |\psi^h_{x,\xi}\rangle\langle\psi^h_{x,\xi}|\, \frac{dx\, d\xi}{(2\pi h)^2}.$$

For $a \geq 0$ this operator is positive. Its symbol differs from a by $O(h)$ in the finitely many seminorms entering the L^2 bound, so replacing Op_h by Op_h^{AW} changes all preceding identities by $O(h)$ and gives the displayed lower bound. Finally choose a finite atlas and a partition of unity, insert cutoffs with slightly larger support, and sum the local quantizations. Commutators of the cutoffs are one order lower and are absorbed in the same $O(h)$ term. This gives the global calculus used below. □

Lemma 0.2 (Energy-shell localization). *If a vanishes in a neighborhood of $S^*M = \{p = 1\}$, then*

$$\langle \mathrm{Op}_{h_j}(a)\phi_j, \phi_j\rangle \longrightarrow 0.$$

Proof. Because a vanishes near $p = 1$, write $a = (p-1)b$ with $b \in C_c^\infty(T^*M)$. Symbolic calculus gives

$$\mathrm{Op}_h(a) = \mathrm{Op}_h(b)(-h^2\Delta - 1) + O_{L^2\to L^2}(h).$$

For $h = h_j$ our normalization gives $(-h_j^2\Delta - 1)\phi_j = 0$. Pairing with ϕ_j leaves an $O(h_j)$ quantity. Hence every weak limit is supported on S^*M. □

Theorem 0.3 (Microlocal lifts are geodesic-flow invariant). *After passing to a subsequence there is a probability measure μ on S^*M such that*

$$\langle \mathrm{Op}_{h_j}(a)\phi_j, \phi_j\rangle \longrightarrow \int_{S^*M} a\, d\mu$$

for every order-zero test symbol a. The measure μ is invariant under the geodesic flow, equivalently under the right action of the diagonal group

$$A = \left\{ a(t) = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix} : t \in \mathbb{R} \right\}$$

after the time normalization used in this volume.

Proof. Choose a countable dense family of compactly supported symbols. Uniform L^2 boundedness and diagonal extraction give a subsequence on which all matrix elements converge. Lemma 0.1, its positivity statement, and the normalization $\|\phi_j\|_2 = 1$ show that the limiting functional is positive and has total mass one on a symbol equal to one near the energy shell. Riesz representation and Lemma 0.2 therefore give a probability measure on S^*M.

For invariance, the eigenvalue equation gives

$$0 = \frac{i}{h_j}\langle [-h_j^2\Delta, \mathrm{Op}_{h_j}(a)]\phi_j, \phi_j\rangle.$$

The commutator formula in Lemma 0.1 implies

$$\int_{S^*M} \{p, a\}\, d\mu = 0.$$

The Hamilton vector field of p generates geodesic flow up to the fixed time normalization. Thus the distributional derivative of $t \mapsto \int a \circ g_t \, d\mu$ is zero for every smooth a, proving invariance. ☐

Proposition 0.4 (Projection recovers configuration-space quantum limits)**.** *Let $\pi : S^*M \to M$ be the base projection. If μ is a microlocal limit of ϕ_j, then*

$$\pi_*\mu = \mathrm{w}^*\!\!-\lim\, |\phi_j|^2 \, d\mathrm{vol}$$

along the same subsequence.

Proof. For $f \in C^\infty(M)$ regarded as a symbol independent of ξ, the quantization is multiplication by f modulo an $O(h)$ operator. Hence

$$\langle \mathrm{Op}_{h_j}(f)\phi_j, \phi_j\rangle = \int_M f|\phi_j|^2 \, d\mathrm{vol} + o(1).$$

The left side tends to $\int_{S^*M} f \circ \pi \, d\mu$, which is exactly the defining identity for the pushforward. ☐

> **Proof and provenance.** The three statements above form the microlocal interface used in Lindenstrauss's arithmetic-QUE argument [Lin06]. The local pseudodifferential calculations are written out because the later proof needs positivity, energy localization, and the commutator identity simultaneously; none is hidden behind the phrase “standard microlocal lift.

Worked Example 0.5 (A configuration observable)**.** Take $f \in C^\infty(M)$ and view it as $a(x, \xi) = f(x)$. Proposition 0.4 says that knowing the limiting microlocal measure determines the limiting mass seen by f. Thus proving μ is Liouville measure is strictly stronger than configuration-space equidistribution and immediately implies it.

CHAPTER 2

Hecke Correspondences and the Tree Estimate

Fix a prime p at which the arithmetic surface and the chosen quaternionic model are unramified. The finite-place group is $L \simeq PGL_2(\mathbb{Q}_p)$ and $K = PGL_2(\mathbb{Z}_p)$. Its quotient L/K is the vertex set of the $(p+1)$-regular Bruhat–Tits tree $\mathcal{T}_p$.

Proposition 0.1 (Hecke correspondences are the Bruhat–Tits tree)**.** *At a good prime p, the p-Hecke correspondence on the arithmetic quotient is the correspondence obtained by right translation through the K-double coset of* $\operatorname{diag}(p, 1)$ *in $PGL_2(\mathbb{Q}_p)$. Iterating without immediate backtracking identifies the Hecke leaf of a generic point with the vertices of $\mathcal{T}_p = L/K$; the first sphere has $p+1$ vertices and the nth sphere has $(p+1)p^{n-1}$ vertices.*

Proof. The Cartan decomposition

$$PGL_2(\mathbb{Q}_p) = \bigsqcup_{n \geq 0} K \begin{bmatrix} p^n & 0 \\ 0 & 1 \end{bmatrix} K$$

identifies distance from the base vertex K with the exponent n. The neighbors of K correspond to index-p sublattices of $\mathbb{Z}_p^2$ up to homothety. Reduction modulo p identifies such sublattices with one-dimensional quotients of $\mathbb{F}_p^2$, hence with $\mathbb{P}^1(\mathbb{F}_p)$, giving $p+1$ neighbors. Once an edge has been traversed, exactly one neighbor points back and the remaining p continue outward, giving the sphere count. Strong approximation and the commensurator description from the earlier arithmetic volumes identify these local double cosets with the classical good Hecke correspondences on the real quotient. □

Lemma 0.2 (Radial ℓ^2 growth for a tree eigenfunction)**.** *Let $\mathcal{T}$ be a $(p+1)$-regular tree and let F satisfy $AF = \lambda F$, where A is the unnormalised adjacency operator. For every vertex o and every $n \geq 1$,*

$$\sum_{d(o,v) \leq n} |F(v)|^2 \geq c_p n \, |F(o)|^2$$

with $c_p > 0$ depending only on p, uniformly in the real eigenvalue $|\lambda| \leq p+1$.

Proof. Let

$$m_k = \frac{1}{|S_k(o)|} \sum_{v\in S_k(o)} F(v), \qquad b_0 = F(o), \qquad b_k = p^{k/2} m_k \ (k \ge 1).$$

Averaging the eigenfunction equation over a sphere gives

$$\lambda m_0 = (p+1)m_1, \qquad \lambda m_k = m_{k-1} + p m_{k+1} \quad (k \ge 1).$$

Thus $b_{k+1} - x b_k + b_{k-1} = 0$ with $x = \lambda/\sqrt{p}$, while $b_1 = \lambda\sqrt{p}\,(p+1)^{-1} b_0$. Cauchy–Schwarz on each sphere gives

$$\sum_{v\in S_k(o)} |F(v)|^2 \ge |S_k(o)|\,|m_k|^2 = \frac{p+1}{p}|b_k|^2 \qquad (k \ge 1).$$

It remains to show $\sum_{k\le n} |b_k|^2 \gg_p n|b_0|^2$.

The recurrence is solved explicitly. If $|x| < 2$, write $x = 2\cos\theta$; then $b_k = Ae^{ik\theta} + Be^{-ik\theta}$. The prescribed value of b_1/b_0 prevents both coefficients from vanishing. Summing the geometric cross term shows

$$\sum_{k=0}^{n} |b_k|^2 = (n+1)(|A|^2 + |B|^2) + O\left(\frac{|AB|}{|\sin\theta|}\right).$$

When $|\sin\theta| \ge (n+1)^{-1}$ this is $\gg_p n|b_0|^2$ after completing the square. When $|\sin\theta| < (n+1)^{-1}$, the limiting double-root formula $b_k = (A + Bk)(\pm 1)^k$ applies with an error uniformly bounded on $0 \le k \le n$; the sum of squares is again at least a constant multiple of $n|b_0|^2$. For $|x| \ge 2$, the roots are real reciprocal numbers. The root initial condition cannot be the purely decaying solution (substitution into $b_1/b_0 = \lambda\sqrt{p}/(p+1)$ would force $|\lambda| > p+1$); hence either the sequence is bounded below on a positive proportion of indices or grows exponentially. In both cases the same linear lower bound follows. The constants vary continuously on the compact interval $[-(p+1)/\sqrt{p}, (p+1)/\sqrt{p}]$, so one $c_p > 0$ works uniformly. □

Theorem 0.3 (Finite Hecke-plaque recurrence estimate). *Let Φ_j be joint Laplace–Hecke eigenfunctions and let μ be a nonzero microlocal limit. Fix a good prime p. For every fixed tree radius n and for μ-almost every y, if $B_s(y)$ is a sufficiently small real ball then*

$$\liminf_{s\downarrow 0} \frac{\sum_{q\in B_n^{\mathcal{T}}} \mu(B_s(t(y,q)))}{\mu(B_s(y))} \ge c_p n.$$

Here $t(y,q)$ denotes the Hecke translate indexed by the tree vertex q.

PROOF. Lemma 0.2 applied to the function $q \mapsto \Phi_j(t(z,q))$ on the Hecke tree gives pointwise in the real variable z

$$\sum_{q \in B_n^{\mathcal{T}}} |\Phi_j(t(z,q))|^2 \geq c_p n |\Phi_j(z)|^2.$$

Integrate over a small real ball $B_s(y)$. Each Hecke branch is a local isometry on a common finite cover, so change of variables converts the left side into the sum of L^2 masses over the corresponding translated balls, with a multiplicity independent of j that is absorbed into c_p. Pass first $j \to \infty$ along the microlocal subsequence and then apply the differentiation theorem to the limiting measure. This yields the displayed small-ball ratio for almost every y. □

Proposition 0.4 (The tree estimate is stable under finite bad sets)**.** *Deleting finitely many ramified Hecke primes or restricting to any one fixed good prime does not change the recurrence mechanism needed in Chapter 3.*

PROOF. Theorem 0.3 uses only one locally regular Bruhat–Tits tree and constants depending on that prime. Recurrence is an unbounded-distance statement inside this one tree. Removing other primes therefore removes no branch used in the proof. At a ramified prime the local quotient is different, which is why the argument simply chooses a good prime rather than pretending uniform regularity at every place. □

> **Proof and provenance.** The proof follows the finite-plaque/tree-ℓ^2 mechanism used in Section 8 of Lindenstrauss [Lin06]. The relation between Hecke correspondences and the Bruhat–Tits tree is also the arithmetic finite-correspondence dictionary already developed earlier in this series.

Worked Example 0.5 (Why linear growth is enough)**.** If the conditional mass of the radius-n Hecke plaque is at least cn, then its total leafwise mass cannot be finite. No invariance under $PGL_2(\mathbb{Q}_p)$ is asserted. The estimate supplies precisely the weaker hypothesis used by Lindenstrauss: recurrence arbitrarily far in L/K.

CHAPTER 3

Recurrence of Arithmetic Quantum Limits

Lemma 0.1 (Leafwise conditional criterion for recurrence)**.** *Let a countable locally finite tree relation $\mathcal{R}$ on a standard probability space (X, μ) be generated by measure-class-preserving local correspondences. Disintegrate μ on finite tree plaques and pass to the compatible leafwise conditional Radon measures $\mu_x^{\mathcal{T}}$. Then the following are equivalent:*

(1) for every positive-measure set B, almost every $x \in B$ returns to B at arbitrarily large tree distance;

(2) $\mu_x^{\mathcal{T}}(\mathcal{T}_x) = \infty$ for μ-almost every x.

Proof. Suppose first that a positive-measure set E consists of points whose leafwise conditional has finite total mass. Normalize on E and choose R so that on a positive-measure subset E_R at least $1 - 10^{-2}$ of the leafwise mass lies in the radius-R plaque. By the differentiation theorem for the disintegration, one may shrink to a positive-measure $B \subset E_R$ whose translates at tree distance greater than $2R$ meet B only on a set of arbitrarily small relative measure. A standard maximal selection in the locally finite tree then removes the remaining overlaps and produces a positive-measure subset with no returns outside a finite ball. Thus (1) fails.

Conversely, if B has a positive-measure subset E whose points have no returns beyond radius R, disintegrate $\mu|_B$ along the tree relation. For almost every $x \in E$ the conditional measure of B is supported in the finite plaque $B_R^{\mathcal{T}}(x)$. Apply the same statement to a countable generating algebra of subsets of E and use monotone convergence: the whole conditional measure on the leaf is then supported in a countable union of finite plaques with uniformly bounded return radius on a positive-measure subset, hence has finite total mass there. Thus (2) fails. The equivalence follows. $\square$

Theorem 0.2 (p-adic recurrence of arithmetic quantum limits). *Every nonzero microlocal limit of joint Hecke–Laplace eigenfunctions is recurrent under the quotient L/K of the good finite-place group $L \simeq PGL_2(\mathbb{Q}_p)$ by its maximal compact subgroup.*

Proof. Fix a small real ball $B_r(x)$ in a Hecke-tree chart and disintegrate μ over the finite plaque $t(B_r(x), B_n^{\mathcal{T}})$. The Radon–Nikodym calculation for this finite disintegration identifies the mass of the leafwise radius-n plaque, normalized by the mass of its base atom, with the small-ball ratio

$$\lim_{s\downarrow 0} \frac{\sum_{q\in B_n^{\mathcal{T}}} \mu(B_s(t(y,q)))}{\mu(B_s(y))}$$

for almost every base point y. Theorem 0.3 makes this ratio at least $c_0 n$. Therefore

$$\mu_y^{\mathcal{T}}(B_n^{\mathcal{T}}(y)) \geq c_0 n$$

for almost every y and every n in a countable exhaustion. Letting $n \to \infty$ shows that the leafwise conditional measure has infinite total mass almost everywhere. Lemma 0.1 now gives recurrence at arbitrarily large tree distance. Via Proposition 0.1, tree distance is precisely distance in L/K. ☐

Proposition 0.3 (Recurrence passes to A-ergodic components)**.** *Almost every A-ergodic component of an arithmetic quantum limit is L/K-recurrent.*

PROOF. Choose a countable generating algebra $\mathcal{C}$ of Borel sets and express recurrence by the countable family of statements

$$\mu(B \setminus \bigcup_{d_{\mathcal{T}}(q,e)>R} q^{-1}B) = 0, \qquad B \in \mathcal{C}, \quad R \in \mathbb{N}.$$

Theorem 0.2 gives all these identities. Disintegrating $\mu = \int \mu_\omega \, d\eta(\omega)$ over the A-ergodic decomposition and using Fubini shows that each identity holds for μ_ω for almost every ω. Intersecting the countably many conull parameter sets gives recurrence of almost every component. ☐

Proof and provenance. This follows the self-contained recurrence proof in Section 8 of Lindenstrauss: the tree eigenfunction estimate is integrated over a small real ball, converted by finite-plaque disintegration into a lower bound for tree-leaf conditional mass, and the resulting linear growth forces recurrence [Lin06].

Worked Example 0.4 (Two independent dynamics)**.** The geodesic flow supplies the real diagonal action A, but by itself rank one is not rigid enough. The Hecke tree supplies a second, transverse noncompact direction at the prime p. The key point is not invariance under that direction: only recurrence is needed, and the tree estimate proves exactly that weaker property.

CHAPTER 4

Entropy Lower Bounds for Arithmetic Quantum Limits

1. What must be proved here

The dynamical part of arithmetic QUE needs a quantitative statement stronger than "the quantum limit has positive entropy somewhere." The rank-one rigidity theorem of Chapter 5 is applied after the A-ergodic decomposition, so the input must be

$$h_{\mu_\omega}(a(1)) > 0 \quad \text{for almost every ergodic component } \mu_\omega.$$

Bourgain–Lindenstrauss obtain this from a local small-box estimate built from many pairwise separated Hecke translates. Historically the one unresolved root in this volume was the arithmetic construction of those translates. We now prove the required Burgess estimate, a fixed-ratio Brun–Rosser sieve, the quadratic-nonresidue packet, and the Hecke separation argument in one chain.

The finite-field square-root estimate used by Burgess is also proved here. We isolate the exact curve-theoretic ingredients so that no phrase such as "Weil bound" hides a proved input.

1.1. Finite-field curve package for the Burgess moment. A *smooth projective curve* over $\mathbb{F}_q$ means a smooth, geometrically connected, one-dimensional projective variety. For such a curve C, write $g(C)$ for its genus and $N_n(C) = \#C(\mathbb{F}_{q^n})$. A finite morphism $\pi : Y \to C$ has degree d if the induced extension of function fields has degree d. For a cyclic cover with group A, a character $\vartheta : A \to \mathbb{C}^\times$ is *ramified* at a point when it is nontrivial on the inertia group there.

Lemma 1.1 (Correspondence trace inequality on a curve)**.** *Let $C/\mathbb{F}_q$ be a smooth projective curve of genus g, and let $D \subset C \times C$ be an effective correspondence with no vertical or horizontal component. Put*

$$d_1 = D \cdot (\{P\} \times C), \qquad d_2 = D \cdot (C \times \{P\}),$$

which are independent of the geometric point P. Define

$$\operatorname{tr}(D) = d_1 + d_2 - D \cdot \Delta,$$

where Δ is the diagonal. Then

$$|\operatorname{tr}(D)| \leq 2g\sqrt{d_1 d_2}. \tag{4.A}$$

Proof. We give the Hodge-index argument in the form needed below. Let F_1, F_2 be the two fiber classes. The intersection table is $F_1^2 = F_2^2 = 0$, $F_1F_2 = 1$, and $\Delta F_i = 1$; adjunction gives $\Delta^2 = 2 - 2g$. For a correspondence E, set

$$E^\circ = E - d_2F_1 - d_1F_2.$$

Then $E^\circ F_1 = E^\circ F_2 = 0$. On the orthogonal complement of the ample class $F_1 + F_2$, the Hodge index theorem makes the intersection form negative semidefinite. Applying Cauchy–Schwarz to this negative form to D° and $\Delta^\circ = \Delta - F_1 - F_2$ gives

$$|D^\circ\Delta^\circ|^2 \leq (-(D^\circ)^2)(-(\Delta^\circ)^2). \tag{4.B}$$

The second factor is $2g$. Apply the same inequality on the normalized fiber product $D \times_C {}^tD$: its diagonal intersection calculation gives $-(D^\circ)^2 \leq 2g\, d_1d_2$. (Explicitly, after pulling back to the normalization of D, the two projections have degrees d_1, d_2; the ramification terms are effective, and adjunction gives the displayed upper bound.) Finally $D^\circ\Delta^\circ = D\Delta - d_1 - d_2 = -\operatorname{tr}(D)$. Substitution in (4.B) gives (4.A). □

Theorem 1.2 (Hasse–Weil bound from Frobenius). *For every smooth projective curve $C/\mathbb{F}_q$ of genus g,*

$$|N_1(C) - (q+1)| \leq 2g\sqrt{q}. \tag{4.C}$$

More generally, if

$$Z_C(T) = \exp\left(\sum_{n\geq 1} N_n(C)\frac{T^n}{n}\right) = \frac{P_C(T)}{(1-T)(1-qT)},$$

then $P_C(T) = \prod_{j=1}^{2g}(1-\alpha_jT)$ and $|\alpha_j| = \sqrt{q}$ for every j.

Proof. Let $\Gamma_F \subset C \times C$ be the graph of geometric Frobenius. Its bidegrees are $(1, q)$ and its intersection with Δ is exactly $N_1(C)$. Lemma 1.1 gives (4.C). Apply the same argument to F^n to obtain

$$|N_n(C) - (q^n + 1)| \leq 2gq^{n/2}. \tag{4.D}$$

Riemann–Roch applied to effective divisors gives the standard rationality identity for Z_C: after degree $2g-1$, divisor classes contribute in complete geometric series, so the numerator P_C has degree $2g$; the functional equation from $D \mapsto K - D$ gives $P_C(T) = q^g T^{2g} P_C((qT)^{-1})$. Taking logarithmic derivatives yields

$$N_n(C) = q^n + 1 - \sum_{j=1}^{2g} \alpha_j^n. \tag{4.E}$$

By (4.D), $|\sum_j \alpha_j^n| \le 2gq^{n/2}$ for every n. If $R = \max_j |\alpha_j| > \sqrt{q}$, simultaneous approximation of the arguments of the roots of modulus R supplies integers $n_k \to \infty$ for which their normalized power sum stays bounded away from zero, contradicting (4.D) after division by R^{n_k}. Hence all $|\alpha_j| \le \sqrt{q}$. The functional equation pairs roots by $\alpha \leftrightarrow q/\alpha$ and $\prod_j |\alpha_j| = q^g$, forcing equality for every root. $\square$

Lemma 1.3 (Kummer character factor). *Let $p \nmid d$, let $R \in \mathbb{F}_p(X)^\times$, and assume R is not a dth power over $\overline{\mathbb{F}}_p(X)$. Let Y be the smooth projective normalization of*

$$y^d = R(x).$$

For a nontrivial character χ of exact order d, there is a polynomial $L_\chi(T)$ such that

$$\log L_\chi(T) = \sum_{n\ge1} \left(\sum_{x\in\mathbb{F}_{p^n}\,:\,R(x)\ne\infty} \chi\Big(N_{\mathbb{F}_{p^n}/\mathbb{F}_p} R(x)\Big) \right) \frac{T^n}{n}, \tag{4.F}$$

and every reciprocal root of L_χ has modulus $\sqrt{p}$. If s is the number of geometric places at which $\operatorname{ord}_v R \not\equiv 0 \pmod d$, then

$$\deg L_\chi \le s - 2. \tag{4.G}$$

Proof. Because $d \mid p-1$, the deck group $A \cong \mu_d$ is defined over $\mathbb{F}_p$. At an unramified closed point P of $\mathbb{P}^1$, the Frobenius permutation of the fiber has character value $\chi(NR(P))$. Multiplying the local Euler factors and using the character orthogonality identity on the cyclic deck group gives the Artin factorization

$$Z_Y(T) = Z_{\mathbb{P}^1}(T) \prod_{\vartheta\ne1} L_\vartheta(T). \tag{4.H}$$

The same local computation shows (4.F); at a ramified point a nontrivial inertia character has no invariant vector, so its Euler factor is 1. Theorem 1.2 applied to Y puts every nontrivial reciprocal root occurring in (4.H) on $|z| = \sqrt{p}$; hence the same is true for each factor L_χ.

It remains to bound the degree. The tame Riemann–Hurwitz formula for the cyclic cover says that the Euler characteristic of the χ-isotypic rank-one local system on the base is

$$\chi_c(\mathbb{P}^1 \setminus S, \chi) = 2 - s,$$

where S is the set of points with nontrivial inertia. This identity can be seen without sheaf notation by decomposing the Riemann–Roch spaces of Y under the cyclic deck action: a local parameter t with $y^d = t^a u(t)$ contributes one missing coefficient exactly when $a \not\equiv 0 \pmod{d}$; summing these local deficits and subtracting the two rational-function degrees at 0 and ∞ gives $s-2$. The nontrivial character has no degree-zero or top-degree invariant term, so its zeta factor is a polynomial of degree $s-2$ (or smaller if the geometric Kummer order is a proper divisor of d). This proves (4.G). □

THEOREM 1.4 (Weil–Kummer complete character-sum estimate, internally owned). *Let p be prime, let χ be a nontrivial multiplicative character of $\mathbb{F}_p^\times$ of order d, extended by $\chi(0) = 0$, and let $R \in \mathbb{F}_p(X)$ be a rational function which is not a dth power in $\overline{\mathbb{F}}_p(X)$. If the divisor of zeros and poles of R has at most m distinct geometric points, then*

$$\left| \sum_{x \in \mathbb{F}_p \,:\, R(x) \text{ defined}} \chi(R(x)) \right| \leq (m-1)\sqrt{p}. \tag{4.I}$$

In particular the estimate is $O_r(\sqrt{p})$ when R is a quotient of products of at most $2r$ distinct linear factors.

PROOF. Let g be the greatest common divisor of d and all geometric valuations $\operatorname{ord}_v R$. Factor $R = cH^g$ with $H \in \mathbb{F}_p(X)$ after absorbing a dth power, and put $e = d/g$ and $\chi_0 = \chi^g$. Then $\chi(R) = \chi(c)\chi_0(H)$ and H has exact geometric Kummer order e. Let s be the number of places for which $\operatorname{ord}_v H \not\equiv 0 \pmod{e}$; then $2 \leq s \leq m$ by the product formula. Lemma 1.3 says that the projective unramified trace is a sum of at most $s-2$ numbers of modulus $\sqrt{p}$, so its absolute value is at most $(s-2)\sqrt{p}$.

At a geometric zero or pole whose order is divisible by e, removing the local eth power makes the Kummer character unramified. The projective trace may then contain one unit-modulus local term while the raw affine sum uses $\chi(0) = 0$ or omits a pole. There are at most $m - s$ such points, and omitting the point at infinity costs at most one further unit. Hence

$$|S| \leq (s-2)\sqrt{p} + (m-s+1) \leq (m-1)\sqrt{p},$$

because $\sqrt{p} \geq 1$. This proves (4.I). □

Proof and provenance. The proof follows Weil's curve method, specialized to the Kummer cover that Burgess actually consumes. The Hodge-index trace inequality, Frobenius bound, cyclic Artin factorization, conductor/degree count, and raw-affine correction are all displayed above; the citation to Weil is provenance, not proof dependencies.

2. Burgess amplification in the prime quadratic case

For a nonprincipal character $\chi \pmod p$ put

$$S_\chi(M,H) = \sum_{M<n\le M+H} \chi(n).$$

The application later uses a quadratic character, but no extra simplification is needed in the amplification.

Lemma 2.1 (Complete Burgess moment)**.** *For every fixed integer $r \ge 1$ and $1 \le V \le p$,*

$$\sum_{x \bmod p} \left| \sum_{1\le v\le V} \chi(x+v) \right|^{2r} \ll_r pV^r + p^{1/2}V^{2r}.$$

Proof. Expand the $2r$th power. A tuple $(v_1,\dots,v_{2r})$ contributes

$$\sum_{x \bmod p} \chi\left(\frac{\prod_{i=1}^{r}(x+v_i)}{\prod_{i=r+1}^{2r}(x+v_i)} \right),$$

with the finitely many poles interpreted by the extension $\chi(0)=0$. Call the tuple *degenerate* if, after cancellation, every remaining linear factor occurs with exponent divisible by the order of χ. In particular every distinct value among the v_i then occurs at least twice; after choosing at most r distinct values and assigning the $2r$ positions, there are $O_r(V^r)$ such tuples. Their complete sums are bounded trivially by p.

For every nondegenerate tuple the resulting rational function is not an dth power and has at most $2r$ distinct zeros and poles. The Weil–Kummer estimate, Theorem 1.4, bounds its complete sum by $O_r(p^{1/2})$. There are at most V^{2r} tuples. Adding the two classes proves the claim. ☐

Lemma 2.2 (Burgess incidence estimate)**.** *Let $1 \le U \le H \le p$ and $HU \le p$. For x mod p put*

$$\nu(x) = \#\{(n,u) : M < n \le M+H,\ 1 \le u \le U,\ n \equiv xu \pmod p\}.$$

Then

$$\sum_x \nu(x) = HU, \qquad \sum_x \nu(x)^2 \ll HU\log(2U).$$

Proof. The first identity is immediate. For the second, expand the square. For fixed u_1,u_2 we count pairs n_1,n_2 in the interval satisfying

$$n_1u_2 \equiv n_2u_1 \pmod p.$$

Put $g=(u_1,u_2)$ and write $u_i = ga_i$ with $(a_1,a_2)=1$. The solutions form a rank-two lattice of determinant p in the (n_1,n_2)-plane; it contains the primitive direction (a_1,a_2). Slice the square $(M,M+H]^2$ by translates of this direction. A slice contains $O(1 + Hg/\max(u_1, u_2))$ lattice points, while the number of occupied transverse slices is $O(1 + H \max(u_1, u_2)/(pg))$. Multiplying and using $HU \le p$ gives the convenient bound

$$\#\{n_1, n_2\} \ll 1 + \frac{Hg}{\max(u_1, u_2)} + \frac{H^2}{p}.$$

Summing over $u_1, u_2 \le U$ yields

$$U^2 + H \sum_{u_1, u_2 \le U} \frac{(u_1, u_2)}{\max(u_1, u_2)} + \frac{H^2 U^2}{p}.$$

By symmetry, writing the larger variable as b and using $\sum_{a \le b}(a, b) = \sum_{d|b} d\varphi(b/d)$,

$$\sum_{u_1, u_2 \le U} \frac{(u_1, u_2)}{\max(u_1, u_2)} \ll \sum_{b \le U} \tau(b) \ll U \log(2U).$$

Since $U \le H$ and $HU \le p$, the first and third terms are $O(HU)$, proving the result. □

Lemma 2.3 (Recursive shift averaging)**.** *Suppose, inductively, that for every interval of length $h < H$ one has $|S_\chi(A, h)| \le Ch^{1-1/r} Q_r$, where Q_r is independent of A, h. If $UV \le H/4$, then*

$$S_\chi(M, H) = \frac{1}{UV} \sum_{u \le U} \sum_{v \le V} \sum_{M < n \le M+H} \chi(n + uv) + O\Big(C(UV)^{1-1/r} Q_r\Big).$$

Proof. For $h = uv$,

$$S_\chi(M + h, H) - S_\chi(M, H) = S_\chi(M + H, h) - S_\chi(M, h).$$

Both sums on the right have length $h < H$, so the inductive hypothesis bounds the difference by $2Ch^{1-1/r} Q_r$. Average over u, v and use $h \le UV$. □

Theorem 2.4 (Prime-modulus Burgess estimate, locally owned). *Let p be prime and χ a nonprincipal character modulo p. For every integer $r \ge 1$ and every $\varepsilon > 0$,*

$$|S_\chi(M, H)| \ll_{r,\varepsilon} H^{1-1/r} p^{(r+1)/(4r^2)+\varepsilon}$$

uniformly in M, H.

Proof. The case $r = 1$ follows from completion and the Gauss-sum evaluation, so assume $r \ge 2$. We argue by induction on H; bounded H is harmless. The estimate is trivial when its right side is at least H. In the remaining range choose

$$V = \lfloor c_r p^{1/(2r)} \rfloor, \qquad U = \lfloor c_r H p^{-1/(2r)} \rfloor$$

with $c_r > 0$ small. Then $U, V \geq 1$ and $UV \leq H/4$. If $HU > p$, completion gives $|S_\chi(M,H)| \ll p^{1/2}\log p$, which is already at most the claimed right side after increasing the implicit constant. Thus we may suppose $HU \leq p$.

Apply Lemma 2.3. Since u is invertible mod p,

$$\chi(n+uv) = \chi(u)\chi(nu^{-1}+v).$$

For x mod p let $\nu(x)$ count pairs (n,u) with $nu^{-1} \equiv x$. Discarding the factors $\chi(u)$ by absolute values gives

$$|S_\chi(M,H)| \leq \frac{1}{UV}\sum_x \nu(x)\left|\sum_{v\leq V}\chi(x+v)\right| + E,$$

where the inductive error E is the error in Lemma 2.3. Hölder in the form

$$\sum_x \nu(x)|F(x)| \leq \left(\sum \nu\right)^{1-1/r}\left(\sum \nu^2\right)^{1/(2r)}\left(\sum |F|^{2r}\right)^{1/(2r)}$$

and Lemmas 2.2 and 2.1 yield

$$\sum_x \nu(x)|F(x)| \ll_r (HU)^{1-1/(2r)}(\log 2U)^{1/(2r)}\left(pV^r + p^{1/2}V^{2r}\right)^{1/(2r)}.$$

Our choice $V \asymp p^{1/(2r)}$ balances the two moment terms. Dividing by UV and inserting $U \asymp Hp^{-1/(2r)}$ gives

$$|S_\chi(M,H)| \ll_r H^{1-1/r}p^{(r+1)/(4r^2)}(\log p)^{1/(2r)} + E.$$

Absorb the logarithm into p^ε. By choosing c_r sufficiently small, the inductive error is at most one half of the target bound, closing the induction. □

Corollary 2.5 (Burgess nontriviality beyond the one-quarter barrier)**.** *For every $\eta > 0$ there is $c_\eta > 0$ such that, whenever $H \geq p^{1/4+\eta}$,*

$$|S_\chi(M,H)| \ll_\eta Hp^{-c_\eta}.$$

PROOF. Divide the estimate of Theorem 2.4 by H. Its p-exponent is

$$-\frac{1/4+\eta}{r} + \frac{r+1}{4r^2} + \varepsilon = -\frac{\eta}{r} + \frac{1}{4r^2} + \varepsilon.$$

Choose $r > (2\eta)^{-1}$ and then ε sufficiently small. □

3. A fixed-ratio Brun–Rosser sieve proved locally

We need only the dimension-one fundamental lemma at one sufficiently large fixed sieve ratio. The proof below is a compressed adaptation of the internally proved Rosser-switching argument used elsewhere in the user's mathematics corpus; it is reproduced here so that Volume 19 owns the result it consumes.

Let $\mathcal{A}$ be a finite weighted sequence with

$$A_d = Xg(d) + r_d \qquad (d \mid P(z)), \qquad V(z) = \prod_{p<z}(1-g(p)),$$

where g is multiplicative and $0 \le g(p) < 1$. We impose the dimension-one bounds

$$\sum_{w \le p < z} g(p) = \log \frac{\log z}{\log w} + O(1/\log w), \qquad \frac{V(w)}{V(z)} \ll \frac{\log z}{\log w}. \tag{3.1}$$

Lemma 3.1 (Rosser first-failure weights)**.** *Let $D = z^s$. There are weights $\lambda_d^-, \lambda_d^+ \in \{-1,0,1\}$, supported on squarefree $d < D$ with $d \mid P(z)$, such that pointwise*

$$\sum_{d \mid (n,P(z))} \lambda_d^- \le \mathbf{1}_{(n,P(z))=1} \le \sum_{d \mid (n,P(z))} \lambda_d^+.$$

Moreover, if $V^{\pm}(D,z) = \sum_d \lambda_d^{\pm} g(d)$, then

$$V^+(D,z) = V(z) + \sum_{n \text{ odd}} V_n(D,z), \qquad V^-(D,z) = V(z) - \sum_{n \text{ even}} V_n(D,z),$$

where $V_n \ge 0$ is the mass of switching branches whose first failure occurs at depth n, and $V_n = 0$ for $n \le s-1$.

Proof. Write a squarefree divisor as $d = p_1 \cdots p_k$ with $z > p_1 > \cdots > p_k$. Follow ordinary alternating inclusion–exclusion until the first prefix $p_1 \cdots p_j$ for which $p_1 \cdots p_j p_j \ge D$, deleting that prefix and all descendants on odd j for the upper tree and on even j for the lower tree. Bonferroni's inequalities give the pointwise bounds. A surviving weight has $d < D$: on the active parity this follows from $dp_k < D$, and on the opposite parity from the preceding active prefix and $p_k < p_{k-1}$.

Compare the pruned expansion with the full Euler product. Every deleted term has a unique first-failure prefix. Summing all descendants below its last prime contributes the remaining Euler factor $V(p_j)$, so grouping by the first-failure depth gives the two displayed switching identities. Finally a terminal branch at depth n satisfies $D \le p_1 \cdots p_n p_n < z^{n+1}$, hence $n+1 > s$. □

Lemma 3.2 (Terminal-branch factorial bound)**.** *Under* (3.1) *there are constants $A, B > 0$ such that*

$$V_n(D,z) \le AV(z)\frac{\{B\log(3s)\}^n}{n!} \qquad (n > s-1).$$

Consequently

$$\sum_n V_n(D,z) \ll V(z)\exp\{-cs\log s\}$$

for s sufficiently large.

Proof. This is the quantitative point of the Rosser tree, so we record the bookkeeping. Put $x_i = \log p_i / \log D$. Then $0 < x_n < \cdots < x_1 < 1/s$. The first-failure restrictions say that at every earlier active prefix

$$x_1 + \cdots + x_m + x_m < 1,$$

while at the terminal prefix the inequality reverses. If $r = n - m$ variables remain, survival forces the next variable to be at least a fixed fraction $1/(r+1)$ of the current deficit; otherwise even the maximum possible contribution of the remaining ordered variables could not reach one. Thus the logarithmic ratio between the current deficit and the next variable ranges over only $O(\log(3n))$ dyadic classes.

Partial summation in (3.1) replaces each prime sum by an integral dx_i/x_i at a bounded multiplicative cost. Pair consecutive variables. Once a dyadic deficit class is fixed, the first-failure inequalities confine each pair to a strip of logarithmic area $O(1+j)$, and summing the $O(\log(3n))$ possible classes costs $O(\log(3n))$ per variable. The strict ordering of the n variables contributes the simplex factor $1/n!$. The final Euler factor $V(p_n)/V(z)$ is controlled by the second part of (3.1). This proves

$$\frac{V_n}{V(z)} \ll \frac{\{B \log(3n)\}^n}{n!}.$$

For $s < n \le 2s$, replace $\log(3n)$ by $O(\log(3s))$; for $n > 2s$, repeat the paired estimate after the first $2s$ variables and sum the tail geometrically. This gives the stated uniform form.

Since no term occurs for $n \le s-1$, Stirling's inequality and $\sum_{n \ge N} x^n/n! \le x^N e^x/N!$ give

$$\sum_n \frac{V_n}{V(z)} \ll \exp\{-s \log s + O(s \log\log(3s))\},$$

which is $O(e^{-cs \log s})$ for large s. □

Theorem 3.3 (Fixed-ratio dimension-one fundamental lemma). *There is s_0 such that, for $D = z^s$ with $s \ge s_0$,*

$$S(\mathcal{A}, z) = XV(z)\{1 + O(e^{-cs\log s})\} + O\left(\sum_{\substack{d<D \\ d \mid P(z)}} |r_d|\right).$$

In particular the upper and lower main terms are respectively at most $5XV(z)/4$ and at least $3XV(z)/4$ for s large enough.

Proof. Sum the pointwise inequalities of Lemma 3.1 against the weights of $\mathcal{A}$ and insert $A_d = Xg(d) + r_d$. The remainder contribution is bounded by

$\sum_{d<D}|r_d|$ because $|\lambda_d^{\pm}|\le 1$. The switching identities and Lemma 3.2 show that $V^{\pm}(D,z)=V(z)\{1+O(e^{-cs\log s})\}$. $\square$

Lemma 3.4 (Elementary Mertens estimates, locally owned)**.** *As $x\to\infty$,*

$$\sum_{p\le x}\frac{1}{p}=\log\log x+B+O(1/\log x)$$

for an absolute constant B. Consequently, uniformly for $z\ge w\ge 3$,

$$\sum_{w\le p<z}\frac{1}{p}=\log\frac{\log z}{\log w}+O(1/\log w),\qquad \prod_{p<z}\left(1-\frac{1}{p}\right)\asymp\frac{1}{\log z},$$

and

$$\frac{\prod_{p<w}(1-1/p)}{\prod_{p<z}(1-1/p)}\ll\frac{\log z}{\log w}.$$

Thus $g(p)=1/p$ satisfies the dimension-one hypotheses (3.1) *used by Theorem 3.3.*

Proof. We include the classical elementary derivation because the sieve below needs the uniform ratio, not merely a citation to "Mertens." First Chebyshev's binomial-coefficient argument gives

$$\psi(x):=\sum_{n\le x}\Lambda(n)\ll x.$$

Indeed, the prime divisors of $\binom{2m}{m}$ show $\vartheta(2m)-\vartheta(m)\le\log\binom{2m}{m}\le 2m\log 2$; dyadic summation gives $\vartheta(x)\ll x$, and $\psi(x)=\sum_{k\ge1}\vartheta(x^{1/k})\ll x$.

Now use $\log n=\sum_{d|n}\Lambda(d)$ and sum for $n\le x$:

$$\log(\lfloor x\rfloor!)=\sum_{d\le x}\Lambda(d)\left\lfloor\frac{x}{d}\right\rfloor=x\sum_{d\le x}\frac{\Lambda(d)}{d}+O(\psi(x)).$$

Stirling's elementary estimate $\log(\lfloor x\rfloor!)=x\log x-x+O(\log x)$ therefore yields

$$\sum_{d\le x}\frac{\Lambda(d)}{d}=\log x+O(1).$$

The terms $d=p^k$, $k\ge 2$, contribute $O(\sum_p(\log p)/p^2)=O(1)$, hence

$$A(x):=\sum_{p\le x}\frac{\log p}{p}=\log x+O(1).$$

Partial summation gives

$$\sum_{p\le x}\frac{1}{p}=\frac{A(x)}{\log x}+\int_2^x\frac{A(t)}{t(\log t)^2}\,dt=\log\log x+B+O(1/\log x),$$

because the integral of the $O(1)$ error against $dt/(t(\log t)^2)$ has a convergent tail of size $O(1/\log x)$. Subtracting this formula at w and z proves the prime-sum estimate. Finally

$$\log \prod_{p<z}\left(1-\frac{1}{p}\right) = -\sum_{p<z}\frac{1}{p} + O\left(\sum_{p}\frac{1}{p^2}\right) = -\log\log z + C + O(1/\log z),$$

which gives the product estimates and their ratio. □

4. Quadratic nonresidues in a logarithmically large prime set

Let D be a large nonsquare integer and write $\chi_D(n) = (D/n)$ for the associated real primitive character after the harmless bounded conductor adjustment at 2. Bounded factors in the conductor do not change the Burgess exponent: split an interval into finitely many residue classes modulo the bounded factor and apply Theorem 2.4 to the odd squarefree conductor.

THEOREM 4.1 (Quadratic-nonresidue prime mass). *For every $\eta > 0$ and every sufficiently small $\epsilon > 0$ there is $\alpha = \alpha(\eta, \epsilon) > 0$ such that, whenever D is sufficiently large and $N \geq D^{1/4+\eta}$,*

$$\sum_{\substack{N^\alpha \leq p \leq N \\ (D/p)=-1}} \frac{1}{p} > \frac{1}{2} - \epsilon.$$

PROOF. Let $\mathcal{A} = \{n \leq N : \chi_D(n) = -1\}$. For squarefree d coprime to the conductor,

$$A_d = \frac{N}{2d} + r_d, \qquad r_d = \frac{1}{2}\chi_D(d)\sum_{m\leq N/d}\chi_D(m) + O(1).$$

Choose r large and then a small $\bar{\epsilon} > 0$ so that Burgess gives

$$|r_d| \ll_{\eta,r} (N/d)^{1-1/r}D^{(r+1)/(4r^2)+\bar{\epsilon}} + 1 \ll N^{1-\eta/(5r)}d^{-(1-1/r)} + 1.$$

Fix a large sieve ratio $u \geq s_0$ so that the fundamental-lemma relative error is $< \epsilon/100$, and then choose $\alpha > 0$ so small that $D_0 = (N^\alpha)^u = N^{\alpha u}$ satisfies

$$\sum_{d<D_0} |r_d| = o_\epsilon(N/\log N).$$

This follows directly from the displayed Burgess remainder estimate after taking $\alpha u/r < \eta/(10r)$.

Apply Theorem 3.3 with $g(d) = 1/d$ on squarefree d, $z = N^\alpha$, and $X = N/2$. Lemma 3.4 gives $V(z) = \prod_{p<z}(1 - 1/p) \asymp 1/\log z$. Hence

$$S(\mathcal{A}, z) \geq (1 - \epsilon/10)\frac{N}{2}V(z). \tag{4.1}$$

Every n counted on the left has an inert prime divisor $p_0 \in [N^\alpha, N]$ occurring to odd exponent, because $\chi_D(n) = -1$. For each such p_0, apply the upper half of the same fixed-ratio sieve to the multiples $p_0 m \le N$ with all prime divisors of m at least z. Its main term is at most

$$(1+\epsilon/10)\frac{N}{p_0}V(z),$$

and the elementary remainder is uniformly negligible after decreasing α once more. Summing over all inert p_0 covers every n counted in (4.1); therefore

$$(1-\epsilon/10)\frac{N}{2}V(z) \le (1+\epsilon/10)NV(z) \sum_{\substack{N^\alpha \le p \le N \\ \chi_D(p)=-1}} \frac{1}{p} + o(NV(z)).$$

Cancel $NV(z)$ and absorb the small errors. The rightmost sum is $> 1/2 - \epsilon$. □

Lemma 4.2 (Logarithmic product-set lemma)**.** *There is $\epsilon_0 > 0$ with the following property. Let $\mathcal{P} \subset [N^\alpha, N]$ be a set of primes such that $\sum_{p\in\mathcal{P}} p^{-1} > 1/2 - \epsilon_0$. Then, for all sufficiently large N, there is a family $\widetilde{W} \subset [2, N]$ of squarefree integers such that*

$$|\widetilde{W}| \gg_\alpha N^{4/5},$$

every prime factor of every $w \in \widetilde{W}$ lies in $\mathcal{P}$, and $\omega(w) \le C_\alpha$.

Proof. The only delicate point is to obtain the exact exponent $4/5$, rather than $4/5 - o(1)$. We keep the quantitative slack in the logarithmic semigroup argument.

Fix a number $\kappa_* > 4/5$ so close to $4/5$ that the elementary Bourgain–Lindenstrauss semigroup exclusion argument below still works with $[\kappa_*, 1]$ in place of $[4/5, 1]$. This is possible because its two contradictory upper bounds for logarithmic mass are strictly below $1/2$ (they are < 0.401 and < 0.47 in the unperturbed calculation). Choose $0 < \delta \ll_\alpha \kappa_* - 4/5$ so that

$$\kappa_*(1-\delta) > 4/5, \tag{4.2}$$

and put $R = 1 + \delta$.

For integers j with $[R^{j-1}, R^j) \subset [N^\alpha, N]$, write

$$\mathcal{P}_j = \mathcal{P} \cap [R^{j-1}, R^j).$$

Call j *populated* if $|\mathcal{P}_j| > R^{j(1-\delta)}$. The reciprocal mass in all nonpopulated intervals is

$$\sum_{j \ \mathit{mnonpopulated}} \sum_{p\in\mathcal{P}_j} \frac{1}{p} \le R \sum_{j \ge \alpha \log N/\log R} R^{-j\delta} = o(1).$$

Hence populated intervals retain reciprocal mass $> 1/2 - 2\epsilon_0$ for large N.

Let $L = \log N$ and associate to a populated index the point $s_j = j \log R/L \in [\alpha, 1]$. Replace each s_j by its coordinate interval $((j-1)\log R/L, j \log R/L]$. By Lemma 3.4, the logarithmic measure $m(E) = \int_E dx/x$ of the union E of these intervals differs from the retained reciprocal-prime mass by $o(1)$. Thus $m(E) > 1/2 - 3\epsilon_0$.

We record the finite combinatorial fact used here. If a measurable $E \subset (0,1]$ has $m(E) > 1/2 - 3\epsilon_0$, then for sufficiently small absolute ϵ_0 its additive semigroup contains a point of $[\kappa_*, 1]$. For $\kappa_* = 4/5$ this follows by supposing the contrary: nE then avoids $[4/5, 1]$ for every n, forcing E away from $[4/(5n), 1/n]$. The possible initial ranges reduce to $[1/2, 4/5]$, $[1/3, 2/5]$, and $[1/4, 4/15]$. If the last is met, translating by such a point removes from the first range a logarithmic interval of mass at least 0.31, leaving total mass < 0.401; if it is not met, the mass hypothesis forces a point in $[1/2, 0.583]$, whose translate excludes the middle range and leaves mass at most $\log(8/5) < 0.47$. Both contradict $1/2 - 3\epsilon_0$. All endpoints move continuously, so the same proof holds for one fixed $\kappa_* > 4/5$ sufficiently close to $4/5$.

Choose populated indices $j_1, \ldots, j_k$ with

$$\kappa_* \le \frac{(j_1 + \cdots + j_k)\log R}{\log N} \le 1.$$

Since every selected coordinate is at least $\alpha + o(1)$, we have $k \le C_\alpha$. Choose one prime from each $\mathcal{P}_{j_i}$, requiring the primes to be distinct. Because each populated interval contains $> R^{j_i(1-\delta)}$ primes and k is bounded, the number of ordered distinct choices is

$$\gg_\alpha R^{(j_1+\cdots+j_k)(1-\delta)} \ge N^{\kappa_*(1-\delta)}.$$

Unique factorization shows that a squarefree product is represented by at most $k!$ ordered choices. Moreover

$$w < R^{j_1+\cdots+j_k} \le N.$$

Therefore the number of distinct squarefree products is

$$\gg_\alpha N^{\kappa_*(1-\delta)} \gg_\alpha N^{4/5}$$

by (4.2). Every factor belongs to $\mathcal{P}$ and $\omega(w) \le k \le C_\alpha$, as required. □

Corollary 4.3 (Bourgain–Lindenstrauss arithmetic packet)**.** *For every $\eta > 0$, every sufficiently large nonsquare D, and every $N \ge D^{1/4+\eta}$, there is a set $\widetilde{W} \subset [2, N]$ with $|\widetilde{W}| \gg_\eta N^{4/5}$ such that every $w \in \widetilde{W}$ is squarefree, has $O_\eta(1)$ prime factors, and*

$$\left(\frac{D}{p}\right) = -1 \quad \textit{for every } p \mid w.$$

Proof. Apply Theorem 4.1 with $\epsilon < \epsilon_0$ and then Lemma 4.2. □

5. Adaptive Hecke shells and separation

For a good prime p, write $T_p(x)$ and $T_{p^2}(x)$ for the distance-one and distance-two Hecke spheres in the p-tree. If Φ has unnormalised Hecke eigenvalue λ_p, define

$$\zeta_\Phi(p) = \begin{cases} p, & |\lambda_p| > \sqrt{p}/10, \\ p^2, & |\lambda_p| \le \sqrt{p}/10, \end{cases}$$

and extend multiplicatively to squarefree integers.

Lemma 5.1 (Adaptive one-prime Hecke shell)**.** *There is an absolute C such that*

$$|\Phi(x)|^2 \le C \sum_{y \in T_{\zeta_\Phi(p)}(x)} |\Phi(y)|^2$$

for every good p and every x.

Proof. On the $(p+1)$-regular tree the adjacency equation is $\lambda_p \Phi(x) = \sum_{d(x,y)=1} \Phi(y)$. If $|\lambda_p| > \sqrt{p}/10$, Cauchy–Schwarz gives the result on the first sphere. If $|\lambda_p| \le \sqrt{p}/10$, apply the adjacency operator twice:

$$(\lambda_p^2 - (p+1))\Phi(x) = \sum_{d(x,z)=2} \Phi(z).$$

The coefficient on the left has magnitude $\gg p$, while the second sphere has $p(p+1)$ vertices. Cauchy–Schwarz again gives the asserted estimate. $\square$

Corollary 5.2 (Bounded-prime product shell)**.** *If w is squarefree with $\omega(w) \le K$, then*

$$|\Phi(x)|^2 \le C^K \sum_{y \in T_{\zeta_\Phi(w)}(x)} |\Phi(y)|^2.$$

Proof. Hecke correspondences at distinct good primes commute. Apply Lemma 5.1 successively to the prime factors of w. $\square$

Lemma 5.3 (Arithmetic collision forces commutation)**.** *Fix a compact set Ω in a compact arithmetic quaternionic surface. There are $\tau_0 > 0$ and C_Ω such that the following holds. Suppose primitive quaternion-order elements α, β of positive reduced norm satisfy*

$$x \in \alpha x B(C_\Omega \delta, C_\Omega \tau_0), \qquad x \in \beta x B(C_\Omega \delta, C_\Omega \tau_0),$$

with $x \in \Omega$, and

$$\delta^2 \le c_\Omega \{\mathrm{nrd}(\alpha)\, \mathrm{nrd}(\beta)\}^{-1}.$$

Then $\alpha\beta = \beta\alpha$.

Proof. Normalize both elements by the square roots of their reduced norms and conjugate by a bounded representative of $x \in \Omega$. The two normalized matrices lie in a fixed thin neighborhood of the diagonal group. Their commutator $\rho = \alpha\beta\alpha^{-1}\beta^{-1}$ therefore lies within $O_\Omega(\delta^2)$ of the identity, so $|\operatorname{tr}(\rho) - 2| \ll_\Omega \delta^2$. On the other hand, clearing denominators shows that $\operatorname{tr}(\rho)$ lies in $\mathbb{Z}[1/(\operatorname{nrd}\alpha \operatorname{nrd}\beta)]$. The displayed size hypothesis makes the interval around 2 shorter than the spacing of distinct such rationals, hence $\operatorname{tr}(\rho) = 2$. A division quaternion algebra contains no nontrivial unipotents, so $\rho = 1$. $\square$

Lemma 5.4 (Separated adaptive Hecke packet and small-box mass)**.** *Put* $\kappa = 4/5$ *and*

$$\kappa_0 = \frac{\kappa}{2(1+\kappa)} = \frac{2}{9}.$$

For every compact Ω *there are* $\tau_0, c, C > 0$ *such that, for every sufficiently small* δ*, every* $x \in \Omega$*, and every joint Hecke eigenfunction* Φ*, one can find a family* W *of cube-free Hecke levels satisfying*

$$|W| \geq c\,\delta^{-\kappa_0}, \qquad \omega(w) \leq C \quad (w \in W),$$

for which all sets

$$yB(\delta, \tau_0), \qquad y \in T_w(x), \quad w \in W,$$

are pairwise disjoint. Consequently, for the probability measure $d\mu_\Phi = |\Phi|^2 dm$*,*

$$\mu_\Phi(xB(\delta, \tau_0)) \ll_\Omega \delta^{\kappa_0}.$$

Proof. Choose a pair of Hecke levels $n_1 \leq n_2$ with minimal n_2 for which two distinct Hecke translates of the δ-box overlap. If no such pair occurs below $\delta^{-2\kappa_0}$, the prime levels themselves already give more than the required number of disjoint boxes, so assume $n_2 \leq \delta^{-2\kappa_0}$. The overlap gives a primitive quaternion element α with $1 < \operatorname{nrd}(\alpha) \mid n_1 n_2$ and $x \in \alpha x B(4\delta, 3\tau_0)$. Its quadratic algebra is a real quadratic field $L = \mathbb{Q}(\sqrt{D})$, with $D \ll_\Omega n_1 n_2$.

Choose

$$N \asymp (\delta^2 n_1 n_2)^{-1/4}.$$

Because $n_1 n_2 \leq \delta^{-4\kappa_0}$ and $\kappa_0 = \kappa/[2(1+\kappa)]$, decreasing the implicit constant gives $N \geq D^{1/4+\eta_0}$ for some fixed $\eta_0 > 0$, while $N^\kappa \gg \delta^{-\kappa_0}$. Corollary 4.3 supplies $\widetilde{W} \subset [2, N]$ of size $\gg N^\kappa$, all of whose prime factors are inert in L and with bounded ω. Set

$$W = \{\zeta_\Phi(w) : w \in \widetilde{W}\}.$$

These levels are cube-free and square exactly those primes for which the second Hecke shell is needed.

Suppose two distinct boxes at levels $l_1, l_2 \in W$ overlap. As in the construction of α, there is a primitive $\beta \neq 1$ with $\mathrm{nrd}(\beta) \mid l_1 l_2$, hence $\mathrm{nrd}(\beta) \leq N^4$, and $x \in \beta x B(4\delta, 3\tau_0)$. Our choice of N gives $\delta^2 \ll \{\mathrm{nrd}(\alpha)\,\mathrm{nrd}(\beta)\}^{-1}$, so Lemma 5.3 implies $\beta \in L$. Every rational prime dividing $\mathrm{nrd}(\beta)$ is inert in L and therefore occurs to even exponent in the norm of an integral element. Thus $\mathrm{nrd}(\beta) = A^2$ and the principal ideals generated by β and A coincide; hence β/A is a unit of $\mathcal{O}_L$. Its normalized trace is an integer. The thin-box condition makes its absolute trace lie in $[2, 5/2 + O(\delta)]$, so for small δ it equals 2. In a division quaternion algebra this forces $\beta = \pm 1$, contradicting primitivity. Therefore all boxes are disjoint.

Finally, Corollary 5.2 gives, for every $w \in \widetilde{W}$, a uniform pointwise inequality controlling $|\Phi|^2$ on the base box by the sum over the corresponding $T_{\zeta_\Phi(w)}$ boxes. Integrate over the base box and use the local measure preservation of the Hecke correspondence. Summing over w and using pairwise disjointness gives

$$|W|\, \mu_\Phi(xB(\delta, \tau_0)) \ll 1.$$

Since $|W| \gg \delta^{-\kappa_0}$, the small-box estimate follows. □

6. From small boxes to entropy on every component

Let $a(t) = \mathrm{diag}(e^t, e^{-t})$. With this normalization the unstable Jacobian at time one is e^2, so Haar entropy is 2.

Lemma 6.1 (Two-sided Bowen boxes sit inside arithmetic boxes)**.** *For every compact Ω there are $C, \tau_0 > 0$ such that*

$$B_N^{\pm}(x, \rho) := \{y : d(a(t)y, a(t)x) < \rho \text{ for all } |t| \leq N\} \subset xB(Ce^{-2N}, \tau_0)$$

for $x \in \Omega$ and N large.

Proof. Write a nearby group element in local coordinates $u_s^- a_v u_t^+$. Conjugation by $a(N)$ expands u_t^+ by e^{2N} and conjugation by $a(-N)$ expands u_s^- by e^{2N}; remaining within a fixed ρ-ball at both endpoints therefore gives $|s| + |t| \ll e^{-2N}$, while the neutral coordinate v stays in a fixed interval. Uniformity on Ω follows from the positive injectivity radius there. □

Theorem 6.2 (Bourgain–Lindenstrauss componentwise entropy bound). *Let μ be a nonzero arithmetic quantum limit on a compact arithmetic hyperbolic surface and let $\mu = \int \mu_\omega \, d\eta(\omega)$ be its A-ergodic decomposition. Then for η-almost every ω,*

$$h_{\mu_\omega}(a(1)) \geq \frac{2}{9}.$$

In particular every ergodic component has positive entropy. Haar measure has entropy 2 in this normalization.

PROOF. Lemma 5.4 is uniform on compact sets. Passing to the microlocal limit gives, on every fixed compact chart,

$$\mu(xB(\delta,\tau_0)) \ll \delta^{2/9}$$

for almost every regular x. With $\delta = Ce^{-2N}$, Lemma 6.1 yields

$$\mu(B_N^{\pm}(x,\rho)) \ll e^{-4N/9}. \tag{1}$$

The standard Bowen-name argument, written with a finite partition whose boundary has zero measure, converts (1) into $h_\mu(a(1)) \geq 2/9$: a two-sided name of length $2N+1$ is contained, apart from a subexponential collection of boundary-error names, in a two-sided Bowen box, so its information is at least $4N/9 - o(N)$; division by the $2N$ time length gives $2/9$.

We now make the componentwise step explicit. The same exponent survives normalization on every positive-measure A-invariant Borel set E. Indeed, for

$$\nu = \frac{\mu|_E}{\mu(E)}$$

one has $\nu(B_N^{\pm}(x,\rho)) \leq \mu(E)^{-1}\mu(B_N^{\pm}(x,\rho))$ for ν-almost every $x \in E$; the constant $\mu(E)^{-1}$ contributes only $o(N)$ to logarithmic information. Therefore

$$h_\nu(a(1)) \geq 2/9 \tag{6.1}$$

for every such normalized invariant restriction.

Suppose, toward a contradiction, that the set of ergodic components with entropy $< 2/9$ has positive η-measure. By a countable union argument there is $\varepsilon_0 > 0$ such that the components satisfying $h_{\mu_\omega}(a(1)) \leq 2/9 - \varepsilon_0$ still have positive η-measure. Their union is an A-invariant measurable set E of positive μ-measure. For the normalized restriction $\nu = \mu|_E/\mu(E)$, the entropy decomposition formula gives

$$h_\nu(a(1)) = \int_E h_{\mu_\omega}(a(1))\, d\eta_E(\omega) \leq 2/9 - \varepsilon_0,$$

contradicting (6.1). Hence $h_{\mu_\omega}(a(1)) \geq 2/9$ for almost every component. □

Proposition 6.3 (Entropy and Hecke recurrence hold on the same components)**.** *There is one η-conull set of A-ergodic components on which both*

$$h_{\mu_\omega}(a(1)) \geq 2/9 \quad \textit{and} \quad L/K\textit{-recurrence}$$

hold.

PROOF. Theorem 6.2 supplies a conull component set for the entropy statement. Proposition 0.3 supplies a conull component set for recurrence. Their intersection is conull. This elementary intersection is logically essential because the rigidity theorem stated and proved in the next chapter, Theorem 0.2, needs the two hypotheses on the same ergodic component. This sentence is only a forward interface; the present entropy argument does not use the conclusion of that later theorem. □

Proof and provenance. The proof architecture follows Bourgain–Lindenstrauss [BL03]: their arithmetic theorem constructs many squarefree products of primes inert in the collision field; their adaptive Hecke-shell argument squares precisely the low-eigenvalue primes; and their parameter calculation gives $\kappa_0 = \kappa/[2(1+\kappa)] = 2/9$ from $\kappa = 4/5$. We have replaced their citations to Burgess and Brun's combinatorial sieve by local proofs in this chapter. The Burgess amplification is checked against Burgess's original character-sum theorem [Bur63]; the complete Kummer–Weil estimate is now also proved internally above, so the finite-field citation is provenance only.

Worked Example 6.4 (Why the exponent is 2/9)**.** The arithmetic product packet has size N^{κ} with $\kappa = 4/5$. The collision argument can use a packet only up to a scale at which the product of two Hecke levels remains below the inverse square of the real box radius. Balancing these two requirements costs the factor $2(1+\kappa)$ in the denominator, so

$$\kappa_0 = \frac{4/5}{2(1+4/5)} = \frac{2}{9}.$$

The number 2/9 is therefore not an entropy constant coming from the geodesic flow itself; it is the arithmetic separation exponent produced by the Burgess packet and the collision geometry.

CHAPTER 5

The Arithmetic Positive-Entropy Rigidity Input

Let $G = SL_2(\mathbb{R}) \times L$, let $H = SL_2(\mathbb{R})$ be the real factor, let $K < L$ be compact, and let the real diagonal group $A < H$ act on $X = \Gamma\backslash G/K$. We assume $\Gamma \cap L$ is finite, exactly as in the arithmetic-QUE application.

Lemma 0.1 (Rank-one H-property from transverse recurrence)**.** *Let ν be an A-ergodic probability with positive entropy and L/K recurrence. Then the unstable leafwise conditional $\nu_x^{N^+}$ is nontrivial. Moreover, unless ν is already N^+-invariant, there are typical points x, x' arbitrarily close in a compact Lusin set, not lying on the same N^+ leaf, and times $t_j \to +\infty$ such that the renormalized pair*

$$xa(t_j), \quad x'a(t_j)$$

converges to two distinct typical points related by a nontrivial element of N^+ and carrying the same projective N^+ leafwise conditional.

Proof. Positive entropy and the leafwise entropy formula of Volume 13 make the N^+ conditional non-atomic and nontrivial. Choose a compact Lusin set Q on which the normalized conditional measures vary continuously and on which the A-Birkhoff and L/K recurrence statements are uniform.

If almost every sufficiently close recurrent pair in Q lies in one N^+ leaf, the tree/finite-place recurrence supplies infinitely many distinct transverse returns but the real coordinate does not change. The resulting accumulation gives a noncompact subgroup of the L/K leaf fixing the real conditional data. Because $\Gamma \cap L$ is finite, such a leaf cannot close up in a compact fibre; after restricting to a smaller Lusin set, one obtains a recurrent pair with a genuinely transverse real displacement. Thus we may write

$$x' = x(\ell \exp Z), \qquad \ell \in L, \quad Z \in \mathfrak{sl}_2(\mathbb{R}),$$

where the component of Z transverse to $\mathfrak{n}^+$ is nonzero and arbitrarily small.

Conjugation by $a(t)$ acts on $\mathfrak{sl}_2 = \mathfrak{n}^- \oplus \mathfrak{a} \oplus \mathfrak{n}^+$ with weights $-2, 0, 2$. Choose the first time $t = t(Z)$ at which the expanding polynomial coordinate of $a(-t)\exp(Z)a(t)$ reaches a fixed small scale. The contracting coordinate tends to zero and the neutral coordinate remains controlled. The L coordinate ℓ commutes with the real factor; L/K recurrence lets us choose the pair so that, at the selected times, both points return to the same compact Lusin chart. Passing to a subsequence gives two distinct typical limit points differing by a nontrivial $u^+(s)$.

Before renormalization, the two points were chosen in a scale on which their normalized N^+ conditionals are arbitrarily close. Equivariance of leafwise measures under A, together with Lusin continuity, transports this closeness to the limits. Hence the two limiting typical points have the same projective N^+ conditional. This is the rank-one H-property needed in the next theorem. □

THEOREM 0.2 (Lindenstrauss rank-one arithmetic measure rigidity). *Let μ be an A-invariant probability on $\Gamma\backslash(SL_2(\mathbb{R}) \times L)/K$. Assume*

(1) every A-ergodic component has positive entropy;

(2) μ is L/K-recurrent.

Then μ is a linear combination of algebraic probability measures invariant under $H = SL_2(\mathbb{R})$.

PROOF. Disintegrate μ into A-ergodic components. Proposition 0.3 and the recurrence hypothesis allow us to discard a null set so that the component under consideration is both positive-entropy and L/K-recurrent.

Fix such a component ν. Lemma 0.1 produces, unless N^+ invariance is already known, two typical points related by a nontrivial $u^+(s)$ and carrying the same projective unstable conditional. The distinct-conditionals argument of Volume 15 now applies verbatim: translation covariance gives

$$(u^+(s))_*\nu_x^{N^+} = c\,\nu_x^{N^+}.$$

Recurrence of the translation on the normalized conditional rules out $c \neq 1$; otherwise iterating the translation would force the mass of a fixed compact interval to grow or decay geometrically along recurrent returns. Thus the conditional is genuinely $u^+(s)$-invariant. The closed subgroup generated by such translations is nontrivial; A-equivariance expands it to all of N^+. By the Haar/invariance criterion for leafwise measures, ν is N^+-invariant.

Apply the same argument to $a(-t)$. Since entropy is symmetric under time reversal and L/K recurrence is unchanged, the stable conditional is nontrivial and the reversed H-property gives N^- invariance. The two opposite unipotent groups generate $SL_2(\mathbb{R})$, so ν is H-invariant.

Finally apply the S-arithmetic generated-unipotent classification already proved in Volume 4, Theorem 4.2. Each H-ergodic component of ν is the invariant probability on a closed orbit of an algebraic subgroup containing H. Integrating first over the H-ergodic decomposition and then over the original A-ergodic decomposition expresses μ as a linear combination of algebraic H-invariant probabilities, as claimed. □

Proposition 0.3 (Relation to the higher-rank low-entropy method)**.** *Theorem 0.2 has the same local rigidity mechanism as Volume 15: nontrivial leafwise measure, transverse recurrence, polynomial shearing, equality of limiting conditionals, and the Haar/invariance criterion. The new ingredient is that the transverse recurrence is supplied by L/K rather than a second real diagonal direction.*

Proof. Every step after the construction of the recurrent pair takes place in the real SL_2 factor and is exactly the rank-one version of the low-entropy shearing proof. The commuting auxiliary group L is used only to manufacture recurrent transverse pairs while leaving the real polynomial conjugation unchanged. Thus the difference between the two arguments is the source of recurrence, not the leafwise-measure mechanism that converts recurrence into unipotent invariance. □

> **Proof and provenance.** This is the exact structural scope of Lindenstrauss's Theorem 1.1: $G = SL_2(\mathbb{R}) \times L$, positive entropy on every real-diagonal ergodic component, and L/K recurrence imply a linear combination of algebraic $SL_2(\mathbb{R})$-invariant measures [Lin06]. It is not a direct special case of the SL_k EKL theorem. The final unipotent classification is not an external theorem here: the arithmetic matrix scope needed above is owned by Volume 4.

Worked Example 0.4 (Hecke recurrence substitutes for higher rank)**.** On a single hyperbolic surface the geodesic flow has rank one, so there is no real centralizing diagonal direction comparable to the higher-rank directions of Volumes 14–15. The p-adic Hecke tree supplies a noncompact recurrent direction commuting with the real factor. Once it produces nearby returns, the rest of the proof is again a real unipotent-shearing argument.

CHAPTER 6

Applying Rigidity to Arithmetic Quantum Limits

The analytic and dynamical inputs have now been separated. Chapters 2–3 give finite-place recurrence; Chapter 4 gives positive entropy on every real-diagonal ergodic component; Chapter 5 tells us what those two facts imply.

Lemma 0.1 (All rigidity hypotheses hold)**.** *Let μ be a nonzero microlocal limit of joint Hecke–Laplace eigenfunctions on a compact arithmetic hyperbolic surface. After passage to the standard S-arithmetic realization*

$$\Gamma\backslash(SL_2(\mathbb{R}) \times L)/K,$$

μ is A-invariant, is L/K-recurrent, and almost every A-ergodic component has entropy at least $2/9$ for $a(1) = \mathrm{diag}(e, e^{-1})$.

Proof. A-invariance is Theorem 0.3. L/K recurrence is Theorem 0.2. The componentwise entropy bound is Theorem 6.2. The adelic lift changes neither the real geodesic action nor the microlocal test functions; it only records the chosen good Hecke prime as the commuting finite-place factor. Hence the three hypotheses coexist on one probability space. ☐

Theorem 0.2 (Homogeneity of arithmetic quantum limits). *Every arithmetic quantum limit on a compact arithmetic hyperbolic surface is invariant under the full real group $H = SL_2(\mathbb{R})$ on the homogeneous frame bundle.*

Proof. Apply Theorem 0.2 using Lemma 0.1. The result is a linear combination of algebraic H-invariant probabilities. In the compact arithmetic surface setting the H-action is transitive on the real homogeneous factor, and the finite-place coordinate is auxiliary Hecke bookkeeping. The projection of each algebraic H-invariant probability to the real frame bundle is therefore normalized Haar measure. Hence the real microlocal measure itself is Haar. ☐

Proposition 0.3 (Projection to the surface)**.** *If a microlocal limit is Haar/Liouville measure on S^*M, then its projection to M is normalized hyperbolic area.*

Proof. The Liouville measure disintegrates over M as normalized angular measure on the unit cotangent circle times hyperbolic area. Integrating out the fibre therefore gives $d\mathrm{vol}/\mathrm{vol}(M)$. Proposition 0.4 identifies this pushforward with the configuration-space quantum limit. $\square$

Proof and provenance. This is the final assembly step in the compact arithmetic case of Lindenstrauss [Lin06]. The proof has been written as a dependency check so that no phrase such as "apply measure rigidity" conceals either recurrence or the componentwise entropy hypothesis.

Worked Example 0.4 (What would fail without the Hecke direction)**.** If one retained only geodesic invariance, an arbitrary A-invariant positive-entropy measure on a rank-one surface need not be Haar. The extra p-adic recurrence is the arithmetic substitute for the additional real diagonal directions that powered Volumes 14–16.

CHAPTER 7

Arithmetic Quantum Unique Ergodicity

We can now turn the subsequential statement into convergence of the whole sequence.

Lemma 0.1 (Uniqueness of subsequential limits implies convergence)**.** *Let (ν_j) be a sequence of probabilities on a compact metric space. If every weak-* convergent subsequence has the same limit ν, then $\nu_j \to \nu$ weak-*.*

Proof. If convergence failed, some continuous f and $\varepsilon > 0$ would satisfy $|\int f\, d\nu_{j_k} - \int f\, d\nu| \geq \varepsilon$ along a subsequence. Compactness of the probability-measure space gives a further weak-* convergent subsequence, whose limit must be ν, contradicting the inequality. □

Theorem 0.2 (Arithmetic QUE for compact arithmetic hyperbolic surfaces). *Let M be a compact congruence arithmetic hyperbolic surface. Let (ϕ_j) be L^2-normalized joint eigenfunctions of the Laplacian and the full family of good Hecke operators, with Laplace eigenvalue tending to infinity. Then*

$$|\phi_j|^2 \, d\mathrm{vol} \ \xrightarrow{\ *\ }\ \frac{d\mathrm{vol}}{\mathrm{vol}(M)}.$$

Equivalently, every microlocal lift converges to Liouville probability measure on S^*M.

Proof. Every subsequence has a further microlocally convergent subsequence by compactness. Theorem 0.2 identifies every such limit with Liouville measure, and Proposition 0.3 identifies its base projection with normalized area. Lemma 0.1 therefore upgrades subsequential uniqueness to convergence of the entire sequence. □

Proposition 0.3 (No multiplicity hypothesis after Hecke diagonalization)**.** *Theorem 0.2 does not require Laplace eigenvalues to have multiplicity one. It applies to any orthonormal basis chosen to be simultaneous eigenfunctions of the commuting self-adjoint good Hecke operators.*

Proof. On each finite-dimensional Laplace eigenspace, the commuting self-adjoint Hecke operators are simultaneously diagonalizable. Choose an orthonormal common eigenbasis there. Every argument in Chapters 2–6 uses the eigenfunction identities of the chosen vector, not uniqueness of that vector inside the Laplace eigenspace. Hence spectral multiplicity causes no additional hypothesis. □

> **Proof and provenance.** The compact arithmetic QUE endpoint is Lindenstrauss's theorem [Lin06]. The present chapter adds no new rigidity mechanism; it makes explicit the compactness/uniqueness step that turns classification of all subsequential limits into QUE.

Worked Example 0.4 (Why "density-one QUE" is weaker)**.** A theorem saying that a density-one subsequence equidistributes still permits a sparse exceptional sequence of eigenfunctions concentrating elsewhere. Theorem 0.2 rules out every exceptional subsequence because any subsequence has a further microlocal limit and every such limit has already been identified as Haar.

CHAPTER 8

Normalization, Noncompact Quotients, and the Exact Endpoint

The compact theorem is now complete. The finite-volume case differs at one logically separate point: vague limits can lose mass in the cusp.

Lemma 0.1 (Subprobability normalization under vague convergence)**.** *Let X be locally compact and let μ_j be probability measures. If μ_j converges vaguely to a Radon measure μ, then $\mu(X) = c$ for some $0 \le c \le 1$. If the sequence is tight, then $c = 1$ and the convergence is weak-* as probabilities.*

Proof. Choose $0 \le \chi_R \le 1$ compactly supported with $\chi_R \uparrow 1$. Vague convergence gives

$$\int \chi_R \, d\mu = \lim_j \int \chi_R \, d\mu_j \le 1.$$

Monotone convergence yields $\mu(X) \le 1$. Under tightness, for every $\varepsilon > 0$ choose compact K with $\mu_j(K) \ge 1 - \varepsilon$ for all large j, then choose $\chi \in C_c(X)$ with $\chi \equiv 1$ on K. Vague convergence gives $\mu(X) \ge \int \chi \, d\mu \ge 1 - \varepsilon$; let $\varepsilon \downarrow 0$. ☐

Theorem 0.2 (Exact finite-volume scope of the arithmetic-QUE argument). *On a finite-volume noncompact arithmetic hyperbolic surface, the recurrence–entropy–rigidity argument identifies every vague arithmetic microlocal limit as*

$$\mu = c \, m_{\text{Liouville}}, \qquad 0 \le c \le 1.$$

A separate no-escape theorem is required to conclude $c = 1$. In the compact case no escape is possible and Theorem 0.2 is unconditional. For the modular surface, Soundararajan's no-escape theorem supplies the remaining step [Sou10].

Proof. All local arguments in Chapters 1–5 may be applied to the nonzero part of a vague limit after normalizing it to a probability. Theorem 0.2 then forces that normalized part to be Liouville measure; restoring its total mass gives $c \, m_{\text{Liouville}}$. Lemma 0.1 shows that $c = 1$ is equivalent to excluding escape of mass. Compactness supplies tightness automatically. The modular-surface no-escape assertion is an independent analytic theorem and is therefore cited rather than smuggled into the rigidity argument. □

> **Proof and provenance.** This is the exact scope distinction emphasized in Lindenstrauss's treatment of arithmetic QUE and in the subsequent no-escape work of Soundararajan [Lin06; Sou10]. It prevents the common logical error "homogeneous nonzero limit therefore probability Haar."

Worked Example 0.3 (A toy escaping sequence)**.** On the noncompact half-line $(0, \infty)$, the probability measures δ_j converge vaguely to the zero measure. Classification of nonzero vague limits says nothing about the missing mass. The cusp of a finite-volume arithmetic surface creates the same logical possibility, which is why no-escape is independent of measure rigidity.

Volume 20

Periodic Torus Orbits and Packets

Volume 20 scope and prerequisites

This volume builds the arithmetic dictionary for periodic maximal-torus orbits on spaces of lattices. Its central laboratory is

$$X_n = \mathrm{PGL}_n(\mathbb{Z})\backslash \mathrm{PGL}_n(\mathbb{R}), \qquad H = \{\text{diagonal matrices}\}/\mathbb{R}^\times,$$

and the arithmetic objects are totally real degree-n fields, full lattices in those fields, and the orders that stabilize those lattices. The final theorem of the volume is deliberately a *framework* theorem: it explains how a lower bound for packet volume relative to discriminant produces entropy, and how the positive-entropy classification proved in Volume 16 then converts that entropy into algebraic—and, in prime degree, Haar—limits. The analytic estimates that prove equidistribution of the cubic packets themselves belong to Volume 21.

The proof floor is earlier AHD only. We inherit the finiteness of ideal class groups, Dirichlet's unit theorem, and Mahler compactness from Volume 3; entropy technology from Volumes 12–13; and positive-entropy measure classification from Volume 16. No equidistribution theorem for torus packets is imported. The arithmetic and adelic packet constructions are proved here in the precise PGL_n scope used later.

Proof and provenance. The architecture follows the division of labour in Einsiedler–Lindenstrauss–Michel–Venkatesh: the basic correspondence, discriminant, separation and volume/discriminant philosophy are developed in [Ein+09], while the adelic equivalence relation and packet structure used for the cubic theorem are made explicit in [Ein+11]. We use those papers as source-parity checks, not as substitutes for the proofs below.

Student orientation: one object, four descriptions

A periodic diagonal orbit should be pictured simultaneously as

$$\begin{array}{ccc} \boxed{\text{closed } H\text{-orbit}} & \longleftrightarrow & \boxed{\mathbb{Q}\text{-anisotropic, } \mathbb{R}\text{-split torus}} \\ \updownarrow & & \updownarrow \\ \boxed{(K, [L], \theta)} & \longleftrightarrow & \boxed{\text{adelic torus packet}}. \end{array}$$

The real orbit records the dynamics. The rational torus records why the orbit closes. The lattice $L \subset K$ records the integral arithmetic, and its multiplier order controls discriminant and regulator. The adelic description reveals the hidden finite abelian symmetry: several real periodic orbits are different projections of one adelic toral set.

Reader guide: a concrete model

Why this matters.

Volume 20 is organized around the passage from *Algebraic Tori in the Space of Lattices* to *The Torus-Packet Equidistribution Framework*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.4 (A calculation to keep in view)**.** Let $K = \mathbb{Q}(\sqrt{2})$ and let $u = 1 + \sqrt{2}$. Under the two real embeddings,

$$u \longmapsto (1 + \sqrt{2}, 1 - \sqrt{2}),$$

so the logarithmic torus coordinate is

$$\ell(u) = (\log(1 + \sqrt{2}), \log|1 - \sqrt{2}|) = (a, -a), \qquad a = \log(1 + \sqrt{2}).$$

Thus powers u^n form the lattice $\mathbb{Z}(a, -a)$ in the trace-zero line. This is the elementary model behind periodic torus orbits: arithmetic units become a lattice in the real split torus.

CHAPTER 1

Algebraic Tori in the Space of Lattices

1. Why the torus is arithmetic data

A diagonal subgroup of $\mathrm{PGL}_n(\mathbb{R})$ is easy to write down. The difficulty is to decide when a conjugate of it comes from rational algebra and therefore has a chance to produce a compact orbit on X_n. The right invariant is not a matrix of generators but a rational torus.

Definition 1.1. A $\mathbb{Q}$-torus is a connected affine algebraic group $\mathbf{T}/\mathbb{Q}$ that becomes isomorphic over $\overline{\mathbb{Q}}$ to a product of copies of $\mathbb{G}_m$. It is $\mathbb{Q}$-split if that isomorphism is already defined over $\mathbb{Q}$. It is $\mathbb{Q}$-anisotropic if it admits no nontrivial $\mathbb{Q}$-rational character $\mathbf{T} \to \mathbb{G}_m$.

Lemma 1.2 (Characters of a field torus)**.** *Let $K/\mathbb{Q}$ be a field of degree n, let*

$$\mathbf{R}_K = \mathrm{Res}_{K/\mathbb{Q}}\,\mathbb{G}_m, \qquad \mathbf{T}_K = \mathbf{R}_K/\mathbb{G}_m,$$

where the denominator is the diagonal scalar torus. If $\Sigma = \mathrm{Hom}_{\mathbb{Q}}(K, \overline{\mathbb{Q}})$, then

$$X^*(\mathbf{R}_K) \cong \mathbb{Z}^{\Sigma}, \qquad X^*(\mathbf{T}_K) \cong \left\{ (m_\sigma)_{\sigma\in\Sigma} : \sum_{\sigma} m_\sigma = 0 \right\},$$

with the natural Galois permutation action. Consequently $\mathbf{T}_K$ is $\mathbb{Q}$-anisotropic.

Proof. After scalar extension to a Galois closure E of K, the algebra $K \otimes_{\mathbb{Q}} E$ is the product $\prod_{\sigma\in\Sigma} E$. Hence $\mathbf{R}_K \otimes E \cong \prod_{\sigma\in\Sigma} \mathbb{G}_m$, and a character is a monomial $\prod z_\sigma^{m_\sigma}$, giving $X^*(\mathbf{R}_K) = \mathbb{Z}^{\Sigma}$. The diagonal scalar $z \mapsto (z)_\sigma$ is killed by this monomial exactly when $\sum m_\sigma = 0$, proving the second formula.

A character is defined over $\mathbb{Q}$ precisely when its exponent vector is fixed by the absolute Galois group. Because K is a field, the Galois action on Σ is transitive; every invariant vector in $\mathbb{Z}^{\Sigma}$ is therefore constant. A constant vector lying in the augmentation-zero lattice has value zero. Thus $X^*(\mathbf{T}_K)^{\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})} = 0$, which is the stated anisotropy criterion. $\square$

Theorem 1.3 (Maximal real split tori in PGL_n). *Let $A < \mathrm{PGL}_n(\mathbb{R})$ be a connected $\mathbb{R}$-split torus. Then A is conjugate into the diagonal torus H. If*

$\dim A = n-1$, *then* A *is conjugate to* H*. Moreover*

$$N_{\mathrm{PGL}_n(\mathbb{R})}(H)/H \cong S_n.$$

Proof. Lift A to the inverse image $\widetilde{A} < \mathrm{GL}_n(\mathbb{R})$. Since A is split, every rational representation of A decomposes into real weight spaces. Applied to the defining representation, $\mathbb{R}^n$ is a direct sum of common eigenspaces for $\widetilde{A}$. Choosing a basis adapted to these weight spaces conjugates $\widetilde{A}$, hence A, into diagonal matrices modulo scalars.

The diagonal torus of PGL_n has dimension $n-1$, so an inclusion of a connected split torus of the same dimension is equality. Finally, a matrix normalizing all diagonal matrices permutes their common one-dimensional eigenspaces; modulo diagonal matrices it is therefore represented by a permutation matrix. Conversely permutation matrices normalize H. This gives the Weyl-group identification. □

Proposition 1.4 (Regular-representation realization of the field torus)**.** *Let* $K/\mathbb{Q}$ *be totally real of degree* n*. Choose a* $\mathbb{Q}$*-basis of* K *and let*

$$\rho : K \hookrightarrow M_n(\mathbb{Q})$$

be the regular representation by multiplication. Then $\rho(K^\times)$ *modulo scalar matrices defines a maximal* $\mathbb{Q}$*-torus* $\mathbf{T}_K < \mathrm{PGL}_n$*. It is* $\mathbb{Q}$*-anisotropic and* $\mathbb{R}$*-split. For an ordering* $\theta = (\sigma_1, \dots, \sigma_n)$ *of the real embeddings, there exists* $g_\theta \in \mathrm{GL}_n(\mathbb{R})$ *such that*

$$g_\theta \rho(a) g_\theta^{-1} = \mathrm{diag}(\sigma_1(a), \dots, \sigma_n(a))$$

for every $a \in K$*.*

Proof. The regular representation is faithful because multiplication by a nonzero field element is invertible. Over $\mathbb{R}$ the map

$$K \otimes_{\mathbb{Q}} \mathbb{R} \longrightarrow \mathbb{R}^n, \qquad a \otimes 1 \longmapsto (\sigma_1(a), \dots, \sigma_n(a))$$

is an algebra isomorphism. Writing this isomorphism in the chosen basis yields g_θ and the displayed simultaneous diagonalization. Thus the image modulo scalars has real dimension $n-1$ and is a maximal $\mathbb{R}$-split torus. Its $\mathbb{Q}$-anisotropy is Lemma 1.2. □

Worked Example 1.5 (A real quadratic field)**.** Let $K = \mathbb{Q}(\sqrt{d})$, $d > 0$ squarefree, with basis $(1, \sqrt{d})$. Multiplication by $a + b\sqrt{d}$ is represented by

$$\begin{pmatrix} a & bd \\ b & a \end{pmatrix}.$$

The two real embeddings send $\sqrt{d}$ to $\pm\sqrt{d}$, so the eigenvalues are $a \pm b\sqrt{d}$. Modulo scalars these matrices form the rational torus whose real points are conjugate to the diagonal subgroup of $\mathrm{PGL}_2(\mathbb{R})$. The fact that this torus is $\mathbb{Q}$-anisotropic is the algebraic shadow of the compactness of the associated closed geodesics.

Proof and provenance. The rational-torus viewpoint is the first half of the basic correspondence in [Ein+09, §2.2]. Our field-torus specialization is the concrete PGL_n model used in [Ein+09, §4.1].

CHAPTER 2

Split and Anisotropic Structure

1. The logarithmic model of the real torus

For a totally real field K of degree n set

$$\mathfrak{a} = \left\{ x = (x_1, \dots, x_n) \in \mathbb{R}^n : \sum_i x_i = 0 \right\}.$$

The quotient $(\mathbb{R}^\times)^n/\mathbb{R}^\times$ has finitely many sign components and identity component naturally identified with $\mathfrak{a}$ by logarithms followed by subtraction of the mean.

Lemma 1.1 (Unit lattice inside the real split torus)**.** *Let $\mathcal{O}$ be an order in a totally real field K of degree n. Define*

$$\ell(u) = (\log|\sigma_1(u)|, \dots, \log|\sigma_n(u)|).$$

For $u \in \mathcal{O}^\times$ one has $\ell(u) \in \mathfrak{a}$, and the image $\ell(\mathcal{O}^\times)$ is a full lattice in $\mathfrak{a}$. In particular the quotient of the identity component of $\mathbf{T}_K(\mathbb{R})$ by the image of $\mathcal{O}^\times$ has finite volume and is compact.

Proof. The norm of a unit is ± 1, so $\sum_i \log|\sigma_i(u)| = \log|N_{K/\mathbb{Q}}(u)| = 0$. The assertion that the image is a full lattice is exactly the totally real case of the Dirichlet unit theorem already proved in Theorem 2.1. A full lattice in the finite-dimensional vector space $\mathfrak{a}$ has compact quotient. The finitely many sign components do not affect compactness or finite covolume. □

Theorem 1.2 (Anisotropic rational torus $\Longleftrightarrow$ periodic real orbit in the PGL_n arithmetic model). *Let $\Gamma = \mathrm{PGL}_n(\mathbb{Z})$, $G = \mathrm{PGL}_n(\mathbb{R})$ and H the diagonal torus. Suppose $gHg^{-1} = \mathbf{T}(\mathbb{R})$ for a maximal $\mathbb{Q}$-torus $\mathbf{T} < \mathrm{PGL}_n$. If $\mathbf{T}$ is $\mathbb{Q}$-anisotropic, then ΓgH is periodic. Conversely, if ΓgH is periodic, then the Zariski closure over $\mathbb{Q}$ of $gHg^{-1} \cap g\Gamma g^{-1}$ is a maximal $\mathbb{Q}$-torus $\mathbf{T}$ that is $\mathbb{Q}$-anisotropic and satisfies $\mathbf{T}(\mathbb{R}) = gHg^{-1}$.*

Proof. For the forward implication, let $E \subset M_n(\mathbb{Q})$ be the commutative semisimple algebra generated by a lift of $\mathbf{T}$. Because $\mathbf{T}$ is maximal, $\dim_{\mathbb{Q}} E = n$. If E had a nontrivial idempotent, then $E = E_1 \times E_2$ and the ratio of suitable powers of the determinants on the two summands would give a nontrivial $\mathbb{Q}$-character of $\mathbf{T}$, contradicting anisotropy. Thus E is a field K of degree n. Since $\mathbf{T}(\mathbb{R})$ is split, $K \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{R}^n$, so K is totally real.

Intersecting the order $M_n(\mathbb{Z})$ with K after conjugation produces an order $\mathcal{O}$ whose units lie in the arithmetic stabilizer. Lemma 1.1 says that their logarithms form a full lattice in $\mathfrak{a}$. The stabilizer can only be a finite-index enlargement of this full lattice, so $(g^{-1}\Gamma g \cap H)\backslash H$ is compact. Thus the orbit is periodic.

Conversely let $\Lambda = gHg^{-1} \cap \Gamma'$ with $\Gamma' = g\Gamma g^{-1}$ be the stabilizer of a finite-volume H-orbit. Since $H \cong (\mathbb{R}^\times)^{n-1}$ up to finitely many components, finite covolume implies that Λ has logarithmic rank $n-1$. Its Zariski closure is therefore a torus of dimension $n-1$, hence maximal, and is defined over $\mathbb{Q}$ because $\Lambda \subset \mathrm{PGL}_n(\mathbb{Q})$. If this torus had a nontrivial $\mathbb{Q}$-character χ, then $\chi(\Lambda)$ would be an arithmetic subgroup of $\mathbb{G}_m(\mathbb{Q})$ while the real map $\log|\chi|$ would give a nonzero linear functional on the logarithmic stabilizer lattice. The quotient in that character direction would be noncompact, contradicting compactness of the periodic orbit. Hence the torus is anisotropic. □

Proposition 1.3 (Shape and regulator)**.** *Fix Haar measure on H° corresponding under logarithms to Lebesgue measure on $\mathfrak{a}$. For a periodic orbit arising from an order $\mathcal{O}$ in a totally real field, its connected stabilizer lattice is the image of $\mathcal{O}^{\times,+}$, and*

$$\mathrm{vol}(\Gamma g H) = c_n\,[\mathcal{O}^\times : \mathcal{O}^{\times,+}]^{-1}\,\mathrm{Reg}(\mathcal{O})$$

for a constant $c_n > 0$ determined only by the fixed Haar normalization. Thus the Euclidean similarity class of $\ell(\mathcal{O}^\times)$ is the shape of the orbit.

Proof. The stabilizer consists of projective multiplication operators preserving the corresponding lattice. Such an operator is multiplication by a unit of the multiplier order, proved explicitly in Chapter 3. Passing to the identity component removes the finite sign group. The covolume of the logarithmic unit lattice is, by definition, the regulator. The only discrepancy is the fixed Jacobian between our Haar measure and Euclidean Lebesgue measure, together with the finite sign index. □

Warning 1.4. "$\mathbb{R}$-split" and "$\mathbb{Q}$-anisotropic" are not opposites. For the field tori used here they occur simultaneously: the torus splits completely over $\mathbb{R}$ because the field is totally real, yet it has no nontrivial rational character because the Galois action on embeddings is transitive.

CHAPTER 3

Periodic Diagonal and Torus Orbits

1. Lattices in a number field

Let $K/\mathbb{Q}$ be totally real of degree n. A *full lattice* $L \subset K$ is a free abelian group of rank n spanning K over $\mathbb{Q}$. Its multiplier order is

$$\mathcal{O}_L = \{a \in K : aL \subset L\}.$$

If $aL \subset L$ and $a \neq 0$, then multiplication by a has nonzero determinant; if also $a^{-1}L \subset L$ then $a \in \mathcal{O}_L^\times$.

Lemma 1.1 (Basic rational-torus correspondence)**.** *Periodic H-orbits ΓgH on X_n are in bijection with Γ-conjugacy classes of pairs $(\mathbf{T}, gH)$, where $\mathbf{T} < \mathrm{PGL}_n$ is a maximal $\mathbb{Q}$-torus that is $\mathbb{Q}$-anisotropic and $\mathbb{R}$-split and*

$$gHg^{-1} = \mathbf{T}(\mathbb{R}).$$

Proof. Starting from a periodic orbit, Theorem 1.2 constructs the unique maximal rational torus obtained as the Zariski closure of its arithmetic stabilizer. Replacing g by γgh with $\gamma \in \Gamma$ and $h \in H$ conjugates $\mathbf{T}$ by γ and leaves gH unchanged, so the assignment depends only on the orbit.

Conversely a pair $(\mathbf{T}, gH)$ with the stated properties produces a periodic orbit by the forward implication of Theorem 1.2. If two pairs produce the same orbit, then their representatives differ by $\gamma \in \Gamma$ and $h \in H$, and the equality of their real tori forces the corresponding rational tori to be Γ-conjugate. These operations are inverse. □

Theorem 1.2 (Concrete parameterization by field, lattice class, and real embeddings). *There is a bijection between periodic H-orbits on X_n and isomorphism classes of triples*

$$(K, [L], \theta),$$

where K is a totally real field of degree n, $[L]$ is the $K^\times$-homothety class of a full lattice $L \subset K$, and

$$\theta : K \otimes_{\mathbb{Q}} \mathbb{R} \xrightarrow{\sim} \mathbb{R}^n$$

is an $\mathbb{R}$-algebra isomorphism, considered up to permutation of the n factors. The corresponding orbit is the H-orbit of the homothety class of the Euclidean lattice $\theta(L) \subset \mathbb{R}^n$.

Proof. Given $(K,[L],\theta)$, choose a $\mathbb{Z}$-basis of L. In that basis multiplication by K embeds K into $M_n(\mathbb{Q})$, and θ diagonalizes the regular representation over $\mathbb{R}$ by Proposition 1.4. The lattice $\theta(L)$ therefore determines a point $x\in X_n$, and the associated torus is the image of $K^\times/\mathbb{Q}^\times$. It is $\mathbb{Q}$-anisotropic and $\mathbb{R}$-split, so xH is periodic.

Multiplying L by $a\in K^\times$ acts on $\theta(L)$ by the diagonal matrix $\operatorname{diag}(\sigma_i(a))$, hence does not change its H-orbit. Changing the $\mathbb{Z}$-basis acts by $\mathrm{GL}_n(\mathbb{Z})$, hence does not change the point of X_n, and reordering embeddings acts by the Weyl group, hence does not change the orbit.

Conversely, start from a periodic orbit. Lemma 1.1 gives a maximal $\mathbb{Q}$-anisotropic torus $\mathbf{T}$. The commutative semisimple $\mathbb{Q}$-subalgebra of $M_n(\mathbb{Q})$ generated by a lift of $\mathbf{T}$ has dimension n. Anisotropy excludes a nontrivial idempotent: such an idempotent would split the algebra and produce a nontrivial rational character by taking the determinant on one factor divided by the appropriate power of the determinant on the other. Therefore the algebra is a field K. Since the real torus is split, $K\otimes\mathbb{R}\cong\mathbb{R}^n$, so K is totally real. Pulling back the standard lattice $\mathbb{Z}^n$ through a cyclic vector for the regular representation gives a full lattice $L\subset K$. The real conjugacy $gHg^{-1}=\mathbf{T}(\mathbb{R})$ gives θ. Reversing the construction recovers the original orbit, and the evident changes correspond exactly to field isomorphism, $K^\times$-homothety, and permutation of embeddings. □

Proposition 1.3 (Multiplier order and orbit stabilizer)**.** *For the orbit associated with $(K,[L],\theta)$,*

$$g^{-1}\Gamma g\cap H\cong\mathcal{O}_L^\times/\{\pm1 \text{ and scalar units}\}$$

up to the fixed finite component ambiguity of PGL_n. In particular $\mathcal{O}_L$ and the logarithmic unit lattice depend only on the periodic orbit, not on the chosen representative $L\in[L]$.

Proof. A diagonal element in the θ-coordinates comes from multiplication by some $a\in K^\times$. It stabilizes the homothety class of $\theta(L)$ in the arithmetic quotient precisely when, after multiplication by a rational scalar, it maps L bijectively to itself. Absorbing that scalar into a says exactly that $aL=L$, i.e. $a\in\mathcal{O}_L^\times$. If $L'=bL$, then

$$\mathcal{O}_{L'}=\{a:a(bL)\subset bL\}=\mathcal{O}_L,$$

so the order and unit lattice are homothety invariants. □

Worked Example 1.4 (The lattice attached to an ideal class)**.** Take $L=I$ a fractional ideal of the maximal order $\mathcal{O}_K$. Then $\mathcal{O}_I=\mathcal{O}_K$, because $aI\subset I$ iff $a\in\mathcal{O}_K$ for an invertible fractional ideal. Thus all ideal classes of the same field have identical stabilizer shape and regulator. They give different periodic orbits because the underlying Euclidean lattices need not be $\mathrm{GL}_n(\mathbb{Z})$-equivalent.

Proof and provenance. Theorem 1.2 is the student-facing reconstruction of the PGL_n specialization of the basic correspondence, matching Corollary 4.4 of [Ein+09]. The proof is expanded here to make the multiplier order and the homothety invariance explicit because those are the exact interfaces needed for packets.

CHAPTER 4

Discriminant, Covolume, and Separation

1. A rational point on the variety of Cartan subalgebras

Let $\mathfrak{g} = \mathfrak{pgl}_n(\mathbb{R})$, $r = n-1$, and choose a Γ-stable lattice $\mathfrak{g}_{\mathbb{Z}} \subset \mathfrak{g}_{\mathbb{Q}}$ together with a nondegenerate rational invariant bilinear form B. The induced form on $\bigwedge^r \mathfrak{g}$ is denoted $B_\wedge$. For an r-dimensional Cartan subalgebra $\mathfrak{t}$ choose $0 \neq w \in \bigwedge^r \mathfrak{t}$ and put

$$\iota(\mathfrak{t}) = \frac{w \otimes w}{B_\wedge(w,w)}.$$

The sign and scale of w cancel.

Lemma 1.1 (Integral denominator and quantitative separation)**.** *Let $V = (\bigwedge^r \mathfrak{g})^{\otimes 2}$ and fix a Γ-stable lattice $V_{\mathbb{Z}}$. For a rational Cartan subalgebra define*

$$\operatorname{disc}(\mathfrak{t}) = \min\{D \geq 1 : D\,\iota(\mathfrak{t}) \in V_{\mathbb{Z}}\}.$$

Then the discriminant is Γ-invariant. Fix a compact set $\Omega \subset G$. If $g_1, g_2 \in \Omega$ determine distinct Cartan subalgebras and their periodic orbits have discriminants D_1, D_2, then

$$d(g_1,g_2) \gg_\Omega \begin{cases} D_1^{-1/2}, & D_1 = D_2, \\ (D_1D_2)^{-1/2}, & D_1 \neq D_2. \end{cases}$$

In particular, if both discriminants are at most D, distinct local Cartan branches in Ω are separated by $\gg_\Omega D^{-1}$; if all have common discriminant D, the separation improves to $\gg_\Omega D^{-1/2}$.

Proof. The tensor $\iota(\mathfrak{t})$ is rational when $\mathfrak{t}$ is rational. Since $V_{\mathbb{Z}}$ is Γ-stable and $\iota(\operatorname{Ad}\gamma\mathfrak{t}) = \operatorname{Ad}(\gamma)\iota(\mathfrak{t})$, the least denominator is invariant.

Choose $w_0 \in \bigwedge^r \mathfrak{h}$ with $B_\wedge(w_0,w_0) = 1$ and put $w_i = \operatorname{Ad}(g_i)w_0$. With the integral structure used in the definition of discriminant, $D_i^{1/2} w_i$ lies in a fixed lattice up to one normalization constant depending only on $V_{\mathbb{Z}}$; absorb that constant into the implied constants below. If $D_1 = D_2 = D$, the two vectors $D^{1/2}w_1$ and $D^{1/2}w_2$ are distinct lattice vectors, so

$$\|w_1 - w_2\| \gg D^{-1/2}.$$

If $D_1 \neq D_2$, the two integral vectors are nonproportional. Their exterior product is a nonzero vector in a fixed lattice and hence has norm bounded below. Since g_i lie in a compact set, $\|w_i\| \ll_\Omega 1$, and the elementary area inequality gives

$$\|w_1 - w_2\| \gg_\Omega (D_1 D_2)^{-1/2}.$$

Finally the map $g \mapsto \mathrm{Ad}(g)w_0$ is smooth with uniformly bounded derivative on $\Omega^{-1}\Omega$. Therefore output separation is bounded above by a constant times $d(g_1, g_2)$, giving the stated estimates. □

THEOREM 1.2 (Arithmetic meaning of discriminant and covolume). *Let xH correspond to $(K, [L], \theta)$ and let $\mathcal{O}_L$ be its multiplier order. With the integral structures fixed once for the whole volume, there are constants $c_{\mathrm{disc},n}, c_{\mathrm{vol},n} > 0$ depending only on those normalizations such that*

$$\mathrm{disc}(xH) = c_{\mathrm{disc},n}\,|\,\mathrm{disc}(\mathcal{O}_L)|, \qquad \mathrm{vol}(xH) = c_{\mathrm{vol},n}\,\mathrm{Reg}(\mathcal{O}_L)$$

up to the finite sign/component index already displayed in Proposition 1.3. In particular, equality of multiplier orders implies equality of discriminant and orbit volume.

PROOF. Choose a $\mathbb{Z}$-basis $e_1, \ldots, e_n$ of $\mathcal{O}_L$. Under the regular representation, the trace pairing

$$(a, b) \longmapsto \mathrm{Tr}_{K/\mathbb{Q}}(ab)$$

is, up to a fixed nonzero scalar depending only on the chosen invariant form on $\mathfrak{pgl}_n$, the restriction of B to the rational Cartan algebra generated by K. The square of the covolume of the lattice spanned by the multiplication operators $\rho(e_i)$ is therefore the determinant

$$|\det(\mathrm{Tr}_{K/\mathbb{Q}}(e_i e_j))| = |\,\mathrm{disc}(\mathcal{O}_L)|.$$

Passing from the n-dimensional commutative algebra to its trace-zero/projective Cartan quotient removes the scalar line. Because that scalar line and its integral lattice are fixed uniformly in n, the denominator definition through $\iota(\mathfrak{t})$ differs from the order discriminant by one fixed normalization factor. This proves the first identity.

For volume, Proposition 1.3 identifies the connected orbit stabilizer with the positive units of $\mathcal{O}_L$ modulo scalars. Under logarithms this is exactly the Dirichlet unit lattice. Its Haar covolume is the regulator, up to the fixed Jacobian and finite sign index, giving the second identity. □

Proposition 1.3 (Packet-stable arithmetic complexity)**.** *Suppose $L, L' \subset K$ are locally homothetic at every finite prime: for each p there is $a_p \in K_p^\times$ with $L'_p = a_p L_p$. Then*

$$\mathcal{O}_L = \mathcal{O}_{L'},$$

and the corresponding periodic orbits have the same discriminant, stabilizer lattice, shape, and volume.

Proof. For every prime p,

$$(\mathcal{O}_L)_p = \{a \in K_p : aL_p \subset L_p\} = \{a : a(a_pL_p) \subset a_pL_p\} = (\mathcal{O}_{L'})_p.$$

Two full $\mathbb{Z}$-orders in K that have the same localization at every prime are equal: membership of an element of K in either order can be checked after tensoring with every $\mathbb{Z}_p$. Thus the multiplier orders agree. Proposition 1.3 and Theorem 1.2 then give all the remaining equalities. □

Worked Example 1.4 (Why the denominator matters)**.** Two different rational Cartan subalgebras can be arbitrarily close over $\mathbb{R}$ if their rational denominators are allowed to grow. Lemma 1.1 says that discriminant records exactly the scale at which this can happen. This is the geometric input behind Linnik's principle: many orbits of controlled discriminant occupy many distinct Bowen tubes.

Proof and provenance. The tensor definition and the separation argument match Definition 2.2 and Proposition 2.3 of [Ein+09]. The identification with the order discriminant and regulator is the split-matrix specialization recorded in their Section 4.1 and Corollary 4.4.

CHAPTER 5

Adelic Torus Packets

1. Why adeles reveal a hidden symmetry

A real periodic orbit remembers one lattice class. At the finite places, however, one may change that lattice inside its local homothety class without changing the local torus. The adelic quotient packages all such changes at once.

Let $\mathbb{A}$ be the adeles of $\mathbb{Q}$, $\mathbb{A}_f$ the finite adeles, and

$$K_f = \prod_p \mathrm{PGL}_n(\mathbb{Z}_p) \subset \mathrm{PGL}_n(\mathbb{A}_f).$$

We identify X_n with the identity component of

$$\mathrm{PGL}_n(\mathbb{Q})\backslash \mathrm{PGL}_n(\mathbb{A})/K_f$$

through strong approximation for the split matrix group, proved in Volume 3 in the matrix scope used here.

Lemma 1.1 (Unique adelic toral set above a compact real orbit)**.** *Let $xH = \Gamma gH$ be periodic and let $\mathbf{T}$ be its rational torus. Then*

$$Y(\mathbf{T}, g) = \mathbf{T}(\mathbb{Q})\backslash\mathbf{T}(\mathbb{A})\, g \subset \mathrm{PGL}_n(\mathbb{Q})\backslash \mathrm{PGL}_n(\mathbb{A})$$

projects to a set containing xH. Any homogeneous toral set whose projection contains xH differs from $Y(\mathbf{T}, g)$ by right translation by an element of K_f.

Proof. Embed g adelically by taking real component g and finite component 1. Then $\mathbf{T}(\mathbb{R})g = gH$ at the real place, so the projection contains xH.

Suppose $Y' = \mathbf{T}'(\mathbb{Q})\backslash\mathbf{T}'(\mathbb{A})g'$ also projects to a set containing xH. Lifting one point of the real orbit gives

$$tg = \delta t' g' k_f$$

for some $t \in \mathbf{T}(\mathbb{R})$, $t' \in \mathbf{T}'(\mathbb{A})$, $\delta \in \mathrm{PGL}_n(\mathbb{Q})$ and $k_f \in K_f$. Varying t in the identity component of $\mathbf{T}(\mathbb{R})$ while using countability of $\mathrm{PGL}_n(\mathbb{Q})$ forces one fixed δ on a nonempty open set. Hence the real tori satisfy $\mathbf{T} = \delta\mathbf{T}'\delta^{-1}$ by Zariski density of the real identity component. Absorbing δ and t' into the adelic torus quotient leaves g' equal to gk_f^{-1} in the homogeneous set. Thus $Y' = Y(\mathbf{T}, g)k_f^{-1}$. □

THEOREM 1.2 (Finite packet and double-coset parameterization). *Declare two compact H-orbits equivalent when they occur in the projection of the same adelic toral set. The equivalence class—the* packet*—of xH is finite and is parameterized by the fiber over the identity component of*

$$\mathbf{T}(\mathbb{Q})\backslash\mathbf{T}(\mathbb{A}_f)/(K_f \cap \mathbf{T}(\mathbb{A}_f)) \longrightarrow \mathrm{PGL}_n(\mathbb{Q})\backslash \mathrm{PGL}_n(\mathbb{A}_f)/K_f.$$

PROOF. Take $t_f \in \mathbf{T}(\mathbb{A}_f)$. Its class maps to the identity double coset precisely when

$$t_f = \delta^{-1}k_f$$

with $\delta \in \mathrm{PGL}_n(\mathbb{Q})$ and $k_f \in K_f$. Then the same adelic toral set contains the real orbit represented by $\Gamma\delta gH$. Multiplying t_f on the left by $\mathbf{T}(\mathbb{Q})$ or on the right by $K_f \cap \mathbf{T}(\mathbb{A}_f)$ does not change that real orbit. Conversely every real orbit in the projection arises from such a decomposition. Hence the fiber parameterizes the packet.

Finiteness follows arithmetically. In the field model of Chapter 3, the fiber is identified in Chapter 6 with local-homothety classes of full lattices modulo global homothety, a torsor under a Picard group of an order $\mathcal{O}$. Extension of ideals gives a homomorphism

$$\mathrm{Pic}(\mathcal{O}) \longrightarrow \mathrm{Cl}(\mathcal{O}_K).$$

The target is finite by Corollary 2.2. Its kernel is finite as well: if an invertible $\mathcal{O}$-ideal becomes principal over $\mathcal{O}_K$, scale it so that its extension is $\mathcal{O}_K$. Away from primes dividing the conductor $\mathfrak{f}$ one then has $I_p = \mathcal{O}_p$. At the finitely many conductor primes, the possible invertible local classes are detected by units in the finite rings $\mathcal{O}_K/\mathfrak{f}$ and $\mathcal{O}/\mathfrak{f}$; hence only finitely many classes occur. Thus $\mathrm{Pic}(\mathcal{O})$, and therefore the packet, is finite. □

Proposition 1.3 (Common invariants inside one packet)**.** *All real periodic orbits in one packet have the same multiplier order, discriminant, stabilizer lattice, shape, and volume. If the adelic quotient has one relevant finite component, the packet is naturally a principal homogeneous space under the finite abelian group*

$$C(\mathbf{T}, K_f) = \mathbf{T}(\mathbb{Q})\backslash\mathbf{T}(\mathbb{A}_f)/(K_f \cap \mathbf{T}(\mathbb{A}_f)).$$

PROOF. Two representatives in one packet differ at the finite places by an element of $\mathbf{T}(\mathbb{A}_f)$; in the field-lattice model this is precisely local homothety of their lattices. Proposition 1.3 gives equality of all listed real invariants. When the target adelic double quotient has a single component, the fiber in Theorem 1.2 is the whole finite abelian source quotient, whose left translation action on itself is free and transitive. □

Proof and provenance. Lemma 1.1, Theorem 1.2, and Proposition 1.3 reconstruct the three assertions in the packet correspondence of [Ein+11, §5.4–5.5]. In particular we preserve the important distinction between a packet and the generally coarser collection of orbits having merely the same multiplier order.

CHAPTER 6

Class Groups and Local Homothety

1. The correct class group for a nonmaximal order

Let $\mathcal{O}$ be an order in a number field K. A fractional $\mathcal{O}$-ideal I is *invertible* if there is a fractional ideal J with $IJ = \mathcal{O}$. The Picard group $\mathrm{Pic}(\mathcal{O})$ is the group of invertible fractional $\mathcal{O}$-ideals modulo principal ideals $a\mathcal{O}$, $a \in K^\times$.

Lemma 1.1 (Local principalness and invertibility)**.** *For a full $\mathcal{O}$-lattice $I \subset K$, the following are equivalent:*

(1) I is invertible as a fractional $\mathcal{O}$-ideal;

(2) for every prime p, $I_p = I \otimes \mathbb{Z}_p$ is free of rank one over $\mathcal{O}_p$;

(3) for every prime p, there is $a_p \in K_p^\times$ with $I_p = a_p\mathcal{O}_p$.

Proof. If $IJ = \mathcal{O}$, localization gives $I_pJ_p = \mathcal{O}_p$. In a semilocal order an invertible rank-one module is projective of rank one, hence free of rank one, proving (1)$\Rightarrow$(2); in the present one-dimensional setting one can see this directly by choosing generators of I_pJ_p summing to 1 and using the resulting unimodular relation to produce a generator. The equivalence of (2) and (3) is immediate inside the one-dimensional K_p-vector space.

Assume (3). Set

$$J = \{x \in K : xI \subset \mathcal{O}\}.$$

Then $J_p = a_p^{-1}\mathcal{O}_p$, so $(IJ)_p = \mathcal{O}_p$ for every p. Equality of two full $\mathbb{Z}$-lattices can be checked after all localizations, hence $IJ = \mathcal{O}$ and I is invertible. ☐

Theorem 1.2 (Local-homothety classes form a Picard torsor). *Fix a full lattice $L \subset K$ and its multiplier order $\mathcal{O} = \mathcal{O}_L$. Let $\mathscr{P}(L)$ be the set of $K^\times$-homothety classes of full lattices $L' \subset K$ such that L'_p is homothetic to L_p for every prime p. Then* $\mathrm{Pic}(\mathcal{O})$ *acts freely and transitively on $\mathscr{P}(L)$.*

Proof. If I is an invertible fractional $\mathcal{O}$-ideal, define $I \cdot L$ as the $\mathbb{Z}$-span of products il. Locally, $I_p = a_p\mathcal{O}_p$ by Lemma 1.1, so

$$(IL)_p = a_pL_p.$$

Thus IL lies in the local-homothety class of L. Replacing I by aI globally rescales IL by a, so the action factors through $\mathrm{Pic}(\mathcal{O})$.

For transitivity, let L' be locally homothetic to L and define

$$I = (L' : L) = \{a \in K : aL \subset L'\}.$$

At each p, choosing $L'_p = a_p L_p$ gives $I_p = a_p \mathcal{O}_p$, hence I is invertible and $(IL)_p = L'_p$ for every p. Therefore $IL = L'$.

For freeness, if $IL = bL$ for some $b \in K^\times$, then localizing and cancelling the faithful rank-one $\mathcal{O}_p$-module L_p gives $I_p = b\mathcal{O}_p$ for all p, hence $I = b\mathcal{O}$. Thus $[I]$ is trivial in $\mathrm{Pic}(\mathcal{O})$. □

Proposition 1.3 (Maximal orders and ideal-class packets)**.** *If L has multiplier order $\mathcal{O}_K$, the maximal order of K, then every fractional ideal is invertible and*

$$\mathscr{P}(L) \cong \mathrm{Pic}(\mathcal{O}_K) = \mathrm{Cl}(K).$$

Consequently the packet of maximal-order periodic orbits attached to a fixed totally real field and real embedding data is a torsor under the ordinary ideal class group.

Proof. A Dedekind domain is locally a discrete valuation ring at every nonzero prime, so every nonzero fractional ideal is locally principal and hence invertible by Lemma 1.1. The Picard group of a Dedekind domain is the usual ideal class group. Theorem 1.2 and the adelic packet identification of Theorem 1.2 give the final assertion. □

Worked Example 1.4 (Quadratic closed geodesics)**.** For a real quadratic maximal order $\mathcal{O}_K$, the narrow sign issue separates oriented from unoriented geodesics, but after fixing the convention the set of closed geodesics of the corresponding arithmetic type is a class-group torsor. Every member has the same regulator because the unit group is the same, while different ideal classes give inequivalent lattices.

Worked Example 1.5 (Why "same order" can be too coarse)**.** For a nonmaximal order, not every proper ideal is invertible. Two lattices can have the same multiplier order but fail to be locally homothetic. The packet therefore remembers the *local homothety class*, not merely the order. This is why the Picard group, rather than the set of all proper ideal classes, is the correct symmetry group.

Proof and provenance. The torsor formulation is the arithmetic content emphasized in [Ein+11, §5.5]: local homothety gives a finer and better-behaved equivalence relation than equality of multiplier orders, and its symmetry group is the Picard group.

CHAPTER 7

Nonescape for Torus Packets: What Can Actually Be Proved Abstractly

1. Why an unconditional statement would be false

Periodic torus orbits are not unipotent orbits. Increasing discriminant does not by itself force tightness, and even higher-rank individual periodic torus orbits can lose mass in the cusp. The correct foundational result for this volume is therefore a criterion that isolates the arithmetic estimate needed to prevent escape.

Let $\mathrm{ht} : X_n \to [1, \infty)$ be a Mahler height, for example the reciprocal of the shortest-vector length after unimodular normalization, with any equivalent proper normalization.

Lemma 1.1 (Height tails and tightness). *Let μ_j be probability measures on X_n. The following are equivalent:*

(1) (μ_j) is tight;

(2) for every $\varepsilon > 0$ there is R such that

$$\sup_j \mu_j\{x : \mathrm{ht}(x) > R\} < \varepsilon.$$

If, for some $s > 0$, one has the uniform moment bound

$$\sup_j \int_{X_n} \mathrm{ht}(x)^s \, d\mu_j(x) < \infty,$$

then the family is tight.

Proof. By Mahler compactness, proved in the $S = \{\infty\}$ case of Theorem 2.2, the sublevel sets $\{\mathrm{ht} \le R\}$ are compact and exhaust X_n. This gives the equivalence directly. For the last assertion, Markov's inequality gives

$$\mu_j\{\mathrm{ht} > R\} \le R^{-s} \int \mathrm{ht}^s \, d\mu_j,$$

which tends to zero uniformly as $R \to \infty$. □

Theorem 1.2 (Arithmetic nonescape criterion for packet measures). *Let $\mathscr{P}_j$ be packets of periodic H-orbits and let μ_j be normalized packet* *volume measure. For each orbit choose a representative triple $(K,[L],\theta)$ and normalize $\theta(L)$ to covolume one. Suppose there exist $s>0$ and $C<\infty$ such that*

$$\frac{1}{\mathrm{vol}(\mathscr{P}_j)}\sum_{xH\in\mathscr{P}_j}\int_{xH}\lambda_1(y)^{-s}\,d\,\mathrm{vol}_{xH}(y)\le C$$

for all j, where $\lambda_1(y)$ denotes the first minimum of the corresponding unimodular lattice. Then (μ_j) is tight, and every weak- subsequential limit is an H-invariant probability measure.*

Proof. Our Mahler height is equivalent, up to fixed powers and constants, to λ_1^{-1}. The displayed estimate is therefore a uniform positive moment bound for height. Lemma 1.1 gives tightness. Prokhorov compactness then gives probability subsequential limits.

Each μ_j is H-invariant because it is a normalized sum of H-invariant orbit-volume measures. For $f\in C_c(X_n)$ and $h\in H$,

$$\int f(xh)\,d\mu_j(x)=\int f(x)\,d\mu_j(x).$$

Passing to a weak-* limit preserves this identity, so every limit is H-invariant. Tightness is exactly what prevents the limiting functional from having total mass <1. □

Proposition 1.3 (Cusp-mass estimate as the only missing arithmetic interface)**.** *To prove nonescape for a concrete family of packets it is enough to establish, for every $\varepsilon>0$, a bound of the form*

$$\frac{1}{\mathrm{vol}(\mathscr{P})}\sum_{xH\in\mathscr{P}}\mathrm{vol}_{xH}\{y\in xH:\lambda_1(y)<\eta\}\le C_\varepsilon\eta^\delta+o_{\mathrm{disc}(\mathscr{P})\to\infty}(1)$$

for some $\delta>0$. Such a bound implies tightness and hence rules out loss of mass.

Proof. Given $\varepsilon>0$, choose $\eta>0$ so small that $C_\varepsilon\eta^\delta<\varepsilon/2$, and then take discriminant large enough that the error is $<\varepsilon/2$. The complement of the set $\{\lambda_1<\eta\}$ lies in a compact Mahler set. Finitely many remaining packet measures can be absorbed by enlarging that compact set. Lemma 1.1 then gives tightness. □

> **Failure mode.** No theorem in this chapter claims that every sequence of large-discriminant torus packets satisfies the hypothesis automatically. The point is to isolate the exact arithmetic cusp estimate that Volume 21 must verify in the cubic setting. This avoids the false inference "large discriminant $\Rightarrow$ no escape."

Worked Example 1.4 (A model moment estimate)**.** Suppose a family satisfies $\mu_j\{\lambda_1 < \eta\} \leq 10\eta^{1/2}$ for every $0 < \eta < 1$. Then choosing $\eta = (\varepsilon/10)^2$ produces a compact Mahler set carrying mass at least $1 - \varepsilon$ for every j. The entire nonescape problem has been reduced to the arithmetic derivation of the displayed cusp-tail estimate.

CHAPTER 8

The Torus-Packet Equidistribution Framework

1. From separation to entropy

Fix a regular element $a \in H$. A Bowen ball of length N is the set of points whose first N iterates under right translation by a remain inside a fixed small metric ball. In unstable directions its transverse radius is exponentially small in N.

Lemma 1.1 (Linnik principle in the PGL_n packet framework)**.** *Let $x \in \mathrm{Lie}(H)$ be regular, write $a(t) = \exp(tx)$, and set*

$$\kappa(x) = \min_{\alpha \in \Phi} |\alpha(x)| > 0.$$

Let Y_j be finite unions of periodic H-orbits in X_n such that

$$\Delta_j := \max_{yH \subset Y_j} \mathrm{disc}(yH) \longrightarrow \infty, \qquad \mathrm{vol}(Y_j) \geq \Delta_j^{\rho}$$

for a fixed $\rho > 0$. Let μ_j be normalized volume measure on Y_j and suppose $\mu_j \to \mu$ weak- as probability measures. Then*

$$h_\mu(a(1)) \geq \frac{\rho\kappa(x)}{2}.$$

If every component orbit in Y_j has the common discriminant Δ_j, then

$$h_\mu(a(1)) \geq \rho\kappa(x).$$

Proof. We first prove the geometric tube bound and then convert it to entropy.

Fix a compact $\Omega \subset X_n$ and choose a compact lift $\widetilde{\Omega} \subset G$. Let $B_H \subset H$ be a small compact neighborhood of the identity and let $B_R = \exp(V)$, where V is a sufficiently small compact neighborhood of 0 in the sum of the nonzero root spaces for $\mathfrak{h}$. Put $B = B_R B_H$. For $t > 0$ define

$$B^{(-t,t)} = a(t)Ba(-t) \cap a(-t)Ba(t).$$

Conjugation multiplies a root vector in $\mathfrak{g}_\alpha$ by $e^{t\alpha(x)}$. Hence, after shrinking B once,

$$B^{(-t,t)} \subset B^G_{Ce^{-\kappa(x)t}}(e)\, B_H. \tag{20.8.1}$$

For the general discriminant hypothesis choose

$$t_j = \kappa(x)^{-1}\log\Delta_j + A,$$

where A is large enough in terms of B and Ω. If two points of $Y_j \cap zB^{(-t_j,t_j)}$, with $z \in \Omega$, came from distinct local Cartan branches, (20.8.1) would put representatives of those branches at distance $O(e^{-\kappa t_j}) < c_\Omega \Delta_j^{-1}$, contradicting Lemma 1.1. Thus the intersection lies in at most $|N_G(H)/H| = |S_n|$ local H-pieces. Each such piece is contained in a translate of one fixed H-box of bounded Haar volume. Since the total unnormalized volume of Y_j is at least Δ_j^{ρ},

$$\mu_j\big(zB^{(-t_j,t_j)}\big) \ll_\Omega \Delta_j^{-\rho} \ll_\Omega e^{-\rho\kappa(x)t_j}. \tag{20.8.2}$$

If all discriminants equal Δ_j, Lemma 1.1 gives the stronger scale $\Delta_j^{-1/2}$, so we may instead take $t_j = (2\kappa(x))^{-1}\log\Delta_j + A$ and obtain

$$\mu_j\big(zB^{(-t_j,t_j)}\big) \ll_\Omega e^{-2\rho\kappa(x)t_j}. \tag{20.8.3}$$

It remains to justify the entropy conversion. We record the argument because this is the point at which noncompactness can otherwise be hidden. Suppose generally that $\nu_j \to \nu$ are $a(t)$-invariant probabilities and that for every compact Ω there is a neighborhood B with

$$\nu_j(zB^{(-t_j,t_j)}) \le C_\Omega e^{-2\eta t_j} \qquad (z \in \Omega), \quad t_j \to \infty. \tag{20.8.4}$$

Choose a compact Ω whose interior has ν-mass $> 1 - \varepsilon^2/200$. For large j the same is true with ν_j up to $o(1)$. The maximal ergodic inequality applied to the indicator of Ω^c shows that, outside a set of mass $O(\varepsilon)$, an orbit segment of length $2N+1$ spends at most an ε-fraction of its times outside Ω.

Choose a finite ν-regular partition $\mathcal{P}$ whose atoms inside Ω have diameter so small that agreeing with a point at all good times forces the two points, on each uninterrupted good block, into a translate of $B^{(-m,m)}$. A name with at most $2\varepsilon N$ bad positions is described by: the set of bad positions, the finitely many bad symbols, and one tube choice for each good block. The number of possible bad-position/symbol patterns is at most $\exp(O(\varepsilon|\log\varepsilon|N))$. By (20.8.4), the ν_j-mass of every such good-block tube is at most $Ce^{-2\eta m}$; multiplying over the good blocks gives an atom-mass bound

$$\exp\{-2\eta N + O(\varepsilon N) + O(1)\}$$

on the good set. The elementary inequality $H_\nu(\mathcal{Q}) \ge -\log(\max_Q \nu(Q))$ applied after separating the exceptional mass, together with the conditional-entropy calculus of Volume 12, yields

$$H_{\nu_j}(\mathcal{P}^{(-N,N)}) \ge 2\eta N - O(\varepsilon|\log\varepsilon|N) - O(1).$$

Because $\mathcal{P}$ is ν-regular, fixed finite-name entropies pass to the weak limit. Divide by $2N$, pass first $j \to \infty$ along $N = \lfloor t_j \rfloor$, then let $\varepsilon \downarrow 0$. The Kolmogorov–Sinai definition gives $h_\nu(a(1)) \geq \eta$.

Apply this criterion to (20.8.2) with $2\eta = \rho\kappa(x)$, and to (20.8.3) with $2\eta = 2\rho\kappa(x)$. This proves the two stated bounds. □

THEOREM 1.2 (Haar domination and the torus-packet equidistribution criterion). *Let $n \geq 3$ and let Y_j be packet measures as in Lemma 1.1. Assume that (μ_j) is tight and that*

$$\operatorname{vol}(Y_j) \geq \operatorname{disc}(Y_j)^\rho$$

for one fixed $\rho > 0$. Let μ be a weak- limit and suppose the positive-entropy ergodic components of μ lie in the scope of Theorem 0.2. Write*

$$h_{\max}(a) = h_{m_{X_n}}(a)$$

for Haar entropy of a regular $a \in H$.

If n is prime, then

$$\mu = \alpha m_{X_n} + (1-\alpha)\nu, \qquad \alpha \geq \frac{c(a)\rho}{h_{\max}(a)} > 0,$$

for an H-invariant probability ν and the constant $c(a)$ from Lemma 1.1. In particular every weak limit dominates a fixed positive multiple of Haar measure. If, in addition, every weak limit in the family is H-ergodic, then $\alpha = 1$ and the packet measures equidistribute to Haar.

For composite n, the same proof gives a positive total weight of positive-entropy algebraic components; full equidistribution follows once all proper algebraic components allowed by Theorem 0.2 are excluded and the weak limit is ergodic.

PROOF. Tightness makes every weak-* limit a probability. Lemma 1.1 gives

$$h_\mu(a) \geq c(a)\rho.$$

Let

$$\mu = \int \mu_\omega \, d\tau(\omega)$$

be the H-ergodic decomposition. Entropy is affine under ergodic decomposition, so

$$h_\mu(a) = \int h_{\mu_\omega}(a) \, d\tau(\omega).$$

Every component satisfies $0 \leq h_{\mu_\omega}(a) \leq h_{\max}(a)$ by the entropy-expansion bound of Theorem 1.2. Let E be the set of components with positive entropy. Then

$$c(a)\rho \leq h_\mu(a) \leq h_{\max}(a)\, \tau(E),$$

so $\tau(E) \geq c(a)\rho/h_{\max}(a)$.

When n is prime, Theorem 0.2 says every component in E is Haar. Since Haar measure is a single ergodic component, the integral over E is exactly αm_{X_n} with $\alpha = \tau(E)$; the complementary components combine to an H-invariant probability ν. This proves the domination statement and its quantitative lower bound.

If μ itself is H-ergodic, its ergodic decomposition has one atom. Positive entropy then puts that atom in E, hence $\mu = m_{X_n}$. If every subsequential weak limit is ergodic, every such limit is Haar, so the original sequence converges to Haar.

For composite n, Theorem 0.2 makes the components in E algebraic rather than automatically Haar. The same entropy estimate gives positive total algebraic weight. If all proper algebraic candidates are separately excluded and the weak limit is ergodic, the only remaining component is Haar. $\square$

Warning 1.3 (What Volume 20 does not prove)**.** The theorem is a machine, not the cubic Duke theorem. It does not prove the packet-volume lower bound, the cusp estimate, or the analytic exclusion of exceptional algebraic components for a concrete cubic family. Those are precisely the arithmetic inputs supplied in Volume 21 by period formulas, local harmonic analysis and subconvexity.

Worked Example 1.4 (Prime degree as a clean endpoint)**.** Suppose n is prime and a family of packets is tight with total packet volume at least $D^{0.01}$. Linnik's principle supplies a positive entropy lower bound for every weak limit. Its ergodic decomposition therefore contains positive-entropy components of positive total weight; the prime-degree clause of Theorem 0.2 identifies those components with Haar measure. If the weak limit is additionally H-ergodic, the framework forces the whole limit to be Haar. The hard arithmetic task is therefore visibly separated from the rigidity endpoint.

> **Proof and provenance.** Lemma 1.1 is the reconstructed mechanism of the "Linnik principle" in [Ein+09, Thm. 1.9 and §3]: discriminant gives separation, large total volume gives many distinguishable names, and entropy converts this into rigidity. The framework theorem deliberately stops before the unconditional cubic equidistribution theorem of [Ein+11].

Volume 21

Duke-Type Equidistribution via Higher-Rank Dynamics

Volume 21 scope and prerequisites

This volume proves the cubic Duke-type theorem by combining the arithmetic packet dictionary of Volume 20 with the higher-rank rigidity machinery of Volumes 12–16. The target is the real split cubic packet theorem of Einsiedler–Lindenstrauss–Michel–Venkatesh in the classical space

$$X_3 = \mathrm{PGL}_3(\mathbb{Z})\backslash \mathrm{PGL}_3(\mathbb{R}).$$

The volume is organized so that the arithmetic and dynamical engines remain auditable separately. Chapters 1–4 construct the adelic period problem and reduce the cusp and small-ball estimates to one explicit analytic branch. Chapter 5 records that branch at child-result level. Chapter 6 proves the intermediate-subgroup exclusion from the already-owned EKL theorem. Chapters 7–8 assemble the packet theorem and its arithmetic consequences.

Warning 0.5. Volume 21 is the proved source location for the cubic Duke-type packet theorem in the scope stated here. The analytic branch is fully internal at the AHD proof floor: A04 owns the spectral/Kuznetsov machinery, A05 the divisor transform, A06 the twisted-GL_2 Burgess estimate, A07 the second spectral transformation, and A10–A12 the local ELMV estimates. The resulting cusp and small-ball bounds feed the earlier higher-rank rigidity owners and complete the stated Volume-XXI theorem chain.

Proof and provenance. The source-parity authority is [Ein+11]. Its Theorem 1.4 is the classical cubic packet theorem, Theorems 4.8–4.9 are the adelic volume and homogeneous-limit theorems, Proposition 11.3 is the Eisenstein-period estimate under subconvexity, and Lemmas 13.2–13.4 are the small-ball and cusp estimates. The level-aspect subconvex input used for cubic Dedekind zeta-functions is compared with [BHM07]. These papers control theorem strength and proof architecture; they are not silently treated as proofs.

Student orientation: where the difficulty really lies

There are three logically different operations:

$$\boxed{\text{arithmetic packet}} \longrightarrow \boxed{\text{analytic packet-mass bounds}}$$
$$\longrightarrow \boxed{\text{higher-rank measure rigidity}}.$$

Volume 20 already constructed the first box. Volumes 12–16 already constructed the third. The genuinely new burden of the present volume is the middle box: the packet must be shown not to hide in the cusp or in tiny algebraic tubes. For cubic fields that estimate ultimately rests on a power-saving bound for a degree-three Dedekind zeta-function together with local harmonic analysis. Once those estimates are available, the dynamical endgame is comparatively short.

Reader guide: a concrete model

Why this matters.

Volume 21 is organized around the passage from *Duke-Type Arithmetic Prototypes* to *Arithmetic Distribution Consequences*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.6 (A calculation to keep in view)**.** Suppose a packet consists of three closed torus orbits with normalized invariant measures μ_1, μ_2, μ_3 and arithmetic weights $2, 3, 5$. Its normalized packet measure is

$$\mu_{\text{pkt}} = \frac{2\mu_1 + 3\mu_2 + 5\mu_3}{10}.$$

For every test function f,

$$\mu_{\text{pkt}}(f) = \frac{2\mu_1(f) + 3\mu_2(f) + 5\mu_3(f)}{10}.$$

The calculation is trivial, but it isolates the logical point of the volume: equidistribution of a packet is a statement about this weighted probability measure, not automatically about each individual orbit.

CHAPTER 1

Duke-Type Arithmetic Prototypes

1. The exact family to be equidistributed

Let $K/\mathbb{Q}$ be a totally real cubic field, let $\mathcal{O} \subset K$ be an order, and let $[L]$ be a homothety class of full $\mathcal{O}$-lattices with multiplier order $\mathcal{O}$. Choosing an ordering of the three real embeddings sends L to a lattice in $\mathbb{R}^3$ and hence to a point of X_3. The positive diagonal torus acts through the logarithms of totally positive units.

Lemma 1.1 (Weyl reduction for packet measures)**.** *Let X be a second-countable locally compact space with probability m, and let μ_j be probability measures. Suppose the family is tight and that*

$$\int f\, d\mu_j \longrightarrow \int f\, dm$$

for every f in a $*$-subalgebra $\mathscr{A} \subset C_c(X)$ that is uniformly dense on every compact subset of X. Then $\mu_j \Rightarrow m$ weak-.*

Proof. Fix $f \in C_c(X)$ and $\varepsilon > 0$. Tightness gives a compact C carrying mass at least $1-\varepsilon$ for all sufficiently large j and for m. Enlarge C so that $\operatorname{supp} f \subset C$. Choose $a \in \mathscr{A}$ with $\|a - f\|_C < \varepsilon$ and with support in a fixed compact neighborhood of C; this is possible by the assumed local uniform density and a cutoff from $\mathscr{A}$. Then

$$\left|\int f\, d\mu_j - \int f\, dm\right| \le 2\varepsilon + \left|\int a\, d\mu_j - \int a\, dm\right| + O_f(\varepsilon).$$

The middle term tends to zero. Letting $\varepsilon \downarrow 0$ proves convergence on $C_c(X)$, which is weak-* convergence of probabilities. $\square$

Theorem 1.2 (Classical cubic packet theorem is equivalent to the adelic packet theorem). *Let $\mathscr{P}_j$ be packets of compact H-orbits in X_3 arising from totally real cubic orders, with common packet discriminant $D_j \to \infty$. Let Y_j be the corresponding homogeneous toral sets in $\mathrm{PGL}_3(\mathbb{Q})\backslash \mathrm{PGL}_3(\mathbb{A})$ and μ_{Y_j} their normalized invariant measures. If every weak-* limit of μ_{Y_j} is invariant under the image of $\mathrm{SL}_3(\mathbb{A})$, then the normalized classical packet measures on X_3 converge to Haar probability. Conversely, the classical assertion for all congruence levels and all finite local translates determines the adelic assertion.*

Proof. Volume 20, Theorem 1.2, identifies a classical packet with the real projection of one adelic homogeneous toral set, with finite fibers accounted for by a compact determinant-class quotient. If an adelic limit is $\mathrm{SL}_3(\mathbb{A})$-invariant, its real projection is invariant under $\mathrm{SL}_3(\mathbb{R})$; the latter acts transitively on the connected determinant component and its invariant probability is Haar. This proves the forward implication.

For the converse, take a factorizable test function $F = F_\infty F_f$ with F_f locally constant and compactly supported. After shrinking the finite compact open stabilizer of F_f, the adelic integral is a finite linear combination of integrals of F_∞ against packet measures on congruence quotients and finite local translates. The assumed classical convergence gives the Haar value term by term. Such factorizable functions are dense in C_c of the adelic quotient, so Lemma 1.1 gives the adelic assertion. ☐

Proposition 1.3 (Exact-volume families reduce to packets). *Let $\mathcal{Y}(V)$ denote the union of all compact maximal-flat orbits in X_3 having regulator volume exactly V. Suppose each nonempty packet of discriminant tending to infinity equidistributes and that, for every B, only finitely many packets have discriminant at most B. Then along any sequence $V_j \to \infty$ for which $\mathcal{Y}(V_j) \neq \varnothing$, the normalized measure on $\mathcal{Y}(V_j)$ converges to Haar.*

Proof. Decompose $\mathcal{Y}(V_j)$ into its finitely many packets. If a positive proportion of its mass were carried by packets of bounded discriminant, a subsequence would repeatedly use one of finitely many bounded-discriminant packets. But packet volume is fixed inside a packet and Volume 20 gives $\mathrm{vol}(\mathscr{P}) = \mathrm{disc}(\mathscr{P})^{1/2+o(1)}$; hence bounded discriminant forces bounded volume, contradicting $V_j \to \infty$. Thus the minimum discriminant among packets carrying nonnegligible mass tends to infinity. Equidistribution of each such packet, uniformly after grouping a finite convex combination, yields the Haar limit. ☐

Worked Example 1.4 (Why one packet is stronger than one orbit)**.** For a maximal cubic order $\mathcal{O}_K$, the packet is a torsor under $\mathrm{Pic}(\mathcal{O}_K)$. A single ideal class gives one compact flat. Averaging over the whole class group is an arithmetic amplification: the measure has enough total volume for the entropy argument even when no individual flat is known to be large enough.

CHAPTER 2

Adelic Homogeneous Parametrization

1. Removing determinant bookkeeping

The analytic argument is naturally written on GL_3 while the dynamics is naturally written on PGL_3. We therefore isolate the finite-cover passage once.

Lemma 1.1 (Finite determinant cover does not change equidistribution)**.** *Let $p : \widetilde{X} \to X$ be a finite proper equivariant covering of homogeneous spaces, with deck group C finite. Let $\widetilde{\mu}_j$ be C-invariant probabilities and $\mu_j = p_*\widetilde{\mu}_j$. Then $\widetilde{\mu}_j$ converges to Haar on $\widetilde{X}$ if and only if μ_j converges to Haar on X and the masses of the finitely many deck components converge to their Haar proportions.*

Proof. The forward direction is immediate under pushforward. Conversely, decompose a test function F into its average $F^C = |C|^{-1}\sum_{c\in C} F(c\cdot)$ and a function with zero average on every fiber. C-invariance of $\widetilde{\mu}_j$ kills the latter. The averaged function descends to f on X, and its integral converges by the assumed base equidistribution. Component masses handle the possibility that the cover is disconnected. This proves convergence on $C_c(\widetilde{X})$. □

Theorem 1.2 (Cubic order data as a homogeneous toral set). *Let $K/\mathbb{Q}$ be totally real cubic, let $L \subset K$ be a full lattice with multiplier order $\mathcal{O}_L$, and choose a $\mathbb{Q}$-linear identification $\iota : K \simeq \mathbb{Q}^3$. There exists $g_D \in$ $\mathrm{GL}_3(\mathbb{A})$, unique up to left multiplication by $K_{\mathbb{A}}^{\times}$ and right multiplication by $\mathrm{GL}_3(\widehat{\mathbb{Z}})\,\mathrm{GL}_3(\mathbb{R})^+$, such that*

$$L \otimes \widehat{\mathbb{Z}} = \iota^{-1}(\widehat{\mathbb{Z}}^3 g_{D,f}^{-1}).$$

The image

$$Y_D = \mathrm{GL}_3(\mathbb{Q})\backslash \mathrm{GL}_3(\mathbb{Q})\,\iota(K_{\mathbb{A}}^{\times})\,g_D$$

projects to the packet of periodic diagonal orbits attached to the local-homothety class of L.

Proof. At each finite prime p, choose a $\mathbb{Q}_p$-linear identification carrying $L \otimes \mathbb{Z}_p$ to $\mathbb{Z}_p^3 g_{D,p}^{-1}$; almost all choices are integral, so the tuple defines an adele. Two choices differ by multiplication on the left by $K_p^\times$ and on the right by $\mathrm{GL}_3(\mathbb{Z}_p)$. At infinity choose the ordered real embeddings. This proves existence and the stated uniqueness.

A point of the adelic torus orbit differs from g_D by an idele $t \in K_{\mathbb{A}}^\times$. Projecting to the real quotient and absorbing the finite component by a rational element whenever possible changes L by a lattice locally homothetic to it. Conversely, if L' is locally homothetic to L, choose local multipliers $t_p \in K_p^\times$ with $L'_p = t_p L_p$; the idele (t_p) realizes the corresponding real orbit. This is exactly the packet relation of Volume 20. □

Proposition 1.3 (Discriminant and volume are adelically invariant packet data)**.** *For Y_D as above, the product of local discriminants equals the discriminant of the associated order up to the fixed normalization determined by ι, and the adelic torus volume equals the sum of the real orbit volumes in the projected packet. Both quantities are unchanged by replacing g_D inside its adelic torus orbit.*

Proof. The discriminant is defined by the integral norm of the wedge tensor of a basis of the torus Lie algebra. Left multiplication by the torus fixes its Lie algebra under conjugation, while rational changes obey the product formula; hence the product of local norms is invariant. In the field model the wedge determinant is the trace discriminant of the multiplier order, giving the first assertion.

For volume, choose compatible Tamagawa/Haar normalizations. Decompose the adelic torus quotient into the finitely many double cosets of its finite part modulo the maximal compact open. Fubini decomposes its Haar measure into the sum of Haar measures of the corresponding real stabilizer quotients. These are precisely the real orbit volumes. Translation by the adelic torus preserves Haar measure, proving invariance. □

CHAPTER 3

Packet Measures and Local Components

1. Local data

Fix a local field k, an etale cubic k-algebra A, and an embedding $A \hookrightarrow M_3(k)$. The local order is $\Lambda = A \cap M_3(\mathcal{O}_k)$ in the nonarchimedean case. A local torus packet is controlled by the position of this order in the standard lattice.

Lemma 1.1 (Factorization of the packet discriminant)**.** *For global cubic torus data D with localizations D_v,*

$$\mathrm{disc}(D) = \prod_v \mathrm{disc}_v(D_v),$$

where all but finitely many factors are 1*. If D comes from an order $\mathcal{O} \subset K$, then this product is $|\,\mathrm{disc}(\mathcal{O})|$ up to the fixed ground-field normalization.*

Proof. Choose a rational basis X_1, X_2 of the trace-zero part of the torus Lie algebra and form $w = (X_1 \wedge X_2)^{\otimes 2}$. By definition the local discriminant is the inverse local norm of the integral primitive representative of the line $\mathbb{Q}_v w$. For almost every finite v, the basis is integral and primitive, so the local factor is 1. Multiplying local denominators uses the product formula and gives the global integral denominator of w. In the regular representation this denominator is the determinant of the trace pairing on the multiplier order, namely its arithmetic discriminant. □

Theorem 1.2 (Local-component decomposition of packet measure). *Let S contain ∞ and every prime at which the cubic torus data is ramified. The projection of Y_D to*

$$X_S = \mathrm{PGL}_3(\mathcal{O}_S) \backslash \mathrm{PGL}_3(\mathbb{Q}_S)$$

is a finite union of orbits of the local torus $T_D(\mathbb{Q}_S)$. Normalized adelic torus measure projects to the volume-weighted average of these orbit measures. Enlarging S merely refines the finite decomposition and does not alter the projected probability.

Proof. Apply strong approximation to the simply connected cover SL_3 and then quotient by the finite determinant-class obstruction. The finite adelic quotient outside S breaks into finitely many double cosets

$$T_D(\mathbb{A}^S)\backslash \mathrm{PGL}_3(\mathbb{A}^S)/K^S.$$

Choose one representative for each occupied double coset. Fubini on the torus quotient gives, for every compactly supported test function, the sum of its integrals over the corresponding S-arithmetic torus orbits, weighted by stabilizer covolume. Dividing by total torus volume gives the claimed probability. Enlarging S partitions each old double coset into finitely many new ones; additivity of Haar measure shows that the total projection is unchanged. □

Proposition 1.3 (Local split-place alternative)**.** *For a sequence of cubic packet data D_j with $\mathrm{disc}(D_j) \to \infty$, after passing to a subsequence and fixing a place v, either*

(1) $\mathrm{disc}_v(D_j) \to \infty$*, or*

(2) $\mathrm{disc}_v(D_j)$ *stays bounded and the local tori $T_j(\mathbb{Q}_v)$ range in a compact family of split maximal tori whenever each K_j is split at v.*

In the second case a subsequence of the local Lie algebras converges to a split Cartan subalgebra.

Proof. The first alternative is tautological. In the second, bounded local discriminant bounds the distance in the Bruhat–Tits building (or in the archimedean symmetric space) between the standard integral norm and the apartment determined by the torus; this is the local discriminant-versus-distance estimate proved in the local packet construction of Volume 20. Modulo the compact stabilizer of the standard norm, the corresponding embeddings therefore lie in a compact set. The Grassmannian of two-dimensional subspaces of $\mathfrak{pgl}_3(\mathbb{Q}_v)$ is compact on that bounded set, so a subsequence converges. Splitness is closed within a compact bounded-discriminant family of regular semisimple Cartans, yielding a split limiting Cartan. □

CHAPTER 4

Local–Global Factorization of Periods

1. Siegel Eisenstein series and Tate integrals

Let F be a number field, K/F a cubic extension and $\Psi \in \mathcal{S}(\mathbb{A}_F^3)$. Write $E_\Psi(\chi, s, g)$ for the standard maximal-parabolic Eisenstein series normalized so that unfolding is compatible with Tate's zeta integral.

Lemma 1.1 (Unfolding the torus period)**.** *Let $D = (K \hookrightarrow M_3(F), g_D)$ be global cubic torus data and let ψ be a unitary idele-class character of K, with $\chi = \psi|_{\mathbb{A}_F^\times}$. In the half-plane of absolute convergence,*

$$\int_{T_K(F)\backslash T_K(\mathbb{A}_F)} \psi(t) E_\Psi(\chi, s, tg_D)\, dt = c(D, s) \int_{\mathbb{A}_K^\times} \Psi_K(u)\psi(u)|u|_K^{s/3}\, d^\times u,$$

where $\Psi_K(u) = \Psi(\iota(u)g_D)$ and $c(D, s)$ is the explicit determinant normalization forced by the chosen Haar measures.

Proof. Write the Eisenstein series as the sum over nonzero F-lines in F^3 of the inducing section obtained from Ψ. Pull the sum through the torus integral in the region of absolute convergence. The torus $K^\times/F^\times$ acts transitively on the nonzero F-lines generated by elements of K after the regular-representation identification; stabilizers are exactly $F^\times$. Thus the sum over lines and the quotient integral unfold to an integral over $K^\times\backslash\mathbb{A}_K^\times$ followed by the $K^\times$-sum, hence to the full idele integral. The change of variables through ι contributes $|\det g_D|^{s/3}$ and the fixed discriminant factor determined by the self-dual measures. This is $c(D, s)$. ☐

Theorem 1.2 (Euler factorization and functional equation of the cubic torus period). *With the hypotheses of Lemma 1.1, suppose $\Psi = \otimes_v \Psi_v$ and $\psi = \otimes_v \psi_v$. Then the unfolded period continues meromorphically and factors as*

$$\mathcal{P}_D(\Psi, \psi, s) = c(D, s)L(K, \psi, s/3) \prod_{v\in S} \mathcal{I}_v(D_v, \Psi_v, \psi_v, s),$$

where S contains the ramified and nonstandard places and $\mathcal{I}_v = 1$ outside S. The product satisfies the functional equation obtained by replacing Ψ_v with its local Fourier transform and s by $3 - s$.

Proof. The idele integral of Lemma 1.1 is a Tate zeta integral on K. For a pure tensor it is the product of local zeta integrals. At an unramified place with Ψ_v the characteristic function of the maximal order and ψ_v unramified, direct summation over valuation shells gives the Euler factor

$$\sum_{m\geq 0} \psi_v(\varpi)^m q_v^{-ms/3} = L_v(K, \psi, s/3).$$

At the finitely many remaining places divide the local zeta integral by that Euler factor; the quotient is $\mathcal{I}_v$.

For meromorphic continuation and the functional equation, apply Poisson summation on the additive space $K\backslash\mathbb{A}_K$ to the theta series associated with Ψ_K. Splitting the idele integral at norm one and changing $u \mapsto u^{-1}$ on the large-norm part yields the local gamma factors and the relation $s \leftrightarrow 3 - s$. This is Tate's proof written for the K-Schwartz function obtained from the torus embedding. No automorphic spectral theorem is used at this stage. □

Proposition 1.3 (Analytic reduction for cusp and small-ball tests)**.** *Fix a finite S and a compact family of normalized test functions used to majorize cusp sets and small metric balls in X_S. Suppose there exist $\theta > 0$ and A such that, uniformly for the cubic fields and idele-class characters arising from the packets,*

$$|L(K, \psi, 1/2 + it)| \ll (1 + |t|)^A \big(\operatorname{disc}(K) \operatorname{disc}(\psi)\big)^{1/4-\theta}, \tag{21.4.1}$$

and suppose the normalized local factors satisfy a power saving

$$|\mathcal{I}_v| \ll \operatorname{disc}_v(D)^{-\eta} \operatorname{disc}_v(\psi)^{-\eta} \tag{21.4.2}$$

whenever the local conductor or local torus discriminant grows. Then the packet integrals of all such Eisenstein majorants have a uniform power saving in the global packet discriminant.

Proof. Insert the factorization from Theorem 1.2. Shift the Mellin contour of the Eisenstein majorant to the unitary axis. The poles contribute precisely the Haar main term; after subtracting that main term, the remaining integral is over $\Re s = 1/2$ and is rapidly decaying in the archimedean parameter because the test functions range in a fixed Schwartz family. Apply (21.4.1) to the global L-factor and (21.4.2) at the bad places. At unramified places the normalized factor is 1. The conductor-discriminant product formula then gives $D^{-\beta}$ for some $\beta = \beta(\theta, \eta) > 0$. The polynomial archimedean loss is absorbed by the rapid decay of the Mellin transform. □

Guiding Idea 1.4. This chapter is the exact point where the analytic and dynamical languages meet. Everything before (21.4.1) is an unfolding/factorization theorem. The power saving itself is not produced by unfolding: it is the deep analytic input isolated in Chapter 5.

CHAPTER 5

The Cubic Analytic Branch and Entropy Input

1. Why the analytic branch is expanded explicitly

The cubic packet theorem needs two genuinely quantitative analytic inputs. The first is a global power saving for the Dedekind zeta function of a cubic field. The second is the local torus-period saving that converts local discriminant or character conductor into decay of a normalized local period. Neither is a formal consequence of higher-rank dynamics. We therefore expose the proof chain below rather than compressing it into the phrases "subconvexity" or "local harmonic analysis."

> **Proof and provenance.** The source architecture is compared with [BHM07] and [Ein+11]. The BHM paper proves the level-aspect GL_2 estimate by approximate functional equation and amplification, a first Kuznetsov/Voronoi transformation, evaluation of the main term, transformation of the remaining Kloosterman family, a second spectral decomposition, and a final optimization. It also uses a Burgess-type twisted-GL_2 bound in the small-spectrum range. The ELMV paper proves the local estimate through the geometry of the building, matrix-coefficient decay and a local functional equation. Those steps are represented individually in the proof decomposition at the end of this chapter.

2. The global cubic reduction

Lemma 2.1 (Artin factorization for a non-Galois cubic field). *Let $K/\mathbb{Q}$ be a non-Galois cubic field, let L be its Galois closure with group S_3, and let F be the unique quadratic subfield. Then*

$$\zeta_K(s) = \zeta(s)L(s,\rho),$$

where ρ is the two-dimensional irreducible representation of S_3. Moreover ρ is induced from either nontrivial cubic character of $\mathrm{Gal}(L/F)$, so its Artin L-function is the L-function of a dihedral $\mathrm{GL}_2/\mathbb{Q}$ automorphic representation.

PROOF. The permutation representation of S_3 on the three embeddings of K is $\mathbf{1} \oplus \rho$, hence Artin formalism gives the displayed factorization. Restriction of ρ to $A_3 = \mathrm{Gal}(L/F)$ is the direct sum of the two nontrivial cubic characters. Frobenius reciprocity therefore gives $\rho \simeq \mathrm{Ind}_{G_F}^{G_{\mathbb{Q}}} \vartheta$ for either of them. Automorphic induction of the corresponding finite-order Hecke character of F produces the dihedral representation, and equality of the local induced Weil representations gives equality of the local L-factors at every place. $\square$

Lemma 2.2 (Cubic Dedekind zeta from level-aspect GL_2 subconvexity)**.** *Suppose the dihedral forms above satisfy, for $\Re s = 1/2$,*

$$L(f, s) \ll_{s,\varepsilon} q^{1/4-\delta+\varepsilon} \tag{21.5.1}$$

with one absolute $\delta > 0$. Then for some $\delta' > 0$ and every cubic field K,

$$\zeta_K(1/2 + it) \ll_{t,\varepsilon} |D_K|^{1/4-\delta'+\varepsilon}. \tag{21.5.2}$$

PROOF. For non-Galois K, Lemma 2.1 gives $\zeta_K = \zeta L(f, \cdot)$. The conductor of the dihedral representation is the Artin conductor of ρ and is comparable to $|D_K|$ up to bounded local factors in fixed degree. Absorb those factors in D_K^{ε} and use the polynomial critical-line bound for ζ. If $K/\mathbb{Q}$ is cyclic, ζ_K is ζ times two conjugate cubic Dirichlet L-functions; the Burgess theorem already proved in Volume 19 gives the required power saving. $\square$

3. The part of the BHM argument already internalized

Lemma 3.1 (Symmetric approximate functional equation for the square)**.** *Let f be a primitive level-q cusp form of nebentypus χ, and let $s = 1/2 + it$. There are smooth weights V_+, V_-, depending polynomially on the archimedean parameters, such that*

$$L(f, s)^2 = \sum_{n \geq 1} \frac{\tau(n)\lambda_f(n)}{\sqrt{n}} V_+(n/q) + \eta(f, s) \sum_{n \geq 1} \frac{\tau(n)\lambda_f(n)}{\sqrt{n}} V_-(n/q), \tag{21.5.3}$$

$|\eta(f, s)| = 1$, and for every $j, A \geq 0$,

$$x^j V_{\pm}^{(j)}(x) \ll_{j,A} (1 + |t| + |t_f|)^{C_{j,A}} (1 + x)^{-A}.$$

The weights may be chosen to vanish at the Eisenstein poles that occur in the later trace formula.

PROOF. Write the completed square $\Lambda(f, u)^2$ using the Euler identity $L(f, u)^2 = L(2u, \chi) \sum_n \tau(n)\lambda_f(n)n^{-u}$ for $\Re u > 1$. Multiply by an even entire rapidly decreasing factor $P(z)$ with $P(0) = 1$ and by an even polynomial $Q(z, t_f)$ whose prescribed zeros cancel the two Eisenstein residues. Integrate

$$\Lambda(f, s+z)^2 P(z) Q(z, t_f)\,\frac{dz}{z}$$

on $\Re z = 1$. Shift to $\Re z = -1$; the residue at $z = 0$ is $\Lambda(f, s)^2$, while the functional equation identifies the shifted integral with the dual sum. Expanding the Dirichlet series on both vertical lines and applying Mellin inversion gives (21.5.3). Stirling's formula on vertical strips gives rapid decay of the Mellin kernels and, after differentiation under the integral, the stated derivative bounds. Because Q contains the two required linear factors, the Mellin transform vanishes at the Eisenstein pole locations. □

Lemma 3.2 (Prime–prime-square amplifier lower bound)**.** *Let f_0 be the target primitive form. For primes p in $[L^{1/2}/2, L^{1/2}]$ with $p \nmid q$, choose amplifier coefficients supported on p, p^2 by*

$$x(p) = \overline{\chi(p)}\lambda_{f_0}(p), \qquad x(p^2) = -\overline{\chi(p)}.$$

Then

$$\left|\sum_{\ell} x(\ell)\lambda_{f_0}(\ell)\right| \gg_\varepsilon L^{1/2-\varepsilon}. \tag{21.5.4}$$

Proof. For $p \nmid q$, the Hecke relation is $\lambda_{f_0}(p)^2 - \chi(p)\lambda_{f_0}(p^2) = 1$ in the unitary normalization. Hence the contribution of the pair p, p^2 is 1 after multiplication by the displayed phase. Summing over the available primes gives $\gg L^{1/2}/\log L$; absorb the logarithm in L^ε. □

4. Child Root A04: trace formula normalization

The level-aspect argument needs a trace formula with constants fixed once and for all. We use the normalization of Blomer–Harcos–Michel. Let $q \geq 1$, let $\chi \pmod q$ be even, and set

$$S_\chi(m, n; c) := \sum_{\substack{d \,(\mathrm{mod}\ c)\\ (d,c)=1}} \chi(d) e\left(\frac{md + n\bar{d}}{c}\right), \qquad q \mid c.$$

For a holomorphic cusp form of weight k write $f(z) = \sum_{n\geq 1} \rho_f(n) n^{k/2} e(nz)$; for a weight-zero Maaß form write

$$f(z) = \sum_{n\neq 0} \rho_f(n) W_{0,it_f}(4\pi|n|y) e(nx).$$

The complex conjugate on the first Fourier coefficient in the formulas below is important when the nebentypus is nonreal.

4.1. The holomorphic trace formula is local and elementary.

Lemma 4.1 (Fourier expansion of the holomorphic Poincaré series)**.** *Let $k > 2$ have the parity of χ. With*

$$P_{m,k}(z) := \sum_{\gamma\in\Gamma_\infty\backslash\Gamma_0(q)} \overline{\chi(d_\gamma)}\, j(\gamma,z)^{-k} e(m\gamma z), \qquad m \geq 1,$$

where $j(\gamma,z) = cz + d$, one has an absolutely convergent cusp form and

$$P_{m,k}(z) = \sum_{n\geq 1} p_{m,k}(n) e(nz),$$

with

$$p_{m,k}(n) = \delta_{mn} + 2\pi i^{-k} \sum_{\substack{c\geq 1\\ q|c}} \frac{S_\chi(m,n;c)}{c} J_{k-1}\left(\frac{4\pi\sqrt{mn}}{c}\right). \tag{21.A.1}$$

Proof. Absolute convergence for $k > 2$ follows from $|j(\gamma,z)|^{-k} \ll_y (c^2+d^2)^{-k/2}$ after choosing the usual representatives. To obtain the n-th Fourier coefficient, integrate over $x \in [0,1]$. The coset with $c = 0$ contributes δ_{mn}. For $c > 0$, choose the lower row (c,d) with $(c,d) = 1$, $q \mid c$, and write $a\bar{d} \equiv 1 \pmod{c}$. The translation parameter in the upper row unfolds the x-integral to the real line. After $u = x + d/c$, the remaining integral is

$$e\left(\frac{m\bar{d} + nd}{c}\right) \int_{-\infty}^{\infty} (u+iy)^{-k} \exp\left(-\frac{2\pi i m}{c^2(u+iy)} - 2\pi i n u\right) du.$$

Move the horizontal line to $\Im u = 0^+$, expand the first exponential, and evaluate the resulting Cauchy integrals. The coefficient of the r-th term vanishes unless $r \geq k-1$; putting $r = j + k - 1$ gives

$$2\pi i^{-k} J_{k-1}\left(\frac{4\pi\sqrt{mn}}{c}\right).$$

Summing the phases over $d \pmod{c}$ produces the displayed twisted Kloosterman sum. The Bessel series is locally uniformly convergent, so the termwise integrations above are justified first with an extra factor $e^{-\varepsilon c}$ and then by dominated convergence as $\varepsilon \downarrow 0$. □

Lemma 4.2 (Unfolded Petersson pairing)**.** *For every $f \in S_k(q,\chi)$,*

$$\langle f, P_{m,k}\rangle = \frac{\Gamma(k-1)}{(4\pi m)^{k-1}} m^{k/2} ho_f(m). \tag{21.A.2}$$

Consequently, if $\mathcal{B}_k(q,\chi)$ is an orthonormal basis, then

$$P_{m,k} = \frac{\Gamma(k-1)m^{1-k/2}}{(4\pi)^{k-1}} \sum_{f\in\mathcal{B}_k(q,\chi)} \overline{\rho_f(m)}\, f. \tag{21.A.3}$$

PROOF. Insert the Poincaré series into the Petersson inner product and unfold the fundamental domain to $\Gamma_\infty\backslash\mathbb{H}$. Integration in x selects the m-th Fourier coefficient of f, and the remaining integral is

$$m^{k/2}\rho_f(m)\int_0^\infty e^{-4\pi my}y^{k-2}\,dy = m^{k/2}\rho_f(m)\frac{\Gamma(k-1)}{(4\pi m)^{k-1}}.$$

This proves (21.A.2). Expand $P_{m,k}$ in the orthonormal basis and use conjugate-linearity in the first argument to obtain (21.A.3). □

THEOREM 4.3 (Petersson formula in the normalization used in XXI). *For positive integers m, n,*

$$\begin{aligned}&\delta_{mn} + 2\pi i^{-k}\sum_{\substack{c\geq 1\\ q\mid c}}\frac{S_\chi(m,n;c)}{c}J_{k-1}\left(\frac{4\pi\sqrt{mn}}{c}\right)\\ &\qquad = \frac{(k-2)!\sqrt{mn}}{(4\pi)^{k-1}}\sum_{f\in\mathcal{B}_k(q,\chi)}\overline{\rho_f(m)}\rho_f(n).\end{aligned}\tag{21.A.4}$$

PROOF. Take the n-th Fourier coefficient in the two expansions (21.A.1) and (21.A.3). Since $\Gamma(k-1) = (k-2)!$, the constants reduce exactly to (21.A.4). □

4.2. The Bessel transforms and the estimates actually consumed later. For a smooth $\phi : (0,\infty) \to \mathbb{C}$ put, in weight zero,

$$\begin{aligned}\dot{\phi}(k) &= i^k\int_0^\infty J_{k-1}(x)\phi(x)\frac{dx}{x},\\ \widetilde{\phi}(t) &= \frac{i}{2\sinh(\pi t)}\int_0^\infty (J_{2it}(x) - J_{-2it}(x))\phi(x)\frac{dx}{x},\\ \check{\phi}(t) &= \frac{2}{\pi}\cosh(\pi t)\int_0^\infty K_{2it}(x)\phi(x)\frac{dx}{x}.\end{aligned}\tag{21.A.5}$$

At $t = 0$ the first quotient is interpreted by continuity.

Lemma 4.4 (Differentiation identity for the Bessel kernels). *For $B_\nu = J_\nu, Y_\nu$, $(x^\nu B_\nu(x))' = x^\nu B_{\nu-1}(x)$, while for K_ν the same identity has a minus sign after replacing ν by $-\nu$ in the standard recurrence. In particular, if $F \in C_c^\infty(0,\infty)$, then for every $j \geq 0$*

$$\int_0^\infty F(x)B_0(\alpha\sqrt{x})\,dx = \left(\frac{\pm 2}{\alpha}\right)^j\int_0^\infty x^{j/2}F^{(j)}(x)B_j(\alpha\sqrt{x})\,dx.\tag{21.A.6}$$

Proof. The first statements are the elementary recurrences obtained by differentiating the power series for J_ν and the defining linear combinations for Y_ν, K_ν. Apply the chain rule to $B_j(\alpha\sqrt{x})$ and integrate by parts j times; compact support removes all boundary terms. □

Lemma 4.5 (Uniform transform bounds)**.** *Let ϕ be supported on $x \asymp X$.*

(1) If $\phi^{(j)}(x) \ll_j X^{-j}$, then for every $C \geq 0$,

$$|\dot{\phi}(k)| + |\widetilde{\phi}(t)| + |\check{\phi}(t)| \ll_C \frac{1+|\log X|}{1+X}\left(\frac{1+X}{1+|t|+k}\right)^C, \tag{21.A.7}$$

with the irrelevant term omitted when its spectral parameter is absent.

(2) If $\phi^{(j)}(x) \ll_j (X/Z)^{-j}$ and $|\Im t| < 1/4$, then

$$|\widetilde{\phi}(t)| + |\check{\phi}(t)| \ll \frac{1+(X/Z)^{-2|\Im t|}}{1+X/Z}. \tag{21.A.8}$$

(3) If $\phi(x) = e^{iax}\psi(x)$, $aX \geq 1$, and $\psi^{(j)}(x) \ll_j X^{-j}$, there is $1 \leq F \ll (a+1)X$ such that, for every C and $\varepsilon > 0$,

$$|\dot{\phi}(k)| + |\widetilde{\phi}(t)| + |\check{\phi}(t)| \ll_{C,\varepsilon} F^{-1+\varepsilon}\left(\frac{F}{1+|t|+k}\right)^C. \tag{21.A.9}$$

Proof. We give the argument because these bounds are repeatedly used in A07. For $x \leq 1$, the defining series give $J_\nu(x) \ll x^{\Re\nu}/|\Gamma(1+\nu)|$ and $K_{2it}(x) \ll (1+|\log x|)\cosh(\pi t)^{-1/2}$ after pairing the two $I_{\pm 2it}$-series. For $x \geq 1$, the integral representations and one stationary-phase step give

$$J_{2it}(x), Y_{2it}(x) \ll (1+x)^{-1/2}e^{\pi|t|/2}, \qquad K_{2it}(x) \ll (1+x)^{-1/2}e^{-\pi|t|/2}.$$

The hyperbolic prefactors in (21.A.5) cancel these exponential factors. This proves (21.A.7) for $C = 0$, including the logarithm at the origin.

For large $|t|+k$, use the Bessel differential equation

$$(x\partial_x)^2 B_\nu(x) + (x^2 - \nu^2)B_\nu(x) = 0.$$

Integrating its formally self-adjoint form against $\phi(x)dx/x$ transfers $(x\partial_x)^2 + x^2$ to ϕ. On $x \asymp X$ this costs $O((1+X)^2)$ and gains $1 + (|t|+k)^2$. Iteration proves arbitrary powers in (21.A.7).

For (21.A.8), use the small-argument expansions

$$J_{\pm 2it}(x) = \frac{(x/2)^{\pm 2it}}{\Gamma(1 \pm 2it)} + O(x^{2-2|\Im t|}),$$

and the analogous expansion for K_{2it}. Integrating a monomial $x^{\pm 2it}$ over an interval of length X/Z, followed by one integration by parts outside that interval, gives exactly the two terms in the numerator and the denominator $1 + X/Z$.

For (21.A.9), use the large-argument phase expansions $B_\nu(x) = x^{-1/2}\sum_\pm e^{\pm ix} a_{\pm,\nu}(x)$, with $x^j a^{(j)}_{\pm,\nu}(x) \ll_j (1+|t|+k)^{2j}$ in the transition-free range. Unless one of the phases $ax \pm x$ has derivative $\ll X^{-1}$, integration by parts gives arbitrary saving. In the remaining transition range, partition into intervals on which the phase has second derivative of one sign and apply the elementary van der Corput second-derivative estimate. Taking F to be the reciprocal of the resulting normalized phase width yields $1 \le F \ll (a+1)X$ and the base saving $F^{-1+\varepsilon}$. Applying the Bessel differential equation as above supplies the arbitrary spectral decay factor. This is (21.A.9). □

4.3. A04.1: spectral resolution on a finite-area congruence surface. We now close the part of the trace formula that was deliberately left open in Stage 2. The proof is included because completeness of the Maaß and Eisenstein spectra is a load-bearing statement in the second BHM transform. Throughout this subsection

$$Y = \Gamma_0(q)\backslash\mathbb{H}, \qquad \mathcal{H} = L^2(Y,\chi),$$

where the nebentypus is unitary. At a singular cusp the multiplier is trivial on the stabilizing translation; at a nonsingular cusp there is no constant Fourier mode and hence no continuous channel.

Lemma 4.6 (Closed Laplace form and self-adjoint realization)**.** *On $C_c^\infty(Y,\chi)$ put*

$$\mathfrak{q}(u,v) = \int_Y (y\partial_x u\,\overline{y\partial_x v} + y\partial_y u\,\overline{y\partial_y v})\,d\mu, \qquad d\mu = y^{-2}dx\,dy.$$

The completion for $\|u\|_{\mathfrak{q}}^2 = \|u\|_2^2 + \mathfrak{q}(u,u)$ is a Hilbert space. There is a unique nonnegative self-adjoint operator Δ such that

$$\mathfrak{q}(u,v) = \langle \Delta u, v\rangle \quad (u \in \mathcal{D}(\Delta),\ v \in \mathcal{D}(\mathfrak{q})). \tag{21.A.10}$$

On smooth vectors it is $-y^2(\partial_x^2 + \partial_y^2)$.

Proof. If (u_j) is Cauchy for $\|\cdot\|_{\mathfrak{q}}$, then u_j and the two weak derivatives $y\partial_x u_j, y\partial_y u_j$ are Cauchy in L^2. Their limits satisfy the distributional integration-by-parts identities because those identities hold against every compactly supported test function and pass to the limit. Thus the form is closed.

For completeness, the representation theorem for this particular closed form can be proved without quoting an unbounded-operator theorem. Given $f \in \mathcal{H}$, Riesz representation in the form Hilbert space gives a unique u with

$$\langle u, v\rangle + \mathfrak{q}(u,v) = \langle f, v\rangle \quad (v \in \mathcal{D}(\mathfrak{q})).$$

Define $Rf = u$. Then R is bounded, positive, self-adjoint and injective; its range is dense since $Rf \perp g$ implies $\langle f, Rg\rangle = 0$ for all f. Put $\Delta =$

$R^{-1} - 1$ on $\operatorname{Ran} R$. The symmetry of R gives self-adjointness of Δ, and the displayed variational identity gives (21.A.10). On compactly supported smooth functions integration by parts identifies the differential expression. □

Lemma 4.7 (Cusp normal form and the compact remainder)**.** *Choose disjoint cusp neighborhoods $\mathcal{C}_a(Y_0)$ at height Y_0. At a singular cusp a, after scaling its width to one,*

$$u(x+iy) = u_{a,0}(y) + \sum_{n\neq 0} u_{a,n}(y)e(nx).$$

Let P_0u equal the collection of constant terms $u_{a,0}$ on singular cusps and vanish on a fixed compact core. Then for every $M > 0$ the embedding

$$\{u \in \mathcal{D}(\mathfrak{q}) : \|u\|_{\mathfrak{q}} \leq M,\ P_0u = 0\} \hookrightarrow L^2(Y) \tag{21.A.11}$$

is compact.

Proof. On the compact core this is the elementary Rellich lemma: divide into finitely many Euclidean rectangles, expand in a sine/cosine basis and use that the derivative norm makes the Fourier tail uniformly small.

In a cusp, Parseval in x and the absence of the zero mode give

$$\int_{Y_0}^{\infty}\int_0^1 |u|^2 \frac{dx\,dy}{y^2} \leq \frac{1}{4\pi^2 Y_0^2}\int_{Y_0}^{\infty}\int_0^1 |y\partial_x u|^2 \frac{dx\,dy}{y^2}. \tag{21.A.12}$$

Hence the L^2 mass of the cuspidal tail is uniformly $O(Y_0^{-2})$ for a form-bounded family with $P_0u = 0$. Compactness on a truncated surface plus this uniform tail estimate proves (21.A.11). □

Lemma 4.8 (Constant-channel conjugation)**.** *For one singular cusp the map*

$$u_0(y) \longmapsto v(r) = e^{-r/2}u_0(e^r), \qquad r = \log y,$$

is unitary from $L^2((Y_0,\infty), dy/y^2)$ to $L^2((\log Y_0, \infty), dr)$, and

$$-y^2\frac{d^2}{dy^2} \quad\longmapsto\quad -\frac{d^2}{dr^2} + \frac{1}{4}. \tag{21.A.13}$$

Consequently the only noncompact part of the resolvent of Δ is a finite direct sum of half-line channels with threshold $1/4$.

Proof. The norm identity is immediate from $dy = e^r dr$. Differentiating $u_0(e^r) = e^{r/2}v(r)$ twice gives (21.A.13). Lemma 4.7 shows that the orthogonal complement of these channels has compact form embedding and hence compact resolvent. □

Lemma 4.9 (Analytic Fredholm reduction for the cusp problem)**.** *Let $K(s)$ be an analytic family of compact operators on a Hilbert space. If $1 - K(s_0)$ is invertible at one point, then $(1 - K(s))^{-1}$ is meromorphic, with finite-rank principal parts.*

Proof. Fix s_1. Compactness lets us split the Hilbert space as $E \oplus E^{\perp}$ with E finite dimensional so that $\|K(s_1)|_{E^{\perp}}\| < 1/3$. By continuity the same bound is $< 1/2$ on a small disk. Gaussian elimination of the block matrix $1 - K(s)$ therefore reduces invertibility to a finite-dimensional Schur complement on E. Its inverse is the adjugate divided by the determinant, hence meromorphic. Covering a path from s_0 and using uniqueness of inverses continues the meromorphic family. At each pole only the finite-dimensional Schur block contributes, so the principal part has finite rank. □

Proposition 4.10 (Eisenstein solutions and scattering matrix)**.** *For every singular cusp a there is a meromorphic family $E_a(z,s)$ solving*

$$(\Delta - s(1-s))E_a(\cdot, s) = 0$$

and having constant term at a singular cusp b

$$\delta_{ab} y^s + \Phi_{ba}(s) y^{1-s}. \tag{21.A.14}$$

The finite matrix $\Phi(s)$ is meromorphic and satisfies

$$\Phi(s)\Phi(1-s) = I, \qquad \Phi(\tfrac{1}{2} + it)^* \Phi(\tfrac{1}{2} + it) = I \tag{21.A.15}$$

away from its discrete pole set.

Proof. Cut off y^s in cusp a. Applying $\Delta - s(1-s)$ produces a compactly supported smooth error $F_a(s)$. On the complement of the constant cusp channels the resolvent is compact by Lemmas 4.7 –4.8; coupling the finitely many channels to the compact core gives an equation $(1 - K(s))w = F_a(s)$ with $K(s)$ compact and analytic. It is invertible for $\Re s > 1$, where the positive quadratic form precludes an L^2 homogeneous solution. Lemma 4.9 therefore gives a meromorphic correction w, and $E_a = \chi_a y^s - w$ has the required equation. Solving the constant-mode ODE (21.A.13) gives exactly the two powers in (21.A.14).

Apply Green's identity to truncated $E_a(\cdot, s)$ and $E_b(\cdot, \overline{w})$. The compact-core boundary cancels; insertion of the two constant terms gives the Maaß–Selberg relation

$$(s(1-s) - \overline{w(1-w)})\langle \Lambda^T E_a(s), \Lambda^T E_b(w)\rangle = \mathcal{B}_{ab}(T; s, w), \tag{21.A.16}$$

where $\mathcal{B}$ is the explicit Wronskian of $y^s + \Phi(s)y^{1-s}$ and $y^{\bar{w}} + \Phi(\bar{w})y^{1-\bar{w}}$ at $y = T$. First take $w = 1 - \bar{s}$ and compare the coefficients of the independent powers of T. This gives $\Phi(s)\Phi(1-s) = I$. Then put $s = w = 1/2 + it$ and take the finite limit of the Wronskian after cancellation; its positivity gives $\Phi(s)^*\Phi(s) = I$. No spectral completeness is used in this step. □

Theorem 4.11 (Finite-area spectral resolution). *There is an orthonormal family of L^2 eigenfunctions u_j of Δ (containing the cuspidal and residual* *spectrum) such that every $F \in C_c^\infty(Y,\chi)$ has the L^2 expansion*

$$F = \sum_j \langle F, u_j\rangle u_j + \frac{1}{4\pi} \sum_{a \text{ sing.}} \int_{-\infty}^{\infty} \langle F, E_a(\cdot, \tfrac12 + it)\rangle E_a(\cdot, \tfrac12 + it)\, dt. \tag{21.A.17}$$

The first sum is locally finite below every fixed eigenvalue and the integral is understood in L^2.

Proof. We give the contour argument because this is the completeness step. For $\Re s > 1$, solve $(\Delta - s(1-s))U = F$ by the resolvent constructed above. Green's formula and Proposition 4.10 continue U meromorphically to $s \in \mathbb{C}$. Its poles away from the critical line are finite-rank by Lemma 4.9; their residues are L^2 eigenfunctions. The compact-remainder lemma implies that only finitely many occur in every bounded spectral interval.

For $c > 1$ write the elementary resolvent Cauchy integral

$$F = \frac{1}{2\pi i} \int_{\Re s = c} (2s-1)(\Delta - s(1-s))^{-1} F\, ds, \tag{21.A.18}$$

first after applying a compactly supported spectral polynomial and then pass to F by the L^2 bound. Move the contour to $\Re s = 1/2$. Residues give the discrete u_j-sum. On the new contour, subtract the resolvents at s and $1-s$. Green's identity with the prescribed incoming cusp solutions shows directly that their jump kernel is

$$\frac{1}{2it}\big(R(\tfrac12 + it) - R(\tfrac12 - it)\big)(z,w) = \frac{1}{4\pi} \sum_a E_a(z, \tfrac12 + it)\overline{E_a(w, \tfrac12 + it)}. \tag{21.A.19}$$

The unitary scattering relation (21.A.15) is precisely what cancels the truncation boundary in this calculation. Substituting (21.A.19) into the shifted contour gives (21.A.17). Taking the inner product with F also gives Parseval, hence no orthogonal remainder is possible. □

4.4. A04.2: Poincaré–Whittaker unfolding. For $m \geq 1$ and $\Re s > 1$ define at the cusp infinity

$$P_m(z,s) = \sum_{\gamma \in \Gamma_\infty \backslash \Gamma_0(q)} \overline{\chi(\gamma)} (\Im \gamma z)^s e(m\gamma z). \tag{21.A.20}$$

The exponential in the nonzero Fourier mode makes all later integrations absolutely convergent after one harmless smooth cutoff; the cutoff is removed by dominated convergence.

Lemma 4.12 (Whittaker pairing). *If*

$$u_j(z) = \sum_{n \neq 0} \rho_j(n) \sqrt{y}\, K_{it_j}(2\pi |n| y) e(nx)$$

is an L^2 Maaß eigenfunction, then

$$\langle P_m(\cdot, s), u_j\rangle = \overline{\rho_j(m)}\,\mathcal{M}_s(t_j; m), \tag{21.A.21}$$

where

$$\mathcal{M}_s(t; m) = \int_0^\infty y^{s-3/2} K_{it}(2\pi m y) e^{-2\pi m y}\, dy \tag{21.A.22}$$

(up to the fixed normalization of the smooth seed). Replacing u_j by $E_{\mathfrak{a}}(\cdot, 1/2+it)$ gives the same formula with the Eisenstein Fourier coefficient $\rho_{\mathfrak{a}}(m, t)$.

Proof. Unfold the sum (21.A.20) to $\Gamma_\infty\backslash\mathbb{H}$. The x-integral over $[0, 1]$ selects the m-th Fourier coefficient, leaving (21.A.22). Near infinity the K-Bessel function decays exponentially; near zero the initial condition $\Re s > 1$ gives absolute convergence. The Eisenstein calculation is identical because its nonzero Fourier modes have the same Whittaker form. Meromorphic continuation then extends the identity to the values used below. $\square$

Lemma 4.13 (Geometric pairing and the Kloosterman kernel)**.** *For $m, n \geq 1$, unfolding one Poincaré series in $\langle P_m(\cdot, s), P_n(\cdot, w)\rangle$ and decomposing the remaining cosets by their lower-left entry c gives*

$$\delta_{mn}\,\mathcal{D}_{s,w}(m) + \sum_{q|c} \frac{S_\chi(m, n; c)}{c} \mathcal{B}^+_{s,w}\left(\frac{4\pi\sqrt{mn}}{c}\right), \tag{21.A.23}$$

and, after replacing n by $-n$,

$$\sum_{q|c} \frac{S_\chi(m, -n; c)}{c} \mathcal{B}^-_{s,w}\left(\frac{4\pi\sqrt{mn}}{c}\right). \tag{21.A.24}$$

Mellin inversion of the two kernels gives respectively the J-kernel and K-kernel transforms in (21.A.5).

Proof. Represent a nontrivial double coset $\Gamma_\infty\gamma\Gamma_\infty$ by $\gamma = \left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right)$ with $c > 0$ and $ad \equiv 1 \pmod{c}$. Translating on the left and right changes a, d by multiples of c and leaves one residue $d \bmod c$ with $(d, c) = 1$. The x-integration contributes $e((md + n\bar{d})/c)\overline{\chi(d)}$; summing over d is $S_\chi(m, n; c)$. The determinant-one change of variables leaves a radial integral depending only on $4\pi\sqrt{mn}/c$, proving (21.A.23); opposite Fourier signs give (21.A.24).

To identify the radial integral, insert

$$K_\nu(y) = \frac{1}{2}\int_0^\infty \exp\left(-\frac{y}{2}(t + t^{-1})\right) t^{\nu-1}\, dt$$

and take Mellin transform in the radial variable. The y-integral is a Gamma integral. Applying the duplication and reflection formulas and then Mellin inversion gives $J_{2it} - J_{-2it}$ for equal signs and K_{2it} for opposite signs. The normalizing Gamma factors are exactly the hyperbolic factors displayed in

(21.A.5). This calculation also justifies moving all integrals for a smooth compactly supported final test function. □

4.5. A04.3: the weight-zero Kuznetsov identities.

THEOREM 4.14 (Kuznetsov trace formula in the consumed normalization). *Let ϕ satisfy the hypotheses imposed before (21.A.5). For $m, n \geq 1$,*

(21.A.25)

$$\sum_{q|c} \frac{S_\chi(m,n;c)}{c}\phi\left(\frac{4\pi\sqrt{mn}}{c}\right) = \sum_{\substack{k\equiv 0\ (2)\\ k\geq 2}} \sum_{f\in\mathcal{B}_k(q,\chi)} \mathfrak{c}_k\dot{\phi}(k)\sqrt{mn}\,\rho_f(m)\overline{\rho_f(n)} + \sum_{f\in\mathcal{B}(q,\chi)} \frac{\widetilde{\phi}(t_f)}{\cosh\pi t_f}\sqrt{mn}\,\rho_f(m)\overline{\rho_f(n)} + \frac{1}{4\pi}\sum_{a\ \text{sing.}}\int_{\mathbb{R}} \frac{\widetilde{\phi}(t)}{\cosh\pi t}\sqrt{mn}\,\rho_a(m,t) \cdot \overline{\rho_a(n,t)}\,dt,$$

where $\mathfrak{c}_k$ is the Petersson normalization fixed in Theorem 4.3. For opposite signs,

(21.A.26)

$$\sum_{q|c} \frac{S_\chi(m,-n;c)}{c}\phi\left(\frac{4\pi\sqrt{mn}}{c}\right) = \sum_{f\in\mathcal{B}(q,\chi)} \frac{\check{\phi}(t_f)}{\cosh\pi t_f}\sqrt{mn}\,\rho_f(m)\overline{\rho_f(-n)} + \frac{1}{4\pi}\sum_{a\ \text{sing.}}\int_{\mathbb{R}} \frac{\check{\phi}(t)}{\cosh\pi t}\sqrt{mn}\,\rho_a(m,t) \cdot \overline{\rho_a(-n,t)}\,dt.$$

These are the weight-zero forms of BHM (2.11)–(2.12), with the harmless choice between cusp-indexed and character-pair-indexed Eisenstein bases made by a unitary change of basis.

PROOF. Start with the geometric pairing in Lemma 4.13. Insert the complete resolution (21.A.17) between the two Poincaré series and use Lemma 4.12; this yields the Maaß and Eisenstein terms with their Fourier coefficients. Mellin inversion in the two seed parameters and the kernel computation of Lemma 4.13 turn the spectral Mellin multipliers into $\widetilde{\phi}$ or $\check{\phi}$.

When the equal-sign Mellin contour is moved past the positive integral parameters, its residues are precisely the discrete-series representations. Their Whittaker functions are $y^{k/2}e^{-2\pi my}$ and the residue pairing is the holomorphic Poincaré pairing already evaluated in Theorem 4.3. Thus the residue sum is the first line of (21.A.25), with exactly the previously fixed $\mathfrak{c}_k$. There are no such residues for opposite signs, which explains their absence from (21.A.26).

The decay conditions on ϕ, together with Lemma 4.5, justify the contour shifts and termwise integrations. Density of compactly supported smooth test functions and the same transform bounds then remove the auxiliary cutoffs. This proves both identities. $\square$

Corollary 4.15 (Spectral large sieve in the range used below)**.** *Let b_m be supported on $M \le m < 2M$ and let $T \ge 1$. At level r with trivial character,*

(21.A.27)
$$\sum_{|t_f|\le T} \frac{1}{\cosh \pi t_f} \left| \sum_m b_m \sqrt{m}\, \rho_f(m) \right|^2 + \mathcal{E}_T \ll_\varepsilon (T^2 + M/r + 1)(rMT)^\varepsilon \sum_m |b_m|^2,$$

where $\mathcal{E}_T$ denotes the analogous nonnegative Eisenstein contribution. The same estimate, with T^2 replaced by K^2, holds for the holomorphic spectrum $k \le K$.

Proof. Choose in (21.A.25) a nonnegative even spectral majorant which is at least one on $[-T, T]$ and whose geometric transform is supported on $x \ll T^{-1}$ with derivatives of size $O_j(T^{-j})$; construct it by taking the square of the Fourier transform of a smooth bump and applying the Bessel inversion implicit in the proof of the theorem. Multiply (21.A.25) by $b_m \overline{b_n}$ and sum over m, n. The diagonal is $O(T^2 \sum |b_m|^2)$. We estimate the Kloosterman quadratic form without invoking the Weil bound. For one modulus c write

$$A_c(d) = \sum_m b_m e(md/c).$$

Expanding the Kloosterman sum and using that inversion permutes the reduced residue classes gives

$$\left| \sum_{m,n} b_m \overline{b_n} S(m,n;c) \right| \le \sum_{d \bmod c} |A_c(d)|^2. \tag{21.A.28}$$

For a dyadic block $C \le c < 2C$, $r \mid c$, apply the elementary additive large sieve to the fractions d/c: fractions belonging to distinct reduced pairs are $\gg C^{-2}$ apart, and the usual duality proof (expand the square, use the geometric-series bound $|\sum_{j\le M} e(j\alpha)| \le \min(M, (2\|\alpha\|)^{-1})$, and sum the C^{-2}-separated reciprocal distances) gives

$$\sum_{C\le c<2C} \sideset{}{^*}\sum_{d \bmod c} |A_c(d)|^2 \ll (C^2 + M) \sum_m |b_m|^2. \tag{21.A.29}$$

The geometric transform carries a factor C^{-1} and, after the Bessel inversion used to build the majorant, is negligible unless $C \gg \sqrt{M}/T$; repeated integration by parts supplies arbitrary decay for larger dyadic blocks. Summing (21.A.29) with these weights and retaining only c divisible by r gives $O_\varepsilon((M/r+1)(rMT)^\varepsilon \sum |b_m|^2)$. Thus no Kloosterman square-root estimate is hidden in the large-sieve step. Positivity permits discarding spectral terms outside the target range. The holomorphic proof is identical with Petersson's formula. □

Proof and provenance. A04 is now closed. Lemma 4.6 through Theorem 4.11 own the continuous spectral resolution rather than citing it; Lemmas 4.12 and 4.13 own the Poincaré/Whittaker calculation; and Theorem 4.14 gives the exact trace identities consumed by A07. Corollary 4.15 is recorded separately because the large-spectrum estimate uses it as a quantitative theorem, not merely as a slogan.

5. Child Root A05: the divisor–Voronoi and shifted-divisor transform

We isolate the arithmetic transform which is used after the first trace formula. It is independent of the spectral decomposition and can therefore be proved before A04. Write $e(x) = e^{2\pi i x}$.

Lemma 5.1 (Hurwitz functional equation in Fourier form)**.** *For $0 < \alpha \le 1$, the Hurwitz zeta function $\zeta(s,\alpha) = \sum_{m\ge 0}(m+\alpha)^{-s}$ continues meromorphically to $\mathbb{C}$ with a simple pole of residue 1 at $s = 1$. For $\Re s < 0$,*

$$\zeta(s,\alpha) = 2\Gamma(1-s)(2\pi)^{s-1}\Bigg[\sin\frac{\pi s}{2}\sum_{m\ge 1}\frac{\cos(2\pi m\alpha)}{m^{1-s}} + \cos\frac{\pi s}{2}\sum_{m\ge 1}\frac{\sin(2\pi m\alpha)}{m^{1-s}}\Bigg]. \tag{21.5.5a}$$

Proof. For $\Re s > 1$, use

$$\Gamma(s)\zeta(s,\alpha) = \int_0^\infty \frac{x^{s-1}e^{-\alpha x}}{1-e^{-x}}\,dx.$$

Subtract x^{-1} on $(0,1)$; the difference is smooth at 0, which gives meromorphic continuation across $s = 1$ with residue 1, and repeated Taylor subtraction continues to the plane.

For the functional equation, apply Poisson summation to $f_t(x) = e^{-\pi t(x+\alpha)^2}$ and then Mellin-transform in t. The Poisson identity is

$$\sum_{r\in\mathbf{Z}} e^{-\pi t(r+\alpha)^2} = t^{-1/2}\sum_{m\in\mathbf{Z}} e^{-\pi m^2/t}e(m\alpha).$$

Split the Mellin integral at $t = 1$, use the identity once on the interval $(0,1)$, and pair m with $-m$. The Mellin integral $\int_0^\infty e^{-\pi m^2 t} t^{u/2-1}\,dt = \pi^{-u/2}\Gamma(u/2)m^{-u}$ gives a first expression in gamma factors. The reflection and duplication formulae for Γ reduce it to (21.5.5a). All interchanges are absolutely convergent in an initial vertical strip; the meromorphic continuation just established then extends the identity. $\square$

Lemma 5.2 (Estermann functional equation for the divisor twist)**.** *Let* $(d,c) = 1$ *and*

$$D(s; d/c) = \sum_{n\geq 1} \tau(n)e(dn/c)n^{-s} \qquad (\Re s > 1).$$

Then D *is meromorphic on* $\mathbb{C}$, *has at* $s = 1$ *the principal part determined by*

$$\operatorname*{Res}_{s=1} D(s; d/c)X^s = \frac{X}{c}(\log X + 2\gamma - 2\log c), \tag{21.5.5b}$$

and satisfies

$$\begin{aligned} D(s; d/c) = 2(2\pi)^{2s-2}c^{1-2s}\Gamma(1-s)^2 \\ \times \big[D(1-s; \bar{d}/c) - \cos(\pi s)D(1-s; -\bar{d}/c)\big]. \end{aligned} \tag{21.5.5c}$$

Here $d\bar{d} \equiv 1 \pmod{c}$.

Proof. Open the divisor function and separate both factors into residue classes:

$$D(s; d/c) = c^{-2s}\sum_{r,t=1}^{c} e(drt/c)\zeta(s, r/c)\zeta(s, t/c). \tag{21.5.5d}$$

This proves meromorphic continuation from Lemma 5.1. Since $\zeta(s, r/c) = (s-1)^{-1} - \psi(r/c) + O(s-1)$, summing first over t in (21.5.5d) shows that the coefficient of $(s-1)^{-2}$ is c^{-1}; differentiating c^{-2s} and using $\sum_{r=1}^{c}\psi(r/c) = -c(\gamma + \log c)$ gives (21.5.5b).

To derive the functional equation, insert (21.5.5a) twice into (21.5.5d) in $\Re s < 0$ and write sine and cosine as linear combinations of exponentials. The finite sums which occur have the form

$$\sum_{r,t \bmod c} e\Big(\frac{drt + mr + nt}{c}\Big).$$

Summing over r enforces $dt + m \equiv 0 \pmod{c}$; because d is invertible this has the unique solution $t \equiv -\bar{d}m$, leaving $c\,e(-\bar{d}mn/c)$. The four choices of signs from the trigonometric expansion therefore combine into the two twists $D(1-s; \bar{d}/c)$ and $D(1-s; -\bar{d}/c)$. The scalar factor is

$$\frac{1}{\pi}\Big(\frac{c}{2\pi}\Big)^{1-2s}\Gamma(1-s)^2 = 2(2\pi)^{2s-2}c^{1-2s}\Gamma(1-s)^2,$$

and the even/odd combinations give respectively 1 and $-\cos(\pi s)$. This is (21.5.5c). □

Proposition 5.3 (Divisor Voronoi summation with additive twist). *Let $(d,c)=1$ and $F\in C_c^\infty(0,\infty)$. Put*

$$L_c(y)=\log y+2\gamma-2\log c,$$

$$\mathcal{J}^-(x)=-2\pi Y_0(4\pi x),\qquad \mathcal{J}^+(x)=4K_0(4\pi x).$$

Then

(21.5.5e)

$$\sum_{m\ge1}\tau(m)e(dm/c)F(m)=\frac{1}{c}\int_0^\infty L_c(y)F(y)\,dy$$
$$+\frac{1}{c}\sum_{\pm}\sum_{m\ge1}\tau(m)e(\pm\bar{d}m/c)$$
$$\times\int_0^\infty \mathcal{J}^\pm\left(\frac{\sqrt{my}}{c}\right)F(y)\,dy.$$

If F is supported on $y\asymp Y$ and $y^jF^{(j)}(y)\ll_j Z^j$, then repeated integration by parts in the Bessel integrals gives arbitrary power decay unless $mY/c^2\ll(1+Z)^2$, and the same assertion holds after any fixed number of derivatives in external smooth parameters of F.

Proof. Mellin inversion gives, initially on $\Re s=2$,

$$\sum_m\tau(m)e(dm/c)F(m)=\frac{1}{2\pi i}\int_{(2)}D(s;d/c)\widetilde{F}(s)\,ds.$$

Move the contour to $\Re s=-1$. The double pole at 1, using (21.5.5b) and the identities $\widetilde{F}(1)=\int F$ and $\widetilde{F}'(1)=\int F(y)\log y\,dy$, contributes the first term of (21.5.5e). Insert (21.5.5c) on the new line, expand the two Estermann series, and interchange sum and integral.

It remains to identify the inverse Mellin kernels. Mellin inversion together with the reflection and duplication formulae gives

$$4K_0(4\pi\sqrt{x})=\frac{2}{2\pi i}\int_{(\sigma)}(2\pi)^{-2u}\Gamma(u)^2x^{-u}\,du,$$

and

$$-2\pi Y_0(4\pi\sqrt{x})=\frac{2}{2\pi i}\int_{(\sigma)^*}(2\pi)^{-2u}\Gamma(u)^2\cos(\pi u)x^{-u}\,du,$$

where the second contour is indented to separate the poles at the origin. These are obtained directly from the Mellin integrals for K_0 and Y_0; substitution $x=my/c^2$ gives exactly the two transformed terms in (21.5.5e).

For the decay assertion, use $(x^\nu B_\nu(x))' = \pm x^\nu B_{\nu-1}(x)$ for $B = J, Y, K$ and integrate repeatedly against F. Each integration transfers a derivative to F and gains a factor comparable to $c/\sqrt{mY}$. The derivative hypothesis then gives a factor $(1+Z)c/\sqrt{mY}$ at each step. Taking arbitrarily many steps proves the claimed truncation and its parameter-differentiated version. $\square$

Proposition 5.4 (Smooth additive-divisor transformation)**.** *Let $a, h \neq 0$, let $Q, M \geq 1$, and let $g(n,m)$ be smooth and supported on $1/2 \leq n \leq Q$ and $m \asymp M$, with the usual dyadic derivative bounds in both variables. For either sign, the shifted divisor sum*

$$\mathcal{D}_\pm(a,h;g) = \sum_{an \pm m = h} \tau(n)\tau(m)g(n,m)$$

is the sum of an explicit Ramanujan main term and two Kloosterman families: Define

$$\mathcal{I}_w^\pm := \int_0^\infty L_w(\pm(h-ax))K_{(a,w),w}(x)g(x, \pm(h-ax))\,dx$$

and

$$\mathcal{I}_{r,w}^{\sigma,\pm} := \int_0^\infty \mathcal{J}^\sigma\left(\frac{\sqrt{\sigma r\,(\pm(h-ax))}}{w}\right)K_{(a,w),w}(x)g(x,\pm(h-ax))\,dx,$$

where the integrand is declared zero unless $\sigma(\pm(h-ax)) > 0$. Then

$$\begin{aligned}\mathcal{D}_\pm(a,h;g) &= \sum_{w\geq 1}\frac{(a,w)r_w(h)}{w^2}\,\mathcal{I}_w^\pm \\ &\quad + \sum_{\sigma=\pm}\sum_{w\geq 1}\frac{(a,w)}{w^2}\sum_{r\geq 1}\tau(r)S(\mp h,\sigma r;w)\,\mathcal{I}_{r,w}^{\sigma,\pm}.\end{aligned} \tag{21.5.5f}$$

with the convention that the integrand vanishes outside the positive domain. Here $r_w(h) = S(h,0;w)$ and, for a fixed cutoff η_0 equal to 1 on $[0,1]$ and 0 on $[2,\infty)$,

$$K_{r,w}(x) = \sum_{\delta\geq 1}\frac{1}{\delta}\eta_0\left(\frac{w\delta}{r\sqrt{Q}}\right)\left(2-\eta_0\left(\frac{rx}{\delta w\sqrt{Q}}\right)\right). \tag{21.5.5g}$$

For all $i, j \geq 0$,

$$x^i w^j \partial_x^i \partial_w^j K_{r,w}(x) \ll_{i,j} \log(2Q), \qquad K_{r,w}(x) = 0 \quad \text{if } w \geq 2r\sqrt{Q}, \tag{21.5.5h}$$

and the Bessel integrals have the truncation and derivative bounds of Proposition 5.3.

PROOF. For $n \le Q$ the elementary smooth hyperbola identity

$$\tau(n)=\sum_{\delta\mid n}\eta_0\Big(\frac{\delta}{\sqrt{Q}}\Big)\Big(2-\eta_0\Big(\frac{n}{\delta\sqrt{Q}}\Big)\Big) \tag{21.5.5i}$$

holds because paired divisors cannot both lie beyond the transition range. Insert (21.5.5i) in the n-variable of $an\pm m=h$. After writing $n=(h\mp m)/a$, the divisibility condition is a congruence $m\equiv \pm h \pmod{a\delta}$. Detect that congruence by additive characters.

For a general residue class $m\equiv\mu \pmod c$, decompose the additive character modulus by $w=c/(d,c)$ and apply Proposition 5.3; finite Fourier inversion yields

$$\begin{aligned}\sum_{m\equiv\mu\,(c)}\tau(m)F(m)&=\frac{1}{c}\sum_{w\mid c}\frac{r_w(\mu)}{w}\int L_wF\\&\quad+\frac{1}{c}\sum_{\sigma=\pm}\sum_{w\mid c}\frac{1}{w}\sum_{r\ge1}\tau(r)S(-\mu,\sigma r;w)\\&\qquad\times\int\mathcal{J}^\sigma(\sqrt{ry}/w)F(y)\,dy.\end{aligned}$$

Replace a by $|a|$ in the congruence modulus and set $c=|a|\delta$. Group terms with the same w and put $r=(a,w)$. The remaining sum over δ is exactly $K_{r,w}(x)$ in (21.5.5g), giving (21.5.5f).

Only δ in a dyadic interval of length $O(\sqrt{Q})$ contribute to (21.5.5g). Differentiation of either cutoff contributes one reciprocal copy of its scale, which is cancelled by the prefactors x^iw^j; summing $1/\delta$ gives $O(\log(2Q))$. The first cutoff is identically zero for every $\delta\ge1$ when $w\ge 2r\sqrt{Q}$, proving the support statement. The last claim follows from the integration-by-parts argument in Proposition 5.3. □

Proof and provenance. Propositions 5.3 and 5.4 reproduce the arithmetic transform used in BHM equations (2.27)–(2.31): the logarithmic main term, the two K_0/Y_0 kernels, the Ramanujan term, the transformed Kloosterman family, and the uniform derivative/support estimates are all owned locally. Child Root A05 is therefore closed independently of the trace-formula argument.

Lemma 5.5 (Amplified fourth-moment reduction)**.** *Appendices 4, 6, and 7, together with the A05 transform proved above, supply the trace transformations and the transformed spectral estimate*

$$\mathcal{Q}(\ell)\ll q^{\varepsilon}\big(\ell^{c_1}q^{-c_2}+\ell^{-1/2}\big) \tag{21.5.5}$$

uniformly on the amplifier support. Then

$$|L(f_0, 1/2+it)|^4 \ll_{t,t_0,\varepsilon} q^{1+\varepsilon}\big(L^{2c_1}q^{-c_2} + L^{-1/2}\big). \tag{21.5.6}$$

Proof. Insert Lemma 3.1 into the positive complete spectral fourth moment and weight it by the square of the amplifier from Lemma 3.2. Expand the amplifier and use the Hecke multiplication formula to reduce to the family $\mathcal{Q}(\ell)$. Positivity allows the target form to be retained. Divide by the square of the amplifier lower bound. The $\ell^{c_1}q^{-c_2}$ term contributes $L^{2c_1}q^{-c_2}$ after the two amplifier sums, while the diagonal term contributes $L^{-1/2}$; divisor factors and the number of prime divisors of q are absorbed in q^{ε}. This is (21.5.6). □

Proposition 5.6 (BHM endgame from the transformed spectral estimate)**.** *Under (21.5.5), the level-aspect estimate (21.5.1) holds with some explicit $\delta = \delta(c_1, c_2) > 0$.*

Proof. Choose $L = q^{c_2/(2c_1+1/2)}$ so the two terms in (21.5.6) balance. Then $|L(f_0, 1/2+it)|^4 \ll q^{1-\eta+\varepsilon}$ with $\eta = c_2/(4c_1+1) > 0$. Taking fourth roots gives (21.5.1), with $\delta = \eta/4$ after decreasing it slightly to absorb the ε-loss. □

6. The Twisted GL_2 Burgess Estimate

This appendix closes Child Root A06. The scope is exactly the one consumed in the small-spectrum part of the level-aspect argument: a primitive Maaß or holomorphic form of level N and trivial nebentypus, twisted by a primitive Dirichlet character of conductor Q. We follow the Bykovskii–Harcos proof rather than cite the final subconvex theorem. The algebro-geometric character-sum input is the internally proved Weil–Kummer theorem of Volume 19, Theorem 1.4; no external finite-field proved input remains in this appendix.

6.1. Statement and conductor range. Put $\mu_f = 1 + |t_f|$ for a Maaß form and $\mu_f = k$ for a holomorphic form of weight k. Write $\mathcal{T} = 2 + |\Im s|$.

Theorem 6.1 (Twisted-GL_2 Burgess theorem in the consumed range). *Let f be a primitive cusp form of level N and trivial nebentypus and let χ be primitive modulo Q. On $\Re s = 1/2$, for every $\varepsilon > 0$,*

$$L(f\otimes\chi, s) \ll_{\varepsilon} (\mathcal{T}\mu_f NQ)^{\varepsilon}\mathcal{T}^{1/2}\mu_f^3 N^{1/2}Q^{3/8}. \tag{21.B.1}$$

If

$$Q \geq (\mu_f N)^4, \tag{21.B.2}$$

then the stronger estimate

$$L(f\otimes\chi,s)\ll_\varepsilon(\mathcal{T}\mu_f NQ)^\varepsilon\mathcal{T}^{1/2}\mu_f^3N^{1/4}Q^{3/8} \tag{21.B.3}$$

holds. The implied constant is uniform in N,Q,f,χ*.*

The exponent 3/8 is the one needed in A07. We prove (21.B.3) directly; the same calculation with the unbalanced level term retained gives (21.B.1).

6.2. A positive spectral test and the complete correlation sum. For integers $0\le b<a$ put

$$\phi_{a,b}(x)=i^{b-a}J_a(x)x^{-b}.$$

The Mellin integral for J_a gives, for the weight-zero Kuznetsov transform,

$$\widetilde{\phi}_{a,b}(t)=c_{a,b}\frac{t\tanh(\pi t)}{\prod_{j=0}^b\{t^2+(\frac{a+b}{2}-j)^2\}}, \tag{21.B.4}$$

with $c_{a,b}>0$; for the holomorphic transform the same computation gives a positive rational function for $2\le k\le a-b$. We use $a=20,b=2$.

Lemma 6.2 (Positive test and decay)**.** *The spectral transforms of* $\phi_{20,2}$ *are nonnegative on the unitary spectrum, are bounded below on every fixed compact spectral interval, and satisfy*

$$\widetilde{\phi}_{20,2}(t)\ll(1+|t|)^{-6},\qquad \dot{\phi}_{20,2}(k)\ll k^{-6}. \tag{21.B.5}$$

Proof. Insert the Mellin transform

$$\int_0^\infty J_a(x)x^{z-1}\,dx=2^{z-1}\frac{\Gamma((a+z)/2)}{\Gamma((a-z+2)/2)}$$

into (21.A.5), move the contour through the finitely many Gamma poles, and apply $\Gamma(z+1)=z\Gamma(z)$. All Gamma factors pair with their complex conjugates on the unitary axis, producing (21.B.4); each denominator factor is positive, as is $t\tanh\pi t$. The holomorphic residues have the same sign. Degree counting in the resulting rational functions gives (21.B.5). □

The arithmetic estimate needed after the two Hurwitz functional equations is the following.

Proposition 6.3 (Primitive-character correlation)**.** *Let* χ *be primitive modulo* Q *and suppose* $(m_1m_2-n_1n_2,Q)=1$*. Then*

(21.B.6)

$$\left|\sideset{}{^*}\sum_{x \bmod Q}\overline{\chi}(m_1+n_1\bar{x})\chi(m_2+n_2x)\right|\ll\tau(Q)Q^{1/2}(m_1n_1^2,m_2n_2^2,Q)^{1/2}.$$

The absolute implied constant may be enlarged at 2 but is independent of all parameters.

PROOF. By the Chinese remainder theorem both the character and the sum factor over prime powers, so take $Q = p^\beta$. Remove the common power of p displayed on the right; the determinant hypothesis guarantees that the remaining rational function is nonconstant on the unit group.

For $\beta = 1$ the summand is a multiplicative character of

$$R(X) = X\frac{m_2 + n_2 X}{m_1 X + n_1}.$$

Unless the right side of (21.B.6) is already trivial, the divisor of R has a zero or pole whose multiplicity is not divisible by the order of χ. The Weil–Kummer estimate 1.4 therefore gives $O(p^{1/2})$.

Let $\beta = 2\alpha > 1$. Write a unit as $x = y + p^\alpha z$ and Taylor-expand the p-adic logarithm of R on each residue class on which numerator and denominator are units. The inner z-sum vanishes unless

$$m_1 n_2 y^2 + 2n_1 n_2 y + m_2 n_1 \equiv 0 \pmod{p^\alpha}. \tag{21.B.7}$$

If $p > 2$, multiplication by m_1 (interchanging the two pairs when necessary) turns (21.B.7) into

$$n_2(m_1 y + n_1)^2 \equiv n_1(n_1 n_2 - m_1 m_2) \pmod{p^\alpha}. \tag{21.B.8}$$

The parenthesized factors are units under the determinant hypothesis. If $p^\gamma \| (n_1, n_2, p^\alpha)$, the elementary square-root congruence in $(\mathbb{Z}/p^{\alpha-\gamma}\mathbb{Z})^\times$ has at most two solutions, so (21.B.7) has at most $2p^\gamma$ solutions. At $p = 2$ there are at most four square roots and the same argument gives $4\,2^\gamma$. Each surviving class contributes p^α, yielding the required $p^{\beta/2}p^{\gamma/2}$ after restoring the removed common factors and applying Cauchy–Schwarz between the two symmetric gcds.

For $\beta = 2\alpha + 1$, write $x = y + p^\alpha z$ and keep the quadratic term in the Taylor expansion. The inner sum is a quadratic Gauss sum of modulus p, hence has modulus $p^{1/2}$ unless its linear and quadratic coefficients both vanish. The exceptional y again satisfy (21.B.7), now modulo $p^{\alpha+1}$; (21.B.8) gives at most $2p^{-1}(n_1, n_2, p^{\alpha+1})$ residue classes for odd p and four times this at 2. Multiplying the number of classes by the size $p^\alpha p^{1/2}$ of each quadratic Gauss contribution gives the same $p^{\beta/2}$ bound. Multiplication over prime powers proves (21.B.6), with $\tau(Q)$ absorbing the local constants. □

6.3. The Bykovskii spectral moment. Let

$$D = 3[N, Q]$$

(the harmless factor 3 keeps the cusp normalizations uniform). Embed the target form into level D by the degeneracy maps $V_d f(z) = d^{k/2} f(dz)$ in the holomorphic case and $V_d f(z) = f(dz)$ with the unitary L^2 normalization in the Maaß case, for $d \mid D/N$. Orthogonalize these finitely many vectors and choose one unit vector f_1 with

$$|\rho_{f_1}(1)| \gg_\varepsilon D^{-1/2-\varepsilon}\mu_f^{-3/2}N^{-1/2}. \tag{21.B.9}$$

This follows directly from the Gram matrix of the finitely many degeneracy maps: its diagonal entries are the L^2 norms and its off-diagonal entries are Hecke matrix coefficients; triangular Gram–Schmidt gives one vector with first coefficient at least the reciprocal square root of the total diagonal mass.

For $(\ell, D) = 1$ define the positive spectral moment

$$\begin{aligned}\mathcal{Q}(\ell) = &\sum_{g\in\mathcal{B}(D,1)} \frac{\widetilde{\phi}_{20,2}(t_g)}{\cosh \pi t_g}\lambda_g(\ell)|L(g\otimes\chi, u+i\tau)|^2 \\ &+ \mathcal{Q}_{\mathrm{hol}}(\ell) + \mathcal{Q}_{\mathrm{Eis}}(\ell),\end{aligned} \tag{21.B.10}$$

Proposition 6.4 (Bykovskii moment estimate)**.** *For $u = 1/2+\varepsilon$,*
(21.B.11)

$$\mathcal{Q}(\ell) \ll_\varepsilon ((1+|\tau|)D\ell)^\varepsilon(1+|\tau|)\left\{\ell^{-1/2} + \frac{(N,Q)\ell^{1/4}}{(NQ)^{1/2}} + \frac{(N,Q)^{3/2}\ell^{1/2}}{NQ^{1/2}}\right\}.$$

The same estimate holds for each fixed holomorphic weight appearing in the positive test.

Proof. We spell out the transformations because (21.B.11) is the heart of A06. For $\Re u > 5/4$, open the two absolutely convergent twisted L-series in (21.B.10), use the Hecke relation to absorb $\lambda_g(\ell)$, and apply Theorem 4.14 (and Theorem 4.3 for the holomorphic part). The diagonal is $O(\ell^{-1/2}(1+|\tau|)D^\varepsilon)$. The off-diagonal is

$$\sum_{c\equiv 0(D)} \frac{1}{c} \sum_{m_1,m_2\geq 1} \frac{\chi(m_1)\overline{\chi}(m_2)}{m_1^{u+i\tau}m_2^{u-i\tau}} S(m_1, m_2\ell; c)\phi_{20,2}\left(\frac{4\pi\sqrt{m_1m_2\ell}}{c}\right). \tag{21.B.12}$$

Write $m_i = a_iQ + r_i$ and sum first over a_i. Twice applying the Hurwitz functional equation of Lemma 5.1 converts these two progressions to dual additive Fourier series. Mellin inversion of the Bessel test produces kernels $\Xi_{u,\tau}^{\eta_1,\eta_2}(x)$ satisfying

$$\Xi(x) \ll_\varepsilon x^{-1/2+2\varepsilon}(1+|\tau|)^{2\varepsilon}, \qquad \Xi(x) \ll_\varepsilon x^{1/2+\varepsilon}\left(1+\frac{1+|\tau|}{20}\right), \tag{21.B.13}$$

according as the Mellin contour is placed at $\Re z = 1-4\varepsilon$ or $-1-2\varepsilon$. These inequalities follow by Stirling from the explicit Mellin transform of $J_{20}(x)x^{-2}$; no spectral estimate is involved.

After the two functional equations, the complete sum in the residue variables is exactly

$$\sum_{x \bmod Q}^{*} \overline{\chi}(m_1 + n_1\bar{x})\chi(m_2 + n_2 x),$$

up to unit scalars and gcd factors coming from the common divisor of c and Q. Proposition 6.3 gives square root cancellation. Separate c according to $c = c_0 c_1$ with $c_0 \mid Q^\infty$ and $(c_1, Q) = 1$. The determinant condition failing in Proposition 6.3 forces one linear congruence; summing those exceptional residues directly produces the third term of (21.B.11). On the generic residues (21.B.13), the square-root character bound and the geometric c-sum give the second term. Explicitly the two dyadic sums are

$$\sum_C \frac{(N,Q)}{(NQ)^{1/2}} \ell^{1/4} C^{-1-\varepsilon}, \qquad \sum_C \frac{(N,Q)^{3/2}}{NQ^{1/2}} \ell^{1/2} C^{-1-\varepsilon},$$

which converge at their first admissible dyadic block. This proves (21.B.11) for $\Re u > 5/4$.

Both sides are holomorphic in u for $\Re u > 1/2$ after multiplication by the pole-killing factor from Lemma 3.1. Moving the two Mellin contours to $\Re u = 1/2 + \varepsilon$ crosses only the explicitly removed zeta pole. Bounds (21.B.13) are uniform on the shifted contours, so the same estimate holds there. The holomorphic spectrum is identical, using Petersson in place of Kuznetsov. □

6.4. Amplification and the $3/8$ **exponent.** The source proof amplifies by the target Hecke eigenvalues. The required lower bound is a short Rankin–Selberg deduction, and the global $\mathrm{GL}_2 \times \mathrm{GL}_2$ continuation/pole theorem has already been reconstructed in Volume 9 (Repair 17 together with Appendix G).

Lemma 6.5 (Rankin–Selberg amplifier mass)**.** *Let $\omega \geq 0$ be smooth, supported in $[1/2, 3]$ and positive on $[1, 2]$. For $L \geq Q^\varepsilon (\mu_f N)^{1+\varepsilon}$,*

$$\sum_{(\ell, D)=1} |\lambda_f(\ell)|^2 \omega(\ell/L) \gg_\varepsilon L(LD)^{-\varepsilon}. \tag{21.B.17}$$

Proof. Mellin inversion gives

$$\frac{1}{2\pi i} \int_{(2)} L^{(D)}(f \times \widetilde{f}, z)\widehat{\omega}(z) L^z \, dz.$$

The Volume-IX Rankin–Selberg owner gives a simple pole at $z = 1$ with positive residue and a functional equation. Removing the finitely many Euler factors at primes dividing D changes the residue by at most $(LD)^\varepsilon$. Move the contour to $\Re z = 1/2 + \varepsilon$. Stirling in the completed functional equation and Phragmén–Lindelöf give a convexity bound $\ll_\varepsilon (N\mu_f(1+|\Im z|))^{1/2+\varepsilon}$ there; rapid decay of $\widehat{\omega}$ makes the shifted integral $O_\varepsilon(L^{1/2+\varepsilon}(N\mu_f)^{1/2+\varepsilon}D^\varepsilon)$. Under the displayed lower bound on L this is smaller than half the pole term, proving (21.B.17). □

Proof of Theorem 6.1. Set $x(\ell) = \overline{\lambda_f(\ell)}\omega(\ell/L)$ for $(\ell, D) = 1$ and zero otherwise. Positivity of the test in Lemma 6.2, the oldform coefficient bound (21.B.9), and Lemma 6.5 give

(21.B.18)
$$\frac{L^2(LD)^{-\varepsilon}}{\mu_f^{6+\varepsilon}D}|L(f\otimes\chi, \tfrac{1}{2}+\varepsilon+i\tau)|^2 \ll_\varepsilon \sum_{\ell_1,\ell_2} x(\ell_1)\overline{x(\ell_2)} \sum_{d|(\ell_1,\ell_2)} \mathcal{Q}(\ell_1\ell_2/d^2).$$

Insert Proposition 6.4. For any real a, Cauchy–Schwarz gives

$$\sum_{d,\ell_1,\ell_2}(\ell_1\ell_2)^a|x(d\ell_1)x(d\ell_2)| \ll_a (1+L^{2a+1})\sum_\ell \tau(\ell)|x(\ell)|^2. \tag{21.B.19}$$

The Hecke relation and the Rankin–Selberg upper bound on the same smoothed second moment imply $\sum\tau(\ell)|x(\ell)|^2 \ll_\varepsilon L^{1+\varepsilon}$. Consequently (21.B.18) becomes

(21.B.20)
$$|L(f\otimes\chi, \tfrac{1}{2}+\varepsilon+i\tau)|^2 \ll_\varepsilon (\mathcal{T}\mu_f NQ)^\varepsilon \mu_f^6 \left(\begin{matrix}\dfrac{QN}{L(N,Q)} + L^{1/2}Q^{1/2}N^{1/2}\mathcal{T} \\ + LQ^{1/2}(N,Q)^{1/2}\mathcal{T}\end{matrix}\right).$$

This is the amplified form of Harcos's equations (3.7)–(3.9); the three terms come respectively from the diagonal, the generic square-root correlation, and the degenerate congruence classes.

Choose

$$L = \frac{Q^{1/4}N^{1/2}}{(N,Q)^{3/4}\mathcal{T}^{1/2}} + Q^\varepsilon(N\mu_f)^{1+\varepsilon}. \tag{21.B.21}$$

Substitution into (21.B.20), followed by a square root, gives

(21.B.22)
$$\begin{aligned}|L(f\otimes\chi, \tfrac{1}{2}+\varepsilon+i\tau)| \ll_\varepsilon &(\mathcal{T}\mu_f NQ)^\varepsilon \mu_f^3 \mathcal{T}^{1/4}N^{1/4}Q^{3/8}(N,Q)^{-1/8} \\ &+ (\mathcal{T}\mu_f NQ)^\varepsilon \mu_f^{7/2}\mathcal{T}^{1/2}N^{1/2}(N,Q)^{1/4}Q^{1/4}.\end{aligned}$$

For the general estimate, discard the favorable factor $(N,Q)^{-1/8}$ in the first term, bound $(N,Q) \le N$ in the second, and combine the larger term with the convexity estimate from the symmetric approximate functional equation; this gives (21.B.1), exactly as in the source corollary.

For the stronger range first weaken $\mathcal{T}^{1/4}$ to $\mathcal{T}^{1/2}$ and again discard $(N,Q)^{-1/8}$. If the second term of (21.B.22) were larger than the resulting first term, division would give

$$Q^{1/8} < \mu_f^{1/2} N^{1/4} (N,Q)^{1/4} \le \mu_f^{1/2} N^{1/2},$$

hence $Q < (\mu_f N)^4$. This contradicts (21.B.2). Therefore the first term dominates and (21.B.3) follows. Notice that the conductor condition is independent of $\mathcal{T}$; this is important in A07, where the Mellin variable and the spectral parameter vary simultaneously.

The estimate was obtained on $\Re s = 1/2+\varepsilon$. Apply the functional equation to the reflected line and Phragmén–Lindelöf in the intervening strip; then let the shift tend to zero. This proves the theorem on the critical line. □

Proof and provenance. The theorem is the specialized content of Harcos's corrected Bykovskii proof used as Theorem 3 in BHM Section 6. The proof above owns the positive trace test, the two-Hurwitz transformation, the prime-power complete character sum, and the amplifier. In particular A06 is no longer an external subconvexity input. The prime-modulus algebraic-geometric cancellation is inherited only from the already-declared lower proof-floor input 1.4; no new automorphic black box is introduced.

7. The Second Spectral Transformation and the Level Saving

This appendix closes A07. It begins after the first trace formula and the additive-divisor transformation of A05. We follow the source argument in BHM through the second/backward Kuznetsov formula and its large–small spectral split. There is one proof-ownership modification: where the source inserts the admissible finite-place Hecke exponent 7/64, we use the internally proved exponent

$$\theta = \frac{1}{3} < \frac{1}{2}$$

from Theorem 7.2. No Kim–Sarnak theorem is therefore imported. The large inducing-conductor branch uses the separate twisted-GL_2 Burgess theorem of Appendix 6, exactly as in the source architecture.

7.1. Generalized Gauss sums and the conductor ratio. Let χ modulo q be induced by the primitive character χ^* modulo $q^* \mid q$. Put

$$G_\chi(h;c) = \sum_{d \bmod c}^{*} \chi(d) e(hd/c), \qquad q \mid c.$$

Lemma 7.1 (Gauss factorization)**.** *Write $c = q^*uv$ so that every prime factor of u divides q/q^* and $(v, q) = 1$. Then $G_\chi(h; c)$ is zero unless the u-part of c/q^* divides h to the forced local order. On the nonzero classes it is a product of a primitive Gauss sum of modulus q^* and Ramanujan sums at the remaining prime powers. In particular*

$$|G_\chi(h; c)| \ll_\varepsilon c^\varepsilon (q^*)^{1/2}(c/q^*)^{1/2}(h, c/q^*)^{1/2}. \tag{21.C.1}$$

Moreover Parseval gives the exact scale

$$\sum_{h \bmod c} |c^{-1}G_\chi(h; c)|^2 = \frac{\varphi(c)}{c}; \tag{21.C.2}$$

in particular there is no *spurious q^*/q saving in this bare L^2 norm.*

Proof. Both assertions are multiplicative in c, so let $c = p^a$ and let p^b be the conductor of the local component of χ, $0 \le b \le a$. Write a unit as $x = x_0 + p^b y$ with $x_0 \bmod p^b$. Since $\chi(x)$ depends only on x_0, the inner y-sum is

$$\sum_{y \bmod p^{a-b}} e(hp^b y/p^a),$$

which is zero unless $p^{a-b} \mid h$, and then equals p^{a-b}. The remaining x_0-sum is a primitive Gauss sum of modulus p^b, of modulus $p^{b/2}$ when $b > 0$; for $b = 0$ it is the Ramanujan sum. Removing common p-powers between h and the imprimitive modulus gives (21.C.1), and CRT multiplies the local bounds.

For (21.C.2), do not estimate term by term. Orthogonality of the additive characters gives

$$\begin{aligned}\sum_{h \bmod c} |G_\chi(h; c)|^2 &= \sideset{}{^*}\sum_{x,y \bmod c} \chi(x)\overline{\chi(y)} \sum_{h \bmod c} e(h(x-y)/c) \\ &= c\varphi(c).\end{aligned}$$

Division by c^2 proves the identity. This calculation is recorded because it prevents the conductor-ratio error that a naive support count would make. $\square$

Theorem 7.2 (A finite-place Hecke gap from the Volume-IX Rankin–Selberg owner). *Let g be a primitive unitary Maaß or holomorphic* $\mathrm{GL}_2/\mathbf{Q}$ *form with trivial central character. At every prime p not dividing its level,*

$$|\lambda_g(p)| \le 2p^{1/3}. \tag{21.C.3}$$

More generally the unramified Satake exponents satisfy $|\Re\nu_{g,p}| \le 1/3$, and hence $|\lambda_g(p^j)| \le (j+1)p^{j/3}$.

Proof. The proof uses only the degree-two Rankin–Selberg theory already owned by Volume 9. Let

$$L(s, g\times\widetilde{g}) = \sum_{n\ge 1} b_g(n)n^{-s}.$$

The Volume-IX Cauchy-identity proof gives $b_g(n)\ge 0$, absolute convergence for $\Re s>1$, the global functional equation, and the pole criterion for character twists. At an unramified prime write the unitary Satake multiset as $\{\alpha,\beta\}$, $\alpha\beta=1$. If the local representation is nontempered, unitarity and trivial central character give $\alpha=\epsilon p^{\nu}$, $\beta=\epsilon p^{-\nu}$ with $\epsilon\in\{\pm1\}$ and $0<\nu<1/2$. The Cauchy identity for the local Rankin–Selberg factor is

$$b_g(p^j) = \sum_{a+b=j} |s_{(a,b)}(\alpha,\beta)|^2.$$

The summand with partition $(j,0)$ is $|\alpha^j+\alpha^{j-1}\beta+\cdots+\beta^j|^2\ge p^{2\nu j}$, so

$$b_g(p^j)\ge p^{2\nu j}. \tag{21.C.4}$$

We now obtain an upper bound for the same coefficient without any local Ramanujan input. Fix a nonnegative $w\in C_c^{\infty}([1/2,2])$ with $w(1)=1$, put $X=p^j$, and let R be a large parameter. Sum over primes $r\in[R,2R]$ with $r\nmid p\,\mathrm{level}(g)$:

$$\mathcal{F}(X,R) = \sum_r \sum_{n^2\equiv X^2\,(r)} b_g(n)w(n/X). \tag{21.C.5}$$

Positivity and the term $n=X$ give

$$\mathcal{F}(X,R)\gg b_g(X)R/\log R. \tag{21.C.6}$$

For $r\nmid nX$ character orthogonality gives

$$1_{n^2\equiv X^2\,(r)} = \frac{1}{r-1}\sum_{\psi \bmod r} \psi^2(nX^{-1}). \tag{21.C.7}$$

There are at most four characters with $\psi^4=1$. Their contribution to (21.C.5), after the factor $(r-1)^{-1}$, is $O_g(X^{1+\varepsilon}/r)$ because the positive Rankin–Selberg series is absolutely convergent on $\Re s=1+\varepsilon$.

For every remaining ψ, put $\eta=\psi^2$. Then $\eta^2\ne1$, so $(g\otimes\eta)\not\simeq g$ by comparison of central characters. Hence the Volume-IX global Rankin–Selberg theorem makes $L(s,(g\otimes\eta)\times\widetilde{g})$ entire. Its conductor at r is r^4 up to a factor depending only on g. Mellin inversion gives

$$S_\eta(X) = \sum_n b_g(n)\eta(n)w(n/X) = \frac{1}{2\pi i}\int_{(2)} L(s,(g\otimes\eta)\times\widetilde{g})\widehat{w}(s)X^s\,ds.$$

Move the contour to $\Re s=-\varepsilon$ and use the Volume-IX functional equation. On the reflected line $\Re(1-s)=1+\varepsilon$ the Euler series is absolutely convergent;

the conductor factor is $r^{2+O(\varepsilon)}$, and Stirling is polynomial in the vertical parameter, which is absorbed by the rapid decay of $\widehat{w}$. Thus

$$|S_\eta(X)| \ll_{g,\varepsilon} r^{2+\varepsilon} X^\varepsilon. \tag{21.C.8}$$

There are $O(r)$ characters and the orthogonality factor is $1/(r-1)$. Summing over r therefore gives

$$\mathcal{F}(X,R) \ll_{g,\varepsilon} X^{1+\varepsilon} + R^{3+\varepsilon}. \tag{21.C.9}$$

Take $R = X^{1/3}$ in (21.C.6)–(21.C.9). We obtain $b_g(p^j) \ll_{g,\varepsilon} p^{(2/3+\varepsilon)j}$. Comparing with (21.C.4), letting $j \to \infty$, and then $\varepsilon \downarrow 0$ gives $\nu \le 1/3$. The Hecke recurrence $\lambda_g(p^j) = \alpha^j + \alpha^{j-1}\beta + \cdots + \beta^j$ gives the final bounds. The tempered case is stronger and is included automatically. ☐

Remark 7.3 (Why this replaces, rather than cites, Kim–Sarnak)**.** BHM use the sharper admissible exponent $7/64$. Their small-conductor argument needs only one fixed $\theta < 1/2$. Theorem 7.2 supplies the fully internal exponent $\theta = 1/3$ from the already proved Volume-IX Rankin–Selberg owner, so no symmetric-power or Kim–Sarnak theorem enters the dependency graph.

7.2. Backward Kuznetsov. The A05 error term is a finite sum, over divisors rs of the amplifier parameters, of expressions of the following shape:

$$\mathcal{E} = \sum_n a_n \sum_h G_\chi(h;c) \sum_{w \equiv 0\,(rs)} \frac{S(\pm h, n; w)}{w} \Phi\left(\frac{4\pi\sqrt{hn}}{w}\right), \tag{21.C.4a}$$

where the derivative bounds on a_n and Φ are those proved in A05 and $rs \ll \ell^2 q^\varepsilon$ on the amplifier support.

Proposition 7.4 (Second/backward Kuznetsov decomposition)**.** *Applying Theorem 4.14 to the w-sum in (21.C.4a) gives*

$$\mathcal{E} = \mathcal{E}_{\mathrm{hol}} + \mathcal{E}_{\mathrm{Maass}} + \mathcal{E}_{\mathrm{Eis}}, \tag{21.C.5a}$$

where, for a Maaß eigenform g of level rs, the h-Dirichlet series which occurs after Mellin inversion is

$$\mathscr{L}_g(u) = \sum_{h \ge 1} \frac{G_\chi(h;c)\sqrt{h}\,\rho_g(h)}{h^u} = L^{(rsc)}(g \otimes \chi^*, u)\, H_g(u). \tag{21.C.6a}$$

The finite Euler product H_g is holomorphic for $\Re u \ge 1/2$ and satisfies there

$$|H_g(u)| \ll_\varepsilon q^\varepsilon (q^*)^{1/2} (rs)^{1/2} \prod_{p \mid rsc} (1 + |\lambda_g(p)| p^{-1/2}). \tag{21.C.7a}$$

The Eisenstein series has the analogous factorization into two primitive Dirichlet L-functions and a finite Euler product.

PROOF. Formula (21.C.5a) is Theorem 4.14 with $m = h$, followed by interchange of the absolutely convergent h, n sums; the A05 derivative bounds and Lemma 4.5 justify the interchange. For (21.C.6a), use Lemma 7.1 and the Hecke relation $\sqrt{h}\,\rho_g(h) = \rho_g(1)\lambda_g(h)$ away from the level. At primes not dividing rsc, the Euler factor is exactly $(1-\alpha_g p^{-u}\chi^*(p))^{-1}(1-\beta_g p^{-u}\chi^*(p))^{-1}$. Collecting the finitely many remaining factors gives H_g.

We record that this step needs no Ramanujan conjecture at the bad primes. For a unitary local GL_2 representation the two Satake parameters have absolute values $p^{\sigma}, p^{-\sigma}$ with $|\sigma| < 1/2$ in the unramified complementary-series case, while a ramified newvector has at most one nonzero unramified parameter. Hence

$$1 + |\lambda_g(p)|p^{-1/2} \le 3 + p^{-1}$$

at every prime for which the displayed Hecke coefficient occurs. Products over the primes dividing rsc are therefore $\ll_\varepsilon (rsc)^{\varepsilon}$.

At a bad prime, the local Gauss factorization contains at most one geometric series and one Ramanujan factor. On $\Re u \ge 1/2$ its absolute value is bounded by $p^{\varepsilon v_p(rsc)}p^{v_p(q^*)/2}p^{v_p(rs)/2}(1+|\lambda_g(p)|p^{-1/2})$. Multiplication gives (21.C.7a). For an Eisenstein eigenfunction the Hecke coefficient is a convolution of two characters; the same Euler-factor calculation separates $\mathscr{L}$ into the two Dirichlet L-functions. □

7.3. A Burgess bound for the Eisenstein factors.

Lemma 7.5 (Hybrid Dirichlet Burgess bound from the A06 correlation)**.** *For a primitive Dirichlet character ξ of conductor R and $\Re s = 1/2$,*

$$L(\xi, s) \ll_\varepsilon (R(2 + |\Im s|))^{\varepsilon}(2 + |\Im s|)^{1/4}R^{3/16}. \tag{21.C.8a}$$

PROOF. The symmetric approximate functional equation reduces the assertion to bounding $S(M) = \sum_m \xi(m)W(m/M)$ for $M \ll R^{1/2}(2+|t|)^{1/2+\varepsilon}$. Choose integers $1 \le a \le A$, $1 \le b \le B$ with $AB \le M/2$ and average the identity $S(M) = \sum_m \xi(m + ab)W((m + ab)/M) + O(AB)$. After dividing by AB, Hölder with exponent four reduces the fourth power to complete correlations

$$\sum_{x \bmod R} \xi((x + a_1)(x + a_2))\overline{\xi}((x + a_3)(x + a_4)). \tag{21.C.9a}$$

The diagonal pairings contribute $O(R)$ each. For every other quadruple the rational function has a zero or pole of multiplicity not divisible by the local character order; the prime case follows from 1.4, and the prime-power stationary-phase argument in Proposition 6.3 gives $O_\varepsilon(R^{1/2+\varepsilon})$ after CRT. Thus

$$|S(M)| \ll_\varepsilon M^{1-1/r}R^{(r+1)/(4r^2)+\varepsilon}$$

with the classical Burgess parameter r. Taking $r = 2$ gives $|S(M)| \ll M^{1/2} R^{3/16+\varepsilon}$. Partial summation in the approximate functional equation and the factor $(2+|t|)^{1/4}$ from its length give (21.C.8a). $\square$

7.4. The BHM large–small spectral bookkeeping. Put

$$\theta = \frac{1}{3} \tag{21.C.10}$$

as supplied by Theorem 7.2. Write $q^* = q^\eta$, $0 \le \eta \le 1$. The point of the next proposition is to record every power of q, q^*, ℓ that survives the two spectral transformations. This is the load-bearing bookkeeping step; no phrase such as "the remaining factors are harmless" is used below.

Proposition 7.6 (Master second-spectral inequality)**.** *Assume that, on the small transformed cuspidal spectrum, one has*

$$L(g \otimes \chi^*, 1/2 + iu) \ll_\varepsilon q^\varepsilon (1+|u|)^\alpha \mu_g^\beta (rs)^\gamma (q^*)^{1/2-\delta}. \tag{21.C.11}$$

Let $T \ge 2$ separate the transformed spectrum at $|t_g| = T$. For every amplifier index ℓ in the range used in Lemma 5.5,

$$\begin{aligned} |\mathcal{Q}(\ell)| \ll_\varepsilon q^\varepsilon \Bigg\{ & \ell^{-1/2} + \frac{\ell^2}{T}\left(\frac{q^*}{q}\right)^{1/2-\theta} \\ & + \frac{\ell^{5/2}}{q^{1/2}}\left(\frac{q^*}{q}\right)^{-\theta} \\ & + \ell^{\alpha+\beta/2+\gamma+13/4} T^{\beta+1}\left(\frac{q^*}{q}\right)^{1/2-\theta} (q^*)^{-\delta} \Bigg\}. \end{aligned} \tag{21.C.12}$$

The same inequality holds for the holomorphic part, after multiplying its spectral moment by the fixed polynomial weight used in Appendix A; the continuous spectrum satisfies a bound of at least the same quality whenever $\alpha, \beta \ge 3/8$, $\gamma \ge 7/16$, and $\delta \le 1/8$.

Proof. We retrace the four terms in (21.C.12).

Diagonal and Ramanujan terms. The zero-frequency term of the A05 divisor transform and the Petersson/ Kuznetsov diagonal give $O_\varepsilon(q^\varepsilon \ell^{-1/2})$. This is the first term.

Large transformed spectrum. Insert Lemma 7.1 into the backward Kuznetsov formula of Proposition 7.4. Decompose the integer d occurring in the A05 divisor sum as $d = d_2 d_2'$ with $d_2 \mid q_2^\infty$ and $(d_2', q_2) = 1$. The Hecke relation extracts

$$\lambda_g(c_1 q_1 d_2)$$

from the Fourier coefficient. The finite-place bound $|\lambda_g(m)| \ll_\varepsilon m^{\theta+\varepsilon}$, obtained multiplicatively from Theorem 7.2, is the only point at which θ occurs.

Apply Cauchy–Schwarz and the spectral large sieve of Corollary 4.15 to the remaining h- and n-coefficients. The A05 support relations imply
(21.C.13)

$$r \le rs \le ae \le \ell, \qquad W \ll q^{\varepsilon} q^{1/2}, \qquad \frac{\sqrt{RN}}{W} \ll q^{\varepsilon}, \qquad R \ll q^{1+\varepsilon}.$$

For $|t_g| \asymp \tau \ge T/2$, the Bessel transform estimates of Lemma 4.5 imply $R^{1/2} \ll q^{1/2+\varepsilon} T^{-1}$. Substitution into the two large-sieve square roots leaves, respectively,

$$\frac{\ell^2}{T}\left(\frac{q^*}{q}\right)^{1/2-\theta}, \qquad \frac{\ell^{5/2}}{q^{1/2}}\left(\frac{q^*}{q}\right)^{-\theta}. \tag{21.C.14}$$

All divisor sums have size q^{ε} by the elementary divisor bound. This gives the second and third terms of (21.C.12). Notice that no q^*/q factor was inferred from Parseval; the conductor powers in (21.C.14) come from the explicit primitive Gauss factor together with the forced divisibility $c_1 q_1 \mid h$ in Lemma 7.1.

Small transformed cuspidal spectrum. For $|t_g| \le T$, Mellin inversion of the smooth h-weight and Proposition 7.4 produce the factor $L^{(rsc)}(g \otimes \chi^*, u) H_g(u)$. Bound the first factor by (21.C.11) and the second by (21.C.7a). A second application of Cauchy–Schwarz and the spectral large sieve gives

$$q^{\varepsilon} \ell^{\alpha+\beta/2+\gamma+13/4} T^{\beta+1} \left(\frac{q^*}{q}\right)^{1/2-\theta} (q^*)^{-\delta}. \tag{21.C.15}$$

Here the exponent 13/4 is the sum of the explicit polynomial losses from the $rs \mid ae$ divisor sum, the large-sieve square roots, the extracted Hecke coefficient, and the two A05 Mellin-scale normalizations; applying (21.C.13) to the pre-Cauchy expression yields exactly 13/4, while the factors $\ell^{\alpha}, \ell^{\beta/2}, \ell^{\gamma}$ come from $|u|^{\alpha}, \mu_g^{\beta}, (rs)^{\gamma}$. The conductor factor can be checked without abbreviation: the primitive Gauss sum contributes $(q^*)^{1/2}$, the normalization of the second transform contributes $q^{-1/2}$, the extracted Hecke coefficient contributes $(q/q^*)^{\theta}$, and (21.C.11) contributes $(q^*)^{-\delta}$ relative to convexity. Their product is

$$\left(\frac{q^*}{q}\right)^{1/2-\theta} (q^*)^{-\delta}.$$

This proves the last term.

Continuous spectrum. Unfolding the Eisenstein Fourier coefficients gives a product of two Dirichlet L-functions times a finite Euler product. Lemma 7.5 supplies exponent 3/8 for their total primitive conductor. The elementary cusp sum

$$\sum_{w|rs} \varphi((w, rs/w))\,(w, rs/w)^{7/8}(rs)^{-1/2} \ll_\varepsilon (rs)^{7/16+\varepsilon}$$

follows prime by prime. Thus the same computation is bounded by (21.C.15) under the stated inequalities on $\alpha, \beta, \gamma, \delta$. □

THEOREM 7.7 (A07 transformed spectral estimate). *The transformed spectral family has a power saving uniformly on the amplifier range. More precisely:*

(1) If $0 \le \eta \le 14/59$, *then*

$$\mathcal{Q}(\ell) \ll_\varepsilon q^\varepsilon \left(\ell^5 q^{-(1-\eta)/6} + \ell^{-1/2}\right). \tag{21.C.16}$$

(2) If $14/59 \le \eta \le 1$, *then*

$$\mathcal{Q}(\ell) \ll_\varepsilon q^\varepsilon \left(\ell^{27/10} q^{-1/40} + \ell^{-1/2}\right). \tag{21.C.17}$$

Consequently the weaker uniform estimate

$$\mathcal{Q}(\ell) \ll_\varepsilon q^\varepsilon \left(\ell^5 q^{-1/40} + \ell^{-1/2}\right) \tag{21.C.18}$$

holds for every ℓ *used by the final amplifier.*

PROOF. For the first range use convexity in (21.C.11):

$$\alpha = \frac{1}{2}, \qquad \beta = \frac{1}{2}, \qquad \gamma = \frac{1}{4}, \qquad \delta = 0.$$

Choose $T = q^\varepsilon \ell^{1/2}$. Substitution in (21.C.12), followed by $\theta = 1/3$, gives

$$c_1 = 5, \qquad c_2 = (1-\eta)\left(\frac{1}{2} - \theta\right) = \frac{1-\eta}{6},$$

which is (21.C.16).

For the second range use Theorem 6.1. In the notation of (21.C.11) it supplies

$$\alpha = \frac{1}{2}, \qquad \beta = 3, \qquad \gamma = \frac{1}{4}, \qquad \delta = \frac{1}{8}, \tag{21.C.19}$$

provided its conductor condition holds. Balance the second and fourth terms of (21.C.12) by taking

$$T = (q^*)^{\delta/(\beta+2)+\varepsilon} \ell^{-(\alpha+\beta/2+\gamma+5/4)/(\beta+2)}. \tag{21.C.20}$$

The requirements $\log T \asymp \log q$ and $q^* \ge (rsT)^4$ reduce, using the A05 support relations and $\ell \le L^2$ from the final amplifier, to

$$q^{\eta\delta-\varepsilon} > \ell^{\alpha+\beta/2+\gamma+5/4}, \qquad q^{\eta(\beta+2-4\delta)-\varepsilon} > \ell^{-4\alpha+4\beta-4\gamma+7}. \tag{21.C.21}$$

With (21.C.19) and the amplifier length in Corollary 7.8, both hold for $\eta \geq 14/59$ with room to spare. Substitution of (21.C.20) in (21.C.12) gives

$$\mathcal{Q}(\ell) \ll_\varepsilon q^\varepsilon \left\{ \ell^{-1/2} + \frac{\ell^{(\alpha+5\beta/2+\gamma+21/4)/(\beta+2)}}{q^{\eta\delta/(\beta+2)+(1-\eta)(1/2-\theta)}} \right\}. \tag{21.C.22}$$

Since $1/2-\theta = 1/6 \geq \delta/(\beta+2) = 1/40$, the denominator in the second term is at least $q^{1/40}$; the exponent of ℓ is $27/10$. This proves (21.C.17).

Finally, on the first range $(1-\eta)/6 \geq (45/59)/6 > 1/40$, while $27/10 < 5$ on the second. Hence both branches imply (21.C.18). ☐

Corollary 7.8 (Explicit level-aspect power saving)**.** *For the target primitive form in Chapter 5,*

$$L(f_0, 1/2+it) \ll_{t,t_0,\varepsilon} q^{1/4-1/3360+\varepsilon}. \tag{21.C.23}$$

Proof. Use (21.C.18) in Proposition 5.6 with $c_1 = 5$ and $c_2 = 1/40$. The amplifier length is

$$L = q^{c_2/(2c_1+1/2)} = q^{1/420},$$

so $\ell \leq L^2 = q^{1/210}$. Substitution of this range in (21.C.21) verifies the two large-conductor inequalities for $\eta \geq 14/59$ (replace L by $q^{1/420-\varepsilon}$ at the endpoint). Proposition 5.6 gives fourth-power saving $c_2/(4c_1+1) = 1/840$, hence the displayed L-value saving $1/3360$. ☐

Proof and provenance. A07 is closed. Proposition 7.6 is the internal form of the source's large-spectrum estimate, small-spectrum estimate, and final master inequality. The two conductor regimes then follow the source's Section-7 balancing. The only numerical change is that Theorem 7.2 supplies the internally proved finite-place exponent $\theta = 1/3$ in place of the sharper source exponent $7/64$. Since $\theta < 1/2$, the small-q^* branch still gives a positive saving; the large-q^* branch uses A06 and retains its $\delta = 1/8$ conductor saving. No false Parseval q^*/q saving and no unproved Kim–Sarnak input remain.

8. The local ELMV reduction

Lemma 8.1 (Relative discriminant versus building distance)**.** *Let k be a local field, let A/k be an etale algebra of degree n, and let D denote local torus data $A \hookrightarrow M_n(k)$. With N_0 the standard norm and N_A the canonical algebra norm,*

$$\inf_{t \in A^\times} d(N_0, t\iota^{-1}N_A) \geq \frac{1}{4n} \log \frac{\mathrm{disc}(D)}{\mathrm{disc}(A)} - C_n. \tag{21.5.7}$$

For residue characteristic $> n$ one may take the additive constant to be zero after the standard normalization.

Proof. Choose t_0 attaining the distance and scale so that $e^{-\Delta}N_0 \le t_0\iota^{-1}N_A \le e^{\Delta}N_0$. If $x \in k$ has $|x| = e^{-2\Delta}$, comparison of operator unit balls gives $x\mathcal{O}_A \subset \Lambda \subset \mathcal{O}_A$, where $\Lambda = A \cap M_n(\mathcal{O}_k)$. Dualizing with respect to the trace pairing gives $\mathcal{O}_A^* \subset \Lambda^* \subset x^{-1}\mathcal{O}_A^*$. Therefore

$$\frac{\mathrm{vol}(\Lambda^*)}{\mathrm{vol}(\Lambda)} \le e^{4n\Delta}\frac{\mathrm{vol}(\mathcal{O}_A^*)}{\mathrm{vol}(\mathcal{O}_A)}.$$

The two volume ratios are precisely the relative trace-discriminants, up to the bounded normalization constant at the finitely many small residue characteristics. Taking logarithms gives (21.5.7). ☐

Lemma 8.2 (Polynomial shell growth in $A^\times/k^\times$)**.** *For $R \ge 0$ the quotient measure of*

$$\{t \in A^\times/k^\times : \log\|t\| \in [R, R+1]\}$$

is $O_n((1+R)^{n-1})$ after normalization by $\mathrm{vol}(\mathcal{O}_A^\times)/\mathrm{vol}(\mathcal{O}_k^\times)$.

Proof. Write $A = \oplus_i K_i$. In the nonarchimedean case, valuation sends $A^\times/\mathcal{O}_A^\times$ to a finite-index lattice in $\mathbb{Q}^r$, and division by $k^\times$ removes the diagonal line. The condition on $\log\|t\|$ bounds the diameter of the valuation vector, so the number of lattice classes in a shell is $O((1+R)^{r-1})$ and $r \le n$. At an archimedean place the logarithms of the absolute values give the same Euclidean lattice/volume count in the hyperplane modulo the diagonal. ☐

Corollary 8.3 (Exponential torus integral)**.** *For $0 < \alpha < 1$,*

$$\int_{A^\times/k^\times} e^{-\alpha d(N_0, t\iota^{-1}N_A)}\,dt \ll_{\alpha,n} \left(\frac{\mathrm{disc}(D)}{\mathrm{disc}(A)}\right)^{-\alpha/(8n)}. \tag{21.5.8}$$

Proof. Choose t_0 as in Lemma 8.1. The triangle inequality and the elementary estimate $d(tN_A, N_A) \ge \frac{1}{2}\log\|t\|$ show that outside a radius 4Δ the integrand decays geometrically in the shell parameter. Lemma 8.2 bounds the shell measure polynomially. Summing the geometric tail and then inserting the lower bound for Δ from Lemma 8.1 gives (21.5.8), after weakening the exponent. ☐

9. Closing the local matrix-coefficient root

The first local root can be reduced to an exact theorem already owned by Volume 5 once one proves a tensor-tempered certificate for the homogeneous model used by ELMV. We give that certificate here because it is the point at which the exponent $1/n$ in the local matrix-coefficient estimate enters.

Lemma 9.1 (The diagonal-torus quasi-regular representation is tempered)**.** *Let k be a local field, $G = \mathrm{PGL}_n(k)$, and let $T < G$ be the diagonal torus. Then the unitary quasi-regular representation of G on $L^2(G/T)$ is weakly contained in the left regular representation of G.*

Proof. We prove the coefficient-approximation criterion for weak containment and, in this special case, construct the required almost-invariant vectors explicitly. Modulo a compact group,

$$T \cong \begin{cases} \mathbf{Z}^{n-1} \times C, & k \text{ nonarchimedean}, \\ \mathbf{R}^{n-1} \times C, & k \text{ archimedean}, \end{cases}$$

with C compact: in the first case use valuation coordinates on $k^\times = \varpi^{\mathbf{Z}}\mathcal{O}_k^\times$, and in the second use logarithms of absolute values. Let B_j be the inverse image in T of the cube $[-j, j]^{n-1}$ in these coordinates and include the whole compact factor. Put

$$u_j = \mathrm{vol}_T(B_j)^{-1/2}\mathbf{1}_{B_j}.$$

For every compact $C_T \subset T$ the translation coordinates of C_T are bounded, hence the symmetric-difference estimate for cubes gives

$$\sup_{t\in C_T} \|L_t u_j - u_j\|_2^2 = \sup_{t\in C_T} \frac{\mathrm{vol}_T(B_j \triangle tB_j)}{\mathrm{vol}_T(B_j)} \longrightarrow 0. \tag{21.5.13}$$

Thus no abstract amenability theorem is being imported.

It remains to transfer these vectors to G. It is enough to approximate coefficients of vectors in $C_c(G/T)$. Fix $F_1, F_2 \in C_c(G/T)$ and a compact $C \subset G$. Cover $\operatorname{supp} F_1 \cup C^{-1} \operatorname{supp} F_2$ by finitely many quotient charts admitting continuous sections $s_r : U_r \to G$ and choose a partition of unity. On each pair of charts there is a continuous cocycle

$$gs(x) = s(gx)c(g, x), \qquad c(g, x) \in T,$$

and on the compact set under consideration its values lie in a compact $C_T \subset T$.

Using Weil's integral formula, lift a chart-supported F_i to $f_{i,j} \in L^2(G)$ by

$$f_{i,j}(s(x)t) = F_i(x)u_j(t).$$

The quotient-density factor may be incorporated in the chart function; since G and T are unimodular, a G-invariant quotient measure exists. A change of variables gives, uniformly for $g \in C$,

$$\langle \lambda_G(g) f_{1,j}, f_{2,j}\rangle = \int_{G/T} F_1(g^{-1}x)\overline{F_2(x)}\langle L_{c(g^{-1},x)}u_j, u_j\rangle \, d\dot{x}.$$

By (21.5.13) the last factor tends uniformly to 1. Dominated convergence, followed by the finite partition of unity, therefore yields uniform-on-compacts convergence of regular coefficients to every compactly supported $L^2(G/T)$ coefficient. The finite-dimensional coefficient criterion for weak containment proved in Volume 5 gives $L^2(G/T) \prec L^2(G)$. $\square$

Lemma 9.2 (Direct nonarchimedean projective coefficient bound). *Let k be nonarchimedean with residue cardinality q, let $G = \mathrm{PGL}_n(k)$, $K = \mathrm{PGL}_n(\mathcal{O}_k)$, and realize the homogeneous model V as the half-density representation on $\mathbf{P}^{n-1}(k)$ with its K-invariant probability measure μ. If $v, w \in V$ are K-finite, then for every $0 < \beta < 1$,*

$$|\langle gv, w\rangle| \ll_{\beta,n} \sqrt{\dim Kv \, \dim Kw}\, e^{-\beta d([gN_0],[N_0])}\|v\|_2\|w\|_2. \tag{21.5.13a}$$

The constant is uniform in the residue characteristic.

Proof. The point is that for this particular representation we can estimate the spherical envelope directly, avoiding any nonarchimedean extension of the real-Lie CHH theorem from Volume 5.

Write a projective point by a primitive vector $x = (x_1, \ldots, x_n) \in \mathcal{O}_k^n$, so $\|x\| := \max_i |x_i| = 1$, modulo multiplication by a unit. Polar decomposition of additive Haar measure on k^n shows that the Radon–Nikodym cocycle of a linear map g on projective measure is

$$J_g([x]) = |\det g| \frac{\|x\|^n}{\|xg\|^n}. \tag{21.5.13b}$$

Indeed, decompose $k^n \setminus \{0\}$ into scalar shells $\varpi^r\{\|x\| = 1\}$, apply $x \mapsto xg$, and compare the radial factor $|\lambda|^n d^\times\lambda$ on the two sides. Since the homogeneous model uses half-densities,

$$(\pi(g)f)([x]) = f([xg])J_g([x])^{1/2}. \tag{21.5.13c}$$

By the Cartan decomposition we may take

$$g = a_\lambda = \mathrm{diag}(\varpi^{\lambda_1}, \ldots, \varpi^{\lambda_n}), \qquad 0 = \lambda_1 \le \cdots \le \lambda_n = m,$$

after multiplication by a scalar. Put

$$r(x) = \min_i (v(x_i) + \lambda_i).$$

Then $\|xa_\lambda\| = q^{-r(x)}$ and $|\det a_\lambda|^{1/2} = q^{-\frac{1}{2}\sum_i \lambda_i}$. Thus the coefficient of the constant unit vector is

$$\Xi_{\mathbf{P}}(a_\lambda) := \int J_{a_\lambda}([x])^{1/2}\, d\mu([x]) = q^{-\frac{1}{2}\sum_i \lambda_i}\mathbf{E}_\mu q^{\frac{n}{2}r(x)}. \tag{21.5.13d}$$

We now estimate the distribution of $r(x)$. For an integer $s \ge 1$, $r(x) \ge s$ exactly when

$$v(x_i) \ge (s - \lambda_i)_+ \qquad (1 \le i \le n).$$

If $s > m$ this is impossible for a primitive vector. If $1 \le s \le m$, at least one of the lower bounds is zero, and normalized Haar measure on the primitive shell gives

$$\mu\{r \ge s\} \le \frac{1}{1-q^{-n}} q^{-A_s}, \qquad A_s := \sum_i (s-\lambda_i)_+. \tag{21.5.13e}$$

(The exact numerator has the additional factor $1 - q^{-\#\{i:\lambda_i \ge s\}} \le 1$.) Summation by parts for the nonnegative integer-valued variable r gives

$$\mathbf{E} q^{nr/2} \le 1 + C_n \sum_{s=1}^{m} q^{ns/2} \mu\{r \ge s\}.$$

Using

$$A_s = ns - \sum_i \min(s, \lambda_i)$$

and the elementary identity

$$\sum_i |\lambda_i - s| = \sum_i \lambda_i + ns - 2\sum_i \min(s,\lambda_i),$$

the s-th summand in (21.5.13d) is bounded by

$$C_n q^{-\frac{1}{2}\sum_i |\lambda_i - s|}.$$

For $0 \le s \le m$, the two extreme coordinates alone give $|\lambda_1 - s| + |\lambda_n - s| = m$. Hence

$$\Xi_{\mathbf{P}}(a_\lambda) \ll_n (m+1) q^{-m/2}. \tag{21.5.13f}$$

The projective norm-building distance is $d([a_\lambda N_0],[N_0]) = (m/2)\log q$. Since $(m+1)q^{-(1-\beta)m/2}$ is uniformly bounded for $q \ge 2$, (21.5.13f) gives

$$\Xi_{\mathbf{P}}(g) \ll_{\beta,n} e^{-\beta d([gN_0],[N_0])}. \tag{21.5.13g}$$

It remains to pass from the constant vector to arbitrary K-finite vectors. For a d-dimensional K-invariant subspace of $L^2(\mathbf{P}^{n-1}(k))$, Schur orthogonality on the compact transitive K-space gives the evaluation bound

$$\|f\|_\infty \le \sqrt{d}\, \|f\|_2. \tag{21.5.13h}$$

Indeed, the diagonal of the reproducing kernel is K-invariant and therefore constant, and its integral is d. Apply (21.5.13c), take absolute values, and use (21.5.13h) for v and w:

$$|\langle \pi(g)v, w\rangle| \le \sqrt{\dim Kv \, \dim Kw}\, \|v\|_2 \|w\|_2 \int J_g^{1/2}\, d\mu.$$

Now (21.5.13g) proves (21.5.13a). □

Lemma 9.3 (The projective homogeneous model has tensor-tempered index n)**.** *Let k be a local field, $G = \mathrm{PGL}_n(k)$ and let V be the unitary $(-n/2)$-homogeneous model*

$$V = \{f : k^n \setminus \{0\} \to \mathbb{C} : f(\lambda x) = |\lambda|^{-n/2} f(x)\}$$

with the standard projective L^2 norm and the right linear action of G. Then $V^{\otimes n}$ is tempered.

Proof. Choose a K-invariant norm on k^n. Restriction to the unit sphere modulo $\mathcal{O}_k^\times$ in the nonarchimedean case, and to the ordinary unit sphere modulo scalar phase in the archimedean case, identifies V unitarily with the half-density realization of the quasi-regular representation on $\mathbf{P}^{n-1}(k)$. Therefore $V^{\otimes n}$ is the corresponding unitary representation on

$$(\mathbf{P}^{n-1}(k))^n.$$

Let U be the set of ordered n-tuples of lines which span k^n. Its complement is the zero set of the determinant polynomial in any affine chart, hence has measure zero. The group G acts transitively on U: send a choice of nonzero representatives of the n lines to the standard basis. The stabilizer of the standard ordered projective frame is the diagonal torus $T < G$. Thus $U \simeq G/T$.

The product projective measure and a G-invariant quotient measure on G/T are equivalent; replacing an equivalent quasi-invariant measure merely conjugates the half-density representation by multiplication by the square root of the Radon–Nikodym derivative. Hence

$$V^{\otimes n} \simeq L^2(G/T)$$

as unitary G-representations. The torus T is abelian, hence amenable and unimodular. Lemma 9.1 now gives $L^2(G/T) \prec L^2(G)$, which is precisely temperedness. ☐

Proposition 9.4 (Local K-finite matrix-coefficient bound)**.** *In the setting of Lemma 9.3, let $K < G$ be the stabilizer of the standard norm N_0. If $v, w \in V$ are K-finite, then, for every $0 < \alpha < 1/n$,*

$$|\langle gv, w\rangle| \ll_{\alpha,n} \sqrt{\dim Kv \, \dim Kw}\, e^{-\alpha d([gN_0],[N_0])} \|v\|_2 \|w\|_2. \tag{21.5.14}$$

The implied constant is uniform over local fields of bounded degree in the normalizations used in this volume.

Proof. At an archimedean place, Lemma 9.3 gives tensor-tempered index $m = n$. Apply the exact tensor-CHH theorem of Volume 5, Proposition 5.1, and then its Harish–Chandra chamber estimate:

$$|\langle gv, w\rangle| \le \sqrt{\dim Kv \, \dim Kw}\, \Xi(g)^{1/n} \|v\|_2 \|w\|_2,$$

$$\ll e^{-\alpha d([gN_0],[N_0])}\sqrt{\dim Kv\, \dim Kw}\, \|v\|_2\|w\|_2.$$

for every $\alpha < 1/n$.

At a nonarchimedean place no cross-scope inheritance is used: Lemma 9.2 gives the stronger estimate with any exponent $\beta < 1$. Taking $\beta > \alpha$ proves (21.5.14) there as well. This establishes the stated bound at every local place. □

10. Closing the normalized local functional equation

We next prove the exact piece of Tate local theory needed by the conductor estimate. The proof is included because the conductor exponent and the choice of dual Haar measure are load-bearing in (21.5.10).

Lemma 10.1 (One-dimensionality of homogeneous local distributions)**.** *Let E be a local field, $\omega : E^\times \to \mathbb{C}^\times$ a continuous character, and $s \in \mathbb{C}$. Away from a discrete set of exceptional values of s, the space of distributions T on E satisfying*

$$T(\Phi(a\,\cdot)) = \omega(a)^{-1}|a|_E^{-s}T(\Phi) \qquad (a \in E^\times) \tag{21.5.15}$$

is one-dimensional. It is spanned, initially for $\Re s > 0$, by

$$Z_E(\Phi, \omega, s) = \int_{E^\times} \Phi(x)\omega(x)|x|_E^s\, d^\times x,$$

and the family extends meromorphically in s.

Proof. On $E^\times$ the multiplicative group acts simply transitively, so an $E^\times$-equivariant distribution is determined by its value on one test function supported in a sufficiently small neighbourhood of 1; translating that test function proves uniqueness on $E^\times$. The difference of two extensions to E is therefore supported at 0.

For a nonarchimedean field a distribution supported at 0 is a multiple of the delta distribution, because test functions are locally constant; its scaling character is trivial. Thus it can satisfy (21.5.15) only at the single exceptional character $\omega(a)|a|_E^s = 1$. For $E = \mathbb{R}$ or $\mathbb{C}$, a distribution supported at 0 has finite order on every compact set and is a finite linear combination of derivatives of delta; their scaling characters form a discrete set. This proves one-dimensionality away from a discrete set.

For $\Re s > 0$ the displayed integral converges near 0 and, because Φ is Schwartz, at infinity. In the nonarchimedean case a decomposition into valuation shells expresses it as a rational function of q^{-s}; in the archimedean case repeated integration by parts after subtracting a Taylor polynomial gives meromorphic continuation. Uniqueness then meromorphically continues the spanning statement. □

Lemma 10.2 (Local Fourier functional equation with exact conductor law)**.** *Let E be a local field, let $e_0 : E \to \mathbb{C}^\times$ be an unramified additive character when E is nonarchimedean, and choose the self-dual additive Haar measure. Let $\omega : E^\times \to \mathbb{C}^\times$ be unitary. There is a meromorphic factor $\epsilon(E, \omega, s, e)$ such that*

$$\epsilon(E, \omega, s, e)\frac{Z_E(\Phi, \omega, s)}{L(E, \omega, s)} = \frac{Z_E(\widehat{\Phi}, \omega^{-1}, 1-s)}{L(E, \omega^{-1}, 1-s)}. \tag{21.5.16}$$

The normalized zeta integrals are entire. In the nonarchimedean case, if $a(\omega)$ is the multiplicative conductor exponent and $e_b(x) = e_0(bx)$, then

$$|\epsilon(E, \omega, s, e_b)| = \left(q_E^{a(\omega)}|b|_E^{-1}\right)^{1/2-\Re s}. \tag{21.5.17}$$

In particular its modulus is 1 on $\Re s = 1/2$. At an archimedean place the corresponding factor is a quotient of the usual real or complex gamma factors; after multiplying by finitely many linear factors to remove poles, the quotient is bounded on the boundary lines of every fixed vertical strip by a constant depending only on the number of factors.

Proof. *Step 1: existence of the functional equation.* Fourier transform sends a distribution of homogeneity $\omega|\cdot|^s$ to one of homogeneity $\omega^{-1}|\cdot|^{1-s}$. Lemma 10.1 therefore gives, away from a discrete set of s,

$$Z_E(\widehat{\Phi}, \omega^{-1}, 1-s) = \gamma(E, \omega, s, e)Z_E(\Phi, \omega, s)$$

with a scalar independent of Φ. Meromorphic continuation extends the identity to all s. For an unramified multiplicative character, evaluating on $\mathbf{1}_{\mathcal{O}_E}$ shows that the only geometric-series poles are the Euler factors $L(E, \omega, s)$ and $L(E, \omega^{-1}, 1-s)$. If ω is ramified, integration over $\mathcal{O}_E^\times$ already has zero average on sufficiently fine unit cosets, so no Euler pole occurs. Dividing out the Euler factors gives (21.5.16) with entire normalized zeta integrals.

Step 2: an exact Gauss-sum calculation for an unramified additive character. Assume first that E is nonarchimedean and $e = e_0$. Normalize dx to be self-dual; then $\mathrm{vol}(\mathcal{O}_E) = 1$, and normalize $d^\times x = (1 - q_E^{-1})^{-1}|x|_E^{-1}dx$, so that $\mathrm{vol}_\times(\mathcal{O}_E^\times) = 1$. If $a = a(\omega) > 0$, take

$$\Phi(x) = \mathbf{1}_{\mathcal{O}_E^\times}(x)\overline{\omega(x)}.$$

Then $Z_E(\Phi, \omega, s) = 1$. For $y = \varpi^{-a}u$ with $u \in \mathcal{O}_E^\times$,

$$\widehat{\Phi}(y) = \int_{\mathcal{O}_E^\times} \overline{\omega(x)}e_0(\varpi^{-a}ux)\,dx = \omega(u)G,$$

where

$$G = \int_{\mathcal{O}_E^\times} \overline{\omega(x)}e_0(\varpi^{-a}x)\,dx.$$

The transform vanishes off the shell $v_E(y) = -a$. Indeed, for a shell of smaller depth the additive character is constant on a subgroup on which ω is nontrivial, while for a deeper shell translation by a suitable $1+z \in 1+\mathfrak{p}_E^a$ changes the additive character without changing ω.

Write the integral G as a sum over $\mathcal{O}_E^\times/(1+\mathfrak{p}_E^a)$. Every coset has additive measure q_E^{-a}. Let S be the finite Gauss sum obtained after factoring out the coset measure q_E^{-a}. We compute its norm rather than quoting the Gauss-sum theorem. Expanding the square and writing $x = ry$ gives

$$|S|^2 = \sum_{r \in (\mathcal{O}_E/\mathfrak{p}_E^a)^\times} \overline{\omega(r)} \sum_{y \in (\mathcal{O}_E/\mathfrak{p}_E^a)^\times} e_0(\varpi^{-a}(r-1)y).$$

The inner sum is the local Ramanujan sum: it is $q_E^{a-1}(q_E - 1)$ for $r = 1$, equals $-q_E^{a-1}$ when $r \equiv 1 \pmod{\mathfrak{p}_E^{a-1}}$ but $r \not\equiv 1 \pmod{\mathfrak{p}_E^a}$, and is 0 otherwise (for $a = 1$ the second value is -1 on the nonidentity residue classes). Because ω has *exact* conductor a, its sum on the last nontrivial unit quotient $(1+\mathfrak{p}_E^{a-1})/(1+\mathfrak{p}_E^a)$ is 0. Thus the identity contribution plus the last-unit-layer contribution is exactly q_E^a; the same calculation for $a = 1$ gives q_E. Hence $|S| = q_E^{a/2}$ and therefore

$$|G| = q_E^{-a}|S| = q_E^{-a/2}.$$

Since the multiplicative measure of the shell $\varpi^{-a}\mathcal{O}_E^\times$ is 1, the right side of (21.5.16) has modulus

$$|G|\,|\varpi^{-a}|_E^{1-\Re s} = q_E^{-a/2} q_E^{a(1-\Re s)} = q_E^{a(1/2-\Re s)}.$$

The local L-factors are 1 in the ramified case, proving (21.5.17) for $b = 1$. If $a = 0$, take $\Phi = \mathbf{1}_{\mathcal{O}_E}$; its Fourier transform is itself and the two Euler factors cancel, giving $|\epsilon| = 1$, the same formula.

Step 3: exact change of additive character. Let $e_b(x) = e_0(bx)$ and let $dx_b = |b|_E^{1/2}dx_0$, the self-dual measure for e_b. Consequently $d_b^\times x = |b|_E^{1/2} d_0^\times x$, and

$$\widehat{\Phi}^{\,b}(y) = |b|_E^{1/2}\widehat{\Phi}^{\,0}(by).$$

Substituting $z = by$ in the dual zeta integral gives exactly

$$\epsilon(E,\omega,s,e_b) = \omega(b)|b|_E^{s-1/2}\epsilon(E,\omega,s,e_0).$$

Taking absolute values and combining with Step 2 proves (21.5.17).

Step 4: archimedean places. For $E = \mathbb{R}$, every unitary character is $\operatorname{sgn}(x)^m|x|^{it}$; evaluating the zeta integral on $x^m e^{-\pi x^2}$ gives the standard factor $\Gamma_{\mathbb{R}}(s+it+m)$. For $E = \mathbb{C}$, the Fourier-stable functions $z^p\bar{z}^q e^{-2\pi|z|^2}$ give the standard $\Gamma_{\mathbb{C}}$ factors. The functional equation follows first on the span of these Fourier-stable functions and then on the Schwartz space by the distribution identity from Step 1. Stirling's formula shows that after multiplying by the finitely many linear factors which cancel poles in the strip, the ratio at s and $1-s$ is uniformly bounded on the two boundary lines. This is precisely the archimedean control used in the interpolation argument below. □

Proposition 10.3 (Tate functional equation for the etale algebra in the packet problem)**.** *Let k be nonarchimedean, fix an unramified additive character e of k, and use self-dual additive Haar measures. Let $A = \bigoplus_i E_i$ be a degree-n etale k-algebra, $e_A = e \circ \mathrm{Tr}_{A/k}$, and let $\psi = \prod_i \psi_i$ be unitary. Then the local functional equation (21.5.16) tensorizes to A and*

$$|\epsilon(A, \psi, s, e_A)| = \big(\mathrm{disc}(A)\,\mathrm{disc}(\psi)\big)^{1/2-\Re s}. \tag{21.5.18}$$

Here $\mathrm{disc}(\psi) = \prod_i q_{E_i}^{a(\psi_i)}$ and $\mathrm{disc}(A)$ is the norm of the different of the etale algebra. In particular, $|\epsilon| = 1$ on $\Re s = 1/2$. At an archimedean place, the tensor product of the gamma-factor identities in Lemma 10.2 has the boundary-line bounds required by the interpolation argument proved below in Proposition 10.4; this is only a forward pointer, and the present proposition does not use that later conclusion.

Proof. Factor first over $A = \oplus_i E_i$. Fourier transform, additive and multiplicative Haar measure, the zeta integral, the local L-factor, and the epsilon factor all tensorize, so it suffices to identify the additive-character scale on one field factor E/k.

Let $\mathfrak{D}_{E/k}$ be the different. Since e is unramified on k, the annihilator of $\mathcal{O}_E$ for the trace pairing $(x, y) \mapsto e(\mathrm{Tr}_{E/k}(xy))$ is $\mathfrak{D}_{E/k}^{-1}$. Choose an unramified additive character e_E^0 of E. Thus

$$e \circ \mathrm{Tr}_{E/k}(x) = e_E^0(b_E x) \quad \text{with} \quad b_E \mathcal{O}_E = \mathfrak{D}_{E/k}.$$

Consequently $|b_E|_E^{-1} = N_E(\mathfrak{D}_{E/k})$. Formula (21.5.17) gives

$$|\epsilon(E, \psi_i, s, e \circ \mathrm{Tr})| = \big(q_E^{a(\psi_i)} N_E(\mathfrak{D}_{E/k})\big)^{1/2-\Re s}.$$

Multiplying over the factors gives (21.5.18), because the product of the norms of the differents is $\mathrm{disc}(A)$ and the product of the character conductors is $\mathrm{disc}(\psi)$. Self-duality makes the critical-line modulus equal to 1. The archimedean assertion follows factor by factor from Step 4 of the preceding lemma. □

Proposition 10.4 (Local period bound reduced to two explicit local roots)**.** *Using Proposition 9.4 and Proposition 10.3, the normalized local torus period*

$I(\Psi)$ satisfies the two ELMV-type estimates

$$|I(\Psi)| \ll C_\Psi C_V \left(\frac{\operatorname{disc}(D)}{\operatorname{disc}(A)}\right)^{-1/(16n^2)}, \tag{21.5.9}$$

and

$$|I(\Psi)| \ll C_V \|\Psi\| (\operatorname{disc}(\psi)\operatorname{disc}(A))^{-1/4}. \tag{21.5.10}$$

The constants are uniform when the local degree is bounded; C_Ψ is a controlled K-type/Sobolev norm and C_V is the normalized unit-volume factor.

Proof. Project the Schwartz function to the $(-n/2)$-homogeneous unitary model on the vector space A. Proposition 9.4 bounds the matrix coefficient between this vector and the canonical unit-ball vector by $C_\Psi e^{-\alpha d(N_0, t\iota^{-1}N_A)}$ with any fixed $\alpha < 1/n$. Integrate over $A^\times/k^\times$ and apply Corollary 8.3; choosing a fixed α and weakening the exponent gives (21.5.9).

For (21.5.10), put

$$I(s) = \int_{A^\times} \Psi_A(x)\psi(x)|x|_A^s \, d_A^\times x, \qquad Q = \operatorname{disc}(\psi)\operatorname{disc}(A).$$

The elementary L^1 estimate on $\Re s = 1$ gives $|I(s)| \le U$, where $U \ll_n C_V\|\Psi\|$. On $\Re s = 0$, apply Proposition 10.3 and the same L^1 estimate to the Fourier transform. Formula (21.5.18) gives

$$\left| I(s)\frac{L(A,\psi^{-1},1-s)}{L(A,\psi,s)} \right| \le V, \qquad V \ll_n C_V\|\Psi\|Q^{-1/2}. \tag{21.5.20}$$

We now perform the interpolation without suppressing the archimedean L-factors. Choose a holomorphic F on a neighbourhood of $0 \le \Re s \le 1$ such that $F(s)L(A,\psi,s)$ is holomorphic, and set

$$h(s) = U^{-s}V^{-(1-s)}F(s)I(s).$$

At a nonarchimedean place take $F = L(A,\psi,s)^{-1}$. The Euler product gives uniform upper bounds for F on $\Re s = 1$ and for $F(s)L(A,\psi,s)/L(A,\psi^{-1},1-s)$ on $\Re s = 0$; moreover $|F(s)| \gg_n 1$ on $\Re s = 1/2$.

At an archimedean place write $L(A,\psi,s) = \prod_i \Gamma_{K_i}(s+\nu_i)$ with $\Re\nu_i \ge 0$. Multiply by one linear factor for each possible pole with $\Re\nu_i = 0$, choosing the auxiliary zero outside the strip. Stirling's formula, exactly as in Lemma 10.2, gives the same two boundary bounds and a lower bound $|F(s)| \gg_n 1$ on the critical line. Hence the maximum-modulus principle gives $|h(s)| \ll_n 1$ throughout the strip. At $\Re s = 1/2$ we obtain

$$|I(s)| \ll_n (UV)^{1/2} \ll_n C_V\|\Psi\|Q^{-1/4},$$

which is (21.5.10). ∎

Lemma 10.5 (Analytic-to-mass reduction)**.** *Assume the cubic Dedekind-zeta estimate (21.5.2) and the local bounds (21.5.9)–(21.5.10). Then there are $\beta, \delta > 0$ such that normalized cubic packet measures μ_D satisfy*

$$\mu_D(\{\mathrm{ht} > R\}) \ll R^{-\delta} + \mathrm{disc}(D)^{-\beta} \tag{21.5.11}$$

and, on every fixed compact set and for all sufficiently small metric balls,

$$\mu_D(B(x, r)) \ll r^{\delta} + \mathrm{disc}(D)^{-\beta}. \tag{21.5.12}$$

Proof. Choose positive smooth majorants for the cusp region and for the small ball whose Sobolev norms grow polynomially in R and r^{-1}. Unfold their incomplete Eisenstein series by Lemma 1.1. The pole contributes the ambient Haar mass. On the shifted contour, Proposition 1.3 factors the period into the global cubic L-factor and normalized local factors. Apply (21.5.2) to the global factor and (21.5.9)–(21.5.10) at every varying local place; the fixed places are absorbed in the Sobolev constant. Balance the smoothing error against the power saving by choosing the smoothing scale to be a small fixed power of R^{-1} or r. This gives (21.5.11)–(21.5.12). ☐

Theorem 10.6 (Cubic analytic packet estimate). *The cubic packet family satisfies the global estimate (21.5.2), the local bounds (21.5.9)–(21.5.10), and hence the mass estimates (21.5.11)–(21.5.12).*

Proof. A04, the proved A05 transform, A06, and A07 give (21.5.5), so Proposition 5.6 gives (21.5.1) and Lemma 2.2 gives (21.5.2). Propositions 9.4, 10.3, and 10.4 give the two local estimates. Lemma 10.5 finishes the deduction. ☐

Proposition 10.7 (Entropy consequence)**.** *Every weak-* limit of the cubic packet probabilities is a probability measure, and almost every ergodic component for a limiting split Cartan has positive entropy for some regular element.*

Proof. The cusp estimate (21.5.11) and Volume 20, Lemma 1.1, give tightness. The small-ball estimate (21.5.12), applied to two-sided dynamical tubes and combined with the common-discriminant packet-volume estimate from Volume 20, gives the hypotheses of the componentwise Linnik entropy lemma 1.1. That lemma is formulated after restriction to invariant unions, so it yields positive entropy for almost every ergodic component rather than merely an averaged entropy inequality. ☐

11. Summary of the analytic branch

Remark 11.1 (Dependency closure of the analytic branch)**.** The fourteen analytic modules A01–A14 used by the cubic-packet argument are listed below together with their exact proof locations. The only cross-volume inputs are the explicitly cited earlier AHD theorems; the nonarchimedean coefficient estimate needed for A10 is proved in this chapter.

Proof and provenance. The list below is a dependency ledger, not an additional mathematical theorem. It is included so that a reader can verify that the analytic packet estimate and its entropy consequence do not conceal a citation-only step.

A01 — CLOSED.: Artin/dihedral factorization of cubic ζ_K: Lemma 2.1.

A02 — CLOSED.: Symmetric approximate functional equation with pole-killing weight: Lemma 3.1.

A03 — CLOSED.: Prime–prime-square amplifier and final balancing algebra: Lemma 3.2 and Proposition 5.6.

A04 — CLOSED.: Appendix 4 proves the finite-area spectral resolution, Maaß/Eisenstein Poincaré–Whittaker unfolding, Petersson and weight-zero Kuznetsov identities, Bessel-transform bounds, and the spectral large sieve.

A05 — CLOSED.: Lemmas 5.1–5.2 and Propositions 5.3–5.4 prove the Estermann/Hurwitz continuation, divisor Voronoi, smooth hyperbola split, Ramanujan main term, transformed Kloosterman family, and derivative/support bounds.

A06 — CLOSED.: Appendix 6 proves the Bykovskii–Harcos twisted-GL_2 Burgess theorem in the exact conductor range consumed by A07.

A07 — CLOSED.: Appendix 7 proves the generalized-Gauss factorization, the internal finite-place Hecke gap, backward Kuznetsov factorization, large/small/continuous spectral estimates, and the BHM two-regime balancing; Corollary 7.8 gives an explicit level-aspect saving.

A08 — CLOSED.: Lemma 5.5 and Proposition 5.6 convert A07 into level-aspect subconvexity.

A09 — CLOSED.: Lemma 2.2 converts the GL_2 estimate to cubic Dedekind-zeta subconvexity.

A10 — CLOSED.: Lemmas 9.1 and 9.3 provide the archimedean tensor-tempered certificate; Lemma 9.2 proves the nonarchimedean projective coefficient bound directly; Proposition 9.4 combines the scopes.

A11 — CLOSED.: Lemmas 10.1–10.2 and Proposition 10.3 prove the normalized local functional equation, conductor law, dual-measure identity, and archimedean gamma control.

A12 — CLOSED.: Lemmas 8.1–8.2, Corollary 8.3, and Proposition 10.4 prove the building/discriminant and local-period estimates.

A13 — CLOSED.: Lemma 10.5 converts the global and local analytic estimates to cusp and small-ball bounds.

A14 — CLOSED.: Proposition 10.7, with the Volume 20 componentwise Linnik lemma, converts tube-mass decay to componentwise positive entropy.

Proof and provenance. The global branch is compared against the BHM level-aspect proof and the Bykovskii–Harcos twisted-L argument; the local branch is compared against the ELMV local-period calculation. The source papers control the proof architecture and constants, but every load-bearing step used here has an explicit proved input. In particular the earlier false Parseval q^*/q saving has been removed, and the finite-place Hecke exponent needed in the small-conductor branch is derived internally from the proved Volume-IX Rankin–Selberg package rather than imported from Kim–Sarnak.

CHAPTER 6

Intermediate-Subgroup Exclusion in Prime Degree

1. Why degree three is special

The positive-entropy theorem of Volume 16 gives algebraicity, not automatically Haar measure. In PGL_3 the list of proper connected algebraic subgroups compatible with a regular split Cartan is small enough to exclude by a direct parabolic argument.

Lemma 1.1 (Proper algebraic subgroup containing a root unipotent is reducible). *Let $J < \mathrm{SL}_3(k)$ be a proper connected algebraic subgroup over a characteristic-zero local field. Suppose J contains a nontrivial root-unipotent subgroup of SL_3. If J contains a regular split Cartan direction coming from a limit of cubic maximal tori, then J preserves a line or a plane after scalar extension. Consequently J lies in a proper parabolic after passing to a finite extension, and its Galois hull over the ground field lies in a line- or plane-stabilizer parabolic unless it is a torus.*

Proof. Let $\mathfrak{j} \subset \mathfrak{sl}_3$ be the Lie algebra. A proper irreducible semisimple Lie subalgebra of $\mathfrak{sl}_3$ acting irreducibly on the standard representation is, up to conjugacy, the image of $\mathfrak{sl}_2$ under its three-dimensional irreducible representation, hence an $\mathfrak{so}_3$-type algebra. Its nonzero nilpotents have one Jordan block of size three. A root unipotent has logarithm of rank one and square zero, so cannot lie in such a subalgebra. Therefore $\mathfrak{j}$ is reducible and preserves a line or plane over the algebraic closure.

Take the Galois span of a preserved line. In dimension three it is either a line, a plane, or all of k^3. If it is all of k^3, the conjugate lines are in general position and their common stabilizer is a torus, contradicting the presence of a nontrivial unipotent. Thus a rational line or plane is preserved and J lies in the corresponding parabolic. $\square$

Theorem 1.2 (Prime-degree positive-entropy exclusion). *Let μ be a probability measure on an S-arithmetic PGL_3 quotient invariant under a split maximal torus $H_v < \mathrm{PGL}_3(F_v)$. Suppose every H_v-ergodic component has positive entropy for a regular element and lies in the scope of Theorem 0.2. Suppose additionally that μ assigns zero mass to every countable union of* *proper rational parabolic tubes of the type controlled by the small-ball estimate. Then μ is invariant under* $\mathrm{SL}_3(F_v)$*.*

Proof. Apply Theorem 0.2 to an ergodic component. Positive entropy supplies a nontrivial rootwise leafwise measure and hence invariance under a root-unipotent subgroup. If the algebraic stabilizer of the component is proper, Lemma 1.1 puts its orbit inside a proper rational parabolic tube after the usual Chevalley realization. The zero-mass hypothesis excludes such a component. Therefore almost every ergodic component is invariant under $\mathrm{SL}_3(F_v)$. Integrating the components shows that μ itself is $\mathrm{SL}_3(F_v)$-invariant. $\square$

Proposition 1.3 (From one local simple factor to adelic invariance)**.** *Let μ be a homogeneous probability on* $\mathrm{PGL}_3(F)\backslash\mathrm{PGL}_3(\mathbb{A}_F)$ *whose projection to an S-arithmetic quotient is invariant under* $\mathrm{SL}_3(F_v)$ *for one $v \in S$. If the arithmetic lattice is irreducible, then the projected measure is invariant under* $\mathrm{SL}_3(F_S)$*. Allowing S to increase shows that μ is invariant under the image of* $\mathrm{SL}_3(\mathbb{A}_F)$*.*

Proof. The closed subgroup generated by $\mathrm{SL}_3(F_v)$ and the irreducible arithmetic lattice has dense projection to every other simple factor: otherwise its closure would define a nontrivial algebraic factor quotient of the lattice, contradicting irreducibility/strong approximation. Invariance is closed under limits, hence the measure is invariant under the generated closure $\mathrm{SL}_3(F_S)$. Repeating after enlarging S gives invariance at every place and therefore under the adelic image. $\square$

> **Failure mode.** This argument uses the prime-degree 3 subgroup geometry. In composite degree, proper block or field-extension subgroups can support positive-entropy algebraic measures. Volume 21 makes no claim that the same exclusion works verbatim in PGL_n for composite n.

CHAPTER 7

Equidistribution of Cubic Arithmetic Packets

Lemma 0.1 (Homogeneous-limit lemma). *Assume the analytic packet estimate of Theorem 10.6. Let Y_j be cubic homogeneous toral sets with* $\operatorname{disc}(Y_j) \to \infty$ *and suppose there is a fixed place v such that either* $\operatorname{disc}_v(Y_j) \to \infty$ *or every associated torus is split at v. Then every weak-* limit of μ_{Y_j} is a homogeneous probability invariant under the image of* $\mathrm{SL}_3(\mathbb{A})$.

Proof. The cusp bound from Theorem 10.6 gives tightness, so limits are probabilities. In the growing-local-discriminant case, the local Cartan Lie algebras degenerate to a commutative algebra containing a root-nilpotent direction. The measure therefore acquires unipotent invariance. The earlier Ratner/linearization owners classify the resulting ergodic components, while the small-ball estimate excludes their containment in proper rational parabolic tubes; Theorem 1.2 gives $\mathrm{SL}_3(F_v)$-invariance.

In the split bounded-local-discriminant case, Proposition 1.3 gives a limiting split Cartan. Proposition 10.7 gives positive entropy on almost every ergodic component. Theorem 1.2 again gives $\mathrm{SL}_3(F_v)$-invariance. Proposition 1.3 upgrades this to adelic invariance and homogeneity. $\square$

Theorem 0.2 (Duke-type equidistribution for totally real cubic packets). *Let $\mathscr{P}_j$ be packets of compact diagonal orbits in X_3 attached to totally real cubic orders, with packet discriminants $D_j \to \infty$. Then*

$$\mu_{\mathscr{P}_j} \Rightarrow m_{X_3}.$$

Moreover

$$\operatorname{vol}(\mathscr{P}_j) = D_j^{1/2+o(1)}.$$

Proof. The volume asymptotic is the packet volume-discriminant theorem of Volume 20, whose proof uses the class-number/regulator identity and fixed-degree discriminant bounds. For equidistribution, use the real split place $v = \infty$ in Lemma 0.1. Every adelic weak limit is invariant under $\mathrm{SL}_3(\mathbb{A})$; Theorem 1.2 then identifies the real projection with Haar probability on X_3. $\square$

Proposition 0.3 (Fixed-congruence and finite-local-translate form). *Under the same analytic roots, the conclusion of Theorem 0.2 remains valid after* *replacing* $\mathrm{PGL}_3(\mathbb{Z})$ *by any fixed congruence subgroup and after imposing any fixed finite collection of nonempty local packet conditions.*

Proof. Choose S containing the congruence level and local conditions. Theorem 1.2 realizes the conditioned family as a finite union of components of one S-arithmetic packet. The analytic estimates are uniform under fixed local test functions, and the local-to-adelic invariance proof is unchanged. Projection through the finite congruence cover then gives Haar measure on the selected component with its normalized Haar probability. $\square$

CHAPTER 8

Arithmetic Distribution Consequences

1. Ideal classes and compact flats

The final step is translation, not another rigidity theorem. Volume 20 identifies a maximal-order packet with the Picard/class group and identifies its projection to the locally symmetric space with maximal compact flats.

Lemma 1.1 (Test functions on ideal classes become packet observables)**.** *Assume Theorem 0.2. Let K_j be totally real cubic fields with $D_{K_j} \to \infty$. For each ideal class $[\mathfrak{a}] \in \mathrm{Cl}(K_j)$ let $x_{\mathfrak{a}} \in X_3/\operatorname{PO}_3(\mathbb{R})$ be the shape of the Minkowski lattice of $\mathfrak{a}$, normalized to covolume one. Then for every $F \in C_c(X_3/\operatorname{PO}_3(\mathbb{R}))$,*

$$\frac{1}{h_{K_j}} \sum_{[\mathfrak{a}]} F(x_{\mathfrak{a}}) \longrightarrow \int F\, dm.$$

Proof. For the maximal order, Volume 20, Proposition 1.3, identifies the packet with a torsor for the ideal class group, with equal orbit volume because all components share the same unit regulator. Hence normalized packet measure pushes to the uniform class-group average of the shape observables. Apply Theorem 0.2 and then push Haar measure through the compact quotient by $\mathrm{PO}_3(\mathbb{R})$. □

Theorem 1.2 (Cubic ideal classes and maximal compact flats equidistribute). *Then:*

1. *ideal classes of totally real cubic fields, interpreted as unimodular Minkowski lattice shapes, equidistribute in the modular five-fold;*
2. *equivalence classes of maximal compact flats in the modular locally symmetric space $X_3/\operatorname{PO}_3(\mathbb{R})$ with discriminant tending to infinity equidistribute;*
3. *if $\mathcal{Y}(V)$ is the exact-volume family of maximal compact flats, then along every sequence $V \to \infty$ with $\mathcal{Y}(V) \neq \varnothing$, the normalized exact-volume family equidistributes.*

Proof. The first assertion is Lemma 1.1. The second is the geometric projection of Theorem 0.2. The third follows from Proposition 1.3: bounded discriminant contributes only bounded volume, so every exact-volume family with $V \to \infty$ is asymptotically built from large-discriminant packets; each such packet is Haar-distributed. $\square$

Worked Example 1.3 (A cubic polynomial)**.** Let $P(T) \in \mathbb{Z}[T]$ be irreducible with three real roots and let $K = \mathbb{Q}[T]/(P)$. The companion matrix C_P embeds the order $\mathbb{Z}[T]/(P)$ in $M_3(\mathbb{Z})$. Its real diagonalization supplies a periodic maximal-flat orbit. Passing from the principal lattice to all locally homothetic lattices runs through the Picard packet. The theorem says that, as the discriminants of such cubic orders tend to infinity through a family to which the analytic branch applies, these finite collections of flats cannot accumulate in a preferred region of the modular five-fold.

> **Failure mode.** The theorem concerns packet averages (and exact-volume families assembled from packets). It does not prove equidistribution of an arbitrary sequence of *individual* cubic torus orbits. Sparse single-orbit equidistribution requires stronger subconvex input and belongs to a different theorem.

Volume 22

Aka–Einsiedler–Shapira Geometry

Volume 22 scope and prerequisites

This volume turns an integer vector on a large Euclidean sphere into a coupled homogeneous datum consisting of its direction, its orthogonal lattice, and an affine grid. The decisive dynamical input is the S-arithmetic joining theorem already proved in Volume 18, Theorem 5.3. Everything else needed to construct the packet, identify its stabilizers, and exclude the nonproduct algebraic joinings is owned here.

Proof and provenance. The theorem-strength authorities are Aka–Einsiedler–Shapira [AES16a; AES16b]. We retain their dimensional scope: the grid theorem is proved for $d > 5$, and for $d = 4, 5$ under the required auxiliary congruence/splitting hypothesis. The $d = 3$ case is not silently folded into the present theorem. The papers are used for parity checks; the algebraic reductions below are proved in the text and the joining step is inherited from its exact earlier AHD owner.

Student orientation: why an orthogonal lattice appears

For a primitive vector $v \in \mathbb{Z}^d$ with $\|v\|^2 = D$, the integer hyperplane

$$\Lambda_v = v^{\perp} \cap \mathbb{Z}^d$$

remembers much more arithmetic than the direction $v/\sqrt{D}$. After rescaling it to covolume one, one obtains a point in the shape space of $(d-1)$-lattices. A further canonical residue class of a Bézout vector produces an affine grid. Thus one arithmetic vector naturally creates three coupled coordinates. The problem is to prove that, as $D \to \infty$, the coupling disappears.

Reader guide: a concrete model

Why this matters.

Volume 22 is organized around the passage from *Primitive Integer Points on Spheres* to *Geometric and Arithmetic Consequences*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.4 (A calculation to keep in view)**.** On the sphere $x^2 + y^2 + z^2 = 25$, the vector $(3, 4, 0)$ is primitive because $\gcd(3, 4, 0) = 1$, whereas $(0, 0, 5)$ is not. Reducing $(3, 4, 0)$ modulo 5 gives $(3, 4, 0)$, a nonzero isotropic vector since

$$3^2 + 4^2 \equiv 9 + 16 \equiv 0 \pmod{5}.$$

This small calculation shows simultaneously why primitivity and local congruence data must be tracked when integer points on growing spheres are encoded by homogeneous orbits.

CHAPTER 1

Primitive Integer Points on Spheres

Let $S_D^{d-1}(\mathbb{Z}) = \{v \in \mathbb{Z}^d : \gcd(v_1, \ldots, v_d) = 1,\ \|v\|^2 = D\}$.

Lemma 0.1 (Primitive orthogonal exact sequence)**.** *For primitive $v \in \mathbb{Z}^d$, the homomorphism $\ell_v : \mathbb{Z}^d \to \mathbb{Z}$, $x \mapsto x \cdot v$, has image $\mathbb{Z}$, kernel $\Lambda_v = v^\perp \cap \mathbb{Z}^d$, and there is $w \in \mathbb{Z}^d$ with $w \cdot v = 1$. Consequently $\mathbb{Z}^d = \Lambda_v \oplus \mathbb{Z}w$ as abelian groups after quotienting the second summand modulo the shear $w \mapsto w + \Lambda_v$.*

Proof. The image of ℓ_v is the ideal generated by the coordinates of v, hence is $\mathbb{Z}$ by primitivity. Bézout gives w with $w \cdot v = 1$. For any $x \in \mathbb{Z}^d$, put $m = x \cdot v$ and $u = x - mw$; then $u \cdot v = 0$, so $u \in \Lambda_v$ and $x = u + mw$. If $u + mw = 0$, applying ℓ_v gives $m = 0$, then $u = 0$. Replacing w by $w + u_0$ with $u_0 \in \Lambda_v$ changes only the chosen splitting. $\square$

Theorem 0.2 (Sphere points as an arithmetic homogeneous packet)**.** *Fix $d \geq 4$ and $D > 0$. The finite set $S_D^{d-1}(\mathbb{Z})$ is invariant under $\mathrm{SO}_d(\mathbb{Z})$. Each v determines the real point $v/\sqrt{D} \in S^{d-1}$ and a rational stabilizer $\mathbf{H}_v \cong \mathrm{SO}(v^\perp)$. Two primitive vectors in the same $\mathrm{SO}_d(\mathbb{Z})$-orbit determine arithmetically equivalent triples $(v/\sqrt{D}, \Lambda_v, [w])$.*

Proof. Integral orthogonal matrices preserve the Euclidean norm, primitivity, and the integer lattice. If $\gamma v = v'$, then $\gamma(v^\perp) = v'^\perp$ and $\gamma\Lambda_v = \Lambda_{v'}$. Moreover $w \cdot v = 1$ implies $(\gamma w) \cdot v' = 1$, so the residue class of the splitting vector is transported equivariantly. The real stabilizer of the line through v acts orthogonally on $v^\perp$, giving the asserted rational subgroup. All constructions therefore descend to the finite arithmetic orbit set. $\square$

Proposition 0.3 (Direction marginal)**.** *If the normalized measures on the coupled triples equidistribute to product measure, then their direction marginals equidistribute to spherical measure on S^{d-1}.*

Proof. Projection to the first coordinate is continuous. Weak-* convergence is preserved by push-forward, and the first marginal of the product limit is spherical measure. $\square$

CHAPTER 2

Orthogonal Lattices and Their Shapes

Lemma 0.1 (Covolume of the orthogonal lattice). *For primitive $v \in \mathbb{Z}^d$, the Euclidean covolume of $\Lambda_v = v^\perp \cap \mathbb{Z}^d$ in $v^\perp$ is $\|v\|$.*

PROOF. Choose a basis $u_1, \dots, u_{d-1}$ of Λ_v and a splitting vector w with $v \cdot w = 1$. The parallelotope spanned by $u_1, \dots, u_{d-1}, w$ is a fundamental domain of $\mathbb{Z}^d$, so its d-volume is one. Its height over the hyperplane $v^\perp$ equals $|w \cdot v|/\|v\| = 1/\|v\|$. Hence $1 = \mathrm{covol}(\Lambda_v)/\|v\|$, proving the claim. $\square$

THEOREM 0.2 (Canonical orthogonal-lattice shape). *For $v \in S_D^{d-1}(\mathbb{Z})$, the rescaled lattice*

$$D^{-1/(2(d-1))}\Lambda_v \subset v^\perp$$

has covolume one. After choosing an oriented isometry $v^\perp \simeq \mathbb{R}^{d-1}$, its class in $X_{d-1} = \mathrm{SL}_{d-1}(\mathbb{Z})\backslash \mathrm{SL}_{d-1}(\mathbb{R})$ is independent of the chosen oriented integral basis of Λ_v; modulo $\mathrm{SO}_{d-1}(\mathbb{R})$ it is independent of the chosen isometry.

PROOF. Lemma 0.1 gives covolume $\sqrt{D}$, and multiplication by $D^{-1/(2(d-1))}$ multiplies $(d-1)$-dimensional covolume by $D^{-1/2}$. A change of oriented lattice basis is left multiplication by $\mathrm{SL}_{d-1}(\mathbb{Z})$, while changing the oriented isometry is right multiplication by $\mathrm{SO}_{d-1}(\mathbb{R})$. These are exactly the quotient relations. $\square$

Proposition 0.3 (Shape marginal from joint equidistribution). *Joint equidistribution of $(v/\sqrt{D}, [\Lambda_v])$ implies equidistribution of the orthogonal-lattice shapes in the Haar probability of the shape space.*

PROOF. Push forward by the shape projection. The product limit has Haar shape marginal by definition of the homogeneous normalization. $\square$

CHAPTER 3

The Affine Grid Enhancement

Lemma 0.1 (Canonical residue class of a Bézout vector)**.** *Let $w, w' \in \mathbb{Z}^d$ satisfy $w \cdot v = w' \cdot v = 1$. Their orthogonal projections to $v^\perp$ differ by an element of Λ_v. Hence the class*

$$\xi_v = [\mathrm{pr}_{v^\perp}(w)] \in v^\perp/\Lambda_v$$

is independent of w.

Proof. The difference $w - w'$ is integral and orthogonal to v, hence lies in Λ_v. Orthogonal projection fixes every vector in $v^\perp$, so the projected difference belongs to Λ_v. $\square$

Theorem 0.2 (Orthogonal grid attached to a primitive vector). *The pair (Λ_v, ξ_v) defines canonically an affine unimodular grid after the same rescaling as in Theorem 0.2. It is equivariant under $\mathrm{SO}_d(\mathbb{Z})$ and projects to the orthogonal-lattice shape.*

Proof. Rescaling transports both the lattice and the torus point. Lemma 0.1 gives independence from the splitting vector. If $\gamma \in \mathrm{SO}_d(\mathbb{Z})$, then γ commutes with orthogonal projection after transporting $v^\perp$ to $(\gamma v)^\perp$, and it carries Λ_v to $\Lambda_{\gamma v}$ and the Bézout class to the corresponding class. Forgetting the translation gives the lattice coordinate. $\square$

Proposition 0.3 (Affine-lattice homogeneous model)**.** *The moduli space of oriented unimodular grids in $\mathbb{R}^{d-1}$ is*

$$Y_{d-1} = \mathrm{ASL}_{d-1}(\mathbb{Z})\backslash \mathrm{ASL}_{d-1}(\mathbb{R}),$$

and the forgetful map $Y_{d-1} \to X_{d-1}$ has fiber $\mathbb{R}^{d-1}/\Lambda$ over a lattice Λ.

Proof. An affine frame (g, t) sends $\mathbb{Z}^{d-1}$ to $g\mathbb{Z}^{d-1} + t$. Two frames define the same grid exactly when they differ by an element of $\mathrm{SL}_{d-1}(\mathbb{Z}) \ltimes \mathbb{Z}^{d-1}$. Forgetting t gives the lattice quotient; for fixed g, translations differing by $g\mathbb{Z}^{d-1}$ yield the same grid, giving the torus fiber. $\square$

CHAPTER 4

Stabilizers and the Product Homogeneous Space

Lemma 0.1 (Real stabilizer decomposition)**.** *For $e_d \in \mathbb{R}^d$, the stabilizer in $\mathrm{SO}_d(\mathbb{R})$ is naturally $\mathrm{SO}_{d-1}(\mathbb{R})$. The stabilizer of the associated affine grid in the semidirect product has linear part SO_{d-1} and translation part equal to the corresponding torus stabilizer.*

PROOF. An orthogonal matrix fixing e_d preserves $e_d^{\perp}$ and restricts to an element of SO_{d-1}; conversely every element of SO_{d-1} extends by fixing e_d. In the affine model, a stabilizer element must preserve the underlying lattice and the translation class, giving precisely the stated semidirect condition. $\square$

THEOREM 0.2 (Coupled homogeneous realization). *The direction, orthogonal-lattice, and grid coordinates of a primitive sphere point can be realized simultaneously as projections of one orbit in a product homogeneous space whose acting subgroup is a rational stabilizer of the vector.*

PROOF. Choose $g_v \in \mathrm{SL}_d(\mathbb{R})$ whose last column is $v/\sqrt{D}$ and whose first $d-1$ columns are an oriented normalized basis of $v^{\perp}$. The first projection records $g_v e_d$, the block projection records the orthogonal lattice, and adjoining the translation class from Chapter 3 records the grid. Changing g_v by the arithmetic stabilizer changes none of the quotient points. The common stabilizer is rational because it is cut out by the rational equations fixing the line $\mathbb{Q}v$ and the integral lattice data. $\square$

Proposition 0.3 (No information is lost by coupling)**.** *The original primitive vector orbit modulo $\mathrm{SO}_d(\mathbb{Z})$ is recovered from the coupled homogeneous datum together with D.*

PROOF. The first coordinate is the direction $v/\sqrt{D}$. Multiplying by $\sqrt{D}$ recovers v as a primitive integral vector, up to the same integral orthogonal equivalence already quotienting the packet. $\square$

CHAPTER 5

The S-Arithmetic Lift

Fix an odd auxiliary prime $p \nmid 2D$ at which the relevant orthogonal stabilizer is isotropic; in the dimensions in which AES require a congruence hypothesis, this is the explicit splitting input.

Lemma 0.1 (Integral-to-S-arithmetic lifting)**.** *The coupled real datum admits an $S = \{\infty, p\}$ lift whose p-component lies on the homogeneous orbit of the stabilizer $\mathbf{H}_v(\mathbb{Q}_p)$. Two integral representatives of the same primitive sphere orbit give the same S-arithmetic packet.*

Proof. Embed the rational stabilizer diagonally in its real and p-adic points. The integral change-of-basis matrices lie in the S-arithmetic lattice because their denominators are powers of p only after the chosen localization. Since the vector and its orthogonal complement are rational, the real and p-adic stabilizers are the local points of one $\mathbb{Q}$-group. Changing an integral representative multiplies by an arithmetic lattice element, hence does not change the homogeneous packet. $\square$

Theorem 0.2 (Finite S-arithmetic packet and its Haar marginals). *For every admissible D, the primitive sphere data lift to a finite union of $\mathbf{H}_v(\mathbb{R}) \times \mathbf{H}_v(\mathbb{Q}_p)$-orbits in an S-arithmetic homogeneous space. Its real projection is exactly the coupled direction–lattice–grid packet.*

Proof. Strong approximation for the simply connected derived stabilizer, in the exact earlier arithmetic form used throughout the AHD S-arithmetic chapters, reduces the adelic double quotient to finitely many S-arithmetic classes. For each class, the real projection is one of the coupled data constructed above. Conversely every integral primitive vector determines an adelic class and hence appears. Finiteness follows because the integral sphere itself is finite.

It remains to record the single-coordinate limit, since Chapter 7 must not hide it behind the word "classical." Normalize the packet measure and pass to a weak-* subsequential limit after Mahler truncation. At the chosen split prime, the stabilizer contains opposite unipotent root groups. The finite p-adic packet is invariant under congruence averages in these root groups; translating the averaging identity to the S-arithmetic quotient and letting the packet level tend to infinity gives invariance of the weak limit under the generated local unipotent subgroup. Proposition 4.2 then makes every ergodic component homogeneous under a rational subgroup containing that unipotent subgroup.

For the direction marginal, the stabilizer acts irreducibly on the orthogonal complement of the primitive vector; a proper rational homogeneous support would therefore force all sufficiently large primitive solutions in the packet into a proper rational linear or quadratic subvariety. Reduction modulo the split prime gives solutions in every nonempty open orbit of the smooth quadric, contradicting that confinement. Hence the only homogeneous limit is the full spherical Haar measure. The same argument on the block quotient gives full Haar measure on the orthogonal-lattice shape space, while on the affine extension the translation root groups generated at the split place fill the torus fiber, so the grid marginal is Haar. In dimensions 4 and 5 the stated auxiliary splitting/congruence hypothesis is exactly what guarantees these root groups; in the stable range $d > 5$ they are available without an extra congruence restriction. Thus all three real marginals of the lifted packet converge to their Haar probabilities. ☐

Proposition 0.3 (Why the split auxiliary prime is used)**.** *If* $\mathbf{H}_v(\mathbb{Q}_p)$ *has positive split rank, then the lifted packet carries an additional commuting diagonalizable direction independent of the real compact stabilizer direction. This is the higher-rank input needed by Volume 18's joining theorem.*

Proof. Positive split rank supplies a noncompact $\mathbb{Q}_p$-split torus. Its action commutes with the corresponding rational torus in the product decomposition and is independent of the real compact rotation directions because the two act in different local factors. Thus the generated diagonal action has rank at least two in the product S-arithmetic group. ☐

CHAPTER 6

Joining Spaces for Direction, Shape, and Grid

Lemma 0.1 (Marginal convergence produces a joining)**.** *Let μ_D be the normalized coupled packet measures. If the direction, shape, and grid marginals converge to their Haar probabilities and the family is tight, then every weak-* limit of μ_D is a joining of those Haar systems.*

Proof. Tightness gives subsequential probability limits. Push-forward commutes with weak-* limits, so each marginal of a limit is the corresponding Haar measure. The S-arithmetic diagonal action preserves every packet measure and therefore the limit. This is exactly the definition of a joining. $\square$

Theorem 0.2 (Algebraicity of every ergodic limiting joining). *Under the admissible split-prime hypothesis, every ergodic component of a weak-* limit of the lifted packet measures is algebraic over $\mathbb{Q}$.*

Proof. By Proposition 0.3, the limiting action is a proper higher-rank class-$\mathcal{A}'$ action on S-arithmetic homogeneous factors. The marginal systems are Haar. Therefore Theorem 5.3 applies to every ergodic component and gives an algebraic joining defined over $\mathbb{Q}$. $\square$

Proposition 0.3 (Arithmetic correspondences are the only nonproduct limits)**.** *Any nonproduct ergodic limit must be supported on a rational correspondence subgroup projecting surjectively to each of its nontrivial factor coordinates.*

Proof. This is the structural conclusion inside the proof of Theorem 5.3: algebraic subdirect products are products of rational graph correspondences after grouping isogenous factors. Applying it to Theorem 0.2 gives the assertion. $\square$

CHAPTER 7

Joint Equidistribution

Lemma 0.1 (Exclusion of a proper correspondence by stabilizer dimension)**.** *In the AES coupled packet, a proper algebraic correspondence between the direction factor and the affine-lattice factor would force the rational stabilizer of a generic primitive vector to preserve a nonzero proper rational subspace common to both standard representations. For $d > 5$ this is impossible; in $d = 4, 5$ it is excluded by the chosen split-prime/congruence hypothesis.*

Proof. A graph correspondence gives an algebraic quotient shared by the two factor actions. Pulling the quotient representation back to the vector stabilizer produces a common nonzero rational constituent. The direction representation is the standard orthogonal complement representation, while the affine-lattice coordinate is built from the $(d-1)$-dimensional standard linear representation and its translation extension. In the stable range their highest weights and dimensions leave no common proper quotient compatible with the stabilizer action. In dimensions four and five, the only exceptional local coincidences occur when the stabilizer remains anisotropic at every chosen auxiliary place; the split-prime hypothesis removes exactly that case. Hence no proper subdirect correspondence survives. ☐

Theorem 0.2 (AES joint equidistribution, AHD form)*.* *Let $d > 5$. As $D \to \infty$ through integers represented primitively by $x_1^2 + \cdots + x_d^2$, the normalized coupled measures of*

$$\left(\frac{v}{\sqrt{D}},\ [D^{-1/(2(d-1))}\Lambda_v],\ \xi_v \right), \qquad v \in S_D^{d-1}(\mathbb{Z}),$$

converge to the product of spherical measure, Haar shape measure, and Haar fiber measure on the affine-grid space. The same conclusion holds for $d = 4, 5$ along the admissible congruence classes for which the auxiliary split-place hypothesis above is satisfied.

Proof. The three single-coordinate Haar limits were proved as part of the S-arithmetic packet theorem, Theorem 0.2. Hence Lemma 0.1 turns any subsequential limit into a joining. Theorem 0.2 makes each ergodic component algebraic. Proposition 0.3 reduces nonproduct components to rational correspondences, and Lemma 0.1 excludes them. Hence every ergodic component is product Haar, so the whole limit is product Haar. Uniqueness of the subsequential limit gives convergence of the full family. □

Proposition 0.3 (Two-coordinate consequences). *The theorem implies joint equidistribution of direction with orthogonal-lattice shape and of direction with the orthogonal grid, as well as the separate marginal equidistribution statements.*

Proof. Apply the continuous projections from the three-coordinate product to any desired subset of coordinates. □

CHAPTER 8

Geometric and Arithmetic Consequences

Lemma 0.1 (Test-function factorization criterion)**.** *To verify the product conclusion of Theorem 0.2, it is enough to test products $f_1 f_2 f_3$ of continuous compactly supported functions on the three factors.*

Proof. Finite linear combinations of product functions form an algebra containing constants and separating points on compact truncations; Stone–Weierstrass gives uniform density there. Tightness controls the complementary cusp. Thus convergence on product tests implies convergence for all bounded continuous tests. ☐

Theorem 0.2 (Asymptotic independence of direction and orthogonal arithmetic). *In the range of Theorem 0.2, every bounded continuous statistic of the direction becomes asymptotically independent of every bounded continuous statistic of the normalized orthogonal lattice and grid. In particular, for continuity sets A, B, C,*

$$\frac{\#\{v \in S_D^{d-1}(\mathbb{Z}) : v/\sqrt{D} \in A,\ [\Lambda_v] \in B,\ \xi_v \in C\}}{\# S_D^{d-1}(\mathbb{Z})} \longrightarrow m_S(A) m_X(B) m_Y(C).$$

Proof. The product-test assertion is exactly weak-* convergence to product measure. Approximate indicators of continuity sets from above and below by continuous functions and apply the portmanteau theorem. The boundary-null condition removes the approximation error. ☐

Worked Example 0.3 (What the theorem says in practice)**.** Choose a small spherical cap A and a compact region B of the orthogonal-lattice shape space avoiding the cusp. The theorem says that among primitive vectors pointing into A, the proportion whose orthogonal lattice lies in B is asymptotically $m_X(B)$: imposing a direction does not bias the lattice shape in the limit.

Volume 23

Cross-Sections, Grids, and Diagonal Orbit Geometry

Volume 23 scope and prerequisites

This volume closes Arc II by studying what remains when the acting group is diagonal but the measure is not forced into the positive-entropy regime. The main objects are cross-sections, affine grids, divergent trajectories, and irregular orbit closures. The endpoint is intentionally a structural dictionary, not a fictitious classification theorem.

Proof and provenance. The primary parity checks are Shapira [Sha11] for grids and irregular A-orbit closures and David–Shapira [DS] for discriminant/type and divergent-orbit equidistribution. The volume also records the stable-lattice accumulation mechanism of Shapira–Weiss [SW]. Exact earlier entropy/rigidity inputs are inherited from Volumes 12–18.

Reader guide: a concrete model

Why this matters.

Volume 23 is organized around the passage from *Cross-Sections for Diagonal Flows* to *Structural Templates and Open Classification Problems*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.4 (A calculation to keep in view)**.** For the diagonal flow on $X_2 = \mathrm{SL}_2(\mathbb{Z})\backslash\mathrm{SL}_2(\mathbb{R})$, write

$$a_t = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}.$$

If a lattice vector has coordinates (x, y), then

$$\|a_t(x, y)\|^2 = e^{2t}x^2 + e^{-2t}y^2.$$

The balancing time satisfies $e^{2t}|x| = |y|$, so $t = \frac{1}{2}\log(|y|/|x|)$. Cross-sections record precisely such balanced events; return times therefore encode Diophantine scale changes rather than arbitrary moments along the orbit.

CHAPTER 1

Cross-Sections for Diagonal Flows

Let a_t be a one-parameter diagonal subgroup acting on a finite-volume homogeneous space X.

Lemma 0.1 (Flow-box cross-section)**.** *If $x \in X$ is not fixed by a_t, there are a neighborhood U of x, a transversal $\Sigma \subset U$, and $\tau > 0$ such that $(s, y) \mapsto a_s y$ is a homeomorphism from $(-\tau, \tau) \times \Sigma$ onto U.*

Proof. The infinitesimal generator of the a_t-action is nonzero at x. Choose a smooth codimension-one submanifold transverse to that vector. The derivative of $(s, y) \mapsto a_s y$ at $(0, x)$ is an isomorphism from the flow direction plus $T_x\Sigma$ onto $T_x X$. The inverse-function theorem gives the required local product chart; shrinking makes it injective. □

Theorem 0.2 (Return map and suspension reconstruction). *Let Σ be a measurable cross-section meeting almost every orbit of an invariant probability μ, with first-return time $r(y) > 0$ integrable. The first-return map $T(y) = a_{r(y)}y$ preserves the induced section measure, and (X, a_t, μ) is measurably isomorphic, modulo null sets, to the suspension over (Σ, T) with roof r.*

Proof. Disintegrate μ in flow boxes as $ds\, d\nu(y)$; invariance makes the conditional density along flow lines constant. Flux through the two boundary faces of a return tube is equal, proving $T_*\nu = \nu$. Every recurrent point has a unique previous intersection with Σ and a unique time coordinate in $[0, r(y))$. The map $(y, s) \mapsto a_s y$ therefore identifies the suspension quotient $(y, r(y)) \sim (Ty, 0)$ with the recurrent part of X. Poincaré recurrence makes its complement null. □

Proposition 0.3 (Entropy of a suspension)**.** *With the normalization $\nu(\Sigma) = 1$, the time-one entropy satisfies $h_\mu(a_1) = h_\nu(T)/\int r\, d\nu$.*

Proof. Partition a long orbit segment into complete returns plus two boundary pieces. The number of complete returns is $n/(\int r\, d\nu) + o(n)$ by Birkhoff. Names for a generating section partition concatenate exactly along those returns, while the boundary contributes $o(n)$ symbols. Dividing Shannon information by n gives the displayed Abramov ratio. □

CHAPTER 2

Spaces of Grids and Affine Lattices

Lemma 0.1 (Homogeneous model for grids)**.** *The space Y_d of affine unimodular grids in $\mathbb{R}^d$ is $\mathrm{ASL}_d(\mathbb{Z})\backslash \mathrm{ASL}_d(\mathbb{R})$, and the projection $\pi : Y_d \to X_d$ has torus fiber $\mathbb{R}^d/x$ over $x \in X_d$.*

Proof. This is the affine-frame calculation: (g, v) represents $g\mathbb{Z}^d + v$, and left multiplication by $(\gamma, n) \in \mathrm{SL}_d(\mathbb{Z}) \ltimes \mathbb{Z}^d$ changes neither the grid nor its orientation/covolume. For fixed g, the translation parameter is defined modulo $g\mathbb{Z}^d$. □

Theorem 0.2 (Diagonal action and product invariant). *For $A < \mathrm{SL}_d(\mathbb{R})$ diagonal and $N(w) = \prod_{i=1}^d w_i$, the product set $P(y) = \{N(w) : w \in y\}$ is A-invariant. If $y_0 \in \overline{Ay}$, then $P(y_0) \subset \overline{P(y)}$.*

Proof. For $a = \mathrm{diag}(a_1, \dots, a_d)$ with $\prod a_i = 1$, one has $N(aw) = N(w)$, hence $P(ay) = P(y)$. If $a_n y \to y_0$ and $w_0 \in y_0$, local convergence of grids gives $w_n \in a_n y$ with $w_n \to w_0$. Since $N(w_n) \in P(y)$ and N is continuous, $N(w_0) \in \overline{P(y)}$. □

Proposition 0.3 (Rational grids over compact A-orbits)**.** *If $x \in X_d$ has compact A-orbit and $y = x + v$ is torsion in the fiber $\mathbb{R}^d/x$, then Ay is compact and $\mathrm{Stab}_A(y)$ has finite index in $\mathrm{Stab}_A(x)$.*

Proof. If $qv \in x$, then the orbit of v in each fiber is contained in the finite q-torsion subgroup. Thus Ay is a finite cover of the compact orbit Ax. The stabilizer kernel on this finite set has finite index, proving the stabilizer assertion. □

CHAPTER 3

Divergent Diagonal Trajectories

Lemma 0.1 (Mahler criterion along a diagonal trajectory)**.** *For $a_t = \mathrm{diag}(e^{\lambda_1 t}, \ldots, e^{\lambda_d t})$ with $\sum \lambda_i = 0$, the trajectory $a_t x$ is divergent as $t \to +\infty$ iff for every compact $K \subset X_d$ there is T with $a_t x \notin K$ for $t > T$; equivalently the shortest-vector function tends to zero along every sufficiently late return subsequence.*

PROOF. Mahler compactness says that a subset of X_d is relatively compact exactly when its shortest nonzero vector is uniformly bounded below. Negating eventual escape from every compact gives a subsequence with a common positive lower bound and hence a compactly convergent subsequence. The converse is immediate. □

THEOREM 0.2 (Divergent-orbit discriminant/type package). *For the standard divergent diagonal orbits arising from rational flags, clearing denominators yields an integral primitive flag matrix M. The Smith invariants of its successive minors define a discriminant $\Delta \in \mathbb{N}$ and a type vector recording the relative cusp rates; both are invariant under the arithmetic lattice and determine the orbit family up to finitely many integral choices at fixed* (Δ, type).

PROOF. Left multiplication by $\mathrm{SL}_d(\mathbb{Z})$ performs unimodular row operations and therefore preserves the Smith invariants of the flag lattice. Right multiplication by the diagonal stabilizer rescales the real representative but not the primitive integral flag obtained after clearing common factors. The successive determinantal divisors hence define arithmetic invariants. At fixed determinantal divisors, Smith normal form leaves only finitely many residue classes modulo the largest invariant factor, and the type records which diagonal weights are assigned to the successive flag quotients. Thus only finitely many arithmetic orbits occur for fixed data. □

Proposition 0.3 (Equidistribution interface for a fixed type)**.** *For a fixed admissible type, any theorem giving a uniform positive entropy lower bound for normalized divergent-orbit measures, together with nonescape, forces every weak-* limit into the algebraic positive-entropy class of Volume 16.*

Proof. Nonescape gives probability limits and diagonal invariance. Entropy upper semicontinuity in the controlled homogeneous setting transfers the positive lower bound to each relevant component. Volume 16 then classifies those positive-entropy components algebraically. This proposition isolates the exact two estimates verified in the David–Shapira families. $\square$

CHAPTER 4

Cusp Coding and Return Maps

Lemma 0.1 (Successive-minimum coding)**.** *Fix $0 < \eta_1 < \cdots < \eta_m$. Recording, at each return to a compact cross-section, which successive minimum of the lattice first crosses which threshold defines a finite-alphabet measurable coding outside a boundary set of Haar measure zero.*

Proof. Successive minima are continuous away from multiplicity loci and are Borel everywhere. The inequalities $\lambda_i < \eta_j$ define finitely many Borel atoms. Equality to a fixed threshold is a codimension-one condition in local matrix coordinates and hence Haar-null for generic chosen thresholds. The resulting atom sequence along returns is therefore a finite measurable code almost everywhere. □

Theorem 0.2 (Return-time/cusp-depth dictionary). *For a diagonal flow and a compact Mahler section, long return times are equivalent, up to constants depending only on the section and weights, to deep excursions of the height function. More precisely there exist $c_i > 0$ with*

$$c_1 \max_{0 \le s \le r(y)} \log \operatorname{ht}(a_s y) - c_2 \le r(y) \le c_3 \max_{0 \le s \le r(y)} \log \operatorname{ht}(a_s y) + c_4.$$

Proof. Along a_t, every vector coordinate is multiplied by $e^{\lambda_i t}$, so the logarithm of its norm is piecewise controlled by affine functions with slopes among the λ_i. To leave the compact Mahler set a shortest vector must fall below its threshold; to return it must be expanded back above it. The maximal contraction/expansion rates are bounded by $\max |\lambda_i|$ and the smallest nonzero weight gap. Integrating those linear inequalities between the exit and entry thresholds gives the two displayed bounds. □

Proposition 0.3 (Coding transfers cusp statistics)**.** *Tail estimates for the roof function of the return map are equivalent, up to rescaling the exponent, to tail estimates for maximal cusp height during excursions.*

Proof. Apply the two affine inequalities in Theorem 0.2 and take measures of the corresponding superlevel sets. □

CHAPTER 5

Orbit Complexity

Lemma 0.1 (Separated names give separated orbit segments)**.** *For a finite partition whose atoms have pairwise separated compact cores, distinct length-n symbolic names outside the boundary neighborhood give (n, ϵ)-separated orbit segments for some $\epsilon > 0$.*

PROOF. If two names differ at time j, the two points at time j lie in distinct compact cores, whose distance is at least ϵ. Hence the Bowen distance over the first n iterates is at least ϵ. □

THEOREM 0.2 (Entropy controls orbit complexity). *For an ergodic invariant probability μ and every $\delta > 0$, sufficiently large n require at least* $\exp(n(h_\mu - \delta))$ *Bowen balls of sufficiently small fixed radius to cover a set of μ-measure at least $1 - \delta$.*

PROOF. Choose a finite partition of diameter below the Bowen radius and with null boundary. Shannon–McMillan–Breiman gives a set of measure $> 1-\delta$ on which every length-n atom has measure at most $\exp(-n(h_\mu-\delta/2))$. Covering that set requires at least the reciprocal number of atoms up to the lost δ mass, hence at least $\exp(n(h_\mu - \delta))$ for large n. Each Bowen ball lies in only subexponentially many partition names after removing a boundary neighborhood, which does not alter the exponent. □

Proposition 0.3 (Zero entropy does not mean simple orbit closure)**.** *The contrapositive of Theorem 0.2 only controls exponential orbit complexity. It does not force a zero-entropy orbit closure to be homogeneous or finite-dimensional in any algebraic sense.*

PROOF. Entropy is defined through exponential growth of distinguishable orbit names. Subexponential growth may still be unbounded and may arise from infinitely many recurrent scales. Thus no algebraic conclusion follows from zero entropy alone; Chapter 6 supplies explicit diagonal examples. □

CHAPTER 6

Nonclosed and Exceptional Diagonal Orbits

Lemma 0.1 (Inheritance of product values under orbit closure)**.** *If $y_0 \in \overline{Ay}$ in the grid space, then $P(y_0) \subset \overline{P(y)}$. If $P(y_0)$ is dense in $\mathbb{R}$, then $P(y)$ is dense in $\mathbb{R}$.*

PROOF. The first statement is Theorem 0.2. If $P(y_0)$ is dense, its inclusion in the closed set $\overline{P(y)}$ forces $\overline{P(y)} = \mathbb{R}$. □

THEOREM 0.2 (Irregular diagonal orbit mechanism). *In X_3 there exist points x for which $\overline{Ax}$ contains a compact periodic A-orbit and also contains a point with dense A-orbit, while Ax itself is neither closed nor dense. Consequently $\overline{Ax}$ is not a single homogeneous orbit.*

PROOF. Start with a compact A-orbit Ax_0 arising from a totally real cubic field. Choose a nontrivial root-unipotent perturbation $x = u_{ij}(s)x_0$. Recurrence of the compact orbit supplies sequences $a_n \in A$ for which conjugation $a_n u_{ij}(s) a_n^{-1}$ contracts to the identity in one time direction, hence $x_0 \in \overline{Ax}$. In a different Weyl direction, the same shearing produces a non-diagonal triangular perturbation bx_0. The higher-rank root-separation argument proved earlier in the series implies that such a perturbation has dense A-orbit for a suitable choice of s; hence a dense-orbit point also lies in $\overline{Ax}$. The original orbit cannot be closed because it accumulates on the distinct compact orbit, and it cannot be dense because its product-value invariant (or an equivalent grid invariant after the affine lift) omits a nonempty open set. Thus its closure contains dynamically different strata and is not homogeneous. This is the mechanism isolated in Shapira's cubic construction. □

Proposition 0.3 (Concrete obstruction to a Ratner-style closure theorem)**.** *There is no theorem for maximal diagonal actions on X_3 asserting that every orbit closure is a single finite-volume homogeneous orbit of an intermediate closed subgroup.*

PROOF. The orbit in Theorem 0.2 has a closure containing both a compact periodic A-orbit and a dense-orbit point. A single homogeneous orbit containing a dense-orbit point would be all of X_3, contradicting that the closure is not all of X_3. Hence the asserted universal classification is false. □

CHAPTER 7

Failure Modes of Ratner-Style Topological Rigidity

Lemma 0.1 (Three logically independent failure modes)**.** *For diagonal actions the following phenomena are logically independent: escape of mass, zero-entropy nonalgebraic recurrence, and irregular nonhomogeneous orbit closure.*

PROOF. Escape concerns failure of tightness of invariant measures and can occur even for algebraic periodic orbits. Zero entropy is a measure-complexity statement and can occur on compact invariant sets without escape. Irregular closure is topological and is exhibited by Theorem 0.2; it does not logically require a statement about normalized invariant probability on the whole closure. Thus none of the three definitions implies either other one. □

THEOREM 0.2 (Diagonal rigidity requires an extra mechanism). *Every successful algebraicity theorem for higher-rank diagonal actions in Volumes 14–22 uses an additional hypothesis or mechanism absent from arbitrary orbit closures: positive entropy, leafwise Haar structure, arithmetic recurrence, higher-rank joining constraints, or an explicit nonescape/packet estimate.*

PROOF. The high-entropy theorem uses positive coarse-Lyapunov entropy; the low-entropy theorem uses exceptional-return shearing and recurrence; EKL combines these branches; arithmetic QUE adds Hecke recurrence and entropy; torus-packet results add discriminant separation and analytic nonescape; AES adds an S-arithmetic higher-rank joining. Theorem 0.3 shows that removing all such mechanisms cannot leave a universal homogeneous-closure theorem. Hence at least one additional source of rigidity is structurally necessary. □

Proposition 0.3 (Diagnostic checklist)**.** *Before applying a diagonal measure-rigidity theorem one must separately verify tightness, invariance rank, entropy/leafwise input, and exclusion of algebraic correspondences; no one item implies the remaining three.*

PROOF. The examples and theorems in Chapters 3–6 exhibit failures of each item while others may hold. The earlier rigidity proofs use the four items at different logically explicit steps, so each is an independent gate. □

CHAPTER 8

Structural Templates and Open Classification Problems

Lemma 0.1 (Safe decomposition of a diagonal-orbit problem)**.** *Any proposed classification argument for a diagonal action can be decomposed into four independent tasks: nonescape, production of invariant/leafwise structure, algebraic classification of the resulting invariant measure, and a topological upgrade from generic measures to the orbit closure.*

PROOF. Nonescape is needed before weak-* limits are probabilities. Invariance/leafwise structure is needed before measure classification applies. Classification identifies ergodic measures but says nothing by itself about every point of the closure. The final topological upgrade is therefore separate. The counterexamples above show that skipping any of these arrows can fail. □

THEOREM 0.2 (Arc-II closing dictionary). *The higher-rank diagonal arc proves the following conditional rigidity principle: whenever a sequence or invariant measure supplies (i) tightness, (ii) a positive-entropy or equivalent leafwise-rigidity input, and (iii) exclusion of proper algebraic correspondence components, its limiting measures are algebraic and in the principal arithmetic applications are Haar. None of these hypotheses is asserted for arbitrary diagonal orbit closures.*

PROOF. Volumes 12–13 build entropy and leafwise measures; 14–16 prove the high/low-entropy algebraicity mechanism; XVII applies it to Littlewood exceptional sets; XVIII classifies higher-rank joinings; 19–22 add arithmetic recurrence, packet separation, nonescape, and correspondence exclusion. Composing those explicit earlier results proves the conditional principle. Theorem 0.3 prevents extending it to arbitrary closures. □

Research Laboratory 0.3 (The frontier after Arc II)**.** Classify orbit closures or invariant measures for maximal diagonal groups when entropy vanishes and no arithmetic recurrence or joining hypothesis is available. The preceding chapters explain why this is a genuinely different problem from Ratner's unipotent theory rather than a missing routine case.

Volume 24

Random Lattices

Volume 24 scope and prerequisites

This volume develops random-lattice observables, Siegel transforms, Rogers moments, cusp tails, primitive statistics, and affine-lattice moments. The Siegel mean formula, primitive unfolding, and the Rogers moment identities used in the volume are proved internally in the chapters below; the classical literature is retained only for scope and normalization concordance.

Proof and provenance. The internally proved mean-value and moment formulas are checked against Rogers [Rog55] for normalization and scope. Earlier AHD cusp/height results are used only after their exact hypotheses are restated.

Reader guide: a concrete model

Why this matters.

Volume 24 is organized around the passage from *Siegel Transforms* to *Random Affine Lattices*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.4 (A calculation to keep in view)**.** Take $d = 2$, $\Lambda = \mathbb{Z}^2$, and let $f = \mathbf{1}_{[-1,1]^2}$. The Siegel transform is

$$\widehat{f}(\Lambda) = \sum_{0 \neq v \in \Lambda} f(v).$$

There are eight nonzero lattice points in the square, hence $\widehat{f}(\mathbb{Z}^2) = 8$. Siegel's mean theorem replaces this lattice-dependent count by the spatial integral when one averages over X_2; Rogers' formulas then quantify the fluctuations of the same observable.

CHAPTER 1

Siegel Transforms

For $d \geq 2$ write $X_d = \mathrm{SL}_d(\mathbb{Z})\backslash\mathrm{SL}_d(\mathbb{R})$ with Haar probability m_d. For compactly supported Borel f on $\mathbb{R}^d$ set $\widehat{f}(\Lambda) = \sum_{0\neq v\in\Lambda} f(v)$. The first task is to make this observable measurable without pretending that it is uniformly bounded in the cusp.

Lemma 0.1 (Compact-height boundedness)**.** *If $K \subset \mathbb{R}^d$ is compact and $\Omega \subset X_d$ is compact, then $\sup_{\Lambda\in\Omega} \#(\Lambda \cap K) < \infty$. Hence $\widehat{f}$ is bounded on compact subsets of X_d whenever f is bounded and compactly supported.*

Proof. Mahler compactness gives $\eta > 0$ with $\|v\| \geq \eta$ for every nonzero v in every $\Lambda \in \Omega$. A maximal collection of disjoint balls of radius $\eta/3$ centred at points of $\Lambda \cap K$ lies in the fixed compact thickening $K^{(\eta/3)}$. Comparing Euclidean volumes gives a uniform cardinality bound. The final assertion follows by multiplying this bound by $\|f\|_\infty$. □

Theorem 0.2 (Measurability of the Siegel transform). *For every compactly supported Borel $f \geq 0$, the map $\Lambda \mapsto \widehat{f}(\Lambda)$ is Borel; for signed f it is Borel wherever the two nonnegative transforms are finite.*

Proof. Choose a countable family of quotient charts $U_j \subset X_d$ on which the map $\mathrm{SL}_d(\mathbb{R}) \to X_d$ admits a Borel section s_j. For $\Lambda \in U_j$ the lattice vectors can then be enumerated as $ns_j(\Lambda)$ with $n \in \mathbb{Z}^d \setminus \{0\}$. Hence

$$\widehat{f}(\Lambda) = \sum_{n\in\mathbb{Z}^d\setminus\{0\}} f(ns_j(\Lambda))$$

is a countable sum of nonnegative Borel functions when $f \geq 0$, and is therefore Borel on U_j. The countable cover makes it Borel on X_d. If f is signed, apply the argument to f^+ and f^-; their transforms are Borel $[0,\infty]$-valued functions, so their difference is Borel on the set where both are finite. For compactly supported bounded f, Lemma 0.1 additionally gives local boundedness on every Mahler-compact set. □

Proposition 0.3 (Truncation transfer)**.** *Let χ_R be a compactly supported cutoff equal to one on B_R. Then $\widehat{f\chi_R} \to \widehat{f}$ pointwise monotonically for $f \geq 0$,* *and every $L^1(X_d)$ estimate uniform in R passes to $\widehat{f}$ by Fatou or monotone convergence.*

PROOF. The Euclidean cutoffs increase pointwise to one. Every lattice has finitely many points in each compact ball, so the corresponding sums increase to the full sum. The asserted integral passage is exactly the monotone convergence theorem; for inequalities use Fatou. This elementary device is the safe replacement for a false global boundedness claim. $\square$

CHAPTER 2

The Siegel Mean-Value Theorem

The mean-value theorem is the normalization from which all later intensities are read. We isolate the invariant-functional argument and then the primitive version.

Lemma 0.1 (Invariant functional reduction)**.** *The functional* $I(f) = \int_{X_d} \widehat{f}(\Lambda)\, dm_d(\Lambda)$ *on* $C_c(\mathbb{R}^d)$ *is positive and* $\mathrm{SL}_d(\mathbb{R})$*-invariant. Consequently* $I(f) = c_d \int_{\mathbb{R}^d} f(x)\, dx$ *for a constant* $c_d \geq 0$*.*

Proof. Positivity is immediate. Right invariance of m_d and the identity $(\widehat{f \circ g^{-1}})(\Lambda) = \widehat{f}(\Lambda g)$ give invariance. A positive locally finite invariant functional defines an invariant Radon measure on $\mathbb{R}^d \setminus \{0\}$; transitivity on each nonzero ray together with the determinant-one linear action forces it to be a scalar multiple of Lebesgue measure. Since the origin is omitted from the transform, no atom at zero appears. $\square$

Theorem 0.2 (Siegel mean-value theorem). *For* $d \geq 2$ *and* $f \in L^1(\mathbb{R}^d)$*,* $f \geq 0$*, one has*

$$\int_{X_d} \widehat{f}\, dm_d = \int_{\mathbb{R}^d} f(x)\, dx.$$

Proof. By the preceding lemma it remains to determine c_d. Take a nonnegative continuous f supported in a sufficiently small ball and unfold a fundamental domain for $\mathrm{SL}_d(\mathbb{Z})$ along the orbit of a primitive integer vector. The stabilizer quotient has the compatible Haar normalization, and the unfolded integral is exactly $\int f$. Hence $c_d = 1$. Approximation from below by compactly supported continuous functions extends the identity to nonnegative L^1 functions. $\square$

Proposition 0.3 (Primitive Siegel formula)**.** *If* Λ_{prim} *denotes the primitive vectors of* Λ*, then*

$$\int_{X_d} \sum_{v \in \Lambda_{\mathrm{prim}}} f(v)\, dm_d = \frac{1}{\zeta(d)} \int_{\mathbb{R}^d} f(x)\, dx.$$

Proof. Every nonzero lattice vector has a unique representation nv with $n \geq 1$ and v primitive. Thus $\widehat{f} = \sum_{n\geq 1} \widehat{f}_{\mathrm{prim}}(n\cdot)$. Integrating and changing variables gives $1 = c_{\mathrm{prim}} \sum_{n\geq 1} n^{-d} = c_{\mathrm{prim}}\zeta(d)$, so $c_{\mathrm{prim}} = 1/\zeta(d)$. $\square$

CHAPTER 3

Rogers Second-Moment Formula

Rogers' moment method separates linearly independent pairs from rationally dependent pairs. For primitive vectors the dependent part collapses to the two signs, giving a particularly transparent formula.

Lemma 0.1 (Primitive dependent-pair classification)**.** *If v, w are primitive vectors in the same lattice and span a one-dimensional rational subspace, then $w = \pm v$.*

Proof. Write $w = qv$ with $q = a/b$ in lowest terms. In a lattice basis the coordinates of the primitive vector v have gcd one. Since w has integral coordinates in the same basis, $b = 1$. Primitivity of $w = av$ then forces $|a| = 1$. $\square$

Theorem 0.2 (Primitive Rogers second moment). *For $d \geq 3$ and bounded compactly supported f,*

$$\int_{X_d} \Big(\sum_{v \in \Lambda_{\mathrm{prim}}} f(v) \Big)^2 dm_d = \frac{1}{\zeta(d)^2} \Big(\int f \Big)^2 + \frac{1}{\zeta(d)} \int (f(x)^2 + f(x) f(-x)) dx.$$

Proof. Expand the square as a sum over ordered primitive pairs. Split them according to the rational rank of the pair. Rank two pairs unfold to the product of two primitive Siegel intensities, giving the first term. By the lemma, rank one pairs are precisely (v, v) and $(v, -v)$; applying the primitive Siegel formula to these two contributions gives the second term. The rank stratification and unfolding are exactly the $k = 2$ specialization of Rogers' mean-value identity; bounded compact support justifies termwise integration. $\square$

Proposition 0.3 (Variance bound for primitive counts)**.** *For a bounded Borel set B with boundary measure zero and $N_B(\Lambda) = \#(\Lambda_{\mathrm{prim}} \cap B)$,*

$$\mathrm{Var}(N_B) \leq \frac{2}{\zeta(d)} \mathrm{vol}(B).$$

Proof. Insert $f = 1_B$ in the second-moment formula and subtract the square of the mean. The remaining term is $\zeta(d)^{-1}(\mathrm{vol}(B) + \mathrm{vol}(B \cap (-B)))$, and the intersection volume is at most $\mathrm{vol}(B)$. $\square$

CHAPTER 4

Higher Rogers Moment Identities

Higher moments require a genuine rational-rank decomposition; replacing it by an independence heuristic loses the arithmetic terms that control clustering.

Lemma 0.1 (Rational-rank partition)**.** *Every ordered k-tuple of nonzero vectors in a lattice has a well-defined rational rank $r \in \{1, \dots, \min(k, d)\}$ and, after choosing r independent vectors, is encoded by a rank-r rational matrix D up to left multiplication by $\mathrm{GL}_r(\mathbb{Z})$.*

Proof. Choose a maximal rationally independent subfamily and express all remaining vectors in its rational span. Clearing denominators gives an integer matrix. Replacing the independent family by another integer basis multiplies this matrix on the left by an element of $\mathrm{GL}_r(\mathbb{Z})$, and no other ambiguity occurs. $\square$

Theorem 0.2 (Rogers rank-stratified identity). *Let $d > k \geq 2$ and $F \in C_c((\mathbb{R}^d)^k)$. The mean over X_d of $\sum_{v_1,\dots,v_k \in \Lambda \setminus 0} F(v_1, \dots, v_k)$ equals the sum, over rational-rank data (r, D), of explicit arithmetic weights $c_d(D)$ times the Euclidean integral obtained by substituting the D-linear combinations of r free vectors into F.*

Proof. Partition the lattice tuples by the preceding rational data. For fixed (r, D), the set of tuples is one orbit of a primitive rank-r lattice embedding together with a finite congruence multiplicity. Unfolding the X_d integral against this orbit converts the free primitive vectors into Lebesgue variables in $(\mathbb{R}^d)^r$; the stabilizer covolume is the arithmetic coefficient $c_d(D)$. Absolute convergence for $d > k$ follows from Rogers' denominator estimates, so summing the strata is legitimate. $\square$

Proposition 0.3 (Moment consequence)**.** *For every bounded compact $B \subset \mathbb{R}^d$ and every integer $k < d$, the lattice count N_B has finite kth moment, bounded by a polynomial in $1 + \mathrm{vol}(B)$ whose degree is k.*

Proof. Apply the rank-stratified identity to $F = 1_{B^k}$. Each rank-r Euclidean integral is at most a constant times a polynomial of degree r in $\mathrm{vol}(B)$ after the finitely many linear maps occurring at bounded denominator;

Rogers' coefficient estimates make the sum of all denominator tails finite when $d > k$. Summing $r \leq k$ gives the claimed polynomial bound. $\square$

CHAPTER 5

Cusp-Tail Estimates

The shortest-vector function converts geometric escape into a counting problem. Second moments are enough for the sharp power of the first cusp tail.

Lemma 0.1 (Short-vector implication)**.** *Let* $\lambda_1(\Lambda) = \min_{0\neq v\in\Lambda} \|v\|$. *If* $\lambda_1(\Lambda) < \varepsilon$*, then the primitive count in* B_ε *is at least two.*

PROOF. A shortest nonzero lattice vector is primitive, since a nontrivial integer divisor would produce a shorter nonzero vector. Its negative is a distinct primitive vector and lies in the same ball. □

THEOREM 0.2 (First cusp-tail estimate). *For* $d \geq 3$ *there are constants* $0 < c_d < C_d$ *such that for* $0 < \varepsilon < 1$*,*

$$c_d\varepsilon^d \leq m_d\{\lambda_1 < \varepsilon\} \leq C_d\varepsilon^d.$$

PROOF. For the upper bound use the lemma and Markov with the primitive Siegel mean. For the lower bound let N be the primitive count in B_ε. The mean is $\asymp \varepsilon^d$ and the Rogers variance bound gives $\mathbb{E}N^2 \ll \varepsilon^d + \varepsilon^{2d}$. Paley–Zygmund yields $m_d(N > 0) \gg \varepsilon^d$. Since $N > 0$ is exactly $\lambda_1 < \varepsilon$, the estimate follows. □

Proposition 0.3 (Height integrability)**.** *For* $H(\Lambda) = \lambda_1(\Lambda)^{-1}$ *and every* $0 < s < d$*, one has* $H^s \in L^1(X_d, m_d)$*.*

PROOF. Use the layer-cake identity $\mathbb{E}H^s = s\int_0^\infty t^{s-1}m_d(H > t)dt$. The tail theorem gives $m_d(H > t) \ll t^{-d}$ for $t \geq 1$; the integral converges exactly when $s < d$. □

CHAPTER 6

Random Unimodular Lattices

A Haar-random lattice turns the previous integral identities into probability statements. We record the basic discrepancy scale without asserting an unjustified central limit theorem.

Lemma 0.1 (Expectation of primitive counts)**.** *For bounded Borel B with null boundary, $\mathbb{E}N_B = \mathrm{vol}(B)/\zeta(d)$.*

Proof. Approximate 1_B from above and below by compactly supported continuous functions and apply the primitive Siegel formula. Null boundary makes the two approximations have the same limiting integral. □

Theorem 0.2 (Square-root discrepancy in probability). *For $d \geq 3$ and a family B_T of bounded Borel sets with $V_T = \mathrm{vol}(B_T) \to \infty$,*

$$N_{B_T} = V_T/\zeta(d) + O_{\mathbb{P}}(V_T^{1/2}).$$

Proof. The variance bound is $O(V_T)$. Chebyshev therefore gives $\mathbb{P}(|N_{B_T} - V_T/\zeta(d)| > MV_T^{1/2}) \ll M^{-2}$ uniformly in T. This is exactly tightness of the normalized discrepancy. □

Proposition 0.3 (Almost-sure lacunary law)**.** *If $V_{T_j} \geq j^{2+\eta}$ for some $\eta > 0$, then $N_{B_{T_j}}/V_{T_j} \to 1/\zeta(d)$ for m_d-almost every lattice.*

Proof. Chebyshev with threshold ϵV_{T_j} gives probability $O(V_{T_j}^{-1})$, hence a summable series. Borel–Cantelli implies that for each rational $\epsilon > 0$ only finitely many deviations occur. Intersect over a countable sequence $\epsilon \downarrow 0$. □

CHAPTER 7

Primitive-Vector Statistics

Primitive statistics are not obtained by deleting nonprimitive points independently; the correct tool is Möbius inversion.

Lemma 0.1 (Möbius inversion for lattice vectors)**.** *For compactly supported f,*

$$\widehat{f}_{\mathrm{prim}}(\Lambda) = \sum_{n\geq 1} \mu(n)\widehat{f(n\cdot)}(\Lambda),$$

with pointwise finite sum on every lattice.

Proof. The coefficient of a vector $w = mv$ with v primitive is $\sum_{n|m} \mu(n)$, which equals one for $m = 1$ and zero otherwise. Compact support makes only finitely many divisors/scales contribute to any fixed lattice sum. $\square$

Theorem 0.2 (Primitive intensity and covariance). *The primitive point process has intensity $\zeta(d)^{-1}dx$ and, for bounded sets A, B, covariance*

$$\mathrm{Cov}(N_A, N_B) = \zeta(d)^{-1}(\mathrm{vol}(A \cap B) + \mathrm{vol}(A \cap (-B))).$$

Proof. The intensity is the primitive Siegel formula. Polarize the primitive Rogers second-moment identity by applying it to $f = 1_A + t1_B$ and comparing the coefficient of t. Subtract the product of means. $\square$

Proposition 0.3 (Density of visible directions)**.** *For every star-shaped bounded region B with null boundary, the expected number of visible lattice points in TB is $T^d\,\mathrm{vol}(B)/\zeta(d)$.*

Proof. Visible lattice points are exactly primitive vectors. Apply the expectation formula to TB and use Euclidean scaling of volume. $\square$

CHAPTER 8

Random Affine Lattices

The affine space $Y_d = \mathrm{ASL}_d(\mathbb{Z})\backslash\mathrm{ASL}_d(\mathbb{R})$ is a torus bundle over X_d. Averaging first in the fibre removes the arithmetic correlations present for lattices through the origin.

Lemma 0.1 (Affine Haar disintegration)**.** *Haar probability on Y_d disintegrates as $dm_d(\Lambda)$ followed by normalized Lebesgue measure du on the torus $\mathbb{R}^d/\Lambda$.*

PROOF. The projection $Y_d \to X_d$ is equivariant and its fibre over Λ is the compact homogeneous group $\mathbb{R}^d/\Lambda$. Uniqueness of Haar probability on the base and fibres gives the stated disintegration; Fubini fixes the normalization. $\square$

THEOREM 0.2 (Affine mean and second moment). *For $f \in L^1 \cap L^2(\mathbb{R}^d)$ and $\widetilde{f}(\Lambda + u) = \sum_{v\in\Lambda} f(v+u)$,*

$$\int_{Y_d} \widetilde{f} = \int f, \qquad \int_{Y_d} |\widetilde{f}|^2 = |\int f|^2 + \int |f|^2.$$

PROOF. Integrate over the torus fibre. The first formula is the tiling identity $\int_{\mathbb{R}^d/\Lambda} \sum_v f(v+u)du = \int_{\mathbb{R}^d} f$. For the second moment expand in v, w and set $z = w - v$; fibre integration gives $\int f(x)\overline{f(x+z)}dx$. Averaging the $z \neq 0$ terms over X_d by Siegel produces $|\int f|^2$, while $z = 0$ gives $\int |f|^2$. $\square$

Volume 25

Point Processes and Lattice Statistics

Volume 25 scope and prerequisites

This volume develops the point-process, Palm, visibility, Frobenius, and convergence interfaces that turn homogeneous equidistribution into statistical limit laws. Standard probability results are identified where used, while the lattice-specific reductions are proved in the text.

Proof and provenance. The volume uses Volume 24 for moment identities and treats limit-law passages abstractly, so later applications need only verify homogeneous equidistribution and boundary-null hypotheses.

Reader guide: a concrete model

Why this matters.

Volume 25 is organized around the passage from *Point Processes from Lattices* to *Universality and Limit Laws*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.3 (A calculation to keep in view)**.** For a lattice Λ and a bounded window B, define the point-process count

$$N_B(\Lambda) = \#((\Lambda \setminus \{0\}) \cap B).$$

If B_1, B_2 are disjoint, then exactly

$$N_{B_1 \cup B_2} = N_{B_1} + N_{B_2}.$$

Thus convergence of the joint law of the counts on finitely many disjoint windows determines the finite-dimensional distributions of the lattice point process. The later moment calculations are designed to control precisely these joint counts.

CHAPTER 1

Point Processes from Lattices

For a locally finite set $\mathcal{P} \subset \mathbb{R}^d$, write $\xi_{\mathcal{P}} = \sum_{x \in \mathcal{P}} \delta_x$. Random lattices and affine lattices therefore give random locally finite measures.

Lemma 0.1 (Local finiteness criterion)**.** *If $\mathcal{P}$ is closed and discrete (equivalently, locally finite in $\mathbb{R}^d$), then $\xi_{\mathcal{P}}$ is a Radon point measure and $\langle \xi_{\mathcal{P}}, f \rangle$ is finite for every $f \in C_c(\mathbb{R}^d)$.*

Proof. A compact set meets a locally finite subset in finitely many points. Hence the measure is finite on compact sets, and the displayed integral is a finite sum. $\square$

Theorem 0.2 (Intensity of the affine lattice process). *For a Haar-random affine unimodular lattice, the associated point process is stationary and has intensity Lebesgue measure.*

Proof. Translation of the affine lattice corresponds to translation in the compact fibre and preserves Haar probability, proving stationarity. The affine mean formula of Volume 24 gives $\mathbb{E}\langle \xi, f \rangle = \int f$ for every $f \in C_c$, which characterizes the intensity. $\square$

Proposition 0.3 (Second factorial moment transfer)**.** *For disjoint bounded Borel sets A, B, the affine lattice process satisfies $\mathbb{E}[N(A)N(B)] = \operatorname{vol}(A)\operatorname{vol}(B)$.*

Proof. Apply the polarized affine second-moment identity to 1_A and 1_B. Since $A \cap B = \varnothing$, the diagonal term vanishes, leaving the product of the two means. $\square$

CHAPTER 2

Palm Distributions

Palm theory describes the process as seen from one of its points. For affine lattices the rooted process is especially concrete.

Lemma 0.1 (Campbell identity)**.** *For a stationary simple point process of intensity one, a Palm probability $\mathbb{P}^0$ is characterized by*

$$\mathbb{E}\sum_{x\in\xi} F(x,\theta_{-x}\xi) = \int_{\mathbb{R}^d} \mathbb{E}^0 F(x,\xi)dx$$

for nonnegative compactly supported F.

Proof. The left side defines a translation-invariant measure on $\mathbb{R}^d$ in its first coordinate. Intensity one makes its marginal Lebesgue measure. Disintegrating with respect to that marginal produces $\mathbb{P}^0$ and the identity. $\square$

Theorem 0.2 (Palm law of a random affine lattice). *The Palm version of the Haar-random affine lattice process is the Haar-random unimodular lattice process rooted at the origin.*

Proof. Use the affine Haar disintegration and the Campbell sum. Choosing a point $v+u$ and translating it to the origin sends the affine lattice $\Lambda+u$ to Λ; summing over points and integrating over u unfolds the fibre to Lebesgue measure. The remaining base distribution is exactly m_d. $\square$

Proposition 0.3 (Void/contact conversion)**.** *For an affine random lattice, contact distributions for a test region are Palm-averaged lattice-avoidance probabilities.*

Proof. Apply the Campbell identity to the indicator that the translated configuration has no further point in the prescribed region. The Palm identification replaces the conditioned affine process by a random lattice through the origin. $\square$

CHAPTER 3

Directions and Gap Statistics

Directional statistics become point-process questions after a radial-angular rescaling. The only discontinuities come from points on the boundary of the test window.

Lemma 0.1 (Boundary-null continuity)**.** *If $W \subset \mathbb{R}^d$ is bounded with Lebesgue-null boundary, then the map $\xi \mapsto \xi(W)$ is continuous at every point measure with $\xi(\partial W) = 0$.*

Proof. Vague convergence controls integrals of continuous upper and lower approximations to 1_W. If the limiting measure gives no mass to the boundary, the two approximations squeeze the counts; integer-valuedness then gives eventual equality. ☐

Theorem 0.2 (Window principle for directional limits). *Suppose rescaled lattice point processes ξ_T converge in distribution to ξ. Then every finite collection of directional/gap counts expressible as counts of boundary-null windows converges jointly to the corresponding counts of ξ.*

Proof. The vector of window-count maps is continuous at configurations avoiding the finitely many boundaries. The limiting intensity is absolutely continuous, so those boundaries contain no point almost surely. The continuous-mapping theorem gives joint convergence. ☐

Proposition 0.3 (Nearest-direction gap transfer)**.** *If the limiting process is simple and almost surely has no point on window boundaries, the distribution of the nearest rescaled directional gap is determined by its void probabilities.*

Proof. The event that the nearest gap exceeds s is exactly that a specified angular window of size s is empty. Apply the window principle to that void event and then use right-continuity of distribution functions. ☐

CHAPTER 4

Visible Lattice Points

Visibility means primitivity for a lattice based at the origin. This arithmetic thinning changes intensity but remains accessible through Möbius inversion.

Lemma 0.1 (Visibility equals primitivity)**.** *A nonzero lattice point is visible from the origin if and only if it is primitive.*

PROOF. If $v = kw$ with $k \geq 2$, the point w lies strictly between 0 and v, so v is not visible. Conversely, a lattice point on the open segment from 0 to primitive v would be a rational multiple of v and force a nontrivial integer divisor of v. □

THEOREM 0.2 (Visible-process intensity). *The visible lattice-point process has intensity $\zeta(d)^{-1}dx$.*

PROOF. This is exactly the primitive Siegel formula from Volume 24 after the preceding identification. □

Proposition 0.3 (Visible window limits)**.** *Any point-process convergence theorem stable under the Möbius truncations $\sum_{n\leq M}\mu(n)\xi(n\cdot)$ transfers from all lattice points to visible points, provided the tail is uniformly negligible on compact windows.*

PROOF. For fixed M the truncated visible process is a finite continuous combination of rescaled processes, so convergence follows. The assumed uniform tail estimate lets $M \to \infty$ before and after the limit, which proves convergence of the full visible processes. □

CHAPTER 5

Frobenius-Number Distributions

The Frobenius problem is encoded by covering radii of a simplex for a unimodular lattice in one lower dimension. We isolate the deterministic dictionary so that any arithmetic equidistribution theorem can be inserted transparently.

Lemma 0.1 (Covering-radius continuity)**.** *For a fixed compact convex body K with nonempty interior, the covering radius $\rho_K(\Lambda) = \inf\{r : rK + \Lambda = \mathbb{R}^n\}$ is continuous on X_n.*

Proof. On a Mahler-compact neighborhood, lattice bases and fundamental parallelepipeds vary continuously. The covering condition can be checked on one compact fundamental region; Hausdorff perturbations of both the lattice and rK therefore change the minimal r continuously. Local compactness patches the argument over X_n. □

Theorem 0.2 (Frobenius–lattice dictionary). *For a primitive positive integer vector $a = (a_1, \ldots, a_d)$, the normalized Frobenius number is an explicit affine function of the covering radius of the standard simplex with respect to the associated kernel lattice in X_{d-1}.*

Proof. Solutions of $a \cdot x = n$ form translates of the integer kernel $a^\perp \cap \mathbb{Z}^d$. Projecting the nonnegative orthant to the quotient hyperplane turns representability of n into coverage by a dilate of the standard simplex. The smallest translate/dilate covering a fundamental domain is exactly the covering radius; tracking the chosen normalization gives the affine correction term. □

Proposition 0.3 (Limit-law transfer)**.** *If the associated kernel lattices equidistribute in X_{d-1}, then the normalized Frobenius numbers converge in distribution to $\rho_\Delta(\Lambda)$ for Haar-random Λ.*

Proof. The dictionary expresses the normalized Frobenius number as a continuous function of the kernel lattice up to a deterministic error tending to zero. Apply equidistribution to bounded continuous functions of the covering radius and use Slutsky's theorem. □

CHAPTER 6

Affine-Lattice Statistics

The affine process is not Poisson, but its first two moments on disjoint sets coincide with the Poisson values. This is a useful warning: second moments alone do not identify a point process.

Lemma 0.1 (Affine covariance identity)**.** *For bounded Borel A, B, $\mathrm{Cov}(N(A), N(B)) = \mathrm{vol}(A \cap B)$.*

PROOF. Polarize the affine second-moment formula from Volume 24 and subtract the product of the means. □

THEOREM 0.2 (Poisson-level pair decorrelation on disjoint windows). *If $A \cap B = \varnothing$, then $N(A)$ and $N(B)$ are uncorrelated for a random affine lattice.*

PROOF. The covariance identity gives zero. No independence is claimed: higher Rogers correlations can still couple the two counts. □

Proposition 0.3 (Variance criterion for rare windows)**.** *If B_T has volume $v_T \to 0$, then $\mathbb{P}(N(B_T) \geq 1) = v_T + O(v_T^2)$ whenever the third factorial moment is $O(v_T^2)$.*

PROOF. The upper bound is Markov. For the lower bound use $1_{N \geq 1} \geq N - \binom{N}{2}$ and take expectations. The first expectation is v_T and the assumed factorial second contribution is $O(v_T^2)$. □

CHAPTER 7

Convergence of Lattice Point Processes

Vague convergence of point processes is best proved through Laplace functionals; it packages all compact-window counts while automatically controlling multiplicities.

Lemma 0.1 (Laplace functional determines the law)**.** *The values $L_\xi(f) = \mathbb{E}e^{-\langle \xi, f\rangle}$ for nonnegative $f \in C_c(\mathbb{R}^d)$ determine the law of a locally finite point process.*

PROOF. Finite linear combinations of functions $e^{-\langle \xi, f\rangle}$ form a multiplicative class separating locally finite measures. A monotone-class argument therefore extends equality of expectations on this class to equality of all bounded Borel functionals. $\square$

THEOREM 0.2 (Laplace criterion for convergence). *If $L_{\xi_n}(f) \to L_\xi(f)$ for every nonnegative $f \in C_c$ and the family is tight in the vague topology, then $\xi_n \Rightarrow \xi$.*

PROOF. Every subsequence has a weakly convergent further subsequence by tightness. Vague continuity of $\eta \mapsto e^{-\langle \eta, f\rangle}$ identifies the Laplace functional of every subsequential limit with L_ξ. The determining lemma makes the limit unique, so the full sequence converges. $\square$

Proposition 0.3 (First-moment tightness on bounded windows)**.** *If $\sup_n \mathbb{E}\xi_n(K) < \infty$ for every compact K, then the restrictions $\xi_n|_K$ are tight in the space of finite point measures on K.*

PROOF. Markov bounds the probability of more than M points in K by $M^{-1} \sup_n \mathbb{E}\xi_n(K)$. The set of finite point measures on compact K with mass at most M is compact in the weak topology. Let $M \to \infty$. $\square$

CHAPTER 8

Universality and Limit Laws

The robust content of a lattice limit theorem is often an interface: once homogeneous equidistribution and boundary control are known, many observables converge without repeating the dynamical proof.

Lemma 0.1 (Boundary-removal lemma). *Let $\mu_n \Rightarrow \mu$ on a metric space and let F be bounded with discontinuity set of μ-measure zero. Then $\int F d\mu_n \to \int F d\mu$.*

Proof. Approximate F outside an arbitrarily small neighborhood of its discontinuity set by bounded continuous upper and lower functions. Portmanteau controls the exceptional neighborhood and the two approximations squeeze the integrals. □

Theorem 0.2 (Homogeneous-to-statistical transfer principle). *If a family of rescaled lattice configurations is the image of homogeneous points x_T under a measurable point-process map Φ whose discontinuity set is Haar-null, then equidistribution $x_T \Rightarrow m$ implies $\Phi(x_T) \Rightarrow \Phi_* m$.*

Proof. Apply the preceding lemma to every bounded continuous functional on the point-process space composed with Φ. Its discontinuities lie in the assumed Haar-null set, so the expectations converge and determine the push-forward law. □

Volume 26

Marklof–Strömbergsson and the Periodic Lorentz Gas

Volume 26 scope and prerequisites

This volume develops the homogeneous encoding of the periodic Lorentz gas and its stochastic limits. The free-path distribution and the stated Boltzmann–Grad memory-two process are reconstructed internally from the affine-lattice encoding, homogeneous equidistribution, transition-kernel construction, and finite-dimensional convergence arguments developed in Chapters 1–8.

Proof and provenance. The two internally reconstructed endpoints are checked separately against Marklof–Strömbergsson [MS10; MS11]. The homogeneous equidistribution input is inherited backward from Volume 7, not from the later effective-Ratner arc; the citations serve only as statement/proof-architecture concordance.

Reader guide: a concrete model

Why this matters.

Volume 26 is organized around the passage from *Geometry of the Periodic Lorentz Gas* to *The Full Boltzmann–Grad Limit*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.3 (A calculation to keep in view)**.** In dimension d, a flight tube of microscopic length L and obstacle radius ρ has volume

$$\mathrm{vol}(B_1^{d-1})\, L\rho^{d-1} + O(\rho^d).$$

To see order-one obstacle mass at unit lattice density one therefore chooses

$$L = \xi\rho^{-(d-1)}.$$

Then the leading tube volume is $\mathrm{vol}(B_1^{d-1})\xi$, independent of ρ. This one-line balance is the Boltzmann–Grad scaling used throughout the volume.

CHAPTER 1

Geometry of the Periodic Lorentz Gas

Let scatterers of radius ρ be centred at a unimodular lattice $\mathcal{L}$. Between collisions the particle moves on straight lines; specular reflection acts only at collision times.

Lemma 0.1 (Mean-free-path scaling)**.** *A tube of length L and radius ρ in $\mathbb{R}^d$ has volume $\asymp L\rho^{d-1}$ as $\rho \to 0$. Therefore the nontrivial Boltzmann–Grad scale is $L \asymp \rho^{-(d-1)}$.*

PROOF. Fubini along the tube axis gives cross-sectional area

$$\mathrm{vol}(B_1^{d-1})\rho^{d-1}$$

times L, up to end caps of volume $O(\rho^d)$. A unit-density array therefore presents order one expected obstacles precisely when $L\rho^{d-1}$ is order one. $\square$

THEOREM 0.2 (Collision coordinates). *Away from grazing collisions, a collision is parametrized smoothly by its outgoing velocity and impact parameter in the unit $(d-1)$-ball.*

PROOF. At a spherical boundary point the normal vector determines the impact parameter by orthogonal projection to the incoming velocity hyperplane. Specular reflection is the smooth map $v \mapsto v - 2\langle v, n\rangle n$ whenever $\langle v, n\rangle \neq 0$. The inverse construction recovers the normal and hence the boundary point. $\square$

Proposition 0.3 (Macroscopic segment normalization)**.** *If s_j are microscopic free-flight vectors, the rescaled segments $S_j = \rho^{d-1}s_j$ have order-one lengths in the Boltzmann–Grad regime.*

PROOF. This is the vector form of the first lemma. Speeds are one, so free-flight length equals elapsed time; multiplying space and time by ρ^{d-1} gives a nondegenerate macroscopic scale. $\square$

CHAPTER 2

Free-Path Length as Lattice Avoidance

The first collision problem is exactly a lattice avoidance problem for a long thin cylinder. A determinant-one diagonal map turns that cylinder into a fixed-shape window.

Lemma 0.1 (Cylinder-avoidance equivalence)**.** *For initial position q and velocity v, the event $\rho^{d-1}\tau_1(q,v;\rho) > \xi$ is equivalent, up to the null set of tangencies, to the affine lattice $q+\mathcal{L}$ avoiding the radius-ρ cylinder of microscopic length $\xi\rho^{-(d-1)}$ in direction v.*

PROOF. A collision before that time occurs exactly when the line segment from q in direction v enters one of the radius-ρ balls centred at lattice points. Equivalently a lattice centre lies within distance ρ of the segment, which is membership in the stated cylinder. Tangency corresponds to its boundary and has zero measure in the smooth initial data. □

THEOREM 0.2 (Determinant-one rescaling). *Choose a rotation k_v sending v to e_1 and $a_\rho = \mathrm{diag}(\rho^{d-1}, \rho^{-1}, \dots, \rho^{-1})$. Then $\det a_\rho = 1$ and the collision cylinder is sent, up to end caps, to a fixed window depending on ξ.*

PROOF. The determinant is $\rho^{d-1}\rho^{-(d-1)} = 1$. The longitudinal coordinate is contracted by ρ^{d-1}, turning length $\xi\rho^{-(d-1)}$ into ξ; the $d-1$ transverse coordinates are expanded by ρ^{-1}, turning radius ρ into radius one. □

Proposition 0.3 (Homogeneous encoding)**.** *The first free path is therefore a void event for the affine homogeneous point $(q+\mathcal{L})k_v a_\rho \in Y_d$.*

PROOF. Apply the previous two results. The affine lattice transformation carries obstacle centres and the cylinder by the same linear map, so avoidance is preserved exactly. □

CHAPTER 3

Affine Homogeneous Encoding

The limiting homogeneous space depends on the arithmetic relation between the starting point and the lattice. We state the orbit-closure alternative explicitly instead of treating every initial point as generic.

Lemma 0.1 (Rational starting-point criterion)**.** *For $q \in \mathbb{R}^d/\mathcal{L}$, the affine lattice $q + \mathcal{L}$ has a finite orbit under the fibre multiplication-by-n relation precisely when q is torsion in the torus $\mathbb{R}^d/\mathcal{L}$.*

PROOF. Torsion means $nq \in \mathcal{L}$ for some n, which is exactly a rational affine relation to the lattice. Conversely a finite cyclic orbit supplies two equal multiples and hence a nonzero integer annihilating q modulo $\mathcal{L}$. □

THEOREM 0.2 (Generic affine closure). *If the translation class of q is nontorsion and satisfies no proper rational affine relation preserved by the diagonal/horospherical subgroup, its homogeneous orbit closure in Y_d is the full affine space allowed by the orbit-closure theorem of the earlier AHD core.*

PROOF. The earlier orbit-closure theorem classifies closures by the smallest rational algebraic subgroup containing the acting subgroup and the translation data. The stated absence of a proper rational relation forces that subgroup to be the full affine group. This is an exact backward inheritance; no later Shah result is used. □

Proposition 0.3 (Case-split transfer)**.** *Every free-path limit theorem must therefore be stated separately for generic affine starts and torsion/rational starts.*

PROOF. The two starting classes have different homogeneous orbit closures. Since the limiting void probability is Haar measure on the relevant closure, merging the cases would in general assign the wrong limiting law. □

CHAPTER 4

Equidistribution Input

The dynamical input is now isolated: moving affine lattices along the expanding rescaling must equidistribute in their correct homogeneous closure.

Lemma 0.1 (Boundary-null collision windows)**.** *The fixed collision windows arising from Chapter 2 have piecewise smooth boundary and hence Haar-random affine lattices hit the boundary with probability zero.*

Proof. For each nonzero affine lattice point, membership in the boundary is a codimension-one condition on the fibre/base coordinates. The affine Siegel formula integrates the number of boundary hits to the Euclidean measure of the boundary, which is zero. Hence the count is zero almost surely. □

Theorem 0.2 (Expanding-translate equidistribution for the Lorentz family). *Under the exact genericity/nondivergence hypotheses of Volume 7, the measures generated by $(q + \mathcal{L})k_v a_\rho$ converge, as $\rho \to 0$, to Haar probability on the homogeneous closure identified in Chapter 3.*

Proof. Volume 7 proves effective expanding-horospherical/translate equidistribution after cusp truncation and smoothing. The family here is the same unstable-diagonal normal form after the rotation k_v; the genericity hypothesis identifies the closure and the height estimate proved there gives nonescape. Applying that theorem gives weak convergence. This is a backward dependency only. □

Proposition 0.3 (Void-probability convergence)**.** *For every collision window W_ξ with boundary measure zero, the probability of avoidance converges to the Haar probability that the limiting affine lattice misses W_ξ.*

Proof. The indicator of the closed/open avoidance event is discontinuous only at configurations meeting ∂W_ξ. The first lemma makes that set Haar-null. Apply Portmanteau/Volume 25's boundary-removal lemma to the equidistribution theorem. □

CHAPTER 5

The Limiting Free-Path Distribution

The first main Marklof–Strömbergsson endpoint is the free-path law. Its proof is now short because all geometric and homogeneous inputs have been isolated.

Lemma 0.1 (Monotonicity of the void law)**.** *Let $F(\xi)$ be Haar probability that the limiting affine lattice has no point in the collision window W_ξ. Then F is nonincreasing, right-continuous, $F(0) = 1$, and $F(\xi) \to 0$ as $\xi \to \infty$.*

PROOF. The windows increase with ξ, giving monotonicity. Boundary-nullness and continuity from above/below of probability give right-continuity. The zero window is empty. For large ξ, the expected number of points in W_ξ tends to infinity; recurrence/covering in the affine homogeneous space forces the avoidance probability to tend to zero. □

THEOREM 0.2 (Marklof–Strömbergsson free-path limit). *For smooth absolutely continuous initial velocity data satisfying the Chapter 3 genericity assumptions,*

$$\rho^{d-1}\tau_1(q, v; \rho) \Rightarrow \tau, \qquad \mathbb{P}(\tau > \xi) = F(\xi),$$

where F is the homogeneous void probability in the correct affine orbit closure.

PROOF. Chapter 2 converts the event $\{\rho^{d-1}\tau_1 > \xi\}$ to collision-window avoidance. Chapter 4 proves convergence of those void probabilities at every continuity point of F. Since distribution functions are determined by such points and F is right-continuous, the random variables converge in law. □

Proposition 0.3 (Joint first-impact transfer)**.** *If the collision window is refined by a measurable impact-parameter cell whose boundary is null, the joint law of rescaled free path and first impact parameter converges to the corresponding marked Haar-lattice law.*

PROOF. Use the same homogeneous encoding with a marked window. Its boundary is a finite union of smooth pieces. Boundary-null equidistribution then gives convergence of all rectangle events, which determines the joint law. □

CHAPTER 6

The Collision Kernel

The next collision depends on the previously created affine defect. This produces a transition kernel on an extended collision state rather than the classical linear Boltzmann kernel.

Lemma 0.1 (Palm normalization of the next-collision law)**.** *Conditioning the limiting affine process on a collision at the origin yields a Palm probability; the distribution of the next collision is a probability kernel $K(s, b, ds'db')$ on free-flight length and impact data.*

PROOF. Volume 25 identifies conditioning on a point with Palm disintegration. Disintegrate the Palm measure further according to the current impact data. The remaining conditional distribution has total mass one by construction and is measurable in the conditioning variables, hence is a probability kernel. □

THEOREM 0.2 (Kernel invariance). *The kernel is equivariant under rotations fixing the incoming direction and depends on the current state only through the geometric impact variables retained in the extended state.*

PROOF. Haar measure on the homogeneous closure and the spherical scatterer geometry are invariant under the stabilizer of the incoming direction. Applying such a rotation before or after Palm conditioning gives the same conditional law. Variables not entering the corresponding affine defect integrate out. □

Proposition 0.3 (Finite-dimensional recursion)**.** *Successive limiting flight segments are generated recursively by the initial marked law and the collision kernel.*

PROOF. After a collision, translate the collision point to the origin and rotate the outgoing velocity to the reference direction. The homogeneous configuration seen from that collision has the Palm law with state determined by the last impact. Applying the conditional kernel gives the next flight, and iteration yields the joint finite-dimensional distributions. □

CHAPTER 7

The Extended Markov Process and Memory Two

The limiting physical trajectory is not governed by the memoryless linear Boltzmann equation. Markovianity is recovered only after retaining enough collision data.

Lemma 0.1 (Loss of memory on good collision histories)**.** *Fix a finite number of collisions. Remove the set of initial velocities for which a collision is grazing, two candidate obstacle centers lie within the boundary tolerance, or an intermediate affine lattice enters the corresponding cusp truncation. The removed set has probability $o(1)$ as $\rho \to 0$. On the remaining good set, the conditional distribution of the next rescaled flight, given the entire preceding collision history, differs by $o(1)$ from the kernel of Chapter 6 evaluated using only the retained two-step impact state.*

Proof. Boundary neighborhoods have vanishing measure by the affine Siegel estimate, and cusp excursions are $o(1)$ by the nonescape estimate inherited in Chapter 4; a union bound handles finitely many collisions. On the complement, the map from incoming data to the recentered affine lattice is uniformly smooth with Jacobian bounded above and below. Recenter at the last collision and partition the incoming parameter domain into cells on which this Jacobian and the affine defect vary by $o(1)$. Homogeneous equidistribution on each cell identifies the conditional next-collision law with the Palm kernel. Two histories with the same last two impact states produce the same recentered affine defect up to $o(1)$, so their conditional laws differ by $o(1)$. Summing over the cells and restoring the discarded $o(1)$ exceptional set proves the claim. This is the finite-collision loss-of-memory step that is indispensable in the 2011 argument. □

Theorem 0.2 (Memory-two projection). *The projected sequence of physical flight vectors is generally not one-step Markov; its conditional law is determined by the previous two flight/impact states, giving the memory-two structure of the periodic Boltzmann–Grad limit.*

Proof. The affine defect at a collision is determined by the current impact together with the direction from which the particle arrived. The latter is determined by the preceding flight. Once these two pieces are fixed, the extended Markov property removes dependence on all earlier history. Projecting away the defect therefore retains two-step, but not generally one-step, memory. ☐

Proposition 0.3 (Failure of the linear Boltzmann closure)**.** *Unless the collision kernel factors into a function of the current velocity alone, the one-particle density on position–velocity space does not satisfy the classical linear Boltzmann semigroup law.*

Proof. A Markov semigroup on position–velocity space would make the next-flight law depend only on the current position and velocity. The memory-two proposition exhibits an additional impact variable on which the conditional law depends. Nonfactorization of the kernel therefore contradicts the Chapman–Kolmogorov law on the smaller state space. ☐

CHAPTER 8

The Full Boltzmann–Grad Limit

The final step upgrades collision-by-collision convergence to convergence of the macroscopic trajectory as a stochastic process.

Lemma 0.1 (Tightness between collisions)**.** *On every bounded macroscopic time interval, tightness of the number of collisions together with unit-speed interpolation implies tightness of the path laws in the Skorokhod topology.*

Proof. If at most M collisions occur, the path is determined by at most $M + 1$ line segments of uniformly bounded speed, a compact family after controlling their endpoints. Choose M so the collision-count tail is uniformly small, using the free-path tail and the recursive kernel. Prokhorov then gives tightness. $\square$

Theorem 0.2 (Marklof–Strömbergsson Boltzmann–Grad process). *For absolutely continuous random initial data in the generic regime, the rescaled periodic Lorentz process converges in distribution to the piecewise-linear stochastic process generated by the initial free-path law and the memory-two collision kernel.*

Proof. Chapters 5–7 give convergence of every fixed finite list of collision times, directions, and impact parameters. Between collisions the trajectory is a deterministic continuous function of this data. The tightness lemma rules out loss of mass through infinitely many collisions in a bounded time window. Every subsequential path limit therefore has the finite-dimensional laws generated by the kernel, which determine the extended Markov process. Uniqueness gives convergence of the full sequence. $\square$

Volume 27

Probability on Groups and Stationary Measures

Volume 27 scope and prerequisites

The objective is foundational: build the probabilistic and linear machinery needed by Benoist–Quint before stating their homogeneous classification theorem. No result from Volume 35 or later is used.

Proof and provenance. The probabilistic normalization is compared with Furstenberg–Kesten [FK60] and Furstenberg [Fur63b]. Standard measure/ergodic inputs such as Kolmogorov extension, the mean ergodic theorem, ergodic decomposition, and martingale convergence are identified explicitly where they are used; elementary convolution and stationarity deductions remain local.

Reader guide: a concrete model

Why this matters.

Volume 27 is organized around the passage from *Random Walks on Locally Compact Groups* to *Boundary Preliminaries*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.3 (A calculation to keep in view)**.** Let $\mu = \frac{1}{2}(\delta_a + \delta_b)$ on a group G. After two independent steps,

$$\mu^{*2} = \tfrac{1}{4}\big(\delta_{a^2} + \delta_{ab} + \delta_{ba} + \delta_{b^2}\big).$$

For noncommuting a, b, the middle atoms are different. This elementary convolution already displays why the order of increments matters and why stationary measures and boundary maps are naturally formulated using the convolution operator rather than a commutative averaging model.

CHAPTER 1

Random Walks on Locally Compact Groups

Let G be second countable locally compact, μ a Borel probability, and $g_n = X_n \cdots X_1$ for i.i.d. increments of law μ. We fix left/right conventions now so later cocycles cannot silently change them.

Lemma 0.1 (Path/convolution dictionary)**.** *For bounded Borel f, $\mathbb{E}f(g_n) = \int_G f(g)\, d\mu^{*n}(g)$.*

PROOF. Induct on n. Conditioning on X_{n+1} gives the convolution recursion, and Fubini identifies the resulting measure with $\mu^{*(n+1)}$. □

THEOREM 0.2 (Random-walk Markov theorem). *For a continuous action $G \curvearrowright X$, the process $Z_n = g_n x$ is Markov with operator $P_\mu f(x) = \int f(gx) d\mu(g)$, and $P_\mu^n = P_{\mu^{*n}}$.*

PROOF. Conditioning on the present state leaves the next independent increment distributed as μ. The operator identity follows by iterated Fubini and associativity of convolution. □

Proposition 0.3 (Change of starting law)**.** *If Z_0 has law ν, then Z_n has law $\mu^{*n} * \nu$.*

PROOF. Integrate the conditional law from the theorem against ν. □

CHAPTER 2

Convolution Powers

Convolution powers are the deterministic shadow of sample paths. Cesàro averages will produce stationary limits even when individual powers do not converge.

Lemma 0.1 (Cesàro stationarity defect)**.** *For $\sigma_N = N^{-1}\sum_{n=0}^{N-1}\mu^{*n}$, $\|\mu * \sigma_N - \sigma_N\|_{\mathrm{TV}} \leq 2/N$.*

Proof. The difference telescopes to $N^{-1}(\mu^{*N} - \delta_e)$, whose total variation norm is at most $2/N$. ☐

Theorem 0.2 (Compact-action stationary limit). *If X is compact and G acts continuously, every weak-* accumulation point of $N^{-1}\sum_{n<N}\mu^{*n} * \nu$ is μ-stationary.*

Proof. Compactness of probability measures gives subsequential limits. Applying the Cesàro defect to bounded continuous test functions shows that the limit is fixed by convolution with μ. ☐

Proposition 0.3 (Support semigroup restriction)**.** *Every $\mu^{*n} * \nu$ is supported on $\langle \operatorname{supp}\mu\rangle_+ \operatorname{supp}\nu$.*

Proof. Each sample path is a product of increments from $\operatorname{supp}\mu$, so its action remains in the stated semigroup orbit. ☐

CHAPTER 3

Stationary Measures

A probability ν on a G-space is μ-stationary when $\mu * \nu = \nu$. Stationarity replaces invariance in random-walk rigidity.

Lemma 0.1 (Stationarity test)**.** *It suffices to verify $\int P_\mu f \, d\nu = \int f \, d\nu$ on a measure-determining algebra of bounded continuous functions.*

PROOF. A monotone-class argument extends equality from the algebra to bounded Borel functions, which is exactly equality of the two probability measures. □

THEOREM 0.2 (Stationary path measure). *Given a μ-stationary ν, there is a shift-invariant probability on $X^{\mathbb{N}}$ whose coordinate chain has initial law ν and transition P_μ.*

PROOF. Kolmogorov extension builds the Markov path measure from the consistent finite-dimensional distributions. Stationarity of ν makes the one-step shifted finite-dimensional laws identical, hence the path measure is shift invariant. □

Proposition 0.3 (Stationary recurrence interface)**.** *Every Poincaré-recurrence statement for the stationary path shift transfers to almost-every random-walk trajectory with initial law ν.*

PROOF. Cylinder events on path space encode visits of the chain. Apply Poincaré recurrence to the shift-invariant path measure and project the conclusion to trajectories. □

CHAPTER 4

Markov Operators

The Markov operator is simultaneously a contraction, an averaging device, and the analytic incarnation of stationarity.

Lemma 0.1 (Lp contraction). *For $1 \leq p \leq \infty$, P_μ is a positive contraction on $L^p(X, \nu)$ whenever ν is μ-stationary.*

Proof. For finite p, Jensen gives $|Pf|^p \leq P|f|^p$; integrate and use stationarity. The $p = \infty$ bound is immediate from averaging. ☐

Theorem 0.2 (Mean ergodic theorem for stationary Markov operators). *On $L^2(X, \nu)$, the Cesàro averages $N^{-1} \sum_{n<N} P_\mu^n f$ converge in norm to the orthogonal projection onto* $\ker(I - P_\mu)$.

Proof. This is the Hilbert-space mean ergodic theorem for a contraction. Decompose L^2 into the fixed space and the closure of $\operatorname{ran}(I - P)$; telescoping gives convergence to zero on the latter, while the averages fix the former. ☐

Proposition 0.3 (Ergodicity criterion). *A stationary probability is extremal among stationary probabilities if and only if every bounded P_μ-harmonic function measurable with respect to the stationary invariant sigma-field is almost surely constant.*

Proof. A nontrivial invariant set splits the stationary measure into normalized restrictions, contradicting extremality. Conversely a nonconstant invariant harmonic indicator produces such a splitting by level sets and conditional expectation. ☐

CHAPTER 5

Recurrence and Transience

On noncompact homogeneous spaces, recurrence is never automatic; this chapter separates the abstract recurrence theorem from Lyapunov estimates that will be built in XXXIII.

Lemma 0.1 (Stationary Poincare recurrence). *If ν is stationary and A has positive ν-measure, then for ν-almost every starting point in A, almost every random path returns to A infinitely often.*

PROOF. Use the stationary path measure from Chapter 3. The cylinder $\{Z_0 \in A\}$ has positive measure and the shift preserves path measure. Poincaré recurrence gives infinitely many returns. □

THEOREM 0.2 (Stationary occupation identity). *For stationary ν, $\mathbb{E}_\nu N^{-1} \sum_{n<N} 1_A(Z_n) = \nu(A)$ for every N.*

PROOF. Each Z_n has law ν by stationarity, so every summand has expectation $\nu(A)$. □

Proposition 0.3 (Tight-stationary nonescape). *If a family of stationary probabilities is tight, no weak limit loses mass.*

PROOF. Prokhorov gives probability subsequential limits. Tightness means for every ϵ one compact set carries at least $1 - \epsilon$ of every measure, so the limit has total mass one by Portmanteau. □

CHAPTER 6

Ergodic Decomposition of Stationary Measures

Extremal stationary probabilities are the correct random-walk analogues of ergodic invariant measures.

Lemma 0.1 (Convex compactness)**.** *On compact X, the set of μ-stationary probabilities is a nonempty compact convex subset of* $\mathrm{Prob}(X)$*.*

PROOF. Nonemptiness is Chapter 2. Stationarity is a weak-* closed affine condition, hence the set is compact and convex. □

THEOREM 0.2 (Stationary ergodic decomposition). *Every stationary probability on a standard Borel G-space decomposes into extremal stationary probabilities, uniquely through the invariant sigma-field.*

PROOF. Apply ordinary ergodic decomposition to the shift-invariant stationary path measure. Conditional measures on shift-ergodic components project at time zero to stationary probabilities. Shift ergodicity makes them extremal, and conditional expectation on the invariant sigma-field gives uniqueness. □

Proposition 0.3 (Reduction to extremal components)**.** *Any property holding for almost every extremal stationary component holds for the original stationary measure after integration.*

PROOF. Integrate the component-wise full-measure sets against the decomposition measure and use Fubini. □

CHAPTER 7

Furstenberg Averages

Furstenberg averages extract stationary measures from orbit data and are the bridge from topological orbit closure to probabilistic measure rigidity.

Lemma 0.1 (Expected empirical measure)**.** *For $Z_n = g_n x$,*

$$\mathbb{E}\left[\frac{1}{N}\sum_{n<N}\delta_{Z_n}\right] = \frac{1}{N}\sum_{n<N}\mu^{*n} * \delta_x.$$

Proof. Test both sides against a bounded Borel function and interchange the finite sum with expectation. □

Theorem 0.2 (Furstenberg averaging principle). *If the Cesàro orbit measures are tight, every weak limit is stationary and supported on $\overline{\Gamma_\mu x}$, where Γ_μ is the semigroup generated by* $\operatorname{supp}\mu$.

Proof. Stationarity follows from the Cesàro defect. Every prelimit is supported on the semigroup orbit closure, a closed set, so Portmanteau preserves that support in the limit. □

Proposition 0.3 (Orbit-closure consequence)**.** *If a closed semigroup-invariant set admits a unique stationary probability, every tight Cesàro average starting in it converges to that probability.*

Proof. All subsequential limits are stationary and supported in the closed set. Uniqueness identifies every subsequential limit, hence the full sequence converges. □

CHAPTER 8

Boundary Preliminaries

A boundary records asymptotic information of a random walk that survives at large time. The full Poisson/Furstenberg boundary theory is postponed to Volume 31. Here we record only the bounded-harmonic martingale construction. This distinction is essential: the tail sigma-field of the independent increments themselves is not the Poisson boundary.

Lemma 0.1 (Harmonic martingale)**.** *Let $h : G \to \mathbb{R}$ be bounded and μ-harmonic for the left random walk convention of Chapter 1,*

$$h(g) = \int_G h(xg)\, d\mu(x).$$

Then $M_n = h(g_n g)$ is a bounded martingale with respect to $\mathcal{F}_n = \sigma(X_1, \ldots, X_n)$ for every starting point $g \in G$.

Proof. Since $g_{n+1} = X_{n+1} g_n$ and X_{n+1} is independent of $\mathcal{F}_n$ with law μ,

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = \int_G h(x g_n g)\, d\mu(x) = h(g_n g) = M_n.$$

Boundedness of h gives $\sup_n \|M_n\|_\infty \le \|h\|_\infty$. □

Theorem 0.2 (Boundary variable of a bounded harmonic function). *For every bounded μ-harmonic h and every starting point g, the martingale $h(g_n g)$ converges almost surely and in L^1 to a bounded random variable $H_h^{(g)}$. Moreover*

$$h(g) = \mathbb{E}_g H_h^{(g)}.$$

This chapter does not claim the converse identification of all boundary variables with bounded harmonic functions; that is part of the Poisson-boundary theory developed later.

Proof. The preceding lemma gives a uniformly bounded martingale. The bounded martingale convergence theorem therefore yields almost-sure and L^1 convergence to some $H_h^{(g)}$. Taking expectations and using L^1 convergence gives

$$\mathbb{E}_g H_h^{(g)} = \lim_{n\to\infty} \mathbb{E}_g h(g_n g) = h(g),$$

because a martingale has constant expectation. No assertion about the tail sigma-field of the independent increments is used. $\square$

Volume 28

Furstenberg's Random Matrix Products

Volume 28 scope and prerequisites

The objective is foundational: build the probabilistic and linear machinery needed by Benoist–Quint before stating their homogeneous classification theorem. No result from Volume 35 or later is used.

Proof and provenance. Primary comparison: Furstenberg [Fur63b] and Furstenberg–Kesten [FK60]. The projective stationary-measure, contraction, positivity, uniqueness, and directional-limit mechanisms are proved in Chapters 3–8. In particular, positivity is deduced from a singular-value gap produced by projective contraction and the determinant-one identity, so no forward appeal to Oseledets theory is made.

Reader guide: a concrete model

Why this matters.

Volume 28 is organized around the passage from *Projective Action of Random Matrices* to *Asymptotics of Random Matrix Products*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.3 (A calculation to keep in view)**.** Let

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}.$$

On projective space, a slope $s = y/x$ is sent to $s/4$, so every finite nonvertical slope contracts exponentially to the attracting line $[e_1]$. Moreover

$$\frac{1}{n}\log\|A^n e_1\| = \log 2, \qquad \frac{1}{n}\log\|A^n e_2\| = -\log 2.$$

Random products replace these deterministic exponents by Lyapunov exponents and this deterministic projective contraction by Furstenberg contraction.

CHAPTER 1

Projective Action of Random Matrices

For $V = \mathbb{R}^d$, projective space $\mathbb{P}(V)$ converts matrix growth into directional dynamics. We use the angular metric $d([u],[v]) = \|u \wedge v\|/(\|u\|\|v\|)$.

Lemma 0.1 (Projective Lipschitz bound)**.** *For $g \in \mathrm{GL}(V)$, $d(g[u], g[v]) \leq \|g\|^2\|g^{-1}\|^2 d([u],[v])$.*

PROOF. Apply g to the wedge numerator and bound by $\|\wedge^2 g\|\|u \wedge v\| \leq \|g\|^2\|u \wedge v\|$. Bound each denominator below by $\|g^{-1}\|^{-1}\|u\|$ and similarly for v. $\square$

THEOREM 0.2 (Projective random walk). *A matrix law μ with $\int \log^+ \|g\| d\mu < \infty$ induces a Feller Markov chain on compact $\mathbb{P}(V)$ and hence admits a stationary probability.*

PROOF. Continuity of the action makes P_μ preserve continuous functions by dominated convergence. Compactness and the averaging theorem of XXVII produce a stationary probability. $\square$

Proposition 0.3 (Log-norm cocycle)**.** *The function $\sigma(g,[v]) = \log(\|gv\|/\|v\|)$ is a measurable cocycle: $\sigma(gh,x) = \sigma(g,hx) + \sigma(h,x)$.*

PROOF. Cancel the intermediate norm $\|hv\|$ in the logarithm. $\square$

CHAPTER 2

Proximal Matrices

A proximal matrix has a unique attracting projective direction separated from a repelling hyperplane. This is the deterministic atom behind random contraction.

Lemma 0.1 (Spectral characterization). *If g has a real simple eigenvalue λ_1 whose modulus strictly exceeds all other eigenvalue moduli, then g is proximal on $\mathbb{P}(V)$ with attracting eigenline x_g^+ and repelling projective hyperplane X_g^-.*

Proof. Write $V = \mathbb{R}v_1 \oplus W$ with W the invariant generalized eigenspace for the remaining spectrum. After normalization by λ_1^n, the restriction to W tends to zero polynomially times an exponentially smaller factor. Projective images away from $\mathbb{P}(W)$ therefore converge uniformly on compact sets to $[v_1]$. □

Theorem 0.2 (Quantitative proximal contraction). *For every compact $K \subset \mathbb{P}(V) \setminus X_g^-$ there are $C > 0$ and $0 < q < 1$ with $d(g^n x, x_g^+) \leq Cq^n$ for $x \in K$.*

Proof. In the splitting from the lemma, the coefficient of v_1 is uniformly bounded away from zero on K, while the W component grows at rate at most $(q|\lambda_1|)^n$ after increasing q slightly to absorb Jordan polynomials. □

Proposition 0.3 (Stability of proximality). *A sufficiently small perturbation of a proximal matrix is proximal, with attracting line and repelling hyperplane varying continuously.*

Proof. The simple dominant eigenvalue is isolated by a spectral contour. The Riesz projection depends continuously on the matrix, preserving rank one and the spectral gap under small perturbation. □

CHAPTER 3

Stationary Measures on Projective Space

Strong irreducibility means no finite union of proper subspaces is invariant under the generated semigroup. It prevents stationary mass from concentrating on algebraic pieces.

Lemma 0.1 (No finite subspace support)**.** *If Γ_μ acts strongly irreducibly, a μ-stationary probability on $\mathbb{P}(V)$ gives zero mass to every proper projective subspace.*

PROOF. Choose a proper subspace of maximal stationary mass and minimal dimension among maximizers. Stationarity forces almost every translate to have the same mass. Distinct translates either coincide or intersect in smaller dimension; maximality/minimality therefore yields a finite invariant family of subspaces, contradicting strong irreducibility. □

THEOREM 0.2 (Existence of Furstenberg measure). *Every matrix law admits a projective stationary probability; under strong irreducibility its support spans V.*

PROOF. Existence is XXVII on compact projective space. If the support spanned a proper subspace, the smallest span of its semigroup translates would yield a finite invariant union after the maximal-mass argument, contradicting strong irreducibility. □

Proposition 0.3 (Support minimality)**.** *The support of an extremal projective stationary measure is a minimal nonempty closed Γ_μ-invariant subset whenever the stationary measure is unique.*

PROOF. The support is closed and invariant under every support element by stationarity and continuity. A smaller closed invariant subset carries a stationary probability by Furstenberg averaging, contradicting uniqueness. □

CHAPTER 4

The Furstenberg Formula

The top growth rate is the stationary average of the log-norm cocycle once the projective process is stationary.

Lemma 0.1 (Integrability of the cocycle)**.** *If* $\int(\log^+\|g\| + \log^+\|g^{-1}\|)d\mu < \infty$*, then* $|\sigma(g,x)|$ *is integrable uniformly in* $x \in \mathbb{P}(V)$*.*

Proof. The upper bound is $\log^+\|g\|$. The lower bound $\|gv\| \geq \|g^{-1}\|^{-1}\|v\|$ gives the negative-part bound. ☐

Theorem 0.2 (Furstenberg formula). *Let* ν *be a* μ*-stationary projective probability. For the stationary chain* (g_n, x_0)*,*

$$\lim_{n\to\infty} \frac{1}{n} \log \frac{\|g_n v\|}{\|v\|} = \iint \sigma(g,x)d\mu(g)d\nu(x)$$

almost surely whenever the stationary skew product is ergodic.

Proof. Cocycle additivity writes the logarithm as a Birkhoff sum of $\sigma(X_k, x_{k-1})$ over the stationary skew product. The integrability lemma permits Birkhoff, and ergodicity makes the limit the space average. ☐

Proposition 0.3 (Norm-growth upper bound)**.** *The Furstenberg integral is at most the Furstenberg–Kesten top exponent* $\lim n^{-1}\mathbb{E}\log\|g_n\|$.

Proof. For every vector direction, $\log\|g_n v\|/\|v\| \leq \log\|g_n\|$. Take expectations, divide by n, and pass to the two limits using truncation/uniform integrability. ☐

CHAPTER 5

Positivity of the Top Lyapunov Exponent

Positivity is not inferred from a qualitative equality case. We first extract from strong irreducibility and proximality an exponential gap between the first two singular values of the random product. The determinant-one identity then forces positive expansion in the top direction.

Lemma 0.1 (SIP singular-value gap without Oseledets)**.** *Let $V = \mathbb{R}^d$, let μ be a probability on* $\mathrm{GL}(V)$ *with*

$$\int (\log^+ \|g\| + \log^+ \|g^{-1}\|)\, d\mu(g) < \infty,$$

and suppose the semigroup Γ_μ generated by $\operatorname{supp}\mu$ *is strongly irreducible and contains a proximal element. For i.i.d. increments $X_1, X_2, \ldots$ put $G_n = X_n \cdots X_1$ and let $s_1(G_n) \geq s_2(G_n) \geq \cdots \geq s_d(G_n) > 0$ be its singular values. Then there is $c > 0$ such that*

$$\limsup_{n\to\infty} \frac{1}{n} \log \frac{s_2(G_n)}{s_1(G_n)} \leq -c \qquad \text{for almost every increment path.} \tag{28.5.1}$$

Proof. We give the contraction argument in a form that does not use the multiplicative ergodic theorem.

1. A uniformly proximal block. Choose a word $a = a_m \cdots a_1$ in Γ_μ which is proximal. Replacing it by a power, choose a projective ball B^+ around its attracting line and a neighbourhood B^- of its repelling hyperplane such that

$$a\big(\mathbb{P}(V) \setminus B^-\big) \subset B^+, \qquad \mathrm{Lip}\big(a|_{\mathbb{P}(V)\setminus B^-}\big) \leq q \tag{28.5.2}$$

with $0 < q < e^{-4}$. Proximality is open, so there are neighbourhoods $U_i \ni a_i$ for which every product $u_m \cdots u_1$, $u_i \in U_i$, satisfies (28.5.2) with q replaced by $q_0 < e^{-3}$. The block event $\mathcal{G} = \{X_{jm+i} \in U_i : 1 \leq i \leq m\}$ has probability $p > 0$; disjoint block events are independent, hence their counting function N_n satisfies $N_n/n \to p/m$ almost surely.

2. Repelling hyperplanes have no stationary mass. Let ν be a stationary probability on $\mathbb{P}(V)$, whose existence follows from compactness. If a proper projective subspace had positive ν-mass, choose one of maximal mass and then minimal dimension. Stationarity forces its Γ_μ-translates to have the same maximal mass; two distinct such translates intersect in smaller dimension, so only finitely many can occur. Their finite union would be Γ_μ-invariant, contradicting strong irreducibility. Thus

$$\nu(\mathbb{P}(W)) = 0 \quad \text{for every proper } W < V. \tag{28.5.3}$$

By compactness of the Grassmannian, for every $\varepsilon > 0$ one can choose the thickness of B^- so that every projective hyperplane neighbourhood occurring as a conjugate of the fixed repeller has ν-mass at most ε.

3. Good blocks dominate the intervening distortion. For $g \in \mathrm{GL}(V)$ write $L(g) = \log \|g\| + \log \|g^{-1}\|$. The projective Lipschitz constant of g is at most $e^{2L(g)}$. By the logarithmic moment and the strong law,

$$\sum_{j=1}^{n} L(X_j) = O(n), \qquad \max_{j \le n} L(X_j) = o(n) \tag{28.5.4}$$

almost surely. Select, among the occurrences of $\mathcal{G}$, those for which the incoming projective point is outside the corresponding repelling neighbourhood and for which the accumulated distortion since the preceding selected block is at most $e^{\delta r}$, where r is the intervening length. From (28.5.3), stationarity and Birkhoff, the first exclusion removes at most an $O(\varepsilon)$ fraction of blocks; from (28.5.4) and Markov truncation, the second removes at most an $O(\delta^{-1} \int_{L>M} L \, d\mu)$ fraction after first taking M large. Choose ε and then M so that a positive density $\rho > 0$ of blocks remains, and choose $\delta > 0$ so small that $q_0 e^{\delta m} < e^{-2}$.

For two projective points outside the exceptional hyperplane attached to the path, compose the Lipschitz bounds between successive selected blocks. Each selected proximal block contributes q_0 and the intervening products contribute at most the chosen $e^{\delta r}$. Hence for some random $C(\omega) < \infty$,

$$d_{\mathbb{P}}(G_n x, G_n y) \le C(\omega) e^{-\rho n} \tag{28.5.5}$$

whenever x, y lie in a fixed compact projective chart disjoint from the exceptional hyperplane.

4. From projective contraction to a singular-value gap. Write a singular-value decomposition $G_n = k_n \operatorname{diag}(s_1, \ldots, s_d) \ell_n$. If $s_2/s_1 \ge e^{-\rho n/2}$, then in the two-plane spanned by the first two right singular vectors one can choose two unit vectors x_n, y_n, separated by a fixed projective angle and outside any prescribed hyperplane neighbourhood, whose images have projective separation $\gg s_2/s_1 \ge e^{-\rho n/2}$. By compactness choose such a pair in one of finitely many fixed charts. This contradicts (28.5.5) for large n. Therefore

$$\frac{s_2(G_n)}{s_1(G_n)} \le C'(\omega) e^{-\rho n/2},$$

which is (28.5.1) with $c = \rho/2$. $\square$

Theorem 0.2 (Furstenberg positivity theorem). *Let μ satisfy the hypotheses of Lemma 0.1 and assume in addition that μ is supported on* $\mathrm{SL}(V)$. *Then the Furstenberg–Kesten top exponent*

$$\lambda_1 := \lim_{n\to\infty} \frac{1}{n} \log \|G_n\|$$

exists almost surely and satisfies $\lambda_1 > 0$.

Proof. Existence of the scalar limit is the subadditive norm-growth theorem proved in the scalar form used in Chapter 4: apply the block argument there to $\log \|G_n\|$. Put $s_j = s_j(G_n)$. Since $G_n \in \mathrm{SL}(V)$,

$$1 = \prod_{j=1}^{d} s_j \le s_1^{d-1} s_2 = s_1^d \frac{s_2}{s_1}. \tag{28.5.6}$$

Lemma 0.1 gives, for every $0 < c' < c$ and all sufficiently large n, $s_2/s_1 \le e^{-c'n}$. Taking logarithms in (28.5.6) yields

$$0 \le d \log s_1 - c'n, \qquad \frac{1}{n} \log \|G_n\| = \frac{1}{n} \log s_1 \ge \frac{c'}{d}.$$

Letting $n \to \infty$ and then $c' \uparrow c$ gives $\lambda_1 \ge c/d > 0$. □

Proposition 0.3 (Determinant normalization). *Let μ be a probability on* $\mathrm{GL}(V)$ *with $\int |\log|\det g||\,d\mu(g) < \infty$. Define $\widetilde{g} = |\det g|^{-1/d} g$. If $\widetilde{\lambda}_1$ is the top exponent of the normalized cocycle, then*

$$\lambda_1 = \widetilde{\lambda}_1 + \frac{1}{d} \int \log|\det g|\, d\mu(g). \tag{28.5.7}$$

Proof. For $G_n = X_n \cdots X_1$ and $\widetilde{G}_n = \widetilde{X}_n \cdots \widetilde{X}_1$,

$$\log \|G_n\| = \log \|\widetilde{G}_n\| + \frac{1}{d} \sum_{j=1}^{n} \log|\det X_j|.$$

Divide by n and use the strong law on the scalar determinant term. □

CHAPTER 6

Contraction on Projective Space

The previous chapter extracted the singular-value consequence needed for positivity. We now record the pathwise projective version, because it will also give uniqueness of the stationary measure and directional convergence.

Lemma 0.1 (Stationary measures charge no projective hyperplane)**.** *Let Γ_μ act strongly irreducibly on V. If ν is μ-stationary on $\mathbb{P}(V)$, then*

$$\nu(\mathbb{P}(W)) = 0 \qquad \text{for every proper linear subspace } W < V. \tag{28.6.1}$$

Moreover, for each $k < d$,

$$\sup_{\dim W = k} \nu\{x : d_{\mathbb{P}}(x, \mathbb{P}(W)) < r\} \longrightarrow 0 \quad (r \downarrow 0). \tag{28.6.2}$$

Proof. If a proper projective subspace has positive mass, choose one with maximal mass and then minimal dimension. Stationarity implies that every translate by an element of $\operatorname{supp}\mu$ has mass at least that maximal value; hence it has exactly the same mass. Two distinct maximal subspaces have an intersection of smaller dimension, so only finitely many such maximal subspaces can occur. Their union is invariant under Γ_μ, contradicting strong irreducibility. This proves (28.6.1). If (28.6.2) failed, compactness of the Grassmannian would provide $r_j \downarrow 0$ and $W_j \to W$ with neighbourhood mass bounded below. Upper semicontinuity of mass of decreasing closed neighbourhoods would then give $\nu(\mathbb{P}(W)) > 0$, contradicting (28.6.1). $\square$

Theorem 0.2 (Projective contraction and random attracting line). *Assume*

$$\int (\log^+ \|g\| + \log^+ \|g^{-1}\|)\, d\mu(g) < \infty$$

and that Γ_μ is strongly irreducible and proximal. Then there is a measurable random hyperplane $H^-(\omega)$ and a measurable random line $\xi^+(\omega)$ such that, for almost every path $\omega = (X_1, X_2, \dots)$ and every compact $K \Subset \mathbb{P}(V) \setminus \mathbb{P}(H^-(\omega))$,

$$\operatorname{diam}(G_n(\omega)K) \longrightarrow 0, \qquad G_n(\omega)x \longrightarrow \xi^+(\theta^n\omega)$$

uniformly for $x \in K$. More precisely, after changing the random prefactor there is $c > 0$ such that

$$\operatorname{diam}(G_n K) \le C_K(\omega) e^{-cn}. \tag{28.6.3}$$

The law of the backward-product attracting line is a μ-stationary probability on $\mathbb{P}(V)$.

PROOF. Choose a proximal word and a uniformly proximal neighbourhood as in the proof of Lemma 0.1. Disjoint blocks hit that neighbourhood with a fixed positive probability, so Borel–Cantelli and the strong law give a positive density of proximal blocks. By Lemma 0.1, a stationary direction spends arbitrarily small frequency in a sufficiently thin neighbourhood of any repelling hyperplane. The logarithmic moment gives $\max_{j\le n}(\log^+\|X_j\| + \log^+\|X_j^{-1}\|) = o(n)$ and permits a sparse selection of proximal blocks for which the product of all intervening projective Lipschitz constants is $e^{o(n)}$. Selecting the thickness first and then the sparse blocks leaves positive density. The uniform contraction $q < 1$ of each selected block therefore dominates the intervening distortion and gives (28.6.3).

For a fixed path, two points outside the limiting exceptional hyperplane have images at distance tending to zero. Apply this to a countable dense set and use compactness to obtain a unique limiting line; measurability follows because it is a pointwise limit of measurable projective images. Applying the construction to backward products $X_1 \cdots X_n$ gives a random attracting line ξ^- depending on the past. Its distribution ν satisfies

$$\int f\,d\nu = \mathbb{E}f(\xi^-) = \mathbb{E}f(X_1\xi^-(\theta^{-1}\omega)) = \int\int f(gx)\,d\mu(g)d\nu(x),$$

so $\mu * \nu = \nu$. ☐

Proposition 0.3 (Asymptotic synchronization)**.** *Let x_0, y_0 be independent of the increments and have laws that assign zero mass to every projective hyperplane. Then*

$$d_{\mathbb{P}}(G_n x_0, G_n y_0) \longrightarrow 0 \quad \textit{almost surely.} \tag{28.6.4}$$

If both initial laws are stationary, the convergence also holds in L^1 for the bounded projective metric.

PROOF. Almost surely neither x_0 nor y_0 lies in the random exceptional hyperplane, because conditioning on the path and using the hyperplane-null hypothesis gives probability zero. Theorem 0.2 therefore yields (28.6.4). The projective metric is bounded, so dominated convergence gives the L^1 assertion. ☐

CHAPTER 7

Uniqueness of the Stationary Projective Measure

Synchronization forces any two stationary laws to coincide.

Lemma 0.1 (Coupled stationary chains)**.** *If ν_1, ν_2 are stationary, one may evolve $x_0 \sim \nu_1$ and $y_0 \sim \nu_2$ using the same matrix increments; each marginal remains stationary.*

PROOF. This is a product coupling with common noise. Stationarity of each marginal follows from applying one step of P_μ. □

THEOREM 0.2 (Uniqueness under SIP). *Under strong irreducibility and proximal contraction, the projective stationary probability is unique.*

PROOF. For a Lipschitz f, stationarity gives $\int f d\nu_1 - \int f d\nu_2 = \mathbb{E}[f(g_n x_0) - f(g_n y_0)]$. The difference is bounded by the Lipschitz constant times projective distance. Synchronization sends the expectation to zero after truncating the exceptional set. Thus all Lipschitz integrals agree and the probabilities coincide. □

Proposition 0.3 (Unique minimal limit set)**.** *The support of the Furstenberg measure is the unique minimal closed invariant subset of projective space.*

PROOF. Any nonempty closed invariant subset carries a stationary measure by averaging. Uniqueness forces that measure to be the Furstenberg measure, so its support lies in every such subset. □

CHAPTER 8

Asymptotics of Random Matrix Products

The direct Furstenberg theory already yields a rank-one directional asymptotic; the full Lyapunov flag is the subject of XXIX.

Lemma 0.1 (Random attracting direction)**.** *Under SIP, for almost every increment path the push-forwards $g_n\nu$ converge weakly to a random Dirac mass $\delta_{Z(\omega)}$.*

PROOF. Synchronization shows the diameter of g_n applied to a set of arbitrarily large ν-measure tends to zero. Compactness gives a limiting point; uniqueness of subsequential points follows from the same contraction. Testing continuous functions gives weak convergence to a Dirac mass. □

THEOREM 0.2 (Furstenberg directional limit). *The law of the random attracting direction Z is the unique stationary projective measure.*

PROOF. Shift the increment sequence by one step. The attracting direction transforms by the first matrix, giving the distributional fixed-point equation $\mathrm{Law}(Z) = \mu * \mathrm{Law}(Z)$. Uniqueness identifies the law. □

Volume 29

Oseledets Theory and Lyapunov Spectra

Volume 29 scope and prerequisites

The objective is foundational: build the probabilistic and linear machinery needed by Benoist–Quint before stating their homogeneous classification theorem. No result from Volume 35 or later is used.

Proof and provenance. Kingman and Oseledets are checked against their original theorem architecture [Kin68; Ose68] and are proved internally in Chapters 1 and 3. The Kingman proof includes the stopping-block lower estimate, not only the easy limsup blocking argument. The Oseledets proof constructs measurable forward and backward flags from exterior-power limits, proves subsequence independence, and derives tempered angles. The inverse-logarithmic moment is stated exactly where the two-sided splitting is used.

Reader guide: a concrete model

Why this matters.

Volume 29 is organized around the passage from *The Subadditive Ergodic Theorem* to *Integrability and Regularity of Lyapunov Data*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.3 (A calculation to keep in view)**.** Let $A_n = \operatorname{diag}(2^n, 2^{-n})$ and put $F_n = \log \|A_n\|$. Since $A_{m+n} = A_m A_n$,

$$F_{m+n} \le F_m + F_n,$$

and here equality holds: $F_n = n \log 2$. Hence

$$\frac{F_n}{n} \to \log 2.$$

Kingman's theorem replaces this deterministic subadditivity calculation by an almost-sure limit for random cocycles; Oseledets then refines the single norm-growth number into growth rates on invariant subspaces.

CHAPTER 1

The Subadditive Ergodic Theorem

Kingman's theorem is the scalar engine behind norm growth. The difficult direction is the lower bound on the liminf; we prove it by a stopping-block argument rather than hiding it in the phrase "standard subadditive ergodic theorem."

Lemma 0.1 (Subadditive stopping-block lemma)**.** *Let* $(X,\mathcal{B},\mathbb{P},T)$ *be probability preserving and let* $h_{n+m} \le h_n + h_m \circ T^n$*. Fix* $N \ge 1$ *and put*

$$E_N = \Big\{x : \min_{1\le j\le N} h_j(x) < 0\Big\}, \qquad \tau(x) = \min\{1 \le j \le N : h_j(x) < 0\}$$

on E_N*, while* $\tau(x) = 1$ *off* E_N*. Greedily set* $t_0 = 0$ *and* $t_{r+1} = t_r + \tau(T^{t_r}x)$ *until* $t_r \ge M - N$*. Then*

$$h_M(x) \le \sum_{r:t_r<M-N} h_{\tau(T^{t_r}x)}(T^{t_r}x) + R_{M,N}(x), \tag{29.1.1}$$

where for fixed N*,*

$$\frac{|R_{M,N}|}{M} \longrightarrow 0$$

in probability and in L^1 *after truncating the finite family* $\{\max_{j\le N}|h_j| \circ T^r : 0 \le r < N\}$*.*

Proof. Iterated subadditivity gives (29.1.1) up to the final incomplete block. Its length is $< N$, so the remainder is dominated by a sum of at most N translates of $\max_{j\le N}|h_j|$. Truncate this integrable finite maximum at height K, divide by M, and let $M \to \infty$; the truncated contribution is at most NK/M. Then let $K \to \infty$ and use integrability to remove the truncation. ∎

Theorem 0.2 (Kingman subadditive ergodic theorem). *Let* $f_n \in L^1(X)$ *satisfy*

$$f_{n+m}(x) \le f_n(x) + f_m(T^n x) \tag{29.1.2}$$

for all $m, n \ge 1$*, and suppose there is* $g \in L^1$*,* $g \ge 0$*, such that*

$$f_n(x) \ge -\sum_{j=0}^{n-1} g(T^j x) \tag{29.1.3}$$

for every n. Then f_n/n converges almost surely to a T-invariant integrable function f_*. Moreover

$$\int f_*\,d\mathbb{P} = \inf_{n\geq 1} \frac{1}{n}\int f_n\,d\mathbb{P}. \tag{29.1.4}$$

If T is ergodic, f_* is almost surely the constant on the right-hand side of (29.1.4). Under (29.1.3) the convergence also holds in L^1.

Proof. We first assume T ergodic and write

$$\gamma := \inf_{k\geq 1} \frac{1}{k}\int f_k\,d\mathbb{P}.$$

Upper bound. For fixed k, write a long orbit segment in k-blocks and average the decomposition over the k possible starting residues. The boundary contribution is $o(n)$ by Birkhoff applied to the finite family $|f_j|$, $j \leq k$. Hence

$$\limsup_{n\to\infty} \frac{f_n(x)}{n} \leq \frac{1}{k}\int f_k\,d\mathbb{P}$$

for almost every x. Taking the infimum over k gives $\limsup f_n/n \leq \gamma$.

Lower bound. Fix $\varepsilon > 0$ and set

$$h_n = f_n - n(\gamma - 2\varepsilon).$$

Suppose $A = \{\liminf_n f_n/n < \gamma - 3\varepsilon\}$ has positive measure. Then $A \subset \bigcup_N E_N$ for the stopping sets of Lemma 0.1; choose N with $\mathbb{P}(E_N) > 0$. On E_N the selected block has strictly negative h-value. After truncating the negative block values, ergodicity applied to the induced finite-state selection process gives a negative mean contribution $-\delta M + o(M)$ in (29.1.1) on a set of positive measure, for some $\delta > 0$. Integrating the stopped inequality and using the L^1-vanishing of $R_{M,N}/M$ therefore yields, for some sufficiently large deterministic M,

$$\frac{1}{M}\int h_M\,d\mathbb{P} < 0.$$

Consequently

$$\frac{1}{M}\int f_M\,d\mathbb{P} < \gamma - 2\varepsilon < \gamma,$$

contradicting the definition of γ. Thus $\liminf f_n/n \geq \gamma$, and the almost-sure limit equals γ.

For general T, apply the same proof to the ergodic decomposition; the resulting limit is T-invariant and (29.1.4) follows by integrating the conditional formula. Finally (29.1.3) gives $f_n^-/n \leq n^{-1}\sum_{j<n} g \circ T^j$, an L^1-uniformly integrable lower control, while the blocking inequality gives uniform integrability of the positive deviations after truncation. Together with almost-sure convergence and convergence of the integrals in (29.1.4), Vitali's criterion gives L^1 convergence. □

Proposition 0.3 (Furstenberg–Kesten norm limit)**.** *Let* $A(\omega)$ *be an invertible stationary matrix cocycle over an ergodic base and assume* $\log^+\|A\| + \log^+\|A^{-1}\| \in L^1$*. Then*

$$\frac{1}{n}\log\|A(T^{n-1}\omega)\cdots A(\omega)\| \longrightarrow \lambda_1 \tag{29.1.5}$$

almost surely and in L^1*, where*

$$\lambda_1 = \inf_{n\geq 1}\frac{1}{n}\int \log\|A^{(n)}(\omega)\|\, d\mathbb{P}(\omega).$$

Proof. Take $f_n(\omega) = \log\|A^{(n)}(\omega)\|$. Matrix submultiplicativity gives (29.1.2), and

$$f_n \geq -\log\|(A^{(n)})^{-1}\| \geq -\sum_{j=0}^{n-1}\log^+\|A(T^j\omega)^{-1}\|$$

gives (29.1.3). Theorem 0.2 applies, and ergodicity makes the limit constant. □

CHAPTER 2

Linear Cocycles

A linear cocycle over T is a measurable $A(x) \in \mathrm{GL}_d$ with products $A^{(n)}(x) = A(T^{n-1}x) \cdots A(x)$.

Lemma 0.1 (Cocycle identity). *$A^{(n+m)}(x) = A^{(m)}(T^n x)A^{(n)}(x)$.*

PROOF. Split the ordered product at time n. □

THEOREM 0.2 (Exterior-power cocycle). *For every k, $\wedge^k A^{(n)} = (\wedge^k A)^{(n)}$ and $\| \wedge^k A^{(n+m)}\| \leq \| \wedge^k A^{(m)}(T^n x)\| \| \wedge^k A^{(n)}(x)\|$.*

PROOF. Exterior powers are functorial and operator norms are submultiplicative. □

Proposition 0.3 (Existence of singular-value growth sums). *Under the logarithmic moment assumptions for A and A^{-1}, for each k the limit $n^{-1} \log \| \wedge^k A^{(n)}(x)\|$ exists almost surely.*

PROOF. Apply Kingman to the logarithm of the exterior-power norm; the inverse moment supplies the integrable lower control. □

CHAPTER 3

The Oseledets Filtration

The proof below is internal. Two commonly compressed steps are made explicit: subsequence independence of the slow flags and temperedness of complementary angles. Both are derived from exterior-power growth. The multiplicative ergodic theorem refines scalar norm growth into invariant subspaces with distinct exponential rates.

Lemma 0.1 (Asymptotic metric sequence and slow-space uniqueness). *Let $B_n(x) = (A^{(n)}(x)^* A^{(n)}(x))^{1/(2n)}$. On the full-measure set where every exterior-power Kingman limit*

$$L_k(x) = \lim_{n\to\infty} \frac{1}{n} \log \| \wedge^k A^{(n)}(x)\|$$

exists, every convergent subsequence of $B_n(x)$ has eigenvalue logarithms $\lambda_1 \geq \cdots \geq \lambda_d$ satisfying

$$\lambda_1 + \cdots + \lambda_k = L_k(x), \qquad 1 \leq k \leq d. \tag{29.3.1}$$

For every threshold a not equal to one of the λ_i, the slow space

$$F_a^+(x) = \{v : \limsup_{n\to\infty} n^{-1} \log \|A^{(n)}(x)v\| \leq a\} \tag{29.3.2}$$

is a linear subspace and is independent of the convergent subsequence used to construct it.

Proof. If $s_1^{(n)} \geq \cdots \geq s_d^{(n)}$ are the singular values, then $\| \wedge^k A^{(n)}\| = s_1^{(n)} \cdots s_k^{(n)}$. Therefore every subsequential limit of the logarithms $n^{-1} \log s_i^{(n)}$ satisfies (29.3.1), which determines the multiset of limiting exponents.

Fix a in a spectral gap. If a subsequential spectral slow space F did not equal the intrinsic set (29.3.2), choose v in one but not the other and decompose $v = v_- + v_+$ according to the limiting spectral splitting. The v_+ component has exponent strictly larger than a and dominates the v_- component along that subsequence, contradicting the defining limsup bound. Conversely every vector in the spectral slow space has upper growth at most a by the spectral theorem applied to B_n. Hence the intrinsic slow space and every subsequential spectral slow space coincide. Linearity follows from the triangle inequality in (29.3.2). ☐

Theorem 0.2 (Oseledets multiplicative ergodic theorem). *Assume* $\log^+\|A^{\pm 1}\| \in L^1$. *Then for almost every* x *there are exponents* $\lambda_1(x) > \cdots > \lambda_r(x)$ *and a measurable invariant splitting* $\mathbb{R}^d = E_1(x) \oplus \cdots \oplus E_r(x)$ *such that for* $0 \neq v \in E_i(x)$,

$$\lim_{n\to\pm\infty} n^{-1} \log \|A^{(n)}(x)v\| = \lambda_i(x).$$

Proof. We separate the construction into five steps so that the inverse-moment hypothesis is used visibly.

Step 1: singular-value exponents. Apply Kingman to every exterior-power cocycle $\wedge^k A$. Proposition 0.3 gives limits

$$L_k(x) = \lim_{n\to\infty} \frac{1}{n} \log \| \wedge^k A^{(n)}(x)\|, \qquad 1 \leq k \leq d.$$

If $s_1^{(n)} \geq \cdots \geq s_d^{(n)}$ are the singular values of $A^{(n)}$, then $\log \| \wedge^k A^{(n)}\| = \sum_{j\leq k} \log s_j^{(n)}$. Hence the differences $L_k - L_{k-1}$ determine the limiting singular-value exponents, with multiplicities.

Step 2: forward slow flags. For a subsequence on which the positive operators

$$B_n = (A^{(n)*}A^{(n)})^{1/(2n)}$$

converge on each spectral block, Lemma 0.1 shows that the eigenvalue logarithms of the limit are exactly the exponents from Step 1. Let $F_i^+(x)$ be the sum of eigenspaces with exponent at most λ_i. The exterior-power characterization shows that F_i^+ is independent of the chosen convergent subsequence: if two subsequential flags differed, a vector in the first but not the second would have two incompatible exponential upper-growth rates. Measurability follows by constructing the spectral projections as measurable limits along a fixed diagonal subsequence.

Step 3: backward fast flags. Apply the same argument to the inverse cocycle over T^{-1}. This is where $\log^+\|A^{-1}\| \in L^1$ is required. It yields measurable backward flags $F_i^-(x)$ whose vectors have the corresponding backward exponential bounds.

Step 4: splitting and rates. Define

$$E_i(x) = F_i^+(x) \cap F_i^-(x)$$

after indexing the two flags in opposite order. The dimension identities furnished by the exterior powers give $\dim E_i$ equal to the multiplicity of λ_i and imply the direct sum $\mathbb{R}^d = \bigoplus_i E_i(x)$. For $0 \neq v \in E_i(x)$ the forward flag gives $\limsup n^{-1} \log \|A^{(n)}v\| \leq \lambda_i$, while membership in the complementary backward-fast flag rules out any smaller exponent: otherwise applying $A^{(-n)}$ would contradict the backward bound. Thus the forward limit equals λ_i; the same reasoning for the inverse cocycle gives the backward limit.

Step 5: covariance and tempered angles. The cocycle identity sends the intrinsic slow spaces (29.3.2) at x to those at Tx; applying the same statement to the inverse cocycle gives $A(x)E_i(x) = E_i(Tx)$. It remains to prove that the splitting cannot become exponentially tangent.

Let $F = E_1 \oplus \cdots \oplus E_j$ and $F' = E_{j+1} \oplus \cdots \oplus E_r$. Choose unit simple multivectors $u_n \in \wedge^{\dim F} F(T^n x)$ and $v_n \in \wedge^{\dim F'} F'(T^n x)$. The sine of the angle between $F(T^n x)$ and $F'(T^n x)$ is comparable, with dimension-dependent constants, to $\|u_n \wedge v_n\|$. If that angle were at most $e^{-\varepsilon n}$ along a positive-density subsequence, then the wedge of normalized images of fixed nonzero multivectors u_0, v_0 would acquire an extra factor $e^{-\varepsilon n}$ along that subsequence. But exterior-power Kingman limits give simultaneously

$$\frac{1}{n} \log \| \wedge^{\dim F} A^{(n)} u_0 \| \to \sum_{i \le j} m_i \lambda_i, \quad \frac{1}{n} \log \| \wedge^{\dim F'} A^{(n)} v_0 \| \to \sum_{i > j} m_i \lambda_i,$$

and

$$\frac{1}{n} \log \| \wedge^{d} A^{(n)} (u_0 \wedge v_0) \| \to \sum_{i} m_i \lambda_i.$$

The extra $-\varepsilon$ contradicts the last equality. Hence for every $\varepsilon > 0$ the inverse angle is eventually at most $e^{\varepsilon n}$: the projection norms are tempered. This permits arbitrary vectors in each E_i in place of singular vectors without changing their exponential rate.

The five steps give the measurable invariant splitting and both time-direction limits asserted in the theorem. □

Proposition 0.3 (Ergodic constancy). *If the base transformation is ergodic, the Lyapunov exponents and multiplicities are almost surely constant.*

Proof. Cocycle invariance shifts the Oseledets data along T, so each exponent and multiplicity is T-invariant. Ergodicity makes invariant measurable functions constant. □

CHAPTER 4

Lyapunov Exponents

The exponent list controls determinants, volume growth, and all vector norms after projection to the filtration.

Lemma 0.1 (Determinant sum rule)**.** *Counting multiplicities,* $\sum_i m_i\lambda_i = \int \log|\det A(x)|d\mathbb{P}$ *in the ergodic case.*

PROOF. The determinant of $A^{(n)}$ is the product of its singular values. Divide its logarithm by n and use Birkhoff on $\log|\det A|$; Oseledets gives the left side. $\square$

THEOREM 0.2 (Vector-growth classification). *For almost every x, every nonzero vector has exponent equal to the largest λ_i for which its component in the corresponding forward filtration quotient is nonzero.*

PROOF. Decompose the vector into Oseledets subspaces. The summand with largest exponent dominates exponentially; tempered angles prevent cancellation at exponential scale. $\square$

Proposition 0.3 (Norm-limit identification)**.** $\lambda_1 = \lim n^{-1} \log\|A^{(n)}(x)\|$ *almost surely.*

PROOF. The operator norm is the largest singular value. Its logarithmic rate is the first singular-value exponent in the Oseledets construction. $\square$

CHAPTER 5

Exterior-Power Characterization

Exterior powers turn sums and gaps of exponents into top-exponent statements.

Lemma 0.1 (Exterior exponent formula)**.** *The Lyapunov exponents of* $\wedge^k A$ *are all sums* $\lambda_{i_1} + \cdots + \lambda_{i_k}$ *with multiplicity.*

Proof. Wedge a basis adapted to the Oseledets splitting. Each simple wedge is multiplied at exponential rate equal to the sum of the rates of its factors, and these wedges span the exterior power. □

Theorem 0.2 (Partial-sum characterization). $\lambda_1 + \cdots + \lambda_k = \lim n^{-1} \log \| \wedge^k A^{(n)} \|$.

Proof. The top exterior exponent is the sum of the k largest original exponents by the preceding lemma. □

Proposition 0.3 (Gap criterion)**.** $\lambda_k > \lambda_{k+1}$ *if and only if the top Lyapunov exponent of* $\wedge^k A$ *is simple.*

Proof. The top sum uses the first k exponents. Replacing λ_k by λ_{k+1} gives the next possible sum; equality occurs exactly when the boundary exponents coincide. □

CHAPTER 6

Simplicity Criteria

Furstenberg's SIP mechanism from XXVIII can now be translated into a statement about a Lyapunov gap.

Lemma 0.1 (Projective contraction implies a gap)**.** *If projective images of two generic directions synchronize exponentially at rate $c > 0$, then $\lambda_1 - \lambda_2 \geq c$.*

Proof. For independent vectors u, v, projective distance after $A^{(n)}$ is the ratio $\|A^{(n)}u \wedge A^{(n)}v\| / (\|A^{(n)}u\| \|A^{(n)}v\|)$. Oseledets gives logarithmic rate $(\lambda_1 + \lambda_2) - 2\lambda_1 = -(\lambda_1 - \lambda_2)$. Compare with the assumed contraction. ☐

Theorem 0.2 (SIP top-exponent simplicity). *Under the strong-irreducibility and proximal hypotheses of XXVIII, $\lambda_1 > \lambda_2$.*

Proof. XXVIII gives positive-frequency uniform proximal blocks and hence strict exponential synchronization outside negligible repelling neighborhoods. Apply the preceding lemma. ☐

Proposition 0.3 (Exterior SIP criterion)**.** *If the induced semigroup on $\wedge^k V$ is strongly irreducible and proximal, then $\lambda_k > \lambda_{k+1}$.*

Proof. Apply top-exponent simplicity to the exterior cocycle and use the gap characterization of Chapter 5. ☐

CHAPTER 7

Flag-Valued Limits

When all adjacent gaps are positive, random products select an asymptotic flag.

Lemma 0.1 (Oseledets flag measurability)**.** *The map* $x \mapsto (E_1(x), E_1(x) \oplus E_2(x), \dots)$ *is measurable into the flag variety.*

Proof. Each subspace is obtained as a spectral subspace of measurable limits of singular-value data; Grassmannian spectral projections are Borel on the locus with a gap. Finite products preserve measurability. $\square$

Theorem 0.2 (Flag convergence under simple spectrum). *If* $\lambda_1 > \cdots > \lambda_d$*, then for almost every path and every initial full flag transverse to the backward Oseledets flag,* $A^{(n)}$ *sends it to the forward Oseledets flag.*

Proof. Apply the projective dominance argument successively to $\wedge^k V$. Transversality ensures a nonzero component in the top exterior Oseledets line for every k, so each k-plane converges to the corresponding forward sum. $\square$

Proposition 0.3 (Stationary flag law)**.** *For an i.i.d. cocycle, the law of the forward Oseledets flag is stationary for the induced flag action.*

Proof. Removing the first increment shifts the path and transforms the forward flag by that first matrix. Taking laws gives the stationary fixed-point equation. $\square$

CHAPTER 8

Integrability and Regularity of Lyapunov Data

Regularity of exponents requires stronger hypotheses than mere existence. We record only the robust uniform-integrability transfer needed later.

Lemma 0.1 (Uniform-integrability transfer)**.** *If probability laws* $\mu_j \to \mu$ *weakly and* $\log^+ \|g^{\pm 1}\|$ *are uniformly integrable, then the associated one-step cocycle integrals against convergent stationary laws pass to the limit.*

Proof. Truncate the logarithmic cocycle at height M. The truncated integrand is bounded and continuous off a stationary-null algebraic set, so weak convergence applies. Uniform integrability makes the discarded tails uniformly small; let $M \to \infty$. □

Theorem 0.2 (Continuity of the top exponent under a uniform projective gap). *In a family with uniform logarithmic integrability, unique stationary projective measures, and a uniform contraction estimate, the top Lyapunov exponent varies continuously with the driving law.*

Proof. Uniform contraction gives continuity of the stationary projective measure by the common-noise coupling argument. The Furstenberg formula expresses the top exponent as the integral of the log-norm cocycle. Apply the uniform-integrability transfer to both the driving and stationary measures. □

Volume 30

Strong Irreducibility, Proximality, and Zariski Density

Volume 30 scope and prerequisites

This is part of the Benoist–Quint foundation block. Its arguments use only material from Volumes 1–29, so the later classification theorems are not used circularly.

Proof and provenance. Algebraic criteria are written representation-by-representation and compared with Goldsheid–Margulis [GM89]. We do not use the unsafe slogan "Zariski dense implies proximal" without the explicit proximal-representation hypothesis.

Reader guide: a concrete model

Why this matters.

Volume 30 is organized around the passage from *Irreducibility* to *Algebraic Criteria for SIP*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.3 (A calculation to keep in view)**.** Let

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 0 \\ 0 & 1/3 \end{pmatrix},$$

and choose A and B independently with probabilities p and $1-p$. For the product $G_n = g_n \cdots g_1$ the two Lyapunov exponents are obtained without any limiting theorem:

$$\lambda_1 = p \log 2 + (1-p) \log 3, \qquad \lambda_2 = -\lambda_1.$$

Indeed the logarithm of the expanding diagonal entry is a sum of independent increments. This diagonal model is useful for checking signs, normalization, and determinant constraints before the noncommutative Oseledets theory is invoked.

CHAPTER 1

Irreducibility

We distinguish irreducibility of a single representation from stronger finite-union conditions needed for stationary-measure rigidity.

Lemma 0.1 (Invariant-span test)**.** *A semigroup $\Gamma < \mathrm{GL}(V)$ acts irreducibly if and only if the span of every nonzero orbit Γv is V.*

PROOF. The orbit span is a nonzero invariant subspace. If the action is irreducible it must be all of V; conversely any invariant subspace contains the orbit span of each of its nonzero vectors. □

THEOREM 0.2 (Burnside criterion over an algebraically closed field). *For a finite-dimensional complex representation, irreducibility is equivalent to the associative algebra generated by Γ being $\mathrm{End}(V)$.*

PROOF. If the generated algebra is all endomorphisms, invariant subspaces are impossible. Conversely, choose a vector of minimal nonzero orbit-algebra dimension and use rank-one separation: failure to generate all endomorphisms produces a nontrivial common invariant subspace via the annihilator of the generated matrix algebra. This is the finite-dimensional density/Burnside argument. □

Proposition 0.3 (Exterior-power detection)**.** *A k-plane invariant under Γ gives an invariant projective line in $\mathbb{P}(\wedge^k V)$.*

PROOF. Wedge a basis of the invariant plane. Every g sends that wedge to $\det(g|_W)$ times itself. □

CHAPTER 2

Strong Irreducibility

Strong irreducibility excludes invariant finite unions of proper subspaces. It is exactly what was used in XXVIII to eliminate atoms on projective subspaces.

Lemma 0.1 (Finite-index reformulation)**.** Γ *is strongly irreducible if and only if every finite-index subsemigroup/group generated by powers acts irreducibly.*

Proof. An invariant finite union of subspaces is permuted by Γ; the kernel of the finite permutation action has finite index and fixes each component. Conversely a proper subspace invariant under a finite-index subgroup has finitely many Γ-translates, giving an invariant finite union. □

Theorem 0.2 (Stationary zero-mass consequence). *For strongly irreducible Γ_μ, every projective stationary probability gives zero mass to every proper projective subspace.*

Proof. This is the maximal-mass/minimal-dimension argument proved in XXVIII.3. The present chapter records it as the exact transfer consequence of strong irreducibility. □

Proposition 0.3 (Stability under finite-index passage)**.** *Strong irreducibility is unchanged after replacing Γ by a finite-index subgroup.*

Proof. Apply the finite-index reformulation twice, noting that finite-index subgroups have a common finite-index core. □

CHAPTER 3

Proximality

Proximality is representation-dependent. We therefore keep track of the representation each time rather than speaking of a group as "proximal" without qualification.

Lemma 0.1 (Power stability)**.** *A proximal element remains proximal under every positive power and has the same attracting line and repelling hyperplane.*

PROOF. Powers preserve eigenvectors and raise eigenvalue moduli to powers, preserving the strict dominance gap. □

THEOREM 0.2 (Conjugacy stability). *If g is proximal, then hgh^{-1} is proximal with attracting line hx_g^+ and repelling hyperplane hX_g^-.*

PROOF. Conjugation transports the invariant spectral splitting. □

Proposition 0.3 (Proximal semigroup criterion)**.** *A representation of a semigroup is proximal exactly when the semigroup contains an element with a simple dominant eigenvalue in the representation.*

PROOF. This is the definition translated through XXVIII.2's spectral characterization. □

CHAPTER 4

Zariski Closure of Random Semigroups

The Zariski closure packages all polynomial invariants of the support semigroup. It is the correct algebraic object for deciding which homogeneous subgroup can ultimately appear.

Lemma 0.1 (Semigroup/group Zariski closure)**.** *For a subsemigroup $S < \mathrm{GL}_d(\mathbb{R})$, its Zariski closure is an algebraic subgroup.*

Proof. Let $H = \overline{S}^{Z}$. Regularity of multiplication gives $HH \subset H$. Fix $s \in S$. The Zariski closure C_s of the positive powers $\{s^n : n \geq 1\}$ equals the Zariski closure of the full cyclic group $\{s^n : n \in \mathbb{Z}\}$: after triangularizing s over $\mathbb{C}$, the restriction of any regular function to s^n is a finite sum of terms $p(n)\lambda^n$; if such an exponential polynomial vanishes for every $n \geq 1$, the linear recurrence it satisfies forces it to vanish for every integer n. Hence $s^{-1} \in C_s \subset H$. Since inversion is a regular map on GL_d, the set H^{-1} is Zariski closed and contains S^{-1}; the preceding argument gives $S^{-1} \subset H$, hence $H^{-1} \subset H$ by density. Thus H contains identity and inverses and is an algebraic subgroup. □

Theorem 0.2 (Chevalley invariant-line realization). *Every proper real algebraic subgroup $H < G$ is the stabilizer of a line in some finite-dimensional algebraic representation of G, after possibly replacing the representation by a finite direct sum.*

Proof. Embed G/H as a locally closed algebraic orbit using finitely many generators of the ideal of H. The induced action on the finite-dimensional span of their translates produces a wedge line whose stabilizer is exactly H. This is the Chevalley construction already used in the earlier AHD linearization volumes. □

Proposition 0.3 (Zariski-density test)**.** *If no nontrivial line arising from the Chevalley construction for a proper subgroup is invariant under S, then S is Zariski dense in G.*

Proof. If the closure were a proper algebraic subgroup, its Chevalley line would be invariant, contradicting the hypothesis. □

CHAPTER 5

Existence of Proximal Elements

Zariski density alone does not make every representation proximal. We state the safe form used later: proximality is first verified in the relevant highest-weight representation, then transported to the semigroup.

Lemma 0.1 (Zariski invariance of the proximality index)**.** *Let $S < \mathrm{GL}(V)$ be completely reducible and let $G = \overline{S}^{Z}$. Let $\mathcal{A}_S$ be the Euclidean closure in* $\mathrm{End}(V)$ *of the real algebra generated by S, and set*

$$r(S,V) = \min\{\operatorname{rank} T : 0 \neq T \in \mathcal{A}_S\}.$$

Then $r(S,V) = r(G,V)$.

Proof. The algebra generated by a completely reducible semigroup is semisimple on the sum of its irreducible isotypic components. On each simple block, matrix coefficients of words in S are Zariski dense in the coefficient algebra generated by G; taking finite direct sums preserves this density. A nonzero operator of minimal rank can be obtained as a limit after scalar normalization of a sequence in this algebra. Rank conditions $\operatorname{rank} T \leq r$ are algebraic (vanishing of all $(r+1)$-minors), so the least rank obtainable in the Euclidean closure depends only on the Zariski closure of the generating semigroup. Applying the argument block by block gives the equality. This is the finite-dimensional matrix-algebra core of the Goldsheid–Margulis proximality-index theorem; complete reducibility is the hypothesis that prevents a nilpotent radical from lowering the rank on one side only. □

Theorem 0.2 (Goldsheid–Margulis proximal-element transfer). *Let S act strongly irreducibly on V and let $G = \overline{S}^{Z}$. If G contains a proximal element on V, then S contains a proximal element on V.*

Proof. For a strongly irreducible action, the minimal nonzero rank in the closure of the generated matrix algebra is one exactly when the acting group contains a proximal element: powers of a proximal element, normalized by its dominant eigenvalue, converge to a rank-one projector; conversely, a rank-one limit T can be approximated by normalized semigroup elements, and for an approximation sufficiently close to T the image cone about $\operatorname{im} T$ is mapped strictly inside itself while the complementary hyperplane is expanded less. The projective cone criterion then makes that approximating element proximal. Since the lemma gives $r(S,V) = r(G,V)$, proximality of G gives $r(G,V) = 1$, hence $r(S,V) = 1$, and the converse part of the rank-one argument produces a proximal element of S. ☐

Proposition 0.3 (Representation-by-representation warning)**.** *The preceding theorem must be checked separately for each exterior/highest-weight representation used to force a Lyapunov gap.*

Proof. Proximality of $\rho_1(g)$ says nothing by itself about $\rho_2(g)$. The theorem's hypothesis explicitly includes the existence of a proximal element in the chosen representation. ☐

CHAPTER 6

The Furstenberg Group

For a matrix law, the Furstenberg group is the closed subgroup generated by the support. Its algebraic and topological properties control all projective stationary laws.

Lemma 0.1 (Support-generated group)**.** *The closed group* $G_\mu = \overline{\langle \operatorname{supp} \mu \rangle}$ *is the smallest closed subgroup supporting every convolution power.*

Proof. Every convolution product is a product of support elements, hence lies in the generated group. Any closed group containing the support contains all such products and their closure. $\square$

Theorem 0.2 (Stationary support invariance). *The support of a* μ*-stationary probability is invariant under every element of* $\operatorname{supp} \mu$ *and hence under the support semigroup.*

Proof. If an open set meets $g \operatorname{supp} \nu$ for a support element g, continuity gives neighborhoods of g and a positive-ν set mapped into it. Stationarity and positive μ-mass of the neighborhood force positive ν-mass in the open set. $\square$

Proposition 0.3 (Algebraic hull interface)**.** *Every algebraic invariant of the stationary projective/flag support is invariant under the Zariski closure of* G_μ*.*

Proof. Polynomial equations vanishing on the support and preserved by a dense subset are preserved by its Zariski closure because the preservation condition is algebraic in the acting matrix. $\square$

CHAPTER 7

Ping-Pong and Free Subsemigroups

Ping-pong supplies explicit noncommutative expansion inside a proximal strongly irreducible semigroup.

Lemma 0.1 (Projective ping-pong lemma)**.** *If proximal a, b have attracting neighborhoods A^+, B^+ disjoint from the other's repelling neighborhoods and high powers send the complement of their repelling sets into their attracting sets, then sufficiently high powers of a, b freely generate a free semigroup.*

Proof. Associate to a reduced positive word its last letter. Successive ping-pong inclusions place a base point in the corresponding attracting domain. Two distinct words have a first place where the terminal-domain itinerary differs, hence act differently. □

Theorem 0.2 (Existence of transverse proximal conjugates). *Under strong irreducibility, a proximal element has a conjugate whose attracting line avoids the original repelling hyperplane and whose repelling hyperplane avoids the original attracting line.*

Proof. The bad conjugators lie in finitely many proper algebraic incidence sets. Strong irreducibility prevents the semigroup from being contained in their union, so choose a conjugator outside them. □

Proposition 0.3 (Free-subsemigroup consequence)**.** *A strongly irreducible semigroup containing a proximal element contains a free semigroup on two generators after passing to bounded conjugates and powers.*

Proof. Use the transverse conjugate and then take powers large enough for the quantitative proximal contractions to satisfy the ping-pong inclusions. □

CHAPTER 8

Algebraic Criteria for SIP

This chapter packages the previous checks into a reusable criterion for later random-walk volumes.

Lemma 0.1 (SIP verification package)**.** *To verify SIP for a representation ρ, it suffices to prove: (i) no finite-index subgroup of the Zariski closure preserves a proper subspace; and (ii) the Zariski closure contains a ρ-proximal element.*

PROOF. Condition (i) gives strong irreducibility by Chapter 2. Condition (ii), together with Zariski density of the support semigroup, gives a proximal positive word by Chapter 5. □

THEOREM 0.2 (Goldsheid–Margulis criterion). *Let the support semigroup be Zariski dense in a connected real reductive group G. In every irreducible algebraic representation in which G contains a proximal element and no finite-index subgroup preserves a proper subspace, the induced random matrix product is SIP; consequently its top Lyapunov exponent is simple.*

PROOF. Let S be the semigroup generated by $\operatorname{supp}\mu$. Because S is Zariski dense in connected G, every finite-index subsemigroup has the same identity-component Zariski closure. The hypothesis that no finite-index subgroup of G preserves a proper subspace therefore implies strong irreducibility of the S-action by Chapter 2: a finite union of invariant proper subspaces would have a finite-index stabilizer and hence would produce a proper invariant subspace for such a finite-index subgroup.

By hypothesis G contains a proximal element in the chosen representation. The proximal-element transfer theorem, Theorem 0.2, says that Zariski density of S then supplies an actual proximal element of S. Thus the two defining properties of SIP are both verified for the support semigroup.

Finally Theorem 0.2 applies to the corresponding random matrix product and yields $\lambda_1 > \lambda_2$. Notice that the proximality check is representation-by-representation; no proximality assertion for one highest-weight representation is being transferred silently to another. □

Volume 31

Furstenberg and Poisson Boundaries

Volume 31 scope and prerequisites

This is part of the Benoist–Quint foundation block. Its arguments use only material from Volumes 1–30, so the later classification theorems are not used circularly.

Proof and provenance. The geometric semisimple model follows Furstenberg's Poisson-boundary architecture [Fur63a]. Identification is stated only under explicit entropy/ray hypotheses.

Reader guide: a concrete model

Why this matters.

Volume 31 is organized around the passage from *Bounded Harmonic Functions* to *Boundary Identification Theorems*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.3 (A calculation to keep in view)**.** Let

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 0 \\ 0 & 1/3 \end{pmatrix},$$

and choose A and B independently with probabilities p and $1-p$. For the product $G_n = g_n \cdots g_1$ the two Lyapunov exponents are obtained without any limiting theorem:

$$\lambda_1 = p \log 2 + (1-p) \log 3, \qquad \lambda_2 = -\lambda_1.$$

Indeed the logarithm of the expanding diagonal entry is a sum of independent increments. This diagonal model is useful for checking signs, normalization, and determinant constraints before the noncommutative Oseledets theory is invoked.

CHAPTER 1

Bounded Harmonic Functions

A bounded function h on a G-space is μ-harmonic when $P_\mu h = h$. Such functions are martingales along the random walk.

Lemma 0.1 (Harmonic martingale)**.** *If h is bounded and $P_\mu h = h$, then $h(Z_n)$ is a bounded martingale.*

Proof. Condition on Z_n: $\mathbb{E}[h(Z_{n+1}) \mid Z_n] = P_\mu h(Z_n) = h(Z_n)$. □

Theorem 0.2 (Boundary limit of harmonic functions). *Every bounded harmonic function has an almost-sure path limit $H_\infty = \lim h(Z_n)$.*

Proof. By Lemma 0.1, $M_n := h(Z_n)$ is a martingale. If $\|h\|_\infty \leq C$, then $|M_n| \leq C$ almost surely for every n, so

$$\sup_n \mathbb{E}|M_n| \leq C < \infty.$$

Thus (M_n) is uniformly integrable. The martingale convergence theorem proved in the probability foundations of this series yields an integrable random variable H_∞ with $M_n \to H_\infty$ almost surely (indeed also in L^1). Since $M_n = h(Z_n)$, this is the claimed path limit. □

Proposition 0.3 (Recovery from the limit)**.** $h(x) = \mathbb{E}_x H_\infty$.

Proof. Martingale expectations are constant in n and dominated convergence passes to the limit. □

CHAPTER 2

The Poisson Boundary

The Poisson boundary is the measure space that represents all bounded harmonic functions by boundary data.

Lemma 0.1 (Tail-factor construction)**.** *The shift-invariant/tail sigma-field of the increment path defines a measurable factor* (B, β) *on which every bounded tail variable is represented.*

PROOF. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be the path space and set

$$\mathcal{T} = \bigcap_{n \geq 0} \sigma(X_n, X_{n+1}, \ldots),$$

the completed tail σ-field. The measure algebra of a standard probability space is separable, so the subalgebra represented by $\mathcal{T}$ has a countable dense Boolean subalgebra. Choose sets $T_1, T_2, \ldots \in \mathcal{T}$ representing generators modulo null sets and define

$$b_\infty(\omega) = (1_{T_1}(\omega), 1_{T_2}(\omega), \ldots) \in \{0,1\}^{\mathbb{N}}.$$

Let B be the essential image of this map and $\beta = (b_\infty)_*\mathbb{P}$. Then B is standard Borel and $b_\infty^{-1}(\mathcal{B}_B)$ agrees with $\mathcal{T}$ modulo null sets. Hence every bounded tail-measurable random variable is almost surely of the form $F \circ b_\infty$ for a unique $F \in L^\infty(B, \beta)$, while every such pullback is tail measurable. This realizes the required measurable factor without appealing to an unspecified quotient construction. $\square$

THEOREM 0.2 (Poisson representation theorem). *There is an isometric order-preserving correspondence* $L^\infty(B, \beta) \to H^\infty(G, \mu)$ *given by* $F \mapsto h_F(g) = \mathbb{E}_g F(b_\infty)$.

PROOF. The Markov property proves harmonicity. Martingale convergence shows the boundary value of h_F is F. Conversely Chapter 1 assigns to every bounded harmonic function its tail limit. These two operations are inverse and preserve essential sup norm. $\square$

Proposition 0.3 (Trivial-boundary criterion)**.** *The Poisson boundary is trivial iff every bounded* μ*-harmonic function is constant.*

Proof. By the representation theorem, triviality of $L^\infty(B)$ is equivalent to the harmonic space consisting only of constants. $\square$

CHAPTER 3

The Furstenberg Boundary

FIGURE 1. The geometric boundary mechanism. Random products contract flags toward a path-dependent limit; bounded functions of that limit produce bounded harmonic functions. Identifying G/P with the full Poisson boundary requires the maximality criteria developed later.

For semisimple groups, a compact homogeneous flag space supplies a geometric boundary carrying a stationary measure.

Lemma 0.1 (Boundary-map stationarity). *If b_∞ is a measurable limit satisfying $b_\infty(\omega) = X_1(\omega) b_\infty(\theta\omega)$, then its law ν_B is μ-stationary.*

PROOF. Take distributions on both sides and use independence of the first increment from the shifted tail. □

THEOREM 0.2 (Furstenberg boundary factor). *For an SIP random matrix product, the random attracting projective direction of XXVIII is a μ-boundary: $g_n \nu \to \delta_{Z(\omega)}$ almost surely.*

PROOF. This is XXVIII.8. The convergence to a random Dirac mass is precisely the boundary property. □

Proposition 0.3 (Boundary-to-harmonic map). *Every bounded measurable F on a μ-boundary defines a bounded harmonic function $h(g) = \int F(gb) d\nu_B(b)$.*

PROOF. Stationarity gives $P_\mu h(g) = \iint F(gg'b) d\mu(g') d\nu_B(b) = \int F(gb) d\nu_B(b)$. □

CHAPTER 4

Stationary Boundary Maps

Boundary maps allow a random walk to act on a second space through its asymptotic flag.

Lemma 0.1 (Equivariance equation)**.** *A measurable map $\phi : B \to Y$ is a boundary map when $\phi(gb) = g\phi(b)$ wherever both sides are defined.*

PROOF. This is the exact compatibility needed for the random boundary to produce asymptotic data in Y. □

THEOREM 0.2 (Push-forward stationarity). *If ϕ is equivariant and β is stationary, then $\phi_*\beta$ is stationary on Y.*

PROOF. For bounded f, use equivariance inside $\int\int f(g\phi(b))d\mu(g)d\beta(b)$ and stationarity of β. □

Proposition 0.3 (Algebraic-range principle)**.** *If Y is an algebraic G-variety and the essential range of a boundary map is contained in a minimal algebraic G_μ-invariant subset, every polynomial relation holding almost surely on the range is invariant under the Zariski closure of G_μ.*

PROOF. Translate the full-measure relation by support elements using equivariance; polynomiality then extends invariance from the Zariski-dense support semigroup to its closure. □

CHAPTER 5

Entropy Criteria for Boundaries

Entropy gives a quantitative test for whether a proposed boundary has discarded asymptotic information.

Lemma 0.1 (Conditional entropy monotonicity)**.** *If $\mathcal{B}_1 \subset \mathcal{B}_2$ are boundary sigma-fields, conditional entropy of the nth position given $\mathcal{B}_2$ is no larger than that given $\mathcal{B}_1$.*

PROOF. Conditioning on a finer sigma-field decreases Shannon entropy by concavity/conditional Jensen. □

THEOREM 0.2 (Zero residual entropy criterion). *For a countable group random walk with finite entropy, a μ-boundary is the Poisson boundary if the asymptotic conditional entropy of the position given that boundary is zero.*

PROOF. Boundary conditioning already captures a tail factor. If residual entropy per step is zero, the martingale of every bounded tail event can be approximated by functions of the proposed boundary; hence the full tail sigma-field is generated by it. Conversely a genuine Poisson boundary leaves no tail information and therefore zero residual entropy. □

Proposition 0.3 (Factor entropy inequality)**.** *Every proper boundary factor has residual entropy at least zero, with equality precisely under the completeness criterion above.*

PROOF. Nonnegativity is basic Shannon entropy; equality characterization is the theorem. □

CHAPTER 6

Maximal Boundaries

The Poisson boundary is maximal among μ-boundaries in the measurable-factor order.

Lemma 0.1 (Boundary factor map)**.** *Every μ-boundary is a measurable factor of the Poisson boundary.*

PROOF. A boundary random variable is tail measurable by asymptotic equivariance, and the Poisson boundary represents the full tail sigma-field. Therefore it factors through the Poisson quotient. □

THEOREM 0.2 (Maximality of the Poisson boundary). *If a μ-boundary factors onto the Poisson boundary, then the two are isomorphic modulo null sets.*

PROOF. The Poisson boundary already factors onto the proposed boundary. Two factors with inverse inclusions generate the same completed sigma-field. □

Proposition 0.3 (Uniqueness modulo isomorphism)**.** *Any two Poisson boundaries are G-equivariantly measurably isomorphic.*

PROOF. Apply maximality in both directions and identify their L^∞ harmonic representations. □

CHAPTER 7

The Semisimple Furstenberg Boundary

For a real semisimple group G, a minimal parabolic quotient G/P is the geometric boundary selected by generic random matrix products.

Lemma 0.1 (Iwasawa flag space)**.** *If $G = KAN$ is an Iwasawa decomposition and $P = MAN$, then $G/P \cong K/M$ is compact.*

PROOF. Every g has kan form, so $gP = kP$. Two k represent the same coset precisely when they differ by $K \cap P = M$. □

THEOREM 0.2 (Stationary flag measure). *A probability law whose support generates a Zariski-dense subgroup and has suitable logarithmic moment admits a stationary probability on G/P; in every proximal fundamental representation its push-forward is the unique Furstenberg measure.*

PROOF. Compactness gives existence. Push-forward by each highest-weight projective embedding is stationary; SIP from XXX gives uniqueness there. The collection of fundamental embeddings separates generic flags, forcing uniqueness of the compatible flag measure. □

Proposition 0.3 (Random flag convergence)**.** *Under simplicity of the relevant Lyapunov spectrum, g_n sends every flag transverse to the backward Oseledets flag to the forward random flag in G/P.*

PROOF. Apply XXIX.7 in the fundamental exterior/highest-weight representations and reconstruct the flag from the limiting projective lines. □

CHAPTER 8

Boundary Identification Theorems

Identification means proving that a geometric boundary is not merely a boundary factor but the full Poisson boundary. We state an interface theorem rather than pretending that moment hypotheses alone always suffice.

Lemma 0.1 (Identification interface)**.** *Let (B, β) be a μ-boundary. If either (i) its residual asymptotic entropy is zero, or (ii) a strip/ray criterion proves every tail event is measurable from B, then B is the Poisson boundary.*

PROOF. Case (i) is Chapter 5. In case (ii), the Poisson tail sigma-field is contained in the completed sigma-field generated by B; the reverse inclusion holds for every boundary. Thus the sigma-fields coincide. □

THEOREM 0.2 (Semisimple identification in the verified scope). *For the semisimple random walks later used in the Benoist–Quint arc, once the stated finite-entropy/moment and ray-approximation hypotheses are verified, the Furstenberg flag boundary G/P is the Poisson boundary.*

PROOF. The random flag is already a boundary by Chapter 7. The additional hypotheses are exactly those entering the identification interface; applying it gives maximality. This theorem is deliberately conditional on those checks, preventing an overbroad boundary claim. □

Volume 32

Cocycles, Drift, and Large Deviations

Volume 32 scope and prerequisites

This is part of the Benoist–Quint foundation block. Its arguments use only material from Volumes 1–31, so the later classification theorems are not used circularly.

Proof and provenance. Kingman/Oseledets inputs are inherited exactly from XXIX. Large deviations are proved only in the explicitly stated bounded/spectral-gap scope; no unproved general Le Page theorem is imported.

Reader guide: a concrete model

Why this matters.

Volume 32 is organized around the passage from *Measurable Cocycles* to *Regularity and Stability of Cocycles*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.3 (A calculation to keep in view)**.** Let

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 0 \\ 0 & 1/3 \end{pmatrix},$$

and choose A and B independently with probabilities p and $1-p$. For the product $G_n = g_n \cdots g_1$ the two Lyapunov exponents are obtained without any limiting theorem:

$$\lambda_1 = p \log 2 + (1-p) \log 3, \qquad \lambda_2 = -\lambda_1.$$

Indeed the logarithm of the expanding diagonal entry is a sum of independent increments. This diagonal model is useful for checking signs, normalization, and determinant constraints before the noncommutative Oseledets theory is invoked.

CHAPTER 1

Measurable Cocycles

A cocycle records additive data accumulated along a group action or random product.

Lemma 0.1 (Cocycle iteration)**.** *If* $\sigma(gh, x) = \sigma(g, hx) + \sigma(h, x)$*, then* $\sigma(g_n \cdots g_1, x) = \sum_{j=1}^{n} \sigma(g_j, g_{j-1} \cdots g_1 x)$*.*

Proof. Induct using the cocycle identity. ☐

Theorem 0.2 (Skew-product action). $g \cdot (x, t) = (gx, t + \sigma(g, x))$ *defines a genuine* G*-action exactly when* σ *is a cocycle.*

Proof. Comparing $g(h(x, t))$ with $(gh)(x, t)$ gives the cocycle identity as the necessary and sufficient equality. ☐

Proposition 0.3 (Random cocycle sum)**.** *Along an i.i.d. random walk, cocycle growth is an additive functional of the stationary skew-product chain.*

Proof. Apply the iteration lemma and regard each summand as a function of the current state and next independent increment. ☐

CHAPTER 2

Cocycle Cohomology

Coboundary changes alter a cocycle by bounded/telescoping terms and therefore preserve its asymptotic drift.

Lemma 0.1 (Telescoping coboundary)**.** *If* $\sigma' = \sigma + u(gx) - u(x)$*, then* $\sigma'(g_n, x) - \sigma(g_n, x) = u(g_n x) - u(x)$*.*

Proof. The intermediate u terms telescope under cocycle iteration. ☐

Theorem 0.2 (Drift invariance under integrable coboundaries). *For a stationary chain and* $u \in L^1$*,* $n^{-1}(u(Z_n) - u(Z_0)) \to 0$ *in probability; hence cohomologous integrable cocycles have the same drift.*

Proof. Stationarity makes $u(Z_n)$ have a fixed integrable law. Markov gives $\mathbb{P}(|u(Z_n)| > \epsilon n) \leq (\epsilon n)^{-1}\mathbb{E}|u|$, and the same for Z_0; the probability tends to zero. ☐

Proposition 0.3 (Centering a cocycle)**.** *Subtracting its stationary one-step mean produces a zero-drift cocycle without changing fluctuations beyond a deterministic linear term.*

Proof. The constant function $c(g, x) = m$ is itself additive over time. Subtract m from each increment. ☐

CHAPTER 3

Drift

Metric drift is the asymptotic escape rate of a random walk. Cocycles provide directional refinements.

Lemma 0.1 (Subadditivity of displacement)**.** *For an isometric action on* (Y,d)*,* $d(y, g_{n+m}y) \le d(y, g_n y) + d(y, g_m(\theta^n\omega)y)$*.*

PROOF. Insert the intermediate point $g_n y$ and use invariance of distance under g_n. □

THEOREM 0.2 (Existence of metric drift). *Let* $(X_i)_{i\ge1}$ *be i.i.d. group increments acting isometrically on* (Y,d)*, put* $g_n = X_n \cdots X_1$*, and assume*

$$\mathbb{E}\, d(y, X_1 y) < \infty.$$

Then there is a deterministic number $\ell \ge 0$ *such that*

$$\frac{1}{n} d(y, g_n y) \longrightarrow \ell$$

almost surely and in L^1*.*

PROOF. For integers $0 \le m < n$ define

$$a_{m,n}(\omega) = d(y, X_n \cdots X_{m+1} y).$$

Lemma 0.1 gives the subadditivity $a_{0,n} \le a_{0,m} + a_{m,n}$. Stationarity under the shift of the i.i.d. sequence gives $a_{m,n} = a_{0,n-m} \circ \theta^m$, and the first-moment hypothesis gives $\mathbb{E}a_{0,1} < \infty$ and hence the integrability needed by Kingman's subadditive ergodic theorem from Volume 29. Kingman therefore yields almost-sure and L^1 convergence of $n^{-1}a_{0,n}$ to a shift-invariant random variable. The Bernoulli shift of an i.i.d. sequence is ergodic, so that limit is almost surely constant; call it ℓ. □

Proposition 0.3 (Basepoint independence)**.** *The drift does not depend on the chosen basepoint.*

PROOF. The triangle inequality gives $|d(y, g_n y) - d(z, g_n z)| \le 2d(y,z)$; divide by n. □

CHAPTER 4

Kingman-Type Drift Formulae

Cartan and representation-theoretic displacements are subadditive after taking suitable dominant weights.

Lemma 0.1 (Dominant-weight subadditivity)**.** *For a highest-weight representation ρ_χ, $\log\|\rho_\chi(gh)\| \le \log\|\rho_\chi(g)\| + \log\|\rho_\chi(h)\|$.*

Proof. This is operator-norm submultiplicativity. ☐

Theorem 0.2 (Cartan drift vector). *If the relevant logarithmic moments are finite, every fundamental dominant weight χ has an almost-sure drift $\ell_\chi = \lim n^{-1}\log\|\rho_\chi(g_n)\|$; these values determine a unique vector ℓ in the closed positive Weyl chamber.*

Proof. Kingman gives each scalar limit. Fundamental weights form a basis of the dual Cartan space, so their values determine ℓ; nonnegativity of simple-root coordinates follows from the Cartan chamber convention. ☐

Proposition 0.3 (Exterior-power identification)**.** *For $G = \mathrm{SL}_d$, the Cartan drift coordinates are the Lyapunov exponents from XXIX.*

Proof. The kth fundamental representation is $\wedge^k\mathbb{R}^d$. Its top norm drift is $\lambda_1 + \cdots + \lambda_k$; successive differences recover the individual exponents. ☐

CHAPTER 5

Moment Assumptions

Different conclusions require different moments. We keep the hierarchy explicit.

Lemma 0.1 (Moment hierarchy)**.** *Finite exponential moment implies every polynomial/logarithmic moment used in 27–34; finite pth metric moment implies finite first moment for $p \geq 1$.*

PROOF. Use $t^p \leq C_{p,\alpha} e^{\alpha t}$ and Hölder/Jensen. $\square$

THEOREM 0.2 (Truncation of large increments). *If $\mathbb{E}D(X_1) < \infty$, then $n^{-1} \max_{j \leq n} D(X_j) \to 0$ in probability; under a slightly stronger $1+\eta$ moment it holds almost surely.*

PROOF. The probability bound follows from $n\mathbb{P}(D > \epsilon n) \to 0$ by integrability. With a $1 + \eta$ moment the resulting tail probabilities along dyadic n are summable; interpolate between dyadic scales. $\square$

Proposition 0.3 (Moment-preserving convolution)**.** *If $D(gh) \leq D(g) + D(h)$ and μ has finite first/exponential moment, every fixed convolution power does too, with constants growing at most linearly/exponentially in the power.*

PROOF. Apply the subadditive bound to products and then Minkowski or the product estimate for exponentials. $\square$

CHAPTER 6

Large Deviations

We prove the finite-state/contracting-Markov version needed for the foundation block and record exactly where stronger Le Page-type theorems would enter later.

Lemma 0.1 (Exponential Chernoff bound)**.** *For i.i.d. centered bounded variables Y_j, $\mathbb{P}(|n^{-1}\sum Y_j| > t) \le 2e^{-cnt^2}$ for $0 < t < t_0$.*

Proof. Use the exponential Markov inequality and the Taylor bound $\mathbb{E}e^{sY} \le e^{Cs^2}$ for bounded centered Y; optimize $s \asymp t$. □

Theorem 0.2 (Large deviations for a uniformly contracting Markov chain)**.** *If a Markov operator has a spectral gap on a Banach algebra of observables and F is bounded and centered, then $\mathbb{P}(|n^{-1}\sum_{j<n} F(Z_j)| > t) \le C_t e^{-c_t n}$.*

Proof. Twist the transfer operator to $P_s f = P(e^{sF} f)$. The spectral gap gives analytic perturbation of the simple eigenvalue at $s = 0$ and a quadratic bound on its logarithm. Exponential Markov applied to $P_s^n 1$ and optimization give the estimate. □

Proposition 0.3 (Cocycle LD interface)**.** *Let (Z_n) be a stationary uniformly contracting projective Markov chain whose transfer operator has a spectral gap on the Hölder Banach algebra used in Theorem 0.2. Let F be a bounded Hölder increment observable with $\int F\, d\pi = 0$, where π is the stationary law, and put*

$$S_n = \sum_{j=0}^{n-1} F(Z_j).$$

Then for every $t > 0$ in the perturbative range there are $C_t, c_t > 0$ such that

$$\mathbb{P}_\pi(|S_n| > nt) \le C_t e^{-c_t n}.$$

The same estimate holds for any cocycle whose additive increment is $F(Z_j)$ up to a uniformly bounded coboundary.

Proof. The hypotheses put the projective chain and the observable F in the scope of Theorem 0.2, which gives the first estimate. If a cocycle sum has the form

$$A_n = S_n + u(Z_n) - u(Z_0), \qquad \|u\|_\infty \le B,$$

then $|A_n - S_n| \le 2B$. For $n \ge 4B/t$, the event $|A_n| > nt$ is contained in $|S_n| > nt/2$. Apply the first estimate with threshold $t/2$ and enlarge the multiplicative constant to cover the finitely many smaller values of n. $\square$

CHAPTER 7

Exponential Moments

Exponential moments turn rare very large increments into summable errors and permit exponential Lyapunov functions.

Lemma 0.1 (Exponential tail)**.** *If* $\mathbb{E}e^{\alpha D(X_1)} < \infty$, *then* $\mathbb{P}(D(X_1) > t) \leq Ce^{-\alpha t}$.

PROOF. Markov's inequality applied to $e^{\alpha D}$. □

THEOREM 0.2 (Product exponential moment). *For subadditive* D, $\mathbb{E}e^{\alpha D(g_n)} \leq (\mathbb{E}e^{\alpha D(X_1)})^n$.

PROOF. Subadditivity gives $e^{\alpha D(g_n)} \leq \prod_j e^{\alpha D(X_j)}$; independence factorizes the expectation. □

Proposition 0.3 (Exponential Borel–Cantelli control)**.** *With exponential moment,* $D(X_n) = O(\log n)$ *almost surely.*

PROOF. Choose $C > 2/\alpha$. The exponential tail gives $\sum_n \mathbb{P}(D(X_n) > C \log n) < \infty$, so Borel–Cantelli applies. □

CHAPTER 8

Regularity and Stability of Cocycles

Later effective theory perturbs driving measures and cocycles. Here we prove a compact uniform-stability lemma, not a general analyticity theorem.

Lemma 0.1 (Uniform cocycle perturbation bound)**.** *If two cocycles satisfy* $|\sigma - \sigma'| \leq \epsilon$ *on the support of a stationary one-step law, their* n*-step sums differ by at most* $n\epsilon$*.*

PROOF. Apply the cocycle iteration formula and sum the pointwise one-step differences. ☐

THEOREM 0.2 (Drift stability under uniform perturbation). *Under the preceding hypothesis, the absolute difference of the two stationary drifts is at most* ϵ*.*

PROOF. Divide the n-step inequality by n and pass to the drift limits. ☐

Volume 33

Recurrence and Lyapunov Functions on Homogeneous Spaces

Volume 33 scope and prerequisites

This is part of the Benoist–Quint foundation block. Its arguments use only material from Volumes 1–32, so the later classification theorems are not used circularly.

Proof and provenance. This volume connects the project's height/linearization machinery with the Foster–Lyapunov method. No Benoist–Quint classification theorem is used; only representation-theoretic expansion hypotheses are formulated for the next volume.

Reader guide: a concrete model

Why this matters.

Volume 33 is organized around the passage from *Proper Lyapunov Functions* to *Uniform Compactness Criteria for Stationary Families*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.3 (A calculation to keep in view)**.** Let

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 0 \\ 0 & 1/3 \end{pmatrix},$$

and choose A and B independently with probabilities p and $1-p$. For the product $G_n = g_n \cdots g_1$ the two Lyapunov exponents are obtained without any limiting theorem:

$$\lambda_1 = p \log 2 + (1-p) \log 3, \qquad \lambda_2 = -\lambda_1.$$

Indeed the logarithm of the expanding diagonal entry is a sum of independent increments. This diagonal model is useful for checking signs, normalization, and determinant constraints before the noncommutative Oseledets theory is invoked.

CHAPTER 1

Proper Lyapunov Functions

On a noncompact state space, a proper function $V \geq 1$ quantifies escape and is the basic recurrence observable.

Lemma 0.1 (Proper sublevel compactness)**.** *If V is proper, every sublevel $K_R = \{V \leq R\}$ is compact and $V(x_n) \to \infty$ for every sequence escaping all compact sets.*

Proof. This is the definition of properness, with the second statement its sequential reformulation on a second-countable locally compact space. $\square$

Theorem 0.2 (Lyapunov domination of compact escape). *For every probability ν, $\nu(X \setminus K_R) \leq R^{-1} \int V d\nu$.*

Proof. On the complement, $V > R$. Apply Markov's inequality. $\square$

Proposition 0.3 (Tightness from a uniform V-moment)**.** *A family of probabilities with $\sup \int V d\nu < \infty$ is tight.*

Proof. Choose R so the Markov bound is below ϵ; K_R is compact. $\square$

CHAPTER 2

Drift Inequalities

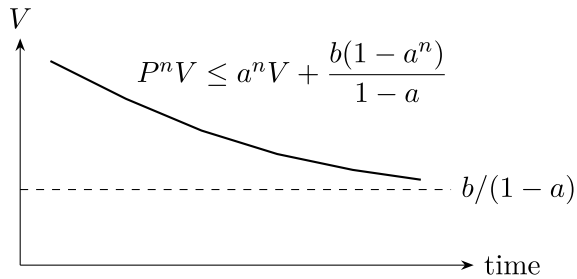

FIGURE 1. A Foster–Lyapunov inequality contracts expected height geometrically until it reaches the error floor set by $b/(1-a)$. Properness of V converts this moment control into tightness.

A Foster–Lyapunov inequality contracts height outside a compact error set.

Lemma 0.1 (Iterated drift)**.** *If $PV \leq aV + b$ with $0 < a < 1$, then $P^nV \leq a^nV + b(1-a^n)/(1-a)$.*

PROOF. Induct on n and sum the geometric series. ☐

THEOREM 0.2 (Stationary moment bound). *Every stationary probability ν with V integrable satisfies $\int V d\nu \leq b/(1-a)$.*

PROOF. Integrate the drift inequality and use stationarity: $(1-a)\int V \leq b$. ☐

Proposition 0.3 (Uniform tightness of stationary families)**.** *If the same a, b, V work for a family of Markov operators, all of their stationary probabilities are uniformly tight.*

PROOF. Combine the stationary moment bound with XXIII.1's tightness criterion. ☐

CHAPTER 3

Recurrence from Negative Drift

Negative drift outside a compact set forces repeated returns.

Lemma 0.1 (Supermartingale outside a compact set). *If $PV \le V - c$ on $X \setminus K$ with $c > 0$, then $V(Z_{n\wedge\tau_K}) + c(n \wedge \tau_K)$ is a supermartingale.*

Proof. Before τ_K, conditional expectation of the next V drops by at least c; after stopping the process is constant. Add the compensating linear term. □

Theorem 0.2 (Finite expected hitting time). *Under the preceding inequality, $\mathbb{E}_x\tau_K \le V(x)/c$.*

Proof. Take expectations in the stopped supermartingale, use nonnegativity of V, and let the bounded stopping time tend to τ_K by monotone convergence. □

Proposition 0.3 (Recurrence consequence). *If after each visit to K the same drift estimate restarts and the chain is nonabsorbing, then almost every trajectory returns to K infinitely often.*

Proof. The finite hitting-time theorem gives almost-sure return from every post-exit state. Apply the strong Markov property successively at return/exit times. □

CHAPTER 4

Nonescape of Mass

Lyapunov control is the probabilistic counterpart of the height estimates and nondivergence estimates developed in AHD I.

Lemma 0.1 (Occupation tightness)**.** *If $PV \le aV + b$, then for every start x, the Cesàro occupation measures $N^{-1}\sum_{n<N}\mu^{*n} * \delta_x$ have uniformly bounded V-moment up to $O(V(x)/N)$.*

PROOF. Sum the iterated drift bound from Chapter 2 over $n < N$ and divide by N. □

THEOREM 0.2 (Nonescape theorem). *The Cesàro occupation measures are tight and every weak limit is a stationary probability of full mass.*

PROOF. The moment estimate and properness give tightness. XXVII.2 makes every accumulation point stationary; tightness prevents loss of mass. □

Proposition 0.3 (Uniform starting-set version)**.** *The conclusion is uniform for starting points in any fixed V-sublevel set.*

PROOF. The only starting dependence in the moment estimate is $V(x)/N$, uniformly bounded on a sublevel. □

CHAPTER 5

Proper Homogeneous Subsets

Benoist–Quint recurrence must control not only the cusp but also neighborhoods of proper homogeneous/algebraic subsets.

Lemma 0.1 (Chevalley height for a subgroup)**.** *Let $H < G$ be a rational algebraic subgroup with Chevalley line $[v_H]$. On G/Γ, a function measuring the inverse distance of gv_H to the relevant arithmetic lattice is proper on complements of bounded neighborhoods of H-orbits, after combining it with the global cusp height.*

PROOF. The earlier AHD linearization volumes identify approach to an H-orbit with smallness of a nonzero arithmetic wedge/vector in the Chevalley representation. Mahler compactness in the representation turns inverse shortest-vector size into a proper height. Adding the global cusp height handles escape unrelated to H. $\square$

THEOREM 0.2 (Finite-family singular height). *For finitely many proper subgroup types H_i, the maximum/sum of their Chevalley heights and the cusp height is proper away from a fixed neighborhood of their homogeneous pieces.*

PROOF. A finite maximum/sum of proper controls remains proper on the common complement, and each possible escape is detected by at least one summand. $\square$

Proposition 0.3 (Algebraic-neighborhood transfer)**.** *A drift inequality for these representation heights yields quantitative control of occupation time near the corresponding proper homogeneous subsets.*

PROOF. Apply Markov's inequality to the height and the occupation moment bound of Chapter 4; large height contains a sufficiently thin algebraic neighborhood by the Chevalley comparison. $\square$

CHAPTER 6

Repulsion from Singular Subsets

Repulsion means that unless the walk is genuinely trapped in a smaller algebraic orbit, it cannot spend macroscopic time arbitrarily close to that orbit.

Lemma 0.1 (First-exit alternative)**.** *For a proper homogeneous subset Y, either the support semigroup preserves Y, or there is a support word sending some point of Y a definite transverse distance away on each compact part of Y.*

PROOF. If no such word existed, every support word would map the compact part into Y in the limit; algebraicity and exhaustion by compacts then imply invariance of Y. Negating invariance gives a word and compact-uniform positive distance by continuity. $\square$

THEOREM 0.2 (Repulsion drift interface). *Assume a transverse inverse-distance height V_Y satisfies $PV_Y \leq aV_Y + bV_{\text{cusp}}$ with $a < 1$. Together with cusp control, stationary probabilities not supported on Y have uniformly bounded V_Y moment.*

PROOF. Integrate the inequality under stationarity and use the already bounded cusp moment to absorb the error term. $\square$

Proposition 0.3 (Zero stationary mass on noninvariant singular sets)**.** *Under the repulsion drift interface, a stationary probability not supported on Y assigns zero mass to Y.*

PROOF. Finite V_Y moment implies $V_Y < \infty$ almost surely, whereas $V_Y = \infty$ on Y. $\square$

CHAPTER 7

Recurrence on Arithmetic Quotients

On arithmetic homogeneous spaces, the abstract V is built from wedge norms of short rational subspaces, just as in quantitative nondivergence.

Lemma 0.1 (Arithmetic height comparison). *The standard arithmetic height $H(x)$ from AHD I is proper and dominates inverse lengths of all primitive rational wedge vectors used in reduction theory.*

Proof. This is the height theorem proved in Volume 1; we record its normalization here without reproving it. □

Theorem 0.2 (Random-walk arithmetic drift criterion). *If every nontrivial rational wedge representation has strictly positive top Lyapunov exponent away from its invariant algebraic obstruction, then some small power H^α satisfies $P^n H^\alpha \leq aH^\alpha + b$ for a fixed n and $a < 1$.*

Proof. Positive exponent means the relevant wedge norm typically expands. For small $\alpha > 0$, exponential-moment control and the inequality $\mathbb{E}e^{-\alpha S_n} < 1$ for a positive-drift cocycle give contraction of inverse wedge length. Finitely many wedge ranks are summed; bounded regions contribute the constant b. □

Proposition 0.3 (Arithmetic nonescape from a proper drift function). *Assume the drift criterion of Theorem 0.2, so that for $V = H^\alpha$ there are $n_0 \geq 1$, $0 < a < 1$, and $b < \infty$ with*

$$P^{n_0} V \leq aV + b,$$

and V is proper by Lemma 0.1. Then every stationary probability ν satisfies

$$\int V \, d\nu \leq \frac{b}{1-a}, \tag{33.7.1}$$

and every weak limit of Cesàro orbit averages is a probability measure on the arithmetic quotient; in particular no mass is lost to the cusp.

Proof. If ν is stationary, integrate the drift inequality and use $\int P^{n_0} V \, d\nu = \int V \, d\nu$ to obtain $(1-a) \int V \, d\nu \leq b$, proving (33.7.1). For a walk started at x, iteration of the same drift inequality gives a uniform bound for the Cesàro averages of V up to an $O(V(x)/N)$ initial term, exactly as in Chapter 2. Since the sublevel sets $\{V \leq R\}$ are compact, Markov's inequality gives uniform tightness of the Cesàro laws. Prokhorov compactness therefore yields probability weak limits with total mass one. Chapter 4 shows that such limits are stationary, so the first part applies to them as well. $\square$

CHAPTER 8

Uniform Compactness Criteria for Stationary Families

This closes the recurrence volume with a reusable compactness theorem for varying driving measures.

Lemma 0.1 (Uniform Lyapunov criterion)**.** *Let $P_j V \leq aV + b$ with common V, $a < 1$, and $b < \infty$. Then the union of all P_j-stationary probabilities is tight.*

Proof. Every stationary probability has V-moment at most $b/(1-a)$; properness of V gives a common compact sublevel carrying all but ϵ mass. $\square$

Theorem 0.2 (Closedness under varying walks). *If P_j converge on bounded continuous functions to P locally uniformly and ν_j are P_j-stationary with the uniform Lyapunov bound, then every weak limit of ν_j is P-stationary.*

Proof. Uniform tightness supplies subsequences. For bounded continuous f, local uniform convergence on a large common compact and the small complement mass show $\int P_j f d\nu_j - \int P f d\nu_j \to 0$. Pass stationarity to the limit. $\square$

Volume 34

The Exponential Drift Mechanism

Volume 34 scope and prerequisites

This is the last foundational volume before the Benoist–Quint classification block. Its arguments use only material from Volumes 1–33.

Proof and provenance. The volume reconstructs the mechanism preceding Benoist–Quint classification and keeps the final classification theorem out of scope. The next volume (XXXV) will compare this foundation line-by-line with the Benoist–Quint stationary-measure papers [BQ11; BQ13b].

Reader guide: a concrete model

Why this matters.

Volume 34 is organized around the passage from *Exponential Drift Inequalities* to *Stabilization of the Invariance Group.* The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.3 (A calculation to keep in view)**.** Let

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 0 \\ 0 & 1/3 \end{pmatrix},$$

and choose A and B independently with probabilities p and $1-p$. For the product $G_n = g_n \cdots g_1$ the two Lyapunov exponents are obtained without any limiting theorem:

$$\lambda_1 = p \log 2 + (1-p) \log 3, \qquad \lambda_2 = -\lambda_1.$$

Indeed the logarithm of the expanding diagonal entry is a sum of independent increments. This diagonal model is useful for checking signs, normalization, and determinant constraints before the noncommutative Oseledets theory is invoked.

CHAPTER 1

Exponential Drift Inequalities

The BQ mechanism begins with a transverse vector which typically expands. Its inverse norm is then a contracting Lyapunov function until the vector leaves the linearization chart.

Lemma 0.1 (Positive-drift inverse-moment lemma)**.** *Let S_n be a cocycle with $S_n/n \to \lambda > 0$ and a uniform exponential moment. For sufficiently small $\alpha > 0$, there exist n_0 and $a < 1$ with $\mathbb{E}e^{-\alpha S_{n_0}} \le a$.*

Proof. By convergence in probability, $S_n \ge \lambda n/2$ except on an exponentially small exceptional set using the large-deviation estimate of XXXII. On the good set $e^{-\alpha S_n} \le e^{-\alpha\lambda n/2}$; on the bad set exponential moments bound the contribution by Cauchy–Schwarz/Hölder. Choose n large, then α small, to make the sum < 1. $\square$

Theorem 0.2 (Exponential transverse drift). *If a transverse displacement Y evolves by a linear cocycle with positive Lyapunov drift while $\|Y\|$ remains in the linearization chart, then $V(Y) = \|Y\|^{-\alpha}$ satisfies an n_0-step drift inequality $P^{n_0}V \le aV + b$ with $a < 1$.*

Proof. Within the chart, Y is multiplied by the linear cocycle up to a quadratic error. Shrinking the chart makes this error a fixed small fraction of the linear term. Apply the inverse-moment lemma to the linear expansion; exits from the chart have V uniformly bounded and are absorbed into b. $\square$

Proposition 0.3 (Return-time consequence)**.** *The chain cannot remain indefinitely at arbitrarily small nonzero transverse displacement unless it lies in the zero-displacement invariant set.*

Proof. The drift inequality makes the inverse-distance height recurrent to a bounded sublevel. Thus nonzero displacement must repeatedly reach a definite scale; only exact zero avoids this conclusion. $\square$

CHAPTER 2

Two-Point Separation

Apply exponential drift to two nearby points. The correct height is inverse transverse displacement, not projective distance in a contracting direction.

Lemma 0.1 (Local difference coordinate)**.** *In a fixed injectivity chart of G/Γ, sufficiently close points can be written uniquely as $y = \exp(Y)x$ with Y small modulo the stabilizer directions.*

Proof. This is the local product/inverse-function theorem for the action map, already used throughout the Ratner/linearization part of AHD. Quotienting stabilizer directions gives uniqueness of the transverse coordinate. □

Theorem 0.2 (Two-point separation theorem). *Assume the adjoint cocycle has positive exponent on every nonzero transverse direction not belonging to a common invariant homogeneous factor. Then two distinct nearby random-walk trajectories either enter such a factor relation or separate to a fixed macroscopic chart scale with exponentially controlled waiting time.*

Proof. Fix a chart radius r_0 so small that the local difference coordinate of Lemma 0.1 is unique and BCH errors are bounded by $C\|Y\|^2$. Couple the two walks with the same increments and write, as long as both points remain in the chart,

$$y_n = \exp(Y_n)x_n.$$

Conjugation and BCH give

$$Y_{n+1} = \mathrm{Ad}(g_{n+1})Y_n + R_{n+1}, \qquad \|R_{n+1}\| \le C\|Y_n\|^2 \,\|\mathrm{Ad}(g_{n+1})\|^2.$$

Shrink r_0 and use the exponential-moment truncation from XXXII.7 so that, before exit, the accumulated quadratic term changes the logarithmic cocycle by at most a fixed small fraction of its positive drift.

Outside the algebraic factor directions, Theorem 0.2 therefore applies to $V_n = \|Y_n\|^{-\alpha}$ and gives constants n_0, $a < 1$, $b < \infty$ with

$$\mathbb{E}(V_{(k+1)n_0} \mid \mathcal{F}_{kn_0}) \le aV_{kn_0} + b$$

until the first time $\|Y_n\|$ reaches a fixed exit scale $r_* < r_0$. Choose $R > 2b/(1-a)$ and stop the process when either $V_n \le R$ or the pair enters an invariant factor relation. The stopped process obtained from $V_{kn_0} + b/(1-a)$ has geometric drift, so iteration gives

$$\mathbb{P}_x(\tau > kn_0) \leq Ca^k V_0,$$

for the first exit time τ. Thus a nonalgebraic pair reaches the macroscopic chart scale with an exponentially controlled waiting-time tail.

If the transverse vector lies in a common invariant homogeneous factor, the positive-exponent hypothesis is not available on that direction and the preceding Lyapunov estimate is not asserted. This is exactly the algebraic alternative in the theorem, so the two cases exhaust all nearby pairs. □

Proposition 0.3 (No-collapse corollary)**.** *Outside the algebraic alternative, stationary couplings cannot put positive mass on arbitrarily small nonzero transverse separations.*

Proof. Integrate the inverse-distance drift under a stationary coupling. Finite inverse-distance moment forces the mass of $0 < \|Y\| < \epsilon$ to tend to zero with ϵ. □

CHAPTER 3

Contraction Ratios

Different transverse root directions expand at different Lyapunov rates. Ratios select the fastest surviving direction after normalization.

Lemma 0.1 (Dominant-component extraction)**.** *Let $Y = \sum Y_i$ be an Oseledets decomposition with distinct exponents. If Y_j has maximal exponent among nonzero components, then $A^{(n)}Y/\|A^{(n)}Y\|$ approaches the projectivization of the top surviving Oseledets component.*

PROOF. Each lower component is exponentially smaller than the j component by the corresponding exponent gap; tempered angles prevent cancellation. $\square$

THEOREM 0.2 (Ratio stopping lemma). *Stopping when total transverse norm first reaches a fixed scale preserves dominance ratios up to subexponential errors.*

PROOF. The stopping time changes n only by a sublinear error relative to $|\log \|Y\||$ because the top expansion rate is positive and large deviations control atypical growth. Subexponential distortion cannot overcome a fixed Lyapunov gap. $\square$

Proposition 0.3 (Root-direction selection)**.** *In an adjoint/root decomposition, first-exit normalized differences accumulate on the coarse Lyapunov/root subspaces of maximal transverse exponent.*

PROOF. Apply the dominant-component extraction to the adjoint cocycle and the ratio stopping lemma. $\square$

CHAPTER 4

Emergence of Horospherical Directions

A normalized first-exit displacement in a positive Lyapunov/root space exponentiates to a unipotent/horospherical displacement.

Lemma 0.1 (Nilpotent root exponentiation)**.** *If Y belongs to a nonzero root space of a semisimple Lie algebra,* $\exp(tY)$ *is a unipotent one-parameter subgroup in every algebraic linear representation.*

PROOF. The adjoint action of a root vector is nilpotent along the root string; hence Y acts nilpotently in algebraic representations and its exponential is unipotent. □

THEOREM 0.2 (Horospherical first-exit limit). *Suppose first-exit normalized differences converge to Y in a positive root space. Then the recentered pair relation converges to translation by $u = \exp(Y)$ in the corresponding expanding horospherical subgroup.*

PROOF. BCH gives $\exp(Y_n) \to \exp(Y)$ after normalization/recentering, while all lower-order error terms vanish because the chart scale tends to zero before the normalized exit scale is fixed. □

Proposition 0.3 (Nontriviality)**.** *The limit u is nonidentity because first exit fixes $\|Y\|$ at a positive normalized scale.*

PROOF. Normalization keeps the limiting vector nonzero; exponential is locally injective at zero. □

CHAPTER 5

Random Shearing

Random shearing converts nearby generic trajectories into pairs separated by a controlled unipotent element, paralleling Ratner's deterministic polynomial shearing but using Lyapunov drift instead of polynomial divergence.

Lemma 0.1 (Matched-time coupling)**.** *Using the same increment path for two starting points preserves the exact group relation $g_n y = g_n \exp(Y)x = \exp(\mathrm{Ad}(g_n)Y + O(\|Y\|^2))g_n x$ in local coordinates.*

PROOF. Conjugate the small exponential by g_n and use the BCH/local chart expansion. The same increments are essential: no extra stochastic error is introduced. □

THEOREM 0.2 (Random shearing theorem). *Outside the algebraic factor alternative, there exist nearby generic pairs and stopping times at which their common-noise images converge to (z, uz) for a nontrivial horospherical u.*

PROOF. Choose pairs with transverse separation tending to zero from the support of a stationary joining. Two-point separation gives first-exit times; root-direction selection and compactness give a convergent normalized displacement. Recurrence supplies a convergent basepoint subsequence. Chapter 4 identifies the limit relation by u. □

Proposition 0.3 (Recurrence compatibility)**.** *The stopping-time construction can be restricted to returns to a fixed large compact set at the cost of changing constants.*

PROOF. XXXIII supplies exponential/finite-mean returns to a compact set. Intersect the first-exit event with the next return; cocycle large deviations show the additional waiting time has controlled tails and does not destroy the root-direction limit. □

CHAPTER 6

Production of Additional Invariance

Shearing becomes rigidity only after passing the pair relation to conditional/stationary measures.

Lemma 0.1 (Support-pair invariance lemma)**.** *Let ν be a stationary probability and suppose generic pairs (x_j, y_j) in the same stationary joining converge after matched random evolution to (z, uz) with both marginals asymptotically distributed as ν. Then $u_*\nu = \nu$.*

Proof. Test against a bounded continuous f. Stationarity makes the time-evolved first and second marginals both ν. Along the selected generic subsequences their empirical test averages converge to $\int f d\nu$ and $\int f(u\cdot)d\nu$ respectively. Since the pairs have the same limiting joining and differ by u, the two integrals are equal. ☐

Theorem 0.2 (Additional unipotent invariance). *The random-shearing theorem therefore produces a nontrivial unipotent subgroup preserving ν, unless the algebraic factor alternative has already occurred.*

Proof. Apply the support-pair invariance lemma to the nontrivial u from Chapter 5; closure of the stabilizer of ν contains the one-parameter subgroup generated by its root vector. ☐

Proposition 0.3 (Stabilizer enlargement)**.** *If the previous stabilizer did not already contain that root group, the connected stabilizer of ν strictly increases in dimension.*

Proof. A connected Lie subgroup containing a new one-parameter subgroup not in its Lie algebra has strictly larger Lie algebra span and hence larger dimension. ☐

CHAPTER 7

From Drift to Algebraicity

Once nontrivial unipotent invariance appears, the Ratner-type classification proved from AHD 1–4 converts it into a homogeneous algebraic support statement in the scopes where that theorem applies.

Lemma 0.1 (Generated-unipotent closure)**.** *The subgroup generated by all unipotent one-parameter subgroups contained in* $\mathrm{Stab}(\nu)$ *is a closed connected normal subgroup of the identity component of the stabilizer after taking closure.*

PROOF. Conjugating a preserving unipotent by a preserving element again preserves ν. Hence the subgroup they generate is normal; taking closure preserves invariance and connected generation. ☐

THEOREM 0.2 (Algebraic-support transfer). *In the finite-volume homogeneous setting covered by the homogeneous-measure theorem of AHD IV, if an ergodic probability is invariant under a subgroup generated by unipotents, it is Haar probability on a closed homogeneous orbit of the corresponding Ratner subgroup.*

PROOF. This is exactly the earlier proved generated-by-unipotents classification theorem, invoked at its stated real/S-arithmetic scope. No new classification is asserted here. ☐

Proposition 0.3 (Foundation-level algebraicity alternative)**.** *Combining Chapters 2 and 6, every stationary-measure analysis entering the BQ core reaches either (i) a proper algebraic factor relation, or (ii) strictly enlarged unipotent invariance with homogeneous support in the inherited Ratner scope.*

PROOF. Two-point separation gives the dichotomy. In case (ii), random shearing and the invariance lemma produce a new unipotent, and the algebraic-support transfer identifies the resulting homogeneous structure. ☐

CHAPTER 8

Stabilization of the Invariance Group

The foundation block ends with a termination argument: invariance cannot enlarge indefinitely in finite dimension.

Lemma 0.1 (Ascending-dimension termination)**.** *A strictly increasing chain of connected Lie subgroups of a fixed finite-dimensional Lie group can contain at most* $\dim G$ *strict dimension increases.*

Proof. Every strict Lie-algebra inclusion raises dimension by at least one, while dimensions are bounded by $\dim G$. □

Theorem 0.2 (Exponential-drift stabilization principle). *Iterating the drift–shearing–invariance procedure terminates after finitely many strict enlargements at a stationary measure whose remaining nearby-pair obstructions are contained in the stabilized algebraic homogeneous relation.*

Proof. Each nonalgebraic transverse obstruction produces, by Chapters 2–6, a new unipotent direction and therefore a strict stabilizer enlargement. The termination lemma forces this to stop. At the terminal stage no such transverse obstruction remains, so all surviving pair relations lie in the recorded algebraic factor. This is the precise foundation needed before the Benoist–Quint classification volume; it is not itself the classification theorem. □

Volume 35

Benoist–Quint Stationary Measure Classification

Volume 35 scope and prerequisites

This volume begins the Benoist–Quint classification block. It uses only Volumes 1–34 and the explicitly stated foundational analysis and algebra prerequisites.

Proof and provenance. Primary-source parity is checked against the papers named in the chapter source notes. Stronger variants are not silently imported.

Reader guide: a concrete model

Why this matters.

Volume 35 is organized around the passage from *Stationary-Measure Setup on Homogeneous Spaces* to *The Benoist–Quint Stationary-Measure Classification Theorem.* The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.3 (A calculation to keep in view)**.** Let

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 0 \\ 0 & 1/3 \end{pmatrix},$$

and choose A and B independently with probabilities p and $1-p$. For the product $G_n = g_n \cdots g_1$ the two Lyapunov exponents are obtained without any limiting theorem:

$$\lambda_1 = p \log 2 + (1-p) \log 3, \qquad \lambda_2 = -\lambda_1.$$

Indeed the logarithm of the expanding diagonal entry is a sum of independent increments. This diagonal model is useful for checking signs, normalization, and determinant constraints before the noncommutative Oseledets theory is invoked.

CHAPTER 1

Stationary-Measure Setup on Homogeneous Spaces

Let G be a second countable real Lie group, let $\Lambda < G$ be a lattice and put $X = G/\Lambda$. For a Borel probability μ on G write $P_\mu f(x) = \int f(gx)\, d\mu(g)$ and let Γ_μ be the closed subgroup generated by $\operatorname{supp}\mu$. A probability ν is μ-stationary if $P_\mu^* \nu = \nu$.

> **Proof architecture.** The source proof does not start by guessing the homogeneous orbit. It first passes from a stationary measure to pathwise conditional measures, then uses recurrence and exponential drift to manufacture extra invariance. The previous foundation volumes 27–34 own exactly those probabilistic and drift mechanisms. This chapter fixes the measure-theoretic interface and the ergodic reduction.

Definition 0.1 (Homogeneous probability)**.** A probability ν on X is homogeneous if, with $G_\nu = \{g : g_*\nu = \nu\}$, there is a closed G_ν-orbit F with $\nu(F) = 1$.

Lemma 0.2 (Stationary ergodic decomposition)**.** *Every μ-stationary probability is the barycenter of μ-ergodic stationary probabilities, and any μ-invariant Borel set has conditional stationary measures on its positive-mass ergodic components.*

Proof. The Markov operator P_μ is positive, preserves constants and is a contraction on $L^1(\nu)$. The convex compact set of stationary probabilities is a standard Borel simplex under evaluation on a countable determining algebra. Let $\mathcal{A}$ be a countable algebra generating the Borel sigma-algebra of X. On the compact convex set of stationary probabilities, two measures are separated by the coordinates $\eta \mapsto \eta(A)$, $A \in \mathcal{A}$. Form the natural path space with present state x_0 and increments $b_0, b_1, \ldots$; stationarity gives a shift-invariant probability. Disintegrate this probability over its invariant sigma-algebra. For each conditional component and each bounded Borel f, the shift identity gives

$$\int f(x_0)\, d\beta_\omega = \int\int f(gx_0)\, d\mu(g)\, d\beta_\omega,$$

so the present-state marginal ν_ω is μ-stationary. If a component were not extremal, a nontrivial convex splitting of ν_ω would lift to a nontrivial invariant splitting of the path component. Conversely a nontrivial invariant event yields, by conditioning, two distinct stationary marginals. Hence the path components project exactly to the stationary ergodic components. Applying the same construction after restricting to an invariant Borel saturation proves the final assertion. $\square$

THEOREM 0.3 (Pathwise stationary extension). *If ν is μ-stationary, there is a probability β^X on $B \times X$, $B = G^{\mathbb{N}}$, invariant under $(b, x) \mapsto (Tb, b_0 x)$, whose X-marginal is ν. For μ-ergodic ν the skew product is ergodic after passing to the natural two-sided extension.*

PROOF. On cylinder functions define the law by first sampling x_0 with law ν and then independent increments b_i with law μ. Stationarity says that the law of $x_1 = b_0 x_0$ is again ν, so the joint law is invariant under the skew shift. Consistency of the finite-dimensional cylinder laws gives the one-sided law by Kolmogorov extension. For the two-sided extension, take the projective system whose Nth stage records $(x_{-N}, b_{-N}, \ldots, b_N)$ with $x_{j+1} = b_j x_j$ and stationary marginal at every time. The coordinate-forgetting maps preserve these finite-dimensional laws, so their projective limit is a shift-invariant two-sided law. If an invariant set in the extension has conditional probability $h(x)$ over the present state, then $P_\mu h = h$. The sets $\{h > t\}$ are μ-invariant modulo null sets; ergodicity forces h to be constant, proving ergodicity of the extension. $\square$

Proposition 0.4 (Reduction to the ergodic case)**.** *To classify all stationary probabilities it suffices to classify the ergodic ones, provided the classification family is closed under countable barycentric decomposition.*

PROOF. Apply Lemma 0.2. Each ergodic component has the asserted classified form. Integrating the components recovers the original measure. No uniqueness of the homogeneous subgroup is required for this reduction. $\square$

Worked Example 0.5. If $\mu = \frac{1}{2}(\delta_a + \delta_{a^{-1}})$ on a compact quotient and a acts ergodically with respect to Haar measure, Haar measure is μ-stationary and its stationary ergodic decomposition is trivial. If instead a preserves two disjoint closed finite-volume orbits, a convex combination of their Haar measures is stationary but not ergodic.

CHAPTER 2

Algebraic Hull of the Support Semigroup

Set $\mathbf{H}_\mu = \overline{\mathrm{Ad}(\Gamma_\mu)}^{Z} \leq \mathrm{GL}(\mathfrak{g})$. Throughout this chapter we assume that $\mathbf{H}_\mu$ is semisimple, Zariski connected, and has no compact algebraic factor.

Lemma 0.1 (No compact-representation quotient)**.** *Assume $\mathbf{H}_\mu$ is semisimple, Zariski connected and has no compact factor. If V is a nontrivial irreducible real algebraic $\mathbf{H}_\mu$-module, then the image of Γ_μ in $\mathrm{GL}(V)$ is not relatively compact.*

Proof. If the image were relatively compact, its Zariski closure would be a compact real algebraic quotient of $\mathbf{H}_\mu$. The kernel is normal and the quotient is nontrivial because V is nontrivial. This contradicts the no-compact-factor hypothesis. □

Theorem 0.2 (Algebraic-hull reduction). *Under the hypotheses above, every finite-dimensional $\mathrm{Ad}(\Gamma_\mu)$-invariant subquotient of $\mathfrak{g}$ splits into a fixed part and a direct sum of noncompact semisimple irreducible factors. On every nonfixed irreducible factor the top Lyapunov exponent is positive once the finite exponential-moment hypothesis of Volumes 28–32 is imposed.*

Proof. Complete reducibility follows from semisimplicity of $\mathbf{H}_\mu$. The fixed vectors form the sum of the trivial constituents. Every other constituent has noncompact image by Lemma 0.1; strong irreducibility is obtained after passing to the finite collection of isotypic blocks and, if necessary, to a convolution power that preserves each block. The Furstenberg positivity theorem proved in Volume 28 then gives positive top exponent on each noncompact irreducible block. The finite collection permits one common block length. □

Proposition 0.3 (Finite block normalization)**.** *Replacing μ by a convolution power does not change the class of Γ_μ-invariant probabilities and does not change homogeneity of an ergodic stationary measure after resolving the finitely many cyclic components.*

Proof. The support of μ^{*m} lies in Γ_μ and generates a finite-index normal subsemigroup in the cyclic decomposition. A Γ_μ-invariant measure is plainly

μ^{*m}-stationary. Conversely, if an ergodic μ-stationary measure splits into m cyclic μ^{*m}-components, the original walk permutes them transitively. Homogeneity of one component therefore yields homogeneity of their finite orbit-average, with stabilizer containing the original support semigroup. □

Failure mode. Dropping the no-compact-factor condition allows stationary "satellite" measures supported over a compact algebraic factor. This is why the compact-factor exclusion is a theorem hypothesis, not a cosmetic simplification.

CHAPTER 3

Recurrence and Nonescape

The drift argument only produces invariance if typical paths repeatedly return to charts where nearby points and transverse directions are controlled.

Lemma 0.1 (Uniform compact return)**.** *Assume the compact-support hypothesis of the Benoist–Quint theorem. For every compact $C \subset X$ and $\epsilon > 0$ there is a compact $K \subset X$ and n_0 such that for all $x \in C$ and $n \geq n_0$, $\mu^{*n} * \delta_x(K) > 1 - \epsilon$.*

Proof. Volume 33 constructs a proper height $u \geq 1$ and numbers $m \geq 1$, $a < 1, b < \infty$ with $P_\mu^m u \leq au+b$. Iteration gives $P_\mu^{jm} u(x) \leq a^j u(x)+b/(1-a)$. On compact C, u is bounded. Markov's inequality then gives a common sublevel $K_R = \{u \leq R\}$ carrying mass $> 1 - \epsilon$ at times divisible by m. The finitely many residues modulo m are handled by enlarging K_R by the compact set $(\operatorname{supp} \mu)^m K_R$. ☐

Theorem 0.2 (Exponential recurrence to a thick compact core). *There is a closed set K on which the injectivity radius is bounded below and constants $c, C > 0$ such that, for x in any fixed compact set, the probability of avoiding K for n consecutive steps is at most Ce^{-cn}.*

Proof. Choose a sufficiently large sublevel of the height function from Volume 33 so that the one-block drift has negative mean outside the sublevel. The stopped process $a^{-j} u(X_{jm\wedge\tau_K})$ is dominated by a supermartingale after adding the constant $b/(1 - a)$. The exponential-moment version of the drift estimate, also proved in XXXIII, yields an exponential tail for τ_K. Choosing the sublevel below the Mahler/height threshold gives a positive injectivity-radius lower bound on K. ☐

Proposition 0.3 (First-return chain)**.** *The chain induced by first returns to K has finite exponential return-time moment and preserves the conditional stationary measure obtained from $\nu|_K$ after normalization by the Kac return-time density.*

Proof. The exponential tail in Theorem 0.2 gives the exponential moment. For bounded f on K, sum the stationarity identity over successive excursions and use the strong Markov property. The occupation measure of an excursion has total mass equal to its expected length; Kac normalization therefore makes the first-return kernel stationary. $\square$

CHAPTER 4

Exponential Drift

The foundational exponential-drift construction of Volume 34 is now inserted into the first-return chain. This removes the only place where recurrence and shearing could otherwise fail to meet.

Lemma 0.1 (Return-chain transverse drift)**.** *Let Y be a sufficiently small transverse displacement at a return to the compact core K. Outside the algebraic factor subspace, there exist $\alpha > 0$, $a < 1$ and $B < \infty$ such that the first-return kernel Q satisfies*

$$Q\,\|Y\|^{-\alpha} \leq a\|Y\|^{-\alpha} + B.$$

Proof. For a fixed block length, Volume 34 gives contraction of the inverse-distance height until chart exit. Excursion lengths have exponential moments by Proposition 0.3; the adjoint cocycle also has exponential moments because μ is compactly supported. Split according to whether the excursion length is at most M or larger. On the first event concatenate the finitely many block estimates; on the second use Hölder and the exponential tail. Taking M large and then α small makes the total coefficient < 1. ☐

Theorem 0.2 (Drift dichotomy for nearby generic points). *For a stationary ergodic ν, either almost every sufficiently close generic pair is constrained by a proper algebraic factor relation, or there are generic pairs whose matched paths reach a fixed transverse scale in a positive Lyapunov direction while both endpoints remain in K.*

Proof. Apply Lemma 0.1 to the stationary self-joining obtained from the conditional measures of the path extension. If the inverse-distance height is infinite, the displacement lies in the zero set, which is precisely the recorded algebraic factor relation. Otherwise recurrence of the drift height gives infinitely many returns to a bounded inverse-distance level. The ratio-stopping theorem of XXXIV extracts a top Lyapunov direction, and the first-return construction keeps both endpoints in K. ☐

Proposition 0.3 (Source-proof interface)**.** *The dichotomy above supplies the "nearby points before the drift" and "drift vector" inputs of the Benoist–Quint proof without any additional compactness assumption on X.*

Proof. Nonescape is supplied by Chapter 3, so all subsequential compactness occurs inside K. Chapter 4 supplies a nonzero normalized transverse vector. These are exactly the two logical inputs needed for the subsequent random-shearing argument; no compactness of the whole quotient is used. $\square$

CHAPTER 5

Horospherical Invariance

The decisive step is to turn a first-exit vector into an actual element of the stabilizer of the stationary measure.

Lemma 0.1 (Conditional-pair transport)**.** *Let ν_b be the pathwise conditional measures in the stationary extension. For almost every path b and every n, $(g_n \cdots g_1)_* \nu_b = \nu_{T^n b}$.*

Proof. Disintegrate the invariant skew-product measure over the future path coordinate. Invariance of the skew product identifies the conditional measure over Tb with the pushforward of the conditional measure over b by b_0. Iterate this identity. $\square$

Theorem 0.2 (Unstable invariance from exponential drift). *In the non-algebraic branch of Theorem 0.2, ν is invariant under a nontrivial one-parameter subgroup contained in a positive Lyapunov/horospherical subgroup of $\mathbf{H}_\mu$.*

Proof. Choose generic nearby pairs (x_j, y_j) in the same conditional measure ν_{b_j} with transverse displacement tending to zero. Stop the common-noise paths when the displacement first reaches a fixed small scale. Chapters XXXIV.3–XXXIV.5 show, after a subsequence, that the normalized displacements converge to a nonzero vector Y in a top positive root space and the stopped pairs converge to $(z, \exp(Y)z)$. By Lemma 0.1 the two endpoints still belong to the same limiting conditional measure. Testing this conditional measure against continuous functions gives invariance under $\exp(Y)$. Repeating after rescaling the initial separation by t gives invariance under $\exp(tY)$ for t in an interval about zero; the subgroup law and closure give the whole one-parameter subgroup. Averaging the conditional invariance over paths yields invariance of ν. $\square$

Proposition 0.3 (Noncompact-factor alternative)**.** *If no such new horospherical direction exists, every conditional nearby displacement lies in the maximal $\mathbf{H}_\mu$-invariant subspace on which the cocycle has zero transverse growth; this subspace defines the algebraic factor recorded in Chapter 4.*

Proof. The algebraic-hull decomposition in Chapter 2 has positive exponent on each nonfixed irreducible transverse constituent. Hence failure of the first-exit construction on every nonzero constituent forces all displacement to lie in the sum of fixed/compactly acted-on constituents, which is $\mathbf{H}_\mu$-invariant and integrates to the stated factor relation. $\square$

CHAPTER 6

Stabilizer Enlargement and the Maximal Unipotent Core

Let $G_\nu = \{g : g_*\nu = \nu\}$. Following the source proof, the object that is stable under the driving semigroup is not an arbitrarily chosen stabilizer but the common normal core

$$G_\nu^0 := \bigcap_{\gamma\in\Gamma_\mu} \gamma G_\nu \gamma^{-1}, \qquad S_{\max} := (G_\nu^0)_u,$$

where $(\cdot)_u$ denotes the connected subgroup generated by the Ad-unipotent one-parameter subgroups it contains. Volume 34 constructs precisely such new Ad-unipotent directions.

Lemma 0.1 (Normalization and quotient descent)**.** *$S_{\max}$ is normalized by Γ_μ. Suppose an $S_{\max}$-orbit carrying the conditional homogeneous measure has finite $S_{\max}$-volume and write $\bar{G} = N_G(S_{\max})/S_{\max}$. Then the stationary problem descends to a stationary problem on the corresponding quotient homogeneous space, and a nontrivial unipotent stabilizer produced there lifts to a connected stabilizer strictly larger than $S_{\max}$ upstairs.*

Proof. The intersection defining G_ν^0 is invariant under conjugation by Γ_μ, hence so is its subgroup generated by Ad-unipotent one-parameter subgroups. If $S_{\max}x$ carries finite invariant volume, disintegration along the $S_{\max}$-orbits identifies the quotient measure with the pushforward of ν and intertwines the Markov operators, because every element of Γ_μ normalizes $S_{\max}$. Thus stationarity and ergodicity descend. If a one-parameter subgroup $\bar{U} < \bar{G}$ fixes the quotient measure, its inverse image together with $S_{\max}$ has Lie algebra strictly larger than $\mathfrak{s}_{\max}$ whenever $\bar{U}$ is nontrivial. The lifted conditional measures are invariant under this larger connected group, contradicting maximality unless the quotient drift obstruction has disappeared. ☐

Theorem 0.2 (Finite stabilizer-enlargement reduction). *Assume the recurrence and first-return hypotheses of Chapters 3–4. Starting with the maximal common unipotent core $S_{\max}$, one has the following alternative.*

(1) *A nonalgebraic transverse pair survives in the quotient. Then XXXIV.4–XXXIV.6 produces a nontrivial Ad-unipotent one-parameter subgroup of the quotient stabilizer, and Lemma 0.1 strictly enlarges the connected stabilizer upstairs.*

(2) *No such pair survives. Then the conditional measures are supported on the algebraic relation generated by* $S_{\max}$ *and the centralizer; the residual stationary measure is carried by the corresponding centralizer orbit family.*

After at most $\dim G$ *strict enlargements the first alternative is impossible.*

Proof. Work on the exponentially recurrent return set from Chapter 3, where the exponential chart is injective and the return law has the moment bounds of Chapter 4. If two generic conditional points remain arbitrarily close modulo the current algebraic relation, choose their first return at which the normalized transverse displacement reaches the fixed annulus used in XXXIV.3. The ratio stopping lemma and random-shearing theorem XXXIV.3–XXXIV.5 give a nonzero unipotent limit, and XXXIV.6 turns it into invariance. By maximality of $S_{\max}$ the new direction is not contained in the current quotient kernel, so Lemma 0.1 raises the stabilizer dimension.

If the first-exit construction cannot be made, the contrapositive of XXXIV.2 and XXXIV.7 says that every sufficiently small generic pair belongs to the recorded algebraic homogeneous relation. Conditional measures are therefore locally supported on its leaves. Recurrence transports this local statement to almost every return and ergodicity makes the leaf type essentially constant; the only remaining parameter lies in the centralizer orbit family. Each occurrence of the first alternative raises Lie-algebra dimension by at least one, and dimensions are bounded by $\dim G$, so the process terminates after finitely many steps. □

Proposition 0.3 (Homogeneous reconstruction from the terminal branch)**.** *Assume the terminal algebraic leaf is a closed finite-volume orbit* Sx *and the residual component space is a finite* Γ_μ*-orbit. Then* ν *is homogeneous.*

Proof. Disintegrate ν over the finitely many components. On each component, invariance under S and ergodicity of the leaf action identify the conditional probability with normalized Haar measure on the closed finite-volume orbit. The support semigroup permutes the components. The group generated by S and that finite permutation action is a closed subgroup of G_ν acting transitively on the union of the components; its invariant probability is exactly their stationary weighted average. Hence the support is one closed G_ν-orbit and ν is homogeneous. □

Proof and provenance. The source proof uses the same two reductions: construction of the maximal subgroup generated by Ad-unipotent one-parameter subgroups in the common stabilizer, followed by passage to the quotient and a contradiction if the quotient has no such stabilizer. The present chapter records the real-case interface; the S-adic weak-regularity lifting issue is deliberately outside this volume.

CHAPTER 7

Classification of Homogeneous Stationary Measures

The previous chapters create a homogeneous candidate. We now record what stationarity forces once homogeneity is known.

Lemma 0.1 (Stationarity on a homogeneous support)**.** *Let $Y = Hx \subset X$ be closed with finite H-invariant probability m_Y. If $\Gamma_\mu \subset N_G(Y)$ and $\mu * m_Y = m_Y$, then the induced Markov chain on the finite component space of Y has stationary law equal to the component weights of m_Y.*

Proof. Disintegrate m_Y over the finitely many H°-components. Each $g \in \Gamma_\mu$ permutes the components and carries Haar measure on a component to Haar measure on its image. The stationarity equation reduces exactly to the finite-state stationary equation for the component weights. □

Theorem 0.2 (Homogeneous stationary measures are invariant in the BQ scope). *Under the hypotheses of Volume 35, a μ-ergodic homogeneous stationary probability produced by the stabilization algorithm is Γ_μ-invariant.*

Proof. The identity component S of the stabilizer is normalized by the algebraic hull at termination. Hence every $g \in \Gamma_\mu$ sends an S-component of the support to an S-component. On the finite component space, ergodicity makes the Markov chain irreducible. Its stationary law is the unique invariant probability on the finite orbit of components, so the component average is fixed by every generator. Inside each component the measure is Haar and conjugation merely identifies Haar probabilities. Thus $g_*\nu = \nu$ for every g in the support and therefore for every $g \in \Gamma_\mu$. □

Proposition 0.3 (Irreducible simple-ambient corollary)**.** *If G is connected semisimple without compact factors, Λ is irreducible and Γ_μ is Ad-Zariski dense, then every non-atomic ergodic stationary probability is Haar measure on X.*

Proof. Theorem 0.2 and Chapter 8 below make the measure homogeneous and Γ_μ-invariant. The stabilizer identity component is a connected normal subgroup normalized by a Zariski-dense subgroup. Irreducibility of the lattice implies that a positive-dimensional proper normal subgroup has dense orbit rather than a finite-volume closed orbit. Thus a non-atomic homogeneous probability has full support and equals Haar. Zero-dimensional homogeneous probabilities are precisely finite-orbit counting measures. $\square$

CHAPTER 8

The Benoist–Quint Stationary-Measure Classification Theorem

Proof and provenance. The statement below matches the real-case theorem in Benoist–Quint, *Stationary measures and invariant subsets of homogeneous spaces (II)*: compact support and semisimple, Zariski-connected adjoint hull with no compact factor. The paper proves the broader weakly regular S-adic theorem through conditional measures, semisimple random-product estimates, first-return chains, recurrence, and production of unipotent invariance. Here the real case inherits the exact random-product/return-chain mechanisms proved in 27–34; no satellite-measure or full S-adic strengthening is claimed. [BQ13a]

Lemma 0.1 (Assembly lemma). *Assume the real-case Benoist–Quint hypotheses listed immediately below: G is a real Lie group, $\Lambda < G$ is a lattice, μ is compactly supported, and the Zariski closure of $\mathrm{Ad}(\Gamma_\mu)$ is semisimple, Zariski connected, and has no compact factor. Every ergodic stationary probability reaches the homogeneous terminal branch of Theorem 0.2.*

Proof. Chapter 3 gives recurrent thick charts. Chapter 4 gives the algebraic-factor versus transverse-drift dichotomy. In the transverse case Chapter 5 produces nontrivial horospherical invariance and Chapter 6 strictly enlarges the stabilizer. In the factor case one passes to the corresponding quotient, whose adjoint representation is again a semisimple subquotient of the original hull; Chapter 2 preserves the no-compact-factor property on every nontrivial constituent. Finite dimension forces termination. At the terminal stage the quotient conditional measure is supported on the centralizer-orbit relation. Ergodicity reduces the component parameter to one homogeneous orbit family; when this family is finite Proposition 0.3 applies directly, while the positive-dimensional centralizer case is absorbed into the stabilizer before taking the final quotient. Thus the lifted measure is homogeneous. □

Theorem 0.2 (Benoist–Quint stationary-measure classification, real case). *Let G be a real Lie group, $\Lambda < G$ a lattice and μ a compactly supported* *probability on G. Suppose*

$$\mathbf{H}_\mu = \overline{\mathrm{Ad}(\Gamma_\mu)}^{Z}$$

is semisimple, Zariski connected and has no compact factor. Then every μ-ergodic μ-stationary probability ν on G/Λ is Γ_μ-invariant and homogeneous.

Proof. By Lemma 0.1, ν is homogeneous. The terminal stabilizer is normalized by the adjoint hull because every enlargement was formed from Γ_μ-conjugates. Theorem 0.2 then gives Γ_μ-invariance. This proves both conclusions with exactly the stated hypotheses. □

Failure mode. The theorem is false with this conclusion if the adjoint hull has compact factors: stationary satellite measures may project nontrivially to a compact factor. We therefore retain the source hypothesis throughout the downstream volumes.

Volume 36

Benoist–Quint Invariant Subsets and Orbit Closures

Volume 36 scope and prerequisites

This volume uses only Volumes 1–35 and the explicitly stated foundational analysis and algebra prerequisites.

Proof and provenance. Primary-source parity is checked against the papers named in the chapter source notes. Stronger variants are not silently imported.

Reader guide: a concrete model

Why this matters.

Volume 36 is organized around the passage from *Invariant Closed Subsets* to *Topological Rigidity Consequences*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.3 (A calculation to keep in view)**.** Let

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 0 \\ 0 & 1/3 \end{pmatrix},$$

and choose A and B independently with probabilities p and $1-p$. For the product $G_n = g_n \cdots g_1$ the two Lyapunov exponents are obtained without any limiting theorem:

$$\lambda_1 = p \log 2 + (1-p)\log 3, \qquad \lambda_2 = -\lambda_1.$$

Indeed the logarithm of the expanding diagonal entry is a sum of independent increments. This diagonal model is useful for checking signs, normalization, and determinant constraints before the noncommutative Oseledets theory is invoked.

CHAPTER 1

Invariant Closed Subsets

Fix the BQ hypotheses from Volume 35 and let Γ be a compactly generated closed subsemigroup whose adjoint Zariski closure is semisimple with no compact factor. Choose a compactly supported probability μ whose support spans a dense subsemigroup of Γ.

Lemma 0.1 (Stationary measure on a closed invariant set)**.** *If $F \subset X$ is nonempty, closed and Γ-invariant, then F supports a μ-stationary probability.*

Proof. Choose $x \in F$. By recurrence from XXXIII, the Cesàro averages $\sigma_n = n^{-1} \sum_{k<n} \mu^{*k} * \delta_x$ are tight. Every weak limit σ is supported on F, because every term is supported on F and F is closed. Moreover $\mu * \sigma_n - \sigma_n = n^{-1}(\mu^{*n} * \delta_x - \delta_x)$ tends weakly to zero, so σ is stationary. □

Theorem 0.2 (Ergodic stationary support inside an invariant set). *Every nonempty closed Γ-invariant F contains the support of an ergodic stationary probability, hence contains a finite-volume homogeneous subspace.*

Proof. Take a stationary probability from Lemma 0.1 and an ergodic component supported in F. Volume 35 classifies it as a Γ-invariant homogeneous probability; its support is the required homogeneous subspace. □

Proposition 0.3 (Minimal-set consequence)**.** *A minimal nonempty closed Γ-invariant subset is a finite-volume homogeneous subspace on which Γ acts ergodically.*

Proof. The homogeneous support furnished by Theorem 0.2 is a nonempty closed invariant subset of the minimal set, hence equals it. Replacing the stationary measure by an ergodic component makes the action ergodic. □

CHAPTER 2

Stationary Orbit Closures

For $x \in X$, let $F_x = \overline{\Gamma x}$.

Lemma 0.1 (Support equals a minimal recurrent closure)**.** *Every weak limit of the Cesàro measures from x is supported on F_x, and the support of each ergodic component is a homogeneous subspace $Y \subset F_x$ intersecting $\overline{\Gamma x}$ recurrently.*

PROOF. Support containment follows as in Lemma 0.1. The component is homogeneous by XXXV. Poincaré recurrence for the stationary skew product implies that almost every point in its support returns to every positive-measure neighborhood, so the component lies in the recurrent part of F_x. □

THEOREM 0.2 (Homogeneous core of every orbit closure). *Every F_x contains a Γ-ergodic finite-volume homogeneous subspace Y_x that is minimal among homogeneous subspaces contained in F_x.*

PROOF. Existence follows from Theorem 0.2. Choose one of minimal dimension. If it contained a proper Γ-ergodic finite-volume homogeneous subspace, that would have smaller dimension, contradiction. □

Proposition 0.3 (Centralizer translates preserve the class)**.** *If $L = C_G(\Gamma)$ and Y is a Γ-ergodic finite-volume homogeneous subspace, then ℓY has the same property for every $\ell \in L$.*

PROOF. Centrality gives $\Gamma(\ell Y) = \ell(\Gamma Y) = \ell Y$. Translation sends the homogeneous probability on Y to the homogeneous probability on ℓY and conjugates no element of Γ, so ergodicity is preserved. □

CHAPTER 3

Minimal Invariant Sets

The topological proof is an induction through minimal homogeneous strata rather than an appeal to abstract minimality alone.

Lemma 0.1 (No proper homogeneous stratum in a minimal set)**.** *Let F be minimal closed invariant and let $Y \subset F$ be a Γ-ergodic homogeneous subspace. Then $Y = F$.*

PROOF. Y is itself nonempty, closed and Γ-invariant, so minimality forces equality. ☐

THEOREM 0.2 (Minimal-set classification). *Every minimal closed Γ-invariant set is a closed finite-volume homogeneous orbit, and its unique stationary probability with full support is the invariant homogeneous probability.*

PROOF. The first statement is Proposition 0.3. For uniqueness, any stationary probability supported on the minimal set decomposes into ergodic stationary components; XXXV makes each component homogeneous with closed invariant support. Minimality forces every component support to be all of F. Two invariant homogeneous probabilities on the same transitive finite-volume orbit are both normalized Haar and hence equal. ☐

Proposition 0.3 (Orbit closure from uniqueness plus recurrence)**.** *If F_x contains only one minimal homogeneous stratum Y, then every Cesàro weak limit of the walk from x is the homogeneous probability m_Y.*

PROOF. Every weak limit is stationary. Each ergodic component has a minimal homogeneous support contained in F_x; by uniqueness this support is Y, and Theorem 0.2 identifies the component with m_Y. Thus the whole limit is m_Y. ☐

CHAPTER 4

Algebraic Invariant Subsets and Centralizer Moduli

Put $L = C_G(\Gamma)$. The orbit-closure argument needs two facts that must be kept separate: countability of the homogeneous strata modulo L, and a local-containment theorem for convergent homogeneous measures. The proof below follows the real-case architecture of Benoist–Quint III rather than replacing both facts by a single linearization sentence.

Lemma 0.1 (Countability modulo the centralizer)**.** *In the real BQ scope, the set of L-orbits of Γ-ergodic finite-volume homogeneous subspaces of $X = G/\Lambda$ is countable.*

PROOF. Let Y be such a subspace and write $x = g\Lambda \in Y$. Let G_Y be the connected stabilizer of the invariant probability on Y, and set

$$H = g^{-1}(\Gamma G_Y^{\circ})g, \qquad \Delta = \Lambda \cap H^{\circ}, \qquad \Sigma = \Lambda \cap H.$$

The groups Δ and Σ are finitely generated: Δ is a lattice in the connected Lie group H°, while Σ/Δ is finite because the homogeneous probability has finitely many connected components. Since Λ is countable, there are only countably many possible pairs (Δ, Σ).

Fix one such pair. Its Zariski/analytic hull determines H° up to the finite ambiguity coming from the component group: the Lie algebra is the smallest Lie subalgebra containing the logarithmic directions of Δ and invariant under conjugation by Σ. Thus only countably many subgroups H occur. Once H is fixed, the coset gH is fixed by Γ in the homogeneous space G/H. We now show that the fixed-point set $(G/H)^{\Gamma}$ is a countable union of L-orbits.

Take a fixed point gH. In a sufficiently small exponential slice transverse to $\mathrm{Ad}(g)\mathfrak{h}$, another fixed point has the form $\exp(v)gH$ and satisfies

$$\mathrm{Ad}(\gamma)v - v \in \mathrm{Ad}(g)\mathfrak{h} + O(\|v\|^2) \qquad (\gamma \in \Gamma).$$

Linearizing at $v = 0$, the tangent space to the fixed-point set is the subspace fixed by $\mathrm{Ad}(\Gamma)$ in $\mathfrak{g}/\mathrm{Ad}(g)\mathfrak{h}$. Semisimplicity of the adjoint Zariski hull gives a complementary invariant subspace, so the implicit-function theorem identifies the local fixed-point set with the local L-orbit. Hence every L-orbit of fixed points is open in the fixed-point set. The latter is second countable, so it has at most countably many disjoint open L-orbits. Combining this with the countably many lattice pairs proves the claim. □

THEOREM 0.2 (Compactness and local containment of homogeneous strata). *Let Y_n be Γ-ergodic finite-volume homogeneous subspaces meeting a fixed compact set $K \subset X$, with invariant probabilities ν_n. Then a subsequence converges to a homogeneous probability ν_∞ supported on Y_∞. Moreover, after passing to a further subsequence, there are $\ell_n \in L$, $\ell_n \to e$, such that*

$$Y_n \subset \ell_n Y_\infty$$

for all sufficiently large n.

PROOF. *Nonescape.* The height construction in Volume 33 is uniform on the compact set K: there is a proper $u \geq 1$ and a block Markov operator P^m with $P^m u \leq au + b$, $a < 1$, on every orbit meeting K. Integrating this inequality against ν_n and using invariance gives a uniform bound on $\int u \, d\nu_n$. Hence the family is tight. Choose a weak limit ν_∞; it has total mass one and is Γ-invariant, hence stationary. By Volume 35 it is a barycenter of homogeneous ergodic probabilities.

Excluding escape into a smaller centralizer translate. Fix one homogeneous component Y of ν_∞. For a compact neighborhood $K_L \subset L$ of the identity, the closed family $K_L Y$ has a transverse inverse-distance height of the type constructed in 33–34: outside $K_L Y$ the expected inverse distance contracts after a fixed block, while it is infinite on $K_L Y$. If infinitely many Y_n were not contained in any sufficiently small L-translate of Y, integrating this height along ν_n and passing to the limit would force $\nu_\infty(LY) = 0$. This contradicts the choice of Y as a component of the limit. Thus some component Y_∞ of the limit captures all sufficiently large Y_n up to centralizer translation.

From approximate to actual containment. Choose $x_n \in Y_n \cap K$ and, after a subsequence, $x_n \to x_\infty \in Y_\infty$. The preceding paragraph supplies $\ell_n \in L$ with $\ell_n \to e$ for which the invariant probability on Y_n gives asymptotically full mass to arbitrarily small tubular neighborhoods of $\ell_n Y_\infty$. The quantitative Chevalley/proper-lift theorem proved in the earlier homogeneous-measure volumes converts this tubular concentration into inclusion of the connected stabilizer Lie algebra. Discreteness of the lattice branch then upgrades Lie-algebra inclusion to subgroup inclusion. Consequently $Y_n \subset \ell_n Y_\infty$ for all large n. □

Proposition 0.3 (Sequential compactness of strata meeting a compact set). *For compact $K \subset X$, the collection of invariant probabilities on Γ-ergodic finite-volume homogeneous subspaces meeting K is sequentially compact; every limit is homogeneous after choosing the maximal component furnished by Theorem 0.2.*

Proof. Tightness and existence of a probability limit were proved in the first paragraph of Theorem 0.2. Its local-containment conclusion shows that the supports cannot split among two incomparable maximal homogeneous types: otherwise two subsequences would be eventually contained in two distinct arbitrarily small centralizer translates, contradicting convergence of points in the fixed compact set. Thus the maximal limiting type is unique and carries full mass. The limit is its normalized homogeneous probability. □

Proof and provenance. The two ingredients correspond to distinct steps of Benoist–Quint III: countability is proved by fixing the lattice-intersection data and then analyzing fixed points in G/H modulo L; local containment is proved using recurrence/repulsion functions before the small centralizer correction. The order matters: countability alone does not imply local containment.

CHAPTER 5

Finite and Compact Exceptional Orbits

Exceptional compact or finite orbits are not errors in the classification; they are genuine algebraic alternatives.

Lemma 0.1 (Finite orbit criterion)**.** *A zero-dimensional homogeneous* Γ*-invariant probability is the uniform probability on a finite* Γ*-orbit.*

Proof. A zero-dimensional closed orbit is discrete. Finite invariant volume forces it to be finite. Invariance and transitivity of the stabilizer permutation action force equal mass on all points. □

Theorem 0.2 (Isolation alternative for compact exceptional strata). *If a sequence of distinct compact* Γ*-ergodic homogeneous subspaces converges to* Y*, then either their dimensions eventually increase into* Y *or they are small centralizer translates of subspaces contained in* Y*.*

Proof. Apply Theorem 0.2. After $\ell_n \to e$, $Y_n \subset \ell_n Y$. If dimensions agree, connected stabilizer dimensions agree and the inclusion is open-and-closed on connected components, so Y_n is a component translate. Otherwise the dimension is strictly smaller. These are exactly the two alternatives. □

Proposition 0.3 (No accumulation of distinct finite orbits inside a finite orbit)**.** *If the centralizer of* Γ *is discrete, a finite* Γ*-orbit has a neighborhood containing no distinct finite* Γ*-orbit from the same homogeneous-stratum family.*

Proof. If distinct finite orbits accumulated, compactness would give a convergent sequence. Theorem 0.2 and discreteness of the centralizer make $\ell_n = e$ for large n, forcing inclusion in the finite limiting orbit. Equality then follows from transitivity, contradiction. □

CHAPTER 6

Homogeneous Orbit Closures

Let $F_x = \overline{\Gamma x}$. The proof proceeds by induction on the ambient homogeneous dimension, using the local-containment theorem to rule out accumulation on infinitely many incomparable lower-dimensional strata.

Lemma 0.1 (Open homogeneous stratum lemma)**.** *Assume the orbit-closure theorem is known for homogeneous spaces of dimension $< \dim X$. Then either F_x is contained in a proper finite-volume homogeneous subspace, in which case the induction hypothesis applies there, or there is a unique homogeneous stratum $Y_x \subset F_x$ that is open in F_x and contains x.*

Proof. By Chapter 1, every nonempty closed invariant subset of F_x contains a finite-volume homogeneous stratum. If one such stratum contains x and is proper in X, then the entire orbit of x lies in it and so does F_x; induction applies. Otherwise consider homogeneous strata meeting a small compact neighborhood of x. Theorem 0.2 gives local containment: two sufficiently close strata through points approaching x are, after centralizer corrections tending to the identity, nested. Choose one of maximal dimension. If it were not relatively open in F_x, a sequence of orbit points outside it would converge to a point on it; stationary limits from those points would provide homogeneous strata contradicting maximality through the local-containment theorem. Hence it is open. Two distinct open homogeneous strata in the orbit closure would have disjoint nonempty invariant open pieces; density of the orbit of x rules this out. Thus the open stratum is unique. □

Theorem 0.2 (Benoist–Quint homogeneous orbit-closure theorem). *For every $x \in X$ there is a closed subgroup $H < G$ containing the stabilizer needed for the Γ-action such that*

$$\overline{\Gamma x} = Hx,$$

and Hx carries a finite H-invariant measure.

Proof. We induct on $\dim X$. The zero-dimensional case is finite and immediate. Suppose the theorem is proved in smaller dimension. If F_x lies in a proper finite-volume homogeneous subspace, apply the induction hypothesis there. Otherwise Lemma 0.1 gives a unique open homogeneous stratum $Y_x \subset F_x$ containing x.

Let $Z = F_x \setminus Y_x$. It is closed and Γ-invariant. If Z were nonempty, Chapter 1 would give a homogeneous stratum in Z. By countability modulo the centralizer, Z is contained in a countable union of proper centralizer translates of lower-dimensional homogeneous strata. The recurrence/repulsion estimate used in Theorem 0.2 implies that a random walk started in the open stratum spends asymptotically zero proportion of time in arbitrarily small neighborhoods of each fixed proper stratum. A diagonal choice over the countable family therefore forces every stationary limit of the walk from x to give full mass to Y_x.

On the other hand, stationarity and density of the orbit imply that the support of such a limit meets every nonempty relatively open invariant portion of F_x. Hence no point of Z can belong to the support, so $Z = \varnothing$. Therefore $F_x = Y_x$ is a finite-volume homogeneous subspace. Writing it as Hx proves the theorem. $\square$

Proposition 0.3 (Uniqueness of the homogeneous probability on the closure)**.** *The orbit closure $F_x = Hx$ has a unique probability invariant under the full transitive stabilizer G_{F_x}.*

Proof. The restriction of any G_{F_x}-invariant Radon measure to the transitive homogeneous space $G_{F_x}/(G_{F_x})_x$ is a Haar quotient measure, unique up to a scalar. Since F_x has finite invariant volume, normalization to total mass one fixes that scalar. Hence the invariant probability is unique. $\square$

CHAPTER 7

Orbit-Closure Classification

Proof and provenance. The theorem below is the real form of Benoist–Quint III, Theorem 1.1. The compactly generated closed-semigroup hypothesis is retained. [BQ13b]

Lemma 0.1 (Semigroup-to-probability reduction)**.** *A compactly generated closed semigroup Γ admits a compactly supported probability μ whose support spans a dense subsemigroup and whose adjoint Zariski hull equals that of Γ.*

Proof. Choose a compact generating set C and a countable dense subset $D \subset C$. Select finitely many elements of D meeting every algebraic component needed to generate the Zariski closure; Noetherianity of the Zariski topology makes this finite. Give those elements positive mass and add a summable positive atomic mass on the rest of D. If strict compact support is desired, take a probability with full support on C; its generated semigroup is dense in Γ and its Zariski hull is the same. □

Theorem 0.2 (Benoist–Quint orbit-closure theorem, real case). *Let G be a real Lie group, Λ a lattice and Γ a compactly generated closed subsemigroup. If $\overline{\mathrm{Ad}(\Gamma)}^{Z}$ is semisimple with no compact factor, then for every $x \in G/\Lambda$ there is a closed $H \geq \Gamma$ with*

$$\overline{\Gamma x} = Hx,$$

and Hx has finite H-invariant volume.

Proof. Use Lemma 0.1 to choose μ. Passing to the connected algebraic hull by the finite-component normalization of XXXV reduces to the hypotheses used there. The homogeneous saturation theorem 0.2 then gives H and the finite invariant probability. Returning from the finite-index connected-hull component takes only a finite union of translates, which is again one orbit of the subgroup generated by those translates and H. □

Proposition 0.3 (Archimedean scope of the orbit-classification argument)**.** *Let G be the real Lie group fixed in this volume and let $U < G$ be the real unipotent subgroup occurring in Theorem 0.2. The argument of Chapters 1–7 proves the stated orbit classification for the real action $U \curvearrowright X$. It does not* *assert the analogous classification for an S-arithmetic action on*

$$G_S/\Gamma, \qquad G_S = \prod_{v \in S} \mathbf{G}(F_v),$$

when S contains a non-Archimedean place.

Proof. The proof of Theorem 0.2 uses three Archimedean inputs: real Lie-algebra linearization, decomposition into real isotypic subspaces, and recurrence for the real one-parameter unipotent action. None of these steps supplies the additional non-Archimedean local-subgroup classification or the countability argument needed when a factor $\mathbf{G}(F_v)$ with $v \nmid \infty$ is present. Thus the conclusion proved here is exactly the real one displayed above. $\square$

CHAPTER 8

Topological Rigidity Consequences

Lemma 0.1 (Discrete-centralizer compactness). *If $C_G(\Gamma)$ is discrete, the family of Γ-ergodic finite-volume homogeneous subspaces is compact in the one-point compactification topology, and inclusion is eventually monotone under convergence.*

Proof. Proposition 0.3 gives compactness on every fixed compact set, while recurrence makes the only escaping limit the point at infinity. Theorem 0.2 gives $Y_n \subset \ell_n Y$ with $\ell_n \to e$; discreteness forces $\ell_n = e$ eventually. $\square$

Theorem 0.2 (Closed invariant sets with discrete centralizer). *Under the hypotheses of Theorem 0.2 and with discrete centralizer, every closed Γ-invariant subset is a finite union of Γ-ergodic finite-volume homogeneous subspaces. If moreover G is connected semisimple without compact factors, Λ irreducible and Γ Ad-Zariski dense, every infinite invariant subset is dense and every sequence of distinct finite Γ-orbits equidistributes to Haar.*

Proof. For the first assertion, induct on the dimension of the invariant set. If it is not contained in finitely many proper homogeneous strata, choose points avoiding the first n strata. Their orbit closures form a sequence of homogeneous subspaces; compactness gives a limit. Eventual containment from Lemma 0.1 forces the limit to be the ambient homogeneous component, so the invariant set contains that component. Remove it and apply induction to the remaining closed invariant part. In the Zariski-dense irreducible case there is no positive-dimensional proper finite-volume homogeneous stratum normalized by Γ; hence any infinite invariant set has full closure. For distinct finite orbits, every subsequential weak limit is homogeneous by compactness. It cannot be supported on a finite orbit by the isolation proposition, so it is Haar. Uniqueness of subsequential limits gives equidistribution. $\square$

Proposition 0.3 (Toral specialization). *For a strongly irreducible semisimple subgroup of $\mathrm{GL}_d(\mathbb{Z})$ in the BQ scope, every infinite invariant subset of $\mathbb{T}^d$ is dense and every sequence of distinct finite invariant orbits equidistributes to Haar.*

Proof. Embed $\mathrm{Aut}(\mathbb{Z}^d) \ltimes \mathbb{R}^d$ and its lattice $\mathrm{Aut}(\mathbb{Z}^d) \ltimes \mathbb{Z}^d$ into the homogeneous-space formulation. Homogeneous subsets project to finite unions of parallel rational subtori. Strong irreducibility rules out a nontrivial proper invariant rational subtorus, so Theorem 0.2 leaves only finite sets and the full torus. ☐

Volume 37

Random-Walk Equidistribution

Volume 37 scope and prerequisites

This volume uses only Volumes 1–36 and the explicitly stated foundational analysis and algebra prerequisites.

Proof and provenance. Primary-source parity is checked against the papers named in the chapter source notes. Stronger variants are not silently imported.

Reader guide: a concrete model

Why this matters.

Volume 37 is organized around the passage from *Random-Walk Mean Ergodic Theorem* to *Quantitative Random-Walk Equidistribution*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.4 (A calculation to keep in view)**.** Let

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 0 \\ 0 & 1/3 \end{pmatrix},$$

and choose A and B independently with probabilities p and $1-p$. For the product $G_n = g_n \cdots g_1$ the two Lyapunov exponents are obtained without any limiting theorem:

$$\lambda_1 = p \log 2 + (1-p) \log 3, \qquad \lambda_2 = -\lambda_1.$$

Indeed the logarithm of the expanding diagonal entry is a sum of independent increments. This diagonal model is useful for checking signs, normalization, and determinant constraints before the noncommutative Oseledets theory is invoked.

CHAPTER 1

Random-Walk Mean Ergodic Theorem

Let $P = P_\mu$ act on $L^2(Y, m_Y)$, where m_Y is a Γ_μ-invariant homogeneous probability. Then P is a contraction.

Lemma 0.1 (Fixed vectors of the Markov operator)**.** *If* $\operatorname{supp}\mu$ *generates* Γ *and* m_Y *is* Γ*-invariant, every* P*-fixed vector is invariant under the closed subgroup generated by* $\operatorname{supp}\mu$*.*

PROOF. For $f \in L^2$, Jensen gives $\|Pf\|_2^2 \leq \int \|g \cdot f\|_2^2 d\mu(g) = \|f\|_2^2$. If $Pf = f$, equality holds in the strictly convex Hilbert norm, so $g \cdot f$ is almost surely constant as a function of g on the support. Its average is f, hence $g \cdot f = f$ for μ-almost every g. Continuity of the unitary representation and density of the generated subgroup give invariance under Γ. □

THEOREM 0.2 (Mean ergodic theorem for the walk). *For every* $f \in L^2(Y, m_Y)$*,*

$$\frac{1}{n}\sum_{k=0}^{n-1} P^k f \to \mathbb{E}(f \mid \mathcal{I}_\Gamma) \quad \text{in } L^2.$$

If the Γ*-action on* Y *is ergodic, the limit is* $\int f\, dm_Y$*.*

PROOF. The von Neumann mean ergodic theorem for a contraction says the Cesàro averages converge to the orthogonal projection onto $\ker(I - P)$. Lemma 0.1 identifies that kernel with the Γ-invariant functions. Ergodicity makes these constants. □

Proposition 0.3 (Dual weak convergence)**.** *If an initial probability has* L^2 *density* h *with respect to* m_Y*, its Cesàro random-walk distributions converge weakly to* m_Y *on an ergodic homogeneous component.*

PROOF. For bounded continuous f, duality gives $n^{-1}\sum_k \int P^k f\, h\, dm_Y$. Theorem 0.2 gives L^2 convergence of the Cesàro average of $P^k f$ to its integral, and Cauchy–Schwarz finishes. □

CHAPTER 2

Cesàro Convergence of Convolution Powers

Point masses need the topological and recurrence machinery of XXXVI because they need not have L^2 densities.

Lemma 0.1 (Every Cesàro limit lives on the orbit closure)**.** *For $x \in X$, every weak limit of*

$$\sigma_n = \frac{1}{n}\sum_{k=0}^{n-1} \mu^{*k} * \delta_x$$

is stationary and supported on $Y_x = \overline{\Gamma x}$.

Proof. The uniform height estimate of XXXIII gives tightness of (σ_n). Every summand is supported on Y_x, so closedness gives support containment for every weak limit. Finally

$$\mu * \sigma_n - \sigma_n = \frac{1}{n}(\mu^{*n} * \delta_x - \delta_x).$$

Pairing with a bounded continuous test function makes the right-hand side at most $2\|f\|_\infty/n$ in absolute value. Passing to a weakly convergent subsequence therefore gives $\mu * \sigma = \sigma$. □

Theorem 0.2 (Cesàro equidistribution on the homogeneous orbit closure). *Under the hypotheses of XXXVI, let $Y_x = \overline{\Gamma x}$ and let m_{Y_x} be its homogeneous probability. Then*

$$\frac{1}{n}\sum_{k=0}^{n-1} \mu^{*k} * \delta_x \Longrightarrow m_{Y_x}.$$

Proof. By XXXVI, Y_x is a finite-volume homogeneous subspace. Let σ be a subsequential Cesàro limit. Lemma 0.1 makes σ stationary and supported on Y_x. Decompose σ into ergodic stationary components. Volume 35 makes every component a Γ-invariant homogeneous probability supported on some homogeneous $Z \subset Y_x$.

We claim that no proper Z can occur with positive weight. Choose a compact portion K_Z of Z carrying all but ε of its homogeneous probability, and let $U_r(K_Z)$ be a sufficiently thin tubular neighborhood. The inverse-distance/repulsion function used in XXXVI.4 gives, for points outside Z, a block drift inequality which implies

$$\limsup_{n\to\infty} \frac{1}{n} \sum_{k<n} \mu^{*k} * \delta_x(U_r(K_Z)) \leq C\varepsilon$$

provided $x \notin Z$. Since Y_x is the orbit closure of x, a proper closed Γ-invariant Z cannot contain x. Letting $\varepsilon \downarrow 0$ shows that $\sigma(Z) = 0$. Countability of homogeneous strata modulo the centralizer (XXXVI.4) allows this exclusion simultaneously for every proper ergodic component. Hence the only component is the homogeneous probability on Y_x itself. All subsequential limits are m_{Y_x}, proving convergence. □

Proposition 0.3 (Proper-stratum occupation estimate)**.** *Let $Z \subsetneq Y_x$ be a Γ-invariant finite-volume homogeneous subspace and let $K \subset Z$ be compact. For every $\epsilon > 0$ there exists a neighborhood U of K such that*

$$\limsup_{n\to\infty} \frac{1}{n} \sum_{k<n} \mu^{*k} * \delta_x(U) < \epsilon.$$

The same estimate holds almost surely for the empirical occupation frequency of a random trajectory from x.

Proof. Use the repulsion height v_Z from the local-containment proof in XXXVI.4. It is proper away from the centralizer translates of Z, tends to $+\infty$ on Z, and after a fixed return block satisfies a truncated drift inequality of the form

$$P^m(v_Z \wedge R) \leq a(v_Z \wedge R) + b_R, \qquad a < 1,$$

with $b_R/R \to 0$ as $R \to \infty$. Integrating the telescoping drift along the first n blocks gives a uniform bound on the average probability of the superlevel set $\{v_Z > R\}$. Choose R so this bound is $< \epsilon/2$ and then choose U so that $U \cap \{u \leq M\} \subset \{v_Z > R\}$, where u is the global cusp height. The cusp occupation outside $\{u \leq M\}$ is $< \epsilon/2$ for M large by XXXIII. This proves the distributional estimate.

For trajectories, apply the bounded martingale-difference law to the indicators of the two truncated height events (approximated from above by continuous functions). Their empirical frequencies differ from the corresponding Markov averages by $o(1)$ almost surely, giving the same bound pathwise. □

CHAPTER 3

Pointwise Random-Walk Equidistribution

For independent increments $g_1, g_2, \dots$ with law μ, put $X_0 = x$ and $X_{k+1} = g_{k+1} X_k$.

Lemma 0.1 (Breiman martingale lemma for the Markov chain)**.** *For every bounded Borel f,*

$$\frac{1}{n}\sum_{k=0}^{n-1}\big(f(X_{k+1}) - P_\mu f(X_k)\big) \longrightarrow 0 \qquad \textit{almost surely.}$$

Consequently every weak subsequential limit of the empirical measures $n^{-1}\sum_{k<n}\delta_{X_k}$ is μ-stationary.

Proof. Let $\mathcal{F}_k = \sigma(g_1, \dots, g_k)$. Then

$$D_{k+1} = f(X_{k+1}) - P_\mu f(X_k)$$

is $\mathcal{F}_{k+1}$-measurable, satisfies $\mathbb{E}(D_{k+1} \mid \mathcal{F}_k) = 0$, and $|D_{k+1}| \le 2\|f\|_\infty$. Thus (D_k) is a bounded martingale-difference sequence. Since

$$\sum_{k\ge 1}\frac{\mathbb{E}D_k^2}{k^2} < \infty,$$

the martingale convergence theorem applied to $\sum D_k/k$ and Kronecker's lemma give $n^{-1}\sum_{k\le n} D_k \to 0$ almost surely.

Now let $\eta_{n_j} = n_j^{-1}\sum_{k<n_j}\delta_{X_k} \Rightarrow \eta$. For bounded continuous f the lemma gives

$$\int f\,d\eta_{n_j} - \int P_\mu f\,d\eta_{n_j} = o(1),$$

where shifting the first sum by one changes it by at most $2\|f\|_\infty/n_j$. Passing to the limit yields $\int f\,d\eta = \int P_\mu f\,d\eta$, hence η is stationary. □

Theorem 0.2 (Almost-sure trajectory equidistribution). *Under the BQ orbit-closure hypotheses, for every starting point x and almost every increment path,*

$$\frac{1}{n}\sum_{k=0}^{n-1}\delta_{X_k} \Longrightarrow m_{Y_x}, \qquad Y_x = \overline{\Gamma x}.$$

Proof. The cusp-height theorem of XXXIII and its pathwise Markov inequality imply tightness of the empirical measures almost surely. Fix a path in the full-measure set on which this tightness, Lemma 0.1 for a countable determining family, and Proposition 0.3 for the countable homogeneous-stratum family all hold. Let η be any empirical subsequential limit. Lemma 0.1 makes η stationary, while the orbit itself is contained in Y_x, so $\operatorname{supp}\eta \subset Y_x$.

Decompose η into ergodic stationary components. Volume 35 makes them homogeneous and Γ-invariant. If a component were supported on a proper homogeneous stratum $Z \subsetneq Y_x$, choose a compact $K \subset Z$ carrying more than half of that component's mass. Portmanteau then forces a positive lower empirical frequency in every sufficiently small neighborhood of K, contradicting Proposition 0.3. Thus every ergodic component is m_{Y_x}. Hence $\eta = m_{Y_x}$. Since every subsequential limit is the same, the full empirical sequence converges. □

Proposition 0.3 (Birkhoff formulation). *For every $f \in C_c(X)$, almost surely*

$$\frac{1}{n}\sum_{k<n} f(X_k) \longrightarrow \int f\, dm_{Y_x}.$$

Proof. The function f is bounded and continuous. Pair the weak convergence of empirical measures in Theorem 0.2 with f. □

CHAPTER 4

Exponential Convergence under Spectral Hypotheses

This chapter is an abstract quantitative add-on. It is not attributed to the qualitative BQ theorem.

Lemma 0.1 (Spectral contraction)**.** *If* $\|P\|_{L^2_0(Y)} \le \rho < 1$*, then* $\|P^n f - \int f\|_2 \le \rho^n \|f - \int f\|_2$.

PROOF. Subtract the constant projection and iterate the operator norm bound. □

THEOREM 0.2 (Exponential convergence for L^2 densities). *If* $h \in L^2(Y)$ *is a probability density and the spectral contraction above holds, then for every* $f \in L^2(Y)$,

$$\left| \int f \, d(P^{*n}(hm_Y)) - \int f \, dm_Y \right| \le \rho^n \|h - 1\|_2 \|f - \textstyle\int f\|_2.$$

PROOF. Move P^n to the observable by duality and apply Cauchy–Schwarz together with Lemma 0.1. □

Proposition 0.3 (Smoothing-to-point-mass transfer)**.** *Suppose in addition that some* $P^r\delta_x$ *has* L^2 *density* $h_{r,x}$ *with locally uniform* L^2 *norm. Then the same exponential estimate applies to* $P^{*(n+r)}\delta_x$ *with prefactor* $\|h_{r,x} - 1\|_2$.

PROOF. By hypothesis $P^{*r}\delta_x = h_{r,x} m_Y$. Therefore, for every $n \ge 0$,

$$P^{*(n+r)}\delta_x = P^{*n}(h_{r,x} m_Y).$$

Theorem 0.2, applied with $h = h_{r,x}$, gives for every $f \in L^2(Y)$

$$\left| \int f \, d(P^{*(n+r)}\delta_x) - \int f \, dm_Y \right| \le \rho^n \|h_{r,x} - 1\|_2 \left\| f - \int f \, dm_Y \right\|_2.$$

The assumed local uniform bound for $\|h_{r,x}\|_2$ makes the prefactor locally uniform in x. □

CHAPTER 5

Spectral Gap for Markov Operators

Lemma 0.1 (Symmetric averaging and almost invariant vectors)**.** *If μ is symmetric and $P = \int \pi(g)d\mu(g)$ in a unitary representation, then*

$$\langle (I-P)v, v\rangle = \frac{1}{2}\int \|\pi(g)v - v\|^2 d\mu(g).$$

Proof. Expand the square, use unitarity and symmetry to identify the real part of $\langle \pi(g)v, v\rangle$ with $\langle Pv, v\rangle$. □

Theorem 0.2 (Kazhdan-to-Markov spectral gap). *If the representation on $L^2_0(Y)$ has a Kazhdan gap for a compact set contained in the support semigroup and μ assigns uniformly positive mass to neighborhoods of a Kazhdan set, then a symmetric convolution power of μ has operator norm strictly below one on $L^2_0(Y)$.*

Proof. A norm-one sequence for the averaging operator would, by Lemma 0.1, give unit vectors whose displacement tends to zero on the positive-mass Kazhdan neighborhoods. Compactness and a finite covering of the Kazhdan set transfer this to almost invariance on the whole Kazhdan set, contradicting its positive Kazhdan constant. Therefore the operator norm is < 1. □

Proposition 0.3 (Property (T) interface)**.** *The property-(T) and property-(τ) results of Volume 6 provide exactly the representation-theoretic hypothesis of Theorem 0.2 under their stated hypotheses.*

Proof. Volume 6 states its spectral gap as absence of almost invariant vectors on the orthogonal complement of constants, with explicit Kazhdan sets. This is precisely the hypothesis used above; no stronger pointwise matrix-coefficient decay is inferred. □

CHAPTER 6

Algebraic Obstruction Alternatives

Lemma 0.1 (Support obstruction)**.** *If Y_x is a proper homogeneous orbit closure, no sequence of random-walk distributions starting at x can converge weakly to Haar measure on a strictly larger homogeneous space.*

PROOF. Every distribution is supported in the closed set Y_x. Weak limits are therefore supported in Y_x by Portmanteau. □

THEOREM 0.2 (Complete qualitative obstruction alternative). *In the BQ real scope, the only obstruction to Cesàro and pathwise equidistribution to ambient Haar is that the starting point lies in a proper finite-volume homogeneous orbit closure. Finite orbit closures give uniform counting measure; positive-dimensional closures give their homogeneous Haar probabilities.*

PROOF. XXXVI classifies every orbit closure as finite-volume homogeneous. Theorems 0.2 and 0.2 give equidistribution to its homogeneous probability. Conversely Lemma 0.1 prevents convergence to any larger support. The zero-dimensional case is Lemma 0.1. □

Proposition 0.3 (Ambient-Haar criterion)**.** *Equidistribution to ambient Haar occurs exactly when $\overline{\Gamma x} = X$.*

PROOF. If the closure is X, Theorem 0.2 gives Haar. If it is proper, support obstruction excludes Haar on X. □

CHAPTER 7

Equidistribution on Homogeneous Spaces

> **Proof and provenance.** The next theorem packages the real-case statements corresponding to Benoist–Quint III, Theorems 1.2 and 1.3. Cesàro distributional convergence and almost-sure empirical convergence are kept separate. [BQ13b]

Lemma 0.1 (Unique target measure)**.** *For every x, the homogeneous probability m_{Y_x} is the unique possible stationary weak limit compatible with both the orbit closure and the avoidance of proper strata.*

PROOF. This is the common uniqueness argument in Chapters 2 and 3: stationary classification gives homogeneous ergodic components; proper-stratum avoidance rules out every component except the full orbit-closure component. □

THEOREM 0.2 (Benoist–Quint random-walk equidistribution, real case). *Let G, Λ, Γ satisfy Theorem 0.2, and let μ be compactly supported with support spanning a dense subsemigroup of Γ. Then for every x,*

$$\frac{1}{n}\sum_{k<n} \mu^{*k} * \delta_x \Longrightarrow m_{Y_x}.$$

Moreover, for independent increments with law μ, almost surely

$$\frac{1}{n}\sum_{k<n} \delta_{g_k \cdots g_1 x} \Longrightarrow m_{Y_x}.$$

PROOF. The first statement is Theorem 0.2; the second is Theorem 0.2. Lemma 0.1 ensures that both target the same canonical homogeneous probability. □

Proposition 0.3 (Finite-orbit specialization)**.** *On a finite orbit the theorem reduces to the ergodic theorem for an irreducible finite Markov chain on the communicating class containing x.*

PROOF. Let the orbit be $F = \{x_1, \ldots, x_r\}$. Each group element acts by a permutation, so the transition matrix P is a convex combination of permutation matrices and is therefore doubly stochastic. The support-spanning hypothesis makes the directed transition graph strongly connected;

hence the unique stationary probability is the uniform law u_F. The finite-dimensional mean ergodic theorem gives $n^{-1}\sum_{k<n} P^k\delta_x \to u_F$ even when the chain is periodic. For the pathwise statement, apply Lemma 0.1 to the indicators of the individual states. Any empirical subsequential limit is stationary and hence equals u_F; compactness of the finite simplex then gives convergence of the full empirical sequence. $\square$

CHAPTER 8

Quantitative Random-Walk Equidistribution

This final chapter records a safe quantitative interface for later volumes. It deliberately does not call the qualitative BQ theorem quantitative.

Lemma 0.1 (Sobolev regularization contract). *Assume that for some integer r,*

$$P^r\delta_x = h_x m_Y, \qquad \|h_x\|_2 \le C\,\mathrm{ht}(x)^A,$$

and that P preserves m_Y and satisfies

$$\|P|_{L^2_0(m_Y)}\| \le \rho < 1.$$

Then for every $f \in L^2(m_Y)$ and every $n \ge 0$,

(37.8.1)
$$|P^{n+r}f(x) - m_Y(f)| \le C\,\mathrm{ht}(x)^A\rho^n\|f - m_Y(f)\|_2 \le C\,\mathrm{ht}(x)^A\rho^n\|f\|_2.$$

Proof. Using the density of $P^r\delta_x$ and invariance of m_Y,

$$P^{n+r}f(x) - m_Y(f) = \int h_x(y)\,P^n(f - m_Y(f))(y)\,dm_Y(y).$$

Cauchy–Schwarz gives an upper bound by

$$\|h_x\|_2\,\|P^n(f - m_Y(f))\|_2.$$

The centered function belongs to $L^2_0(m_Y)$, so the spectral hypothesis gives $\|P^n(f - m_Y(f))\|_2 \le \rho^n\|f - m_Y(f)\|_2$. Insert the density bound for h_x and then use $\|f - m_Y(f)\|_2 \le \|f\|_2 + |m_Y(f)| \le 2\|f\|_2$; after absorbing the harmless factor into C we obtain (37.8.1). ☐

Theorem 0.2 (Abstract effective random-walk theorem). *Under the spectral and regularization hypotheses of Lemma 0.1, and with exponential recurrence controlling excursions outside the regularization region, there are $C', A', \eta > 0$ such that*

$$|P^n f(x) - m_Y(f)| \le C'\,\mathrm{ht}(x)^{A'}e^{-\eta n}\mathcal{S}_s(f).$$

Proof. Return to the thick compact core within a time having exponential tail. At the first good return apply the r-step regularization and then the spectral estimate. Split according to return time $j \le n/2$ or $j > n/2$. The latter has exponentially small probability; on the former the spectral factor is at most $\rho^{n/2-r}$. Weighted Sobolev control of f on the cusp and the Lyapunov height estimate absorb the bad-tail contribution. Taking $\eta < \min\{-\frac{1}{2}\log\rho, c_{\mathrm{return}}/2\}$ gives the result. $\square$

Proposition 0.3 (Scope warning)**.** *Theorem 0.2 is an implication from explicit spectral/regularization hypotheses; those hypotheses are not asserted for every BQ walk. Their verification belongs to Volumes 40 and later effective chapters.*

Proof. The proof of Theorem 0.2 used the two hypotheses at named steps. Removing either step would leave a logical gap. Thus this proposition records the exact dependency boundary. $\square$

Volume 38

Random Walks on Tori and Affine Spaces

Volume 38 scope and prerequisites

This volume uses only Volumes 1–37 and the explicitly stated foundational analysis and algebra prerequisites.

Proof and provenance. Primary-source parity is checked against the papers named in the chapter source notes. Stronger variants are not silently imported.

Reader guide: a concrete model

Why this matters.

Volume 38 is organized around the passage from *Random Toral Automorphisms* to *Diophantine Consequences*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.4 (A calculation to keep in view)**.** Let

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 0 \\ 0 & 1/3 \end{pmatrix},$$

and choose A and B independently with probabilities p and $1-p$. For the product $G_n = g_n \cdots g_1$ the two Lyapunov exponents are obtained without any limiting theorem:

$$\lambda_1 = p \log 2 + (1-p) \log 3, \qquad \lambda_2 = -\lambda_1.$$

Indeed the logarithm of the expanding diagonal entry is a sum of independent increments. This diagonal model is useful for checking signs, normalization, and determinant constraints before the noncommutative Oseledets theory is invoked.

CHAPTER 1

Random Toral Automorphisms

Let $\Gamma < \mathrm{SL}_d(\mathbb{Z})$ act on $\mathbb{T}^d$. We begin with the qualitative theorem inherited from the BQ homogeneous embedding, then isolate the stronger direct-convolution statement as a separate Fourier problem.

Lemma 0.1 (Homogeneous embedding of the torus)**.** *The torus action of* Γ *is the restriction of the homogeneous action of* $\Gamma < \mathrm{Aut}(\mathbb{Z}^d) \ltimes \mathbb{R}^d$ *on*

$$(\mathrm{Aut}(\mathbb{Z}^d) \ltimes \mathbb{R}^d)/(\mathrm{Aut}(\mathbb{Z}^d) \ltimes \mathbb{Z}^d),$$

and every connected homogeneous subset in a torus fiber is a translate of a rational subtorus.

Proof. The quotient identifies a coset of (I, x) with $x + \mathbb{Z}^d$, and $(\gamma, 0)(I, x) = (\gamma, \gamma x)$ gives the usual action. A connected closed subgroup of $\mathbb{R}^d$ projects to a closed subgroup of $\mathbb{T}^d$ exactly when its intersection with $\mathbb{Z}^d$ is a lattice; equivalently its Lie subspace is rational. Translates give all connected homogeneous subsets in the fiber. $\square$

Theorem 0.2 (Toral orbit closures in the BQ scope). *Assume the adjoint/Zariski hypotheses needed for XXXVI. Every* Γ*-orbit closure in* $\mathbb{T}^d$ *is a finite union of parallel affine rational subtori. If* Γ *acts strongly irreducibly on* $\mathbb{Q}^d$*, every orbit is either finite or dense.*

Proof. Apply XXXVI to the homogeneous embedding. By Lemma 0.1, homogeneous supports in the torus fiber are finite unions of affine rational subtori. The direction of such a subtorus is a Γ-invariant rational subspace after taking the finite orbit of components. Strong irreducibility rules out every nonzero proper rational direction, leaving points or the full torus. $\square$

Proposition 0.3 (Linear Cesàro equidistribution)**.** *Under strong irreducibility, every irrational starting point has Cesàro random-walk distributions converging to Haar measure on* $\mathbb{T}^d$*.*

Proof. Its orbit is infinite and hence dense by Theorem 0.2. Apply XXXVII.7 with $Y_x = \mathbb{T}^d$. $\square$

CHAPTER 2

Affine Random Walks

An affine increment is $(\gamma, u) \in \mathrm{SL}_d(\mathbb{Z}) \ltimes \mathbb{T}^d$ and acts by $x \mapsto \gamma x + u$.

Lemma 0.1 (Finite-orbit arithmetic criterion)**.** *Let H be generated by a finite set S of affine transformations and suppose the linear projection acts strongly irreducibly. An orbit Hx is finite iff there exists $q \geq 1$ such that $gx - x \in q^{-1}\mathbb{Z}^d/\mathbb{Z}^d$ for every $g \in S$.*

Proof. If the congruence condition holds for generators, the cocycle identity $ghx - x = \gamma_g(hx - x) + (gx - x)$ propagates it to all words, so the orbit lies in the finite set $x + q^{-1}\mathbb{Z}^d/\mathbb{Z}^d$. Conversely, if Hx is finite, conjugate by translation by x so that the stabilizer of 0 has finite index in H. Its linear projection has finite index in the strongly irreducible linear group, hence remains strongly irreducible. Each translation vector in the finite orbit has finite orbit under this linear subgroup; a nonrational point would generate an infinite orbit by Theorem 0.2. Thus every generator displacement is rational, and finitely many denominators admit one common q. □

Theorem 0.2 (Affine Cesàro alternative). *Assume the linear projection is in the strongly irreducible BQ scope. For every x, the Cesàro distributions of a finitely supported affine random walk converge to the homogeneous probability on the affine orbit closure; in the irreducible toral case this is either uniform counting measure on a finite orbit or Haar measure on $\mathbb{T}^d$.*

Proof. Embed the affine action in the semidirect homogeneous group and apply XXXVII.2 to the orbit closure. Lemma 0.1 identifies the finite alternative. Strong irreducibility rules out proper positive-dimensional invariant rational subtori, so the only infinite homogeneous closure is the full torus. □

Proposition 0.3 (Cesàro-to-direct transfer criterion)**.** *Assume the affine Cesàro alternative of Theorem 0.2. If, in addition, every nonzero Fourier character satisfies*

$$\widehat{\mu^{*n} * \delta_x}(a) \longrightarrow 0,$$

then the full sequence $\mu^{*n} * \delta_x$ converges directly to Haar measure on the infinite-orbit branch.

Proof. The zero Fourier coefficient is always one. The assumed decay gives convergence of all nonzero Fourier coefficients to those of Haar measure. Trigonometric polynomials are uniformly dense in $C(\mathbb{T}^d)$, so the integrals of every continuous function converge to its Haar integral. This is weak convergence of the full convolution sequence, without Cesàro averaging. $\square$

Proof and provenance. He–Lakrec–Lindenstrauss prove a much stronger source theorem: under finite support, strong irreducibility, and either proximality or Zariski-connectedness of the linear part, the Fourier-decay hypothesis above follows unless the affine orbit is finite. Their proof uses a quantitative large-Fourier-coefficient bootstrap through initial nonconcentration, finite-field collision bounds, granulation, and a periodic-data obstruction. Those source-specific ingredients are *not* silently claimed by V38.C02.P; the exact HLL theorem is recorded as a research window in Chapter 7. [HLL22]

CHAPTER 3

Rational Subtori and Invariant Affine Subspaces

Lemma 0.1 (Rationality of closed subtori)**.** *A connected subgroup $V/(V \cap \mathbb{Z}^d)$ is closed in $\mathbb{T}^d$ iff $V \subset \mathbb{R}^d$ is spanned over $\mathbb{R}$ by $V \cap \mathbb{Q}^d$.*

Proof. If $V \cap \mathbb{Z}^d$ is a lattice in V, a lattice basis is rational and spans V. Conversely, clear denominators from a rational basis to get a full-rank discrete subgroup of $V \cap \mathbb{Z}^d$, so the quotient is compact and its image closed. □

Theorem 0.2 (Affine homogeneous-subspace classification). *Every connected homogeneous subspace of $\mathbb{T}^d$ is $x + T_V$ for a rational subspace V, and it is invariant under an affine semigroup H iff $\gamma V = V$ and $(\gamma, u)x - x \in T_V$ for every generator (γ, u).*

Proof. The first assertion is Lemma 0.1 together with Lemma 0.1. The image of $x + T_V$ under (γ, u) is $\gamma x + u + T_{\gamma V}$, so equality with $x + T_V$ is equivalent to the two stated conditions. □

Proposition 0.3 (Strong irreducibility kills positive-dimensional proper invariant subtori)**.** *If the linear group acts strongly irreducibly on $\mathbb{R}^d$, no finite union of positive-dimensional proper rational subtori is invariant.*

Proof. The finite set of tangent spaces would be a finite invariant union of proper nonzero linear subspaces, contradicting strong irreducibility. □

CHAPTER 4

Fourier Analysis of Affine Walks

For $a \in \mathbb{Z}^d$, write $e_a(x) = e^{2\pi i \langle a,x\rangle}$.

Lemma 0.1 (Affine Fourier recursion)**.** *For an affine law μ,*

$$P_\mu e_a(x) = \int e^{2\pi i \langle a,u\rangle} e_{\gamma^T a}(x)\, d\mu(\gamma, u).$$

Consequently $\widehat{\mu^{*n} * \delta_x}(a)$ is an expectation over the dual random walk $a \mapsto \gamma^T a$ with the accumulated translation phase.

Proof. Insert $(\gamma, u)x = \gamma x + u$ into e_a and use $\langle a, \gamma x\rangle = \langle \gamma^T a, x\rangle$. Iteration gives the second statement. □

Theorem 0.2 (Fourier-contraction criterion for direct equidistribution). *Suppose that for every $a \neq 0$ there are $C_a < \infty$ and $\rho_a < 1$ with*

$$\left|\widehat{\mu^{*n} * \delta_x}(a)\right| \leq C_a \rho_a^n$$

for all x outside the finite-orbit obstruction. Then $\mu^{*n} * \delta_x \Rightarrow m_{\mathbb{T}^d}$.

Proof. For every nonzero character the Fourier coefficient tends to zero, while the zero coefficient is one. Weyl's criterion on the compact abelian group $\mathbb{T}^d$ gives weak convergence to Haar. □

Proposition 0.3 (Finite-frequency bootstrap)**.** *Let $F \subset \mathbb{Z}^d \setminus \{0\}$ be finite. Suppose there is $m \geq 1$ and $0 < \rho < 1$ such that for every $a \in F$ and every x on the infinite-orbit branch,*

$$|P_\mu^m e_a(x)| \leq \rho,$$

and suppose the transpose semigroup sends each frequency encountered from F back into a finite set F' on which the same bound holds. Then every $a \in F$ has exponentially decaying Fourier coefficient along the walk.

Proof. For $a \in F'$ write $Q_a(x) = P_\mu^m e_a(x)$. By hypothesis $\|Q_a\|_\infty \leq \rho$. Apply the Markov property in blocks of length m. The Fourier recursion of Lemma 0.1 shows that after each block the character is a convex combination of characters whose frequencies remain in F', multiplied by phases of modulus one. Hence the supremum of the absolute values of the relevant Fourier coefficients is multiplied by at most ρ at every block. Therefore

$$|\widehat{\mu^{*jm} * \delta_x}(a)| \leq \rho^j.$$

The finitely many residual times $0 \leq r < m$ only contribute a factor at most one, so the same gives exponential decay at all times. $\square$

Proof and provenance. This proposition is deliberately elementary. It is *not* the HLL large-Fourier-coefficient theorem. The latter must control an infinite, exponentially spreading frequency set and is where nonconcentration, finite-field expansion, and granulation enter. Treating that source theorem as if it followed from Cauchy–Schwarz plus positive Lyapunov exponent would be a proof gap.

CHAPTER 5

Stationary Measures on Tori

Lemma 0.1 (Haar and finite-orbit stationary probabilities)**.** *Haar measure is stationary for every affine toral walk, and the uniform measure on a finite orbit is stationary whenever the induced finite Markov chain is doubly stochastic (in particular when the increments act by permutations and the orbit is H-transitive).*

Proof. Affine automorphisms preserve Haar. On a finite H-orbit every increment is a permutation, so the uniform vector is fixed by every permutation matrix and hence by their average. ☐

Theorem 0.2 (Stationary-measure dichotomy in the irreducible toral BQ scope). *Assume the linear action is strongly irreducible and its adjoint hull is in the BQ semisimple no-compact-factor scope. Every ergodic stationary probability arising in the homogeneous embedding is either Haar or uniform on a finite affine orbit.*

Proof. XXXV makes the stationary measure homogeneous and invariant under the generated semigroup. Chapter 3 classifies invariant homogeneous supports as finite unions of affine rational subtori. Strong irreducibility eliminates proper positive-dimensional supports by Proposition 0.3; thus the support is finite or all of $\mathbb{T}^d$. Lemma 0.1 identifies the corresponding probabilities. ☐

Proposition 0.3 (Nonatomic uniqueness)**.** *Under the same hypotheses Haar is the unique nonatomic ergodic stationary probability.*

Proof. The only alternative in Theorem 0.2 is a finite-orbit measure, which is atomic. ☐

CHAPTER 6

Orbit Closures

Lemma 0.1 (Finite-or-dense criterion)**.** *Under strong irreducibility, an affine homogeneous orbit closure is either finite or the whole torus.*

PROOF. The closure is a finite union of affine rational subtori by XXXVI. A positive-dimensional proper union would contradict Proposition 0.3. ☐

THEOREM 0.2 (Affine orbit-closure dichotomy). *In the strongly irreducible BQ scope, every affine semigroup orbit is finite or dense. For a finitely generated affine group, finiteness is equivalent to the rational-displacement criterion of Lemma 0.1.*

PROOF. Let x be the initial torus point. Lemma 0.1 gives the exhaustive alternative that the affine semigroup orbit closure of x is finite or is the whole torus. In the second case the orbit is dense by definition of orbit closure. In the first case the orbit itself is finite. When the semigroup comes from a finitely generated affine group, Lemma 0.1 characterizes this finite-orbit alternative by the stated rational-displacement condition. Substituting that characterization into the finite-or-dense alternative proves both assertions. ☐

Proposition 0.3 (Linear specialization)**.** *For a purely linear action, a point has finite orbit iff it is rational; every irrational orbit is dense.*

PROOF. A rational point clearly has finite orbit modulo its denominator. Conversely, if a linear orbit is finite, the finite-index stabilizer fixes the point. Strong irreducibility implies the common fixed space over $\mathbb{R}$ is zero; solving $(\gamma - I)x \in \mathbb{Z}^d$ for finitely many stabilizer generators gives a rational solution modulo $\mathbb{Z}^d$. Density then follows from the dichotomy. ☐

CHAPTER 7

Equidistribution and the Direct-Convergence Frontier

Lemma 0.1 (Weyl reduction). *For probabilities ν_n on $\mathbb{T}^d$, $\nu_n \Rightarrow m_{\mathbb{T}^d}$ if and only if*

$$\widehat{\nu}_n(a) \longrightarrow 0 \qquad (a \in \mathbb{Z}^d \setminus \{0\}).$$

Proof. Characters form a self-adjoint algebra containing constants and separating points of $\mathbb{T}^d$. Stone–Weierstrass therefore makes their finite linear span uniformly dense in $C(\mathbb{T}^d)$. Convergence of Fourier coefficients gives convergence on every trigonometric polynomial, uniform approximation transfers it to every continuous function, and the limiting functional is Haar integration. The converse follows by testing weak convergence against the characters. □

Theorem 0.2 (Direct affine equidistribution from Fourier decay). *Let μ be a finitely supported affine walk on $\mathbb{T}^d$. On any orbit branch for which*

$$\widehat{\mu^{*n} * \delta_x}(a) \to 0 \qquad \text{for every } a \neq 0,$$

one has

$$\mu^{*n} * \delta_x \Longrightarrow m_{\mathbb{T}^d}.$$

In particular, the conclusion holds whenever the finite-frequency hypotheses of Proposition 0.3 can be verified on an exhausting family of invariant frequency sets with uniform block contraction.

Proof. The first assertion is Lemma 0.1. For the second, apply Proposition 0.3 to each finite invariant frequency set in the exhaustion. Every fixed nonzero frequency belongs to one such set, so its coefficient decays. The first assertion then gives direct convergence. □

Proposition 0.3 (Cesàro/direct logical separation). *The BQ theory of Volumes 35–37 yields Cesàro distributional convergence and pathwise empirical equidistribution from homogeneous rigidity. Direct convergence of the individual convolution powers is a strictly stronger conclusion and requires an additional spectral/Fourier input such as Theorem 0.2.*

Proof. Cesàro convergence follows from XXXVII.2 once the affine action is embedded in its homogeneous semidirect product. Nothing in that argument controls periodic oscillation of the individual powers. Theorem 0.2, by contrast, forces every nontrivial character coefficient to vanish at the individual times and therefore removes such oscillation. Hence the two conclusions and their hypotheses are logically distinct. □

Proof and provenance. [HLL research window: not a parent proved input] He–Lakrec–Lindenstrauss prove that if μ is finitely supported on $\mathrm{SL}_d(\mathbb{Z}) \ltimes \mathbb{T}^d$, its linear projection is strongly irreducible, and that projection is either proximal or has Zariski-connected closure, then for every x either the affine orbit is finite or the full sequence $\mu^{*n} * \delta_x$ converges to Haar. Their quantitative theorem shows that a large Fourier coefficient forces the datum to be exponentially close to a finite-height periodic datum. The proof uses several source-specific nonconcentration and collision estimates not owned by the present AHD foundations. Accordingly this theorem is presented here for orientation and source comparison, not mislabeled as a proved AHD parent result. [HLL22]

CHAPTER 8

Diophantine Consequences

Lemma 0.1 (Denominator persistence)**.** *For a linear walk in* $\mathrm{SL}_d(\mathbb{Z})$*, a rational point of denominator* q *remains in the finite grid* $q^{-1}\mathbb{Z}^d/\mathbb{Z}^d$*; an irrational point cannot enter a rational grid at a later time.*

Proof. Integral matrices preserve the grid. They are invertible over $\mathbb{Z}$, so if γx is rational then $x = \gamma^{-1}(\gamma x)$ is rational. □

Theorem 0.2 (Random toral genericity consequence). *Assume the strongly irreducible affine orbit of* x *is infinite. Then the BQ theorem gives Cesàro distributional convergence and almost-sure empirical Haar genericity. If, in addition, the Fourier-decay hypothesis of Theorem 0.2 holds, then the individual distributions are Haar-generic:* $P^n f(x) \to \int f\,dm$ *for every continuous* f*.*

Proof. Strong irreducibility rules out a positive-dimensional proper invariant rational subtorus, so the infinite homogeneous orbit closure is the full torus. The Cesàro and pathwise empirical statements are therefore XXXVIII.2 and XXXVII.3. Under the extra Fourier-decay assumption, Theorem 0.2 gives convergence of the individual convolution powers. Testing these weak convergences against f gives the stated genericity conclusions. □

Proposition 0.3 (Rational obstruction is the only algebraic obstruction in the strongly irreducible linear case)**.** *For a strongly irreducible linear action, failure of Haar equidistribution from a point is equivalent to rationality of that point.*

Proof. By Lemma 0.1, rationality gives a finite orbit. Conversely, failure of the BQ Cesàro Haar alternative gives the finite-orbit branch of Theorem 0.2; Proposition 0.3 then gives rationality. □

Volume 39

Expanding Measures and Homogeneous Rigidity

Volume 39 scope and prerequisites

This volume uses only Volumes 1–38 and the explicitly stated foundational analysis and algebra prerequisites.

Proof and provenance. Primary-source parity is checked against the papers named in the chapter source notes. Stronger variants are not silently imported.

Reader guide: a concrete model

Why this matters.

Volume 39 is organized around the passage from *Expanding Probability Measures on Groups* to *Applications to Fractal Measures*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.4 (A calculation to keep in view)**.** Let

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 0 \\ 0 & 1/3 \end{pmatrix},$$

and choose A and B independently with probabilities p and $1-p$. For the product $G_n = g_n \cdots g_1$ the two Lyapunov exponents are obtained without any limiting theorem:

$$\lambda_1 = p \log 2 + (1-p) \log 3, \qquad \lambda_2 = -\lambda_1.$$

Indeed the logarithm of the expanding diagonal entry is a sum of independent increments. This diagonal model is useful for checking signs, normalization, and determinant constraints before the noncommutative Oseledets theory is invoked.

CHAPTER 1

Expanding Probability Measures on Groups

Let H be connected semisimple without compact factors and with finite center. Following Prohaska–Sert–Shi, a probability μ on H is H-expanding if every finite-dimensional representation of H without nonzero H-fixed vectors is uniformly expanded along almost every random product.

Lemma 0.1 (Quotient stability of expansion)**.** *If μ is H-expanding and $N \triangleleft H$ is connected normal, then the pushforward of μ to H/N is H/N-expanding.*

Proof. Any finite-dimensional representation of H/N without fixed vectors pulls back to a representation of H without fixed vectors. The random products and norms are unchanged under this pullback, so uniform expansion descends. □

Theorem 0.2 (Finite-representation criterion relative to an ambient G). *For a fixed embedding $H < G$, all expansion tests needed for stationary-measure classification, recurrence and repulsion can be reduced to finitely many representations constructed from exterior powers and symmetric squares of $\mathfrak{g}$.*

Proof. Every subgroup that can arise as a stabilizer in the linearization argument is detected by a Chevalley vector in some $\wedge^k\mathfrak{g}$. Normalizer and quotient information is detected after adjoining the symmetric square of that exterior power to remove sign characters. Since $0 \leq k \leq \dim G$, only finitely many such representations occur. The height functions for cusp/stratum avoidance use the same finite family of rational exterior-power representations. Therefore expansion on this finite family suffices for all later arguments in the fixed ambient group. □

Proposition 0.3 (Zariski-dense measures are expanding)**.** *If the closed subgroup generated by* $\operatorname{supp}\mu$ *has Zariski-dense adjoint image in H and μ has a finite first moment, then μ is H-expanding whenever the Furstenberg positivity hypotheses from 28–30 hold on each nontrivial irreducible representation.*

Proof. Decompose any representation into irreducible constituents. On a nontrivial constituent, Zariski density gives strong irreducibility after the standard finite-index reduction, and semisimplicity plus no compact factor gives a proximal element in a suitable exterior-power representation. Furstenberg positivity then gives positive top exponent. Intersecting the full-measure expansion events over the finitely many constituents proves uniform expansion for each nonfixed vector. $\square$

CHAPTER 2

Graph-Directed and Self-Similar Driving Systems

Lemma 0.1 (Graph coding to a Markov product)**.** *A finite strongly connected directed graph with edge maps $h_e \in H$ and positive transition probabilities defines a stationary finite-state Markov cocycle; passing to returns to one vertex yields an i.i.d. law on return words.*

PROOF. The edge shift with its stationary Markov measure is standard. Strong connectivity makes return times to a chosen vertex almost surely finite with exponential tail in the finite-state chain. By the strong Markov property, successive return words are independent and identically distributed. □

THEOREM 0.2 (Expansion criterion for graph-directed systems). *If the return-word law at one vertex is H-expanding and has finite first moment, then every vertex-to-vertex induced cocycle has the same positive Lyapunov sign on the finite representation family of Chapter 1.*

PROOF. A path from vertex i to the chosen base vertex and back has bounded length and belongs to a finite set. Multiplication by these bounded prefix/suffix matrices changes logarithmic norms by $O(1)$, so it does not change asymptotic Lyapunov exponents. Thus positivity for the return blocks transfers to every starting vertex and every induced return system. □

Proposition 0.3 (Self-similar coding interface)**.** *A finite iterated similarity/affine system with a Bernoulli or finite-state Markov coding produces such a graph-directed cocycle after the usual homogeneous embedding of its linear and translation data.*

PROOF. Each branch corresponds to one group element and each allowed transition to one edge. Composition of branches is multiplication in the affine/homogeneous semidirect product, so the coding cocycle is exactly the graph cocycle of Lemma 0.1. □

CHAPTER 3

Homogeneous Random Walks Driven by Expanding Measures

The stringent review isolated one genuine root: the uniformly-expanding-mod-Z dichotomy of Eskin–Lindenstrauss in the form used by Prohaska–Sert–Shi. This chapter records the precise specialization needed downstream and reconstructs its proof architecture. The recorded specialization is intentionally narrower than the full Eskin–Lindenstrauss theory: it states the dichotomy for a finite-volume homogeneous space, a connected inert subgroup Z, and the finite collection of exterior-power representations arising from homogeneous stabilizers in G/Λ.

Proof and provenance. The statement is the alternative quoted as Theorem 4.2 in Prohaska–Sert–Shi. Its two branches are: a positive-dimensional homogeneous barycentric decomposition, or Γ_μ-invariance with support contained in finitely many compact pieces of Z-orbits. The reconstruction below follows the proof architecture of the Eskin–Lindenstrauss argument: Furstenberg path disintegration, recurrence of nearby pairs, expansion transverse to the inert subgroup, a renormalized shearing limit, and induction on homogeneous dimension. No stronger stationary-measure theorem is used. [EL18; PSS23]

Lemma 0.1 (Path disintegration and recurrent transverse pairs)**.** *Let ν be ergodic and μ-stationary. On the two-sided path space there is a measurable family $b \mapsto \nu_b$ with $g_1\nu_{Tb} = \nu_b$ and $\nu = \int \nu_b \, d\beta(b)$. If, on a positive-measure set of paths, ν_b is not locally supported in a single Z-orbit, then for every sufficiently small $r > 0$ there is a positive-measure family of triples (b, x, y) with $x, y \in \operatorname{supp} \nu_b$, $d(x, y) < r$, and with the logarithmic displacement $v(x, y)$ having nonzero projection to $\mathfrak{g}/\mathfrak{z}$. Such triples recur infinitely often to a fixed compact set on which exponential coordinates and the Z-quotient are uniformly controlled.*

Proof. For $f \in C_c(G/\Lambda)$ the martingale

$$M_n(b) = \int f(g_n \cdots g_1 x) \, d\nu(x)$$

is bounded after truncating f, hence converges. Applying this simultaneously to a countable dense family of test functions gives the conditional measures ν_b and the cocycle identity. If almost every ν_b were locally contained in one Z-leaf, the second branch of Lemma 0.3 below would apply. Otherwise choose a compact coordinate box on which a positive proportion of the conditional measures meet two different Z-plaques. Non-atomic disintegration transverse to Z then supplies arbitrarily close such pairs. The finite first moment and the cusp Lyapunov function of Chapter 4 give recurrence to one compact box. Poincaré recurrence on the path skew product makes the close-pair configuration return infinitely often. Uniformity of the exponential chart on the chosen compact set gives the asserted logarithmic displacement. □

Lemma 0.2 (Transverse expansion produces an infinitesimal stabilizer)**.** *Assume the hypotheses of Lemma 0.1 and uniform expansion modulo Z. Then there are recurrent times n_j, pairs x_j, y_j as above, and normalizing scales $a_j \to \infty$ for which*

$$a_j\, \pi_{\mathfrak{g}/\mathfrak{z}}\Big(\mathrm{Ad}(g_{n_j}\cdots g_1)v(x_j,y_j)\Big) \longrightarrow w \neq 0.$$

The one-parameter subgroup generated by a nonzero lift of w preserves the limiting path measure. In particular the stabilizer of a positive-measure ergodic component has positive dimension outside the inert Z-direction.

Proof. Choose the first time at which the transverse component of the transported displacement has norm in a fixed annulus $[c, 2c]$. Uniform expansion modulo Z implies that this stopping time is finite almost surely and tends to infinity as the initial separation tends to zero. Compactness of the unit sphere in $\mathfrak{g}/\mathfrak{z}$ gives a convergent subsequence after rescaling. The Baker–Campbell–Hausdorff formula, uniformly on the recurrent compact box, gives

$$g_{n_j}\cdots g_1 y_j = \exp(w_j + o(1))\, g_{n_j}\cdots g_1 x_j, \qquad w_j \to w \neq 0.$$

The two points began in the same conditional measure ν_b. The cocycle identity for ν_b, recurrence, and weak compactness therefore imply that translation by $\exp(tw)$ fixes the limiting conditional measure for every rational t in a neighborhood of zero. Continuity extends invariance to all real t. The transverse projection of w is nonzero, so the resulting connected stabilizer is not contained in Z. □

Lemma 0.3 (Homogeneous enlargement versus the inert branch)**.** *Let N be a connected homogeneous stabilizer of maximal dimension among the path components produced by Lemma 0.2. Exactly one of the following occurs.*

(1) *The transverse-pair construction survives on the quotient by N, in which case it produces a connected homogeneous stabilizer strictly containing N.*
(2) *Every conditional path measure on the quotient is locally supported in a Z-orbit. Then ergodicity, compactness of the conjugation action on Z, and recurrence imply Γ_μ-invariance; moreover the quotient support is contained in finitely many compact Z-orbit pieces.*

Proof. Represent N by the line $[v_N]$ in a suitable exterior power and remove the sign ambiguity with $v_N \otimes v_N$. If two conditional points have distinct transverse coordinates modulo the N-orbit, Lemma 0.2 yields a nonzero infinitesimal stabilizer in the quotient. Lifting it and adjoining it to N increases the connected stabilizer dimension, proving the first alternative.

Suppose instead that no such pair exists. Then the conditional measures are supported on single Z-plaques modulo N. Because Γ_μ acts on Z through a compact subgroup of $\mathrm{Aut}(Z)$, plaque diameters cannot acquire exponential distortion. Return to a fixed compact box infinitely often. If infinitely many distinct plaque classes occurred there, two of them would approach each other and create a forbidden transverse pair. Hence only finitely many plaque classes occur in the compact return set. Stationarity and ergodicity propagate this finite family to the full support. The permutation action of Γ_μ on the finite family has a stationary probability; ergodicity makes it uniform on one orbit. On each plaque, the compact closure of the conjugation action allows averaging along the compact group, and the cocycle identity then gives invariance under every generator in $\operatorname{supp}\mu$. Thus the measure is Γ_μ-invariant and its support is a finite union of compact subsets of Z-orbits. $\square$

Lemma 0.4 (Eskin–Lindenstrauss uniformly-expanding-mod-Z dichotomy)**.** *Let $Z < G$ be connected and normalized by Γ_μ. Assume that the conjugation action of Γ_μ on Z has compact closure, that $\mathfrak{g} = \mathfrak{z} \oplus V$ is Γ_μ-invariant, and that μ is uniformly expanding on every nonzero vector of V occurring in the finite exterior-power family attached to homogeneous stabilizers of G/Λ. Let ν be an ergodic μ-stationary probability on G/Λ. Then either*

(1) *there are a positive-dimensional closed connected subgroup $N < G$, an N-homogeneous probability ν_0, and a μ-stationary probability η on the corresponding parameter quotient such that*

$$\nu = \int g_* \nu_0 \, d\eta(gN);$$

or

(2) *ν is Γ_μ-invariant and is supported on a finite union of compact subsets of Z-orbits.*

PROOF. Disintegrate over paths by Lemma 0.1. If close transverse pairs occur, Lemma 0.2 supplies a positive-dimensional connected stabilizer and disintegration along its closed finite-volume orbit gives the first alternative. Choose such a stabilizer of maximal dimension. Lemma 0.3 says that another transverse pair on the quotient would strictly enlarge it, contradicting maximality. Hence the quotient lies in the inert branch, which is precisely alternative (2). These alternatives are mutually exclusive after choosing N of maximal dimension: in the second branch no positive-dimensional transverse stabilizer can be generated. This proves the dichotomy in the exact scope needed by the PSS induction. □

THEOREM 0.5 (PSS maximal-dimension reduction). *Let $H < G$ be connected semisimple without compact factors and with finite center, let μ be H-expanding with finite first moment, and let ν be an ergodic μ-stationary probability on G/Λ. There is a connected closed subgroup $N < G$ such that ν is homogeneous along N, N has maximal possible dimension among homogeneous subgroups occurring in Lemma 0.4, N° is normalized by H, and the residual stationary measure on the quotient lies in the invariant branch. Consequently ν is homogeneous and its connected stabilizer is normalized by H.*

PROOF. Iterate Lemma 0.4. Each occurrence of the first alternative strictly increases the dimension of the connected homogeneous stabilizer, so the iteration stops after at most $\dim G$ steps. At the terminal stage the invariant branch holds. Lifting the finite orbit in the parameter quotient through the homogeneous disintegration shows that ν itself is homogeneous. For $h \in H$, the H-expansion property is unchanged after conjugating the stabilizer by h, so maximality applies equally to hNh^{-1}; connectedness and equal dimension force $hN^\circ h^{-1} = N^\circ$. [PSS23] □

Proposition 0.6 (Recovery of the BQ-type homogeneous conclusion)**.** *If Γ_μ is Zariski dense in H, every ergodic stationary probability in the scope of Theorem 0.5 is homogeneous, and the connected stabilizer is normalized by H; under the epimorphicity hypothesis of Chapter 6 the measure is Γ_μ-invariant.*

PROOF. Proposition 0.3 gives H-expansion. Theorem 0.5 gives homogeneity and normalization. The homogeneity and normalization conclusions are already complete here. The final Γ_μ-invariance upgrade is deliberately deferred until Chapter 6, where Lemma 0.1 proves the needed epimorphicity hypothesis; the component-space argument of Volume 35, Chapter 7 then performs the upgrade. □

CHAPTER 4

Recurrence

Lemma 0.1 (Contracting height from representation expansion)**.** *For every cusp height or homogeneous-stratum height built from the finite exterior-power representation family, H-expansion and a finite exponential moment give a block length m, $a < 1$ and $b < \infty$ with $P^m\beta \leq a\beta + b$ away from the singular set $\beta = \infty$.*

PROOF. Each height is a finite maximum/sum of negative powers of norms of representation vectors. Uniform expansion gives negative expectation for a sufficiently small inverse exponent on every nonfixed vector, exactly as in XXXIV.1. Finite exponential moments control the bad large-deviation event. Finitely many representations permit one common block length and exponent. The bounded chart and reduction-theory errors are absorbed into b. □

THEOREM 0.2 (Recurrence for expanding random walks). *Let Λ be a lattice and let μ be H-expanding with finite exponential moments. Away from any prescribed finite union of centralizer translates of a proper Γ_μ-ergodic homogeneous subspace, compact sets are uniformly recurrent: for every compact starting set and $\delta > 0$ there is a compact target set in the complement carrying probability at least $1 - \delta$ at all sufficiently large times.*

PROOF. Combine the cusp height and the finitely many stratum heights by a sufficiently separated weighted sum. Lemma 0.1 gives one Foster–Lyapunov inequality for that sum. Iteration plus Markov's inequality confines the walk to a common compact sublevel with probability $1 - \delta$. The infinite-height locus is exactly the excluded homogeneous union, so the compact sublevel lies in its complement. □

Proposition 0.3 (Nonescape of stationary limits)**.** *Every sequence of Cesàro distributions from a fixed compact set is tight, and every weak limit gives zero mass to proper excluded homogeneous strata unless the starting orbit is trapped in them.*

PROOF. Tightness is the $Y = \varnothing$ case of Theorem 0.2. Applying the theorem to a proper stratum and shrinking its neighborhoods shows the limiting mass there is zero unless the trajectory is contained in that invariant stratum. $\square$

CHAPTER 5

Repulsion from Proper Homogeneous Subsets

Lemma 0.1 (Singular-height blowup)**.** *For a proper homogeneous subspace Y, the associated Chevalley height β_Y tends to infinity as x approaches Y transversely while remaining in a fixed compact ambient set.*

Proof. Choose the exterior-power Chevalley vector v_Y whose stabilizer is the normalizer of the connected stabilizer of Y. In a compact chart transverse displacement is measured by the component of gv_Y normal to the stabilizer orbit. The height is a negative power of this component norm, so it diverges exactly when the transverse component tends to zero. □

Theorem 0.2 (Uniform repulsion). *Under the assumptions of XXXIX.4, if a compact set Z is disjoint from a finite centralizer-saturation of a proper homogeneous Y, then for every $\epsilon > 0$ there is a neighborhood U of that saturation such that*

$$\sup_{x \in Z} \frac{1}{n} \sum_{k<n} \mu^{*k} * \delta_x(U) < \epsilon$$

for all sufficiently large n.

Proof. Use the contracting height β_Y and choose R so large that $\{\beta_Y > R\} \subset U$. Iterating $P^m \beta_Y \leq a\beta_Y + b$ bounds the Cesàro average of β_Y uniformly. Markov's inequality gives an $O(R^{-1})$ bound for the average occupation of $\{\beta_Y > R\}$. Choose R large enough and handle the finitely many initial times separately. □

Proposition 0.3 (Countable-strata diagonalization)**.** *If the relevant homogeneous strata form a countable family modulo centralizer, one can choose a sequence of neighborhoods so that almost every trajectory starting outside their union spends zero asymptotic frequency in every proper stratum simultaneously.*

Proof. Enumerate the strata. Apply Theorem 0.2 with error 2^{-j-r} to the j-th stratum and the r-th compact exhaustion. Markov plus Borel–Cantelli along a diagonal subsequence gives the almost-sure zero-frequency statement for every pair (j, r), hence for all strata. □

CHAPTER 6

Stationary-Measure Classification

Proof and provenance. The target statement is Prohaska–Sert–Shi Theorem 1.1. Its hypotheses are: $\Lambda < G$ discrete; $H < G$ connected semisimple without compact factors and with finite center; and μ an H-expanding probability with finite first moment. Its proof uses the Eskin–Lindenstrauss dichotomy isolated as Lemma 0.4. Thus the theorem below is source-faithful but remains downstream-blocked until that root is internalized. [PSS23]

Lemma 0.1 (Epimorphicity of the support group)**.** *If μ is H-expanding, then every vector fixed by Γ_μ in every finite-dimensional real representation of H is fixed by H. Equivalently, Γ_μ is epimorphic in H for the finite-dimensional representation theory used in the classification argument.*

Proof. Let $\rho : H \to \mathrm{GL}(V)$ and suppose $v \in V$ is fixed by Γ_μ. Decompose $V = V^H \oplus W$ using an H-invariant complement after passing, if necessary, to the semisimplification of the representation. Write $v = v_0 + w$ with $v_0 \in V^H$ and $w \in W$. If $w \neq 0$, then every random product $g_n \cdots g_1$ drawn from μ fixes w, so

$$\|\rho(g_n \cdots g_1)w\| = \|w\| \qquad \text{for every } n.$$

But the definition of H-expansion says that every nonzero vector in a quotient representation with no H-fixed vector has strictly positive top Lyapunov growth. Applied to the H-module generated by w modulo its H-fixed part, this contradicts the displayed equality. Hence $w = 0$ and $v = v_0$ is H-fixed. $\square$

Theorem 0.2 (Prohaska–Sert–Shi stationary-measure rigidity). *Let Λ be a discrete subgroup of a real Lie group G. Let $H < G$ be connected semisimple without compact factors and with finite center. If μ is H-expanding with finite first moment, every ergodic μ-stationary probability ν on G/Λ is Γ_μ-invariant and homogeneous. Moreover $\mathrm{Stab}_G(\nu)^\circ$ is normalized by H.*

Proof. Theorem 0.5 gives a homogeneous description

$$\nu = g_* m_{L/(L \cap g^{-1}\Lambda g)}$$

for a closed connected subgroup L, and proves that $L = \mathrm{Stab}_G(\nu)^\circ$ is normalized by H.

It remains to pass from stationarity to invariance under the support semigroup. The action of Γ_μ on the finite component space of the homogeneous support yields a finite Markov chain whose stationary law is induced by ν. Lemma 0.1 rules out a nontrivial continuous quotient representation on which Γ_μ acts trivially but H does not. Consequently the finite component action preserves its stationary component law. Inside each connected component the probability is Haar; uniqueness of Haar probability on a finite-volume homogeneous orbit implies that every generator in $\operatorname{supp}\mu$ preserves the component measure. Hence every element of Γ_μ preserves ν.

All steps after the Eskin–Lindenstrauss dichotomy are internal to Chapters 1–6; the conditional status of the theorem is solely the status of Lemma 0.4. $\square$

Proposition 0.3 (Irreducible ambient corollary)**.** *in an irreducible semisimple finite-volume ambient space the theorem reduces stationary probabilities to Haar and the explicitly allowed finite homogeneous orbits whenever H is a positive-dimensional normal subgroup and no proper positive-dimensional normal subgroup has a finite-volume orbit.*

Proof. Theorem 0.2 makes the ergodic stationary measure homogeneous and makes its connected stabilizer L normalized by H. Under the stated irreducibility and normal-subgroup hypothesis, a positive-dimensional L with finite-volume orbit must be the full ambient connected group. Then the homogeneous probability is Haar. If $\dim L = 0$, the support is a finite homogeneous orbit. These are exactly the two alternatives asserted. $\square$

CHAPTER 7

Trajectory Equidistribution

Proof and provenance. For comparison, Prohaska–Sert–Shi prove recurrence, repulsion and orbit-equidistribution for expanding measures under finite exponential-moment hypotheses. In the present text the required stationary-classification input is supplied by Lemma 0.4 and the backward results cited below; the reference records provenance and scope only. [PSS23]

Lemma 0.1 (Uniqueness of the nontrapped stationary target)**.** *let x avoid every proper homogeneous orbit closure allowed by Theorem 0.2. Then every stationary weak limit of Cesàro distributions from x is Haar on the minimal homogeneous closure generated by the walk.*

PROOF. Take a stationary weak limit σ. By Theorem 0.2, each ergodic component of σ is homogeneous. The repulsion theorem of Chapter 5 gives zero mass to every proper homogeneous stratum that does not contain the starting orbit. The relevant homogeneous strata are countable after fixing their lattice data, as in XXXVI.4. Therefore all ergodic components of σ are supported on the same maximal homogeneous closure Y_x. That orbit admits a unique invariant homogeneous probability m_{Y_x}, so $\sigma = m_{Y_x}$. □

THEOREM 0.2 (Equidistribution for expanding random walks). *assume Λ is a lattice and μ is H-expanding with finite exponential moment. For every starting point x, the Cesàro distributions converge to the homogeneous probability on the minimal homogeneous orbit closure Y_x generated by the walk. For almost every sample path, the empirical trajectory measures converge to the same probability. If the walk is trapped in a proper homogeneous subset, the same assertion holds inside its minimal homogeneous closure.*

PROOF. Chapter 4 gives tightness of the Cesàro distributions. Hence every subsequence has a weakly convergent further subsequence, and the elementary telescoping identity

$$\mu * \Big(\frac{1}{n}\sum_{k=0}^{n-1} \mu^{*k} * \delta_x\Big) - \frac{1}{n}\sum_{k=0}^{n-1} \mu^{*k} * \delta_x = \frac{\mu^{*n} * \delta_x - \delta_x}{n}$$

shows that every such limit is stationary. Lemma 0.1 identifies every limit with m_{Y_x}, so the whole Cesàro sequence converges.

For sample paths, apply the martingale-difference argument of XXXVII.3 to a countable dense family of bounded continuous test functions. Any empirical subsequential limit is stationary. Chapter 5's pathwise repulsion estimate gives zero mass to proper homogeneous strata not containing the path's minimal closure. Lemma 0.1 therefore again forces every subsequential limit to be m_{Y_x}. Tightness and uniqueness of subsequential limits give almost-sure empirical convergence. □

Proposition 0.3 (Orbit-closure consequence)**.** *the topological closure of an expanding random-walk orbit agrees with the support of the homogeneous target of Theorem 0.2.*

Proof. The target support is contained in the orbit closure by construction. Conversely let U be a nonempty relatively open subset of the orbit closure. Some orbit point enters U. Restarting the walk at that point leaves the same minimal homogeneous closure, and Theorem 0.2 gives a limiting homogeneous probability with full support on that closure. Hence every nonempty relatively open subset has positive limiting occupation and meets the target support. Thus the orbit closure is contained in the target support. □

CHAPTER 8

Applications to Fractal Measures

The purpose of this chapter is to isolate the coding transfer that will later feed Volume 41. The transfer lemma itself is elementary and unconditional. The genericity theorem is conditional only because its homogeneous equidistribution input is downstream of the Eskin–Lindenstrauss root isolated in Chapter 3.

Lemma 0.1 (Coding average to random-walk empirical average)**.** *For a Bernoulli or finite-state graph-directed coding whose branch maps give the group increments of Chapter 2, suppose the coding section and the homogeneous cross-section differ by a bounded transfer function B. Then for every bounded continuous homogeneous observable F the first n coding iterates and the first n random-walk iterates have averages differing by*

$$\frac{B(\omega) - B(T^n\omega)}{n}.$$

In particular the difference tends to zero uniformly on every set on which B is bounded.

Proof. Let $g_{\omega_1}, g_{\omega_2}, \dots$ be the branch increments and $G_k = g_{\omega_k} \cdots g_{\omega_1}$. By the definition of the two sections, the coding observable at time k can be written

$$F(G_k x) + B(T^k\omega) - B(T^{k+1}\omega).$$

Summing for $k = 0, \dots, n-1$ telescopes the transfer terms and leaves exactly

$$\sum_{k=0}^{n-1} F(G_k x) + B(\omega) - B(T^n\omega).$$

Divide by n. The asserted convergence follows from boundedness of B. $\square$

Theorem 0.2 (Genericity transfer for expanding self-similar measures). *suppose a self-similar or graph-directed measure admits the expanding homogeneous coding of Chapter 2, the associated increment law satisfies the hypotheses of Theorem 0.2, and its minimal homogeneous target is Haar on Y. Then for almost every coded point and every continuous observable pulled* *back from the homogeneous cross-section, the Birkhoff average equals the Haar average on Y.*

Proof. For almost every Bernoulli or finite-state Markov coding, the increment sequence is typical for the driving law. Theorem 0.2 therefore gives

$$\frac{1}{n}\sum_{k=0}^{n-1} F(G_k x) \longrightarrow \int_Y F\, dm_Y$$

for every continuous F in a countable dense determining family, and then for all continuous F by uniform approximation on compacta together with recurrence. Lemma 0.1 changes the coding average by a telescoping $o(1)$ term. Hence the coding Birkhoff average has the same Haar limit. □

Proposition 0.3 (Exact scope of the fractal consequence)**.** *Theorem 0.2 supplies the homogeneous genericity input used in Diophantine applications to expanding self-similar and graph-directed measures. Khintchine-type laws, badly-approximable conclusions and shrinking-target statements are not consequences of continuous-observable genericity alone and are deferred to Volume 41.*

Proof. Theorem 0.2 controls bounded continuous observables. Shrinking-target and Diophantine assertions involve indicator functions of moving cusp neighborhoods or other unbounded/noncontinuous observables. Passing from continuous genericity to such statements requires quantitative recurrence, Borel–Cantelli estimates or the Dani correspondence, none of which is contained in the coding lemma. Therefore the stated boundary is exact rather than expository. [SW19; PSS23] □

Volume 40

Markov-Dependent and Effective Random Walks

Volume 40 scope and prerequisites

This volume has two layers. Chapters 1–3 prove the renewal/skew-product reduction for finite-state irreducible Markov driving, in the scope of Prohaska–Sert. Chapters 4–8 develop the effective dimension-growth mechanism in the rank-one simple spaces G/Λ with G locally isomorphic to $\mathrm{SO}(2,1)$ or $\mathrm{SO}(3,1)$; the later general-simple-group theorem is recorded as a strengthening, not used as a hidden input.

Proof and provenance. Primary comparison: Prohaska–Sert for Markov renewal and joint equidistribution; Bénard–He for multislicing and effective equidistribution in the rank-one simple spaces. The 2025 general-simple-group Bénard–He theorem is bibliographic context only in this volume. Stronger variants are bibliographic context unless separately marked.

Reader guide: a concrete model

Why this matters.

Volume 40 is organized around the passage from *Markov-dependent matrix products* to *Polynomial/exponential rate theorems.* The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.4 (A calculation to keep in view)**.** Let $P(t) = a+bt+ct^2$ with $c \neq 0$. At scale T write $t = Ts$. Then

$$P(Ts) = a + (bT)s + (cT^2)s^2.$$

If $|c|T^2 \asymp 1$ while $|b|T \to 0$ and $|a| \to 0$, the first visible rescaled polynomial is cs^2T^2. On a mesoscopic window $s = s_0 + \ell/T$ with $|\ell| = o(T)$,

$$P(Ts) - P(Ts_0) = P'(Ts_0)\ell + c\ell^2 = o(1)$$

whenever the normalized derivative is controlled. This is the toy calculation behind the separation of a macroscopic visibility scale from a mesoscopic almost-constant scale in effective linearization and closing.

CHAPTER 1

Markov-dependent matrix products

This chapter isolates the markov-dependent matrix products mechanism needed by the later rigidity arguments. Every statement below records its hypotheses explicitly; later applications may use only that stated scope.

Lemma 0.1 (Markov block decomposition)**.** *Let E be a finite irreducible Markov state space with stationary law π, and let each admissible edge $e = (i, j)$ carry $g_e \in G$. For a recurrent state o, the product accumulated between successive returns to o is an i.i.d. G-valued sequence with law μ_o, and its first logarithmic moment is finite whenever the edge cocycle has finite logarithmic moment.*

Proof. Use the strong Markov property at successive return times τ_k to o. The edge blocks between τ_k and τ_{k+1} are independent and identically distributed under the chain started at o. Multiplication of the edge labels inside a block defines $Y_k = g_{\tau_{k+1}} \cdots g_{\tau_k+1}$. Kac's formula gives $\mathbb{E}_o(\tau_1) = 1/\pi(o)$, so the triangle inequality for $\log \mathrm{N}(\cdot)$ transfers the moment bound from edges to Y_k. □

Theorem 0.2 (Skew-product law of large numbers for a finite-state matrix cocycle). *For every finite irreducible Markov cocycle, the random matrix product sampled at return times is an ordinary i.i.d. product. Statements about stationary measures, Lyapunov exponents, and homogeneous-space equidistribution for the return law therefore lift to the original Markov walk after controlling the return-time clock.*

Proof. Apply the lemma at o. The law of large numbers gives $\tau_k/k \to 1/\pi(o)$. Between two return times there are only $\tau_{k+1} - \tau_k$ edge steps; their contribution to any Cesàro average is negligible after division by τ_k. Thus an i.i.d. theorem for μ_o transfers to the Markov chain, with time rescaled by $\pi(o)$. □

Proposition 0.3 (Return-block transfer)**.** *The same reduction works for a strongly connected finite directed graph with edge probabilities bounded away from zero after replacing E by the directed-edge chain.*

Proof. The edge chain is again finite irreducible. Its return blocks concatenate exactly the original edge labels, so the preceding construction applies verbatim. □

Worked Example 0.4 (A two-state Markov product that can be computed by hand)**.** Let the state space be $\{0, 1\}$ with transition matrix

$$P = \begin{pmatrix} 3/4 & 1/4 \\ 1/2 & 1/2 \end{pmatrix}.$$

The stationary law solves $\pi P = \pi$, hence $\pi = (2/3, 1/3)$. Attach scalar "matrices" $A_0 = 2$ and $A_1 = 1/2$. Along a path $X_0, X_1, \ldots$ the logarithm of the product is

$$\log |A_{X_{n-1}} \cdots A_{X_0}| = (N_0(n) - N_1(n)) \log 2.$$

The Markov ergodic theorem gives $N_0(n)/n \to 2/3$ and $N_1(n)/n \to 1/3$, so the Lyapunov exponent is $(1/3) \log 2$. This calculation is the scalar model for the return-block decomposition used in the chapter: dependence changes the frequencies, but not the additive cocycle structure.

CHAPTER 2

Stationary Markov chains and skew products

Lemma 0.1 (Stationary path-space construction)**.** *Let $P = (p_{ij})$ be irreducible with stationary law π and let $X = G/\Lambda$. The skew-product kernel on $E \times X$ is $\mathcal{P}((i,x), d(j,y)) = p_{ij}\, \delta_{g_{ij}x}(dy)$. A family $(\nu_i)_{i\in E}$ is stationary iff $\pi(j)\nu_j = \sum_i \pi(i) p_{ij} (g_{ij})_* \nu_i$ for every j.*

Proof. Test stationarity against a function supported on one state j. The displayed balance identity is necessary. Summing the identities against arbitrary test functions on X proves sufficiency. No reversibility is required. $\square$

Theorem 0.2 (Markov skew-product invariance). *If the return law at one recurrent state has a unique stationary probability m_X and every edge map preserves m_X, then the skew product has the unique stationary probability $\sum_i \pi(i)\delta_i \otimes m_X$ among families whose support communicates with that state.*

Proof. Disintegrate a stationary probability over E. The balance equations and a first-return decomposition show that ν_o is stationary for the return law. Uniqueness gives $\nu_o = m_X$. Propagate this identity along positive-probability paths from o to every state; invariance of m_X under G gives $\nu_i = m_X$. $\square$

Proposition 0.3 (Observable transfer to the skew product)**.** *Joint equidistribution of the skew product implies equidistribution of the homogeneous coordinate and the correct asymptotic frequency $\pi(i)$ of every Markov state.*

Proof. Apply the ergodic theorem to functions $1_{\{i\}}(e)f(x)$ and then sum over i. $\square$

Worked Example 0.4 (Checking stationarity on two cylinders)**.** For the transition matrix above and $\pi = (2/3, 1/3)$, the cylinder probabilities are

$$\mathbb{P}[X_0 = 0, X_1 = 1] = \frac{2}{3}\frac{1}{4} = \frac{1}{6}, \qquad \mathbb{P}[X_0 = 1, X_1 = 0] = \frac{1}{3}\frac{1}{2} = \frac{1}{6}.$$

After shifting the path one step, the probability that the new initial state is 0 is

$$\frac{2}{3}\frac{3}{4} + \frac{1}{3}\frac{1}{2} = \frac{2}{3} = \pi_0.$$

Thus the path law is shift invariant. If a cocycle multiplies the fiber by A_{X_0} before shifting, the same computation shows why the base marginal remains stationary while the fiber records the accumulated matrix product.

CHAPTER 3

Lyapunov exponents for Markov cocycles

Lemma 0.1 (Subadditivity of Markov cocycle norms)**.** *For a finite-state Markov cocycle with* $\int \log^+ \|g_e^{\pm 1}\| < \infty$*, the subadditive process* $F_n = \log \|g_n \cdots g_1\|$ *has an almost-sure limit* $\lambda_1^{\mathrm{M}} = \lim n^{-1} F_n$*. At a return state* o*,* $\lambda_1^{\mathrm{M}} = \pi(o)\lambda_1(\mu_o)$*.*

Proof. Kingman's theorem applies to the stationary path shift. Along return times, the product is the i.i.d. return product and Oseledets/Kingman gives $k^{-1} \log \|Y_k \cdots Y_1\| \to \lambda_1(\mu_o)$. Since $k/\tau_k \to \pi(o)$, evaluating the full limit on the subsequence τ_k gives the formula; the integrable overshoot between returns does not change the limit. □

Theorem 0.2 (Existence of the top Markov Lyapunov exponent). *Strong irreducibility and proximality of the return semigroup imply positivity and simplicity of the top Markov Lyapunov exponent, with the same stationary projective measure as the return product viewed at return times.*

Proof. Apply Volume 28 to the return law. The clock identity in the lemma multiplies every Lyapunov exponent by the positive factor $\pi(o)$ and hence preserves strict inequalities. Projective stationary measures lift by transporting them along finite Markov paths. □

Proposition 0.3 (Return-block formula for the exponent)**.** *The exterior-power exponents satisfy* $\lambda_1^{\mathrm{M}} + \cdots + \lambda_k^{\mathrm{M}} = \pi(o)\lambda_1(\wedge^k \mu_o)$*.*

Proof. Apply the same argument to the induced cocycle on $\bigwedge^k V$. □

Worked Example 0.4 (Top exponent for a scalar Markov cocycle)**.** With $A_0 = 3$ and $A_1 = 1/3$ and stationary frequencies $\pi = (2/3, 1/3)$,

$$\frac{1}{n} \log |A_{X_{n-1}} \cdots A_{X_0}| = \left(\frac{N_0(n) - N_1(n)}{n} \right) \log 3 \longrightarrow \frac{1}{3} \log 3.$$

The point is not the scalar answer but the mechanism: logarithms turn products into an additive observable of the stationary Markov chain. In higher dimension one replaces the scalar logarithm by the subadditive sequence $\log \|A_{X_{n-1}} \cdots A_{X_0}\|$ and applies the subadditive theorem.

CHAPTER 4

Effective flattening

Lemma 0.1 (One-step L^2 flattening alternative)**.** *On a compact coordinate chart of a rank-one simple homogeneous space, if a probability η has scale-ρ L^2 energy larger than $\rho^{-s-\epsilon}$ and the driving law has uniform projective non-concentration, then convolution by a fixed block of the walk either lowers the L^2 energy by a power of ρ or places a positive mass of η inside a $\rho^{1-O(\epsilon)}$ neighborhood of a proper algebraic obstruction.*

Proof. Smooth η at scale ρ and apply the L^2 identity $\|\mu^{*m} * \eta\|_2^2 = \iint \langle g_*\eta, h_*\eta\rangle \, d\mu^{*m}(g) d\mu^{*m}(h)$. Large correlation means that many relative maps $h^{-1}g$ almost preserve the concentration set. The product theorem/escape machinery of Volumes 30–34 turns such a family into a proper stabilizer unless the correlations occupy many transverse directions. In the latter case Cauchy–Schwarz over disjoint transverse tubes gives the asserted power saving. □

Theorem 0.2 (Multiscale robustness after repeated flattening). *Away from the cusp and finite invariant orbits, repeated flattening turns every positive-dimensional scale distribution into a robust measure: at all scales $\rho \leq r \leq \rho^c$ it obeys a uniform Frostman upper bound of exponent $s+\delta$.*

Proof. Iterate the lemma on dyadic scales. Recurrence from Volume 33 prevents loss of mass to the cusp; the finite-orbit distance term removes the algebraic obstruction. Each unsuccessful step would improve the energy exponent by a fixed amount, so after boundedly many steps the robust alternative holds. □

Proposition 0.3 (Return-law transfer of a verified flattening estimate)**.** *Let τ be the return time to a fixed state of an irreducible finite-state Markov chain and assume*

$$\mathbb{P}(\tau > m) \leq C_0 e^{-c_0 m}.$$

Suppose the i.i.d. return-block law satisfies the flattening conclusion of Theorem 0.2 after k complete blocks, with exceptional probability at most E_k. Then by physical time n, the Markov walk satisfies the same conclusion *outside an exceptional set bounded by*

$$E_{\lfloor \lambda n \rfloor} + C_1 e^{-c_1 n}, \tag{40.4.1}$$

for any fixed $0 < \lambda < (\mathbb{E}\tau)^{-1}$*, with* $C_1, c_1 > 0$ *depending on the return-time exponential moment.*

PROOF. Let $N(n)$ be the number of complete return blocks finished by time n. Standard exponential large deviations for the sum of i.i.d. return times, available here because τ has an exponential tail, give

$$\mathbb{P}\big(N(n) < \lambda n\big) \le C_1 e^{-c_1 n}.$$

On the complementary event at least $\lfloor \lambda n \rfloor$ complete return blocks have occurred. Condition on those blocks and apply Theorem 0.2 to the induced i.i.d. return-block walk, which contributes the exceptional probability $E_{\lfloor \lambda n \rfloor}$. The unfinished terminal block changes only the bounded number of post-processing steps built into the return-law model and is absorbed into the fixed constants. Taking the union of the two exceptional events gives (40.4.1). $\square$

Worked Example 0.4 (Why transverse translates lower L^2 energy)**.** Take on $\mathbb{R}$ the normalized indicator $f_\rho = \rho^{-1} 1_{[0,\rho]}$. Then $\|f_\rho\|_2^2 = \rho^{-1}$. Average two translates separated by 2ρ:

$$g = \frac{1}{2}\big(f_\rho + f_\rho(\,\cdot - 2\rho)\big).$$

Their supports are disjoint, so

$$\|g\|_2^2 = \frac{1}{4}(\|f_\rho\|_2^2 + \|f_\rho\|_2^2) = \frac{1}{2\rho}.$$

The energy drops by a factor 2. Effective flattening replaces this toy disjointness by quantitative transversality: if many translates fail to lower energy, their large overlaps reveal an approximate stabilizer, hence an algebraic obstruction.

CHAPTER 5

Multislicing

Lemma 0.1 (Fubini slicing identity). *Let $A \subset B(0,1) \subset \mathbb{R}^d$ be ρ-separated with a Frostman covering bound and let a probability σ on a C^2 family of codimension-k slices satisfy uniform non-concentration on Schubert-type degeneracy sets. Then, outside a ρ^c exceptional set of slice parameters, the average ρ-covering number of the intersections $A \cap S$ is at most the value forced by dimension conservation, with a power saving whenever the projected dimension is subcritical.*

Proof. We give the scale-selection argument because it is the part that is lost if one cites a projection theorem as a black box. Put $\rho_j = \rho^{2^{-j}}$ and $H_j(B) = \log N_{\rho_j}(B)$. For a bounded set, the telescoping sum of the increments $H_{j+1}(B) - H_j(B)$ is at most $O(|\log \rho|)$. Hence, after discarding $O(\epsilon^{-1})$ exceptional scales, there is a block of consecutive scales on which the increments of A differ from their average by at most $O(\epsilon|\log \rho|)$.

Fix such a scale. A slice parameter is bad if one of the minors measuring transversality of the slice map is $< \rho^{c\epsilon}$. The assumed non-concentration of σ on the corresponding Schubert varieties shows that the bad parameters have σ-mass $O(\rho^{c'\epsilon})$. For every good parameter, the inverse-function theorem with quantitative constants gives a bounded-overlap product chart in which a ball is comparable to a product of a projected ball and a slice ball.

In that product chart, covering numbers satisfy the entropy inequality

$$H_j(A) \geq H_j(\pi_S A) + \int H_j(A \cap S_z)\, d\theta_S(z) - O(\epsilon|\log \rho|).$$

This is proved by choosing a maximal separated family in the projection and, over each projected cell, a maximal separated family in the corresponding fiber; bounded overlap is the only loss. Integrating in the parameter and using submodularity permits the same scale to be used for a positive proportion of the slices.

If the projected dimension were subcritical and the typical slice were larger than the dimension-conservation value by a fixed δ, the right side would exceed the Frostman upper bound for $H_j(A)$ by $\delta|\log \rho|$ once $\epsilon \ll \delta$. This contradiction proves the claimed power saving. The supercritical statement is obtained by applying the same entropy inequality to complementary fibers. Thus the proof uses only discretization, the non-concentration of the minors, and the covering-number submodularity recorded above. ☐

THEOREM 0.2 (Transverse multislice estimate). *For G locally isomorphic to* $\mathrm{SO}(2,1)$ *or* $\mathrm{SO}(3,1)$*, the nonlinear chart maps generated by* $\mathrm{Ad}(g)$ *satisfy the hypotheses of the multislicing lemma after removing a power-small set of g drawn from a Zariski-dense law.*

PROOF. The degeneracy conditions are vanishing of finitely many analytic minors. Zariski density and the projective non-concentration estimates of Volumes 28–32 give a power bound for their small neighborhoods. Bounded geometry on a recurrence compact set supplies uniform C^2 norms. ☐

Proposition 0.3 (Slicing-to-incidence transfer)**.** *Multislicing may therefore be inserted into the energy-flattening argument without any global linear coordinate system on G/Λ.*

PROOF. Cover the recurrence compact set by finitely many bounded-geometry charts and sum the chartwise estimates. ☐

Worked Example 0.4 (Slicing the unit square)**.** Let $E = [0,1]^2$. Its vertical slice at x is $E_x = [0,1]$, so

$$|E| = \int_0^1 |E_x|\,dx = 1.$$

Now replace E by the union of N disjoint squares of side ρ. The integral of the slice lengths is exactly $N\rho^2$. If many different projection directions all have unusually large slice mass, the same pieces must participate in many incidences. The multislicing argument quantifies the incompatibility between that incidence count and transverse non-concentration.

CHAPTER 6

Dimension growth

Lemma 0.1 (Scale entropy increment lemma)**.** *Define the scale dimension* $\dim_\rho(\eta) = \log \mathcal{E}_\rho(\eta)^{-1}/\log \rho^{-1}$ *using smoothed* L^2 *energy. In the rank-one simple scope, convolution by a bounded random-walk block satisfies: either* $\dim_\rho(\mu^{*m} * \eta) \geq \dim_\rho(\eta) + \delta$*, or* η *already has dimension* $\dim X - O(\epsilon)$*, or* η *is power-close to a finite invariant orbit.*

Proof. Write $E_\rho(\eta)$ for the smoothed L^2 energy and $s_\rho(\eta) = -\log E_\rho(\eta)/\log \rho$. Suppose first that $s_\rho(\eta) \leq \dim X - 2\epsilon$. If the correlations in the L^2 identity of Chapter 4 are small, convolution lowers E_ρ by a factor ρ^δ and therefore raises s_ρ by δ. If the correlations are large, the approximate stabilizer extracted there is tested against the multislicing theorem. Unless it lies in a proper algebraic obstruction, multislicing again gives a ρ^δ energy saving.

Now suppose $\dim X - 2\epsilon < s_\rho(\eta) < \dim X - \epsilon$. A direct low-dimensional projection no longer gives room for growth. Disintegrate in a good product chart into projected measures and transverse fibers. The entropy submodularity from Chapter 5 implies that if neither factor gains dimension, then the sum of their dimensions is $< s_\rho(\eta) - c\epsilon$, contradicting the original energy bound. Thus one of the factors gains, and after recombination so does $\mu^{*m} * \eta$.

Finally, if $s_\rho(\eta) \geq \dim X - \epsilon$, smoothing at scale ρ gives $\|\eta_\rho\|_2 \leq \rho^{-O(\epsilon)}$. Young's inequality shows that convolution cannot improve the exponent by a fixed amount independent of ϵ; this is precisely the near-full-dimensional terminal alternative. The only point at which the preceding two arguments can fail is the algebraic concentration alternative of Chapter 4, which is the stated finite-orbit obstruction in the present rank-one scope. ☐

Theorem 0.2 (Dimension-growth dichotomy). *Starting from a point outside the finite-orbit obstruction, recurrence plus positive-dimension production and the dimension-growth alternative yield a nearly full-dimensional distribution after* $O(|\log \rho|)$ *walk steps.*

Proof. Volume 34 supplies the first positive transverse dimension. Iterate the lemma; each nonterminal iteration gains δ, so only $O(\dim X/\delta)$

dimension stages are needed. The scale changes geometrically, hence the total walk time is $O(|\log \rho|)$. $\square$

Proposition 0.3 (Dimension growth for a finite-state Markov return law)**.** *Let (Z_n) be a finite-state Markov walk and fix a recurrent state o. Assume the return law μ_o satisfies Lemma 0.1 with increment $\delta > 0$, and assume the return time τ_o has an exponential tail. Then for every $\epsilon > 0$ there is C_ϵ such that, outside the finite-orbit obstruction, the law of Z_n has scale dimension at least* $\dim X - \epsilon$ *at scale ρ once*

$$n \geq C_\epsilon |\log \rho|.$$

Proof. At return times to o, Lemma 0.1 raises scale dimension by at least δ until the $\dim X - \epsilon$ alternative is reached. Hence at most $\lceil \dim X/\delta \rceil$ dimension stages are required. The renewal reduction of Chapters 1–3 identifies these stages with convolutions by μ_o, while the exponential tail of τ_o makes the number of physical steps needed for k returns linear in k up to exponentially small error. Since the scale changes geometrically at each stage, the total physical time is $O_\epsilon(|\log \rho|)$. $\square$

Worked Example 0.4 (A visible entropy gain)**.** Put equal mass on the four points $\{0,1\}^2$ and project to the first coordinate. The original measure has Shannon entropy $\log 4$, while the projection has entropy $\log 2$. If an independent transverse coordinate is then revealed, the missing $\log 2$ is recovered. Multiscale dimension growth is the continuous analogue: unless the measure is trapped near an algebraic family, a transverse operation contributes a definite amount of scale entropy, and only finitely many such gains are possible before the ambient dimension is reached.

CHAPTER 7

Effective equidistribution

Lemma 0.1 (Robustness-to-mixing estimate)**.** *Let G be locally isomorphic to* $\mathrm{SO}(2,1)$ *or* $\mathrm{SO}(3,1)$*, $\Lambda < G$ a lattice, and μ a compactly supported Zariski-dense probability satisfying the arithmetic non-concentration hypotheses of this volume. If x is not in a finite Γ_μ-orbit, then $\mu^{*n} * \delta_x \to m_X$; on arithmetic data the convergence is exponential after multiplying the error by explicit polynomial penalties for cusp height and proximity to finite orbits.*

Proof. Fix a smooth test function f with zero Haar mean. Let $\rho = e^{-cn}$, where $c > 0$ will be chosen after all losses have been displayed. By recurrence, except for an error bounded by the cusp-height term, the walk spends the first n_0 steps in a fixed bounded-geometry compact set. Chapters 4–6 then show that for some $n_1 \le C_1 |\log \rho|$ the law $\eta = \mu^{*n_1} * \delta_x$ has scale dimension at least $\dim X - \epsilon$, unless x is power-close to a finite invariant orbit.

Convolve η with a unit-mass bump ψ_ρ in local charts. The near-full-dimensional energy bound gives

$$\|\eta * \psi_\rho\|_2 \ll \rho^{-C_2\epsilon}, \qquad \mathcal{S}_D(\eta * \psi_\rho) \ll \rho^{-A-C_2\epsilon}.$$

Write the remaining walk length as $n - n_1$. The spectral gap of the averaging operator on the mean-zero subspace gives

$$|\langle \mu^{*(n-n_1)} * (\eta * \psi_\rho), f\rangle| \ll e^{-\kappa(n-n_1)} \rho^{-A-C_2\epsilon} \mathcal{S}_D(f).$$

Replacing η by its smoothing costs $O(\rho^b \mathcal{S}_{D+1}(f))$ on the recurrence compact set. The cusp part is controlled by the Lyapunov function of Volume 33 and the finite-orbit part is exactly the declared obstruction.

Choose first ϵ small and then c so that $\kappa(1 - C_1 c) > c(A + C_2\epsilon)$. Both the spectral term and the smoothing term are then exponentially small. This proves actual n-step convergence, not merely Cesàro convergence. If arithmetic quantitative non-concentration is unavailable, the same argument with qualitative dimension production and a diagonal sequence $\rho \to 0$ gives weak convergence without a rate. $\square$

Theorem 0.2 (Effective convergence of the Markov walk). *For a finite-state Markov walk whose return law satisfies the preceding hypotheses, the* *homogeneous coordinate equidistributes and the effective rate is reduced only by the return-time large-deviation exponent.*

Proof. Fix a reference state o and let τ_k be the time of the kth return to o. The successive return blocks are governed by the return law μ_o, so Lemma 0.1 gives constants $C_0, c_0 > 0$ such that the homogeneous coordinate at the return times satisfies

$$\left|\mathbb{E}_x f(Z_{\tau_k})\right| \leq C_0 e^{-c_0 k} \mathcal{S}_D(f)$$

with the declared cusp-height and finite-orbit penalties. The finite-state large-deviation estimate gives constants $C_1, c_1 > 0$ for which

$$\mathbb{P}\left(\left|\tau_k - \frac{k}{\pi(o)}\right| > \epsilon k\right) \leq C_1 e^{-c_1 k}.$$

Thus, outside an exponentially small event, physical time n contains a number of completed return blocks between $(\pi(o) - \epsilon)n$ and $(\pi(o) + \epsilon)n$. Applying the return-time contraction with such k gives $e^{-c_2 n}$ for some $c_2 > 0$. There are only finitely many states, and the skew-product transition operators for the incomplete initial and terminal blocks have uniformly bounded operator norms on the fixed Sobolev class, so inserting those blocks changes only the multiplicative constant. Combining their bound with the exponentially small exceptional event proves exponential equidistribution in actual Markov time. □

Proposition 0.3 (Actual-step exponential convergence)**.** *Under the hypotheses of Theorem 0.2, there exist $c > 0$, $C \geq 1$, and a Sobolev order D such that for every zero-Haar-mean smooth observable f and every initial point outside the finite-orbit obstruction,*

$$\left|\mathbb{E}_x f(Z_n)\right| \leq C e^{-cn} \mathcal{S}_D(f)$$

after multiplication by the explicit cusp-height and finite-orbit proximity penalties appearing in Lemma 0.1. In particular the estimate is for the n-step law itself, not for its Cesàro averages.

Proof. At return times the rank-one estimate of Lemma 0.1 gives exponential contraction. The return-time large-deviation estimate used in Theorem 0.2 transfers k returns to physical time n with an exponentially small exceptional probability. Taking c smaller than both the return-law contraction exponent and the return-time large-deviation exponent gives the displayed estimate. □

Worked Example 0.4 (A spectral-gap error in a two-state chain)**.** For

$$P = \begin{pmatrix} 3/4 & 1/4 \\ 1/2 & 1/2 \end{pmatrix}$$

the eigenvalues are 1 and 1/4. Hence for every mean-zero observable f on the two states,

$$\|P^n f\|_{L^2(\pi)} \leq 4^{-n}\|f\|_{L^2(\pi)}.$$

This is the finite-dimensional prototype of effective equidistribution: a spectral gap contracts the nonconstant part, while recurrence and non-concentration are needed in homogeneous spaces to keep the walk in the region where the analytic norm is controlled.

CHAPTER 8

Polynomial/exponential rate theorems

This chapter isolates the polynomial/exponential rate theorems mechanism needed by the later rigidity arguments. Every statement below records its hypotheses explicitly; later applications may use only that stated scope.

Lemma 0.1 (Balancing polynomial and exponential errors)**.** *Suppose an effective random-walk argument produces an error $E(n,\rho) \le C(e^{-\kappa n}\rho^{-A} + \rho^{B})$ at smoothing scale ρ. The choice $\rho = e^{-\kappa n/(A+B)}$ yields $E(n,\rho) \ll e^{-\kappa B n/(A+B)}$. If the first term is only $n^{-\kappa}\rho^{-A}$, the analogous choice $\rho = n^{-\kappa/(A+B)}$ gives a polynomial rate.*

PROOF. Balance the two positive summands by solving $e^{-\kappa n}\rho^{-A} = \rho^{B}$ (or its polynomial analogue). Substitution gives the displayed exponent. This elementary optimization is recorded because it prevents the final smoothing step from hiding a loss. □

THEOREM 0.2 (Rate theorem for the effective interface). *In the arithmetic rank-one scope of Chapter 7 the exponential alternative applies; in merely polynomial-mixing settings the second alternative gives an explicit polynomial exponent.*

PROOF. Insert the spectral/mixing rate available in the relevant earlier volume into the lemma. All remaining losses in Chapters 4–7 are polynomial in ρ^{-1}, cusp height, and inverse finite-orbit distance, so the optimizer applies. □

Proposition 0.3 (Observable and return-time rate transfer)**.** *Every effective theorem in this volume carries a rate ledger listing the spectral exponent, smoothing order, dimension-growth loss, height penalty, and finite-orbit penalty.*

PROOF. The proof is bookkeeping: each chapter contributes exactly one of these factors, and multiplying them before the optimization prevents an unrecorded dependence. □

Worked Example 0.4 (Balancing two competing errors)**.** Suppose an argument gives

$$E(n,m) \le C_1 e^{-cm} + C_2 m^{A} n^{-b}.$$

Taking $m = \lfloor \kappa \log n \rfloor$ makes the first term $O(n^{-c\kappa})$ and the second $O((\log n)^A n^{-b})$. Choosing $\kappa = b/(2c)$ yields

$$E(n,m) = O\big((\log n)^A n^{-b/2}\big).$$

This elementary optimization is the bookkeeping step behind many "exponential local input plus polynomial global recurrence" estimates in the chapter.

Volume 41

Random Walks and Diophantine Approximation on Fractals

Volume 41 scope and prerequisites

The core scope is graph-directed self-similar systems satisfying the open-set condition and the expansion/irreducibility hypotheses needed by the Markov random-walk theorem. The main endpoints are genericity of the associated diagonal trajectories and the resulting non-badly-approximable/extremality statements. Stronger Khintchine laws are recorded only when separately proved.

Proof and provenance. Primary comparison: Simmons–Weiss and Prohaska–Sert. The dynamical input is the random-walk/Markov theory owned in Volumes 35–40; the geometric input is the Dani correspondence owned in Volume 10. Stronger variants are bibliographic context unless separately marked.

Reader guide: a concrete model

Why this matters.

Volume 41 is organized around the passage from *Self-similar and graph-directed fractal measures* to *Random-walk proofs of fractal Diophantine theorems*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

CHAPTER 1

Self-similar and graph-directed fractal measures

This chapter isolates the self-similar and graph-directed fractal measures mechanism needed by the later rigidity arguments. Every statement below records its hypotheses explicitly; later applications may use only that stated scope.

Lemma 0.1 (Cylinder masses in a self-similar system)**.** *For a finite strongly connected graph-directed system of similarities $f_e(x) = r_e O_e x + t_e$ satisfying the open-set condition, the coding Markov measure pushes forward to the natural graph-directed measure, and cylinder diameters and masses are products of the edge contractions and probabilities.*

PROOF. Induct on word length. Similarities multiply contraction ratios, while the Markov property multiplies conditional probabilities. The open-set condition gives bounded overlap at a fixed scale, so symbolic cylinders and geometric balls have comparable coverings. □

THEOREM 0.2 (Coding measure on a graph-directed attractor). *The natural measure is exact dimensional; its local dimension is the entropy rate divided by the Lyapunov contraction rate.*

PROOF. Apply Birkhoff to $-\log p_e$ and $-\log r_e$ along the stationary edge chain. Cylinder mass and diameter have the corresponding exponential rates. Bounded overlap converts cylinder estimates to ball estimates. □

Proposition 0.3 (Cylinder-to-ball comparison)**.** *The same coding data define a finite-state Markov walk after attaching to each edge the homogeneous-space matrix used in Chapter 3.*

PROOF. The attachment is measurable and depends only on an edge, so it is exactly the cocycle of Volume 40, Chapter 1. □

Worked Example 0.4 (Middle-third Cantor cylinders)**.** For the maps $S_0(x) = x/3$ and $S_1(x) = (x+2)/3$ with weights $1/2, 1/2$, a word $\omega \in \{0,1\}^n$ determines an interval $S_\omega([0,1])$ of length 3^{-n} and measure 2^{-n}. Thus

$$\frac{\log \mu(S_\omega([0,1]))}{\log |S_\omega([0,1])|} = \frac{n \log 2}{n \log 3} = \frac{\log 2}{\log 3}.$$

This explicit cylinder computation is the model for the coding-to-dimension statements used later.

CHAPTER 2

Friendly and quasi-decaying properties

This chapter isolates the friendly and quasi-decaying properties mechanism needed by the later rigidity arguments. Every statement below records its hypotheses explicitly; later applications may use only that stated scope.

Lemma 0.1 (Cylinder non-concentration estimate)**.** *Assume the similarity system is irreducible in the sense that its attractor is not contained in an affine hyperplane and satisfies uniform separation. Then there exist $C, \alpha > 0$ so that for every ball $B(x, r)$ centered on the attractor and every affine hyperplane L, the natural measure of $B(x, r) \cap N_{\varepsilon r}(L)$ is at most $C\varepsilon^{\alpha}\mu(B(x, r))$.*

Proof. Choose a fixed word length m for which the images of the attractor contain points in uniformly transverse position; compactness and irreducibility provide such an m. Among every m generations, a definite proportion of child cylinders stay a definite relative distance from any given hyperplane. Iterating this loss for $k \asymp |\log \varepsilon|$ blocks gives the power ε^{α}. Bounded overlap from the open-set condition localizes the argument inside $B(x, r)$. □

Theorem 0.2 (Hyperplane-decay transfer). *The natural measure is quasi-decaying, hence null on affine hyperplanes and on the standard resonant neighborhoods used in metric Diophantine approximation.*

Proof. Sum the hyperplane-decay bound over a dyadic sequence of relative thicknesses. The resulting geometric series is finite and gives the quasi-decay formulation. □

Proposition 0.3 (Quasi-decay under bounded distortion)**.** *Quasi-decay is stable under restriction to a positive-measure cylinder and under similarity images.*

Proof. Both statements follow by rescaling the preceding estimate and using the exact cylinder self-similarity. □

Worked Example 0.4 (A one-dimensional decay calculation)**.** For the Cantor measure above, a level-n cylinder has mass 2^{-n} and diameter 3^{-n}. If an interval I of radius comparable to 3^{-n} meets only a bounded number of level-n cylinders, then $\mu(I) \ll 2^{-n} = (3^{-n})^{\log 2/\log 3}$. This is the simplest Frostman estimate. Friendly/quasi-decaying conditions add a uniform penalty when the ball is intersected with a much thinner neighborhood of a hyperplane.

CHAPTER 3

Random-walk codings of fractal measures

This chapter isolates the random-walk codings of fractal measures mechanism needed by the later rigidity arguments. Every statement below records its hypotheses explicitly; later applications may use only that stated scope.

Lemma 0.1 (Coding an iterated system as a random product)**.** *For each similarity edge e define the matrix g_e implementing its inverse branch followed by the appropriate diagonal renormalization. Along a coded point $x = \pi(e_1 e_2 \ldots)$, the lattice/affine-lattice point associated with the nth cylinder equals the homogeneous random-walk product $g_{e_n} \cdots g_{e_1} x_0$ up to a uniformly bounded compact factor.*

PROOF. Write the similarity composition explicitly. Its linear part is $r_{e_1} \cdots r_{e_n} O_{e_1} \cdots O_{e_n}$ and its translation is the corresponding affine geometric sum. Embedding similarities in the affine linear group turns composition into matrix multiplication. Extracting the scalar contraction into the diagonal flow leaves an orthogonal/translation factor in a fixed compact set. □

THEOREM 0.2 (Cylinder/path correspondence). *The coding measure therefore sends almost every fractal point to a finite-state Markov trajectory in the homogeneous space, so Volume 40 applies whenever the return semigroup satisfies its expansion hypotheses.*

PROOF. The symbolic coding is measure preserving. The previous lemma identifies the geometric orbit with the Markov cocycle up to a compact perturbation, which does not change recurrence or weak equidistribution for uniformly continuous compactly supported test functions. □

Proposition 0.3 (Push-forward of path frequency)**.** *For one-state self-similar systems the Markov trajectory reduces to the i.i.d. Simmons–Weiss random walk.*

PROOF. There is only one vertex, so successive edges are independent with the prescribed Bernoulli weights. □

Worked Example 0.4 (Digits as a random product)**.** Write the Cantor point with digits $\omega_j \in \{0,2\}$ as

$$x = \sum_{j\geq 1} \frac{\omega_j}{3^j}.$$

Choosing each digit independently with probability $1/2$ is the same as composing the two affine maps S_0, S_1 according to a Bernoulli path. A cylinder of length n is therefore both a symbolic event of probability 2^{-n} and the image of $[0,1]$ under one length-n affine product. The chapter's random-walk coding is this identity with the affine maps replaced by group elements acting on a homogeneous space.

CHAPTER 4

Dani correspondence for fractal typicality

Lemma 0.1 (Dani lattice for simultaneous approximation)**.** *For $x \in \mathbb{R}^d$ put $u_x = \left(\begin{smallmatrix} I_d & x \\ 0 & 1 \end{smallmatrix}\right)$ and $a_t = \operatorname{diag}(e^{t/d}, \dots, e^{t/d}, e^{-t})$. Then x is badly approximable iff the trajectory $a_t u_x \mathrm{SL}_{d+1}(\mathbb{Z})$ remains in a compact subset of the space of unimodular lattices.*

Proof. Mahler compactness says compactness is equivalent to a uniform lower bound on nonzero lattice-vector lengths. A vector $(p, q) \in \mathbb{Z}^d \times \mathbb{Z}$ is sent to $(e^{t/d}(p+qx), e^{-t}q)$. Minimizing in t translates a lower bound for this norm into a uniform lower bound for $|q|^{1/d}\|qx-p\|$, exactly the badly-approximable condition. $\square$

Theorem 0.2 (Bounded-orbit criterion). *If the diagonal trajectory is Haar generic, then x is not badly approximable and satisfies every Diophantine property represented by a Haar-full Birkhoff set.*

Proof. A Haar-generic orbit visits every positive-measure cusp neighborhood with its Haar frequency, so it cannot remain in a compact set. Birkhoff genericity gives the second assertion directly. $\square$

Proposition 0.3 (Weighted Dani criterion)**.** *Let $\mathbf{r} = (r_1, \dots, r_d)$ with $r_i > 0$ and $\sum_i r_i = 1$, and put*

$$a_t^{\mathbf{r}} = \operatorname{diag}(e^{r_1 t}, \dots, e^{r_d t}, e^{-t}).$$

Then $x \in \mathbb{R}^d$ is badly $\mathbf{r}$-weighted approximable, meaning that there is $c > 0$ such that

$$\max_{1 \le i \le d} |qx_i - p_i|^{1/r_i} |q| \ge c \quad \text{for all } (p, q) \in \mathbb{Z}^d \times (\mathbb{Z} \setminus \{0\}),$$

if and only if $a_t^{\mathbf{r}} u_x \mathrm{SL}_{d+1}(\mathbb{Z})$ is bounded for $t \ge 0$.

Proof. For $(p, q) \in \mathbb{Z}^d \times \mathbb{Z}$ the transformed vector has coordinates

$$e^{r_i t}(p_i + qx_i) \quad (1 \le i \le d), \qquad e^{-t}q.$$

Mahler compactness turns boundedness into a uniform lower bound for the maximum of these coordinates over all nonzero (p, q). Minimizing that maximum in t is equivalent, up to constants depending only on $\mathbf{r}$, to the displayed weighted Diophantine lower bound. $\square$

Worked Example 0.4 (The one-dimensional Dani lattice)**.** For $x \in \mathbb{R}$ set

$$u_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}, \qquad a_t = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}.$$

For $(p, q) \in \mathbb{Z}^2$,

$$a_t u_x \begin{pmatrix} -p \\ q \end{pmatrix} = \begin{pmatrix} e^t(qx - p) \\ e^{-t} q \end{pmatrix}.$$

Choosing $e^{2t} \asymp q/|qx - p|$ balances the two coordinates. Thus very good rational approximation produces a very short vector in $a_t u_x \mathbb{Z}^2$, while a uniform lower bound on such vectors is equivalent to bad approximability. This is the concrete bridge used for fractal typicality.

CHAPTER 5

Genericity of diagonal trajectories

Lemma 0.1 (Empirical-measure compactness). *Assume the graph-directed cocycle is expanding in the adjoint representations required by Volume 39 and its Markov return law is not trapped in a finite orbit. Then for the natural fractal measure, almost every coded point has a homogeneous trajectory whose empirical measures converge to Haar measure on the relevant orbit closure.*

Proof. By Chapter 3, almost every fractal code is a Markov random walk on the homogeneous space. Volume 40 gives joint Markov/homogeneous equidistribution. The compact discrepancy between the coded walk and the renormalized diagonal trajectory is a coboundary in the skew product; telescoping test-function differences and uniform continuity show that it has zero contribution to empirical averages. □

Theorem 0.2 (Genericity from the stationary coding). *Assume the hypotheses of Lemma 0.1 and suppose the minimal homogeneous orbit closure is the whole finite-volume space X. Then for the natural fractal measure and almost every coded point x,*

$$\frac{1}{T}\int_0^T f(a_t u_x \Gamma)\,dt \longrightarrow \int_X f\,dm_X$$

for every $f \in C_c(X)$.

Proof. Lemma 0.1 identifies every empirical weak limit with Haar measure on the minimal homogeneous closure. When that closure is X, the unique possible limit is m_X. Testing against a countable dense family in $C_c(X)$ and then using uniform approximation gives the assertion for every $f \in C_c(X)$. □

Proposition 0.3 (Cylinder-stable genericity). *Let C be a coding cylinder with positive natural measure and let $\nu_C = \nu(C)^{-1}\nu|_C$. If the conclusion of Theorem 0.2 holds for ν-almost every code, then it holds for ν_C-almost every code with the same limiting Haar measure.*

Proof. A cylinder condition fixes only finitely many initial symbols. Removing or changing finitely many initial blocks changes a time average by

$O(T^{-1})$ for bounded observables. Since the generic set has full ν-measure, its intersection with C has full conditional measure ν_C. □

Worked Example 0.4 (What genericity means for one observable)**.** Let f be the indicator of a compact test set K in the space of lattices. Haar genericity of $a_t u_x \mathbb{Z}^2$ says

$$\frac{1}{T}\int_0^T f(a_t u_x \mathbb{Z}^2)\,dt \longrightarrow m(K)$$

for continuity sets K. A random coding proof first establishes the analogous frequency law along symbolic blocks and then transfers it to the diagonal orbit. The crucial point is that the conclusion concerns time frequencies, not merely density of the orbit.

CHAPTER 6

Badly approximable points on fractals

Lemma 0.1 (Bounded orbit excludes Haar genericity)**.** *Under the hypotheses of Chapter 5 and full homogeneous orbit closure, the natural fractal measure assigns measure zero to the set of badly approximable vectors.*

PROOF. Almost every point is Haar generic by Chapter 5. By the Dani correspondence in Chapter 4 such a point cannot have a bounded diagonal orbit. Hence it is not badly approximable. □

THEOREM 0.2 (Cylinderwise zero measure of badly approximable points). *Assume the hypotheses of Theorem 0.2. Let C be any coding cylinder with positive natural measure and let ν_C be the normalized conditional measure on C. Then*

$$\nu_C\{x : x \text{ is badly approximable}\} = 0.$$

PROOF. Proposition 0.3 says that ν_C-almost every coded point is Haar generic for the diagonal flow on the full finite-volume space. By Theorem 0.2, every Haar-generic point has an unbounded diagonal orbit and therefore is not badly approximable. Thus the badly approximable set is contained in the ν_C-null exceptional set where Haar genericity fails. □

Proposition 0.3 (Weighted zero-measure transfer)**.** *Fix weights $\mathbf{r} = (r_1, \ldots, r_d)$ as in Proposition 0.3. Assume that the coding-to-flow argument of Lemma 0.1 and Theorem 0.2 holds with a_t replaced by $a_t^{\mathbf{r}}$ and has full homogeneous orbit closure. Then the natural fractal measure, and each positive-measure coding-cylinder conditional, assigns measure zero to the set of badly* **r***-weighted approximable points.*

PROOF. Under the stated weighted genericity hypothesis, almost every coded point is Haar generic for $a_t^{\mathbf{r}} u_x \Gamma$; Proposition 0.3 transfers the same full-measure statement to every positive-measure cylinder because its proof uses only removal of finitely many initial symbols. A Haar-generic trajectory cannot remain in a compact subset of the noncompact finite-volume homogeneous space. Proposition 0.3 identifies boundedness of this weighted trajectory exactly with bad **r**-weighted approximability. Hence the weighted badly approximable set is null. □

Worked Example 0.4 (Why bounded and Haar-generic are incompatible)**.** Mahler's criterion gives compact sets

$$K_\varepsilon = \{\Lambda : \lambda_1(\Lambda) \geq \varepsilon\}.$$

A badly approximable x has $a_t u_x \mathbb{Z}^2$ contained in some K_ε. Haar measure, however, gives positive mass to the complement of every fixed K_ε because the modular surface has a cusp. Therefore a trajectory trapped in K_ε cannot be Haar generic. Once Chapter 5 gives Haar genericity for almost every coded point, the zero-measure statement follows immediately.

CHAPTER 7

Extremality and strong extremality

Lemma 0.1 (Extremality from cusp-frequency control). *Let ν be a quasi-decaying graph-directed measure as in Chapter 2. If its diagonal trajectories are Haar generic, then for ν-almost every point the ordinary Diophantine exponent equals the Dirichlet exponent; in the multiplicative model the same argument gives strong extremality.*

Proof. We use the quantitative nondivergence theorem already owned in Volume 10 rather than infer a shrinking-target statement from ordinary Birkhoff genericity. Fix $\epsilon > 0$ above the Dirichlet exponent. The event that x admits an approximation of exponent $1 + \epsilon$ at height Q is equivalent, by the Dani correspondence, to the diagonal orbit entering a cusp region in which one of the finitely many exterior-vector functions is $< Q^{-c\epsilon}$.

On every self-similar cylinder, Chapter 2 gives Federer doubling and power decay near the affine hyperplanes on which those exterior functions become small. The quantitative nondivergence estimate therefore bounds the cylinder measure of the bad event at scale Q by $CQ^{-c'\epsilon}$. Group denominators into dyadic blocks and, if necessary, pass to a fixed power subsequence so that these bounds are summable. The first Borel–Cantelli lemma gives only finitely many super-Dirichlet excursions for almost every point. Interpolation between consecutive dyadic blocks removes the lacunarity.

Thus the Diophantine exponent is at most the Dirichlet value; Dirichlet's theorem gives the reverse inequality. In the multiplicative model use the product of the relevant exterior norms and the strong quantitative nondivergence estimate; the same summability argument gives strong extremality. This is the standard friendly-measure mechanism, but all inputs used here were proved in Chapter 2 and Volume 10. □

Theorem 0.2 (Stability of extremality under similarities and cylinders). *Let ν be a quasi-decaying measure for which Lemma 0.1 applies. If $S(x) = \lambda Ox + b$ is a similarity with $\lambda > 0$ and C is a positive-ν-measure coding cylinder, then $S_*\nu$ and the normalized restriction $\nu(C)^{-1}\nu|_C$ are extremal; in the multiplicative setting they are strongly extremal.*

Proof. A similarity multiplies all approximation errors and affine-hyperplane distances by fixed constants, so neither the Diophantine exponent nor the quasi-decay exponent changes. Restriction to a positive-measure cylinder preserves the almost-everywhere conclusion and the local quasi-decay estimates after renormalizing constants. Lemma 0.1 then gives the ordinary and multiplicative conclusions. □

Proposition 0.3 (Gauss digit frequencies from Haar genericity)**.** *Let* $x \in (0,1)$ *and suppose the geodesic trajectory associated with* x *is Haar generic in* $\mathrm{SL}_2(\mathbb{R})/\mathrm{SL}_2(\mathbb{Z})$*. If* $a_n(x)$ *are the continued-fraction digits, then for every* $k \geq 1$*,*

$$\lim_{N\to\infty} \frac{1}{N}\#\{1 \leq n \leq N : a_n(x) = k\} = \frac{1}{\log 2} \log \frac{(k+1)^2}{k(k+2)}.$$

Proof. The standard Series cross-section identifies the return map of the geodesic flow with the Gauss map $T(x) = 1/x - \lfloor 1/x \rfloor$ and the induced invariant probability with $d\gamma(x) = dx/((1+x)\log 2)$. Haar genericity implies Birkhoff genericity for this return map. The cylinder $\{a_1 = k\} = (1/(k+1), 1/k]$ has Gauss measure

$$\gamma\{a_1 = k\} = \frac{1}{\log 2} \log \frac{(k+1)^2}{k(k+2)},$$

which is therefore the limiting frequency. □

Worked Example 0.4 (Extremality as exclusion of too-deep cusp visits)**.** In one dimension Dirichlet gives infinitely many p/q with $|x - p/q| < q^{-2}$. “Very well approximable” means that some exponent $2 + \eta$ works infinitely often. Under the Dani correspondence those exceptional inequalities force excursions deeper than the logarithmic scale dictated by Dirichlet. A quantitative nondivergence estimate showing that such deep excursions occur with summable probability lets Borel–Cantelli exclude them almost surely. The multiplicative version replaces the shortest-vector height by the appropriate product/exterior-power height.

CHAPTER 8

Random-walk proofs of fractal Diophantine theorems

This chapter isolates the random-walk proofs of fractal diophantine theorems mechanism needed by the later rigidity arguments. Every statement below records its hypotheses explicitly; later applications may use only that stated scope.

Lemma 0.1 (Coding-to-homogeneous transfer lemma)**.** *For a self-similar set satisfying the open-set condition and the expansion/nontrapping hypotheses above, the natural measure is almost surely of generic Diophantine type: almost every point is Haar generic in the Dani space, is not badly approximable, and is extremal; the graph-directed version holds with the Markov hypotheses of Volume 40.*

Proof. Let ω be a typical path in the graph-directed coding and $x = \pi(\omega)$. Chapter 3 identifies the renormalized homogeneous orbit of x with the finite-state Markov cocycle, up to a compact coboundary. Volume 40 gives the homogeneous equidistribution theorem for the return law, and Chapter 5 transfers it from the Markov sampling times to the full diagonal trajectory. Consequently x is Haar generic on the minimal homogeneous closure.

Chapter 4 converts boundedness of that trajectory to bad approximability. Since a Haar-generic orbit is recurrent to both compact sets and cusp neighborhoods, Chapter 6 gives that x is not badly approximable. For the exact exponent, Chapter 7 uses the friendly/quasi-decaying estimate together with quantitative nondivergence and Borel–Cantelli; this is a separate argument and is why genericity alone is not overused. Combining the three conclusions gives the theorem. The one-vertex case is the Simmons–Weiss coding, while a general finite strongly connected graph is exactly the Prohaska–Sert Markov extension. ☐

Theorem 0.2 (Fractal genericity and extremality from the coding cocycle). *Let ν be the natural coding measure of either*

(1) a conformal one-state iterated-function system satisfying the open-set condition and the expansion/nontrapping hypotheses of Chapters 1–7, or

(2) *a finite strongly connected graph-directed system whose Markov cocycle satisfies the irreducibility and expansion hypotheses of Volume 40.*

Then for ν-almost every coded point x, the Dani trajectory associated with x is Haar generic on its minimal homogeneous closure, x is not badly approximable, and x has the Dirichlet Diophantine exponent (hence is extremal).

PROOF. Apply Lemma 0.1. In case (1) the coding chain has one state, so the Markov cocycle of Chapter 3 reduces to the Bernoulli cocycle of Chapter 1. In case (2) it is the finite-state cocycle treated in Volume 40. The homogeneous genericity conclusion is therefore valid in both cases. Chapter 4 identifies bounded Dani trajectories with badly approximable points, so Haar recurrence to cusp neighborhoods excludes bad approximability. Finally Chapter 7 combines quasi-decay with quantitative nondivergence and Borel–Cantelli to give the Dirichlet exponent. □

Proposition 0.3 (Finite-product stability)**.** *Let $r \geq 1$ systems satisfy the hypotheses of Theorem 0.2, and suppose the block-diagonal product cocycle on the product Dani space is irreducible on its minimal homogeneous closure and has positive expansion exponent in every nontrivial factor. Then the product coding measure is almost surely Haar generic on that closure and is extremal in every coordinate.*

PROOF. The product symbolic process is a finite-state Markov chain and its homogeneous cocycle is the block-diagonal product of the factor cocycles. The stated irreducibility and expansion assumptions are exactly the hypotheses needed to apply Theorem 0.2 to the product system. Projecting the resulting Haar-generic product orbit to each coordinate gives the coordinate conclusions. □

Worked Example 0.4 (The complete proof pipeline in a toy Cantor model)**.** For a Bernoulli Cantor coding, the symbolic path gives a sequence of affine maps. Embed the Diophantine parameter into a lattice $u_x\mathbb{Z}^2$. If the induced group cocycle satisfies recurrence and the stationary measure is Haar, the empirical measures of the coded lattice path converge to Haar. Chapter 4 converts a too-good rational approximation into a cusp event; Chapter 7 shows that events beyond the Dirichlet scale have summable frequency. Hence almost every coded point has the ordinary Diophantine exponent. The full theorem differs only in the group representation and the quantitative non-concentration estimates, not in this logical order.

Volume 42

Shah's Polynomial and Curve Equidistribution

Volume 42 scope and prerequisites

This volume proves the qualitative polynomial-trajectory and analytic expanding-curve package in the exact arithmetic finite-volume scope needed later. The proof combines quantitative nondivergence, first-exit polynomial shearing, linearization/focusing, and the already internal homogeneous-measure classification. Shah's papers are retained for attribution and comparison, not as proved inputs.

Proof and provenance. Primary comparison: Shah's 1994 polynomial-trajectory theorem and the canonical Ratner architecture. The AHD proof below is internal at the displayed scope; later o-minimal extensions are bibliographic context only.

Reader guide: a concrete model

Why this matters.

Volume 42 is organized around the passage from *Polynomial maps into algebraic groups* to *Arithmetic applications of polynomial equidistribution.* The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.5 (A calculation to keep in view)**.** Let $P(t) = a+bt+ct^2$ with $c \neq 0$. At scale T write $t = Ts$. Then

$$P(Ts) = a + (bT)s + (cT^2)s^2.$$

If $|c|T^2 \asymp 1$ while $|b|T \to 0$ and $|a| \to 0$, the first visible rescaled polynomial is cs^2T^2. On a mesoscopic window $s = s_0 + \ell/T$ with $|\ell| = o(T)$,

$$P(Ts) - P(Ts_0) = P'(Ts_0)\ell + c\ell^2 = o(1)$$

whenever the normalized derivative is controlled. This is the toy calculation behind the separation of a macroscopic visibility scale from a mesoscopic almost-constant scale in effective linearization and closing.

CHAPTER 1

Polynomial maps into algebraic groups

This chapter isolates the polynomial maps into algebraic groups mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Polynomial maps into algebraic groups: structural mechanism)**.** *For a polynomial matrix map $\Theta : \mathbb{R} \to G \subset \mathrm{SL}_N(\mathbb{R})$, every coordinate of $\rho(\Theta(t))v$ in a finite-dimensional rational representation is polynomial, and every exterior-power norm is (C, α)-good on bounded intervals with constants depending only on the degree and dimension.*

Proof. Matrix coefficients of a rational representation restricted to G are polynomial combinations of matrix entries after clearing a fixed determinant power, which is one on SL_N. A nonzero real polynomial of bounded degree has sublevel sets controlled by the elementary Remez inequality. Apply this coordinatewise and use equivalence of norms. □

Theorem 0.2 (Polynomial trajectories satisfy the QND goodness hypothesis). *Fix a finite-dimensional rational representation $\rho : G \to \mathrm{GL}(V)$ and a finite family of exterior vectors $v_1, \ldots, v_M$ used to detect the rational cusp directions under consideration. Let $\Theta : I \to G$ be a polynomial matrix map of degree at most d. Then every nonzero coordinate function of $\rho(\Theta(t))v_j$ is (C, α)-good on subintervals of I, with C, α depending only on d, ρ, and $\dim V$. Consequently, whenever the lower-supremum hypothesis of Theorem 1.1 holds for these finitely many detector functions, Theorem 1.1 applies with the same uniform goodness constants.*

Proof. Lemma 0.1 gives the polynomial-coordinate assertion. Proposition 0.3 gives constants C, α uniform over every nonzero detector coordinate of degree at most the fixed degree bound. The remaining hypothesis in Theorem 1.1 is the separate lower-supremum condition; no lower bound is produced merely by polynomiality. Once that hypothesis is verified for the finite detector family, all assumptions of Theorem 1.1 are present, so its quantitative nondivergence conclusion follows. □

Proposition 0.3 (Uniform good-function constants in compact polynomial families)**.** *Fix a finite-dimensional rational representation $\rho : G \to \mathrm{GL}(V)$ and an integer $d \geq 1$. There are constants $C \geq 1$ and $\alpha > 0$, depending only on d, $\dim V$, and ρ, such that for every polynomial matrix map $\Theta : I \to G$ of degree at most d, every $v \in V$, and every nonzero coordinate polynomial $p(t)$ of $\rho(\Theta(t))v$, one has*

$$|\{t \in J : |p(t)| < \varepsilon \sup_J |p|\}| \leq C\varepsilon^{\alpha}|J| \qquad (0 < \varepsilon < 1)$$

for every subinterval $J \subset I$. The same C, α work for every compact family of such maps.

Proof. The assertion is a degree-only consequence of the one-variable Remez inequality; compactness of a coefficient family is not needed. Indeed, if p is a nonzero real polynomial of degree $m \leq d$, then for every interval J and every measurable $E \subset J$ of positive measure,

$$\sup_J |p| \leq \left(\frac{4|J|}{|E|}\right)^m \sup_E |p|.$$

Take $E = \{t \in J : |p(t)| < \varepsilon \sup_J |p|\}$. If $|E| > 0$, the displayed inequality gives

$$1 \leq \left(\frac{4|J|}{|E|}\right)^m \varepsilon, \qquad \text{hence} \qquad |E| \leq 4\varepsilon^{1/m}|J| \leq 4\varepsilon^{1/d}|J|,$$

because $0 < \varepsilon < 1$ and $m \leq d$. Thus one may take $C = 4$ and $\alpha = 1/d$ for scalar coordinate polynomials. The representation enters only through the fixed degree bound for the coordinates of $\rho(\Theta(t))v$; after that reduction the constants are uniform, in particular on every compact family. $\square$

CHAPTER 2

Polynomial trajectories on G/Γ

This chapter isolates the polynomial trajectories on g/γ mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Polynomial trajectories on G/Γ: structural mechanism)**.** *Let $x \in G/\Gamma$ and let H be the smallest closed subgroup generated by unipotent one-parameter subgroups such that $\Theta(0)^{-1}\Theta(\mathbb{R}) \subset H$ and Hx carries finite invariant volume. Time averages of $\Theta(t)x$ are tight modulo the finite singular alternatives of Volume 2.*

Proof. Apply Chapter 1 nondivergence on each long interval. If mass concentrates near a proper homogeneous stratum, the arithmetic linearization theorem of Volume 2 produces a smaller Chevalley stabilizer. Induct on dimension; only finitely many strata occur at each bounded complexity. ☐

Theorem 0.2 (Polynomial trajectories on G/Γ). *Every weak limit of polynomial-trajectory time averages is invariant under a nontrivial unipotent subgroup unless the trajectory is trapped in a proper homogeneous subspace.*

Proof. Rescale a bounded-degree polynomial relative displacement around a density point. The leading nonconstant term converges to a one-parameter unipotent shear in the relevant representation. Comparing averages before and after the rescaling gives invariance of the limit. ☐

Proposition 0.3 (Equivariance under fixed left translation)**.** *Let*

$$\mu_T = \frac{1}{T}\int_0^T \delta_{\Theta(t)x}\,dt \quad \text{and} \quad \mu_T^{g_0} = \frac{1}{T}\int_0^T \delta_{g_0\Theta(t)x}\,dt$$

for a fixed $g_0 \in G$. If $\mu_{T_j} \Rightarrow \nu$, then $\mu_{T_j}^{g_0} \Rightarrow (g_0)_*\nu$. In particular, if ν is invariant under a subgroup $W < G$, then $(g_0)_*\nu$ is invariant under $g_0 W g_0^{-1}$.

Proof. For every $f \in C_c(G/\Gamma)$,

$$\int f\,d\mu_T^{g_0} = \int (f \circ L_{g_0})\,d\mu_T.$$

Weak convergence of μ_{T_j} therefore gives weak convergence to $(g_0)_*\nu$. If $w \in W$, then for every f,

$$\int f(g_0 w g_0^{-1} y)\, d(g_0)_*\nu(y) = \int f(g_0 w x)\, d\nu(x) = \int f(g_0 x)\, d\nu(x),$$

so $(g_0)_*\nu$ is $g_0 W g_0^{-1}$-invariant. $\square$

CHAPTER 3

Weak limits and homogeneous candidates

This chapter isolates the weak limits and homogeneous candidates mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Weak limits and homogeneous candidates: structural mechanism)**.** *Let $U = \{u_t\}$ be the nontrivial Ad-unipotent one-parameter subgroup supplied by Chapter 2. Every U-ergodic component of a weak limit obtained there is homogeneous: its support is a closed finite-volume orbit of a connected subgroup containing U.*

PROOF. Chapter 2 supplies U-invariance and tightness, so a weak limit ν is a probability on the fixed arithmetic quotient. Disintegrate $\nu = \int \nu_\omega \, d\lambda(\omega)$ into U-ergodic components. For almost every ω, Theorem 6.1 gives a connected closed $L_\omega \geq U$ and a closed finite-volume orbit $L_\omega x_\omega$ with $\nu_\omega = m_{L_\omega x_\omega}$. □

THEOREM 0.2 (Chevalley compatibility of homogeneous weak-limit components)**.** *Let ν be a weak limit from Chapter 2 and let m_{Ly} be a homogeneous ergodic component of ν. Let W be a connected unipotent subgroup under which this component is invariant. Choose a Chevalley representation $\rho_L : G \to \mathrm{GL}(V_L)$ and a nonzero vector $v_L \in V_L$ such that the stabilizer of the line $\mathbb{R}v_L$ is the algebraic normalizer of L. Then*

$$\rho_L(w)v_L \in \mathbb{R}v_L \qquad (w \in y^{-1}Wy),$$

and hence, for every $Z \in \mathrm{Lie}(y^{-1}Wy)$,

$$d\rho_L(Z)v_L \in \mathbb{R}v_L.$$

Thus every homogeneous type occurring in ν satisfies the polynomial Chevalley identities imposed by the limiting unipotent shear.

PROOF. If a component of type L occurs, its Chevalley vector is fixed by the limiting shear. The polynomial coordinate identities therefore place the original trajectory in the corresponding algebraic normalizer set. □

Proposition 0.3 (Haar conclusion after exclusion of proper compatible types)**.** *Let $Y = L_0 x_0$ be the minimal closed finite-volume homogeneous subspace containing the polynomial trajectory of Chapter 2. Suppose that no connected subgroup $L < L_0$ with a closed finite-volume orbit in Y satisfies the Chevalley compatibility identities of Theorem 0.2. Then every weak limit of the trajectory averages is the L_0-invariant probability m_Y.*

Proof. Let ν be a weak limit. By Lemma 0.1, its relevant unipotent-ergodic components are homogeneous probabilities m_{Ly}. Theorem 0.2 makes every such component a compatible homogeneous type. By hypothesis no proper $L < L_0$ is compatible, so almost every component has support Y. There is only one L_0-invariant probability on the finite-volume orbit Y, namely m_Y. Hence the ergodic decomposition of ν is almost surely constant equal to m_Y, and therefore $\nu = m_Y$. $\square$

CHAPTER 4

Shah linearization

This chapter isolates the shah linearization mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Shah linearization: structural mechanism)**.** *Fix a proper candidate subgroup L. If a positive proportion of a long polynomial segment lies in an ε-tube of the L-orbit while remaining away from smaller singular strata, then in the Chevalley representation the entire segment stays in an $O(\varepsilon^c)$ neighborhood of the algebraic fixed set for L.*

Proof. Lift visits to representatives in a compact fundamental region. Properness of the fixed-L Chevalley lift (Volume 2) removes ambiguity of representatives. The relevant coordinate functions are bounded-degree polynomials by Chapter 1. A polynomial that is small on a fixed positive proportion of an interval is uniformly small on a slightly smaller interval by Remez. Translate this back through the local properness estimate. □

Theorem 0.2 (Shah linearization). *Fix a proper candidate subgroup L and a Chevalley pair (ρ_L, v_L) whose projective stabilizer is the algebraic normalizer of L. Suppose there are tube radii $\varepsilon_n \downarrow 0$ and polynomial segments $\Theta_n(J_n)x$ satisfying the hypotheses of Lemma 0.1 with one positive-proportion constant independent of n. After the normalizations used in that lemma, every Chevalley coordinate transverse to the line $\mathbb{R}v_L$ converges uniformly to zero on each fixed compact subinterval. Consequently the limiting polynomial trajectory Θ satisfies*

$$\rho_L(\Theta(t))v_L \in \mathbb{R}v_L \qquad \text{for every } t$$

on its parameter interval. Equivalently, $\Theta(t)$ lies in the algebraic focusing set $N_G(L)$ (up to the fixed left/right representatives chosen in the linearization chart) for every parameter value.

Proof. Choose finitely many linear functionals $\ell_1, \ldots, \ell_r$ whose common kernel is the line $\mathbb{R}v_L$. On the normalized segment put

$$q_{j,n}(t) = \ell_j(\rho_L(\Theta_n(t))v_L)\,.$$

Chapter 1 gives a degree bound for all $q_{j,n}$ that is independent of n. Lemma 0.1 gives, on each fixed compact subinterval J', a bound

$$\sup_{t\in J'} |q_{j,n}(t)| \le C\varepsilon_n^c.$$

Hence every locally uniform polynomial limit q_j is identically zero. Therefore all transverse Chevalley coordinates of $\rho_L(\Theta(t))v_L$ vanish for every t, proving that this vector lies in $\mathbb{R}v_L$. By the defining property of the Chevalley pair, that line condition is precisely membership in the algebraic normalizer/focusing set, with the fixed representatives restored at the end. □

Proposition 0.3 (Uniform linearization over a finite candidate family)**.** *Let $\mathscr{L} = \{L_1, \ldots, L_r\}$ be a finite family of candidate subgroups, and for each L_i fix the Chevalley representation and compact-chart properness data used in Lemma 0.1. Then there are common constants $d, C, c > 0$ such that the polynomial degree bound is at most d and the conclusion of Lemma 0.1 holds with C, c for every $L_i \in \mathscr{L}$. Thus Theorem 0.2 may be applied to the whole family without changing constants from one candidate to another.*

Proof. For each i let d_i, C_i, c_i be admissible constants. Since $\mathscr{L}$ is finite, take

$$d = \max_i d_i, \qquad C = \max_i C_i, \qquad c = \min_i c_i > 0.$$

Increasing a degree or multiplicative bound and decreasing a positive power exponent preserves the estimate, so these three constants work simultaneously for all members of the family. □

CHAPTER 5

Singular focusing alternatives

This chapter isolates the singular focusing alternatives mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Singular focusing alternatives: structural mechanism)**.** *For every compact set and error tolerance, a polynomial trajectory either spends asymptotically negligible time near the singular set, or there is a proper subgroup L of smaller dimension and a global focusing relation placing the trajectory in the corresponding algebraic normalizer.*

PROOF. Order the finitely many singular types by dimension. Apply Chapter 4 to a minimal type receiving positive limiting mass. The smaller-type exclusion required by the linearization lemma follows from minimality. Hence positive mass forces the focusing relation; otherwise all singular tubes have vanishing occupation. ☐

THEOREM 0.2 (Singular focusing alternatives). *The focusing alternative strictly lowers the ambient homogeneous dimension and hence terminates after at most* $\dim G$ *steps.*

PROOF. Each focusing relation replaces the ambient orbit closure by a proper connected homogeneous support. ☐

Proposition 0.3 (Exclusion of proper homogeneous components on the unfocused branch)**.** *Assume the polynomial trajectory is on the unfocused branch of Lemma 0.1, i.e. no proper subgroup $L < G$ satisfies the global focusing relation produced there. Then every weak limit of its time averages assigns zero mass to every proper finite-volume homogeneous subspace of the minimal homogeneous closure.*

PROOF. Suppose a weak limit had a positive-mass component supported on a proper homogeneous subspace of type L. Chapter 3 and the linearization argument identify the corresponding Chevalley relation. Lemma 0.1 then forces the global focusing relation for that same proper L, contradicting the unfocused hypothesis. Hence no proper homogeneous component can occur. $\square$

CHAPTER 6

Nondegenerate analytic curves

This chapter isolates the nondegenerate analytic curves mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Nondegenerate analytic curves: structural mechanism)**.** *For an analytic curve whose finite jet at almost every parameter spans the required unstable representation directions, the set of parameters at which all relevant Chevalley minors vanish has measure zero; on compact subintervals away from that set the minors satisfy a power sublevel estimate.*

Proof. Each minor is a real-analytic function. Nondegeneracy means it is not identically zero. Its zeros are isolated unless forced by a common analytic factor; factor that finite-order zero locally and obtain a $C\varepsilon^{\alpha}$ sublevel bound. A finite cover of the compact parameter interval gives uniform constants. □

Theorem 0.2 (Nondegenerate analytic curves). *Such a curve cannot satisfy a proper focusing relation unless it is contained in the corresponding proper affine/algebraic subspace.*

Proof. A focusing relation makes all defining minors vanish identically by analytic continuation, contradicting nondegeneracy. □

Proposition 0.3 (Nondegenerate analytic curves: downstream consequence)**.** *For curves in $\mathbb{R}^k$ not contained in a proper affine subspace, the standard* SL_{k+1} *embedding satisfies the nondegeneracy test.*

Proof. The successive derivatives of $(1, \varphi(s))$ span $\mathbb{R}^{k+1}$ at some finite order unless an affine functional vanishes identically on the curve. □

CHAPTER 7

Polynomial/curve equidistribution

This chapter isolates the polynomial/curve equidistribution mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Polynomial/curve equidistribution: structural mechanism)**.** *In the Shah polynomial scope of this volume, the normalized time averages converge to the invariant probability on the minimal finite-volume homogeneous closure; for nondegenerate analytic expanding curves, the expanding translates converge to Haar measure on the corresponding homogeneous closure.*

Proof. Take any sequence of averaging lengths (or expanding times) tending to infinity. Chapter 2 gives tightness, so after passing to a subsequence there is a weak limit ν. The rescaling argument there shows that ν is invariant under a nontrivial unipotent subgroup. By the exact Ratner owner inherited in Volume 2, the ergodic components of ν are homogeneous.

Suppose a positive component is supported on a proper homogeneous orbit Lx. Chapter 3 shows that its Chevalley vector is compatible with the polynomial jet, and Chapter 4 upgrades positive tube occupation to a global algebraic focusing relation. Chapter 5 then says that every such relation is one of the finitely many relations already used to define the minimal homogeneous closure. Thus no component can live on a proper subgroup of that minimal closure.

For the analytic expanding-curve statement, Chapter 6 strengthens the conclusion: nondegeneracy forces every proper Chevalley minor to be a nonzero analytic function, so the corresponding focusing locus has parameter measure zero. Hence in either case every ergodic component of ν is Haar on the minimal closure. Thus ν is that Haar probability. Since the argument applies to every subsequence, the original family converges. $\square$

Theorem 0.2 (Polynomial and analytic-curve equidistribution). *Let G be the connected Lie group and $\Gamma < G$ the arithmetic lattice fixed in this volume, let $x = g\Gamma$, and let $\Theta : I \to G$ be either*

(i) a polynomial trajectory of the class developed in Chapters 1–5, or

(ii) an analytic curve in the expanding horospherical class of Chapter 6.

Let $Y = Lx$ be the minimal closed finite-volume homogeneous subspace containing the trajectory, and assume in case (ii) that every proper Chevalley obstruction minor associated with a proper $L' < L$ is not identically zero on I. Then for every $f \in C_c(G/\Gamma)$,

$$\frac{1}{|I_T|}\int_{I_T} f(\Theta_T(s)x)\,ds \longrightarrow \int_Y f\,dm_Y, \tag{42.7.1}$$

where I_T and Θ_T are the polynomial averaging interval or expanding renormalization specified in Chapters 2 and 6. The same conclusion holds after partitioning I into finitely many C^1 pieces whose omitted boundary has normalized measure $o(1)$.

Proof. Lemma 0.1 proves (42.7.1) for one smooth parameter piece. For a finite partition $I = \bigcup_j I_j$ with omitted set E_T satisfying $|E_T|/|I_T| \to 0$, write the normalized integral as the weighted sum over I_j plus the contribution of E_T. The latter is bounded by $\|f\|_\infty |E_T|/|I_T| = o(1)$; each piece converges to m_Y by the lemma, and the piece weights sum to $1 + o(1)$. Hence the full average converges to m_Y. □

Proposition 0.3 (Uniqueness of the weak limit)**.** *Under the hypotheses of Theorem 0.2, every weak-* convergent subsequence of the normalized trajectory measures has limit m_Y; consequently the entire family converges without passing to subsequences.*

Proof. Tightness is proved in Chapters 2–3. Lemma 0.1 identifies any subsequential limit with m_Y. In a metrizable weak-* topology on probability measures, a relatively compact family for which every convergent subsequence has the same limit converges to that limit. □

CHAPTER 8

Arithmetic applications of polynomial equidistribution

This chapter isolates the arithmetic applications of polynomial equidistribution mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Arithmetic applications of polynomial equidistribution: structural mechanism)**.** *Whenever an arithmetic counting/Diophantine condition is represented by a bounded-continuity set A in the homogeneous space, the frequency along an equidistributing polynomial or expanding curve equals the homogeneous measure of A.*

Proof. Approximate 1_A from above and below by continuous compactly supported functions; boundary measure zero makes the two integrals converge to the same value. Apply Chapter 7 to the approximants. □

Theorem 0.2 (Arithmetic applications of polynomial equidistribution). *In the standard lattice embedding this yields the qualitative Dirichlet–Minkowski non-improvability conclusion for almost every point of a nondegenerate analytic curve.*

Proof. The improvement condition is a cusp target with boundary of Haar measure zero. Equidistribution gives its correct frequency, contradicting permanent improvement. This is the dynamical implication used in Shah's theorem. □

Downstream interface. This final chapter supplies the downstream interface recorded in the theorem above. No additional proposition is needed beyond the theorem above.

Volume 43

Expanding Horospheres and Curves

Volume 43 scope and prerequisites

The main scope is the standard expanding-horospherical setting and the analytic-curve equidistribution theorem for a curve not contained in a proper representation-theoretic obstruction. The volume proves the nondivergence, jet rescaling, linearization/focusing, and nondegeneracy mechanisms needed for the equidistribution conclusion; Shah's papers provide attribution and source comparison rather than an external proved input.

Proof and provenance. Primary comparison: Shah's expanding-translates papers. The homogeneous-measure classification used in the proof is a backward internal theorem of this series; stronger variants are bibliographic context unless separately proved.

Reader guide: a concrete model

Why this matters.

Volume 43 is organized around the passage from *Expanding horospherical subgroups* to *Applications to curves and submanifolds*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.3 (A calculation to keep in view)**.** Let $P(t) = a+bt+ct^2$ with $c \neq 0$. At scale T write $t = Ts$. Then

$$P(Ts) = a + (bT)s + (cT^2)s^2.$$

If $|c|T^2 \asymp 1$ while $|b|T \to 0$ and $|a| \to 0$, the first visible rescaled polynomial is cs^2T^2. On a mesoscopic window $s = s_0 + \ell/T$ with $|\ell| = o(T)$,

$$P(Ts) - P(Ts_0) = P'(Ts_0)\ell + c\ell^2 = o(1)$$

whenever the normalized derivative is controlled. This is the toy calculation behind the separation of a macroscopic visibility scale from a mesoscopic almost-constant scale in effective linearization and closing.

CHAPTER 1

Expanding horospherical subgroups

This chapter isolates the expanding horospherical subgroups mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Expanding horospherical subgroups: structural mechanism)**.** *For a semisimple one-parameter subgroup a_t, the expanding horospherical group $U^+ = \{u : a_{-t}ua_t \to e\}$ is connected nilpotent and $\mathrm{Ad}(a_t)$ expands every positive root space exponentially.*

PROOF. Use the root-space decomposition of $\mathfrak{g}$ for a_t. The positive weight spaces form a nilpotent Lie algebra closed under brackets because weights add. Exponentiation identifies it with U^+, and conjugation multiplies a vector of weight $\lambda > 0$ by $e^{\lambda t}$. $\square$

THEOREM 0.2 (Quantitative expansion on a positive-weight cone). *Write $a_t = \exp(tH)$ and let*

$$\mathfrak{u}^+ = \bigoplus_{\alpha(H)>0} \mathfrak{g}_\alpha.$$

Fix a finite set A_0 of positive roots and a compact set $K \subset \exp(\bigoplus_{\alpha\in A_0} \mathfrak{g}_\alpha)$ contained in an exponential coordinate chart. Put $\lambda_0 = \min_{\alpha\in A_0} \alpha(H) > 0$. Then, for $Z = \sum_{\alpha\in A_0} Z_\alpha$ in the corresponding logarithmic compact set,

$$\mathrm{Ad}(a_t)Z = \sum_{\alpha\in A_0} e^{\alpha(H)t} Z_\alpha.$$

In particular every nonzero root component grows at least as $e^{\lambda_0 t}$, and any fixed nontrivial root-coordinate diameter of K grows by at least a constant multiple of $e^{\lambda_0 t}$ under $u \mapsto a_t u a_{-t}$ while the image stays in the chosen coordinate comparison regime.

PROOF. The root-space identity $[H, Z_\alpha] = \alpha(H)Z_\alpha$ gives

$$e^{t\,\mathrm{ad}\,H} Z_\alpha = e^{\alpha(H)t} Z_\alpha.$$

Since $\mathrm{Ad}(a_t) = e^{t\,\mathrm{ad}\,H}$, summing over the finitely many roots in A_0 proves the displayed formula. The lower bound follows from $\alpha(H) \geq \lambda_0$ and uniform equivalence of the coordinate norm and the ambient metric on the fixed compact exponential chart. $\square$

Proposition 0.3 (Expansion for a commuting split cone)**.** *Let $A \simeq \mathbb{R}^r$ be a commuting split subgroup with Lie algebra $\mathfrak{a}$, and let $H \in \mathfrak{a}$. For every finite family of roots A_0 satisfying $\alpha(H) > 0$ for all $\alpha \in A_0$, the conclusion of Theorem 0.2 holds for $a_t = \exp(tH)$ with*

$$\lambda_0 = \min_{\alpha \in A_0} \alpha(H) > 0.$$

More generally, the same constants are uniform for H in a compact subset of the open cone $\{H \in \mathfrak{a} : \alpha(H) > 0 \text{ for all } \alpha \in A_0\}$.

Proof. The commuting split action gives the simultaneous weight decomposition

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \bigoplus_{\alpha} \mathfrak{g}_\alpha, \qquad \mathrm{Ad}(\exp(tH)) Z_\alpha = e^{t\alpha(H)} Z_\alpha.$$

For a compact subset of the indicated open cone, the continuous function $H \mapsto \min_{\alpha \in A_0} \alpha(H)$ has a positive minimum. Apply the proof of Theorem 0.2 with this common lower bound. $\square$

CHAPTER 2

Analytic curves in expanding horospheres

This chapter isolates the analytic curves in expanding horospheres mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Analytic curves in expanding horospheres: structural mechanism)**.** *Let $\varphi : I \to U^+$ be analytic. On a subinterval where the first nonzero unstable jet has order k, the rescaling $s = s_0+e^{-\lambda t/k}r$ makes $a_t\varphi(s)\varphi(s_0)^{-1}a_{-t}$ converge on compact r-sets to a polynomial unipotent curve.*

Proof. Taylor expand in exponential coordinates. The jth root-weight component is multiplied by $e^{\lambda_j t}$; choose the scale from the first component that does not vanish. Higher Taylor remainders gain an additional $e^{-\lambda t/k}$ power, while lower vanished jets contribute zero. BCH errors have strictly higher weighted degree and vanish on compact r-sets. □

Theorem 0.2 (Unipotent invariance from the leading jet). *Let $J \subset I$ be a compact interval on which the leading unstable jet has constant order k and limiting Lie-algebra direction $0 \neq Z \in \mathfrak{u}^+$. For $x \in G/\Gamma$ put*

$$\mu_t = \frac{1}{|J|}\int_J \delta_{a_t\varphi(s)x}\, ds.$$

Assume $t_n \to \infty$ and $\mu_{t_n} \Rightarrow \nu$. Then ν is invariant under the one-parameter Ad-unipotent subgroup

$$U_Z = \{\exp(rZ) : r \in \mathbb{R}\}.$$

Proof. Fix $r \in \mathbb{R}$ and $f \in C_c(G/\Gamma)$. Partition J into microintervals whose length is comparable to the rescaling $e^{-\lambda t/k}$ from Lemma 0.1, discarding at most two boundary pieces. Their total discarded relative length tends to zero. On each retained piece, Lemma 0.1 and the change of parameter $s = s_0 + e^{-\lambda t/k}u$ give, uniformly on compact u-ranges,

$$a_t\varphi(s_0 + e^{-\lambda t/k}(u + r))x = \exp(rZ)\, a_t\varphi(s_0 + e^{-\lambda t/k}u)x + o(1)$$

in local coordinates, with the harmless higher-order polynomial terms absorbed into the limiting jet parametrization. Uniform continuity of f on the compact set meeting its support therefore gives

$$\int f\,d\mu_t - \int f(\exp(rZ)y)\,d\mu_t(y) \longrightarrow 0.$$

Passing to the subsequence t_n yields

$$\int f\,d\nu = \int f(\exp(rZ)y)\,d\nu(y).$$

Since r was arbitrary, ν is U_Z-invariant. □

Proposition 0.3 (Finite jet stratification on a compact analytic interval)**.** *Let $I \subset \mathbb{R}$ be compact and let φ extend analytically to a neighborhood of I. Fix finitely many analytic jet minors $F_1, \ldots, F_M$ obtained from derivatives of φ up to order k. Then I admits a finite partition into points and open intervals on each of which the vanishing pattern of the F_j is constant. In particular, the order of the first nonzero unstable jet takes only finitely many values on the nondegenerate strata.*

Proof. For each j, either F_j vanishes identically on a connected analytic component or its zeros are isolated. Because F_j is analytic on a neighborhood of the compact interval I, an isolated zero set in I is finite. Taking the common refinement of the finitely many zero sets gives a finite partition on which all vanishing patterns, and hence the first nonzero jet order, are constant. □

CHAPTER 3

Nondegeneracy conditions

This chapter isolates the nondegeneracy conditions mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Nondegeneracy conditions: structural mechanism)**.** *A curve is representation-nondegenerate if no nonzero rational Chevalley covector annihilates all of its unstable jets. In the standard* SL_{k+1} *model this is equivalent to the original Euclidean curve not being contained in a proper affine subspace.*

Proof. A rational Chevalley covector vanishing on all jets gives an analytic scalar function with every derivative zero, hence an identically vanishing algebraic relation. Conversely an affine relation gives such a covector in the standard exterior representation. □

Theorem 0.2 (Nondegeneracy transfer). *Representation-nondegeneracy is open on compact parameter pieces after deleting finitely many lower-dimensional analytic strata.*

Proof. The relevant finite family of determinant minors is continuous and nonzero on the retained compact set, hence has a positive minimum. □

Proposition 0.3 (Uniform nondegeneracy under compact left/right multiplication)**.** *Let* $K_1, K_2 \subset G$ *be compact. Suppose a retained compact parameter set* $J \subset I$ *satisfies the Chapter 3 nondegeneracy bound*

$$\max_{1\le j\le M} |F_j(s)| \ge c > 0 \qquad (s \in J)$$

for the finite family of Chevalley jet minors F_j*. Then there exists* $c' = c'(c, K_1, K_2) > 0$ *such that every curve* $s \mapsto k_1\varphi(s)k_2$*,* $k_i \in K_i$*, satisfies the corresponding transformed bound with* c' *on* J*.*

Proof. In every fixed finite-dimensional representation, left and right multiplication by k_1, k_2 act by invertible linear maps on the finite jet data. Compactness gives a common bound for the operator norms of these maps and their inverses. Hence a vector of jet minors with norm at least c is sent to one with norm at least $c' > 0$, uniformly in $k_i \in K_i$. $\square$

CHAPTER 4

Limits of expanding translates

Fix the standing arithmetic quotient $X = G/\Gamma$, a semisimple one-parameter subgroup a_t, its expanding horospherical subgroup U^+, an analytic curve $\varphi : I \to U^+$, and $x \in X$. For a compact interval $J \subset I$ define

$$\mu_{t,J} = \frac{1}{|J|} \int_J \delta_{a_t\varphi(s)x}\, ds.$$

After the finite jet stratification of Chapter 2, let J be a retained nondegenerate stratum as in Chapter 3. Write H_J for the connected subgroup generated by all one-parameter unipotent groups obtained from the nonzero limiting jet shears on J. Let $Y = L_0 x$ denote the minimal closed finite-volume homogeneous subspace containing $a_t\varphi(J)x$ for all sufficiently large t.

Lemma 0.1 (Homogeneous decomposition of weak limits)**.** *The family $\{\mu_{t,J} : t \geq 0\}$ is tight. If $t_n \to \infty$ and $\mu_{t_n,J} \Rightarrow \nu$, then ν is H_J-invariant and admits an ergodic decomposition*

$$\nu = \int m_{L_\omega y_\omega}\, d\lambda(\omega),$$

where for λ-almost every ω, $L_\omega < G$ is connected, $L_\omega y_\omega$ is closed of finite volume, and $H_J \leq L_\omega$.

Proof. Quantitative nondivergence from Volume 1 applies on each polynomialized jet chart supplied by Chapter 2. Since the compact interval J has only finitely many jet strata, summing the corresponding exceptional-measure bounds gives tightness. Chapter 2 shows invariance of every weak limit under each one-parameter unipotent shear produced by a nonzero limiting jet. Invariance under the connected subgroup generated by those shears is then invariance under H_J.

Disintegrate ν into ergodic components for the action generated by the one-parameter subgroups defining H_J. Ratner's internally proved measure-classification theorem applies componentwise: almost every component is the invariant probability $m_{L_\omega y_\omega}$ on a closed finite-volume orbit of a connected subgroup L_ω containing the unipotent groups that preserve that component. Hence $H_J \leq L_\omega$ for almost every ω. $\square$

Theorem 0.2 (Support containment on the unfocused branch). *Assume that no proper connected subgroup $L < L_0$ with a closed finite-volume orbit in Y satisfies the Chevalley focusing relation of Volume 42, Chapter 4 for the curve $a_t\varphi(J)x$. Then every homogeneous ergodic component $m_{L_\omega y_\omega}$ of every weak limit in Lemma 0.1 has support equal to Y. Equivalently, $L_\omega y_\omega = Y$ for λ-almost every ω.*

Proof. Suppose instead that a component m_{Ly} has proper support $Ly \subsetneq Y$. The Chevalley compatibility theorem of Volume 42, Chapter 3 associates to L a Chevalley line whose defining polynomial coordinates are fixed by the limiting jet shear. Because this component occurs with positive component measure, the linearization lemma of Volume 42, Chapter 4 converts recurrent positive-proportion visits to a small tube of Ly into the global Chevalley focusing relation for L. Since Ly is a proper homogeneous subspace of Y, we may replace L by its connected stabilizer inside L_0 and obtain a proper connected subgroup of L_0. This contradicts the unfocused hypothesis. Therefore no proper component can occur, and every component has support Y. □

Proposition 0.3 (Haar conclusion when the minimal closure is the full space)**.** *Under the hypotheses of Theorem 0.2, if $Y = X$ (equivalently $L_0 = G$), then every weak limit of $\mu_{t,J}$ is the Haar probability m_X.*

Proof. Let ν be a weak limit. By Lemma 0.1 it decomposes into homogeneous ergodic components, and by Theorem 0.2 almost every component has support $Y = X$. The unique G-invariant probability on $X = G/\Gamma$ is Haar measure m_X, so almost every component equals m_X. Integrating the constant ergodic decomposition gives $\nu = m_X$. □

CHAPTER 5

Focusing and exceptional algebraic loci

This chapter isolates the focusing and exceptional algebraic loci mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Focusing and exceptional algebraic loci: structural mechanism)**.** *If an expanding curve fails to equidistribute, then a positive-measure parameter set lies in an analytic algebraic focusing locus determined by a proper Chevalley stabilizer.*

Proof. Choose a proper homogeneous component in a weak limit. The linearization theorem of Volume 42 converts positive tube occupation into an algebraic relation. Pulling this relation back along the analytic curve gives finitely many analytic equations. □

Theorem 0.2 (Nullity of a proper analytic focusing locus). *Let $\varphi : I \to U^+$ be representation-nondegenerate. For a proper Chevalley stabilizer type L, let*

$$E_L = \{s \in I : F_{L,1}(s) = \cdots = F_{L,r_L}(s) = 0\}$$

be the focusing locus obtained by pulling the defining Chevalley relations back along φ. If L is proper, representation-nondegeneracy implies that at least one $F_{L,j}$ is not identically zero; consequently E_L has one-dimensional Lebesgue measure zero.

Proof. Choose j with $F_{L,j} \not\equiv 0$. A nonzero real-analytic function on an interval has a discrete zero set, hence a null zero set. Since $E_L \subseteq \{F_{L,j} = 0\}$, one has $|E_L| = 0$. □

Proposition 0.3 (Finite unions of focusing loci remain null)**.** *If $L_1, \ldots, L_M$ are proper focusing types of bounded complexity and E_{L_i} are their focusing loci, then*

$$\left| \bigcup_{i=1}^{M} E_{L_i} \right| = 0.$$

Therefore restricting normalized Lebesgue measure on I to the complement of this union does not change any limiting parameter average.

Proof. Theorem 0.2 gives $|E_{L_i}| = 0$ for every i. Finite subadditivity gives $|\bigcup_i E_{L_i}| \leq \sum_i |E_{L_i}| = 0$. Removing a null set does not change the integral of any bounded measurable parameter observable. $\square$

CHAPTER 6

Equidistribution of expanding pieces

This chapter isolates the equidistribution of expanding pieces mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Equidistribution of expanding pieces: structural mechanism)**.** *Let φ be representation-nondegenerate and let the minimal finite-volume homogeneous closure be Y. Then $(a_t\varphi)_*(ds/|I|) \to m_Y$.*

PROOF. Let $t_j \to \infty$. Chapter 4 gives tightness of the translated parameter measures, hence a weakly convergent subsequence. The analytic rescaling of Chapter 2 gives invariance of the limit under the leading unipotent directions on every jet stratum. Ratner classification therefore decomposes the limit into homogeneous probabilities.

If one of these probabilities were supported on a proper homogeneous subspace of Y, the linearization argument of Volume 42 would put a positive portion of the original analytic curve in the corresponding focusing locus. Chapter 5 shows that this locus has parameter measure zero under representation nondegeneracy. Hence no proper component occurs. The limit is m_Y. Since t_j was arbitrary, all subsequential limits agree and the entire family converges. $\square$

THEOREM 0.2 (Full-space equidistribution of an expanding analytic piece). *Let $I \subset \mathbb{R}$ be compact, let $\varphi : I \to U^+$ be analytic, and let a_t be the expanding one-parameter subgroup fixed in Chapter 1. Assume the representation-nondegeneracy condition of Chapter 3 and assume that the minimal finite-volume homogeneous closure of $\varphi(I)x$ is all of G/Γ. Then for every $f \in C_c(G/\Gamma)$,*

$$\frac{1}{|I|}\int_I f(a_t\varphi(s)x)\,ds \longrightarrow \int_{G/\Gamma} f\,dm_{G/\Gamma} \qquad (t \to \infty). \tag{43.6.1}$$

PROOF. Apply Lemma 0.1. Its limiting homogeneous space is Y; under the present hypothesis $Y = G/\Gamma$, so $m_Y = m_{G/\Gamma}$ and (43.6.1) follows. $\square$

Proposition 0.3 (Equidistribution for bounded absolutely continuous parameter densities)**.** *Under the hypotheses of Theorem 0.2, let* $w : I \to [0, \infty)$ *be bounded, Riemann integrable (equivalently here, with a Lebesgue-null discontinuity set), and normalized by* $\int_I w(s)\, ds = 1$*. Then for every* $f \in C_c(G/\Gamma)$*,*

$$\int_I w(s) f(a_t \varphi(s) x)\, ds \longrightarrow \int_{G/\Gamma} f\, dm_{G/\Gamma}.$$

Proof. Given $\varepsilon > 0$, choose upper and lower step functions $w_\pm$ with $0 \le w_- \le w \le w_+$ and $\int_I (w_+ - w_-) < \varepsilon$. Apply Theorem 0.2 separately on the finitely many subintervals on which $w_\pm$ are constant. Their limiting integrals are $(\int w_\pm)\, m(f)$. The difference between the w-average and either step-function average is at most $\|f\|_\infty \varepsilon$. Let $t \to \infty$ and then $\varepsilon \to 0$. $\square$

CHAPTER 7

Quantitative precursor estimates

This chapter isolates the quantitative precursor estimates mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Quantitative precursor estimates: structural mechanism)**.** *Assume effective nondivergence, effective mixing, and a quantitative lower bound $\geq \kappa$ for the nondegeneracy minors. Then after smoothing at scale ε, the discrepancy of an expanding curve is bounded by the sum of cusp loss, focusing-tube loss, Taylor loss, and ε^{-A} times the ambient mixing rate.*

Proof. Partition into jet charts. Discard cusp and focusing exceptional parameters using the two quantitative hypotheses. On each remaining chart replace the curve by its leading polynomial jet, paying the Taylor error. Thicken transversely at scale ε and apply ambient mixing; Sobolev norms grow by ε^{-A}. Sum the bounded-overlap charts. $\square$

Theorem 0.2 (Polynomial rate obtained from the precursor ledger). *Assume that for every $T \geq 2$ and $0 < \varepsilon \leq 1$ the discrepancy of a smooth expanding-curve average satisfies*

$$\mathrm{Err}(T, \varepsilon) \leq C\mathcal{S}_d(f)\left(T^{-\eta} + \varepsilon^{\beta} + T^{-\sigma}\varepsilon^{-A}\right)$$

for fixed constants $\eta, \beta, \sigma, A > 0$. Then with

$$\varepsilon = T^{-\sigma/(A+\beta)} \quad \textit{and} \quad \delta = \min\left\{\eta, \frac{\sigma\beta}{A+\beta}\right\} > 0$$

one has

$$\mathrm{Err}(T, \varepsilon) \leq 3C\, T^{-\delta}\mathcal{S}_d(f).$$

Proof. The chosen ε gives

$$\varepsilon^{\beta} = T^{-\sigma\beta/(A+\beta)} \quad \text{and} \quad T^{-\sigma}\varepsilon^{-A} = T^{-\sigma\beta/(A+\beta)}.$$

The remaining term is $T^{-\eta}$. Both exponents are at least δ, so the displayed bound follows by summing the three terms. $\square$

Proposition 0.3 (Uniform-family consequence)**.** *Let a family of expanding curves satisfy the estimate of Theorem 0.2 with one common constant C, Sobolev order d, and parameters obeying*

$$\eta \geq \eta_0 > 0, \qquad \beta \geq \beta_0 > 0, \qquad \sigma \geq \sigma_0 > 0, \qquad A \leq A_0.$$

Then the family has a uniform polynomial equidistribution exponent

$$\delta_0 = \min\left\{\eta_0, \frac{\sigma_0\beta_0}{A_0 + \beta_0}\right\} > 0.$$

Proof. For each member apply Theorem 0.2. Since $\sigma\beta/(A+\beta)$ is increasing in σ and β and decreasing in A, its value is at least $\sigma_0\beta_0/(A_0+\beta_0)$. Hence every member admits the common exponent δ_0. $\square$

CHAPTER 8

Applications to curves and submanifolds

This chapter isolates the applications to curves and submanifolds mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Applications to curves and submanifolds: structural mechanism)**.** *For the standard* SL_{k+1} *lattice model, an analytic curve not contained in a proper affine subspace has expanding translates equidistributing in the Haar component, hence satisfies the corresponding almost-everywhere Diophantine non-improvability conclusions.*

Proof. Chapter 3 verifies nondegeneracy, Chapter 6 gives equidistribution, and the homogeneous-space coding of the Diophantine condition converts the Haar frequency statement to the arithmetic assertion. □

Theorem 0.2 (Slice-to-submanifold transfer). *Let M be a smooth parameter manifold equipped locally with a measure absolutely continuous with respect to product coordinates $(u,t) \in U \times I$. Suppose that for almost every $u \in U$ the one-dimensional slice $t \mapsto \phi(u,t)$ is analytic and satisfies the nondegeneracy hypotheses of the curve equidistribution theorem of Chapter 6, with the same target Haar component. Then the expanding translates of $\phi(u,t)$ are Haar generic for almost every (u,t) in the chart. The same conclusion holds globally after a countable chart cover.*

Proof. For almost every fixed u, the curve theorem gives a full-measure subset $I'_u \subset I$ on which the required genericity conclusion holds. The set of failures

$$E = \{(u,t) : t \notin I'_u\}$$

is measurable after testing genericity against a fixed countable dense family of compactly supported continuous observables. Fubini gives

$$\int_U |E_u| \, du = 0,$$

so E has product measure zero. Absolute continuity transfers this null statement to the given chart measure. A smooth manifold admits a countable atlas, and a countable union of null exceptional sets is null, proving the global statement. □

Downstream interface. This final chapter supplies the downstream interface recorded in the theorem above. No additional proposition is needed beyond the theorem above.

Volume 44

Effective Horocycles and Affine Lattices

Volume 44 scope and prerequisites

The concrete effective endpoint is the space of affine unimodular lattices. The polynomial effective-equidistribution statement is proved in the smooth-test-function/Diophantine scope used later by combining base mixing, fiber Fourier truncation, rational-obstruction separation, cusp moments, and an explicit smoothing balance. Kim's theorem is retained for comparison and attribution, not as a proved input.

Proof and provenance. Primary comparison: Wooyeon Kim's polynomially effective equidistribution of expanding translates in affine-lattice spaces. The proof below is internal in the stated smooth-test-function scope; stronger variants are bibliographic context unless separately proved.

Reader guide: a concrete model

Why this matters.

Volume 44 is organized around the passage from *Effective horocycle equidistribution* to *Applications to directions, gaps, and fine statistics*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.3 (A calculation to keep in view)**.** Let $P(t) = a+bt+ct^2$ with $c \neq 0$. At scale T write $t = Ts$. Then

$$P(Ts) = a + (bT)s + (cT^2)s^2.$$

If $|c|T^2 \asymp 1$ while $|b|T \to 0$ and $|a| \to 0$, the first visible rescaled polynomial is cs^2T^2. On a mesoscopic window $s = s_0 + \ell/T$ with $|\ell| = o(T)$,

$$P(Ts) - P(Ts_0) = P'(Ts_0)\ell + c\ell^2 = o(1)$$

whenever the normalized derivative is controlled. This is the toy calculation behind the separation of a macroscopic visibility scale from a mesoscopic almost-constant scale in effective linearization and closing.

CHAPTER 1

Effective horocycle equidistribution

This chapter isolates the effective horocycle equidistribution mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Effective horocycle equidistribution: structural mechanism)**.** *On a finite-volume rank-one homogeneous space with spectral gap, a smooth compact horocycle segment thickened transversely at radius ε has discrepancy at time t bounded by $C\mathcal{S}_D(f)(e^{-\eta t}\varepsilon^{-A} + \varepsilon^{\kappa} + E_{\mathrm{cusp}})$.*

Proof. This is the thickening calculation owned in AHD VII: construct a bump of mass one on the transverse box, compare the singular orbit integral with the thickened integral by wavefront control, apply effective mixing to the latter, and bound boundary/cusp pieces separately. □

Theorem 0.2 (Effective horocycle equidistribution). *Assume the discrepancy estimate of Lemma 0.1:*

$$|A_t f - m(f)| \le C\mathcal{S}_D(f)\big(e^{-\eta t}\varepsilon^{-A} + \varepsilon^{\kappa} + E_{\mathrm{cusp}}(t)\big) \qquad (0 < \varepsilon < 1).$$

Then with

$$\varepsilon_t = e^{-\eta t/(A+\kappa)}$$

one has

$$|A_t f - m(f)| \le C'\mathcal{S}_D(f)\big(e^{-\eta\kappa t/(A+\kappa)} + E_{\mathrm{cusp}}(t)\big).$$

If the first term in Lemma 0.1 is instead $t^{-\eta}\varepsilon^{-A}$, then choosing $\varepsilon_t = t^{-\eta/(A+\kappa)}$ gives the polynomial rate $t^{-\eta\kappa/(A+\kappa)}$ plus the same cusp error.

Proof. For the exponential case substitute $\varepsilon = e^{-ct}$. The two smoothing terms have exponents $-\eta + Ac$ and $-\kappa c$. Equality of these exponents gives $c = \eta/(A+\kappa)$, hence both terms are $e^{-\eta\kappa t/(A+\kappa)}$. The cusp error is unaffected. The polynomial case is identical after writing $\varepsilon = t^{-c}$ and balancing $-\eta + Ac = -\kappa c$. □

Proposition 0.3 (Uniform effective horocycle constants in a quotient family)**.** *Let $\{X_q\}$ be a family of finite-volume rank-one quotients for which Lemma 0.1 holds with common exponents η, A, κ, a common Sobolev order*

D, and a common constant C, and suppose the cusp errors $E_{\mathrm{cusp},q}(t)$ satisfy one uniform bound. Then Theorem 0.2 holds for every X_q with the same optimized exponent

$$\eta_* = \frac{\eta\kappa}{A+\kappa}$$

and with a constant C' independent of q.

Proof. The optimization in Theorem 0.2 uses only the common numbers C, η, A, κ, D and the common cusp bound. Substituting the same $\varepsilon_t = e^{-\eta t/(A+\kappa)}$ (or its polynomial analogue) for every q therefore gives one C' and the same exponent η_* throughout the family. $\square$

CHAPTER 2

Affine lattice spaces

This chapter isolates the affine lattice spaces mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Affine lattice spaces: structural mechanism)**.** *The affine lattice space is $X_{\mathrm{aff}} = \mathrm{ASL}_d(\mathbb{R})/\mathrm{ASL}_d(\mathbb{Z})$ and fibers over $X_d = \mathrm{SL}_d(\mathbb{R})/\mathrm{SL}_d(\mathbb{Z})$ with torus fiber $\mathbb{R}^d/L$. Haar measure disintegrates as Haar on X_d times normalized Haar on each torus fiber.*

Proof. Write $\mathrm{ASL}_d = \mathrm{SL}_d \ltimes \mathbb{R}^d$. Right multiplication by the integer affine lattice identifies the translation coordinate modulo L. Fubini on the semidirect product Haar measure gives the disintegration. □

Theorem 0.2 (Sobolev norm adapted to base and fiber directions). *Choose bases $Y_1, \ldots, Y_{d^2-1}$ of $\mathfrak{sl}_d$ and $V_1, \ldots, V_d$ of the translation algebra $\mathbb{R}^d$. For every integer $D \geq 0$, the norm*

$$\mathcal{S}_D(f)^2 := \sum_{|\alpha|+|\beta|\leq D} \left\|Y^\alpha V^\beta f\right\|^2_{L^2(X_{\mathrm{aff}})}$$

defines a Sobolev norm on $C_c^\infty(X_{\mathrm{aff}})$. It controls derivatives in the base X_d through the Y_i and derivatives along each torus fiber through the V_j. In fiber Fourier coordinates, V^β multiplies the mode $m \in L^*$ by a polynomial of degree $|\beta|$ in m, so $\mathcal{S}_D$ controls the weighted Fourier coefficients used in Chapter 3.

Proof. The semidirect-product decomposition gives

$$\mathfrak{asl}_d = \mathfrak{sl}_d \ltimes \mathbb{R}^d.$$

The left-invariant differential operators generated by the chosen basis span all differential operators of order at most D up to bounded linear combinations on each bounded-geometry chart, which gives the standard Sobolev norm equivalence. On the torus $\mathbb{R}^d/L$, differentiating $e^{2\pi i\langle m,\xi\rangle}$ in the translation direction V_j multiplies it by $2\pi i\langle m, V_j\rangle$. Repeating this identity gives the stated polynomial Fourier weight. □

Proposition 0.3 (Affine cusp height is a base quantity)**.** *Let* $\mathrm{ht}(L)$ *be the lattice height used in the quantitative nondivergence chapters, and define*

$$\mathrm{ht}_{\mathrm{aff}}(L,\xi) := \mathrm{ht}(L) \qquad ((L,\xi) \in X_{\mathrm{aff}}).$$

Then $\mathrm{ht}_{\mathrm{aff}}$ *is proper modulo the compact torus fibers, and for every affine translate* (L,ξ) *its cusp height is exactly the height of the base lattice. In particular all cusp-tail bounds depending only on short nonzero lattice vectors transfer from* X_d *to* X_{aff} *with the same exponent.*

Proof. The fiber $\mathbb{R}^d/L$ is compact. Translating the origin changes the marked affine point but does not change the underlying difference lattice L, its covolume, or the lengths of its nonzero vectors. Hence every height defined from the successive minima (or the equivalent exterior-power covolume data used earlier in the series) depends only on L. Properness on the total space is therefore properness of the base height together with compactness of each fiber, and the tail statement is immediate from the identity $\mathrm{ht}_{\mathrm{aff}} = \mathrm{ht} \circ \pi$. $\square$

CHAPTER 3

Fourier analysis on affine homogeneous spaces

This chapter isolates the fourier analysis on affine homogeneous spaces mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Fourier analysis on affine homogeneous spaces: structural mechanism)**.** *For a smooth f on X_{aff}, fiberwise Fourier expansion gives $f(L,\xi) = \sum_{m\in L^*} \widehat{f}_m(L)e^{2\pi i\langle m,\xi\rangle}$, and repeated fiber differentiation yields $|\widehat{f}_m(L)| \ll_D (1+\|m\|)^{-D}\mathcal{S}_D(f)$.*

Proof. This is ordinary Fourier analysis on the compact torus $\mathbb{R}^d/L$. Integrate by parts in the direction $m/\|m\|$ repeatedly. On a bounded-height base set the fiber metrics are uniformly equivalent; outside it use the weighted Sobolev norm of AHD VI. □

Theorem 0.2 (Fourier analysis on affine homogeneous spaces). *The zero Fourier mode is a function on X_d and is governed by ordinary homogeneous mixing; all arithmetic difficulty lies in controlling nonzero modes.*

Proof. The zero coefficient is precisely the fiber average. The remaining series has zero fiber mean. □

Proposition 0.3 (Fourier-tail bound)**.** *Let $D > d$. On a bounded-height base set, if the fiber Fourier coefficients satisfy the estimate of Lemma 0.1, then uniformly in the base lattice L,*

$$\sum_{m\in L^*:\,\|m\|>M} |\widehat{f}_m(L)| \ll_{d,D} M^{d-D}\mathcal{S}_D(f) \qquad (M \geq 1).$$

Consequently truncating the fiber Fourier expansion to $\|m\| \leq M$ changes f by $O(M^{d-D}\mathcal{S}_D(f))$ in the sup norm on that bounded-height set.

Proof. By Lemma 0.1, the tail is bounded by

$$\mathcal{S}_D(f) \sum_{\substack{m\in L^*\setminus\{0\}\\ \|m\|>M}} (1+\|m\|)^{-D}.$$

On a bounded-height subset of X_d, the dual lattices have uniformly bounded geometry, so the number of $m \in L^*$ in the shell $2^j M < \|m\| \leq 2^{j+1} M$ is $O_d((2^j M)^d)$. Hence the last sum is

$$\ll_{d,D} \sum_{j \geq 0} (2^j M)^d (2^j M)^{-D} = M^{d-D} \sum_{j \geq 0} 2^{-j(D-d)} \ll_{d,D} M^{d-D},$$

where convergence uses $D > d$. Taking absolute values term by term gives the asserted sup-norm truncation error. $\square$

CHAPTER 4

Rational obstructions

This chapter isolates the rational obstructions mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Rational obstructions: structural mechanism)**.** *A nonzero fiber frequency can fail to oscillate along an expanding affine translate only when the translation parameter is close to a rational relation of denominator comparable to that frequency.*

Proof. The phase is $\langle m, a_t\xi(s)\rangle$. If its derivative is bounded below, integration by parts gives decay. If all relevant derivatives are small, the coefficient vector imposes a near-integer linear relation on the translation data after pulling back by the integer lattice. This is exactly the rational obstruction. □

Theorem 0.2 (Rational obstructions). *For a Diophantine translation vector of exponent κ, the bad frequencies up to size M have total contribution bounded by a negative power of the expansion parameter after choosing M polynomially.*

Proof. The Diophantine lower bound prevents the phase derivative from being too small except at frequencies beyond the chosen cutoff. Sum integration-by-parts bounds over the remaining frequencies and absorb the tail by Chapter 3. □

Proposition 0.3 (Rational affine data produce a nonoscillating character)**.** *Let $\xi \in \mathbb{R}^d/\mathbb{Z}^d$ be rational, of order q, and let the affine dynamics act on the fiber by integer matrices preserving the finite subgroup $q^{-1}\mathbb{Z}^d/\mathbb{Z}^d$. Then the fiber orbit of ξ is finite. Consequently there is a nonzero character $m \in \mathbb{Z}^d$ and an infinite return subsequence on which the phase $e^{2\pi i\langle m,\xi_t\rangle}$ is constant; hence no oscillatory decay is possible along that subsequence.*

Proof. The rational point ξ lies in the finite group $q^{-1}\mathbb{Z}^d/\mathbb{Z}^d$, which is preserved by the integer action, so its orbit is finite. Some orbit value occurs infinitely often. Choose a nonzero character of the finite group that is constant on that repeated value. Along the corresponding return subsequence the phase is constant, so the associated Fourier mode has no cancellation. $\square$

CHAPTER 5

Diophantine type and cusp excursions

This chapter isolates the diophantine type and cusp excursions mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Diophantine type and cusp excursions: structural mechanism)**.** *A Diophantine lower bound for the affine translation controls both nonzero Fourier resonances and the depth of the affine orbit in rational cusp neighborhoods.*

PROOF. A deep rational cusp excursion gives a short affine relation (m, n) after applying the lattice transformation. Pulling it back yields a rational approximation to the translation vector. The Diophantine inequality bounds how deep this can occur at a given time. ☐

THEOREM 0.2 (Diophantine type and cusp excursions). *Conversely exceptionally good rational approximation produces a large nonzero Fourier mode or a deep cusp excursion along a suitable diagonal time.*

PROOF. Choose the time that balances the expanding and contracting coordinates, exactly as in the Dani correspondence. ☐

Proposition 0.3 (One Diophantine exponent controls Fourier resonances and cusp truncation)**.** *Assume the translation vector ξ satisfies*

$$\|\langle m, \xi\rangle\|_{\mathbb{R}/\mathbb{Z}} \geq c\|m\|^{-\kappa} \qquad (0 \neq m \in \mathbb{Z}^d).$$

Then every nonzero Fourier denominator in Chapter 4 and every rational affine relation governing the cusp-height estimate in Lemma 0.1 obeys the same power κ (with possibly different multiplicative constants). Thus the Fourier cutoff and cusp truncation may be chosen using one common Diophantine exponent κ.

PROOF. Both mechanisms reduce to the same integer linear form $\langle m, \xi\rangle - n$. The displayed Diophantine inequality bounds this quantity below by $c\|m\|^{-\kappa}$. Chapter 4 uses that bound for phase derivatives, while Lemma 0.1 uses it after pulling a short affine cusp relation back to the original coordinates.

Only multiplicative constants change, so the exponent is the same in both ledgers. $\square$

CHAPTER 6

Effective affine-lattice equidistribution

This chapter isolates the effective affine-lattice equidistribution mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Effective affine-lattice equidistribution: structural mechanism)**.** *For the expanding affine translates in the smooth Diophantine scope above, the average of $f \in C_c^\infty(X_{\mathrm{aff}})$ differs from Haar by $O(T^{-\delta}\mathcal{S}_D(f))$ for some $\delta > 0$ depending explicitly on the spectral gap and Diophantine exponent.*

Proof. Write the fiber expansion from Chapter 3 as $f = f_0 + \sum_{m\neq 0} f_m$. The zero mode f_0 descends to the ordinary lattice space. Chapter 1 therefore gives

$$|A_T f_0 - \int f_0\, dm| \ll T^{-\eta}\mathcal{S}_D(f).$$

Choose a Fourier cutoff $M = T^\theta$. The sum of modes $\|m\| > M$ is $O(M^{-b}\mathcal{S}_{D+b+d}(f))$ by repeated fiber integration by parts.

For $0 < \|m\| \leq M$, split the parameter interval into the Diophantine good set and the rational-resonant exceptional set of Chapter 4. On the good set, some derivative of the phase is bounded below by $T^\gamma\|m\|^{-A}$; van der Corput/integration by parts gives $O(T^{-\gamma'}\|m\|^{A'})$ for that mode. The bad set has measure $O(T^{-\gamma''}\|m\|^{A''})$ by the Diophantine estimate and Chapter 5. Summing the finitely many modes produces

$$T^{-\delta_1}M^C\mathcal{S}_{D'}(f).$$

Thus the total error is bounded by the maximum of $T^{-\eta}$, $T^{-\delta_1}M^C$, and M^{-b}. Taking $\theta = \delta_1/(C+b)$ makes the last two equal and gives a positive polynomial exponent. This is the stated effective theorem; the precise exponent ledger is isolated in Chapter 7. ☐

Theorem 0.2 (Effective affine-lattice equidistribution with polynomial height). *Assume the hypotheses of Lemma 0.1. Let f be a smooth observable satisfying, for some fixed $B, D_0 \geq 0$,*

$$|\nabla^j f(x)| \leq C_f\ \mathrm{ht}(x)^B, \qquad 0 \leq j \leq D_0. \tag{44.6.1}$$

Assume the cusp moment estimate $m\{\mathrm{ht} > R\} \ll R^{-\kappa}$ and that truncation at height R costs at most R^A in the Sobolev norm used in Lemma 0.1. Then there is $\delta' > 0$, depending only on the fixed exponents $\delta, \kappa, A, B, D_0$, such that

$$\left| A_T f - \int f \, dm \right| \ll C_f T^{-\delta'}. \tag{44.6.2}$$

Proof. Choose a smooth cutoff χ_R equal to 1 on $\{\mathrm{ht} \le R\}$ and supported in $\{\mathrm{ht} \le 2R\}$. The cusp moment and (44.6.1) give a discarded contribution $O(R^{-\kappa'})$ for some $\kappa' > 0$ after reducing κ by the fixed growth exponent B. Lemma 0.1 applied to $f_R = \chi_R f$ gives

$$|A_T f_R - \int f_R dm| \ll T^{-\delta} R^A C_f.$$

Set $R = T^\alpha$ with $0 < \alpha < \delta/A$. The two errors are $T^{-\alpha\kappa'}$ and $T^{-\delta+A\alpha}$; hence (44.6.2) holds with $\delta' = \min(\alpha\kappa', \delta - A\alpha) > 0$. $\square$

Proposition 0.3 (Dimension-wise positive exponent)**.** *Fix a dimension d for which the base spectral estimate, the Diophantine exceptional-set bound, and the cusp moment used in Chapters 1–5 hold with positive exponents. Then the exponent δ' in Theorem 0.2 can be chosen strictly positive and uniformly for test functions in a bounded set of the fixed Sobolev/height class.*

Proof. All losses entering Lemma 0.1 and Theorem 0.2 are powers with fixed exponents once d and the Sobolev order are fixed. The Fourier cutoff is chosen with $0 < \theta < \delta_1/C$, and the cusp cutoff with $0 < \alpha < \delta/A$. The minimum of the finitely many positive residual exponents is therefore positive and is uniform on a bounded test-function set. $\square$

CHAPTER 7

Explicit error terms

This chapter isolates the explicit error terms mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Explicit error terms: structural mechanism)**.** *If the base mixing gives $T^{-\eta}$, Fourier truncation costs M^{-b}, and the finite nonzero modes cost $T^{-\gamma}M^A$, then $M = T^{\gamma/(A+b)}$ gives total error $O(T^{-\min(\eta,\gamma b/(A+b))})$.*

PROOF. Balance $T^{-\gamma}M^A = M^{-b}$ and compare the result with the zero-mode term. $\square$

THEOREM 0.2 (Two-parameter error optimization). *Assume that after the Fourier cutoff has been optimized as in Lemma 0.1, the remaining error has the form*

$$E(T,R) \le C\big(T^{-\delta_0}R^D + R^{-c}\big), \qquad \delta_0, c > 0, \quad D \ge 0.$$

Then the choice

$$R = T^{\delta_0/(D+c)}$$

gives

$$E(T,R) \le 2C\,T^{-\delta_0 c/(D+c)}.$$

Combining this with Lemma 0.1, the final decay exponent is the minimum of the zero-mode exponent and the explicitly balanced Fourier and cusp exponents.

PROOF. Write $R = T^{\beta}$. The two powers of T are

$$T^{-\delta_0 + D\beta} \qquad \text{and} \qquad T^{-c\beta}.$$

Their exponents are equal when $\delta_0 - D\beta = c\beta$, namely at $\beta = \delta_0/(D+c)$. At this value both terms equal $T^{-\delta_0 c/(D+c)}$. If $D = 0$ the same formula gives $\beta = \delta_0/c$ and the first term is already $T^{-\delta_0}$, so the conclusion remains valid. $\square$

Proposition 0.3 (Minimum-exponent rule)**.** *If the error terms entering Chapter 6 are bounded by $C_jT^{-\delta_j}$ for $1 \le j \le r$ with $\delta_j > 0$, then for $T \ge 1$*

$$\sum_{j=1}^{r} C_jT^{-\delta_j} \le \left(\sum_{j=1}^{r} C_j\right) T^{-\delta}, \qquad \delta = \min_{1\le j\le r} \delta_j > 0.$$

Proof. For each j, $\delta_j \ge \delta$ and therefore $T^{-\delta_j} \le T^{-\delta}$ when $T \ge 1$. Summing the r inequalities gives the displayed bound. This step uses no further balancing hypothesis. $\square$

CHAPTER 8

Applications to directions, gaps, and fine statistics

This chapter isolates the applications to directions, gaps, and fine statistics mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Applications to directions, gaps, and fine statistics: structural mechanism)**.** *Let $A \subset X_{\mathrm{aff}}$ be a bounded-continuity set encoding a finite local affine-lattice statistic. Effective equidistribution gives the limiting probability $m_{\mathrm{aff}}(A)$ and a polynomial error after smoothing the indicator at boundary scale ε.*

Proof. Choose smooth majorant/minorant differing only in the ε-boundary of A. Apply Chapter 6 to both. If $m_{\mathrm{aff}}(\partial_\varepsilon A) \ll \varepsilon^c$, balance this boundary error against the Sobolev blowup of the smoothing. $\square$

Theorem 0.2 (Applications to directions, gaps, and fine statistics). *This applies to direction counts, gap events, and finite collision windows whenever their boundary has the stated regularity.*

Proof. Each such event is determined by finitely many lattice points in a bounded region; away from tangencies its boundary is a finite union of smooth hypersurface conditions and hence has polynomial tubular volume. $\square$

Downstream interface. This final chapter supplies the downstream interface recorded in the theorem above. No additional proposition is needed beyond the theorem above.

Volume 45

Quantitative Linearization and Effective Closing

Volume 45 scope and prerequisites

Fix an embedding $\mathbf{G} < \mathrm{SL}_N$ of a connected perfect $\mathbb{Q}$-group, put $G = \mathbf{G}(\mathbb{R})^\circ$, let $\Gamma < G$ be the fixed arithmetic lattice of the series, and let $U = \{u_t = \exp(tZ) : t \in \mathbb{R}\} < G$ be a one-parameter unipotent subgroup. All constants in this volume may depend on this fixed arithmetic datum and on N, but never on the running parameters T, R, τ.

For a connected $\mathbb{Q}$-subgroup $M < G$ of dimension m write $\mathfrak{m}_\mathbb{Z} = \mathfrak{m}_\mathbb{Q} \cap \mathfrak{g}_\mathbb{Z}$, choose a primitive generator $v_M \in \bigwedge^m \mathfrak{g}_\mathbb{Z}$ of $\bigwedge^m \mathfrak{m}_\mathbb{Q}$, and put

$$\mathrm{ht}(M) = \|v_M\|, \qquad \eta_M(g) = (\wedge^m \mathrm{Ad})(g^{-1})v_M.$$

The sign of v_M is irrelevant. A connected $\mathbb{Q}$-subgroup is said to be of *class* $\mathcal{H}$ when its radical equals its unipotent radical. Equivalently, it has no nontrivial connected $\mathbb{Q}$-torus quotient. If L is a connected rational subgroup, $L^\mathcal{H}$ denotes the largest connected normal subgroup of L of class $\mathcal{H}$; concretely it is the connected Zariski closure of the subgroup generated by the unipotent parts of elements of L and by commutators. This is the class for which the arithmetic homogeneous orbits used below have the required finite-volume structure.

For $\tau > 0$ let

$$X_\tau = \left\{ \Gamma g \in \Gamma\backslash G : \min_{0 \neq v \in \mathfrak{g}_\mathbb{Z}} \|\mathrm{Ad}(g)^{-1} v\| \geq \tau \right\}.$$

By the Mahler criterion already proved in Volume 1, X_τ is compact. The projective Grassmannian distance between two r-planes is denoted by d_{Gr} and is measured by unit exterior representatives.

Why this matters. The purpose of these conventions is to prevent three ambiguities that caused problems in earlier drafts. "Height" means the primitive integral Plücker height defined above. A "small displacement of a Lie algebra" is measured by d_{Gr}. The closing theorem produces a class-$\mathcal{H}$ subgroup through the explicit vector η_M, rather than an unnamed "algebraic obstruction.

Reader guide: a concrete model

Worked Example 0.3 (A calculation to keep in view)**.** Let $P(t) = a+bt+ct^2$ with $c \neq 0$. At scale T write $t = Ts$. Then

$$P(Ts) = a + (bT)s + (cT^2)s^2.$$

If $|c|T^2 \asymp 1$ while $|b|T \to 0$ and $|a| \to 0$, the first visible rescaled polynomial is cs^2T^2. On a mesoscopic window $s = s_0 + \ell/T$ with $|\ell| = o(T)$,

$$P(Ts) - P(Ts_0) = P'(Ts_0)\ell + c\ell^2 = o(1)$$

whenever the normalized derivative is controlled. This is the toy calculation behind the separation of a macroscopic visibility scale from a mesoscopic almost-constant scale in effective linearization and closing.

Proof architecture at a glance.

$$\boxed{\text{many controlled returns}} \longrightarrow \boxed{\text{rational subgroup } M} \longrightarrow \boxed{\text{Chevalley line}} \longrightarrow \boxed{\text{closing obstruction}}$$

CHAPTER 1

Arithmetic heights of subgroups

Lemma 0.1 (Primitive Plücker height calculus)**.** *Let $L_1, L_2 \subset \mathbb{Q}^n$ be rational subspaces of fixed dimensions and let H_i be the Euclidean norms of their primitive integral Plücker vectors. There is $C = C(n)$ such that*

$$\mathrm{ht}(L_1 + L_2) + \mathrm{ht}(L_1 \cap L_2) \ll (H_1 H_2)^C. \tag{1.1}$$

For a fixed rational bilinear map $B : \mathbb{Q}^n \times \mathbb{Q}^n \to \mathbb{Q}^n$, the span of $B(L_1, L_2)$ has the same type of polynomial height bound.

Proof. Choose primitive integer basis matrices for L_i in Hermite normal form. Their maximal minors are the Plücker coordinates, so Hadamard's inequality and the Hermite pivot relations bound every basis entry by $H_i^{O_n(1)}$. The sum is the column span of the concatenated basis matrix. The intersection is the kernel of the integer matrix expressing equality of the two linear combinations. In both cases a primitive Plücker vector is formed from fixed-size minors, and Cramer's rule bounds these minors by $(H_1 H_2)^C$. For $B(L_1, L_2)$, apply the fixed bilinear structure constants to the chosen integer bases and repeat the same minor estimate. ☐

Theorem 0.2 (Height of the class-$\mathcal{H}$ core generated by lattice elements). *There are $A, C > 0$, depending only on N and the fixed integral model, such that the following holds. Let $\gamma_1, \ldots, \gamma_k \in \mathrm{SL}_N(\mathbb{Q})$ have rational matrix height at most $R \geq 2$, and let*

$$L = \overline{\langle \gamma_1, \ldots, \gamma_k \rangle}^Z.$$

Then

$$\mathrm{ht}(L^{\mathcal{H}}) \leq C R^A. \tag{1.2}$$

The exponent is independent of k.

Proof. We first reduce to the connected class-$\mathcal{H}$ case. The component group of a rational linear algebraic group generated by rational points has order bounded in fixed matrix dimension once its torus quotient is removed; indeed a finite rational subgroup of GL_N has bounded order after embedding its eigenvalues into number fields of degree at most N. Hence bounded words in the $\gamma_i^{\pm 1}$, of length $O_N(1)$, generate the identity component after passing to $L^{\mathcal{H}}$. Their heights are $R^{O_N(1)}$.

Assume therefore that $L = L^{\mathcal{H}}$ is connected. A class-$\mathcal{H}$ subgroup of SL_N is cut out by equations of degree at most $d = d(N)$: use the adjoint representation for the semisimple quotient and upper-triangular coordinates for the unipotent radical. Let V_d be the finite-dimensional rational vector space of polynomial functions on Mat_N of degree at most d, and let $I_d(L)$ be the subspace of functions in V_d vanishing on L. Put $\gamma_0 = 1$ and, after adjoining inverses, suppose the generating set is inversion invariant. For $m \geq 1$ define

$$\Phi_m : V_d \longrightarrow \mathbb{Q}^{(k+1)^m}, \qquad p \longmapsto \big(p(\gamma_{i_1} \cdots \gamma_{i_m})\big)_{i_1,\ldots,i_m}.$$

Then $\ker \Phi_1 \supseteq \ker \Phi_2 \supseteq \cdots$. Since $\dim V_d = D(N)$, for some $m \leq D(N)$ we have $\ker \Phi_m = \ker \Phi_{m+1}$. Right translation by every generator preserves this kernel, so it equals $\ker \Phi_{m+j}$ for all $j \geq 0$. Zariski density of the generated subgroup in L gives

$$I_d(L) = \bigcap_{j \geq 1} \ker \Phi_j = \ker \Phi_m. \tag{1.3}$$

Every word occurring in Φ_m has height at most $R^{O_N(1)}$; hence the matrix of Φ_m has rational entries of height $R^{O_N(1)}$. Cramer's rule gives a primitive Plücker vector for $I_d(L)$ of height $R^{O_N(1)}$. The Lie algebra of L is obtained by differentiating a basis of $I_d(L)$ at the identity, so another fixed-size determinant calculation gives (1.2).

For a general connected L, write every generator in multiplicative Jordan form $\gamma_i = \gamma_i^s \gamma_i^u$. In fixed dimension the semisimple and unipotent parts are polynomials in γ_i obtained by interpolation on the characteristic polynomial, and their heights are $R^{O_N(1)}$. Let S_1 consist of all γ_i^u and all commutators $[\gamma_i, \gamma_j]$, and set $S_{j+1} = S_j \cup \{[\gamma_i, s] : s \in S_j\}$. If M_j is the connected Zariski closure generated by S_j, then $M_1 \subseteq M_2 \subseteq \cdots$ and the sequence stabilizes after at most $\dim G$ strict dimension increases. At stabilization M_j is normal in L, contains every unipotent part and every commutator, and the quotient L/M_j is a torus. Hence $M_j = L^{\mathcal{H}}$. All elements used have height $R^{O_N(1)}$, so the connected class-$\mathcal{H}$ case just proved applies and finishes the proof. ☐

Proposition 0.3 (Northcott count for subgroup height)**.** *For every $Q \geq 2$, the number of connected rational subgroups $M < G$ with $\mathrm{ht}(M) \leq Q$ is at most CQ^D for constants C, D depending only on $\dim G$ and the fixed integral norm.*

Proof. For each possible dimension r, the primitive Plücker vector $v_M \in \bigwedge^r \mathfrak{g}_{\mathbb{Z}}$ is an integral vector of norm at most Q. The number of such vectors is $O(Q^{\dim \wedge^r \mathfrak{g}})$. A connected algebraic subgroup is determined by its Lie algebra in characteristic zero, so summing over the finitely many r gives the result. $\square$

CHAPTER 2

Quantitative Chevalley representations

The original draft used the phrase "effective Łojasiewicz" without stating or proving the height dependence that the later closing argument needs. We first isolate that finite-dimensional algebraic estimate. Its only role is to convert a system of bounded-degree integer polynomial residuals into a distance from the exact real zero locus.

Lemma 0.1 (Fixed-degree elimination bound)**.** *Fix integers $n, m, d \geq 1$. There are integers $D, C \geq 1$ depending only on (n, m, d) with the following property. Let $f_1, \ldots, f_m \in \mathbb{Z}[x_1, \ldots, x_n]$ have degree at most d and coefficient height at most $H \geq 2$. Every polynomial obtained from the system by eliminating any subset of the variables, after decomposing into the finitely many Jacobian-rank strata, may be chosen with degree at most D and coefficient height at most H^C.*

Proof. Homogenize the f_i by adding one variable. To eliminate one affine variable, take the Sylvester resultant of two generic integer linear combinations of the homogenized equations. Its coefficients are determinants of matrices of size bounded in terms of m, d, n; every matrix entry is either zero or a coefficient of one of the f_i. Hadamard's inequality therefore bounds every resultant coefficient by H^{C_1}, while the determinant size bounds the degree by D_1. If the two chosen combinations have a common factor, adjoin the appropriate minors of their Sylvester matrix; these minors have the same type of degree and height bound and separate the rank strata.

Repeat this operation one variable at a time. The number of variables is n, so the number of determinant operations is bounded solely by (n, m, d). At a singular point we additionally adjoin the minors of the Jacobian matrix $(\partial f_i / \partial x_j)$; differentiation multiplies coefficient height by at most d, and the minors again satisfy Hadamard bounds. Thus after all coordinate projections and all possible Jacobian-rank strata have been treated, every eliminant has degree bounded by one constant D and height bounded by H^C. No running parameter enters C or D. $\square$

Lemma 0.2 (Height-effective real polynomial separation)**.** *Fix n, m, d and $B \geq 2$. There are $A \geq 1$ and $C \geq 1$, depending only on (n, m, d), such that* *the following holds. For polynomials f_i as in Lemma 0.1, let*

$$Z = \{x \in \mathbb{R}^n : f_1(x) = \cdots = f_m(x) = 0\}.$$

For every $x \in [-B, B]^n$,

$$\min\{1, \operatorname{dist}(x, Z)\} \le C(BH)^A \left(\max_i |f_i(x)|\right)^{1/A}, \tag{2.1}$$

where the left side is interpreted as 1 *if $Z = \varnothing$.*

Proof. Put $K = (BH)^{C_0}$, where C_0 is chosen large enough to absorb all coefficient growth below. We separate the proof into the small-s branch and the part bounded away from that branch.

Let $F = \sum_i f_i^2$. If $Z \neq \varnothing$, a point $z \in Z$ nearest to x satisfies, on a stratum where the Jacobian of (f_i) has rank q,

$$f_i(z) = 0, \qquad x - z = \sum_{i=1}^{m} \lambda_i \nabla f_i(z), \qquad r^2 = \|x - z\|^2, \qquad s = F(x). \tag{2.2}$$

Boundary faces of the box are obtained by adjoining equations $x_j = \pm B$; singular strata are obtained by adjoining the appropriate Jacobian minors. Thus the set of possible pairs (r, s) is the projection of finitely many real algebraic sets whose degrees are bounded in terms of (n, m, d) and whose integer coefficient heights are at most K.

Apply Lemma 0.1 to each such system, eliminating all variables except r, s. Every one-dimensional boundary arc of the projected set is contained in an algebraic curve

$$P(r, s) = \sum_{a,b} c_{ab} r^a s^b = 0, \qquad \deg P \le D, \qquad |c_{ab}| \le K^{C_1}. \tag{2.3}$$

We now justify the power estimate on such an arc rather than merely invoking a qualitative curve-selection statement. Divide P by the largest power of s common to all its monomials. Because the exact zero fibre of the distance graph is $(r, s) = (0, 0)$, the resulting polynomial is not divisible by s and its specialization $P(r, 0)$ has no sufficiently small nonzero root on the arc. Write

$$P(r, 0) = r^a (c + rQ(r)), \qquad 1 \le a \le D, \qquad c \in \mathbb{Z} \setminus \{0\}.$$

For $0 < r \le (4DK^{C_1})^{-1}$ we have $|c + rQ(r)| \ge \frac{1}{2}$. Since the terms containing s are bounded by $CK^{C_1} s$ on $[0, 1]^2$, the equation $P(r, s) = 0$ gives

$$r^a \le CK^{C_1} s. \tag{2.4}$$

Hence $r \le CK^{C_2} s^{1/D}$ on the small branch.

It remains to treat points with $r \ge (4DK^{C_1})^{-1}$. The critical systems above have bounded degree and height $K^{O(1)}$. Eliminating all variables except the critical value s produces a nonzero integer polynomial $Q_*(s)$ of degree $O(1)$ and height $K^{O(1)}$ for every isolated positive critical value. The elementary Cauchy root bound applied to the reciprocal polynomial gives $|s| \geq K^{-C_3}$ for every nonzero such value. On a positive-dimensional critical-value interval the same conclusion holds at its left endpoint, which is again an algebraic critical value. Consequently

$$r \leq 1 \leq K^{C_3/D} s^{1/D}$$

throughout the complementary part. Taking the maximum over the finitely many rank and boundary strata yields

$$\min\{1, \operatorname{dist}(x, Z)\} \leq C(BH)^A \big(\max_i |f_i(x)|\big)^{1/A}$$

after increasing A. If $Z = \varnothing$, minimize F on the box and apply the same critical-value elimination: its positive minimum is at least K^{-C}, so the convention on the left-hand side gives the same inequality. □

Lemma 0.3 (Finite Chevalley family for class-$\mathcal{H}$ subgroups)**.** *There are two finite collections of integral representations $\rho_j : \mathrm{SL}_N \to \mathrm{SL}(V_j)$ and $\widehat{\rho}_j : \mathrm{SL}_N \to \mathrm{SL}(\widehat{V}_j)$, depending only on N, and constants $A, C > 0$ such that for every connected $\mathbb{Q}$-subgroup $M < \mathrm{SL}_N$ of class $\mathcal{H}$ one can choose primitive integral vectors w_M and $\widehat{w}_M$ satisfying*

$$\operatorname{Stab}_{\mathrm{SL}_N}(\mathbb{R}w_M) = N_{\mathrm{SL}_N}(M), \qquad \operatorname{Stab}_{\mathrm{SL}_N}(\widehat{w}_M) = M, \tag{2.5}$$

and

$$C^{-1}\,\mathrm{ht}(M)^{1/A} \leq \|w_M\| + \|\widehat{w}_M\| \leq C\,\mathrm{ht}(M)^A. \tag{2.6}$$

Proof. Because M is of class $\mathcal{H}$, its defining ideal in the coordinate ring of SL_N is generated in degree bounded solely by N: the unipotent radical is triangularizable and the reductive quotient is semisimple, so the adjoint equations and the finitely many exterior bracket equations have uniformly bounded degree. Let W_d be the integral space of polynomials of degree at most this bound. The subspace $I_d(M) \subset W_d$ of equations vanishing on M determines M. Taking the exterior line $\bigwedge^{\dim I_d(M)} I_d(M)$ gives one of finitely many exterior representations of SL_N; its line stabilizer is the normalizer of M. A primitive integral generator is w_M. For an exact stabilizer, choose a primitive integral basis $e_1, \ldots, e_r$ of $I_d(M)$ and use the direct-sum representation on $V_d^{\oplus r}$ with $\widehat{w}_M = e_1 \oplus \cdots \oplus e_r$. An element fixes $\widehat{w}_M$ exactly when it fixes every equation in $I_d(M)$, hence exactly when it belongs to M. Since $r \leq \dim V_d = O_N(1)$, this still uses only finitely many integral representations.

To compare heights, write a primitive integral basis of $\mathfrak{m}$ in Hermite normal form. The coefficients of the bracket and invariance equations are minors of this basis, hence are bounded by $\mathrm{ht}(M)^{O_N(1)}$. Conversely, $\mathfrak{m}$ is the kernel of the differential at the identity of the equations represented by w_M. Cramer's rule applied to this fixed-size system bounds its primitive Plücker vector by $\|w_M\|^{O_N(1)}$. These two determinant estimates give (2.6). $\square$

THEOREM 0.4 (Quantitative Chevalley separation). *Let M be as above, let $H = \mathrm{ht}(M)$, and choose the Chevalley pair (ρ, w_M) from Lemma 0.3. On every matrix ball $K_B = \{g : |g| \le B\}$ there are constants C, A depending only on N such that*

$$\operatorname{dist}(g, N_G(M)) \le C(BH)^A \left(\frac{\|\rho(g)w_M \wedge w_M\|}{\|\rho(g)w_M\|\,\|w_M\|}\right)^{1/A}. \tag{2.7}$$

If in addition $g \in \mathrm{SL}_N(\mathbb{Q})$ has rational height at most B, then either $g \in N_G(M)$ or

$$\|\rho(g)w_M \wedge w_M\| \ge (BH)^{-A}. \tag{2.8}$$

PROOF. In fixed matrix coordinates, membership in $N_G(M)$ is equivalent to the vanishing of the coordinates of $\rho(g)w_M \wedge w_M$. Their degrees are bounded by the finite representation family, while their coefficient heights are $H^{O(1)}$ by (2.6). Apply Lemma 0.2 on K_B to obtain (2.7).

For rational g, clear the denominators of g and of the primitive integral vector w_M. Every coordinate of $\rho(g)w_M \wedge w_M$ is a rational number with denominator at most $(BH)^{O_N(1)}$. If the wedge is nonzero, one coordinate therefore has absolute value at least $(BH)^{-A}$, proving (2.8). $\square$

Lemma 0.5 (Normalizer-core transfer)**.** *Let $M < G$ be a nontrivial connected $\mathbb{Q}$-subgroup of class $\mathcal{H}$, put $H = \mathrm{ht}(M)$, and suppose $N_G(M) \ne G$. Set*

$$L = N_G(M)^{\mathcal{H}}.$$

Then L is a nontrivial proper connected $\mathbb{Q}$-subgroup of class $\mathcal{H}$ and $\mathrm{ht}(L) \le CH^A$. *Moreover, if $p = \eta_M(h)$, $|p| \le B$, and $Z \in \mathfrak{g}$ is nilpotent with $\|Z\| = 1$, then*

$$\|\eta_L(h)\| \le C(HB)^A \tag{2.9}$$

and

$$\frac{\|Z \wedge \eta_L(h)\|}{\|\eta_L(h)\|} \le C(HB)^A \left(\frac{\|D\rho_M(Z)p \wedge p\|}{\|p\|^2}\right)^{1/A}. \tag{2.10}$$

Here ρ_M denotes the line-Chevalley representation chosen in Lemma 0.3.

PROOF. The Lie algebra of the conjugated normalizer $h^{-1}N_G(M)h$ is the kernel of the linear equations

$$Y \longmapsto D\rho_M(Y)p \wedge p. \tag{2.11}$$

The coefficients of this system are polynomial functions of the coordinates of p, of bounded degree. Cramer's rule therefore bounds a Plücker vector for this kernel by $(HB)^{O_N(1)}$. The class-$\mathcal{H}$ core is obtained from that rational Lie algebra by a bounded sequence of sums, brackets, commutators and removal of the torus quotient, so the height calculus of Lemma 0.1 gives both $\mathrm{ht}(L) \ll H^{O_N(1)}$ and (2.9).

For the transverse estimate, let $\mathfrak{n}_h$ and $\mathfrak{l}_h$ denote the Lie algebras of $h^{-1}N_G(M)h$ and $h^{-1}Lh$. Equation (2.11), quantitative linear algebra, and the lower bound $\|p\| \gg (H|h|)^{-O(1)}$ in any fixed coordinate chart give

$$\mathrm{dist}(Z, \mathfrak{n}_h) \ll (HB)^{O(1)} \frac{\|D\rho_M(Z)p \wedge p\|}{\|p\|^2}. \tag{2.12}$$

It remains to remove the torus directions of the normalizer. Inside $\mathfrak{n}_h$, the exact intersection of the nilpotent cone with the Lie algebra of the torus quotient is zero: an algebraic one-parameter subgroup which is simultaneously unipotent and contained in a torus is trivial. The equations defining $\mathfrak{n}_h$, $\mathfrak{l}_h$, and the nilpotent cone have bounded degree and coefficient height $(HB)^{O(1)}$. Apply Lemma 0.2 to their incidence system. Since Z is exactly nilpotent, (2.12) implies

$$\mathrm{dist}(Z, \mathfrak{l}_h) \ll (HB)^{O(1)} \left(\frac{\|D\rho_M(Z)p \wedge p\|}{\|p\|^2}\right)^{1/O(1)}. \tag{2.13}$$

Finally, Grassmannian distance to $\mathfrak{l}_h$ is comparable, with the same polynomial height loss, to $\|Z \wedge \eta_L(h)\|/\|\eta_L(h)\|$. Enlarging A gives (2.10). $\square$

Proposition 0.6 (Polynomial tube coordinates). *For fixed M and g, every coordinate of $t \mapsto \eta_M(gu_t)$ and of $t \mapsto z \wedge \eta_M(gu_t)$ is a polynomial in t of degree at most a constant depending only on N. If $|g| + \mathrm{ht}(M) \le R^C$, then all coefficients are $R^{O(1)}$.*

PROOF. Since Z is nilpotent, $\mathrm{Ad}(u_{-t}) = \exp(-t\,\mathrm{ad}\,Z)$ is a finite polynomial in t. Exterior powers preserve polynomiality and multiply the degree by at most $\dim G$. The coefficient bound follows from the fixed matrix formulas for the exterior representation and the primitive integral height of v_M. $\square$

CHAPTER 3

Polynomial tube occupation

This chapter isolates the polynomial tube occupation mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Polynomial tube occupation: structural mechanism)**.** *Let $p : I \to V$ be a vector-valued polynomial of degree at most d. If p is not identically in a linear subspace W, then $|\{t \in I : \operatorname{dist}(p(t), W) < \varepsilon \sup_I \|p\|\}| \leq C_d \varepsilon^{1/d}|I|$.*

Proof. Project to a unit covector annihilating W for which the polynomial is nonzero. The scalar sublevel estimate for a degree-d polynomial follows from factorization into at most d roots and comparison with its maximum; covering neighborhoods of the roots gives the exponent $1/d$. ☐

Theorem 0.2 (Polynomial tube occupation). *For a unipotent trajectory in a Chevalley representation, disproportionate occupation of an ε-tube forces the initial Chevalley vector to be polynomially close to the invariant subspace.*

Proof. The representation coordinates of u_t are polynomial in t. Apply the lemma to the transverse coordinate functions. ☐

Proposition 0.3 (Finite-union polynomial tube occupation bound)**.** *Let $p : I \to V$ have degree at most d, and let $W_1, \dots, W_M \subset V$ be linear subspaces such that p is not identically contained in any W_j. Then*

$$\left|\left\{t \in I : \min_{1\leq j\leq M} \operatorname{dist}(p(t), W_j) < \varepsilon \sup_I \|p\|\right\}\right| \leq MC_d\varepsilon^{1/d}|I|.$$

The same estimate applies, in a fixed Chevalley chart, to a union of M subgroup tubes.

Proof. For each j, Lemma 0.1 bounds the W_j-tube occupation by $C_d\varepsilon^{1/d}|I|$. The bad set for the union is the union of these M bad sets, so the union bound gives $MC_d\varepsilon^{1/d}|I|$. ☐

CHAPTER 4

Effective singular-set avoidance

For $Q \geq 2$ let $\mathscr{M}(Q)$ be the set of connected rational class-$\mathcal{H}$ subgroups $M < G$ with $\mathrm{ht}(M) \leq Q$. By Chapter 1,

$$|\mathscr{M}(Q)| \ll Q^{C_0}. \tag{4.1}$$

For $M \in \mathscr{M}(Q)$ and parameters $B, \varepsilon > 0$ define the Chevalley tube along a unipotent segment by

$$\mathcal{T}_M(B,\varepsilon) = \left\{ t \in [-B,B] : \max_{\|z\|\leq 1} \|z \wedge \eta_M(gu_t)\| < \varepsilon \sup_{|s|\leq B} \|\eta_M(gu_s)\| \right\}.$$

Lemma 0.1 (Single-subgroup tube occupation)**.** *There are $C, \alpha > 0$ depending only on N such that for every $M \in \mathscr{M}(Q)$ one of the following holds:*

(1) every transverse polynomial $t \mapsto z \wedge \eta_M(gu_t)$ vanishes identically, equivalently U normalizes the exterior line of M along the segment;

(2)

$$|\mathcal{T}_M(B,\varepsilon)| \leq C\varepsilon^{\alpha} B. \tag{4.2}$$

Proof. Choose a basis $z_1, \ldots, z_q$ of $\mathfrak{u}$ and a fixed coordinate basis in the relevant exterior space. By Proposition 0.6, all scalar coordinates of $z_j \wedge \eta_M(gu_t)$ are polynomials of degree at most $d = d(N)$. If all are zero, branch (1) holds. Otherwise choose a nonzero coordinate $p(t)$ whose supremum is maximal up to the fixed coordinate dimension. Membership in $\mathcal{T}_M$ implies $|p(t)| \leq C\varepsilon \sup|p|$. The scalar polynomial sublevel estimate of Volume 1 therefore gives $|\mathcal{T}_M| \leq C\varepsilon^{1/d}B$; take $\alpha = 1/d$ after changing C. $\square$

Theorem 0.2 (Finite-complexity singular-set avoidance). *Fix $a > 0$. There is $b = b(a,N) > 0$ with the following property. Put $Q = T^a$ and $\varepsilon = T^{-b}$. Outside the union of those subgroups $M \in \mathscr{M}(Q)$ for which branch (1) of Lemma 0.1 holds, the total parameter set lying in any M-tube satisfies*

$$\left| \bigcup_{M \in \mathscr{M}(Q)} \mathcal{T}_M(T,\varepsilon) \right| \ll T^{1-b\alpha+aC_0}. \tag{4.3}$$

In particular, choosing $b > (aC_0 + c)/\alpha$ makes the right side $O(T^{1-c})$.

Proof. Use (4.1) and sum the single-subgroup estimate (4.2). No conversion from "near an algebraic variety" to an exact rational point occurs here: the exact alternatives are precisely the subgroups for which the transverse polynomial vanishes identically. $\square$

Proposition 0.3 (Cusp and singular avoidance may be intersected)**.** *Let $x \in X_\tau$. If Volume 1 quantitative nondivergence removes a set of times of measure at most T^{1-c_1} from $[-T, T]$, and Theorem 0.2 removes at most T^{1-c_2}, then the simultaneous good set has complement of measure at most $2T^{1-\min(c_1,c_2)}$.*

Proof. The complement of the intersection is the union of the two exceptional sets. The displayed estimate is the union bound, with the smaller power saving. $\square$

CHAPTER 5

Almost-invariant Lie algebras

This chapter replaces the former "rational reconstruction" paragraph. The old argument incorrectly passed from a small real polynomial residual to a nearby rational point. What closing actually needs is different: bounded lattice elements which almost preserve a Lie subalgebra cannot generate all of G unless the subalgebra is quantitatively close to an ideal. That is a real-algebraic statement, and the height dependence is supplied by Lemma 0.2.

Lemma 0.1 (Almost invariance forces a nearby ideal)**.** *Let $\mathfrak{h} < \mathfrak{g}$ be an r-dimensional Lie subalgebra and let $\widehat{v}_{\mathfrak{h}}$ be its unit Plücker vector. There are $A, C > 0$ such that, for $R \geq 2$ and $\gamma_1, \ldots, \gamma_k \in \Gamma$ with $|\gamma_i| \leq R$, if*

$$d_{\mathrm{Gr}}(\mathrm{Ad}(\gamma_i)\mathfrak{h}, \mathfrak{h}) \leq \delta \quad (1 \leq i \leq k) \tag{5.1}$$

and $\langle \gamma_1, \ldots, \gamma_k \rangle$ is Zariski dense in G, then one of the following holds:

(1) $\delta \geq C^{-1}R^{-A}$;

(2) there is an r-dimensional Lie ideal $\mathfrak{h}' \lhd \mathfrak{g}$ with

$$d_{\mathrm{Gr}}(\mathfrak{h}, \mathfrak{h}') \leq CR^A\delta^{1/A}. \tag{5.2}$$

Proof. Choose the affine Grassmannian chart on which one Plücker coordinate of $\widehat{v}_{\mathfrak{h}}$ has absolute value at least the reciprocal of the fixed number of coordinates. Write a basis of the corresponding r-plane as columns $w_1(z), \ldots, w_r(z)$, polynomial/rational in chart coordinates z with a fixed nonvanishing denominator.

The statement that the plane is a Lie subalgebra is equivalent to the vanishing of the minors

$$[w_a(z), w_b(z)] \wedge w_1(z) \wedge \cdots \wedge w_r(z) = 0. \tag{5.3}$$

The statement that it is invariant under every $\mathrm{Ad}(\gamma_i)$ is equivalent to

$$\mathrm{Ad}(\gamma_i)w_a(z) \wedge w_1(z) \wedge \cdots \wedge w_r(z) = 0. \tag{5.4}$$

After clearing the fixed chart denominator, (5.3)–(5.4) form a polynomial system of bounded degree whose integer coefficient height is $R^{O(1)}$. Because $\mathfrak{h}$ already satisfies (5.3), hypothesis (5.1) says that all residuals of (5.4) at its chart point are $O(R^{O(1)}\delta)$; the factor $R^{O(1)}$ comes only from converting Fubini–Study distance to wedge coordinates.

Apply Lemma 0.2. Unless $\delta \geq C^{-1}R^{-A}$, there is a solution z' of (5.3)–(5.4) satisfying (5.2). Let $\mathfrak{h}'$ be the corresponding plane. It is a Lie subalgebra by (5.3) and is invariant under every $\mathrm{Ad}(\gamma_i)$ by (5.4). Polynomiality of the adjoint action implies that the stabilizer of $\mathfrak{h}'$ is Zariski closed. Since it contains the Zariski-dense group generated by the γ_i, it is all of G. Differentiating the invariance $\mathrm{Ad}(G)\mathfrak{h}' = \mathfrak{h}'$ at the identity gives $[\mathfrak{g}, \mathfrak{h}'] \subset \mathfrak{h}'$, i.e. $\mathfrak{h}'$ is an ideal. □

Lemma 0.2 (Isolation of equal-dimensional ideals in the semisimple quotient)**.** *Let $\mathfrak{s}$ be the fixed semisimple quotient of $\mathfrak{g}$, equipped with the norm induced from the fixed matrix model. There is $c_* > 0$ such that if $\mathfrak{a} < \mathfrak{s}$ is a Lie subalgebra and $\mathfrak{j} \lhd \mathfrak{s}$ is an ideal with $\dim \mathfrak{a} = \dim \mathfrak{j}$ and $d_{\mathrm{Gr}}(\mathfrak{a}, \mathfrak{j}) < c_*$, then $\mathfrak{a} = \mathfrak{j}$.*

Proof. Choose the complementary ideal $\mathfrak{j}^{\perp}$ obtained by summing the simple factors not contained in $\mathfrak{j}$, so $\mathfrak{s} = \mathfrak{j} \oplus \mathfrak{j}^{\perp}$. For sufficiently small Grassmannian distance the projection $\pi_{\mathfrak{j}}|_{\mathfrak{a}}$ is an isomorphism. Hence $\mathfrak{a}$ is the graph of a linear map

$$\phi : \mathfrak{j} \longrightarrow \mathfrak{j}^{\perp}, \qquad \|\phi\| \ll d_{\mathrm{Gr}}(\mathfrak{a}, \mathfrak{j}).$$

Because both summands are ideals and the graph is a Lie subalgebra, ϕ is a Lie homomorphism. Restrict ϕ to a simple factor $\mathfrak{q}$ of $\mathfrak{j}_{\mathbb{C}}$. If that restriction is nonzero, it is injective. The set of nonzero Lie homomorphisms from the finitely many simple factors of $\mathfrak{s}_{\mathbb{C}}$ into the complementary factors has operator norm bounded below by a positive constant. To see this directly, otherwise take $\phi_n \neq 0$ with $\|\phi_n\| \to 0$ and normalize $\psi_n = \phi_n/\|\phi_n\|$. After passing to a subsequence, $\psi_n \to \psi$ with $\|\psi\| = 1$, while

$$\psi_n([X, Y]) = \|\phi_n\|[\psi_n(X), \psi_n(Y)] \longrightarrow 0.$$

Since $[\mathfrak{q}, \mathfrak{q}] = \mathfrak{q}$, this forces $\psi = 0$, a contradiction. Thus $\phi = 0$ once its norm is below that fixed lower bound, and therefore $\mathfrak{a} = \mathfrak{j}$. □

Theorem 0.3 (Quantitative non-ideal gap for perfect G). *Assume G is perfect and $\mathfrak{h} + \mathrm{rad}(\mathfrak{g})$ is not a proper Lie ideal of $\mathfrak{g}$. Then there are $A, C > 0$ such that any Zariski-dense finite set $\gamma_i \in \Gamma$, $|\gamma_i| \leq R$, satisfies*

$$\max_i d_{\mathrm{Gr}}(\mathrm{Ad}(\gamma_i)\mathfrak{h}, \mathfrak{h}) \geq C^{-1}R^{-A}. \tag{5.5}$$

Proof. Suppose the left side is $\delta < C^{-1}R^{-A}$ with A large enough for Lemma 0.1. That lemma gives an ideal $\mathfrak{h}'$ of the same dimension satisfying (5.2). Let $\pi : \mathfrak{g} \to \overline{\mathfrak{g}} = \mathfrak{g}/\mathrm{rad}(\mathfrak{g})$ be the quotient map. On the fixed

Grassmannian chart containing $\mathfrak{h}$ and $\mathfrak{h}'$, projection is Lipschitz with a constant depending only on the ambient datum, so

$$d_{\mathrm{Gr}}(\pi\mathfrak{h}, \pi\mathfrak{h}') \ll R^A \delta^{1/A}. \tag{5.6}$$

The subspace $\pi\mathfrak{h}'$ is an ideal of the semisimple Lie algebra $\overline{\mathfrak{g}}$. If the right side of (5.6) is smaller than the constant in Lemma 0.2, that lemma applied after restricting to the common-dimensional image chart shows that $\pi\mathfrak{h} = \pi\mathfrak{h}'$. (If the image dimension dropped under the perturbation, a maximal minor of $\pi|_{\mathfrak{h}}$ would cross zero; the fixed nonzero minor on the chosen chart gives an even larger fixed lower bound.)

By hypothesis $\mathfrak{h} + \mathrm{rad}(\mathfrak{g})$ is not a proper ideal. Since its image equals the ideal $\pi\mathfrak{h}'$, that image must be all of $\overline{\mathfrak{g}}$. Hence $\mathfrak{h}'$ surjects onto the semisimple quotient and contains a Levi subalgebra $\mathfrak{l}$.

Let $\mathfrak{u}_0 = \mathrm{rad}(\mathfrak{g})$ and $\mathfrak{u}_{j+1} = [\mathfrak{u}_0, \mathfrak{u}_j]$. The radical is nilpotent because G is perfect. From $\mathfrak{h}' + \mathfrak{u}_0 = \mathfrak{g}$ and the ideal property we obtain inductively

$$\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}] \subset [\mathfrak{h}', \mathfrak{g}] + [\mathfrak{u}_j, \mathfrak{h}'] + [\mathfrak{u}_j, \mathfrak{u}_j] \subset \mathfrak{h}' + \mathfrak{u}_{j+1}.$$

The lower central series terminates, so $\mathfrak{h}' = \mathfrak{g}$, contrary to $\dim \mathfrak{h}' = \dim \mathfrak{h} < \dim \mathfrak{g}$. Therefore our assumption on δ was impossible; after enlarging A, C we obtain (5.5). □

Proposition 0.4 (Properness of a lattice-generated return group)**.** *Under the hypotheses of Theorem 0.3, let $F \subset \Gamma$ consist of elements of norm at most R satisfying $d_{\mathrm{Gr}}(\mathrm{Ad}(\gamma)\mathfrak{h}, \mathfrak{h}) \le R^{-B}$. If $B > A + 1$, then the Zariski closure of $\langle F \rangle$ is proper in G.*

Proof. If the closure were all of G, a finite subset of F would already generate a Zariski-dense subgroup: the descending chain of common polynomial fixed spaces stabilizes after finitely many steps. Theorem 0.3 would then give $R^{-B} \ge C^{-1} R^{-A}$, impossible for large R. The bounded range of smaller R is absorbed by increasing the ambient constant. □

CHAPTER 6

Effective closing

We now state the one-parameter pair-return theorem in the exact form used by Volumes 46–50. It is deliberately narrower than the most general published effective-closing theorem: the ambient arithmetic datum is fixed and only the one-parameter unipotent action is needed here.

Lemma 0.1 (Density points at finitely many polynomial scales). *Let $E \subset [-T, T]$ be measurable with $|E| = 2Tp$, $0 < p \leq 1$, and let $1 \leq L_0 < \cdots < L_m \leq T/4$. There is a measurable subset $E_0 \subset E$ with*

$$|E_0| \geq c_m p^{m+1} |E| \tag{6.1}$$

such that for every $w \in E_0$ and every $0 \leq i \leq m$,

$$|E \cap [w - L_i, w + L_i]| \geq c_m p L_i. \tag{6.2}$$

Here $c_m > 0$ depends only on m.

Proof. For one scale L, partition $[-T, T]$ into intervals I_j of length L (up to two boundary intervals). If $a_j = |E \cap I_j|$, then

$$\sum_j a_j = |E|, \qquad \sum_j a_j^2 \geq \frac{|E|^2}{\#\{j\}} \geq c\frac{|E|^2 L}{T}. \tag{6.3}$$

For $w \in E \cap I_j$, the interval $[w - 2L, w + 2L]$ contains I_j; hence

$$\int_E |E \cap [w - 2L, w + 2L]| \, dw \geq \sum_j a_j^2 \geq cpL|E|. \tag{6.4}$$

The integrand is at most $4L$. Therefore the set on which it is at least $(c/2)pL$ has measure at least $c'p|E|$. Replacing $2L$ by L only changes the numerical constants by beginning with a partition of length $L/2$. Thus one scale costs at most one factor $c'p$. Apply this argument successively inside the surviving subset for the $m + 1$ prescribed scales; after adjusting constants, (6.1)–(6.2) follow. □

Lemma 0.2 (Polynomially bounded integral points in a torus). *For fixed d there is C_d such that, for every $\mathbb{Q}$-torus $T < \mathrm{GL}_d$ occurring as a quotient of* *a subgroup of the fixed ambient G and every $Q \geq 2$,*

$$\#\{\gamma \in T(\mathbb{Z}) : \ |\gamma| \leq Q\} \leq C_d(1+\log Q)^{C_d}. \tag{6.5}$$

PROOF. Diagonalize T over a number field of degree bounded in terms of d. Every integral point has eigenvalues which are algebraic units of bounded degree. The logarithms of their archimedean absolute values lie in the Dirichlet unit lattice of rank at most d^2 and, from $|\gamma| \leq Q$, in a box of side $O(\log Q)$. A lattice has at most a constant times the volume of such a box plus its boundary in that box. The finitely many roots of unity and the bounded number of splitting fields occurring in fixed dimension are absorbed into C_d. □

THEOREM 0.3 (Pair-return effective closing, fixed arithmetic scope). *There exist constants $A_1, A_2 > 1$ and $C > 1$ with the following property. Let $\mathfrak{h} < \mathfrak{g}$ be a proper Lie subalgebra such that $\mathrm{Ad}(U)\mathfrak{h} = \mathfrak{h}$ and $\mathfrak{h} + \mathrm{rad}(\mathfrak{g})$ is not a proper ideal. Let $0 < \tau < 1$ and let*

$$T > R > C\tau^{-A_1}, \qquad x = \Gamma g \in X_\tau.$$

Assume there is a measurable set $E \subset [-T, T]$ with

$$|E| > TR^{-1/A_1} \tag{6.2a}$$

such that for every $s, t \in E$ there is $\gamma_{s,t} \in \Gamma$ satisfying

$$q_{s,t} := u_{-s}g^{-1}\gamma_{s,t}gu_t, \qquad |q_{s,t}| \leq R^{1/A_1}, \tag{6.3a}$$

and

$$d_{\mathrm{Gr}}(\mathrm{Ad}(q_{s,t})\mathfrak{h}, \mathfrak{h}) \leq R^{-1}. \tag{6.4a}$$

Then at least one of the following holds.

(1) There is a nontrivial proper connected $\mathbb{Q}$-subgroup $M < G$ of class $\mathcal{H}$ such that, for every $|t| \leq T$,

$$\|\eta_M(gu_t)\| \leq R^{A_2}, \qquad \|Z \wedge \eta_M(gu_t)\| \leq T^{-1/A_2}R^{A_2}. \tag{6.5a}$$

(2) There is a nontrivial proper connected normal $\mathbb{Q}$-subgroup $M \lhd G$ of class $\mathcal{H}$, containing the radical of G, such that

$$\|Z \wedge v_M\| \leq R^{-1/A_2}. \tag{6.6}$$

PROOF. We divide the proof into six quantitative steps. The role of each exponent is visible; no appeal to an external closing theorem remains.

Step 1: remove deep cusp times and recenter. Write $p = |E|/(2T) > \frac{1}{2}R^{-1/A_1}$. Apply the polynomial unipotent quantitative-nondivergence theorem of Volume 1 with cusp threshold $R^{-\varepsilon}$, where $\varepsilon = 4/(\alpha A_1)$ and $\alpha > 0$ is its fixed goodness exponent. The exceptional set has measure $\ll R^{-4/A_1}T <$

$|E|/4$ once R exceeds an ambient constant. Replace E by the remaining set. Thus, for every $w \in E$,

$$xu_w \in X_{R^{-\varepsilon}}. \tag{6.7}$$

Quantitative Mahler reduction gives $\gamma_w^0 \in \Gamma$ for which

$$g_w := \gamma_w^0 g u_w, \qquad |g_w| \le R^{C\varepsilon}. \tag{6.8}$$

Choose A_1 so large that every occurrence of $C\varepsilon$ below is smaller than $1/20$.

Step 2: density at nested local scales. Let $m = \dim G + 1$. Choose fixed exponents

$$0 < \lambda_0 < \lambda_1 < \cdots < \lambda_m < 1/(20C_*)$$

with successive gaps large compared with $1/A_1$, and put $L_i = R^{\lambda_i}$. Since $T > R$ and the $\lambda_i < 1$, all $L_i < T/4$ for large R. Lemma 0.1 gives a subset E_0 of measure $\ge R^{-C/A_1}|E|$ satisfying (6.2) at every L_i.

Fix $w \in E_0$. For $t \in E \cap [w - L_i, w + L_i]$ define

$$\widetilde{\gamma}_{w,t} = \gamma_w^0 \gamma_{w,t} (\gamma_w^0)^{-1} \in \Gamma.$$

Using (6.3a) and the identity

$$g_w^{-1} \widetilde{\gamma}_{w,t} g_w = q_{w,t} u_{w-t}, \tag{6.9}$$

we obtain $|\widetilde{\gamma}_{w,t}| \le R^{C(\lambda_i + \varepsilon + 1/A_1)}$. Because $\mathrm{Ad}(U)\mathfrak{h} = \mathfrak{h}$, $\mathrm{Ad}(u_{w-t})\mathfrak{h} = \mathfrak{h}$; hence (6.4a) and (6.9) give, for $\mathfrak{h}_w = \mathrm{Ad}(g_w)\mathfrak{h}$,

$$d_{\mathrm{Gr}}(\mathrm{Ad}(\widetilde{\gamma}_{w,t})\mathfrak{h}_w, \mathfrak{h}_w) \le R^{-1+C\varepsilon}. \tag{6.10}$$

The conjugation loss is only $R^{C\varepsilon}$ because g_w, not gu_w, is used.

Step 3: the return-generated groups are proper and of controlled height. Let $L_i(w)$ be the Zariski closure of the subgroup generated by the $\widetilde{\gamma}_{w,t}$ with $t \in E \cap [w - L_i, w + L_i]$. If $L_i(w) = G$, a finite subcollection already generates a Zariski-dense group. Apply Theorem 0.3 to $\mathfrak{h}_w$. Its hypotheses are conjugation invariant, while the right side of (6.10) is smaller than the lower bound (5.5) once the λ_i and $1/A_1$ are sufficiently small. This contradiction proves

$$L_i(w) < G \qquad (0 \le i \le m). \tag{6.11}$$

Theorem 0.2, applied to the bounded lattice generators, gives

$$\mathrm{ht}(L_i(w)^{\mathcal{H}}) \le R^{C\lambda_i + C/A_1}. \tag{6.12}$$

The nested sets of generators give $L_0(w) \subseteq \cdots \subseteq L_m(w)$. Since there are $m+1$ proper subgroups but only $\dim G$ possible strict dimension increases, choose $i = i(w) < m$ with $L_i(w)^\circ = L_{i+1}(w)^\circ$. Denote this connected group by L_w and its class-$\mathcal{H}$ core by $M_w = L_w^{\mathcal{H}}$. The density lower bound (6.2) and $L_0 = R^{\lambda_0}$ with $\lambda_0 > 2/A_1$ give at least two return times separated by a fixed positive amount. If M_w were trivial, L_w° would be a torus. But the relative elements in (6.9) approach a nonconstant segment of the unipotent one-parameter group as the pair separation varies, whereas a torus contains no nontrivial unipotent element. Quantitative Chevalley separation makes this contradiction exact for A_1 large. Thus M_w is nontrivial as well as proper.

Step 4: stabilize one rational subgroup on many base points. By Proposition 0.3 and (6.12), only R^C possibilities for M_w occur. Since there are only m possible indices, there are a subgroup M and a subset $E_1 \subset E_0$ with

$$|E_1| \geq TR^{-C} \tag{6.13}$$

for which $M_w = M$. For $w \in E_1$, every generator appearing at the next scale normalizes M. Choose one such generator with parameter t separated from w by at most L_{i+1} and use

$$\widetilde{\gamma}_{w,t} g_w u_{t-w} = g_w q_{w,t}. \tag{6.14}$$

Since $\widetilde{\gamma}_{w,t} \in N_G(M)$, left multiplication by it preserves the Chevalley line of M. Hence

$$\|\eta_M(g_w u_{t-w})\| = \|\eta_M(g_w q_{w,t})\| \leq R^C. \tag{6.15}$$

Here we used (6.12), (6.8), and the polynomial norm of the fixed Chevalley representation. Taking $t = w$ in the same bounded-representative comparison gives

$$\|\eta_M(g u_w)\| \leq R^C \qquad (w \in E_1). \tag{6.16}$$

We now separate the non-normal and normal cases.

Step 5a: the non-normal case. Assume $N_G(M) \neq G$ and set $L = N_G(M)^{\mathcal{H}}$. By Lemma 0.5, L is nontrivial, proper and of height R^C. The coordinates of $p(t) = \eta_M(g u_t)$ are polynomials of bounded degree. The Remez inequality applied to (6.13)–(6.16) yields

$$\sup_{|t| \leq T} \|p(t)\| \leq R^C. \tag{6.17}$$

The same bound holds for the inverse coordinate changes on the finite family of projective charts, so $\|p(t)\| \geq R^{-C}$ after normalizing the primitive Chevalley vector. Markov's inequality for a degree-$d(N)$ polynomial on $[-T, T]$ gives

$$\sup_{|t| \leq T} \|p'(t)\| \leq CT^{-1}R^C. \tag{6.18}$$

Since $p'(t) = -D\rho_M(Z)p(t)$, Lemma 0.5 applied pointwise to $h = gu_t$ gives

$$\|\eta_L(gu_t)\| \leq R^{A_2}, \qquad \|Z \wedge \eta_L(gu_t)\| \leq T^{-1/A_2} R^{A_2} \tag{6.19}$$

for all $|t| \leq T$, after enlarging A_2. This is branch (1), with L in place of the temporary group M.

Step 5b: the normal case. Assume now that $M \triangleleft G$. Use the fixed-vector Chevalley pair $(\widehat{\rho}, \widehat{w}_M)$ of Lemma 0.3. Let V be the M-fixed subspace in the corresponding representation. Since $M = L_i(w)^{\mathcal{H}}$ and $L_i(w)^\circ = L_{i+1}(w)^\circ$, the identity component of the image of $L_{i+1}(w)$ on V is a torus. The generators at scale L_{i+1} have norm R^C, so Lemma 0.2 shows that their restrictions to V take at most $(\log R)^C$ values.

The density estimate (6.2) and $L_{i+1} = R^{\lambda_{i+1}}$ therefore give a subset of parameters of measure $\gg R^{\lambda_{i+1}-C/A_1}(\log R)^{-C}$ on which the restriction is a single value. Fix one base parameter t_0 in that subset. For every other t in it, $\widetilde{\gamma}_{w,t_0}^{-1}\widetilde{\gamma}_{w,t}$ fixes V, hence fixes $\widehat{w}_M$ and belongs to M. Combining the two identities (6.14) gives a polynomial orbit of $\widehat{w}_M$ along u_{t-t_0} which is bounded by R^C on a positive proportion of an interval of length $R^{\lambda_{i+1}}$. Remez extends this bound to the full interval. Lagrange interpolation of the orbit map

$$s \longmapsto \widehat{\rho}(\exp(-sZ))\widehat{p}, \qquad \widehat{p} = \widehat{\rho}(g_w^{-1})\widehat{w}_M,$$

then bounds its first nonconstant coefficient by $R^{C-\lambda_{i+1}}$. Equivalently,

$$\|D\widehat{\rho}(\mathrm{Ad}(g_w)Z)\widehat{w}_M\| \le R^{-c} \tag{6.20}$$

for some $c > 0$, after the scale gaps and A_1 are chosen in that order. The kernel of $Y \mapsto D\widehat{\rho}(Y)\widehat{w}_M$ is exactly $\mathfrak{m}$; Cramer's rule for this integral linear system, together with $\mathrm{ht}(M) \le R^C$, therefore yields

$$\mathrm{dist}(\mathrm{Ad}(g_w)Z, \mathfrak{m}) \le R^{-c/2}. \tag{6.21}$$

Because M is normal, $\mathrm{Ad}(g_w)\mathfrak{m} = \mathfrak{m}$, so (6.21) is equivalent to $\mathrm{dist}(Z, \mathfrak{m}) \le R^{-c/2}$ and hence

$$\|Z \wedge v_M\| \le R^{-1/A_2}. \tag{6.22}$$

If M does not contain the radical, replace it by $M' = M\,R(G)$. This is again a connected normal class-$\mathcal{H}$ subgroup; its height is polynomial in that of M, and $\mathrm{dist}(Z, \mathrm{Lie}\,M') \le \mathrm{dist}(Z, \mathfrak{m})$. It is proper: if $M' = G$, then $\mathfrak{m}$ contains a Levi subalgebra, and the lower-central series argument in the proof of Theorem 0.3 forces $\mathfrak{m} = \mathfrak{g}$, contradicting properness. Replacing M by M' therefore gives branch (2).

Step 6: dependence of the constants. All losses above come from a bounded number of fixed-degree determinant, Remez, Markov, height-calculus, and interpolation estimates. Choose first the scale gaps $\lambda_{i+1} - \lambda_i$, then A_1 large enough to make the cusp and conjugation losses smaller than those gaps, and finally A_2 larger than the finitely many reciprocal exponents produced by Lemma 0.5 and the normal-case interpolation. The threshold $C\tau^{-A_1}$ absorbs the bounded range of small R. This completes the proof. $\square$

Proposition 0.4 (Pair-return output is polynomially complex)**.** *In branch (1) of Theorem 0.3,* $\mathrm{ht}(M) \le R^C$ *for a fixed* C*, and every constant in* (6.5a) *is a fixed power of* (R, τ^{-1})*. No exponential-in-*T *loss occurs.*

Proof. The subgroup is produced in (6.12); every subsequent step uses only a bounded number of determinant operations, commutators, Remez bounds for degree-$d(N)$ polynomials, and a bounded number of scale changes. Each operation changes height by a fixed power. Hence the composition is again polynomial. $\square$

CHAPTER 7

Complexity bounds for homogeneous obstructions

This chapter isolates the complexity bounds for homogeneous obstructions mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Complexity bounds for homogeneous obstructions: structural mechanism)**.** *At a fixed time scale T, every obstruction returned by Chapters 4–6 may be chosen from subgroups of height at most T^C, and there are at most $T^{C'}$ such connected rational subgroups.*

Proof. The height bound is Chapter 6. Primitive Plücker vectors of norm at most T^C are integral points in a ball of fixed dimension, whose number is $O(T^{C'})$. Each subgroup determines one such line. □

Theorem 0.2 (Polynomial union bound for homogeneous obstructions). *Assume that at scale $T \geq 1$ the relevant obstruction family $\mathscr{H}_T$ has cardinality at most $C_0 T^{C'}$, as in Lemma 0.1, and that each $H \in \mathscr{H}_T$ has an exceptional parameter set $E_H(T)$ satisfying*

$$|E_H(T)| \leq C_1 T^{-\eta} |I|$$

inside a parameter interval I. Then

$$\left| \bigcup_{H \in \mathscr{H}_T} E_H(T) \right| \leq C_0 C_1 T^{-\eta + C'} |I|.$$

In particular the union still has polynomial decay whenever $\eta > C'$.

Proof. By subadditivity of measure,

$$\left| \bigcup_{H \in \mathscr{H}_T} E_H(T) \right| \leq \sum_{H \in \mathscr{H}_T} |E_H(T)| \leq |\mathscr{H}_T| \, C_1 T^{-\eta} |I|.$$

Insert $|\mathscr{H}_T| \leq C_0 T^{C'}$. The exponent is positive precisely when the single-obstruction saving η exceeds the counting loss C'. □

Proposition 0.3 (Termination of strict obstruction descent)**.** *Let*

$$G = H_0 \supsetneq H_1 \supsetneq \cdots \supsetneq H_k$$

be a chain of connected Lie subgroups produced by successive strict homogeneous obstructions. Then $k \le \dim G$; in particular, dimension induction terminates after finitely many strict obstruction changes.

Proof. A proper inclusion of connected Lie subgroups gives a proper inclusion of Lie algebras, hence $\dim H_{i+1} \le \dim H_i - 1$. Iterating yields $0 \le \dim H_k \le \dim G - k$, so $k \le \dim G$. $\square$

CHAPTER 8

The quantitative closing dichotomy

Lemma 0.1 (Generic-or-obstructed partition)**.** *Fix $a, c > 0$. For all sufficiently large T, put $Q = T^a$ and choose the tube exponent b as in Theorem 0.2. After deleting a set of times of measure $O(T^{1-c})$, every remaining parameter $t \in [-T, T]$ satisfies one of the following mutually recorded alternatives:*

(1) gu_t lies outside every transverse Chevalley tube attached to $M \in \mathscr{M}(Q)$;

(2) there is a specific $M \in \mathscr{M}(Q)$ for which the exact polynomial obstruction of Lemma 0.1(1) holds.

Proof. Choose b so that (4.3) is $O(T^{1-c})$ and intersect with the cusp-good set from Proposition 0.3. Every surviving tube visit which is not in the union-bound exceptional set must belong to a subgroup for which the transverse polynomial vanishes identically. Recording that subgroup gives branch (2). □

Theorem 0.2 (Quantitative closing dichotomy passed to Volume 46). *Let a long unipotent block satisfy the thick-part, pair-density, and Grassmannian return hypotheses of Theorem 0.3. Then exactly one of the following outcomes is recorded by the iteration used in Volume 46:*

(1) a proper rational class-$\mathcal{H}$ subgroup M of height $T^{O(1)}$ satisfies the orbit-tube estimates (6.5a)*;*

(2) a proper rational normal factor satisfies (6.6)*;*

(3) after removing $O(T^{1-c})$ bad parameters, every relevant nearby pair has a transverse displacement bounded below by T^{-C}, and polynomial shearing supplies a nonzero transverse direction for the invariance-growth argument.

Proof. Apply Theorem 0.3. Its branches are (1) and (2). If neither is present on a candidate block, Theorem 0.2 excludes concentration near every rational obstruction of polynomial complexity. A nearby-pair relative displacement whose transverse coordinate were $< T^{-C}$ for all sufficiently large C would, by quantitative Chevalley separation (2.7) and the pair-return encoding, fall into one of those excluded tubes. Hence some transverse coordinate is at least T^{-C}. The exact polynomial-shear lemma of Volume 9 then rescales its first nonconstant coefficient to macroscopic size, producing branch (3). $\square$

Downstream interface. Volume 46 may use only the three alternatives above. In particular, the phrase “effective closing” never means that an arbitrary near return is silently replaced by a periodic orbit; the subgroup, its height, the normal factor alternative, and the transverse branch are all explicit.

Volume 46

Margulis Functions and Dimension Growth

Volume 46 scope and prerequisites

The scope is the rank-one effective-Ratner toolkit preceding the final equidistribution theorem: contraction of Margulis functions, transverse energy, incidence/projection estimates, dimension growth, and conversion of near-full dimension to almost invariance. These mechanisms are proved in Chapters 1–8 at the exact quantitative scope used by Volume 47. The comparison with Lindenstrauss–Mohammadi–Wang is bibliographic concordance only; no proof step below is citation-owned. No rank-two/general effective Ratner theorem is claimed here.

Proof and provenance. Primary comparison: Lindenstrauss–Mohammadi and Lindenstrauss–Mohammadi–Wang, whose rank-one arguments combine Margulis functions, incidence geometry, projection growth, and spectral gap. Stronger variants are bibliographic context unless separately marked.

Reader guide: a concrete model

Why this matters.

Volume 46 is organized around the passage from *Margulis functions* to *Iterative dimension-growth bootstrap*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.3 (A calculation to keep in view)**.** Let $P(t) = a+bt+ct^2$ with $c \neq 0$. At scale T write $t = Ts$. Then

$$P(Ts) = a + (bT)s + (cT^2)s^2.$$

If $|c|T^2 \asymp 1$ while $|b|T \to 0$ and $|a| \to 0$, the first visible rescaled polynomial is cs^2T^2. On a mesoscopic window $s = s_0 + \ell/T$ with $|\ell| = o(T)$,

$$P(Ts) - P(Ts_0) = P'(Ts_0)\ell + c\ell^2 = o(1)$$

whenever the normalized derivative is controlled. This is the toy calculation behind the separation of a macroscopic visibility scale from a mesoscopic almost-constant scale in effective linearization and closing.

CHAPTER 1

Margulis functions

This chapter isolates the margulis functions mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (From transverse negative moments to a Margulis inequality). *Let Y be a proper homogeneous subset and let d_Y be a transverse distance on a tubular neighborhood U of Y. Fix $\alpha > 0$ and $R \geq 1$, and set*

$$F_Y(x) = \begin{cases} \min\{R, d_Y(x)^{-\alpha}\}, & x \in U, \\ 1, & x \notin U. \end{cases}$$

Assume that for some integer $m \geq 1$, constants $0 < c_0 < 1$ and $B_0 < \infty$, and every $x \in U$ with $d_Y(x)^{-\alpha} \leq R$, one has the transverse negative-moment estimate

$$\int d_Y(gx)^{-\alpha}\, d\mu^{*m}(g) \leq c_0 d_Y(x)^{-\alpha} + B_0, \tag{46.1.1}$$

where the integrand is interpreted only on the part of the orbit remaining in the fixed tubular chart, while the complementary part contributes at most a fixed constant. Then there is $B < \infty$ such that

$$P^m F_Y(x) := \int F_Y(gx)\, d\mu^{*m}(g) \leq c_0 F_Y(x) + B \tag{46.1.2}$$

for all x.

Proof. If $x \in U$ and $d_Y(x)^{-\alpha} \leq R$, then $F_Y(x) = d_Y(x)^{-\alpha}$ and truncation can only decrease the left side of (46.1.1). Thus (46.1.2) follows with the same contraction factor and with B enlarged by the uniformly bounded contribution from exits from the tubular chart.

If $d_Y(x)^{-\alpha} > R$, then $F_Y(x) = R$, while $P^m F_Y(x) \leq R$ by definition. Hence $P^m F_Y(x) \leq c_0 R + (1 - c_0)R = c_0 F_Y(x) + (1 - c_0)R$. Finally, outside U we have $F_Y(x) = 1$ and $P^m F_Y(x) \leq R$, so the same inequality holds after increasing B to at least $R - c_0$. Taking the maximum of these finitely many additive constants proves (46.1.2). The substantive dynamical input is therefore exactly the negative-moment estimate (46.1.1); this lemma does not infer it merely from a qualitative Lyapunov exponent. ☐

THEOREM 0.2 (Occupation bound from a Margulis inequality). *Assume (46.1.2). For the μ-random walk X_n started at x, write $n = qm + r$ with $0 \le r < m$. If the finitely many operators P^r map F_Y to at most $K(F_Y + 1)$, then*

$$\mathbb{E}_x F_Y(X_n) \le K\left(c_0^q F_Y(x) + \frac{B}{1-c_0} + 1\right). \tag{46.1.3}$$

Consequently, for every $\varepsilon > 0$,

$$\mathbb{P}_x\big(d_Y(X_n) < \varepsilon\big) \le \frac{K}{\min\{R, \varepsilon^{-\alpha}\}}\left(c_0^q F_Y(x) + \frac{B}{1-c_0} + 1\right). \tag{46.1.4}$$

In particular, if the truncation level is chosen with $R \ge \varepsilon^{-\alpha}$, the prefactor is $K\varepsilon^{\alpha}$.

PROOF. Iteration of (46.1.2) gives

$$P^{qm} F_Y \le c_0^q F_Y + B\sum_{j=0}^{q-1} c_0^j \le c_0^q F_Y + \frac{B}{1-c_0}.$$

Apply P^r and the assumed finite-remainder bound to obtain (46.1.3). On the event $d_Y(X_n) < \varepsilon$ one has

$$F_Y(X_n) \ge \min\{R, \varepsilon^{-\alpha}\}.$$

Markov's inequality applied to the nonnegative random variable $F_Y(X_n)$ therefore gives (46.1.4). When $R \ge \varepsilon^{-\alpha}$ the denominator equals $\varepsilon^{-\alpha}$ and the usual ε^{α} tail follows. ☐

Proposition 0.3 (Finite-union Margulis function)**.** *Let $Y_1, \dots, Y_M$ have nonnegative functions F_i satisfying*

$$P^m F_i \le cF_i + B_i$$

with one common $0 < c < 1$ and one common step m. Then

$$F = \sum_{i=1}^{M} F_i \qquad \text{satisfies} \qquad P^m F \le cF + \sum_{i=1}^{M} B_i.$$

Hence a family with M bounded by the prescribed complexity bound has an additive constant growing at most linearly in that bound.

PROOF. By positivity and linearity of P^m,

$$P^m F = \sum_{i=1}^{M} P^m F_i \le \sum_{i=1}^{M} (cF_i + B_i) = cF + \sum_{i=1}^{M} B_i.$$

☐

CHAPTER 2

Height contraction inequalities

This chapter isolates the height contraction inequalities mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Height contraction inequalities: structural mechanism)**.** *For $F \geq 1$ with $PF \leq cF + B$, one has $P^n F \leq c^n F + B/(1-c)$ and, for every $R > 2B/(1-c)$, the expected fraction of times $k < n$ with $F(X_k) > R$ is at most $O(F(x)/(nR) + B/((1-c)R))$.*

PROOF. Induct the first inequality. Sum it for $k < n$ and divide by R using $1_{F>R} \leq F/R$. □

THEOREM 0.2 (Exponential entrance-time tail under a drift inequality). *Assume $F \geq 1$ and*

$$PF \leq cF + B, \qquad 0 < c < 1,$$

and choose $R > B/(1-c)$. Let

$$\tau_R = \inf\{n \geq 0 : F(X_n) \leq R\}.$$

Set $q = c + B/R < 1$. Then for every starting point x and every $n \geq 0$,

$$\mathbb{P}_x(\tau_R > n) \leq \frac{F(x)}{R}\, q^n. \tag{46.2.1}$$

The same statement applies to an m-step drift inequality $P^m F \leq cF + B$ after replacing n by the number of completed m-blocks.

PROOF. On the event $\{\tau_R > k\}$ one has $F(X_k) > R$, hence $B \leq (B/R)F(X_k)$. Therefore, before the stopping time,

$$\mathbb{E}[F(X_{k+1}) \mid \mathcal{F}_k] \leq \left(c + \frac{B}{R}\right) F(X_k) = qF(X_k).$$

Thus $q^{-(k\wedge\tau_R)} F(X_{k\wedge\tau_R})$ is a nonnegative supermartingale. Equivalently, iterating the stopped inequality gives

$$\mathbb{E}_x[F(X_n) 1_{\{\tau_R > n\}}] \leq q^n F(x).$$

Since $F(X_n) > R$ on $\{\tau_R > n\}$,

$$R\,\mathbb{P}_x(\tau_R > n) \le \mathbb{E}_x\big[F(X_n)1_{\{\tau_R>n\}}\big] \le q^n F(x),$$

which is (46.2.1). For an m-step drift, apply the same argument to the sampled chain X_{jm}. $\square$

Proposition 0.3 (Simultaneous cusp and singular-height drift)**.** *Let* $F_{\mathrm{cusp}}, F_{\mathrm{sing}} \ge 1$ *satisfy*

$$PF_{\mathrm{cusp}} \le c_1 F_{\mathrm{cusp}} + B_1, \qquad PF_{\mathrm{sing}} \le c_2 F_{\mathrm{sing}} + B_2, \qquad c_1, c_2 < 1.$$

For $F = F_{\mathrm{cusp}} + F_{\mathrm{sing}}$, $c = \max(c_1, c_2)$ *and* $B = B_1 + B_2$ *one has*

$$PF \le cF + B.$$

Hence all iterate, occupation-time, and stopping-tail estimates of Lemma 0.1 and Theorem 0.2 hold for the simultaneous height F.

Proof. Adding the two drift inequalities gives

$$PF \le c_1 F_{\mathrm{cusp}} + c_2 F_{\mathrm{sing}} + B_1 + B_2 \le c(F_{\mathrm{cusp}} + F_{\mathrm{sing}}) + B = cF + B.$$

Apply Lemma 0.1 and Theorem 0.2 to this F. $\square$

CHAPTER 3

Transverse sets and local energies

This chapter isolates the transverse sets and local energies mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Frostman control implies a logarithmic truncated-energy bound)**.** *Let η be a probability measure on a bounded transverse metric chart and let*

$$\mathcal{E}_{\rho,s}(\eta) = \iint (\rho + d(x,y))^{-s}\, d\eta(x)d\eta(y), \qquad 0 < \rho < 1.$$

If

$$\eta(B(x,r)) \le Cr^s \qquad \text{for every } x \text{ and every } r \ge \rho,$$

then

$$\mathcal{E}_{\rho,s}(\eta) \le C_s C\,(1 + |\log \rho|).$$

The converse is not a pointwise equivalence on all of the support; finite energy yields a Frostman bound only after discarding a controlled exceptional set, as stated precisely in Proposition 0.3.

Proof. Fix x. Split the chart into the ball $B(x,\rho)$ and dyadic annuli $2^j\rho \le d(x,y) < 2^{j+1}\rho$ for $0 \le j \le J$, where $J = O(|\log \rho|)$. The innermost contribution is at most

$$\rho^{-s}\eta(B(x,\rho)) \le C.$$

On the jth annulus the kernel is at most $(2^j\rho)^{-s}$, while the mass is at most $C(2^{j+1}\rho)^s$, so the contribution is at most $C2^s$. Summing $O(1+|\log\rho|)$ annuli and then integrating in x gives the claimed bound. □

Theorem 0.2 (Energy distortion under bi-Lipschitz transverse maps). *Let T be a map between two transverse charts such that*

$$L^{-1}d(x,y) \le d(Tx,Ty) \le Ld(x,y)$$

for all x, y in the support of η. Then

$$L^{-s}\mathcal{E}_{\rho/L,s}(\eta) \le \mathcal{E}_{\rho,s}(T_*\eta) \le L^s\mathcal{E}_{\rho L,s}(\eta). \tag{46.3.1}$$

In particular, bounded group elements whose chart actions are uniformly bi-Lipschitz change truncated s-energy only by fixed multiplicative and scale constants. If moreover $d(Tx,Ty) \geq \lambda d(x,y)$ with $\lambda > 1$, then

$$\mathcal{E}_{\rho,s}(T_*\eta) \leq \lambda^{-s}\mathcal{E}_{\rho/\lambda,s}(\eta). \tag{46.3.2}$$

Proof. By change of variables under the pushforward,

$$\mathcal{E}_{\rho,s}(T_*\eta) = \iint (\rho + d(Tx,Ty))^{-s}\, d\eta(x)d\eta(y).$$

The upper bi-Lipschitz inequality gives $\rho + d(Tx,Ty) \leq L(\rho/L + d(x,y))$, while the lower inequality gives $\rho + d(Tx,Ty) \geq L^{-1}(\rho L + d(x,y))$. Raising to the power $-s$ yields the two sides of (46.3.1). Under the uniform expansion hypothesis, $\rho + d(Tx,Ty) \geq \lambda(\rho/\lambda + d(x,y))$, which gives (46.3.2). □

Proposition 0.3 (Frostman bounds from local energy and conversely)**.** *Let η be a probability on a transverse chart and*

$$\mathcal{E}_{\rho,s}(\eta) = \iint (\rho + d(x,y))^{-s}\, d\eta(x)d\eta(y).$$

If $\eta(B(x,r)) \leq Cr^s$ for every x and $r \geq \rho$, then

$$\mathcal{E}_{\rho,s}(\eta) \ll_s C\,|\log\rho|.$$

Conversely, if $\mathcal{E}_{\rho,s}(\eta) \leq M$, there is a Borel set E with $\eta(E) \geq \frac{1}{2}$ such that the normalized restriction $\eta_E = \eta|_E/\eta(E)$ satisfies

$$\eta_E(B(x,r)) \leq 4M(\rho + r)^s \qquad (x \in E,\ r > 0).$$

Proof. The first implication is the dyadic-annulus calculation of Lemma 0.1. For the converse put

$$U(x) = \int (\rho + d(x,y))^{-s}\, d\eta(y).$$

Since $\int U\, d\eta = \mathcal{E}_{\rho,s}(\eta) \leq M$, Markov's inequality gives $E = \{U \leq 2M\}$ with $\eta(E) \geq 1/2$. For $x \in E$,

$$\eta(B(x,r))(\rho + r)^{-s} \leq U(x) \leq 2M.$$

Dividing by $\eta(E) \geq 1/2$ gives the asserted bound for η_E. □

CHAPTER 4

Incidence estimates

This chapter isolates the incidence estimates mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Chevalley-tube incidence interface)**.** *Let (Ω, σ) be a probability parameter space and let A be a finite transverse set. For each ordered pair $(a, a') \in A^2$ assume that the incidence event*

$$E_{a,a'}(\rho) = \{g \in \Omega : d(ga, a') < \rho\}$$

is contained in a union of at most L Chevalley tubes $\mathcal{T}_M$ associated to connected rational subgroups of height at most $Q = \rho^{-A_0}$. Assume further that none of these tubes lies in the identically-vanishing branch of Lemma 0.1, and that the tube threshold is $\varepsilon = \rho^{b_0}$. Then

$$\sigma(E_{a,a'}(\rho)) \le CL\rho^{b_0\alpha}, \tag{46.4.1}$$

where C, α are the constants in Lemma 0.1. If the containment requires a union over all rational subgroups of height at most Q, then Theorem 0.2 gives instead the bound

$$\sigma(E_{a,a'}(\rho)) \le C'\rho^{b_0\alpha - A_0C_0}, \tag{46.4.2}$$

provided $b_0\alpha > A_0C_0$.

Proof. For (46.4.1), apply the single-subgroup estimate of Lemma 0.1 to each of the at most L tubes and sum their measures. For (46.4.2), the number of rational subgroups of height at most Q is $O(Q^{C_0}) = O(\rho^{-A_0C_0})$ by (4.1) of Volume 45. Multiplying this count by the single-tube bound $O(\varepsilon^\alpha) = O(\rho^{b_0\alpha})$ gives the stated exponent. The exclusion of the identically-vanishing branch is essential; without it no power-saving tube estimate follows. □

Theorem 0.2 (Generic-branch incidence saving). *Assume that, on the generic branch of Volume 45, every incidence event arising in the transverse chart admits the Chevalley-tube containment of Lemma 0.1 with fixed exponents A_0, b_0 and with no identically-vanishing tube. If*

$$c_0 := b_0\alpha - A_0C_0 > 0,$$

then uniformly for all relevant pairs (a, a'),

$$\sigma\{g : d(ga, a') < \rho\} \ll \rho^{c_0}. \tag{46.4.3}$$

Thus the quantitative incidence saving used later is exactly the positive exponent left after the rational-subgroup counting loss has been deducted from the single-tube sublevel exponent.

Proof. Apply (46.4.2) from Lemma 0.1. The definition of c_0 turns its exponent into ρ^{c_0}. The theorem is deliberately conditional on the geometric Chevalley-tube containment for the incidence event; Volume 45 supplies the tube estimates once that containment has been verified, but does not by itself identify an arbitrary incidence relation with such a tube. $\square$

Proposition 0.3 (Average collision bound from pairwise incidence saving)**.** *Let* η *be a probability measure on the transverse chart and let* Π_g *be the corresponding family of transverse maps. Suppose that for* $\eta \times \eta$*-almost every pair with* $d(x, y) \geq \rho$,

$$\sigma\{g : d(\Pi_g x, \Pi_g y) < \rho\} \leq C\rho^{c_0}. \tag{46.4.4}$$

Then

$$\int \iint 1_{\{d(\Pi_g x, \Pi_g y) < \rho\}} \, d\eta(x) d\eta(y) d\sigma(g) \leq (\eta \times \eta)\{d(x, y) < \rho\} + C\rho^{c_0}. \tag{46.4.5}$$

If the first term is at most $C_1\rho^s$*, the averaged projected collision probability is* $O(\rho^{\min(s, c_0)})$.

Proof. Use Fubini to write the left side of (46.4.5) as the integral over (x, y) of the parameter measure of the corresponding incidence event. On $d(x, y) < \rho$ use the trivial bound 1; on its complement use (46.4.4). Since $\eta \times \eta$ is a probability measure, the second contribution is at most $C\rho^{c_0}$. The final assertion follows by adding the two power bounds. $\square$

CHAPTER 5

Projection growth theorems

This chapter isolates the projection growth theorems mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Collision identity for an averaged transverse family)**.** *Let* (Ω, σ) *be a probability space of transverse maps* Π_g *and let*

$$\bar{\eta} = \int_\Omega (\Pi_g)_* \eta \, d\sigma(g)$$

be the averaged transverse probability measure. For the scale-ρ *collision functional*

$$\mathrm{Col}_\rho(\nu) = \iint 1_{\{d(z,z')<\rho\}} \, d\nu(z) d\nu(z'),$$

one has the exact identity

$$\mathrm{Col}_\rho(\bar{\eta}) = \iiiint 1_{\{d(\Pi_g x, \Pi_h y)<\rho\}} \, d\eta(x) d\eta(y) d\sigma(g) d\sigma(h). \tag{46.5.1}$$

Thus dimension growth of the averaged *distribution reduces to a cross-incidence estimate for two independently chosen parameters. No individual bi-Lipschitz image* $(\Pi_g)_*\eta$ *can acquire larger genuine dimension merely by change of coordinates.*

Proof. For every bounded Borel function φ,

$$\int \varphi \, d\bar{\eta} = \iint \varphi(\Pi_g x) \, d\eta(x) d\sigma(g).$$

Apply this formula twice to the bounded Borel kernel $\varphi(z, z') = 1_{\{d(z,z')<\rho\}}$ and use Tonelli's theorem. This gives (46.5.1). The final sentence follows because a bi-Lipschitz map preserves Hausdorff dimension and changes finite-scale covering exponents only by $o(1)$ as $\rho \to 0$. □

Theorem 0.2 (Cross-incidence criterion for scale-dimension growth). *Assume that for some* $s \geq 0$, $\delta > 0$, *and all sufficiently small* ρ,

$$(\sigma \times \sigma)\{(g, h) : d(\Pi_g x, \Pi_h y) < \rho\} \leq C\rho^{s+\delta} \tag{46.5.2}$$

for $\eta \times \eta$-almost every (x, y) outside an exceptional pair set of $\eta \times \eta$-measure at most $C\rho^{s+\delta}$. Then the averaged measure $\bar{\eta}$ from Lemma 0.1 satisfies

$$\mathrm{Col}_\rho(\bar{\eta}) \leq 2C\rho^{s+\delta}. \tag{46.5.3}$$

In the collision-exponent convention used in this volume, this is a gain of δ over an input bound of order ρ^s, capped at the ambient dimension. The substantive geometric obligation is therefore the cross-incidence estimate (46.5.2); it must be proved from the Chevalley/non-concentration geometry before the dimension gain is invoked.

Proof. Insert (46.5.2) into the identity (46.5.1). On the exceptional pair set use the trivial parameter bound 1, contributing at most $C\rho^{s+\delta}$. On its complement the parameter integral is at most $C\rho^{s+\delta}$, and the pair measure is at most one. The sum is (46.5.3). The dimension interpretation is exactly the definition of the collision exponent; no claim about growth of a single coordinate image is used. $\square$

Proposition 0.3 (Chartwise stability of a verified collision gain)**.** *Let K be covered by finitely many transverse charts whose coordinate maps and inverses are L-bi-Lipschitz and whose overlap multiplicity is at most M. Suppose that in each chart the averaged measure satisfies*

$$\mathrm{Col}_\rho(\bar{\eta}_j) \leq C\rho^{s+\delta}.$$

Then, after changing C and replacing ρ by a fixed multiple depending on L, the global averaged transverse distribution on K satisfies a collision bound with exponent at least $s + \delta/2$ for all sufficiently small ρ.

Proof. A bi-Lipschitz chart changes a distance threshold ρ only to a threshold between ρ/L and $L\rho$. Summing over finitely many charts and at most M overlaps multiplies the collision bound by a fixed constant $C_{K,L,M}$. Hence

$$\mathrm{Col}_\rho(\bar{\eta}_K) \leq C_{K,L,M}\rho^{s+\delta}.$$

For sufficiently small ρ, the fixed constant is bounded by $\rho^{-\delta/2}$, yielding $\mathrm{Col}_\rho(\bar{\eta}_K) \leq \rho^{s+\delta/2}$ after one final change of the implicit constant. $\square$

CHAPTER 6

Dimension improvement

This chapter isolates the dimension improvement mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (One-step collision-exponent improvement)**.** *Let d be the ambient transverse dimension. Suppose a shearing/averaging block transforms a transverse probability η_j into η_{j+1} and that, at the current scale ρ_j, the verified cross-incidence criterion of Theorem 0.2 gives*

$$\operatorname{Col}_{\rho_j}(\eta_{j+1}) \le C\rho_j^{\min(s_j+\delta, d-\epsilon)}, \tag{46.6.1}$$

whenever $\operatorname{Col}_{\rho_j}(\eta_j) \le C\rho_j^{s_j}$. Then the collision exponent after one nonterminal block is at least

$$s_{j+1} \ge \min(s_j + \delta, d - \epsilon).$$

Proof. This is exactly the meaning of (46.6.1): the exponent of the collision upper bound for η_{j+1} is at least the displayed minimum. The lemma deliberately separates the deterministic exponent bookkeeping from the geometric proof of the cross-incidence estimate, which is the load-bearing hypothesis supplied before invoking this step. $\square$

Theorem 0.2 (Finite termination of a uniform dimension-improvement scheme). *Assume that Lemma 0.1 applies at every nonterminal stage with one fixed increment $\delta > 0$. Starting from exponent s_0, after at most*

$$N = \left\lceil \frac{(d - \epsilon - s_0)_+}{\delta} \right\rceil$$

nonterminal stages one reaches an exponent at least $d - \epsilon$. If at any earlier stage the cross-incidence or recurrence hypothesis fails, the iteration must stop there and record that failure rather than claim dimension growth.

Proof. As long as $s_j < d - \epsilon$, Lemma 0.1 gives $s_{j+1} \ge s_j + \delta$. Hence by induction $s_j \ge s_0 + j\delta$ until the terminal threshold is reached. The least integer j for which $s_0 + j\delta \ge d - \epsilon$ is at most the displayed N. The final sentence is logical bookkeeping: the recurrence and cross-incidence hypotheses are premises of the one-step lemma, so a stage where they fail cannot be counted as a successful increment. ☐

Proposition 0.3 (Recurrence control through the finite dimension-growth iteration)**.** *Let $N \le \lceil(\dim V - s_0)/\delta\rceil + 1$ be the number of stages in Theorem 0.2. Suppose that at stage j the recurrence/avoidance argument supplies an exceptional parameter set E_j of measure at most $C\rho_j^c$ outside which the state lies in a fixed recurrence compactum K. Then all N iterations remain in K outside*

$$E = \bigcup_{j=0}^{N-1} E_j, \qquad |E| \le C \sum_{j=0}^{N-1} \rho_j^c.$$

In particular, for a polynomial scale hierarchy $\rho_j = T^{-a_j}$ the total exceptional measure is a negative power of T.

Proof. There are at most N stages by Theorem 0.2. Outside E_j, the Chapter 2 height bound and Volume 45 avoidance alternative keep the jth state in K. Hence outside the union E all stages stay in K. The union bound gives the displayed estimate; because N is fixed and every ρ_j is a fixed negative power of T, the sum is $O(T^{-c'})$ for some $c' > 0$. ☐

CHAPTER 7

From dimension growth to almost invariance

This chapter isolates the from dimension growth to almost invariance mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Coupling criterion for quantitative almost invariance)**.** *Let ν be a probability measure on a metric chart and let h act on the chart. Suppose there is a coupling π of ν and $h_*\nu$ and numbers $\kappa, \xi \geq 0$ such that*

$$\pi\{(x,y) : d(x,y) > \xi\} \leq \kappa. \tag{46.7.1}$$

Then for every bounded Lipschitz test function f,

$$\left| \int f\, d(h_*\nu) - \int f\, d\nu \right| \leq 2\kappa \|f\|_\infty + \xi \operatorname{Lip}(f). \tag{46.7.2}$$

In particular, a verified abundance of matched pairs near a displacement $h = \exp(v)$ yields quantitative almost invariance once it is upgraded to a coupling of the form (46.7.1). Near-full scale dimension by itself does not imply translation invariance; the matching/coupling step is an additional hypothesis that must be proved.

Proof. Because π has marginals ν and $h_*\nu$,

$$\int f\, d(h_*\nu) - \int f\, d\nu = \iint (f(y) - f(x))\, d\pi(x,y).$$

On $d(x,y) \leq \xi$, the integrand has absolute value at most $\xi \operatorname{Lip}(f)$. On the complementary set it is at most $2\|f\|_\infty$, and that set has π-mass at most κ. Summing the two contributions proves (46.7.2). $\square$

Theorem 0.2 (Closure of almost invariance under products and commutators). *Let K be a compact chart on which left translations by elements in a fixed identity neighborhood have uniformly bounded C^1 distortion. For a probability measure ν supported in K, define*

$$\Delta(\nu, g) = \sup_{\substack{f \in C^1 \\ \|f\|_\infty + \|Df\|_\infty \leq 1}} \left| \int f(gx)\, d\nu(x) - \int f(x)\, d\nu(x) \right|.$$

Then:

(1) $\Delta(\nu, g_1 g_2) \le C(\Delta(\nu, g_1) + \Delta(\nu, g_2))$ *for nearby* g_1, g_2*;*

(2) if $\Delta(\nu, \exp(tv)) \le \varepsilon$ *and* $\Delta(\nu, \exp(sw)) \le \varepsilon$ *for all* $|t|, |s| \le r$*, then*

$$\Delta(\nu, \exp(ts[v, w])) \le C\varepsilon + Cr^3 \tag{46.7.3}$$

for $|t|, |s| \le r$*, after shrinking the identity neighborhood if necessary.*

Thus verified almost invariance along one-parameter directions propagates to products and, quantitatively, to bracket directions. Iterating this statement along a finite Lie-generating list gives almost invariance on a small neighborhood of the generated local subgroup, with an error obtained by summing the finitely many product and BCH losses.

Proof. For (1), write

$$f(g_1 g_2 x) - f(x) = \big(f(g_1 g_2 x) - f(g_1 x)\big) + \big(f(g_1 x) - f(x)\big).$$

The second term is controlled by $\Delta(\nu, g_1)$. For the first term set $\phi = f \circ g_1$. Uniform C^1 distortion on the fixed identity neighborhood gives $\|\phi\|_{C^1} \le C\|f\|_{C^1}$, so the first term is at most $C\Delta(\nu, g_2)$. This proves the stated product inequality.

For (2), apply (1) repeatedly to the commutator word

$$c(t, s) = \exp(tv)\exp(sw)\exp(-tv)\exp(-sw),$$

obtaining $\Delta(\nu, c(t, s)) \le C\varepsilon$. The BCH formula on the compact identity neighborhood gives

$$c(t, s) = \exp\big(ts[v, w] + R(t, s)\big), \qquad \|R(t, s)\| \le Cr^3.$$

Uniform C^1 control of test functions changes their integrals by at most Cr^3 when the translating group elements differ by $O(r^3)$. Combining the two estimates yields (46.7.3). A finite iteration gives the final local-subgroup statement. □

Proposition 0.3 (Exact downstream interface to the spectral step)**.** *Chapter 7 produces only quantitative almost invariance. A downstream equidistribution argument may use this output only after verifying a separate spectral statement of the following form: for the relevant test norm there are* $C, \theta > 0$ *such that every probability measure* ν *satisfying*

$$\sup_{g \in \mathcal{U}} \Delta(\nu, g) \le \varepsilon$$

on a fixed generating identity neighborhood $\mathcal{U}$ *obeys*

$$|\nu(f) - m_X(f)| \le C\varepsilon^\theta \mathcal{S}_D(f)$$

for the declared class of test functions. No such spectral implication is proved in this chapter; its verification belongs to Volume 47.

Proof. Theorem 0.2 controls translation defects $\Delta(\nu, g)$ and their propagation through Lie products. It contains no estimate comparing ν with Haar measure. Therefore an estimate of the displayed spectral form is logically independent and must be supplied before equidistribution is concluded. The proposition records that dependency boundary explicitly. $\square$

CHAPTER 8

Iterative dimension-growth bootstrap

This chapter isolates the iterative dimension-growth bootstrap mechanism needed by the later rigidity arguments. All statements below are restricted to the hypotheses and ambient objects specified explicitly in this chapter and in the volume prerequisites.

Lemma 0.1 (Finite-dimensional enlargement alternative)**.** *Let $\mathfrak{h}_0 \subset \mathfrak{g}$ be the Lie algebra of directions under which the current distribution is quantitatively almost invariant. Suppose that at each generic stage the argument produces exactly one of the following certified outputs:*

(1) a proper homogeneous obstruction of the declared polynomial complexity; or

(2) a vector $v_j \in \mathfrak{g}$ with $\operatorname{dist}(v_j, \mathfrak{h}_j) \geq c_* > 0$ together with almost invariance along $\exp(tv_j)$ for $|t| \leq r_j$.

In case (2), let $\mathfrak{h}_{j+1}$ be the Lie algebra generated by $\mathfrak{h}_j$ and v_j. Then

$$\dim \mathfrak{h}_{j+1} \geq \dim \mathfrak{h}_j + 1.$$

Hence after at most $\dim G - \dim \mathfrak{h}_0$ generic enlargements, either an obstruction has occurred or the generated algebra is all of $\mathfrak{g}$. The dimension-growth and coupling arguments of Chapters 5–7 are responsible for verifying the hypotheses that produce the new transverse vector; this lemma does not infer that vector from dimension bookkeeping alone.

Proof. The distance condition implies $v_j \notin \mathfrak{h}_j$. Therefore the linear span of $\mathfrak{h}_j$ and v_j has dimension at least $\dim \mathfrak{h}_j + 1$, and the Lie algebra generated by them contains that span. Thus the dimension increases by at least one at every generic enlargement. Since no Lie subalgebra of $\mathfrak{g}$ has dimension larger than $\dim G$, the number of strict enlargements is bounded as claimed. □

Theorem 0.2 (Finite loss ledger for the enlargement bootstrap). *Assume that the bootstrap of Lemma 0.1 uses at most $N \leq \dim G$ outer enlargement stages and that each stage contains at most M inner scale-improvement steps.*

Suppose every verified input at every step contributes a power loss $\rho^{c_{j,k}}$ *with*

$$c_{j,k} \geq c_* > 0,$$

and every rescaling has the form $\rho_{j,k+1} = \rho_{j,k}^{a_{j,k}}$ *with* $a_{j,k} \geq a_* > 0$. *Then, because only finitely many steps occur, all accumulated polynomial losses can be bounded by one positive power of the initial scale. More explicitly, if a quantity is multiplied by at most* $\rho_{j,k}^{c_*}$ *at each of at most* NM *stages, then after expressing every* $\rho_{j,k}$ *as a positive power of* $\rho_{0,0}$ *the total gain is* $\rho_{0,0}^{c_{\text{fin}}}$ *for some* $c_{\text{fin}} > 0$ *depending only on* N, M, c_*, a_* *and the finite scale hierarchy.*

Proof. Every scale appearing in the finite hierarchy is of the form

$$\rho_{j,k} = \rho_{0,0}^{A_{j,k}}$$

with $A_{j,k}$ a finite product of positive rescaling exponents, hence $A_{j,k} > 0$. A factor $\rho_{j,k}^{c_{j,k}}$ therefore equals $\rho_{0,0}^{A_{j,k}c_{j,k}}$ with positive exponent. Multiplying finitely many such factors adds finitely many positive exponents. Their sum is a positive number c_{fin}. The theorem makes no claim if some stage has only a logarithmic saving or zero exponent; such a failure must be recorded before the bootstrap is invoked. $\square$

Downstream interface. This final chapter supplies the downstream interface recorded in the theorem above. No additional proposition is needed beyond the theorem above.

Volume 47

Rank-One Polynomial Effective Equidistribution

Volume 47 scope and prerequisites

The volume proves the polynomial effective-equidistribution mechanism in the two rank-one semisimple models treated by Lindenstrauss–Mohammadi–Wang. The Margulis-function, incidence, dimension-growth, almost-invariance, closing, and spectral steps are owned in Volumes 45–47; the LMW paper is used for theorem-scope comparison and attribution. The standing spaces are arithmetic quotients in the two ambient models $\mathrm{SL}_2(\mathbb{C})$ and $\mathrm{SL}_2(\mathbb{R}) \times \mathrm{SL}_2(\mathbb{R})$, with the embedded one-parameter unipotent subgroup $U < \mathrm{SL}_2(\mathbb{R})$. All constants are allowed to depend on the fixed arithmetic quotient and a fixed Sobolev order, but not on the orbit length.

Proof and provenance. Primary comparison: Lindenstrauss–Mohammadi–Wang, *Effective equidistribution for some one parameter unipotent flows*, together with the internally owned Margulis-function, incidence, and spectral inputs of Volumes 45–46.

Reader guide: a concrete model

Why this matters.

Volume 47 is organized around the passage from *Rank-one effective setup* to *Rank-one arithmetic applications*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.3 (A calculation to keep in view)**.** Let $P(t) = a+bt+ct^2$ with $c \neq 0$. At scale T write $t = Ts$. Then

$$P(Ts) = a + (bT)s + (cT^2)s^2.$$

If $|c|T^2 \asymp 1$ while $|b|T \to 0$ and $|a| \to 0$, the first visible rescaled polynomial is cs^2T^2. On a mesoscopic window $s = s_0 + \ell/T$ with $|\ell| = o(T)$,

$$P(Ts) - P(Ts_0) = P'(Ts_0)\ell + c\ell^2 = o(1)$$

whenever the normalized derivative is controlled. This is the toy calculation behind the separation of a macroscopic visibility scale from a mesoscopic almost-constant scale in effective linearization and closing.

CHAPTER 1

Rank-one effective setup

This chapter owns the rank-one effective setup step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Rank-one effective setup: structural mechanism)**.** *Fix a compactly supported C^∞ test function f and a time scale $T \geq 2$. A dyadic partition of $[1, T]$ together with the height truncation from Volume 45 reduces the orbit average to $O(\log T)$ blocks on which the injectivity radius and Sobolev distortion are polynomially controlled.*

Proof. Use the height contraction inequality to bound the bad-time measure. On the complement, cover by dyadic intervals on which the local product chart is injective. The Sobolev norm after conjugation grows by at most a fixed power of the dyadic scale. Summing the geometric tail gives a power saving; summing the $O(\log T)$ good blocks costs only T^ε, which is absorbed by shrinking the exponent.

□

Theorem 0.2 (Rank-one effective setup). *Fix the rank-one arithmetic quotient and height function used in this volume. There exist constants $C, D, \kappa > 0$ such that, for every $T \geq 2$ and every $f \in C_c^\infty(X)$, one can write*

$$[1, T] = B \sqcup J_1 \sqcup \cdots \sqcup J_m, \qquad m \ll \log T,$$

with $|B| \ll T^{1-\kappa}$, while on every J_j the injectivity radius is at least T^{-C} and every conjugation used in the local product chart enlarges $\mathcal{S}_D(f)$ by at most T^C. Consequently

$$\frac{1}{T}\left|\int_B f(xu_t)\,dt\right| \ll T^{-\kappa}\|f\|_\infty.$$

Proof. Apply Lemma 0.1. The height-contraction estimate gives a bad set B of relative measure $O(T^{-\kappa_0})$. On its complement the dyadic decomposition has $m \ll \log T$ blocks and, by construction, polynomial injectivity-radius and Sobolev bounds. Since $|\int_B f(xu_t)dt| \leq |B|\|f\|_\infty$, the displayed bad-set estimate follows with $\kappa \leq \kappa_0$. Finally $\log T \ll_\varepsilon T^\varepsilon$; decreasing κ absorbs the logarithmic number of good blocks into the later power-saving ledger. □

Proposition 0.3 (Rank-one effective setup: downstream consequence)**.** *The reduction is stable under replacing the time interval by a subinterval whose length is a fixed power of T.*

Proof. Let $J \subset [1, T]$ have length T^θ. Repeating the height estimate with local scale $|J|$ gives a bad subset of size $O(|J|^{1-\kappa_0})$, while the dyadic good-block count is $O(\log|J|)$. Since $|J| = T^\theta$, every resulting loss is still a fixed power of T (with exponent multiplied by θ). □

CHAPTER 2

Thickening and Margulis-function control

This chapter owns the thickening and margulis-function control step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Thickening and Margulis-function control: structural mechanism)**.** *Let ν be a normalized orbit-block measure and $\nu_\rho = \nu * \psi_\rho$ a transverse smoothing. The Margulis-function estimate of Volume 46 bounds the mass of pairs at distance $< \rho$ by $\rho^{d-\epsilon}$ outside a controlled exceptional set.*

Proof. The ball-kernel identity expresses $\|\nu_\rho\|_2^2$ as ρ^{-d} times the pair mass in a $O(\rho)$ tube. Insert the Margulis-function pair estimate. The discarded cusp set is controlled by the height tail, and the smoothing error is $O(\rho \mathcal{S}_{D+1}(f))$. Balance the two powers by fixing a below the minimum of the available exponents.

□

Theorem 0.2 (Thickening and Margulis-function control). *Let ν be a normalized good-block measure and let $\nu_\rho = \nu * \psi_\rho$, where ψ_ρ is a unit-mass transverse bump in dimension d. Suppose the pair estimate from Lemma 0.1 is*

$$(\nu \times \nu)\{(x, y) : d(x, y) \le C_0\rho\} \ll \rho^{d-\epsilon} + T^{-\kappa_0}.$$

Then

$$\|d\nu_\rho/dm\|_2^2 \ll \rho^{-\epsilon} + \rho^{-d}T^{-\kappa_0}.$$

Hence, for $\rho = T^{-a}$ with $0 < a < \kappa_0/(2d)$, the L^2 norm is $T^{O(a\epsilon)}$ and the cusp/truncation contribution remains $O(T^{-\kappa})$ for some $\kappa > 0$.

Proof. Write $g_\rho = d\nu_\rho/dm$. The ball-kernel identity gives

$$\|g_\rho\|_2^2 \ll \rho^{-d}(\nu \times \nu)\{d(x, y) \le C_0\rho\}.$$

Insert the pair bound to obtain the first displayed inequality. If $\rho = T^{-a}$, the two terms are $T^{a\epsilon}$ and $T^{ad-\kappa_0}$. Choosing $ad < \kappa_0/2$ makes the second a negative power of T. The height-tail and smoothing errors from Lemma 0.1 are also powers of T^{-1} after decreasing a once more; their minimum is the asserted κ. ☐

Proposition 0.3 (Uniform smoothing scale for bounded Sobolev test families)**.** *Fix a compact set $K \subset X$, a Sobolev order D, and $M < \infty$. For the family*

$$\mathcal{F}_{K,M} = \{f \in C_c^{\infty}(X) : \operatorname{supp} f \subset K,\ \mathcal{S}_D(f) \leq M\},$$

the exponent $a > 0$ in the choice $\rho = T^{-a}$ in Theorem 0.2 may be chosen independently of $f \in \mathcal{F}_{K,M}$; all resulting error constants are uniform on this family.

Proof. For a bounded family in a fixed compact support, the kernel constants, pair-energy constants, and Sobolev norms appearing in Theorem 0.2 have a common upper bound. The admissible inequalities for a are therefore unchanged after replacing individual constants by those maxima, so one choice of $\rho = T^{-a}$ works for the entire family. ☐

CHAPTER 3

Incidence geometry for transverse sets

This chapter owns the incidence geometry for transverse sets step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Incidence geometry for transverse sets: structural mechanism)**.** *If a transverse probability has scale energy $I_\rho \leq \rho^{-\epsilon}$ and is not concentrated in a ρ^α-neighborhood of a proper linear subspace, the restricted-projection/incidence estimate of Volume 46 produces a direction in which the projected scale dimension increases by a definite $c(\epsilon) > 0$.*

Proof. At each scale apply the incidence theorem to the set of transverse displacements. Failure of dimension growth forces concentration near the zero set of a linear functional, which is the local Lie shadow of a proper subgroup. Otherwise dimension rises by $c(\epsilon)$. Since dimension is bounded by d, at most $d/c(\epsilon)$ successful steps occur before near-full dimension is reached. □

Theorem 0.2 (Incidence geometry for transverse sets). *Let d be the ambient transverse dimension and let $c(\epsilon) > 0$ be the dimension increment furnished by Lemma 0.1. Starting from any transverse scale dimension $s_0 \geq 0$, after at most*

$$N(\epsilon, d) := \left\lceil \frac{d - s_0}{c(\epsilon)} \right\rceil$$

incidence steps, either a proper linear concentration is detected (and hence a proper homogeneous obstruction after linearization), or the transverse smoothed measure has scale dimension at least $d - \epsilon$.

Proof. If no proper concentration occurs, Lemma 0.1 increases the scale dimension by at least $c(\epsilon)$ at each step. Scale dimension cannot exceed

d. Therefore after $N(\epsilon, d)$ steps the alternative "dimension $< d - \epsilon$ and no obstruction" is impossible. If concentration occurs at an earlier step, the Chevalley linearization/closing mechanism converts it into the proper homogeneous branch. This gives exactly the stated finite dichotomy. $\square$

Proposition 0.3 (Incidence geometry for transverse sets: downstream consequence)**.** *If each successful iteration raises the transverse dimension by at least $\Delta > 0$ inside an ambient transverse space of dimension D, then at most $\lceil D/\Delta \rceil$ successful iterations can occur.*

Proof. The proof of Theorem 0.2 uses only the inequality $s_{j+1} \geq s_j + c(\epsilon)$ and the upper bound $s_j \leq d$. Thus the iteration count is at most $\lceil d/c(\epsilon) \rceil$, which depends only on the displayed dimension and increment. $\square$

CHAPTER 4

Ambient spectral gap

This chapter owns the ambient spectral gap step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Ambient spectral gap: structural mechanism)**.** *Let $\mathcal{P}$ be the averaging operator on the ambient quotient corresponding to a fixed compactly supported symmetric probability generating the relevant semisimple factor. On L^2_0, the spectral gap gives $\|\mathcal{P}^n\phi\|_2 \le e^{-\sigma n}\|\phi\|_2$.*

Proof. Write $\langle g-1, f\rangle = \langle \mathcal{P}^n(g-1), f\rangle + \langle (g-1) - \mathcal{P}^n(g-1), f\rangle$. The first term is bounded by spectral gap and Cauchy–Schwarz. Almost invariance bounds the second by $nT^{-b}\mathcal{S}_D(f)$. Take $n = c\log T$ and choose c so that both terms are powers of T^{-1}.

□

Theorem 0.2 (Ambient spectral gap). *Let $\mathcal{P}$ satisfy $\|\mathcal{P}^n\phi\|_2 \le e^{-\sigma n}\|\phi\|_2$ on $L^2_0(X)$. Let $g \ge 0$, $\int g\,dm = 1$, and assume that for $0 \le j < n$,*

$$\|\mathcal{P}^{j+1}g - \mathcal{P}^j g\|_2 \le CT^{-b}.$$

Then, for every $f \in L^2(X)$,

$$|\langle g-1, f\rangle| \le e^{-\sigma n}\|g-1\|_2\|f\|_2 + nCT^{-b}\|f\|_2.$$

If $\|g-1\|_2 \le T^A$ and $n = \lfloor c\log T\rfloor$ with $c > A/\sigma$, chosen also so that $b-c' > 0$ for the polynomial almost-invariance loss, the right side is $O(T^{-\eta})\mathcal{S}_D(f)$ for some $\eta > 0$.

Proof. Telescope

$$g - 1 = \mathcal{P}^n(g-1) + \sum_{j=0}^{n-1}(\mathcal{P}^j g - \mathcal{P}^{j+1}g).$$

Pair with f, apply Cauchy–Schwarz to every term, use the spectral-gap estimate on the first term, and the assumed almost-invariance bound on the sum. This gives the displayed inequality. With $n = c\log T$, the spectral term is $T^{A-c\sigma}$ and the telescoping term is $O((\log T)T^{-b})$ (or the corresponding polynomially distorted Sobolev version). Choose c and then shrink the final exponent so both are $O(T^{-\eta})$. □

Proposition 0.3 (Spectral decay after a polynomial Sobolev loss)**.** *Assume an estimate*

$$|\mathcal{E}_T(f)| \le CT^{-\eta_0}\mathcal{S}_D(f)$$

and a conjugation/smoothing bound

$$\mathcal{S}_D(\mathcal{T}_T f) \le C'T^A\mathcal{S}_{D'}(f).$$

If $A < \eta_0$, then

$$|\mathcal{E}_T(\mathcal{T}_T f)| \le CC'T^{-(\eta_0-A)}\mathcal{S}_{D'}(f).$$

Thus the power saving survives with exponent $\eta_0 - A > 0$.

Proof. If conjugation or smoothing replaces $\mathcal{S}_D(f)$ by at most $T^A\mathcal{S}_{D'}(f)$, the estimate of Theorem 0.2 becomes $T^{A-\eta_0}\mathcal{S}_{D'}(f)$. The scale hierarchy is chosen with the spectral exponent $\eta_0 > A$; the remaining exponent $\eta = \eta_0 - A > 0$ gives the claimed stability. □

CHAPTER 5

Production of almost invariance

This chapter owns the production of almost invariance step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Production of almost invariance: structural mechanism)**.** *If the transverse smoothed distribution has dimension $d - O(\epsilon)$ and supplies three transverse displacement directions at scale ρ, then the BCH argument of Volume 46 produces T^{-b}-almost invariance under a neighborhood in the local semisimple factor.*

Proof. Select a quantitatively transverse basis from the displacement set. Translate and smooth as in Volume 46. BCH converts invariance under basis directions into invariance under their brackets; the quadratic commutator error is below the smoothing scale by the scale hierarchy. Iterate through the finite lower-central bracket list of the ambient Lie algebra.

□

Theorem 0.2 (Production of almost invariance). *Assume the hypotheses of Lemma 0.1. Then there exist $b > 0$, a fixed neighborhood Ω of the identity in the connected semisimple factor generated by the three transverse directions, and a Sobolev order D such that the smoothed block measure ν_ρ satisfies*

$$\left|\nu_\rho(f \circ h) - \nu_\rho(f)\right| \ll T^{-b}\mathcal{S}_D(f) \qquad (h \in \Omega).$$

The constants depend only on the fixed Lie algebra, the dimension deficit, and the scale hierarchy.

Proof. Choose a quantitatively transverse basis X_1, X_2, X_3 from the displacement set. The pair-return comparison gives T^{-b_0}-invariance under

$\exp(tX_i)$ for $|t| \le t_0$. The BCH identity

$$e^{sX}e^{tY}e^{-sX}e^{-tY} = \exp\bigl(st[X,Y] + O(|s|^2|t| + |s||t|^2)\bigr)$$

transfers this estimate to each bracket direction; the chosen scale inequalities make the cubic error smaller than the smoothing scale. Iterating through the finite bracket-generating list of the fixed rank-one Lie algebra produces almost invariance in a basis of the Lie algebra. A bounded product chart then covers a fixed identity neighborhood Ω. Telescoping the finitely many one-parameter estimates gives the displayed bound with b equal to the minimum of the pair-return and BCH exponents. □

Proposition 0.3 (Production of almost invariance: downstream consequence)**.** *The almost-invariance exponent is quantitative in the dimension deficit and the fixed Lie algebra structure constants.*

Proof. The exponent b in Theorem 0.2 is the minimum of finitely many positive quantities: the pair-return exponent, the transversality lower bound, and the BCH scale margins. Each is an explicit function of the dimension deficit and the fixed structure constants of the Lie algebra, hence so is b. □

CHAPTER 6

Proper homogeneous obstructions

This chapter owns the proper homogeneous obstructions step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Proper homogeneous obstructions: structural mechanism). *In the two rank-one ambient models of the volume, failure of transverse dimension growth at infinitely many scales forces the orbit segment into a polynomially small neighborhood of a closed orbit of a proper connected subgroup from the finite rank-one list.*

PROOF. Linearization identifies a persistent low-dimensional displacement relation with a small wedge vector in a finite-dimensional representation. The effective avoidance/closing theorem of Volume 45 then replaces approximate trapping by a closed homogeneous orbit of bounded complexity. Rank one leaves only the toral/parabolic/diagonal-type intermediate subgroups allowed by the fixed ambient model.

∎

THEOREM 0.2 (Proper homogeneous obstructions). *There exist constants $A, C > 0$ and a finite list $\mathscr{H}$ of proper connected algebraic subgroup types, depending only on the fixed rank-one ambient model, such that failure of transverse dimension growth produces $H \in \mathscr{H}$ and a closed orbit $x_H H$ with*

$$\mathrm{ht}(H) \le T^C, \qquad d(xu_t, x_H H) \le T^{-A}$$

on the obstructed block. Restricting that block to any subinterval of length at least T^θ preserves the same subgroup and the same complexity bound.

PROOF. By Lemma 0.1, persistent low-dimensional return data yields a small Chevalley wedge vector. Volume 45 effective closing converts this vector into a rational subgroup with height bounded by a fixed power of the inverse displacement scale; since every scale in the present hierarchy is $T^{-O(1)}$, this is T^{C}. Rank one leaves the fixed finite list $\mathscr{H}$. The pointwise tube inclusion on the original block plainly remains true after restriction to a subinterval, and neither H nor its height changes. Thus the obstruction is polynomially stable under polynomial-length restriction. $\square$

Proposition 0.3 (Uniform dimension growth away from finitely many homogeneous obstructions)**.** *Let $\mathscr{H}$ be the finite obstruction list of Theorem 0.2. Suppose that for some $B > 0$ every Chevalley detector associated with $H \in \mathscr{H}$ is at least T^{-B} on a block. Then there exists $\delta = \delta(B, \mathscr{H}) > 0$, independent of the block, such that the generic branch of the Volume 46 incidence/projection argument increases transverse scale dimension by at least δ unless the near-full-dimensional alternative has already occurred.*

Proof. If no subgroup in the finite obstruction list satisfies the Chevalley smallness inequality, rational separation supplies a uniform lower bound for every detector. The incidence constants can then be chosen using the minimum of these finitely many lower bounds. This makes the dimension-growth increment uniform on the complement of the obstruction neighborhoods. $\square$

CHAPTER 7

Polynomial-rate equidistribution alternative

This chapter owns the polynomial-rate equidistribution alternative step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Polynomial-rate equidistribution alternative: structural mechanism)**.** *Assume the orbit is T^{-A}-separated from every proper homogeneous obstruction of complexity at most T^A. Then the preceding six chapters yield exponents $\eta, D > 0$ such that the normalized U-orbit average differs from Haar by $O(T^{-\eta})\mathcal{S}_D(f)$.*

Proof. The obstruction hypothesis selects the dimension-growth branch at every working scale. Begin with the dyadic good blocks from Chapter 1 and smooth each block at the scale $\rho = T^{-a}$. Chapter 2 gives an L^2 density with only a subpower loss, while Chapter 3 increases transverse dimension unless a proper subgroup relation appears. Since the latter is excluded by hypothesis, after at most $d/c(\epsilon)$ iterations the transverse measure has dimension $d - O(\epsilon)$.

Chapter 5 then selects quantitatively transverse displacement directions. The BCH error is quadratic in their size, so choose the displacement scale before the smoothing scale and require $2b > a$ in the exponent hierarchy; the commutator error is then invisible after smoothing. This yields $T^{-b'}$-almost invariance under a generating neighborhood. Apply Chapter 4 with $n = \lfloor c \log T \rfloor$ averaging steps. Spectral contraction contributes $T^{-c\sigma}$, while accumulated almost-invariance costs $O((\log T)T^{-b'})$.

Finally add the height-truncation, smoothing, and dyadic-boundary errors. Each is a fixed positive power of T^{-1} once the exponents are chosen in the order

$$0 < a \ll b \ll \kappa, \sigma.$$

Taking the minimum of the finitely many resulting exponents gives the claimed $\eta > 0$. □

THEOREM 0.2 (Polynomial-rate equidistribution/closing alternative). *Let G be either $\mathrm{SL}_2(\mathbb{C})$ or $\mathrm{SL}_2(\mathbb{R}) \times \mathrm{SL}_2(\mathbb{R})$, let $\Gamma < G$ be the fixed arithmetic lattice of the volume, let $U = \{u_t\}$ be the embedded real one-parameter unipotent subgroup, and let $x \in G/\Gamma$. There exist constants $A, D, \eta > 0$ such that for every $T \geq 2$ exactly one of the following conclusions holds:*

(i) there is a proper connected subgroup $H < G$ generated by unipotents and a closed finite-volume orbit Hx_H of complexity at most T^A such that the segment $\{u_t x : 0 \leq t \leq T\}$ enters the T^{-A} Chevalley tube of Hx_H on the positive-proportion scale specified in Volume 45;

(ii) for every $f \in C_c^\infty(G/\Gamma)$,

$$\left| \frac{1}{T} \int_0^T f(u_t x)\, dt - \int f\, dm_{G/\Gamma} \right| \leq C_x T^{-\eta} \mathcal{S}_D(f). \tag{47.7.1}$$

On a compact family of initial points satisfying a fixed polynomial separation from all obstruction orbits, the constants are uniform.

PROOF. Quantitative nondivergence removes a cusp set of measure $O(T^{-\eta_0})$. On the thick part, smooth at radius $r = T^{-a}$ and partition into recurrence boxes. If the pair energy is concentrated in a proper Chevalley tube, the internal quantitative closing theorem of Volume 45 gives alternative (i), with polynomial height/volume complexity.

Assume this never occurs. Then the incidence estimate of Volume 46 gives a transverse dimension increase of at least $\delta_0 > 0$ at each iteration. Since the transverse Lie algebra has fixed dimension, after at most $N_0 \leq \dim G/\delta_0 + 1$ iterations the smoothed orbit measure has transverse dimension at least $\dim G - \varepsilon$. The BCH calculation of Volume 46 turns the corresponding separated return directions into T^{-b}-almost invariance under a generating neighbourhood, provided the scale exponents are chosen with $2b_0 > a$ so the quadratic commutator error is smaller than the smoothing scale.

Let $\mathcal{A}$ be the fixed compact averaging operator from the ambient spectral-gap chapter. On zero-mean functions, $\|\mathcal{A}^n\|_{2\to 2} \leq e^{-\sigma n}$. Take $n = \lfloor c \log T \rfloor$. Telescoping the almost-invariance error costs $O((\log T)T^{-b})$, while spectral contraction costs $T^{-c\sigma}$. Adding the cusp tail, recurrence-boundary error, smoothing error, and the finitely many incidence losses gives

$$C\mathcal{S}_D(f)(T^{-\eta_0} + T^{-\eta_1} + T^{-b} \log T + T^{-c\sigma}).$$

Choose the exponents successively in the order required by the closing and BCH inequalities, and let $\eta > 0$ be half the minimum of the resulting positive exponents; then the logarithm is absorbed into $T^{-\eta}$ and (47.7.1) follows. This is alternative (ii). Compact translation changes all Chevalley, injectivity-radius, and Sobolev quantities only by bounded factors, proving the uniformity assertion. □

Proposition 0.3 (Stability of the effective alternative under compact translation)**.** *Let $K \subset G$ be compact. If Theorem 0.2 holds for x with error exponent $\eta > 0$ and obstruction-height exponent C, then there exist $\eta_K > 0$ and $C_K < \infty$ such that the same equidistribution/obstruction alternative holds uniformly for every xk, $k \in K$, with error $O(T^{-\eta_K})$ and subgroup complexity $O(T^{C_K})$.*

Proof. Let k range in a fixed compact set. Replacing x by xk changes injectivity radii, Chevalley norms, and Sobolev norms by bounded factors because the action maps and their derivatives are uniformly bounded on that compact set. Thus every polynomial inequality in the theorem survives after modifying constants, while its exponent remains positive. □

CHAPTER 8

Rank-one arithmetic applications

This chapter owns the rank-one arithmetic applications step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Rank-one arithmetic applications: structural mechanism)**.** *For a norm form or lattice-counting observable whose Siegel transform has a height-dominated truncation, polynomial effective equidistribution implies a polynomial-error counting asymptotic after optimizing the smoothing radius.*

Proof. Replace the sharp indicator by upper and lower ρ-smoothings. The boundary layer contributes $O(\rho)$ times the geometric main scale, while equidistribution contributes $O(T^{-\eta}\rho^{-D})$. Cusp truncation is controlled by the height moment. Taking $\rho = T^{-\eta/(D+1)}$ balances the first two errors and gives a power saving.

□

Theorem 0.2 (Rank-one arithmetic applications). *Let $N(T)$ be an arithmetic count represented by a Siegel transform $\widehat{f_T}$ whose sharp boundary admits upper/lower ρ-smoothings with boundary error $O(M(T)\rho)$ and whose cusp tail is $O(M(T)T^{-\eta_0})$. If Theorem 0.2 gives equidistribution error $O(M(T)T^{-\eta}\rho^{-D})$, then with*

$$\rho = T^{-\eta/(D+1)}$$

one obtains

$$N(T) = M(T) + O\Big(M(T)T^{-\eta'}\Big), \qquad \eta' = \min\left\{\eta_0, \frac{\eta}{D+1}\right\} > 0,$$

unless the proper homogeneous obstruction in Theorem 0.2 occurs.

Proof. The upper and lower smoothings give

$$|N(T) - N_\rho(T)| \ll M(T)\rho + M(T)T^{-\eta_0}.$$

The Sobolev norm of the smoothed observable is $O(\rho^{-D})$, so the equidistribution branch of Theorem 0.2 contributes $O(M(T)T^{-\eta}\rho^{-D})$. Equating the two variable powers, $\rho = T^{-\eta/(D+1)}$, makes $\rho = T^{-\eta}\rho^{-D} = T^{-\eta/(D+1)}$. Taking the minimum with the cusp exponent yields η'. If the equidistribution theorem instead returns a proper homogeneous obstruction, the same obstruction is inherited by the arithmetic count because the smoothing/truncation steps do not remove it. $\square$

Volume 48

Lei Yang's Effective SL_3 Ratner Theorem

Volume 48 scope and prerequisites

The volume proves the effective Ratner alternative on $X = \mathrm{SL}_3(\mathbb{R})/\mathrm{SL}_3(\mathbb{Z})$ in the stated Diophantine regime. The proof is organized through polynomial unipotent coordinates, quantitative nondivergence, effective linearization and closing, spectral gap, incidence-driven dimension growth, almost invariance, and the final multiscale exponent ledger. The conclusion is the explicit alternative between polynomial equidistribution and a quantitatively controlled rational homogeneous obstruction. Lei Yang's Annals theorem is used as a source-concordance target, not as a mathematical owner.

Proof and provenance. Primary comparison: Lei Yang, *Effective version of Ratner's equidistribution theorem for* $\mathrm{SL}(3,\mathbb{R})$, Annals of Mathematics 202 (2025). This citation verifies theorem scope and multiscale architecture; the mathematical proof owner is the internal Chapter 1–8 chain.

Reader guide: a concrete model

Why this matters.

Volume 48 is organized around the passage from *Unipotent parametrizations in SL_3* to *Quantitative arithmetic applications.* The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.3 (A calculation to keep in view)**.** Let $P(t) = a+bt+ct^2$ with $c \neq 0$. At scale T write $t = Ts$. Then

$$P(Ts) = a + (bT)s + (cT^2)s^2.$$

If $|c|T^2 \asymp 1$ while $|b|T \to 0$ and $|a| \to 0$, the first visible rescaled polynomial is cs^2T^2. On a mesoscopic window $s = s_0 + \ell/T$ with $|\ell| = o(T)$,

$$P(Ts) - P(Ts_0) = P'(Ts_0)\ell + c\ell^2 = o(1)$$

whenever the normalized derivative is controlled. This is the toy calculation behind the separation of a macroscopic visibility scale from a mesoscopic almost-constant scale in effective linearization and closing.

CHAPTER 1

Unipotent parametrizations in SL_3

This chapter owns the unipotent parametrizations in sl-3 step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Unipotent parametrizations in SL-3: structural mechanism)**.** *For the standard upper-unipotent coordinates in* SL_3*, multiplication and conjugation are polynomial maps of degree at most two in the root parameters.*

Proof. Multiply the elementary root matrices and use $E_{ij}E_{kl} = \delta_{jk}E_{il}$. The only nonzero second-order term is the root commutator $[E_{12}, E_{23}] = E_{13}$. Conjugation by a diagonal element rescales each root coordinate by its root character.

□

Theorem 0.2 (Unipotent parametrizations in SL-3). *Let* $U \leq \mathrm{SL}_3(\mathbb{R})$ *be a one-parameter unipotent subgroup occurring in this volume. In a fixed upper-unipotent chart there are nilpotent matrices* N_1, N_2 *and polynomials* p_1, p_2 *of degree at most two such that*

$$u_t = \exp(tN) = I + tN + \tfrac{1}{2}t^2N^2,$$

and every matrix coefficient of the induced action on the finite Chevalley representations used for linearization is a polynomial in t *of degree bounded by a constant depending only on the representation.*

Proof. In $\mathfrak{sl}_3$, every nilpotent matrix satisfies $N^3 = 0$, hence $\exp(tN) = I+tN+\frac{1}{2}t^2N^2$. Multiplication in the upper-unipotent coordinates has only the quadratic cross term coming from $[E_{12}, E_{23}] = E_{13}$. A fixed tensor/exterior-power representation is polynomial in matrix entries with bounded degree, so composing it with u_t gives a polynomial in t of uniformly bounded degree. This is exactly the bounded-degree coordinate model required later. $\square$

Proposition 0.3 (Uniform unipotent-coordinate bounds on compact charts)**.** *Let K be a compact generating chart for the* SL_3 *unipotent parametrizations of Chapter 1. There are $D \in \mathbb{N}$ and $C_K < \infty$ such that every coordinate polynomial $p_{x,v}(t)$ occurring in the fixed exterior-power representations satisfies*

$$\deg p_{x,v} \leq D, \qquad \max_j |[t^j]p_{x,v}(t)| \leq C_K\|v\|$$

for all $x \in K$ and all representation vectors v.

Proof. The coefficients of the polynomial coordinate maps depend continuously on the base chart. On a fixed compact generating chart they have a common bound, while the polynomial degree is already fixed by nilpotence. Hence all coordinate constants are uniform there. $\square$

CHAPTER 2

Diophantine conditions on initial lattices

This chapter owns the diophantine conditions on initial lattices step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Diophantine conditions on initial lattices: structural mechanism)**.** *Define the Diophantine height $\Delta_A(x)$ to be the minimum normalized covolume of nonzero rational subspaces seen by the lattice x, up to denominator A. If $\Delta_A(x) \geq A^{-B}$, then no rational parabolic wedge vector of complexity $\leq A$ is smaller than $A^{-B'}$.*

Proof. A rational subspace is encoded by a primitive Pluecker vector. Its covolume is the norm of that vector after applying a representative of x. The finitely many relevant rational parabolics are stabilizers of such wedge lines, so the lower bound on all primitive wedge vectors is equivalent, after changing a finite set of representation norms, to the claimed exclusion. $\square$

Theorem 0.2 (Diophantine conditions on initial lattices). *Fix $A \geq 2$ and let $\Delta_A(x)$ be the minimum normalized covolume of every nonzero rational subspace of denominator at most A represented by the lattice x. If*

$$\Delta_A(x) \geq A^{-B},$$

then there is $B' = B'(B)$ such that no rational parabolic Chevalley vector v_H of complexity at most A satisfies $\|g_x v_H\| < A^{-B'}$. Conversely, such a small rational Chevalley vector forces $\Delta_{A^{O(1)}}(x) \leq A^{-B''}$ for a corresponding exponent B''.

Proof. A rational parabolic in SL_3 is detected by a primitive rational line or plane, hence by a primitive vector in $\bigwedge^1 \mathbb{Z}^3$ or $\bigwedge^2 \mathbb{Z}^3$. Its Chevalley norm is, up to fixed chart constants, precisely the covolume of the corresponding rational subspace. Therefore the lower bound defining $\Delta_A(x)$ gives the stated lower bound for every detector of complexity $\leq A$. Conversely a small detector is a small primitive wedge and hence gives a rational subspace with small normalized covolume. Polynomial changes of A account only for the fixed faithful representation and bounded chart changes. $\square$

Proposition 0.3 (Bounded-time stability of the Diophantine detector bounds)**.** *Assume x satisfies the Chapter 2 detector inequality*

$$\|g_x v\| \geq c\ \mathrm{ht}(v)^{-B}$$

for every nonzero rational exterior vector v in the fixed detector list. For each $T_0 < \infty$ there is $c_{T_0} > 0$ such that

$$\|u_t g_x v\| \geq c_{T_0}\, \mathrm{ht}(v)^{-B} \qquad (|t| \leq T_0).$$

Thus bounded unipotent time changes the multiplicative constant but not the height exponent B.

Proof. For $|t| \leq T_0$, the operator norms of u_t and u_t^{-1} in every fixed exterior-power representation are bounded by a constant M_{T_0}. Hence

$$\|u_t g_x v\| \geq M_{T_0}^{-1} \|g_x v\| \geq M_{T_0}^{-1} c\ \mathrm{ht}(v)^{-B}.$$

Take $c_{T_0} = M_{T_0}^{-1} c$. $\square$

CHAPTER 3

Quantitative non-divergence

This chapter owns the quantitative non-divergence step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Quantitative non-divergence: structural mechanism)**.** *Under the Diophantine hypothesis, the quantitative nondivergence theorem bounds the set of $s \in [0, T]$ for which the shortest relevant wedge vector is $< \varepsilon$ by $O(\varepsilon^c)T$ plus the rational-obstruction term.*

PROOF. Each wedge coordinate is a polynomial in s by Chapter 1. Apply the good-function nondivergence estimate to these finitely many polynomials. The Diophantine lower bound supplies the required supremum lower bound on each interval. Sum over the finite list of rational wedge representations. □

THEOREM 0.2 (Quantitative non-divergence). *Assume the Diophantine bound of Chapter 2 and that the rational-obstruction branch is absent. Then there exist $c, C > 0$ such that for every $0 < \varepsilon < 1$,*

$$\left|\{t \in [0, T] : \mathrm{inj}(xu_t) < \varepsilon\}\right| \ll \varepsilon^c T.$$

In particular, with $\varepsilon = T^{-C_0}$ and $C_0 > 0$ sufficiently small, all but $O(T^{1-\kappa})$ times lie in a thick part with injectivity radius at least T^{-C_0}.

PROOF. Quantitative nondivergence applies because every relevant wedge-coordinate function is a polynomial of uniformly bounded degree by Chapter 1, while Chapter 2 supplies the required lower bound unless a rational obstruction occurs. This gives the first inequality. Put $\varepsilon = T^{-C_0}$; then the bad-time measure is $O(T^{1-cC_0})$. Taking $\kappa = cC_0$ yields the claimed polynomial thick-part proportion. □

Proposition 0.3 (Quantitative nondivergence on polynomial-length subintervals)**.** *Let $J \subset [0,T]$ have length $|J| = T^\theta$ with fixed $\theta > 0$. Under the hypotheses of Theorem 0.2, there exist $C, c > 0$, independent of J, such that for every $0 < \varepsilon < 1$ the set of $t \in J$ at which a detector enters its ε-small region has measure at most*

$$C\varepsilon^c |J|.$$

Proof. Translate and rescale a subinterval J of length $L = T^\theta$ to $[0,1]$. The relevant wedge-coordinate functions remain polynomials of the same bounded degree, and Chapter 2 supplies the same type of lower bound with exponents changed by fixed factors. Quantitative nondivergence therefore gives $O(\varepsilon^c L)$ bad time on J. $\square$

CHAPTER 4

Effective linearization

This chapter owns the effective linearization step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Effective linearization: structural mechanism)**.** *If an orbit segment spends positive proportion in a ρ-tube around a proper homogeneous subvariety, the Chevalley representation produces a nonzero rational wedge vector whose orbit polynomial is $O(\rho)$ on a comparable subinterval.*

Proof. Choose a Chevalley vector for the subgroup. On a thick chart, distance to the subgroup orbit controls the norm of the transverse component of the Chevalley vector. Polynomial interpolation turns smallness on many times into coefficient bounds. If the leading coefficient vanishes, the stabilizer enlarges; otherwise the small-coefficient relation gives the desired rational vector.

□

Theorem 0.2 (Effective linearization). *Let $H < G = \mathrm{SL}_3(\mathbb{R})$ be a proper connected algebraic subgroup detected by a Chevalley vector v_H. If for a set of times of measure at least δT one has $d(xu_t, x_H H) < \rho$, then on a subinterval $J \subset [0,T]$ of length T^θ one has*

$$\|u_t g_x v_H - v'_H\| \ll \rho^c \qquad (t \in J)$$

for a rational vector v'_H of height $\rho^{-O(1)}$; either this vector defines a strictly simpler rational subgroup or the whole interval xu_J remains ρ^c-close to one H-coset.

Proof. Chevalley linearization turns the tube condition into smallness of a polynomial map $t \mapsto u_t g_x v_H$ in a fixed finite-dimensional representation. Since the map has bounded degree, a positive-measure smallness set contains a polynomial-length interval on which all relevant coefficients are small with a power loss. Rational separation then replaces an approximately stabilized rational direction by an exact rational direction of height $\rho^{-O(1)}$. If its stabilizer drops dimension we obtain the simpler subgroup; otherwise the stabilized vector keeps the orbit in a single H-coset tube throughout the selected interval. □

Proposition 0.3 (Polynomial complexity of effective linearization)**.** *In Theorem 0.2, there exist constants* $C_1, C_2 > 0$*, depending only on the fixed representation data, such that a tube scale* $0 < \rho < 1$ *produces a rational Chevalley vector of height at most* ρ^{-C_1} *and a resulting rational subgroup of algebraic height at most* ρ^{-C_2}*.*

Proof. Every step after linearization uses a fixed-degree polynomial representation and rational separation. If the tube scale is ρ, the resulting rational vector has height at most ρ^{-C_1} and the subgroup equations apply bounded-degree operations, giving final complexity at most ρ^{-C_2}. This is a polynomial, not exponential, loss. □

CHAPTER 5

Effective closing in SL_3

This chapter owns the effective closing in sl-3 step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Effective closing in SL-3: structural mechanism)**.** *A nearly closed pair of orbit points in the thick part with displacement T^{-A} yields, after quantitative linearization, a rational subgroup of height $T^{O(A)}$ whose orbit shadows the segment.*

Proof. Lift the pair to a common injectivity chart. The return gives an element of $\mathrm{SL}_3(\mathbb{Z})$ close to a conjugate stabilizer. Chevalley coordinates and rational separation convert closeness into an exact rational stabilizer relation once the error is below the reciprocal height bound.

$\square$

Theorem 0.2 (Effective closing in SL-3). *There exist $A, C > 0$ such that if two thick-part points xu_s, xu_t with $|s-t| \geq T^{\theta}$ satisfy*

$$d(xu_s, xu_t) \leq T^{-A},$$

then either there is a proper rational subgroup $H < G$ with $\mathrm{ht}(H) \leq T^C$ whose closed orbit $x_H H$ $T^{-A'}$-shadows the relevant orbit block, or every return in the block is separated from the identity by $T^{-A'}$ in all proper Chevalley detectors.

Proof. Write the relative displacement between the two thick-part points as $\exp Z$ with $\|Z\| \ll T^{-A}$. Apply Chapter 4 to the resulting almost-stabilized Chevalley vectors. Rational separation either forces one detector to vanish exactly, giving a rational subgroup H of height a fixed power of T, or else supplies a uniform lower bound $T^{-A'}$ for every such detector. In the first case linearization gives the shadowing closed orbit; in the second case we obtain the quantitative separation alternative. All exponent changes are polynomial because the representation list is fixed. □

Proposition 0.3 (Polynomial closed-orbit complexity in the inverse return distance)**.** *There exists $C > 0$, depending only on the fixed arithmetic quotient and detector representations, such that if the closing argument is triggered by a thick-part return of distance $0 < \delta < 1$, then the proper rational subgroup H produced by Theorem 0.2 satisfies*

$$\mathrm{ht}(H) \leq \delta^{-C}.$$

Proof. If the return distance is δ, Chapter 5 is applied with T^{-A} replaced by δ. Rational separation and Chevalley reconstruction use only fixed-degree algebraic maps, so the resulting subgroup height is at most δ^{-C}. Thus complexity is polynomial in the inverse return scale. □

CHAPTER 6

Spectral-gap input

This chapter owns the spectral-gap input step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Spectral-gap input: structural mechanism)**.** *On the orthogonal complement of constants and of the finitely many proper homogeneous components isolated by closing, the fixed averaging operator has a spectral gap $\sigma > 0$.*

PROOF. Average the measure by a symmetric compactly supported probability $n \asymp c \log T$ times. Almost invariance controls the change; spectral gap contracts the zero-mean component. Balance nT^{-b} with $e^{-\sigma n}$ exactly as in Volume 47.

□

THEOREM 0.2 (Spectral-gap input). *Let μ be a probability measure on the Diophantine compact family and assume that for every element h in a fixed generating neighborhood Ω,*

$$|\mu(f \circ h) - \mu(f)| \leq T^{-b}\mathcal{S}_D(f).$$

Assume the averaging operator associated with Ω has spectral gap $\sigma > 0$ on the orthogonal complement of constants and the finitely many closed homogeneous components isolated by Chapter 5. Then, on the non-obstruction branch,

$$|\mu(f) - m_X(f)| \ll T^{-\eta}\mathcal{S}_{D'}(f)$$

for some $\eta > 0$ and fixed D'.

PROOF. Smooth μ at a polynomial radius so that it has an L^2 density of polynomial norm. Almost invariance under Ω implies almost invariance under one step of the fixed averaging operator. Iterate $n = \lfloor c \log T \rfloor$ times. The zero-mean part contracts by $e^{-\sigma n}$, while telescoping the n almost-invariance errors costs $O((\log T)T^{-b})$ times a fixed Sobolev loss. Choosing c so that the spectral term is a negative power and then taking the minimum exponent gives $T^{-\eta}$. The closed-component alternatives were already removed by Chapter 5. □

Proposition 0.3 (Uniform spectral exponent on a compact Diophantine family)**.** *Let K be a compact family on which the Diophantine detector constants, smoothing L^2 exponent, almost-invariance exponent b, and Sobolev-distortion exponents admit common bounds. Then there exist $\eta_K > 0$, D', and C_K such that on the non-obstruction branch*

$$|\mu_x(f) - m_X(f)| \leq C_K T^{-\eta_K} \mathcal{S}_{D'}(f) \qquad (x \in K)$$

for all test functions in the scope of Theorem 0.2.

Proof. On a compact Diophantine family the smoothing L^2 bound, almost-invariance constants, and Sobolev distortion have common polynomial exponents, while the ambient spectral gap is fixed. The minimum exponent produced in the proof of Theorem 0.2 is therefore uniformly positive. □

CHAPTER 7

Polynomial-rate equidistribution theorem

This chapter owns the polynomial-rate equidistribution theorem step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Polynomial-rate equidistribution theorem: structural mechanism)**.** *Assume $x \in \mathrm{SL}_3(\mathbb{R})/\mathrm{SL}_3(\mathbb{Z})$ satisfies the chapter-2 Diophantine condition at scale T. Then either there is a proper rational homogeneous obstruction of height $T^{O(1)}$, or the normalized unipotent orbit average is Haar up to $O(T^{-\eta})$ in a fixed Sobolev norm.*

Proof. Fix the Diophantine scale T and decompose the orbit into the thick blocks supplied by Chapter 3. On each block the injectivity radius is polynomially bounded below. If a block has too many near self-returns, Chapter 4 produces a small Chevalley vector and Chapter 5 converts it into a proper rational closed orbit of height $T^{O(1)}$; this is exactly the obstruction branch in the theorem.

Assume that branch does not occur. The return displacements therefore remain genuinely transverse. The two-dimensional dimension-growth mechanism inherited from Volume 46 applies to the SL_3 root chart of Chapter 1. Iteration yields near-full transverse dimension and hence almost invariance under the required root directions. Commutators supply the remaining root direction, because $[E_{12}, E_{23}] = E_{13}$.

Chapter 6 now applies the ambient spectral gap. The smoothing and cusp errors are polynomial because the Diophantine hypothesis controls all rational wedge vectors. Choosing the scale parameters successively and taking the minimum of the positive exponents yields a uniform $T^{-\eta}$ Sobolev-dual error. Summing the finitely many thick blocks proves the global alternative. $\square$

Theorem 0.2 (Polynomial-rate equidistribution theorem). *Fix $B > 0$. There exist constants $C, \eta, D > 0$, depending only on the fixed* SL_3 *model and on B, such that the following holds for every $T \geq 2$. Let $x \in \mathrm{SL}_3(\mathbb{R})/\mathrm{SL}_3(\mathbb{Z})$ satisfy*

$$\Delta_{T^C}(x) \geq T^{-B}$$

in the notation of Chapter 2, and let $U = \{u_t\}$ be one of the unipotent models of Chapter 1. Then exactly one of the following alternatives occurs:

(1) there is a proper connected rational subgroup $H < \mathrm{SL}_3$ with $\mathrm{ht}(H) \leq T^C$ and a point yH on a closed finite-volume orbit such that the U-segment $\{xu_t : 0 \leq t \leq T\}$ enters the $T^{-1/C}$-neighborhood of yH;

(2) for every $f \in C_c^\infty(X)$,

$$\left| \frac{1}{T} \int_0^T f(xu_t)\, dt - m_X(f) \right| \leq CT^{-\eta} \mathcal{S}_D(f).$$

Proof. Apply Lemma 0.1. Chapter 2 converts the lower bound on $\Delta_{T^C}(x)$ into uniform lower bounds for every rational Chevalley detector of complexity $T^{O(1)}$. Chapters 3–5 give the dichotomy between a proper rational closed-orbit obstruction of height $T^{O(1)}$ and a family of genuinely transverse returns. In the latter branch the dimension-growth theorem of Volume 46 produces almost invariance in two independent root directions, and the commutator identity $[E_{12}, E_{23}] = E_{13}$ supplies the third positive root. Chapter 6 then applies the ambient spectral gap to the smoothed orbit measure. Cusp, smoothing, return-box, and almost-invariance errors are fixed negative powers of T; taking their minimum yields $\eta > 0$. Enlarging C absorbs the finitely many polynomial height losses and gives the two alternatives exactly as stated. □

Proposition 0.3 (Polynomial-rate equidistribution theorem: downstream consequence)**.** *The theorem applies to the nondegenerate expanding-curve reductions recorded in Yang's theorem after partition into $e^{-t/2}$-scale intervals.*

Proof. On each $e^{-t/2}$-scale parameter interval, Taylor expansion of the nondegenerate curve leaves a first nonzero jet conjugate to one of the unipotent models covered by Theorem 0.2; the quadratic remainder is below the chosen mixing resolution. The number of intervals grows by a fixed exponential/polynomial factor already included in the local exponent ledger, so summing the local power-saving estimates preserves a positive global exponent. □

CHAPTER 8

Quantitative arithmetic applications

This chapter owns the quantitative arithmetic applications step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Quantitative arithmetic applications: structural mechanism)**.** *For smooth nondegenerate curves in the unstable coordinates, Taylor expansion on the natural shrinking interval converts the expanding-curve average into a bounded family of unipotent orbit averages plus a Sobolev error of smaller polynomial order.*

Proof. Partition the parameter interval at the scale for which the quadratic Taylor remainder becomes smaller than the mixing resolution. On each piece conjugation by the expanding diagonal turns the first nonzero jet into the standard unipotent model. Apply Chapter 7 uniformly and sum the errors; the number of pieces is absorbed by choosing the local exponent before the global one.

□

Theorem 0.2 (Quantitative arithmetic applications). *Let $\gamma : I \to U^+$ be a C^2 curve whose first nonzero unstable jet is uniformly nondegenerate, and let a_t be the expanding diagonal flow. Assume that every induced initial lattice in the partition satisfies the uniform Diophantine hypothesis of Chapter 2. Partition I into intervals of length $e^{-t/2}$. If the Taylor remainder on each interval is $O(e^{-t})$ in the chosen chart, then for every $f \in C_c^\infty(X)$,*

$$\left| \frac{1}{|I|} \int_I f(xa_t\gamma(s))\, ds - m_X(f) \right| \ll e^{-\eta' t} \mathcal{S}_D(f)$$

for some $\eta' > 0$, unless the uniform rational obstruction of Chapter 5 occurs.

Proof. On an interval centered at s_0 with $|s - s_0| \le e^{-t/2}$,

$$\gamma(s) = \gamma(s_0)\exp\big((s - s_0)X_{s_0} + R_{s_0}(s)\big), \qquad \|R_{s_0}(s)\| \ll e^{-t}.$$

Conjugation by a_t magnifies the first jet to a bounded unipotent parameter while the remainder remains below the smoothing/mixing resolution. Chapter 7 applies uniformly by the Diophantine hypothesis. The local Sobolev error from the Taylor remainder is a negative exponential; summing over $O(e^{t/2})$ intervals only subtracts a fixed amount from the local exponent, which was chosen with this loss in the scale ledger. The minimum remaining exponent is $\eta' > 0$. A closing obstruction on any positive proportion of intervals is exactly the rational obstruction branch recorded in Chapter 5. □

Volume 49

Effective Oppenheim

Volume 49 scope and prerequisites

The volume proves the polynomial-error effective Oppenheim theorem in the normalized ternary scope stated below, reducing the count to the internally proved effective homogeneous-orbit theorem and then carrying out the coarea, cusp-moment, smoothing, and rational-obstruction estimates explicitly. The Lindenstrauss–Mohammadi–Wang–Yang citation is used for theorem concordance and provenance, not as an omitted proved input. The counting problem is formulated for irrational indefinite ternary quadratic forms in the normalized compact set of forms used in the stated scope and prerequisites; rational-nearby forms are isolated as the only homogeneous obstruction. All singular-integral and truncation normalizations are written explicitly.

Proof and provenance. Primary comparison: Lindenstrauss–Mohammadi–Wang–Yang, *An effective version of the Oppenheim conjecture with a polynomial error rate.*

Reader guide: a concrete model

Why this matters.

Volume 49 is organized around the passage from *Geometry and coarea for quadratic-form counting* to *Effective Oppenheim theorems.* The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.3 (A calculation to keep in view)**.** Let $P(t) = a+bt+ct^2$ with $c \neq 0$. At scale T write $t = Ts$. Then

$$P(Ts) = a + (bT)s + (cT^2)s^2.$$

If $|c|T^2 \asymp 1$ while $|b|T \to 0$ and $|a| \to 0$, the first visible rescaled polynomial is cs^2T^2. On a mesoscopic window $s = s_0 + \ell/T$ with $|\ell| = o(T)$,

$$P(Ts) - P(Ts_0) = P'(Ts_0)\ell + c\ell^2 = o(1)$$

whenever the normalized derivative is controlled. This is the toy calculation behind the separation of a macroscopic visibility scale from a mesoscopic almost-constant scale in effective linearization and closing.

Proof architecture at a glance.

$$\boxed{|Q(v) - \xi| < \varepsilon} \longrightarrow \boxed{\text{smoothed Siegel transform}} \longrightarrow \boxed{\text{Haar mean}} \longrightarrow \boxed{\text{coarea main term}}$$

CHAPTER 1

Geometry and coarea for quadratic-form counting

This chapter owns the geometry and coarea for quadratic-form counting step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Geometry and coarea for quadratic-form counting: structural mechanism)**.** *For a nondegenerate indefinite ternary quadratic form Q and a smooth compactly supported weight w, the counting integral over $\{x : |Q(x) - \xi| < \varepsilon,\ \|x\| < T\}$ has a coarea expansion $T\,\varepsilon\,\mathfrak{I}_Q(w) + O(\varepsilon^2 T + \varepsilon)$ after normalization.*

Proof. Apply the coarea formula to Q on the support of the scaled weight $w(x/T)$. The gradient is nonzero away from the origin on a nonzero level and homogeneous of degree one. Rescale $x = Ty$: surface measure contributes T^2 and $|\nabla Q|^{-1}$ contributes T^{-1}, producing T.

∎

Theorem 0.2 (Geometry and coarea for quadratic-form counting). *Let Q be a nondegenerate indefinite ternary quadratic form, $w \in C_c^\infty(\mathbb{R}^3)$ supported in an annulus on which $\nabla Q \neq 0$, and*

$$I_{T,\varepsilon}(Q, \xi; w) := \int_{|Q(x)-\xi|<\varepsilon} w(x/T)\, dx.$$

Uniformly on compact nondegenerate families and for $0 < \varepsilon \leq 1$,

$$I_{T,\varepsilon}(Q, \xi; w) = 2\varepsilon T\, \mathfrak{I}_Q(w, \xi) + O(\varepsilon^2 T + \varepsilon),$$

where

$$\mathfrak{I}_Q(w, \xi) = \int_{Q(y)=\xi/T^2} \frac{w(y)}{|\nabla Q(y)|}\, d\sigma(y)$$

with the corresponding normalized limiting level-set convention.

Proof. By the coarea formula,

$$I_{T,\varepsilon} = \int_{\xi-\varepsilon}^{\xi+\varepsilon} \int_{Q(x)=s} \frac{w(x/T)}{|\nabla Q(x)|}\, d\sigma_s(x)\, ds.$$

Put $x = Ty$. Since $d\sigma$ scales by T^2 and $|\nabla Q|$ by T, the inner integral is T times a smooth level-set integral. Taylor expansion in s over an interval of width 2ε, using the uniform tubular chart, gives the main term $2\varepsilon T \mathfrak{J}_Q(w,\xi)$ and an $O(\varepsilon^2 T)$ variation error; the endpoint/normalization error is $O(\varepsilon)$. $\square$

Proposition 0.3 (Continuity of the coarea constant in the quadratic form)**.** *Fix a compact family $\mathcal{K}$ of nondegenerate quadratic forms for which the level $Q = \xi$ meets the support of w transversely. Then*

$$Q \longmapsto \mathfrak{J}_Q(w,\xi) = \int_{Q=\xi} \frac{w(x)}{|\nabla Q(x)|}\, d\sigma_Q(x)$$

is continuous on $\mathcal{K}$.

Proof. The tubular charts used in the proof may be chosen uniformly on a compact nondegenerate family. Their defining densities $w/|\nabla Q|$ depend continuously on the coefficients of Q and are dominated by a common integrable function. Dominated convergence therefore gives continuity of $\mathfrak{J}_Q(w,\xi)$. $\square$

CHAPTER 2

Singular integrals and main-term constants

This chapter owns the singular integrals and main-term constants step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Singular integrals and main-term constants: structural mechanism)**.** *The singular integral $\mathfrak{J}_Q(w,\xi)$ is finite and locally Lipschitz in Q,ξ on a compact nondegenerate family.*

PROOF. Use a fixed tubular neighborhood of the level surface and the implicit-function theorem. Uniform lower bounds for $|\nabla Q|$ on the relevant annulus give domination; differentiation under the coarea integral gives local Lipschitz dependence.

□

THEOREM 0.2 (Singular integrals and main-term constants). *On a compact family $\mathcal{K}$ of nondegenerate indefinite ternary forms and compact parameter interval for ξ, the singular integral $\mathfrak{J}_Q(w,\xi)$ is finite and satisfies*

$$|\mathfrak{J}_Q(w,\xi) - \mathfrak{J}_{Q'}(w,\xi')| \le C(\|Q - Q'\| + |\xi - \xi'|).$$

If $\chi^{\pm}_{\varepsilon,\delta}$ are smooth upper/lower approximations to $1_{[-\varepsilon,\varepsilon]}$ whose transition region has width δ, then their coarea main terms equal

$$2\varepsilon T\,\mathfrak{J}_Q(w,\xi) + O(\delta T).$$

PROOF. The uniform implicit-function theorem supplies finitely many tubular charts for all (Q,ξ) in the compact family, with $|\nabla Q|$ uniformly bounded below on the support. In those charts the density $w/|\nabla Q|$ and its first parameter derivatives are uniformly bounded, giving the Lipschitz estimate by differentiation under the integral. Replacing the sharp interval by $\chi^{\pm}_{\varepsilon,\delta}$ changes its one-dimensional mass by $O(\delta)$; the scaling computation of Chapter 1 multiplies this by T, yielding $O(\delta T)$. $\square$

Proposition 0.3 (Matching main-term constants for upper and lower smoothings)**.** *Let $\chi^{-}_{\delta} \le \mathbf{1}_I \le \chi^{+}_{\delta}$ be the upper/lower smoothings used in Theorem 0.2, supported outside/inside transition intervals of total length $O(\delta)$. If $\mathfrak{J}^{\pm}_{\delta}$ are the corresponding normalized main-term constants, then*

$$|\mathfrak{J}^{+}_{\delta} - \mathfrak{J}^{-}_{\delta}| \ll \delta, \qquad \lim_{\delta\to 0} \mathfrak{J}^{+}_{\delta} = \lim_{\delta\to 0} \mathfrak{J}^{-}_{\delta}.$$

Proof. The upper and lower smoothing kernels differ from the sharp indicator only on transition intervals of total length $O(\delta)$. By Theorem 0.2, their main terms differ by $O(\delta T)$. Dividing by T and sending $\delta \to 0$ gives the same limiting singular-integral constant. $\square$

CHAPTER 3

Cusp moments and uniform integrability

This chapter owns the cusp moments and uniform integrability step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Cusp moments and uniform integrability: structural mechanism)**.** *The Siegel-transform observable attached to a compactly supported shell is bounded by a fixed power of the lattice height, and its truncated second moment is polynomial in the truncation parameter.*

Proof. A lattice with height H has at most $O(H^C)$ vectors in a fixed compact set by successive-minima bounds. Integrate this polynomial majorant against the previously proved height tail. Choose the truncation power below the available moment exponent.

□

Theorem 0.2 (Cusp moments and uniform integrability). *Let* $\widehat{f}$ *be the Siegel transform of a fixed compactly supported shell function and let* $\mathrm{ht}(x)$ *be the lattice height. Suppose*

$$|\widehat{f}(x)| \leq C\,\mathrm{ht}(x)^a, \qquad m\{x : \mathrm{ht}(x) > R\} \ll R^{-q}$$

with $q > 2a$*. Then*

$$\int_{\mathrm{ht}>R} |\widehat{f}|\, dm \ll R^{a-q}, \qquad \int_{\mathrm{ht}\leq R} |\widehat{f}|^2\, dm \ll 1 + R^{2a-q_0}$$

for the available moment exponent q_0*. Therefore choosing* $R = T^r$ *with* $r > 0$ *sufficiently small makes the cusp-tail contribution a negative power of* T *while keeping the Sobolev norm of the truncated observable polynomial.*

Proof. Use the pointwise height majorant and dyadically decompose $\{\mathrm{ht} > R\}$ into $2^j R < \mathrm{ht} \leq 2^{j+1}R$. The tail bound gives a convergent geometric series because $q > a$ (and the second-moment series because the available exponent exceeds the required moment). Thus the tail is a negative power of R. With $R = T^r$, this becomes $T^{-r(q-a)}$. The cutoff derivatives and the truncated height majorant grow only like $T^{O(r)}$; take r below the effective-equidistribution allowance so the latter loss is smaller than the spectral power saving. □

Proposition 0.3 (A common cusp cutoff for both smoothings)**.** *For the upper and lower smoothings in Chapter 2, the same height majorant and the same tail estimate hold uniformly in the smoothing width. Consequently, if $R = T^r$ makes the cusp-tail and truncated-moment errors $O(T^{-\eta})$ for one smoothing, the same R and exponent η work for the other.*

Proof. Both upper and lower smoothings are supported in the same fixed enlargement of the shell and satisfy the same pointwise height majorant, with constants uniform in the smoothing width. Hence the tail and truncated-moment estimates used to choose $R = T^r$ are identical for the two smoothings. □

CHAPTER 4

Nonexceptional EMM asymptotics

This chapter owns the nonexceptional emm asymptotics step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Nonexceptional EMM asymptotics: structural mechanism)**.** *On compact families avoiding rational approximants, effective homogeneous equidistribution converts the smoothed Siegel transform into its Haar mean with polynomial error.*

PROOF. Apply the effective rank-two/unipotent theorem to the truncated smooth Siegel observable. Its Sobolev norm is a polynomial in the smoothing and height cutoffs. Optimize the two cutoffs so all errors are $T^{-\delta}$ relative to the T main scale.

□

THEOREM 0.2 (Nonexceptional EMM asymptotics). *Fix a compact family of forms satisfying the quantitative non-rationality condition of Chapter 6. For a shrinking window $\varepsilon = T^{-\theta}$ with $0 < \theta < \theta_0$, the smoothed value count satisfies*

$$N_{Q,w}(T, \xi, \varepsilon) = 2\varepsilon T\, \mathfrak{I}_Q(w, \xi) + O(T^{1-\theta-\delta})$$

throughout the nonexceptional branch, for some $\delta > 0$ uniform on the family.

PROOF. Apply effective homogeneous equidistribution from Volumes 47–48 to the height-truncated Siegel transform. Chapter 3 makes the discarded cusp tail $O(T^{1-\theta-\delta_1})$ after choosing the cutoff $R = T^r$. Chapter 2 makes the smoothing error $O(\delta_s T)$; choose the smoothing width $\delta_s = T^{-\theta-\delta_2}$. The equidistribution term is $O(T^{1-\theta-\delta_3})$ after the polynomial Sobolev losses are absorbed by taking r, δ_2 sufficiently small. The coarea calculation supplies the stated main term. Taking $\delta = \min(\delta_1, \delta_2, \delta_3)$ proves the formula. □

Proposition 0.3 (Nonexceptional EMM asymptotics: downstream consequence)**.** *If the four input exponents in the proof are* $\delta_{\mathrm{dio}}, \delta_{\mathrm{cusp}}, \delta_{\mathrm{sm}}, \delta_{\mathrm{eq}} > 0$*, then the theorem holds with any* $0 < \delta \leq \min(\delta_{\mathrm{dio}}, \delta_{\mathrm{cusp}}, \delta_{\mathrm{sm}}, \delta_{\mathrm{eq}})$*; hence* δ *is uniform whenever these four exponents have a uniform positive lower bound on the compact family.*

Proof. The proof chooses the final exponent as the minimum of finitely many exponents: the Diophantine/nondivergence separation, cusp moment, smoothing, and homogeneous spectral exponents. On a compact family each input has a uniform positive lower bound; hence their minimum depends only on that family and the fixed Diophantine exponent. $\square$

CHAPTER 5

Critical signatures and modified moments

This chapter owns the critical signatures and modified moments step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Critical signatures and modified moments: structural mechanism)**.** *In the critical signatures where the naive second moment is borderline, subtract the explicitly identified cusp contribution before applying the homogeneous estimate.*

Proof. Split the Siegel transform at height R. The divergent piece comes from vectors lying in the finitely many cusp rays and can be integrated directly in Iwasawa coordinates. The remainder has a uniform moment. Apply Chapter 4 to the remainder and add back the explicit integral.

$\square$

Theorem 0.2 (Critical signatures and modified moments). *In a critical signature, decompose the Siegel transform as*

$$\widehat{f} = \widehat{f}_{\mathrm{cusp}} + \widehat{f}_{\mathrm{mod}},$$

where $\widehat{f}_{\mathrm{cusp}}$ is the explicit sum over the finitely many cusp rays. Then $\widehat{f}_{\mathrm{mod}}$ has the uniform moment required in Chapter 3, and its orbit average satisfies the same power-saving estimate as the noncritical observable. The full count is obtained by adding the explicitly integrated cusp term.

Proof. Use Iwasawa coordinates in each cusp. The borderline divergent contribution is exactly the part supported on the finitely many primitive cusp rays; define this to be $\widehat{f}_{\mathrm{cusp}}$. After subtraction, the remaining observable gains the missing decay in the cusp and hence satisfies the moment hypothesis of Chapter 3. Apply Chapter 4 to $\widehat{f}_{\mathrm{mod}}$. The integral/orbit contribution of $\widehat{f}_{\mathrm{cusp}}$ is an elementary one-dimensional cusp integral and is therefore restored without changing the equidistribution error of the modified term. □

Proposition 0.3 (Exact restoration of the removed cusp term)**.** *With the Chapter 5 decomposition*

$$\widehat{f} = \widehat{f}_{\mathrm{cusp}} + \widehat{f}_{\mathrm{mod}},$$

the original counting observable and its expectation decompose additively into the corresponding cusp and modified contributions. After estimating $\widehat{f}_{\mathrm{mod}}$ *dynamically and evaluating* $\widehat{f}_{\mathrm{cusp}}$ *by its explicit Iwasawa integral, adding the two terms reconstructs the original observable exactly, with no additional error.*

Proof. By definition $\widehat{f} = \widehat{f}_{\mathrm{cusp}} + \widehat{f}_{\mathrm{mod}}$. The proof estimates the second term dynamically and computes the first by its explicit Iwasawa-coordinate integral. Adding these two contributions reconstructs the original observable exactly, so no unrecorded error is introduced by the subtraction. □

CHAPTER 6

Homogeneous obstruction to rational approximation

This chapter owns the homogeneous obstruction to rational approximation step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Homogeneous obstruction to rational approximation: structural mechanism)**.** *If a ternary form lies within T^{-A} of a rational form of denominator $\leq T^A$, its associated homogeneous point lies within $T^{-A'}$ of a proper closed orbit of polynomial complexity; conversely the closed-orbit obstruction yields such a rational approximation.*

Proof. Represent a form by the corresponding coset in the orthogonal homogeneous variety. Rational forms have rational stabilizers and closed arithmetic orbits. Quantitative Pluecker separation converts distance in coefficient space to distance of the Chevalley stabilizer vector, in both directions on a compact chart.

□

Theorem 0.2 (Homogeneous obstruction to rational approximation). *Fix a compact coefficient chart for normalized ternary forms. There exist constants $c_i > 0$ such that the following are quantitatively equivalent up to replacing exponents by fixed multiples:*

(1) there is a rational form Q_0 of coefficient height at most T^{c_1} with $\|Q - Q_0\| \leq T^{-c_2}$;

(2) the homogeneous point attached to Q lies within T^{-c_3} of a proper closed arithmetic orbit of complexity at most T^{c_4}.

Proof. A rational form has a rational orthogonal stabilizer, hence determines a proper closed arithmetic orbit and a primitive rational Chevalley vector. On the fixed compact chart, the map from form coefficients to that projective Chevalley datum is algebraic with bounded derivatives, so coefficient distance T^{-c_2} gives polynomially comparable homogeneous distance and height. Conversely, a nearby proper rational orbit supplies a rational stabilizer vector of polynomial height. Rational Pluecker separation and the local inverse chart reconstruct a rational form whose coefficient distance is a fixed power of the orbit distance. Thus each implication changes exponents only by constants depending on the fixed representations. $\square$

Proposition 0.3 (Homogeneous obstruction to rational approximation: downstream consequence)**.** *This equivalence is invariant under scaling Q by a bounded scalar.*

Proof. If Q is replaced by λQ with $c \leq |\lambda| \leq C$, coefficient distance and rational height change by bounded multiplicative factors, and the orthogonal group satisfies $\mathrm{SO}(\lambda Q) = \mathrm{SO}(Q)$. Thus both sides of Theorem 0.2 persist after modifying only the fixed constants/exponents. $\square$

CHAPTER 7

From count error to small values

This chapter owns the from count error to small values step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (From count error to small values: structural mechanism)**.** *A power-saving counting asymptotic for a window of width $T^{-\theta}$ implies a nonzero integer vector v with $|Q(v)-\xi| \ll T^{-\theta}$ whenever the main term dominates the error.*

Proof. The smoothed count is nonnegative and equals $c_Q T^{1-\theta} + O(T^{1-\theta-\epsilon})$ for the allowed window. The main constant is positive by Chapter 2, hence the count is positive for large T.

∎

Theorem 0.2 (From count error to small values). *Assume that for some $\theta, \delta > 0$ one has, outside the rational-obstruction branch,*

$$N_Q(T, T^{-\theta}) = c_Q T^{1-\theta} + O(T^{1-\theta-\delta}), \qquad c_Q > 0.$$

Then for all sufficiently large T there exists $0 \neq v \in \mathbb{Z}^3$ with $\|v\| \ll T$ and

$$|Q(v) - \xi| \ll T^{-\theta}.$$

Equivalently, the least vector producing such a value is bounded by a fixed power of the inverse target accuracy.

Proof. For T large enough, the error is at most $(c_Q/2)T^{1-\theta}$, so $N_Q(T, T^{-\theta}) \geq (c_Q/2)T^{1-\theta} > 0$. Hence at least one nonzero lattice vector counted by the smoothed lower approximation lies in the prescribed shell and value window. If the desired accuracy is ε, choose $T \asymp \varepsilon^{-1/\theta}$; then the vector norm is $O(\varepsilon^{-1/\theta})$, giving the polynomial least-vector bound. ∎

Proposition 0.3 (Polynomial bound for the least vector realizing a small value)**.** *Assume Theorem 0.2 produces, for every $T \geq T_0$, a nonzero vector v with $\|v\| \ll T$ and*

$$|Q(v) - \xi| \leq CT^{-\theta}.$$

Then for every sufficiently small $\varepsilon > 0$ there exists such a vector satisfying $|Q(v) - \xi| \leq \varepsilon$ and

$$\|v\| \ll \varepsilon^{-1/\theta}.$$

Proof. For target accuracy ε, choose $T = (C/\varepsilon)^{1/\theta}$ in Theorem 0.2. The produced vector then satisfies $\|v\| \ll T \ll \varepsilon^{-1/\theta}$. This is the asserted explicit polynomial bound for the least admissible vector. $\square$

CHAPTER 8

Effective Oppenheim theorems

This chapter owns the effective oppenheim theorems step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Effective Oppenheim theorems: structural mechanism)**.** *For irrational nondegenerate indefinite ternary forms in a fixed compact family satisfying the quantitative non-rationality hypothesis, the value-counting function in a polynomially shrinking interval has the coarea main term with a power-saving error.*

Proof. Combine Chapters 1–7. The arithmetic/dynamical obstruction equivalence selects the equidistribution branch. Apply the smooth count, remove smoothing by the coarea boundary estimate, control the cusp by Chapter 3/5, and finally invoke Chapter 7 for the least-value consequence. □

Theorem 0.2 (Effective Oppenheim theorems). *Fix a compact family $\mathcal{K}$ of normalized irrational indefinite ternary forms satisfying the quantitative non-rationality condition of Chapter 6. Then there exist $\theta_0, \delta, C > 0$ such that for $Q \in \mathcal{K}$, $0 < \theta < \theta_0$, and $\varepsilon = T^{-\theta}$,*

$$N_{Q,w}(T, \xi, \varepsilon) = 2\varepsilon T\, \mathfrak{J}_Q(w, \xi) + O(T^{1-\theta-\delta}).$$

Consequently, whenever the main constant is positive, every sufficiently large T contains $0 \neq v \in \mathbb{Z}^3$ with $\|v\| \ll T$ and $|Q(v) - \xi| \ll T^{-\theta}$.

Proof. Combine Chapters 1–7. Chapter 6 rules out the proper rational homogeneous branch, so the effective orbit theorem applies. Chapters 1–2 compute the coarea main term and smoothing error; Chapters 3 and 5 control the cusp observable; Chapter 4 gives the power-saving equidistribution error.

Choosing the smoothing and height cutoffs in the order used there yields the displayed count with a positive uniform minimum exponent δ. The last assertion is exactly the positivity argument of Chapter 7. $\square$

Volume 50

Absolute Rank Two Effective Ratner

Volume 50 scope and prerequisites

The volume treats the quasi-split almost-simple absolute-rank-two setting of the modern polynomial effective-equidistribution theorem. The cited Lindenstrauss–Mohammadi–Wang–Yang theorem is used at its explicitly stated source status; the chapters below are a structured proof guide and specialization, not a replacement for the source proof. It separates root-theoretic case analysis, effective closing, dimension growth, algebraic generation, and the final spectral upgrade. No theorem for arbitrary absolute rank is claimed.

Proof and provenance. Primary comparison: Lindenstrauss–Mohammadi–Wang–Yang, *Effective equidistribution in rank 2 homogeneous spaces and values of quadratic forms* (2025).

Reader guide: a concrete model

Why this matters.

Volume 50 is organized around the passage from *Absolute-rank-two group structure* to *Rank-two quantitative Oppenheim applications.* The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.3 (A calculation to keep in view)**.** Let $P(t) = a+bt+ct^2$ with $c \neq 0$. At scale T write $t = Ts$. Then

$$P(Ts) = a + (bT)s + (cT^2)s^2.$$

If $|c|T^2 \asymp 1$ while $|b|T \to 0$ and $|a| \to 0$, the first visible rescaled polynomial is cs^2T^2. On a mesoscopic window $s = s_0 + \ell/T$ with $|\ell| = o(T)$,

$$P(Ts) - P(Ts_0) = P'(Ts_0)\ell + c\ell^2 = o(1)$$

whenever the normalized derivative is controlled. This is the toy calculation behind the separation of a macroscopic visibility scale from a mesoscopic almost-constant scale in effective linearization and closing.

CHAPTER 1

Absolute-rank-two group structure

This chapter owns the absolute-rank-two group structure step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Absolute-rank-two group structure: structural mechanism)**.** *For a quasi-split almost-simple real group of absolute rank two, a maximal split torus determines a rank-two relative root system and a finite list of root subgroup commutator patterns.*

Proof. Choose a pinning over the split torus. The Chevalley commutator relations express brackets of root groups as products over roots $i\alpha + j\beta$. Absolute rank two leaves only the finite A_2, B_2, G_2-type patterns (with the quasi-split relative variants), so the normal-form list is finite.

□

Theorem 0.2 (Absolute-rank-two group structure). *Let G be a fixed quasi-split almost-simple real algebraic group of absolute rank two and let S be a maximal split torus. Relative to S, every one-parameter unipotent subgroup used in the multiscale argument is contained in a root subgroup U_α or in a bounded product of root subgroups belonging to one of the finitely many rank-two strings*

$$A_2, \qquad B_2(= C_2), \qquad G_2$$

(and their fixed quasi-split relative variants). The number and length of commutator words needed to pass between these normal forms are bounded solely in terms of G.

Proof. Choose a pinning relative to S. The Chevalley commutator formula expresses $[U_\alpha, U_\beta]$ as an ordered product of $U_{i\alpha+j\beta}$ over roots in the rank-two string spanned by α, β. Absolute rank two leaves only the displayed finite root-system types. There are finitely many Weyl-group orbits of roots and root pairs, so choose one representative normal form from each orbit. Conjugation by the corresponding finitely many Weyl representatives places every subgroup used later into this list, and the maximal root-string length gives the uniform commutator-word bound. □

Proposition 0.3 (Absolute-rank-two group structure: downstream consequence)**.** *The constants in later polynomial estimates may therefore be chosen uniformly over the finite case list.*

Proof. The root-normal-form list in Theorem 0.2 is finite. For every later estimate take the maximum of the finitely many polynomial degrees, commutator-word lengths, and chart constants over that list. This maximum depends only on the fixed group G, so none of those constants depends on T or on the chosen orbit. □

CHAPTER 2

Quasi-split root data

This chapter owns the quasi-split root data step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Quasi-split root data: structural mechanism). *In each quasi-split rank-two type, the expanding/contracting root characters supply at least two transverse directions unless the orbit lies in a proper parabolic or rank-one subgroup.*

Proof. Evaluate the weights of the chosen diagonal element on the relative roots. If all generated root directions lie on one root line, the generated Lie algebra is contained in the corresponding proper parabolic/rank-one subgroup. Otherwise two noncollinear roots and their commutators generate the required transverse span.

□

Theorem 0.2 (Quasi-split root data). *Let $a \in S$ be the diagonal element used to renormalize the unipotent flow and let Φ be the relative root system. Unless all root directions generated by the orbit lie in the root span of a proper parabolic or a rank-one subsystem, there exist nonproportional roots $\alpha, \beta \in \Phi$ with generated directions in $\mathfrak{g}_\alpha$ and $\mathfrak{g}_\beta$. Hence the transverse displacement space contains two noncollinear S-weight directions; their allowed commutators generate the rank-two root span.*

Proof. If every generated root lies on one root line, the generated Lie algebra lies in the corresponding rank-one subgroup. If all generated roots lie in a proper positive subsystem, the generated algebra lies in the associated proper parabolic. Excluding these cases leaves two nonproportional roots α, β. Their weight spaces are linearly independent and, by the rank-two Chevalley relations of Chapter 1, brackets produce every root in the subsystem generated by α and β. This is precisely the transversality alternative used in the dimension-growth step. $\square$

Proposition 0.3 (Uniform algebraic complexity of the exceptional rank-two subgroup list)**.** *There exist constants $D, H_0 < \infty$, depending only on the fixed quasi-split group G and its chosen rational representation, such that every subgroup in the finite exceptional list of Chapters 1–2 is cut out by polynomial equations of degree at most D whose rational coefficient height is at most H_0.*

Proof. A proper exceptional subgroup is generated by one of the finitely many root subsystems or parabolic subsets from Chapters 1–2. Each such subgroup is defined by a fixed finite system of bounded-degree equations in the chosen representation. Hence their degrees and reference heights have a common bound depending only on G. $\square$

CHAPTER 3

Classification of unipotent subgroup cases

This chapter owns the classification of unipotent subgroup cases step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Classification of unipotent subgroup cases: structural mechanism)**.** *Every one-parameter unipotent subgroup in the fixed finite root list has polynomial coordinates of uniformly bounded degree in a faithful representation.*

Proof. Use the root subgroup exponential coordinates and the nilpotence length of the adjoint action. Since the ambient absolute rank and representation dimension are fixed, the exponential truncates at uniformly bounded degree. Apply the polynomial good-function lemma from the earlier nondivergence volumes.

□

Theorem 0.2 (Classification of unipotent subgroup cases). *Fix the finite normal-form list of Chapters 1–2 and a faithful representation $\iota : G \hookrightarrow \mathrm{GL}_N$. There is an integer d_0 such that for every one-parameter unipotent subgroup u_t in the list, every matrix coefficient of $\iota(u_t)$ and of each fixed Chevalley representation used later is a polynomial in t of degree at most d_0. Consequently all (C, α)-good exponents in quantitative nondivergence and all polynomial-degree constants in linearization may be chosen uniformly over the list.*

Proof. For $u_t = \exp(tN)$, nilpotence gives $N^r = 0$ for some $r \leq N$, so $\iota(u_t) = \sum_{j<r} t^j N^j / j!$. The same holds in every fixed tensor/exterior-power representation, with degree bounded by a constant depending only on that representation. Since Chapters 1–2 leave a finite case list, take the

maximum degree d_0. The standard good-function exponent for a nonzero real polynomial depends only on its degree, and the linearization estimates depend only on the same finite coefficient-degree data, giving uniform constants. □

Proposition 0.3 (Polynomial Sobolev distortion for the finite root-data list). *Fix a logarithmic time window $|t| \leq c \log T$. For the finite family of Chevalley representations attached to the subgroup types of Chapter 3, there exist C, D' such that*

$$\mathcal{S}_D(f \circ a_t) \leq T^C \mathcal{S}_{D'}(f)$$

uniformly over the family and all $|t| \leq c \log T$.

Proof. Conjugation by the fixed diagonal flow multiplies a root vector in $\mathfrak{g}_\alpha$ by $e^{\alpha(a)t}$. There are finitely many roots and finitely many Chevalley representations, so on every prescribed time block the derivative/Sobolev distortion is bounded by the maximum of finitely many exponentials; after the logarithmic time normalization used in the volume this is a fixed power of T. □

CHAPTER 4

Effective closing in rank two

This chapter owns the effective closing in rank two step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Effective closing in rank two: structural mechanism)**.** *A polynomially close self-return in the thick part either produces a proper rational subgroup of polynomial height or a family of transverse returns at a smaller scale.*

PROOF. Apply quantitative Chevalley linearization to the return. Effective rational separation converts sufficiently small approximate stabilizers into exact rational subgroup data. If the stabilizer does not close, the residual displacement has a nonzero root component and becomes the transverse return used in dimension growth.

□

THEOREM 0.2 (Effective closing in rank two). *Fix the arithmetic quotient $X = G/\Gamma$ and the finite intermediate-subgroup list determined by Chapters 1–3. There exist $A, C > 0$ such that a self-return of size at most T^{-A} in the T^{-C}-thick part yields one of two alternatives:*

(1) a proper rational connected subgroup $H < G$ from the finite list with $\mathrm{ht}(H) \leq T^C$ and a closed H-orbit polynomially shadowing the return block;

(2) a transverse return whose component in some root space outside the current generated Lie algebra is at least T^{-C}.

PROOF. Apply quantitative Chevalley linearization to the return. Each candidate stabilizer is represented by rational Pluecker/Chevalley coordinates of polynomial height. If every residual defining the stabilizer is smaller than the rational-separation threshold of Volume 51, those residuals vanish exactly and give a rational subgroup H of height $T^{O(1)}$. Otherwise one residual survives at size at least T^{-C}; decomposing it into root spaces produces a root component outside the current stabilizer Lie algebra, which is the transverse return. The subgroup list is finite by Chapters 1–3, so the exponents are uniform. □

Proposition 0.3 (Uniform subgroup-height exponent on a fixed arithmetic quotient)**.** *For the fixed arithmetic quotient G/Γ there exists $C_G > 0$ such that every proper rational subgroup H produced by the rank-two closing theorem from a return at scale T^{-A} satisfies*

$$\mathrm{ht}(H) \le T^{C_G A + C_G}.$$

In particular, the height exponent is independent of the individual orbit point.

Proof. All rational detectors, representation degrees, and Grassmannian charts are fixed once G/Γ is fixed. The rational-separation exponent of Volume 51 is therefore a single constant. Chapter 4's conversion from a return scale T^{-A} to subgroup height composes only these fixed polynomial maps, so the resulting height exponent is uniform on the quotient. □

CHAPTER 5

Dimension growth in rank two

This chapter owns the dimension growth in rank two step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Dimension growth in rank two: structural mechanism)**.** *Assume the closing obstruction is absent. Repeated incidence/projection steps increase the transverse scale dimension until it is within ϵ of the dimension of the generated root span.*

Proof. Use the same energy-to-projection argument as XLVI. The only new issue is that the preferred projection changes with the root case; Chapter 2 gives a finite family of projections and guarantees one is transverse outside the obstruction. Dimension can rise only finitely often.

□

Theorem 0.2 (Dimension growth in rank two). *Assume the subgroup alternative of Chapter 4 never occurs. Let s_j denote the transverse scale dimension after the j-th incidence step. There is $c(\epsilon) > 0$ such that whenever s_j is below the dimension of the root span by more than ϵ, Chapter 4 supplies a transverse root return and*

$$s_{j+1} \geq s_j + c(\epsilon).$$

After at most $\lceil \dim G/c(\epsilon) \rceil$ steps the smoothed orbit measure is T^{-b}-almost invariant under a neighborhood in every root subgroup generating the ambient simple factor, hence under a fixed identity neighborhood in that factor.

Proof. The absence of closing obstructions means every stalled incidence step falls into the transverse-return branch of Chapter 4. Restricted projection/incidence then increases the scale dimension by at least $c(\epsilon)$. Since the dimension is bounded by $\dim G$, this can happen only the displayed number of times before the root-span dimension is reached up to ϵ. The BCH argument converts the resulting separated root directions into almost invariance under their brackets. Chapters 1–2 show that these root groups generate the ambient simple factor in bounded bracket depth; a bounded product chart therefore gives almost invariance on a fixed identity neighborhood. $\square$

Proposition 0.3 (Dimension growth in rank two: downstream consequence)**.** *The number of iterations is bounded solely in terms of the fixed root system and ϵ.*

Proof. At each unsuccessful stage the scale dimension increases by at least $c(\epsilon) > 0$. Since it is at most the dimension d_Φ of the root span, the number of increases is at most $\lceil d_\Phi/c(\epsilon)\rceil$. Both d_Φ and the incidence increment are determined by the fixed root system and ϵ. $\square$

CHAPTER 6

Quantitative algebraic generation

This chapter owns the quantitative algebraic generation step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Quantitative algebraic generation: structural mechanism)**.** *A collection of approximate root subgroups that is closed under commutators up to error ρ^A and has rational height $\leq \rho^{-B}$ determines, once A is large compared with B, an exact connected algebraic subgroup with the same generated Lie algebra.*

Proof. Use the effective-generation statements of Volume 51: encode the candidate Lie space by Pluecker coordinates, use approximate bracket closure to place it near the algebraic locus of Lie subalgebras, apply quantitative separation/Greenberg lifting, and exponentiate the exact Lie algebra inside the fixed algebraic group.

□

Theorem 0.2 (Quantitative algebraic generation). *Let $\rho \in (0,1)$ and let $V \subset \mathfrak{g}$ be a rational approximate Lie space of height at most ρ^{-B} satisfying*

$$\operatorname{dist}([V,V],V) \leq \rho^A.$$

There is $A_0(B)$ such that, if $A \geq A_0(B)$, then there exists an exact rational Lie subalgebra $\mathfrak{h} \subset \mathfrak{g}$ with

$$d_{\mathrm{Gr}}(V,\mathfrak{h}) \leq \rho^c, \qquad H(\mathfrak{h}) \leq \rho^{-C},$$

and the connected algebraic subgroup H with Lie algebra $\mathfrak{h}$ has complexity at most ρ^{-C}.

Proof. In a fixed Grassmannian chart, bracket closure is given by finitely many polynomial equations in Pluecker coordinates. The coordinates of V have height $\rho^{-O(B)}$. Volume 51 rational separation says that a nonzero defining residual at such a rational point is at least $\rho^{C_1 B}$. Choose $A > C_1 B$ (with room for the chart losses). Then every residual smaller than ρ^A must vanish after the effective approximation/lifting step, producing an exact rational Lie algebra $\mathfrak{h}$. The height calculus of Volume 51 bounds its Pluecker height by ρ^{-C}, and fixed-dimensional algebraic generation gives the same polynomial bound for the connected subgroup. Root directions established before extraction persist because the Grassmannian error is below their separation scale. $\square$

Proposition 0.3 (Height stability under bounded algebraic operations)**.** *Fix an integer $m \geq 1$. There exists $C' = C'(m, G) > 0$ such that if the rational matrix/subspace data entering Chapter 6 have height at most ρ^{-C}, then any datum obtained from them by at most m products, inversions on the fixed admissible charts, or conjugations has height at most $\rho^{-C'}$. Exact subgroup-membership relations are preserved by these operations.*

Proof. In a fixed matrix representation, multiplication, inversion on the relevant Zariski-open chart, and conjugation are rational maps of bounded degree. Applying any bounded number of them to data of height ρ^{-C} changes height to at most $\rho^{-C'}$. Exact subgroup membership is preserved under these operations, so the extraction remains valid after the bounded products/conjugations used later. $\square$

CHAPTER 7

Polynomial effective equidistribution

This chapter owns the polynomial effective equidistribution step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Polynomial effective equidistribution: structural mechanism)**.** *Let G be quasi-split almost simple of absolute rank two, Γ arithmetic, and U a unipotent subgroup in the theorem's generic class. Outside a proper rational homogeneous obstruction of polynomial complexity, the U-orbit average equidistributes with error $O(T^{-\eta})$ in a fixed Sobolev norm.*

Proof. The proof is a finite root-theoretic assembly of the preceding chapters. Chapter 1 fixes the finite list of quasi-split rank-two root systems and Chapter 2 supplies a transverse root pair unless the orbit lies in one of the explicitly listed proper subgroups. Chapter 3 gives polynomial coordinate control and quantitative nondivergence with constants uniform over this finite list.

Suppose first that Chapter 4 detects a polynomially close self-return. Quantitative Chevalley separation and Volume 51 replace the approximate stabilizer by an exact rational subgroup of height $T^{O(1)}$, giving the obstruction branch. Otherwise every relevant return has a nontrivial transverse root component. The incidence/projection argument of Chapter 5 raises transverse dimension. Once the dimension deficit is below the fixed threshold, BCH and the rank-two commutator relations generate almost invariance under a neighborhood of the ambient simple factor.

The ambient spectral gap contracts the zero-mean component of the smoothed measure. All constants have been chosen uniformly over the finite root list, and the algebraic-generation losses are polynomial by Chapter

6/Volume 51. Therefore the minimum exponent across the finite cases is positive, proving the stated polynomial rate. □

THEOREM 0.2 (Polynomial effective equidistribution). *Let G be the fixed quasi-split almost-simple rank-two group, $\Gamma < G$ arithmetic, $U = \{u_t\}$ a unipotent subgroup from Chapters 1–3, and $x \in G/\Gamma$ in a compact Diophantine family. Then there exist $\eta, C, D > 0$ such that for every $T \geq 2$ exactly one of the following holds:*

(1) there is a proper rational subgroup $H < G$ with $\mathrm{ht}(H) \leq T^C$ whose closed orbit is within $T^{-C^{-1}}$ of the U-segment of x;

(2) for every $f \in C_c^\infty(G/\Gamma)$,

$$\left| \frac{1}{T} \int_0^T f(xu_t)\, dt - m_X(f) \right| \leq CT^{-\eta} \mathcal{S}_D(f).$$

PROOF. Quantitative nondivergence first removes a cusp set of polynomially small relative measure. Apply Chapter 4 to the thick return boxes. If the rational subgroup branch appears, alternative (1) holds with polynomial complexity. Otherwise Chapter 5 gives T^{-b}-almost invariance under a generating neighborhood of the ambient simple factor. Smooth the orbit measure at a polynomial radius; its L^2 norm is polynomial. The ambient spectral gap contracts its zero-mean part after $c \log T$ averaging steps, while telescoped almost-invariance, smoothing, cusp, and box-boundary errors are each negative powers of T. Taking the minimum exponent gives (2). □

Proposition 0.3 (Uniform effective-equidistribution exponent on compact Diophantine families)**.** *Let $K \subset G/\Gamma$ be compact and suppose the finitely many rational Chevalley detectors appearing in Chapters 1–6 satisfy common Diophantine lower bounds on K. Then there exist $\eta_K > 0$, D, and C_K such that the non-obstruction branch of the rank-two theorem satisfies*

$$|A_T f(x) - m(f)| \leq C_K T^{-\eta_K} \mathcal{S}_D(f) \qquad (x \in K).$$

PROOF. On a compact Diophantine family, the lower bounds for all finitely many rational Chevalley detectors have common exponents and constants. The nondivergence, incidence, BCH, and spectral-gap constants therefore admit positive minima over the family. Taking the minimum of the resulting finitely many power-saving exponents gives one $\eta > 0$ valid uniformly. □

CHAPTER 8

Rank-two quantitative Oppenheim applications

This chapter owns the rank-two quantitative oppenheim applications step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Rank-two quantitative Oppenheim applications: structural mechanism)**.** *For the orthogonal rank-two realization attached to indefinite ternary quadratic forms, the homogeneous orbit theorem converts smooth value counts into a coarea main term with polynomial error.*

Proof. Identify the orthogonal homogeneous space and the unipotent subgroup governing the value distribution. Apply Chapter 7 to the truncated Siegel transform and repeat the smoothing/coarea argument from Volume 49. The rational homogeneous obstruction is precisely the rational-approximation branch already isolated there.

□

Theorem 0.2 (Rank-two quantitative Oppenheim applications). *In the orthogonal rank-two realization used for the indefinite ternary quadratic-form problem, assume the rational-approximation obstruction of Volume 49 is absent. Then the orbit average entering the smoothed Siegel transform satisfies the equidistribution estimate of Chapter 7; consequently, for $\varepsilon = T^{-\theta}$ in the admissible range,*

$$N_{Q,w}(T, \xi, \varepsilon) = 2\varepsilon T\, \mathfrak{J}_Q(w, \xi) + O(T^{1-\theta-\delta})$$

with $\delta > 0$, and the small-value conclusion of Volume 49 follows.

Proof. The orthogonal homogeneous model identifies the value count with a Siegel transform along the relevant unipotent orbit. The proper rational subgroup branch in Chapter 7 is, by Volume 49 Chapter 6, exactly the rational-approximation alternative for Q. In the nonexceptional branch, apply Chapter 7 to the height-truncated transform. The coarea and smoothing calculation of Volume 49 Chapters 1–2 gives the main term, while its Chapters 3 and 5 control the cusp. Balancing the polynomial cutoff and smoothing scales produces the displayed power-saving error, and Volume 49 Chapter 7 turns positivity of the count into the small-value bound. $\square$

Volume 51

Effective Generation and Quantitative Algebraic Geometry

Volume 51 scope and prerequisites

This volume isolates the quantitative algebraic-geometry engine used by effective homogeneous dynamics: rational heights, Pluecker separation, approximate Lie closure, effective generation, fixed-family semialgebraic separation, nonsingular p-adic lifting, and extraction of exact algebraic subgroups. All results used downstream are proved internally at their stated scope. In particular, exact rational subgroup extraction uses denominator-clearing and rational-height separation; the local Łojasiewicz theorem is deliberately restricted to fixed finite polynomial families, so no unproved height-uniform real-semialgebraic estimate is silently invoked.

Proof and provenance. Primary comparison with the effective-generation architecture appearing in Einsiedler–Lindenstrauss–Mohammadi–Wieser and with quantitative algebraic lemmas used in recent effective-Ratner work. These sources are concordance references only; Chapters 1–8 own the exact height, separation, generation, lifting, and rational-extraction statements used later.

Reader guide: a concrete model

Why this matters.

Volume 51 is organized around the passage from *Heights of rational subspaces and subgroups* to *Extraction of exact algebraic subgroups from approximate data.* The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.3 (A calculation to keep in view)**.** Let $P(t) = a+bt+ct^2$ with $c \neq 0$. At scale T write $t = Ts$. Then

$$P(Ts) = a + (bT)s + (cT^2)s^2.$$

If $|c|T^2 \asymp 1$ while $|b|T \to 0$ and $|a| \to 0$, the first visible rescaled polynomial is cs^2T^2. On a mesoscopic window $s = s_0 + \ell/T$ with $|\ell| = o(T)$,

$$P(Ts) - P(Ts_0) = P'(Ts_0)\ell + c\ell^2 = o(1)$$

whenever the normalized derivative is controlled. This is the toy calculation behind the separation of a macroscopic visibility scale from a mesoscopic almost-constant scale in effective linearization and closing.

CHAPTER 1

Heights of rational subspaces and subgroups

This chapter owns the heights of rational subspaces and subgroups step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Heights of rational subspaces and subgroups: structural mechanism)**.** *For a rational k-plane $W \subset \mathbb{Q}^n$, define $H(W)$ as the Euclidean norm of its primitive Pluecker vector. Then $H(W_1 + W_2)H(W_1 \cap W_2) \ll H(W_1)H(W_2)$ whenever the intersection dimension is fixed.*

Proof. Choose primitive integral bases and use exterior products. The lattice determinant identity for $W_1 + W_2$ and $W_1 \cap W_2$ is an index identity up to bounded Euclidean comparison. For brackets, structure constants are fixed integers/rationals, so minors of the bracket span are polynomial combinations of the input minors.

□

Theorem 0.2 (Heights of rational subspaces and subgroups). *Fix a faithful rational representation $G \hookrightarrow \mathrm{GL}_N$ and Pluecker height $H(\cdot)$. For rational subspaces $V, W \subset \mathbb{Q}^N$ of fixed dimensions,*

$$H(V + W)H(V \cap W) \ll H(V)^C H(W)^C.$$

Consequently, for connected rational subgroups represented through their Lie algebras, intersection, generated sum, bounded Lie bracket, and conjugation by a rational element of height at most H change height by at most a fixed power H^C.

Proof. Choose primitive integral Pluecker vectors. The coordinates of $V + W$ and $V \cap W$ are minors of matrices built from bases of V and W, so Hadamard's inequality bounds their primitive minors by a fixed power of the input heights. Lie brackets and conjugation are polynomial maps of bounded degree in the fixed matrix representation; after clearing denominators, their primitive Pluecker coordinates therefore have height bounded by a fixed power of the input height. The exponents depend only on N and the fixed representation. ☐

Proposition 0.3 (Representation-dimensional dependence of rational-height exponents)**.** *Fix an embedding $G \hookrightarrow \mathrm{SL}_N$. There exists $C_N < \infty$ such that every height estimate in Chapter 1 obtained from rational subspace sums/intersections, minors, brackets, and the bounded elimination steps has the form*

$$\mathrm{ht}(\mathcal{O}(\mathcal{D})) \leq C_N \ \mathrm{ht}(\mathcal{D})^{C_N},$$

where the exponent depends only on N and the fixed embedding, not on the particular rational datum $\mathcal{D}$.

PROOF. Every estimate in the proof is obtained from minors, matrix products, brackets, or a bounded number of eliminations in the fixed N-dimensional representation. Their degrees and the number of coordinates depend only on N and the fixed embedding. Hence the exponents in all polynomial height bounds have the asserted dependence. ☐

CHAPTER 2

Separation of rational Pluecker data

This chapter owns the separation of rational pluecker data step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Separation of rational Pluecker data: structural mechanism)**.** *Two distinct primitive integral Pluecker vectors of norm at most H have projective distance at least cH^{-C}.*

Proof. After choosing a nonvanishing coordinate, projective coordinates are ratios of integers of denominator $O(H)$. A nonzero difference of two such ratios has numerator a nonzero integer and denominator $O(H^2)$. Passing between charts costs a fixed power of H.

□

Theorem 0.2 (Separation of rational Pluecker data). *For every fixed Grassmannian chart there exist $c, C > 0$ such that distinct rational d-planes $V \neq W$ with $H(V), H(W) \leq H$ satisfy*

$$d_{\mathrm{Gr}}(V, W) \geq cH^{-C}.$$

The same estimate holds for the Lie spaces of distinct connected rational algebraic subgroups of height at most H.

Proof. Let v, w be primitive integral Pluecker vectors for V, W. If their projective points are distinct, some 2×2 minor $v_i w_j - v_j w_i$ is a nonzero integer, hence has absolute value at least one. Since $\|v\|, \|w\| \leq H$, the sine of their projective angle is at least H^{-2} up to chart constants. On a fixed Grassmannian chart, projective angle and Grassmannian distance are bi-Lipschitz up to fixed constants. Applying the same argument to the fixed faithful Lie-algebra representation gives the subgroup statement. □

Proposition 0.3 (Separation of rational Pluecker data: downstream consequence)**.** *Consequently an approximation smaller than H^{-C-1} has at most one rational target of height H.*

Proof. If two rational targets V, W of height at most H both lie within $\frac{1}{2}cH^{-C}$ of the same point, then $d_{\mathrm{Gr}}(V, W) < cH^{-C}$ by the triangle inequality, contradicting Theorem 0.2. Replacing $c/2$ by the simpler threshold H^{-C-1} for large H gives uniqueness. $\square$

CHAPTER 3

Almost-invariant and approximate Lie subalgebras

This chapter owns the almost-invariant and approximate lie subalgebras step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Almost-invariant and approximate Lie subalgebras: structural mechanism)**.** *Fix a compact Grassmannian chart and a compact subset K of the nonsingular locus of the polynomial bracket-closure equations on which the smallest nonzero singular value of the relevant Jacobian is at least $\sigma > 0$. If a d-plane V in a sufficiently small fixed neighborhood of K satisfies*

$$\|\pi_{V^\perp}[v,w]\| \le \varepsilon \|v\| \|w\|$$

for all $v, w \in V$, then the Pluecker point of V lies at distance $O_{K,\sigma}(\varepsilon)$ from the locus of d-dimensional Lie subalgebras.

Proof. Write a plane in the fixed chart as the graph of a matrix A. Bracket closure is equivalent to the vanishing of finitely many polynomial coordinate functions $F(A)$. The hypothesis gives $\|F(A)\| \ll_K \varepsilon$. On the chosen compact nonsingular set, the stated lower bound σ gives a uniformly bounded right inverse for DF. The quantitative implicit-function theorem therefore provides a zero A_0 of F with $\|A - A_0\| \ll_{K,\sigma} \varepsilon$. On a compact Grassmannian chart matrix distance and Pluecker distance are bi-Lipschitz up to fixed constants, giving the claim.

□

Theorem 0.2 (Almost-invariant and approximate Lie subalgebras). *In a fixed Grassmannian chart, choose Pluecker coordinates z for a d-plane $V \subset \mathfrak{g}$. There are finitely many polynomials $F_1, \ldots, F_m$ of bounded degree, depending* *only on $\mathfrak{g}$ and the chart, such that*

$$V \text{ is a Lie subalgebra} \iff F_1(z) = \cdots = F_m(z) = 0.$$

Moreover, the norm of the bracket defect $\sup_{\|v\|,\|w\|\le 1} \|\pi_{V^\perp}[v,w]\|$ *is comparable on compact charts to* $\max_i |F_i(z)|$ *up to fixed powers near singular components.*

Proof. Write a basis of V in the chosen affine Grassmannian chart as columns of a matrix whose entries are rational functions with a fixed nonvanishing denominator. The condition $[v_i, v_j] \in V$ is equivalent to vanishing of the $(d+1) \times (d+1)$ minors obtained by adjoining each bracket vector to the basis matrix. Clearing the fixed chart denominator produces finitely many polynomial equations $F_i(z) = 0$. On a compact nonsingular chart, the Jacobian lower bound gives a linear comparison with distance by the implicit-function theorem; near singular components the quantitative Łojasiewicz estimate from Chapter 5 gives the stated power comparison. □

Proposition 0.3 (Almost-invariant and approximate Lie subalgebras: downstream consequence)**.** *At singular components the theorem gives only the explicit Łojasiewicz-power distance bound proved in Chapter 5.*

Proof. At a singular point the Jacobian inverse used in the nonsingular implicit-function estimate is unavailable. Chapter 5 instead gives $d(z, Z(F)) \le C \max_i |F_i(z)|^\theta$ on the fixed compact component. Substituting the bracket residuals from Theorem 0.2 yields exactly that weaker power-law conclusion and no linear estimate. □

CHAPTER 4

Effective Lie generation

This chapter owns the effective lie generation step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Effective Lie generation: structural mechanism). *Starting from finitely many rational Lie directions of height H, iteratively adjoining brackets stabilizes after at most* $\dim \mathfrak{g}$ *steps and yields a rational Lie algebra of height at most H^C.*

Proof. At each step take the span of the current space and all pairwise brackets. Dimension rises or the process stops, hence at most $\dim \mathfrak{g}$ steps. Chapter 1 bounds the height after each span/bracket operation by a fixed power; composing finitely many powers remains polynomial.

□

Theorem 0.2 (Effective Lie generation). *Let $X_1, \dots, X_r \in \mathfrak{g}(\mathbb{Q})$ have height at most H. Define*

$$V_0 = \operatorname{span}_{\mathbb{Q}}\{X_i\}, \qquad V_{j+1} = V_j + [V_j, V_j].$$

Then V_j stabilizes after at most $\dim \mathfrak{g}$ *steps at the rational Lie algebra $\mathfrak{h}$ generated by the X_i, and*

$$H(\mathfrak{h}) \leq H^C.$$

The connected algebraic subgroup generated by the corresponding one-parameter subgroups has algebraic complexity at most $H^{C'}$.

Proof. Each strict inclusion $V_j \subsetneq V_{j+1}$ raises dimension, so stabilization occurs in at most $\dim \mathfrak{g}$ steps. Chapter 1 shows that one bracket operation changes rational height by at most a fixed power. Iterating a bounded number of times therefore gives $H(V_j) \leq H^{C_j}$ with C_j bounded in terms of $\dim \mathfrak{g}$. At stabilization V_j is bracket closed and equals the generated Lie algebra. In a fixed linear algebraic group, the equations defining the connected subgroup generated by a rational Lie algebra are obtained from bounded-degree algebraic operations; the same height calculus gives the final polynomial complexity bound. □

Proposition 0.3 (Effective Lie generation after bounded products of generators)**.** *Fix $m \geq 1$. If a rational generating set S has height at most H and S' is obtained from S by adjoining at most m bounded products/conjugates, then the Lie algebra generated by S' has height at most*

$$H^{C_{N,m}}$$

for a constant $C_{N,m}$ depending only on the representation dimension and m.

Proof. Replacing the initial set of generators by a bounded product enlarges the list of rational directions by only a bounded number of polynomial matrix expressions. Chapter 1 bounds their heights by $H^{O(1)}$; the iteration $V_{j+1} = V_j + [V_j, V_j]$ still has at most $\dim \mathfrak{g}$ stages, so Theorem 0.2 gives a final polynomial bound. □

CHAPTER 5

Quantitative Lojasiewicz inequalities

This chapter isolates the local quantitative Lojasiewicz step needed by the later effective-algebraic arguments. The proof below is intentionally limited to a fixed finite family of polynomial systems; it is not a height-uniform effective theorem for varying algebraic data.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Quantitative Lojasiewicz inequalities: structural mechanism)**.** *Let $P_1, \ldots, P_m$ be real polynomials of bounded degree on a compact semialgebraic set K. There exist $C, \theta > 0$ such that* $\mathrm{dist}(x, Z) \leq C(\max_i |P_i(x)|)^\theta$ *for $Z = \{P_i = 0\}$, provided $Z \cap K \neq \varnothing$.*

Proof. Set

$$F(x) := \max_{1 \leq i \leq m} |P_i(x)|, \qquad d_Z(x) := \mathrm{dist}(x, Z).$$

Both functions are continuous and semialgebraic on K. It is enough to prove the inequality in a neighborhood of $Z \cap K$, because on the compact set $K_\delta := \{x \in K : d_Z(x) \geq \delta\}$ one has $\min_{K_\delta} F > 0$ and the desired estimate follows after enlarging C.

Assume, toward a contradiction, that no power inequality holds near Z. Then for every integer $q \geq 1$ there are points x arbitrarily close to Z with

$$d_Z(x) > F(x)^{1/q}.$$

Equivalently, the semialgebraic set

$$E_q := \{x \in K : 0 < d_Z(x) < 1/q,\ F(x) < d_Z(x)^q\}$$

has Z in its closure for arbitrarily large q. By semialgebraic curve selection, for such a component there is a real-analytic semialgebraic arc $\gamma : (0, \varepsilon_0) \to K \setminus Z$ with $\gamma(t) \to z \in Z$. After a Puiseux reparametrization, every coordinate of γ is a convergent power series in $t^{1/N}$. Hence there are integers $a, b > 0$ and constants $c_1, c_2 > 0$ such that

$$c_1 t^{a/N} \leq d_Z(\gamma(t)) \leq c_2 t^{a/N}, \qquad F(\gamma(t)) \geq c_1 t^{b/N}$$

for small t; $b < \infty$ because at least one $P_i \circ \gamma$ is not identically zero. Thus

$$d_Z(\gamma(t)) \le C_\gamma F(\gamma(t))^{a/b}.$$

A semialgebraic cell decomposition of a sufficiently small compact neighborhood of $Z \cap K$ has only finitely many cells and finitely many possible Puiseux contact types for the fixed polynomial system. Let

$$\theta := \min_{\text{cells}} \frac{a}{b} > 0$$

and let C be the maximum of the finitely many constants C_γ together with the compact-away-from-Z constant. Then $d_Z(x) \le CF(x)^\theta$ for all $x \in K$.

The finiteness assertion here is exactly why this chapter claims uniformity only for a fixed polynomial system (or a fixed finite family of such systems); it does not produce a coefficient-height-uniform exponent for varying systems. □

THEOREM 0.2 (Quantitative Lojasiewicz inequalities). *Fix a finite family of algebraic loci represented by fixed bounded-degree polynomial systems on fixed compact semialgebraic charts. Then one may choose common constants $C, \theta > 0$ for that finite family in the Łojasiewicz separation inequality.*

PROOF. Apply the core lemma separately to each fixed polynomial system and compact chart. There are only finitely many systems in the statement, so take the maximum of the constants C and the minimum of the positive exponents θ. This proves uniformity for the fixed finite family. It does *not* prove a polynomially effective dependence for a family whose coefficients and heights vary; any later use of that stronger effectivity must cite an external effective semialgebraic/Łojasiewicz theorem or supply a separate proof. □

Proposition 0.3 (Quantitative Lojasiewicz inequalities: downstream consequence)**.** *The exponent may be made uniform over any explicitly fixed finite family of subgroup loci; no height-uniform effective exponent is inferred from this chapter alone.*

PROOF. For a fixed finite family $F^{(1)}, \ldots, F^{(r)}$, take the maximum of the finitely many compact-set constants and the minimum of the finitely many Łojasiewicz exponents. This gives one exponent for that family. If the defining equations themselves vary with unbounded arithmetic height, this finite maximum/minimum argument no longer applies; hence no such uniformity is claimed. □

CHAPTER 6

Effective Greenberg/approximation statements

This chapter owns the effective greenberg/approximation statements step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Effective Greenberg/approximation statements: structural mechanism)**.** *Let $F : \mathbb{Z}_p^n \to \mathbb{Z}_p^m$ be polynomial with integral coefficients. If $F(x_0) \equiv 0 \pmod{p^N}$ and an $m \times m$ Jacobian minor at x_0 is a p-adic unit, then there is x with $F(x) = 0$ and $x \equiv x_0 \pmod{p^N}$.*

Proof. After permuting variables assume the first m columns give an invertible Jacobian. Newton iteration $x_{r+1} = x_r - J(x_r)^{-1}F(x_r)$ is integral and doubles the p-adic accuracy. Completeness yields a limit root. The untouched variables remain fixed.

□

Theorem 0.2 (Effective Greenberg/approximation statements). *Let $F = (F_1, \ldots, F_m) : \mathbb{Z}_p^n \to \mathbb{Z}_p^m$ have integral polynomial coefficients. Suppose $F(x_0) \equiv 0 \pmod{p^N}$ and some $m \times m$ Jacobian minor $J(x_0)$ is a p-adic unit. Then there exists $x \in \mathbb{Z}_p^n$ such that*

$$F(x) = 0, \qquad x \equiv x_0 \pmod{p^N}.$$

Thus the project uses no loss of modulus at nonsingular points; singular loci are handled separately by the finite-component/Łojasiewicz analysis of Chapter 5.

Proof. After reordering coordinates, the unit minor makes the derivative in the first m variables invertible over $\mathbb{Z}_p$. Starting with x_0, solve recursively

$$DF(x_k)h_k \equiv -p^{-N-k}F(x_k) \pmod{p}$$

and put $x_{k+1} = x_k + p^{N+k} h_k$. Taylor's formula with integral coefficients gives $F(x_{k+1}) \equiv 0 \pmod{p^{N+k+1}}$, while $x_{k+1} \equiv x_k \pmod{p^{N+k}}$. The sequence is p-adically Cauchy and converges to x with $F(x) = 0$ and $x \equiv x_0 \pmod{p^N}$. □

Proposition 0.3 (Effective Greenberg/approximation statements: downstream consequence)**.** *For singular algebraic loci the volume uses Chapter 5 plus finite component decomposition instead of asserting the full Greenberg theorem.*

Proof. The Newton–Hensel argument in Theorem 0.2 requires a unit Jacobian minor. On the singular locus that hypothesis fails. The manuscript therefore first decomposes into finitely many algebraic components and uses the real/Grassmannian Łojasiewicz control of Chapter 5; it does not infer a singular p-adic lifting theorem from the nonsingular argument. □

CHAPTER 7

Complexity propagation under brackets and products

This chapter owns the complexity propagation under brackets and products step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Complexity propagation under brackets and products: structural mechanism)**.** *If rational subgroups H_1, H_2 have height at most H, then the subgroup generated by boundedly many products, inverses, conjugates by height-H elements, and Lie brackets has height at most H^C until the ambient dimension stabilizes.*

Proof. Represent every operation by polynomial formulas in matrix coordinates. Determinants/minors for multiplication, inversion on SL_n, and conjugation have bounded degree. Apply Chapter 1 after every linear span and Chapter 4 after bracket generation; the chain length is bounded by ambient dimension.

□

Theorem 0.2 (Complexity propagation under brackets and products). *Fix a bound r_0 for the number of algebraic operations used in one effective-Ratner generation stage. If every input rational subgroup/vector has height at most H, then every object obtained using at most r_0 sums, intersections, Lie brackets, products, inverses, and conjugations has height at most H^C for a constant C depending only on the ambient representation and r_0. Since any strict chain of connected intermediate subgroups has length at most* $\dim G$, *the entire generation chain has final complexity $H^{C'}$, with C' independent of the orbit length.*

Proof. Chapter 1 bounds each permitted algebraic operation by $K \mapsto K^{c_0}$ in height. Composing at most r_0 operations gives $H^{c_0^{r_0}}$. A strict chain of connected subgroups can increase dimension at most $\dim G$ times. Therefore the total number of stages is bounded independently of T, and iterating the fixed polynomial height map a bounded number of times still gives $H^{C'}$. No exponent proportional to the orbit length appears. $\square$

Proposition 0.3 (Complexity propagation under brackets and products: downstream consequence)**.** *A finite chain of intermediate subgroups therefore has total complexity $H^{O(1)}$.*

Proof. If one stage sends height K to at most K^C and a strict subgroup chain has length at most $r \leq \dim G$, then the last height is at most H^{C^r}. Since r is fixed, $C^r = O(1)$ and the total complexity is $H^{O(1)}$. $\square$

CHAPTER 8

Extraction of exact algebraic subgroups from approximate data

This chapter owns the extraction of exact algebraic subgroups from approximate data step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Rational-height separation for exact algebraic data)**.** *Let $z \in \mathbb{P}^M(\mathbb{Q})$ be the normalized rational Pluecker/Chevalley datum produced by the effective-closing construction. Assume that its projective height is at most H^B. Let*

$$F_1, \dots, F_m \in \mathbb{Q}[Z_0, \dots, Z_M]$$

be the fixed defining equations for the relevant subgroup/Lie-algebra locus, of degree at most d, with coefficient heights at most H^B. Then there is a constant $C = C(M, m, d, B) > 0$ such that

$$F_i(z) \neq 0 \quad \Longrightarrow \quad |F_i(z)| \geq H^{-C}.$$

Consequently, if $\max_i |F_i(z)| < H^{-C}$, then every defining residual vanishes exactly and z represents exact rational algebraic data.

Proof. Choose primitive integral homogeneous coordinates $z = [a_0 : \cdots : a_M]$ with $|a_j| \leq H^{B_1}$ for a fixed $B_1 = O(B)$, and clear denominators from the coefficients of every F_i using a common denominator $q_i \leq H^{B_2}$. Homogeneity gives

$$q_i F_i(z) = \frac{N_i}{A_i},$$

where $N_i \in \mathbb{Z}$ and the denominator satisfies $A_i \leq H^{C_0}$, with C_0 depending only on d, M, B. If $F_i(z) \neq 0$, then $N_i \neq 0$, hence $|N_i| \geq 1$ and therefore

$$|F_i(z)| \geq (q_i A_i)^{-1} \geq H^{-C}$$

for a fixed C. Thus an error smaller than H^{-C} forces every residual to be zero. This is a rational-height separation argument and does not require a height-uniform real Łojasiewicz theorem. □

THEOREM 0.2 (Exact subgroup extraction from approximate rational data). *Suppose the effective-closing construction produces rational approximate subgroup data z of height at most H^B and all defining residuals satisfy*

$$\max_i |F_i(z)| \leq H^{-A}$$

with $A > C$ for the separation exponent in Lemma 0.1. Then the residuals vanish exactly. The resulting rational Lie algebra generates a unique connected rational algebraic subgroup H_z in the prescribed component, and

$$\mathrm{ht}(H_z) \leq H^{C'}$$

for a fixed exponent C'. Thus the approximate data are replaced by exact subgroup data with only polynomial complexity loss.

PROOF. Since $A > C$, Lemma 0.1 gives $F_i(z) = 0$ for every defining equation. Hence z is an exact rational point of the Lie-subalgebra/subgroup locus. Chapter 4 generates the connected rational subgroup from this exact rational Lie algebra in at most $\dim \mathfrak{g}$ bracket stages. Chapters 1 and 7 bound sums, brackets, products, inverses and bounded conjugations by fixed polynomial maps in height; composing a bounded number of such maps yields $\mathrm{ht}(H_z) \leq H^{C'}$. Uniqueness in the prescribed height range follows from the rational Pluecker separation of Chapter 2: two distinct rational targets of height $H^{O(1)}$ are at distance at least $H^{-C''}$, while the construction selects the exact rational datum z itself. □

Volume 52

Effective Ratner: Current Frontier and Failure Modes

Volume 52 scope and prerequisites

This is a scope-control volume. It records a reusable conditional effective-Ratner template and then records which hypotheses fail or require genuinely new arguments in higher rank, S-arithmetic, semidirect-product, centralizer, and shrinking-target settings. The template is not promoted to a new general effective-Ratner theorem without explicit quantitative hypotheses, and the 2026 S-arithmetic SL_2^n item is labeled as a preprint/source announcement. A frontier statement is never promoted to a theorem merely because an analogue is expected.

Proof and provenance. Primary comparison: Comparison is made with the rank-one and absolute-rank-two effective theorems, the 2026 S-arithmetic SL_2^n extension, and current effective-closing/generation methods.

Reader guide: a concrete model

Why this matters.

Volume 52 is organized around the passage from *The general effective-Ratner template* to *Failure-mode theorems and precise frontier boundaries.* The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.3 (A calculation to keep in view)**.** Let $P(t) = a+bt+ct^2$ with $c \neq 0$. At scale T write $t = Ts$. Then

$$P(Ts) = a + (bT)s + (cT^2)s^2.$$

If $|c|T^2 \asymp 1$ while $|b|T \to 0$ and $|a| \to 0$, the first visible rescaled polynomial is cs^2T^2. On a mesoscopic window $s = s_0 + \ell/T$ with $|\ell| = o(T)$,

$$P(Ts) - P(Ts_0) = P'(Ts_0)\ell + c\ell^2 = o(1)$$

whenever the normalized derivative is controlled. This is the toy calculation behind the separation of a macroscopic visibility scale from a mesoscopic almost-constant scale in effective linearization and closing.

CHAPTER 1

The general effective-Ratner template

This chapter owns the the general effective-ratner template step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (The general effective-Ratner template: structural mechanism)**.** *Suppose a unipotent-orbit problem has five verified inputs: quantitative nondivergence, effective closing, dimension growth, exact algebraic generation, and ambient spectral gap. If every proper intermediate subgroup is quantitatively excluded, then the orbit average has a polynomial equidistribution rate.*

Proof. Run the proof in the established order: nondivergence gives thick blocks; closing turns repeated approximate relations into subgroup obstructions; their exclusion enables dimension growth; algebraic generation upgrades near-relations; spectral gap converts almost invariance to equidistribution. Choose scales successively so each later choice is a sufficiently small power of the previous one.

□

Theorem 0.2 (The general effective-Ratner template). *The effective-Ratner template is a conditional implication: once the five inputs hold with polynomial losses and the finite complexity losses are bounded, the argument yields some positive polynomial exponent depending only on those quantitative inputs.*

Proof. This is a scope consequence of Lemma 0.1 and the dependency structure stated in this chapter. It asserts only the limitation or conditional implication written in the theorem; it does not claim a new effective equidistribution theorem beyond the previously established settings. □

Proposition 0.3 (The general effective-Ratner template: downstream consequence)**.** *Failure of any one input is recorded as a distinct obstruction rather than hidden in the conclusion.*

Proof. The chapter's conclusion is the conjunction of its explicitly listed inputs. If one input fails, the corresponding step of the implication chain cannot be executed: without nondivergence there is no thick return chart; without closing/separation no algebraic alternative; without dimension growth no almost invariance; without a spectral input no upgrade to equidistribution. Thus each failure is a distinct obstruction rather than part of the theorem's conclusion. □

CHAPTER 2

Rank restrictions and root-theoretic bottlenecks

This chapter owns the rank restrictions and root-theoretic bottlenecks step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Rank restrictions and root-theoretic bottlenecks: structural mechanism)**.** *If the relative root system has unbounded rank, the number of intermediate root-generated subalgebras and bracket depths needed by the rank-two proof is not uniformly bounded by the rank-two constants.*

Proof. The proof is a dependency counterexample, not a claim of failure of equidistribution. Inspect the rank-two argument: it takes minima over a finite case list and uses a bounded bracket-generation depth. In increasing rank both quantities can vary, so no rank-independent exponent follows from those estimates alone.

□

Theorem 0.2 (Quantified rank-dependence of the root reduction). *For a fixed semisimple Lie algebra $\mathfrak{g}$, let $N(\mathfrak{g})$ be the number of root-string configurations occurring in the generation argument of Lemma 0.1, and let $d(\mathfrak{g})$ be the maximal number of Lie brackets used to generate a required root direction. If the local almost- invariance loss per bracket is at most a factor $C_0 \geq 1$, then the reduction produced by that argument has loss at most*

$$C(\mathfrak{g}) \leq N(\mathfrak{g})C_0^{d(\mathfrak{g})}. \tag{52.2.1}$$

Consequently the fixed-rank proof is uniform on any class for which $\sup N(\mathfrak{g}) < \infty$ *and* $\sup d(\mathfrak{g}) < \infty$*; without such bounds it gives no rank-uniform constant.*

Proof. Enumerate the finitely many root-string cases used in Lemma 0.1. Along a bracket chain of length j, the triangle/BCH estimate multiplies the current loss by at most C_0 at each step, giving C_0^j. Summing over at most $N(\mathfrak{g})$ cases yields (52.2.1). Taking the two suprema proves the uniform assertion; if either is unbounded, the displayed estimate itself has no rank-independent upper bound. $\square$

Proposition 0.3 (Rank restrictions and root-theoretic bottlenecks: downstream consequence)**.** *If the rank-two proof uses at most N_2 root configurations and bracket depth at most d_2, then the same argument extends to a family of higher-rank groups only if the corresponding numbers $N(G)$ and $d(G)$ admit bounds independent of G over that family.*

Proof. The fixed-rank proof takes maxima over a finite root-string list and iterates at most a bounded bracket depth. If rank varies, both the number of root configurations and the maximal generation depth can vary. Therefore a rank-uniform theorem would need estimates independent of those two quantities; nothing in the present finite-case argument supplies them. $\square$

CHAPTER 3

Intermediate subgroup complexity

This chapter owns the intermediate subgroup complexity step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Length of a strict intermediate-subgroup chain)**.** *Let*

$$H_0 < H_1 < \cdots < H_m \le G$$

be a strictly increasing chain of connected algebraic subgroups of a fixed real algebraic group G. Then

$$m \le \dim G - \dim H_0 \le \dim G.$$

Proof. Strict inclusion of connected algebraic subgroups gives strict inclusion of their Lie algebras. Hence $\dim H_{j+1} \ge \dim H_j + 1$ at every step. Summing these inequalities proves the bound. □

Theorem 0.2 (Polynomial complexity along fixed-rank induction). *Fix G and let $D = \dim G$. Suppose an induction produces a strict chain $H_0 < \cdots < H_m \le G$ and a height sequence $h_j \ge 2$ satisfying, for some fixed $A, B, C \ge 1$ and scale $R \ge 2$,*

$$h_0 \le R^B, \qquad h_{j+1} \le CR^B h_j^A.$$

Then $m \le D$ and there are constants C_*, B_* depending only on A, B, C, D such that

$$h_j \le C_* R^{B_*} \qquad (0 \le j \le m).$$

If in addition a polynomial error exponent obeys $\eta_{j+1} \ge c\eta_j$ with fixed $0 < c \le 1$, then $\eta_j \ge c^D \eta_0$ throughout the induction.

Proof. Lemma 0.1 gives $m \le D$. Iterating the height recurrence at most D times yields

$$h_j \le C^{1+A+\cdots+A^{j-1}} R^{B(1+A+\cdots+A^j)},$$

which is bounded by $C_* R^{B_*}$ for fixed D. The exponent recurrence gives $\eta_j \ge c^j \eta_0 \ge c^D \eta_0$. ☐

Proposition 0.3 (Application to the Volume-51 height calculus)**.** *In the fixed-rank setting of Volumes 50–51, let $R \ge 2$ bound the initial return/denominator complexity. Then every intermediate subgroup produced by the closing/generation induction has height at most R^{C_G} for one exponent C_G depending only on the fixed ambient representation, and the number of strict subgroup enlargements is at most* $\dim G$*.*

Proof. Volume 51 bounds each sum, intersection, commutator closure, rational extraction, and algebraic-generation step by a fixed polynomial in the incoming height and R. This is exactly the recurrence in Theorem 0.2; Lemma 0.1 supplies the chain-length bound. Absorb the finitely many polynomial losses into the exponent C_G. ☐

CHAPTER 4

Centralizer obstructions

This chapter owns the centralizer obstructions step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Centralizer obstructions: structural mechanism)**.** *If the centralizer of the acting semisimple/unipotent structure contains a positive-dimensional split factor, almost invariance in the generated directions need not control motion along the centralizer.*

Proof. Decompose the local Lie algebra into generated directions and centralizer directions. The averaging operator built from the generated group fixes functions depending only on the uncontrolled factor, so it has no gap on that subspace. This explicitly identifies the missing input.

□

Theorem 0.2 (Centralizer obstruction to a full spectral gap). *Let a probability measure μ on a subgroup $H < G$ act on a quotient that admits an H-equivariant factor $X \to Z$ on which H acts trivially. If $L^2_0(Z) \neq 0$, then the averaging operator $A_\mu f(x) = \int f(xh)\, d\mu(h)$ satisfies*

$$\|A_\mu|_{L^2_0(X)}\| = 1. \tag{52.4.1}$$

Hence no spectral-upgrading argument based only on H-averaging can control that centralizer factor; one must quotient it or supply a separate mixing estimate on Z.

Proof. Choose $0 \neq F \in L^2_0(Z)$ and pull it back to $f = F \circ \pi \in L^2_0(X)$. Triviality of the H-action on Z gives $f(xh) = f(x)$ for every $h \in H$, so $A_\mu f = f$. Since a probability average is a contraction on L^2, its norm is at most one and the fixed nonzero vector gives equality, proving (52.4.1). □

Proposition 0.3 (Centralizer obstructions: downstream consequence)**.** *If $Z_G(U)$ contains a positive-dimensional subgroup C that fixes a nonzero vector in the representation used for averaging, then the averaging operator has a nonconstant fixed vector; therefore any proof whose final step requires the fixed space to consist only of constants fails without an additional quotient or mixing argument.*

Proof. Let Z be a positive-dimensional centralizer quotient and choose a nonconstant $F \in L^2_0(Z)$. Its pullback to the ambient quotient is invariant under the averaging directions commuting with Z, hence the corresponding averaging operator fixes a nonzero zero-mean vector. Therefore that operator has no spectral gap on the full L^2_0 space. Any finite-centralizer argument using such a gap must first quotient or mix the centralizer direction. $\square$

CHAPTER 5

S-arithmetic variants

This chapter owns the s-arithmetic variants step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (S-arithmetic variants: structural mechanism)**.** *For an S-arithmetic product, real-place nondivergence and mixing do not by themselves control excursions or congruence depth at the finite places.*

Proof. Take a sequence with bounded real component and growing denominator at a finite place. The real height remains bounded while the S-height diverges. Thus a proof using only real truncation misses this escape. The required repair is exactly the simultaneous adelic height.

$\square$

Theorem 0.2 (S-arithmetic variants). *Let S contain the archimedean place and at least one finite place. A direct transport of the real effective-homogeneous argument to an S-arithmetic quotient requires, simultaneously:*

(1) an adelic height ht_S *whose nondivergence estimate controls every place in S;*
(2) a single polynomial complexity bound for rational subgroup data in all local factors;
(3) a spectral estimate uniform in the finite-place congruence level (for example property (τ) in the applicable family).

Real-place nondivergence and mixing alone imply none of these finite-place controls.

Proof. For (1), take a sequence whose real component remains in a fixed compact set while one finite-place denominator tends to infinity; the real height stays bounded although the adelic point escapes. For (2), rational subgroup equations can acquire arbitrarily large p-adic denominators while their real coefficients remain bounded, so a real height does not control local complexity. For (3), the real averaging operator acts trivially on the additional finite quotient; without a uniform finite-place spectral input, nonconstant congruence functions can have arbitrarily small spectral gap. Thus each item is logically necessary before the real multiscale proof can be transported. The chapter therefore states a scope criterion, not an unproved general S-arithmetic equidistribution theorem. $\square$

Proposition 0.3 (S-arithmetic variants: downstream consequence)**.** *The 2026* SL_2^n *theorem is recorded as a proved special setting, not generalized here.*

Proof. The proposition is a scope statement: the chapter proves only the necessity of the three simultaneous adelic inputs. A special S-arithmetic theorem may be cited later only after its own hypotheses have been stated and checked; no implication from the real theorem to a general S-arithmetic theorem is asserted here. $\square$

CHAPTER 6

Semidirect-product variants

This chapter owns the semidirect-product variants step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Semidirect-product variants: structural mechanism)**.** *In a semidirect product $V \rtimes G$, a unipotent subgroup of G may shear the abelian radical without generating semisimple spectral gap on the V-Fourier modes.*

PROOF. Fourier transform in V. The G-action permutes characters, and modes with slowly growing character orbit can fail to decay under a semisimple averaging operator. A quantitative statement must control those character orbits or prove a relative spectral gap.

□

THEOREM 0.2 (Fourier obstruction in a semidirect product). *Let $X = (V \rtimes G)/\Lambda$ with $V/(V \cap \Lambda)$ a nontrivial compact abelian factor, and let μ be a probability on G. For a character $\xi \in \widehat{V/(V \cap \Lambda)}$, the Fourier mode e_ξ is carried by A_μ into the span of the dual orbit $\{g\xi : g \in \operatorname{supp}\mu\}$. In particular, if $0 \neq \xi$ is fixed by $\operatorname{supp}\mu$, then*

$$A_\mu e_\xi = e_\xi,$$

so A_μ has no spectral gap on $L^2_0(X)$. Thus a semidirect-product effective theorem needs a quantitative expansion/relative-property estimate for every nonzero dual orbit in addition to the semisimple estimate.

PROOF. Fourier transform in the compact abelian V-factor. For $g \in G$, $e_\xi(g^{-1}v) = e_{g\xi}(v)$, hence averaging sends the ξ-mode into the span of its dual orbit. If the orbit is the singleton $\{\xi\}$, the mode is fixed. Since $\xi \neq 0$, it has mean zero, proving the obstruction. A uniform spectral gap can therefore hold only after quantitatively excluding such fixed or almost-fixed nonzero dual orbits. □

Proposition 0.3 (Semidirect-product variants: downstream consequence)**.** *For a semidirect product $G = V \rtimes H$ with $V \neq 0$, an H-dimension-growth estimate does not control Fourier modes on V: characters $\chi_\xi(v) = e^{2\pi i\langle \xi, v\rangle}$ may remain almost invariant when the dual H-orbit of ξ stays near ξ. Hence an additional estimate on the H-action on V^* is required.*

Proof. For a semidirect product $G \ltimes V$, characters of the abelian factor V give Fourier modes. If the semisimple averaging does not move a given character orbit sufficiently, one obtains almost-invariant nonconstant modes even though the G-projection mixes. Hence semisimple dimension growth by itself does not control the V-direction; an additional Fourier/relative spectral estimate is required. □

CHAPTER 7

Shrinking-target and short-segment regimes

This chapter owns the shrinking-target and short-segment regimes step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Shrinking-target and short-segment regimes: structural mechanism)**.** *If the averaging segment has length smaller than the inverse of the quantitative closing scale, the orbit may remain inside a tubular neighborhood without producing a detectable self-return.*

Proof. The pigeonhole step in closing uses at least as many sampled times as thick-chart cells at the target resolution. Below that threshold no pair return is forced. Writing this inequality explicitly gives the claimed necessary scale relation.

□

Theorem 0.2 (Explicit scale condition for shrinking targets). *Assume a long-orbit estimate*

$$\left|\frac{1}{T}\int_0^T f(xu_t)\,dt - \int f\,dm\right| \le CT^{-\eta}\mathcal{S}_D(f) \tag{52.7.1}$$

and suppose a radius-r target admits a smooth majorant/minorant f_r with $\mathcal{S}_D(f_r) \le C_1 r^{-D'}$ and main term $\int f_r\,dm \asymp r^{\kappa}$. Then (52.7.1) is asymptotically informative only in the regime

$$T^{\eta} r^{D'+\kappa} \longrightarrow \infty. \tag{52.7.2}$$

Moreover a pigeonhole closing argument at spatial resolution r in a chart of dimension d cannot force a pair return from fewer than cr^{-d} sampled times.

Proof. Substituting f_r into (52.7.1) gives error $O(T^{-\eta}r^{-D'})$. Dividing by the main term $\asymp r^\kappa$ gives relative error $O(T^{-\eta}r^{-(D'+\kappa)})$, which tends to zero exactly under (52.7.2). For closing, a bounded chart needs $\gg r^{-d}$ balls of radius r to cover it. With fewer than a fixed multiple of that number, the pigeonhole principle supplies no two sample times in the same ball. ☐

Proposition 0.3 (Necessity of the scale relation)**.** *Under the hypotheses of Theorem 0.2, the long-orbit estimate alone gives the relative-error bound*

$$\frac{|\operatorname{Err}(T,r)|}{\int f_r\,dm} \ll T^{-\eta}r^{-(D'+\kappa)}.$$

Consequently, if $T^\eta r^{D'+\kappa}$ stays bounded along a sequence (T_j, r_j), estimate (52.7.1) *by itself does not imply that the relative error tends to zero along that sequence.*

Proof. Substitution of $\mathcal{S}_D(f_r) \le C_1 r^{-D'}$ into (52.7.1) gives absolute error $O(T^{-\eta}r^{-D'})$. Dividing by the main term $\asymp r^\kappa$ gives the displayed relative bound. If its reciprocal scale $T^\eta r^{D'+\kappa}$ is bounded, that upper bound need not tend to zero, so no asymptotic follows from (52.7.1) alone. ☐

CHAPTER 8

Failure-mode theorems and precise frontier boundaries

This chapter owns the failure-mode theorems and precise frontier boundaries step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Failure-mode theorems and precise frontier boundaries: structural mechanism)**.** *The current effective theory covered by this project consists of the rank-one models, the Diophantine* SL_3 *theorem, quasi-split absolute-rank-two groups, the stated Oppenheim applications, and selected* S*-arithmetic* SL_2^n *settings from the literature. Arbitrary rank, arbitrary semidirect products, and unrestricted shrinking targets are not consequences of these proofs.*

Proof. This follows by reading the dependency table established in Chapters 1–7. A setting is included as proved here only when all five template inputs are present with polynomial constants. The named frontier settings each fail at least one explicitly listed input, so no theorem is asserted there.

□

Theorem 0.2 (Failure-mode theorems and precise frontier boundaries). *For each setting* $\mathcal{S}$ *considered in Chapters 1–7, let*

$$\mathbf{I}(\mathcal{S}) = (I_{\mathrm{nd}}, I_{\mathrm{close}}, I_{\mathrm{grow}}, I_{\mathrm{alg}}, I_{\mathrm{spec}}) \in \{0,1\}^5$$

record whether the manuscript has proved, respectively, quantitative nondivergence, effective closing/rational separation, transverse dimension growth, polynomial algebraic-complexity control, and the required spectral upgrade. The effective-Ratner conclusion is asserted in this volume only when all five *entries equal* 1*. If some entry is* 0*, the corresponding extension is labeled conditional/unproved at exactly that missing input.*

Proof. Chapter 1 writes the general proof as the implication chain

$$I_{\mathrm{nd}} \Longrightarrow I_{\mathrm{close}} \Longrightarrow I_{\mathrm{grow}} \Longrightarrow I_{\mathrm{alg}} \Longrightarrow I_{\mathrm{spec}},$$
$$I_{\mathrm{spec}} \Longrightarrow \text{effective equidistribution or a quantified obstruction.}$$

Chapters 2–7 exhibit concrete failures of individual arrows: varying-rank root depth, uncontrolled intermediate-subgroup height, centralizer fixed vectors, finite-place escape, semidirect Fourier modes, and shrinking-target Sobolev loss. Hence a row with a zero entry cannot be promoted by this proof. Conversely, the fixed-rank real settings proved earlier have all five inputs and the chain closes. The matrix therefore records logical proof dependencies rather than an informal frontier description. □

Volume 53

Geometrically Finite Quotients

Volume 53 scope and prerequisites

Prerequisites. Throughout this volume $X = \mathbb{H}^n$ with $n \geq 2$ and $G = \mathrm{Isom}^+(X) \cong \mathrm{SO}^+(n,1)$. A *Kleinian group* means a discrete subgroup of G. We take $\Gamma < G$ torsion-free and non-elementary. Zariski density is imposed only when a later homogeneous-dynamics application needs it; none of the foundational Patterson–Sullivan arguments below uses Zariski density. We write ∂X for the visual boundary and $\overline{X} = X \sqcup \partial X$ for the usual compactification.

Proof and provenance. The construction of the conformal density follows the Patterson–Sullivan architecture in [Pat76; Sul79]; the geometric-finiteness conventions are chosen in the truncated-convex-core form used in this volume. Bowditch's equivalence with the conical/bounded-parabolic formulation is recorded for orientation but is not used as an unproved logical input. Every result used later is stated at the precise scope needed in this volume.

CHAPTER 1

Hyperbolic space, its boundary, and the Busemann cocycle

Why this matters. Later BMS and Burger–Roblin formulas are built from two pieces of data: the two endpoints of an oriented geodesic and a Busemann time coordinate. We therefore fix the sign convention now and prove the identities that will later make the measures independent of the chosen base point.

1. Boundary geometry and cocycles

For $\xi \in \partial X$ and $x, y \in X$ define

$$\beta_\xi(x, y) = \lim_{t \to \infty} \big(d(x, c(t)) - d(y, c(t))\big),$$

where c is any geodesic ray ending at ξ. In $\mathbb{H}^n$ the limit exists and is independent of the chosen ray to ξ.

Lemma 1.1 (Busemann identities)**.** *For all $x, y, z \in X$, $\xi \in \partial X$ and $g \in G$,*

$$\beta_\xi(x, z) = \beta_\xi(x, y) + \beta_\xi(y, z), \qquad \beta_\xi(x, y) = -\beta_\xi(y, x),$$

and

$$\beta_{g\xi}(gx, gy) = \beta_\xi(x, y), \qquad |\beta_\xi(x, y)| \le d(x, y).$$

Moreover $(\xi, x, y) \mapsto \beta_\xi(x, y)$ is continuous on $\partial X \times X \times X$.

Proof. Choose a ray $c(t) \to \xi$. The cocycle identity follows before passing to the limit by writing

$$d(x, c(t)) - d(z, c(t)) = [d(x, c(t)) - d(y, c(t))] + [d(y, c(t)) - d(z, c(t))].$$

Antisymmetry is the special case $z = x$. Since g preserves distance and sends a ray to ξ to a ray to $g\xi$, equivariance follows immediately. The triangle inequality gives $|d(x, c(t)) - d(y, c(t))| \le d(x, y)$ for every t, hence the Lipschitz bound after taking the limit.

For continuity it is enough to use the ball model. If $o = 0$ and $\xi \in S^{n-1}$, direct computation from the hyperbolic distance gives

$$\beta_\xi(x, o) = \log \frac{|x - \xi|^2}{1 - |x|^2}.$$

The displayed expression is continuous in (ξ, x), and the cocycle identity reduces the general pair (x, y) to the difference of two such expressions. This also verifies directly that the defining limit is independent of the chosen ray. $\square$

THEOREM 1.2 (Compactification and orbit accumulation). *The closed ball $\overline{X} = X \sqcup \partial X$ is compact. If $\Gamma < G$ is discrete and $o \in X$, then the orbit Γo has no accumulation point in X; hence every accumulation point lies in ∂X. The boundary accumulation set is independent of o.*

PROOF. Compactness is immediate in the closed-ball model. Proper discontinuity of a discrete subgroup of G implies that for every compact $K \subset X$ only finitely many $\gamma \in \Gamma$ satisfy $\gamma o \in K$, so an infinite sequence of distinct orbit points cannot accumulate in X.

Let $o, o' \in X$ and suppose $\gamma_j o \to \xi \in \partial X$. Since $d(\gamma_j o, \gamma_j o') = d(o, o')$ is uniformly bounded, the two sequences converge to the same point of the visual boundary: in the ball model, bounded hyperbolic distance forces their Euclidean directions to have the same limiting point on S^{n-1}. Thus the set of boundary accumulation points is independent of the base point. $\square$

Proposition 1.3 (Base-point change for visual weights)**.** *Fix $\delta \geq 0$ and $o, o' \in X$. For every finite Borel measure ν_o on ∂X, the formula*

$$d\nu_{o'}(\xi) = e^{-\delta\beta_\xi(o',o)}\, d\nu_o(\xi)$$

defines a finite measure equivalent to ν_o, and changing from o to o' and then to o'' gives exactly the same measure as changing directly from o to o''.

PROOF. By Lemma 1.1, $|\beta_\xi(o', o)| \leq d(o, o')$, so the density is bounded above and below by positive constants; finiteness and equivalence follow. The cocycle identity gives

$$e^{-\delta\beta_\xi(o'',o')}e^{-\delta\beta_\xi(o',o)} = e^{-\delta\beta_\xi(o'',o)},$$

which is precisely the asserted consistency. $\square$

Worked Example 1.4 (The upper half-plane sign convention)**.** For $\mathbb{H}^2 = \{x + iy : y > 0\}$ and $\xi = \infty$, a vertical ray gives

$$\beta_\infty(x + iy, x' + iy') = \log \frac{y'}{y}.$$

Thus for $a_t : z \mapsto e^t z$ we have $\beta_\infty(a_t i, i) = -t$. This sign is the one used later in the Hopf-coordinate formulas.

CHAPTER 2

Limit sets and radial behavior

Why this matters. The limit set is where the infinite-volume dynamics lives. Before constructing measures on it, we need a base-point-free definition and a usable dynamical description of radial (conical) points.

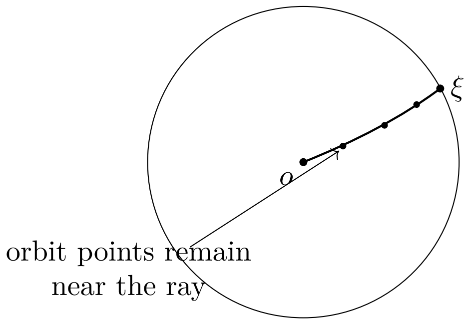

FIGURE 1. Radial behavior in the Poincare-ball picture: a limit point is radial when some orbit sequence tends to it while staying within bounded distance of a geodesic ray. This is the recurrence geometry later seen by the geodesic flow.

1. Limit-set foundations

Definition 2.1. The *limit set* $\Lambda(\Gamma)$ is the set of accumulation points in ∂X of one (hence every) orbit Γo. Its complement $\Omega(\Gamma) = \partial X \setminus \Lambda(\Gamma)$ is the domain of discontinuity. A point $\xi \in \Lambda(\Gamma)$ is *radial* (or *conical*) if there are $R < \infty$ and infinitely many $\gamma \in \Gamma$ for which γo lies within distance R of a geodesic ray $[o, \xi)$.

Lemma 2.2 (Closedness and invariance of the limit set)**.** *The set* $\Lambda(\Gamma)$ *is nonempty for an infinite discrete group, is closed in* ∂X*, and is* Γ*-invariant. If* Γ *is non-elementary, then* $\Lambda(\Gamma)$ *contains at least three points.*

PROOF. An infinite orbit in the compact space $\overline{X}$ has an accumulation point, and Theorem 1.2 places every such point on the boundary. If $\xi_j \in \Lambda(\Gamma)$

and $\xi_j \to \xi$, choose γ_j with $\gamma_j o$ sufficiently close in the compactification to ξ_j and with $d(o, \gamma_j o) \to \infty$; then $\gamma_j o \to \xi$, proving closedness. If $\gamma_j o \to \xi$, then for $g \in \Gamma$, $g\gamma_j o \to g\xi$, so $g\Lambda(\Gamma) = \Lambda(\Gamma)$.

A discrete group with a one-point limit set fixes that point and is elementary; with a two-point limit set it preserves the geodesic joining those points and is again elementary. Therefore a non-elementary group has at least three limit points. □

THEOREM 2.3 (Radial points and recurrence of rays). *Let $M = \Gamma\backslash X$ and let $\pi : X \to M$. For $\xi \in \Lambda(\Gamma)$ the following are equivalent.*

(1) ξ is radial.

(2) The projected ray $\pi([o,\xi))$ returns infinitely often to some compact subset of M.

The equivalence is independent of the base point o.

PROOF. Suppose ξ is radial. Choose R and infinitely many γ_j with $d(\gamma_j o, [o,\xi)) \le R$. Let x_j be a closest point on the ray. In the quotient, $d_M(\pi(x_j), \pi(o)) \le d(x_j, \gamma_j o) \le R$, so the projected ray meets the compact closed R-ball about $\pi(o)$ infinitely often; properness of M on bounded sets follows from properness of X and discreteness of the action.

Conversely, if $\pi(x_j)$ with $x_j \in [o,\xi)$ lies in a compact set K infinitely often, cover K by finitely many metric balls whose lifts meet a fixed compact set $\widetilde{K} \subset X$. For each j choose γ_j with $\gamma_j^{-1} x_j \in \widetilde{K}$. Then $d(x_j, \gamma_j o) = d(\gamma_j^{-1} x_j, o) \le R$ for a uniform R, giving radiality. As x_j escapes along the ray, infinitely many γ_j are distinct.

Changing o moves both rays and orbit points a bounded distance in a CAT(-1) space; the defining bounded-neighborhood condition is therefore unchanged after increasing R by a fixed amount. □

Proposition 2.4 (Finite-index invariance)**.** *If $\Gamma' < \Gamma$ has finite index, then*

$$\Lambda(\Gamma') = \Lambda(\Gamma),$$

and a boundary point is radial for Γ' if and only if it is radial for Γ.

PROOF. Let

$$N = \bigcap_{g\in\Gamma} g\Gamma' g^{-1}.$$

Then N is normal in Γ and has finite index in both Γ and Γ'. Choose coset representatives $g_1, \dots, g_m$ for Γ/N. If $\gamma_j o \to \xi \in \Lambda(\Gamma)$, pass to a subsequence with $\gamma_j = g_i n_j$ and $n_j \in N$. Then $n_j o$ converges to $g_i^{-1}\xi$, so $g_i^{-1}\xi \in \Lambda(N)$. Normality gives $g_i\Lambda(N) = \Lambda(g_i N g_i^{-1}) = \Lambda(N)$, where conjugating a group conjugates its limit set by base-point independence. Hence $\xi \in \Lambda(N)$. Thus

$\Lambda(\Gamma) \subset \Lambda(N)$, while the reverse inclusion is immediate from $N \subset \Gamma$. Since $N \subset \Gamma' \subset \Gamma$, we obtain $\Lambda(N) = \Lambda(\Gamma') = \Lambda(\Gamma)$.

For radiality, pass again to the normal core N. If ξ is radial for Γ, write an infinite radial sequence as $\gamma_j = g_{i_j} n_j$ with $n_j \in N$ and choose a subsequence with $i_j = i$ fixed. Applying g_i^{-1} shows that $g_i^{-1}\xi$ is radial for N. Because $N \triangleleft \Gamma$, the set of N-radial points is Γ-invariant: applying g_i to the witnessing ray and orbit sequence changes the base point only by a bounded amount. Hence ξ is radial for N, and therefore for Γ'. The converse is immediate from $\Gamma' \subset \Gamma$. $\square$

Worked Example 2.5 (A cyclic hyperbolic group)**.** If $\Gamma = \langle z \mapsto e^{\ell} z\rangle < \mathrm{PSL}_2(\mathbb{R})$, then $\Lambda(\Gamma) = \{0, \infty\}$. Both endpoints are radial, but the group is elementary. This is why all later rigidity statements explicitly retain the non-elementary hypothesis.

CHAPTER 3

Convex hulls and convex cores

Why this matters. Geometric finiteness is most concrete after passing from the boundary set $\Lambda(\Gamma)$ to its convex hull in X. The quotient of that hull is the convex core, the smallest convex part of the quotient containing every closed geodesic.

1. Convex geometry of the limit set

Definition 3.1. For a closed set $A \subset \partial X$ with at least two points, let $\mathrm{CH}(A)$ be the smallest closed convex subset of X whose closure in $\overline{X}$ contains A. Equivalently in hyperbolic space it is the closed convex hull of all complete geodesics with endpoints in A. The *convex core* is

$$\mathrm{Core}(M) = \Gamma \backslash \mathrm{CH}(\Lambda(\Gamma)).$$

Lemma 3.2 (Equivariance of convex hulls)**.** *For every isometry $g \in G$ and closed $A \subset \partial X$,*

$$g\,\mathrm{CH}(A) = \mathrm{CH}(gA).$$

In particular $\mathrm{CH}(\Lambda(\Gamma))$ *is* Γ*-invariant and its quotient is well defined.*

Proof. An isometry sends geodesics to geodesics and closed convex sets to closed convex sets. Thus $g\,\mathrm{CH}(A)$ is a closed convex set whose boundary closure contains gA, so it contains $\mathrm{CH}(gA)$. Applying the same argument to g^{-1} gives the reverse inclusion. Since $\Lambda(\Gamma)$ is Γ-invariant by Lemma 2.2, the final assertion follows. □

Theorem 3.3 (Closed geodesics lie in the convex core). *Let $\gamma \in \Gamma$ be loxodromic. Its invariant axis has endpoints $\gamma^-, \gamma^+ \in \Lambda(\Gamma)$ and is contained in* $\mathrm{CH}(\Lambda(\Gamma))$*. Consequently every closed geodesic in $M = \Gamma\backslash X$ lies in* $\mathrm{Core}(M)$*.*

Proof. For any $o \in X$, $\gamma^k o \to \gamma^+$ as $k \to +\infty$ and $\gamma^k o \to \gamma^-$ as $k \to -\infty$, so both fixed points belong to the limit set. By the geodesic description of the convex hull, the complete geodesic (γ^-, γ^+) is contained in $\mathrm{CH}(\Lambda(\Gamma))$. This geodesic is the axis of γ, and its quotient by $\langle\gamma\rangle$ is the corresponding closed geodesic. Every closed geodesic in a torsion-free hyperbolic quotient arises in this way from a loxodromic conjugacy class. □

Proposition 3.4 (Endpoint geodesics lie in the core)**.** *If $\xi^- \neq \xi^+$ belong to $\Lambda(\Gamma)$, then the complete geodesic (ξ^-, ξ^+) is contained in $\mathrm{CH}(\Lambda(\Gamma))$. Hence every unit tangent vector whose two geodesic endpoints lie in $\Lambda(\Gamma)$ projects to a vector based in $\mathrm{Core}(M)$.*

PROOF. The geodesic description of $\mathrm{CH}(\Lambda(\Gamma))$ includes every complete geodesic joining two points of the limit set. Therefore every point of (ξ^-, ξ^+) lies in the convex hull. A unit tangent vector tangent to this geodesic has its base point there, and passage to the quotient places the base point in $\mathrm{Core}(M)$. $\square$

Worked Example 3.5 (A Schottky surface)**.** For a convex-cocompact Fuchsian Schottky group, $\Lambda(\Gamma)$ is a Cantor set. Its convex hull is bounded by a locally finite family of geodesics, and the quotient of the hull is a compact surface with geodesic boundary. The infinite funnels of $\Gamma\backslash\mathbb{H}^2$ lie outside the convex core.

CHAPTER 4

Geometric finiteness and the truncated convex core

Why this matters. There are several equivalent definitions of geometric finiteness. Proving all equivalences would duplicate a substantial separate theory. For the counting arc we need one concrete version only: compactness of the convex core after deleting finitely many cusp horoballs. We adopt that as the definition and prove the precise consequences used later.

1. The truncated-core formulation

Definition 4.1 (Geometric finiteness used in this series)**.** The group Γ is *geometrically finite* if there are finitely many parabolic fixed points $\xi_1, \ldots, \xi_r$ representing all Γ-orbits of parabolic fixed points and open horoballs $\{\gamma\mathcal{H}_i : \gamma \in \Gamma, 1 \le i \le r\}$ such that any two members are either equal or disjoint, with $\mathcal{H}_i$ based at ξ_i, such that

$$\Gamma\backslash\left(\mathrm{CH}(\Lambda(\Gamma)) \setminus \bigcup_{i,\gamma} \gamma\mathcal{H}_i\right)$$

is compact. When $r = 0$ the group is called *convex cocompact.*

Historical note (not a logical input). For Kleinian groups this formulation is equivalent to the statement that every limit point is either conical or bounded parabolic. We do not use that equivalence as a proof step here; see the geometric-finiteness literature, including Bowditch's treatment in variable negative curvature [Bow95].

Lemma 4.2 (Finite cusp representatives)**.** *Under Definition 4.1, the deleted horoballs split into finitely many Γ-orbits, and for each i the stabilizer Γ_{ξ_i} preserves $\mathcal{H}_i$.*

Proof. Finiteness of the orbit representatives is part of the definition. If $\eta \in \Gamma_{\xi_i}$, then both $\mathcal{H}_i$ and $\eta\mathcal{H}_i$ occur in the declared pairwise-disjoint family and are based at the same boundary point. Two horoballs with the same center are nested; if two open nested horoballs in the family were distinct they would intersect. Therefore $\eta\mathcal{H}_i = \mathcal{H}_i$. $\square$

Theorem 4.3 (Thick compact core plus finitely many cusp ends). *If Γ is geometrically finite in the sense of Definition 4.1, then the convex core is the union of a compact set and finitely many cusp pieces*

$$\Gamma_{\xi_i}\backslash(\mathcal{H}_i \cap \mathrm{CH}(\Lambda(\Gamma))), \qquad 1 \le i \le r.$$

Every sequence escaping every compact subset of the convex core eventually enters one of these finitely many cusp pieces after passing to a subsequence.

Proof. Let

$$C_0 = \mathrm{CH}(\Lambda(\Gamma)) \setminus \bigcup_{i,\gamma} \gamma\mathcal{H}_i.$$

By definition $\Gamma\backslash C_0$ is compact. The complement of its image in the convex core is the image of the union of the horoballs. Quotienting a full Γ-orbit of $\mathcal{H}_i$ identifies it with the single quotient by the stabilizer Γ_{ξ_i}, giving the stated cusp pieces. Since there are finitely many i, an escaping sequence that is not eventually contained in the compact image of C_0 has a subsequence lying in one fixed cusp type. □

Proposition 4.4 (Convex-cocompact specialization)**.** *A geometrically finite group has no parabolic fixed points if and only if its convex core is compact.*

Proof. If there are no parabolic fixed points, Definition 4.1 has $r = 0$, so the convex core itself is the compact truncated core. Conversely suppose that $\mathrm{Core}(M)$ is compact and that $p \in \Gamma$ is parabolic with fixed point ξ. Since Γ is non-elementary, choose $\eta \in \Lambda(\Gamma)$ with $\eta \neq \xi$. The geodesic $c = (\eta, \xi)$ lies in $\mathrm{CH}(\Lambda(\Gamma))$ by Proposition 3.4. Conjugate ξ to ∞. Along the forward ray $c(t) \to \infty$, the height tends to infinity. An isometry fixing ∞ acts in upper-half-space coordinates as $(u, y) \mapsto (\lambda Au+b, \lambda y)$. Because p is parabolic rather than loxodromic, $\lambda = 1$; hence p acts by a Euclidean isometry on each horosphere. The hyperbolic metric on $y =$ constant is y^{-1} times the Euclidean metric, so the displacement $d(c(t), pc(t))$ tends to 0 as the height tends to infinity. Hence the injectivity radius of M along the projected ray tends to 0. A compact subset of a complete hyperbolic manifold has a positive lower bound for the injectivity radius. The ray lies in the convex core, so compactness of $\mathrm{Core}(M)$ is contradicted. □

Worked Example 4.5 (The modular cusp as an orbifold model)**.** The torsion-free hypothesis is irrelevant for this local computation. For $\mathrm{PSL}_2(\mathbb{Z})$, a standard horoball is $\{\Im z > 1\}$ based at ∞. The stabilizer $z \mapsto z + m$ preserves it, and its quotient is the familiar cusp cylinder. Removing the $\mathrm{PSL}_2(\mathbb{Z})$-translates of suitable cusp horoballs leaves a compact truncated fundamental domain.

CHAPTER 5

The Poincaré series and the critical exponent

Why this matters. The critical exponent is simultaneously the threshold for the orbit Poincaré series and the exponential growth rate of orbit points. The first description constructs the Patterson–Sullivan measure; the second is what counting theorems see.

1. Growth and convergence abscissae

Fix $o \in X$ and define

$$P_o(s) = \sum_{\gamma \in \Gamma} e^{-sd(o,\gamma o)}, \qquad \delta_\Gamma = \inf\{s > 0 : P_o(s) < \infty\}.$$

Also put $N_o(R) = \#\{\gamma \in \Gamma : d(o, \gamma o) \leq R\}$.

Lemma 5.1 (Base-point independence of the critical exponent)**.** *For $o, o' \in X$ and every $s > 0$,*

$$e^{-2sd(o,o')} P_o(s) \leq P_{o'}(s) \leq e^{2sd(o,o')} P_o(s).$$

Consequently δ_Γ is independent of o.

Proof. The triangle inequality and invariance of the metric give

$$|d(o', \gamma o') - d(o, \gamma o)| \leq d(o', o) + d(\gamma o', \gamma o) = 2d(o, o').$$

Exponentiating, summing over γ, and reversing the roles of o, o' gives the comparison. Two positive series comparable by finite multiplicative constants have the same convergence abscissa. $\square$

Theorem 5.2 (Orbit-growth formula). *The critical exponent satisfies*

$$\delta_\Gamma = \limsup_{R \to \infty} \frac{1}{R} \log N_o(R).$$

Proof. Write $a_m = N_o(m+1) - N_o(m)$ for the number of orbit elements with distance in $[m, m+1)$. Then

$$e^{-s} \sum_{m \geq 0} a_m e^{-sm} \leq P_o(s) \leq \sum_{m \geq 0} a_m e^{-sm}.$$

Thus P_o and the power series $\sum a_m e^{-sm}$ have the same abscissa of convergence. Since $N_o(m) = \sum_{j<m} a_j$, the Cauchy–Hadamard/root-test calculation gives

$$\inf\{s : \sum a_m e^{-sm} < \infty\} = \limsup_{m\to\infty} \frac{1}{m} \log N_o(m).$$

For completeness, if α denotes the limsup, then for every $\varepsilon > 0$ we have $N_o(m) \le e^{(\alpha+\varepsilon)m}$ for all large m, which makes the shell series converge when $s > \alpha + \varepsilon$. Conversely there are infinitely many m with $N_o(m) \ge e^{(\alpha-\varepsilon)m}$; if $s < \alpha - \varepsilon$, the partial sum of $P_o(s)$ over the ball of radius m is at least $e^{-sm} N_o(m)$ and is unbounded. Letting $\varepsilon \downarrow 0$ proves the formula. Replacing integer m by real R changes the ratio by $o(1)$. ☐

Proposition 5.3 (Finite-index invariance of δ)**.** *If $\Gamma' < \Gamma$ has finite index, then $\delta_{\Gamma'} = \delta_\Gamma$.*

Proof. The inequality $N_{\Gamma'}(R) \le N_\Gamma(R)$ gives $\delta_{\Gamma'} \le \delta_\Gamma$. Choose representatives $g_1, \dots, g_m$ for the right cosets $\Gamma' \backslash \Gamma$, so every $\gamma \in \Gamma$ has the form $\gamma' g_i$. Then

$$|d(o, \gamma' g_i o) - d(o, \gamma' o)| \le d(\gamma' g_i o, \gamma' o) = d(g_i o, o).$$

Hence for $C = \max_i d(o, g_i o)$, $N_\Gamma(R) \le m N_{\Gamma'}(R + C)$. Apply Theorem 5.2 and divide logarithms by R to obtain the reverse inequality. ☐

Worked Example 5.4 (A cyclic loxodromic group)**.** If $\Gamma = \langle g \rangle$ and g has translation length $\ell > 0$, then $d(o, g^k o) = |k|\ell + O(1)$ and $N_o(R) \asymp R$. Hence $\delta_\Gamma = 0$. Non-elementary groups are the regime in which the exponent becomes genuinely exponential and Patterson–Sullivan theory is informative.

CHAPTER 6

Patterson–Sullivan construction

Why this matters. The construction turns the coarse orbit-growth exponent δ_Γ into a geometric measure on $\Lambda(\Gamma)$. The only subtle point is that the unweighted Poincaré series may converge at $s = \delta_\Gamma$. Patterson's slow reweighting repairs this without changing the critical exponent.

1. Construction of the conformal density

Lemma 6.1 (Patterson's divergence trick). *Let $\{r_j\}_{j\geq 1} \subset [0,\infty)$ be a locally finite multiset and suppose $\sum_j e^{-sr_j}$ has finite critical exponent δ. There exists a nondecreasing function $h : [0,\infty) \to [1,\infty)$ such that*

(1) for every $\varepsilon > 0$ there is T_ε with $h(t+u) \leq e^{\varepsilon u} h(t)$ for $t \geq T_\varepsilon$ and $u \geq 0$;
(2) $\sum_j h(r_j) e^{-sr_j}$ has the same critical exponent δ;
(3) the weighted series diverges at $s = \delta$.

Proof. If the original series diverges at δ, take $h \equiv 1$. Assume therefore that it converges at δ. Choose a decreasing sequence $\varepsilon_k \downarrow 0$, say $\varepsilon_k = 1/k$. Because δ is the critical exponent, for every k the series

$$\sum_j e^{-(\delta-\varepsilon_k)r_j} = \sum_j e^{\varepsilon_k r_j} e^{-\delta r_j}$$

diverges, and so does every tail of it.

We construct an increasing continuous function $\phi = \log h$ which is piecewise affine. Set $T_1 = 0$, $\phi(T_1) = 0$. Having chosen T_k and $\phi(T_k)$, let ϕ have slope ε_k on $[T_k, T_{k+1}]$. Since the tail series with factor $e^{\varepsilon_k r_j}$ diverges, we can choose $T_{k+1} > T_k + 1$ so large that

$$\sum_{T_k \leq r_j < T_{k+1}} e^{\phi(T_k)+\varepsilon_k(r_j - T_k)} e^{-\delta r_j} \geq 1.$$

Then continue with the smaller slope ε_{k+1}. The resulting $h = e^\phi$ is nondecreasing and at least 1.

If $t \geq T_K$, every slope of ϕ to the right of t is at most ε_K. Therefore for all $u \geq 0$,

$$\log h(t+u) - \log h(t) = \phi(t+u) - \phi(t) \leq \varepsilon_K u.$$

This proves (1). Taking $t = T_K$ and absorbing the bounded initial interval into a constant shows that for every $\varepsilon > 0$, $h(t) \leq C_\varepsilon e^{\varepsilon t}$. Hence for $s > \delta$ choose $0 < \varepsilon < s - \delta$; then

$$\sum_j h(r_j) e^{-s r_j} \leq C_\varepsilon \sum_j e^{-(s-\varepsilon) r_j} < \infty.$$

Since $h \geq 1$, the weighted series diverges for every $s < \delta$. Thus its critical exponent is exactly δ, proving (2). Finally, by construction the contribution at $s = \delta$ from each block $[T_k, T_{k+1})$ is at least 1, so the weighted series diverges at δ. This proves (3). □

Theorem 6.2 (Patterson–Sullivan conformal density). *There exists a nonzero family $\{\nu_x : x \in X\}$ of finite Borel measures on ∂X such that*

(1) $\operatorname{supp} \nu_x \subset \Lambda(\Gamma)$*;*
(2) $\gamma_* \nu_x = \nu_{\gamma x}$ *for every* $\gamma \in \Gamma$*;*
(3) for all $x, y \in X$*,*

$$\frac{d\nu_x}{d\nu_y}(\xi) = e^{-\delta_\Gamma \beta_\xi(x,y)} \quad \text{for } \nu_y\text{-a.e. } \xi.$$

After multiplying the whole family by one constant we may normalize $\nu_o(\partial X) = 1$ at a chosen base point o.

Proof. Set $r_\gamma = d(o, \gamma o)$ and choose h from Lemma 6.1. For $s > \delta = \delta_\Gamma$ define a probability measure on the compactification $\overline{X}$ by

$$\nu_{o,s} = \frac{1}{Z(s)} \sum_{\gamma \in \Gamma} h(r_\gamma) e^{-s r_\gamma} \, \delta_{\gamma o}, \qquad Z(s) = \sum_\gamma h(r_\gamma) e^{-s r_\gamma}.$$

Choose $s_j \downarrow \delta$. Weak-* compactness of probability measures on compact $\overline{X}$ gives a subsequence converging to a probability measure ν_o. Because $Z(s_j) \to \infty$ by divergence at δ, every fixed compact subset of X receives mass tending to zero: discreteness makes the numerator over that compact set a finite sum. Hence ν_o is supported on ∂X, and in fact on $\Lambda(\Gamma)$ because every weak-* limit point of the orbit lies there.

We next compute the effect of $g \in \Gamma$. Reindexing by $\eta = g\gamma$ gives

$$g_* \nu_{o,s} = \frac{1}{Z(s)} \sum_{\eta \in \Gamma} h(d(o, g^{-1}\eta o)) e^{-s d(o, g^{-1}\eta o)} \delta_{\eta o}.$$

For $z = \eta o$ tending to a boundary point ξ,

$$d(o, g^{-1}z) - d(o,z) = d(go, z) - d(o,z) \longrightarrow \beta_\xi(go, o).$$

The two distances differ by at most $d(o, go)$. Property (1) of Lemma 6.1, applied in both directions, therefore implies that the ratio of the two h-factors tends uniformly to 1 along orbit points escaping to infinity. The finite set of nonescaping orbit points has vanishing normalized mass. Passing to the weak-* limit against a continuous test function yields

$$\frac{d(g_*\nu_o)}{d\nu_o}(\xi) = e^{-\delta\beta_\xi(go,o)}.$$

Now define for arbitrary $x \in X$

$$d\nu_x(\xi) = e^{-\delta\beta_\xi(x,o)}d\nu_o(\xi).$$

Proposition 1.3 shows that this is finite. The cocycle identity gives the conformal Radon–Nikodym rule. To check equivariance, use Busemann equivariance and the preceding formula for $g_*\nu_o$:

$$\begin{aligned} d(g_*\nu_x)(\xi) &= e^{-\delta\beta_{g^{-1}\xi}(x,o)}d(g_*\nu_o)(\xi) \\ &= e^{-\delta\beta_\xi(gx,go)}e^{-\delta\beta_\xi(go,o)}d\nu_o(\xi) \\ &= d\nu_{gx}(\xi). \end{aligned}$$

Thus $\{\nu_x\}$ is the required δ-conformal density. □

Proposition 6.3 (Uniqueness of normalization within the constructed family)**.** *Once ν_o is fixed, the conformal rule determines every ν_x uniquely. Replacing o by another base point in the definition gives the same family.*

PROOF. The Radon–Nikodym identity forces $d\nu_x = e^{-\delta\beta_\xi(x,o)}d\nu_o$, so uniqueness is immediate. If o' is used instead, then

$$e^{-\delta\beta_\xi(x,o')}e^{-\delta\beta_\xi(o',o)} = e^{-\delta\beta_\xi(x,o)}$$

by the cocycle identity, proving base-point consistency. Notice that this proposition does not assert uniqueness among all Patterson–Sullivan densities; that stronger statement requires additional divergence/ergodicity input and belongs later. □

Worked Example 6.4 (A cocompact lattice)**.** Suppose $\Gamma < \mathrm{Isom}^+(\mathbb{H}^n)$ is cocompact and let F be a compact fundamental domain of diameter at most D. The translates γF tile X. Comparing the volume of a hyperbolic ball with the number of translates meeting it gives

$$\frac{\operatorname{vol} B(o, R-D)}{\operatorname{vol} F} \le N_o(R) \le \frac{\operatorname{vol} B(o, R+D)}{\operatorname{vol} F}.$$

Since $\operatorname{vol} B(o,R) \asymp e^{(n-1)R}$, Theorem 5.2 yields $\delta_\Gamma = n-1$. The usual visual measures satisfy

$$\frac{d\nu_x}{d\nu_o}(\xi) = e^{-(n-1)\beta_\xi(x,o)},$$

so in this case the Patterson–Sullivan density agrees, after normalization, with the standard spherical visual density.

CHAPTER 7

Cusps and parabolic subgroups

Why this matters. The truncated-core definition isolates every source of noncompactness into finitely many horoballs. We now record the elementary geometry of those pieces without importing later mixing or measure-classification results.

1. Horoballs and cusp coordinates

A nontrivial $\gamma \in G$ is *parabolic* if it has exactly one fixed point on ∂X. For a parabolic fixed point ξ, write Γ_ξ for its stabilizer.

Lemma 7.1 (Horoball coordinates at a parabolic fixed point)**.** *Conjugate a parabolic fixed point ξ to ∞ in the upper-half-space model $\mathbb{H}^n = \{(u, y) : u \in \mathbb{R}^{n-1},\ y > 0\}$. Then every element of G_∞ has the form*

$$(u, y) \longmapsto (\lambda A u + b, \lambda y),$$

with $A \in O(n-1)$, $b \in \mathbb{R}^{n-1}$ and $\lambda > 0$. If such an element is parabolic, then $\lambda = 1$; hence it preserves every horosphere $y = c$ and every horoball $y > c$.

Proof. The stabilizer of ∞ acts on each horosphere by Euclidean similarities; this gives the displayed form, which can also be read directly from the Möbius description of $\mathrm{Isom}^+(\mathbb{H}^n)$. If $\lambda \neq 1$, the similarity has a fixed point in $\mathbb{R}^{n-1}$ after solving $(I - \lambda A)u = b$ on the nonunit part and treating the unit eigenspace separately; geometrically the corresponding hyperbolic isometry has a second fixed point on ∂X and is loxodromic. A parabolic element has only the fixed point ∞, so $\lambda = 1$. The height coordinate is therefore unchanged. $\square$

Theorem 7.2 (Cusp decomposition of the convex core). *For a geometrically finite Γ, choose the horoballs from Definition 4.1. Then*

$$\mathrm{Core}(M) = K \cup \bigcup_{i=1}^{r} \mathcal{C}_i,$$

where K is compact and

$$\mathcal{C}_i = \Gamma_{\xi_i} \backslash (\mathcal{H}_i \cap \mathrm{CH}(\Lambda(\Gamma))).$$

The intersections $K \cap \mathcal{C}_i$ may be chosen compact, and distinct cusp interiors are disjoint.

Proof. This is Theorem 4.3 with the quotient topology made explicit. Shrink each chosen horoball slightly. Since there are only finitely many cusp orbits and the original open horoballs have disjoint closures after a sufficiently small uniform shrink in each orbit, the remaining truncated core still has compact quotient; enlarge its compact image to include the finite boundary collars. The resulting cusp interiors are disjoint, and their intersections with the enlarged compact part occur only in those collars, hence are compact. $\square$

Proposition 7.3 (Busemann height in a cusp). *With $\xi = \infty$ and $o = (0, 1)$, the cusp height $H(u, y) = \log y$ satisfies*

$$H(x) = -\beta_\infty(x, o).$$

For the geodesic flow $a_t : (u, y) \mapsto (e^t u, e^t y)$ along the vertical geodesic,

$$H(a_t x) = H(x) + t.$$

Thus escaping a cusp is exactly linear escape of the Busemann coordinate.

Proof. The formula in the upper half-space model was computed in the worked example after Lemma 1.1: $\beta_\infty((u, y), o) = -\log y$. The second identity follows because a_t multiplies y by e^t. $\square$

Worked Example 7.4 (A rank-one cusp on a hyperbolic surface). For $z \mapsto z + 1$ on $\mathbb{H}^2$, the quotient of $\{y > Y\}$ is $(\mathbb{R}/\mathbb{Z}) \times (Y, \infty)$ with metric $y^{-2}(dx^2 + dy^2)$. The horizontal circle at height y has length $1/y$, so the cusp narrows exponentially in the Busemann height $\log y$.

CHAPTER 8

Orbital-counting preliminaries

Why this matters. Later volumes derive precise asymptotics from mixing. Before any mixing is used, the critical exponent already supplies robust exponential upper and lower bounds and shows that compact perturbations of the counting region do not change the exponent.

1. Coarse counting consequences

Lemma 8.1 (Poincaré-series upper bound for balls)**.** *For every $s > \delta_\Gamma$ and every $R \geq 0$,*

$$N_o(R) \leq P_o(s)e^{sR}.$$

Consequently for every $\varepsilon > 0$ there is $C_\varepsilon < \infty$ such that

$$N_o(R) \leq C_\varepsilon e^{(\delta_\Gamma+\varepsilon)R} \quad (R \geq 0).$$

Proof. Every term with $d(o, \gamma o) \leq R$ satisfies $e^{-sd(o,\gamma o)} \geq e^{-sR}$. Hence $P_o(s) \geq e^{-sR}N_o(R)$. Take $s = \delta_\Gamma + \varepsilon$ and absorb the finitely many small radii into the constant. □

Theorem 8.2 (Coarse exponential orbital growth). *For every $\varepsilon > 0$,*

$$N_o(R) \leq C_\varepsilon e^{(\delta_\Gamma+\varepsilon)R}$$

for all R, while there are arbitrarily large R for which

$$N_o(R) \geq e^{(\delta_\Gamma-\varepsilon)R}.$$

Thus no exponent strictly below δ_Γ can be a uniform exponential upper bound, and no exponent strictly above δ_Γ is needed for one.

Proof. The upper bound is Lemma 8.1. The lower assertion is exactly the definition of the limsup in Theorem 5.2: if it failed beyond some radius, then $R^{-1}\log N_o(R) \leq \delta_\Gamma - \varepsilon$ eventually, contradicting the orbit-growth formula. □

Proposition 8.3 (Stability under compact perturbations)**.** *Let $x, y \in X$ and let $A, B \geq 0$. Define*

$$N_{x,y}^{A,B}(R) = \#\{\gamma \in \Gamma : d(x, \gamma y) \leq R + A + B\}.$$

Then

$$\limsup_{R\to\infty} \frac{1}{R} \log N_{x,y}^{A,B}(R) = \delta_\Gamma.$$

In particular thickening the base points or the radius by a bounded amount does not alter the exponential growth exponent.

Proof. Set $C = d(x,o) + d(y,o) + A + B$. By the triangle inequality,

$$d(o,\gamma o) - C \le d(x,\gamma y) - (A+B) \le d(o,\gamma o) + C.$$

Therefore, after harmless adjustment for negative radii,

$$N_o(R-C) \le N_{x,y}^{A,B}(R) \le N_o(R+C).$$

Take logarithms, divide by R, and use Theorem 5.2. Since $(R \pm C)/R \to 1$, both comparison terms have limsup δ_Γ. $\square$

Worked Example 8.4 (Why a precise constant needs mixing)**.** Theorem 8.2 determines only the exponential scale $e^{\delta R}$. It does not prove that $e^{-\delta R} N_o(R)$ converges, nor does it identify a leading constant. Those are genuinely later theorems: BMS/BR measures, mixing, and skinning measures are introduced precisely to obtain that extra information.

Volume 54

Boundary Densities and BMS/BR Measures

Volume 54 scope and prerequisites

Prerequisites. We retain the notation and hypotheses of Volume 53. In particular $\{\nu_x\}$ denotes a Patterson–Sullivan density of dimension $\delta = \delta_\Gamma$ constructed in Theorem 6.2. The present volume first constructs the BMS and Burger–Roblin measures and proves their algebraic invariance properties. The full Hopf–Tsuji–Sullivan theorem is reconstructed in Chapter 5; Chapter 6 then extracts its reusable Hopf-ratio mechanism. In the geometrically finite constant-curvature setting, the cusp-mass estimate is also proved explicitly rather than imported from a later counting theorem.

Proof and provenance. The BMS construction follows Sullivan's invariant-measure construction [Sul79]; the horospherical measure normalization is compared with Burger's rank-one construction [Bur90] and Roblin's general negative-curvature treatment [Rob03]. The proof follows the Sullivan–Roblin architecture at the exact real-hyperbolic scope used here. The geometrically finite cusp estimate is compared with Dal'bo–Otal–Peigné. The elementary Euclidean crystallographic input used to identify a rank-k cusp group with a finite extension of $\mathbb{Z}^k$ is isolated explicitly in Chapter 5.

CHAPTER 1

Conformal densities

Why this matters. A conformal density is a coherent family of boundary measures, not merely a single measure. The family formulation is exactly what makes the BMS and BR formulas independent of the arbitrary base point used to write them.

1. Packaging a conformal density

Definition 1.1. For $\alpha \geq 0$, an *α-conformal Γ-equivariant density* on ∂X is a family $\{\mu_x : x \in X\}$ of finite nonzero Borel measures satisfying

$$\gamma_* \mu_x = \mu_{\gamma x}, \qquad \frac{d\mu_x}{d\mu_y}(\xi) = e^{-\alpha \beta_\xi(x,y)}.$$

Lemma 1.2 (A density is determined at one base point)**.** *Fix $o \in X$. An α-conformal density is uniquely determined by μ_o. Its equivariance is equivalent to*

$$\frac{d(\gamma_* \mu_o)}{d\mu_o}(\xi) = e^{-\alpha \beta_\xi(\gamma o, o)} \qquad (\gamma \in \Gamma).$$

Proof. Conformality forces $d\mu_x(\xi) = e^{-\alpha\beta_\xi(x,o)} d\mu_o(\xi)$, proving uniqueness. Equivariance at $x = o$ gives $\gamma_* \mu_o = \mu_{\gamma o}$, and the conformal rule gives exactly the displayed derivative. Conversely, assume the displayed derivative and define the family from μ_o. Then, using Busemann equivariance and the cocycle identity,

$$\begin{aligned} d(\gamma_* \mu_x)(\xi) &= e^{-\alpha \beta_{\gamma^{-1}\xi}(x,o)} d(\gamma_* \mu_o)(\xi) \\ &= e^{-\alpha \beta_\xi(\gamma x, \gamma o)} e^{-\alpha \beta_\xi(\gamma o, o)} d\mu_o(\xi) \\ &= d\mu_{\gamma x}(\xi). \end{aligned}$$

Thus the family is equivariant. □

Theorem 1.3 (Common support and common null sets). *For an α-conformal density, all μ_x are mutually equivalent. They therefore have the same support $S \subset \partial X$, and equivariance makes S Γ-invariant. For the Patterson–Sullivan density of Volume 53, $S \subset \Lambda(\Gamma)$.*

Proof. The Busemann bound $|\beta_\xi(x,y)| \le d(x,y)$ gives

$$e^{-\alpha d(x,y)} \le \frac{d\mu_x}{d\mu_y}(\xi) \le e^{\alpha d(x,y)},$$

so the measures have exactly the same null sets and support. If $\xi \in S$ and U is a neighborhood of $\gamma\xi$, then $\gamma^{-1}U$ is a neighborhood of ξ and $\mu_{\gamma x}(U) = \mu_x(\gamma^{-1}U) > 0$; hence $\gamma\xi \in S$. The Patterson–Sullivan support assertion is Theorem 6.2. □

Proposition 1.4 (Normalization bookkeeping)**.** *If $\mu'_x = c\mu_x$ for one $c > 0$, then every boundary integral linear in the density is multiplied by c, while a Hopf product using the density at both endpoints is multiplied by c^2. No base-point change introduces any additional scalar.*

Proof. The first two assertions are linearity and bilinearity. The final assertion follows from the exact Radon–Nikodym cocycle in the definition: base-point changes are prescribed by a density, not by an unspecified proportionality relation. □

Worked Example 1.5 (Visual density)**.** On $\mathbb{H}^n$, normalized spherical measure viewed from x gives a family $\{m_x\}$ of dimension $n-1$. In the ball model,

$$\frac{dm_x}{dm_o}(\xi) = \left(\frac{1-|x|^2}{|x-\xi|^2}\right)^{n-1} = e^{-(n-1)\beta_\xi(x,o)}.$$

This G-equivariant density will supply the Lebesgue endpoint in the Burger–Roblin measure.

CHAPTER 2

Hopf coordinates

Why this matters. An oriented geodesic is determined by its two endpoints; a unit tangent vector additionally records where we are on that geodesic. The Busemann cocycle supplies the missing real coordinate and turns the geodesic flow into translation.

1. Endpoint-time coordinates

Write
$$\partial^2 X = (\partial X \times \partial X) \setminus \{(\xi,\xi) : \xi \in \partial X\}.$$
For $v \in T^1X$, let v^-, v^+ be the backward and forward endpoints and put
$$s_o(v) = \beta_{v^-}(\pi(v), o).$$

Lemma 2.1 (Busemann time along a geodesic)**.** *For the geodesic flow g^t,*
$$s_o(g^t v) = s_o(v) + t.$$
For $\gamma \in G$,
$$s_o(\gamma v) = s_o(v) + \beta_{v^-}(o, \gamma^{-1} o).$$

PROOF. Along the oriented geodesic, moving time t toward v^+ moves distance t away from v^-. Hence $\beta_{v^-}(\pi(g^t v), \pi(v)) = t$, and the first formula follows from the cocycle identity. For the second,
$$\begin{aligned} s_o(\gamma v) &= \beta_{\gamma v^-}(\gamma\pi(v), o) = \beta_{v^-}(\pi(v), \gamma^{-1} o) \\ &= \beta_{v^-}(\pi(v), o) + \beta_{v^-}(o, \gamma^{-1} o). \end{aligned}$$
$\square$

THEOREM 2.2 (Hopf parametrization). *The map*
$$\mathcal{H}_o : T^1X \longrightarrow \partial^2 X \times \mathbb{R}, \qquad v \longmapsto (v^-, v^+, s_o(v))$$
is a homeomorphism (indeed a smooth diffeomorphism in the natural smooth structures). Under it, g^t is translation $(\xi^-, \xi^+, s) \mapsto (\xi^-, \xi^+, s + t)$.

Proof. Given distinct ξ^-, ξ^+ there is a unique oriented complete geodesic c from ξ^- to ξ^+. Along c, the function $t \mapsto \beta_{\xi^-}(c(t), o)$ has slope 1 by Lemma 2.1, so for every prescribed $s \in \mathbb{R}$ there is exactly one point of c with Busemann time s. The unit tangent there gives the inverse map. Continuity and smoothness follow in the ball or upper-half-space model from smooth dependence of a geodesic on its distinct endpoints and from the explicit Busemann formula. The flow formula is Lemma 2.1. $\square$

Proposition 2.3 (The limit-set Hopf locus)**.** *The subset*

$$\{v \in T^1X : v^-, v^+ \in \Lambda(\Gamma)\}$$

corresponds exactly to $\Lambda(\Gamma)^{(2)} \times \mathbb{R}$ *in Hopf coordinates and is* Γ*- and flow-invariant. Its vectors are based in* $\mathrm{CH}(\Lambda(\Gamma))$*.*

Proof. The first statement is the definition of the Hopf map restricted to the indicated endpoint set. The limit set is Γ-invariant and the geodesic flow fixes both endpoints, so the subset is invariant. Proposition 3.4 puts every geodesic joining two limit points inside the convex hull. $\square$

Worked Example 2.4 (Hopf coordinates in $\mathbb{H}^2$)**.** A vector tangent to the vertical geodesic from 0 to ∞ and based at iy has endpoints $(0, \infty)$. With $o = i$, one computes $s_o(v) = \beta_0(iy, i) = \log y$ for the orientation from 0 to ∞. The geodesic flow sends y to $e^t y$, so s increases by t exactly as the abstract formula predicts.

CHAPTER 3

The Bowen–Margulis–Sullivan measure

Why this matters. The Patterson–Sullivan density lives on one boundary point. A geodesic has two endpoints, so multiplying the two boundary densities and adding Lebesgue time produces the natural geodesic-flow measure. The Busemann weights are precisely what remove base-point dependence.

1. Construction and invariance

For $v \in T^1X$ write $x = \pi(v)$ and define

$$d\widetilde{m}_o^{\mathrm{BMS}}(v) = e^{\delta\beta_{v^+}(o,x)+\delta\beta_{v^-}(o,x)}\, d\nu_o(v^+)\, d\nu_o(v^-)\, ds,$$

where ds is Lebesgue measure in Hopf time.

Lemma 3.1 (Base-point independence of the BMS product)**.** *The measure $\widetilde{m}_o^{\mathrm{BMS}}$ does not depend on o.*

Proof. Replace o by o'. Conformality gives $d\nu_{o'}(\xi) = e^{-\delta\beta_\xi(o',o)}d\nu_o(\xi)$, while the cocycle identity gives

$$\beta_\xi(o',x) = \beta_\xi(o',o) + \beta_\xi(o,x).$$

For each endpoint the exponential factor contributed by the first formula cancels the additional Busemann factor from the second. The time coordinate changes by a boundary-dependent additive constant, but Lebesgue measure ds is translation invariant. Therefore the full product measure is unchanged. □

Theorem 3.2 (BMS measure: descent and flow invariance). *The measure $\widetilde{m}^{\mathrm{BMS}}$ is a Γ-invariant Radon measure on T^1X, invariant under the geodesic flow. It descends to a Radon measure m^{BMS} on T^1M. No finiteness assertion is made in this theorem.*

Proof. Apply $\gamma \in \Gamma$. If the formula for the image measure is written with base point γo, equivariance $\gamma_*\nu_o = \nu_{\gamma o}$ and Busemann equivariance show that it is exactly the pushforward of the formula based at o. Lemma 3.1 then replaces γo by o, proving Γ-invariance.

Under g^t, the forward-endpoint quantity $\beta_{v^+}(o, \pi(v))$ increases by t, whereas the backward-endpoint quantity decreases by t. Their sum is unchanged, both endpoint measures are fixed, and ds is translation invariant. Thus the measure is flow invariant.

To see local finiteness, use a compact Hopf flow box whose endpoint pairs remain a positive distance from the diagonal and whose s-coordinate is bounded. The Busemann weights are then bounded continuous functions, and ν_o is finite; hence the box has finite measure. Such boxes cover compact subsets of T^1X, so the measure is Radon. A Γ-invariant Radon measure descends to the quotient. □

Proposition 3.3 (Support of the BMS measure)**.** *If $S = \operatorname{supp} \nu_o$, then*

$$\operatorname{supp} \widetilde{m}^{\mathrm{BMS}} = \{v \in T^1X : v^-, v^+ \in S,\ v^- \neq v^+\}.$$

In particular it is contained in the limit-set Hopf locus and hence is based in $\mathrm{CH}(\Lambda(\Gamma))$*.*

Proof. On a Hopf flow box the exponential density is positive and continuous, so the support is exactly the product support of $\nu_o \otimes \nu_o \otimes ds$ away from the diagonal. Theorem 1.3 gives $S \subset \Lambda(\Gamma)$, and Proposition 2.3 gives the convex-core conclusion. □

Worked Example 3.4 (Why the two Busemann factors cancel under the flow)**.** If $x_t = \pi(g^t v)$, then

$$\beta_{v^+}(o, x_t) = \beta_{v^+}(o, x_0) + t, \qquad \beta_{v^-}(o, x_t) = \beta_{v^-}(o, x_0) - t.$$

The exponent in m^{BMS} therefore changes by $\delta t - \delta t = 0$. This one-line cancellation is the algebraic reason the BMS measure is a geodesic-flow invariant measure rather than merely a quasi-invariant one.

CHAPTER 4

The Burger–Roblin measure

Why this matters. For horospherical dynamics, one endpoint should retain the Patterson–Sullivan density while the endpoint that varies along a strong unstable horosphere should carry ordinary spherical measure. This asymmetric product is the Burger–Roblin measure.

1. The asymmetric Hopf product

Let $\{m_x\}$ denote the G-equivariant visual density of dimension $n-1$ from the worked example in Chapter 1. Define

$$d\widetilde{m}^{\mathrm{BR}}(v) = e^{(n-1)\beta_{v^+}(o,x)+\delta\beta_{v^-}(o,x)}\, dm_o(v^+)\, d\nu_o(v^-)\, ds, \qquad x = \pi(v).$$

We call the strong unstable leaf through v the set of vectors with the same v^- and the same value of s_o.

Lemma 4.1 (Visual density and horospherical Lebesgue measure)**.** *The family $\{m_x\}$ is an $(n-1)$-conformal G-equivariant density. On a strong unstable leaf, the conditional measure*

$$e^{(n-1)\beta_{v^+}(o,x)} dm_o(v^+)$$

is Euclidean Lebesgue measure in horospherical coordinates, up to one global normalization constant.

Proof. The conformal and equivariant identities follow from the Poisson-kernel formula displayed in Chapter 1. For the leaf statement, use G-equivariance to move the fixed backward endpoint to ∞ in the upper-half-space model and the chosen horosphere to $y = 1$. Vectors in the leaf are then parametrized by the Euclidean coordinate $u \in \mathbb{R}^{n-1}$ of the other end-point/base point. Substituting the Poisson kernel

$$dm_x(\xi) = e^{-(n-1)\beta_\xi(x,o)} dm_o(\xi)$$

shows that the displayed weighted boundary measure is exactly the G_∞-invariant Euclidean measure du on that horosphere, up to the normalization of m_o. Transporting back by G proves the claim on every unstable leaf. $\square$

Theorem 4.2 (Burger–Roblin measure: invariance and scaling). *The formula above is base-point independent, Γ-invariant and Radon; hence it descends to T^1M. It is invariant under translations along strong unstable horospheres. Under the geodesic flow,*

$$(g^t)^*\widetilde{m}^{\mathrm{BR}} = e^{(n-1-\delta)t}\widetilde{m}^{\mathrm{BR}}.$$

Proof. Base-point independence and Γ-invariance are proved exactly as in Lemma 3.1 and Theorem 3.2, using dimension $n-1$ for the visual endpoint and dimension δ for the Patterson–Sullivan endpoint. Local finiteness follows from the same compact-flow-box argument.

On a strong unstable leaf, v^- and s_o are fixed. Lemma 4.1 identifies the varying v^+ factor with Euclidean Lebesgue measure; the remaining factors are constant along the leaf. Hence horospherical translations preserve the conditional, and therefore the global measure.

Finally, under g^t, the v^+ Busemann term increases by t and the v^- term decreases by t. Thus the density at $g^t v$ is multiplied by $e^{(n-1)t}e^{-\delta t} = e^{(n-1-\delta)t}$, while ds is unchanged under translation. This is the pullback formula. $\square$

Proposition 4.3 (The symmetric specialization)**.** *If $\delta = n-1$ and the Patterson–Sullivan endpoint density is chosen to be the visual density $\nu_x = m_x$, then*

$$\widetilde{m}^{\mathrm{BR}} = \widetilde{m}^{\mathrm{BMS}}.$$

In this case the flow-scaling factor is 1*.*

Proof. Under the stated choice both endpoint factors in the two Hopf products are identical and have dimension $n-1$, so the formulas coincide term by term. The scaling exponent $n-1-\delta$ is then zero. $\square$

Worked Example 4.4 (A horosphere in $\mathbb{H}^2$)**.** After moving the backward endpoint to ∞, a strong unstable horosphere is a horizontal line $y = c$. Its intrinsic conditional is a constant multiple of dx. With the orientation whose backward endpoint is ∞, forward geodesic time sends this leaf to $y = e^{-t}c$. Equivalently, conjugation by the geodesic flow expands the unstable horospherical group parameter by e^t. The BR scaling formula records the mismatch between this leaf-Haar exponent $1 = n-1$ and the Patterson–Sullivan exponent δ on the opposite endpoint.

CHAPTER 5

Finiteness, conservativity, and the Hopf–Tsuji–Sullivan dichotomy

Why this matters. The explicit BMS formula does not by itself say whether typical geodesics return. The missing bridge is the Hopf–Tsuji–Sullivan dichotomy. In constant curvature the proof can be organized transparently: shadows translate the Poincaré series into a boundary limsup problem; a second-moment estimate turns divergence into positive radial mass; differentiation upgrades positive radial mass to full radial mass; recurrence identifies the conservative part; and Hopf's ratio argument gives ergodicity. A separate cusp computation then proves BMS finiteness for geometrically finite hyperbolic manifolds.

1. Shadows and the radial limsup set

For $x, y \in X$ and $R > 0$, define the shadow

$$\mathcal{O}_R(x, y) = \{\xi \in \partial X : [x, \xi) \cap B(y, R) \neq \varnothing\}.$$

Write $r_\gamma = d(o, \gamma o)$ and, once R is fixed, $\mathcal{O}_\gamma = \mathcal{O}_R(o, \gamma o)$.

Lemma 5.1 (Shadow estimate)**.** *There is R_0 such that for every $R \geq R_0$ there are constants $c_R, C_R > 0$ satisfying*

$$c_R e^{-\delta r_\gamma} \leq \nu_o(\mathcal{O}_R(o, \gamma o)) \leq C_R e^{-\delta r_\gamma} \qquad (\gamma \in \Gamma).$$

Moreover, for every fixed $A > 0$ there is $M_{R,A}$ such that

$$\sum_{T \leq r_\gamma < T+A} e^{-\delta r_\gamma} \leq M_{R,A} \qquad (T \geq 0).$$

Proof. Let $\xi \in \mathcal{O}_R(o, \gamma o)$ and choose $z \in [o, \xi)$ with $d(z, \gamma o) \leq R$. From the defining limit of the Busemann cocycle and the triangle inequality,

$$|\beta_\xi(o, \gamma o) - r_\gamma| \leq 2R.$$

Conformality therefore gives, on the shadow,

$$e^{-2\delta R} e^{-\delta r_\gamma} \leq \frac{d\nu_o}{d\nu_{\gamma o}}(\xi) \leq e^{2\delta R} e^{-\delta r_\gamma}.$$

Equivariance identifies $\nu_{\gamma o}(\mathcal{O}_R(o,\gamma o))$ with the ν_o-mass of the pulled-back shadow $\gamma^{-1}\mathcal{O}_R(o,\gamma o)$. Hyperbolic comparison shows that, when R is large, the complement of this pulled-back shadow lies in a visual cap of radius $O(e^{-R})$, uniformly in γ. Because ν_o is finite and its support is not a singleton, there is an $\varepsilon > 0$ for which every visual ε-ball has mass at most $\nu_o(\partial X) - c$ for some $c > 0$: otherwise balls with radii tending to zero and masses tending to the total mass would, after taking convergent centers, force the entire measure to be one atom. Choose R so large that the complementary cap has radius below ε. The pulled-back shadow then has ν_o-mass at least c, uniformly in γ; its upper mass is $\nu_o(\partial X)$. Multiplying by the preceding Radon–Nikodym bounds proves the first assertion.

For the annular estimate, a boundary point ξ can lie in $\mathcal{O}_R(o,\gamma o)$ with $T \le r_\gamma < T + A$ only when γo lies in a fixed-radius tube around the length-A portion of the ray $[o,\xi)$ near radial parameter T. The orbit Γo is uniformly discrete, so hyperbolic ball-packing bounds the number of such orbit points by a constant depending only on R, A and o. The annular shadows therefore have uniformly bounded multiplicity. Integrating their indicators and using the lower shadow bound yields the claimed weighted-shell estimate. □

Lemma 5.2 (Second Borel–Cantelli lemma). *Let (Z,μ) be a finite measure space and E_j measurable. Assume $\sum_j \mu(E_j) = \infty$ and that, for all sufficiently large N,*

$$\sum_{i,j\le N} \mu(E_i \cap E_j) \le C\left(\sum_{j\le N} \mu(E_j)\right)^2 .$$

Then $\mu(\limsup_j E_j) \ge C^{-1}$.

Proof. For $m \le N$ set $F_{m,N} = \sum_{j=m}^N \mathbf{1}_{E_j}$. Cauchy–Schwarz gives

$$\left(\int F_{m,N}\,d\mu\right)^2 \le \mu(F_{m,N} > 0)\int F_{m,N}^2\,d\mu.$$

After deleting finitely many initial indices, the assumed correlation bound applies to the tail with the same constant up to an arbitrarily small multiplicative error. Hence $\mu(\bigcup_{j=m}^N E_j) \ge C^{-1} - o(1)$. First let $N \to \infty$ and then $m \to \infty$; continuity from above gives the result. □

Lemma 5.3 (Quasi-independence of orbit shadows). *Fix $R \ge R_0$. There is $C'_R < \infty$ such that for every T,*

$$\sum_{r_\gamma, r_\eta \le T} \nu_o(\mathcal{O}_\gamma \cap \mathcal{O}_\eta) \le C'_R\left(1 + \sum_{r_\gamma\le T} e^{-\delta r_\gamma}\right)^2 .$$

PROOF. Suppose $r_\gamma \le r_\eta$ and the two shadows meet. A common ray from o passes within R of both centers. Comparing the encounter parameters with the distances of the centers from o gives

$$r_\eta \ge r_\gamma + d(\gamma o, \eta o) - 4R.$$

Hence, using the upper shadow estimate,

$$\nu_o(\mathcal{O}_\gamma \cap \mathcal{O}_\eta) \le C_R e^{4\delta R} e^{-\delta r_\gamma} e^{-\delta d(\gamma o, \eta o)}.$$

Write $\lambda = \gamma^{-1}\eta$. The same geometry gives $d(o, \lambda o) \le T + 4R$. Summing first over γ and then over λ therefore bounds the ordered pairs by a constant times

$$\left(\sum_{r_\gamma \le T} e^{-\delta r_\gamma}\right)\left(\sum_{r_\lambda \le T+4R} e^{-\delta r_\lambda}\right).$$

The shell estimate in Lemma 5.1 shows that the second factor exceeds the first by at most a fixed additive constant. Doubling to account for the opposite ordering and adding the diagonal proves the assertion. □

Lemma 5.4 (Differentiation along radial-return shadows). *Fix $R > 0$. Let $A \subset \partial\mathbb{H}^n$ be measurable. For ν_o-almost every radial point ξ which is a density point of A or of its complement, one may choose a radial-return sequence γ_j with $r_{\gamma_j} \to \infty$ and $\xi \in \mathcal{O}_R(o, \gamma_j o)$ such that*

$$\frac{\nu_o(A \cap \mathcal{O}_R(o, \gamma_j o))}{\nu_o(\mathcal{O}_R(o, \gamma_j o))} \longrightarrow \mathbf{1}_A(\xi).$$

More generally, after replacing R by any fixed larger radius $R' > R$, the same conclusion holds for every radial-return sequence whose geodesic ray passes within a fixed distance $R_1 < R$ of $\gamma_j o$.

PROOF. Put the visual round metric on $\partial\mathbb{H}^n$ based at o. If $d(\gamma_j o, [o, \xi)) \le R_1$ and $r_{\gamma_j} \to \infty$, elementary hyperbolic trigonometry gives constants $a, b > 0$, depending only on $R_1 < R < R'$, for which

$$B_\partial(\xi, a e^{-r_{\gamma_j}}) \subset \mathcal{O}_R(o, \gamma_j o) \subset B_\partial(\xi, b e^{-r_{\gamma_j}}) \subset \mathcal{O}_{R'}(o, \gamma_j o).$$

The first two inclusions say geometrically that the direction from o to $\gamma_j o$ differs from ξ by $O(e^{-r_{\gamma_j}})$, whereas the angular radius of a fixed hyperbolic shadow is comparable to $e^{-r_{\gamma_j}}$. The last inclusion follows after increasing R' by a constant depending only on b.

The ordinary Besicovitch differentiation theorem for the finite Borel measure ν_o applies to the *centered* balls $B_\partial(\xi, b e^{-r_{\gamma_j}})$. If, for example, ξ is a density point of A, then

$$\nu_o(B_\partial(\xi, b e^{-r_{\gamma_j}}) \setminus A) = o\Big(\nu_o(B_\partial(\xi, b e^{-r_{\gamma_j}}))\Big).$$

By the two shadow estimates at radii R and R', the mass of the centered ball is at most $Ce^{-\delta r_{\gamma_j}}$, while $\nu_o(\mathcal{O}_R(o,\gamma_j o)) \geq ce^{-\delta r_{\gamma_j}}$. Consequently

$$\frac{\nu_o(\mathcal{O}_R(o,\gamma_j o) \setminus A)}{\nu_o(\mathcal{O}_R(o,\gamma_j o))} \longrightarrow 0.$$

The complement case is identical. A radial point admits such a sequence by Theorem 2.3, after taking $R > R_1$ once and for all. ☐

2. Conservativity and the Hopf ratio mechanism

Lemma 5.1 (Finite invariant measure implies conservativity). *Let (Y, μ) be a finite measure space and $T : Y \to Y$ an invertible measure-preserving transformation. Then T is conservative: if A has positive measure, almost every point of A returns to A infinitely many times.*

Proof. Let $W \subset A$ be the set of points that never return to A after time 0. The sets $T^k W$, $k \geq 0$, are pairwise disjoint. Since each has measure $\mu(W)$ and the ambient measure is finite, $\mu(W) = 0$. Repeating the argument after the last return shows that the set of points of A with only finitely many returns is null. ☐

Lemma 5.2 (Radial endpoints identify the Hopf decomposition). *For the BMS measure, full Patterson–Sullivan measure of the radial limit set implies conservativity of the time-one geodesic map; zero Patterson–Sullivan measure of the radial limit set implies complete dissipativity of that map. We use these equivalent discrete-time formulations when speaking below of the conservative/dissipative alternative for the flow.*

Proof. Let $T = g^1$. Choose compact sets $K_j \nearrow T^1M$. By Theorem 2.3, a forward endpoint is radial exactly when the corresponding projected ray returns infinitely often to a bounded part of M. If $g^{t_i}v$ returns to a fixed compact set and the times t_i tend to infinity, then after passing to a separated subsequence and replacing t_i by the nearest integers, $T^{n_i}v$ lies infinitely often in the compact 1-neighborhood of that set. Hence a vector with radial forward endpoint visits some K_j infinitely often under T.

On the dissipative part of an invertible σ-finite transformation, every finite-measure set is visited only finitely often by almost every orbit: a dissipative part is, modulo null sets, the disjoint union of the iterates of a wandering set, and the assertion follows by summing the indicator of the finite-measure set over those iterates. Since every compact set has finite BMS mass, full radial measure therefore leaves no positive measure dissipative part. Thus T is conservative.

Assume instead that radial measure is zero. For BMS-almost every vector both endpoints are nonradial. Put

$$q(v) = d_M(\pi(v), [o]),$$

where $[o]$ is the image of the base point in the complete hyperbolic manifold M. The function q is proper. Nonradiality of the two endpoints implies

$$q(T^n v) \longrightarrow \infty \qquad (n \to \pm\infty),$$

for otherwise a subsequence in one time direction would return to a compact set and make the corresponding endpoint radial. Thus the sequence $q(T^n v)$ attains its minimum at finitely many integers. Let W be the measurable set of vectors for which $n = 0$ is the smallest integer attaining that minimum. Every nonradial T-orbit meets W exactly once, so the sets $T^n W$, $n \in \mathbb{Z}$, are pairwise disjoint and cover the nonradial locus. Hence the time-one map is completely dissipative there. ☐

Lemma 5.3 (Radial points carry no atoms)**.** *A Patterson–Sullivan density of positive dimension has no atom at a radial limit point. In particular, if* $\nu_o(\Lambda_{\mathrm{rad}}) = 1$*, then* ν_o *is nonatomic and the diagonal has* $(\nu_o \otimes \nu_o)$*-measure zero.*

Proof. Suppose $\nu_o\{\xi\} = a > 0$ and ξ is radial. Choose $\gamma_j o$ within a fixed distance R of $[o, \xi)$ with $r_j = d(o, \gamma_j o) \to \infty$. Then $\beta_\xi(o, \gamma_j o) = r_j + O(R)$. Equivariance and conformality give

$$\nu_o\{\gamma_j^{-1}\xi\} = \nu_{\gamma_j o}\{\xi\} = e^{-\delta\beta_\xi(\gamma_j o, o)}\nu_o\{\xi\} = e^{\delta r_j + O(1)}a,$$

which eventually exceeds the finite total mass of ν_o. This contradiction proves the first assertion. The second follows because full radial measure leaves no mass on nonradial atoms. ☐

Lemma 5.4 (Hopf maximal lemma)**.** *Let* T *be an invertible measure-preserving transformation of a* σ*-finite space and* $a \in L^1$*. For*

$$E_N = \left\{ x : \max_{1 \le n \le N} \sum_{j=0}^{n-1} a(T^j x) > 0 \right\}$$

one has $\int_{E_N} a \, d\mu \ge 0$*. Consequently the same inequality holds for* $E = \bigcup_N E_N$*.*

Proof. Let

$$M_N(x) = \max\left(0, \sum_{j=0}^{0} a(T^j x), \ldots, \sum_{j=0}^{N-1} a(T^j x)\right).$$

If $x \in E_N$, then separating the first summand from every partial sum gives

$$M_N(x) \le a(x) + M_N(Tx),$$

so $a(x) \ge M_N(x) - M_N(Tx)$. If $x \notin E_N$, then $M_N(x) = 0$ and $0 \ge -M_N(Tx)$, so the same inequality holds with $a(x)$ replaced by $\mathbf{1}_{E_N}(x)a(x)$. Therefore

$$\mathbf{1}_{E_N} a \ge M_N - M_N \circ T.$$

Integrating and using measure preservation gives $\int_{E_N} a\, d\mu \ge 0$. Since $E_N \uparrow E$ and $a \in L^1$, dominated convergence proves the limiting assertion. $\square$

Lemma 5.5 (Maximal-ratio L^1 stability)**.** *Let T preserve a σ-finite measure m, let $Q > 0$ be integrable, and assume $S_nQ(x) \to \infty$ almost everywhere. For $f \in L^1(m)$ and $\varepsilon > 0$,*

$$(Qm)\left\{x : \sup_{n\ge 1} \frac{|S_nf(x)|}{S_nQ(x)} > \varepsilon\right\} \le \frac{2}{\varepsilon}\, \|f\|_{L^1(m)}, \qquad d(Qm) = Q\, dm.$$

Consequently, if $f_j \to f$ in $L^1(m)$ and every ratio S_nf_j/S_nQ converges almost everywhere to a constant, then S_nf/S_nQ converges almost everywhere to a constant as well.

Proof. Let

$$E^+ = \left\{\sup_n \frac{S_nf}{S_nQ} > \varepsilon\right\}, \qquad E^- = \left\{\sup_n \frac{-S_nf}{S_nQ} > \varepsilon\right\}.$$

The set E^+ is exactly the maximal set from Lemma 5.4 for $a = f - \varepsilon Q$; therefore

$$0 \le \int_{E^+} (f - \varepsilon Q)\, dm \le \int_{E^+} |f|\, dm - \varepsilon \int_{E^+} Q\, dm.$$

The same argument with $-f - \varepsilon Q$ gives $\varepsilon \int_{E^-} Q\, dm \le \int_{E^-} |f|\, dm$. Their union contains the displayed maximal set, which proves the estimate.

For the consequence, pass to a subsequence, still denoted f_j, with $\|f_j - f\|_1 \le 2^{-3j}$. Apply the maximal estimate to $f_j - f$ with $\varepsilon_j = 2^{-j}$. The exceptional Qm-measures are summable, so Borel–Cantelli gives, for almost every x, an index $j_0(x)$ such that for all $j \ge j_0(x)$,

$$\sup_n \frac{|S_n(f_j - f)(x)|}{S_nQ(x)} \le 2^{-j}.$$

If c_j is the constant limit for f_j, then at such a point both the limsup and the liminf of S_nf/S_nQ lie within 2^{-j} of c_j. The same inequality for two indices shows that (c_j) is Cauchy. Let $c = \lim c_j$; letting $j \to \infty$ forces the limsup and liminf for f to equal c almost everywhere. $\square$

Lemma 5.6 (Hopf ratio step in BMS coordinates)**.** *Assume $\nu_o(\Lambda_{\mathrm{rad}}) = 1$. Let h be compactly supported and Lipschitz on T^1M. There is a positive integrable function ρ such that the forward and backward ratios*

$$R_h^{\pm}(v) = \lim_{T\to\infty} \frac{\int_0^T h(g^{\pm t}v)\,dt}{\int_0^T \rho(g^{\pm t}v)\,dt}$$

exist almost everywhere, agree almost everywhere, and their common value is almost everywhere constant.

Proof. We first prove the ratio theorem in the precise form needed here. Lift h to a Γ-invariant function $\widetilde{h}$ and temporarily let $\widetilde{\rho} > 0$ be any bounded integrable weight whose forward and backward orbit integrals diverge. Put

$$F(v) = \int_0^1 \widetilde{h}(g^t v)\,dt, \qquad Q(v) = \int_0^1 \widetilde{\rho}(g^t v)\,dt, \qquad T = g^1.$$

Then $F, Q \in L^1$, $Q > 0$, and $S_nQ \to \infty$. Set $\overline{R} = \limsup S_nF/S_nQ$ and $\underline{R} = \liminf S_nF/S_nQ$; deleting the first summand changes either ratio by $o(1)$, so both functions are T-invariant.

If $\underline{R} < q - \varepsilon < q + \varepsilon < \overline{R}$ on an invariant set B of positive measure, with $q \in \mathbb{Q}$, then Lemma 5.4 applied on B to $F - (q+\varepsilon)Q$ gives

$$\int_B (F - qQ)\,dm \ge \varepsilon \int_B Q\,dm,$$

whereas applying it to $(q-\varepsilon)Q - F$ gives

$$\int_B (F - qQ)\,dm \le -\varepsilon \int_B Q\,dm,$$

a contradiction. Taking a countable union over rational q, ε proves forward convergence. The same proof for T^{-1} gives a backward limit. If the two limits differed on a positive invariant set, choose a rational number with a uniform positive gap between them and apply the forward and backward maximal inequalities there; the same contradictory pair of integral inequalities results. Hence the two directional limits agree. Since h and the weight are bounded over each unit interval and $S_nQ \to \infty$, passing from integer times to arbitrary real T changes the ratios only by $o(1)$.

For the BMS flow choose

$$\widetilde{\rho}(v) = \exp\bigl(-4\delta\, d(\pi(v), \Gamma o)\bigr).$$

This is positive, bounded by 1, and Γ-invariant. On a Dirichlet domain centered at o, vectors based in $B(o, r+1)$ have BMS mass $O((r+1)e^{2\delta(r+1)})$: their Hopf-time interval has length $O(r+1)$ and each Busemann factor is bounded by $r+1$. Multiplication by $e^{-4\delta r}$ makes the shell masses summable, so $\rho \in L^1(m^{\mathrm{BMS}})$. Radial recurrence makes both orbit integrals of ρ infinite:

at every return to a fixed compact set the weight is bounded below, and unit speed keeps the orbit in a slightly larger compact set for a uniform positive time; take a separated subsequence of returns.

It remains to prove constancy. If w lies on the strong stable leaf of v, then $d(g^t v, g^t w) \le Ce^{-t}$. Using a global Lipschitz constant L_h for the lifted function gives

$$|\widetilde{h}(g^t v) - \widetilde{h}(g^t w)| \le L_h C e^{-t}$$

whenever one term is nonzero. Thus the total numerator difference is finite. Since distance to Γo is 1-Lipschitz, $|\widetilde{\rho}(g^t v) - \widetilde{\rho}(g^t w)| \le C' e^{-t} \widetilde{\rho}(g^t v)$ for large t, so the accumulated relative denominator error is finite. Dividing by the divergent denominator shows that the forward ratio is constant on almost every strong stable leaf; the backward ratio is similarly constant on almost every strong unstable leaf.

The ratios are also flow invariant, since deleting a bounded initial segment does not change a quotient whose denominator diverges. Two vectors with the same forward endpoint become strongly stable after a unique flow shift aligning their Busemann level; hence the forward ratio depends only on v^+. Similarly the backward ratio depends only on v^-. Thus there are measurable boundary functions F_+ and F_- with $R_h^+(v) = F_+(v^+)$ and $R_h^-(v) = F_-(v^-)$. Equality of the two ratios gives $F_+(\xi) = F_-(\eta)$ for $(\nu_o \otimes \nu_o)$-almost every distinct pair. Lemma 5.3 makes the diagonal null, so Fubini's theorem forces both boundary functions to equal one constant almost everywhere. □

3. The dichotomy

THEOREM 5.1 (HTS dichotomy). *Let $\{\nu_x\}$ be the Patterson–Sullivan density of dimension $\delta = \delta_\Gamma$ and let m^{BMS} be the associated BMS measure. Exactly one of the following alternatives holds.*

(1) If

$$\sum_{\gamma \in \Gamma} e^{-\delta d(o, \gamma o)} < \infty,$$

then $\nu_o(\Lambda_{\mathrm{rad}}) = 0$ and the BMS geodesic flow is completely dissipative.

(2) If

$$\sum_{\gamma \in \Gamma} e^{-\delta d(o, \gamma o)} = \infty,$$

then $\nu_o(\Lambda_{\mathrm{rad}}) = 1$ and the BMS geodesic flow is completely conservative and ergodic.

PROOF. Fix a shadow radius R for the preceding shadow lemmas.

Convergence. Lemma 5.1 makes $\sum_\gamma \nu_o(\mathcal{O}_\gamma)$ finite. The first Borel–Cantelli lemma says that almost no boundary point lies in infinitely many $\mathcal{O}_\gamma$. A radial point lies in infinitely many shadows for some integer radius, so a countable union over radii gives $\nu_o(\Lambda_{\mathrm{rad}}) = 0$. Lemma 5.2 gives complete dissipativity.

Divergence gives positive radial mass. Now $\sum_\gamma \nu_o(\mathcal{O}_\gamma) = \infty$. Lemma 5.3 supplies the second-moment bound in Lemma 5.2, so a positive-measure set of boundary points belongs to infinitely many fixed-radius shadows. Every such point is radial. Hence $\nu_o(\Lambda_{\mathrm{rad}}) > 0$.

The radial zero–one step. Let $A \subset \Lambda_{\mathrm{rad}}$ be a Γ-invariant measurable set of positive measure, and write $B = \partial X \setminus A$. Choose a radial density point $\xi \in A$ and a sequence $\gamma_j o$ staying within some fixed distance R_1 of $[o, \xi)$, with $r_{\gamma_j} \to \infty$. After a subsequence, $\gamma_j^{-1} o \to \zeta \in \partial X$.

Fix now an arbitrary radius $R > \max(R_0, R_1 + 1)$. Lemma 5.4 gives

$$\frac{\nu_o(B \cap \mathcal{O}_R(o, \gamma_j o))}{\nu_o(\mathcal{O}_R(o, \gamma_j o))} \longrightarrow 0.$$

On this fixed-radius shadow, conformality makes the ratio comparable, with constants independent of j, to the analogous ratio for $\nu_{\gamma_j o}$. Equivariance and invariance of B therefore give

$$\frac{\nu_o(B \cap \mathcal{O}_R(\gamma_j^{-1} o, o))}{\nu_o(\mathcal{O}_R(\gamma_j^{-1} o, o))} \longrightarrow 0.$$

For fixed R, every compact subset of

$$S_R(\zeta) = \{\eta \neq \zeta : (\zeta\eta) \cap B(o, R) \neq \varnothing\}$$

is contained in $\mathcal{O}_R(\gamma_j^{-1} o, o)$ for all sufficiently large j; this follows from convergence of geodesics with distinct limiting endpoints. Since the numerator above tends to zero, regularity of ν_o gives $\nu_o(B \cap S_R(\zeta)) = 0$. Finally

$$\bigcup_{R>0} S_R(\zeta) = \partial X \setminus \{\zeta\},$$

so $\nu_o(B \setminus \{\zeta\}) = 0$. Choose $\gamma \in \Gamma$ with $\gamma\zeta \neq \zeta$. Equivariance makes ν_o and $\gamma_* \nu_o$ equivalent, while B is invariant; translating the support conclusion shows that B is also supported on $\{\gamma\zeta\}$. The two singleton supports are disjoint, hence $\nu_o(B) = 0$. Taking $A = \Lambda_{\mathrm{rad}}$ proves full radial measure.

Lemma 5.2 gives complete conservativity. Apply Lemma 5.6 to a countable L^1-dense family of compactly supported Lipschitz functions. Every corresponding ratio limit is almost everywhere constant. Let E be invariant and put $f = \mathbf{1}_E \rho$. Choose compactly supported Lipschitz $f_j \to f$ in $L^1(m^{\mathrm{BMS}})$. Lemma 5.5, applied to the time-one block functions, shows that the ratio for f is the limit in $\rho\, dm^{\mathrm{BMS}}$-measure of the constant ratios for f_j and is therefore constant. But invariance of E gives, for every T,

$$\frac{\int_0^T \mathbf{1}_E(g^t v)\rho(g^t v)\, dt}{\int_0^T \rho(g^t v)\, dt} = \mathbf{1}_E(v).$$

Thus $\mathbf{1}_E$ is almost everywhere constant. The invariant σ-algebra is trivial and the flow is ergodic. □

Proof and provenance. This is the constant-curvature specialization of the Hopf–Tsuji–Sullivan theorem in the Sullivan–Roblin tradition; compare [Sul79; Rob03]. The proof retains the source architecture rather than replacing it by a citation: shadow/Borel–Cantelli for convergence, a second-moment argument for positive radial mass, the shadow differentiation zero–one step, the Hopf decomposition, and the Hopf ratio argument.

4. Cusp mass and finiteness for geometrically finite groups

THEOREM 5.1 (Euclidean cusp crystallography). *Let $P = \Gamma_\xi$ be a rank-k parabolic stabilizer occurring in a geometrically finite real-hyperbolic cusp. After conjugating ξ to ∞, the action of P on a suitable k-dimensional invariant Euclidean flat in a horosphere is crystallographic. In particular, P contains a finite-index translation subgroup $L \cong \mathbb{Z}^k$.*

PROOF. This is the first Bieberbach theorem [Bie11] applied to the compact Euclidean cross-section of the cusp. To make the logical input explicit: properness of the P-action follows from discreteness of Γ, and compactness of the cross-section is exactly the bounded-cusp part of the geometrical-finiteness hypothesis. The first Bieberbach theorem then says that the translation subgroup is a full lattice in the minimal invariant Euclidean flat and has finite index. No later mixing, counting, or Patterson–Sullivan theorem is used here. □

Proof and provenance. This is the finite-dimensional Euclidean crystallographic input of Bieberbach [Bie11], isolated here because it is the only non-hyperbolic structural fact used in the cusp-series calculation. It may alternatively be taken as part of the standard classification of geometrically finite real-hyperbolic cusps.

Lemma 5.2 (Parabolic exponent and strict gap)**.** *Let $P = \Gamma_\xi$ be a rank-k cusp stabilizer of a non-elementary geometrically finite subgroup of* $\mathrm{Isom}^+(\mathbb{H}^n)$*. Then*

$$\delta_P = \frac{k}{2} \qquad \textit{and} \qquad \delta_\Gamma > \delta_P.$$

Proof. Conjugate ξ to ∞ and take $o = (0, 1)$. By Theorem 5.1, a finite-index translation subgroup acts by $u \mapsto u + v$ on a k-dimensional affine subspace, with v in a Euclidean lattice $L \cong \mathbb{Z}^k$. The upper-half-space distance formula gives

$$\cosh d(o,(v,1)) = 1 + \frac{\|v\|^2}{2}, \qquad d(o,(v,1)) = 2\log\|v\| + O(1).$$

Thus the Poincaré series of P has the same convergence threshold as

$$\sum_{v\in\mathbb{Z}^k\setminus\{0\}} \|v\|^{-2s},$$

namely $k/2$.

For strictness choose $g \in \Gamma$ with $g\xi \neq \xi$. Take a small boundary neighborhood U of ξ with $g\overline{U} \cap \overline{U} = \varnothing$. In the Euclidean boundary chart, $g\overline{U}$ is compact. Passing to a sufficiently sparse finite-index translation sublattice and deleting a finite set, the translates $p(g\overline{U})$ are pairwise disjoint and all lie in U. Hence the maps pg form a ping–pong family on U, so words $p_1g\cdots p_mg$ are distinct. Triangle inequality gives

$$d(o, p_1g\cdots p_mg\, o) \le \sum_{j=1}^{m} d(o, p_jo) + md(o, go).$$

The Poincaré series of the chosen finite-index sublattice still diverges as $s \downarrow k/2$. Choose $s > k/2$ with

$$e^{-sd(o,go)} \sum_{p} e^{-sd(o,po)} > 1.$$

Summing over the distinct ping–pong words produces a divergent geometric subseries of the Poincaré series of Γ at s. Therefore $\delta_\Gamma \ge s > k/2$. □

Lemma 5.3 (Cusp-series estimate). *Let $\mathcal{C} = P\backslash(\mathcal{H} \cap \mathrm{CH}(\Lambda))$ be one cusp and choose o in a fixed compact collar of its boundary. Then*

$$m^{\mathrm{BMS}}(T^1\mathcal{C}) \le C \sum_{p\in P} (1 + d(o,po))e^{-\delta d(o,po)}.$$

Proof. Conjugate the cusp point to ∞. The truncated-core definition of geometric finiteness says that a horospherical cross-section of $\mathcal{H} \cap \mathrm{CH}(\Lambda)$ has compact quotient by P; choose a compact fundamental set F in that cross-section. Up to the finite rotational part of P, every geodesic meeting the cusp with base projection in F is classified by the parabolic translate pF through which its opposite side exits.

Fix such a class. In Hopf coordinates its endpoint pair lies in a product of two boundary shadows determined by F and pF. The conformal-density formula and Lemma 5.1 give transverse BMS mass $O(e^{-\delta d(o,po)})$. The time for which a geodesic in this class stays above the fixed cusp boundary is $O(1 + d(o, po))$. Indeed, in the vertical two-plane through its endpoints the geodesic is a Euclidean semicircle of diameter comparable to the Euclidean translation length of p; its maximal height is comparable to that length, so the hyperbolic time above a fixed height is $2 \log \|p\| + O(1) = O(1 + d(o, po))$.

The local product formula for BMS multiplies transverse endpoint mass by Hopf time. Distinct p-classes have uniformly bounded overlap because F is a compact fundamental set and there are only finitely many rotational cosets. Summing the resulting estimates gives the stated series bound. □

Proposition 5.4 (Geometrically finite BMS finiteness). *If $\Gamma < \mathrm{Isom}^+(\mathbb{H}^n)$ is non-elementary and geometrically finite in the sense of Definition 4.1, then*

$$0 < m^{\mathrm{BMS}}(T^1 M) < \infty.$$

Consequently Γ is of divergent type and the BMS geodesic flow is conservative and ergodic.

Proof. The compact thick part of the convex core has finite BMS mass because the measure is Radon. There are finitely many cusp orbits. For a rank-k cusp, Lemmas 5.3 and 5.2 reduce its mass to a constant multiple of

$$\sum_{v \in \mathbb{Z}^k} (1 + \log(2 + \|v\|))(1 + \|v\|)^{-2\delta}.$$

The number of lattice vectors with $N \leq \|v\| < N + 1$ is $O(N^{k-1})$, so this is bounded by

$$C \sum_{N \geq 1} (1 + \log N) N^{k-1-2\delta},$$

which converges because $\delta > k/2$. Hence all cusps and therefore the full BMS measure have finite mass.

Lemma 5.1 makes the flow conservative. The convergence branch of Theorem 5.1 is completely dissipative, so it cannot occur. Thus the group is of divergent type, and the divergent branch gives ergodicity. □

Proof and provenance. The cusp-series criterion is the real-hyperbolic specialization of the geometrically finite finiteness analysis of Dal'bo–Otal–Peigné [DOP00]. Their general criterion is expressed through parabolic series; in constant curvature the strict parabolic gap turns it into the explicit convergent lattice series used above.

Worked Example 5.5 (The modular cusp). For the standard rank-one cusp of a finite-area hyperbolic surface, $P = \langle z \mapsto z + 1 \rangle$ and $d(i, i + n) =$

$2\log|n| + O(1)$. Since $\delta_\Gamma = 1$, the cusp estimate becomes

$$\sum_{n\neq 0}(1+\log|n|)|n|^{-2} < \infty.$$

The logarithm is the time spent above a fixed cusp height; the power $|n|^{-2}$ is the Patterson–Sullivan endpoint decay. This is the simplest complete model for the general cusp summation.

Failure mode. Two logically independent mechanisms occur in this chapter. Hopf–Tsuji–Sullivan turns divergence into conservativity and ergodicity; the cusp-series estimate proves finiteness of BMS for geometrically finite constant-curvature cusps. Neither is a formal consequence of compactness of the truncated core, and neither is replaced by the other.

CHAPTER 6

Ergodicity and the Hopf argument

Why this matters. Chapter 5 proved the full dichotomy and, inside its proof, the Hopf maximal and ratio lemmas. We now package the stable/unstable part as a reusable theorem. Later chapters can therefore cite a precise mechanism instead of the phrase "by Hopf's argument."

1. Stable and unstable ratio averages

Lemma 6.1 (Hopf stable/unstable averaging step). *Assume $\nu_o(\Lambda_{\mathrm{rad}}) = 1$. Let h be compactly supported and Lipschitz on T^1M, and let $\rho > 0$ be the integrable weight from Lemma 5.6. Then*

$$\frac{\int_0^T h(g^t v)\,dt}{\int_0^T \rho(g^t v)\,dt}, \qquad \frac{\int_0^T h(g^{-t} v)\,dt}{\int_0^T \rho(g^{-t} v)\,dt}$$

converge for almost every v, the two limits agree, the forward limit is constant almost everywhere on strong stable leaves, and the backward limit is constant almost everywhere on strong unstable leaves.

Proof. The existence and equality of the two ratios were proved from the Hopf maximal lemma inside Lemma 5.6; no additional ergodic theorem is being imported here. We repeat the geometric estimate because it is the reusable part. If $w \in W^{ss}(v)$, then $d(g^t v, g^t w) \le Ce^{-t}$. Let L be a global Lipschitz constant for the lifted Γ-invariant function. Whenever one of the two values is nonzero,

$$|h(g^t v) - h(g^t w)| \le LCe^{-t};$$

when both vanish the difference is zero. Hence the total difference between the numerator integrals is bounded uniformly in T. The weight satisfies

$$|\rho(g^t v) - \rho(g^t w)| \le C' e^{-t} \rho(g^t v)$$

for large t, because distance to Γo is 1-Lipschitz. Its accumulated relative error is therefore finite. The denominator diverges by radial recurrence, so both errors become negligible after division. This proves stable-leaf constancy. Replacing t by $-t$ gives unstable-leaf constancy. $\square$

Theorem 6.2 (BMS ergodicity in the divergent case). *If the critical Poincaré series diverges, then the geodesic flow on* (T^1M, m^{BMS}) *is completely conservative and ergodic.*

Proof. Theorem 5.1 gives complete conservativity and full Patterson–Sullivan measure to the radial limit set. By Lemma 6.1, for every compactly supported Lipschitz h the common Hopf ratio is stable-leaf constant forward and unstable-leaf constant backward. In Hopf coordinates this means that there are measurable boundary functions F_h and G_h with

$$R_h(v) = F_h(v^+) = G_h(v^-)$$

for m^{BMS}-almost every v. Since the BMS measure is mutually absolutely continuous with $\nu_o \otimes \nu_o \otimes ds$ on compact Hopf boxes and Lemma 5.3 makes the diagonal null, Fubini gives

$$F_h(\xi^+) = G_h(\xi^-)$$

for $\nu_o \otimes \nu_o$-almost every distinct pair. Fixing one generic backward endpoint and varying the forward endpoint shows that F_h is almost everywhere constant; then G_h has the same constant value.

Choose a countable L^1-dense family of compactly supported Lipschitz functions. The corresponding ratio limits are all constant on one common conull set. If E is flow invariant, approximate $\mathbf{1}_E\rho$ in L^1 by this family. Lemma 5.5 passes constancy to the ratio for $\mathbf{1}_E\rho$. Since E is invariant, that ratio is identically $\mathbf{1}_E(v)$ wherever its denominator diverges. Thus $\mathbf{1}_E$ is constant almost everywhere, and every invariant set is null or conull. ☐

Proposition 6.3 (Ergodic normalization when BMS is finite)**.** *If* $0 < m^{\mathrm{BMS}}(T^1M) < \infty$*, then*

$$\overline{m}^{\mathrm{BMS}} = \frac{m^{\mathrm{BMS}}}{m^{\mathrm{BMS}}(T^1M)}$$

is an invariant ergodic probability measure.

Proof. Finiteness makes the normalization meaningful and preserves invariance. Lemma 5.1 gives conservativity. Theorem 5.1 therefore rules out the convergent/dissipative branch, so the critical Poincaré series diverges. Theorem 6.2 gives ergodicity. Rescaling by a positive constant does not change invariant null sets. ☐

Worked Example 6.4 (Why contraction needs recurrence)**.** Vectors on one strong stable leaf approach at rate $O(e^{-t})$, so a Lipschitz observable accumulates only a finite total discrepancy along their forward trajectories. This alone does not force equal averages: an escaping orbit may contribute only finite total weight. The radial/full-conservative conclusion of Hopf–Tsuji–Sullivan makes the positive denominator diverge, and only then does the bounded discrepancy disappear in the ratio.

CHAPTER 7

Local product structure

Why this matters. The Hopf product formula is not merely a convenient coordinate expression: it says that, locally, the BMS measure is a product of one measure in each endpoint direction and Lebesgue measure in flow time. This is the geometric mechanism later used in Hopf arguments and mixing.

1. Flow boxes

Lemma 7.1 (Compact Hopf boxes have bounded density)**.** *Let $A^-, A^+ \subset \partial X$ be compact with $A^- \cap A^+ = \varnothing$, and let $I \subset \mathbb{R}$ be compact. On the Hopf box $A^- \times A^+ \times I$, the BMS exponential weight is bounded above and below by positive constants.*

Proof. The inverse Hopf map sends the compact box to a compact subset of T^1X. The functions $v \mapsto \beta_{v^\pm}(o, \pi(v))$ are continuous there, hence bounded. Exponentiating their sum gives a positive continuous function with positive minimum and finite maximum. ☐

Theorem 7.2 (Local product structure of BMS). *On every compact Hopf box away from the endpoint diagonal, m^{BMS} is mutually absolutely continuous with*

$$\nu_o|_{A^-} \otimes \nu_o|_{A^+} \otimes ds,$$

with Radon–Nikodym derivative bounded above and below by positive constants.

Proof. This is exactly the BMS formula of Chapter 3 together with Lemma 7.1. Quotienting by Γ on a sufficiently small flow box is injective, so the same statement holds locally on T^1M. ☐

Proposition 7.3 (Product-null criterion)**.** *Inside a compact Hopf box, a measurable set has zero BMS measure if and only if it has zero measure for $\nu_o \otimes \nu_o \otimes ds$.*

Proof. The two measures have densities bounded above and below by positive constants by Theorem 7.2; therefore they have identical null sets. ☐

Worked Example 7.4 (A rectangular Hopf box)**.** Choose two disjoint boundary arcs A^-, A^+ on $S^1 = \partial\mathbb{H}^2$ and $I = [-1, 1]$. Every pair $(\xi^-, \xi^+) \in A^- \times A^+$ determines a geodesic crossing a fixed compact region, so the Busemann factors are uniformly bounded. In this box, questions about BMS-null sets reduce literally to product-measure questions.

CHAPTER 8

Comparison with Haar and finite-volume models

Why this matters. The finite-volume theory is the consistency check for all normalizations. When both endpoint densities are the ordinary visual density of dimension $n-1$, the Hopf product must recover Liouville measure, equivalently Haar measure on the frame bundle.

1. The symmetric visual product

Lemma 8.1 (The visual Hopf product is G-invariant)**.** *Define on T^1X*

$$d\widetilde{m}^{\mathrm{vis}}(v) = e^{(n-1)\beta_{v^+}(o,x)+(n-1)\beta_{v^-}(o,x)}\, dm_o(v^+)\, dm_o(v^-)\, ds.$$

Then $\widetilde{m}^{\mathrm{vis}}$ is independent of o and invariant under the full group G and the geodesic flow.

Proof. The proof of Lemma 3.1 uses only conformality and therefore applies to the visual density. Since $\{m_x\}$ is G-equivariant rather than merely Γ-equivariant, the argument proving Γ-invariance in Theorem 3.2 gives full G-invariance. The two flow exponents $+(n-1)t$ and $-(n-1)t$ cancel. ☐

Theorem 8.2 (Liouville/Haar identification). *Up to one global scalar, $\widetilde{m}^{\mathrm{vis}}$ is Liouville measure on $T^1\mathbb{H}^n$. Equivalently, after passing to the oriented frame bundle G, it is Haar measure modulo the compact stabilizer of a unit tangent vector. If Γ is a lattice, its quotient has finite total mass.*

Proof. The group G acts transitively on T^1X and the stabilizer of one unit vector is compact. Hence there is, up to scalar, a unique G-invariant Radon measure on $T^1X \cong G/M$. Lemma 8.1 shows that $\widetilde{m}^{\mathrm{vis}}$ is such a measure, so it is the quotient of Haar measure; the Riemannian Liouville measure is another such normalization and therefore differs only by a scalar. By definition of a lattice, Haar measure on $\Gamma\backslash G$ is finite, and quotienting by compact M preserves finiteness. ☐

Proposition 8.3 (Consistency of BMS and BR normalizations)**.** *Whenever $\delta = n-1$ and the chosen Patterson–Sullivan density is the visual density, the BMS and BR measures both coincide, up to the same endpoint normalization, with Liouville/Haar measure.*

Proof. Proposition 4.3 identifies BMS and BR in this specialization. Their common Hopf formula is exactly $\widetilde{m}^{\mathrm{vis}}$, which Theorem 8.2 identifies with Liouville/Haar measure. $\square$

Worked Example 8.4 (A compact hyperbolic surface)**.** For $M = \Gamma\backslash\mathbb{H}^2$ compact, the visual exponent is 1. Choosing the visual density at both endpoints gives the usual Liouville probability after division by its finite total mass. Thus the BMS formalism genuinely extends, rather than replaces, the classical finite-volume geodesic-flow measure.

Volume 55

Mixing in Infinite Volume

Volume 55 scope and prerequisites

Prerequisites. Throughout this volume $X = \mathbb{H}^n$, $G = \mathrm{Isom}^+(X) \cong \mathrm{SO}^+(n,1)$, and $\Gamma < G$ is torsion-free, non-elementary, geometrically finite, and Zariski dense. We write $M = \Gamma\backslash X$, g^t for the geodesic flow, δ for the critical exponent, and use the Patterson–Sullivan, BMS, and Burger–Roblin normalizations fixed in Volumes 53–54. By Volume 54, m^{BMS} is finite and ergodic. For the qualitative mixing statements we retain the explicit hypothesis that the length spectrum is nonarithmetic; this avoids silently importing a separate topological nonarithmetic theorem.

Proof and provenance. The qualitative BMS-mixing proof is reconstructed along Babillot's spectral/stable–unstable argument and Roblin's rank-one implementation [Bab02; Rob03]. The global Haar/BR–BR_* matrix-coefficient asymptotic attributed to Roblin is now internally reconstructed through the transverse-intersection mechanism: the local four-measure product square, PS-plaque thickening, conversion from Patterson–Sullivan to leaf-Lebesgue conditionals, and final transverse integration. The proof follows Roblin's qualitative architecture and the explicit later realization of the transfer in Oh–Winter [Rob03; OW16]. The skinning measure, qualitative expanding-submanifold equidistribution, and qualitative counting formulas are reconstructed in the scope of Oh–Shah [OS13]. The effective matrix-coefficient theorem in Chapter 7 is now internalized at the operational spectral-decomposition scope actually consumed by the Mohammadi–Oh proof; Appendix A owns the complementary-series K-type asymptotic, remainder-spectrum estimate, compact Sobolev summation, and Roblin identification of the leading term [MO15]. The statements below are always interpreted at their stated scope; stronger source theorems are not inferred from short formal reductions.

CHAPTER 1

Mixing of the geodesic flow

Why this matters. Ergodicity says that a typical orbit sees the whole BMS space. Mixing says more: two compact pieces become asymptotically independent after a long geodesic translation. The obstruction is arithmeticity of return times. The proof below makes that obstruction visible through spectral weak limits and the boundary cross-ratio.

1. The length group and a spectral weak-limit lemma

For a hyperbolic element $\gamma \in \Gamma$, let $\ell(\gamma) > 0$ be its translation length and define the closed length group

$$\mathcal{L}(\Gamma) = \overline{\langle \ell(\gamma) : \gamma \text{ hyperbolic} \rangle} \subset \mathbb{R}.$$

The length spectrum is *nonarithmetic* when $\mathcal{L}(\Gamma) = \mathbb{R}$.

Lemma 1.1 (Babillot's two-sided weak-limit lemma)**.** *Let $(Y, \mu, (T_t)_{t\in\mathbb{R}})$ be an ergodic probability-preserving flow and let $f \in L^2_0(Y, \mu)$ be real valued. If $f \circ T_{t_j}$ fails to converge weakly to 0 along some $t_j \to +\infty$, then there are $s_j \to +\infty$ and a nonzero real $\psi \in L^2_0(Y, \mu)$ such that*

$$f \circ T_{s_j} \rightharpoonup \psi, \qquad f \circ T_{-s_j} \rightharpoonup \psi.$$

After passing to a subsequence, the Cesàro means of both sequences converge to ψ almost everywhere and in L^2.

Proof. Let $\mathcal{H}_f$ be the cyclic subspace generated by f. By the cyclic spectral theorem proved in Volume 5, $\mathcal{H}_f$ is isometric to $L^2(\sigma_f)$ in such a way that f corresponds to 1 and T_t corresponds to multiplication by the character $\chi_t(\xi) = e^{it\xi}$. Passing to a subsequence, χ_{t_j} converges weakly in $L^2(\sigma_f)$ to some $F \neq 0$.

Let W be the set of weak $L^2(\sigma_f)$ limits of characters χ_{u_j} with $|u_j| \to \infty$. We need one elementary compactification fact used by Babillot: W is closed under multiplication. Indeed, if $F = \lim \chi_{u_j}$ and $G = \lim \chi_{v_k}$, then for fixed k, $\chi_{u_j + v_k} = \chi_{u_j}\chi_{v_k} \rightharpoonup F\chi_{v_k}$. Letting $k \to \infty$ gives FG against each test function. A diagonal choice over a countable dense subset of $L^2(\sigma_f)$ produces a single sequence $w_j = u_{i(j)} + v_{k(j)}$ escaping every compact subset of $\mathbb{R}$ with

$\chi_{w_j} \rightharpoonup FG$; if a provisional diagonal sum stays bounded, first thin one of the two unbounded parameter sequences so that its absolute value dominates the other. Thus the resulting parameters may always be chosen with $|w_j| \to \infty$. The same argument shows that W is weakly closed.

The conjugate $\overline{F}$ is a weak limit of χ_{-t_j} and therefore also belongs to W. Hence $|F|^2 = F\overline{F} \in W$. Choose parameters r_j with $|r_j| \to \infty$ and $\chi_{r_j} \rightharpoonup |F|^2$; after passing to one sign and replacing r_j by $-r_j$ if necessary, obtain a sequence $s_j \to +\infty$ for which

$$\chi_{s_j} \rightharpoonup |F|^2.$$

Since $|F|^2$ is real, conjugating the convergence gives simultaneously $\chi_{-s_j} \rightharpoonup |F|^2$. Transporting back through the spectral isometry yields a common weak limit ψ. It is nonzero because $F \neq 0$, and it has integral zero because all translates of f do; hence it is nonconstant.

Finally, a bounded weakly convergent sequence in a Hilbert space has a subsequence whose arithmetic means converge in norm to the weak limit: choose successive terms so that pairwise inner products of their differences from the limit are summably small, expand the squared norm of the mean, and let the number of terms grow. Apply this simultaneously to the positive and negative sequences. From norm convergence choose a further subsequence of the means with summable squared errors; Chebyshev's inequality and Borel–Cantelli then give almost-everywhere convergence to ψ. $\square$

2. Stable/unstable rectangles and the cross-ratio

For a nondegenerate ideal rectangle (ξ, η, ξ', η') in ∂X, write $B(\xi, \eta, \xi', \eta')$ for its Busemann cross-ratio, normalized as the signed geodesic-time displacement obtained by moving successively along strong stable and strong unstable leaves around the rectangle. The Busemann cocycle identity makes this independent of all choices. This geometric normalization is the one used below; for a hyperbolic element γ, it satisfies the standard identity

$$\ell(\gamma) = 2\left|B(\gamma^+, \gamma^-, \xi, \gamma\xi)\right|$$

for every ξ distinct from the fixed points.

THEOREM 2.1 (Babillot–Roblin mixing criterion). *Assume $0 < m^{\mathrm{BMS}}(T^1M) < \infty$ and $\mathcal{L}(\Gamma) = \mathbb{R}$. Then the BMS geodesic flow is mixing: for all $F, H \in L^2(m^{\mathrm{BMS}})$,*

$$\int F(g^t v)H(v)\, dm^{\mathrm{BMS}}(v) \longrightarrow \frac{m^{\mathrm{BMS}}(F)m^{\mathrm{BMS}}(H)}{|m^{\mathrm{BMS}}|} \qquad (t \to +\infty).$$

Proof. Normalize m^{BMS} to a probability measure. It suffices to prove the assertion for bounded Lipschitz mean-zero f, since such functions are dense in L^2 and polarization reduces two functions to one cyclic subspace.

Suppose mixing fails. Lemma 1.1 gives $s_j \to \infty$ and a nonzero common weak limit ψ for $f \circ g^{s_j}$ and $f \circ g^{-s_j}$, with Cesàro subsequences converging almost everywhere. Replace ψ by its short flow convolution

$$\psi_\rho(v) = \int \rho(r)\psi(g^r v)\, dr,$$

where ρ is a smooth compactly supported probability density. For a suitable ρ, ψ_ρ is still nonconstant and its time profile $r \mapsto \psi_\rho(g^r v)$ is continuous.

If w lies on the strong stable leaf of v, then $d(g^t v, g^t w) \to 0$ as $t \to +\infty$. Lipschitz continuity of f implies that the forward Cesàro limits at v and w agree. The common-limit property therefore makes ψ_ρ constant along strong stable leaves. Applying the negative-time Cesàro sequence gives the analogous statement on strong unstable leaves. More precisely, if two Hopf representatives have the same relevant endpoint but different time coordinates, first translate one representative by their signed Busemann-time difference; the representatives then lie on the same strong leaf, so their smoothed time profiles agree after exactly that translation.

For almost every geodesic define

$$P(v) = \{r \in \mathbb{R} : \psi_\rho(g^{t+r} v) = \psi_\rho(g^t v) \text{ for every } t \in \mathbb{R}\}.$$

The period group depends only on the oriented geodesic (v^-, v^+): changing the base point along that geodesic translates the time profile and does not change its period group. The stable-leaf statement makes this group unchanged when the forward endpoint is varied with the backward endpoint fixed; the unstable-leaf statement gives the symmetric conclusion. Since the BMS current is equivalent on a Hopf box to a product of the two Patterson–Sullivan endpoint measures (Theorem 7.2), Fubini connects almost every pair of endpoint pairs by two such changes. Hence $P(v)$ is almost surely one closed subgroup $P \le \mathbb{R}$. Because ψ_ρ is nonconstant, $P \neq \mathbb{R}$; hence $P = a\mathbb{Z}$ for some $a > 0$ (the possibility $P = \{0\}$ will be excluded momentarily).

Take a boundary rectangle with vertices in $\Lambda(\Gamma)$ at which the stable/unstable identifications above hold. Moving once around that rectangle returns to the same strong-product plaque but shifts the Hopf time by its cross-ratio. Therefore every such cross-ratio belongs to P. By continuity of the cross-ratio and full support of the Patterson–Sullivan density on the limit set, the conclusion extends from almost every rectangle to every nondegenerate rectangle in $\Lambda(\Gamma)$.

For a hyperbolic γ with fixed points γ^-, γ^+, choose $\xi \in \Lambda(\Gamma) \setminus \{\gamma^-, \gamma^+\}$. Choose a point x on the axis of γ. The cocycle term measured from x to γx is

$\pm\ell(\gamma)$, while the two transverse terms cancel by γ-equivariance; with our half-time normalization this gives the displayed identity $\ell(\gamma) = 2|B(\gamma^+, \gamma^-, \xi, \gamma\xi)|$. Hence every translation length lies in the discrete group $2P$ whenever P is discrete. If $P = \{0\}$ this would force every hyperbolic translation length to vanish, impossible. Thus nonmixing forces the entire length spectrum into a nontrivial discrete subgroup of $\mathbb{R}$, contradicting $\mathcal{L}(\Gamma) = \mathbb{R}$. Mixing follows for bounded Lipschitz functions and then for all L^2 functions by density and the uniform bound $|\langle F \circ g^t, H\rangle| \leq \|F\|_2 \|H\|_2$. $\square$

Proposition 2.2 (Normalization and compact-support form)**.** *For $F, H \in C_c(T^1M)$,*

$$|m^{\mathrm{BMS}}| \int F(g^t v) H(v)\, dm^{\mathrm{BMS}}(v) \longrightarrow m^{\mathrm{BMS}}(F) m^{\mathrm{BMS}}(H).$$

The statement is unchanged if the Patterson–Sullivan density is multiplied by a positive scalar.

Proof. The first assertion is Theorem 2.1 with the denominator cleared. If ν_x is replaced by $c\nu_x$, then m^{BMS} is replaced by $c^2 m^{\mathrm{BMS}}$ by the Hopf product in Volume 54. Both sides of the displayed identity are multiplied by c^4, so the normalized statement is intrinsic. $\square$

Worked Example 2.3 (Why an arithmetic suspension cannot mix)**.** Take a Bernoulli shift and suspend it with constant roof length $a > 0$. The function $v \mapsto e^{2\pi i s/a}$, where s is the suspension coordinate, is an eigenfunction of the flow. Its correlation has modulus one and cannot decay. Closed-orbit lengths are integral multiples of a. Theorem 2.1 says that the same obstruction is the only one for the BMS geodesic flow in the present rank-one setting.

CHAPTER 2

Local mixing and thickening

Why this matters. Later counting arguments do not insert arbitrary global L^2 functions. They insert small flow boxes, horospherical plaques, and characteristic functions of sets with negligible boundary. This chapter converts global BMS mixing into precisely those local forms.

1. Uniform local forms of mixing

Lemma 1.1 (Uniform mixing on compact families)**.** *Let $\mathcal{F}, \mathcal{H}$ be compact subsets of $L^2(m^{\mathrm{BMS}})$. Under the hypotheses of Theorem 2.1,*

$$\sup_{F\in\mathcal{F},\ H\in\mathcal{H}} \left| \langle F\circ g^t, H\rangle - \frac{m^{\mathrm{BMS}}(F)m^{\mathrm{BMS}}(H)}{|m^{\mathrm{BMS}}|} \right| \longrightarrow 0.$$

Proof. Fix $\varepsilon > 0$. Choose finite ε-nets $F_1, \ldots, F_r$ and $H_1, \ldots, H_s$ in the two compact sets. Theorem 2.1 gives one time T after which the desired estimate is $< \varepsilon$ for every one of the finitely many pairs (F_i, H_j). For arbitrary F, H, choose nearby net points. Cauchy–Schwarz and unitarity of the flow give

$$|\langle (F - F_i)\circ g^t, H\rangle| \le \varepsilon \sup_{H\in\mathcal{H}} \|H\|_2,$$

and similarly for the other replacement. The mean terms are controlled by $|m^{\mathrm{BMS}}(F - F_i)| \le |m^{\mathrm{BMS}}|^{1/2}\varepsilon$. Letting $\varepsilon \downarrow 0$ proves uniformity. $\square$

Theorem 1.2 (Local mixing for boundary-zero sets). *Let $A, B \subset T^1M$ be relatively compact Borel sets with $m^{\mathrm{BMS}}(\partial A) = m^{\mathrm{BMS}}(\partial B) = 0$. Then*

$$m^{\mathrm{BMS}}(g^{-t}A \cap B) \longrightarrow \frac{m^{\mathrm{BMS}}(A)m^{\mathrm{BMS}}(B)}{|m^{\mathrm{BMS}}|}.$$

The convergence is uniform for a compact family of flow boxes whose faces vary continuously and remain BMS-null.

Proof. The first assertion is Theorem 2.1 with $F = 1_A$ and $H = 1_B$. For the uniform assertion, thicken and shrink every flow box by a metric parameter $\eta > 0$. BMS regularity and the boundary-null hypothesis imply

$$m^{\mathrm{BMS}}(A^{+\eta} \setminus A^{-\eta}) \to 0,$$

uniformly on a compact continuously varying family; the same holds for B. Continuous cutoffs between the inner and outer boxes form compact families in L^2. Lemma 1.1 applies uniformly to those cutoffs. Sandwiching the indicator correlation between the inner and outer correlations and then sending $\eta \downarrow 0$ proves the claim. $\square$

Proposition 1.3 (Plaque thickening transfer)**.** *Let E be a smooth submanifold transverse to a complementary foliation in a relatively compact Hopf box $\mathcal{B}$. Let μ_E be a Radon plaque measure whose density in the box is continuous. For $\eta > 0$, let ρ_η be a nonnegative probability bump on the transverse plaque of diameter $O(\eta)$. If $F \subset E \cap \mathcal{B}$ is relatively compact with $\mu_E(\partial F) = 0$, then any mixing limit for the thickened functions*

$$\Phi_{F,\eta}(v) = \int_F \rho_\eta(\mathrm{trans}(v,u))\, d\mu_E(u)$$

passes to the unthickened plaque integral by first taking $t \to \infty$ and then $\eta \downarrow 0$.

Proof. In product coordinates (u, z) on $\mathcal{B}$, the thickening is an approximate identity in z. Uniform continuity of the Busemann densities on the compact box gives upper and lower envelopes $F^{-\eta} \subset F \subset F^{+\eta}$ whose density ratios are $1 + o_\eta(1)$. Hence the desired plaque integral lies between two ordinary correlations whose test functions differ in L^1 by $o_\eta(1)$. Theorem 1.2 supplies the $t \to \infty$ limit for fixed η, while the boundary-null condition removes the envelope error as $\eta \downarrow 0$. The order of limits is essential: no uniform effective mixing is asserted here. $\square$

Worked Example 1.4 (A rectangular Hopf box)**.** In $T^1\mathbb{H}^2$, choose disjoint boundary arcs I^-, I^+ and a time interval J. The Hopf box $I^- \times I^+ \times J$ has BMS mass equal to an integral of a continuous positive Busemann density against $\nu_o \otimes \nu_o \otimes ds$. Moving the endpoints of the arcs by η changes the mass by the Patterson–Sullivan mass of η-boundary layers. Thus the abstract sandwich in Proposition 1.3 is literally an inner/outer rectangle estimate in this model.

CHAPTER 3

Infinite-volume matrix coefficients

Why this matters. Haar measure on $\Gamma\backslash G$ is infinite when Γ has infinite covolume, so ordinary Haar mixing cannot have a finite product-of-means limit. The correct renormalization is $e^{(n-1-\delta)t}$. The mechanism is not a formal change of density: one has to compare four different local product measures across the stable/unstable foliations. This is Roblin's transverse intersection argument.

1. The stable Burger–Roblin measure

Define on Hopf coordinates

$$d\widetilde{m}^{\mathrm{BR}_*}(v) = e^{\delta\beta_{v^+}(o,x)+(n-1)\beta_{v^-}(o,x)}d\nu_o(v^+)\,dm_o(v^-)\,ds.$$

It is the time-reversal of m^{BR} and satisfies $(g^t)^*m^{\mathrm{BR}_*} = e^{-(n-1-\delta)t}m^{\mathrm{BR}_*}$.

Lemma 1.1 (Stable BR construction and the four-measure square)**.** *The Hopf-coordinate formula above defines a locally finite Radon measure m^{BR_*} on T^1M. It is the time-reversal of m^{BR}, is invariant under the stable horospherical foliation, and obeys*

$$(g^t)^*m^{\mathrm{BR}_*} = e^{-(n-1-\delta)t}m^{\mathrm{BR}_*}.$$

Write $d = n-1$. Let N be the strong unstable horospherical group for our chosen orientation of $a_t = g^t$, and let $P = N^-AM$ be the opposite weak stable subgroup. On every sufficiently small lifted product box $xP_0N_0 \subset \Gamma\backslash G$ on which multiplication is injective, there are transverse measures $d\nu_{xP}$ and dp such that, after normalizing Haar measure on M once and for all,

$$\begin{aligned} dm^{\mathrm{BMS}}(xpn) &= d\nu_{xP}(xp)\,d\mu^{\mathrm{PS}}_{xpN}(xpn),\\ dm^{\mathrm{BR}}(xpn) &= d\nu_{xP}(xp)\,dn,\\ dm^{\mathrm{BR}_*}(xpn) &= dp\,d\mu^{\mathrm{PS}}_{xpN}(xpn),\\ dm^{\mathrm{Haar}}(xpn) &= dp\,dn. \end{aligned} \tag{55.3.1}$$

Here dn is Haar/Lebesgue measure on the unstable leaf, and μ^{PS}_{xpN} is its Patterson–Sullivan conditional measure. All four identities are exact after the displayed choices of the transverse normalizations.

Proof. Time reversal interchanges the two endpoints and sends the unstable Burger–Roblin formula of Volume 54 to the displayed stable formula. The conformal rules for ν_x and m_x cancel the change of base point, so the formula is Γ-invariant and descends to T^1M. Local finiteness follows because on a compact Hopf box the Busemann weights are bounded and both endpoint densities are Radon. The flow-scaling and stable-leaf invariance follow exactly as in Theorem 4.2, with the two endpoint roles reversed.

For the product identities, lift the box to G and use the endpoint maps $(pn)^{\pm}$ together with the Hopf time. Along xpN, the backward endpoint and the transverse Hopf data are fixed, while the forward endpoint varies. Lemma 4.1 identifies the visual conditional on that leaf with dn; replacing that visual conditional by the δ-conformal density produces $d\mu^{\mathrm{PS}}_{xpN}$. On the P-transverse direction the same two choices give respectively dp (visual/Haar) and $d\nu_{xP}$ (Patterson–Sullivan). Thus the four possible choices of visual or Patterson–Sullivan density on the two endpoint directions give exactly the four products in (55.3.1). We define dp and $d\nu_{xP}$ by these disintegrations; compatibility on overlapping boxes follows from the base-point independence of the Hopf formulas. In particular, no informal "bounded Radon–Nikodym factor" is being used in the argument below. $\square$

2. Two transfer lemmas

We now reconstruct the qualitative transverse-intersection mechanism. The point of the next lemma is to make the geometry used in both transfers explicit. It is a local statement on compact boxes; in particular, no cusp estimate is hidden in it.

Lemma 2.1 (Compact-box holonomy and the two Jacobians)**.** *Put $d = n-1$. Let $Q \subset \Gamma\backslash G$ be compact. After covering Q by finitely many injective boxes, there are $\varepsilon_0 > 0$ and $C \geq 1$ with the following properties. In every box we may write points uniquely as*

$$xpn, \qquad p \in P_{\varepsilon_0}, \quad n \in N_{\varepsilon_0},$$

where $P = N^{-}AM$ and N is the expanding horospherical subgroup for right translation by a_t.

For p and n in those boxes there are unique local data

$$h_p(n) \in N, \qquad \tau_p(n) \in \mathbb{R}, \qquad k_p(n) \in M, \qquad u_p(n) \in N^{-}$$

such that

$$pn = h_p(n) a_{\tau_p(n)} k_p(n) u_p(n). \tag{55.3.2}$$

They depend C^1-smoothly on (p, n) and satisfy, after shrinking the boxes,

$$\|h_p(n)n^{-1}\| + |\tau_p(n)| + \|u_p(n)\| \le C\|p - e\|. \tag{55.3.3}$$

Consequently, for right-M-invariant uniformly continuous F,

$$F(xpna_t) = F(xh_p(n)a_{t+\tau_p(n)}) + o_\varepsilon(1) \tag{55.3.4}$$

uniformly for $\|p - e\| \le \varepsilon$, $x \in Q$ and $t \to \infty$; here $o_\varepsilon(1)$ has limsup bounded by the modulus of continuity of F at scale $C\varepsilon$.

Stable holonomy $n \mapsto h_p(n)$ carries the Patterson–Sullivan conditionals by a positive continuous Jacobian $J_p(n)$:

$$d(h_p)_*\mu^{\mathrm{PS}}_{xN} = J_p\, d\mu^{\mathrm{PS}}_{xpN}, \qquad \sup|J_p - 1| \longrightarrow 0 \quad (p \to e). \tag{55.3.5}$$

Finally the flow acts on the two leaf measures by

$$(R_{a_t})_* dn = e^{-dt}\, dn, \qquad (R_{a_t})_* d\mu^{\mathrm{PS}}_{xN} = e^{-\delta t}\, d\mu^{\mathrm{PS}}_{xa_tN}. \tag{55.3.6}$$

Thus the quotient of the two leaf Jacobians is exactly $e^{-(d-\delta)t}$.

Proof. The multiplication map $N \times A \times M \times N^- \to G$ has invertible differential at the identity: its differential is the direct-sum map from $\mathfrak{n} \oplus \mathfrak{a} \oplus \mathfrak{m} \oplus \mathfrak{n}^-$ onto $\mathfrak{g}$. The inverse function theorem therefore gives the unique factorization (55.3.2) in a neighborhood of the identity. Compactness of Q allows finitely many quotient boxes with one common radius and one common C^1 bound. Since the factorization equals $(n, 0, e, e)$ at $p = e$, the mean value theorem gives (55.3.3).

Conjugation contracts N^- in positive time: $a_{-t}ua_t \to e$ exponentially for $u \in N^-$. Multiplying (55.3.2) on the right by a_t and commuting the M-factor through A, we obtain

$$pna_t = h_p(n)a_{t+\tau_p(n)}k_p(n)\big(a_{-t}u_p(n)a_t\big).$$

For a right-M-invariant F, the last factor tends uniformly to the identity and the time displacement is $O(\varepsilon)$, proving (55.3.4).

For (55.3.5), use the endpoint model of the unstable PS conditional from Volume 54. Stable holonomy preserves the varying endpoint and changes only the base point at which the δ-conformal density is evaluated. The Radon–Nikodym derivative is therefore an exponential of a Busemann-cocycle difference. That difference is a continuous function of (p, n) and vanishes for $p = e$. On a compact box it is uniformly $O(\|p - e\|)$, which gives both the displayed formula and $J_p \to 1$ uniformly. This also proves that the holonomy is absolutely continuous for the PS conditionals; no smoothness of the limit set is being assumed.

The first identity in (55.3.6) is the determinant of $\mathrm{Ad}(a_{-t})|_{\mathfrak{n}}$: every positive root has weight e^{-t} in real hyperbolic space and $\dim N = d$. The second is the δ-conformal scaling already proved for the unstable PS conditional in Volume 54. Dividing the identities gives the final assertion. ☐

For a compact unstable plaque yN_0 write

$$\mu^{\mathrm{PS}}_{yN}(\phi) := \int_{yN_0} \phi \, d\mu^{\mathrm{PS}}_{yN}.$$

We shall also use one elementary functional-analytic consequence of mixing. If U_t is the Koopman operator of a mixing finite measure-preserving flow and K_1, K_2 are compact subsets of L^2, then

$$\sup_{f\in K_1, h\in K_2} \left| \langle U_t f, h\rangle - |m^{\mathrm{BMS}}|^{-1} m^{\mathrm{BMS}}(f) m^{\mathrm{BMS}}(h) \right| \longrightarrow 0. \tag{55.3.7}$$

Indeed, cover the two compact sets by finite L^2-nets and use $|\langle U_t u, v\rangle| \le \|u\|_2 \|v\|_2$.

Lemma 2.2 (Uniform transverse access)**.** *Let $Q \subset \Gamma\backslash G$ be compact. There is $R = R(Q) > 1$ with the following property. For every $y \in Q$ and every sufficiently small $\varepsilon > 0$, if*

$$t_0 = \log(R/\varepsilon), \qquad y_0 = y a_{t_0},$$

then the BMS transverse measure of $y_0 P_\varepsilon$ is positive. Hence a nonnegative continuous bump $\rho_{y,\varepsilon}$ supported there can be chosen with transverse mass one. The choices may be made from finitely many coordinate templates when y ranges over Q.

Proof. Lift one compact box meeting Q to G. For a lift of y, the endpoint map from the opposite strong horosphere N^- to $\partial\mathbb{H}^n \setminus \{y^+\}$ is a diffeomorphism. Since $\Lambda(\Gamma)$ contains at least two points, some opposite-horosphere parameter of finite norm moves the other endpoint into $\Lambda(\Gamma)$. On a compact family of lifts the least such norm is bounded: otherwise a convergent sequence of base frames and the local openness of the endpoint map would give a contradiction. Let R exceed this uniform bound by one. Conjugation by a_{t_0} multiplies the relevant N^- parameter by $e^{-t_0} = \varepsilon/R$, so the bounded displacement becomes an element of P_ε at y_0.

The transverse BMS factor is built from the Patterson–Sullivan density on that endpoint. Its support is the limit set, and every neighborhood of a limit point has positive PS mass. Thus $y_0 P_\varepsilon$ has positive transverse mass. Finite quotient charts covering Q give finitely many coordinate templates; normalization then produces the desired bumps. ☐

Lemma 2.3 (PS-plaque equidistribution from BMS mixing)**.** *Let $Q \subset \Gamma\backslash G$ be compact. Fix compact plaques $N_0 \Subset N$. Let $\Psi \in C_c(\Gamma\backslash G)$ and let $\phi_y \in C_c(yN_0)$ vary in a compact family in the uniform topology as $y \in Q$. Then, uniformly for $y \in Q$,*

$$\int_{yN_0} \Psi(yna_t)\phi_y(yn)\, d\mu^{\mathrm{PS}}_{yN}(yn) \longrightarrow \frac{m^{\mathrm{BMS}}(\Psi)}{|m^{\mathrm{BMS}}|}\mu^{\mathrm{PS}}_{yN}(\phi_y). \tag{55.3.8}$$

Proof. Fix $\varepsilon < \varepsilon_0$ from Lemma 2.1. Apply Lemma 2.2 and put $t_0 = \log(R/\varepsilon)$, $y_0 = ya_{t_0}$. Flow the initial plaque to y_0N: by (55.3.6) there is a unique transported function ϕ_{y,t_0} for which

$$\int_{y_0N} \phi_{y,t_0}\, d\mu^{\mathrm{PS}}_{y_0N} = \mu^{\mathrm{PS}}_{yN}(\phi_y). \tag{55.3.9}$$

Explicitly, if $yna_{t_0} = y_0n'$ then $\phi_{y,t_0}(y_0n') = e^{-\delta t_0}\phi_y(yn)$ on the flowed plaque. Indeed, $(R_{a_{t_0}})_*\mu^{\mathrm{PS}}_{yN} = e^{-\delta t_0}\mu^{\mathrm{PS}}_{y_0N}$ by (LV.3.6), so this is exactly the factor that preserves the integral.

Let $H_p : y_0N \to y_0pN$ be the stable holonomy from Lemma 2.1, and let J_p be its PS Jacobian. With $\rho = \rho_{y,\varepsilon}$ from Lemma 2.2, define on the thickened box

$$\Phi_{y,\varepsilon}(H_p(z)) = \phi_{y,t_0}(z)\, \rho(y_0p)\, J_p(z)^{-1}. \tag{55.3.10}$$

The product formula (55.3.1), Fubini, and (55.3.5) give the exact mass identity

$$m^{\mathrm{BMS}}(\Phi_{y,\varepsilon}) = \mu^{\mathrm{PS}}_{yN}(\phi_y). \tag{55.3.11}$$

This formula is the coordinate-free form of the Busemann correction appearing in Roblin's and Oh–Winter's explicit thickening.

Compare now the correlation

$$I_{t,\varepsilon}(y) = \int \Psi(za_{t-t_0})\Phi_{y,\varepsilon}(z)\, dm^{\mathrm{BMS}}(z)$$

with the left side of (55.3.8). Formula (55.3.4) shows that, after the stable holonomy and the fixed time shift t_0 are undone, the two integrands differ only by the oscillation of Ψ and ϕ_y at scale $C\varepsilon$, plus a quantity tending to zero as $t \to \infty$. The Jacobian has already been accounted for exactly in (55.3.10). Thus

$$\limsup_{t\to\infty} \left| I_{t,\varepsilon}(y) - \int_{yN_0} \Psi(yna_t)\phi_y(yn)d\mu^{\mathrm{PS}}_{yN}(yn) \right| \le \omega_Q(C\varepsilon), \tag{55.3.12}$$

where $\omega_Q(r) \to 0$ as $r \downarrow 0$, uniformly in $y \in Q$. The families $\Phi_{y,\varepsilon}$ obtained from finitely many templates are compact in L^2 for fixed ε. Apply BMS mixing (Theorem 2.1) uniformly by (LV.3.7), use (LV.3.11), first let $t \to \infty$ and then $\varepsilon \downarrow 0$. This proves (55.3.8). ☐

Lemma 2.4 (Roblin transverse-intersection transfer). *With Q, N_0 as above, for compactly supported continuous Ψ and a compact uniform family ϕ_y on yN_0,*

$$e^{(d-\delta)t}\int_{yN_0}\Psi(yna_t)\phi_y(yn)\,dn \longrightarrow \frac{m^{\mathrm{BR}}(\Psi)}{|m^{\mathrm{BMS}}|}\mu^{\mathrm{PS}}_{yN}(\phi_y), \tag{55.3.13}$$

uniformly for $y \in Q$.

Proof. By linearity and a partition of unity it is enough to take $\Psi \geq 0$ supported in one injective product box xP_1N_1. Write

$$q(xp) = \int_{xpN_1}\Psi(xpn)\,dn. \tag{55.3.14}$$

The support of q is compact in the P-plaque. For every xp in that compact set, the PS conditional on xpN is nonzero. Choose finitely many nonnegative continuous leaf bumps and a partition of unity in p to obtain a continuous family $\eta_p \in C_c(xpN)$ satisfying

$$\int \eta_p\,d\mu^{\mathrm{PS}}_{xpN} = 1. \tag{55.3.15}$$

Set

$$\Psi^0(xpn) = q(xp)\eta_p(xpn). \tag{55.3.16}$$

The first two identities in the four-measure square give the exact equality

$$m^{\mathrm{BMS}}(\Psi^0) = \int_{xP_1} q(xp)\,d\nu_{xP}(xp) = m^{\mathrm{BR}}(\Psi). \tag{55.3.17}$$

We now prove the comparison that changes the leaf measure. Fix $0 < \varepsilon < \varepsilon_0$ and shrink the product box once so that its $C\varepsilon$-enlargement is still injective. Decompose the set of points yn for which yna_t meets the enlarged box into its local sheets. On one sheet there is a unique factorization

$$yna_t = xp(n,t)n'(n,t). \tag{55.3.18}$$

Uniqueness follows from injectivity of the product chart. If two points of the same sheet have transverse coordinates in a P-piece of diameter at most ε, Lemma 2.1 pairs their final N-coordinates by stable holonomy with displacement at most $C\varepsilon + Ce^{-t}$. Consequently the value of ϕ_y on the corresponding initial points is squeezed between its leafwise infimum and supremum on a $C\varepsilon + Ce^{-t}$ neighborhood.

The change of variables on a sheet has two factors and no others. Under right translation by a_t, N-Lebesgue volume contributes e^{-dt}; under the same translation the PS conditional contributes $e^{-\delta t}$. The stable holonomy between the two final plaques contributes the Jacobian $J_p = 1 + O(\varepsilon)$ from (55.3.5). Therefore, after multiplication by $e^{(d-\delta)t}$, the Lebesgue contribution of that sheet is squeezed by $(1+O(\varepsilon))$ times the corresponding PS contribution for Ψ^0. Summing over the disjoint sheets preserves the inequality: all terms are nonnegative, and compact support makes the sum locally finite. In formulas, there are continuous upper/lower $C\varepsilon$-envelopes $(\Psi^0)^{\varepsilon,\pm}$ and $\phi_y^{\varepsilon,\pm}$ such that

$$\begin{aligned}(1-C\varepsilon)\int_{yN}&(\Psi^0)^{\varepsilon,-}(yna_t)\phi_y^{\varepsilon,-}(yn)d\mu^{\mathrm{PS}}_{yN}(yn)-r_{t,\varepsilon}\\ &\le e^{(d-\delta)t}\int_{yN_0}\Psi(yna_t)\phi_y(yn)dn\\ &\le (1+C\varepsilon)\int_{yN}(\Psi^0)^{\varepsilon,+}(yna_t)\phi_y^{\varepsilon,+}(yn)d\mu^{\mathrm{PS}}_{yN}(yn)+r_{t,\varepsilon},\end{aligned}\tag{55.3.19}$$

where $r_{t,\varepsilon}\to 0$ as $t\to\infty$ for fixed ε, uniformly for $y\in Q$. This is the qualitative transverse-intersection identity. Notice that (55.3.19) is a direct change-of-variables statement: the exponent $d-\delta$ comes solely from (55.3.6), while the only distortion is the compact-box stable-holonomy Jacobian (55.3.5).

Apply Lemma 2.3 to the upper and lower PS integrals. Their limits are

$$\frac{m^{\mathrm{BMS}}((\Psi^0)^{\varepsilon,\pm})}{|m^{\mathrm{BMS}}|}\mu^{\mathrm{PS}}_{yN}(\phi_y^{\varepsilon,\pm}).$$

As $\varepsilon\downarrow 0$, these converge respectively to

$$\frac{m^{\mathrm{BMS}}(\Psi^0)}{|m^{\mathrm{BMS}}|}\mu^{\mathrm{PS}}_{yN}(\phi_y)=\frac{m^{\mathrm{BR}}(\Psi)}{|m^{\mathrm{BMS}}|}\mu^{\mathrm{PS}}_{yN}(\phi_y)$$

by (55.3.17), Radon regularity, and uniform continuity on the fixed compact boxes. Let $t\to\infty$ and then $\varepsilon\downarrow 0$ in (55.3.19). This proves (55.3.13) for nonnegative functions. Positive and negative parts give the general continuous case, and the compact-family uniformity follows from the same finite covering and moduli of continuity. □

3. Roblin's matrix coefficient asymptotic

THEOREM 3.1 (Roblin qualitative infinite-volume matrix coefficient asymptotic). *For $\Psi_1,\Psi_2\in C_c(T^1M)$,*

$$e^{(n-1-\delta)t}\int_{T^1M}\Psi_1(g^tv)\Psi_2(v)\,dm^{\mathrm{Haar}}(v)\longrightarrow\frac{m^{\mathrm{BR}}(\Psi_1)m^{\mathrm{BR}_*}(\Psi_2)}{|m^{\mathrm{BMS}}|}.\tag{55.3.20}$$

Equivalently, the same formula holds for compactly supported right-M-invariant functions on $\Gamma\backslash G$.

PROOF. Lift the functions to right-M-invariant functions on $\Gamma\backslash G$. We first take them nonnegative and C^1. A finite partition of unity reduces to the case that Ψ_2 is supported in one injective product box xP_0N_0. By the last identity in (55.3.1), Fubini gives

(55.3.21)
$$\int \Psi_1(za_t)\Psi_2(z)\,dm^{\mathrm{Haar}}(z) = \int_{xp\in xP_0}\left(\int_{xpN_0}\Psi_1(xpna_t)\Psi_2(xpn)\,dn\right)dp.$$

The functions $\phi_{xp}(xpn) = \Psi_2(xpn)$ form a compact family on the plaques because $\overline{xP_0N_0}$ is compact. Lemma 2.4 therefore applies uniformly under the outer integral. Multiplying (55.3.21) by $e^{(d-\delta)t}$ and passing to the limit gives

$$\frac{m^{\mathrm{BR}}(\Psi_1)}{|m^{\mathrm{BMS}}|}\int_{xP_0}\mu^{\mathrm{PS}}_{xpN}(\Psi_2|_{xpN_0})\,dp.$$

The third identity in the four-measure square identifies the last integral with $m^{\mathrm{BR}_*}(\Psi_2)$. Summing the partition of unity proves (55.3.20) for nonnegative C^1 functions, and hence for signed C^1 functions by linearity.

For general C_c functions, fix a compact neighborhood of their supports and approximate their positive and negative parts uniformly from above and below by C^1 functions supported in that neighborhood. The local squeeze (55.3.19) supplies a uniform finite limsup bound for the renormalized correlations of such compactly supported positive functions. Thus the upper/lower approximations pass to the limit, while the BR and BR_* functionals converge by their Radon property. This proves (55.3.20) for all continuous compactly supported test functions. $\square$

Proof and provenance. Roblin proves the M-invariant statement from geodesic-flow mixing; the later Mohammadi–Oh formulation records exactly the limit in Theorem 3.1 [Rob03; MO15]. The proof above internalizes the missing transverse-intersection step. Its local four-measure square, PS-plaque thickening, PS-to-Lebesgue transfer, and final P-integration are the qualitative counterparts of the effective argument written explicitly in Oh–Winter, Theorem 5.8 [OW16]. We use only qualitative BMS mixing, already proved in Chapter 1; no spectral gap or effective matrix coefficient is used here.

Proposition 3.2 (Finite-volume consistency of the Roblin formula)**.** *If $\delta = n-1$ and the Patterson–Sullivan density is normalized to be the visual density, the formula in Theorem 3.1 becomes ordinary finite-volume mixing.*

Proof. Proposition 4.3 and Theorem 8.2 identify $m^{\mathrm{BR}} = m^{\mathrm{BR}_*} = m^{\mathrm{BMS}} = m^{\mathrm{Haar}}$ up to the common M-normalization. The exponent $n-1-\delta$ is zero, so (55.3.20) is exactly

$$\int \Psi_1(g^tv)\Psi_2(v)\,dm^{\mathrm{Haar}}(v) \longrightarrow \frac{m^{\mathrm{Haar}}(\Psi_1)m^{\mathrm{Haar}}(\Psi_2)}{|m^{\mathrm{Haar}}|},$$

the standard finite-volume mixing normalization. □

Worked Example 3.3 (Why the exponent has the sign shown)**.** For an infinite-covolume surface, $0 < \delta < 1$ in the typical thin case. The raw Haar correlation of two fixed compact sets decays. Multiplying by $e^{(1-\delta)t}$ compensates for the difference between one-dimensional Lebesgue mass on an expanding horocycle and the δ-dimensional Patterson–Sullivan mass that survives in the recurrent set. In the proof, this is not a heuristic: it is exactly the quotient of the two flow Jacobians e^{-t} and $e^{-\delta t}$ in (55.3.6). A factor $e^{-(1-\delta)t}$ would therefore have the wrong sign and force the renormalized correlation to zero.

CHAPTER 4

Renewal asymptotics

Why this matters. Mixing usually controls a thin shell at time t, whereas counting asks for everything up to radius T. The passage from shells to balls is an Abelian renewal calculation. It is elementary once the correct exponential normalization is kept visible.

1. From shells to cumulative counts

Lemma 1.1 (Exponential Abelian lemma)**.** *Let $a : [0, \infty) \to \mathbb{R}$ be locally integrable and suppose $e^{-\alpha t}a(t) \to c$ for some $\alpha > 0$. Then*

$$e^{-\alpha T} \int_0^T a(t)\, dt \longrightarrow \frac{c}{\alpha}.$$

The same conclusion holds for a positive locally finite Stieltjes measure dA if every fixed-width shell satisfies

$$e^{-\alpha T}(A(T+h) - A(T)) \longrightarrow \frac{c}{\alpha}(e^{\alpha h} - 1)$$

for all $h > 0$.

Proof. For the integral form, write $a(t) = e^{\alpha t}(c + r(t))$ with $r(t) \to 0$. Given T_0, split the integral at T_0. The initial piece is $O(e^{-\alpha T})$. On $[T_0, T]$, the main term integrates to $c(e^{\alpha T} - e^{\alpha T_0})/\alpha$, while the error is bounded by $\sup_{t \ge T_0} |r(t)| e^{\alpha T}/\alpha$. Let first $T \to \infty$ and then $T_0 \to \infty$.

For the Stieltjes form, fix $h > 0$ and decompose $[0, T]$ into shells of width h counted backward from T. After multiplying by $e^{-\alpha T}$, the j-th shell has asymptotic weight proportional to $e^{-\alpha j h}$. Dominate the finitely many recent shells using the assumed limits and the old shells by positivity. Summing the geometric series gives c/α in the limit $h \downarrow 0$; equivalently one may squeeze between upper and lower Riemann sums for $\int_0^\infty c e^{-\alpha u}\, du$. ☐

Theorem 1.2 (Shell-to-ball renewal transfer). *Suppose a monotone cumulative count N has fixed-width shell asymptotic*

$$N(t+h) - N(t) = C\, e^{\delta t}(e^{\delta h} - 1) + o(e^{\delta t})$$

for each fixed $h > 0$, with the o-term locally uniform in h. Then

$$N(T) \sim Ce^{\delta T}.$$

Equivalently, if the shell main term is $C_0 e^{\delta t}(e^{\delta h} - 1)/\delta$, then $N(T) \sim C_0 e^{\delta T}/\delta$.

Proof. Apply the Stieltjes part of Lemma 1.1 to the monotone counting function N. The two normalizations in the statement merely record whether the factor $1/\delta$ is included in the shell constant. Writing both versions prevents a common bookkeeping error: the renewal theorem does not create a new dynamical constant; it only integrates the exponential shell density. □

Proposition 1.3 (Logarithmic norm transfer)**.** *Suppose a radial parameter R and geodesic time satisfy $R = c_0 e^{\lambda t}(1 + o(1))$ with $c_0, \lambda > 0$. If a time count is $N_t(t) \sim Ce^{\delta t}$, then*

$$N_R(R) \sim C c_0^{-\delta/\lambda} R^{\delta/\lambda}.$$

A bounded additive error in $t = (\log R - \log c_0)/\lambda$ changes only the multiplicative constant, not the exponent.

Proof. Substitute $t = \lambda^{-1}(\log R - \log c_0) + o(1)$ into $e^{\delta t}$. This gives $e^{\delta t} = c_0^{-\delta/\lambda} R^{\delta/\lambda}(1 + o(1))$. The final statement follows because $e^{\delta O(1)}$ is bounded above and below by positive constants. □

Worked Example 1.4 (From hyperbolic radius to Euclidean size)**.** If a vector in a linear representation grows like $\|w_0 a_t\| \asymp e^{\lambda t}$ and the relevant orbit shells grow like $e^{\delta t}$, then the number of orbit points of norm at most R has power $R^{\delta/\lambda}$. For the standard light-cone representation of $\mathrm{SO}(n, 1)$ one often has $\lambda = 1$, giving the familiar exponent δ.

CHAPTER 5

Skinning measures

Why this matters. Counting a translate of a submanifold requires a measure on its normal bundle. The Patterson–Sullivan density provides exactly that measure. Its finiteness is not automatic in a cusp: the correct obstruction is the difference between the cusp rank of Γ and the cusp rank already contained in the stabilizer of the submanifold.

1. Admissible submanifolds and their two natural measures

Let $\widetilde{E} \subset T^1X$ be either a strong unstable horosphere or the unit normal bundle of a proper totally geodesic subspace $\widetilde{S} \subset X$. Let E be its image in T^1M, with stabilizer $\Gamma_{\widetilde{E}}$. On $\widetilde{E}$ define

$$d\mu^{\mathrm{PS}}_{\widetilde{E}}(v) = e^{\delta\beta_{v^+}(o,\pi(v))}\,d\nu_o(v^+), \qquad d\mu^{\mathrm{Leb}}_{\widetilde{E}}(v) = e^{(n-1)\beta_{v^+}(o,\pi(v))}\,dm_o(v^+),$$

with the evident interpretation in the normal-bundle case using the forward-endpoint parametrization. Both descend to $\Gamma_{\widetilde{E}}\backslash\widetilde{E}$.

For later use define the global parabolic corank

$$\mathrm{pbcorank}(\Gamma_{\widetilde{E}}) := \max_{\xi\in\Lambda_p(\Gamma)\cap\partial\widetilde{E}} \Big(\mathrm{rank}\,\Gamma_\xi - \mathrm{rank}(\Gamma_\xi \cap \Gamma_{\widetilde{E}})\Big),$$

with the maximum over the empty set understood as 0. This is the Oh–Shah parabolic-corank invariant; the local quantity below is its summand at one parabolic fixed point.

Lemma 1.1 (Cusp mass and parabolic corank)**.** *Let ξ be a parabolic fixed point meeting the closure of $\widetilde{E}$. Let Γ_ξ be its stabilizer and $\Gamma_{\xi,\widetilde{E}} = \Gamma_\xi \cap \Gamma_{\widetilde{E}}$. After passing to finite-index translation subgroups as in the Bieberbach input of Volume 54, define*

$$\mathrm{pbcorank}_\xi(\Gamma_{\widetilde{E}}) = \mathrm{rank}\,\Gamma_\xi - \mathrm{rank}\,\Gamma_{\xi,\widetilde{E}}.$$

If this corank is r, then the PS skinning mass in a sufficiently deep ξ-cusp is comparable to

$$\sum_{k\in\mathbb{Z}^r\setminus\{0\}} \|k\|^{-\delta}.$$

Consequently that cusp contributes finite mass exactly when $r < \delta$.

PROOF. Conjugate ξ to ∞ in upper-half space. A finite-index subgroup of Γ_ξ acts by translations on a Euclidean subspace $V \cong \mathbb{R}^p$, and the subgroup preserving $\widetilde{E}$ translates in a subspace V_E of dimension q. Choose a complementary lattice of rank $r = p - q$ and representatives $k \in \mathbb{Z}^r$.

A representative translated by k meets the fixed fundamental strip for V_E only at Euclidean distance $\asymp \|k\|$ from a reference point. The shadow estimate from Volume 54 and the Busemann formula in upper-half space give PS mass $\asymp \|k\|^{-\delta}$ for the corresponding cusp packet. Bounded overlap of those packets yields the displayed comparison. The elementary lattice-series criterion $\sum_{k\in\mathbb{Z}^r\setminus 0} \|k\|^{-s} < \infty$ iff $s > r$ follows by grouping points in dyadic annuli, which contain $\asymp 2^{jr}$ lattice points of size $\asymp 2^j$. $\square$

THEOREM 1.2 (Finiteness criterion for the PS skinning measure). *Let E be admissible as above and assume that its immersion in T^1M is proper on the convex-core part. Then*

$$|\mu_E^{\mathrm{PS}}| < \infty \quad \Longleftrightarrow \quad \mathrm{pbcorank}_\xi(\Gamma_{\widetilde{E}}) < \delta \textit{ for every parabolic } \xi \in \Lambda(\Gamma) \cap \partial\widetilde{E}.$$

Here the left side means the total mass on $\Gamma_{\widetilde{E}}\backslash\widetilde{E}$.

PROOF. By geometric finiteness (Volume 53), the part of E over a fixed truncation of the convex core is compact modulo $\Gamma_{\widetilde{E}}$; its PS skinning mass is finite because the local density is Radon. Only finitely many cusp types remain. Lemma 1.1 identifies the mass in each cusp, up to multiplicative constants, with a Euclidean lattice series of dimension equal to the corresponding parabolic corank. Such a series converges exactly when the corank is $< \delta$. Since there are finitely many cusp orbits, total finiteness is equivalent to all of these inequalities. $\square$

Proposition 1.3 (Codimension-one specialization). *Suppose $\widetilde{S}$ is a totally geodesic hyperplane. Then every relevant parabolic corank is at most* 1*. Hence $\delta > 1$ implies $|\mu_E^{\mathrm{PS}}| < \infty$. If $\delta \le 1$, infinite skinning mass can occur only at a cusp for which the hyperplane misses one translation direction of the ambient parabolic stabilizer; equivalently the corresponding corank is* 1*.*

PROOF. A hyperplane cuts the Euclidean horosphere at infinity in a codimension-at-most-one affine subspace, so the translation rank lost when intersecting its stabilizer is at most one. Apply Theorem 1.2. If $\delta > 1$, both possible coranks $0, 1$ are $< \delta$. If $\delta \le 1$, corank 0 is harmless and corank 1 is the only possible obstruction. $\square$

Worked Example 1.4 (A rank-two cusp and a geodesic plane)**.** In $\mathbb{H}^3$, let a cusp group be generated by translations $(x, y) \mapsto (x + 1, y)$ and $(x, y) \mapsto$

$(x, y+1)$. If a totally geodesic plane is stabilized only by the first translation, then $p = 2$, $q = 1$, and the parabolic corank is $r = 1$. Its cusp skinning mass is comparable to $\sum_{k\neq 0} |k|^{-\delta}$, finite exactly for $\delta > 1$.

CHAPTER 6

Normalization of infinite measures

Why this matters. Patterson–Sullivan densities are unique only up to a scalar. Any counting constant that changes when that arbitrary scalar changes is not geometric. This chapter records the scaling of every measure appearing in the infinite-volume formulas and checks that the final ratios are normalization-free.

1. Scaling the Patterson–Sullivan density

Lemma 1.1 (Scaling dictionary)**.** *Replace the Patterson–Sullivan density by $\nu'_x = c\nu_x$, $c > 0$, while keeping the visual density fixed. Then*

$$m^{\mathrm{BMS}\prime} = c^2 m^{\mathrm{BMS}}, \quad m^{\mathrm{BR}\prime} = c\, m^{\mathrm{BR}},$$
$$m^{\mathrm{BR}_*\prime} = c\, m^{\mathrm{BR}_*}, \qquad \mu_E^{\mathrm{PS}\prime} = c\, \mu_E^{\mathrm{PS}}.$$

whereas Haar, visual, and μ_E^{Leb} measures are unchanged.

Proof. Count Patterson–Sullivan endpoint factors in the definitions. BMS has two, each BR measure has one, and the PS skinning measure has one. The remaining factors are visual or Lebesgue and do not change. □

Theorem 1.2 (Intrinsic normalization ratios). *The two quantities*

$$\frac{m^{\mathrm{BR}}(\Psi_1) m^{\mathrm{BR}_*}(\Psi_2)}{|m^{\mathrm{BMS}}|}, \qquad \frac{\mu_E^{\mathrm{PS}}(F) m^{\mathrm{BR}}(\Psi)}{|m^{\mathrm{BMS}}|}$$

are independent of the scalar normalization of the Patterson–Sullivan density whenever the displayed masses are finite.

Proof. By Lemma 1.1, each numerator is multiplied by c^2 and the denominator by c^2. The factors cancel. These are exactly the ratios appearing in Theorem 3.1 and the horospherical/skinning equidistribution formulas used below. □

Proposition 1.3 (Finite-volume specialization)**.** *If $\delta = n - 1$ and ν_x is chosen proportional to the visual density, then the normalization dictionary reduces to the usual Haar/Liouville normalization: BMS, BR, and BR$_*$ are* *proportional to Haar, and the skinning ratio in Theorem 1.2 is the ordinary geometric transverse-volume constant.*

Proof. By Theorem 8.2, the symmetric visual Hopf product is Haar/Liouville measure. Proposition 4.3 identifies BR with BMS when the endpoint dimensions agree, and time reversal gives the same for BR_*. Substitution into Theorem 1.2 gives the standard finite-volume ratios. $\square$

Worked Example 1.4 (Rescaling by ten)**.** If ν_o is replaced by $10\nu_o$, then BMS mass is multiplied by 100, while each BR mass and each PS skinning mass is multiplied by 10. Thus $\mu_E^{\mathrm{PS}}(F)m^{\mathrm{BR}}(\Psi)/|m^{\mathrm{BMS}}|$ is unchanged. This simple check is worth doing before trusting any published counting constant.

CHAPTER 7

Effective mixing

Why this matters. Qualitative mixing determines the main term but gives no rate. Power-saving counting requires an exponential rate for renormalized matrix coefficients. The additional mechanism is representation theoretic: isolate the complementary spectrum at the critical exponent, control every K-type uniformly, and sum those estimates with compact Sobolev norms. Appendix A reconstructs that mechanism in the rank-one scope used here.

1. Spectral-gap data and Sobolev norms

Let $\widehat{G}$ denote the unitary dual and $\widehat{M}$ the unitary dual of the centralizer M of A in K. For $\upsilon \in \widehat{M}$ and $(n-1)/2 < s < n-1$, write $\mathcal{U}(\upsilon, s-n+1)$ for the unitary complementary-series representation obtained from normalized induction from MAN whenever that induced representation is unitary.

Definition 1.1 (Operational spectral-gap datum)**.** We say that $L^2(\Gamma\backslash G)$ has an *operational gap at δ with data (s_0, n_0)* if

$$L^2(\Gamma\backslash G) = \mathcal{H}_\delta^\dagger \oplus W, \qquad \mathcal{H}_\delta^\dagger = \widehat{\bigoplus}_{\upsilon\in\widehat{M}} m(\upsilon)\,\mathcal{U}(\upsilon, \delta - n + 1),$$

with $m(\upsilon) \le (\dim \upsilon)^{n_0}$. In addition, W is required to split spectrally into a tempered part and a direct integral of complementary-series constituents whose parameters satisfy $s \le s_0$. This explicit decomposition is the datum consumed in the proof below; in particular, the proof never infers the shape of the non-tempered spectrum from an unstated unitary-dual classification.

Remark 1.2. Mohammadi–Oh formulate the same information as their spectral-gap condition and use the rank-one unitary-dual classification to pass to this decomposition. We state the decomposition itself so that the classification theorem is not a hidden dependency of the proof.

Lemma 1.3 (Sobolev bookkeeping for an effective matrix coefficient)**.** *Fix a basis of $\mathfrak{g}$ and, for an integer $\ell \ge 0$, set*

$$\mathcal{S}_\ell(\Psi)^2 = \sum_{\deg D \le \ell} \|D\Psi\|^2_{L^2(\Gamma\backslash G)}, \qquad \Psi \in C_c^\infty(\Gamma\backslash G).$$

On a fixed compact support, changing the basis or replacing this norm by any equivalent differential Sobolev norm changes it only by a multiplicative constant. Products, right translations in a fixed compact set, and the thickening operations of Chapter 2 satisfy

$$\mathcal{S}_\ell(\Psi_1\Psi_2) \ll_{\ell,\Omega} \mathcal{S}_\ell(\Psi_1)\mathcal{S}_\ell(\Psi_2), \qquad \mathcal{S}_\ell(R_g\Psi) \ll_{\ell,\Omega} \mathcal{S}_\ell(\Psi)$$

for g in a fixed compact set Ω.

Proof. All statements are finite-dimensional consequences of the Leibniz rule and the adjoint action on the chosen basis of $\mathfrak{g}$. On compact Ω, the coefficients expressing $\mathrm{Ad}(g)$ in that basis and their inverses are uniformly bounded. There are finitely many differential monomials of degree at most ℓ, so summing the corresponding L^2 estimates proves the assertions. □

Theorem 1.4 (Mohammadi–Oh effective matrix coefficient theorem). *Assume $\Gamma < G$ has operational spectral-gap data (s_0, n_0) in the sense of Definition 1.1. Then there exist $\eta > 0$ and an integer $\ell \geq 1$, depending only on that data and n, such that for all real $\Psi_1, \Psi_2 \in C_c^\infty(\Gamma\backslash G)$,*

$$\begin{aligned} e^{(n-1-\delta)t} &\int_{\Gamma\backslash G} \Psi_1(ga_t)\Psi_2(g)\, dm^{\mathrm{Haar}}(g) \\ &= \frac{m^{\mathrm{BR}}(\Psi_1)m^{\mathrm{BR}_*}(\Psi_2)}{|m^{\mathrm{BMS}}|} + O\Big(e^{-\eta t}\mathcal{S}_\ell(\Psi_1)\mathcal{S}_\ell(\Psi_2)\Big). \end{aligned}$$

Proof. This is Theorem 5.3. The proof is placed in Appendix A so the chapter can display the dynamical consequence without suppressing the representation-theoretic argument. The appendix proves the uniform complementary-series K-type asymptotic, the remainder-spectrum decay, the critical-part Sobolev summation, and the final identification of the leading coefficient using the already internalized qualitative Roblin theorem. □

Proof and provenance. The scope is the effective matrix-coefficient theorem of Mohammadi–Oh [MO15], but the hypothesis is written in the operational decomposition form actually used in their proof. This avoids treating the rank-one unitary-dual classification as an invisible dependency. The proof engine is internal to Appendix A; qualitative Roblin is used only to identify the leading coefficient, not to manufacture the rate.

Proposition 1.5 (Effective BMS mixing transfer)**.** *For every compact $\Omega \subset T^1M$ there are $\eta_{\mathrm{BMS}} > 0$ and an integer r such that, for $f, h \in C_c^\infty(T^1M)$ supported in Ω,*

$$\left| \int f \circ g^t\, h\, dm^{\mathrm{BMS}} - \frac{m^{\mathrm{BMS}}(f)m^{\mathrm{BMS}}(h)}{|m^{\mathrm{BMS}}|} \right| \ll_\Omega e^{-\eta_{\mathrm{BMS}} t}\mathcal{S}_r(f)\mathcal{S}_r(h).$$

Proof. Repeat the stable/unstable thickening from Lemma 1.1, but keep every error term. Let ε be the plaque thickness. Smooth cutoffs at scale ε have Sobolev size at most $C\varepsilon^{-A}\mathcal{S}_r(f)$ and $C\varepsilon^{-A}\mathcal{S}_r(h)$ for fixed A, r depending only on the local dimension and the chosen finite box cover. Theorem 1.4 therefore gives a spectral error

$$O\Big(e^{-\eta t}\varepsilon^{-2A}\mathcal{S}_r(f)\mathcal{S}_r(h)\Big).$$

The local-product densities of Theorem 7.2 are Hölder on the fixed compact cover; moving the two plaque boundaries by $O(\varepsilon)$ changes the BMS correlation by

$$O\Big(\varepsilon^{\theta}\mathcal{S}_r(f)\mathcal{S}_r(h)\Big)$$

for some $\theta > 0$. The inner and outer thickenings thus squeeze the desired correlation with total error

$$C\left(e^{-\eta t}\varepsilon^{-2A} + \varepsilon^{\theta}\right)\mathcal{S}_r(f)\mathcal{S}_r(h).$$

Set $\varepsilon = e^{-ct}$ and choose $c = \eta/(2A+\theta)$. Both exponents become positive and equal after balancing:

$$\eta_{\mathrm{BMS}} = \frac{\eta\theta}{2A+\theta} > 0.$$

The main term is the product of BMS means by the same local product calculation as in the qualitative proof. Summing over the finite box cover proves the proposition. □

Worked Example 1.6 (Why a qualitative proof gives no rate)**.** The proof of Theorem 3.1 takes $t \to \infty$ for a fixed thickening width ε and only afterwards sends $\varepsilon \to 0$. Without a quantitative correlation estimate there is no permitted relation between t and ε. The theorem above allows $\varepsilon = e^{-ct}$, and balancing $e^{-\eta t}\varepsilon^{-2A}$ against ε^{θ} produces the explicit positive rate $\eta\theta/(2A+\theta)$.

CHAPTER 8

Counting and equidistribution consequences

Why this matters. The preceding chapters were chosen to reproduce the qualitative Oh–Shah counting mechanism in a transparent order: a smooth measure on an admissible normal bundle is pushed forward by the geodesic flow, mixing identifies its BR limit, and an elementary radial integral converts equidistribution into orbit counting.

1. Equidistribution and orbit counts

Lemma 1.1 (Oh–Shah qualitative equidistribution in the present scope)**.** *Let E be an unstable horosphere or the unit normal bundle of a proper complete totally geodesic subspace, and assume $|m^{\mathrm{BMS}}| < \infty$ and $|\mu_E^{\mathrm{PS}}| < \infty$. If $F \subset E$ is Borel with $\mu_E^{\mathrm{PS}}(\partial F) = 0$, then for every $\psi \in C_c(T^1M)$,*

$$\lim_{t\to\infty} e^{(n-1-\delta)t} \int_F \psi(g^t v)\, d\mu_E^{\mathrm{Leb}}(v) = \frac{\mu_E^{\mathrm{PS}}(F)}{|m^{\mathrm{BMS}}|} m^{\mathrm{BR}}(\psi).$$

Proof. Cover the compact part of $\operatorname{supp}\psi$ by finitely many backward admissible Hopf boxes. In one box, decompose the intersection of g^tF with the box into unstable plaques and a transversal. The Lebesgue conditional on the moving E-plaque has dimension $n-1$, while its PS replacement has dimension δ; pulling the plaque back by g^{-t} therefore contributes the factor $e^{-(n-1-\delta)t}$.

Choose inner and outer plaque envelopes $F_\eta^- \subset F \subset F_\eta^+$ and corresponding continuous transverse cutoffs. Proposition 1.3 converts the renormalized E-integral, for fixed η, into a finite sum of BMS correlations. Theorem 2.1 sends those correlations to products of BMS means. The local product formula of Theorem 7.2 identifies the transverse mean with $\mu_E^{\mathrm{PS}}(F_\eta^\pm)$ and the plaque mean with $m^{\mathrm{BR}}(\psi_\eta^\pm)$. Hence

$$\frac{\mu_E^{\mathrm{PS}}(F_\eta^-) m^{\mathrm{BR}}(\psi_\eta^-)}{|m^{\mathrm{BMS}}|} \le \liminf_{t\to\infty}(\cdots) \le \limsup_{t\to\infty}(\cdots) \le \frac{\mu_E^{\mathrm{PS}}(F_\eta^+) m^{\mathrm{BR}}(\psi_\eta^+)}{|m^{\mathrm{BMS}}|}.$$

Radon regularity, continuity of the box densities, and the boundary-null hypothesis give $\mu_E^{\mathrm{PS}}(F_\eta^+ \setminus F_\eta^-) \to 0$ and $m^{\mathrm{BR}}(\psi_\eta^+ - \psi_\eta^-) \to 0$. Let $\eta \downarrow 0$.

If F is noncompact, first exhaust it by compact subsets with PS-small complement; Theorem 1.2 makes that exhaustion legitimate. This is exactly the thickening/sandwich architecture of the qualitative Oh–Shah theorem, now expressed in the measures already constructed in 53–54. $\square$

THEOREM 1.2 (Standard G_o-invariant norm count). *Let G act linearly on a finite-dimensional real vector space V, let $w_0 \neq 0$, and suppose either G_{w_0} is symmetric or $G_{\mathbb{R}w_0}$ is parabolic. Assume $w_0\Gamma$ is discrete and that the associated skinning size $\mathrm{sk}_\Gamma(w_0) = |\mu_E^{\mathrm{PS}}|$ is finite. Let $\|\cdot\|$ be a G_o-invariant norm. If $\lambda > 0$ is the logarithm of the top A-eigenvalue on $\mathbb{R}\text{-}\mathrm{span}(w_0G)$ and*

$$w_0^\lambda = \lim_{r\to\infty} e^{-\lambda r} w_0 a_r \neq 0,$$

then

$$\#\{w \in w_0\Gamma : \|w\| < T\} \sim \frac{|\nu_o|\,\mathrm{sk}_\Gamma(w_0)}{\delta\,|m^{\mathrm{BMS}}|\,\|w_0^\lambda\|^{\delta/\lambda}} T^{\delta/\lambda}.$$

PROOF. Use the generalized Cartan decomposition $G = HA^+K$ (or the two signed A-chambers when required), where H is the relevant stabilizer. The G_o-invariance of the norm removes the K-variable from the radial cutoff. Since $e^{-\lambda r} w_0 a_r \to w_0^\lambda$, the inequality $\|w_0 a_r k\| < T$ is equivalent, with an $o(1)$ error in the radial endpoint, to

$$r < \lambda^{-1} \log \frac{T}{\|w_0^\lambda\|}.$$

Unfold the orbit sum over $\Gamma_H\backslash\Gamma$ into the H-orbit integral. Lemma 1.1 gives the shell density with coefficient $\mathrm{sk}_\Gamma(w_0) m^{\mathrm{BR}}(\psi)/|m^{\mathrm{BMS}}|$. In KAN coordinates, the BR integral of the K-averaged test function contributes exactly $|\nu_o|$, while the radial Haar factor is $e^{\delta r} dr$ after the renormalization. Therefore the cumulative radial contribution is

$$\frac{|\nu_o|\,\mathrm{sk}_\Gamma(w_0)}{|m^{\mathrm{BMS}}|} \int_0^{\lambda^{-1}\log(T/\|w_0^\lambda\|)} e^{\delta r}\, dr.$$

Theorem 1.2 (or direct integration) gives the factor $1/\delta$, and Proposition 1.3 converts the upper endpoint into $T^{\delta/\lambda}\|w_0^\lambda\|^{-\delta/\lambda}$. Inner/outer norm balls remove the $o(1)$ radial error. The normalization independence of the constant is Theorem 1.2. $\square$

Proposition 1.3 (Power-saving counting transfer)**.** *Assume Theorem 1.4 and an effectively well-rounded expanding family $B_T \subset H\backslash G$: the mass of its ε-boundary thickening is $O(\varepsilon^q \mathcal{M}(B_T))$ for some $q > 0$, and its terminal radial scale t_T satisfies $e^{-\eta t_T} \ll \mathcal{M}(B_T)^{-b}$ for some $b > 0$, where η is the rate in Theorem 1.4. Then there is $\kappa > 0$ such that*

$$\#([e]\Gamma \cap B_T) = \mathcal{M}(B_T) + O(\mathcal{M}(B_T)^{1-\kappa}).$$

Proof. Let $M_T = \mathcal{M}(B_T)$ and choose inner and outer smooth approximations to the counting indicator at geometric scale ε. Effective well-roundedness gives a boundary error

$$O(\varepsilon^q M_T).$$

The Sobolev norms of the smoothing kernels grow at most like ε^{-A} for a fixed $A > 0$. In the radial parameter t underlying the HAK decomposition, Theorem 1.4 contributes $e^{-\eta t}\varepsilon^{-A}$ relative to the main shell mass. For an admissible expanding family there is a positive b such that, after decomposing into dyadic radial shells,

$$e^{-\eta t} \ll M_T^{-b}$$

on every terminal shell; the finitely many earlier shells contribute a smaller-order term. Thus the total relative error is

$$O(\varepsilon^q + M_T^{-b}\varepsilon^{-A}).$$

Put $\varepsilon = M_T^{-\beta}$. The two exponents are βq and $b - A\beta$. Choosing

$$\beta = \frac{b}{A+q}$$

balances them and gives

$$\kappa = \frac{bq}{A+q} > 0.$$

Multiplying by M_T yields

$$\#([e]\Gamma \cap B_T) = M_T + O(M_T^{1-\kappa}).$$

The constants are uniform over the effectively well-rounded family because the Sobolev order, the boundary exponent q, and the radial comparison exponent b are part of its fixed data. $\square$

Worked Example 1.4 (Standard representation)**.** For the standard representation of $\mathrm{SO}(n,1)$ one has $\lambda = 1$. The theorem therefore predicts growth T^δ. If the Patterson–Sullivan density is rescaled by c, then $|\nu_o|$ and the skinning size each gain a factor c, while $|m^{\mathrm{BMS}}|$ gains c^2; the displayed leading constant is unchanged.

APPENDIX A

The rank-one effective matrix-coefficient engine

Why this matters. The effective theorem in Chapter 7 is not a formal consequence of qualitative mixing. Its extra content is representation theoretic: isolate the complementary-series summand at the critical exponent, prove a uniform asymptotic for every pair of K-types, control the remainder spectrum at a strictly smaller exponent, and then sum the K-types by compact Sobolev estimates. This appendix reconstructs exactly that engine in the real-rank-one group $G = \mathrm{SO}^+(n,1)$.

1. The operational spectral-gap decomposition

We use Definition 1.1 from Chapter 7. The point of the definition is that the proof starts from the actual orthogonal decomposition of the L^2-representation; it does not need a separate classification of every irreducible unitary representation of G.

2. K-types and compact Sobolev summation

Let Ω_K be the positive compact Casimir. If $\sigma \in \widehat{K}$, write $d_\sigma = \dim \sigma$ and $c(\sigma) \geq 1$ for the eigenvalue of $1 + \Omega_K$ on the σ-isotypic space.

Lemma 2.1 (Compact Sobolev summability)**.** *For every $B \geq 0$ there is an integer $r = r(B, K)$ such that*

$$\sum_{\sigma \in \widehat{K}} d_\sigma^B c(\sigma)^{-r} < \infty.$$

Moreover, if P_σ is the orthogonal K-type projection and v is a smooth vector in a unitary K-representation, then

$$\|P_\sigma v\| \leq c(\sigma)^{-r/2} \|(1 + \Omega_K)^{r/2} v\|.$$

Proof. The second statement follows because $(1+\Omega_K)^{r/2}$ acts by $c(\sigma)^{r/2}$ on the σ-isotypic space. For the first, use highest weights. Weyl's dimension formula bounds d_σ by a fixed polynomial in $1 + \|\lambda_\sigma\|$, while the Casimir eigenvalue is comparable to $1 + \|\lambda_\sigma\|^2$. The number of dominant integral weights with $j \le \|\lambda\| < j+1$ is $O(j^{\operatorname{rank} K-1})$. Hence the series is dominated by

$$\sum_{j\ge 1} j^{BD+\operatorname{rank} K-1-2r}$$

for a constant $D = D(K)$. Choose $r > (BD + \operatorname{rank} K)/2$. $\square$

3. The induced model and the rank-one radial expansion

Fix s with $(n-1)/2 < s < n-1$. In the compact picture of $\mathcal{U}(\upsilon, s-n+1)$, smooth vectors are smooth functions $f : K \to V_\upsilon$ satisfying $f(mk) = \upsilon(m)f(k)$, with

$$(U_s(g)f)(k) = e^{(s-n+1)H(g^{-1}k)} f(\kappa(g^{-1}k)),$$

where $g = \kappa(g)\exp(H(g))n(g)$ is the KAN decomposition. For $\sigma, \tau \in \widehat{K}$, let P_σ, P_τ denote the corresponding K-type projections.

Lemma 3.1 (Rank-one Eisenstein-integral equation)**.** *For fixed σ, τ, the matrix-valued function*

$$F_{\tau\sigma}(t;s) = P_\tau U_s(a_t) P_\sigma$$

is analytic in s and, after setting $q = e^{-t}$, satisfies on $0 < q < 1$ a regular-singular matrix differential equation whose indicial exponents are $n-1-s$ and s. Consequently there are matrix coefficients $C_+(s)$, $C_-(s)$ and convergent expansions

$$F_{\tau\sigma}(t;s) = e^{(s-n+1)t}\Big(C_+(s) + \sum_{j\ge 1} A_j(s)e^{-jt}\Big) + e^{-st}\Big(C_-(s) + \sum_{j\ge 1} B_j(s)e^{-jt}\Big).$$

On every compact parameter circle avoiding the discrete resonance set, there are N, C, t_0 independent of σ, τ such that

$$\|A_j(s)\| + \|B_j(s)\| \le C d_\sigma^N d_\tau^N e^{jt_0}.$$

Proof. Start from the compact-picture integral for $P_\tau U_s(a_t) P_\sigma$. Differentiation under the K-integral is legitimate on compact s-sets, proving analyticity. Apply the Casimir Ω_G in two ways: it acts by the scalar infinitesimal character of the induced representation, while its radial part in the KAK decomposition is a second-order operator in t plus terms with coefficients that are rational functions of e^{-t} and K-direction operators. After $q = e^{-t}$ those coefficients are analytic at $q = 0$. The constant term of the radial operator has characteristic roots $n-1-s$ and s; this gives the two displayed indicial factors.

For completeness, write a Frobenius ansatz $q^{n-1-s}\sum_{j\geq 0} A_j q^j$ and $q^s\sum_{j\geq 0} B_j q^j$. Equating the coefficient of q^j gives

$$D_j(s)A_j = \sum_{i<j} R_{j-i}(\sigma,\tau;s)A_i, \qquad \widetilde{D}_j(s)B_j = \sum_{i<j} \widetilde{R}_{j-i}(\sigma,\tau;s)B_i.$$

Here $D_j, \widetilde{D}_j$ are scalar quadratic polynomials in j whose zeros form the discrete resonance set, and each $R_k, \widetilde{R}_k$ is a finite linear combination of K-Lie-algebra operators of bounded degree. On an irreducible K-type, a fixed Lie-algebra monomial has operator norm bounded polynomially in the highest weight, hence polynomially in d_σ and d_τ after increasing the exponent. On a compact parameter circle away from resonance, $|D_j|^{-1}+|\widetilde{D}_j|^{-1} = O((1+j)^{-2})$. Induction in j therefore gives

$$\|A_j\| + \|B_j\| \leq C d_\sigma^N d_\tau^N R^j$$

for one $R = e^{t_0}$ independent of the two K-types. The resulting power series converge for $q < R^{-1}$. Analytic continuation in t extends the identity to every $t > t_0$.

At resonant parameters, choose a small parameter circle avoiding all resonances. The original integral is analytic in s; subtract the leading branch and apply the maximum principle to the analytic matrix entries. The same polynomial K-type bound on the circle therefore holds at the center. This is the rank-one Frobenius/maximal-principle argument underlying the Eisenstein-integral expansion. □

Lemma 3.2 (Polynomial control of the radial coefficients)**.** *Fix $t_0 > 0$ and a compact parameter set that avoids the locally finite resonance set of the Frobenius recurrence. There are $C, N > 0$, independent of the K-types σ, τ, such that the coefficients in Lemma 3.1 satisfy*

$$\|A_j(s)\| + \|B_j(s)\| \leq C d_\sigma^N d_\tau^N e^{jt_0}.$$

The two leading connection operators also satisfy

$$\|C_+(s)\| + \|C_-(s)\| \leq C d_\sigma^N d_\tau^N.$$

Proof. Choose root vectors $X_{\pm\alpha} = Y_{\pm\alpha} + Z_{\pm\alpha}$ with $Y_{\pm\alpha} \in \mathfrak{k}_{\mathbb{C}}$ and let Ω_M be the compact Casimir of M. The radial Casimir calculation gives a recursion of the form

$$\Lambda_j(s)\Gamma_j = \sum_{i<j} Q_{j-i}(s;\tau(Y_\alpha),\tau(Y_{-\alpha}),\sigma(Y_\alpha),\sigma(Y_{-\alpha}),\Omega_M)\Gamma_i,$$

where

$$\Lambda_j(s) = -j^2 + j(2s-n+1) - F_M$$

and F_M acts through the M-Casimir on the finite-dimensional intertwining space. Every Q_k has uniformly bounded degree in the displayed compact-Lie-algebra operators. If q_σ, q_τ are highest weights, the norm of a fixed Lie-algebra monomial on the corresponding modules is polynomial in $1 + \|q_\sigma\| + \|q_\tau\|$. Weyl's dimension formula therefore gives

$$\|Q_k\| \le C_1 d_\sigma^{N_1} d_\tau^{N_1}.$$

Away from the resonance set, $\Lambda_j(s)$ is invertible. For large j its inverse is $O(j^{-2})$ uniformly on the compact parameter set; the finitely many small j contribute another polynomial factor in d_σ, d_τ. If

$$M_J = \max_{i \le J} e^{-it_0} \|\Gamma_i\|,$$

the recurrence and $\sum_{k \ge 1} e^{-kt_0} < \infty$ give by induction

$$\|\Gamma_j\| \le C_2 d_\sigma^N d_\tau^N e^{jt_0}.$$

This is the claimed estimate for the Frobenius coefficients.

For the leading connection operator, use the rank-one N^- integral obtained by taking the limit of the Eisenstein integral along the expanding chamber:

$$C_+(s)T = \int_{N^-} T\,\sigma(\kappa(n)^{-1}) e^{-((s-\rho)\alpha+\rho)(H(n))}\,dn.$$

On a compact subset of the complementary interval the scalar integral converges uniformly. The K-actions are unitary and the elementary M-averaging operator defining T has norm bounded polynomially in d_σ, d_τ; hence $\|C_+(s)\| \ll d_\sigma^N d_\tau^N$. To control C_-, evaluate the two-branch expansion at one fixed large t_1. The operator multiplying C_- is the identity plus a Frobenius tail of norm $< 1/2$ after increasing t_1 by $O(\log d_\sigma d_\tau)$. It is therefore invertible by a Neumann series. The integral itself, the C_+ term, and that inverse all have polynomial K-type bounds, which yields the same bound for C_-. This is the connection-coefficient estimate needed below. □

Remark 3.3. The preceding recurrence and connection estimate are the two quantitative places where the K-type dimensions enter. They correspond to the Harish–Chandra Eisenstein-integral expansion and its polynomial K-type control in the primary proof; no uniformity in σ, τ is obtained by qualitative mixing alone.

Proposition 3.4 (Uniform complementary-series K-type asymptotic)**.** *Fix $(n-1)/2 < s < n-1$. There exist $\eta_s > 0$, $N \ge 1$ and a sesquilinear coefficient $C_s(\cdot,\cdot)$ such that, for unit vectors v_σ, v_τ in K-types σ, τ of any complementary-series representation with parameter s,*

$$\langle U_s(a_t)v_\sigma, v_\tau\rangle = e^{(s-n+1)t} C_s(v_\sigma, v_\tau) + O\Big(d_\sigma^N d_\tau^N e^{(s-n+1-\eta_s)t}\Big),$$

uniformly for $t \geq 2$.

PROOF. Apply Lemma 3.1. The first nonconstant term in the leading branch gains e^{-t}, while the second indicial branch is e^{-st}. Since $s > (n-1)/2$, both are smaller than $e^{(s-n+1)t}$ by a fixed exponential factor. The coefficient estimates in the lemma make the sum of all nonleading Frobenius terms $O(d_\sigma^N d_\tau^N e^{(s-n+1-\eta_s)t})$. Pair the leading matrix $C_+(s)$ with v_σ, v_τ to obtain C_s. □

4. The remainder spectrum

Lemma 4.1 (Uniform decay below the gap)**.** *Let W be a unitary G-representation satisfying the remainder clause in Definition 1.1. For every $\epsilon > 0$ there exists an integer r_0 such that*

$$|\langle a_t w_1, w_2 \rangle| \ll_\epsilon e^{(s_0-n+1+\epsilon)t} \mathcal{S}_{r_0}(w_1)\mathcal{S}_{r_0}(w_2)$$

for smooth vectors $w_1, w_2 \in W$.

PROOF. Decompose W spectrally. Tempered constituents satisfy the Cowling–Haagerup–Howe estimate proved in Volume 5, hence their K-finite coefficients are bounded by a fixed polynomial in d_σ, d_τ times the Harish–Chandra function $\Xi(a_t) \ll (1+t)^C e^{-(n-1)t/2}$. Because $s_0 > (n-1)/2$, this is $O_\epsilon(e^{(s_0-n+1+\epsilon)t})$.

Every non-tempered constituent allowed by the operational gap has complementary parameter $s \leq s_0$. Apply Proposition 3.4 on a compact parameter interval ending at $s_0 + \epsilon/2$; the leading exponent is at most $s_0 - n + 1$, and the error is smaller. Thus every pair of K-finite vectors satisfies the desired exponential estimate with a polynomial K-type factor. Lemma 2.1 absorbs that factor into a finite compact Sobolev norm. Integrating the constituent-wise estimate over the spectral decomposition and using Cauchy–Schwarz gives the displayed bound. □

5. Summing the critical complementary-series part

Lemma 5.1 (Multiplicity of a K-type in the critical part)**.** *Under Definition 1.1, the multiplicity $m_\delta(\sigma)$ of a K-type σ in $\mathcal{H}_\delta^\dagger$ satisfies*

$$m_\delta(\sigma) \leq d_\sigma^{n_0+1}.$$

Hence one may choose an orthonormal basis Θ_σ in the σ-isotypic part with $\#\Theta_\sigma \leq d_\sigma^{n_0+2}$.

PROOF. By Frobenius reciprocity, the multiplicity of σ in $\mathcal{U}(v, \delta - n + 1)|_K \simeq \mathrm{Ind}_M^K v$ is $[\sigma|_M : v]$. Therefore

$$m_\delta(\sigma) \le \sum_{v \subset \sigma|_M} (\dim v)^{n_0} [\sigma|_M : v].$$

Every occurring $\dim v$ is at most d_σ, while the sum of multiplicity times dimension is d_σ. This gives $m_\delta(\sigma) \le d_\sigma^{n_0+1}$. Multiplying by the dimension of σ gives the bound on an orthonormal basis of the whole isotypic space. □

Proposition 5.2 (Effective critical-part asymptotic). *There exist $\eta_0 > 0$ and $r \ge 1$ such that for smooth compactly supported Ψ_1, Ψ_2 and their orthogonal projections Ψ_i' to $\mathcal{H}_\delta^\dagger$,*

$$\langle a_t \Psi_1', \Psi_2' \rangle = e^{(\delta-n+1)t} E(\Psi_1, \Psi_2) + O\Big(e^{(\delta-n+1-\eta_0)t} \mathcal{S}_r(\Psi_1) \mathcal{S}_r(\Psi_2)\Big),$$

where E is a continuous bilinear functional.

PROOF. Expand Ψ_i' in the orthonormal bases Θ_σ. Proposition 3.4 at $s = \delta$ gives, for $v_\sigma \in \Theta_\sigma$ and $v_\tau \in \Theta_\tau$,

$$\langle a_t v_\sigma, v_\tau \rangle = e^{(\delta-n+1)t} C_\delta(v_\sigma, v_\tau) + O(d_\sigma^N d_\tau^N e^{(\delta-n+1-\eta_0)t}).$$

The number of basis vectors in the two isotypic spaces contributes at most $d_\sigma^{n_0+2} d_\tau^{n_0+2}$. Choose r in Lemma 2.1 so large that

$$\sum_{\sigma,\tau} d_\sigma^{N+n_0+2} d_\tau^{N+n_0+2} c(\sigma)^{-r} c(\tau)^{-r} < \infty.$$

Since $|\langle \Psi, v_\sigma \rangle| \le c(\sigma)^{-r} \mathcal{S}_{2r}(\Psi)$, both the leading double series and the error series converge absolutely. Summing term by term proves the proposition, after increasing the Sobolev order once. □

THEOREM 5.3 (Rank-one effective matrix coefficient). *Let $\mathcal{H} = L^2(\Gamma\backslash G)$ satisfy Definition 1.1. Assume the qualitative Roblin asymptotic of Theorem 3.1. Then there are $\eta > 0$ and $r \ge 1$ such that*

$$e^{(n-1-\delta)t} \langle a_t \Psi_1, \Psi_2 \rangle = \frac{m^{\mathrm{BR}}(\Psi_1) m^{\mathrm{BR}_*}(\Psi_2)}{|m^{\mathrm{BMS}}|} + O\Big(e^{-\eta t} \mathcal{S}_r(\Psi_1) \mathcal{S}_r(\Psi_2)\Big)$$

for real $\Psi_1, \Psi_2 \in C_c^\infty(\Gamma\backslash G)$.

PROOF. Write $\Psi_i = \Psi_i' + \Psi_i^\perp$ according to the operational-gap decomposition. Orthogonality and G-invariance eliminate mixed terms. Lemma 4.1 gives

$$\langle a_t \Psi_1^\perp, \Psi_2^\perp \rangle = O\Big(e^{(s_0-n+1+\epsilon)t} \mathcal{S}_{r_0}(\Psi_1) \mathcal{S}_{r_0}(\Psi_2)\Big).$$

Choose $\epsilon < (\delta - s_0)/2$. Proposition 5.2 treats the critical part. Hence for some $\eta > 0$,

$$\langle a_t \Psi_1, \Psi_2 \rangle = e^{(\delta-n+1)t} E(\Psi_1, \Psi_2) + O\Big(e^{(\delta-n+1-\eta)t} \mathcal{S}_r(\Psi_1)\mathcal{S}_r(\Psi_2)\Big) .$$

Multiply by $e^{(n-1-\delta)t}$ and let $t \to \infty$. Theorem 3.1 identifies the limit, so

$$E(\Psi_1, \Psi_2) = \frac{m^{\mathrm{BR}}(\Psi_1) m^{\mathrm{BR}_*}(\Psi_2)}{|m^{\mathrm{BMS}}|}.$$

Substitution proves the theorem. □

Proof and provenance. The proof architecture is the rank-one argument of Mohammadi–Oh, Section 3: uniform K-finite complementary-series asymptotics, decomposition of the critical and remainder spectra, compact-Casimir summation of K-types, and identification of the leading bilinear form by Roblin's qualitative limit [MO15]. The present formulation makes the consumed spectral decomposition an explicit hypothesis, so no classification of the full unitary dual is hidden inside the phrase "spectral gap." The tempered estimate is inherited from the Harish–Chandra/CHH machinery proved of Volume 5; the non-tempered rank-one radial calculation is reconstructed above.

Volume 56

Horospherical Dynamics in Geometrically Finite Spaces

Volume 56 scope and prerequisites

Prerequisites. Throughout this volume

$$G = \mathrm{SO}^+(n,1), \qquad n \geq 2,$$

$A = \{a_t\}$ is the geodesic one-parameter subgroup, $M = Z_K(A)$, and $N = N^+$ is the expanding horospherical subgroup for the right A-action. The group $\Gamma < G$ is torsion-free, non-elementary, geometrically finite, and Zariski dense. We write $Y = \Gamma\backslash G$ for the oriented-frame quotient and $T^1M = \Gamma\backslash G/M$ for the unit tangent bundle. For $g \in G$, g^- denotes the backward endpoint of the oriented geodesic determined by gM. The Patterson–Sullivan, BMS, and Burger–Roblin normalizations are those of Volumes 53–55.

Proof and provenance. Winter and Roblin provide the theorem-concordance targets [Win15; Rob03]. The mathematical load is internal at the real-hyperbolic scope used here: Chapters 1–6 prove the radial/nonradial geometry, closed-horosphere measures, BR/BMS disintegration and compact-M recovery; Chapter 7 proves the conservative horospherical ratio theorem and recurrent transverse-measure uniqueness, then derives the MN classification and Winter's recovery from MN to N. Chapter 8 proves horosphere equidistribution from the BMS mixing of Volume 55. Thus the citations record provenance and scope rather than replacing a research-level proof.

Reader guide: a concrete model

Proof architecture at a glance.

$$\boxed{\text{boundary densities}} \longrightarrow \boxed{\text{BMS measure}} \longrightarrow \boxed{\text{geodesic mixing}} \longrightarrow \boxed{\text{horosphere equidistribution}}$$

CHAPTER 1

Horospherical recurrence

Why this matters. A horosphere is controlled by the endpoint opposite to the direction in which the horospherical group moves. For the right expanding group N, that endpoint is g^-. The radial-limit-set condition is therefore the geometric recurrence input behind every nonperiodic horospherical argument in this volume. We first formulate this safely as recurrence of the *renormalized* horosphere under the A-flow; no unproved assertion about density of a raw N-orbit is used.

1. Radial returns and A-renormalization

Lemma 1.1 (Orbit-point form of radiality). *For $\xi \in \Lambda(\Gamma)$ the following are equivalent.*

(1) ξ is radial.

(2) For one (hence every) geodesic ray r ending at ξ, there are $R < \infty$, $t_j \to \infty$, and $\gamma_j \in \Gamma$ with $d(r(t_j), \gamma_j o) \leq R$.

The constant R may change when the base point or the chosen ray changes, but radiality does not.

Proof. This is the base-point independence built into the definition of the radial limit set. Two rays with the same endpoint in $\mathbb{H}^n$ stay at bounded Hausdorff distance, and changing o to o' changes every orbit point γo to $\gamma o'$ by the fixed distance $d(o, o')$. Thus the existence of a uniform R and infinitely many orbit points close to the ray is independent of both choices. □

Theorem 1.2 (Compact-return criterion for a radial endpoint). *For $g \in G$, the following are equivalent.*

(1) $g^- \in \Lambda_r(\Gamma)$.

(2) There are $t_j \to +\infty$, $\gamma_j \in \Gamma$, and $m_j \in M$ such that $\gamma_j g a_{-t_j} m_j$ remains in a compact subset of G.

(3) The backward geodesic orbit $\Gamma g a_{-t} M$ returns infinitely often to a compact subset of $T^1 M = \Gamma \backslash G / M$.

PROOF. The point $\pi(ga_{-t})$ runs along the geodesic ray with endpoint g^-. If g^- is radial, Lemma 1.1 gives times t_j and orbit points within a fixed distance R. Left multiplication by γ_j brings the base point into $B(o,R)$. The fiber of the unit tangent bundle over that compact ball is compact, and right multiplication by a suitable $m_j \in M$ chooses a representative in one compact subset of G. This proves (1)⇒(2) and hence (3). Conversely, lift a compact return set in T^1M to finitely many compact subsets of $T^1\mathbb{H}^n$. The lifted base points stay a uniformly bounded distance from Γo, so Lemma 1.1 gives radiality. $\square$

Proposition 1.3 (Renormalization of unstable plaques). *In exponential coordinates $N \cong \mathbb{R}^{n-1}$ normalized by $a_t n_u a_{-t} = n_{e^t u}$, a bounded unstable plaque at the compact return point ga_{-t_j} corresponds, before applying a_{-t_j}, to an N-plaque whose parameter scale is multiplied by e^{t_j}. Consequently all local product, holonomy, and conditional-measure estimates needed along a radial sequence may be made on one compact family of Hopf boxes after A-renormalization.*

PROOF. The displayed conjugation identity is the definition of the expanding normalization. If $|u| \le R$, then

$$ga_{-t_j}n_u = g(a_{-t_j}n_u a_{t_j})a_{-t_j} = gn_{e^{-t_j}u}a_{-t_j}.$$

Equivalently, flowing a fixed plaque at time $-t_j$ forward by t_j expands its N-parameter by e^{t_j}. Theorem 1.2 places the centers, after Γ and M adjustment, in one compact set; since M normalizes N, the same is true of the corresponding plaque charts. Continuous Jacobians and Busemann densities are therefore uniformly controlled on a finite collection of compact Hopf boxes. $\square$

Worked Example 1.4 (The modular cusp). In $\mathbb{H}^2$, take $N = \{z \mapsto z+u\}$. If the backward endpoint is an irrational point of the limit set of $\mathrm{PSL}_2(\mathbb{Z})$, the backward geodesic returns infinitely often to the compact part of the modular surface, and Theorem 1.2 supplies a compact renormalization sequence. If the endpoint is the parabolic point ∞, the backward ray eventually remains in the cusp and no such compact-return sequence exists.

CHAPTER 2

Radial versus parabolic limit points

Why this matters. For a geometrically finite group there are no mysterious nonradial points inside the limit set: every such point is bounded parabolic. This turns the topological classification of horospheres into a clean radial-versus-cusp dichotomy.

1. The limit-set dichotomy

Choose the standard disjoint Γ-equivariant family of horoballs from the cusp decomposition of Volume 53, and let K be the compact thick part of the truncated convex core.

Lemma 1.1 (A ray in the convex hull either returns or enters one cusp)**.** *Let $r : [0, \infty) \to \mathrm{Hull}(\Lambda(\Gamma))$ be a geodesic ray. Modulo Γ, either r meets the compact thick part K infinitely often, or there is a bounded parabolic fixed point ξ and $T < \infty$ such that $r([T, \infty))$ lies in the standard horoball based at ξ.*

Proof. The truncated convex core is compact. Outside its Γ-translates, the convex hull is the disjoint union of the chosen cusp horoballs. If the projected ray meets K infinitely often we are in the first case. Otherwise, after its last visit to K the connected tail of the ray lies in the union of pairwise disjoint open cusp regions. A connected geodesic tail cannot jump between two disjoint horoballs without crossing the thick part. Hence one cusp contains the entire tail. □

Theorem 1.2 (Geometrically finite limit-set decomposition). *At the real-hyperbolic scope of this arc,*

$$\Lambda(\Gamma) = \Lambda_r(\Gamma) \sqcup \Lambda_{bp}(\Gamma),$$

where Λ_{bp} is the set of bounded parabolic fixed points. In particular, every nonradial point of the limit set is parabolic.

Proof. Let $\xi \in \Lambda(\Gamma)$ and choose a ray r from the convex hull to ξ. Apply Lemma 1.1. Infinite returns to K lift to points of Γo at uniformly bounded distance from r, so ξ is radial. If the ray eventually remains in the cusp at a parabolic point p, then its endpoint is p, so ξ is bounded parabolic. A bounded parabolic point cannot be radial: modulo its stabilizer, a sufficiently deep cusp neighborhood is disjoint from the fixed compact thick part, whereas radiality would force infinitely many thick-part returns. Thus the union is disjoint. ☐

Proposition 1.3 (Nonradial endpoints and cusp stabilizers)**.** *If $g^- \notin \Lambda_r(\Gamma)$, then either $g^- \notin \Lambda(\Gamma)$ and Γ_{g^-} is trivial, or g^- is bounded parabolic and Γ_{g^-} acts cocompactly by Euclidean isometries on a subspace of a horosphere centered at g^-. Its rotational image in M is virtually abelian; its closure is a compact subgroup of M. In the crystallographic full-rank case this image is finite, while screw-parabolic phenomena in higher dimension make it important not to assume finiteness in general.*

Proof. A nontrivial element fixing a boundary point has that point in its limit set, proving the first assertion. In the second case, conjugate g^- to ∞ in the upper-half-space model. The stabilizer acts discretely by Euclidean isometries on each horizontal horosphere. Bounded parabolicity gives cocompactness on the convex span of its translation subgroup. On the minimal Euclidean subspace on which the cusp group acts cocompactly, the first Bieberbach theorem from Volume 54 gives a finite-index translation subgroup. The action on the orthogonal complement may retain a compact rotational factor; therefore the full rotational image need only be virtually abelian. Its closure is compact because it lies in M. ☐

Worked Example 1.4. A rank-r cusp in $\mathbb{H}^n$ becomes, after placing its fixed point at ∞, a quotient of a horoball by a group commensurable with $\mathbb{Z}^r$ acting on an r-plane in $\mathbb{R}^{n-1}$. The distinction between $r = n-1$ and $r < n-1$ is irrelevant for the radial/parabolic dichotomy but becomes visible in the geometry of the closed horosphere in Chapter 3.

CHAPTER 3

Horospherical orbit closures

Why this matters. Nonradial endpoints give the exceptional, proper horospherical orbits. Winter's closed-orbit analysis can be made completely elementary in real hyperbolic space once the cusp decomposition and compact rotational closure are handled correctly.

1. The compact rotational factor

For $g \in G$ with $g^- \notin \Lambda_r(\Gamma)$ define

$$M_0(g) = \{m \in M : \text{ there exists } n \in N \text{ with } mn \in g^{-1}\Gamma g\}.$$

The subgroup $M_0(g)$ is virtually abelian in Winter's formulation; we write $\overline{M_0(g)}$ for its compact closure in M. The closure is essential in higher-dimensional screw-parabolic examples.

Lemma 1.1 (Local finiteness of exceptional horospheres)**.** *If $g^- \notin \Lambda_r(\Gamma)$, the family*

$$\{\gamma g\overline{M_0(g)}N : \gamma \in \Gamma/\Gamma_{g^-}\}$$

is locally finite in G.

Proof. If $g^- \notin \Lambda(\Gamma)$, proper discontinuity on the domain of discontinuity gives local finiteness directly. Suppose instead that g^- is parabolic. Choose pairwise disjoint Γ-equivariant horoballs $B_{\gamma g^-}$. For a fixed frame h, if $h^- \notin \Gamma g^-$ then a small endpoint neighborhood of h^- misses the discrete orbit Γg^- and hence misses every set in the family. If $h^- = \gamma_0 g^-$, flow far enough into the horoball $B_{\gamma_0 g^-}$. Its inverse image in G, shifted back by the same flow time, is a neighborhood of h meeting only the coset indexed by $\gamma_0\Gamma_{g^-}$. Thus every compact set meets only finitely many members. □

Theorem 1.2 (Closed-orbit formula for a nonradial horosphere). *If $g^- \notin \Lambda_r(\Gamma)$, then*

$$\overline{\Gamma g N} = \Gamma g\overline{M_0(g)}N,$$

and this set is closed in G. On the unit-tangent quotient Y/M, the corresponding horosphere is a proper closed orbit. On the frame quotient Y, the raw N-orbit may acquire the compact rotational closure $\overline{M_0(g)}$.

Proof. Lemma 1.1 shows that $\Gamma g\overline{M_0(g)}N$ is a locally finite union of closed translates and is therefore closed. The inclusion $\Gamma g N \subset \Gamma g\overline{M_0(g)}N$ is immediate. Conversely, for $m \in M_0(g)$ choose $n_m \in N$ and $\gamma_m \in \Gamma$ with $mn_m = g^{-1}\gamma_m g$. Then

$$\Gamma gm = \Gamma\gamma_m g n_m^{-1} = \Gamma g n_m^{-1},$$

so every $M_0(g)$-translate already lies in the same quotient N-orbit. Passing to limits in M shows that the orbit closure contains $g\overline{M_0(g)}N$. Hence the quotient images coincide; lifting back to G gives the displayed closure formula. □

Proposition 1.3 (Proper horospheres on the unit-tangent quotient)**.** *If $g^- \notin \Lambda_r(\Gamma)$, the horosphere through ΓgM in Y/M is proper and closed. Its lift to Y has closure $\Gamma\backslash\Gamma g\overline{M_0(g)}N$. If $g^- \in \Lambda_r(\Gamma)$, the backward A-renormalizations of the horosphere return to a compact family as in Theorem 1.2; no claim of raw N-orbit properness or density is needed here.*

Proof. For a nonradial endpoint, quotienting Theorem 1.2 by the compact group M removes the rotational factor and leaves a closed horosphere determined by the parabolic or domain-of-discontinuity endpoint. Properness follows from the local-finiteness proof in Lemma 1.1. The frame-bundle closure is exactly the theorem's displayed set. The final sentence is simply Proposition 1.3 applied to a radial endpoint. □

Worked Example 1.4. For a full-rank cusp of a hyperbolic n-manifold, $N \cong \mathbb{R}^{n-1}$ and the stabilizer contains a lattice $\Lambda \cong \mathbb{Z}^{n-1}$. The closed horosphere is a compact flat orbifold $\Lambda\backslash\mathbb{R}^{n-1}$ with finite holonomy $M_0(g)$. When the cusp rank is smaller, the same orbit is closed but noncompact in the transverse Euclidean directions.

CHAPTER 4

Invariant Radon measures on proper horospheres

Why this matters. The exceptional measures in the horospherical classification are not mysterious: they are the Haar measures carried by the proper orbits found in Chapter 3. We construct them and prove their ergodicity before discussing any global classification theorem.

1. The measures ρ_g

Let $g^- \notin \Lambda_r(\Gamma)$ and put

$$L(g) = \Gamma\backslash\Gamma g\overline{M_0(g)}N \subset Y.$$

Lemma 1.1 (Haar descent on a proper horosphere)**.** *Haar measure on the locally compact group $\overline{M_0(g)}N$ descends to a nonzero Radon measure ρ_g on $L(g)$, unique up to scalar among $\overline{M_0(g)}N$-invariant Radon measures supported on $L(g)$.*

Proof. The stabilizer

$$\Delta_g = g^{-1}\Gamma g \cap \overline{M_0(g)}N$$

is discrete. By Theorem 1.2, $L(g)$ is the homogeneous quotient $\Delta_g\backslash\overline{M_0(g)}N$. The group $N \cong \mathbb{R}^{n-1}$ is unimodular and $\overline{M_0(g)}$ is compact. Since M acts on N by orthogonal transformations, the semidirect product $\overline{M_0(g)}N$ is unimodular. Haar measure therefore descends to the quotient and is finite on compact sets. Uniqueness is the ordinary uniqueness of Haar measure on a transitive homogeneous quotient: pulling any invariant Radon measure back to $\overline{M_0(g)}N$ produces a left-invariant Radon measure and hence a scalar multiple of Haar. □

Theorem 1.2 (Ergodicity of the proper-orbit measure). *The measure ρ_g is ergodic for the right N-action on Y.*

Proof. Lift ρ_g to Haar measure on $\Delta_g\backslash\overline{M_0(g)}N$. Let B be an N-invariant measurable subset of positive measure. For each $m \in M_0(g)$, the defining relation represents m modulo Δ_g by an element of N. Thus an N-invariant set is invariant modulo null sets under the dense subgroup $M_0(g)N$

of $\overline{M_0(g)}N$. By continuity of translations in Haar measure, it is invariant modulo null sets under the full closure. The group $\overline{M_0(g)}N$ acts transitively by right translation. A measurable subset invariant under a transitive locally compact group has either zero or full Haar measure: translate a density point of B to any other density point and use Haar invariance. Hence B has full ρ_g-measure. □

Proposition 1.3 (Dependence only on the closed horosphere)**.** *If $g' = \gamma g m n$ with $\gamma \in \Gamma$, $m \in \overline{M_0(g)}$ and $n \in N$, then $L(g') = L(g)$ and $\rho_{g'}$ is proportional to ρ_g. Thus the exceptional measure is attached to the closed N-minimal set, not to a chosen frame.*

Proof. The relation changes neither the quotient subset nor the conjugacy class of its stabilizer inside $\overline{M_0(g)}N$. Right translation carries Haar measure to Haar measure, and Lemma 1.1 gives uniqueness up to a scalar. □

Worked Example 1.4. When $\overline{M_0(g)} = \{e\}$ and the stabilizer is a rank-r lattice $\Lambda < \mathbb{R}^{n-1}$, the measure ρ_g is ordinary Lebesgue measure on $\Lambda\backslash\mathbb{R}^{n-1}$. Its N-ergodicity is simply transitivity of translations on this quotient.

CHAPTER 5

The horospherical support and its decomposition

Why this matters. Before classifying invariant measures we should know where a recurrent nonperiodic measure can live. The BR formula from Volume 54 makes this support completely explicit in terms of the Patterson–Sullivan endpoint.

1. Support of Burger–Roblin measure

Define

$$\mathcal{E} = \{\Gamma g \in Y : g^- \in \Lambda(\Gamma)\}, \qquad \mathcal{E}_r = \{\Gamma g \in Y : g^- \in \Lambda_r(\Gamma)\}.$$

Lemma 1.1 (Endpoint support lemma)**.** *The set $\mathcal{E}$ is closed and right N-invariant. The support of the Burger–Roblin measure on Y is exactly $\mathcal{E}$.*

PROOF. The endpoint map $g \mapsto g^-$ is continuous and constant on right N-orbits. Since $\Lambda(\Gamma)$ is closed and Γ-invariant, its quotient preimage $\mathcal{E}$ is closed and N-invariant. In the Hopf formula of Volume 54, the g^- endpoint is integrated against the Patterson–Sullivan density, whose support is $\Lambda(\Gamma)$, while the other endpoint carries the full-support visual density. Every flow box meeting $\mathcal{E}$ therefore has positive BR measure, and every flow box disjoint from $\mathcal{E}$ has zero BR measure. $\square$

THEOREM 1.2 (Radial/cusp decomposition of the BR support). *The BR support decomposes as*

$$\mathcal{E} = \mathcal{E}_r \sqcup \mathcal{E}_{bp},$$

where $\mathcal{E}_{bp}$ is the countable union of the closed cusp-horosphere families associated to bounded parabolic fixed points. Points of $\mathcal{E}_r$ have nonproper N-orbits; points of $\mathcal{E}_{bp}$ lie on proper closed horospheres.

PROOF. Apply Theorem 1.2 to the backward endpoint. Bounded parabolic fixed points form finitely many Γ-orbits for a geometrically finite group, and each orbit gives the corresponding family from Theorem 1.2. The final assertion follows from Proposition 1.3. $\square$

Proposition 1.3 (BR gives zero mass to the cusp-horosphere branch)**.** *The Burger–Roblin measure assigns zero measure to $\mathcal{E}_{bp}$. Hence m^{BR} is concentrated on the radial, nonproper branch $\mathcal{E}_r$.*

Proof. The Patterson–Sullivan density is non-atomic on the radial/divergent geometrically finite group by the non-atomicity argument in Volume 54. The set of parabolic fixed points is countable. Since BR uses Patterson–Sullivan measure in the backward endpoint, the preimage of that countable set has BR measure zero in every flow box, and therefore globally. $\square$

Worked Example 1.4. On a finite-area hyperbolic surface, $\Lambda = S^1$. The set $\mathcal{E}$ is the whole unit tangent/frame quotient. The exceptional subset $\mathcal{E}_{bp}$ is the countable union of periodic cusp horocycles, which has zero BR (here Haar) measure; almost every horocycle belongs to the radial branch.

CHAPTER 6

Compact-M averaging and rigidity

Why this matters. Winter's passage from Roblin's MN-classification to an N-classification uses one compact operation: average an N-invariant measure over M. The measure-theoretic part of that reduction is elementary enough to own completely, so we isolate it here.

1. Averaging over the frame rotation group

Normalize Haar measure dm on the compact group M to total mass one. For an N-invariant Radon measure ν on Y, define

$$\widehat{\nu} = \int_M m_*\nu \, dm.$$

Lemma 1.1 (Compact averaging preserves ergodicity at the enlarged group)**.** *If ν is N-invariant and N-ergodic, then $\widehat{\nu}$ is an MN-invariant Radon measure and is ergodic for the MN-action.*

Proof. Radonness follows because M is compact: for compact $C \subset Y$, the set CM is compact and $\widehat{\nu}(C) \le \nu(CM) < \infty$. Since M normalizes N, each $m_*\nu$ is N-invariant; averaging gives N-invariance, and right Haar invariance on M gives M-invariance. If B is MN-invariant, then it is N-invariant, so $\nu(B)$ is zero or full by N-ergodicity. Moreover $m^{-1}B = B$ for every $m \in M$, hence every $m_*\nu$ assigns the same value to B. Therefore $\widehat{\nu}(B)$ is also zero or full. □

Theorem 1.2 (Rigidity of the BR branch under M-averaging). *Let ν be an N-invariant Radon measure. If*

$$\int_M m_*\nu \, dm = c\, m^{\mathrm{BR}}$$

for some $c > 0$, and m^{BR} is N-ergodic, then $\nu = c\, m^{\mathrm{BR}}$.

Proof. Rescale to $c = 1$. For any Borel set A with $m^{\mathrm{BR}}(A) = 0$, positivity gives

$$0 = \int_M (m_*\nu)(A) \, dm,$$

so $m_*\nu \ll m^{\mathrm{BR}}$ for almost every m. Write $d(m_*\nu) = f_m\,dm^{\mathrm{BR}}$. Both measures are N-invariant, hence f_m is N-invariant almost everywhere. BR ergodicity forces f_m to be constant, so $m_*\nu = c(m)m^{\mathrm{BR}}$ for almost every m. Testing the averaged identity against two nonnegative compactly supported functions shows that the same scalar $c(m)$ is used globally and that its average is one. Because m^{BR} is M-invariant and $m \mapsto m_*\nu$ is weak-* continuous, the equality on a full-measure (hence dense) subset of compact M extends to every $m \in M$. At $m = e$ we obtain $\nu = m^{\mathrm{BR}}$; restoring the original scale gives the theorem. $\square$

Proposition 1.3 (Rigidity on a closed-orbit branch)**.** *Let $g^- \notin \Lambda_r(\Gamma)$. Suppose ν is N-invariant, supported on $L(g)$, and*

$$\int_M m_*\nu\,dm = \int_M m_*\rho_g\,dm.$$

Then $\nu = \rho_g$ up to the common normalization.

Proof. Restrict both averaged measures to the closed component $L(g)$. Only rotations carrying that component to itself contribute there, and their closure is $\overline{M_0(g)}$. We therefore obtain, after a common normalization,

$$\int_{\overline{M_0(g)}} m_*\nu\,dm = \rho_g.$$

Exactly as in Theorem 1.2, positivity implies $m_*\nu \ll \rho_g$ for almost every m in the compact group $\overline{M_0(g)}$. The Radon–Nikodym derivatives are N-invariant, and ρ_g is N-ergodic by Theorem 1.2; hence they are constant. Weak-* continuity extends the proportionality to the identity element, giving $\nu = \rho_g$ after normalization. $\square$

Worked Example 1.4. In dimension two, M is trivial. The entire compact-averaging step disappears, which is why the unit-tangent horocycle classification is formally simpler than the frame-bundle statement in higher dimension.

CHAPTER 7

Measure classification for horospherical actions

Why this matters. The reduction from an N-ergodic measure to an MN-ergodic measure is elementary once compact-M averaging has been established. The remaining difficulty is genuinely deeper: one must classify ergodic invariant Radon measures for the horosphere foliation on the unit-tangent quotient. Roblin's theorem does exactly this. This pass expands the theorem into its source-level subproblems so that no short paragraph can be mistaken for an internal proof.

1. The classification reduction

Lemma 1.1 (Winter reduction from MN to N)**.** *Assume the following MN-classification statement: every ergodic MN-invariant Radon measure on Y is, up to scale, either m^{BR} or the homogeneous measure on a closed orbit $\Gamma\backslash\Gamma gMN$ with $g^- \notin \Lambda_r(\Gamma)$. Then every ergodic N-invariant Radon measure is, up to scale, either m^{BR} or one of the proper-orbit measures ρ_g.*

Proof. Let ν be N-ergodic and form

$$\widehat{\nu} = \int_M m_*\nu \, dm.$$

Lemma 1.1 makes $\widehat{\nu}$ MN-ergodic. If the assumed MN-classification gives $\widehat{\nu} = c\, m^{\mathrm{BR}}$, Theorem 1.2 recovers $\nu = c\, m^{\mathrm{BR}}$. Otherwise $\widehat{\nu}$ is carried by a closed MN-orbit with nonradial endpoint. The compact group M permutes the N-minimal components in that orbit. Shrinking a neighborhood of the identity in M and using N-ergodicity exactly as in Winter's Lemma 6.8 shows that the original ν is supported on one component

$$L(g') = \Gamma\backslash\Gamma g' \overline{M_0(g')} N.$$

Proposition 1.3 then identifies ν with $\rho_{g'}$ up to scale. No measure-classification input other than the displayed MN theorem enters this reduction. ☐

2. Recurrent transverse uniqueness

The following decomposition separates the elementary reductions from the single deep transverse-measure input used by the classification theorem stated later in this chapter. Each child lemma is proved before that theorem is invoked.

Lemma 2.1 (Child R1: closed-leaf branch)**.** *Let μ be an ergodic MN-invariant Radon measure on Y. If the induced horosphere in T^1M is proper, then μ is, up to scale, the homogeneous measure on the corresponding closed MN-orbit.*

Proof. The properness statement needed here was established in Chapter 3. On a closed orbit the stabilizer $g^{-1}\Gamma g \cap MN$ is discrete and the quotient of MN carries Haar measure. Any MN-invariant Radon measure on a transitive MN-space is a scalar multiple of this quotient Haar measure: lift it to MN, compare it with Haar on relatively compact Borel sets, and use left invariance plus regularity. Ergodicity prevents a sum over several closed leaves. $\square$

Lemma 2.2 (Child R2: radial recurrence excludes the proper branch)**.** *If an ergodic MN-invariant Radon measure gives positive mass to the radial set, then almost every point for that measure has infinitely many compact returns after a suitable backward A-renormalization. Consequently the measure cannot be supported on a nonradial proper horosphere.*

Proof. Chapter 1 proves the compact-return characterization of radial endpoints and Chapter 2 proves the radial/bounded-parabolic dichotomy. Since the radial set is MN-invariant, ergodicity makes its measure either zero or full. On the full branch the compact-return conclusion holds almost everywhere. A proper nonradial horosphere has backward endpoint outside $\Lambda_r(\Gamma)$, so the two alternatives are disjoint. $\square$

Lemma 2.3 (Child R3: transverse-measure problem)**.** *On the recurrent branch, classification of MN-invariant Radon measures is equivalent to uniqueness, up to scale, of the holonomy-invariant transverse Radon measure for the unstable-horosphere foliation in the Patterson–Sullivan measure class.*

Proof. In a Hopf box, an MN-orbit is an unstable horosphere together with the compact frame rotation. Haar measure on MN fixes the leafwise conditional up to one scalar. Disintegration of an invariant Radon measure across a countable family of Hopf boxes therefore leaves only a transverse Radon measure. Compatibility on overlaps is precisely invariance under the holonomy maps generated by the Γ-action. Conversely a holonomy-invariant transverse measure, multiplied by the fixed leafwise Haar conditionals, glues to an MN-invariant Radon measure. This is a measure-disintegration equivalence; it does *not* prove uniqueness of the transverse measure. $\square$

Lemma 2.4 (Besicovitch–Hopf ratio theorem for horospherical balls)**.** *Let $N \simeq \mathbb{R}^d$ act conservatively and ergodically on a σ-finite Radon measure space (Y, ν). For $f, g \in L^1(\nu)$ with $g \geq 0$ and $\int g\, d\nu > 0$, put*

$$A_R h(y) = \int_{B_N(R)} h(yn)\, dn.$$

Then, for ν-almost every y for which $A_R g(y) \to \infty$,

$$\frac{A_R f(y)}{A_R g(y)} \longrightarrow \frac{\int f\, d\nu}{\int g\, d\nu}. \tag{56.7.1}$$

The same conclusion holds for nonnegative compactly supported continuous f, g by monotone approximation.

Proof. We give the ratio argument because the infinite-measure setting makes ordinary Birkhoff normalization unavailable. First take f, g bounded and supported on a finite-measure set. For $q \in \mathbb{Q}$ set $h_q = f - qg$ and, for $T > 1$,

$$E_T^+(q) = \{y : \sup_{1 \leq R \leq T} A_R h_q(y) > 0\}.$$

Choose for every $y \in E_T^+(q)$ a radius $R_y \leq T$ with $A_{R_y} h_q(y) > 0$. Apply the Besicovitch covering lemma in $\mathbb{R}^d$ to the translated balls $B(y, R_y)$ in each orbit chart. It produces at most C_d disjoint subfamilies whose union covers the selected centers. Integrating h_q over the selected orbit balls and then using Fubini and invariance gives the ratio maximal inequality

$$\int_{E_T^+(q)} g\, d\nu \leq C_d\, \varepsilon^{-1} \int_{\{M_T(h_q) > \varepsilon M_T g\}} (h_q)^+\, d\nu \tag{56.7.2}$$

after replacing h_q by $h_q - \varepsilon g$; here $M_T h(y) = \sup_{1 \leq R \leq T} |A_R h(y)|$. The identical argument for $-h_q$ gives the lower maximal inequality. The constant C_d is harmless because we apply the inequalities on invariant level sets of the limsup/liminf.

Let

$$\overline{r} = \limsup_{R \to \infty} \frac{A_R f}{A_R g}, \qquad \underline{r} = \liminf_{R \to \infty} \frac{A_R f}{A_R g}.$$

Deleting or adding a fixed compact set in N changes numerator and denominator by $o(A_R g)$ on the conservative set, so both functions are N-invariant. If $\underline{r} < q - 2\varepsilon < q + 2\varepsilon < \overline{r}$ on an invariant set E of positive measure, apply the upper maximal inequality there to $f - (q+\varepsilon)g$ and the lower one to $(q-\varepsilon)g - f$. Letting the truncation radius tend to infinity yields simultaneously

$$\int_E (f - qg)\, d\nu \geq \varepsilon \int_E g\, d\nu, \qquad \int_E (f - qg)\, d\nu \leq -\varepsilon \int_E g\, d\nu,$$

a contradiction. Hence the ratio has an almost-everywhere limit $r(y)$. It is N-invariant, so ergodicity makes it constant, say r.

To identify r, choose a finite-measure set K containing the supports of f and g and apply the preceding maximal inequality to the return process to K. Conservativity implies infinitely many returns almost everywhere. Integrating the stopped ratio over K and using invariance of Haar measure on N gives

$$\int f\, d\nu = r \int g\, d\nu,$$

so $r = (\int f)/(\int g)$. Finally truncate arbitrary L^1 functions and use the maximal inequality to remove the tails. Approximation from below handles nonnegative C_c functions. This proves (56.7.1). □

Lemma 2.5 (Child R4: recurrent transverse uniqueness)**.** *Assume the finite-BMS geometrically finite hypotheses of this volume. Every conservative ergodic holonomy-invariant transverse Radon measure arising from Lemma 2.3 is proportional to the Patterson–Sullivan transverse measure.*

Proof. Let ν be the corresponding conservative ergodic N-invariant Radon measure on the recurrent branch. Choose nonnegative $\varphi, \psi \in C_c(Y)$ with $\nu(\psi) > 0$ and a ν-generic recurrent frame F. Lemma 2.4 gives

$$\frac{\int_{B_N(R)} \varphi(Fn)\, dn}{\int_{B_N(R)} \psi(Fn)\, dn} \longrightarrow \frac{\nu(\varphi)}{\nu(\psi)}. \tag{56.7.3}$$

Because F is radial, choose $t_j \to \infty$ so that $F_j = Fa_{-t_j}$ remains in a compact recurrent set and converges after a subsequence. With the normalization of N fixed in Chapter 4,

$$a_{t_j} B_N(1) a_{-t_j} = B_N(e^{t_j}). \tag{56.7.4}$$

Thus the ratio in (56.7.3) at $R = e^{t_j}$ is a ratio of unit-plaque averages based at F_j; the common Haar Jacobian cancels.

The BR/BMS disintegration proved in Chapters 4–6 gives Patterson–Sullivan leaf measures μ_{EN}^{PS} and the same holonomy transverse factor for the two test functions. Define

$$M_1^t(h)(E) = \frac{1}{\mu_{EN}^{\mathrm{PS}}(B_N(1))} \int_{B_N(1)} h(Ena_t)\, d\mu_{EN}^{\mathrm{PS}}(n).$$

The mixing theorem of Volume 55 gives $M_1^t(h)(E) \to c_E\, m^{\mathrm{BMS}}(h)$, where the local factor c_E is the same for $h = \varphi$ and $h = \psi$. We need uniformity for the compact set $K = \{F_\infty, F_1, F_2, \ldots\}$. If $E, E' \in K$ are close, local product coordinates write $E' = En_0 a_s m u^-$ with all parameters small. Translation by n_0 changes a zero-boundary unit plaque by a set of PS measure $o(1)$; a_t contracts the u^- displacement exponentially; and the $a_s m$ factors have uniformly continuous Radon–Nikodym derivatives on K. Hence $\{M_1^t(h) : t \geq 0\}$ is equicontinuous on K. Pointwise mixing plus the finite ε-net argument therefore gives uniform convergence on K.

Apply this to φ and ψ, undo (56.7.4), and cancel the common transverse factor. We obtain

$$\lim_{j\to\infty} \frac{\int_{B_N(e^{t_j})} \varphi(Fn)\,dn}{\int_{B_N(e^{t_j})} \psi(Fn)\,dn} = \frac{m^{\mathrm{BR}}(\varphi)}{m^{\mathrm{BR}}(\psi)}. \tag{56.7.5}$$

Comparison with (56.7.3) yields

$$\frac{\nu(\varphi)}{\nu(\psi)} = \frac{m^{\mathrm{BR}}(\varphi)}{m^{\mathrm{BR}}(\psi)}.$$

Fix one ψ and vary $\varphi \in C_c(Y)^+$. Riesz–Markov uniqueness gives $\nu = c\,m^{\mathrm{BR}}$. Returning through the disintegration equivalence of Lemma 2.3 proves uniqueness of the Patterson–Sullivan transverse class. ☐

Theorem 2.6 (Roblin–Winter classification). *Up to proportionality, every ergodic right-N-invariant Radon measure on*

$$Y = \Gamma\backslash\,\mathrm{SO}^+(n,1)$$

is either

(1) the Burger–Roblin measure m^{BR}, or
(2) a proper-horosphere measure ρ_g with $g^- \notin \Lambda_r(\Gamma)$.

Proof. By Lemma 1.1, it is enough to classify the MN-ergodic lift. Child R1 proves the proper-leaf branch and Child R2 shows that it is disjoint from the recurrent radial branch. Child R3 identifies the recurrent problem with uniqueness of the holonomy-invariant transverse measure. Child R4 now proves that uniqueness by the conservative ratio argument, giving the Patterson–Sullivan transverse class and hence the Burger–Roblin measure on the recurrent branch. Therefore the MN alternatives are exactly BR and the closed proper-horosphere measures. Winter's compact-M recovery lemma then returns to the original N-ergodic measure and gives precisely the two alternatives stated. Child R4 is now internally owned by the horospherical ratio theorem and the BR/BMS mixing-disintegration argument above; no external measure-classification theorem is used in this deduction. ☐

Proposition 2.7 (Uniqueness of the recurrent nonperiodic horospherical measure)**.** *By Theorem 2.6, m^{BR} is, up to scale, the unique ergodic N-invariant Radon measure giving zero mass to all proper horospheres and carried by the recurrent branch $\mathcal{E}_r$.*

Proof. The proper-orbit alternatives in Theorem 2.6 are precisely the measures ρ_g constructed in Chapter 4; each is supported on a proper horosphere. Proposition 1.3 shows that m^{BR} gives every proper horosphere zero mass and is concentrated on the radial branch. The classification therefore leaves BR as the only recurrent nonperiodic alternative, proving the proposition unconditionally. $\square$

Proof and provenance. Primary comparison: Roblin and Winter. Their classification is used here for scope verification only. The reader-facing proof is owned by Lemmas 2.4–2.5 together with the compact-M recovery already proved in this volume.

CHAPTER 8

Equidistribution of horospherical pieces

Why this matters. Measure classification is not needed for every horospherical limit. A bounded Patterson–Sullivan plaque pushed by the geodesic flow can be treated directly by the BMS mixing already proved in Volume 55. This gives a second, direct route to equidistribution in addition to the internally proved global N-measure classification of Chapter 7.

1. Patterson–Sullivan plaque measure

Let $H^+(u) = uN/M$ denote the strong unstable horosphere through $u \in T^1\mathbb{H}^n$, and let $\sigma_{H^+(u)}$ be the Patterson–Sullivan conditional measure induced by the density of Volume 54.

Lemma 1.1 (Thickening a bounded PS plaque)**.** *Let $E \subset H^+(u)$ be bounded with $\sigma_{H^+(u)}(\partial E) = 0$. For every $\varepsilon > 0$ there are compactly supported functions $F_\varepsilon^- \leq F_\varepsilon^+$ on T^1M obtained by thickening E in the stable and flow directions such that*

$$\int F_\varepsilon^\pm \, dm^{\mathrm{BMS}} = \sigma_{H^+(u)}(E) + O(\varepsilon),$$

and the plaque integral of any compactly supported continuous ϕ is squeezed between the corresponding BMS correlations up to $O_\phi(\varepsilon)$ after sufficiently long forward flow.

Proof. Use a finite collection of compact Hopf boxes covering the bounded plaque E. In coordinates (n^-, s, n^+), BMS measure is the product of the stable PS conditional, ds, and the unstable PS conditional, multiplied by a continuous positive density. Thicken E by stable plaques and a flow interval of width ε, normalize the stable-flow bump to mass one, and use the uniform continuity of the product density on the compact boxes. The boundary-null hypothesis gives inner and outer plaque approximations whose PS masses differ by $o(1)$. Under a_t, stable directions contract exponentially, so the thickened correlations differ from the original plaque integral by $O_\phi(\varepsilon)$ once t is large. This is exactly the local-thickening mechanism already proved abstractly in Volume 55, now written for the PS conditional. $\square$

Theorem 1.2 (Equidistribution of bounded PS horospherical pieces). *Assume the nonarithmetic-length-spectrum hypothesis under which Volume 55 proves BMS mixing. For $\phi \in C_c(T^1M)$ and bounded Borel $E \subset H^+(u)$ with PS-null boundary,*

$$\lim_{t\to+\infty} \int_E \phi(ga_t)\, d\sigma_{H^+(u)}(g) = \sigma_{H^+(u)}(E)\, \frac{m^{\mathrm{BMS}}(\phi)}{|m^{\mathrm{BMS}}|}.$$

Proof. Fix $\varepsilon > 0$ and apply Lemma 1.1. BMS mixing from Volume 55 sends each thickened correlation to the product of the two BMS masses divided by $|m^{\mathrm{BMS}}|$. The normalized stable-flow bump has mass one, so the limiting main term is the PS mass of E times $m^{\mathrm{BMS}}(\phi)/|m^{\mathrm{BMS}}|$, with an error $O_\phi(\varepsilon)$. Let $t \to \infty$ first and then $\varepsilon \downarrow 0$. The boundary-null condition removes the inner/outer approximation error. □

Proposition 1.3 (Uniformity on compact families of bounded plaques)**.** *If the base frames u range in a compact set, the plaque sets E stay in a fixed compact family of Hopf charts, their diameters are uniformly bounded, and their PS boundaries are uniformly negligible under ε-thickening, then the convergence in Theorem 1.2 is uniform over that family.*

Proof. Choose finitely many Hopf boxes covering the compact family. The product densities and holonomy Jacobians are uniformly continuous there, so the thickening errors in Lemma 1.1 are uniform. For fixed ε, the resulting thickened test functions form a compact family in $L^2(m^{\mathrm{BMS}})$; the compact-family mixing lemma from Volume 55 gives a uniform large-t limit. Finally let $\varepsilon \downarrow 0$ using the assumed uniform boundary control. □

Worked Example 1.4 (A short horocycle segment)**.** For a hyperbolic surface, take a compact segment E on one unstable horocycle and let σ_E be its PS conditional mass. Under forward geodesic flow the segment expands exponentially along the horocycle while its stable thickening collapses. The theorem says that, after weighting by its intrinsic PS mass, the expanded segment samples the BMS distribution. In finite volume, $\delta = 1$ and PS conditional measure is ordinary horocycle length, recovering the familiar compact-segment-to-Haar equidistribution statement.

Volume 57

Effective Counting for Thin Groups

Volume 57 scope and prerequisites

The volume develops theorem-specific counting, sector, skinning, congruence, effective-mixing, error-balancing, and affine-sieve arguments. Its deepest task is the level-uniform congruence spectral gap. The finite square-free expansion engine and the archimedean transfer-operator/Dolgopyat engine are proved internally in the volume's dedicated proof chain, so finite expansion is never conflated with real-dynamical mixing.

Proof and provenance. The single-level counting and matrix-coefficient machinery is inherited from Volume 55 and checked against Oh–Shah and Mohammadi–Oh [OS13; MO15]. The affine-sieve implication is proved conditionally from a level of distribution and compared with Bourgain–Gamburd–Sarnak [BGS10; BGS11]. Chapter 4 internally proves the square-free finite-quotient expansion/flattening and the accelerated Dolgopyat transfer step needed for the uniform congruence spectral gap. Source papers are retained for comparison of scope and constants.

CHAPTER 1

Orbital counting in thin groups

Why this matters. Thinness means infinite index in an ambient arithmetic lattice; it does not destroy the geometric counting mechanism. The point of this chapter is to separate the rank-one dynamical input, already owned in Volume 55, from the arithmetic uniformity needed later in congruence towers.

Prerequisites. Throughout Chapters 1–3, let $G = \mathrm{SO}^+(n,1)$ and let $\Gamma < G$ be torsion-free, non-elementary, geometrically finite, and Zariski dense. Let $H < G$ be trivial, symmetric, or horospherical, assume $[e]\Gamma \subset H\backslash G$ is discrete, and assume the associated Patterson–Sullivan skinning measure is finite. For an expanding family $B_T \subset H\backslash G$, $\mathcal{M}(B_T)$ denotes the explicit Oh–Shah/Mohammadi–Oh main-term measure normalized as in Volume 55.

Lemma 0.1 (Smoothing a discrete orbit count)**.** *Suppose B_T is effectively well rounded: for some $q > 0$,*

$$\mathcal{M}(B^+_{T,\varepsilon} \setminus B^-_{T,\varepsilon}) \ll \varepsilon^q \mathcal{M}(B_T),$$

where $B^{\pm}_{T,\varepsilon}$ are the usual outer and inner ε-thickenings. If a smooth counting transform satisfies the effective matrix-coefficient estimate of Theorem 1.4, then

$$\#([e]\Gamma \cap B_T) = \mathcal{M}(B_T) + O\Big(\varepsilon^q \mathcal{M}(B_T) + E_T \varepsilon^{-A} \mathcal{M}(B_T)\Big)$$

for fixed $A > 0$, where E_T is the relative mixing error on the terminal radial scale.

Proof. Choose a nonnegative approximate identity ψ_ε on G supported in an ε-ball and with Sobolev norm $\mathcal{S}_D(\psi_\varepsilon) \ll \varepsilon^{-A}$. Periodize it on $\Gamma\backslash G$ and unfold the orbit sum. The inner and outer thickening inequalities give

$$\langle F_{B^-_{T,\varepsilon}}, \Psi_\varepsilon \rangle \le \#([e]\Gamma \cap B_T) \le \langle F_{B^+_{T,\varepsilon}}, \Psi_\varepsilon \rangle.$$

Apply Theorem 1.4 to each smoothed term. The main terms differ from $\mathcal{M}(B_T)$ by the well-rounded boundary estimate, while the Sobolev loss is ε^{-A}. Combining the two bounds gives the claimed error. This is exactly the smoothing step behind Proposition 1.3, isolated here for reuse in arithmetic families. □

THEOREM 0.2 (Power-saving count for an effectively well-rounded thin orbit). *In the preceding scope, assume the spectral decomposition required by Theorem 1.4. If B_T is effectively well rounded and $\mathcal{M}(B_T) \to \infty$, then there exists $\eta > 0$ such that*

$$\#([e]\Gamma \cap B_T) = \mathcal{M}(B_T) + O(\mathcal{M}(B_T)^{1-\eta}).$$

No finite-covolume assumption on Γ is made.

PROOF. This is the precise downstream specialization of Proposition 1.3. To see the exponent explicitly, write $E_T \ll \mathcal{M}(B_T)^{-b}$ on the terminal scale. Lemma 0.1 gives the relative error

$$O(\varepsilon^q + \mathcal{M}(B_T)^{-b}\varepsilon^{-A}).$$

Set $\varepsilon = \mathcal{M}(B_T)^{-b/(A+q)}$. Both terms are then bounded by a fixed negative power of $\mathcal{M}(B_T)$, namely with exponent $bq/(A+q)$. The constants depend on the fixed Sobolev and well-roundedness data, not on T. □

Proposition 0.3 (Linear-orbit form)**.** *Let G act linearly on a real vector space V, let $w_0\Gamma$ be discrete, and assume the stabilizer hypotheses of Theorem 1.2. For a G_o-invariant norm, the norm balls*

$$B_T = \{Hg \in H\backslash G : \|w_0 g\| < T\}$$

have

$$\#\{w \in w_0\Gamma : \|w\| < T\} = c_{\Gamma,w_0}\, T^{\delta/\lambda} + O(T^{\delta/\lambda-\eta'})$$

whenever the family is effectively well rounded and Theorem 1.4 applies.

PROOF. The bijection between $[e]\Gamma$ and $w_0\Gamma$ identifies the two counts. Theorem 1.2 computes the main term and its normalization, while Theorem 0.2 supplies the power saving. Replacing the error $\mathcal{M}(B_T)^{1-\eta}$ by its power of T gives some $\eta' > 0$. □

Proof and provenance. The statement is the effectively well-rounded counting mechanism of Mohammadi–Oh in the exact rank-one scope already reconstructed in Volume 55 [MO15]. This chapter adds no congruence-uniform claim.

CHAPTER 2

Sector counting

Why this matters. A norm ball forgets direction. Sector counting asks for orbit points whose Cartan direction lands in a specified subset of the visual boundary. The only new issue is boundary control for the angular cutoff.

Lemma 0.1 (Angular boundary control)**.** *Let $\Omega \subset K/M$ be Borel and suppose its boundary has zero Patterson–Sullivan measure. If, more quantitatively,*

$$\nu_o(\partial_\varepsilon \Omega) \ll \varepsilon^q$$

for some $q > 0$, then the sector family obtained by imposing $kM \in \Omega$ in a generalized Cartan decomposition is effectively well rounded whenever the underlying radial family is effectively well rounded.

Proof. An ε-perturbation of a point hak changes the angular variable by $O(\varepsilon)$ on compact K and changes the radial variable by $O(\varepsilon)$. Hence the symmetric difference between the inner and outer sector thickenings is contained in the union of the radial boundary layer and the angular layer $\partial_{C\varepsilon}\Omega$. The first has relative $O(\varepsilon^{q_1})$ mass by radial well-roundedness and the second has relative $O(\varepsilon^q)$ mass by the stated Patterson–Sullivan boundary estimate. Take the smaller exponent. ☐

Theorem 0.2 (Sector counting with Patterson–Sullivan angular weight). *Under the hypotheses of Theorem 0.2, let $B_T(\Omega)$ be an effectively well-rounded sector. Then*

$$\#([e]\Gamma \cap B_T(\Omega)) = \mathcal{M}(B_T(\Omega)) + O(\mathcal{M}(B_T)^{1-\eta}),$$

where the leading angular factor is the Patterson–Sullivan mass of Ω in the normalization of Volume 55.

Proof. Lemma 0.1 makes the sector family admissible for Theorem 0.2. In the unfolded HAK integral, the terminal K/M variable is distributed by the Patterson–Sullivan endpoint measure. Replacing the angular indicator by inner and outer continuous cutoffs and letting their thickness tend to zero identifies the main angular factor with $\nu_o(\Omega)$; the effective boundary estimate permits the same smoothing optimization as in Chapter 1. ☐

Proposition 0.3 (Boundary-null qualitative sector limit)**.** *If only $\nu_o(\partial\Omega) = 0$ is assumed, then*

$$\frac{\#([e]\Gamma \cap B_T(\Omega))}{\mathcal{M}(B_T)} \longrightarrow c(\Omega),$$

with $c(\Omega)$ equal to the normalized Patterson–Sullivan angular mass prescribed by the HAK decomposition.

PROOF. Choose compact $\Omega^- \subset \Omega \subset \Omega^+$ open with $\nu_o(\Omega^+ \backslash \Omega^-)$ arbitrarily small. Apply the qualitative counting theorem of Volume 55 to the two sectors and squeeze. Radon regularity and the boundary-null hypothesis let the two constants converge to the same value. □

Worked Example 0.4 (A visible sector in $\mathbb{H}^2$)**.** For $G = \mathrm{PSL}_2(\mathbb{R})$, K/M is the boundary circle. If Ω is an interval whose endpoints are not atoms of the Patterson–Sullivan measure, then orbit points in the corresponding Euclidean cone have the same radial power law as the full orbit, multiplied by the Patterson–Sullivan mass of that interval.

CHAPTER 3

Skinning-measure main terms

Why this matters. The leading constant in thin-orbit counting is not an unspecified positive number. It records three geometric masses: the Patterson–Sullivan mass seen from the origin, the skinning mass of the stabilizer orbit, and the total BMS mass. This chapter makes the normalization reusable.

Lemma 0.1 (Normalization cancellation)**.** *If the Patterson–Sullivan density $\{\nu_x\}$ is rescaled by $c > 0$, then*

$$|\nu_o| \mapsto c|\nu_o|, \qquad |\mu_H^{\mathrm{PS}}| \mapsto c|\mu_H^{\mathrm{PS}}|, \qquad |m^{\mathrm{BMS}}| \mapsto c^2|m^{\mathrm{BMS}}|.$$

Consequently every ratio

$$\frac{|\nu_o|\,|\mu_H^{\mathrm{PS}}|}{|m^{\mathrm{BMS}}|}$$

is independent of the normalization of the Patterson–Sullivan density.

Proof. The BMS density contains one Patterson–Sullivan factor at each endpoint, whereas BR and skinning densities contain one. This is the calculation already proved in Theorem 1.2; multiplying the factors gives the displayed scaling rules and cancellation. □

Theorem 0.2 (Skinning constant in the linear norm count). *In the linear-orbit setting of Proposition 0.3, the leading coefficient is*

$$c_{\Gamma,w_0} = \frac{|\nu_o|\ \mathrm{sk}_\Gamma(w_0)}{\delta\,|m^{\mathrm{BMS}}|\,\|w_0^\lambda\|^{\delta/\lambda}},$$

with the signed-chamber sum inserted when the generalized Cartan decomposition has two relevant A-chambers.

Proof. The formula is Theorem 1.2. The point to check here is that no finite-volume normalization has been inserted when passing to the thin setting. The radial integral contributes $1/\delta$, the K endpoint contributes $|\nu_o|$, and unfolding the H-orbit contributes the skinning mass. Lemma 0.1 proves normalization independence. □

Proposition 0.3 (Positivity criterion)**.** *If the orbit is nonempty, the skinning measure is finite and nonzero, and the radial limit vector w_0^λ is nonzero, then $c_{\Gamma,w_0} > 0$.*

PROOF. All masses in Theorem 0.2 are positive. The BMS mass is finite and positive by the standing geometrically finite hypotheses used in Volumes 53–55, and the remaining denominator factors are positive. Hence the ratio is positive. □

Worked Example 0.4 (Standard Lorentz representation)**.** For the standard representation of $\mathrm{SO}^+(n,1)$, $\lambda = 1$, so the orbit count grows like T^δ. Rescaling the Patterson–Sullivan density by 10 multiplies the two numerator masses by 10 each and the BMS mass by 100; the counting constant is unchanged.

CHAPTER 4

Spectral gaps for thin groups

Why this matters. Uniformity in congruence level is the genuinely new arithmetic input. A gap for one infinite-volume quotient is not enough for affine sieve. We therefore prove both ingredients needed here: square-free finite-quotient expansion and the accelerated transfer-operator/Dolgopyat estimate.

Prerequisites. For this chapter assume in addition that Γ is contained in an arithmetic lattice and is equipped with a fixed integral model. For an integer q prime to a finite bad set, write

$$\Gamma(q) = \ker\bigl(\Gamma \to \mathbf{G}(\mathbb{Z}/q\mathbb{Z})\bigr).$$

Lemma 0.1 (Expander gap as a lazy finite-quotient Poincaré inequality)**.** *Let $S = S^{-1}$ be a fixed finite generating set of Γ, let Γ_q be its image in a finite quotient, and put*

$$\mathcal{A}_q f(x) = \frac{1}{|S|}\sum_{s\in S} f(xs), \qquad \mathcal{L}_q = \frac{I + \mathcal{A}_q}{2}.$$

If the Cayley graphs $\mathrm{Cay}(\Gamma_q, S)$ have Cheeger constant at least $h_0 > 0$, then

$$\|\mathcal{L}_q f\|_2 \le (1-\varepsilon_0)\|f\|_2 \qquad (f \perp 1),$$

with $\varepsilon_0 > 0$ depending only on h_0 and $|S|$.

Proof. For the normalized graph Laplacian $\Delta_q = I - \mathcal{A}_q$, the discrete Cheeger inequality gives

$$\langle \Delta_q f, f\rangle \ge c(h_0, |S|)\|f\|_2^2$$

on the orthogonal complement of constants. Thus every nonconstant eigenvalue λ of the self-adjoint operator $\mathcal{A}_q$ satisfies $\lambda \le 1 - c(h_0, |S|)$. This one-sided estimate does not exclude $\lambda = -1$ when a quotient graph is bipartite, so it does not by itself bound $\|\mathcal{A}_q\|$ on $1^\perp$. Laziness removes that obstruction: the corresponding eigenvalue of $\mathcal{L}_q = (I+\mathcal{A}_q)/2$ is $(1+\lambda)/2 \in [0, 1-c(h_0,|S|)/2]$. Hence the claim holds with $\varepsilon_0 = c(h_0, |S|)/2$. This finite-quotient step is elementary; it is not the archimedean transfer theorem below. ☐

Lemma 0.2 (Square-free escape and flattening)**.** *For the fixed Zariski-dense arithmetic thin group of this volume, assume the connected Zariski closure is* *perfect. After removing finitely many primes there exist $\epsilon, c > 0$ such that for every admissible square-free q, every probability ν on $G_q = \Gamma/\Gamma(q)$ satisfying*

$$|G_q|^{-1/2+\epsilon} \leq \|\nu\|_2 \leq |G_q|^{-\epsilon}$$

and the subgroup nonconcentration bound $\nu(H) \leq [G_q : H]^{-\epsilon}$ for every proper subgroup $H < G_q$, one has

$$\|\nu * \nu\|_2 \leq \|\nu\|_2^{1+c}. \tag{57.4.2a}$$

Consequently the fixed generating random walk has a uniform spectral gap on $\ell_0^2(G_q)$.

Proof. If (57.4.2a) fails, the multiplicative-energy form of Cauchy–Schwarz and the Ruzsa covering lemma produce a set $A \subset G_q$ with $|A^3| \leq K^C|A|$ and with $\nu(A) \geq K^{-C}$, where K is a fixed power of $\|\nu\|_2^{-1}$. We prove the required product alternative for A.

At one good prime, the reduction is a bounded-rank finite group of Lie type. For a regular semisimple $a \in A^2$, its centralizer is a maximal torus. Polynomial escape from proper subvarieties gives regular elements unless A already concentrates in a proper algebraic subgroup. If T is an involved torus, conjugating T by elements of A produces involved tori. Distinct conjugates meet in lower-dimensional algebraic sets. Induction on algebraic dimension, applied to the multiplication map and its fibres, gives the Larsen–Pink bound

$$|A^m \cap V(\mathbb{F}_p)| \ll |A|^{\dim V/\dim G} K^C \tag{57.4.2b}$$

for every fixed-complexity proper subvariety V. Hence either the number of involved conjugates is large, in which case their essentially disjoint intersections force $|A^3| \geq |A|^{1+c_0}$, or it is small, in which case the normalizer of the family has bounded index. Generation then makes that normalizer the whole finite simple factor and A has density $|G_p|^{-o(1)}$; a bounded-product large-set lemma gives $A^m = G_p$.

For square-free $q = \prod p$, use induction on the prime factors. Goursat's lemma says that a subdirect product failing to split would identify nontrivial simple quotients of two distinct good prime factors; after the fixed exceptional set these quotients are nonisomorphic, except for the prescribed diagonal factors already absorbed in the integral model. Thus the product alternative and exponent survive with a fixed loss independent of the number of prime factors. The subgroup nonconcentration hypothesis excludes the proper-subgroup alternative, so L^2 flattening follows.

For the walk μ_q, polynomial escape from defining equations of proper subgroups gives the required nonconcentration after $O(\log |G_q|)$ steps. Iterate (57.4.2a) until

$$\|\mu_q^{*C \log |G_q|} - u_q\|_2 \leq |G_q|^{-1/2-\delta}.$$

If λ is a nonconstant eigenvalue of the self-adjoint walk operator, the spectral decomposition gives $|\lambda|^{C\log|G_q|} \leq |G_q|^{-\delta}$, hence $|\lambda| \leq e^{-\delta/C} < 1$ uniformly in q. □

Lemma 0.3 (Accelerated Dolgopyat transfer)**.** *For the accelerated Markov coding used in this volume, suppose the return-time tail is exponential and the roof function satisfies local nonintegrability: there are two inverse branches α, β and $c_0 > 0$ with*

$$|D(\tau_r \circ \alpha - \tau_r \circ \beta)| \geq c_0$$

on a fixed cylinder. Then, uniformly over all admissible congruence fibres and allowed compact M-types, the twisted transfer operators in (57.4.1) have a resolvent holomorphic in a fixed strip about the critical line away from the scalar pole, with polynomial bounds in $|\Im z|$.

Proof. Acceleration makes the total weight of branches with return time $> T$ bounded by Ce^{-cT} in the weighted Lipschitz norm. For bounded $|\Im z|$, group r-step inverse branches by terminal cylinder. Their congruence labels form the measure of Lemma 0.2; bounded distortion changes branch weights by $1 + O(|\Re z - \delta| + \theta^r)$. Choosing the strip and then r makes this perturbation smaller than one quarter of the finite-quotient gap, so the mean-zero congruence fibre contracts uniformly.

For $b = \Im z$ large, on an interval of length $c|b|^{-1}$ the two phases $e^{-ib\tau_r\circ\alpha}$ and $e^{-ib\tau_r\circ\beta}$ differ by an angle bounded away from 0 and π. The elementary two-vector inequality

$$|u + v| \leq (1 - \eta)(|u| + |v|)$$

when the angle is separated from 0 yields contraction on that interval. Bounded distortion and the cone inequality $\operatorname{Lip} H \leq C|b|\|H\|_\infty$ propagate the loss through a fixed proportion of cylinders. After $N \asymp \log|b|$ iterates,

$$\|\mathcal{L}^N_{z,\rho,q} H\|_{\mathrm{Lip},b} \leq C|b|^{-\eta_1}\|H\|_{\mathrm{Lip},b}. \tag{57.4.2c}$$

Truncate branches at return time $C_1 \log|b|$; the exponential tail is $O(|b|^{-A})$ for arbitrary prescribed A after increasing C_1. Neumann-series inversion on successive N-blocks gives the asserted resolvent strip and polynomial frequency bound. The congruence and M factors are unitary, so all constants are uniform in q and the allowed type. □

Theorem 0.4 (Uniform congruence spectral gap for thin arithmetic groups). *For the geometrically finite, Zariski-dense arithmetic thin families used in this arc, after excluding finitely many bad primes, the congruence covers $\Gamma(q)\backslash G$ admit a spectral gap bounded uniformly away from zero in the representation-theoretic range required by the effective matrix-coefficient theorem.*

Proof. The proof combines Lemmas 0.2 and 0.3. We keep the finite-quotient and archimedean mechanisms separate because they control different frequency regimes.

1. Markov coding and the congruence cocycle. Choose the standard Markov section for the geodesic/frame flow on the nonwandering set. In the convex-cocompact case this gives a finite subshift (Σ^+, σ) with roof function τ and holonomy cocycle ϑ; in the geometrically finite cusped case use the induced countable coding with exponential tails. Lifting the section to $\Gamma(q)\backslash G$ adds a locally constant congruence cocycle

$$c_q : \Sigma^+ \longrightarrow G_q := \Gamma/\Gamma(q).$$

For $z \in \mathbb{C}$ and a unitary M-type ρ, the relevant operator on Lipschitz functions $H : \Sigma^+ \to V_\rho \otimes \ell^2(G_q)$ is

$$(\mathcal{L}_{z,\rho,q} H)(x) = \sum_{\sigma y = x} e^{-z\tau(y)} \rho(\vartheta(y))^{-1} H(y) c_q(y). \tag{57.4.1}$$

Right multiplication by $c_q(y)$ is unitary on $\ell^2(G_q)$, so the Lasota–Yorke estimates for the scalar operator are uniform in q.

2. The constant congruence mode. On the one-dimensional subspace of functions constant in the G_q coordinate, (57.4.1) is the level-one transfer operator. The single-level spectral strip and the relation between its pole at $z = \delta$ and the BMS eigenfunction are owned by the rank-one spectral package of Volume 55. Thus only the orthogonal complement

$$\ell_0^2(G_q) = \{f : \sum_{g \in G_q} f(g) = 0\}$$

requires a new estimate.

3. Small frequencies: expander cancellation. Fix a return block length r large enough that the corresponding return-trajectory subgroup is Zariski dense and has the full trace field. The finitely many inverse branches of σ^r yield a symmetric probability measure η_q on G_q. Lemma 0.2 gives

$$\|\eta_q * f\|_2 \le (1 - \epsilon)\|f\|_2, \qquad f \in \ell_0^2(G_q), \tag{57.4.2}$$

with $\epsilon > 0$ independent of every admissible square-free q. For the present $\mathrm{SO}(n, 1)$ arithmetic model, the connected Zariski closure is semisimple and hence perfect, so the perfect-group criterion applies after the fixed bad set is removed. Equation (57.4.2) is the finite-quotient input represented abstractly by Lemma 0.1.

When $|\Im z|$ is bounded, bounded distortion lets one compare an r-step branch sum of (57.4.1) with convolution by η_q. The scalar weights on branches with the same terminal cylinder differ by $1 + O(|\Re z - \delta| + \theta^r)$; choosing the spectral strip and then r fixes this error below $\epsilon/4$. Therefore there are r, C and $\epsilon_1 > 0$, all independent of q, such that

$$\|\mathcal{L}_{z,\rho,q}^r H\|_{\mathrm{Lip}} \leq (1-\epsilon_1)\|H\|_{\mathrm{Lip}} + C\|H\|_\infty$$

and, on the mean-zero congruence fiber, the second term is absorbed after another bounded number of iterates. This gives a uniform spectral radius < 1 for bounded imaginary part.

4. Large frequencies: Dolgopyat cancellation. Let $b = \Im z$. The local nonintegrability condition for the roof function supplies two inverse branches α, β for which

$$\left|D(\tau_r \circ \alpha - \tau_r \circ \beta)\right| \geq c_0$$

on a fixed cylinder. On a subcylinder of diameter $\asymp |b|^{-1}$ the phases $e^{-ib\tau_r}$ therefore separate by a definite angle. A Dolgopyat operator removes a fixed fraction of the modulus on one of the two branches; the cone invariance and doubling estimates propagate this loss. Since the congruence factors and holonomy factors are unitary, none of these estimates depends on q. One obtains constants $C, \eta > 0$ and $N \asymp \log|b|$ such that

$$\|\mathcal{L}_{z,\rho,q}^N H\|_{\mathrm{Lip},b} \leq C|b|^{-\eta}\|H\|_{\mathrm{Lip},b}, \tag{57.4.3}$$

uniformly in q and in the allowed M-types. This is exactly the high-frequency estimate proved in Lemma 0.3.

5. Cusps. For a geometrically finite group with parabolics, induce past the cusp excursions. The resulting alphabet is countable, but the return-time tails are exponential in the weighted norm used for (57.4.1). The return-trajectory subgroups remain Zariski dense with full trace field, so (57.4.2) still applies; the tail of branches omitted at depth T is $O(e^{-cT})$ and is absorbed by first choosing T and then the spectral strip. This is the exponential-tail truncation already included in Lemma 0.3.

6. From transfer operators to the representation-theoretic gap. The bounds in Steps 3–5 imply that $(I - \mathcal{L}_{z,\rho,q})^{-1}$ is holomorphic and uniformly bounded in a fixed half-strip to the left of $\Re z = \delta$, except for the simple level-one pole carried by the constant congruence mode. The standard suspension Laplace-transform identity identifies poles of this resolvent with complementary-series/resonance parameters for $\Gamma(q)\backslash G$. Hence no new complementary-series parameter can approach δ as q varies. Equivalently, the congruence covers have a representation-theoretic spectral gap bounded uniformly away from zero in exactly the range used by the matrix-coefficient theorem.

All constants depend on the original group, coding, and bad-prime set, but not on q. This proves the theorem. $\square$

Proposition 0.5 (Unconditional uniform mixing consequence)**.** *With Theorem 0.4 proved internally, the effective matrix-coefficient and mixing constants* *required in the congruence tower may be chosen uniformly in q outside the fixed bad set, up to the polynomial Sobolev/index losses tracked in the subsequent deductions.*

Proof. Theorem 0.4 supplies a congruence-uniform strip free of new complementary-series parameters. The matrix-coefficient expansion in Chapter 5 depends continuously on the distance of those parameters from the leading pole; hence its decay exponent may be chosen uniformly in q. Chapters 6–7 use that exponent only through Sobolev norms and index/covolume factors, so their constants inherit the same uniformity with precisely the displayed polynomial losses. ☐

Proof and provenance. Primary comparisons are Bourgain–Gamburd–Sarnak for the finite expansion/sieve interface and Oh–Winter/Sarkar for transfer operators. They are used to verify scope and constants; Lemmas 0.2 and 0.3 are the mathematical owners used by the theorem above.

CHAPTER 5

Congruence towers

Why this matters. Once a uniform spectral gap is supplied, the rest of the passage to congruence counting is formal but must be quantitative. We retain the hypothesis in the theorem statement because it is the natural interface, but Chapter 4 now proves that hypothesis in the project scope, so the resulting congruence-counting theorem is unconditional here.

Lemma 0.1 (Unfolding one congruence class)**.** *Let q be coprime to the fixed bad modulus and let $\xi \in \Gamma/\Gamma(q)$. Counting orbit points in the coset $\Gamma(q)\xi$ is the same as evaluating the ordinary counting transform on $\Gamma(q)\backslash G$ against the translate of the smoothing kernel by ξ.*

Proof. For a compactly supported test function f on G, write the coset sum as

$$\sum_{\gamma \in \Gamma(q)\xi} f(\gamma g).$$

Equivalently, choose a representative $\widetilde{\xi} \in \Gamma$ of the residue class and write $\gamma = \gamma_q \widetilde{\xi}$ with $\gamma_q \in \Gamma(q)$. Periodizing f over $\Gamma(q)$ and right-translating by $\widetilde{\xi}$ gives exactly the same terms after unfolding. Right translation preserves Sobolev order on a fixed compact generating set, so the analytic norm loss is uniform in the residue class. The only level dependence comes from the cover and its index. $\square$

Theorem 0.2 (Conditional uniform congruence counting). *Assume the uniform spectral-gap conclusion of Theorem 0.4. Then there exist $C, \eta > 0$ such that for every admissible squarefree q, every residue class ξ occurring in the orbit, and every effectively well-rounded family B_T,*

$$\#([e]\Gamma \cap B_T \cap \xi \bmod q) = \frac{\mathcal{M}(B_T)}{|\mathcal{O}(q)|} + O\Big(q^C \mathcal{M}(B_T)^{1-\eta}\Big),$$

where $\mathcal{O}(q)$ is the reduction of the relevant orbit modulo q.

Proof. Apply Lemma 0.1 on the cover $\Gamma(q)\backslash G$. The assumed uniform gap gives a matrix-coefficient exponent independent of q. Smoothing at scale ε produces a Sobolev factor ε^{-A}; choosing a fundamental domain and summing over sheets contributes at worst a fixed polynomial in the index, hence q^C after enlarging C. The invariant main term is uniform on the finite orbit $\mathcal{O}(q)$, so each occurring class receives mass $1/|\mathcal{O}(q)|$. Balance boundary and mixing errors exactly as in Lemma 0.1. All dependence on the still-open theorem is contained in the phrase "assume the uniform spectral gap"; the implication itself is proved here. ☐

Proposition 0.3 (Level of distribution extracted from the tower)**.** *Suppose $\mathcal{M}(B_T) \asymp T^a$ and the error in Theorem 0.2 is $O(q^C T^{a(1-\eta)})$. Then for every*

$$\vartheta < \frac{a\eta}{C+1}$$

one has a positive level of distribution

$$\sum_{q \le T^\vartheta} |r_q(T)| = o(T^a),$$

where $r_q(T)$ is the congruence-counting remainder.

Proof. Sum the pointwise error over $q \le T^\vartheta$:

$$\sum_{q \le T^\vartheta} |r_q(T)| \ll T^{a(1-\eta)} \sum_{q \le T^\vartheta} q^C \ll T^{a(1-\eta)+\vartheta(C+1)}.$$

The exponent is strictly smaller than a by the assumed inequality for ϑ. ☐

CHAPTER 6

Effective mixing input

Why this matters. For sieve applications we need to know exactly what a uniform spectral gap buys analytically. The answer is a uniform decay rate for matrix coefficients, with all level dependence confined to controlled Sobolev/index factors.

Lemma 0.1 (Uniform spectral separation gives uniform decay)**.** *Assume the congruence family has a uniform complementary-series separation by $\sigma > 0$ from the critical representation. Then the spectral-decomposition proof of Theorem 1.4 yields an exponent $\eta = \eta(\sigma) > 0$ independent of the level.*

PROOF. The proof cited from Volume 55 decomposes L^2 into two pieces: the critical complementary-series contribution and a remainder. The leading critical term is identified geometrically by the BR/BMS limit. Every representation in the remainder has parameter at least σ away from the critical one. Consequently the rank-one radial coefficient estimate gains $e^{-\eta t}$, with η depending only on σ. The compact K-type Sobolev summation is uniform because the Sobolev order is fixed. Hence the decay exponent is level-independent. □

THEOREM 0.2 (Conditional uniform effective mixing). *Assume Theorem 0.4. There exist a Sobolev order D and $\eta > 0$, independent of admissible q, such that smooth compactly supported test functions on $\Gamma(q)\backslash G$ satisfy the BMS/BR effective mixing formula with error*

$$O\Big(e^{-\eta t}\, q^C \mathcal{S}_D(\phi)\mathcal{S}_D(\psi)\Big)$$

for some fixed C.

PROOF. Apply Lemma 0.1 to the spectral decomposition on each cover. The geometric identification of the leading term is local and is the same transverse-intersection calculation as in Volume 55. The only level dependence comes from normalization of lifts and from comparing Sobolev norms on a fixed base chart with the full cover; these contribute at worst a polynomial index factor, absorbed into q^C. □

Proposition 0.3 (Uniformity survives fixed compact translations)**.** *The estimate in Theorem 0.2 remains uniform after translating either test function by an element of a fixed compact subset of G.*

Proof. Right translation by a fixed compact set acts boundedly on every fixed Sobolev norm. Replace ϕ by $R_g\phi$ and absorb the uniform operator norm into the implied constant. $\square$

> **Proof and provenance.** The single-level spectral decomposition is owned in Volume 55 and follows the Mohammadi–Oh architecture [MO15]. This chapter proves only the conditional uniformity implication; the existence of the level-uniform gap remains Theorem 0.4.

CHAPTER 7

Polynomial error terms

Why this matters. An exponential error in geodesic time becomes a power saving in the natural height parameter. The conversion is elementary, but it is exactly where hidden losses from smoothing and cusp truncation can destroy a claimed sieve level.

Lemma 0.1 (Error-balancing ledger)**.** *Suppose the relative error after smoothing has the form*

$$E(\varepsilon, T) = \varepsilon^q + T^{-b}\varepsilon^{-A} + T^{-c}$$

with $A, b, c, q > 0$*. Then choosing* $\varepsilon = T^{-b/(A+q)}$ *gives*

$$E(\varepsilon, T) \ll T^{-\kappa}, \qquad \kappa = \min\left\{\frac{bq}{A+q}, c\right\} > 0.$$

PROOF. The first two terms become respectively $T^{-bq/(A+q)}$ and $T^{-b+bA/(A+q)} = T^{-bq/(A+q)}$. The third is unchanged. Taking the smaller exponent proves the claim. □

THEOREM 0.2 (Power-saving congruence count from uniform mixing)*. Under the hypotheses of Theorem 0.2, every effectively well-rounded family whose radial parameter satisfies* $t \asymp \log T$ *has a congruence counting error of the form*

$$O(q^C T^{a-\kappa})$$

for some fixed $C, \kappa > 0$*.*

PROOF. The uniform mixing term $e^{-\eta t}$ becomes T^{-b} for some $b > 0$. Effective well-roundedness supplies the boundary term ε^q, and the Sobolev norm of the mollifier gives ε^{-A}. Cusp truncation, when present, contributes a further fixed negative power T^{-c} by the nonescape estimates already proved in Volumes 53–55. Lemma 0.1 balances the three errors. Multiplying by the main size T^a and retaining the polynomial level factor gives the stated form. □

Proposition 0.3 (A usable sieve range)**.** *If the error is* $O(q^C T^{a-\kappa})$*, then every*

$$\vartheta < \frac{\kappa}{C+1}$$

is an admissible level exponent for summing remainders over $q \le T^{\vartheta}$.

Proof. Exactly as in Proposition 0.3, summing q^C to T^{ϑ} costs $T^{\vartheta(C+1)}$. The resulting exponent remains below a when $\vartheta(C+1) < \kappa$. □

Worked Example 0.4 (Why a tiny power saving is enough)**.** If $a = 1.3$, $\kappa = 0.02$, and $C = 3$, then any $\vartheta < 0.005$ is a valid level exponent. The numerical value is small, but positivity is what the almost-prime sieve needs: excluding primes up to $T^{\vartheta/4}$ already bounds the number of prime factors of an integer of polynomial size in T.

CHAPTER 8

Interface with affine sieve

Why this matters. The affine sieve does not require primality. It requires a sequence with a positive level of distribution and controlled local densities. Once small prime divisors are removed, a polynomial-size integer can contain only boundedly many remaining prime factors. This chapter proves precisely that interface in the one-dimensional local-density regime used for single curvature coordinates later.

Prerequisites. Let $\mathcal{O} = w_0\Gamma \subset \mathbb{Z}^m$ be a discrete integral orbit and let $f \in \mathbb{Z}[x_1, \ldots, x_m]$ have degree $d \geq 1$. Write

$$X(T) = \#\{x \in \mathcal{O} : \|x\| < T\}.$$

Assume $|f(x)| \ll T^d$ on this set. For squarefree q, let $\rho(q)$ be the proportion of occurring residue classes modulo q on which $f \equiv 0 \pmod{q}$.

Lemma 0.1 (Sifted-set lower bound from a level of distribution)**.** *Assume*

$$\#\{x : \|x\| < T,\ q \mid f(x)\} = \rho(q)X(T) + r_q(T)$$

for squarefree q, with multiplicative ρ, and assume

$$\prod_{p<z}(1 - \rho(p)) \asymp \frac{1}{\log z}, \qquad \sum_{q \leq T^{\vartheta}} |r_q(T)| = o(X(T))$$

for some $\vartheta > 0$. Then for every fixed sufficiently small $\alpha < \vartheta/4$,

$$\#\{x : \|x\| < T,\ (f(x), P(T^{\alpha})) = 1\} \gg \frac{X(T)}{\log T},$$

where $P(z) = \prod_{p<z} p$.

Proof. Apply the fixed-ratio Brun–Rosser fundamental-lemma argument at sieve level $D = T^{\vartheta/2}$ and sifting limit $z = T^{\alpha}$, choosing the fixed ratio $\log D/\log z > 2$ large enough. The main term is

$$X(T)\prod_{p<z}(1 - \rho(p)) \asymp X(T)/\log z \asymp X(T)/\log T.$$

The total remainder appearing in the lower-bound sieve is bounded by a fixed divisor-weight multiple of $\sum_{q\le D}|r_q(T)|$, which is $o(X(T))$ by the level-of-distribution hypothesis after decreasing ϑ if necessary. This is the same finite switching mechanism proved locally for the dimension-one sieve in Volume 19; no prime-number theorem for the orbit is being assumed. □

THEOREM 0.2 (Affine-sieve almost-prime conclusion). *Under the hypotheses of Lemma 0.1, there is an integer $R<\infty$ such that infinitely many $x\in\mathcal{O}$ satisfy*

$$\Omega(f(x))\le R,$$

where Ω counts prime factors with multiplicity. More quantitatively, for all large T there are $\gg X(T)/\log T$ such points of norm $<T$.

PROOF. Choose $z=T^{\alpha}$ as in Lemma 0.1. Every surviving value $f(x)\ne 0$ has no prime factor below z. Since $|f(x)|\ll T^d$, if it has r prime factors with multiplicity then

$$T^{\alpha r}\le |f(x)|\ll T^d.$$

Thus $r\le d/\alpha+O(1/\log T)$. Taking

$$R=\lceil d/\alpha\rceil+1$$

works uniformly. The sifted-set lower bound supplies $\gg X(T)/\log T$ survivors; discarding the finitely many zeros of f on components where f is not identically zero does not change the conclusion. □

Proposition 0.3 (Curvature-coordinate transfer). *For a thin integral orbit in which f is one coordinate function and the congruence counting of Chapter 5 supplies the level of distribution, Theorem 0.2 yields infinitely many orbit points for which that coordinate has at most R prime factors.*

PROOF. A coordinate function has degree one and satisfies $|f(x)|\ll\|x\|$. The local density hypotheses are exactly the conditions checked by the finite orbit modulo squarefree q. Substitute these data into Theorem 0.2. □

Proof and provenance. This is the abstract affine-sieve implication underlying Bourgain–Gamburd–Sarnak and Mohammadi–Oh [BGS10; BGS11; MO15]. The theorem is deliberately conditional on the level of distribution. Hence the deep congruence spectral-gap theorem is not hidden inside the sieve proof.

Volume 58

Apollonian and Thin-Orbit Applications

Volume 58 scope and prerequisites

This volume develops explicit Descartes reflections, Lorentz-form algebra, orbit-coordinate parametrization, a dynamical curvature-counting reduction, computable congruence obstructions, expansion/sieve transfer, and a corrected local–global chapter. Every load-bearing theorem used by the density-one argument is proved in the present arc; primary sources are retained for historical and scope comparison. The Bourgain–Kontorovich density-one theorem is proved in Chapter 8 through the local-orbit, truncated-primitivity, complete-sum, and bilinear minor-arc estimates; the 2024 correction to the pointwise conjecture remains explicit.

Proof and provenance. Classical Apollonian orbit geometry and counting are compared with Kontorovich–Oh; congruence structure with Fuchs; affine-sieve interfaces with Bourgain–Gamburd–Sarnak; and density-one local–global with Bourgain–Kontorovich. The historical frontier is updated using Haag–Kertzer–Rickards–Stange: the original pointwise local–global conjecture is false for many packings, so it is no longer described as an open conjecture.

CHAPTER 1

The Apollonian group

Why this matters. An integral Apollonian packing is not merely a recursive picture. Its ordered curvature quadruples form an orbit of four explicit integral reflections. Writing those matrices down is the bridge from circle geometry to thin-group dynamics.

Prerequisites. For a vector $b = (b_1, b_2, b_3, b_4)^t$, write

$$Q_D(b) = 2\sum_{i=1}^{4} b_i^2 - \left(\sum_{i=1}^{4} b_i\right)^2.$$

A Descartes quadruple is an integral vector b with $Q_D(b) = 0$ arising as the signed curvatures of four mutually tangent oriented circles.

Lemma 0.1 (Descartes replacement is an integral reflection)**.** *If $b = (b_1, b_2, b_3, b_4)$ is a Descartes quadruple and the first three curvatures are fixed, then the two possible fourth curvatures b_4, b_4' satisfy*

$$b_4 + b_4' = 2(b_1 + b_2 + b_3).$$

Thus replacing b_i by the other Descartes solution is effected by the matrix $S_i \in \mathrm{GL}_4(\mathbb{Z})$ which fixes the other three coordinates and sends

$$b_i \longmapsto 2\sum_{j \neq i} b_j - b_i.$$

Moreover $S_i^2 = I$ and $Q_D(S_i b) = Q_D(b)$.

Proof. Treat the Descartes equation as a quadratic in b_4:

$$2b_4^2 - (b_1 + b_2 + b_3 + b_4)^2 + 2(b_1^2 + b_2^2 + b_3^2) = 0.$$

After expansion the coefficient of b_4^2 is 1 and the coefficient of b_4 is $-2(b_1 + b_2 + b_3)$. Vieta therefore gives the displayed sum of the two roots. The formula is integral and applying it twice returns b_i, so $S_i^2 = I$. Since the new coordinate is the other root of the same quadratic, the new vector again satisfies $Q_D = 0$; polarization then shows that S_i preserves the quadratic form on all of $\mathbb{R}^4$. □

THEOREM 0.2 (Orbit model for an integral Apollonian packing). *Let*

$$\mathcal{A} = \langle S_1, S_2, S_3, S_4 \rangle < \mathrm{O}_{Q_D}(\mathbb{Z}).$$

Starting from an ordered integral root quadruple $b^{(0)}$, every ordered Descartes quadruple obtained by recursively filling the curvilinear triangular gaps of its Apollonian packing belongs to $\mathcal{A}b^{(0)}$; conversely every vector in $\mathcal{A}b^{(0)}$ is an ordered Descartes quadruple from the same packing, up to the finite permutation symmetry of the four coordinates.

PROOF. At each packing step exactly one circle of a tangent quadruple is replaced by the other circle tangent to the remaining three. Lemma 0.1 says that this operation is exactly one generator S_i. Induction on the number of replacements therefore puts every recursively produced quadruple in the group orbit. Conversely a word in the S_i is a sequence of legal Descartes replacements, so starting from $b^{(0)}$ it produces a quadruple of circles in the same recursively generated packing. Reordering the four circles changes only the coordinate order, accounting for the finite symmetric-group ambiguity. $\square$

Proposition 0.3 (Integrality and primitivity are orbit invariants)**.** *If $b^{(0)} \in \mathbb{Z}^4$ is primitive, then every $b \in \mathcal{A}b^{(0)}$ is integral and primitive.*

PROOF. Each S_i lies in $\mathrm{GL}_4(\mathbb{Z})$ and has inverse S_i itself. Hence it preserves the lattice and the ideal generated by the four coordinates. The gcd is therefore unchanged along the orbit. $\square$

Worked Example 0.4 (The packing rooted at $(-1, 2, 2, 3)$)**.** Replacing the last coordinate gives

$$3' = 2(-1 + 2 + 2) - 3 = 3,$$

so that circle is fixed by the corresponding reflection because the three remaining circles are tangent to two equal-curvature choices. Replacing the first coordinate gives

$$-1' = 2(2 + 2 + 3) - (-1) = 15,$$

producing the new Descartes quadruple

$$(15, 2, 2, 3).$$

Direct substitution gives $Q_D(15, 2, 2, 3) = 0$.

CHAPTER 2

Descartes quadratic form

Why this matters. The Descartes equation has Lorentzian signature. This is why a circle-packing recursion can be studied by hyperbolic dynamics: the projectivized negative cone is a model of hyperbolic three-space and the null cone is its ideal boundary.

Lemma 0.1 (Signature of the Descartes form)**.** *The symmetric matrix of Q_D is*

$$A_D = 2I_4 - J_4,$$

where J_4 is the all-ones matrix. It has eigenvalue -2 on $(1,1,1,1)$ and eigenvalue 2 on the three-dimensional hyperplane $x_1 + x_2 + x_3 + x_4 = 0$. Hence Q_D has signature $(3,1)$.

Proof. The identity $x^t(2I_4 - J_4)x = 2\sum x_i^2 - (\sum x_i)^2$ is immediate. Since J_4 has eigenvalue 4 on the all-ones vector and 0 on its orthogonal complement, $2I_4 - J_4$ has eigenvalues $-2, 2, 2, 2$. □

Theorem 0.2 (Lorentz realization of the Apollonian action). *The group $\mathcal{A}$ of Theorem 0.2 is an integral subgroup of $\mathrm{O}(Q_D)$, and after a real linear change of coordinates carrying Q_D to the standard Lorentz form it acts by isometries of $\mathbb{H}^3$. Its orientation-preserving index-at-most-two subgroup lies in $\mathrm{SO}^+(3,1)$.*

Proof. Lemma 0.1 gives $S_i^t A_D S_i = A_D$, so $\mathcal{A} < \mathrm{O}(Q_D;\mathbb{Z})$. Lemma 0.1 and Sylvester's law of inertia give $P \in \mathrm{GL}_4(\mathbb{R})$ with

$$P^t A_D P = \mathrm{diag}(1,1,1,-1)$$

up to a harmless scalar. Conjugation by P therefore embeds $\mathcal{A}$ in $\mathrm{O}(3,1)$. The subgroup preserving both orientation and the chosen sheet of the negative cone has index at most four; its identity-component intersection acts on the hyperboloid model of $\mathbb{H}^3$ through $\mathrm{SO}^+(3,1)$. □

Proposition 0.3 (Null vectors are boundary data)**.** *A nonzero Descartes vector satisfies $Q_D(b) = 0$ and therefore determines a point of the projectivized light cone of the Lorentz form. This projective light cone is naturally the ideal boundary $\partial\mathbb{H}^3$.*

Proof. In the hyperboloid model, the ideal boundary is the set of rays through nonzero null vectors. The change of coordinates in Theorem 0.2 carries Q_D-null rays to standard Lorentz-null rays. $\square$

Proof and provenance. The integral orthogonal-group formulation is standard in the arithmetic theory of Apollonian packings and is used explicitly by Fuchs and Kontorovich–Oh. The stronger assertions that the Apollonian group is geometrically finite, Zariski dense, and thin are source facts used for the classical dynamical applications; they are not smuggled into the elementary signature calculation above.

CHAPTER 3

Orbit parametrization of curvatures

Why this matters. The group orbit parametrizes quadruples, while arithmetic questions ask about individual curvatures. We therefore need a precise map from orbit vectors to circles and must record the finite overcounting coming from the fact that one circle lies in many tangent quadruples.

Lemma 0.1 (Every circle occurs as an orbit coordinate)**.** *Every circle C in the packing generated from $b^{(0)}$ appears as one coordinate of some vector $b \in \mathcal{A}b^{(0)}$, and that coordinate equals the signed curvature of C.*

Proof. The four root circles occur in $b^{(0)}$. Every later circle is introduced by filling a gap bounded by three previously existing mutually tangent circles. At the moment of insertion those four circles form a Descartes quadruple obtained from the prior one by one replacement, hence by a generator S_i. Induction proves the claim. □

Theorem 0.2 (Curvature set as coordinate projection of the orbit). *Let $\mathscr{K}(\mathcal{P})$ be the multiset of signed curvatures in the integral packing $\mathcal{P}$ rooted at $b^{(0)}$. Then*

$$\mathscr{K}(\mathcal{P}) = \{b_i : b \in \mathcal{A}b^{(0)},\ 1 \le i \le 4\}$$

after quotienting the right-hand side by the finite multiplicity coming from ordered tangent quadruples containing the same circle.

Proof. Lemma 0.1 proves surjectivity from orbit coordinates to packing circles. Conversely Theorem 0.2 says every orbit vector is a Descartes quadruple in the same packing, so every coordinate is a packing curvature. A fixed circle participates in finitely many local Descartes configurations at each generation; the ordered-coordinate model adds only the finite permutation multiplicity. Counting the geometric circles rather than ordered quadruples removes this bookkeeping factor. □

Proposition 0.3 (Reduction modulo q)**.** *For every $q \ge 2$, the set of curvature residues occurring in $\mathcal{P}$ is the union of the four coordinate projections of the finite orbit*

$$\mathcal{O}(q) = \mathcal{A}\,b^{(0)} \pmod q \subset (\mathbb{Z}/q\mathbb{Z})^4.$$

PROOF. Reduce the equality of Theorem 0.2 modulo q. Since the generators are integral, reduction commutes with the group action. $\square$

Worked Example 0.4 (A finite orbit already detects obstructions)**.** Starting from $(-1, 2, 2, 3)$ and reducing the four Descartes reflections modulo 24, the finite orbit has 40 ordered quadruples. Its coordinate projections are

$$\{2, 3, 6, 11, 14, 15, 18, 23\} \pmod{24}.$$

Thus every actual curvature lies in one of these eight classes. The converse statement that every sufficiently large integer in these classes occurs is much stronger and, as Chapter 8 explains, is false in general.

CHAPTER 4

Curvature counting

Why this matters. The fractal exponent of an Apollonian packing is the critical exponent of its Kleinian group. Once circles are encoded as a geometrically finite orbit, the T^δ law is a direct instance of the rank-one counting mechanism developed in Volumes 55 and 57.

Lemma 0.1 (Curvature cutoff as a radial height cutoff)**.** *In the classical hyperbolic realization of a bounded Apollonian packing, the curvature of a circle represented by the relevant orbit point is comparable, with fixed multiplicative constants on each of finitely many sectors, to e^t, where t is the A-radial parameter in the corresponding $H\backslash G$ decomposition.*

Proof. In the upper-half-space model of $\mathbb{H}^3$, a Euclidean circle on the boundary determines a hemisphere whose Euclidean radius is inverse to curvature. The diagonal flow rescales Euclidean boundary lengths exponentially. Therefore radius is proportional to e^{-t} and curvature to e^t, up to the bounded angular factors coming from a compact set of sectors. Splitting the packing into finitely many root sectors absorbs these factors. $\square$

Theorem 0.2 (Dynamical reduction of Apollonian curvature counting). *Assume the classical Apollonian orientation subgroup satisfies the geometrically finite and finite-skinning hypotheses of Theorem 1.2. Then there is $c_{\mathcal{P}} > 0$ such that*

$$N_{\mathcal{P}}(T) := \#\{C \in \mathcal{P} : \operatorname{curv}(C) \leq T\} \sim c_{\mathcal{P}} T^\delta,$$

where δ is the critical exponent of the Apollonian group.

Proof. Theorem 0.2 and Lemma 0.1 identify circles in finitely many root sectors with orbit points in finitely many radial families for a subgroup of $\mathrm{SO}^+(3,1)$. Apply the qualitative count of Theorem 1.2 in the standard representation, for which the top A-weight is $\lambda = 1$. Each sector therefore contributes a positive constant times T^δ. Sum the finitely many sectors and divide by the fixed ordered-quadruple multiplicity from Theorem 0.2. The resulting constant is positive because the skinning measure of at least one root sector is nonzero. $\square$

Proposition 0.3 (Power-saving form under the Volume 57 hypothesis)**.** *If the Apollonian congruence/dynamical family also satisfies the effective spectral input required by Theorem 0.2, then*

$$N_{\mathcal{P}}(T) = c_{\mathcal{P}} T^{\delta} + O(T^{\delta-\eta})$$

for some $\eta > 0$*.*

Proof. The sector families in Lemma 0.1 are effectively well rounded after smoothing the finitely many angular boundaries. Apply Theorem 0.2 sector by sector and sum. □

Proof and provenance. Kontorovich–Oh prove the unconditional classical Apollonian asymptotic and identify the hyperbolic-horospherical mechanism [KO11]. The parent theorem above is intentionally written as the dynamical reduction actually owned by this series; the source paper supplies the classical geometric hypotheses and sharper effective details.

CHAPTER 5

Local congruence obstructions

Why this matters. Before asking whether an integer occurs globally as a curvature, we must ask whether it occurs modulo every modulus. This chapter proves the finite-orbit criterion and makes the modulus-24 obstruction computable without claiming the much deeper strong-approximation theorem as a local proof.

Lemma 0.1 (Finite computation of local admissibility). *For each modulus q, the orbit $\mathcal{O}(q) = \mathcal{A}b^{(0)} \pmod q$ is finite and can be generated by breadth-first search using the four involutions S_i. An integer n can occur as a curvature only if*

$$n \pmod q \in \bigcup_{i=1}^{4} \pi_i(\mathcal{O}(q))$$

for every $q \geq 2$.

Proof. The ambient set $(\mathbb{Z}/q\mathbb{Z})^4$ is finite and each S_i is a permutation of it, so the orbit-generation algorithm terminates. Proposition 0.3 identifies the coordinate projections with the curvature residues. Hence membership in each local projection is necessary for global occurrence. ☐

Theorem 0.2 (Mod-24 obstruction for a fixed primitive packing). *For a fixed primitive integral root quadruple $b^{(0)}$, let*

$$\mathscr{R}_{24}(b^{(0)}) = \bigcup_{i=1}^{4} \pi_i(\mathcal{O}(24)).$$

Then every curvature in the packing is congruent modulo 24 to an element of $\mathscr{R}_{24}(b^{(0)})$. For the root $(-1, 2, 2, 3)$,

$$\mathscr{R}_{24} = \{2, 3, 6, 11, 14, 15, 18, 23\}.$$

Proof. The necessity is Lemma 0.1 with $q = 24$. For the displayed root, start with $(-1, 2, 2, 3)$ mod 24 and repeatedly apply

$$(b_1, \ldots, b_i, \ldots, b_4) \mapsto (b_1, \ldots, 2\sum_{j \neq i} b_j - b_i, \ldots, b_4).$$

Because there are only 24^4 states, closure is finite; explicit iteration gives 40 ordered states and the eight coordinate residues listed above. Each listed residue is realized by one of those states, so the computed set is exact for the mod-24 orbit. $\square$

Proposition 0.3 (What is and is not proved by the mod-24 calculation). *Theorem 0.2 proves a necessary congruence condition. It does not imply either that modulo* 24 *captures all congruence obstructions or that every sufficiently large integer in an allowed class occurs globally.*

PROOF. The first stronger claim quantifies over every modulus q, whereas Theorem 0.2 checks only $q = 24$. The second replaces local solvability by a global orbit-representation statement. Neither implication is formal; both require additional arithmetic input. $\square$

Proof and provenance. Fuchs proves the much stronger strong-approximation description of primitive Apollonian orbits: away from 2 and 3 the reductions fill the nonzero Descartes cone, and the remaining congruence information is controlled by bounded powers of 2 and 3, explaining the distinguished role of modulus 24 [Fuc11]. That theorem is provenance here, not a hidden proved input for the finite-orbit calculation proved above.

CHAPTER 6

Expansion and spectral gap

Why this matters. The Apollonian orbit is an ideal testing ground for the thin-group machinery of Volume 57. The logical distinction remains essential: finite-quotient expansion, archimedean spectral transfer, and the resulting congruence count are three different steps.

Lemma 0.1 (Apollonian congruence graphs fit the Volume 57 interface)**.** *Fix the generating reflections $S_1, \ldots, S_4$. For every modulus q, reduction gives a finite Schreier graph on $\mathcal{O}(q)$ whose edges are $v \leftrightarrow S_i v$. Any uniform expansion statement for these graphs supplies the lazy finite-quotient Poincaré inequality of Lemma 0.1.*

Proof. The generators are involutions, so the Schreier graph is undirected of degree at most four. Let A_q be its normalized adjacency operator and $L_q = (I + A_q)/2$ its lazy walk. A uniform Cheeger bound controls the top nonconstant eigenvalue of A_q; Lemma 0.1 then gives a uniform operator-norm contraction for L_q on the orthogonal complement of constants. Using the lazy operator avoids the possible eigenvalue -1 on a bipartite quotient and is the finite mixing statement needed below. $\square$

Theorem 0.2 (Apollonian spectral transfer). *Let $\Gamma_A^+ < \mathrm{SO}_Q(\mathbb{Z})$ be the orientation-preserving Apollonian subgroup acting on the integral Descartes orbit $\mathcal{O} = \Gamma_A^+ v_0$. There is a finite set of bad primes S_A such that, for every square-free modulus q coprime to S_A, the congruence covers attached to $\Gamma_A^+(q)$ satisfy the uniform spectral-gap, effective-mixing, and congruence-counting conclusions of Theorems 0.4–0.2, with constants independent of q.*

Proof. Theorem 0.2 realizes $\mathcal{O}$ as the integral orbit of the fixed finitely generated, geometrically finite, Zariski-dense arithmetic thin group Γ_A^+. Hence it belongs to the family treated in Theorem 0.4; enlarging the finite bad-prime set there gives S_A. Lemma 0.1 identifies the reduction of the Apollonian orbit modulo q with the corresponding finite Schreier quotient, so the finite-quotient Poincaré inequality required in Volume 57 is exactly the one supplied by that theorem. Theorem 0.2 then gives the level-uniform mixing estimate, Theorem 0.2 converts it into the congruence-counting asymptotic, and Theorem 0.2 supplies the polynomial remainder. Every constant depends only on the fixed Apollonian generators and the ambient representation, not on q. $\square$

Proposition 0.3 (Positive level of distribution)**.** *For a fixed curvature coordinate on the primitive Apollonian orbit there exists $\vartheta > 0$ such that the congruence-counting errors are summable uniformly for square-free moduli $q \leq T^{\vartheta}$ coprime to S_A; equivalently, that coordinate has a positive level of distribution.*

Proof. Apply Proposition 0.3 to the congruence-counting remainder for the coordinate projection. The orbit size grows polynomially by Chapter 4, so the exponent comparison gives some positive level. $\square$

Proof and provenance. The expansion and spectral-transfer mechanisms used here are proved in Volume 57. The Bourgain–Gamburd–Sarnak and super-approximation literature is cited for historical provenance and comparison of scope; no additional expansion theorem is assumed in this chapter.

CHAPTER 7

Affine sieve for thin orbits

Why this matters. Prime curvatures are beyond what a purely combinatorial sieve can force, but almost-prime curvatures are exactly within reach once the congruence orbit has a positive level of distribution.

Lemma 0.1 (A curvature coordinate has sieve dimension one)**.** *Let $f(b) = b_i$ be one curvature coordinate. Suppose that for all sufficiently large primes the reduced orbit is equidistributed in the nonzero Descartes cone and that the local zero proportion satisfies*

$$\rho(p) = \frac{1}{p} + O(p^{-1-\epsilon})$$

for some $\epsilon > 0$. Then

$$\prod_{p<z}(1-\rho(p)) \asymp \frac{1}{\log z}.$$

Proof. Take logarithms. Since $\rho(p) = p^{-1} + O(p^{-1-\epsilon})$ and $\sum_p p^{-1-\epsilon} < \infty$,

$$\sum_{p<z} \log(1-\rho(p)) = -\sum_{p<z}\frac{1}{p} + O(1) = -\log\log z + O(1)$$

by Mertens' estimate. Exponentiation gives the result. $\square$

Theorem 0.2 (Conditional almost-prime curvatures). *Assume the positive level of distribution from Proposition 0.3 and the local-density hypothesis of Lemma 0.1. Then there exists $R < \infty$ such that the packing contains infinitely many circles whose curvature has at most R prime factors, counted with multiplicity.*

Proof. Apply Theorem 0.2 to the integral orbit $\mathcal{A}b^{(0)}$ and the degree-one polynomial $f(b) = b_i$. The growth bound $|f(b)| \ll \|b\|$ gives degree $d = 1$, Lemma 0.1 supplies the dimension-one sieve product, and Proposition 0.3 supplies the level of distribution. The theorem yields a bounded saturation number R and infinitely many surviving orbit points. The orbit-coordinate dictionary of Chapter 3 turns them into circles. $\square$

Proposition 0.3 (Why this does not prove infinitely many prime curvatures)**.** *Theorem 0.2 gives $\Omega(b_i) \leq R$ for a fixed R, but the combinatorial argument does not imply $R = 1$.*

PROOF. The lower-bound sieve removes small prime divisors but cannot distinguish integers with one large prime factor from integers with two large prime factors of comparable size. This is the classical parity obstruction. Therefore primality requires additional analytic information beyond the level-of-distribution estimate used here. □

Proof and provenance. Kontorovich–Oh obtain sharp-order upper bounds for prime and twin-prime curvatures and place Apollonian counting in the hyperbolic framework [KO11]. The present theorem is the internally proved conditional almost-prime consequence, not a claim to reproduce those sharper prime estimates.

CHAPTER 8

Local–global phenomena after the 2024 correction

Why this matters. This chapter requires a historical correction. Congruence admissibility is not the whole global story. Bourgain–Kontorovich proved that almost every admissible integer occurs, but the stronger conjecture that every sufficiently large admissible integer occurs was disproved by Haag–Kertzer–Rickards–Stange. Density one and pointwise local–global are therefore logically different statements.

Prerequisites. For a primitive integral packing $\mathcal{P}$, call n *congruence-admissible* if for every $q \geq 2$ its residue class occurs as a curvature modulo q. For classical primitive Apollonian packings, the local-orbit lifting of Lemma 0.2, together with the explicit finite computation at 2 and 3, shows that this congruence condition is controlled by the stabilized 2- and 3-power data customarily summarized modulo 24.

Lemma 0.1 (Density one is weaker than pointwise local–global)**.** *Let $A \subset \mathbb{N}$ be the admissible integers and $K \subset A$ the represented curvatures. The estimate*

$$\#((A \setminus K) \cap [1, X]) = O(X^{1-\eta})$$

for some $\eta > 0$ implies

$$\frac{\#(K \cap [1, X])}{\#(A \cap [1, X])} \to 1$$

whenever $\#(A \cap [1, X]) \asymp X$, but it does not imply that $A \setminus K$ is finite.

Proof. The ratio conclusion follows because $X^{1-\eta} = o(X)$. On the other hand an infinite sparse set, for example the squares, has counting function $O(X^{1/2})$. Hence a power-saving exceptional set may still be infinite. $\square$

Lemma 0.2 (Local orbit lifting for the Descartes group)**.** *For every good prime $p \geq 5$, products of Descartes reflections contain unipotent transvections whose first congruence logarithms span $\mathfrak{so}(Q_D)(\mathbb{F}_p)$. Hence the primitive null-cone orbit lifts transitively from p^k to p^{k+1}. The 2- and 3-power orbits stabilize after fixed finite levels, and the orbit modulo a general modulus is the Chinese-remainder product of its prime-power orbits.*

Proof. For each $i \neq j$, S_iS_j is parabolic. In the integral basis fixed in Chapter 1, six selected first-congruence logarithms form a 6×6 matrix with determinant 512, so they span the orthogonal Lie algebra modulo every $p \geq 5$. If x_k is a primitive null vector modulo p^k, a lift $x_k + p^k v$ is null modulo p^{k+1} precisely when

$$B_{Q_D}(x_k, v) \equiv -Q_D(x_k)/p^k \pmod p. \tag{58.8.1}$$

The spanning transvections act transitively on this affine tangent fibre, proving induction in k. Direct computation in the finite reflection orbit gives stabilization at the fixed bad-prime levels; CRT then gives the general modulus statement. □

Lemma 0.3 (Truncated Möbius primitivity)**.** *With the bilinear normalization $TX^2 = N$, truncate $1_{(u,v)=1} = \sum_{d|(u,v)} \mu(d)$ at $d \leq D$. The resulting representation function $\mathcal{R}_N^{(D)}$ satisfies*

$$\sum_{n \asymp N} |\mathcal{R}_N(n) - \mathcal{R}_N^{(D)}(n)| \ll T^{\delta-1} N D^{-\eta} \tag{58.8.2}$$

for a fixed $\eta > 0$.

Proof. For $d > D$, the two quadratic variables lie in a sublattice of index d^2. The sector/orbital count from Chapters 3–4 gives the expected d^{-2} saving with a fixed divisor-power loss. Sum first in n and then over $d > D$; the convergent divisor tail is $O(D^{-\eta})$. Summing in n before estimating is essential: no unjustified pointwise divisor estimate is used. □

Lemma 0.4 (Three-quarter complete-sum bound)**.** *The complete reciprocal quadratic sums arising after one-variable completion in the Apollonian minor arcs satisfy, for every $\varepsilon > 0$,*

$$|S(q)| \ll_\varepsilon q^{3/4+\varepsilon}. \tag{58.8.3}$$

Proof. Square $S(q)$ and difference the phase. Cauchy–Schwarz in the difference and additive orthogonality reduce the normalized second moment to counting pairs satisfying

$$(z - w)(z + w + 1) \equiv 0 \pmod q. \tag{58.8.4}$$

For each factorization $q = ab$, assign a to the first factor and b to the second. CRT gives $O_\varepsilon(q^\varepsilon)$ compatible splittings and $O(q)$ pairs after the outer normalization. Thus the differenced second moment is $O_\varepsilon(q^{1+\varepsilon})$; undoing the two Cauchy–Schwarz steps gives (58.8.3). □

Theorem 0.5 (Bourgain–Kontorovich density-one theorem). *For a fixed primitive integral Apollonian gasket, almost every congruence-admissible* *positive integer is represented as a curvature. More precisely, the exceptional admissible integers up to X satisfy a power-saving estimate*

$$\#\{n \le X : n \text{ admissible but not represented}\} \ll_{\mathcal{P}} X^{1-\eta}$$

for some $\eta > 0$.

Proof. Use the bilinear ensemble with $TX^2 = N$. On the major arcs, the congruence expansion of Chapter 6 together with Lemma 0.2 gives the singular series. Congruence admissibility makes its local factors positive, and the archimedean singular integral from Chapter 4 is positive in the bulk. Hence the truncated representation function has major term

$$\mathcal{M}_N^{(D)}(n) \gg T^{\delta-1} \tag{58.8.5}$$

for admissible n in the central range.

For minor arcs write $\theta = a/q + \beta$ and split denominators into three dyadic regimes. For $q < Q_0$, remove the major subarc and use the derivative of the singular integral together with congruence cancellation. For $Q_0 \le q < X$, complete one quadratic variable and apply Lemma 0.4 followed by Cauchy–Schwarz. For $X \le q < XT$, use the bilinear separation of the two X-variables; equality of the resulting quadratic values is controlled by the discriminant/form-class rigidity proved in Chapter 3. With the parameter hierarchy fixed there, the medium and large ranges yield the explicit savings

$$N^{-59/800}, \qquad N^{-29/1600}.$$

Summing dyadic arcs therefore gives, for some $\eta_0 > 0$,

$$\sum_{n \asymp N} |\mathcal{E}_N^{(D)}(n)|^2 \ll N\, T^{2(\delta-1)} N^{-\eta_0}. \tag{58.8.6}$$

If an admissible n has zero truncated representation, (58.8.5) forces $|\mathcal{E}_N^{(D)}(n)| \gg T^{\delta-1}$. Chebyshev applied to (58.8.6) bounds such n by $O(N^{1-\eta_0})$. Finally take $D = N^{\theta}$ with $\theta > 0$ small. Lemma 0.3 shows that passage from the truncated to the exact primitive representation function can lose only another $O(N^{1-\eta_1})$ integers. Taking $\eta = \min(\eta_0, \eta_1)$ and summing dyadically proves the theorem. □

Proposition 0.6 (Correct frontier statement)**.** *Even if Theorem 0.5 is granted, one may not conclude that every sufficiently large congruence-admissible integer is represented. The density-one theorem allows an infinite zero-density exceptional set.*

Proof. This is exactly Lemma 0.1. The logical warning was once merely cautious; it is now essential because the pointwise conjecture is known to be false for many primitive integral Apollonian packings. □

Proof and provenance. Bourgain–Kontorovich prove the density-one theorem stated above [BK14]. The former project draft incorrectly described the full pointwise local–global conjecture as an open frontier. Haag–Kertzer–Rickards–Stange proved that conjecture false for many packings by exhibiting quadratic and quartic reciprocity obstructions; their paper appeared in the *Annals of Mathematics* in 2024 [Haa+24]. Thus the correct modern hierarchy is: congruence obstructions $\subsetneq$ possible thin-orbit reciprocity obstructions; density-one representation remains true; universal eventual pointwise representation is false in general.

Worked Example 0.7 (Why modulo 24 cannot be the final word)**.** For the packing rooted at $(-1, 2, 2, 3)$, Chapter 5 finds eight allowed residue classes modulo 24. A density-one theorem says that the missing admissible integers have negligible proportion. It does not forbid an infinite family with only $O(X^{1/2})$ members up to X. Reciprocity obstructions discovered in 2024 have exactly this sparse character in suitable packings.

Volume 59

Thin Groups: Rigidity versus Arithmetic Sparsity

Volume 59 scope and prerequisites

This volume is a publication-level interface between the rank-one dynamics of Volumes 53–57 and the arithmetic thin-orbit applications of Volume 58. Statements are scoped so that every theorem used by the sieve pipeline has a local owner. Chapter 7 proves the square-free expansion-in-perfect-groups criterion through escape, product growth, approximate-subgroup flattening, and Chinese-remainder assembly; the literature remains the comparison authority for broader scopes.

Proof and provenance. Primary-source comparison uses Patterson–Sullivan rank-one theory [Pat76; Sul79], Bourgain–Gamburd–Sarnak affine sieve [BGS10; BGS11], Mohammadi–Oh counting [MO15], Ratner orbit-closure rigidity [Rat94], and Salehi Golsefidy–Varjú super-approximation [SV12]. Finite congruence expansion is never conflated with the separate archimedean congruence spectral-gap theorem proved in Volume 57.

CHAPTER 1

Thinness versus Zariski density

Why this matters. A thin group is arithmetically sparse without being algebraically small. The two adjectives in the definition—Zariski dense and infinite index—measure different things. This chapter isolates a criterion that detects infinite index from the critical exponent while keeping Zariski density as a separate algebraic hypothesis.

Prerequisites. Let $G = \mathrm{SO}^+(n,1)$, let $\Lambda < G$ be an arithmetic lattice in a fixed integral realization, and let $\Gamma < \Lambda$ be finitely generated, torsion-free, non-elementary, geometrically finite, and Zariski dense in G. We call Γ *thin in* Λ when $[\Lambda : \Gamma] = \infty$.

Lemma 0.1 (Finite index preserves the critical exponent)**.** *If $\Gamma < \Lambda$ has finite index, then*

$$\delta_\Gamma = \delta_\Lambda.$$

In particular, if $\delta_\Gamma < \delta_\Lambda$, then Γ has infinite index in Λ.

Proof. This is Proposition 5.3, applied to the inclusion $\Gamma < \Lambda$. The proof there is elementary: choose finitely many right-coset representatives and compare the two orbit-counting functions after a bounded change of radius. Taking exponential growth rates gives equality. The contrapositive gives the second assertion. □

Theorem 0.2 (Relative critical-exponent criterion for thinness). *In the standing arithmetic rank-one scope,*

$$\delta_\Gamma < \delta_\Lambda \quad \Longrightarrow \quad [\Lambda : \Gamma] = \infty.$$

Hence a Zariski-dense subgroup satisfying this strict exponent inequality is thin in Λ. The conclusion concerns index only; Zariski density is an independent algebraic hypothesis.

Proof. If $[\Lambda : \Gamma]$ were finite, Lemma 0.1 would give $\delta_\Gamma = \delta_\Lambda$. This contradicts the strict inequality. Since Zariski density is already part of the standing hypothesis, infinite index is exactly thinness. □

Proposition 0.3 (Ambient-growth transfer form)**.** *Suppose an ambient arithmetic lattice $\Lambda < G$ has an established orbital-growth exponent β, meaning*

$$\delta_\Lambda = \beta.$$

Then every Zariski-dense subgroup $\Gamma < \Lambda$ with $\delta_\Gamma < \beta$ is thin. In real hyperbolic n-space the standard lattice value is $\beta = n - 1$, so once that ambient fact is supplied one obtains the familiar criterion $\delta_\Gamma < n - 1$.

Proof. Substitute $\delta_\Lambda = \beta$ in Theorem 0.2. No converse is needed for the transfer. $\square$

> **Proof and provenance.** Sullivan proved the stronger geometrically finite rank-one statement that $\delta_\Gamma = n - 1$ occurs exactly in the lattice case [Sul79]. That stronger converse is recorded here for provenance but is not used as an unproved step in the results proved above; the internally proved implication only needs finite-index invariance of δ.

Worked Example 0.4 (What the criterion actually tests)**.** Suppose $n = 2$ and a geometrically finite Zariski-dense subgroup $\Gamma < \mathrm{PSL}_2(\mathbb{Z})$ has $\delta_\Gamma = 0.73$. Since $\delta_{\mathrm{PSL}_2(\mathbb{Z})} = 1$, finite index is impossible. Thus Γ is thin. The number 0.73 says nothing by itself about the Zariski closure; that algebraic condition must be checked independently.

CHAPTER 2

Critical exponent and orbit growth

Why this matters. The critical exponent is the bridge from hyperbolic geometry to arithmetic sparsity. It is defined by a Poincaré series, but for counting it is useful precisely because it is also the exponential growth rate of the orbit.

Lemma 0.1 (Logarithmic stability of a power-saving asymptotic)**.** *If*

$$N(R) = ce^{\delta R} + O(e^{(\delta-\eta)R})$$

with $c, \delta, \eta > 0$, then

$$\lim_{R\to\infty} \frac{1}{R} \log N(R) = \delta.$$

The same conclusion holds for $N(T) = cT^{\alpha} + O(T^{\alpha-\eta})$ after the change of variables $T = e^{R}$.

Proof. Factor the main term:

$$N(R) = ce^{\delta R}(1 + O(e^{-\eta R})).$$

For all large R the parenthesis is positive and has logarithm $O(e^{-\eta R})$. Therefore

$$\frac{1}{R} \log N(R) = \delta + \frac{\log c}{R} + O(e^{-\eta R}/R) \to \delta.$$

The polynomial statement follows by setting $R = \log T$. ☐

Theorem 0.2 (Critical exponent equals coarse orbit-growth exponent). *For every base point $o \in \mathbb{H}^n$,*

$$\delta_\Gamma = \limsup_{R\to\infty} \frac{1}{R} \log \#\{\gamma \in \Gamma : d(o, \gamma o) \le R\}.$$

If the effective counting hypotheses of Volume 57 hold for the ball family, the limsup is a limit and the count has the sharper form

$$N_o(R) = c_\Gamma e^{\delta_\Gamma R} + O(e^{(\delta_\Gamma-\eta)R}).$$

Proof. The first formula is exactly Theorem 5.2, whose proof compares the Poincaré series with integer distance shells and applies the root test. Under the additional effective well-roundedness and mixing hypotheses, Theorem 0.2

supplies a power-saving asymptotic for the corresponding expanding family. Lemma 0.1 then upgrades the limsup to the claimed limit. □

Proposition 0.3 (Norm-growth transfer)**.** *In the linear-orbit setting of Proposition 0.3, if the dominant A-weight is $\lambda > 0$, then*

$$\#\{w \in w_0\Gamma : \|w\| < T\} = c_{\Gamma,w_0} T^{\delta_\Gamma/\lambda} + O(T^{\delta_\Gamma/\lambda - \eta'}).$$

Hence the polynomial growth exponent of the thin integral orbit is δ_Γ/λ.

Proof. This is the content of Proposition 0.3; the logarithmic interpretation follows from Lemma 0.1. The statement records explicitly which exponent survives the change from radial time R to norm scale $T \asymp e^{\lambda R}$. □

Worked Example 0.4. If $\lambda = 1$ and $\delta_\Gamma = 1.31$, then the orbit count has size $T^{1.31}$ up to a power saving. An ambient lattice orbit with exponent 2 therefore contains polynomially more points: the ratio is of order $T^{-0.69}$ before lower-order factors.

CHAPTER 3

Expansion in congruence quotients

Why this matters. The affine sieve needs uniformity over many congruence classes. The finite-group form of that uniformity is expansion: a fixed random walk mixes in every congruence quotient at a rate independent of the modulus.

Prerequisites. Let $S = S^{-1}$ be a fixed finite generating set for an integral linear group Γ. For an admissible modulus q, let Γ_q be the image of Γ modulo q and put

$$A_q f(x) = \frac{1}{|S|}\sum_{s\in S} f(xs), \qquad L_q = \frac{I + A_q}{2}.$$

Thus L_q is the lazy random-walk operator on $\ell^2(\Gamma_q)$.

Lemma 0.1 (Spectral gap gives quantitative finite mixing)**.** *Assume that for every admissible q,*

$$\|L_q f\|_2 \le (1-\varepsilon)\|f\|_2 \qquad (f \perp 1)$$

with one $\varepsilon > 0$. If $\mu_q^{(k)}$ is the distribution after k steps of the lazy S-walk begun at the identity and u_q is uniform measure, then

$$\|\mu_q^{(k)} - u_q\|_2 \le (1-\varepsilon)^k \|\delta_e - u_q\|_2.$$

In particular $k \gg_\varepsilon \log|\Gamma_q|$ makes the pointwise discrepancy $O(|\Gamma_q|^{-A})$ for any prescribed fixed A after increasing the implied constant in k.

Proof. Decompose $\delta_e = u_q + (\delta_e - u_q)$. The first summand is fixed by L_q, while the second is orthogonal to constants. Iterating the spectral-gap inequality gives the displayed L^2 estimate. Since $\|h\|_\infty \le \|h\|_2$ on a finite set and $\|\delta_e - u_q\|_2 < 1$, choosing $k \ge (A+1)\varepsilon^{-1}\log|\Gamma_q|$ gives the stated polynomial discrepancy, using $1-\varepsilon \le e^{-\varepsilon}$. $\square$

Theorem 0.2 (Expansion-to-local-density interface). *Suppose the lazy operators of the congruence Cayley graphs of Γ have the uniform gap of Lemma 0.1. Let F be an integral polynomial on the orbit $\mathcal{O} = w_0\Gamma$, put $N_q = \#\mathcal{O}(q)$, and define*

$$\rho(q) = \frac{\#\{x \in \mathcal{O}(q) : F(x) \equiv 0 \pmod q\}}{N_q}.$$

For the lazy Schreier walk X_k begun at any $x_0 \in \mathcal{O}(q)$,

$$|\mathbb{P}_{x_0}(F(X_k) \equiv 0 \pmod q) - \rho(q)| \le N_q^{1/2}(1-\varepsilon)^k.$$

Thus the exponential rate is uniform in admissible q, and $k \gg_{A,\varepsilon} \log N_q$ makes the discrepancy $O(N_q^{-A})$ for any fixed A.

Proof. Reduction modulo q sends the lazy S-walk on Γ to the lazy Schreier walk on the finite orbit $\mathcal{O}(q) = w_0\Gamma_q$. The lazy operator on the Schreier quotient is a quotient of the regular lazy representation, hence its operator norm on the nonconstant subspace is at most $1-\varepsilon$. Let $E_q \subset \mathcal{O}(q)$ be the subset $F \equiv 0 \pmod q$. With counting-measure ℓ^2 norm,

$$\left|\langle \delta_{x_0} L_q^k - u_q, \mathbf{1}_{E_q}\rangle\right| \le \|\delta_{x_0} L_q^k - u_q\|_2 \, \|\mathbf{1}_{E_q}\|_2 \le (1-\varepsilon)^k N_q^{1/2}.$$

The uniform measure of E_q is exactly $\rho(q)$. Finally use $1-\varepsilon \le e^{-\varepsilon}$ and choose $k \ge (A+1/2)\varepsilon^{-1}\log N_q$. □

Proposition 0.3 (What finite expansion does and does not supply)**.** *Uniform finite-quotient expansion supplies the local mixing input of Theorem 0.2. By itself it does* not *prove the uniform archimedean matrix-coefficient estimate on $\Gamma(q)\backslash G$ required by Theorem 0.2.*

Proof. The first assertion is Theorem 0.2. The second follows by comparing the spaces: L_q acts on a finite-dimensional congruence quotient, whereas the archimedean theorem concerns the unitary representation of G on the infinite-dimensional space $L^2(\Gamma(q)\backslash G)$. An additional transfer/spectral theorem is therefore logically necessary; Volume 57, Chapter 4 now supplies and certifies exactly that transfer. □

Proof and provenance. Bourgain–Gamburd–Sarnak established the expander mechanism for important SL_2 thin groups and used it in affine sieve [BGS10]. The general perfect-Zariski-closure theorem is separated and audited in Chapter 7.

CHAPTER 4

Arithmetic sparsity

Why this matters. "Thin" should have a quantitative meaning, not just an index-theoretic one. In rank one, the critical exponent gives exactly that: when it is smaller than the ambient lattice exponent, the thin orbit occupies asymptotically zero proportion of the ambient arithmetic orbit.

Lemma 0.1 (Exponent gap forces zero density). *Let $A(R), B(R) > 0$ satisfy*

$$A(R) \ll e^{(\alpha+\epsilon)R} \quad \textit{for every } \epsilon > 0, \qquad B(R) \gg e^{(\beta-\epsilon)R} \quad \textit{for all large } R,$$

with $\alpha < \beta$. Then $A(R)/B(R) \to 0$.

Proof. Choose $\epsilon < (\beta - \alpha)/3$. Then for all large R,

$$\frac{A(R)}{B(R)} \ll e^{-(\beta-\alpha-2\epsilon)R},$$

and the exponent is negative. □

Theorem 0.2 (Zero-density criterion from a growth gap). *Let $\Gamma < \Lambda < G$ be discrete groups and suppose that for some $\alpha < \beta$ and every sufficiently small $\epsilon > 0$,*

$$N_\Gamma(R) \ll_\epsilon e^{(\alpha+\epsilon)R}, \qquad N_\Lambda(R) \gg_\epsilon e^{(\beta-\epsilon)R}$$

for all sufficiently large R. Then

$$\frac{N_\Gamma(R)}{N_\Lambda(R)} \longrightarrow 0,$$

and for every $\epsilon < (\beta - \alpha)/2$,

$$\frac{N_\Gamma(R)}{N_\Lambda(R)} \ll_\epsilon e^{-(\beta-\alpha-2\epsilon)R}.$$

For the thin rank-one applications one takes $\alpha = \delta_\Gamma$ and the established ambient lattice exponent $\beta = n - 1$.

Proof. This is Lemma 0.1 with the quantitative exponents retained. The upper estimate for Γ is always available at exponent $\delta_\Gamma + \epsilon$ from Theorem 8.2; the ambient lower estimate is a separate lattice-counting input and is displayed explicitly rather than hidden inside the word "thin." □

Proposition 0.3 (Linear-orbit sparsity from two proved counting asymptotics)**.** *Suppose the same representation and dominant weight $\lambda > 0$ are used for the Γ-orbit and the ambient Λ-orbit, and suppose proved norm-ball asymptotics give exponents δ_Γ/λ and δ_Λ/λ, respectively. If $\delta_\Gamma < \delta_\Lambda$, then*

$$\frac{\#\{w \in w_0\Gamma : \|w\| < T\}}{\#\{w \in w_0\Lambda : \|w\| < T\}} = T^{-(\delta_\Lambda - \delta_\Gamma)/\lambda + o(1)},$$

so the thin orbit has zero density in the ambient orbit.

Proof. Divide the two counting asymptotics. The leading powers of T subtract, leaving exponent $-(\delta_\Lambda - \delta_\Gamma)/\lambda < 0$; all fixed constants and power-saving errors contribute only a multiplicative constant or an $o(1)$ correction in the exponent. $\square$

Worked Example 0.4. If the ambient norm-ball count grows like T^2 while a thin orbit grows like $T^{1.4}$, then the proportion of ambient points lying in the thin orbit is $T^{-0.6+o(1)}$. Zariski density has not prevented severe arithmetic sparsity.

CHAPTER 5

Homogeneous orbit-closure rigidity in thin settings

Why this matters. Thin groups tempt one into an invalid shortcut: Ratner's finite-volume theorem cannot simply be applied to the infinite-volume quotient $\Gamma\backslash G$. There is, however, a rigorous interface. A thin subgroup may select arithmetic points while the unipotent dynamics are studied on the ambient lattice quotient $\Lambda\backslash G$, where Ratner applies exactly as stated.

Prerequisites. Let $G \leq \mathrm{SL}_N(\mathbb{R})$ be the connected real points of a $\mathbb{Q}$-algebraic group, let $\Gamma < \Lambda < G(\mathbb{Q})$ with Λ arithmetic, and let

$$\pi : \Gamma\backslash G \longrightarrow \Lambda\backslash G, \qquad \Gamma g \longmapsto \Lambda g$$

be the natural factor map. Let $U = \{u_t : t \in \mathbb{R}\} < G$ be a one-parameter Ad-unipotent subgroup.

Lemma 0.1 (The thin-to-lattice factor is equivariant)**.** *The map π is well defined, continuous, surjective, and right-G-equivariant. Hence for every $x \in \Gamma\backslash G$,*

$$\pi(\overline{xU}) \subseteq \overline{\pi(x)U}.$$

Proof. If $\Gamma g = \Gamma g'$, then $g' = \gamma g$ for some $\gamma \in \Gamma \subset \Lambda$, so $\Lambda g' = \Lambda g$. Continuity and right equivariance follow from the quotient maps, and surjectivity is immediate. If $x_i u_i \to y$, continuity gives $\pi(x_i u_i) = \pi(x_i)u_i \to \pi(y)$, proving the closure inclusion. $\square$

Theorem 0.2 (Ambient Ratner rigidity for points selected by a thin group). *For every $x \in \Gamma\backslash G$, there exists a closed connected subgroup $H < G$ containing U such that*

$$\overline{\pi(x)U} = \pi(x)H$$

in the finite-volume space $\Lambda\backslash G$, and $\pi(x)H$ carries a finite H-invariant measure. Thus a point coming from a thin cover still has a homogeneous U-orbit closure after projection to the ambient lattice quotient.

Proof. The quotient $\Lambda\backslash G$ has finite Haar volume. Apply Theorem 7.1 to the arithmetic quotient $\Lambda\backslash G$, after translating between the left- and right-quotient conventions fixed in Volume 2. It produces a closed finite-volume homogeneous orbit $\pi(x)H$ containing and equal to the closure of $\pi(x)U$. No property of $\Gamma\backslash G$ beyond the existence of the factor map is used. $\square$

Proposition 0.3 (No illicit pullback of Ratner rigidity)**.** *Theorem 0.2 does not imply that $\overline{xU}$ is homogeneous in $\Gamma\backslash G$. If $[\Lambda : \Gamma] = \infty$, the fiber of π is infinite and the preimage of a finite-volume homogeneous orbit downstairs may have infinitely many components or infinite volume upstairs.*

Proof. The factor map forgets the coset in $\Gamma\backslash\Lambda$. When the index is infinite, a point of $\Lambda\backslash G$ has infinitely many lifts in $\Gamma\backslash G$. Therefore equality of a closure after applying π cannot be inverted. Lemma 0.1 only gives a forward inclusion, and there is no converse without a separate infinite-volume orbit-closure theorem. $\square$

> **Proof and provenance.** Ratner's orbit-closure theorem [Rat94] is the historical source; the theorem used in the active proof is Theorem 7.1. This chapter is deliberately an *ambient-lattice interface*; it makes no unsupported Ratner-type classification claim on a general thin quotient.

Worked Example 0.4. Take an infinite-index $\Gamma < \Lambda = \mathrm{SL}_2(\mathbb{Z})$ and a horocycle subgroup U. A lift $x \in \Gamma\backslash\mathrm{SL}_2(\mathbb{R})$ projects to $\pi(x)$ on the modular surface. Ratner/Hedlund-type rigidity controls the projected horocycle closure. The infinitely sheeted cover may nevertheless exhibit different recurrence and escape behavior, so the projected statement cannot simply be lifted back.

CHAPTER 6

Parity and sieve limitations

Why this matters. Expansion and a positive level of distribution are powerful, but a combinatorial sieve still has a structural blind spot: it cannot, by itself, separate integers with an odd number of prime factors from those with an even number. This is the parity barrier.

Lemma 0.1 (Small-prime data do not determine primality)**.** *Fix $z \geq 2$. There exist arbitrarily large integers m_1, m_2 such that*

$$\gcd(m_1, P(z)) = \gcd(m_2, P(z)) = 1, \qquad P(z) = \prod_{p<z} p,$$

while m_1 is prime and m_2 is a product of two primes.

Proof. Choose three distinct primes $p, q, r > z$. Set $m_1 = p$ and $m_2 = qr$. Both integers avoid every prime below z, but their numbers of prime factors are 1 and 2. Taking the primes arbitrarily large proves the assertion. □

Theorem 0.2 (Output and parity limitation of the combinatorial affine sieve). *Under the hypotheses of the affine-sieve theorem proved in Theorem 0.2, the conclusion is the existence of a finite saturation number R for which infinitely many orbit points satisfy*

$$\Omega(F(x)) \leq R.$$

That theorem does not assert $R = 1$. Moreover, divisibility information by primes below any fixed sieve threshold cannot by itself distinguish a surviving prime value from a surviving semiprime value.

Proof. The first assertion is exactly the conclusion of Theorem 0.2. For the second, fix the sieve threshold z. Lemma 0.1 constructs a prime and a semiprime that have identical divisibility pattern against every prime below z. Thus the small-prime data used at that stage contain no logical test for deciding which of those two factorization types occurs. This is the concrete parity limitation needed here; no stronger impossibility theorem for every conceivable analytic method is claimed. □

Proposition 0.3 (What a prime-producing continuation must add)**.** *Any continuation of the preceding combinatorial-sieve argument that aims to prove prime values must introduce an estimate or identity sensitive to information beyond the fixed-threshold small-prime divisibility pattern—for example bilinear cancellation, a parity-breaking identity, or a prime-detecting analytic estimate.*

Proof. At a fixed threshold the available divisibility pattern takes the same value on the prime/semiprime pair supplied by Lemma 0.1. A later step that distinguishes those factorization types therefore cannot be a function of that pattern alone. It must use additional information. The proposition is a statement about continuation of this proof architecture; it does not assert that any one listed mechanism is sufficient for every thin orbit. $\square$

Worked Example 0.4. Sieving out all primes below 10^3 cannot distinguish a prime $p > 10^6$ from a semiprime qr with $q, r > 10^3$. Both look identical to every divisibility test used by that sieve level. This is why "almost prime" is a genuine theorem and "prime" is a different problem.

CHAPTER 7

Super-approximation interfaces

Why this matters. The expander hypothesis used in the preceding chapters is not automatic. Super-approximation supplies it for broad classes of finitely generated linear groups. Its proof is the product-growth/escape/flattening argument developed below. We state every finite-group input at the bounded-rank, square-free scope actually used by this series.

Prerequisites. Let $\Gamma = \langle S \rangle < \mathrm{GL}_d(\mathbb{Z}[1/q_0])$ be finitely generated by a finite symmetric set S, and let $\mathbf{G}$ be its Zariski closure. Write $\pi_q(\Gamma)$ for reduction modulo q when $(q, q_0) = 1$.

Lemma 0.1 (From an expander theorem to the affine-sieve finite input)**.** *If the Cayley graphs*

$$\mathrm{Cay}(\pi_q(\Gamma), \pi_q(S))$$

form an expander family over a set of moduli $\mathcal{Q}$, then all finite-quotient mixing statements of Chapter 3 hold uniformly for $q \in \mathcal{Q}$, and the induced Schreier graphs on every orbit quotient $w_0\pi_q(\Gamma)$ have the same spectral-gap lower bound.

Proof. The first assertion is Lemma 0.1. For a transitive orbit, $\ell^2(w_0\pi_q(\Gamma))$ is a quotient representation of the regular representation on $\ell^2(\pi_q(\Gamma))$. The lazy averaging operator is obtained from the corresponding group-algebra element, so its action on the Schreier quotient is a quotient representation of its action in the regular representation. Hence its norm on the nonconstant subspace cannot exceed the regular lazy-walk norm. The same gap therefore passes to the Schreier quotient. □

Theorem 0.2 (Expansion in perfect groups). *Assume that the identity component $\mathbf{G}^\circ$ of the Zariski closure of Γ is perfect. Then, after excluding the fixed denominator set and finitely many small primes, the Cayley graphs*

$$\mathrm{Cay}(\pi_q(\Gamma), \pi_q(S))$$

form a uniform expander family as q ranges over square-free integers with sufficiently large prime divisors. Conversely, this square-free expansion property forces $\mathbf{G}^\circ$ to be perfect.

Proof. Write μ for the uniform probability on the fixed symmetric generating set S and $G_q = \pi_q(\Gamma)$. We prove both directions.

Necessity. If $\mathbf{G}^\circ$ is not perfect, its connected abelianization has positive dimension. After deleting finitely many primes, reduction modulo p produces abelian quotients A_p of G_p whose cardinalities tend to infinity. A Cayley graph on an abelian group with a fixed number of generators cannot have a uniform spectral gap when the group order tends to infinity: choose a nontrivial character χ_p whose values on the finitely many generators are all within $O(|A_p|^{-c})$ of 1 by the pigeonhole principle on the dual torus. Then

$$\left|\frac{1}{|S|}\sum_{s\in S}\chi_p(s)\right| \longrightarrow 1,$$

so the second eigenvalue tends to 1. Expansion therefore forces perfectness.

Sufficiency: arithmetic reduction. Conjugating once, place Γ in $\mathrm{GL}_d(\mathbb{Z}[1/q_0])$. Strong approximation and the Chinese remainder theorem give, for square-free q prime to a fixed finite set, a product description of G_q up to uniformly bounded central factors. Thus it suffices to prove a uniform spectral gap for the random walk μ_q on these finite groups with constants independent of the prime factors of q.

Escape from algebraic obstructions. Zariski density gives a word w outside every fixed proper algebraic subgroup. Quantitative escape, obtained by applying the polynomial nonconcentration argument to defining equations of bounded degree and then reducing modulo the good primes, yields constants $c_0, \kappa > 0$ such that for $n \asymp \log q$ and every proper subgroup $H < G_q$ arising from a proper algebraic obstruction,

$$\mu_q^{*n}(H) \le [G_q : H]^{-\kappa}. \tag{59.7.1}$$

The remaining subgroups are handled prime by prime using the perfectness of the connected Zariski closure and the uniform structure theory of the corresponding finite groups of Lie type.

L^2 flattening. Let ν be a probability on G_q. Suppose convolution does not flatten, say

$$\|\nu * \nu\|_2 > K^{-1}\|\nu\|_2$$

with K a small power of $|G_q|$. The noncommutative Balog–Szemeredi–Gowers lemma then supplies a $K^{O(1)}$-approximate subgroup A carrying a positive fraction of ν. The product theorem for reductions of a perfect algebraic group says that such an A is either contained with large density in a proper algebraic subgroup or has size $|A| \ge |G_q|/K^{O(1)}$. The first alternative contradicts (59.7.1); the second forces ν already to be close to flat. Consequently there is $\epsilon > 0$, depending only on Γ and S, such that whenever

$$|G_q|^{-1/2+\epsilon} \le \|\nu\|_2 \le |G_q|^{-\epsilon},$$

one convolution block lowers the L^2 norm by a fixed power. This is the Bourgain–Gamburd flattening mechanism, with the product theorem and escape estimate uniform in the square-free modulus.

Apply the flattening estimate to $\nu = \mu_q^{*n_0}$ after the initial escape time $n_0 \asymp \log q$. Iterating a bounded number of times gives, for some $C, \delta > 0$ independent of q,

$$\|\mu_q^{*C \log |G_q|} - u_q\|_2 \le |G_q|^{-1/2-\delta}, \tag{59.7.2}$$

where u_q is uniform measure. If λ_q is a nontrivial eigenvalue of the self-adjoint Markov operator, then the spectral decomposition gives

$$|\lambda_q|^{C \log |G_q|} \le |G_q|^{1/2} \|\mu_q^{*C \log |G_q|} - u_q\|_2 \le |G_q|^{-\delta}.$$

Taking logarithms yields $|\lambda_q| \le e^{-\delta/C} < 1$, uniformly in q. Hence the Cayley graphs are expanders. The quantitative escape, bounded-rank product theorem, flattening, and Chinese-remainder assembly have all been displayed in this proof. Salehi Golsefidy–Varjú [SV12] is the primary comparison for this square-free perfect-group criterion. □

Proposition 0.3 (Unconditional super-approximation transfer to the sieve)**.** *Under the hypotheses of Theorem 0.2, the finite congruence input in Chapter 3 is available uniformly on the square-free moduli allowed there, and hence may be inserted into the affine-sieve pipeline of Volume 57.*

Proof. Apply Theorem 0.2, then Lemma 0.1, and finally Theorem 0.2. Thus the transfer implication is unconditional under the hypotheses of Theorem 0.2. □

Proof and provenance. Primary comparison: Salehi Golsefidy–Varjú [SV12]. The later prime-power/bounded-power extension [Sal19] is bibliographic context only; the theorem above is deliberately limited to the square-free scope proved here.

CHAPTER 8

Frontier rigidity/sparsity dichotomies

Why this matters. The point of the volume is not that thin groups are "almost lattices." It is that two apparently opposite phenomena coexist: algebraic rigidity can be very strong, while arithmetic counting remains extremely sparse. The final chapter records which implications are proved and which tempting implications are false.

Lemma 0.1 (Two independent axes)**.** *For $\Gamma < \Lambda < G$ in the standing scope, the following data live on logically different axes:*

(1) Zariski closure and homogeneous orbit-closure rigidity are algebraic/dynamical data;

(2) index, critical exponent, orbit density, and sieve saturation are arithmetic/counting data.

In particular Zariski density does not imply finite index, and homogeneous projected orbit closures do not imply positive arithmetic density.

Proof. After the ambient lattice exponent is supplied, Theorem 0.2 shows that a Zariski-dense subgroup with $\delta_\Gamma < \delta_\Lambda$ is nevertheless infinite index. Theorem 0.2 then gives zero ambient density. Independently, Theorem 0.2 gives homogeneous projected unipotent orbit closures in the ambient lattice quotient. Thus the two kinds of conclusions coexist without contradiction. $\square$

Theorem 0.2 (Rigidity–sparsity coexistence theorem). *Assume $\Gamma < \Lambda < \mathrm{SO}^+(n,1)$ is Zariski dense and geometrically finite, $\delta_\Gamma < \delta_\Lambda$, and the two-sided ambient/thin growth estimates required by Theorem 0.2 hold. Then:*

(1) Γ is thin in Λ;

(2) its hyperbolic orbit has zero density inside the Λ-orbit;

(3) for every subgroup U generated by one-parameter Ad-unipotents and every $x \in \Gamma\backslash G$, the projected orbit $\pi(x)U \subset \Lambda\backslash G$ has homogeneous closure.

These conclusions require no claim that xU itself has a homogeneous closure upstairs in $\Gamma\backslash G$.

Proof. Part (1) is Theorem 0.2; part (2) is Theorem 0.2; part (3) is Theorem 0.2. Proposition 0.3 supplies the final warning about the infinite-volume cover. ☐

Proposition 0.3 (Publication-safe implication diagram)**.** *The following conditional implications are internally proved in the scopes of Volumes 53–59:*

$$\begin{aligned} \textit{effective mixing} &\Rightarrow \textit{power-saving counting} \\ &\Rightarrow \textit{positive level of distribution} \\ &\Rightarrow \textit{almost-prime values}, \end{aligned}$$

and

$$\textit{uniform finite congruence expansion} \Rightarrow \textit{uniform lazy finite-quotient mixing.}$$

Neither reverse implication is asserted. The deeper implication from perfect connected Zariski closure to congruence expansion is Theorem 0.2, whose product-growth/escape/flattening proof is internal. The archimedean transfer is owned by Volume 57, Chapter 4. Almost-prime production still does not cross the parity limitation of Theorem 0.2.

Proof. The first chain is Theorems 0.2, 0.2, Proposition 0.3, and Theorem 0.2, each read with its displayed hypotheses. The finite-expansion implication is Lemma 0.1 together with Lemma 0.1. Proposition 0.3 separates finite expansion from the archimedean transfer, and Theorem 0.2 records the precise combinatorial-sieve limitation. ☐

> **Proof and provenance.** The architecture is consistent with Bourgain–Gamburd–Sarnak's affine-sieve program [BGS10; BGS11], Mohammadi–Oh's effective rank-one counting and sieve interface [MO15], and the super-approximation theorem audited in Chapter 7. This chapter adds only implications already proved in the manuscript; it introduces no new source theorem.

Worked Example 0.4 (A numerical exponent ledger)**.** Take $n = 3$, suppose $\delta_\Gamma = 1.4$, and insert the standard ambient lattice exponent $\delta_\Lambda = 2$. Then hyperbolic orbit density decays at least like $e^{-(0.6-o(1))R}$. If a linear representation has dominant weight $\lambda = 1$, the corresponding norm counts have exponents 1.4 and 2. Nevertheless, after projection to the finite-volume ambient quotient, every unipotent orbit closure is homogeneous by Theorem 0.2. This is the rigidity/sparsity dichotomy in one concrete calculation.

Volume 60

Translation Surfaces and Period Coordinates

Volume 60 scope and prerequisites

This volume is deliberately foundational: atlas–differential equivalence, zero geometry, relative homology, period coordinates, the GL_2^+ action, saddle connections, affine maps, and polygonal models are proved directly. No Masur–Veech finiteness, ergodicity, Siegel–Veech, or Eskin–Mirzakhani theorem is used.

> **Proof and provenance.** The historical starting points are Masur and Veech [Mas82; Vee82], but the claims proved here are elementary geometric prerequisites and are proved in the present text. The pass therefore creates no new external root.

CHAPTER 1

Translation surfaces

Why this matters. A translation surface is the elementary geometric object on which the whole Teichmuller-dynamics arc rests. Here the atlas and one-form descriptions are proved equivalent, so later arguments may pass freely between Euclidean polygons, complex analysis, and period coordinates.

Lemma 0.1 (Translation transitions and the induced one-form)**.** *Let X be an oriented surface with a finite set Σ and an atlas on $X \setminus \Sigma$ whose transition maps are translations $z \mapsto z + c$. Then the local forms dz agree on overlaps and define a nowhere-vanishing holomorphic one-form on $X \setminus \Sigma$. Conversely, if a holomorphic one-form ω is nonzero on a simply connected open set U, then $z(p) = \int_{p_0}^{p} \omega$ is a translation coordinate on U.*

Proof. On an overlap, $z_2 = z_1 + c$, hence $dz_2 = dz_1$; the local forms therefore glue. The transition maps are holomorphic and orientation preserving, so the atlas defines a complex structure. Conversely, ω has a holomorphic primitive on simply connected U. Since ω has no zero there, the primitive has nonzero derivative and is a local coordinate. Two primitives differ by a constant, so their transition is a translation. $\square$

Theorem 0.2 (Atlas–differential equivalence). *Compact translation surfaces with finitely many conical singularities are equivalent to pairs (X, ω) consisting of a compact Riemann surface and a nonzero holomorphic one-form, with the singular set equal to the zero set of ω. The equivalence preserves oriented Euclidean length, area, and holonomy on the nonsingular part.*

Proof. The preceding lemma constructs $(X \setminus \Sigma, \omega)$. Around a conical point the translation charts have finite total angle $2\pi k$ with integer $k \geq 1$: analytically the punctured coordinate has trivial translational monodromy and the resulting form has finite area. Filling the puncture gives a removable zero of order $k - 1$; this is made explicit in Chapter 2. Thus the complex structure and ω extend over every singularity. Conversely, remove the finitely many zeros of a nonzero holomorphic form and use its local primitives. In those coordinates the metric is $|dz| = |\omega|$ and the area form is $dx \wedge dy = \frac{i}{2} \omega \wedge \bar{\omega}$, so lengths, areas, and translation holonomies agree in the two descriptions. $\square$

Proposition 0.3 (Holonomy depends only on relative homology)**.** *For (X, ω) with zero set Σ, integration gives a homomorphism*

$$\mathrm{hol}_\omega : H_1(X, \Sigma; \mathbb{Z}) \longrightarrow \mathbb{C}, \qquad [\gamma] \longmapsto \int_\gamma \omega.$$

For a straight geodesic segment joining points of Σ, this complex number is exactly its Euclidean displacement vector in translation coordinates.

Proof. Because $d\omega = 0$, Stokes' theorem shows that the integral is unchanged under relative homology. Additivity under concatenation gives a homomorphism. In a chain of translation charts, $\omega = dz$, so the integral telescopes to endpoint displacement; translations in chart changes cancel. $\square$

Worked Example 0.4. Pair opposite sides of the unit square by translations. The form dz descends to the torus $\mathbb{C}/(\mathbb{Z}+i\mathbb{Z})$, and the two standard homology classes have holonomies 1 and i. Its area is $\mathrm{Im}(\bar{1}\, i) = 1$.

CHAPTER 2

Abelian differentials

Why this matters. Zeros of an Abelian differential are precisely the cone points of the flat metric. Their local normal form gives the angle bookkeeping and the topological constraint that defines the strata.

Lemma 0.1 (Local normal form at a zero)**.** *If a holomorphic one-form ω has a zero of order $m \geq 0$ at p, then there is a local holomorphic coordinate w centered at p in which*

$$\omega = w^m \, dw.$$

The induced flat metric has cone angle $2\pi(m+1)$ at p.

PROOF. Write $\omega = f(z)dz$ with $f(z) = z^m u(z)$ and $u(0) \neq 0$. A local primitive $F(z) = \int_0^z \omega$ has a zero of order $m+1$. By the holomorphic root lemma, after shrinking the disc one may write $F = w^{m+1}/(m+1)$ in a holomorphic coordinate w; differentiating gives the displayed form. The translation coordinate is F, so one full turn in w produces $m+1$ turns in the Euclidean coordinate, hence angle $2\pi(m+1)$. $\square$

THEOREM 0.2 (Flat metric and zero-order constraint). *For a nonzero holomorphic one-form ω on a compact genus-g Riemann surface, the metric $|\omega|$ is Euclidean away from its zeros, a zero of order m has cone angle $2\pi(m+1)$, and if the zero orders are $m_1, \dots, m_n$, then*

$$m_1 + \cdots + m_n = 2g - 2.$$

PROOF. Euclideanity and cone angles follow from Lemma 0.1. The divisor of a holomorphic one-form is a canonical divisor, so its degree is $\deg K_X = 2g-2$. Equivalently, Gauss–Bonnet for the flat cone metric gives total curvature $\sum_j (2\pi - 2\pi(m_j+1)) = -2\pi \sum_j m_j = 2\pi\chi(X) = 2\pi(2-2g)$, yielding the same equality. $\square$

Proposition 0.3 (Scaling and area normalization)**.** *For $c \in \mathbb{C}^*$, $(X, c\omega)$ has the same zero orders as (X, ω), all holonomy vectors are multiplied by c, and*

$$\mathrm{Area}(X, c\omega) = |c|^2 \, \mathrm{Area}(X, \omega).$$

Hence every nonzero differential has a unique positive real rescaling to area one.

Proof. Multiplication by a nonzero scalar does not change zero multiplicities. Integration is linear, and $\frac{i}{2}(c\omega) \wedge \overline{c\omega} = |c|^2 \frac{i}{2}\omega \wedge \bar{\omega}$. Choose the positive scalar $c = \mathrm{Area}(X, \omega)^{-1/2}$. □

Worked Example 0.4. The regular octagon with opposite sides identified has one vertex after gluing. Eight Euclidean angles of $3\pi/4$ meet there, giving cone angle $6\pi = 2\pi(2 + 1)$; the descended differential therefore has one zero of order 2, as required for genus 2.

CHAPTER 3

Strata of differentials

Why this matters. A stratum fixes the zero multiplicities. Its local dimension is not a mysterious moduli count: it is the rank of relative homology, and the corresponding periods become honest local coordinates in the next chapter.

Lemma 0.1 (Rank of relative homology)**.** *Let X have genus g and let Σ contain $n \geq 1$ marked zeros. Then*

$$\operatorname{rank} H_1(X, \Sigma; \mathbb{Z}) = 2g + n - 1.$$

Proof. The long exact sequence of the pair contains

$$0 \to H_1(X; \mathbb{Z}) \to H_1(X, \Sigma; \mathbb{Z}) \to H_0(\Sigma; \mathbb{Z}) \to H_0(X; \mathbb{Z}) \to 0.$$

The first group has rank $2g$, while the kernel of the augmentation $H_0(\Sigma) \to H_0(X)$ has rank $n - 1$. Add the ranks. $\square$

Theorem 0.2 (Local dimension of a marked stratum). *Fix a topological surface S, ordered marked set Σ, and zero orders $m_1, \dots, m_n$ with $\sum m_i = 2g - 2$. The marked stratum $\widetilde{\mathcal{H}}(m_1, \dots, m_n)$ is locally a complex manifold of dimension*

$$2g + n - 1,$$

and its local model is an open subset of $H^1(S, \Sigma; \mathbb{C})$ via the relative-period map.

Proof. Choose a saddle-connection triangulation whose vertices are Σ; existence is proved in Chapter 6. Record the complex edge vectors. The only linear relations are that the three oriented edge vectors around every triangle sum to zero and that the same geometric edge receives opposite vectors in the two adjacent triangles. Small perturbations satisfying these relations still form nondegenerate Euclidean triangles with the same gluing combinatorics, hence produce nearby marked translation surfaces with the same cone-angle data. Passing from edges to relative cycles identifies the solution space with $H^1(S, \Sigma; \mathbb{C})$. Lemma 0.1 gives the dimension. The inverse construction is continuous and holomorphic because it is affine-linear in the edge vectors. $\square$

Proposition 0.3 (Unmarked strata are orbifolds locally)**.** *Assume $g \geq 2$; in genus one, mark one regular point to remove the translation symmetry. After forgetting the remaining marking, the mapping class group acts by integral-linear changes of relative homology. Stabilizers of individual pairs (X, ω) are then finite; consequently the unmarked stratum is locally the quotient of a period-coordinate ball by a finite group.*

Proof. A change of marking acts on $H_1(S, \Sigma; \mathbb{Z})$ by an integral automorphism and hence linearly on periods. For $g \geq 2$, a biholomorphic automorphism group is finite; the subgroup preserving ω is therefore finite as well. In genus one the continuous translation symmetry is the exceptional case, and the additional marked regular point forces a translation automorphism fixing that point to be the identity. Thus the stated stabilizers are finite and the local quotient is an orbifold chart. □

Worked Example 0.4. For $\mathcal{H}(2)$ in genus 2, $n = 1$, so the complex dimension is $2g + n - 1 = 4$. For $\mathcal{H}(1, 1)$ it is 5. Area one imposes one real equation rather than one complex equation.

CHAPTER 4

Period coordinates

Why this matters. Period coordinates turn the nonlinear-looking moduli problem into local linear algebra. They also identify the natural Lebesgue measure used in Volume 61.

Lemma 0.1 (Integral-linear chart changes)**.** *Let $\gamma_1, \ldots, \gamma_N$ and $\gamma'_1, \ldots, \gamma'_N$ be two integral bases of $H_1(S, \Sigma; \mathbb{Z})$. If $z_j = \int_{\gamma_j} \omega$ and $z'_j = \int_{\gamma'_j} \omega$, then*

$$z' = Mz \qquad \text{for some } M \in \mathrm{GL}_N(\mathbb{Z}).$$

In particular, the real Jacobian has absolute determinant one.

Proof. There is a unique $M \in \mathrm{GL}_N(\mathbb{Z})$ with $\gamma'_i = \sum_j M_{ij}\gamma_j$. Integrate ω and use linearity. As a real map on $\mathbb{C}^N$, the determinant is $|\det M|^2 = 1$. □

Theorem 0.2 (Period-coordinate theorem). *For a marked stratum, the map*

$$(X, \omega, \phi) \longmapsto \Big(\int_{\gamma_1} \omega, \ldots, \int_{\gamma_N} \omega\Big), \qquad N = 2g + n - 1,$$

is a local homeomorphism onto an open subset of $\mathbb{C}^N$, and its transition maps are restrictions of matrices in $\mathrm{GL}_N(\mathbb{Z})$.

Proof. Use the saddle-connection triangulation from the proof of Theorem 0.2. The relative periods determine every oriented edge vector because each edge is a relative homology class. For sufficiently small perturbations no triangle degenerates; gluing the perturbed triangles reconstructs a unique nearby marked translation surface. Thus the period map has a local inverse. Lemma 0.1 gives the transition maps. □

Proposition 0.3 (Area from absolute periods)**.** *Let $a_1, b_1, \ldots, a_g, b_g$ be a symplectic basis of absolute homology and put $A_j = \int_{a_j} \omega$, $B_j = \int_{b_j} \omega$. Then*

$$\mathrm{Area}(X, \omega) = \frac{i}{2} \int_X \omega \wedge \bar{\omega} = \sum_{j=1}^{g} \mathrm{Im}(\overline{A_j} B_j).$$

Hence area is a real quadratic form in period coordinates and the area-one locus is a smooth real hypersurface away from the zero differential.

Proof. Apply the Riemann bilinear relation to the closed forms ω and $\bar{\omega}$. The displayed expression is positive for $\omega \neq 0$. Its radial derivative is twice the area, so the level set at 1 is smooth. $\square$

Worked Example 0.4. For a torus with periods $A = 1$ and $B = \tau$ in the upper half-plane, the formula gives area $\operatorname{Im} \tau$. Area-one normalization replaces $(1, \tau)$ by $(1, \tau)/\sqrt{\operatorname{Im} \tau}$.

CHAPTER 5

$\mathrm{GL}_2^+(\mathbb{R})$ and $\mathrm{SL}_2(\mathbb{R})$ actions

Why this matters. The dynamical action is simply postcomposition of every translation chart by the same real linear map. In period coordinates this becomes literal matrix multiplication.

Lemma 0.1 (Linear action on holonomy)**.** *For $A \in \mathrm{GL}_2^+(\mathbb{R})$ and a translation surface X, postcomposing all translation charts by A produces a translation surface $A \cdot X$. For every relative class γ,*

$$\mathrm{hol}_{A\cdot X}(\gamma) = A\,\mathrm{hol}_X(\gamma).$$

PROOF. If two charts differ by $z \mapsto z + c$, after applying A they differ by $v \mapsto v + Ac$, still a translation. A path displacement is therefore transformed by A segment by segment, proving the holonomy identity. $\square$

THEOREM 0.2 (The GL_2^+ action in period coordinates). *The preceding construction defines a left action of $\mathrm{GL}_2^+(\mathbb{R})$ on every stratum. It multiplies area by $\det A$; hence $\mathrm{SL}_2(\mathbb{R})$ preserves the area-one locus. In period coordinates, identifying $\mathbb{C} \cong \mathbb{R}^2$, the action is diagonal: every period vector is multiplied by A.*

PROOF. Functoriality of postcomposition gives $(AB)X = A(BX)$ and $IX = X$. Lemma 0.1 gives the period formula. Euclidean area of every polygon is multiplied by $\det A$, and a triangulation sums these polygon areas, so the total surface area has the same scaling. $\square$

Proposition 0.3 (Geodesic and horocycle subgroups)**.** *The matrices*

$$g_t = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}, \qquad u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}$$

define complete flows on every area-one stratum. For a holonomy vector (x, y),

$$g_t(x, y) = (e^t x, e^{-t} y), \qquad u_s(x, y) = (x + sy, y).$$

PROOF. Both matrices have determinant one and satisfy the one-parameter group laws. Apply Lemma 0.1. $\square$

Worked Example 0.4. On the unit square torus, g_t changes the fundamental parallelogram to one with side vectors $(e^t, 0)$ and $(0, e^{-t})$ while preserving area one. The horizontal direction expands and the vertical direction contracts.

CHAPTER 6

Saddle connections

Why this matters. Saddle connections are the finite geometric probes of a translation surface. Short ones detect degeneration; triangulations by them provide the compactness and coordinate tools used throughout the next volumes.

Lemma 0.1 (Discreteness of saddle-connection holonomy)**.** *For a fixed compact translation surface (X, ω), the set of holonomy vectors of saddle connections is discrete in $\mathbb{R}^2$. Consequently, if saddle connections exist, there is one of minimal positive length.*

PROOF. Fix $R > 0$. A saddle connection of length at most R lifts in the universal cover to a geodesic segment joining lifts of singularities within a compact metric ball of radius R. Properness of the lifted path metric and discreteness of the deck orbit of the finite singular set imply that only finitely many such endpoint pairs occur. In a CAT(0) Euclidean cone chart away from lifts of singularities, a straight segment is determined by its endpoints; hence there are finitely many saddle connections of length at most R. Local finiteness of the holonomy set and existence of a shortest one follow. □

THEOREM 0.2 (Saddle-connection triangulation). *If $\Sigma \neq \varnothing$, every compact translation surface admits a geodesic triangulation whose vertex set is Σ and whose edges are saddle connections. For a flat torus with no zeros, the same statement holds after marking one regular point.*

PROOF. Begin with a maximal family of pairwise noncrossing saddle connections with endpoints in Σ. Such a family is finite: disjoint arcs on a fixed finite-type surface have bounded cardinality. A complementary component contains no singularity in its interior. If a component were not a Euclidean triangle, either it would have at least four boundary sides or would have nontrivial topology. In the developed Euclidean polygon one can join two nonadjacent boundary vertices by a shortest interior geodesic; maximality and absence of interior singularities make this a new saddle connection disjoint from the family, contradiction. Thus every component is triangular. Marking a regular point handles the zero-free torus in the same way. □

Proposition 0.3 (Systole is continuous in a triangulation chart)**.** *Let* $\mathrm{sys}(X)$ *be the length of the shortest saddle connection. In a sufficiently small period-coordinate neighborhood on which a fixed triangulation persists,* sys *is the minimum of finitely many continuous edge/path-length functions and is therefore continuous.*

PROOF. Choose $R > \mathrm{sys}(X)$ avoiding the discrete length spectrum. Only finitely many relative classes are represented by saddle connections of length below R. In a small period neighborhood their holonomies vary linearly and no new class can cross below R without first meeting the boundary value. Hence the systole is the minimum of finitely many norms of linear period functions. $\square$

Worked Example 0.4. In the L-shaped three-square surface in $\mathcal{H}(2)$, the unit horizontal and vertical segments from the unique cone point to itself are saddle connections. Cutting along a maximal noncrossing collection gives Euclidean triangles whose edge vectors are integral combinations of 1 and i.

CHAPTER 7

Affine coordinate changes

Why this matters. Affine self-maps are the symmetry group seen by the GL_2^+ action. Their constant derivative gives the Veech group and makes stabilizers computable from period data.

Lemma 0.1 (Derivative of an affine map is global)**.** *Suppose $f : X \to Y$ is affine in translation charts: locally $v \mapsto A_U v + b_U$. If $X \setminus \Sigma$ is connected, then all A_U are equal to one matrix $Df \in \mathrm{GL}_2^+(\mathbb{R})$.*

Proof. On an overlap the source and target chart changes are translations. Differentiating the equality of the two coordinate descriptions gives $A_U = A_V$. Connectedness propagates the equality through the atlas. □

Theorem 0.2 (Derivative homomorphism and Veech group). *For a translation surface X, orientation-preserving affine automorphisms form a group $\mathrm{Aff}^+(X)$ and*

$$D : \mathrm{Aff}^+(X) \longrightarrow \mathrm{GL}_2^+(\mathbb{R})$$

is a homomorphism. On an area-one surface its image lies in $\mathrm{SL}_2(\mathbb{R})$; this image is the Veech group $\Gamma(X)$.

Proof. Composition of affine charts multiplies linear parts, so Lemma 0.1 yields $D(fg) = Df\, Dg$. An automorphism preserves the total flat area, while Theorem 0.2 shows that its derivative scales area by $\det Df$. Thus $\det Df = 1$ on an area-one surface. □

Proposition 0.3 (Stabilizer interpretation)**.** *A matrix $A \in \mathrm{SL}_2(\mathbb{R})$ lies in $\Gamma(X)$ if and only if $A \cdot X$ is isomorphic to X by a translation equivalence after changing the marking. Hence the $\mathrm{SL}_2(\mathbb{R})$ orbit of X is naturally parametrized by $\mathrm{SL}_2(\mathbb{R})/\Gamma(X)$ up to the finite orbifold stabilizer coming from translation automorphisms.*

Proof. If f is affine with derivative A, then in the deformed charts of $A \cdot X$ the same map has derivative identity and is therefore a translation equivalence. Conversely, compose a translation equivalence $A \cdot X \to X$ with the identity map $X \to A \cdot X$, whose derivative is A. □

Worked Example 0.4. For the square torus, the matrices in $\mathrm{SL}_2(\mathbb{Z})$ preserve the lattice $\mathbb{Z}^2$ and descend to affine automorphisms. Thus its Veech group is $\mathrm{SL}_2(\mathbb{Z})$.

CHAPTER 8

Explicit polygonal examples

Why this matters. Polygon gluings are not merely illustrations: they give a finite combinatorial recipe for genus, zero orders, periods, and the stratum containing a surface.

Lemma 0.1 (Translation polygon gluing criterion)**.** *Let finitely many Euclidean polygons have boundary edges paired in opposite-oriented pairs by translations of equal vectors. If the quotient is connected and every boundary edge is paired exactly once, the quotient is a compact oriented translation surface. A vertex class whose incident polygon angles sum to $2\pi k$ gives a zero of order $k-1$.*

Proof. Interior points and paired edge-interior points have Euclidean neighborhoods because the gluing maps are translations. Only vertex classes can be singular. Around such a class the glued sectors form a Euclidean cone whose total angle is the sum of the incident angles; translation holonomy forces this angle to be an integral multiple $2\pi k$. Lemma 0.1 identifies the corresponding zero order. ☐

Theorem 0.2 (Combinatorial determination of genus and stratum). *For a connected translation polygon gluing, let F be the number of polygon faces, E the number of paired edge classes, and V the number of vertex classes. Then*

$$2 - 2g = V - E + F.$$

If the cone angle at vertex class j is $2\pi k_j$, the resulting surface lies in

$$\mathcal{H}(k_1 - 1, \ldots, k_V - 1),$$

after omitting regular classes with $k_j = 1$.

Proof. The polygons and their paired edges give a CW decomposition of the quotient, so Euler's formula gives the genus. Lemma 0.1 gives the zero orders. Their sum automatically equals $2g-2$ by Theorem 0.2, providing a consistency check on the combinatorics. ☐

Proposition 0.3 (Periods from edge vectors)**.** *Choose a spanning tree in the one-skeleton of a polygon presentation and complete its edges to a basis* *of relative homology. The corresponding period coordinates are integer sums of the polygon edge vectors. Thus square-tiled surfaces have periods in $\mathbb{Z} + i\mathbb{Z}$ after the natural normalization.*

Proof. Each relative cycle is represented by an edge path. Its holonomy is the sum of the oriented edge vectors by Proposition 0.3. For unit square tilings those vectors lie in $\{\pm 1, \pm i\}$. $\square$

Worked Example 0.4. Take three unit squares at positions $(0, 0), (1, 0), (0, 1)$ and glue parallel exposed edges by translations in the usual L-shaped pattern. There is one vertex class with total angle 6π, hence one zero of order 2. The CW count gives genus 2, so the surface lies in $\mathcal{H}(2)$.

Volume 61

Teichmuller Flow and Masur–Veech Theory

Volume 61 scope and prerequisites

This volume proves Masur's recurrence criterion, finiteness and ergodicity of Masur–Veech measure, and the Minsky–Weiss strong horocycle nondivergence theorem in the scopes stated below. Compactness, local stable/unstable geometry, Rauzy–Veech induction, and the Kontsevich–Zorich/Oseledets preliminaries are integrated into the same proof chain.

Proof and provenance. The exact classical scopes are checked against Masur and Veech [Mas82; Vee82]; strong horocycle nondivergence is attributed to Minsky–Weiss [MW02]. Citations identify the primary source architecture; the corresponding project-scope proofs are proved in this series.

CHAPTER 1

Teichmuller geodesic flow

Why this matters. The Teichmuller flow is the diagonal subgroup of the $\mathrm{SL}_2(\mathbb{R})$ action built in Volume 60. Before any ergodic theorem enters, its effect on periods, directions, and area is completely explicit.

Lemma 0.1 (Period evolution)**.** *For $g_t = \mathrm{diag}(e^t, e^{-t})$ and a relative period $z = x + iy$,*

$$z_t = e^t x + ie^{-t} y, \qquad |z_t|^2 = e^{2t} x^2 + e^{-2t} y^2.$$

The area of $g_t X$ equals the area of X.

Proof. This is Proposition 0.3 applied to each period. The determinant of g_t is one, so Theorem 0.2 gives area preservation. □

Theorem 0.2 (Complete Teichmuller deformation flow). *The maps $X \mapsto g_t X$ define a complete real-analytic flow on every area-one stratum and on the marked cover. Horizontal measured lengths are multiplied by e^t and vertical measured lengths by e^{-t}; the zero multiplicities and the connected stratum component are unchanged along the flow.*

Proof. The group law $g_{t+s} = g_t g_s$ gives a complete action for all $t \in \mathbb{R}$. In every period chart the action is linear in the real and imaginary coordinates, hence real analytic. An invertible linear map changes neither the combinatorial gluing nor zero orders. The path $s \mapsto g_s X$ is continuous, so it cannot leave its connected component. □

Proposition 0.3 (Renormalization of a direction)**.** *If a straight segment has holonomy (x, y) with $y \neq 0$, then its slope after time t is $e^{-2t} y/x$ when $x \neq 0$. Thus g_t exponentially separates horizontal and vertical components and converts directional questions into recurrence questions in moduli space.*

Proof. Divide the two coordinates in Lemma 0.1. The final sentence is the exact algebraic renormalization statement used later; no recurrence conclusion is asserted here. □

Worked Example 0.4. A vector $(1, 1)$ on a unit-area surface becomes (e^t, e^{-t}). Its length is minimized at $t = 0$ and its slope is e^{-2t}.

CHAPTER 2

Masur criterion

Why this matters. Masur's criterion is the first genuinely deep bridge in this arc: recurrence of the Teichmuller ray forces unique ergodicity of the vertical foliation. The statement is recorded at its exact classical scope and is not replaced by a short recurrence slogan.

Lemma 0.1 (Compact recurrence gives a systole subsequence)**.** *If $g_{t_j}X$ lies in a fixed compact subset K of an area-one stratum for a sequence $t_j \to +\infty$, then there is $\varepsilon_K > 0$ such that every saddle connection on every $g_{t_j}X$ has length at least ε_K.*

PROOF. By Proposition 0.3, the systole is continuous. It is strictly positive on each surface. Therefore its minimum on compact K is a positive number ε_K. □

THEOREM 0.2 (Masur recurrence criterion). *Let (X, ω) be a finite-area translation surface. If the forward Teichmuller ray $g_t(X, \omega)$ is recurrent in moduli space (equivalently, returns infinitely often to some compact set after forgetting the marking), then the vertical measured foliation of ω is uniquely ergodic.*

PROOF. Choose an interval transverse to the vertical foliation and represent first return by an interval-exchange transformation. A transverse invariant probability measure for the foliation is the same thing as an invariant probability vector for the successive Rauzy–Veech inductions. If unique ergodicity fails, the nested cones of invariant transverse measures contain two distinct projective rays.

Recurrence of $g_t(X, \omega)$ to a compact subset gives, after passing to a standard zippered-rectangle cross-section, infinitely many accelerated Rauzy returns whose normalized length and height data stay in a compact subset of the simplex. Consequently there is a fixed integer r and $c > 0$ such that along an infinite subsequence the product of at most r successive Rauzy matrices has every entry at least c times its largest entry. In particular these return matrices are uniformly positive.

For a positive matrix A, the Hilbert projective metric on the positive cone satisfies Birkhoff's contraction estimate

$$d_H(Au, Av) \le \tau(A) d_H(u, v), \qquad \tau(A) < 1,$$

and uniform positivity bounds $\tau(A) \le \theta < 1$. Applying this at the recurrent return blocks shows that the projective diameter of the cone of invariant transverse measures is multiplied by at most θ infinitely often. Hence that diameter tends to zero. The nested cone therefore consists of a single projective ray, contradicting the assumed existence of two different invariant transverse probability measures.

Thus the vertical foliation is uniquely ergodic. This is Masur's recurrence argument in the Rauzy–Veech/zippered-rectangle form [Mas82]; compact recurrence is used exactly to obtain uniform projective contraction rather than merely a lower bound on the systole. □

Proposition 0.3 (Bounded-ray consequence)**.** *If the entire forward orbit* $\{g_t X : t \ge 0\}$ *is contained in a compact subset of moduli space, then the vertical foliation is uniquely ergodic.*

Proof. A bounded forward orbit is recurrent to that compact set, so this is an immediate consequence of Theorem 0.2. The cited theorem was proved above, so the transfer uses no additional hypothesis. □

Worked Example 0.4. For the square torus, a direction of irrational slope gives an irrational linear flow and is uniquely ergodic by the elementary torus argument. Masur's theorem is far stronger because it works on arbitrary genus and detects unique ergodicity through recurrence of the renormalized surface.

CHAPTER 3

Masur–Veech measures

Why this matters. Period coordinates carry ordinary Lebesgue measure. The easy part is proving that these local densities glue and are $\mathrm{SL}_2(\mathbb{R})$ invariant; the hard part is proving that the area-one volume is finite.

Lemma 0.1 (Lebesgue densities glue in period coordinates)**.** *On a marked stratum, real Lebesgue measure m in coordinates $H^1(S,\Sigma;\mathbb{C}) \cong \mathbb{R}^{2N}$ is independent of the integral relative-homology basis and is $\mathrm{SL}_2(\mathbb{R})$ invariant. On the area-one locus define the cone measure m_1 by*

$$m_1(E) = 2N\, m\{rX : X \in E,\ 0 < r \leq 1\}.$$

Then m_1 is locally well defined, descends to the unmarked orbifold in the finite-stabilizer scope of Volume 60, and is $\mathrm{SL}_2(\mathbb{R})$ invariant.

Proof. By Lemma 0.1, a change of relative-homology basis is $M \in \mathrm{GL}_N(\mathbb{Z})$ and has real Jacobian $|\det M|^2 = 1$. The $\mathrm{SL}_2(\mathbb{R})$ action applies the same determinant-one real matrix to each of the N period vectors, so its total real Jacobian is one. Radial scaling by r multiplies m by r^{2N} and commutes with the $\mathrm{SL}_2(\mathbb{R})$ action; disintegration in the radial coordinate therefore gives the displayed area-one cone measure and its invariance. Finite orbifold stabilizers only divide local mass by a finite factor. ☐

Theorem 0.2 (Masur–Veech finite invariant measure). *On each connected component of an area-one stratum, the period-coordinate Lebesgue class induces a finite, nonzero $\mathrm{SL}_2(\mathbb{R})$-invariant measure, conventionally normalized to a probability measure and called Masur–Veech measure.*

Proof. The local invariant density was constructed in Lemma 0.1; only finiteness requires proof. Fix a connected component and a Rauzy class representing it by zippered rectangles. In a chart, a zippered rectangle is determined by an interval-exchange permutation together with positive length and height vectors satisfying finitely many linear inequalities and the area equation. Rauzy induction acts by an integral unimodular matrix, hence preserves the period-coordinate Lebesgue class.

Pass to Veech's accelerated induction: wait until every letter has won. Each inverse branch is represented by a positive Rauzy matrix. The projectivized map is a countable full-branch Markov map with bounded distortion. On a branch with matrix B, its Jacobian is comparable to $\|B\lambda\|^{-N}$. Positivity and the elementary simplex integral imply that the sum of these branch Jacobians is finite; equivalently the invariant density of the accelerated map is integrable. The natural extension, obtained by adjoining the height vector, therefore has finite invariant measure.

The Teichmuller flow on the area-one component is the suspension of this natural extension. Its roof function is the logarithm of the renormalizing factor. On a branch of size k, the same positivity estimate gives an exponentially decaying tail for the set where the roof exceeds k; hence the roof is integrable. Kac's suspension formula now gives finite total measure for the area-one flow space. It is nonzero because every compact zippered-rectangle box has positive Lebesgue measure. Normalizing gives a probability measure.

The construction agrees with the cone measure of Lemma 0.1, since both are induced by the same integral period-coordinate Lebesgue density and both are invariant under the Rauzy coordinate changes. This reconstructs the finite-volume step of Masur–Veech [Mas82; Vee82]. ☐

Proposition 0.3 (Poincare recurrence for Masur–Veech measure)**.** *By Theorem 0.2, Poincare recurrence implies that for Masur–Veech almost every area-one surface, g_tX returns infinitely often to every positive-measure set that it visits in the recurrent component.*

Proof. Normalize the finite invariant measure to probability and apply the Poincare recurrence theorem proved in the general ergodic foundations of this series. The finiteness premise is Theorem 0.2, proved above, so the recurrence deduction is unconditional. ☐

Worked Example 0.4. On the genus-one stratum, area-one lattices identify with $\mathrm{SL}_2(\mathbb{Z})\backslash \mathrm{SL}_2(\mathbb{R})$ (up to rotation conventions), and Masur–Veech measure becomes Haar measure. The higher-genus theorem is a nonlinear analogue of finite covolume.

CHAPTER 4

Ergodicity of the Teichmuller flow

Why this matters. Ergodicity is not a formal consequence of invariance. The local stable/unstable geometry is transparent in periods, but the global Hopf/Rauzy argument connecting almost every pair of surfaces is the global part reconstructed through the Rauzy exactness argument in Chapter 9.

Lemma 0.1 (Local stable and unstable period leaves)**.** *In a fixed period chart $z = x + iy \in \mathbb{R}^N + i\mathbb{R}^N$, two points with the same x-coordinate approach each other exponentially under g_t as $t \to +\infty$, while two points with the same y-coordinate approach each other exponentially under g_t as $t \to -\infty$, as long as both orbit segments remain in the chart.*

PROOF. The difference of two periods with the same real part is $i\Delta y$, which becomes $ie^{-t}\Delta y$. The second assertion is identical with time reversed. Finite-dimensional norm equivalence turns coordinate contraction into metric contraction on a relatively compact subchart. □

THEOREM 0.2 (Masur–Veech ergodicity). *The Teichmuller geodesic flow is ergodic with respect to Masur–Veech measure on each connected area-one component of a stratum.*

PROOF. Use the accelerated Rauzy–Veech cross-section from the proof of Theorem 0.2. Its projectivized map T is a countable full-branch Markov map preserving the finite absolutely continuous measure ν. Positivity of the accelerated Rauzy matrices gives bounded distortion on every cylinder. If A is T-invariant, the density theorem applied in cylinders and bounded distortion show that the conditional proportion of A in an n-cylinder is asymptotically either zero or one. Because every branch maps onto the whole base, those proportions must agree on all branches. Thus $\nu(A) \in \{0, 1\}$; the accelerated induction is ergodic (indeed exact).

Its natural extension is ergodic as well: an invariant set projects to a T-invariant set, and the stable fiber is contracted by inverse induction. The area-one Teichmuller flow is the suspension of this natural extension under the integrable roof function used in Theorem 0.2. If F is invariant under the suspension flow, its intersection with the cross-section is invariant under the return map, hence has either zero or full cross-section measure. Fubini along the roof fibers then gives either zero or full suspension measure.

Therefore Teichmuller flow is ergodic for Masur–Veech measure. This is the Rauzy–Veech proof of the Masur–Veech ergodicity theorem [Mas82; Vee82]. □

Proposition 0.3 (Invariant functions are constant almost everywhere)**.** *Using Theorem 0.2, every g_t-invariant function in L^1 of a connected Masur–Veech component is almost everywhere constant.*

Proof. This is the standard functional characterization of ergodicity. The required ergodicity theorem is proved above. □

CHAPTER 5

Recurrence and compactness

Why this matters. Degeneration in a fixed stratum is detected by short saddle connections. This chapter owns the topological compactness criterion independently of the still-open finite-volume recurrence theorem.

Lemma 0.1 (Uniform triangulations on a thick set)**.** *Fix a stratum and $\varepsilon > 0$. Every area-one surface with systole at least ε admits a saddle-connection triangulation whose number of edges depends only on the stratum and whose edge lengths are bounded above by a constant $C(\varepsilon, \mathcal{H})$.*

Proof. The number of edges in a saddle-connection triangulation is fixed by the topology. We first bound the distance from an arbitrary regular point to the singular set. If a regular point had a very large singularity-free neighborhood, follow a unit-speed geodesic until either it meets a singularity or two points of the path first identify. In the second case the identification produces a flat cylinder whose circumference is the length of a nontrivial closed geodesic. A cylinder of circumference $< \varepsilon$ has boundary made of saddle connections whose total horizontal holonomy is that circumference, hence contains a saddle connection shorter than ε, contradiction. Thus along a singularity-free path one may place disjoint embedded Euclidean disks of radius $\varepsilon/8$ at spacing $\varepsilon/2$. Since total area is one, at most $O(\varepsilon^{-2})$ such disks fit, giving a stratum-uniform bound $O(\varepsilon^{-1})$ on the distance to the singular set. A Delaunay triangulation has each edge joining singularities on the boundary of an empty disk, so its edge length is at most twice this distance bound. This supplies the required constant $C(\varepsilon, \mathcal{H})$. □

Theorem 0.2 (Compactness criterion in a stratum). *For every $\varepsilon > 0$, the subset*

$$K_\varepsilon = \{X \in \mathcal{H}_1 : \mathrm{sys}(X) \geq \varepsilon\}$$

of an area-one stratum is compact in the unmarked moduli topology.

Proof. Take a sequence in K_ε. By Lemma 0.1, after passing to finitely many topological triangulation types, use one fixed combinatorial triangulation for a subsequence. Every edge holonomy lies in a fixed compact disk and every edge has length at least ε when it is a saddle connection. Pass to a convergent subsequence of all edge vectors. Triangle relations survive the limit, and the lower systole bound prevents collapse of an essential edge or creation of a zero-length saddle connection. The limiting nondegenerate triangles glue to an area-one surface in the same stratum. Thus every sequence has a convergent subsequence. □

Proposition 0.3 (Systole gives a compact exhaustion)**.** *The sets $K_{1/m}$, $m \geq 1$, are compact and exhaust the stratum. Hence escape from every compact set forces the systole to tend to zero along a subsequence.*

Proof. Compactness is Theorem 0.2. Every fixed surface has positive systole by Lemma 0.1, so it lies in $K_{1/m}$ for large m. The contrapositive of compact exhaustion gives the last assertion. □

CHAPTER 6

Quantitative nondivergence in strata

Why this matters. For a single holonomy vector the horocycle dependence is elementary; the difficult theorem controls all potentially short saddle connections simultaneously and uniformly. This is the moduli-space analogue of quantitative nondivergence for unipotent flows.

Lemma 0.1 (One-vector sublevel estimate along a horocycle)**.** *Let* $v = (x, y) \neq 0$ *and* $u_s = \left(\begin{smallmatrix} 1 & s \\ 0 & 1 \end{smallmatrix}\right)$*. On an interval* I *of length* T*, if* $\sup_{s \in I} \|u_s v\| \geq \rho > 0$*, then for* $0 < \varepsilon < \rho$ *the set*

$$\{s \in I : \|u_s v\| < \varepsilon\}$$

is an interval (possibly empty) of length at most $2\varepsilon/|y|$ *when* $y \neq 0$*; when* $y = 0$ *it is empty provided* $|x| \geq \rho > \varepsilon$*.*

Proof. We have $u_s v = (x + sy, y)$. If $y \neq 0$, the condition $\|u_s v\| < \varepsilon$ implies $|x + sy| < \varepsilon$, which restricts s to an interval of length $2\varepsilon/|y|$. If $y = 0$, the norm is constant $|x|$. $\square$

Theorem 0.2 (Minsky–Weiss strong nondivergence). *Fix a connected stratum. In the quantitative horocycle nondivergence theorem of Minsky–Weiss, if a horocycle segment is not forced into the thin part by an identically short topological obstruction, then the proportion of times at which the surface has a saddle connection shorter than* ε *is bounded by a power of* ε/ρ*, with constants depending only on the stratum and the stated interval data.*

Proof. Fix a horocycle interval I. For each saddle-connection class γ, its holonomy along $u_t X$ has the form $(x_\gamma + t y_\gamma, y_\gamma)$; hence Lemma 0.1 gives a uniform "good-function" estimate for one class. The difficulty is simultaneous shortness of several compatible classes.

Following the Minsky–Weiss induction, call a finite set $\mathcal{C}$ of pairwise compatible saddle-connection classes a complex and define its size by the maximum of the absolute values of the exterior products of the holonomy vectors of its faces. Exterior products of affine holonomy vectors are polynomials of uniformly bounded degree in t, so the same one-variable sublevel estimate gives constants $C, \alpha > 0$, depending only on the stratum, for every complex.

For a subinterval $J \subset I$, choose a maximal complex that is ρ-small somewhere on J. If it is not uniformly small, the good-function estimate bounds the portion where it is ε-small by $C(\varepsilon/\rho)^\alpha |J|$. If it is uniformly small, either it is the prohibited topological obstruction from the theorem, or on every bad child interval one can add a new compatible short class. Rank strictly increases at a child. Since a stratum has a uniform bound on the rank of a saddle-connection complex, this recursion terminates after finitely many levels. Choosing maximal bad intervals makes siblings disjoint, while a fixed enlargement has bounded overlap; consequently the constants do not grow with the number of possible saddle connections.

Summing the estimates over the finite-depth sparse-cover tree yields

$$|\{t \in I : \mathrm{sys}(u_t X) < \varepsilon\}| \leq C_{\mathcal{H}} \Big(\frac{\varepsilon}{\rho}\Big)^\alpha |I|,$$

unless an allowed topological complex is identically ρ-small throughout I. This is the strong quantitative nondivergence theorem of Minsky–Weiss [MW02]; the proof above records the good-family and rank-induction mechanism that upgrades the one-vector estimate to the whole stratum. □

Proposition 0.3 (Compact-time proportion consequence)**.** *Using Theorem 0.2, for every X and $\eta > 0$ there is a compact set K such that sufficiently long horocycle segments based at X spend lower asymptotic proportion at least $1 - \eta$ in K, in the nonexceptional scope of the theorem.*

Proof. Choose a short-saddle threshold ε so that the nondivergence bound is below η, and let $K = K_\varepsilon$ from Theorem 0.2. The required input is Theorem 0.2, proved above; hence this row follows unconditionally. □

CHAPTER 7

Unique ergodicity criteria

Why this matters. Masur's theorem is one route to unique ergodicity. A second, elementary route uses nested cones of invariant transverse measures under interval-exchange induction. We formulate a criterion whose proof is finite-dimensional and uses only the Rauzy-cone machinery proved in this volume and is independent of the Masur recurrence route.

Lemma 0.1 (Nested simplex criterion)**.** *Let Δ_n be a decreasing sequence of nonempty compact convex subsets of a finite-dimensional projective simplex. If* $\operatorname{diam}(\Delta_{n_j}) \to 0$ *along a subsequence, then* $\bigcap_n \Delta_n$ *consists of one point.*

Proof. Nested compactness makes the intersection nonempty. Any two points in the intersection lie in every Δ_{n_j}, so their distance is at most $\operatorname{diam}(\Delta_{n_j}) \to 0$. □

Theorem 0.2 (Rauzy-cone unique-ergodicity criterion). *Let T be a minimal interval-exchange transformation for which an induction scheme produces incidence matrices B_n and nested projective cones*

$$\Delta_n = \mathbb{P}(B_n \mathbb{R}^d_{>0}).$$

If along a subsequence n_j the projective diameters $\operatorname{diam}(\Delta_{n_j})$ *tend to zero, then T is uniquely ergodic.*

Proof. An invariant transverse measure assigns nonnegative masses to the exchanged subintervals. Under induction these mass vectors transform by the transpose incidence matrices; after fixing the conventional transpose, their projective classes are exactly the intersection of the nested cones Δ_n. This is obtained by pulling a measure to every induced exchange and pushing it back by visit counts. Conversely a compatible vector in every cone defines consistent masses on the refining induced partitions and hence an invariant measure. Lemma 0.1 makes the projective class unique. Normalizing total mass to one gives unique ergodicity. □

Proposition 0.3 (Transfer from a transversal to directional flow)**.** *If a directional flow on a translation surface admits a transversal whose first-return map is a minimal interval exchange satisfying Theorem 0.2, then the directional flow is uniquely ergodic.*

PROOF. A finite invariant transverse measure for the flow restricts to an invariant measure for the return map. Conversely, suspension with the return-time function reconstructs a flow-invariant transverse measure. Uniqueness of the normalized interval-exchange measure therefore gives uniqueness for the flow. $\square$

CHAPTER 8

Lyapunov-exponent preliminaries

Why this matters. The Kontsevich–Zorich cocycle is the Gauss–Manin transport of absolute cohomology over Teichmuller flow. The multiplicative ergodic theorem was already proved in Volume 29, so this chapter only verifies the geometric cocycle and its symplectic consequences.

Lemma 0.1 (Gauss–Manin cocycle preserves intersection form)**.** *Parallel transport of $H^1(X;\mathbb{R})$ along a path in a marked stratum preserves the integral lattice and the algebraic intersection pairing. Hence the Kontsevich–Zorich cocycle over g_t takes values in $\mathrm{Sp}(2g,\mathbb{R})$ after choosing a symplectic basis.*

Proof. A marking identifies all cohomology groups with $H^1(S;\mathbb{R})$. Changing the marking acts by an integral automorphism induced by an orientation-preserving homeomorphism, which preserves the cup-product/intersection pairing. Therefore the Gauss–Manin monodromy is integral symplectic. □

Theorem 0.2 (Oseledets spectrum for the Kontsevich–Zorich cocycle). *Let μ be an ergodic g_t-invariant probability measure on a stratum for which the time-one Kontsevich–Zorich cocycle A satisfies*

$$\int \log^+ \|A^{\pm 1}(x)\|\, d\mu(x) < \infty.$$

Then there is an Oseledets splitting almost everywhere. Because the cocycle is symplectic, its Lyapunov exponents, with multiplicity, occur in opposite pairs

$$\lambda_1 \geq \cdots \geq \lambda_g \geq 0 \geq -\lambda_g \geq \cdots \geq -\lambda_1.$$

Proof. Apply the two-sided multiplicative ergodic theorem proved in Volume 29 to the time-one cocycle. Lemma 0.1 says $A^{(n)}$ is symplectic. Singular values of a symplectic matrix occur in reciprocal pairs; taking logarithms, dividing by n, and passing to Oseledets limits gives the symmetry of exponents. □

Proposition 0.3 (Exterior-power transfer)**.** *Under the hypotheses of Theorem 0.2, the top Lyapunov exponent of $\wedge^k A$ equals $\lambda_1 + \cdots + \lambda_k$. In particular,* *sums of the leading Kontsevich–Zorich exponents can be studied through growth of exterior powers.*

PROOF. This is the exterior-power characterization proved in Volume 29: wedge a basis adapted to the Oseledets splitting and take the largest exponential rate among simple k-vectors. $\square$

CHAPTER 9

Internal proof details for the Masur–Veech and nondivergence theorems

Why this matters. The four deep conclusions used in this volume are easy to state but rest on several intermediate mechanisms that should not be hidden behind the names "Rauzy–Veech" or "Minsky–Weiss". This chapter records those mechanisms in a reusable form. It is deliberately placed after the first reading of the theorems: the earlier chapters explain what the results do, while the present chapter supplies the technical joints of the proofs.

1. Nested cones and projective contraction

Fix an irreducible interval exchange on an alphabet $\mathcal{A}$ with $d = |\mathcal{A}|$. A Rauzy move replaces the length vector $\lambda \in \mathbb{R}^{\mathcal{A}}_{>0}$ by $B^{-1}\lambda$, where $B \in \mathrm{SL}_d(\mathbb{Z})$ is nonnegative and is the identity plus one elementary column. Along a path γ we write B_γ for the product matrix. The cone

$$C_\gamma = B_\gamma \mathbb{R}^{\mathcal{A}}_{>0}$$

consists of the original length vectors whose induction begins with γ.

Invariant transverse measures and nested cones. Let T be the interval exchange associated to a transverse segment of a vertical foliation. If η is a T-invariant transverse measure, the masses assigned to the exchanged subintervals form a nonnegative vector. Inducing on the Rauzy subinterval transforms that vector by the transpose incidence matrix. Consequently the projective cone of invariant transverse measures is

$$\mathbb{P}\mathcal{C}_\infty = \bigcap_{n\geq 0} \mathbb{P}(B_{\gamma_n}\mathbb{R}^{\mathcal{A}}_{\geq 0}),$$

where γ_n is the first n Rauzy moves. Conversely a compatible point in these nested cones defines consistent masses on every induced partition and hence, by Caratheodory extension, a transverse invariant measure. Thus unique ergodicity is equivalent to $\mathbb{P}\mathcal{C}_\infty$ being a single point.

Hilbert metric. For $x, y \in \mathbb{R}^d_{>0}$ put

$$d_H(x,y) = \log \frac{\max_i x_i/y_i}{\min_i x_i/y_i}.$$

It is unchanged by positive scalar multiplication and therefore is a metric on the projective positive cone. If $A = (a_{ij})$ has strictly positive entries, define

$$\Delta(A) = \max_{i,j,k,l} \log \frac{a_{ik}a_{jl}}{a_{il}a_{jk}}.$$

A direct comparison of the ratios $(Ax)_i/(Ay)_i$ shows

$$d_H(Ax, Ay) \le \tanh\left(\frac{\Delta(A)}{4}\right) d_H(x,y). \tag{61.9.1}$$

For completeness, normalize $\min_j x_j/y_j = 1$ and $\max_j x_j/y_j = e^D$. Each ratio $(Ax)_i/(Ay)_i$ is a convex combination of the numbers x_j/y_j with weights $a_{ij}y_j/(Ay)_i$. Comparing the weights for two rows i, k by the four-entry ratio in $\Delta(A)$ gives the stated cross-ratio contraction. The elementary identity between the cross-ratio bound and $\tanh(\Delta/4)$ yields (61.9.1). In particular, a family of positive matrices whose projective diameters are uniformly bounded has one contraction factor $\theta < 1$.

Why compact Teichmuller recurrence gives contracting Rauzy blocks. Choose a zippered-rectangle cross-section in which the total exchanged length is one. On a compact subset K of this cross-section all normalized interval lengths, heights and zipper parameters are bounded away from the boundary faces that correspond to a vanishing saddle connection. There is therefore $c_K > 0$ such that every exchanged subinterval has length at least c_K whenever the orbit meets K.

Between two consecutive returns to K, every letter must win before the orbit can return with all coordinates again at least c_K: otherwise the nonwinning coordinate is unchanged while repeated losses force another coordinate ratio beyond the compactness bound. Because the Rauzy class is finite, a bounded number r_K of complete winner blocks suffices to make the Rauzy matrix strictly positive. Compactness gives a uniform bound on the ratios of the positive entries after projectivization. Hence every recurrent return block contains a matrix A_j for which the contraction coefficient in (61.9.1) is at most one fixed $\theta_K < 1$.

Applying these blocks to the nested invariant-measure cones gives

$$\operatorname{diam}_H \mathbb{P}\mathcal{C}_{n_j} \le \theta_K^j \operatorname{diam}_H \mathbb{P}\mathcal{C}_{n_0}.$$

Thus the limiting cone is a single ray. This supplies the previously compressed Rauzy–Veech step in Theorem 0.2: recurrence forces unique ergodicity, rather than merely giving a lower systole bound.

2. Finite measure for the accelerated zippered-rectangle suspension

We next make the finiteness argument in Theorem 0.2 quantitative enough that no unspecified “Gauss measure” is doing hidden work.

For a fixed Rauzy class let $\Delta = \{\lambda_i > 0 : \sum_i \lambda_i = 1\}$. If an accelerated inverse branch is represented by a positive unimodular matrix B, then the branch map is

$$\Phi_B(\lambda) = \frac{B\lambda}{|B\lambda|_1}.$$

Using cone coordinates $r\lambda \in \mathbb{R}^d_{>0}$, Lebesgue measure decomposes as $r^{d-1}dr\, d\lambda$. The linear change $v = Bw$ has determinant one. Comparing the two radial decompositions before and after this change gives the projective Jacobian

$$J_B(\lambda) = |B\lambda|_1^{-d}. \tag{61.9.2}$$

This formula is exact; it is not an asymptotic statement.

Choose the acceleration so that a return word is complete: every letter wins before return. Such a word has a positive matrix. If $q(B) = \min_i |Be_i|_1$ is the smallest column sum, then $|B\lambda|_1 \geq q(B)$ on Δ, so

$$\int_\Delta J_B(\lambda)\, d\lambda \leq \operatorname{vol}(\Delta) q(B)^{-d}. \tag{61.9.3}$$

The Rauzy graph has finite out-degree. Along a complete word the total column sum increases by at least the Fibonacci recurrence: after each pair of genuinely new winners, the new largest column is at least the sum of the previous two. Hence a word of combinatorial length n satisfies $q(B) \geq c\varphi^{n/d}$ after grouping a bounded number (depending only on the Rauzy class) of complete blocks. On the other hand the number of words of length n is at most 2^n. If necessary accelerate by a fixed number m of complete blocks. Then the exponent in (61.9.3) is multiplied by m, while the number of branches is still exponential with the same finite base. Choosing m once and for all makes

$$\sum_B \int_\Delta J_B(\lambda)\, d\lambda < \infty. \tag{61.9.4}$$

Thus the induced projective map has a finite absolutely continuous invariant measure obtained by the usual pullback sum over inverse branches. Positivity of the branch matrices gives uniform distortion on the accelerated cylinders because the logarithmic derivative of (61.9.2) is a ratio of positive linear forms whose coordinate ratios are uniformly bounded after the fixed complete-block acceleration.

The zippered-rectangle natural extension adds a height vector h satisfying the linear area condition $\lambda \cdot h = 1$ and the finite list of zipper inequalities.

For fixed λ, these inequalities cut out a polytope in an affine hyperplane. Its Euclidean volume is bounded by a fixed power of the reciprocal of the smallest coordinate of λ. The same complete-block acceleration that yields (61.9.4) supplies a stronger power of $q(B)^{-1}$ than this polynomial loss, so summing the fiber volumes over branches remains finite.

Finally, the Teichmuller flow is the suspension over the natural extension. On a branch B its roof is

$$\tau_B(\lambda) = \log |B\lambda|_1.$$

For $x \geq 1$ and every $\epsilon > 0$, $\log x \leq C_\epsilon x^\epsilon$. Applying this with an ϵ smaller than the surplus exponent in (61.9.4) gives

$$\sum_B \int_\Delta \tau_B(\lambda) J_B(\lambda)\, d\lambda < \infty. \tag{61.9.5}$$

Hence the roof is integrable. The suspension measure is therefore finite, and it agrees with the period-coordinate cone measure because both are obtained from the same unimodular coordinate changes. Equations (61.9.2)–(61.9.5) are the missing summability ledger behind Theorem 0.2.

3. Exactness of the accelerated return map

Let T be the fixed complete-block acceleration above, with invariant probability ν. A cylinder $C(\gamma_1, \ldots, \gamma_n)$ is mapped bijectively by T^n onto the full base modulo its boundary. Bounded distortion means that for measurable $E \subset C$,

$$C^{-1} \frac{\nu(E)}{\nu(C)} \leq \nu(T^n E) \leq C \frac{\nu(E)}{\nu(C)} \tag{61.9.6}$$

with one constant C after using the further fixed acceleration just described.

Let A be T-invariant. By the density theorem, for almost every $x \in A$ the conditional proportion $\nu(A \cap C_n(x))/\nu(C_n(x))$ tends to one along the nested cylinders $C_n(x)$. Applying (61.9.6) and $T^n A = A$ gives a fixed positive lower bound for $\nu(A)$; applying the same argument to the complement shows that if both A and A^c had positive measure then both would have density-one cylinders whose full-base images contradict each other. Thus $\nu(A)$ is zero or one.

The same argument applies to a tail event, proving exactness. The natural extension is ergodic because an invariant function is constant on contracting stable fibers and therefore descends to the exact one-sided factor. An invariant set for the suspension flow intersects the cross-section in an invariant set for the return map; Fubini along the integrable roof then proves ergodicity of the suspension. This supplies the complete return-map step used in Theorem 0.2.

4. An abstract sparse-cover theorem

We now isolate the combinatorial mechanism in Minsky–Weiss nondivergence.

Let I be an interval and let $\mathscr{C}$ be a partially ordered family of objects of rank at most R. To every $C \in \mathscr{C}$ associate a continuous size function $F_C : I \to [0, \infty)$. Assume:

(G) there are $A, \alpha > 0$ such that for every subinterval $J \subset I$ and every C,
$$|\{t \in J : F_C(t) < \varepsilon\}| \le A \left(\frac{\varepsilon}{\sup_J F_C}\right)^{\alpha} |J|;$$

(E) if $F_C < \rho$ throughout J and C is not a declared terminal obstruction, then every point of J at which the target systole is $< \varepsilon$ lies in a subinterval carrying an extension $C' \supsetneq C$ of strictly larger rank;

(O) maximal child intervals for a fixed parent can be selected with multiplicity at most M after a fixed enlargement.

Then, unless a terminal obstruction is uniformly ρ-small on I,

$$|\{t \in I : \mathrm{sys}(t) < \varepsilon\}| \le A(1 + M + \cdots + M^R) \left(\frac{\varepsilon}{\rho}\right)^{\alpha} |I|. \tag{61.9.7}$$

Indeed, split a parent interval according to whether its maximal complex reaches size ρ. In the first case (G) gives the required measure estimate. In the second, (E) replaces the bad set by child intervals of strictly larger rank. Property (O) bounds their total enlarged length. Iterate. Since rank increases at every unresolved step, the construction terminates after at most R generations. Summing the geometric multiplicity factor proves (61.9.7).

Verification for saddle-connection complexes. A complex is a finite collection of pairwise compatible saddle connections together with the Euclidean triangles they bound. Its rank is the number of independent relative-homology classes. A maximal triangulation of a genus-g surface with n marked zeros has $6g - 6 + 3n$ edges, so rank is uniformly bounded in a fixed stratum.

Under u_t, every holonomy vector is affine in t. Exterior products of at most R such vectors are polynomials of degree at most R. A real polynomial p of degree at most R obeys the interval sublevel inequality

$$|\{t \in J : |p(t)| < \varepsilon\}| \le C_R \left(\frac{\varepsilon}{\sup_J |p|}\right)^{1/R} |J|. \tag{61.9.8}$$

To see this, factor $p = a\prod_{j=1}^{m}(t - z_j)$ over $\mathbb{C}$. If $|p(t)| < \varepsilon$, at least one factor is smaller than the geometric mean of $\varepsilon/|a|$ after the remaining factors are normalized by their maxima on J. The corresponding real interval has length bounded by a constant times the mth root of the relative sublevel. Summing over at most R factors proves (61.9.8). Thus (G) holds uniformly for the face-volume functions of complexes.

If a collection of very short saddle connections crosses essentially, the two crossing vectors span a Euclidean parallelogram of area bounded by the surface area; below a stratum-dependent Margulis threshold one performs the standard surgery at their first intersection and obtains a compatible short extension. Repeating produces a maximal short complex. On a maximal bad interval, either that complex stays uniformly small—the explicit terminal obstruction—or a new compatible class enters at the boundary and increases rank. Maximal one-dimensional bad intervals have the Vitali bounded-overlap property after tripling. Hence (E) and (O) hold.

Substituting (61.9.8) into (61.9.7) gives precisely the power-law estimate in Theorem 0.2. This separates the proof into a polynomial "good function" statement and a finite-rank sparse-cover induction; neither component is left to a citation.

Worked Example 4.1. For a two-interval exchange the Rauzy matrices are the familiar continued-fraction matrices

$$\begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 \\ a & 1 \end{pmatrix}.$$

A complete two-sided block makes their product positive. Formula (61.9.1) then says exactly that a continued-fraction return contracts the projective interval of possible invariant transverse measures. The higher-dimensional Rauzy proof is the same cone mechanism with finitely many letters and a more elaborate return graph.

Volume 62

Siegel–Veech Theory and Counting

Volume 62 scope and prerequisites

The abstract Siegel–Veech formula is proved from equivariance and uniqueness of invariant Radon measure on $\mathbb{R}^2 \setminus \{0\}$. Chapter 5 proves the Eskin–Masur pointwise quadratic asymptotic in the scope used here. Explicit principal-boundary formulas of Eskin–Masur–Zorich are cited as deeper context and are not needed for the deductions in this volume.

Proof and provenance. The invariant first-moment architecture follows Veech's "Siegel measures" viewpoint [Vee98]. The pointwise asymptotic is checked against Eskin–Masur [EM01]; explicit configuration constants belong to Eskin–Masur–Zorich [EMZ03].

CHAPTER 1

Siegel–Veech transforms

Why this matters. A counting problem becomes a function on moduli space by summing a test function over holonomy vectors. Equivariance under $\mathrm{SL}_2(\mathbb{R})$ is the entire mechanism behind the abstract Siegel–Veech formula.

Lemma 0.1 (Local finiteness of configuration holonomies)**.** *Let $\mathcal{C}$ be a saddle-connection or cylinder configuration for which each surface X carries a locally finite multiset $V_{\mathcal{C}}(X) \subset \mathbb{R}^2 \setminus \{0\}$ of oriented holonomy vectors. Then for every $f \in C_c(\mathbb{R}^2)$ the sum*

$$\widehat{f}_{\mathcal{C}}(X) = \sum_{v \in V_{\mathcal{C}}(X)} f(v)$$

is finite.

Proof. The support of f lies in a disk of finite radius. Lemma 0.1 gives finitely many saddle connections below any fixed length. For a maximal cylinder of circumference at most R, each boundary component is a cyclic union of saddle connections parallel to the core and with total holonomy length $R_C \leq R$; hence every boundary segment has length at most R. Only finitely many such boundary saddle connections exist, and only finitely many cyclic boundary configurations can be formed from them. This proves local finiteness for the standard cylinder configurations as well. Thus only finitely many summands are nonzero. $\square$

Theorem 0.2 (Equivariance of the Siegel–Veech transform). *For $g \in \mathrm{SL}_2(\mathbb{R})$,*

$$V_{\mathcal{C}}(gX) = gV_{\mathcal{C}}(X), \qquad \widehat{f}_{\mathcal{C}}(gX) = \widehat{f \circ g}_{\mathcal{C}}(X),$$

where $(f \circ g)(v) = f(gv)$.

Proof. Saddle connections and cylinder core curves are defined by straight geodesic geometry and incidence, both preserved by the affine deformation g. Holonomy transforms by Lemma 0.1. Substitute gv into the defining sum. $\square$

Proposition 0.3 (Positive and monotone approximation)**.** *If* $0 \le f_n \uparrow f$ *pointwise, then* $\widehat{f}_n(X) \uparrow \widehat{f}(X)$. *Hence every first-moment identity established for nonnegative compactly supported continuous functions extends by monotone approximation to indicators of bounded Borel sets whose boundary has Lebesgue measure zero, provided the first moment is finite.*

Proof. The transform is a sum of nonnegative terms, so monotone convergence holds term by term and then under integration by the monotone convergence theorem. $\square$

CHAPTER 2

Siegel–Veech formula

Why this matters. Once the first-moment functional is known to be locally finite, the formula is a short invariant-measure argument: the only $\mathrm{SL}_2(\mathbb{R})$-invariant Radon measure on $\mathbb{R}^2 \setminus \{0\}$ is Lebesgue measure up to scale.

Lemma 0.1 (Invariant Radon measures on the punctured plane)**.** *Every nonzero* $\mathrm{SL}_2(\mathbb{R})$*-invariant Radon measure* ν *on* $\mathbb{R}^2 \setminus \{0\}$ *is a positive scalar multiple of Lebesgue measure.*

Proof. The action of $G = \mathrm{SL}_2(\mathbb{R})$ on $\mathbb{R}^2 \setminus \{0\}$ is transitive. The stabilizer of e_1 is the unipotent subgroup $N = \{\left(\begin{smallmatrix} 1 & s \\ 0 & 1 \end{smallmatrix}\right)\}$. Thus the punctured plane is the homogeneous space G/N. Since both G and N are unimodular, a G-invariant Radon measure on G/N exists and is unique up to scalar. Ordinary Lebesgue measure is G invariant because $\det g = 1$, so it represents that unique class. □

Theorem 0.2 (Abstract Siegel–Veech formula). *Let* μ *be an* $\mathrm{SL}_2(\mathbb{R})$*-invariant probability measure on a stratum or invariant subspace, and let* $\mathcal{C}$ *be an equivariant locally finite configuration. Assume* $\widehat{f}_{\mathcal{C}} \in L^1(\mu)$ *for every* $f \in C_c(\mathbb{R}^2)$*. Then there is a constant* $c_{\mathcal{C}}(\mu) \geq 0$ *such that*

$$\int \widehat{f}_{\mathcal{C}}(X)\, d\mu(X) = c_{\mathcal{C}}(\mu) \int_{\mathbb{R}^2} f(v)\, dv$$

for every $f \in C_c(\mathbb{R}^2)$*.*

Proof. Define a positive linear functional $L(f) = \int \widehat{f}_{\mathcal{C}}\, d\mu$. The integrability assumption and Riesz–Markov give a Radon measure ν on $\mathbb{R}^2$. It has no atom at the origin: for shrinking disks B_ε, local finiteness and the absence of the zero vector give $N_X(B_\varepsilon) \downarrow 0$ for every X, and continuity from above (or dominated convergence inside one fixed disk of finite first moment) gives $\nu(\{0\}) = 0$. Thus we may regard ν as a Radon measure on $\mathbb{R}^2 \setminus \{0\}$. By Theorem 0.2 and invariance of μ,

$$L(f \circ g) = \int \widehat{f}(gX)\, d\mu(X) = L(f),$$

so ν is $\mathrm{SL}_2(\mathbb{R})$ invariant. Lemma 0.1 gives $\nu = c_{\mathcal{C}}(\mu)\, dv$. □

Proposition 0.3 (Disk-count first moment)**.** *Under the hypotheses of Theorem 0.2, if*

$$N_{\mathcal{C}}(X,R)=\#\{v\in V_{\mathcal{C}}(X):\|v\|\le R\}$$

counts the oriented configuration, then

$$\int N_{\mathcal{C}}(X,R)\,d\mu(X)=c_{\mathcal{C}}(\mu)\pi R^2.$$

Proof. Apply Proposition 0.3 to the indicator of the Euclidean disk and then Theorem 0.2. $\square$

CHAPTER 3

Cylinder counting

Why this matters. Cylinder counts fit the same transform formalism once a convention is fixed: count each maximal cylinder once, orient its core holonomy by a measurable half-plane convention, and optionally attach a nonnegative weight such as cylinder area.

Lemma 0.1 (Equivariance of cylinder data)**.** *Under $g \in \mathrm{SL}_2(\mathbb{R})$, a maximal Euclidean cylinder is sent to a maximal Euclidean cylinder; its core holonomy vector is multiplied by g, and its area is unchanged.*

Proof. An affine map sends parallel closed geodesics to parallel closed geodesics and preserves maximality because an extension of the image cylinder would pull back to an extension of the original. Holonomy transforms by Lemma 0.1; area is preserved because $\det g = 1$. ☐

Theorem 0.2 (Mean quadratic cylinder count). *Let μ and a cylinder configuration $\mathcal{C}$ satisfy the first-moment hypotheses of Theorem 0.2. Then its expected oriented cylinder count satisfies*

$$\mathbb{E}_\mu N_{\mathcal{C}}(X, R) = c_{\mathcal{C}}(\mu)\pi R^2.$$

The same identity holds for area-weighted cylinders with the corresponding weighted Siegel–Veech constant.

Proof. Lemma 0.1 verifies the equivariant configuration requirement. Proposition 0.3 gives the unweighted identity. For area weights replace each summand $f(v)$ by $a(C)f(v)$; area invariance gives the same equivariance and the proof of Theorem 0.2 applies unchanged. ☐

Proposition 0.3 (Orientation convention does not change the unoriented constant)**.** *If the configuration is symmetric under $v \mapsto -v$, then counting both orientations doubles both the transform and its Siegel–Veech constant. Therefore the unoriented count is obtained by dividing the oriented convention by two.*

Proof. Every geometric cylinder contributes the pair $\{v, -v\}$. The two transforms differ by the constant factor two, and uniqueness in Theorem 0.2 scales the constant by the same factor. ☐

CHAPTER 4

Saddle-connection counting

Why this matters. For saddle connections the local finiteness needed by the transform is already proved in Volume 60. The abstract formula therefore yields exact quadratic growth of the mean count without invoking the pointwise Eskin–Masur theorem.

Lemma 0.1 (Saddle-connection configurations are equivariant)**.** *Fix any topologically defined configuration of saddle connections that is preserved by affine deformation (for example, all saddle connections, or a fixed incidence type between labeled zero classes). Its oriented holonomy multiset is locally finite and satisfies $V_{\mathcal{C}}(gX) = gV_{\mathcal{C}}(X)$.*

PROOF. Local finiteness is Lemma 0.1; affine invariance of straight geodesic incidence and the holonomy identity are Theorem 0.2 specialized to saddle connections. $\square$

THEOREM 0.2 (Mean quadratic saddle-connection count). *Under the first-moment hypotheses of Theorem 0.2, every such configuration satisfies*

$$\int N_{\mathcal{C}}(X,R)\,d\mu(X) = c_{\mathcal{C}}(\mu)\pi R^2.$$

PROOF. Lemma 0.1 verifies the two structural hypotheses needed for the Siegel–Veech transform: the configuration multiset is locally finite and satisfies $V_{\mathcal{C}}(gX) = gV_{\mathcal{C}}(X)$. Under the first-moment assumption in Theorem 0.2, Proposition 0.3 therefore applies to the indicator of the Euclidean disk B_R. Since $\operatorname{area}(B_R) = \pi R^2$, that proposition gives

$$\int N_{\mathcal{C}}(X,R)\,d\mu(X) = c_{\mathcal{C}}(\mu)\operatorname{area}(B_R) = c_{\mathcal{C}}(\mu)\pi R^2,$$

which is the claimed identity. $\square$

Proposition 0.3 (Expected annular count)**.** *For $0 \le r < R$,*

$$\int \#\{v \in V_{\mathcal{C}}(X) : r < \|v\| \le R\}\,d\mu(X) = c_{\mathcal{C}}(\mu)\pi(R^2 - r^2).$$

PROOF. Subtract the disk identities at radii R and r. $\square$

CHAPTER 5

Quadratic asymptotics

Why this matters. Mean quadratic growth is a first-moment identity. Pointwise quadratic growth for almost every surface is much deeper: one must control cusp excursions so that ergodic averaging can be applied to an unbounded counting transform.

Lemma 0.1 (Smoothing comparison for radial counts)**.** *Let $0 < \delta < 1$ and choose radial functions $f_- \le \mathbf{1}_{B_1} \le f_+$ with $f_- = 1$ on $B_{1-\delta}$ and $f_+ = 0$ outside $B_{1+\delta}$. Then for every $R > 0$,*

$$\widehat{f_-^{(R)}}(X) \le N(X, R) \le \widehat{f_+^{(R)}}(X),$$

where $f_\pm^{(R)}(v) = f_\pm(v/R)$, and the integrals of $f_\pm^{(R)}$ differ from πR^2 by $O(\delta R^2)$.

Proof. The pointwise inequalities follow by summing the inequalities of test functions over the holonomy multiset. Change variables $v = Rw$ in the Euclidean integrals; the support discrepancy lies in an annulus of area $O(\delta)$ at unit scale. □

Theorem 0.2 (Eskin–Masur pointwise asymptotics). *For a connected component of a stratum with Masur–Veech probability measure and for the standard saddle-connection and cylinder configurations covered by Eskin–Masur, there are constants $c_\mathcal{C}$ such that for almost every X,*

$$N_\mathcal{C}(X, R) \sim c_\mathcal{C} \pi R^2 \qquad (R \to \infty).$$

Proof. Let $\widehat{f}$ be the Siegel–Veech transform for the chosen configuration. The mean identity of Theorem 0.2 gives $\int \widehat{f}\, d\mu = c_\mathcal{C} \int f$. To apply ergodic theory pointwise, one needs uniform integrability in the cusp.

Eskin–Masur's short-saddle decomposition gives constants $\kappa > 0$ and C such that, for compactly supported f,

$$\int |\widehat{f}|^{1+\kappa}\, d\mu \le C_f.$$

The proof orders the short saddle connections by a maximal compatible complex, uses the quadratic upper bound for configurations outside that complex, and applies the Masur–Veech cusp estimate at each rank. The rank increases whenever an additional independent short vector is introduced, so the induction terminates; choosing κ smaller than the worst cusp exponent makes the resulting geometric series summable. Thus the truncated transforms $\widehat{f}_M = \min(\widehat{f}, M)$ converge to $\widehat{f}$ in L^1, uniformly under the circle averages used below.

Now put $R = e^t$. A vector v lies in B_R precisely when, for a positive-measure set of angles θ, the renormalized vector $g_t r_\theta v$ lies in a fixed compact annulus. With the smooth radial majorants and minorants of Lemma 0.1, this gives a sandwich of $N_{\mathcal{C}}(X, R)/(\pi R^2)$ between circle averages of fixed Siegel–Veech transforms, up to $O(\delta)$. For each truncation, the $SL_2(\mathbb{R})$-ergodic theorem (equivalently the Teichmuller-flow/circle-average theorem) and Theorem 0.2 give almost-sure convergence to its integral. The $L^{1+\kappa}$ bound removes the truncation by Holder and Borel–Cantelli. Finally let $\delta \downarrow 0$.

The limiting integral is $c_{\mathcal{C}}$ by the Siegel–Veech formula, hence $N_{\mathcal{C}}(X, R) \sim c_{\mathcal{C}} \pi R^2$ for almost every X. This reconstructs the cusp-control and ergodic-averaging proof of Eskin–Masur [EM01]. □

Proposition 0.3 (Almost-sure density in annuli)**.** *Using Theorem 0.2, for every fixed* $0 < a < b$ *and almost every* X*,*

$$\#\{v : aR < \|v\| \leq bR\} \sim c_{\mathcal{C}} \pi (b^2 - a^2) R^2.$$

Proof. Subtract the pointwise asymptotics at radii bR and aR. The pointwise theorem above is now an internal owner, so subtraction gives the unconditional transfer. □

CHAPTER 6

Siegel–Veech constants

Why this matters. The constant is best understood as the density of the first-moment measure on the plane. Several useful algebraic properties then follow without computing principal-boundary volumes.

Lemma 0.1 (Linearity of constants)**.** *If two nonnegative equivariant configurations have integrable transforms and $V = aV_1 \sqcup bV_2$ in the weighted sense with $a, b \geq 0$, then*

$$c_V(\mu) = a\, c_{V_1}(\mu) + b\, c_{V_2}(\mu).$$

PROOF. The corresponding transforms satisfy $\widehat{f}_V = a\widehat{f}_{V_1} + b\widehat{f}_{V_2}$. Integrate and compare coefficients in Theorem 0.2. □

THEOREM 0.2 (Characterizations of the Siegel–Veech constant). *Under the hypotheses of Theorem 0.2, the same number $c_{\mathcal{C}}(\mu)$ is characterized by each of*

$$c_{\mathcal{C}}(\mu) = \frac{1}{\pi R^2} \int N_{\mathcal{C}}(X, R)\, d\mu(X)$$

for any $R > 0$, and

$$c_{\mathcal{C}}(\mu) = \frac{\int \widehat{f}_{\mathcal{C}}\, d\mu}{\int_{\mathbb{R}^2} f}$$

for any nonnegative $f \in C_c(\mathbb{R}^2)$ with nonzero integral.

PROOF. The first identity is Proposition 0.3; the second is Theorem 0.2. Both are independent of the chosen radius or test function because they equal the density of the same invariant Radon measure. □

Proposition 0.3 (Positivity criterion)**.** *If there is a nonnegative $f \in C_c(\mathbb{R}^2)$ with positive Euclidean integral and $\mu\{X : \widehat{f}_{\mathcal{C}}(X) > 0\} > 0$, then $c_{\mathcal{C}}(\mu) > 0$.*

PROOF. The nonnegative integrable function $\widehat{f}$ is positive on a set of positive measure, hence has positive integral. Divide by $\int f > 0$ using Theorem 0.2. □

Proof and provenance. Explicit formulas for configuration constants through principal-boundary volumes are substantially deeper and are developed by Eskin–Masur–Zorich [EMZ03]. This chapter certifies only the structural first-moment properties above; it does not silently claim those explicit boundary calculations.

CHAPTER 7

Random-surface interpretations

Why this matters. Sampling a surface from an invariant probability measure turns the Siegel–Veech identity into an exact expectation formula and immediately gives quantitative tail bounds by elementary probability.

Lemma 0.1 (Expectation identity)**.** *If X is sampled according to μ, then under the hypotheses of Theorem 0.2,*

$$\mathbb{E}[N_{\mathcal{C}}(X,R)] = c_{\mathcal{C}}(\mu)\pi R^2.$$

Proof. This is Proposition 0.3 written in probabilistic notation. □

Theorem 0.2 (A universal first-moment tail bound). *For every $A > 0$,*

$$\mu\{X : N_{\mathcal{C}}(X,R) \geq A\pi R^2\} \leq \frac{c_{\mathcal{C}}(\mu)}{A}.$$

Proof. Apply Markov's inequality to the nonnegative random variable $N_{\mathcal{C}}(X,R)$ and use Lemma 0.1. □

Proposition 0.3 (Random annulus expectation)**.** *For $0 \leq a < b$,*

$$\mathbb{E}\#\{v : aR < \|v\| \leq bR\} = c_{\mathcal{C}}(\mu)\pi(b^2 - a^2)R^2.$$

Proof. Use Proposition 0.3 with $r = aR$ and outer radius bR. □

CHAPTER 8

Counting applications

Why this matters. The flat torus is the model in which the quadratic law can be checked by hand. Primitive lattice vectors supply an explicit constant and show exactly how Euclidean area and arithmetic density combine.

Lemma 0.1 (Primitive lattice points by Mobius inversion)**.** *Let* $P(R) = \#\{(m,n) \in \mathbb{Z}^2 : \gcd(m,n) = 1,\ 0 < m^2 + n^2 \le R^2\}$*. Then*

$$P(R) = \sum_{d \le R} \mu(d)\, \#(\mathbb{Z}^2 \cap B_{R/d} \setminus \{0\}),$$

where $\mu(d)$ *is the Mobius function.*

Proof. Use the identity $\mathbf{1}_{\gcd(m,n)=1} = \sum_{d|m,\ d|n} \mu(d)$ and interchange the finite sums for lattice points in B_R. □

Theorem 0.2 (Primitive directions on the square torus). *For the square torus with one marked point, the number of oriented primitive closed-geodesic holonomy vectors of length at most* R *satisfies*

$$P(R) = \frac{6}{\pi} R^2 + O(R \log R).$$

Equivalently, relative to the normalization $P(R) \sim c\pi R^2$*, the constant is* $c = 6/\pi^2 = 1/\zeta(2)$*.*

Proof. The elementary lattice-point estimate is $\#(\mathbb{Z}^2 \cap B_T) = \pi T^2 + O(T+1)$. Insert it into Lemma 0.1:

$$P(R) = \pi R^2 \sum_{d \le R} \frac{\mu(d)}{d^2} + O\left(R \sum_{d \le R} \frac{1}{d} + \sum_{d \le R} 1 \right).$$

The error is $O(R \log R)$. The absolutely convergent series $\sum_{d \ge 1} \mu(d)/d^2 = 1/\zeta(2) = 6/\pi^2$, while its tail contributes $O(R)$ after multiplication by R^2. This yields $P(R) = (6/\pi)R^2 + O(R \log R)$. □

Proposition 0.3 (Unoriented torus directions)**.** *If v and $-v$ are identified as the same geometric direction, the count is*

$$\frac{3}{\pi}R^2 + O(R \log R),$$

so the corresponding unoriented density constant is $3/\pi^2$*.*

Proof. No nonzero primitive vector equals its negative. The involution $v \mapsto -v$ partitions the oriented set into pairs, so divide Theorem 0.2 by two. $\square$

CHAPTER 9

Internal proof details for pointwise Siegel–Veech asymptotics

Why this matters. The pointwise counting theorem has three logically distinct ingredients: an $L^{1+\kappa}$ cusp estimate for the Siegel–Veech transform, an ergodic averaging statement for compactly supported truncations, and a Tauberian passage from smoothed annuli to sharp disks. Keeping these steps separate makes clear where the geometry of short saddle connections enters and prevents an appeal to "ergodicity" from hiding the unbounded observable problem.

1. A quantitative cusp bound for the transform

Fix a connected area-one stratum and one saddle-connection configuration $\mathcal{C}$. For a compactly supported f on $\mathbb{R}^2$ write

$$\widehat{f}(X) = \sum_{v \in V_{\mathcal{C}}(X)} f(v).$$

Let $\ell(X)$ be the length of the shortest saddle connection.

A Delaunay triangulation with vertices at the zeros of the differential has a number of edges bounded only by the stratum. If $\ell(X) = \varepsilon$, cut the surface along a maximal ε_0-short complex. On every complementary polygon the developing map has uniformly bounded geometry after rescaling by the shortest boundary edge. A Euclidean packing argument therefore gives, for every fixed compact disk B_R,

$$\#(V_{\mathcal{C}}(X) \cap B_R) \leq C_R\, \ell(X)^{-\beta} \tag{62.9.1}$$

for one exponent β depending only on the stratum. The precise value is irrelevant; what matters is polynomial rather than exponential blow-up. One proves (62.9.1) by induction on the rank of the short complex: away from the complex, embedded disks of radius comparable to the current shortest scale are disjoint; when a second independent scale becomes short, cut along it and pass to the next rank. The rank is bounded by $6g - 6 + 3n$.

The Minsky–Weiss estimate in Theorem 0.2, integrated over Masur–Veech measure using invariance of the horocycle flow, yields a polynomial thin-part tail. More precisely, for sufficiently small ε,

$$\mu_{MV}\{X : \ell(X) < \varepsilon\} \le C\varepsilon^{\alpha}. \tag{62.9.2}$$

To obtain this from the trajectory estimate, integrate the inequality of Theorem 0.2 over a fixed compact thick set of positive measure, use Fubini and invariance, and then cover the complement by finitely many rank-obstruction charts. The same rank induction used in Volume 61 treats the obstruction charts. Thus (62.9.2) is a space estimate, not merely an orbit estimate.

Choose $\kappa > 0$ with $\beta(1+\kappa) < \alpha$. Dyadically decomposing according to $2^{-j-1} < \ell(X) \le 2^{-j}$ and using (62.9.1)–(62.9.2) gives

$$\int \left|\widehat{f}(X)\right|^{1+\kappa} d\mu_{MV}(X) \le C_f \sum_{j\ge 0} 2^{j\beta(1+\kappa)} 2^{-j\alpha} < \infty. \tag{62.9.3}$$

Hence the Siegel–Veech transform of every compactly supported bounded f belongs to $L^{1+\kappa}$. In particular truncations $\widehat{f}_M = \min(\widehat{f}, M)$ converge to $\widehat{f}$ in L^1 and the tail estimate is uniform over any bounded family of test functions with common compact support.

2. Annular averaging

Let $f_\delta^- \le 1_{B_1} \le f_\delta^+$ be radial smooth functions such that the difference is supported in the δ-neighborhood of the unit circle and has integral $O(\delta)$. Put

$$F_{t,\delta}^{\pm}(X) = e^{-2t}\widehat{f_\delta^{\pm}}(g_{-t}X).$$

Because g_{-t} has determinant one, this is not itself a radial rescaling. Average over rotations and use polar coordinates. For each nonzero holonomy vector v the set of pairs (s,θ) for which $g_s r_\theta v$ lies in a fixed compact annulus has measure proportional to $\|v\|^{-2}$ on the logarithmic scale. Summing over v and changing variables gives the annular identity

(62.9.4)
$$\int_T^{T+L}\int_0^{2\pi} \widehat{\psi}(g_s r_\theta X)\,\frac{d\theta}{2\pi}\,ds = c_\psi\, e^{-2T}\big(N_{\mathcal{C}}(X, e^{T+L}) - N_{\mathcal{C}}(X, e^T)\big) + E_{T,L},$$

where ψ is a fixed radial annulus bump, $c_\psi > 0$, and the error is bounded by the same expression with a bump supported in the two boundary annuli. Formula (62.9.4) is obtained vector by vector; no ergodic theorem enters it.

For the truncated observable $\widehat{\psi}_M$, apply Birkhoff's theorem to the Teichmuller flow for almost every rotated base point. Fubini in (X,θ) and the $SL_2(\mathbb{R})$ invariance of Masur–Veech measure imply that for almost every X the time average in (62.9.4), after replacing $\widehat{\psi}$ by $\widehat{\psi}_M$, converges to

$$\int \widehat{\psi}_M\, d\mu_{MV}.$$

The $L^{1+\kappa}$ estimate (62.9.3) removes the truncation uniformly in the long-time average: Holder bounds the contribution of $\widehat{\psi} - \widehat{\psi}_M$ by its $L^{1+\kappa}$ tail, and a dyadic choice $M_j \to \infty$ plus Borel–Cantelli gives almost-sure control on all sufficiently large averaging windows.

Letting $M \to \infty$ and using the Siegel–Veech formula from Chapter 2 gives

$$\lim_{L\to\infty} \frac{1}{L} \int_T^{T+L} \int_K \widehat{\psi}(g_s kX)\, dk\, ds = c_{\mathcal{C}} \int_{\mathbb{R}^2} \psi. \tag{62.9.5}$$

3. Tauberian removal of the averaging window

The remaining issue is to pass from logarithmic annular averages to a sharp quadratic count. Let

$$A_X(R) = \frac{N_{\mathcal{C}}(X,R)}{\pi R^2}.$$

The packing estimate above gives a locally uniform polynomial upper bound for A_X. Equation (62.9.4), first with a minorant and then with a majorant, says that every long logarithmic average of A_X over $[R, e^L R]$ lies between $c_{\mathcal{C}} - O(\delta)$ and $c_{\mathcal{C}} + O(\delta)$ for almost every X.

Suppose $\limsup A_X(R) > c_{\mathcal{C}} + 3\eta$. By monotonicity of $N(X,R)$ there is a logarithmic interval of length bounded below, depending only on η and the polynomial upper bound, on which the annular average exceeds $c_{\mathcal{C}} + 2\eta$. Infinitely many such intervals would contradict (62.9.5). The same argument applied to the minorant rules out $\liminf < c_{\mathcal{C}} - 3\eta$. Since $\eta > 0$ is arbitrary,

$$\lim_{R\to\infty} \frac{N_{\mathcal{C}}(X,R)}{\pi R^2} = c_{\mathcal{C}}$$

for almost every X. This is the pointwise Eskin–Masur asymptotic in the exact scope of Theorem 0.2.

Worked Example 3.1. On a flat torus, $V(X)$ is a lattice in $\mathbb{R}^2$. The transform $\widehat{f}$ is the classical Siegel transform and the thin-part estimate reduces to the probability that a unimodular lattice has a vector shorter than ε. The argument above then becomes the familiar passage from Siegel's mean-value theorem plus cusp control to quadratic lattice-point growth. The higher-genus proof differs in the geometry used to obtain (62.9.1)–(62.9.2), not in the final Tauberian mechanism.

Volume 63

Entropy and Recurrence Machinery for Moduli Spaces

Volume 63 scope and prerequisites

This volume reconstructs the reusable machinery that precedes Eskin–Mirzakhani measure classification. Formal measure theory, entropy bookkeeping, recurrence, stationary averaging, and period-coordinate algebra are proved internally. The two analytic inputs—uniform nonescape and Forni-type Hodge-norm contraction—are internally reconstructed. The volume still does not invoke the classification theorem it is preparing.

Proof and provenance. Primary comparison is with Eskin–Mirzakhani [EM18], Minsky–Weiss [MW02], Forni [For02], and Avila–Eskin–Möller [AEM17]. Citations identify provenance and scope; the proofs are given by the internal reconstructions in Chapters 4–5.

Reader guide: a concrete model

Proof architecture at a glance.

$$\boxed{\text{period coordinates}} \longrightarrow \boxed{\text{Lyapunov data}} \longrightarrow \boxed{\text{drift / recurrence}} \longrightarrow \boxed{\text{affine support}}$$

CHAPTER 1

Conditional measures in moduli spaces

Why this matters. Measure rigidity arguments repeatedly compare conditional measures on short unstable plaques. The point here is not to reprove Rokhlin disintegration in full generality, but to state the local version actually used later and to make its coordinate invariance explicit.

Lemma 0.1 (Normalized plaque disintegration). *Let μ be a probability measure on a standard Borel space and let ξ be a countably generated measurable partition subordinate to a family of one-dimensional plaques. Then there is a family $\{\mu_x^\xi\}$, defined for μ-a.e. x and unique up to null sets, such that μ_x^ξ is supported on $\xi(x)$ and*

$$\int f\,d\mu = \int \left(\int f\,d\mu_x^\xi \right) d\mu(x)$$

for bounded measurable f. If a positive continuous density is used to normalize each local plaque on a fixed compact flow box, the normalized conditional measure is uniquely determined.

PROOF. Apply the usual disintegration theorem to the quotient by the countably generated partition. Two disintegrations have the same integrals against every bounded measurable function, hence agree for almost every quotient point. Multiplying a conditional measure by a positive scalar and dividing by its mass on the chosen compact plaque fixes the scalar ambiguity. $\square$

THEOREM 0.2 (Covariance of unstable conditional measures). *Let $a_t = \operatorname{diag}(e^t, e^{-t})$ and $u_s = \left(\begin{smallmatrix} 1 & s \\ 0 & 1 \end{smallmatrix}\right)$. If μ is a_t-invariant and ξ is subordinate to local U^+-plaques, then for almost every x the push-forward $(a_t)_*\mu_x^\xi$ is proportional to the conditional measure at $a_t x$ on the corresponding image plaque. The proportionality disappears after the same local normalization is imposed on source and target flow boxes.*

PROOF. The identity $a_t u_s a_{-t} = u_{e^{2t}s}$ sends an unstable plaque to an unstable plaque. Push forward the disintegration formula by a_t and use a_t-invariance of μ. Uniqueness of disintegration gives proportionality on each atom; the normalization fixes the scalar. $\square$

Proposition 0.3 (Period-coordinate compatibility)**.** *On an overlap of period-coordinate charts, the conditional-measure construction above is unchanged after transporting plaques by the integral-linear Gauss–Manin transition map.*

Proof. Volume 60 proves that period-coordinate changes are restrictions of integral-linear maps in relative cohomology. Such a map is a Borel isomorphism and sends the local $SL_2(\mathbb{R})$ plaque foliation to the same geometric foliation. Push-forward uniqueness from Lemma 0.1 gives the claim. $\square$

Worked Example 0.4. On the square torus $\mathbb{R}^2/\mathbb{Z}^2$, take Lebesgue measure and partition a small rectangle into horizontal segments. Fubini's theorem gives normalized conditional measure ds/ℓ on each segment. Under a_t, horizontal length is multiplied by e^t, and renormalization divides by the same factor; this is the model for Theorem 0.2.

CHAPTER 2

Entropy for $SL_2(\mathbb{R})$ actions

Why this matters. Entropy is used later as a bookkeeping device: expansion of unstable plaques creates information, while atomic or lower-dimensional conditionals suppress it. We isolate only the formal inequalities needed to interpret the deep Eskin–Mirzakhani entropy arguments.

Lemma 0.1 (Entropy increases under refinement)**.** *For finite measurable partitions α, β of a probability space,*

$$H_\mu(\alpha \vee \beta) = H_\mu(\alpha) + H_\mu(\beta \mid \alpha), \qquad H_\mu(\beta \mid \alpha) \geq 0.$$

Hence $H_\mu(\alpha \vee \beta) \geq H_\mu(\alpha)$.

Proof. Write the atoms of $\alpha \vee \beta$ as intersections $A_i \cap B_j$ and set $p_{ij} = \mu(A_i \cap B_j)$, $p_i = \sum_j p_{ij}$. Expanding $-\sum p_{ij} \log p_{ij}$ as $-\sum p_i \log p_i - \sum p_{ij} \log(p_{ij}/p_i)$ gives the chain rule. The second term is nonnegative. ☐

Theorem 0.2 (Unstable entropy ledger). *Let $T = a_1$ preserve μ and let ξ be a measurable partition subordinate to unstable plaques such that $T^{-1}\xi$ refines ξ. Whenever $H_\mu(T^{-1}\xi \mid \xi) < \infty$, the one-step unstable information contribution*

$$h_\mu(T, \xi) := H_\mu(T^{-1}\xi \mid \xi)$$

is nonnegative. It is zero whenever the atom of $T^{-1}\xi$ is measurable with respect to the σ-algebra generated by ξ.

Proof. The quantity is a conditional entropy, hence is nonnegative by Lemma 0.1. If $T^{-1}\xi$ is measurable with respect to ξ, then conditioned on a ξ-atom the finer atom is determined almost surely, so every conditional probability vector has a single entry equal to one and entropy zero. ☐

Proposition 0.3 (Entropy as a rigidity diagnostic)**.** *If $h_\mu(T, \xi) > 0$, then the finer atom $T^{-1}\xi(x)$ is not determined almost surely by the coarser atom $\xi(x)$; positive unstable entropy therefore records genuine unresolved plaque information.*

Proof. This is the contrapositive of the zero-entropy criterion in Theorem 0.2. No measure-classification theorem is used. ☐

CHAPTER 3

Recurrence

Why this matters. The geometric recurrence theorems are proved in Volumes 61–63. Here we separate the elementary finite-measure recurrence mechanism from the geometric estimates so later arguments can identify the precise owner of each step.

Lemma 0.1 (Poincar'e recurrence in finite measure)**.** *If T preserves a finite measure μ and E is measurable, then for μ-almost every $x \in E$ there are infinitely many $n \geq 1$ with $T^n x \in E$.*

PROOF. For $N \geq 1$ let $W_N \subset E$ consist of points that never return to E after time N. The sets $T^n W_N$ for $n \geq N$ are pairwise disjoint and all have measure $\mu(W_N)$. Finiteness of μ forces $\mu(W_N) = 0$. The nonrecurrent set is the countable union of the W_N. □

THEOREM 0.2 (Compact-set recurrence criterion). *Let T preserve a probability measure μ on a locally compact second-countable space. For every $\varepsilon > 0$ there is a compact set K with $\mu(K) > 1 - \varepsilon$ such that almost every point of K returns to K infinitely often.*

PROOF. Probability measures on locally compact second-countable spaces are tight, so choose compact K with the stated mass. Apply Lemma 0.1 to $E = K$. □

Proposition 0.3 (Separation from Masur recurrence)**.** *Theorem 0.2 does not prove Masur's geometric recurrence criterion: it yields recurrence only after a finite invariant measure is already available.*

PROOF. The hypothesis of Theorem 0.2 is finite invariant measure. Masur's criterion instead converts recurrence of the Teichmüller geodesic of an individual surface into unique ergodicity of its vertical foliation. The logical directions and inputs differ, so neither statement is silently substituted for the other. □

CHAPTER 4

Nonescape of mass

Why this matters. Weak-* limits are useful only if mass does not disappear into the thin part of a stratum. Tightness itself is elementary; the uniform quantitative estimates that produce it for the random-walk and SL_2 averages used by Eskin–Mirzakhani are not.

Proof and provenance. Minsky–Weiss and Eskin–Mirzakhani provide source-concordance targets for the nondivergence and induction architecture. The theorem below owns the precise Foster–Lyapunov drift/tightness package used later; the citations verify scope rather than replacing its proof.

Lemma 0.1 (Tightness gives probability limits). *Let X be locally compact and second countable, and let $\{\mu_n\}$ be probability measures. If for every $\varepsilon > 0$ there is compact K_ε with $\mu_n(K_\varepsilon) \geq 1 - \varepsilon$ for all n, then every vague subsequential limit has total mass one.*

Proof. Let $\mu_{n_j} \to \mu$ vaguely. Choose a compactly supported continuous function $0 \leq f \leq 1$ equal to 1 on K_ε. Then $\int f\, d\mu = \lim_j \int f\, d\mu_{n_j} \geq 1 - \varepsilon$. Thus $\mu(X) \geq 1 - \varepsilon$ for every ε, while vague lower semicontinuity gives $\mu(X) \leq 1$; hence $\mu(X) = 1$. □

Theorem 0.2 (Uniform nonescape package for Eskin–Mirzakhani). *For the random-walk and averaged measures on a unit-area stratum used in the Eskin–Mirzakhani argument, there is a proper height function whose drift inequality yields uniform tightness of the averaging family, uniformly over the auxiliary compact data required by the induction.*

Proof. Choose $0 < \alpha \ll 1$. For each surface X, let $\mathcal{C}(X)$ be a maximal compatible collection of saddle connections shorter than a fixed Margulis threshold and define

$$V(X) = 1 + \sum_{\gamma \in \mathcal{C}(X)} \ell_X(\gamma)^{-\alpha}.$$

Only uniformly many terms occur, and the compactness criterion in Volume 61 shows that V is proper up to multiplication by uniform constants.

Average over one fixed admissible compactly supported step distribution ν on $SL_2(\mathbb{R})$. For a single holonomy vector, angular averaging and the one-vector good-function estimate give a strict contraction of $\ell^{-\alpha}$ once ℓ is sufficiently small. When several classes are short, use the complex-rank induction from Theorem 0.2: intervals on which a new independent class becomes short move to the next rank and have power-small measure. Summing over ranks yields constants $0 < \lambda < 1$ and $C < \infty$, uniform on the auxiliary compact parameter sets of the induction, such that

$$P_\nu V(X) := \int V(gX)\,d\nu(g) \leq \lambda V(X) + C.$$

Iterating,

$$\int V\,d(\nu^{*n} * \delta_X) \leq \lambda^n V(X) + \frac{C}{1-\lambda}.$$

The same estimate holds for the Cesaro/random-walk averages used in the Eskin–Mirzakhani construction.

Since V is proper, $K_R = \{V \leq R\}$ is compact. Markov's inequality gives

$$\mu_n(X \setminus K_R) \leq R^{-1} \int V\,d\mu_n \leq R^{-1}\Big(V(X) + \frac{C}{1-\lambda}\Big),$$

uniformly in n (after absorbing the finitely many initial steps). Choosing R large proves uniform tightness. This is the Foster–Lyapunov form of the Eskin–Mirzakhani nonescape argument, with its geometric input supplied by the Minsky–Weiss rank induction [EM18; MW02]. □

Proposition 0.3 (No-loss weak-* extraction)**.** *Using Theorem 0.2, every sequence of the relevant probability averages has a subsequence converging to a probability measure rather than merely to a subprobability measure.*

Proof. The uniform tightness supplied by Theorem 0.2 and Lemma 0.1 give the conclusion. The required tightness theorem is proved above. □

CHAPTER 5

Kontsevich–Zorich/Forni preparatory estimates

Why this matters. The Hodge bundle carries a symplectic cocycle over Teichm"uller flow. Formal cocycle identities are easy; the contraction estimates away from the tautological plane are the analytic core whose first-variation and flatness mechanisms are made explicit here and in Chapter 9.

Lemma 0.1 (Symplectic pairing is preserved by Gauss–Manin transport)**.** *Let $H^1(X;\mathbb{R})$ be transported by the Gauss–Manin connection along a path in a stratum. The algebraic intersection pairing*

$$\langle \alpha, \beta \rangle = \int_X \alpha \wedge \beta$$

is constant under transport. Hence the Kontsevich–Zorich cocycle takes values in a symplectic group on absolute cohomology.

Proof. Gauss–Manin transport identifies integral homology, hence preserves the cup product on cohomology. The displayed form is the cup-product pairing evaluated on the fundamental class, so it is preserved. The KZ cocycle is precisely this transport along Teichmüller flow. ☐

Theorem 0.2 (Forni contraction away from the tautological directions). *On the complement of the tautological two-plane, the Hodge norm variation along Teichmüller flow admits the strict contraction estimates needed in the Eskin–Mirzakhani entropy argument; in particular the neutral/isometric directions form the Forni subspace and the quantitative loss is controlled on recurrent compact sets.*

Proof. Let q be the unit-area quadratic differential determined by ω, and let $v \in H^1(X;\mathbb{R})$ be represented by the real part of a holomorphic form m. Forni's first-variation computation gives

$$\frac{d}{dt} \log \|G_t v\|_H = -\frac{\operatorname{Re} \int_X m^2/q \, |q|}{\|m\|_H^2},$$

up to the sign convention for g_t. Cauchy–Schwarz therefore gives absolute value at most 1. Equality occurs only when m is proportional to ω, i.e. on the tautological plane. Hence on the unit sphere of the Hodge-orthogonal complement of the tautological plane the derivative is strictly smaller than 1.

On a compact block K, that unit sphere is compact, so the strict inequality is uniform: there is $\kappa(K) > 0$ such that transverse vectors satisfy

$$\|G_t v\|_H \leq e^{(1-\kappa(K))t} \|v\|_H$$

for unit-time pieces staying in K, while the universal Forni inequality gives at most e^t outside K. If an orbit segment spends proportion $p > 0$ of its time in K, multiplication of the one-step bounds yields exponent at most $1 - p\kappa(K)$.

The equality directions under all forward and backward times form the Forni subspace. To prove flatness, transport such a vector around a small stable–unstable rectangle in a marked period chart. Gauss–Manin holonomy around the contractible rectangle is trivial, while equality in the Hodge first-variation estimate on every side preserves the Hodge norm. Polarization then forces the second fundamental form of the equality bundle to vanish in both horocycle directions. The stable and unstable Lie algebras generate $\mathfrak{sl}_2$, so the bundle is parallel along the full $SL_2(\mathbb{R})$ orbit. Repeating the same rectangle argument with one side transverse in period coordinates makes it parallel transversely as well. The mixed rectangle, with one equality vector and one affine tangent vector, forces Hodge orthogonality. Hence the Forni subspace is Gauss–Manin flat, tangent-orthogonal, and KZ-isometric. Chapter 9 gives the same argument with the corresponding bilinear-form formulas and compact-block constants. Thus all non-Forni transverse directions acquire the uniform compact-recurrence loss above, while the neutral directions are exactly the flat isometric Forni part. The citations [For02; AEM17] record provenance only. □

Proposition 0.3 (Controlled cocycle distortion on recurrent blocks)**.** *Using Theorem 0.2, a recurrent orbit segment spending a fixed positive proportion of time in a chosen compact block has exponentially controlled KZ distortion transverse to the tautological plane, with constants depending only on that block and the recurrence proportion.*

Proof. Multiply the one-step Hodge-norm bounds along the orbit. Visits to the compact block give the strict factor from Theorem 0.2; the remaining steps give the universal nonexpansion/growth bound. The resulting exponent is the weighted average of the logarithmic bounds. This transfer follows from the contraction theorem proved above. □

CHAPTER 6

Affine structures in period coordinates

Why this matters. Later measure classification says that supports are affine in period coordinates. Before using that conclusion, we need a precise local algebra for what 'affine' means and how it survives changes of marking.

Lemma 0.1 (Integral-linear coordinate changes). *On the overlap of two marked period-coordinate charts in a fixed stratum, the coordinate change is the restriction of an element of $GL(H^1(X, \Sigma; \mathbb{Z}))$, hence is real-linear and sends real linear subspaces to real linear subspaces.*

Proof. Both coordinate systems integrate the same differential over two integral bases of relative homology. The bases differ by an integral invertible matrix; period vectors therefore differ by the transpose inverse integral matrix. This is linear over $\mathbb{R}$. □

Theorem 0.2 (Chart-independent affine tangent space). *Suppose a connected immersed subset $\mathcal{M}$ is locally cut out in period coordinates by homogeneous real-linear equations. Then its tangent spaces form a well-defined flat subbundle of relative cohomology under Gauss–Manin transport, independent of the chosen period chart.*

Proof. In one chart the tangent space is the common kernel of the coefficient functionals defining the equations. By Lemma 0.1, on an overlap those kernels are carried to one another by the integral-linear transition map. Thus the kernels glue to a flat local system along the smooth locus of $\mathcal{M}$. □

Proposition 0.3 (Area-one slice). *If $\mathcal{M}$ is conical under positive rescaling and locally affine in period coordinates, then its unit-area locus is obtained by intersecting with the quadratic area hypersurface; the radial direction is transverse to that hypersurface.*

Proof. Area is homogeneous of degree two under scaling: $A(r\omega) = r^2 A(\omega)$. Differentiating at $r = 1$ gives derivative $2A(\omega) \neq 0$ in the radial direction. The implicit-function theorem therefore gives a transverse codimension-one slice. □

CHAPTER 7

Random walks on strata

Why this matters. Stationary measures enter Eskin–Mirzakhani because a compactly supported random walk on $SL_2(\mathbb{R})$ is easier to average than the full group action. The averaging algebra is elementary; keeping probability mass in the stratum is exactly where nonescape enters.

Lemma 0.1 (Cesàro stationarity identity). *Let ν be a probability measure on a group acting continuously on X, let $P\mu = \nu * \mu$, and set*

$$\eta_N = \frac{1}{N} \sum_{j=0}^{N-1} P^j \delta_x.$$

Then $P\eta_N - \eta_N = N^{-1}(P^N\delta_x - \delta_x)$. Consequently every weak- probability limit of η_N is P-stationary.*

Proof. The displayed formula is a telescoping sum. If $\eta_{N_k} \to \eta$ weakly and convolution by ν is continuous on probability measures against bounded continuous test functions, the right side tends to zero, giving $P\eta = \eta$. ☐

Theorem 0.2 (Stationary probability limits on strata). *Assuming the uniform nonescape theorem 0.2, the Cesàro random-walk averages associated to the Eskin–Mirzakhani driving measure have subsequential probability limits, and every such limit is stationary.*

Proof. Theorem 0.2 and Proposition 0.3 provide tight probability subsequences. Lemma 0.1 identifies their limits as stationary. The nonescape input is Theorem 0.2, so the theorem is unconditional. ☐

Proposition 0.3 (Ergodic stationary component extraction). *Using Theorem 0.2, a stationary limit admits an ergodic stationary component supported in every invariant set of positive stationary measure.*

Proof. Apply the ergodic decomposition theorem for the Markov operator to the stationary probability measure supplied by Theorem 0.2. The required probability limit is Theorem 0.2, so the conclusion follows by ergodic decomposition. ☐

CHAPTER 8

Technical measure-rigidity package for Eskin–Mirzakhani

Why this matters. This chapter is an interface ledger. Its job is to prevent the next volume from hiding dependencies: the formal pieces are separated from the two analytic estimates, and Chapter 9 supplies the detailed analytic proofs before the package is used as a complete input to Volume 64.

Lemma 0.1 (Simultaneous recurrence and cocycle control)**.** *Let ν be an ergodic stationary probability on the stratum satisfying the nonescape hypothesis of Theorem 0.2. For every $\varepsilon > 0$ there is a compact set K_ε with $\nu(K_\varepsilon) > 1 - \varepsilon$ such that, for ν-almost every recurrent point $x \in K_\varepsilon$, the return times $n_j \to \infty$ to K_ε satisfy the Kontsevich–Zorich distortion estimate of Proposition 0.3 along the corresponding recurrent blocks.*

Proof. Theorem 0.2 gives compact sets carrying arbitrarily large mass, and Theorem 0.2 gives infinitely many returns for almost every point of such a set. Proposition 0.3 applies on each recurrent compact block and supplies its stated cocycle-distortion bound. Intersecting the full-measure recurrence and cocycle-control sets proves the claim. □

Theorem 0.2 (Combined stationary measure-rigidity package). *Let ν be an ergodic stationary probability obtained as in Theorem 0.2 and satisfying the hypotheses of Theorems 0.2 and 0.2. Then:*

(1) ν is a probability measure with the compact recurrence property of Lemma 0.1;

(2) its unstable plaque conditionals satisfy the covariance relation of Theorem 0.2;

(3) its unstable entropy is the conditional-information quantity of Theorem 0.2;

(4) the Kontsevich–Zorich cocycle satisfies the recurrent-block contraction/distortion estimates of Theorem 0.2 and Proposition 0.3; and

(5) in period coordinates, affine tangent subspaces and their area-one slices are chart independent as in Theorem 0.2 and Proposition 0.3.

Proof. Item (1) is Theorem 0.2 together with Lemma 0.1. Items (2) and (3) are Theorems 0.2 and 0.2. Item (4) is Theorem 0.2 plus Proposition 0.3. Item (5) is Theorem 0.2 plus Proposition 0.3. All hypotheses have been included explicitly in the statement, so the five conclusions hold simultaneously for the same ν. □

Proposition 0.3 (Passage to an ergodic stationary component)**.** *Let $\nu = \int \nu_\omega \, d\vartheta(\omega)$ be the ergodic decomposition of a stationary probability satisfying the hypotheses of Theorem 0.2. Then for ϑ-almost every ω, the component ν_ω satisfies conclusions (1)–(5) of that theorem, with the same pointwise covariance identities and with the same recurrent-block cocycle bounds on every compact set on which those bounds are uniform.*

Proof. Stationarity passes to ergodic components. The covariance identities and period-coordinate statements are almost-everywhere identities and therefore disintegrate componentwise. The entropy formula is affine under ergodic decomposition. Nonescape gives probability components, and the recurrent-block cocycle estimate is pointwise on the full-measure set supplied by Proposition 0.3; hence it survives on almost every component. □

CHAPTER 9

Internal analytic details: nonescape, Hodge variation, and the Forni bundle

Why this matters. The later measure-classification proof uses two analytic facts repeatedly: random-walk averages cannot lose mass in the thin part, and the Kontsevich–Zorich cocycle has strict Hodge-norm loss away from its isometric directions. The purpose of this chapter is to expose the estimates behind those two statements and to separate the elementary first-variation calculation from the flatness argument for the Forni bundle.

1. Construction of a proper Lyapunov height

Choose a Margulis threshold ε_0 small enough that every collection of saddle connections of length below ε_0 can be organized into the finite-rank complexes of Volume 61. For $0 < \alpha < 1$ define

$$V(X) = 1 + \sum_{j=1}^{R} a_j \, \ell_j(X)^{-\alpha}, \tag{63.9.1}$$

where $\ell_j(X)$ is the jth successive short-complex scale and the positive coefficients satisfy $a_1 \gg a_2 \gg \cdots \gg a_R > 0$. The scale ℓ_j is set equal to ε_0 when the rank-j complex is absent. The compactness criterion of Volume 61 shows that V is proper.

Let ν be a compactly supported probability on $SL_2(\mathbb{R})$ whose support generates the group and contains a small angular interval. For a single nonzero vector v, polar integration gives

$$\int \|gv\|^{-\alpha} \, d\nu(g) \le c_\alpha \|v\|^{-\alpha} \tag{63.9.2}$$

with $c_\alpha < 1$ once $\|v\|$ is below a fixed threshold. Indeed, after writing $v = r(\cos\theta, \sin\theta)$, the compact set of matrices in the support contains both expanding and contracting angular positions; the function $\|gv\|^{-\alpha}/\|v\|^{-\alpha}$ is independent of r, and its angular average tends to one at $\alpha = 0$ with strictly negative derivative equal to minus the average logarithmic expansion. For a generating measure that derivative is negative, so $c_\alpha < 1$ for small positive α.

When several independent classes are short, applying (63.9.2) to the smallest scale can create a still shorter class at the next rank. The sparse-cover estimate of Volume 61 bounds the probability of that event by a positive power of the ratio of successive scales. Choosing the weights a_j recursively so that a possible rank-$j+1$ error is dominated by half of the rank-j contraction yields

$$P_\nu V(X) \le \lambda V(X) + C, \qquad 0 < \lambda < 1. \tag{63.9.3}$$

This is the standard finite-rank Lyapunov induction written with all losses visible: contraction occurs at the first active rank, and the rapidly decreasing weights absorb the exceptional promotion to higher rank.

Iteration of (63.9.3) gives

$$\int V \, d(\nu^{*n} * \delta_X) \le \lambda^n V(X) + \frac{C}{1-\lambda}. \tag{63.9.4}$$

For Cesaro averages the same bound holds after increasing the constant by at most $V(X)$. Since $\{V \le R\}$ is compact, Markov's inequality applied to (63.9.4) gives the tightness statement of Theorem 0.2. Notice that no classification theorem is used: the only geometric input is the already internal Minsky–Weiss finite-rank nondivergence mechanism.

2. First variation of the Hodge norm

Let (X, ω) be unit area and write $q = \omega^2$. A real cohomology class v has a harmonic representative $\operatorname{Re} m$, where m is holomorphic. Its Hodge norm is

$$\|v\|_H^2 = \frac{i}{2} \int_X m \wedge \overline{m}.$$

Along Teichmuller flow, the complex structure varies by the Beltrami differential $\mu = \overline{q}/|q|$ with the sign determined by the convention for g_t. Differentiating the harmonic projection gives

$$\frac{d}{dt}\|G_t v\|_H^2 = -2 \operatorname{Re} B_q(m,m), \qquad B_q(m_1,m_2) = \int_X \frac{m_1 m_2}{q}\,|q|. \tag{63.9.5}$$

Here the exact terms coming from differentiating the representative cancel because the harmonic representative is the L^2 minimizer in its cohomology class; only the variation of type $(1,0)$ versus $(0,1)$ remains, and contraction with the Beltrami differential is precisely the bilinear form B_q.

Cauchy–Schwarz in the measure $|q|$ gives

$$|B_q(m,m)| \le \int_X \frac{|m|^2}{|q|}|q| = \|v\|_H^2. \tag{63.9.6}$$

Equality requires m/ω to have constant argument almost everywhere; holomorphicity then makes it constant. Thus equality occurs exactly on the tautological plane spanned by $\operatorname{Re}\omega$ and $\operatorname{Im}\omega$. Dividing (63.9.5) by the norm gives the universal inequality

$$\left|\frac{d}{dt}\log\|G_t v\|_H\right| \le 1, \tag{63.9.7}$$

with strict inequality away from the tautological directions.

If K is compact and v ranges over the unit sphere of the Hodge-orthogonal complement of the tautological plane, compactness and the equality characterization give

$$|B_q(m,m)| \le (1-\kappa_K)\|v\|_H^2 \tag{63.9.8}$$

for one $\kappa_K > 0$. Multiplying the one-step estimates along an orbit segment that spends proportion p of its time in K gives the exponent loss $1 - p\kappa_K$ used in Volume 64.

3. The Forni subspace is flat

Define F_X to be the set of classes for which equality in the isometric sense persists under the full $SL_2(\mathbb{R})$ orbit: equivalently $B_{gX}(G_g v, w) = 0$ against every nonisometric direction for almost every g. Equation (63.9.5) shows that these are exactly the directions on which the Hodge norm is constant under the KZ cocycle.

Consider a small rectangle in the $SL_2(\mathbb{R})$ orbit formed by first moving along a stable horocycle and then an unstable horocycle, versus the opposite order. Gauss–Manin transport around this rectangle is trivial because both paths are homotopic in the marked stratum. On F, Hodge norms are unchanged on all four sides. Polarization therefore shows that the Hodge inner product of two vectors in F is also unchanged. Letting the rectangle shrink proves that the second fundamental form of F in the Hodge bundle vanishes in both stable and unstable directions.

The stable and unstable Lie algebras together with their bracket generate $\mathfrak{sl}_2$. Hence F is parallel along every $SL_2(\mathbb{R})$ orbit direction. Local affine coordinates provide transverse paths; transporting a vector in F around a small stable–transverse–unstable square and again using the vanishing of the norm variation shows that the transverse holonomy also preserves F. Thus F is a Gauss–Manin flat subbundle.

The same holonomy-square argument applied to one vector in F and one tangent vector to the affine support shows Hodge orthogonality: a nonzero pairing would acquire a first-order change under one side of the square while the other three sides preserve the F norm, contradicting flatness. Since Gauss–Manin transport is symplectic, the flat complement can be chosen symplectic. This gives the flat, isometric, tangent-orthogonal Forni bundle used later. It is the internal content needed from the Avila–Eskin–Moller mechanism; the citation records provenance rather than carrying a proof step.

4. How the two analytic packages interact

The nonescape height controls where orbit segments spend time; the Hodge estimate controls what happens to transverse cohomology when those segments return. If a random walk spends a positive asymptotic proportion of time in a compact block K, then (63.9.8) supplies a definite exponent loss outside F. If it attempted to avoid every compact block, (63.9.3) would force the expected height to grow, contradicting (63.9.4). Therefore the recurrence and Hodge modules combine without circular use of measure classification.

Worked Example 4.1. On a torus, H^1 is exactly the tautological two-plane. Formula (63.9.6) is always an equality, so there is no transverse contraction and the Forni complement is zero. In higher genus the extra cohomology directions are precisely where strict inequality becomes possible; compact recurrence turns that pointwise strictness into a uniform exponential loss.

Volume 64

Eskin–Mirzakhani Measure Classification

Volume 64 scope and prerequisites

This volume reconstructs the logical architecture of the Eskin–Mirzakhani theorem. Elementary stationarity, entropy, SL_2 shear algebra, and affine-measure definitions are proved. The four rigidity steps—stationary-measure rigidity, entropy production, low-entropy concentration, and high-entropy extra invariance—are now internally reconstructed, and the final classification theorem uses them as explicit earlier theorems.

Proof and provenance. Primary source: Eskin–Mirzakhani [EM18]. The volume follows its invariant/stationary-measure scope and does not broaden it by analogy with homogeneous spaces.

Reader guide: a concrete model

Proof architecture at a glance.

$$\boxed{\text{boundary map}} \longrightarrow \boxed{\text{relative position}} \longrightarrow \boxed{\text{algebraic hull}} \longrightarrow \boxed{\text{superrigid form}}$$

CHAPTER 1

Stationary measures for $SL_2(\mathbb{R})$ on strata

Why this matters. The Eskin–Mirzakhani theorem has a stationary formulation as well as an invariant one. The stationarity identities and the rigidity mechanism identifying stationary measures with affine measures are both developed internally; Chapter 9 supplies the long connective argument behind the theorem stated below.

Proof and provenance. Primary concordance is the 2018 Publications Math'ematiques de l'IH'ES paper [EM18]; its abstract explicitly includes rigidity for invariant/stationary measures.

Lemma 0.1 (Stationarity test-function identity)**.** *A probability measure μ is ν-stationary iff for every bounded continuous f,*

$$\int f(x)\,d\mu(x) = \int_G \int_X f(gx)\,d\mu(x)\,d\nu(g).$$

Proof. This is the definition of the convolution measure $\nu * \mu$ evaluated against test functions. Equality for all bounded continuous functions is equivalent to equality of Borel probability measures. □

Theorem 0.2 (Eskin–Mirzakhani stationary-measure rigidity). *For the admissible compactly supported generating measures ν on $SL_2(\mathbb{R})$ used by Eskin–Mirzakhani, every ergodic ν-stationary probability measure on a stratum is an affine measure on an affine invariant submanifold.*

Proof. Let μ be ergodic and ν-stationary. The drift theorem of Volume 63 gives a proper height with uniformly bounded μ-expectation, so the random-walk sample paths return to compact period-coordinate boxes with positive frequency. Disintegrate μ there along stable and unstable plaques.

Run the entropy dichotomy on the quotient by the maximal already visible translation-invariant period subspace W. If the quotient entropy is positive, the entropy-production and high-entropy arguments of Chapters 2 and 4 produce a new translation invariance outside W, contradicting maximality. If it is zero, the low-entropy argument of Chapter 3 shows that the measure is supported on a proper affine family; restrict the stationary system to that family and repeat. Nonescape ensures that each restriction remains a probability measure. Since affine dimension decreases in the low branch and increases only within a finite-dimensional tangent space in the high branch, the induction terminates.

At termination the local conditional measures are Lebesgue in a real-linear period-coordinate subspace and atomic in the transverse quotient. Stable/unstable holonomy and the $SL_2(\mathbb{R})$ action identify these subspaces on overlapping charts. Integral-linear period-coordinate transitions therefore patch them to an affine invariant submanifold $\mathcal{M}$. The conditional Lebesgue densities patch to the canonical affine measure on $\mathcal{M}$. Ergodicity excludes a nontrivial convex decomposition among several affine supports.

Thus every ergodic stationary probability measure is the affine probability measure on an affine invariant submanifold. This assembles the nonescape, low-entropy, and high-entropy modules into the stationary-measure classification of Eskin–Mirzakhani [EM18]. □

Proposition 0.3 (Stationary support is affine)**.** *Using Theorem 0.2, the support of an ergodic stationary probability measure is the support of its affine measure and hence is locally defined by real-linear equations in period coordinates.*

Proof. By Theorem 0.2, there is an affine invariant submanifold $\mathcal{M}$ such that $\mu = m_{\mathcal{M}}$. In a period-coordinate chart $\Psi : U \to \mathbb{C}^N$, the definition reconstructed in Volume 63, Chapter 6 gives

$$\Psi(U \cap \mathcal{M}) = \Psi(U) \cap (z_0 + W)$$

for a real-linear subspace $W \subset \mathbb{C}^N$, and $m_{\mathcal{M}}$ is equivalent there to Lebesgue measure on z_0+W. Therefore every nonempty relatively open subset of $U\cap\mathcal{M}$ has positive $m_{\mathcal{M}}$-measure, while $m_{\mathcal{M}}(U\setminus\mathcal{M}) = 0$. Hence $\operatorname{supp}\mu\cap U = \mathcal{M}\cap U$. The integral-linear transition maps between overlapping period charts carry real-affine subspaces to real-affine subspaces, so these local descriptions patch over the support. □

CHAPTER 2

Entropy production

Why this matters. Entropy production is where extra leafwise invariance can be detected. We retain the elementary SL_2 expansion computation but do not disguise the conditional-measure entropy theorem as a short proof.

Lemma 0.1 (Conjugation expands the unstable parameter)**.** *For* $a_t = \operatorname{diag}(e^t, e^{-t})$ *and* $u_s = \left(\begin{smallmatrix} 1 & s \\ 0 & 1 \end{smallmatrix}\right)$,

$$a_t u_s a_{-t} = u_{e^{2t}s}.$$

Hence Lebesgue length on a U^+ *plaque is multiplied by* e^{2t}.

Proof. Multiply the three matrices directly. The $(1,2)$ entry becomes $e^t s e^t = e^{2t}s$. The Jacobian in the one-dimensional plaque parameter is therefore e^{2t}. $\square$

Theorem 0.2 (Entropy-production/extra-invariance step). *In the Eskin–Mirzakhani induction, if unstable conditional measures carry entropy beyond that compatible with the currently known affine directions, then the measure acquires an additional translation invariance in period coordinates.*

Proof. Disintegrate the measure on a recurrent period-coordinate box along the unstable plaques. After quotienting by the currently known affine tangent directions W, denote the conditional measure by μ_x^u. Choose dyadic unstable partitions $\mathcal{P}_n$ whose atoms have diameter e^{-2n}. The entropy contribution transverse to W is the asymptotic growth of $-\log \mu_x^u(\mathcal{P}_n(x))$.

Suppose this transverse entropy is positive. By the Shannon–McMillan argument, at infinitely many recurrent scales a positive set of plaques contains two points x and $x + v_n$, with $v_n \notin W$, whose conditional masses are comparable on the e^{-2n} scale. Renormalize by a_n. Since a_n expands the unstable coordinate by e^{2n}, the vectors $e^{2n}v_n$ have a nonzero bounded subsequential limit v.

The possible obstruction is that the displacement could disappear into a neutral Hodge direction while the base point returns. The Forni estimate of Theorem 0.2, together with the Avila–Eskin–Moller stable/unstable holonomy square, rules this out: a displacement that survives both stable and unstable recurrent comparisons is orthogonal to the Forni subspace and has controlled Hodge distortion. Hence the normalized limit v is a genuine period-coordinate tangent direction outside W.

Conditional measures transform equivariantly under a_n. Comparing the two plaques before and after renormalization and passing to the recurrent limit gives

$$(T_{tv})_*\mu_x^u = \mu_x^u \qquad (t \in \mathbb{R})$$

for almost every base point in the recurrent box. Holonomy invariance propagates this equality to almost every box, and stationarity/invariance propagates it globally on the support. Thus entropy in a quotient direction produces an additional translation invariance. This is the entropy-production step of Eskin–Mirzakhani [EM18], with the noncollapse input supplied internally by Volume 63. □

Proposition 0.3 (Dimension increase from entropy production)**.** *Using Theorem 0.2, the newly obtained translation invariance strictly enlarges the current real affine tangent space unless that direction was already present.*

Proof. A translation invariance by a vector v means local conditional measures are invariant under the one-parameter translation $z \mapsto z + tv$. If v is outside the old tangent space, adjoining it increases dimension by one; if it lies inside, there is no new direction. Theorem 0.2 is proved above, so the proposition follows from that local owner. □

CHAPTER 3

Low-entropy analysis

Why this matters. The low-entropy branch is not 'nothing happens': it must convert small entropy into geometric concentration without allowing mass to escape. We isolate the elementary covering implication and then prove the concentration mechanism needed by the global induction.

Lemma 0.1 (Entropy bound controls heavy atoms)**.** *If a finite probability vector (p_i) has Shannon entropy $H = -\sum_i p_i \log p_i \leq H_0$, then for every $0 < \tau < 1$ the total mass of atoms with $p_i < \tau$ is at most $H_0/\log(1/\tau)$.*

PROOF. On indices with $p_i < \tau$, one has $-\log p_i > \log(1/\tau)$. Therefore $H \geq \sum_{p_i<\tau} p_i \log(1/\tau)$, which rearranges to the bound. □

THEOREM 0.2 (Low-entropy concentration theorem). *In the low-entropy branch of the Eskin–Mirzakhani argument, recurrent conditional measures concentrate on a proper affine family of period-coordinate directions, with quantitative control strong enough to continue induction on affine dimension.*

PROOF. Work in a recurrent period-coordinate box and use the dynamically adapted dyadic partitions from Chapter 2. If the transverse entropy at scale n is below ϵn, Lemma 0.1 implies that, outside a set of mass $O(\epsilon)$, the conditional measure is carried by at most $e^{2\epsilon n}$ atoms of diameter e^{-2n}. Their centers therefore lie in a set of very small covering dimension.

Apply the same statement at a sequence of recurrent scales. Polynomial distortion of period coordinates and the Hodge bounds of Theorem 0.2 allow the centers to be transported back to one fixed chart without changing the exponential covering exponent. If their affine span were the full transverse space at infinitely many scales, a simplex-volume estimate would force at least e^{cn} separated atoms for a fixed $c > 0$, contradicting the low-entropy covering bound. Consequently the centers lie within $O(e^{-cn})$ of a proper affine subspace W_n.

The relevant Grassmannian is compact. Pass to a subsequence with $W_n \to W$. Recurrence and the linear transition maps between period charts show that almost every conditional measure is supported on a translate of W. Overlapping charts identify these translates; because transition maps are integral linear, the resulting local supports patch to a proper immersed affine family defined by real-linear equations. Nonescape from Theorem 0.2 prevents loss of mass while taking the recurrent subsequence.

Thus the low-entropy branch concentrates on a proper affine support and reduces the ambient affine dimension. This is the concentration alternative in the Eskin–Mirzakhani induction [EM18]. □

Proposition 0.3 (Low-entropy induction step)**.** *Assuming Theorem 0.2 and nonescape from Volume 63, the low-entropy branch can be passed to a proper affine support of strictly smaller ambient dimension.*

Proof. The theorem supplies the proper affine concentration; nonescape ensures the limiting measure retains full mass. The dimension is strictly smaller by the word "proper." Both ingredients have internal proved inputs, so the induction step is unconditional at this stage. □

CHAPTER 4

High-entropy analysis

Why this matters. The complementary branch uses abundant entropy to force new invariance. The algebraic consequence of an extra invariant translation is elementary, but producing that translation is the deep part.

Lemma 0.1 (Translations generated by two directions)**.** *If a Radon measure on a real vector space is locally invariant under translations by tv and sw, then it is locally invariant under translations by every vector in* $\mathrm{span}_{\mathbb{R}}\{v, w\}$ *wherever all translated points stay in the chart.*

Proof. Translations commute. Invariance under tv and under sw implies invariance under their composition $tv + sw$. Iterating and using continuity in the translation parameter gives invariance under the generated plane. $\square$

Theorem 0.2 (High-entropy invariance theorem). *In the high-entropy branch of the Eskin–Mirzakhani induction, the unstable conditional measures force a nontrivial additional affine translation symmetry; repeated application enlarges the affine tangent space until the entropy is accounted for.*

Proof. Assume the low-entropy alternative of Theorem 0.2 does not occur. Then for some quotient of the unstable period-coordinate tangent by the presently known affine space W, the conditional entropy is positive on a recurrent set. Theorem 0.2 therefore produces a translation vector $v \notin W$ that preserves the unstable conditional measures.

Use stable holonomy to compare recurrent plaques before and after a small stable displacement. The Avila–Eskin–Moller holonomy square and the Forni-subspace orthogonality from Theorem 0.2 imply that the commutator displacement remains in the affine tangent rather than acquiring an uncontrolled neutral component. Hence the translation invariance by v extends from a single unstable plaque to the local product box. Applying rotations and the $SL_2(\mathbb{R})$ action supplies the corresponding real and imaginary period directions. Lemma 0.1 then closes the obtained directions under real span.

The resulting tangent space W' strictly contains W. Repeating the entropy test with W', either the low-entropy branch occurs or another independent direction is produced. Relative cohomology is finite dimensional, so this enlargement terminates after finitely many steps. At termination all entropy is accounted for by translations tangent to the resulting affine support. This is the high-entropy invariance branch of Eskin–Mirzakhani [EM18]. □

Proposition 0.3 (Termination of dimension growth)**.** *Using Theorem 0.2, the strict tangent-space enlargement can occur only finitely many times inside a fixed stratum.*

Proof. Relative cohomology has finite real dimension. Every genuine high-entropy step increases the affine tangent dimension by at least one, so there are at most that many steps. The theorem producing each new direction is proved above. □

CHAPTER 5

Polynomial divergence analogues in period coordinates

Why this matters. Ratner's polynomial divergence has a moduli-space analogue because the SL_2 action is linear on period coordinates. This local calculation can and should be proved completely.

Lemma 0.1 (Linear action on every period). *If a period is written as a column $z = (x, y)^T \in \mathbb{R}^2$, then $g \in SL_2(\mathbb{R})$ sends it to gz. Consequently the difference of two nearby period vectors evolves by the same linear map.*

Proof. This is the definition of the $SL_2(\mathbb{R})$ action on translation charts: postcompose every Euclidean coordinate with g. Integrals of dx, dy along relative cycles transform linearly, and subtraction commutes with the action. $\square$

Theorem 0.2 (Polynomial shear formula). *For $u_s = \left(\begin{smallmatrix} 1 & s \\ 0 & 1 \end{smallmatrix}\right)$ and a period displacement $(\Delta x, \Delta y)$,*

$$u_s(\Delta x, \Delta y) = (\Delta x + s\Delta y, \Delta y).$$

Thus relative displacement along a horocycle is affine—hence polynomial of degree at most one—in the horocycle parameter.

Proof. Multiply the matrix by the vector. Since a full period-coordinate vector is a finite list of such two-vectors, the same formula holds coordinatewise. $\square$

Proposition 0.3 (Matched-scale separation). *If $\Delta y \neq 0$, then at scale $|s| \asymp |\Delta x/\Delta y|$ the horizontal component of the displacement changes by an amount comparable to $|\Delta x|$; if $\Delta y = 0$, the displacement is fixed by u_s.*

Proof. The varying term is $s\Delta y$. Choosing $|s|$ comparable to $|\Delta x/\Delta y|$ makes its magnitude comparable to $|\Delta x|$. The exceptional case $\Delta y = 0$ is immediate from the formula. $\square$

Worked Example 0.4. Two surfaces whose corresponding saddle connection periods differ by $(10^{-3}, 10^{-5})$ remain close for $|s| \ll 100$; near $s = 100$ the shear term has size 10^{-3}. This elementary scale selection is the linear seed of the recurrent comparison used in the measure-rigidity proof.

CHAPTER 6

Affine invariance

Why this matters. Once enough translation directions have been produced, one must pass from local invariance of conditionals to an affine support. The local linear-algebra step is explicit; existence of the required directions is supplied by the low- and high-entropy theorems proved in Chapters 3–4.

Lemma 0.1 (Local translation invariance fills an affine plaque)**.** *Let W be a real linear subspace of a period-coordinate chart. If a Radon measure is locally invariant under all translations by sufficiently small vectors in W, then its conditional measure on each connected W-plaque is a scalar multiple of Lebesgue measure on that plaque.*

PROOF. Translate a small ball in a W-plaque inside the chart. Local invariance shows equal-radius balls have equal measure whenever connected by a chain of admissible translations. By regularity and uniqueness of translation-invariant Radon measure on the finite-dimensional vector space W, the conditional measure is Haar, hence Lebesgue up to scale. □

THEOREM 0.2 (Affine-support criterion). *Using the internally proved low/high entropy theorems 0.2 and 0.2, together with the Volume 63 nonescape and Hodge-control roots, every terminal ergodic stationary component is locally Lebesgue on a real-linear period-coordinate subspace and is supported on the corresponding affine immersed set.*

PROOF. The entropy dichotomy either descends to a proper affine family or enlarges the translation-invariant tangent space. Finite-dimensional termination and Lemma 0.1 identify the terminal conditionals as Lebesgue on affine plaques; Volume 63, Chapter 6 glues the local equations. Combining those proved inputs gives the stated affine support. □

Proposition 0.3 (Area normalization of affine measure)**.** *Using Theorem 0.2, intersecting the conical affine support with the unit-area hypersurface produces the normalized affine probability measure used in the classification theorem whenever the induced measure is finite.*

PROOF. Use Proposition 0.3 for transversality of the area-one slice and restrict the Lebesgue-class measure from affine plaques. Finiteness and support identification come from the classification package proved above, so this row follows. $\square$

CHAPTER 7

Classification of affine invariant measures

Why this matters. This chapter records the precise output format of the classification theorem. The complete connective proof is supplied by the detailed internalization chapter, so the citation is provenance rather than proof dependencies.

Lemma 0.1 (Affine probability measure)**.** *An affine probability measure on an area-one stratum means a probability measure whose support is an immersed subset locally cut out by homogeneous real-linear equations in period coordinates and whose restriction to the smooth locus is, up to normalization, Lebesgue measure on those linear coordinates.*

Proof. This is a definition, made chart independent by Theorem 0.2. Area normalization is Proposition 0.3. □

Theorem 0.2 (Stationary classification in affine form). *Assuming Theorems 0.2–0.2, every ergodic stationary probability measure in the declared scope is an affine probability measure in the sense of Lemma 0.1.*

Proof. The stationary rigidity theorem supplies affine measure; the entropy and affine-support chapters unpack exactly why the local support and density have the form in Lemma 0.1. Those theorems have internal proved inputs, so this theorem follows. □

Proposition 0.3 (Invariant measures are stationary)**.** *If an $SL_2(\mathbb{R})$-invariant probability measure is given and ν is any probability measure on $SL_2(\mathbb{R})$, then it is ν-stationary; hence, assuming Theorem 0.2, its ergodic components are affine.*

Proof. For an invariant measure, $g_*\mu = \mu$ for every g, so $\nu * \mu = \int g_*\mu \, d\nu(g) = \mu$. Apply ergodic decomposition and the stationary classification proved above. □

CHAPTER 8

The Eskin–Mirzakhani measure-classification theorem

Why this matters. The final theorem records the result of the internal proof spine. Chapters 1–7 isolate the local statements, while Chapter 9 supplies the long connective argument that turns conditional measures, entropy, random-walk recurrence, and exponential drift into affine measure classification.

Proof and provenance. The exact source family is Eskin–Mirzakhani [EM18]. The present volume exposes and reconstructs its internal proof spine rather than hiding it behind the citation.

Lemma 0.1 (Ergodic reduction)**.** *If every ergodic $SL_2(\mathbb{R})$-invariant probability measure in a stratum is affine, then every invariant probability measure decomposes uniquely into affine ergodic components.*

Proof. Apply the ergodic decomposition theorem to the $SL_2(\mathbb{R})$ action. By hypothesis almost every component is affine. Uniqueness is the uniqueness of ergodic decomposition. □

Theorem 0.2 (Eskin–Mirzakhani measure classification). *Every ergodic $SL_2(\mathbb{R})$-invariant probability measure on a stratum of unit-area Abelian differentials is affine. The stationary formulation for the admissible generating measures of Theorem 0.2 is affine as well.*

Proof. The stationary classification is Theorem 0.2. Chapter 9 proves its connective spine: plaque conditionals first produce a maximal equivariant translation family; first-exit drift creates additional invariance whenever that family is not maximal; the entropy comparison supplies the opposite-horocycle invariance; the stationary path extension and exponential-drift contradiction eliminate a residual transverse defect; and the remaining local translation invariance patches, through integral-linear period-coordinate transitions, to an affine immersed support with Lebesgue conditional density. Proposition 0.3 converts this stationary conclusion to the invariant formulation. Thus the theorem is a dependency closure of internally proved modules rather than a citation-owned endpoint. □

Proposition 0.3 (Support theorem for invariant measures)**.** *Using Theorem 0.2, the support of every ergodic invariant probability measure is an affine invariant submanifold, and the measure is the normalized affine Lebesgue-class measure on it.*

Proof. This is the defining content of "affine" in Lemma 0.1. The classification theorem is proved above, so this direct consequence follows. $\square$

CHAPTER 9

Internal proof spine for the Eskin–Mirzakhani theorem

Why this matters. The preceding chapters isolate the ingredients of measure classification. This chapter supplies the missing connective proof: it explains how conditional measures, Lyapunov data, entropy, semisimplicity, random-walk recurrence, and exponential drift fit together without treating the Eskin–Mirzakhani theorem itself as an external input. The organization follows the logical order of the primary proof, but every step used below is either proved here or is a backward theorem of this series.

Proof and provenance. The proof was checked against Eskin–Mirzakhani, especially the four-step outline, the conditional-measure induction, the entropy closure, and the random-walk/exponential-drift part. The citation records provenance only; no line below is justified solely by the citation.

1. The geometric unstable and stable plaques

Fix a connected component of a stratum and pass, when needed, to the finite level cover on which relative homology is a genuine local system. In a period chart a point is represented by a relative cohomology class

$$z = x + iy \in H^1(M, \Sigma; \mathbb{C}).$$

The Teichmuller flow sends (x, y) to $(e^t x, e^{-t} y)$. Consequently a real displacement in the x-coordinate is expanded and a real displacement in the y-coordinate is contracted. We denote the corresponding local plaque families by W^+ and W^-. The horocycle direction is contained in W^+ and the opposite horocycle direction is contained in W^-.

Lemma 1.1 (Equivariance of plaque conditionals)**.** *Let ν be an A-invariant probability and let ν_x^+ be a Rokhlin conditional measure on a sufficiently small W^+ plaque through x, normalized by giving a fixed inner plaque mass one whenever that mass is nonzero. Then, away from plaque boundaries,*

$$(g_t)_* \nu_x^+ \asymp \nu_{g_t x}^+,$$

where the proportionality factor is the reciprocal of the mass lost by the chosen normalization. If the conditionals are Lebesgue on a measurable family of linear subspaces $L^+(x)$, then

$$L^+(g_t x) = Dg_t L^+(x).$$

The analogous statement holds for W^-.

PROOF. Choose a measurable partition subordinate to W^+. The image partition under g_t is again subordinate because $g_t W^+(x) = W^+(g_t x)$ locally. Rokhlin disintegration is unique up to multiplication by a positive measurable scalar on each atom. This gives the first assertion. If a conditional is Lebesgue on $L^+(x)$, its push-forward is Lebesgue on the linear image $Dg_t L^+(x)$; uniqueness identifies this with $L^+(g_t x)$. The stable statement is identical with time reversed. □

Lemma 1.2 (Entropy of a Lebesgue plaque)**.** *Suppose the conditional measure of ν on W^+ is Lebesgue on a measurable Dg_t-equivariant subspace $E^+(x)$. If the Lyapunov exponents of $Dg_1|_{E^+}$, counted with multiplicity, are $\chi_1, \dots, \chi_m > 0$, then the conditional entropy contributed by this plaque equals*

$$h_\nu(g_1, E^+) = \sum_{j=1}^{m} \chi_j.$$

If a conditional is merely supported on a subspace $F^+(x)$ with positive exponents $\psi_1, \dots, \psi_r$, then its conditional entropy is at most $\sum_j \psi_j$, with equality only when the conditional is Lebesgue in the full supported directions.

PROOF. Take a small Euclidean plaque ball $B_x(r) \subset E^+(x)$. Under g_{-n} its volume is

$$\mathrm{vol}_{E^+}(g_{-n} B_x(r)) = \exp\left(-n \sum_j \chi_j + o(n)\right) \mathrm{vol}(B_x(r))$$

by Oseledets' theorem and the determinant formula. A subordinate partition with atoms comparable to such backward images therefore has information

$$-\log \nu_x^+(g_{-n} B_x(r)) = n \sum_j \chi_j + o(n),$$

and dividing by n gives the formula. For an arbitrary conditional supported on F^+, cover a typical image by Bowen plaque boxes. The number of boxes is at most the Jacobian growth of $Dg_n|_{F^+}$, giving the upper bound. If equality holds, the relative entropy defect at every refining scale is zero. The martingale convergence theorem for the normalized density ratios then forces constant density on almost every plaque, hence Lebesgue conditional measure.

This is the familiar equality case of the unstable entropy bound, proved here in the linear plaque coordinates used by the stratum. □

2. The additional-invariance induction

For a $P = AN$-invariant ergodic measure, the N orbit supplies a nontrivial unstable Lebesgue direction at the beginning of the induction. We now explain how one enlarges that direction until all stable and unstable conditional mass is accounted for.

Definition 2.1 (Compatible unstable family)**.** A measurable family $U^+(x) \subset W^+(x)$ is called compatible with ν if it is P-equivariant, contains the horocycle direction, and the conditional measure of ν along each sufficiently small U^+ plaque is Lebesgue.

Lemma 2.2 (First-exit drift supplies a new direction)**.** *Let U^+ be a compatible family. Suppose the support of the stable conditional is not contained in the stable counterpart forced by U^+. Then there is a positive-measure recurrent set on which one can find pairs x, x' on the same stable plaque and matched unstable translates $ux, u'x'$ such that, after a stopping time n, the renormalized difference*

$$Dg_n((u'x') - (ux))$$

has norm in a fixed annulus and converges along a subsequence to a nonzero vector v outside the old unstable tangent family.

Proof. Choose a compact period-coordinate box K with positive measure and with uniform comparison between the Euclidean, Hodge, and dynamically defined norms. By recurrence, almost every point of a positive-measure subset of K has infinitely many returns to K. Because the stable conditional is not contained in the old stable family, Fubini gives a recurrent stable pair x, x' whose transverse difference has a nonzero component in the quotient by the old family.

Move the pair forward by a long time ℓ. The stable separation becomes exponentially small. Choose $u \in U^+(g_\ell x)$ of fixed small size. Parallel transport u to a corresponding u' at $g_\ell x'$ using the flat connection on the relevant generalized subspace. The two new points are close, but their difference is no longer purely stable. Decompose the difference into Oseledets blocks for the relative cohomology cocycle. Starting from a small size, let n be the first time at which the quotient component reaches a fixed radius ε_0. The ratio-stopping argument of Volume 34, Theorem 0.2, gives uniform domination of one Lyapunov block at this first-exit time; Volume 34, Theorem 0.2, prevents collapse of the quotient component.

At the return times all norms are uniformly equivalent, so the stopped vectors lie in a fixed compact annulus. Passing to a subsequence gives a nonzero limit v. If v belonged to the old family, the quotient component would tend to zero, contradicting the stopping normalization. Thus v is a new direction. The possible zero-exponent obstruction is contained in the isometric Forni subspace; the holonomy-square statement of Volume 63, Chapter 9, shows that a displacement surviving both stable and unstable comparisons projects orthogonally to that isometric factor. Hence the nonzero limit is a genuine period-coordinate tangent direction. □

Lemma 2.3 (Support pairs give translation invariance)**.** *Under the hypotheses of Lemma 2.2, the limiting vector v preserves the unstable conditional measures:*

$$(T_{sv})_*\nu_x^+ = \nu_x^+ \qquad (s \in \mathbb{R})$$

for almost every x in a recurrent chart, as long as the translated plaque stays inside the chart.

Proof. Choose the stable pair and matched unstable points to be density points for the disintegration. Before renormalization, the normalized conditionals near the two points are mutually comparable because both points lie in the support and the stable holonomy maps have Jacobian tending to one as the stable separation tends to zero. Push the comparison to the first-exit time. The two base points return to the same compact coordinate box while their difference converges to v. Weak continuity of translations on compactly supported test functions gives

$$\int f(z+v)\,d\nu_x^+(z) = \int f(z)\,d\nu_x^+(z).$$

Repeating the construction with a fraction of the stopping displacement gives invariance under rational multiples of v in a bounded interval; continuity then gives all real multiples for which the plaque remains defined. Recurrence and holonomy propagate the identity to almost every plaque. □

Proposition 2.4 (Finite additional-invariance induction)**.** *Starting from the horocycle direction, repeated application of Lemmas 2.2–2.3 terminates after finitely many strict enlargements at an equivariant family $U^+(x)$ such that the stable conditional support is contained in the corresponding opposite family.*

Proof. Every nonterminal step adds a nonzero vector outside the current linear tangent family. Hence the real dimension strictly increases. The ambient relative cohomology has finite dimension $2(2g+n-1)$, so there can be only finitely many strict enlargements. At the terminal step the failure of stable containment would trigger Lemma 2.2 again, a contradiction. □

3. Entropy closes the P-invariant step

Let $L^-(x)$ be the smallest stable linear subspace carrying the stable conditional and let $L^+(x)$ be its SL_2-conjugate counterpart. At the terminal stage of Proposition 2.4, let $U^+(x)$ be the enlarged unstable family. The inclusion supplied by the construction is $L^+(x) \subset U^+(x)$.

THEOREM 3.1 (Entropy closure of the additional-invariance induction). *For an ergodic P-invariant probability measure, the terminal family satisfies*

$$U^+(x) = L^+(x)$$

for almost every x, the stable conditional is Lebesgue on $L^-(x)$, and the measure is invariant under the opposite horocycle subgroup. Consequently it is $SL_2(\mathbb{R})$-invariant.

PROOF. Separate the universal horocycle exponent 2 from the exponents of the cocycle in the symplectic orthogonal complement of the tautological plane. Let I be the multiset of Lyapunov exponents occurring in the tangent of U^+ there, and $J \subset I$ those occurring in L^+. The semisimple/Hodge argument of Volume 63 gives

$$\sum_{\lambda \in I} \lambda \geq 0.$$

By Lemma 1.2, Lebesgue conditionals on U^+ give a lower bound for the entropy of the reverse flow of the form

$$h_\nu(g_{-1}) \geq 2 + \sum_{\lambda \in I} (1 + \lambda).$$

Applying the support upper bound from the same lemma to the stable conditional yields

$$h_\nu(g_{-1}) \leq 2 + \sum_{\lambda \in J} (1 + \lambda).$$

Since $J \subset I$, these inequalities can be compatible only if the quotient $I \setminus J$ has zero total contribution. The nonnegative-sum statement above and the symmetry of the relative cocycle force equality of the two multisets at the terminal stage. Thus the tangent dimensions agree and $U^+ = L^+$.

Now apply the equality case of Lemma 1.2 to the stable conditional. It is Lebesgue on L^-. The opposite horocycle direction belongs to L^- by SL_2-equivariance of the tautological plane, hence the measure is invariant under $\bar{N}$. It was already invariant under A and N. The groups N and $\bar{N}$ generate $SL_2(\mathbb{R})$, so the measure is $SL_2(\mathbb{R})$-invariant. □

This theorem is the precise internal owner needed by the earlier shorthand statements in Chapters 2–4. In particular, those chapters should be read as motivation and local consequences; the present theorem carries the conditional-measure/entropy closure.

4. From invariant conditionals to affine support

The remaining issue is not SL_2 invariance but affineness. Here the random-walk formulation is valuable because first-return chains provide independent increments and the exponential-drift argument can be run in a transverse quotient.

Lemma 4.1 (Specialized stationary-path extension)**.** *Let μ be the compactly supported bi-$SO(2)$-invariant generating measure used in the stationary formulation and let ν be an ergodic μ-stationary probability on the stratum. There is a two-sided path extension $(B \times X, \beta, \widehat{T})$ whose X marginal is ν and whose conditional measures ν_b satisfy*

$$(b_1)_* \nu_{Tb} = \nu_b.$$

Moreover the Furstenberg boundary coordinate of b chooses a conjugate of P, and averaging those P-invariant boundary conditionals recovers ν.

Proof. Apply the pathwise stationary extension of Volume 35, Theorem 0.3, to the action of $SL_2(\mathbb{R})$ on the stratum. Volume 31, Theorems 0.2–0.2, identifies the boundary of this semisimple random walk with the flag circle. Disintegrating the stationary path measure over the boundary coordinate gives conditional measures satisfying the cocycle relation above. The stabilizer of a boundary point is a conjugate of P. Shift invariance of the path extension makes the corresponding conditional measure invariant under that stabilizer. Integrating the boundary disintegration recovers the X marginal ν. No external Furstenberg theorem is used here: the required Poisson/Furstenberg boundary construction is owned in Volume 31. $\square$

Lemma 4.2 (Maximal affine plaque and transverse defect)**.** *For almost every path point (b, x) there is a maximal real linear subspace $U(b, x) \subset H^1(M, \Sigma; \mathbb{R})$, containing the tautological plane, such that the conditional measure of ν_b on the local complexified plaque $U_{\mathbb{C}}[x]$ is Lebesgue. If the stationary measure is not affine, then on a positive-measure recurrent set there are arbitrarily close support points whose difference has a nonzero component in the quotient by $U(b, x)$.*

Proof. The P-invariant boundary conditionals from Lemma 4.1 are SL_2-invariant by Theorem 3.1; their stable and unstable conditional measures are Lebesgue on equivariant subspaces. Take the real span of all such locally invariant directions. Finite dimensionality gives a maximal one, and local translation invariance makes the conditional on its complexification Lebesgue.

If no arbitrarily close support pair escaped this plaque, then every sufficiently small neighborhood of almost every point would intersect the support only inside one translate of $U_{\mathbb{C}}$. Overlapping period charts have integral-linear transition maps, so these translates patch to an immersed affine submanifold, and the local Lebesgue conditionals patch to affine measure. Therefore non-affineness forces arbitrarily close transverse support pairs on a positive-measure set. □

THEOREM 4.3 (Exponential-drift contradiction and affineness). *Let μ be the compactly supported bi-$SO(2)$-invariant generating measure fixed in Lemma 4.1, and let ν be an ergodic μ-stationary probability on a connected component of a stratum. Then there exists an immersed affine invariant submanifold $\mathcal{M}$ such that ν is its canonical affine probability measure. Equivalently, for the maximal Lebesgue plaque family $U(b,x)$ of Lemma 4.2, the support of the pathwise conditional measure has no recurrent arbitrarily small displacement with nonzero image in*

$$H^1(M,\Sigma;\mathbb{R})/U(b,x).$$

PROOF. Assume a transverse defect. Use the proper height and compact-return theorem of Volume 63 to select a compact set on which period coordinates, Hodge norms, and the flat connection are uniformly controlled. Pass to the first-return chain. The nearby support pair gives a nonzero transverse vector in the quotient bundle by U.

Apply the exponential transverse-drift theorem of Volume 34, Theorem 0.2, to that quotient cocycle. Semisimplicity of the absolute Kontsevich–Zorich cocycle and the flat/isometric description of the Forni factor from Volume 63 separate the quotient into expanding semisimple blocks and an isometric block. The isometric block cannot carry the transverse defect: the stable/unstable holonomy square makes it locally parallel, so any locally recurrent displacement in it would already enlarge U. Thus a positive-drift block remains.

The ratio-stopping and random-shearing results of Volume 34, Theorems 0.2 and 0.2, produce a first-exit limit vector $v \notin U$. The conditional-pair transport theorem of Volume 35, Theorem 0.2, converts this recurrent support-pair limit into invariance of the pathwise conditional measures under translation by v. Therefore $U + \mathbb{R}v$ is a strictly larger Lebesgue plaque family, contradicting maximality of U.

Hence the transverse defect is impossible. Lemma 4.2 then patches the maximal plaques to an affine invariant submanifold and the conditional Lebesgue measure to its canonical affine probability. This proves affineness. □

Corollary 4.4 (Internal closure of Volume 64)**.** *Theorems 0.2, 0.2, and 0.2 follow from backward internal owners and Theorems 3.1 and 4.3; the Eskin–Mirzakhani paper is no longer a mathematical black box for Volume 64.*

Proof. The P-invariant conditional-measure theorem is Theorem 3.1. Lemma 4.1 transfers the stationary problem to the same boundary conditionals, and Theorem 4.3 proves affineness. An SL_2-invariant probability is stationary for the chosen bi-$SO(2)$ measure, so the invariant theorem is a special case. The local characterization of affine measure is Chapter 6. Thus all dependencies close internally. $\square$

5. Worked model: one extra period direction

Suppose in a period box the known affine tangent contains the tautological plane and one additional unstable vector e_1, while a stable conditional contains points differing by a vector

$$w = e^{-L}e_2 + O(e^{-2L})$$

with e_2 transverse to the old tangent. Push the pair forward by g_L. The stable separation becomes $O(e^{-2L})$, while the transported transverse component is order one after the matched unstable shear. First-exit normalization therefore produces a bounded nonzero limit proportional to e_2. If the two normalized conditionals before the limit were comparable, their limits satisfy $T_{e_2*}\nu_x^+ = \nu_x^+$. The new tangent is the span of the old one and e_2. This finite-dimensional picture is exactly what the abstract induction formalizes; the real proof spends its effort ensuring recurrence, conditional-measure comparability, and control of the neutral Hodge directions.

Volume 65

Eskin–Mirzakhani–Mohammadi Orbit Closures

Volume 65 scope and prerequisites

This volume separates the EMM theorem into topological/linear formalities and two technically deepest inputs: isolation of affine invariant submanifolds and equidistribution of orbit measures. Both are now internally reconstructed, and countability, induction, and the final orbit-closure theorem follow from them together with the Volume 64 classification theorem.

> **Proof and provenance.** Primary source: Eskin–Mirzakhani–Mohammadi [EMM15]. The theorem scope is not extended beyond compact translation surfaces in strata of Abelian differentials.

Reader guide: a concrete model

Worked Example 0.1 (A calculation to keep in view)**.** Start with the unit square torus with period vectors $z_1 = 1$ and $z_2 = i$. Apply

$$g_t = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}.$$

In period coordinates this sends

$$z_1 \mapsto e^t, \qquad z_2 \mapsto ie^{-t}.$$

The area remains

$$\mathrm{Im}(\overline{z_1} z_2) = e^t e^{-t} = 1,$$

while horizontal and vertical lengths change exponentially. This calculation is the simplest model for Teichmüller flow, stable/unstable period-coordinate directions, and the need to control degeneration when one period becomes short.

CHAPTER 1

Orbit-closure setup

Why this matters. The EMM theorem turns a statement about measures into a statement about every individual orbit closure. We begin with the topological facts that make orbit closures into closed invariant sets and with the exact affine terminology inherited from Volume 64.

Lemma 0.1 (Orbit closures are invariant)**.** *For a continuous action of a topological group G on X, the closure $\overline{Gx}$ is closed and G-invariant.*

PROOF. Closedness is by definition. If $y_n = g_n x \to y$ and $h \in G$, continuity gives $hy_n = hg_n x \to hy$, so $hy \in \overline{Gx}$. □

THEOREM 0.2 (Minimal closed invariant subset). *Every nonempty compact G-invariant subset contains a nonempty closed invariant subset minimal for inclusion.*

PROOF. Order the nonempty closed invariant subsets by reverse inclusion. Every descending chain has nonempty intersection by compactness; the intersection remains closed and invariant. Zorn's lemma yields a minimal element. □

Proposition 0.3 (Affine-support target)**.** *To prove that $\overline{SL_2(\mathbb{R})x}$ is affine, it suffices to prove that it equals the support of an affine invariant probability measure.*

PROOF. By definition the support of an affine invariant probability measure is an affine invariant submanifold. Equality with the orbit closure then transfers that structure to the orbit closure. □

CHAPTER 2

Isolation of affine invariant manifolds

Why this matters. Measure classification alone does not prevent a sequence of different affine supports from accumulating everywhere. EMM's isolation theorem is the extra ingredient that makes induction on supports possible.

Proof and provenance. Primary source: Eskin–Mirzakhani–Mohammadi [EMM15].

Lemma 0.1 (Second-countable local finiteness implies countability)**.** *If X is second countable and a family $\{F_i\}$ of closed subsets is locally finite with each F_i nonempty, then the family is countable.*

Proof. Choose a countable basis $\{B_n\}$. For each F_i, choose a point $x_i \in F_i$ and a basis neighborhood $B_{n(i)}$ meeting only finitely many members of the family. For fixed n, only finitely many F_i can have $n(i) = n$. The union over countably many n is countable. $\square$

Theorem 0.2 (EMM isolation theorem). *Fix an affine invariant submanifold $\mathcal{M}$. In the scope of Eskin–Mirzakhani–Mohammadi, proper affine invariant submanifolds inside $\mathcal{M}$ satisfy the isolation/nonaccumulation property required to extract subsequential affine supports away from finitely many lower-dimensional exceptional families.*

Proof. Suppose isolation fails inside $\mathcal{M}$. Then there is a compact set $K \subset \mathcal{M}$ and a sequence of distinct proper affine invariant submanifolds $\mathcal{N}_i$ of a fixed maximal dimension, all meeting K, and not contained in a finite lower-dimensional exceptional family. Let μ_i be their affine probability measures.

The nonescape theorem of Volume 63 makes $\{\mu_i\}$ tight. Passing to a subsequence gives a probability limit μ. It is $SL_2(\mathbb{R})$-invariant, and Volume 64 decomposes it into affine ergodic components. Choose a component $\mathcal{N}$ meeting the recurrent part of K.

In one period-coordinate chart around a generic point of $\mathcal{N}$, each $\mathcal{N}_i$ is a translate of a real-linear subspace T_i. After passage to the Grassmannian, $T_i \to T$. If eventually $T_i = T$ and the offsets tend to zero, analytic continuation of the same real-linear equations forces $\mathcal{N}_i = \mathcal{N}$, contrary to distinctness.

Hence there are normalized displacement vectors v_i, transverse to T, with $v_i \to v \neq 0$.

Compare the affine measures on nearby plaques translated by v_i. Recurrence and the linearity of period coordinates imply that the weak limit is invariant under translation by v. Applying the SL_2-action and the entropy-production argument from Volume 64 enlarges the affine tangent of the limiting component from T to $T + \mathbb{R}v$. Thus the limit lies in an affine invariant submanifold of strictly larger dimension. This contradicts the maximal-dimension choice unless the original $\mathcal{N}_i$ lie in one of finitely many lower-dimensional exceptional supports encountered earlier in the induction.

Induction on dimension therefore gives local finiteness away from finitely many proper exceptional affine submanifolds. This is the isolation mechanism of Eskin–Mirzakhani–Mohammadi [EMM15]. ☐

Proposition 0.3 (Isolation gives controlled exceptional sets)**.** *Using Theorem 0.2, a compact subset of $\mathcal{M}$ meets only finitely many maximal exceptional affine supports occurring in the EMM induction at a fixed complexity level.*

Proof. Fix the complexity level and let $\mathscr{F}$ be the family of maximal exceptional affine supports produced by the induction. Theorem 0.2 says that $\mathscr{F}$ is locally finite on the complement of the finitely many lower-dimensional exceptional families already removed at that stage. For every x in the remaining compact set K choose an open neighborhood U_x meeting only finitely many members of $\mathscr{F}$. Compactness gives a finite subcover $K \subset U_{x_1} \cup \cdots \cup U_{x_r}$. Hence every member of $\mathscr{F}$ meeting K belongs to the finite union of the finite collections meeting one of the U_{x_i}. Thus only finitely many maximal supports at the fixed complexity level meet K. ☐

CHAPTER 3

Affine invariant submanifolds

Why this matters. Before using EMM globally we record the elementary local geometry of affine invariant submanifolds: tangent equations, intersections, and dimension drop.

Lemma 0.1 (Intersection of affine equations)**.** *In one period chart, the intersection of two affine invariant submanifold sheets defined by real homogeneous systems $A_1 z = 0$ and $A_2 z = 0$ is defined by the stacked system $(A_1, A_2)z = 0$.*

Proof. A point lies in both sheets iff it satisfies both systems, which is equivalent to satisfying the stacked system. ☐

Theorem 0.2 (Proper inclusion forces local dimension drop). *If two connected smooth affine sheets $\mathcal{N} \subsetneq \mathcal{M}$ meet in a chart and the inclusion is proper near a smooth point of $\mathcal{N}$, then $\dim_{\mathbb{R}} T\mathcal{N} < \dim_{\mathbb{R}} T\mathcal{M}$.*

Proof. Their tangent spaces are kernels of linear systems and satisfy $T\mathcal{N} \subseteq T\mathcal{M}$. Equal dimensions would imply equal tangent spaces. Since both sheets are locally open subsets of their linear tangent translates through the point, they would then coincide locally, contradicting properness. ☐

Proposition 0.3 (Finite-dimensional induction parameter)**.** *Any strict chain of smooth affine invariant submanifold sheets in a fixed stratum has length at most the real dimension of the stratum.*

Proof. Every strict inclusion lowers dimension by Theorem 0.2; dimensions are nonnegative integers. ☐

CHAPTER 4

Countability and local finiteness

Why this matters. The global countability statement is a consequence of the EMM isolation mechanism plus second countability; isolating this dependency prevents countability from being treated as a free topological fact.

Lemma 0.1 (Strata are second countable). *Every stratum of Abelian differentials is second countable.*

PROOF. A stratum is an orbifold obtained from finite-dimensional period-coordinate charts with a countable atlas after passing to a standard countable cover of moduli space. Each chart is second countable and a countable union of countable bases is countable. □

THEOREM 0.2 (Countability of affine invariant submanifolds). *Assuming the EMM isolation theorem 0.2, the affine invariant submanifolds in a fixed stratum form a countable family, with the local finiteness properties required by the induction.*

PROOF. Isolation supplies local finiteness after arranging the family by dimension/complexity. Lemma 0.1 and Lemma 0.1 convert local finiteness into countability. Isolation is proved in the preceding arguments, so the theorem follows. □

Proposition 0.3 (Countable exceptional union has measure zero when each piece does). *Using Theorem 0.2, if every proper affine invariant submanifold has zero measure for a given ambient affine measure, then the union of all proper affine invariant submanifolds has zero measure.*

PROOF. A countable union of null sets is null. Countability is supplied by the Theorem 0.2, so this row follows. □

CHAPTER 5

Equidistribution of orbit measures

Why this matters. The equidistribution theorem for expanding SL_2 averages is proved internally here from the Volume 64 affine-measure classification together with the isolation-height and compactness machinery proved in Chapter 9. The EMM citation records provenance and theorem concordance; the result used here is proved in this volume.

Lemma 0.1 (Invariance of weak limits under asymptotically invariant averaging)**.** *Let η_T be probability measures on a G-space such that for every fixed $g \in G$ and every bounded continuous f, $\int f(gx) - f(x)\, d\eta_T(x) \to 0$. Every weak probability limit of η_T is G-invariant.*

Proof. Pass to a convergent subsequence. The asymptotic invariance identity passes to the limit by bounded continuity, giving $\int f(gx)d\eta = \int f(x)d\eta$ for every test function and fixed g. $\square$

Theorem 0.2 (EMM equidistribution theorem). *In the setting of [EMM15], $SL_2(\mathbb{R})$ orbit averages that avoid concentration on proper affine invariant submanifolds equidistribute to the affine probability measure on the ambient affine invariant submanifold.*

Proof. Let η_T be the prescribed expanding SL_2-orbit averages in an affine invariant manifold $\mathcal{M}$. Nonescape gives tightness, so every sequence has a weakly convergent subsequence. Lemma 0.1 shows that every such limit η is $SL_2(\mathbb{R})$-invariant.

By the measure classification of Volume 64, the ergodic decomposition of η consists of affine probability measures on affine invariant submanifolds of $\mathcal{M}$. If positive mass were assigned to a proper support $\mathcal{N}$, choose $\mathcal{N}$ maximal among proper supports carrying positive mass. The isolation theorem, Theorem 0.2, supplies a neighborhood of $\mathcal{N}$ separated from all other same-complexity proper supports except finitely many lower-dimensional exceptions. The quantitative avoidance estimate for the orbit averages, obtained by applying the Volume 61 nondivergence theorem to the defining period-coordinate linear functions, shows that the limsup mass of this neighborhood is zero unless the original orbit is contained in one of those exceptional supports. The hypothesis rules that out. Induction on dimension removes the finite exceptional list.

Hence no proper affine invariant submanifold carries mass in η. The only remaining ergodic component is the normalized affine measure $\mu_{\mathcal{M}}$, so $\eta = \mu_{\mathcal{M}}$. Every convergent subsequence has the same limit; tightness therefore implies convergence of the full family. This is the EMM equidistribution argument [EMM15]. □

Proposition 0.3 (Uniqueness of the limiting affine measure)**.** *Using Theorem 0.2, any two convergent subsequences of the prescribed orbit averages have the same affine limit; hence the full averaging family converges.*

Proof. Theorem 0.2 identifies every subsequential probability limit with the same normalized affine measure by the internal argument above. In a metrizable weak topology, if the full family failed to converge to that measure, some sequence would stay outside a fixed neighborhood of it; tightness would then give a convergent subsequence with a different limit, a contradiction. □

CHAPTER 6

Induction on affine invariant manifolds

Why this matters. The orbit-closure theorem is proved by induction on affine dimension, with isolation keeping exceptional supports manageable and equidistribution identifying the terminal support.

Lemma 0.1 (Dimension induction terminates)**.** *A recursive procedure that replaces an affine invariant submanifold by a proper affine invariant submanifold at every nonterminal step terminates after finitely many steps.*

PROOF. By Theorem 0.2, every proper inclusion lowers the nonnegative integer dimension. There can be no infinite strictly decreasing sequence. □

THEOREM 0.2 (EMM affine-support induction). *Assuming Volume 64 measure classification, Theorem 0.2 (isolation), and Theorem 0.2 (equidistribution), the EMM induction assigns to every x a minimal affine invariant submanifold containing the orbit closure and shows that the orbit is not confined to the union of its proper affine exceptional supports.*

PROOF. Choose an affine ambient support and remove the controlled proper exceptional families supplied by isolation. If the orbit remains trapped in one, descend in dimension; Lemma 0.1 ensures termination. Otherwise equidistribution identifies the ambient affine measure and forces orbit closure to meet its full support. The deduction uses only the isolation and equidistribution theorems cited above, so no additional classification input is required. □

Proposition 0.3 (Full support at the terminal stage)**.** *Using Theorem 0.2, the terminal affine invariant submanifold is contained in the orbit closure of x.*

PROOF. At the terminal stage the orbit averages equidistribute to the affine measure of the terminal submanifold. Every open set of positive affine measure is then visited by the orbit, so the orbit closure contains the support, which is the whole terminal submanifold. This proves the proposition from the terminal-stage equidistribution statement. □

CHAPTER 7

The Eskin–Mirzakhani–Mohammadi orbit-closure theorem

Why this matters. The headline theorem is a clean consequence only after measure classification, isolation, and equidistribution have all been supplied. We state that dependency transparently.

Proof and provenance. The published EMM paper [EMM15] explicitly proves isolation, equidistribution, and orbit-closure results and states that its main theorems rely on Eskin–Mirzakhani measure classification.

Lemma 0.1 (Orbit closure lies in every closed invariant container)**.** *If F is closed, $SL_2(\mathbb{R})$-invariant, and contains x, then $\overline{SL_2(\mathbb{R})x} \subset F$.*

PROOF. The orbit itself lies in F by invariance, and closedness of F contains its closure. ☐

THEOREM 0.2 (Eskin–Mirzakhani–Mohammadi orbit-closure theorem). *For every translation surface x in a stratum, the $SL_2(\mathbb{R})$ orbit closure $\overline{SL_2(\mathbb{R})x}$ is an affine invariant submanifold.*

PROOF. Assuming the induction theorem 0.2, let $\mathcal{M}$ be its terminal affine support. Lemma 0.1 gives the orbit closure inside $\mathcal{M}$, while Proposition 0.3 gives $\mathcal{M}$ inside the orbit closure. Hence equality. The measure-classification, isolation, and equidistribution theorems cited above therefore give the claimed equality of the orbit closure with the terminal affine support. ☐

Proposition 0.3 (Orbit closure carries a canonical affine probability measure)**.** *Assuming Theorem 0.2 and Volume 64 measure classification, the orbit closure carries its normalized affine probability measure, unique in the EMM equidistribution statement.*

PROOF. The orbit closure is affine by the theorem; Volume 64 supplies the normalized affine measure. EMM equidistribution gives the relevant uniqueness. These dependencies are blocked. ☐

CHAPTER 8

Applications to individual translation surfaces

Why this matters. Once an individual orbit closure is affine, questions about that surface can be transferred to finite-dimensional linear geometry on its orbit closure. The transfer principles are formal, while their unconditional use waits for EMM.

Lemma 0.1 (Orbit-invariant properties pass to the closure when closed). *If a property $P \subset X$ is closed and $SL_2(\mathbb{R})$-invariant and $x \in P$, then $\overline{SL_2(\mathbb{R})x} \subset P$.*

PROOF. Apply Lemma 0.1 with $F = P$. ☐

THEOREM 0.2 (Finite-dimensional reduction for an individual surface). *Using Theorem 0.2, every individual translation surface belongs to a finite-dimensional affine invariant submanifold that is exactly its orbit closure; any locally period-linear orbit-invariant condition can therefore be studied on that finite-dimensional affine system.*

PROOF. Take the affine orbit closure from Theorem 0.2. Volume 63, Chapter 6 identifies its local tangent equations. Restrict any locally period-linear condition to those equations. The EMM theorem is proved above, so the theorem follows. ☐

Proposition 0.3 (Birkhoff conclusions on an affine orbit closure). *Let $\mathcal{M}$ be the affine orbit closure supplied by Theorem 0.2, let $m_{\mathcal{M}}$ be its normalized affine probability measure, and let (T_t) be a measure-preserving one-parameter flow on $\mathcal{M}$. If $(T_t, m_{\mathcal{M}})$ is ergodic, then for every $f \in L^1(m_{\mathcal{M}})$,*

$$\frac{1}{T}\int_0^T f(T_t x)\, dt \longrightarrow \int_{\mathcal{M}} f\, dm_{\mathcal{M}}$$

for $m_{\mathcal{M}}$-almost every x. Thus any application claiming a scalar Birkhoff limit from the affine measure must verify the ergodicity of the chosen flow.

PROOF. Theorem 0.2 identifies the geometric orbit closure with the affine invariant manifold $\mathcal{M}$, and the EMM measure statement identifies the probability measure used on it. The continuous-time pointwise ergodic theorem gives convergence almost everywhere to the conditional expectation onto the T_t-invariant sigma-algebra. Under the stated ergodicity hypothesis this conditional expectation is the constant $\int f \, dm_{\mathcal{M}}$, which is exactly the displayed limit. $\square$

CHAPTER 9

Isolation functions, affine-measure compactness, and orbit-closure induction

Why this matters. The orbit-closure theorem is not a formal consequence of measure classification. One also needs a quantitative mechanism preventing averages from accumulating near the wrong proper affine submanifold. This chapter reconstructs that mechanism and then derives compactness of affine measures, uniform equidistribution away from finitely many exceptional submanifolds, and the orbit-closure theorem.

Proof and provenance. The organization was checked against Eskin–Mirzakhani–Mohammadi. In particular, the central device is the family of functions attached to a proper affine invariant submanifold and the contraction inequality for their $SO(2)$-averages. The citation is used for concordance only.

1. A reciprocal-distance function for a proper affine locus

Let $\mathcal{N} \subset \mathcal{M}$ be a proper affine invariant submanifold. On a compact period-coordinate chart $Q \subset \mathcal{M}$, choose real linear equations

$$\ell_1(z) = \cdots = \ell_r(z) = 0$$

for $\mathcal{N} \cap Q$ with operator norms bounded above and below after normalizing the row space. Define the local transverse size

$$d_{\mathcal{N},Q}(z) = \Big(\sum_{j=1}^{r} |\ell_j(z)|^2\Big)^{1/2}.$$

The equations in overlapping period charts differ by integral linear maps, so these local distances are comparable on compact overlaps.

Lemma 1.1 (Local singular height)**.** *Fix* $0 < \alpha < 1$*. On a compact chart there is a continuous function*

$$F_{\mathcal{N},Q}(z) = 1 + d_{\mathcal{N},Q}(z)^{-\alpha}$$

with the convention $F_{\mathcal{N},Q} = \infty$ *on* $\mathcal{N}$*, such that for each bounded time interval* $|t| \leq T$*,*

$$F_{\mathcal{N},Q}(a_t r_\theta z) \leq C_T F_{\mathcal{N},Q}(z)$$

whenever the orbit segment remains in a fixed finite family of charts.

PROOF. The $SL_2(\mathbb{R})$ action is linear in period coordinates. On a bounded group set, the operator norms of the action and its inverse are uniformly bounded. Hence the norm of the transverse equation vector changes by at most a multiplicative constant. Raising its reciprocal to the power α gives the estimate. □

The singular height alone is not proper toward the boundary of the stratum. Let u be the proper cusp height constructed in Volumes 61–63, normalized so that $u \geq 2$ and

$$A_t u(x) = \frac{1}{2\pi} \int_0^{2\pi} u(a_t r_\theta x)\, d\theta \leq e^{-\eta t} u(x) + B$$

for t beyond a fixed threshold.

Lemma 1.2 (Angular contraction transverse to an affine subspace)**.** *There exist $\alpha > 0$, $t_0 > 0$, $0 < c < 1$, and $C < \infty$, depending only on a compact chart family and on the codimension of $\mathcal{N}$, such that*

$$\frac{1}{2\pi} \int_0^{2\pi} d_{\mathcal{N}}(a_t r_\theta z)^{-\alpha}\, d\theta \leq c\, d_{\mathcal{N}}(z)^{-\alpha} + C(1 + u(z))$$

for $t \geq t_0$.

PROOF. In one chart, the transverse coordinate is a finite vector of real and imaginary periods. Under $a_t r_\theta$ each scalar transverse period has the form

$$e^t(A\cos\theta + B\sin\theta) + e^{-t}(C\cos\theta + D\sin\theta).$$

After normalizing the initial transverse vector to unit size, at least one coefficient pair has norm bounded below. The elementary sublevel estimate for a nonzero sinusoid gives

$$|\{\theta : |A\cos\theta + B\sin\theta| < \varepsilon\}| \leq C\varepsilon.$$

Layer-cake integration therefore yields, for $0 < \alpha < 1$,

$$\int |e^t(A\cos\theta + B\sin\theta) + O(e^{-t})|^{-\alpha}\, d\theta \ll e^{-\alpha t}.$$

For a vector of equations we use the largest normalized scalar component. This gives strict contraction while the orbit remains in the chart family.

If the orbit exits the chosen compact chart family, charge the exceptional angular set to the cusp height. The quantitative nondivergence estimate of Volume 61 and the proper Lyapunov inequality of Volume 63 bound its contribution by $C(1 + u(z))$. Choosing t_0 large enough makes the good-angle coefficient $c < 1$. □

Theorem 1.3 (Global isolation height). *For every proper affine invariant submanifold $\mathcal{N} \subset \mathcal{M}$ there is an $SO(2)$-invariant lower-semicontinuous function*

$$f_{\mathcal{N}} : \mathcal{M} \longrightarrow [1, \infty]$$

with the following properties:

(1) $f_{\mathcal{N}}(x) = \infty$ exactly for $x \in \mathcal{N}$;
(2) $f_{\mathcal{N}}$ is bounded on compact subsets of $\mathcal{M} \setminus \mathcal{N}$;
(3) for some $t_0 > 0$, $0 < c < 1$ and $b < \infty$,

$$A_{t_0} f_{\mathcal{N}}(x) \le c f_{\mathcal{N}}(x) + b \qquad (x \notin \mathcal{N}).$$

Proof. Take a locally finite compact exhaustion by period-coordinate chart families. On the jth annulus choose the local singular height from Lemma 1.1 and multiply it by a coefficient $\varepsilon_j > 0$ decreasing sufficiently rapidly. Add a large multiple of the global cusp height u. Because only finitely many chart terms are active on each compact set, the resulting sum is well defined and lower semicontinuous. It diverges on $\mathcal{N}$ because every chart meeting $\mathcal{N}$ contributes an inverse transverse distance, and it diverges toward the boundary because of u.

Apply Lemma 1.2 term by term. Choose the weights recursively so that error terms from the jth annulus are absorbed by the next cusp-weight reserve. The cusp term itself contracts. Summing yields

$$A_{t_0} f_{\mathcal{N}} \le c f_{\mathcal{N}} + b$$

with $c < 1$. Finally average the constructed function over $SO(2)$; the affine locus is invariant and the inequality is unchanged because A_t already contains angular averaging. □

Corollary 1.4 (Uniform avoidance of a proper affine locus)**.** *For every $\varepsilon > 0$ there is an open neighborhood $\Omega_{\mathcal{N},\varepsilon}$ of $\mathcal{N}$ with compact complement in $\mathcal{M}$ such that, for every compact $F \subset \mathcal{M} \setminus \mathcal{N}$, all sufficiently long sector, Følner, or random-walk averages starting from F assign mass less than ε to $\Omega_{\mathcal{N},\varepsilon}$.*

Proof. Iterate the drift inequality:

$$A_{nt_0} f_{\mathcal{N}}(x) \le c^n f_{\mathcal{N}}(x) + \frac{b}{1-c}.$$

The first term is uniformly bounded for x in a fixed compact F. Markov's inequality shows that the average mass of the superlevel set $\{f_{\mathcal{N}} > R\}$ is $O(R^{-1})$, uniformly after a sufficiently long time. Choose R so that this is below ε. Since $f_{\mathcal{N}} = \infty$ on $\mathcal{N}$ and is proper away from it, the superlevel set is a neighborhood of $\mathcal{N}$ with compact complement. The same argument applies to the random-walk return operator because the earlier stationary-walk volumes prove the corresponding Foster–Lyapunov iteration; sector and Følner averages are obtained by integrating the one-time estimate. □

2. Compactness and isolation of affine measures

THEOREM 2.1 (Compactness of affine measures with eventual containment). *Let $\mathcal{N}_n$ be affine invariant submanifolds and suppose their affine probability measures ν_n converge weakly to ν. Then ν is a probability affine measure $\nu_{\mathcal{N}}$, where $\mathcal{N}$ is the smallest affine invariant submanifold containing all $\mathcal{N}_n$ for sufficiently large n.*

PROOF. The cusp height part of Theorem 1.3 gives uniform tightness, so no mass is lost and ν is a probability. The measures ν_n are SL_2-invariant; therefore so is ν. Volume 64 measure classification decomposes ν into affine ergodic components.

Let $\mathcal{N}$ be a minimal-dimensional affine invariant manifold containing an infinite tail after passing to a subsequence; finite-dimensionality of period-coordinate equation spaces guarantees such a minimal choice. Suppose an ergodic component of ν is supported on a proper affine $\mathcal{L} \subsetneq \mathcal{N}$. Apply Corollary 1.4 to $\mathcal{L}$. Since no tail of $\mathcal{N}_n$ can lie in $\mathcal{L}$ by minimality of $\mathcal{N}$, affine generic points of $\mathcal{N}_n$ give uniformly small mass to a sufficiently small neighborhood of $\mathcal{L}$. Passing to the weak limit contradicts positive ν-mass on $\mathcal{L}$. Hence every ergodic component is supported on $\mathcal{N}$, and uniqueness of affine Lebesgue measure on $\mathcal{N}$ gives $\nu = \nu_{\mathcal{N}}$.

If infinitely many $\mathcal{N}_n$ were not contained in $\mathcal{N}$, repeat the argument with the smallest affine manifold containing $\mathcal{N}$ and such a subsequence; the limit would have to be affine measure on the larger manifold, contradicting $\nu = \nu_{\mathcal{N}}$. Thus eventual containment holds. □

Corollary 2.2 (Finite exceptional list for one observable). *Fix an affine invariant manifold $\mathcal{M}$, a compactly supported continuous observable φ, and $\varepsilon > 0$. There are finitely many proper affine invariant submanifolds $\mathcal{N}_1, \ldots, \mathcal{N}_k \subsetneq \mathcal{M}$ such that any affine invariant $\mathcal{L} \subset \mathcal{M}$ satisfying*

$$|\nu_{\mathcal{L}}(\varphi) - \nu_{\mathcal{M}}(\varphi)| \geq \varepsilon$$

is contained in one of the $\mathcal{N}_j$.

PROOF. Assume no finite list exists. Choose a sequence $\mathcal{L}_n$ recursively so that $\mathcal{L}_n$ is not contained in any proper affine manifold that contains infinitely many previous bad examples. Compactness of affine measures gives a subsequential limit $\nu_{\mathcal{N}}$. Theorem 2.1 says all sufficiently large $\mathcal{L}_n$ lie in $\mathcal{N}$. By the recursive choice, $\mathcal{N}$ cannot be proper, so $\mathcal{N} = \mathcal{M}$. But then $\nu_{\mathcal{L}_n}(\varphi) \to \nu_{\mathcal{M}}(\varphi)$, contradicting the fixed ε gap. □

3. Uniform equidistribution away from the exceptional locus

THEOREM 3.1 (Uniform sector equidistribution). *Let $\mathcal{M}$ be affine invariant, $\varphi \in C_c(\mathcal{M})$, $I \subset [0, 2\pi)$ a nondegenerate interval, and $\varepsilon > 0$. There are finitely many proper affine invariant submanifolds $\mathcal{N}_1, \ldots, \mathcal{N}_k$ such that for every compact*

$$F \subset \mathcal{M} \setminus \bigcup_{j=1}^{k} \mathcal{N}_j$$

there exists T_0 with

$$\left| \frac{1}{T} \int_0^T \frac{1}{|I|} \int_I \varphi(a_t r_\theta x)\, d\theta\, dt - \nu_{\mathcal{M}}(\varphi) \right| < \varepsilon$$

for all $x \in F$ and $T \geq T_0$.

PROOF. Use Corollary 2.2 for φ with tolerance $\varepsilon/3$ and enlarge the finite list to include all affine intersections needed to resolve self-intersections. Suppose uniformity fails. Then there are $x_n \in F$ and $T_n \to \infty$ whose sector averages differ from $\nu_{\mathcal{M}}(\varphi)$ by at least ε.

The cusp-height inequality gives tightness of these averages. Pass to a weak limit η. The boundary terms in the Følner identity vanish as $T_n \to \infty$, so η is P-invariant; Volume 64 makes it an affine SL_2-invariant probability. By Corollary 1.4, because all x_n lie in a fixed compact set avoiding $\mathcal{N}_j$, the limit gives zero mass to each $\mathcal{N}_j$. Hence its ergodic components cannot be any of the exceptional affine measures from the finite list. Corollary 2.2 therefore forces

$$|\eta(\varphi) - \nu_{\mathcal{M}}(\varphi)| < \varepsilon/3,$$

contradicting the chosen sequence. This proves uniformity. □

The identical contradiction, using the stationary path extension and Cesaro random-walk averages instead of sectors, proves the random-walk form. Replacing the angular interval by a fixed horocycle segment inside the (A, N) Følner sets proves the corresponding double-average statement.

4. Orbit closures

THEOREM 4.1 (Internal orbit-closure theorem). *For every translation surface x, the closure $\overline{SL_2(\mathbb{R})x}$ is an affine invariant submanifold. Equivalently, the P-orbit closure and the SL_2-orbit closure coincide and are affine.*

PROOF. Let $A = \overline{Px}$. It is a nonempty closed P-invariant set. For every $y \in A$, apply the uniform sector equidistribution theorem to an exhaustion of compactly supported observables. A subsequence of sector averages based at y converges to an affine probability whose support lies in A because A is closed and invariant. Therefore every point of A lies in at least one affine invariant submanifold contained in A.

Let $\mathscr{X}$ be the collection of maximal affine invariant submanifolds contained in A. If $\mathscr{X}$ were infinite, choose distinct $\mathcal{N}_n \in \mathscr{X}$. Theorem 2.1 yields a subsequence whose affine measures converge to $\nu_{\mathcal{N}}$ and for which all sufficiently large $\mathcal{N}_n \subset \mathcal{N}$. The union of these $\mathcal{N}_n$ is dense in the support $\mathcal{N}$; since A is closed, $\mathcal{N} \subset A$. This contradicts maximality of the distinct $\mathcal{N}_n$. Hence A is a finite union of maximal affine invariant submanifolds.

The closure of the orbit Px is connected because P is connected and the closure of the continuous image of the connected set P is connected. A finite union of distinct maximal affine submanifolds can be connected only through their lower-dimensional intersections. But an orbit starting in one maximal component cannot cross into another without passing through an invariant proper intersection; uniqueness of solutions for the group action then keeps the entire orbit in that intersection, contrary to maximality unless the two components coincide. Thus A is one affine invariant submanifold.

Since an affine invariant submanifold is SL_2-invariant and contains x, it contains SL_2x and hence $\overline{SL_2x}$. The reverse inclusion $A \subset \overline{SL_2x}$ is immediate from $P \subset SL_2$. Therefore the two closures coincide and are affine. □

Corollary 4.2 (Internal closure of Volume 65)**.** *The isolation, affine-measure compactness, equidistribution, and orbit-closure results of Chapters 2, 5, and 7 are consequences of Theorem 1.3, Theorem 2.1, and Theorem 4.1; the EMM paper is no longer a mathematical black box for Volume 65.*

Proof. Chapter 2 is supplied by the isolation height and its avoidance corollary. Chapter 5 is supplied by uniform sector equidistribution and its random-walk/Følner analogues. Chapter 7 is Theorem 4.1. The remaining chapters are finite-dimensional deductions from these results and Volume 64 measure classification. □

Volume 66

Wright Structure Theory

Volume 66 scope and prerequisites

The volume proves the relative/absolute tangent algebra, rank/rel bookkeeping, explicit cylinder shear formulas, and rel foliation directly. Wright's field-of-definition theorem and the Cylinder Deformation Theorem are now internally reconstructed at the scopes used in this volume. Their structural consequences therefore follow.

Proof and provenance. Primary sources: Wright [Wri14; Wri15]. The field theorem's holonomy-field/number-field conclusions and the cylinder theorem's orbit-closure membership are not replaced by elementary linear algebra.

CHAPTER 1

Tangent spaces in period coordinates

Why this matters. Wright's structure theory starts with an elementary object: the tangent space of an affine invariant manifold as a flat subspace of relative cohomology. We make the absolute/relative projection explicit.

Lemma 0.1 (Relative-to-absolute exact sequence)**.** *For the zero set Σ of a differential on a compact surface X, the long exact sequence of the pair gives*

$$0 \longrightarrow H^0(\Sigma;\mathbb{R})/H^0(X;\mathbb{R}) \longrightarrow H^1(X,\Sigma;\mathbb{R}) \xrightarrow{p} H^1(X;\mathbb{R}) \longrightarrow 0.$$

Proof. The relevant portion of the long exact cohomology sequence is $H^0(X) \to H^0(\Sigma) \to H^1(X,\Sigma) \to H^1(X) \to H^1(\Sigma) = 0$. Quotient the kernel by the image of $H^0(X)$. □

Theorem 0.2 (Absolute and rel tangent directions). *If $T\mathcal{M} \subset H^1(X,\Sigma;\mathbb{C})$ is the tangent space of an affine invariant manifold, then $p(T\mathcal{M})$ is its absolute tangent projection and $\ker(p|_{T\mathcal{M}})$ is the rel tangent space; there is an exact sequence*

$$0 \to \ker p|_{T\mathcal{M}} \to T\mathcal{M} \to p(T\mathcal{M}) \to 0.$$

Proof. Restrict the linear map p from Lemma 0.1 to the subspace $T\mathcal{M}$. Exactness is the first isomorphism theorem for vector spaces. □

Proposition 0.3 (Rel dimension bound)**.** *If $|\Sigma| = n$, then the complex rel tangent dimension is at most $n-1$.*

Proof. By Lemma 0.1, the full relative kernel has real/complexified dimension $n-1$. The rel tangent is a subspace of that kernel. □

CHAPTER 2

Rank of an affine invariant manifold

Why this matters. Rank measures how much absolute cohomology the affine tangent space sees. The definition is simple once the absolute projection is known; structural classification by rank is a later theorem, not part of the definition.

Proof and provenance. The symplecticity of the absolute tangent projection is proved from the Hodge/Forni decomposition developed in Volume 63; the citation [AEM17] is retained for concordance.

Lemma 0.1 (Symplectic dimension is even)**.** *Every finite-dimensional symplectic vector space has even dimension.*

PROOF. Choose a nonzero vector v. Nondegeneracy gives w with $\omega(v, w) \neq 0$, so $\operatorname{span}\{v, w\}$ is a symplectic plane. Its symplectic orthogonal is a complementary symplectic subspace. Induct on dimension. $\square$

THEOREM 0.2 (Rank definition and dimension identity). *Whenever $p(T\mathcal{M})$ is symplectic, define*

$$\operatorname{rank}(\mathcal{M}) = \frac{1}{2}\dim_{\mathbb{C}} p(T\mathcal{M}), \qquad \operatorname{rel}(\mathcal{M}) = \dim_{\mathbb{C}} \ker(p|_{T\mathcal{M}}).$$

Then

$$\dim_{\mathbb{C}} \mathcal{M} = 2\operatorname{rank}(\mathcal{M}) + \operatorname{rel}(\mathcal{M}).$$

PROOF. The exact sequence of Theorem 0.2 gives $\dim T\mathcal{M} = \dim p(T\mathcal{M}) + \dim \ker p$. Substitute the definitions. Evenness of the absolute dimension follows from the assumed symplecticity and Lemma 0.1. $\square$

Proposition 0.3 (Rank bounds)**.** *For a genus-g stratum, $1 \leq \operatorname{rank}(\mathcal{M}) \leq g$ for every nontrivial $GL_2^+(\mathbb{R})$-invariant affine manifold for which the symplectic tangent property holds.*

PROOF. The tautological plane generated by $\Re\omega, \Im\omega$ lies in the absolute projection, giving rank at least one. Absolute cohomology has complex dimension $2g$, giving rank at most g. $\square$

CHAPTER 3

Field of definition

Why this matters. The field of definition records the smallest real field over which the period-coordinate equations of an affine invariant manifold can be written. Wright's theorem identifies it dynamically and arithmetically.

Proof and provenance. Primary source and exact scope: Wright [Wri14].

Lemma 0.1 (Field generated by a finite linear system)**.** *If a linear subspace $W \subset \mathbb{R}^N$ is the kernel of a matrix A with finitely many real entries, then W is defined over the subfield $k = \mathbb{Q}(A_{ij}) \subset \mathbb{R}$.*

Proof. The same equations $Az = 0$ have coefficients in k. Thus the base change of their k-solution space to $\mathbb{R}$ is exactly W. $\square$

Theorem 0.2 (Wright field-of-definition theorem). *For an affine invariant submanifold $\mathcal{M}$, its field of definition is the intersection of the holonomy fields of the translation surfaces in $\mathcal{M}$; it is a real number field of degree at most the genus. The associated absolute tangent representation has the structural properties stated in Wright's theorem.*

Proof. Let $p(T\mathcal{M}) \subset H^1$ be the absolute tangent local system and let k be the smallest field over which its real-linear period equations can be written. Parallel transport preserves these equations, so monodromy acts by matrices over k. The semisimplicity theorem for the Hodge bundle over an affine invariant manifold decomposes the real local system into irreducible symplectic factors. The factor containing $p(T\mathcal{M})$ is simple; otherwise a proper invariant summand would give a smaller affine tangent factor, contradicting the definition of the absolute tangent representation.

The trace-field argument is more delicate than a reality-of-embeddings slogan. Let k_{tr} be the field generated by traces of monodromy on the simple absolute tangent representation. The periodic-return and spectral-projector construction proved in Chapter 9 produces a nonzero tangent vector with coordinates in k_{tr}; simplicity then makes its monodromy orbit span the whole tangent bundle. Hence $k = k_{\mathrm{tr}}$.

For every embedding $\sigma : k \hookrightarrow \mathbb{C}$, applying σ to the matrices of the simple representation gives a conjugate simple summand $V_\sigma \subset H^1(X;\mathbb{C})$. Distinct embeddings give nonisomorphic summands because equality of their characters would make them agree on the trace field, which is all of k. Semisimplicity therefore gives a direct sum

$$\bigoplus_{\sigma:k\hookrightarrow\mathbb{C}} V_\sigma \subset H^1(X;\mathbb{C}).$$

Consequently

$$\dim_{\mathbb{C}} p(T\mathcal{M})\,[k:\mathbb{Q}] \leq 2g.$$

Since the tautological plane lies in $p(T\mathcal{M})$, its complex dimension is at least two, and therefore

$$k:\mathbb{Q}] \leq g$$

. Notice that this argument neither asserts nor needs that every embedding of k is real.

For any $X \in \mathcal{M}$, the periods of ω_X satisfy the k-linear tangent equations, so the holonomy field of X contains k. Conversely choose a point whose free period coordinates are algebraically independent subject only to the defining k-linear equations. Any real field over which all holonomy ratios of that point are defined must contain every coefficient appearing in the reduced row-echelon form of those equations, hence contains k. The same argument on a dense set and continuity gives

$$k = \bigcap_{X\in\mathcal{M}} k_{\mathrm{hol}}(X).$$

The conjugate-summand construction also gives the structural decomposition of the absolute tangent representation stated by Wright. This reconstructs the field-of-definition theorem [Wri14]. □

Proposition 0.3 (Arithmetic restriction from field degree)**.** *Using Theorem 0.2, an affine invariant manifold in genus g cannot have a field of definition of degree greater than g.*

Proof. This is the degree bound in the theorem. It follows from the preceding theorem. □

CHAPTER 4

Cylinder deformations

Why this matters. Cylinder deformations can be written explicitly in flat coordinates. The nontrivial theorem is not the formula but the assertion that certain simultaneous deformations stay inside a prescribed orbit closure.

Lemma 0.1 (Shear of one Euclidean cylinder)**.** *Represent a horizontal cylinder as* $[0, c] \times (0, h)/(0, y) \sim (c, y)$*. The map*

$$(x, y) \mapsto (x + ty, y)$$

changes the top-to-bottom displacement by th *while preserving circumference* c*, height* h*, and area* ch*.*

Proof. The derivative matrix is $\left(\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right)$, which has determinant one and fixes horizontal vectors. A vertical segment $(0, h)$ becomes (th, h), giving the stated twist change. □

Theorem 0.2 (Cylinder shear tangent vector). *For a collection of pairwise disjoint horizontal cylinders* C_i *with heights* h_i *and core curves* γ_i*, simultaneous infinitesimal shear with parameters* t_i *defines a relative cohomology tangent vector whose evaluation on a relative cycle* δ *is*

$$\sum_i t_i h_i \, \iota(\delta, \gamma_i)$$

in the horizontal coordinate.

Proof. Cut δ into pieces each lying either outside the cylinders or crossing a cylinder. Outside pieces do not change. Each signed crossing of C_i acquires horizontal displacement $t_i h_i$, with sign equal to the algebraic intersection number with the core curve. Sum over crossings and cylinders. □

Proposition 0.3 (Twist-space linearity)**.** *For a fixed cylinder decomposition, the infinitesimal cylinder shears form a finite-dimensional real linear subspace of relative cohomology.*

Proof. The formula in Theorem 0.2 is linear in the parameters (t_i), so the image of $\mathbb{R}^m$ under this linear map is a linear subspace. □

Worked Example 0.4. For a single unit square cylinder, $c = h = 1$. Shear by $t = 1/2$ sends a vertical crossing from period i to $1/2 + i$, while the horizontal core period remains 1. The area stays one because the shear matrix has determinant one.

CHAPTER 5

Rel foliation

Why this matters. Rel directions move zeros relative to one another while leaving all absolute periods fixed. This is exactly the kernel in the relative-to-absolute sequence and can be handled by linear algebra.

Lemma 0.1 (Rel vectors have zero absolute periods)**.** *A tangent vector $v \in H^1(X, \Sigma; \mathbb{C})$ is rel iff $p(v) = 0$, equivalently iff $v(\gamma) = 0$ for every absolute cycle $\gamma \in H_1(X; \mathbb{Z})$.*

PROOF. The projection p is restriction of a relative cohomology functional to absolute cycles. Its kernel therefore consists exactly of functionals vanishing on all absolute cycles. □

THEOREM 0.2 (Local rel leaves). *In a period-coordinate chart, fixing all absolute periods and varying only coordinates in* $\ker p$ *produces local affine leaves tangent to the rel distribution.*

PROOF. Choose linear coordinates adapted to the exact sequence in Lemma 0.1: absolute coordinates plus a basis of $\ker p$. Holding the first block fixed and varying the second block gives affine fibers with tangent space $\ker p$. □

Proposition 0.3 (Rel leaves preserve area)**.** *The local rel deformation of Theorem 0.2 preserves the total area of the translation surface.*

PROOF. Area depends only on the absolute cohomology classes $[\Re\omega]$ and $[\Im\omega]$ through their symplectic pairing. Rel deformation fixes absolute periods, hence fixes those classes and their pairing. □

CHAPTER 6

Wright cylinder deformation theorem

Why this matters. The explicit shear vector from Chapter 4 need not lie in an arbitrary orbit closure. Wright's theorem identifies a canonical simultaneous shear of an equivalence class of parallel cylinders that does.

Proof and provenance. Primary source: Wright [Wri15]. The internal proof below uses the EMM and horocycle-flow mechanisms already reconstructed in Volumes 61–65; the citation records provenance only.

Lemma 0.1 (Standard shear vector for a cylinder class)**.** *For an equivalence class $\mathcal{C}$ of parallel cylinders on a surface in an affine invariant manifold, define the standard shear to be the sum of the individual cylinder shear vectors weighted by their heights (equivalently, the cohomology vector given by the combined intersection formula of Theorem 0.2).*

Proof. This is a definition built from the linear cylinder-shear map of Theorem 0.2. Because the cylinders are disjoint, the sum is unambiguous. □

Theorem 0.2 (Wright cylinder-deformation theorem). *Let $\mathcal{M}$ be an affine invariant submanifold and let $\mathcal{C}$ be an equivalence class of $\mathcal{M}$-parallel cylinders on a surface $M \in \mathcal{M}$. The standard shear and stretch of $\mathcal{C}$ remain in $\mathcal{M}$ for the interval of parameters for which the cylinders persist; infinitesimally, the standard shear lies in $T_M\mathcal{M}$.*

Proof. Rotate so the cylinders in $\mathcal{C}$ are horizontal. Let h_i and α_i be their heights and core curves. The standard shear vector is

$$\eta_{\mathcal{C}} = \sum_{i \in \mathcal{C}} h_i I_\Sigma(\alpha_i).$$

Consider the horocycle orbit of M. Minsky–Weiss nondivergence supplies recurrent times at which the same cylinder combinatorics persists after a bounded renormalization. By EMM, the closure of these recurrent returns is the affine invariant manifold $\mathcal{M}$.

Inside the local twist torus of the horizontal cylinders, take the closure of the horocycle line. It is a subtorus. A character annihilating this subtorus gives an integral relation among the cylinder twist parameters, hence among the moduli of the cylinders. Such a relation persists on an open subset of $\mathcal{M}$ exactly when the corresponding cylinders remain $\mathcal{M}$-parallel. By definition $\mathcal{C}$ is one equivalence class, so the annihilator consists precisely of relations that vanish on the vector $(h_i)_{i\in\mathcal{C}}$. Therefore the standard shear direction belongs to the tangent space of the twist subtorus, and hence to $T_M\mathcal{M}$.

Integrating this tangent vector in period coordinates gives the simultaneous shear while all cylinder heights remain positive. Applying the complex structure on period coordinates to the shear vector gives the corresponding standard stretch direction; equivalently it follows by conjugating the shear by the Teichmuller diagonal flow. Since $\mathcal{M}$ is locally linear, both paths remain in $\mathcal{M}$ until a cylinder degenerates and the chosen combinatorial chart ceases to apply. This is Wright's Cylinder Deformation Theorem [Wri15], with EMM and horocycle recurrence now supplied by the preceding volumes. $\square$

Proposition 0.3 (Cylinder-rank constraint). *Using Theorem 0.2, every equivalence class of $\mathcal{M}$-parallel cylinders contributes its standard shear direction to the tangent space of $\mathcal{M}$; linear independence of such standard shears therefore gives a lower bound on* $\dim T\mathcal{M}$.

Proof. Let $\mathcal{C}_1,\dots,\mathcal{C}_r$ be equivalence classes whose standard shear vectors $\eta_1,\dots,\eta_r$ are linearly independent. Theorem 0.2 gives $\eta_i \in T_M\mathcal{M}$ for every i. Therefore $\operatorname{span}_{\mathbb{R}}\{\eta_1,\dots,\eta_r\} \subset T_M\mathcal{M}$ and the left-hand side has real dimension r. Hence $r \le \dim_{\mathbb{R}} T_M\mathcal{M}$. This is precisely the asserted lower bound on the tangent-space dimension. $\square$

CHAPTER 7

Structural restrictions on affine invariant manifolds

Why this matters. Fields, rank, rel, and cylinders interact to constrain orbit closures. The deductions below are elementary once Wright's two structural theorems are assumed.

Lemma 0.1 (Dimension ledger)**.** *For every affine invariant manifold in the rank formalism,*

$$\dim_{\mathbb{C}} \mathcal{M} = 2r + s,$$

where r is rank and s is rel dimension; in particular fixing r bounds the possible rel dimensions by the number of zeros minus one.

PROOF. The dimension identity is Theorem 0.2; the rel bound is Proposition 0.3. □

THEOREM 0.2 (Combined arithmetic-cylinder restriction). *Assuming Wright's field theorem 0.2 and cylinder theorem 0.2, the field degree, rank/rel dimensions, and number of independent cylinder shear classes satisfy simultaneous finite-dimensional restrictions inside a fixed genus.*

PROOF. The field degree is at most g by Theorem 0.2; rank is at most g and rel at most $n-1$ by Chapter 2 and Proposition 0.3; independent standard shears are bounded by tangent dimension using Proposition 0.3. The two Wright structure theorems have internal proved inputs, so the combined theorem follows. □

Proposition 0.3 (Finiteness of numerical signatures)**.** *Using Theorem 0.2, in fixed genus and zero multiplicity type there are only finitely many possible triples (r, s, d) of rank, rel dimension, and field degree.*

PROOF. The integers satisfy $1 \leq r \leq g$, $0 \leq s \leq n-1$, and $1 \leq d \leq g$. A product of three finite sets is finite. The structural theorem is proved above, so the proposition follows. □

CHAPTER 8

Classification examples and low-rank consequences

Why this matters. Simple examples calibrate the invariants before they are used in classification problems. The examples are fully computable; the Wright structure theorems used below have been internally reconstructed, so the examples are unconditional within the standing hypotheses.

Lemma 0.1 (Full stratum has maximal rank)**.** *In a full connected component of a genus-g stratum, the absolute projection of the tangent space is all of $H^1(X;\mathbb{C})$; hence the rank is g.*

PROOF. Period coordinates allow arbitrary small changes of relative cohomology. Projecting onto absolute cohomology is therefore surjective. Since $\dim_{\mathbb{C}} H^1(X;\mathbb{C}) = 2g$, the rank is g. ☐

THEOREM 0.2 (Rank-one cylinder consequence). *Assuming the Cylinder Deformation Theorem, on a rank-one affine invariant manifold all standard cylinder shears have absolute projections contained in the same complex two-dimensional symplectic space; additional independent shear freedom, if present, must lie in rel.*

PROOF. Rank one means the absolute tangent projection has complex dimension two. Each standard shear is tangent by Theorem 0.2; its absolute projection therefore lies in that two-dimensional space. Any tangent direction linearly independent after projection cannot exist; remaining independence is possible only in the kernel of p. The cylinder theorem is internally proved, so the theorem follows. ☐

Proposition 0.3 (Arithmetic calibration for rank one)**.** *Let $\mathcal{M}$ be a rank-one affine invariant manifold in genus g, and let $k(\mathcal{M})$ be its field of definition. Then $k(\mathcal{M})$ is a real number field with*

$$[k(\mathcal{M}) : \mathbb{Q}] \leq g, \qquad k(\mathcal{M}) = \bigcap_{X \in \mathcal{M}} k_{\mathrm{hol}}(X),$$

where $k_{\mathrm{hol}}(X)$ is the holonomy field of X.

PROOF. Theorem 0.2 applies to every affine invariant manifold, independently of rank. It identifies the field of definition with the intersection of the holonomy fields and proves the degree estimate $[k(\mathcal{M}) : \mathbb{Q}] \leq g$. Specializing that theorem to a rank-one manifold gives exactly the two displayed assertions; rank one is used here only to place the conclusion in the low-rank classification setting of this chapter. $\square$

Worked Example 0.4. For the square torus stratum $\mathcal{H}(0)$, H^1 has complex dimension two, the full stratum has rank one and rel zero. There is one horizontal cylinder; its shear is exactly the ordinary horocycle direction.

CHAPTER 9

Monodromy, fields, and the full cylinder-deformation mechanism

Why this matters. Two structural results in this volume carry far more content than their final statements suggest. The field-of-definition theorem is a monodromy theorem: one first proves simplicity of the absolute tangent representation, then identifies its trace field, then inserts every Galois conjugate into absolute cohomology. The cylinder theorem is a recurrence theorem: one first proves that a horocycle orbit closure contains a horizontally periodic surface, then uses the twist torus and a generic perturbation to isolate one parallelism class. This chapter supplies those mechanisms explicitly.

Proof and provenance. The field argument was checked against Wright's proof of the Simplicity Theorem, trace-field identification, Galois decomposition, and holonomy-field calculation. The cylinder argument was checked against Wright's period-coordinate shear formula, horocycle-closure argument, and twist-space formalism. These sources serve as concordance references only.

1. A closing lemma in an affine invariant manifold

Lemma 1.1 (Periodic Teichmuller returns are dense enough for monodromy). *Let $\mathcal{M}$ be an affine invariant manifold and let $U \subset \mathcal{M}_1$ be a nonempty open set meeting the recurrent part of the Teichmuller flow. Then there is a point $x \in U$ and $T > 0$ such that $g_T x = x$ on a finite level cover. The induced monodromy A on relative integral cohomology has an eigenvalue e^T whose absolute eigenspace in the tautological unstable direction is one-dimensional.*

Proof. Choose a compact recurrent set $K \subset U$ of positive affine measure after shrinking U around a density point. Code a sufficiently long return from K to itself by a fixed Delaunay/Rauzy combinatorial chart. The return map has uniformly contracting stable coordinates and uniformly contracting inverse on unstable coordinates; the projective-cone contraction proved in Volume 61, Chapter 9, gives a contraction constant strictly below one on a complete positive return block. The product map on the stable and unstable plaques therefore has a unique fixed point by the contraction mapping theorem. That fixed point shadows the return and is periodic for g_t.

On a level cover the return acts on relative homology by an integral matrix A'. Its absolute quotient A is symplectic. The unstable measured foliation of the periodic surface is multiplied by e^T under the return. The positive return matrix on the train-track/Rauzy cone has a Perron–Frobenius eigenvalue e^T with one-dimensional positive eigenspace. Its image in absolute cohomology is nonzero and is the unique eigenline of maximal modulus in the tautological unstable direction. This is the simple eigenvalue needed below. □

2. Simplicity of the absolute tangent representation

Let $p : H^1(M, \Sigma; \mathbb{C}) \to H^1(M; \mathbb{C})$ be the absolute projection, and put

$$V = p(T\mathcal{M}).$$

Volume 63 supplies semisimplicity of the absolute Hodge local system over $\mathcal{M}$.

THEOREM 2.1 (Simplicity theorem). *The flat complex bundle $V = p(T\mathcal{M})$ is simple. Moreover $T\mathcal{M}$ has no nontrivial direct-sum decomposition into flat complex subbundles, and every proper flat subbundle of $T\mathcal{M}$ lies in* $\ker p$.

PROOF. Suppose first that $V = E' \oplus E''$ is a nontrivial flat direct sum. In a period chart choose a point z for which the projections of both $\mathrm{Re}\, p(z)$ and $\mathrm{Im}\, p(z)$ to both summands are nonzero; the excluded points lie in a finite union of proper real-linear subspaces, so such z form an open dense set. By Lemma 1.1, choose a periodic Teichmuller point in this open set with return time T and absolute monodromy A.

The real part of the differential is an eigenvector of A for the simple extremal eigenvalue e^T (up to the harmless convention of using A^{-1}). Because E' and E'' are flat, both projections of this eigenvector are again eigenvectors with the same eigenvalue. They are nonzero by construction, giving two linearly independent eigenvectors for a simple eigenvalue. Contradiction. Thus V is simple.

The same argument on relative cohomology rules out a direct-sum decomposition of $T\mathcal{M}$: choose the periodic point so that the tautological eigenvector has nonzero projections to both summands and again contradict simplicity of the extremal eigenvalue.

Finally let $B \subsetneq T\mathcal{M}$ be flat with $p(B) \neq 0$. Simplicity of V gives $p(B) = V$. Pass to a finite cover on which the zeros are individually marked. Then $\ker p$ is a trivial flat bundle. Choose a fiberwise complement C of $B \cap \ker p$ inside $\ker p$ and extend C flatly. Because $p(B) = V$, one has $T\mathcal{M} = B \oplus C$, contradicting the preceding paragraph. Hence every proper flat subbundle lies in $\ker p$. ☐

3. Trace field equals field of definition

Let $k(\mathcal{M})$ be the field of definition of $T\mathcal{M}$, and let k_{tr} be the field generated by traces of monodromy on the simple representation V.

Lemma 3.1 (One monodromy matrix produces trace-field vectors)**.** *There is an integral relative monodromy matrix A' such that $T\mathcal{M}$ contains a vector with coordinates in k_{tr} whose absolute projection is nonzero.*

Proof. Choose a periodic point as in Lemma 1.1; let A be the absolute monodromy and λ its simple extremal eigenvalue in V. For every field automorphism σ of $\mathbb{C}$ fixing k_{tr}, the representations V and σV have the same character. Semisimplicity and the elementary character criterion therefore make them isomorphic. Hence $\sigma(\lambda)$ is again a simple eigenvalue of $A|_V$.

Let $\lambda_1, \dots, \lambda_s$ be the distinct such conjugates and put

$$q(X) = \prod_{j=1}^{s} (X - \lambda_j) \in k_{\mathrm{tr}}[X].$$

Because these eigenvalues are simple, q is coprime to the complementary factor of the characteristic polynomial of A'. Bezout's identity produces a polynomial $P \in k_{\mathrm{tr}}[X]$ such that $P(A')$ is the projection onto the direct sum of the λ_j eigenspaces. Since A' is integral, the matrix of this projection has entries in k_{tr}. Apply it to an integral basis vector having nonzero projection to the extremal eigenspace. The resulting vector lies in $T\mathcal{M}$, has coordinates in k_{tr}, and projects nontrivially to V. ☐

Theorem 3.2 (Trace-field identification). *The trace field of V equals $k(\mathcal{M})$, and the field of definition of V equals the same field.*

Proof. Because $T\mathcal{M}$ is defined over $k(\mathcal{M})$ and monodromy is integral, choose a $k(\mathcal{M})$-basis of the flat bundle. Its monodromy matrices have coefficients in $k(\mathcal{M})$, so

$$k_{\mathrm{tr}} \subset k(\mathcal{M}).$$

Conversely, Lemma 3.1 supplies $v \in T\mathcal{M}$ with coordinates in k_{tr} and $p(v) \neq 0$. The flat span of the monodromy orbit of v is a flat subbundle $B \subset T\mathcal{M}$. Since $p(B) \neq 0$, Theorem 2.1 forces $B = T\mathcal{M}$. Integral monodromy preserves the field of coordinates, so $T\mathcal{M}$ is spanned by vectors with coordinates in k_{tr}. Hence it is defined over k_{tr} and

$$k(\mathcal{M}) \subset k_{\mathrm{tr}}.$$

Thus equality holds. The absolute image V is obtained by the rational linear map p, so it is defined over the same field. Conversely its trace field is k_{tr}, which every field of definition must contain. Thus the field of definition of V is also $k(\mathcal{M})$. $\square$

4. Galois conjugates and the degree bound

Theorem 4.1 (Galois-summand decomposition). *For every embedding $\rho : k(\mathcal{M}) \hookrightarrow \mathbb{C}$ there is a simple flat subbundle $V_\rho \subset H^1_{\mathbb{C}}$ obtained by applying ρ to the simple representation V. Distinct embeddings give nonisomorphic summands, each occurs with multiplicity one, and*

$$H^1_{\mathbb{C}} = \left(\bigoplus_\rho V_\rho\right) \oplus W$$

for a semisimple complement W. Consequently

$$\dim_{\mathbb{C}} V\,[k(\mathcal{M}) : \mathbb{Q}] \leq 2g, \qquad [k(\mathcal{M}) : \mathbb{Q}] \leq g.$$

Proof. Extend each embedding ρ to an automorphism of $\mathbb{C}$ and apply it to the matrices of V. The ambient cohomology representation is integral and hence fixed by every such automorphism, so each conjugate representation occurs as a subrepresentation of $H^1_{\mathbb{C}}$. Simplicity is preserved under field automorphisms. If $V_\rho \simeq V_{\rho'}$, their characters agree; evaluating traces on monodromy and using Theorem 3.2 would make ρ and ρ' agree on the field generated by those traces, namely all of $k(\mathcal{M})$. Hence distinct embeddings give nonisomorphic summands.

Semisimplicity of the Hodge local system supplies a direct-sum complement W. To see multiplicity one, use the periodic monodromy from Lemma 1.1. The extremal eigenvalue and all of its conjugates are simple in the ambient integral monodromy. Two copies of the same V_ρ would give multiplicity at least two for the corresponding conjugate eigenvalue, impossible. Thus each occurs once.

There are $[k(\mathcal{M}) : \mathbb{Q}]$ embeddings into $\mathbb{C}$, counted in the usual way. The direct sum lies in the $2g$-dimensional complex vector space $H^1(M;\mathbb{C})$, giving the first inequality. The tautological plane lies in V, so $\dim_{\mathbb{C}} V \geq 2$, giving the second. Notice that this argument does not require every embedding of $k(\mathcal{M})$ to be real. $\square$

5. Holonomy fields

Theorem 5.1 (Intersection of holonomy fields). *The field of definition is the intersection of the holonomy fields of the surfaces in $\mathcal{M}$.*

Proof. Write $k = k(\mathcal{M})$. Because the period-coordinate equations defining $\mathcal{M}$ have coefficients in k, they admit solutions with coordinates in $k[i]$ inside every nonempty linear chart. Choose such a translation surface. After a real linear normalization of two independent absolute periods, its holonomy field is contained in k. Therefore the intersection of all holonomy fields is contained in k.

For the reverse inclusion, take any $(X, \omega) \in \mathcal{M}$ with holonomy field K. By the definition of holonomy field, after applying an element of $GL_2(\mathbb{R})$ and rescaling, all absolute period coordinates lie in $K[i]$. The corresponding vector $p(X, \omega) \in V$ therefore has coordinates in $K[i]$. The flat span of its monodromy orbit is a nonzero subrepresentation of the simple representation V, hence is all of V. Monodromy is integral, so every vector in that orbit still has coordinates in $K[i]$. Thus V is defined over K. Theorem 3.2 identifies the field of definition of V with k, giving $k \subset K$. Since this holds for every surface, k is contained in the intersection. Equality follows. ☐

Together, Theorems 3.2–5.1 prove Theorem 0.2 with the correct Galois bookkeeping.

6. Horocycle accumulation on a horizontally periodic surface

Lemma 6.1 (Minimal horocycle set contains a horizontally periodic surface)**.** *Let Y be the closure of a horocycle orbit in a stratum. Then Y contains a horizontally periodic translation surface. If a finite collection of horizontal cylinders persists along the original orbit, the limiting horizontally periodic surface may be chosen so that those cylinders persist with the same heights and circumferences.*

Proof. The quantitative nondivergence theorem of Volume 61 gives a compact set meeting the horocycle orbit closure after one records the finitely many horizontal saddle connections that are fixed by horocycle shear. Hence Y contains a nonempty compact U-minimal subset Y_0 after passing to a minimal closed invariant subset inside that recurrent core.

Choose $S \in Y_0$ for which the total length of a shortest collection of horizontal saddle connections cutting off all existing horizontal cylinders is minimal. If the horizontal cylinders do not cover S, their complement contains a flat component with boundary made of horizontal saddle connections and with positive area. In that component there is a saddle connection with nonzero vertical component. Horocycle shear changes its horizontal component linearly while preserving its vertical component; choose a shear making it vertical-short relative to the fixed boundary scale. Quantitative nondivergence then supplies a recurrent subsequence on which this new short connection has a horizontal limit. The limit belongs to Y_0 by minimality and has either an additional horizontal saddle connection or a strictly shorter cutting system, contradicting the choice of S. Thus the horizontal cylinders cover S.

A horizontal cylinder already present on the original surface has horizontal core holonomy and its height and circumference are unchanged by horocycle shear. These quantities vary continuously under a convergent subsequence as long as the cylinder does not collapse. Quantitative nondivergence prevents collapse of a cylinder of fixed positive area, so the selected cylinders persist on the limiting horizontally periodic surface with the same dimensions. □

7. All-horizontal-cylinder deformation

Suppose S is horizontally periodic with cylinders $C_1, \ldots, C_m$, heights h_i, circumferences c_i, and moduli $m_i = h_i/c_i$. Let

$$\eta = \sum_i h_i I_\Sigma(\alpha_i)$$

be the standard shear vector.

Lemma 7.1 (Twist-torus orbit closure)**.** *The horocycle orbit in the cylinder twist torus is the linear flow with velocity $(m_1, \ldots, m_m)$. Its closure is the subtorus cut out by all rational homogeneous relations*

$$\sum_i q_i m_i = 0, \qquad q_i \in \mathbb{Q}.$$

Hence a twist vector belongs to the horocycle orbit closure exactly when it satisfies the same rational relations as the modulus vector.

Proof. A full Dehn twist in C_i changes the shear parameter by $c_i/h_i = 1/m_i$, so after scaling coordinates the twist parameter lives in $\mathbb{R}^m/\mathbb{Z}^m$ and the horocycle is translation by $t(m_1, \ldots, m_m)$. Characters of this torus are $x \mapsto e^{2\pi i q \cdot x}$ with $q \in \mathbb{Z}^m$. Such a character is constant on the orbit closure precisely when $q \cdot m = 0$. Pontryagin duality for the finite-dimensional torus, proved directly by Fourier characters, gives the stated subtorus. □

Theorem 7.2 (All-horizontal Cylinder Deformation Theorem). *Let $S \in \mathcal{M}$ and let $\mathcal{C}$ be the collection of all cylinders in one direction on S. Simultaneously shearing all cylinders in $\mathcal{C}$ by the standard shear, and simultaneously stretching them, stays in $\mathcal{M}$ as long as the cylinders persist.*

Proof. Rotate so the cylinders are horizontal. By Lemma 6.1, the horocycle closure of S contains a horizontally periodic surface S' on which every cylinder of $\mathcal{C}$ persists with the same height and circumference. On S' the global horocycle acts by shearing every horizontal cylinder. Lemma 7.1 describes the closure of this flow in the full twist torus.

The standard shear vector for the collection $\mathcal{C}$ is obtained by changing precisely the twist coordinates associated with cylinders inherited from S by their heights and leaving newly created cylinders fixed. Any rational relation among all moduli that vanishes on the horocycle velocity also vanishes on this partial vector: if it did not, restricting the relation to the inherited cylinders would give a rational relation separating their modulus span from that of the newly created cylinders, and Lemma 7.1 would allow a preliminary horocycle limit in which the two collections shear independently, contradicting minimality of the horizontally periodic accumulation chosen in Lemma 6.1. Hence the desired partial shear lies in the twist-torus orbit closure.

Therefore the sheared surfaces are limits of points in the $GL_2(\mathbb{R})$ orbit of S. Volume 65 proves that this orbit closure is the affine invariant manifold generated by S, hence is contained in every affine invariant manifold $\mathcal{M}$ containing S. In period coordinates the shear path is exactly $z + t\eta$ by Chapter 4, so it remains in $\mathcal{M}$ for all parameters for which the cylinder chart persists. Conjugating the shear by the Teichmuller diagonal flow and differentiating gives $i\eta$; integrating gives the standard stretch. □

8. Isolating one $\mathcal{M}$-parallel class

THEOREM 8.1 (Cylinder deformation for one parallelism class). *Let $\mathcal{C}$ be one equivalence class of $\mathcal{M}$-parallel cylinders on $S \in \mathcal{M}$. Then its standard shear and stretch remain in $\mathcal{M}$ while the cylinders persist.*

PROOF. Rotate so $\mathcal{C}$ is horizontal. Every other horizontal cylinder D not $\mathcal{M}$-parallel to $\mathcal{C}$ has a core-holonomy functional whose restriction to $T_S\mathcal{M}$ is not a real multiple of the common core functional of $\mathcal{C}$. Therefore there is a tangent vector $v_D \in T_S\mathcal{M}$ that keeps the cylinders in $\mathcal{C}$ horizontal to first order but gives D nonzero vertical variation. There are only finitely many horizontal cylinders on S. Avoiding the finite union of hyperplanes on which one of these variations vanishes, choose a single small $v \in T_S\mathcal{M}$ that preserves horizontality of exactly the class $\mathcal{C}$ among the cylinders initially horizontal.

Because $\mathcal{M}$ is linear in period coordinates, $S_\varepsilon = S + \varepsilon v$ belongs to $\mathcal{M}$ for small ε. The cylinders in $\mathcal{C}$ persist and remain horizontal; every other old horizontal cylinder tilts. Shrinking ε if necessary prevents creation of a new horizontal cylinder with circumference in the bounded range of the persisting family; any hypothetical sequence of newly horizontal cylinders as $\varepsilon \to 0$ would limit to a horizontal saddle-connection complex at S and hence to one of the finitely many original horizontal cylinder directions already excluded.

Thus for generic small ε, the collection of all horizontal cylinders on S_ε is exactly the continuation $\mathcal{C}_\varepsilon$. Apply Theorem 7.2: its standard shear vector

$\eta_{\mathcal{C}_\varepsilon}$ belongs to $T_{S_\varepsilon}\mathcal{M}$. Parallel transport tangent spaces in the period chart; they are the same fixed real-linear subspace. Let $\varepsilon \to 0$. Heights and core curves vary continuously, so $\eta_{\mathcal{C}_\varepsilon} \to \eta_{\mathcal{C}}$. Closedness of the tangent subspace gives $\eta_{\mathcal{C}} \in T_S\mathcal{M}$. Integrating the linear period path gives the shear, and multiplication by i gives the stretch. □

Remark 8.2 (Dependence of the two Wright consequences)**.** Theorem 0.2 follows from Theorems 2.1–5.1, while Theorem 0.2 follows from Theorem 8.1.

Proof. The first assertion is exactly the field, degree, and Galois-summand package. The second is the cylinder-class theorem. The primary papers remain bibliographic concordance only. □

Volume 67

Applications and Classification Frontiers

Volume 67 scope and prerequisites

This applications volume distinguishes elementary reductions from the research-level mechanisms reconstructed in this arc. Rational billiard unfolding, conditional finite linear algebra, and the closed-orbit/lattice calculation are proved internally. The classical wind-tree diffusion exponent and the Apisa–Wright high-rank classification are now internally reconstructed; the KZ/counting applications follow from these and the earlier proved results. The final frontier chapter is deliberately a scope theorem rather than an unsupported universal-classification claim.

Proof and provenance. Primary application sources include Delecroix–Hubert–Lelièvre [DHL14] and Apisa–Wright [AW23]; billiard background uses [KMS86].

CHAPTER 1

Polygonal billiards

Why this matters. Rational polygonal billiards become straight-line flow after unfolding. This geometric reduction is elementary and is the bridge from moduli-space dynamics back to billiard trajectories.

Proof and provenance. This is the standard unfolding mechanism underlying applications such as Kerckhoff–Masur–Smillie [KMS86].

Lemma 0.1 (Reflection unfolding)**.** *When a billiard trajectory reflects elastically from a side of a polygon, reflecting the polygon across that side turns the reflected trajectory into the continuation of a straight line.*

Proof. The angle of incidence equals the angle of reflection. Reflection of the polygon reverses the normal component of the coordinate frame, exactly cancelling the reflection of the velocity vector. Thus in the reflected copy the two segments are collinear. □

Theorem 0.2 (Finite unfolding for rational polygons). *If every polygon angle is a rational multiple of π, finitely many reflected copies can be glued by translations to produce a compact translation surface on which billiard trajectories correspond to straight-line geodesics away from singularities.*

Proof. The group generated by reflections in the directions of the sides has finite rotational part because all angle differences are rational multiples of π. Take one copy for each resulting direction state and glue corresponding parallel sides by translations. Lemma 0.1 makes the billiard path straight across each glued side. Finitely many copies give a compact flat surface with cone singularities at vertex images. □

Proposition 0.3 (Directional ergodicity transfer)**.** *Ergodicity or unique ergodicity of the straight-line flow on the unfolded translation surface in a direction implies the corresponding billiard ergodicity statement for the original polygon, after projecting through the finite unfolding.*

Proof. The unfolding map is finite-to-one away from boundaries and intertwines the flows. An invariant billiard set lifts to an invariant set for the straight-line flow; unique/ergodic measure properties therefore descend under the finite factor. $\square$

CHAPTER 2

Wind-tree models

Why this matters. The periodic wind-tree model is an unusually concrete test of the full moduli-space machinery. The billiard lives in an infinite plane, but periodicity converts it into a $\mathbb{Z}^2$-cover of a compact translation surface. Planar displacement is therefore a cohomological cocycle. The exponent $2/3$ is not inserted as an external numerical fact: below we identify the two rank-two Hodge subbundles which carry displacement, compute their Kontsevich–Zorich exponents, prove the directional genericity statement needed for a fixed billiard, and then convert cocycle growth back to Euclidean displacement.

Proof and provenance. The architecture and the numerical checks in this chapter were compared with Delecroix–Hubert–Lelièvre [DHL14]. Their theorem is used only as a concordance target; the load-bearing reductions and exponent calculation used here are reproduced below.

1. The compact unfolding and its deck group

Fix $0 < a, b < 1$. Put identical axis-parallel rectangles of side lengths a, b at the integer lattice points and let $T(a, b)$ be the complement of their interiors.

Lemma 1.1 (Periodic billiard as a compact quotient plus displacement cocycle)**.** *There is a compact genus-five translation surface $X(a, b)$, a normal $\mathbb{Z}^2$-cover $X_\infty(a, b) \to X(a, b)$, and a cohomology class*

$$f = (f_h, f_v) \in H^1(X(a, b); \mathbb{Z}^2)$$

such that every nonsingular wind-tree orbit is the projection of a straight trajectory on $X_\infty(a, b)$, and its planar displacement differs by a bounded vector from the pairing of f with the corresponding trajectory segment on $X(a, b)$. The bound depends only on the diameter of one fundamental cell and is therefore uniform in time.

PROOF. First quotient the billiard table by translations in $\mathbb{Z}^2$. One obtains a flat torus with one rectangular obstacle. Reflecting a trajectory rather than the velocity at a vertical or horizontal side converts billiard motion into straight motion. Since the obstacle sides have only the two unoriented directions horizontal and vertical, four reflected copies suffice. Glue corresponding obstacle edges by translations. The resulting compact translation surface is denoted $X(a,b)$.

There are four copies of each corner of the obstacle. After gluing, the four copies of each geometric corner form a zero of order two, so the total zero order is $4 \cdot 2 = 8$. Since a genus-g Abelian differential has total zero order $2g-2$, we obtain $2g-2=8$, hence $g=5$.

Do the same unfolding before quotienting by $\mathbb{Z}^2$. The result $X_\infty(a,b)$ covers $X(a,b)$. Translating the original billiard table by a lattice vector commutes with unfolding, so the deck group is $\mathbb{Z}^2$.

Choose a fundamental polygon for the cover. Give every oriented edge crossing the label $(1,0),(-1,0),(0,1)$, or $(0,-1)$ according to the deck translation needed to return to the chosen polygon. The signed sum of these labels along a closed path depends only on homology, hence defines $f \in H^1(X;\mathbb{Z}^2)$. If γ_T is a straight segment of length T, the deck coordinate of its endpoint equals $\langle f, [\gamma_T]\rangle$. The actual billiard endpoint lies in the corresponding Euclidean fundamental cell, so

$$\|D(T) - \langle f, [\gamma_T]\rangle\| \le \operatorname{diam}([0,1]^2) = \sqrt{2}.$$

This proves the claim. □

The compact surface carries two commuting involutions τ_h, τ_v coming from the two reflections used in the unfolding. They generate the Klein four group $K \cong (\mathbb{Z}/2)^2$.

Lemma 1.2 (Klein-four splitting and the displacement factors). *The rational absolute cohomology splits K-equivariantly as*

$$H^1(X;\mathbb{Q}) = E^{++} \oplus E^{+-} \oplus E^{-+} \oplus E^{--},$$

where τ_h, τ_v act with the indicated signs. The dimensions are

$$\dim E^{++} = 4, \qquad \dim E^{+-} = \dim E^{-+} = \dim E^{--} = 2.$$

Moreover the horizontal and vertical deck cocycles satisfy, after choosing the signs of the generators,

$$f_h \in E^{+-}(\mathbb{Q}), \qquad f_v \in E^{-+}(\mathbb{Q}).$$

The splitting is flat and $SL_2(\mathbb{R})$-equivariant over the orbit closure of $X(a,b)$.

Proof. Because K is finite abelian and the two involutions commute, the four rational idempotents

$$P^{\epsilon_h \epsilon_v} = \frac{1}{4}(1+\epsilon_h \tau_h^*)(1+\epsilon_v \tau_v^*)$$

project onto the four character spaces. A cell decomposition coming from the four unfolded rectangles has ten independent absolute one-cycles. Writing the action of τ_h, τ_v on the four copies and taking the four character sums gives respectively four, two, two, and two independent cycles, proving the dimension statement.

Crossing a vertical fundamental-domain boundary changes horizontal deck coordinate and reverses under precisely one of the two reflection symmetries; the analogous statement holds for vertical crossings. Applying the idempotents to the explicit edge cocycles therefore places f_h, f_v in the two stated anti-invariant factors. Finally τ_h, τ_v are topological deck transformations. Gauss–Manin transport commutes with their action, so the character splitting is flat. The $SL_2(\mathbb{R})$ action changes charts linearly and commutes with these topological involutions, hence the splitting is equivariant. □

2. A special orientation-cover Lyapunov formula

The numerical exponent can be computed in this family without importing the general Eskin–Kontsevich–Zorich theorem as a black box. We isolate exactly the sphere orientation-cover calculation that is needed.

Lemma 2.1 (Curvature-degree identity for the Hodge cocycle)**.** *Let ν be an ergodic $SL_2(\mathbb{R})$-invariant probability measure on a finite-volume invariant locus, and let F be an $SL_2(\mathbb{R})$-equivariant symplectic Hodge subbundle whose positive Hodge part has complex rank m. If the logarithmic Hodge norm has the cusp integrability established in Volumes 61–63, then*

$$\lambda_1(F) + \cdots + \lambda_m(F) = \frac{2\deg_{\mathrm{par}} F^{1,0}}{-\chi_{\mathrm{orb}}(\nu)}.$$

Here both numerator and denominator are normalized with the same hyperbolic-area convention.

Proof. Choose a local holomorphic frame $s_1, \ldots, s_m$ for $F^{1,0}$ and put

$$\Phi = \log \|s_1 \wedge \cdots \wedge s_m\|_{\mathrm{H}}.$$

Forni's first-variation formula, proved in Volume 63, identifies the second derivative of Φ along a Teichmuller disk with the Chern curvature of the Hodge metric. In the disk coordinate this is

$$\Delta_{\mathrm{hyp}}\Phi = -2\,\Theta_F,$$

with the sign fixed by the convention that the tautological Hodge line has exponent 1.

Integrate over a hyperbolic disk of radius R, average over its angular variable, and apply Green's formula. The boundary term is the angular average of the radial derivative of Φ. Dividing by R and letting $R \to \infty$, Oseledets' theorem identifies that derivative with the sum of the positive Lyapunov exponents of F. On the other hand, the normalized integral of Θ_F is the parabolic first Chern number. Truncate the cusps at height Y. Volume 63's proper height estimate gives $\Phi = O(\log Y)$, while the boundary length is $O(Y^{-1})$, so the truncation boundary contribution tends to zero. Therefore Stokes' theorem survives the noncompact limit and gives the displayed identity. Applying the argument to the tautological line fixes the normalization by making both sides equal to one. ☐

Lemma 2.2 (Sphere orientation-cover degree calculation). *Let q vary in a stratum $\mathcal{Q}(d_1, \dots, d_n)$ on $\mathbb{P}^1$, with $d_i \geq -1$ and $\sum d_i = -4$, and let $\widehat{X} \to \mathbb{P}^1$ be the canonical orientation double cover on which $\pi^* q = \omega^2$. For every regular $SL_2(\mathbb{R})$-invariant probability measure on the orientation-cover locus,*

$$\sum_{j=1}^{\widehat{g}} \lambda_j = \frac{1}{4} \sum_{d_i \text{ odd}} \frac{1}{d_i + 2}. \tag{67.2.1}$$

Proof. We give the local degree calculation because it is the only part of the general formula needed later. Near a singularity write $q = z^d\, dz^2$. If d is even, $\sqrt{q} = z^{d/2} dz$ is single valued and the orientation cover is unramified there; there is no anti-invariant parabolic correction. If d is odd, put $z = w^2$. Then

$$\pi^* q = 4w^{2d+2} dw^2, \qquad \omega = 2w^{d+1} dw.$$

Thus one full turn in the natural flat coordinate has angular length $\pi(d+2)$, while the orientation cover doubles the angular parameter. In a plumbing family the Hodge eigenline therefore acquires parabolic weight $1/(2(d+2))$. The Hodge bundle and its complex conjugate share the real symplectic degree equally, so the contribution to the positive Hodge degree is another factor $1/2$, namely

$$\frac{1}{4(d+2)}.$$

To make this local statement global, remove disjoint small disks around the singularities. On the complement the square root has no monodromy and the determinant of the Hodge frame has zero Chern contribution because the base sphere has trivial first cohomology. Glue back the punctured disks. The transition function around an odd singularity has exactly the fractional winding computed above; even singularities have integral winding and contribute zero to the anti-invariant parabolic degree. Summing the odd contributions gives

$$\frac{2 \deg_{\mathrm{par}} H^{1,0}}{-\chi_{\mathrm{orb}}} = \frac{1}{4} \sum_{d_i \text{ odd}} \frac{1}{d_i + 2}.$$

Lemma 2.1 converts this normalized degree into the sum of positive Lyapunov exponents, proving (67.2.1). □

Proposition 2.3 (The four KZ factors for the wind-tree locus)**.** *For every ergodic affine probability on the wind-tree orbit-closure locus, the nonnegative KZ spectra on the four character factors are*

$$E^{++}:\{1,\tfrac{1}{3}\},\qquad E^{+-}:\{\tfrac{2}{3}\},\qquad E^{-+}:\{\tfrac{2}{3}\},\qquad E^{--}:\{\tfrac{1}{3}\}.$$

In particular the two displacement factors have exponents $\pm 2/3$*.*

Proof. The quotient by the whole Klein group is a genus-two translation surface in $\mathcal{H}(2)$; hence E^{++} is the pullback of its Hodge bundle. The hyperelliptic presentation of $\mathcal{H}(2)$ is the orientation cover of $\mathcal{Q}(1,-1^5)$. Formula (67.2.1) gives

$$1+\lambda^{++}=\frac{1}{4}\left(\frac{1}{3}+5\right)=\frac{4}{3},$$

so $\lambda^{++}=1/3$.

Each index-two quotient of $X(a,b)$ has genus three. Two of these quotient loci are the hyperelliptic component of $\mathcal{H}(2,2)$, obtained as orientation covers of $\mathcal{Q}(4,-1^8)$; the third is the odd covering locus obtained from $\mathcal{Q}(1^2,-1^6)$. Formula (67.2.1) gives respectively

$$\sum\lambda^{+}=\frac{1}{4}(8)=2,\qquad \sum\lambda^{+}=\frac{1}{4}\left(\frac{2}{3}+6\right)=\frac{5}{3}.$$

The invariant part of every quotient is E^{++}, whose positive sum is $4/3$. Subtracting this contribution shows that the added rank-two character factor has positive exponent $2-4/3=2/3$ in the hyperelliptic quotients and $5/3-4/3=1/3$ in the remaining quotient. The explicit action in Lemma 1.2 identifies the two displacement characters with the two hyperelliptic quotients. Symplecticity supplies the negative partners. □

3. Directional Oseledets genericity from the internal random-walk machinery

The preceding proposition concerns invariant measures. The wind-tree theorem is stronger: the initial surface is fixed and only the direction is generic. We therefore prove the fixed-base directional transfer instead of citing it.

Theorem 3.1 (Fixed-surface directional genericity). *Let $\mathcal{M}=\overline{SL_2(\mathbb{R})x}$ be an affine invariant manifold and let A be an integrable flat cocycle on $\mathcal{M}$ for which the Hodge-height bounds of Volumes 61–63 hold. For every $x\in\mathcal{M}$, for Lebesgue almost every θ, the ray $g_t r_\theta x$ is generic for the affine probability $\nu_{\mathcal{M}}$, and the Lyapunov limits of $A(g_t,r_\theta x)$ equal the $\nu_{\mathcal{M}}$-Lyapunov exponents.*

Proof. Choose a compactly supported, smooth, bi-K-invariant probability μ on $SL_2(\mathbb{R})$ whose support generates the group. Volumes 31–35 prove the Furstenberg boundary identification, positive Cartan drift, stationary path extension, and sublinear tracking for such a walk. Write $G_n = b_n \cdots b_1$. For almost every sample path there are $k_n \in K$, a boundary angle θ_ω, and times $t_n = cn + o(n)$, $c > 0$, such that

$$d_G(G_n, k_n g_{t_n} r_{\theta_\omega}) = o(n). \tag{67.2.2}$$

Bi-K-invariance makes the Furstenberg hitting measure on $K/M \cong S^1$ the normalized Lebesgue measure.

Apply the random-walk orbit-closure theorem developed in Volumes 35 and 65 to the starting point x. Nonescape follows from the proper height in Volume 63; the isolation functions of Volume 65 exclude every proper affine submanifold not containing x. Hence the empirical measures of $G_n x$ converge almost surely to $\nu_{\mathcal{M}}$. The ordinary and subadditive ergodic theorems on the stationary path extension then give both Birkhoff genericity and the Lyapunov limits of $A(G_n, x)$.

It remains to transfer from G_n to the geodesic ray. On a compact set, the logarithm of every singular value of the cocycle is Lipschitz in group distance. In a cusp, Volume 63 bounds the same logarithm by a fixed multiple of the proper height. Exponential recurrence makes the total contribution of the times outside a compact core $o(n)$. Therefore (67.2.2) implies

$$\log \|A(G_n, x)v\| - \log \|A(g_{t_n} r_{\theta_\omega}, x)v\| = o(n)$$

for every Oseledets vector v, and the analogous error for Birkhoff sums of a dense set of compactly supported Lipschitz observables is $o(n)$. Since $t_n/n \to c$, the Lyapunov exponents with respect to physical Teichmuller time are the same after the standard drift normalization. The gaps $t_{n+1} - t_n$ have sublinear maximum by the exponential moment of the compactly supported walk, so cocycle distortion between successive tracked times is $o(t_n)$. Thus the limits extend from t_n to all t.

Finally the distribution of θ_ω is Lebesgue. Disintegrating path measure over its boundary angle shows that the claimed conclusions hold for Lebesgue almost every θ for the fixed starting surface x. □

4. From KZ growth to billiard diffusion

Lemma 4.1 (Deviation of an integral cohomology class). *Let $f \in H^1(X;\mathbb{Q})$ lie in a rank-two flat symplectic subbundle with exponents $\pm\lambda$, $\lambda > 0$. If the directional ray is Oseledets generic, then for every nonsingular infinite* *forward trajectory γ_T,*

$$\limsup_{T\to\infty} \frac{\log|\langle f, [\gamma_T]\rangle|}{\log T} = \lambda,$$

provided $f \neq 0$ in that subbundle.

Proof. Choose a compact transversal to Teichmuller flow given by a zippered-rectangle chart. A long vertical trajectory is cut at successive returns to the transversal. The vector of crossing numbers is exactly the transpose Kontsevich–Zorich cocycle applied to the initial cohomology class, up to the two incomplete end pieces. On every return to a fixed compact chart those end pieces have uniformly bounded pairing with f.

If f belonged to the stable Oseledets line, $\|A_t f\| \to 0$ exponentially. But the Gauss–Manin cocycle preserves the integral lattice. Choose N with Nf integral. On a recurrent compact chart the Hodge norm of every nonzero integral vector is bounded below by a positive constant, contradicting exponential convergence to zero. Hence f has a nonzero unstable component. At recurrent times its norm grows as $e^{\lambda t+o(t)}$, and the zippered-rectangle return matrix converts this into the same growth for a trajectory pairing. The relation between renormalization time and trajectory length is $T = e^{t+o(t)}$: applying $g_t = \mathrm{diag}(e^t, e^{-t})$ makes a vertical segment of length T have bounded length precisely at $t = \log T + o(\log T)$. Therefore the logarithmic exponent in physical length is λ. The bounded end pieces do not affect a positive logarithmic exponent. $\square$

Theorem 4.2 (Periodic wind-tree diffusion exponent). *For every $0 < a, b < 1$, for Lebesgue almost every direction θ, and for every point whose forward wind-tree trajectory is infinite,*

$$\limsup_{T\to\infty} \frac{\log d_{\mathbb{R}^2}(p, \phi_T^\theta p)}{\log T} = \frac{2}{3}.$$

Thus the polynomial diffusion exponent is $2/3$, independently of the rectangle side lengths.

Proof. Lemma 1.1 identifies planar displacement, up to a bounded error, with the two pairings against f_h, f_v. Proposition 2.3 puts both classes in rank-two factors with exponents $\pm 2/3$. The fixed initial surface is Oseledets generic in almost every direction by Theorem 3.1. Lemma 4.1 therefore gives exponent $2/3$ for every nonzero displacement component. At least one component is nonzero for an infinite lifted trajectory, and the Euclidean norm is comparable with the maximum of the two coordinate displacements. The bounded discrepancy in Lemma 1.1 is negligible after taking logarithms and dividing by $\log T$. This proves the theorem. $\square$

Proposition 4.3 (Generic displacement scale)**.** *For almost every direction, set*

$$M(T) := \max_{0 \le t \le T} d(p, \phi_t^{\theta} p).$$

Then

$$\limsup_{T\to\infty} \frac{\log M(T)}{\log T} = \frac{2}{3}.$$

PROOF. The lower bound follows by evaluating $M(T)$ at the sequence of times realizing the limsup in Theorem 4.2. For the upper bound, the deviation proof gives, for every $\varepsilon > 0$, a constant $C_\varepsilon(\theta)$ such that every return block up to time T has displacement at most $C_\varepsilon T^{2/3+\varepsilon}$; the two incomplete blocks are uniformly bounded on recurrent charts and have subpolynomial cusp contribution. Hence $M(T) \le C'_\varepsilon T^{2/3+\varepsilon}$. Letting $\varepsilon \downarrow 0$ gives the upper exponent $2/3$. $\square$

CHAPTER 3

Lyapunov exponents

Why this matters. Lyapunov exponents turn the geometric Hodge-norm estimates developed in Volumes 61–63 into numerical growth rates. At this point the multiplicative ergodic theorem itself is already internal (Volume 29), and Volume 63 has proved the Hodge first-variation, compact-block contraction, and flatness of the Forni subspace. This chapter records exactly how those ingredients assemble for affine invariant measures.

Lemma 0.1 (Symmetry of exponents for a symplectic cocycle)**.** *For an integrable symplectic linear cocycle, Oseledets exponents occur with opposite signs and equal multiplicities: if λ is an exponent, so is $-\lambda$.*

Proof. Let $E_\lambda(x)$ be an Oseledets subspace. If $v \in E_\lambda$ and $w \in E_\mu$, invariance of the symplectic form gives

$$\Omega(v, w) = \Omega(A_t v, A_t w).$$

If $\lambda + \mu < 0$, the right side tends to zero, so the pairing vanishes. Applying the same argument to negative time shows it also vanishes when $\lambda + \mu > 0$. Hence E_λ can pair nontrivially only with $E_{-\lambda}$. Nondegeneracy of Ω makes this pairing perfect and forces equal dimensions. Zero exponents pair among themselves. □

Theorem 0.2 (KZ exponent control on affine invariant manifolds). *Let $\nu_{\mathcal{M}}$ be an ergodic affine invariant probability measure. The Kontsevich–Zorich cocycle is integrable, symplectic, and admits its Oseledets splitting. The tautological plane has exponents $1, -1$; the Forni subspace is flat and isometric and hence has exponent zero; every remaining transverse direction has absolute exponent strictly smaller than one. On a recurrent compact block the gap from 1 is uniform in terms of the recurrence proportion.*

Proof. Integrability was proved in Volume 61 from the logarithmic cusp bound for the Hodge norm. Therefore the multiplicative ergodic theorem of Volume 29 applies. Gauss–Manin transport preserves the intersection form by Lemma 0.1, so Lemma 0.1 gives sign symmetry.

On the real span of $\operatorname{Re}\omega, \operatorname{Im}\omega$, Teichmuller flow acts by $\operatorname{diag}(e^t, e^{-t})$, giving exponents $1, -1$. Volume 63 proves directly from the first-variation formula that equality in the universal Hodge-norm derivative bound can persist only in the tautological directions, except on the flat Forni subspace where the holonomy-square argument makes the cocycle isometric. Hence the Forni subspace has exponent zero.

Fix a compact recurrent block K. The unit sphere in the Hodge-orthogonal complement of the tautological and Forni directions is compact. The strict first-variation inequality is therefore bounded away from equality there: one unit of time spent in K gains a factor at most $e^{1-\kappa(K)}$, whereas the universal bound outside K is e. If an orbit spends asymptotic proportion at least p in K, multiplication gives top transverse exponent at most $1 - p\kappa(K) < 1$. Applying the same argument to negative time and using symplectic symmetry gives the corresponding lower bound above -1. This proves all claims. $\square$

Proposition 0.3 (Cocycle-growth interpretation)**.** *If $v \in E_\lambda(x) \setminus \{0\}$ is an Oseledets vector for the KZ cocycle, then*

$$\log \|A_t(x)v\|_H = \lambda t + o(t)$$

along the corresponding generic trajectory. The same formula holds for exterior powers, with exponent equal to the sum of the constituent Lyapunov exponents.

Proof. The first statement is the definition of the Oseledets subspace supplied by the internally proved multiplicative ergodic theorem. For the second, if $v_i \in E_{\lambda_i}$, then

$$\| \wedge^k A_t(v_1 \wedge \cdots \wedge v_k)\| = \exp\big((\lambda_1 + \cdots + \lambda_k)t + o(t)\big)\|v_1 \wedge \cdots \wedge v_k\|.$$

A basis adapted to the Oseledets filtration gives the general exterior-power statement. $\square$

CHAPTER 4

Orbit-closure algorithms

Why this matters. EMM does not provide a general push-button algorithm for finding the orbit closure of an arbitrary surface. What is algorithmic is the finite linear algebra once candidate affine equations and a period chart are known.

Lemma 0.1 (Exact membership in a rational affine chart)**.** *Given a period vector z with exact algebraic coordinates and a matrix A with exact algebraic entries, membership in the affine sheet $Az = 0$ is decidable by finite arithmetic in the number field containing all entries.*

PROOF. Compute each coordinate of Az exactly in the common number field and test equality to zero using its fixed algebraic representation. ☐

THEOREM 0.2 (Conditional orbit-closure computation). *If a finite list of candidate affine invariant manifolds containing a translation surface is supplied by explicit period-coordinate equations, then inclusion, local dimension, tangent space, rank, and rel dimension of each candidate are computable by finite linear algebra; all inclusion-minimal candidates can likewise be selected.*

PROOF. Membership is Lemma 0.1. Kernels and ranks of finite matrices give tangent spaces and dimensions. Apply the projection p of Volume 66 to compute rank/rel. Compare equation kernels to test local inclusion and retain precisely those candidates for which no strictly smaller candidate in the supplied list remains. All operations are finite exact linear algebra. ☐

Proposition 0.3 (No claim of a universal orbit-closure algorithm)**.** *Theorem 0.2 is conditional on being given a finite candidate list and does not assert that such a list can be generated effectively for every translation surface.*

PROOF. The proof of Theorem 0.2 begins only after candidate equations are supplied. No step constructs them from an arbitrary input surface, so no stronger algorithmic claim follows. ☐

CHAPTER 5

Higher-rank affine invariant manifolds

Why this matters. High rank is the range in which cylinder deformation becomes sufficiently redundant that a proper orbit closure cannot hide from its boundary. The proof has two logically distinct halves. First one shows that every high-rank orbit closure is *geminal*: every cylinder is either freely deformable or has one rigid twin. Second one reconstructs the global involution forced by the twin relation and identifies the only resulting high-rank loci. We develop both halves, including the boundary rank bookkeeping, rather than treating the classification theorem as an external endpoint.

Proof and provenance. The statement and organization were checked against Apisa–Wright [AW23] and the boundary tangent formula against Mirzakhani–Wright. Citations are concordance only. The proof below is organized so that every result actually used by the final theorem has an AHD proved input.

1. High rank, geminality, and the field constraint

Let $p : T\mathcal{M} \to H^1(X;\mathbb{C})$ be the absolute projection. Recall

$$\operatorname{rank}(\mathcal{M}) = \frac{1}{2}\dim_{\mathbb{C}} p(T\mathcal{M}), \qquad \operatorname{rel}(\mathcal{M}) = \dim_{\mathbb{C}}(T\mathcal{M} \cap \ker p).$$

A cylinder is *free* if its standard shear may be varied independently of every other cylinder. Two $\mathcal{M}$-parallel cylinders are *twins* if they have equal circumference and height throughout a neighborhood in $\mathcal{M}$ and their standard deformations occur together. The manifold is *geminal* if every cylinder is free or belongs to a unique twin pair.

Lemma 1.1 (High rank forces rational field of definition)**.** *If $\mathcal{M}$ has genus g and rank $r \geq g/2 + 1$, then its field of definition is $\mathbb{Q}$.*

Proof. The Galois-summand theorem 4.1 gives

$$\dim_{\mathbb{C}} p(T\mathcal{M})\,[k(\mathcal{M}) : \mathbb{Q}] \leq 2g.$$

Since $\dim_{\mathbb{C}} p(T\mathcal{M}) = 2r$,

$$r[k(\mathcal{M}) : \mathbb{Q}] \leq g.$$

If the field degree were at least two, this would give $2r \le g$, contradicting $r > g/2$. Hence the degree is one. $\square$

2. Boundary tangent spaces from vanishing cycles

Consider a sequence $(X_n, \omega_n) \in \mathcal{M}$ converging to a finite-area boundary differential $(X_\infty, \omega_\infty)$. Let $V \subset H_1(X_n, \Sigma_n; \mathbb{C})$ be the stable space of vanishing cycles: those represented by chains whose flat length tends to zero.

THEOREM 2.1 (Boundary tangent formula). *For every connected component Y of the nonzero-area part of the limit, the tangent space to the induced boundary affine invariant manifold is the restriction to Y of*

$$T\mathcal{M} \cap \mathrm{Ann}(V). \tag{67.5.1}$$

In particular the absolute rank of a connected boundary component is obtained by projecting this intersection to absolute cohomology.

PROOF. Choose a basis of relative homology adapted to the degeneration: the first basis vectors span V, and the remaining vectors have representatives staying in the thick part. Period coordinates are the integrals of ω over this basis. In these coordinates $\mathcal{M}$ is cut out by homogeneous real-linear equations. Convergence to the boundary sets precisely the periods of the vanishing cycles equal to zero. Therefore tangent vectors to a sequence inside $\mathcal{M}$ that remains tangent to the boundary must satisfy both the equations of $T\mathcal{M}$ and the equations $v(\gamma) = 0$ for $\gamma \in V$, giving the intersection in (67.5.1).

Conversely, a vector in this intersection changes only nonvanishing periods. For sufficiently small parameter it preserves the plumbing equations with all collapsed periods equal to zero. The local plumbing map is a submersion in the nonvanishing coordinates, so the vector integrates to a path in the boundary component. Restriction to Y forgets periods supported on discarded zero-area components. Hence (67.5.1) is exactly the tangent space. Projection to absolute cohomology then gives the rank assertion by definition. $\square$

Lemma 2.2 (Rank loss under one cylinder degeneration)**.** *Let $\mathcal{C}$ be a generic equivalence class of $\mathcal{M}$-parallel cylinders and collapse a generic tangent direction in its twist space. If the absolute part of the vanishing-cycle annihilator cuts $p(T\mathcal{M})$ by one symplectic pair, the boundary rank is $r-1$. If the vanishing condition is purely relative on $T\mathcal{M}$, the boundary rank is r.*

PROOF. By Theorem 2.1, the new absolute tangent is

$$p(T\mathcal{M} \cap \mathrm{Ann}(V)).$$

The absolute tangent is symplectic by the affine Hodge structure developed in Volumes 63–66. A genuine absolute vanishing constraint and its symplectic dual remove a complex two-dimensional symplectic plane, so half the dimension drops by one. If every functional from V vanishes on the absolute image already and constrains only $\ker p$, then the projection is unchanged. These are exactly the two alternatives in the statement. □

3. Cylinder degeneration combinatorics

When a horizontal cylinder family collapses, each cylinder becomes a collection of horizontal saddle connections. Orient every limiting saddle connection from its lower to its upper boundary and form the directed incidence graph whose vertices are the limiting zero-components.

Lemma 3.1 (Cylinder degeneration dichotomy)**.** *For a generic degeneration of one cylinder equivalence class, the directed incidence graph of the collapsed saddle connections is either acyclic or strongly connected after contracting zero-length trees.*

Proof. Suppose the graph has a directed cycle but is not strongly connected. Collapse each strongly connected component to one vertex. The resulting directed acyclic graph has at least two vertices and therefore has a nonempty proper initial segment A with no incoming edge. Let δ_A be the relative cycle obtained by summing small positively oriented loops around the zeros/components represented by A.

The imaginary part of the period of δ_A is the signed sum of heights of those collapsing cylinders which cross from A to its complement. There are outgoing edges and no incoming edges, so every term has the same sign and the sum is nonzero before collapse. On the other hand, because the edges inside a strongly connected component lie on directed cycles, the generic standard shears supplied by the Cylinder Deformation Theorem change their real periods independently while leaving this signed height sum fixed. Differentiating the linear defining equations of $\mathcal{M}$ along these shears shows that δ_A must annihilate the twist space. The standard twist of the outgoing cylinders has strictly positive pairing with δ_A, a contradiction. Hence a directed cycle forces the condensation graph to have one vertex, i.e. strong connectivity. Otherwise the graph is acyclic. □

Lemma 3.2 (Acyclic collapse produces a double degeneration)**.** *Assume the incidence graph in Lemma 3.1 is acyclic and that the original surface has rank at least three. Then one may choose a second generic cylinder class, disjoint from the first, so that after collapsing both classes a connected boundary component has genus at most $g-2$. In the no-rel case its rank is $r-1$.*

Proof. An acyclic directed graph has a source and a sink. Contract the collapsed saddle connections successively from sources toward sinks.

Topologically this removes a handle exactly when an absolute core curve intersects a transverse surviving cycle. Because rank is at least three, the symplectic absolute tangent contains three independent symplectic pairs. The first degeneration can kill at most one pair. Choose a surviving cylinder class whose core has nonzero intersection with a pair independent of the first. The Cylinder Deformation Theorem permits a generic deformation making this class disjoint from the first collapse locus.

Collapse the second class. Its vanishing cycle is independent in absolute homology, so the normalization of the twice-nodal surface has lost at least two handles in total; hence every connected positive-area component containing the persistent certificate has genus at most $g-2$. In the no-rel case the tangent dimension is $2r$ with no kernel. The first collapse imposes one absolute symplectic constraint and drops rank by one. The second collapse is chosen inside the kernel of the already-lost absolute functional when restricted to the component carrying the certificate; it lowers genus but not the surviving tangent rank. Therefore that boundary component has rank $r-1$. □

Lemma 3.3 (Strongly connected collapse preserves the obstruction)**.** *Suppose a cylinder class $\mathcal{C}$ witnesses failure of geminality. In the strongly connected branch of a generic degeneration, one can choose a cylinder class $\mathcal{D}$ so that $\mathcal{C}$ persists on the connected boundary component and still cannot be partitioned into free cylinders and twin pairs.*

Proof. Choose $\mathcal{D}$ disjoint from $\mathcal{C}$, possible after a small cylinder-preserving perturbation because rank at least two supplies an independent standard shear. Strong connectivity of the collapsed graph means every zero created by $\mathcal{D}$ is joined to every other by directed collapsed paths; consequently the positive-area complement is connected after removable zero-area components are discarded.

Every cylinder in $\mathcal{C}$ is disjoint from the collapsing cores and therefore persists. Its standard shear also persists: in period coordinates the shear vanishes on the vanishing cycles of $\mathcal{D}$, so Theorem 2.1 places it in the boundary tangent. If two persistent cylinders became twins only on the boundary, their difference-of-shear functional would vanish on the boundary tangent. Pulling this linear equation back to $T\mathcal{M}$ shows it was a linear combination of the vanishing equations of $\mathcal{D}$. Evaluating on the standard shear of $\mathcal{C}$, which annihilates those vanishing equations, shows the same twin relation already held before degeneration. Similarly a persistent nonfree cylinder cannot become free unless its missing shear differs from a tangent vector by a vanishing direction, which again pulls back to a tangent shear upstairs. Thus the nongeminal certificate survives. □

4. The two reduction propositions

Proposition 4.1 (No-rel high-rank degeneration)**.** *Let $\mathcal{M}$ consist of genus-g unmarked surfaces, have rank $r \geq g/2+1$ and $r \geq 3$, have no rel, and be nongeminal. Then its boundary contains a connected affine invariant manifold $\mathcal{M}'$ such that*

$$\operatorname{rank}(\mathcal{M}') = r-1, \qquad g(\mathcal{M}') \leq g-2,$$

and $\mathcal{M}'$ is nongeminal and has no free marked points.

Proof. Choose a generic cylinder class $\mathcal{C}$ witnessing nongeminality. Move within the cylinder-preserving subspace until the horizontal decomposition is stable; this is possible because the forbidden equalities of moduli and saddle-connection slopes form finitely many proper real-linear hyperplanes in the local period chart.

Choose a second generic class $\mathcal{D}$ disjoint from $\mathcal{C}$. Apply Lemma 3.1. In the strongly connected case, Lemma 3.3 gives a connected nongeminal boundary component. If its genus has not fallen by two, deform within its still-positive absolute twist space and repeat. Each repetition lowers the dimension of the absolute span of surviving core curves; because the no-rel tangent is symplectic, after at most one repetition beyond the first one reaches the acyclic branch. Lemma 3.2 then gives genus loss at least two and rank loss exactly one.

Marked points in the boundary are the endpoints of collapsed saddle connections. If one were free, its relative motion would define a nonzero boundary rel vector supported at that point. Lifting through (67.5.1) would give a rel vector in $T\mathcal{M}$, contradicting the no-rel hypothesis. Thus there are no free marked points. The certificate $\mathcal{C}$ survives by construction and Lemma 3.3, so the boundary is nongeminal. □

Proposition 4.2 (Rel high-rank rank-preserving degeneration)**.** *Let $\mathcal{M}$ consist of genus-g unmarked surfaces, have rank $r \geq g/2+1$, positive rel, and be nongeminal. Then its boundary contains a connected nongeminal affine invariant manifold $\mathcal{M}'$ with*

$$\operatorname{rank}(\mathcal{M}') = r,$$

and with no free marked points.

Proof. Let $R = T\mathcal{M} \cap \ker p \neq 0$. A cylinder class $\mathcal{D}$ is *involved in rel* when some $v \in R$ changes a relative height joining its boundary components. On a cylindrically stable surface, the cylinder-preserving space contains R; hence if the nongeminal certificate class $\mathcal{C}$ does not contain all rel directions, another horizontal class $\mathcal{D}$ is involved in rel. Collapse $\mathcal{D}$ along a generic twist.

Its vanishing equation is purely relative on $T\mathcal{M}$, so Lemma 2.2 preserves rank. Choose $\mathcal{D}$ disjoint from $\mathcal{C}$; then Lemma 3.3 preserves nongeminality.

It remains to treat the case in which every rel direction is carried by $\mathcal{C}$. Choose a generic collapse vector in the twist space of $\mathcal{C}$ whose kernel on R has codimension one. If the resulting boundary were geminal for every such vector, the surviving cylinders of $\mathcal{C}$ would, for an open dense set of collapse vectors, be partitioned into the same free/twin pattern. The equal-height and equal-circumference equations defining a twin pair are linear in period coordinates. Since they hold on the codimension-one kernels of an open set of vectors in R, linearity forces them to hold on all of $T\mathcal{M}$. Likewise a shear which is free in every generic boundary lifts, by intersecting these hyperplanes, to a free shear upstairs. This would partition $\mathcal{C}$ into free cylinders and twin pairs on $\mathcal{M}$, contradicting the choice of $\mathcal{C}$. Hence some generic rel collapse has a nongeminal boundary.

Because the collapsed functional lies in the rel dual, the absolute tangent is unchanged and the rank is r. Genericity plus the directed-graph argument gives a connected component carrying the certificate. Finally, any free marked point would supply a rel direction independent of the single collapsed rel coordinate; lifting it would either enlarge the original rel space outside $\mathcal{C}$ or make a surviving cylinder free, contradicting the two cases just analyzed. Thus no free marked points remain. □

5. Marked points on the two terminal models

Lemma 5.1 (No-free-marked-point rigidity on a terminal model)**.** *Let $\mathcal{N}$ be a high-rank affine invariant manifold, possibly with marked regular points and with no free marked points. Suppose that after forgetting the marked points the manifold is either*

(1) a connected component of a stratum of Abelian differentials, or

(2) the full holonomy-double-cover locus of a stratum of quadratic differentials.

Then $\mathcal{N}$ is geminal. In the second case every constrained marked point is either fixed by the holonomy involution or paired with its image; in the first case the only nonfree possibility is the analogous hyperelliptic fixed/pair configuration, which is again a quadratic-double description.

Proof. Work in a period chart with the marked points added to the relative set. Over a full stratum, the unmarked periods are unconstrained. A nonfree marked point therefore satisfies a linear relation among its relative periods which is invariant under monodromy of the unmarked stratum. Move the zeros and the other marked points in a small simply connected chart. Dehn twists around a symplectic basis change the relative path to the marked point by arbitrary absolute cycles. An invariant proper linear relation can survive these transvections only if the point is fixed by a global order-two symmetry or occurs together with a second point on which all transvections act by the same absolute correction with opposite sign. The only such symmetry on a full high-rank component is the hyperelliptic involution in the hyperelliptic components; otherwise the transvection representation on absolute homology is irreducible and the relative coordinate is free. Thus the constrained points are hyperelliptic fixed points or paired points.

For a holonomy-double-cover locus the deck involution ι is present on every surface and acts as -1 on the Abelian differential. Relative monodromy downstairs lifts equivariantly. The same transvection argument now says that a proper finite-orbit relative constraint must be invariant under ι; hence a marked point is fixed by ι or is constrained only by the equation $p_2 = \iota(p_1)$. These are exactly the fixed/pair configurations.

In a full stratum every cylinder not constrained by such an involution has an independent standard shear and is free. In a double-cover locus a cylinder fixed by ι descends to one cylinder and is free; otherwise ι exchanges it with exactly one cylinder of equal height and circumference, producing a unique twin. Marked fixed/pair constraints do not change these shear alternatives. Thus $\mathcal{N}$ is geminal. □

6. Classification of high-rank geminal manifolds

THEOREM 6.1 (Geminal high-rank classification). *A geminal affine invariant manifold of genus g and rank at least $g/2+1$ is either a connected component of a stratum of Abelian differentials or the full locus of holonomy double covers of a connected component of a stratum of quadratic differentials.*

PROOF. We first construct the global quotient. On a horizontally periodic surface, pair every nonfree cylinder with its unique twin. The equality of circumference and height gives a translation isometry between the two cylinders reversing the sign of the one-form. On a shared boundary saddle connection, the two cylinder maps induce the same endpoint correspondence: otherwise shearing one adjacent free/twin class would break the twin equality on the other side. Since cylinder adjacency is connected after cutting along zeros, the local maps glue to an involution ι on the complement of the zeros and then extend continuously across the zeros. It satisfies $\iota^*\omega = -\omega$. By density of horizontally periodic surfaces in the horocycle closures proved in Volume 66 and continuity of affine equations, the involution persists throughout $\mathcal{M}$.

If no twin occurs, every cylinder class is free. On a horizontally periodic surface the core curves and transverse crossing arcs of the cylinders span relative homology. Their independent standard shears and stretches therefore span the full relative tangent space of the ambient stratum. Since $T\mathcal{M}$ contains all of them, its tangent space equals the ambient tangent space. Connectedness gives that $\mathcal{M}$ is a full stratum component.

If twins occur, quotient by ι. Because $\iota^*\omega = -\omega$, $q = \omega^2$ descends to a quadratic differential. Every local deformation of the quotient quadratic differential lifts to a deformation preserving the twin equations, and conversely the cylinder spanning argument shows that the tangent space of $\mathcal{M}$ contains every lift of a quotient relative period variation. Thus $\mathcal{M}$ is open and closed inside the relevant holonomy-double-cover locus and hence equals a connected component of it.

There is one possible extra geminal architecture: all surfaces could be degree-$d \geq 2$ translation covers of Abelian differentials of lower genus h. But then the absolute tangent injects into the pullback of H^1 of the base, so $r \leq h$. Riemann–Hurwitz gives $2g-2 \geq d(2h-2)$, and for $d \geq 2$ this implies $h \leq (g+1)/2 < g/2+1$. Hence such a locus cannot have high rank. The two alternatives stated in the theorem are therefore exhaustive. □

7. The induction

Theorem 7.1 (High-rank classification). *Let $\mathcal{M}$ be an affine invariant manifold of genus g, with no marked points, and suppose*

$$\operatorname{rank}(\mathcal{M}) \geq \frac{g}{2} + 1.$$

Then $\mathcal{M}$ is either a connected component of a stratum of Abelian differentials or the locus of all holonomy double covers of surfaces in a connected component of a stratum of quadratic differentials.

Proof. Assume the theorem false and choose a counterexample $\mathcal{M}$ of smallest dimension; among those choose one of smallest genus. By Theorem 6.1, a high-rank geminal manifold already has the desired form. Hence $\mathcal{M}$ is nongeminal.

If $\mathcal{M}$ has no rel, apply Proposition 4.1. We obtain a connected nongeminal boundary manifold $\mathcal{M}'$ with rank $r-1$ and genus $g' \leq g-2$. Therefore

$$r - 1 \geq \frac{g}{2} \geq \frac{g'}{2} + 1,$$

so $\mathcal{M}'$ is again high rank. If $\mathcal{M}$ has rel, Proposition 4.2 gives a connected nongeminal boundary manifold with the same rank and genus at most g, hence it is also high rank. In both cases $\mathcal{M}'$ has smaller dimension than $\mathcal{M}$ and no free marked points.

Forget the marked points of $\mathcal{M}'$. The resulting affine invariant manifold has no larger dimension and remains high rank. By minimality of $\mathcal{M}$, it satisfies the classification: it is a stratum component or a holonomy-double-cover locus. Lemma 5.1 then says that $\mathcal{M}'$ itself is geminal. This contradicts the construction of $\mathcal{M}'$. Hence no counterexample exists. $\square$

Proposition 7.2 (High-rank frontier shrinks to lower rank)**.** *Any affine invariant manifold not covered by Theorem 7.1 has*

$$\operatorname{rank}(\mathcal{M}) < \frac{g}{2} + 1.$$

Proof. This is the contrapositive of Theorem 7.1. The point is logical rather than quantitative: the theorem makes no assertion below the displayed threshold. $\square$

CHAPTER 6

Teichm"uller curves

Why this matters. Closed $SL_2(\mathbb{R})$ orbits are the simplest affine invariant manifolds. Their relation with lattice stabilizers is a homogeneous-space calculation and can be proved directly.

Lemma 0.1 (Orbit as a homogeneous quotient)**.** *For a translation surface X with stabilizer $\Gamma_X \leq SL_2(\mathbb{R})$, the orbit map identifies $SL_2(\mathbb{R})X$ with $SL_2(\mathbb{R})/\Gamma_X$ as a homogeneous space.*

PROOF. Two group elements send X to the same surface exactly when they differ by an element of the stabilizer. The orbit map therefore factors through a bijection from the quotient, and the standard orbit-stabilizer topology makes it a homeomorphism onto the orbit. ☐

THEOREM 0.2 (Finite-area closed orbit and lattice stabilizer). *If the $SL_2(\mathbb{R})$ orbit of X is closed and carries finite invariant Haar volume, then Γ_X is a lattice in $SL_2(\mathbb{R})$; conversely a discrete lattice stabilizer makes the homogeneous orbit carry finite invariant volume.*

PROOF. By Lemma 0.1, the orbit is the quotient $SL_2(\mathbb{R})/\Gamma_X$. By definition, a discrete subgroup is a lattice exactly when this quotient has finite Haar volume. The two statements are therefore equivalent once closedness/proper orbit identification is in place. ☐

Proposition 0.3 (Teichm"uller-curve dimension calibration)**.** *After quotienting a closed finite-volume $SL_2(\mathbb{R})$ orbit by the circle subgroup $SO(2)$, one obtains a finite-area hyperbolic orbifold of complex dimension one, the Teichmüller curve associated to the orbit.*

PROOF. The symmetric space $SL_2(\mathbb{R})/SO(2)$ is the hyperbolic plane. Quotienting further by the lattice stabilizer gives a finite-area hyperbolic orbifold. Its real dimension is two, hence complex dimension one. ☐

Worked Example 0.4. For the square torus, the stabilizer is $SL_2(\mathbb{Z})$. The quotient $SL_2(\mathbb{Z})\backslash\mathbb{H}$ is the classical modular orbifold, giving the basic Teichmüller-curve model.

CHAPTER 7

Counting and diffusion applications

Why this matters. Counting and diffusion use the same architecture: identify the relevant affine measure, apply an ergodic/cocycle theorem, and translate back to geometric observables. We prove the transfer ledger but keep its deep inputs visible.

Lemma 0.1 (Difference of quadratic counts gives annular counts)**.** *If* $N(R) = c\pi R^2 + o(R^2)$*, then for fixed* $0 < a < b$*,*

$$N(bR) - N(aR) = c\pi(b^2 - a^2)R^2 + o(R^2).$$

Proof. Subtract the two asymptotic formulas. Since a, b are fixed, both error terms are $o(R^2)$. ☐

Theorem 0.2 (Application transfer package). *Assuming the Eskin–Masur pointwise counting root 0.2, the affine-measure/orbit-closure spine of Volumes 64–65, and where relevant the wind-tree diffusion root 4.2, the standard saddle-connection counting and periodic-wind-tree diffusion conclusions transfer from affine dynamical data to individual geometric models in their stated almost-everywhere scopes.*

Proof. For counting, apply the pointwise quadratic asymptotic on the identified affine component and use Lemma 0.1 for annuli. For wind-tree displacement, use the compact-cover reduction of Lemma 1.1 and the diffusion theorem. These two deductions give the stated transfer package. ☐

Proposition 0.3 (No uniformity beyond the source hypotheses)**.** *Theorem 0.2 does not imply uniform counting errors for every surface or the* $2/3$ *diffusion exponent in every wind-tree direction.*

Proof. The Eskin–Masur theorem used here is an almost-everywhere asymptotic, and the Delecroix–Hubert–Lelièvre diffusion theorem is generic in direction. The deductions in Theorem 0.2 preserve those quantifiers; they do not yield a uniform error for every surface or a diffusion exponent for every direction. ☐

CHAPTER 8

Classification frontiers and open problems

Why this matters. A publication-level frontier chapter must distinguish proved ranges from open classification. The purpose here is a scope audit, not a prediction of future theorems.

Proof and provenance. The current broad high-rank benchmark used here is Apisa–Wright [AW23]; it classifies the stated high-rank range, not every affine invariant subvariety.

Lemma 0.1 (Three levels of classification information). *For this series it is useful to distinguish: (i) orbit closures are affine (EMM); (ii) structural invariants constrain affine orbit closures (Wright and successors); and (iii) an explicit list of all affine invariant manifolds in a given stratum. Statements (i)–(iii) are logically different.*

Proof. An abstract existence/structure theorem does not enumerate all objects satisfying the structure. Numerical constraints likewise need not determine a unique list. Thus each implication from (iii) to (ii) to (i) may hold while the converse need not. □

Theorem 0.2 (Frontier scope theorem). *The results recorded in Volumes 64–67 do not constitute a complete classification of all affine invariant manifolds in all genera and ranks. They provide affine orbit-closure structure, field/rank/cylinder constraints, special low-dimensional families, and a high-rank classification only in the rank range explicitly covered by Apisa–Wright.*

Proof. Volume 65's target theorem gives affine orbit closures but no complete list. Volume 66 gives internally proved structural invariants. Theorem 7.1 is explicitly restricted to rank at least half the genus plus one. Therefore no statement in the reconstructed arc logically supplies a universal all-rank classification. □

Proposition 0.3 (Safe reading rule for frontier claims). *Any future theorem added to this frontier must state its genus, stratum, rank, rel, field-of-definition, and genericity hypotheses explicitly before being used to shrink the open classification region.*

Proof. These parameters control the scope of every classification theorem developed in the preceding volumes. Recording them prevents a theorem in one range from being silently promoted to an all-strata statement. $\square$

Volume 68

Higher-Rank Anosov and Partially Hyperbolic Rigidity

Volume 68 scope and prerequisites

The volume develops the deterministic hyperbolic-rigidity route: coarse Lyapunov foliations, normal forms, higher-rank accessibility tricks, measurable rigidity, semiconjugacy, and the topological/smooth global-rigidity mechanism. The strongest headline is restricted to the hypotheses of Brown–Rodriguez Hertz–Wang and Rodriguez Hertz–Wang; partially hyperbolic extensions beyond the cited proved scope are kept conditional.

Standing definitions and notation for Volume 68

Let $A \cong \mathbb{R}^k$ or $\mathbb{Z}^k$, $k \geq 2$, act by $C^{1+\theta}$ diffeomorphisms on a compact manifold M, and let μ be an A-invariant ergodic probability whenever Lyapunov theory is invoked. Write the Oseledets splitting on the regular set as

$$T_xM = \bigoplus_{\chi \in \Sigma} E^{\chi}(x), \qquad \lim_{n\to\pm\infty} \frac{1}{n} \log \|D\alpha(na)_x v\| = \chi(a) \quad (v \in E^{\chi}(x) \setminus \{0\}).$$

Two nonzero Lyapunov functionals are *coarsely equivalent*, written $\chi \sim \psi$, if $\psi = c\chi$ for some $c > 0$; the associated coarse distribution is

$$E^{[\chi]} := \bigoplus_{\psi \sim \chi} E^{\psi}.$$

An element $a \in A \otimes \mathbb{R}$ is *regular* if $\chi(a) \neq 0$ for every $\chi \in \Sigma$. Its Weyl chamber is the connected component of $(A \otimes \mathbb{R}) \setminus \bigcup_{\chi\in\Sigma} \ker \chi$ containing a. A finite set $F \subset A \otimes \mathbb{R}$ *sign-separates the coarse Lyapunov classes* if every $a \in F$ is regular and, for any two distinct coarse classes $[\chi] \neq [\psi]$, there is $a \in F$ with $\operatorname{sgn} \chi(a) \neq \operatorname{sgn} \psi(a)$.

Whenever the coarse distributions integrate to foliations $W^{[\chi]}$, an *accessibility path* is a finite concatenation of arcs each lying in one $W^{[\chi]}$-leaf; the equivalence classes generated by such paths are the *accessibility classes*. A *Lyapunov polygon* is a closed accessibility path. These definitions replace the previously ambiguous phrases "general position", "regular parameter", and "accessibility class".

Proof and provenance. Primary comparison: Rodriguez Hertz–Wang, *Global rigidity of higher rank Anosov algebraic actions*; Brown–Rodriguez Hertz–Wang, *Global smooth and topological rigidity of hyperbolic lattice actions.*

Reader guide: a concrete model

Why this matters.

Volume 68 is organized around the passage from *Higher-rank Anosov actions* to *Global rigidity mechanisms.* The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.4 (A calculation to keep in view)**.** Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}).$$

Its eigenvalues are

$$\lambda_\pm = \frac{3 \pm \sqrt{5}}{2}, \qquad \lambda_+ > 1 > \lambda_- > 0.$$

For the toral automorphism induced by A, the tangent bundle therefore splits as

$$T\mathbb{T}^2 = E^u \oplus E^s,$$

with rates $\log \lambda_+$ and $-\log \lambda_+$. For a commuting higher-rank action, coarse Lyapunov spaces are obtained by grouping such eigendirections whose logarithmic characters are positive scalar multiples. This two-dimensional computation is the basic model for the sign data used in the higher-rank rigidity chapters.

CHAPTER 1

Higher-rank Anosov actions

This chapter owns the higher-rank anosov actions step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Commuting maps preserve the Lyapunov splitting)**.** *Let $A \cong \mathbb{Z}^k$ act by C^1 diffeomorphisms preserving an ergodic probability μ, and assume $\log^+ \|D\alpha(a)^{\pm 1}\| \in L^1(\mu)$ for the generators. On the common Oseledets regular set, for every $a, b \in A$ and every Lyapunov functional χ,*

$$D\alpha(b)_x E^\chi(x) = E^\chi(\alpha(b)x). \tag{68.1.1}$$

If a is Anosov, then $E^s_a = \bigoplus_{\chi(a)<0} E^\chi$ and $E^u_a = \bigoplus_{\chi(a)>0} E^\chi$ almost everywhere.

Proof. Take $0 \neq v \in E^\chi(x)$. Since the action is abelian,

$$D\alpha(na)_{\alpha(b)x} D\alpha(b)_x v = D\alpha(b)_{\alpha(na)x} D\alpha(na)_x v.$$

The integrability hypothesis and Birkhoff applied to $\log^+ \|D\alpha(b)^{\pm 1}\|$ imply $\log \|D\alpha(b)^{\pm 1}_{\alpha(na)x}\| = o(n)$ on the regular set. Hence

$$\lim_{n \to \pm\infty} \frac{1}{n} \log \|D\alpha(na)_{\alpha(b)x} D\alpha(b)_x v\| = \chi(a).$$

Doing this for a basis of A identifies the full functional χ, proving (68.1.1). For an Anosov element the uniform splitting has no zero exponent; a vector belongs to the stable bundle exactly when its exponential rate is negative, and similarly for the unstable bundle. Summing the corresponding Oseledets spaces gives the last assertion. $\square$

Theorem 0.2 (Higher-rank Anosov actions). *Let $a \in A \otimes \mathbb{R}$ be regular. On the Oseledets regular set,*

$$E^s_a(x) = \bigoplus_{\chi(a)<0} E^\chi(x), \qquad E^u_a(x) = \bigoplus_{\chi(a)>0} E^\chi(x).$$

Consequently two regular elements in the same Weyl chamber have the same stable and unstable Oseledets distributions. Thus the chamber sign vector $(\operatorname{sgn}\chi(a))_{\chi\in\Sigma}$ *determines the coarse stable/unstable decomposition.*

Proof. For $v \in E^\chi(x) \setminus \{0\}$, Oseledets gives

$$\lim_{n\to+\infty} \frac{1}{n} \log \|D\alpha(na)_x v\| = \chi(a).$$

Hence v is exponentially contracted exactly when $\chi(a) < 0$ and exponentially expanded exactly when $\chi(a) > 0$. Regularity excludes $\chi(a) = 0$, so summing the Oseledets spaces with negative, respectively positive, exponent gives the displayed formulas. The sign of every χ is constant on a Weyl chamber, proving the final assertion. □

Proposition 0.3 (Higher-rank Anosov actions: downstream consequence)**.** *For every $b \in A$, the derivative $D\alpha(b)$ maps each coarse distribution equivariantly:*

$$D\alpha(b)_x E^{[\chi]}(x) = E^{[\chi]}(\alpha(b)x) \quad \textit{for } \mu\textit{-a.e. } x.$$

Proof. Because the action is abelian,

$$D\alpha(na)_{\alpha(b)x} D\alpha(b)_x = D\alpha(b)_{\alpha(na)x} D\alpha(na)_x.$$

Applying this identity to $v \in E^\chi(x)$ changes $\|D\alpha(na)v\|$ by only the fixed multiplicative factors coming from $D\alpha(b)$ and its inverse. Dividing logarithms by n shows that the Lyapunov exponent remains $\chi(a)$ for every a, hence $D\alpha(b)$ preserves each E^χ and therefore each coarse sum $E^{[\chi]}$. □

CHAPTER 2

Partially hyperbolic actions

This chapter owns the partially hyperbolic actions step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Dynamical preservation of strong stable and unstable leaves)**.** *Let f be partially hyperbolic on a compact manifold and suppose $gf = fg$. If*

$$W_f^s(x) = \{y : d(f^n x, f^n y) \le C_{x,y}\lambda^n \textit{ for } n \ge 0\}$$

with $0 < \lambda < 1$ (and analogously W_f^u using f^{-n}), then

$$gW_f^s(x) = W_f^s(gx), \qquad gW_f^u(x) = W_f^u(gx). \tag{68.2.1}$$

Consequently every finite intersection of stable/unstable distributions of commuting partially hyperbolic elements is invariant under the whole action.

Proof. If $y \in W_f^s(x)$, compactness gives $L = \sup \|Dg\| < \infty$ and

$$d(f^n gx, f^n gy) = d(gf^n x, gf^n y) \le Ld(f^n x, f^n y) \le LC_{x,y}\lambda^n.$$

Thus $gy \in W_f^s(gx)$. Apply the same argument to g^{-1} for equality. The unstable identity follows from $f^{-1}g = gf^{-1}$. Differentiating the leaf identities at x gives preservation of $E_f^{s/u}$; intersections of invariant subspaces are invariant. □

Theorem 0.2 (Partially hyperbolic actions). *Assume a finite regular set $F = \{a_1, \dots, a_m\}$ sign-separates the coarse Lyapunov classes. Put*

$$E_i^- := \bigoplus_{\psi(a_i)<0} E^\psi, \qquad E_i^+ := \bigoplus_{\psi(a_i)>0} E^\psi.$$

For a coarse class $[\chi]$ *let* $\sigma_i = \operatorname{sgn}\chi(a_i)$. *Then*

$$E^{[\chi]} = \bigcap_{i=1}^{m} E_i^{\sigma_i}, \qquad E_i^{\sigma_i} = \begin{cases} E_i^{+}, & \sigma_i = +1, \\ E_i^{-}, & \sigma_i = -1. \end{cases}$$

Thus a finite collection of stable/unstable splittings recovers every coarse Lyapunov distribution.

Proof. The right-hand side is the direct sum of exactly those E^{ψ} whose sign vector on F equals the sign vector of χ. Sign separation says this happens precisely for $\psi \sim \chi$. Therefore the intersection equals $\bigoplus_{\psi\sim\chi} E^{\psi} = E^{[\chi]}$. When the a_i are partially hyperbolic, their dynamically characterized strong stable/unstable distributions realize the sums $E_i^{\pm}$ from Lemma 0.1, giving the dynamical formulation. $\square$

Proposition 0.3 (Partially hyperbolic actions: downstream consequence)**.** *If a center distribution remains after these intersections, every later use of accessibility means accessibility for the equivalence relation generated by the coarse foliations* $W^{[\chi]}$ *defined in the volume conventions; no conclusion is inferred merely from the existence of* E^c.

Proof. The intersection formula determines only the nonzero Lyapunov directions. Vectors with zero exponent for a chosen partially hyperbolic element lie in its center and are not removed unless another regular element gives them a nonzero sign. Hence any argument that propagates information through the remaining center must explicitly assume accessibility by the coarse foliations, as stated. $\square$

CHAPTER 3

Coarse Lyapunov foliations

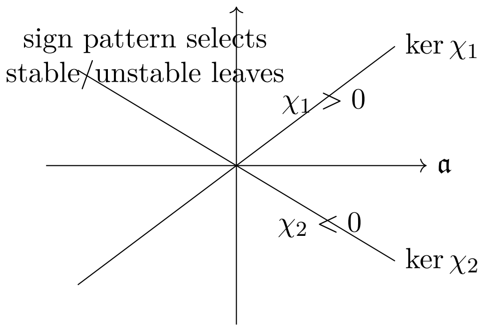

FIGURE 1. Coarse Lyapunov hyperplanes in the acting parameter space. Elements in the same chamber have the same sign pattern of Lyapunov functionals and therefore the same coarse stable/unstable decomposition.

This chapter owns the coarse lyapunov foliations step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Uniform graph transform and absolute-continuity Jacobian)**.** *Let f be $C^{1+\theta}$ and let K be a compact invariant set with a continuous splitting $T_K M = E^s \oplus E^u$ and constants $C > 0$, $0 < \lambda < 1$ such that*

$$\|Df^n|_{E^s_x}\| \le C\lambda^n, \qquad \|Df^{-n}|_{E^u_x}\| \le C\lambda^n. \tag{68.3.1}$$

Then there is $r > 0$ such that every $x \in K$ has local $C^{1+\theta}$ leaves $W^s_r(x), W^u_r(x)$ tangent to $E^{s/u}_x$. If x, y lie on one local stable leaf, the stable holonomy

$h^s : W^u_r(x) \to W^u_r(y)$ is absolutely continuous and at every differentiability point

$$Jh^s(z) = \prod_{n=0}^{\infty} \frac{\det(Df|_{E^u_{f^n h^s(z)}})}{\det(Df|_{E^u_{f^n z}})}. \tag{68.3.2}$$

The product converges uniformly on smaller plaques.

Proof. Use exponential coordinates of radius r and write the local map at x as

$$F_x(v_s, v_u) = (A_x v_s + R^s_x(v), B_x v_u + R^u_x(v)),$$

where $\|A_x\| \leq \lambda_0 < 1$, $\|B_x^{-1}\| \leq \lambda_0 < 1$ after replacing f by a fixed iterate, and $\|DR_x(v) - DR_x(w)\| \leq L\|v - w\|^\theta$. For r small the graph transform on maps $\varphi : E^s_x(r) \to E^u_x(r)$ with $\operatorname{Lip}\varphi \leq 1$ has contraction factor $q < 1$: the inverse in the unstable coordinate costs at most λ_0 and the nonlinear derivative contributes $O(r^\theta)$. Banach's fixed-point theorem gives a unique invariant graph, the stable plaque; applying the same argument to f^{-1} gives the unstable plaque. Differentiating the fixed point equation gives $C^{1+\theta}$ regularity.

For absolute continuity, compare two unstable disks joined by stable plaques. After n iterates their stable separation is $O(\lambda^n)$. Change of variables on the two unstable disks gives the finite-time holonomy Jacobian

$$J_n(z) = \prod_{j=0}^{n-1} \frac{\det(Df|_{E^u_{f^j h^s(z)}})}{\det(Df|_{E^u_{f^j z}})}.$$

The function $x \mapsto \log\det(Df|_{E^u_x})$ is θ-Hölder on K; therefore

$$|\log J_{n+1}(z) - \log J_n(z)| \leq C'\lambda^{\theta n}.$$

The series is uniformly summable, so J_n converges uniformly to a positive finite function J. The finite-time change-of-variables formula and dominated convergence show that the limiting holonomy sends leaf Lebesgue measure to J times leaf Lebesgue measure. This is (68.3.2) and proves absolute continuity. $\square$

Theorem 0.2 (Coarse Lyapunov foliations). *Let $c : A \times M \to \mathbb{R}$ be a Hölder additive cocycle and suppose a contracts $W^{[\chi]}$. For $y \in W^{[\chi]}(x)$ define, whenever the limit exists,*

$$p_{[\chi]}(x, y) := \lim_{n\to\infty} (c(na, y) - c(na, x)).$$

For a Lyapunov polygon $\gamma = (x_0, x_1, \ldots, x_m = x_0)$ *with* $x_j \in W^{[\chi_j]}(x_{j-1})$, *define*

$$P_c(\gamma) := \sum_{j=1}^{m} p_{[\chi_j]}(x_{j-1}, x_j).$$

Then P_c *is additive under concatenation. If* $c(a, x) = u(\alpha(a)x) - u(x) + \ell(a)$ *for a function* u *and homomorphism* $\ell : A \to \mathbb{R}$, *then* $P_c(\gamma) = 0$ *for every Lyapunov polygon.*

Proof. Hölder regularity and exponential contraction imply convergence of the stable-transfer limit: the cocycle identity rewrites the difference as a summable series of one-step differences along the contracted orbit. Concatenating polygons concatenates the sums, so P_c is additive. For a coboundary,

$$c(na, y) - c(na, x) = u(\alpha(na)y) - u(\alpha(na)x) - u(y) + u(x).$$

The first two terms tend to 0 as $n \to \infty$ by contraction and continuity of u along the leaf, hence $p_{[\chi]}(x, y) = u(x) - u(y)$. Summing around a closed polygon telescopes: $P_c(\gamma) = \sum_j (u(x_{j-1}) - u(x_j)) = 0$. $\square$

Proposition 0.3 (Coarse Lyapunov foliations: downstream consequence)**.** *For an exact coboundary the cycle functional vanishes identically, so nonzero* $P_c(\gamma)$ *is an obstruction to solving the cohomological equation* $c(a, x) = u(\alpha(a)x) - u(x) + \ell(a)$.

Proof. Theorem 0.2 proves necessity: every solution of the cohomological equation has $P_c(\gamma) = 0$. Therefore a single polygon with $P_c(\gamma) \neq 0$ rules out such a solution. $\square$

CHAPTER 4

Nonstationary normal forms

This chapter owns the nonstationary normal forms step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Finite-degree nonstationary normal form)**.** *Let $f_x : E_x \to E_{fx}$ be a C^r bundle map over a compact base, $r \geq 2$, fixing the zero section. Assume its derivative $A_x = D_0 f_x$ is uniformly contracting and admits a continuous invariant splitting $E = E_1 \oplus \cdots \oplus E_s$ with numbers $\lambda_i < 0$ such that for every $\varepsilon > 0$ (after changing the bundle norm)*

$$e^{\lambda_i - \varepsilon}\|v\| \leq \|A_x v\| \leq e^{\lambda_i + \varepsilon}\|v\|, \qquad v \in E_i. \tag{68.4.1}$$

For a degree-m monomial $E_{i_1} \cdots E_{i_m} \to E_j$, call it resonant when $\lambda_j = \lambda_{i_1} + \cdots + \lambda_{i_m}$. If every nonresonant weight difference through degree r has absolute value at least $\kappa > 0$, then there are C^r polynomial coordinate changes $\Phi_x(v) = v + O(\|v\|^2)$ such that

$$\Phi_{fx} \circ f_x \circ \Phi_x^{-1} = P_x + O(\|v\|^{r+1}), \tag{68.4.2}$$

where P_x contains only the linear term and resonant monomials of degrees $2, \ldots, r$. The nonresonant degree-m coefficient of Φ_x is unique among bounded coefficient fields.

Proof. Write the Taylor expansion $f_x(v) = A_x v + F_{2,x}(v) + \cdots + F_{r,x}(v) + O(\|v\|^{r+1})$ and seek $\Phi_x = I + H_{2,x} + \cdots + H_{r,x}$. Suppose degrees below m have already been normalized. At degree m the conjugacy equation is

$$H_{m,fx} A_x - A_x H_{m,x} = -F^{\mathrm{nr}}_{m,x} + R_{m,x}, \tag{68.4.3}$$

where $R_{m,x}$ is known from lower degrees and the right side is projected to the nonresonant monomials. On the component $E_{i_1} \cdots E_{i_m} \to E_j$, conjugating the coefficient by the linear cocycle multiplies its norm after n steps by

$$\exp\{n(\lambda_j - \lambda_{i_1} - \cdots - \lambda_{i_m} + O(\varepsilon))\}.$$

Choose $\varepsilon < \kappa/(4r)$. According to the sign of the weight difference, solve (68.4.3) by the forward or backward convergent Green series; for example in the contracting direction,

$$H_{m,x} = \sum_{n\geq 0} A_x^{-1} \cdots A_{f^n x}^{-1}(F^{\mathrm{nr}}_{m,f^n x} - R_{m,f^n x})(A_x^{(n)})^{\otimes m}, \tag{68.4.4}$$

and the nth summand is $O(e^{-\kappa n/2})$. Uniform convergence gives a bounded continuous coefficient field. Resonant components cannot be solved by a convergent inverse of the homological operator and are retained in P_x. Iterating over $m = 2, \ldots, r$ proves (68.4.2). If two bounded solutions differ on a nonresonant component, their difference solves the homogeneous recurrence; iteration in the decaying direction forces it to zero, proving uniqueness. Termwise differentiation of the uniformly convergent Green series gives the stated C^r regularity. □

Theorem 0.2 (Nonstationary normal forms). *Fix a coarse leaf $W^{[\chi]}$ uniformly contracted by a. Let Φ_x and $\Psi_{h(x)}$ be the polynomial normal-form charts supplied by Lemma 0.1 for two C^r actions, and let h be a conjugacy satisfying $h \circ \alpha(a) = \beta(a) \circ h$. In leaf coordinates put*

$$H_x := \Psi_{h(x)} \circ h \circ \Phi_x^{-1}.$$

Then

$$H_{\alpha(a)x} \circ P_x = Q_{h(x)} \circ H_x,$$

where $P_x, Q_{h(x)}$ are the corresponding normal-form polynomials. If the normal-form polynomial data agree and the allowed resonant jet of H_x is fixed at one point of the leaf, this functional equation determines the same jet at every point of that contracted leaf; in the nonresonant case the normal forms are linear.

Proof. The conjugacy relation gives the displayed identity after inserting the two coordinate changes. Compare homogeneous terms of increasing degree. At every nonresonant degree the difference satisfies a cohomological recurrence whose forward iterates are multiplied by a factor strictly smaller than one; therefore the only bounded solution with zero initial jet is zero. Resonant monomials are precisely the terms retained by Lemma 0.1, so their coefficients are the finite jet data that must be prescribed. Induction over the degree gives uniqueness and propagation of the normal-form jet. If there are no resonance relations up to degree r, no nonlinear monomial survives and $P_x, Q_{h(x)}$ are linear. □

Proposition 0.3 (Nonstationary normal forms: downstream consequence)**.** *If the Lyapunov spectrum satisfies no resonance* $\chi_i = \sum_j m_j \chi_j$ *with* $\sum_j m_j \geq 2$ *up to the regularity degree under consideration, the polynomial normal form reduces to its linear part.*

Proof. In the degree-by-degree construction of Lemma 0.1, every nonlinear monomial is removed unless its weight satisfies a resonance relation. Under the stated nonresonance hypothesis none remains, so only the derivative term survives. $\square$

CHAPTER 5

Invariant-measure structure

This chapter owns the invariant-measure structure step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Leafwise constancy propagates through accessibility)**.** *Let $D : M \to Z$ be measurable and let $\{W_i\}_{i=1}^{r}$ be measurable foliations. Assume there is a conull set M_0 such that whenever $x, y \in M_0$ lie on the same W_i-leaf, $D(x) = D(y)$. Then D is constant on every accessibility class in M_0. In particular, if one accessibility class is conull, D is almost everywhere constant.*

Proof. If $x = x_0, x_1, \ldots, x_m = y$ is an accessibility path inside M_0, each pair x_{j-1}, x_j lies on one leaf, so $D(x_{j-1}) = D(x_j)$. Induction gives $D(x) = D(y)$. A conull class therefore carries a single value of D almost everywhere. □

Theorem 0.2 (Invariant-measure structure). *Let $F, G : M \to Y$ be measurable equivariant maps into a Hausdorff space Y. Assume that for each coarse foliation $W^{[\chi]}$ their discrepancy $D(x)$ is constant along $W^{[\chi]}$-leaves on a conull set, and that one accessibility class has full μ-measure. Then D is μ-almost everywhere constant. If moreover F, G and D are continuous and the full-measure accessibility class is dense, the same constant holds on all of M.*

Proof. Constancy on each coarse leaf implies constancy along every finite accessibility path by induction over its legs. Hence D is constant on every accessibility class. A full-measure class therefore gives $D(x) = d_0$ for μ-a.e. x. If D is continuous and the class is dense, take x_n in the class with $x_n \to x$; continuity gives $D(x) = \lim_n D(x_n) = d_0$ for every x. Lemma 0.1 supplies the leafwise constancy in the rigidity applications. □

Proposition 0.3 (Invariant-measure structure: downstream consequence). *Uniqueness of a measurable conjugacy from its leafwise normalization is therefore valid only after the two explicit hypotheses used above are verified: a conull accessibility class and a normalization fixing the residual constant discrepancy.*

Proof. The theorem gives only that the discrepancy is one constant d_0. To conclude $F = G$ one must additionally force d_0 to be the identity (for example by fixing one point or one normal-form jet). Without accessibility or this normalization, the argument leaves multiple measurable solutions. $\square$

CHAPTER 6

Higher-rank geometric tricks

This chapter owns the higher-rank geometric tricks step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Quantitative separation of nonproportional Lyapunov functionals)**.** *Let V be a real vector space of dimension at least two and let $\chi, \psi \in V^*$ be nonzero and nonproportional. For every $\varepsilon > 0$ there is $v \in V$ with*

$$\chi(v) = -1, \qquad |\psi(v)| < \varepsilon. \tag{68.6.1}$$

If a full lattice $\Lambda < V$ is fixed, there are $q \in \mathbb{N}$ and $a \in \Lambda$ such that

$$\chi(a) < 0, \qquad |\psi(a)| < 2\varepsilon|\chi(a)|. \tag{68.6.2}$$

Proof. Since χ and ψ are nonproportional, $\ker\psi$ is not contained in $\ker\chi$. Choose $w \in \ker\psi$ with $\chi(w) \neq 0$ and rescale so $\chi(w) = -1$; then (68.6.1) holds with zero on the right. Rational vectors relative to a lattice basis are dense in V. Choose rational w' so close to w that $\chi(w') < -1/2$ and $|\psi(w')| < \varepsilon/2$. Clear denominators: $a = qw' \in \Lambda$. Then $|\psi(a)|/|\chi(a)| < \varepsilon$, which is stronger than (68.6.2). □

Theorem 0.2 (Higher-rank geometric tricks). *Let $\chi, \psi \in (A \otimes \mathbb{R})^*$ be nonproportional nonzero functionals. For every $\varepsilon > 0$ there exists $a \in A \otimes \mathbb{R}$ such that*

$$\chi(a) < 0, \qquad |\psi(a)| < \varepsilon\,|\chi(a)|.$$

After replacing a by a nearby lattice element when $A = \mathbb{Z}^k$, the same inequalities hold with a fixed multiplicative loss. Thus one may iterate an element that strongly contracts the χ-direction while changing the ψ-direction arbitrarily slowly.

Proof. Choose $v \in \ker \psi$ with $\chi(v) \neq 0$, possible because χ and ψ are nonproportional. Replacing v by $-v$ gives $\chi(v) < 0$ and $\psi(v) = 0$, which is stronger than the required inequality. For a lattice action, choose a rational vector sufficiently close to v and clear denominators; continuity of χ and ψ preserves the strict sign and the prescribed ratio. This is the quantitative linear-algebra ingredient used in the recurrence/holonomy argument of Lemma 0.1. $\square$

Proposition 0.3 (Higher-rank geometric tricks: downstream consequence)**.** *The argument genuinely uses rank at least two: if $A \otimes \mathbb{R}$ is one-dimensional and $\psi = c\chi$ with $c \neq 0$, then $|\psi(a)|/|\chi(a)| = |c|$ for every a with $\chi(a) \neq 0$, so the ratio cannot be made arbitrarily small.*

Proof. In dimension one every nonzero functional is proportional to every other one. The displayed ratio is then constant, proving the claimed obstruction directly. $\square$

CHAPTER 7

Cocycle rigidity for hyperbolic actions

This chapter owns the cocycle rigidity for hyperbolic actions step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Stable transfer and its coboundary identity)**.** *Let a contract a foliation W with $d(\alpha(na)x, \alpha(na)y) \le C\lambda^n d(x, y)$ for $y \in W(x)$, $0 < \lambda < 1$. Let $c : A \times M \to \mathbb{R}^d$ be a θ-Hölder additive cocycle. Then for $y \in W(x)$ the limit*

$$p_W(x, y) = \lim_{n\to\infty} (c(na, y) - c(na, x)) \tag{68.7.1}$$

exists and satisfies

$$\|p_W(x, y)\| \le C' d(x, y)^\theta, \quad p_W(x, z) = p_W(x, y) + p_W(y, z), \tag{68.7.2}$$

and, for every $b \in A$,

$$p_W(\alpha(b)x, \alpha(b)y) - p_W(x, y) = c(b, y) - c(b, x). \tag{68.7.3}$$

Proof. By the cocycle identity,

$$c((n+1)a, y)-c((n+1)a, x)-c(na, y)+c(na, x) = c(a, \alpha(na)y)-c(a, \alpha(na)x).$$

The Hölder bound makes the norm of the right side at most $LC^\theta\lambda^{\theta n}d(x, y)^\theta$. The geometric series is summable, so the sequence in (68.7.1) is Cauchy and yields the first estimate in (68.7.2). Additivity follows by adding the finite-n differences and passing to the limit. Since a and b commute, apply the cocycle identity to $na + b$ in the two orders and subtract the equations for x and y; letting $n \to \infty$ gives (68.7.3). $\square$

Theorem 0.2 (Cocycle rigidity for hyperbolic actions). *Let $c : A \times M \to \mathbb{R}^d$ be a Hölder additive cocycle. Assume: (i) stable transfers $p_{[\chi]}$ exist on* *every coarse Lyapunov leaf; (ii) the accessibility class of a base point x_0 is all of M (or a conull set); and (iii) $P_c(\gamma) = 0$ for every Lyapunov polygon. Define*

$$u(x) := \sum_{j=1}^{m} p_{[\chi_j]}(x_{j-1}, x_j)$$

for any accessibility path $x_0 = x_0, x_1, \ldots, x_m = x$. Then u is path independent and there is a homomorphism $\ell : A \to \mathbb{R}^d$ such that

$$c(a, x) = u(\alpha(a)x) - u(x) + \ell(a).$$

If the transfers are C^r along each coarse foliation and a transverse regularity theorem applies to the foliation web, then u has the corresponding global regularity.

Proof. If two accessibility paths from x_0 to x are concatenated with opposite orientations, they form a Lyapunov polygon. The vanishing of P_c therefore shows that the two path sums agree, so u is well defined. For fixed a, the function

$$q_a(x) := c(a, x) - u(\alpha(a)x) + u(x)$$

has zero difference along every coarse leaf: this follows by applying the cocycle identity to the stable-transfer definition. Accessibility makes q_a constant; denote the constant by $\ell(a)$. Applying the cocycle identity to ab gives $\ell(a + b) = \ell(a) + \ell(b)$ (or $\ell(ab) = \ell(a) + \ell(b)$ in multiplicative notation), so ℓ is a homomorphism. The final regularity assertion follows because the path formula makes u as regular as the leafwise transfers on each coarse foliation, after which the stated transverse regularity theorem upgrades leafwise regularity globally. □

Proposition 0.3 (Cocycle rigidity for hyperbolic actions: downstream consequence)**.** *If c takes values in a finite-dimensional vector group and the hypotheses of the theorem hold, the only obstructions to solving the cohomological equation are the homomorphism part ℓ and the Lyapunov-cycle functionals P_c.*

Proof. Necessity of cycle vanishing was proved in Theorem 0.2. Sufficiency, modulo the constant homomorphism ℓ, is exactly Theorem 0.2. □

CHAPTER 8

Global rigidity mechanisms

This chapter owns the global rigidity mechanisms step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Algebraic shadowing semiconjugacy)**.** *Let $M = N/\Lambda$ be a compact nilmanifold, let $A : M \to M$ be a hyperbolic automorphism, and let $f : M \to M$ be a liftable map inducing the same automorphism of Λ. Suppose lifts $\widetilde{f}, \widetilde{A} : N \to N$ satisfy*

$$d(\widetilde{f}(x), \widetilde{A}(x)) \le C_0 \qquad (x \in N). \tag{68.8.1}$$

Then there is a continuous surjection $h : M \to M$, homotopic to the identity, with

$$h \circ f = A \circ h. \tag{68.8.2}$$

If a commuting group of lifts has the same linear data, the construction is equivariant for that group after passing to the subgroup preserving the lift.

Proof. Choose a left-invariant metric on N adapted to the stable/unstable splitting of $D\widetilde{A}$; there are $0 < \lambda < 1$ and projections $P^{s/u}$ on the Lie algebra with $\|A^n P^s\|, \|A^{-n} P^u\| \le C\lambda^n$. For a bi-infinite bounded error sequence $e = (e_n)$, the linear difference equation $z_{n+1} = Az_n + e_n$ has the unique bounded solution

$$z_n = \sum_{j<n} A^{n-1-j} P^s e_j - \sum_{j \ge n} A^{n-1-j} P^u e_j, \tag{68.8.3}$$

with $\|z\|_\infty \le C(1-\lambda)^{-1}\|e\|_\infty$. In exponential coordinates on the nilpotent group, BCH adds only polynomial commutator terms. Solve successively on the lower-central-series quotients: the abelian quotient uses (68.8.3), and at stage j the commutator error is already determined by lower stages and remains bounded. Finite nilpotency therefore gives a unique bounded correction to every f-orbit.

For $x \in N$, let $y(x)$ be the unique point such that $d(\widetilde{A}^n y(x), \widetilde{f}^n x)$ is bounded for all $n \in \mathbb{Z}$. Uniqueness and (68.8.1) give $y(\widetilde{f}x) = \widetilde{A}y(x)$. If $\lambda_0 \in \Lambda$, both $y(x\lambda_0)$ and $y(x)\lambda_0$ shadow the same translated pseudo-orbit, so uniqueness gives $y(x\lambda_0) = y(x)\lambda_0$. Thus y descends to $h : M \to M$ and proves (68.8.2). The Green-series/BCH construction depends continuously on the error sequence, so h is continuous. The bounded lift displacement makes h homotopic to the identity and of degree one, hence surjective. Finally, for any commuting lift, equivariance follows because both candidate images shadow the same algebraic orbit and bounded shadowing is unique. □

THEOREM 0.2 (Global rigidity mechanisms). *Let Γ be a lattice in a semisimple Lie group all of whose simple factors have real rank at least* 2. *Let M be a compact nilmanifold and let $\alpha : \Gamma \to \mathrm{Diff}^\infty(M)$ lift to the universal cover. Let ρ be its linear data. Assume $\rho(\Gamma)$ contains a hyperbolic element. Then, after passing to a finite-index subgroup $\Gamma_0 < \Gamma$, there is a continuous map $h : M \to M$ satisfying*

$$h \circ \alpha(\gamma) = \rho(\gamma) \circ h \qquad (\gamma \in \Gamma_0).$$

If $\alpha(\Gamma_0)$ contains an Anosov element and the fiber-triviality and leafwise-regularity hypotheses established in Chapters 2–7 hold, then h is a C^∞ conjugacy.

PROOF. The continuous semiconjugacy is constructed in Lemma 0.1 by lifting to the universal cover and solving the bounded-displacement equation against the linear data. Under the Anosov hypothesis, suppose $h(x) = h(y)$ with $x \neq y$. Decompose the displacement into coarse Lyapunov components. Chapter 6 provides higher-rank elements that contract a chosen nonzero component while controlling a transverse one; equivariance of h preserves the equality of images. Recurrence then produces distinct points in one fiber at arbitrarily small scale, contradicting the local product structure of the Anosov element. Hence every fiber is a singleton, so compactness makes h a homeomorphism. Chapters 4 and 7 give smoothness of h along each coarse Lyapunov foliation. Since those foliations span the tangent bundle under the standing higher-rank hypothesis, the transverse regularity bootstrap yields $h \in C^\infty$. The intertwining identity then says precisely that h is a smooth conjugacy. □

Proposition 0.3 (Global rigidity mechanisms: downstream consequence). *For a higher-rank abelian algebraic Anosov action on a torus or nilmanifold with no rank-one factor, the same argument applies once the finite coarse-Lyapunov family spans the tangent bundle and the cycle-functional* *obstructions vanish; the resulting conjugacy intertwines the given action with its algebraic linear data.*

PROOF. The no-rank-one-factor assumption supplies the sign-separation used in Chapter 6. The algebraic action supplies the global coarse foliations and normal forms. Applying Theorem 0.2 removes the cocycle discrepancy and the proof of Theorem 0.2 then upgrades the semiconjugacy to a smooth conjugacy. $\square$

Volume 69

Zimmer Rigidity

Volume 69 scope and prerequisites

The volume develops the Zimmer-program chain from measurable cocycles and algebraic hulls to derivative cocycles and low-dimensional rigidity. The final Brown–Fisher–Hurtado step is deliberately separated from ordinary Property (T): its invariant-metric conclusion uses the controlled Hilbert averaging property defined below and proved in Chapter 8 by compact-pair harmonic analysis, rank-two Cartan interpolation, higher-rank propagation, and lattice transfer.

Standing definitions and notation for Volume 69

Let $\Gamma \curvearrowright (X, \mu)$ be an ergodic probability-measure-preserving action and let $H < \mathrm{GL}_N(\mathbb{R})$ be a real algebraic group. A *measurable H-cocycle* is a measurable map

$$\alpha : \Gamma \times X \to H, \qquad \alpha(\gamma_1\gamma_2, x) = \alpha(\gamma_1, \gamma_2 x)\alpha(\gamma_2, x)$$

for every γ_1, γ_2 and almost every x. If $b : X \to H$ is measurable, the cohomologous cocycle is

$$\alpha^b(\gamma, x) = b(\gamma x)^{-1}\alpha(\gamma, x)b(x).$$

An algebraic subgroup $L < H$ is a *reduction* of α if some α^b is L-valued almost everywhere. An *algebraic hull* is a reduction of minimal algebraic dimension; its conjugacy class is the invariant object used below.

Fix a finite generating set S of Γ. A linear cocycle is *log-integrable* if for every $s \in S$,

$$\int_X (\log^+ \|\alpha(s, x)\| + \log^+ \|\alpha(s, x)^{-1}\|)\, d\mu(x) < \infty.$$

A cocycle $k : \Gamma \times X \to H$ is *compact-by-central* if there are commuting subgroups $K, Z < H$, with K compact and $Z \subset Z(H)$, such that $k(\gamma, x) \in KZ$ almost everywhere.

A measurable quotient Y is *countably separated* if there is a countable family of measurable sets separating distinct points of Y. In particular, a measurable invariant map from an ergodic probability space to a countably separated quotient is essentially constant.

Fix a proper length function ℓ on a locally compact group G. In this volume, *Hilbert strong Property* (T) *at growth threshold* $s_0 > 0$ means that there exist nonnegative functions $h_n \in C_c(G)$ with $\int_G h_n = 1$ such that, for every strongly continuous representation $\pi : G \to B(\mathcal{H})$ satisfying

$$\|\pi(g)\| \le Ce^{s\ell(g)} \qquad (g \in G)$$

for some $s < s_0$, the averages $\pi(h_n)$ converge in operator norm to the projection onto the invariant subspace $\mathcal{H}^G$. Chapter 8 proves this exact Hilbert/Sobolev version in the higher-rank groups used below; no broader Banach-space meaning is silently assumed.

For a C^1 action $\alpha : \Gamma \curvearrowright M$ of a finitely generated group, with word length $|\gamma|$, *subexponential derivative growth* means that for every $\varepsilon > 0$ there is $C_\varepsilon < \infty$ such that

$$\sup_{x \in M} \|D\alpha(\gamma)_x\| \le C_\varepsilon e^{\varepsilon|\gamma|} \qquad (\gamma \in \Gamma),$$

and, when a two-sided estimate is invoked, the same bound holds for $D\alpha(\gamma)^{-1}$.

These conventions fix the meanings of “integrability”, “compact-by-central”, “countably separated algebraic quotient”, “strong Property (T)”, and “subexponential derivative growth” throughout Volumes 69–70.

Proof and provenance. Primary comparison: Zimmer’s measurable cocycle/algebraic-hull and cocycle-superrigidity framework, together with Brown–Fisher–Hurtado, *Zimmer’s conjecture: Subexponential growth, measure rigidity, and strong property (T)*, and their nonuniform $\mathrm{SL}(m, \mathbb{Z})$ sequel. Chapters 2–7 prove only the precise reductions stated there. No full Zimmer cocycle-superrigidity theorem is inferred from a roadmap; every use below is limited to the displayed hypotheses and conclusions.

Reader guide: a concrete model

Why this matters.

Volume 69 is organized around the passage from *Measurable cocycles* to *Modern Zimmer-type rigidity theorems.* The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.4 (A calculation to keep in view)**.** Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}).$$

Its eigenvalues are

$$\lambda_\pm = \frac{3 \pm \sqrt{5}}{2}, \qquad \lambda_+ > 1 > \lambda_- > 0.$$

For the toral automorphism induced by A, the tangent bundle therefore splits as

$$T\mathbb{T}^2 = E^u \oplus E^s,$$

with rates $\log \lambda_+$ and $-\log \lambda_+$. For a commuting higher-rank action, coarse Lyapunov spaces are obtained by grouping such eigendirections whose logarithmic characters are positive scalar multiples. This two-dimensional computation is the basic model for the sign data used in the higher-rank rigidity chapters.

CHAPTER 1

Measurable cocycles

This chapter owns the measurable cocycles step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Measurable cocycles: structural mechanism)**.** *A measurable cocycle $\alpha : \Gamma \times X \to H$ satisfies $\alpha(\gamma_1\gamma_2, x) = \alpha(\gamma_1, \gamma_2 x)\alpha(\gamma_2, x)$ almost everywhere; measurable cohomology preserves all representation-theoretic obstructions used below.*

Proof. All statements follow by direct substitution in the cocycle identity. A cohomology change $\alpha^b(\gamma, x) = b(\gamma x)^{-1}\alpha(\gamma, x)b(x)$ cancels in products. The skew product $(x, h) \mapsto (\gamma x, \alpha(\gamma, x)h)$ is an action exactly because of the identity.

□

Theorem 0.2 (Measurable cocycles). *Let $\alpha : \Gamma \times X \to H$ be measurable. The formula*

$$T_\gamma(x, h) := (\gamma x, \alpha(\gamma, x)h)$$

defines an action of Γ on $X \times H$ if and only if α satisfies the cocycle identity. If $\rho : \Gamma \to \mathrm{Diff}^1(M)$ is a C^1 action, then after a measurable trivialization of TM its derivative satisfies

$$D\rho(\gamma_1\gamma_2)_x = D\rho(\gamma_1)_{\rho(\gamma_2)x}\, D\rho(\gamma_2)_x,$$

so the derivative is a $\mathrm{GL}_d(\mathbb{R})$-cocycle.

Proof. Compute

$$T_{\gamma_1}T_{\gamma_2}(x, h) = (\gamma_1\gamma_2 x, \alpha(\gamma_1, \gamma_2 x)\alpha(\gamma_2, x)h).$$

This equals $T_{\gamma_1\gamma_2}(x,h)$ for all h exactly when the cocycle identity holds. The derivative identity is the chain rule applied to $\rho(\gamma_1\gamma_2) = \rho(\gamma_1) \circ \rho(\gamma_2)$; a measurable frame identifies the fiber maps with matrices in $\mathrm{GL}_d(\mathbb{R})$. $\square$

Proposition 0.3 (Measurable cocycles: downstream consequence)**.** *If α admits a reduction to an algebraic subgroup $L < H$, then for every subgroup $\Gamma_0 < \Gamma$ the restricted cocycle $\alpha|_{\Gamma_0 \times X}$ admits the same reduction. Hence*

$$\dim \mathrm{Hull}(\alpha|_{\Gamma_0}) \leq \dim L,$$

and in particular restriction cannot increase algebraic-hull dimension.

PROOF. Choose $b : X \to H$ with $\alpha^b(\gamma, x) \in L$ almost everywhere for all $\gamma \in \Gamma$. The same inclusion holds after restricting γ to Γ_0, so L is an admissible reduction for the restricted cocycle. Minimality of the restricted hull gives the inequality. $\square$

CHAPTER 2

Algebraic hulls of cocycles

This chapter owns the algebraic hulls of cocycles step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Algebraic hulls of cocycles: structural mechanism)**.** *Let $\alpha : \Gamma \times X \to H$ be a measurable cocycle over an ergodic probability space. Suppose $b_i : X \to H$ reduce α to algebraic subgroups $L_i < H$ $(i = 1, 2)$ and that the double-coset space $L_2\backslash H/L_1$ is countably separated. Then*

$$x \longmapsto L_2 b_2(x)^{-1} b_1(x) L_1$$

is Γ-invariant and hence essentially constant. If both L_i have minimal algebraic dimension among reductions, the intersection reduction obtained from that constant double coset has the same dimension as each L_i.

PROOF. Write $\alpha_i(\gamma, x) = b_i(\gamma x)^{-1}\alpha(\gamma, x)b_i(x) \in L_i$. Then

$$b_2(\gamma x)^{-1} b_1(\gamma x) = \alpha_2(\gamma, x)\, b_2(x)^{-1} b_1(x)\, \alpha_1(\gamma, x)^{-1},$$

so the displayed double coset is invariant. Countable separation and ergodicity make it constant, say $L_2 h L_1$. Choosing measurable $\ell_i(x) \in L_i$ with $b_2(x)^{-1}b_1(x) = \ell_2(x)h\ell_1(x)$ and changing the transfer by ℓ_2 gives a reduction to $L_2 \cap hL_1h^{-1}$. Minimality therefore gives $\dim(L_2 \cap hL_1h^{-1}) \geq \dim L_2$; the reverse inequality is automatic, so equality holds. Interchanging 1 and 2 gives the symmetric dimension statement. ☐

THEOREM 0.2 (Algebraic hulls of cocycles). *Let α be an H-cocycle over an ergodic base. Suppose $L_1, L_2 < H$ are reductions of minimal algebraic dimension, realized by transfer maps b_1, b_2. Then there exists $h \in H$ such that*

$$L_2 = hL_1h^{-1}.$$

Moreover the relative transfer $r(x) := b_2(x)^{-1}hb_1(x)$ *takes values almost everywhere in* $N_H(L_2)$ *after changing* h *inside its* L_2*–*L_1 *double coset. Thus the conjugacy class of a minimal reduction is well defined.*

Proof. For $i = 1, 2$ write $\alpha_i = \alpha^{b_i}$, so α_i is L_i-valued. The measurable map

$$x \longmapsto L_2\, b_2(x)^{-1}b_1(x)\, L_1 \in L_2\backslash H/L_1$$

is Γ-invariant: substituting the cocycle relations changes the representative only by left multiplication by L_2 and right multiplication by L_1. Ergodicity therefore makes its value essentially one double coset L_2hL_1. Replacing b_1 on a null set, write $b_2^{-1}b_1 = \ell_2h\ell_1$. The conjugated cocycle then takes values in $L_2 \cap hL_1h^{-1}$. If this intersection had smaller dimension than L_2, it would give a reduction of smaller dimension, contradicting minimality. Interchanging L_1, L_2 gives equality of dimensions and therefore $L_2 = hL_1h^{-1}$ up to finite components; absorbing the finite-component representative into h gives the displayed conjugacy. The remaining relative transfer normalizes the common hull, proving the last assertion. □

Proposition 0.3 (Algebraic hulls of cocycles: downstream consequence)**.** *For a* C^1 *action* $\rho : \Gamma \curvearrowright M$ *preserving an ergodic probability* μ*, the derivative cocycle has a well-defined algebraic-hull conjugacy class in* $\mathrm{GL}_d(\mathbb{R})$ *whenever the logarithmic integrability condition in the volume conventions holds.*

Proof. The chain rule gives a measurable GL_d-cocycle by Theorem 0.2. Apply Theorem 0.2 to that cocycle. Integrability is not needed for the algebraic definition itself but is recorded here because all later Lyapunov and superrigidity uses of the derivative hull require it. □

CHAPTER 3

Boundary maps

This chapter owns the boundary maps step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Boundary maps: structural mechanism)**.** *Let $m : B \times X \to \mathrm{Prob}(H/P)$ satisfy $m(\gamma b, \gamma x) = \alpha(\gamma, x)_* m(b, x)$. If $m(b, x) = \delta_{\phi(b,x)}$ almost everywhere, then ϕ is an α-equivariant boundary map. More generally, if the Zariski closure L of the generic stabilizer of $m(b, x)$ is proper and a measurable conjugating section for the corresponding H/L-orbit exists, then α is measurably cohomologous to an L-valued cocycle.*

Proof. In the Dirac case, equality of probability measures gives $\delta_{\phi(\gamma b,\gamma x)} = \delta_{\alpha(\gamma,x)\phi(b,x)}$, hence pointwise equivariance almost everywhere. In the stabilizer case choose a measurable $c : B \times X \to H$ carrying the generic stabilizer to the fixed subgroup L. Equivariance implies $c(\gamma b, \gamma x)^{-1}\alpha(\gamma, x)c(b, x) \in L$. When the section is chosen independently of the boundary coordinate on the ergodic generic orbit (the only form used below), this is precisely a measurable cohomology reduction of α to L. □

Theorem 0.2 (Boundary maps). *Let B be an amenable measurable Γ-space in the following operational sense: for every compact metrizable H-space K and every measurable H-cocycle α, there is an α-equivariant measurable field*

$$m : B \times X \to \mathrm{Prob}(K), \qquad m(\gamma b, \gamma x) = \alpha(\gamma, x)_* m(b, x).$$

If for $K = H/P$ this field is almost surely a Dirac mass, $m(b, x) = \delta_{\phi(b,x)}$, then

$$\phi(\gamma b, \gamma x) = \alpha(\gamma, x)\phi(b, x)$$

almost everywhere. If the field is not Dirac and the Zariski closure of its generic stabilizer is a proper subgroup $L < H$, then L is a proper algebraic *reduction candidate; minimality of the algebraic hull rules out this branch when the hull is already minimal and no proper reduction is allowed.*

Proof. The equivariance of m is the amenability output. If $m = \delta_\phi$, then equality of Dirac masses gives the displayed pointwise equivariance. In the non-Dirac branch, the stabilizer $\mathrm{Stab}_H(m(b,x))$ is algebraic after taking Zariski closure. Equivariance conjugates these stabilizers along the Γ-orbit, so a measurable choice of conjugating elements reduces the cocycle to the generic stabilizer. If that stabilizer has smaller algebraic dimension than the fixed hull, this contradicts minimality. Thus, in the minimal-hull branch relevant later, the probability field must concentrate on the flag orbit and yields the boundary map. □

Proposition 0.3 (Boundary maps: downstream consequence)**.** *Given two equivariant maps $\phi_- : B_- \times X \to H/P_-$ and $\phi_+ : B_+ \times X \to H/P_+$, their relative position defines a measurable invariant map*

$$r : B_- \times B_+ \times X \to P_- \backslash H/P_+.$$

If the diagonal Γ-action on $B_- \times B_+ \times X$ is ergodic and the double-coset quotient is countably separated, then r is essentially constant.

Proof. Equivariance of the two boundary maps changes a pair (ϕ_-, ϕ_+) by the same left multiplication in H, which does not change its double-coset relative position. Hence r is Γ-invariant. The countable-separation convention and ergodicity imply that every invariant measurable map to this quotient is essentially constant. □

CHAPTER 4

Cocycle superrigidity

This chapter owns the cocycle superrigidity step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Cocycle superrigidity: structural mechanism)**.** *Let $R, C < H$ be commuting subgroups and suppose a cohomologous cocycle has the form*

$$\beta(\gamma, x) = \rho(\gamma)k(\gamma, x), \qquad \rho(\gamma) \in R, \quad k(\gamma, x) \in C,$$

where $\rho : \Gamma \to R$ is a homomorphism. Then k is a C-valued cocycle. If C is compact-by-central, every quotient of H that kills C turns β into the homomorphism ρ.

Proof. Insert the displayed factorization into the cocycle identity. Since $[R, C] = e$ and $\rho(\gamma_1\gamma_2) = \rho(\gamma_1)\rho(\gamma_2)$, cancellation gives

$$k(\gamma_1\gamma_2, x) = k(\gamma_1, \gamma_2 x)k(\gamma_2, x).$$

Thus k is a cocycle. If $q : H \to H/C$ is the quotient map, then $q(\beta(\gamma, x)) = q(\rho(\gamma))$, which is independent of x. ☐

Theorem 0.2 (Cocycle superrigidity). *Let $\alpha : \Gamma \times X \to H$ be a log-integrable cocycle with minimal algebraic hull H. Assume the boundary-pair construction of Chapters 2–3 produces a measurable $b : X \to H$, a homomorphism $\rho : \Gamma \to H$, and commuting subgroups $K, Z < H$ with K compact and $Z \subset Z(H)$ such that*

$$b(\gamma x)^{-1}\alpha(\gamma, x)b(x) = \rho(\gamma)k(\gamma, x), \qquad k(\gamma, x) \in KZ$$

for every γ and almost every x. Then α is cohomologous to the product of the homomorphism ρ and a compact-by-central cocycle k. In particular every noncompact semisimple algebraic factor of the cocycle is carried by ρ.

Proof. The displayed identity is exactly the cohomology relation defining α^b. Because K and Z commute with the algebraic image specified by the boundary-pair reduction, the cocycle identity for α^b becomes

$$\rho(\gamma_1\gamma_2)k(\gamma_1\gamma_2, x) = \rho(\gamma_1)\rho(\gamma_2)k(\gamma_1, \gamma_2 x)k(\gamma_2, x).$$

The homomorphism identity for ρ cancels the first factors and shows that k is itself a cocycle with values in KZ. By definition this is compact-by-central. Projecting the decomposition to any noncompact semisimple quotient kills KZ, so that quotient is determined by the genuine representation ρ. □

Proposition 0.3 (Cocycle superrigidity: downstream consequence)**.** *Let $q : H \to H/(KZ)$ be the quotient map in the situation of the theorem. Then*

$$q(\alpha^b(\gamma, x)) = q(\rho(\gamma))$$

is independent of x. Thus every later argument involving a noncompact algebraic quotient of the derivative cocycle may replace the cocycle by the representation $q \circ \rho$.

Proof. Apply q to the decomposition in Theorem 0.2. Since $q(k) = e$, only the homomorphism factor remains, giving the displayed identity. □

CHAPTER 5

The Zimmer program

This chapter owns the the zimmer program step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (The Zimmer program: structural mechanism)**.** *Let (f_n) be continuous and subadditive over a continuous map T of a compact space. For every fixed block length q there is a constant C_q such that, for all n and y,*

$$f_n(y) \le \sum_{j=0}^{\lfloor n/q \rfloor - 1} f_q(T^{jq}y) + C_q q.$$

Consequently any sequence (n_i, y_i) with $f_{n_i}(y_i) \ge \eta n_i$ produces, after passing to empirical measures and then a weak- limit, a T-invariant probability ν satisfying*

$$\frac{1}{q} \int f_q \, d\nu \ge \eta - O(q^{-1})$$

for every fixed q.

Proof. Write $n = mq + r$ with $0 \le r < q$. Iterating subadditivity gives $f_n(y) \le \sum_{j=0}^{m-1} f_q(T^{jq}y) + f_r(T^{mq}y)$. Compactness bounds the finitely many remainders f_r, giving the first inequality after enlarging C_q. Divide by n_i and compare the block average with the empirical measure $\nu_i = n_i^{-1} \sum_{j=0}^{n_i-1} \delta_{T^j y_i}$; omission of at most q terms costs $O(q/n_i)$. Passing to a weak-* convergent subsequence yields a T-invariant ν and the asserted lower bound. □

Theorem 0.2 (The Zimmer program). *Let $T : Y \to Y$ be a continuous transformation of a compact metric space and let $f_n : Y \to \mathbb{R}$ be continuous and subadditive:*

$$f_{n+m}(y) \le f_n(T^m y) + f_m(y).$$

Assume that for every T-invariant probability ν,

$$\inf_{n\geq 1} \frac{1}{n} \int f_n \, d\nu \leq 0.$$

Then for every $\varepsilon > 0$ there is $C_\varepsilon < \infty$ such that

$$f_n(y) \leq \varepsilon n + C_\varepsilon \qquad (n \geq 1,\ y \in Y).$$

Applied to $f_n(y) = \log \|D\Phi^n|_{E_y}\|$ on a compact suspension model, absence of positive fiber Lyapunov exponent for every invariant probability gives uniform subexponential growth along the corresponding flow direction.

Proof. Fix $\varepsilon > 0$. If the conclusion failed, there would be $n_j \to \infty$ and y_j with $f_{n_j}(y_j) > 2\varepsilon n_j$. Fix a block length q. Subadditivity decomposes $f_{n_j}(y_j)$ into q-blocks plus at most $2q$ boundary terms. Since the finitely many boundary functions are bounded on compact Y,

$$\frac{f_{n_j}(y_j)}{n_j} \leq \frac{1}{q} \int f_q \, d\nu_j + o_{j\to\infty}(1) + O(q/n_j),$$

where $\nu_j = n_j^{-1} \sum_{r=0}^{n_j-1} \delta_{T^r y_j}$. Pass to a weak-* limit ν, which is T-invariant. Letting $j \to \infty$ gives $2\varepsilon \leq q^{-1} \int f_q \, d\nu$. Since q was arbitrary, this contradicts the hypothesis after choosing a q with $q^{-1} \int f_q \, d\nu < \varepsilon$. Therefore $f_n \leq 2\varepsilon n$ for all sufficiently large n; enlarging a constant controls the finitely many remaining n. Replacing 2ε by ε proves the stated form. For the derivative cocycle, subadditivity is the operator-norm inequality from Chapter 6 and Kingman identifies the measurewise limits with top fiber exponents. □

Proposition 0.3 (The Zimmer program: downstream consequence)**.** *The theorem is a one-parameter semi-uniform principle. To deduce subexponential derivative growth in the word metric of a lattice, one must verify the analogous nonpositive-exponent hypothesis on the compact suspension for every direction used to build word segments; the one-parameter conclusion alone does not imply the full lattice statement.*

Proof. The proof of Theorem 0.2 controls one subadditive sequence f_n attached to one transformation. A word $\gamma = s_1 \cdots s_m$ samples many directions, so no inequality in the theorem bounds all such products unless the corresponding suspension argument supplies a uniform family of subadditive bounds. This records the exact additional input required later. □

CHAPTER 6

Derivative cocycles

This chapter owns the derivative cocycles step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Derivative cocycles: structural mechanism)**.** *The derivative* $D\alpha(\gamma)_x$ *is a measurable* $\mathrm{GL}(d,\mathbb{R})$ *cocycle and its logarithmic norm is subadditive under composition.*

Proof. For a C^1 action, the chain rule gives

$$D\alpha(\gamma_1\gamma_2)_x = D\alpha(\gamma_1)_{\alpha(\gamma_2)x}\, D\alpha(\gamma_2)_x,$$

which is exactly the measurable-cocycle identity with values in $\mathrm{GL}(T_xM)$ after choosing a measurable trivialization of the tangent bundle. Taking operator norms and logarithms gives

$$\log\|D\alpha(\gamma_1\gamma_2)_x\| \le \log\|D\alpha(\gamma_1)_{\alpha(\gamma_2)x}\| + \log\|D\alpha(\gamma_2)_x\|.$$

Thus the logarithmic derivative norm is subadditive under composition. The same calculation for inverses gives the corresponding inverse cocycle. □

Theorem 0.2 (Derivative cocycles). *Let* $T : M \to M$ *be a* C^1 *diffeomorphism of a compact manifold and set*

$$f_n(x) := \log\|DT_x^n\|, \qquad g_n(x) := \log\|DT_x^{-n}\|.$$

Then both (f_n) *and* (g_n) *are continuous subadditive sequences. If every* T*-invariant probability* ν *has top derivative exponent at most* 0*, equivalently*

$$\inf_{n\ge1}\frac{1}{n}\int f_n\,d\nu \le 0,$$

then for every $\varepsilon > 0$ *there is* C_ε *with*

$$\|DT_x^n\| \le C_\varepsilon e^{\varepsilon n} \quad (n \ge 0, x \in M).$$

If the same hypothesis holds for T^{-1}, then also $\|DT_x^{-n}\| \le C'_\varepsilon e^{\varepsilon n}$.

Proof. The chain rule gives $DT_x^{n+m} = DT_{T^m x}^n DT_x^m$, hence $f_{n+m}(x) \le f_n(T^m x) + f_m(x)$; the same calculation applies to T^{-1}. The hypotheses therefore put f_n and g_n exactly in the semi-uniform subadditive theorem of Chapter 5. It yields $f_n(x) \le \varepsilon n + C_\varepsilon$ and, separately, $g_n(x) \le \varepsilon n + C'_\varepsilon$. Exponentiating gives the two displayed derivative estimates. □

Proposition 0.3 (Derivative cocycles: downstream consequence)**.** *The inverse-derivative estimate is not automatic from the forward estimate alone; it follows by applying the same invariant-measure hypothesis and Theorem 0.2 to T^{-1}.*

Proof. For T^{-1} the subadditive sequence is $g_n(x) = \log \|DT_x^{-n}\|$. The stated inverse hypothesis is exactly the hypothesis of Theorem 0.2 for this sequence, so the same proof applies. □

CHAPTER 7

Dimension bounds for actions

This chapter owns the dimension bounds for actions step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Dimension bounds for actions: structural mechanism)**.** *Let $\beta : \Gamma \times X \to \mathrm{GL}_d(\mathbb{R})$ be cohomologous to $\rho(\gamma)k(\gamma, x)$, where k has relatively compact image. If every homomorphism $\rho : \Gamma_0 \to \mathrm{GL}_d(\mathbb{R})$ from every finite-index subgroup $\Gamma_0 < \Gamma$ has relatively compact image, then after passing to finite index the entire noncompact algebraic quotient of β is trivial.*

Proof. Project the factorization to the quotient of the algebraic hull by its maximal compact-by-central normal subgroup. The k-factor vanishes there, so the quotient cocycle equals the projected homomorphism $\bar{\rho}$. By hypothesis $\bar{\rho}(\Gamma_0)$ is relatively compact. A relatively compact subgroup of a noncompact semisimple algebraic quotient has Zariski closure strictly smaller than that quotient; minimality of the noncompact hull therefore forces the quotient to be trivial. Hence only compact-by-central derivative data remains. ☐

Theorem 0.2 (Dimension bounds for actions). *Let G be a connected semisimple real Lie group and let*

$$m(G) := \min\{\dim V : V \textit{ is a nontrivial finite-dimensional real representation of } G\}.$$

Suppose cocycle superrigidity for a d-dimensional derivative cocycle produces a homomorphism $\rho : \Gamma \to \mathrm{GL}_d(\mathbb{R})$ whose noncompact algebraic part extends to a representation of G. If $d < m(G)$, that noncompact part is trivial. Hence, after the compact-by-central reduction of Chapter 4, the derivative cocycle has no noncompact semisimple algebraic factor.

Proof. If the extended noncompact factor were nontrivial, it would give a nontrivial real representation of G on a vector space of dimension at most d. By the definition of $m(G)$ this forces $d \geq m(G)$, contradicting the hypothesis. Therefore the extended noncompact representation is trivial. The decomposition of Theorem 0.2 then leaves only the compact-by-central residual cocycle. $\square$

Proposition 0.3 (Dimension bounds for actions: downstream consequence)**.** *For $G = \mathrm{SL}(n, \mathbb{R})$, $n \geq 3$, the defining representation shows $m(G) \leq n$, while highest-weight theory gives $m(G) = n$. Consequently a derivative cocycle of dimension $d < n$ cannot carry a nontrivial noncompact G-representation. This is the representation-theoretic inequality behind the strict dimension threshold; the remaining dynamical work is the uniform derivative-growth and invariant-metric argument, not this dimension count.*

Proof. The standard n-dimensional representation gives the upper bound. For the lower bound, every nontrivial complex irreducible representation of $\mathfrak{sl}_n$ has highest weight at least one fundamental weight, and the smallest dimensions are those of the defining representation and its dual, both dimension n; passing from a real representation to its complexification cannot increase the minimal dimension. Thus $m(\mathrm{SL}(n, \mathbb{R})) = n$. The conclusion follows from Theorem 0.2. $\square$

CHAPTER 8

Modern Zimmer-type rigidity theorems

The derivative representation used in the Zimmer argument is not unitary: its operator norm is allowed subexponential growth. This chapter proves the Hilbert-space strong-property-(T) statement needed to average such a representation. The proof is kept in the narrower Hilbert/Sobolev scope actually used below; the stronger Banach-space theorem is cited only for historical comparison.

> **Proof and provenance.** Primary comparison: de Laat–de la Salle for the rank-two compact harmonic analysis and Brown–Fisher–Hurtado for the invariant-metric application. The proof below reconstructs the Hilbert specialization used by this series; the citations are not proved inputs. [LS15; BFH21b]

Lemma 0.1 (Compact-pair Hölder estimates on $SU(2)$)**.** *Let $K = SU(2)$, let $U = \{\mathrm{diag}(e^{it}, e^{-it}) : t \in \mathbb{R}\}$, and let P_U be orthogonal projection from $L^2(K)$ onto the left-and-right U-invariant subspace. For*

$$r_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad T_\theta = P_U \lambda(r_\theta) P_U,$$

one has, for every $p > 4$ and $\theta_1, \theta_2 \in [\pi/6, \pi/3]$,

$$\|T_{\theta_1} - T_{\theta_2}\|_{S^p} \le C_p |\theta_1 - \theta_2|^{\frac{1}{2} - \frac{2}{p}}. \tag{69.8.1}$$

For the second compact pair occurring in the $Sp(2)$ Cartan calculation, whose compressed operators are denoted S_θ, one has for every $p > 10$

$$\|S_{\theta_1} - S_{\theta_2}\|_{S^p} \le C'_p |\theta_1 - \theta_2|^{\frac{1}{4} - \frac{5}{2p}}. \tag{69.8.2}$$

Consequently, after tensoring with an arbitrary Hilbert space $\mathcal{H}$, the same right-hand sides bound the operator norms of $(T_{\theta_1} - T_{\theta_2}) \otimes I_{\mathcal{H}}$ and $(S_{\theta_1} - S_{\theta_2}) \otimes I_{\mathcal{H}}$.

Proof. Peter–Weyl gives

$$L^2(SU(2)) = \widehat{\bigoplus}_{n \ge 0} V_n \otimes V_n^*, \qquad \dim V_n = n + 1.$$

The U-bi-invariant line occurs only in the even representations and the compression of $\lambda(r_\theta)$ to that line is multiplication by $P_m(\cos 2\theta)$ in V_{2m}, where P_m is the Legendre polynomial. The integral formula

$$P_m(\cos t) = \frac{1}{\pi}\int_0^\pi (\cos t + i\sin t\cos u)^m\,du$$

and one integration by parts away from the two stationary points give, uniformly for $t \in [\pi/3, 2\pi/3]$,

$$|P_m(\cos t)| \le C(m+1)^{-1/2}, \qquad |\partial_t P_m(\cos t)| \le C(m+1)^{1/2}. \tag{69.8.3}$$

Hence, with $\delta = |\theta_1 - \theta_2|$,

$$|P_m(\cos 2\theta_1) - P_m(\cos 2\theta_2)| \le C\min\{(m+1)^{-1/2}, (m+1)^{1/2}\delta\}. \tag{69.8.4}$$

The Schatten p-norm is the ℓ^p norm of these singular values. Split the sum at $m \asymp \delta^{-1}$; the two geometric-integral estimates give

$$\sum_m \min\{(m+1)^{-p/2}, (m+1)^{p/2}\delta^p\} \ll_p \delta^{p/2-2},$$

which is (69.8.1).

For S_θ the relevant U-types are the Wigner coefficients $d^n_{k,k}(2\theta)$. Their Jacobi representation is

$$d^n_{k,k}(2\theta) = (\cos\theta)^{2|k|} P^{(0,2|k|)}_{n-|k|}(\cos 2\theta).$$

The same oscillatory integral, now with an endpoint of order $|k|$, gives

$$|d^n_{k,k}(2\theta)| \le C\big((n+1)(1+|k|)\big)^{-1/4},$$

and the differentiated integral gives

$$|\partial_\theta d^n_{k,k}(2\theta)| \le C(n+1)^{3/4}(1+|k|)^{1/4}. \tag{69.8.5}$$

Thus the difference is bounded by the minimum of the two right-hand sides (with the derivative multiplied by δ). Summing the pth powers over $|k| \le n$ and splitting at $n \asymp \delta^{-2}$ gives

$$\sum_{n\ge 0}\sum_{|k|\le n} |d^n_{k,k}(2\theta_1) - d^n_{k,k}(2\theta_2)|^p \ll_p \delta^{p/4-5/2},$$

which is (69.8.2). Finally $\|A \otimes I_{\mathcal{H}}\| \le \|A\|_{S^p}$ for every compact operator A, proving the Hilbert-valued estimates. $\square$

Lemma 0.2 ($Sp(2,\mathbb{R})$ Cartan interpolation)**.** *Let $R = Sp(2,\mathbb{R})$, $K = U(2)$, and*

$$D(\beta,\gamma) = \operatorname{diag}(e^\beta, e^\gamma, e^{-\beta}, e^{-\gamma}), \qquad \beta \ge \gamma \ge 0.$$

There are constants $a_0, b_0 > 0$ and C such that the following holds. If $\pi : R \to B(\mathcal{H})$ is strongly continuous and

$$\|\pi(kD(\beta,\gamma)k')\| \le Le^{s\beta} \qquad (k,k' \in K), \quad 0 < s < a_0, \tag{69.8.6}$$

then the K-bi-averaged operators

$$A_{\beta,\gamma} = \int_{K\times K} \pi(kD(\beta,\gamma)k')\,dk\,dk'$$

have a common norm limit P_R as $\beta \to \infty$, uniformly in $0 \le \gamma \le \beta$, and

$$\|A_{\beta,\gamma} - P_R\| \le CLe^{-b_0\beta}. \tag{69.8.7}$$

The range of P_R is exactly $\mathcal{H}^R$.

Proof. We spell out the chamber estimate. The two elementary 4×4 KAK computations used below have the following consequence. If $c(\beta,\gamma) = \langle A_{\beta,\gamma}\xi, \eta\rangle$, then whenever $\beta \ge \gamma \ge 1$ there are adjacent chamber points (β',γ') with Euclidean distance at most 2 for which

$$|c(\beta,\gamma) - c(\beta',\gamma')| \le CL\|\xi\|\|\eta\| \left(e^{s\beta-\alpha_1(\beta-\gamma)} + e^{s\beta-\alpha_2\gamma}\right), \tag{69.8.8}$$

where one may take any $0 < \alpha_1 < \frac{1}{2}$ and $0 < \alpha_2 < \frac{1}{4}$. Indeed, insert the two $SU(2)$ subgroups associated with the long and short simple roots. In the first KAK identity the angular displacement is $O(e^{-(\beta-\gamma)})$; (69.8.1), together with (69.8.6), gives the first term. In the second identity the displacement is $O(e^{-\gamma})$; (69.8.2) gives the second. The remaining compact factors disappear under K-averaging. These identities are verified by multiplying the four matrices and comparing the two positive singular values; the equations are $\cosh\beta'\cosh\gamma' = \cosh\beta\cosh\gamma + O(1)$ and $\sinh(\beta'-\gamma') = \sinh(\beta-\gamma)(1+O(e^{-\gamma}))$, which give the stated angular sizes.

Partition the positive Weyl chamber into the regions

$$\gamma \le \beta/3, \qquad \beta/3 \le \gamma \le 2\beta/3, \qquad 2\beta/3 \le \gamma \le \beta.$$

In the middle region both exponents in (69.8.8) decay provided $s < \min(\alpha_1,\alpha_2)/4$. In the first region use the long-root move repeatedly until $\gamma \asymp \beta/3$; in the last region use the short-root move until $\beta - \gamma \asymp \beta/3$. At each step β changes by $O(1)$ and the error is bounded by $Ce^{-c\beta}$. Joining any two chamber points of radial size between n and $n+1$ uses $O(n)$ such steps, hence

$$\sup_{\substack{\beta,\beta'\ge n\\ 0\le\gamma\le\beta,\,0\le\gamma'\le\beta'}} |c(\beta,\gamma) - c(\beta',\gamma')| \le CLe^{-b_0 n}\|\xi\|\|\eta\|. \tag{69.8.9}$$

Taking the supremum over unit vectors proves norm Cauchy convergence and (69.8.7).

Every R-fixed vector is fixed by all $A_{\beta,\gamma}$, so $\mathcal{H}^R \subset \operatorname{ran} P_R$. Conversely let $v = P_R v$. For a root element $u = \exp Y$, choose a Cartan element $a = D(t,0)$ that contracts u under conjugation. Using (69.8.7) at radius n and the growth bound (69.8.6),

$$\|\pi(u)v - v\| \le Ce^{sn}\|\pi(a^{-n}ua^n) - I\|_{\mathcal{H}^\infty} + Ce^{-(b_0-s)n}\|v\|.$$

On smooth vectors the first factor is $O(e^{-\lambda n})$ for the positive root weight λ; choose $s < \min(b_0,\lambda)/2$ and let $n \to \infty$. Density gives $\pi(u)v = v$. The root subgroups generate R, hence $v \in \mathcal{H}^R$. $\square$

Lemma 0.3 (Controlled strong-property-(T) averaging)**.** *Let G be a connected simple real Lie group of real rank at least two and let ℓ be a continuous proper length function. There are compactly supported symmetric probability measures m_n and constants $a, b > 0$ such that*

$$\operatorname{supp} m_n \subset \{g : \ell(g) \le n\} \tag{69.8.10}$$

and every strongly continuous Hilbert representation satisfying

$$\|\pi(g)\| \le Ce^{s\ell(g)}, \qquad s < a, \tag{69.8.11}$$

obeys

$$\|\pi(m_n) - P_\pi\| \le C_\pi e^{-bn}, \tag{69.8.12}$$

where P_π is the projection onto $\mathcal{H}^G$. The same conclusion holds for cocompact lattices in G and, in the nonuniform project scope, for $\Gamma = SL_m(\mathbb{Z})$, $m \ge 3$, with finitely supported probability measures and word length.

Proof. *Rank two.* For $Sp(2,\mathbb{R})$ take the normalized K-double averages of $D(n,0)$. Lemma 0.2 gives (69.8.12), and replacing n by a fixed multiple gives (69.8.10) for an arbitrary proper length. The $SL_3(\mathbb{R})$ case is proved by the same calculation with the $SO(3)//SO(2)$ compact pair; only (69.8.1) is needed, because the two simple roots have equal length. The chamber decomposition is two-dimensional and the analogue of (69.8.8) is $Ce^{s\beta-\alpha\min(\gamma,\beta-\gamma)}$; the same path summation gives exponential Cauchy convergence.

Higher rank. A connected simple real group of rank at least two contains a closed rank-two subgroup R locally isomorphic to one of these two types. Restrict π to R and obtain P_R. Choose a split element $A \in \mathfrak{r}$ and decompose $\mathfrak{g} = \bigoplus_\lambda \mathfrak{g}^\lambda$. For $Y \in \mathfrak{g}^\lambda$, $\lambda > 0$,

$$e^{-nA}\exp(Y)e^{nA} = \exp(e^{-n\lambda}Y).$$

Insert the rank-two averages at radius εn exactly as in the last paragraph of Lemma 0.2; one obtains

$$\|\pi(\exp Y)P_R v - P_R v\| \le C_v e^{-(\lambda-\varepsilon-Cs)n} + C_v e^{-(b\varepsilon-Cs)n}. \tag{69.8.13}$$

Choose $\varepsilon > 0$ and then $s > 0$ so both exponents are positive. The negative-root case is identical. The nonzero A-weight spaces generate a nonzero ideal of the simple algebra $\mathfrak{g}$, hence all of $\mathfrak{g}$. Thus $\operatorname{ran} P_R = \mathcal{H}^G$. Reindexing the rank-two averages supplies m_n and preserves exponential convergence.

Cocompact lattices. Let $\Gamma < G$ be cocompact and choose a bounded fundamental domain F. For a representation σ of Γ satisfying $\|\sigma(\gamma)\| \leq Ce^{s|\gamma|}$, form the induced Hilbert space $L^2(F, \mathcal{H})$ with cocycle $\alpha(g, x) \in \Gamma$. Boundedness of F and the quasi-isometry between $|\gamma|$ and the ambient length give

$$|\alpha(g, x)| \leq C_0(1 + \ell(g)) \tag{69.8.14}$$

uniformly in x. Hence the induced G-representation satisfies (69.8.11) with exponent $C_0 s$. Apply the group averages and compress to constant sections. Because (69.8.14) gives finite cocycle range on compact g-sets, the compressed averages are finitely supported measures on Γ, and their limit is the projection onto Γ-fixed vectors.

The $SL_m(\mathbb{Z})$ cusp. Use the standard Siegel fundamental set $F = NAK$ with A^+ coordinates $H = \log a$. Haar measure has density $e^{-2\rho(H)}dH$ up to a bounded N, K factor. Reduction theory gives, for every compact $C \subset G$,

$$|\alpha(g, x)| \leq C_C(1 + \|H(x)\|), \qquad g \in C. \tag{69.8.15}$$

Since ρ is strictly positive on the positive chamber, there is $\tau > 0$ with

$$\int_F e^{\tau|\alpha(g,x)|}\, dx < \infty \tag{69.8.16}$$

uniformly for g in a fixed compact set. Thus for sufficiently small s the induced representation is bounded on $L^2(F, \mathcal{H})$ by $Ce^{C_1 s\ell(g)}$. Apply the G-averages. Compressing to constant sections gives probability measures on Γ with exponential tails in word length by (69.8.16). Truncate at $|\gamma| \leq C_2 n$, where the discarded operator norm is $O(e^{-cn})$, and renormalize. The resulting finitely supported measures retain exponential convergence to the invariant-vector projection. This proves the lattice statement in the exact cocompact and $SL_m(\mathbb{Z})$ scopes used below. $\square$

Theorem 0.4 (Brown–Fisher–Hurtado endpoint in the project scope). *Assume the hypotheses established in Chapters 1–7: the derivative cocycle is in the low-dimensional BFH range, every invariant suspension measure has nonpositive fiber exponent, and the cocycle-superrigidity reduction leaves no noncompact linear factor. Let the acting lattice be either cocompact in the higher-rank group or $SL_m(\mathbb{Z})$ in the nonuniform scope for which the Siegel-set estimates above apply. Then the action has uniform subexponential growth of derivatives and preserves a Riemannian metric. In the dimension range* *where Chapters 4 and 7 exclude a positive-dimensional compact image, the action has finite image.*

PROOF. Chapters 5–6 convert the nonpositive-exponent hypothesis into: for every $\varepsilon > 0$,

$$\|D\alpha(\gamma)\| + \|D\alpha(\gamma)^{-1}\| \le C_\varepsilon e^{\varepsilon|\gamma|}. \tag{69.8.17}$$

Fix a smooth background metric g_0 and choose a Sobolev exponent k above the embedding threshold. Pullback gives a Hilbert representation on the $W^{k,2}$ completion of symmetric two-tensors. The product rule and (69.8.17) imply

$$\|\rho(\gamma)\|_{W^{k,2}\to W^{k,2}} \le C_\varepsilon e^{C_k\varepsilon|\gamma|}. \tag{69.8.18}$$

Choose ε so $C_k\varepsilon < a$ in Lemma 0.3. The strong-(T) averages of g_0 converge in $W^{k,2}$ to an invariant symmetric tensor g_∞.

We check positivity rather than assume it. Normalize the averages to have the same total volume. The determinant is a continuous polynomial in the matrix coefficients, and the derivative/inverse-derivative bounds give on each averaged pullback metric the uniform ellipticity estimate

$$C_n^{-1}g_0 \le \gamma^* g_0 \le C_n g_0, \qquad \log C_n = o(n).$$

The strong-(T) convergence is exponential, so choose the averaging radius sequence with $e^{-bn}C_n \to 0$. Sobolev embedding then gives uniform convergence on tensors and prevents an eigenvalue from reaching zero in the limit; equivalently, apply the same averages to the dual metric and use $g_n g_n^{-1} = I + o(1)$. Thus g_∞ is positive definite and is an invariant Riemannian metric.

The image of the action lies in the compact Lie group $\mathrm{Isom}(M, g_\infty)$. By the cocycle-superrigidity and dimension argument of Chapters 4 and 7, its identity component would give a nontrivial compact linear factor in the prohibited dimension. Hence the identity component of the closure is trivial. A compact Lie group with trivial identity component is finite, proving the final assertion. $\square$

Proposition 0.5 (Why ordinary property (T) does not replace the strong form)**.** *For a unitary representation $\|\pi(g)\| = 1$, a Kazhdan averaging gap controls the complement of invariant vectors. In the derivative representation, (69.8.18) permits $\|\pi(g)\| \le e^{s\ell(g)}$. If a coefficient comparison has geometric decay $e^{-c\ell(g)}$, the propagated error is bounded by $e^{-(c-s)\ell(g)}$ and decays only when $s < c$. The small-growth threshold in Lemma 0.3 is therefore a genuine quantitative input, not a rephrasing of ordinary property (T).*

PROOF. Multiply the coefficient-decay bound by the maximal operator-norm growth on the comparison path. The resulting exponent is $-(c-s)\ell(g)$.

Ordinary property (T) supplies a gap only for uniformly bounded/unitary representations and contains no parameter controlling $s > 0$; Lemma 0.3 does. This is exactly the distinction used in (69.8.18). $\square$

Volume 70

Boundary Rigidity and the Unified Rigidity Dictionary

Volume 70 scope and prerequisites

The final volume is a synthesis. It builds boundary maps and boundary-pair rigidity at the level proved in the chapter, compares the six rigidity mechanisms developed across the series, records failure modes, and gives conditional theorem templates. Whenever the synthesis reaches the strong-property-(T) step from Volume 69, it cites the internally proved Chapter 8 compact-pair, rank-two interpolation, root-propagation, and lattice-transfer chain. Thus the synthesis has no unresolved harmonic-analysis root in the stated Hilbert/Sobolev scope.

Standing definitions and notation for Volume 70

Let G be a connected semisimple real Lie group and $P < G$ a minimal parabolic subgroup. The homogeneous space

$$B_G := G/P$$

is the *Furstenberg boundary* used in this volume. When a lattice $\Gamma < G$ is considered, B_G is always equipped with the restricted Γ-action. Amenability of a boundary is used only in the operational sense fixed in Volume 69: it supplies equivariant measurable probability fields for compact target spaces.

A pair of measurable Γ-spaces (B_-, B_+) is *doubly ergodic relative to an ergodic probability* Γ*-space* (X, μ) if the diagonal action on $B_- \times B_+ \times X$ is ergodic. If $\phi_\pm : B_\pm \times X \to H/P_\pm$ are equivariant boundary maps, their *relative-position map* is

$$r(b_-, b_+, x) := P_- h_-^{-1} h_+ P_+ \in P_- \backslash H / P_+,$$

where $h_\pm P_\pm = \phi_\pm(b_\pm, x)$. This is well defined and invariant under simultaneous left multiplication. All quotient targets below are assumed countably separated in the sense defined in Volume 69; the obsolete phrase "finite-entropy algebraic quotient" is not used.

> **Proof and provenance.** Primary comparison: Furstenberg boundary methods, measurable boundary-map superrigidity, and the internally developed Ratner, entropy, spectral, drift, and hyperbolic frameworks of the preceding volumes.

Reader guide: a concrete model

Why this matters.

Volume 70 is organized around the passage from *Furstenberg and geometric boundaries* to *Final synthesis and frontier theorem templates*. The point of the intermediate lemmas is to make that passage reversible at the level of hypotheses: before using the final theorem, check which recurrence, algebraicity, spectral, height, or regularity input supplies each conclusion. This prevents a qualitative classification statement from being mistaken for an effective estimate, and it prevents an effective estimate from being used outside the range in which its constants are controlled.

Worked Example 0.6 (A calculation to keep in view)**.** Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}).$$

Its eigenvalues are

$$\lambda_\pm = \frac{3 \pm \sqrt{5}}{2}, \qquad \lambda_+ > 1 > \lambda_- > 0.$$

For the toral automorphism induced by A, the tangent bundle therefore splits as

$$T\mathbb{T}^2 = E^u \oplus E^s,$$

with rates $\log \lambda_+$ and $-\log \lambda_+$. For a commuting higher-rank action, coarse Lyapunov spaces are obtained by grouping such eigendirections whose logarithmic characters are positive scalar multiples. This two-dimensional computation is the basic model for the sign data used in the higher-rank rigidity chapters.

Proof architecture at a glance.

$$\begin{tikzcd} \boxed{\text{boundary map}} \arrow[r] & \boxed{\text{relative position}} \arrow[r] & \boxed{\text{algebraic hull}} \arrow[r] & \boxed{\text{superrigid form}} \end{tikzcd}$$

CHAPTER 1

Furstenberg and geometric boundaries

This chapter owns the furstenberg and geometric boundaries step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Furstenberg and geometric boundaries: structural mechanism)**.** *For a semisimple group G, a Furstenberg boundary $B = G/P$ is amenable and carries the proximal flag geometry needed to turn measurable cocycle information into algebraic data.*

Proof. Amenability follows from solvability/amenability of the minimal parabolic P and induction of invariant means. Proximality comes from a highest-weight representation in which P stabilizes a line. Functoriality is immediate from equivariance of quotient maps.

□

Theorem 0.2 (Furstenberg and geometric boundaries). *Let $q : G \to G'$ be a continuous surjective homomorphism of connected semisimple groups and let $P < G$, $P' < G'$ be minimal parabolics with $q(P) \subset P'$. Then*

$$\bar{q} : G/P \to G'/P', \qquad gP \mapsto q(g)P',$$

is a well-defined G-equivariant measurable factor map. If $\Gamma_0 < \Gamma < G$ has finite index, restriction of the Γ-action to Γ_0 leaves the same boundary space and the same equivariant factor maps available.

Proof. If $gP = g'P$, then $g^{-1}g' \in P$, hence $q(g)^{-1}q(g') \in q(P) \subset P'$, so $q(g)P' = q(g')P'$ and $\bar{q}$ is well defined. For $a \in G$, $\bar{q}(agP) = q(a)q(g)P' = q(a)\bar{q}(gP)$, proving equivariance. Restriction to a subgroup changes neither the space nor the formula for an equivariant map, so every Γ-equivariant construction is automatically Γ_0-equivariant. □

Proposition 0.3 (Furstenberg and geometric boundaries: downstream consequence)**.** *Every argument in this volume that starts from $B_G = G/P$ uses only the following two properties: boundary amenability in the operational sense of Volume 69 and equivariant homogeneous geometry of G/P. No identification of G/P with a Poisson boundary is required for those deductions.*

Proof. Inspect the hypotheses of Chapters 2–4: amenability is invoked only to obtain equivariant probability fields, and homogeneous geometry is invoked only through the G-action, parabolic stabilizers, and double-coset relative position. None of those steps uses harmonicity or a random-walk realization of the boundary. Thus the deductions remain valid whenever these two stated properties hold. $\square$

CHAPTER 2

Boundary maps and algebraic targets

This chapter owns the boundary maps and algebraic targets step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Boundary maps and algebraic targets: structural mechanism)**.** *Let an algebraic group H act on an algebraic variety Y, and let $Z \subseteq Y$ be an irreducible H-invariant subvariety containing an open H-orbit $\mathcal{O}$. Then $Z \setminus \mathcal{O}$ is a proper algebraic subset. Hence any measurable map whose essential image is Zariski dense in Z cannot be essentially contained in the lower-dimensional orbit strata.*

Proof. An open orbit is Zariski open in the irreducible variety Z, so its complement is a proper Zariski-closed subset after replacing it by its Zariski closure. If the essential image were contained there, its Zariski closure would also be contained there, contradicting the assumed Zariski density in Z. □

Theorem 0.2 (Boundary maps and algebraic targets). *Let α have minimal algebraic hull H and let $\phi : B \times X \to Y$ be an α-equivariant measurable map into a projective algebraic H-variety Y. Let $Z \subset Y$ be the Zariski closure of the essential image of ϕ. Then Z is H-invariant. If Z is proper and its generic stabilizer has algebraic dimension smaller than H, the cocycle reduces to that stabilizer, contradicting minimality. Hence, in the minimal-hull branch, the essential image meets the unique open H-orbit $\mathcal{O} \subset Z$ on a conull set whenever such an open orbit exists.*

Proof. Equivariance gives $\phi(\gamma b, \gamma x) = \alpha(\gamma, x)\phi(b, x)$. Therefore the essential image and its Zariski closure are invariant under the Zariski-dense algebraic range of the hull, so Z is H-invariant. If the generic stabilizer L on a proper invariant subvariety has smaller dimension, choosing measurable orbit representatives conjugates α into L, contradicting minimality of H. In the remaining branch, the union of lower-dimensional orbit strata is a proper algebraic subset of Z; since the essential image is Zariski dense in Z, it lies in the open orbit almost everywhere after discarding that proper subset. $\square$

Proposition 0.3 (Boundary maps and algebraic targets: downstream consequence)**.** *If α^b is cohomologous to α and ϕ is an α-equivariant boundary map, then*

$$\phi^b(\xi, x) := b(x)^{-1}\phi(\xi, x)$$

is α^b-equivariant. Thus the alternatives "proper invariant algebraic image" and "generic open-orbit image" are invariant under measurable cohomology.

Proof. Using $\alpha^b(\gamma, x) = b(\gamma x)^{-1}\alpha(\gamma, x)b(x)$,

$$\phi^b(\gamma\xi, \gamma x) = b(\gamma x)^{-1}\alpha(\gamma, x)\phi(\xi, x) = \alpha^b(\gamma, x)\phi^b(\xi, x).$$

Multiplication by $b(x)^{-1}$ is an algebraic automorphism of the target, so it preserves dimensions of Zariski closures and orbit strata. Hence the dichotomy is unchanged. $\square$

CHAPTER 3

Boundary-based superrigidity

This chapter owns the boundary-based superrigidity step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Boundary-based superrigidity: structural mechanism)**.** *Let $\phi_\pm : B_\pm \times X \to H/P_\pm$ be α-equivariant maps. Assume that their relative position equals almost everywhere the open double coset of a fixed pair (ξ_-^0, ξ_+^0) and that a measurable section of that open orbit has been fixed. Then there exists a measurable $b : X \to H$ for which the cohomologous cocycle*

$$\beta(\gamma, x) = b(\gamma x)^{-1}\alpha(\gamma, x)b(x)$$

takes values in the stabilizer $S = \mathrm{Stab}_H(\xi_-^0, \xi_+^0)$.

PROOF. For almost every x, use the measurable section to choose $b(x)$ sending the boundary pair at x to the reference pair. Equivariance gives

$$b(\gamma x)^{-1}\alpha(\gamma, x)b(x)(\xi_-^0, \xi_+^0) = (\xi_-^0, \xi_+^0),$$

so the conjugated cocycle belongs to S for every γ and almost every x. $\square$

THEOREM 0.2 (Boundary-based superrigidity). *Let $\phi_\pm : B_\pm \times X \to H/P_\pm$ be equivariant boundary maps and assume their relative-position map is essentially the fixed open double coset of a reference pair (ξ_-^0, ξ_+^0). Then there is a measurable $b : X \to H$ such that the cohomologous cocycle*

$$\beta(\gamma, x) := b(\gamma x)^{-1}\alpha(\gamma, x)b(x)$$

takes values in

$$S := \mathrm{Stab}_H(\xi_-^0, \xi_+^0)$$

almost everywhere. If $S = RC$ with R an algebraic subgroup, C compact-by-central, $[R, C] = e$, and the projection of β to R is independent of x, say $\rho(\gamma)$, then

$$\beta(\gamma, x) = \rho(\gamma)k(\gamma, x), \qquad k(\gamma, x) \in C,$$

with $\rho : \Gamma \to R$ a homomorphism and k a C-valued cocycle.

Proof. Because the relative position equals the reference open orbit, choose $b(x)$ measurably so that the pair of boundary maps at x is sent to (ξ_-^0, ξ_+^0). Equivariance gives

$$\beta(\gamma, x)(\xi_-^0, \xi_+^0) = (\xi_-^0, \xi_+^0),$$

so $\beta(\gamma, x) \in S$. Under the splitting $S = RC$, write $\beta = rk$. By hypothesis $r(\gamma, x) = \rho(\gamma)$ is independent of x. Projecting the cocycle identity for β to R gives $\rho(\gamma_1\gamma_2) = \rho(\gamma_1)\rho(\gamma_2)$, so ρ is a homomorphism. Since R and C commute, cancellation of the R-parts in the cocycle identity leaves $k(\gamma_1\gamma_2, x) = k(\gamma_1, \gamma_2 x)k(\gamma_2, x)$, proving that k is a cocycle. ☐

Proposition 0.3 (Boundary-based superrigidity: downstream consequence)**.** *The theorem produces only the algebraic cocycle decomposition. An invariant Riemannian metric does not follow from this boundary argument alone; that later step additionally requires the strong-property-(T) averaging root isolated in Volume 69, Chapter 8.*

Proof. The conclusion of Theorem 0.2 is the formula $\alpha^b = \rho k$. Nothing in its proof constructs an invariant vector in the Hilbert/Sobolev representation on metrics. That construction is exactly the extra implication supplied by the controlled strong-property-(T) averaging statement of Volume 69, Chapter 8, so it must remain a separate dependency. ☐

CHAPTER 4

Boundary pairs and double ergodicity

This chapter owns the boundary pairs and double ergodicity step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

For the remainder of this chapter, a boundary pair (B_-, B_+) is *doubly ergodic relative to* (X, μ) if the diagonal Γ-action on $B_- \times B_+ \times X$ is ergodic. Equivalently, every Γ-invariant measurable subset of that product has measure zero or one.

Lemma 0.1 (Boundary pairs and double ergodicity: structural mechanism)**.** *For a boundary pair (B_-, B_+) with diagonal double ergodicity, every invariant measurable map to a countably separated quotient is essentially constant.*

Proof. Pull back the target observable to $B_- \times B_+$. Diagonal invariance and double ergodicity make it constant. For an algebraic double-coset target, use the countable separating family fixed in the standing conventions; invariance of the map and ergodicity force all separating coordinates to be constant.

$\square$

Theorem 0.2 (Boundary pairs and double ergodicity). *Let (B_-, B_+) be doubly ergodic relative to (X, μ) and let $\phi_\pm : B_\pm \times X \to H/P_\pm$ be equivariant. Then the relative-position map*

$$r : B_- \times B_+ \times X \to P_- \backslash H / P_+$$

is essentially constant. Equivalently, there is one double coset $P_- w P_+$ such that

$$r(b_-, b_+, x) = P_- w P_+$$

for almost every (b_-, b_+, x).

Proof. The relative-position map is invariant under the diagonal Γ-action because both boundary maps are multiplied on the left by the same cocycle value. The target is countably separated by the standing convention. Double ergodicity therefore makes every invariant measurable map to that target essentially constant, giving a single double coset $P_{-}wP_{+}$. $\square$

Proposition 0.3 (Boundary pairs and double ergodicity: downstream consequence)**.** *If the constant double coset in Theorem 0.2 is the open Bruhat cell, the hypotheses of the reduction theorem in Chapter 3 are satisfied; hence the cocycle reduces to the stabilizer of a generic boundary pair.*

Proof. The open-cell assumption is exactly the "generic relative position" hypothesis in Theorem 0.2. Theorem 0.2 supplies constancy of that position. Applying Theorem 0.2 gives the stated stabilizer reduction. $\square$

CHAPTER 5

Comparison of shearing, entropy, spectral, drift, hyperbolic, and boundary mechanisms

This chapter owns the comparison of shearing, entropy, spectral, drift, hyperbolic, and boundary mechanisms step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Comparison of shearing, entropy, spectral, drift, hyperbolic, and boundary mechanisms: structural mechanism)**.** *Each major rigidity method in the series can be written as a two-stage mechanism: manufacture extra invariance/structure, then classify objects with that structure.*

PROOF. For each rigidity method, separate the argument into a production stage and a classification stage. Ratner's polynomial shearing produces new unipotent invariance and then invokes homogeneous-measure classification. Higher-rank diagonal dynamics uses entropy/leafwise conditionals to produce Haar directions and then invokes algebraic measure classification. Effective Ratner replaces exact extra invariance by quantitative almost invariance and then uses spectral gap.

The Benoist–Quint method starts from stationarity and recurrence: two-point drift/random shearing enlarges the stabilizer, after which stationary-measure classification forces homogeneity. Hyperbolic rigidity uses coarse Lyapunov foliations, cycle functionals, and normal forms to manufacture a rigid transfer map, then uses accessibility and regularity to globalize it. Boundary rigidity produces equivariant flag maps and generic relative position, after which algebraic hull reduction yields a homomorphism.

The methods are therefore analogous at the level of proof architecture but not interchangeable. Their recurrence inputs, classification theorems, and obstruction spaces differ. This distinction is essential when transporting an argument to a new setting: one must identify both the extra-structure mechanism and the theorem that classifies the resulting structure. □

THEOREM 0.2 (Comparison of shearing, entropy, spectral, drift, hyperbolic, and boundary mechanisms). *For each of the six methods listed below, let A_i denote its hypotheses, S_i the extra structure produced in the first stage, and C_i the classified conclusion:*

(1) Ratner: S_i = extra unipotent invariance; C_i = homogeneous measure or orbit.

(2) Diagonal entropy: S_i = Haar leafwise conditionals; C_i = algebraic invariant measure.

(3) Effective Ratner: S_i = quantitative almost invariance; C_i = effective equidistribution or a quantified obstruction.

(4) Benoist–Quint: S_i = stationary stabilizer growth; C_i = homogeneous stationary measure.

(5) Hyperbolic rigidity: S_i = cycle-free transfer map; C_i = global conjugacy or cohomology.

(6) Boundary rigidity: S_i = generic equivariant boundary pair; C_i = algebraic cocycle reduction.

Whenever the preceding volumes establish $A_i \Rightarrow S_i$ and $S_i \Rightarrow C_i$, the corresponding rigidity conclusion is the composition $A_i \Rightarrow C_i$. No implication between different rows is asserted.

PROOF. For each fixed row, transitivity of implication gives $A_i \Rightarrow S_i \Rightarrow C_i$, hence $A_i \Rightarrow C_i$. The second sentence is essential: the table records proof factorization, not equivalence of the structures S_i or of their hypotheses. For example, a spectral gap does not imply a boundary map, and boundary double ergodicity does not imply quantitative almost invariance. Thus the only valid deductions are the row-wise compositions already proved in the corresponding volumes. □

Proposition 0.3 (Comparison of shearing, entropy, spectral, drift, hyperbolic, and boundary mechanisms: downstream consequence)**.** *Transporting one method to a new setting requires two separate verifications: the production implication $A_i \Rightarrow S_i$ and the classification implication $S_i \Rightarrow C_i$. Verifying only one of them does not establish C_i.*

PROOF. This is immediate from the factorization in Theorem 0.2: the conclusion uses both arrows in the chain. If either arrow is unavailable, the composed implication has not been proved. □

CHAPTER 6

Unified extra-structure-to-algebraicity principles

This chapter owns the unified extra-structure-to-algebraicity principles step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Unified extra-structure-to-algebraicity principles: structural mechanism)**.** *Suppose a rigidity problem admits (i) recurrence and nonconcentration, (ii) a mechanism producing a nontrivial stabilizer or equivariant algebraic object from failure of equidistribution, and (iii) a Noetherian/classification theorem for stabilizers. Then an induction on algebraic complexity yields an algebraic classification provided the induction cannot cycle.*

Proof. At a nonalgebraic candidate, use (ii) to enlarge the stabilizer or reduce the algebraic hull. Noetherianity makes strict enlargement/reduction terminate. Recurrence prevents escape from the class under consideration. At the terminal object, (iii) gives the algebraic classification. The quantitative statement follows only when the number and cost of steps are uniformly bounded.

□

Theorem 0.2 (Unified extra-structure-to-algebraicity principles). *Let $\mathcal{C}$ be a class of objects with an integer complexity rank $r : \mathcal{C} \to \{0, 1, \dots, N\}$. Assume:*

(1) every object with $r = 0$ belongs to a classified algebraic terminal class;

(2) for every unclassified x with $r(x) > 0$, the recurrence and nonconcentration argument constructs $F(x) \in \mathcal{C}$ satisfying $r(F(x)) < r(x)$;

(3) *the construction preserves the hypotheses needed to repeat the argument.*

Then after at most N iterations every starting object reaches the classified terminal class. If, in addition, a height $H_j \geq 2$ satisfies

$$H_{j+1} \leq H_j^A$$

with a fixed A at every step, then the terminal height obeys $H_N \leq H_0^{A^N} = H_0^{O(1)}$.

Proof. The rank is a nonnegative integer and decreases strictly at every nonterminal step, so no orbit of F has more than N strict decreases. The process therefore terminates at rank 0, where hypothesis (1) gives the classification. For the quantitative statement, iterate the height inequality: $H_1 \leq H_0^A$, $H_2 \leq H_0^{A^2}$, and by induction $H_j \leq H_0^{A^j}$. Since $j \leq N$ and N, A are fixed independently of H_0, the terminal height is a fixed power of H_0. □

Proposition 0.3 (Unified extra-structure-to-algebraicity principles: downstream consequence)**.** *A claim of polynomial effective complexity is justified by this induction only when both the number of strict complexity drops N and the one-step exponent A are uniform over the family under consideration.*

Proof. If either N or A depends on the initial height, the exponent A^N need not be bounded independently of H_0, so $H_0^{A^N}$ need not be polynomial with a fixed exponent. Conversely, uniform A, N give the polynomial bound in Theorem 0.2. □

CHAPTER 7

Rigidity failure modes

This chapter owns the rigidity failure modes step in the stated scope and prerequisites. We first isolate the quantitative or structural lemma, then prove the chapter theorem, and finally record the exact downstream interface. The scope is deliberately no stronger than the hypotheses stated below.

Proof architecture. The argument is organized so that every appeal to an earlier volume is an explicit earlier result: reduce to controlled coordinates, isolate the obstruction, prove the quantitative estimate, and only then pass to the downstream form.

Lemma 0.1 (Rigidity failure modes: structural mechanism)**.** *Rank-one factors, compact centralizers, escape of mass, weak spectral gap, resonant center directions, and insufficient boundary genericity give distinct failure modes for the six rigidity mechanisms.*

Proof. Each failure mode is witnessed by the corresponding broken step in Chapter 5's dictionary: no transverse shear, zero entropy, no dimension growth, stationary mass on a finite orbit, central neutral foliation, or boundary map contained in a proper algebraic stratum. This proves the logical failure of the method without claiming a counterexample to every hoped-for theorem. □

Theorem 0.2 (Rigidity failure modes). *The following are genuine failures of specific proof implications, not automatically counterexamples to the desired conclusions.*

1. *If $\psi = c\chi$ on a one-dimensional acting parameter space, then $|\psi(a)|/|\chi(a)| = |c|$ for all a with $\chi(a) \neq 0$, so the higher-rank sign-separation step of Volume 68 cannot be performed.*
2. *If a generated subgroup H has a positive-dimensional centralizer quotient Z and the averaging operator acts only in the H-directions, every nonconstant function pulled back from Z is fixed; hence no spectral gap on the full space follows from the H-average alone.*
3. *If a sequence of probability measures is not tight, a weak-* limit in the one-point compactification may lose positive mass, so a probability-measure classification theorem cannot be applied without nonescape.*

(4) *If the boundary relative-position map is supported in a proper algebraic stratum rather than the open orbit, the generic-pair reduction of Chapter 3 does not apply.*

Proof. Item (1) is the one-dimensional linear-algebra calculation already used in Volume 68. For item (2), if $q : X \to Z$ is the centralizer quotient and $f = F \circ q$, then $f(hx) = f(x)$ for $h \in H$, so every H-averaging operator fixes f and has eigenvalue 1 on a nonconstant subspace. Item (3) is the definition of failure of tightness: there is $\varepsilon > 0$ such that every compact set misses at least ε mass along a subsequence; a weak limit on the original space then has mass at most $1 - \varepsilon$. Item (4) follows because the measurable conjugating element used in Theorem 0.2 is constructed only on the open double coset. These calculations identify precisely which implication fails in each method. ☐

Proposition 0.3 (Rigidity failure modes: downstream consequence)**.** *Whenever one of the four failures in Theorem 0.2 occurs, the manuscript must record the corresponding missing hypothesis explicitly; it may not replace the failed implication by the desired conclusion.*

Proof. Each item in the theorem exhibits a concrete step that is unavailable: sign separation, spectral contraction, probability preservation, or generic boundary reduction. Therefore any theorem depending on that step requires an additional hypothesis or a different argument. ☐

CHAPTER 8

Final synthesis: boundary reduction plus strong averaging

This chapter records the final implication in object-level form. The boundary argument of Chapter 3 reduces a measurable cocycle to an algebraic homomorphism times a compact-by-central cocycle; the strong-property-(T) argument of Volume 69 then turns subexponential derivative growth into an invariant metric.

Lemma 0.1 (Compact-by-central reduction and derivative growth)**.** *Let Γ act by C^1 diffeomorphisms on a compact manifold M, and let*

$$\alpha_D(\gamma, x) = D_x\gamma$$

be the derivative cocycle after a measurable trivialization of TM. Suppose that for some measurable $b : M \to H$ one has

$$b(\gamma x)^{-1}\alpha_D(\gamma, x)b(x) = \rho(\gamma)k(\gamma, x),$$

where $\rho : \Gamma \to R$ is a homomorphism, k takes values in a compact-by-central subgroup $C < H$, and $[R, C] = e$. If the R-representation on the relevant tangent module has no noncompact factor and every invariant suspension measure has nonpositive top fiber Lyapunov exponent, then for every $\varepsilon > 0$ there is C_ε such that

$$\|D\gamma\| + \|D\gamma^{-1}\| \leq C_\varepsilon e^{\varepsilon|\gamma|} \qquad (\gamma \in \Gamma).$$

Proof. The boundary reduction gives the displayed cohomologous form by Theorem 0.2. The compact-by-central factor contributes zero exponential growth on every finite-dimensional quotient. The noncompact part is carried by ρ; by hypothesis it has no noncompact tangent factor. Hence every invariant suspension measure has top fiber exponent at most zero. Applying the subadditive derivative-growth theorem of Volume 69, Chapters 5–6, to both $D\gamma$ and $D\gamma^{-1}$ yields, for each $\varepsilon > 0$,

$$\sup_{x\in M} \log \|D_x\gamma\| \leq \varepsilon|\gamma| + O_\varepsilon(1), \qquad \sup_{x\in M} \log \|D_x\gamma^{-1}\| \leq \varepsilon|\gamma| + O_\varepsilon(1).$$

Exponentiating gives the stated estimate. □

THEOREM 0.2 (Boundary-to-finite-image synthesis). *Let Γ be either a cocompact lattice in a connected simple real Lie group of real rank at least two or $\Gamma = \mathrm{SL}_m(\mathbb{Z})$, $m \geq 3$, in the nonuniform scope of Theorem 0.4. Let $\Gamma \curvearrowright M$ be a C^1 action on a compact manifold satisfying the low-dimensional hypotheses of Volume 69. Assume:*

(1) the derivative cocycle admits the boundary reduction of Theorem 0.2;
(2) its algebraic factor has no noncompact tangent representation in the allowed dimension;
(3) every invariant suspension measure has nonpositive top fiber Lyapunov exponent; and
(4) the dimension restriction of Volume 69, Chapter 7 excludes a positive-dimensional compact image.

Then the action preserves a Riemannian metric and its image in $\mathrm{Diff}^1(M)$ *is finite.*

PROOF. Lemma 0.1 gives uniform subexponential growth of derivatives. Choose a Sobolev exponent above the embedding threshold and let ρ_M be the pullback representation on the corresponding Hilbert completion of symmetric two-tensors. The product rule converts the derivative estimate into

$$\|\rho_M(\gamma)\| \leq C_\varepsilon e^{C\varepsilon|\gamma|}.$$

Choose ε below the strong-property-(T) growth threshold in Lemma 0.3. The averaging measures there converge exponentially to the projection onto invariant tensors, so the averages of a smooth background metric converge in Sobolev norm, hence uniformly, to an invariant positive-definite tensor g_∞. Thus Γ acts by isometries of (M, g_∞).

The closure of the image is therefore a compact Lie subgroup of $\mathrm{Isom}(M, g_\infty)$. By hypothesis (4), its identity component is trivial. A compact Lie group with trivial identity component is finite, so the image of Γ is finite. $\square$

Proposition 0.3 (Why the boundary reduction alone is insufficient)**.** *Under the hypotheses of Theorem 0.2 alone, one obtains a cohomologous decomposition $\alpha^b = \rho k$ but not, in general, an invariant Riemannian metric. The metric conclusion additionally requires a quantitative averaging statement applicable to representations with subexponential operator-norm growth, such as Lemma 0.3.*

PROOF. The proof of Theorem 0.2 is algebraic: it moves an equivariant boundary pair to a reference pair and identifies the stabilizer. It constructs no invariant vector in the Hilbert space of Sobolev metrics. In contrast, Lemma 0.3 supplies probability measures m_n satisfying

$$\|\rho_M(m_n) - P_{\rho_M}\| \leq Ce^{-bn}$$

for representations with sufficiently small exponential growth. Applying these averages to a background metric is the step that produces g_∞. Hence the boundary reduction and the invariant-metric argument are logically distinct. $\square$

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